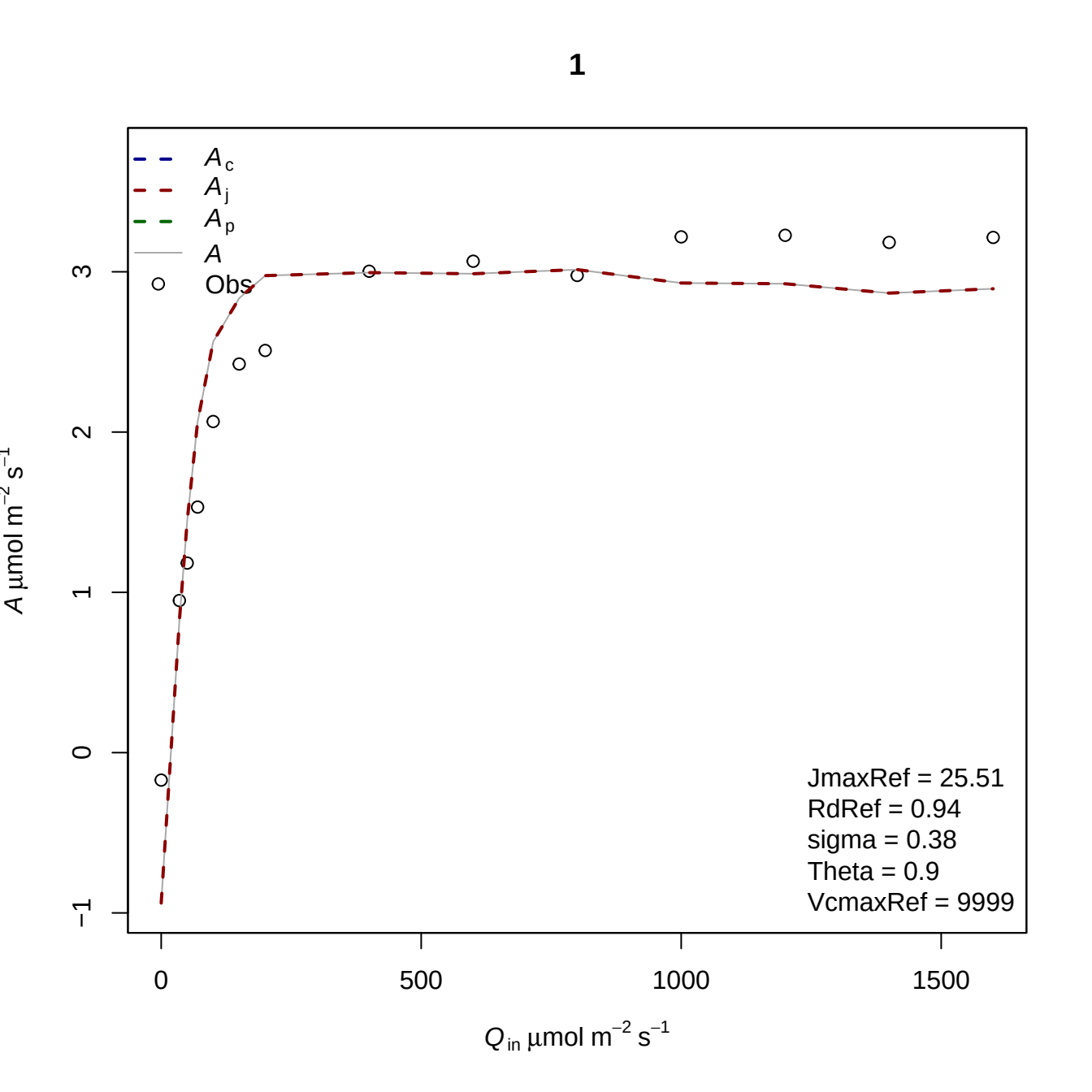
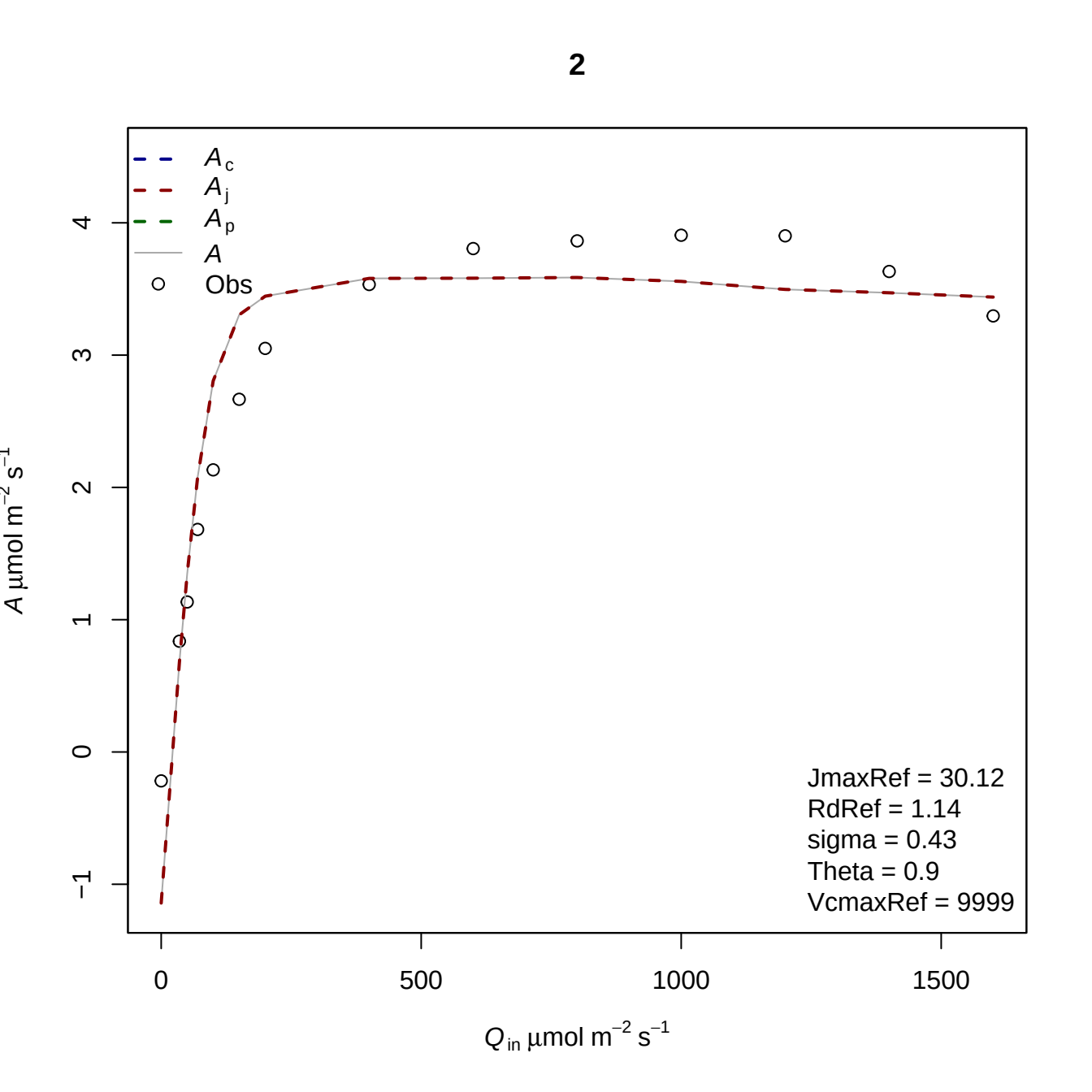
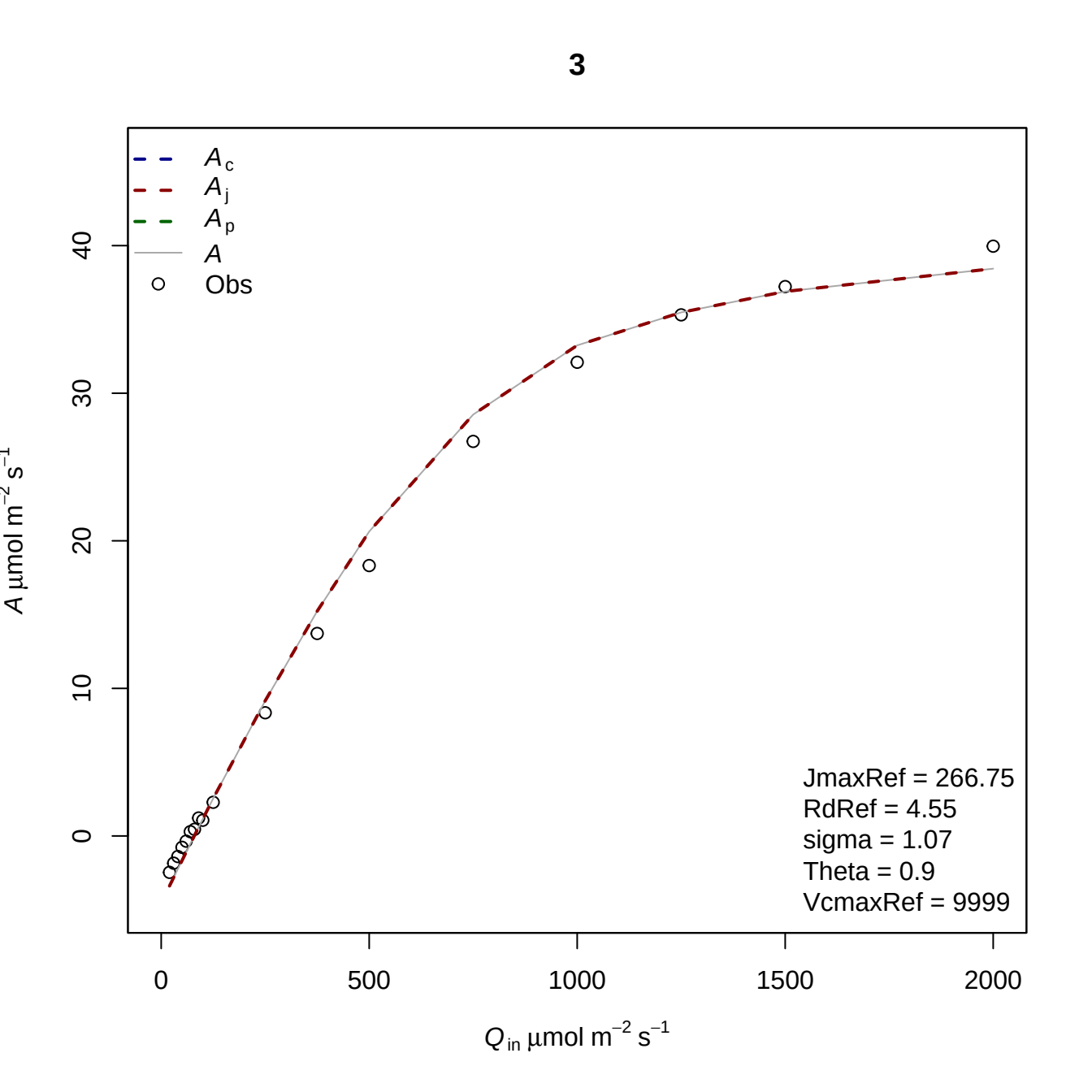
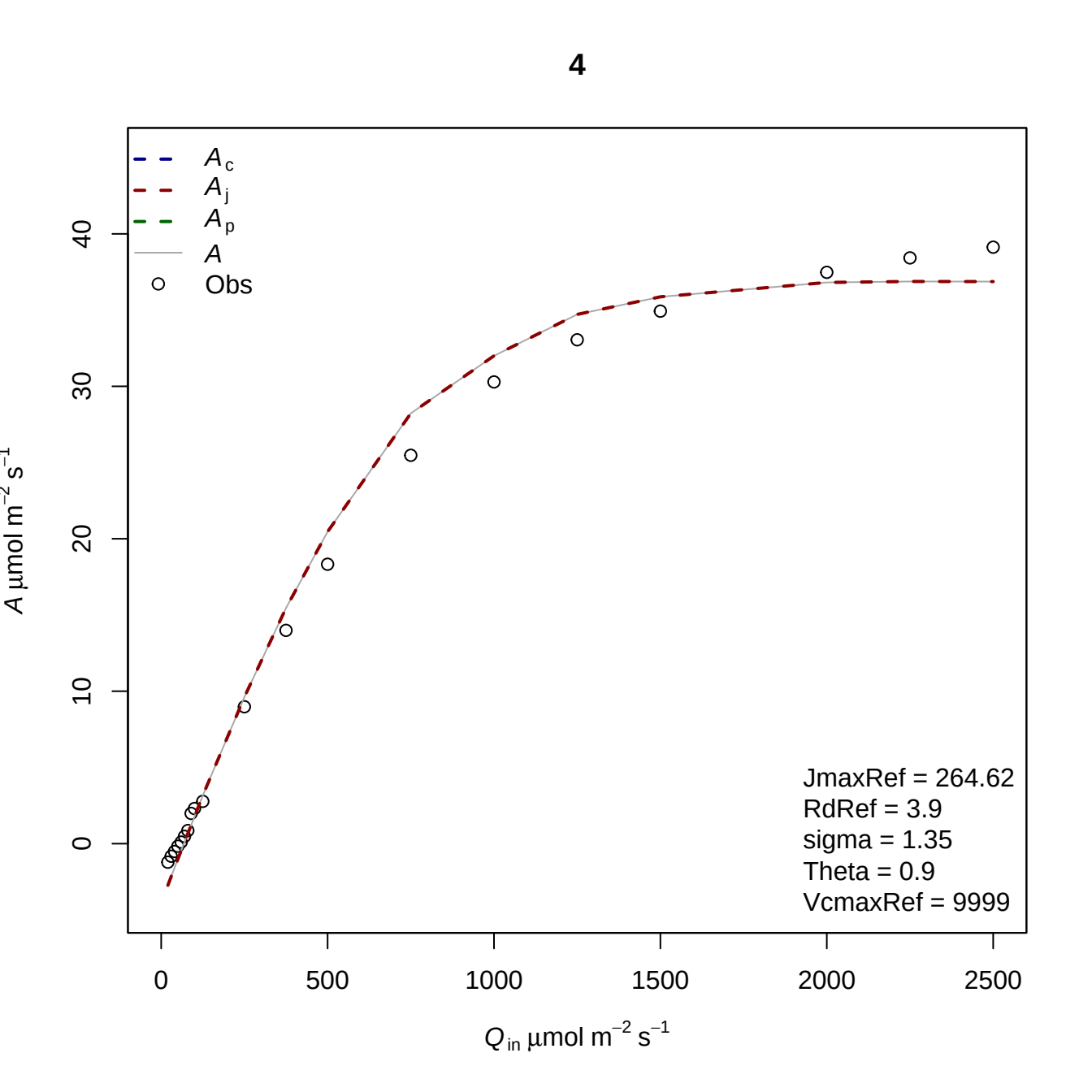
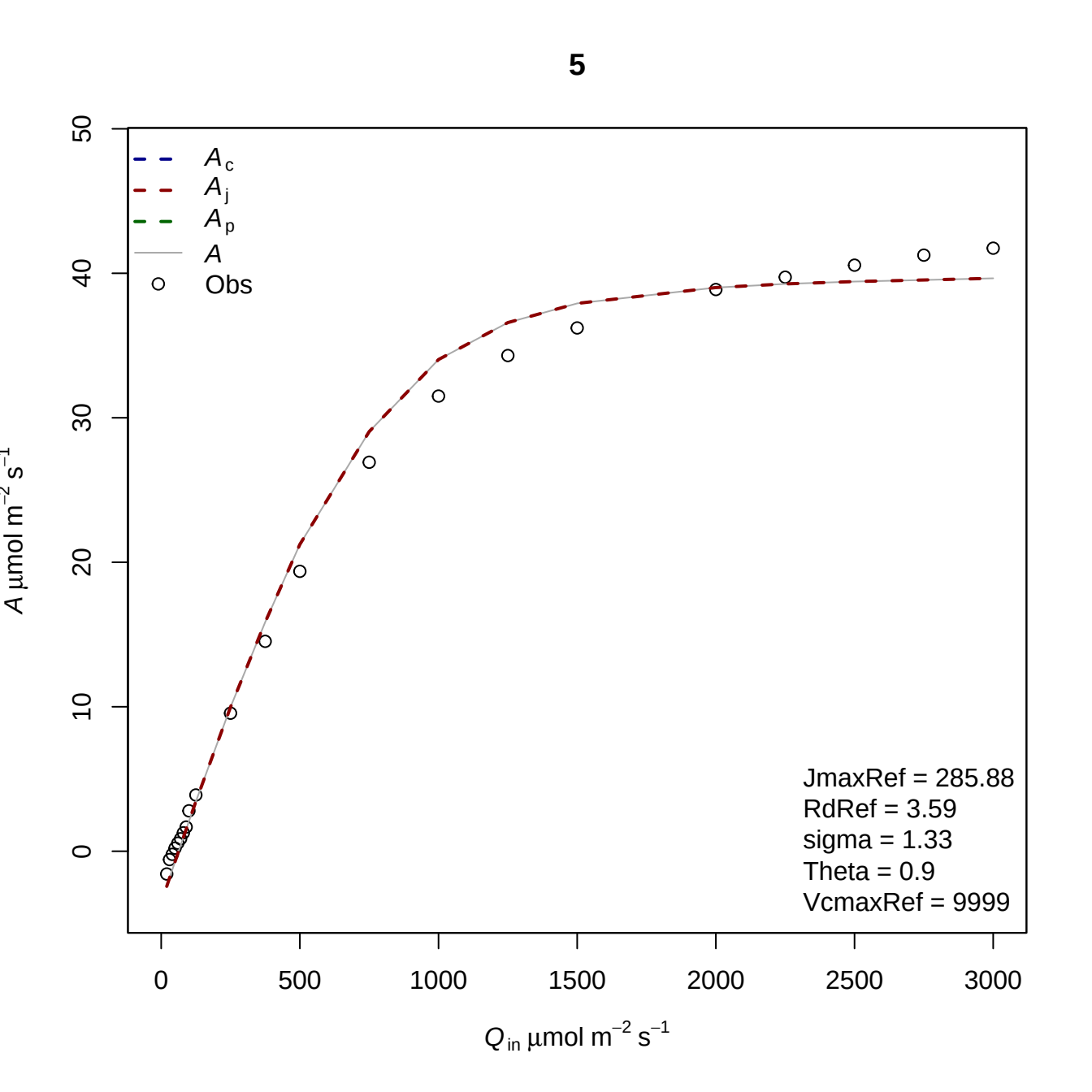
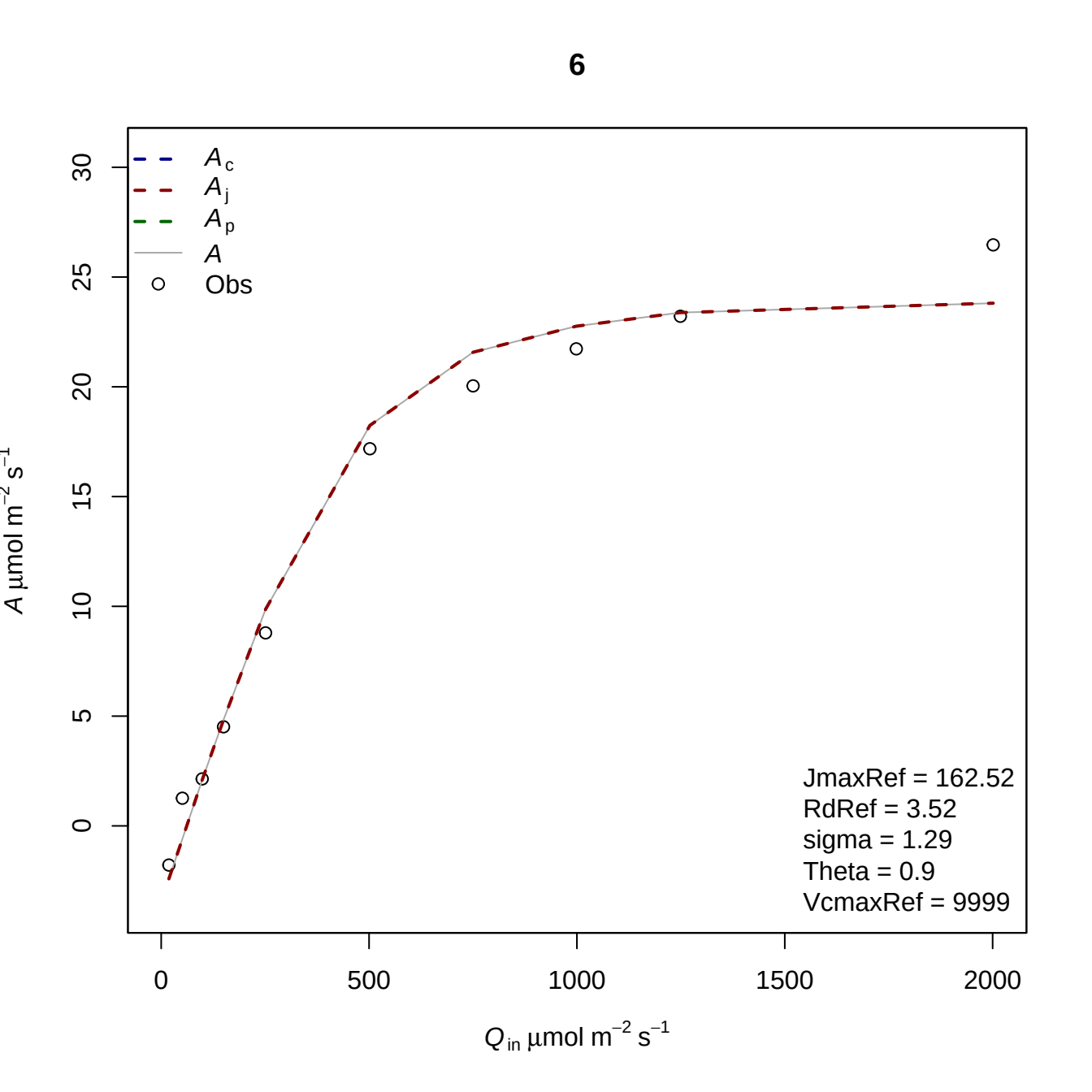

1
A_c
A_j
A_p
A
Obs
JmaxRef = 25.51
RdRef = 0.94
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

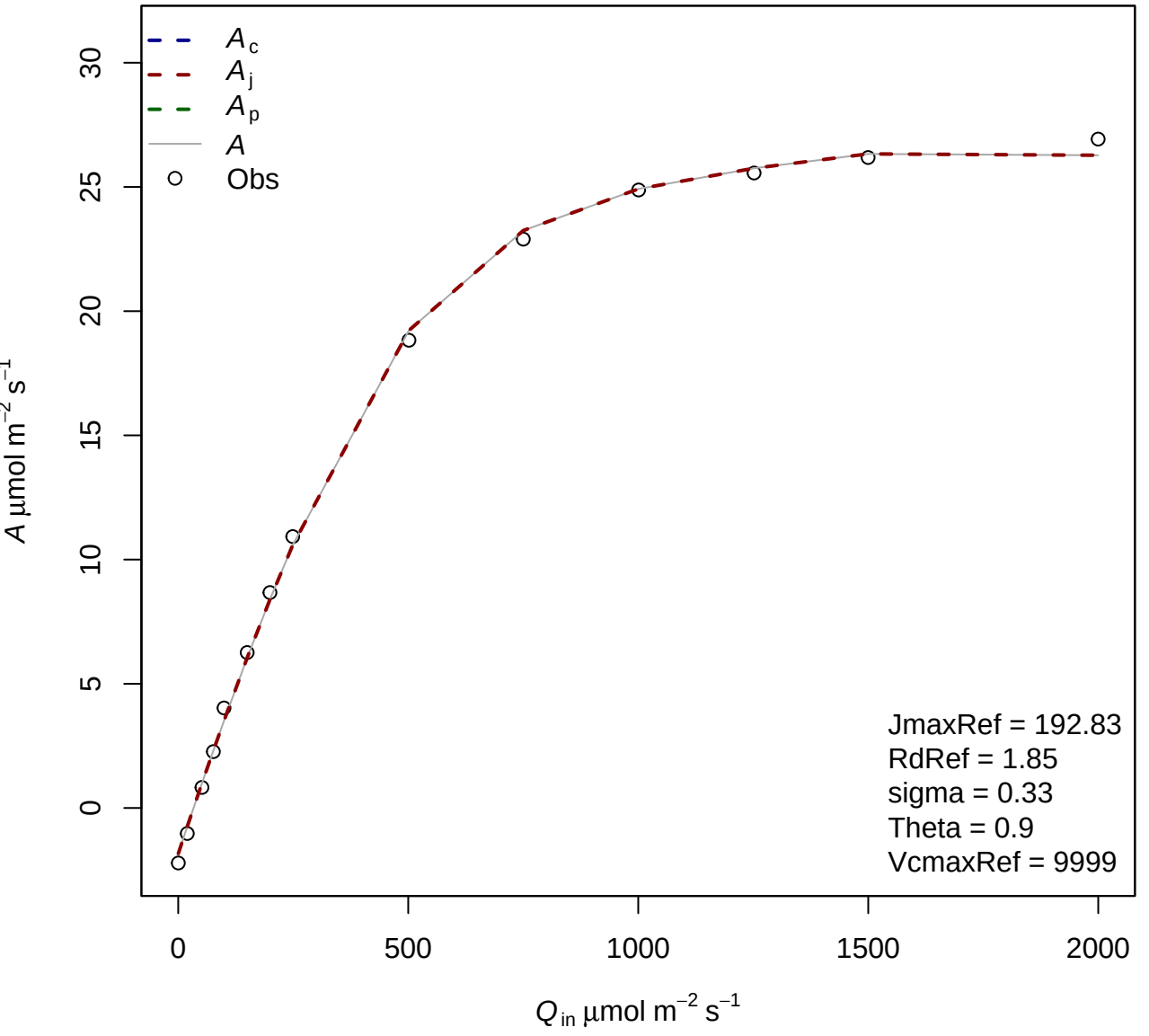

2
A_c
A_j
A_p
A
Obs
JmaxRef = 30.12
RdRef = 1.14
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

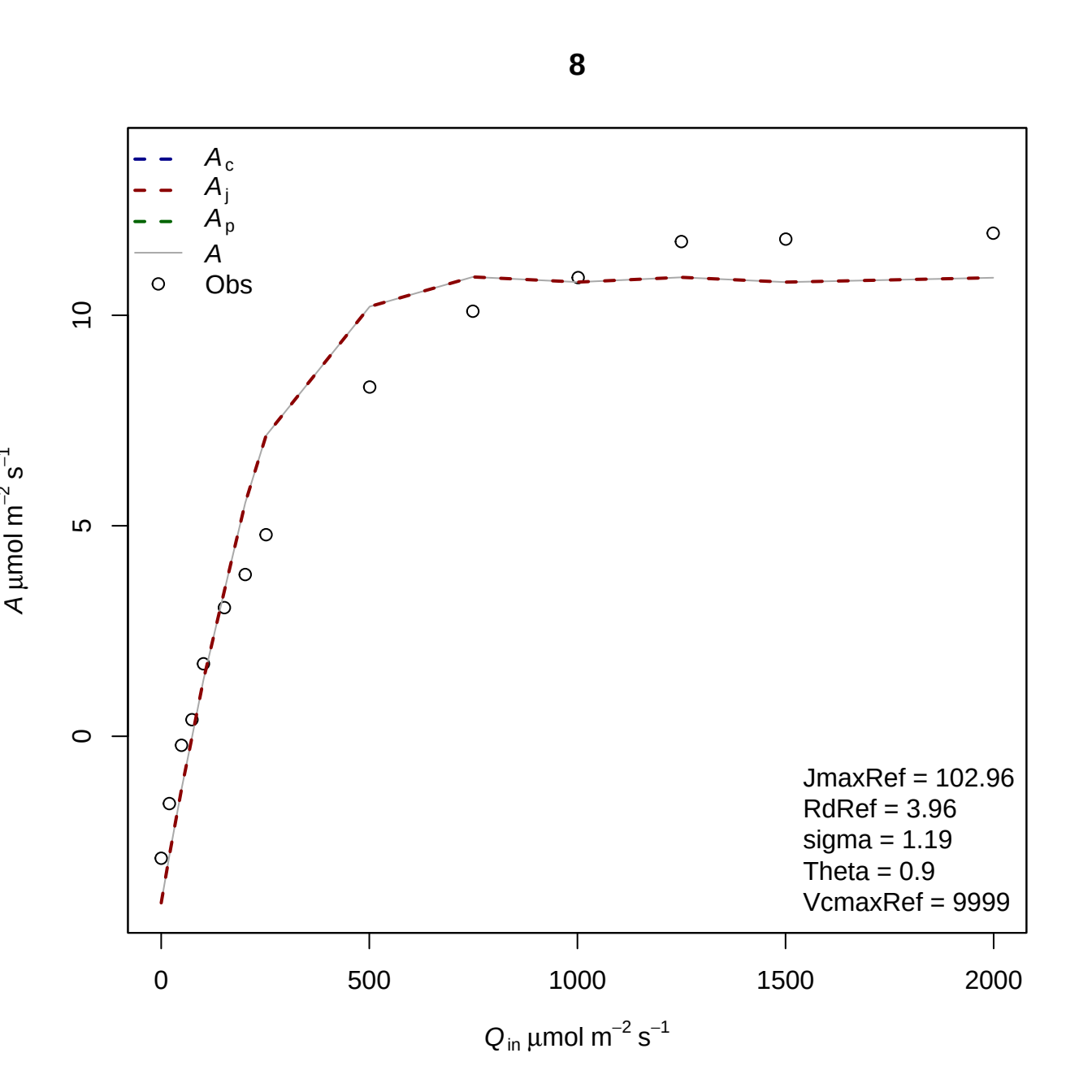

3
A_c
A_j
A_p
A
Obs
JmaxRef = 266.75
RdRef = 4.55
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

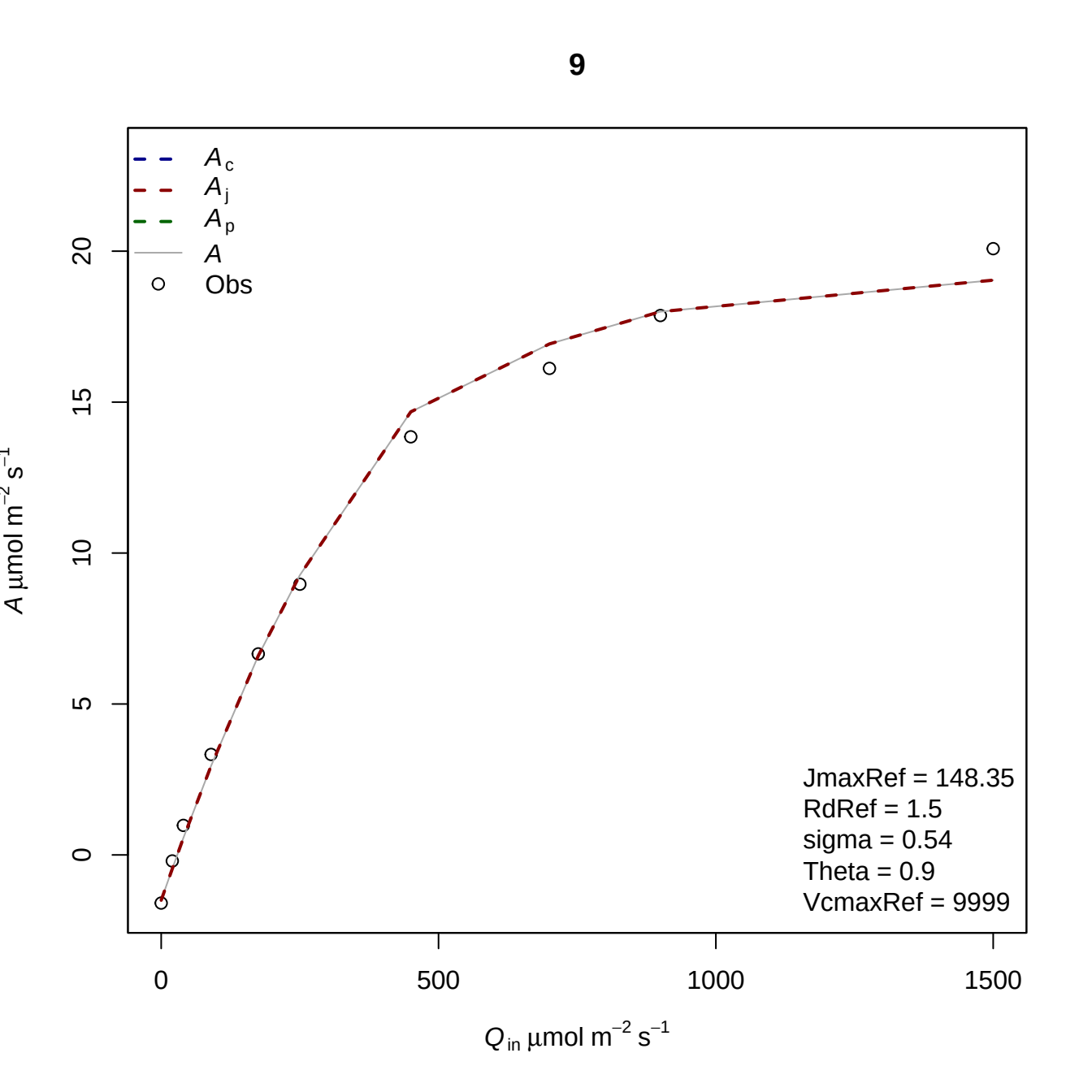

4
A_c
A_j
A_p
A
Obs
JmaxRef = 264.62
RdRef = 3.9
sigma = 1.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

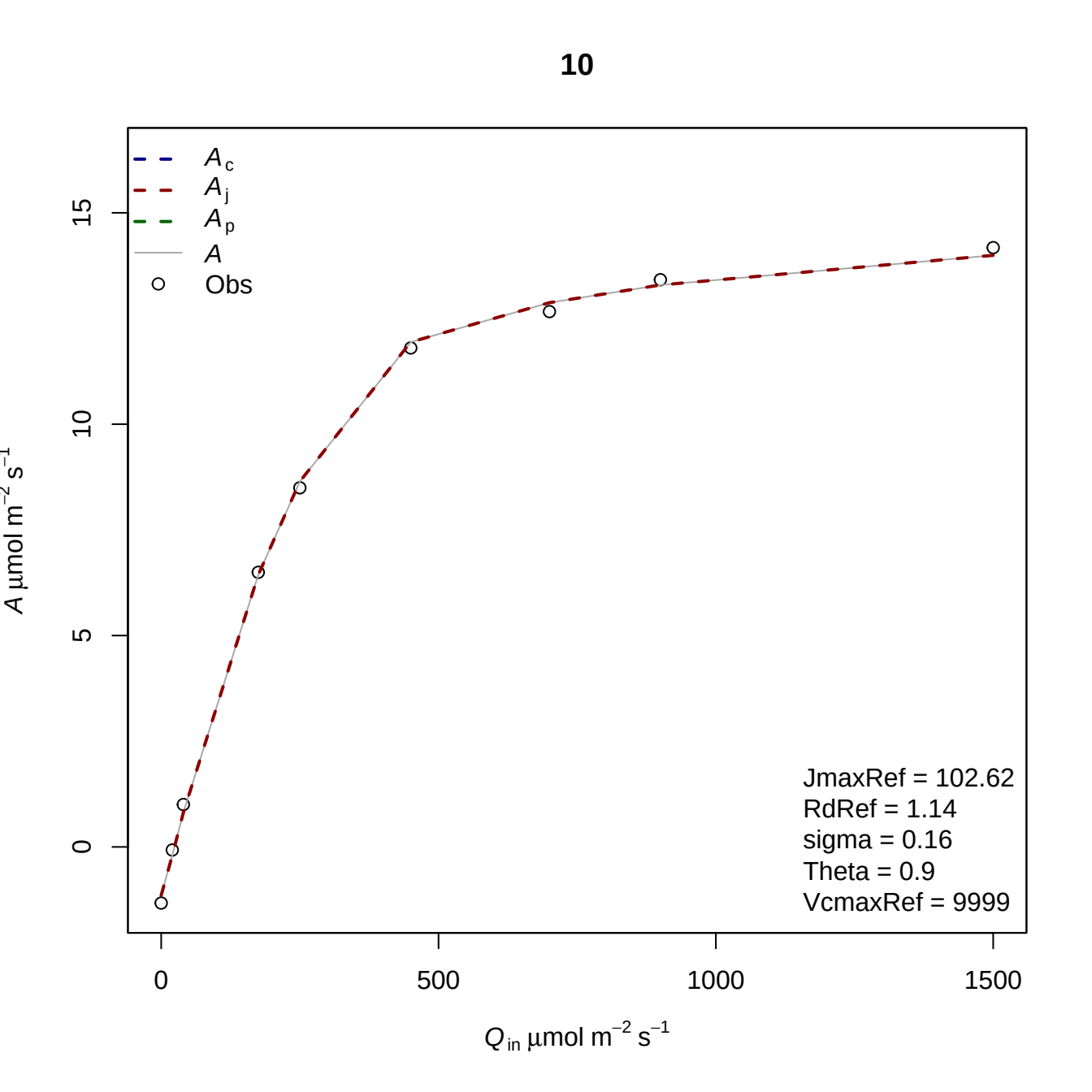

5
A_c
A_j
A_p
A
Obs
JmaxRef = 285.88
RdRef = 3.59
sigma = 1.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

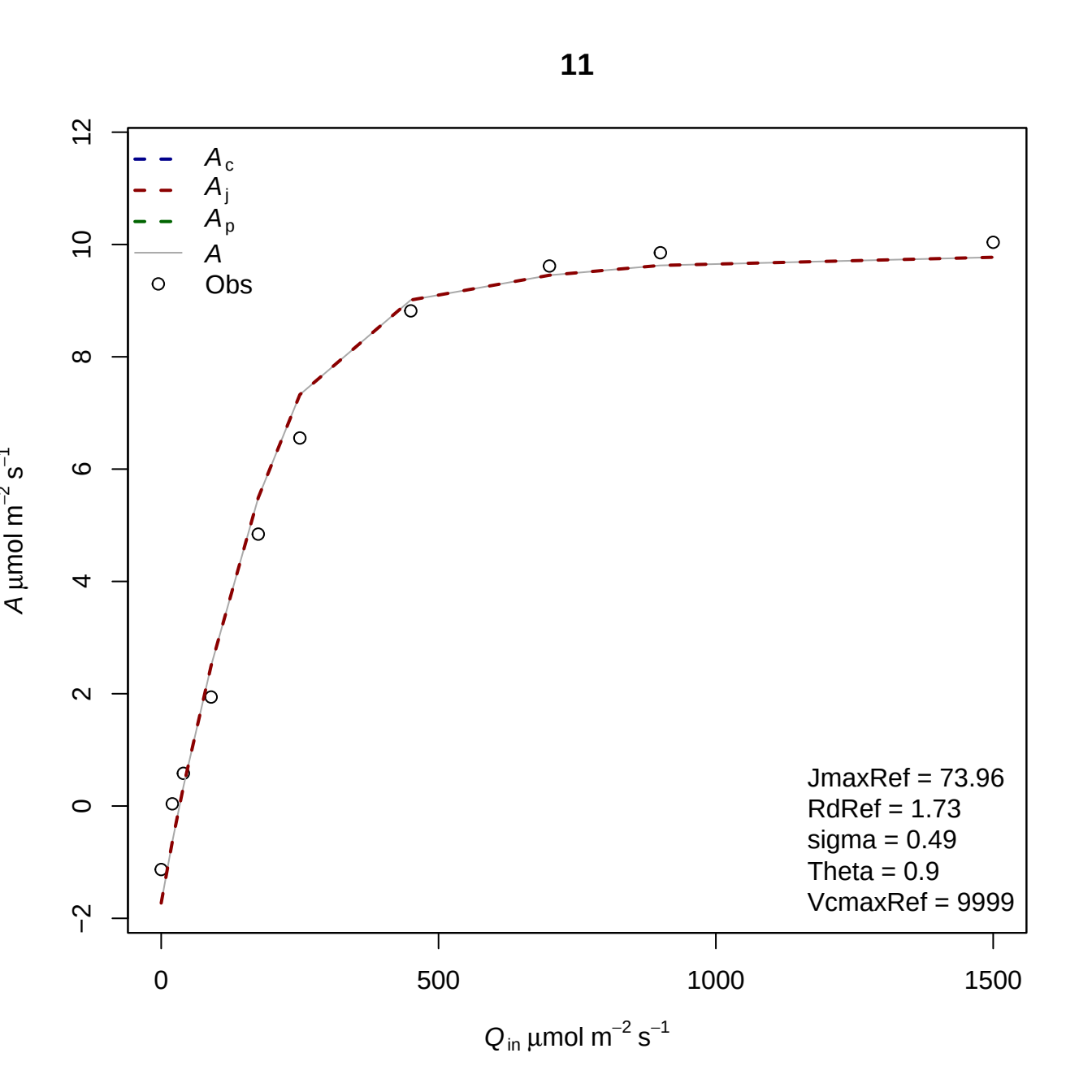

6
A_c
A_j
A_p
A
Obs
JmaxRef = 162.52
RdRef = 3.52
sigma = 1.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

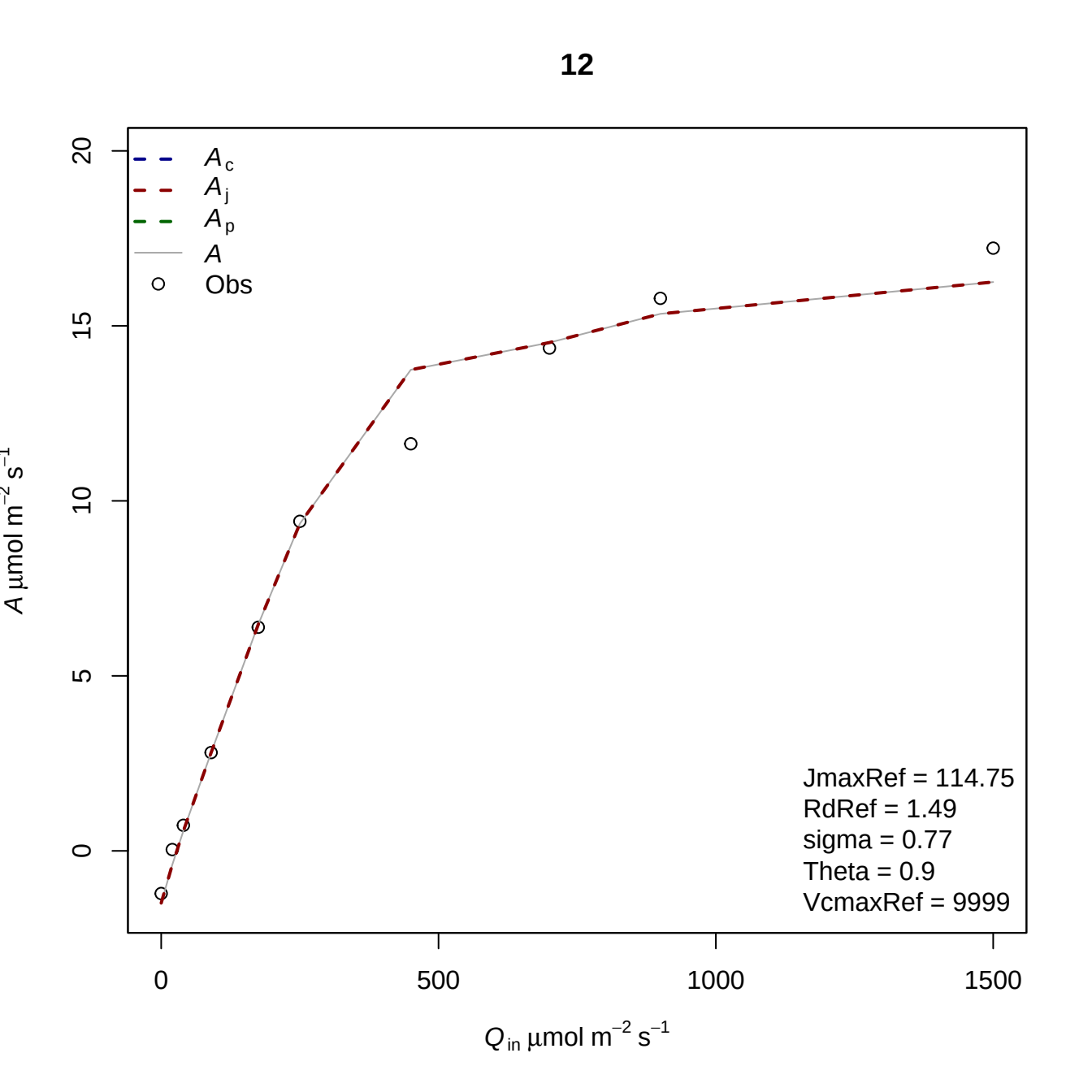

A_c
A_j
A_p
A
Obs
JmaxRef = 192.83
RdRef = 1.85
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

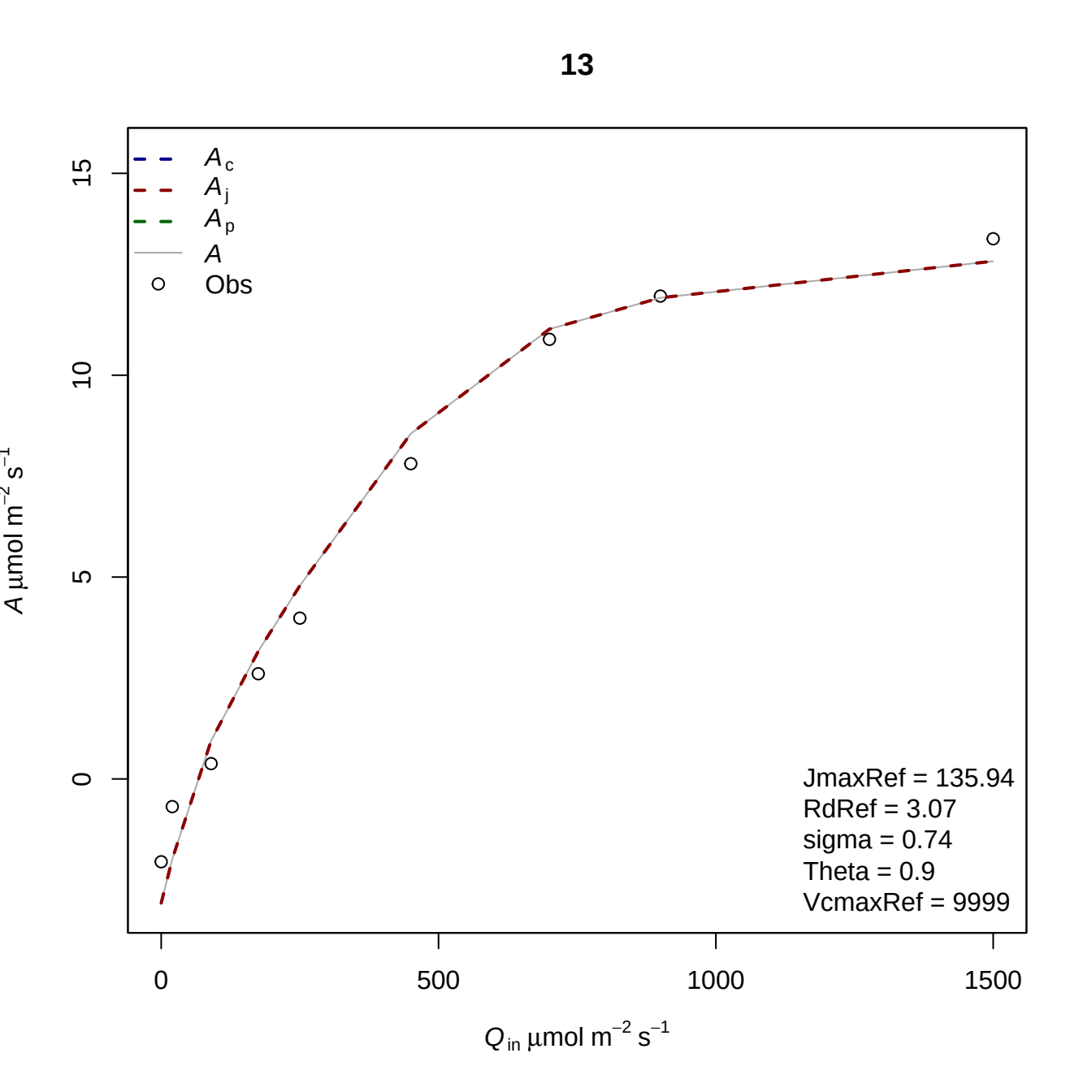

8
A_c
A_j
A_p
A
Obs
JmaxRef = 102.96
RdRef = 3.96
sigma = 1.19
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

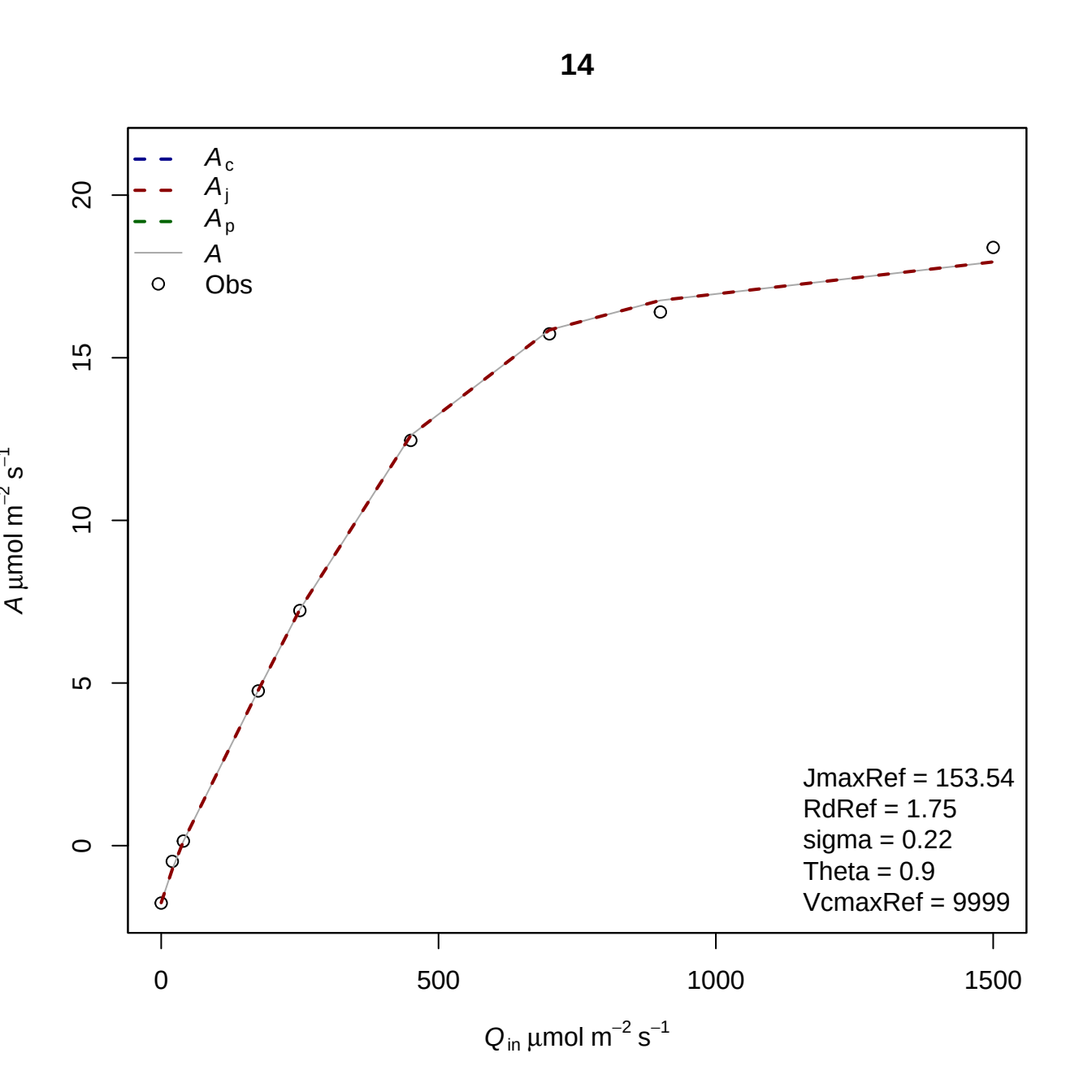

9
A_c
A_j
A_p
A
Obs
JmaxRef = 148.35
RdRef = 1.5
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

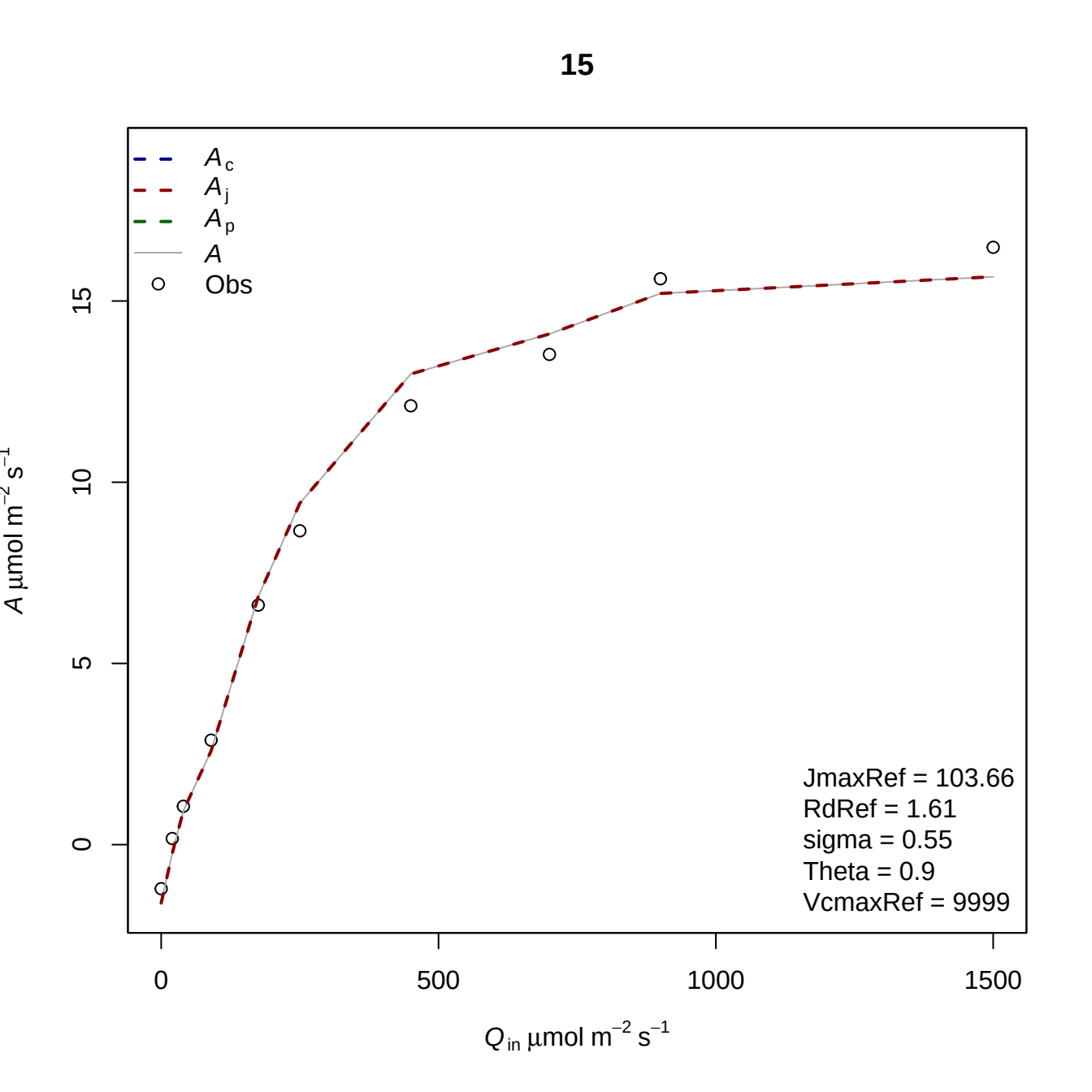

10
A_c
A_j
A_p
A
Obs
JmaxRef = 102.62
RdRef = 1.14
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

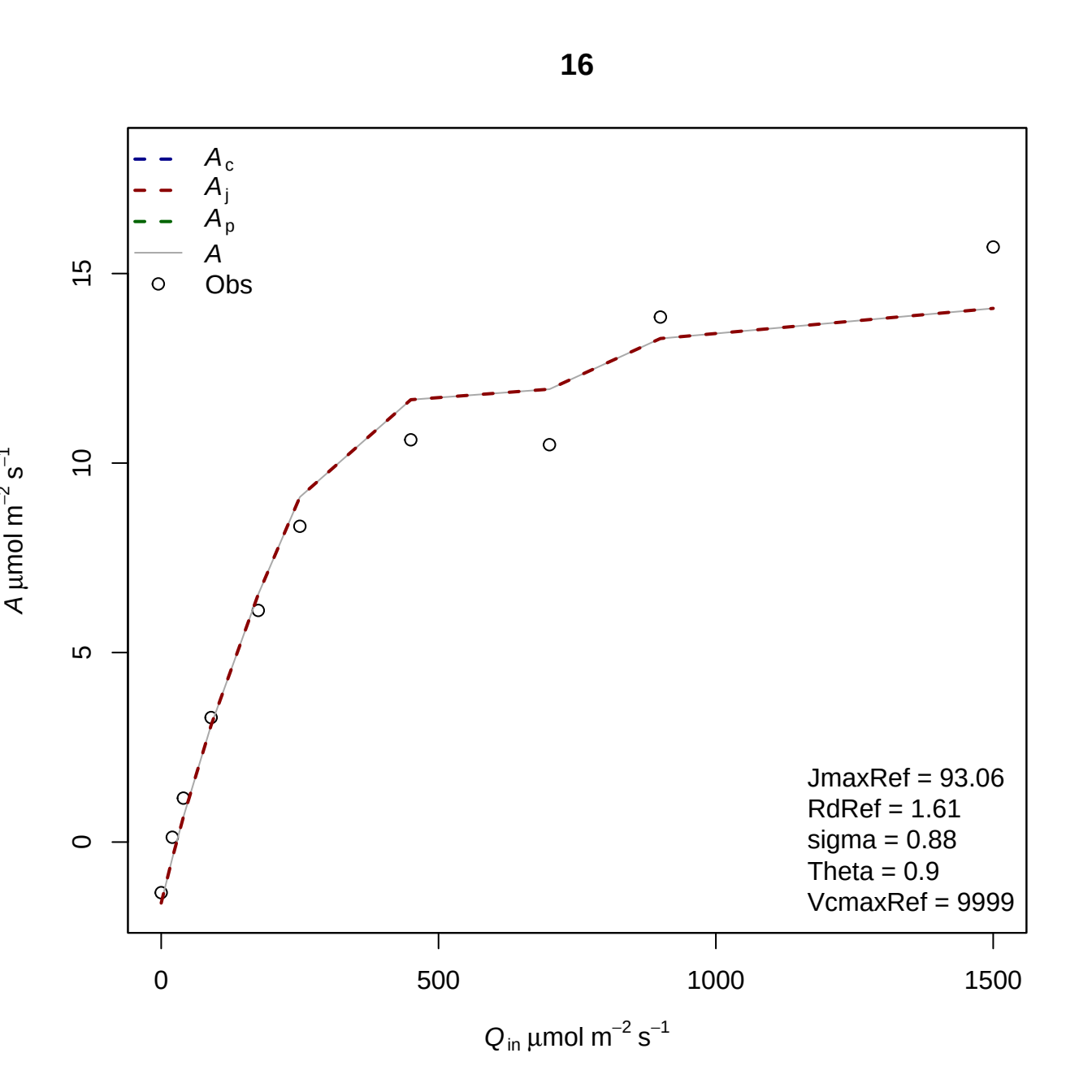

11
A_c
A_j
A_p
A
Obs
JmaxRef = 73.96
RdRef = 1.73
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

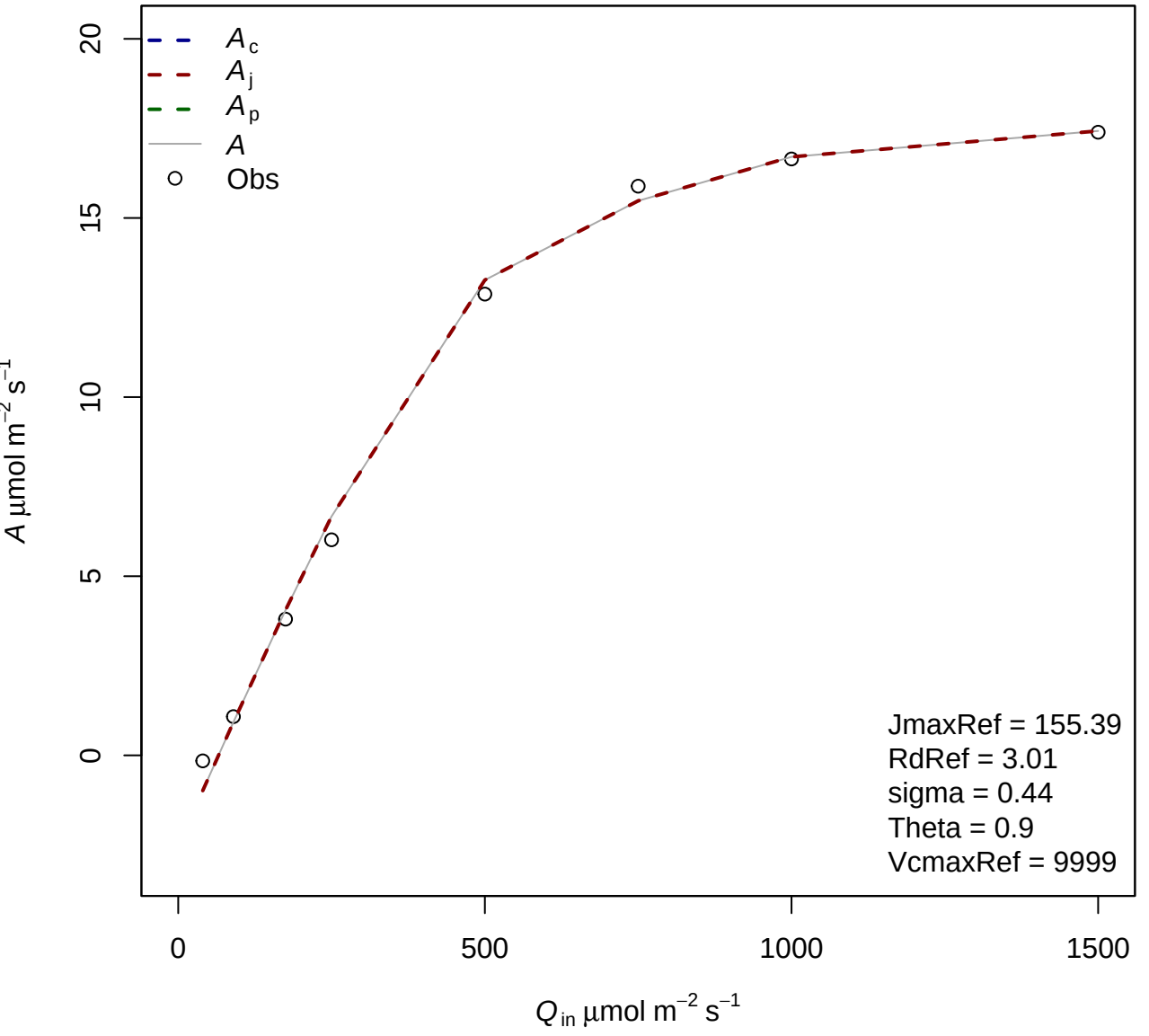

12
A_c
A_j
A_p
A
Obs
JmaxRef = 114.75
RdRef = 1.49
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

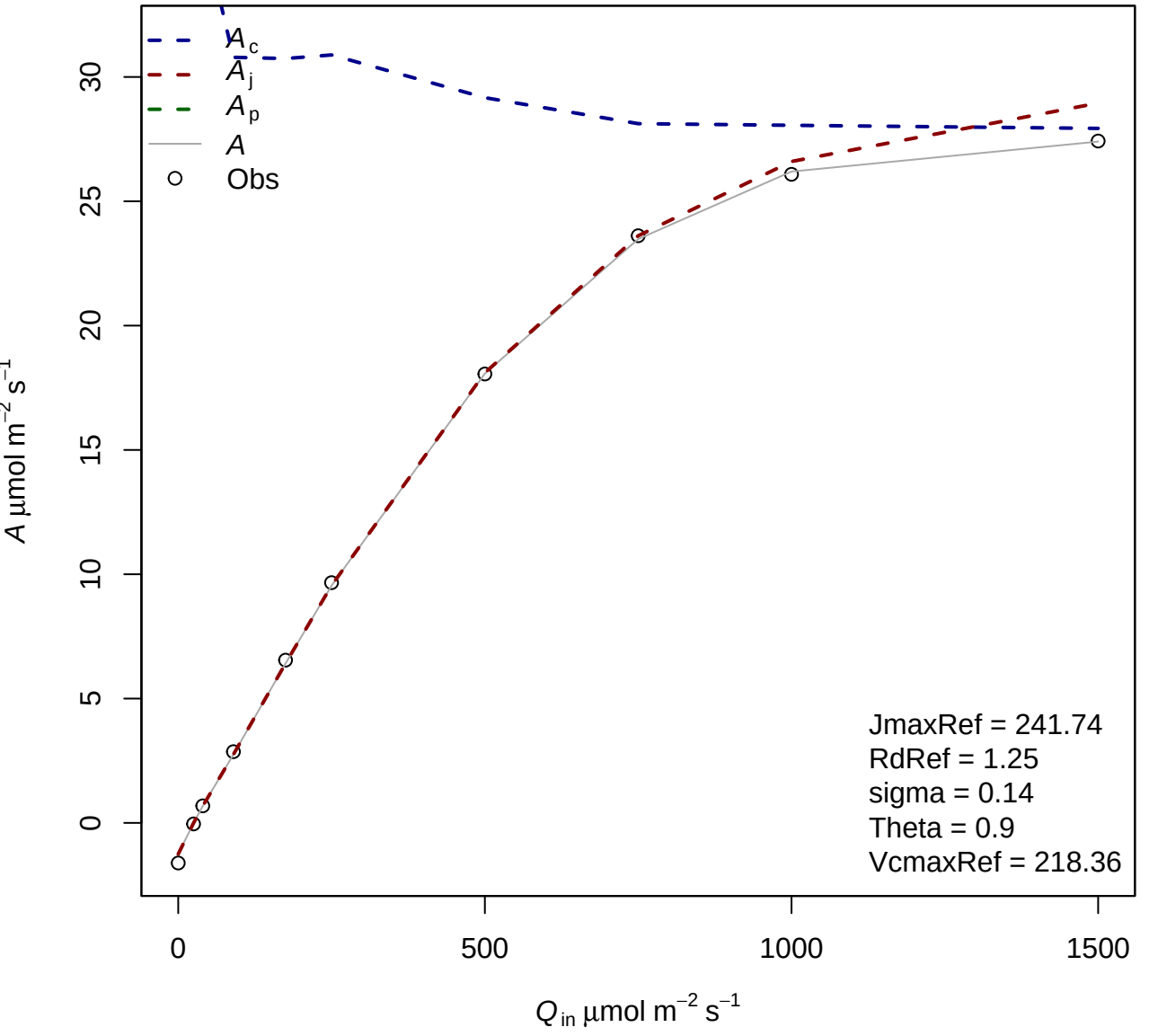

13
A_c
A_j
A_p
A
Obs
JmaxRef = 135.94
RdRef = 3.07
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

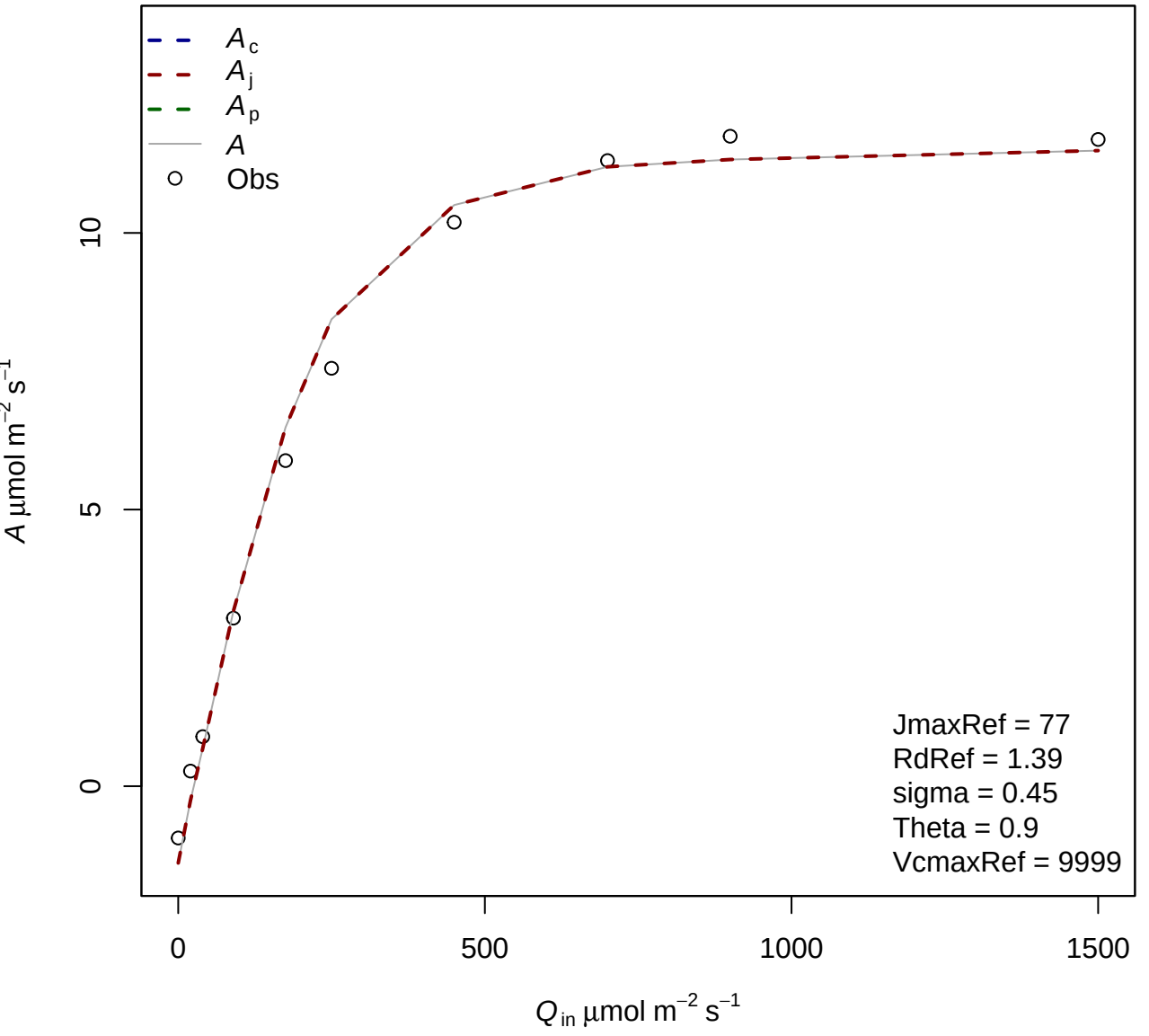

14
A_c
A_j
A_p
A
Obs
JmaxRef = 153.54
RdRef = 1.75
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

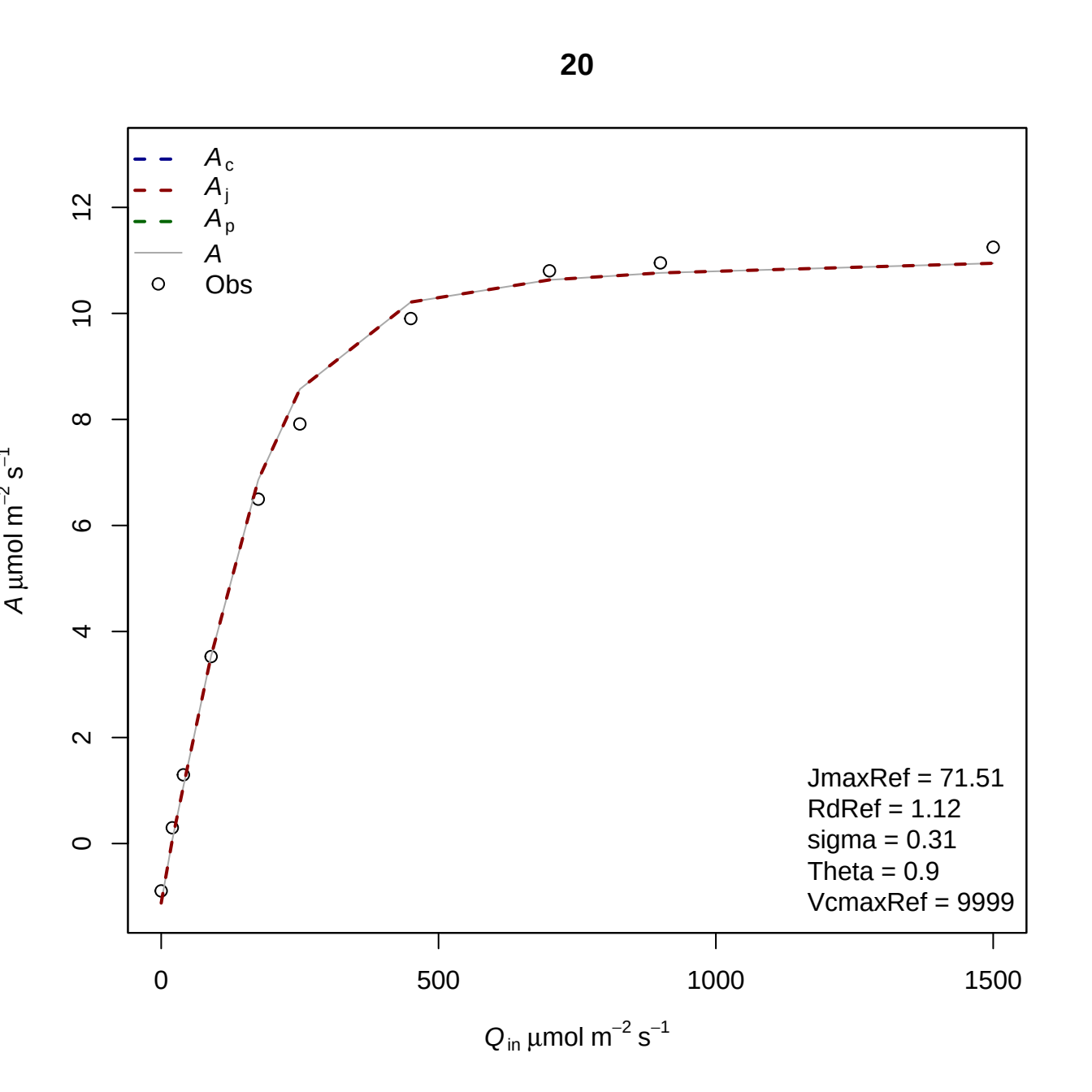

15
A_c
A_j
A_p
A
Obs
JmaxRef = 103.66
RdRef = 1.61
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

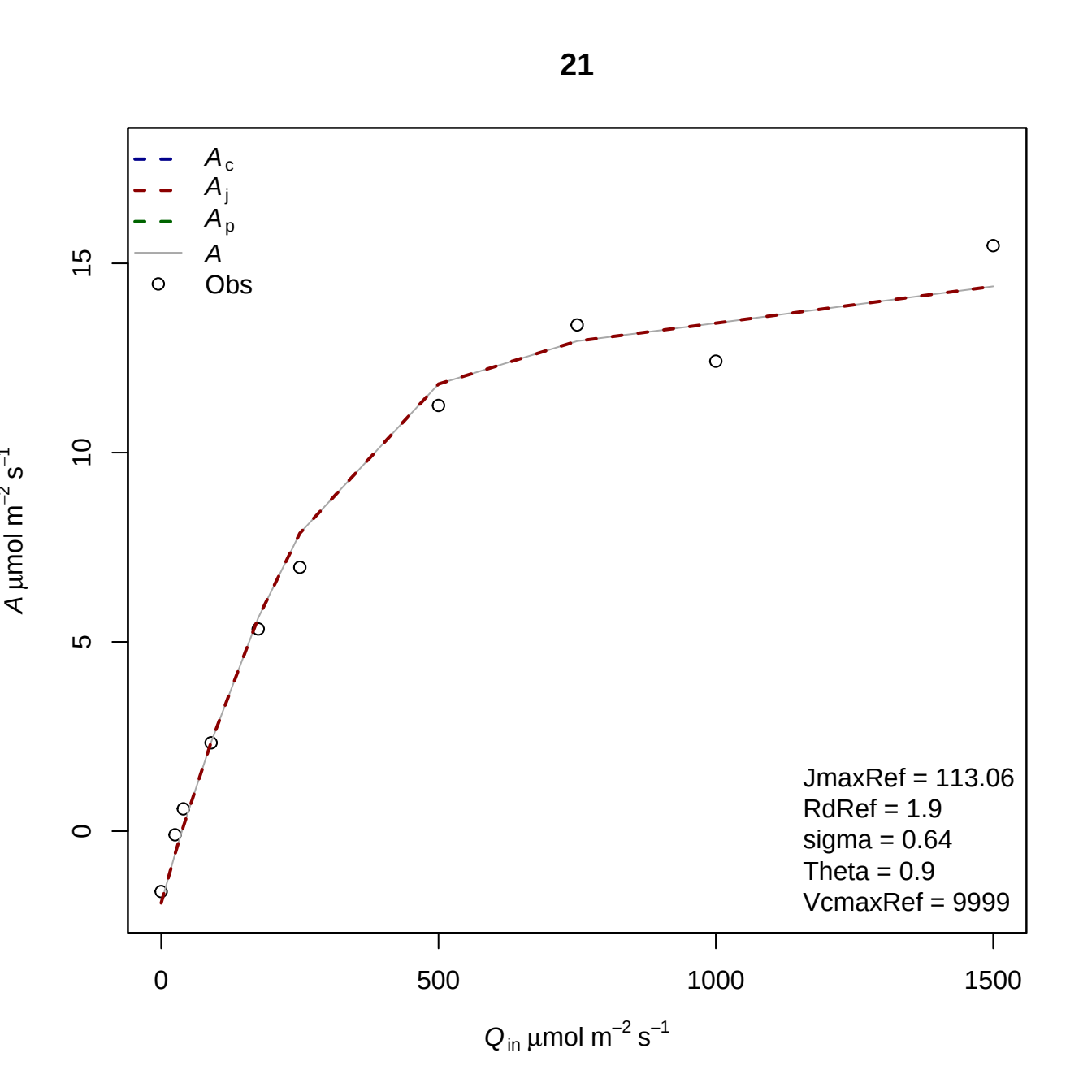

16
A_c
A_j
A_p
A
Obs
JmaxRef = 93.06
RdRef = 1.61
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

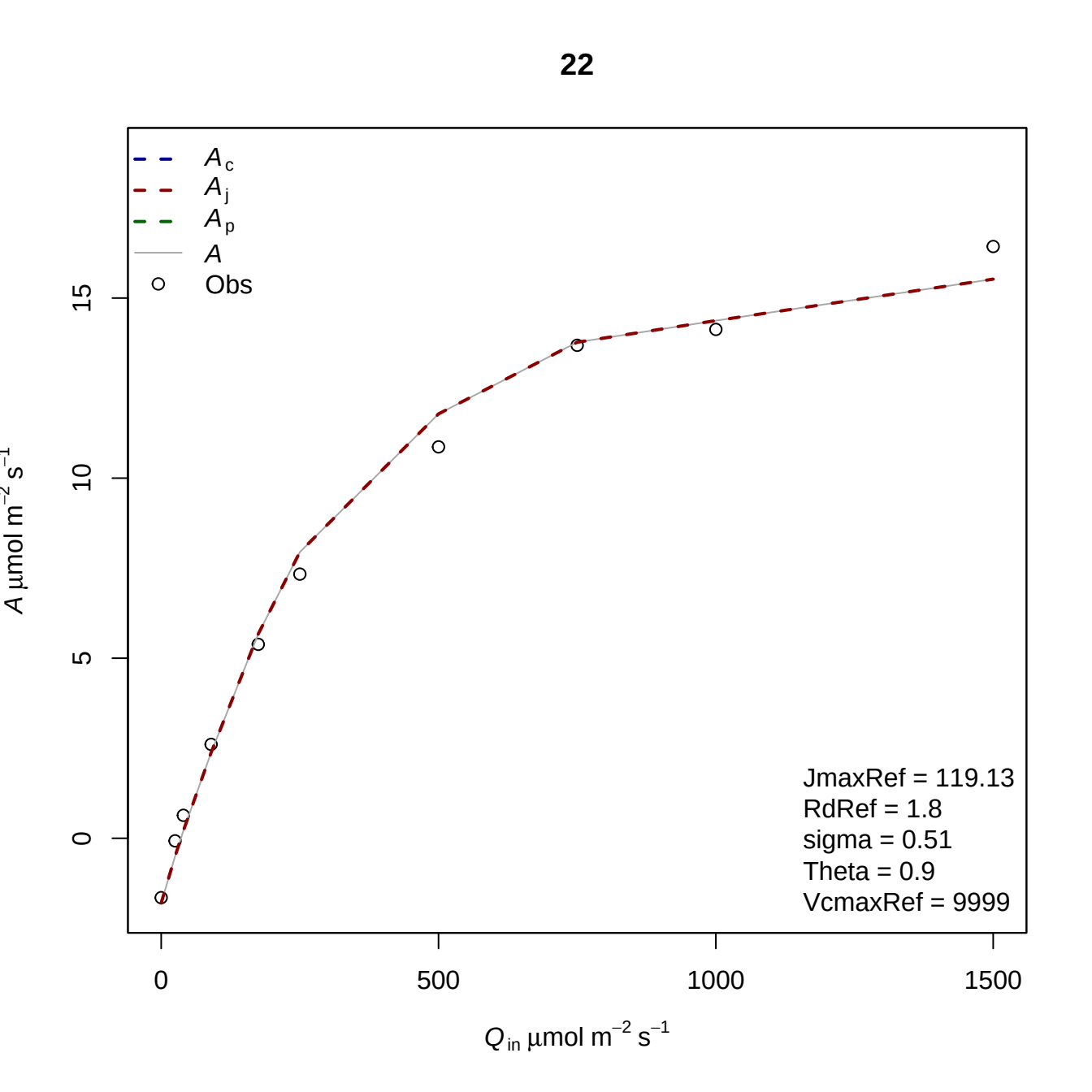

A_c
A_j
A_p
A
Obs
JmaxRef = 155.39
RdRef = 3.01
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

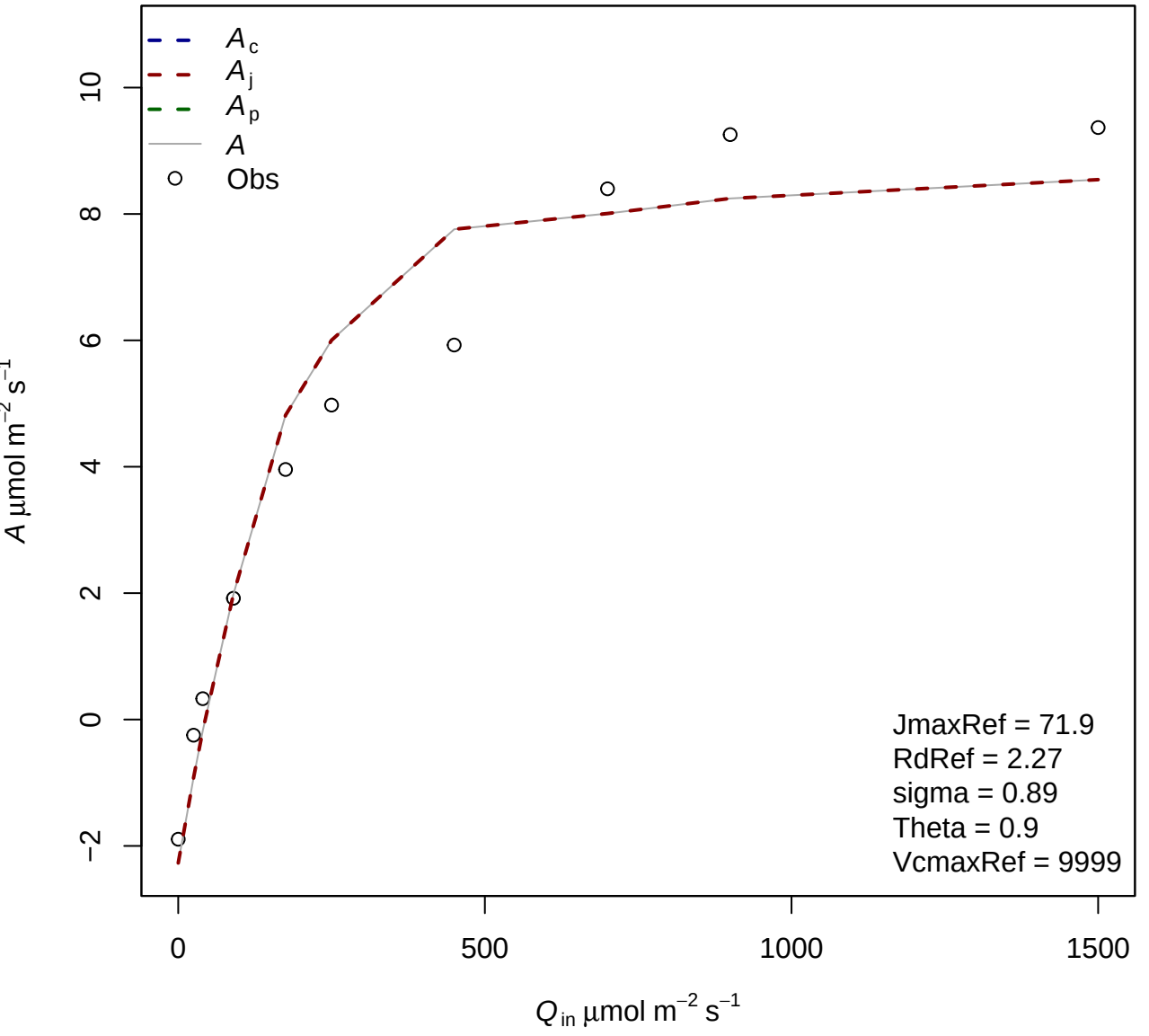

A_c
A_j
A_p
A
Obs
JmaxRef = 241.74
RdRef = 1.25
sigma = 0.14
Theta = 0.9
VcmaxRef = 218.36
0
5
10
15
20
25
30
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

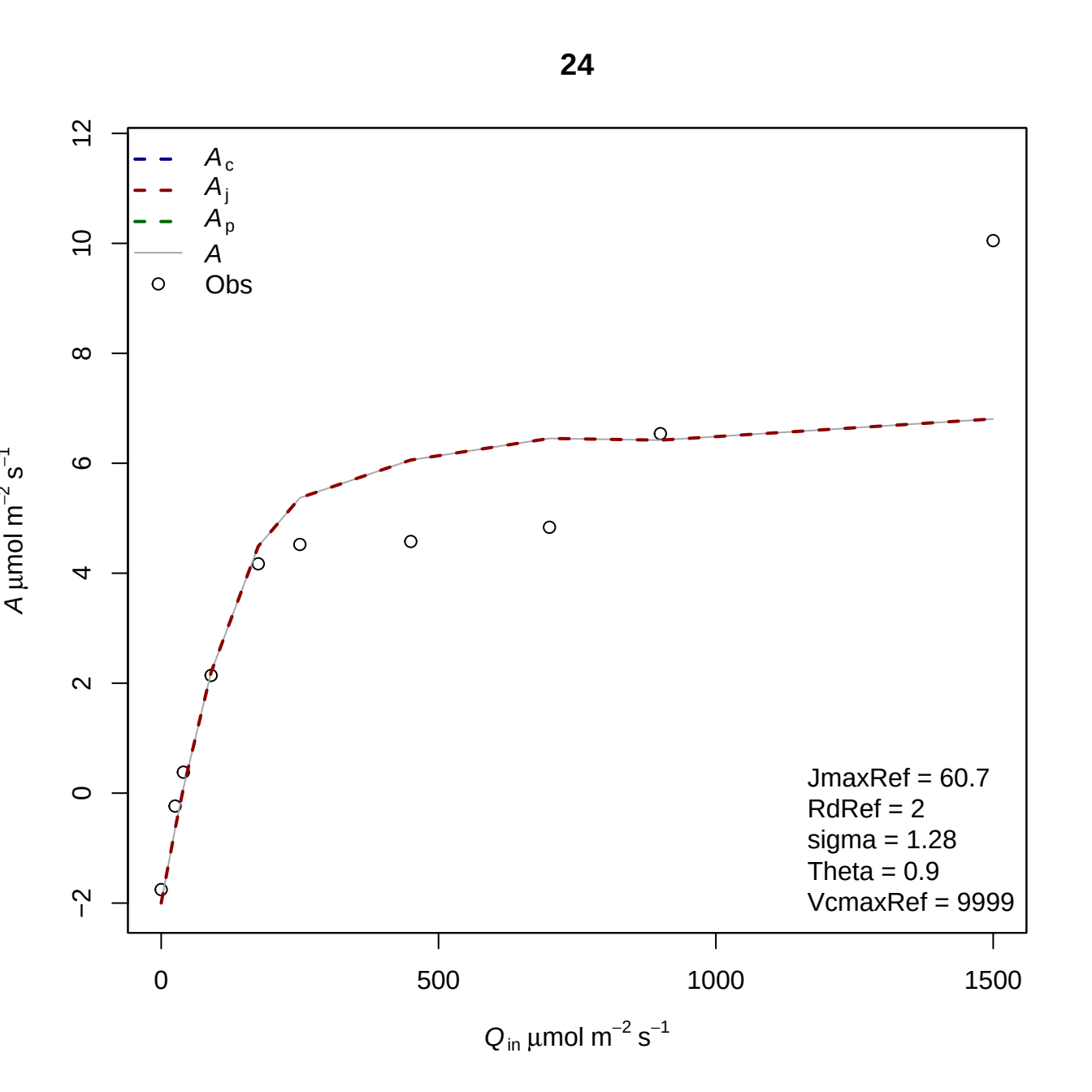

A_c
A_j
A_p
A
Obs
JmaxRef = 77
RdRef = 1.39
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

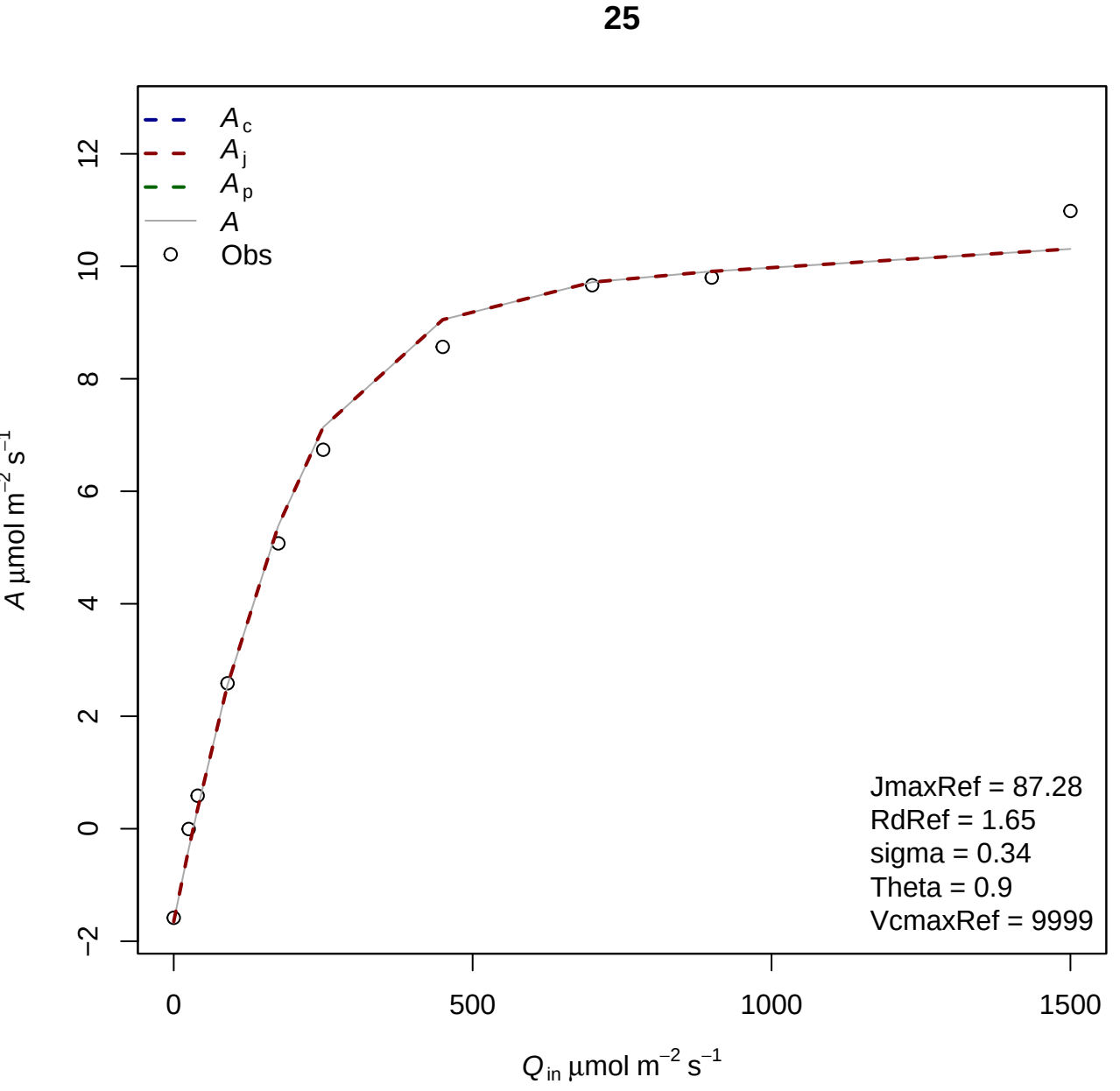

20
A_c
A_j
A_p
A
Obs
JmaxRef = 71.51
RdRef = 1.12
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

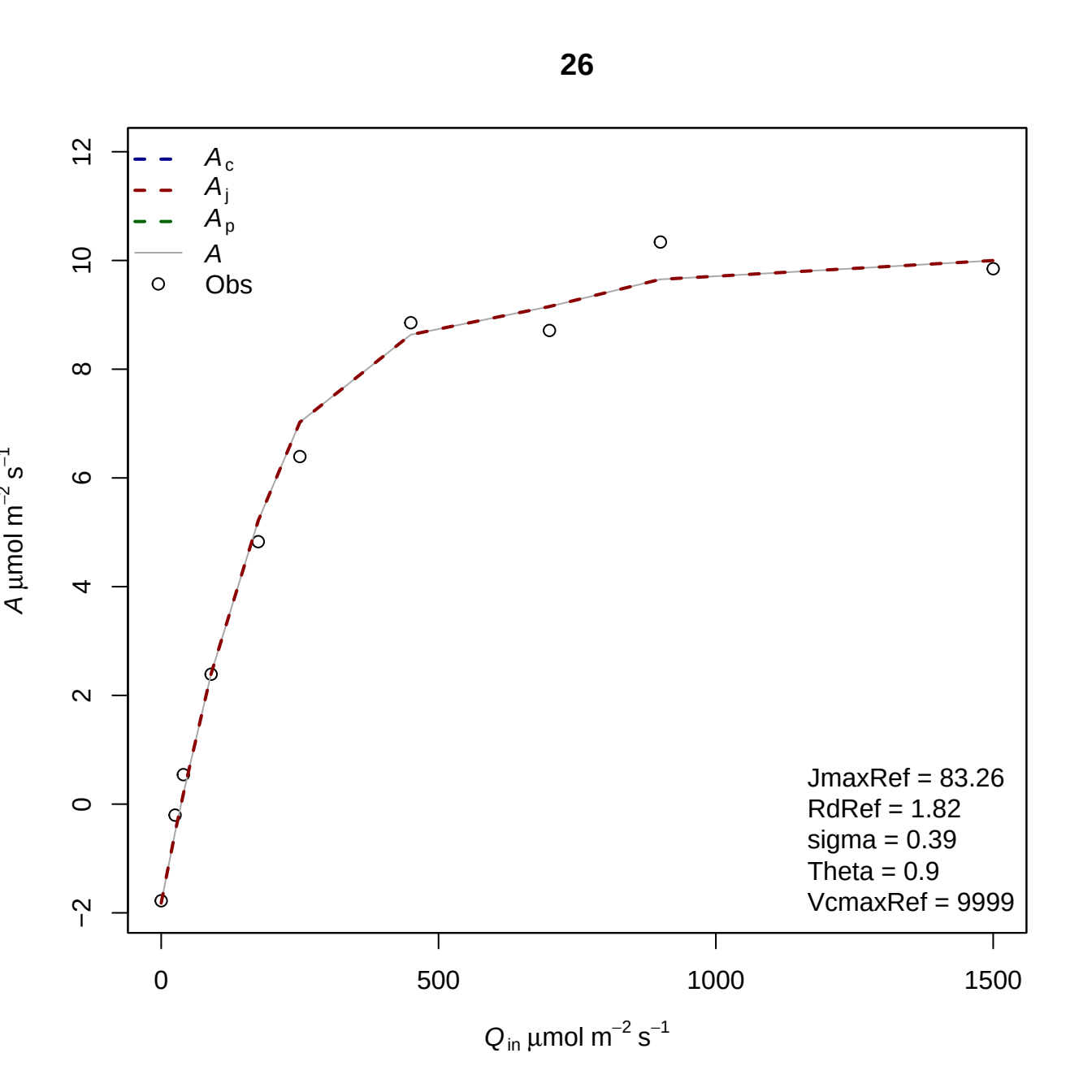

21
A_c
A_j
A_p
A
Obs
JmaxRef = 113.06
RdRef = 1.9
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

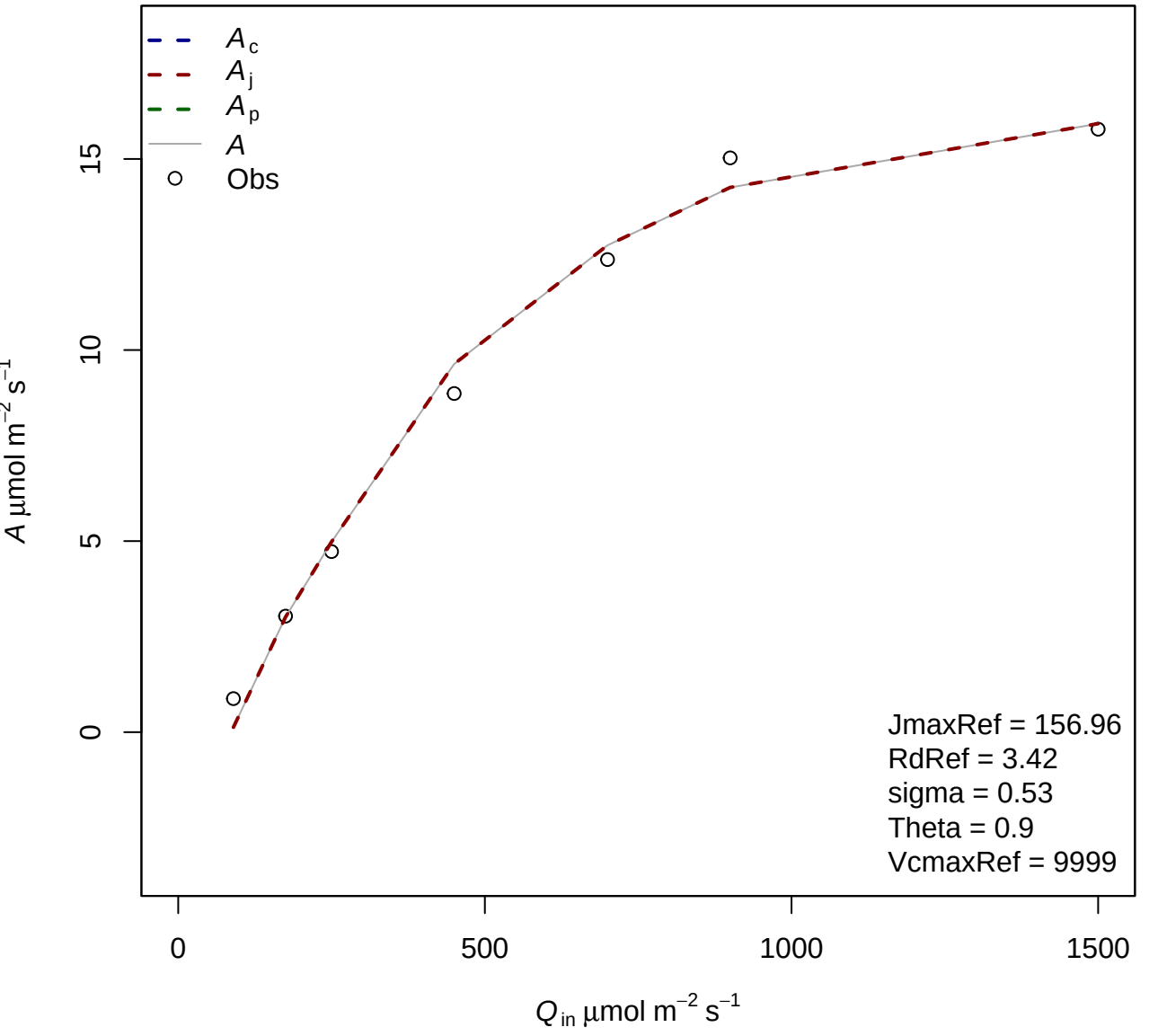

22
A_c
A_j
A_p
A
Obs
JmaxRef = 119.13
RdRef = 1.8
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

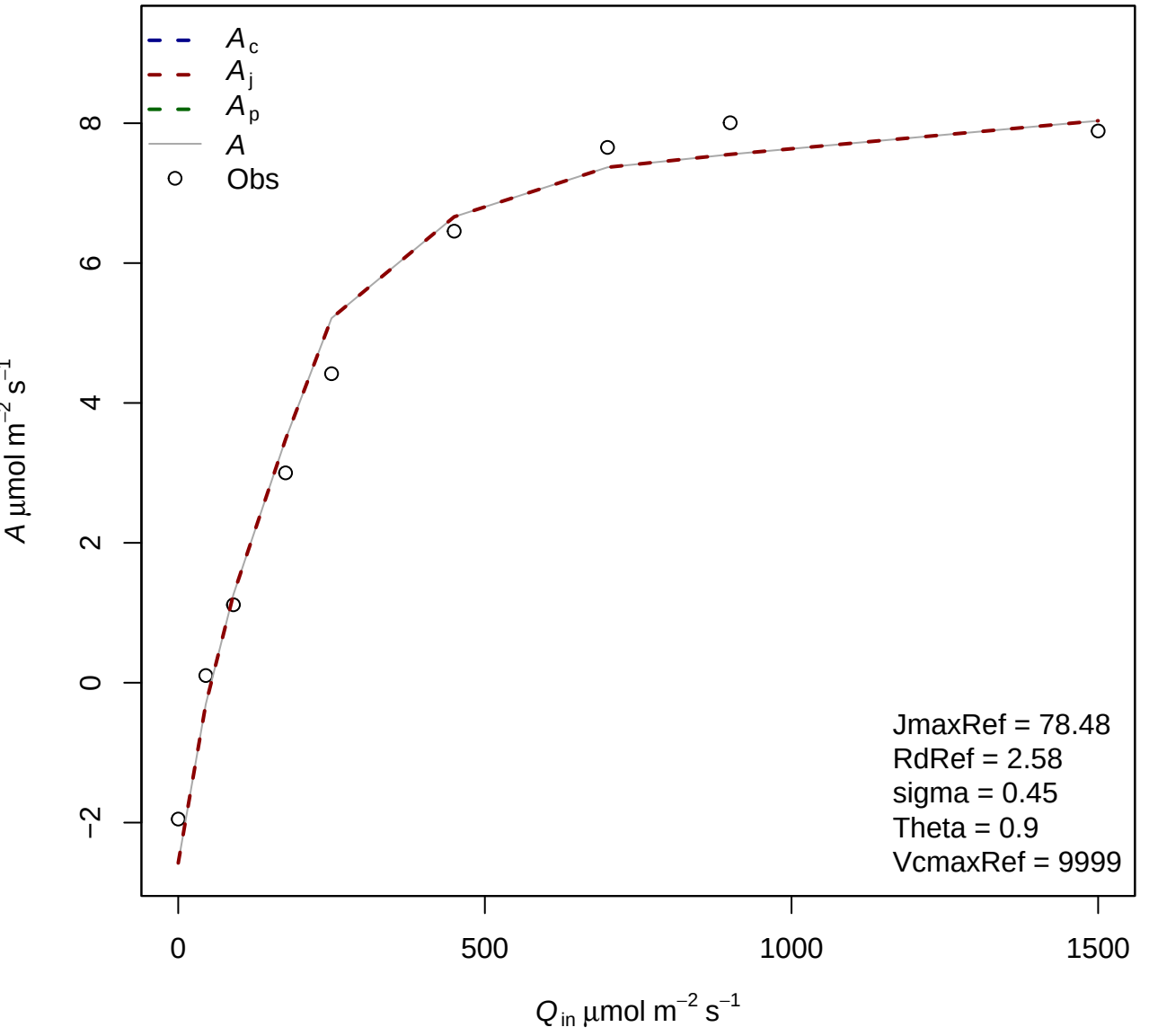

A_c
A_j
A_p
A
Obs
JmaxRef = 71.9
RdRef = 2.27
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

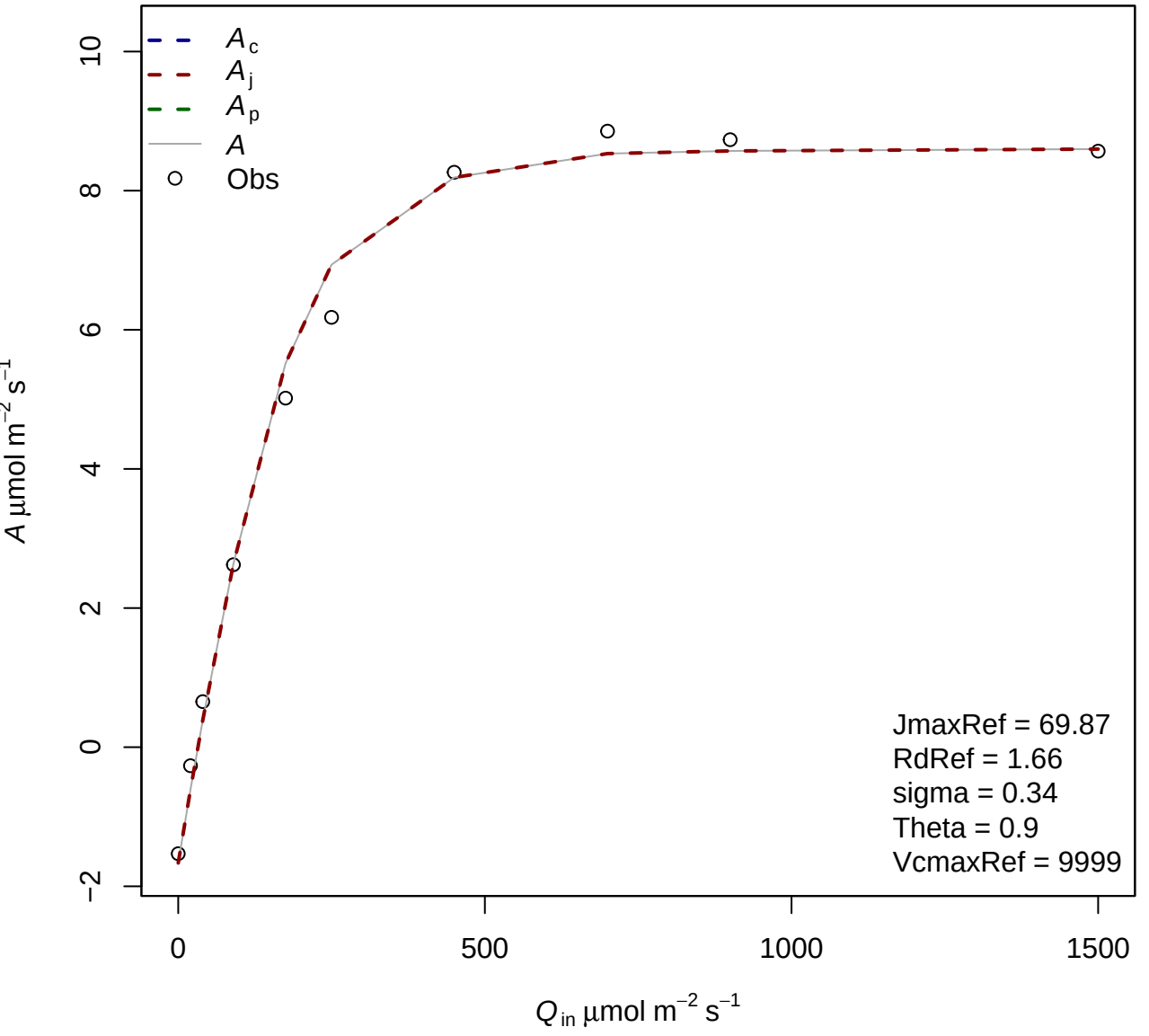

24
A_c
A_j
A_p
A
Obs
JmaxRef = 60.7
RdRef = 2
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

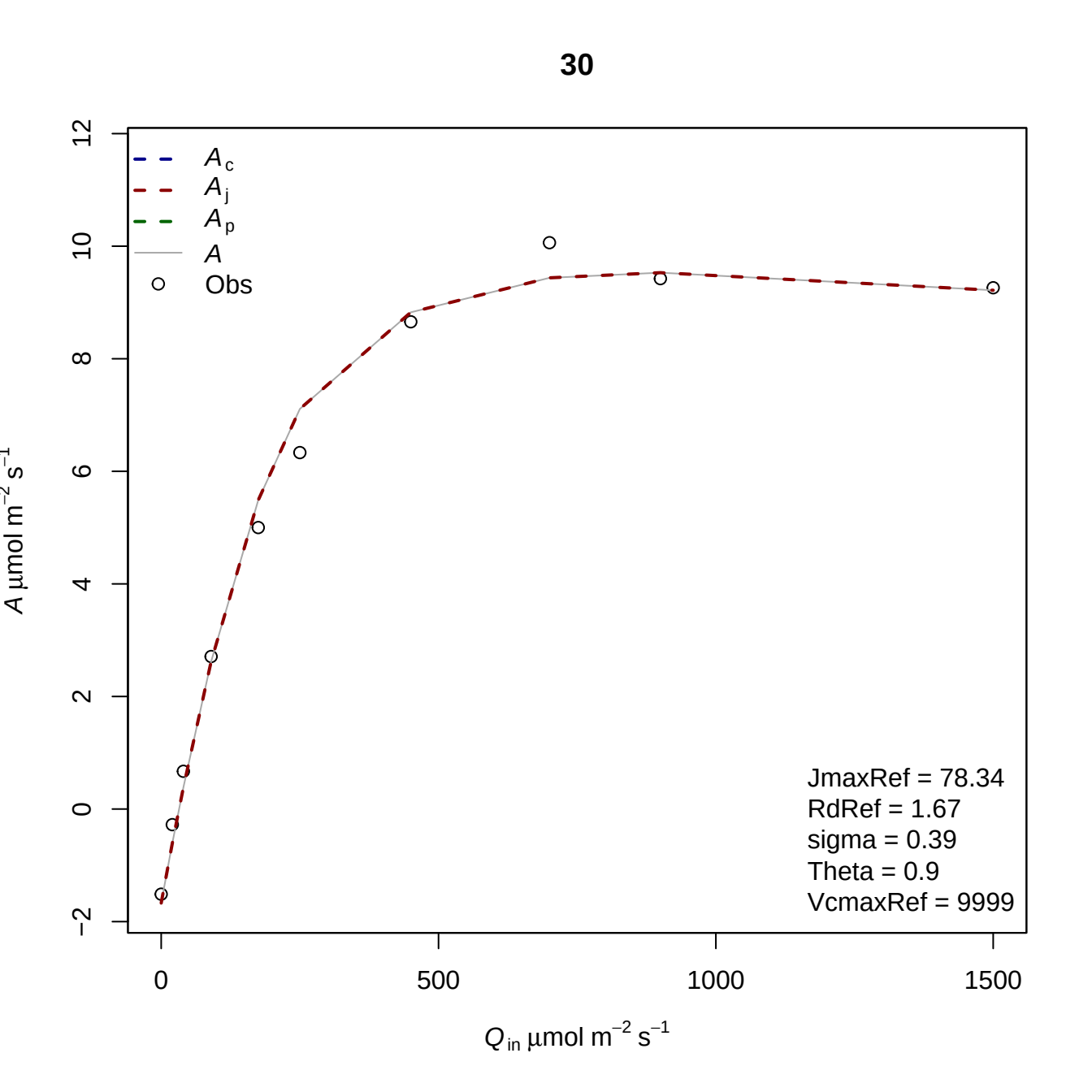

25
A_c
A_j
A_p
A
Obs
JmaxRef = 87.28
RdRef = 1.65
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

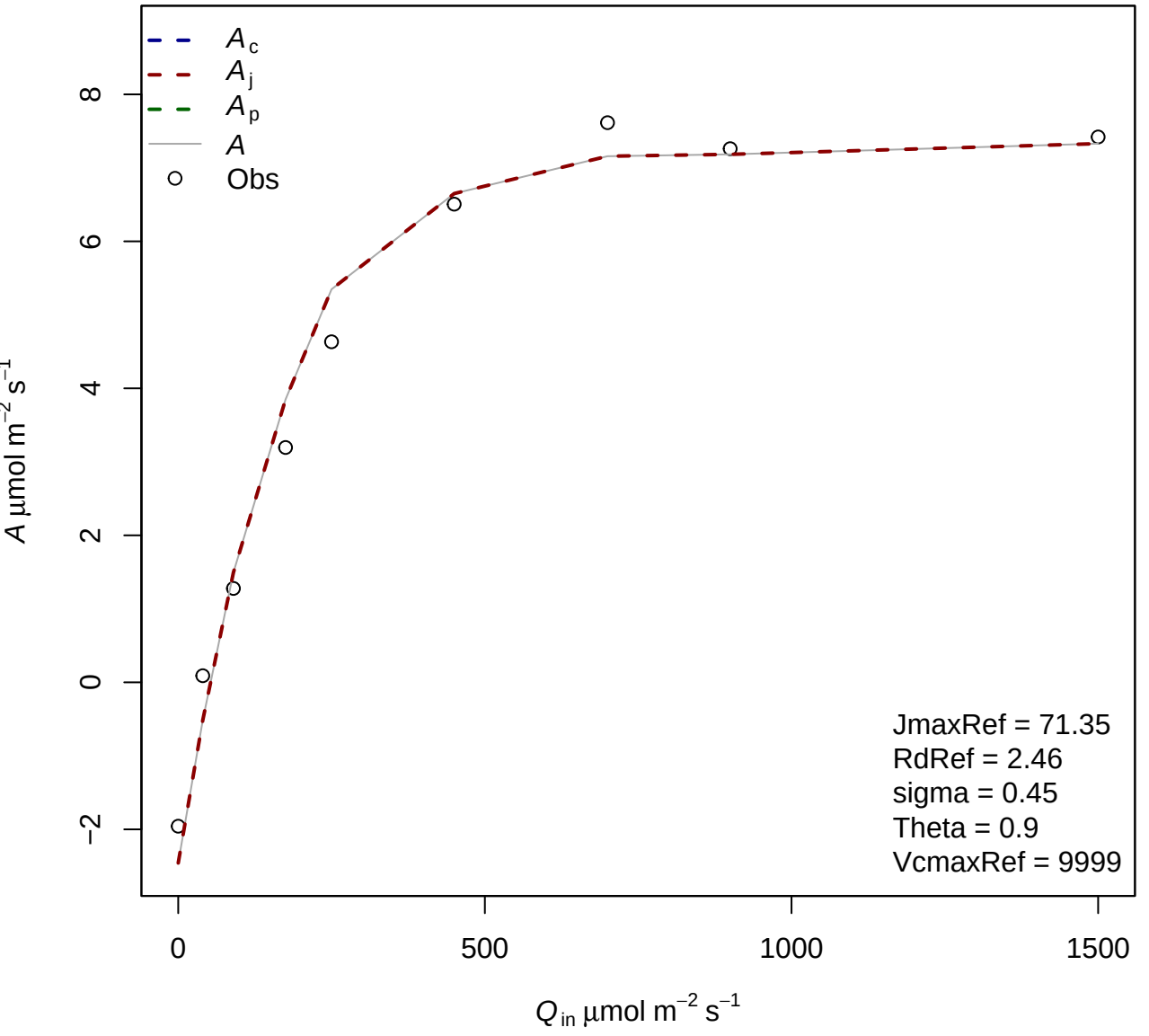

26
A_c
A_j
A_p
A
Obs
JmaxRef = 83.26
RdRef = 1.82
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

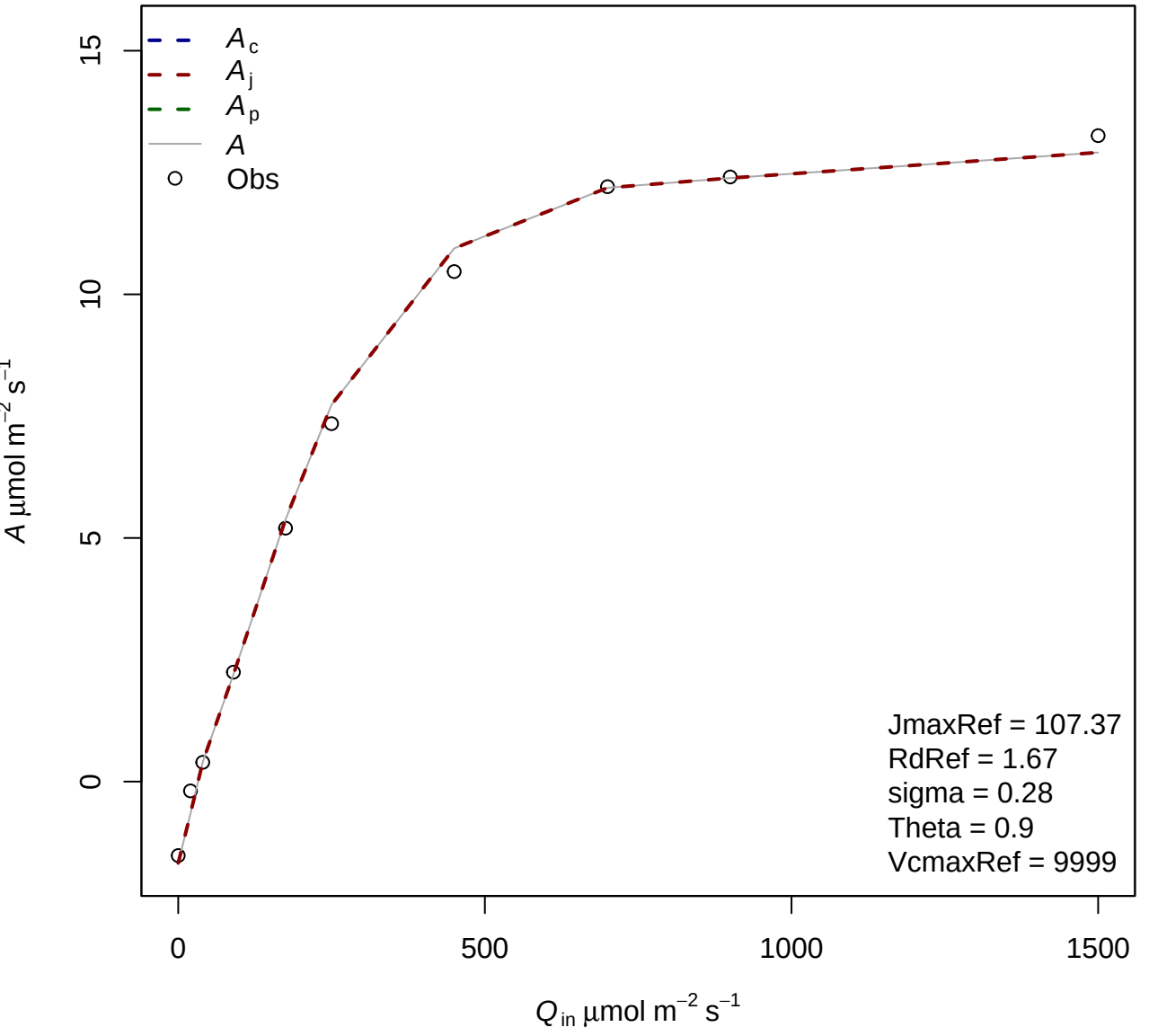

A_c
A_j
A_p
A
Obs
JmaxRef = 156.96
RdRef = 3.42
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

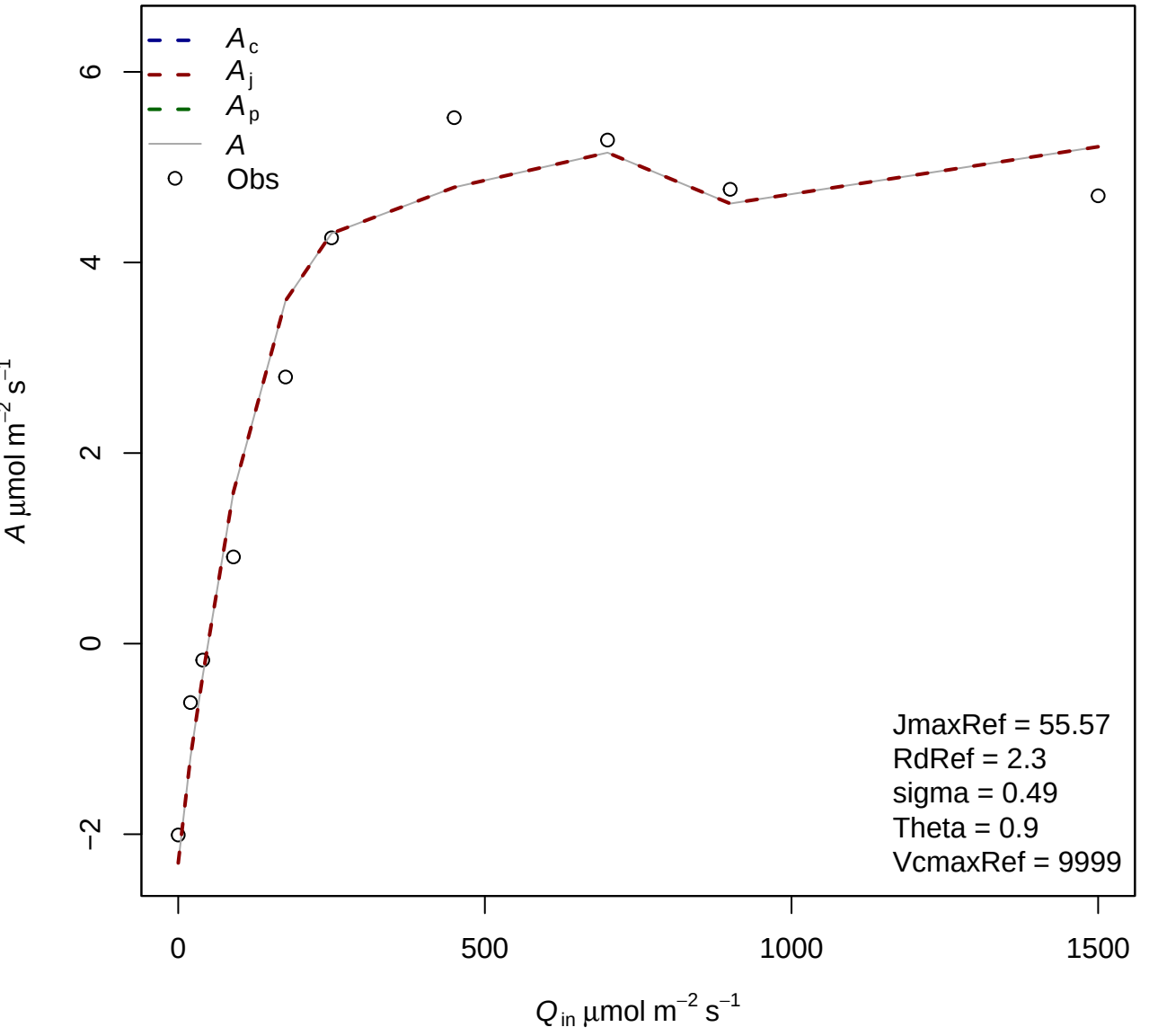

A_c
A_j
A_p
A
Obs
JmaxRef = 78.48
RdRef = 2.58
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

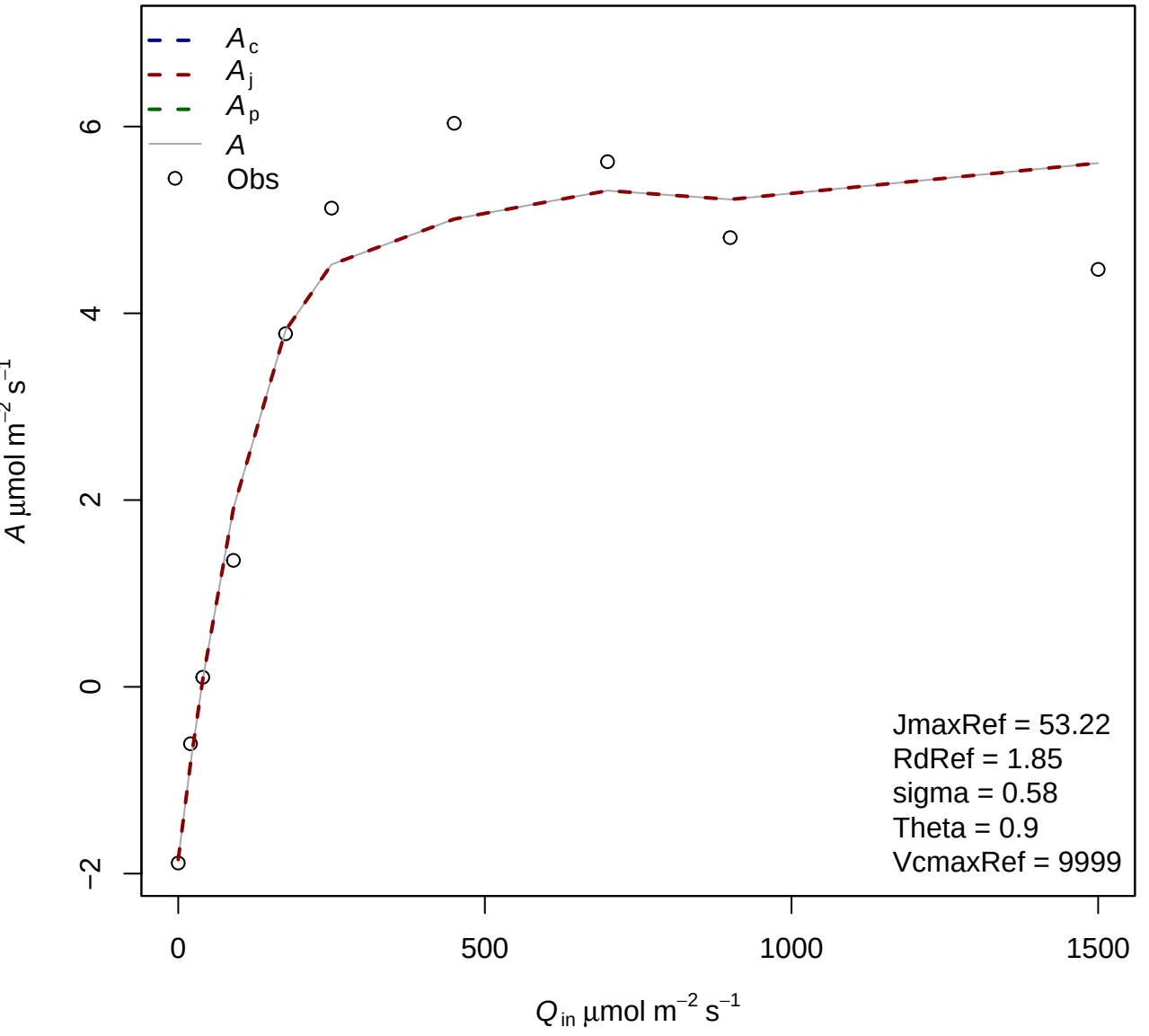

A_c
A_j
A_p
A
Obs
JmaxRef = 69.87
RdRef = 1.66
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

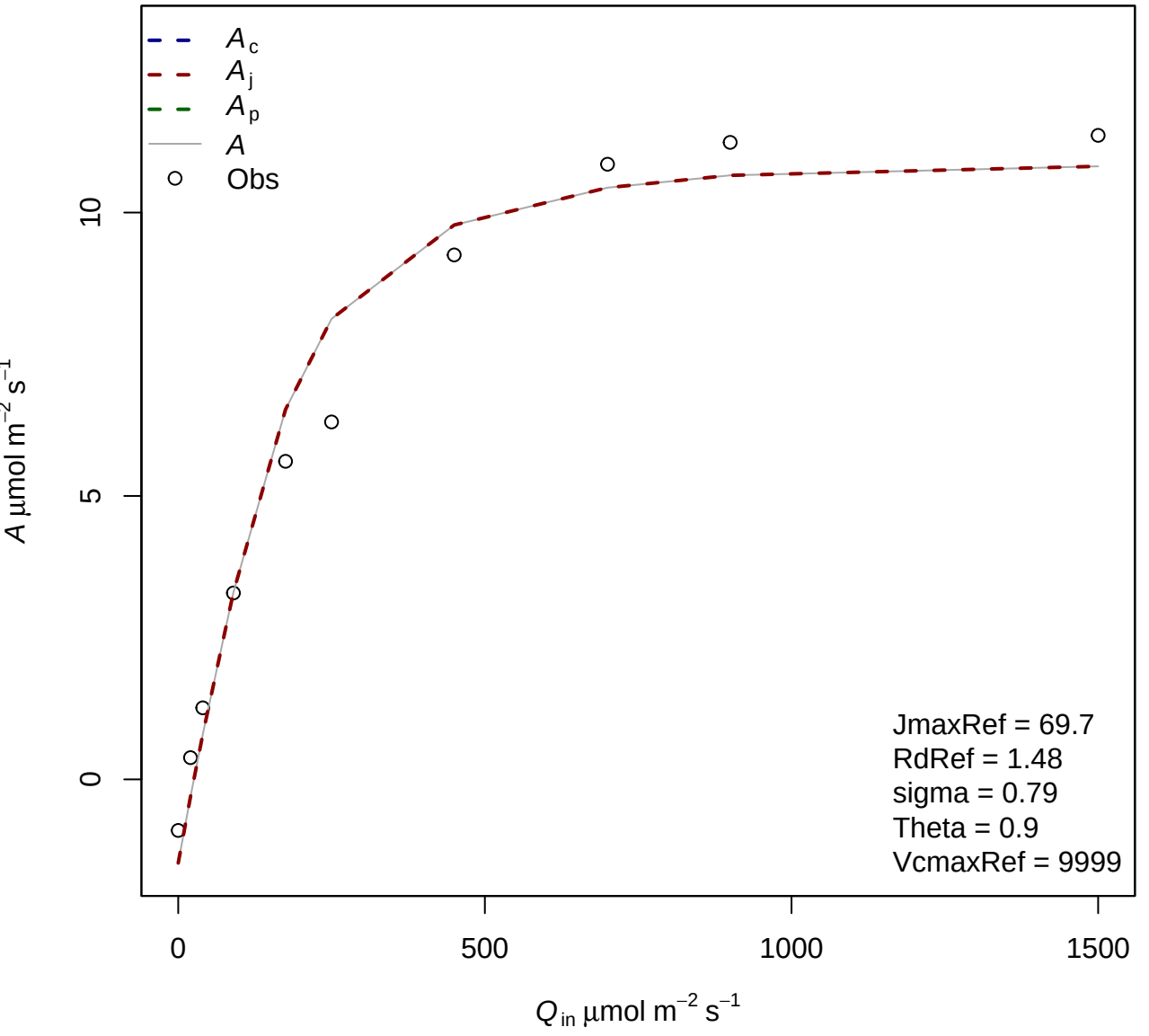

30
A_c
A_j
A_p
A
Obs
JmaxRef = 78.34
RdRef = 1.67
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

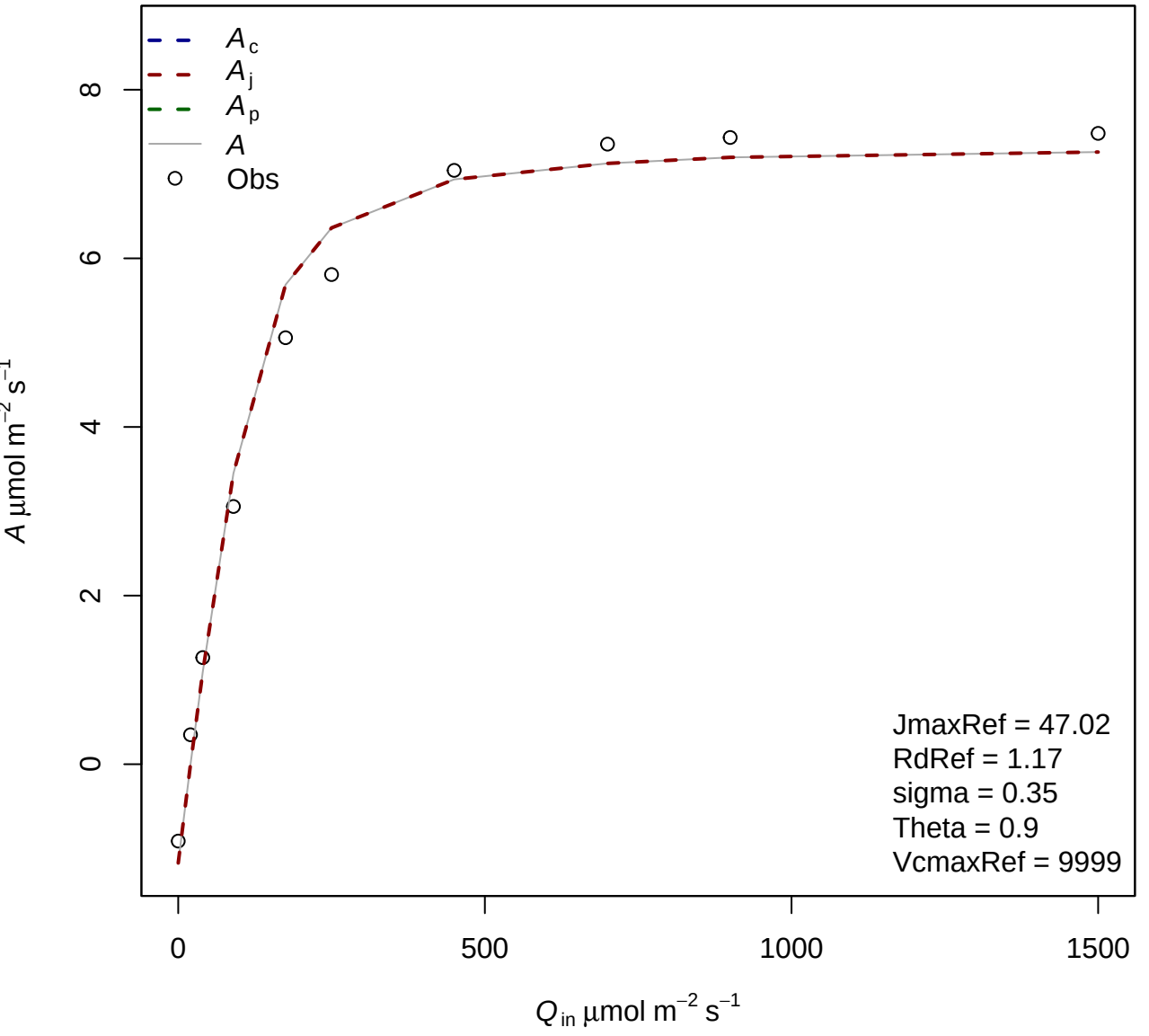

A_c
A_j
A_p
A
Obs
JmaxRef = 71.35
RdRef = 2.46
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

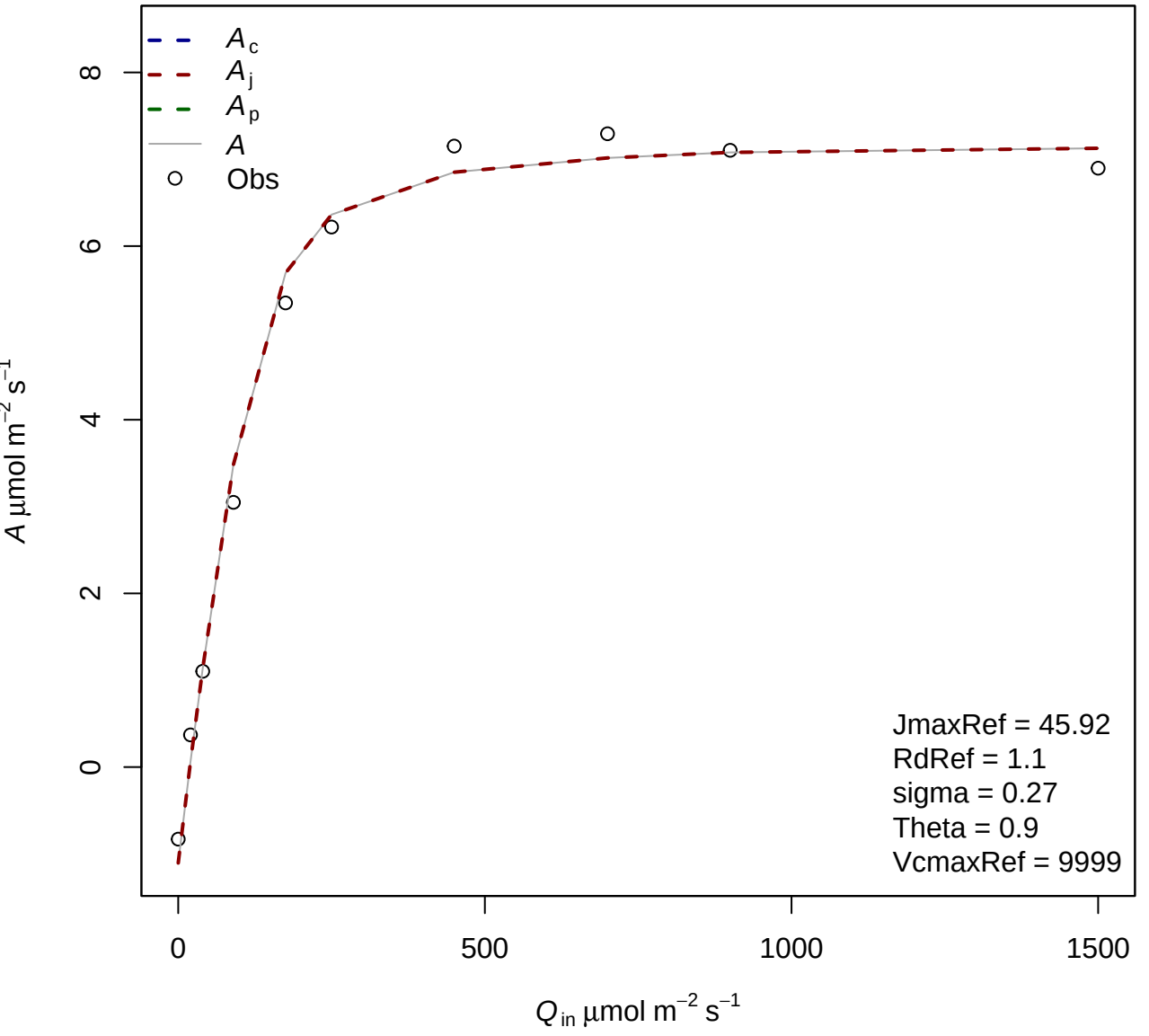

A_c
A_j
A_p
A
Obs
JmaxRef = 107.37
RdRef = 1.67
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

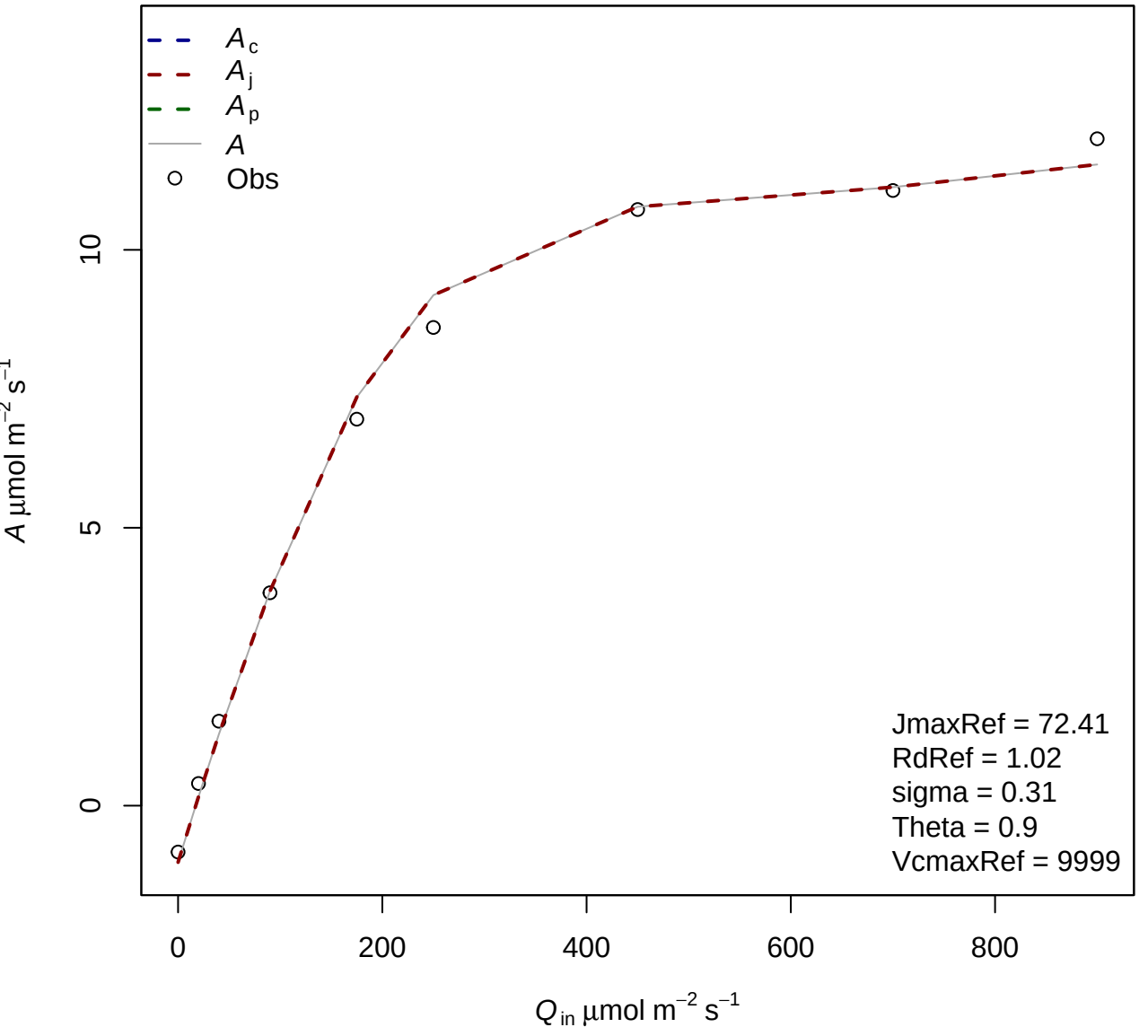

A_c
A_j
A_p
A
Obs
JmaxRef = 55.57
RdRef = 2.3
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

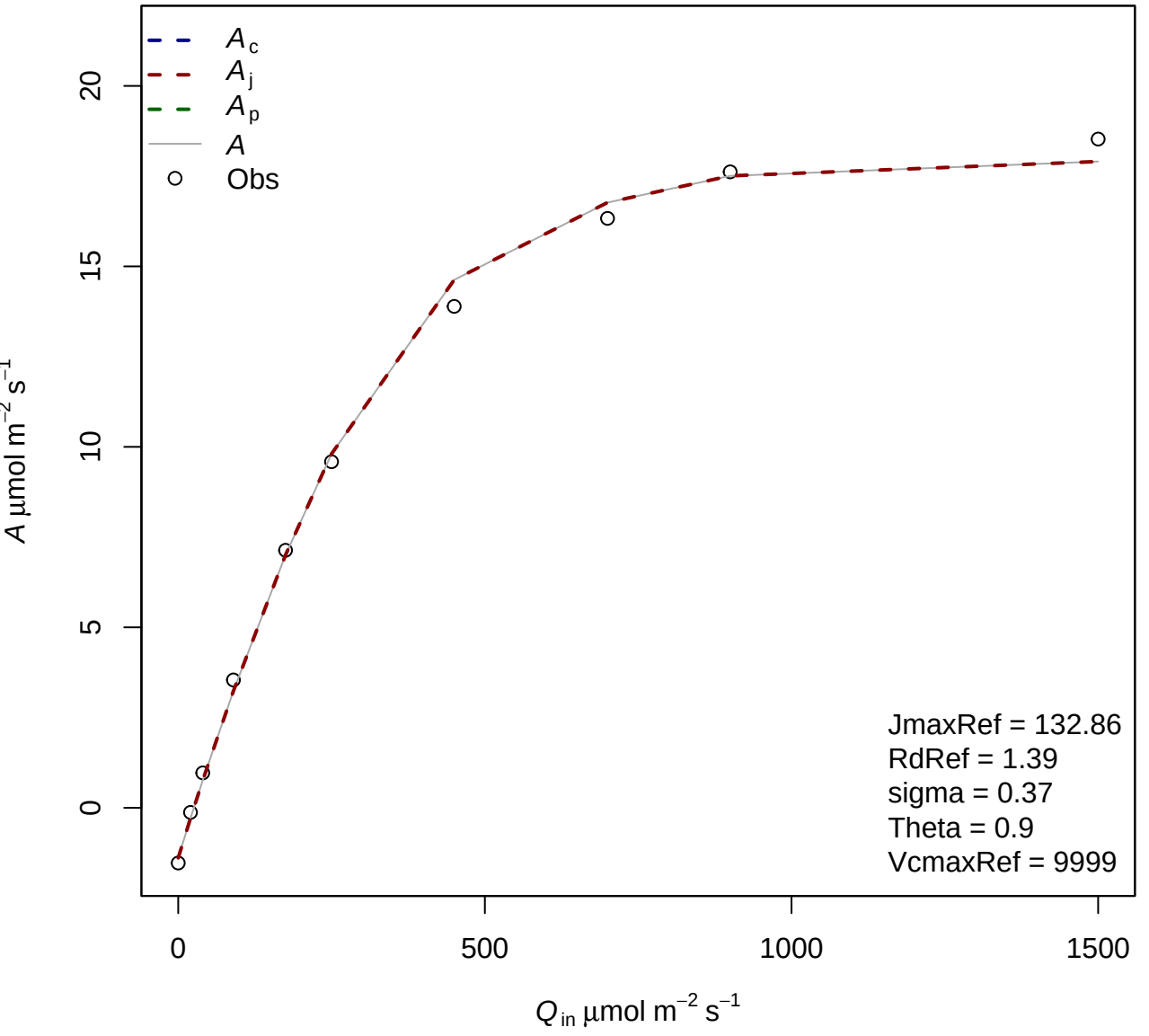

A_c
A_j
A_p
A
Obs
JmaxRef = 53.22
RdRef = 1.85
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

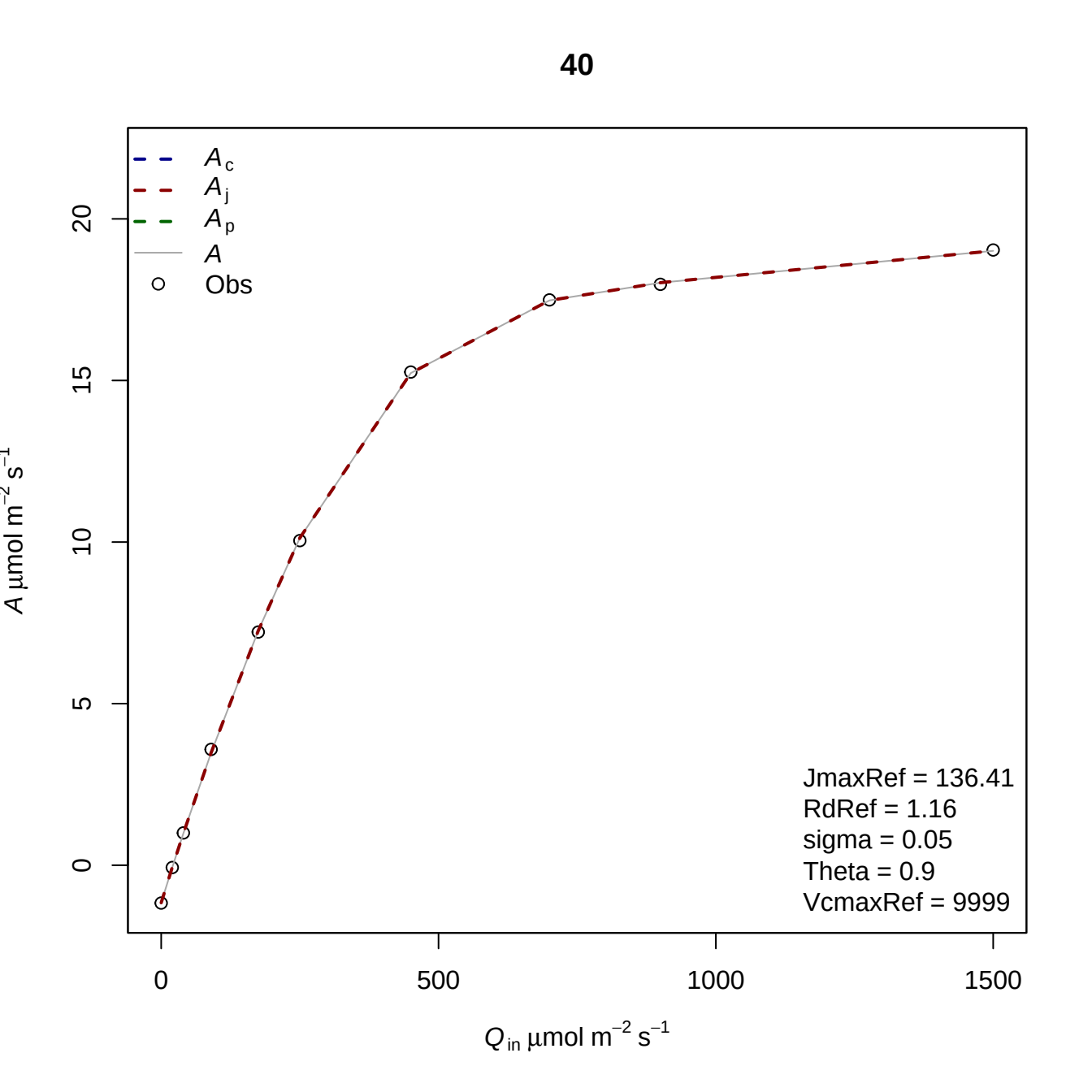

A_c
A_j
A_p
A
Obs
JmaxRef = 69.7
RdRef = 1.48
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

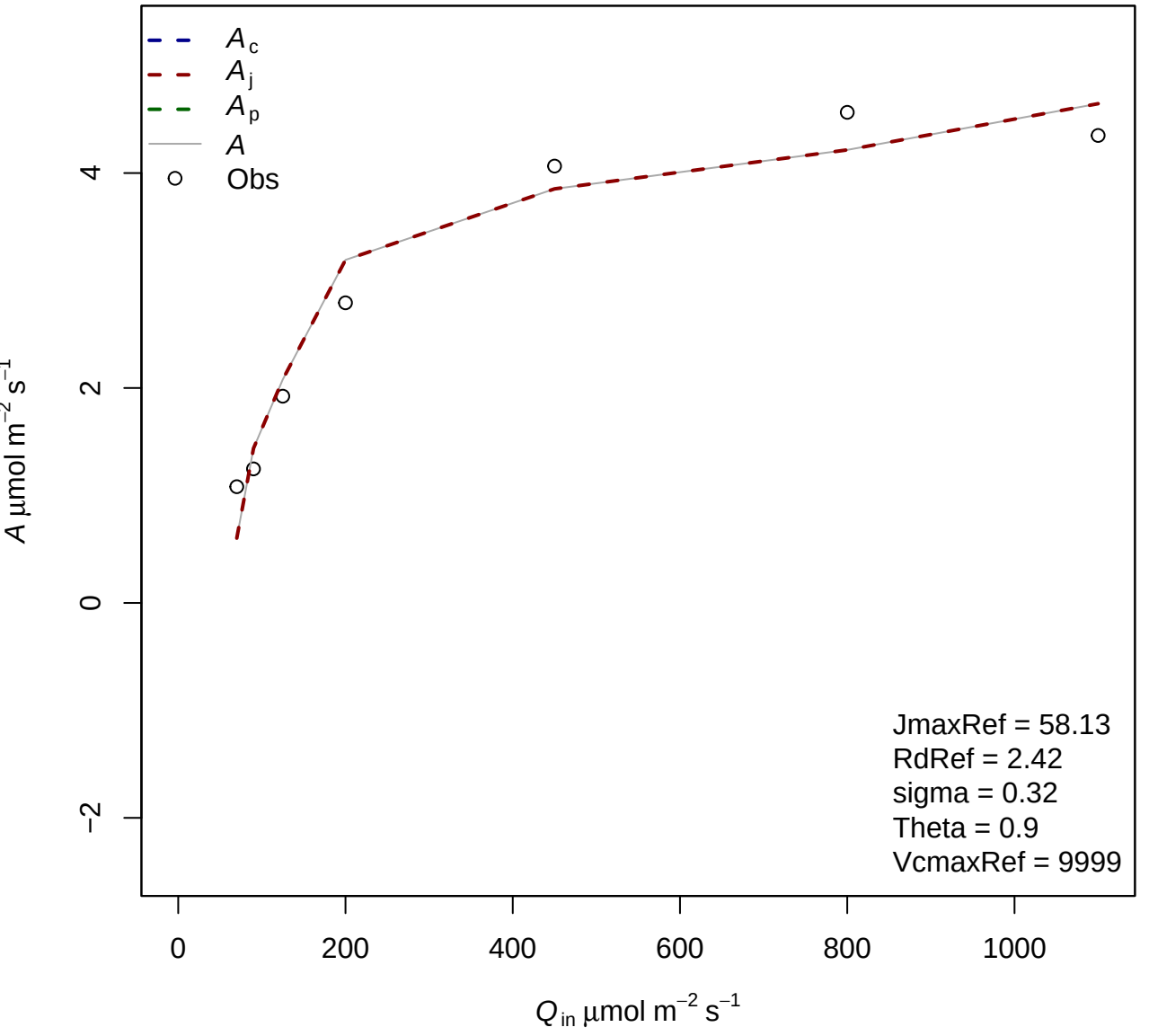

A_c
A_j
A_p
A
Obs
JmaxRef = 47.02
RdRef = 1.17
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

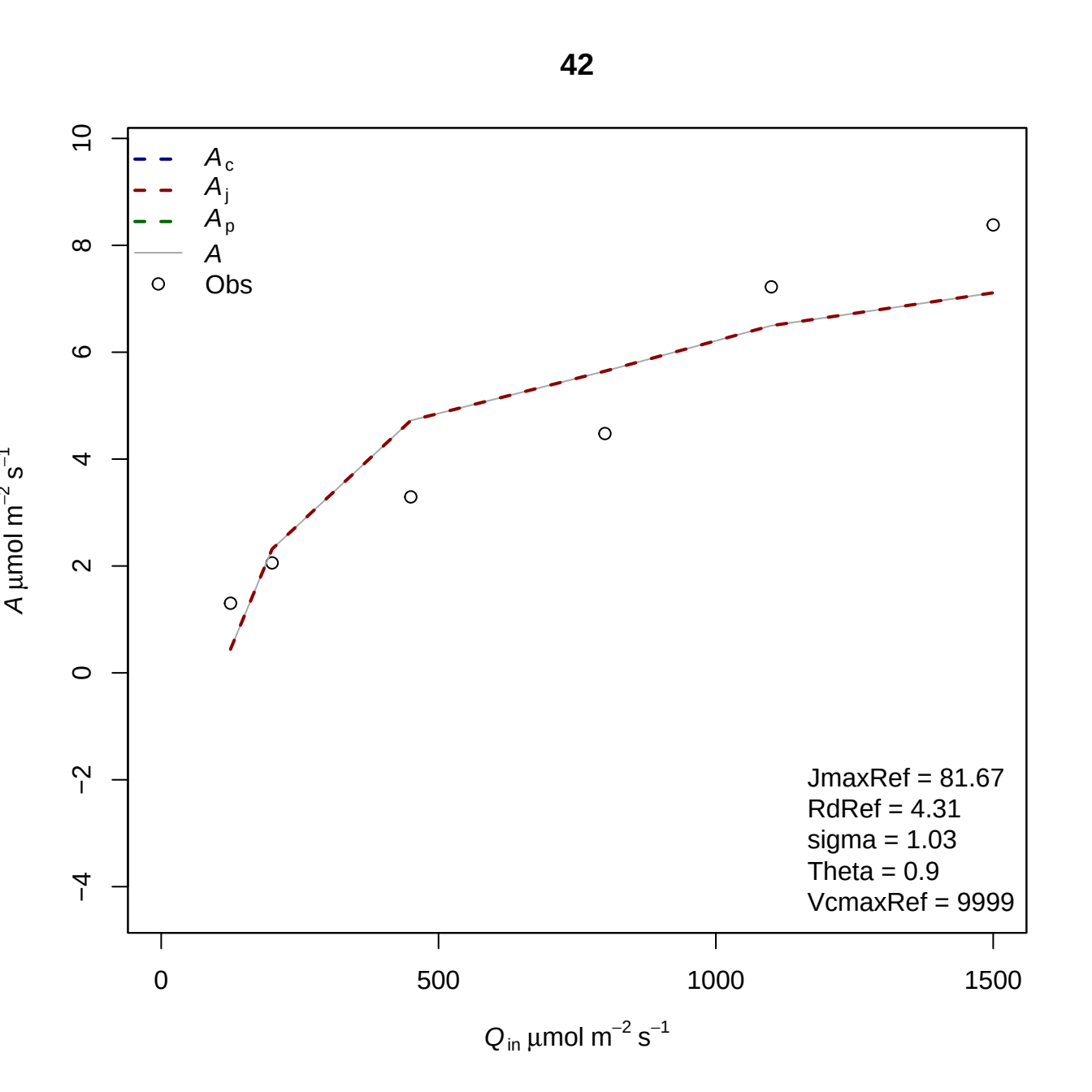

A_c
A_j
A_p
A
Obs
JmaxRef = 45.92
RdRef = 1.1
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

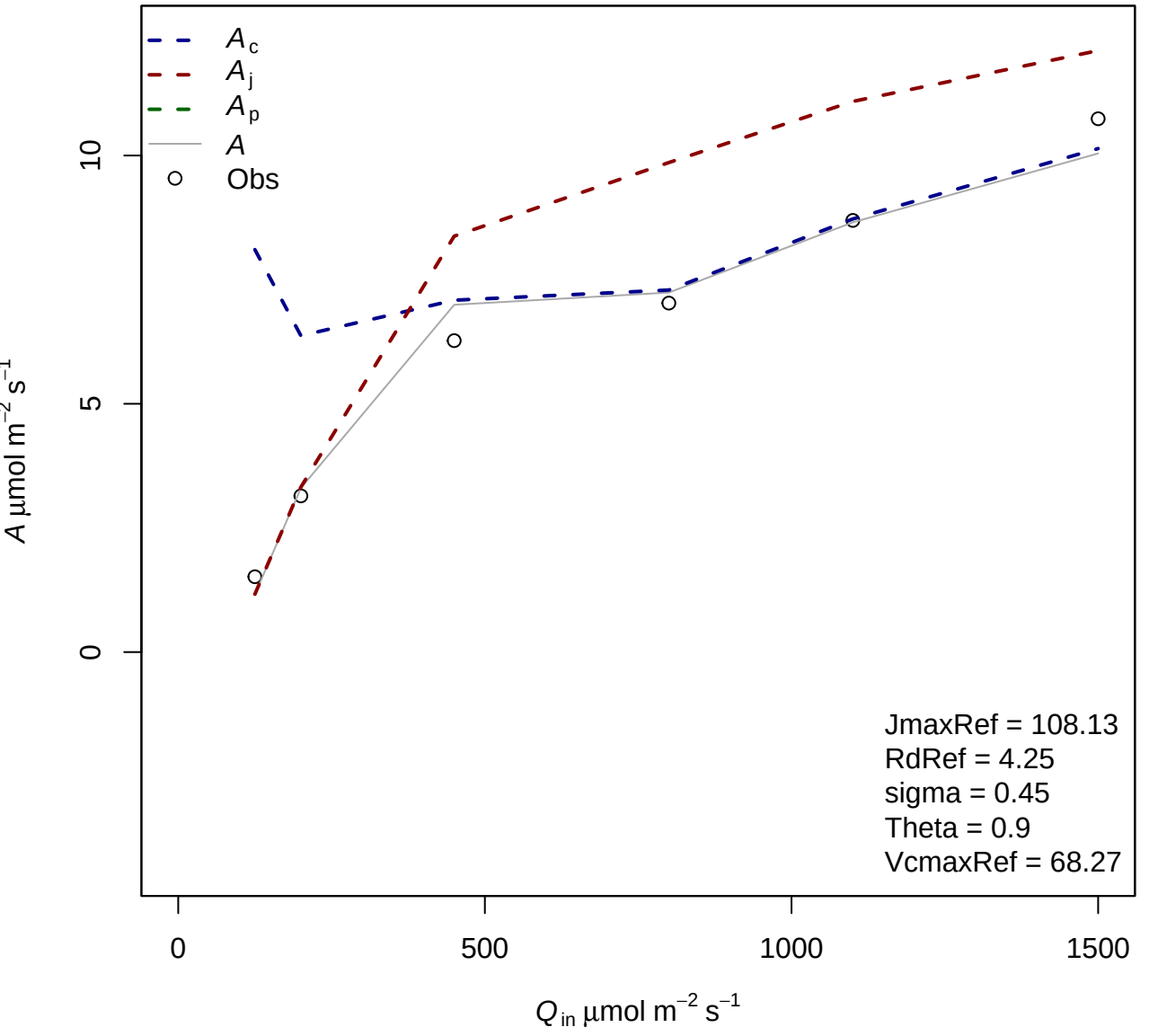

A_c
A_j
A_p
A
Obs
JmaxRef = 72.41
RdRef = 1.02
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
Q_{in} µmol m^{-2} s^{-1}

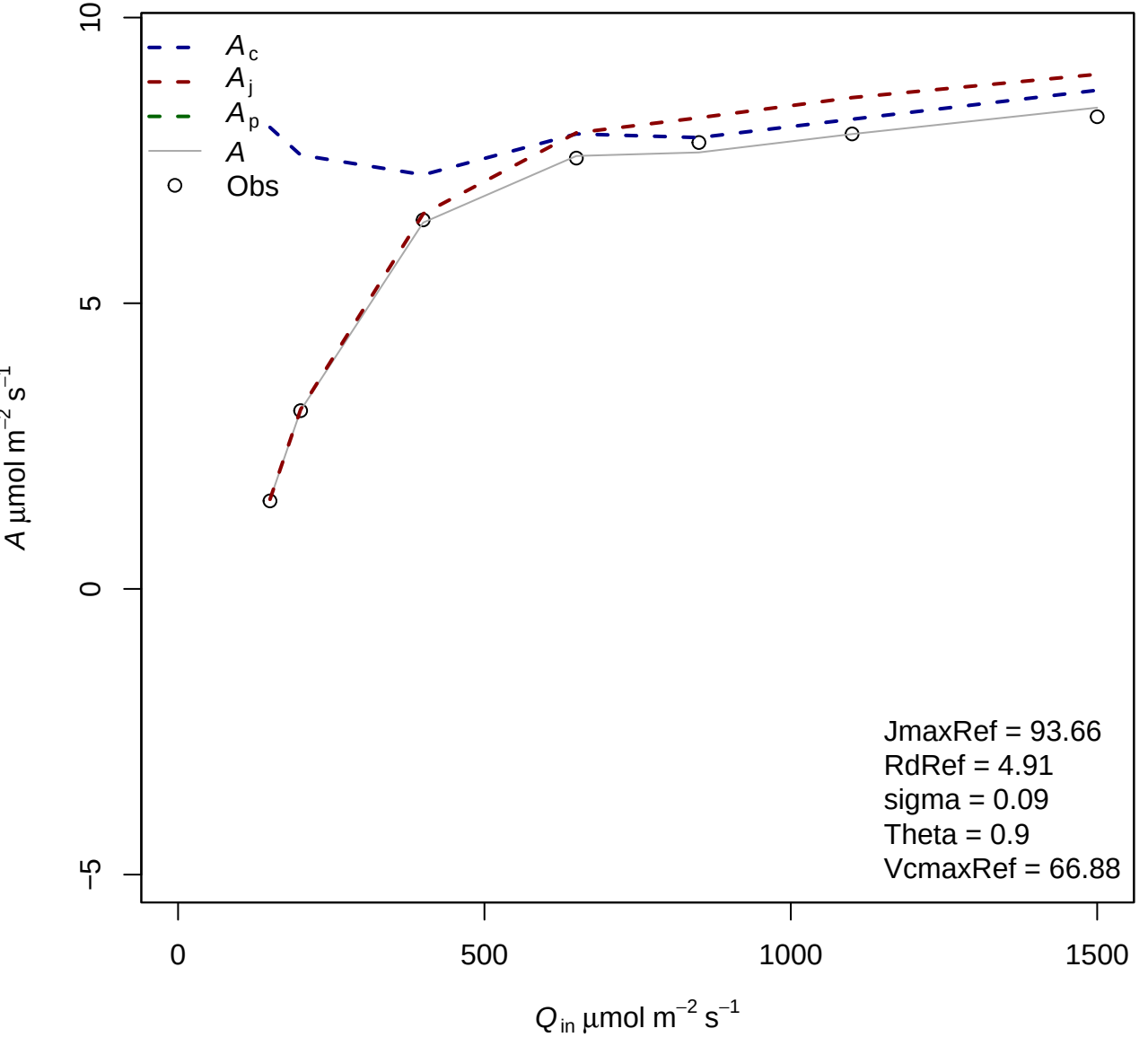

A_c
A_j
A_p
A
Obs
JmaxRef = 132.86
RdRef = 1.39
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

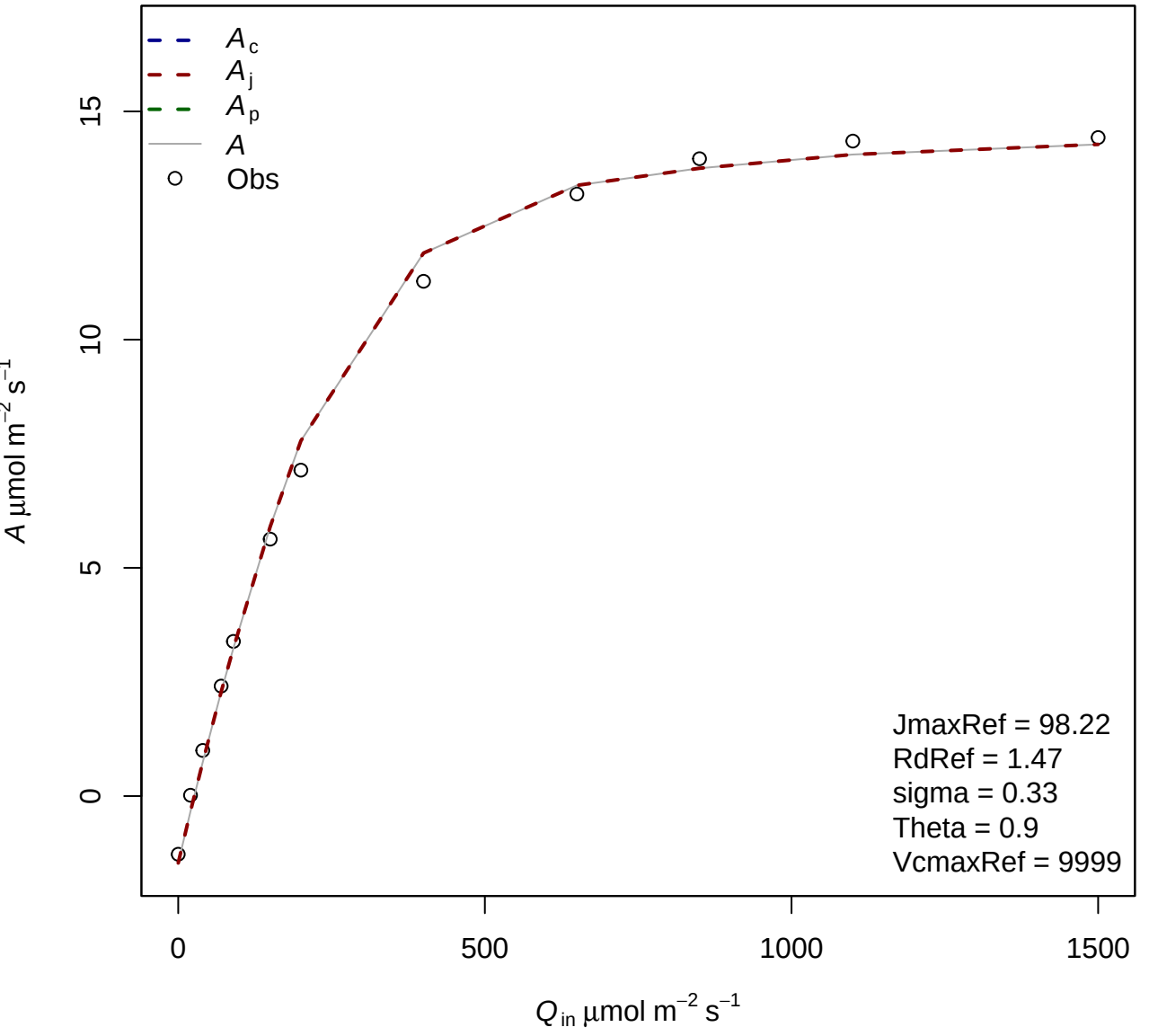

40
A_c
A_j
A_p
A
Obs
JmaxRef = 136.41
RdRef = 1.16
sigma = 0.05
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

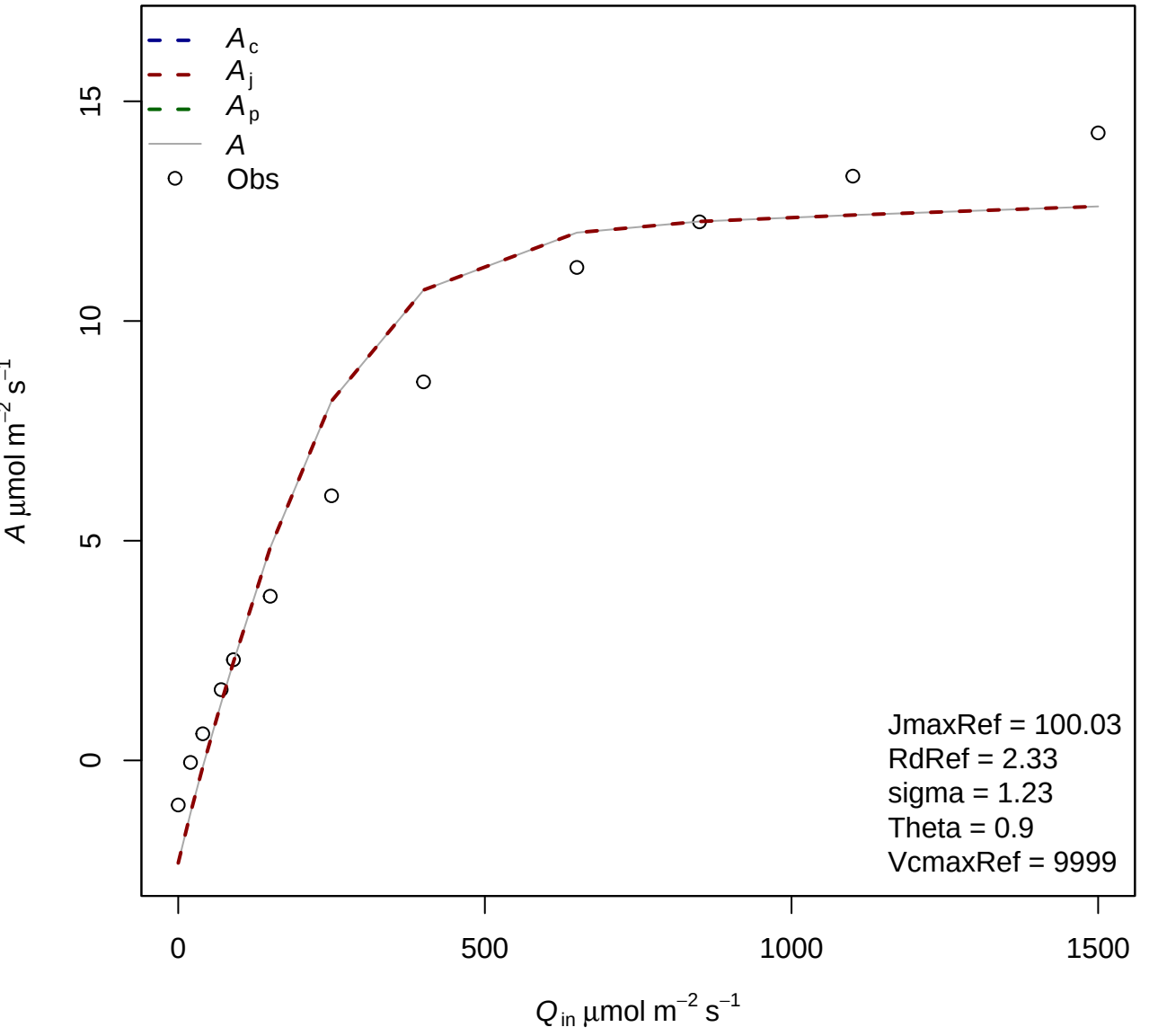

A_c
A_j
A_p
A
Obs
JmaxRef = 58.13
RdRef = 2.42
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

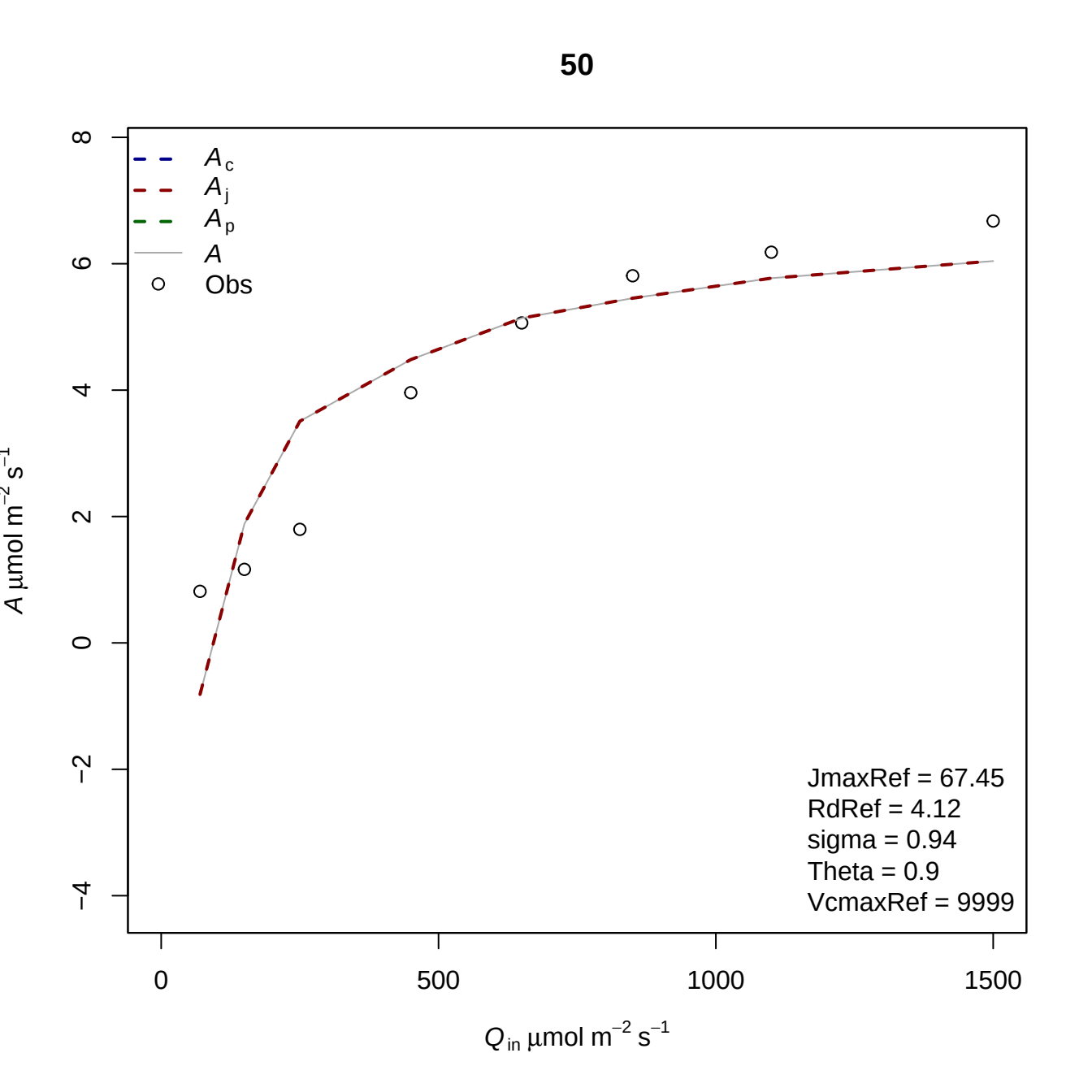

42
A_c
A_j
A_p
A
Obs
JmaxRef = 81.67
RdRef = 4.31
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

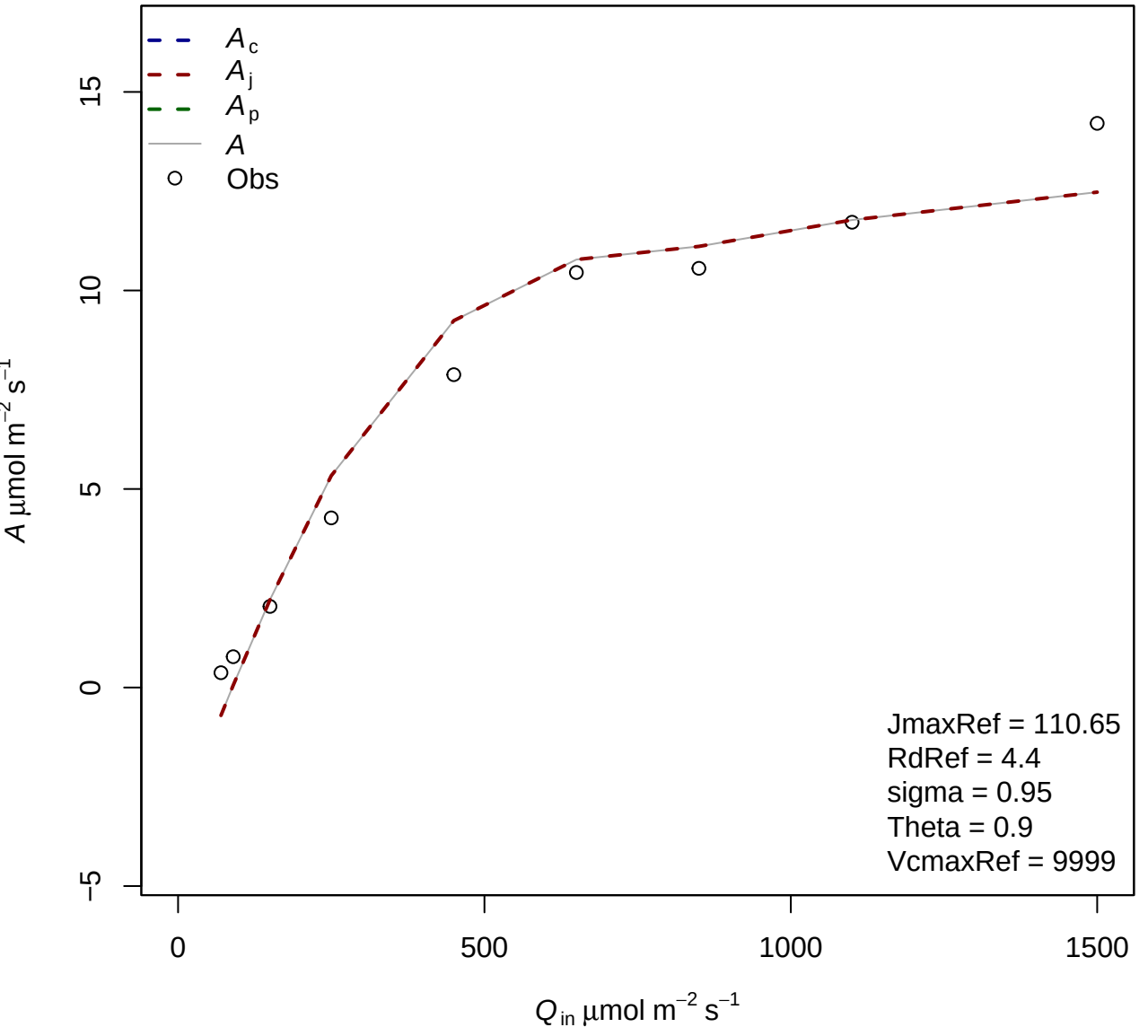

A_c
A_j
A_p
A
Obs
JmaxRef = 108.13
RdRef = 4.25
sigma = 0.45
Theta = 0.9
VcmaxRef = 68.27
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

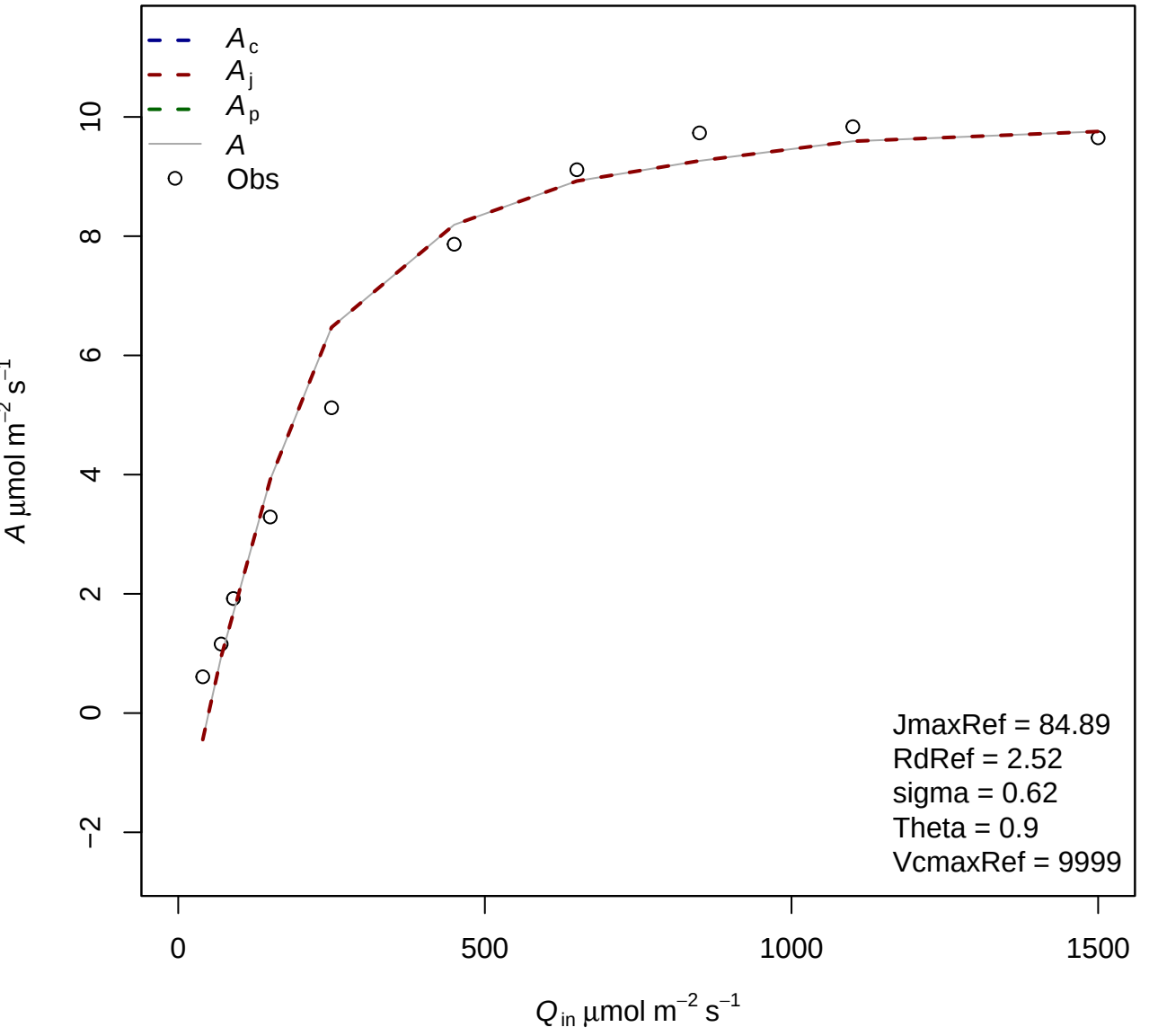

A_c
A_j
A_p
A
Obs
JmaxRef = 93.66
RdRef = 4.91
sigma = 0.09
Theta = 0.9
VcmaxRef = 66.88
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

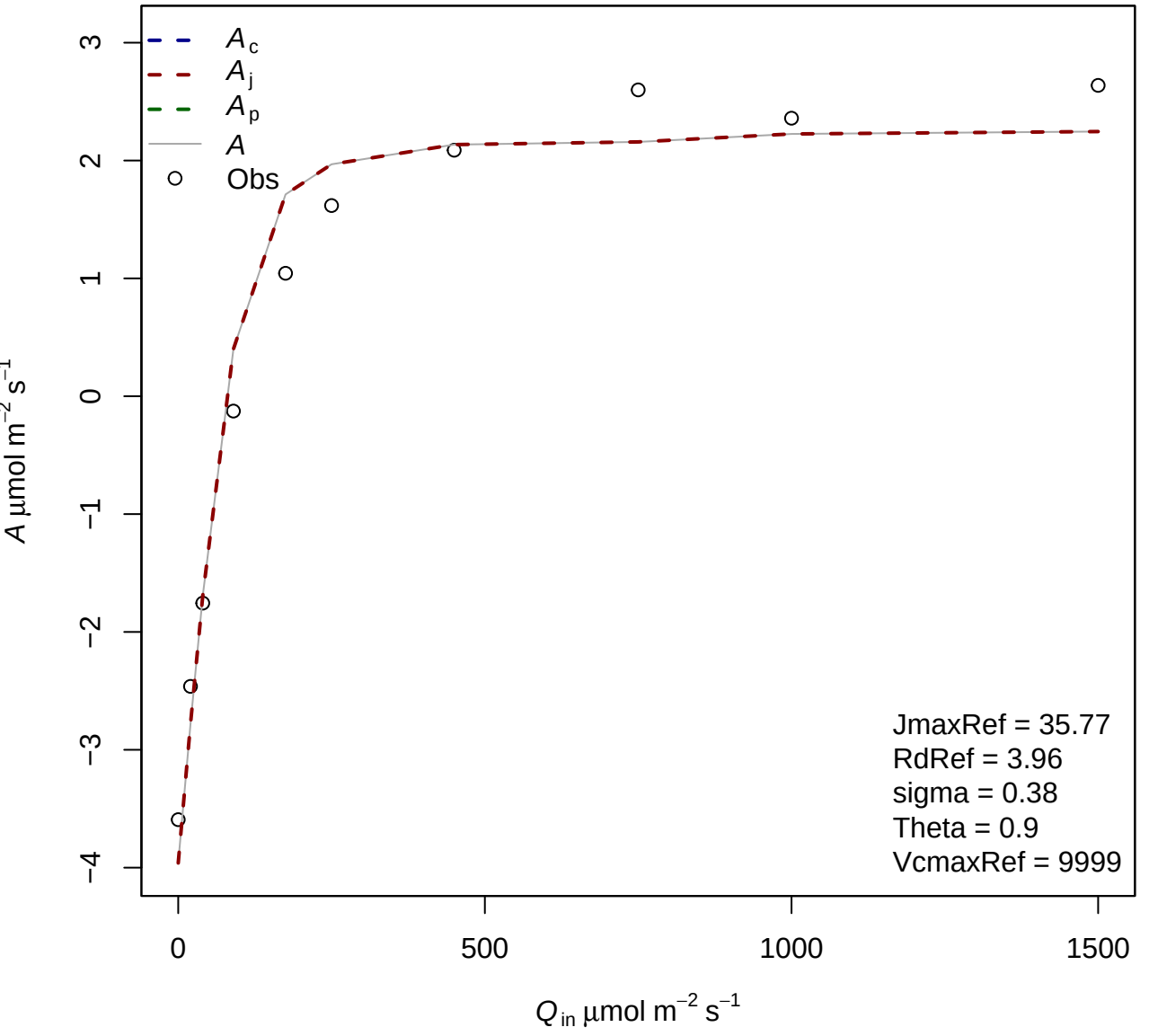

A_c
A_j
A_p
A
Obs
JmaxRef = 98.22
RdRef = 1.47
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

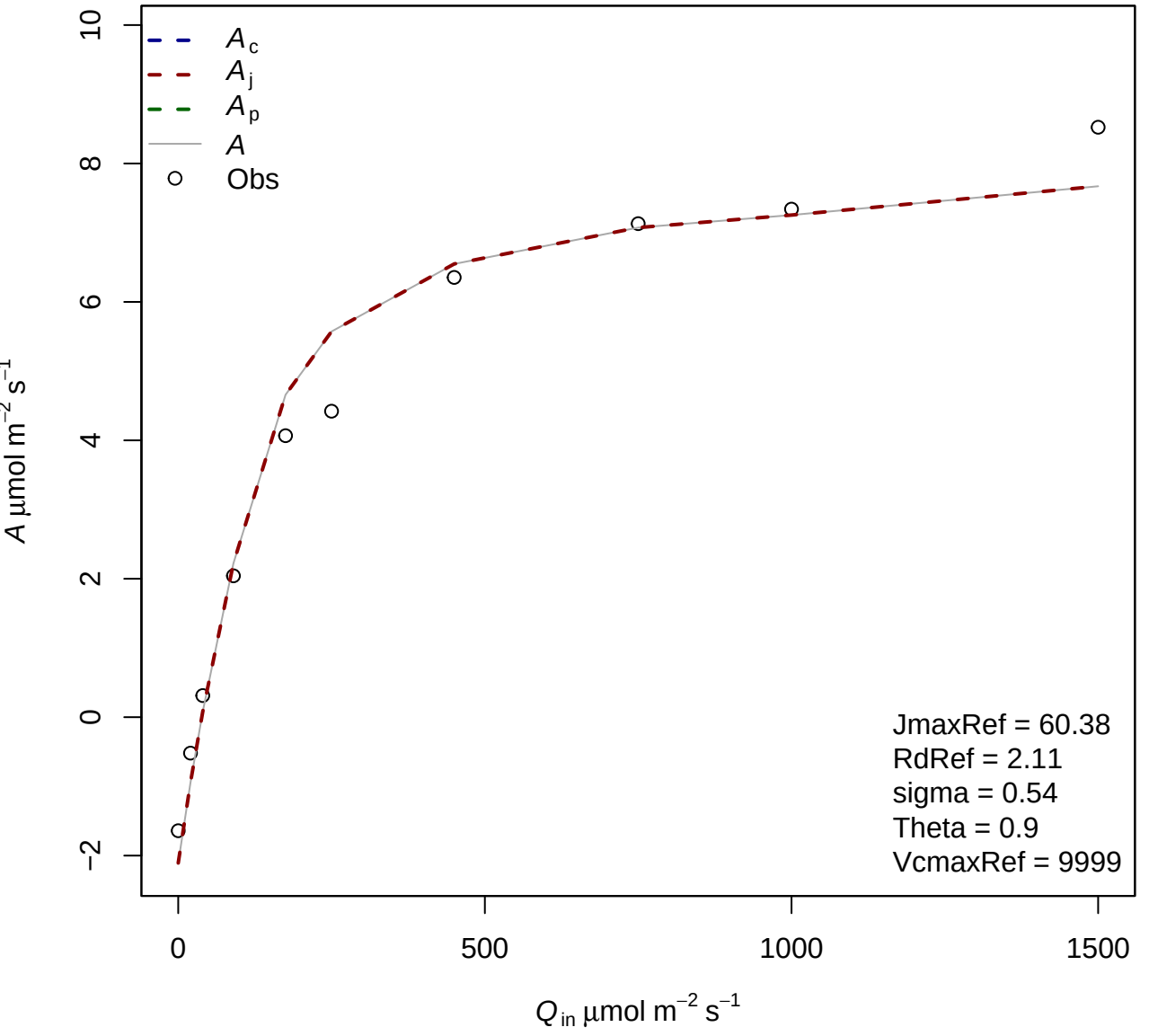

A_c
A_j
A_p
A
Obs
JmaxRef = 100.03
RdRef = 2.33
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

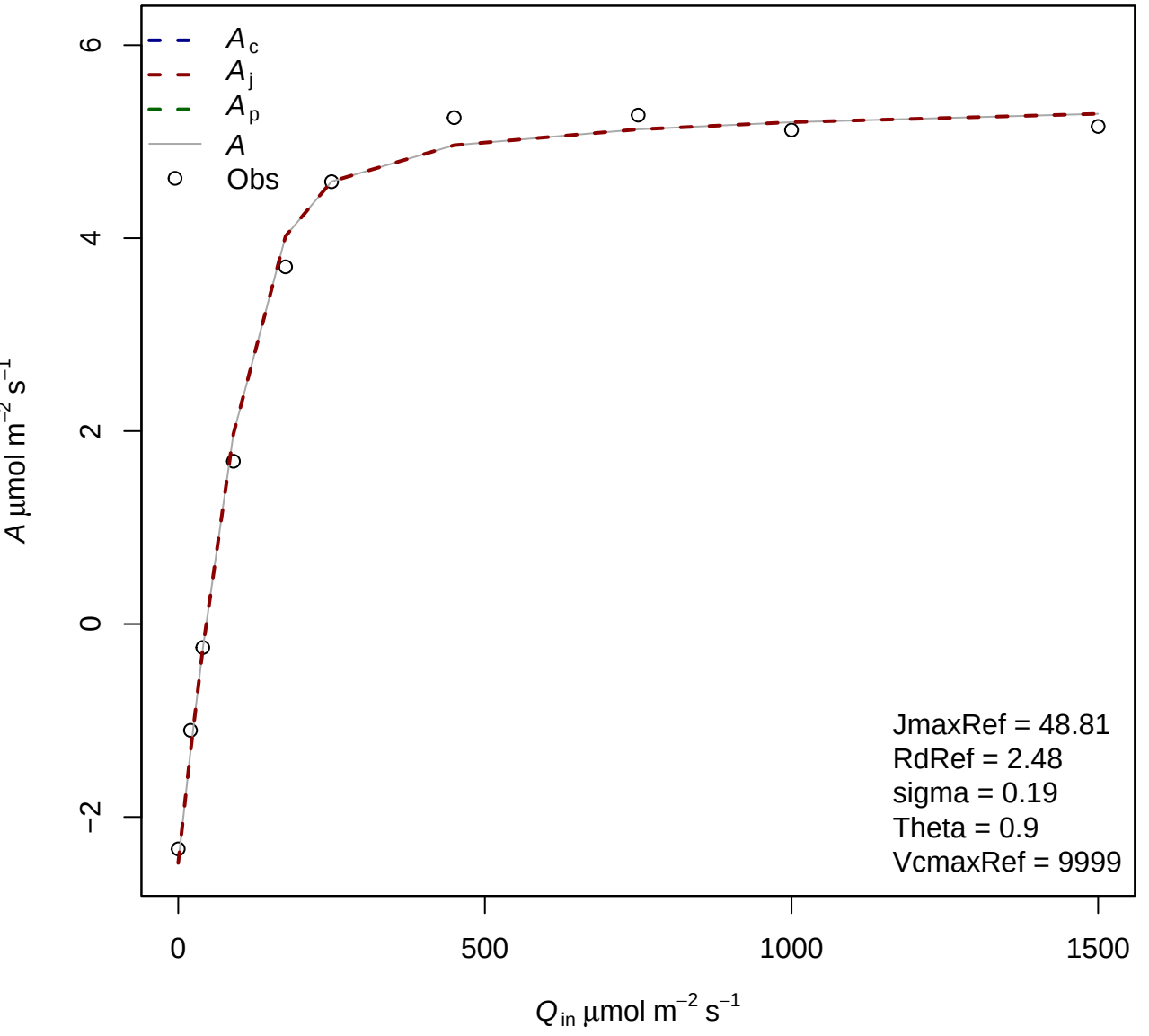

50
A_c
A_j
A_p
A
Obs
JmaxRef = 67.45
RdRef = 4.12
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

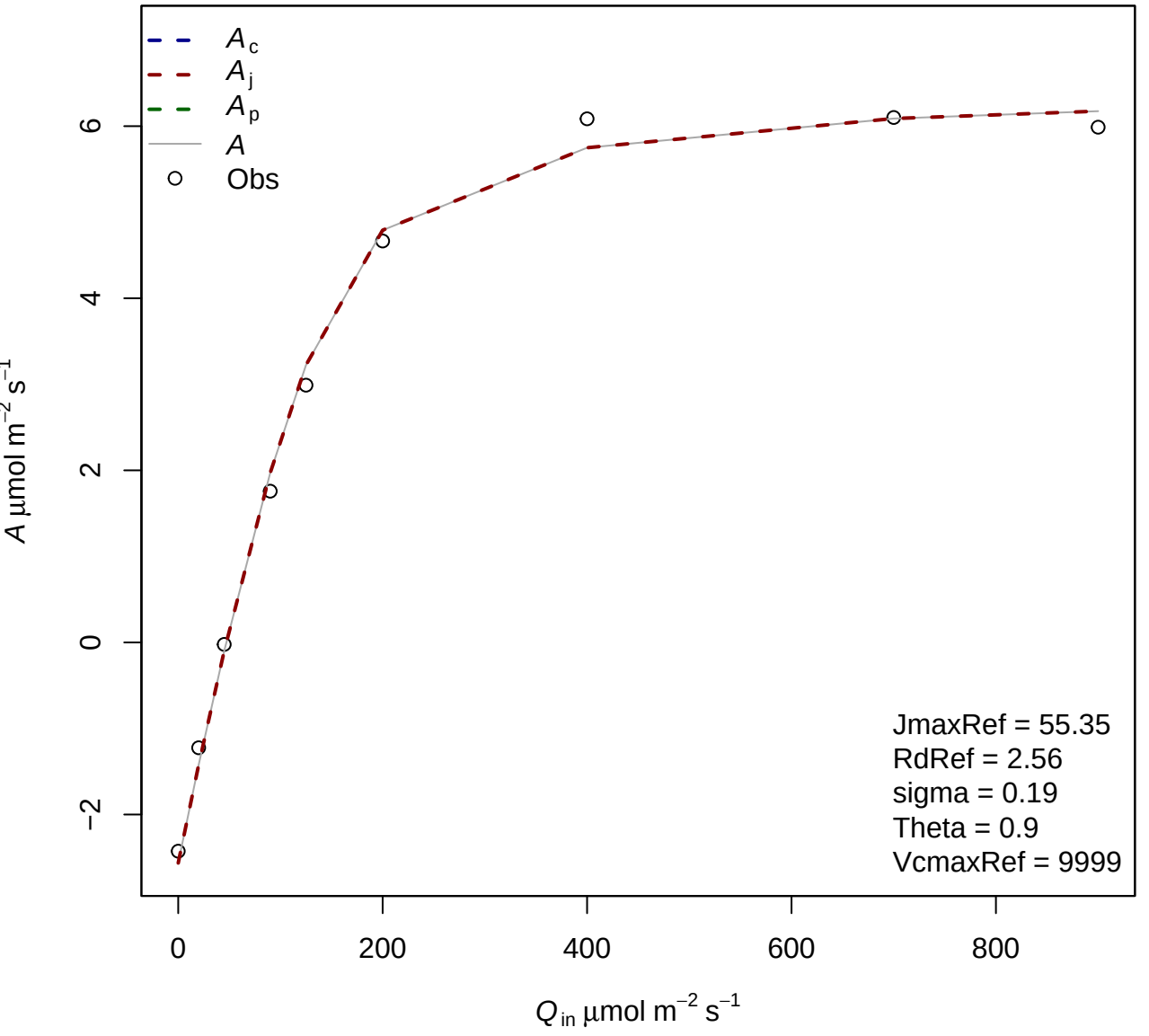

A_c
A_j
A_p
A
Obs
JmaxRef = 110.65
RdRef = 4.4
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

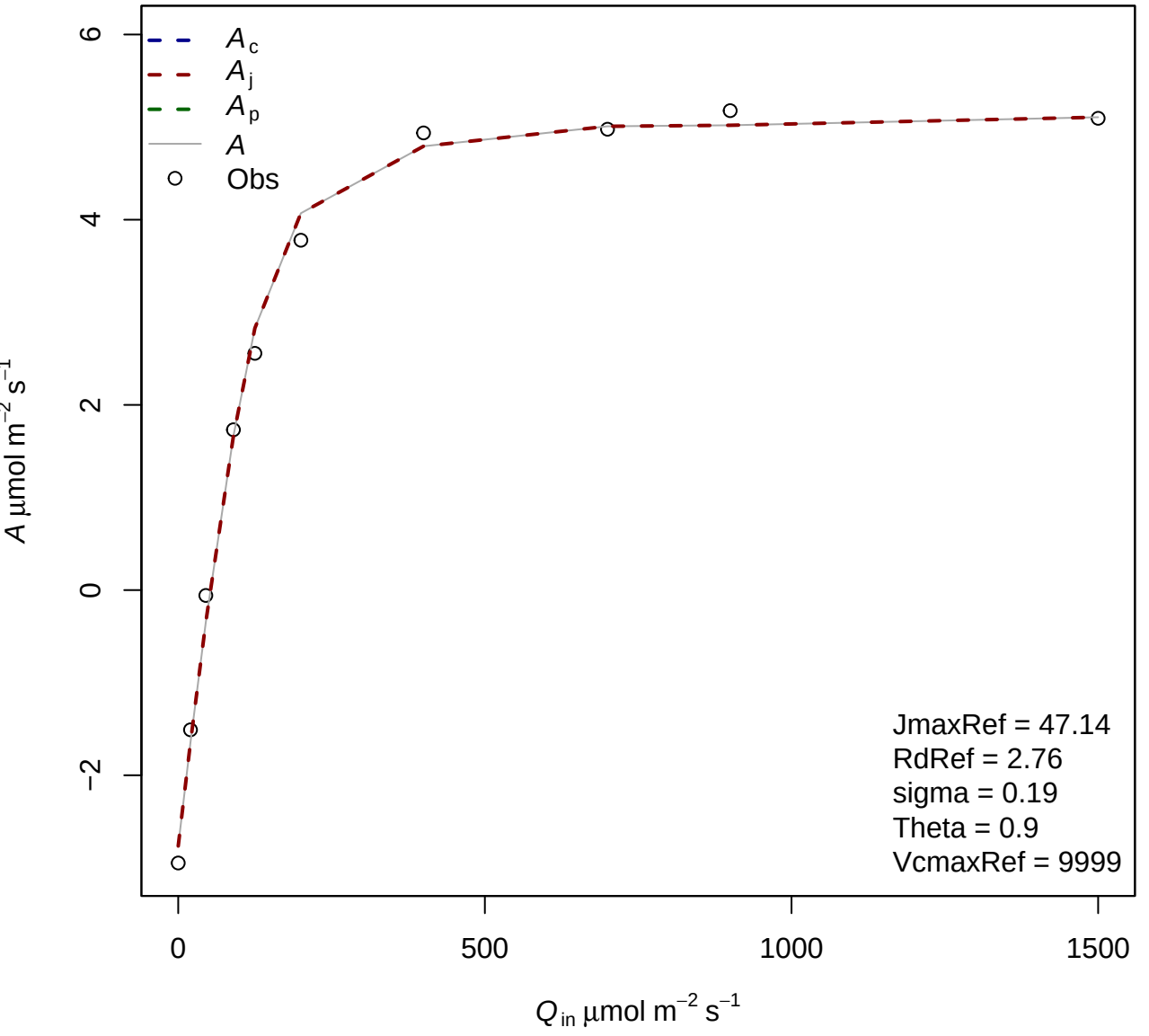

A_c
A_j
A_p
A
Obs
JmaxRef = 84.89
RdRef = 2.52
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

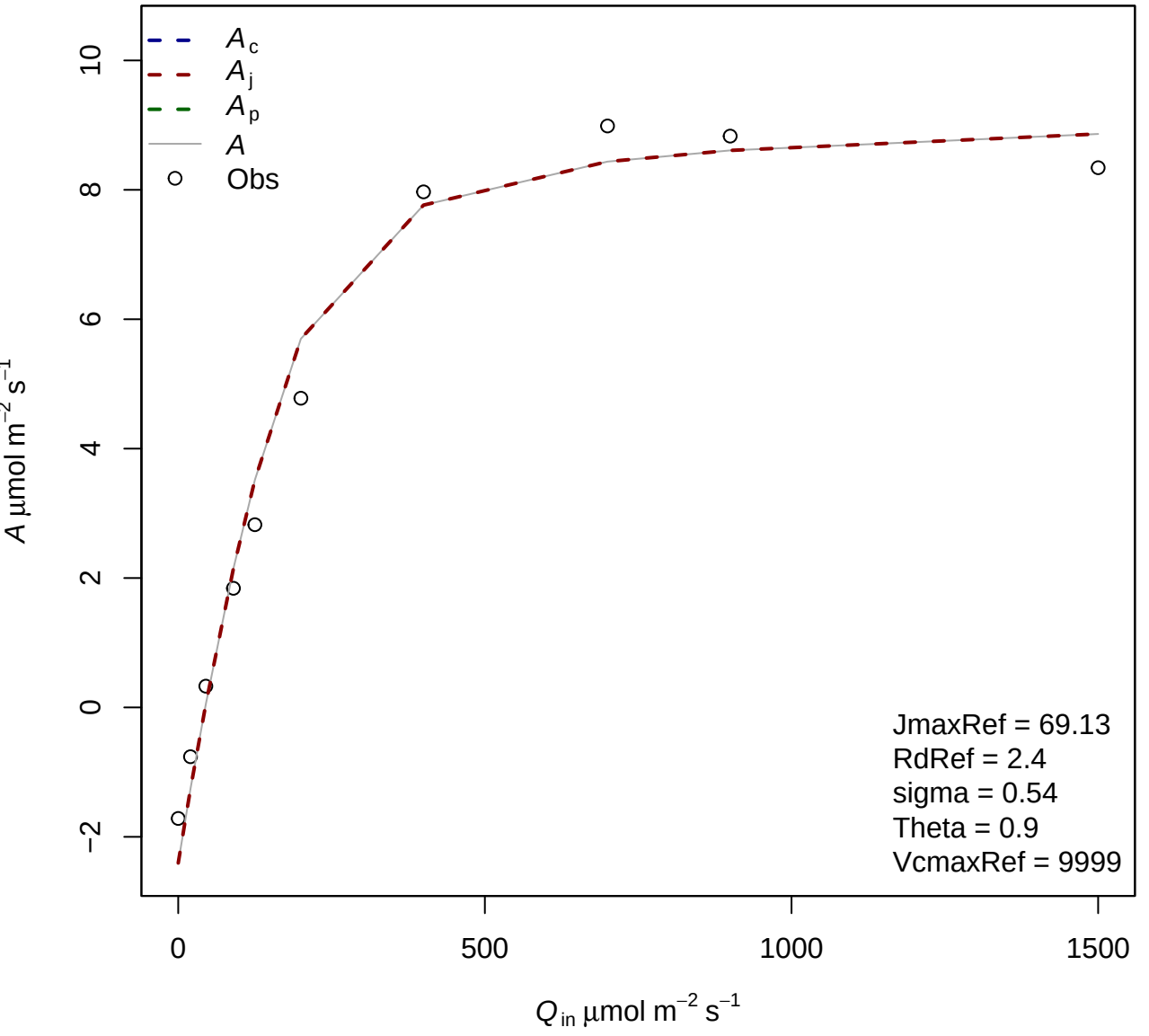

A_c
A_j
A_p
A
Obs
JmaxRef = 35.77
RdRef = 3.96
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
−4
0
500
1000
1500

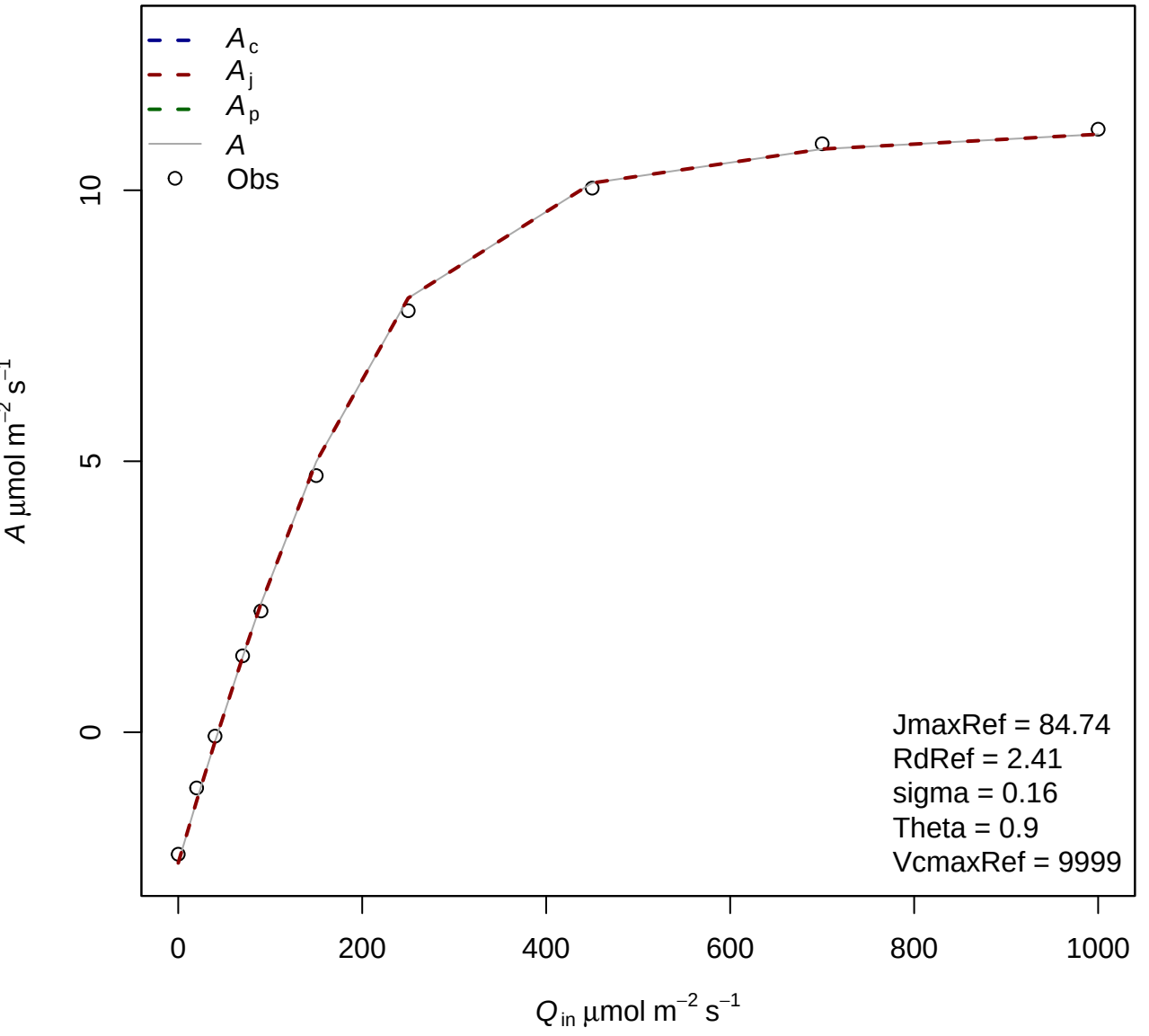

A_c
A_j
A_p
A
Obs
JmaxRef = 60.38
RdRef = 2.11
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

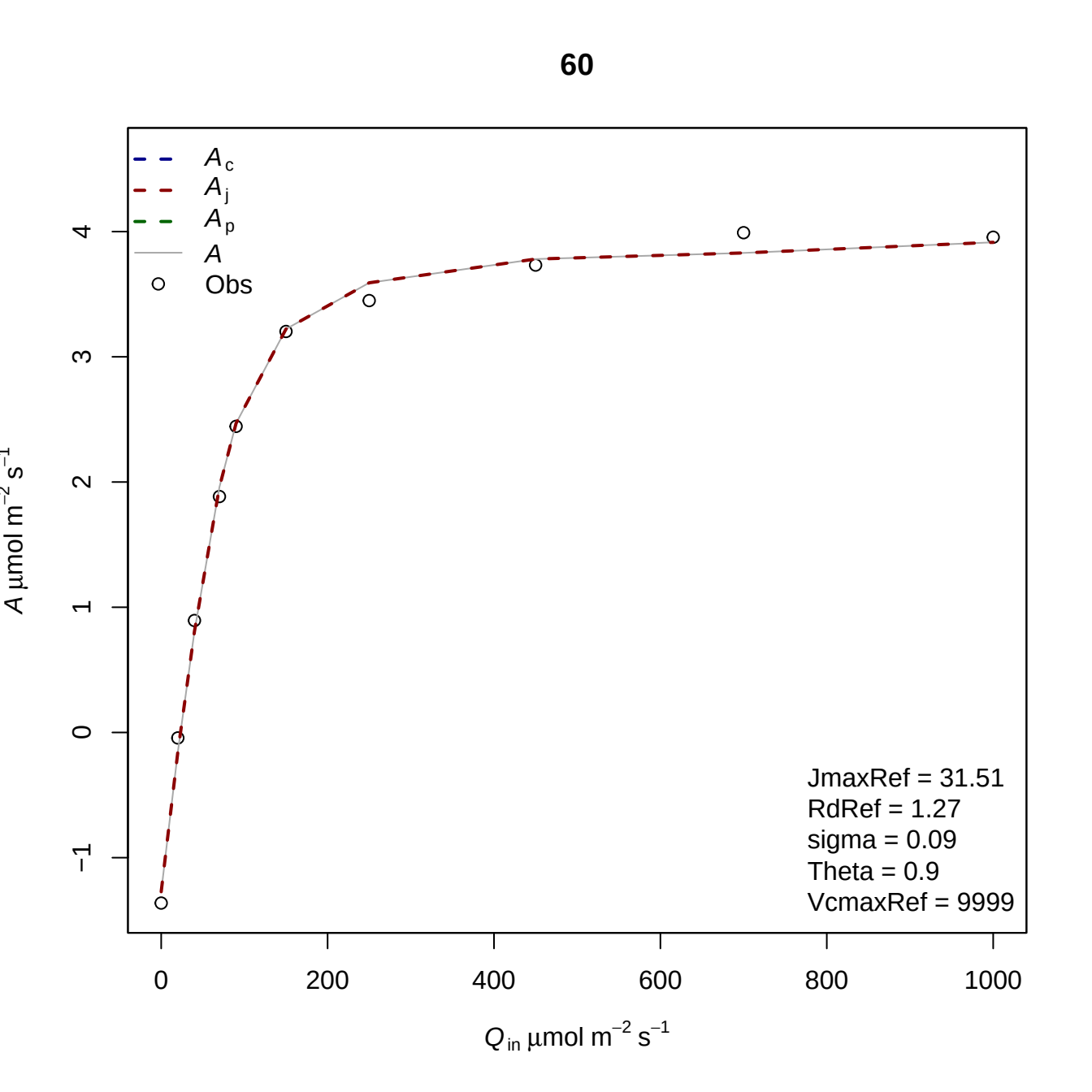

A_c
A_j
A_p
A
Obs
JmaxRef = 48.81
RdRef = 2.48
sigma = 0.19
Theta = 0.9
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

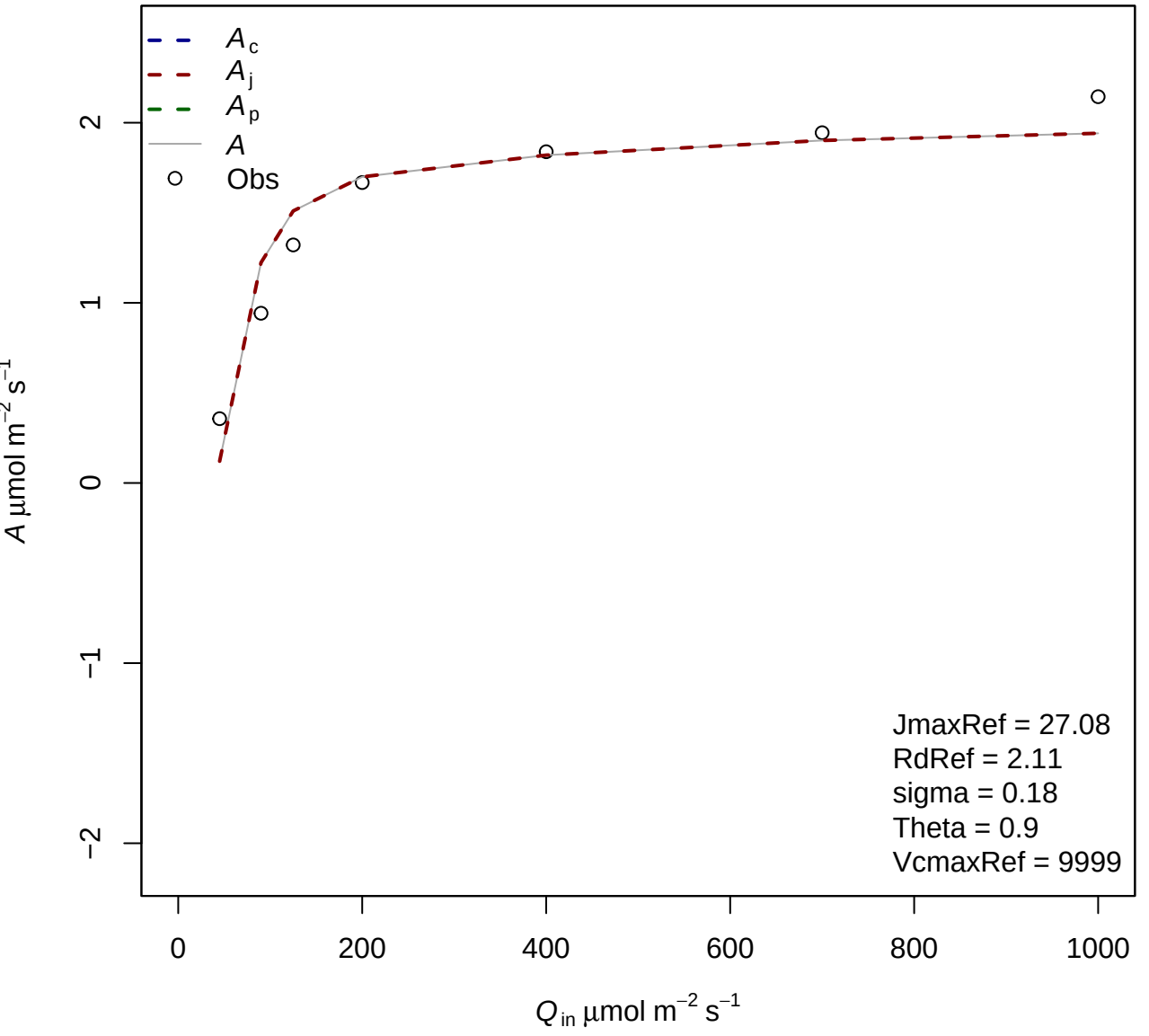

A_c
A_j
A_p
A
Obs
JmaxRef = 55.35
RdRef = 2.56
sigma = 0.19
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

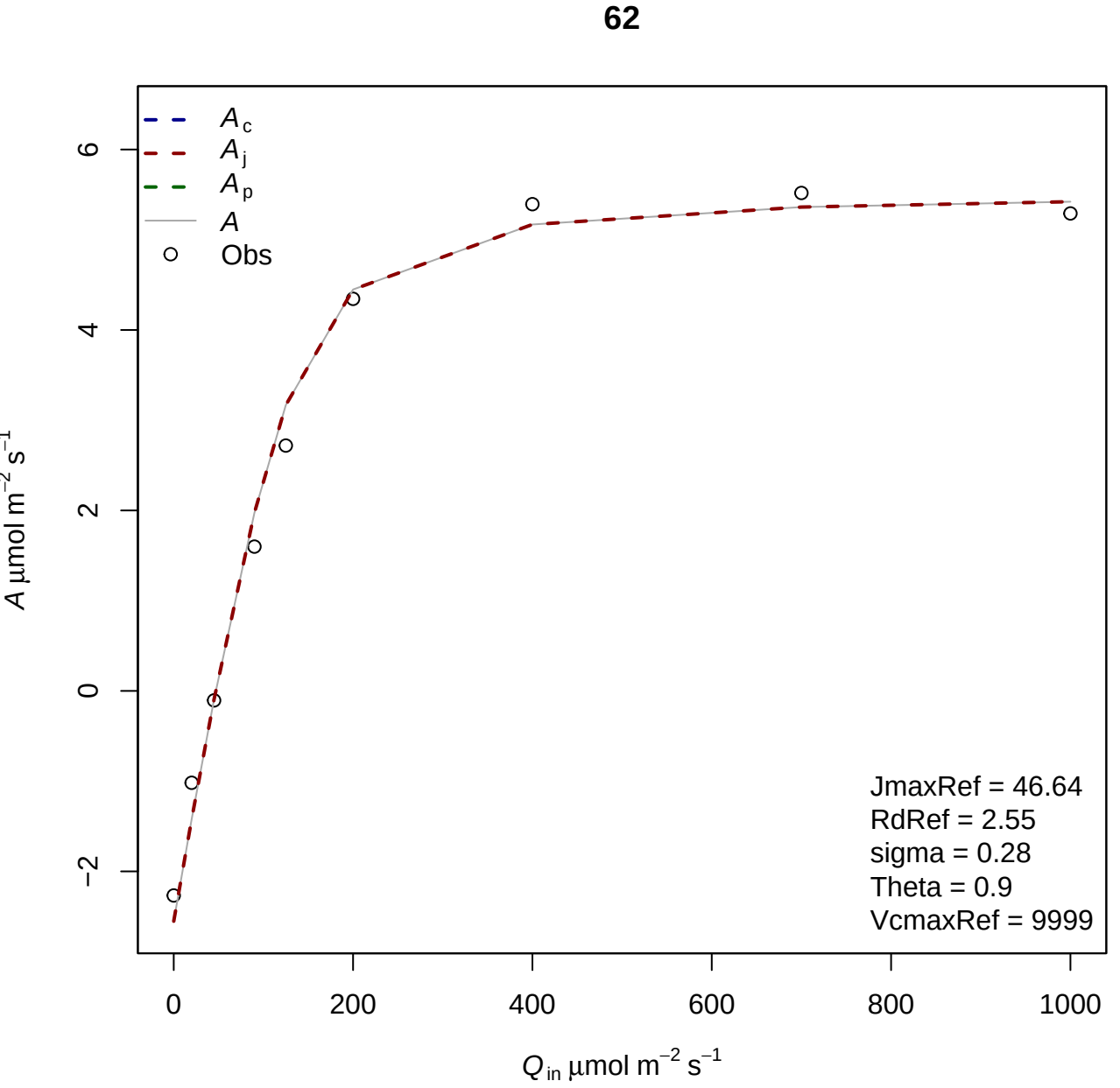

A_c
A_j
A_p
A
Obs
JmaxRef = 47.14
RdRef = 2.76
sigma = 0.19
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

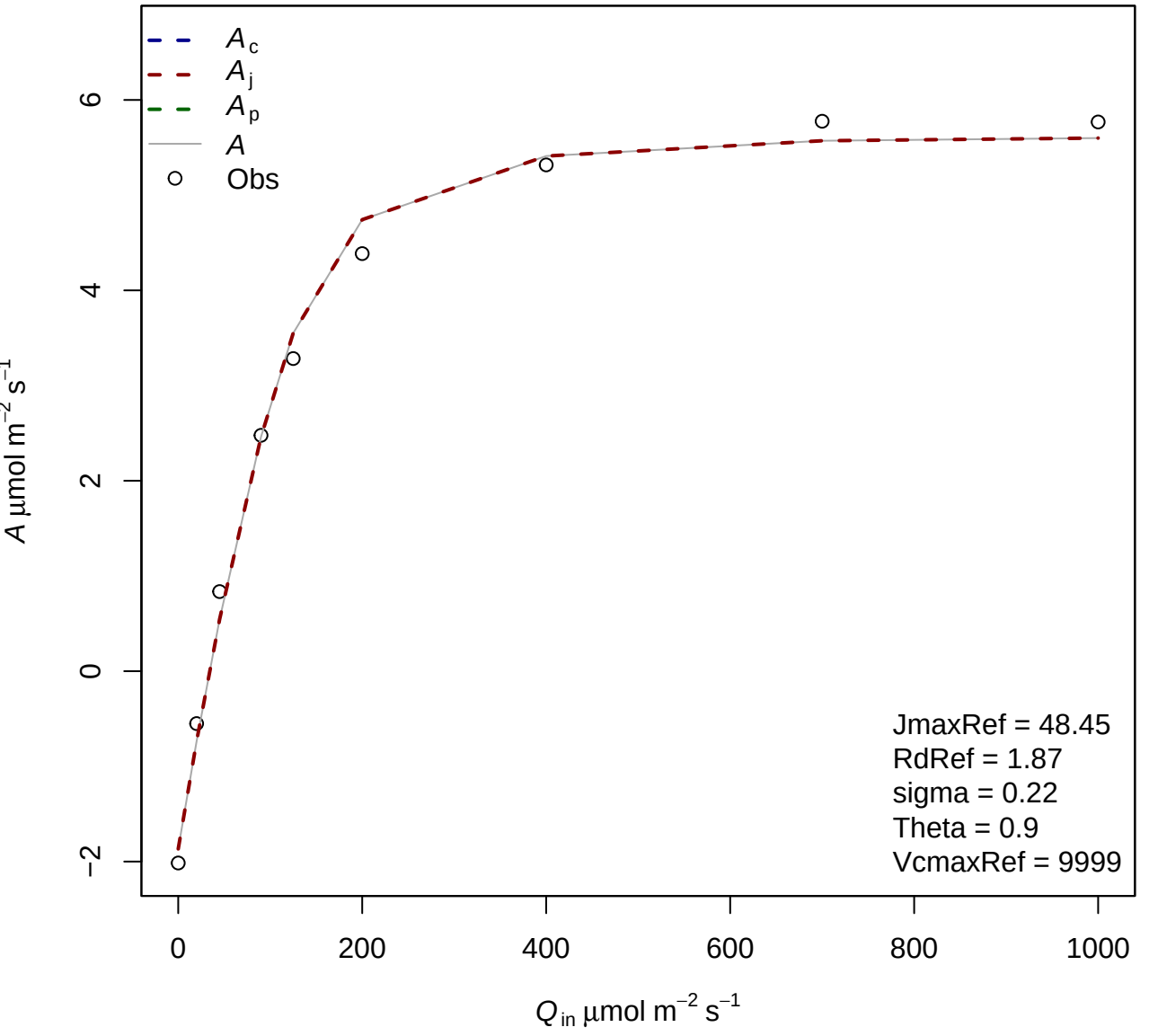

A_c
A_j
A_p
A
Obs
JmaxRef = 69.13
RdRef = 2.4
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

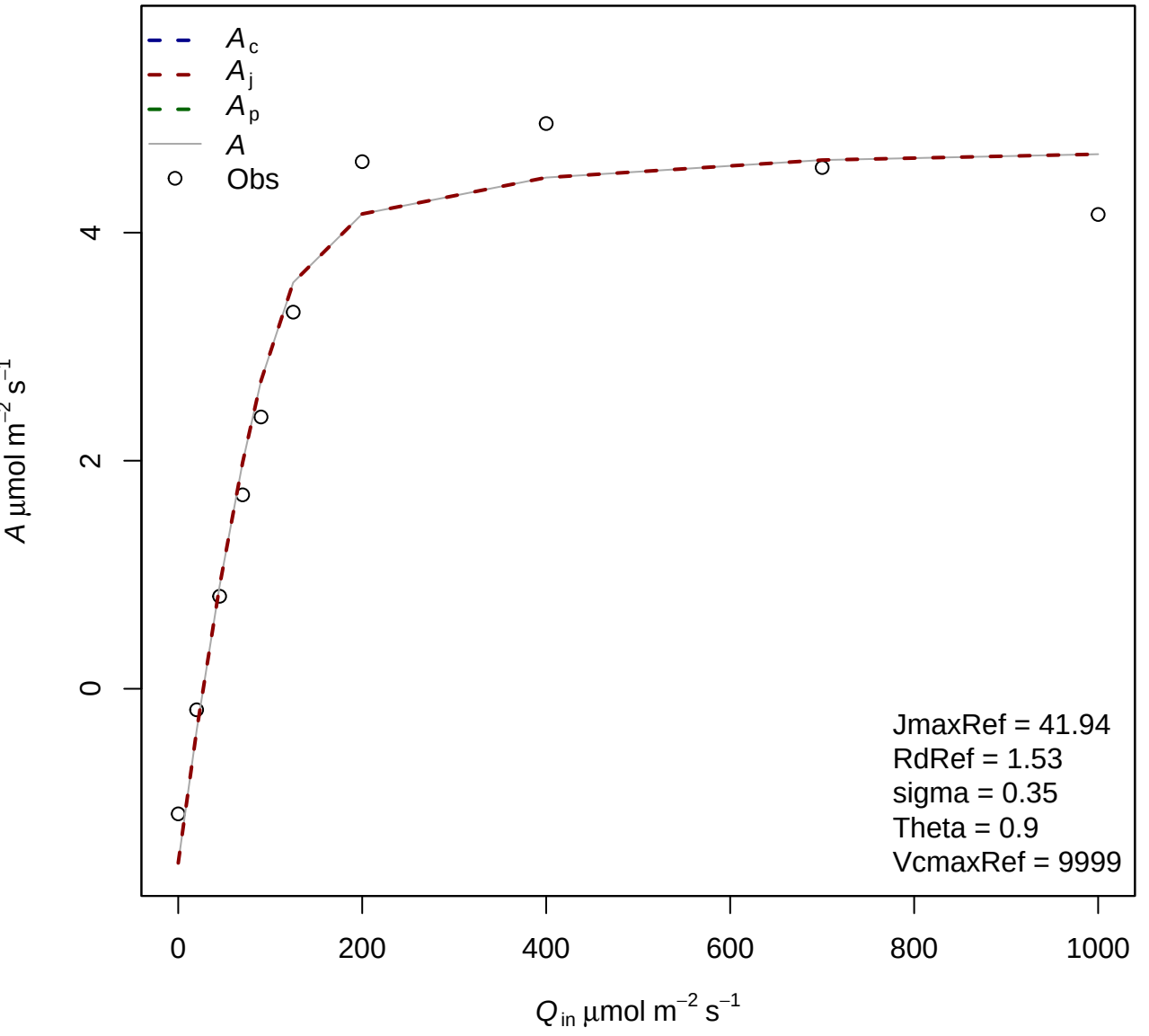

A_c
A_j
A_p
A
Obs
JmaxRef = 84.74
RdRef = 2.41
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

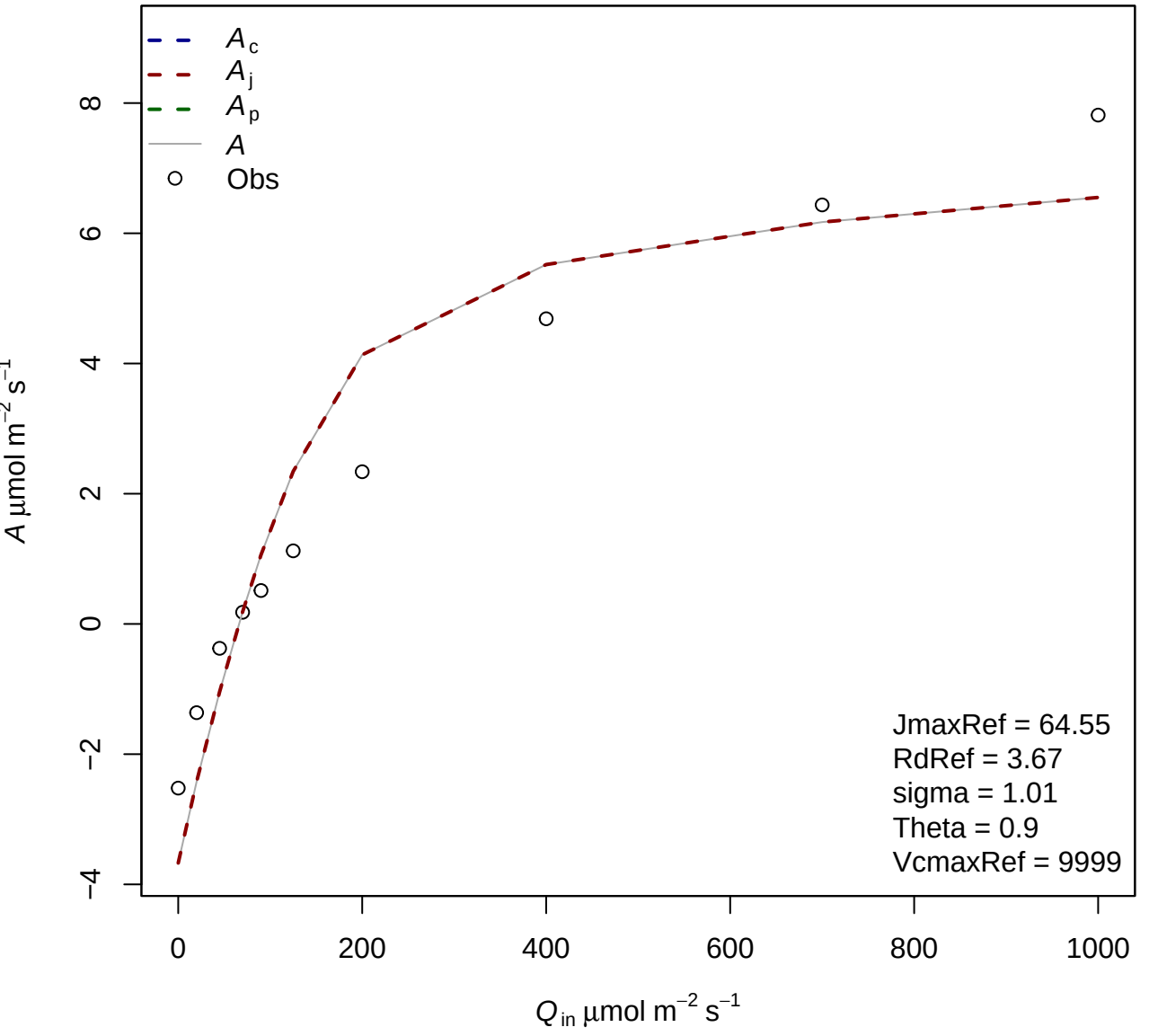

60
A_c
A_j
A_p
A
Obs
JmaxRef = 31.51
RdRef = 1.27
sigma = 0.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

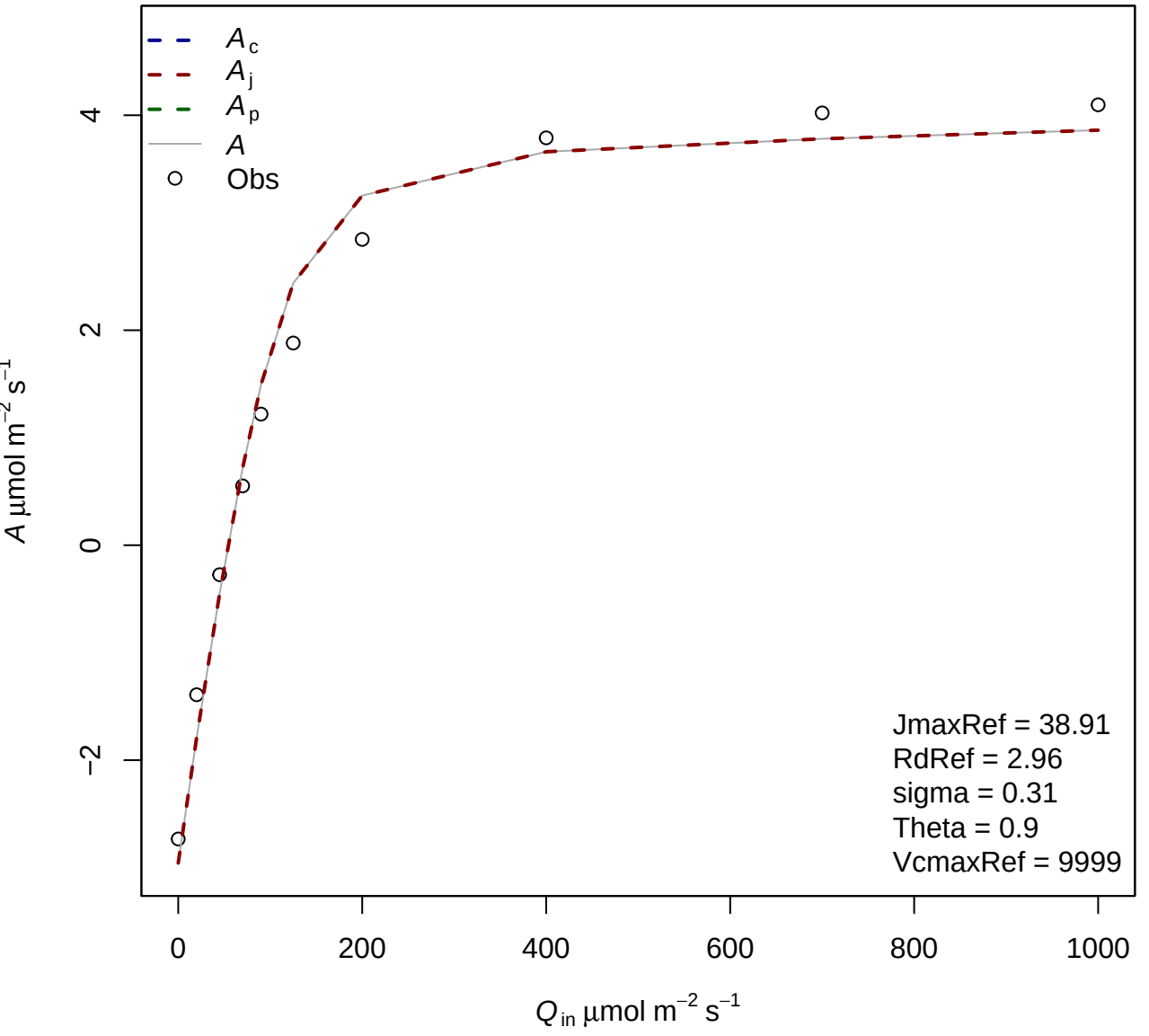

A_c
A_j
A_p
A
Obs
JmaxRef = 27.08
RdRef = 2.11
sigma = 0.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
200
400
600
800
1000

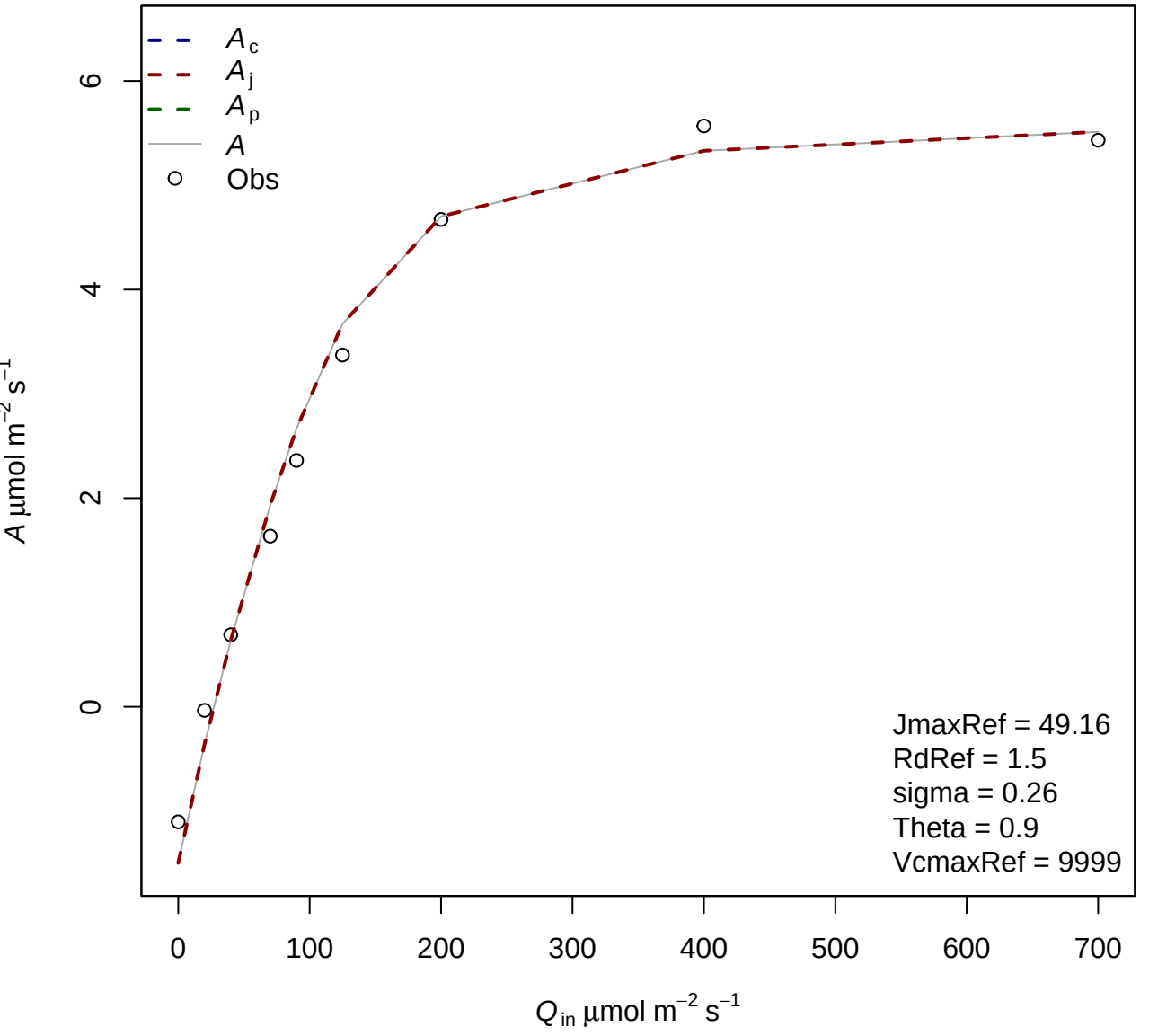

62
A_c
A_j
A_p
A
Obs
JmaxRef = 46.64
RdRef = 2.55
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

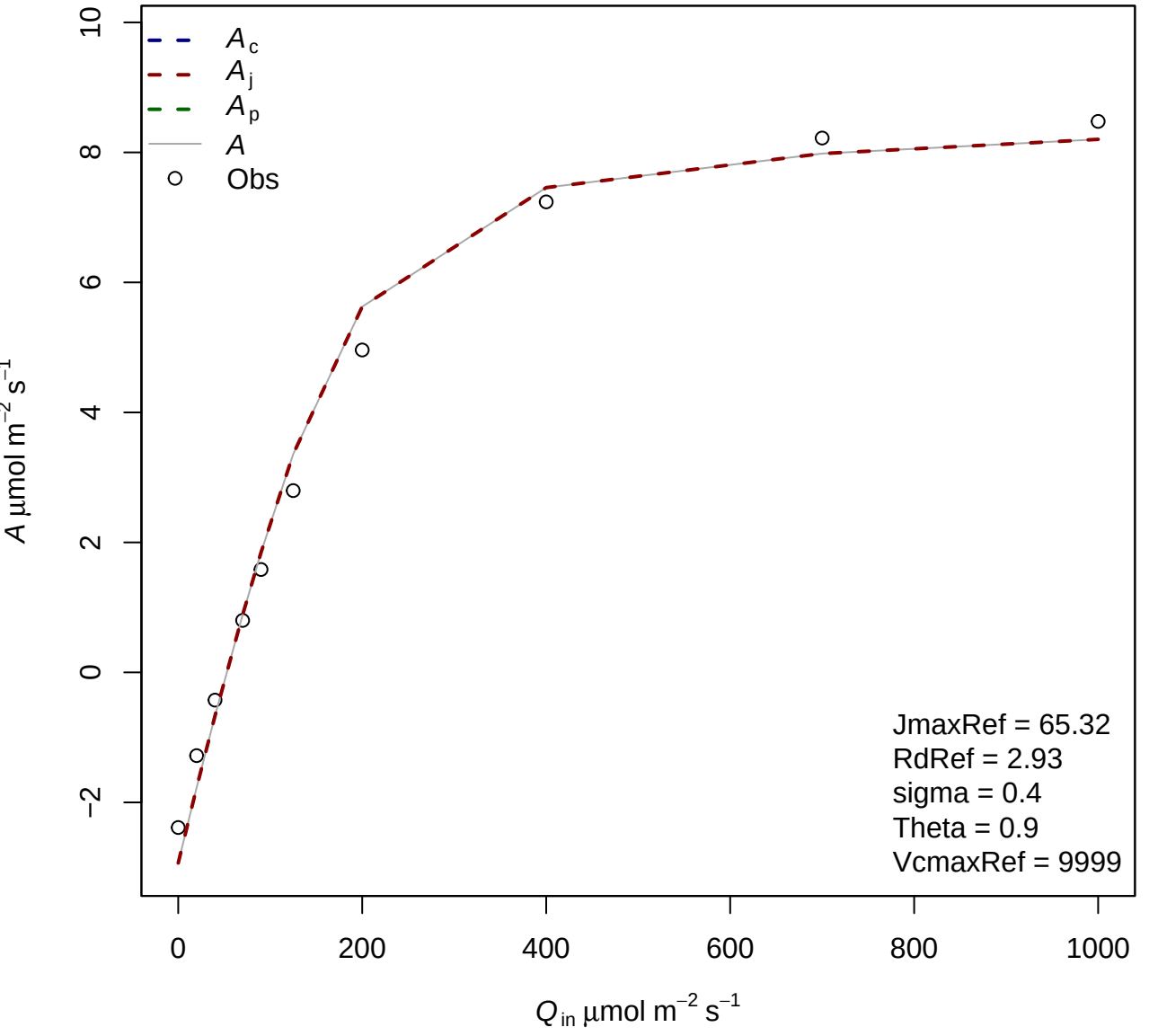

A_c
A_j
A_p
A
Obs
JmaxRef = 48.45
RdRef = 1.87
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

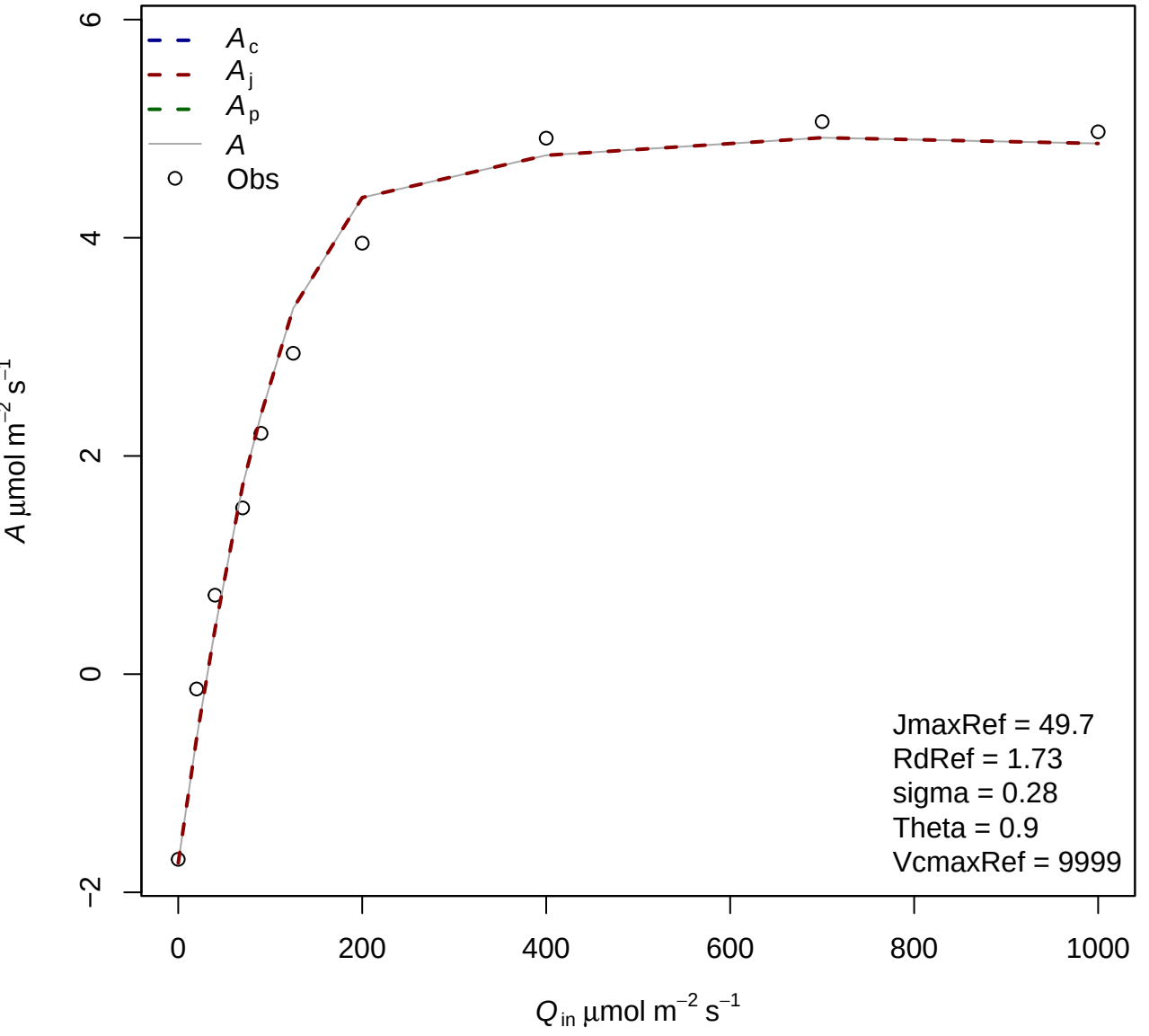

A_c
A_j
A_p
A
Obs
JmaxRef = 41.94
RdRef = 1.53
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
0
2
4
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

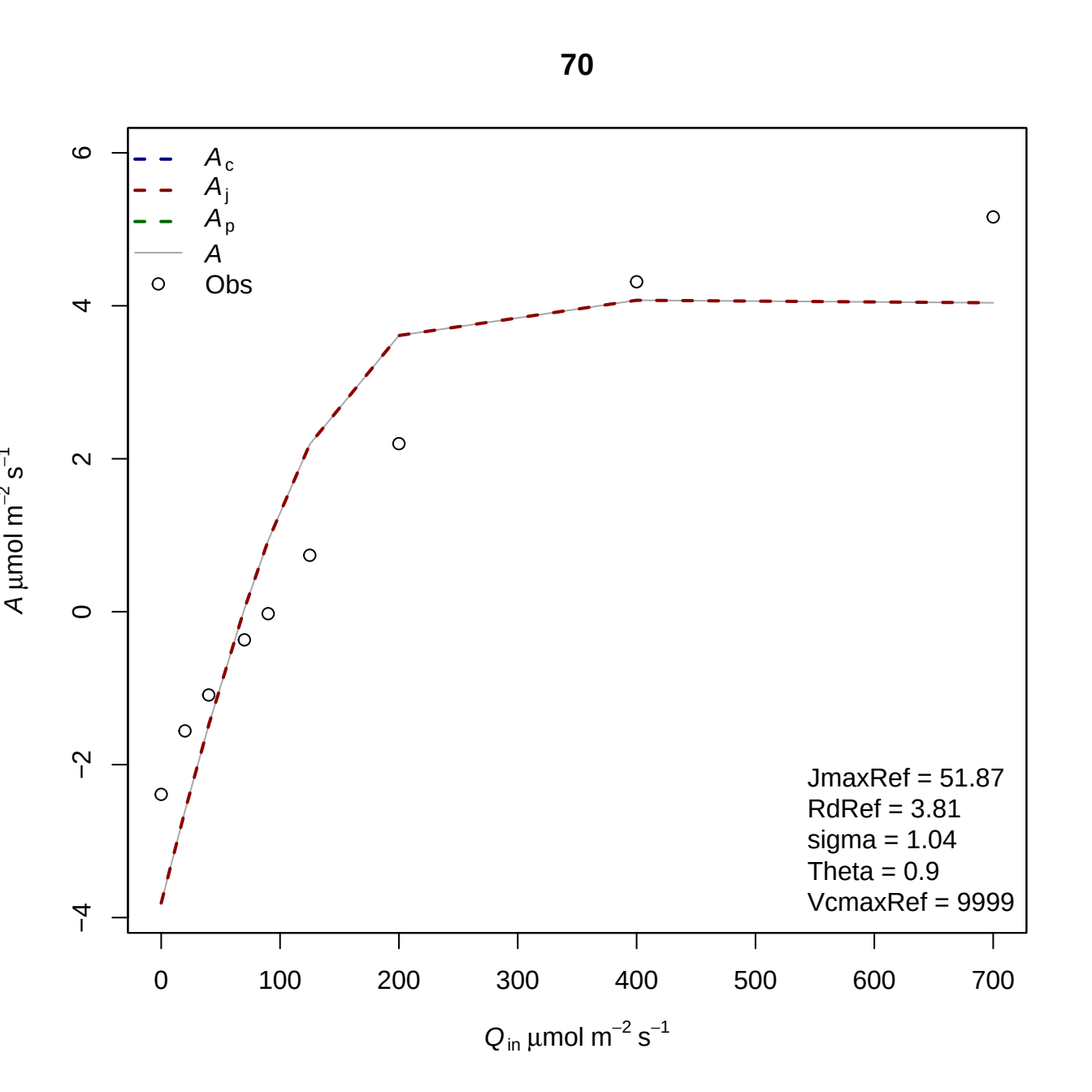

A_c
A_j
A_p
A
Obs
JmaxRef = 64.55
RdRef = 3.67
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000

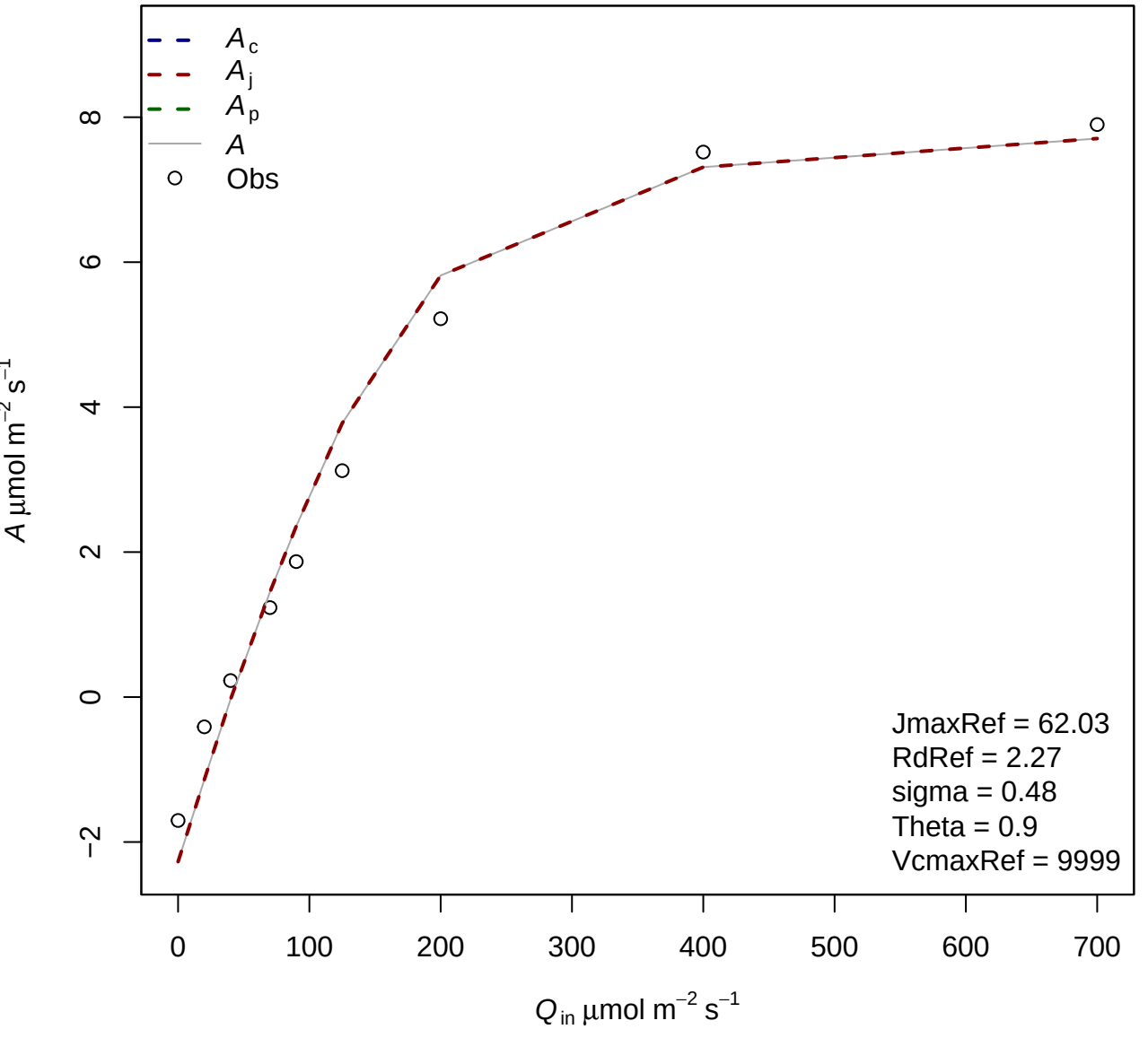

A_c
A_j
A_p
A
Obs
JmaxRef = 38.91
RdRef = 2.96
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

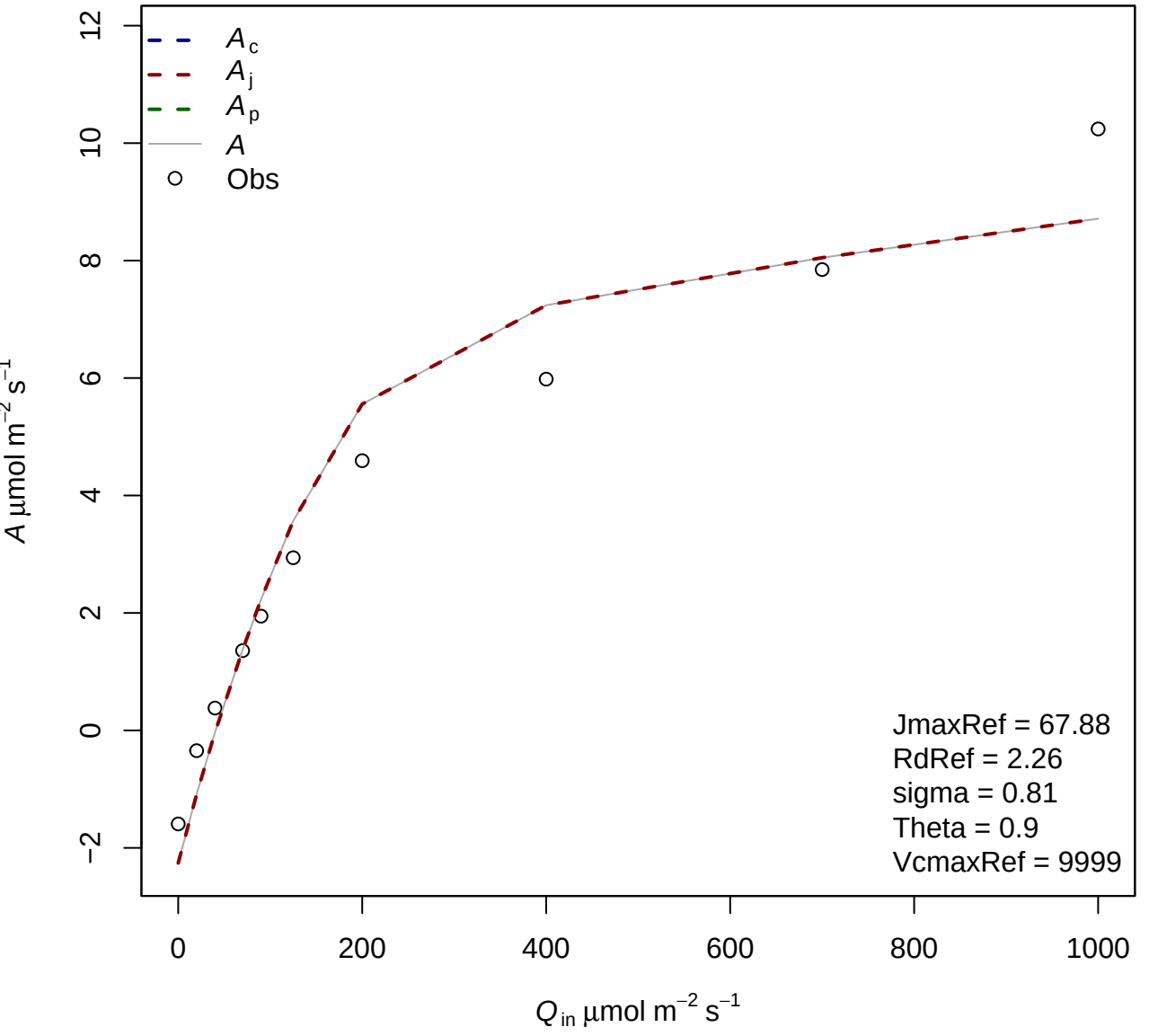

A_c
A_j
A_p
A
Obs
JmaxRef = 49.16
RdRef = 1.5
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
100
200
300
400
500
600
700

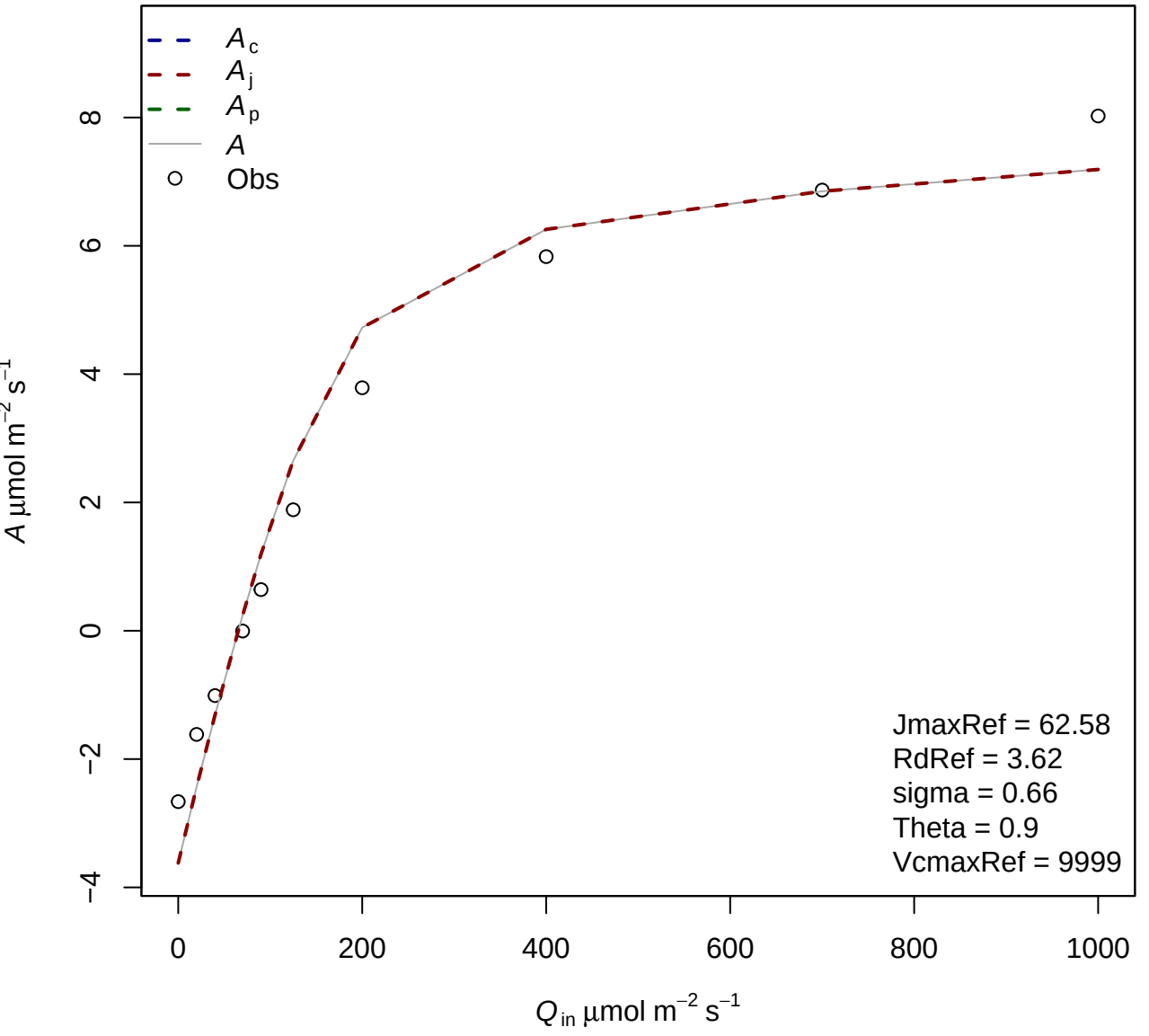

A_c
A_j
A_p
A
Obs
JmaxRef = 65.32
RdRef = 2.93
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

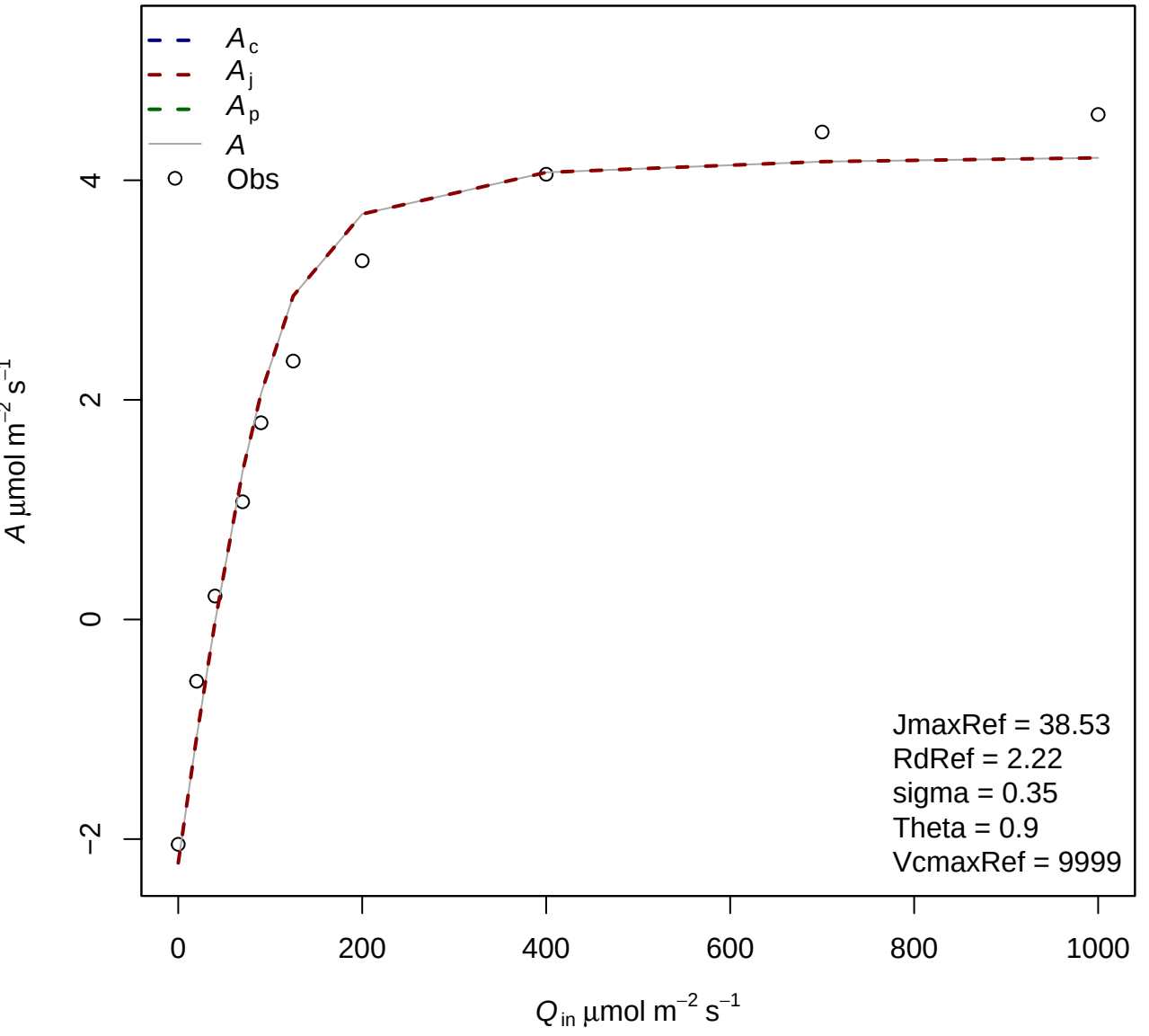

A_c
A_j
A_p
A
Obs
JmaxRef = 49.7
RdRef = 1.73
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

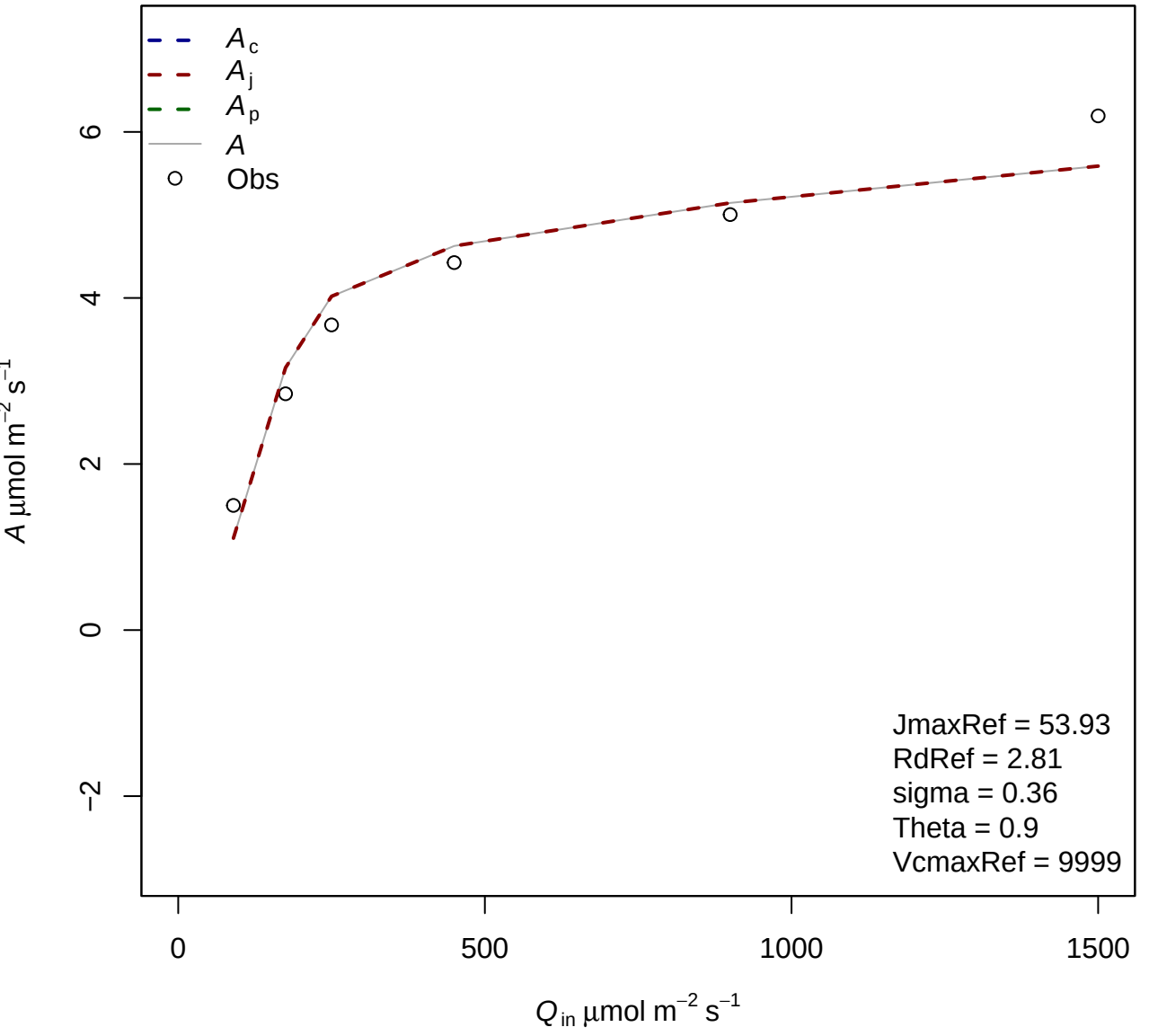

70
A_c
A_j
A_p
A
Obs
JmaxRef = 51.87
RdRef = 3.81
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
100
200
300
400
500
600
700

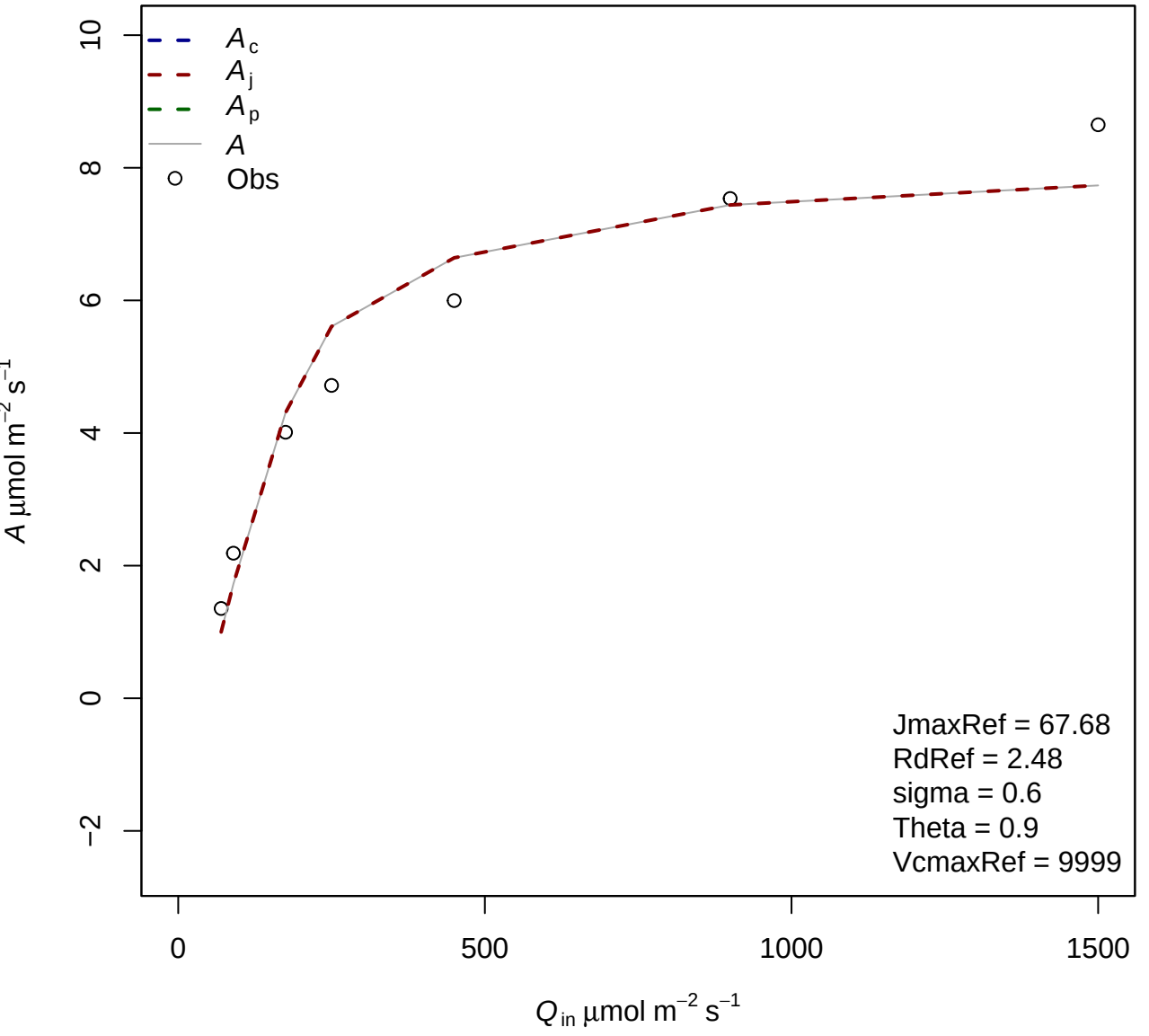

A_c
A_j
A_p
A
Obs
JmaxRef = 62.03
RdRef = 2.27
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

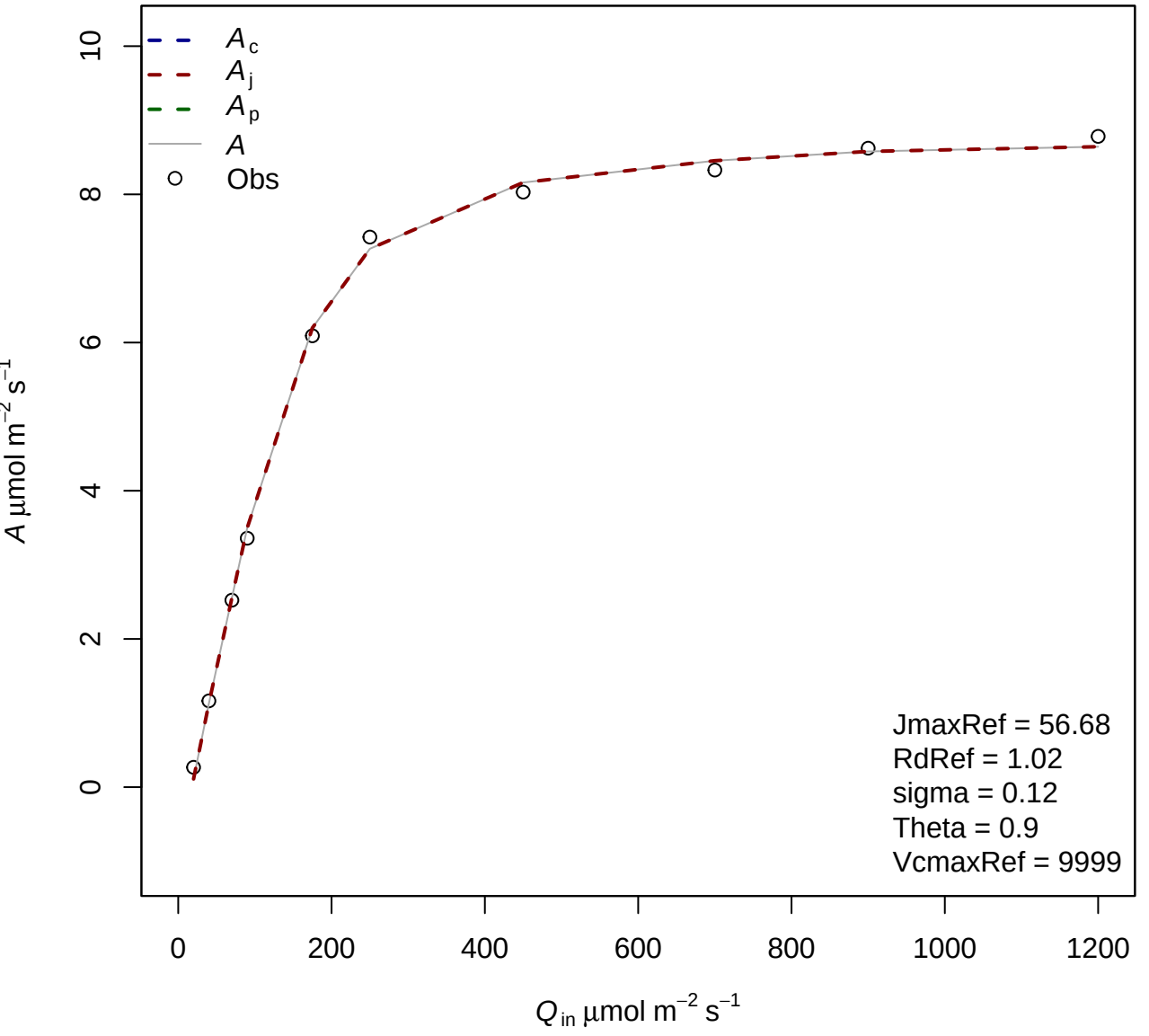

A_c
A_j
A_p
A
Obs
JmaxRef = 67.88
RdRef = 2.26
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

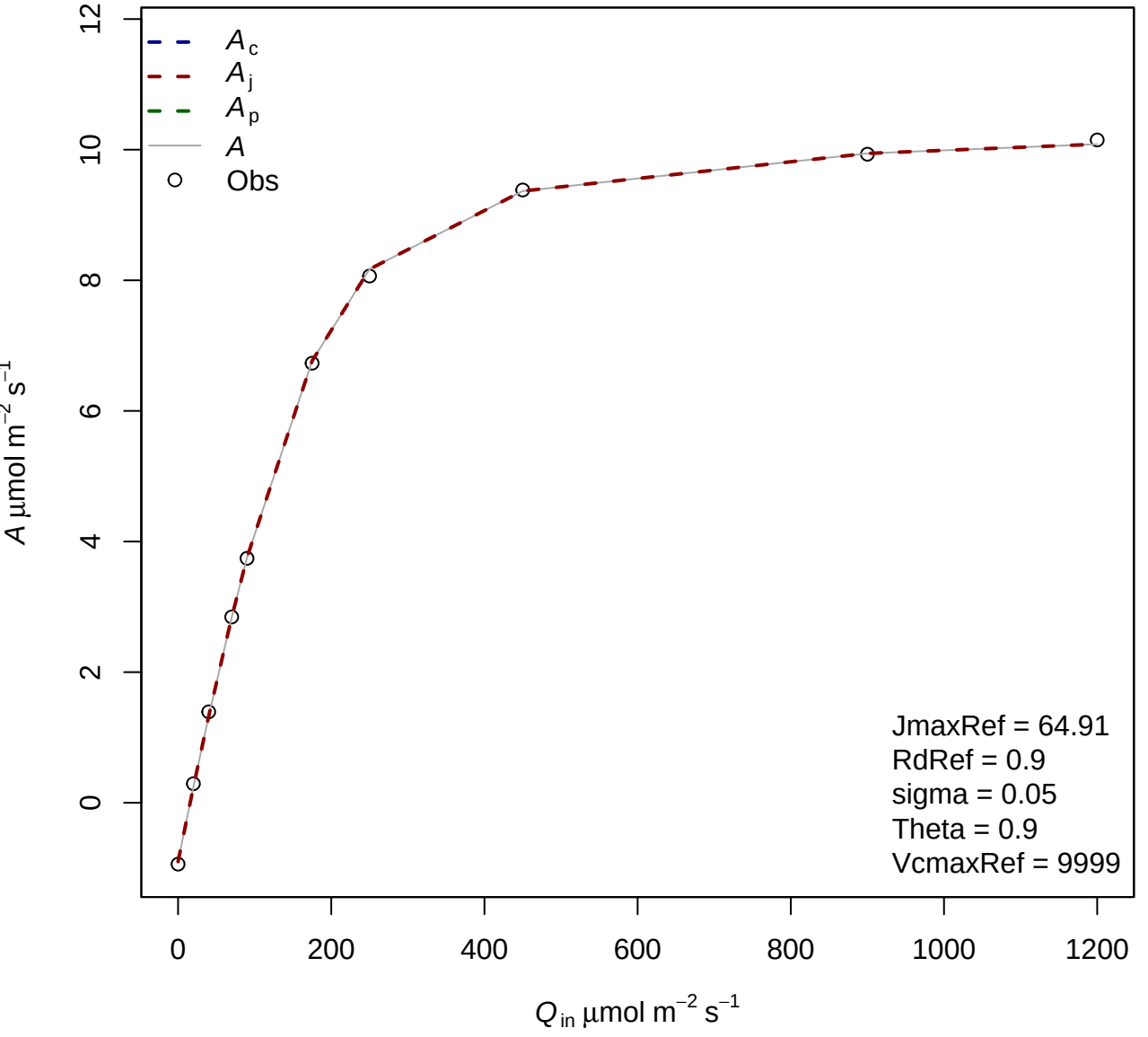

A_c
A_j
A_p
A
Obs
JmaxRef = 62.58
RdRef = 3.62
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
200
400
600
800
1000

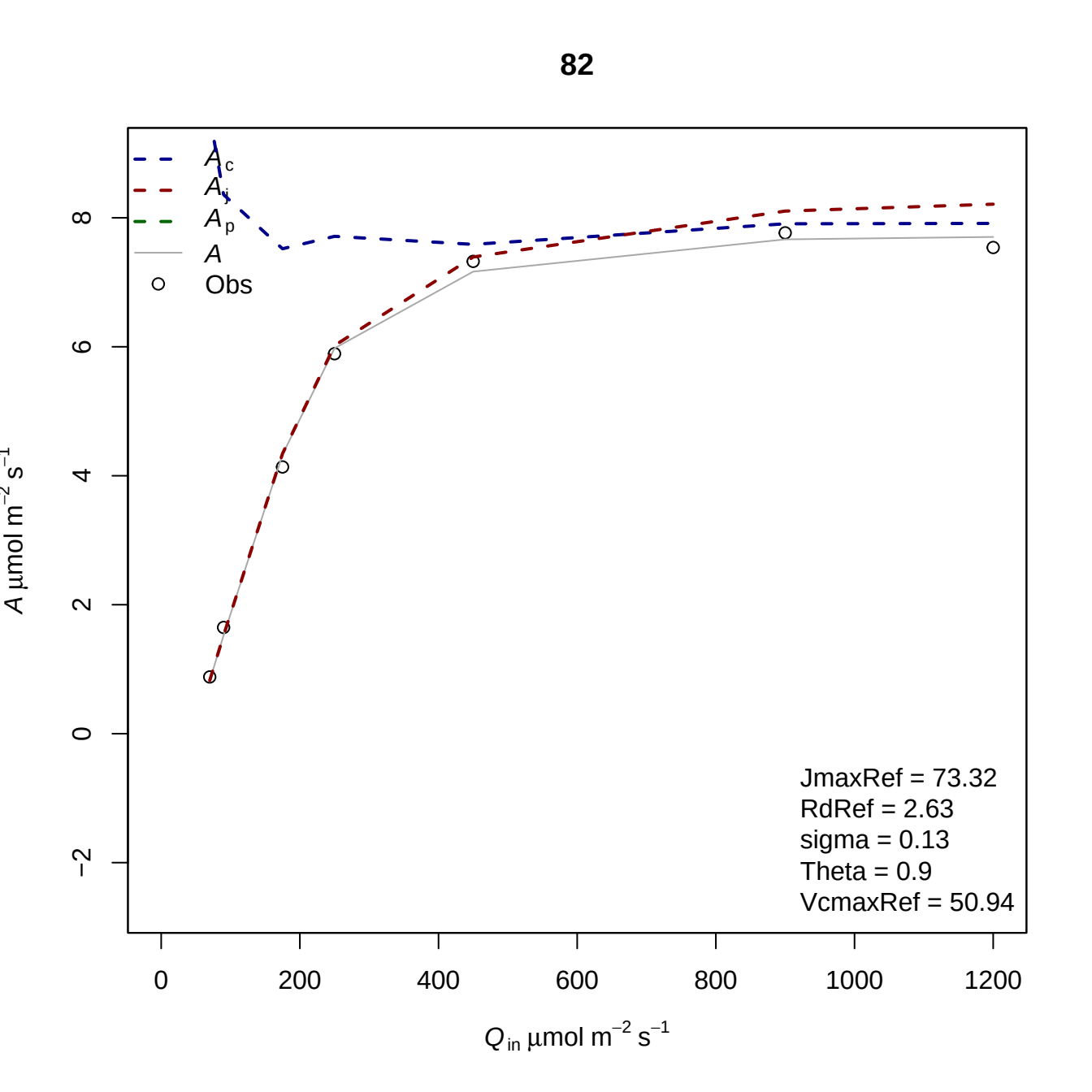

A_c
A_j
A_p
A
Obs
JmaxRef = 38.53
RdRef = 2.22
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

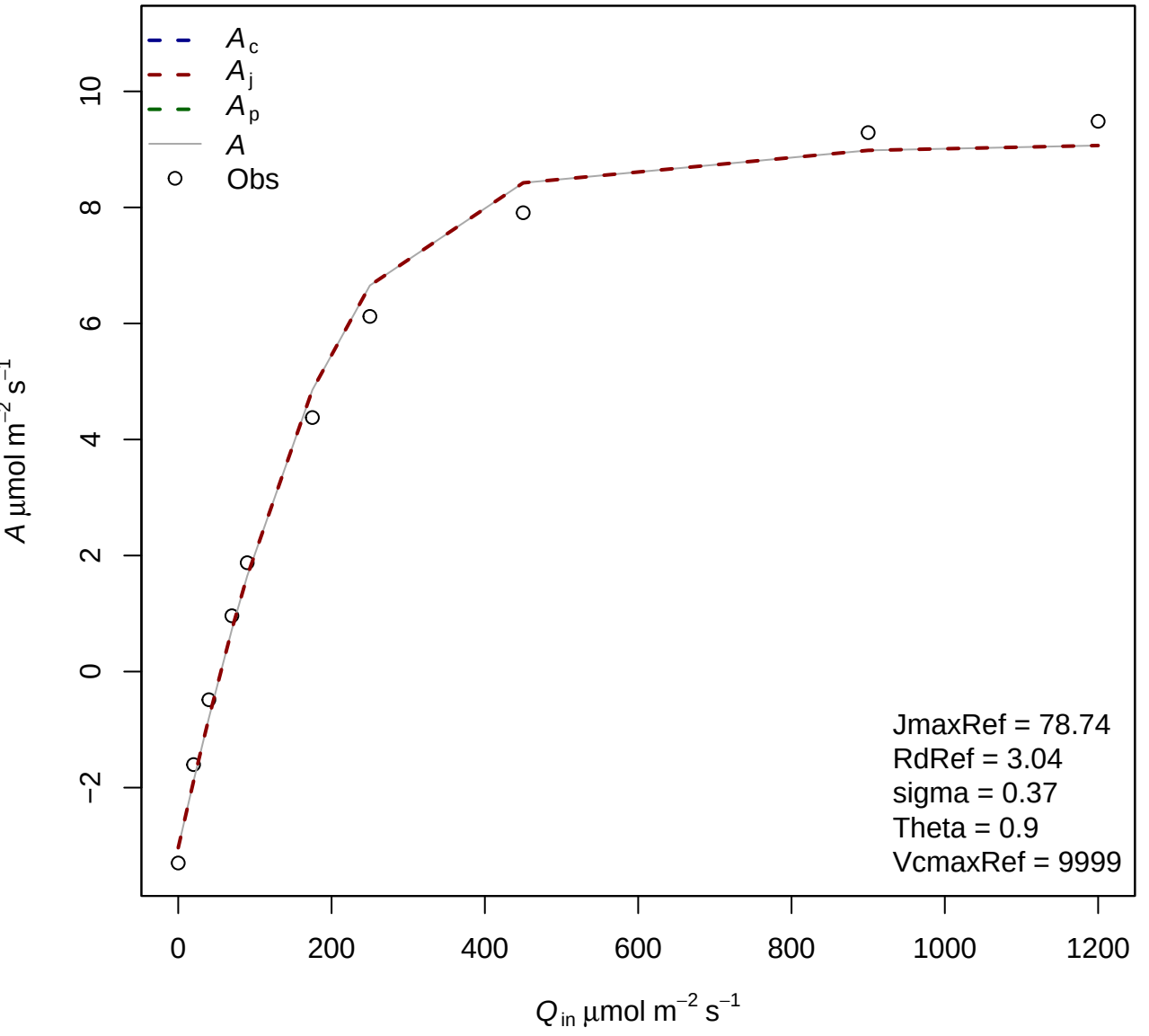

A_c
A_j
A_p
A
Obs
JmaxRef = 53.93
RdRef = 2.81
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

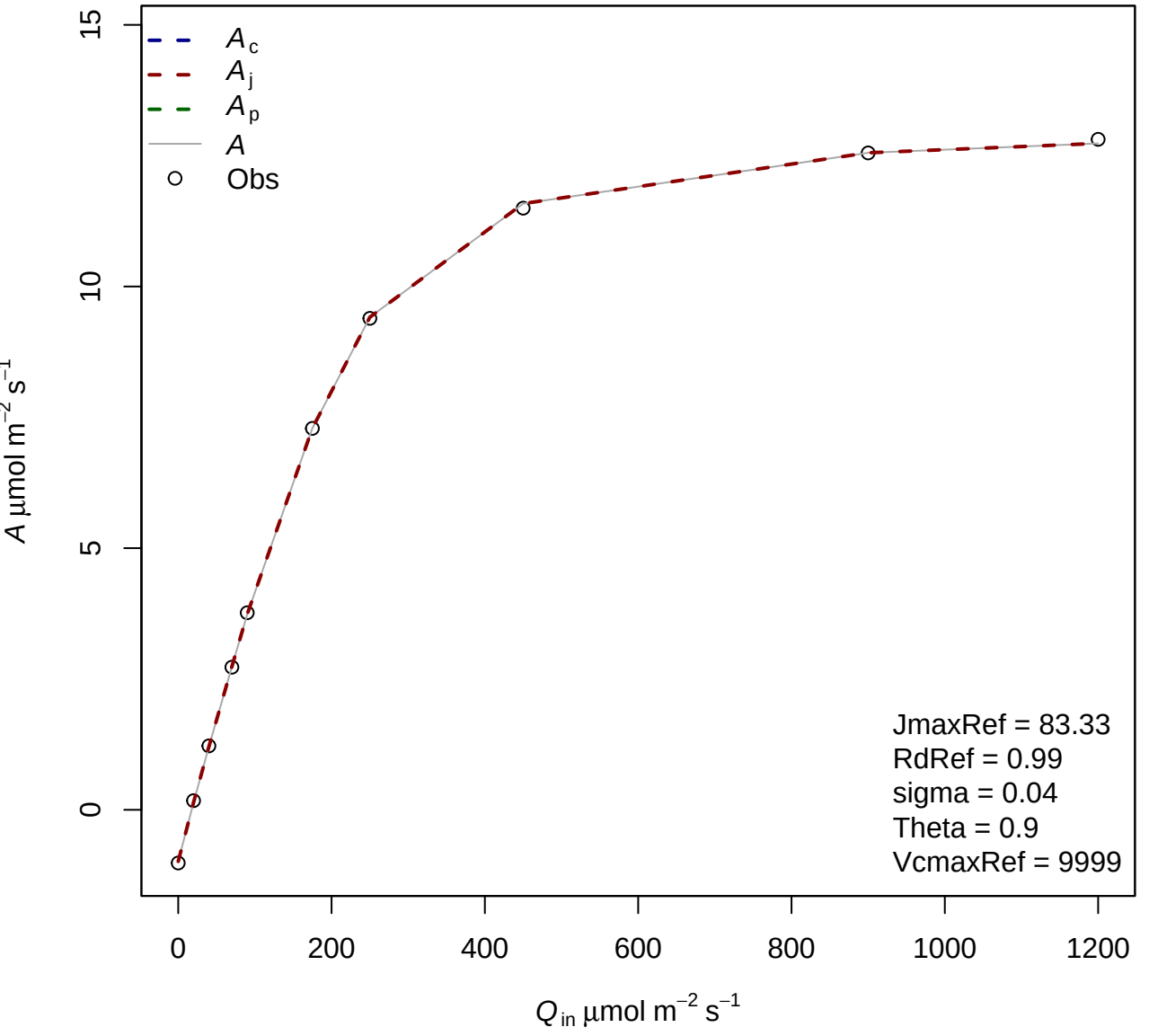

A_c
A_j
A_p
A
Obs
JmaxRef = 67.68
RdRef = 2.48
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

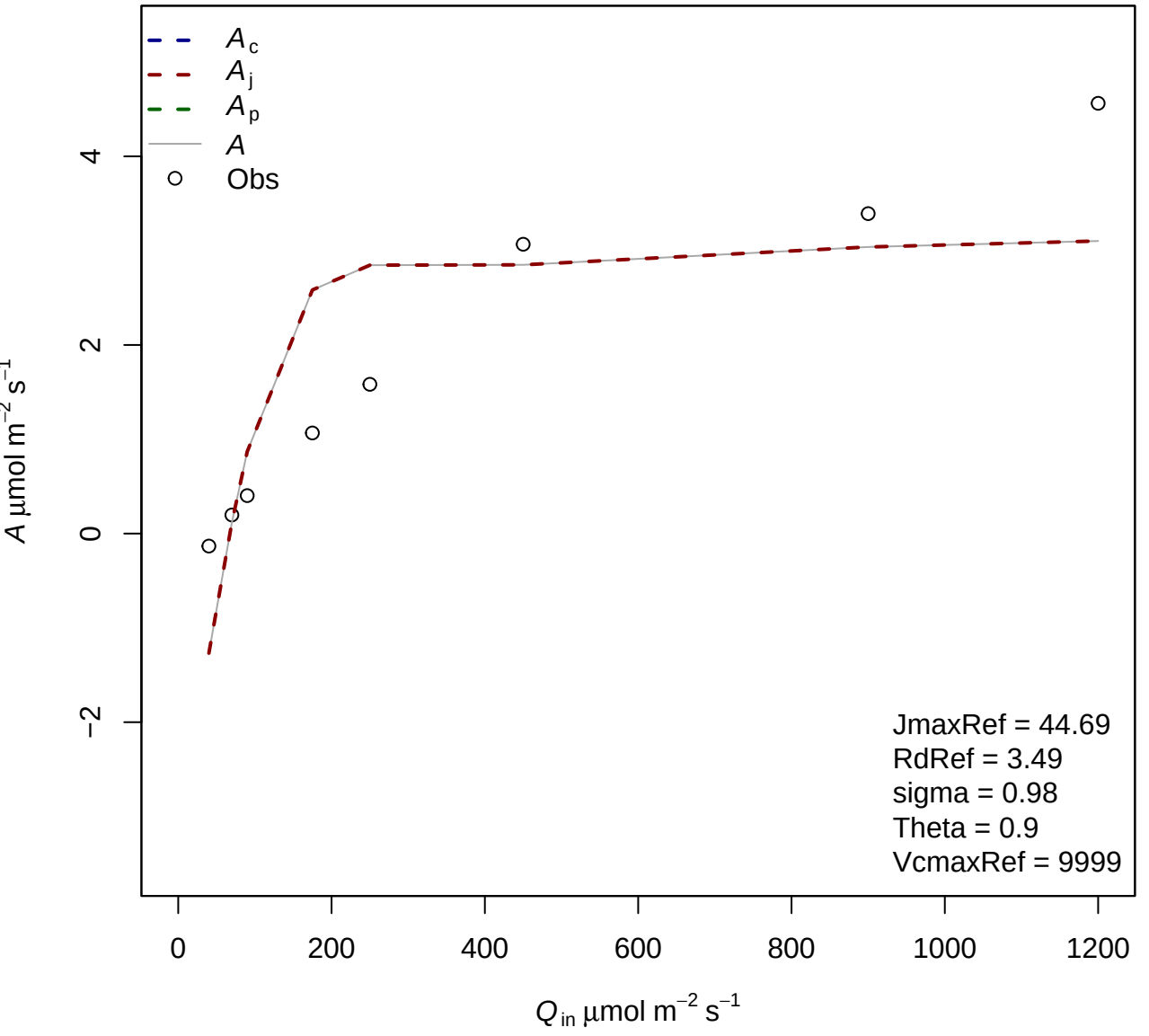

A_c
A_j
A_p
A
Obs
JmaxRef = 56.68
RdRef = 1.02
sigma = 0.12
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

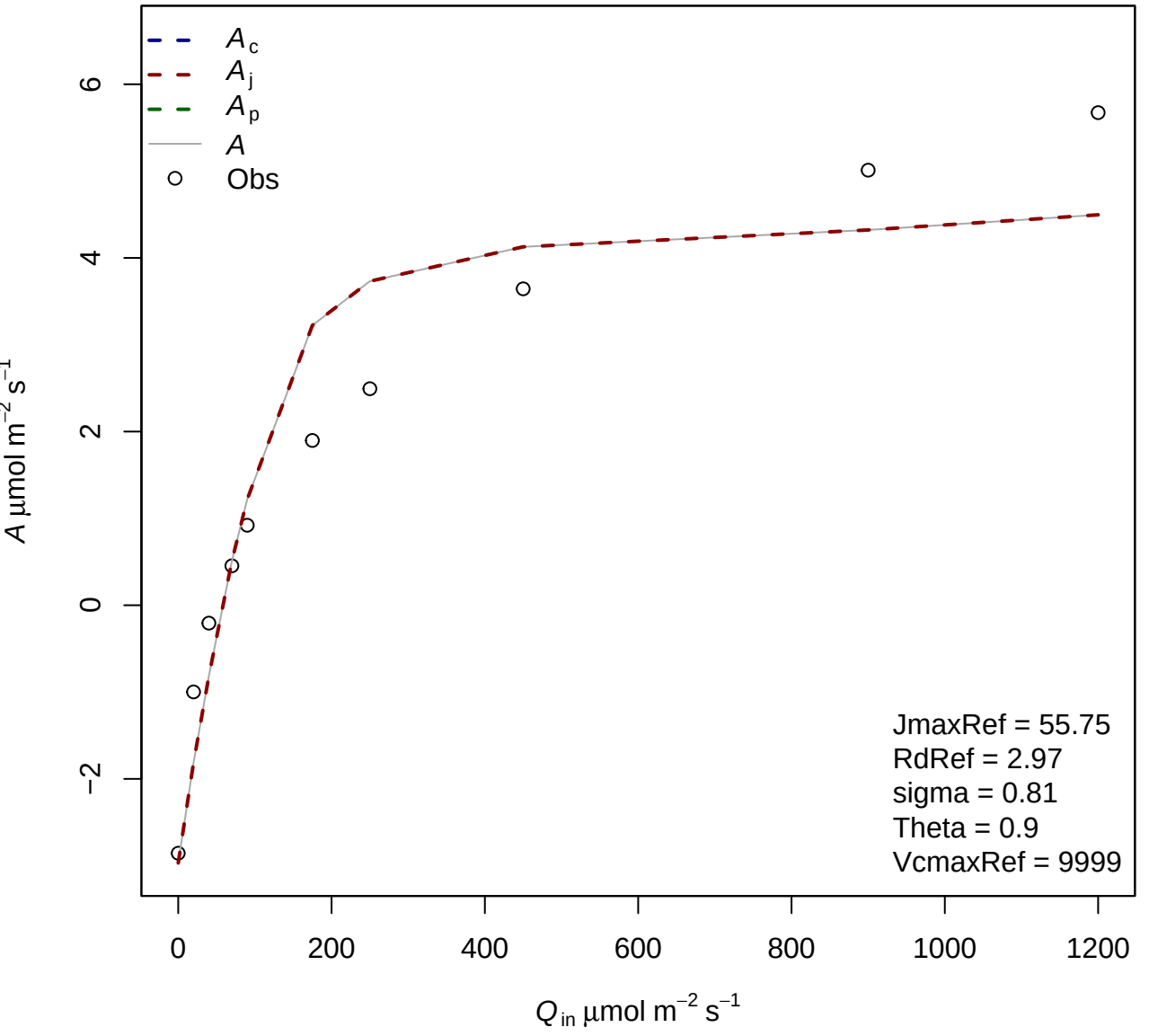

A_c
A_j
A_p
A
Obs
JmaxRef = 64.91
RdRef = 0.9
sigma = 0.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

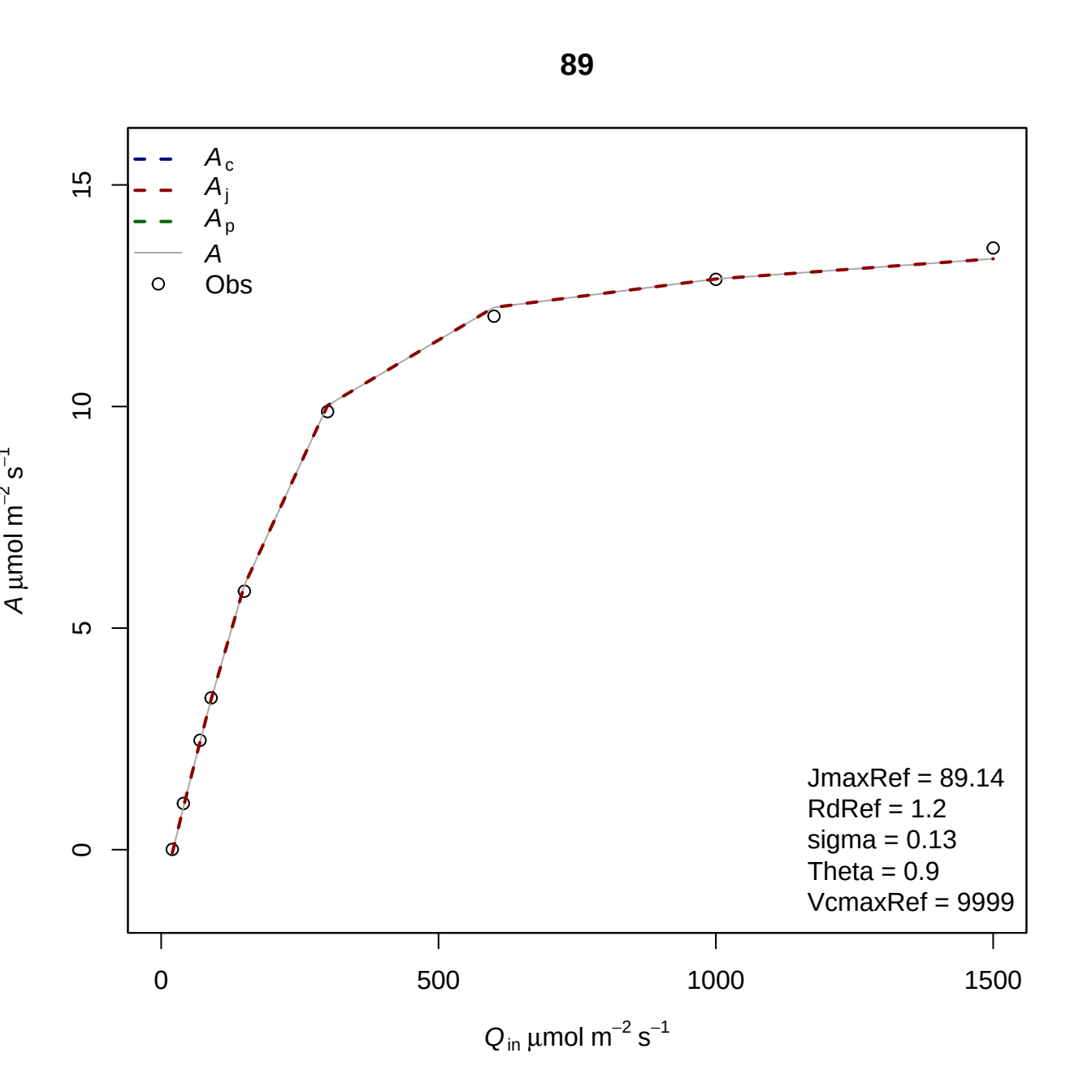

82
A_c
A_j
A_p
A
Obs
JmaxRef = 73.32
RdRef = 2.63
sigma = 0.13
Theta = 0.9
VcmaxRef = 50.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

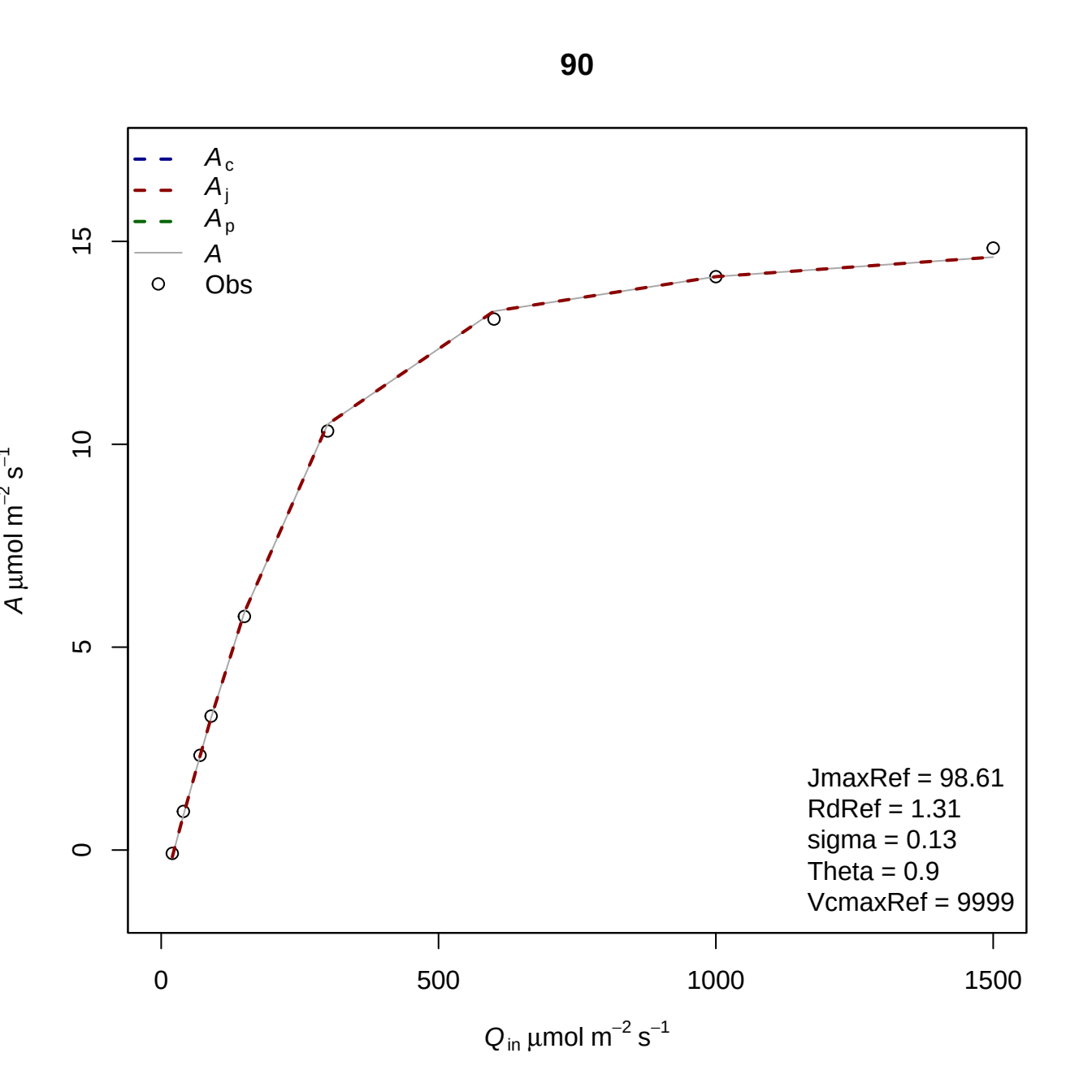

A_c
A_j
A_p
A
Obs
JmaxRef = 78.74
RdRef = 3.04
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

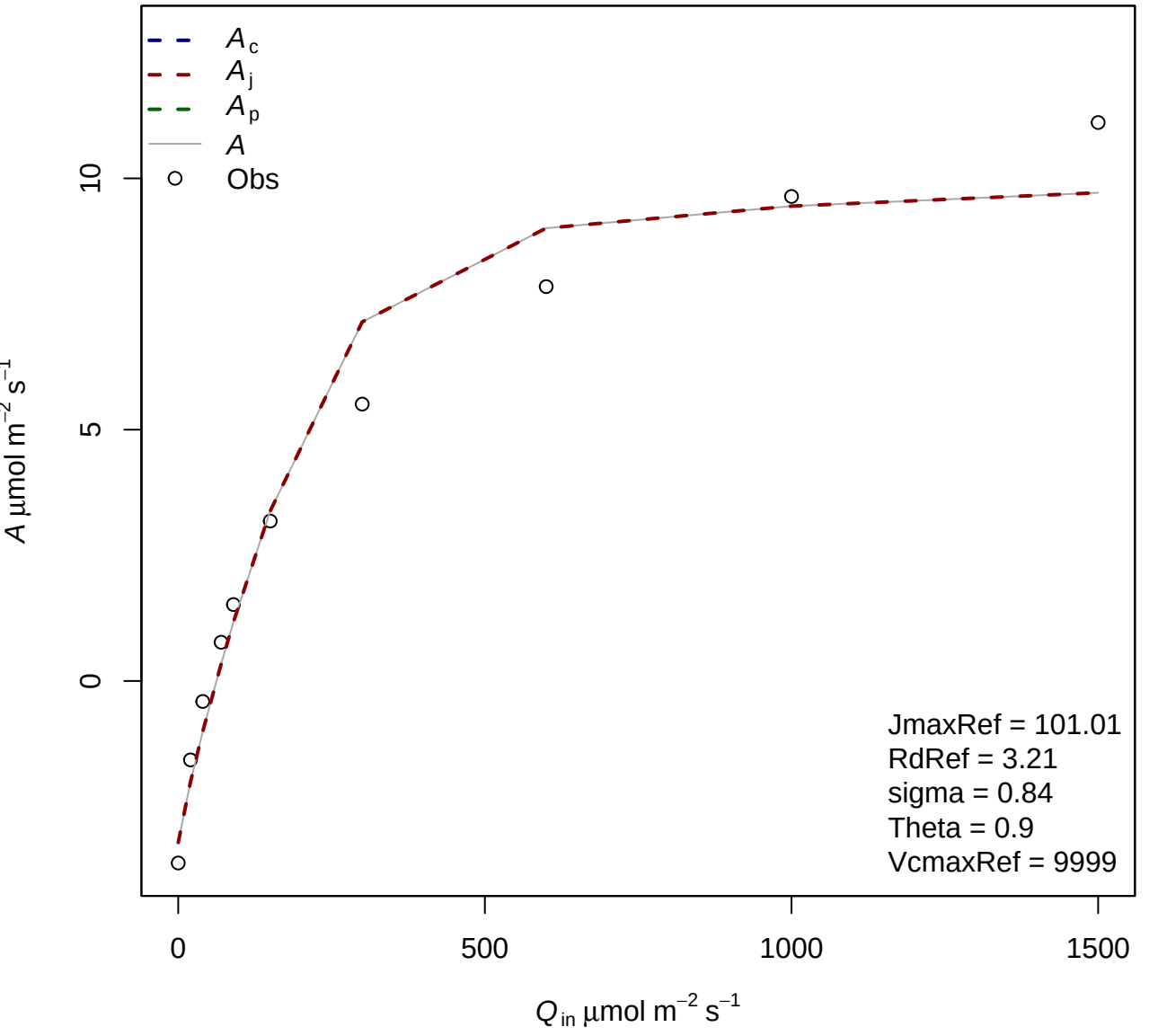

A_c
A_j
A_p
A
Obs
JmaxRef = 83.33
RdRef = 0.99
sigma = 0.04
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

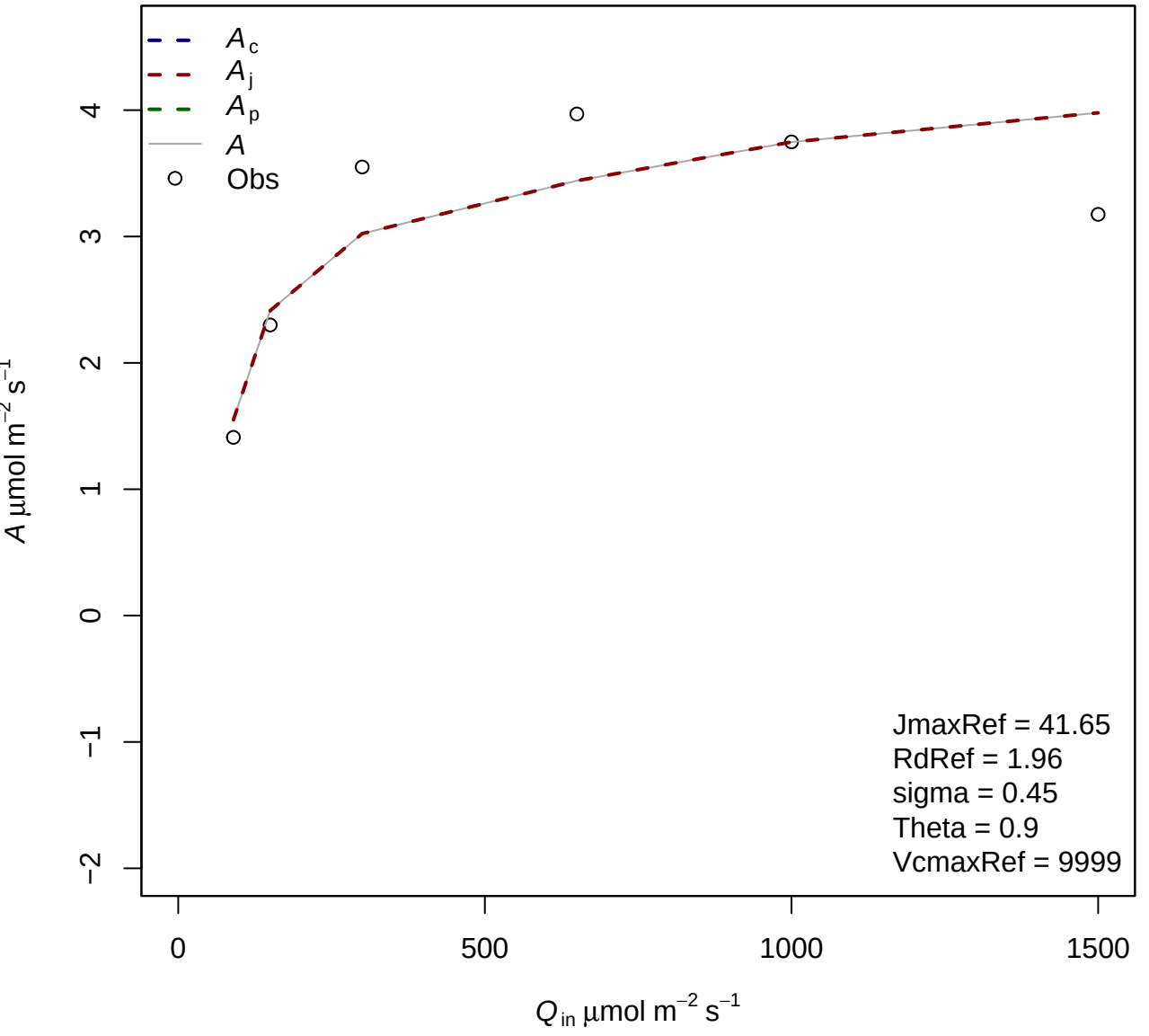

A_c
A_j
A_p
A
Obs
JmaxRef = 44.69
RdRef = 3.49
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

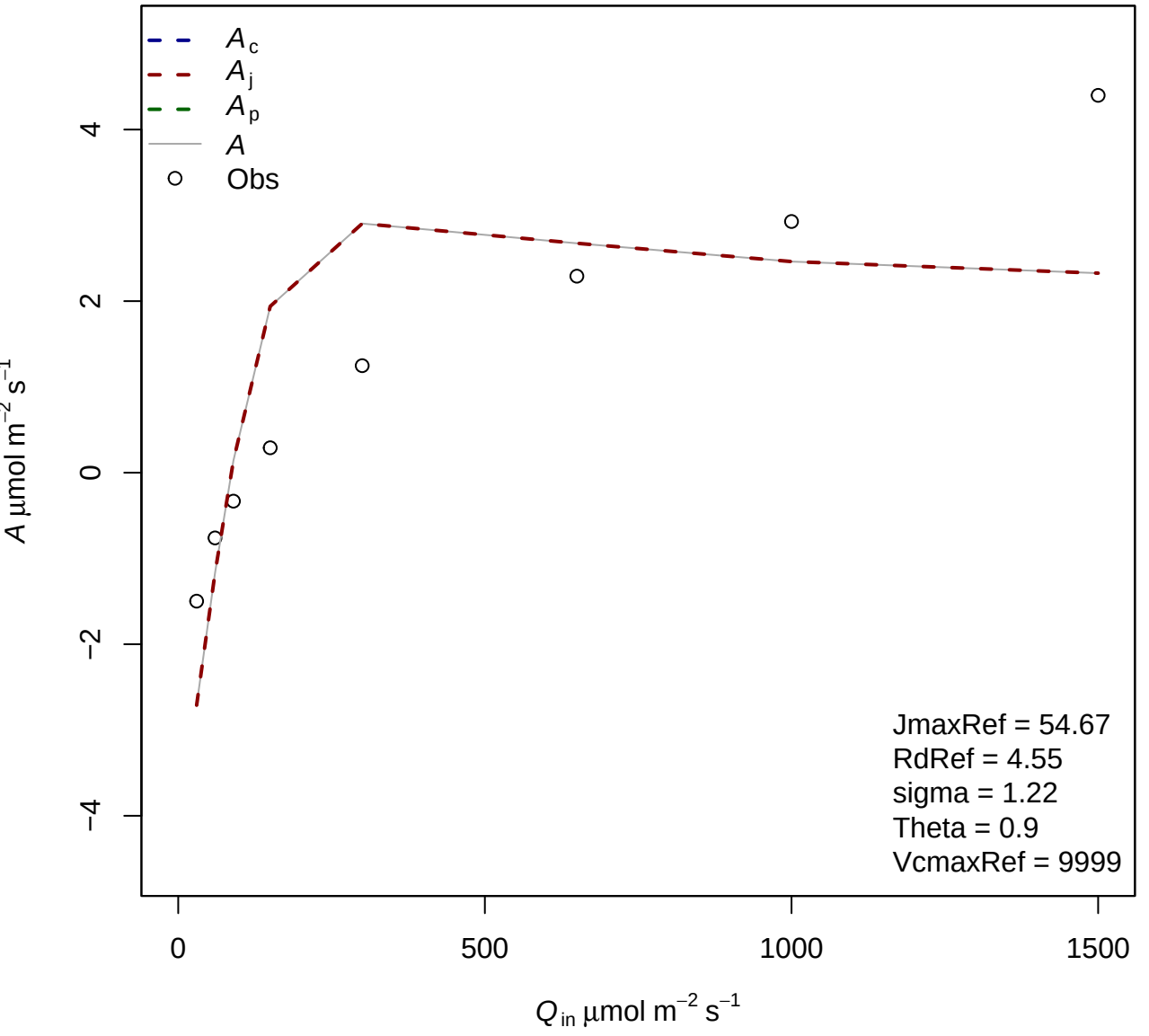

A_c
A_j
A_p
A
Obs
JmaxRef = 55.75
RdRef = 2.97
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

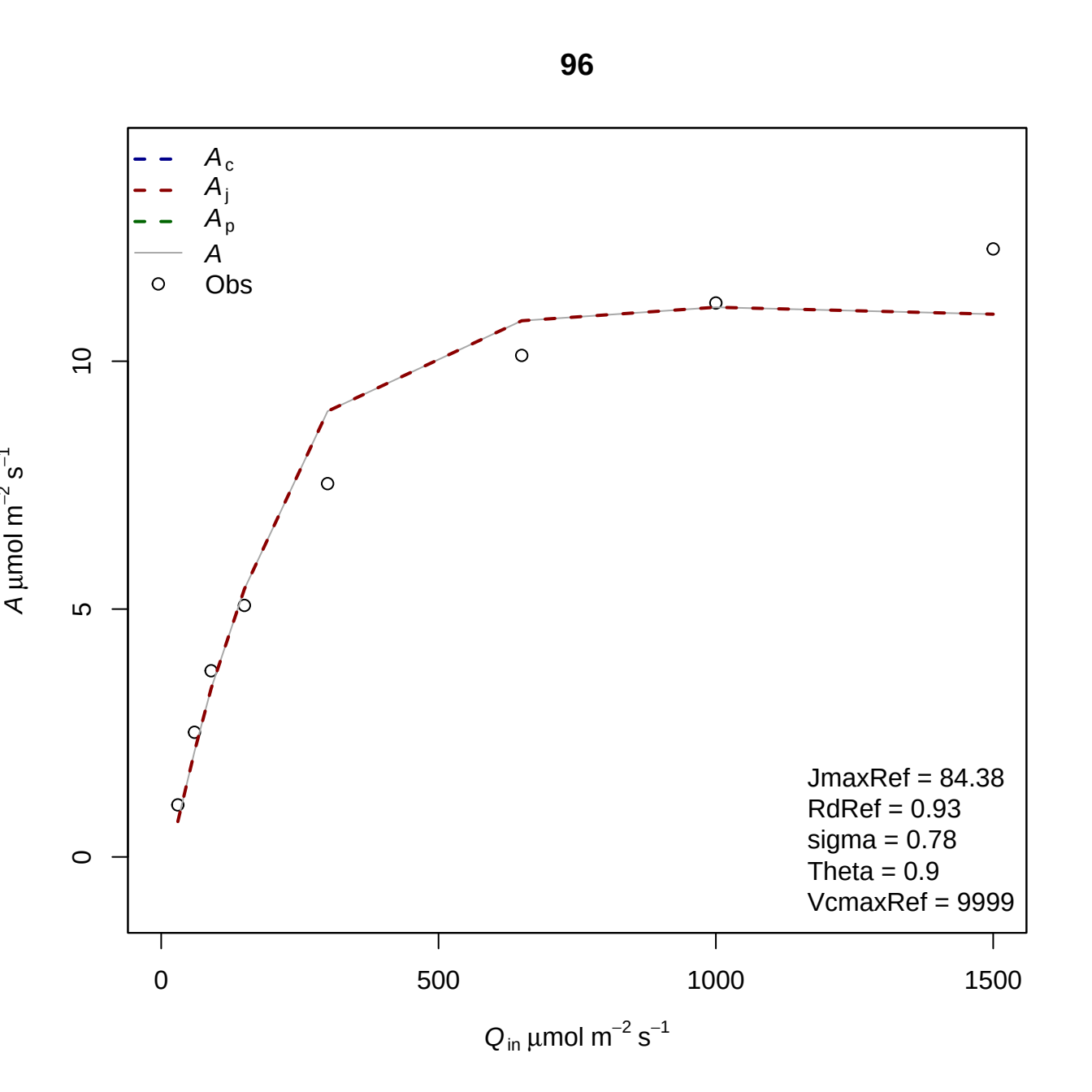

89
A_c
A_j
A_p
A
Obs
JmaxRef = 89.14
RdRef = 1.2
sigma = 0.13
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

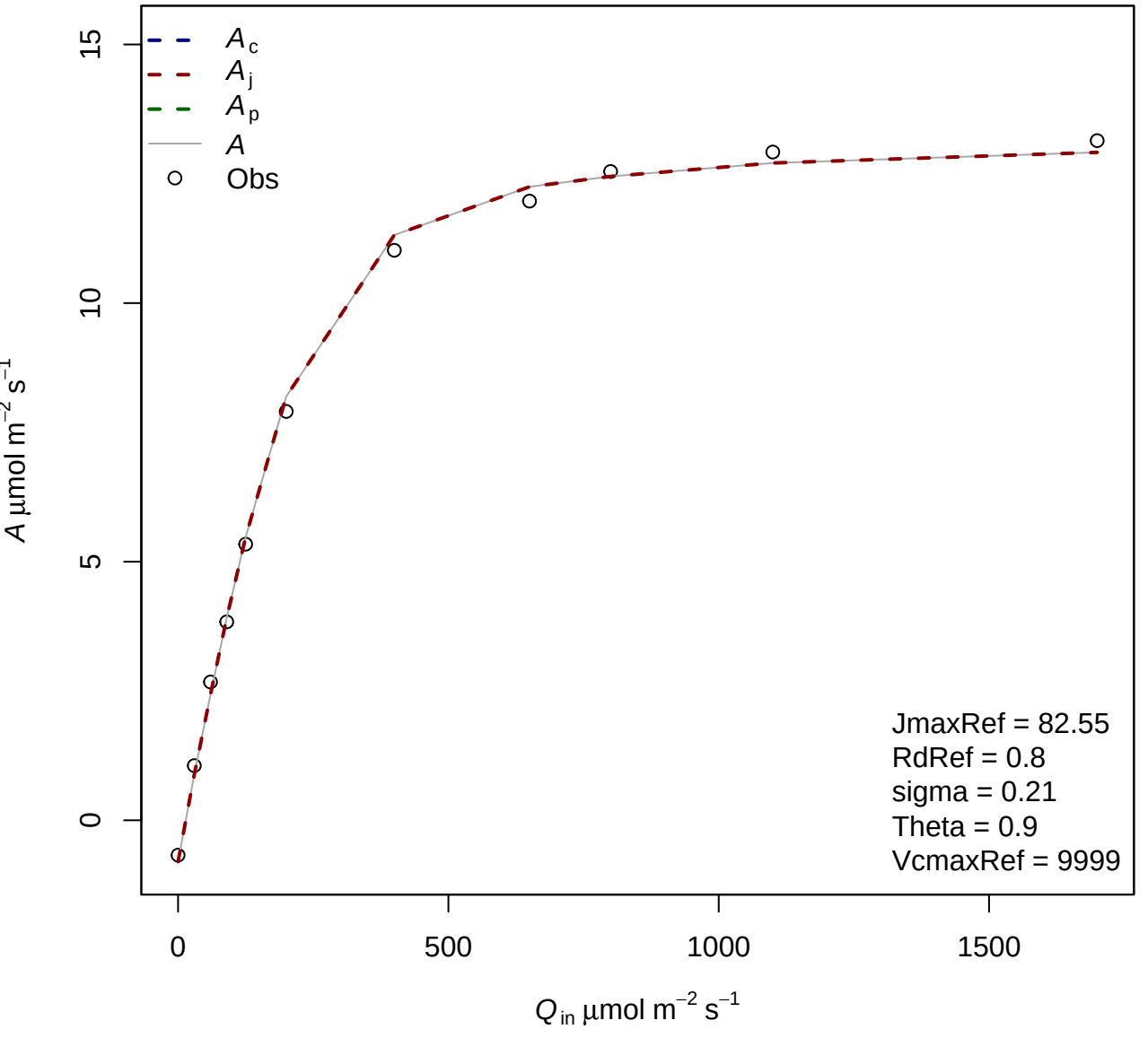

90
A_c
A_j
A_p
A
Obs
JmaxRef = 98.61
RdRef = 1.31
sigma = 0.13
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

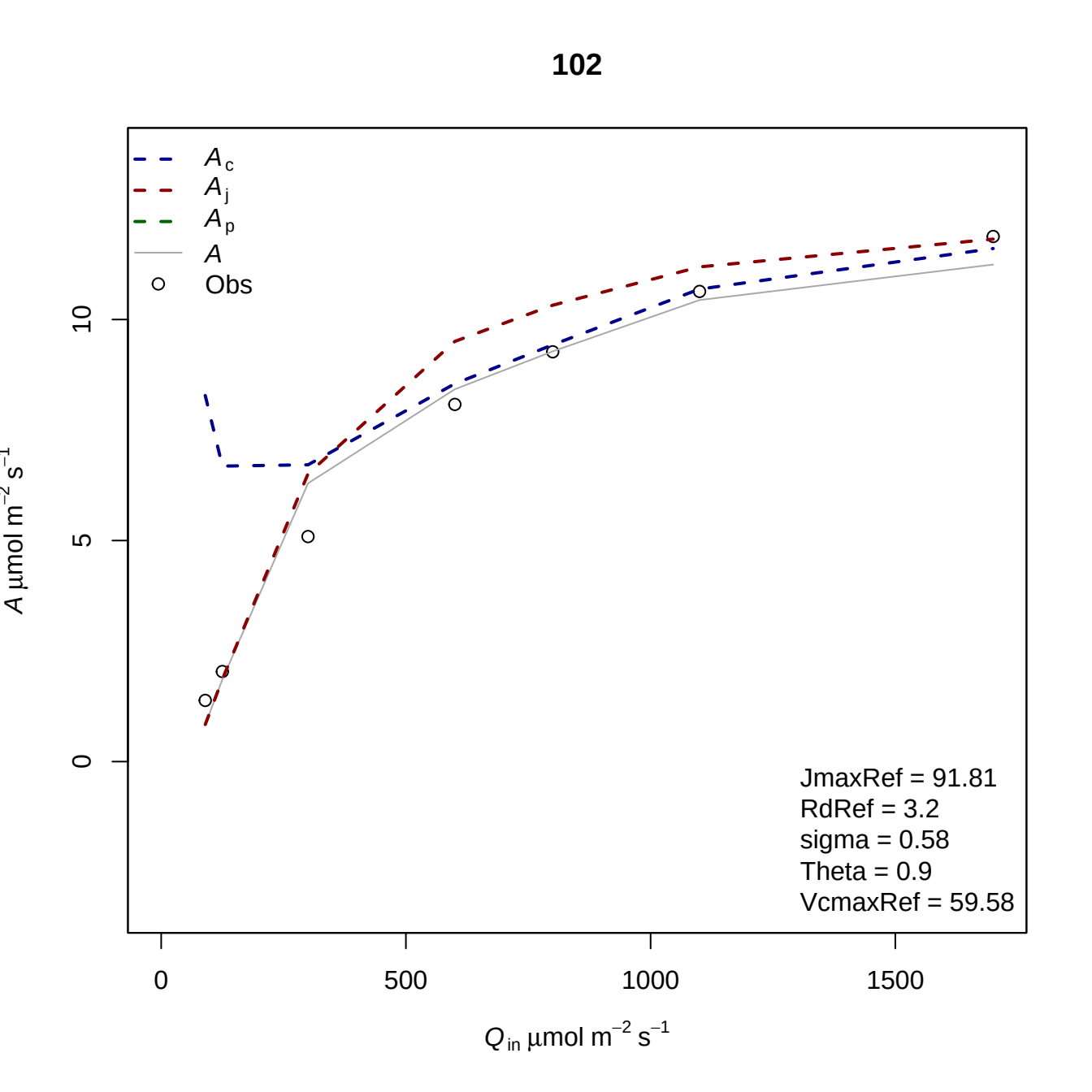

A_c
A_j
A_p
A
Obs
JmaxRef = 101.01
RdRef = 3.21
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

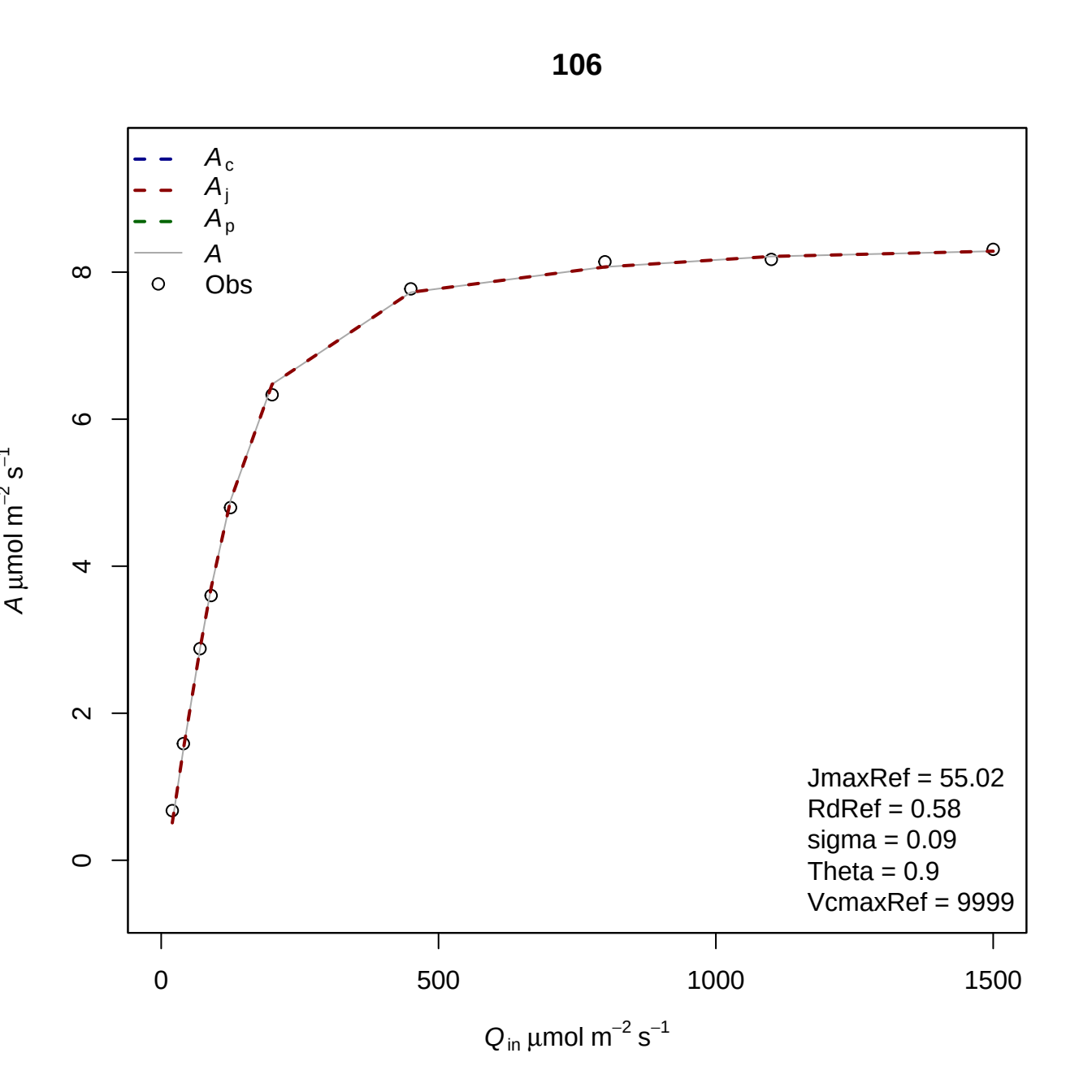

A_c
A_j
A_p
A
Obs
JmaxRef = 41.65
RdRef = 1.96
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

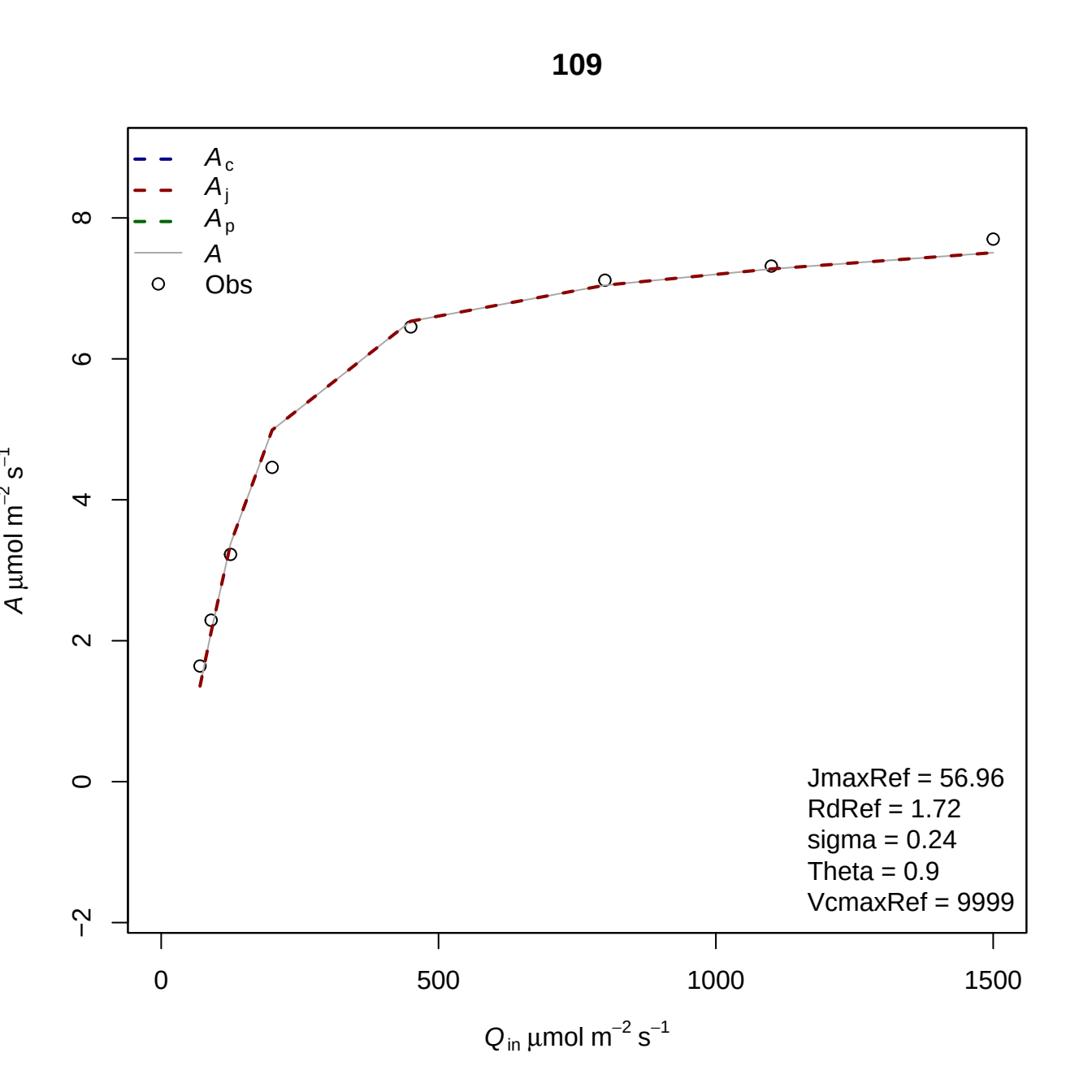

A_c
A_j
A_p
A
Obs
JmaxRef = 54.67
RdRef = 4.55
sigma = 1.22
Theta = 0.9
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

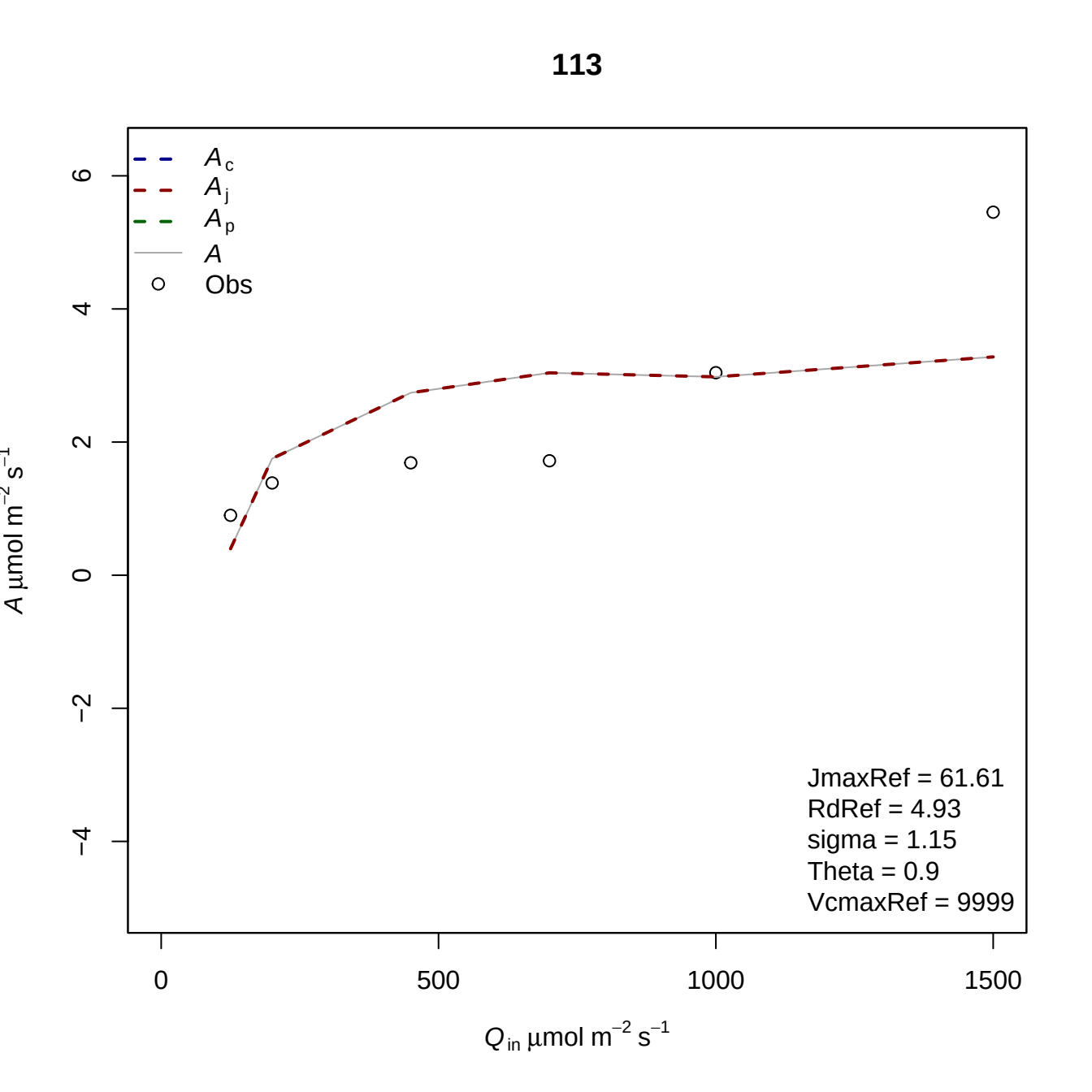

96
A_c
A_j
A_p
A
Obs
JmaxRef = 84.38
RdRef = 0.93
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

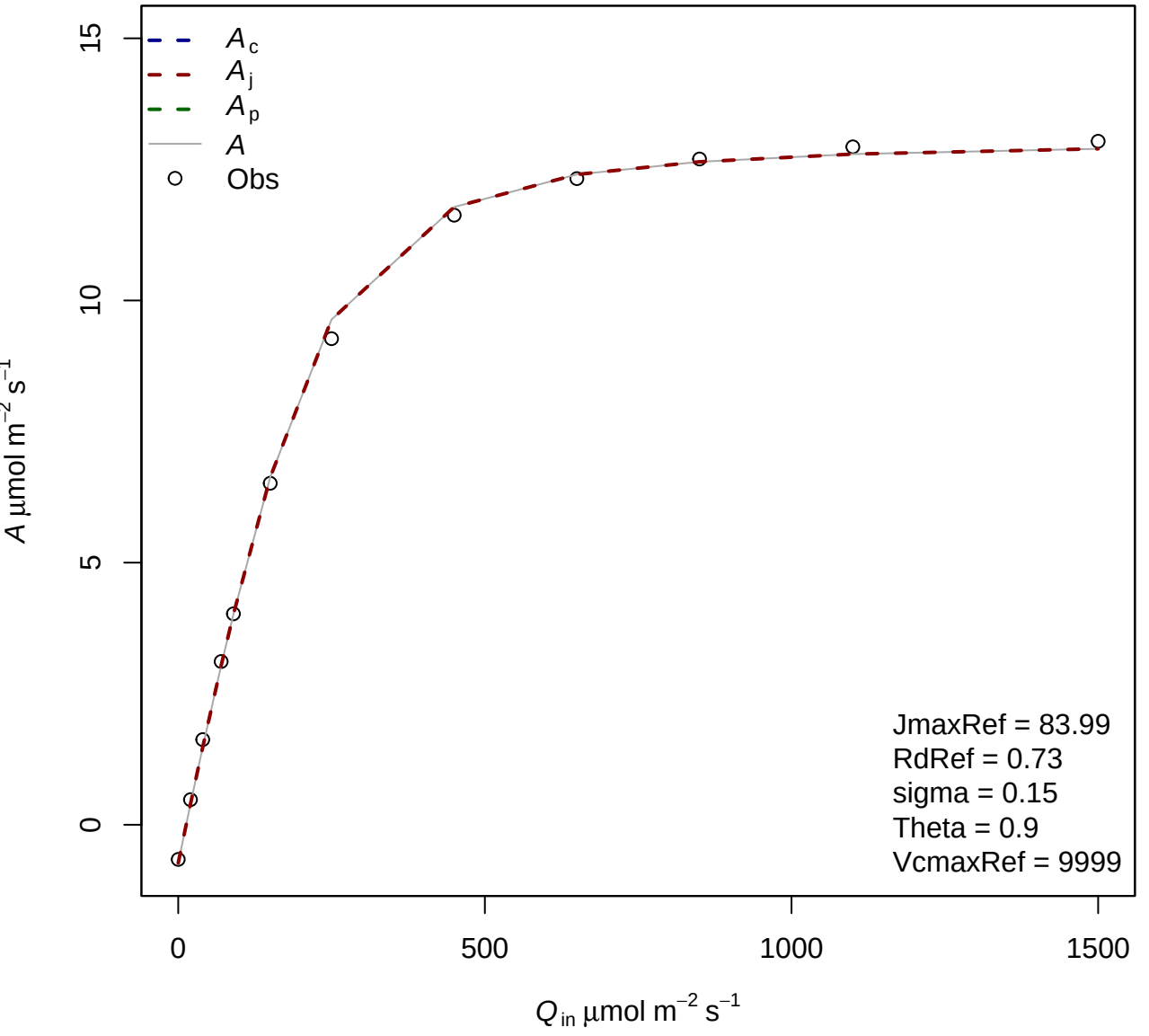

A_c
A_j
A_p
A
Obs
JmaxRef = 82.55
RdRef = 0.8
sigma = 0.21
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

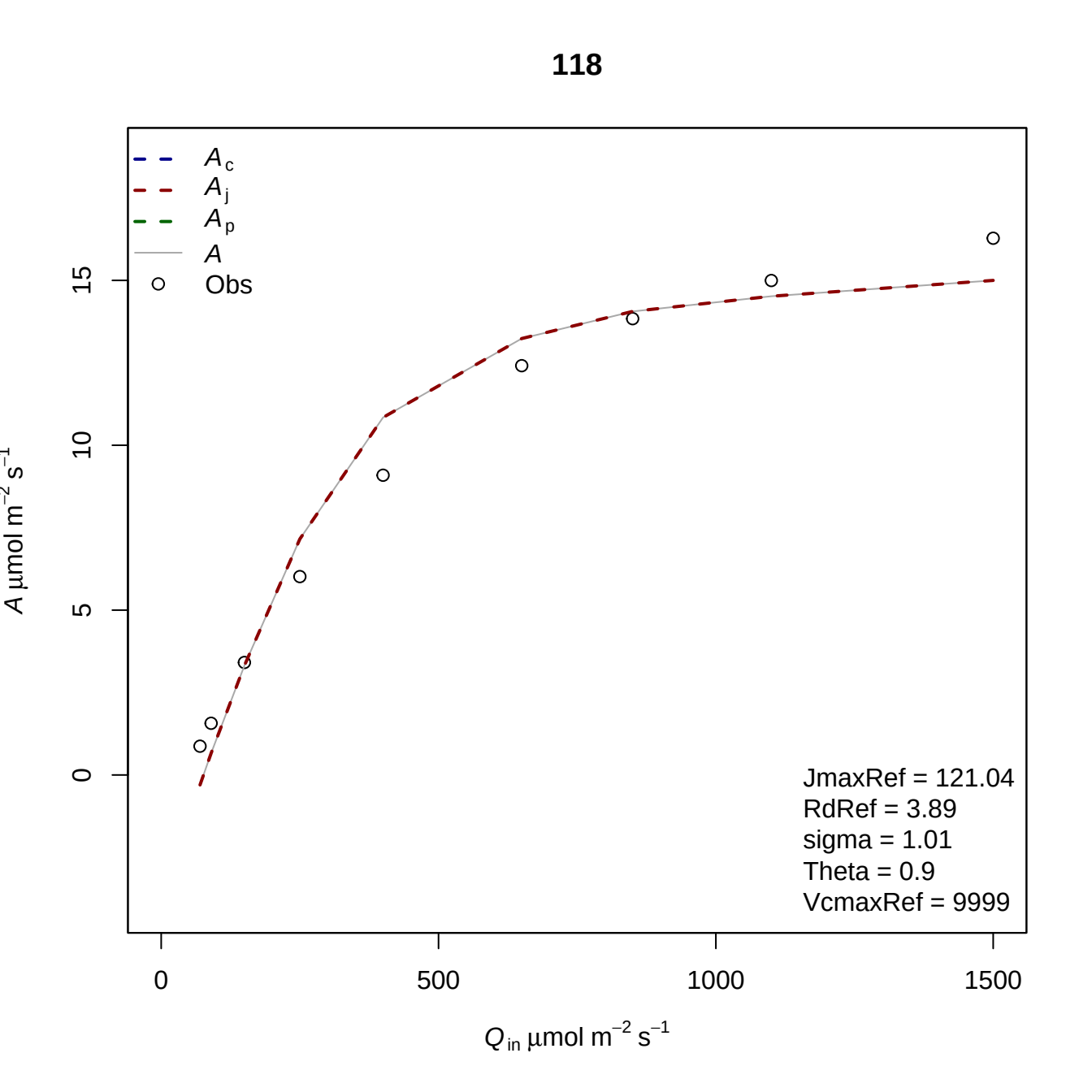

102
A_c
A_j
A_p
A
Obs
JmaxRef = 91.81
RdRef = 3.2
sigma = 0.58
Theta = 0.9
VcmaxRef = 59.58
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

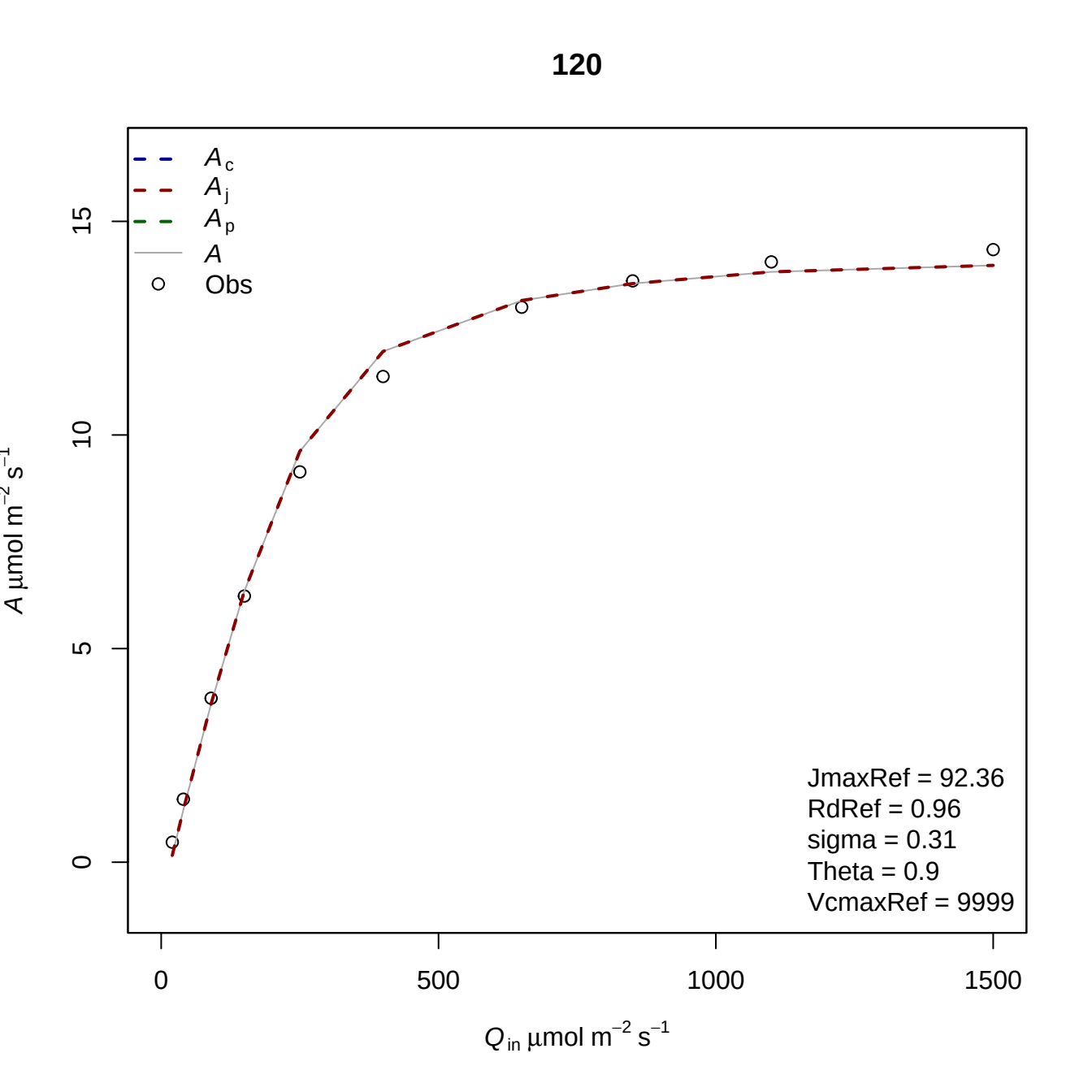

106
A_c
A_j
A_p
A
Obs
JmaxRef = 55.02
RdRef = 0.58
sigma = 0.09
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

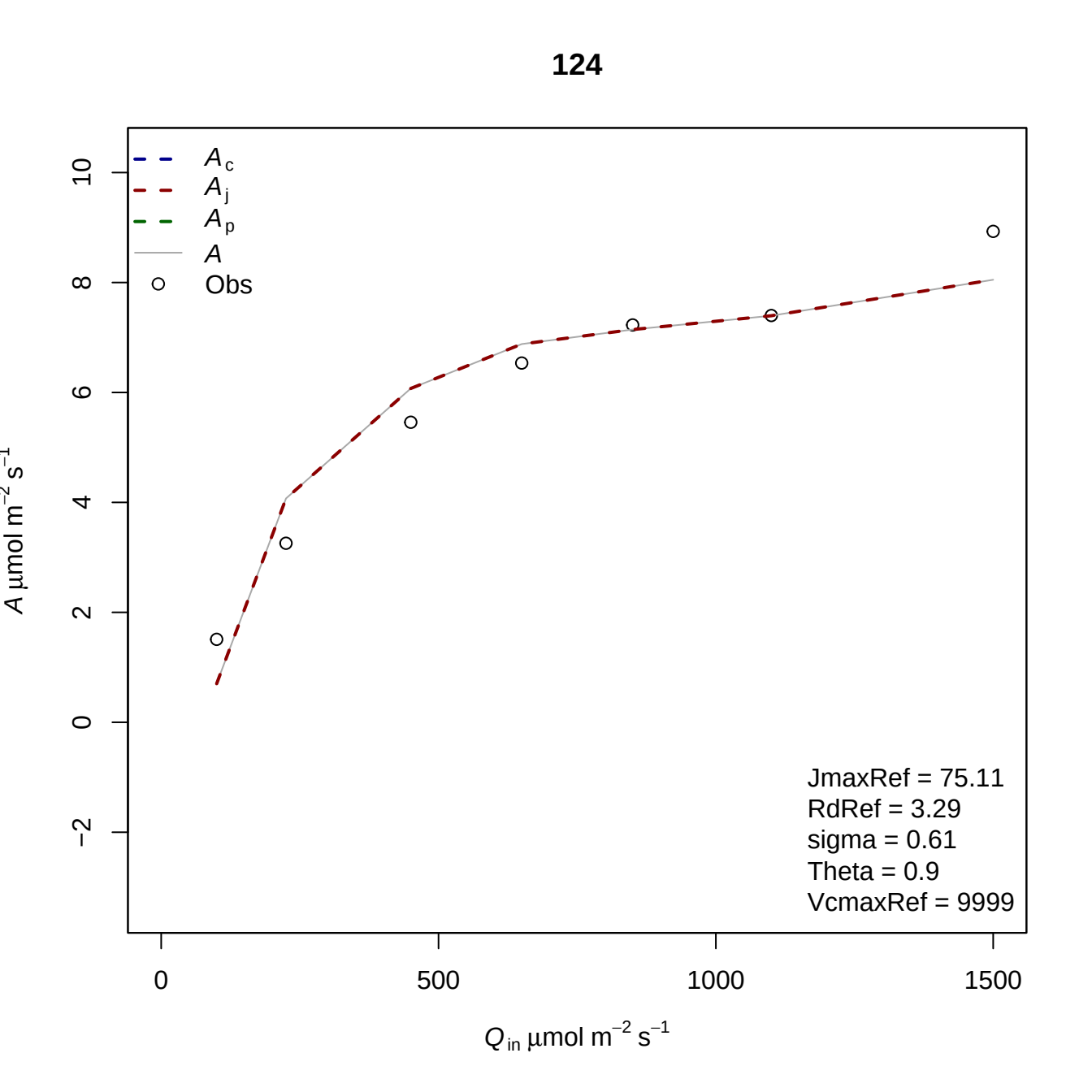

109
A_c
A_j
A_p
A
Obs
JmaxRef = 56.96
RdRef = 1.72
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

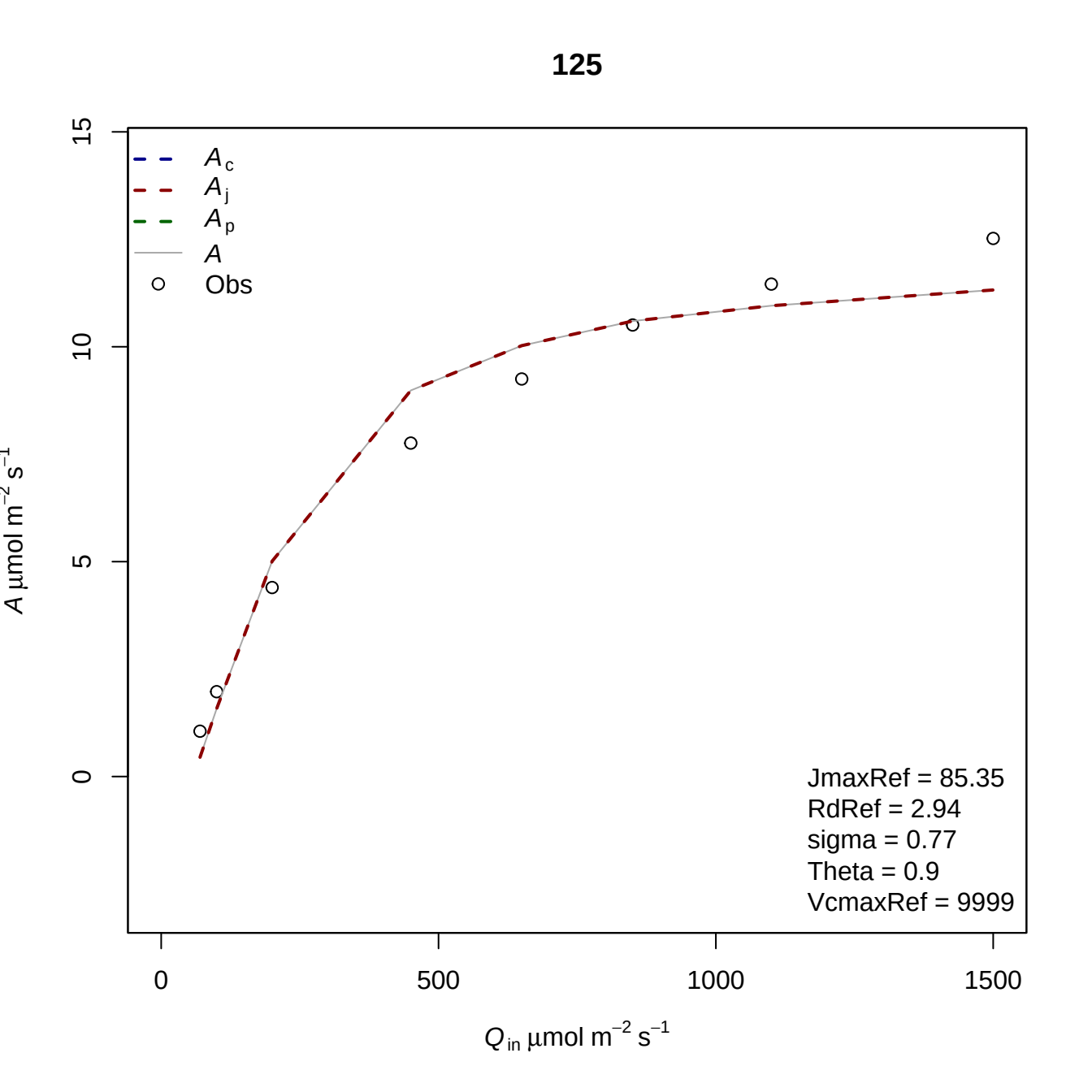

113
A_c
A_j
A_p
A
Obs
JmaxRef = 61.61
RdRef = 4.93
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

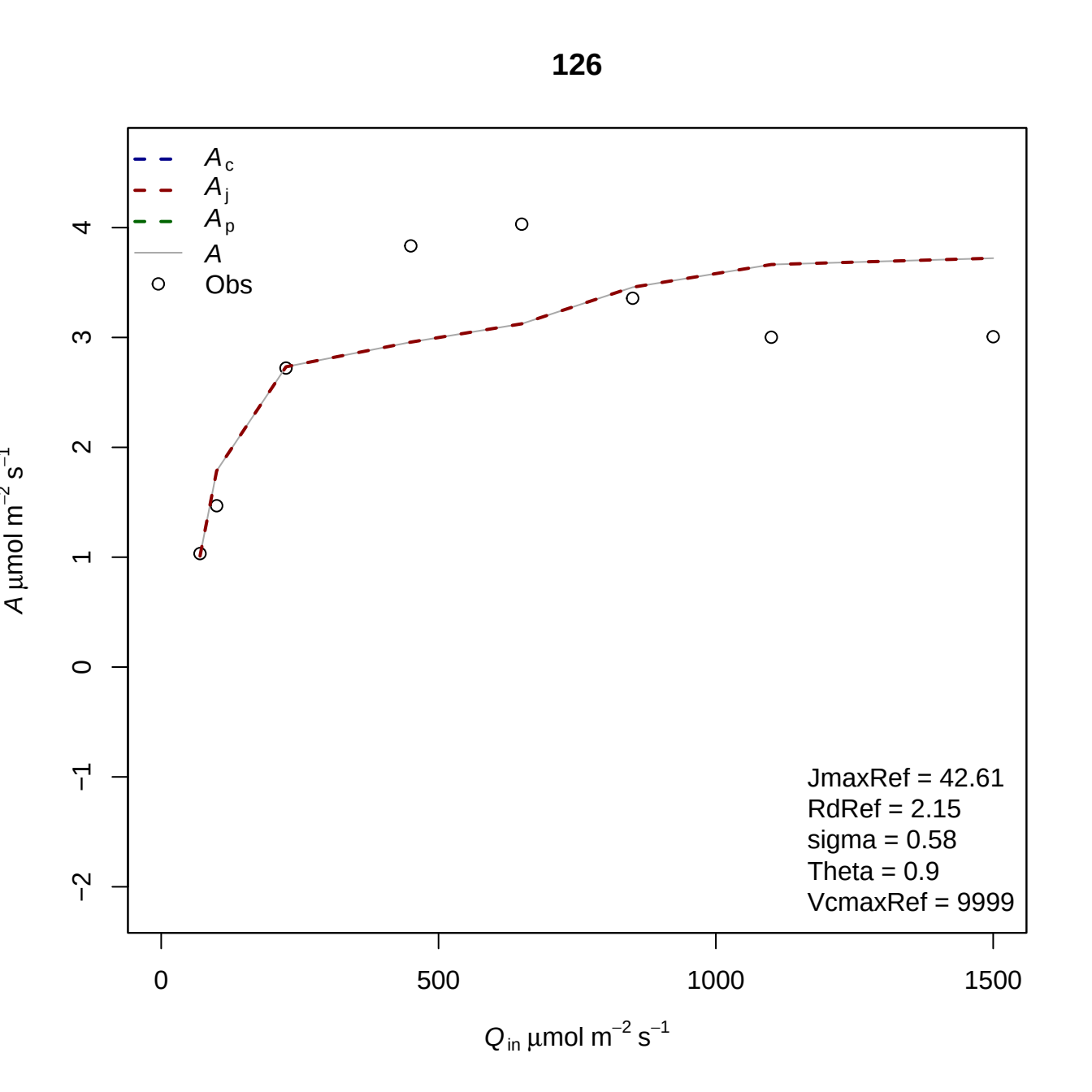

A_c
A_j
A_p
A
Obs
JmaxRef = 83.99
RdRef = 0.73
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

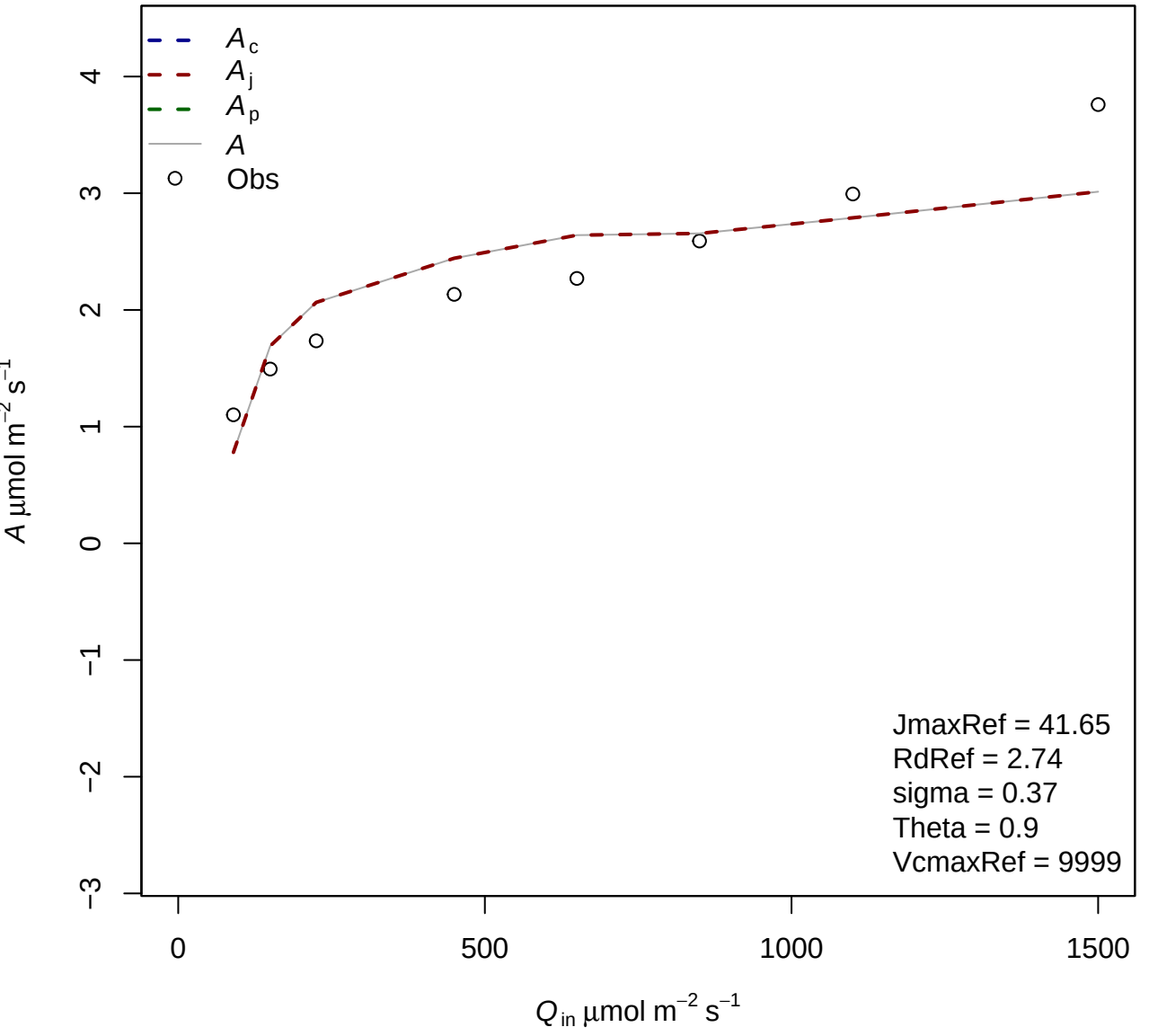

118
A_c
A_j
A_p
A
Obs
JmaxRef = 121.04
RdRef = 3.89
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

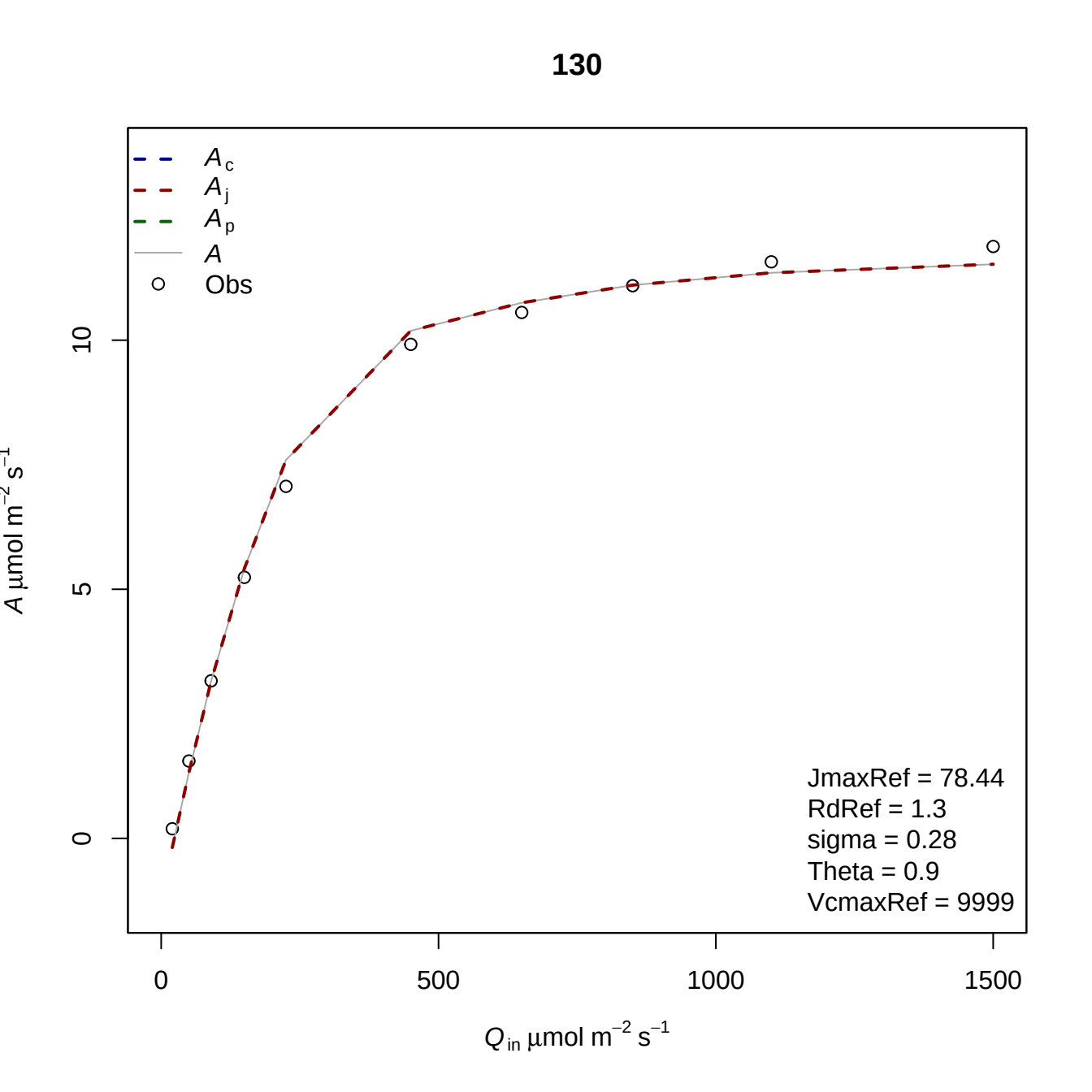

120
A_c
A_j
A_p
A
Obs
JmaxRef = 92.36
RdRef = 0.96
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

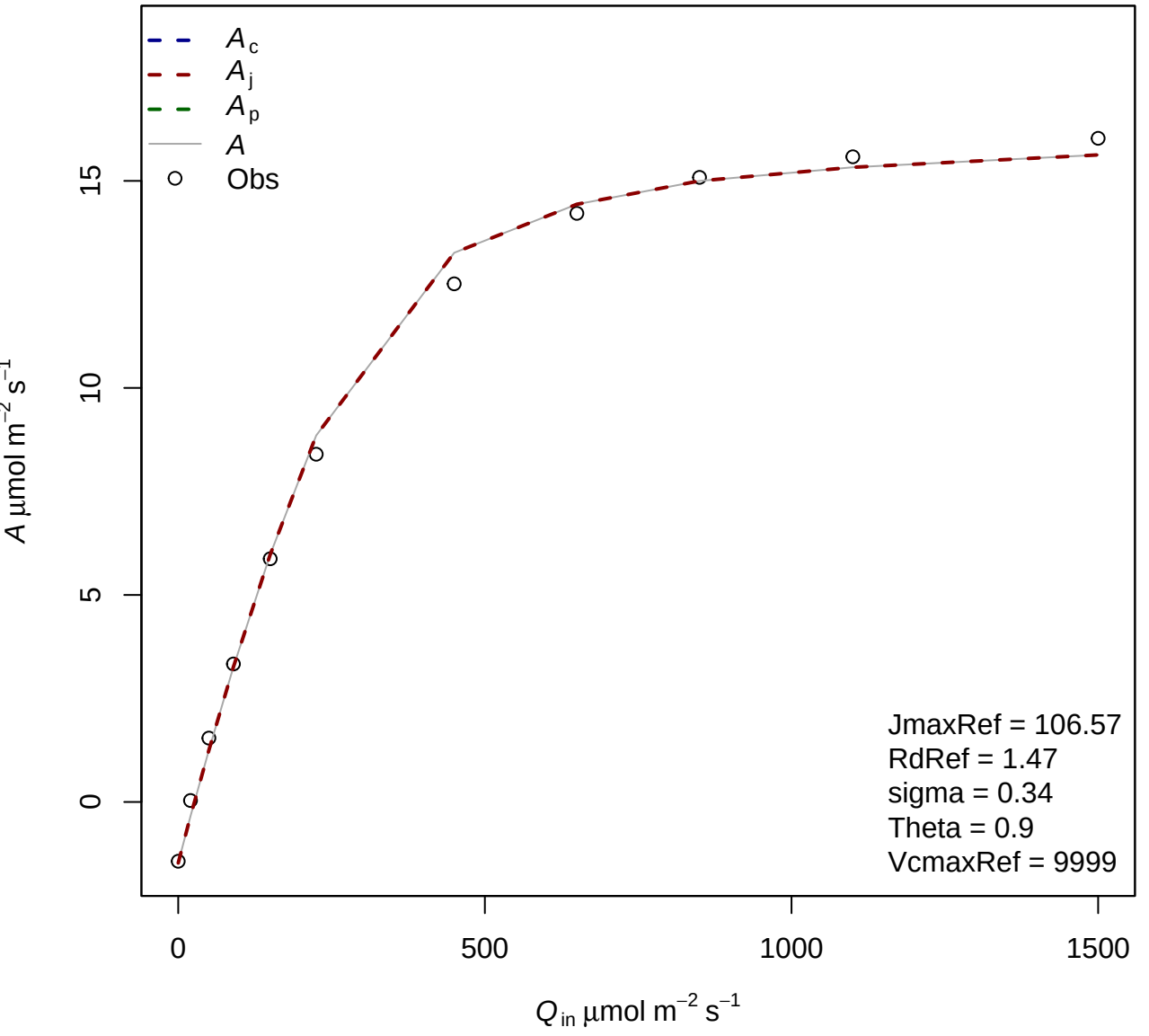

124
A_c
A_j
A_p
A
Obs
JmaxRef = 75.11
RdRef = 3.29
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

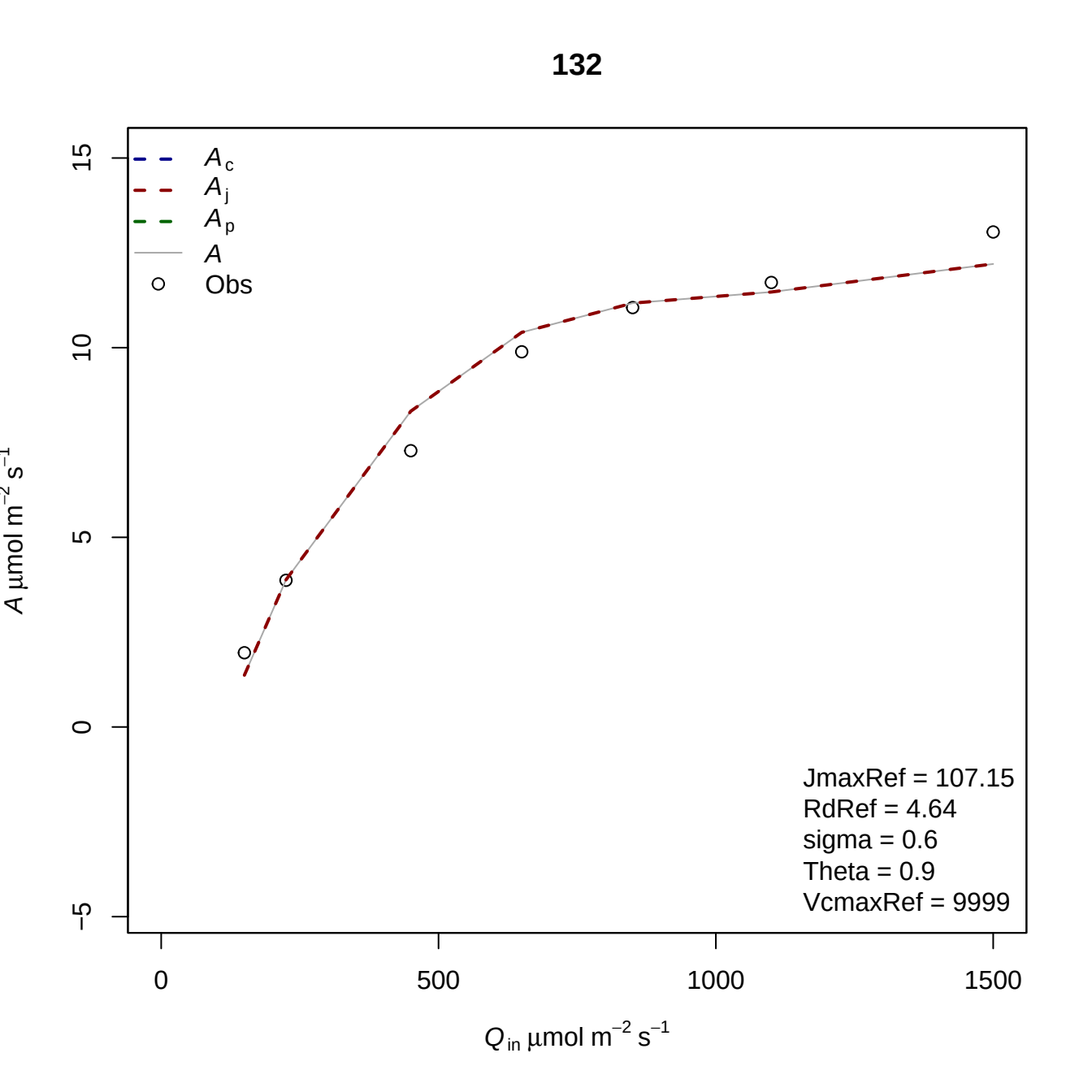

125
A_c
A_j
A_p
A
Obs
JmaxRef = 85.35
RdRef = 2.94
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

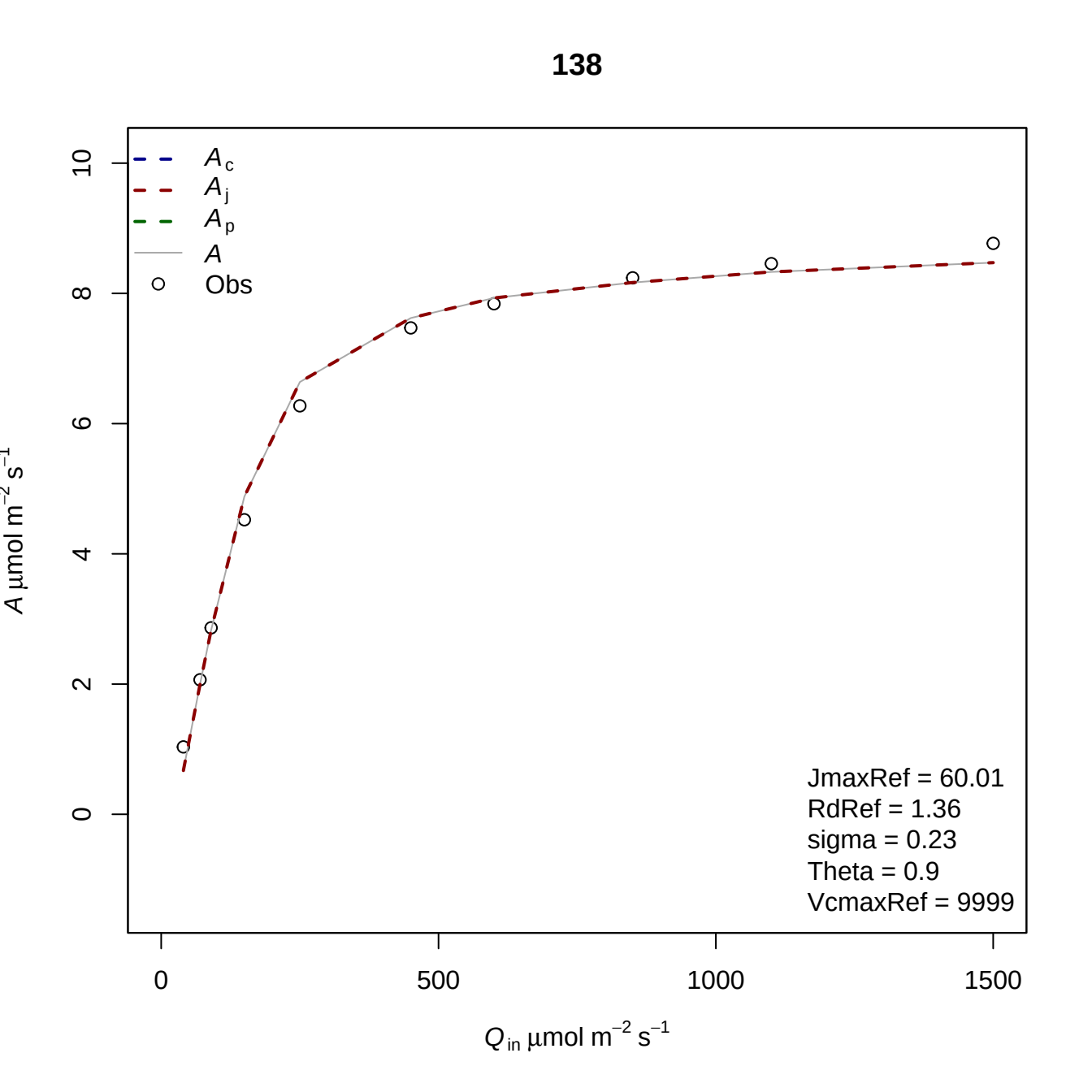

126
A_c
A_j
A_p
A
Obs
JmaxRef = 42.61
RdRef = 2.15
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

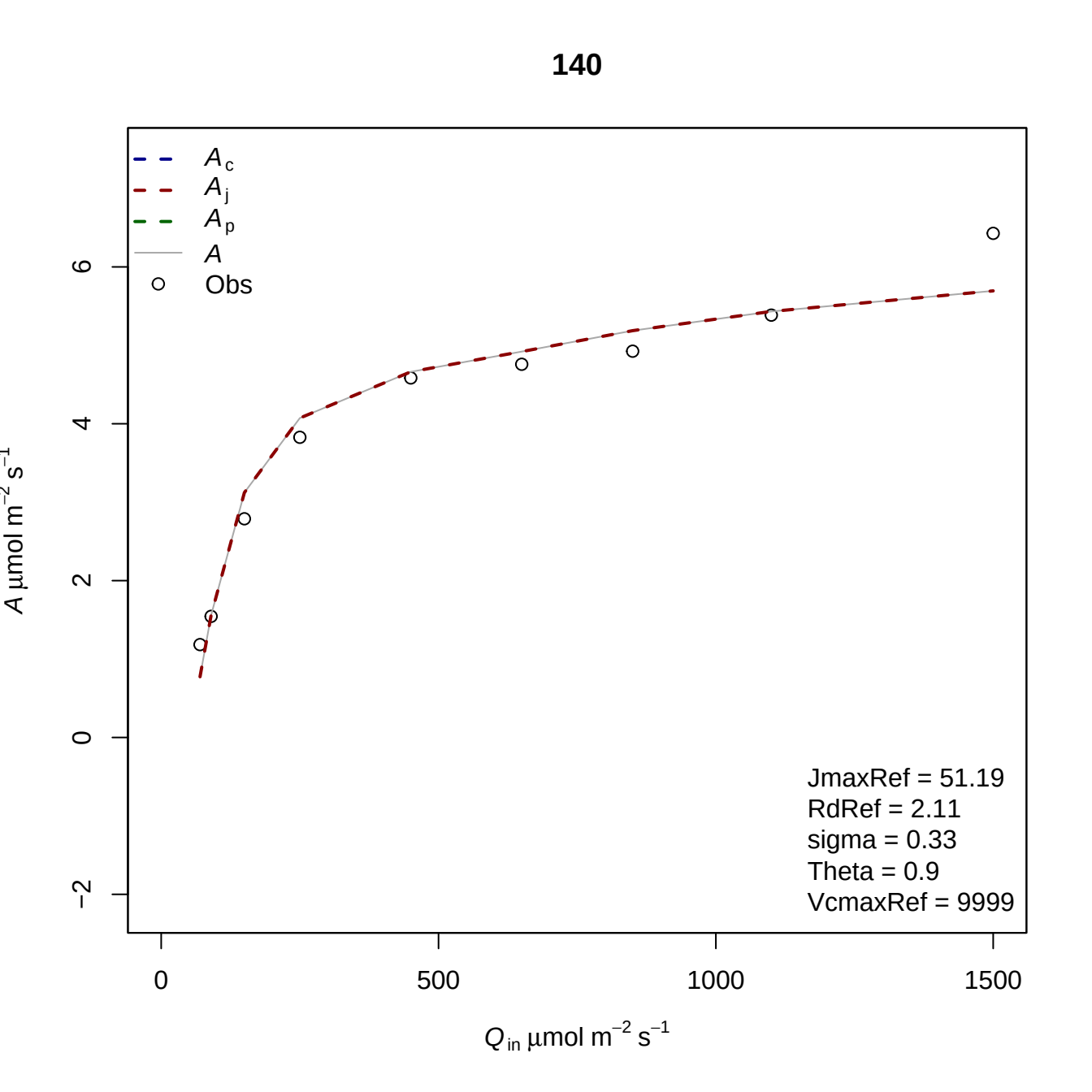

A_c
A_j
A_p
A
Obs
JmaxRef = 41.65
RdRef = 2.74
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

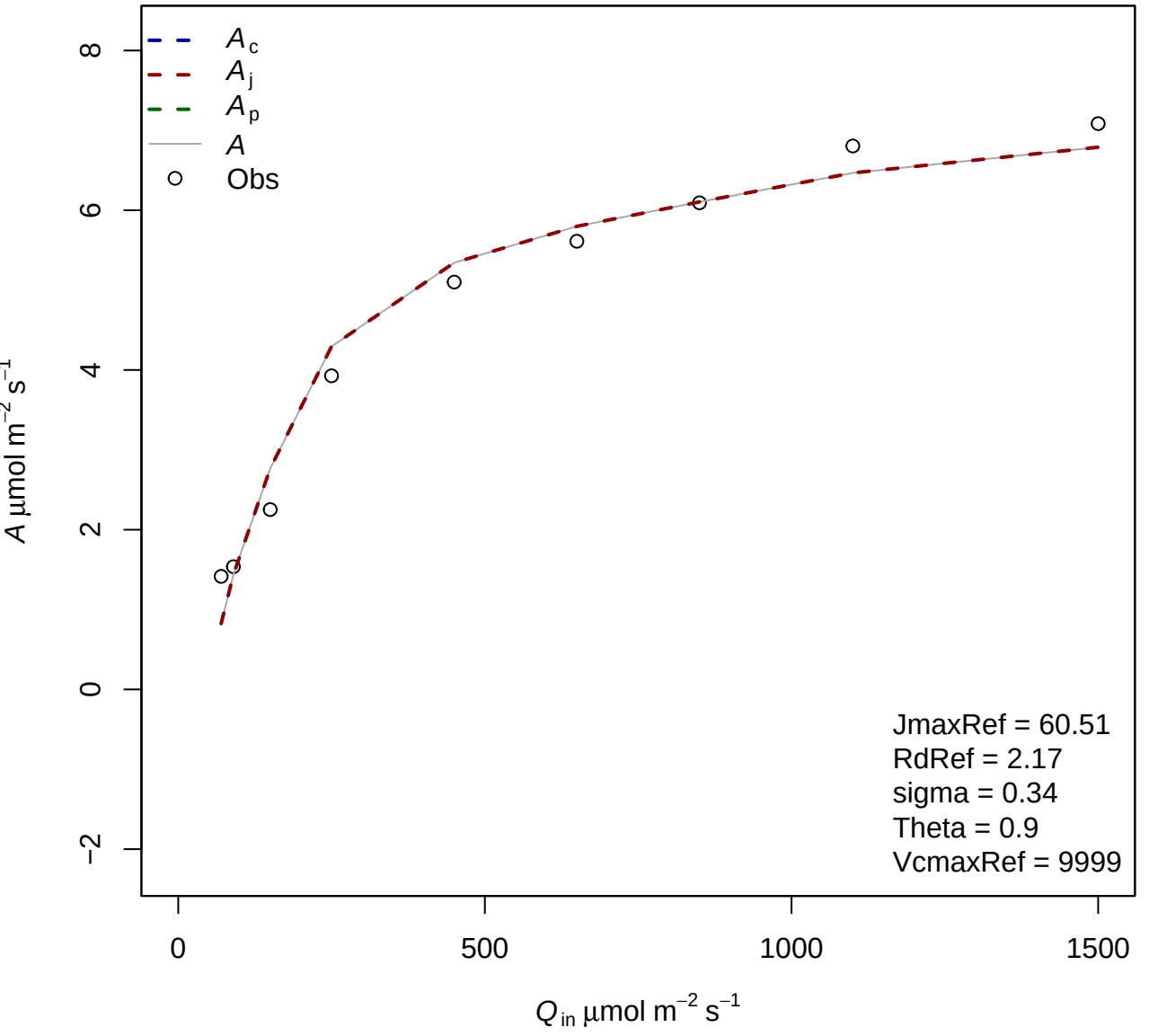

130
A_c
A_j
A_p
A
Obs
JmaxRef = 78.44
RdRef = 1.3
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

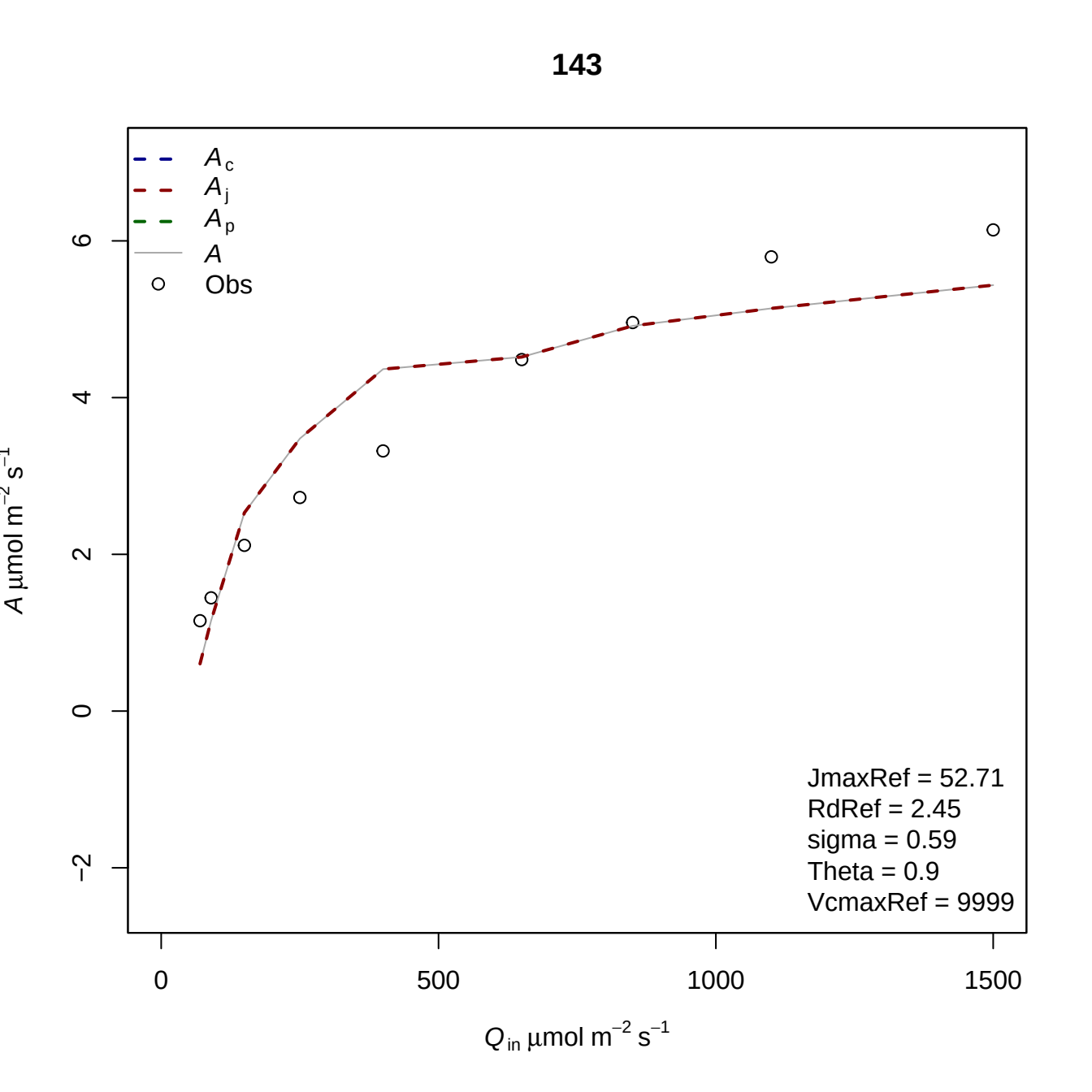

A_c
A_j
A_p
A
Obs
JmaxRef = 106.57
RdRef = 1.47
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

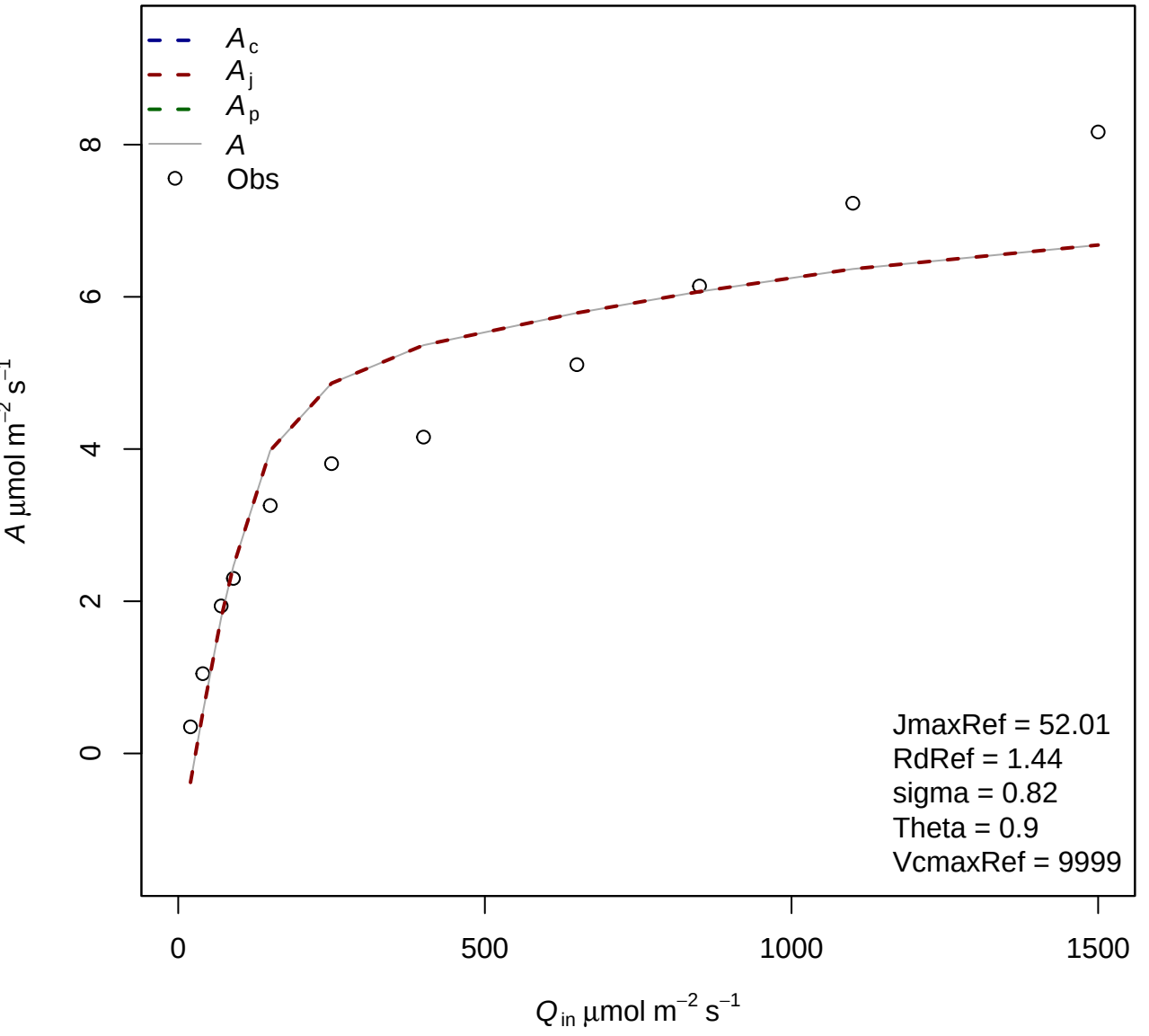

132
A_c
A_j
A_p
A
Obs
15
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 107.15
RdRef = 4.64
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999

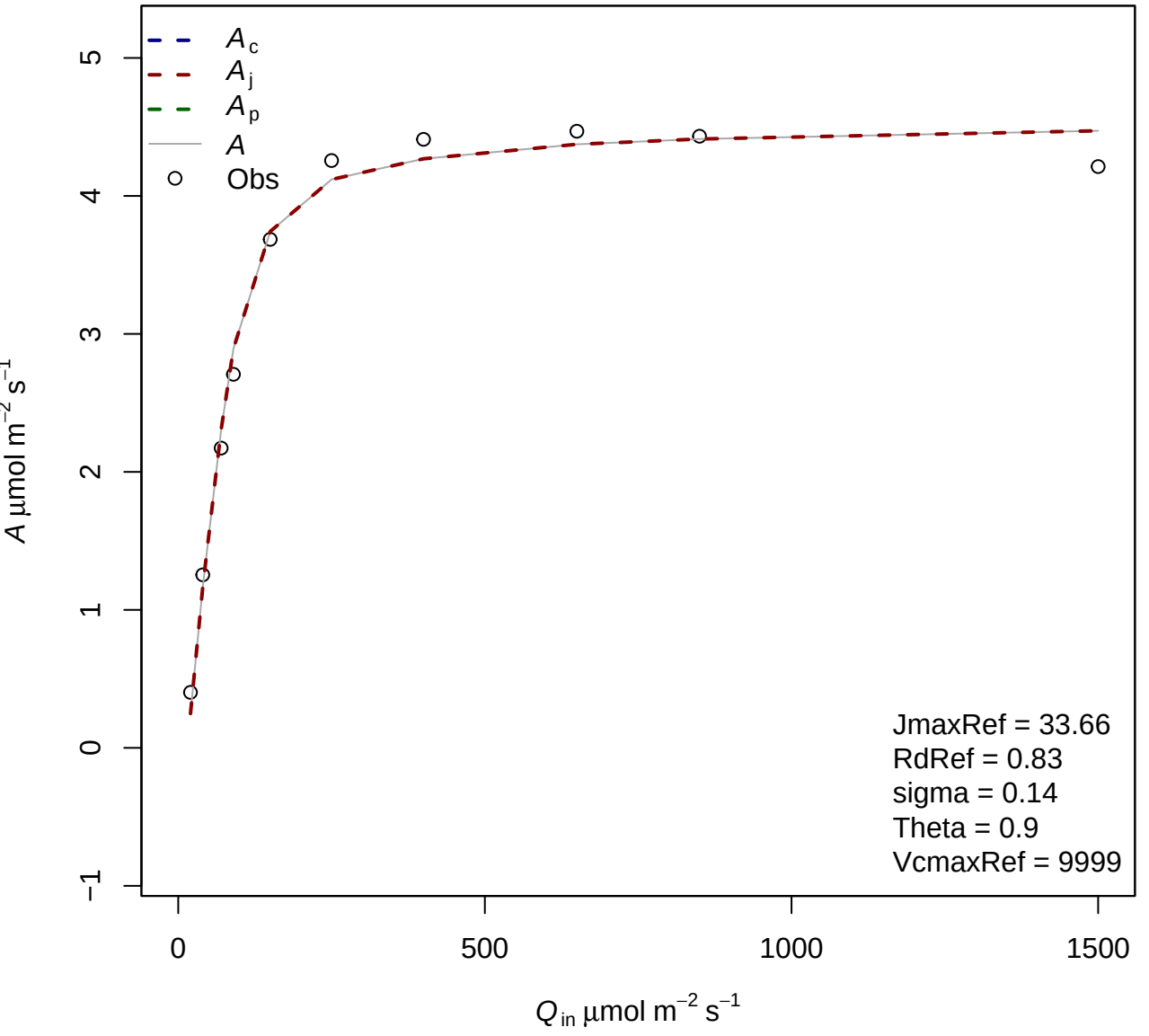

138
A_c
A_j
A_p
A
Obs
JmaxRef = 60.01
RdRef = 1.36
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

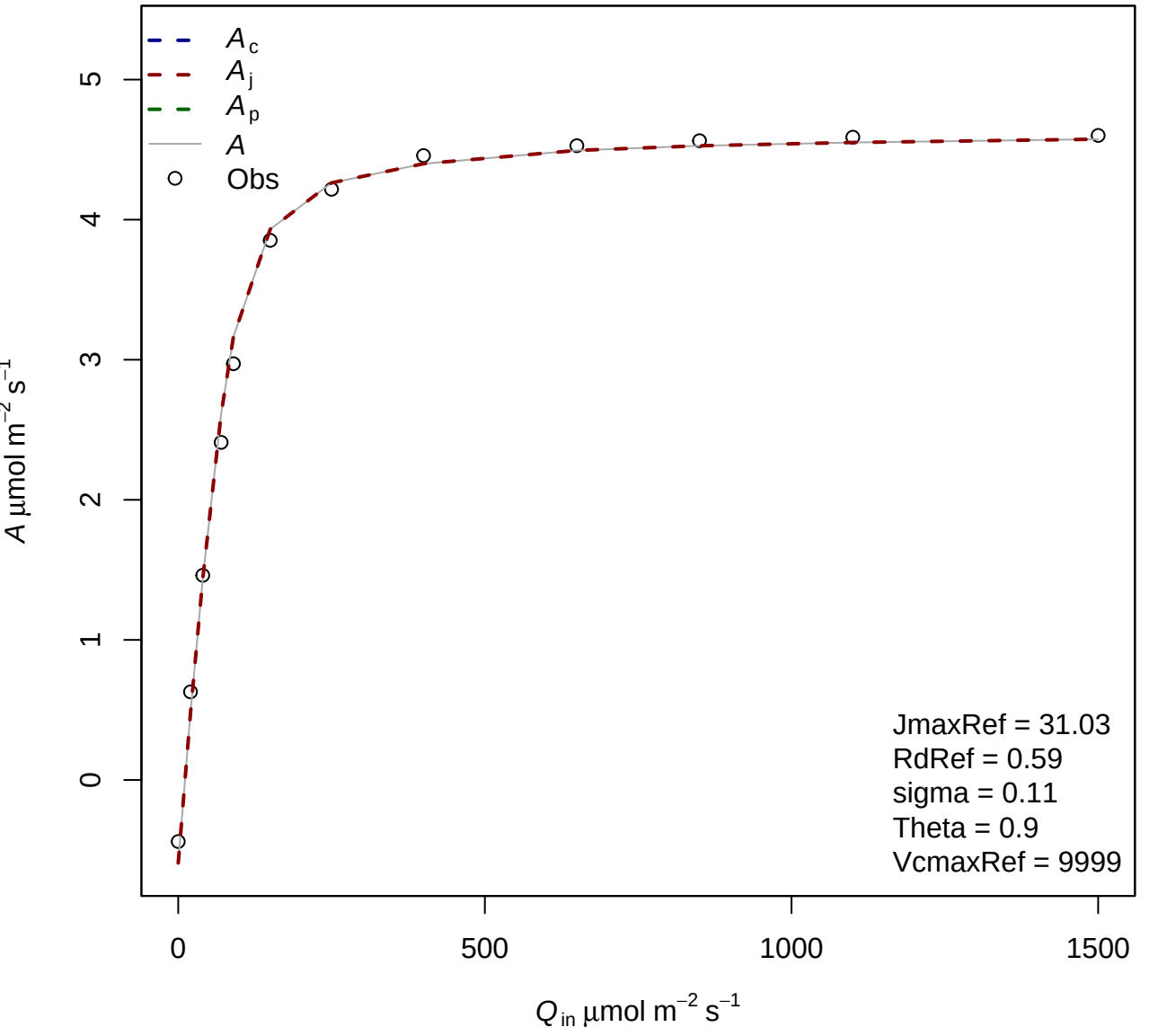

140
A_c
A_j
A_p
A
Obs
JmaxRef = 51.19
RdRef = 2.11
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

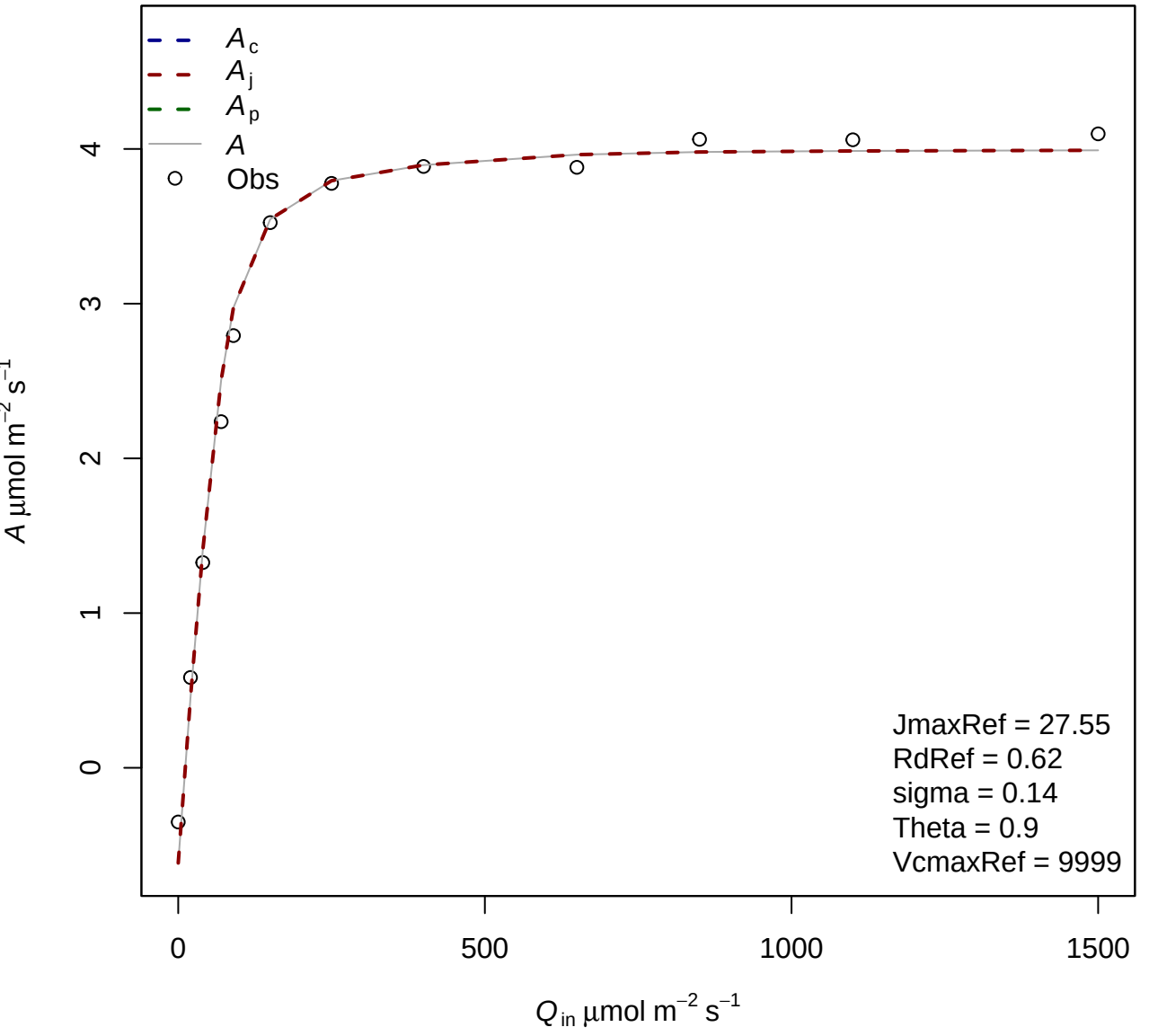

A_c
A_j
A_p
A
Obs
JmaxRef = 60.51
RdRef = 2.17
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

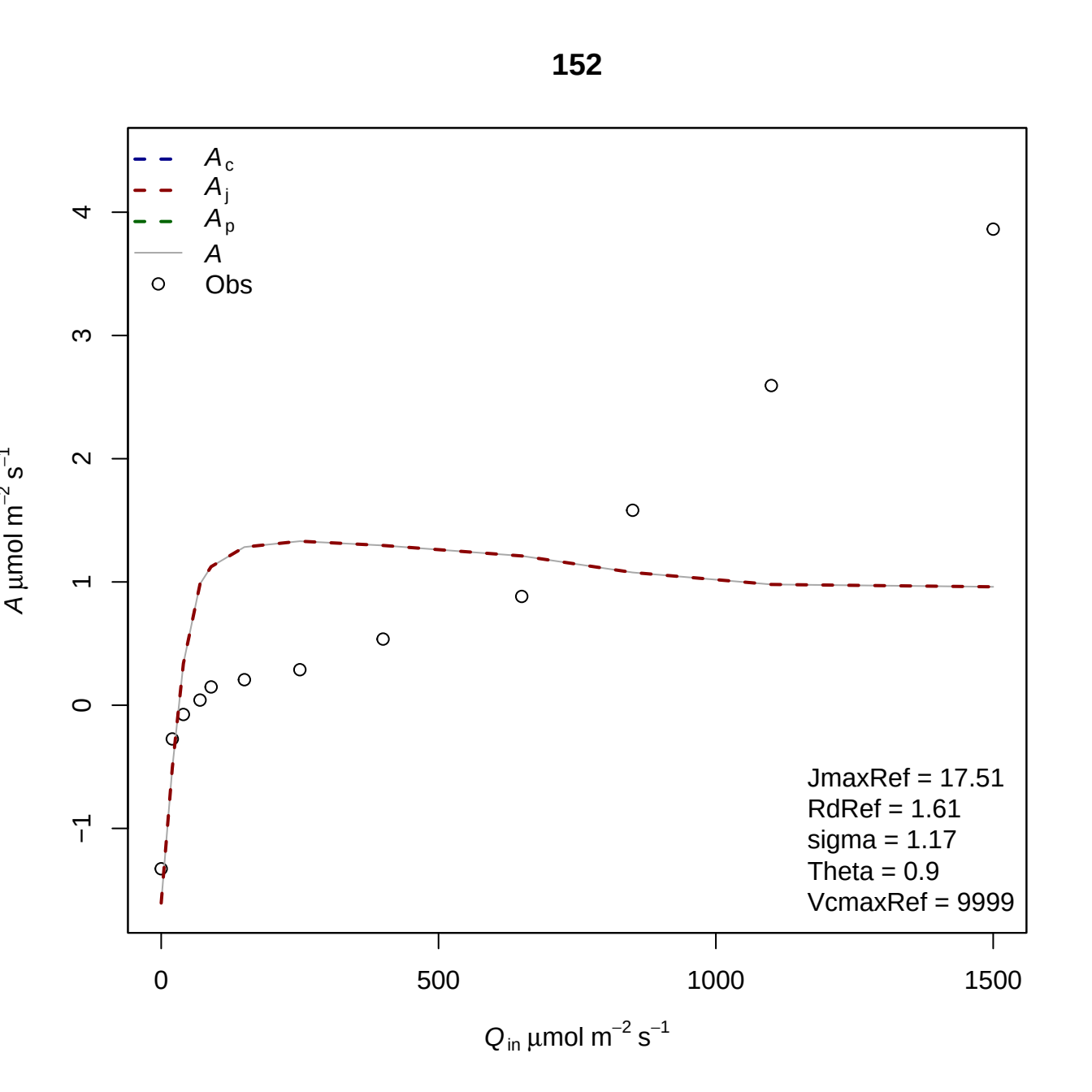

143
A_c
A_j
A_p
A
Obs
JmaxRef = 52.71
RdRef = 2.45
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

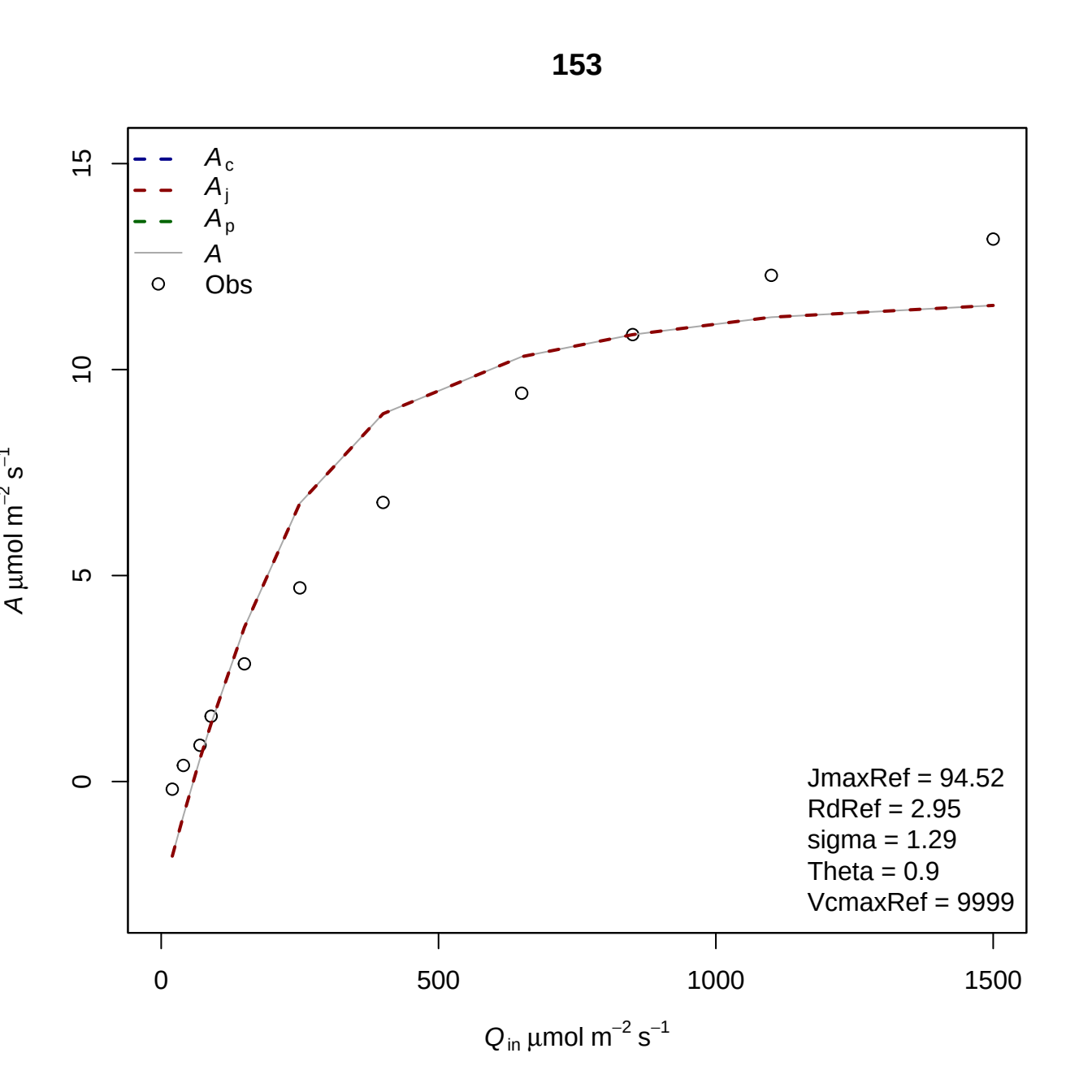

A_c
A_j
A_p
A
Obs
JmaxRef = 52.01
RdRef = 1.44
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

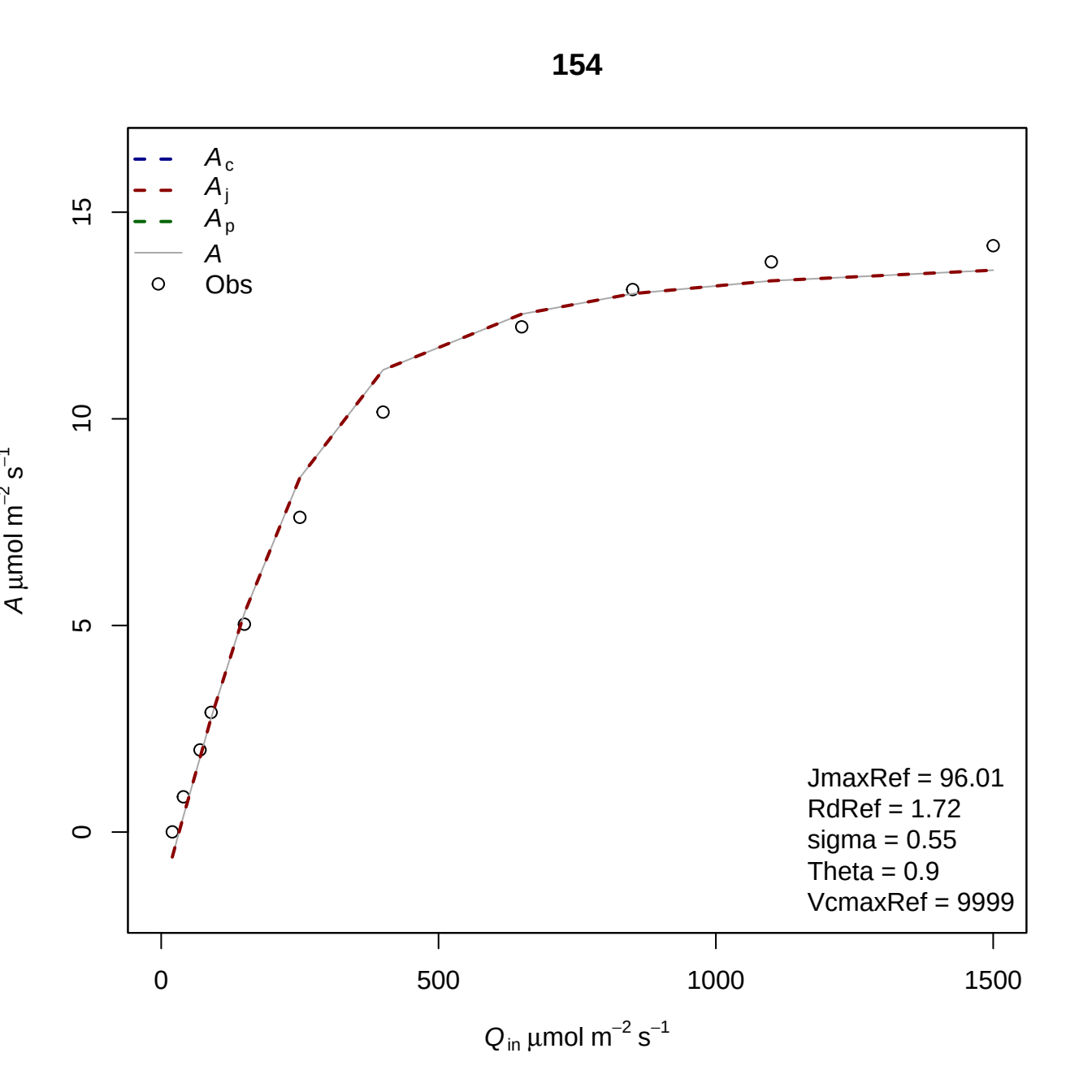

A_c
A_j
A_p
A
Obs
JmaxRef = 33.66
RdRef = 0.83
sigma = 0.14
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

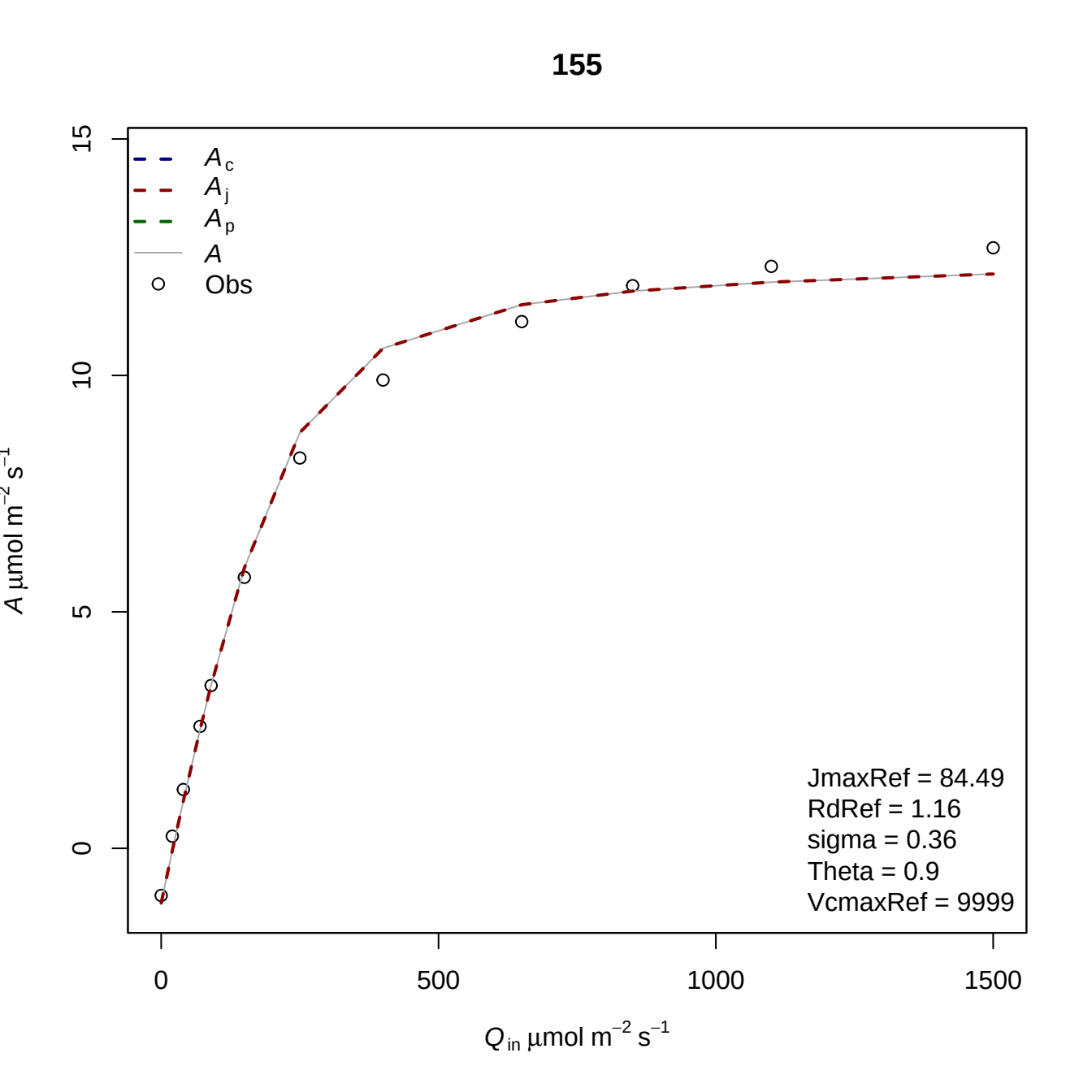

A_c
A_j
A_p
A
Obs
JmaxRef = 31.03
RdRef = 0.59
sigma = 0.11
Theta = 0.9
VcmaxRef = 9999
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

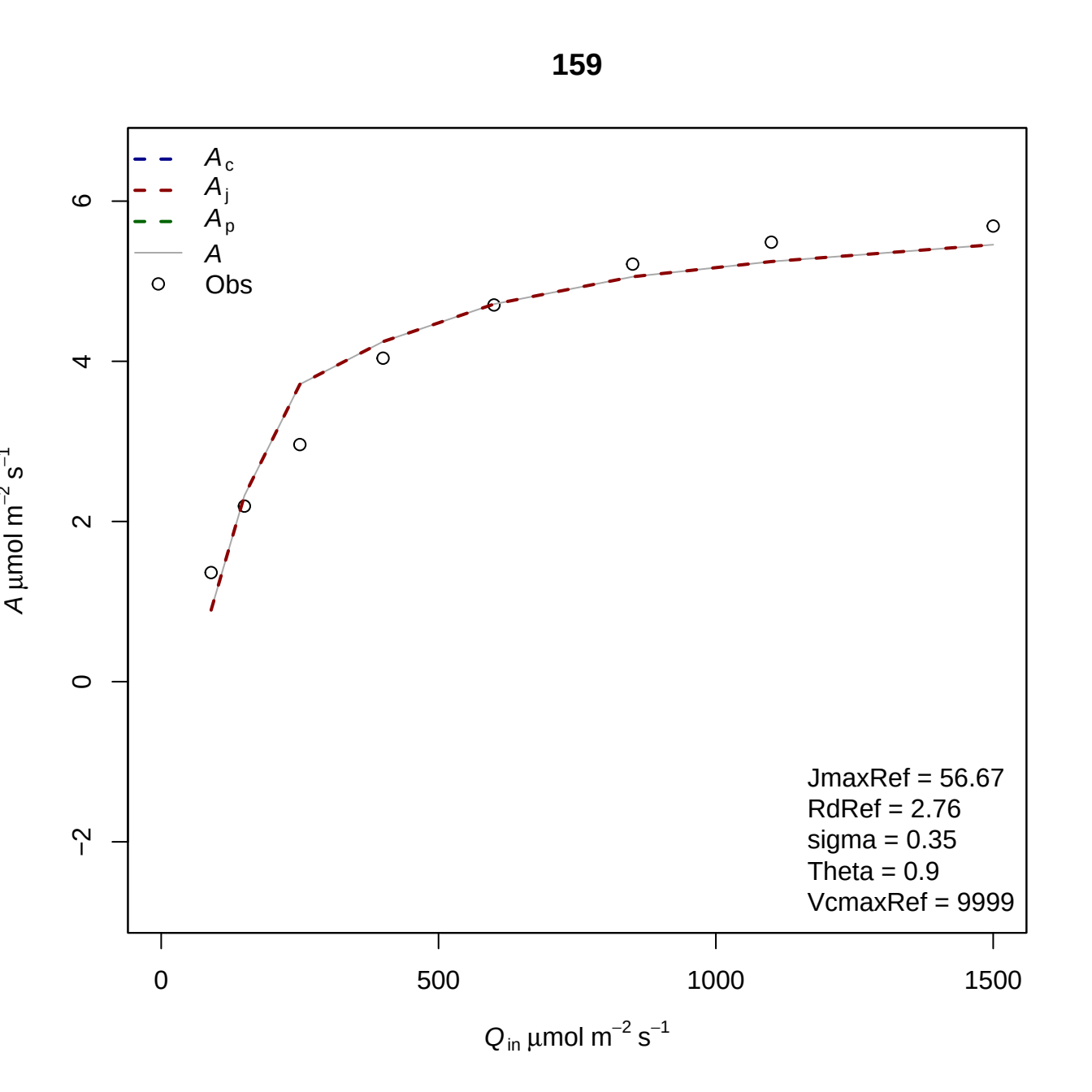

A_c
A_j
A_p
A
Obs
JmaxRef = 27.55
RdRef = 0.62
sigma = 0.14
Theta = 0.9
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

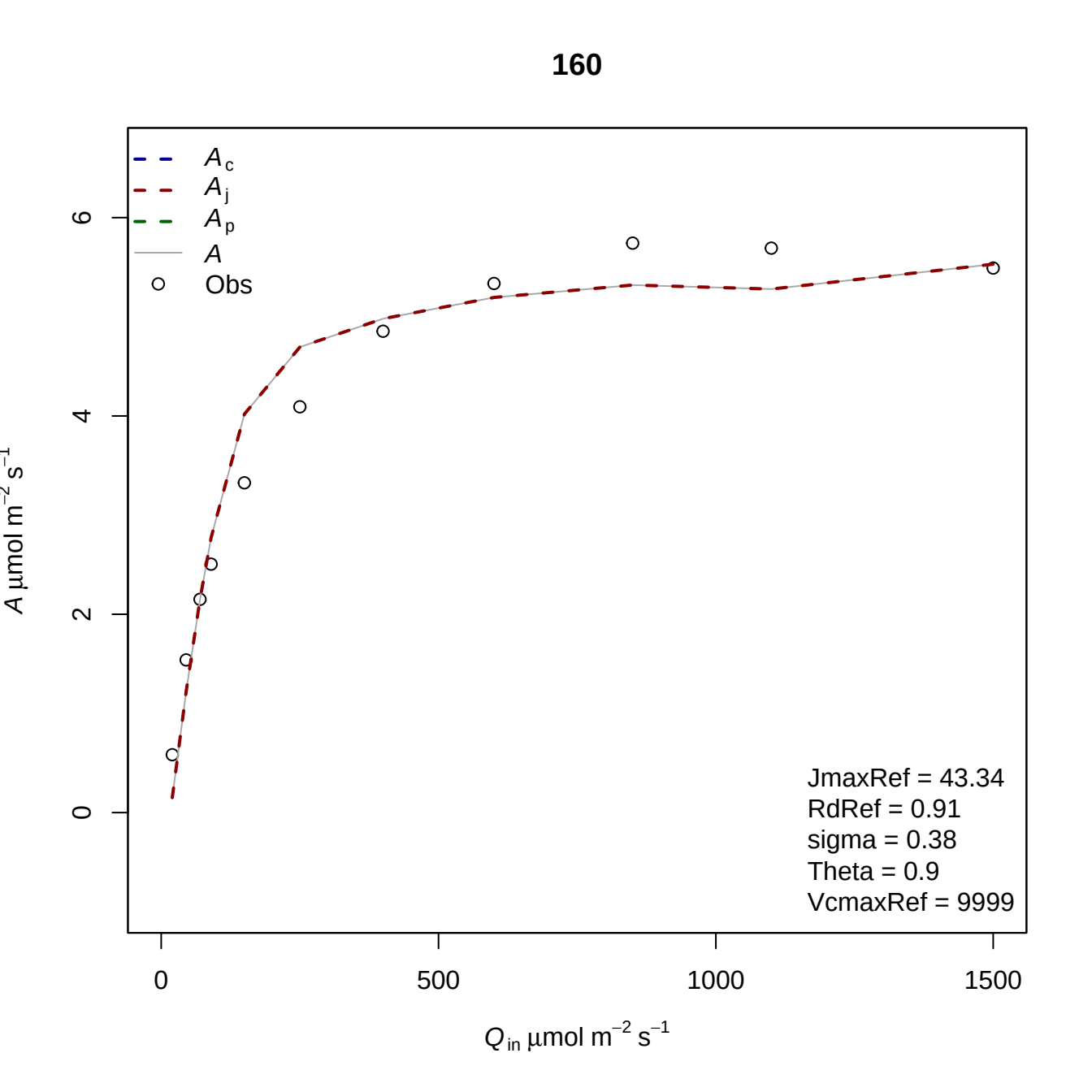

152
A_c
A_j
A_p
A
Obs
JmaxRef = 17.51
RdRef = 1.61
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

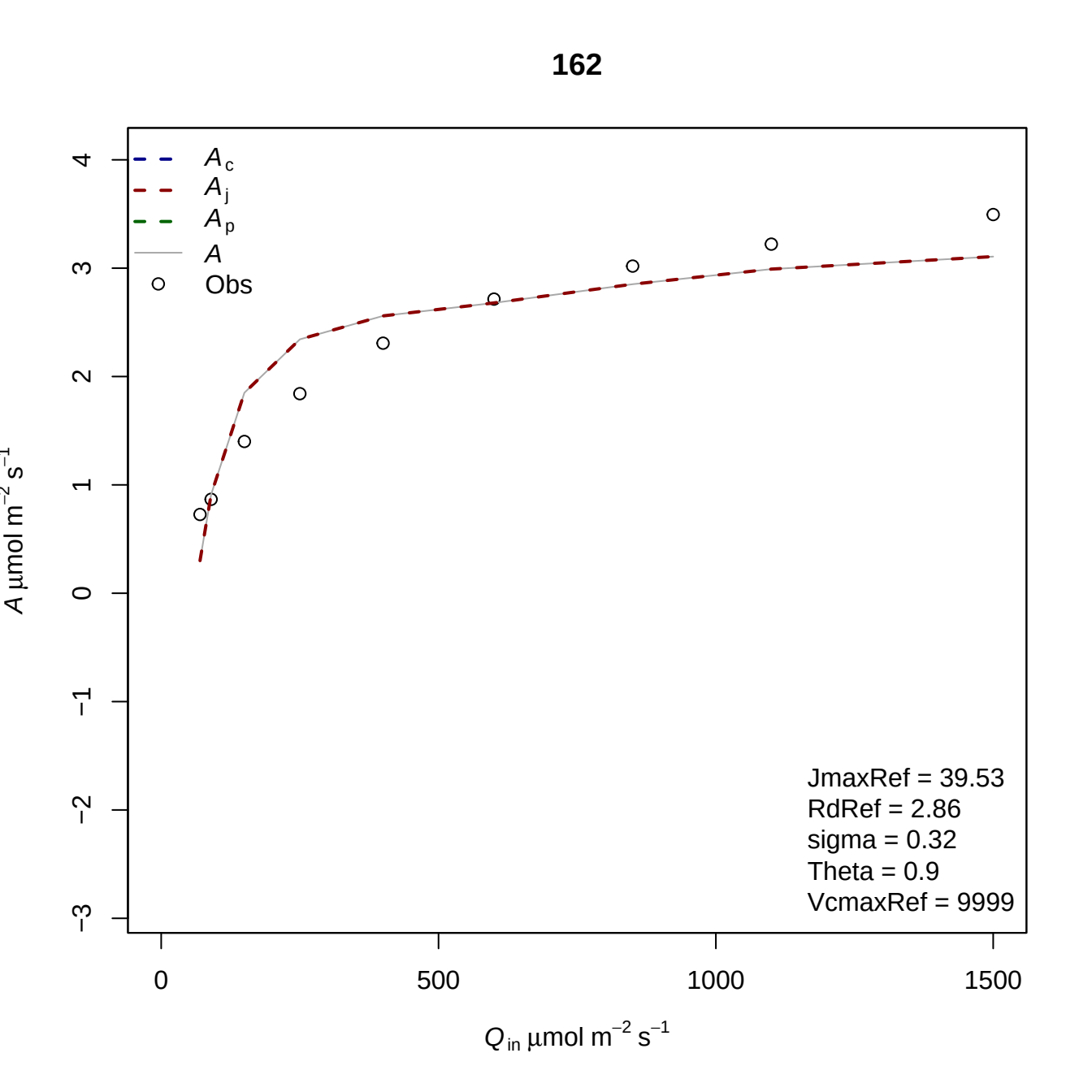

153
A_c
A_j
A_p
A
Obs
JmaxRef = 94.52
RdRef = 2.95
sigma = 1.29
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

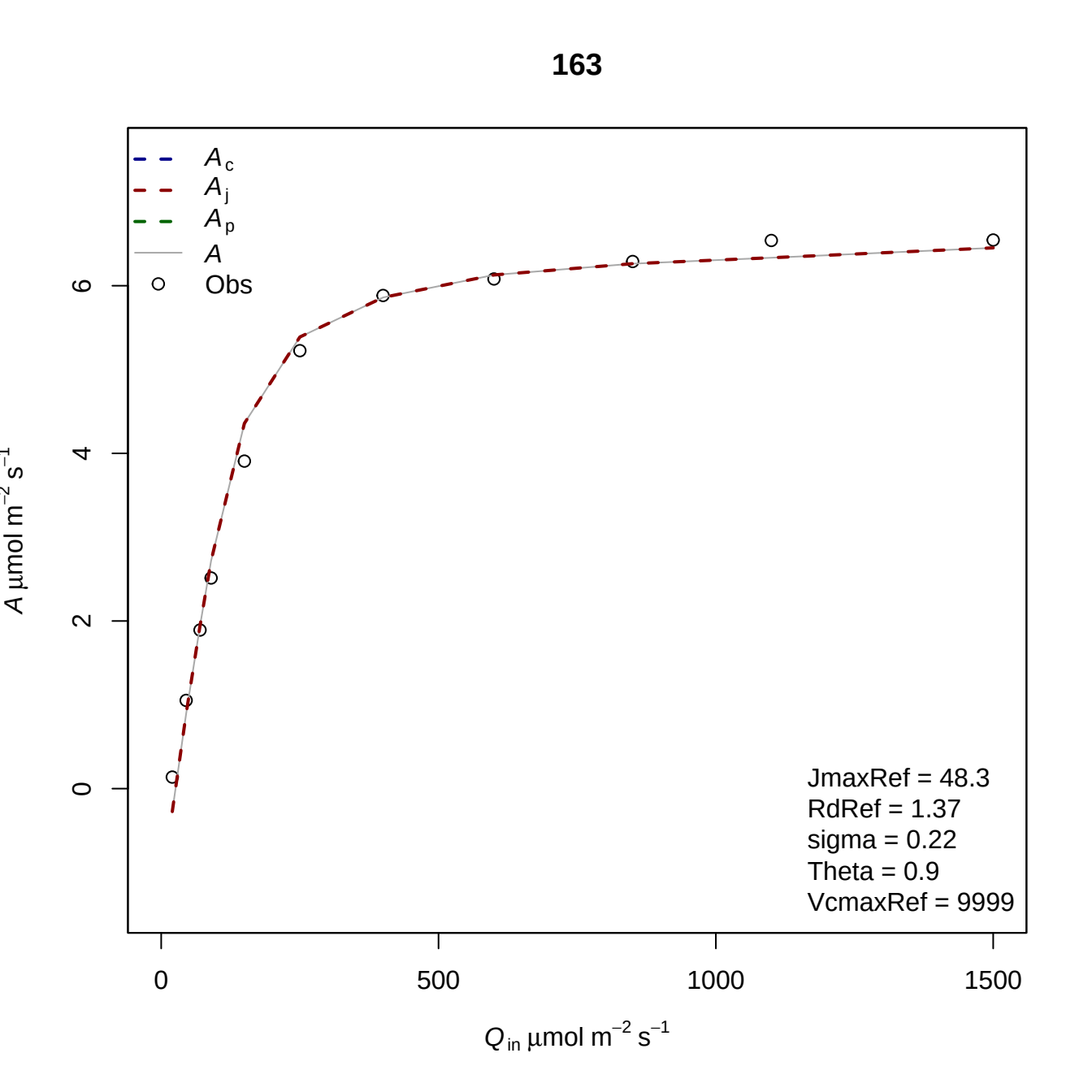

154
A_c
A_j
A_p
A
Obs
JmaxRef = 96.01
RdRef = 1.72
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

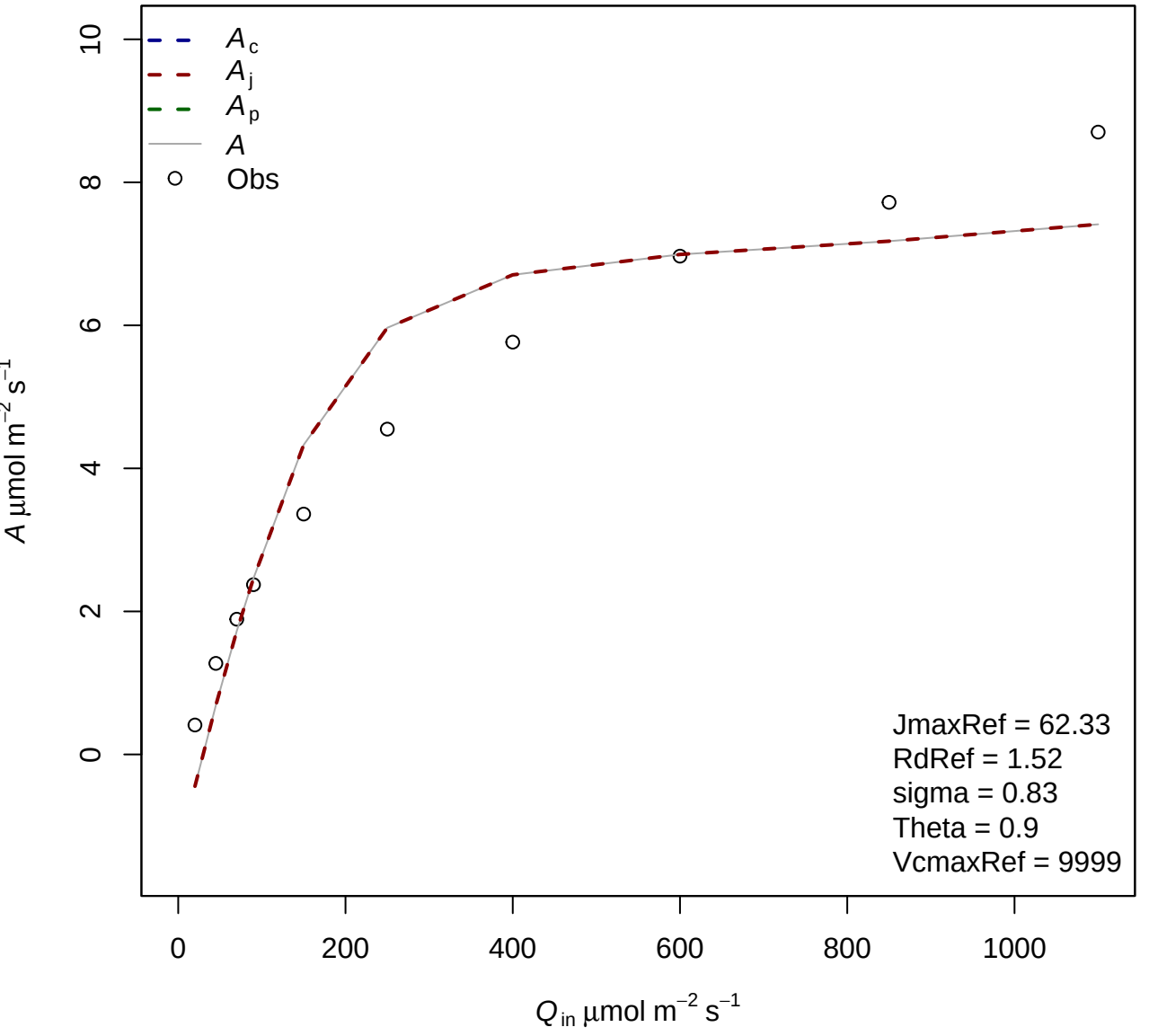

155
A_c
A_j
A_p
A
Obs
JmaxRef = 84.49
RdRef = 1.16
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

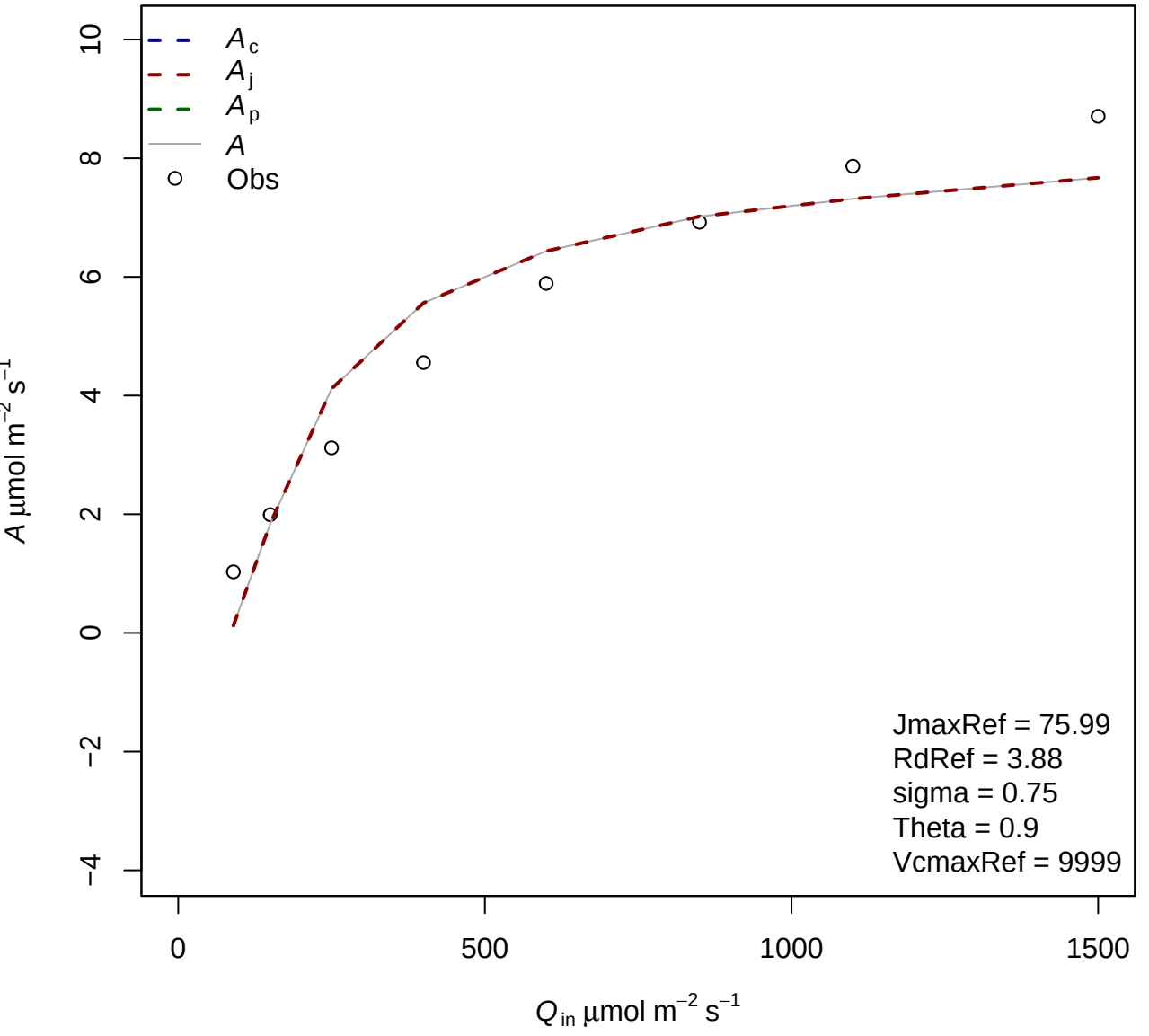

159
A_c
A_j
A_p
A
Obs
JmaxRef = 56.67
RdRef = 2.76
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

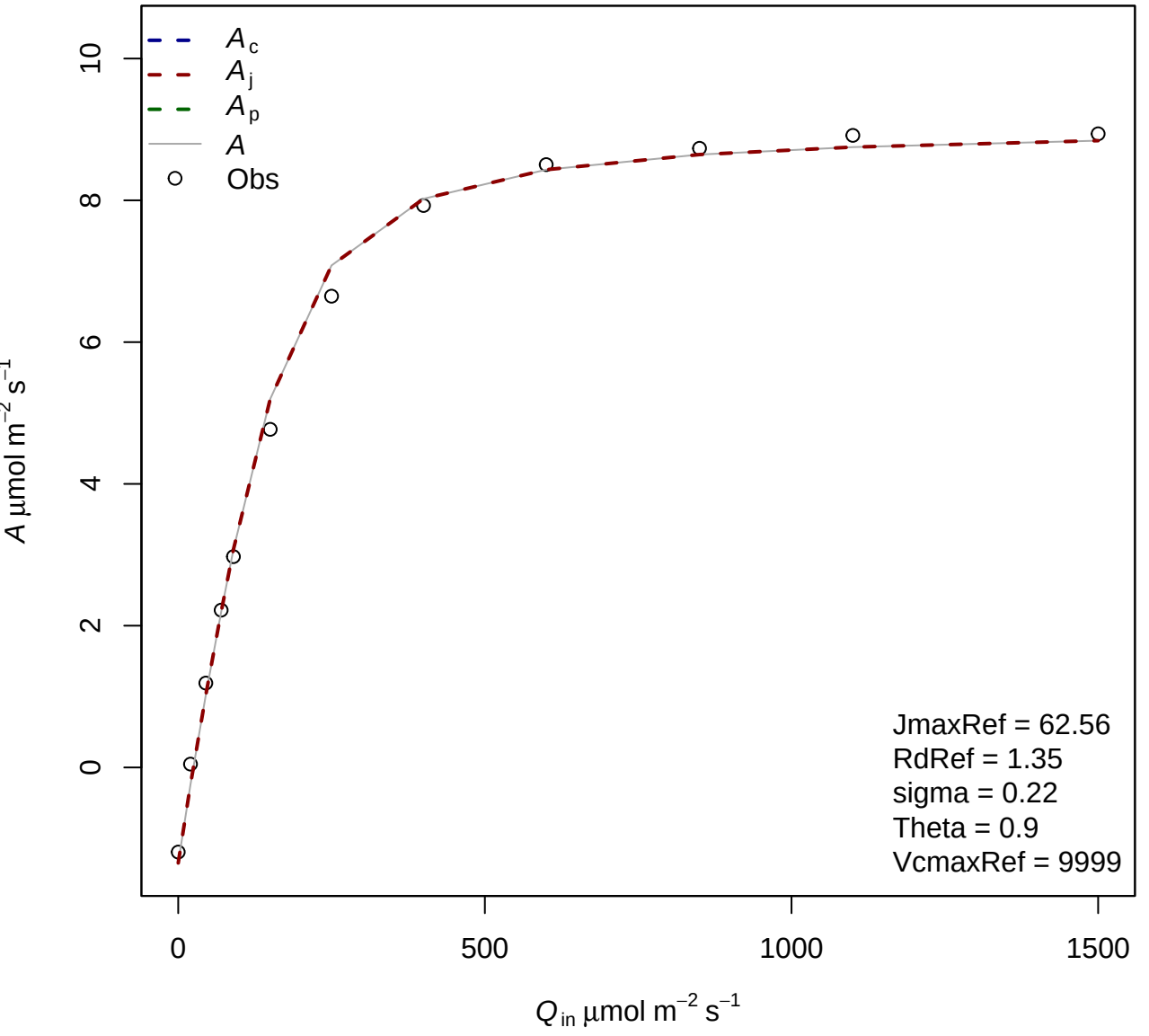

160
A_c
A_j
A_p
A
Obs
JmaxRef = 43.34
RdRef = 0.91
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

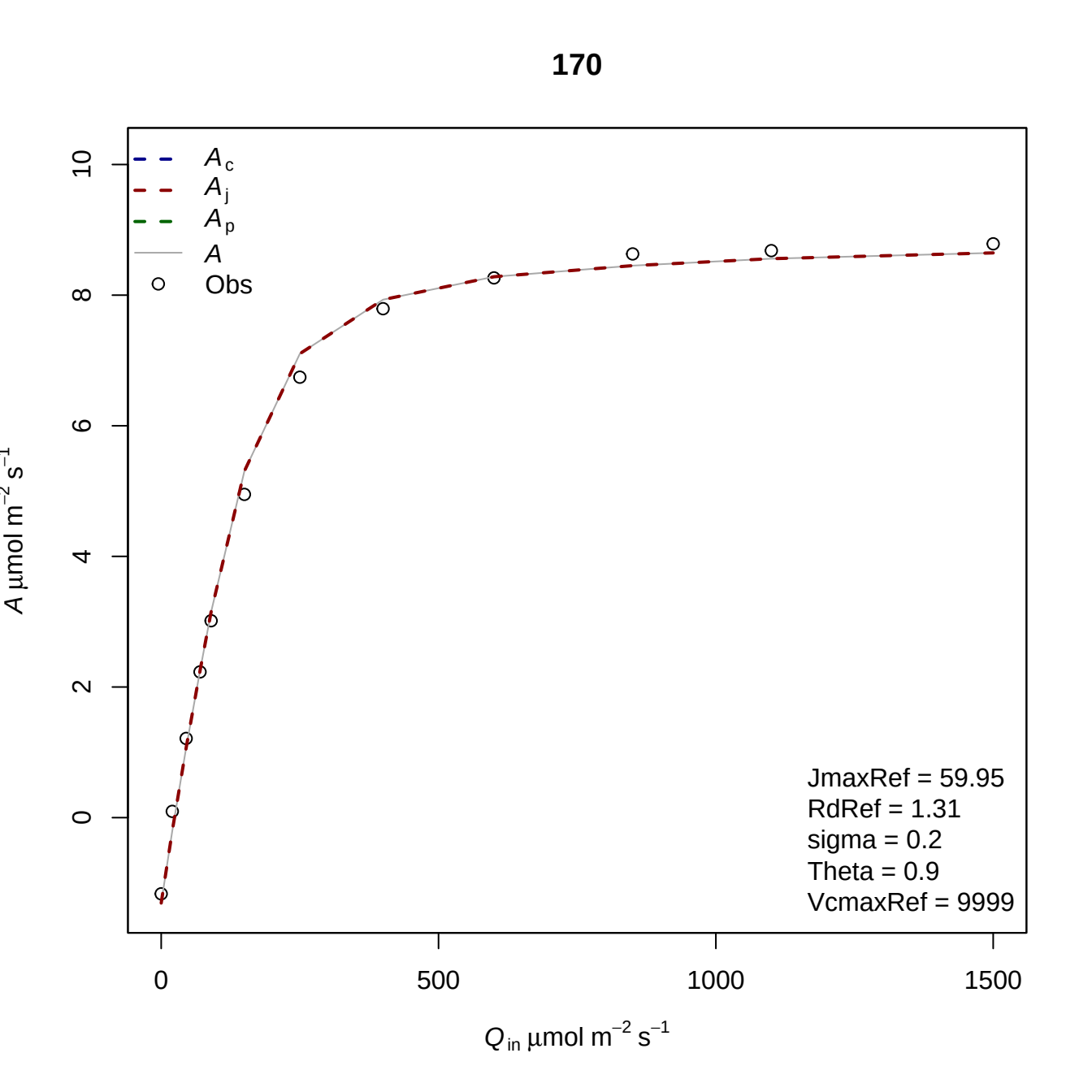

162
A_c
A_j
A_p
A
Obs
JmaxRef = 39.53
RdRef = 2.86
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

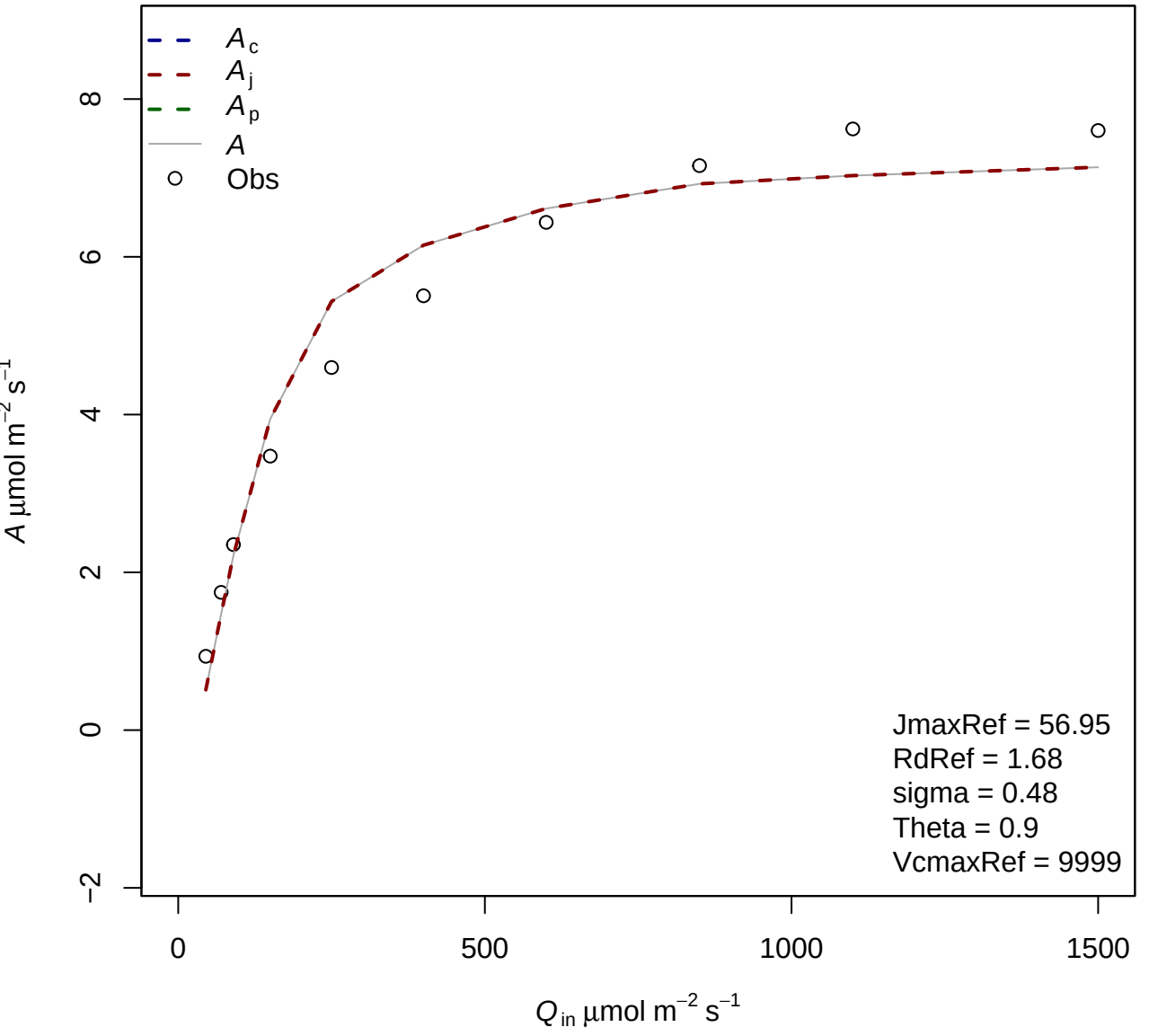

163
A_c
A_j
A_p
A
Obs
JmaxRef = 48.3
RdRef = 1.37
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

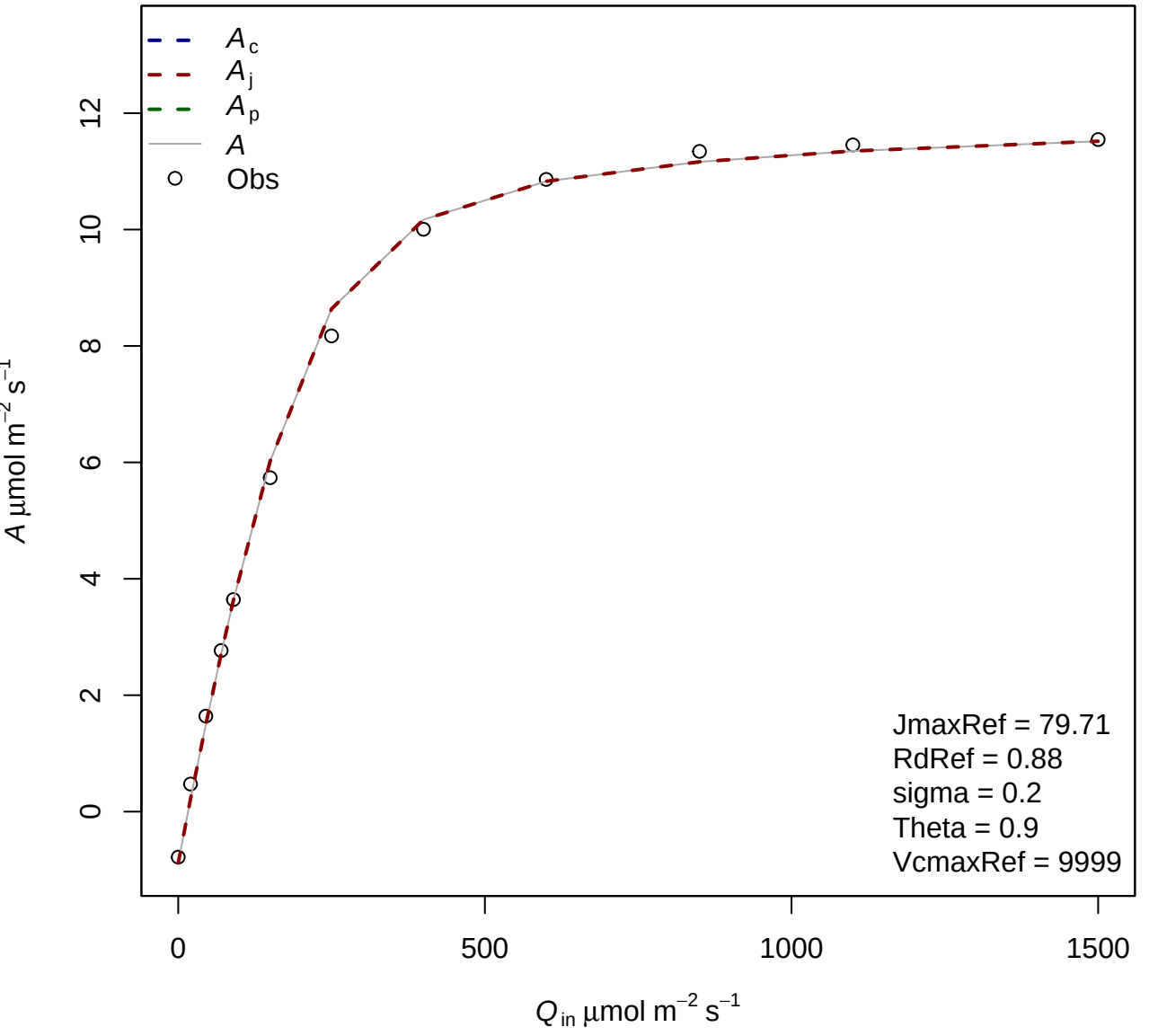

A_c
A_j
A_p
A
Obs
JmaxRef = 62.33
RdRef = 1.52
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

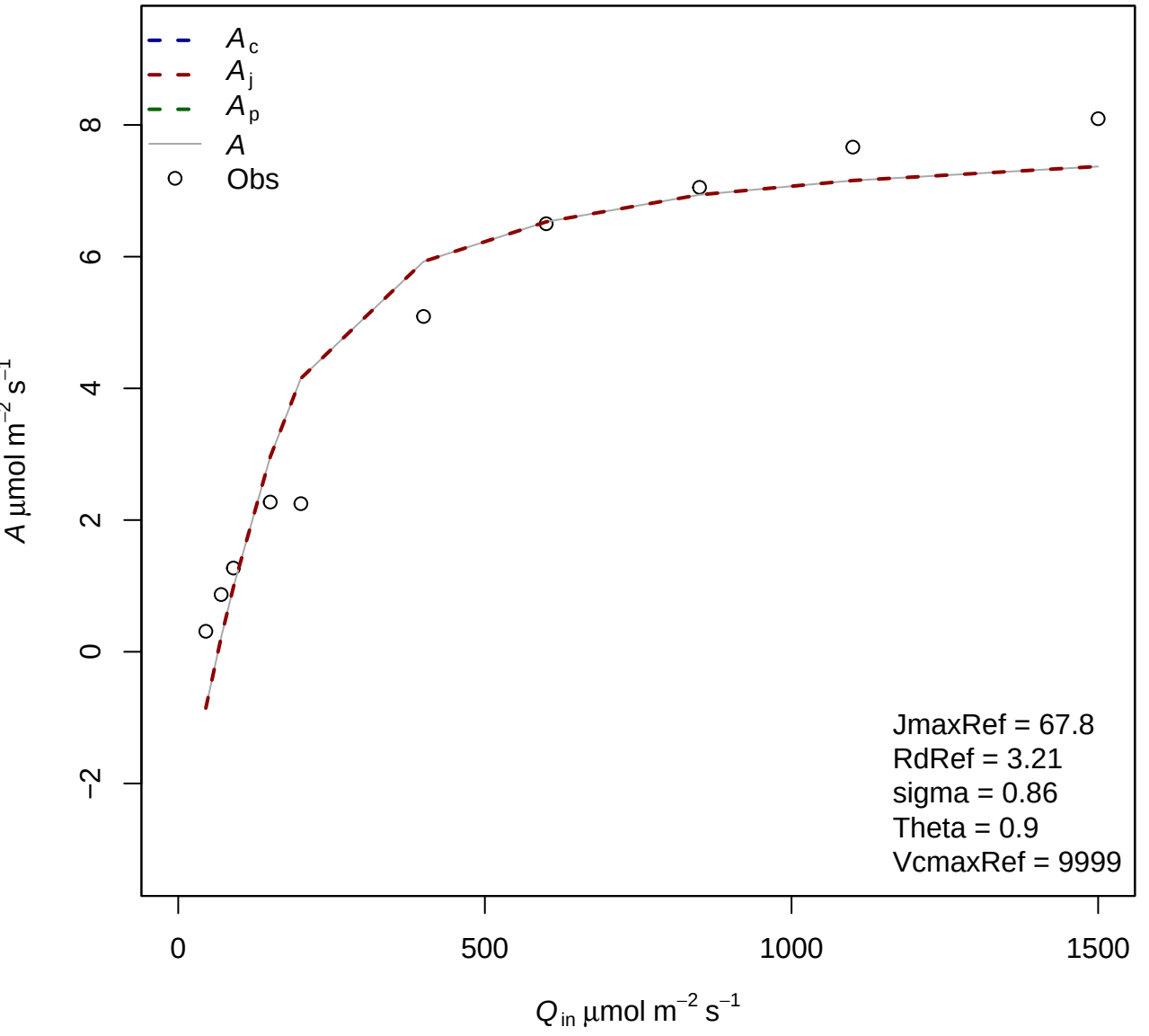

A_c
A_j
A_p
A
Obs
JmaxRef = 75.99
RdRef = 3.88
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

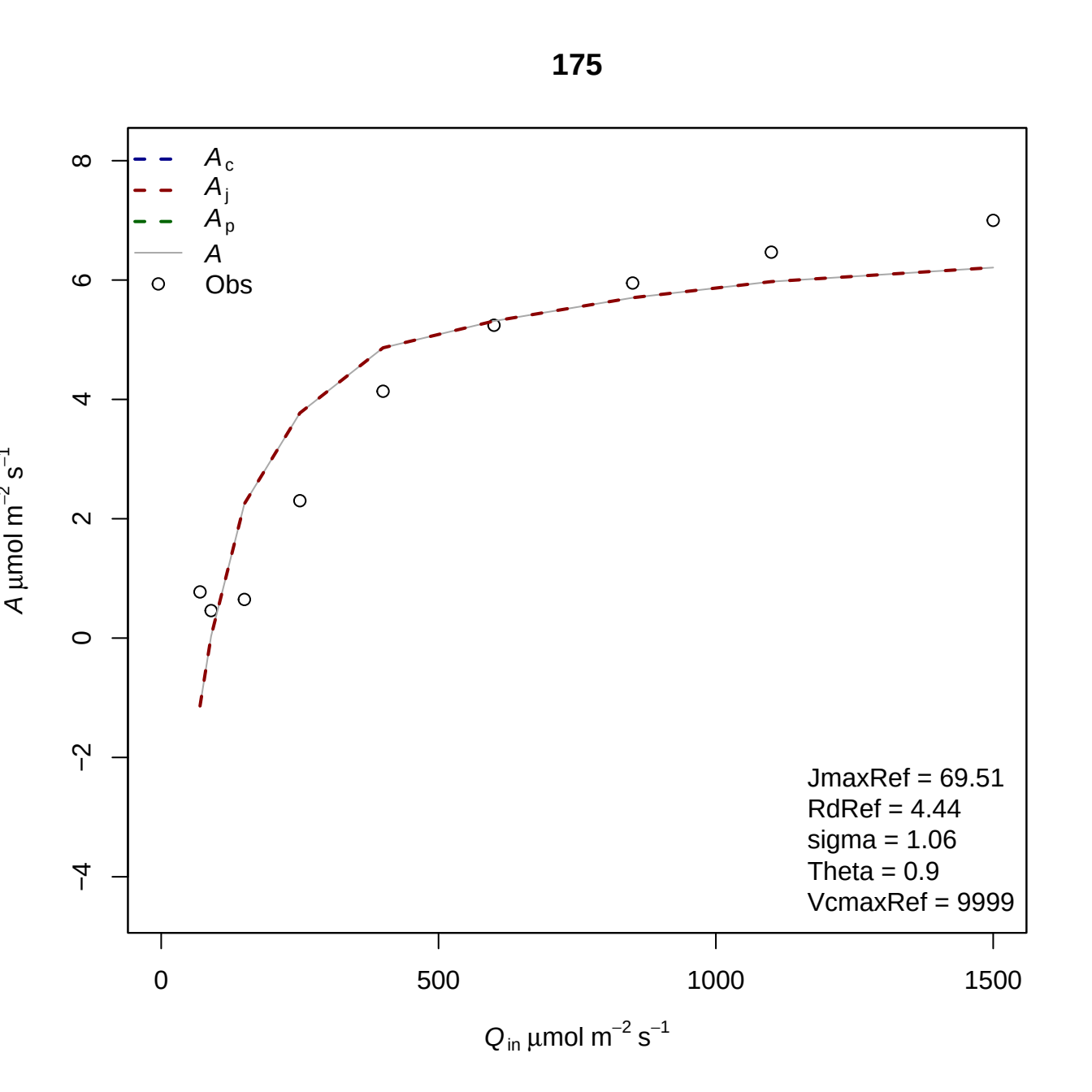

A_c
A_j
A_p
A
Obs
JmaxRef = 62.56
RdRef = 1.35
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

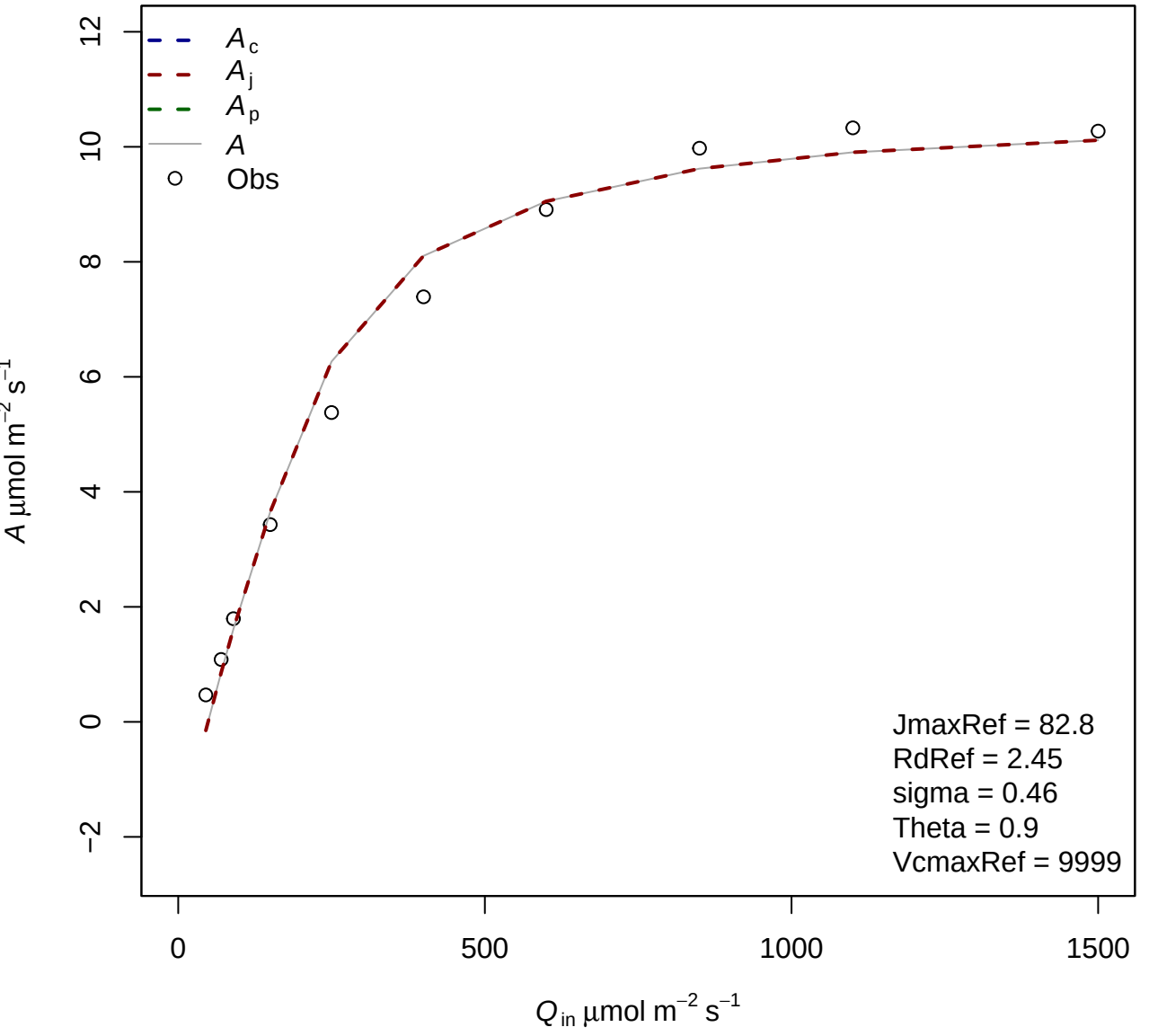

170
A_c
A_j
A_p
A
Obs
JmaxRef = 59.95
RdRef = 1.31
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

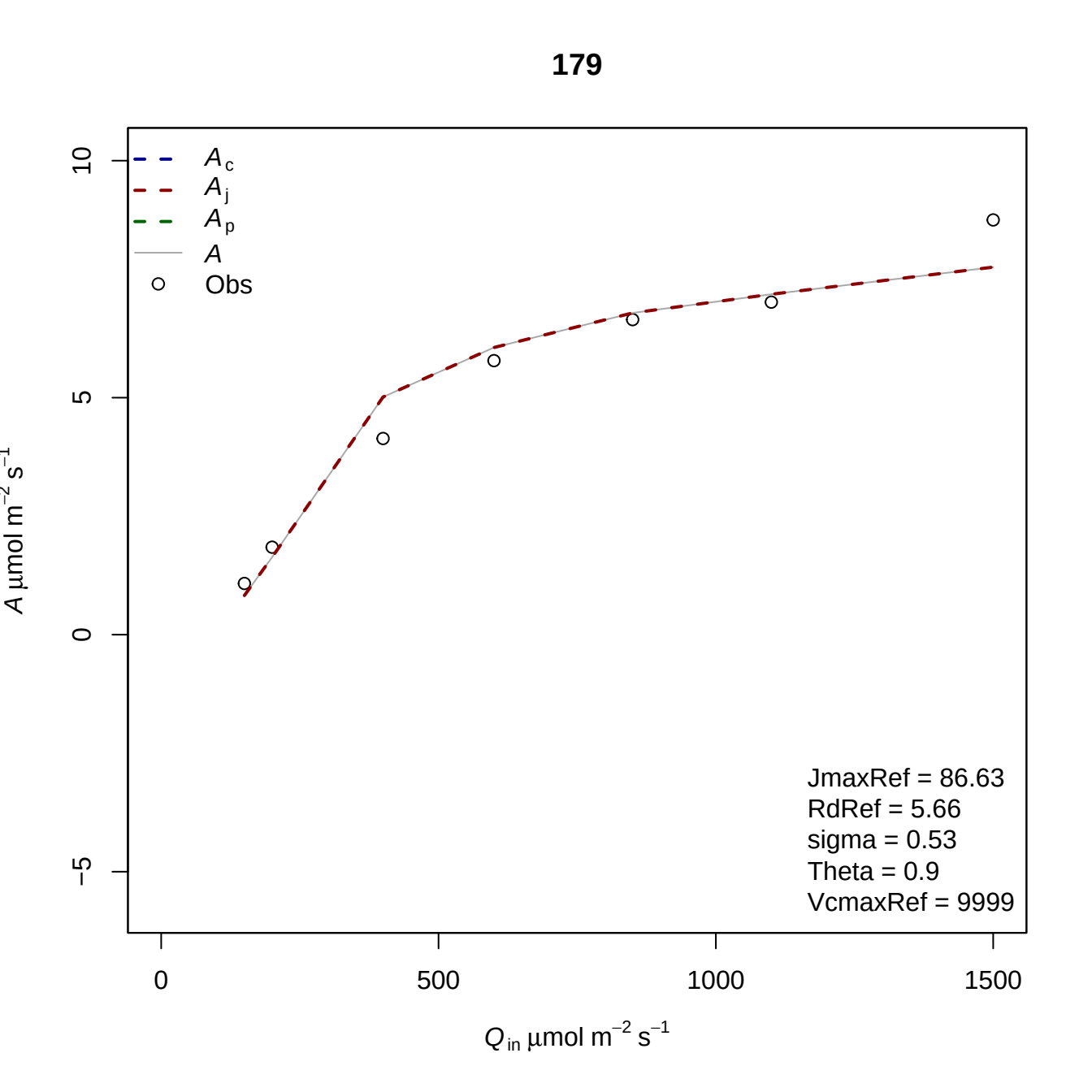

A_c
A_j
A_p
A
Obs
JmaxRef = 56.95
RdRef = 1.68
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

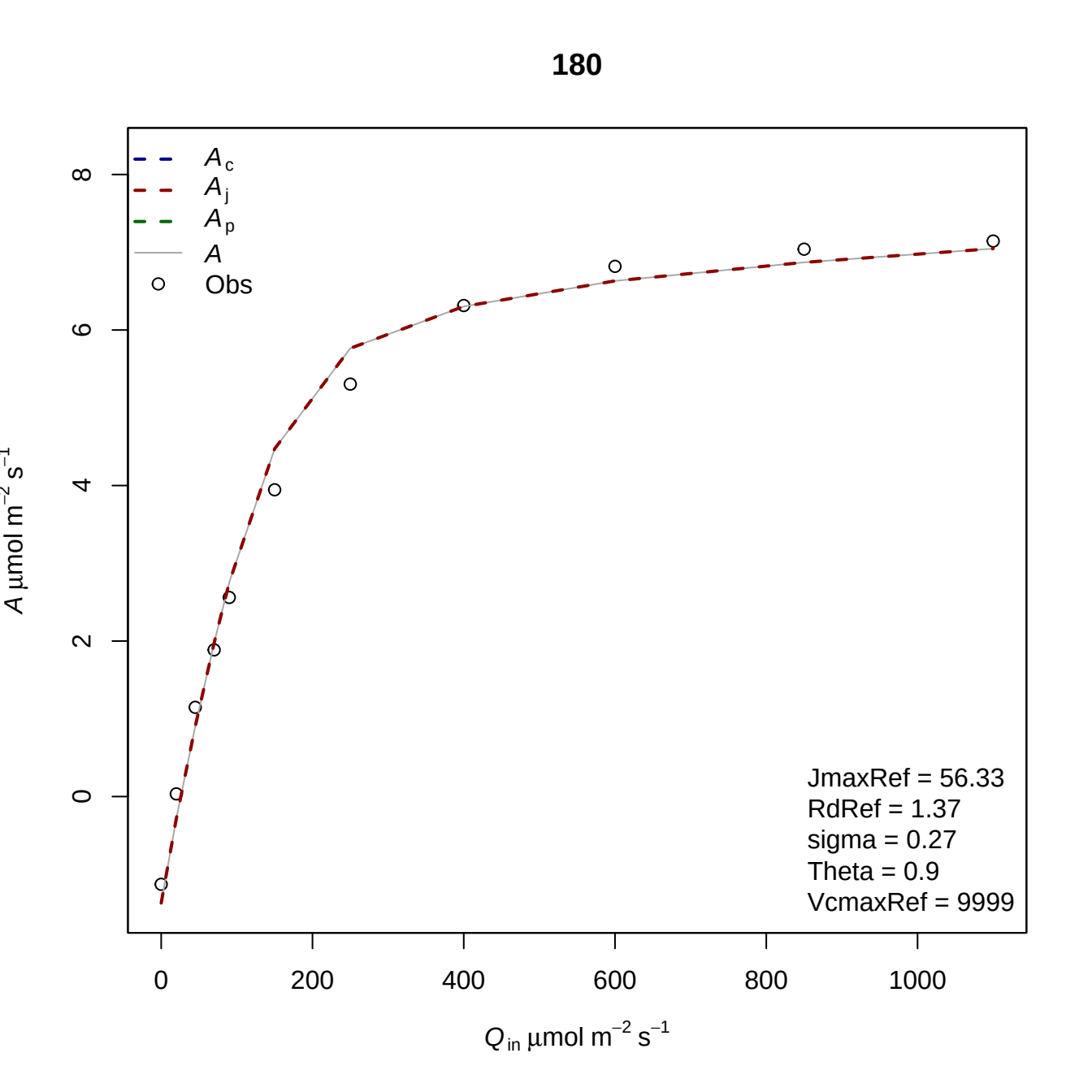

A_c
A_j
A_p
A
Obs
JmaxRef = 79.71
RdRef = 0.88
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

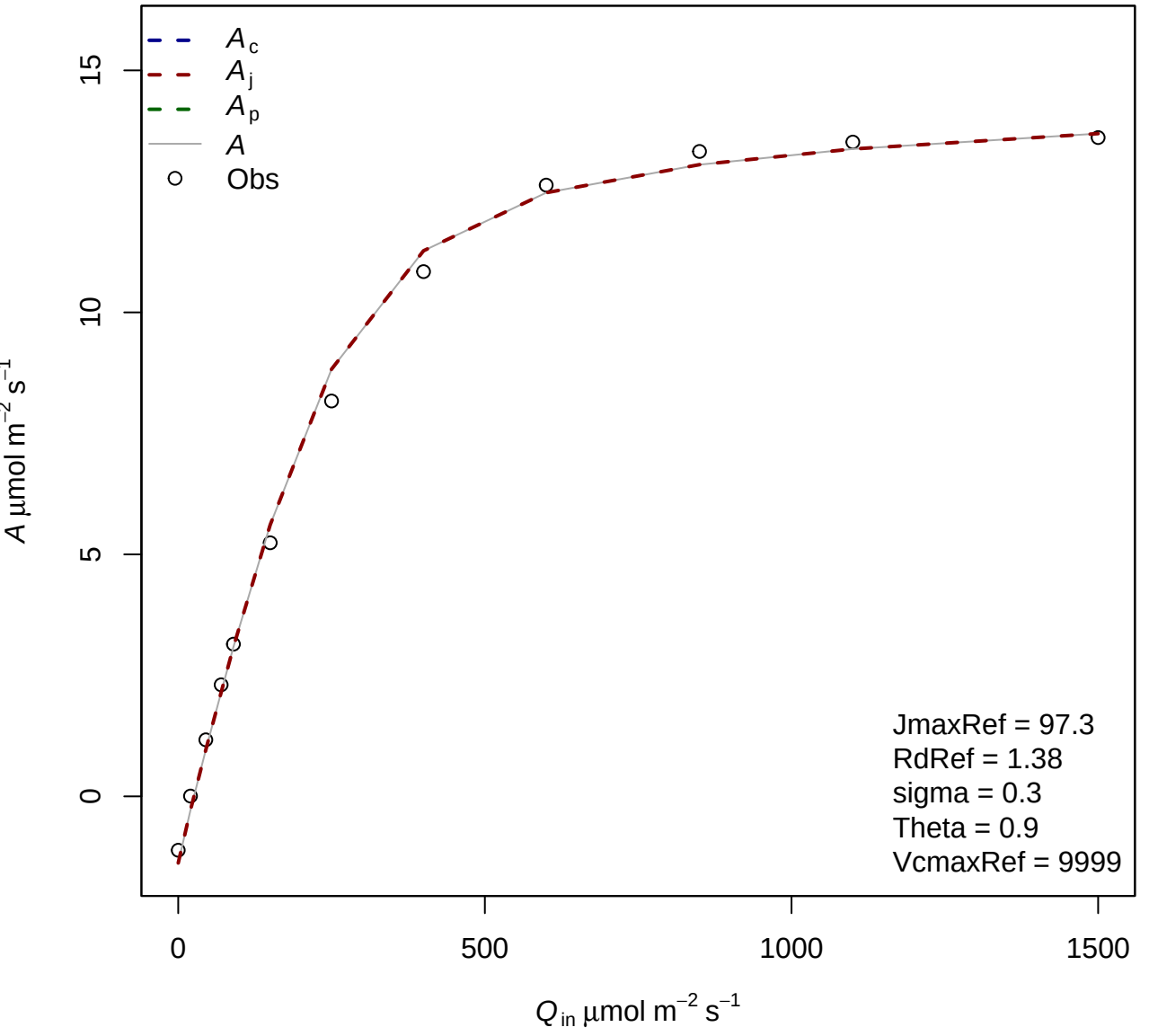

A_c
A_j
A_p
A
Obs
JmaxRef = 67.8
RdRef = 3.21
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

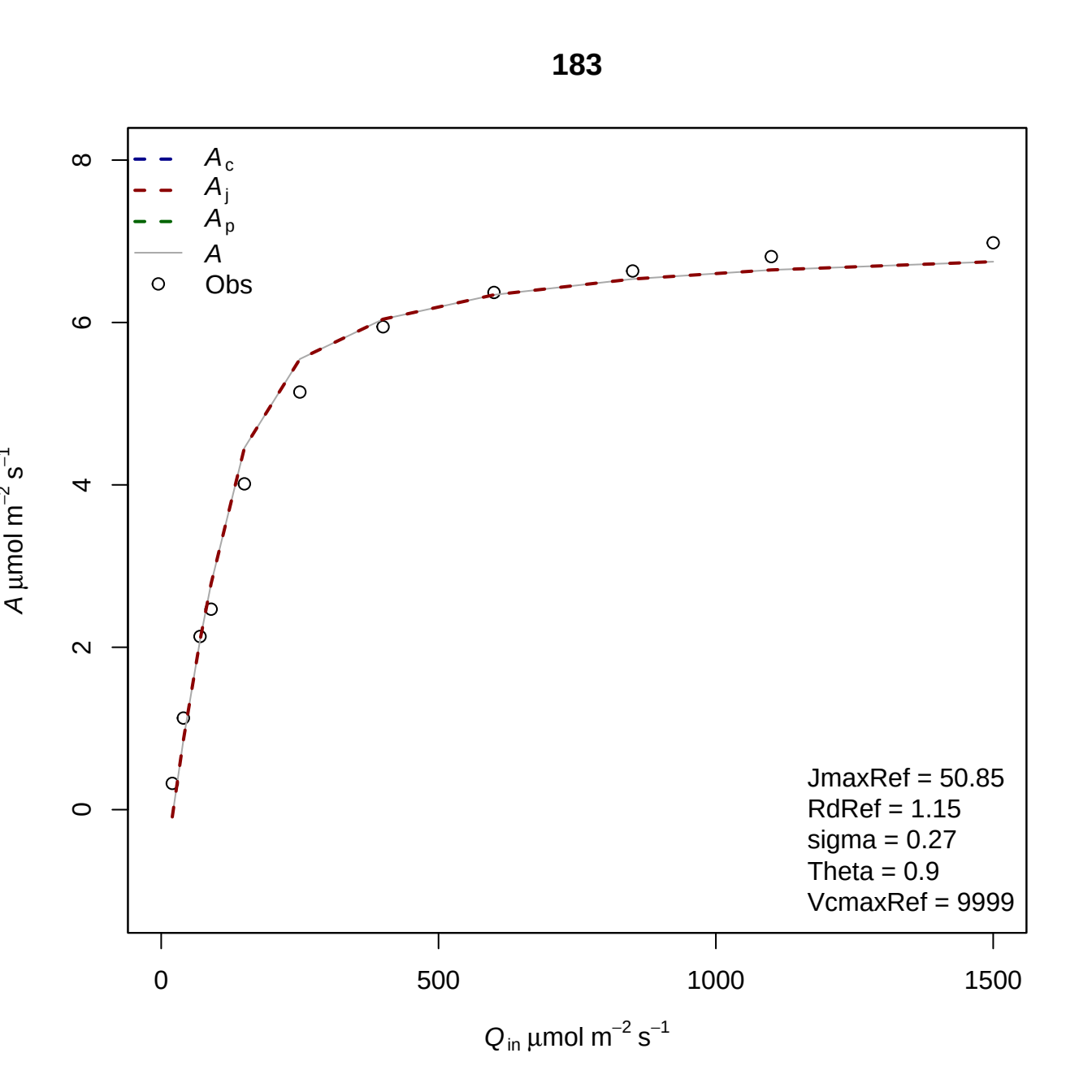

175
A_c
A_j
A_p
A
Obs
JmaxRef = 69.51
RdRef = 4.44
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

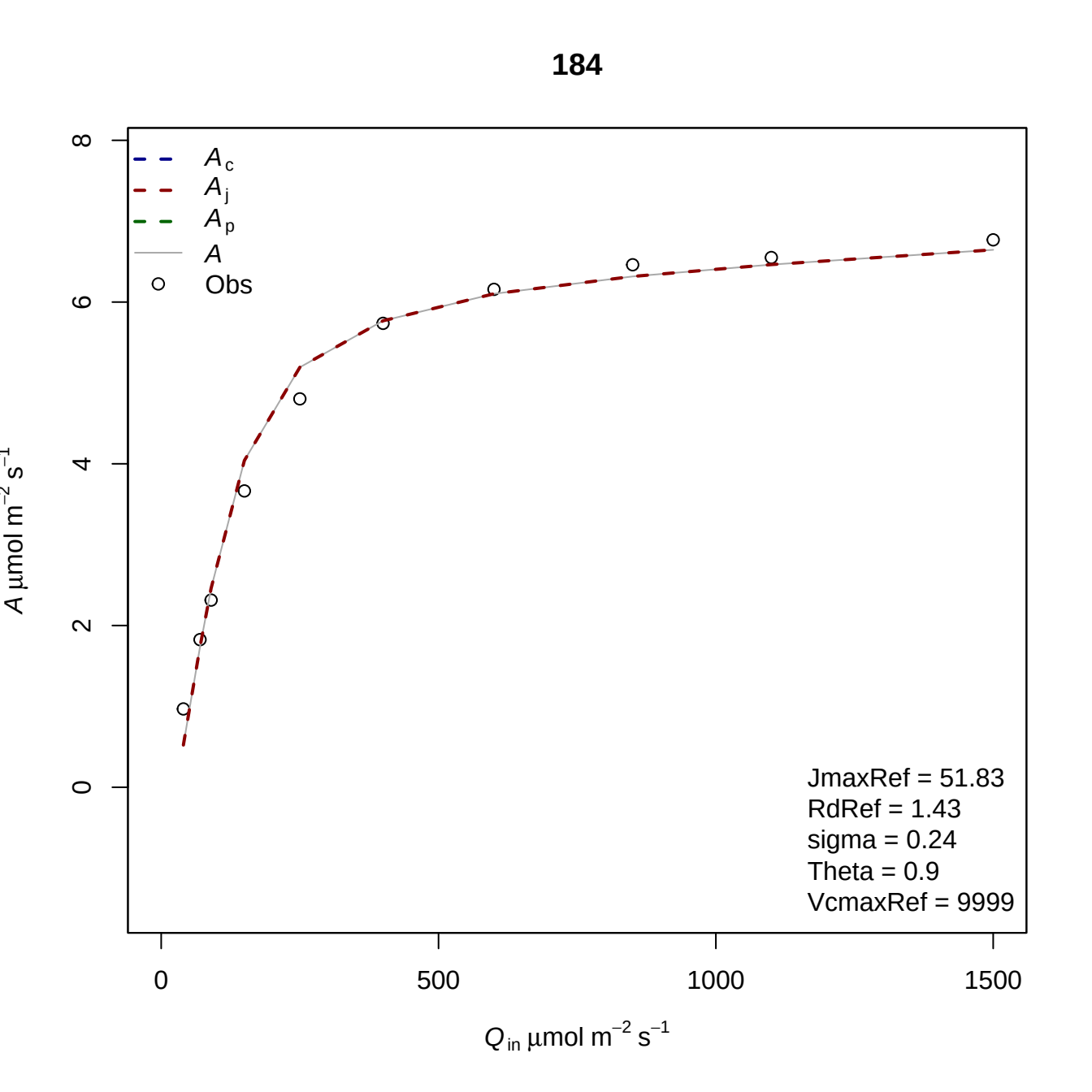

A_c
A_j
A_p
A
Obs
JmaxRef = 82.8
RdRef = 2.45
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

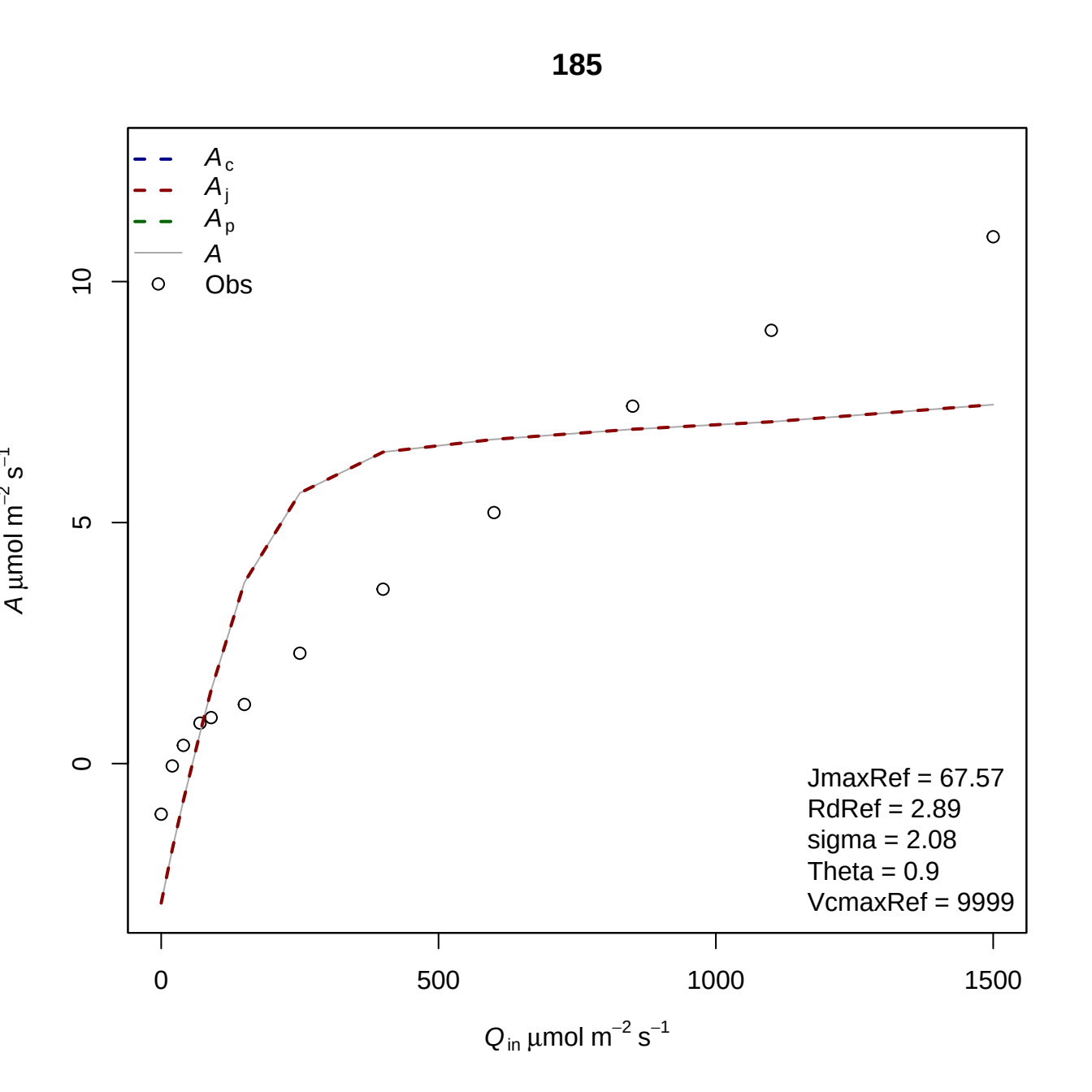

179
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 86.63
RdRef = 5.66
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999

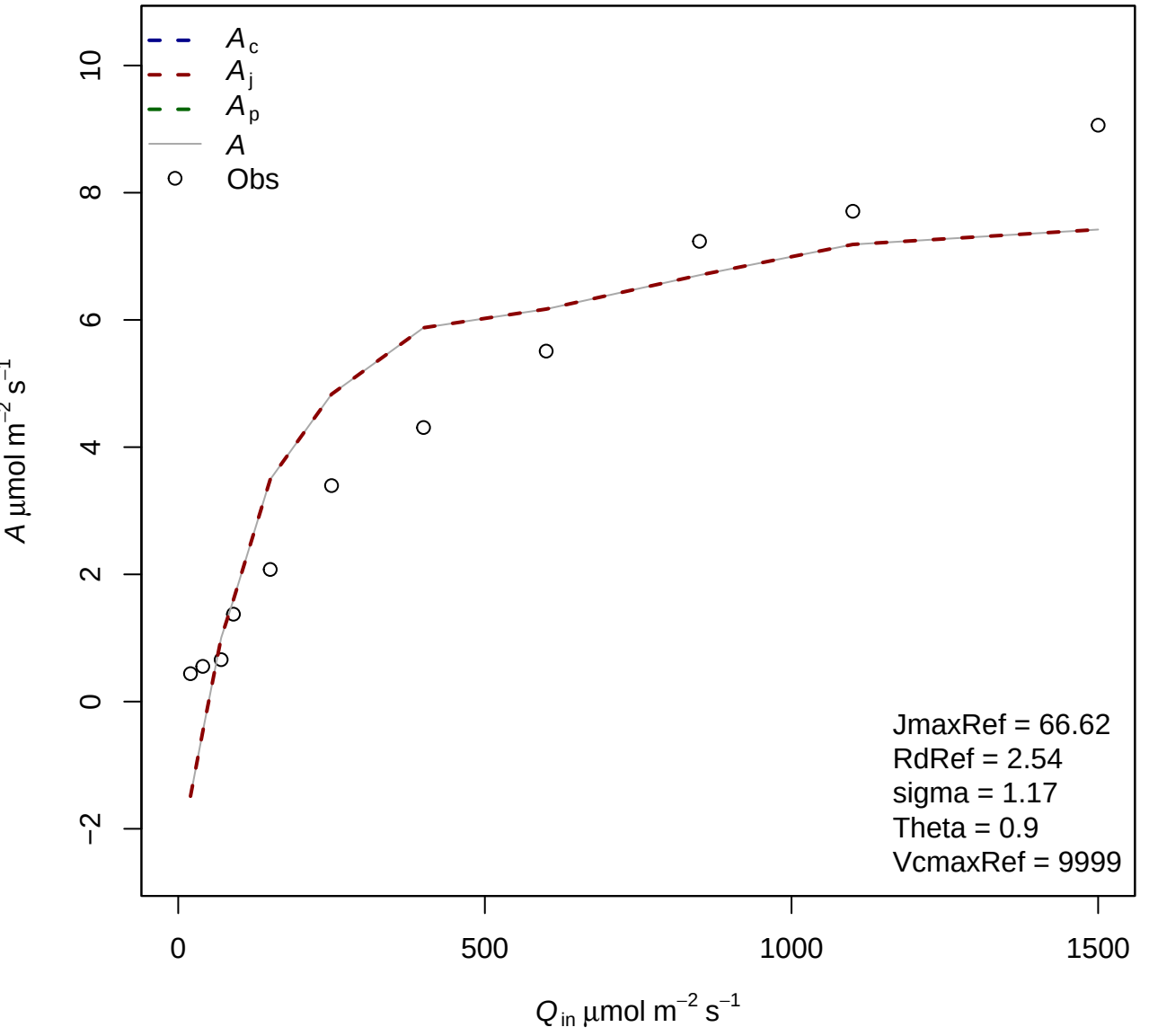

180
A_c
A_j
A_p
A
Obs
JmaxRef = 56.33
RdRef = 1.37
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

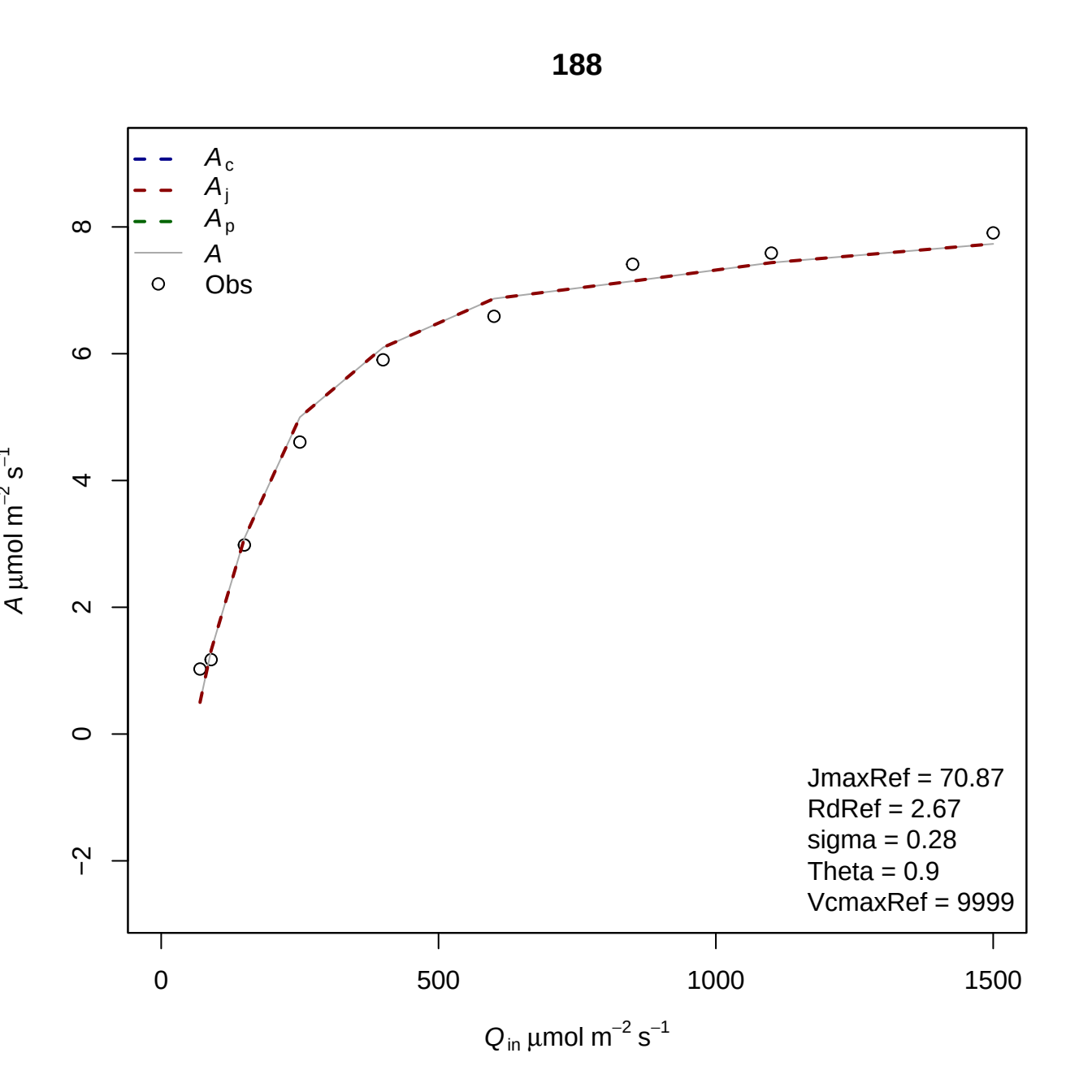

A_c
A_j
A_p
A
Obs
JmaxRef = 97.3
RdRef = 1.38
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

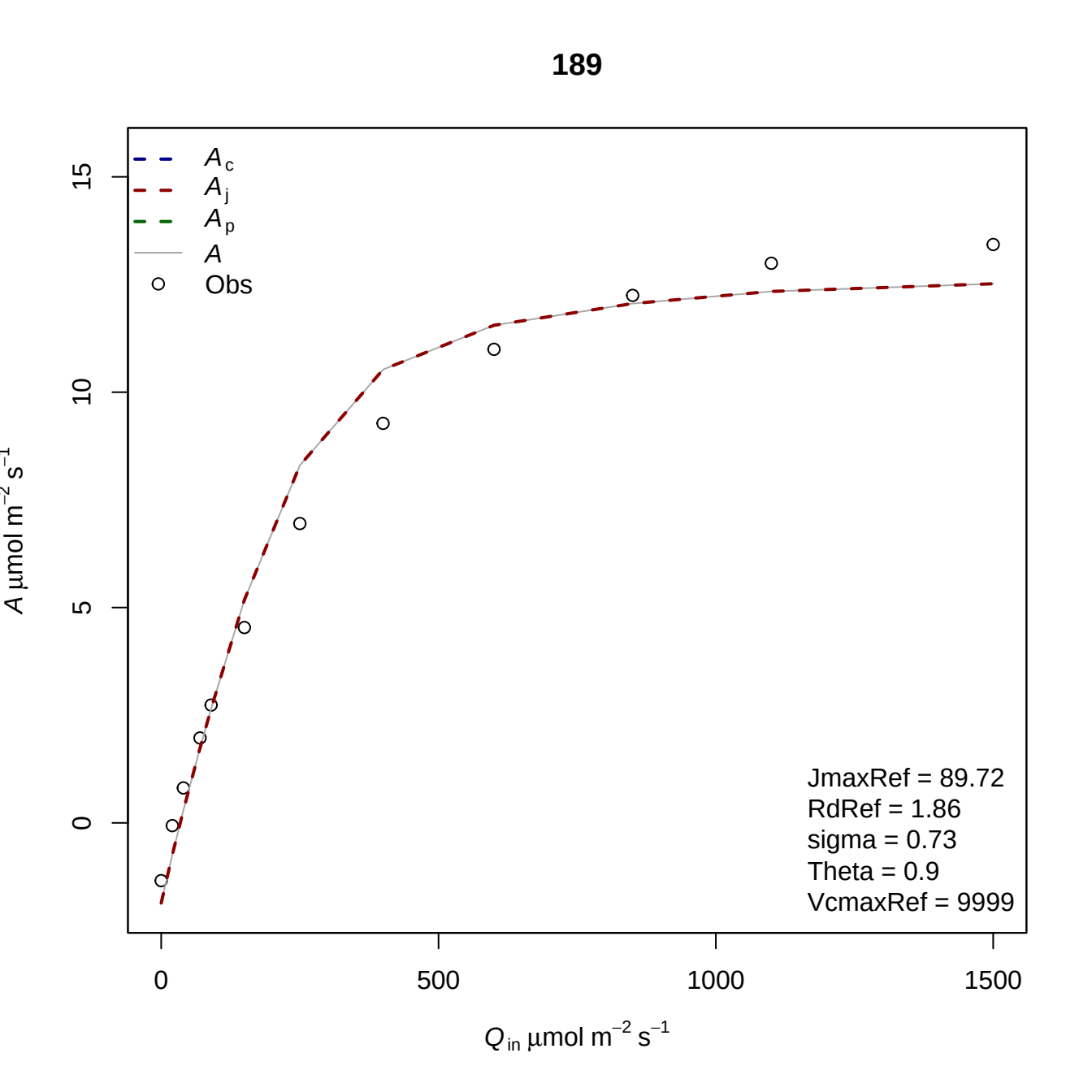

183
A_c
A_j
A_p
A
Obs
JmaxRef = 50.85
RdRef = 1.15
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

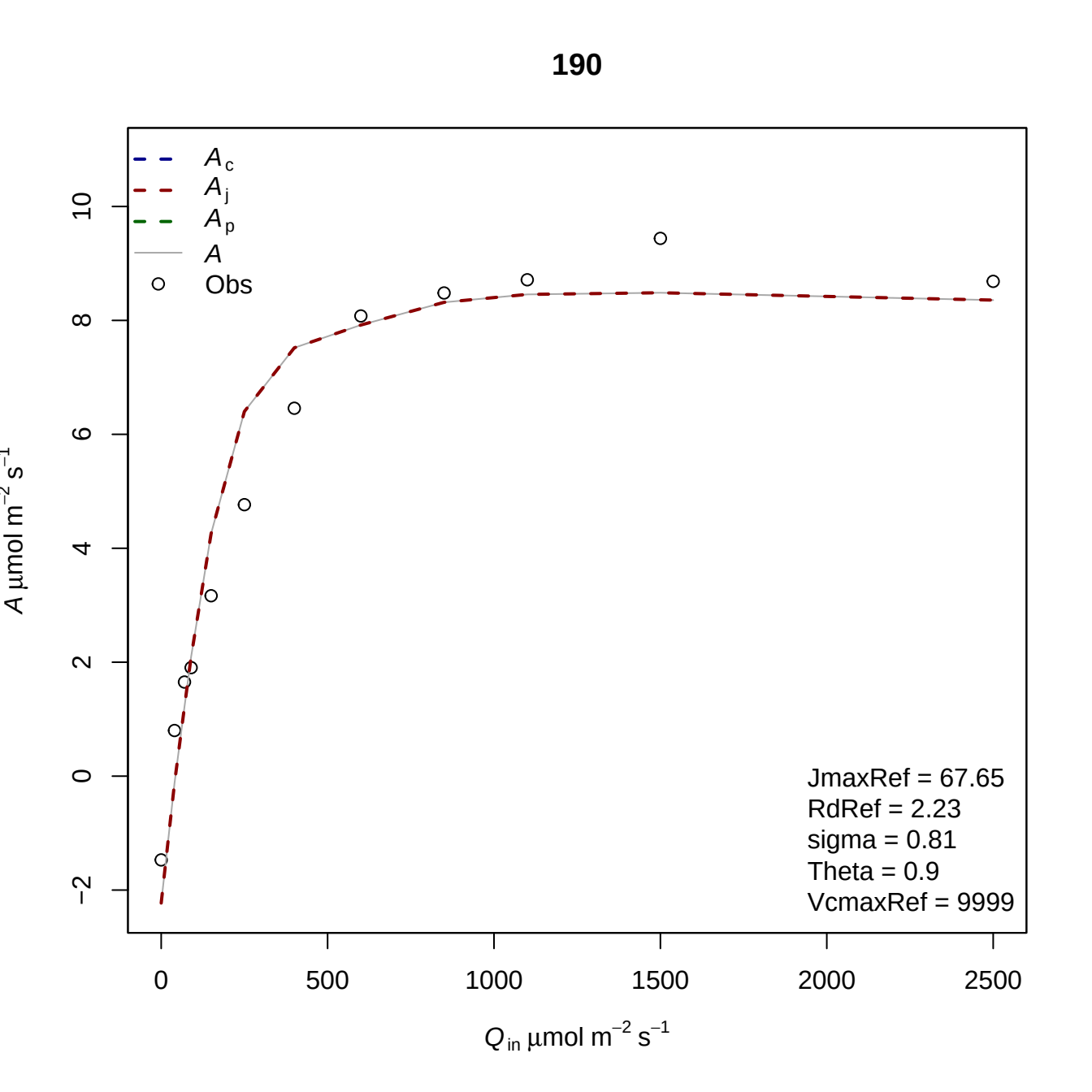

184
A_c
A_j
A_p
A
Obs
JmaxRef = 51.83
RdRef = 1.43
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

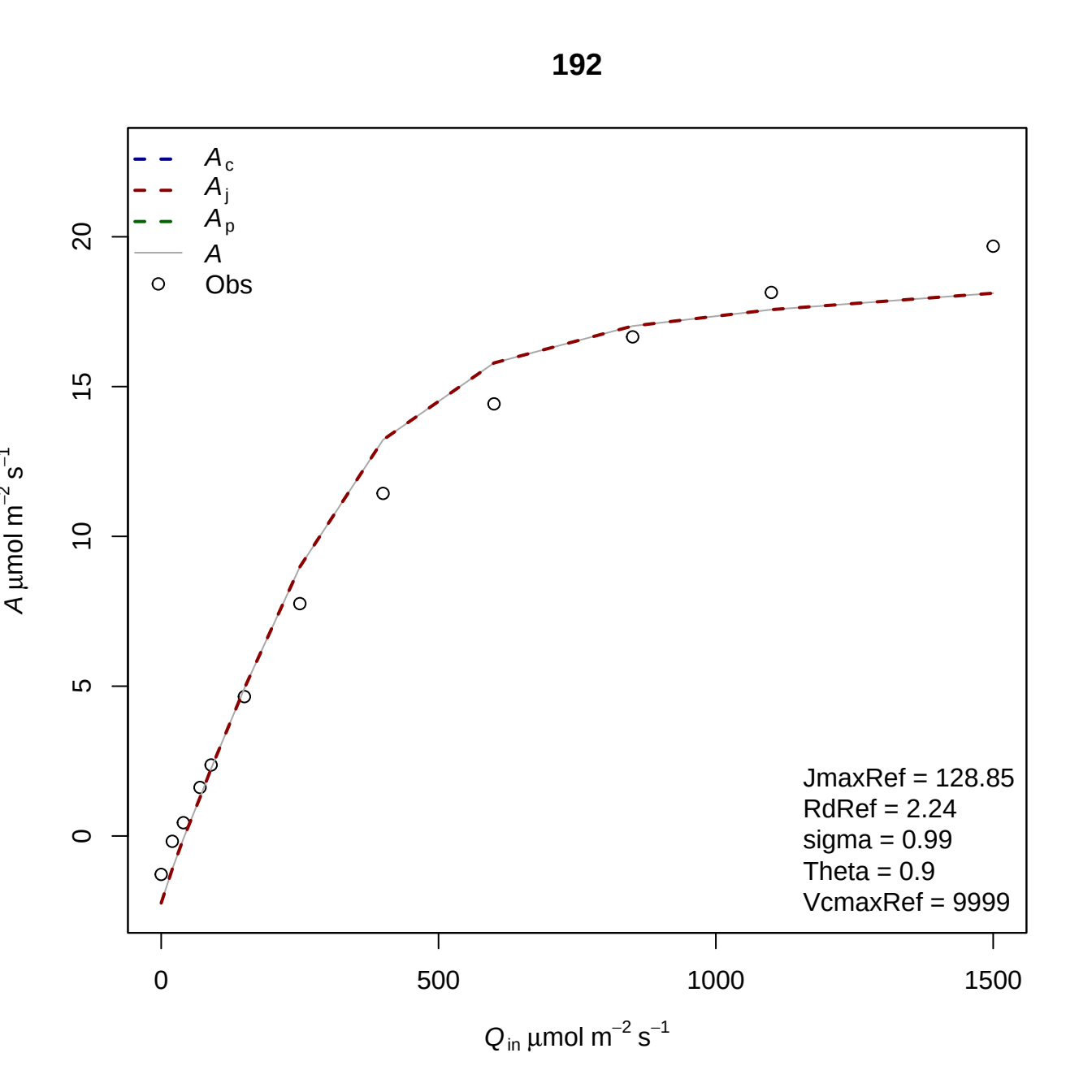

185
A_c
A_j
A_p
A
Obs
JmaxRef = 67.57
RdRef = 2.89
sigma = 2.08
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

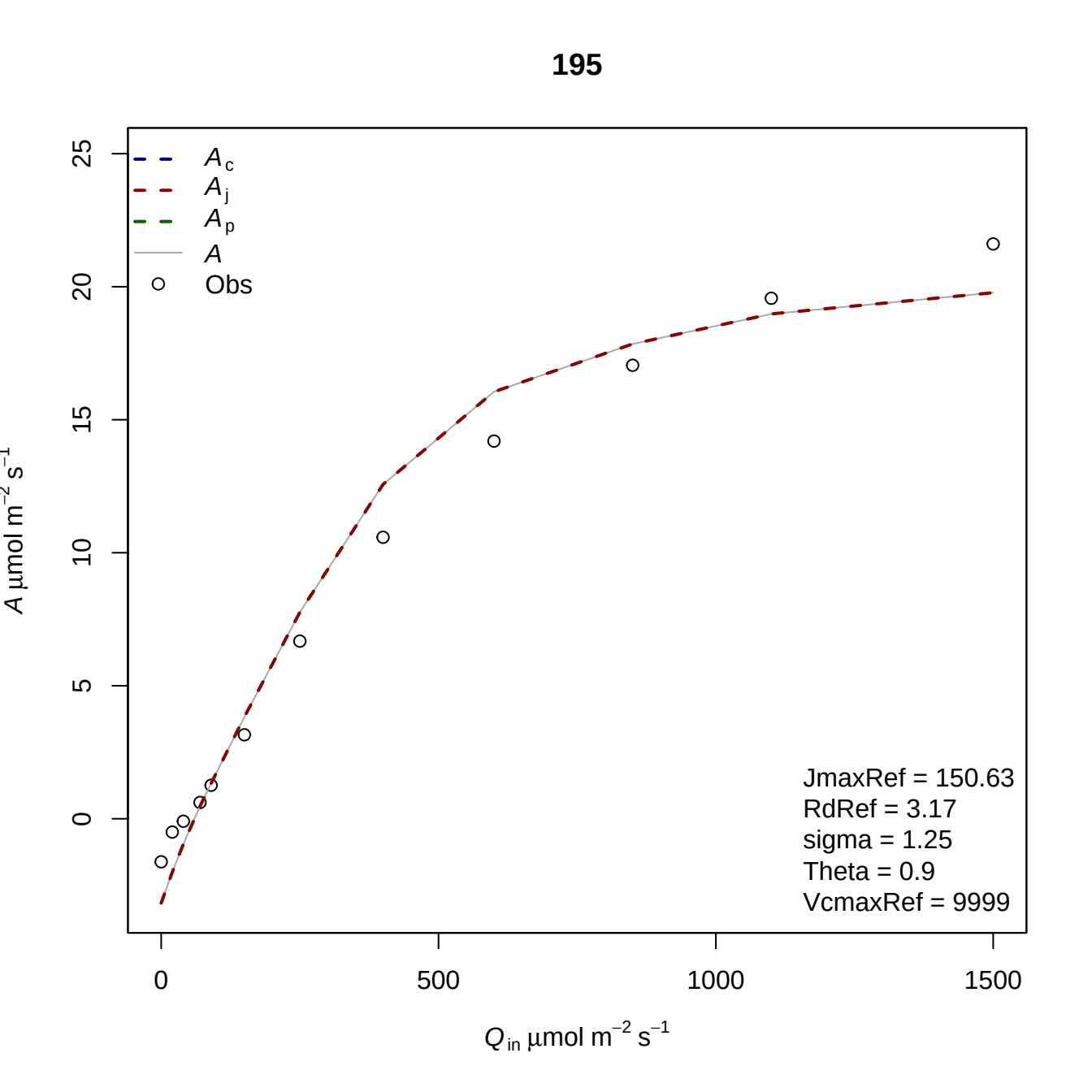

A_c
A_j
A_p
A
Obs
JmaxRef = 66.62
RdRef = 2.54
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

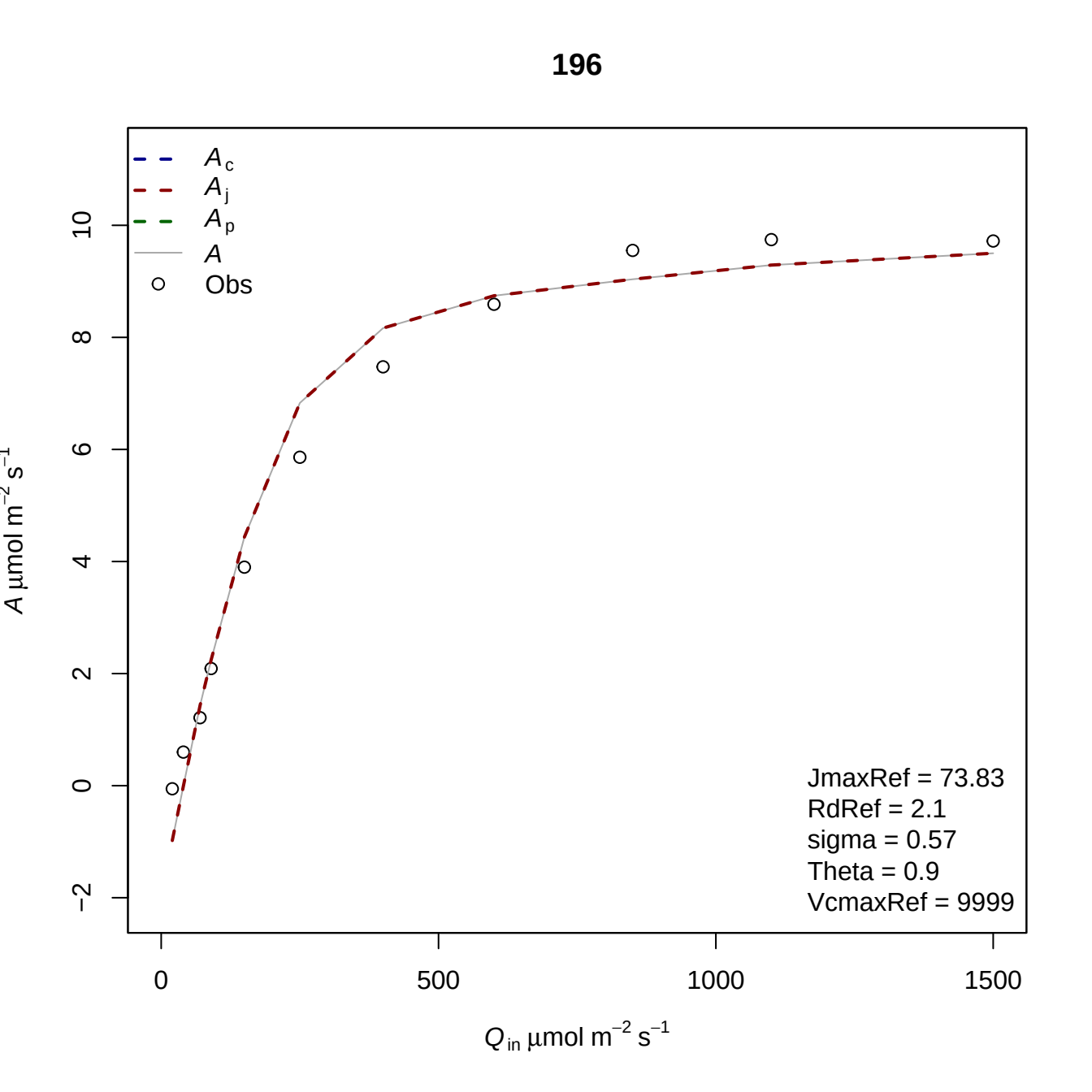

188
A_c
A_j
A_p
A
Obs
JmaxRef = 70.87
RdRef = 2.67
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

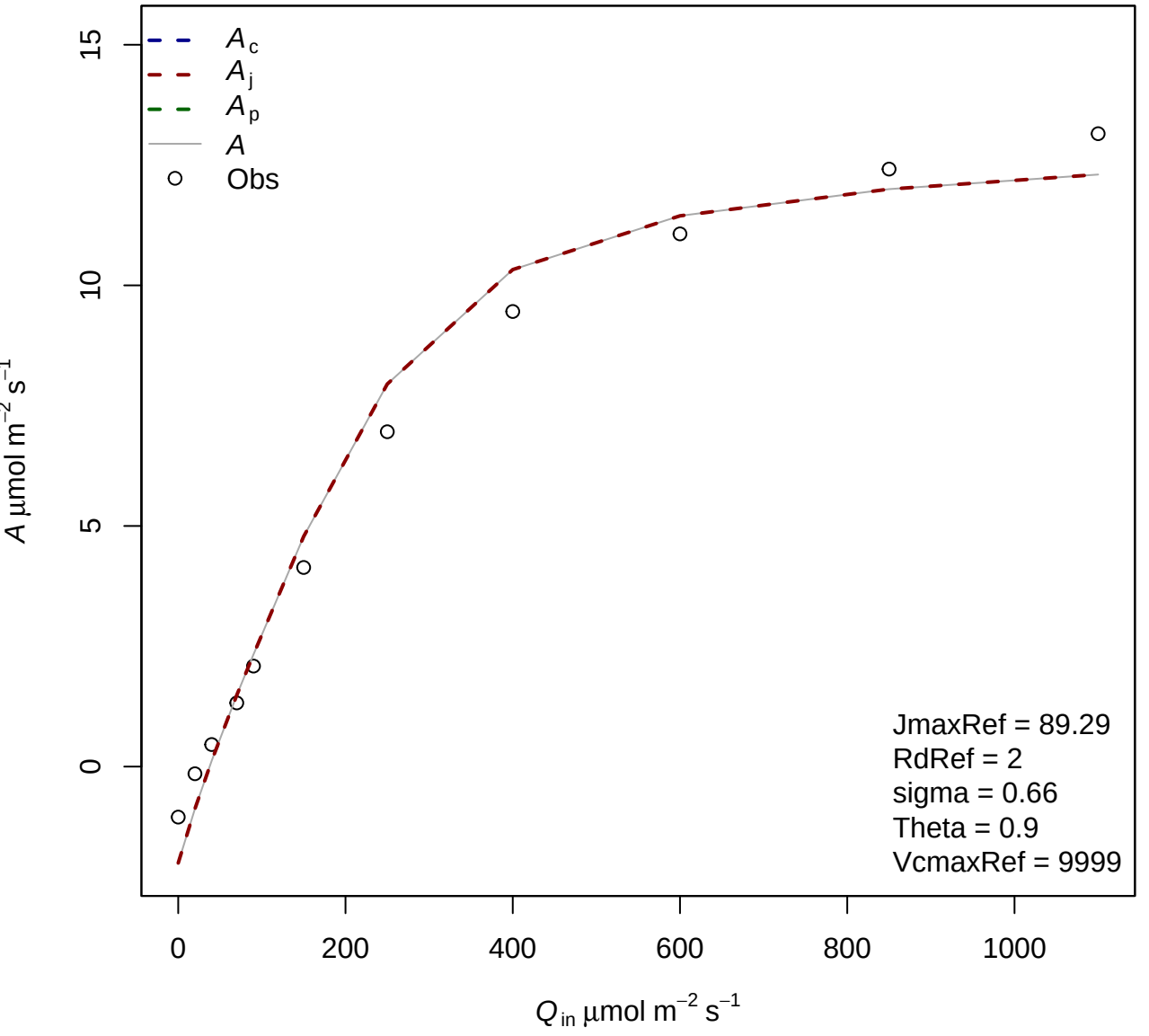

189
A_c
A_j
A_p
A
Obs
JmaxRef = 89.72
RdRef = 1.86
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

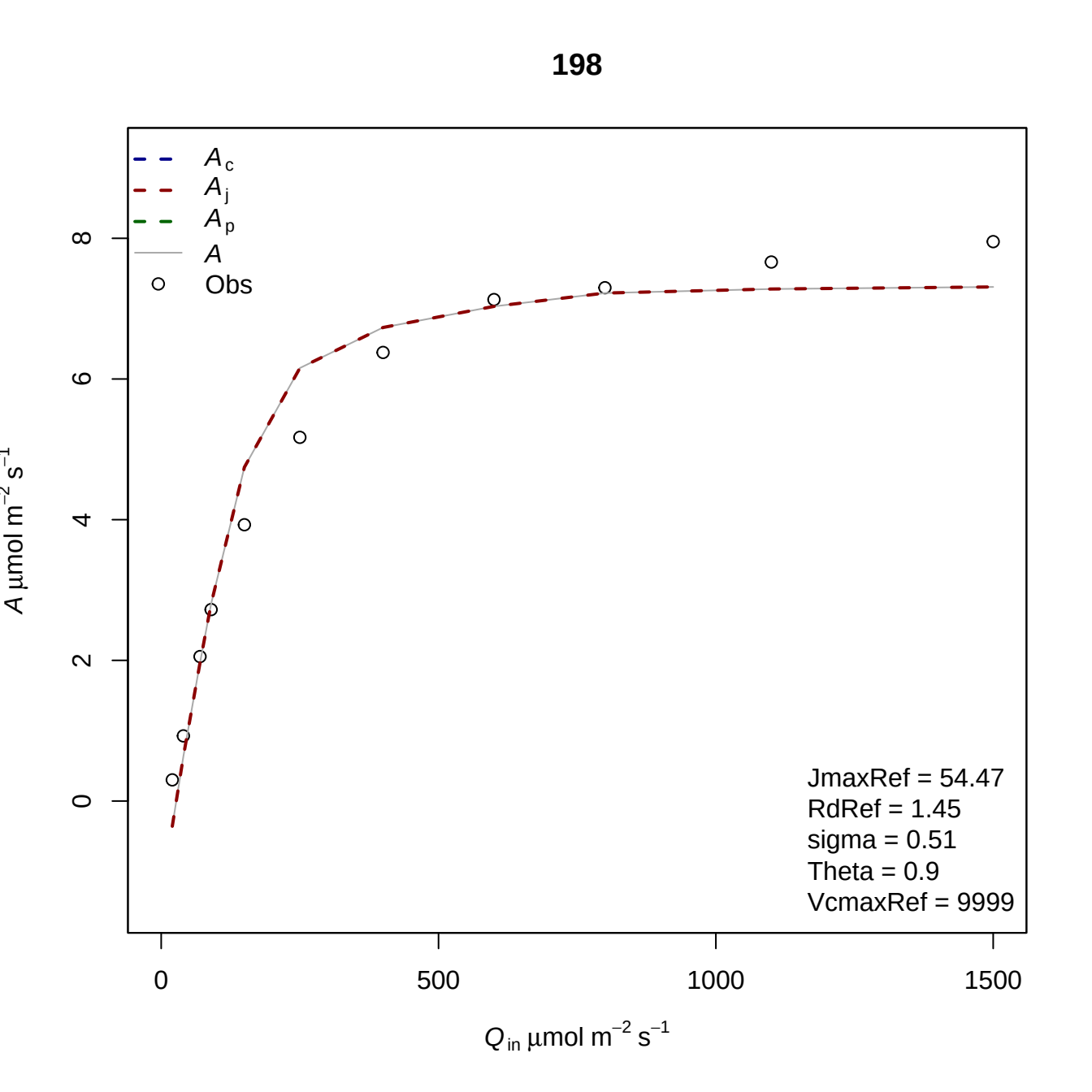

190
A_c
A_j
A_p
A
Obs
JmaxRef = 67.65
RdRef = 2.23
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000
2500

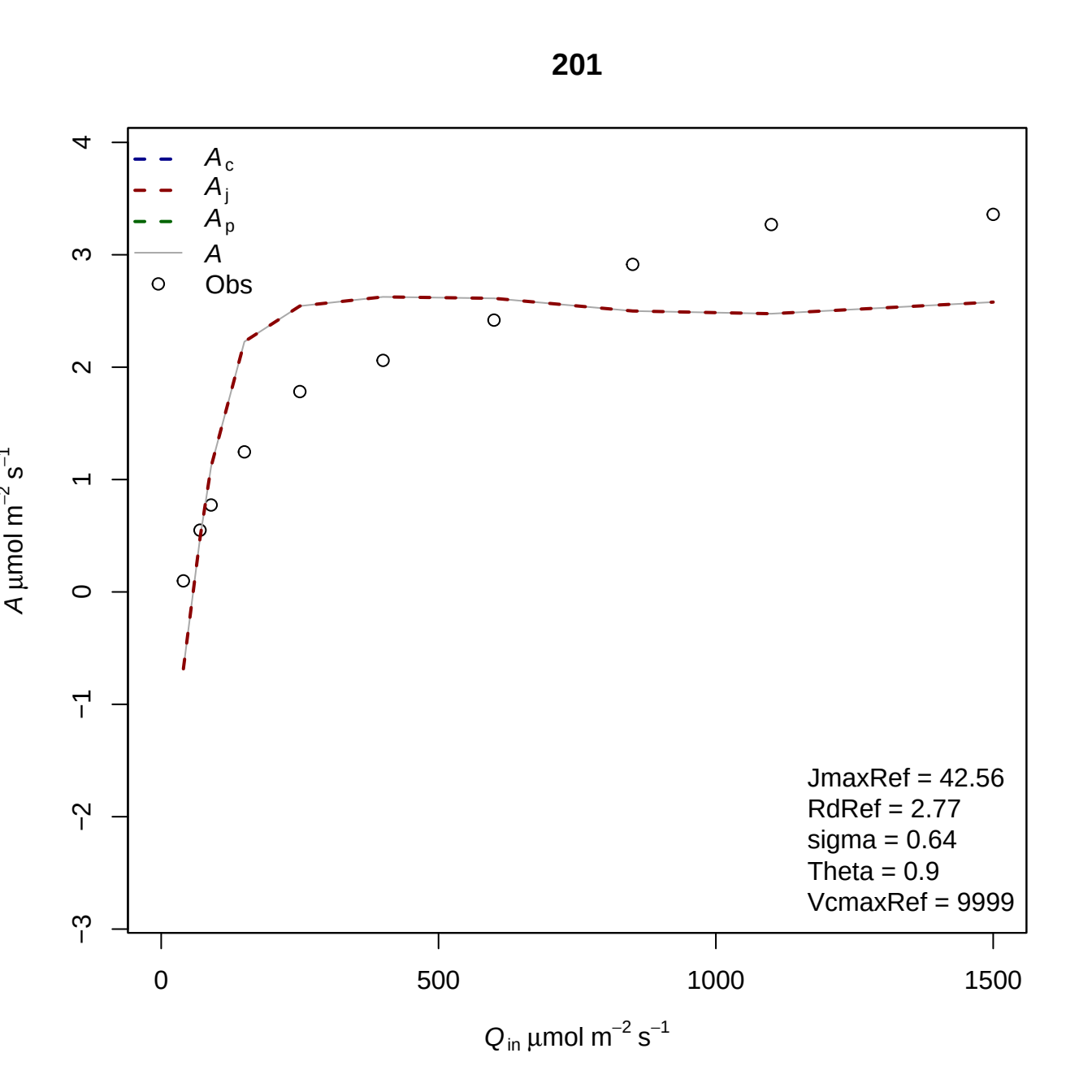

192
A_c
A_j
A_p
A
Obs
JmaxRef = 128.85
RdRef = 2.24
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

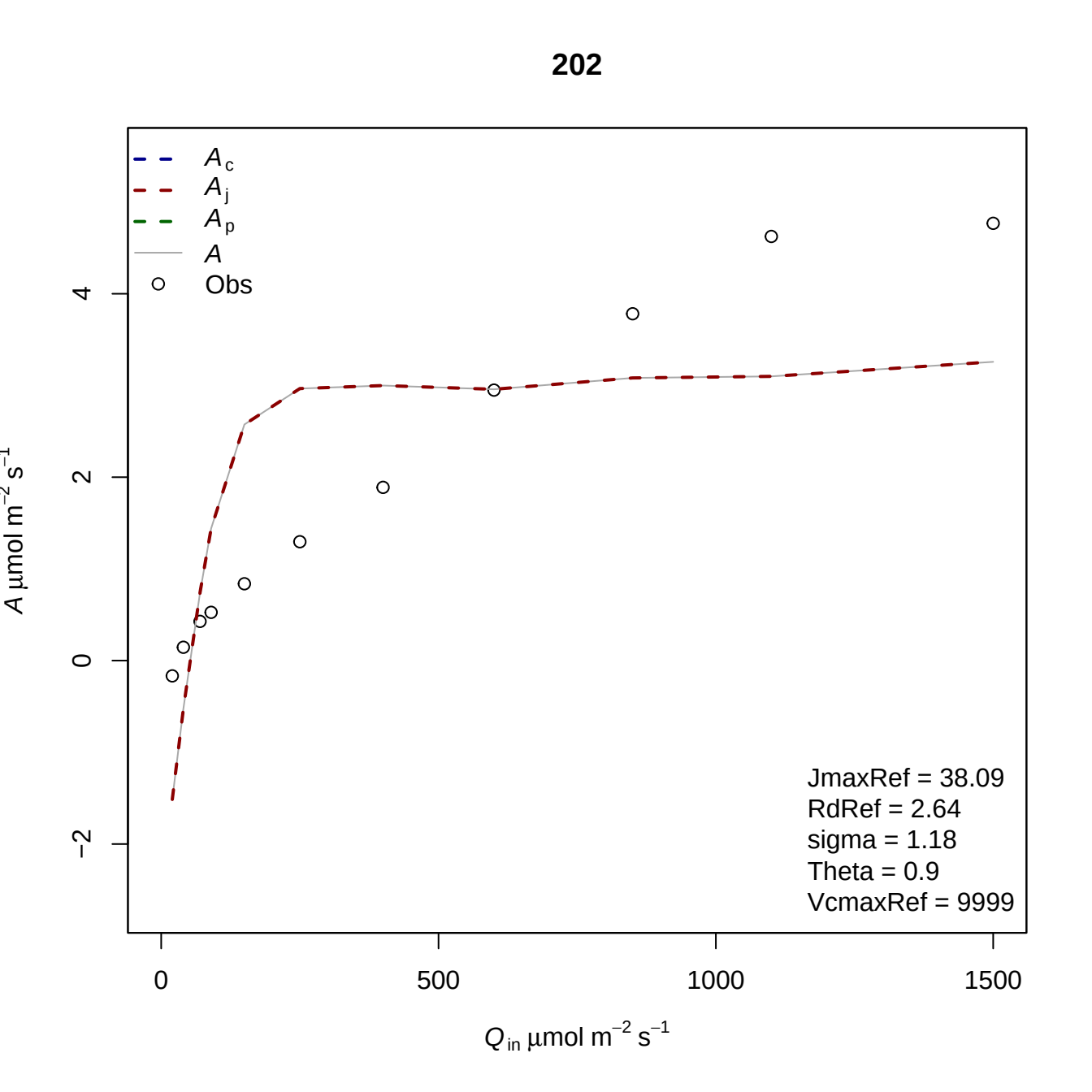

195
A_c
A_j
A_p
A
Obs
JmaxRef = 150.63
RdRef = 3.17
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

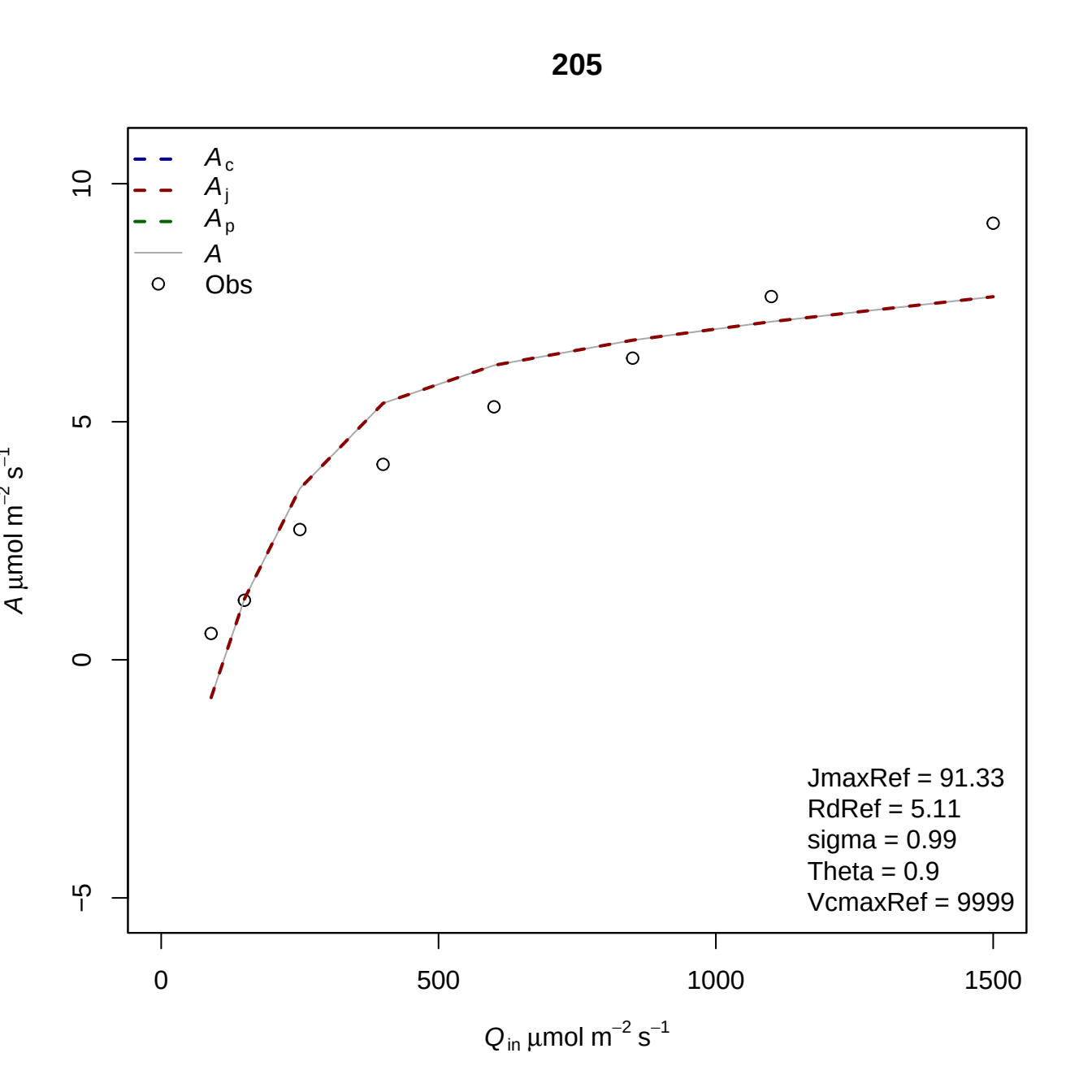

196
A_c
A_j
A_p
A
Obs
JmaxRef = 73.83
RdRef = 2.1
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

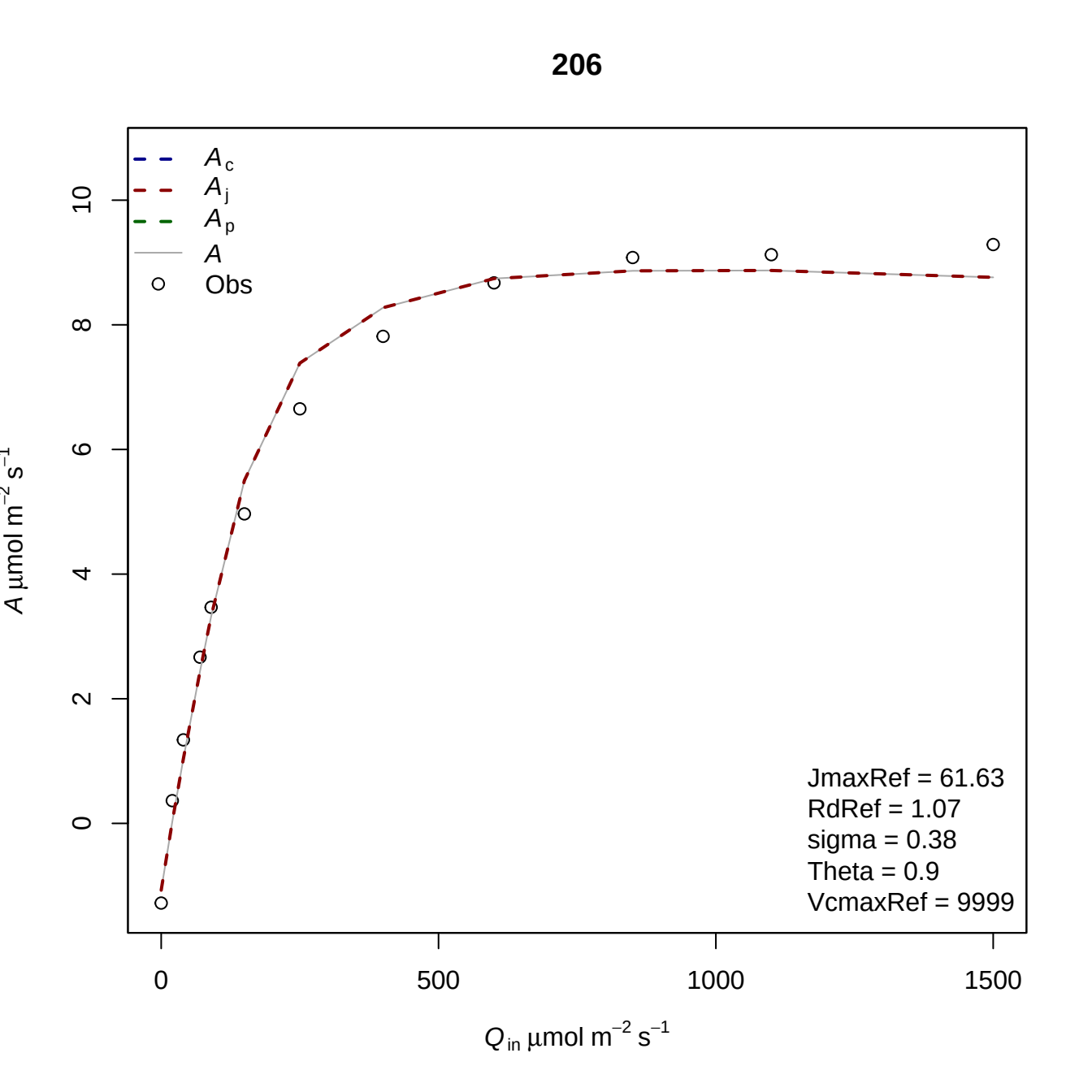

A_c
A_j
A_p
A
Obs
JmaxRef = 89.29
RdRef = 2
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

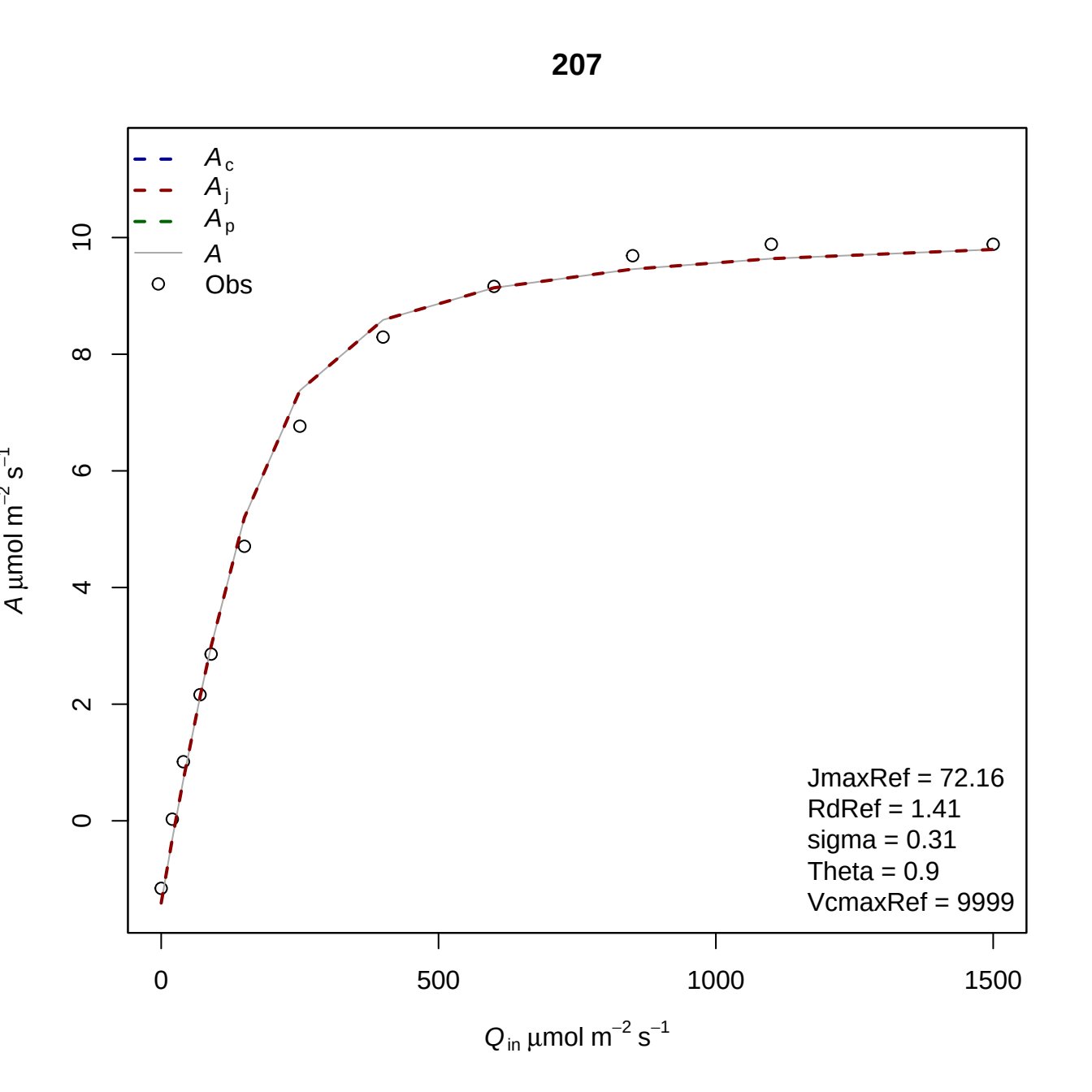

198
A_c
A_j
A_p
A
Obs
JmaxRef = 54.47
RdRef = 1.45
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

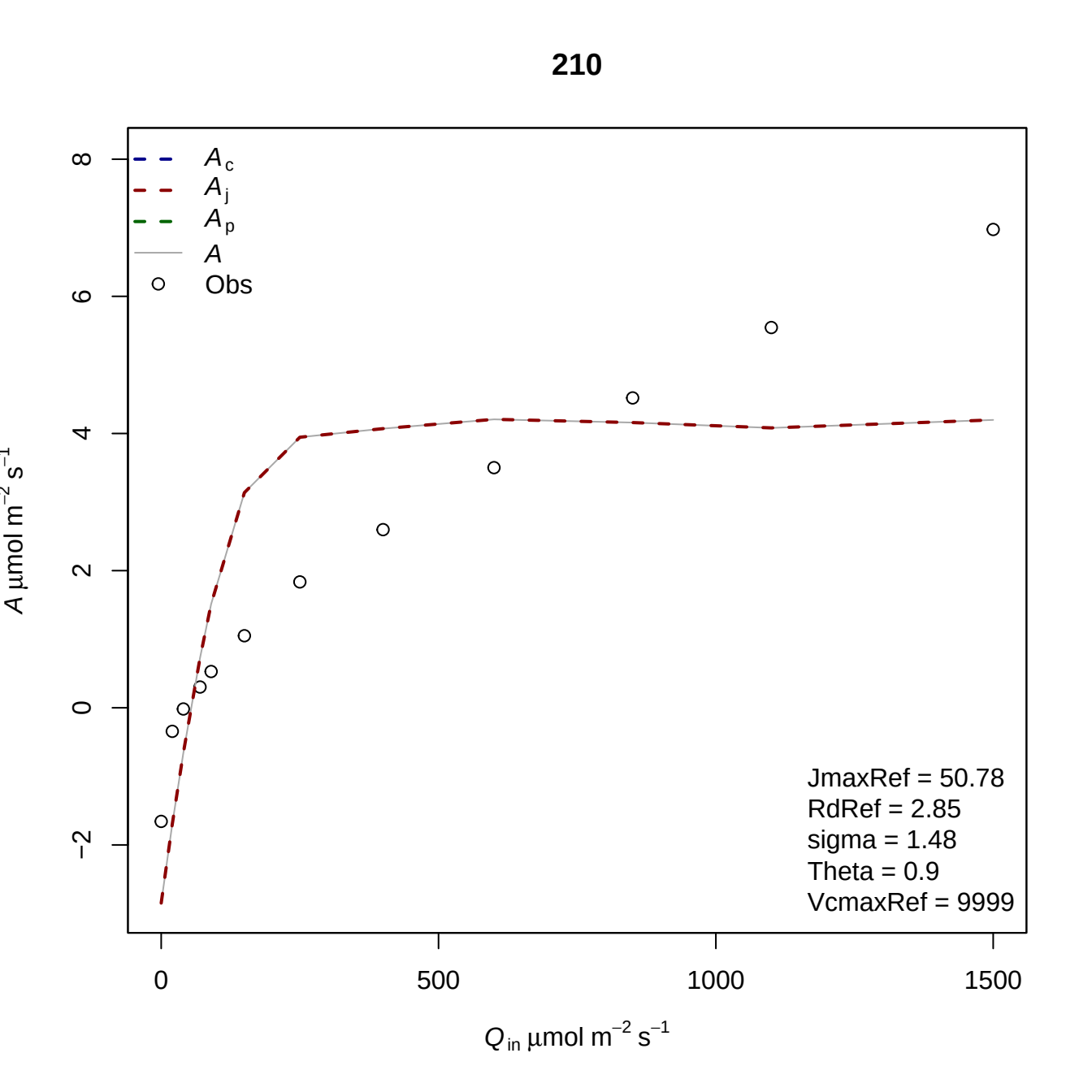

201
A_c
A_j
A_p
A
Obs
JmaxRef = 42.56
RdRef = 2.77
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

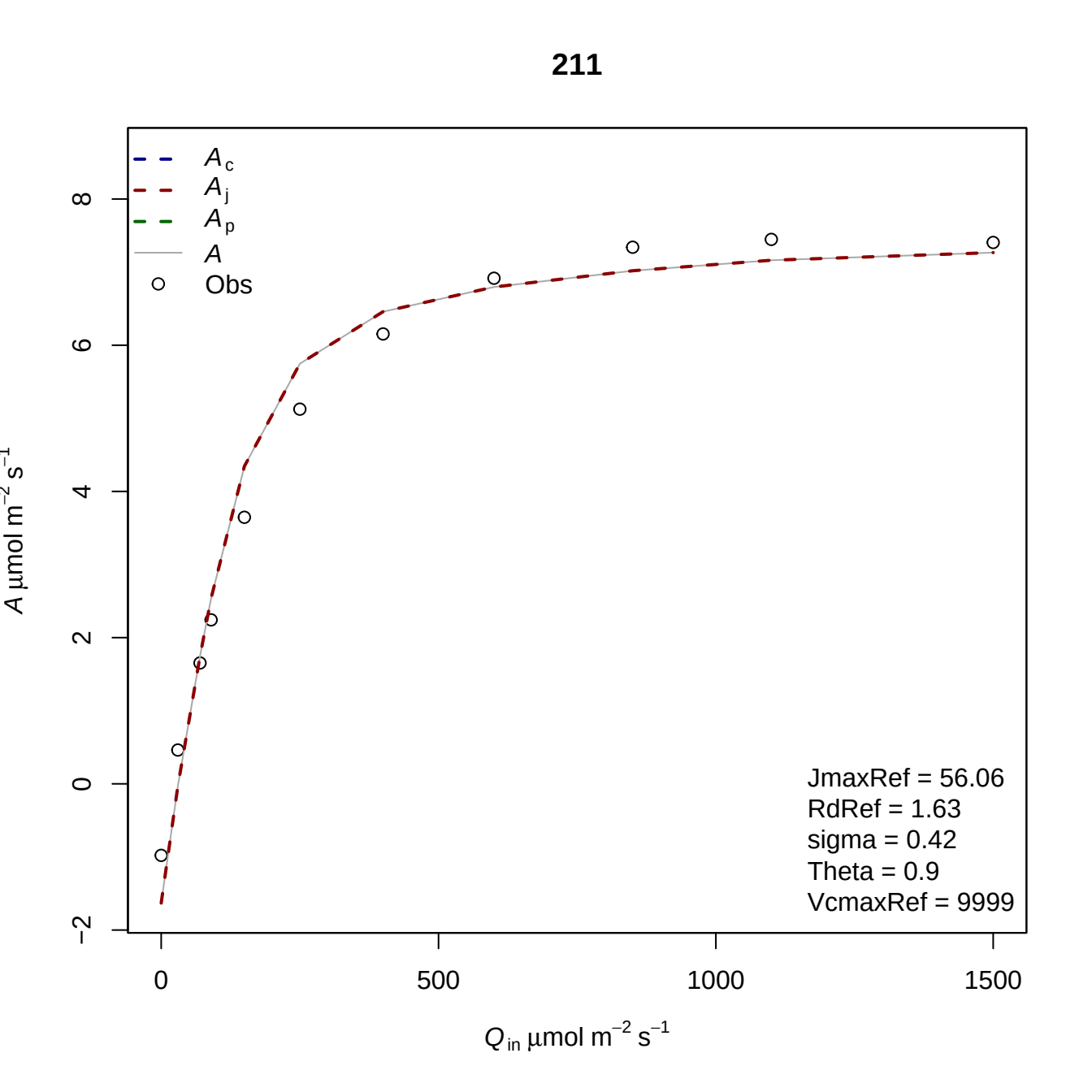

202
A_c
A_j
A_p
A
Obs
JmaxRef = 38.09
RdRef = 2.64
sigma = 1.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

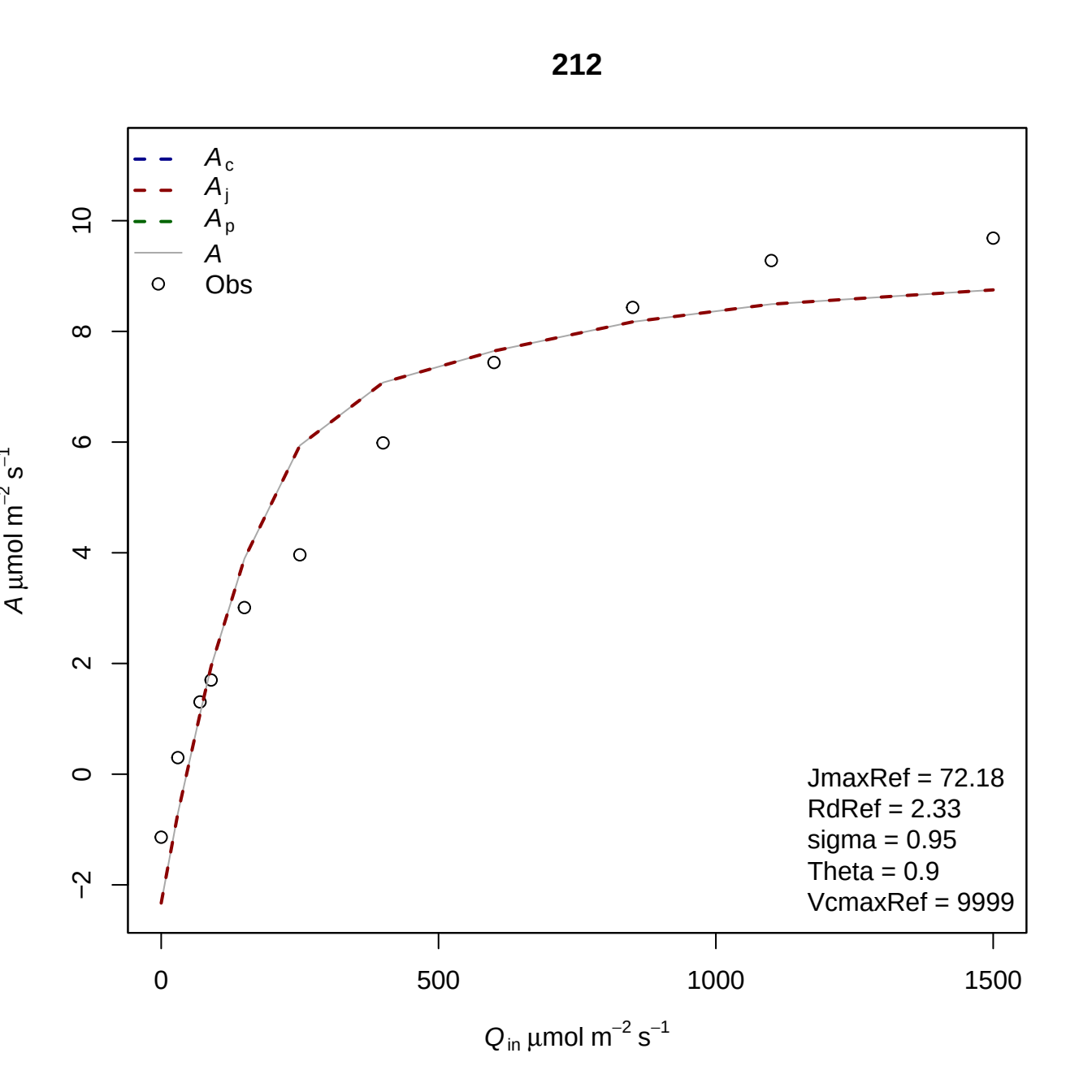

205
A_c
A_j
A_p
A
Obs
JmaxRef = 91.33
RdRef = 5.11
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

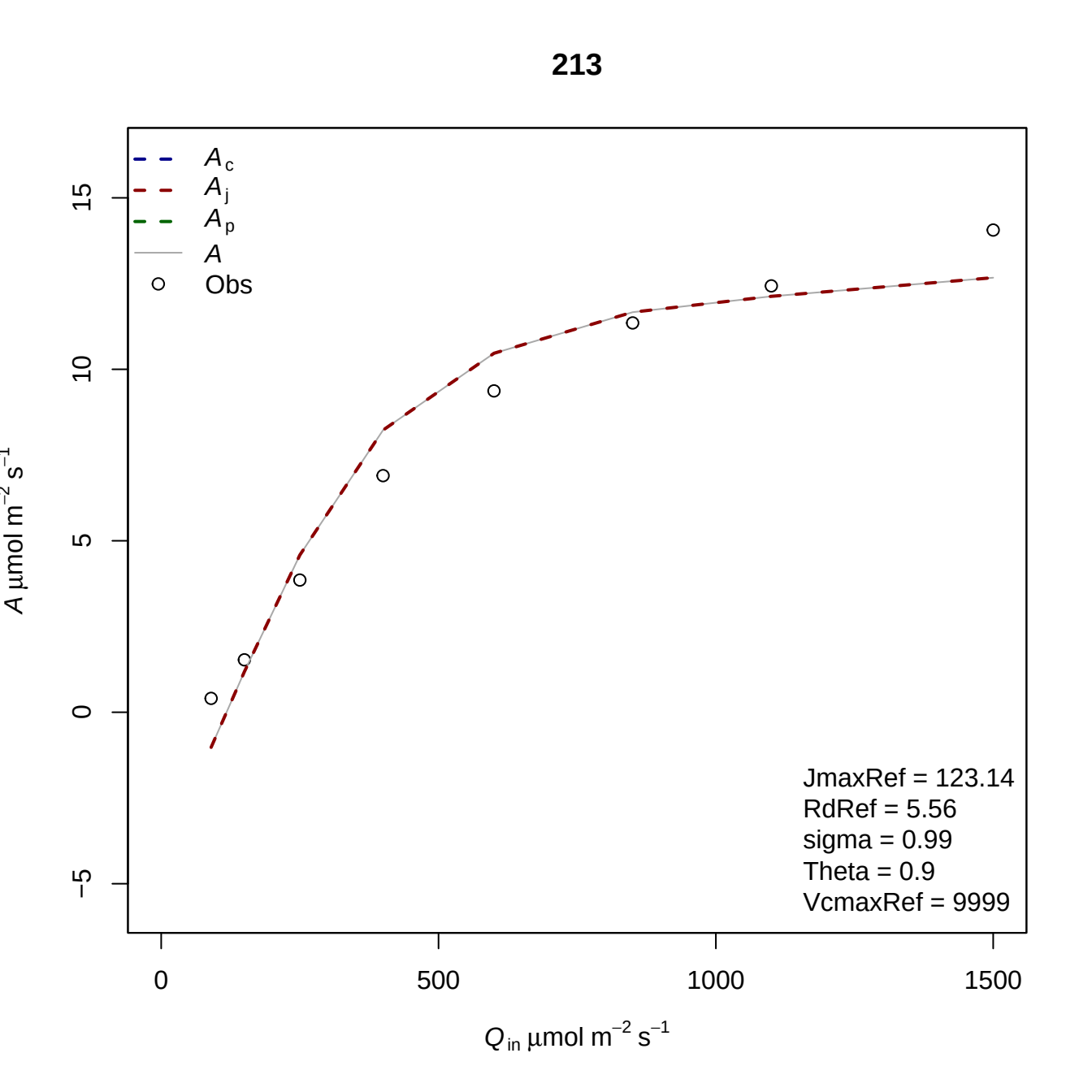

206
A_c
A_j
A_p
A
Obs
JmaxRef = 61.63
RdRef = 1.07
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

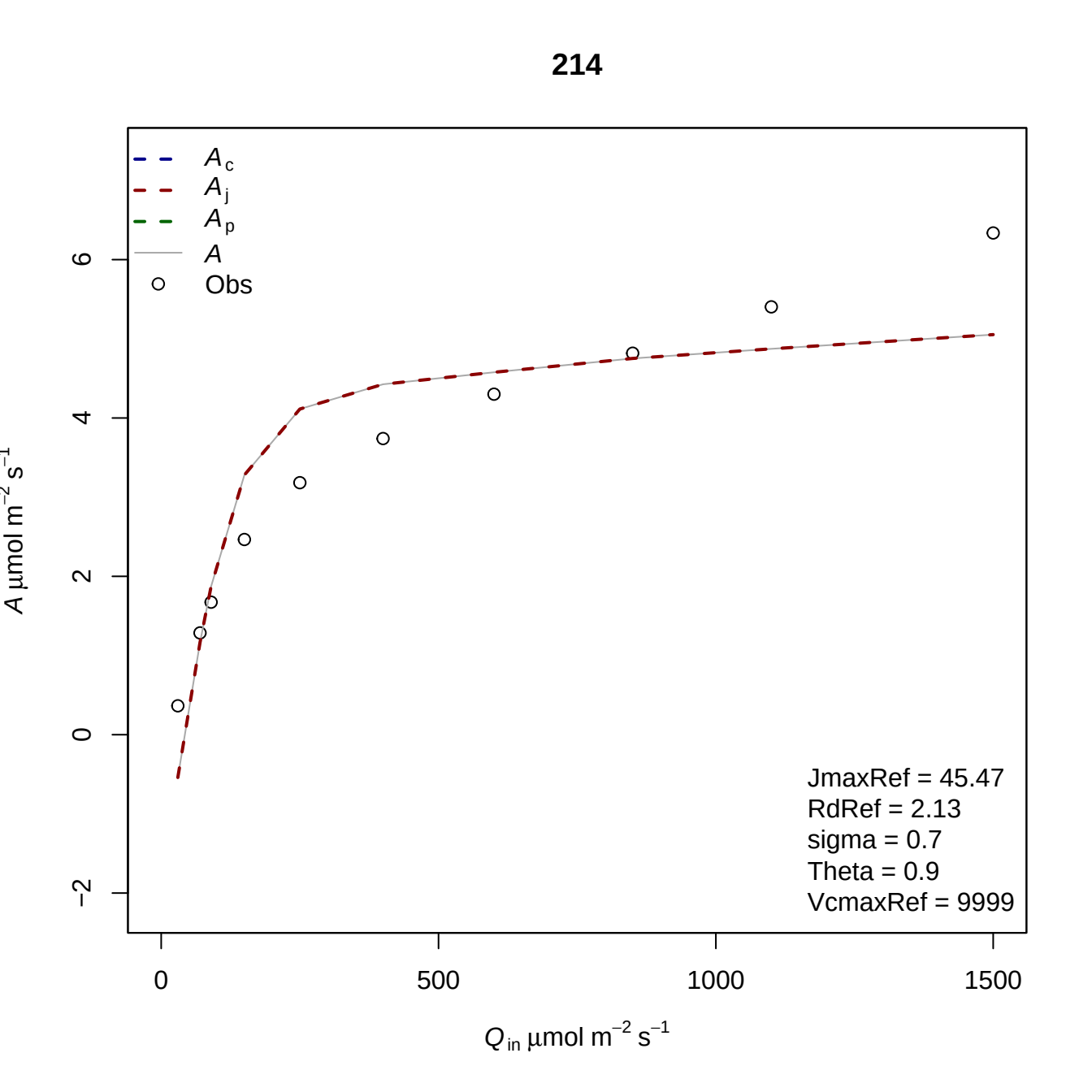

207
A_c
A_j
A_p
A
Obs
JmaxRef = 72.16
RdRef = 1.41
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

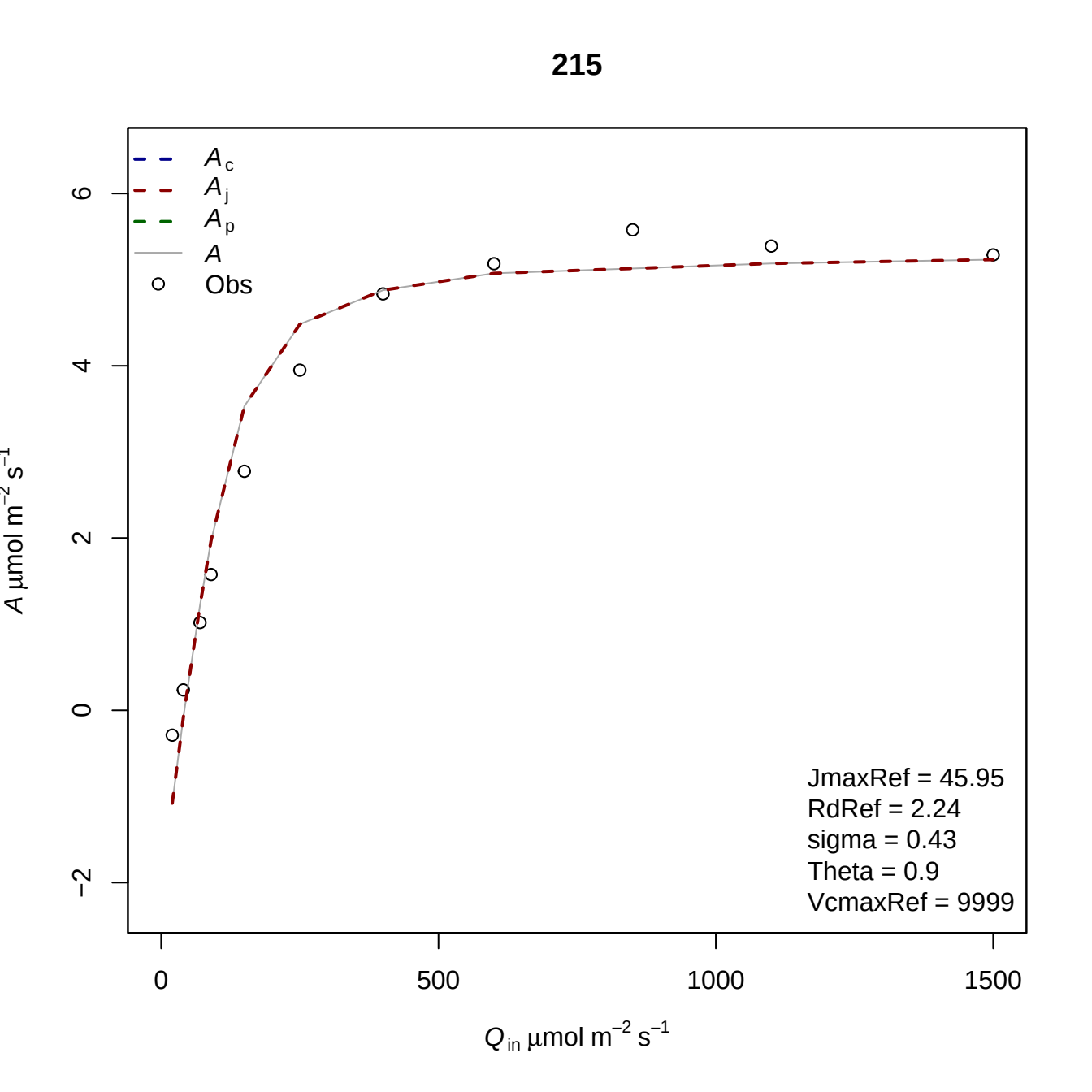

210
A_c
A_j
A_p
A
Obs
JmaxRef = 50.78
RdRef = 2.85
sigma = 1.48
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

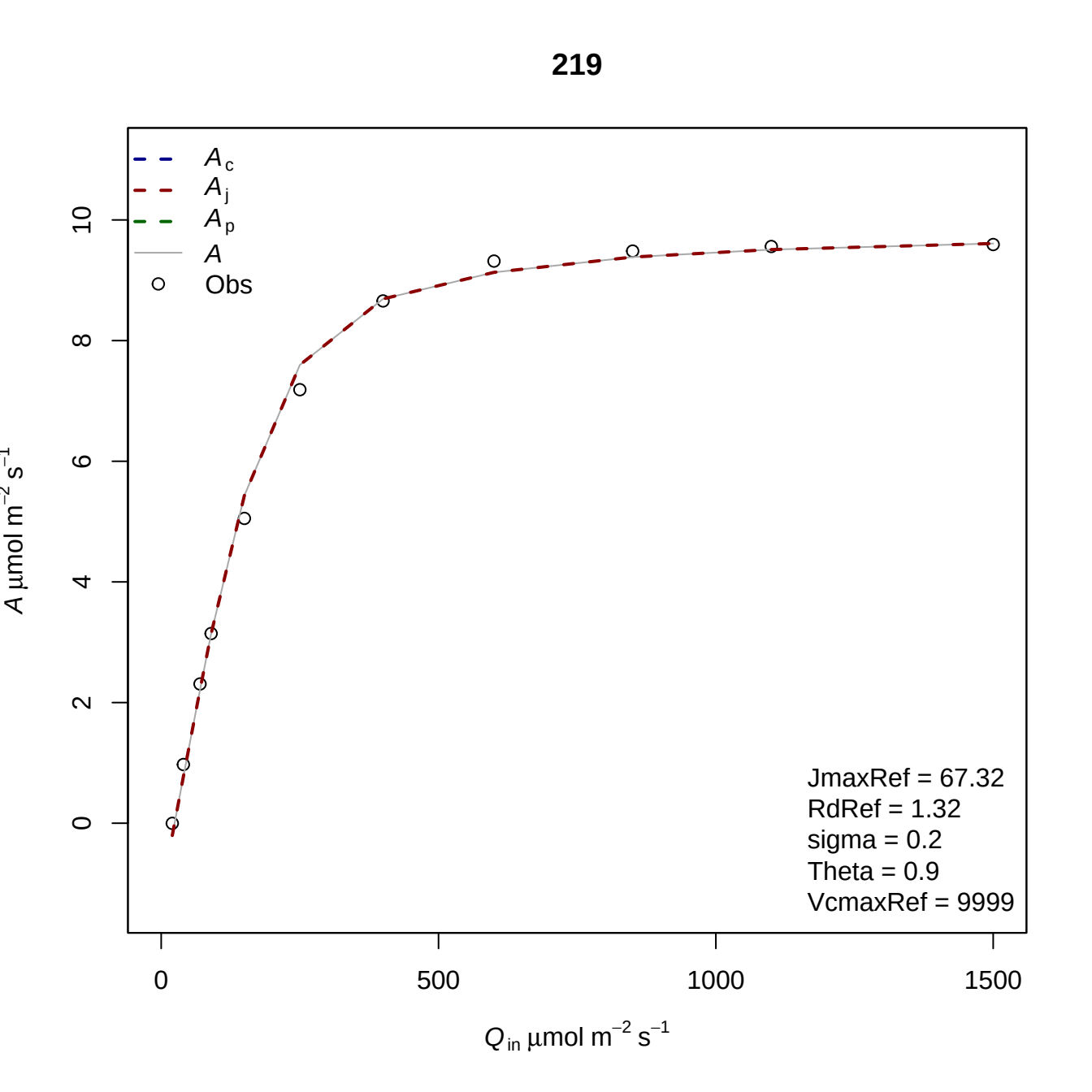

211
A_c
A_j
A_p
A
Obs
JmaxRef = 56.06
RdRef = 1.63
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

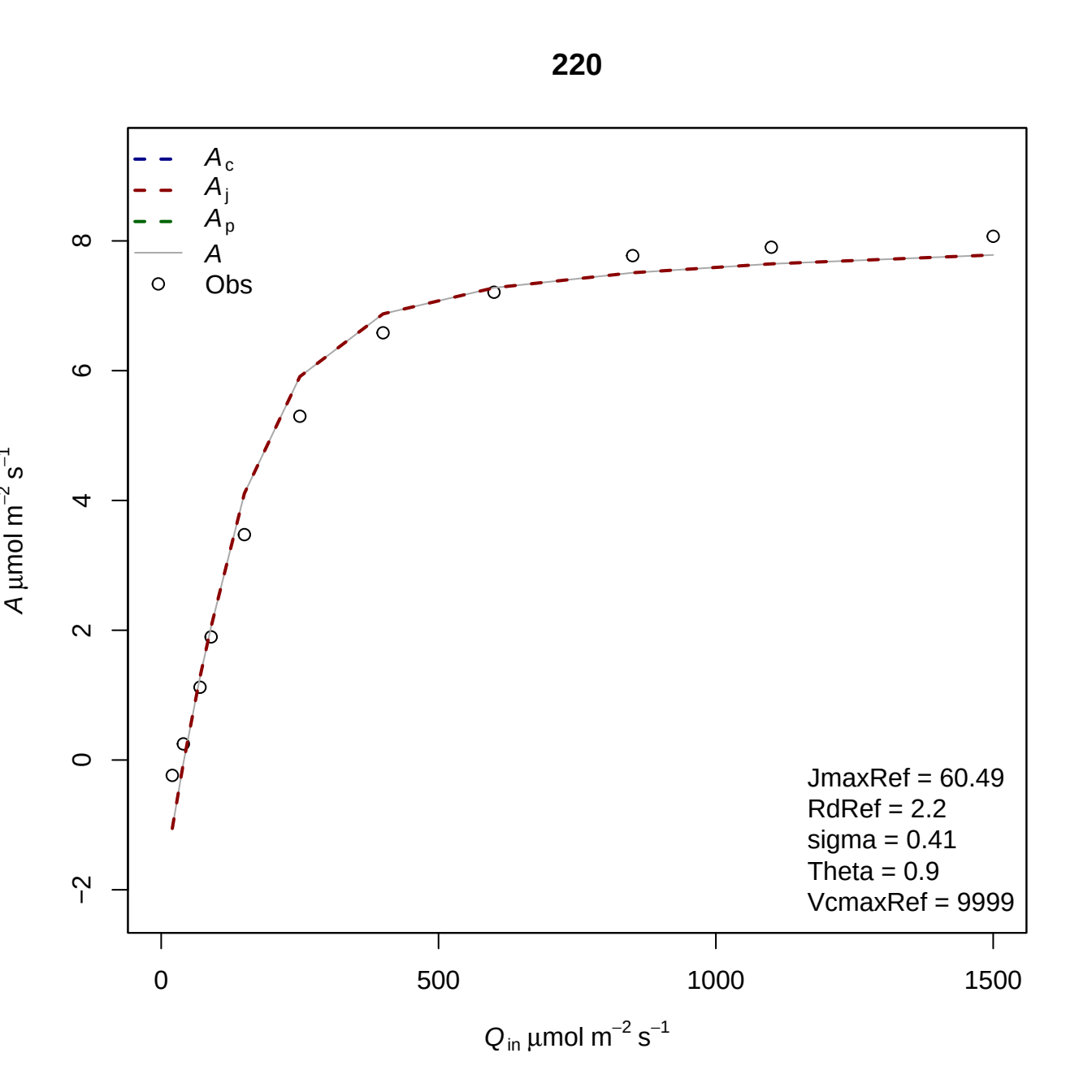

212
A_c
A_j
A_p
A
Obs
JmaxRef = 72.18
RdRef = 2.33
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

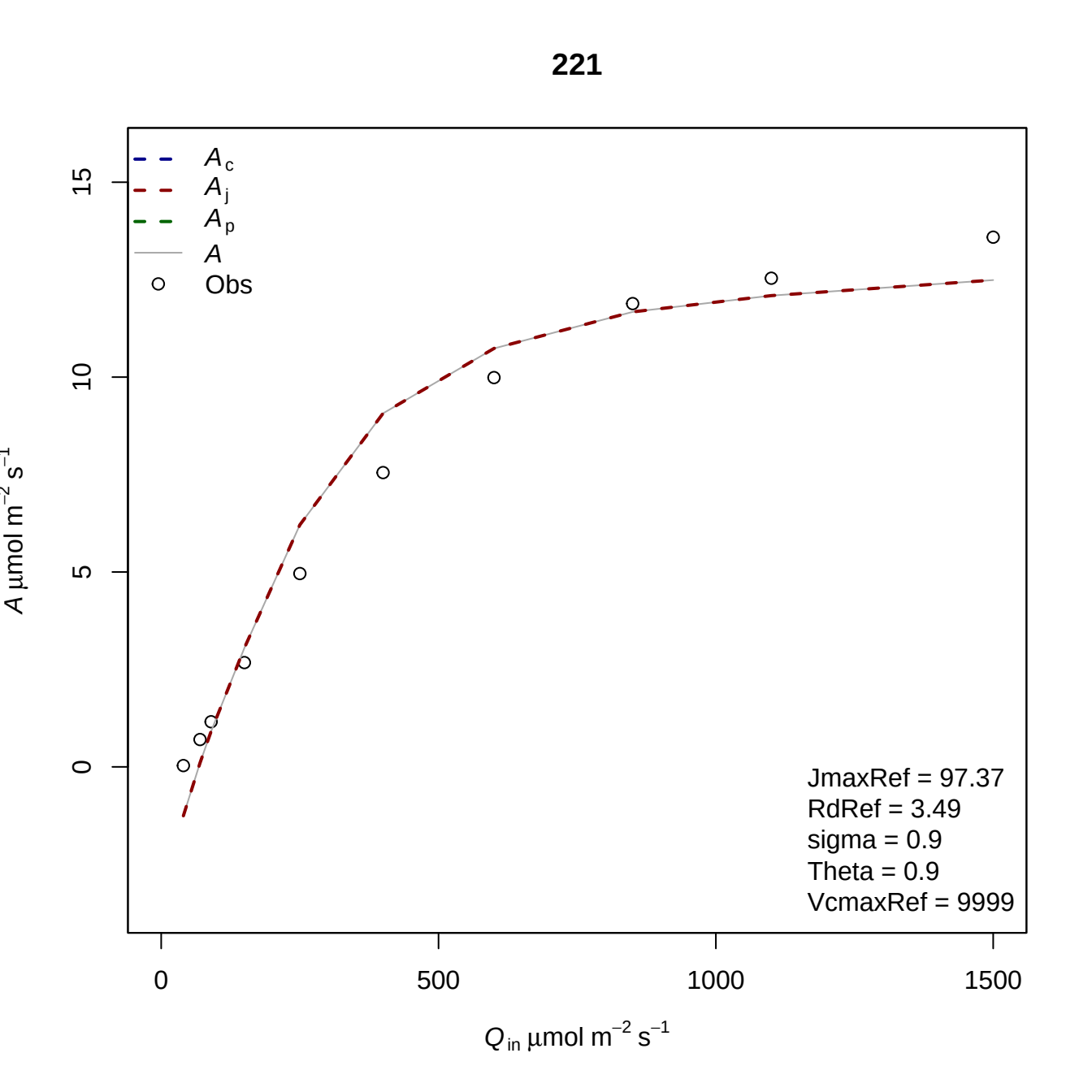

213
A_c
A_j
A_p
A
Obs
JmaxRef = 123.14
RdRef = 5.56
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

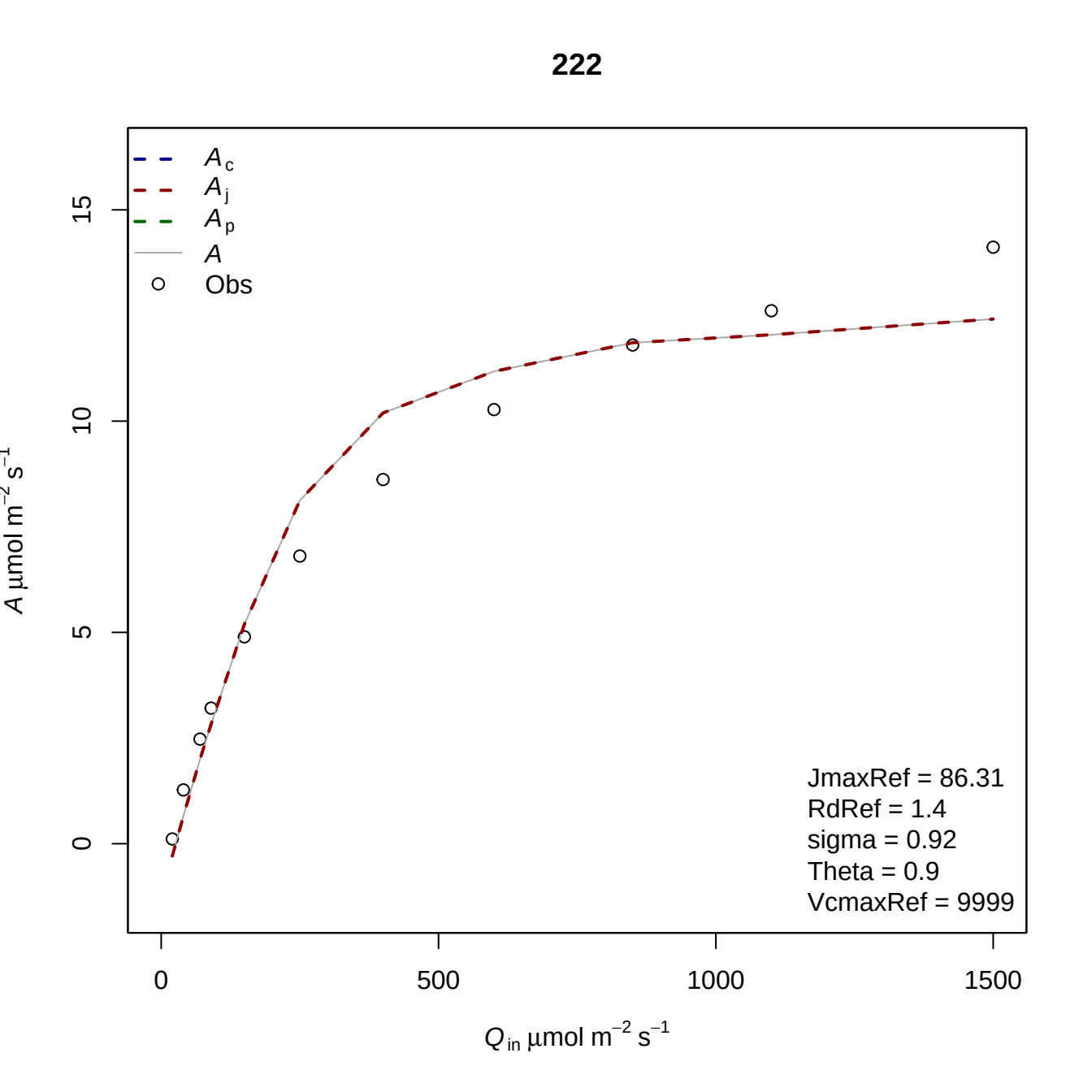

214
A_c
A_j
A_p
A
Obs
JmaxRef = 45.47
RdRef = 2.13
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

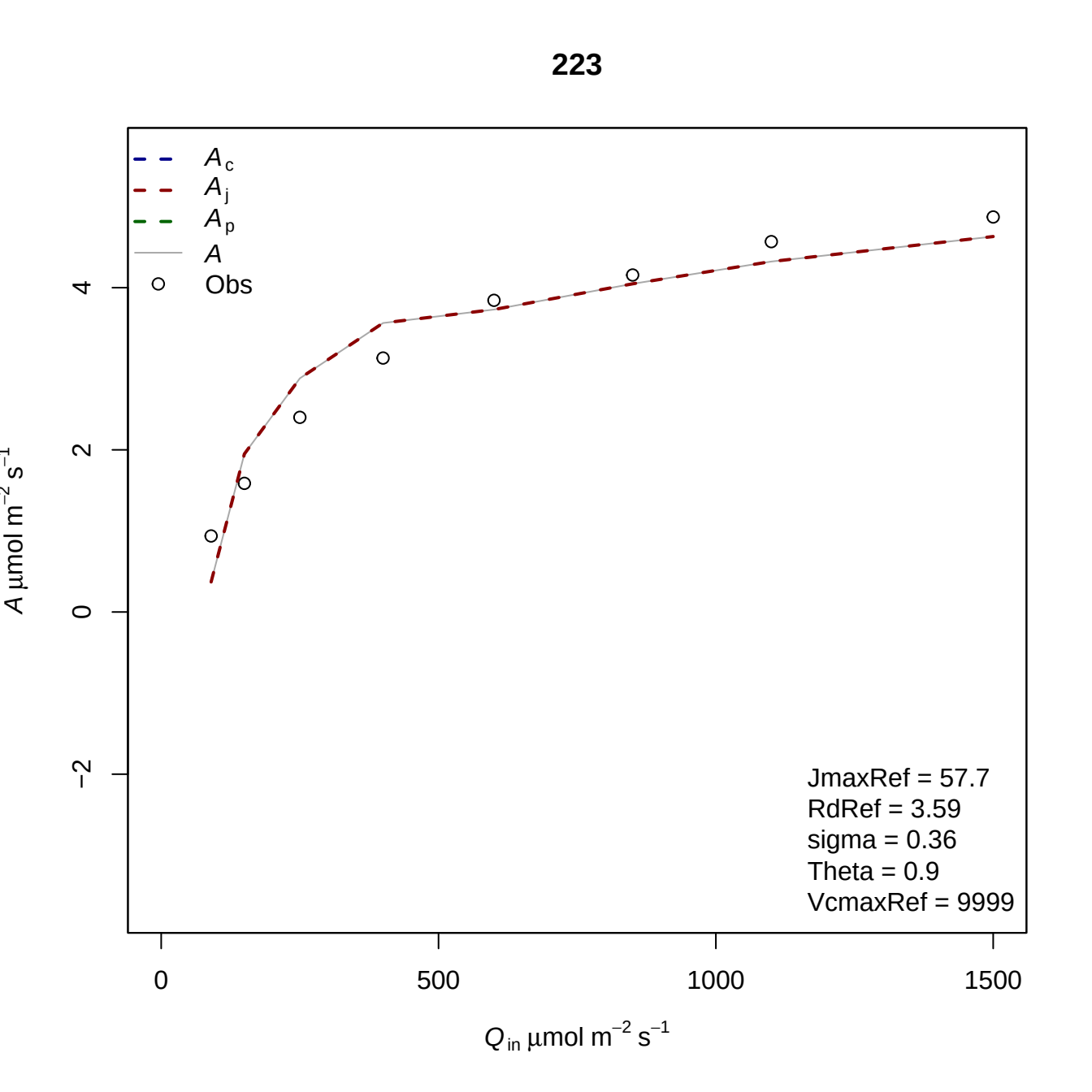

215
A_c
A_j
A_p
A
Obs
JmaxRef = 45.95
RdRef = 2.24
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

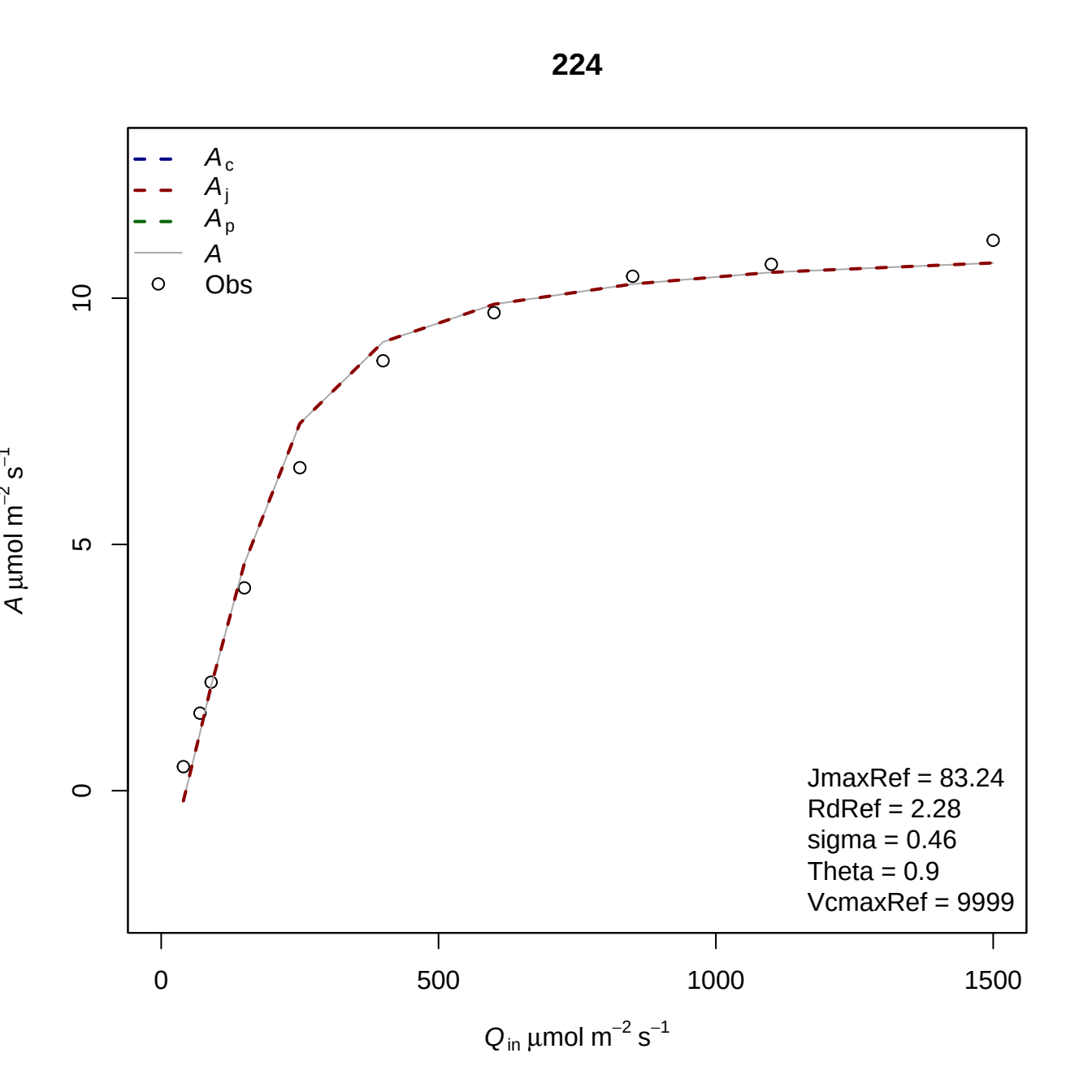

219
A_c
A_j
A_p
A
Obs
JmaxRef = 67.32
RdRef = 1.32
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

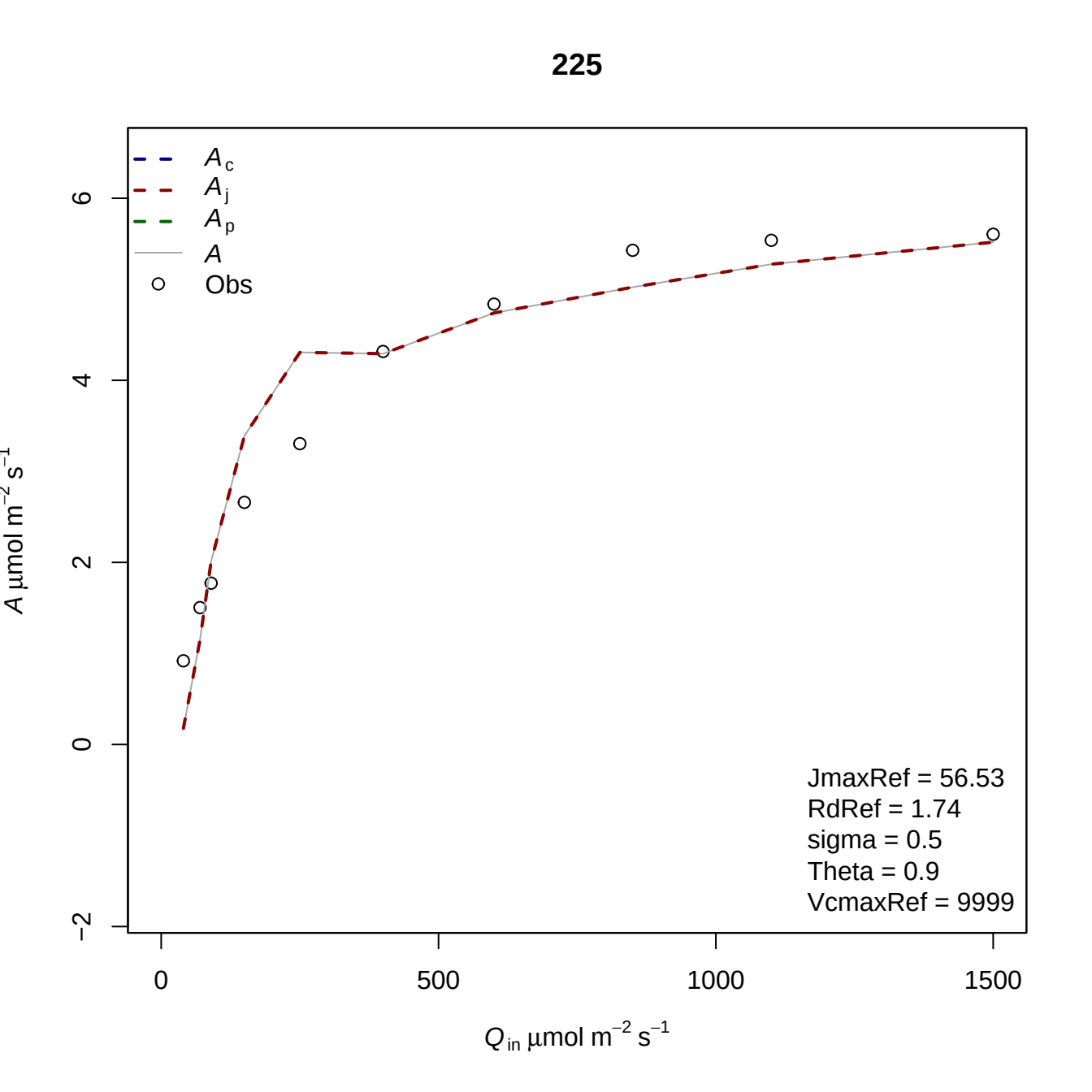

220
A_c
A_j
A_p
A
Obs
JmaxRef = 60.49
RdRef = 2.2
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

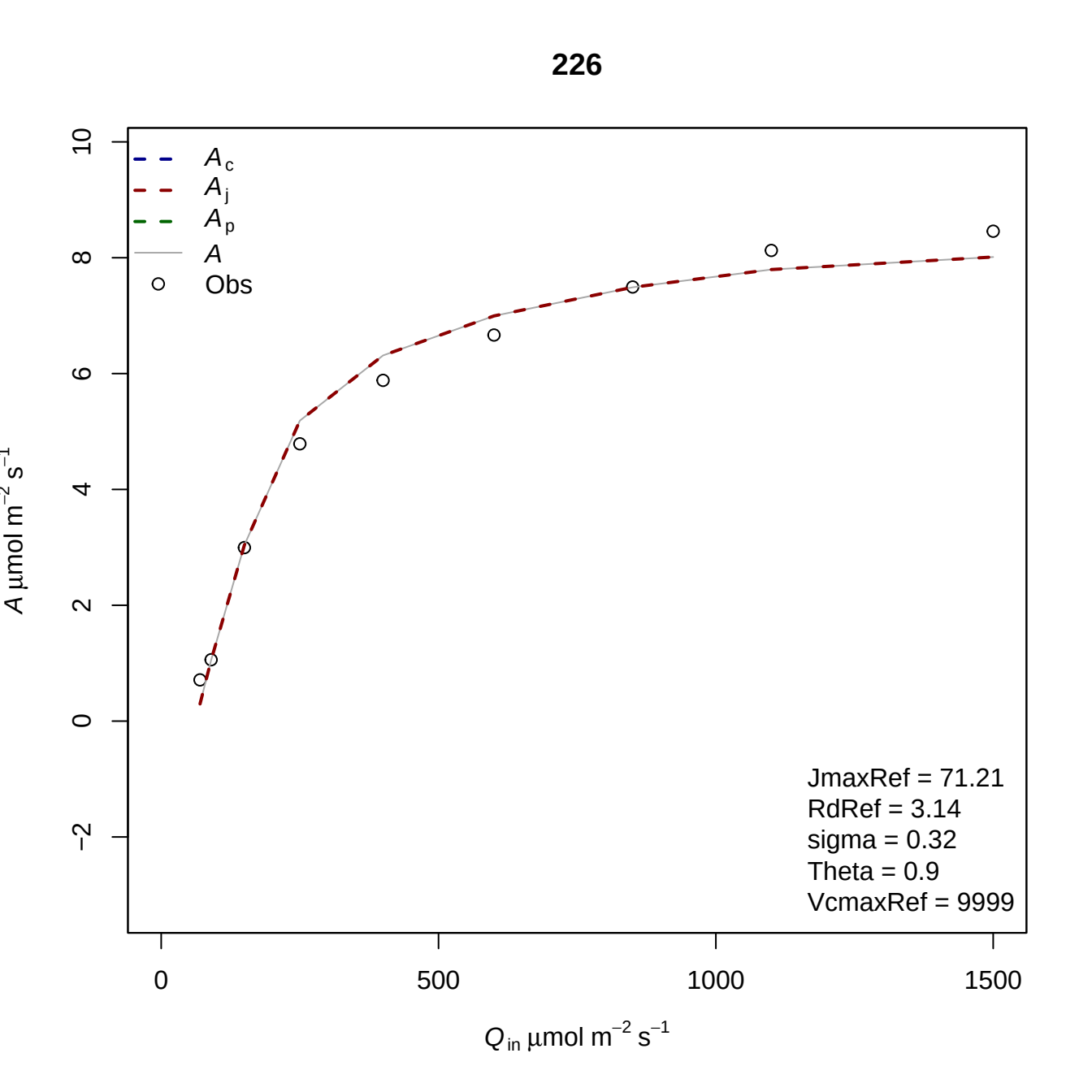

221
A_c
A_j
A_p
A
Obs
JmaxRef = 97.37
RdRef = 3.49
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

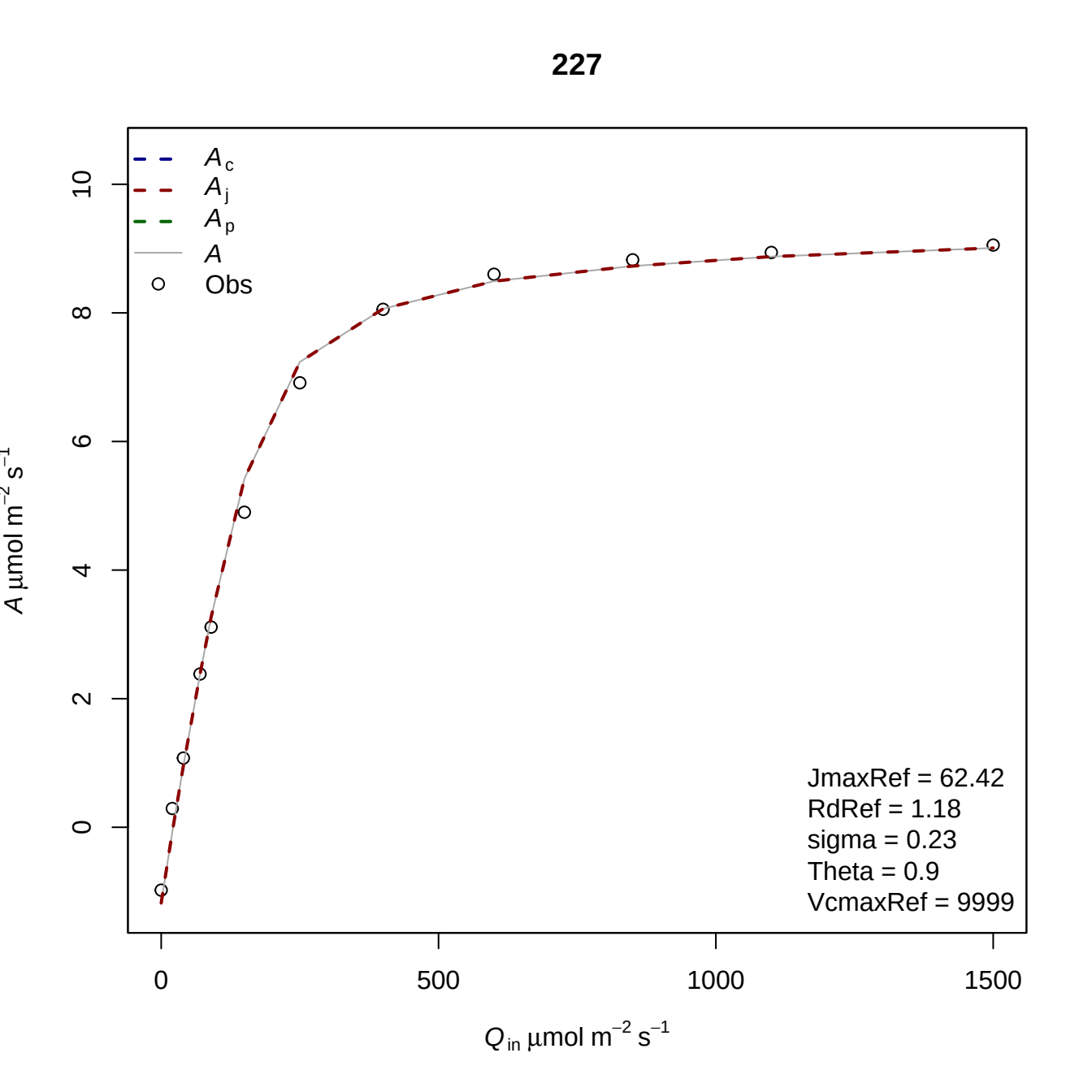

222
A_c
A_j
A_p
A
Obs
JmaxRef = 86.31
RdRef = 1.4
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

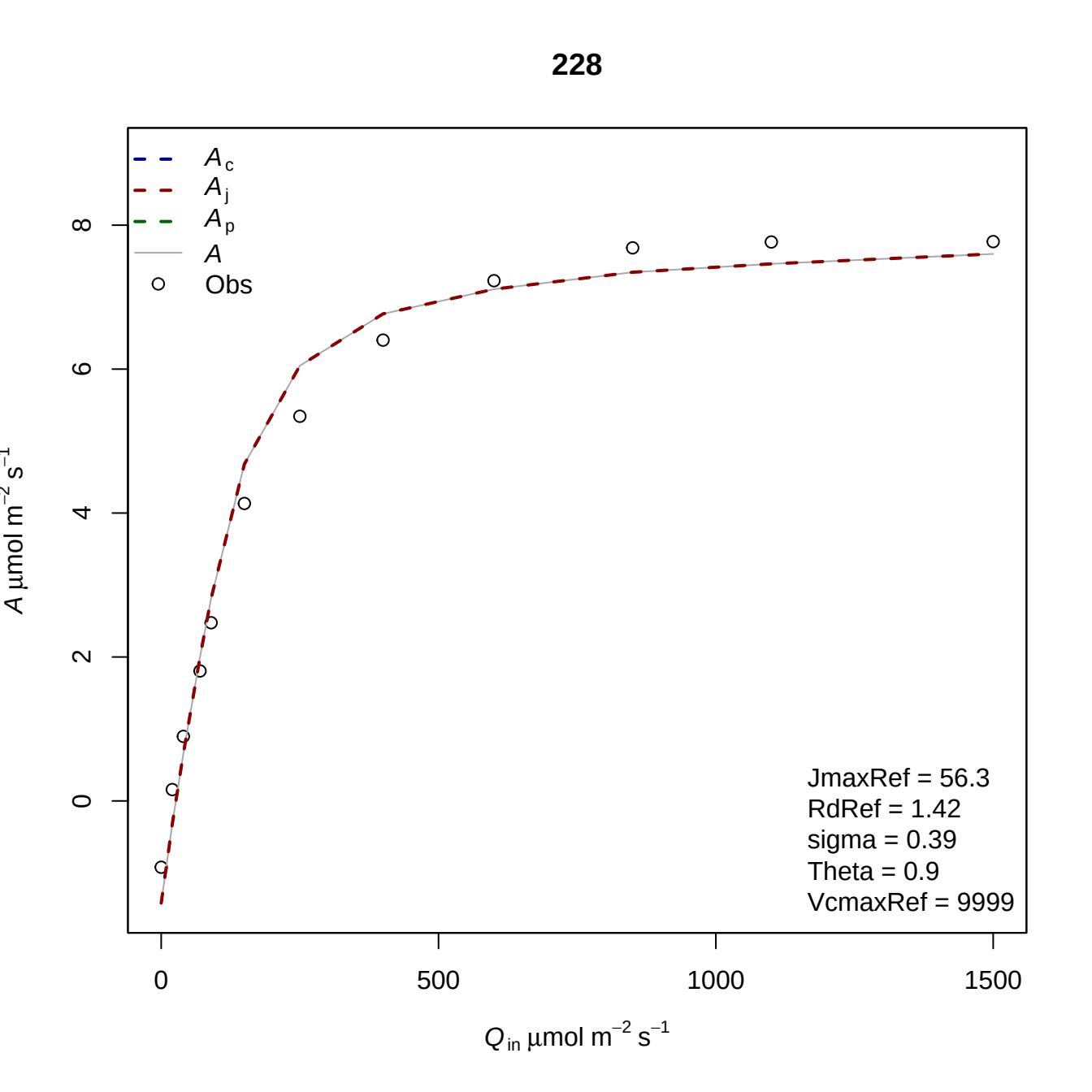

223
A_c
A_j
A_p
A
Obs
JmaxRef = 57.7
RdRef = 3.59
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

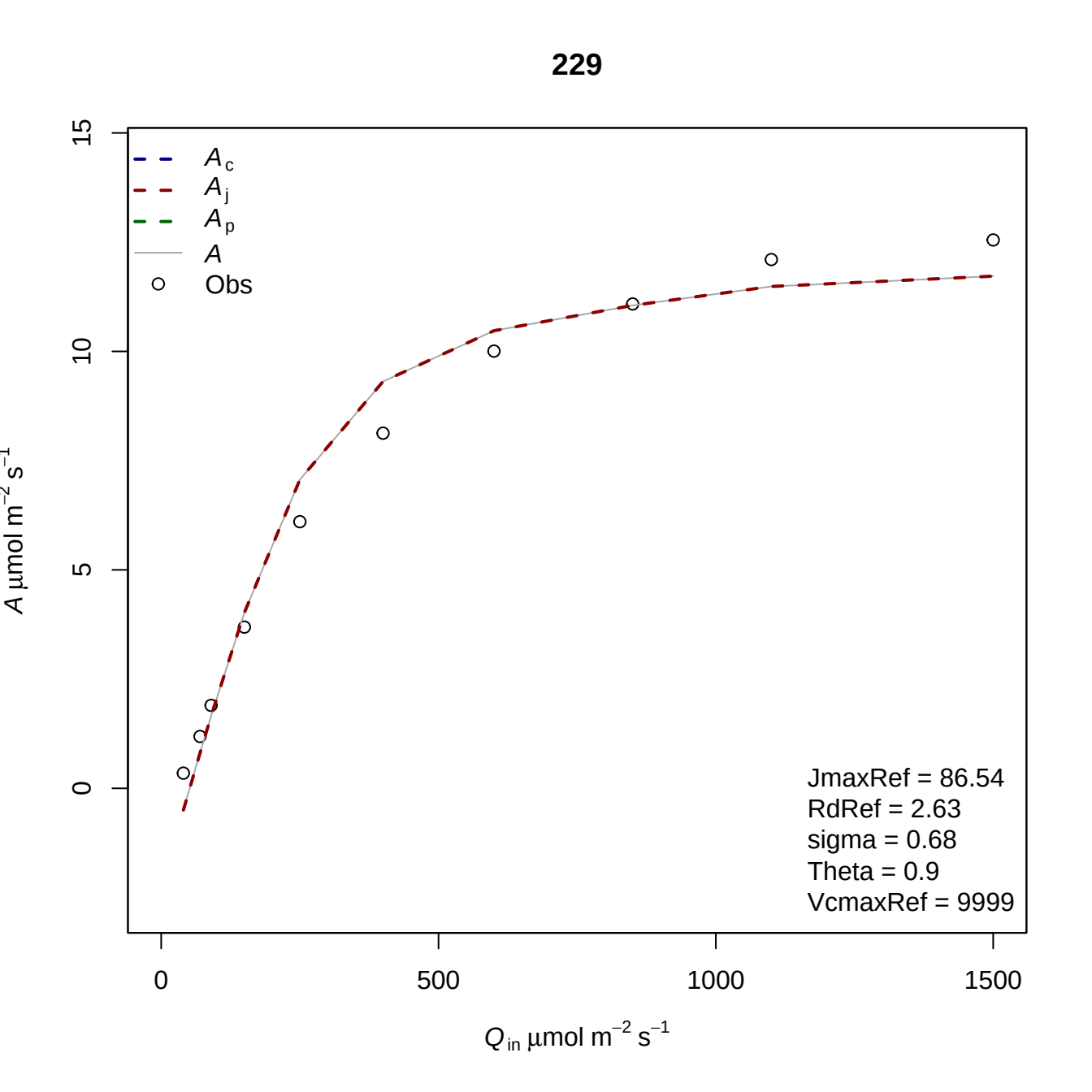

224
A_c
A_j
A_p
A
Obs
JmaxRef = 83.24
RdRef = 2.28
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

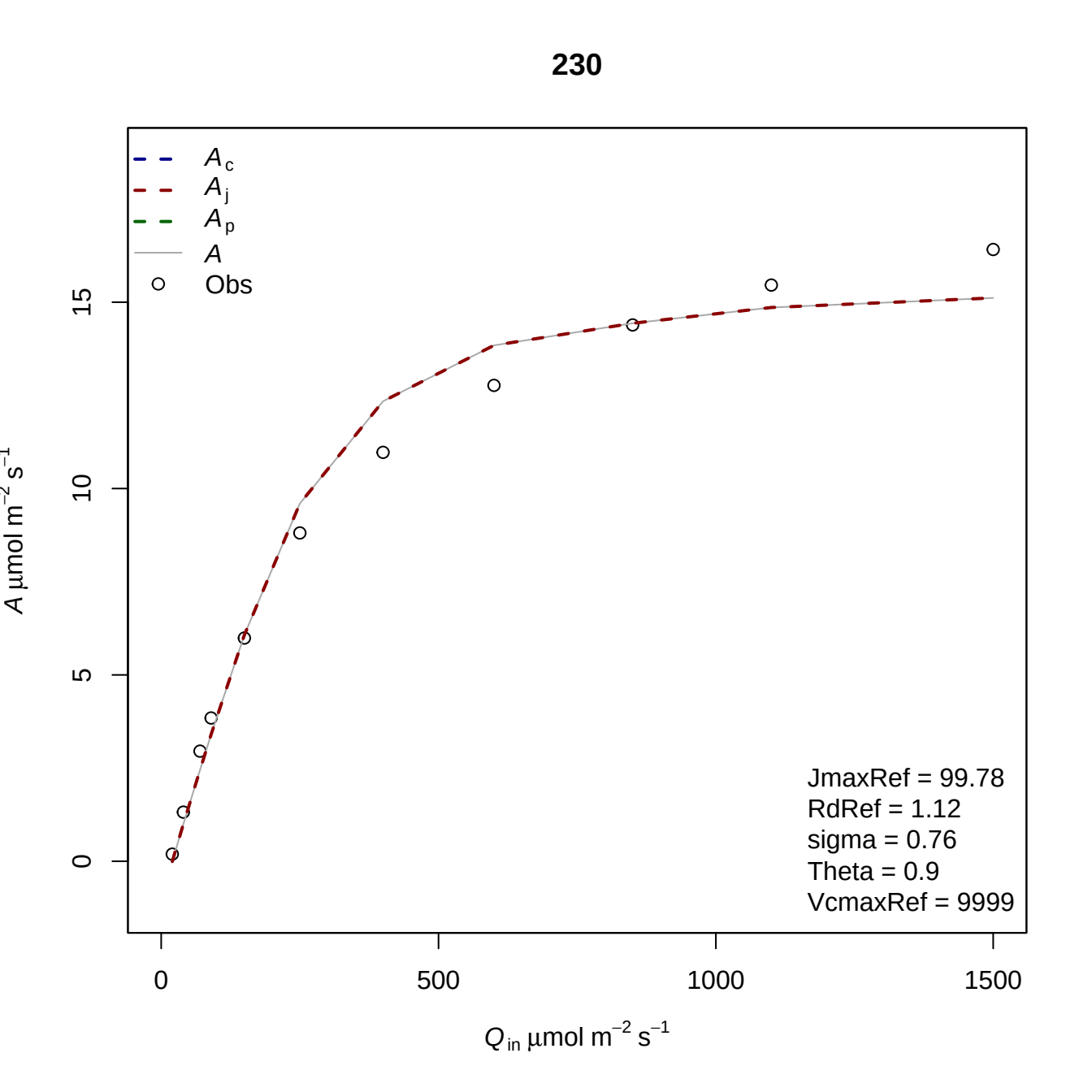

225
A_c
A_j
A_p
A
Obs
JmaxRef = 56.53
RdRef = 1.74
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

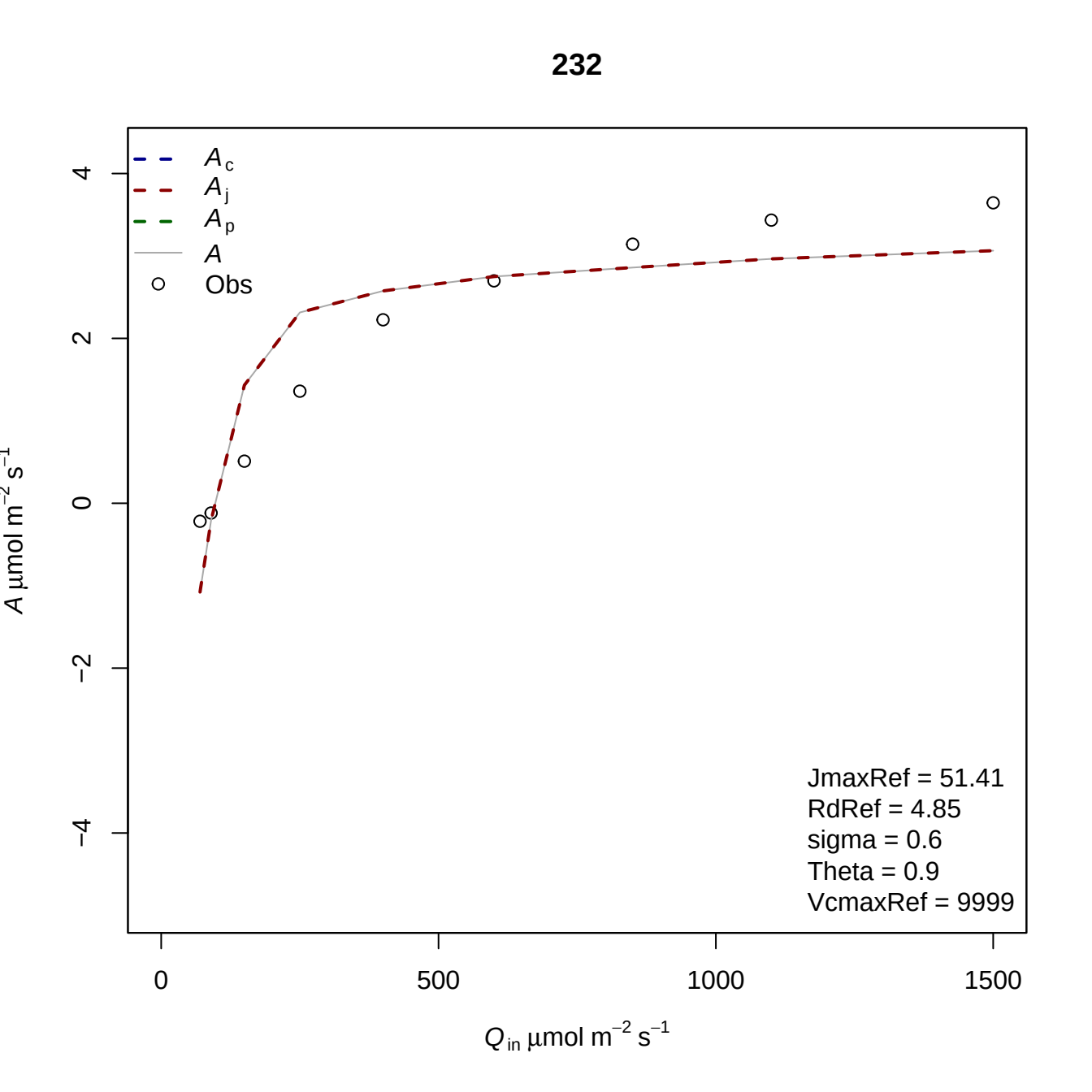

226
A_c
A_j
A_p
A
Obs
JmaxRef = 71.21
RdRef = 3.14
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

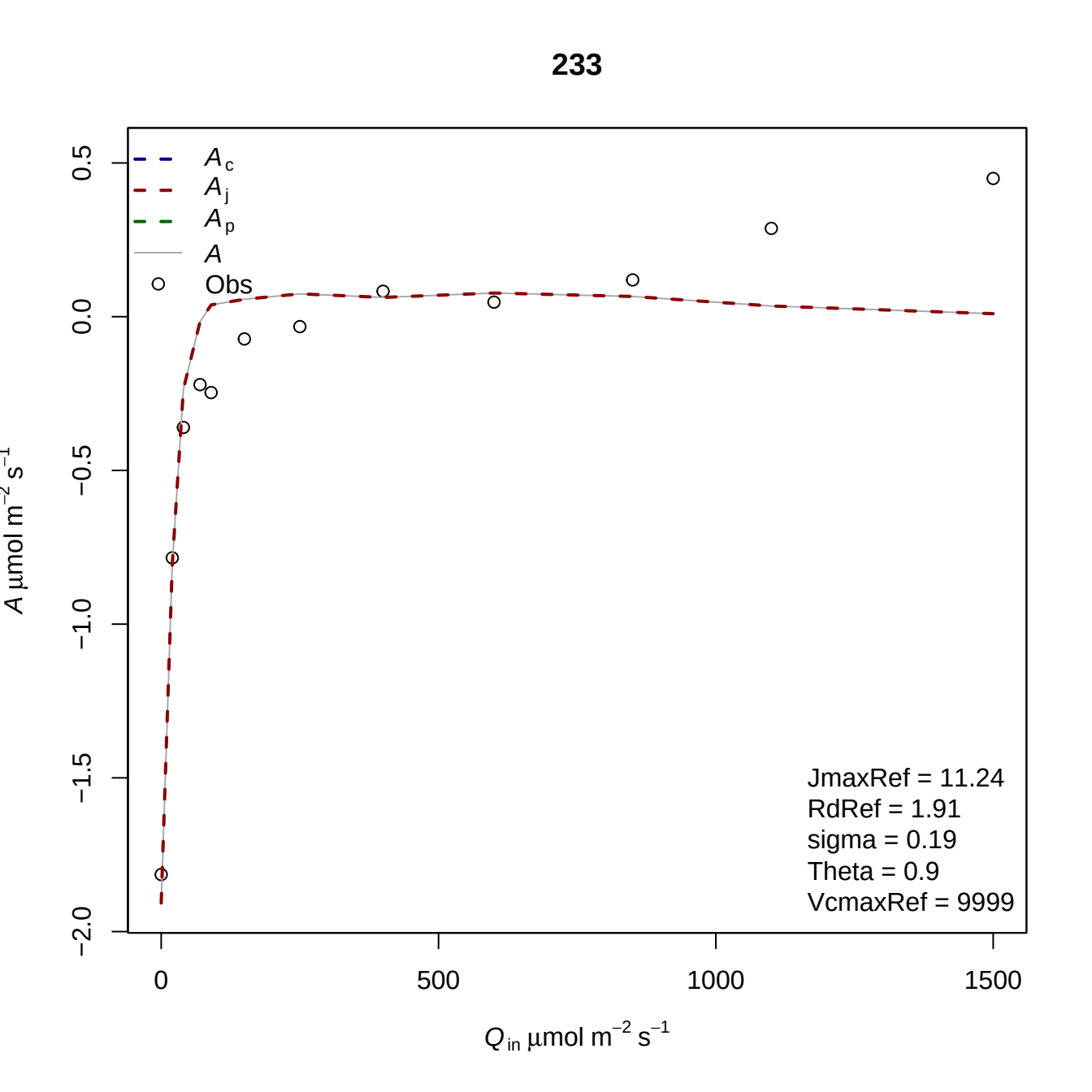

227
A_c
A_j
A_p
A
Obs
JmaxRef = 62.42
RdRef = 1.18
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

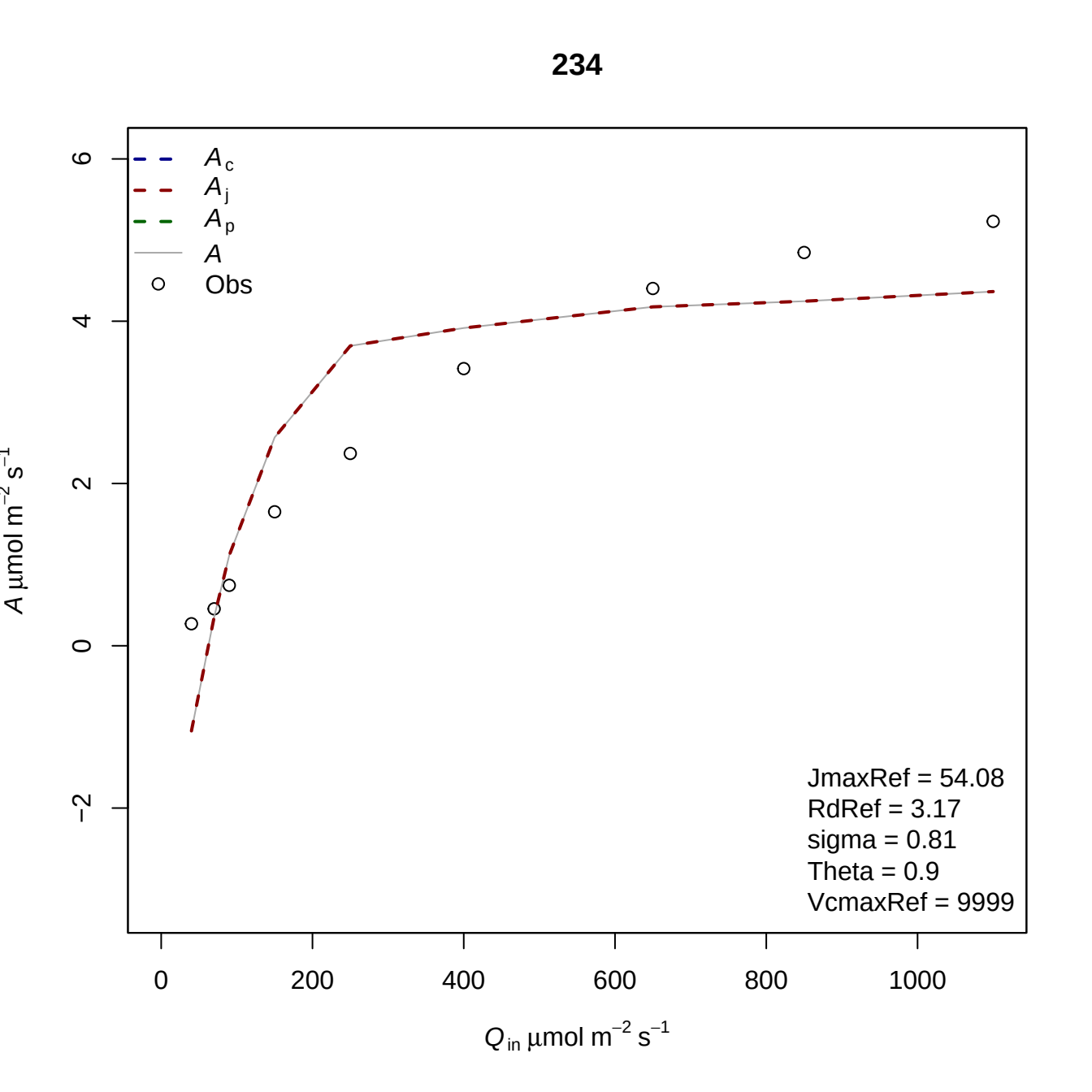

228
A_c
A_j
A_p
A
Obs
JmaxRef = 56.3
RdRef = 1.42
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

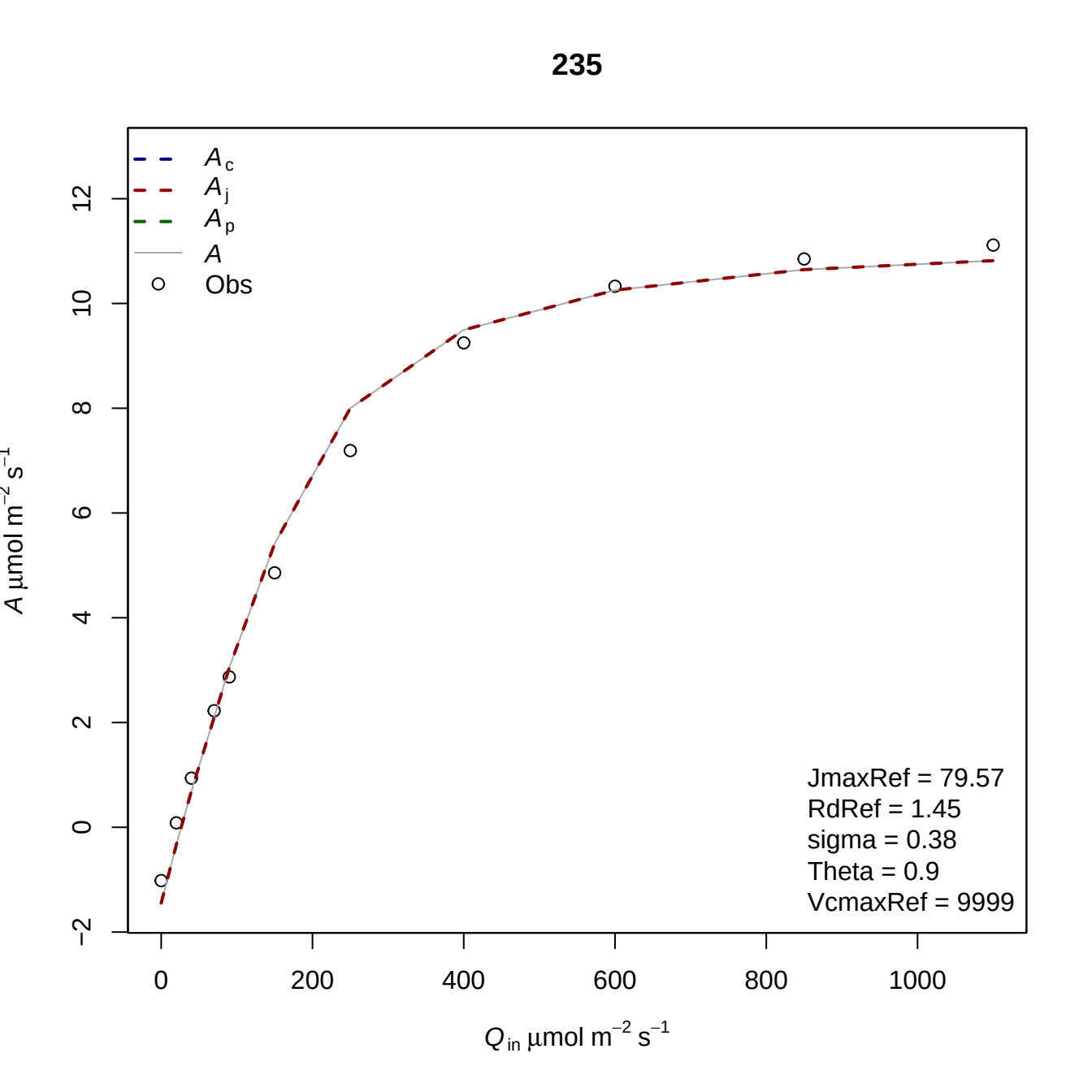

229
A_c
A_j
A_p
A
Obs
JmaxRef = 86.54
RdRef = 2.63
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

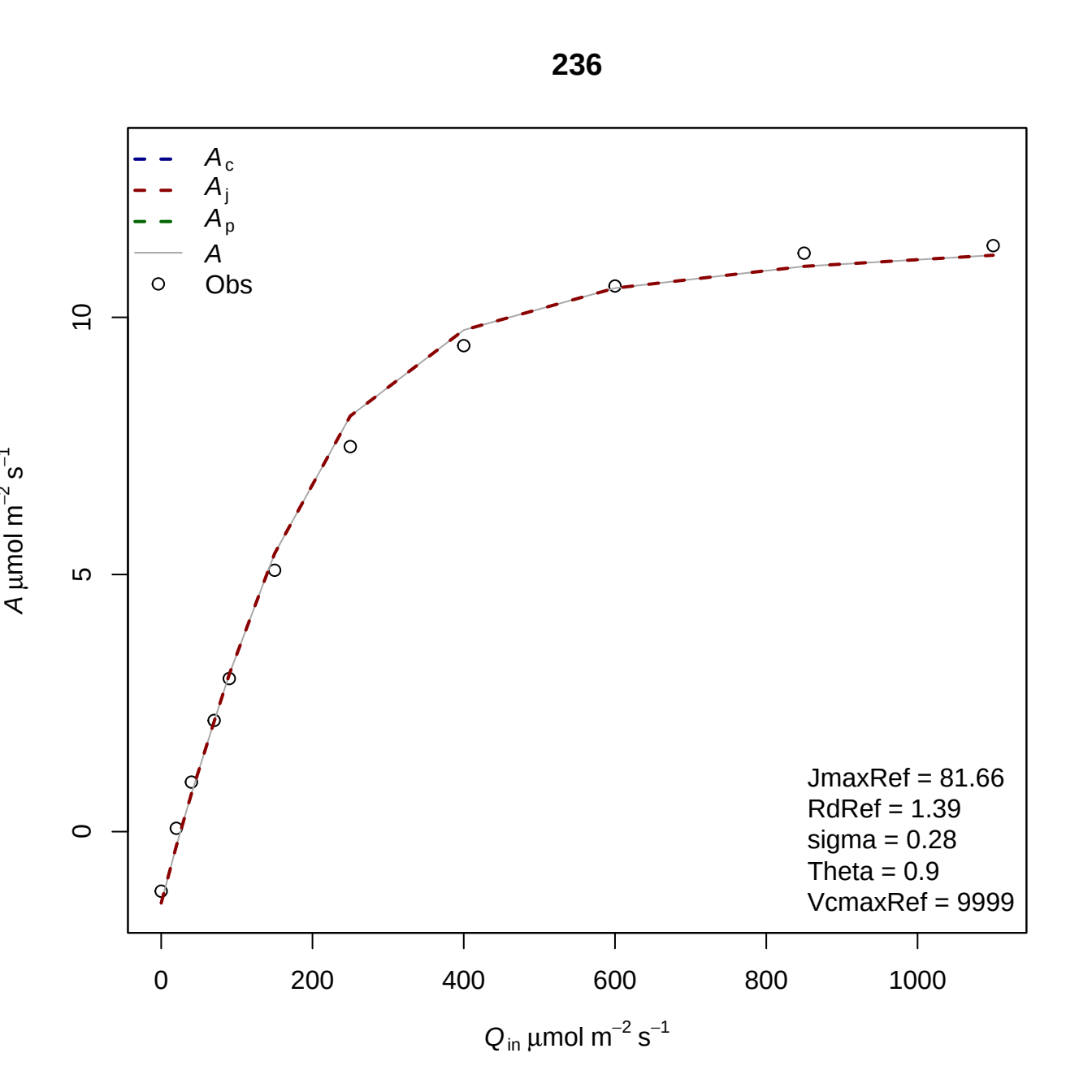

230
A_c
A_j
A_p
A
Obs
JmaxRef = 99.78
RdRef = 1.12
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

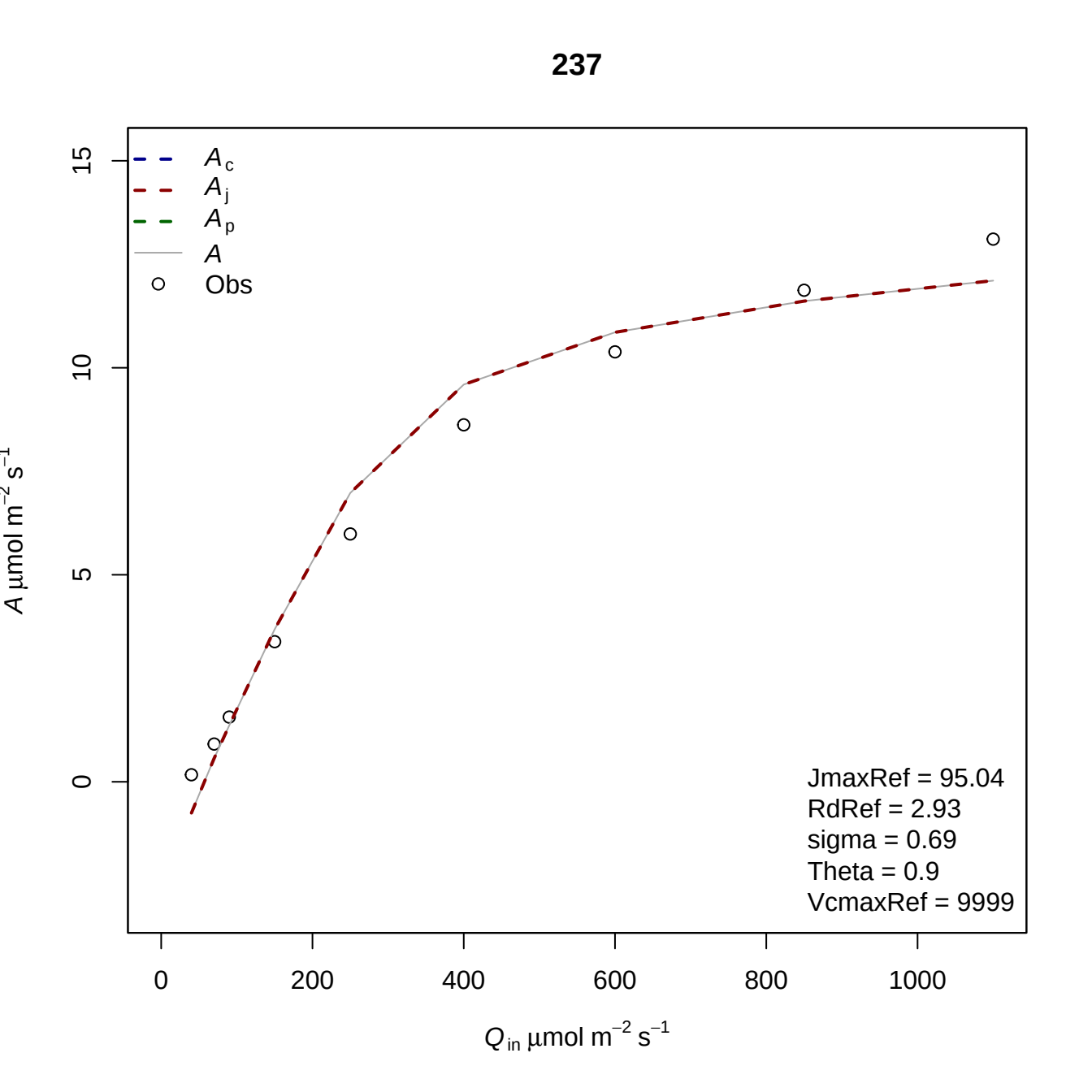

232
A_c
A_j
A_p
A
Obs
JmaxRef = 51.41
RdRef = 4.85
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

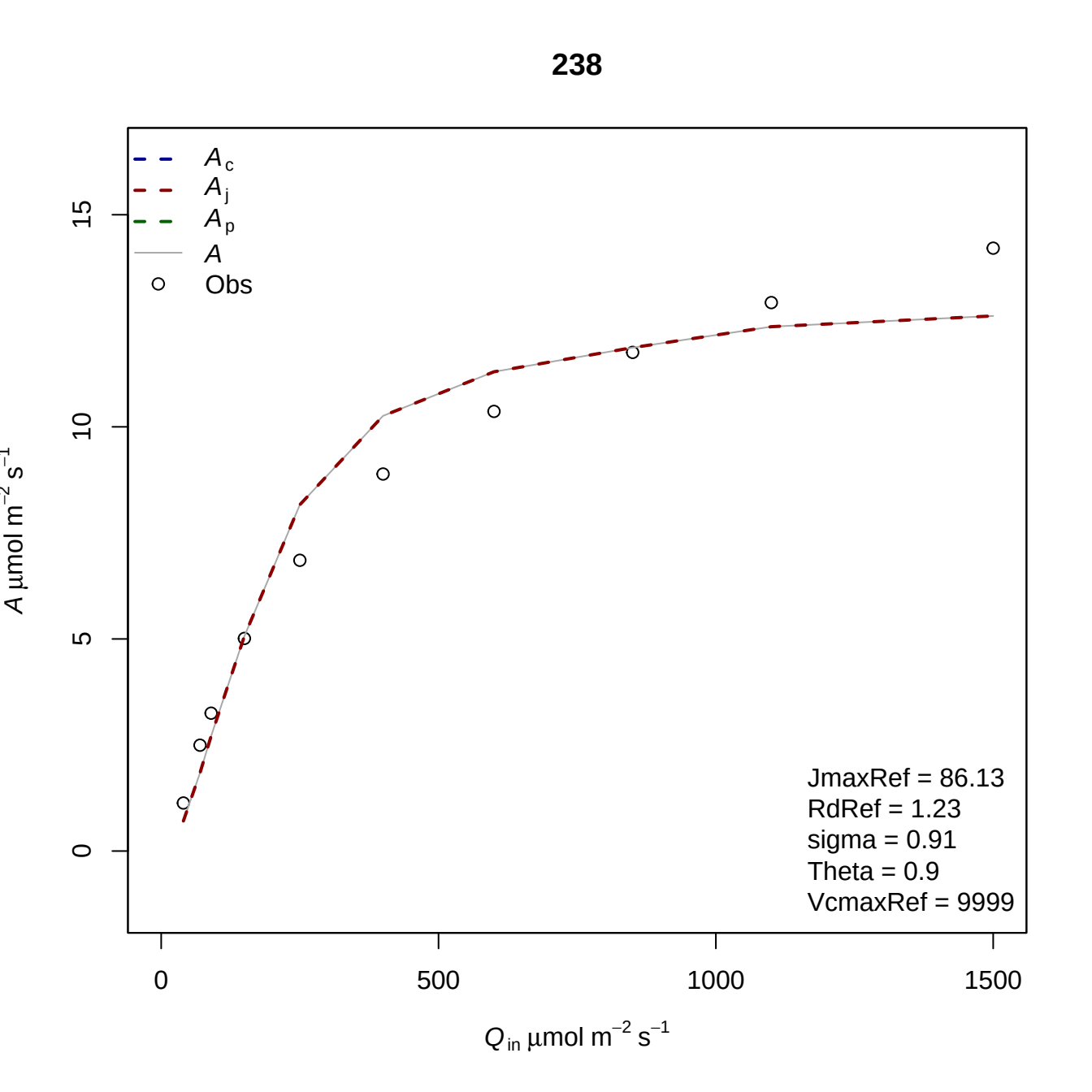

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.24
RdRef = 1.91
sigma = 0.19
Theta = 0.9
VcmaxRef = 9999
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

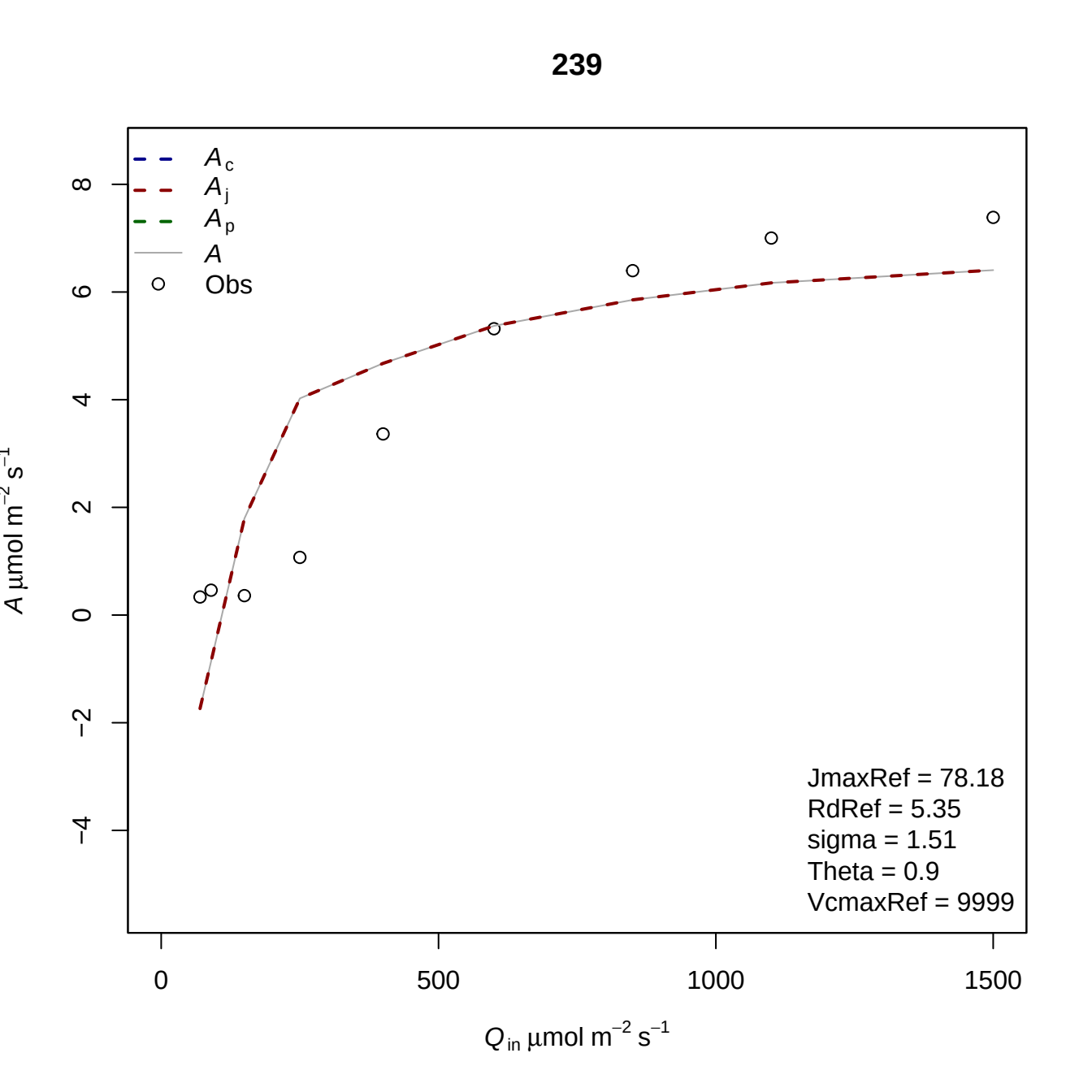

234
A_c
A_j
A_p
A
Obs
JmaxRef = 54.08
RdRef = 3.17
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

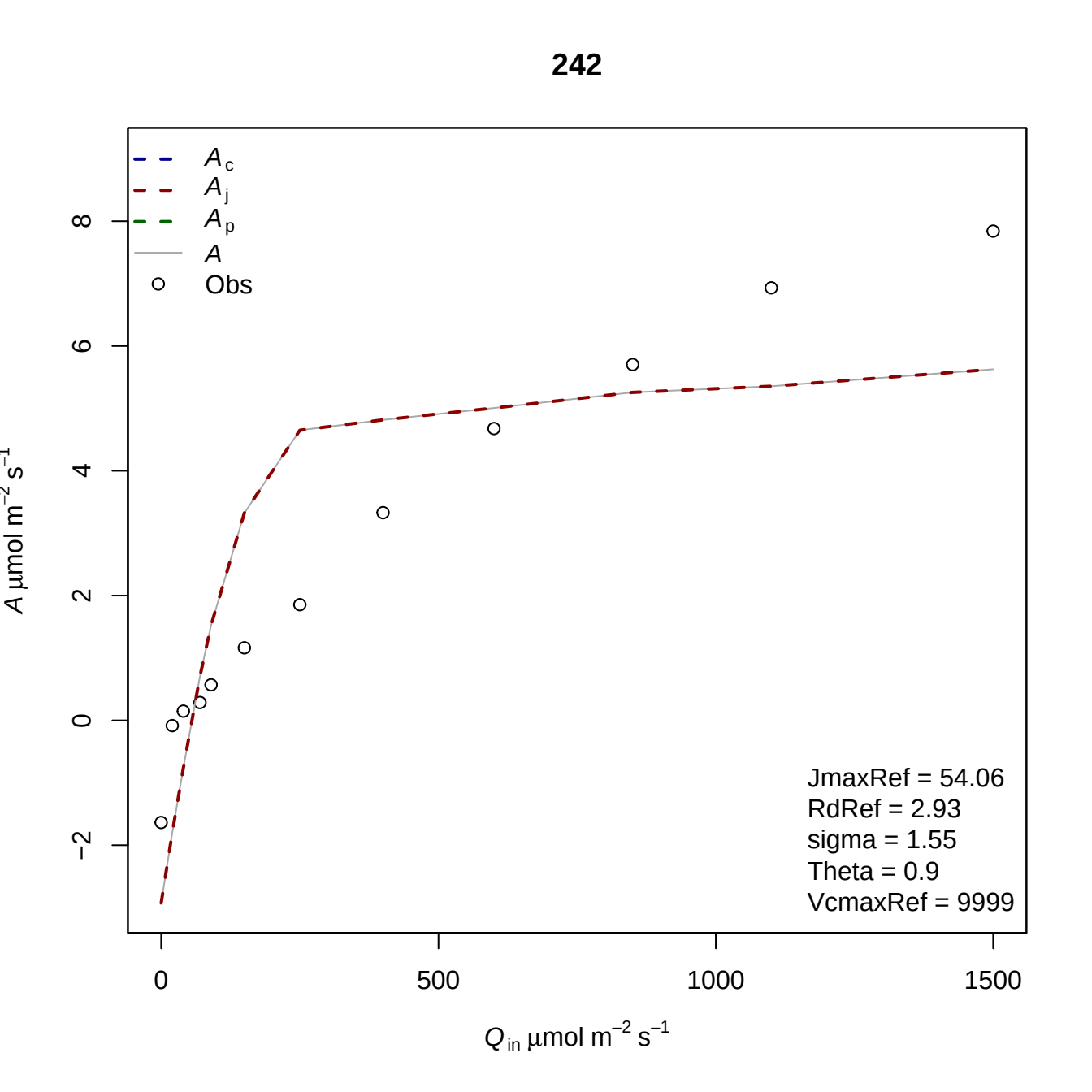

235
A_c
A_j
A_p
A
Obs
JmaxRef = 79.57
RdRef = 1.45
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

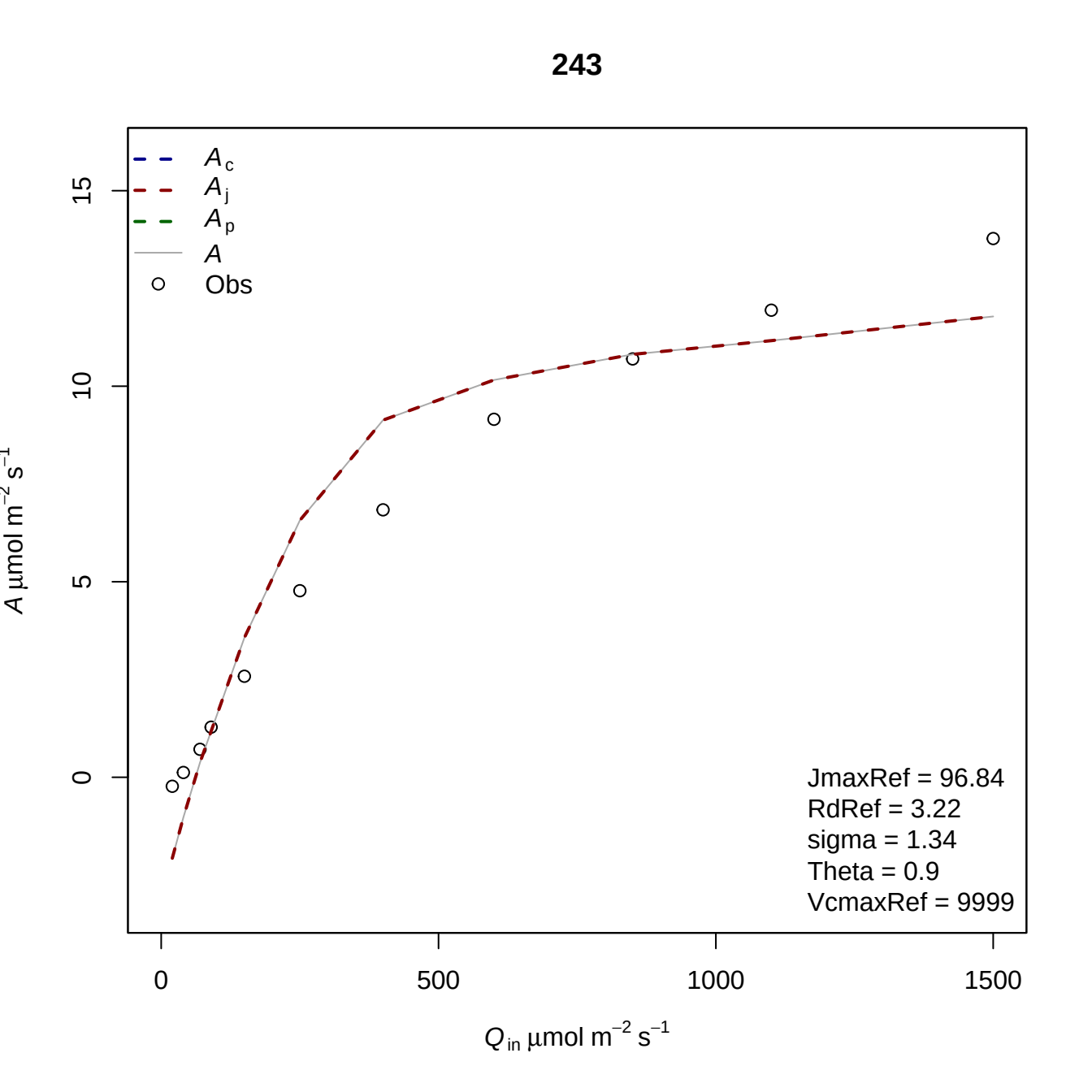

236
A_c
A_j
A_p
A
Obs
JmaxRef = 81.66
RdRef = 1.39
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

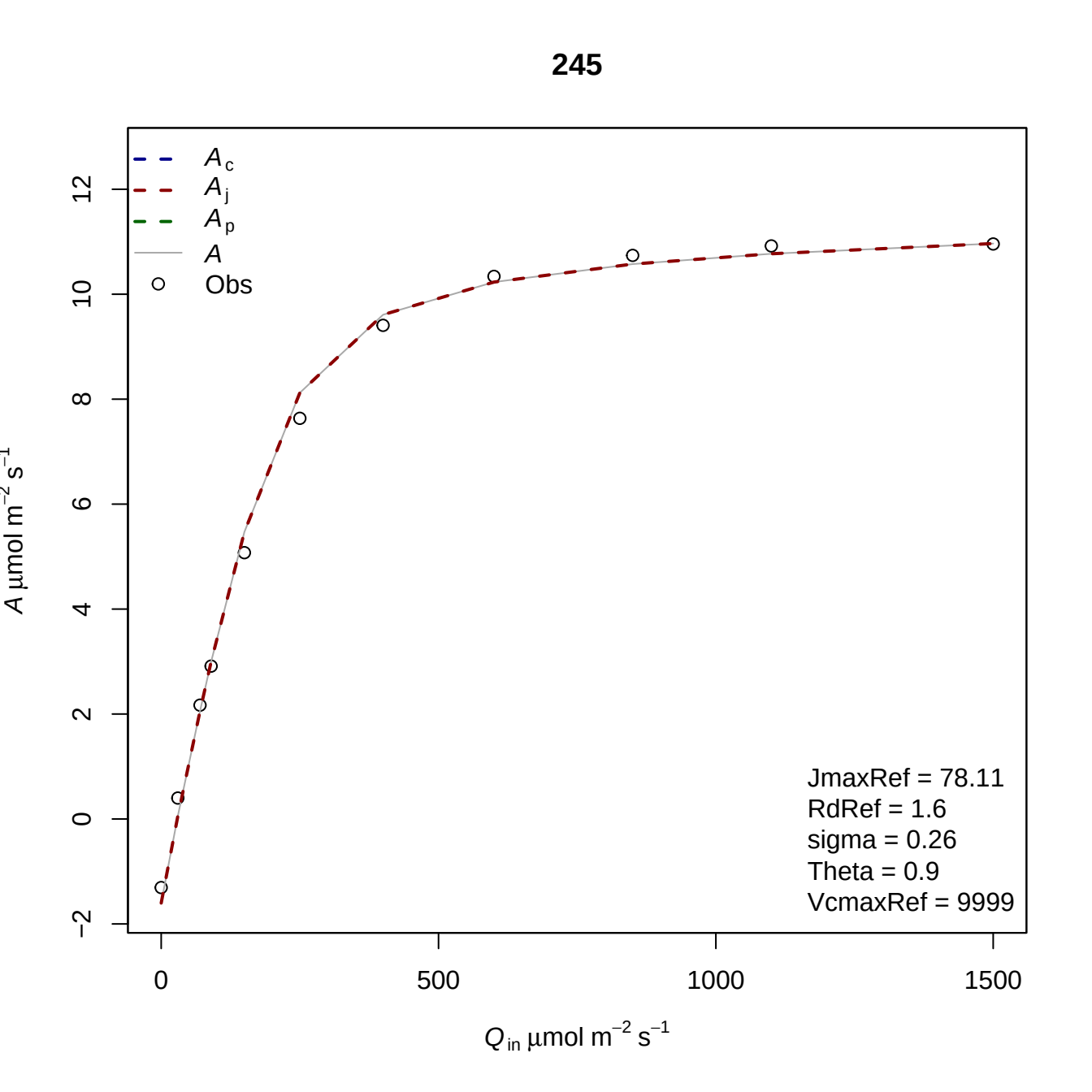

237
A_c
A_j
A_p
A
Obs
JmaxRef = 95.04
RdRef = 2.93
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

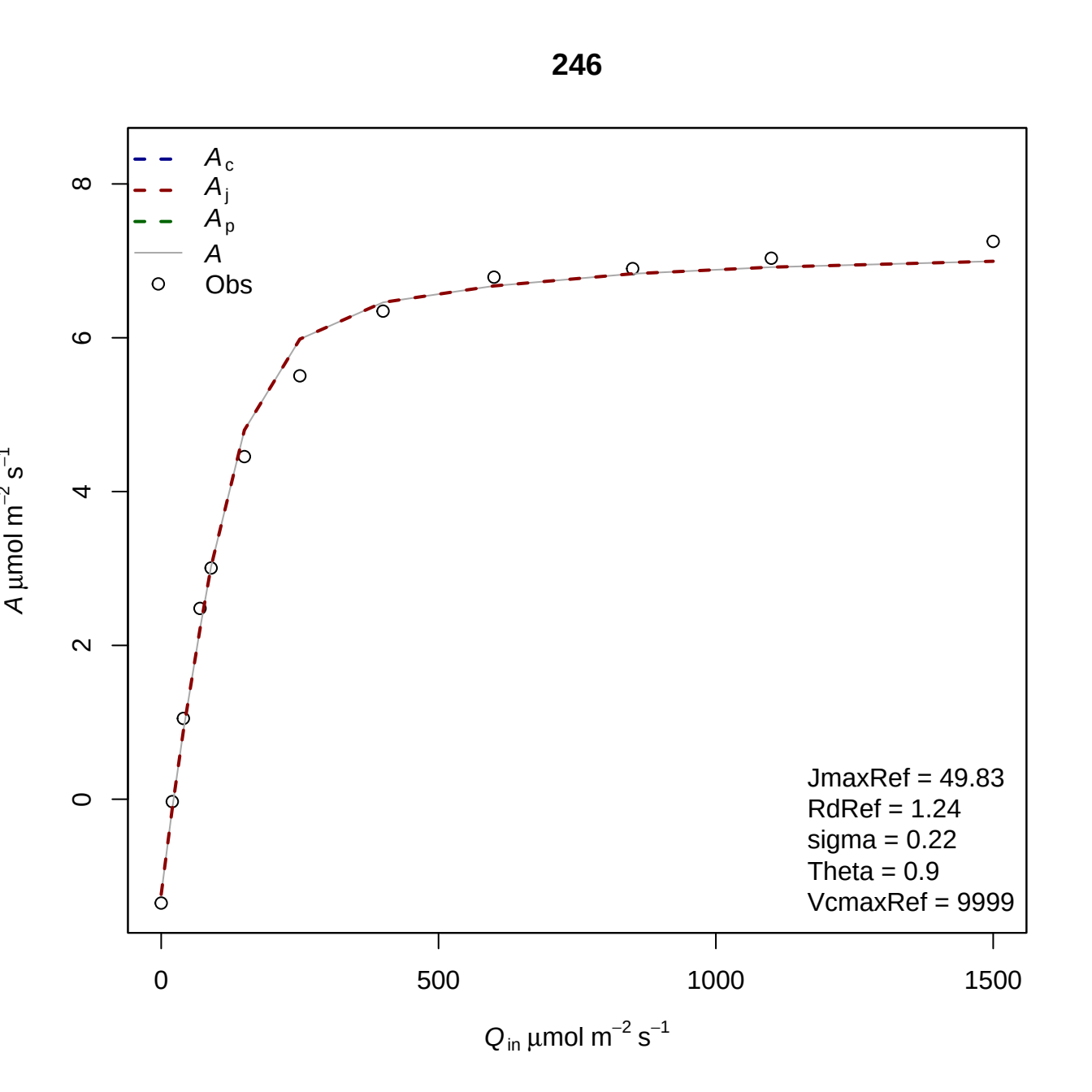

238
A_c
A_j
A_p
A
Obs
JmaxRef = 86.13
RdRef = 1.23
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

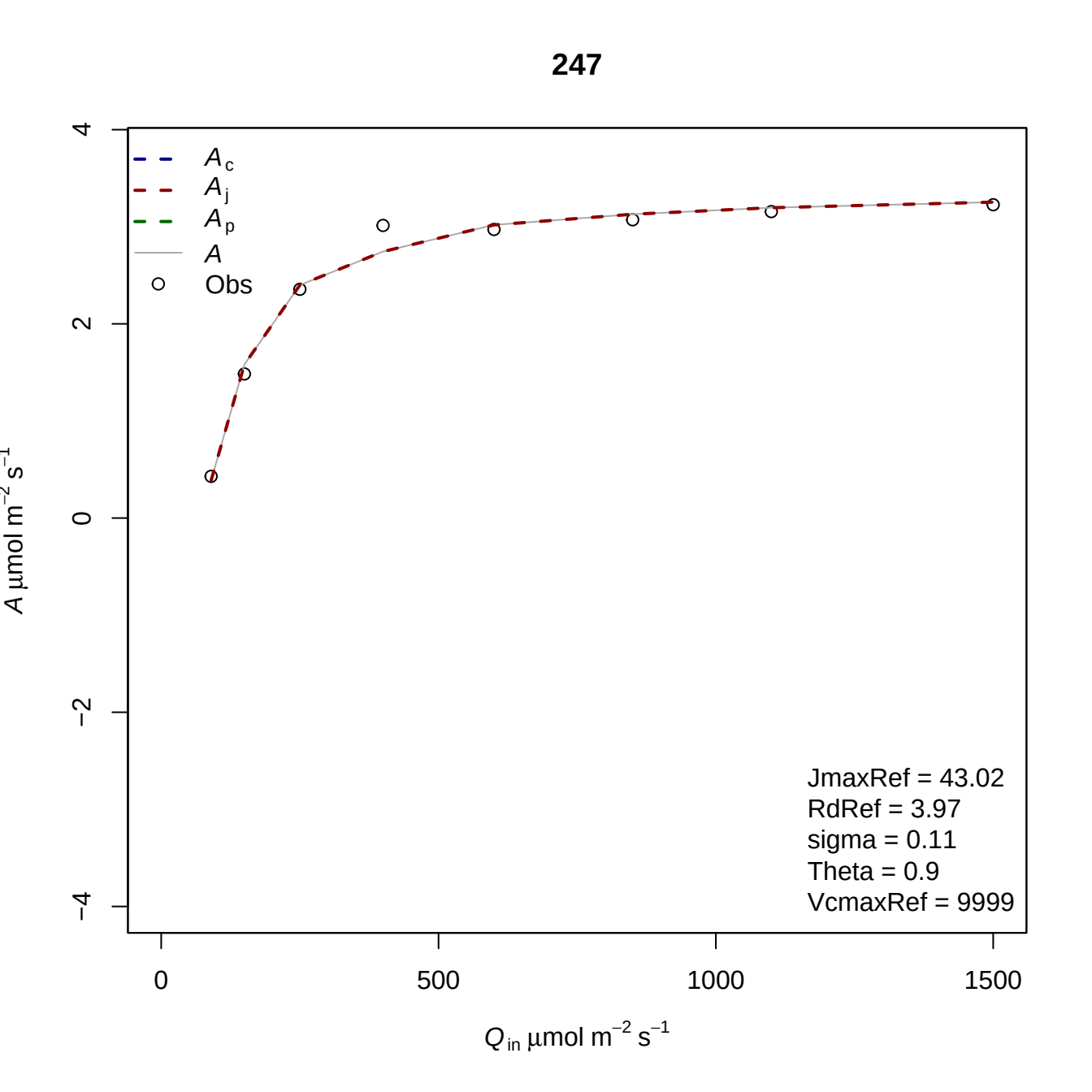

239
A_c
A_j
A_p
A
Obs
JmaxRef = 78.18
RdRef = 5.35
sigma = 1.51
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

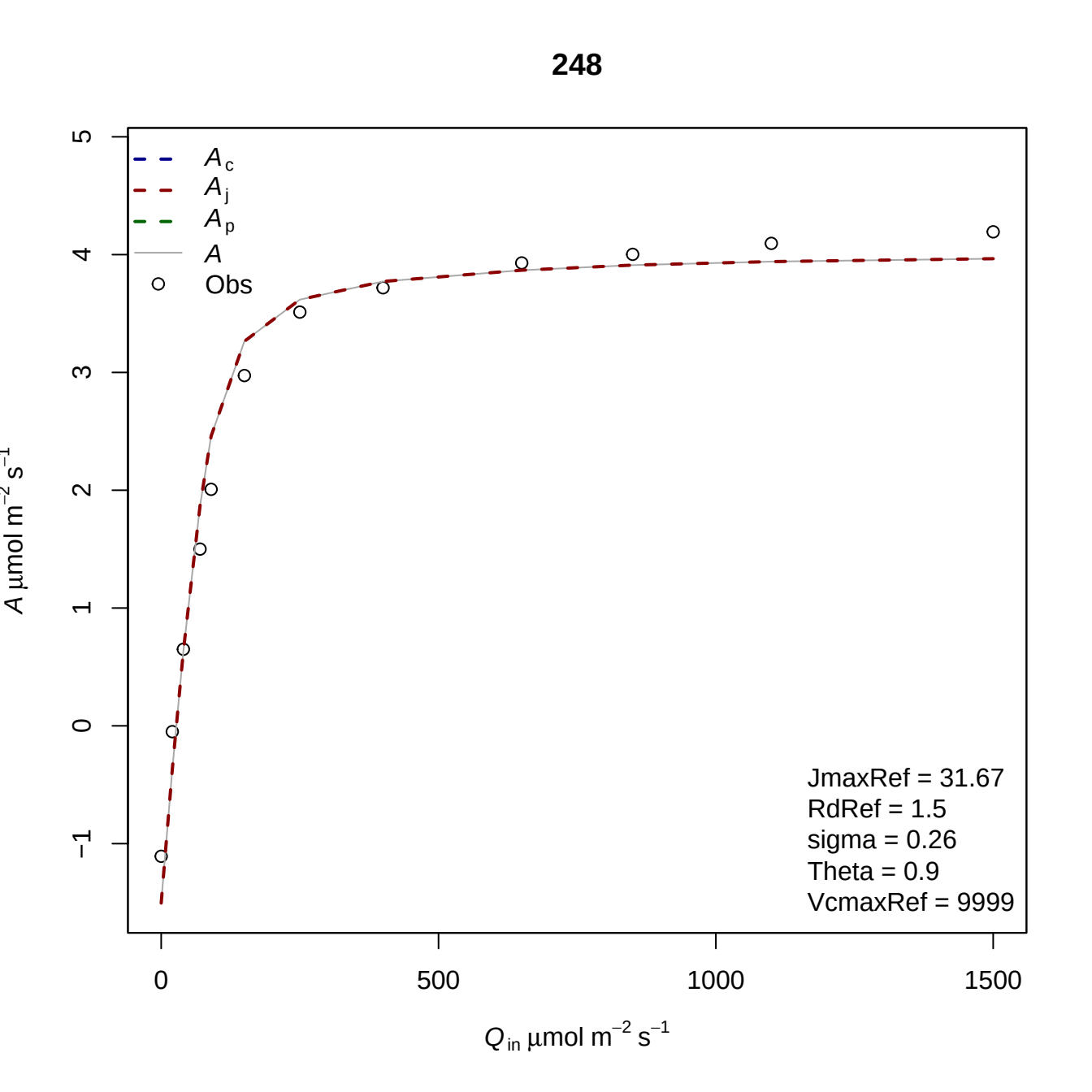

242
A_c
A_j
A_p
A
Obs
JmaxRef = 54.06
RdRef = 2.93
sigma = 1.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

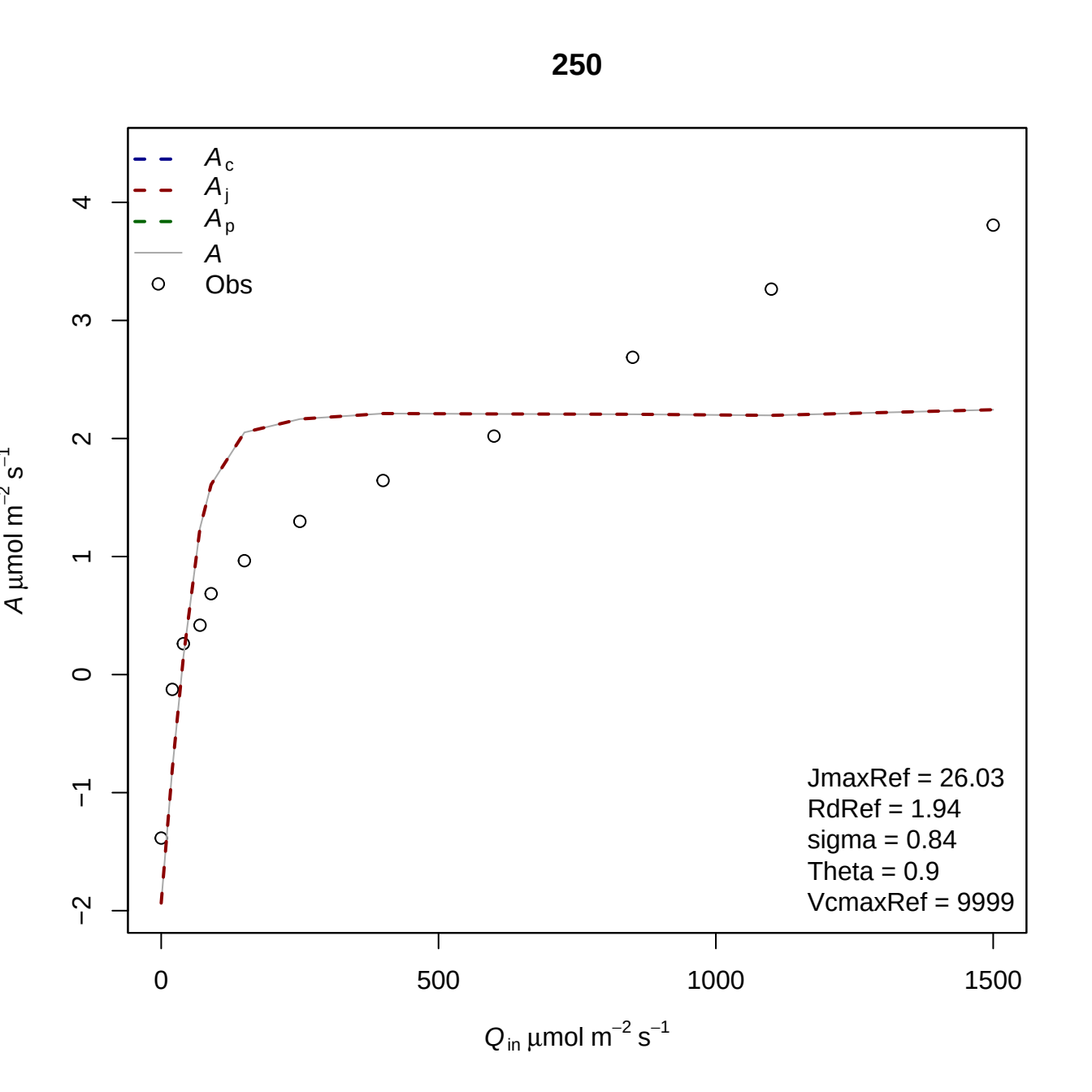

243
A_c
A_j
A_p
A
Obs
JmaxRef = 96.84
RdRef = 3.22
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

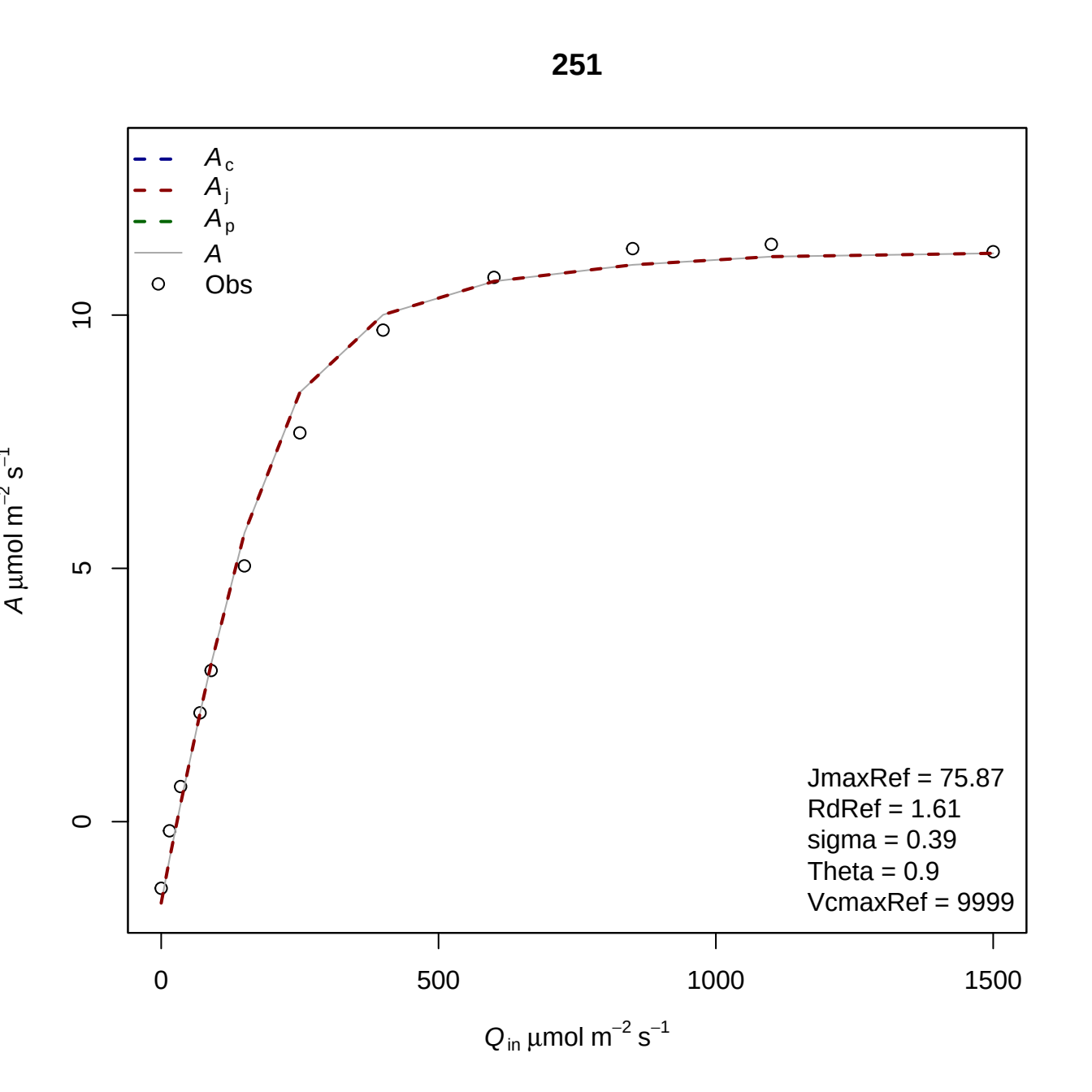

245
A_c
A_j
A_p
A
Obs
JmaxRef = 78.11
RdRef = 1.6
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

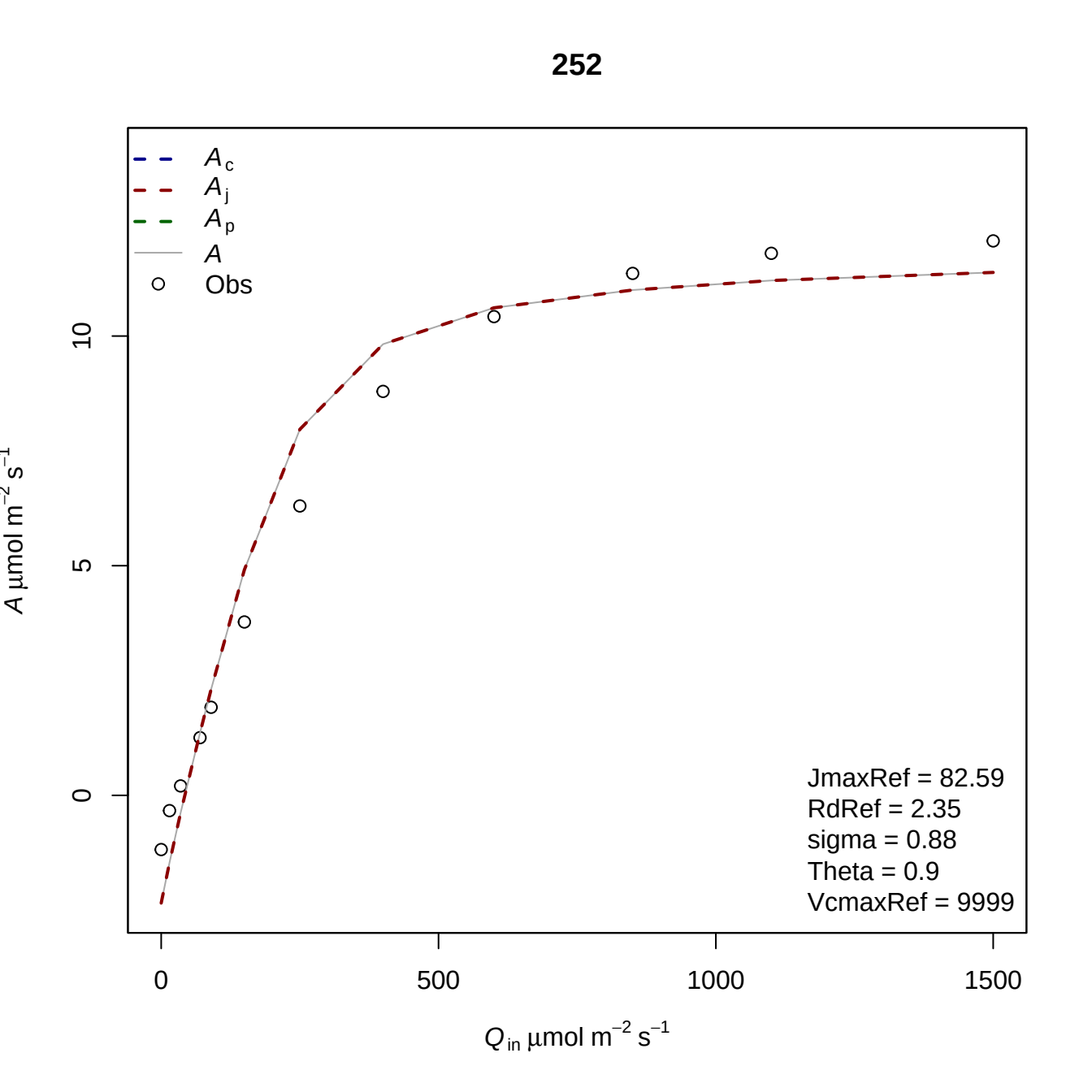

246
A_c
A_j
A_p
A
Obs
JmaxRef = 49.83
RdRef = 1.24
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

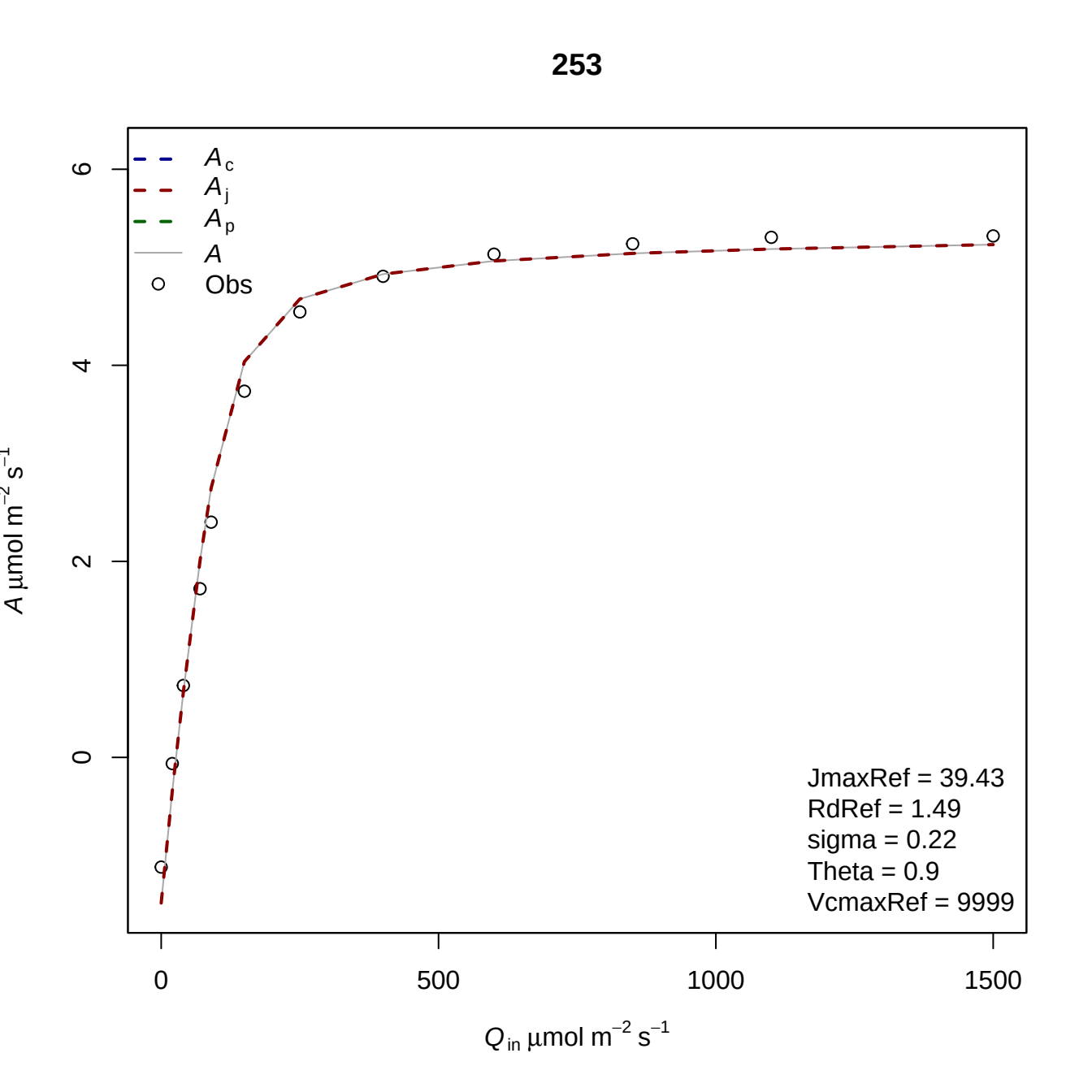

247
A_c
A_j
A_p
A
Obs
JmaxRef = 43.02
RdRef = 3.97
sigma = 0.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

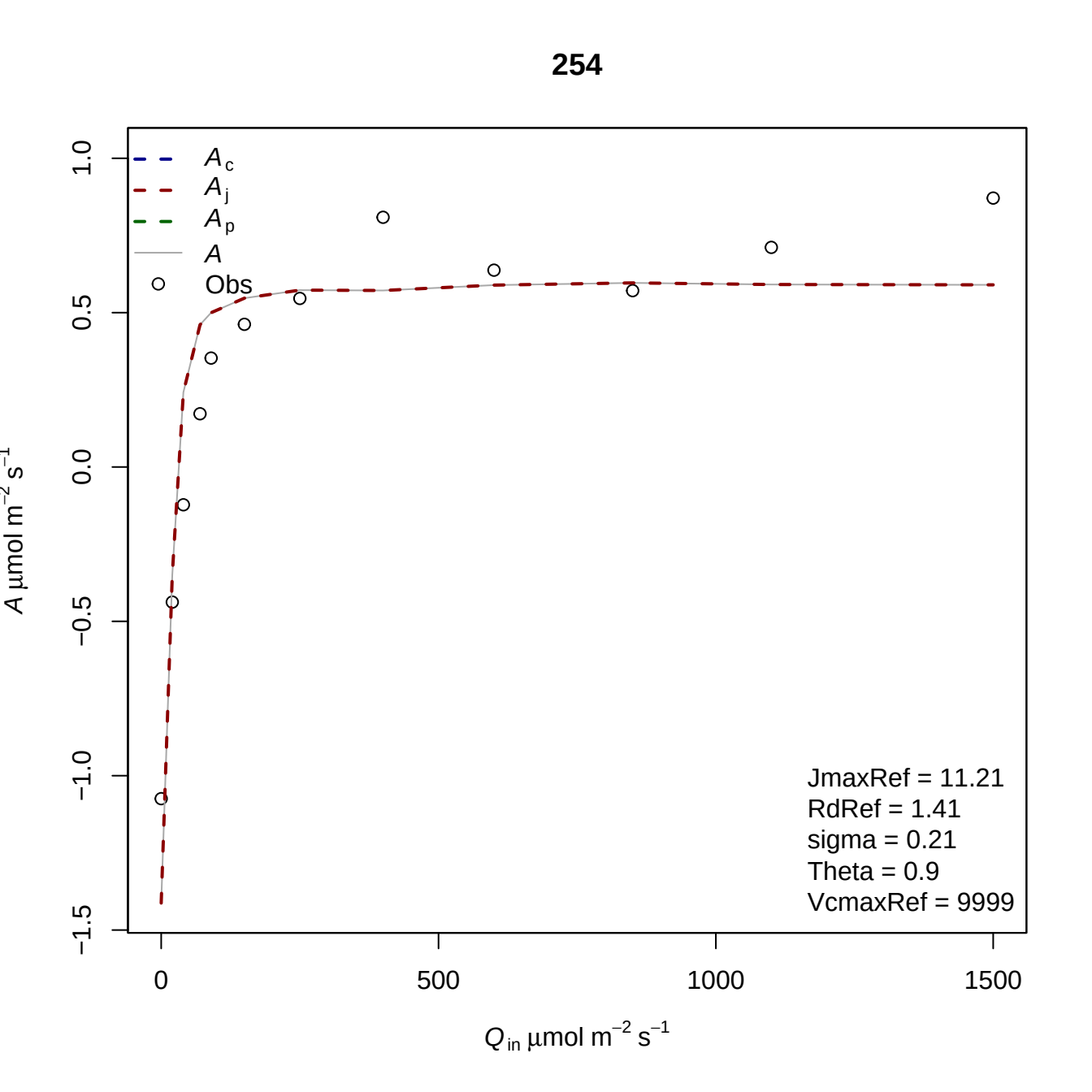

248
A_c
A_j
A_p
A
Obs
JmaxRef = 31.67
RdRef = 1.5
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

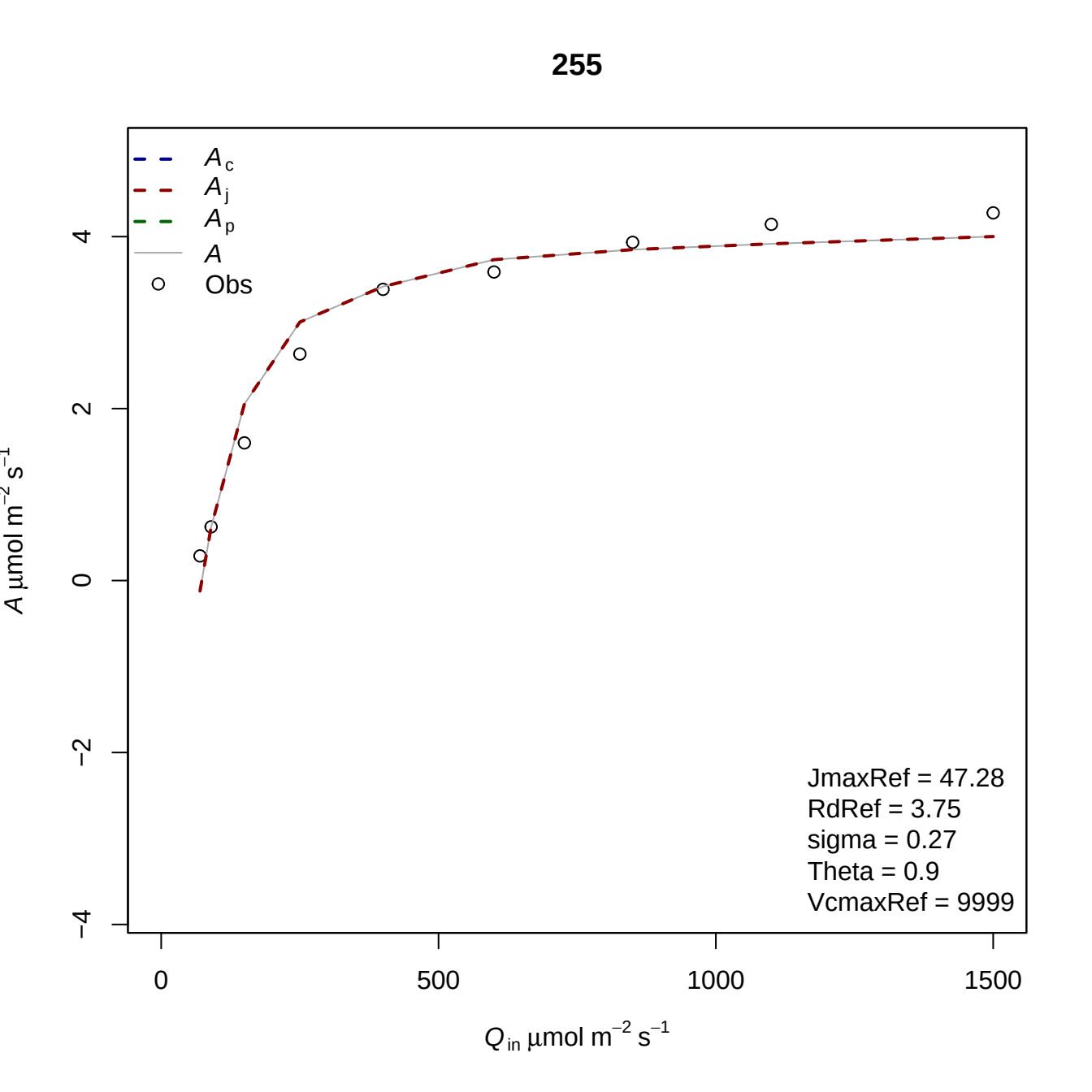

250
A_c
A_j
A_p
A
Obs
JmaxRef = 26.03
RdRef = 1.94
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

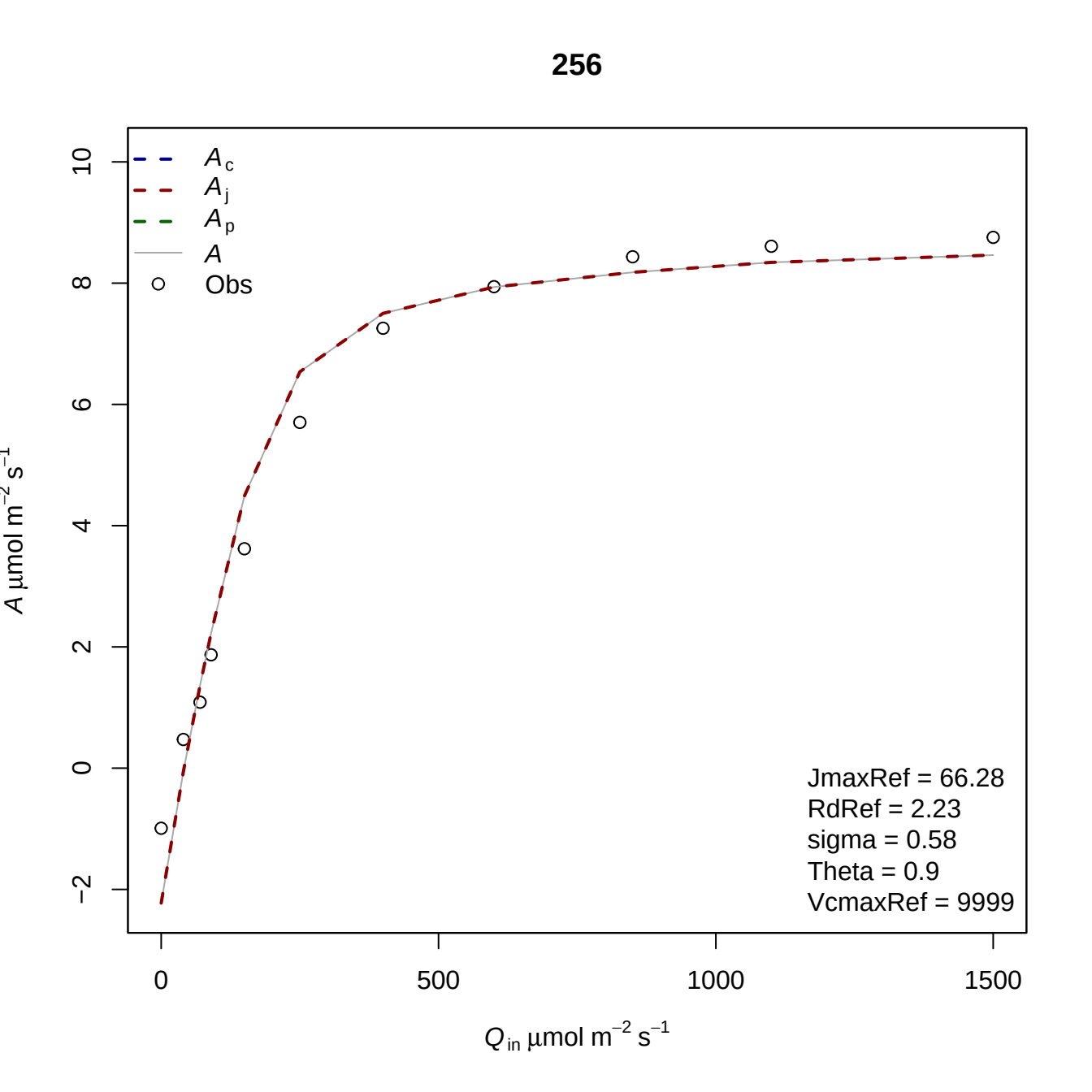

251
A_c
A_j
A_p
A
Obs
JmaxRef = 75.87
RdRef = 1.61
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

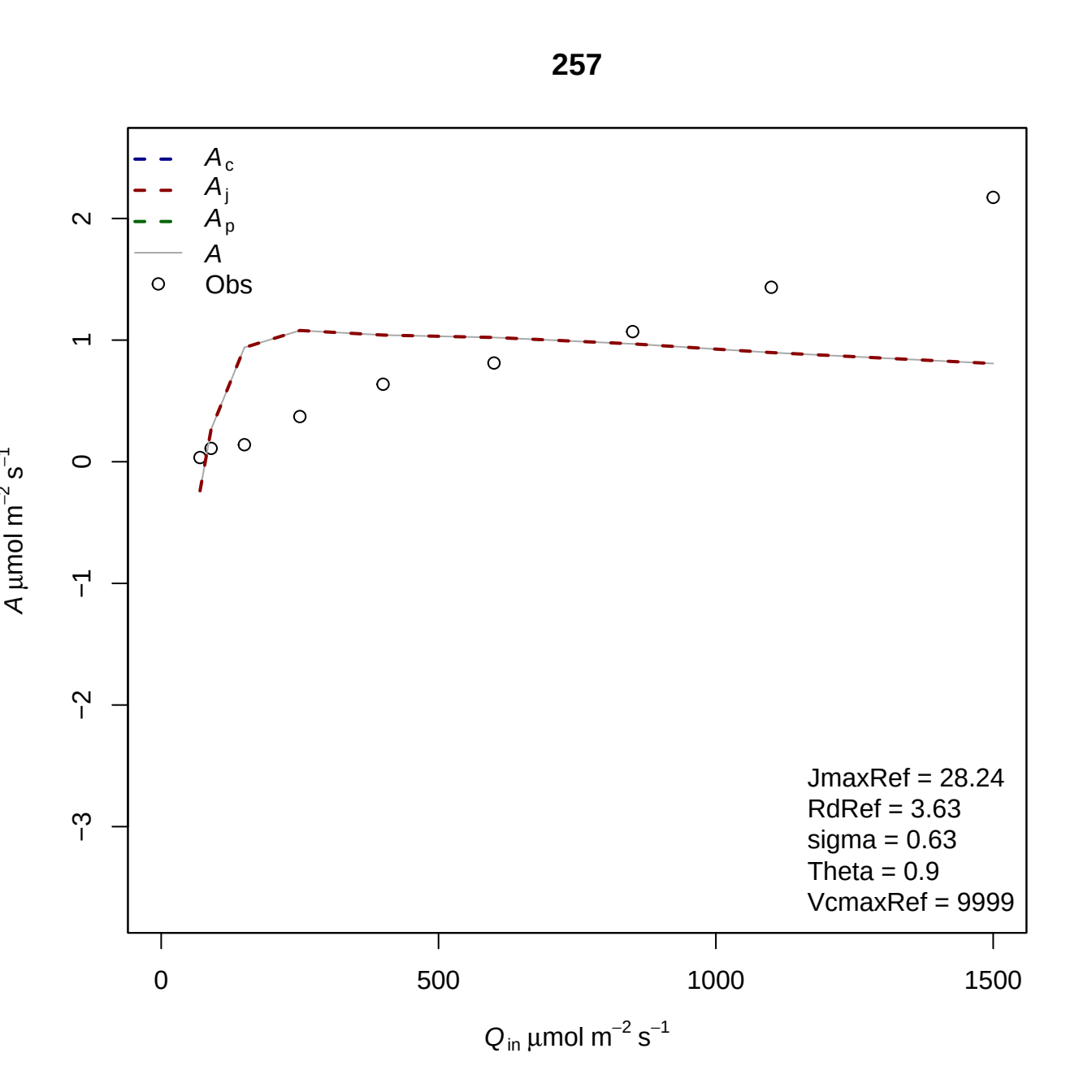

252
A_c
A_j
A_p
A
Obs
JmaxRef = 82.59
RdRef = 2.35
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

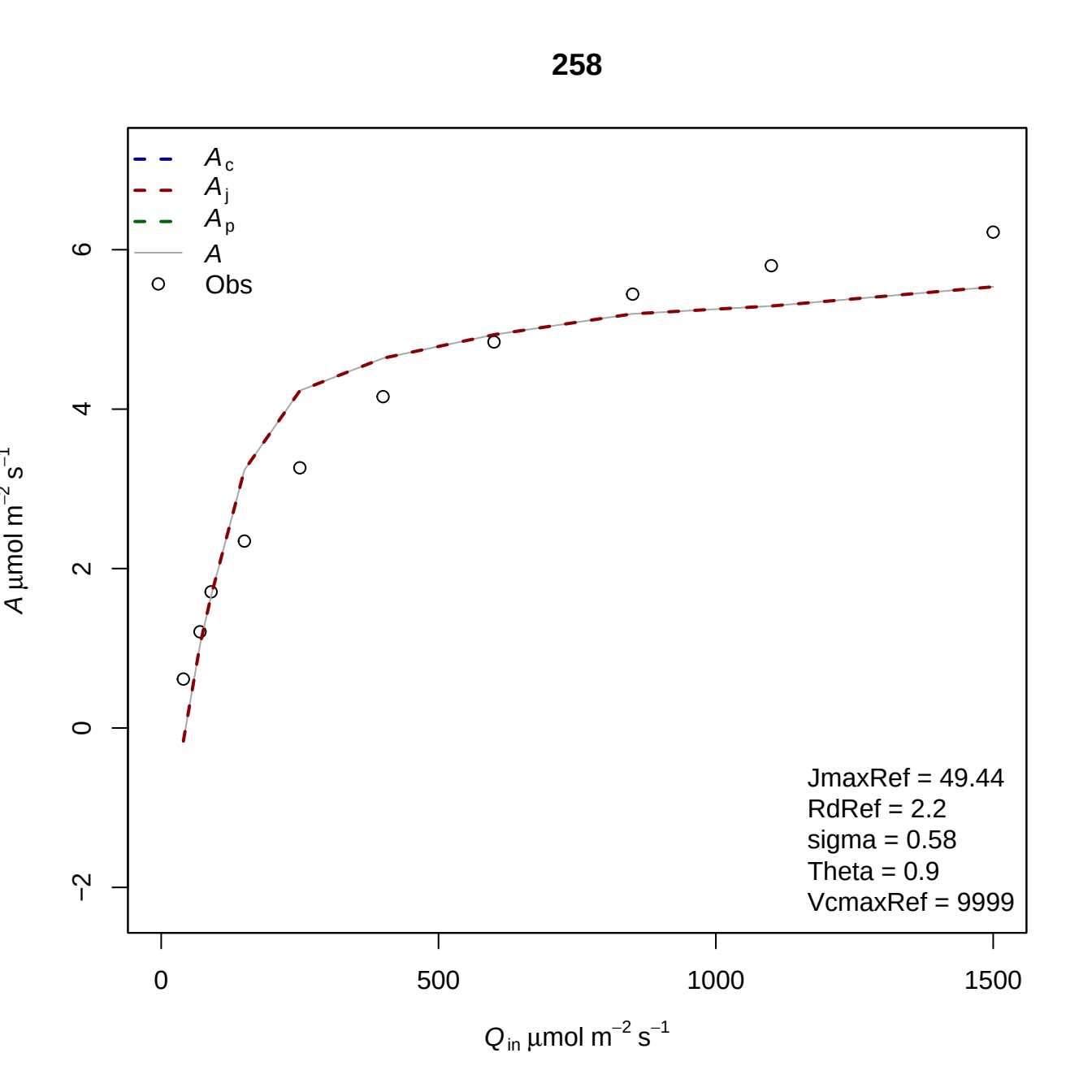

253
A_c
A_j
A_p
A
Obs
JmaxRef = 39.43
RdRef = 1.49
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

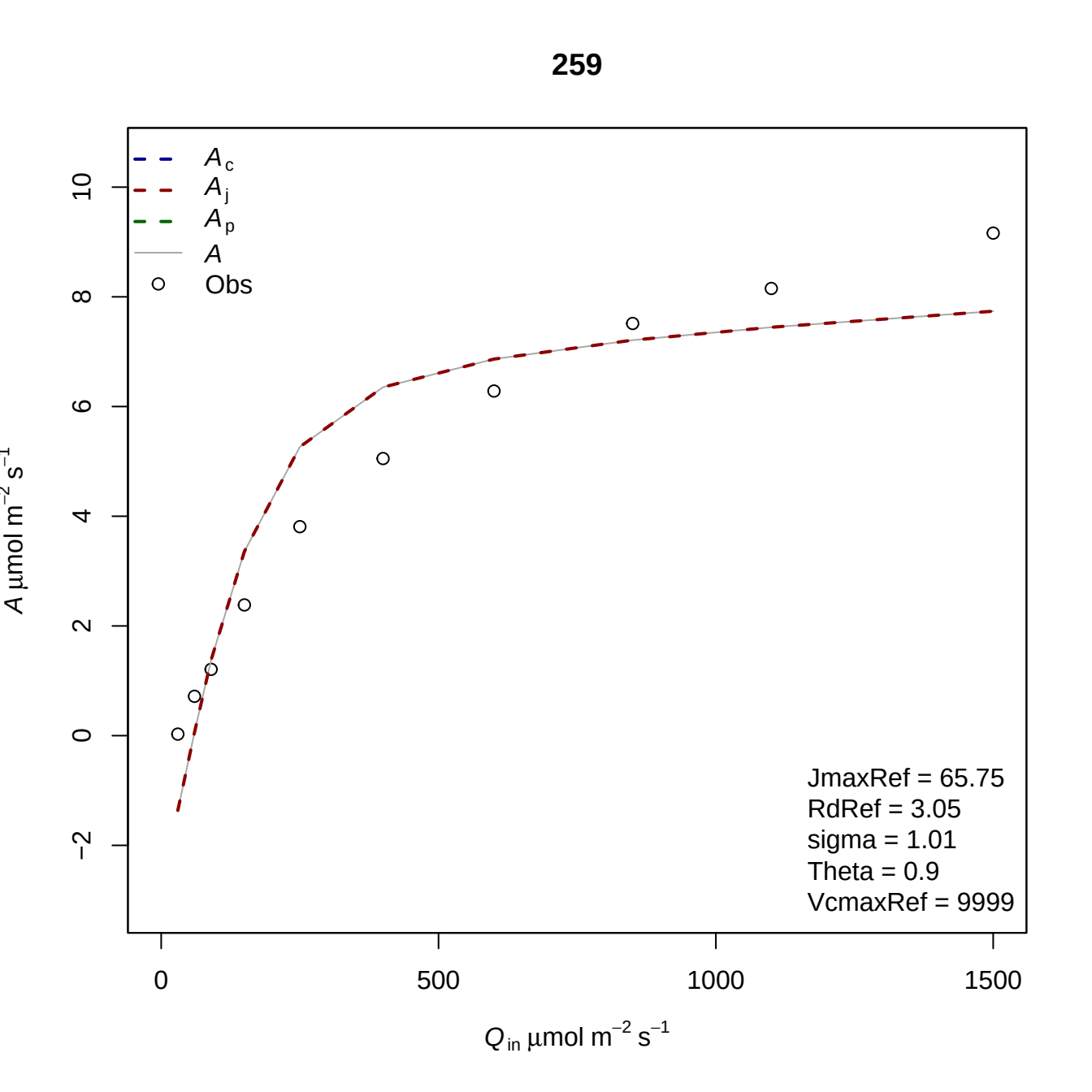

254
A_c
A_j
A_p
A
Obs
JmaxRef = 11.21
RdRef = 1.41
sigma = 0.21
Theta = 0.9
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

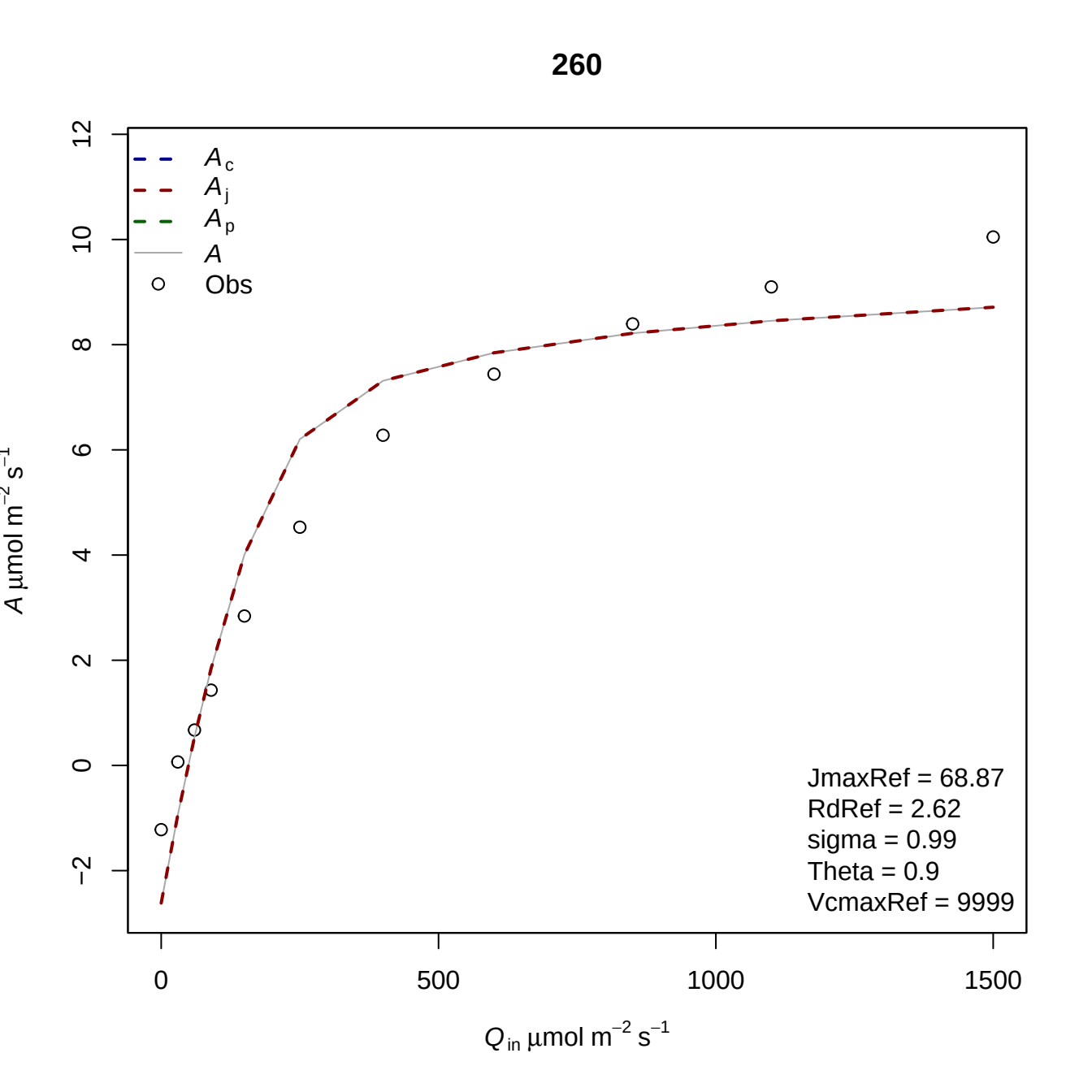

255
A_c
A_j
A_p
A
Obs
JmaxRef = 47.28
RdRef = 3.75
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

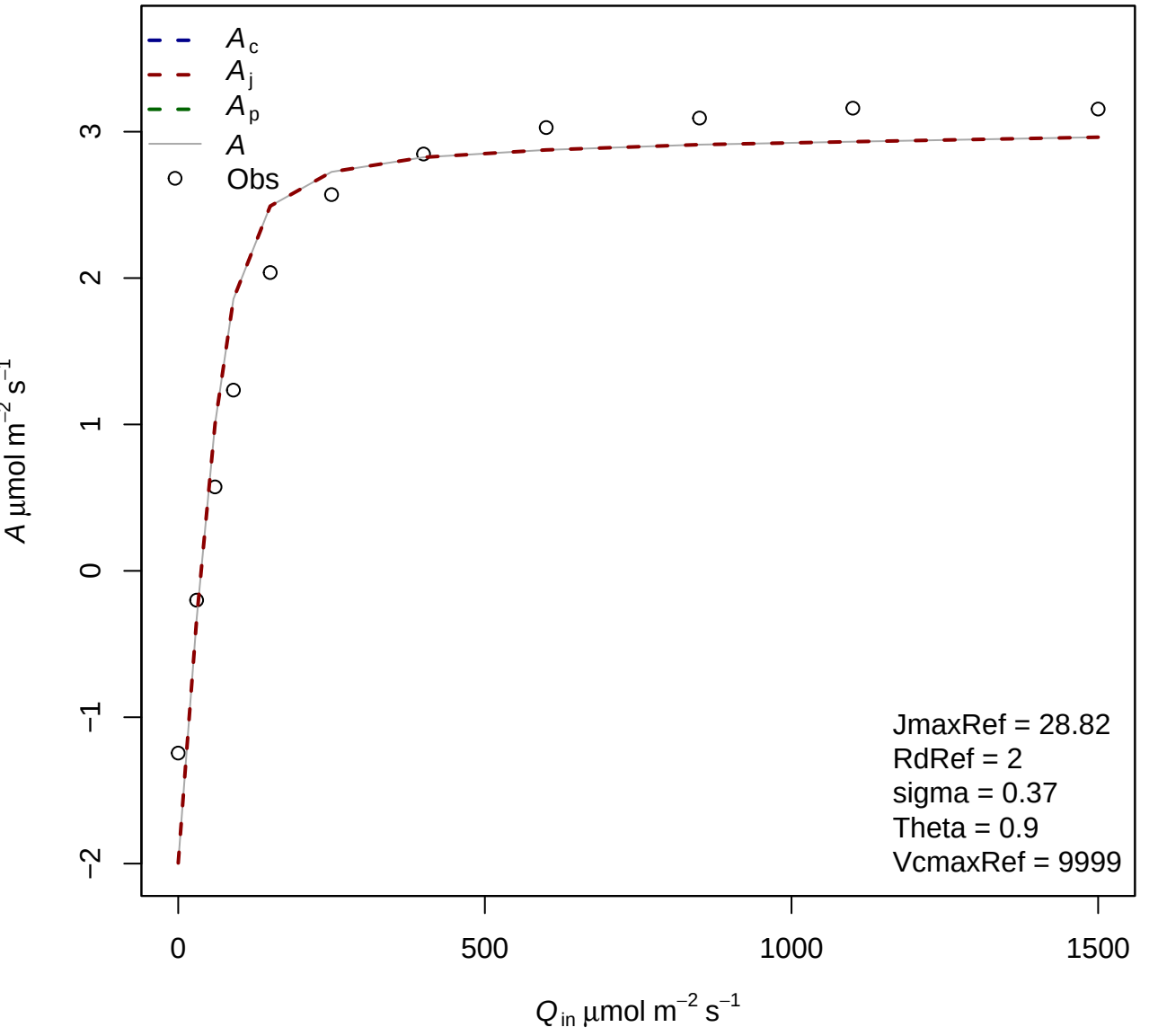

256
A_c
A_j
A_p
A
Obs
JmaxRef = 66.28
RdRef = 2.23
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

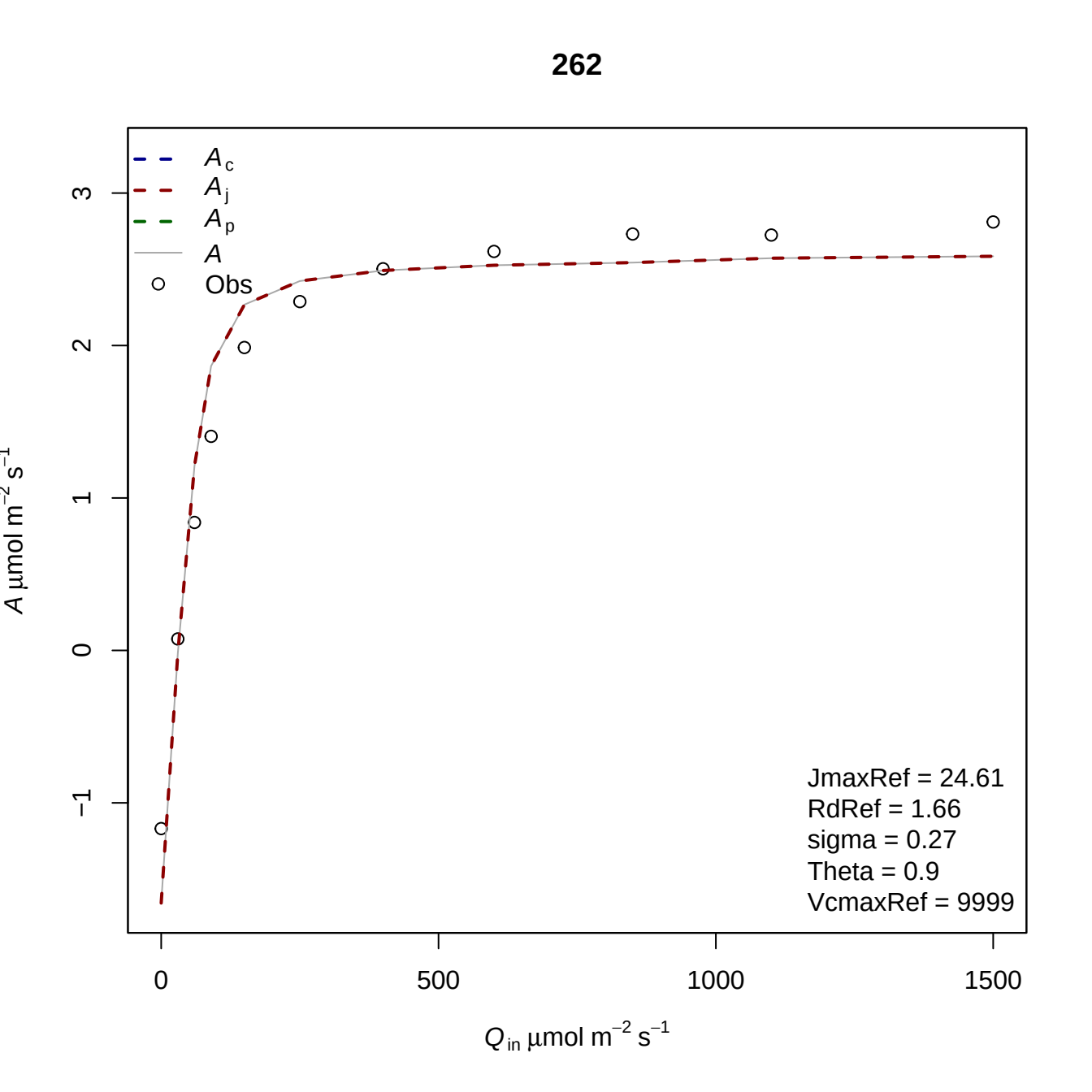

257
A_c
A_j
A_p
A
Obs
JmaxRef = 28.24
RdRef = 3.63
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500

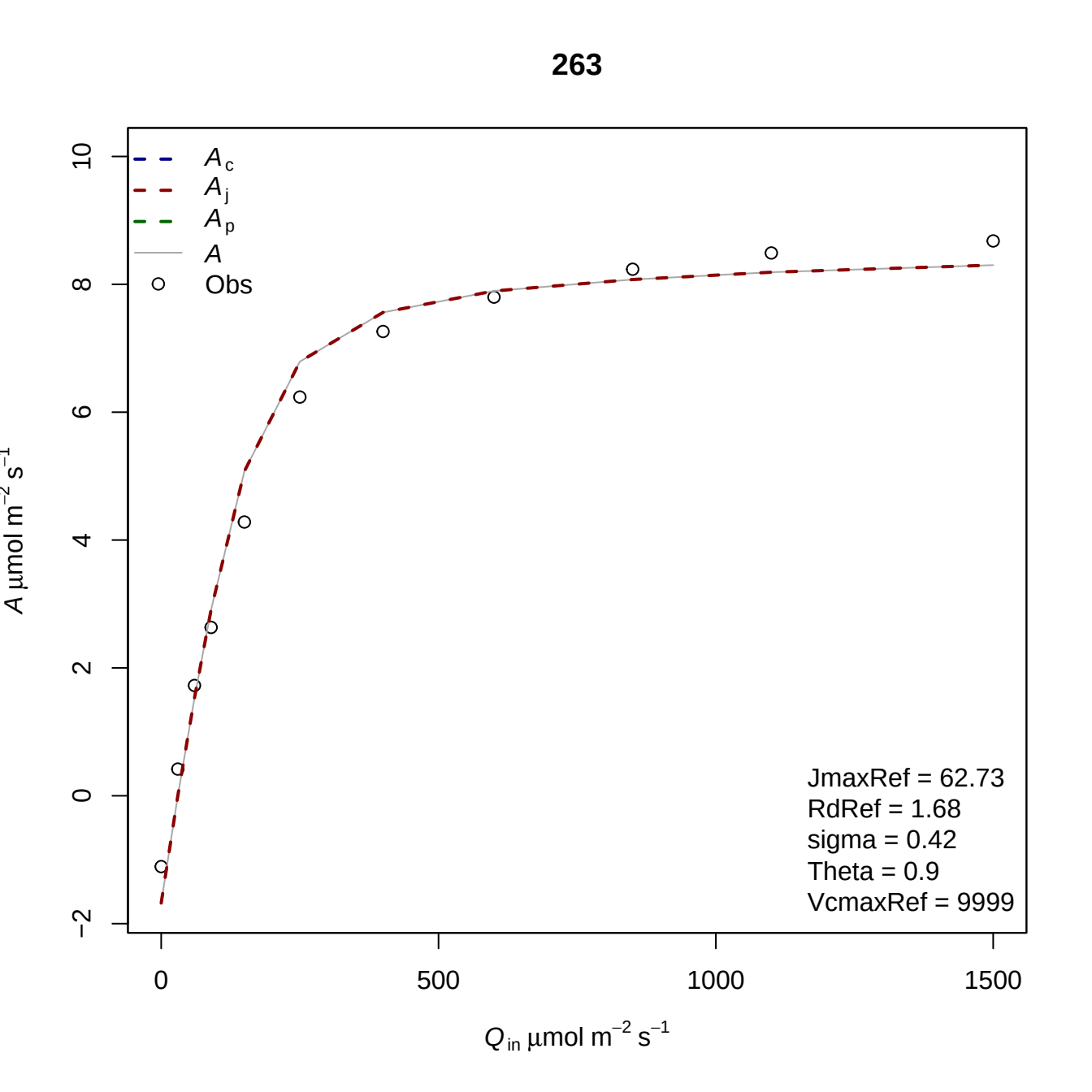

258
A_c
A_j
A_p
A
Obs
JmaxRef = 49.44
RdRef = 2.2
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

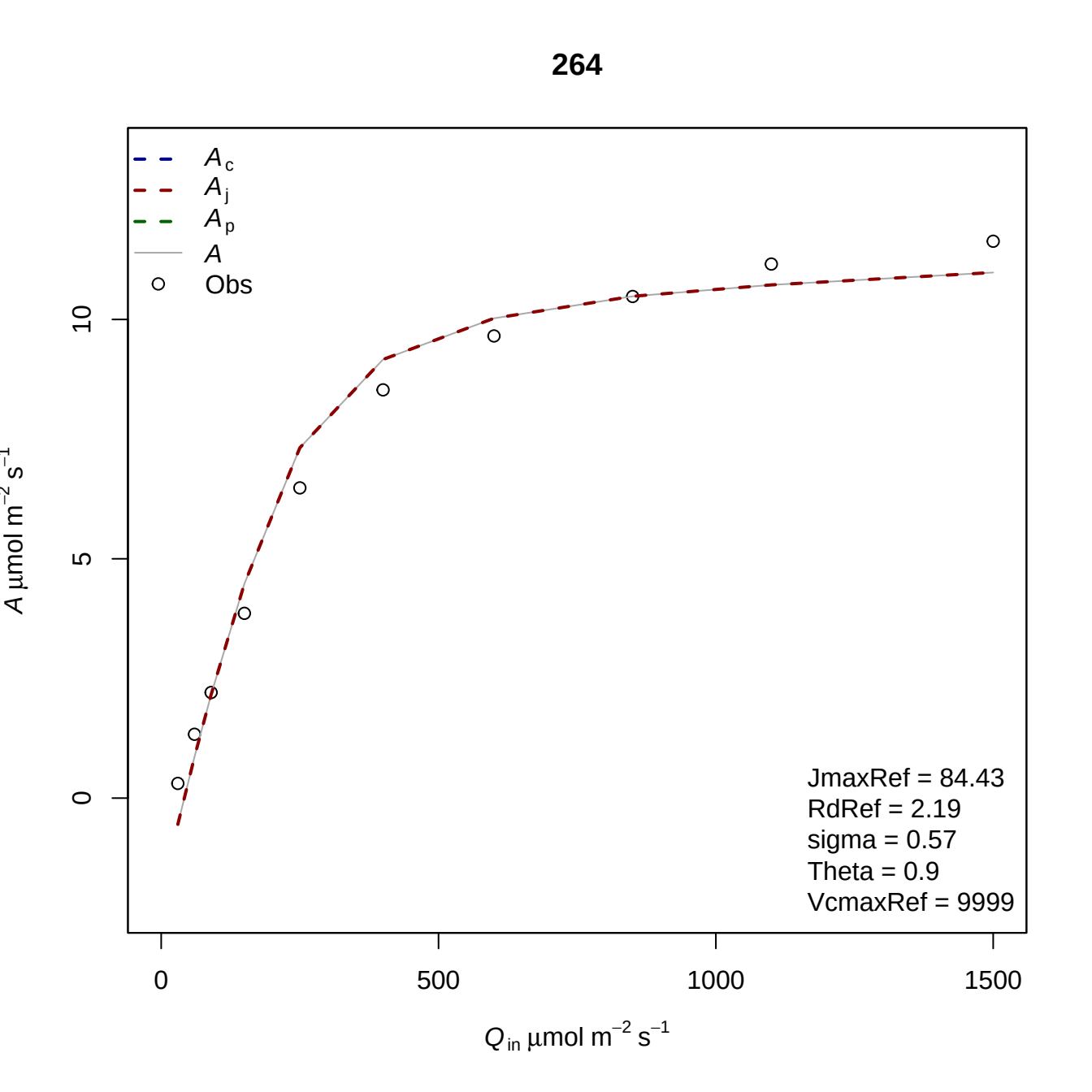

259
A_c
A_j
A_p
A
Obs
JmaxRef = 65.75
RdRef = 3.05
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

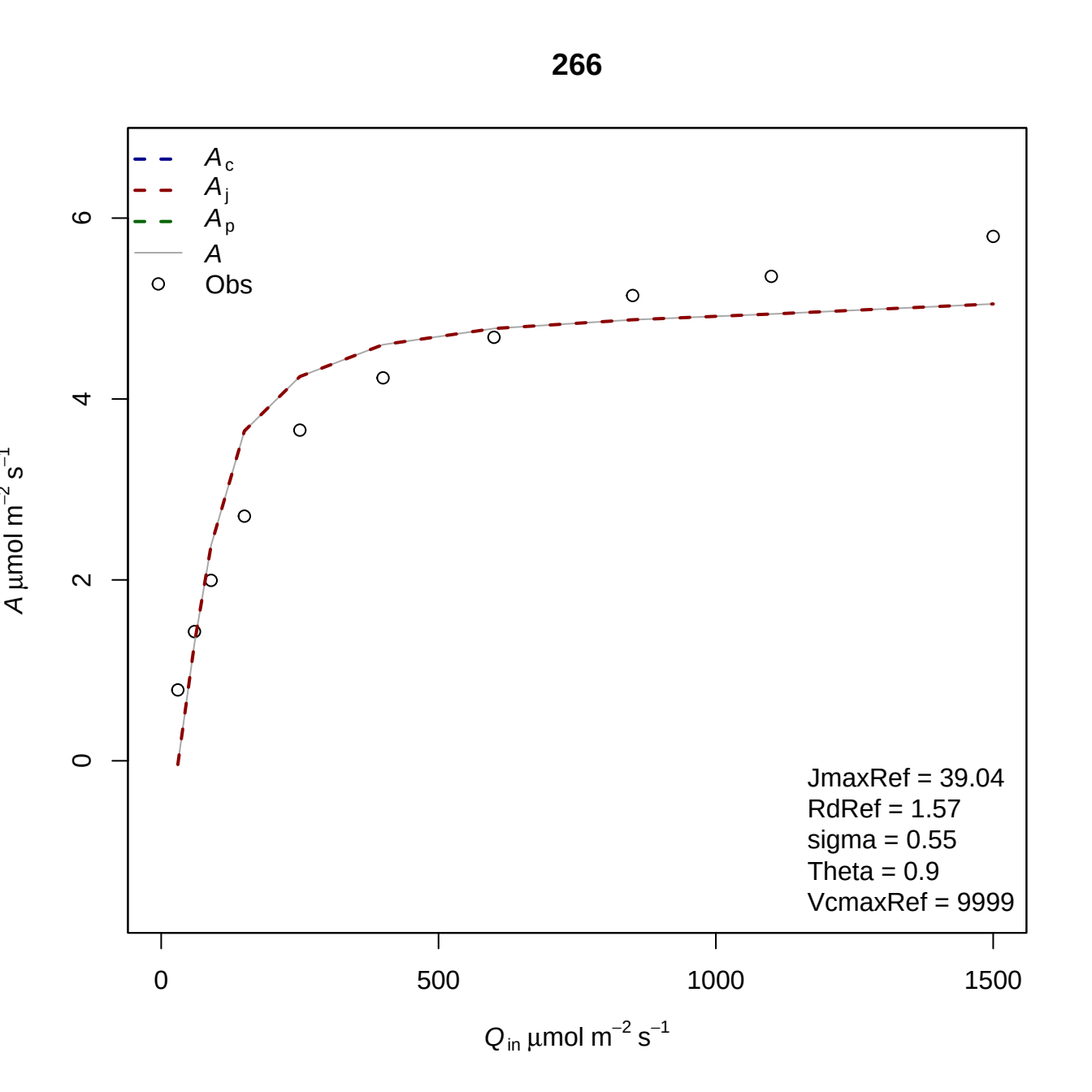

260
A_c
A_j
A_p
A
Obs
JmaxRef = 68.87
RdRef = 2.62
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

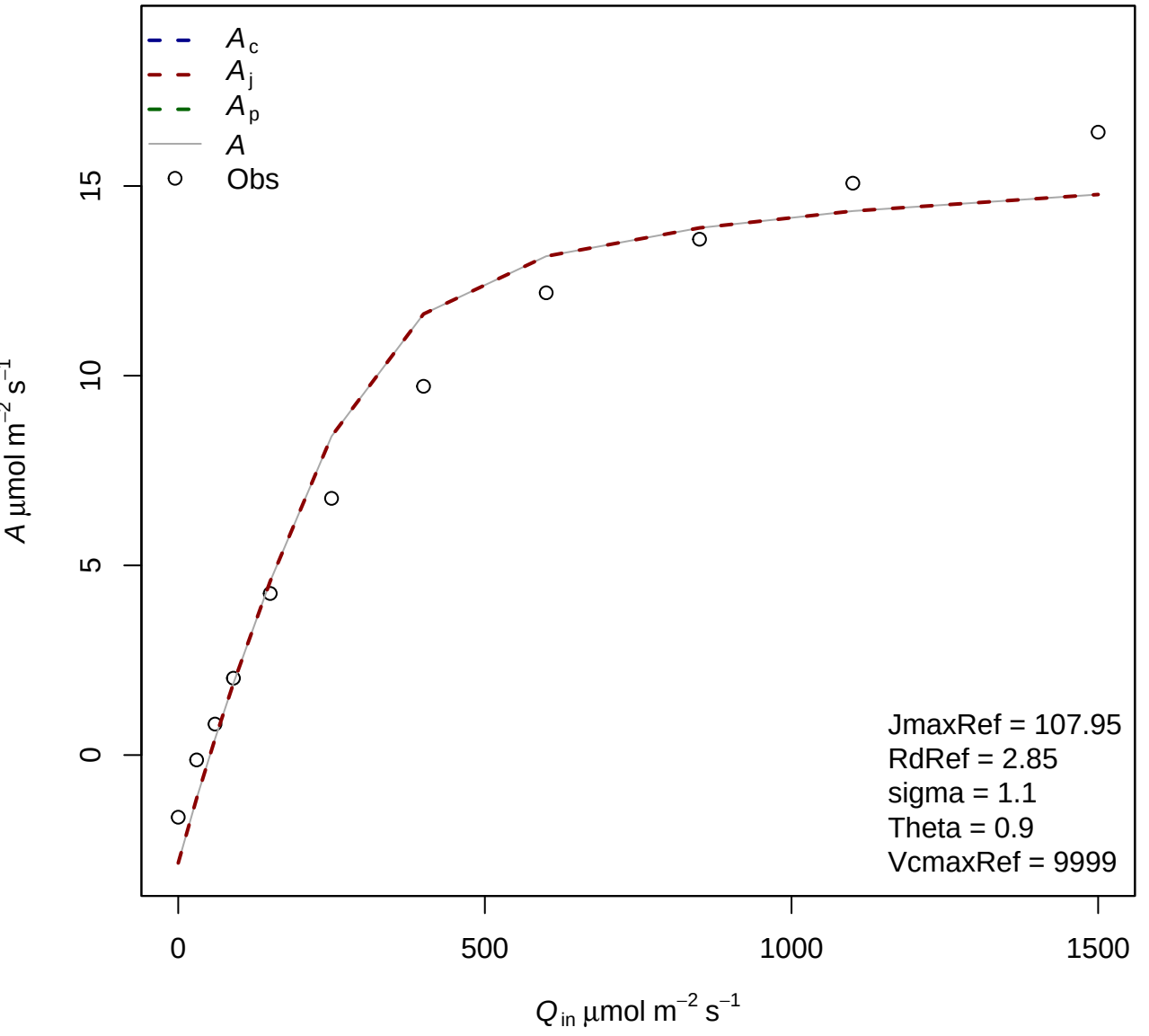

A_c
A_j
A_p
A
Obs
JmaxRef = 28.82
RdRef = 2
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
500
1000
1500

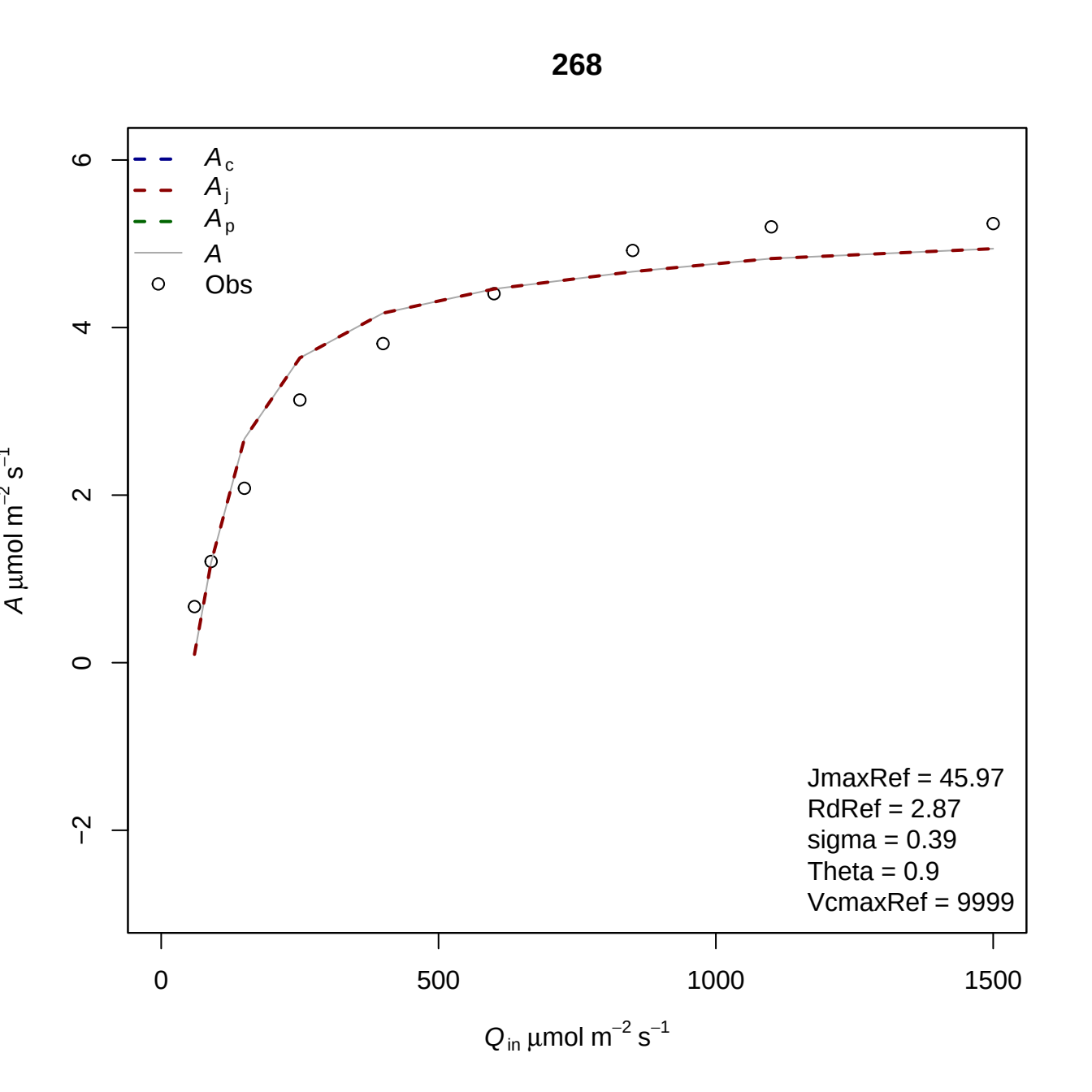

262
A_c
A_j
A_p
A
Obs
JmaxRef = 24.61
RdRef = 1.66
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

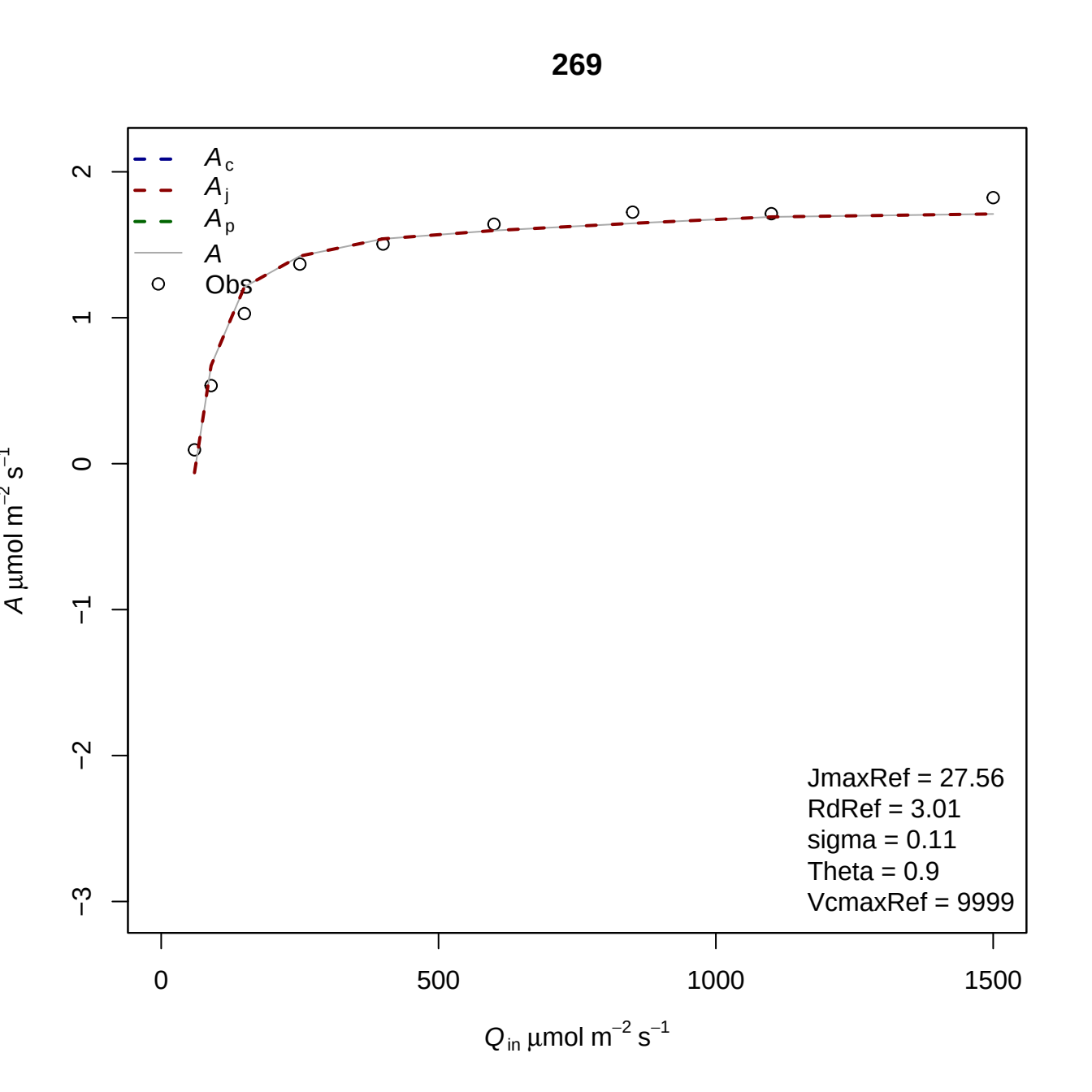

263
A_c
A_j
A_p
A
Obs
JmaxRef = 62.73
RdRef = 1.68
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

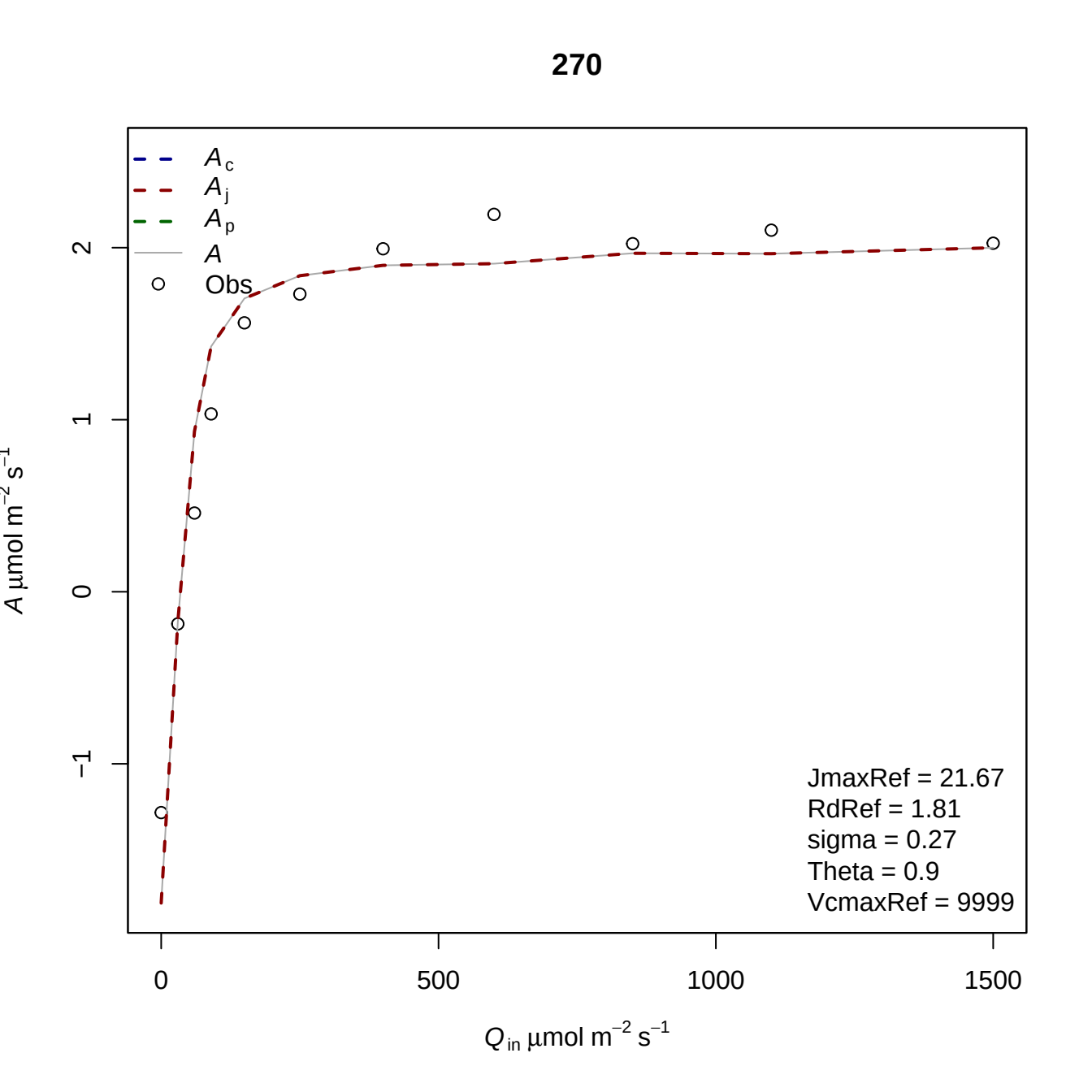

264
A_c
A_j
A_p
A
Obs
JmaxRef = 84.43
RdRef = 2.19
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

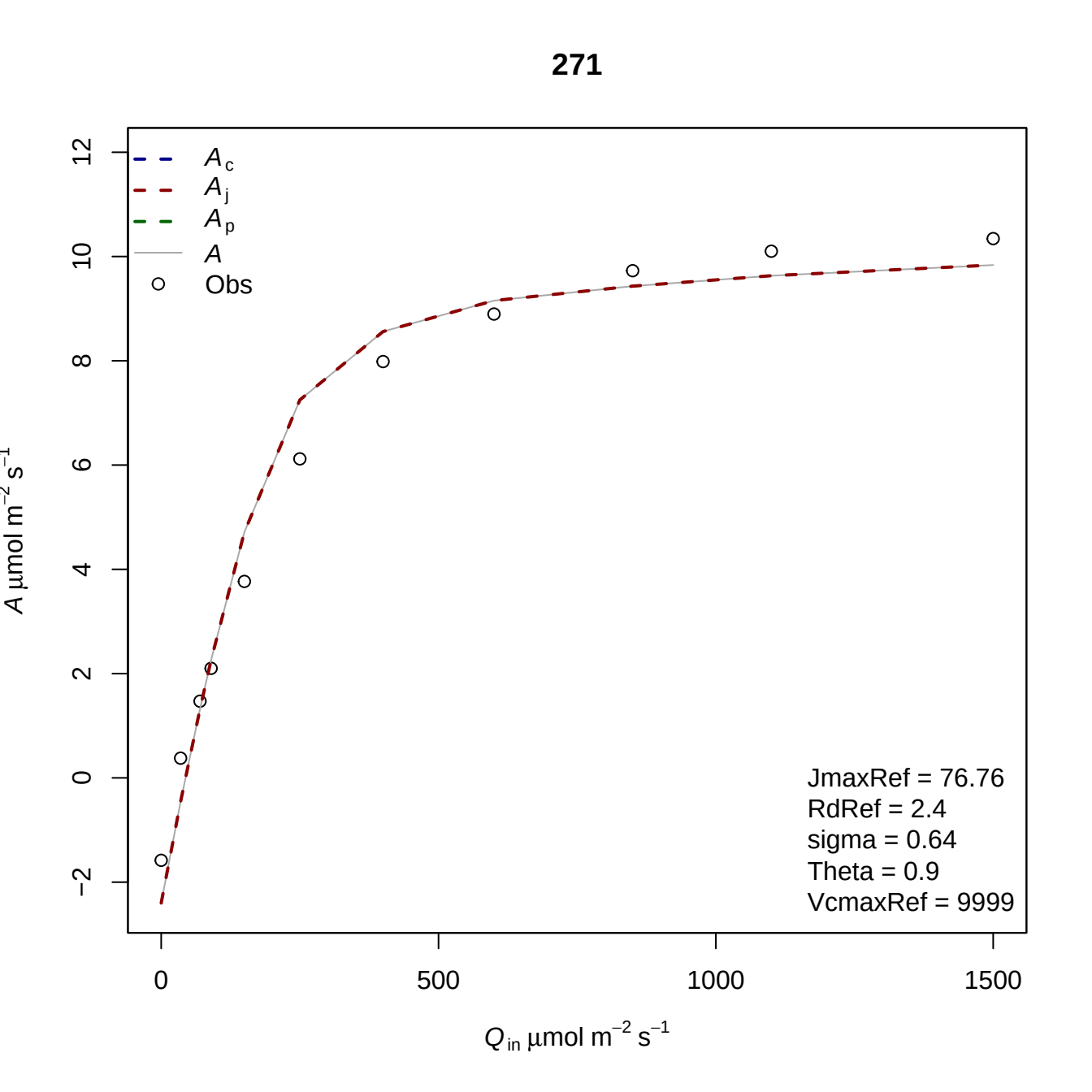

266
A_c
A_j
A_p
A
Obs
JmaxRef = 39.04
RdRef = 1.57
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

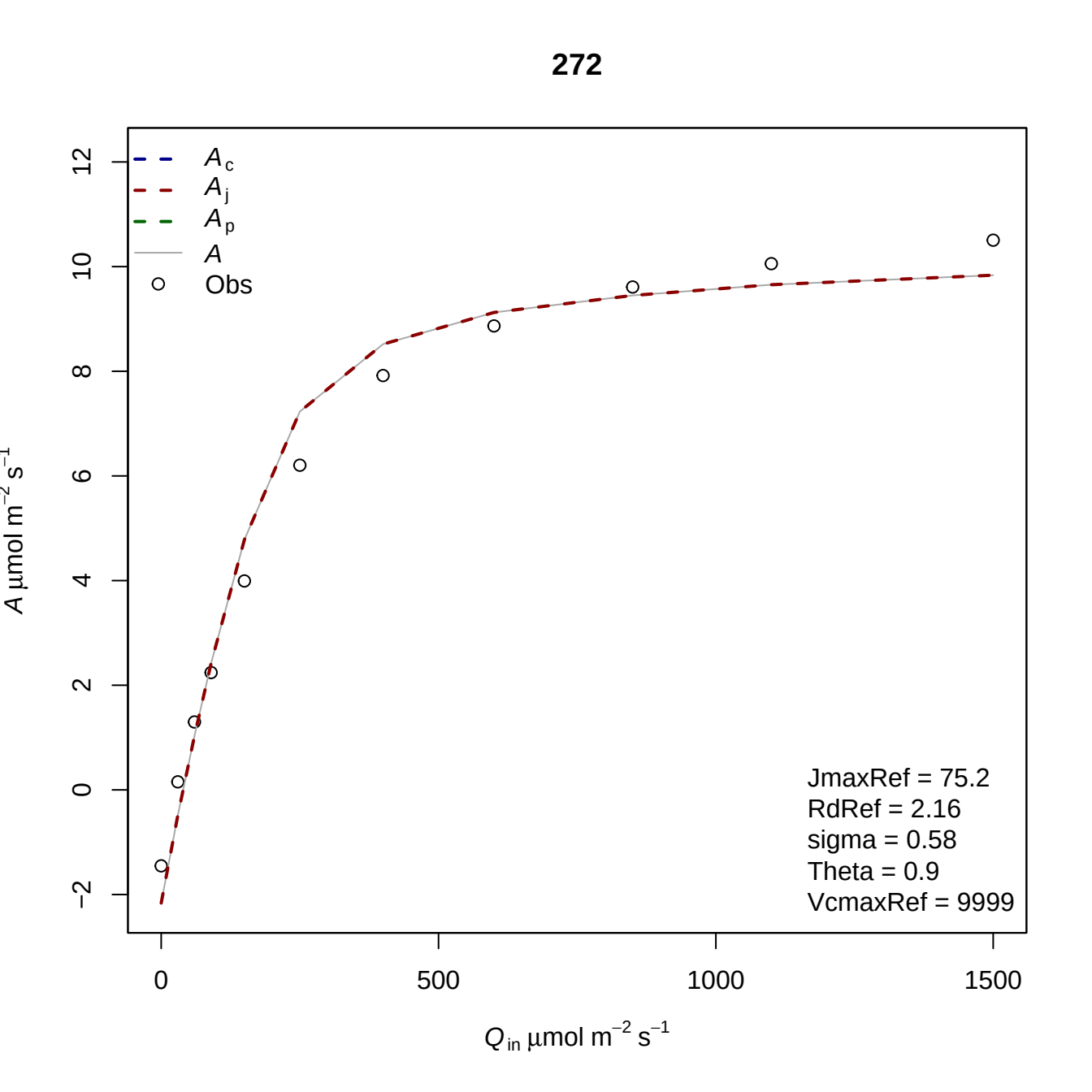

A_c
A_j
A_p
A
Obs
JmaxRef = 107.95
RdRef = 2.85
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

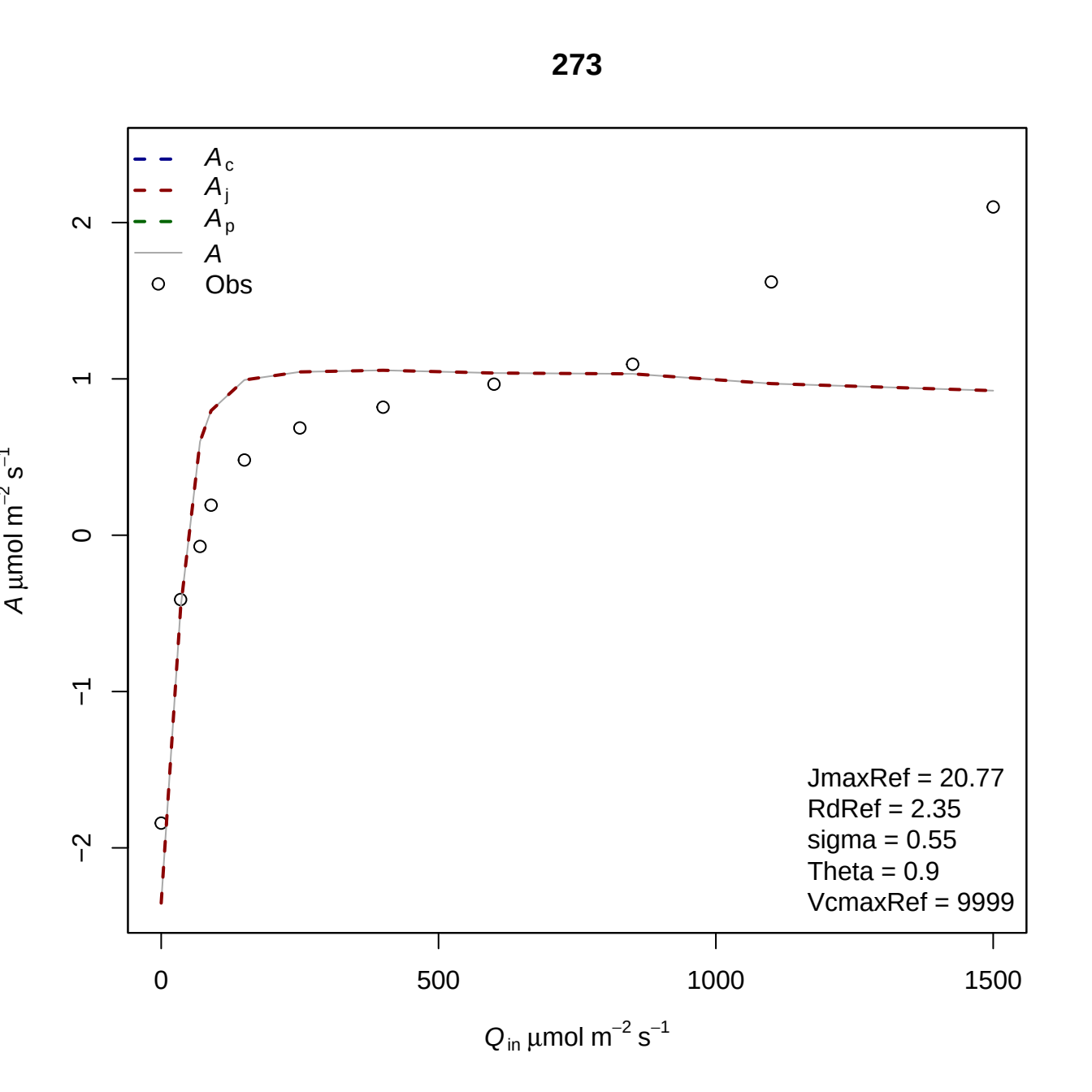

268
A_c
A_j
A_p
A
Obs
JmaxRef = 45.97
RdRef = 2.87
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

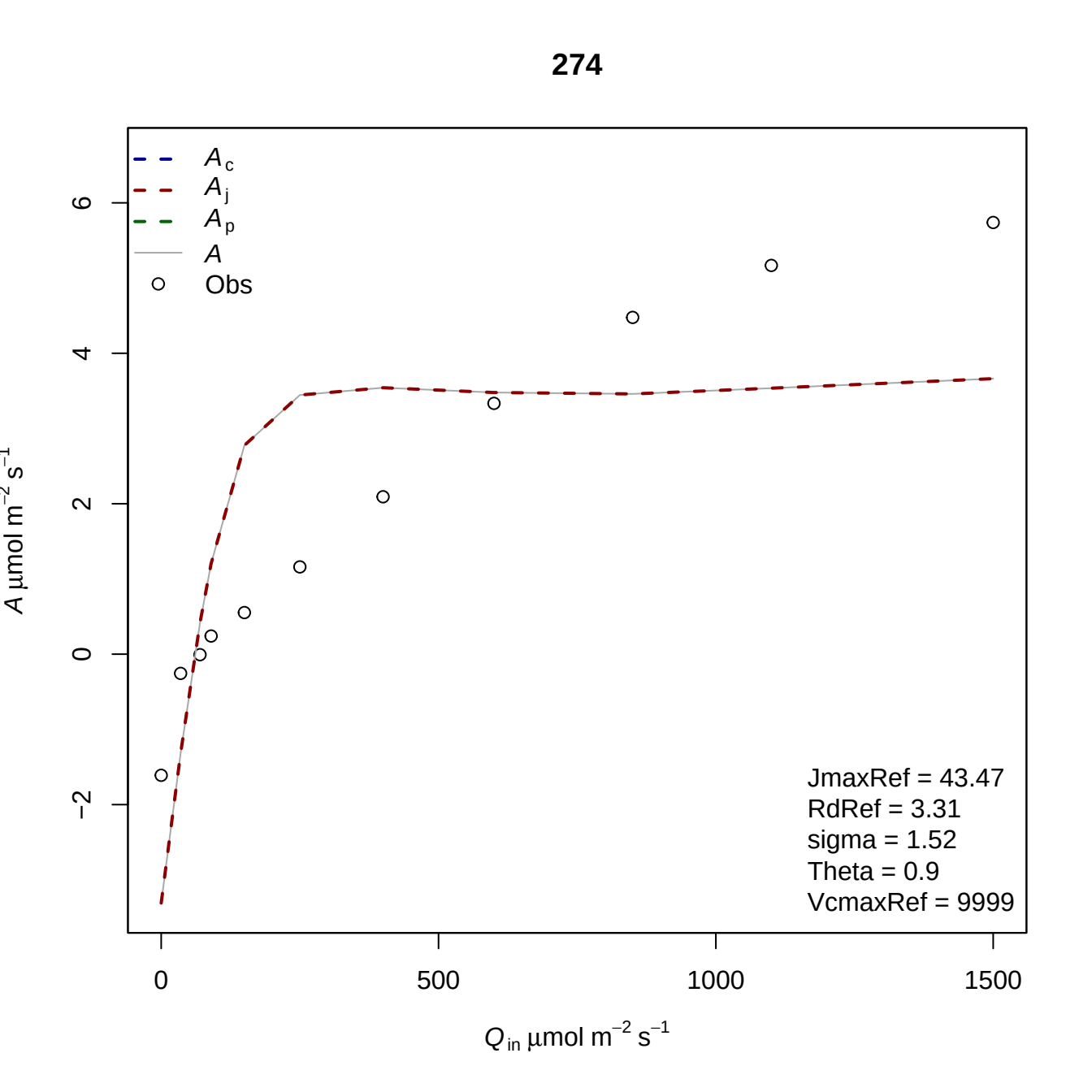

269
A_c
A_j
A_p
A
Obs
JmaxRef = 27.56
RdRef = 3.01
sigma = 0.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500

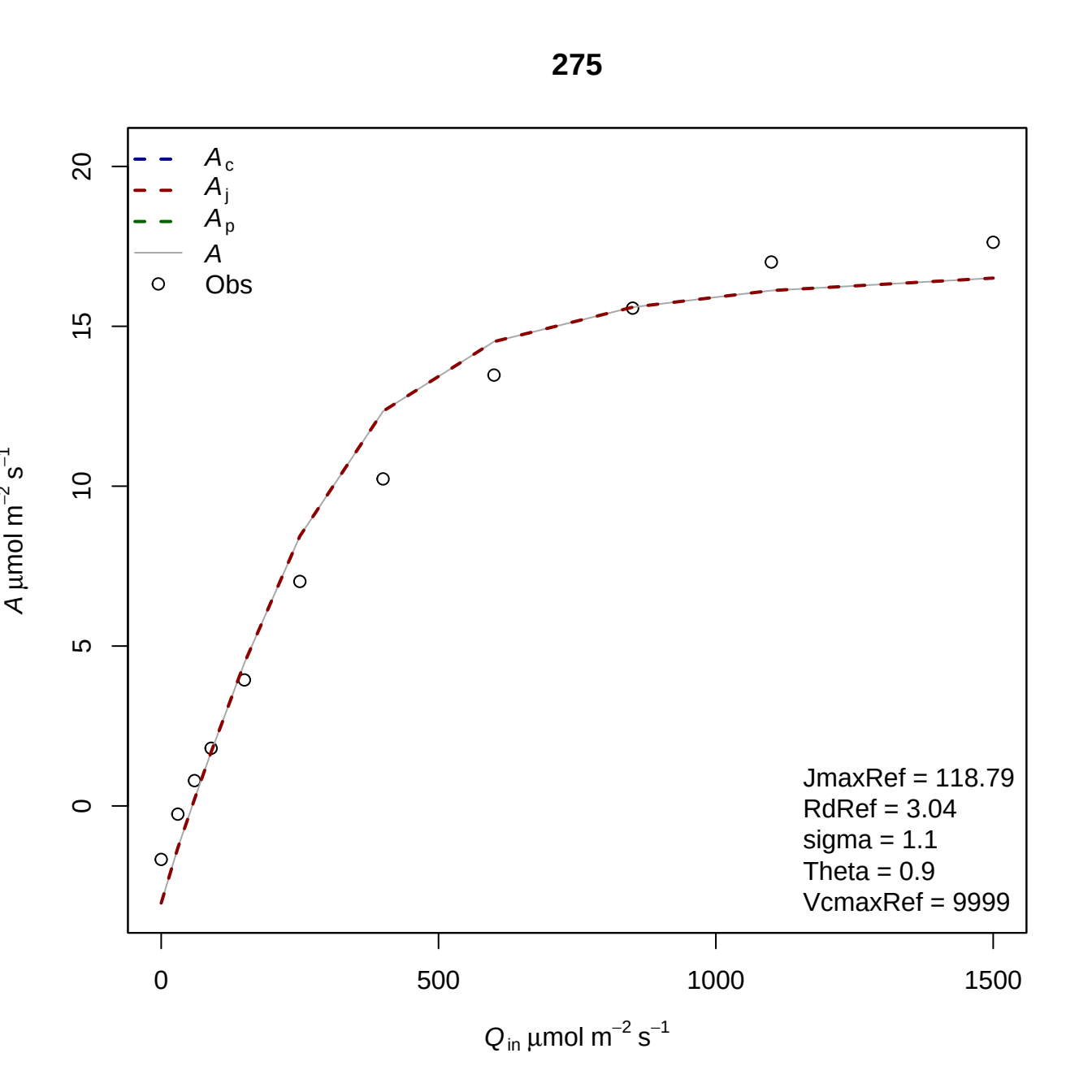

270
A_c
A_j
A_p
A
Obs
JmaxRef = 21.67
RdRef = 1.81
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
0
500
1000
1500

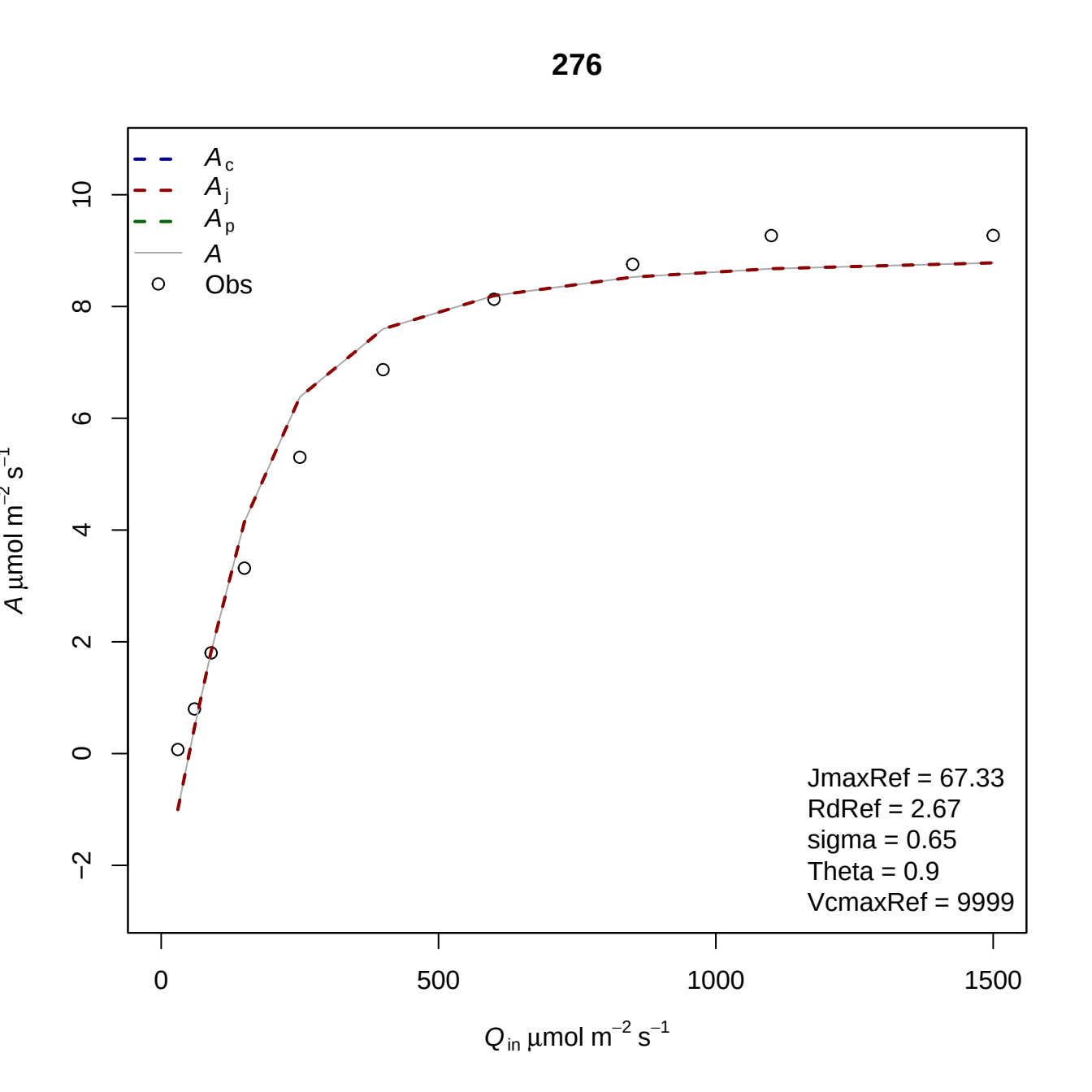

271
A_c
A_j
A_p
A
Obs
JmaxRef = 76.76
RdRef = 2.4
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

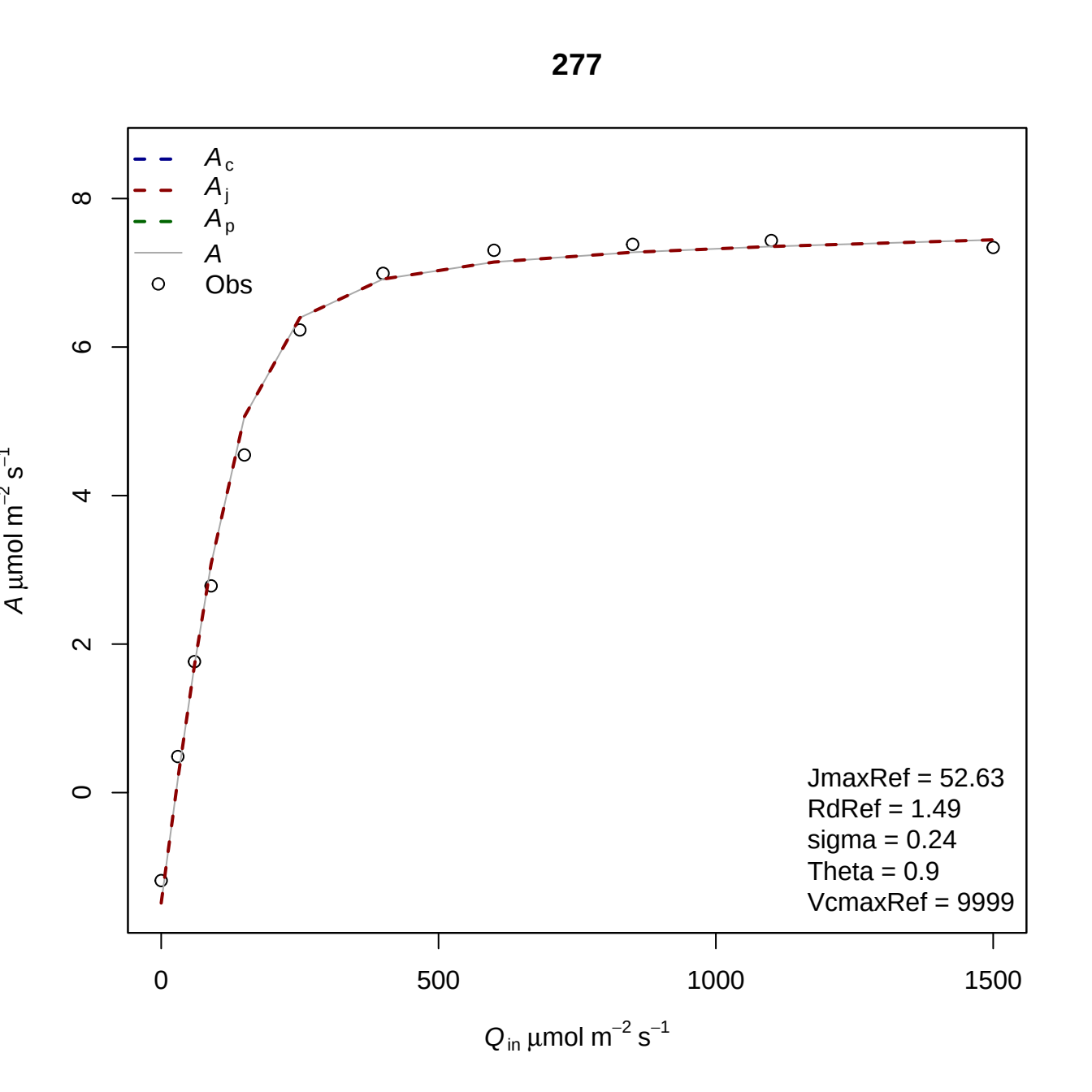

272
A_c
A_j
A_p
A
Obs
JmaxRef = 75.2
RdRef = 2.16
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

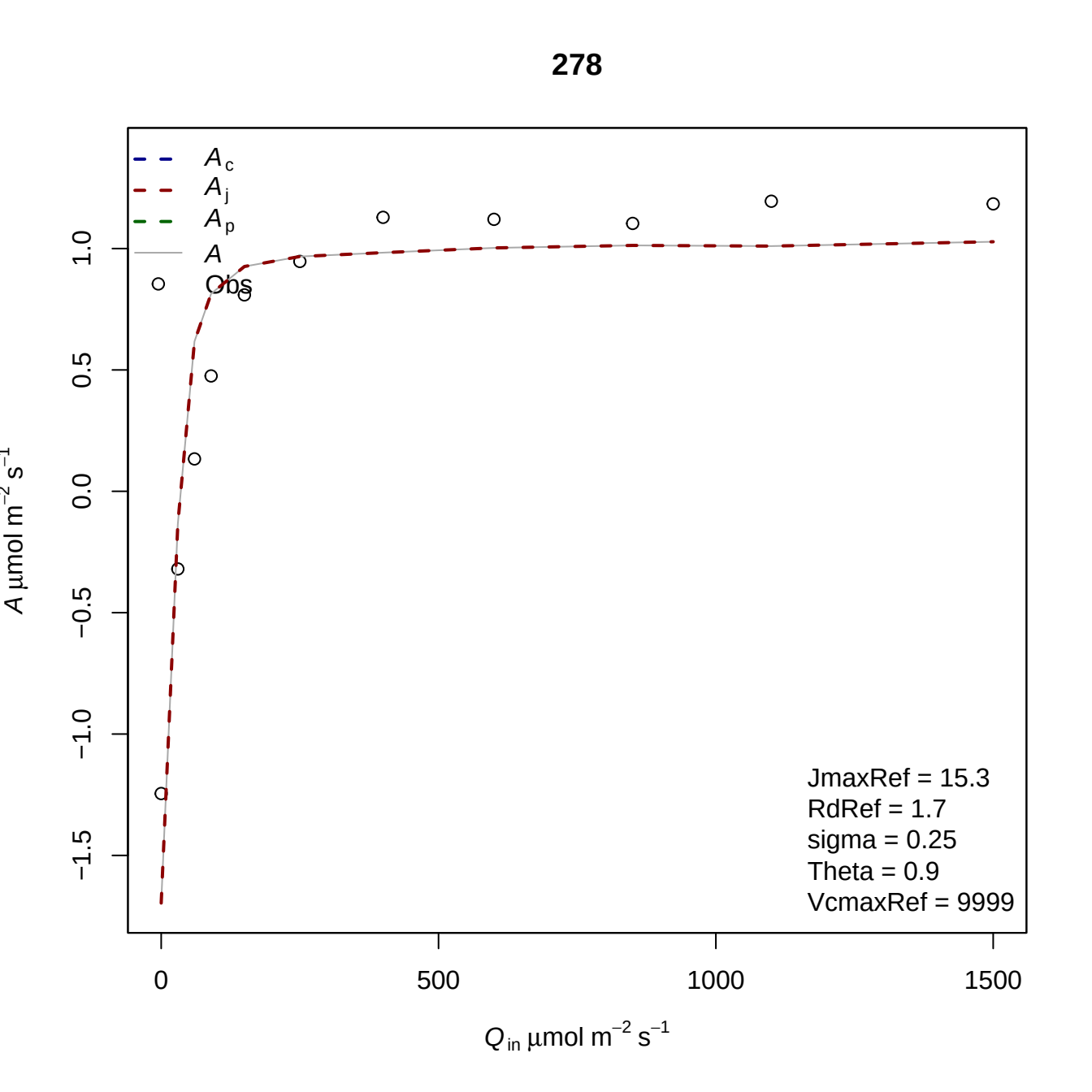

273
A_c
A_j
A_p
A
Obs
JmaxRef = 20.77
RdRef = 2.35
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

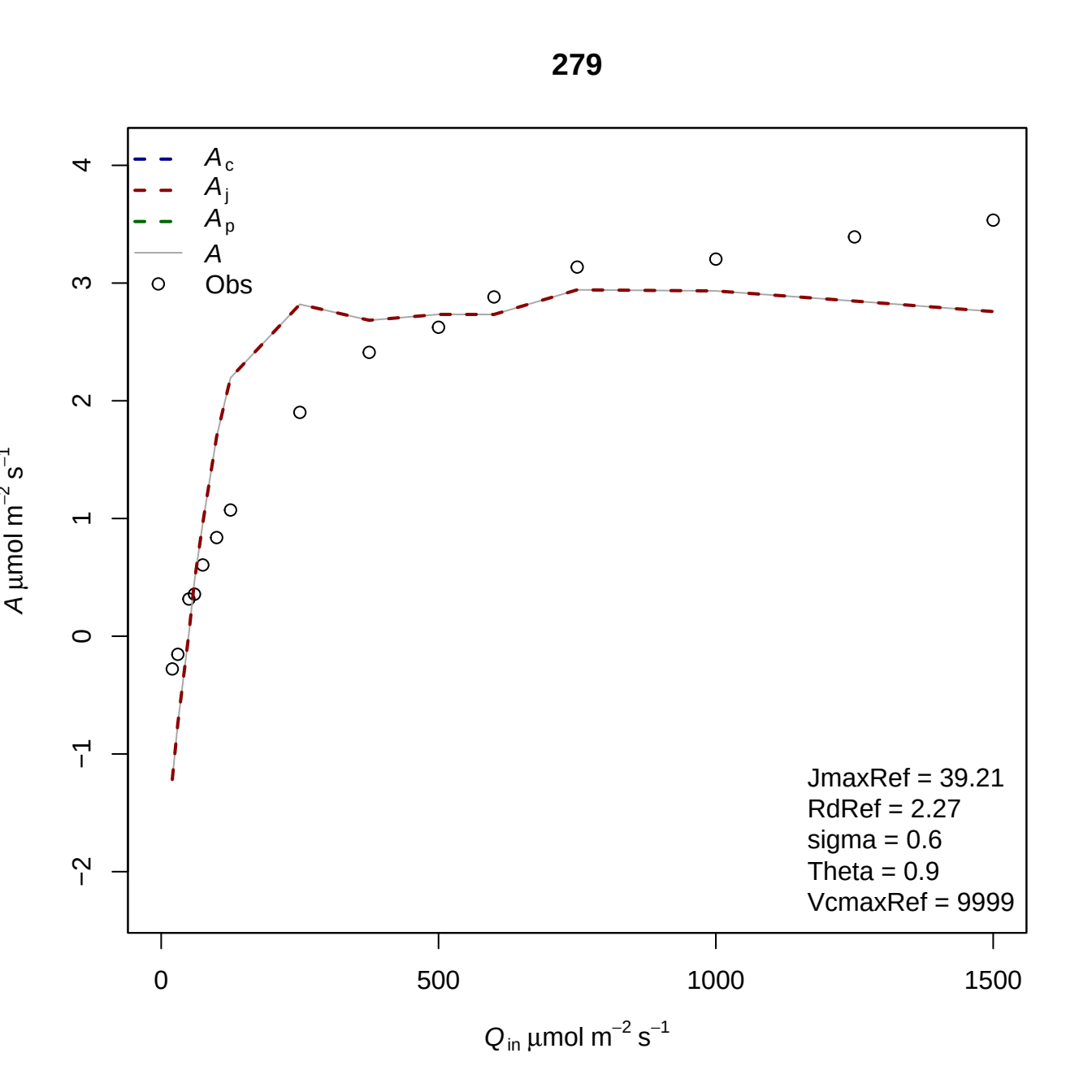

274
A_c
A_j
A_p
A
Obs
JmaxRef = 43.47
RdRef = 3.31
sigma = 1.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

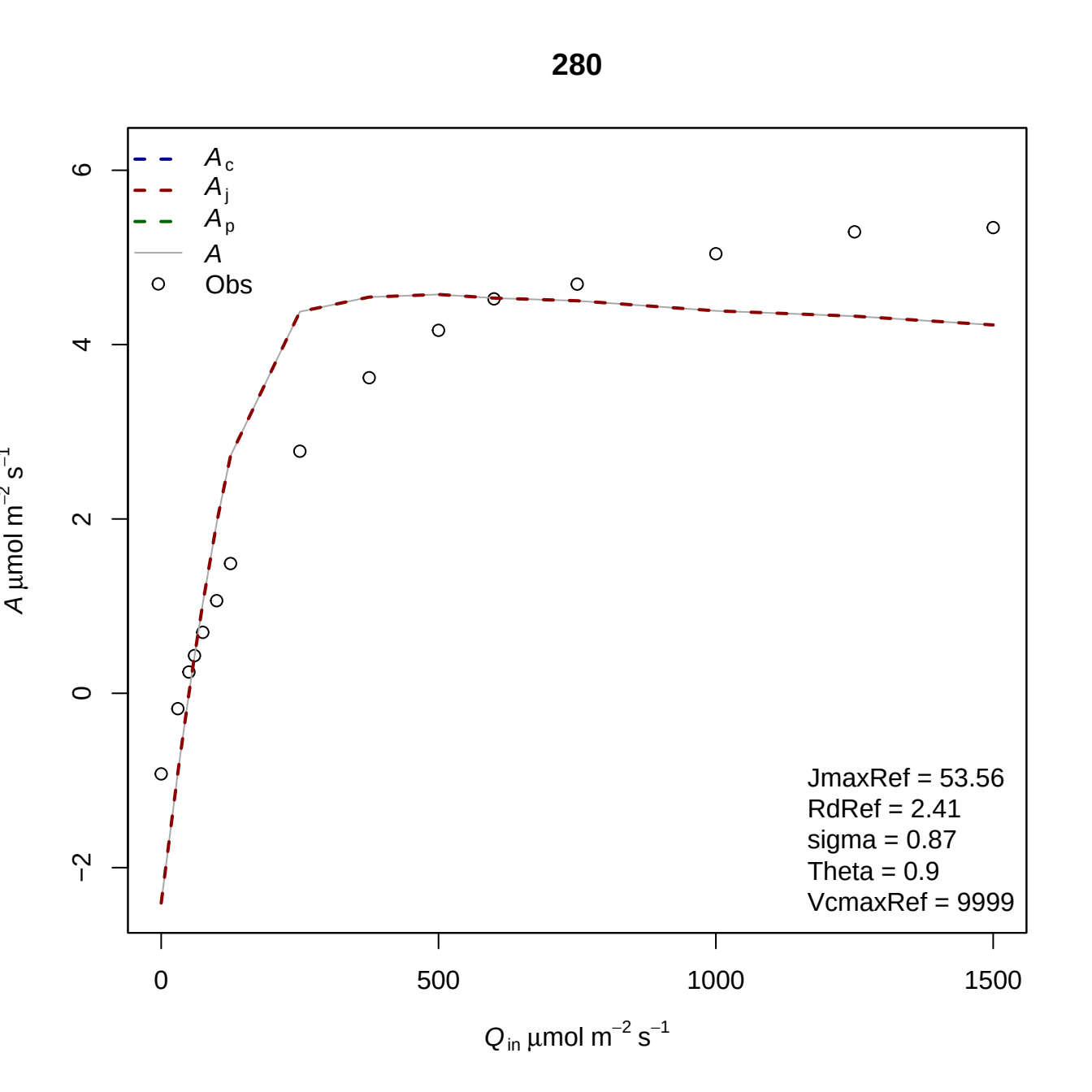

275
A_c
A_j
A_p
A
Obs
JmaxRef = 118.79
RdRef = 3.04
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

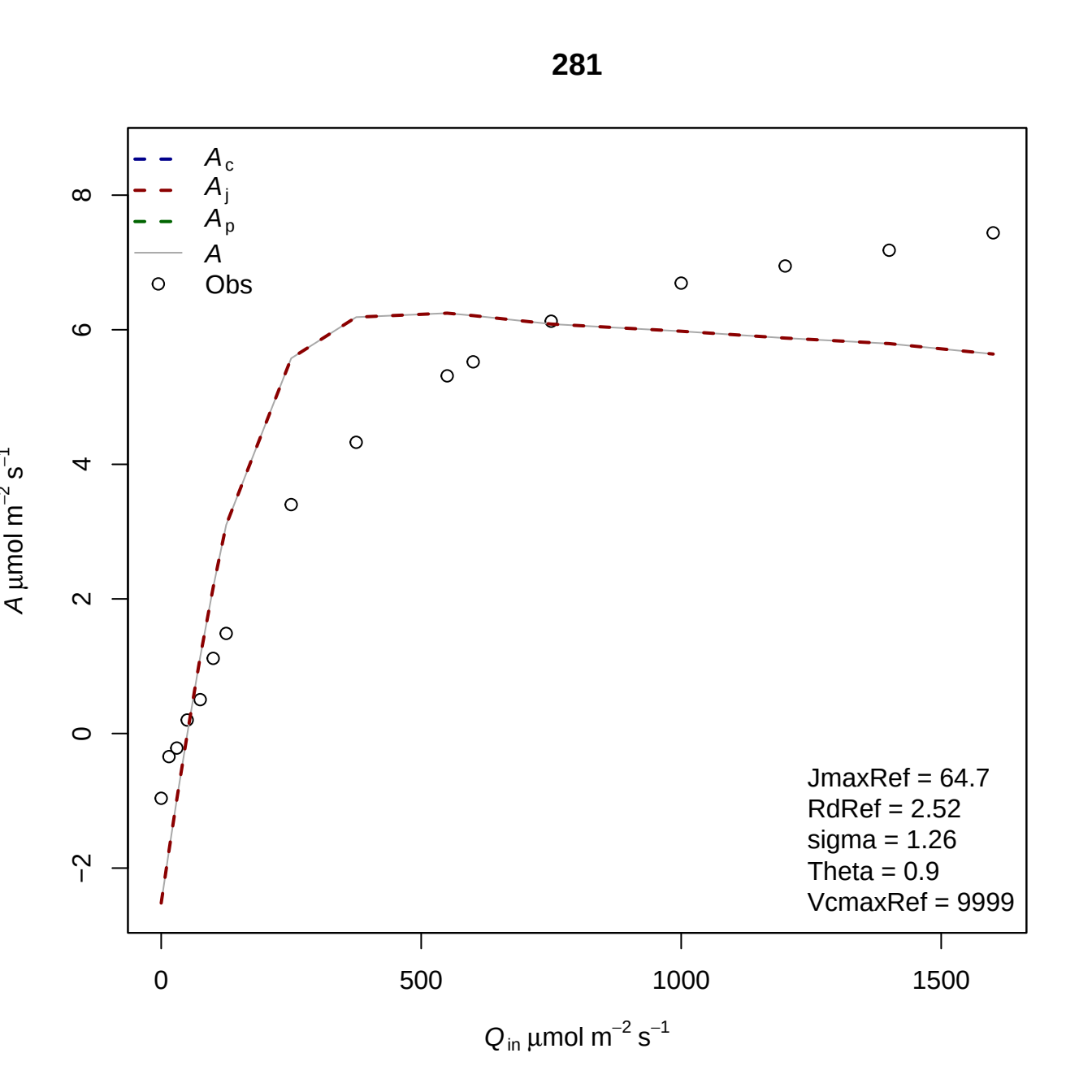

276
A_c
A_j
A_p
A
Obs
JmaxRef = 67.33
RdRef = 2.67
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

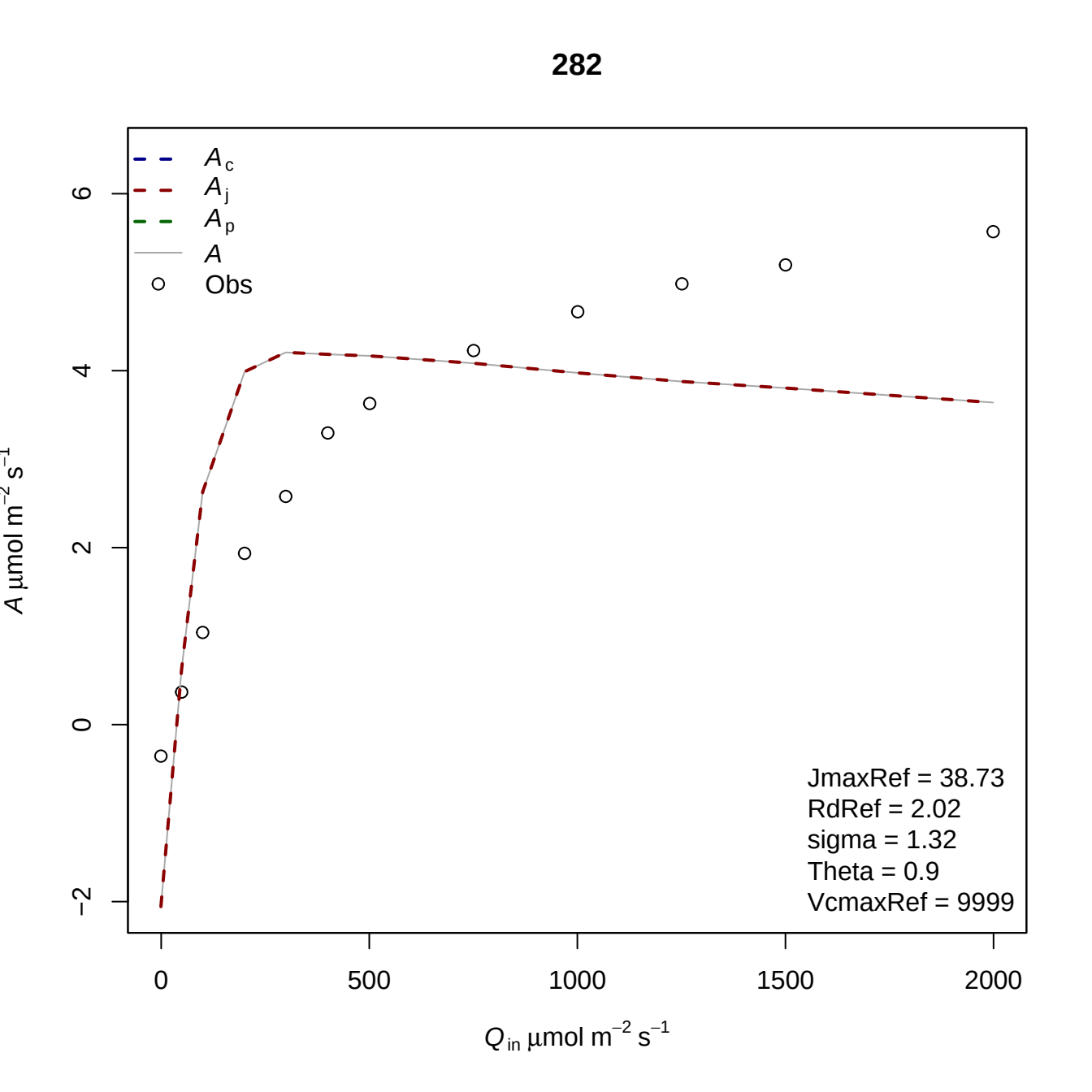

277
A_c
A_j
A_p
A
Obs
JmaxRef = 52.63
RdRef = 1.49
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

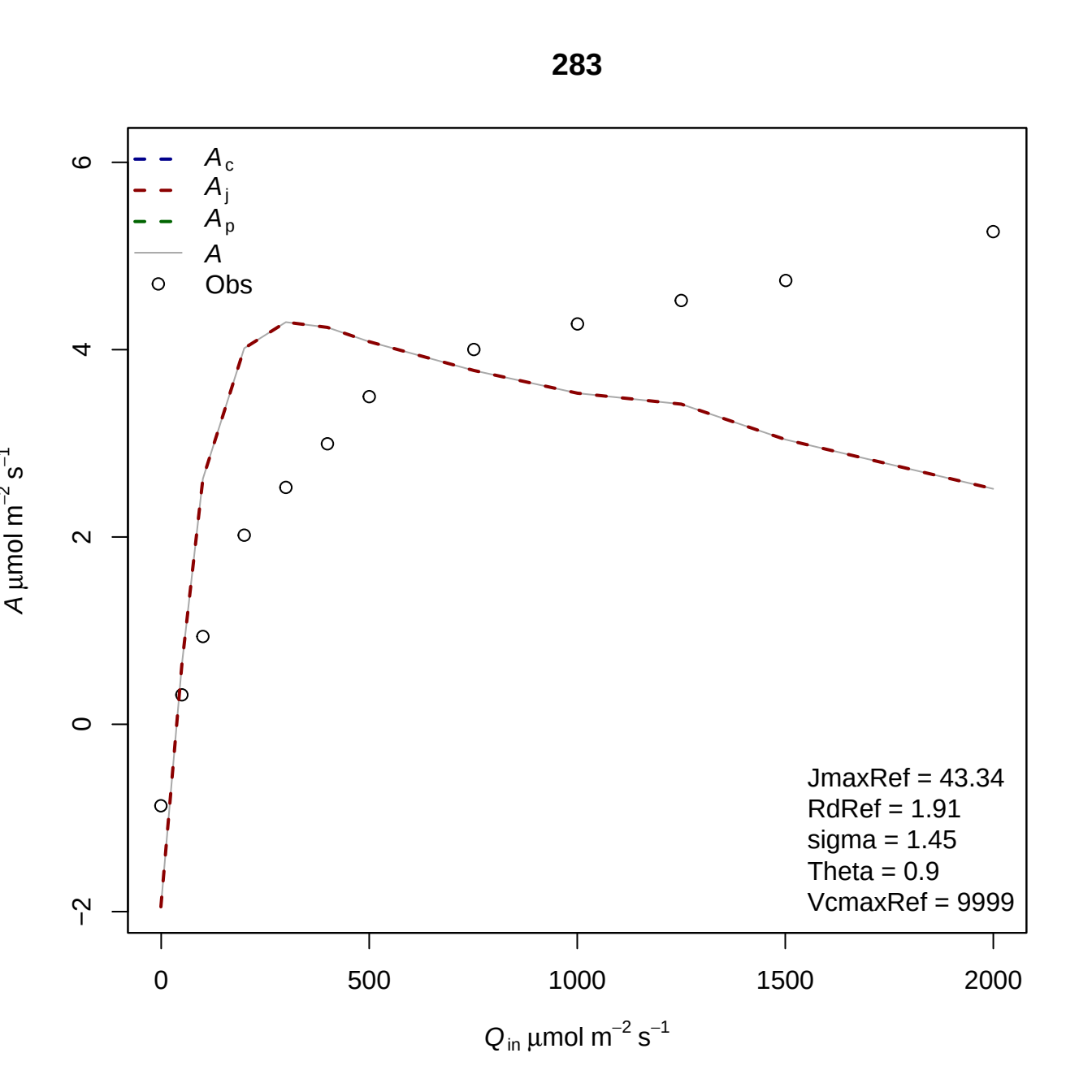

278
A_c
A_j
A_p
A
Obs
JmaxRef = 15.3
RdRef = 1.7
sigma = 0.25
Theta = 0.9
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

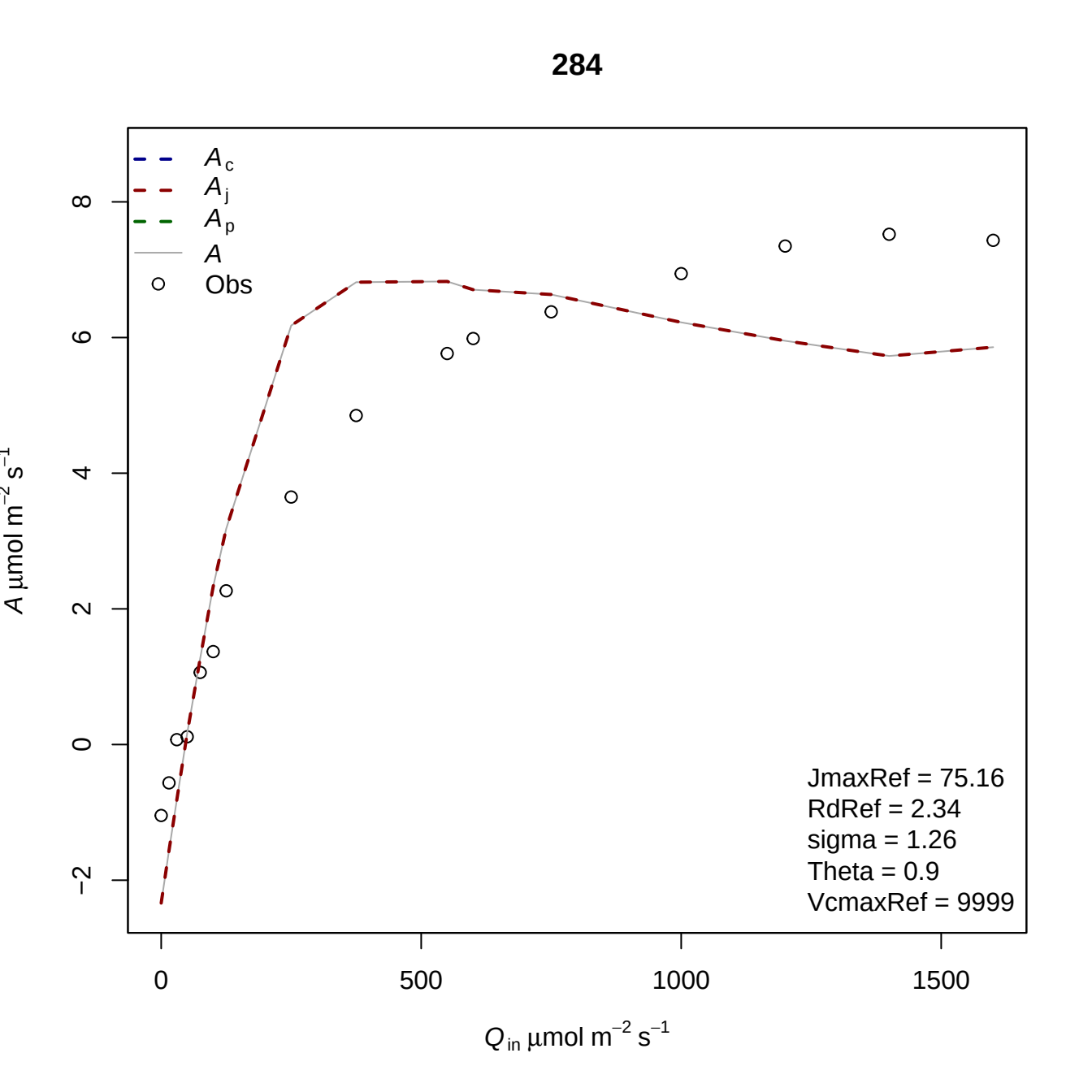

279
A_c
A_j
A_p
A
Obs
JmaxRef = 39.21
RdRef = 2.27
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

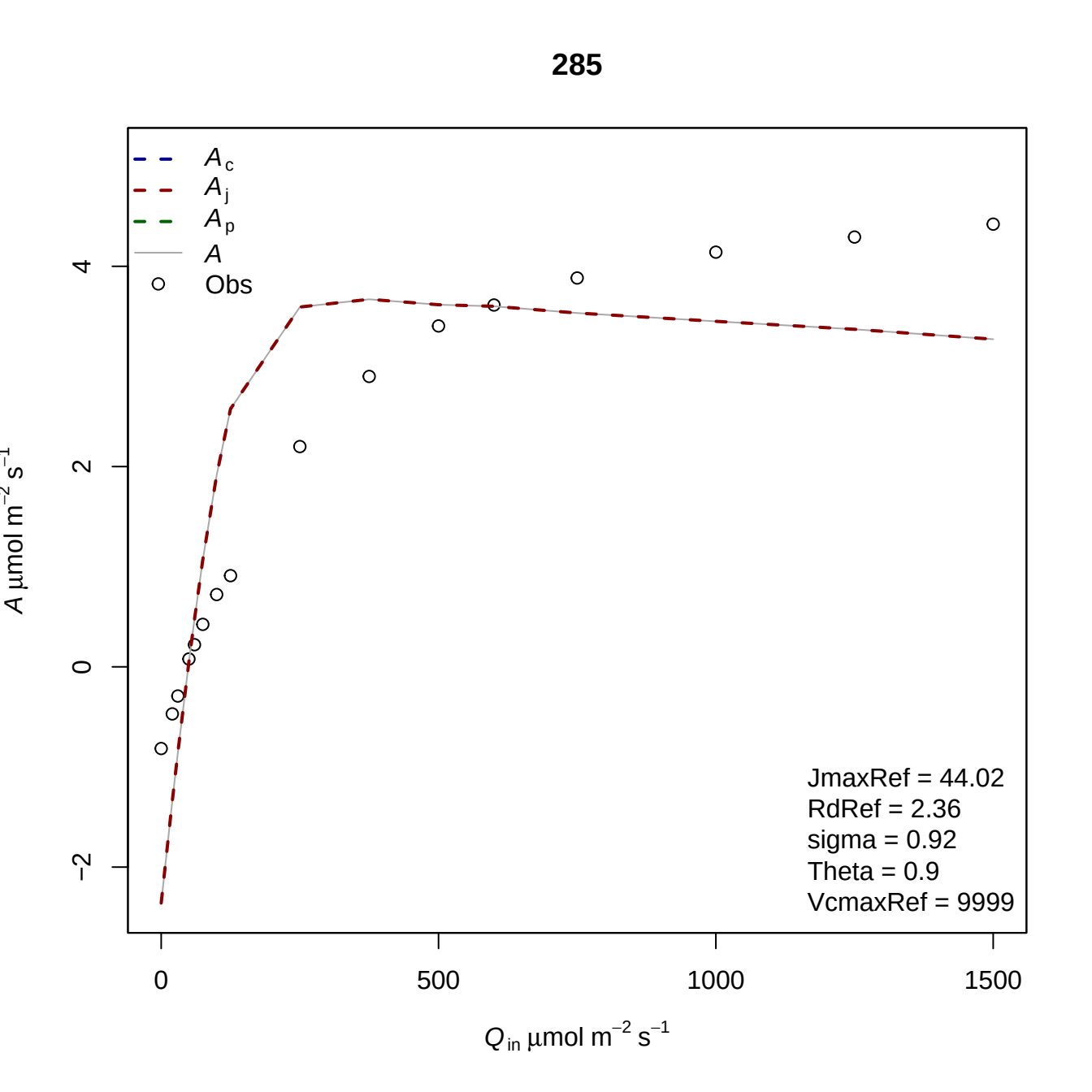

280
A_c
A_j
A_p
A
Obs
JmaxRef = 53.56
RdRef = 2.41
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

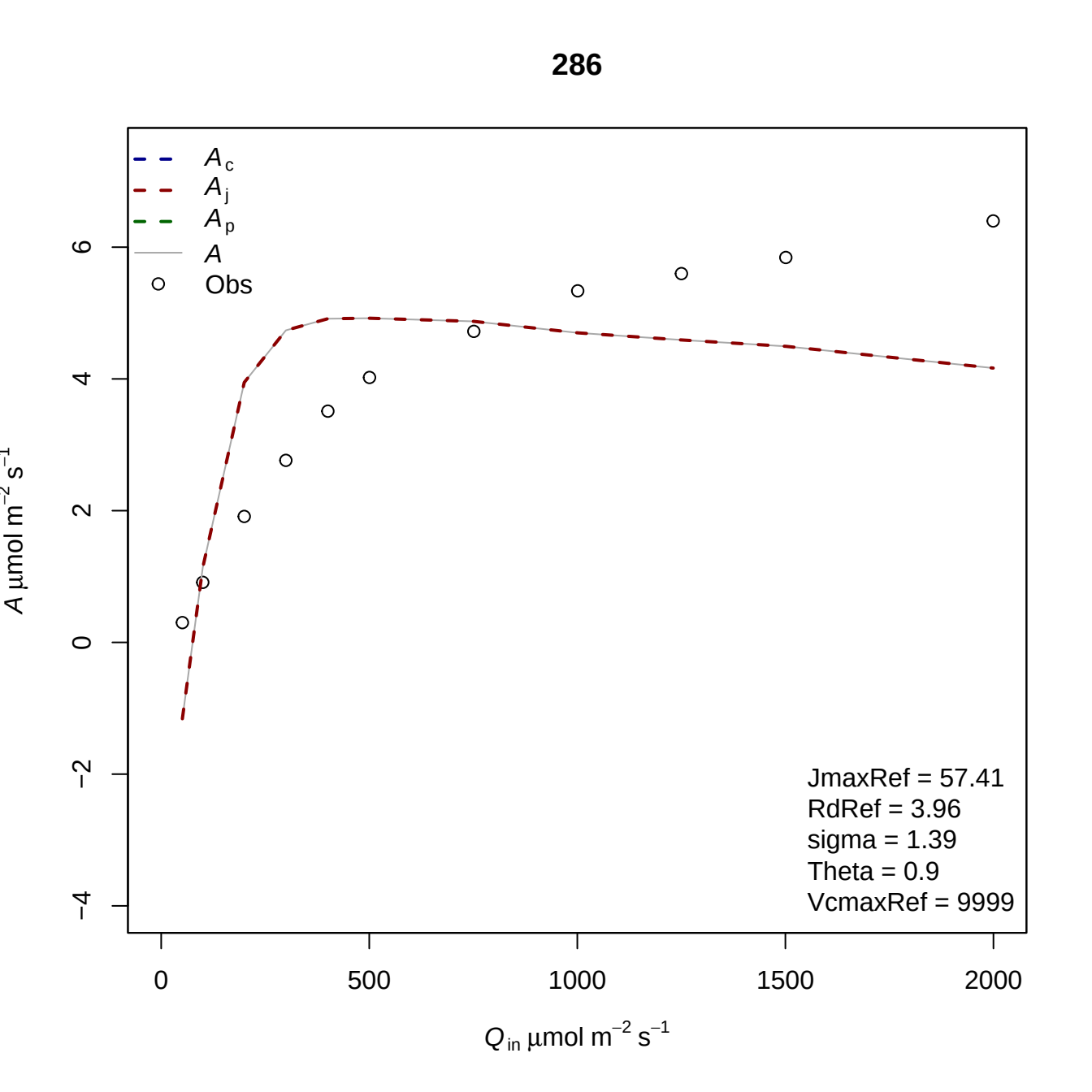

281
A_c
A_j
A_p
A
Obs
JmaxRef = 64.7
RdRef = 2.52
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

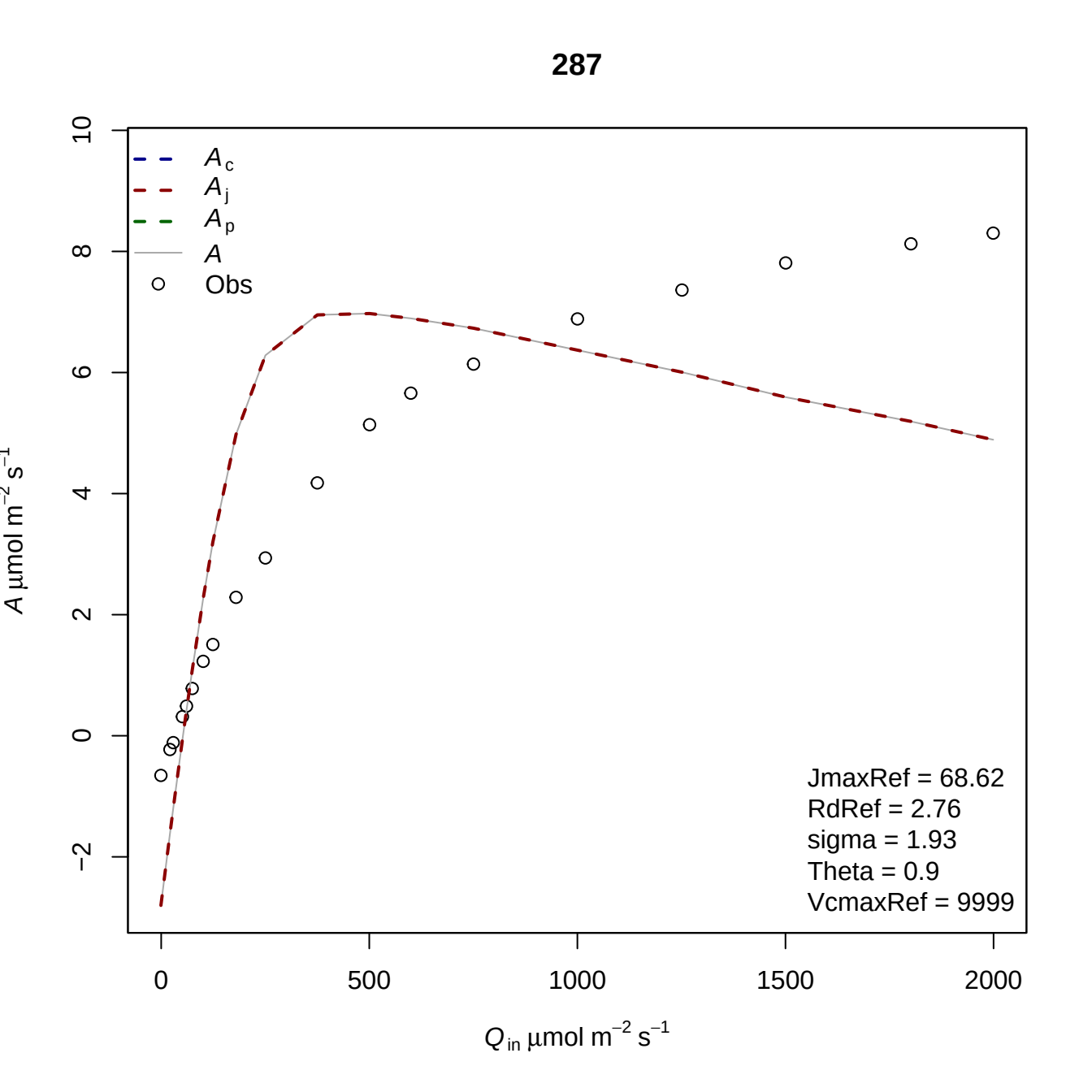

282
A_c
A_j
A_p
A
Obs
JmaxRef = 38.73
RdRef = 2.02
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

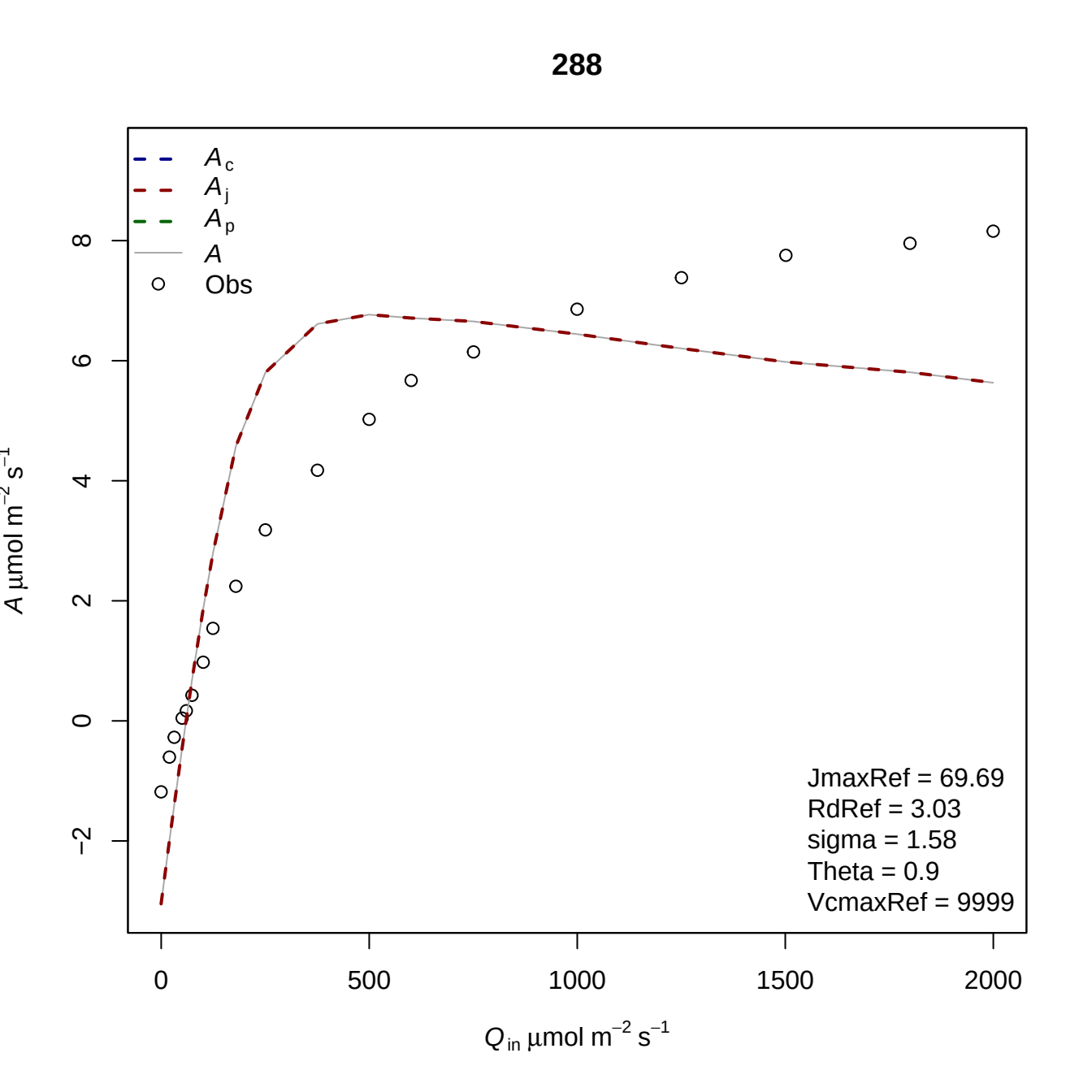

283
A_c
A_j
A_p
A
Obs
JmaxRef = 43.34
RdRef = 1.91
sigma = 1.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

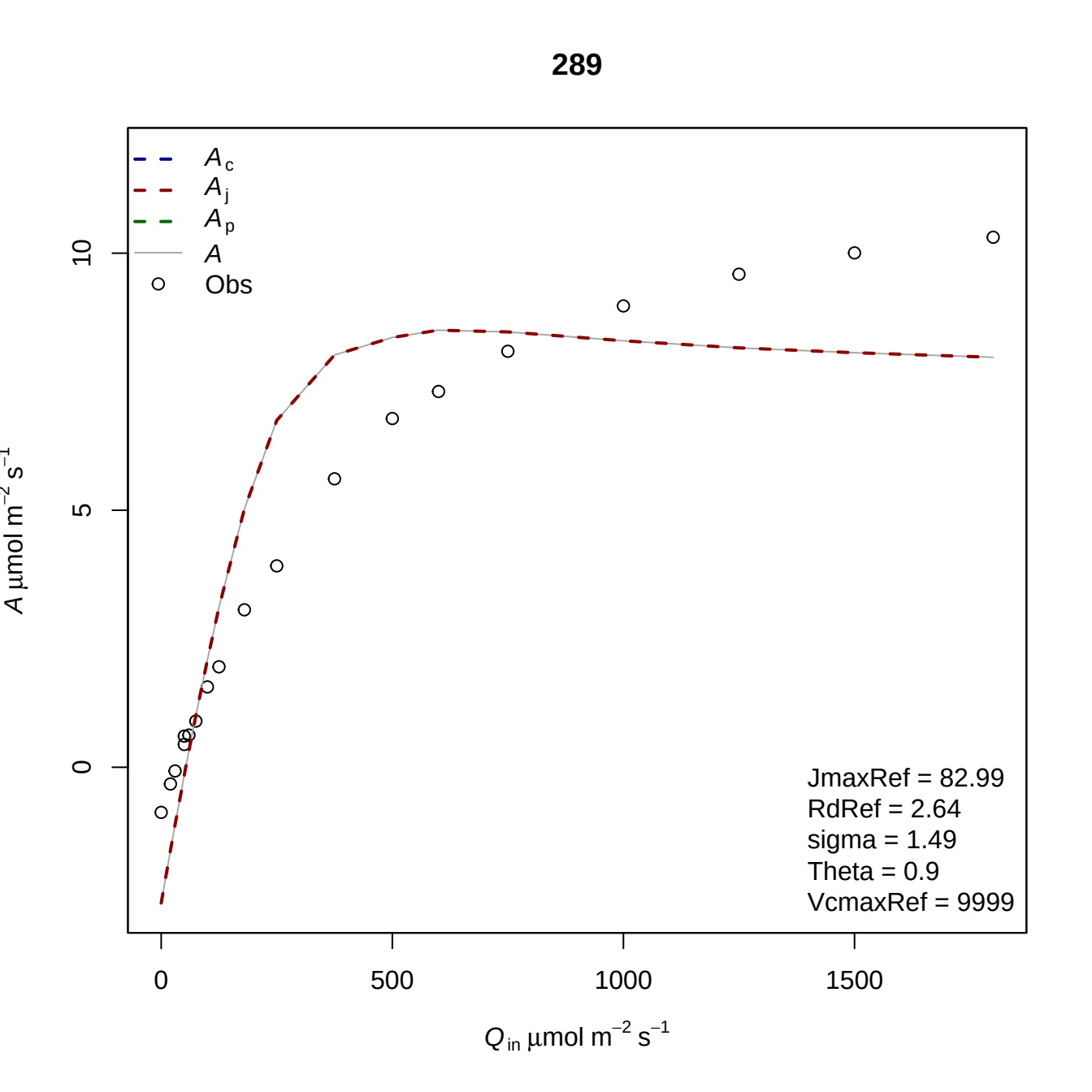

284
A_c
A_j
A_p
A
Obs
JmaxRef = 75.16
RdRef = 2.34
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

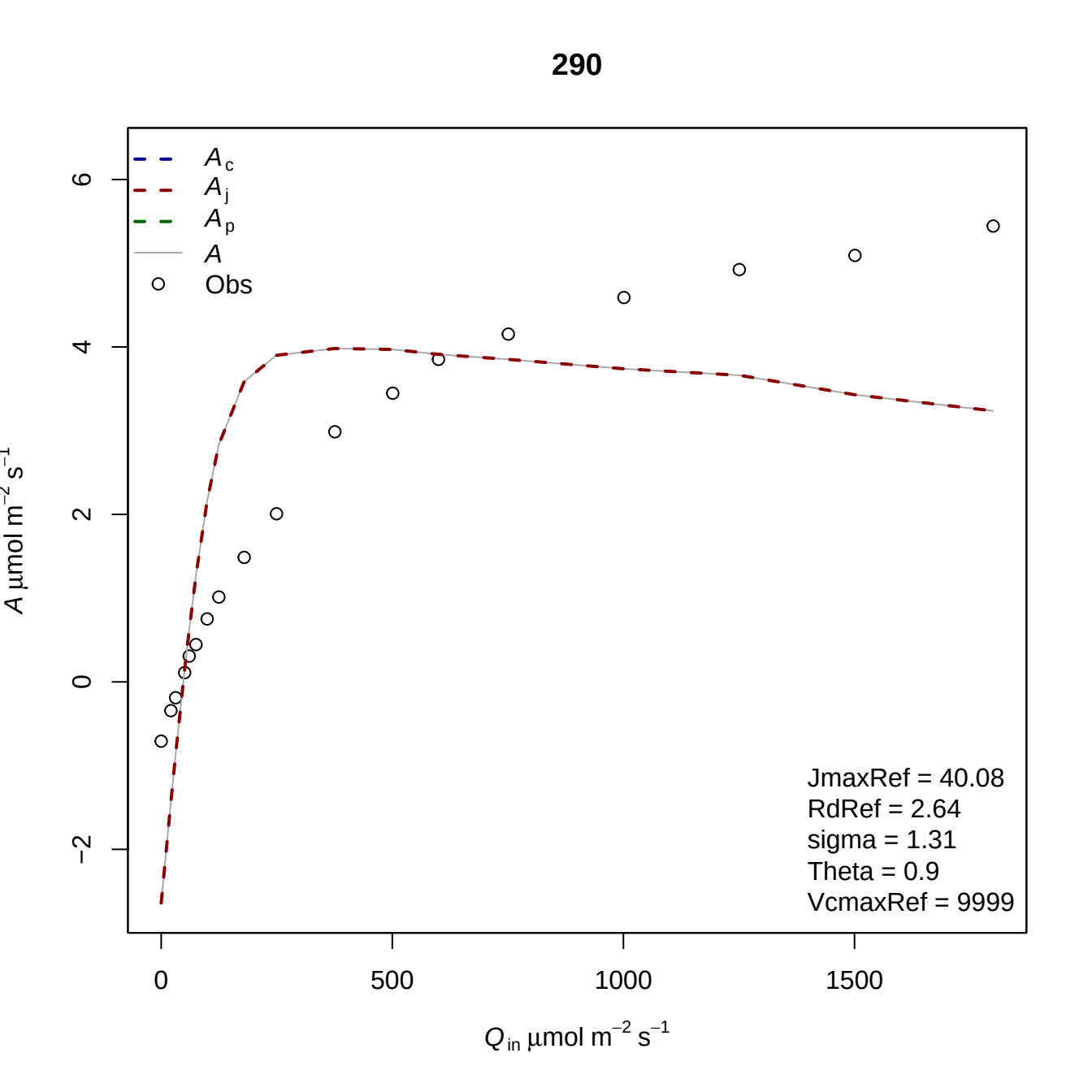

285
A_c
A_j
A_p
A
Obs
JmaxRef = 44.02
RdRef = 2.36
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

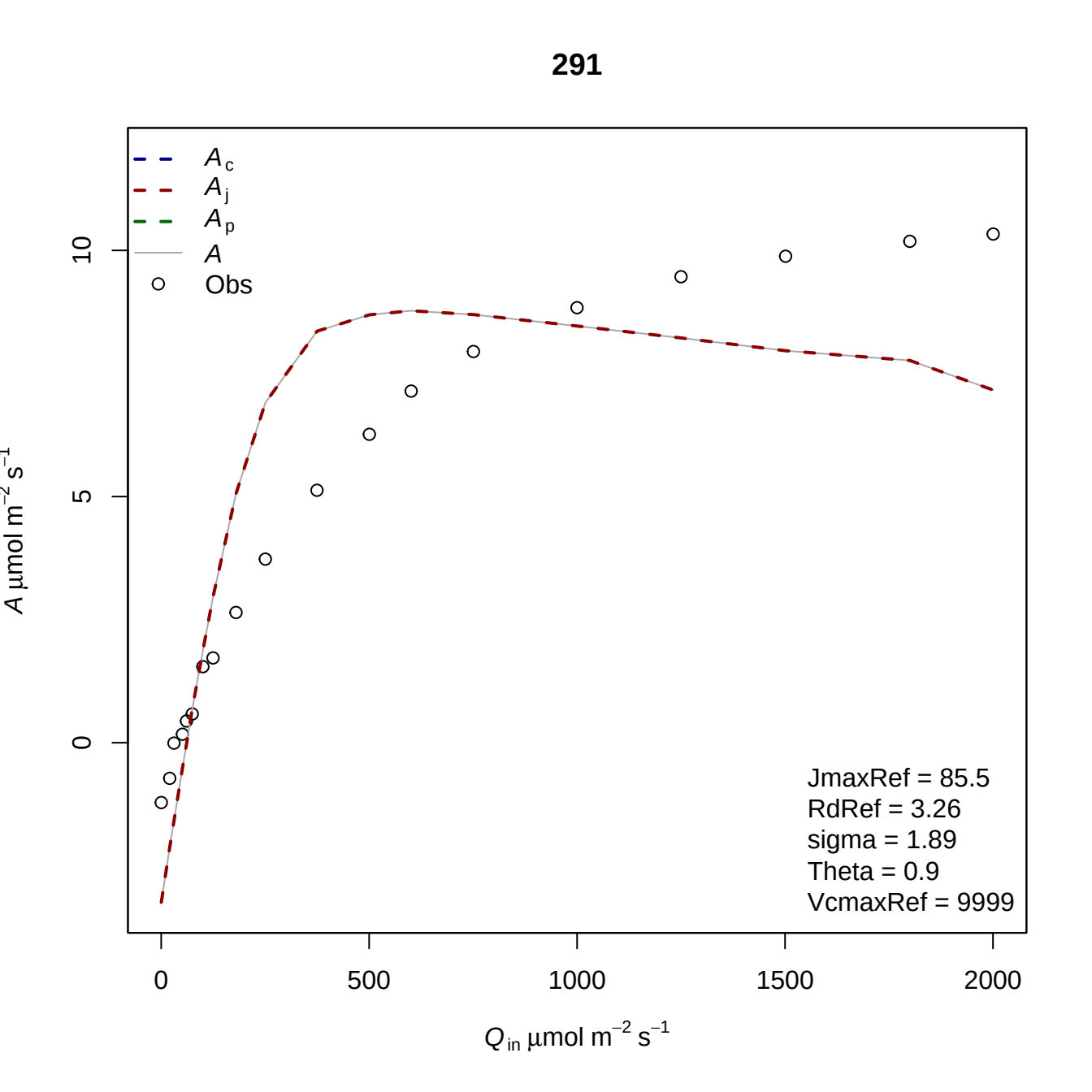

286
A_c
A_j
A_p
A
Obs
JmaxRef = 57.41
RdRef = 3.96
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

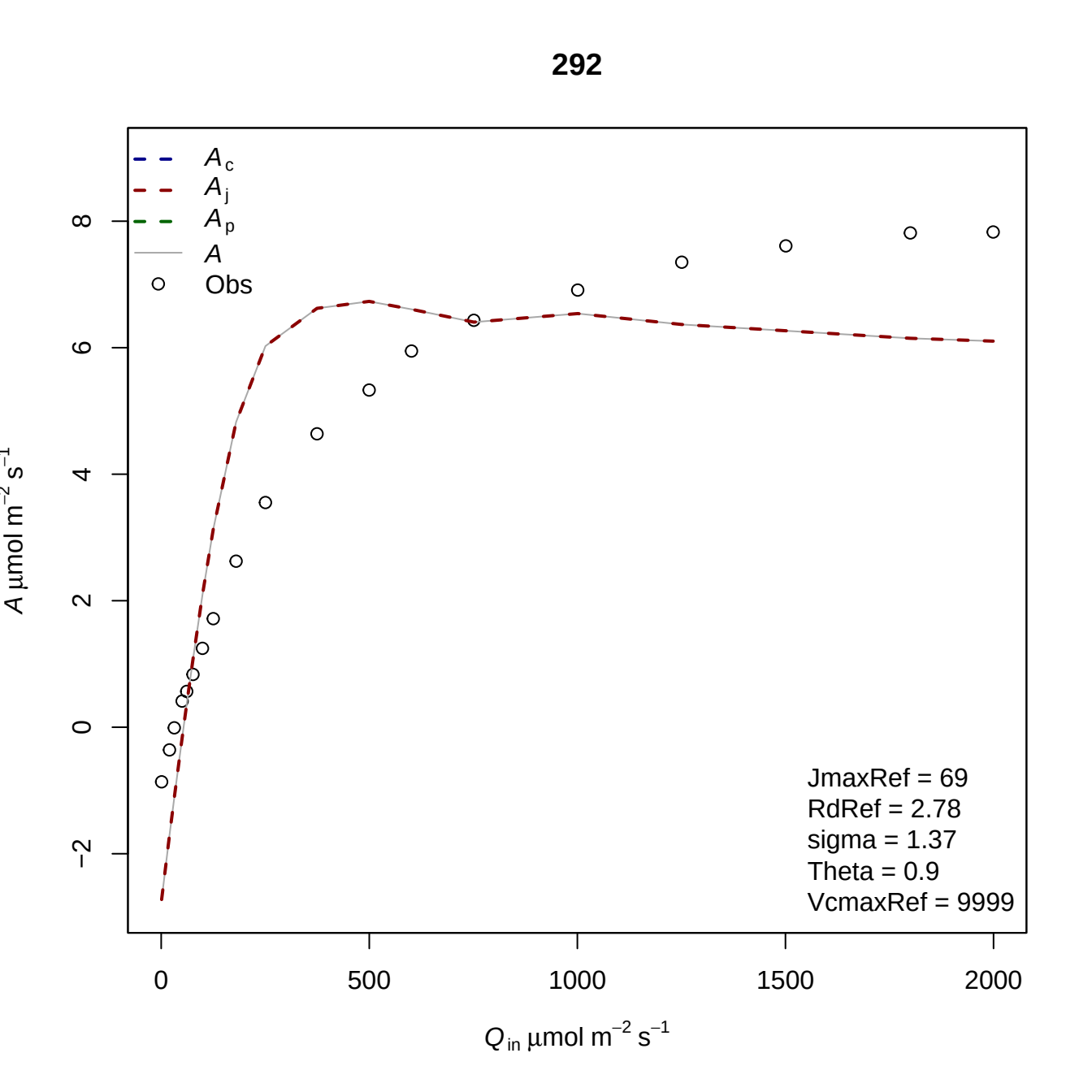

287
A_c
A_j
A_p
A
Obs
JmaxRef = 68.62
RdRef = 2.76
sigma = 1.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

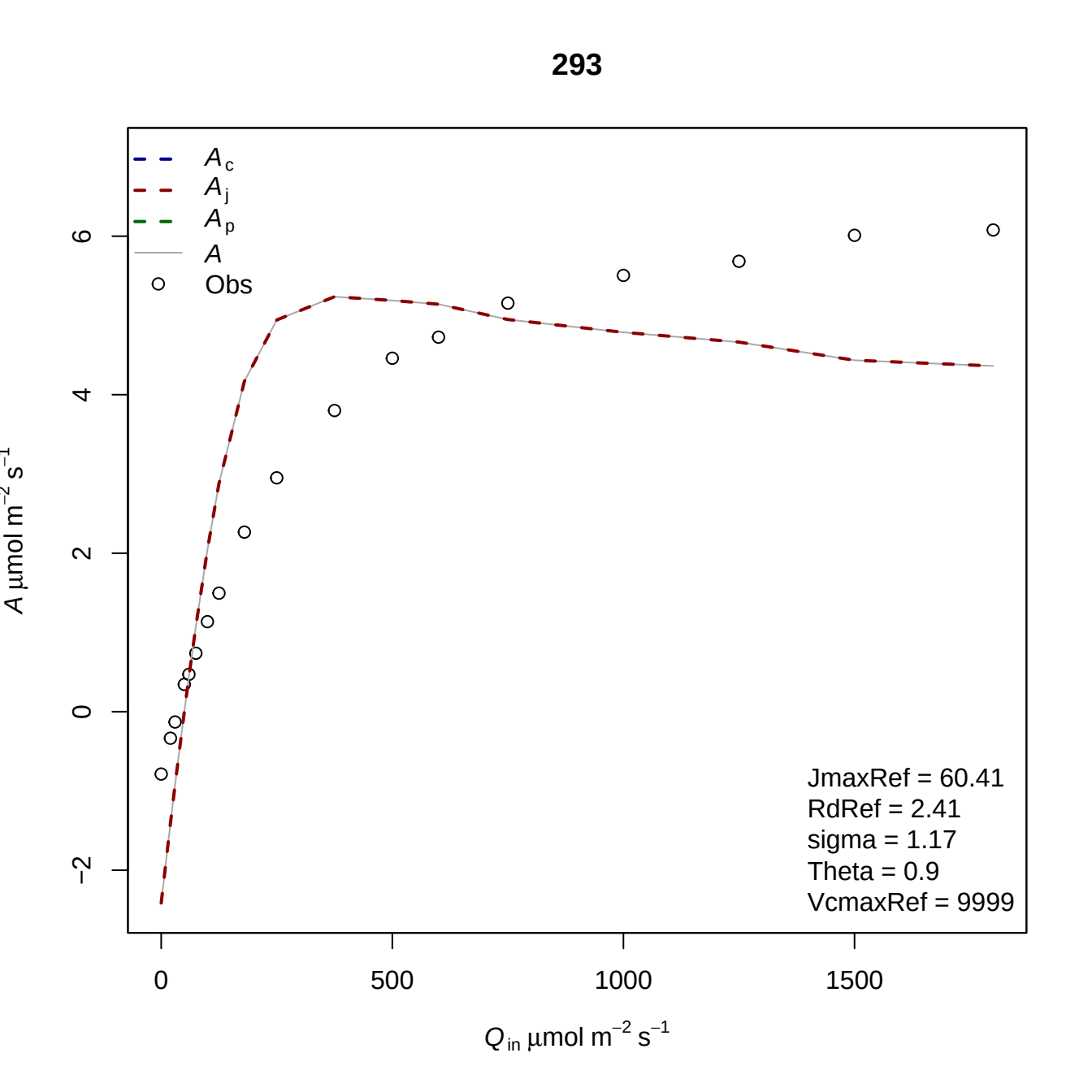

288
A_c
A_j
A_p
A
Obs
JmaxRef = 69.69
RdRef = 3.03
sigma = 1.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

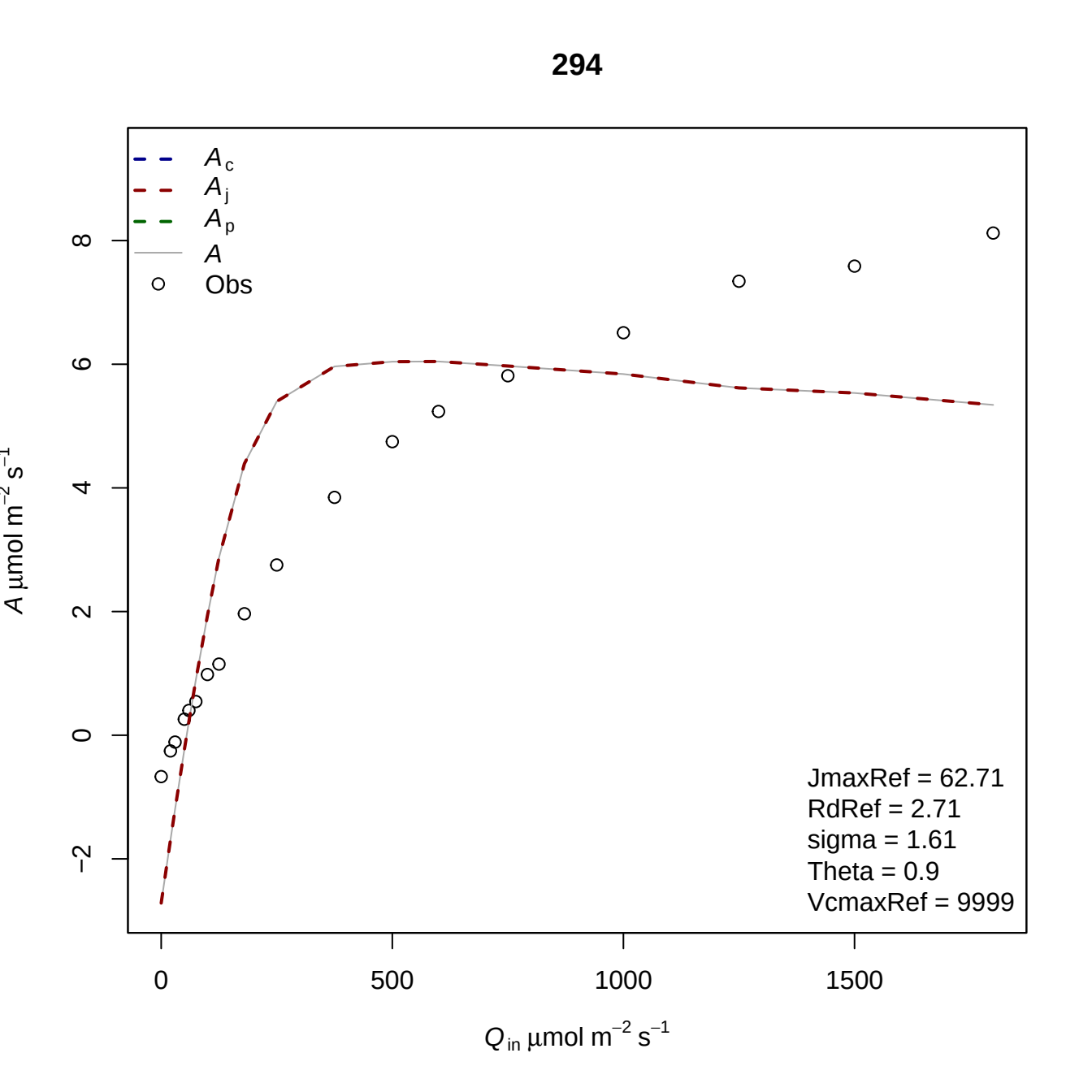

289
A_c
A_j
A_p
A
Obs
JmaxRef = 82.99
RdRef = 2.64
sigma = 1.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

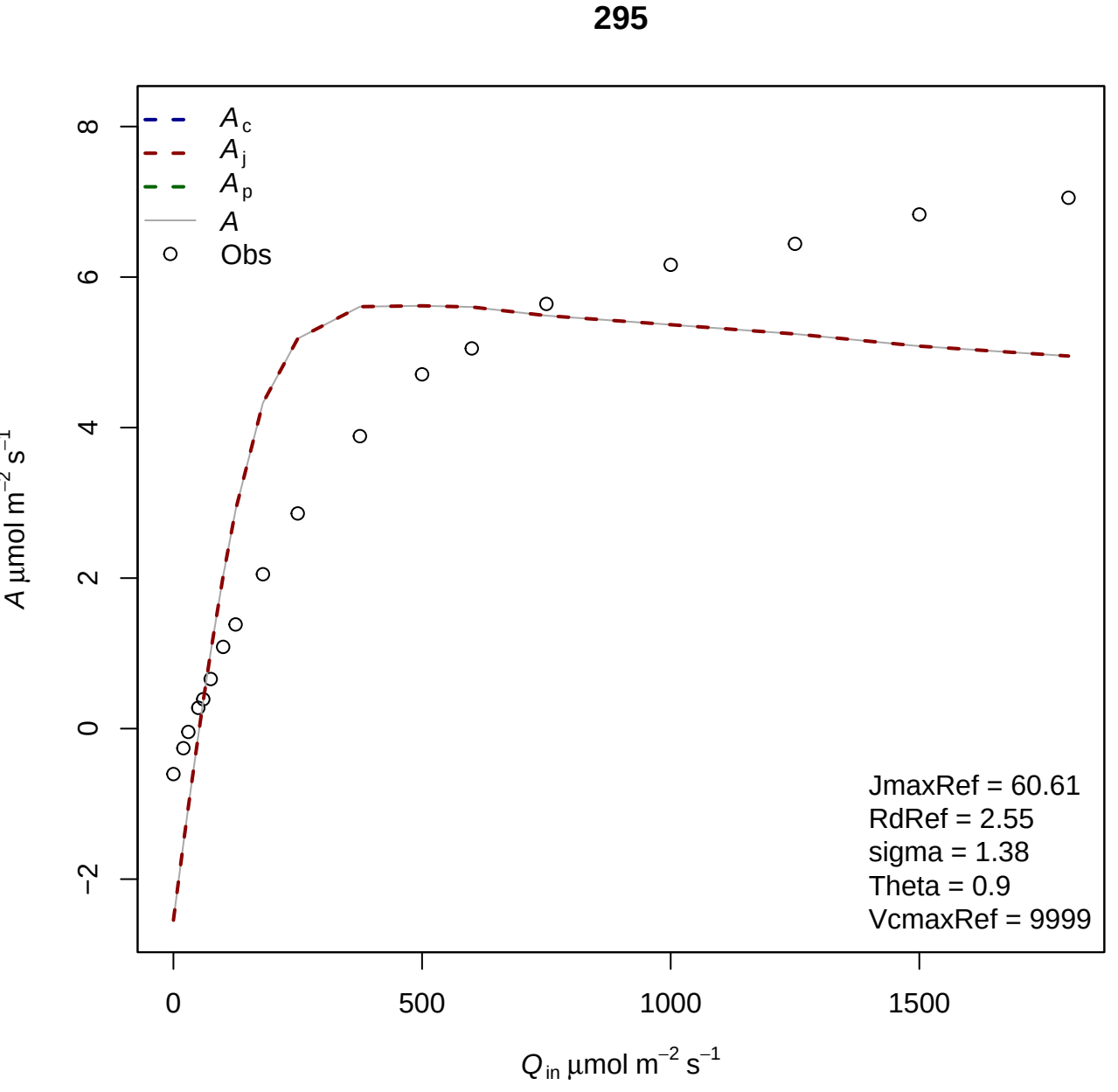

290
A_c
A_j
A_p
A
Obs
JmaxRef = 40.08
RdRef = 2.64
sigma = 1.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

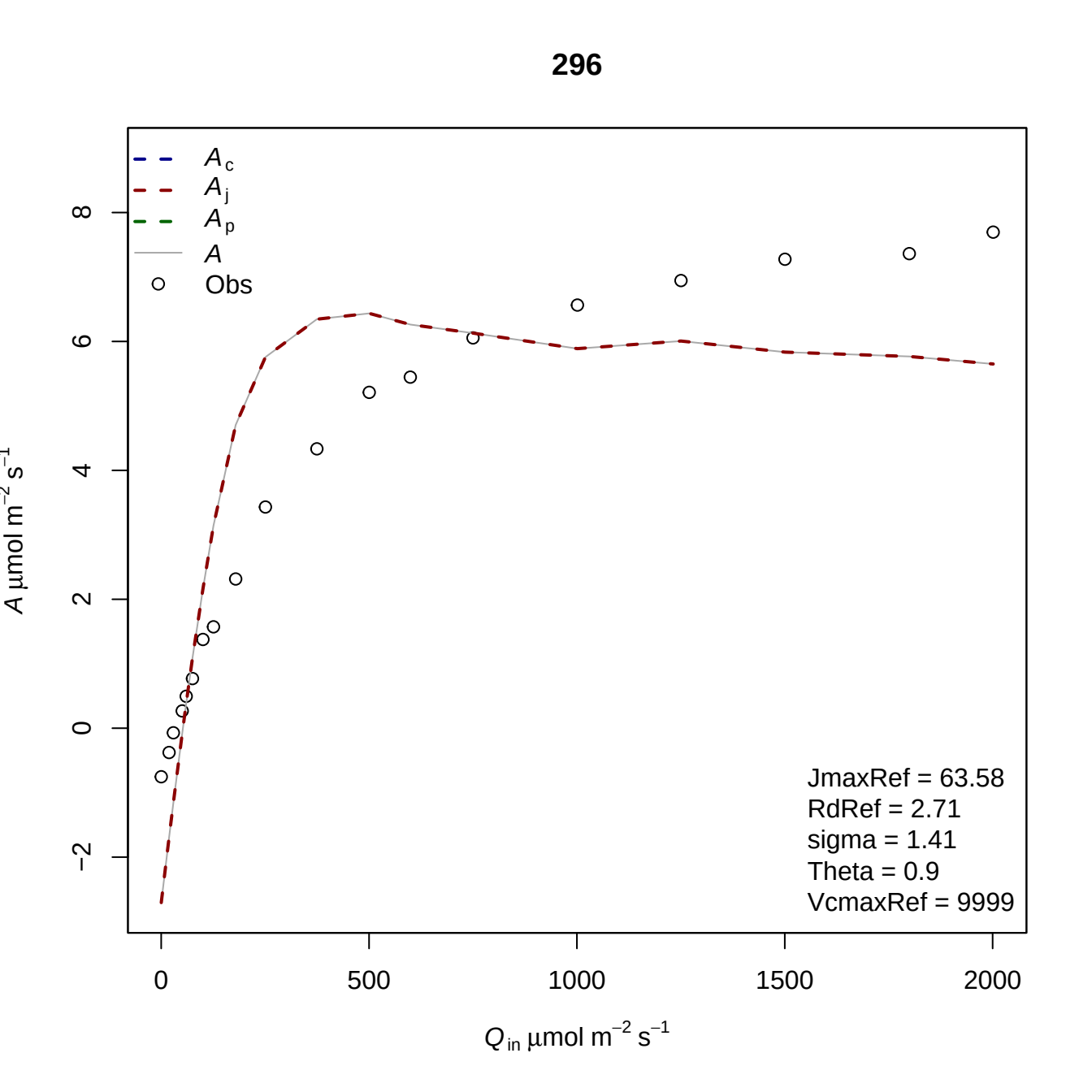

291
A_c
A_j
A_p
A
Obs
JmaxRef = 85.5
RdRef = 3.26
sigma = 1.89
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

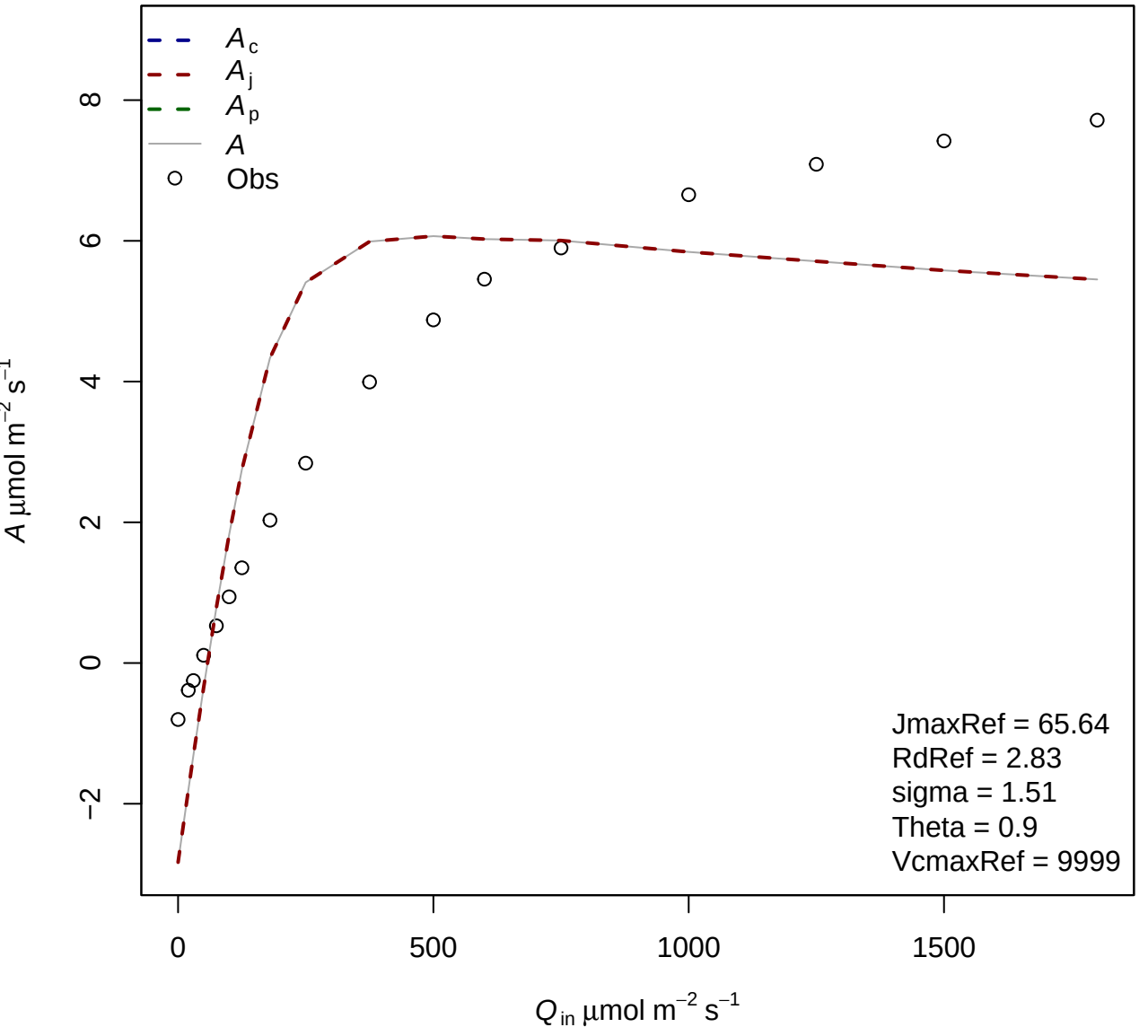

292
A_c
A_j
A_p
A
Obs
JmaxRef = 69
RdRef = 2.78
sigma = 1.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

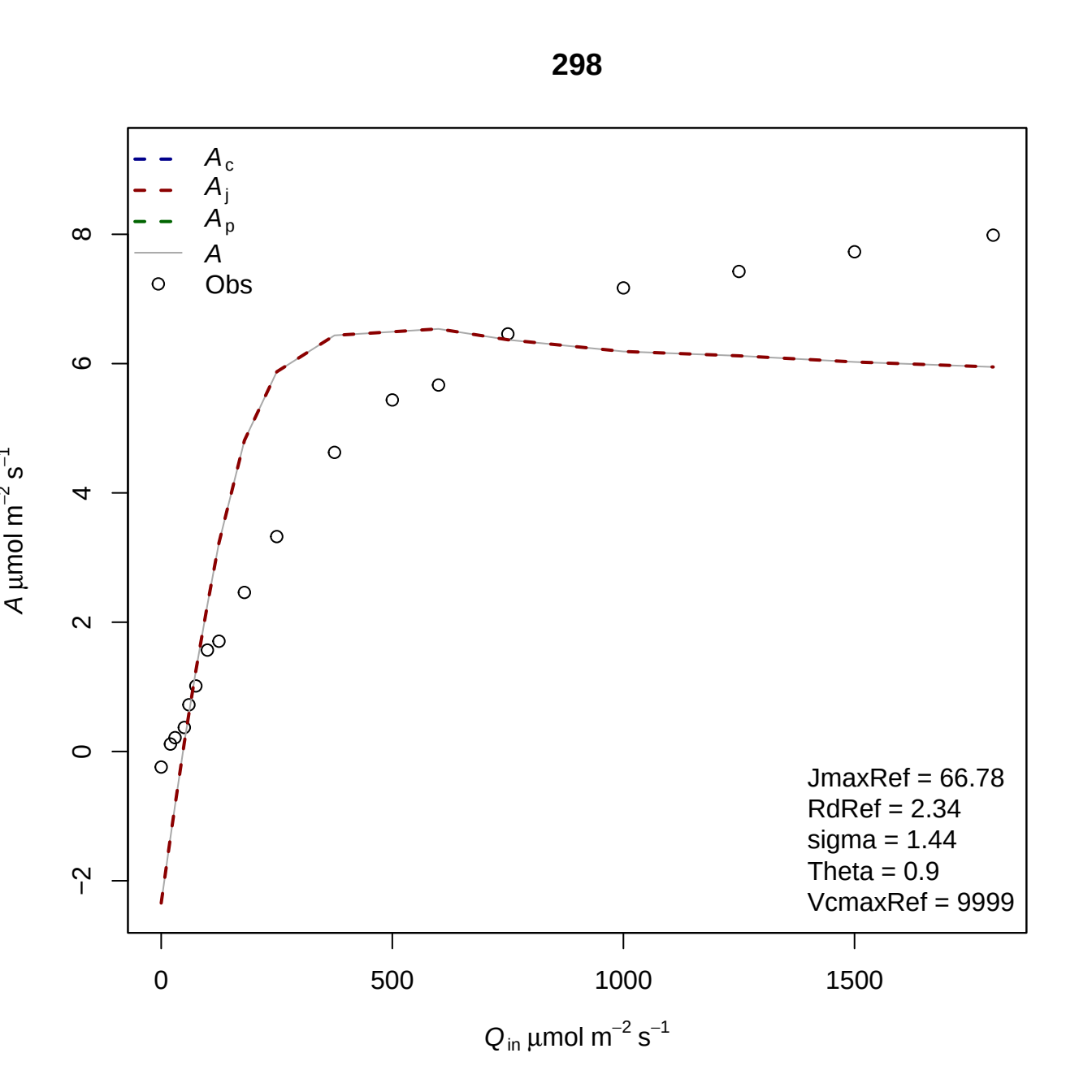

293
A_c
A_j
A_p
A
Obs
JmaxRef = 60.41
RdRef = 2.41
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

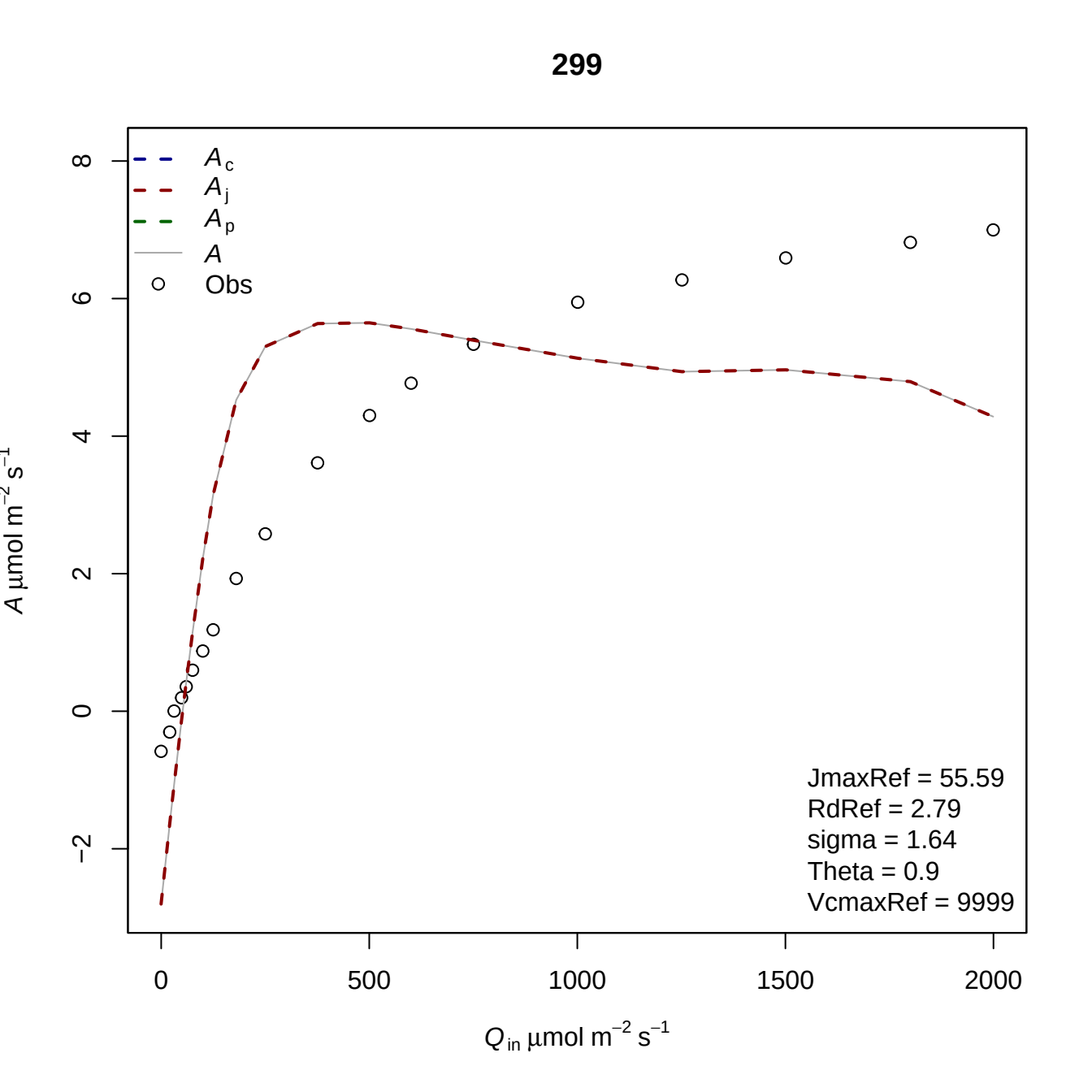

294
A_c
A_j
A_p
A
Obs
JmaxRef = 62.71
RdRef = 2.71
sigma = 1.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

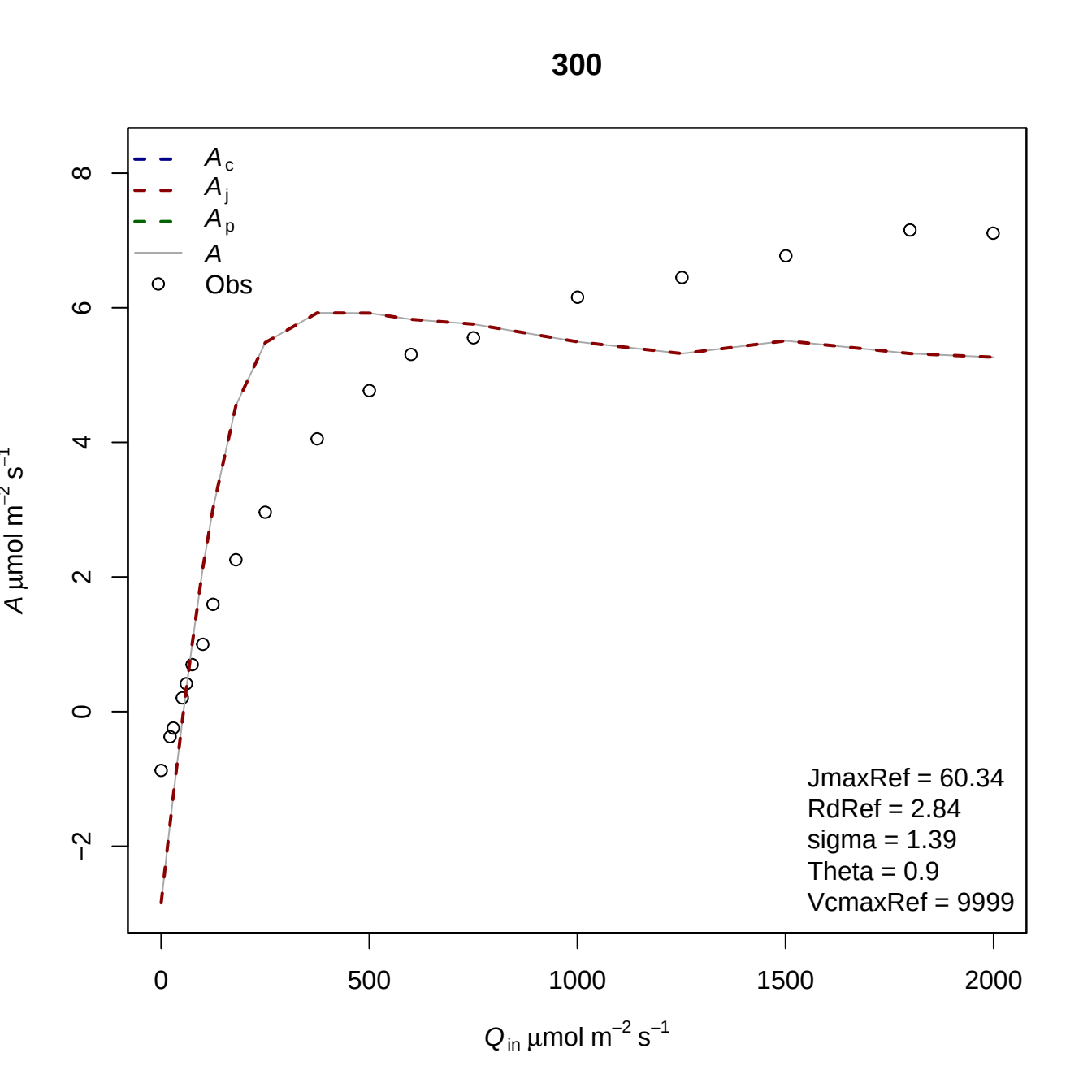

295
A_c
A_j
A_p
A
Obs
JmaxRef = 60.61
RdRef = 2.55
sigma = 1.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

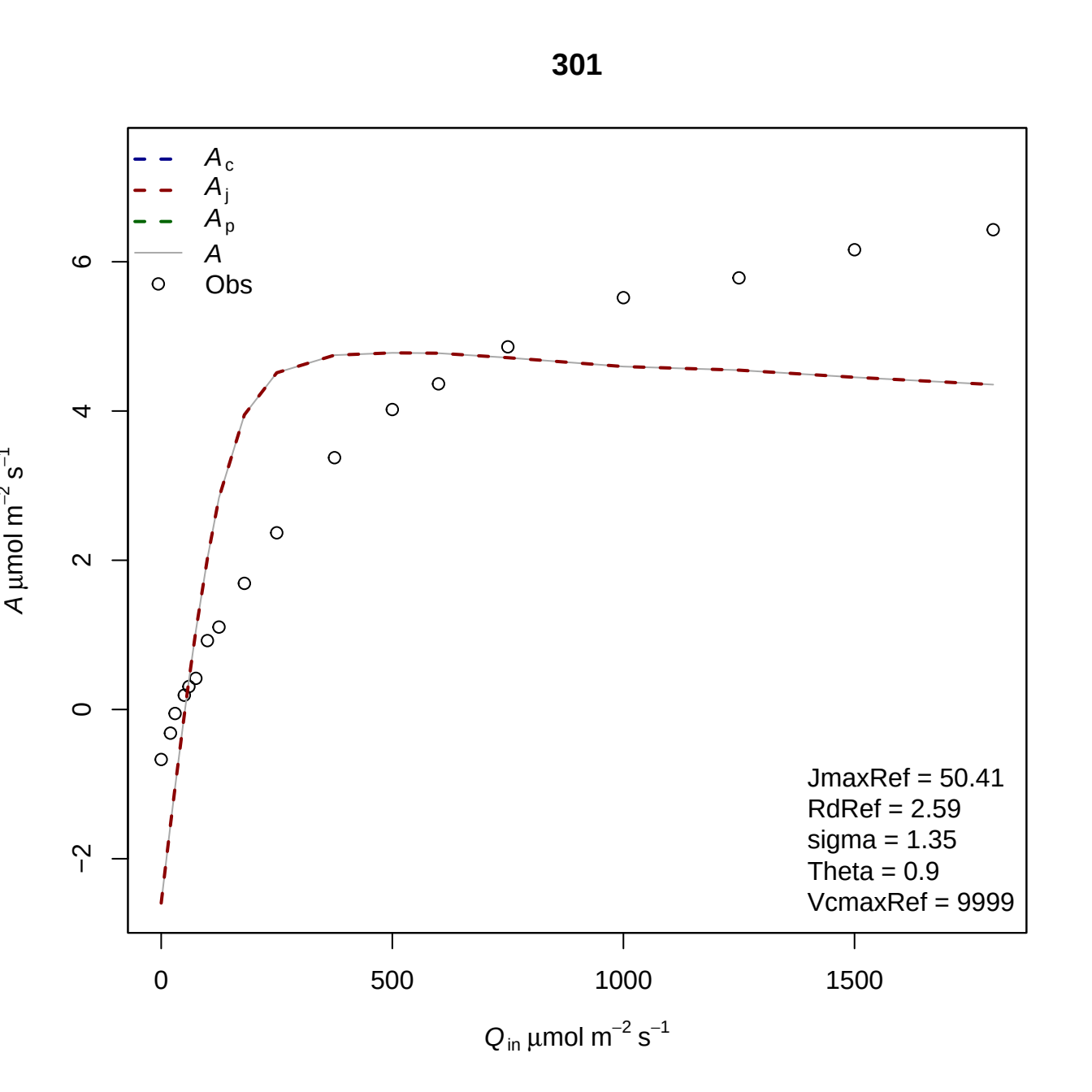

296
A_c
A_j
A_p
A
Obs
JmaxRef = 63.58
RdRef = 2.71
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

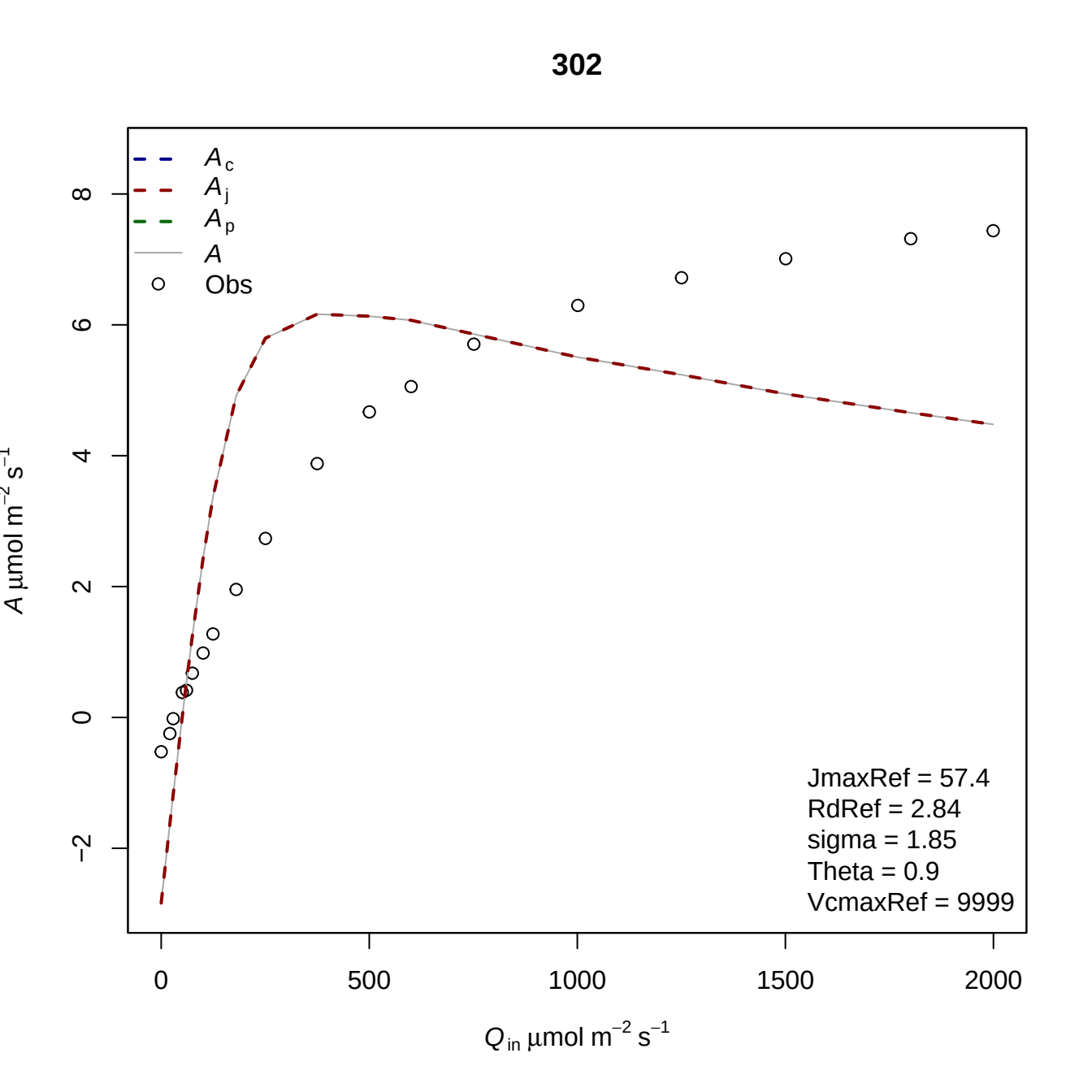

A_c
A_j
A_p
A
Obs
JmaxRef = 65.64
RdRef = 2.83
sigma = 1.51
Theta = 0.9
VcmaxRef = 9999
−2
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

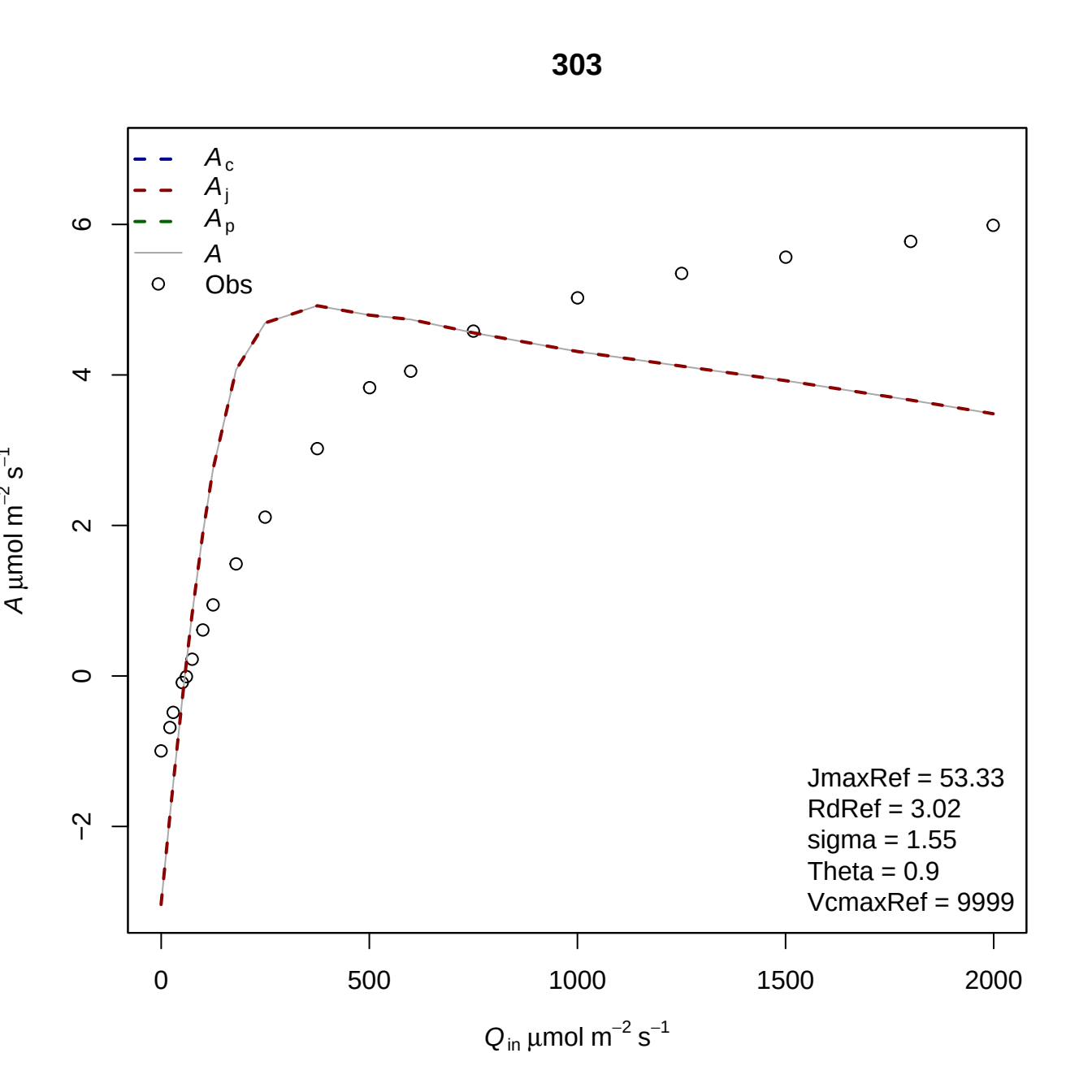

298
A_c
A_j
A_p
A
Obs
JmaxRef = 66.78
RdRef = 2.34
sigma = 1.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

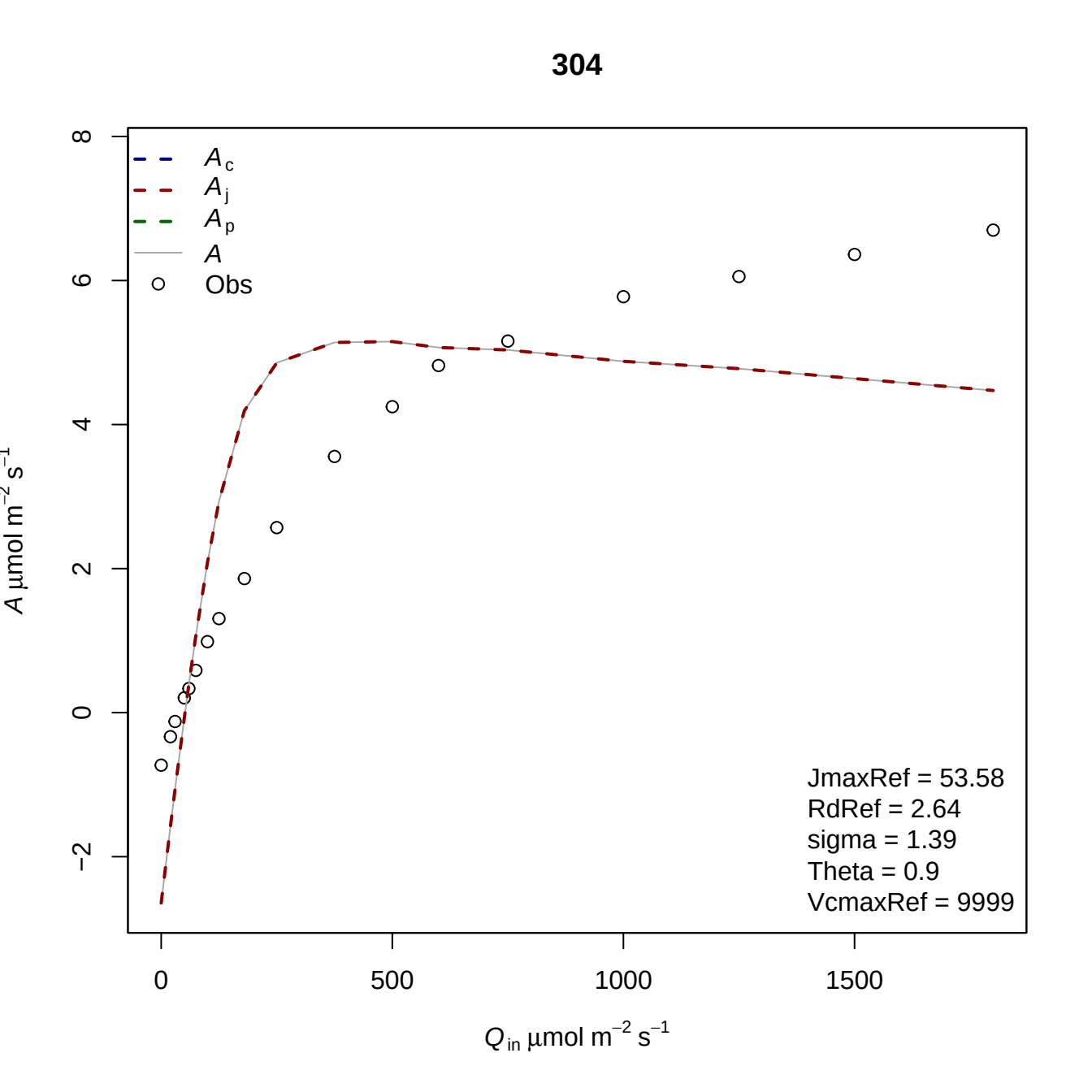

299
A_c
A_j
A_p
A
Obs
JmaxRef = 55.59
RdRef = 2.79
sigma = 1.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

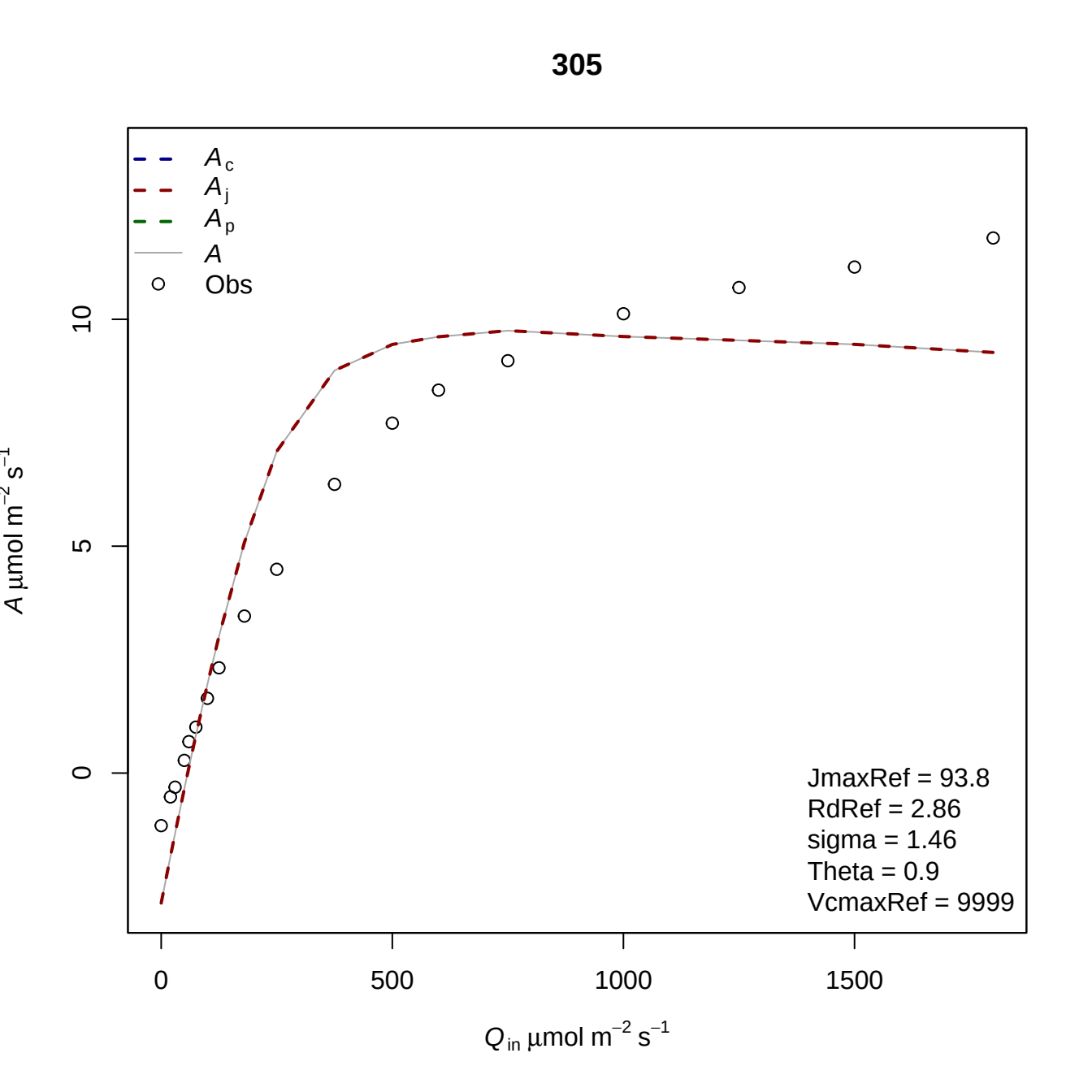

300
A_c
A_j
A_p
A
Obs
JmaxRef = 60.34
RdRef = 2.84
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

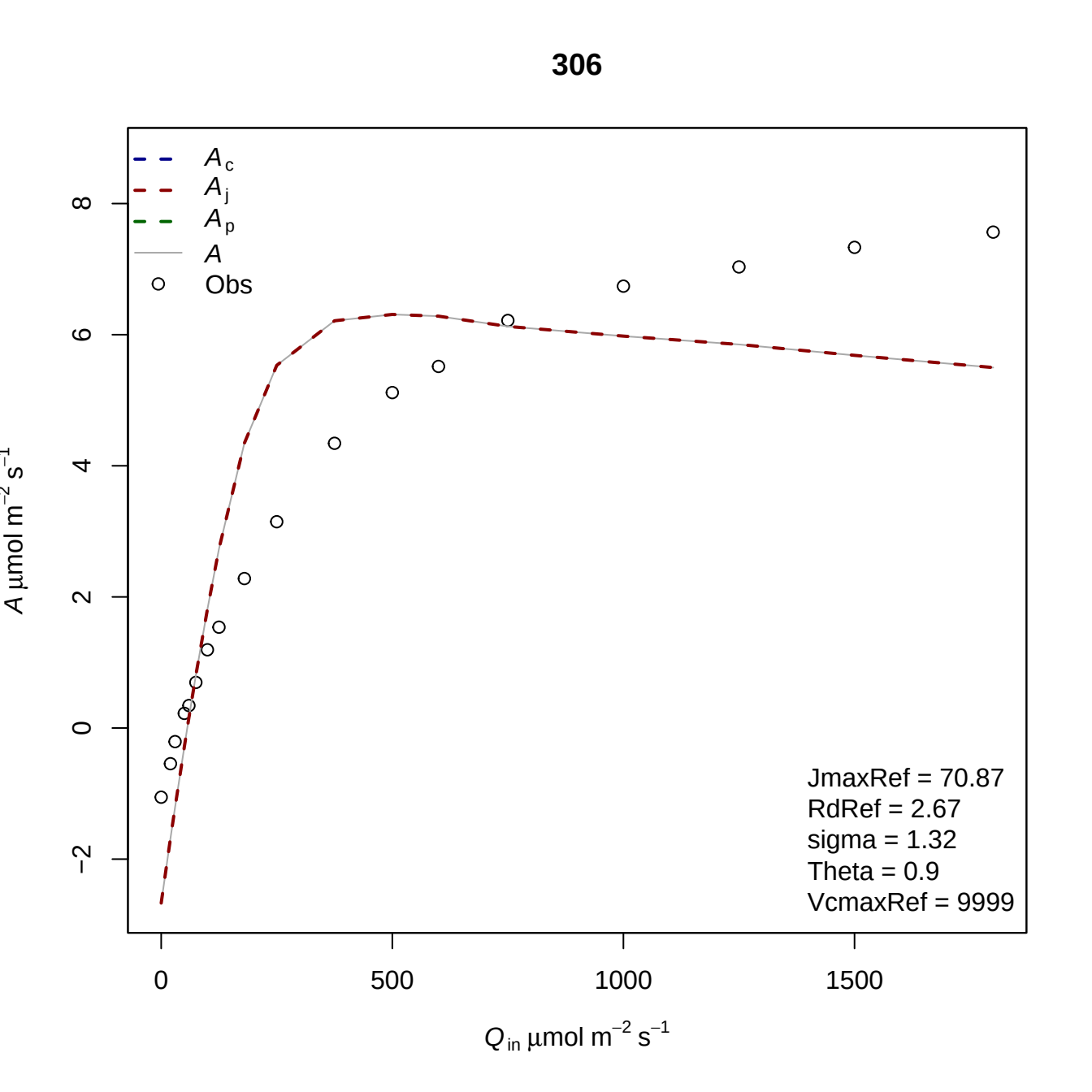

301
A_c
A_j
A_p
A
Obs
JmaxRef = 50.41
RdRef = 2.59
sigma = 1.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

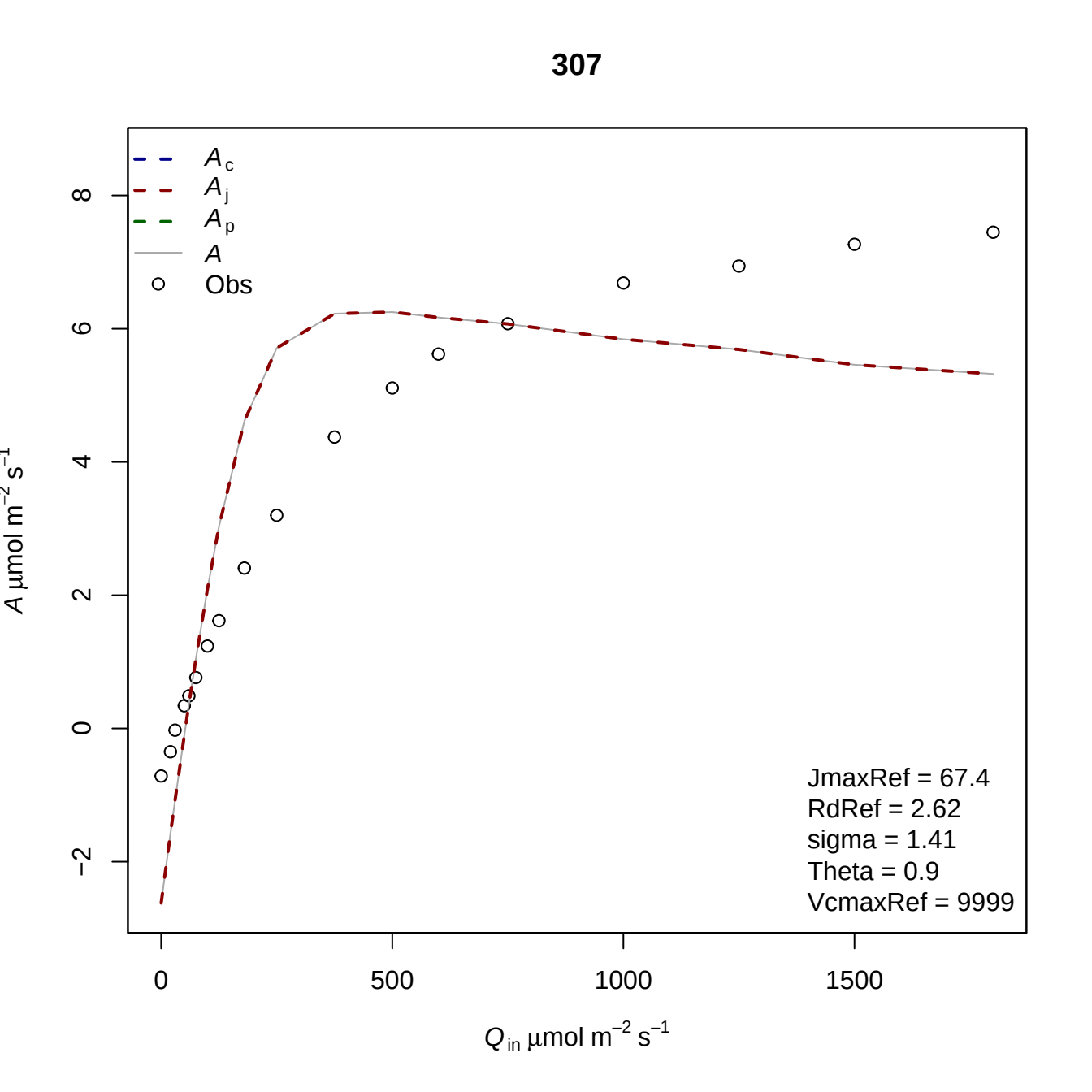

302
A_c
A_j
A_p
A
Obs
JmaxRef = 57.4
RdRef = 2.84
sigma = 1.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

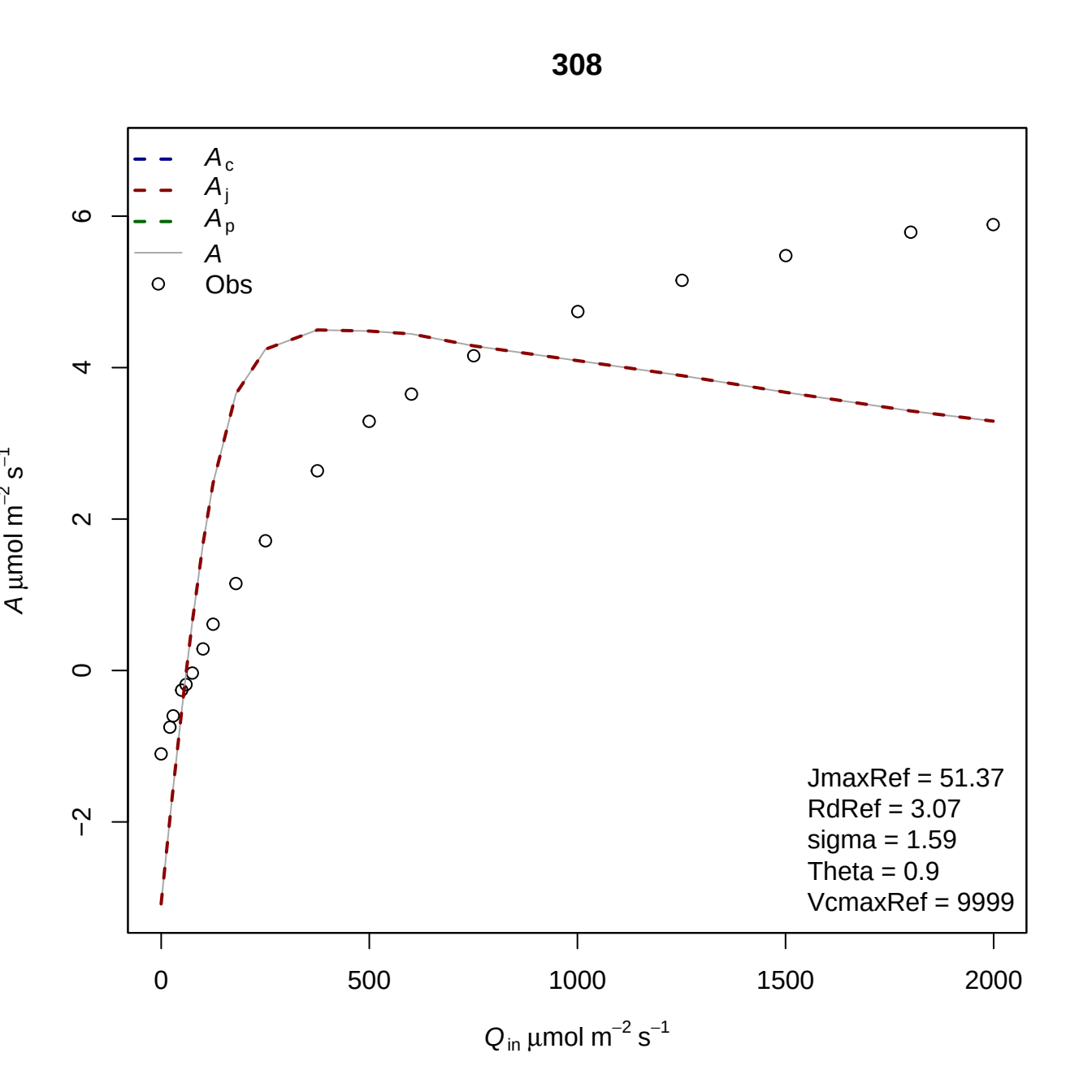

303
A_c
A_j
A_p
A
Obs
JmaxRef = 53.33
RdRef = 3.02
sigma = 1.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

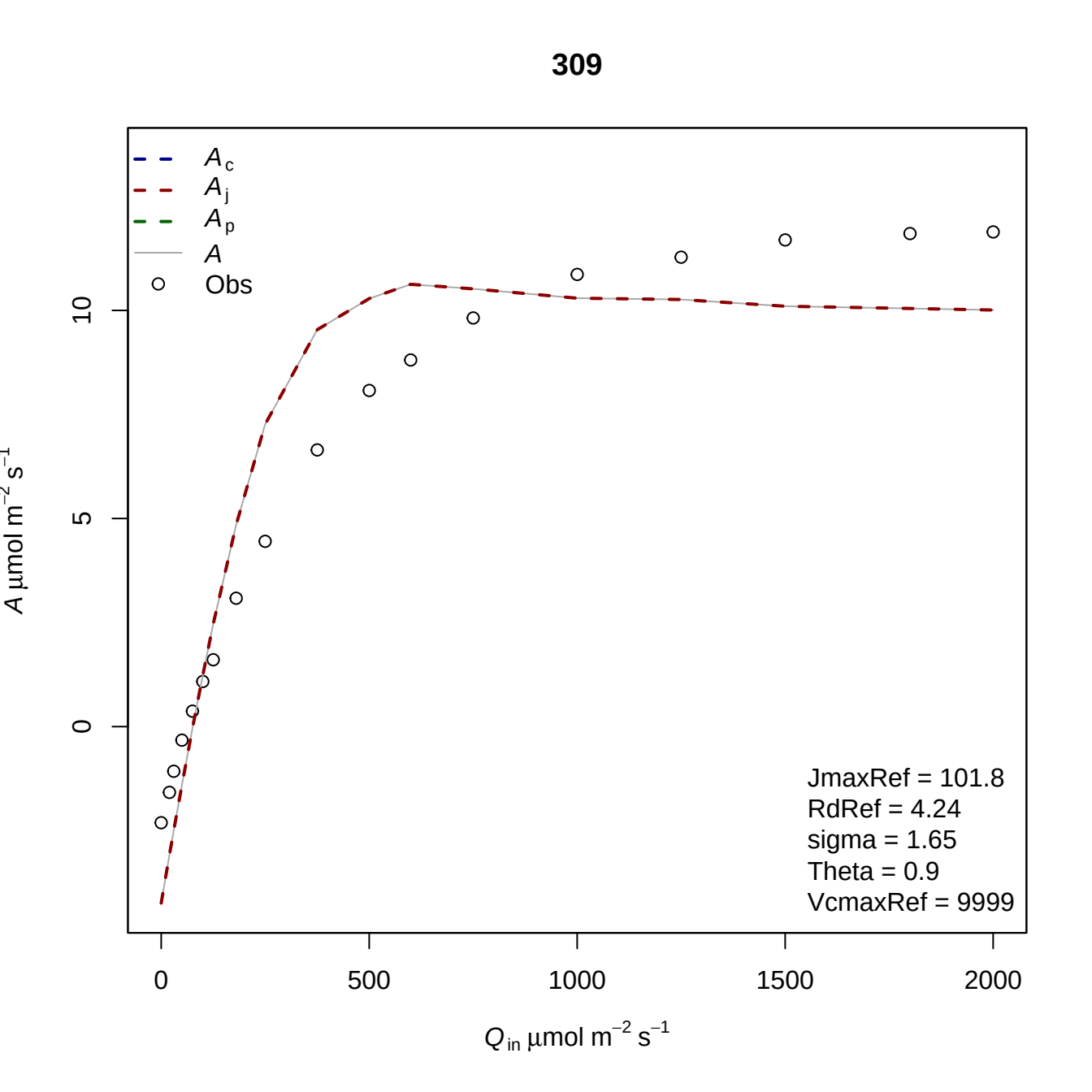

304
A_c
A_j
A_p
A
Obs
JmaxRef = 53.58
RdRef = 2.64
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

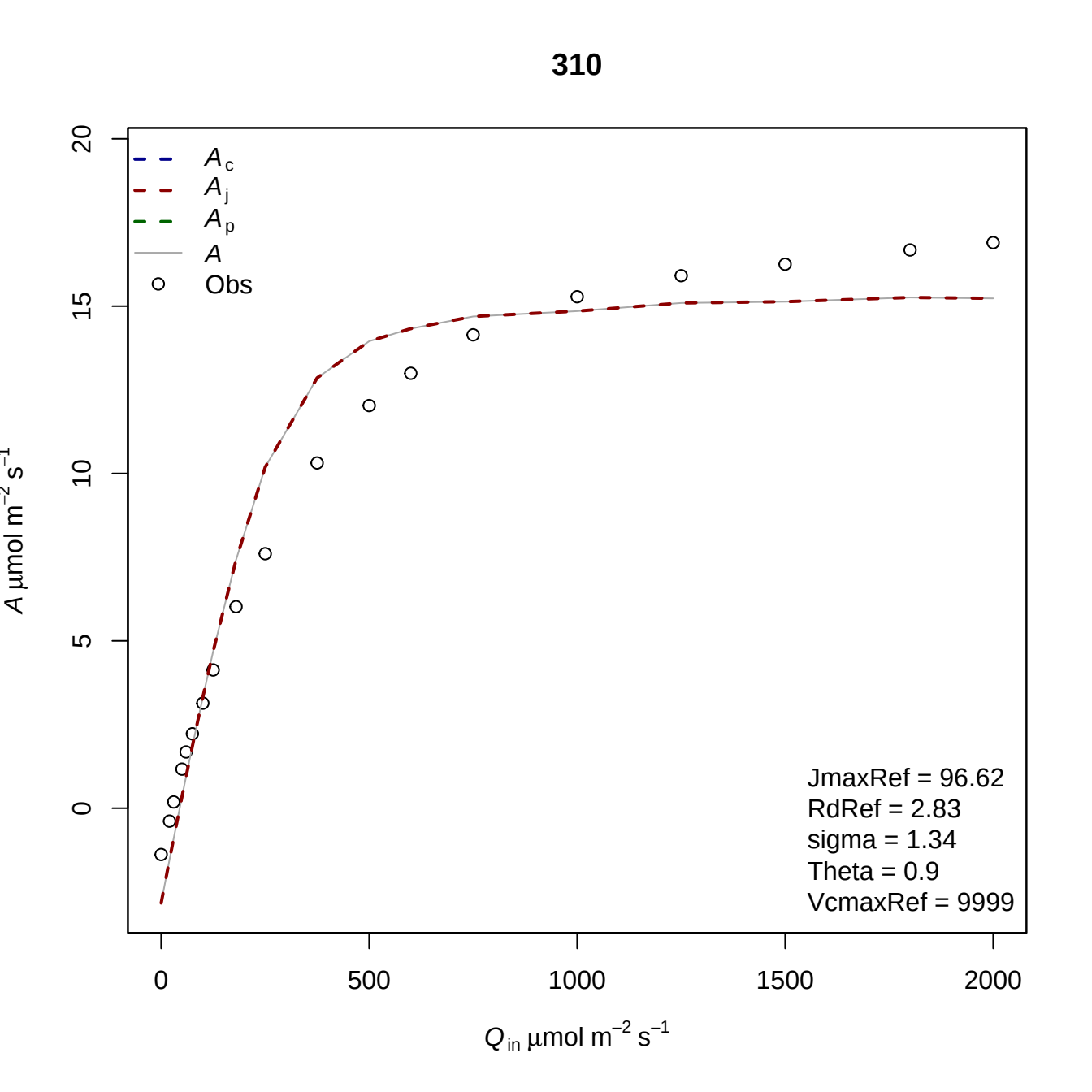

305
A_c
A_j
A_p
A
Obs
JmaxRef = 93.8
RdRef = 2.86
sigma = 1.46
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

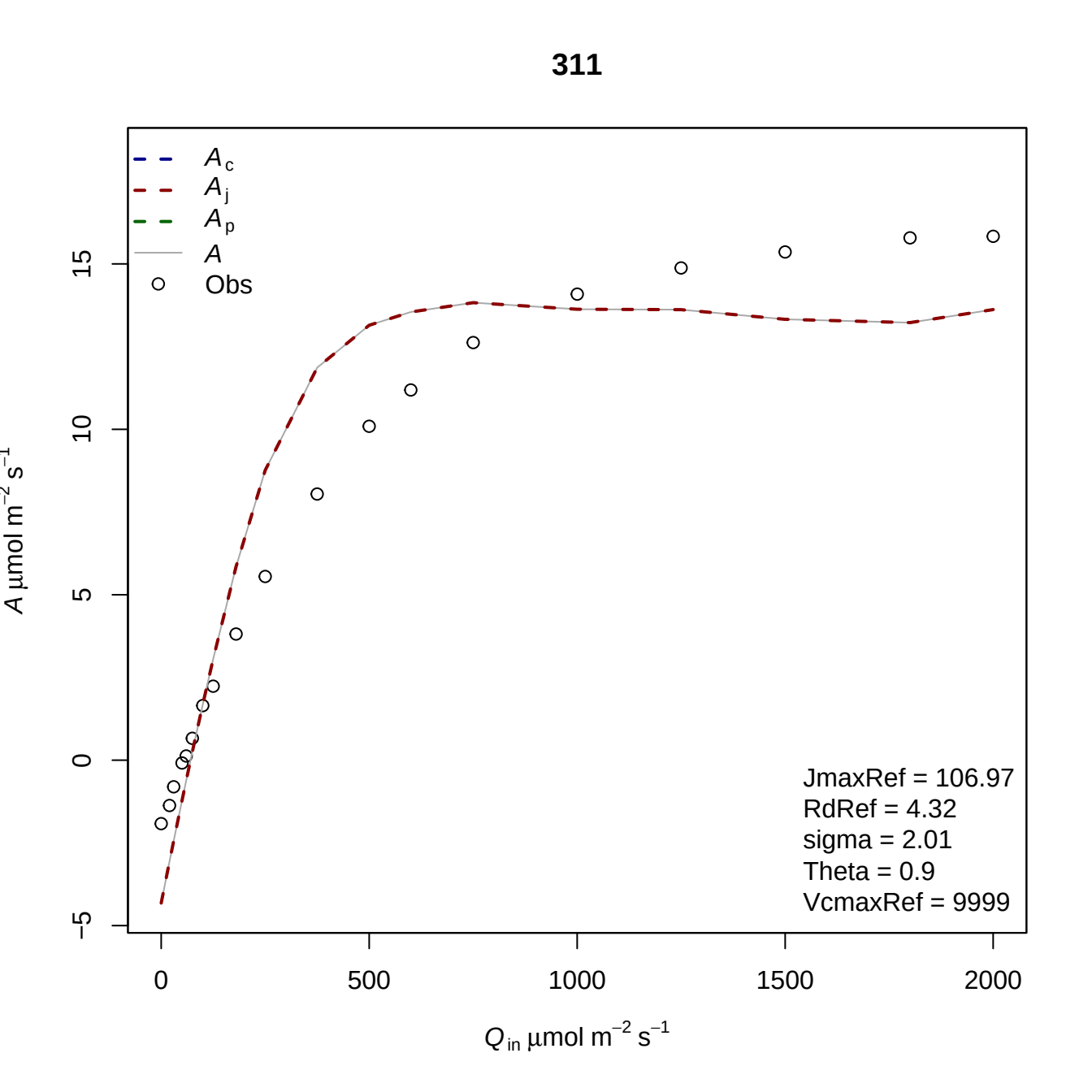

306
A_c
A_j
A_p
A
Obs
JmaxRef = 70.87
RdRef = 2.67
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

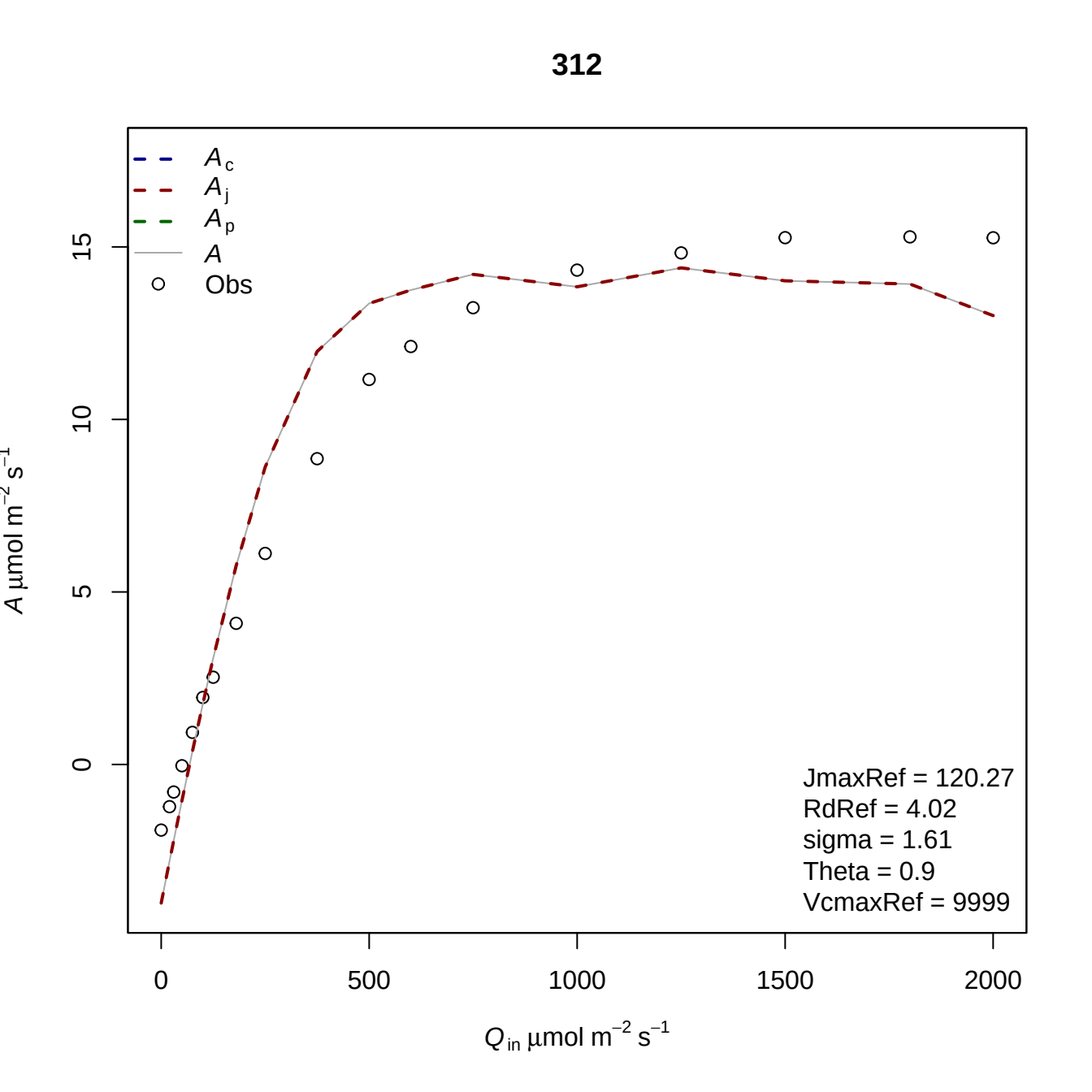

307
A_c
A_j
A_p
A
Obs
JmaxRef = 67.4
RdRef = 2.62
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

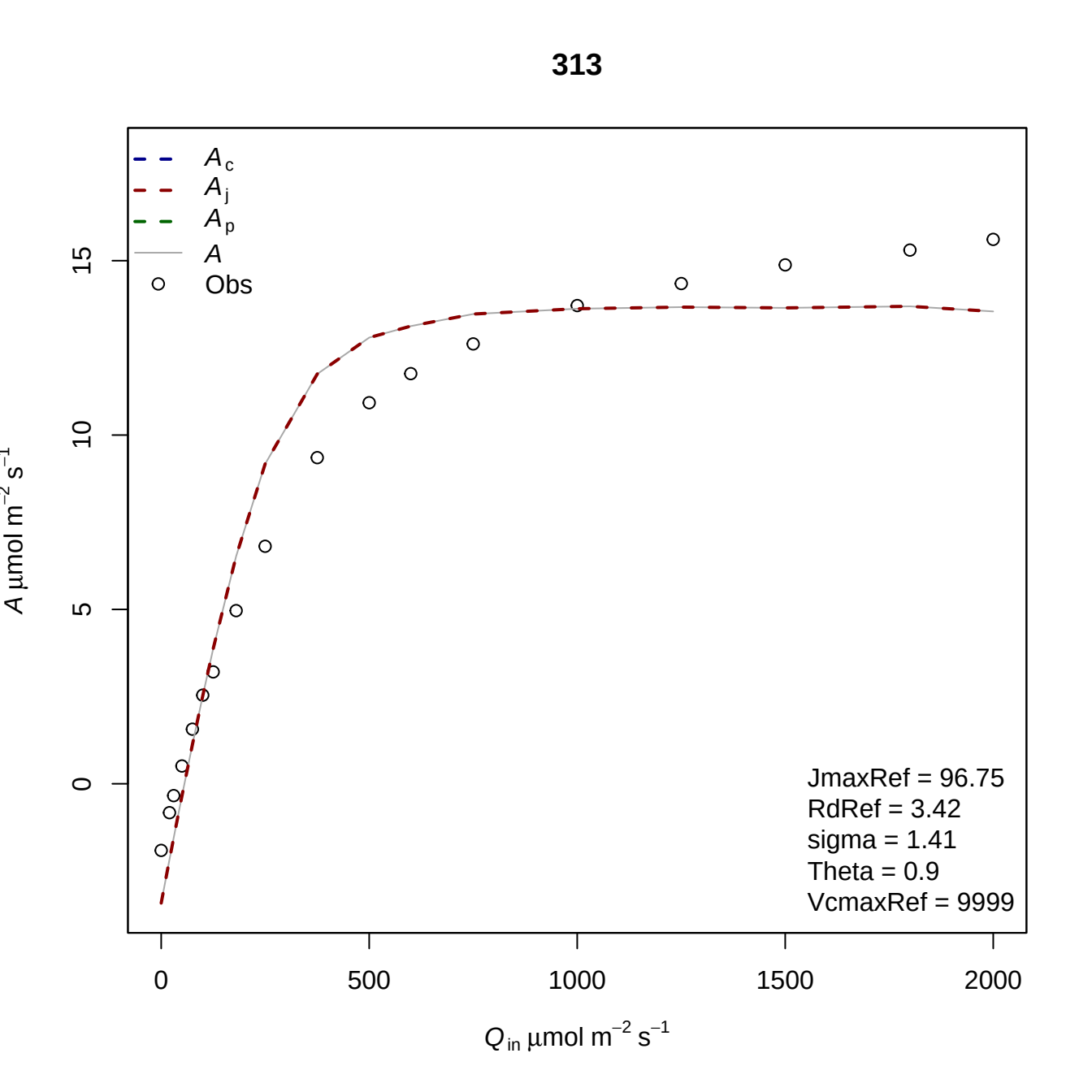

308
A_c
A_j
A_p
A
Obs
JmaxRef = 51.37
RdRef = 3.07
sigma = 1.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

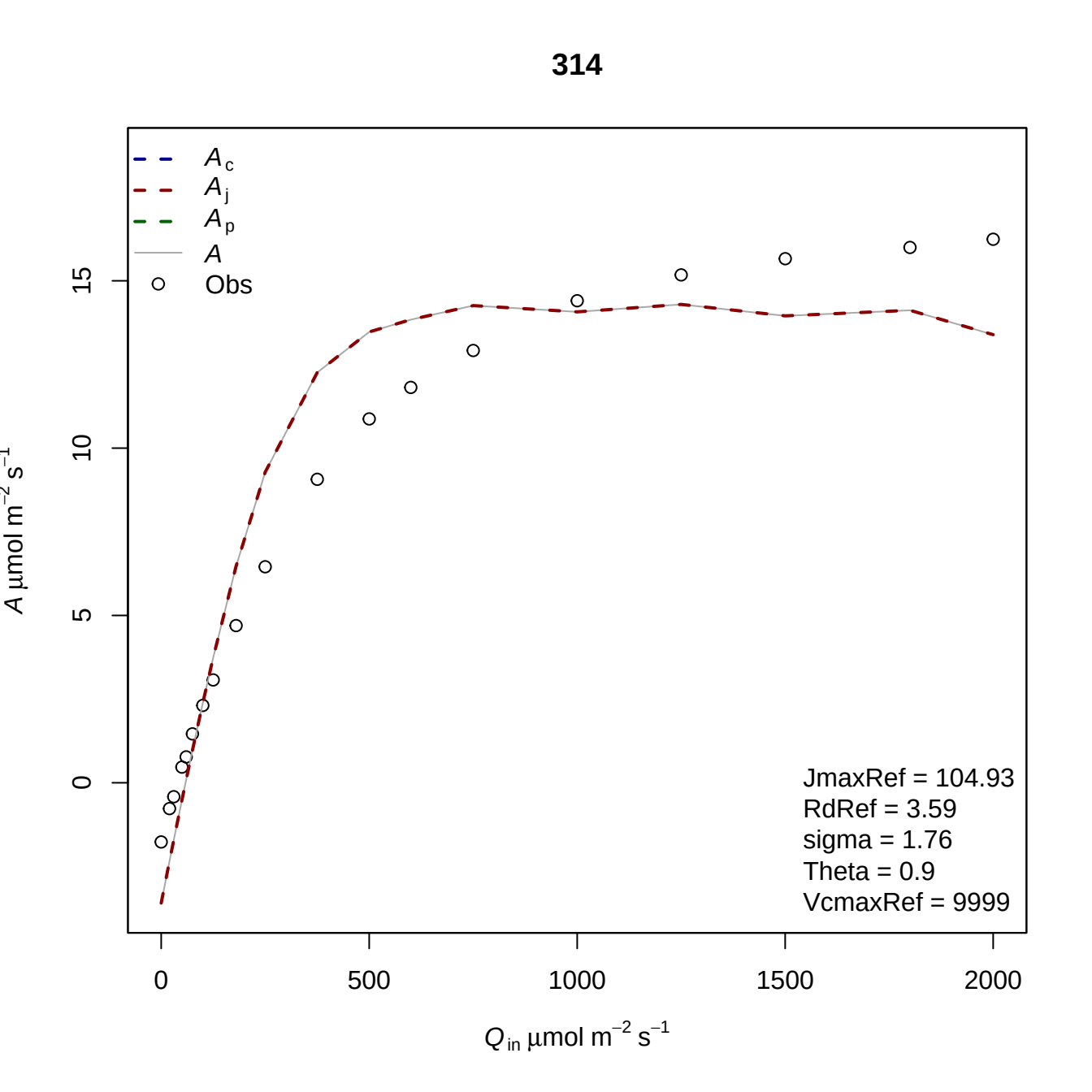

309
A_c
A_j
A_p
A
Obs
JmaxRef = 101.8
RdRef = 4.24
sigma = 1.65
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

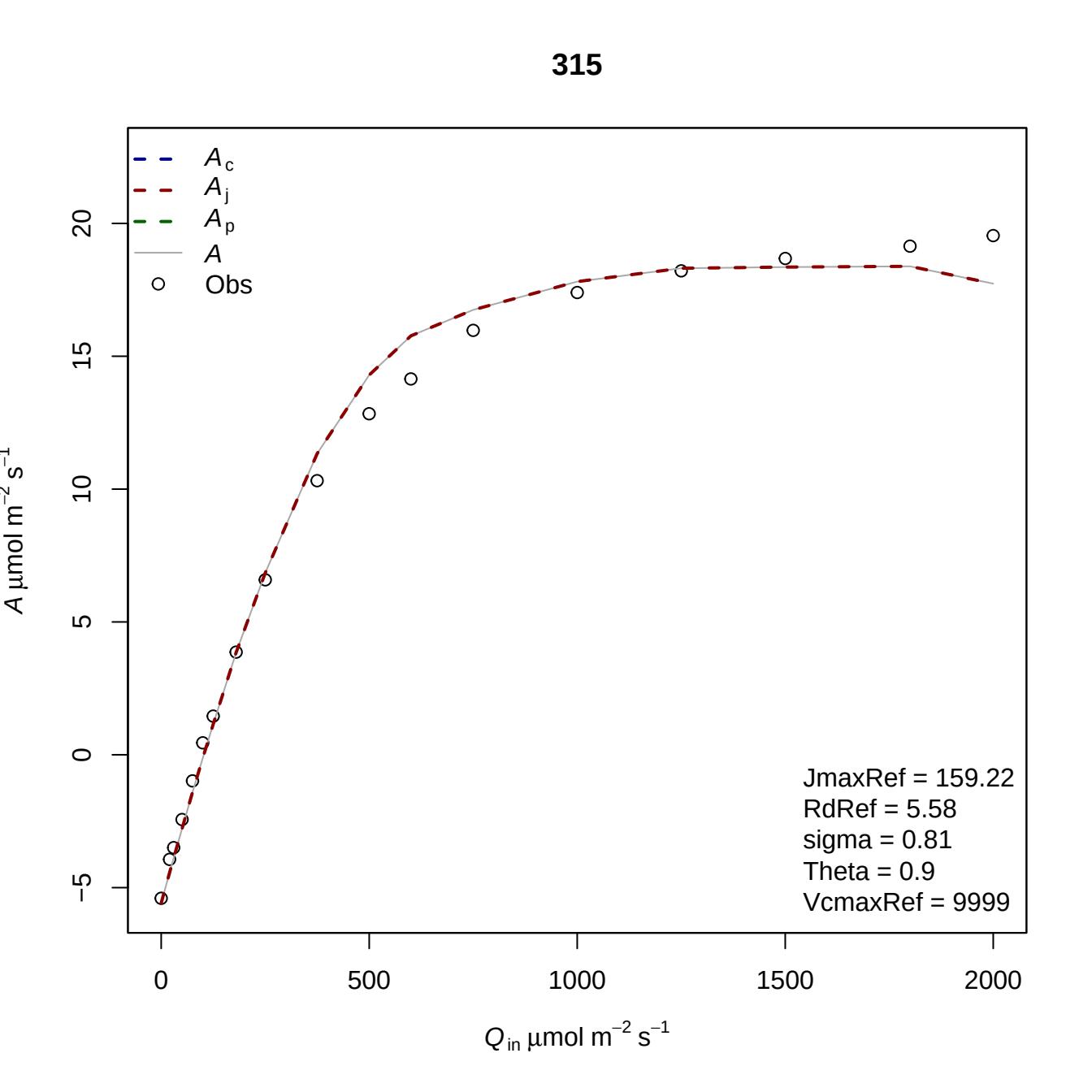

310
A_c
A_j
A_p
A
Obs
JmaxRef = 96.62
RdRef = 2.83
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

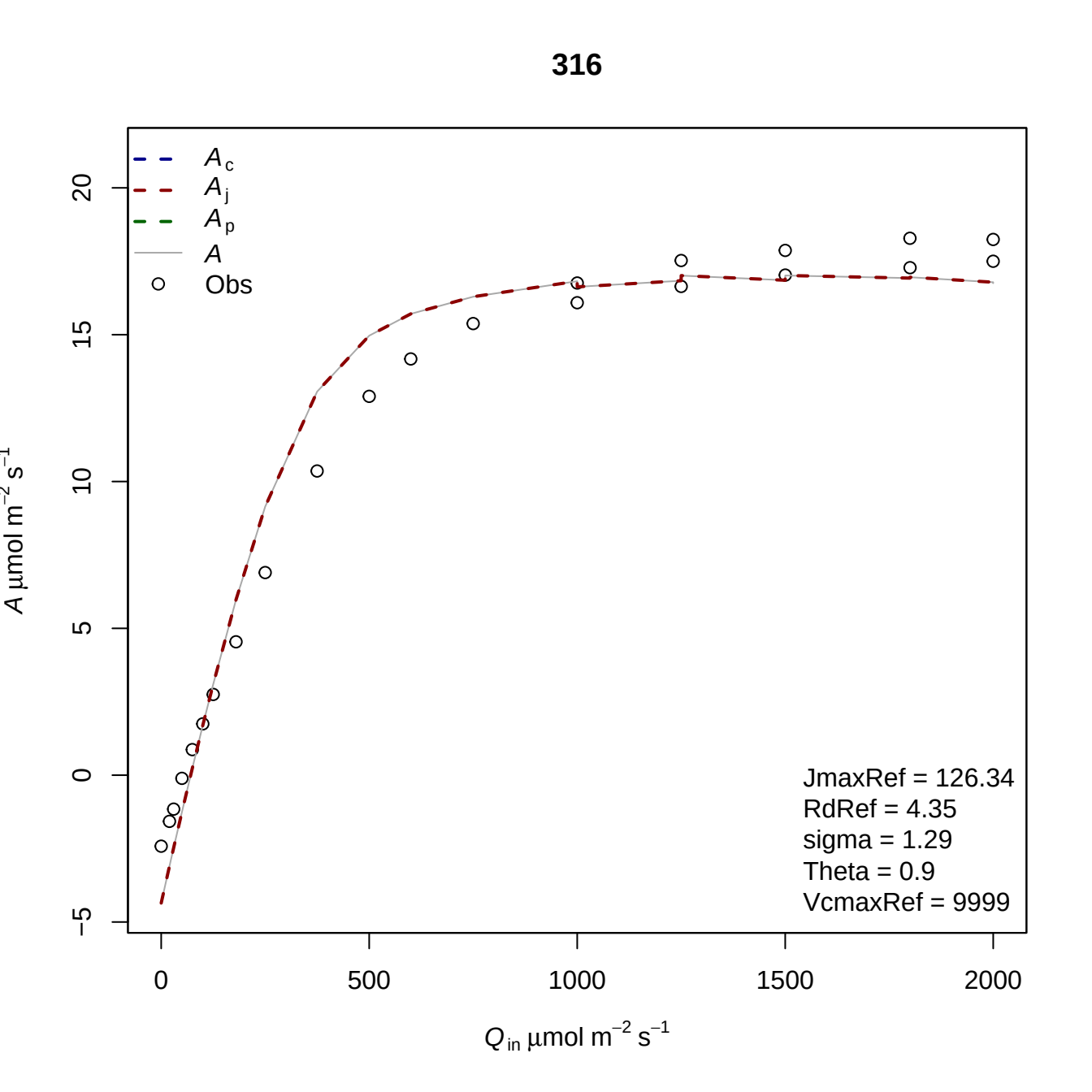

311
A_c
A_j
A_p
A
Obs
JmaxRef = 106.97
RdRef = 4.32
sigma = 2.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

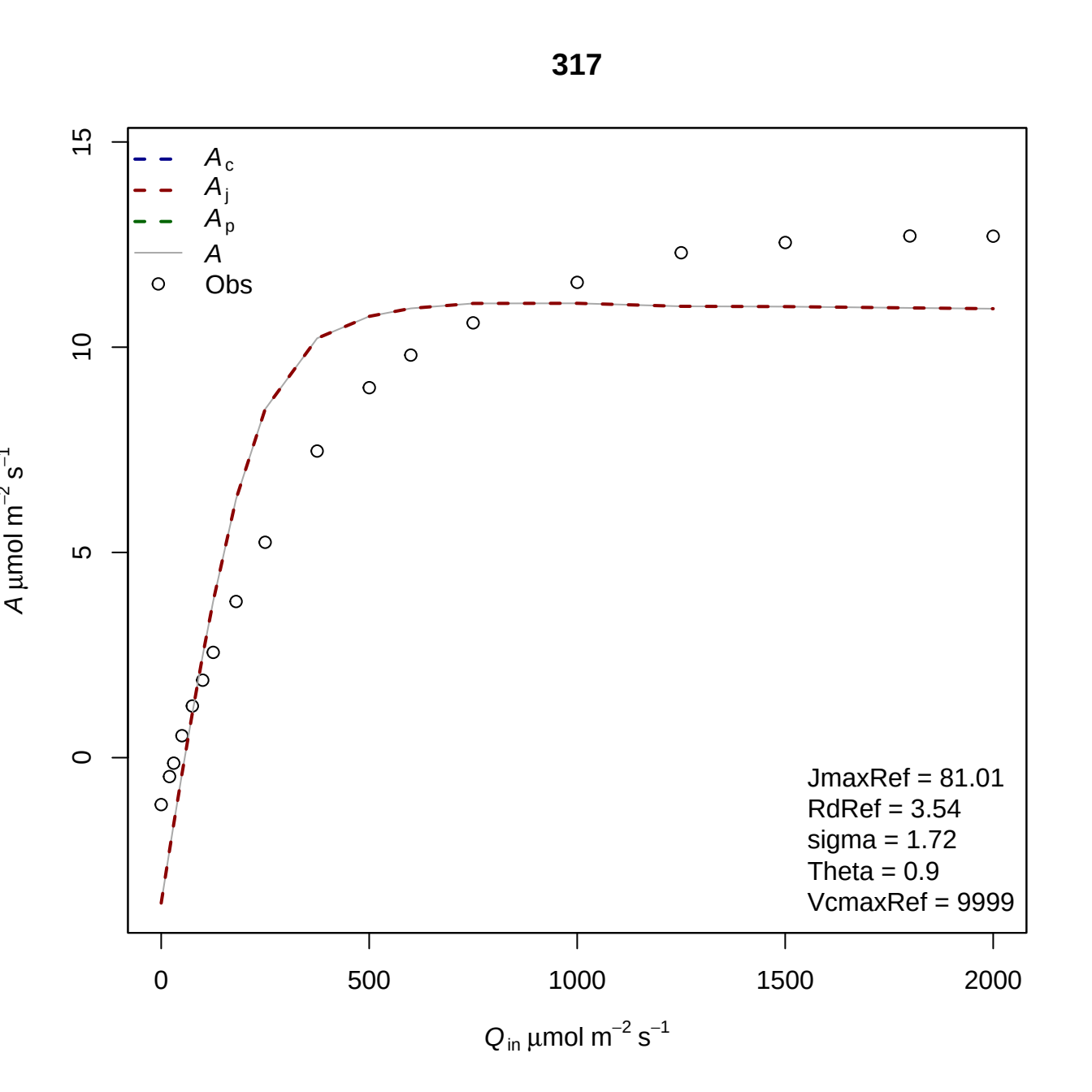

312
A_c
A_j
A_p
A
Obs
JmaxRef = 120.27
RdRef = 4.02
sigma = 1.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

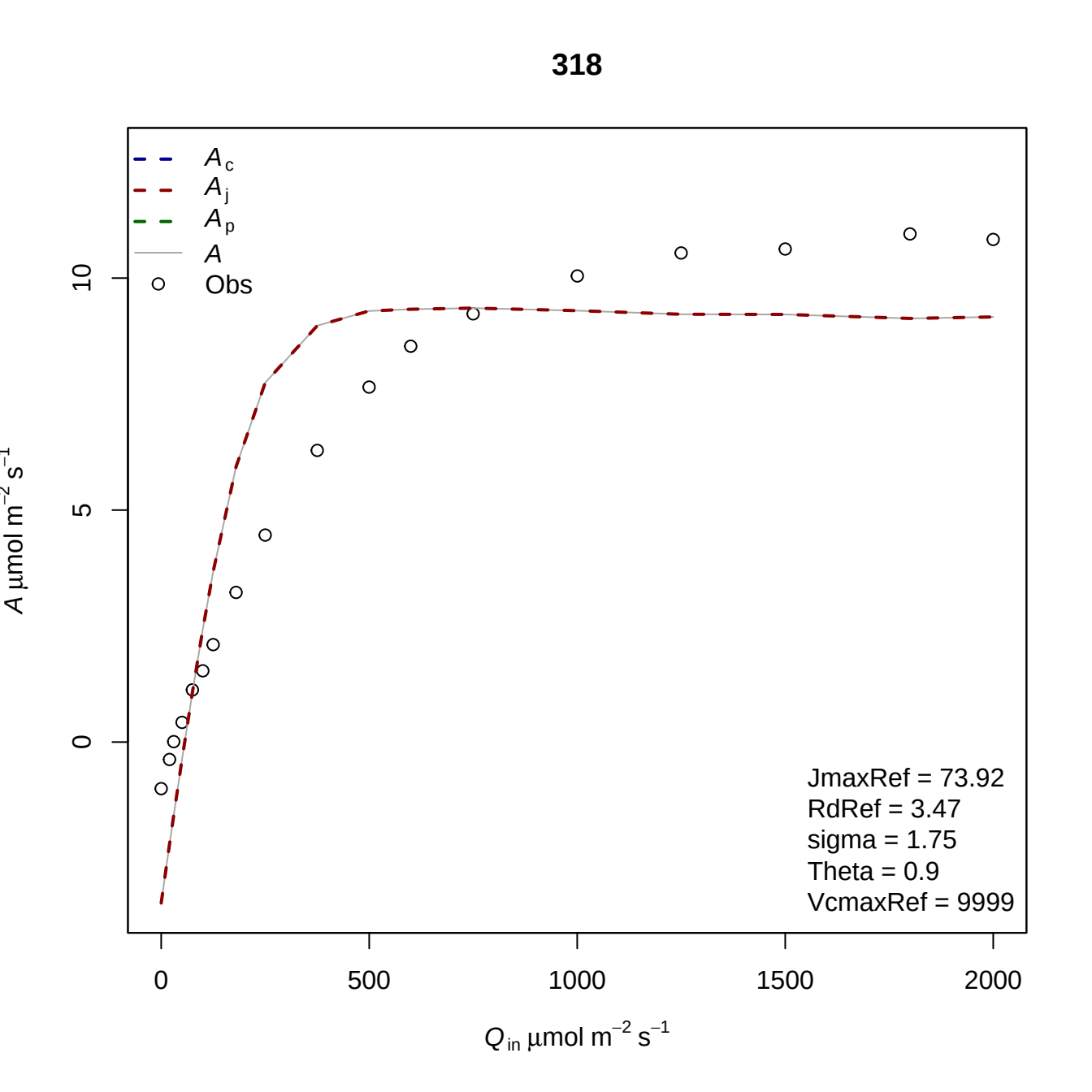

313
A_c
A_j
A_p
A
Obs
JmaxRef = 96.75
RdRef = 3.42
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

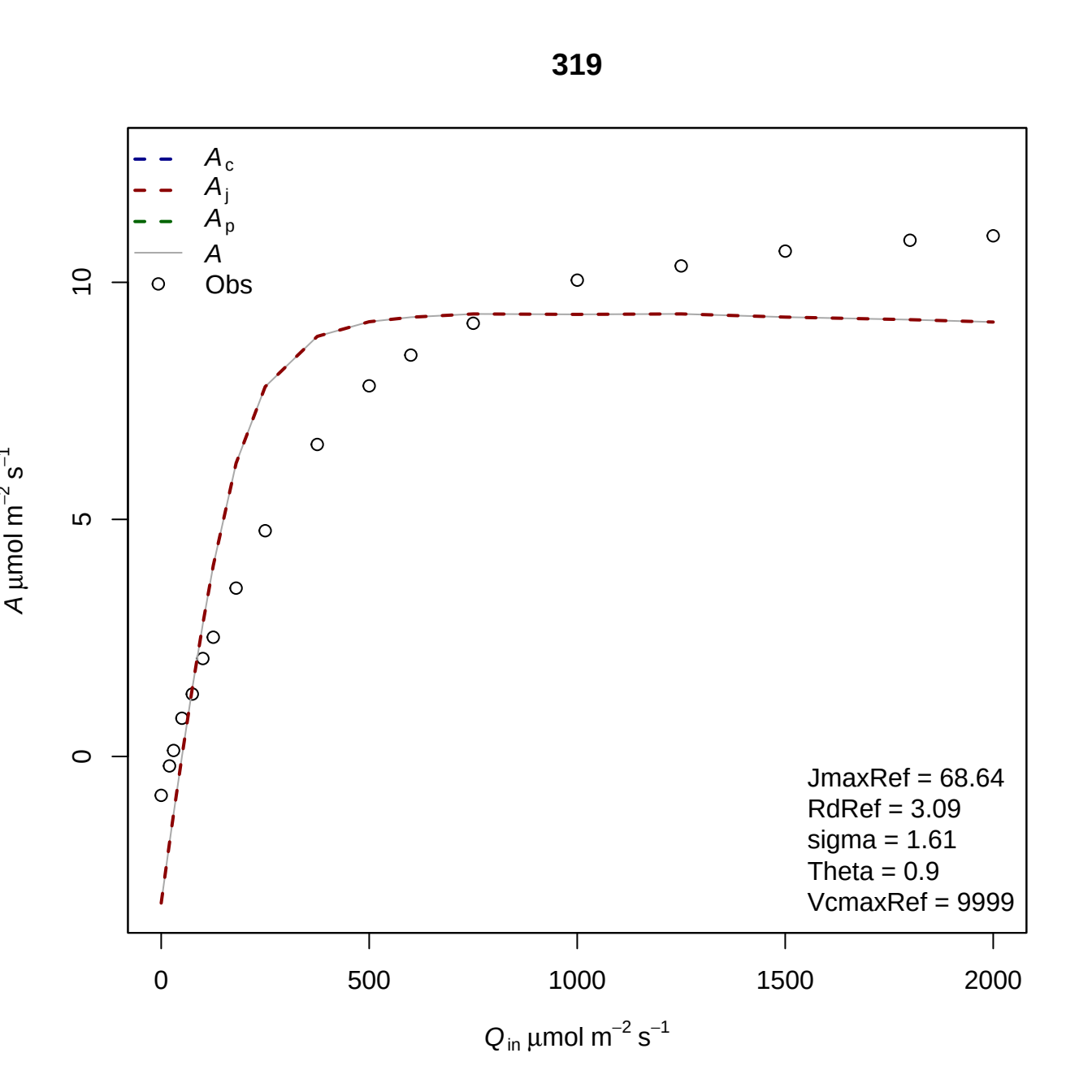

314
A_c
A_j
A_p
A
Obs
JmaxRef = 104.93
RdRef = 3.59
sigma = 1.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

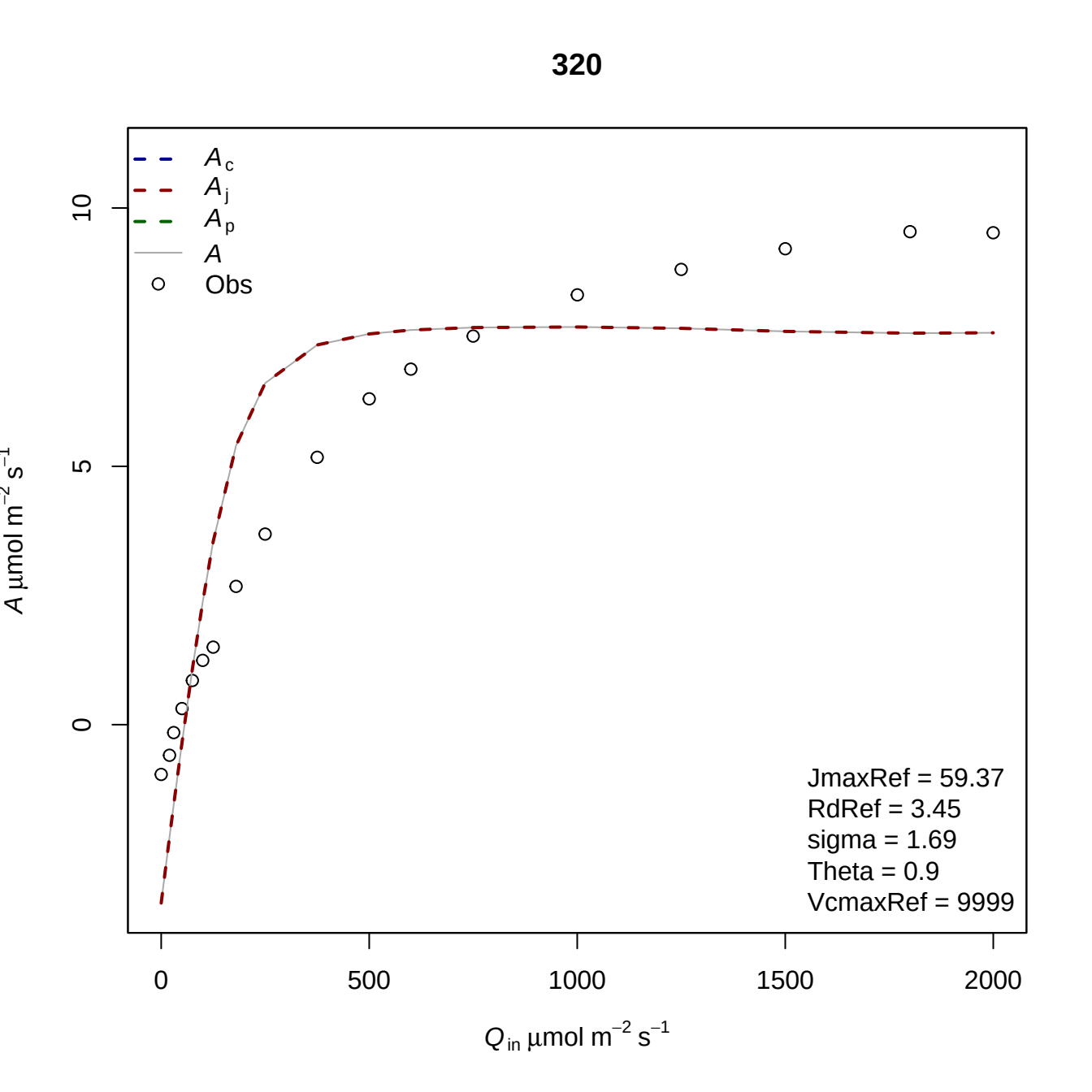

315
A_c
A_j
A_p
A
Obs
JmaxRef = 159.22
RdRef = 5.58
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

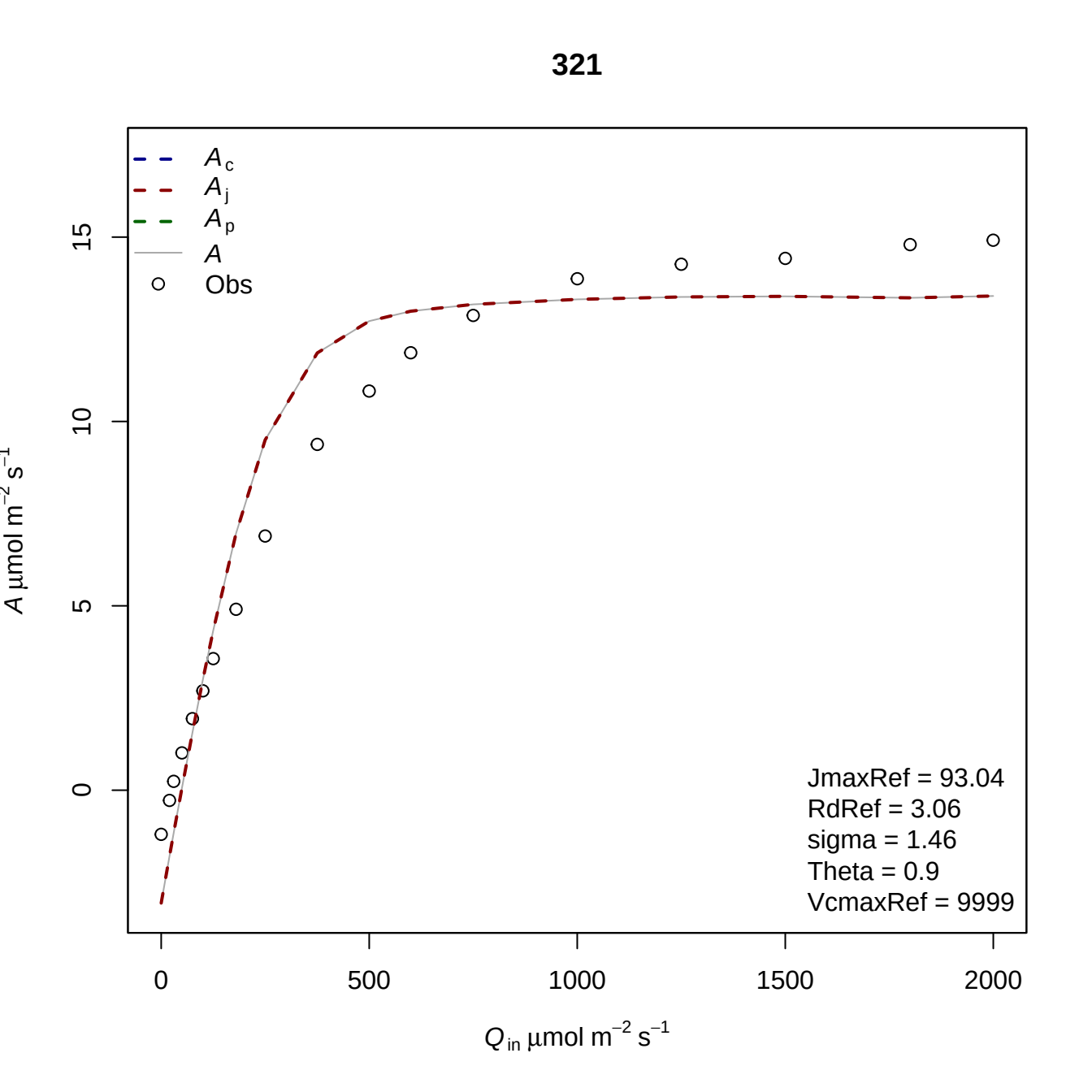

316
A_c
A_j
A_p
A
Obs
JmaxRef = 126.34
RdRef = 4.35
sigma = 1.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

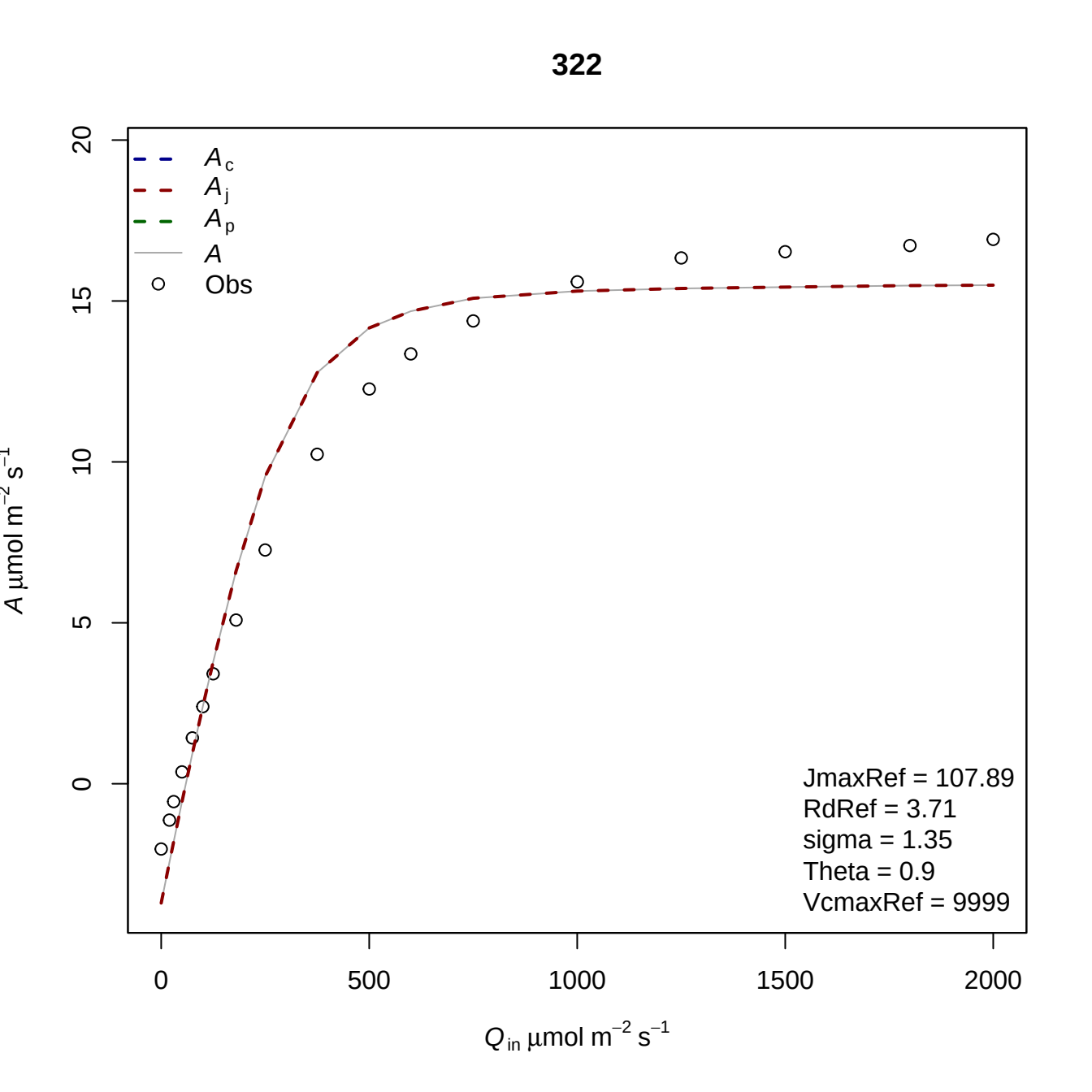

317
A_c
A_j
A_p
A
Obs
JmaxRef = 81.01
RdRef = 3.54
sigma = 1.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

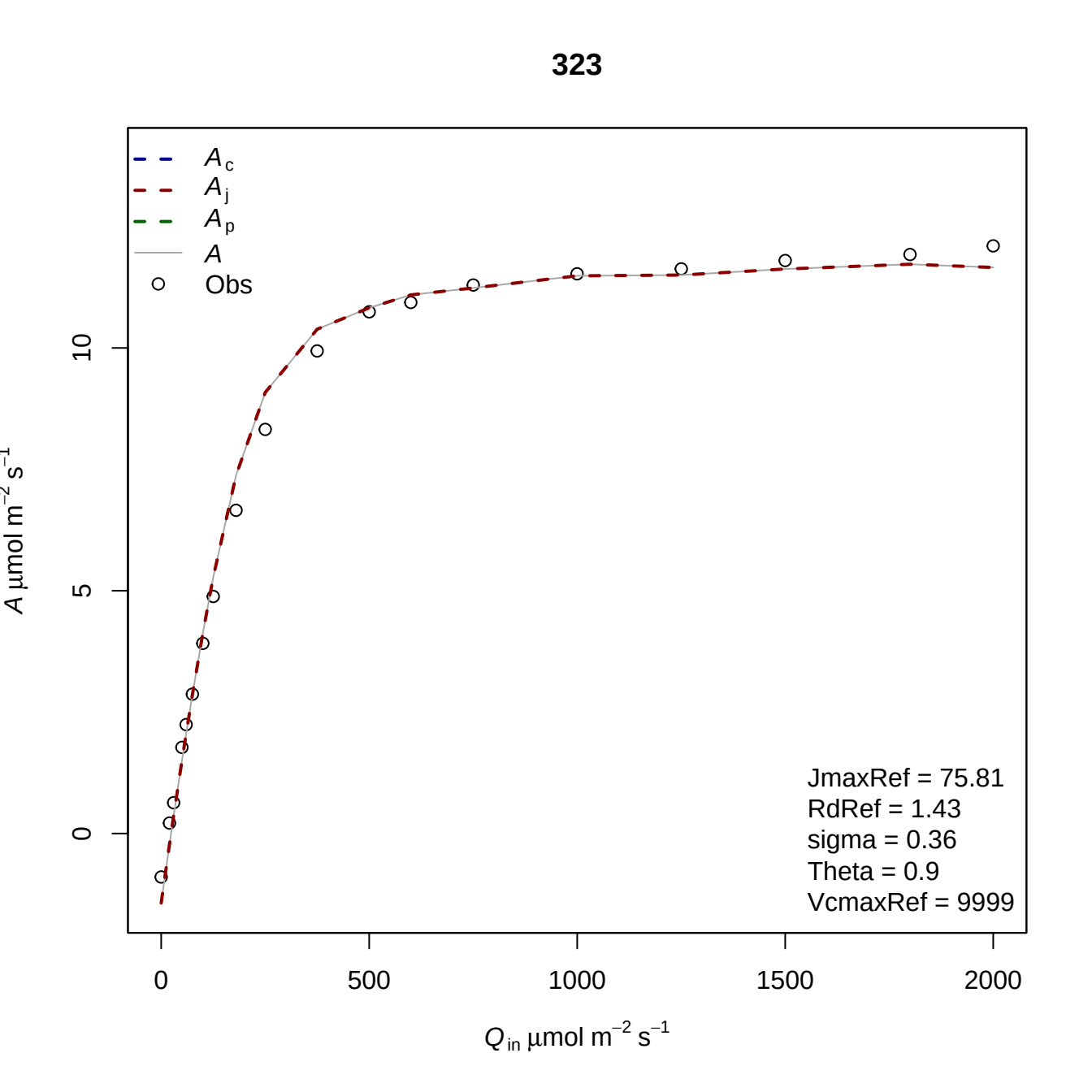

318
A_c
A_j
A_p
A
Obs
JmaxRef = 73.92
RdRef = 3.47
sigma = 1.75
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

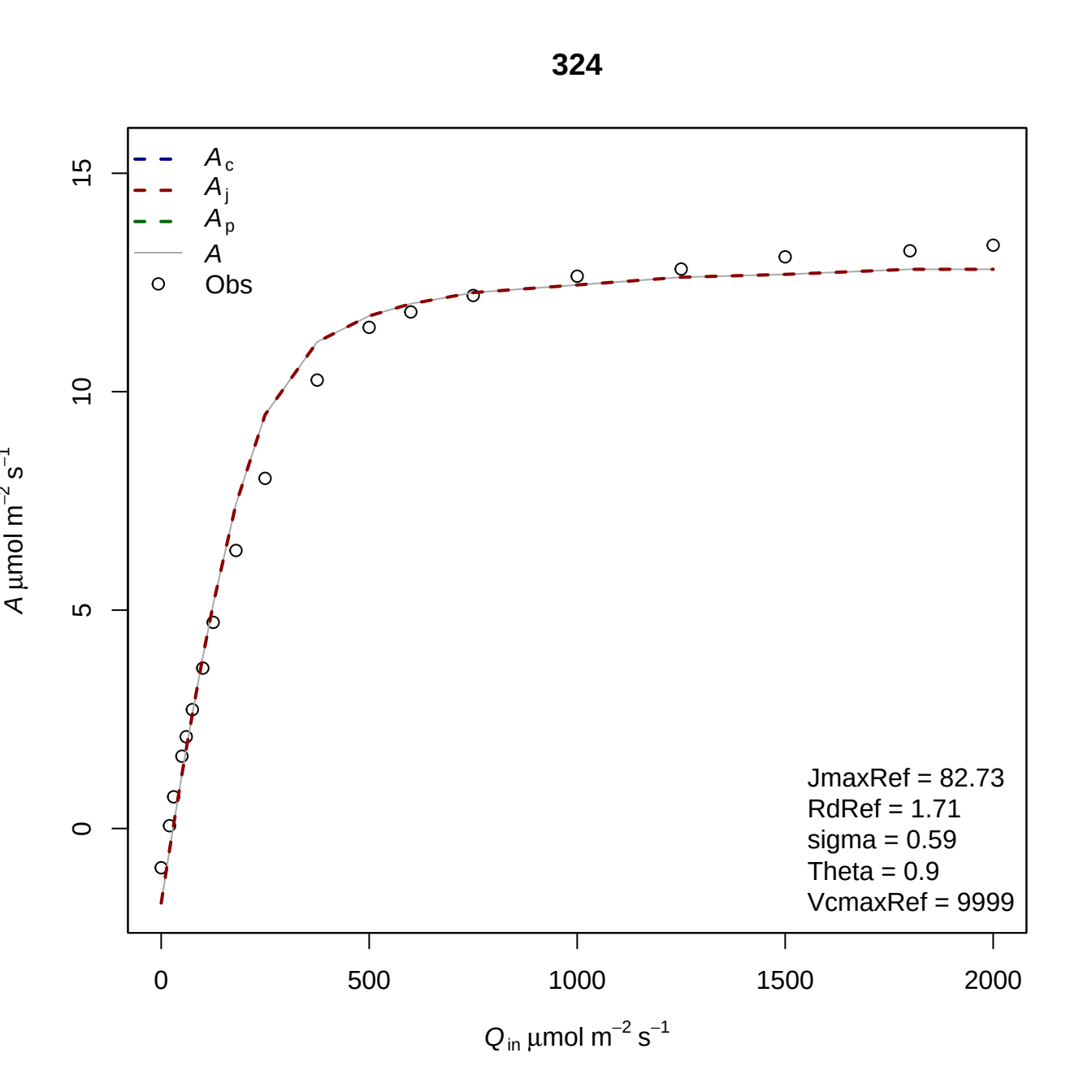

319
A_c
A_j
A_p
A
Obs
JmaxRef = 68.64
RdRef = 3.09
sigma = 1.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

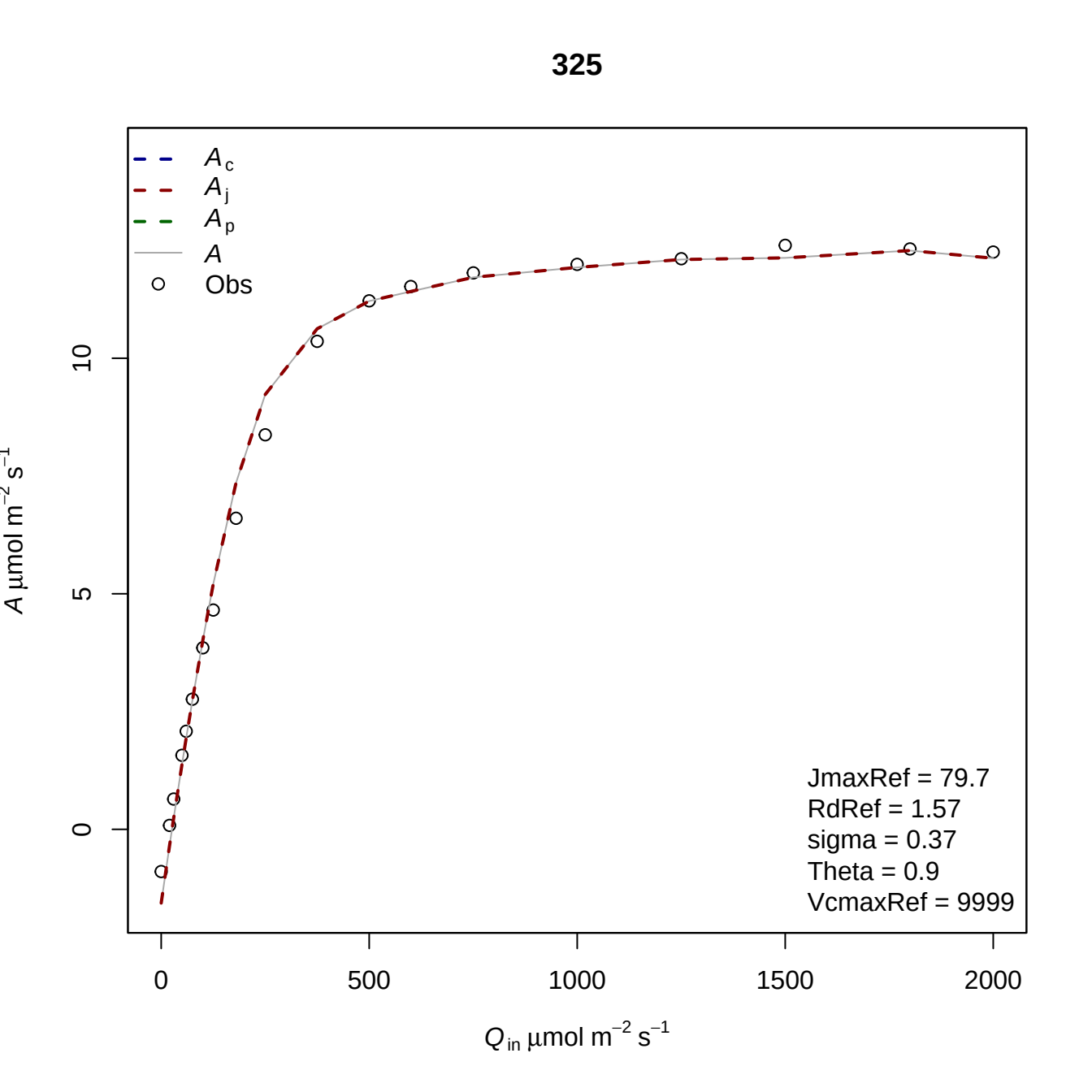

320
A_c
A_j
A_p
A
Obs
JmaxRef = 59.37
RdRef = 3.45
sigma = 1.69
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

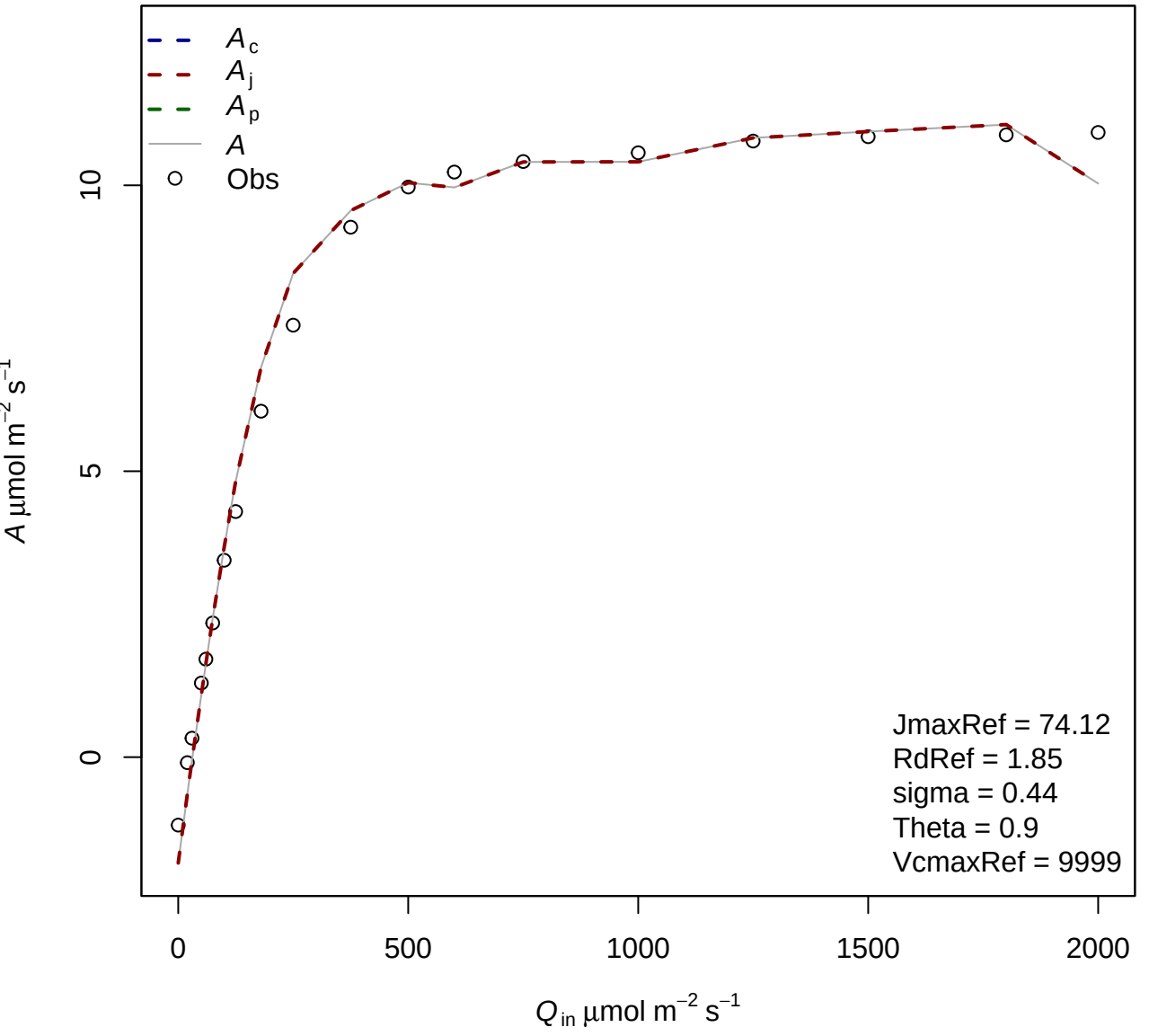

321
A_c
A_j
A_p
A
Obs
JmaxRef = 93.04
RdRef = 3.06
sigma = 1.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

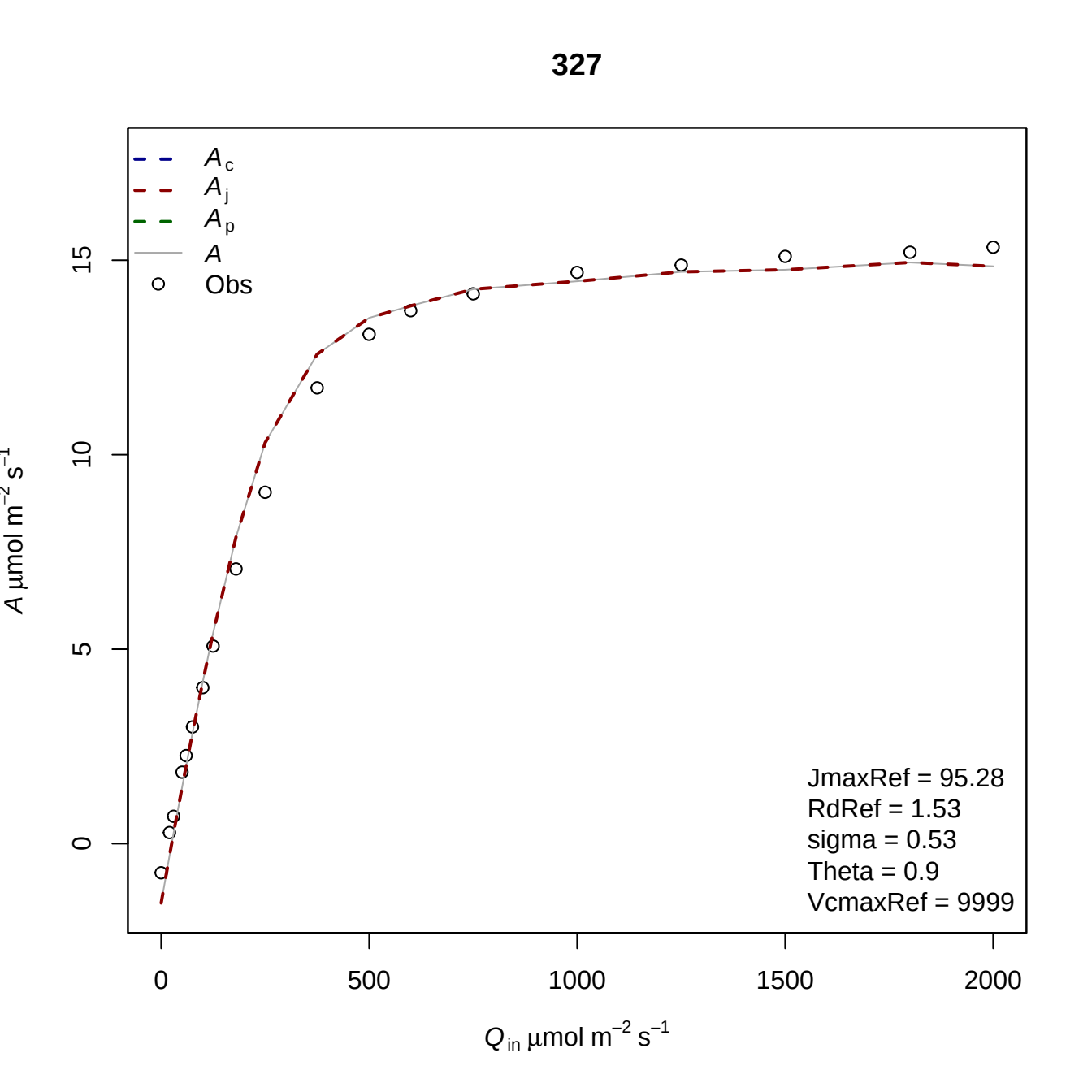

322
A_c
A_j
A_p
A
Obs
JmaxRef = 107.89
RdRef = 3.71
sigma = 1.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

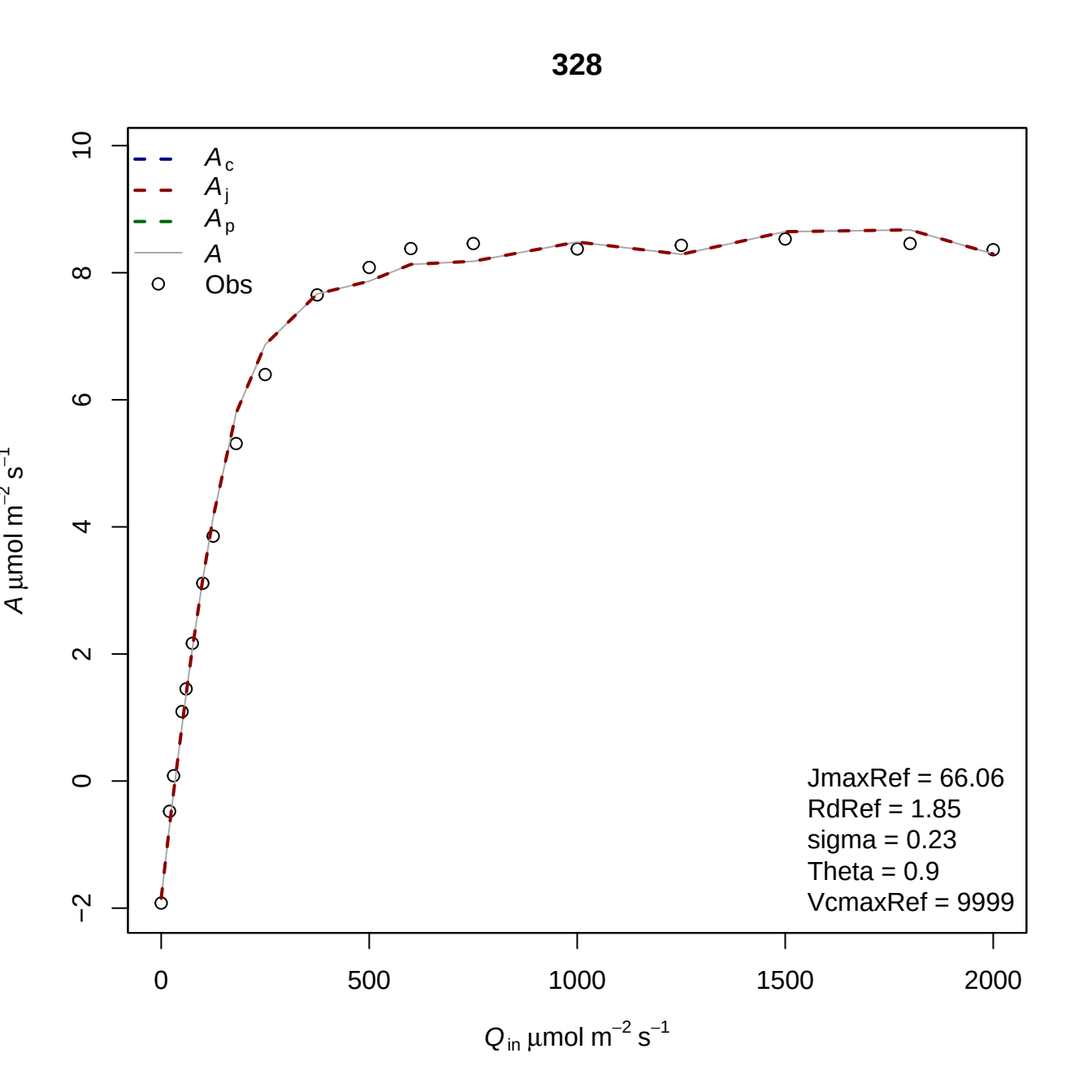

323
A_c
A_j
A_p
A
Obs
JmaxRef = 75.81
RdRef = 1.43
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

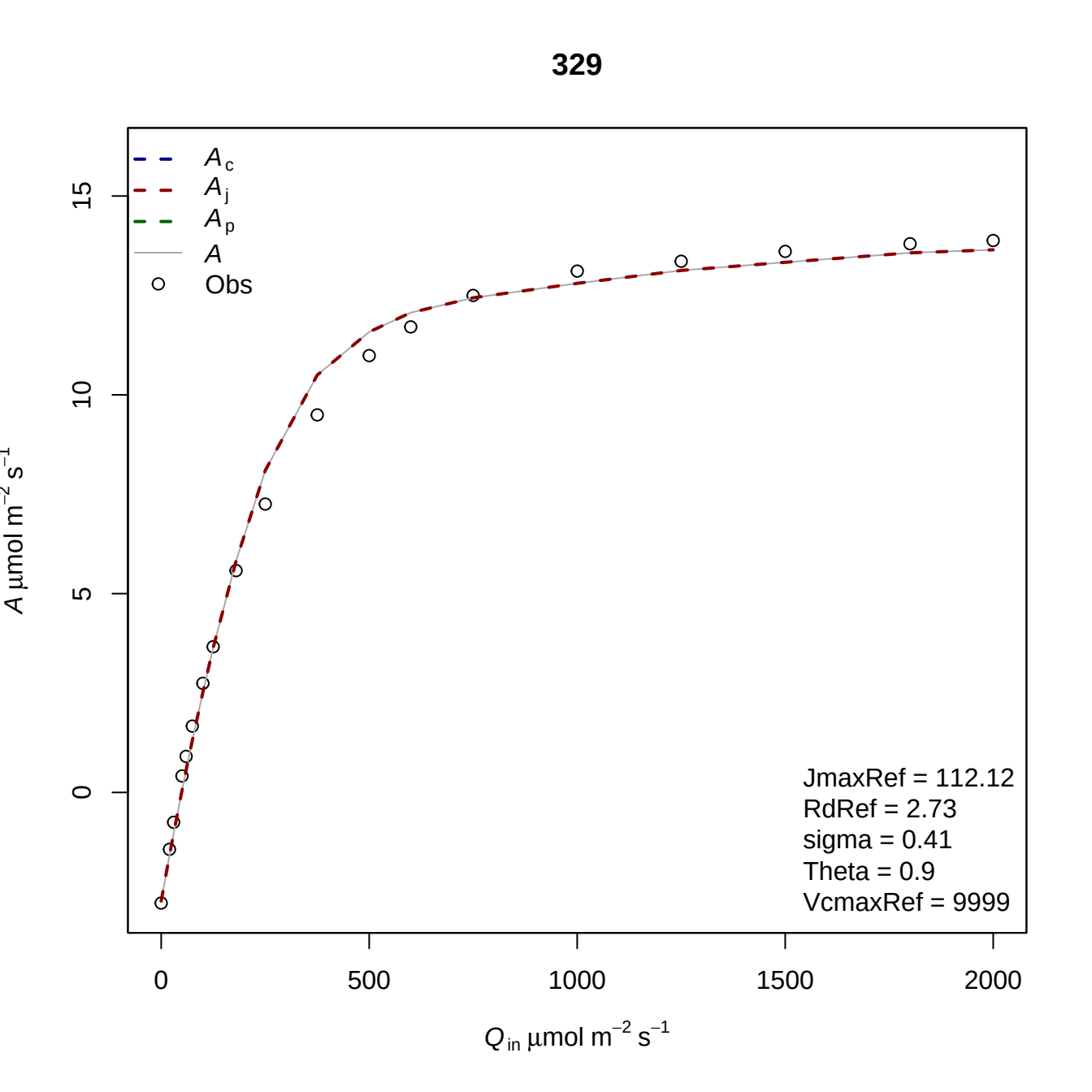

324
A_c
A_j
A_p
A
Obs
JmaxRef = 82.73
RdRef = 1.71
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

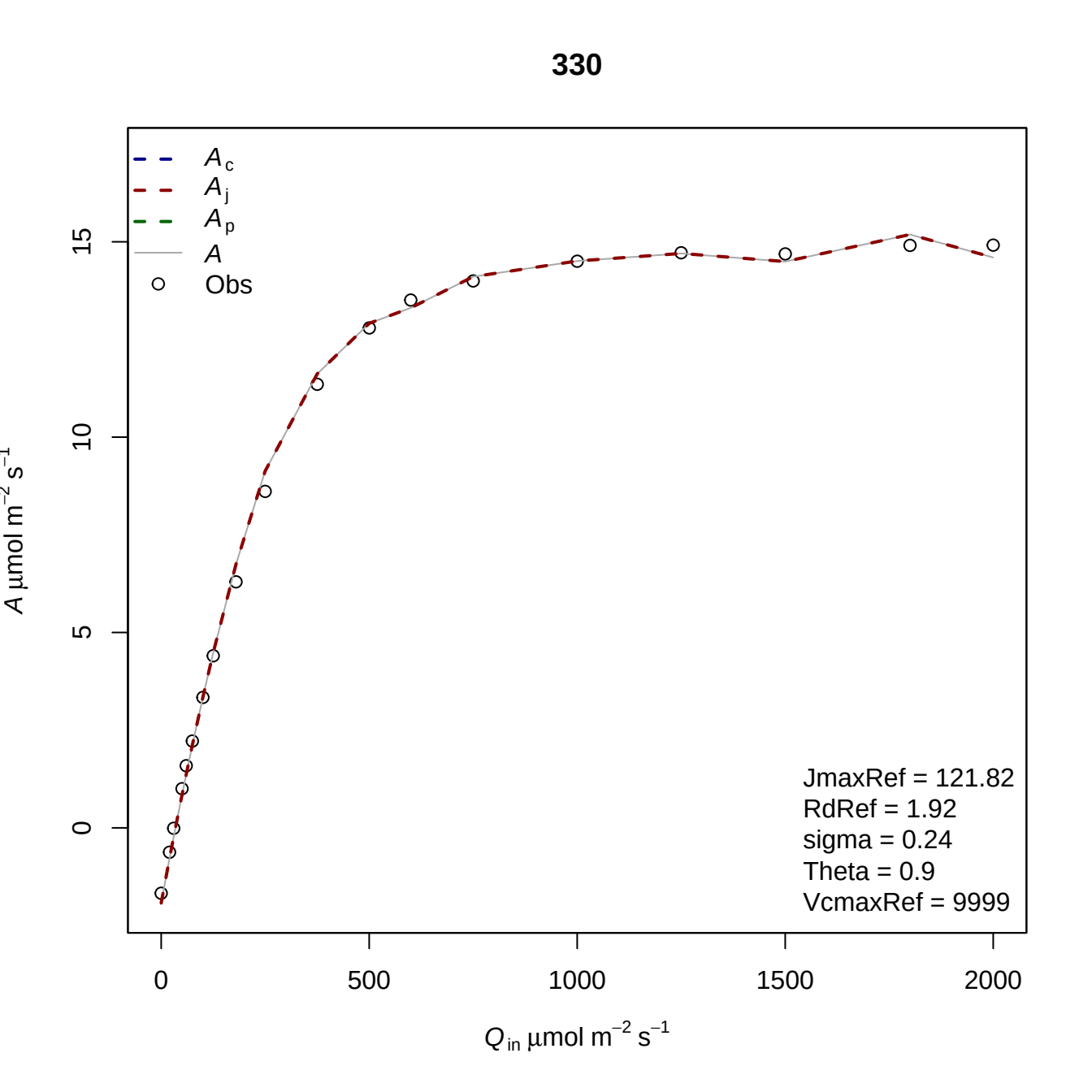

325
A_c
A_j
A_p
A
Obs
JmaxRef = 79.7
RdRef = 1.57
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

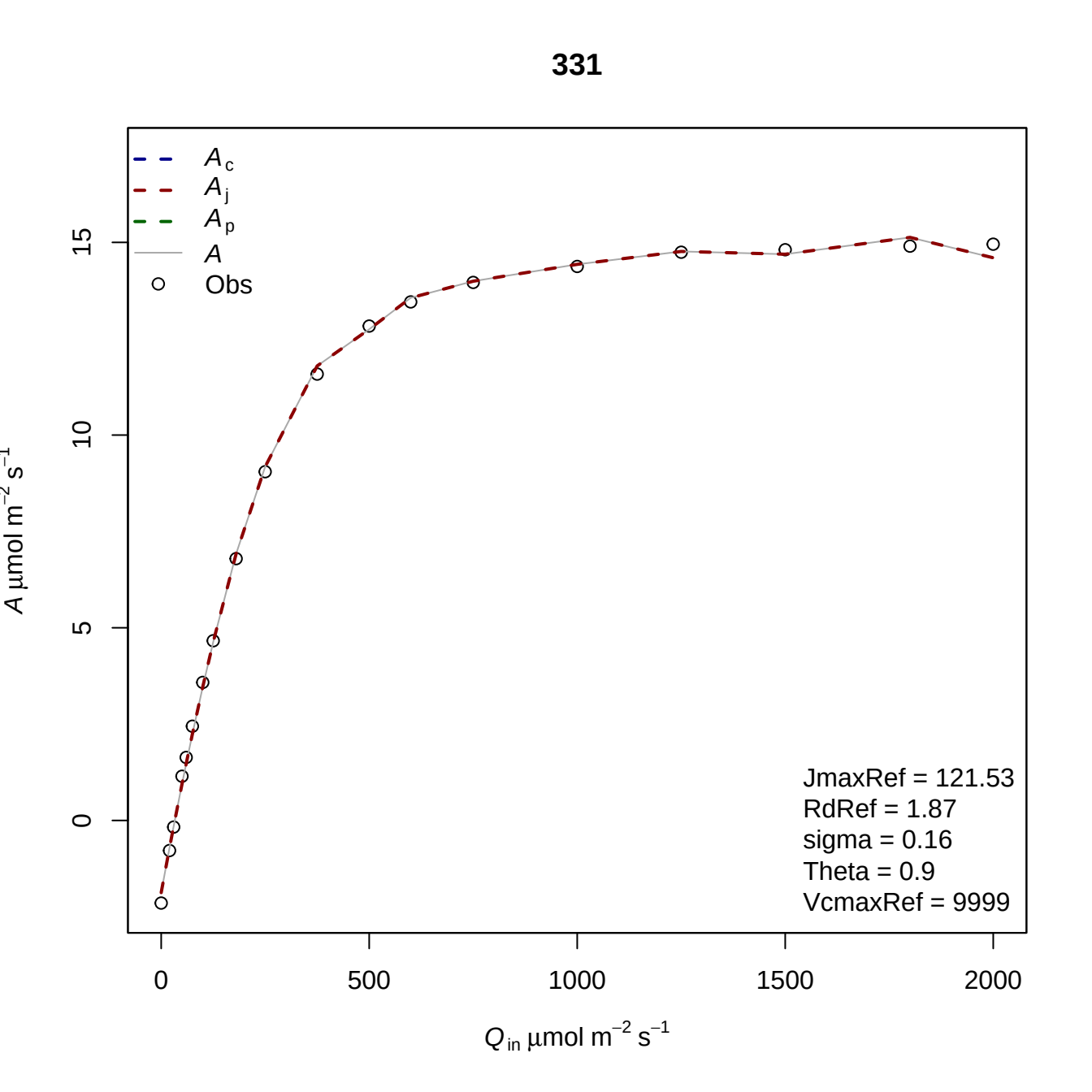

A_c
A_j
A_p
A
Obs
JmaxRef = 74.12
RdRef = 1.85
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

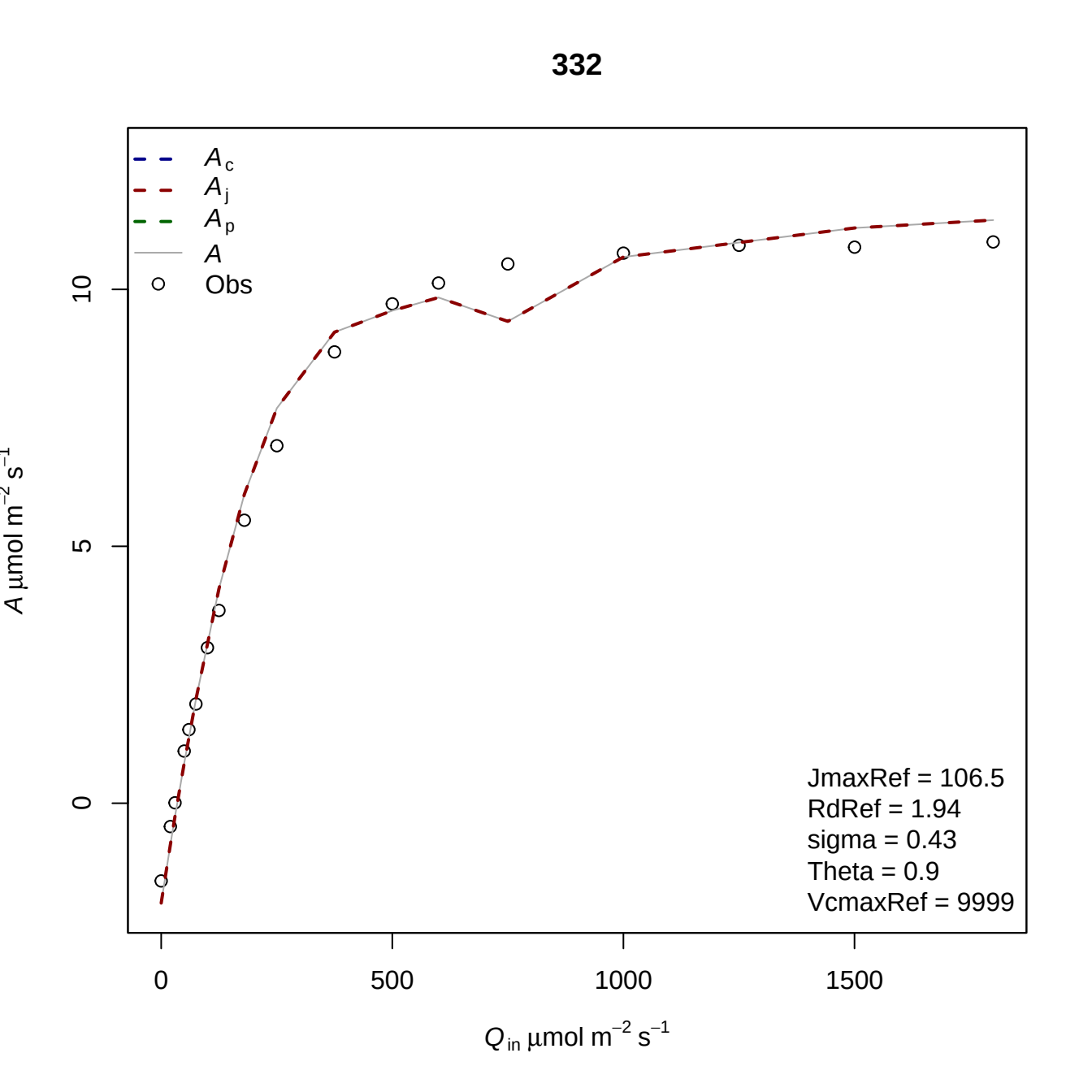

327
A_c
A_j
A_p
A
Obs
JmaxRef = 95.28
RdRef = 1.53
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

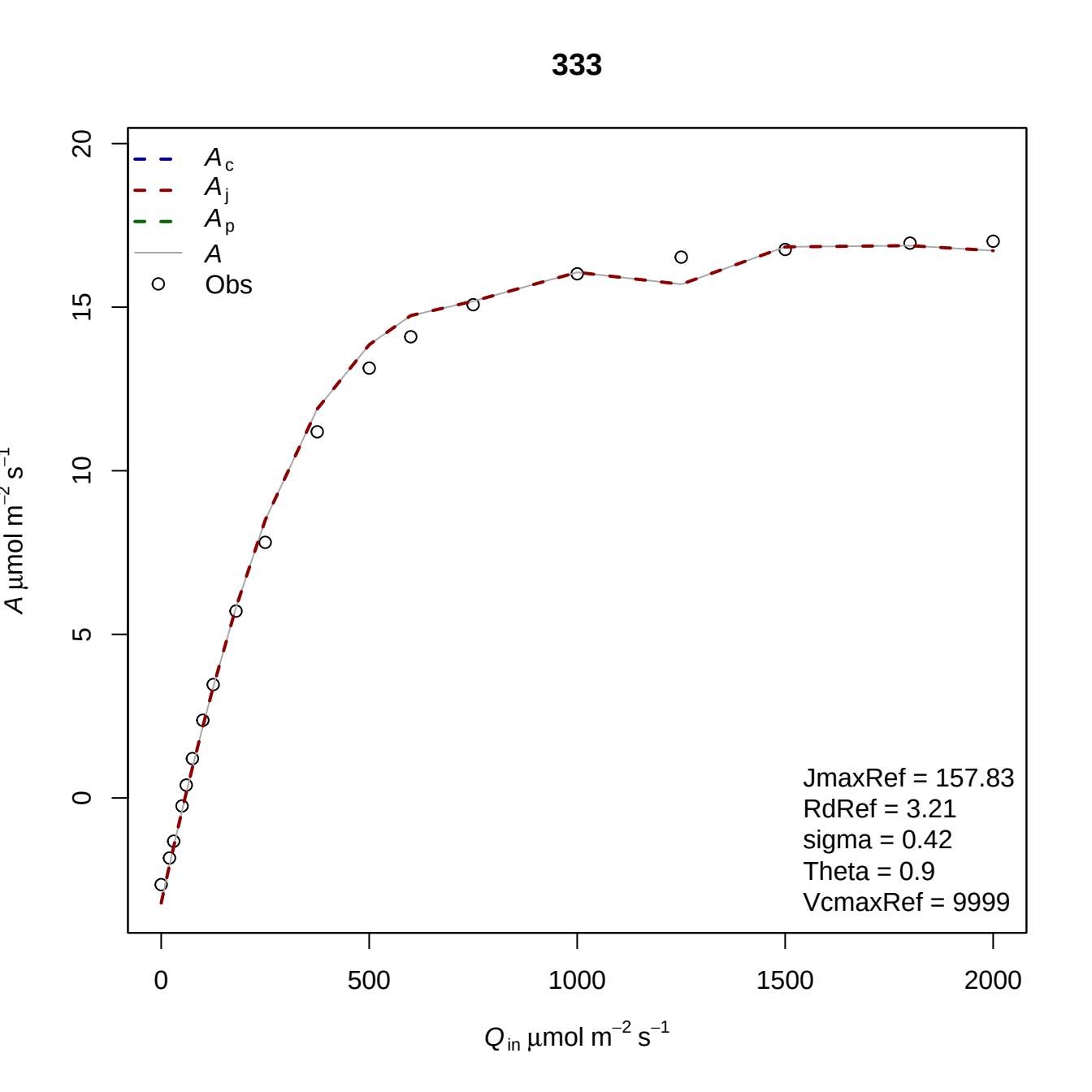

328
A_c
A_j
A_p
A
Obs
JmaxRef = 66.06
RdRef = 1.85
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

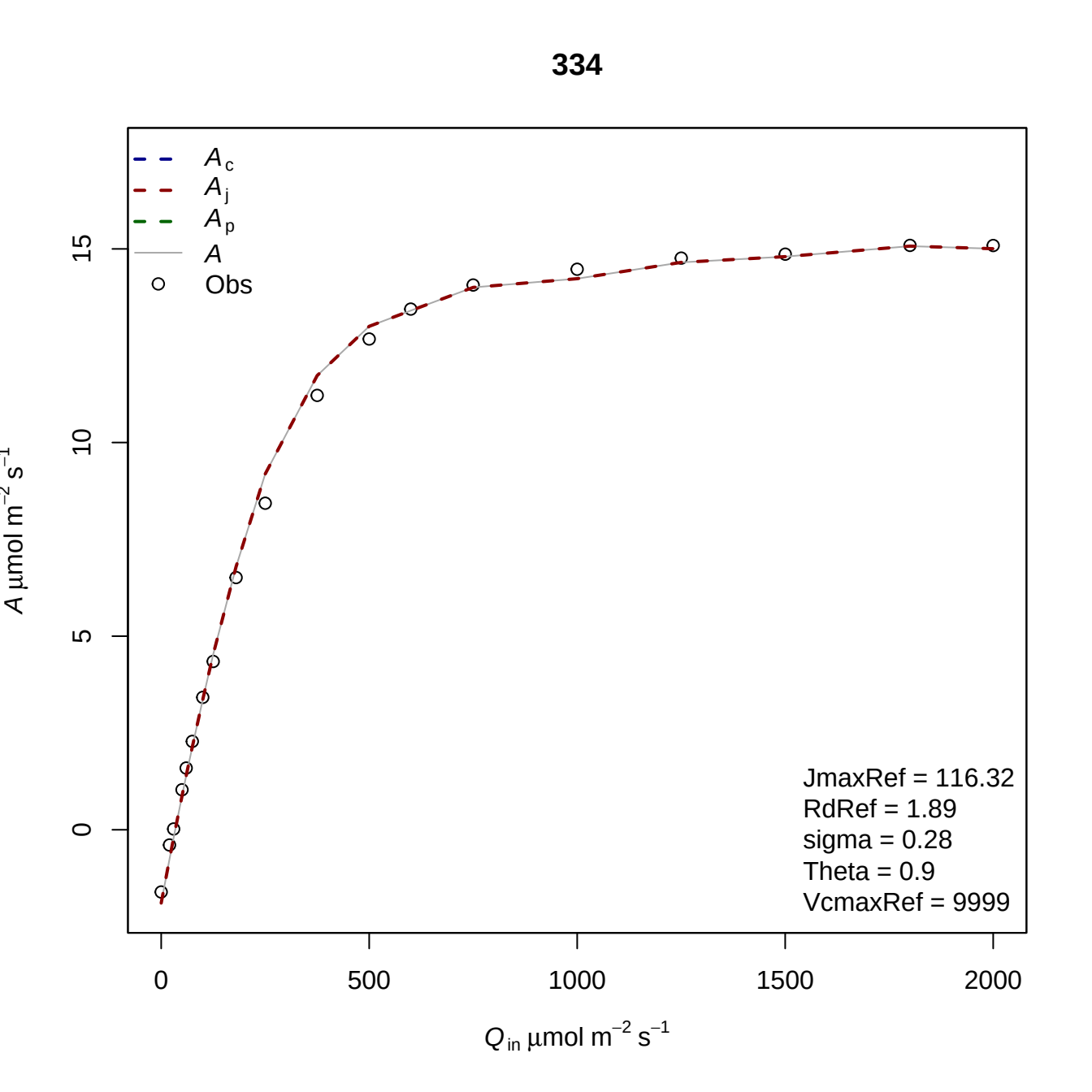

329
A_c
A_j
A_p
A
Obs
JmaxRef = 112.12
RdRef = 2.73
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

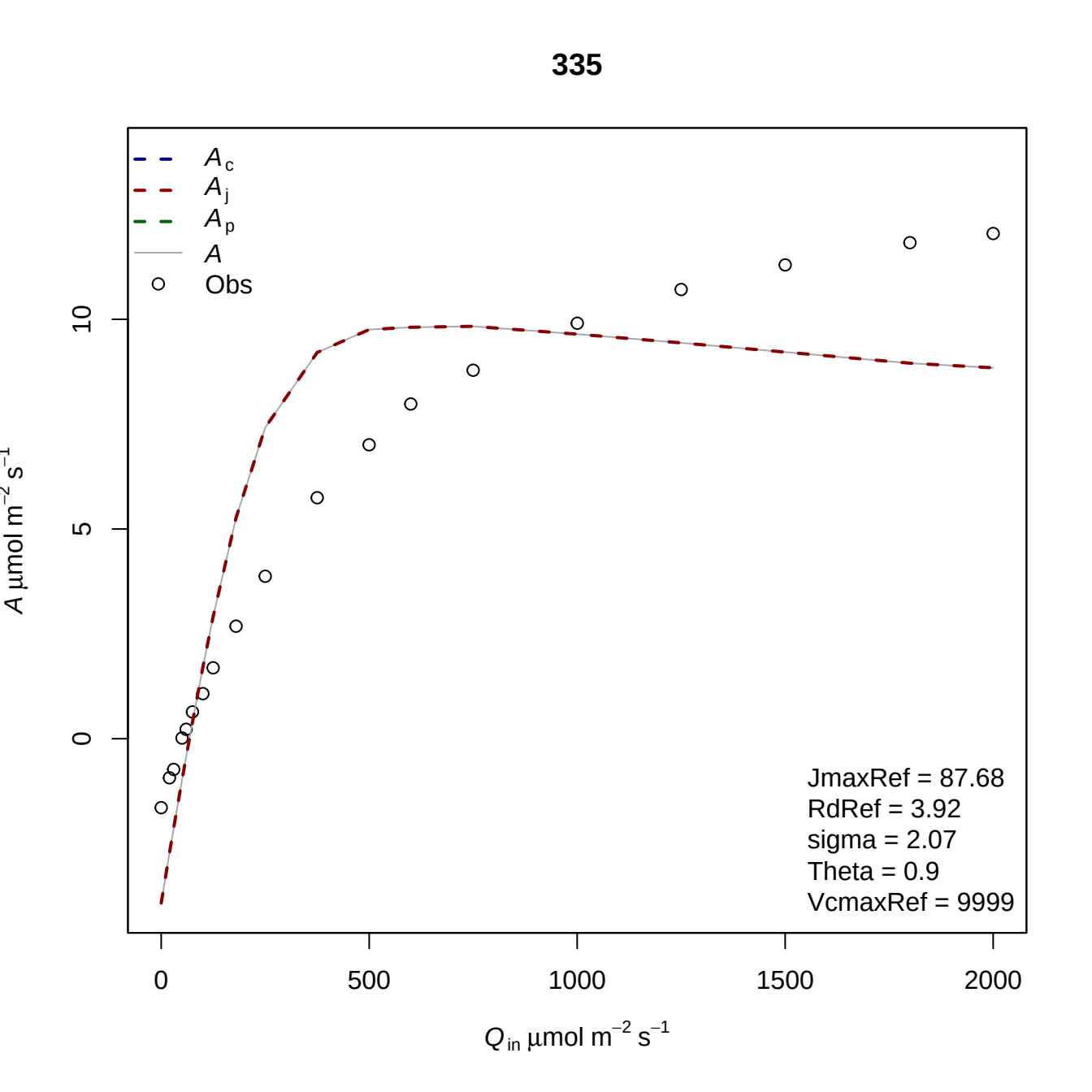

330
A_c
A_j
A_p
A
Obs
JmaxRef = 121.82
RdRef = 1.92
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

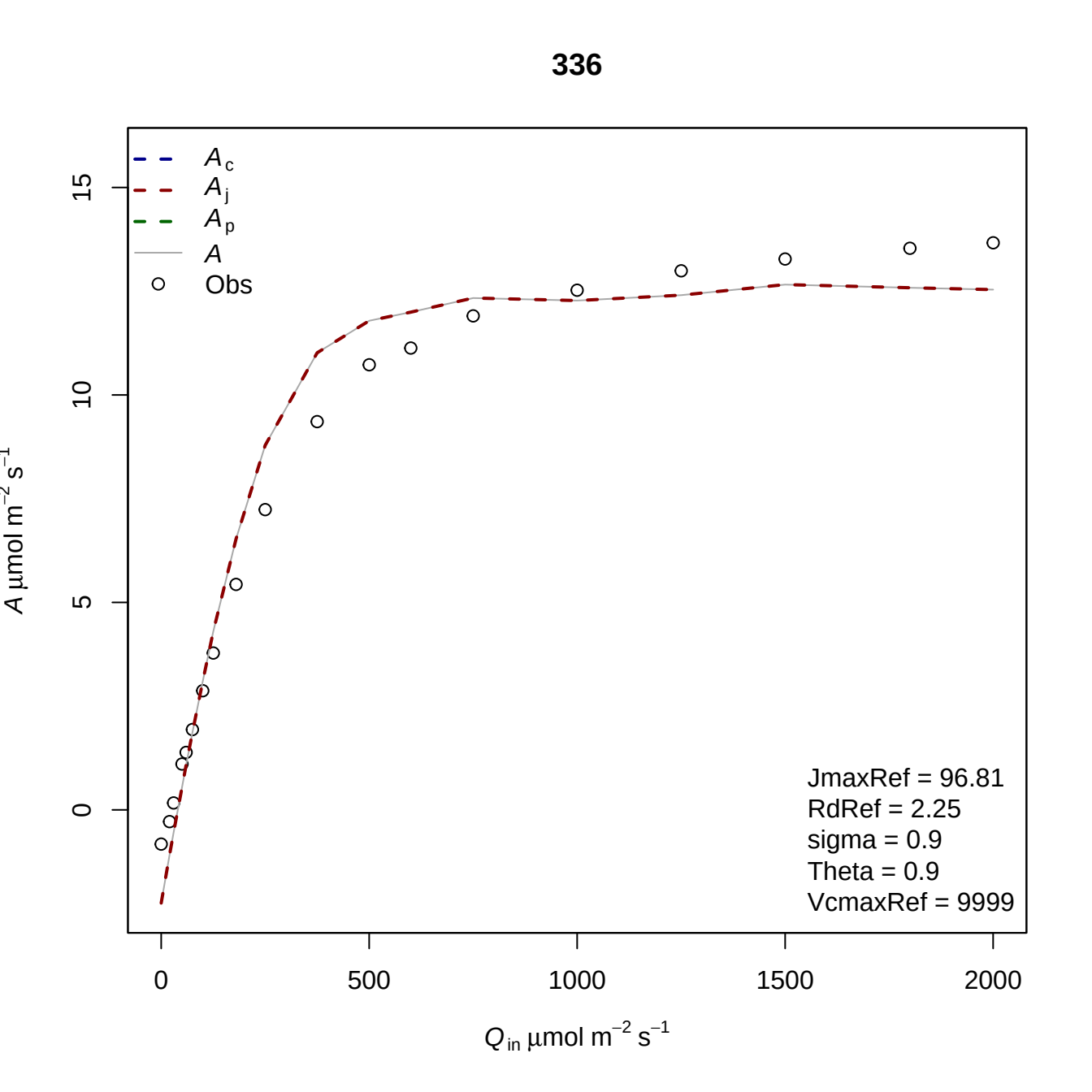

331
A_c
A_j
A_p
A
Obs
JmaxRef = 121.53
RdRef = 1.87
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

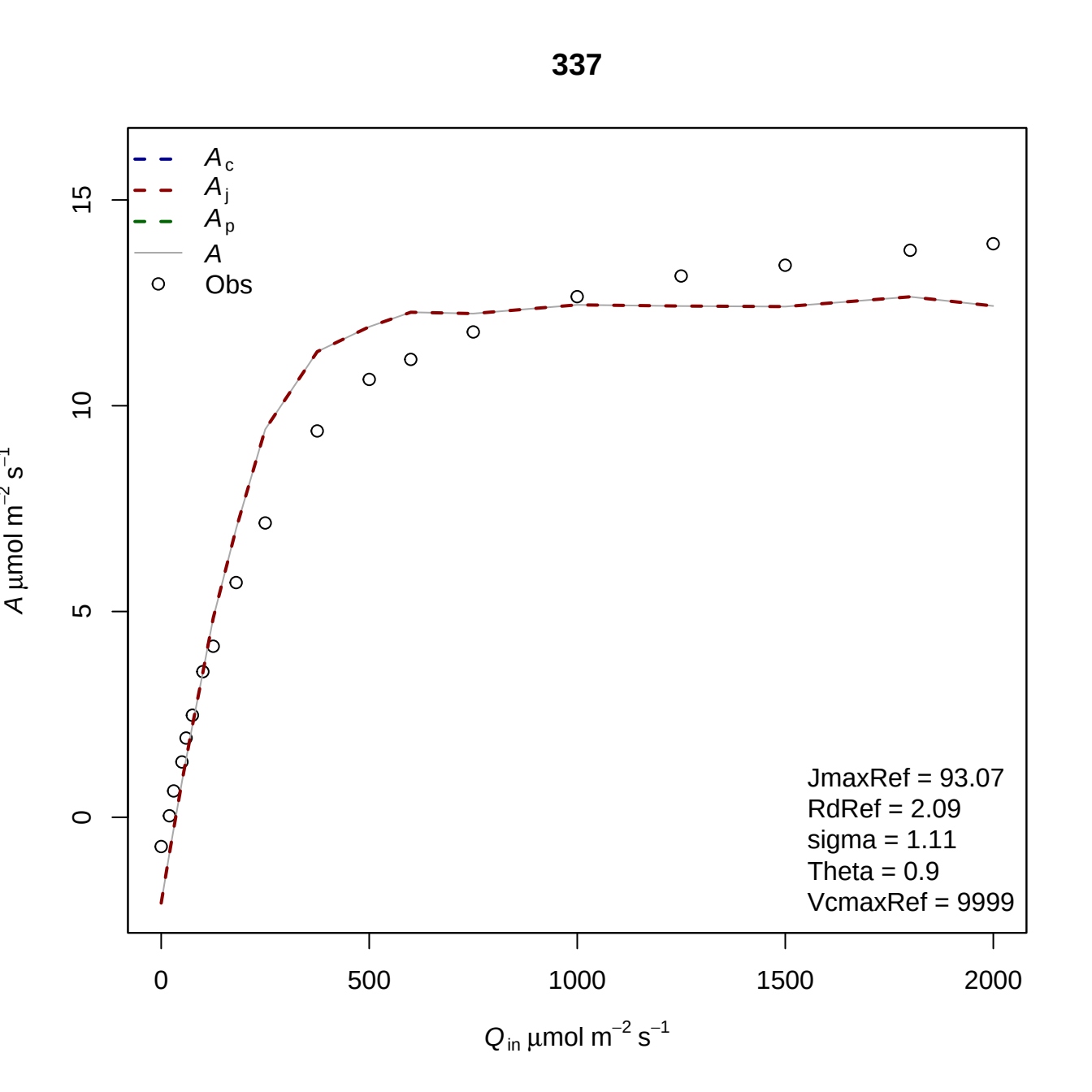

332
A_c
A_j
A_p
A
Obs
JmaxRef = 106.5
RdRef = 1.94
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

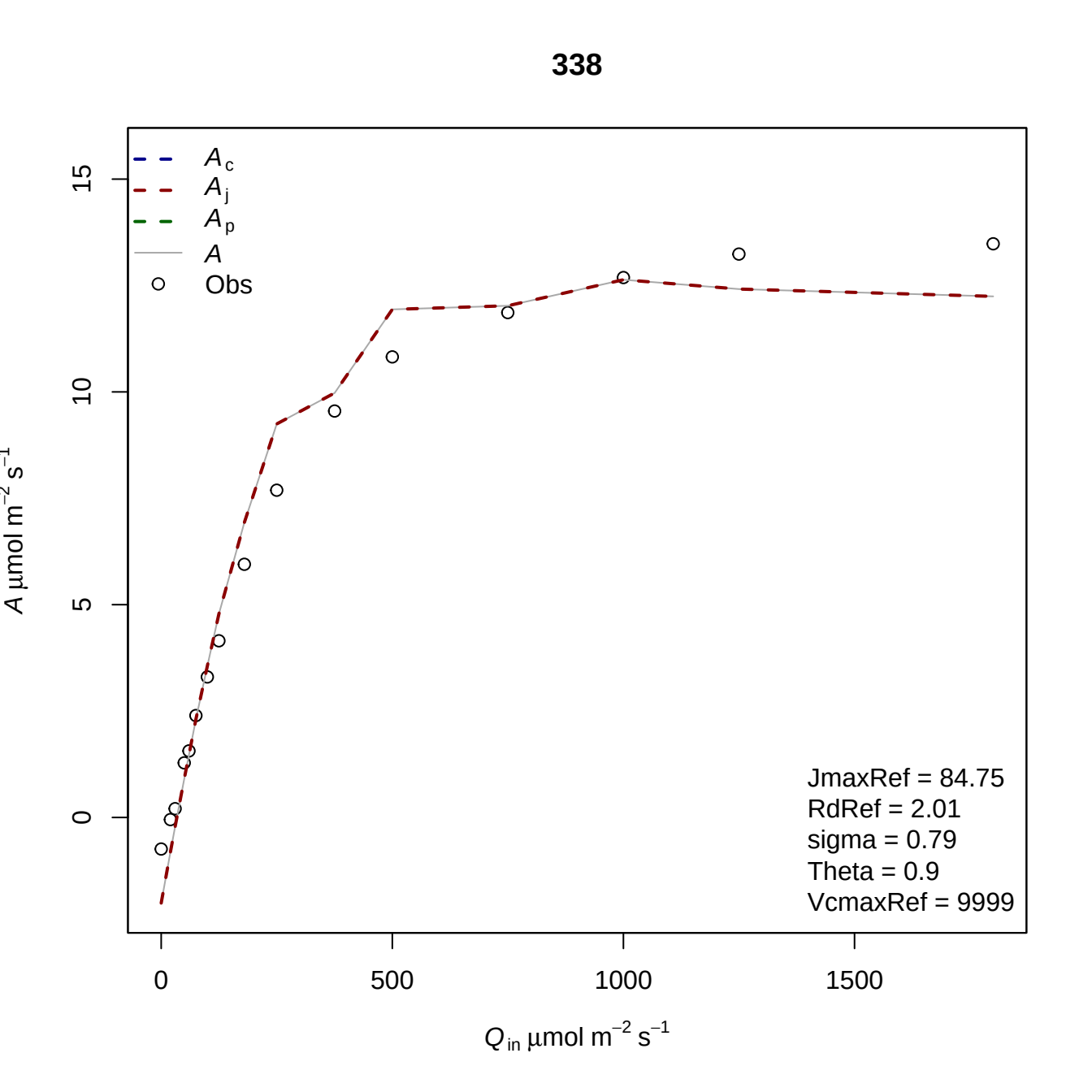

333
A_c
A_j
A_p
A
Obs
JmaxRef = 157.83
RdRef = 3.21
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

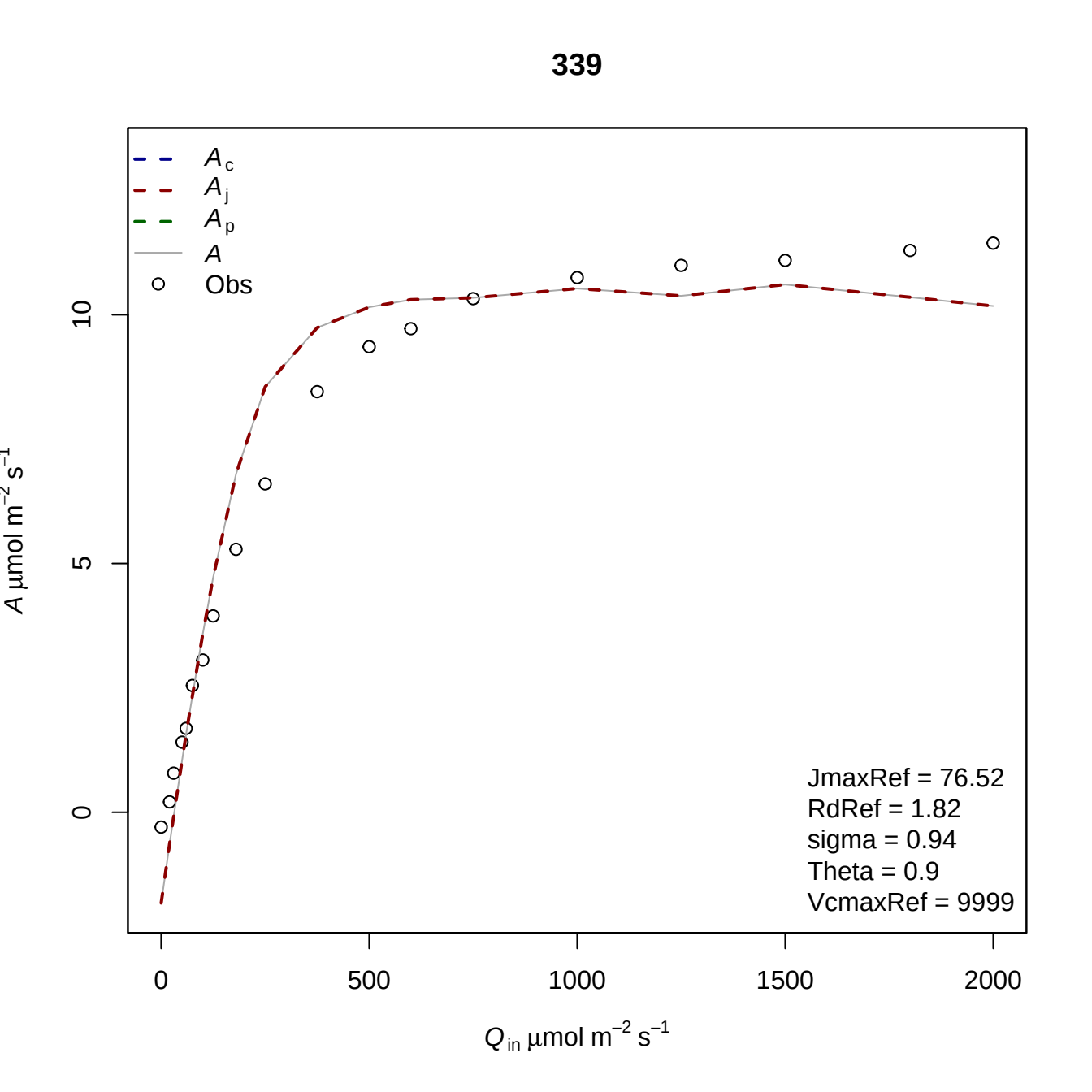

334
A_c
A_j
A_p
A
Obs
JmaxRef = 116.32
RdRef = 1.89
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

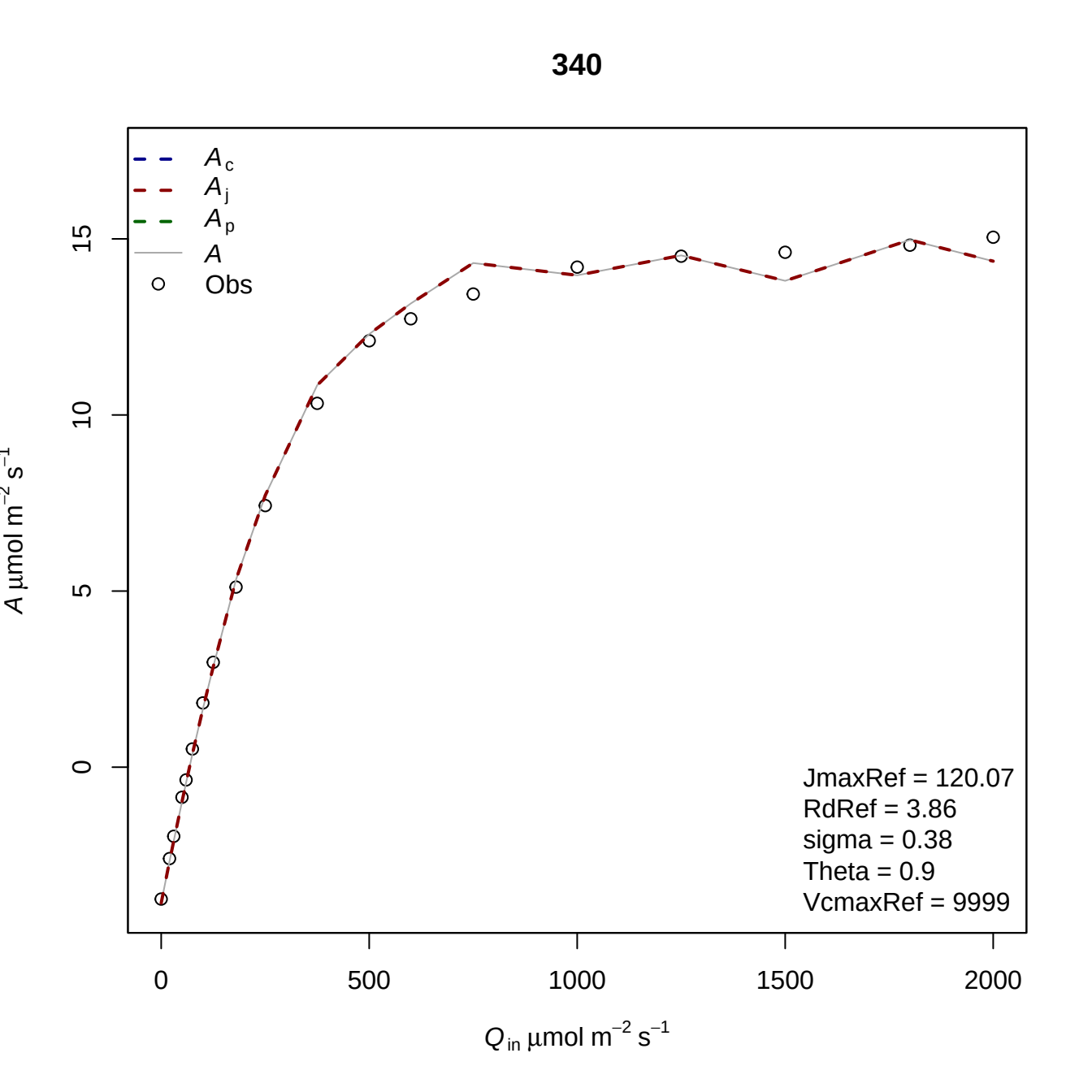

335
A_c
A_j
A_p
A
Obs
JmaxRef = 87.68
RdRef = 3.92
sigma = 2.07
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

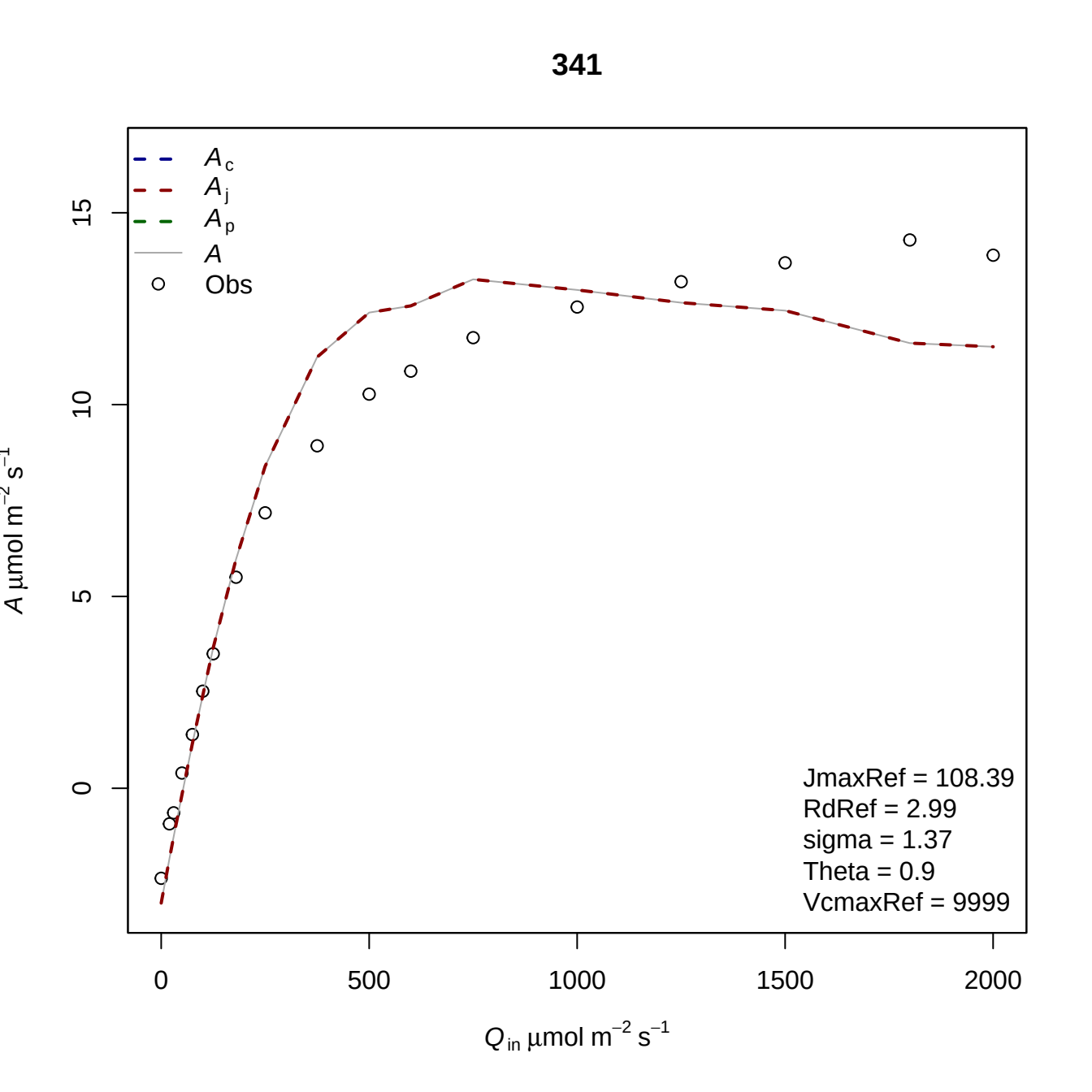

336
A_c
A_j
A_p
A
Obs
JmaxRef = 96.81
RdRef = 2.25
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

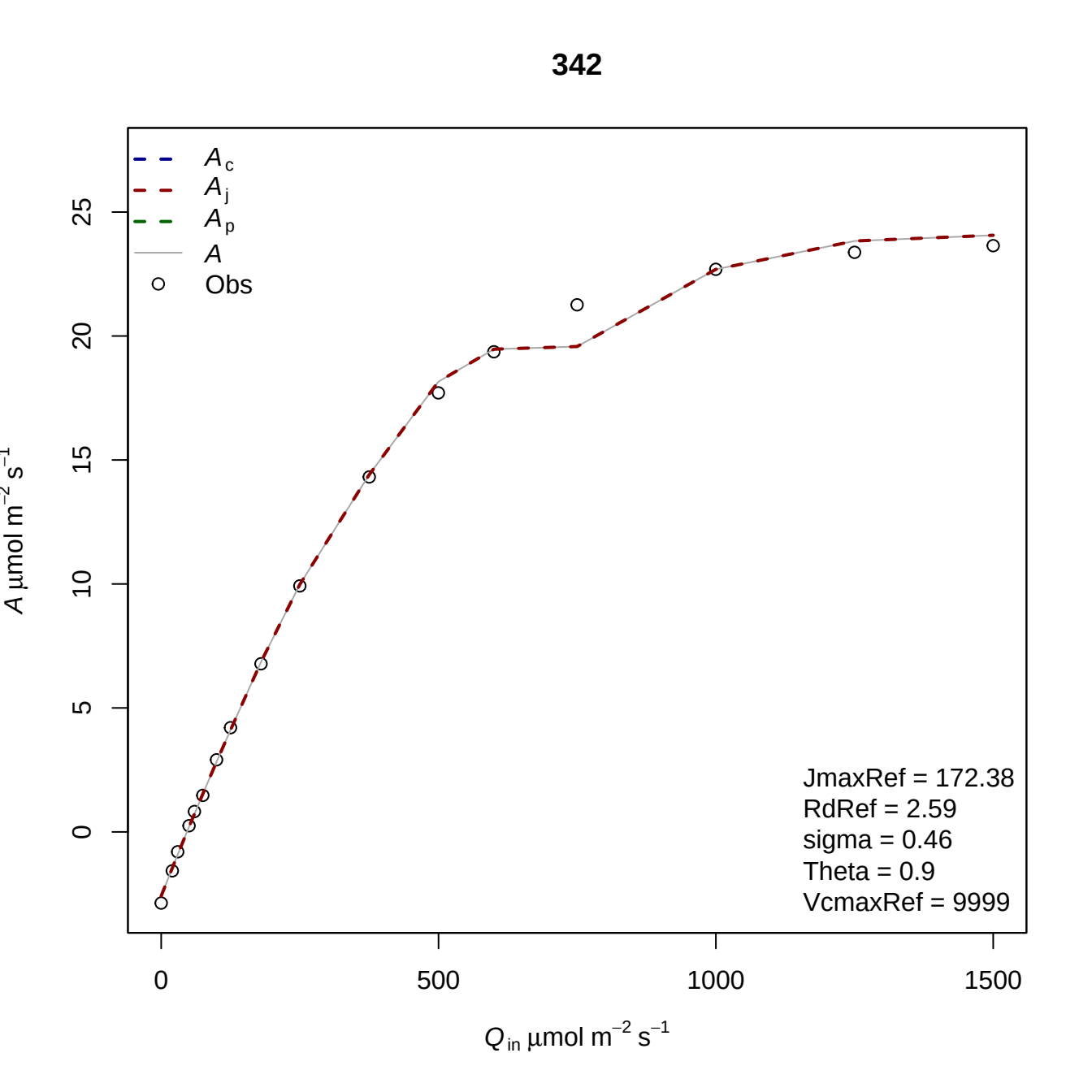

337
A_c
A_j
A_p
A
Obs
JmaxRef = 93.07
RdRef = 2.09
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

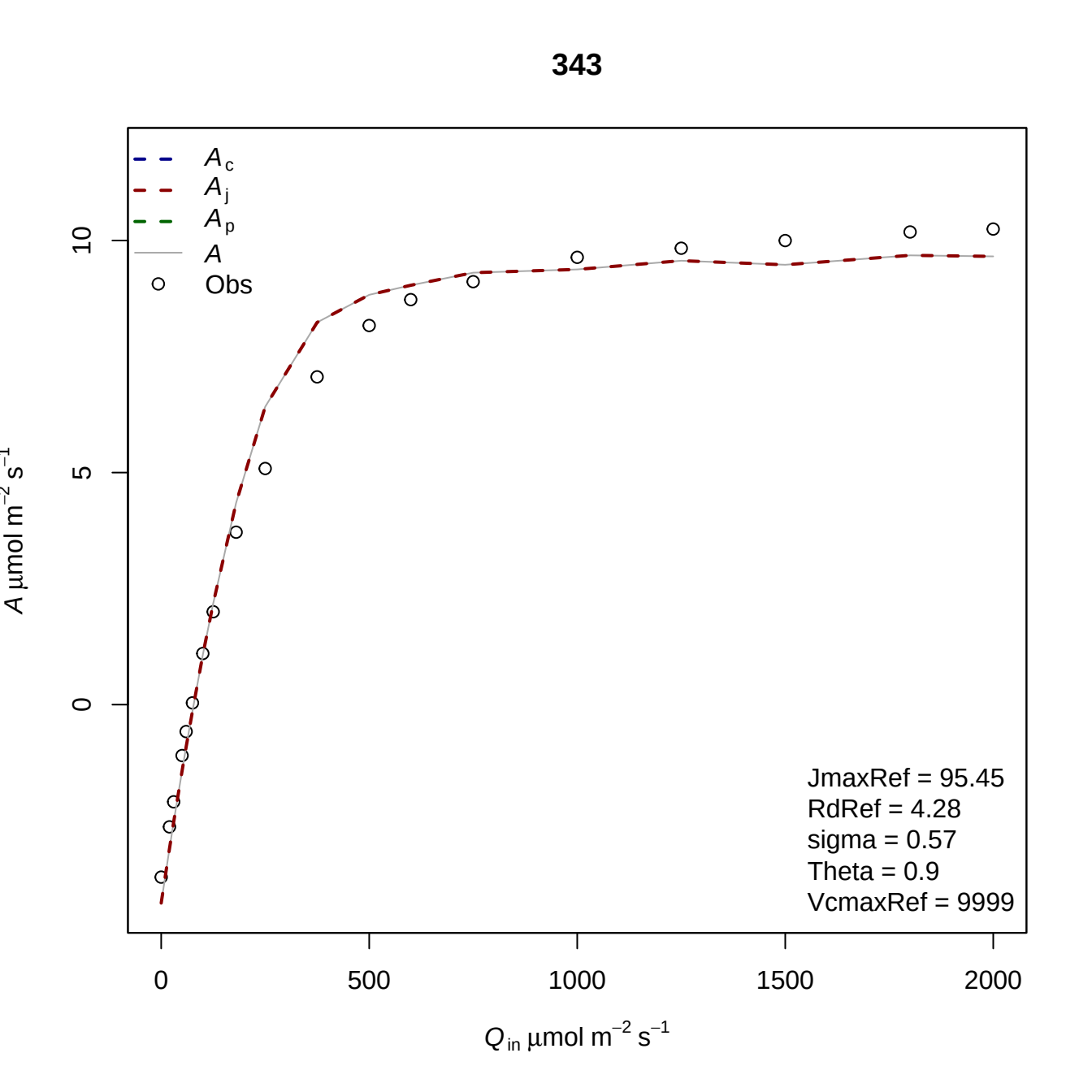

338
A_c
A_j
A_p
A
Obs
JmaxRef = 84.75
RdRef = 2.01
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

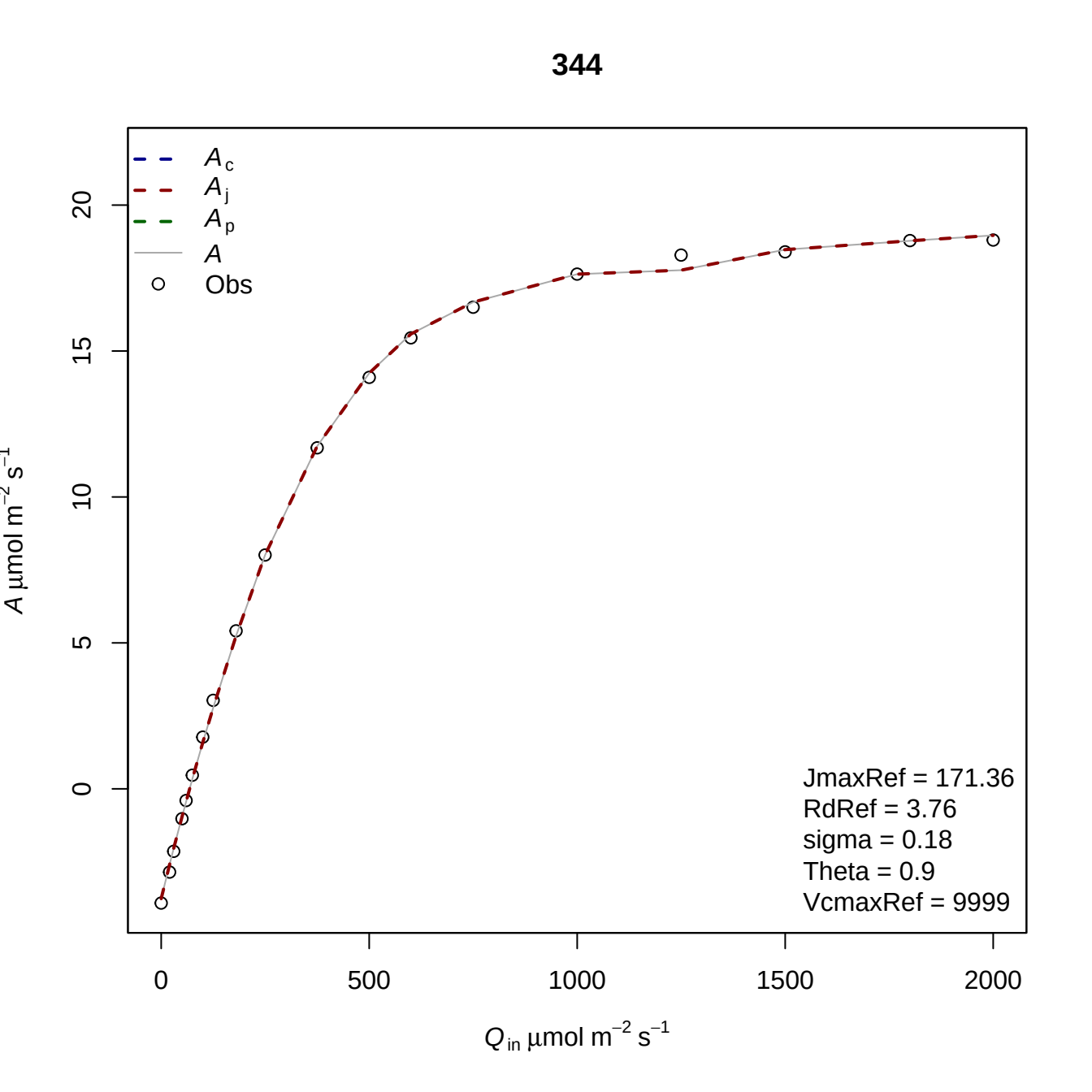

339
A_c
A_j
A_p
A
Obs
JmaxRef = 76.52
RdRef = 1.82
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

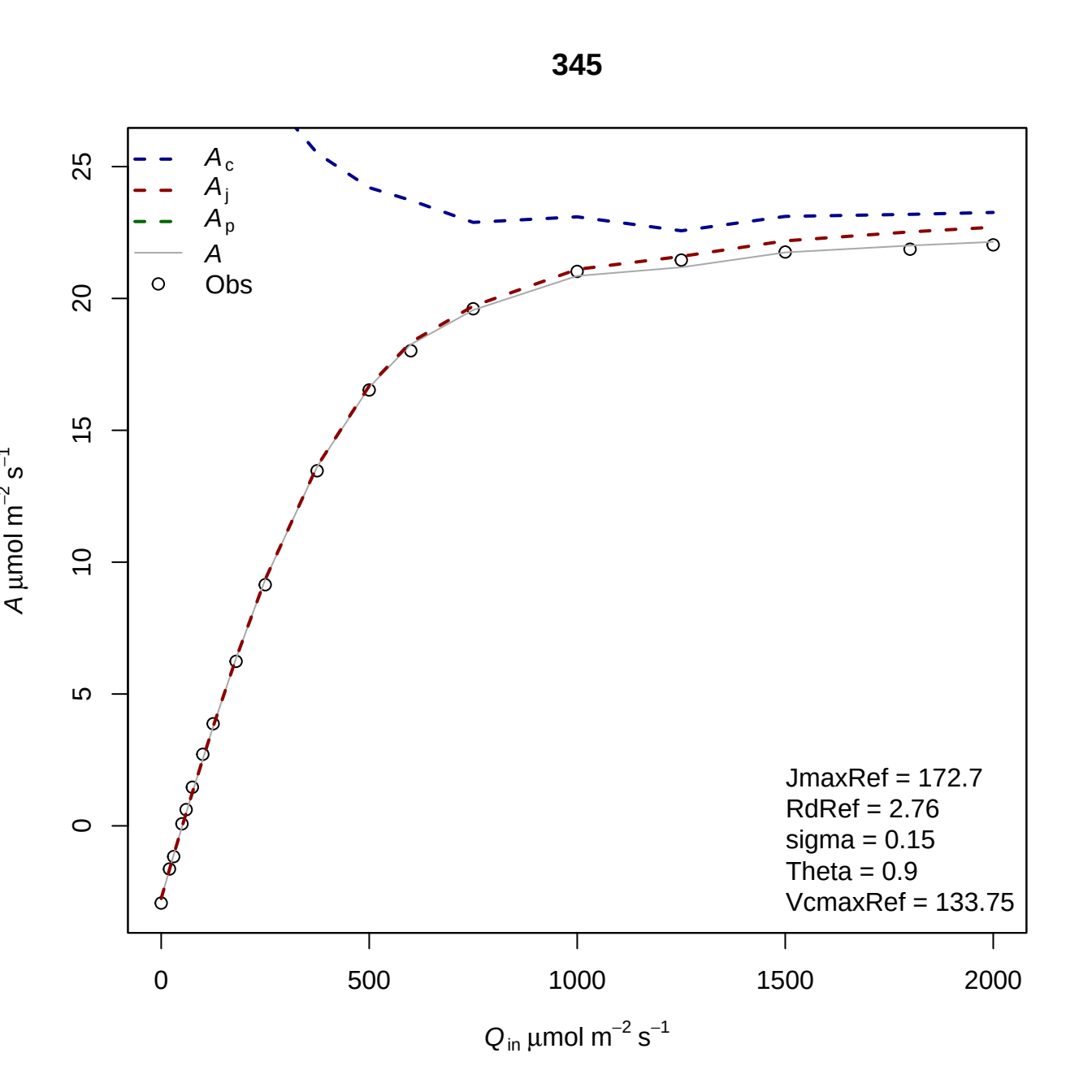

340
A_c
A_j
A_p
A
Obs
JmaxRef = 120.07
RdRef = 3.86
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

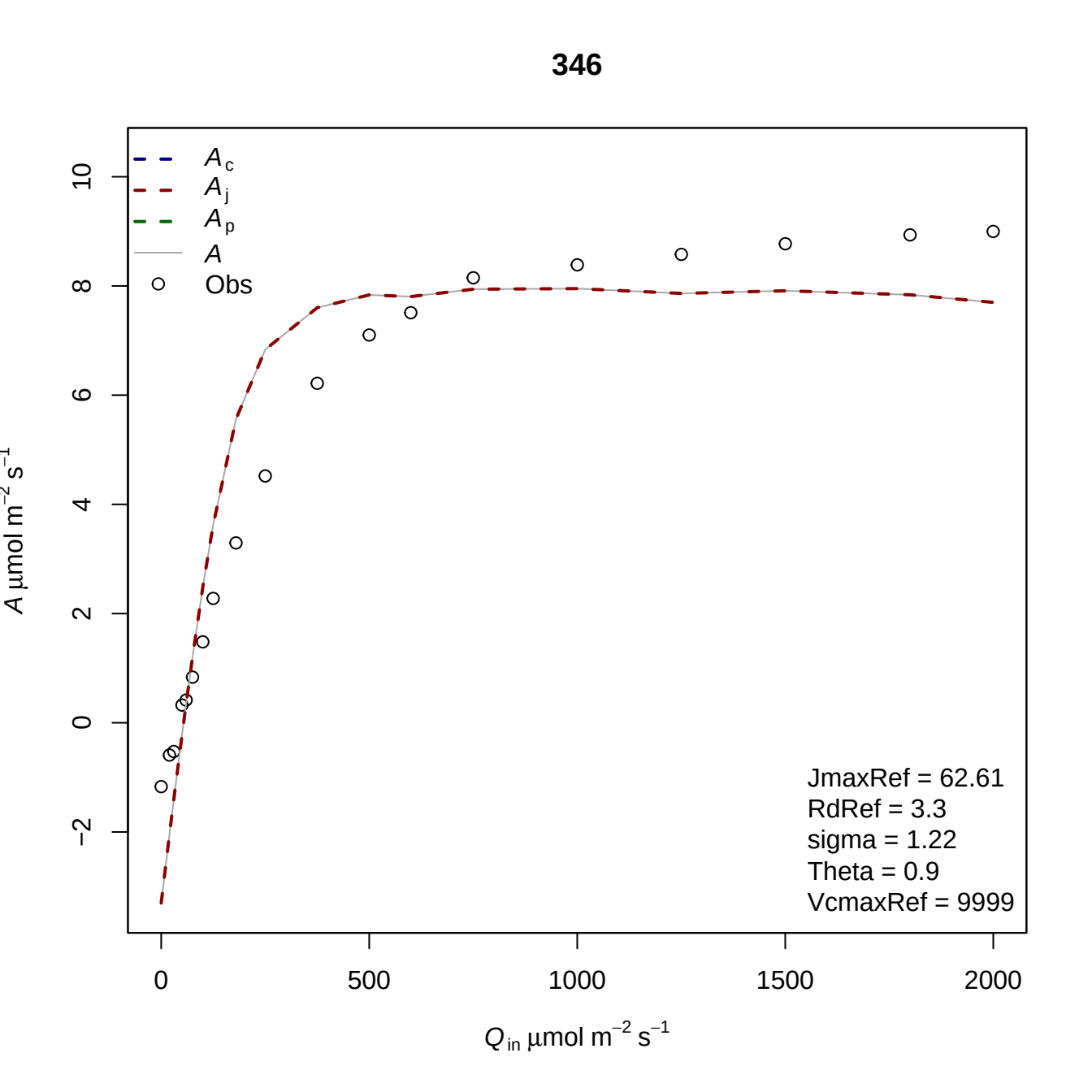

341
A_c
A_j
A_p
A
Obs
JmaxRef = 108.39
RdRef = 2.99
sigma = 1.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

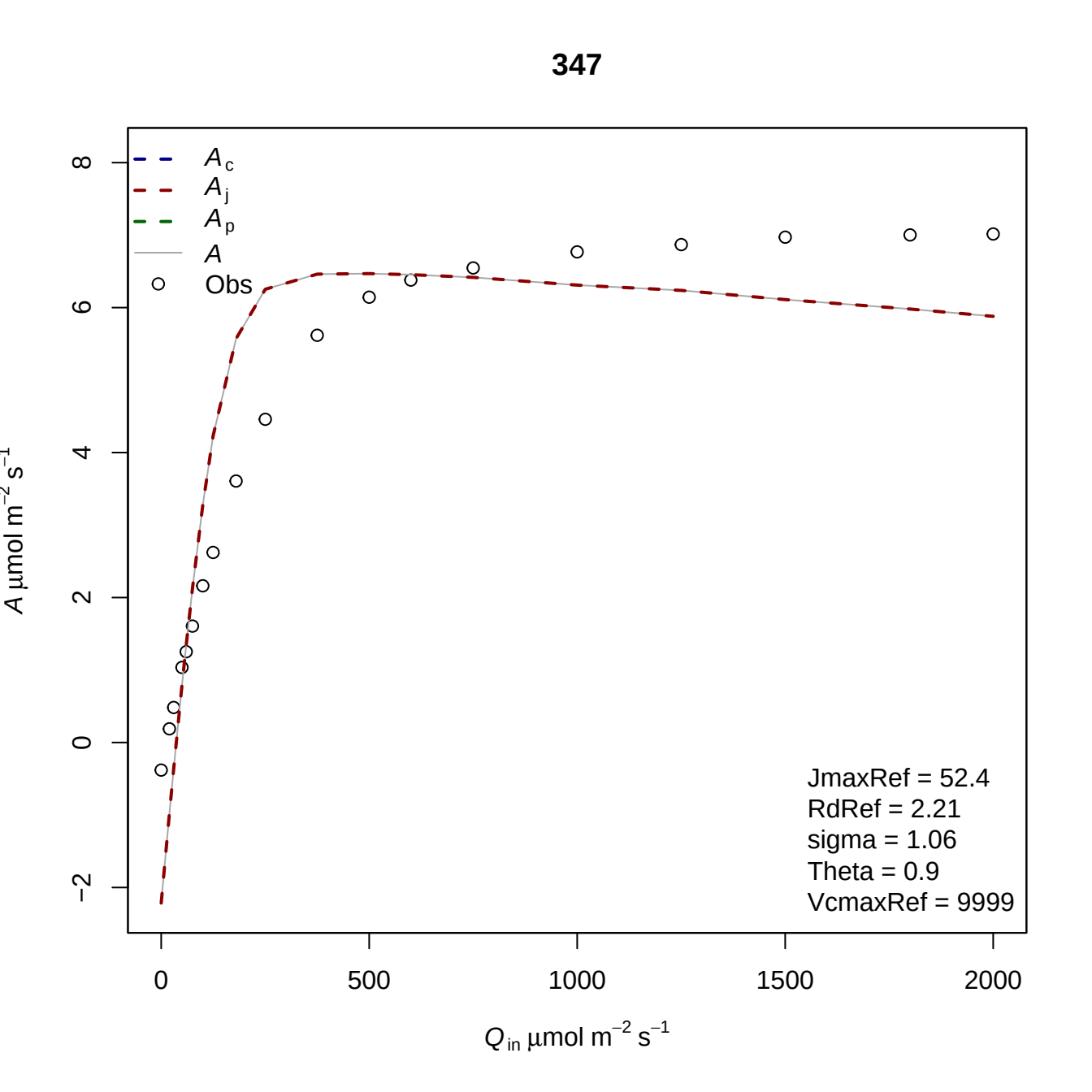

342
A_c
A_j
A_p
A
Obs
JmaxRef = 172.38
RdRef = 2.59
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

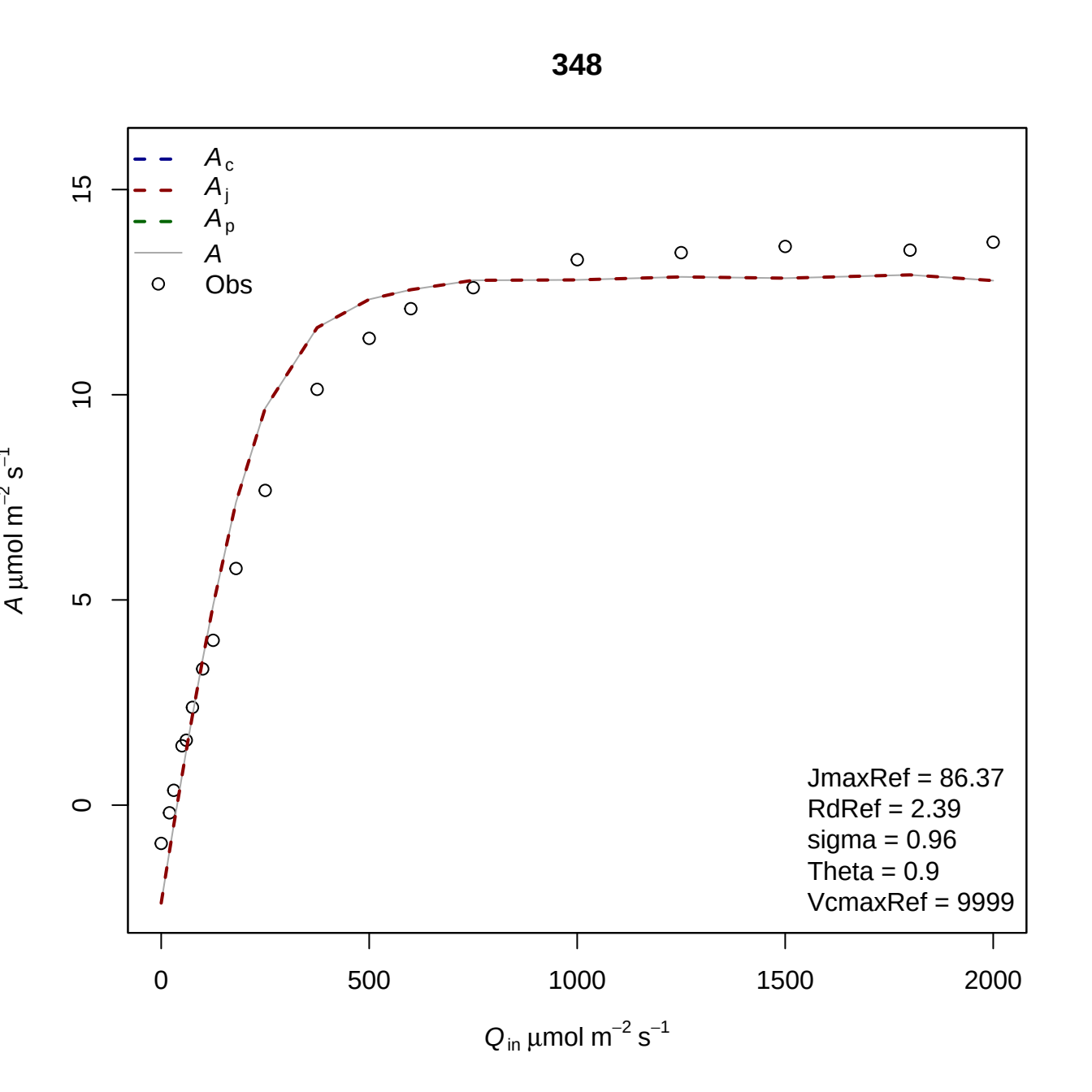

343
A_c
A_j
A_p
A
Obs
JmaxRef = 95.45
RdRef = 4.28
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

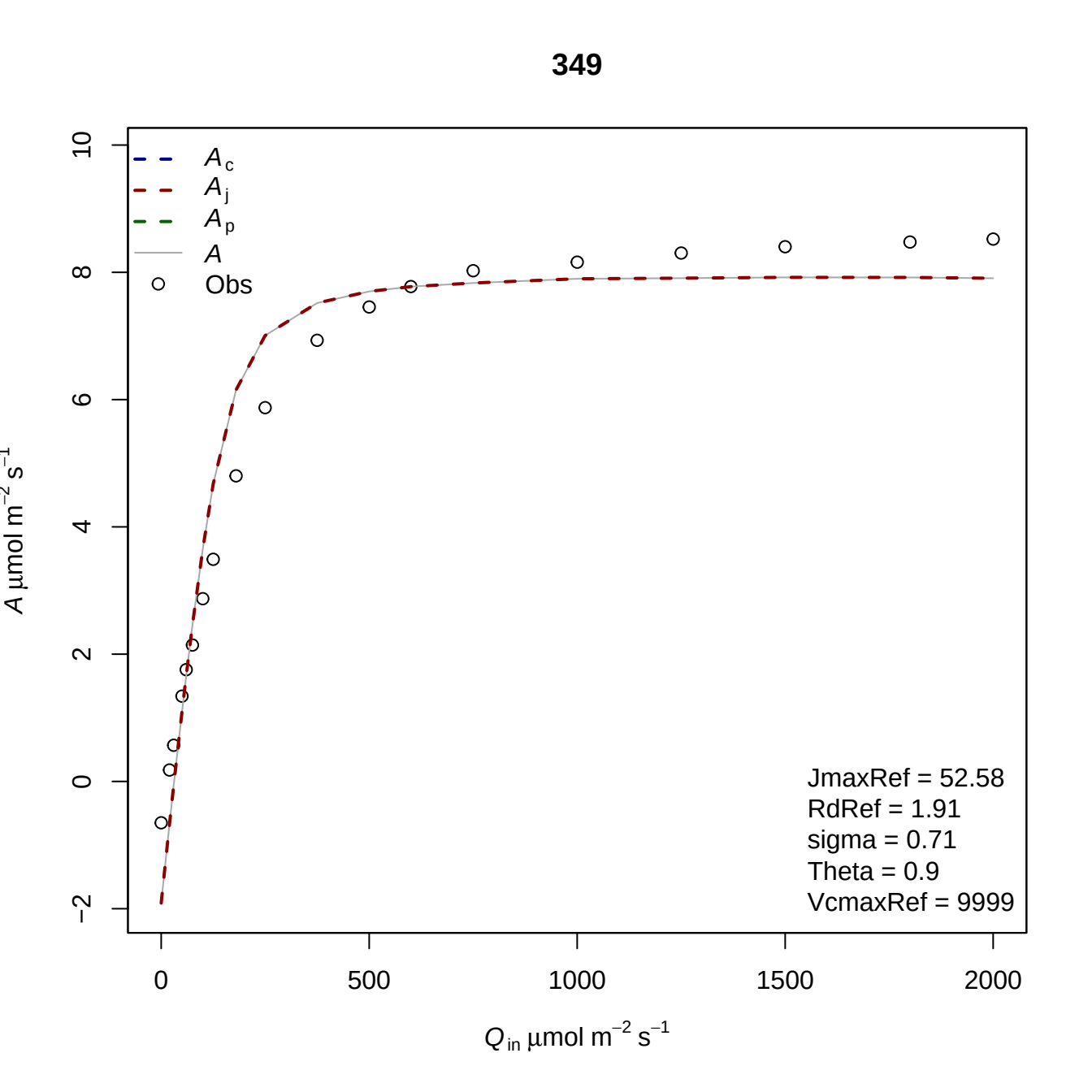

344
A_c
A_j
A_p
A
Obs
JmaxRef = 171.36
RdRef = 3.76
sigma = 0.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

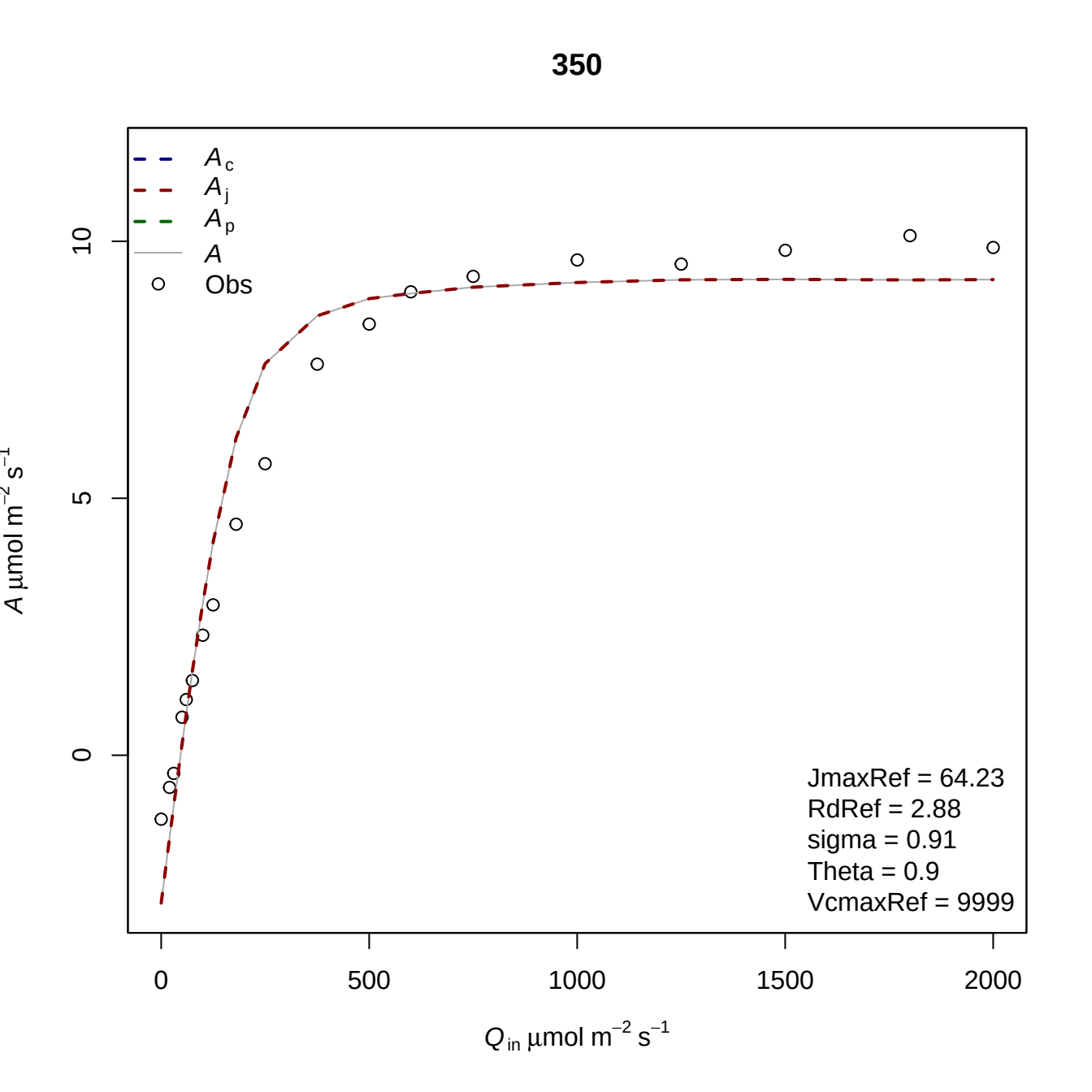

345
A_c
A_j
A_p
A
Obs
JmaxRef = 172.7
RdRef = 2.76
sigma = 0.15
Theta = 0.9
VcmaxRef = 133.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

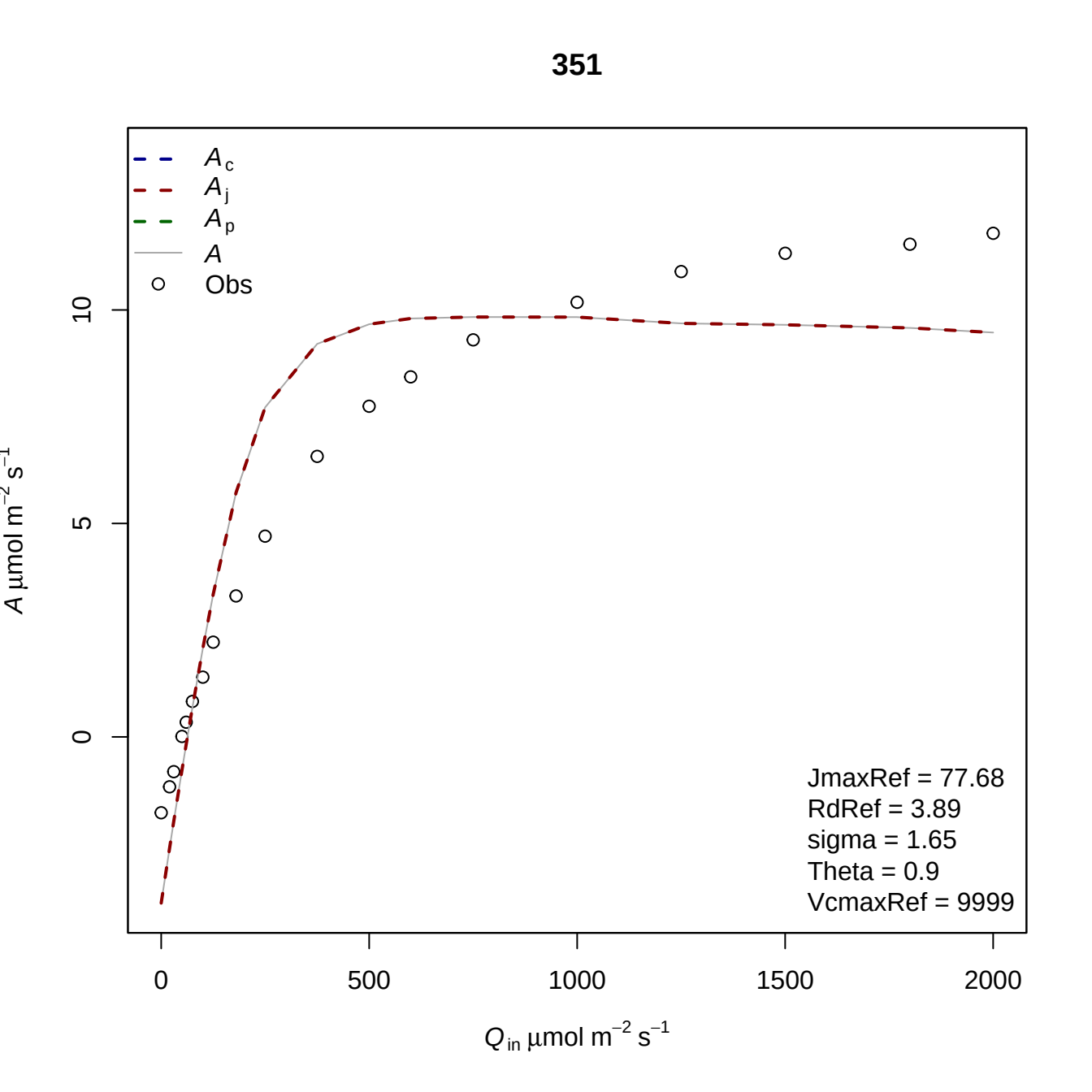

346
A_c
A_j
A_p
A
Obs
JmaxRef = 62.61
RdRef = 3.3
sigma = 1.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

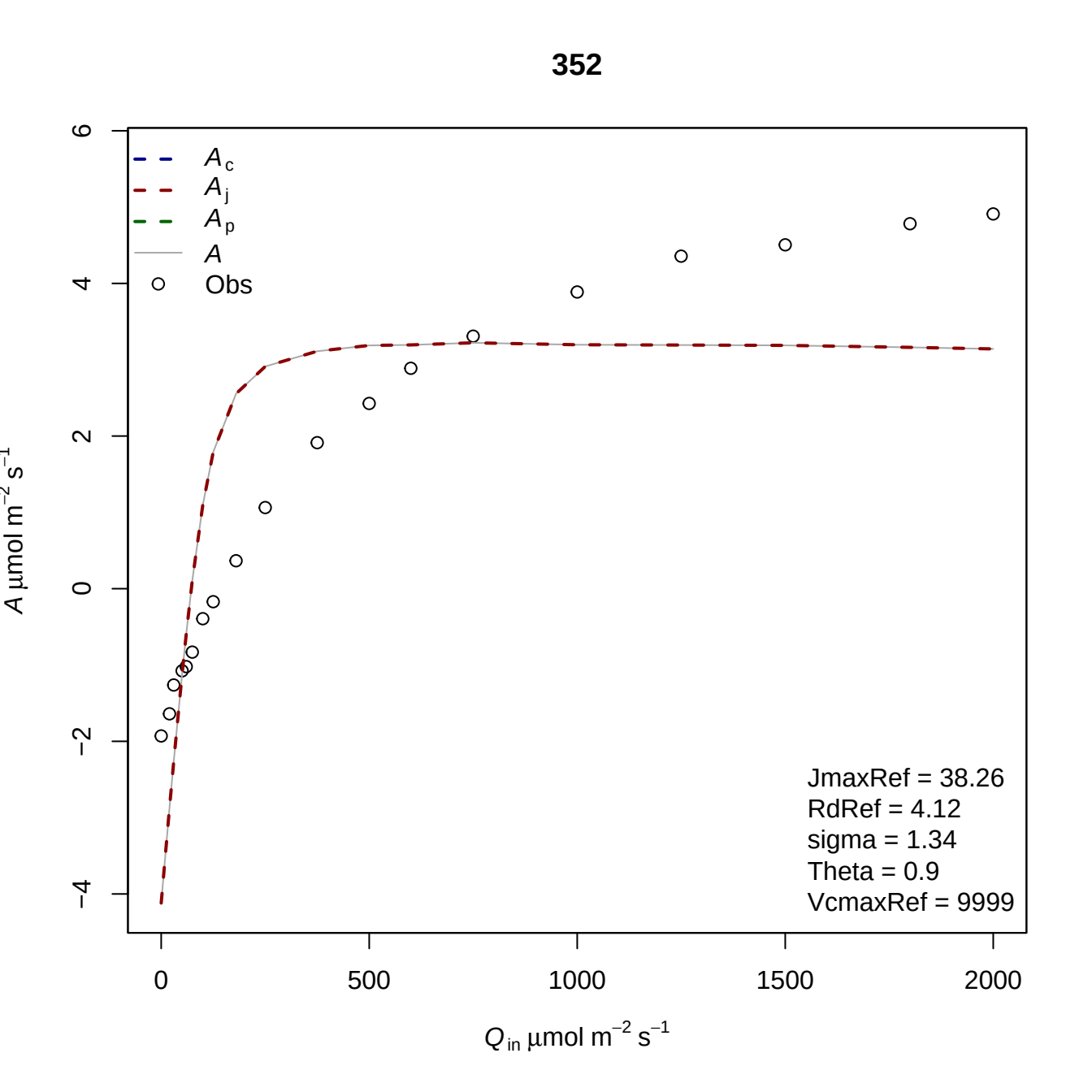

347
A_c
A_j
A_p
A
Obs
JmaxRef = 52.4
RdRef = 2.21
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

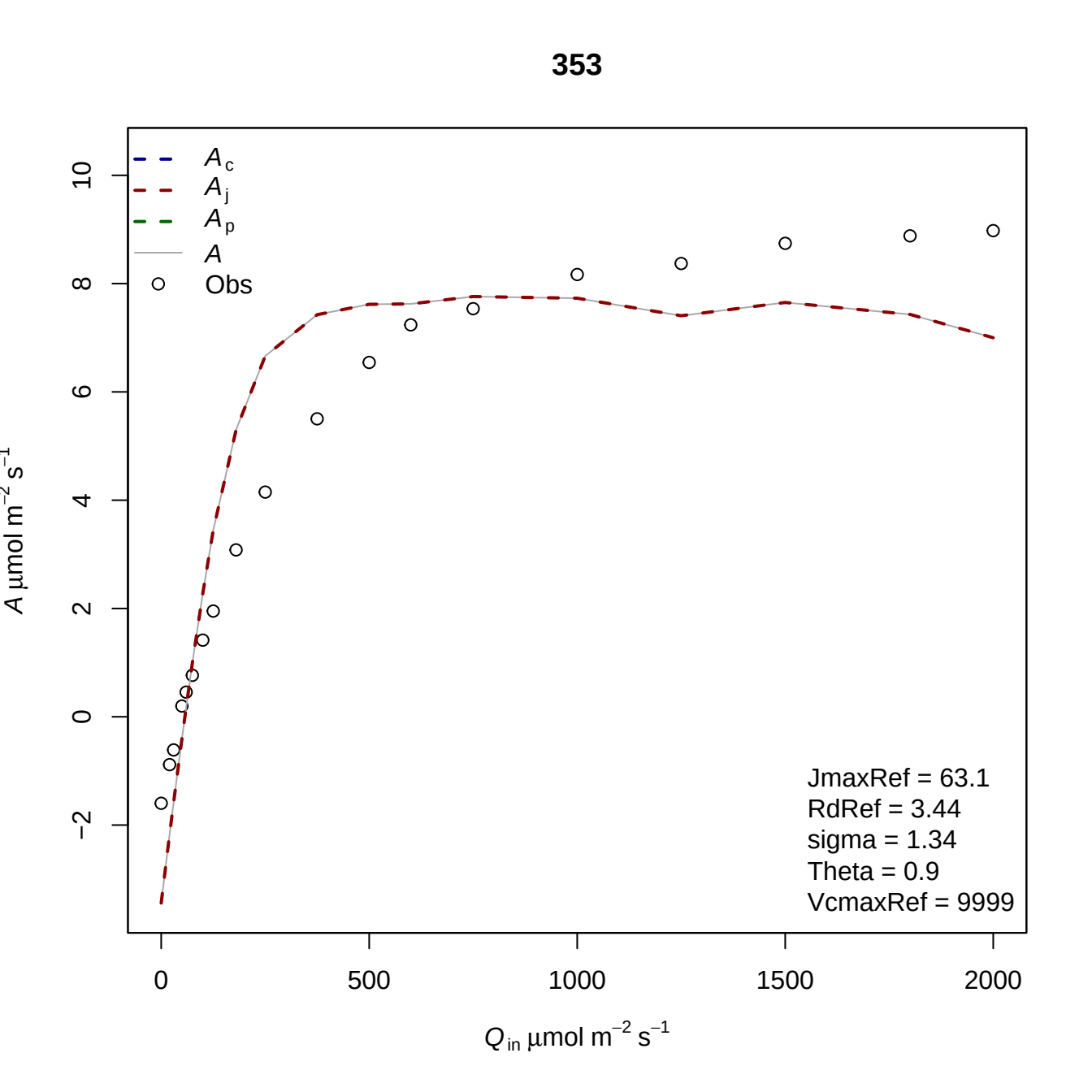

348
A_c
A_j
A_p
A
Obs
JmaxRef = 86.37
RdRef = 2.39
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

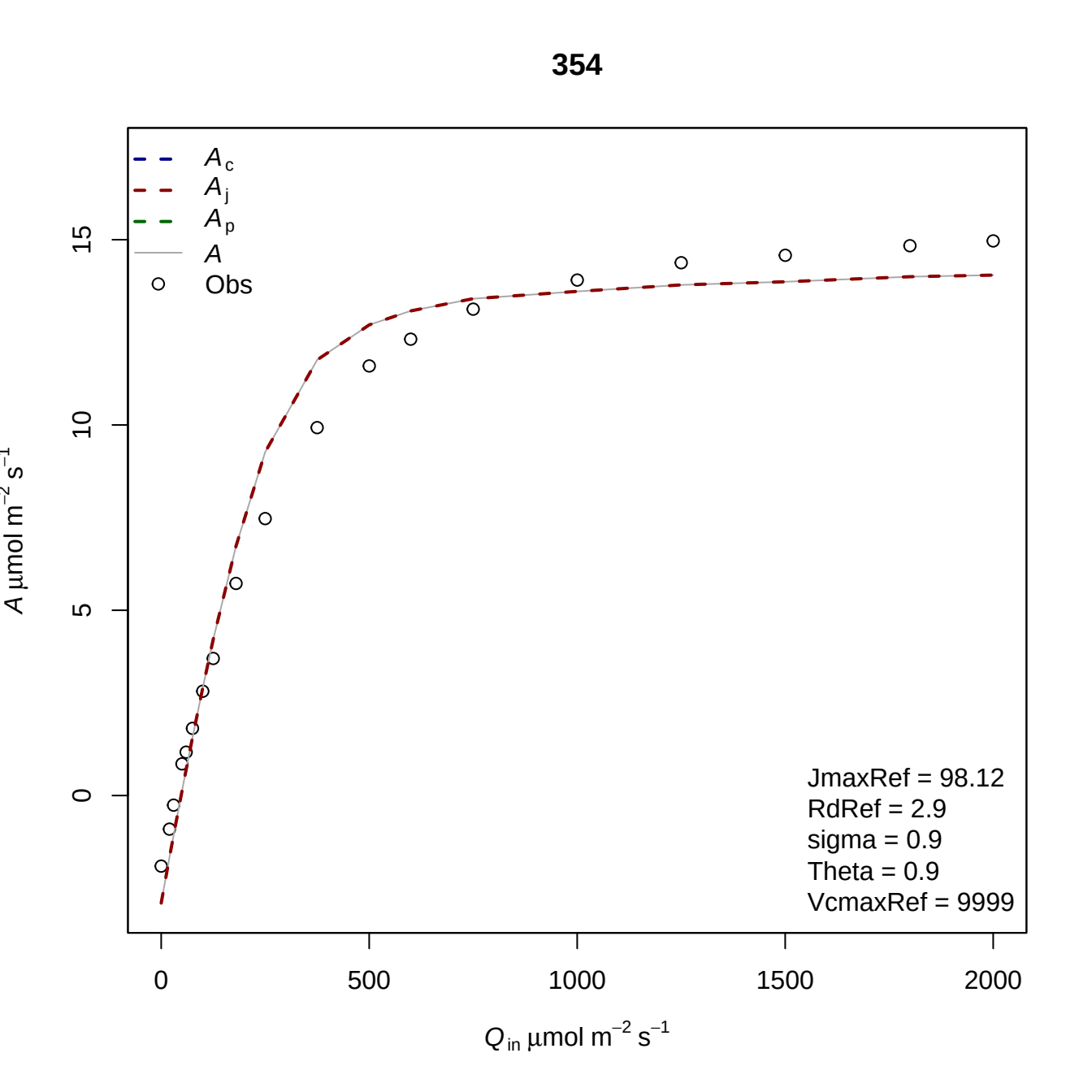

349
A_c
A_j
A_p
A
Obs
JmaxRef = 52.58
RdRef = 1.91
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

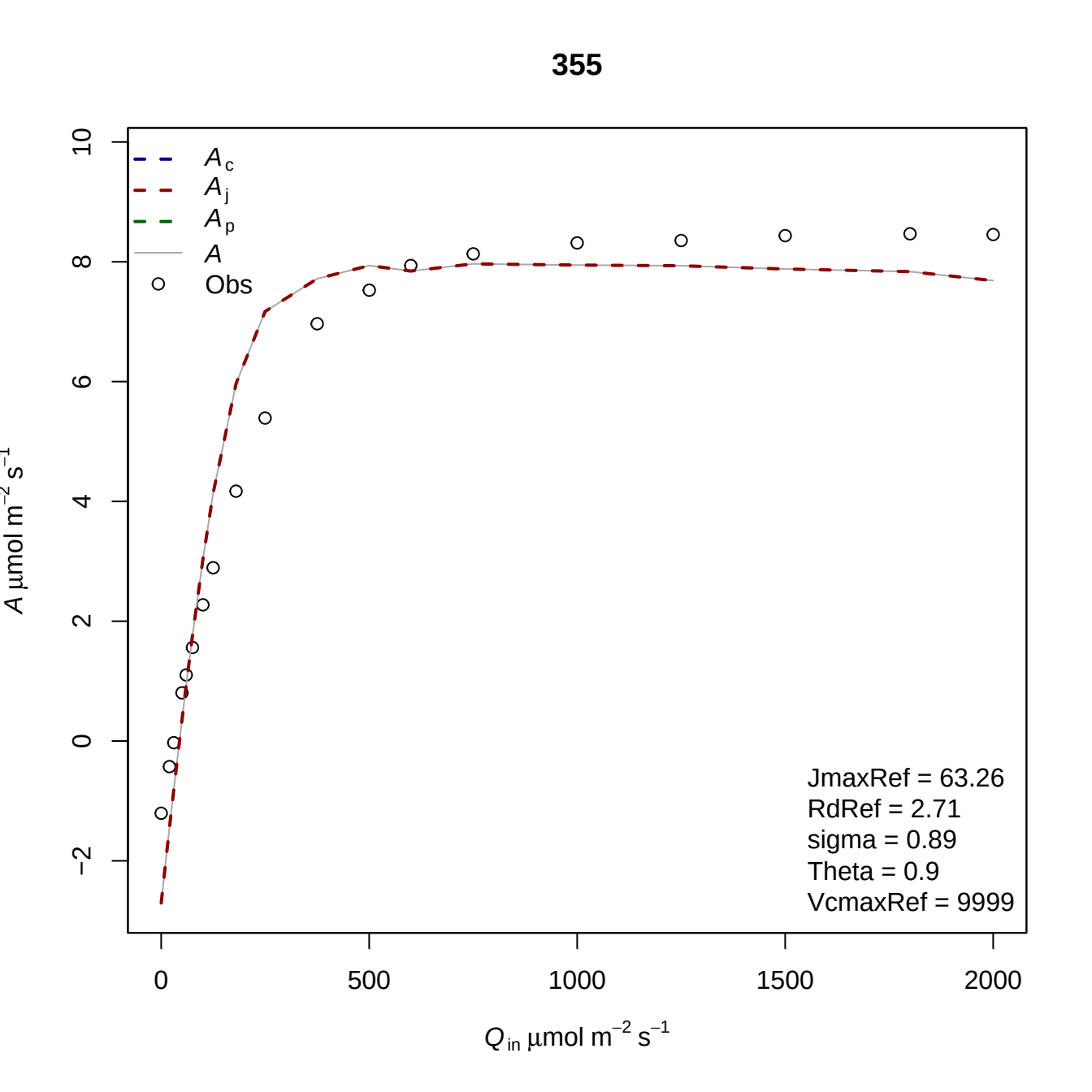

350
A_c
A_j
A_p
A
Obs
JmaxRef = 64.23
RdRef = 2.88
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

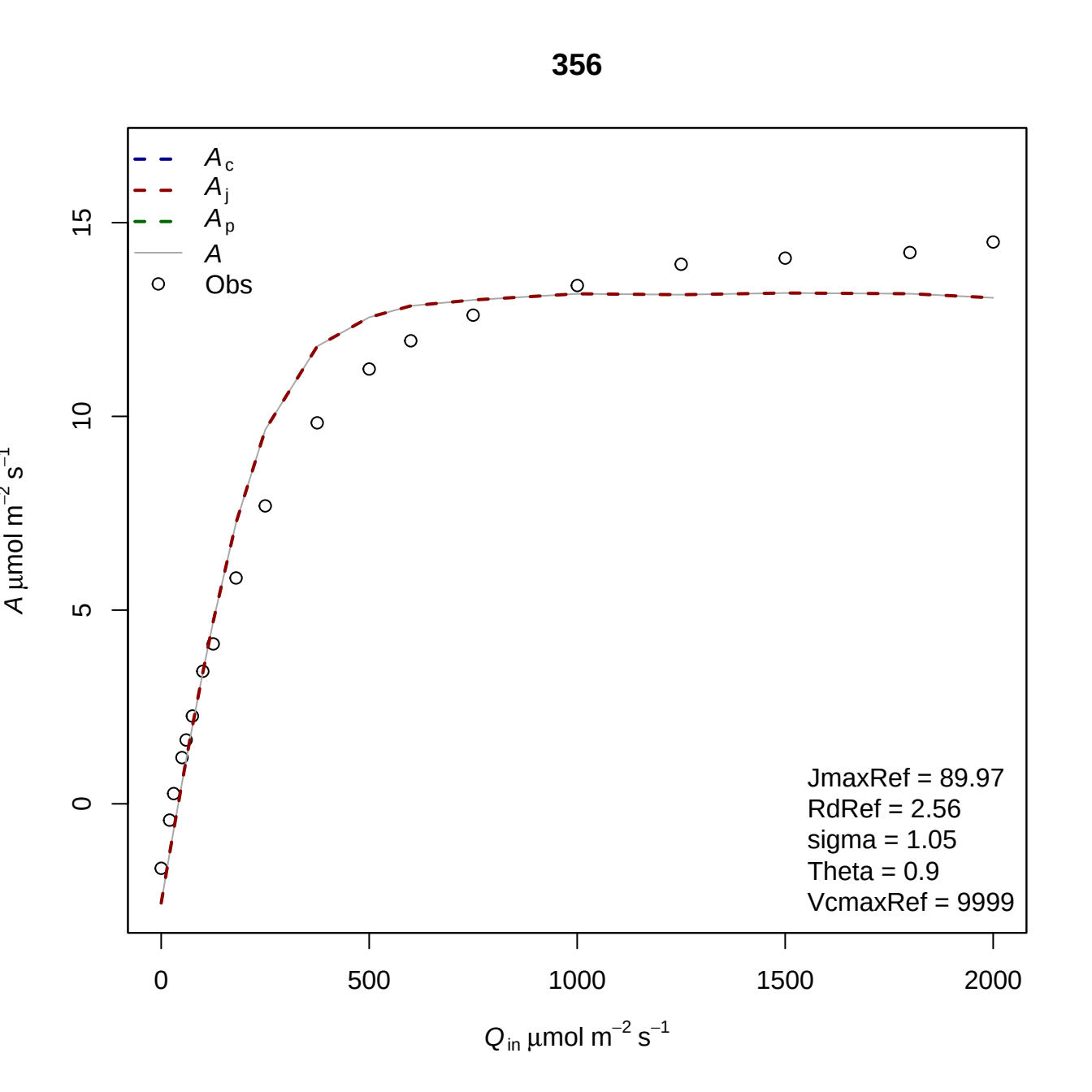

351
A_c
A_j
A_p
A
Obs
JmaxRef = 77.68
RdRef = 3.89
sigma = 1.65
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

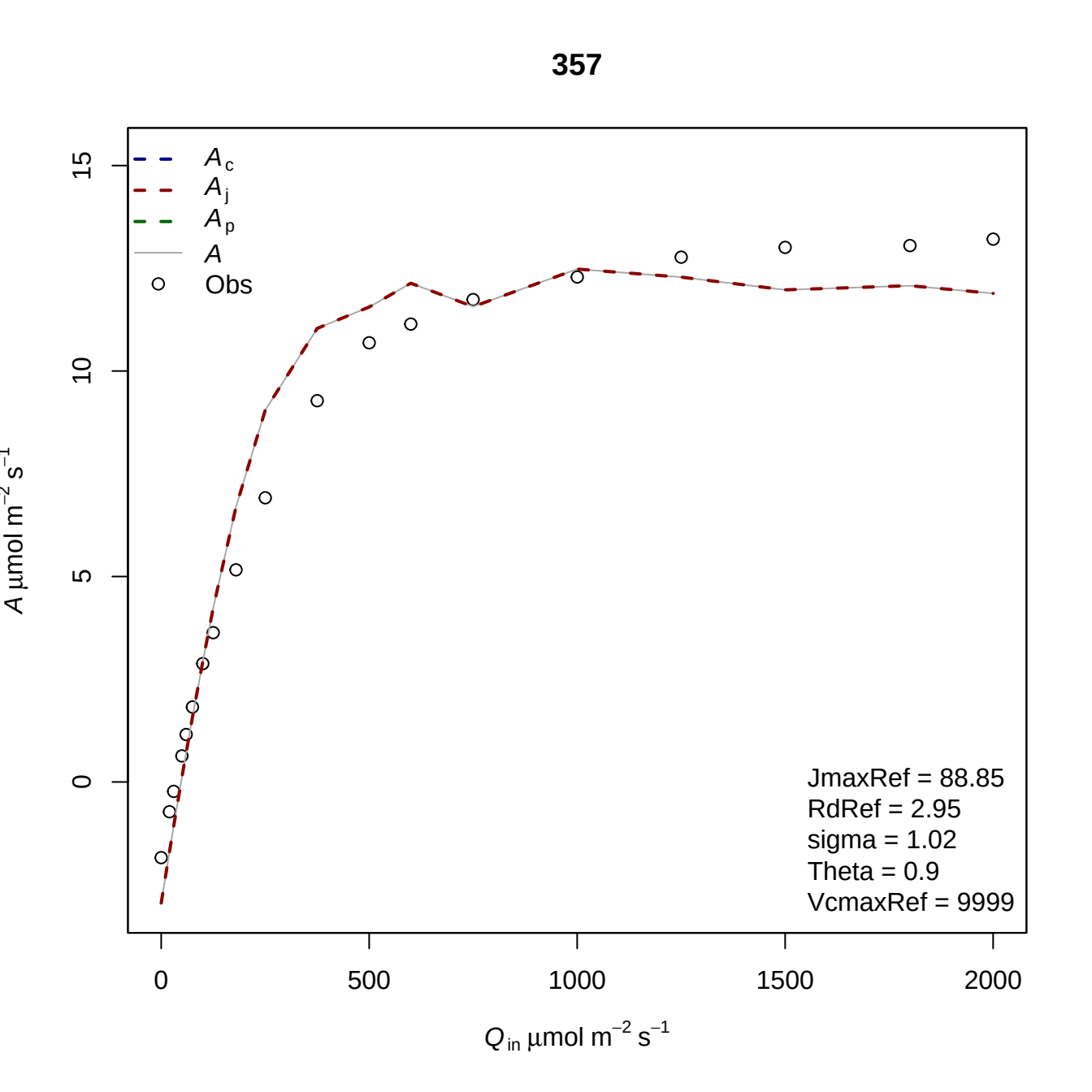

352
A_c
A_j
A_p
A
Obs
JmaxRef = 38.26
RdRef = 4.12
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

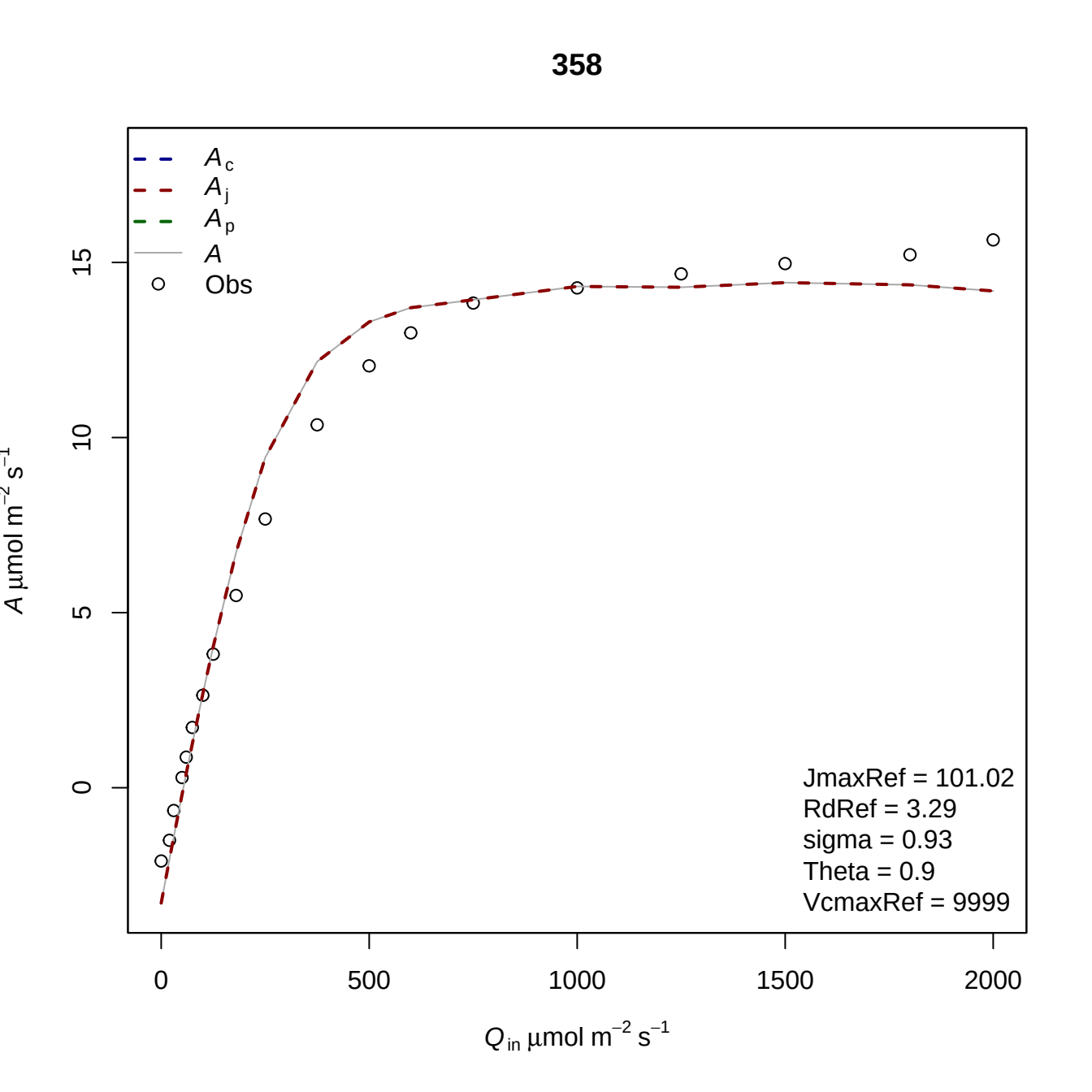

353
A_c
A_j
A_p
A
Obs
JmaxRef = 63.1
RdRef = 3.44
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

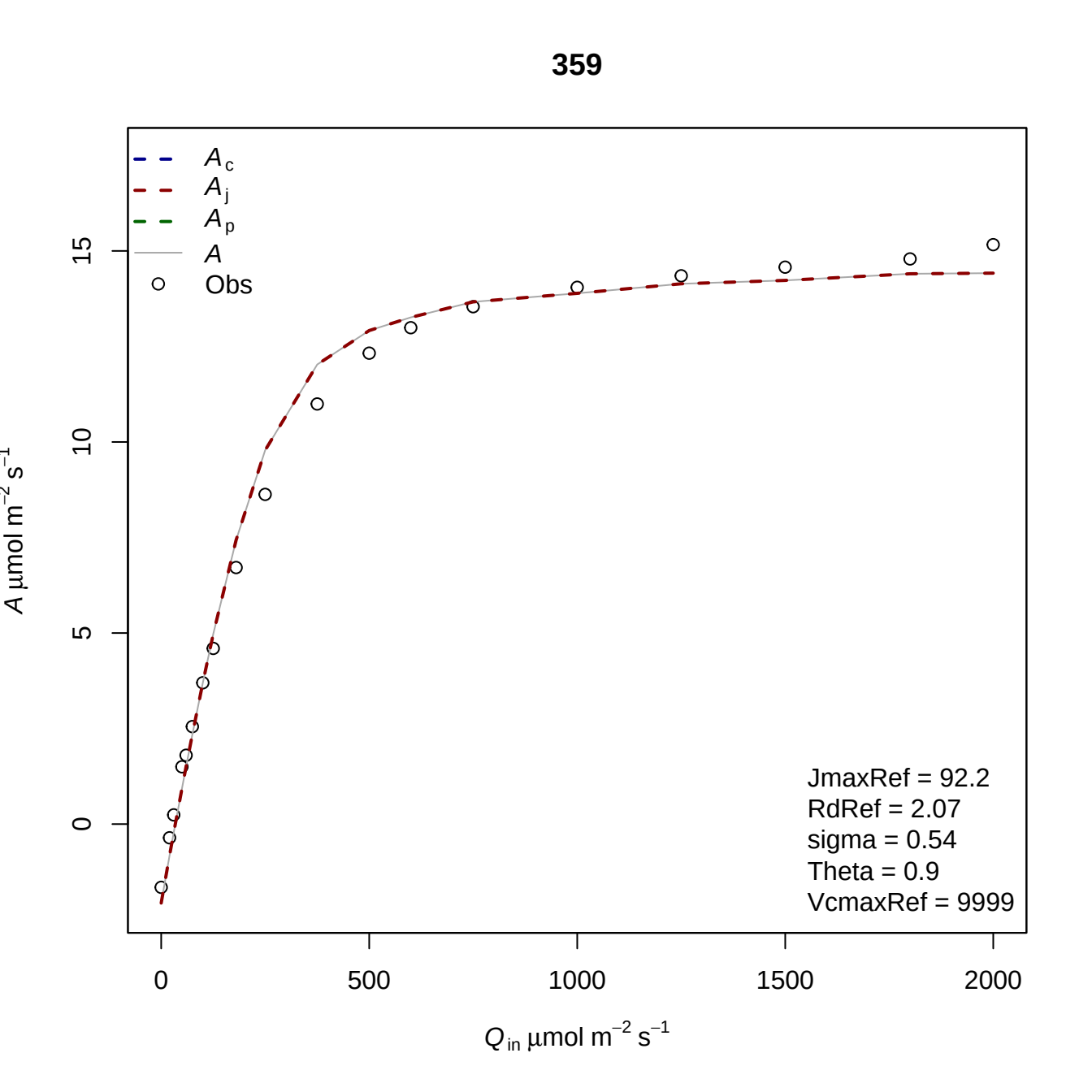

354
A_c
A_j
A_p
A
Obs
JmaxRef = 98.12
RdRef = 2.9
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

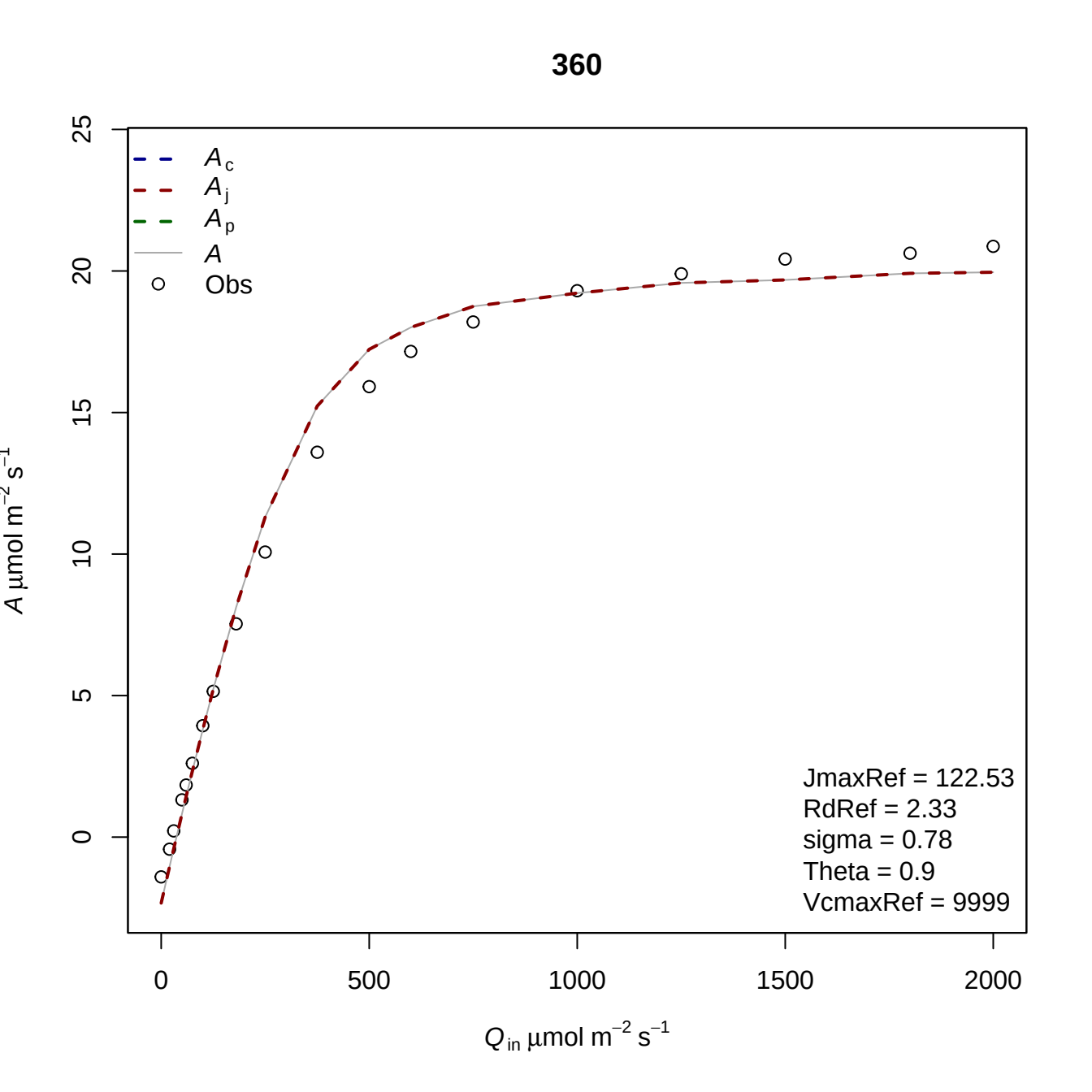

355
A_c
A_j
A_p
A
Obs
JmaxRef = 63.26
RdRef = 2.71
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

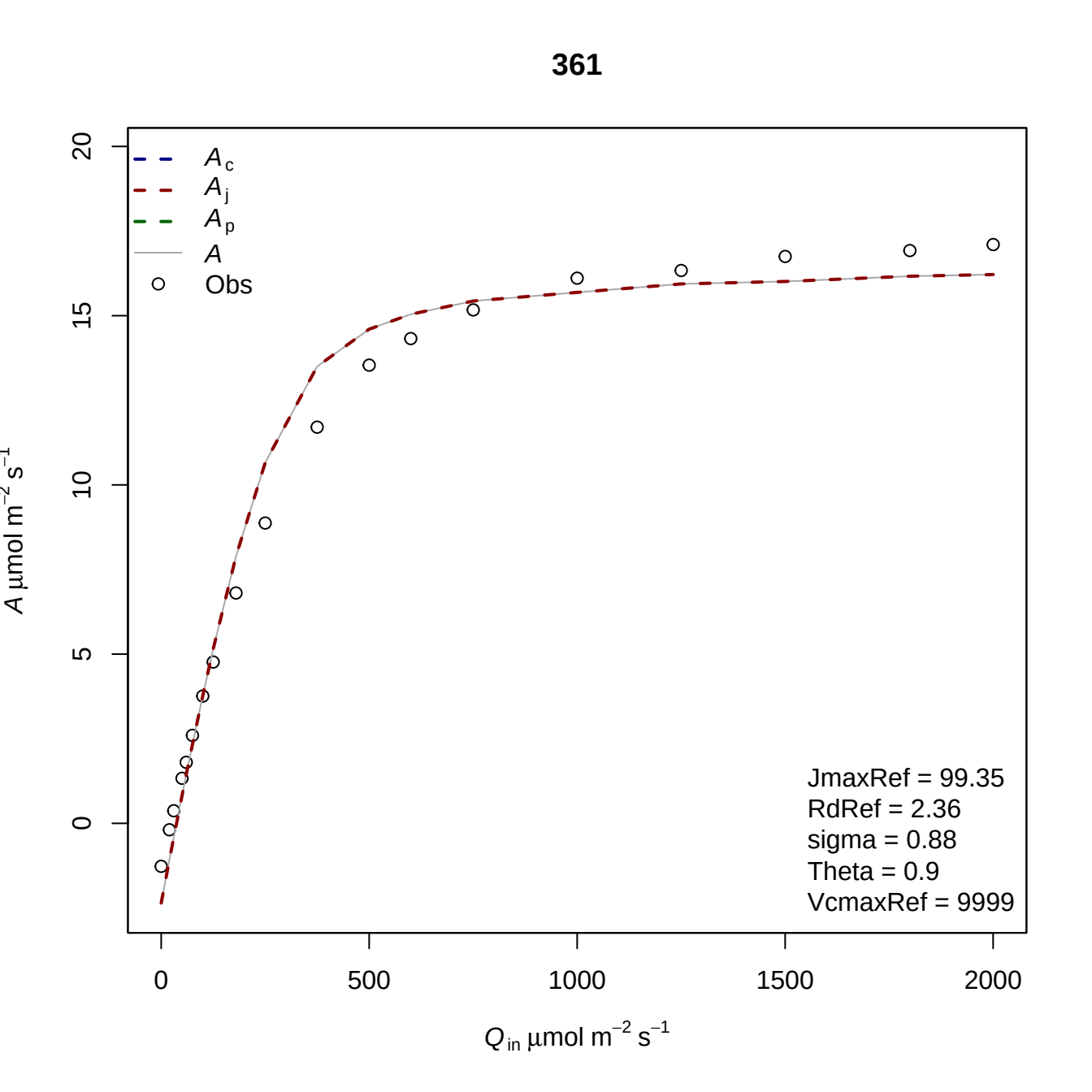

356
A_c
A_j
A_p
A
Obs
JmaxRef = 89.97
RdRef = 2.56
sigma = 1.05
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

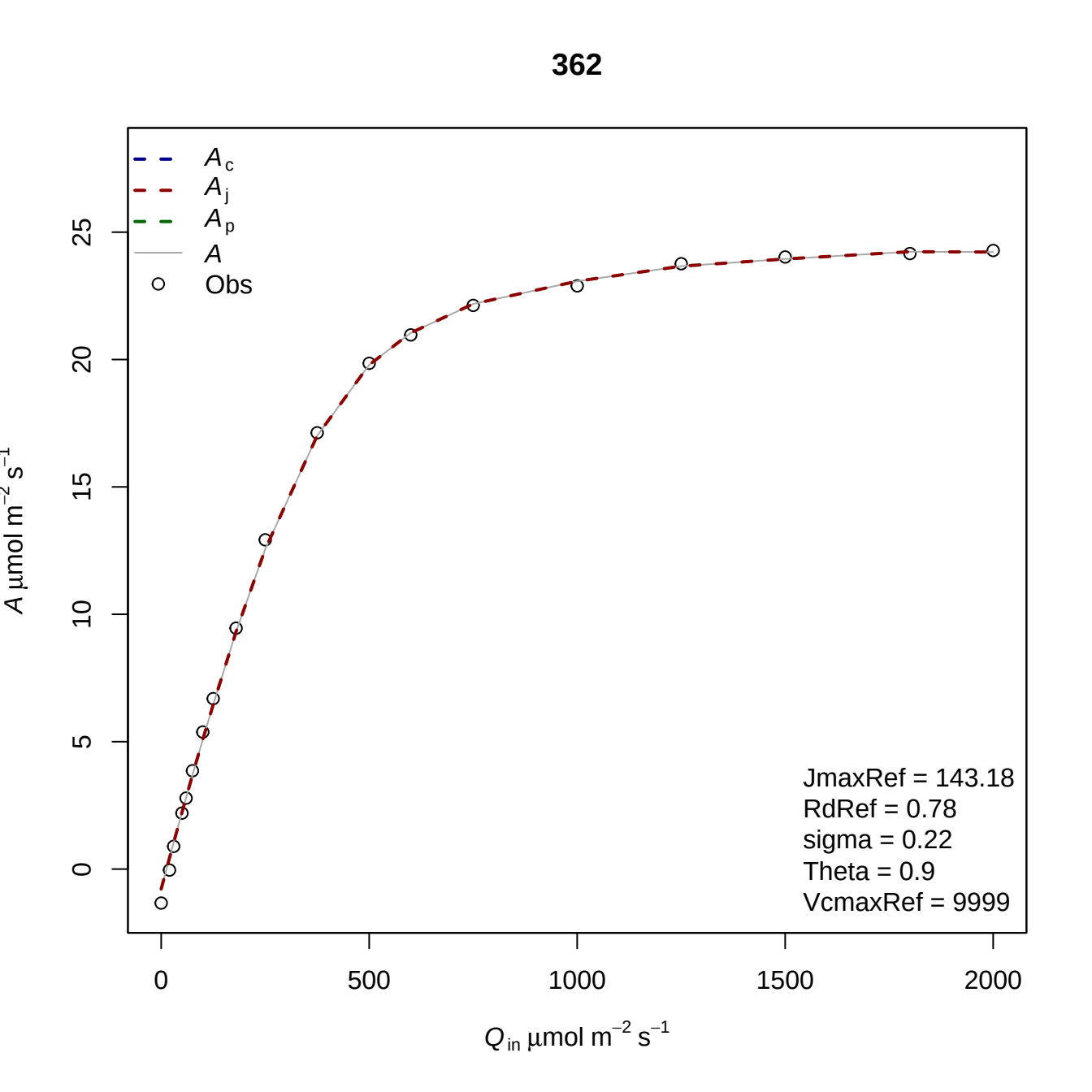

357
A_c
A_j
A_p
A
Obs
JmaxRef = 88.85
RdRef = 2.95
sigma = 1.02
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

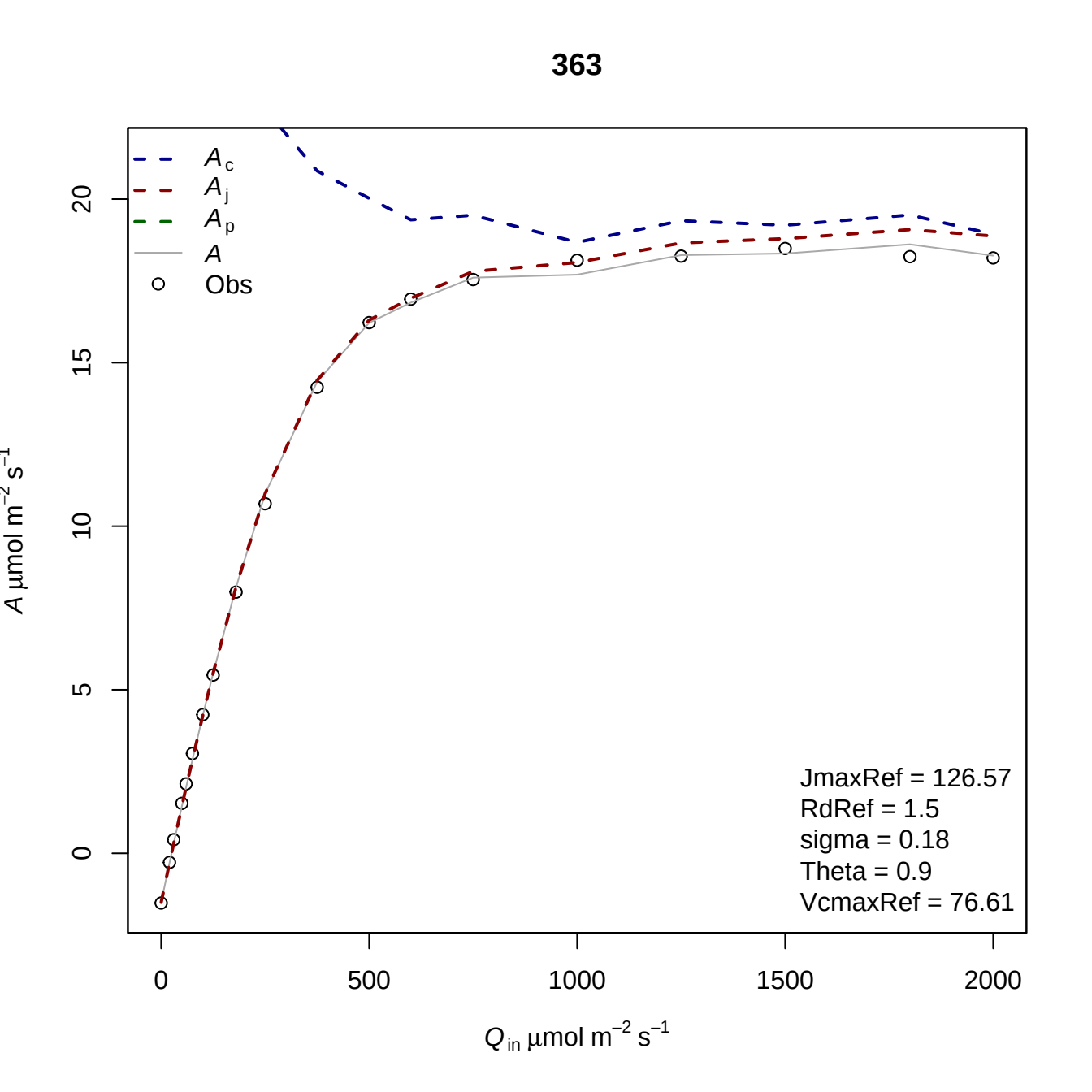

358
A_c
A_j
A_p
A
Obs
JmaxRef = 101.02
RdRef = 3.29
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

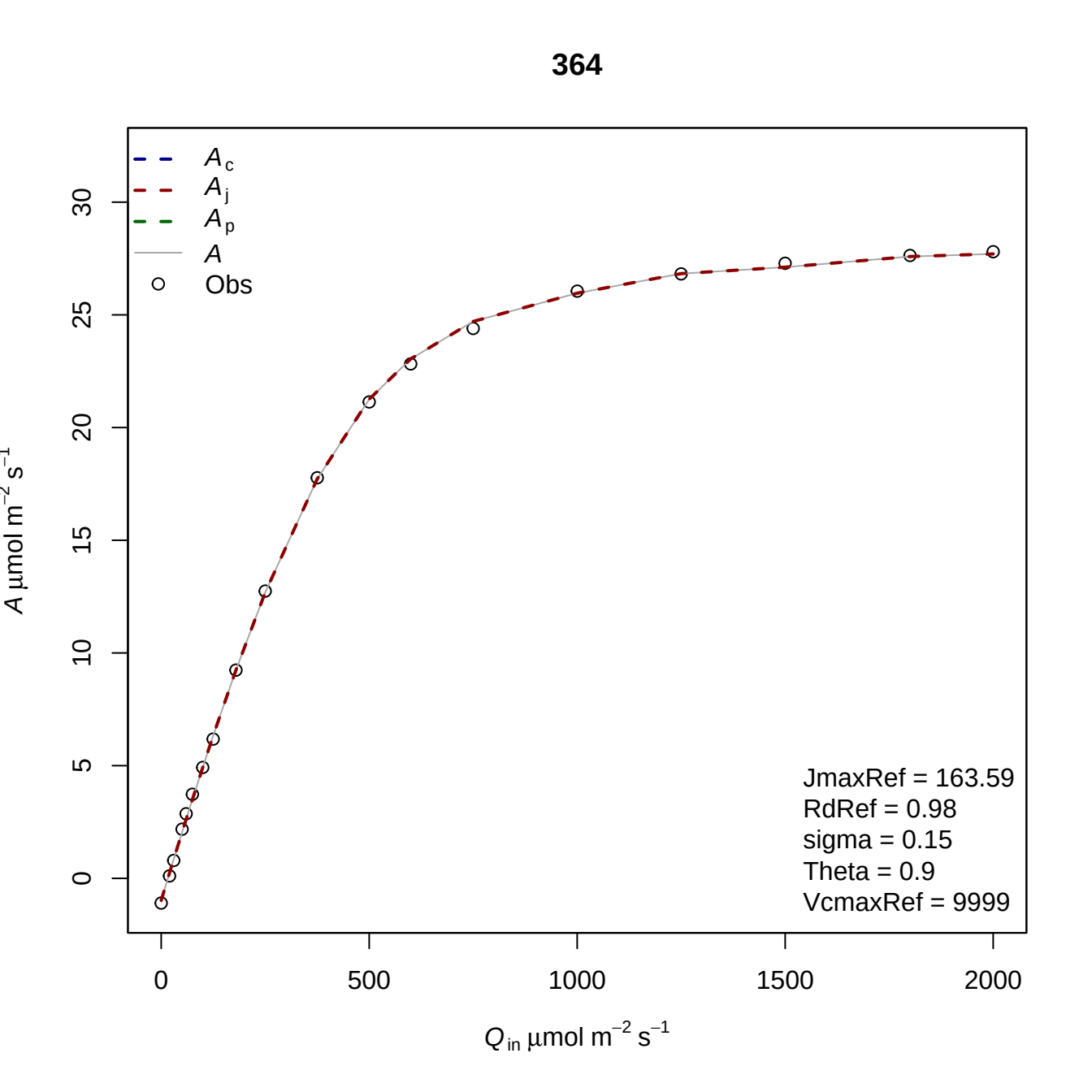

359
A_c
A_j
A_p
A
Obs
JmaxRef = 92.2
RdRef = 2.07
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

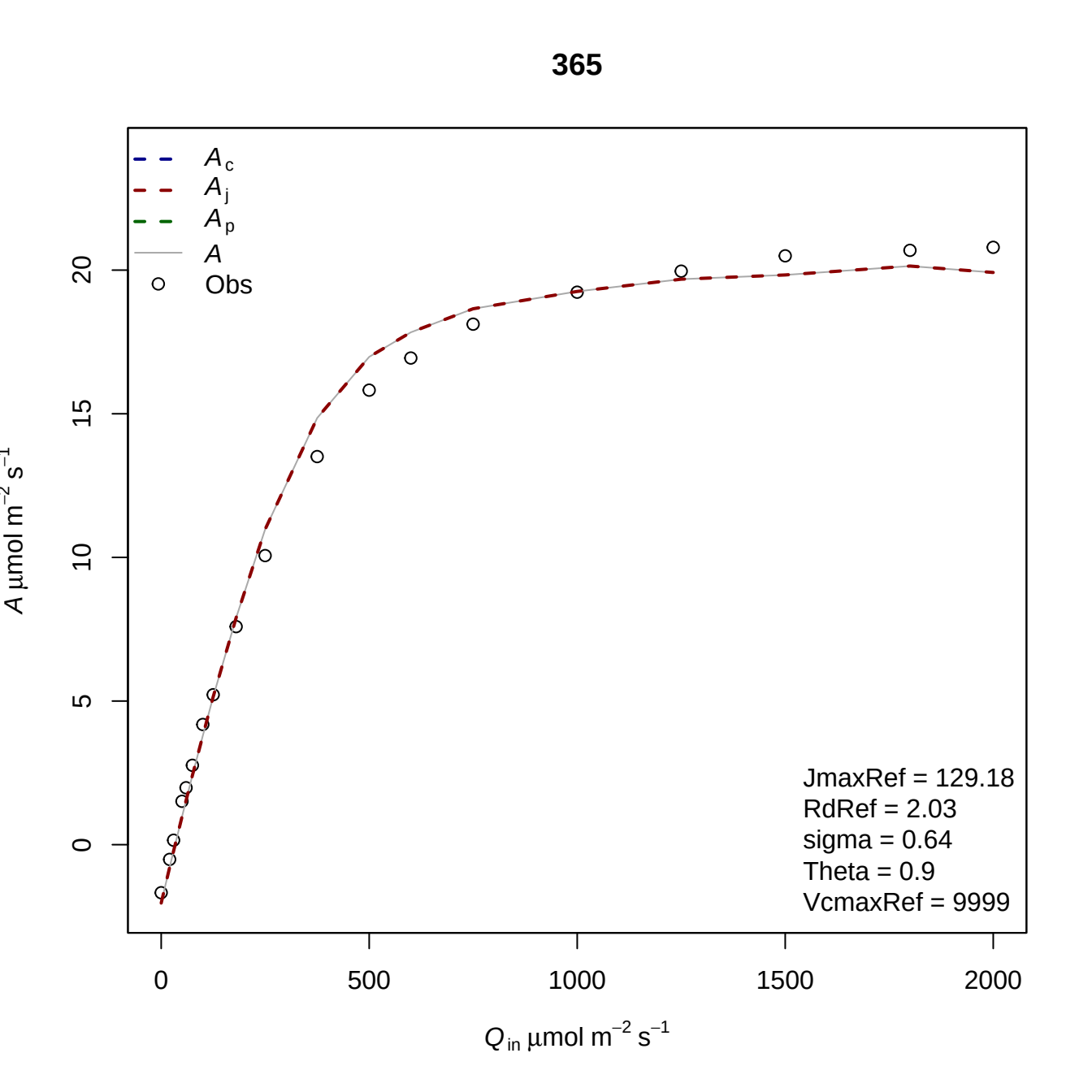

360
A_c
A_j
A_p
A
Obs
JmaxRef = 122.53
RdRef = 2.33
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

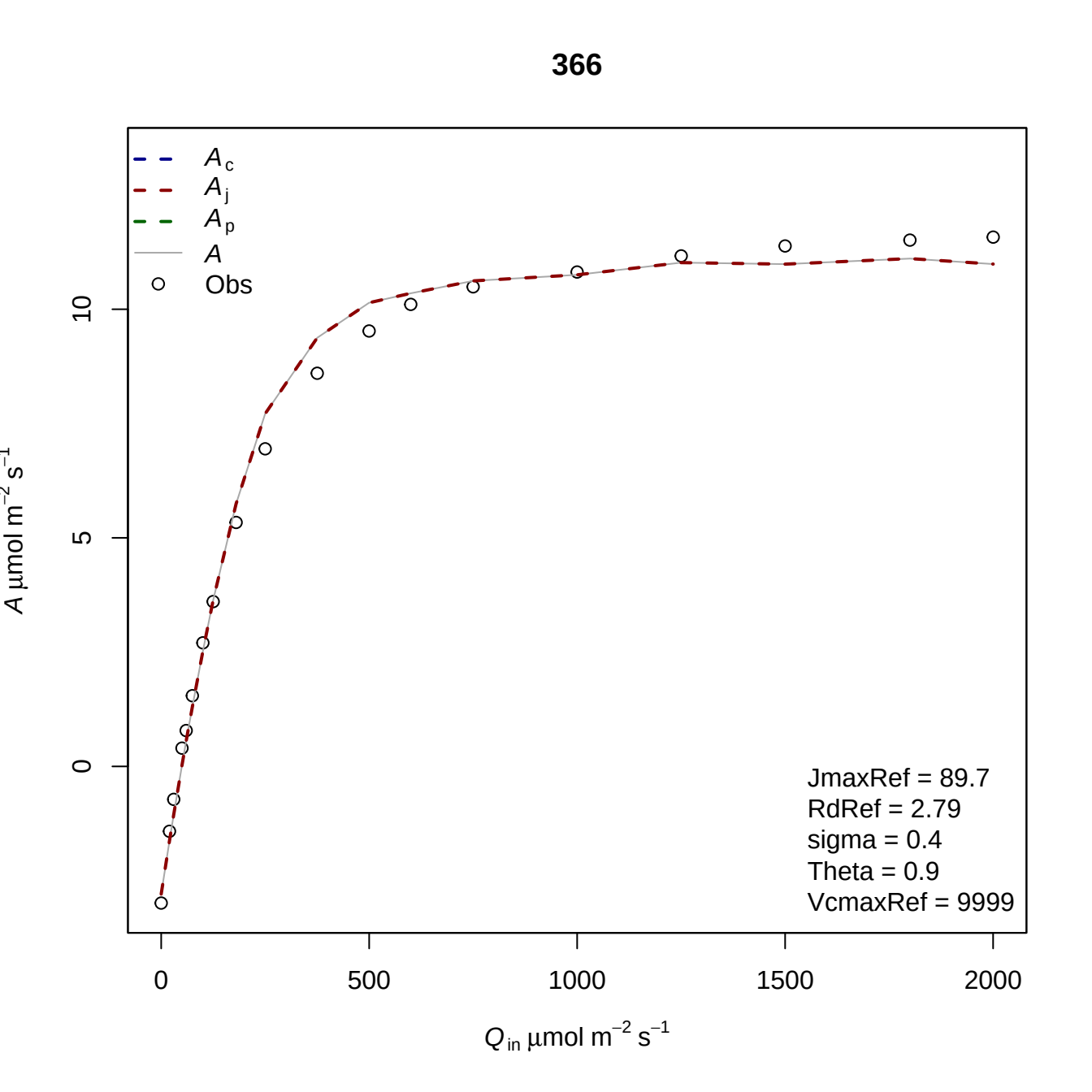

361
A_c
A_j
A_p
A
Obs
JmaxRef = 99.35
RdRef = 2.36
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

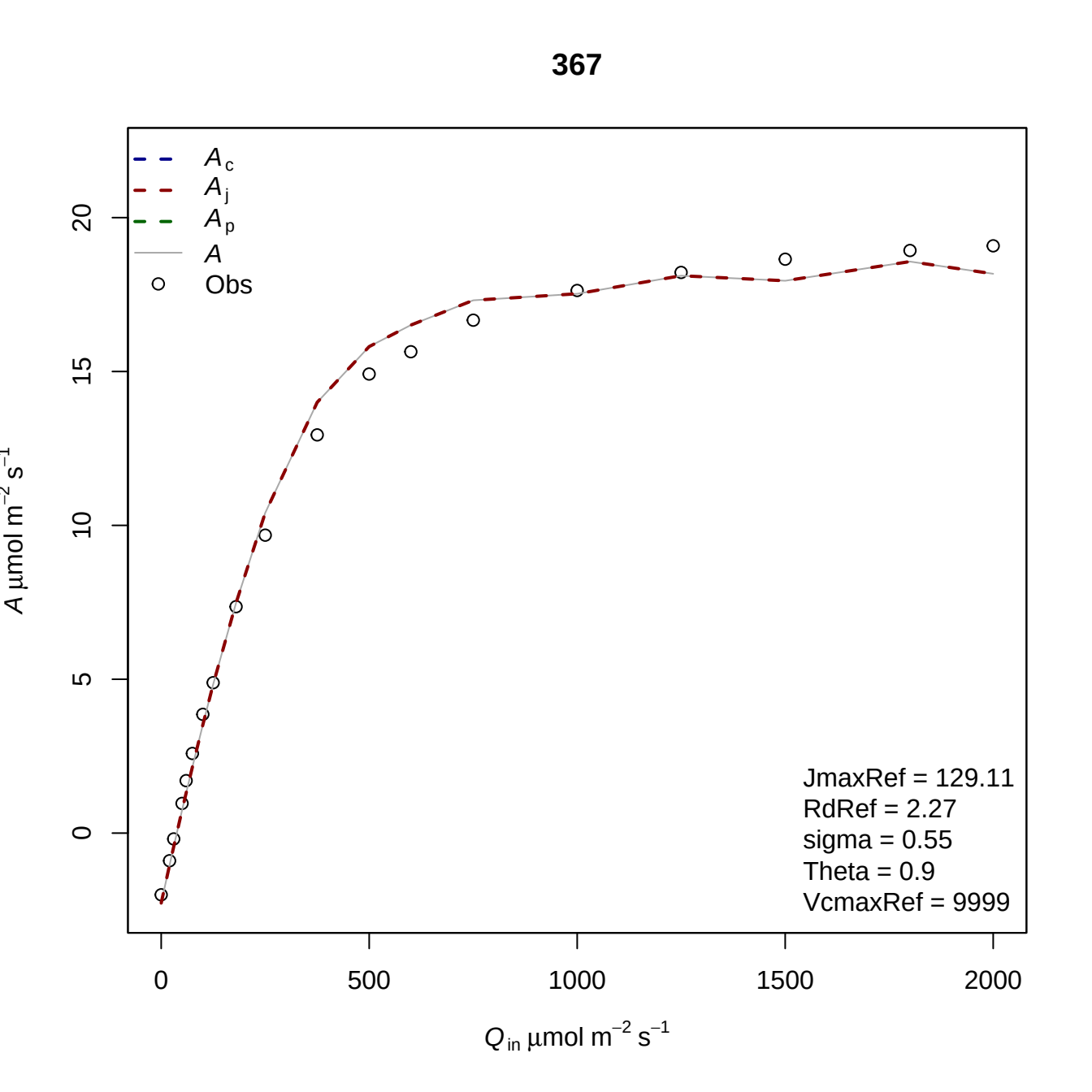

362
A_c
A_j
A_p
A
Obs
JmaxRef = 143.18
RdRef = 0.78
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

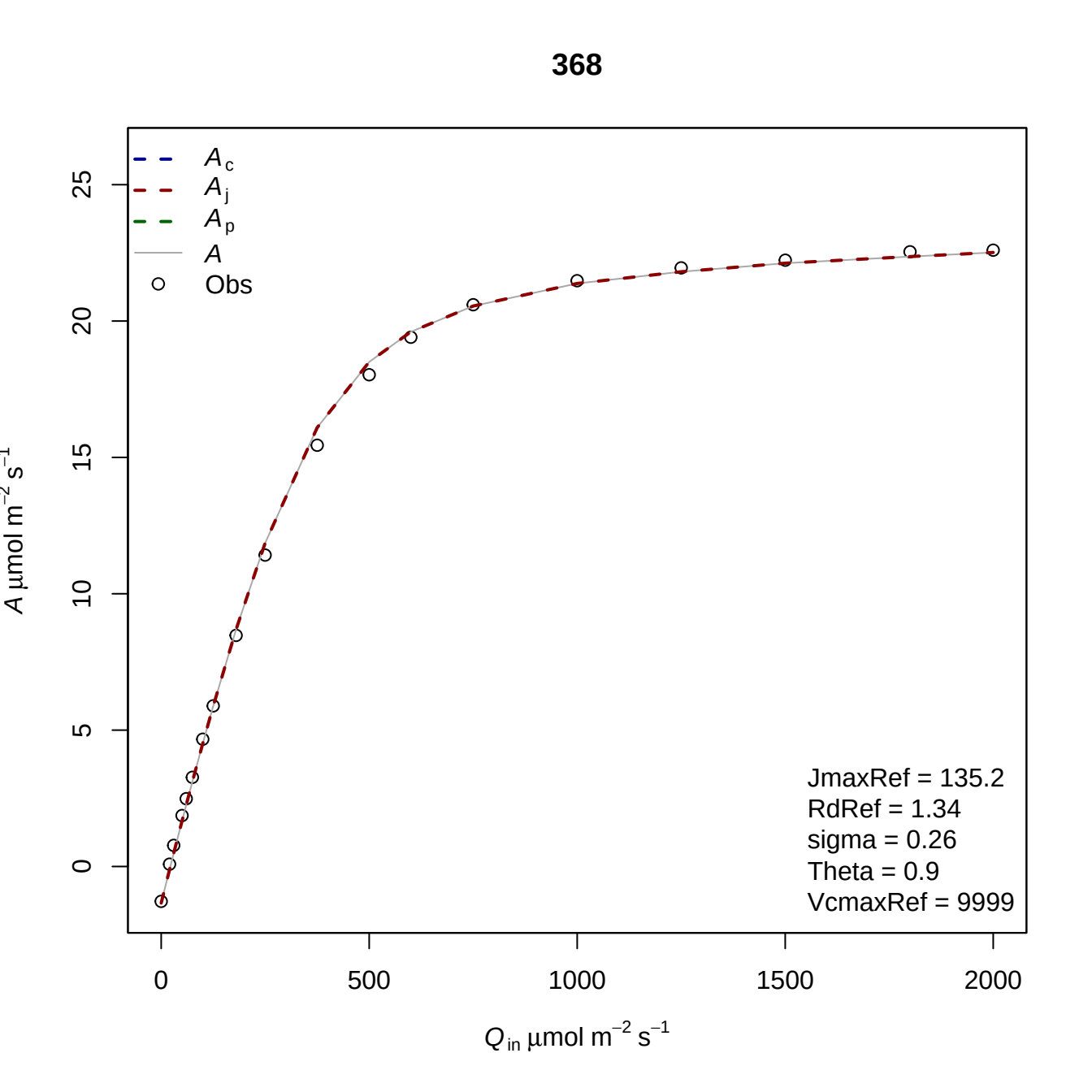

363
A_c
A_j
A_p
A
Obs
JmaxRef = 126.57
RdRef = 1.5
sigma = 0.18
Theta = 0.9
VcmaxRef = 76.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

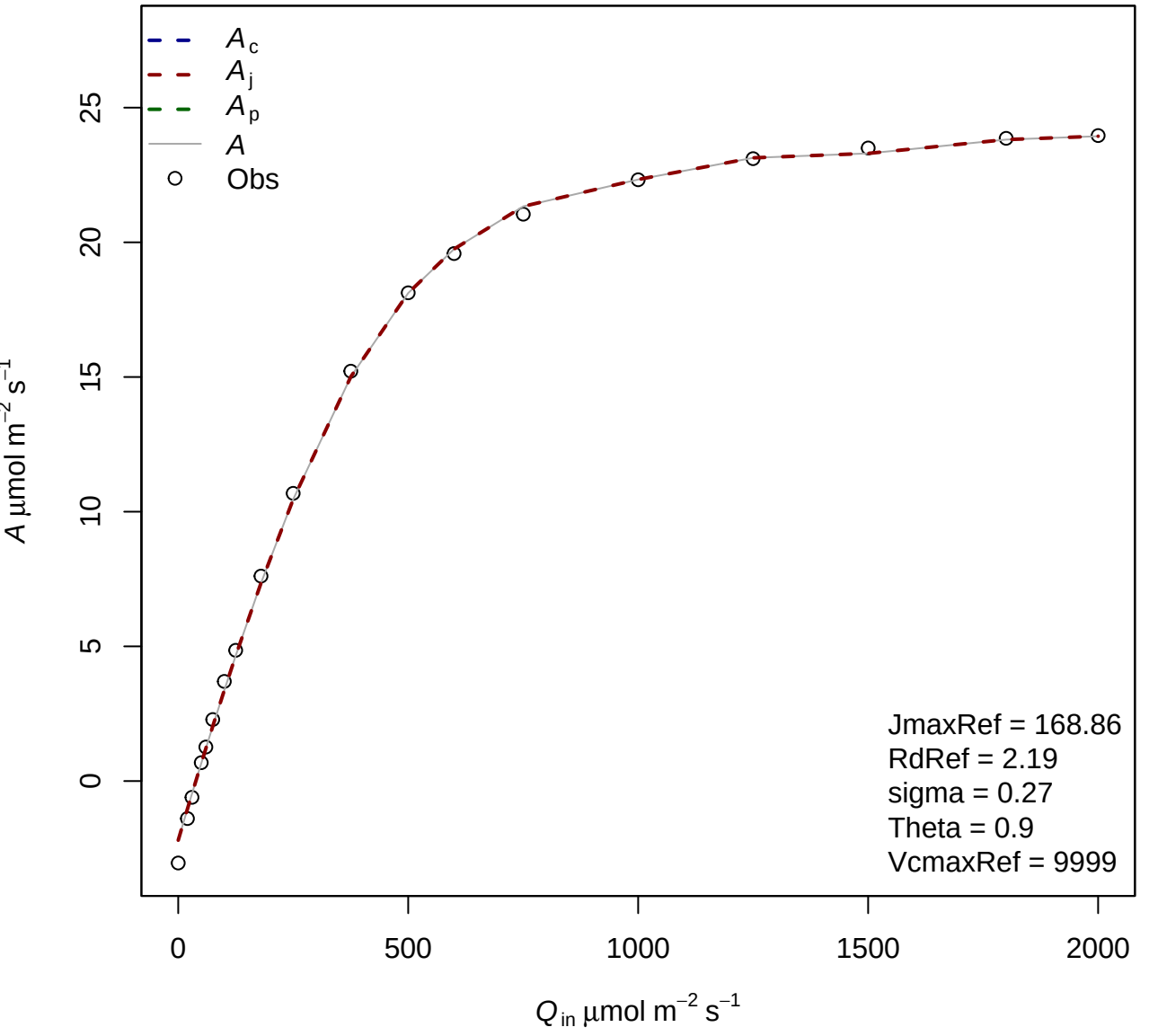

364
A_c
A_j
A_p
A
Obs
JmaxRef = 163.59
RdRef = 0.98
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

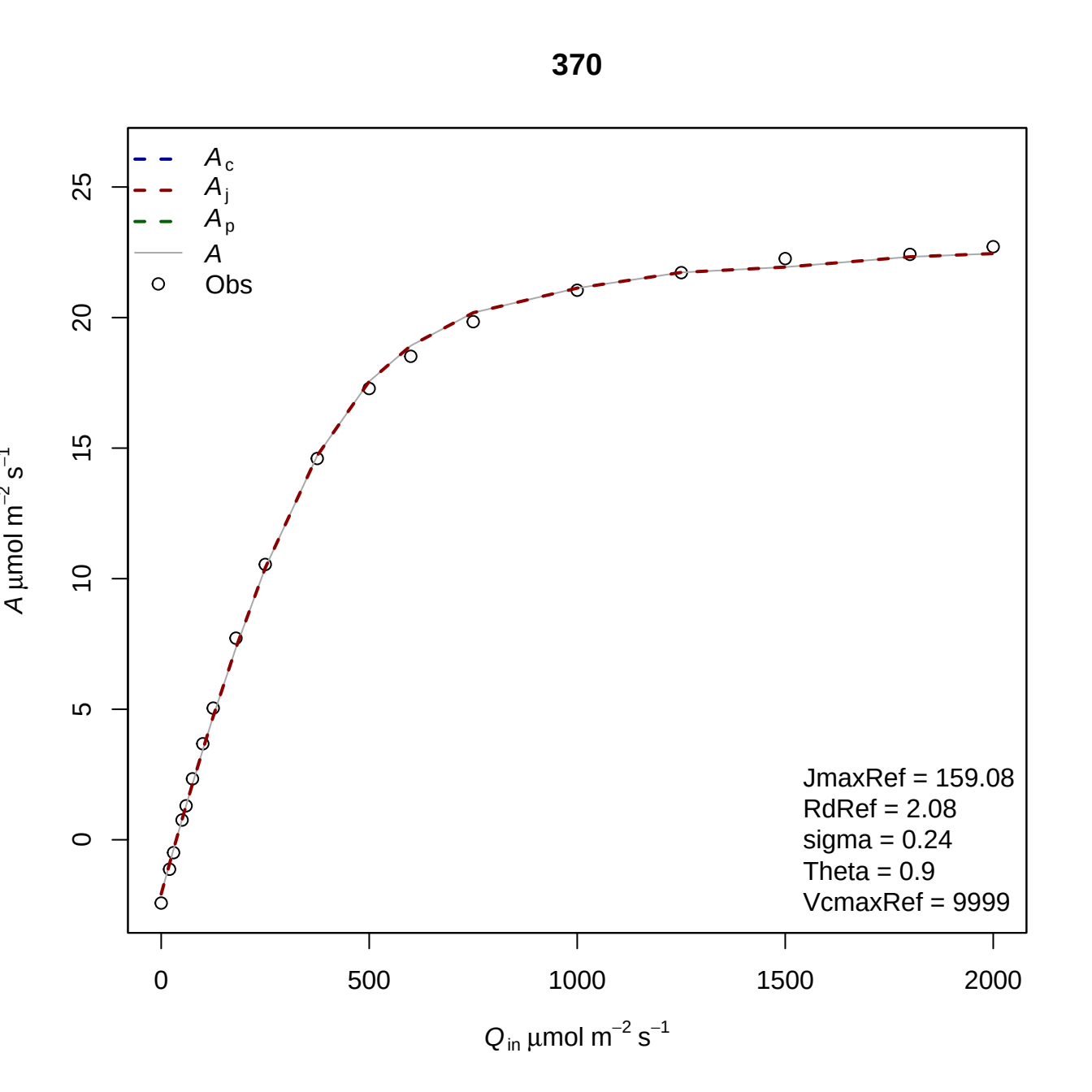

365
A_c
A_j
A_p
A
Obs
JmaxRef = 129.18
RdRef = 2.03
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

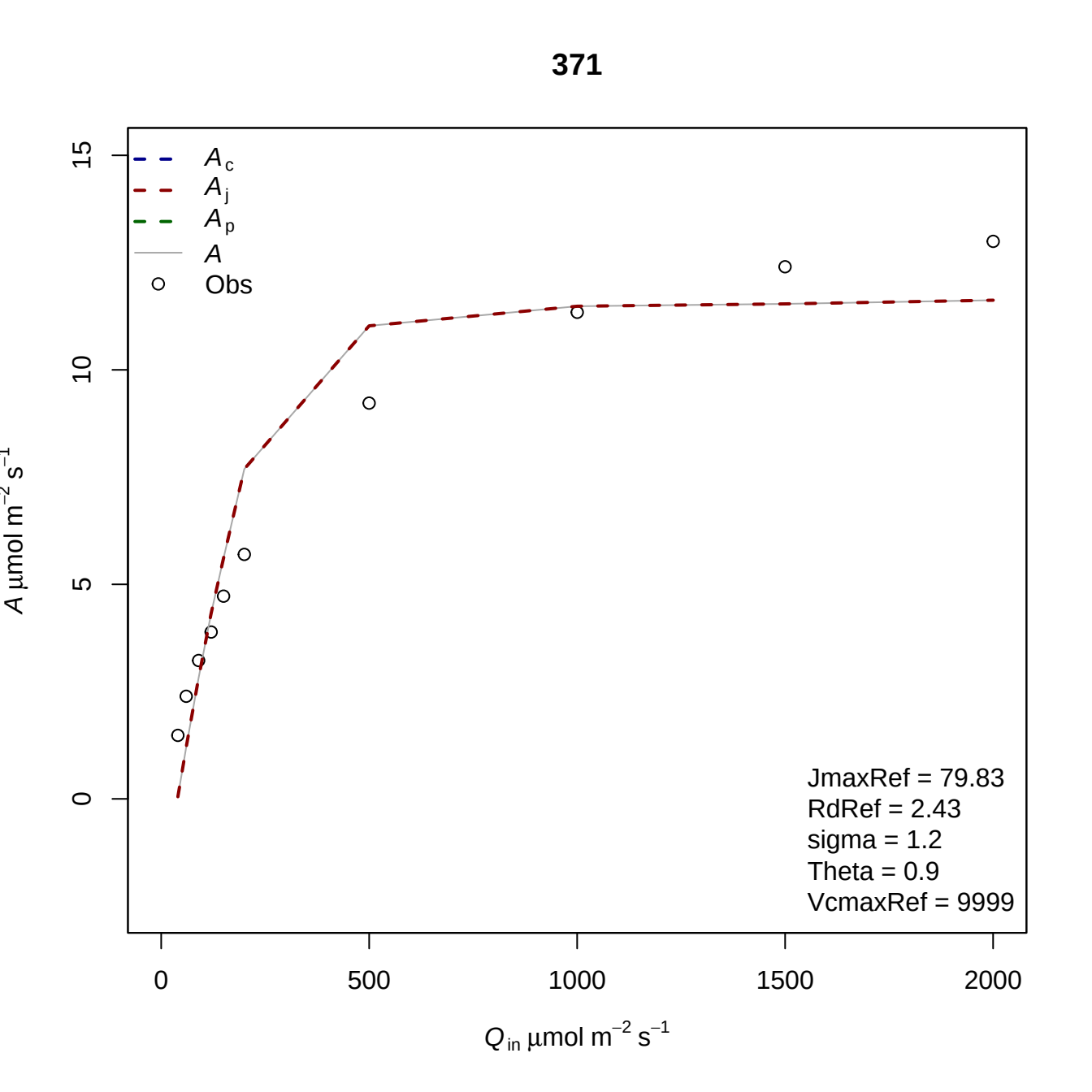

366
A_c
A_j
A_p
A
Obs
JmaxRef = 89.7
RdRef = 2.79
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

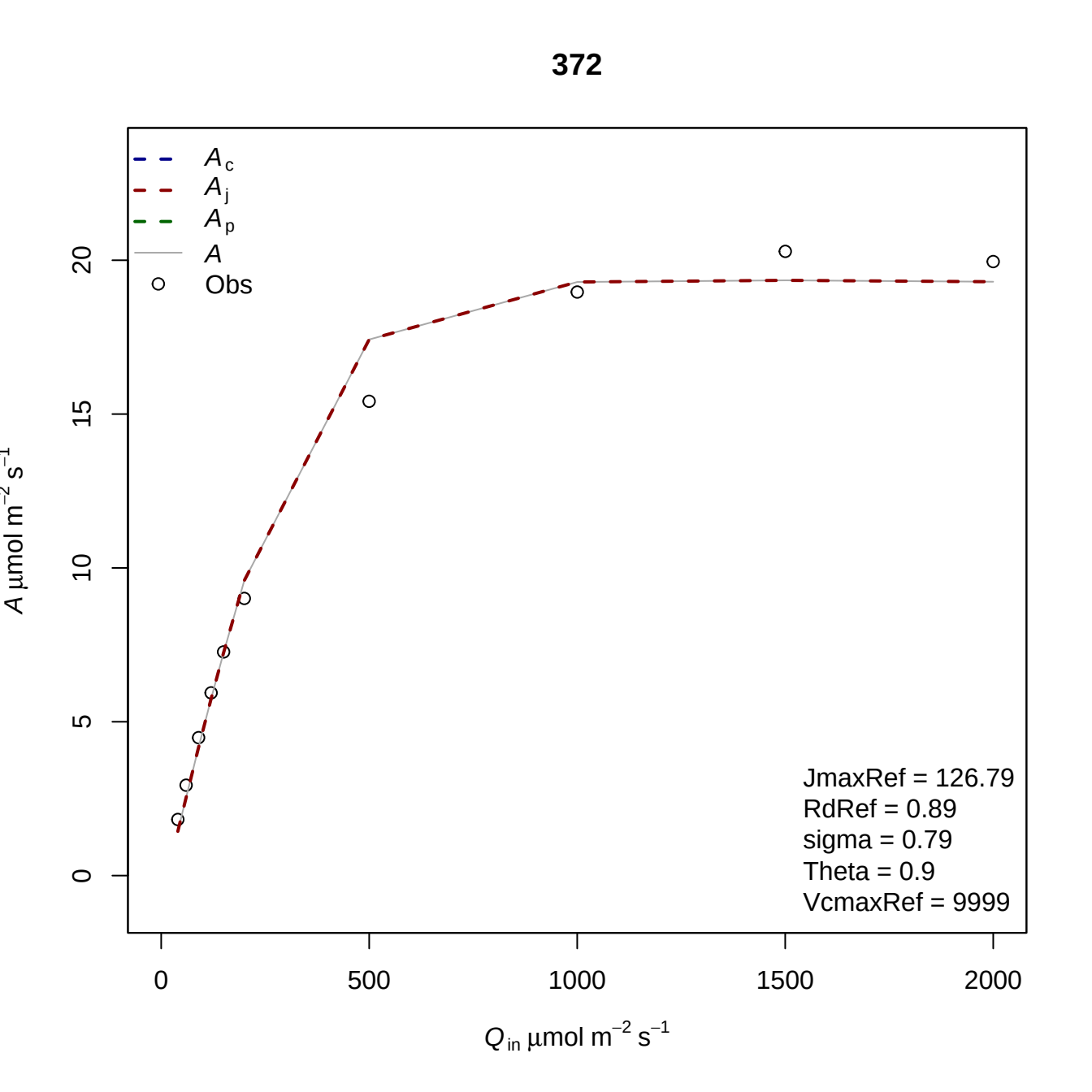

367
A_c
A_j
A_p
A
Obs
JmaxRef = 129.11
RdRef = 2.27
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

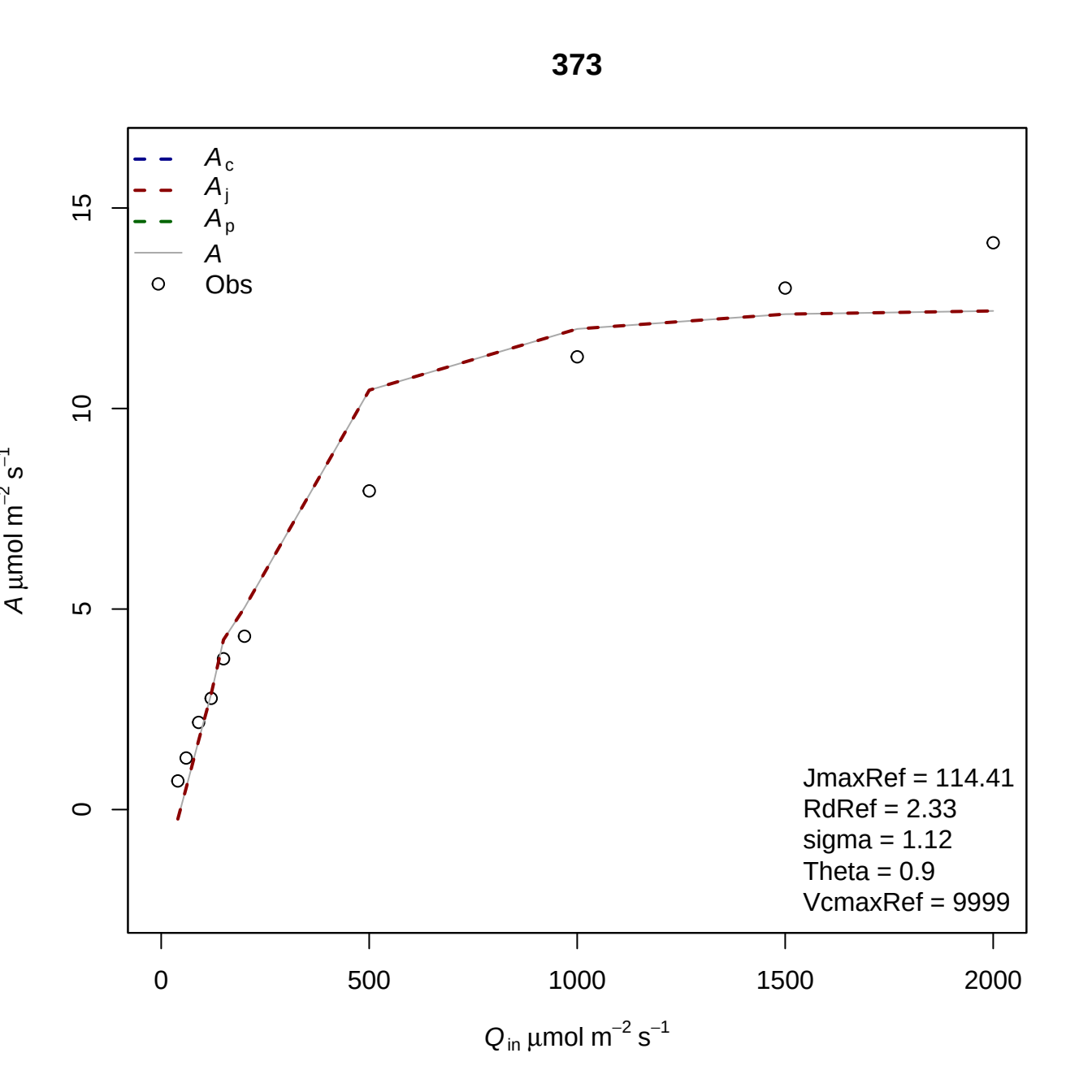

368
A_c
A_j
A_p
A
Obs
JmaxRef = 135.2
RdRef = 1.34
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

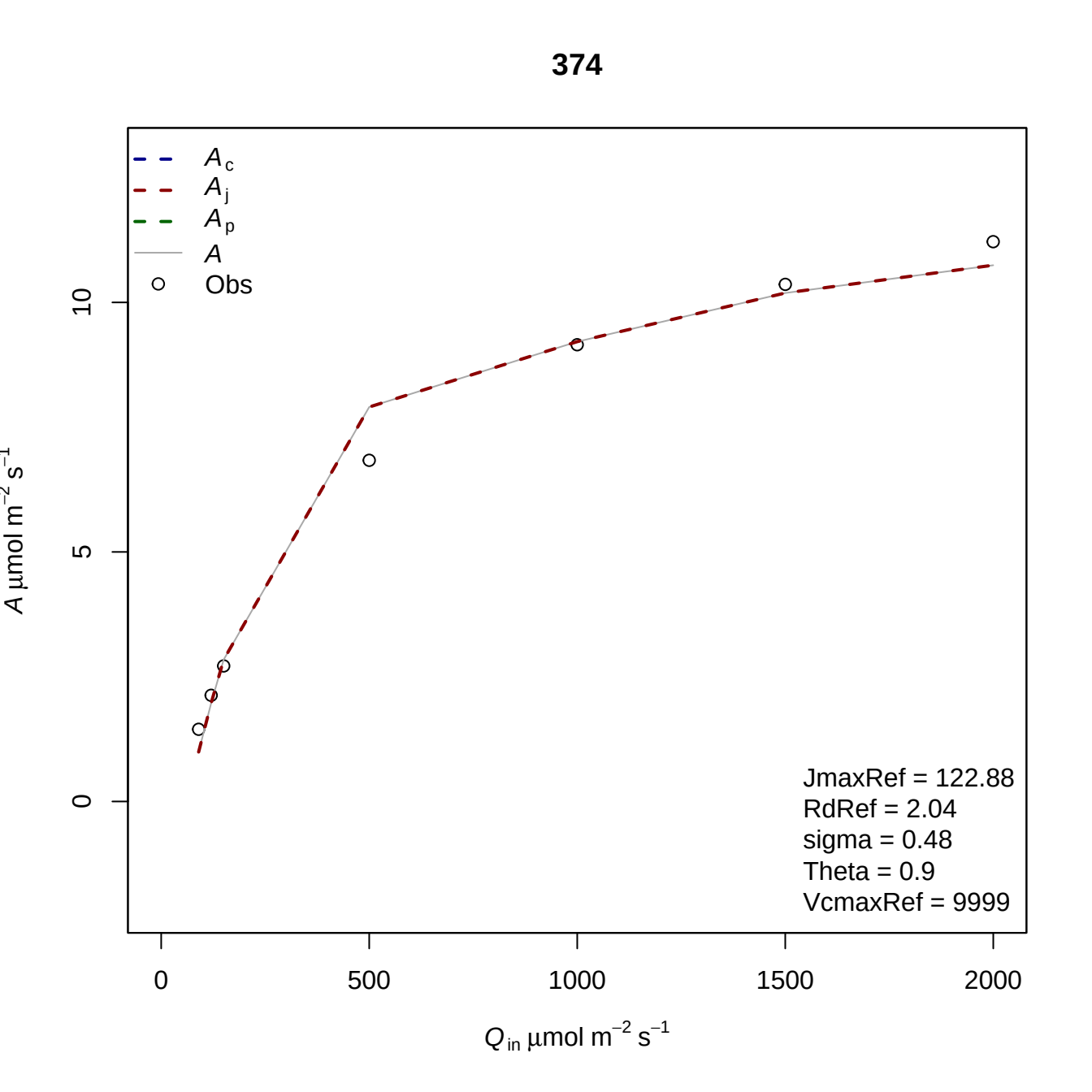

A_c
A_j
A_p
A
Obs
JmaxRef = 168.86
RdRef = 2.19
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

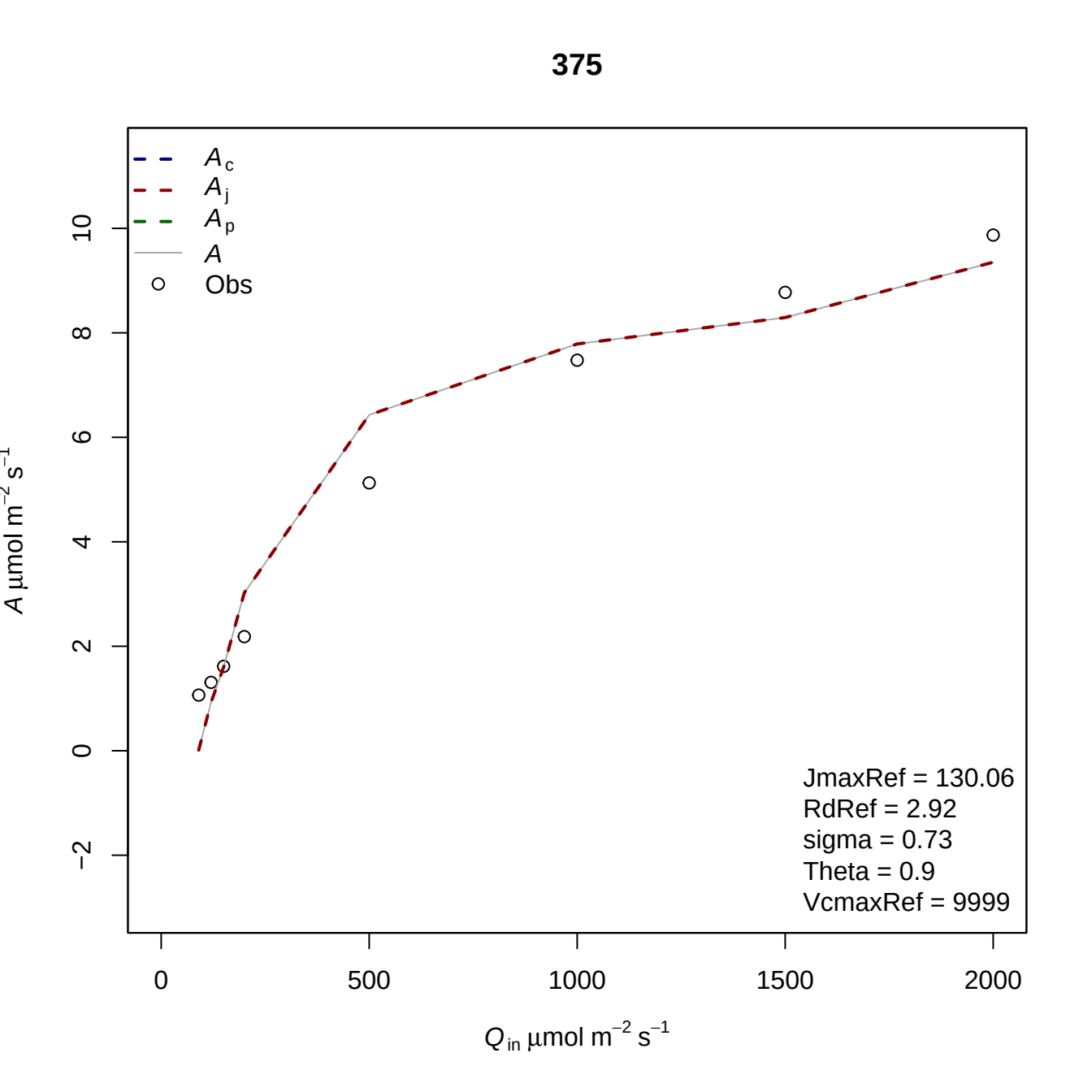

370
A_c
A_j
A_p
A
Obs
JmaxRef = 159.08
RdRef = 2.08
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

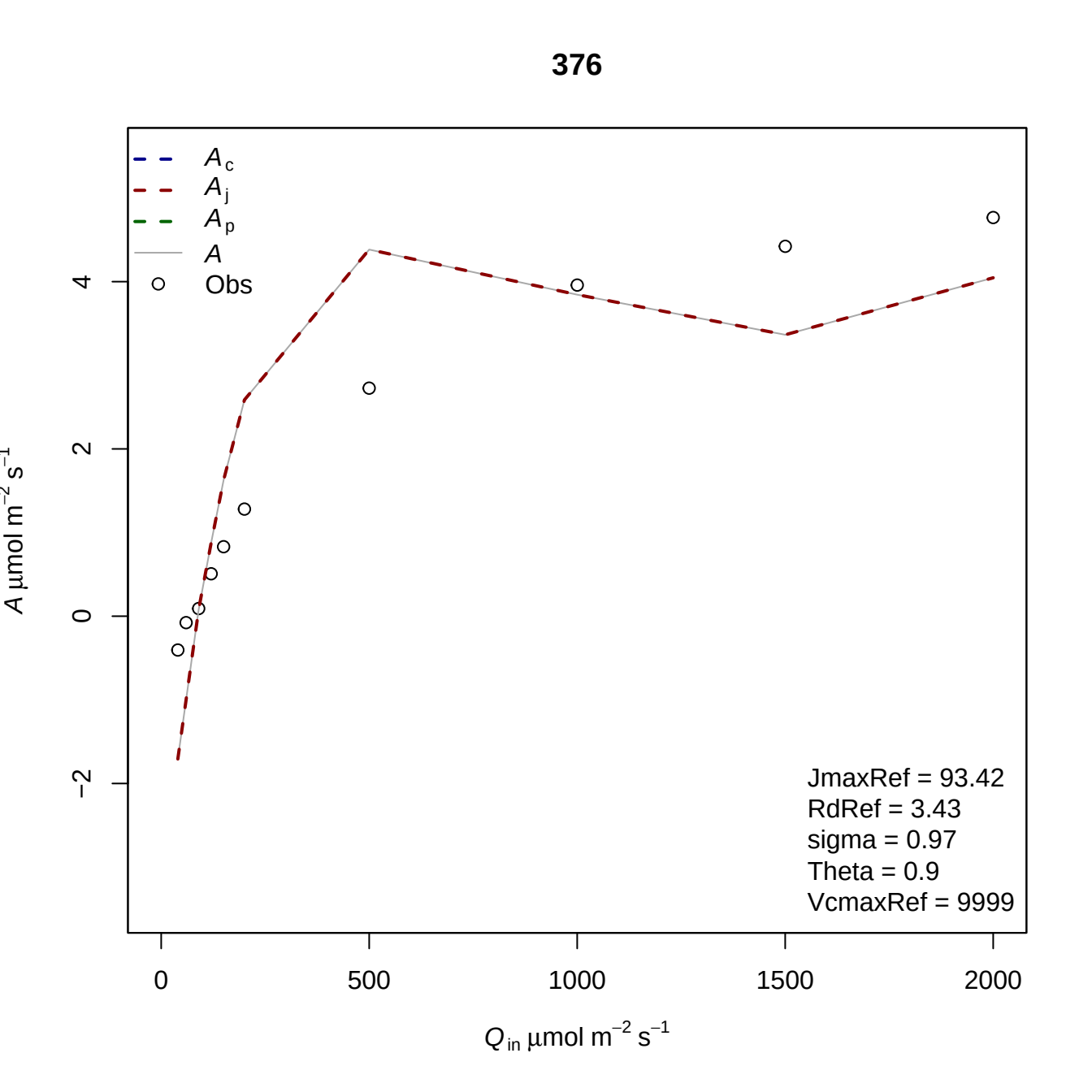

371
A_c
A_j
A_p
A
Obs
JmaxRef = 79.83
RdRef = 2.43
sigma = 1.2
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

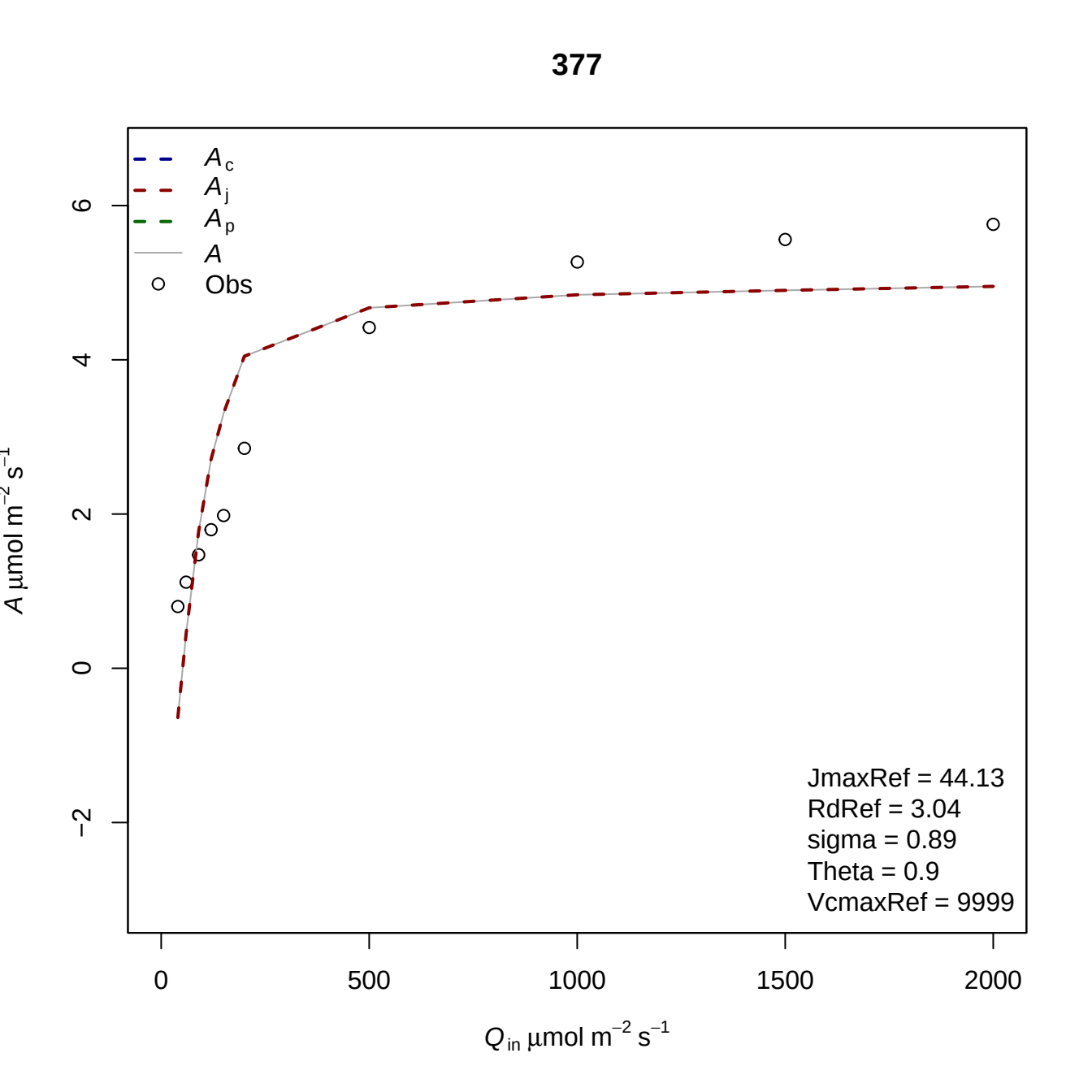

372
A_c
A_j
A_p
A
Obs
JmaxRef = 126.79
RdRef = 0.89
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

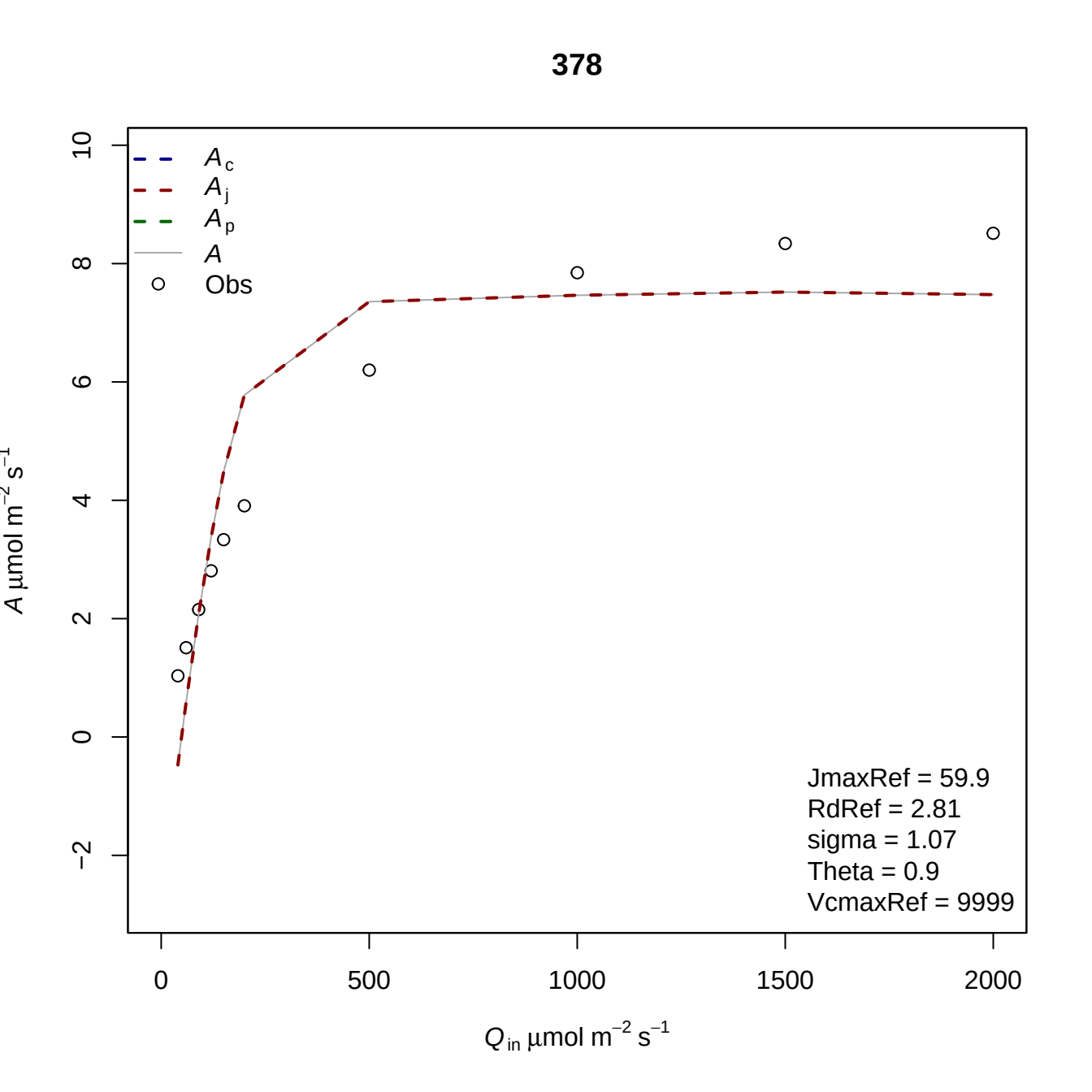

373
A_c
A_j
A_p
A
Obs
JmaxRef = 114.41
RdRef = 2.33
sigma = 1.12
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

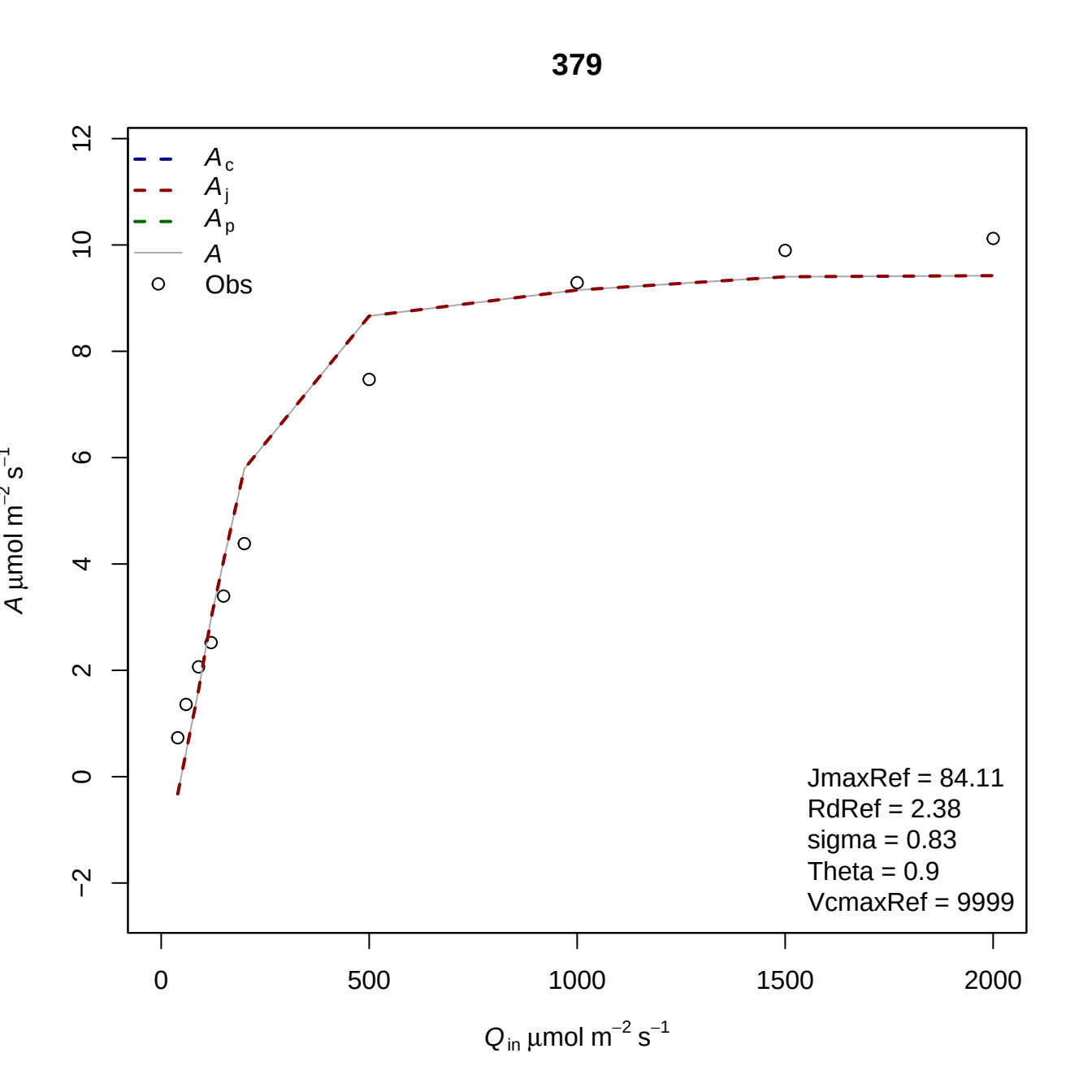

374
A_c
A_j
A_p
A
Obs
JmaxRef = 122.88
RdRef = 2.04
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

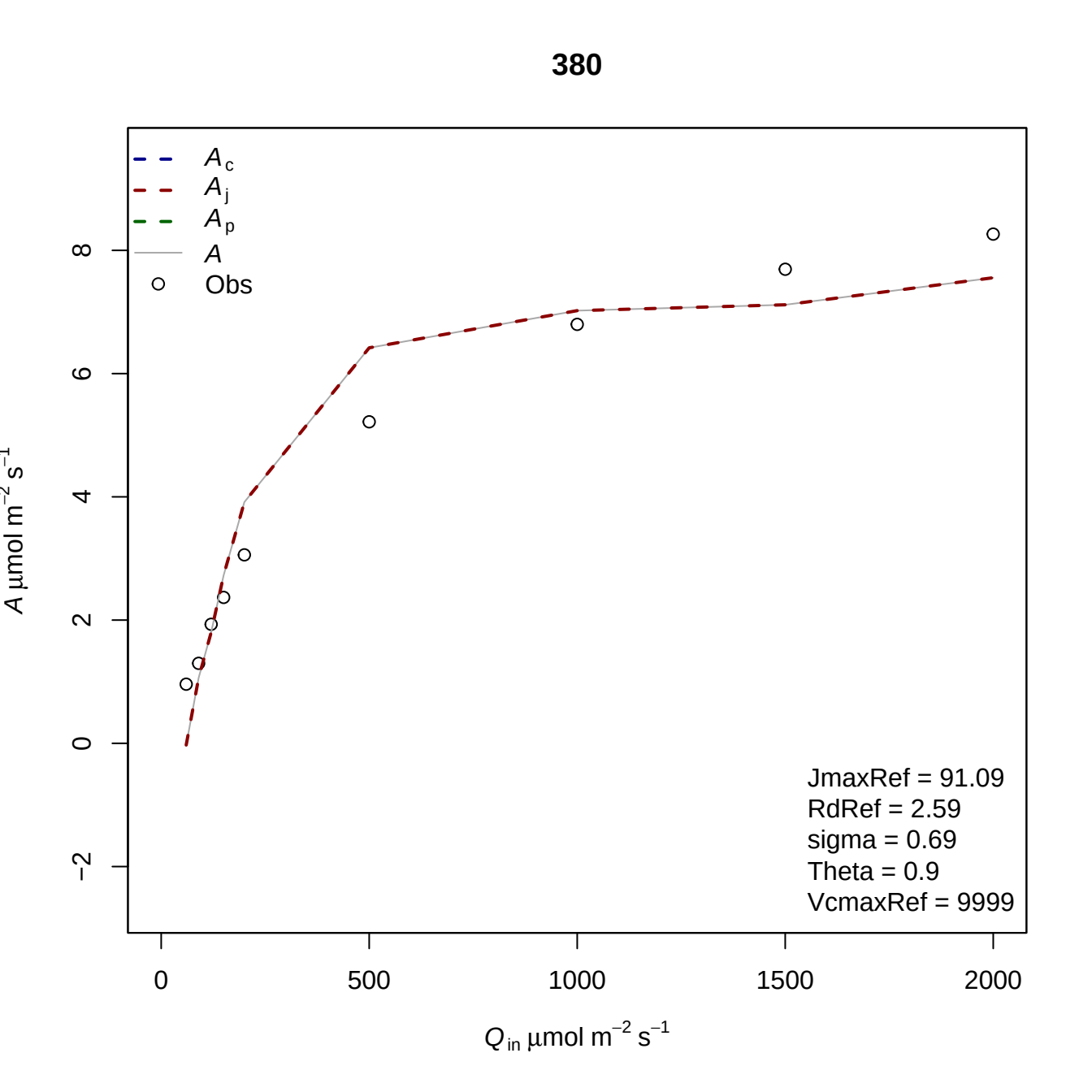

375
A_c
A_j
A_p
A
Obs
JmaxRef = 130.06
RdRef = 2.92
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

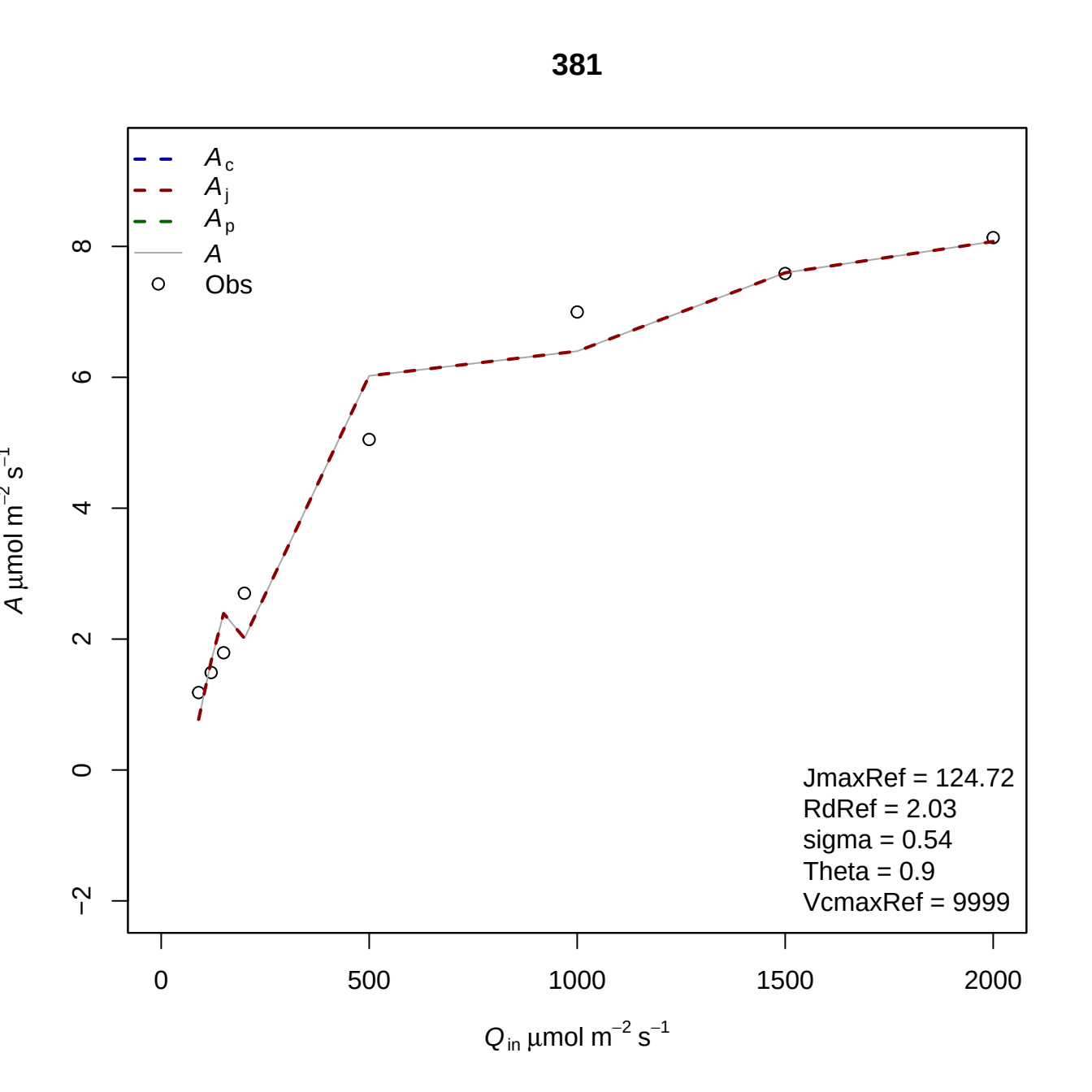

376
A_c
A_j
A_p
A
Obs
JmaxRef = 93.42
RdRef = 3.43
sigma = 0.97
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

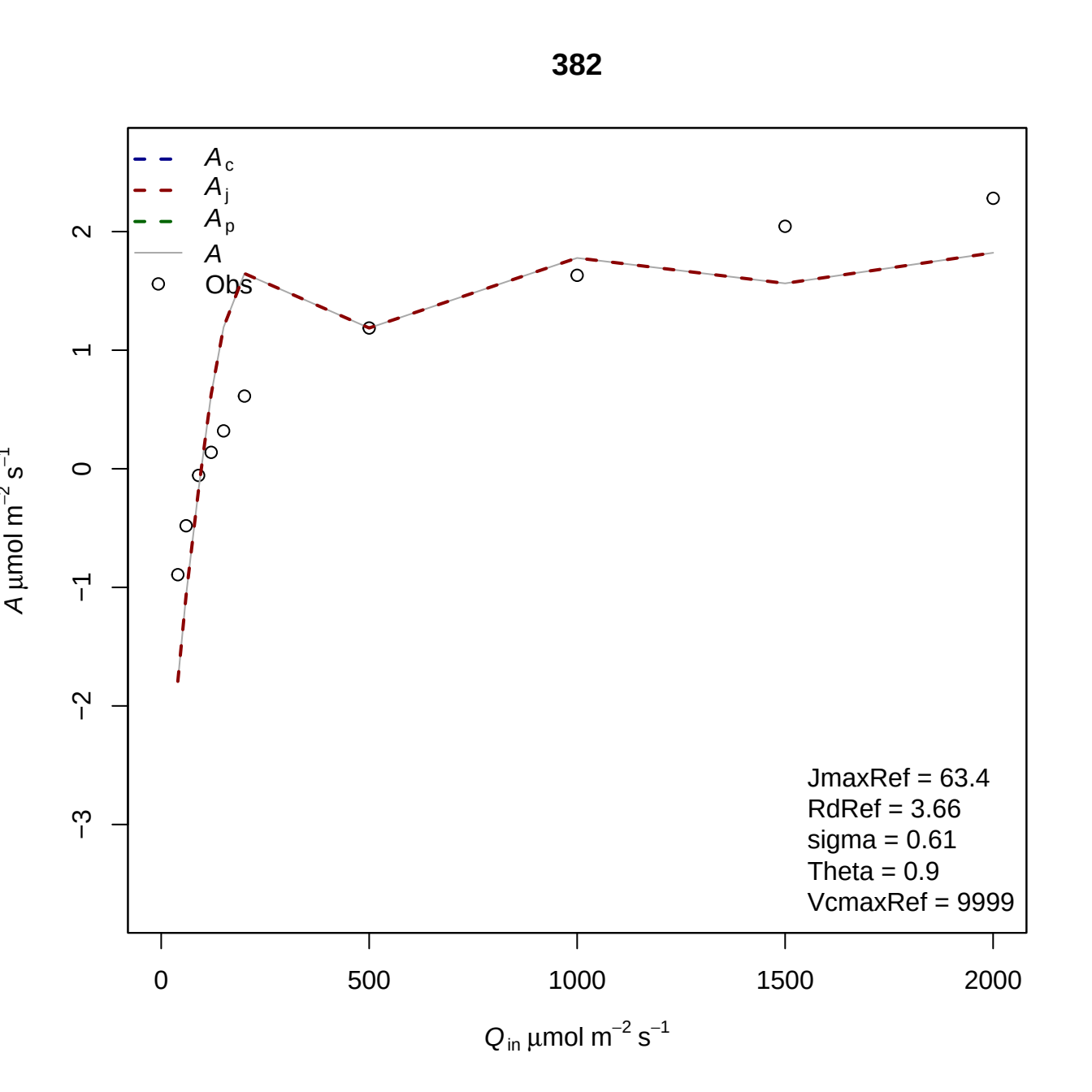

377
A_c
A_j
A_p
A
Obs
JmaxRef = 44.13
RdRef = 3.04
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

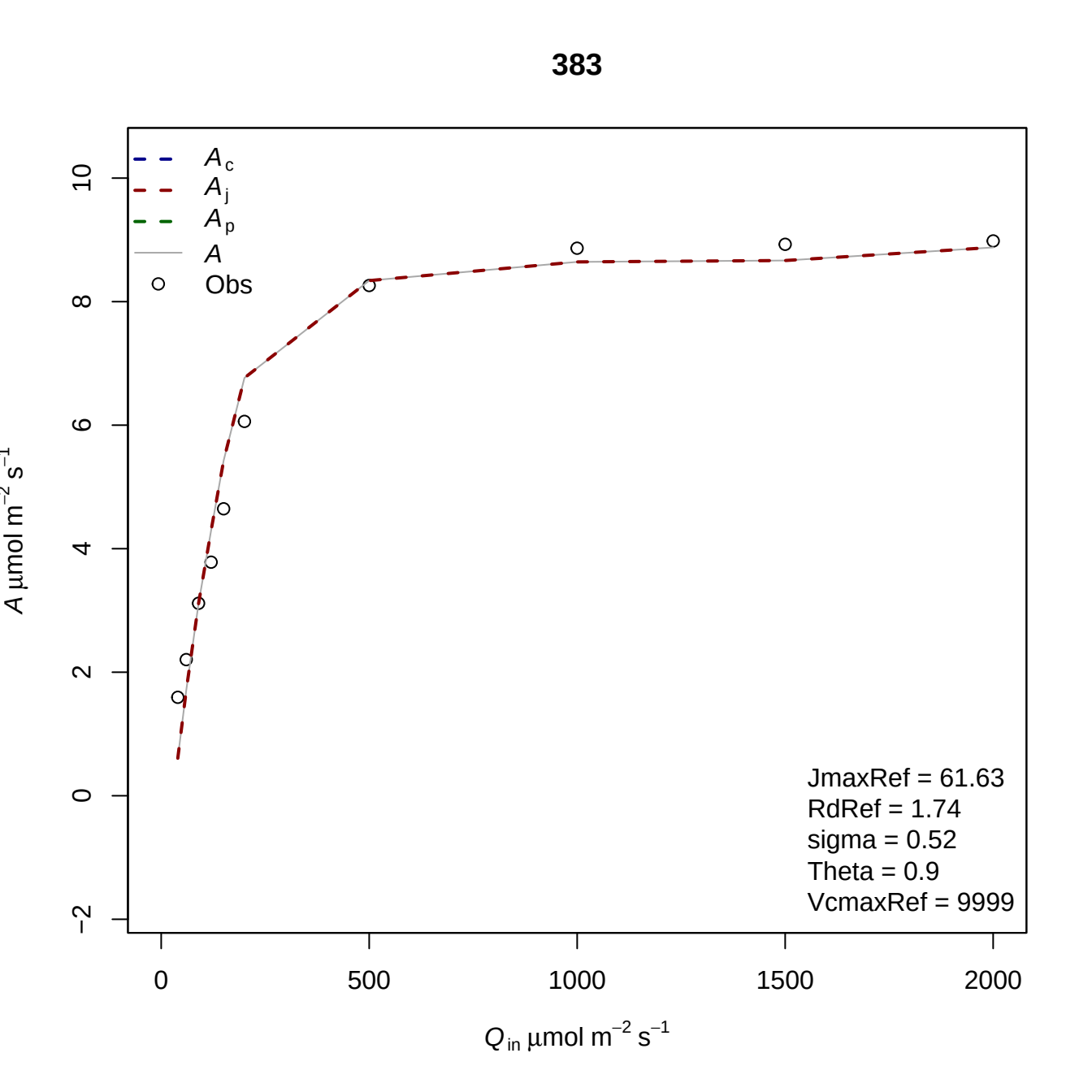

378
A_c
A_j
A_p
A
Obs
JmaxRef = 59.9
RdRef = 2.81
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

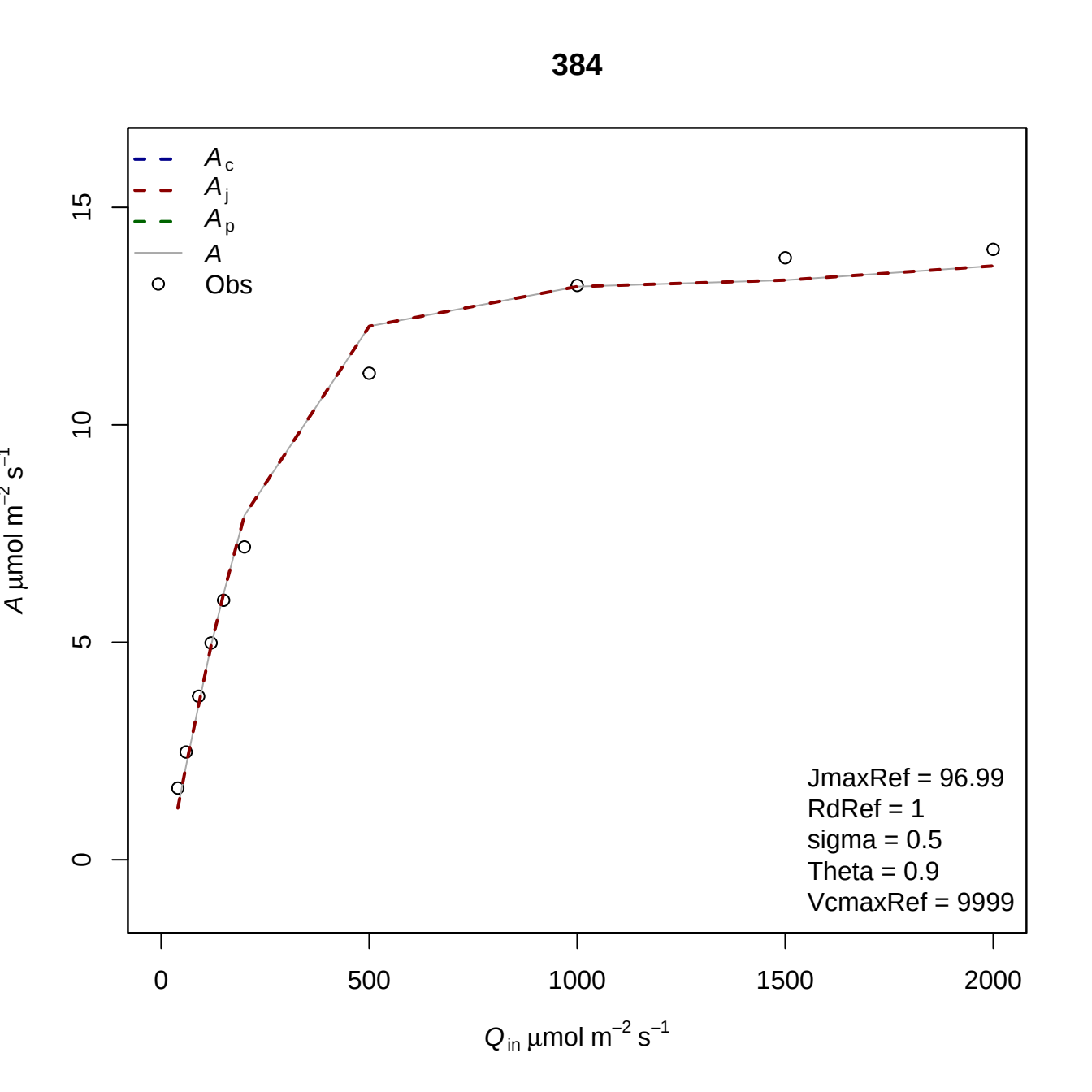

379
A_c
A_j
A_p
A
Obs
JmaxRef = 84.11
RdRef = 2.38
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

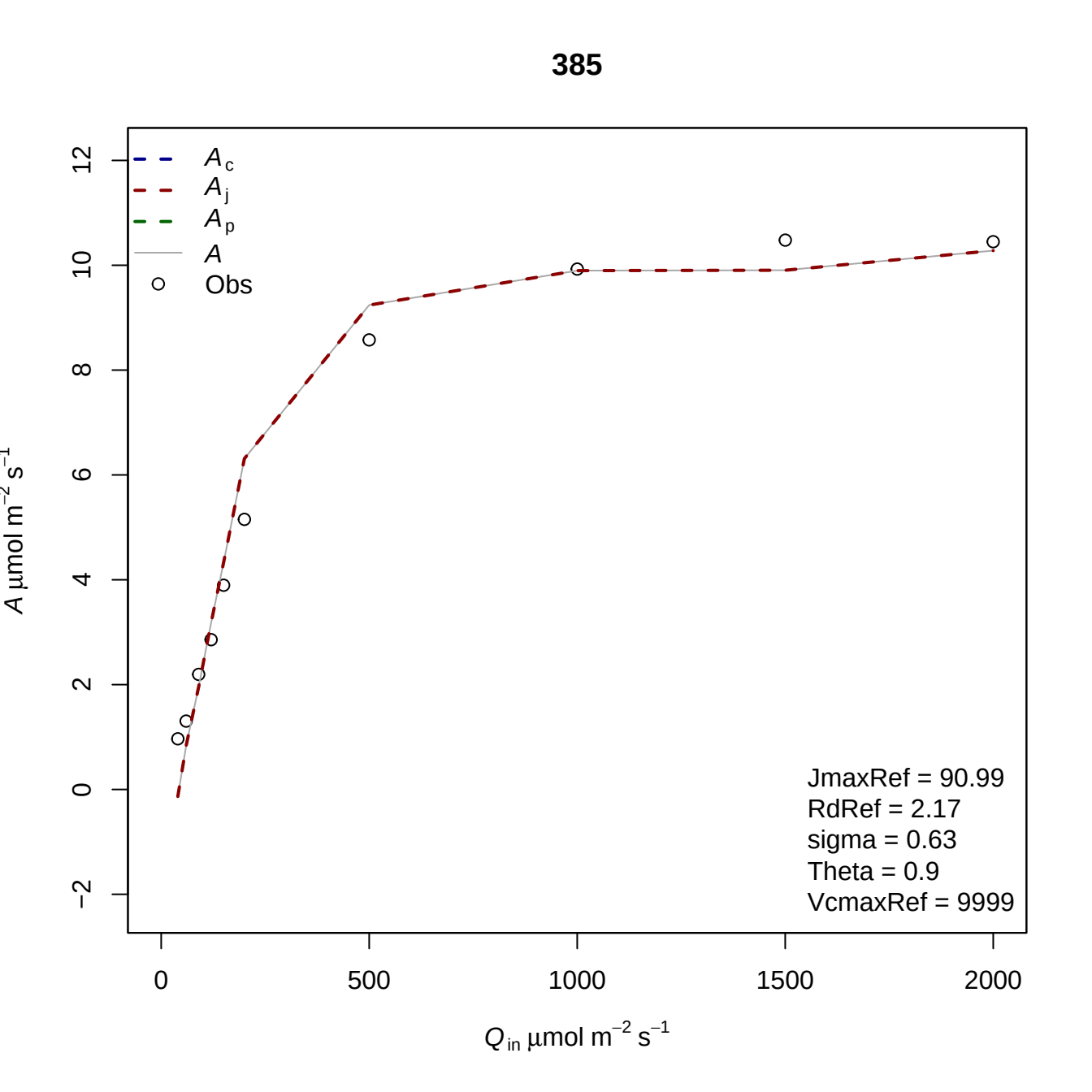

380
A_c
A_j
A_p
A
Obs
JmaxRef = 91.09
RdRef = 2.59
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

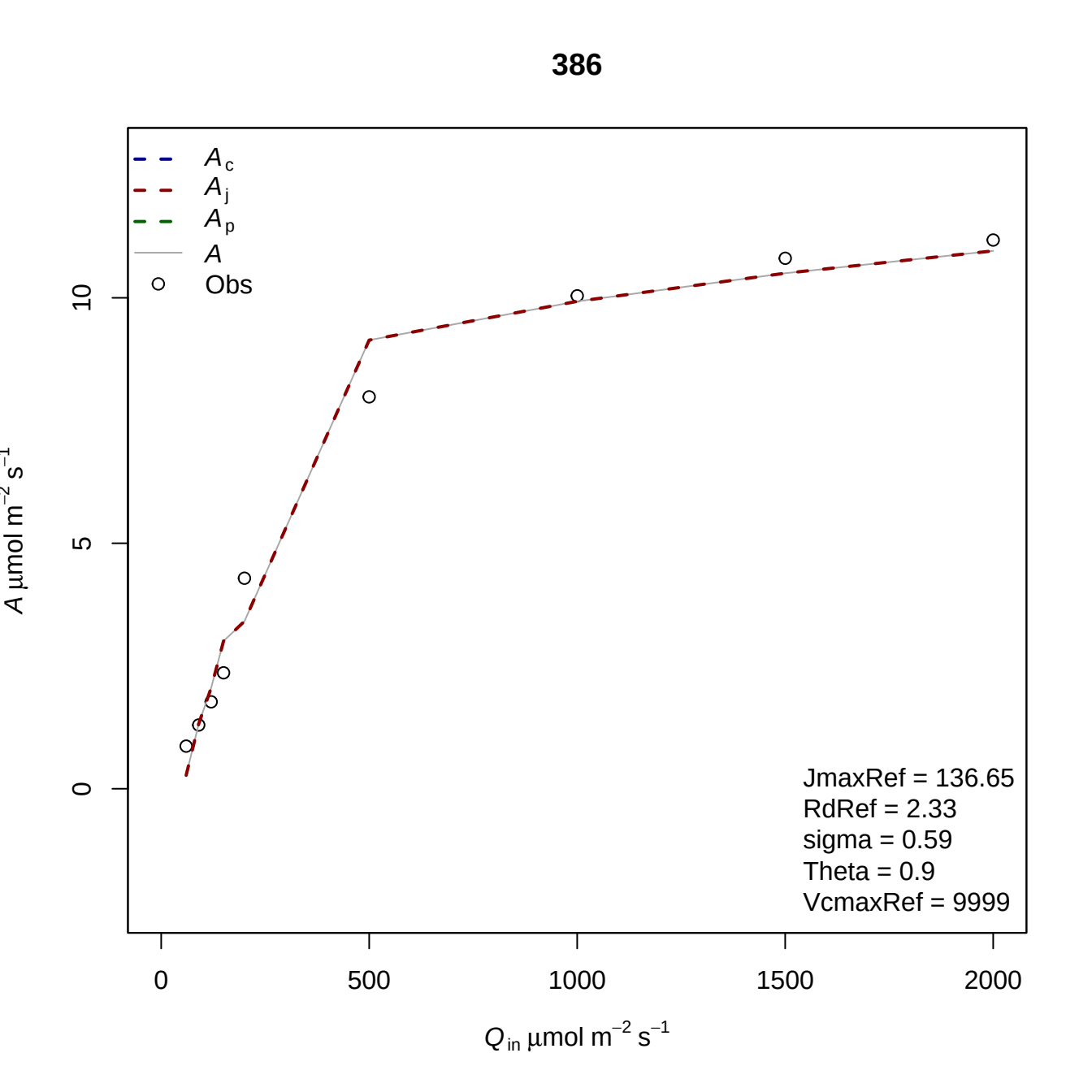

381
A_c
A_j
A_p
A
Obs
JmaxRef = 124.72
RdRef = 2.03
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

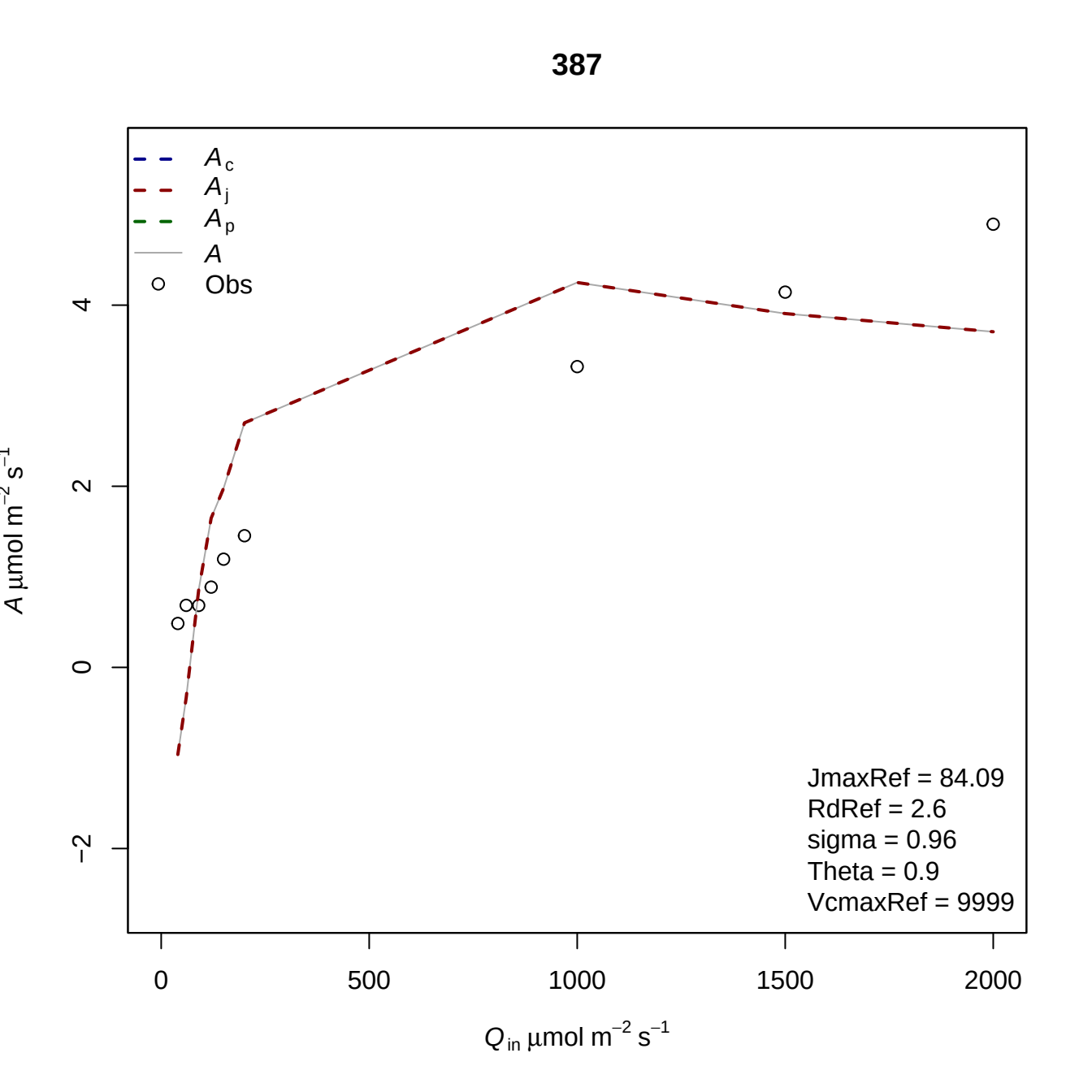

382
A_c
A_j
A_p
A
Obs
JmaxRef = 63.4
RdRef = 3.66
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500
2000

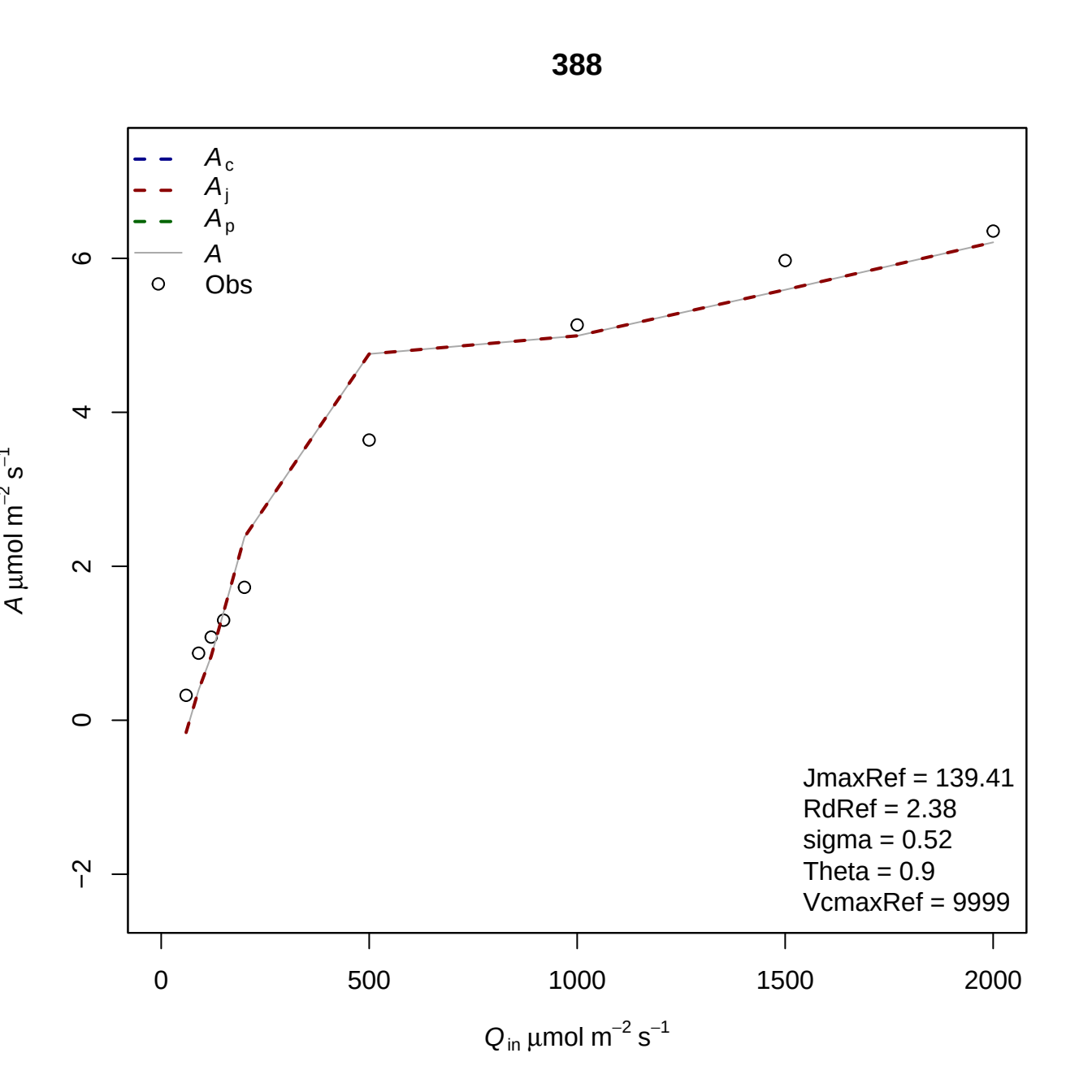

383
A_c
A_j
A_p
A
Obs
JmaxRef = 61.63
RdRef = 1.74
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

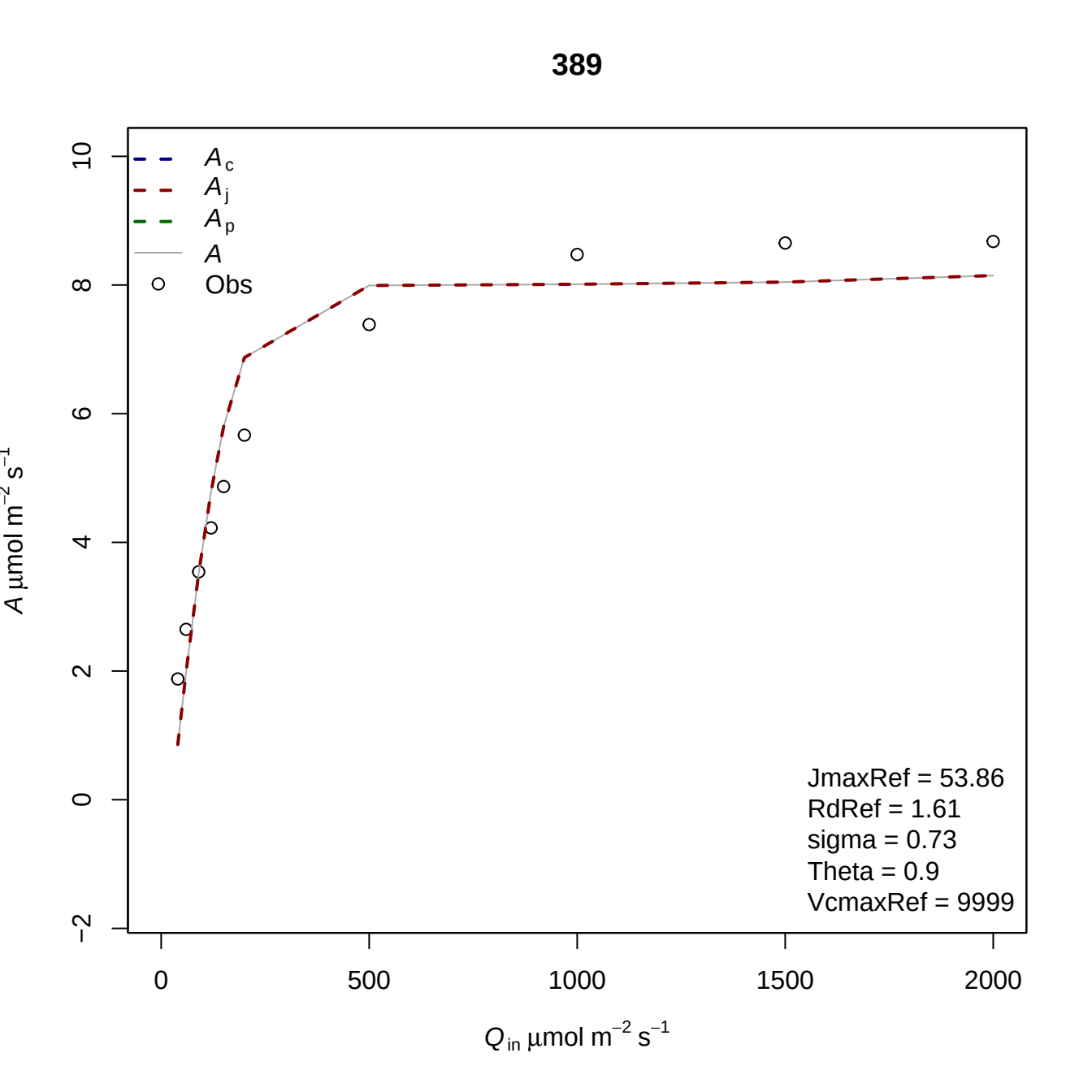

384
A_c
A_j
A_p
A
Obs
JmaxRef = 96.99
RdRef = 1
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

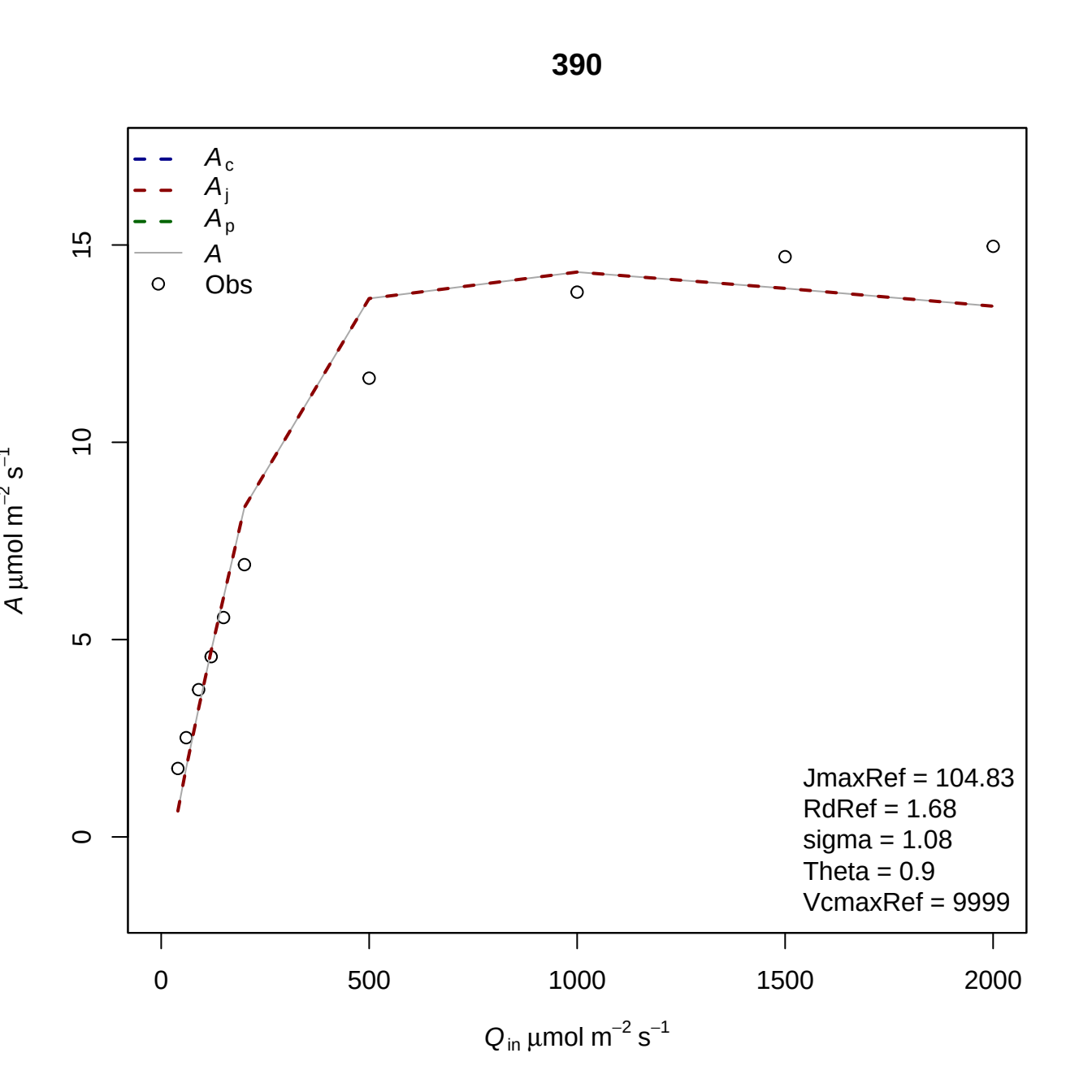

385
A_c
A_j
A_p
A
Obs
JmaxRef = 90.99
RdRef = 2.17
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

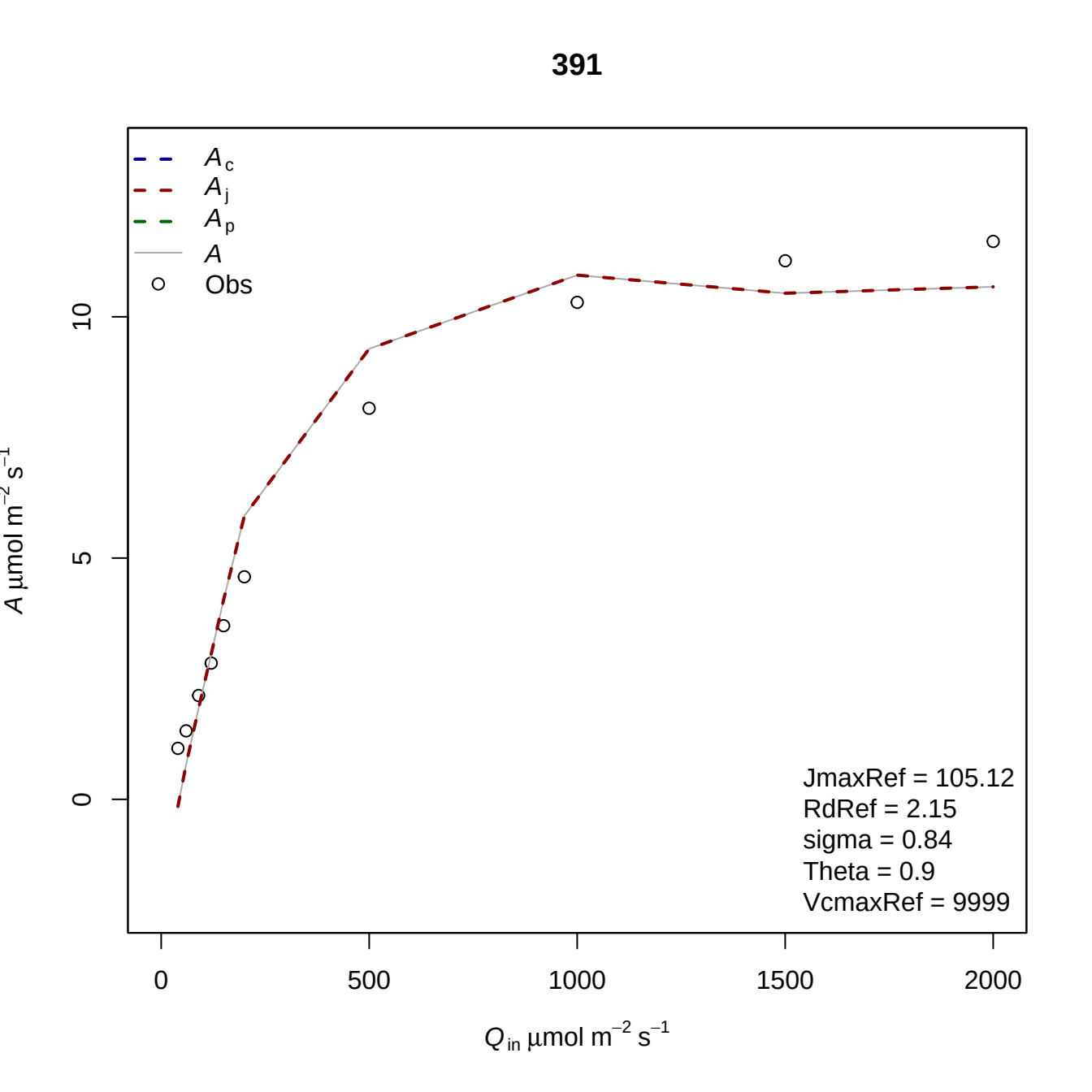

386
A_c
A_j
A_p
A
Obs
JmaxRef = 136.65
RdRef = 2.33
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

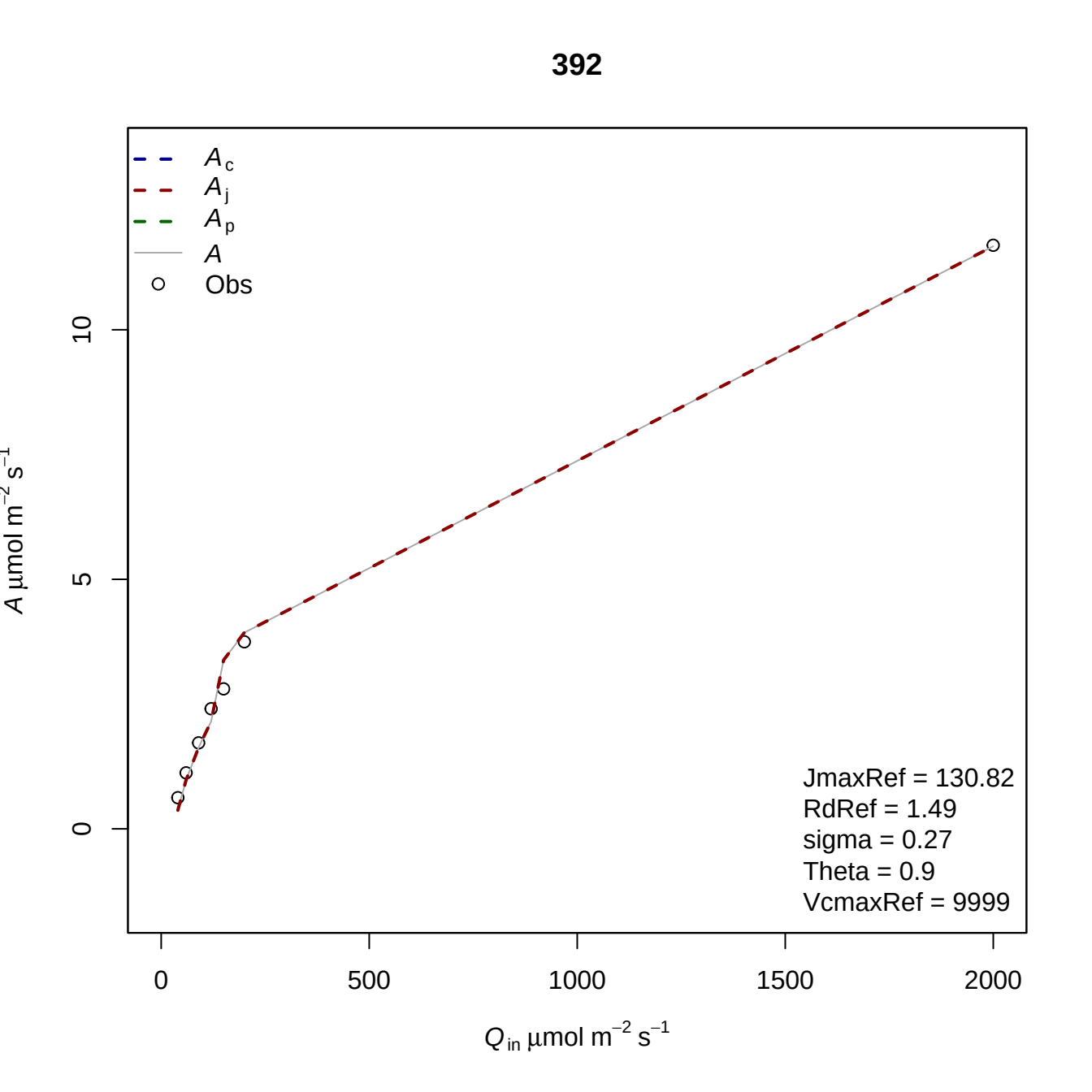

387
A_c
A_j
A_p
A
Obs
JmaxRef = 84.09
RdRef = 2.6
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

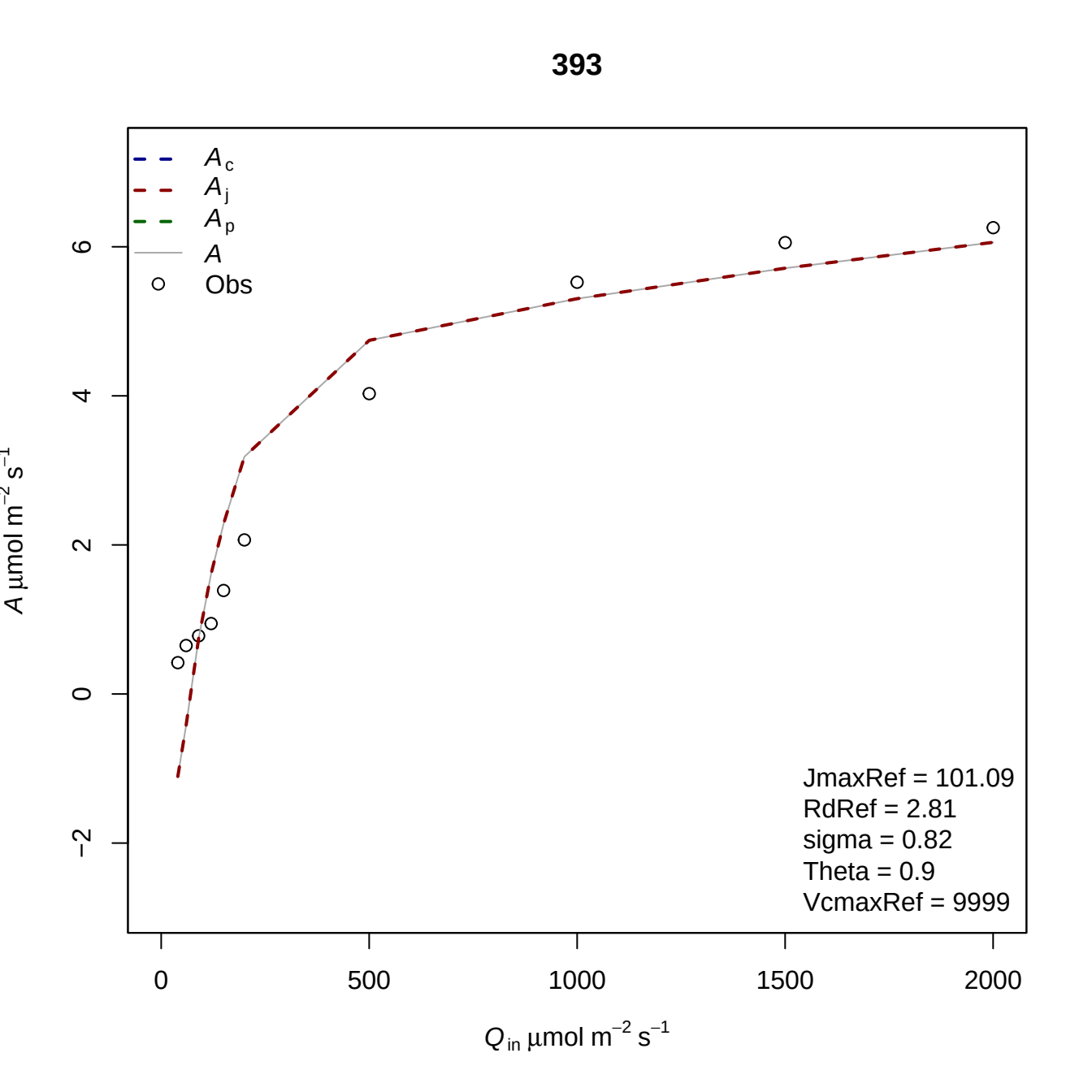

388
A_c
A_j
A_p
A
Obs
JmaxRef = 139.41
RdRef = 2.38
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

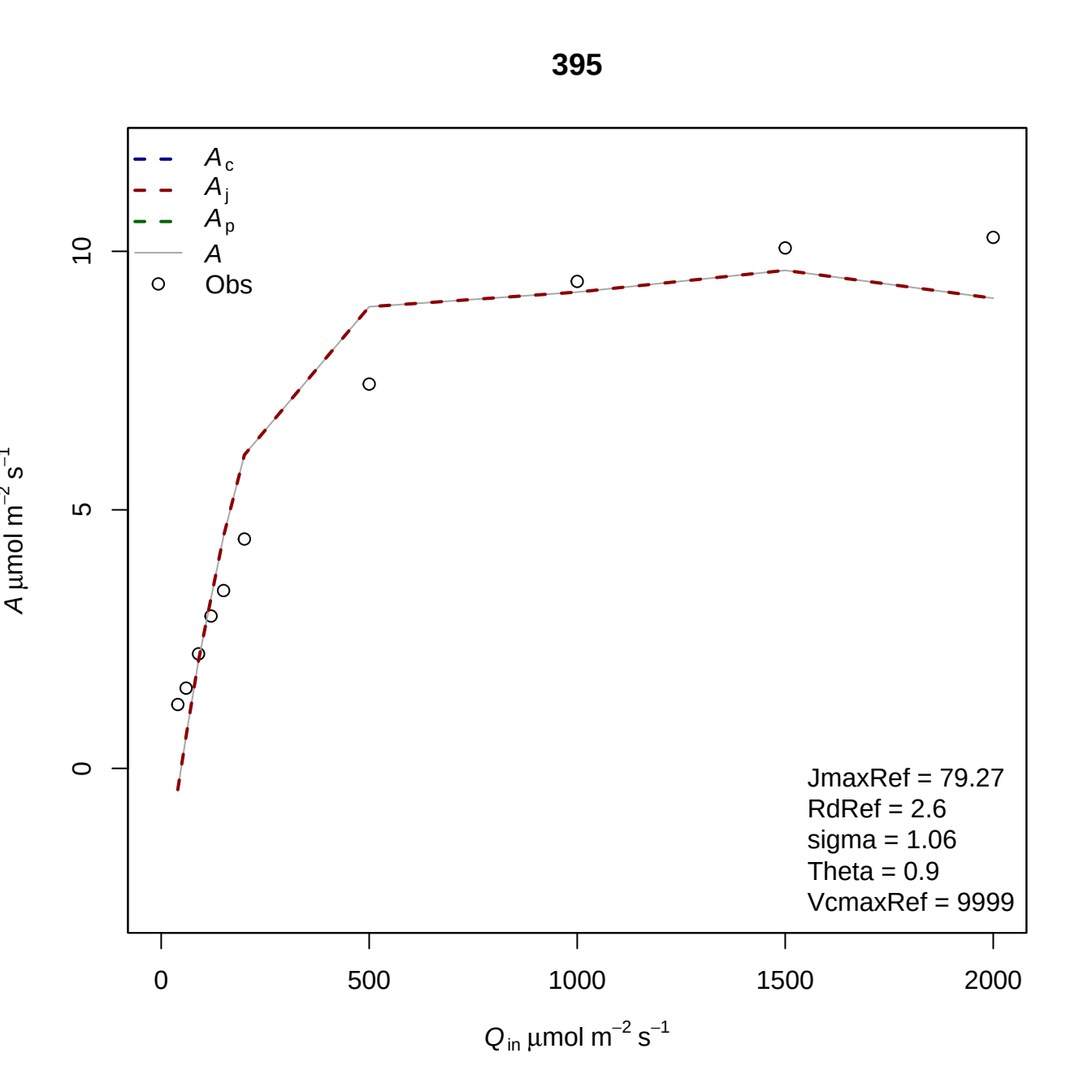

389
A_c
A_j
A_p
A
Obs
JmaxRef = 53.86
RdRef = 1.61
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

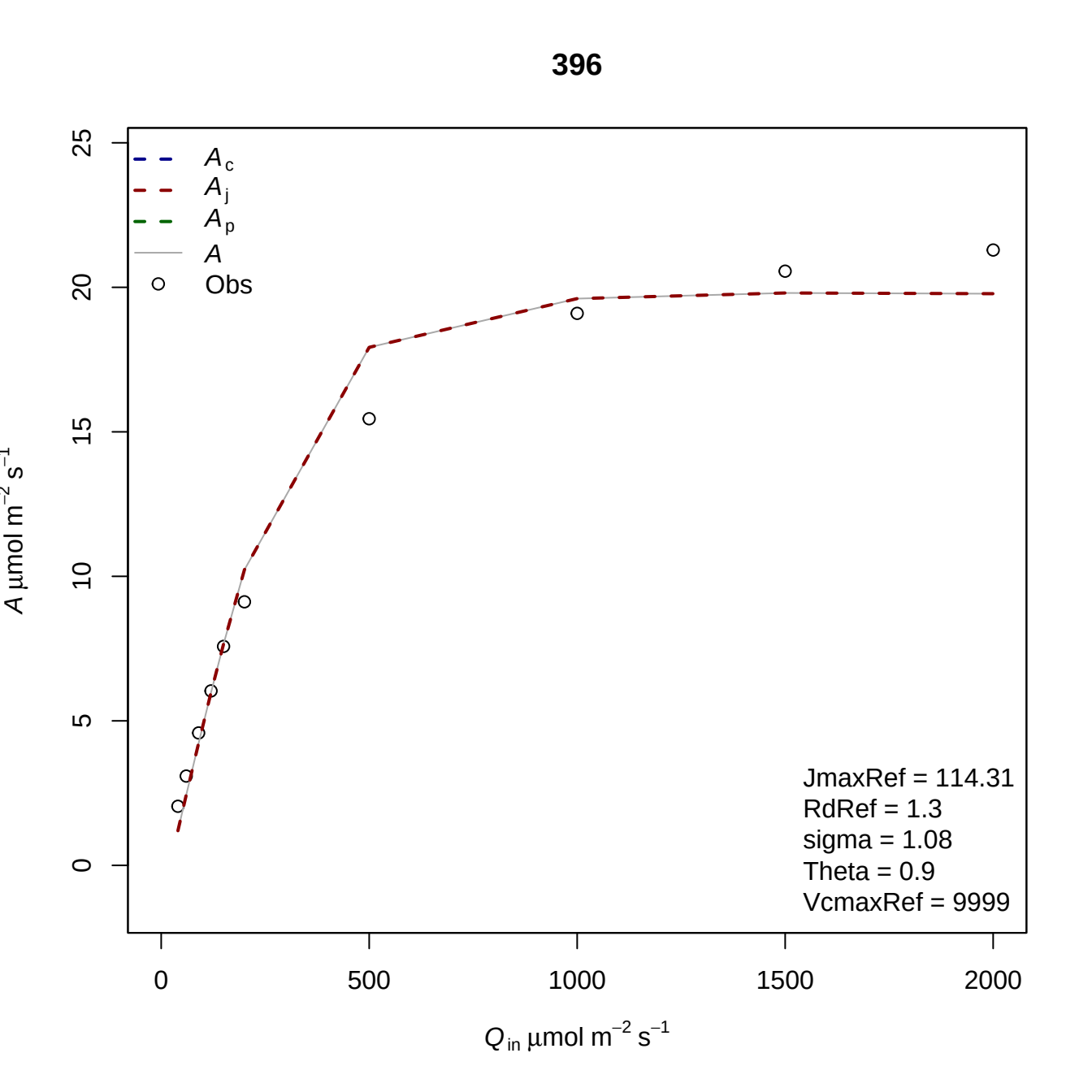

390
A_c
A_j
A_p
A
Obs
JmaxRef = 104.83
RdRef = 1.68
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

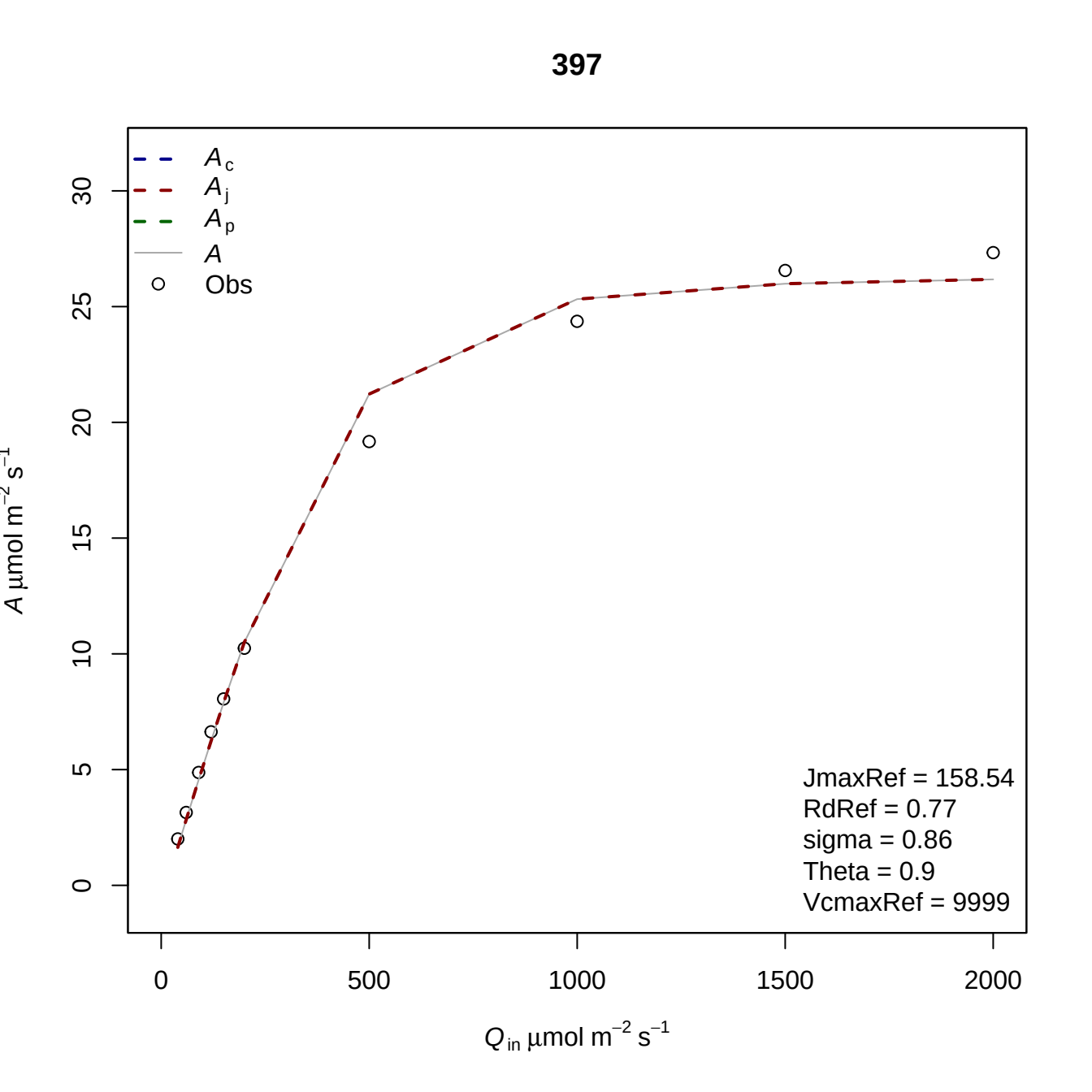

391
A_c
A_j
A_p
A
Obs
JmaxRef = 105.12
RdRef = 2.15
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

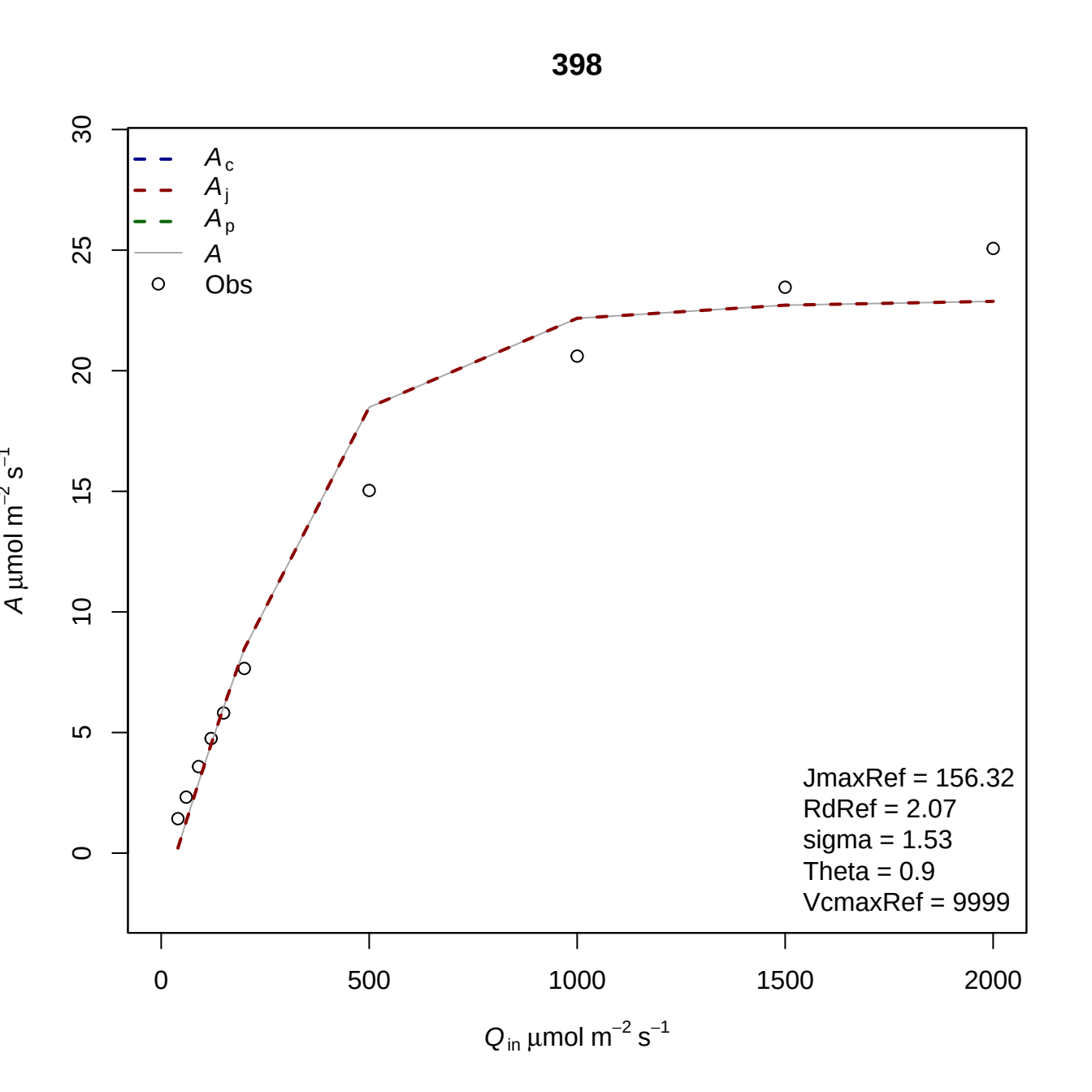

392
A_c
A_j
A_p
A
Obs
JmaxRef = 130.82
RdRef = 1.49
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

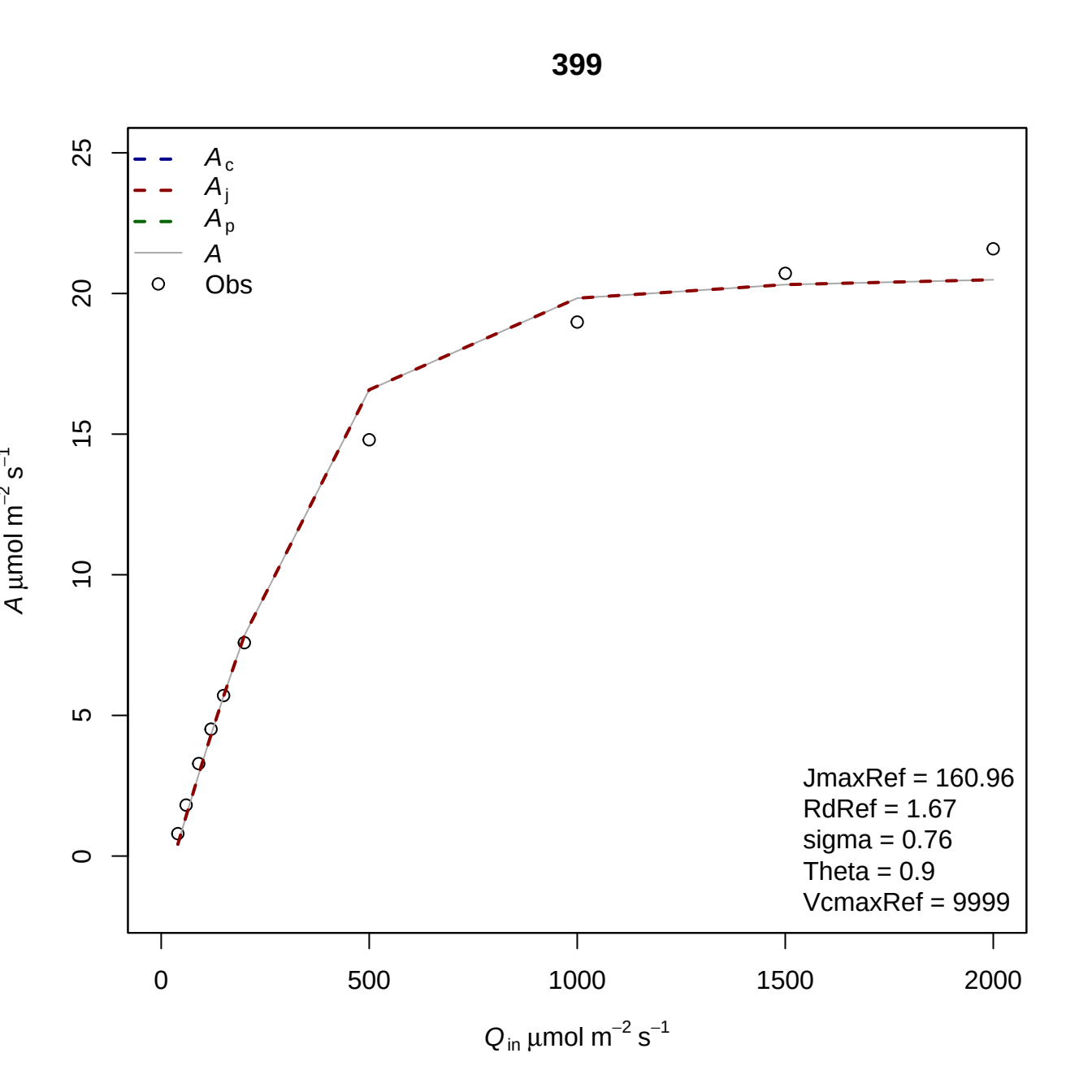

393
A_c
A_j
A_p
A
Obs
JmaxRef = 101.09
RdRef = 2.81
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

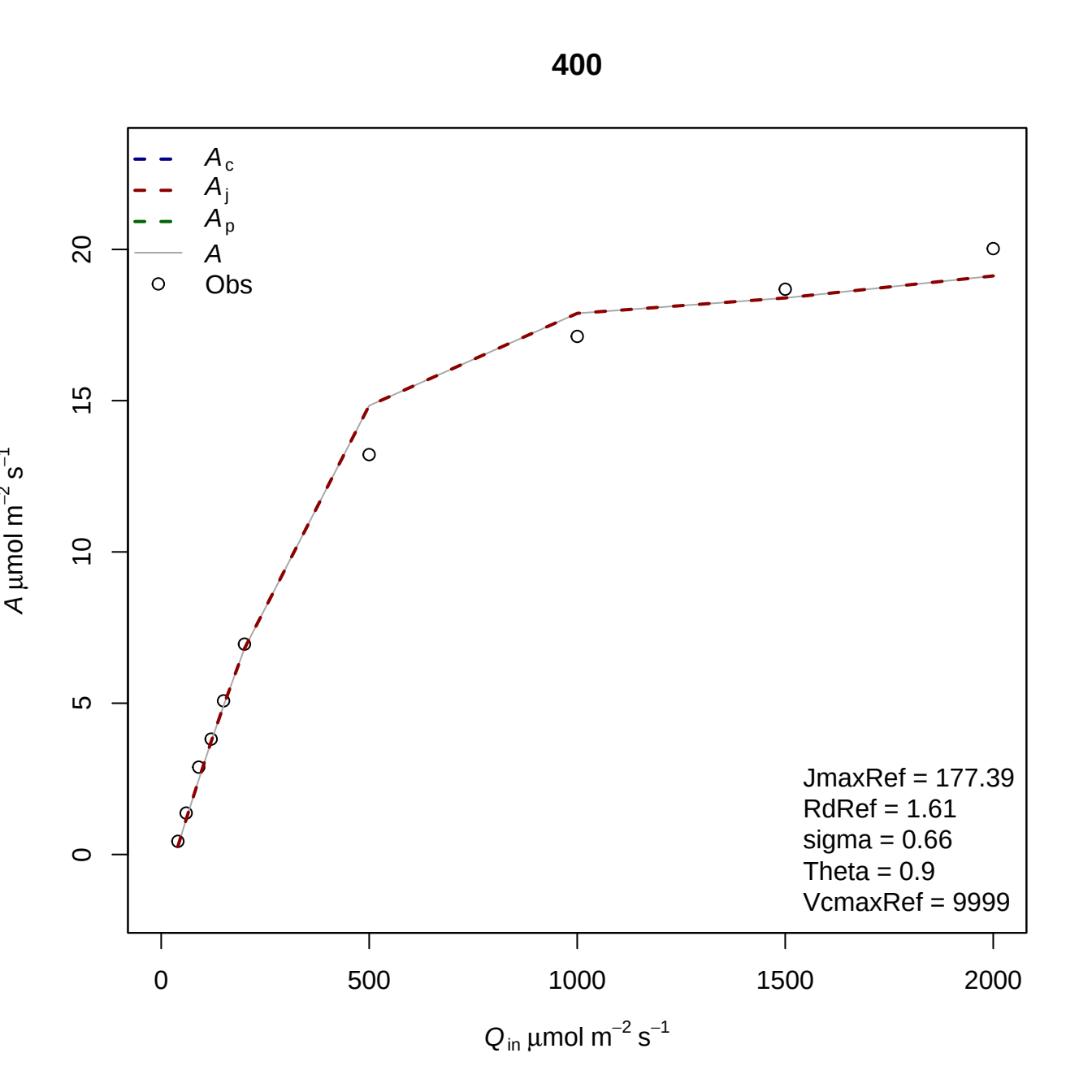

395
A_c
A_j
A_p
A
Obs
JmaxRef = 79.27
RdRef = 2.6
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

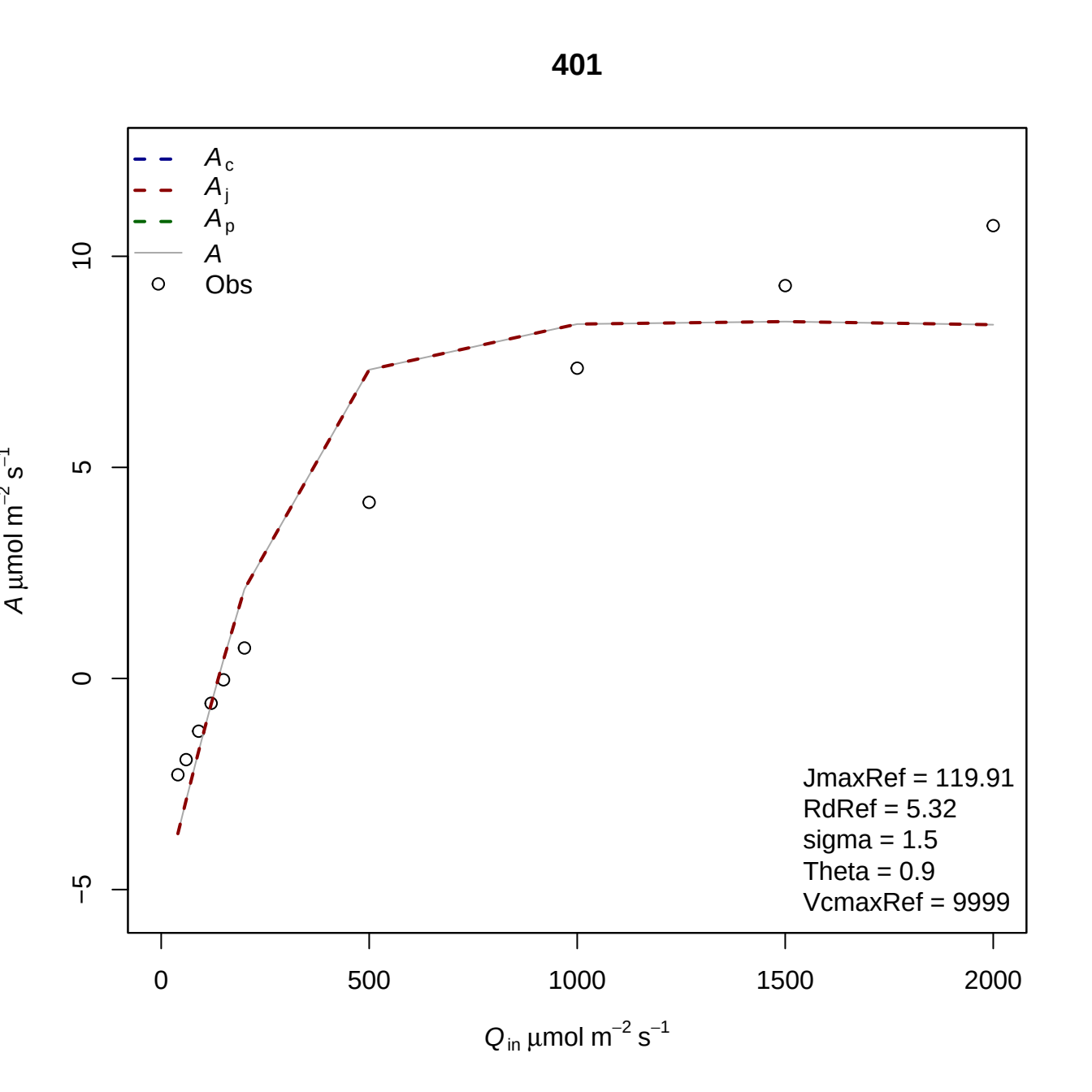

396
A_c
A_j
A_p
A
Obs
JmaxRef = 114.31
RdRef = 1.3
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

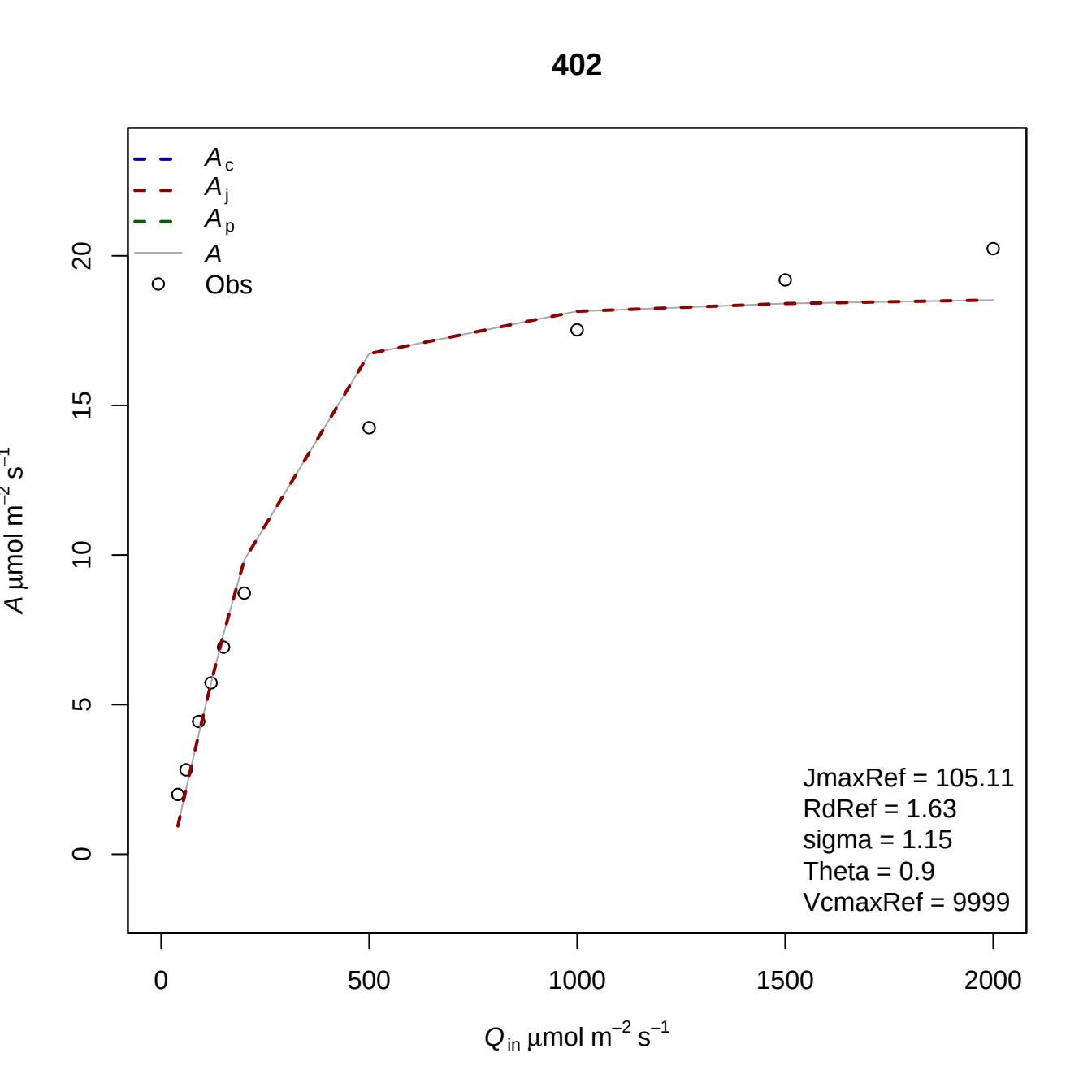

397
A_c
A_j
A_p
A
Obs
JmaxRef = 158.54
RdRef = 0.77
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

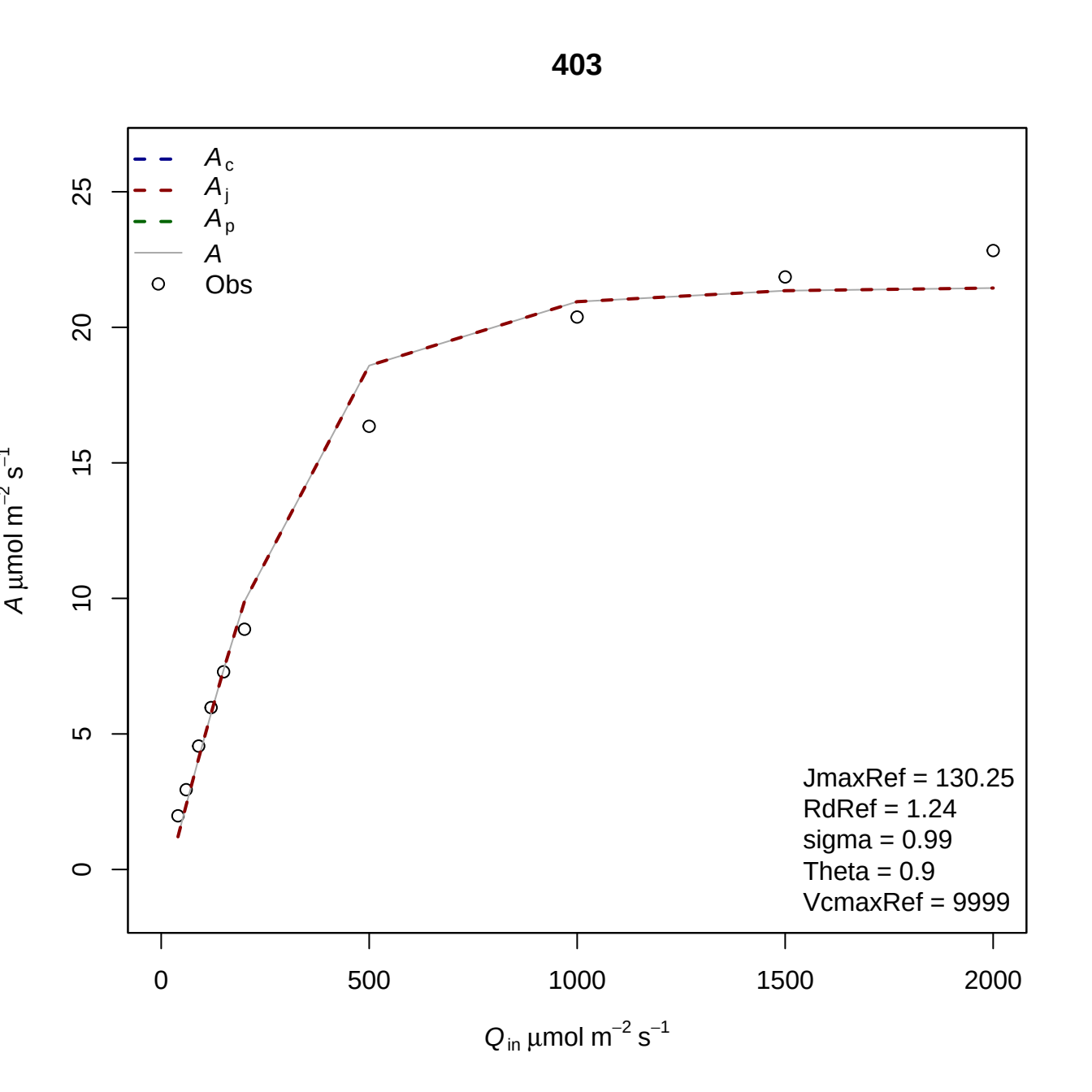

398
A_c
A_j
A_p
A
Obs
JmaxRef = 156.32
RdRef = 2.07
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

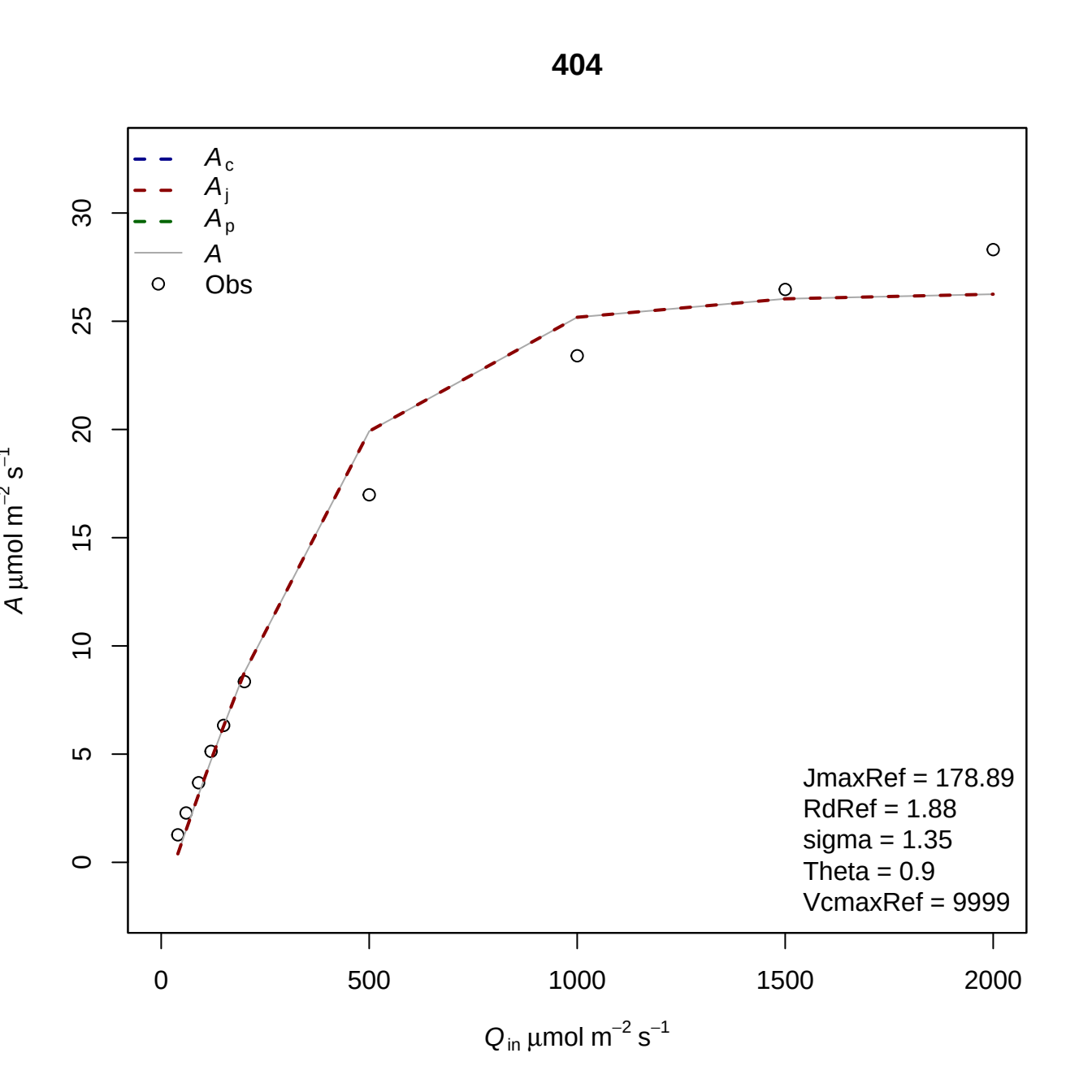

399
A_c
A_j
A_p
A
Obs
JmaxRef = 160.96
RdRef = 1.67
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

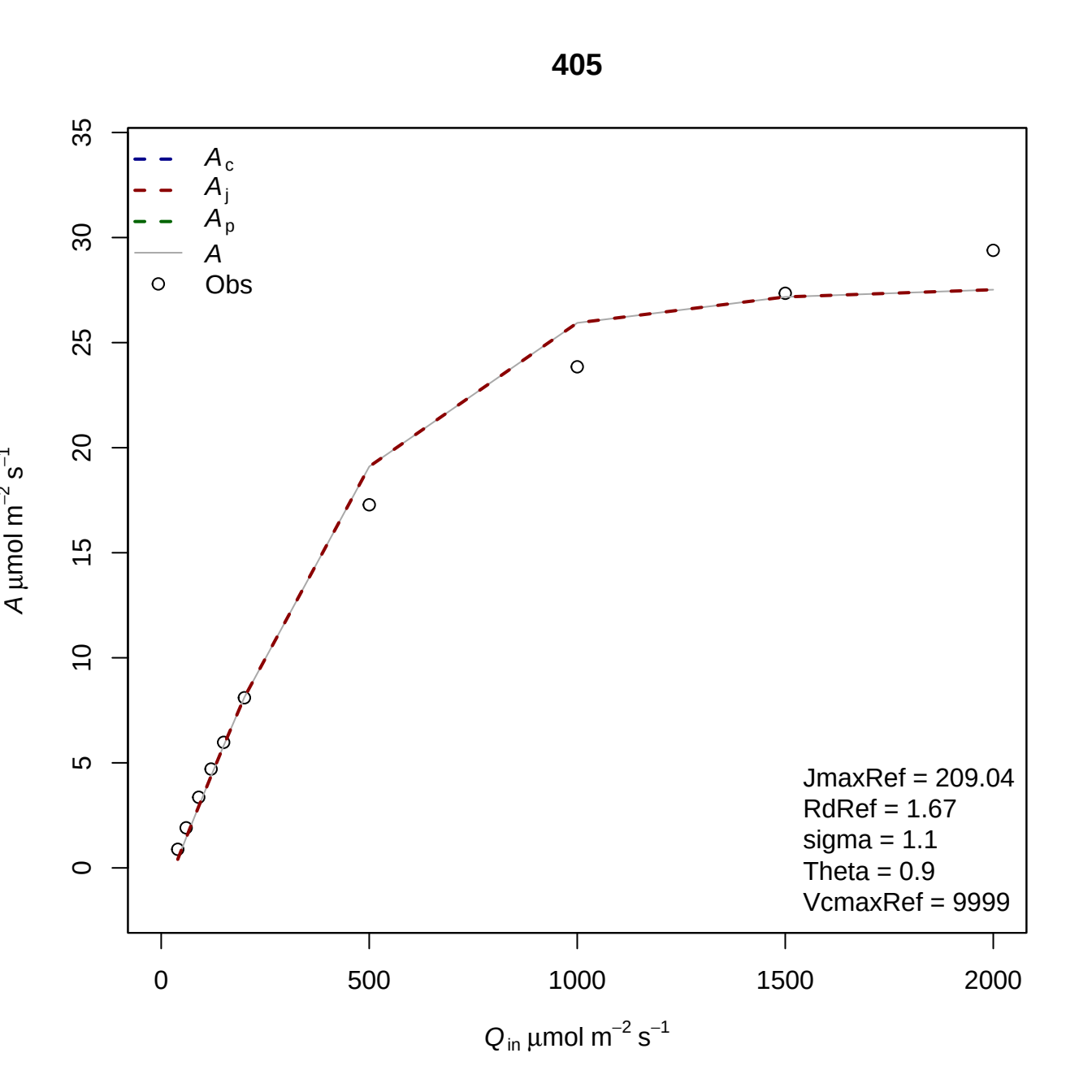

400
A_c
A_j
A_p
A
Obs
JmaxRef = 177.39
RdRef = 1.61
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

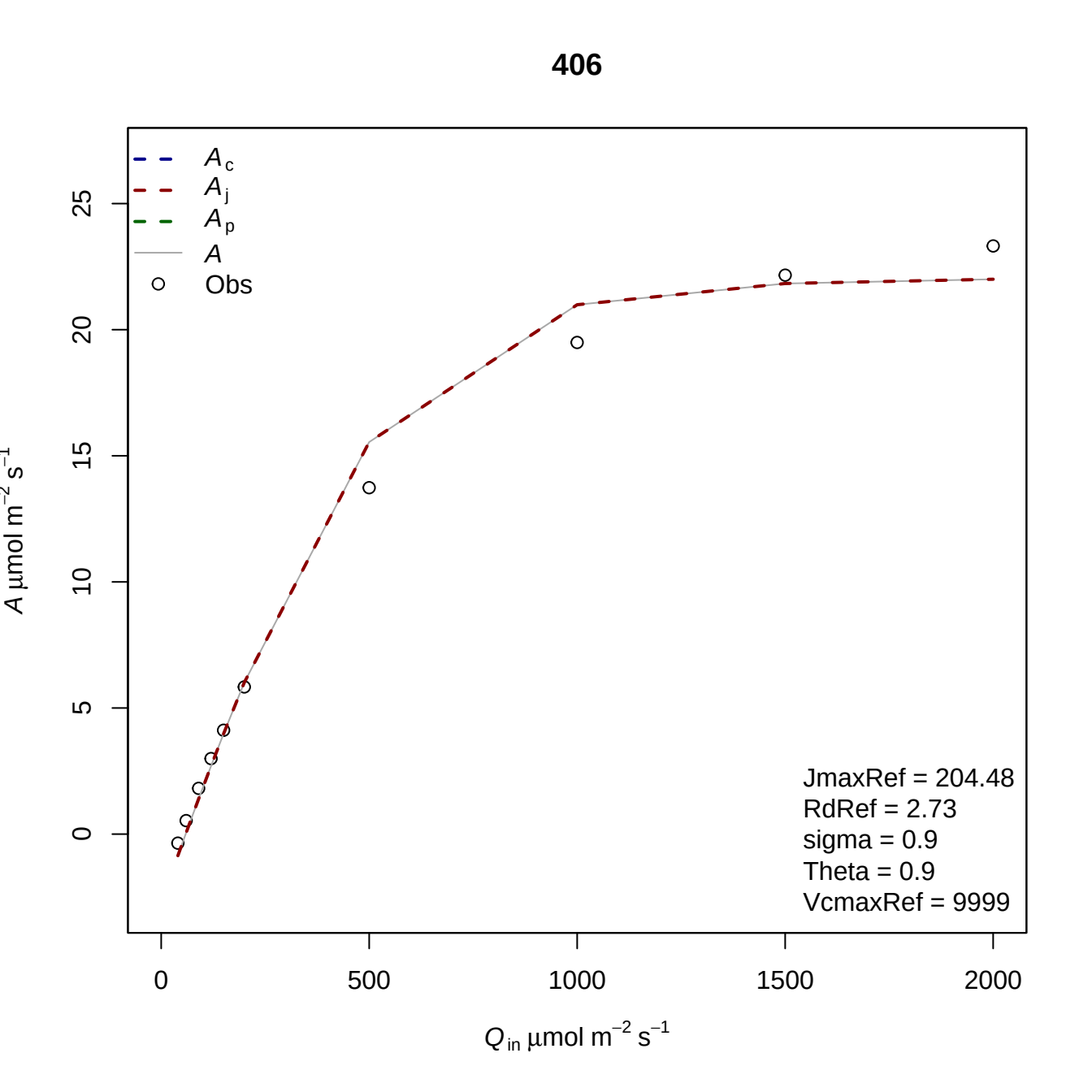

401
A_c
A_j
A_p
A
Obs
JmaxRef = 119.91
RdRef = 5.32
sigma = 1.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

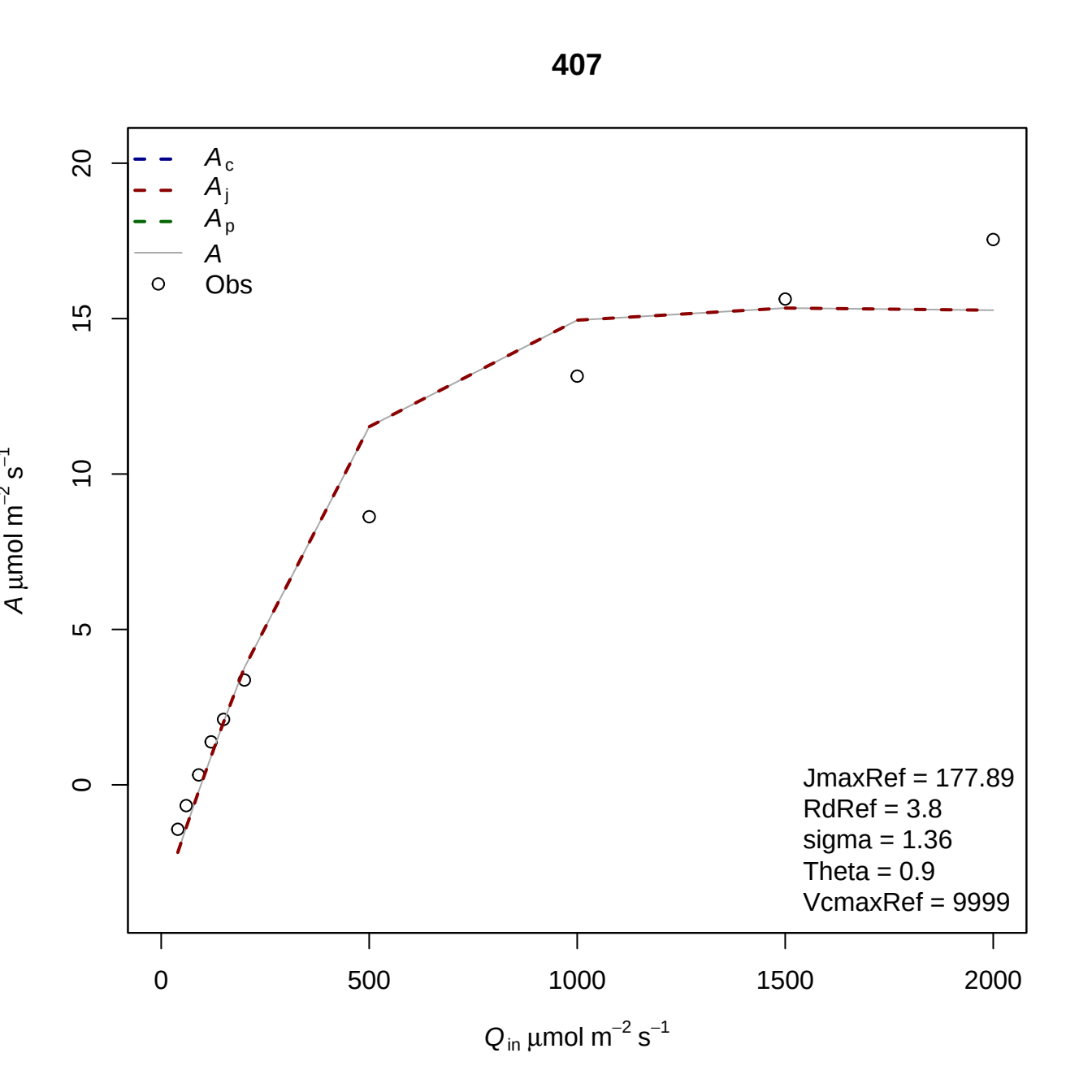

402
A_c
A_j
A_p
A
Obs
JmaxRef = 105.11
RdRef = 1.63
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

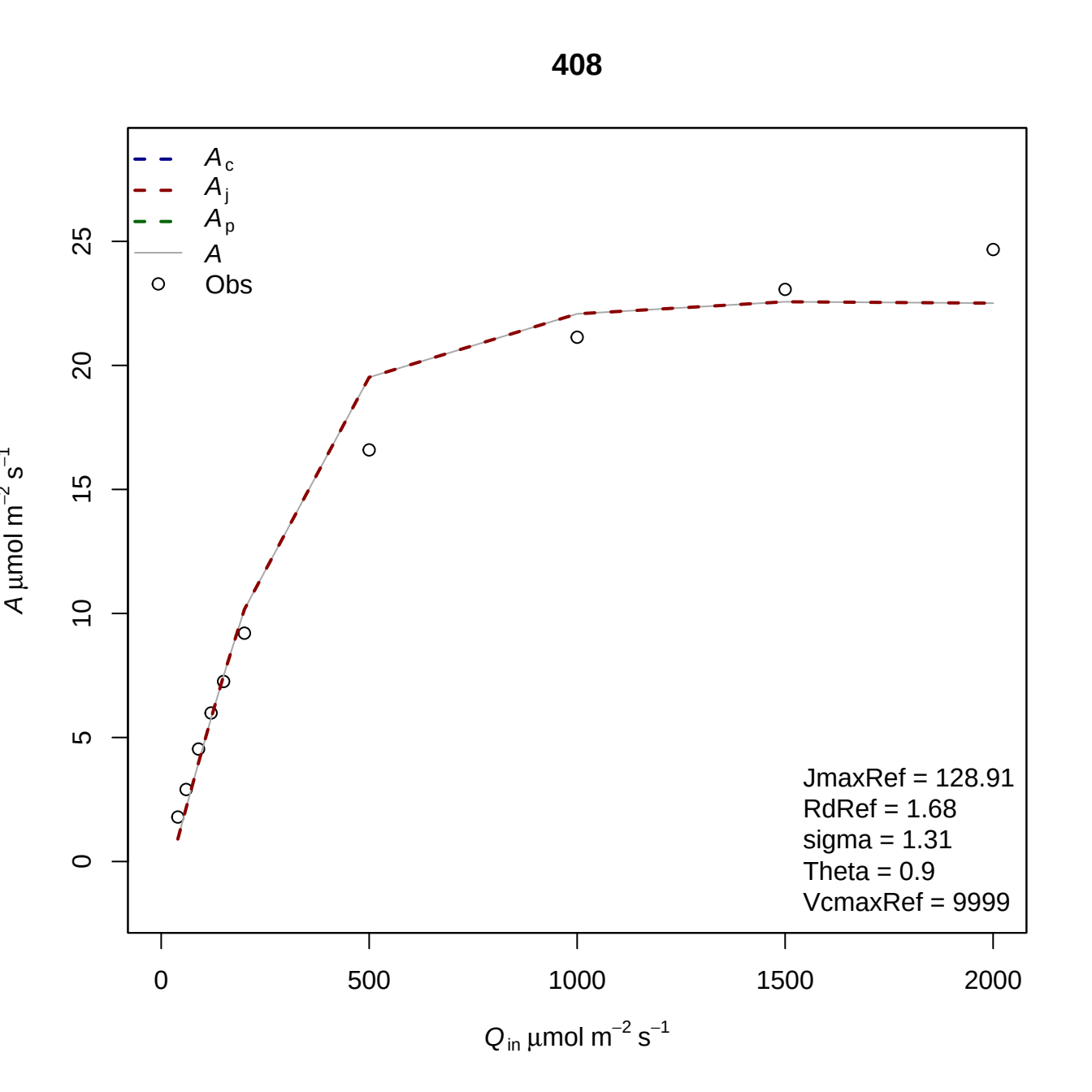

403
A_c
A_j
A_p
A
Obs
JmaxRef = 130.25
RdRef = 1.24
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

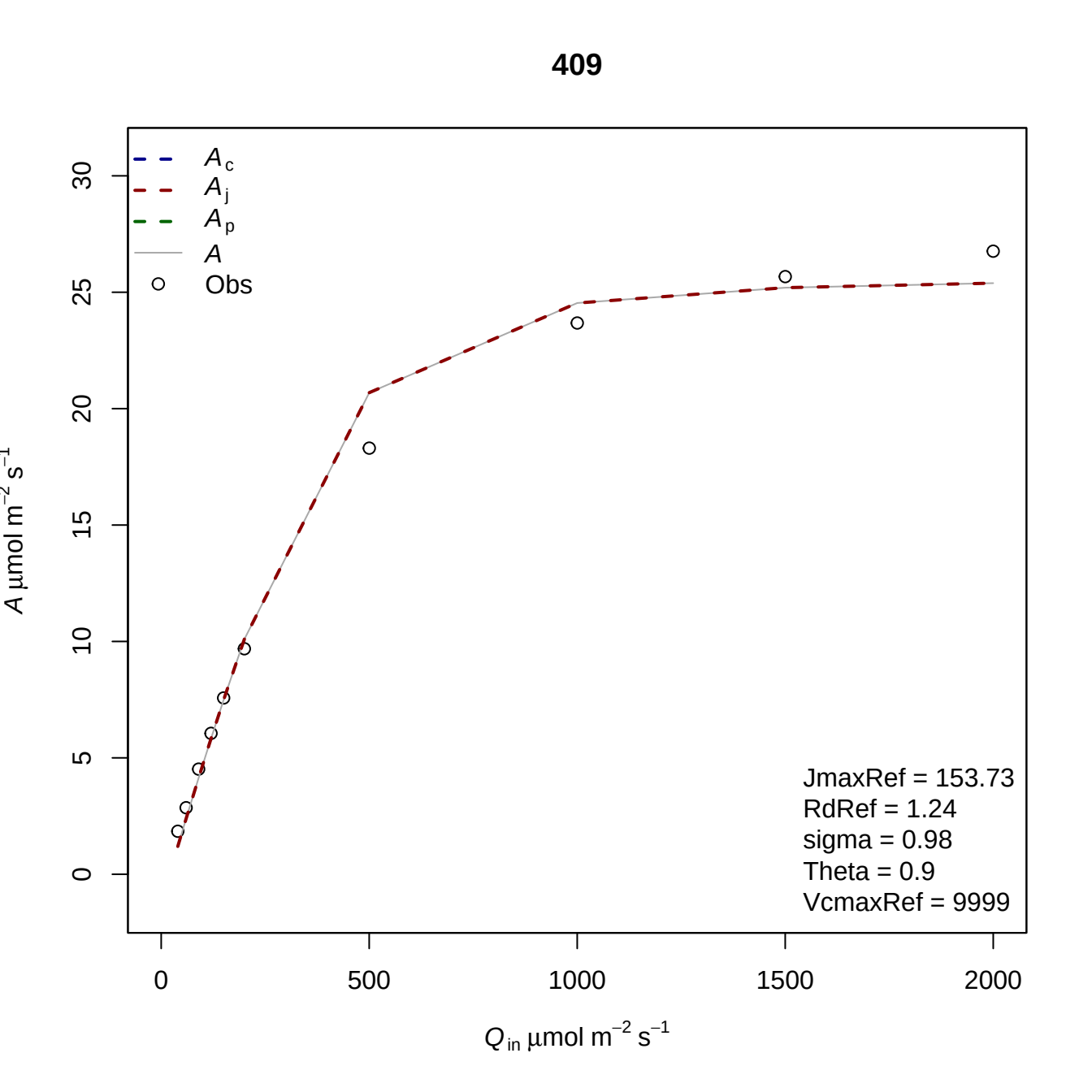

404
A_c
A_j
A_p
A
Obs
JmaxRef = 178.89
RdRef = 1.88
sigma = 1.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

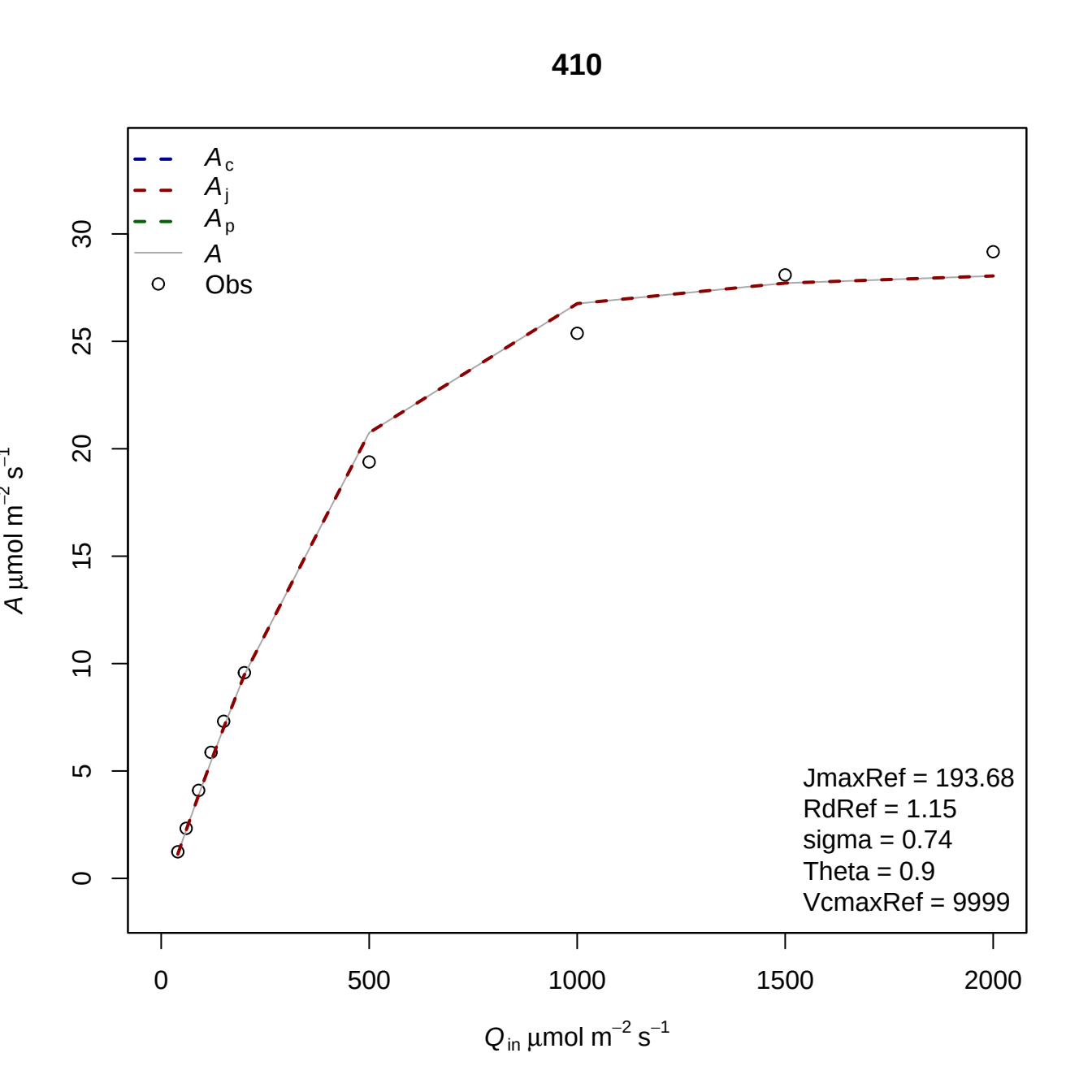

405
A_c
A_j
A_p
A
Obs
JmaxRef = 209.04
RdRef = 1.67
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

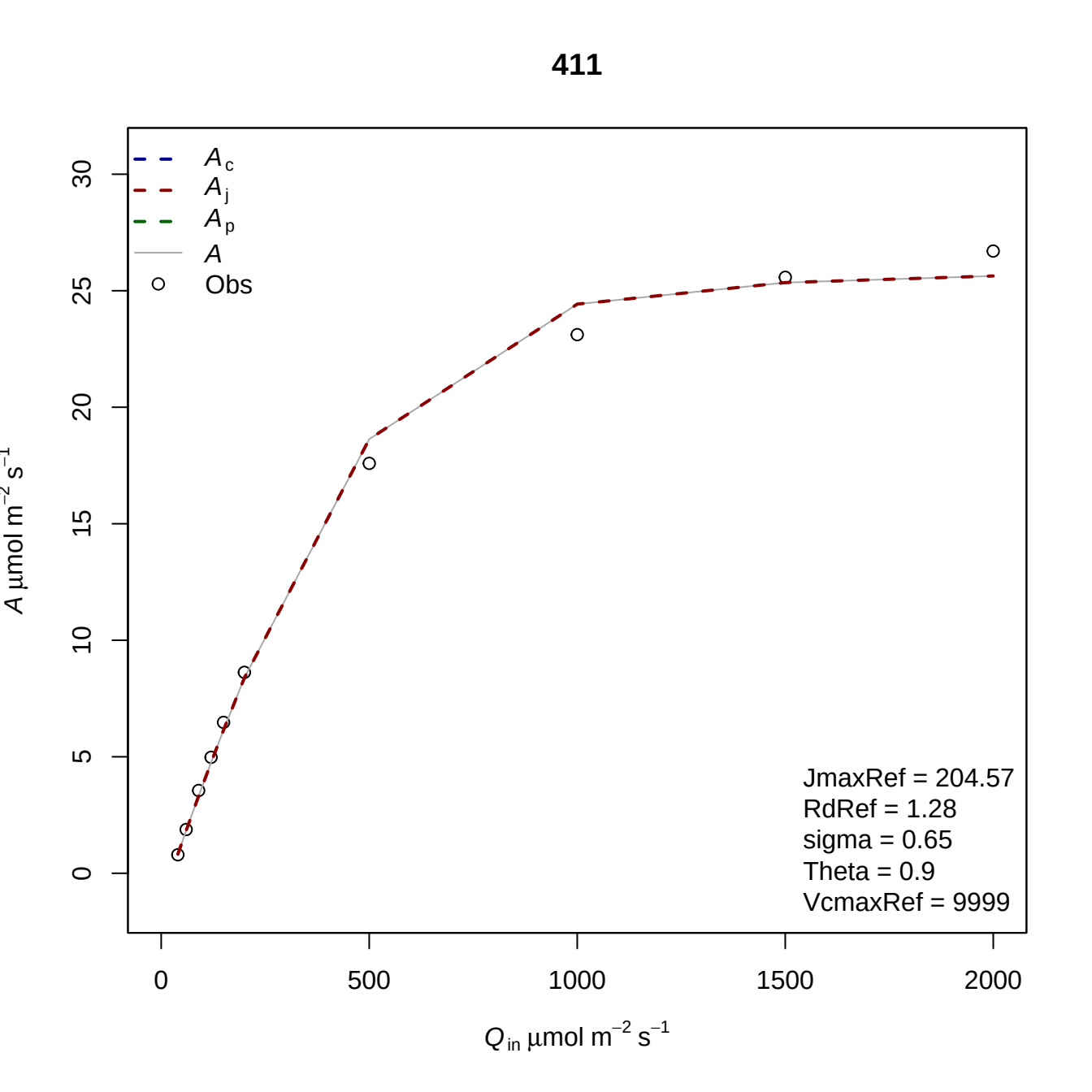

406
A_c
A_j
A_p
A
Obs
JmaxRef = 204.48
RdRef = 2.73
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

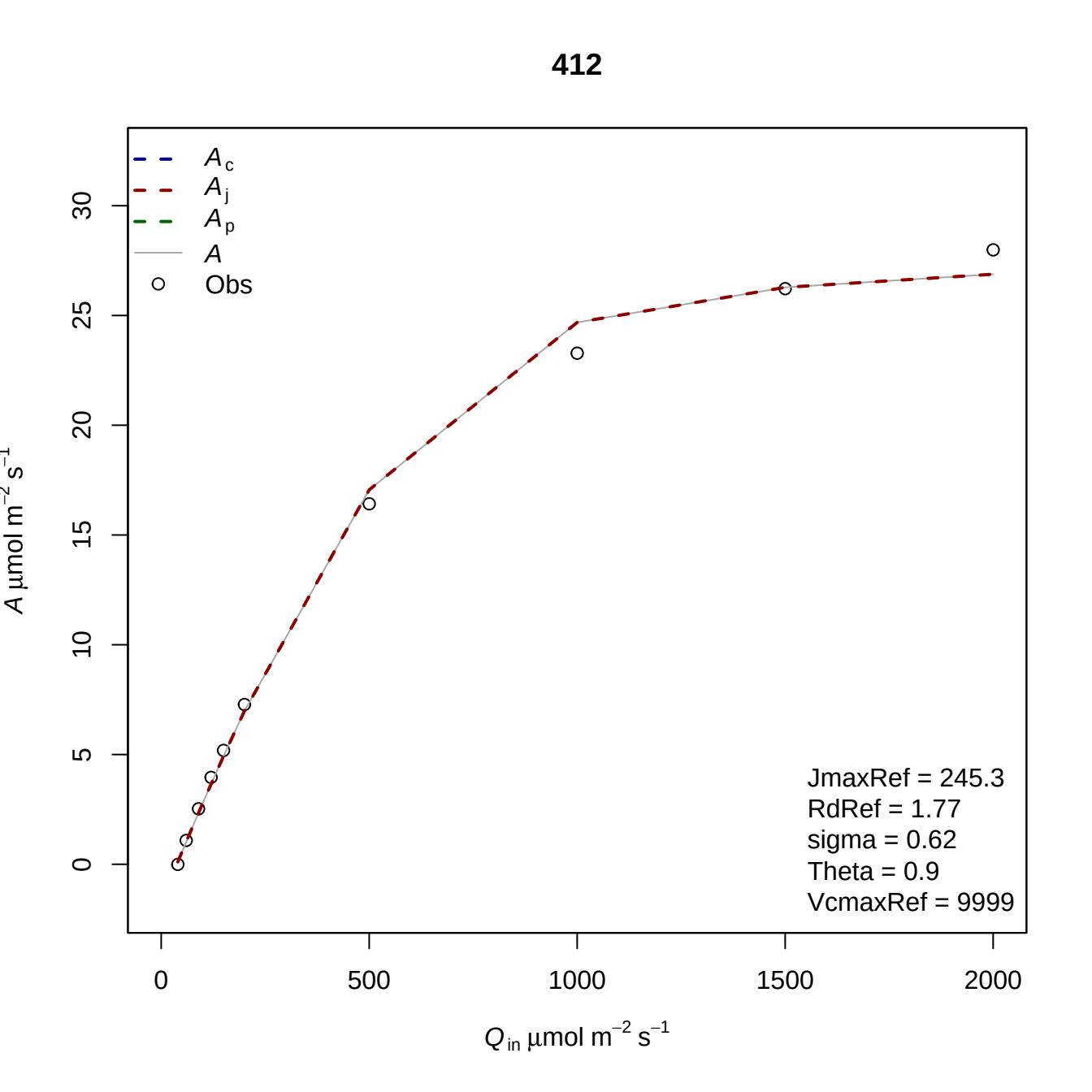

407
A_c
A_j
A_p
A
Obs
JmaxRef = 177.89
RdRef = 3.8
sigma = 1.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

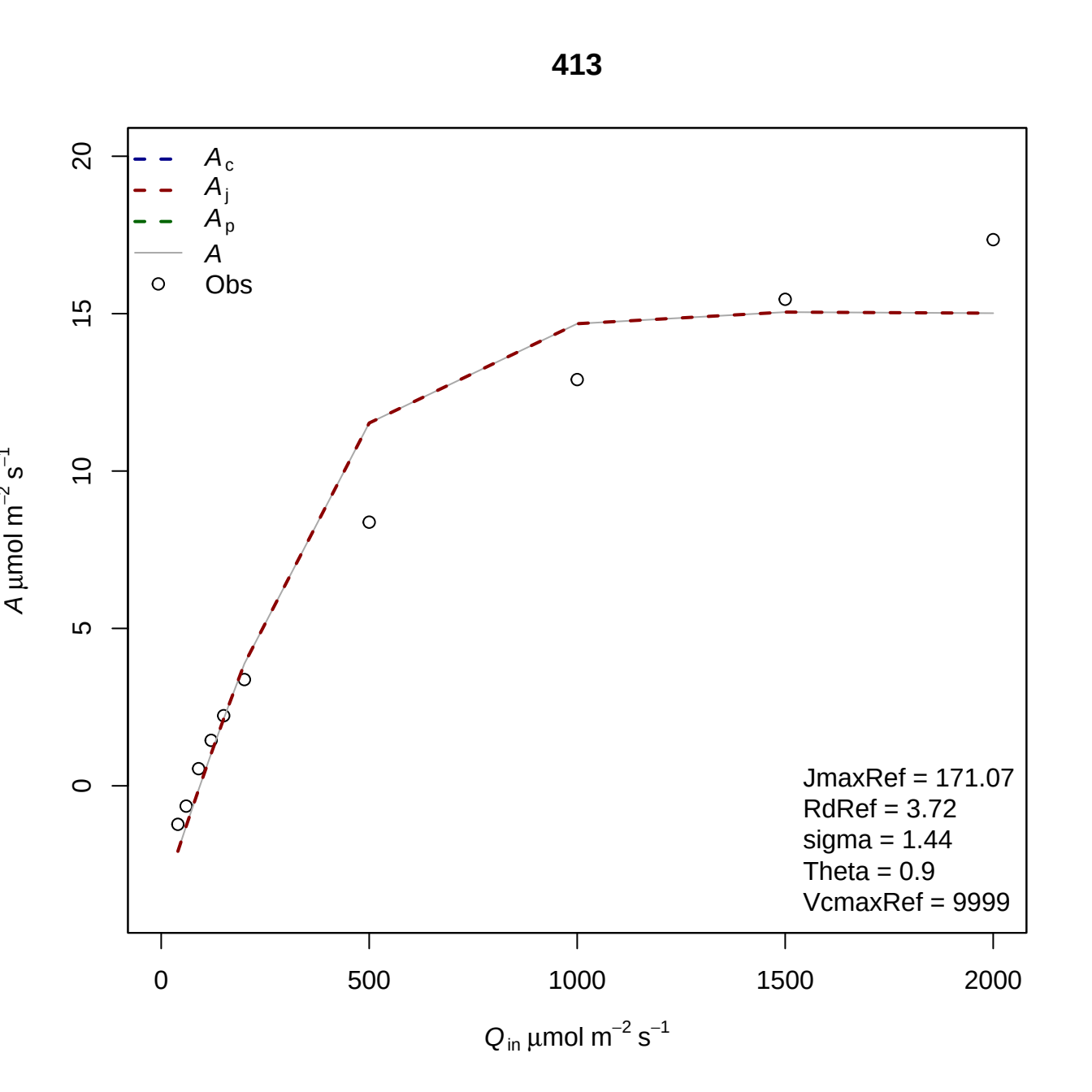

408
A_c
A_j
A_p
A
Obs
JmaxRef = 128.91
RdRef = 1.68
sigma = 1.31
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

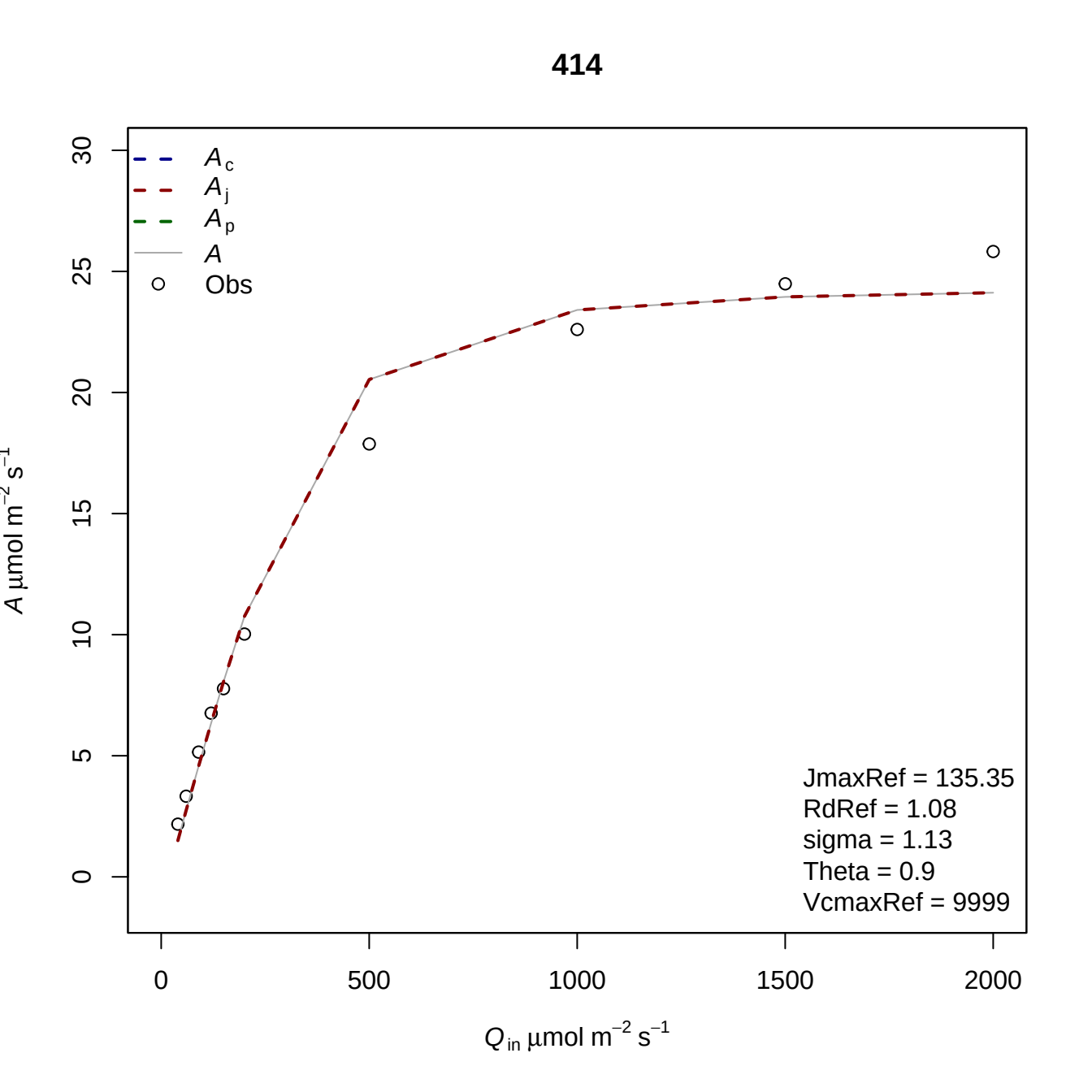

409
A_c
A_j
A_p
A
Obs
JmaxRef = 153.73
RdRef = 1.24
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

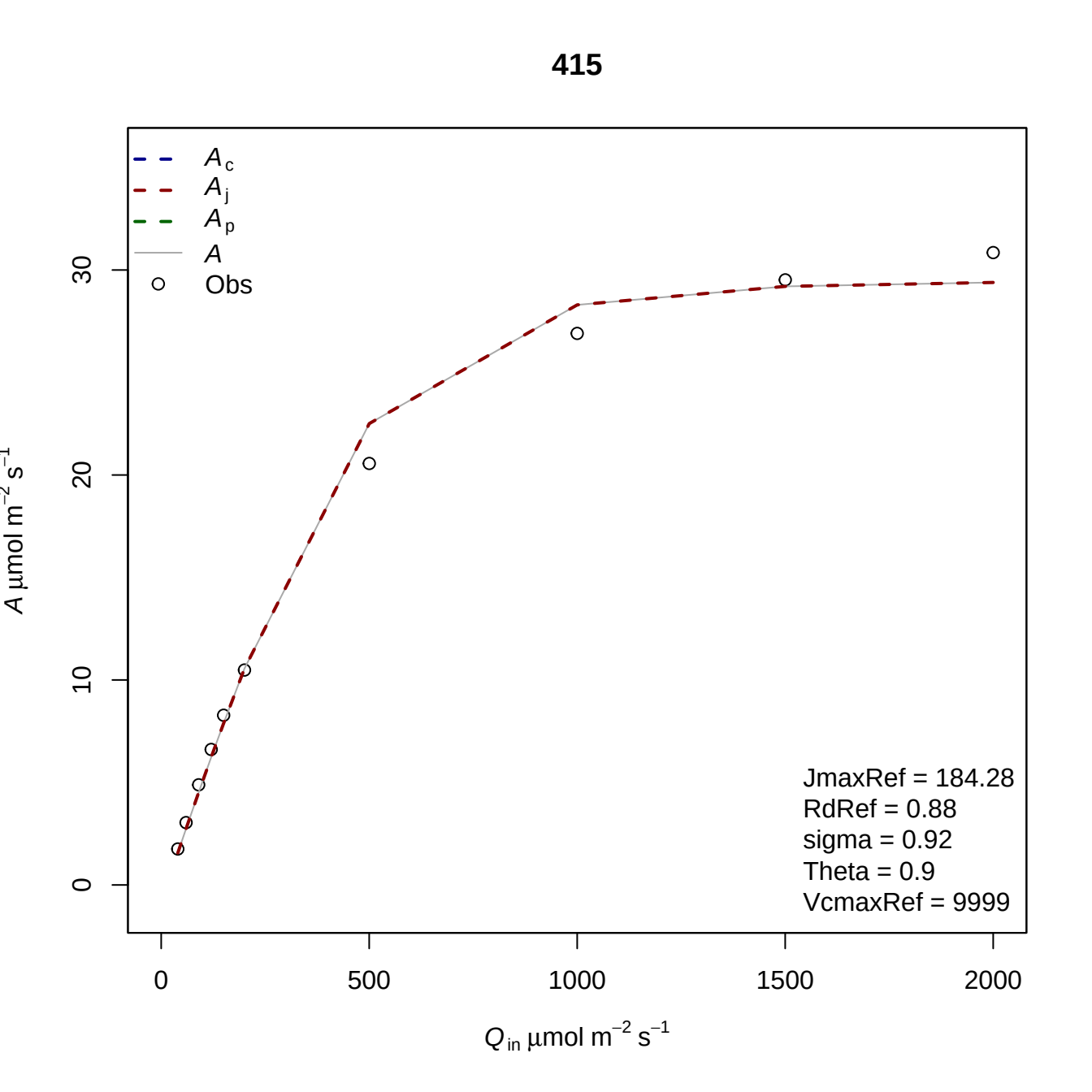

410
A_c
A_j
A_p
A
Obs
JmaxRef = 193.68
RdRef = 1.15
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

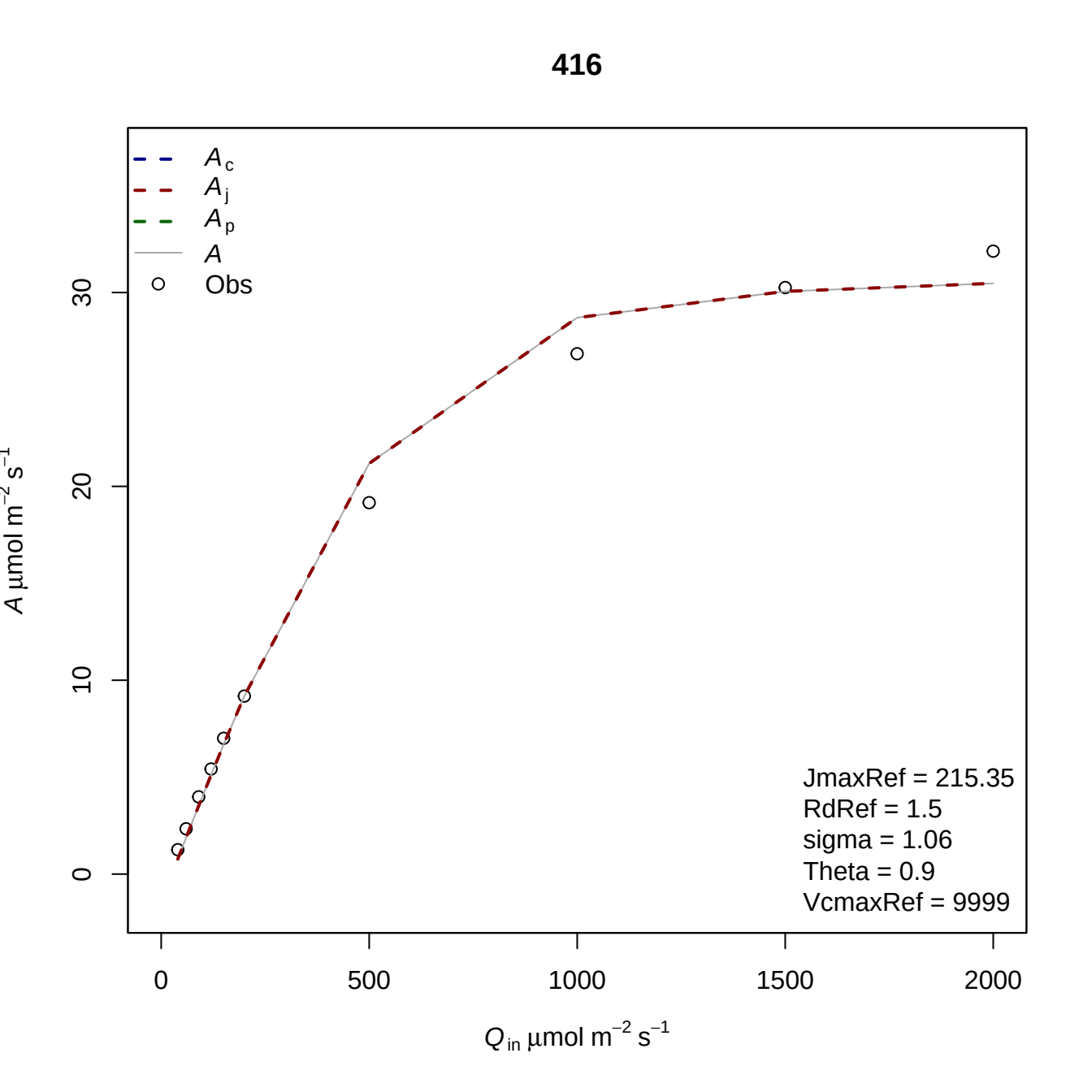

411
A_c
A_j
A_p
A
Obs
JmaxRef = 204.57
RdRef = 1.28
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

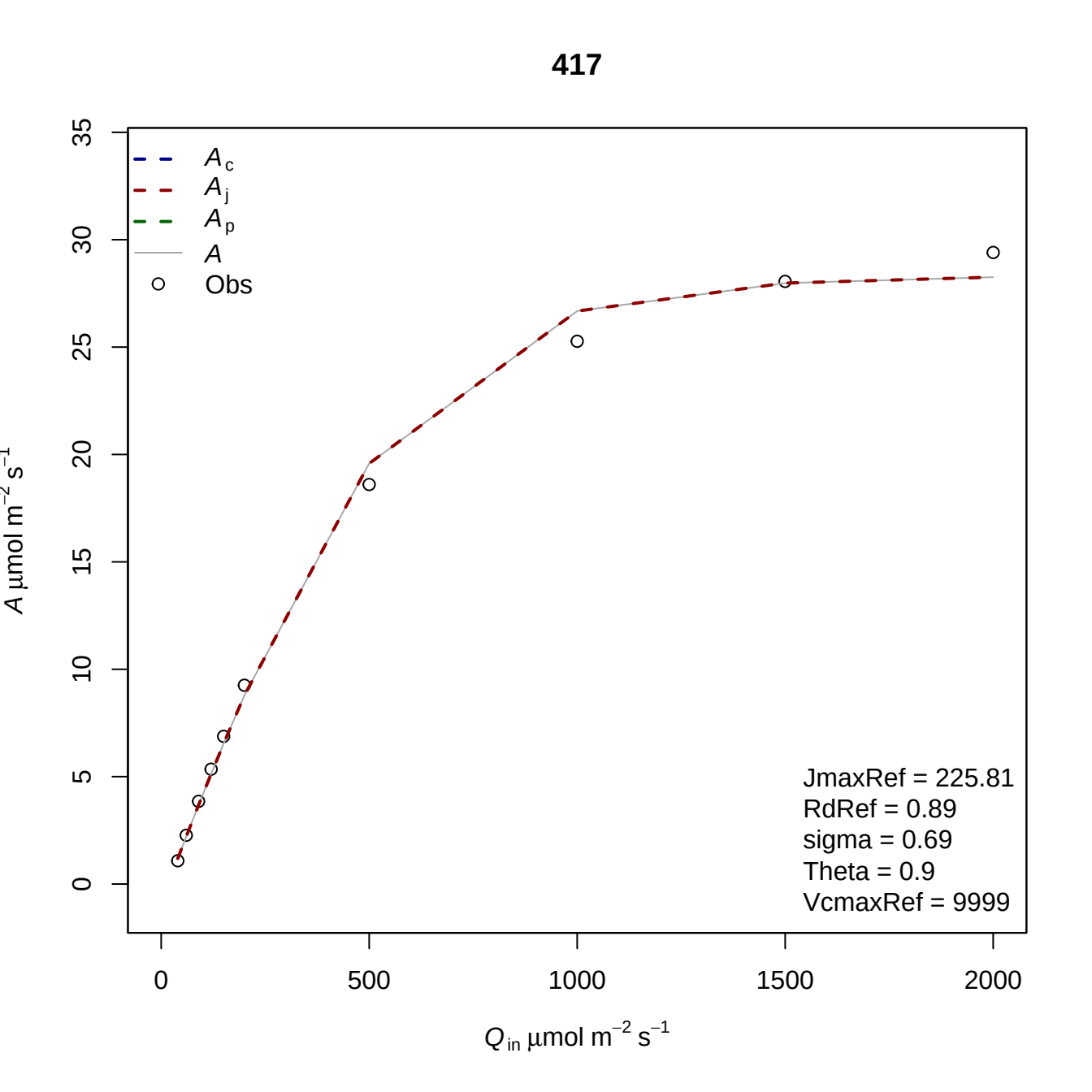

412
A_c
A_j
A_p
A
Obs
JmaxRef = 245.3
RdRef = 1.77
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

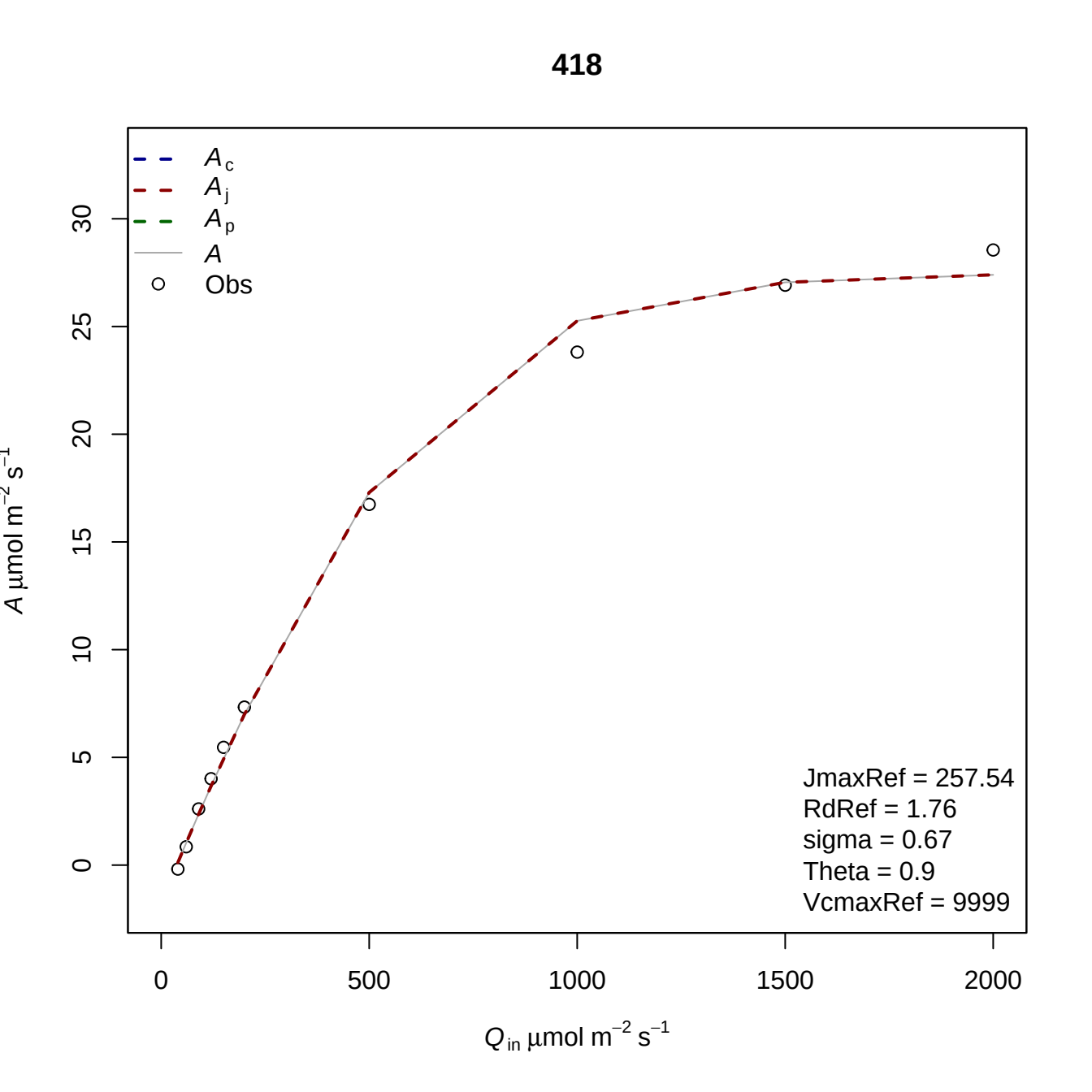

413
A_c
A_j
A_p
A
Obs
JmaxRef = 171.07
RdRef = 3.72
sigma = 1.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

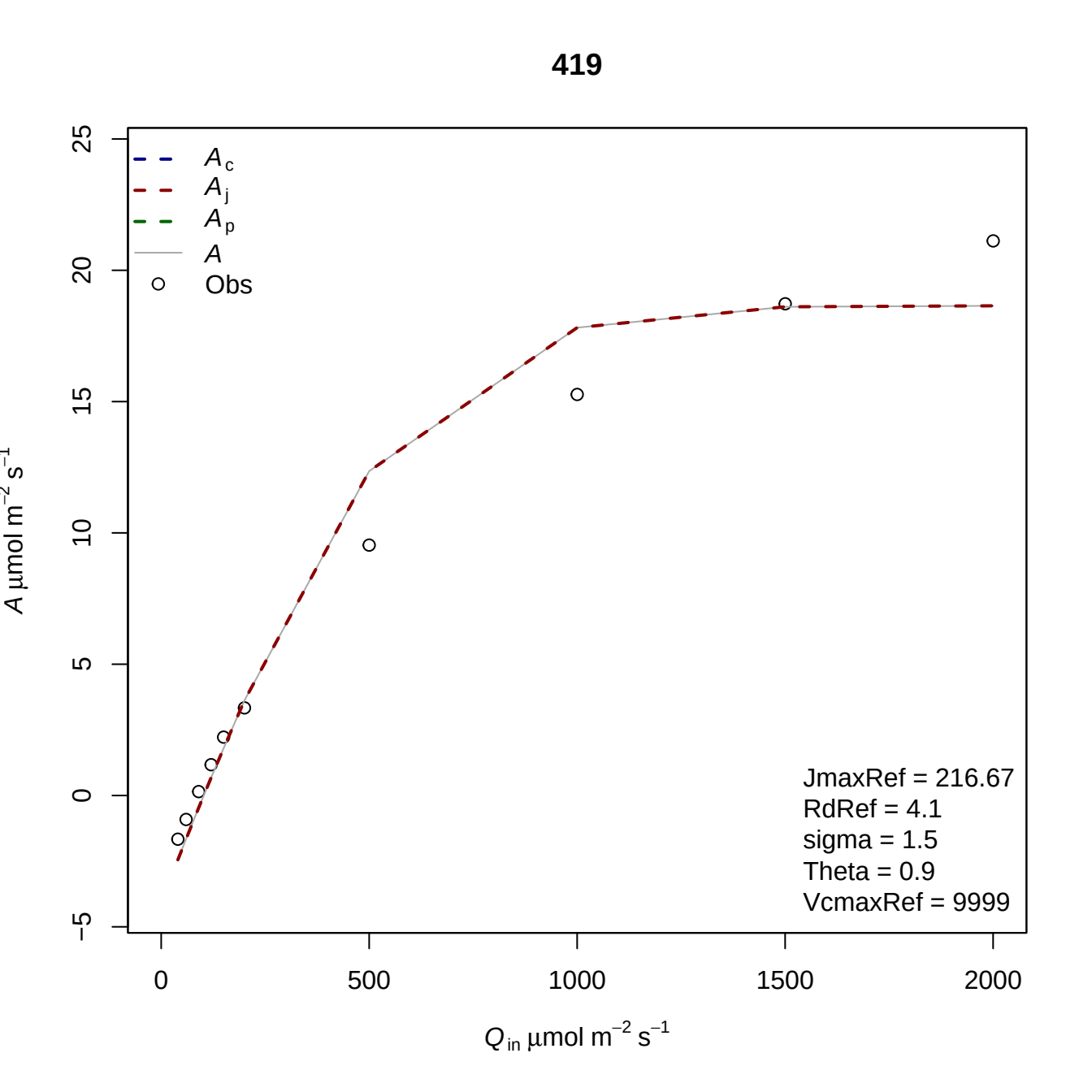

414
A_c
A_j
A_p
A
Obs
JmaxRef = 135.35
RdRef = 1.08
sigma = 1.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

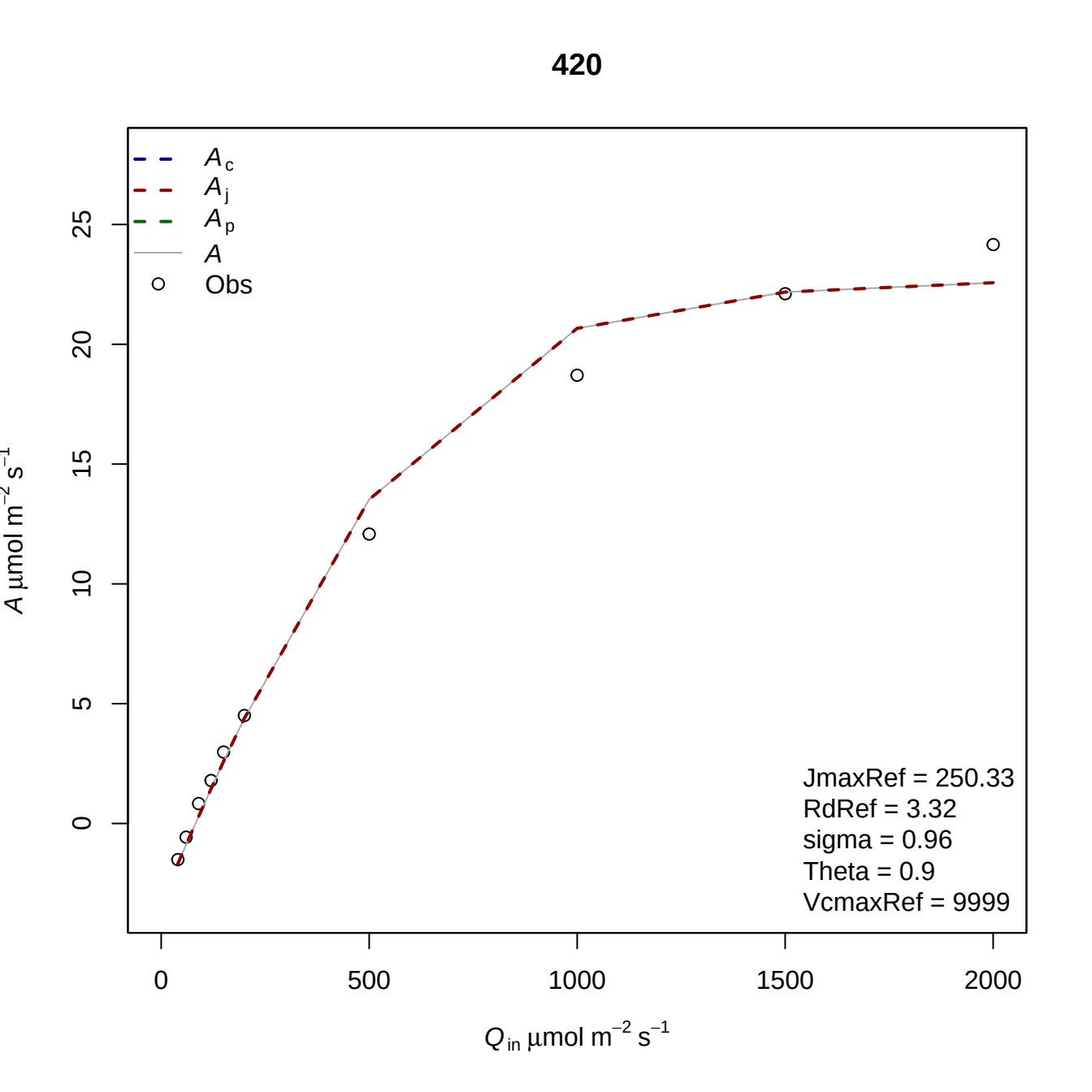

415
A_c
A_j
A_p
A
Obs
JmaxRef = 184.28
RdRef = 0.88
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

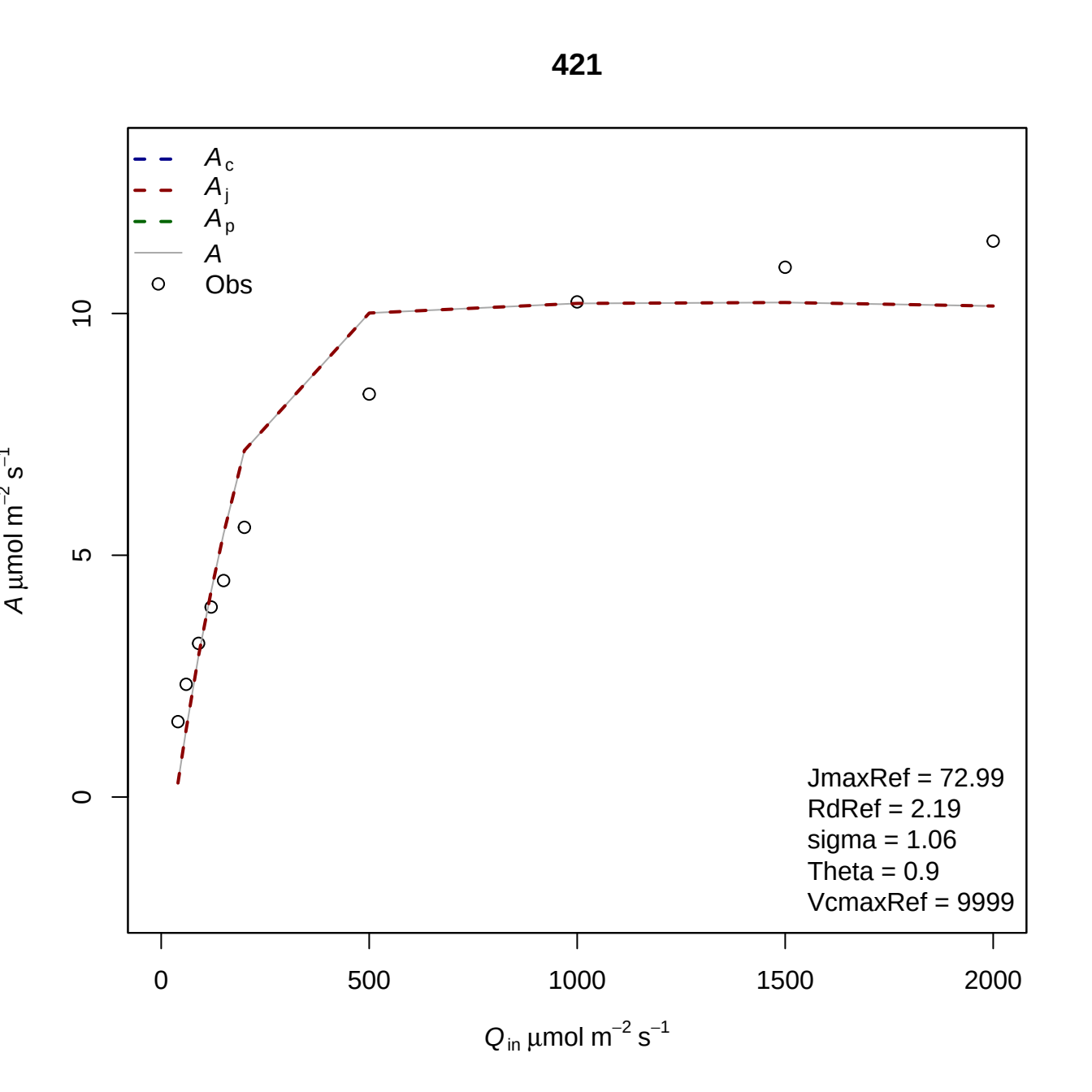

416
A_c
A_j
A_p
A
Obs
JmaxRef = 215.35
RdRef = 1.5
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

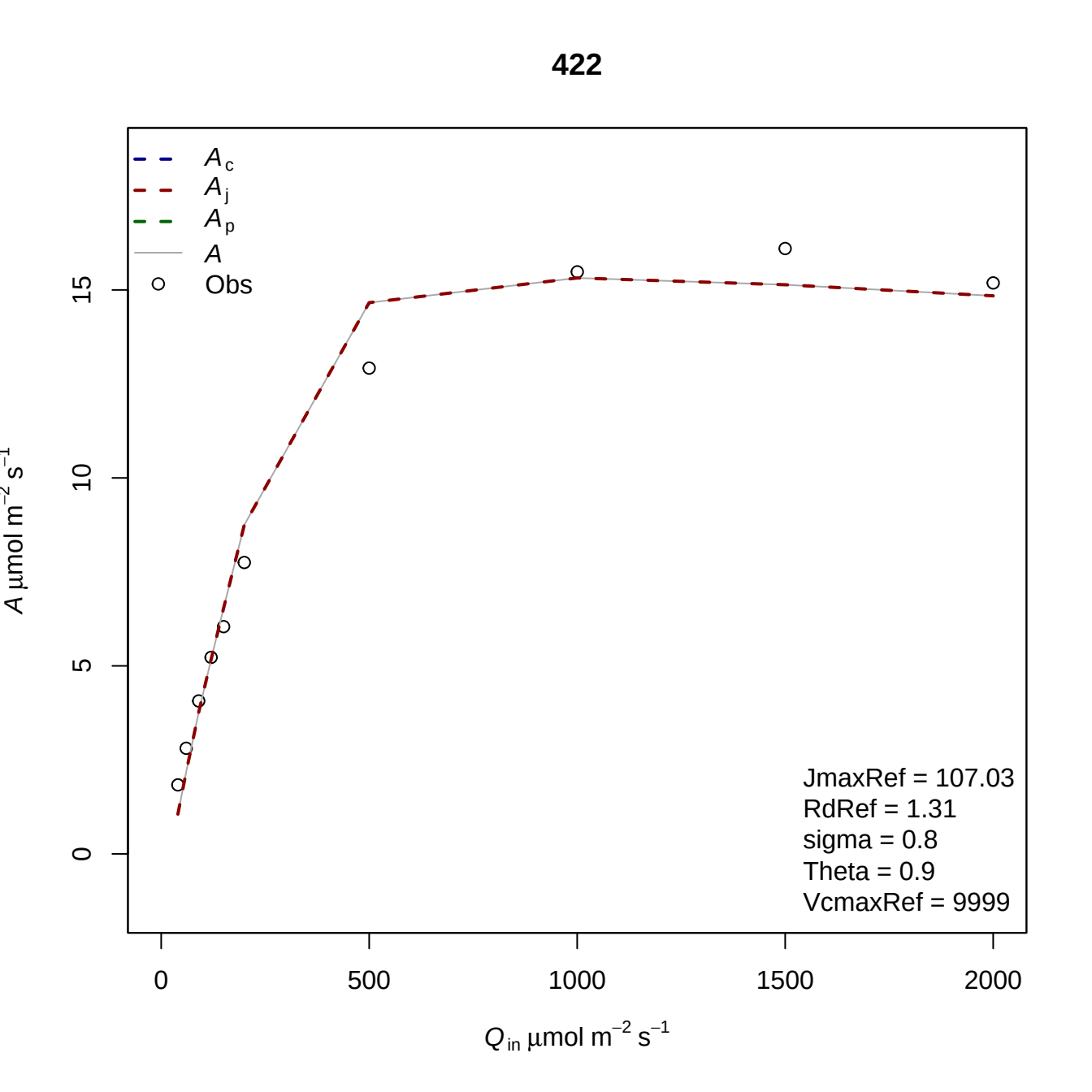

417
A_c
A_j
A_p
A
Obs
JmaxRef = 225.81
RdRef = 0.89
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

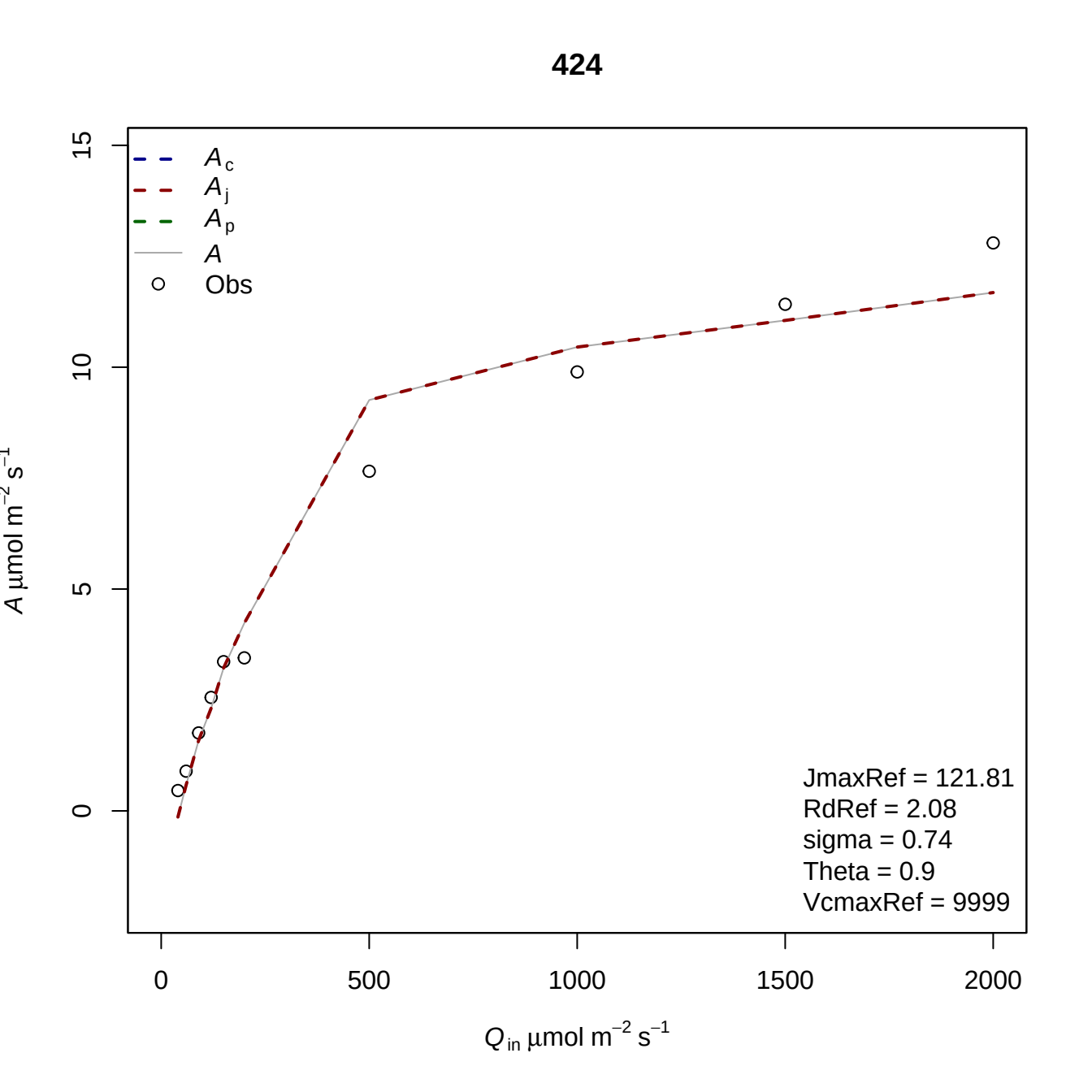

418
A_c
A_j
A_p
A
Obs
JmaxRef = 257.54
RdRef = 1.76
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

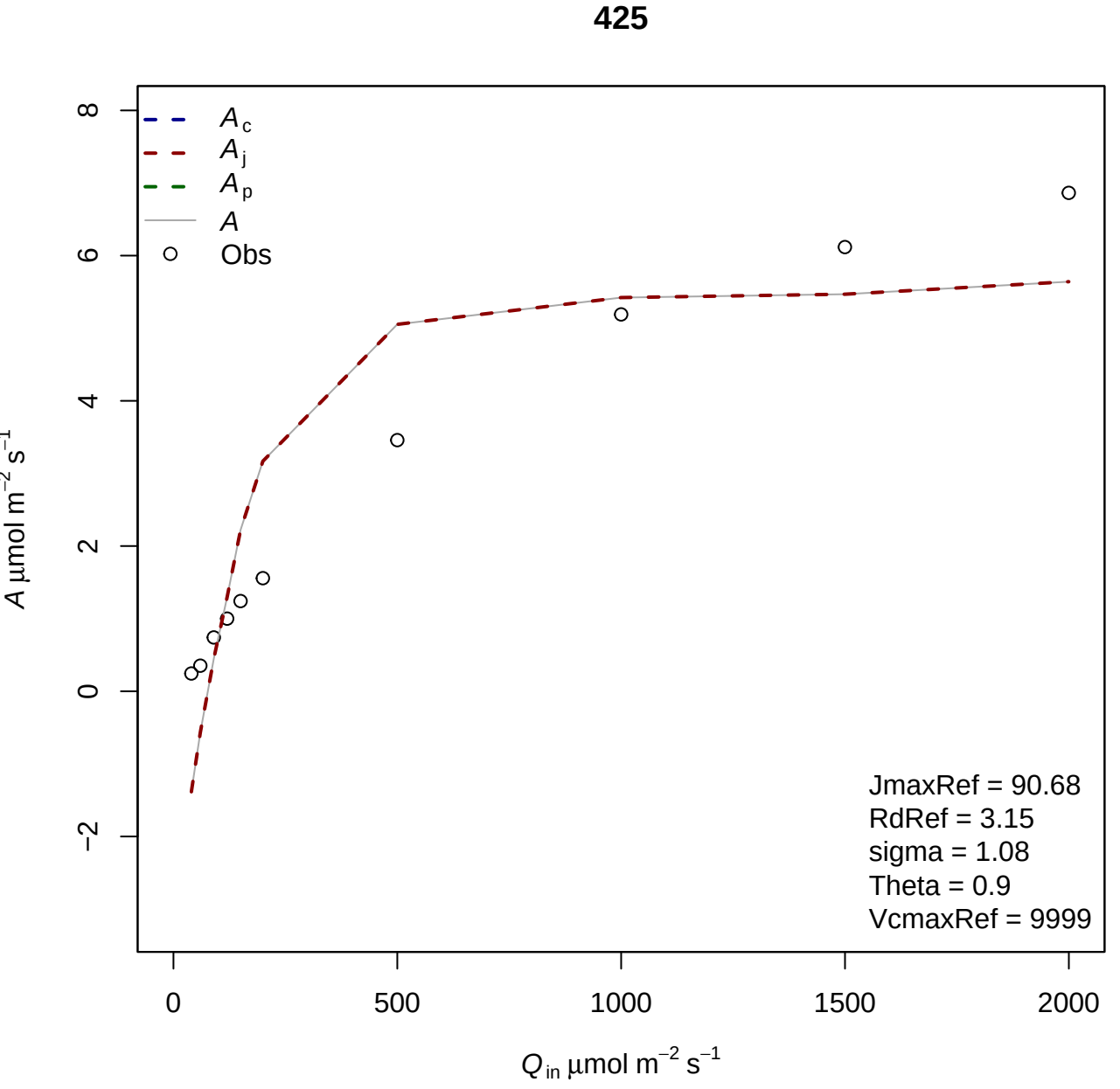

419
A_c
A_j
A_p
A
Obs
JmaxRef = 216.67
RdRef = 4.1
sigma = 1.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

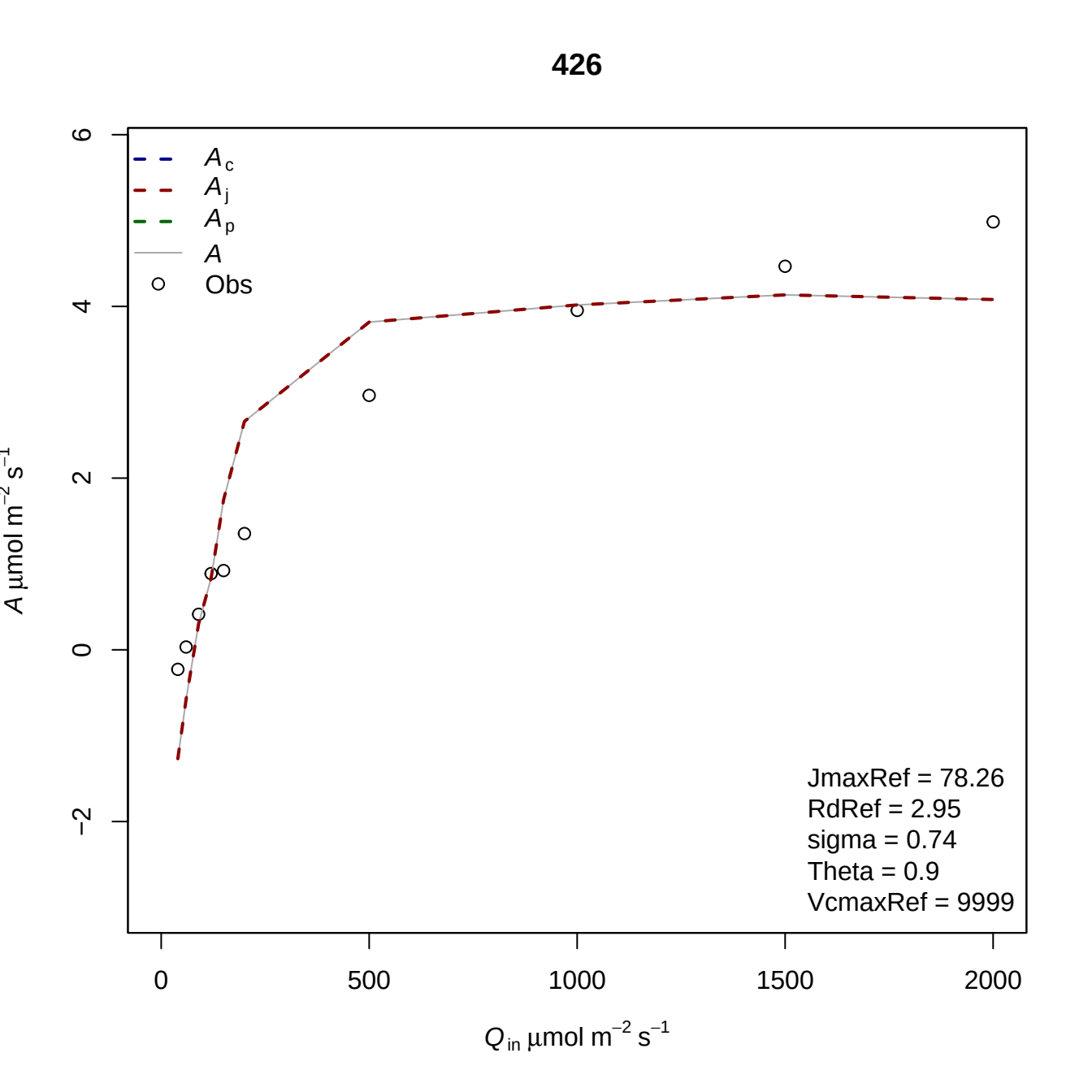

420
A_c
A_j
A_p
A
Obs
JmaxRef = 250.33
RdRef = 3.32
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

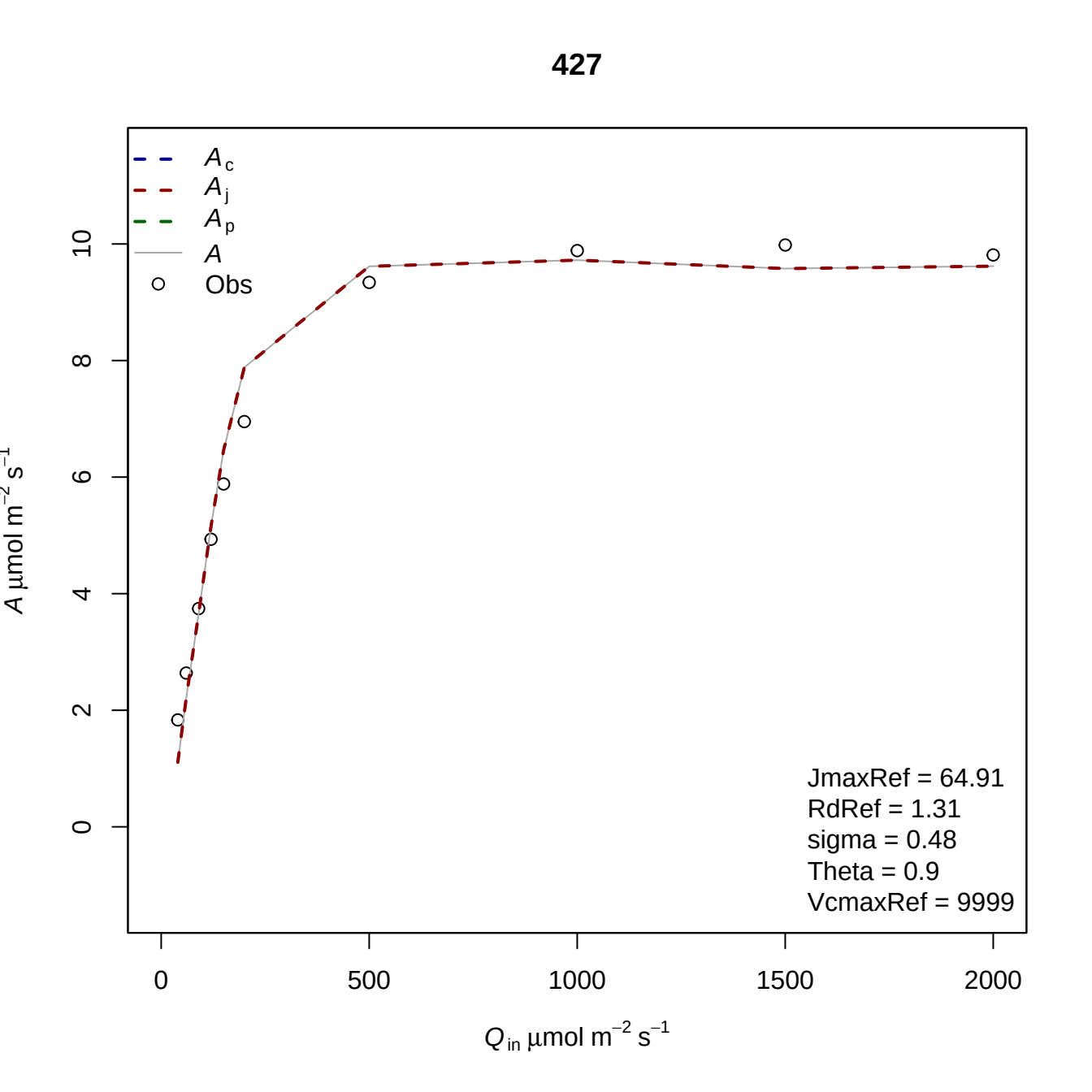

421
A_c
A_j
A_p
A
Obs
JmaxRef = 72.99
RdRef = 2.19
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

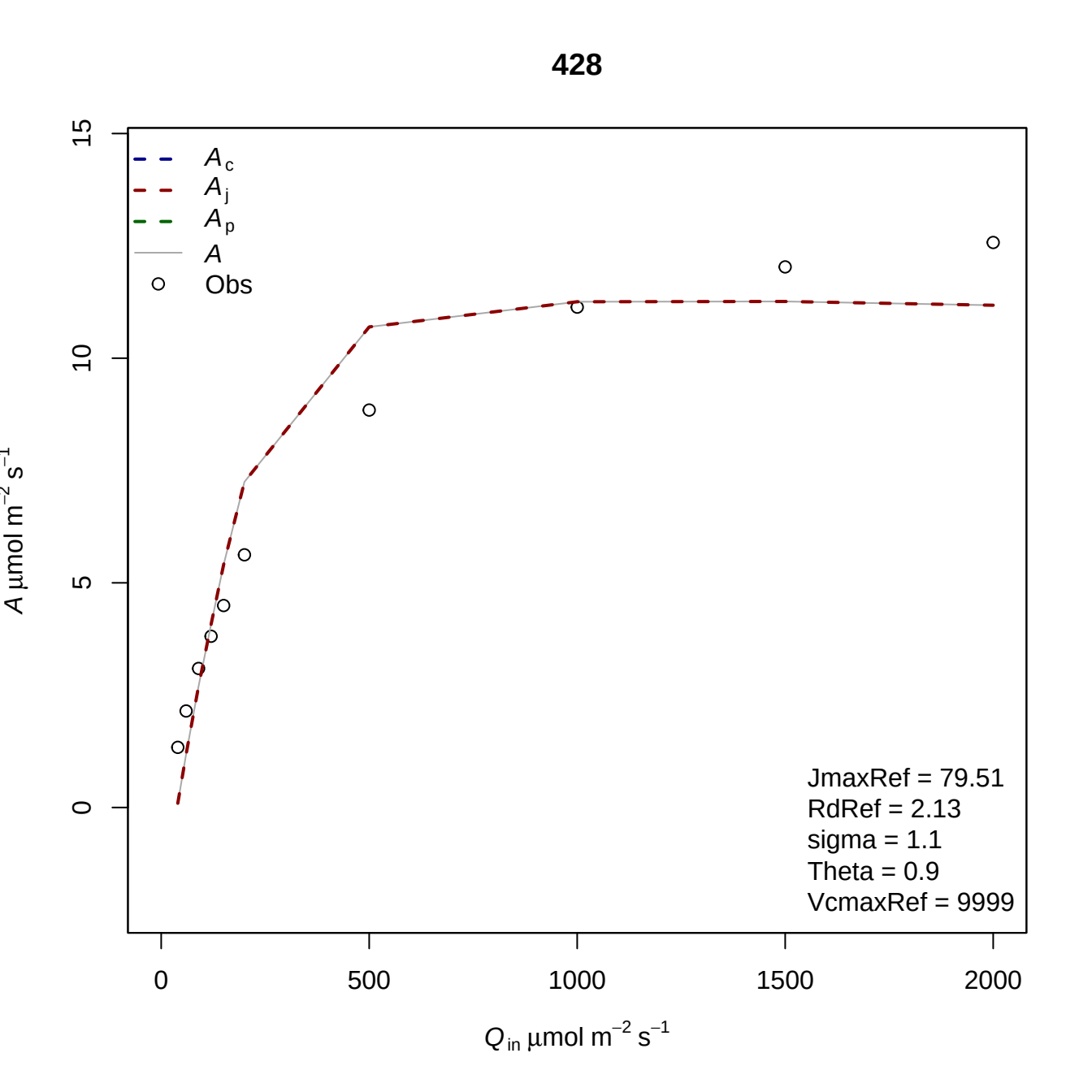

422
A_c
A_j
A_p
A
Obs
JmaxRef = 107.03
RdRef = 1.31
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

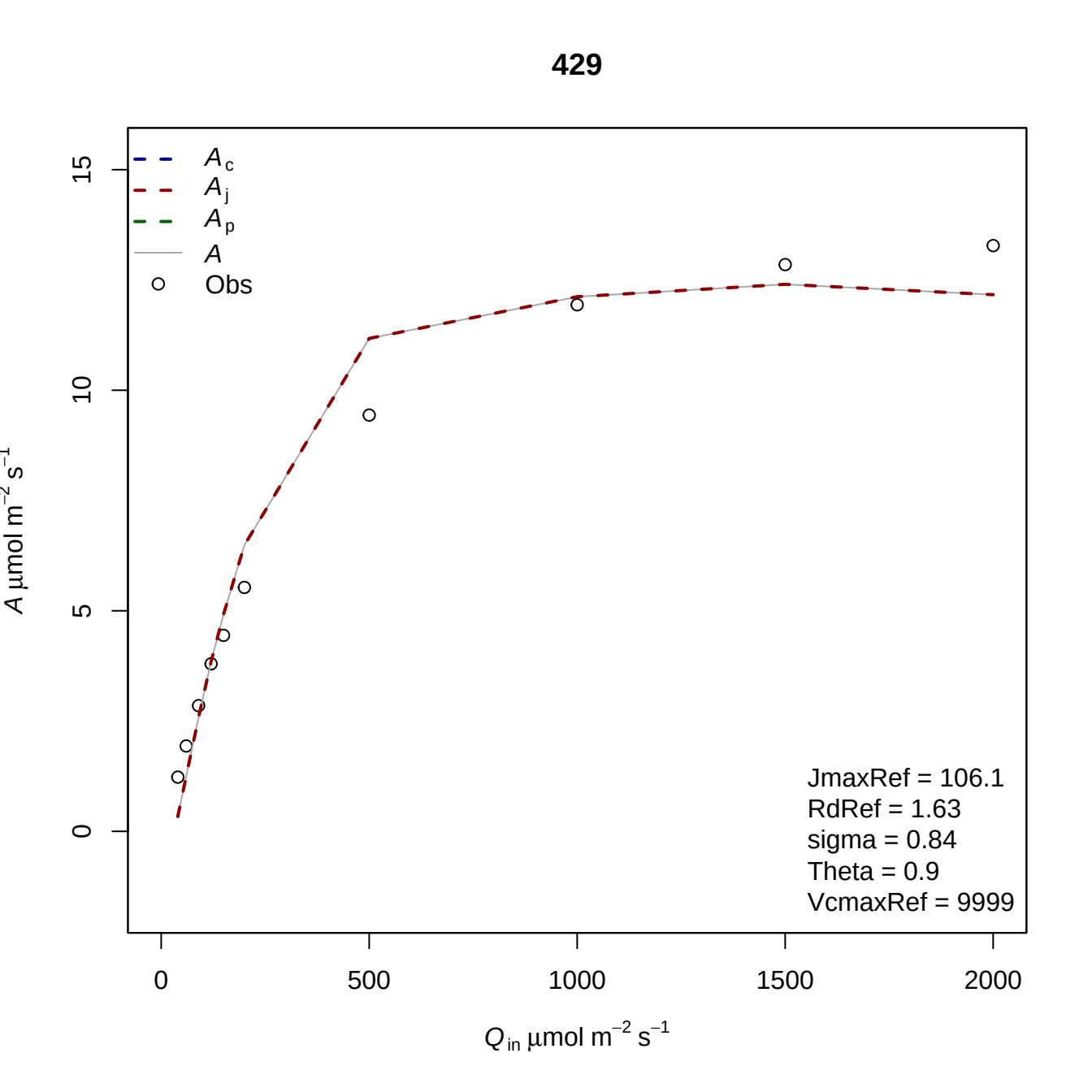

424
A_c
A_j
A_p
A
Obs
JmaxRef = 121.81
RdRef = 2.08
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

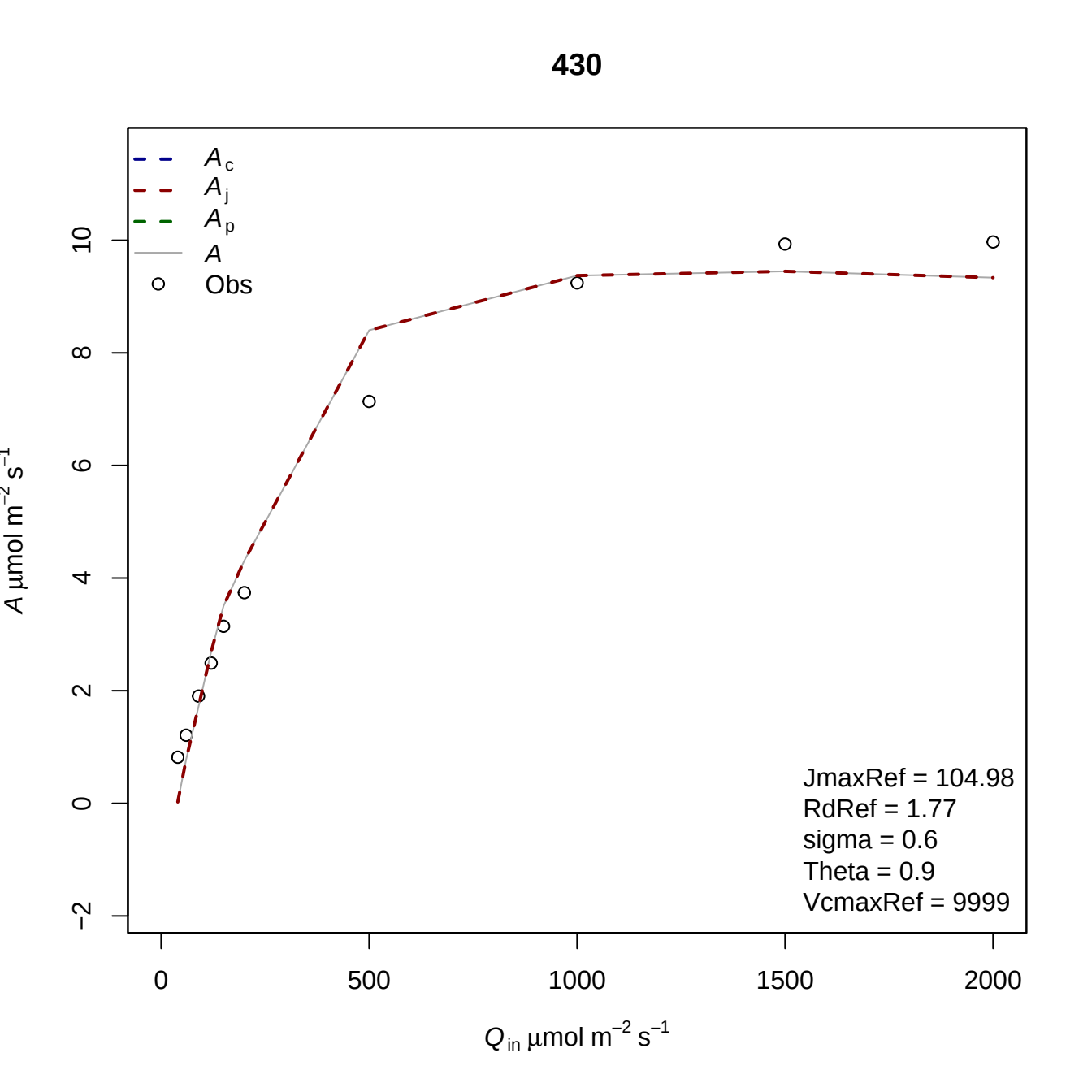

425
A_c
A_j
A_p
A
Obs
JmaxRef = 90.68
RdRef = 3.15
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

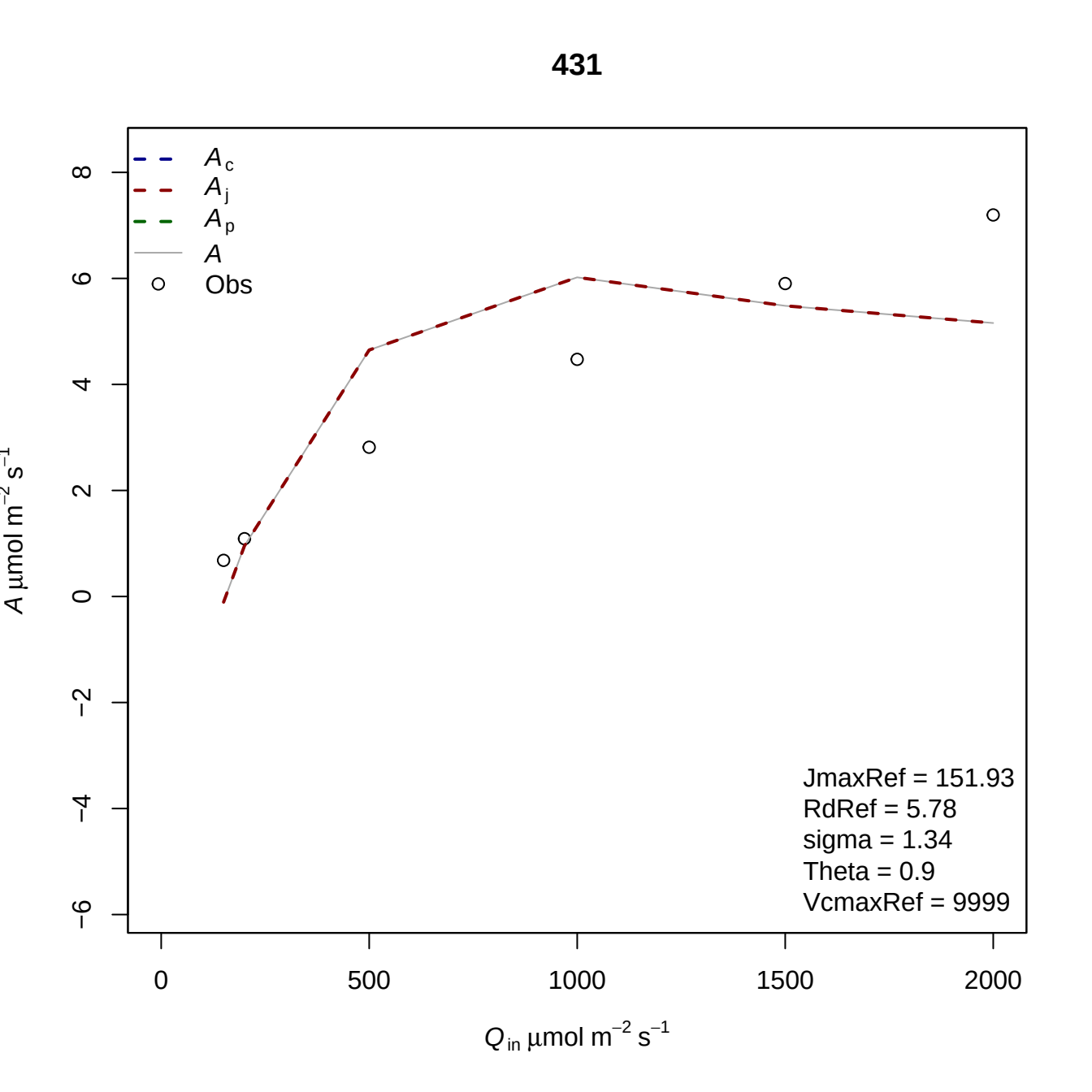

426
A_c
A_j
A_p
A
Obs
JmaxRef = 78.26
RdRef = 2.95
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

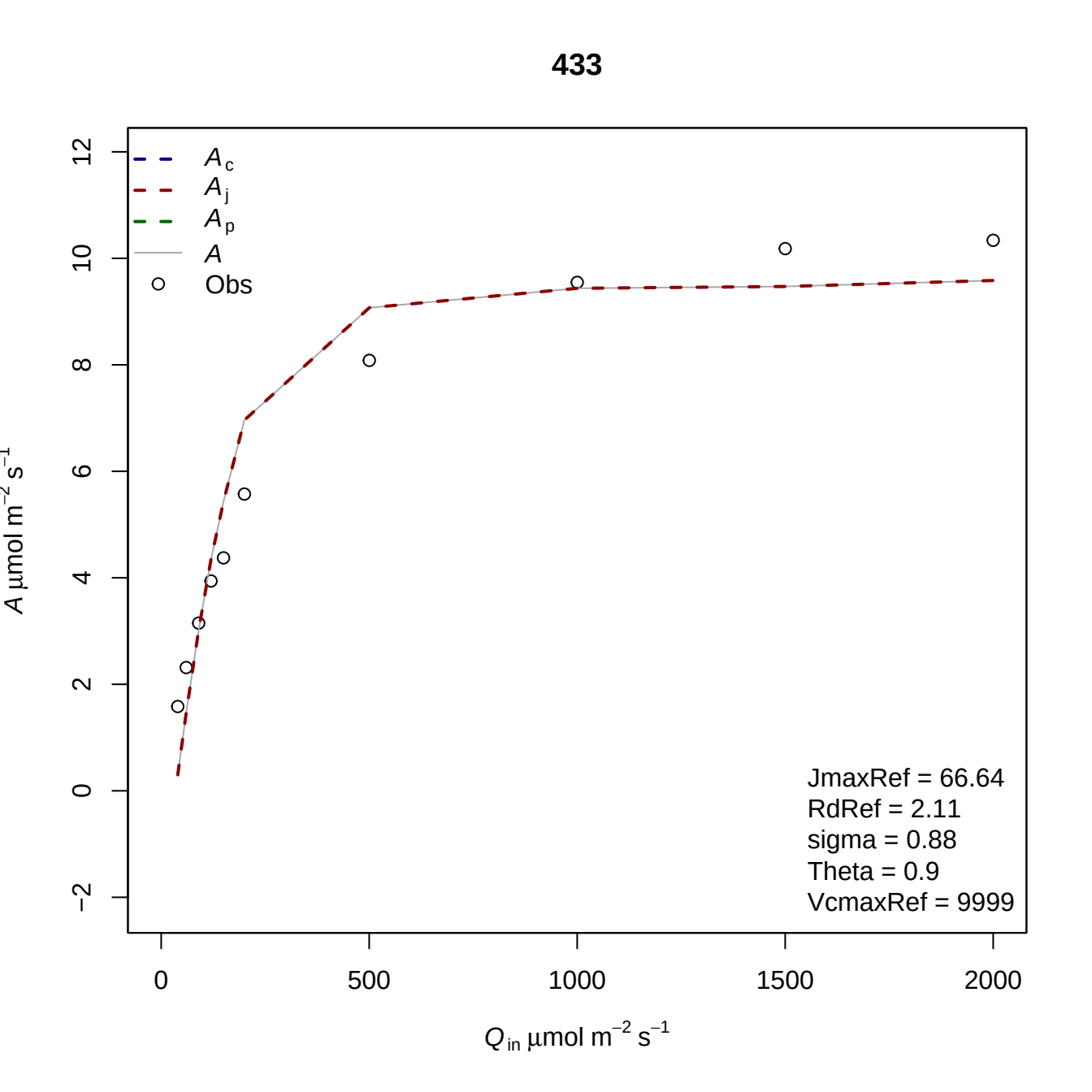

427
A_c
A_j
A_p
A
Obs
JmaxRef = 64.91
RdRef = 1.31
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

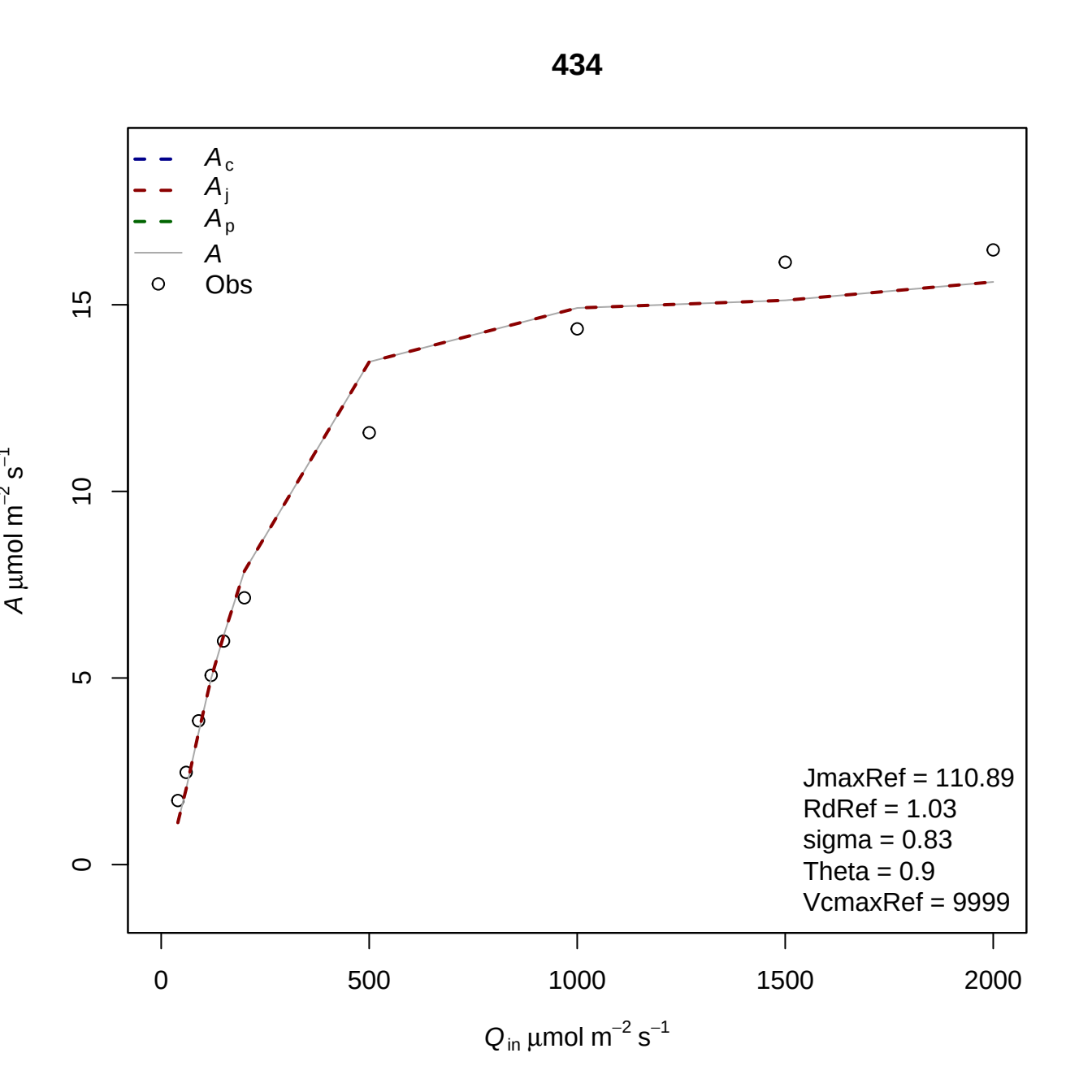

428
A_c
A_j
A_p
A
Obs
JmaxRef = 79.51
RdRef = 2.13
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

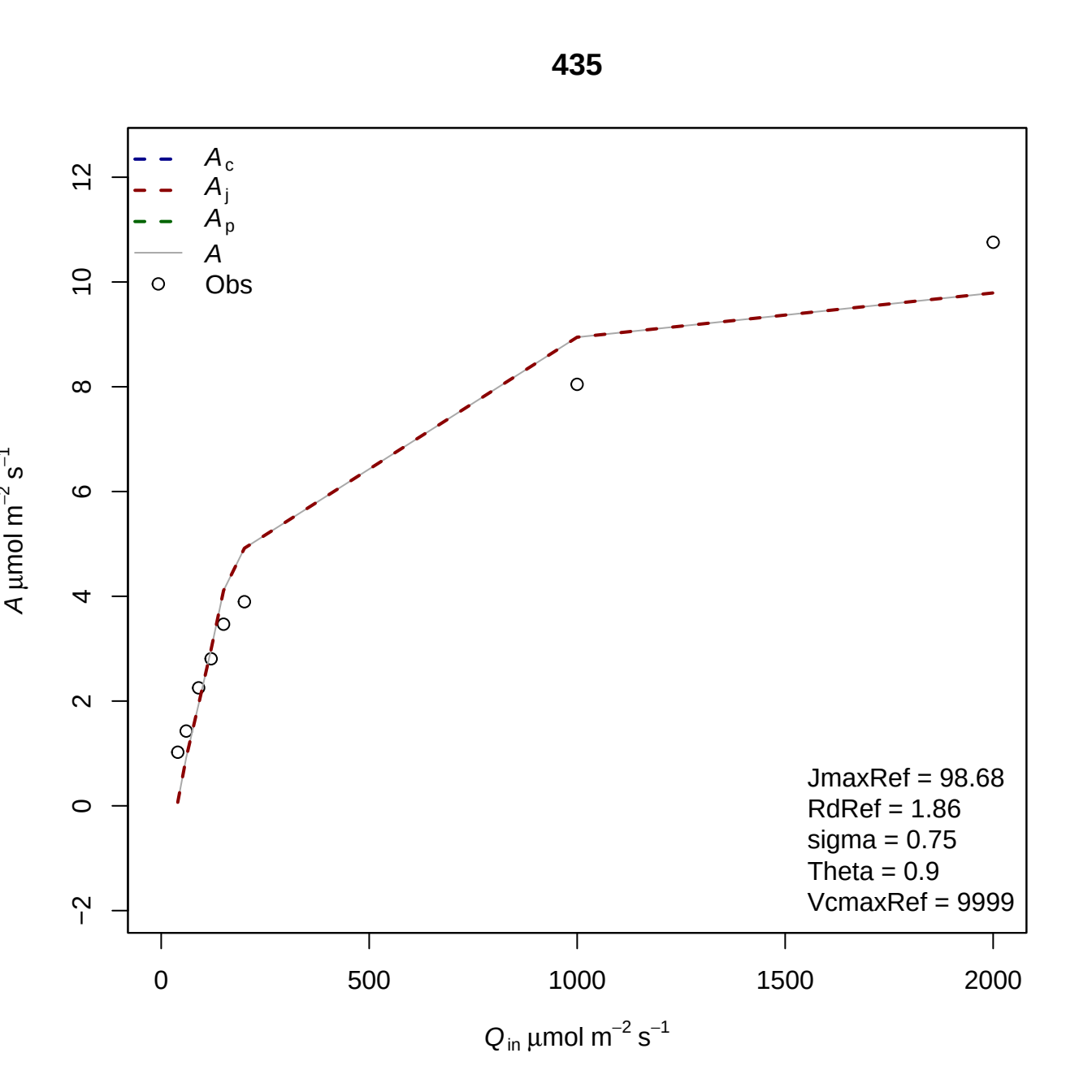

429
A_c
A_j
A_p
A
Obs
JmaxRef = 106.1
RdRef = 1.63
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

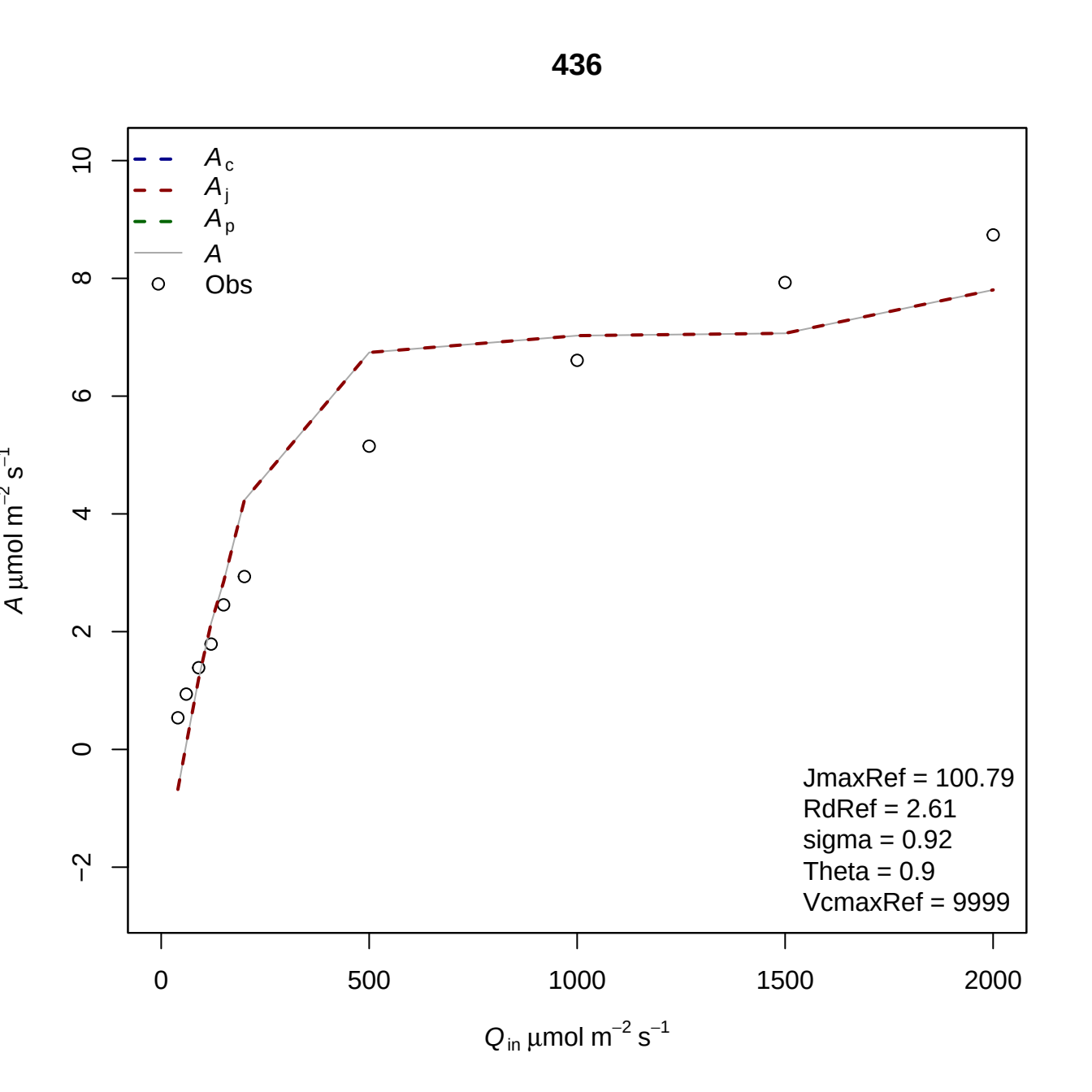

430
A_c
A_j
A_p
A
Obs
JmaxRef = 104.98
RdRef = 1.77
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

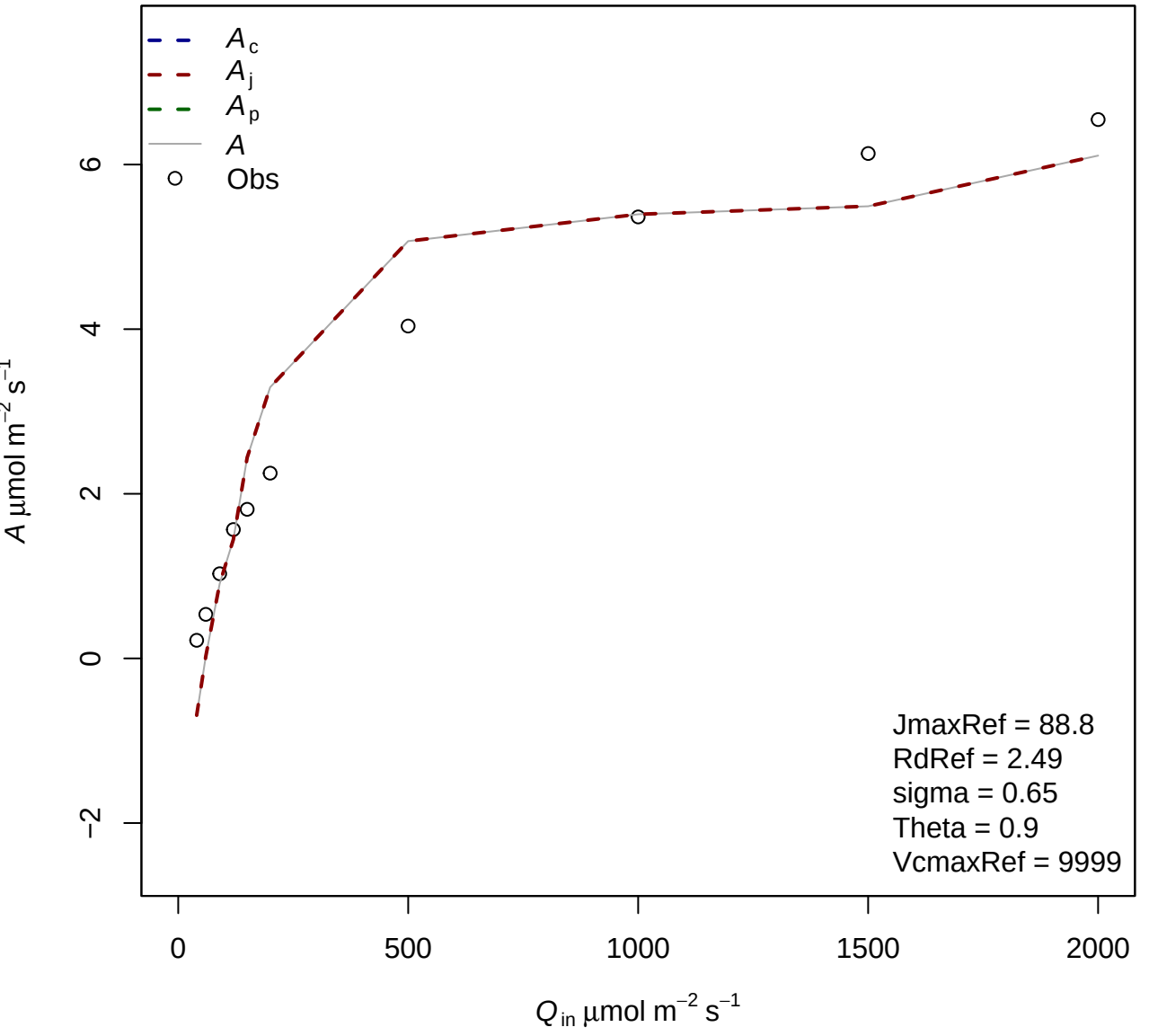

431
A_c
A_j
A_p
A
Obs
JmaxRef = 151.93
RdRef = 5.78
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
−6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

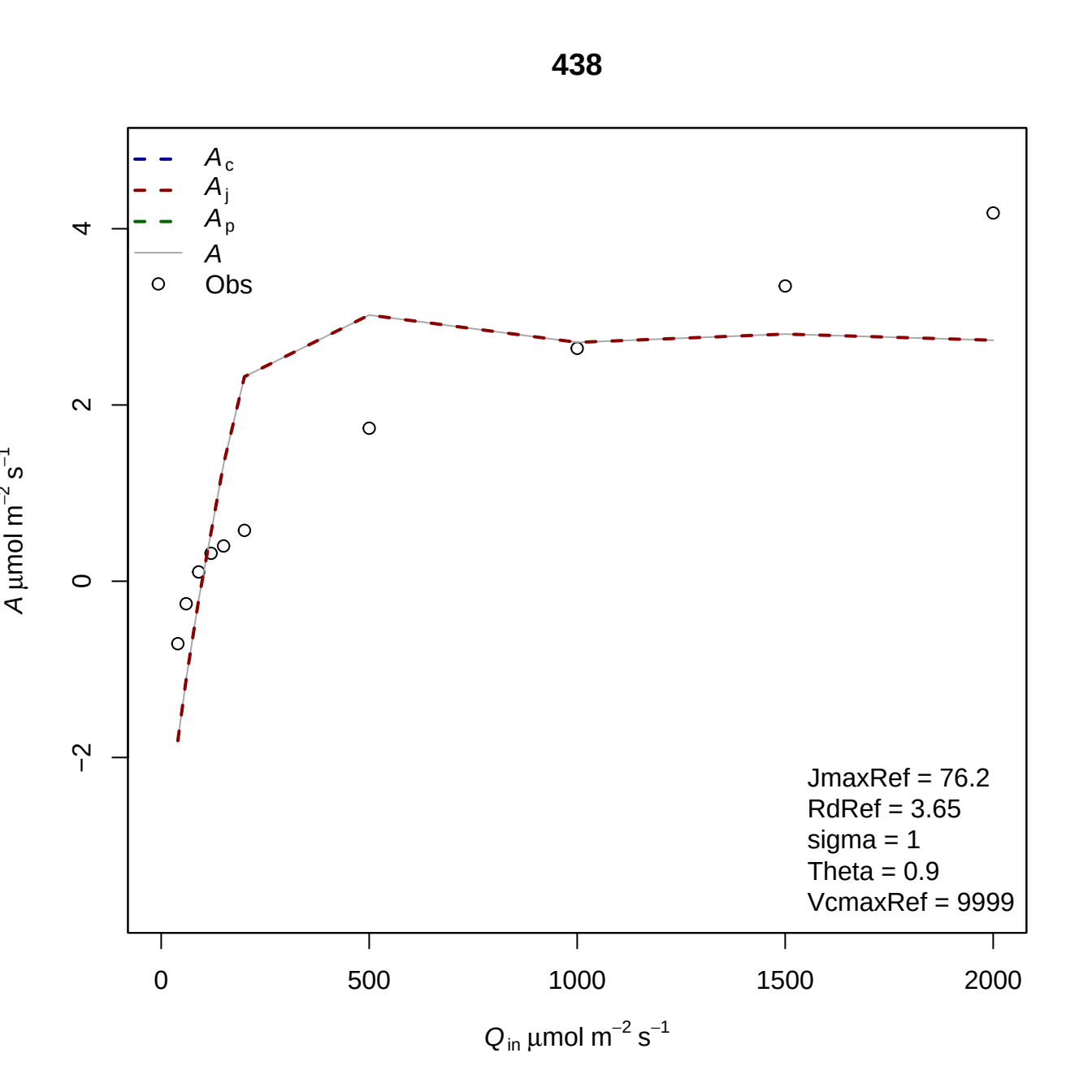

433
A_c
A_j
A_p
A
Obs
JmaxRef = 66.64
RdRef = 2.11
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

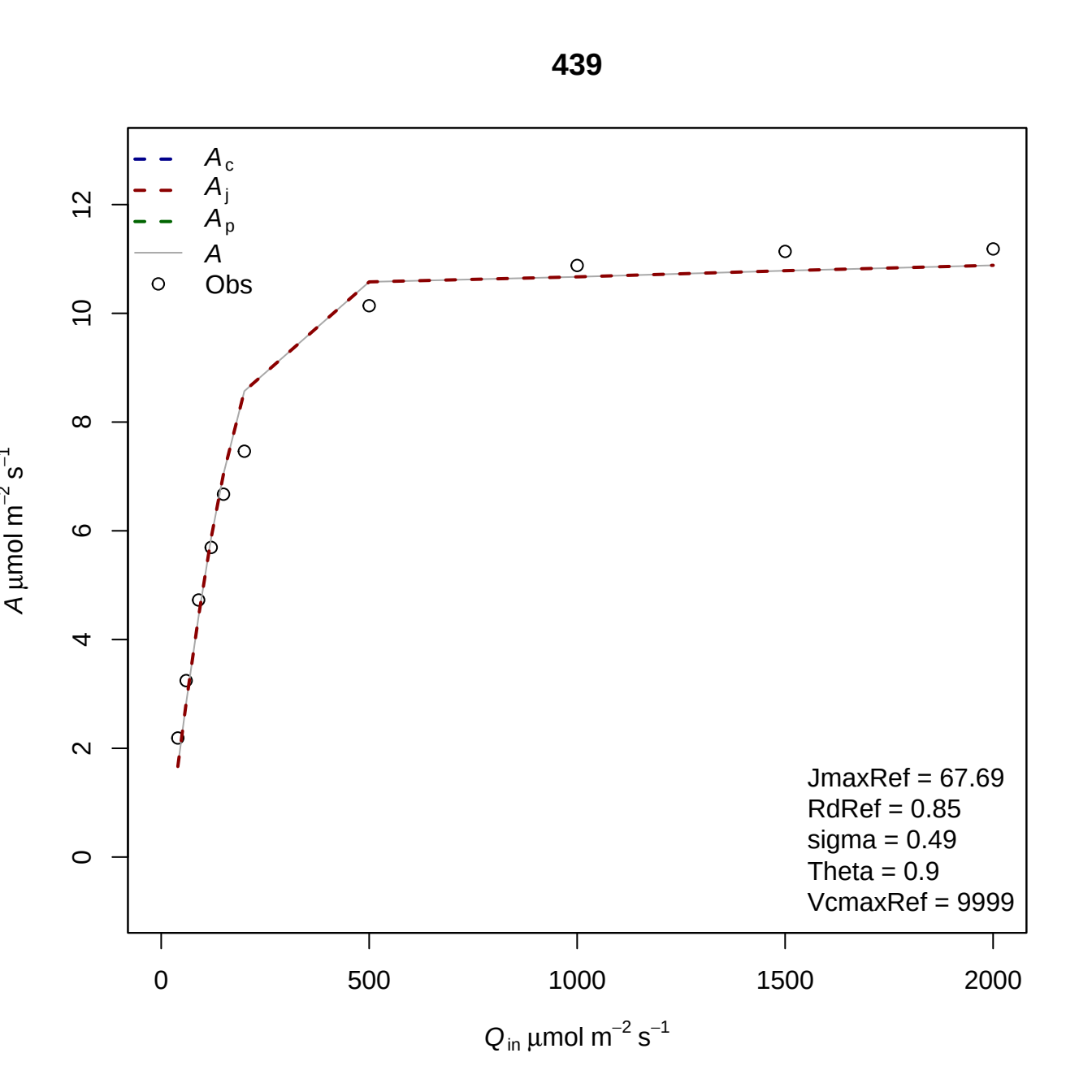

434
A_c
A_j
A_p
A
Obs
JmaxRef = 110.89
RdRef = 1.03
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

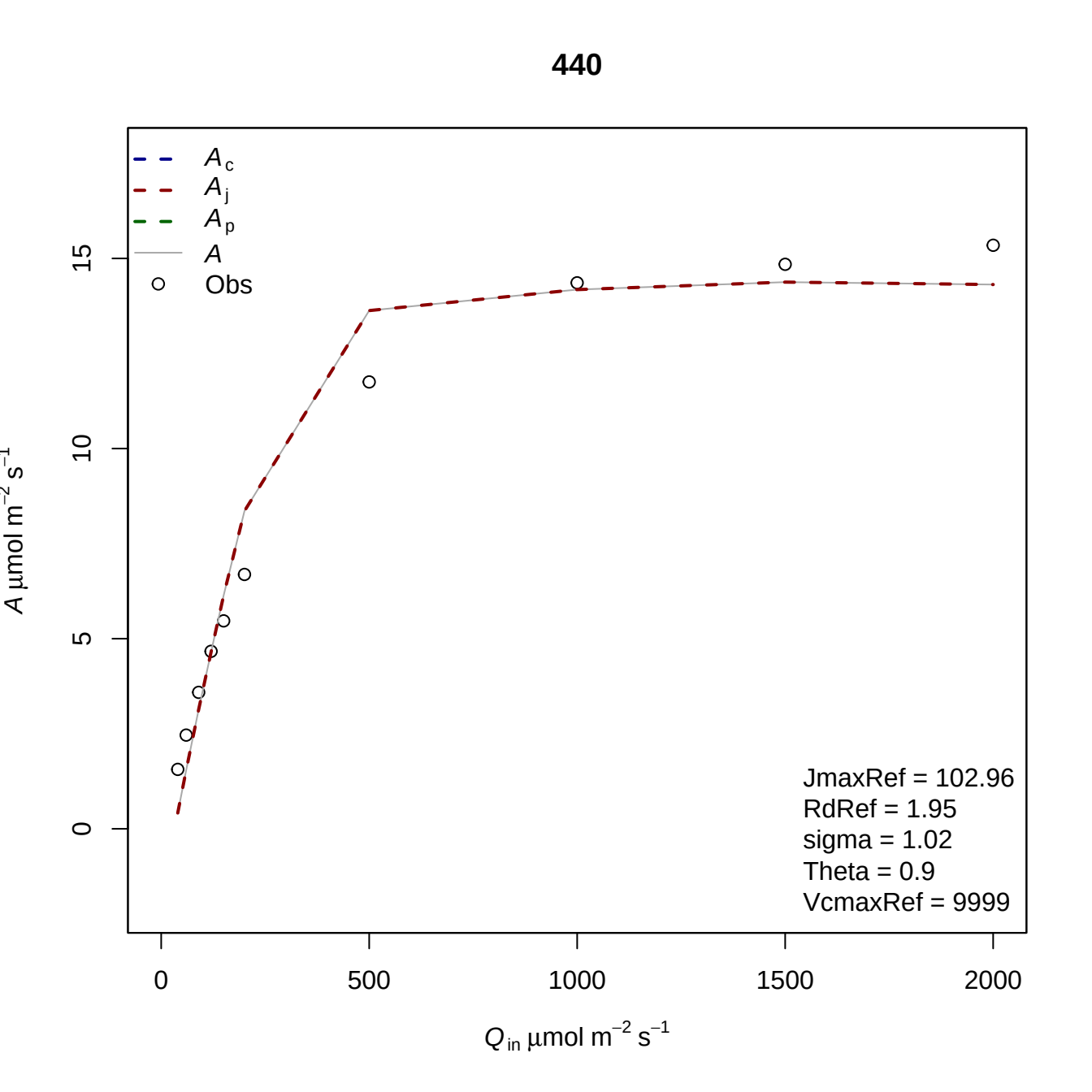

435
A_c
A_j
A_p
A
Obs
JmaxRef = 98.68
RdRef = 1.86
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

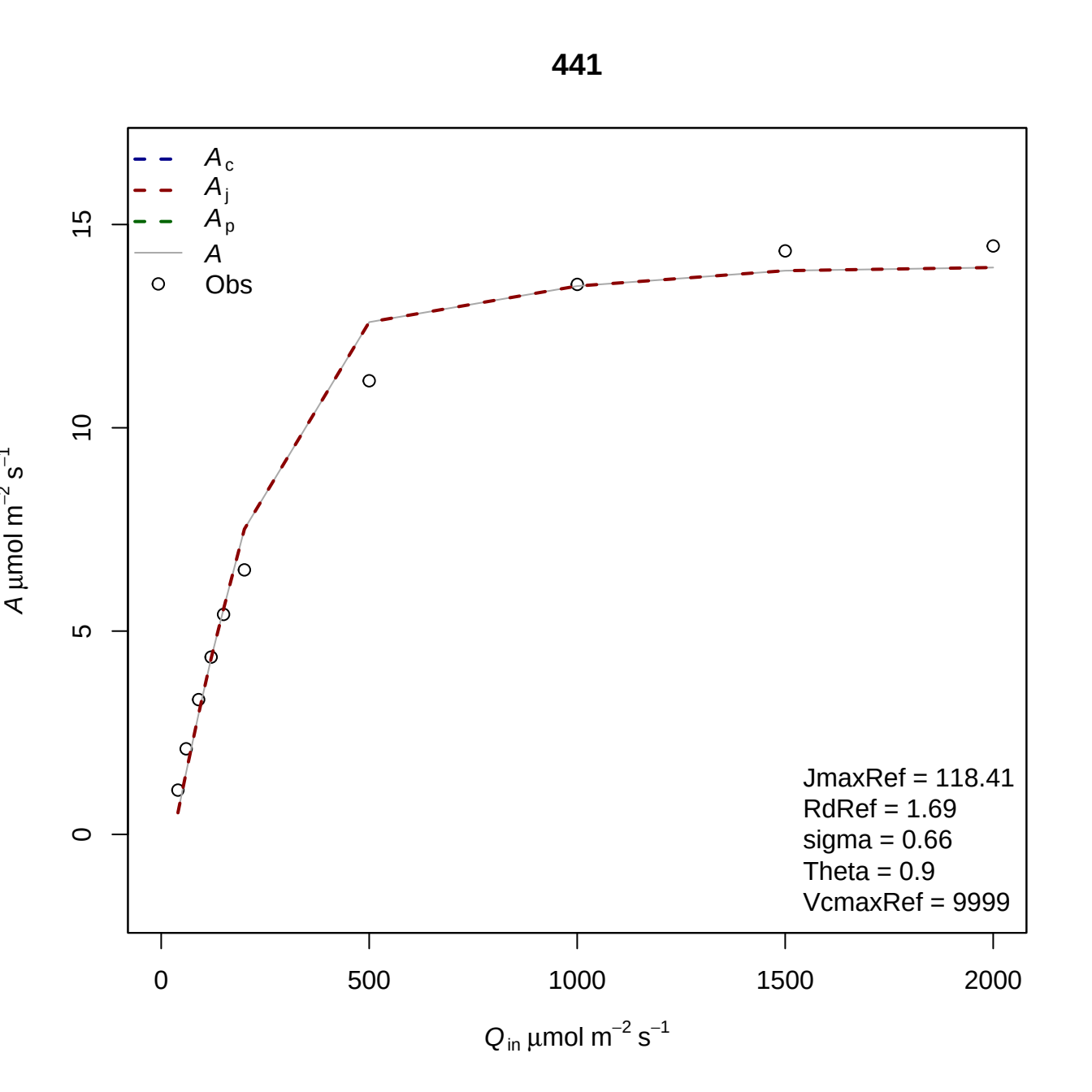

436
A_c
A_j
A_p
A
Obs
JmaxRef = 100.79
RdRef = 2.61
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

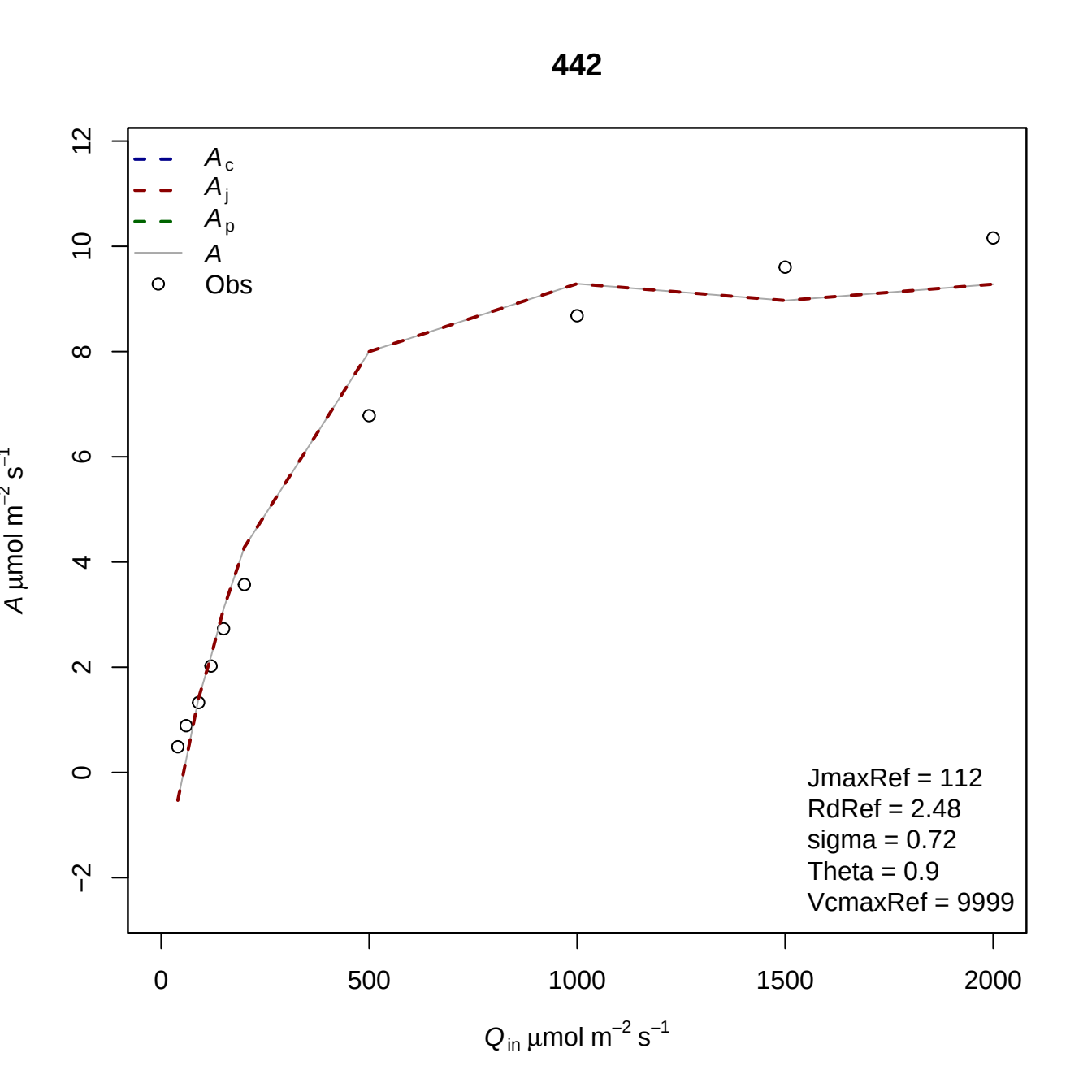

A_c
A_j
A_p
A
Obs
JmaxRef = 88.8
RdRef = 2.49
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

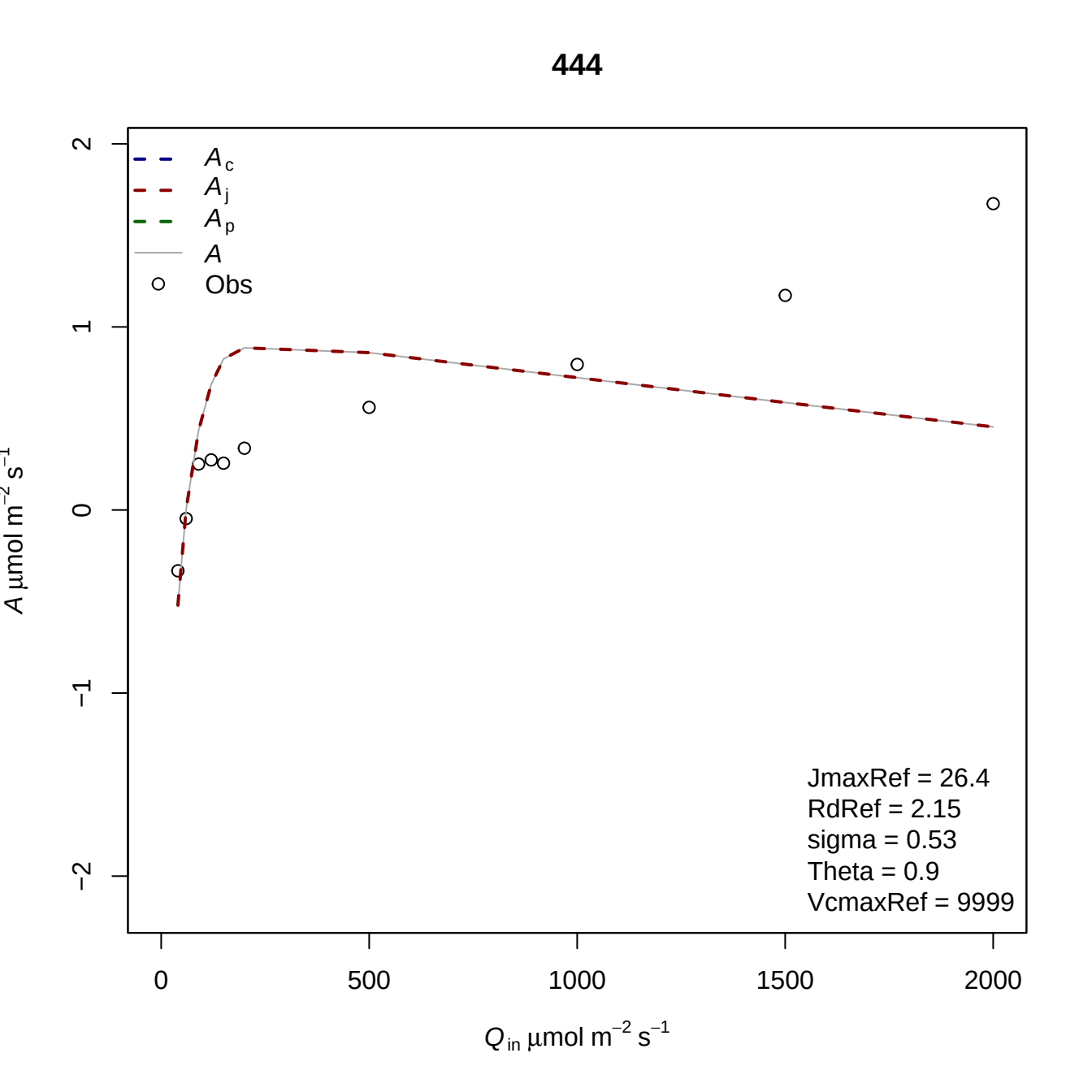

438
A_c
A_j
A_p
A
Obs
JmaxRef = 76.2
RdRef = 3.65
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

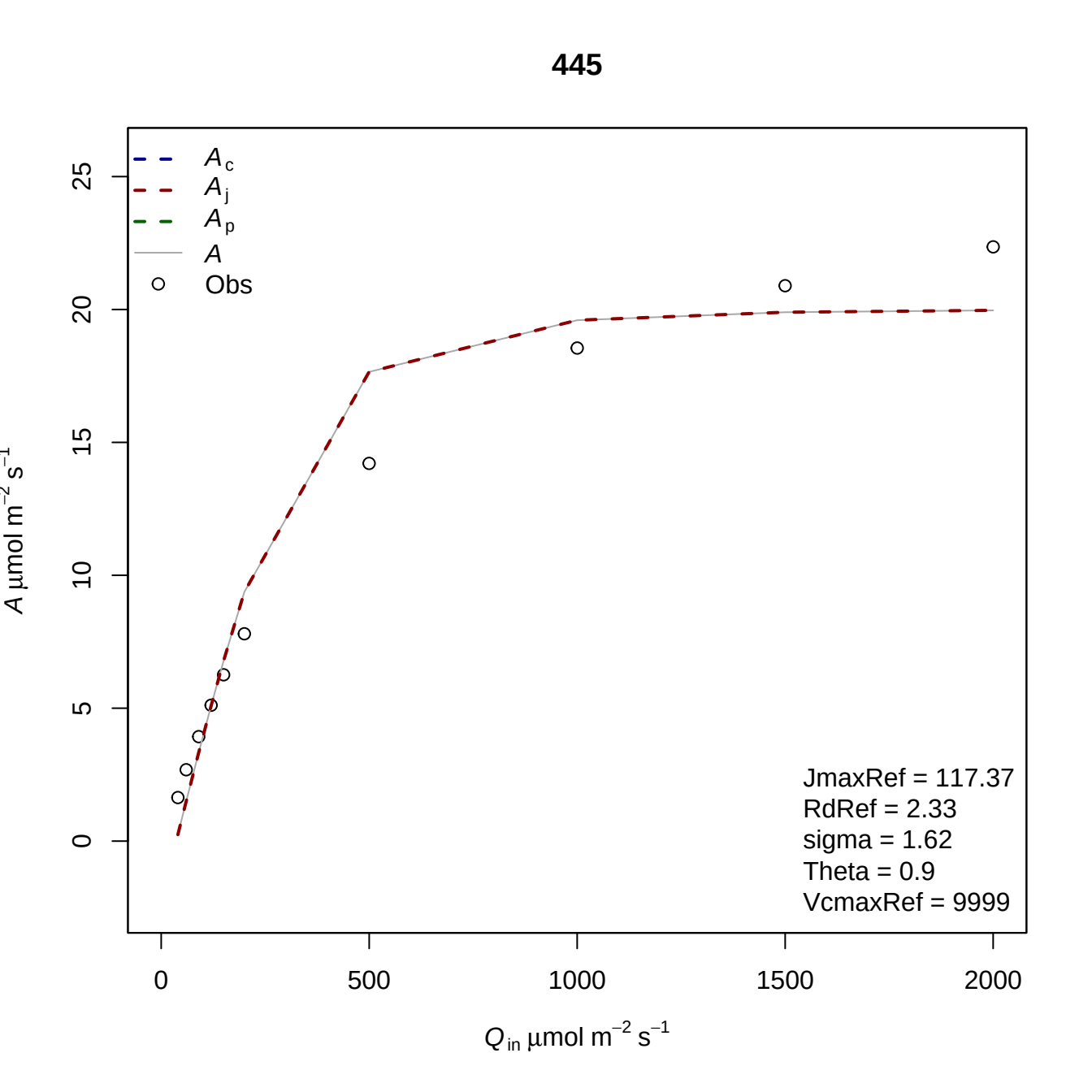

439
A_c
A_j
A_p
A
Obs
JmaxRef = 67.69
RdRef = 0.85
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

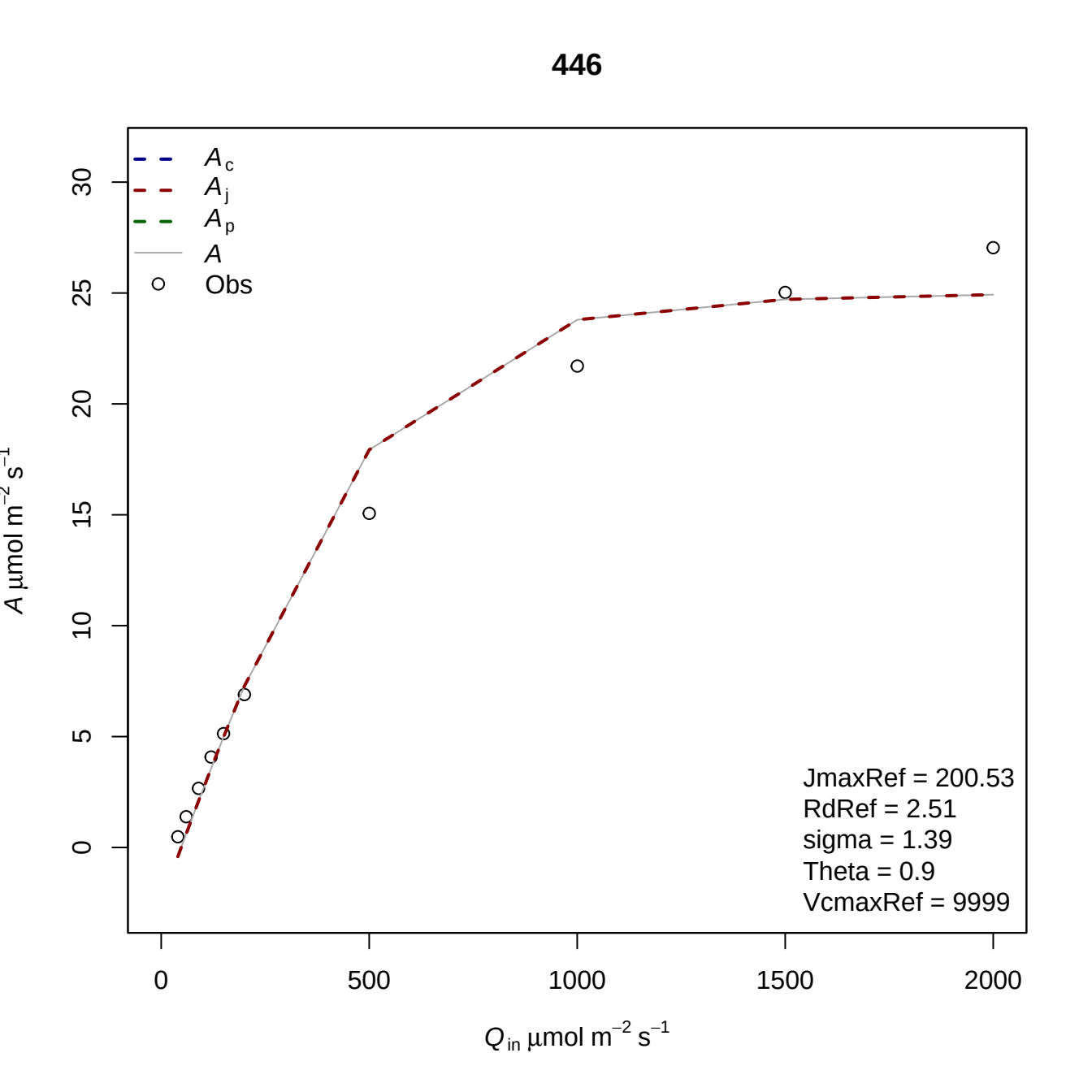

440
A_c
A_j
A_p
A
Obs
JmaxRef = 102.96
RdRef = 1.95
sigma = 1.02
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

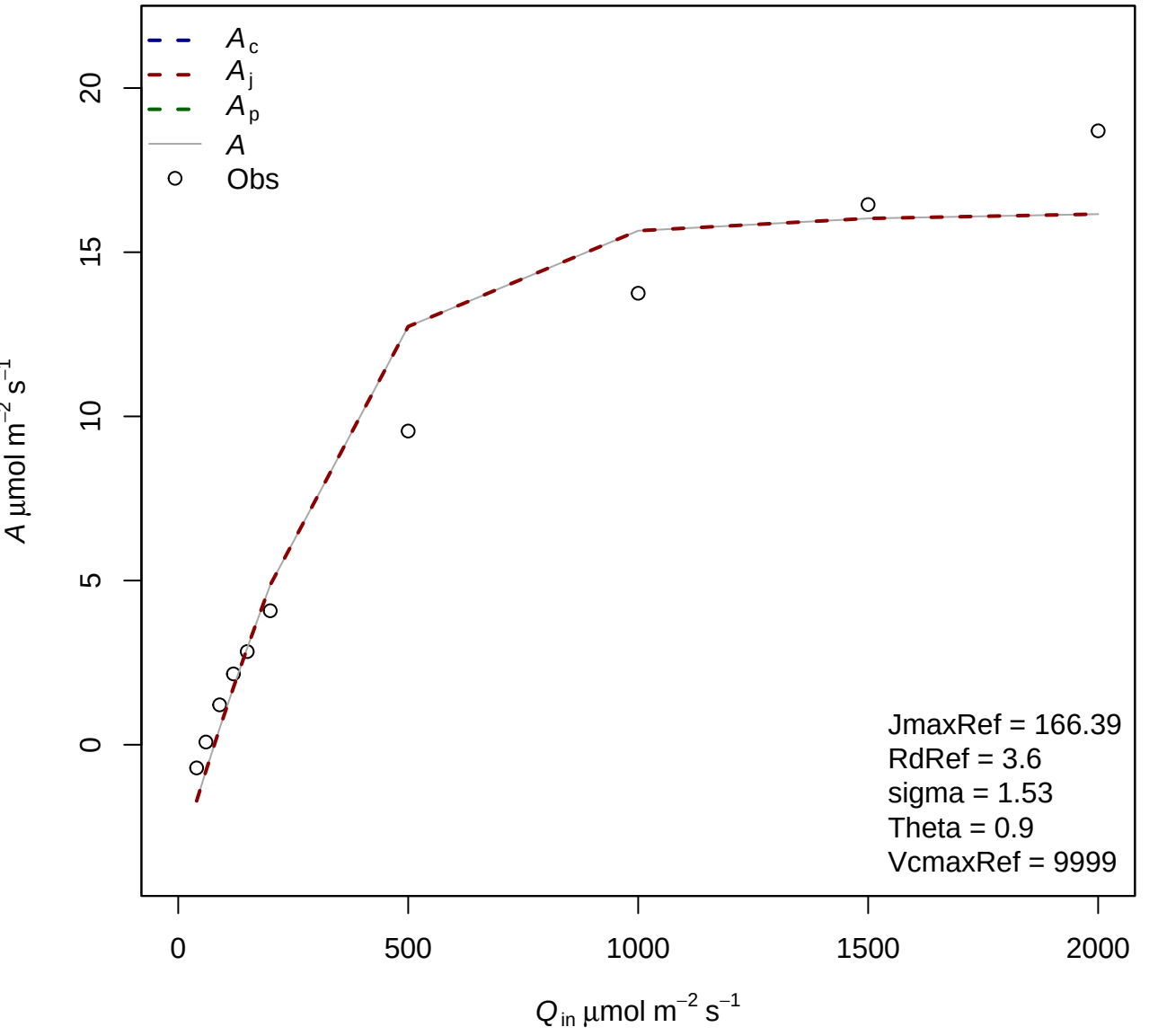

441
A_c
A_j
A_p
A
Obs
JmaxRef = 118.41
RdRef = 1.69
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

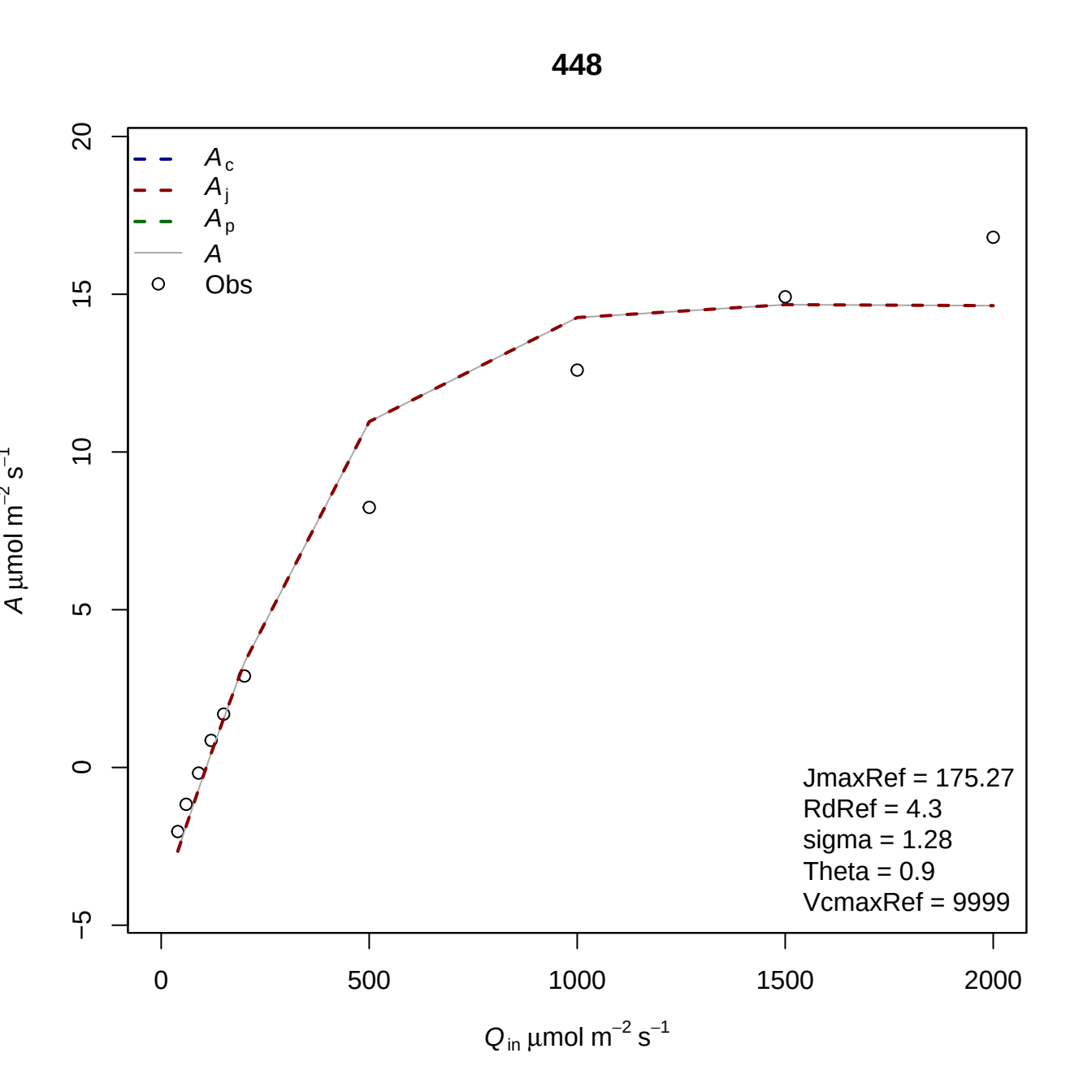

442
A_c
A_j
A_p
A
Obs
JmaxRef = 112
RdRef = 2.48
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

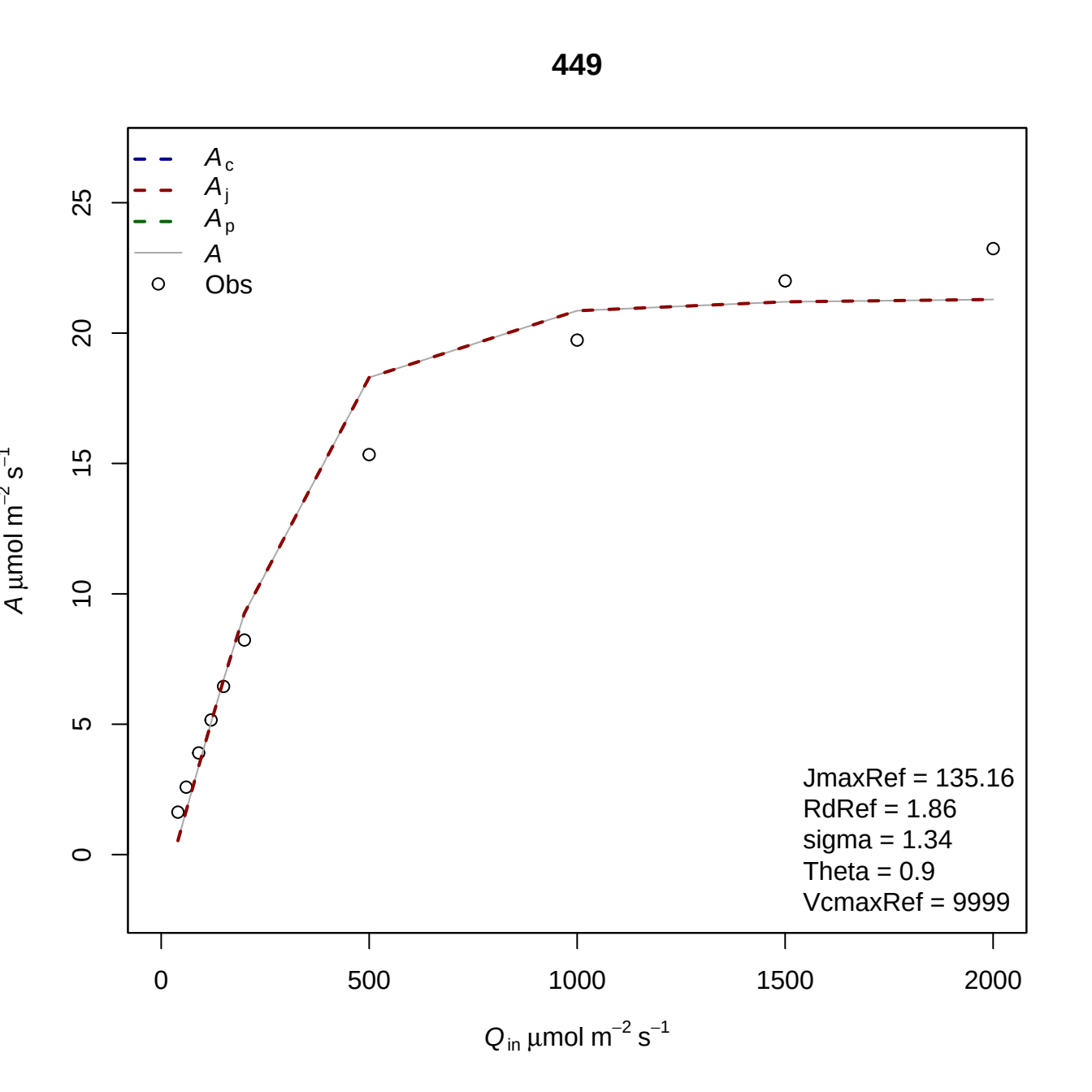

444
A_c
A_j
A_p
A
Obs
JmaxRef = 26.4
RdRef = 2.15
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

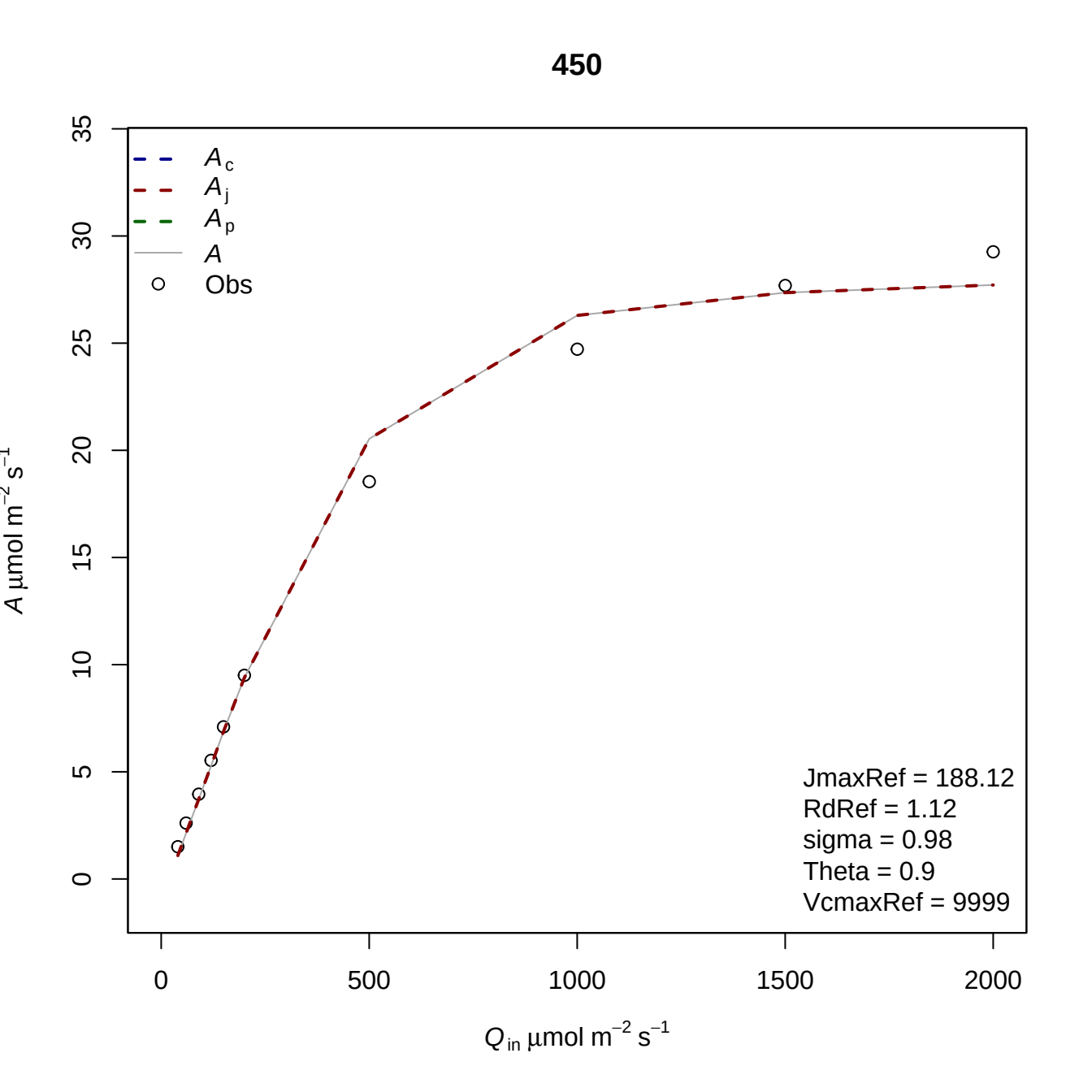

445
A_c
A_j
A_p
A
Obs
JmaxRef = 117.37
RdRef = 2.33
sigma = 1.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

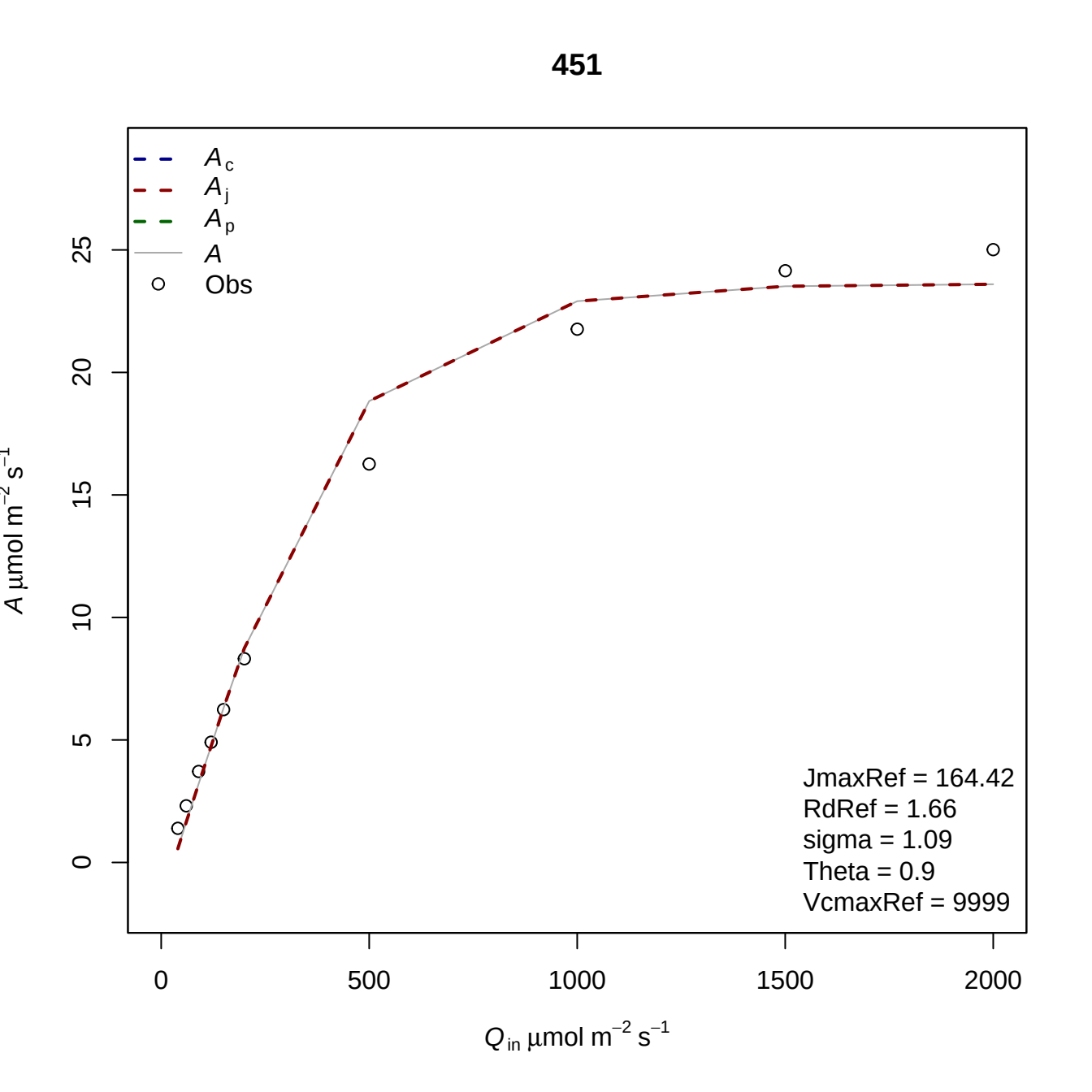

446
A_c
A_j
A_p
A
Obs
JmaxRef = 200.53
RdRef = 2.51
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

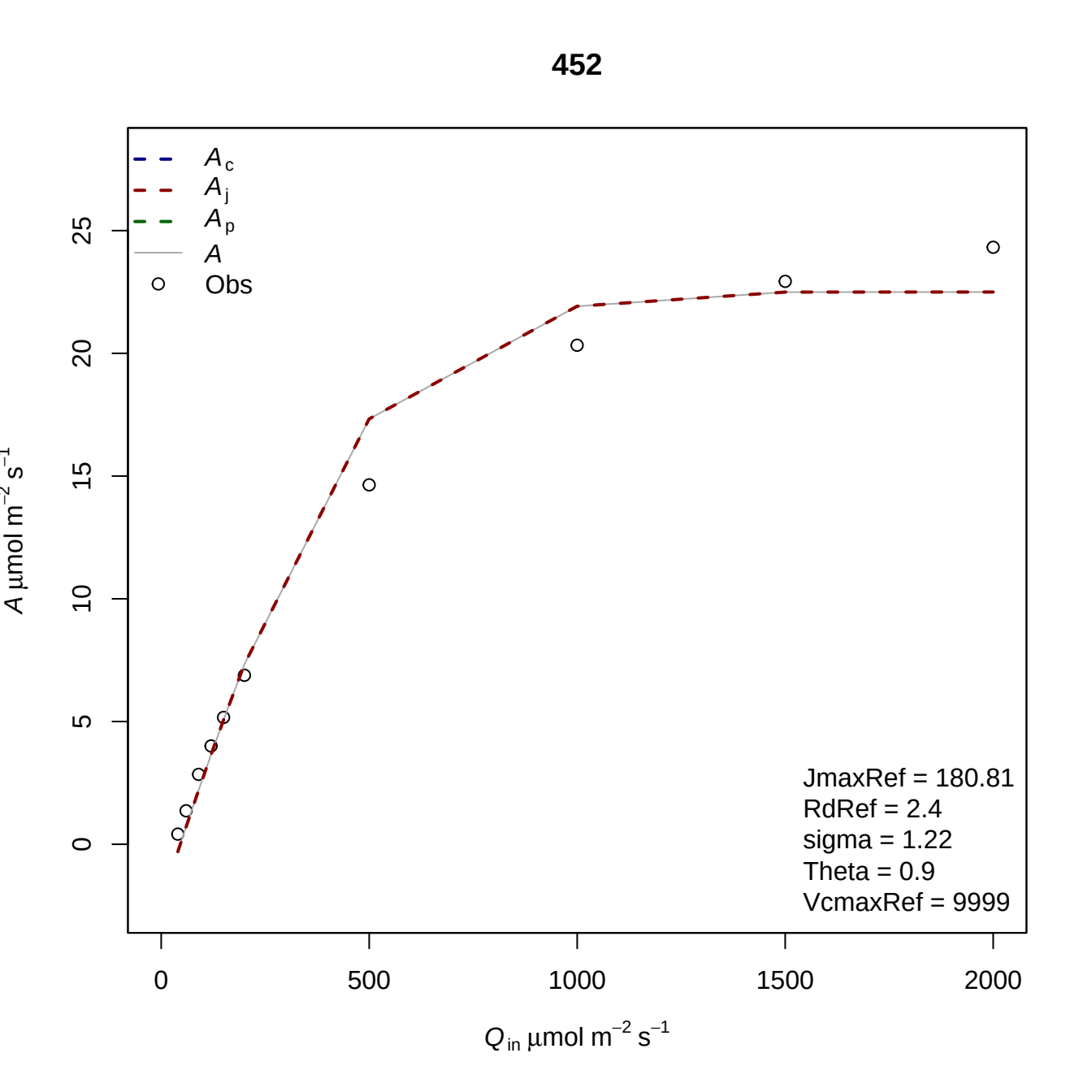

A_c
A_j
A_p
A
Obs
JmaxRef = 166.39
RdRef = 3.6
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

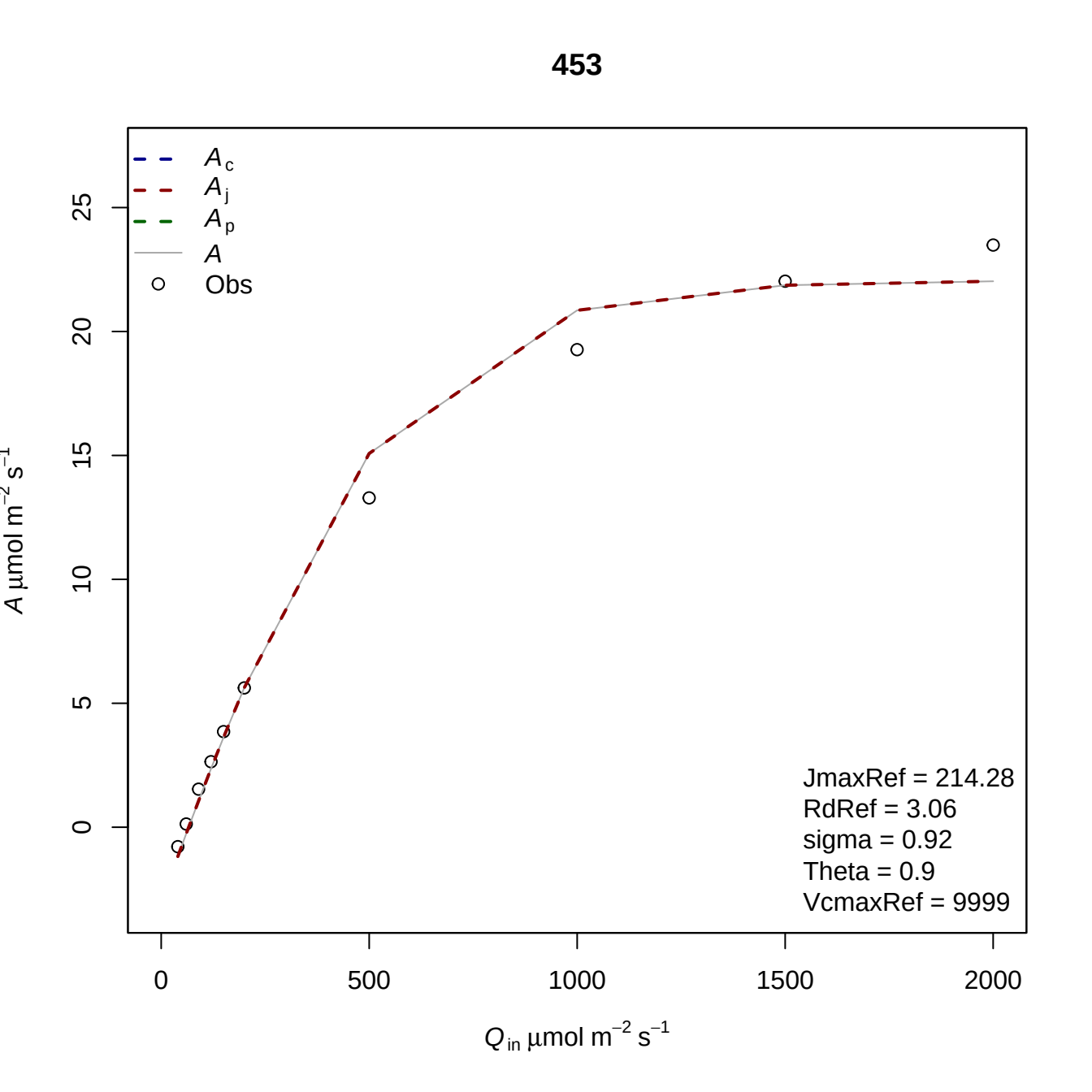

448
A_c
A_j
A_p
A
Obs
JmaxRef = 175.27
RdRef = 4.3
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

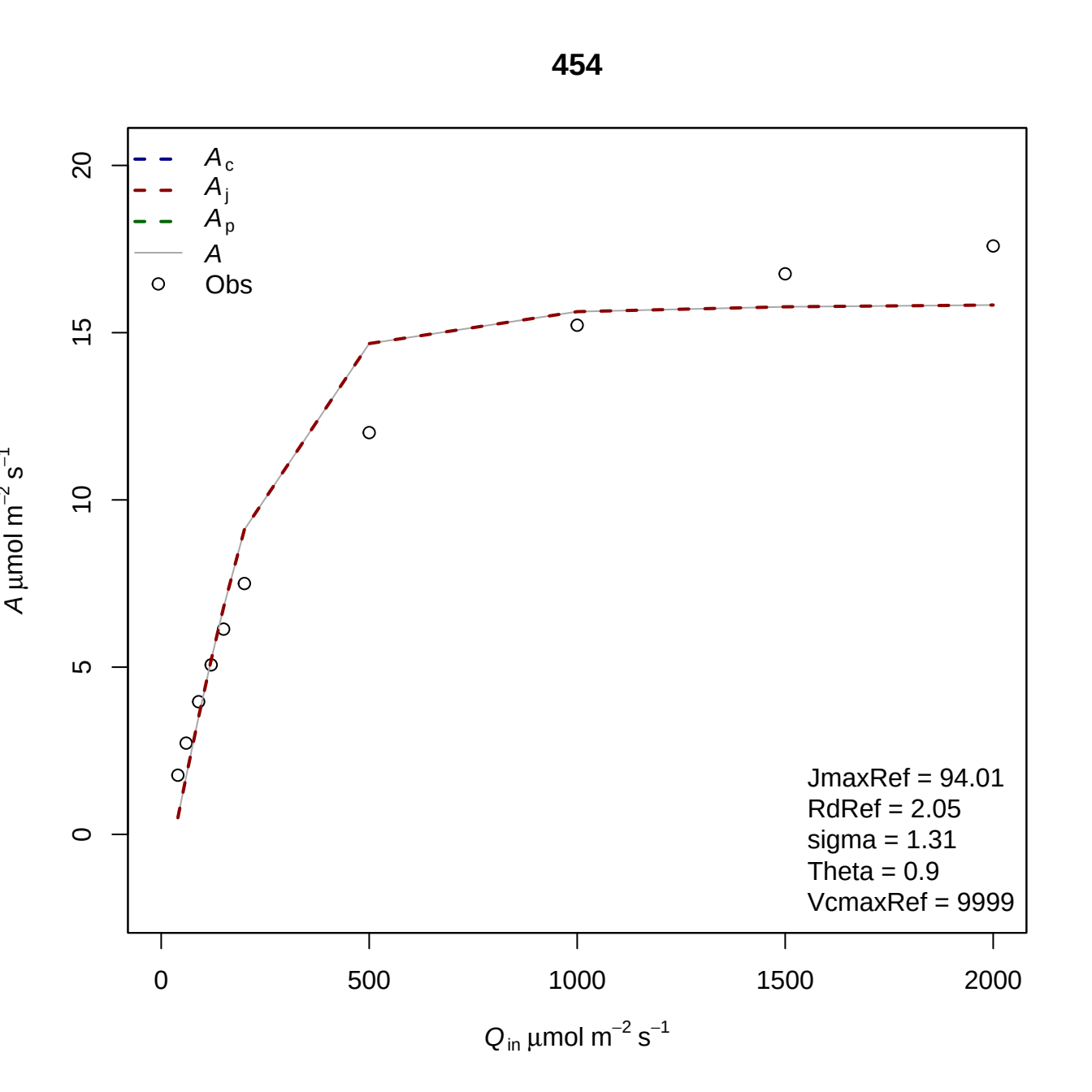

449
A_c
A_j
A_p
A
Obs
JmaxRef = 135.16
RdRef = 1.86
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

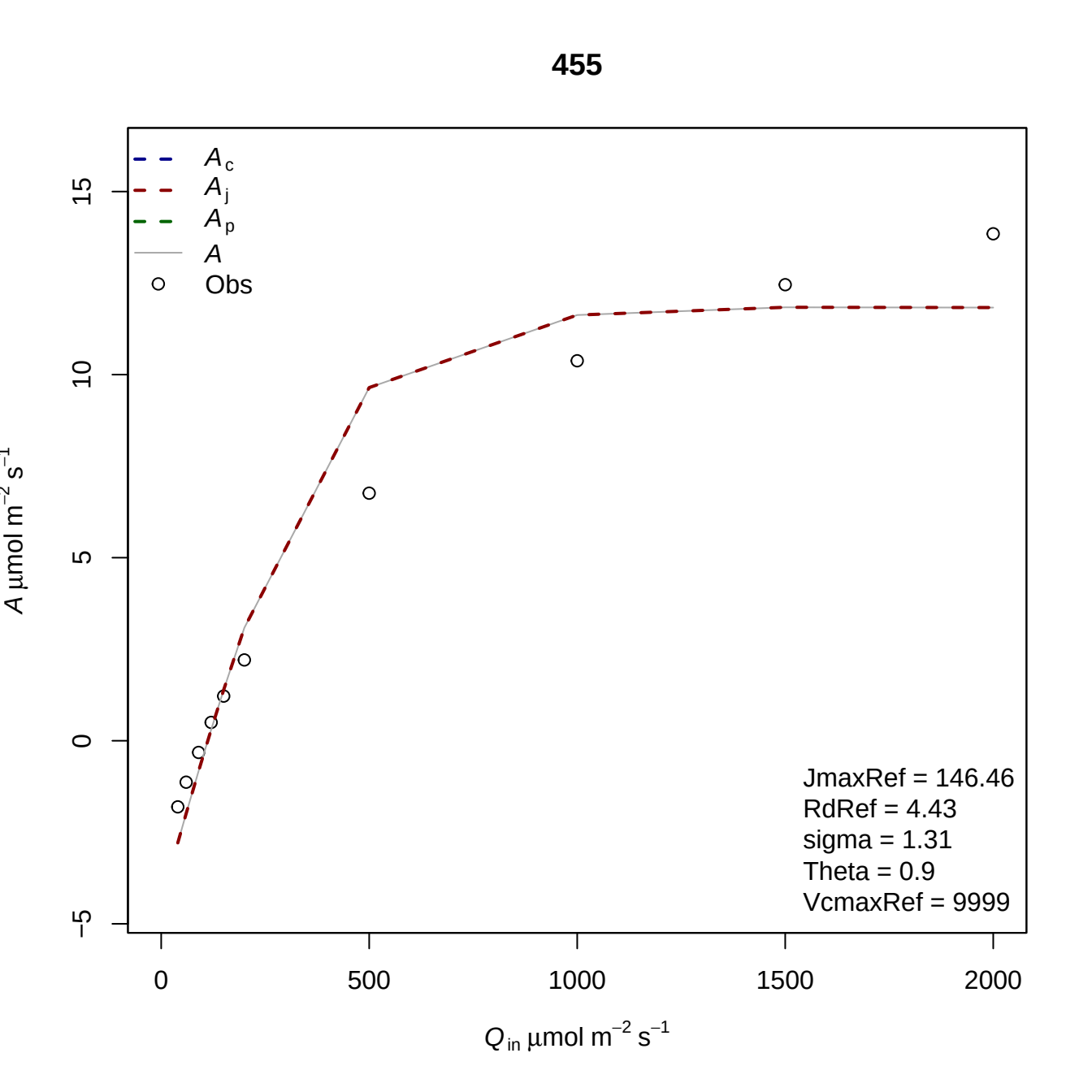

450
A_c
A_j
A_p
A
Obs
JmaxRef = 188.12
RdRef = 1.12
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
35
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

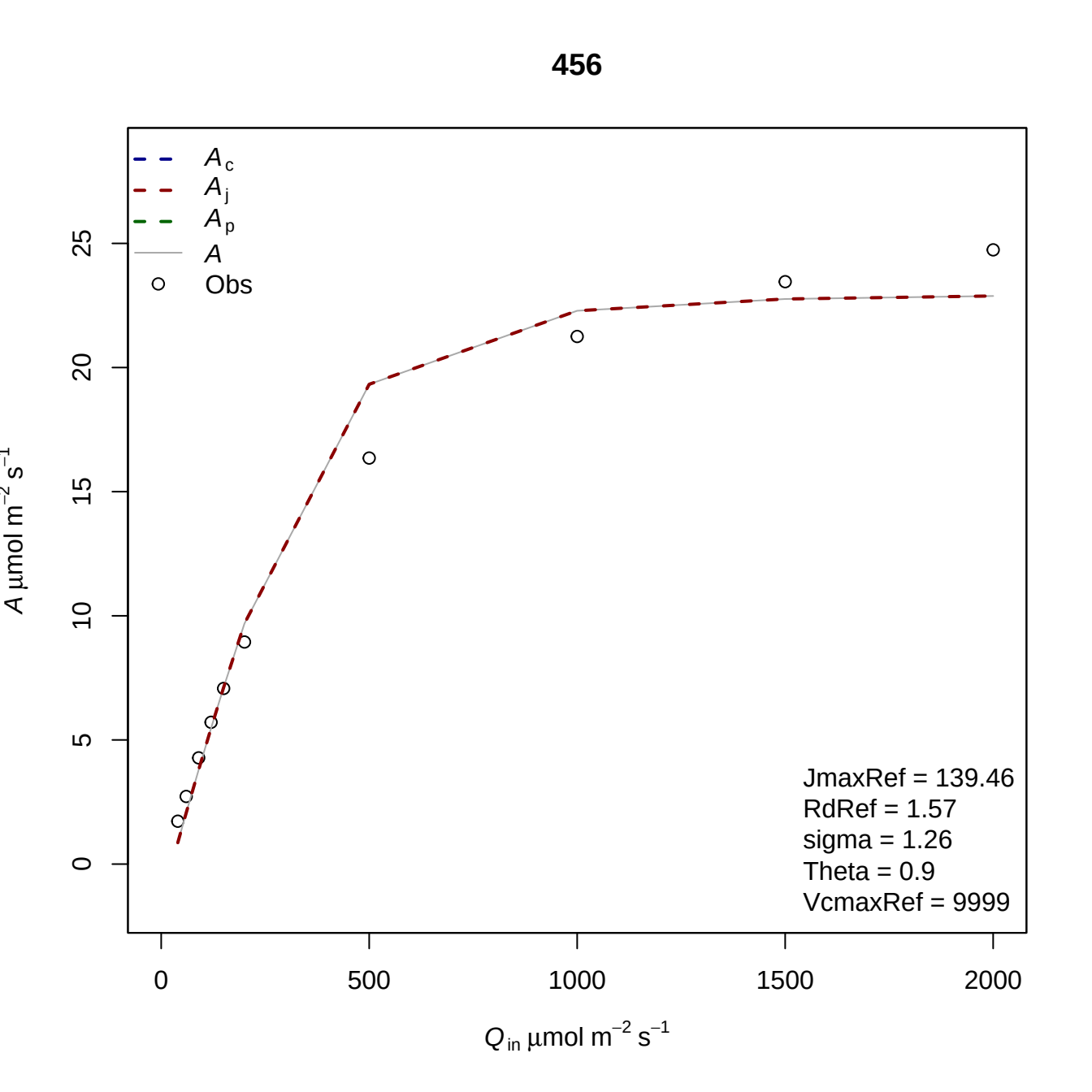

451
A_c
A_j
A_p
A
Obs
JmaxRef = 164.42
RdRef = 1.66
sigma = 1.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

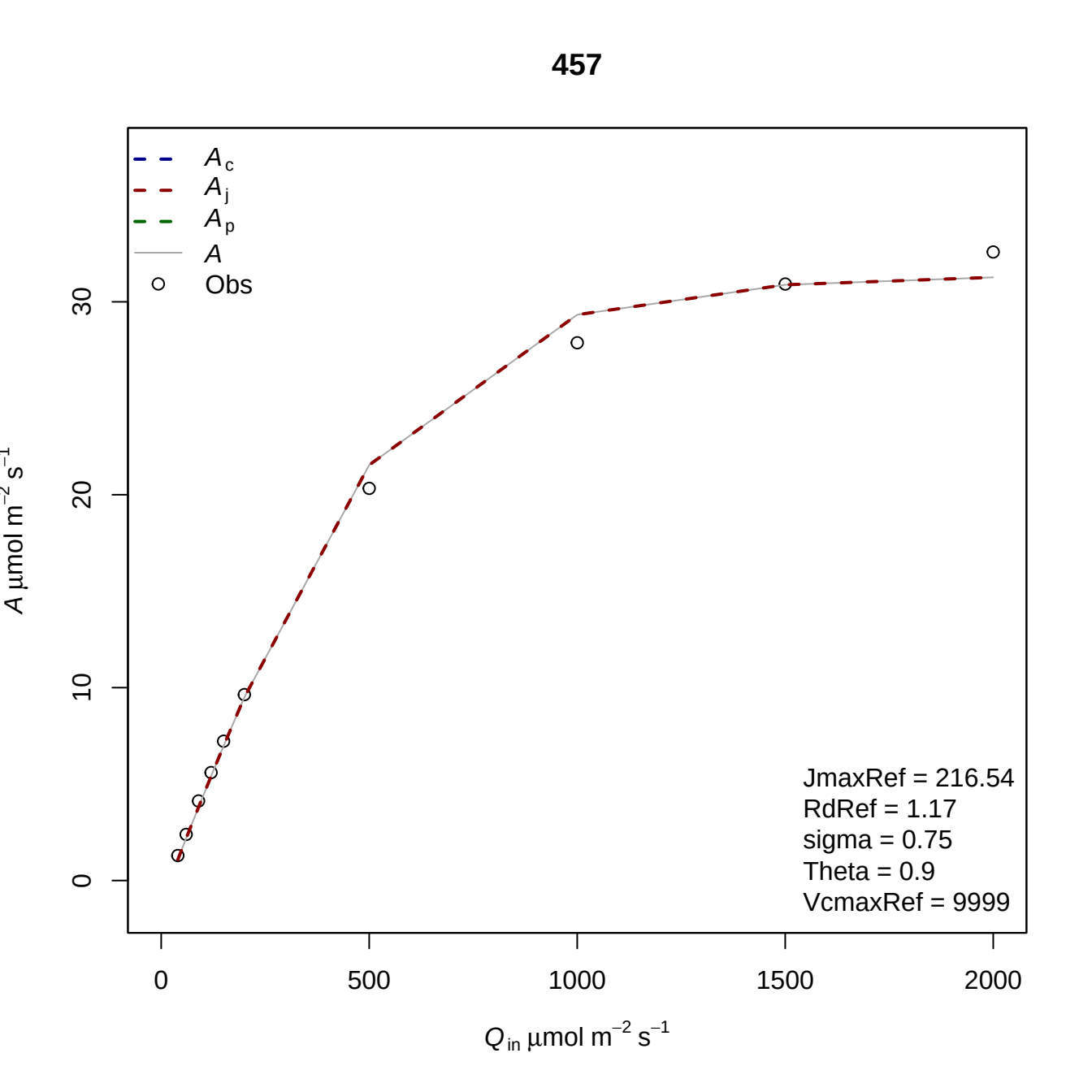

452
A_c
A_j
A_p
A
Obs
JmaxRef = 180.81
RdRef = 2.4
sigma = 1.22
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

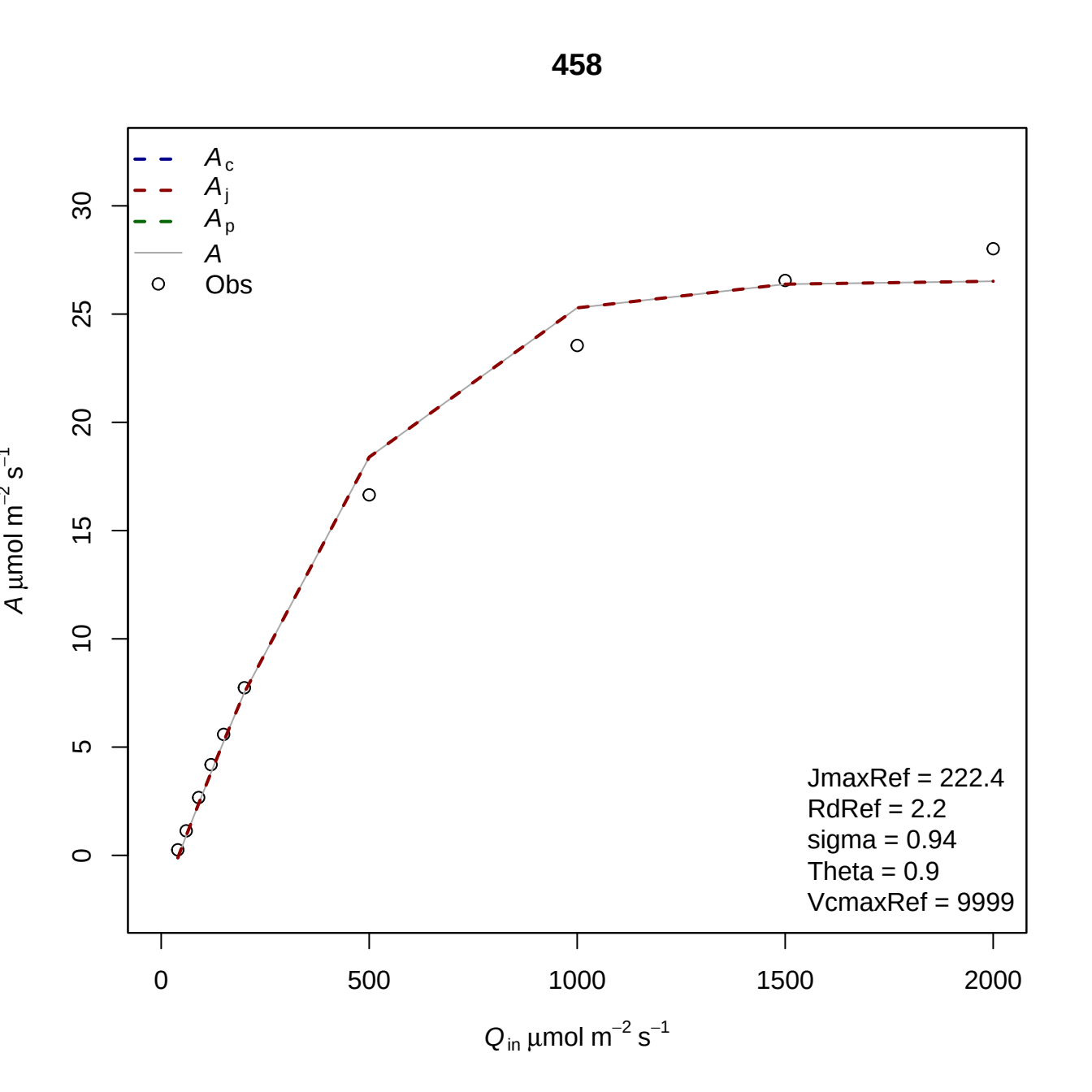

453
A_c
A_j
A_p
A
Obs
JmaxRef = 214.28
RdRef = 3.06
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

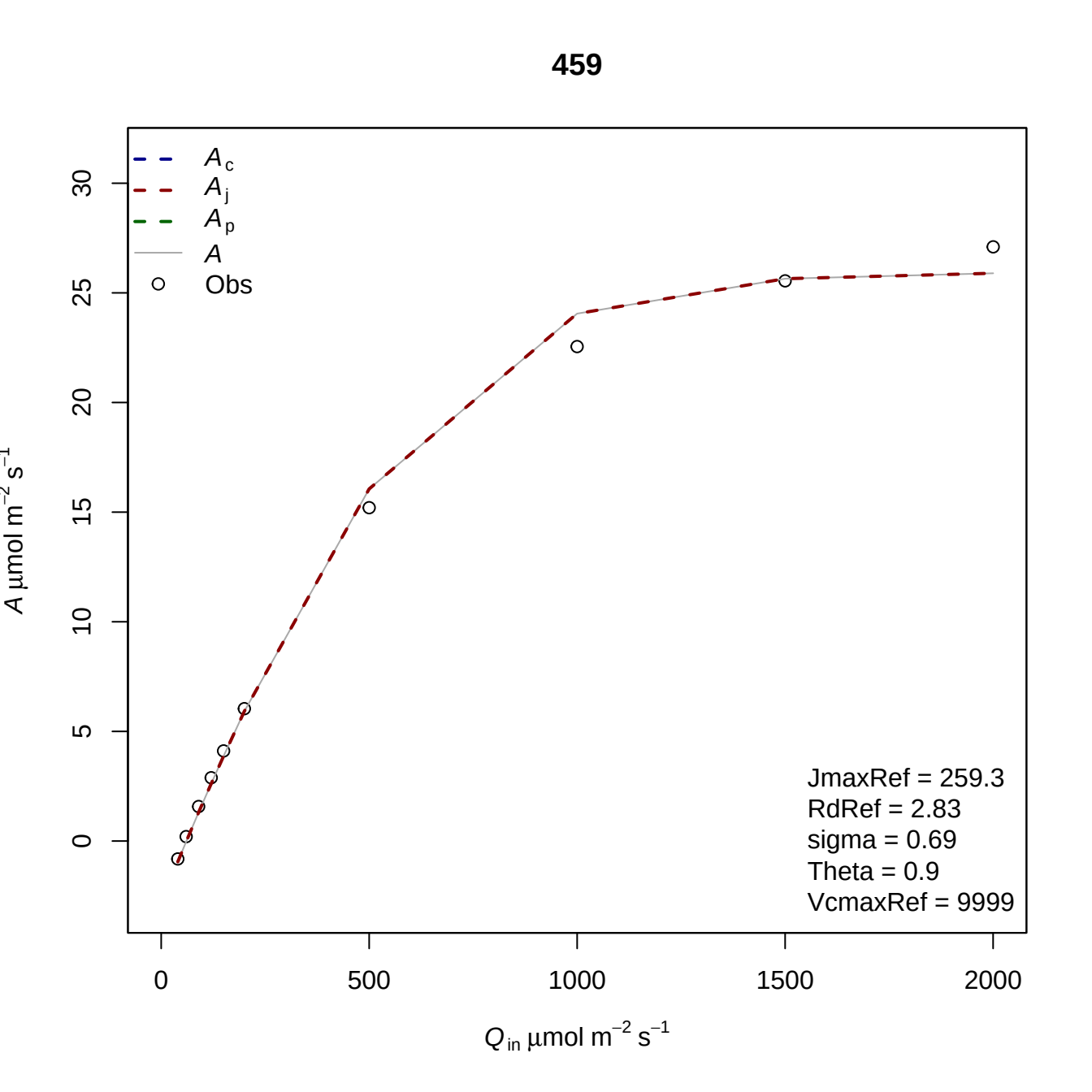

454
A_c
A_j
A_p
A
Obs
JmaxRef = 94.01
RdRef = 2.05
sigma = 1.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

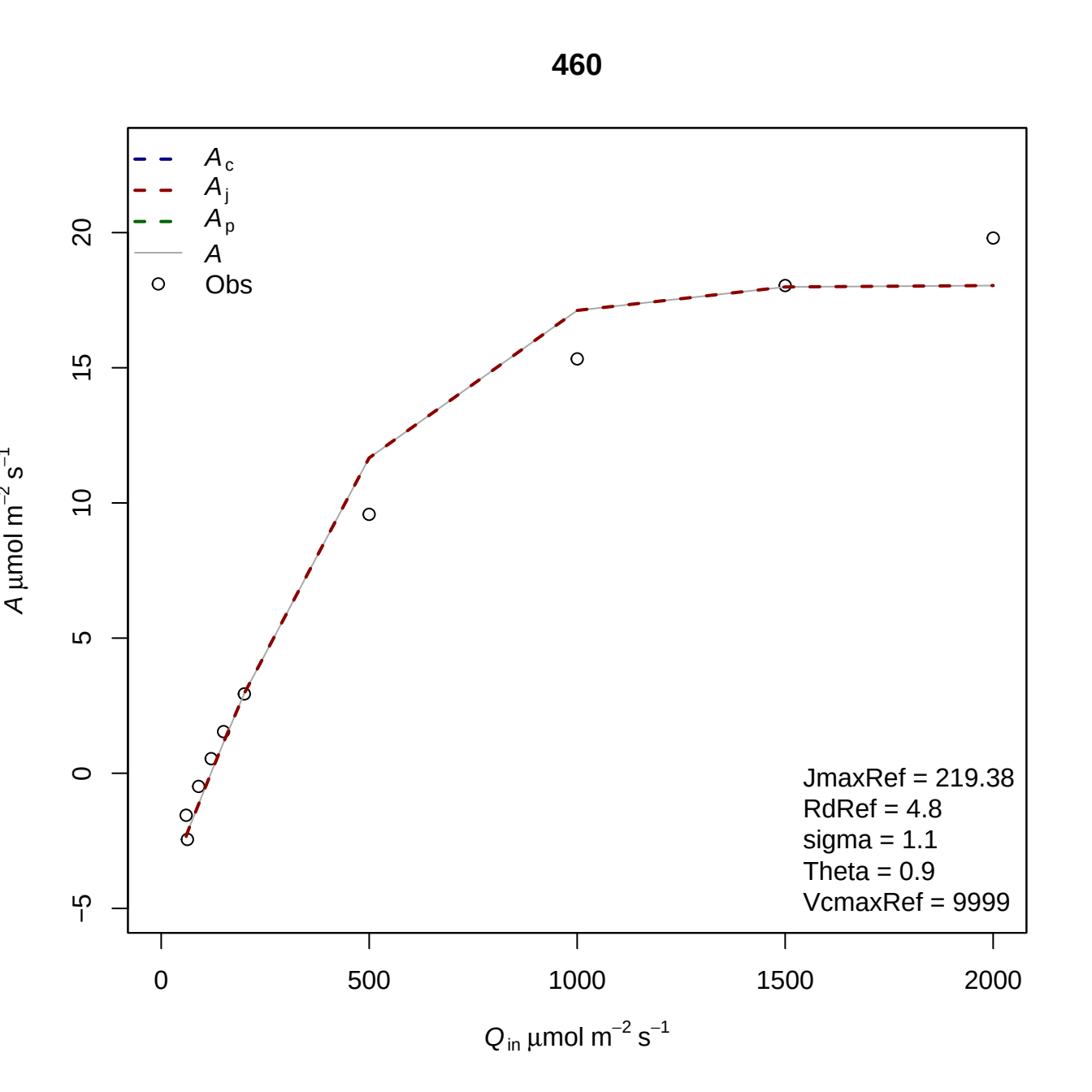

455
A_c
A_j
A_p
A
Obs
JmaxRef = 146.46
RdRef = 4.43
sigma = 1.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

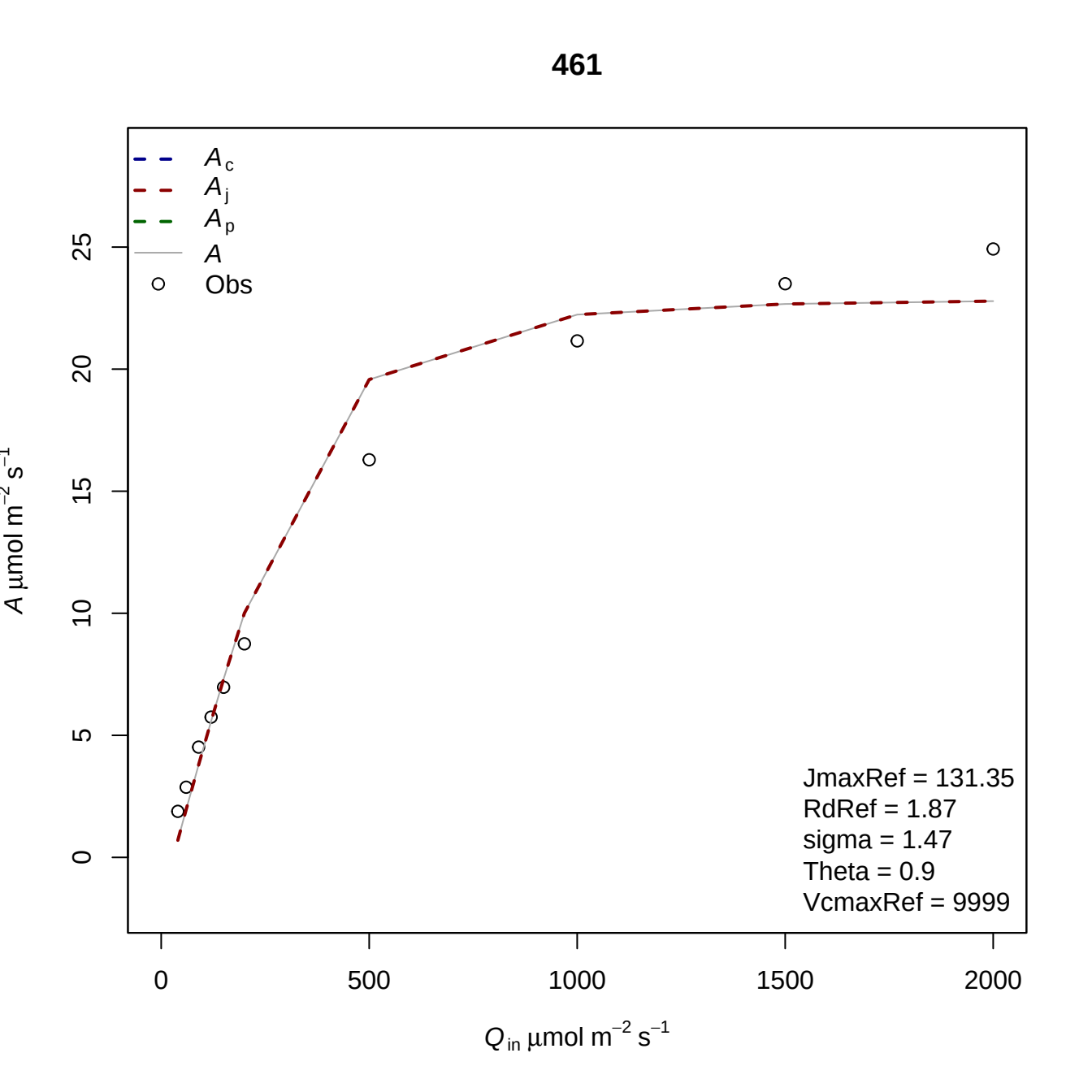

456
A_c
A_j
A_p
A
Obs
JmaxRef = 139.46
RdRef = 1.57
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

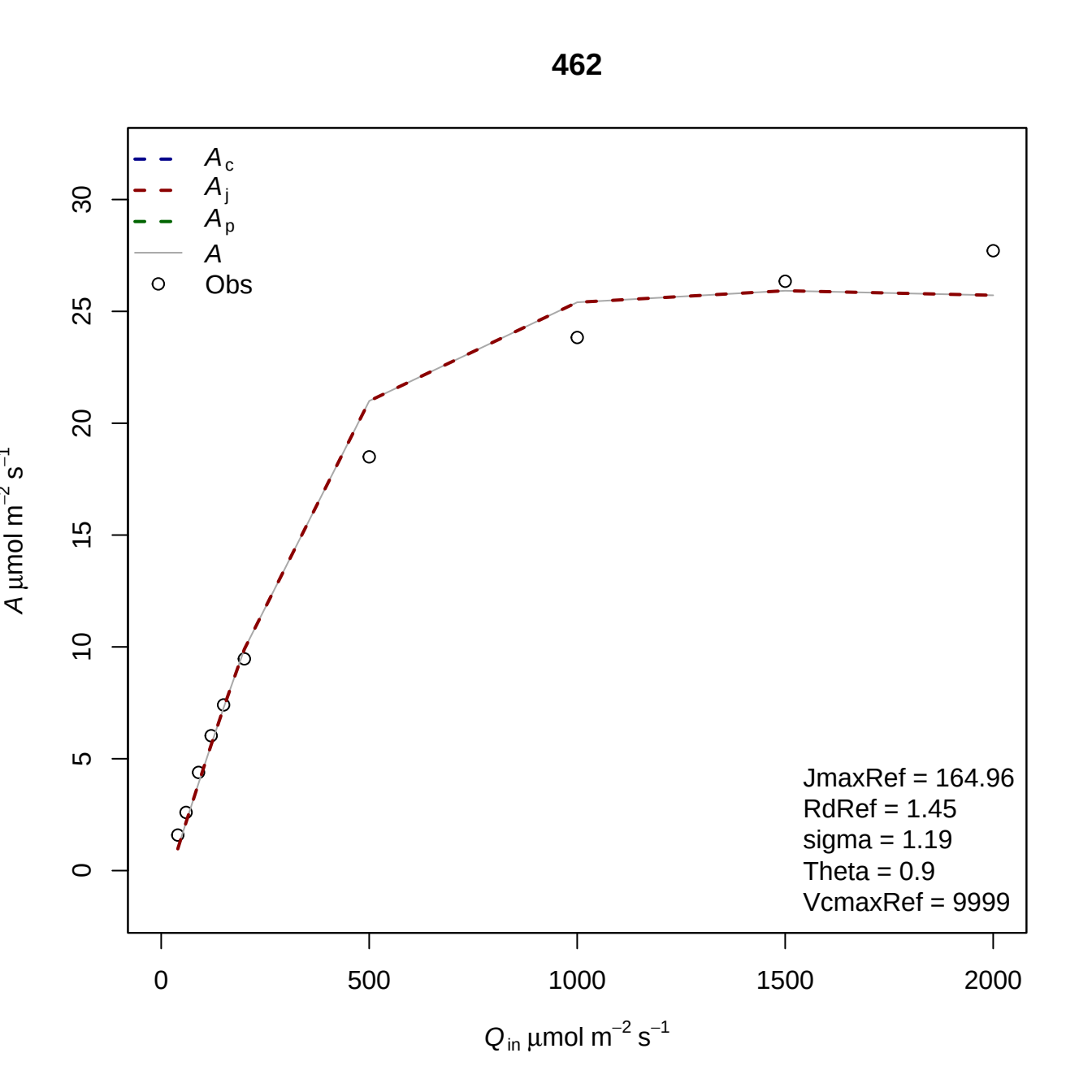

457
A_c
A_j
A_p
A
Obs
JmaxRef = 216.54
RdRef = 1.17
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

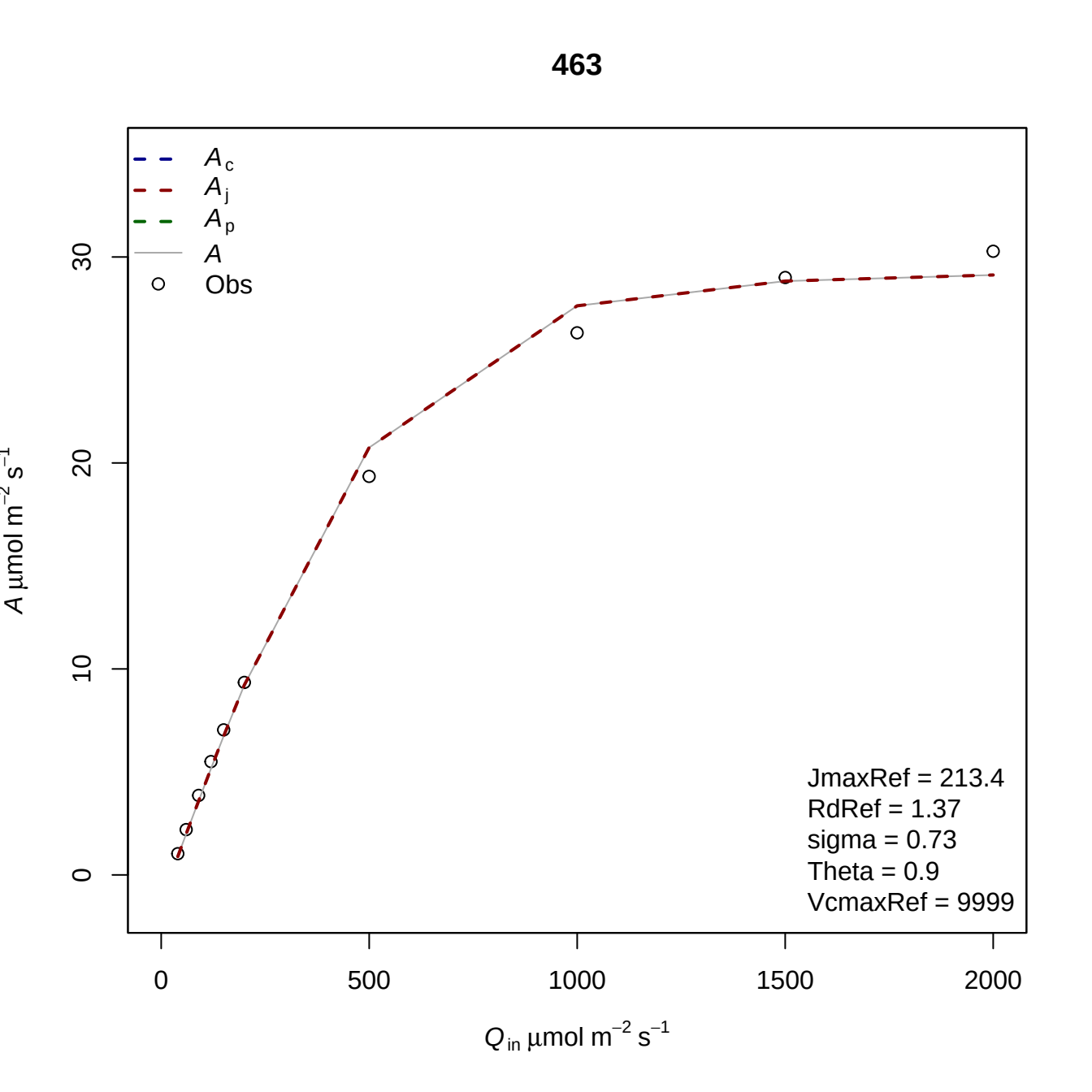

458
A_c
A_j
A_p
A
Obs
JmaxRef = 222.4
RdRef = 2.2
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

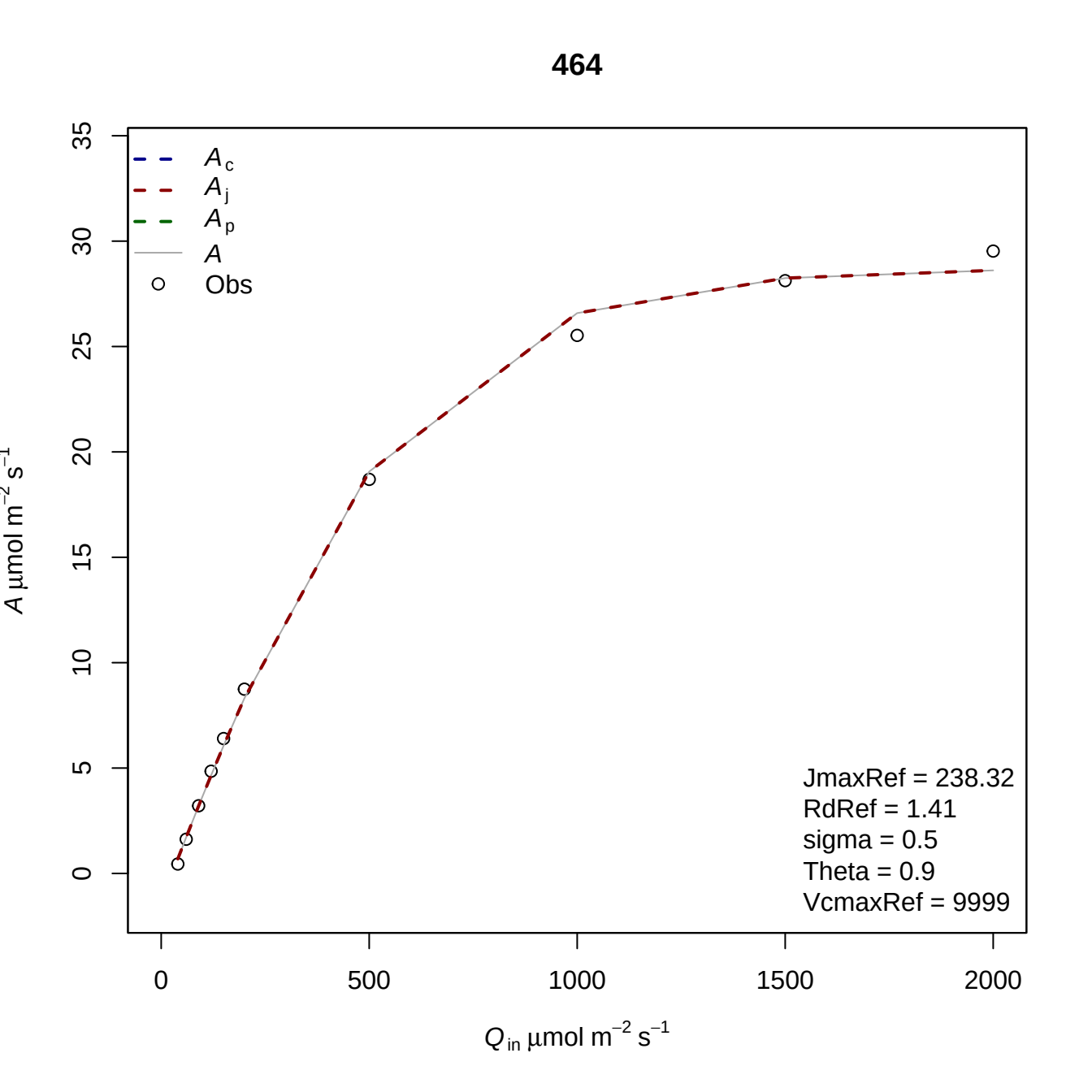

459
A_c
A_j
A_p
A
Obs
JmaxRef = 259.3
RdRef = 2.83
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

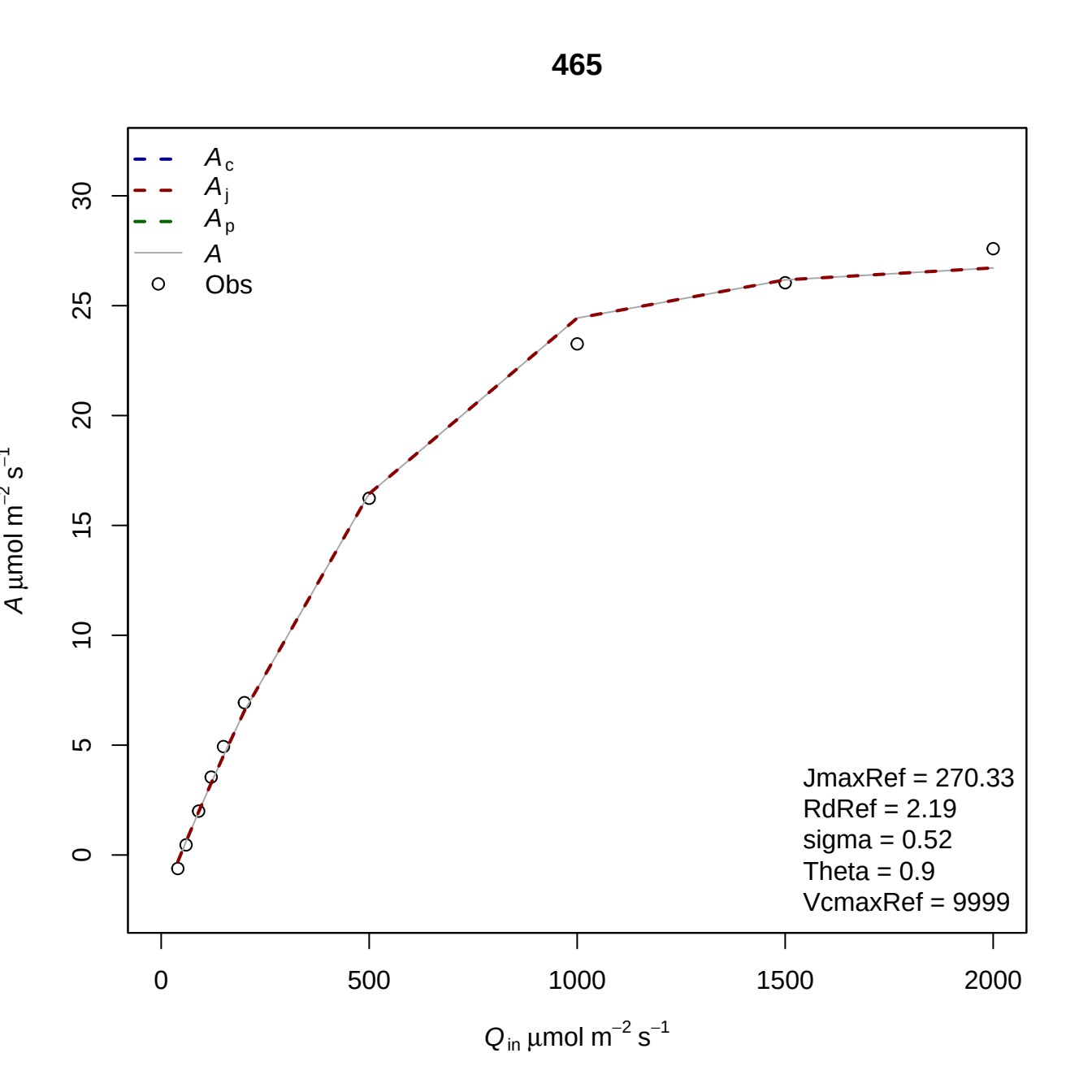

460
A_c
A_j
A_p
A
Obs
JmaxRef = 219.38
RdRef = 4.8
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

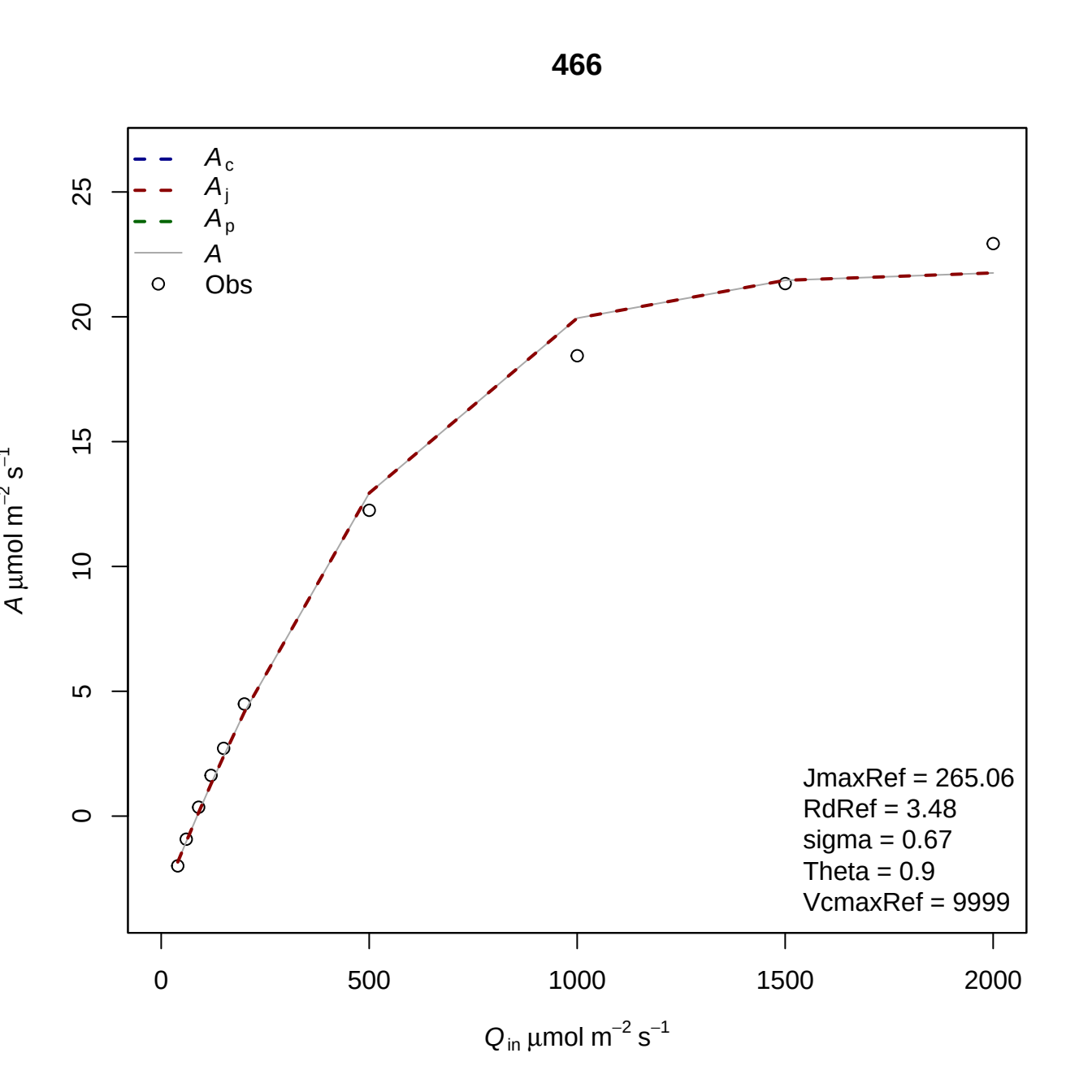

461
A_c
A_j
A_p
A
Obs
JmaxRef = 131.35
RdRef = 1.87
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

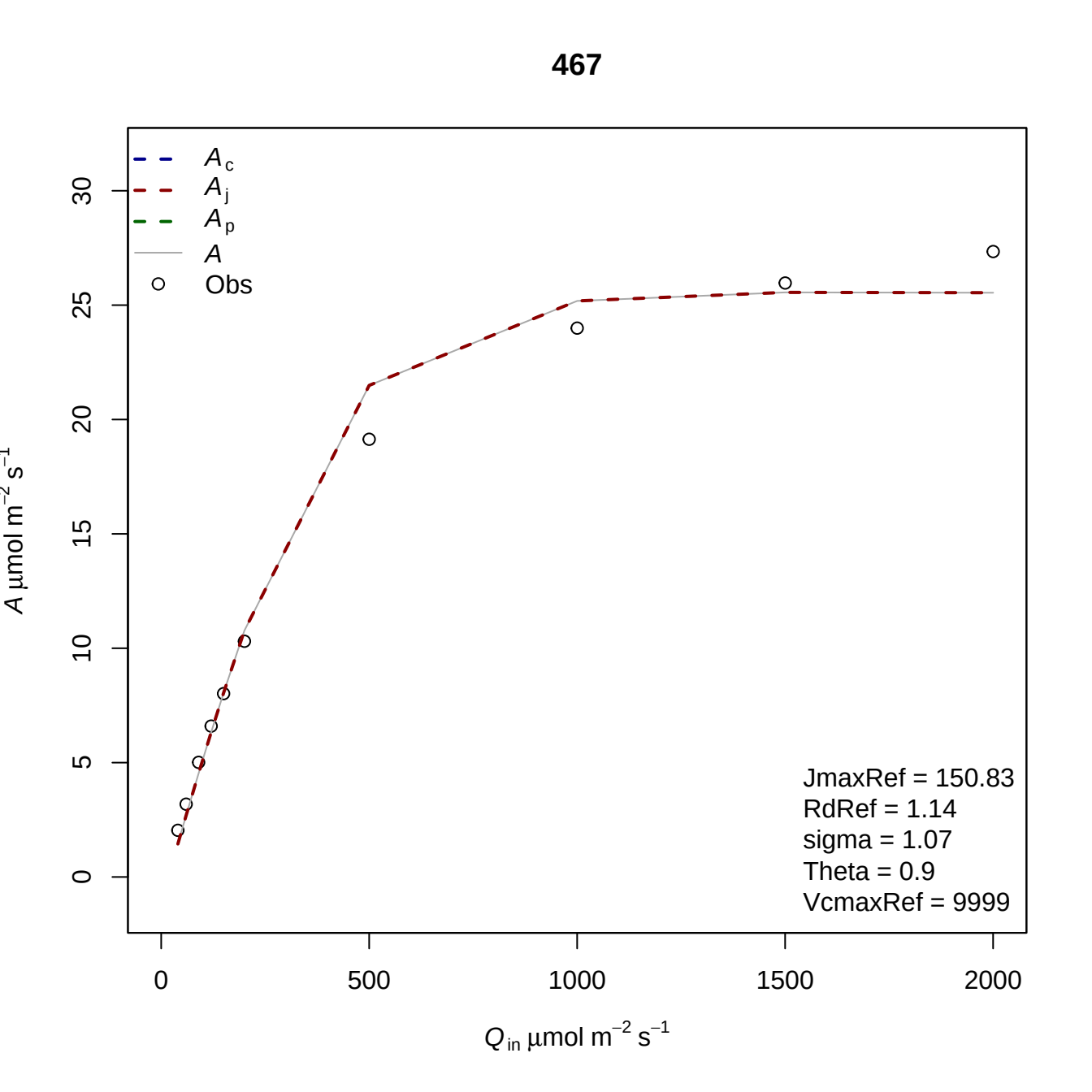

462
A_c
A_j
A_p
A
Obs
JmaxRef = 164.96
RdRef = 1.45
sigma = 1.19
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

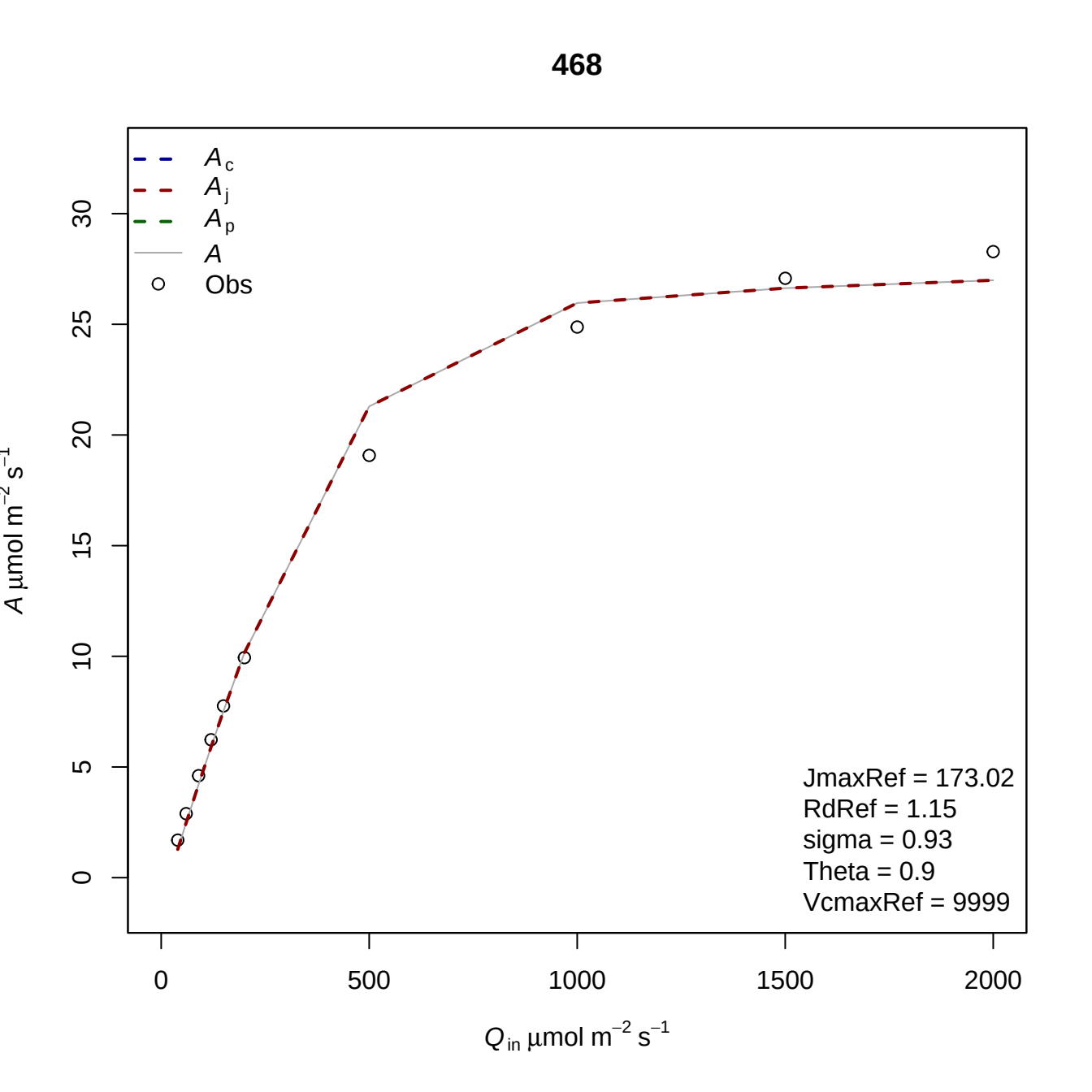

463
A_c
A_j
A_p
A
Obs
JmaxRef = 213.4
RdRef = 1.37
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

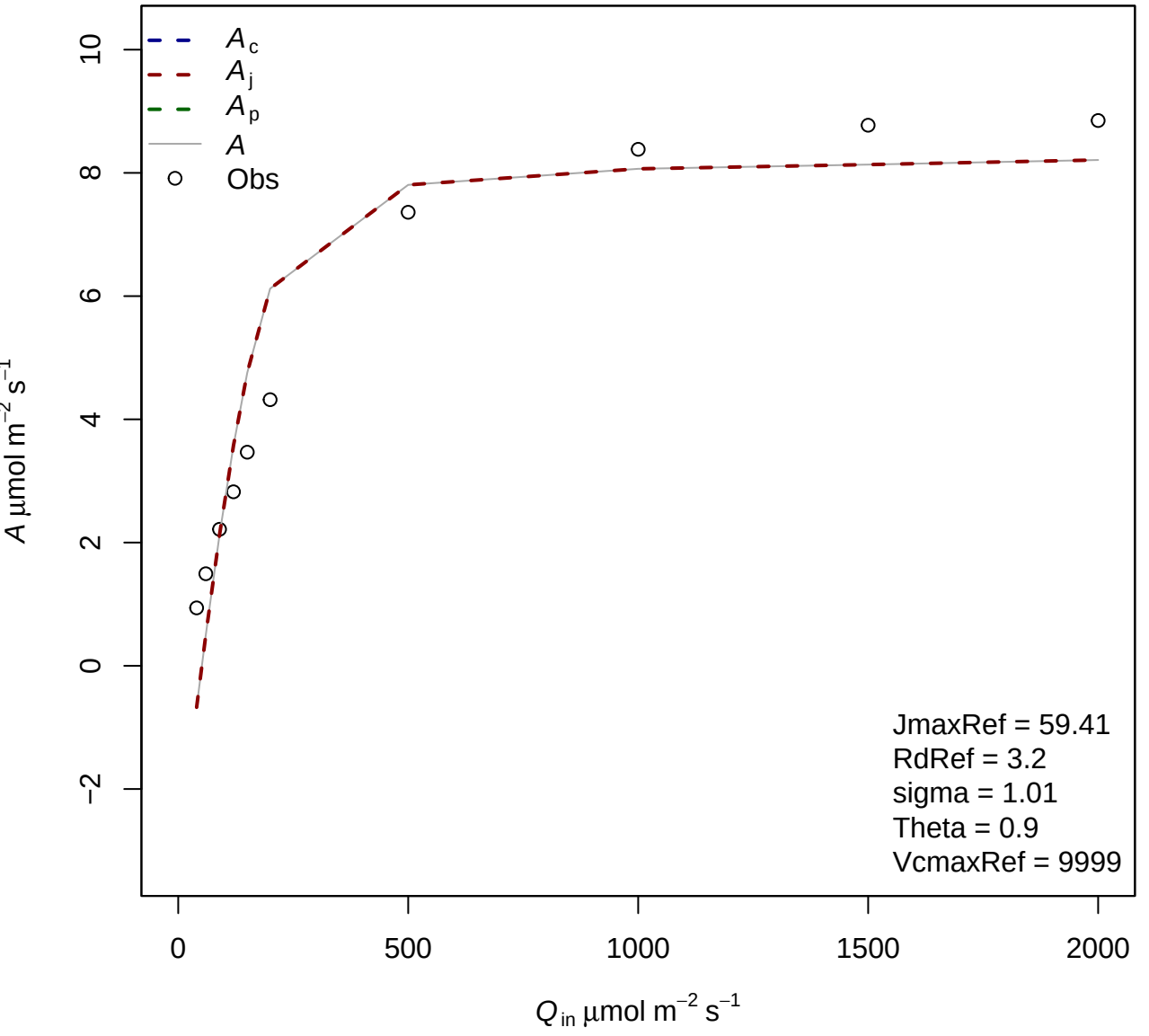

464
A_c
A_j
A_p
A
Obs
JmaxRef = 238.32
RdRef = 1.41
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

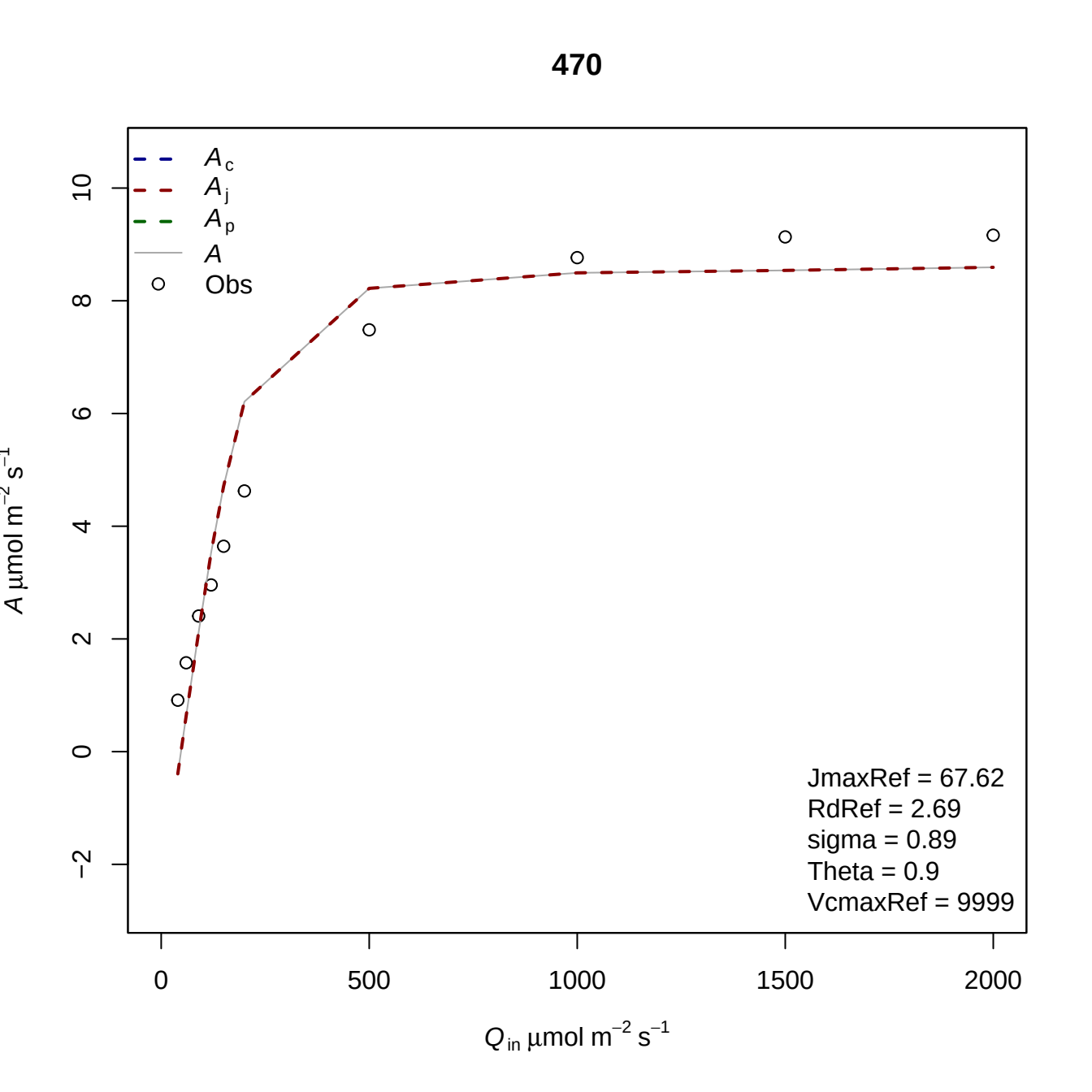

465
A_c
A_j
A_p
A
Obs
JmaxRef = 270.33
RdRef = 2.19
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

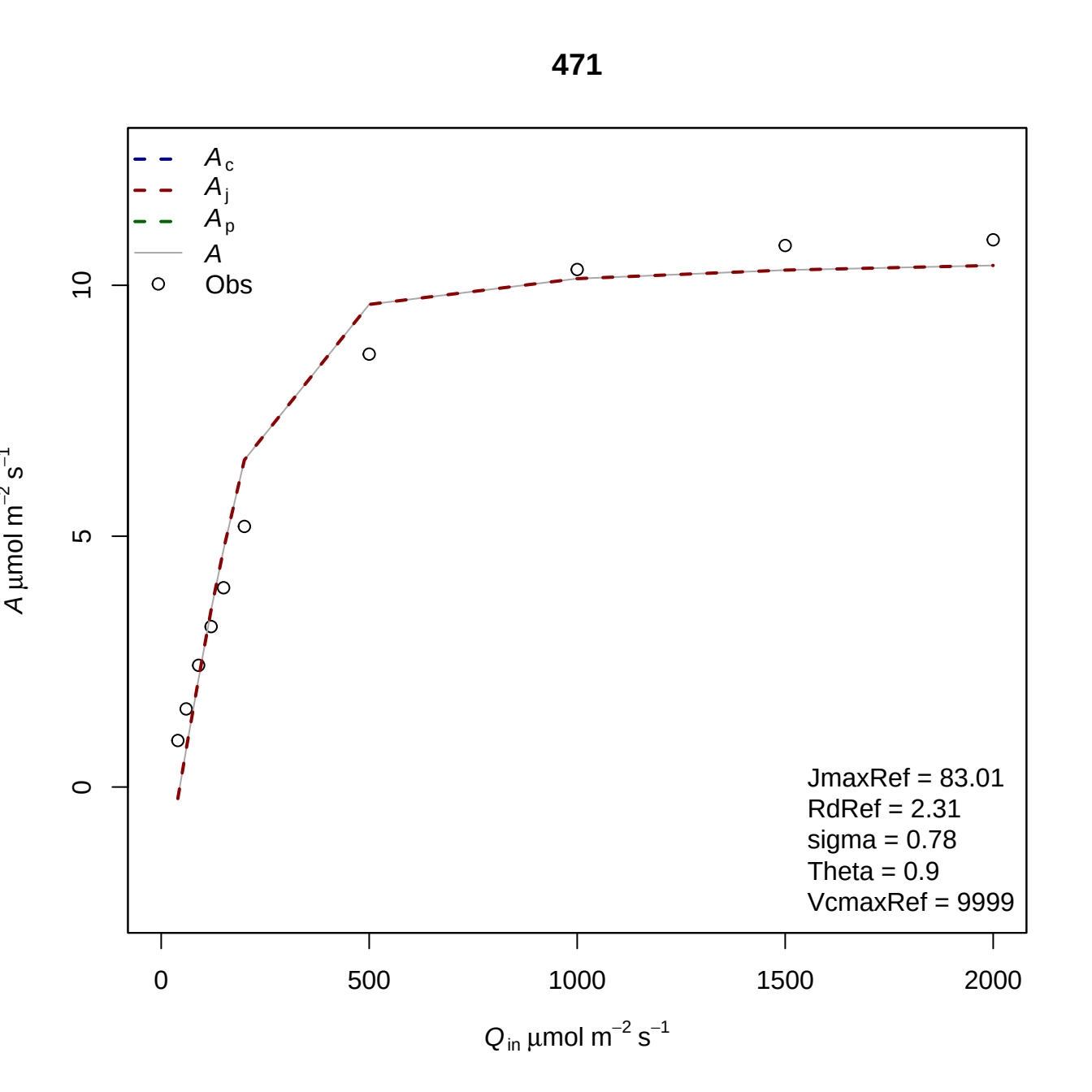

466
A_c
A_j
A_p
A
Obs
JmaxRef = 265.06
RdRef = 3.48
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

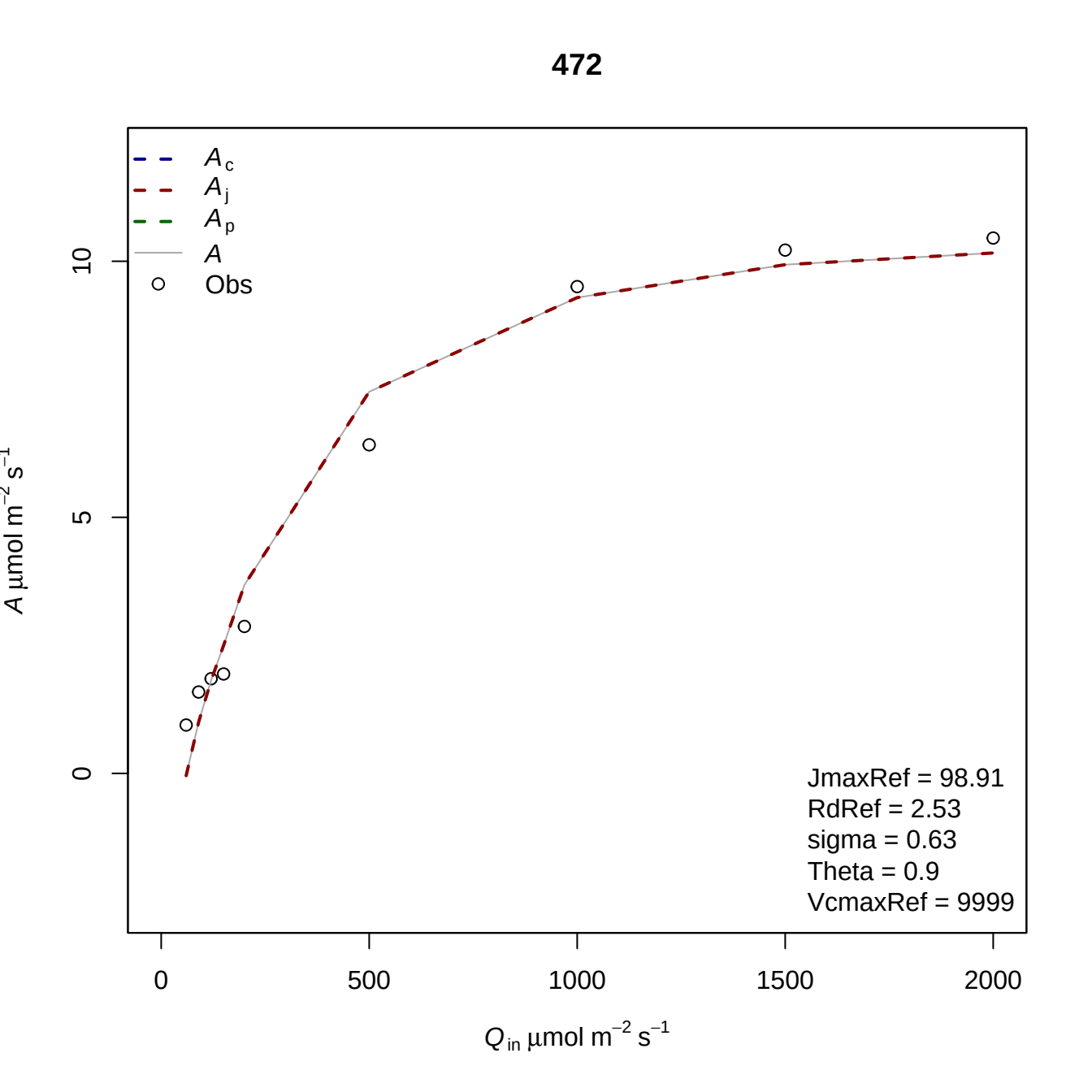

467
A_c
A_j
A_p
A
Obs
JmaxRef = 150.83
RdRef = 1.14
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

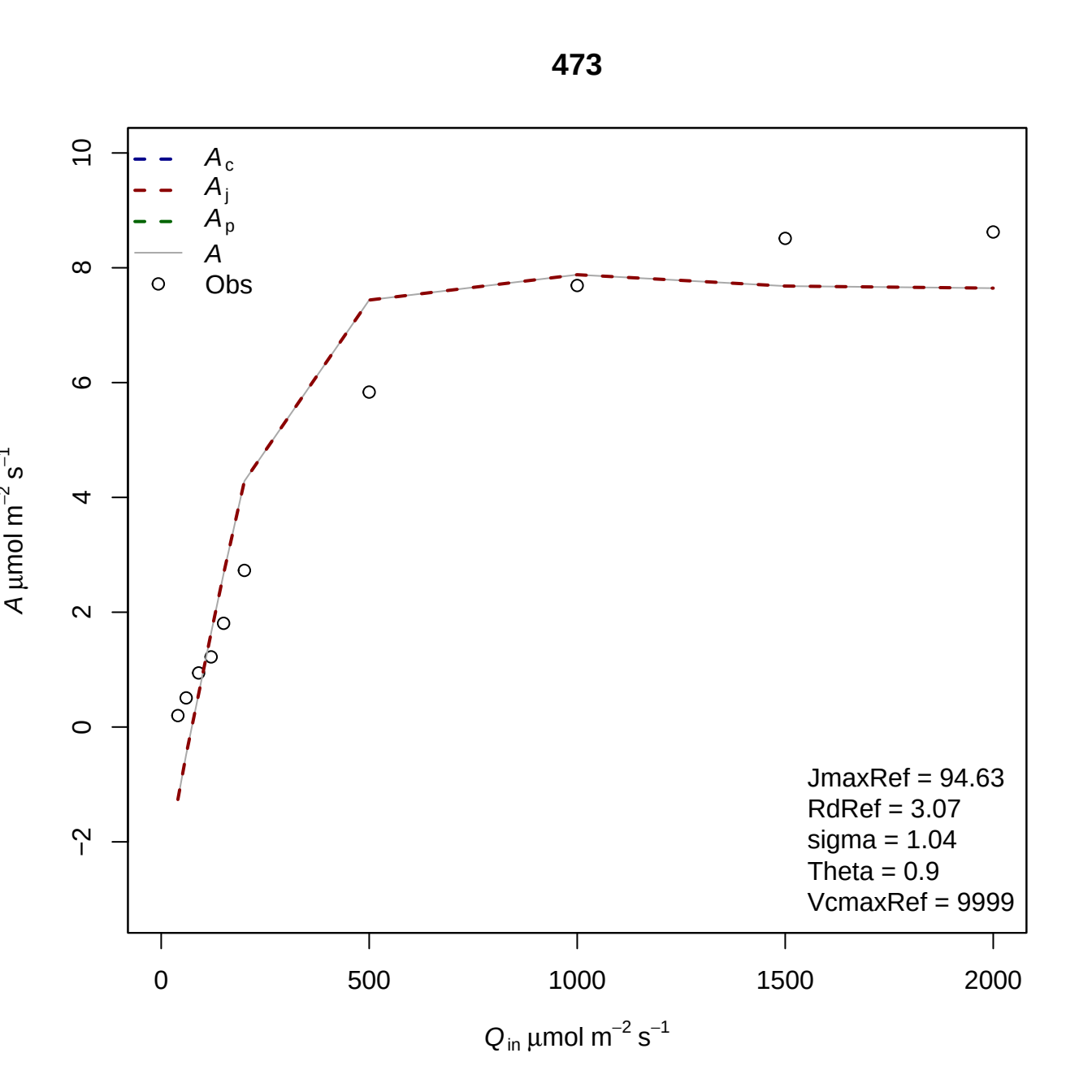

468
A_c
A_j
A_p
A
Obs
JmaxRef = 173.02
RdRef = 1.15
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

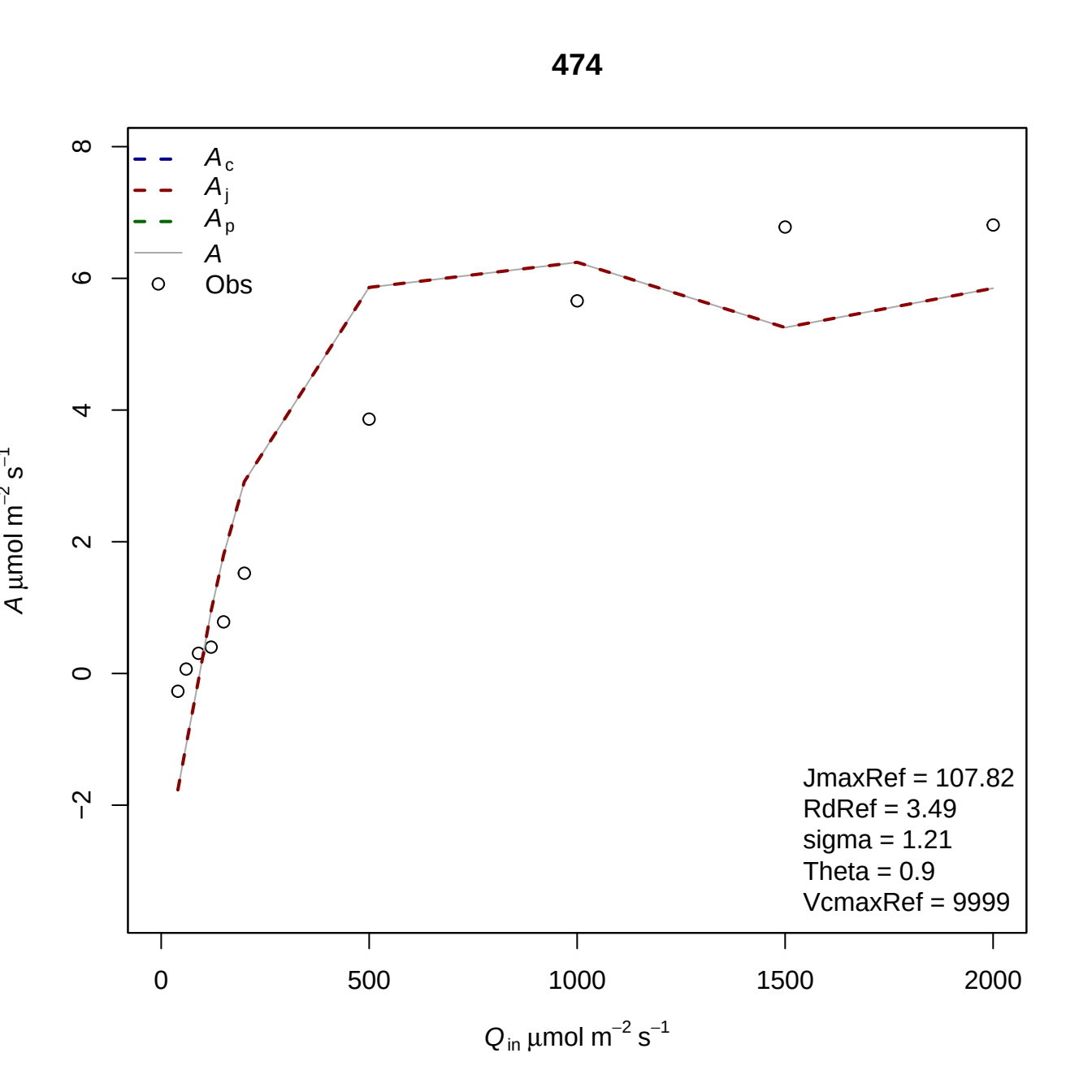

A_c
A_j
A_p
A
Obs
JmaxRef = 59.41
RdRef = 3.2
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

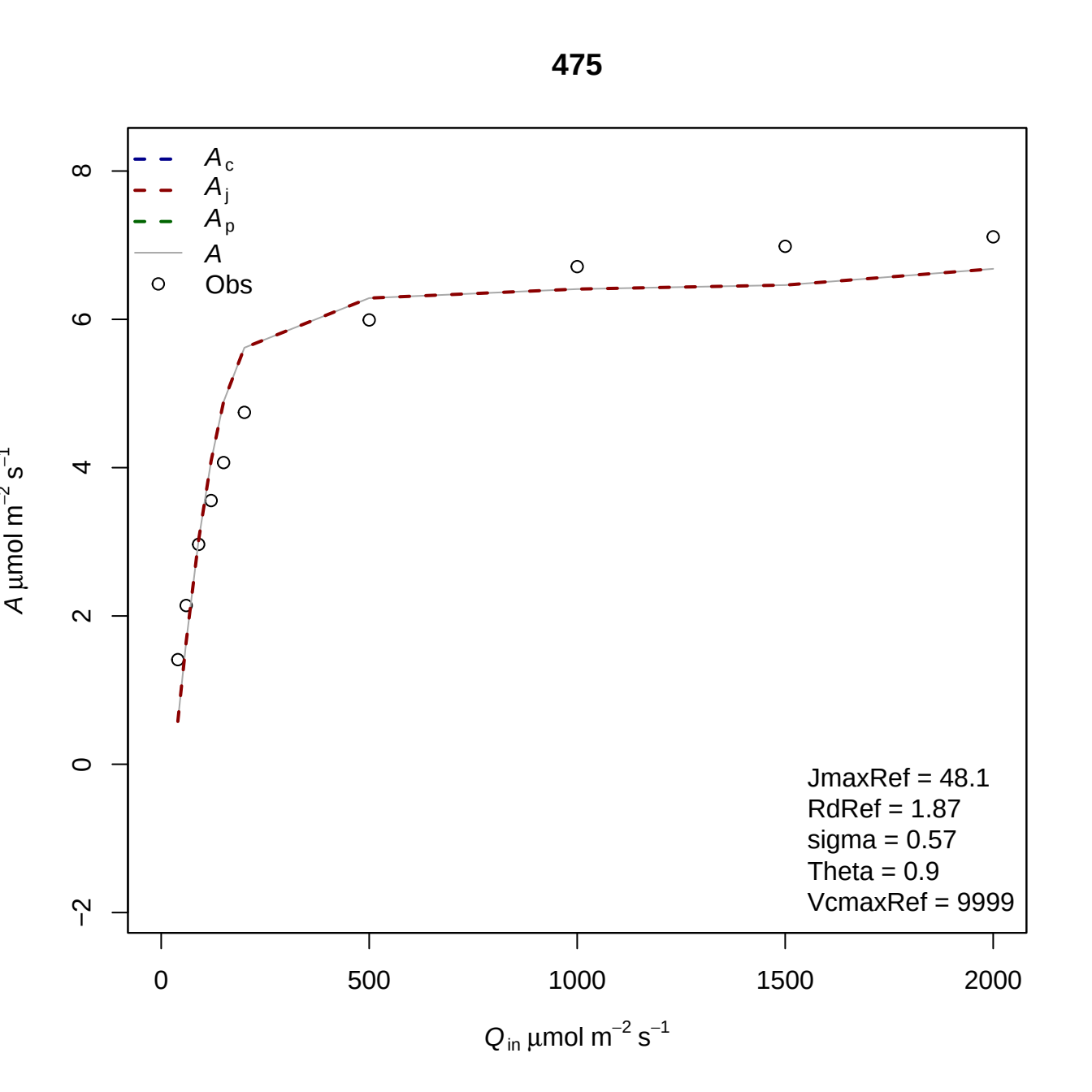

470
A_c
A_j
A_p
A
Obs
JmaxRef = 67.62
RdRef = 2.69
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

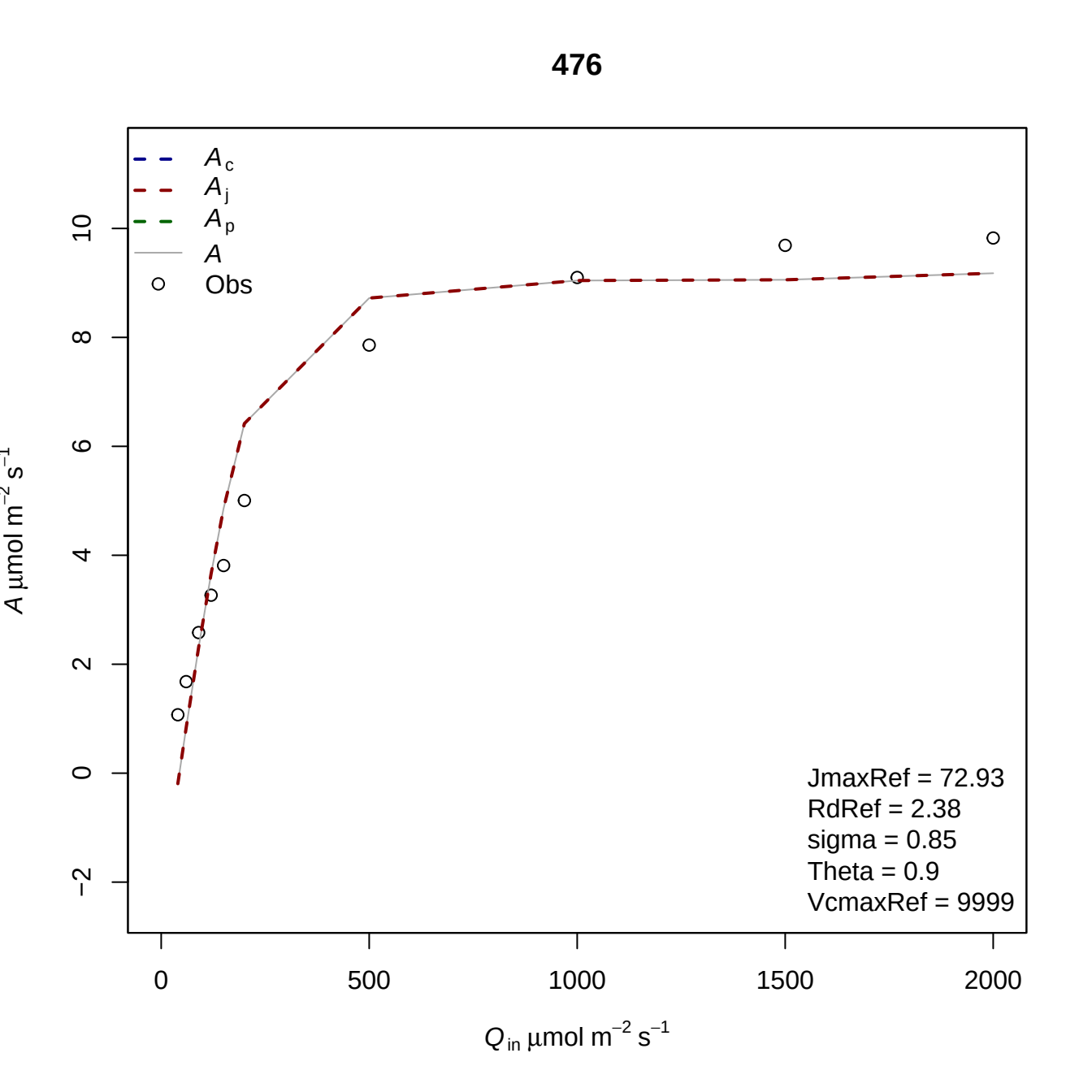

471
A_c
A_j
A_p
A
Obs
JmaxRef = 83.01
RdRef = 2.31
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

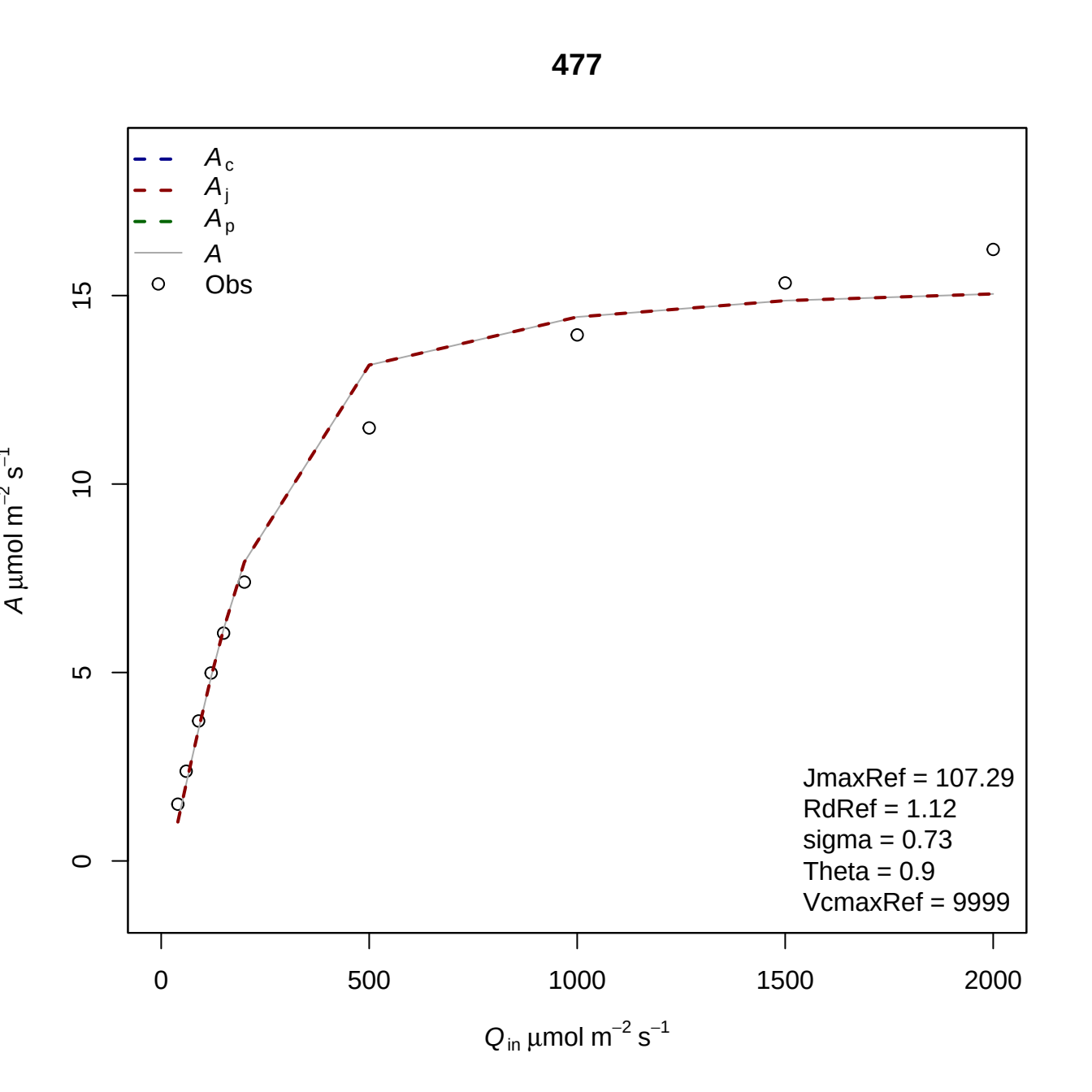

472
A_c
A_j
A_p
A
Obs
JmaxRef = 98.91
RdRef = 2.53
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

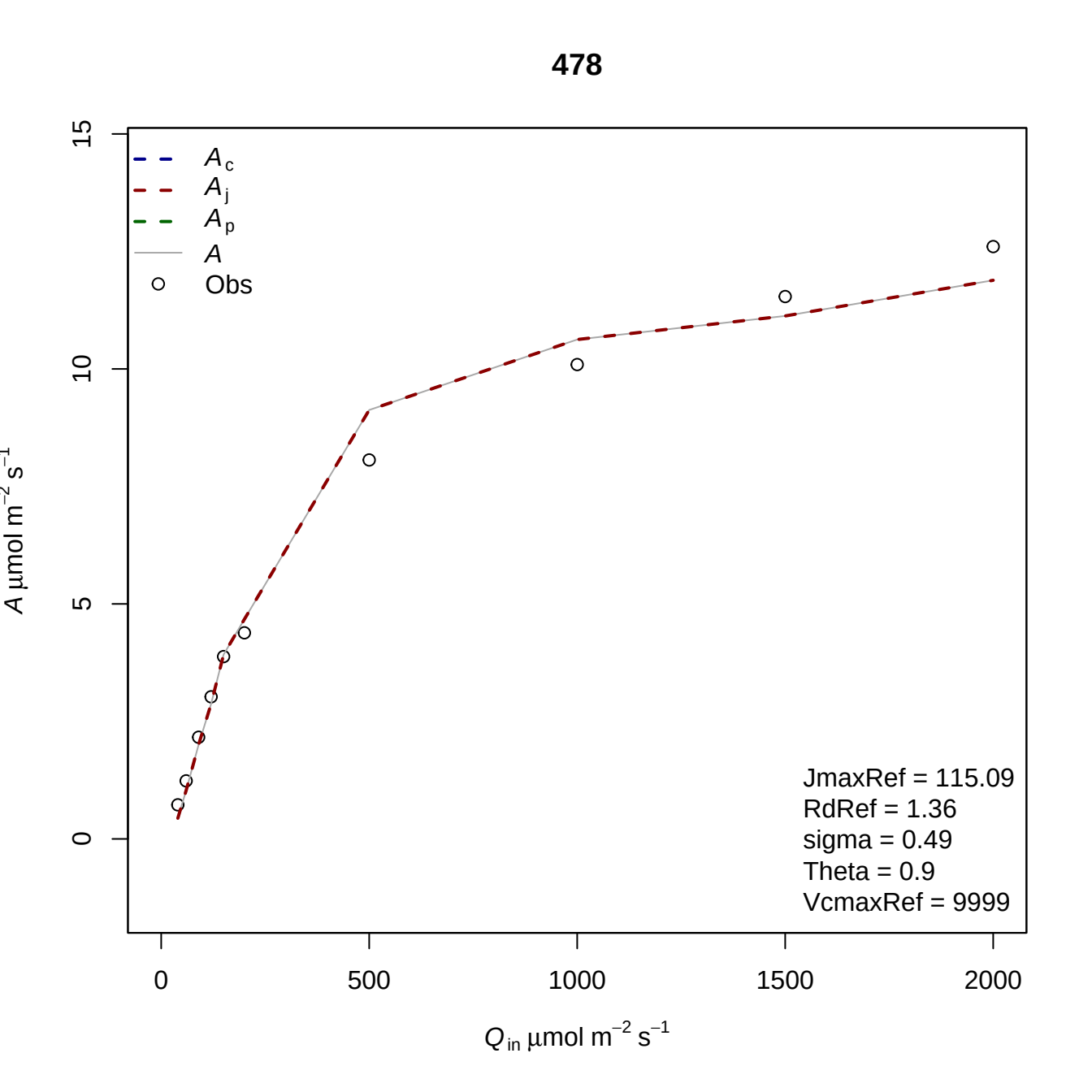

473
A_c
A_j
A_p
A
Obs
JmaxRef = 94.63
RdRef = 3.07
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

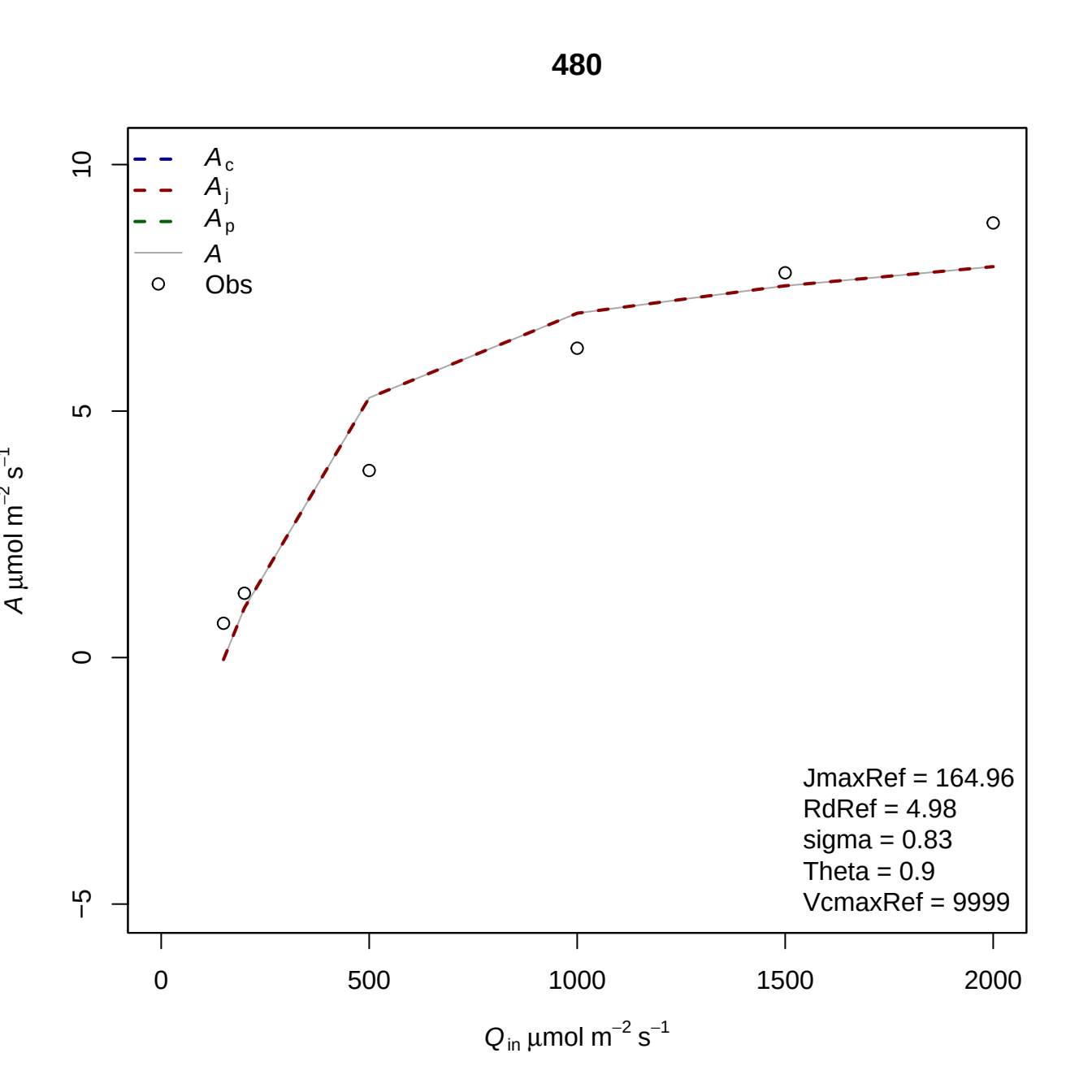

474
A_c
A_j
A_p
A
Obs
JmaxRef = 107.82
RdRef = 3.49
sigma = 1.21
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

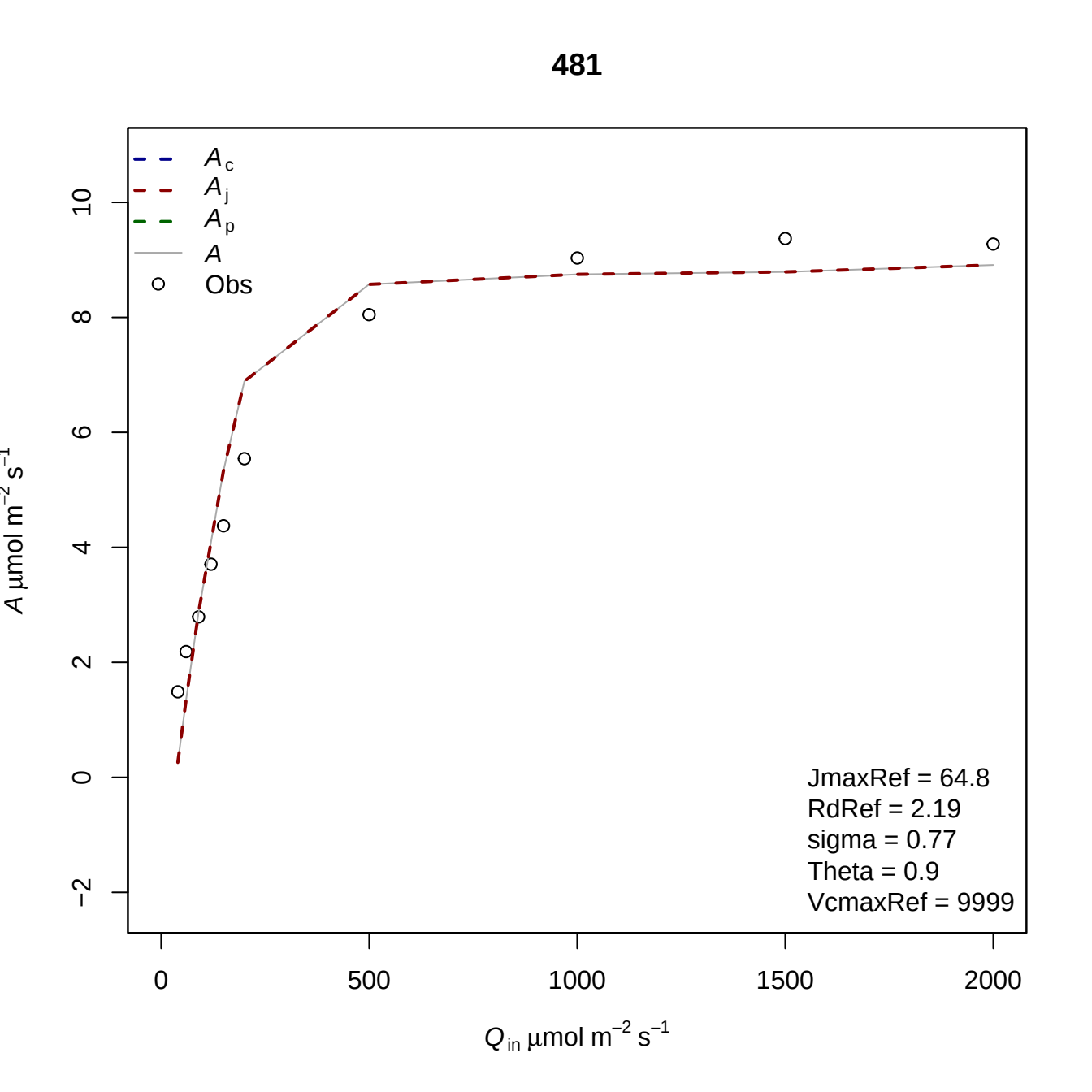

475
A_c
A_j
A_p
A
Obs
JmaxRef = 48.1
RdRef = 1.87
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

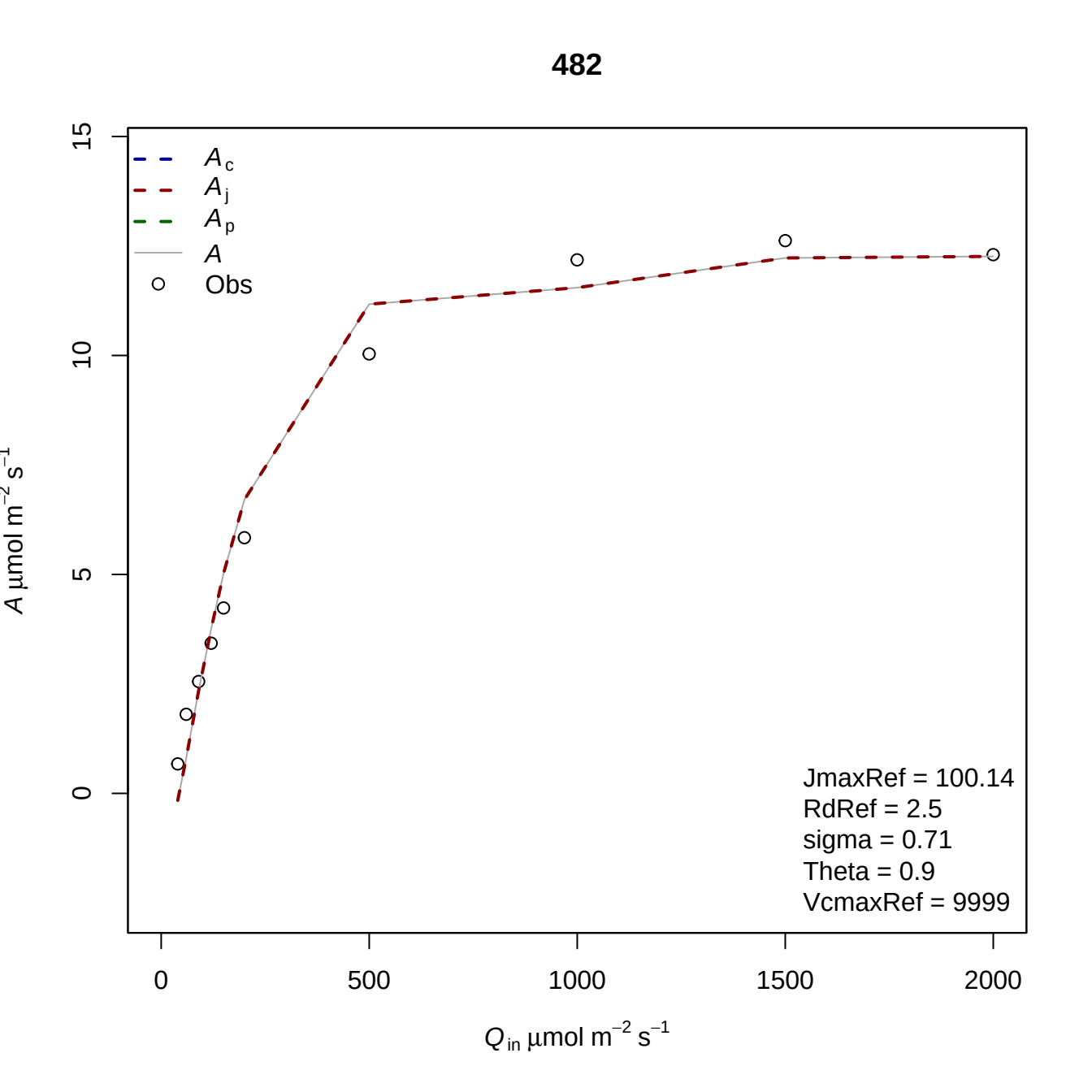

476
A_c
A_j
A_p
A
Obs
JmaxRef = 72.93
RdRef = 2.38
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

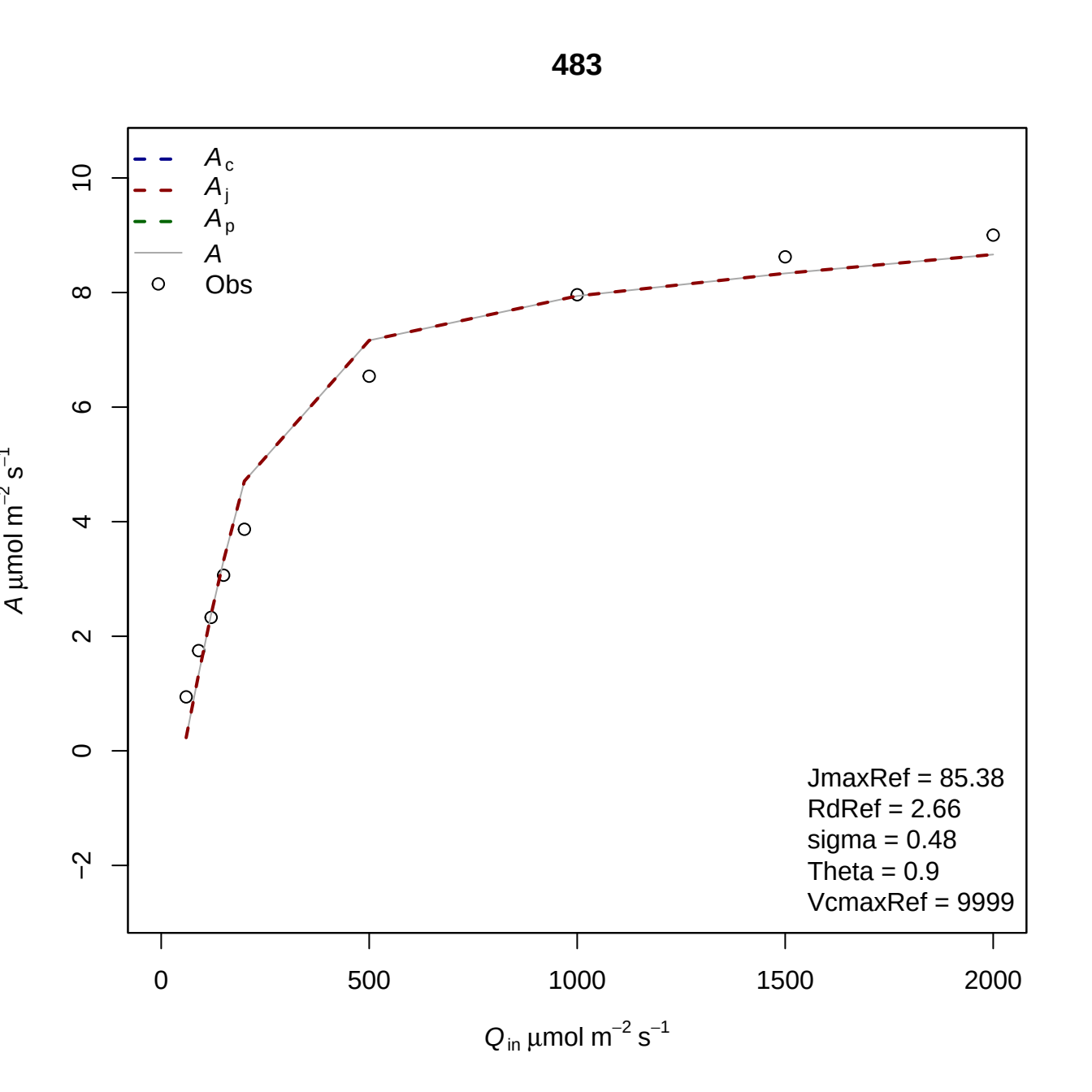

477
A_c
A_j
A_p
A
Obs
JmaxRef = 107.29
RdRef = 1.12
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

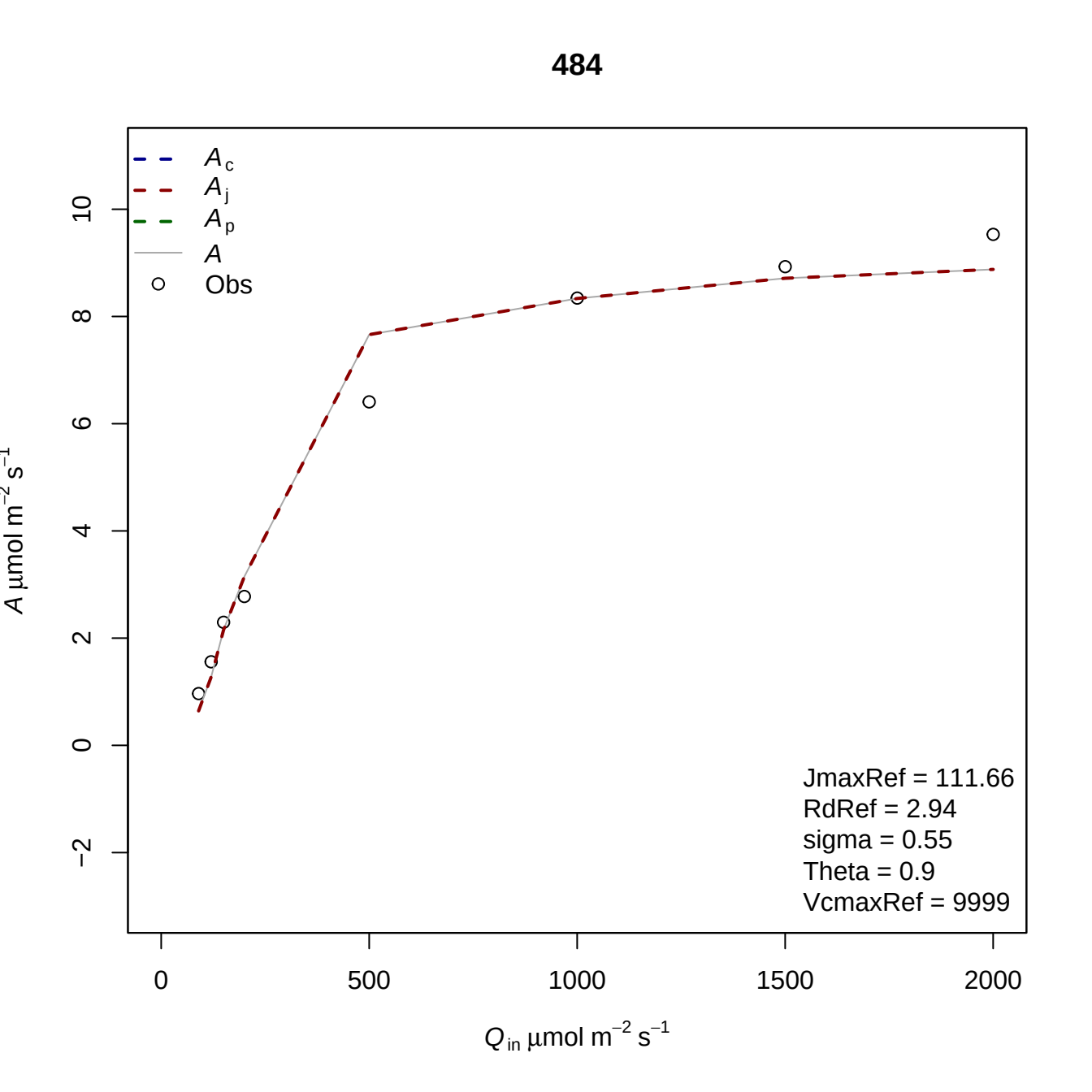

478
A_c
A_j
A_p
A
Obs
JmaxRef = 115.09
RdRef = 1.36
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

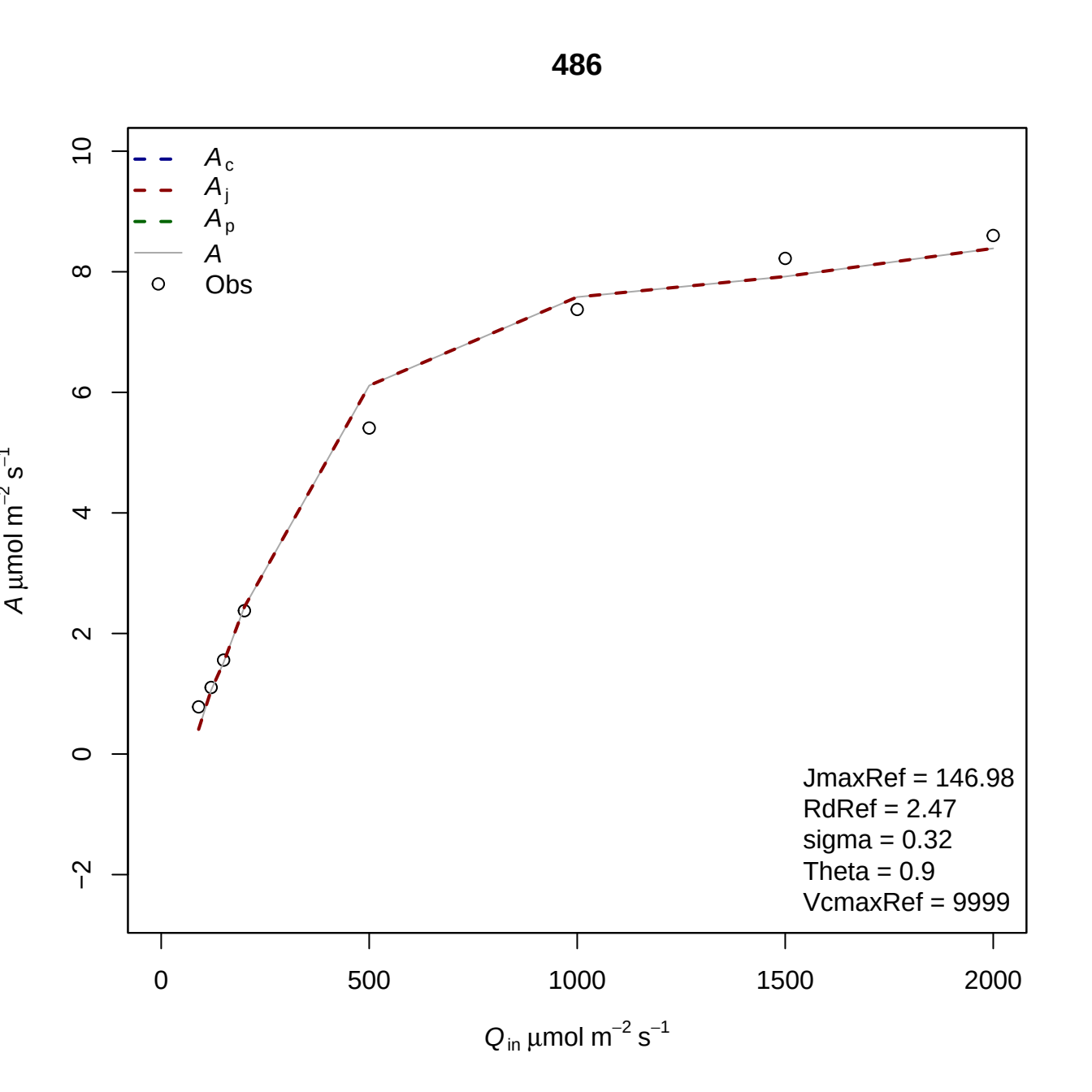

480
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 164.96
RdRef = 4.98
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999

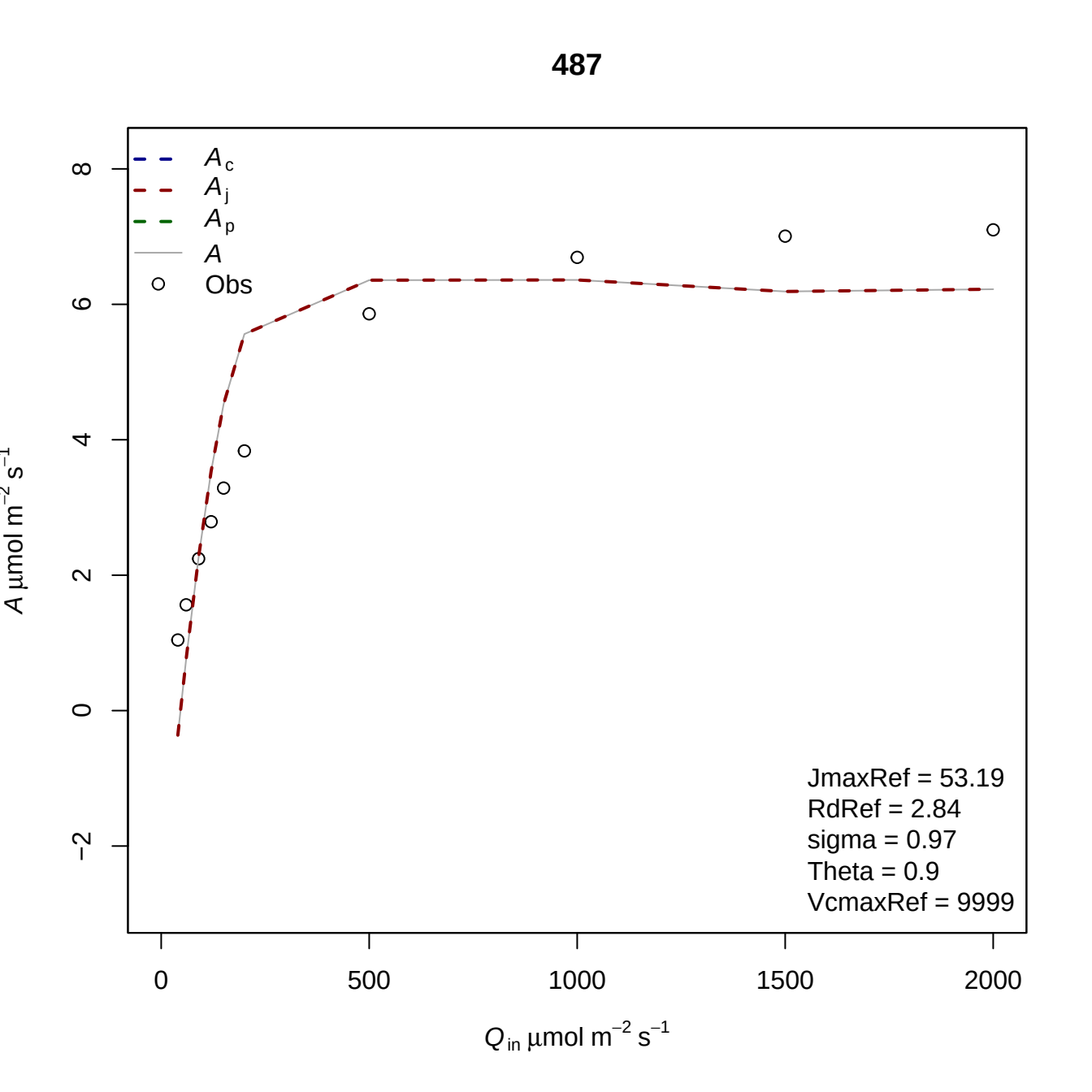

481
A_c
A_j
A_p
A
Obs
JmaxRef = 64.8
RdRef = 2.19
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

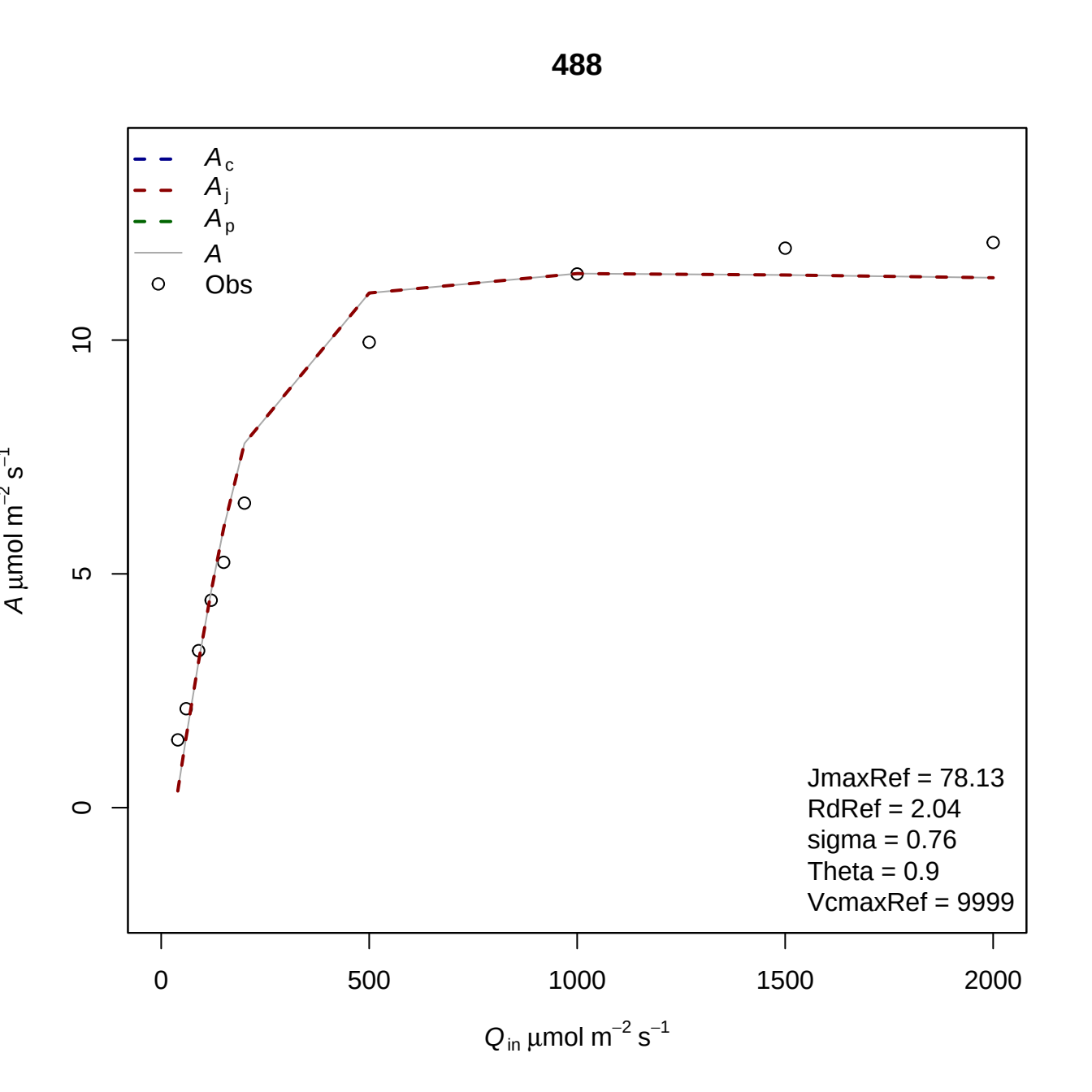

482
A_c
A_j
A_p
A
Obs
JmaxRef = 100.14
RdRef = 2.5
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

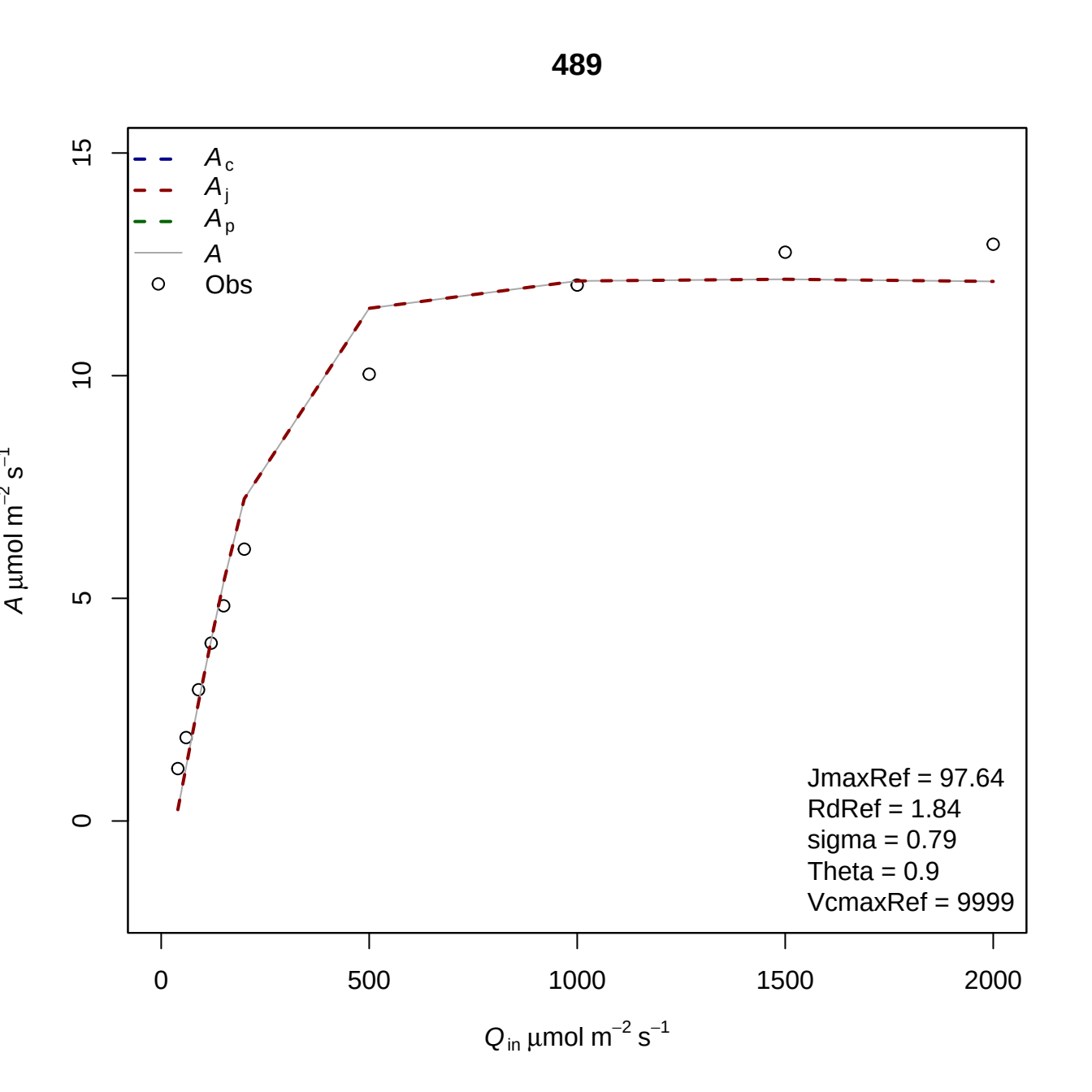

483
A_c
A_j
A_p
A
Obs
JmaxRef = 85.38
RdRef = 2.66
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

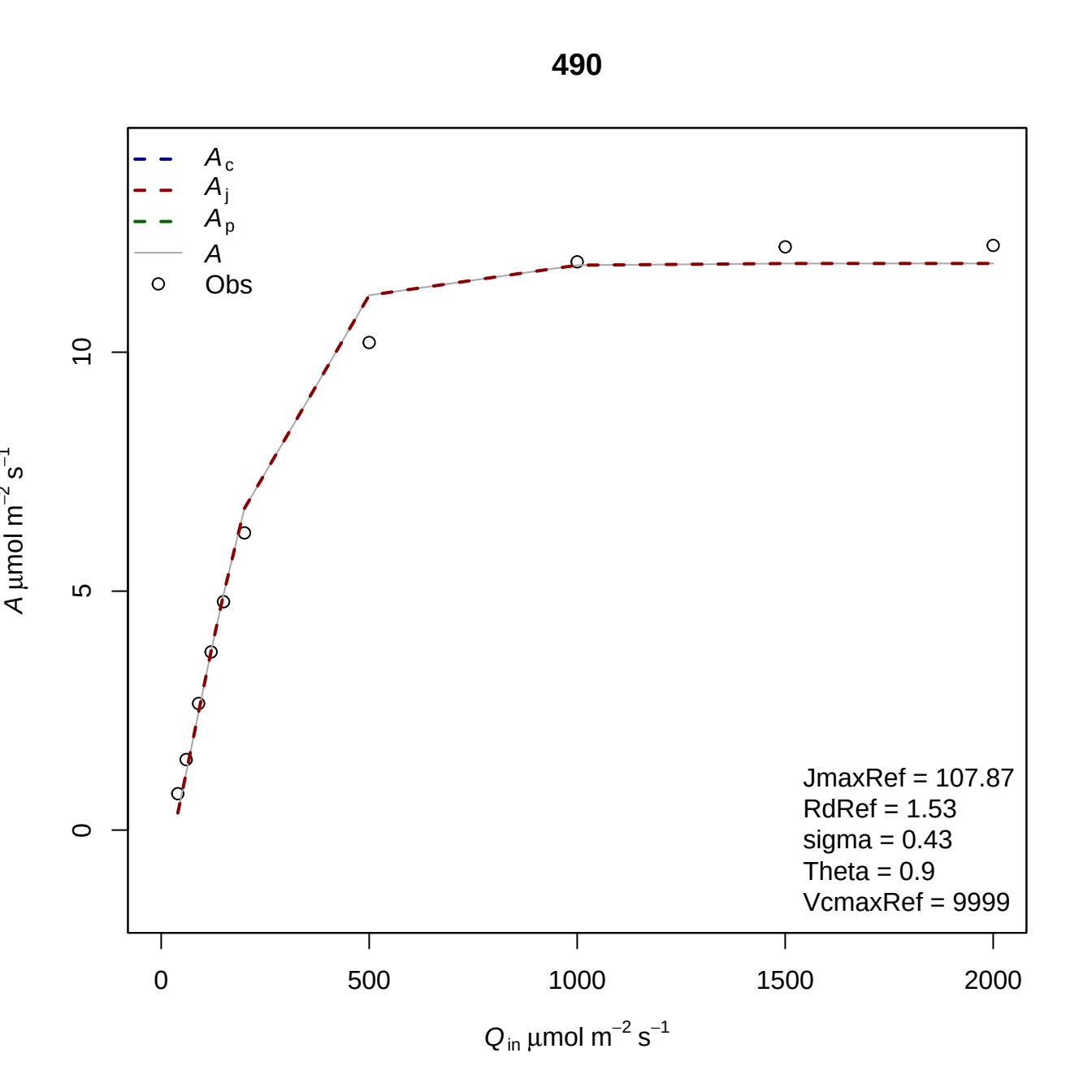

484
A_c
A_j
A_p
A
Obs
JmaxRef = 111.66
RdRef = 2.94
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

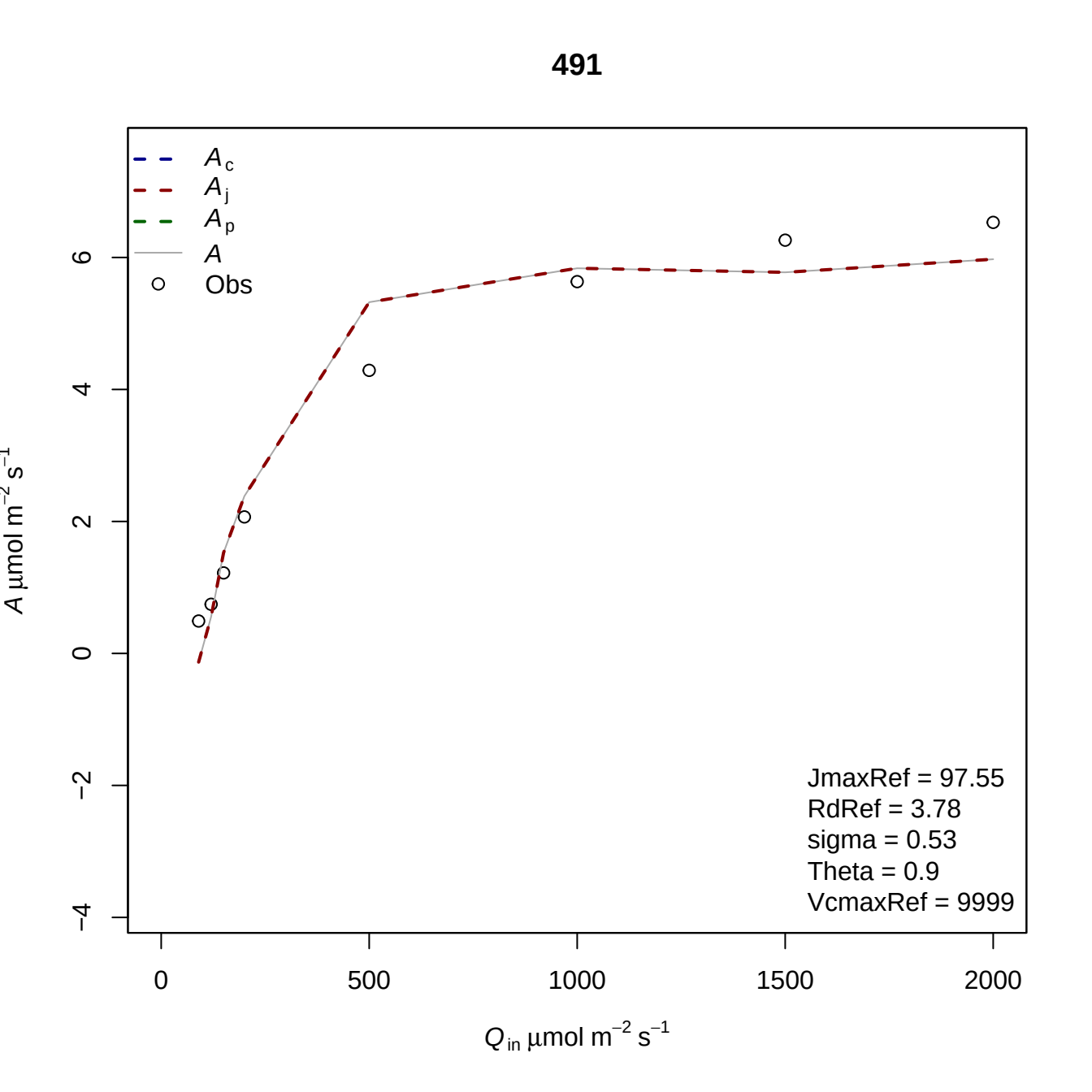

486
A_c
A_j
A_p
A
Obs
JmaxRef = 146.98
RdRef = 2.47
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

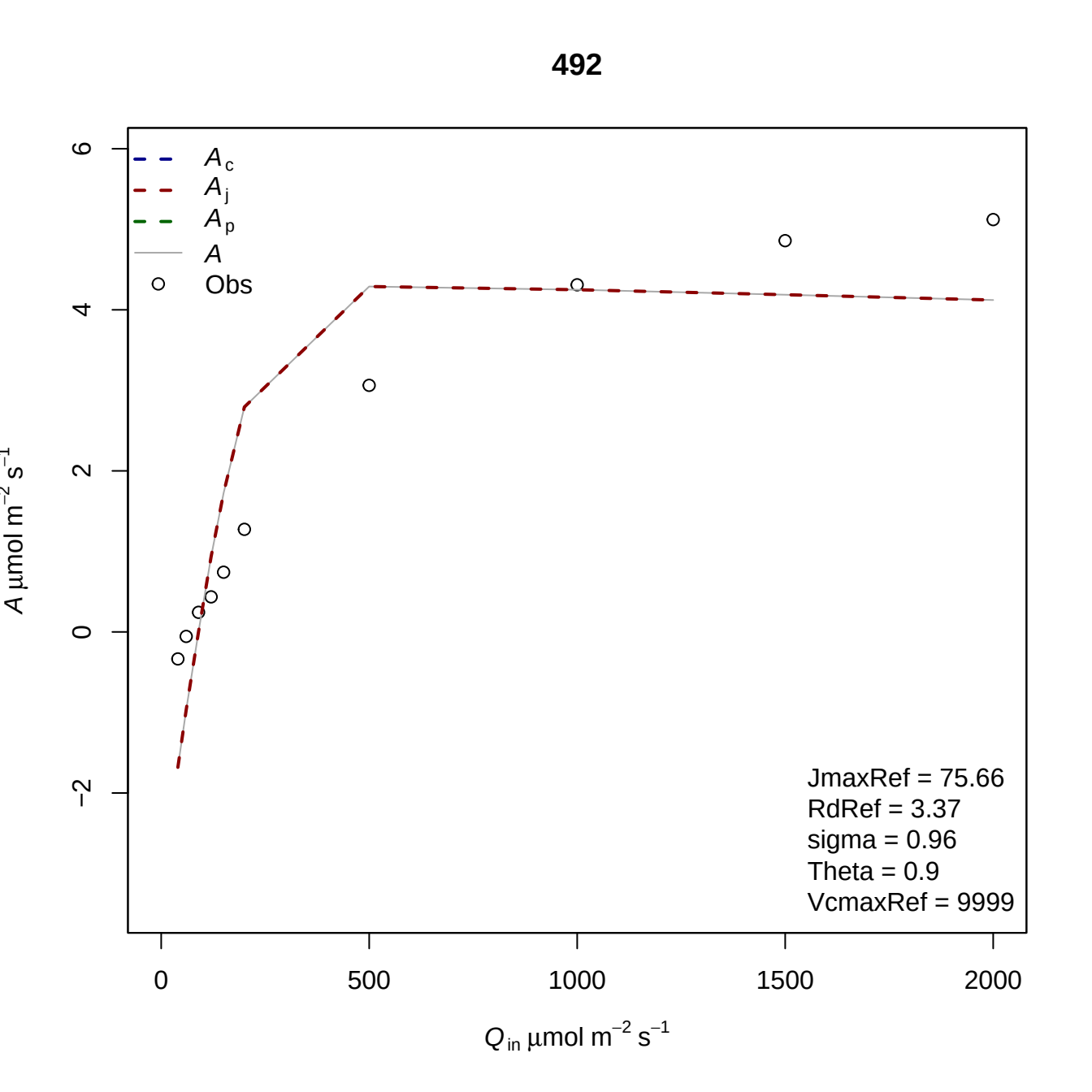

487
A_c
A_j
A_p
A
Obs
JmaxRef = 53.19
RdRef = 2.84
sigma = 0.97
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

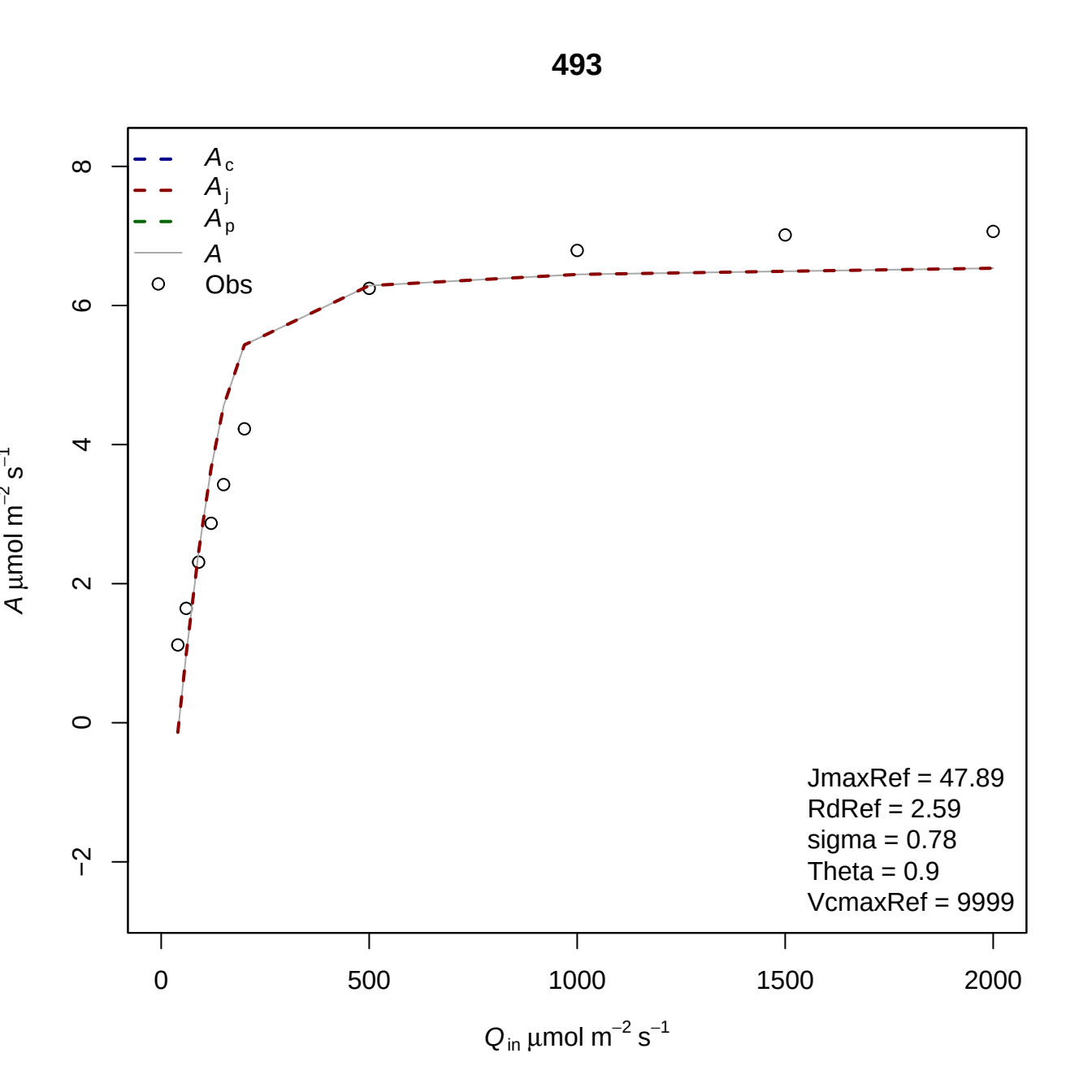

488
A_c
A_j
A_p
A
Obs
JmaxRef = 78.13
RdRef = 2.04
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

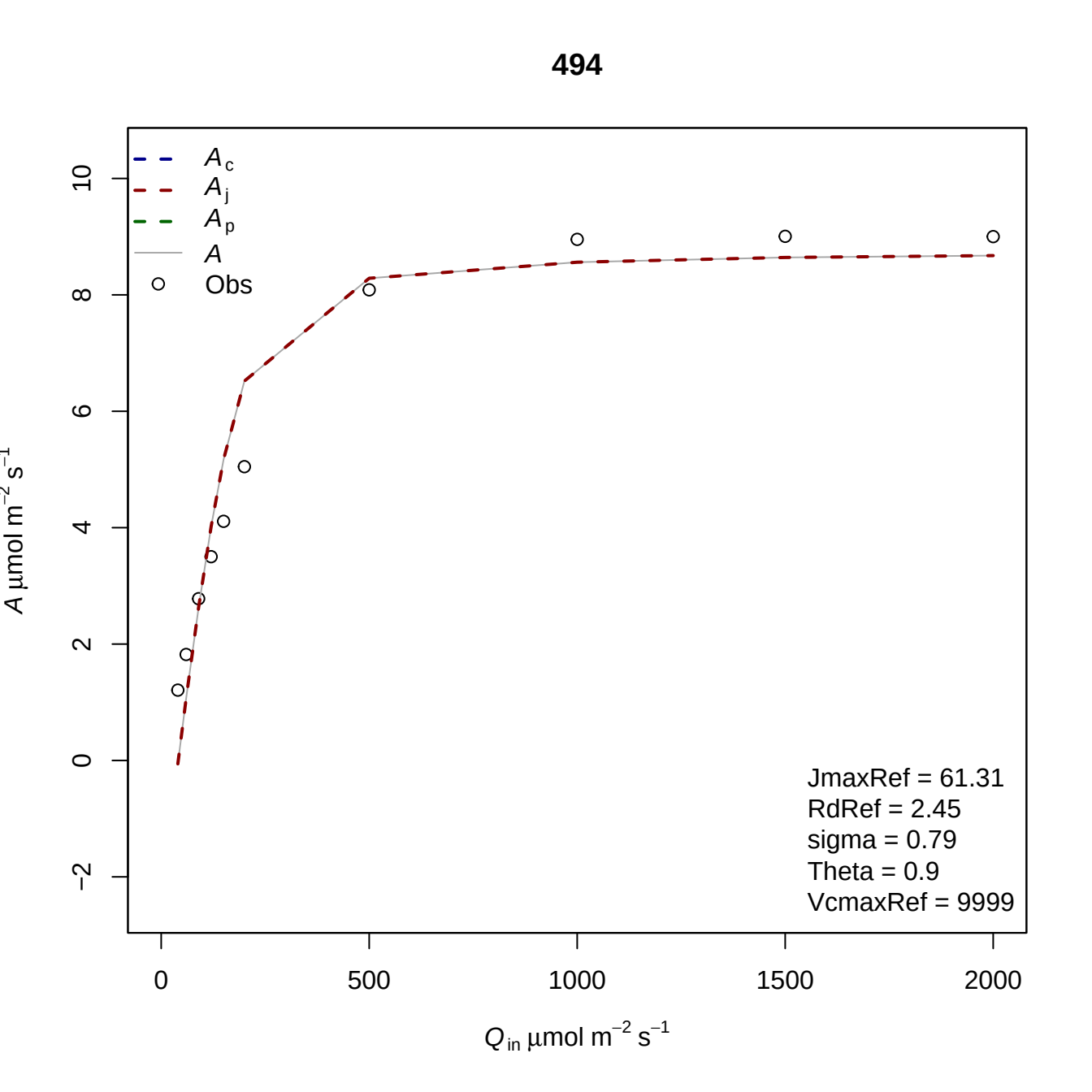

489
A_c
A_j
A_p
A
Obs
JmaxRef = 97.64
RdRef = 1.84
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

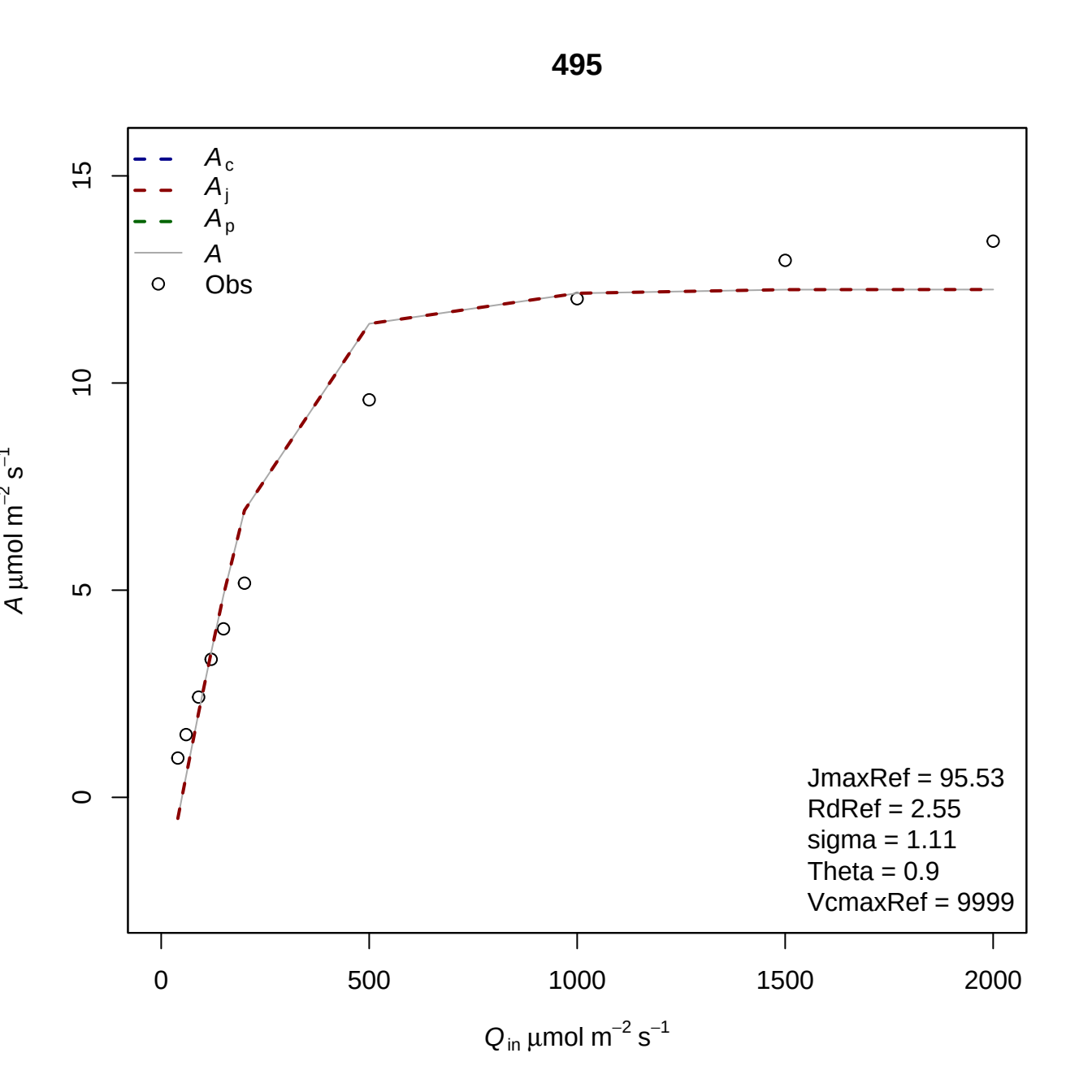

490
A_c
A_j
A_p
A
Obs
JmaxRef = 107.87
RdRef = 1.53
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

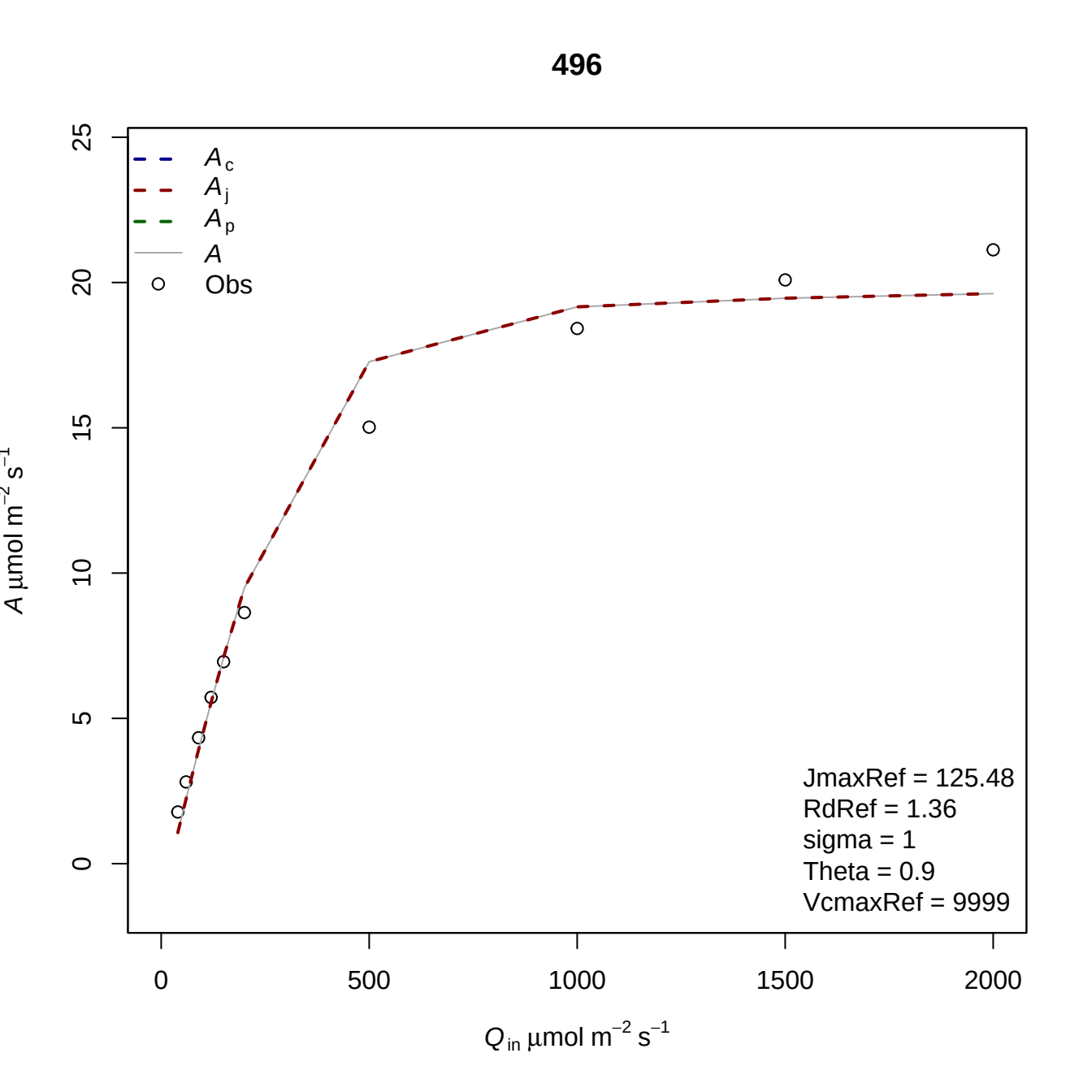

491
A_c
A_j
A_p
A
Obs
JmaxRef = 97.55
RdRef = 3.78
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
0
500
1000
1500
2000

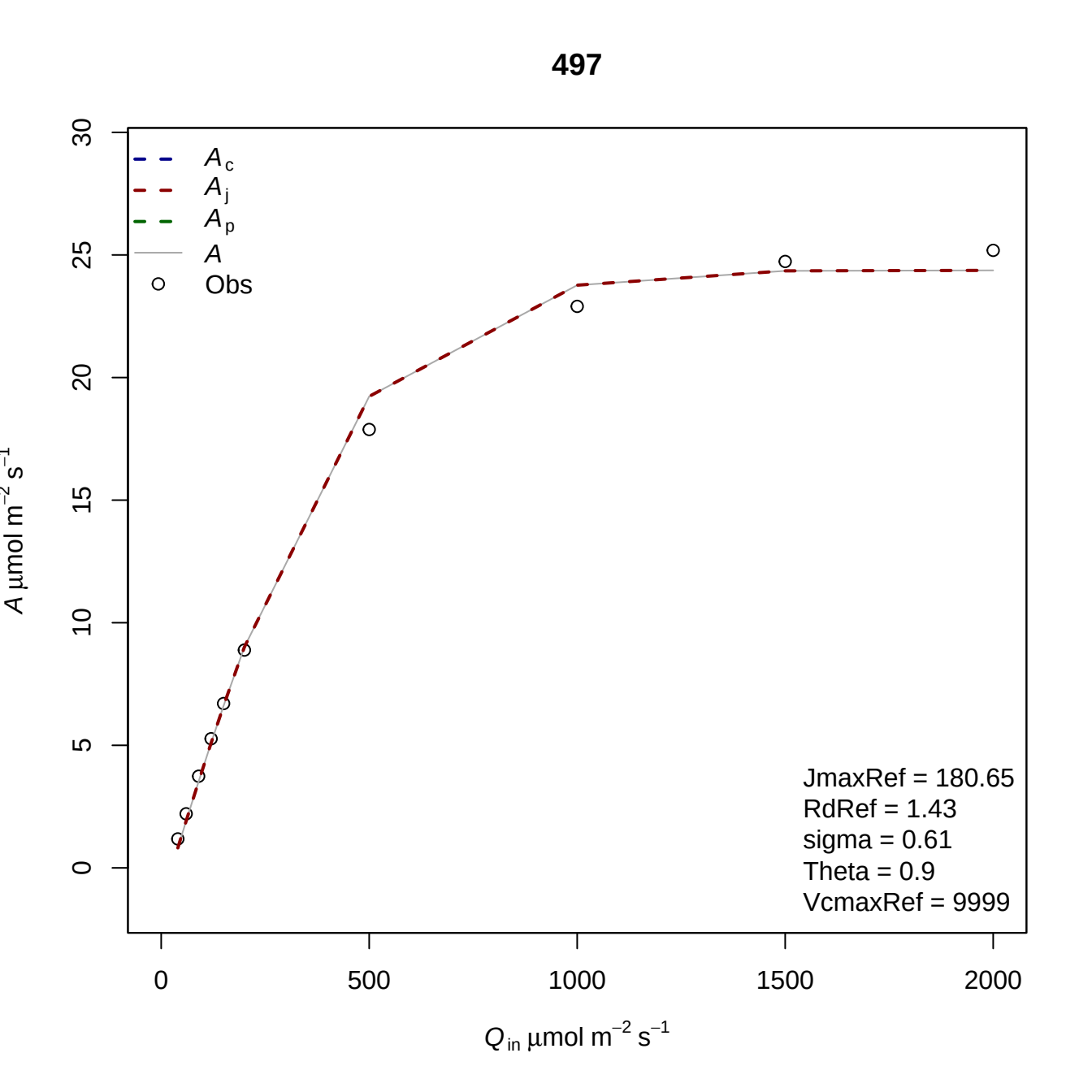

492
A_c
A_j
A_p
A
Obs
JmaxRef = 75.66
RdRef = 3.37
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

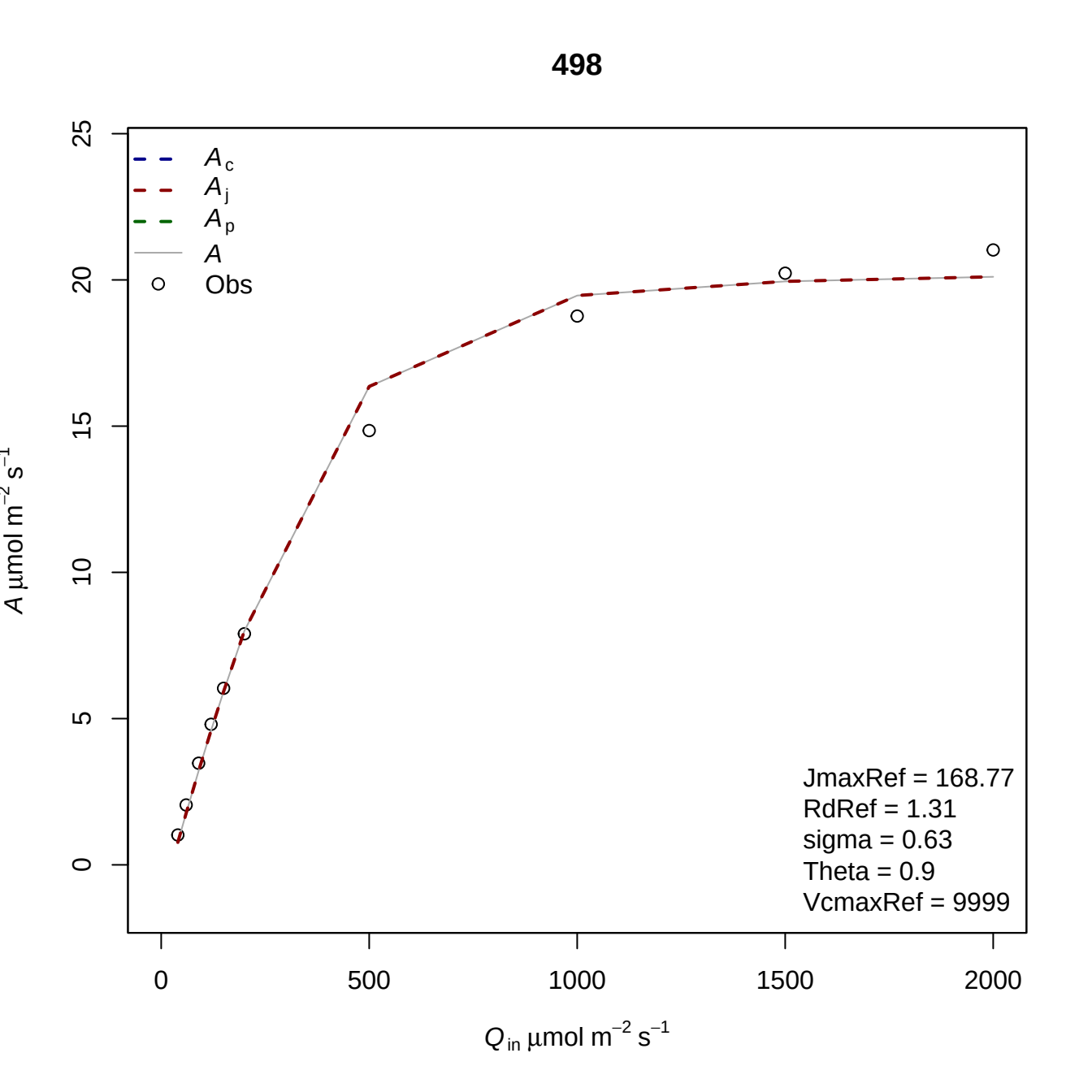

493
A_c
A_j
A_p
A
Obs
JmaxRef = 47.89
RdRef = 2.59
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

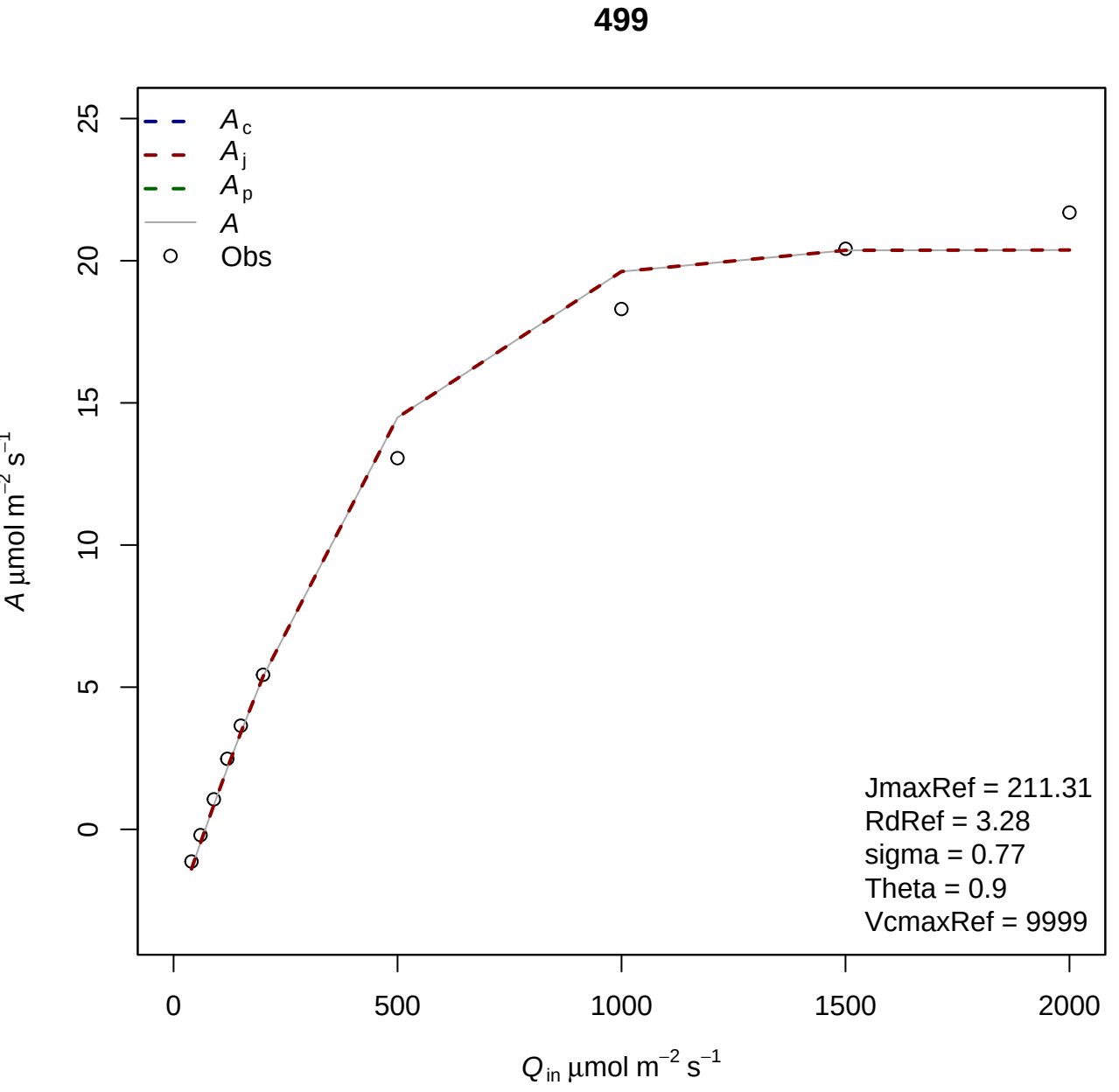

494
A_c
A_j
A_p
A
Obs
JmaxRef = 61.31
RdRef = 2.45
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

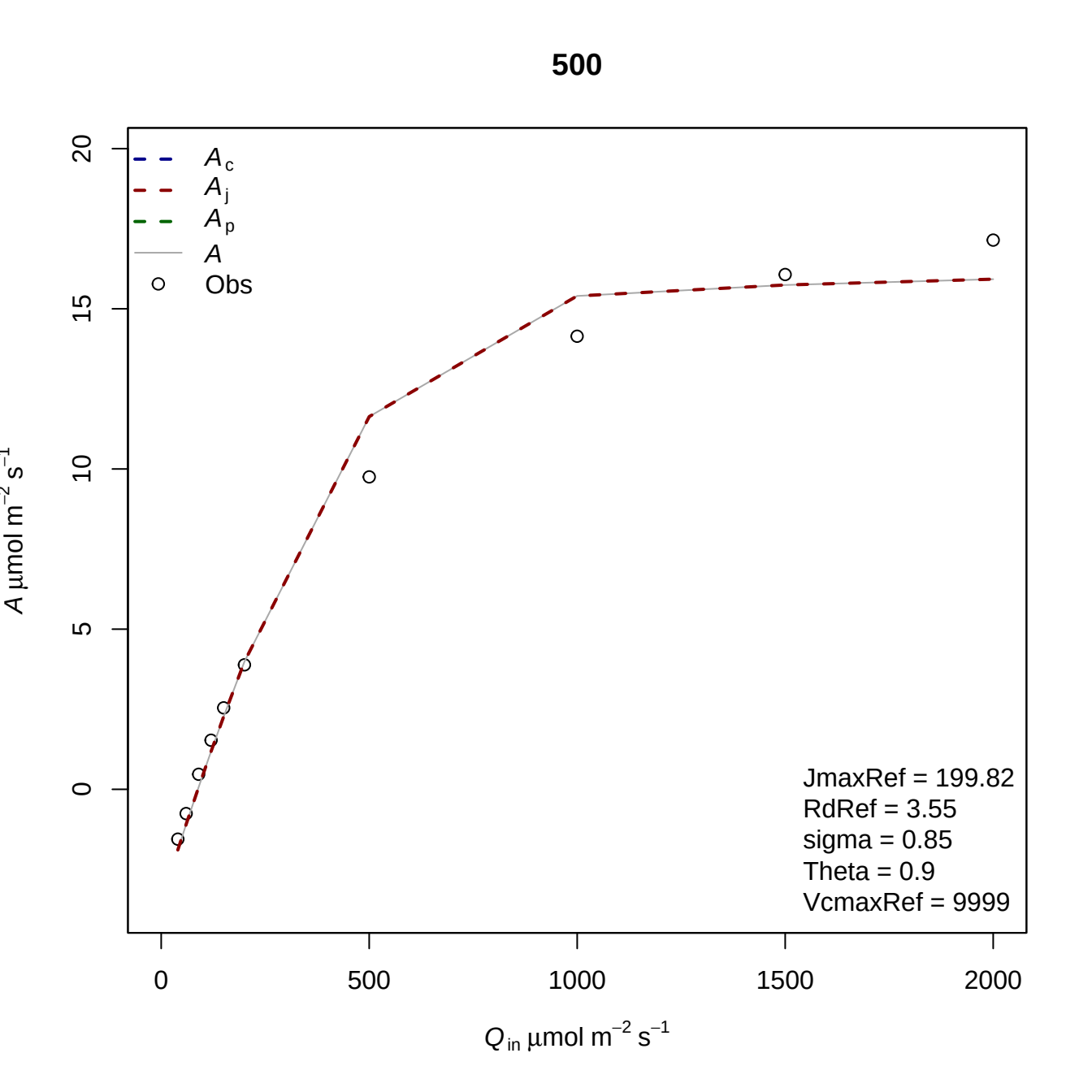

495
A_c
A_j
A_p
A
Obs
JmaxRef = 95.53
RdRef = 2.55
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

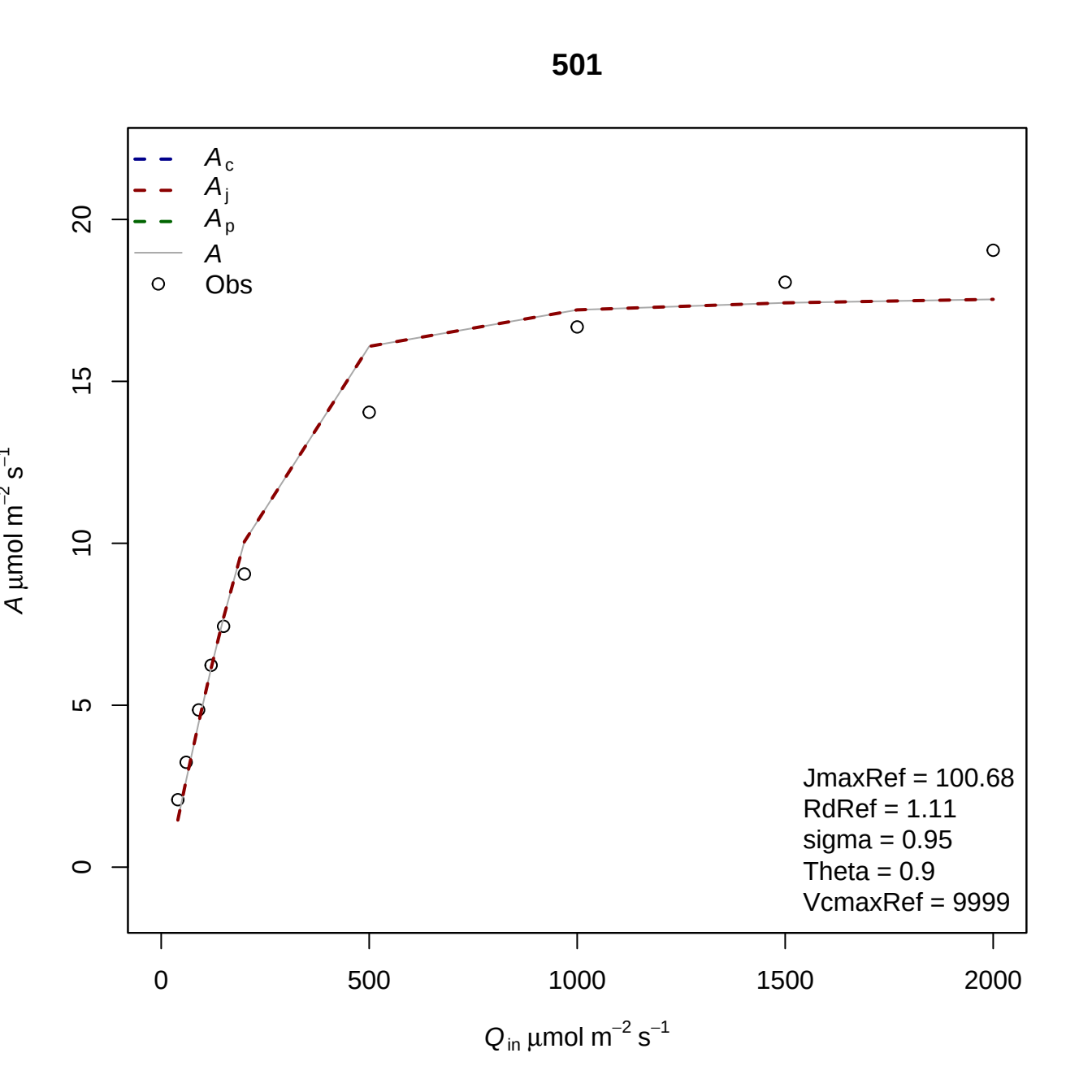

496
A_c
A_j
A_p
A
Obs
JmaxRef = 125.48
RdRef = 1.36
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

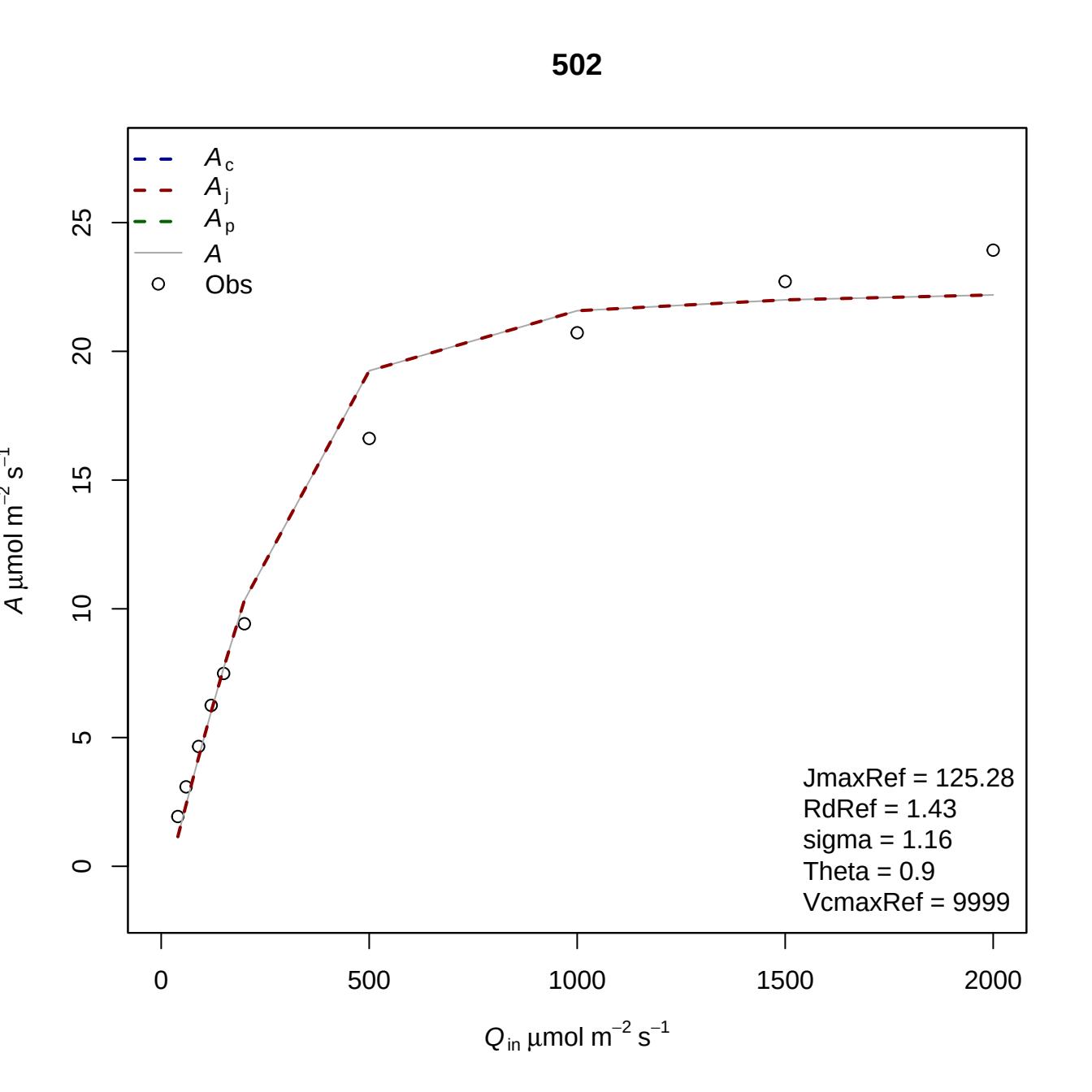

497
A_c
A_j
A_p
A
Obs
JmaxRef = 180.65
RdRef = 1.43
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

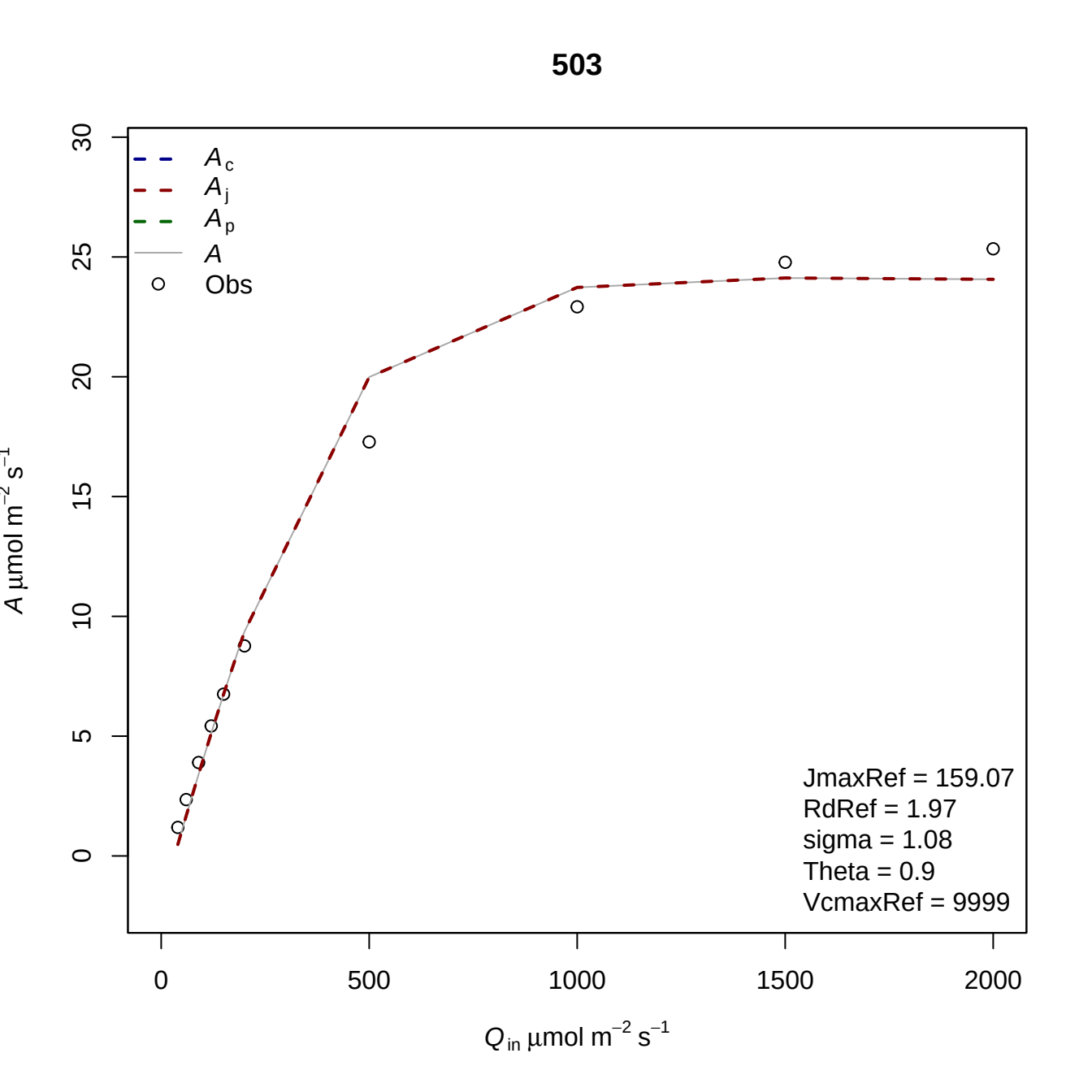

498
A_c
A_j
A_p
A
Obs
JmaxRef = 168.77
RdRef = 1.31
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

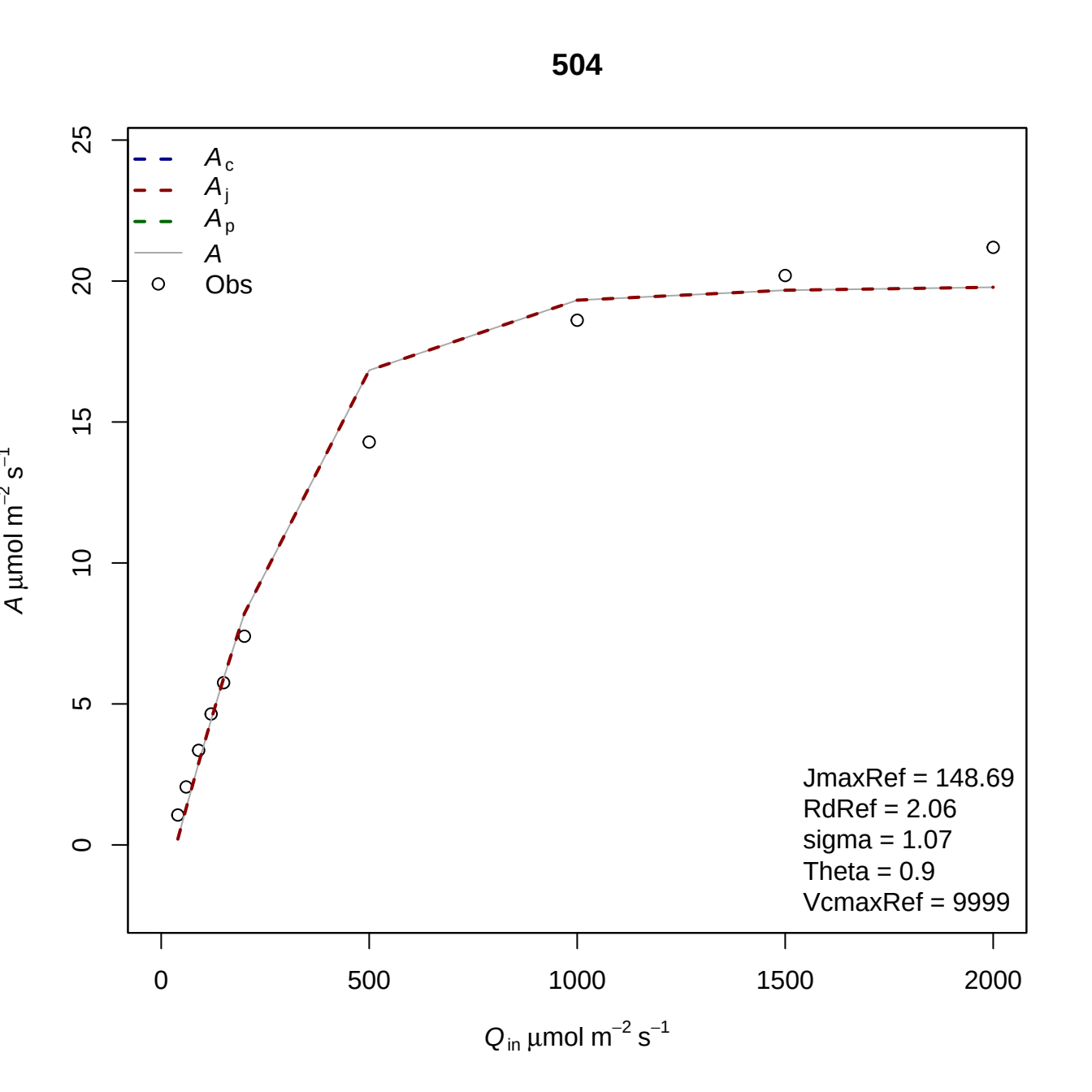

499
A_c
A_j
A_p
A
Obs
JmaxRef = 211.31
RdRef = 3.28
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

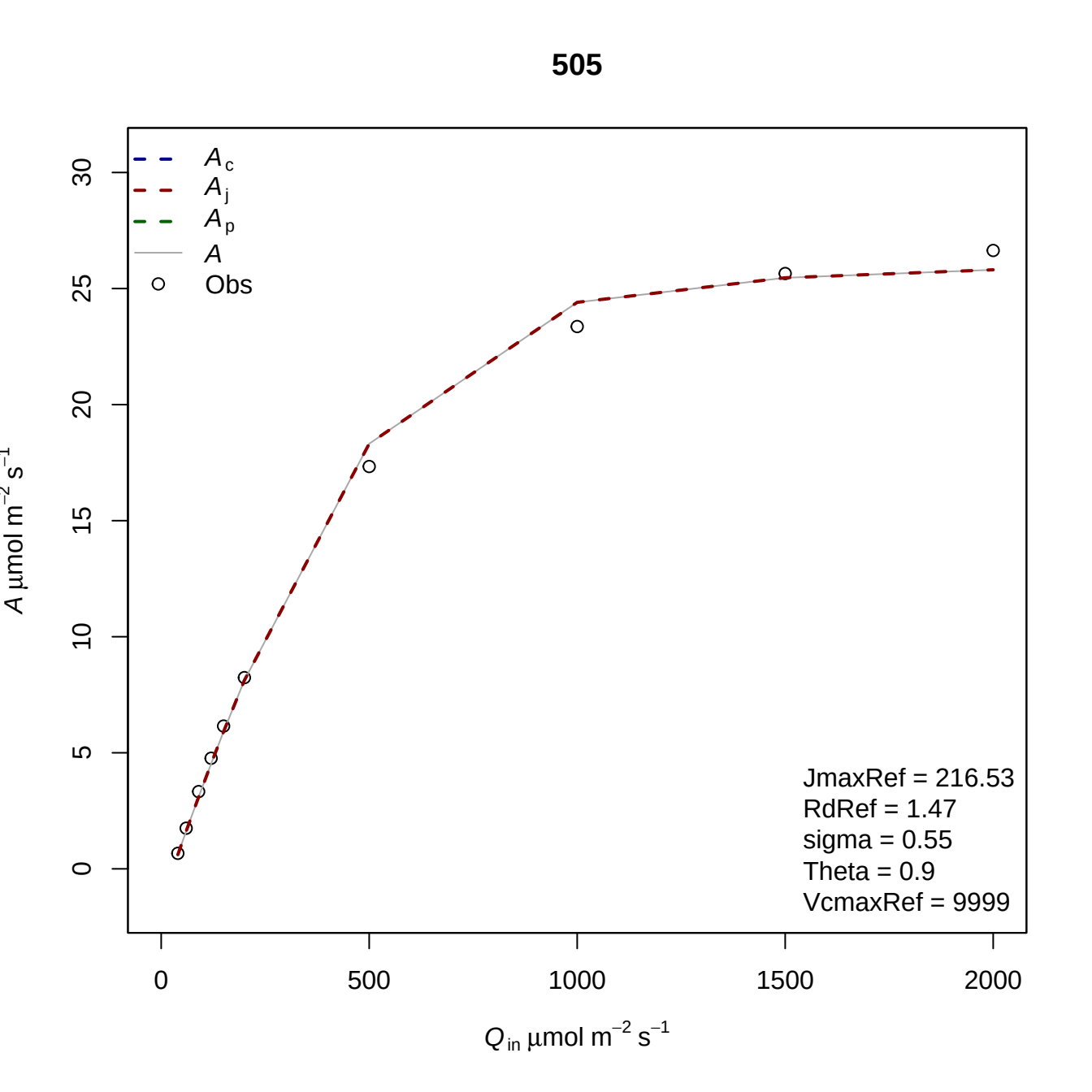

500
A_c
A_j
A_p
A
Obs
JmaxRef = 199.82
RdRef = 3.55
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

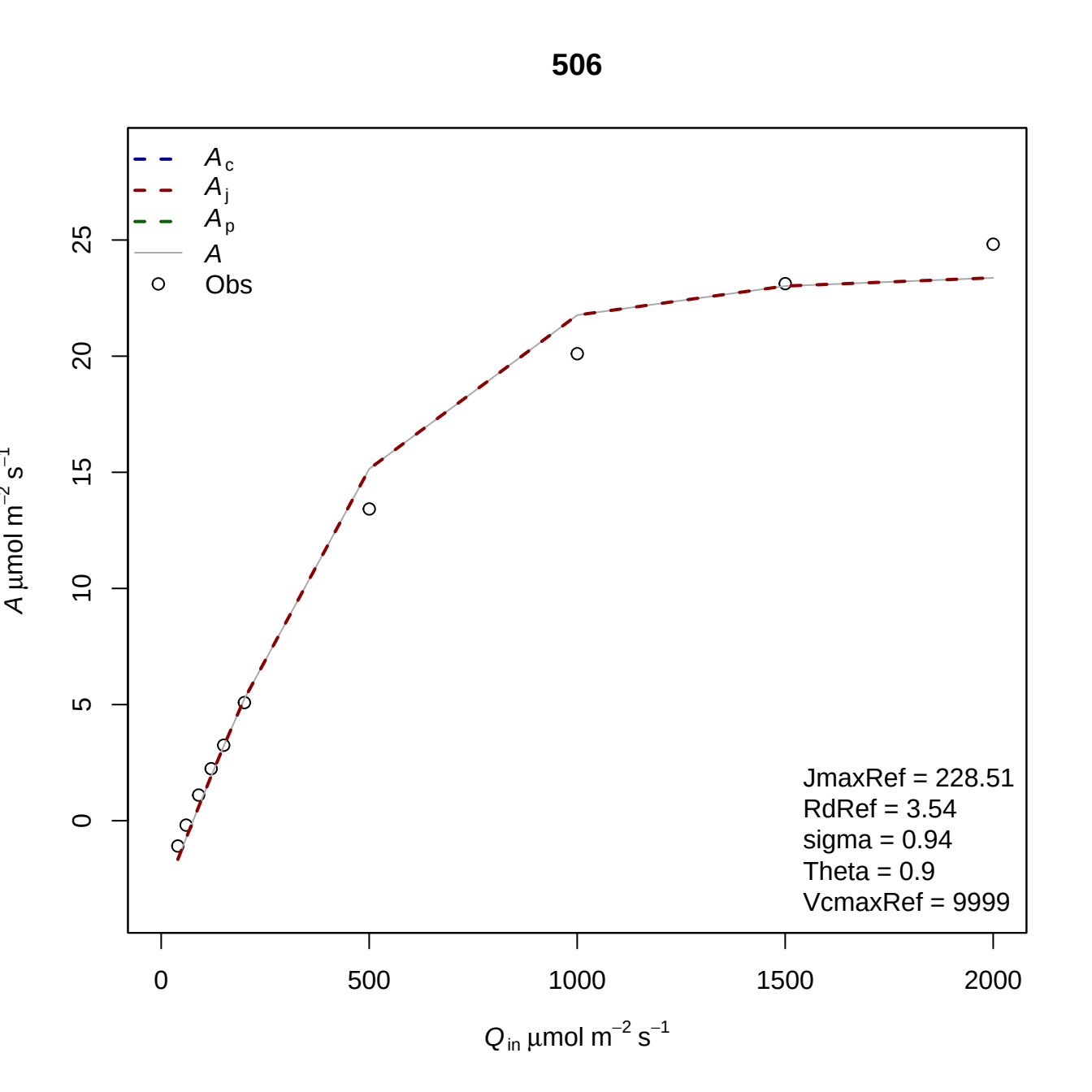

501
A_c
A_j
A_p
A
Obs
JmaxRef = 100.68
RdRef = 1.11
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

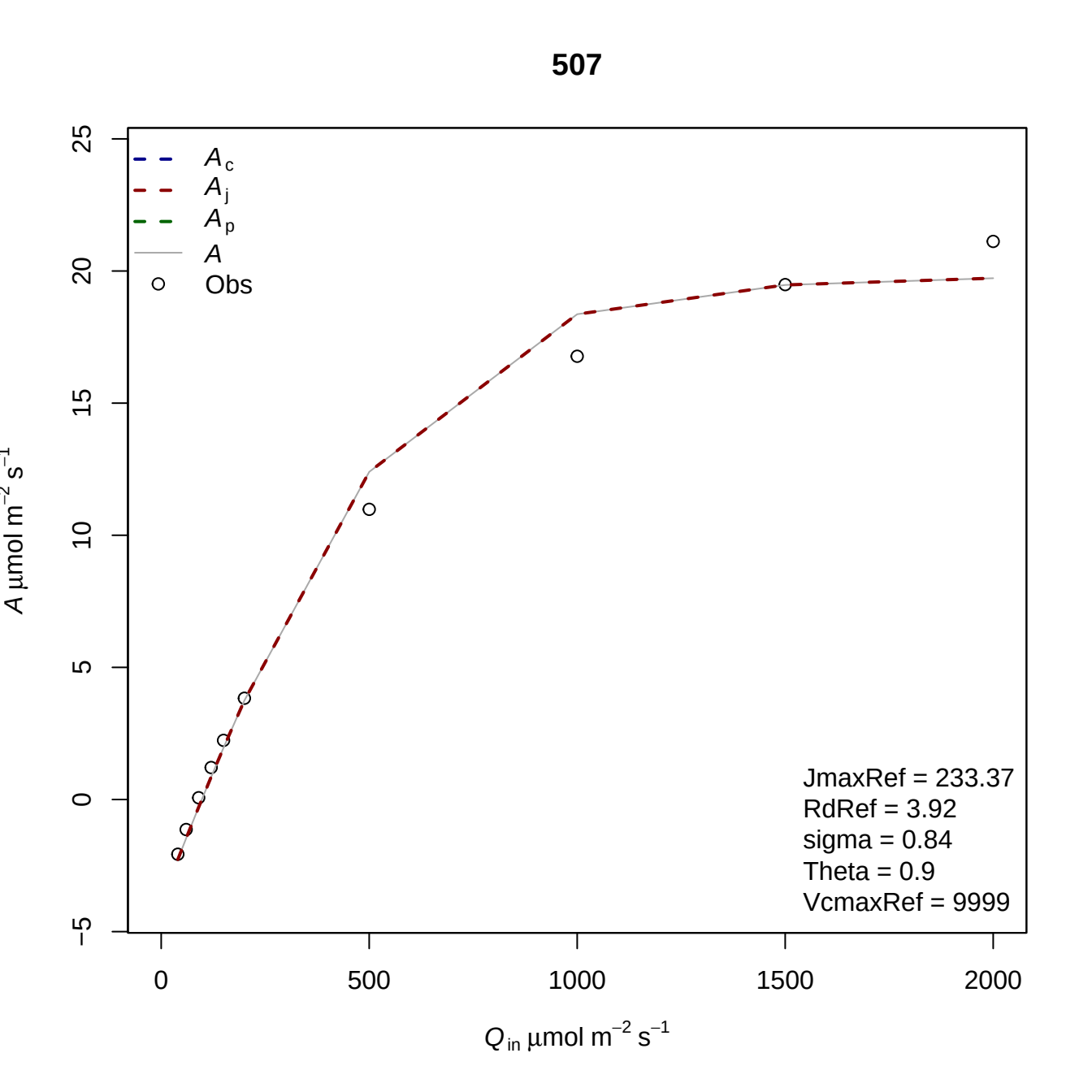

502
A_c
A_j
A_p
A
Obs
JmaxRef = 125.28
RdRef = 1.43
sigma = 1.16
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

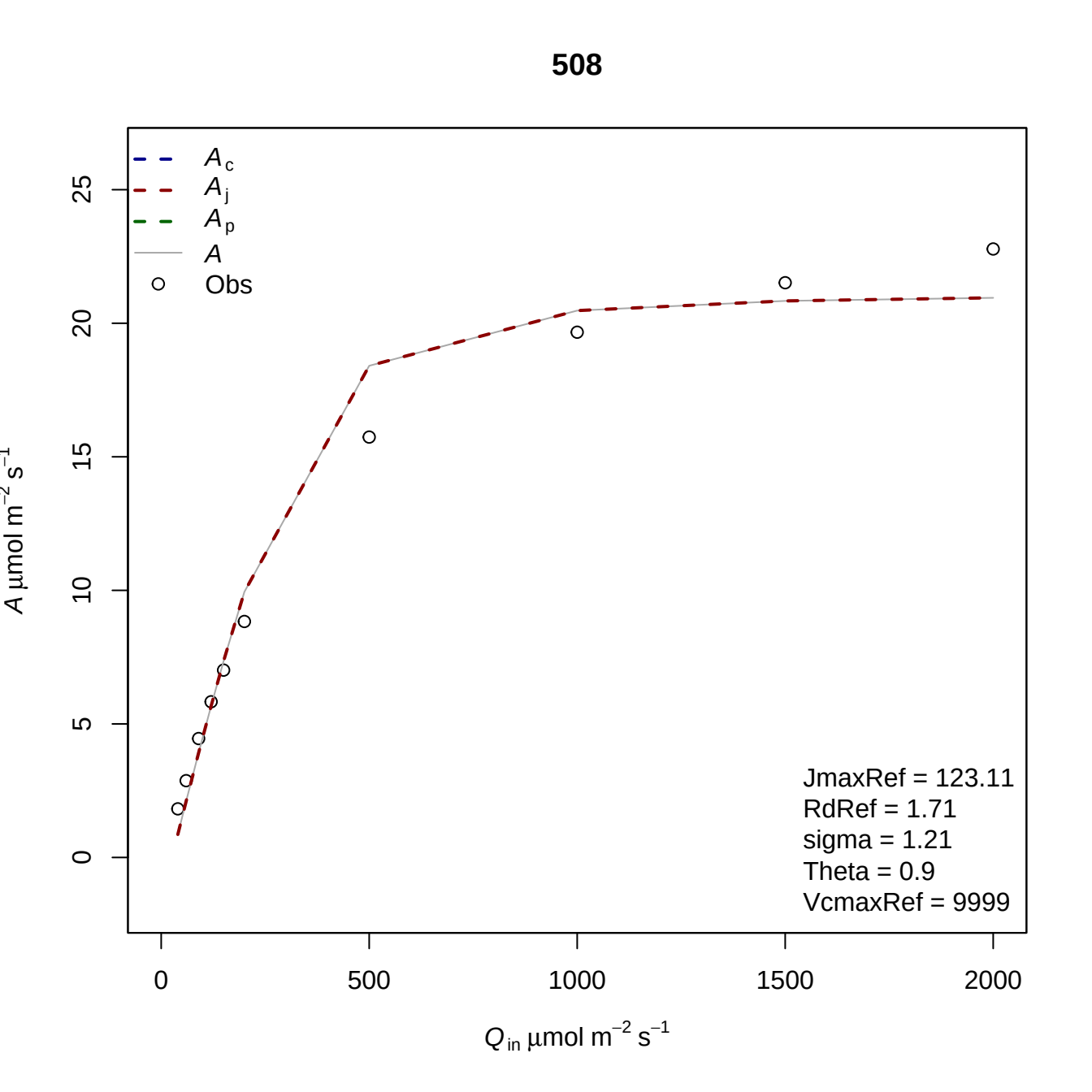

503
A_c
A_j
A_p
A
Obs
JmaxRef = 159.07
RdRef = 1.97
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

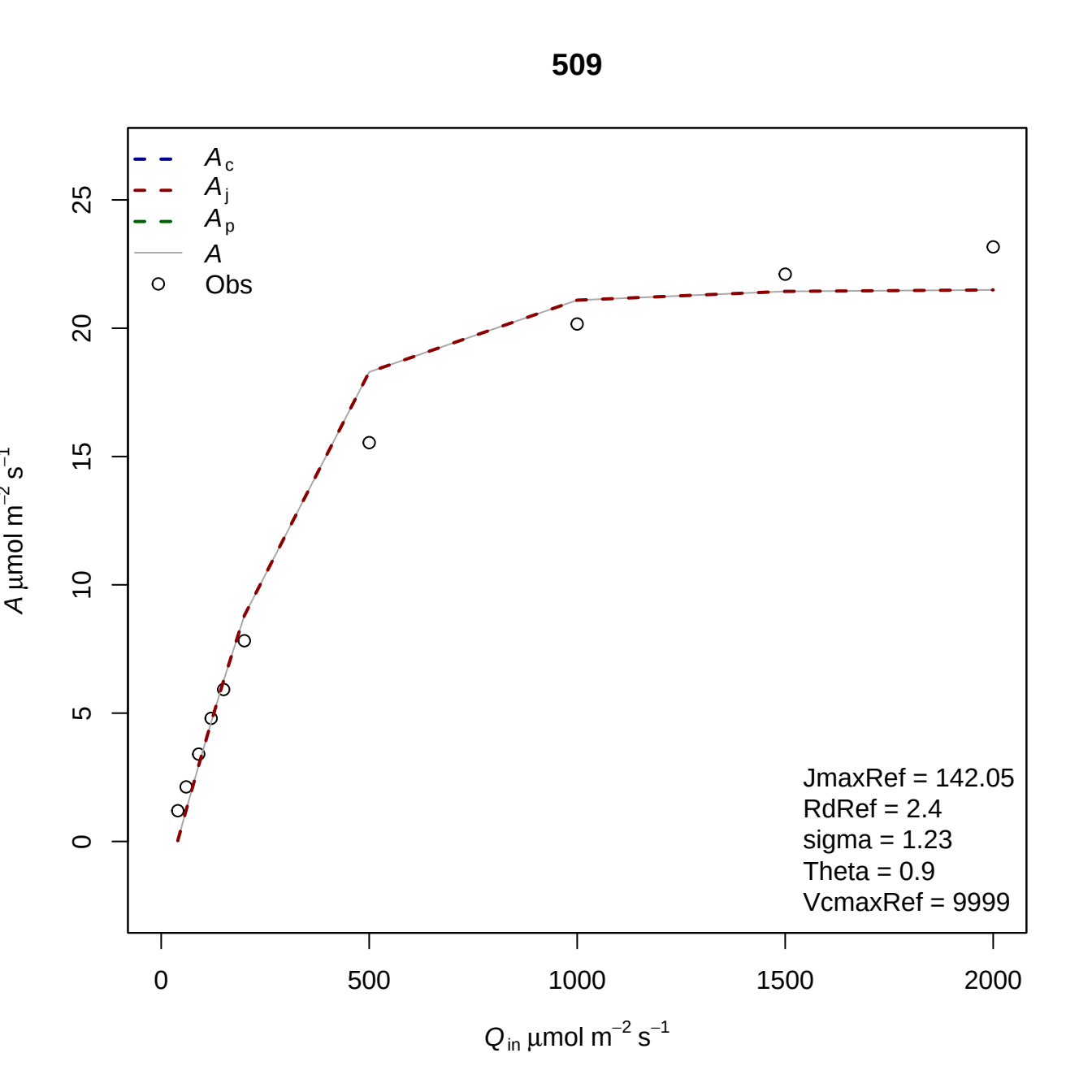

504
A_c
A_j
A_p
A
Obs
JmaxRef = 148.69
RdRef = 2.06
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

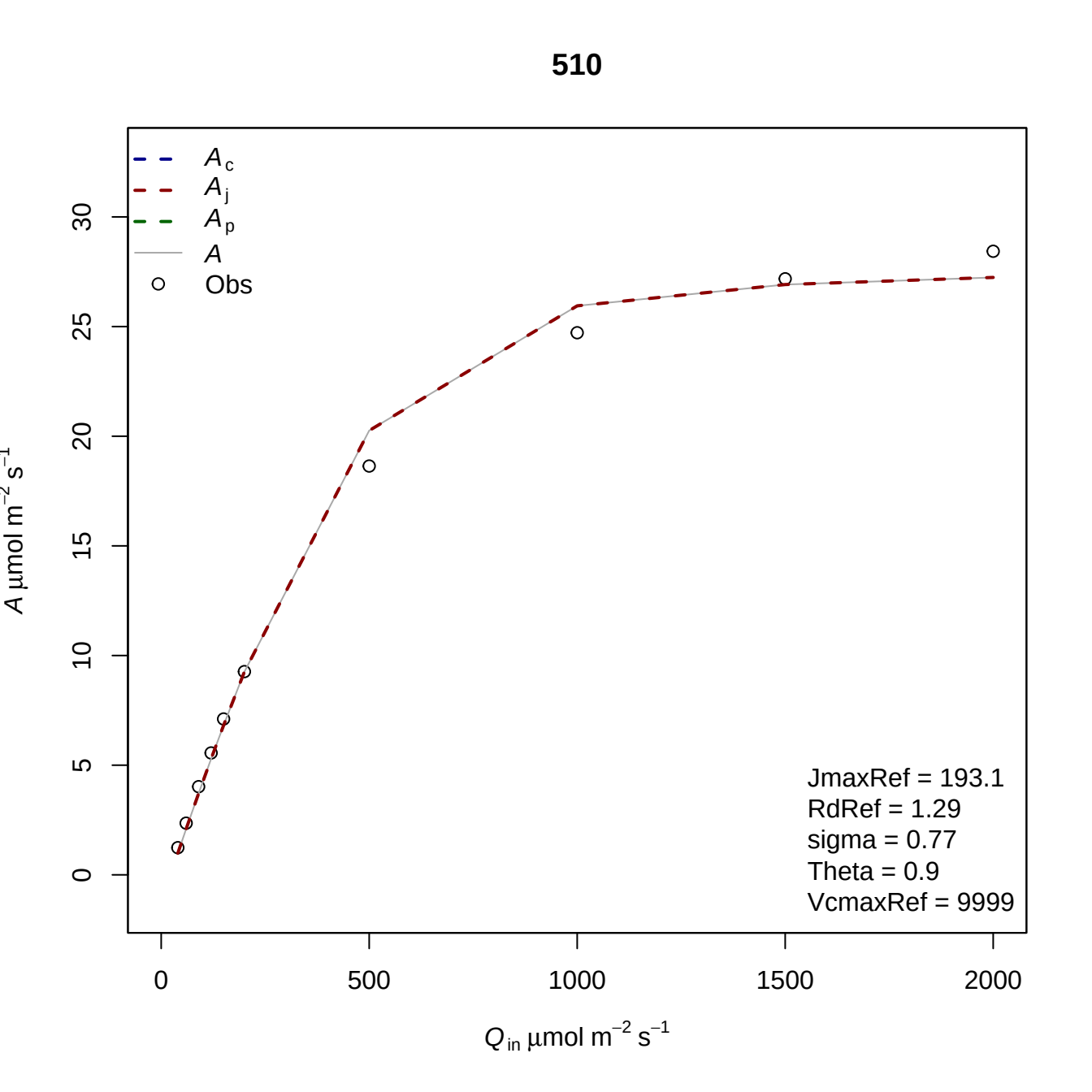

505
A_c
A_j
A_p
A
Obs
JmaxRef = 216.53
RdRef = 1.47
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

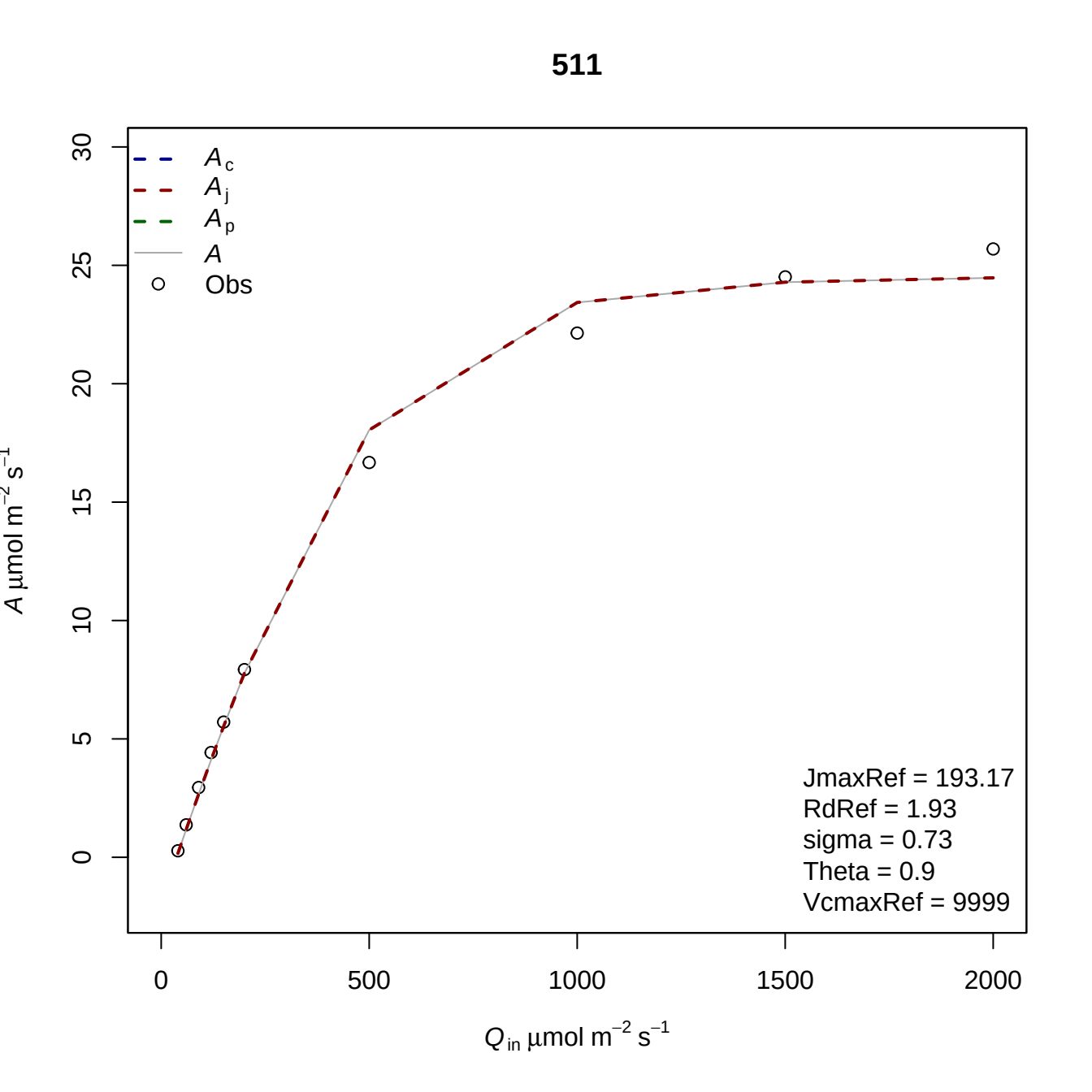

506
A_c
A_j
A_p
A
Obs
JmaxRef = 228.51
RdRef = 3.54
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

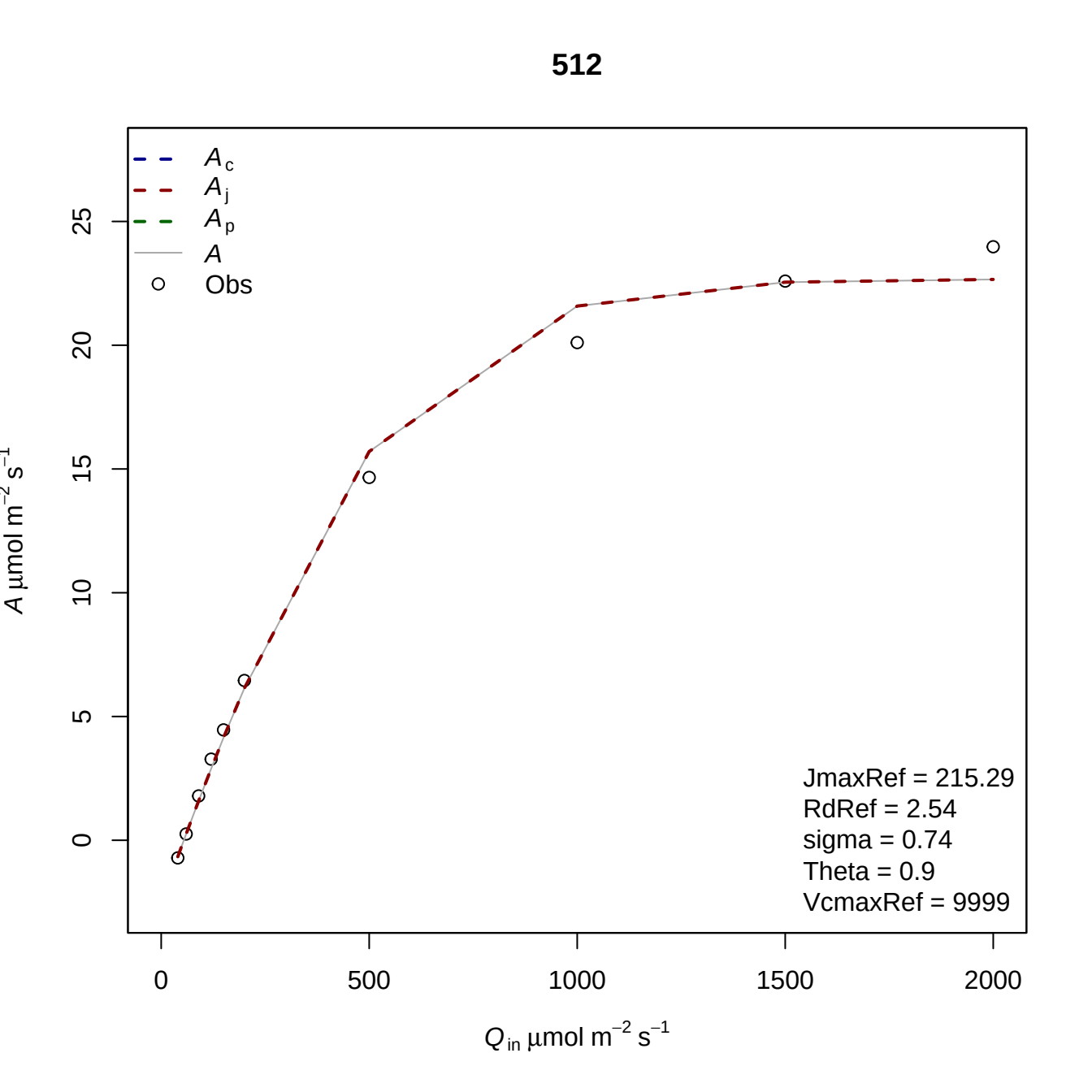

507
A_c
A_j
A_p
A
Obs
JmaxRef = 233.37
RdRef = 3.92
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

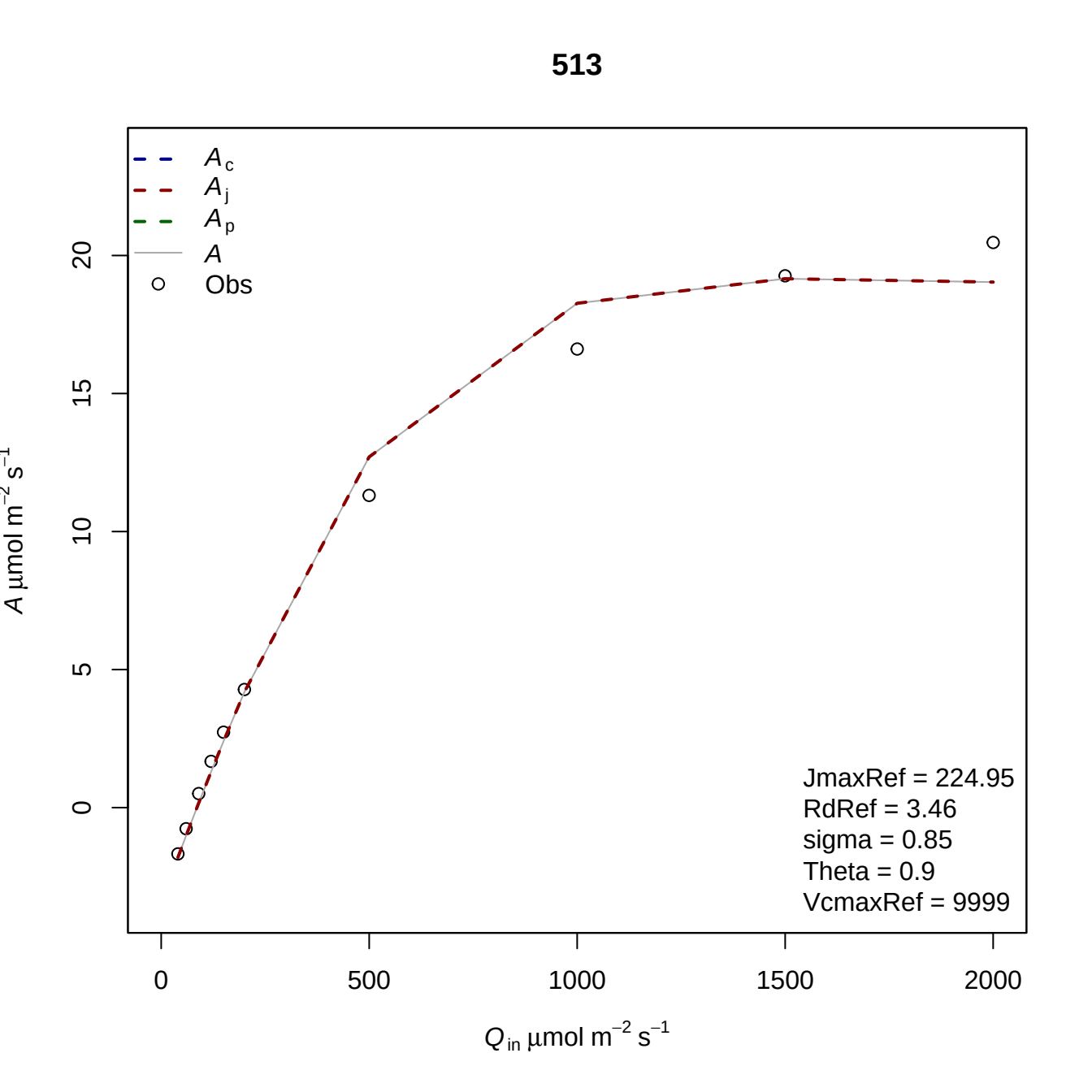

508
A_c
A_j
A_p
A
Obs
JmaxRef = 123.11
RdRef = 1.71
sigma = 1.21
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

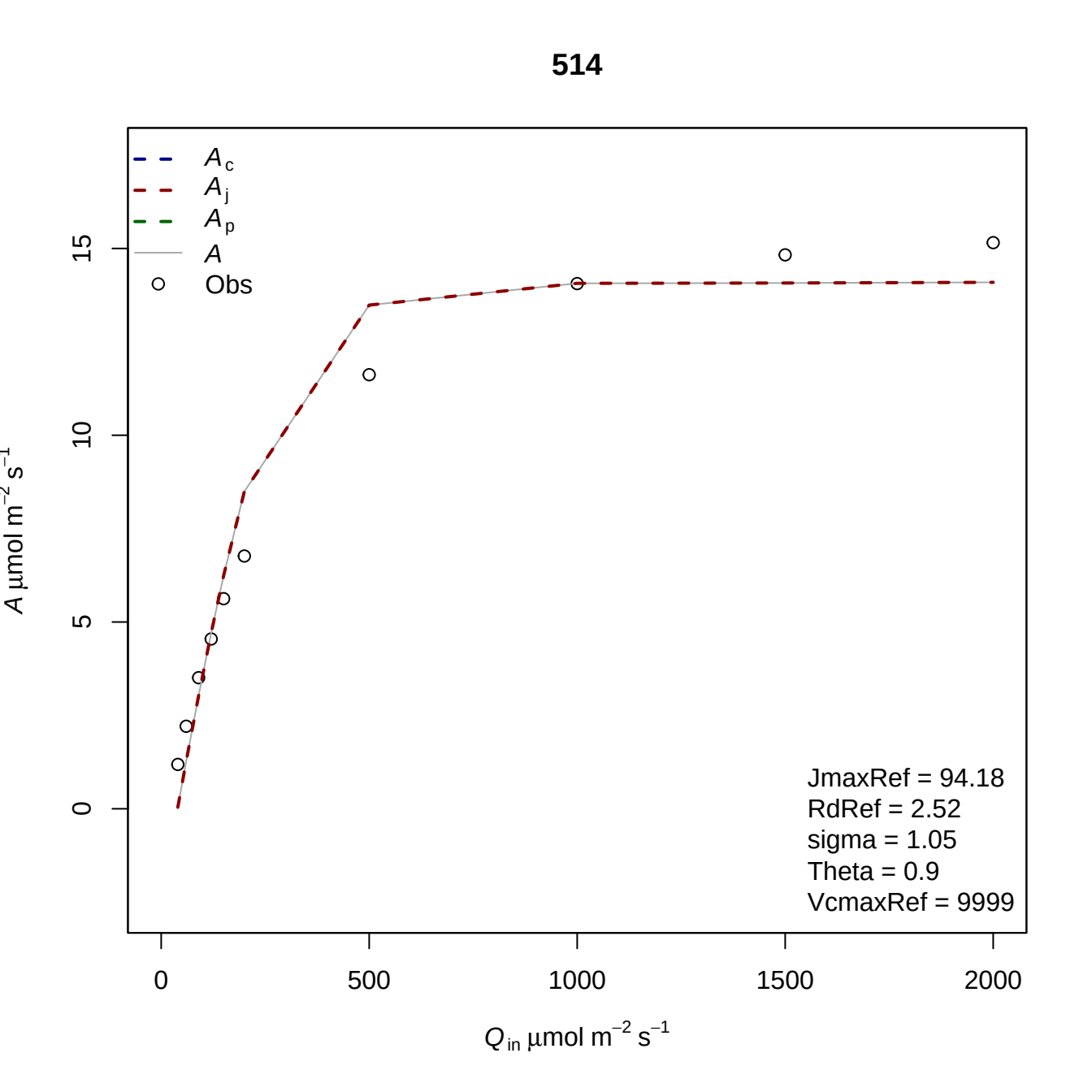

509
A_c
A_j
A_p
A
Obs
JmaxRef = 142.05
RdRef = 2.4
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

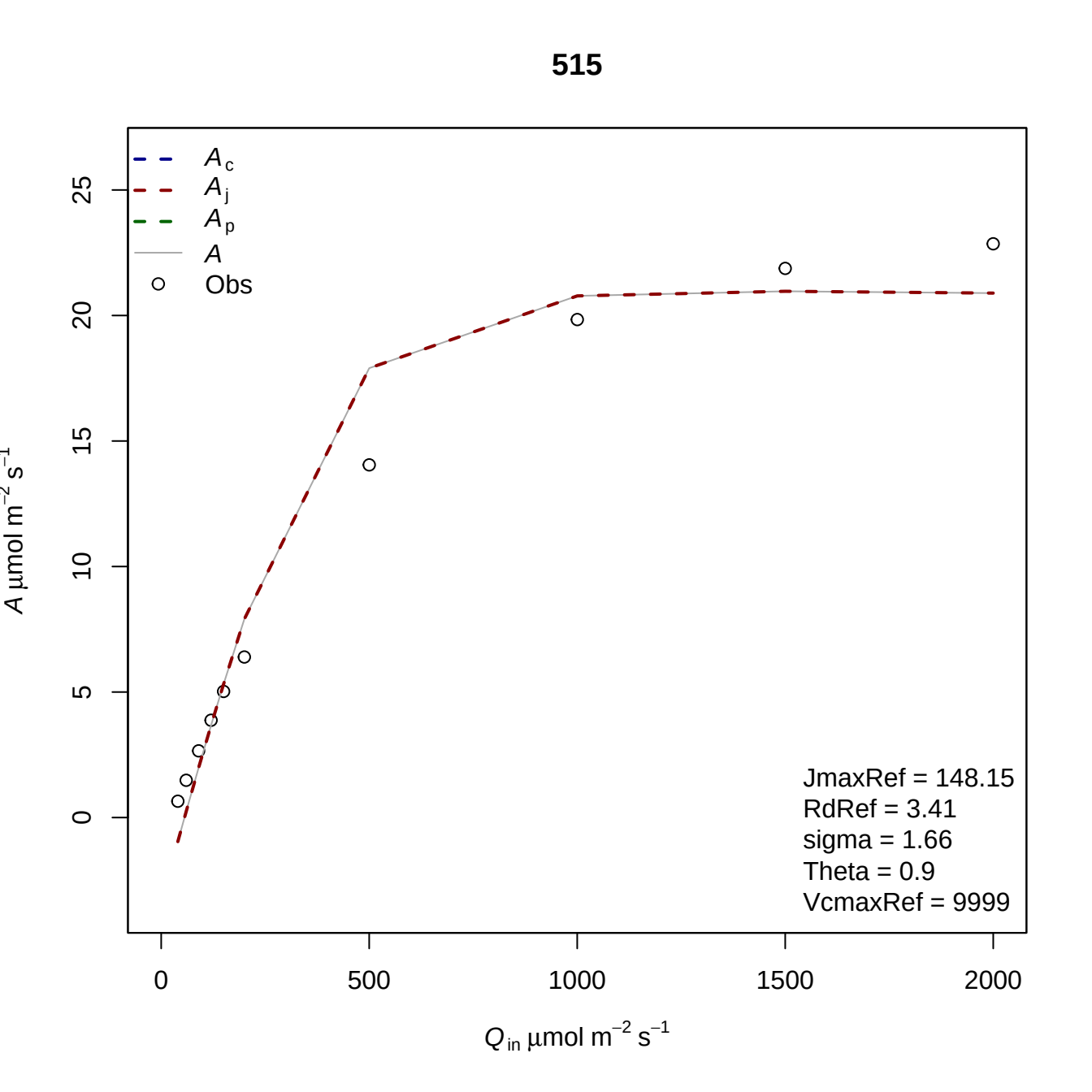

510
A_c
A_j
A_p
A
Obs
JmaxRef = 193.1
RdRef = 1.29
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

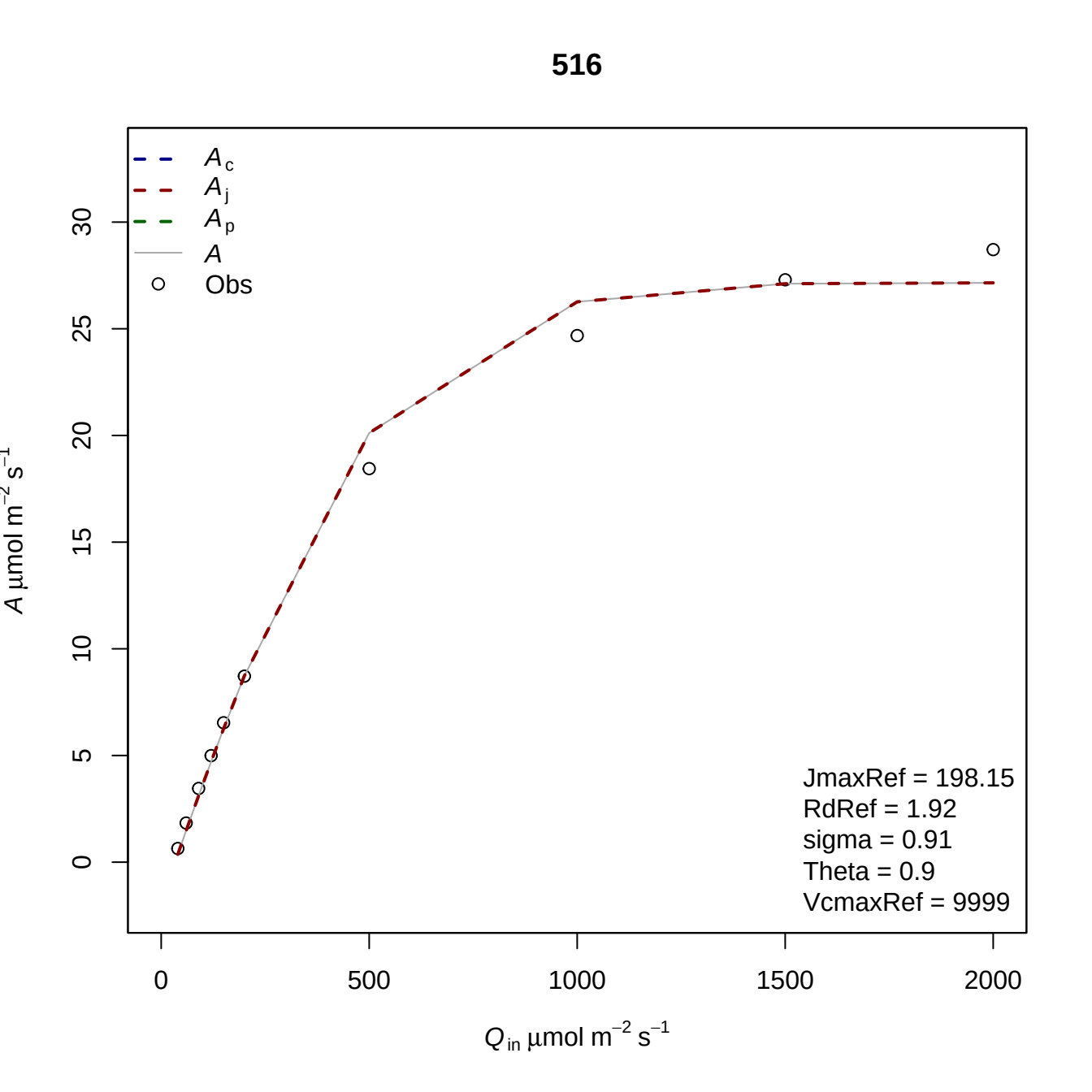

511
A_c
A_j
A_p
A
Obs
JmaxRef = 193.17
RdRef = 1.93
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

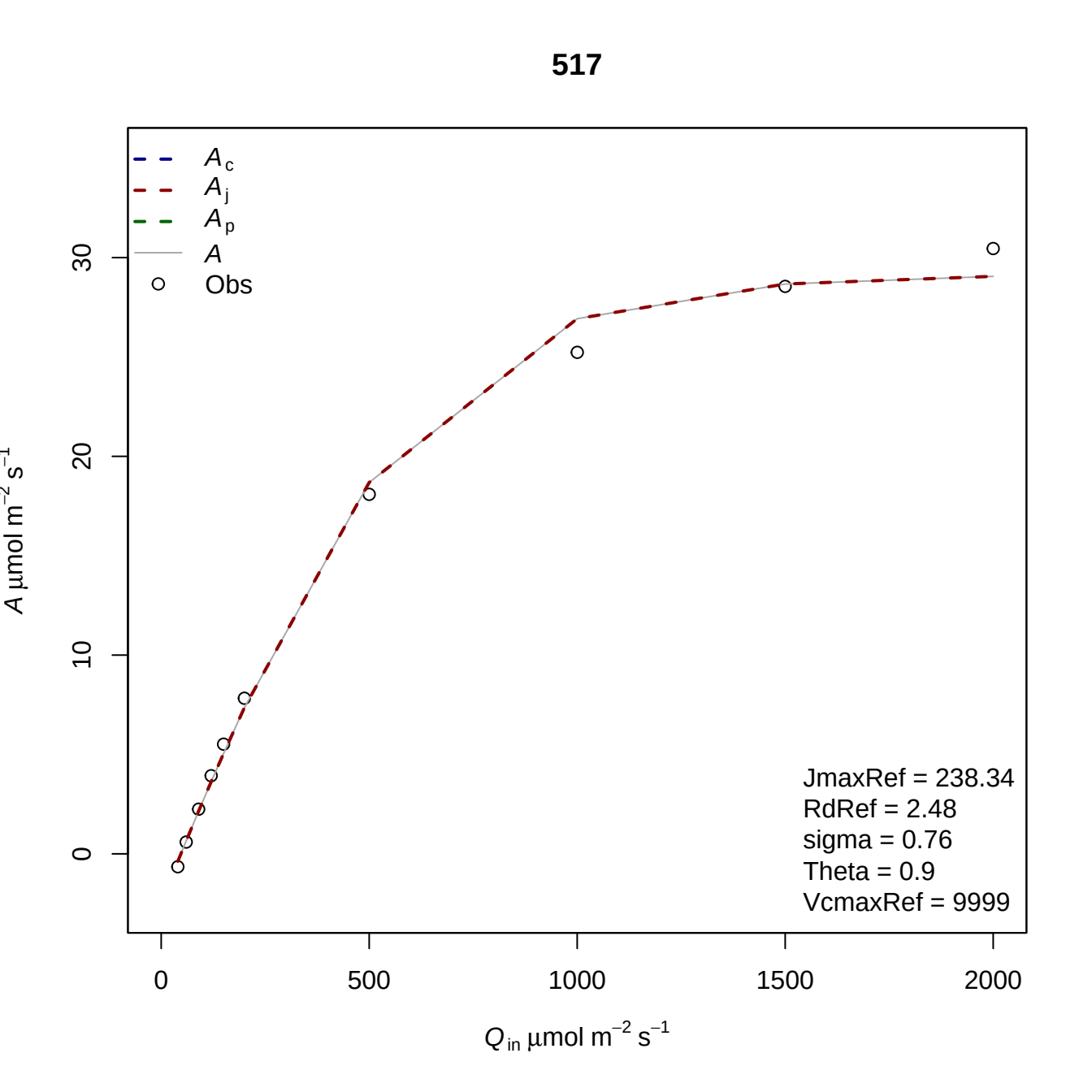

512
A_c
A_j
A_p
A
Obs
JmaxRef = 215.29
RdRef = 2.54
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

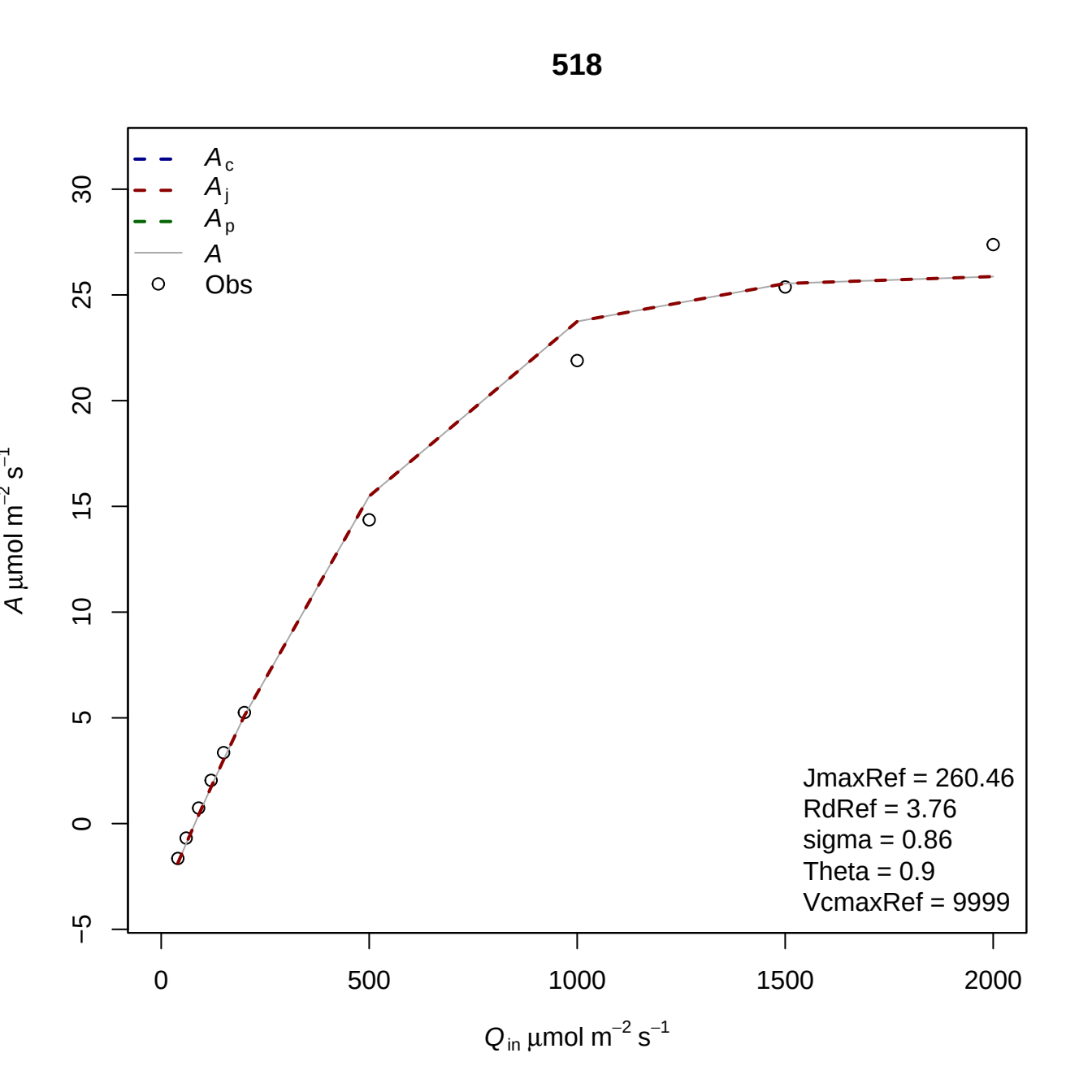

513
A_c
A_j
A_p
A
Obs
JmaxRef = 224.95
RdRef = 3.46
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

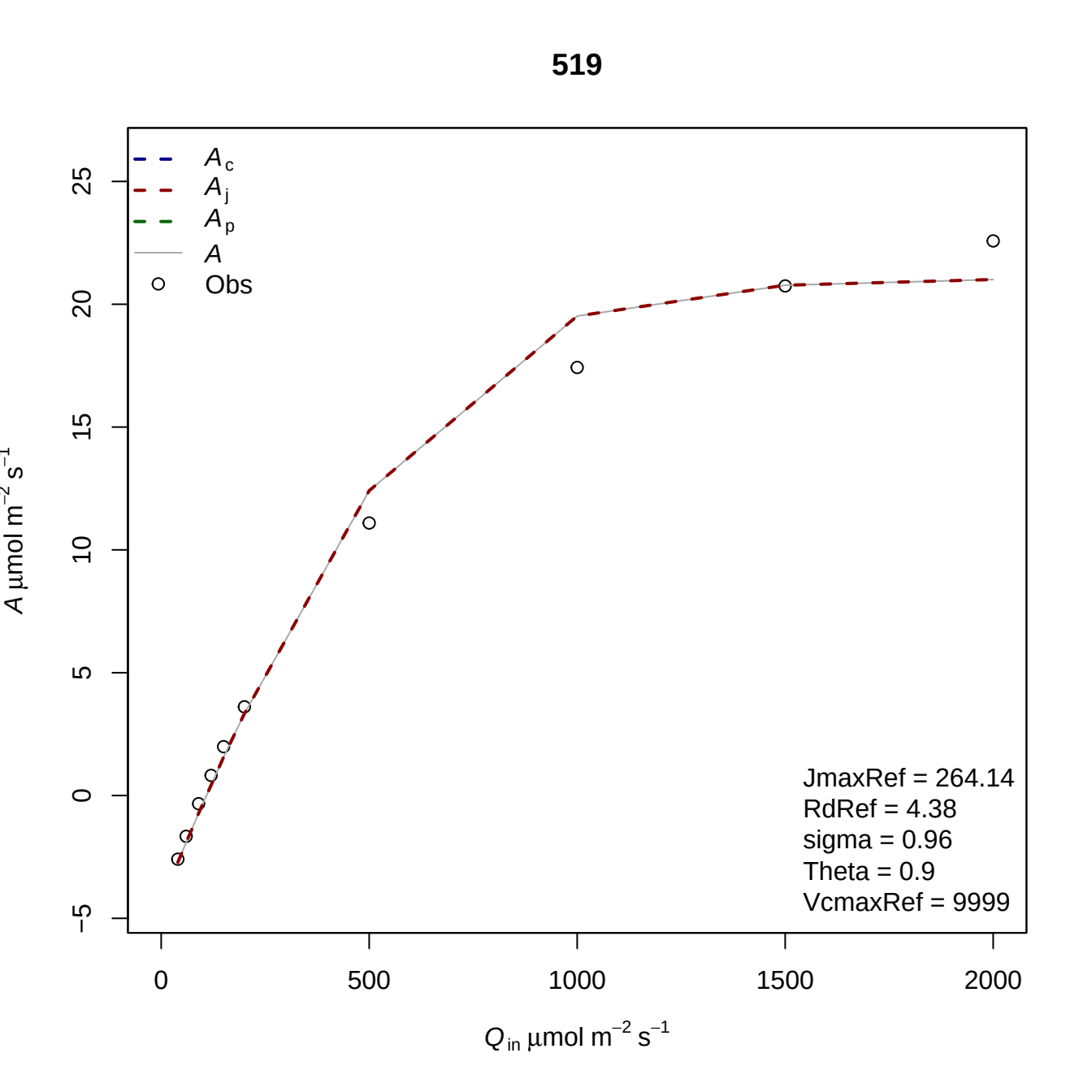

514
A_c
A_j
A_p
A
Obs
JmaxRef = 94.18
RdRef = 2.52
sigma = 1.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

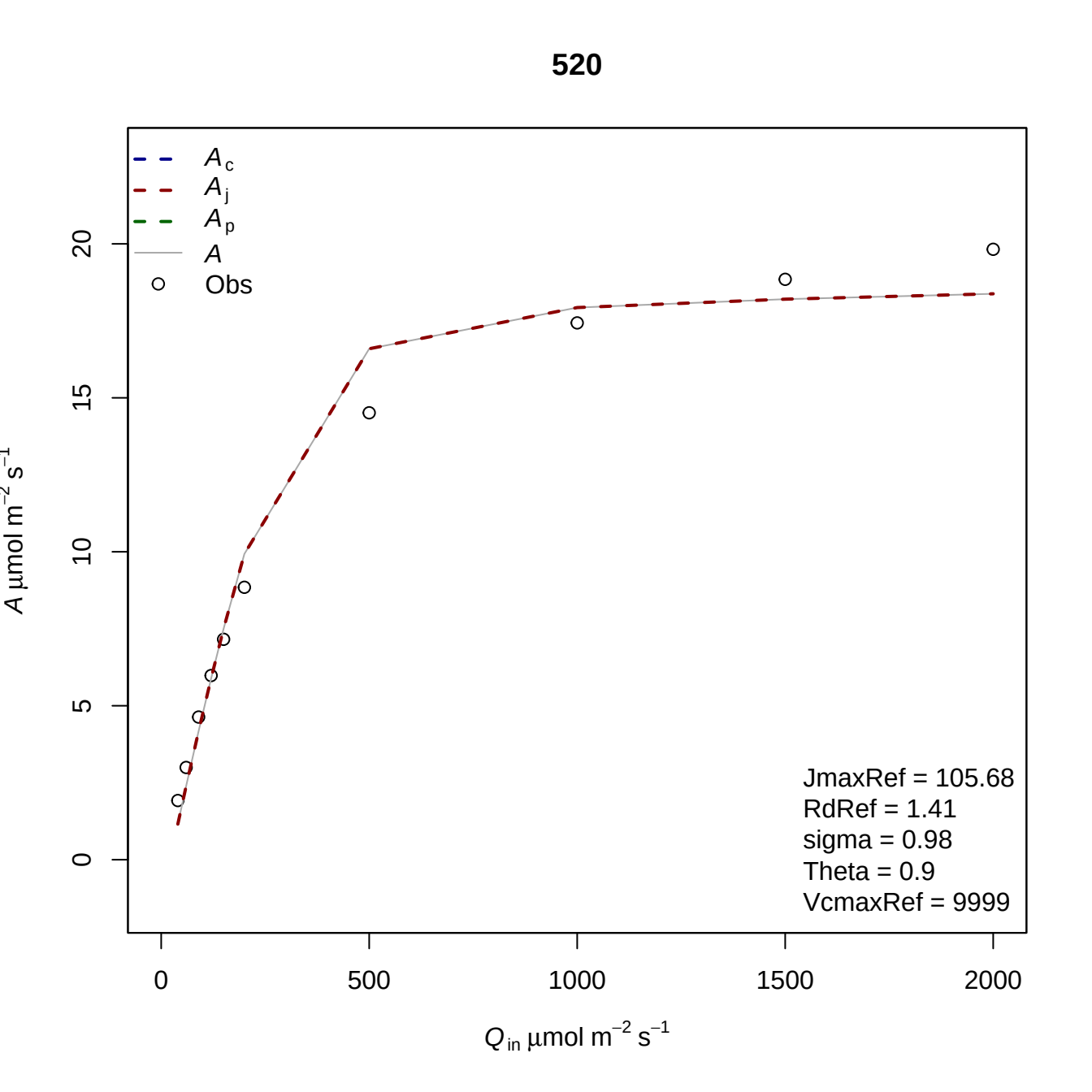

515
A_c
A_j
A_p
A
Obs
JmaxRef = 148.15
RdRef = 3.41
sigma = 1.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

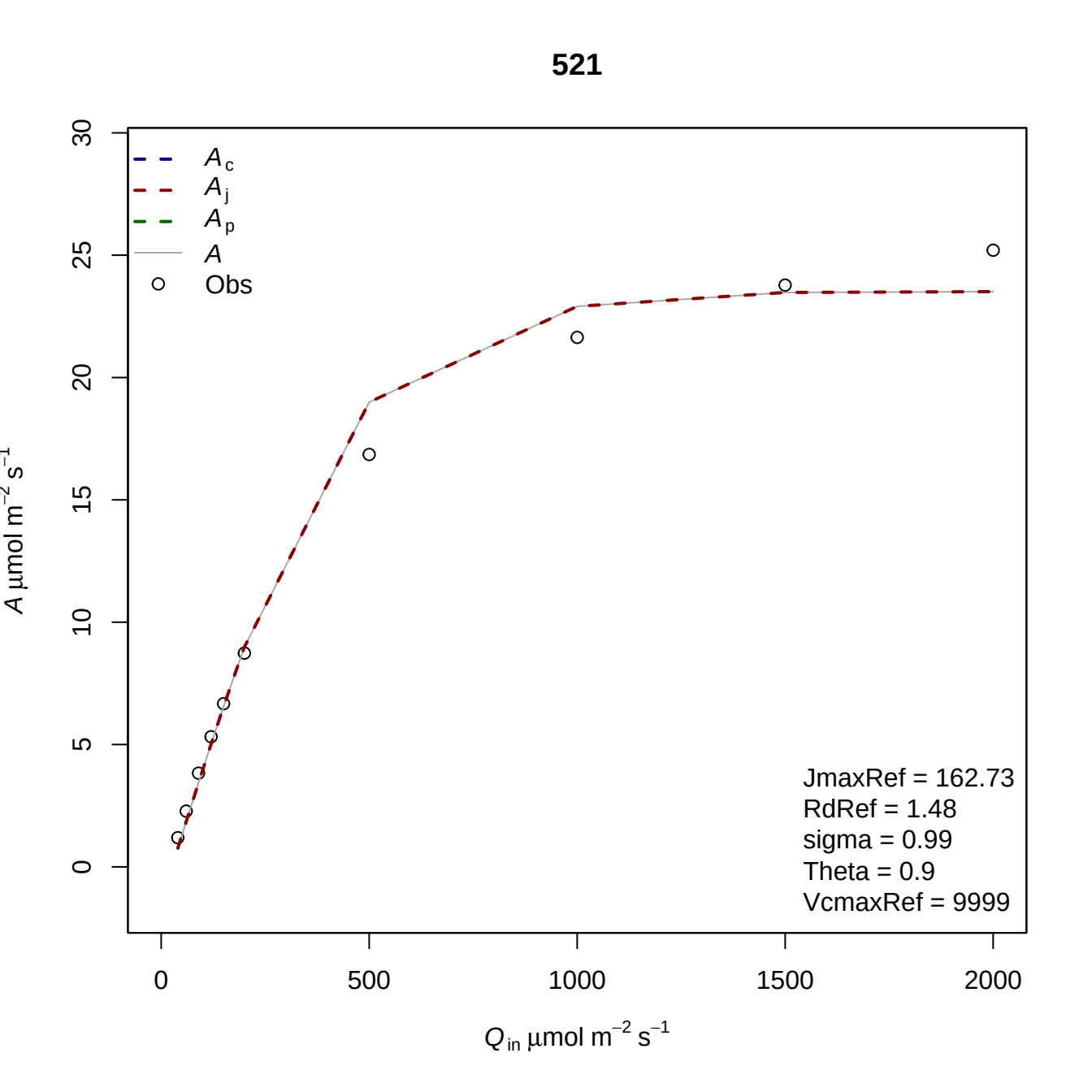

516
A_c
A_j
A_p
A
Obs
JmaxRef = 198.15
RdRef = 1.92
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

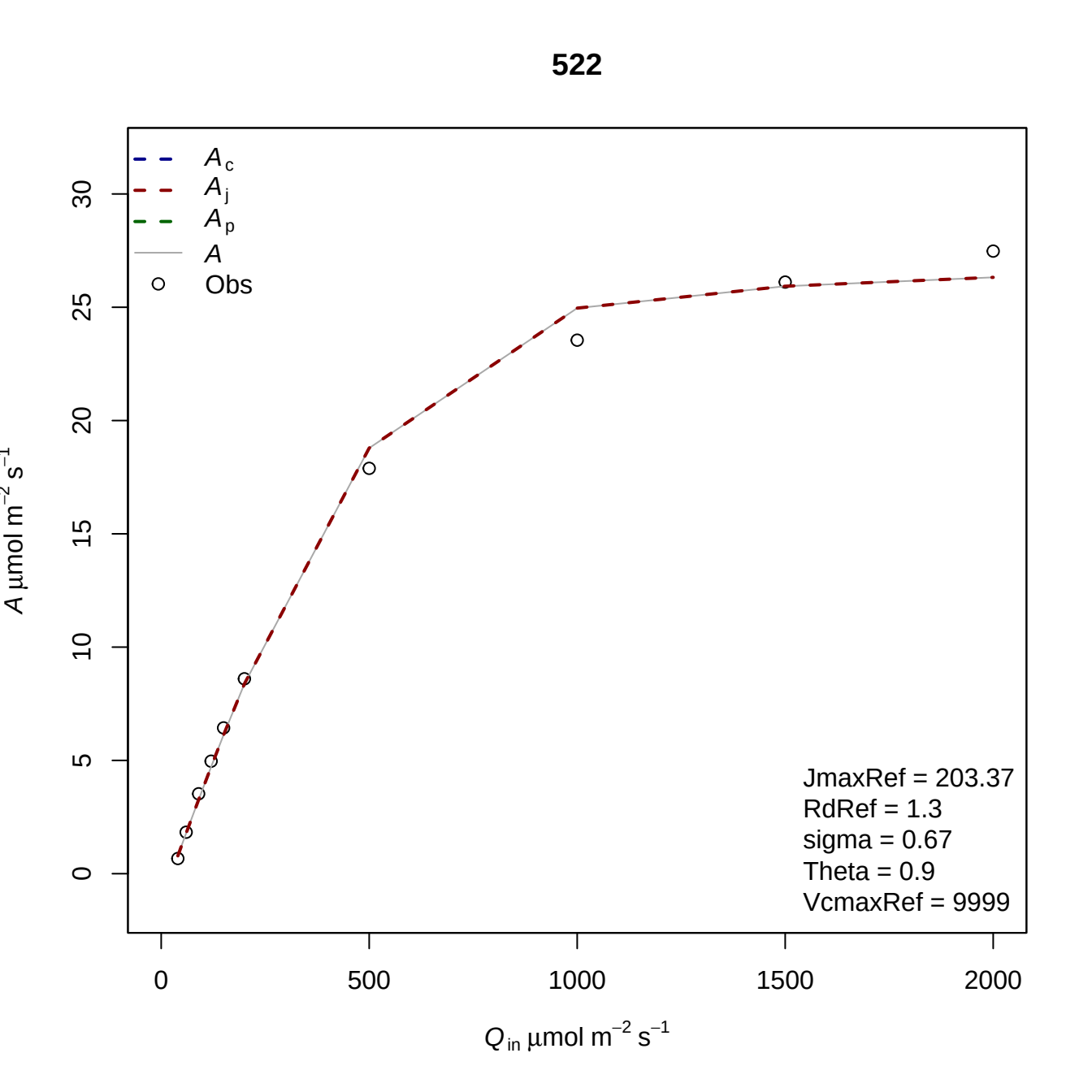

517
A_c
A_j
A_p
A
Obs
JmaxRef = 238.34
RdRef = 2.48
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

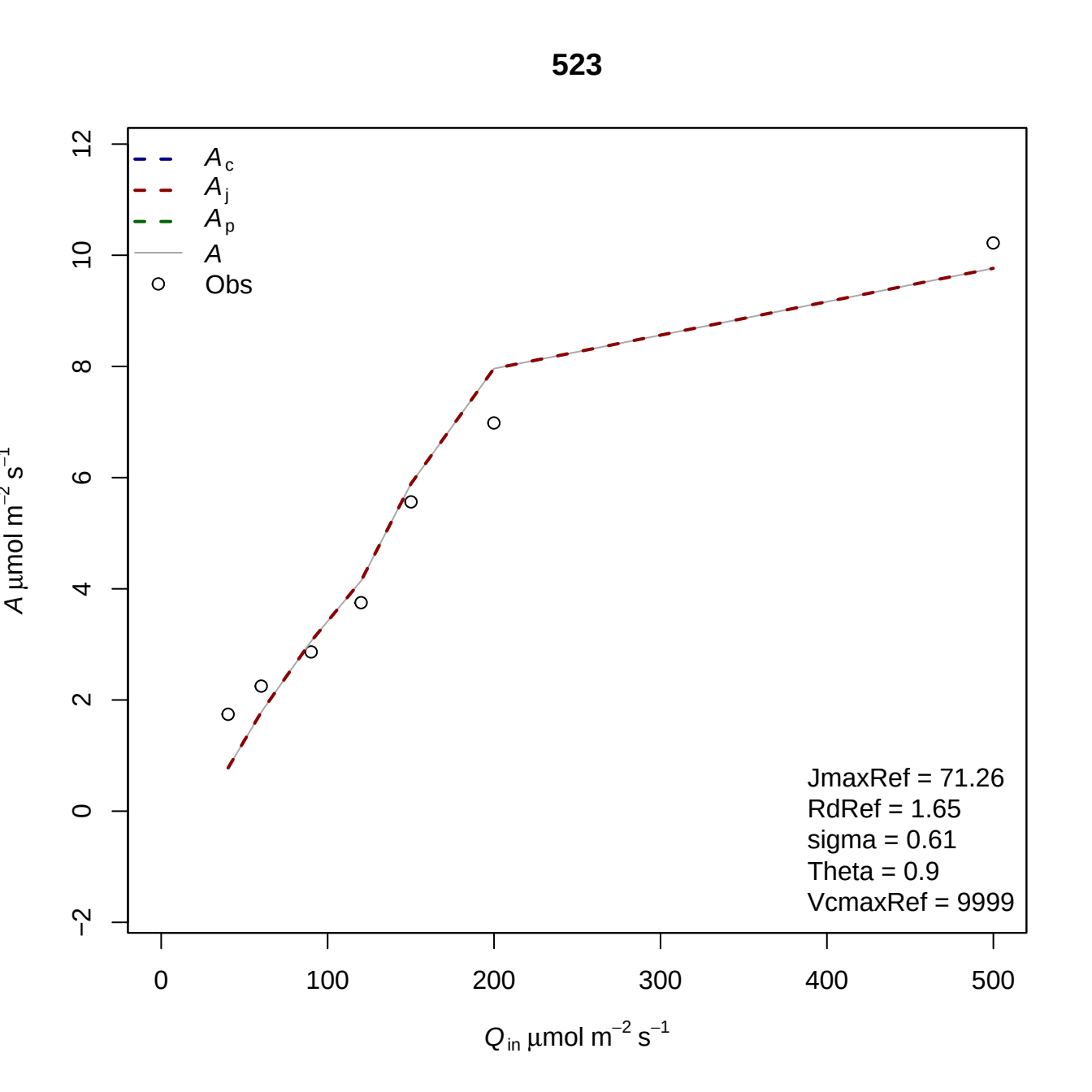

518
A_c
A_j
A_p
A
Obs
JmaxRef = 260.46
RdRef = 3.76
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

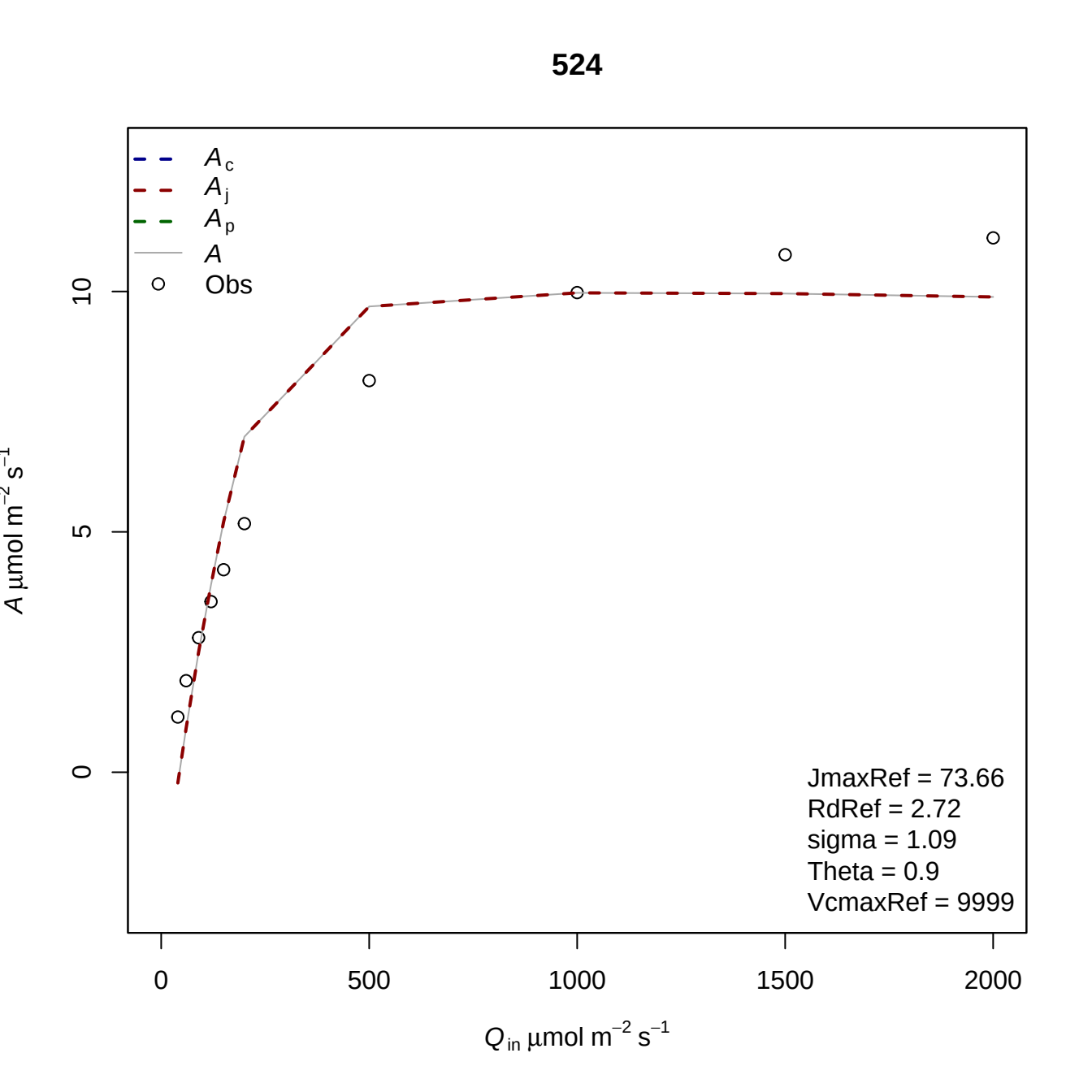

519
A_c
A_j
A_p
A
Obs
JmaxRef = 264.14
RdRef = 4.38
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

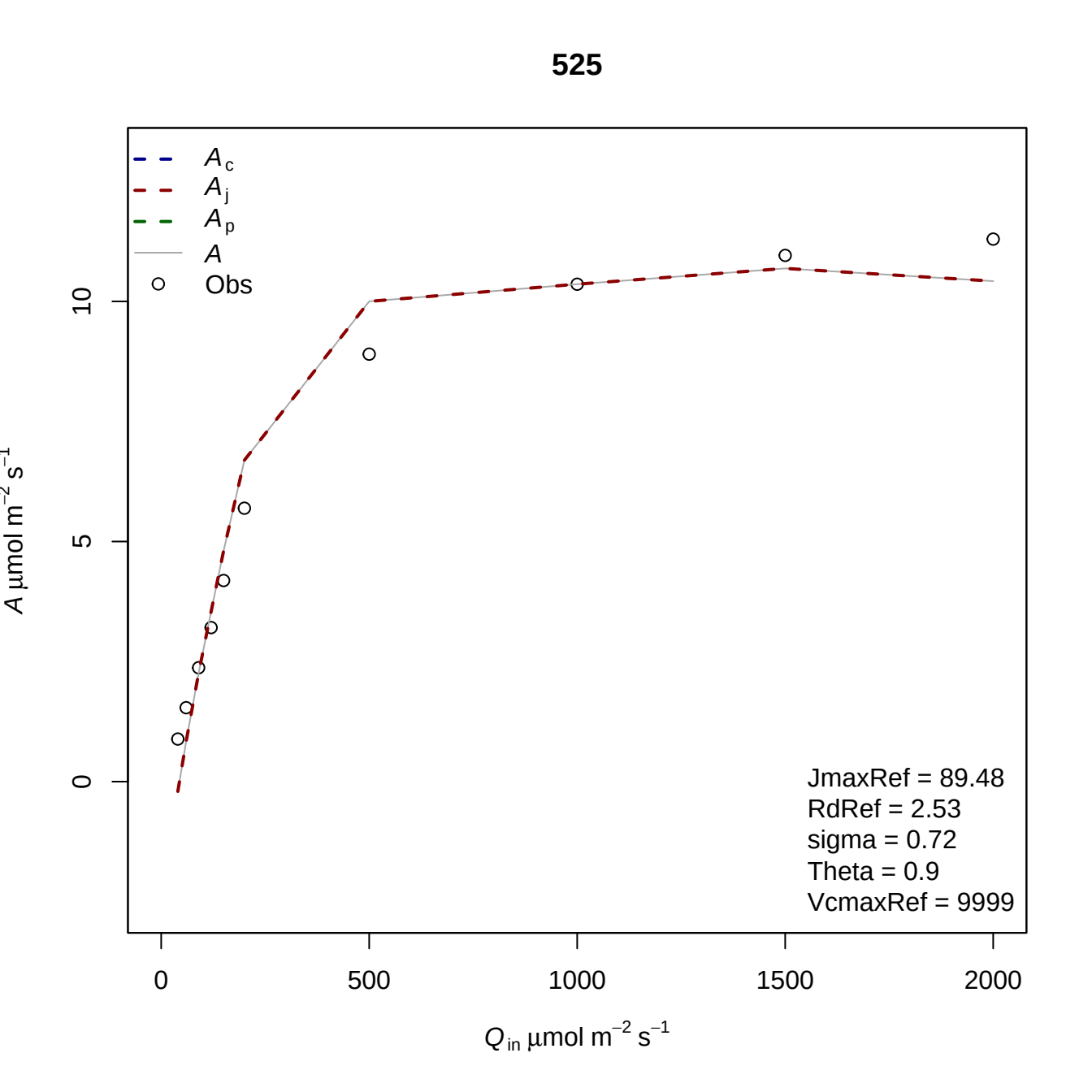

520
A_c
A_j
A_p
A
Obs
JmaxRef = 105.68
RdRef = 1.41
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

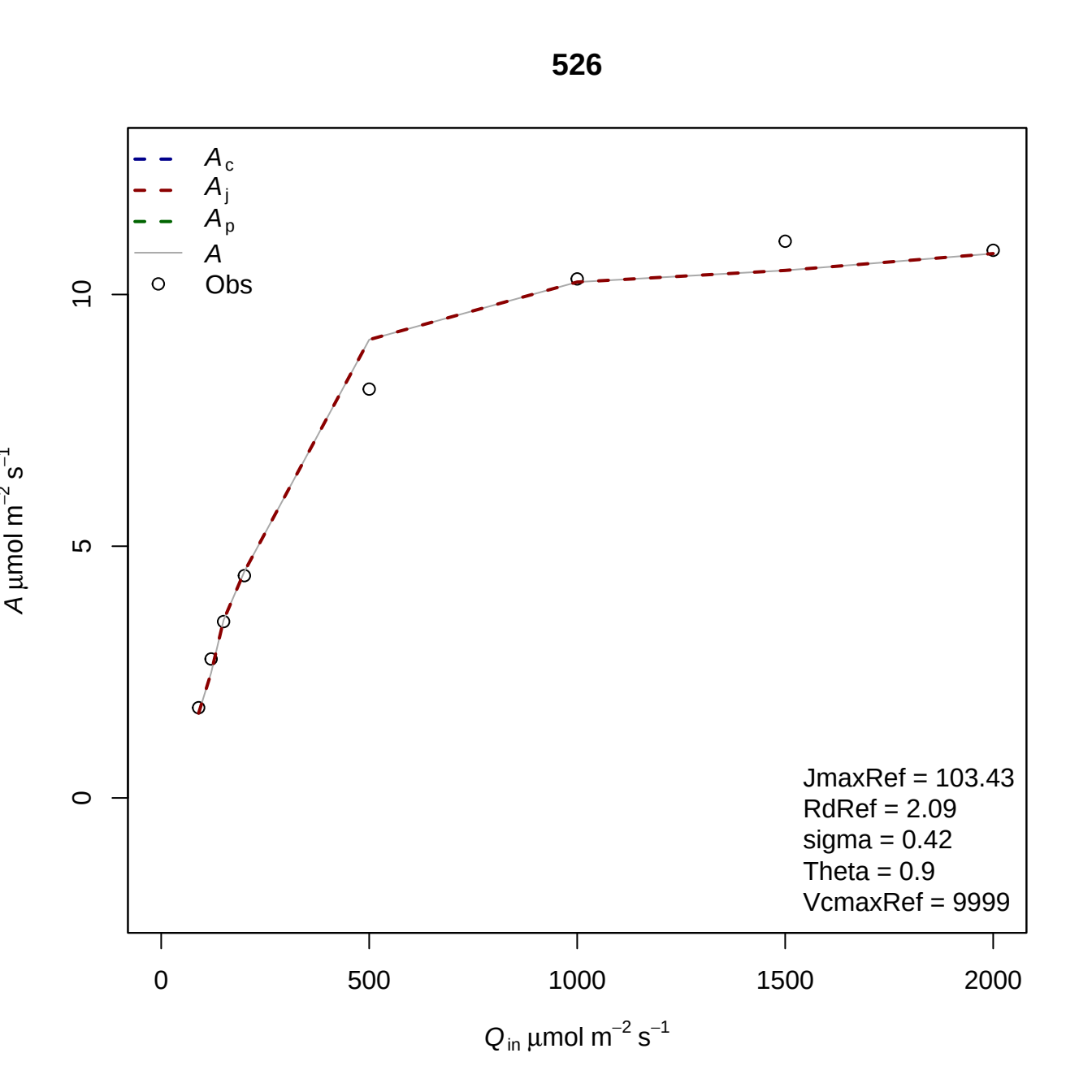

521
A_c
A_j
A_p
A
Obs
JmaxRef = 162.73
RdRef = 1.48
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

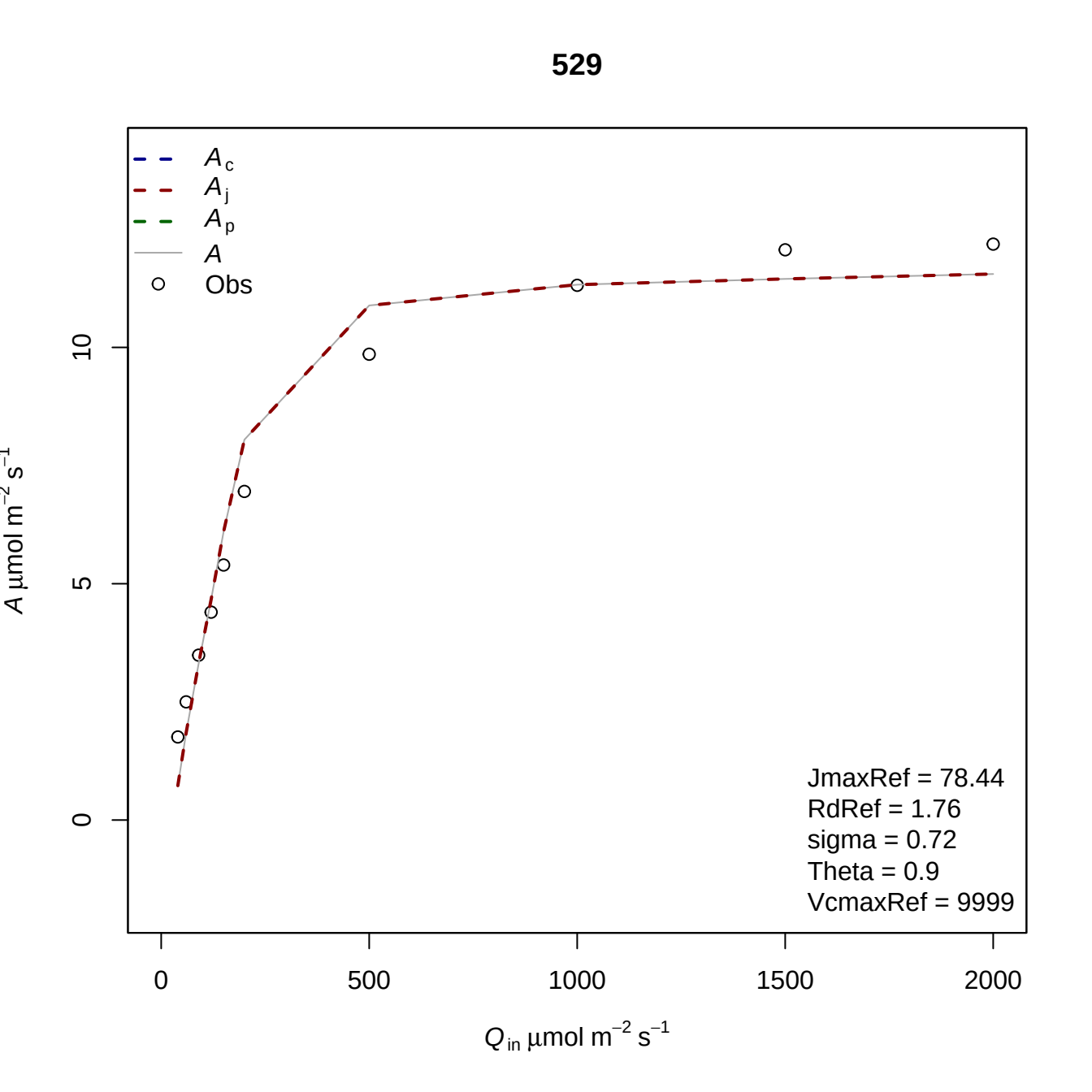

522
A_c
A_j
A_p
A
Obs
JmaxRef = 203.37
RdRef = 1.3
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

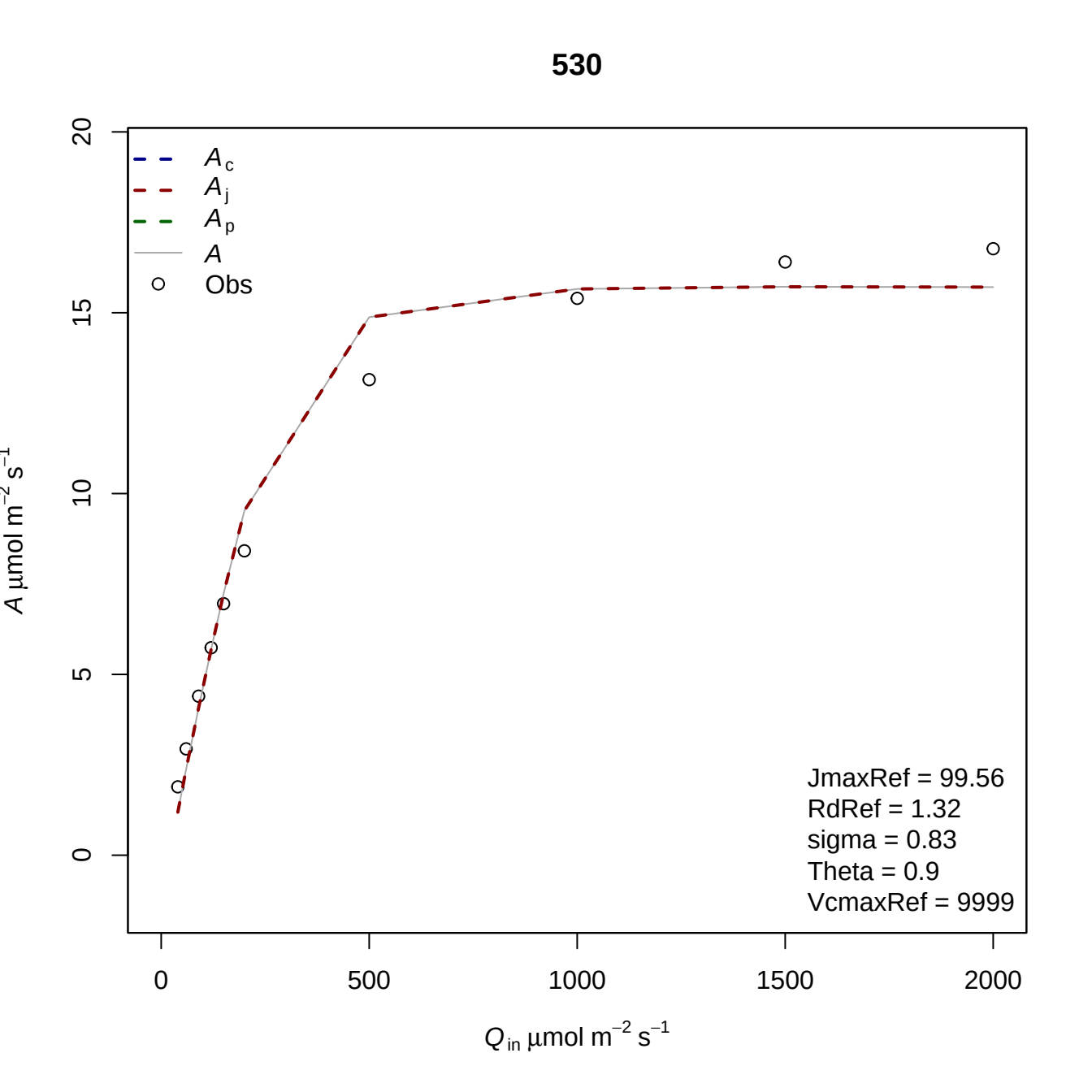

523
A_c
A_j
A_p
A
Obs
JmaxRef = 71.26
RdRef = 1.65
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
100
200
300
400
500

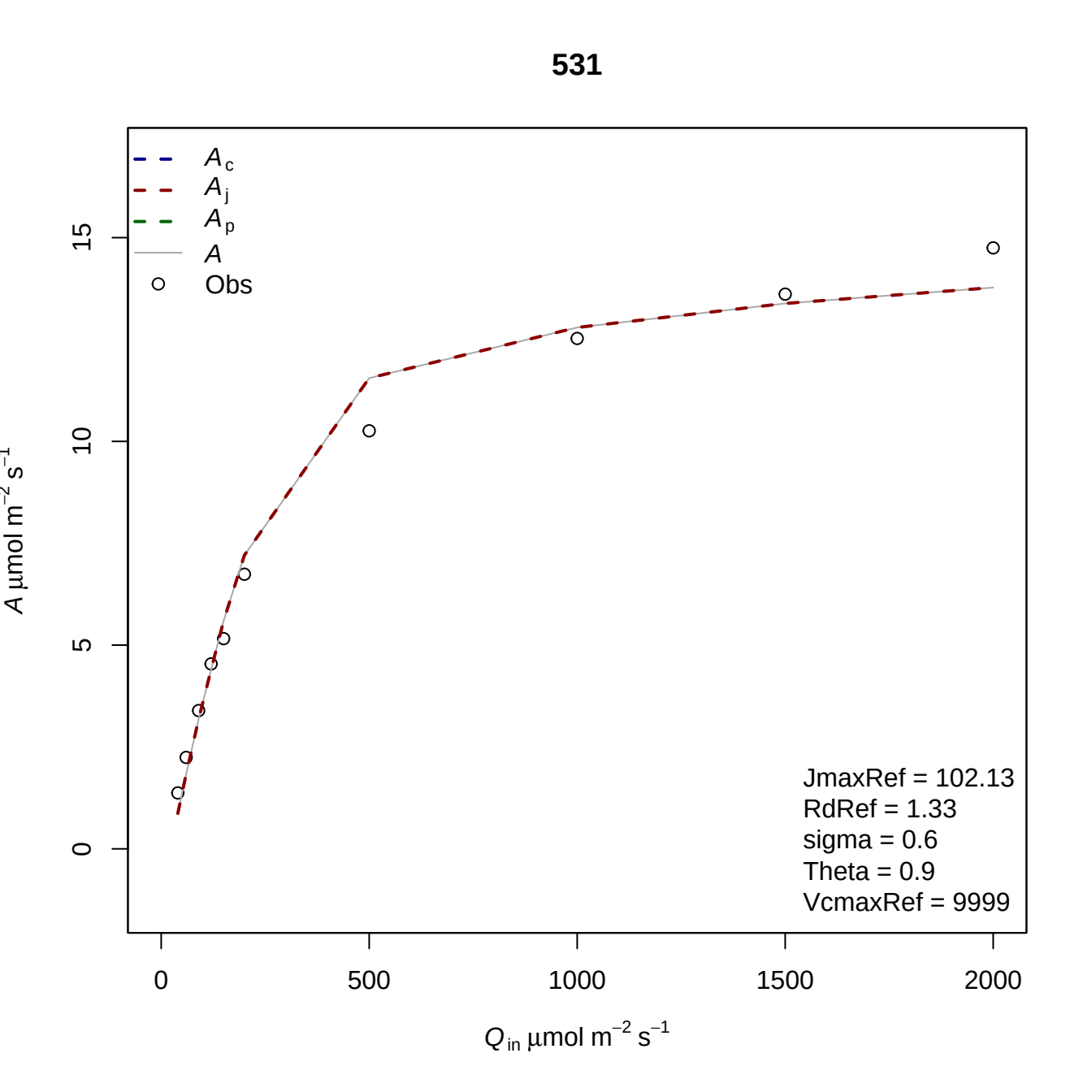

524
A_c
A_j
A_p
A
Obs
JmaxRef = 73.66
RdRef = 2.72
sigma = 1.09
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

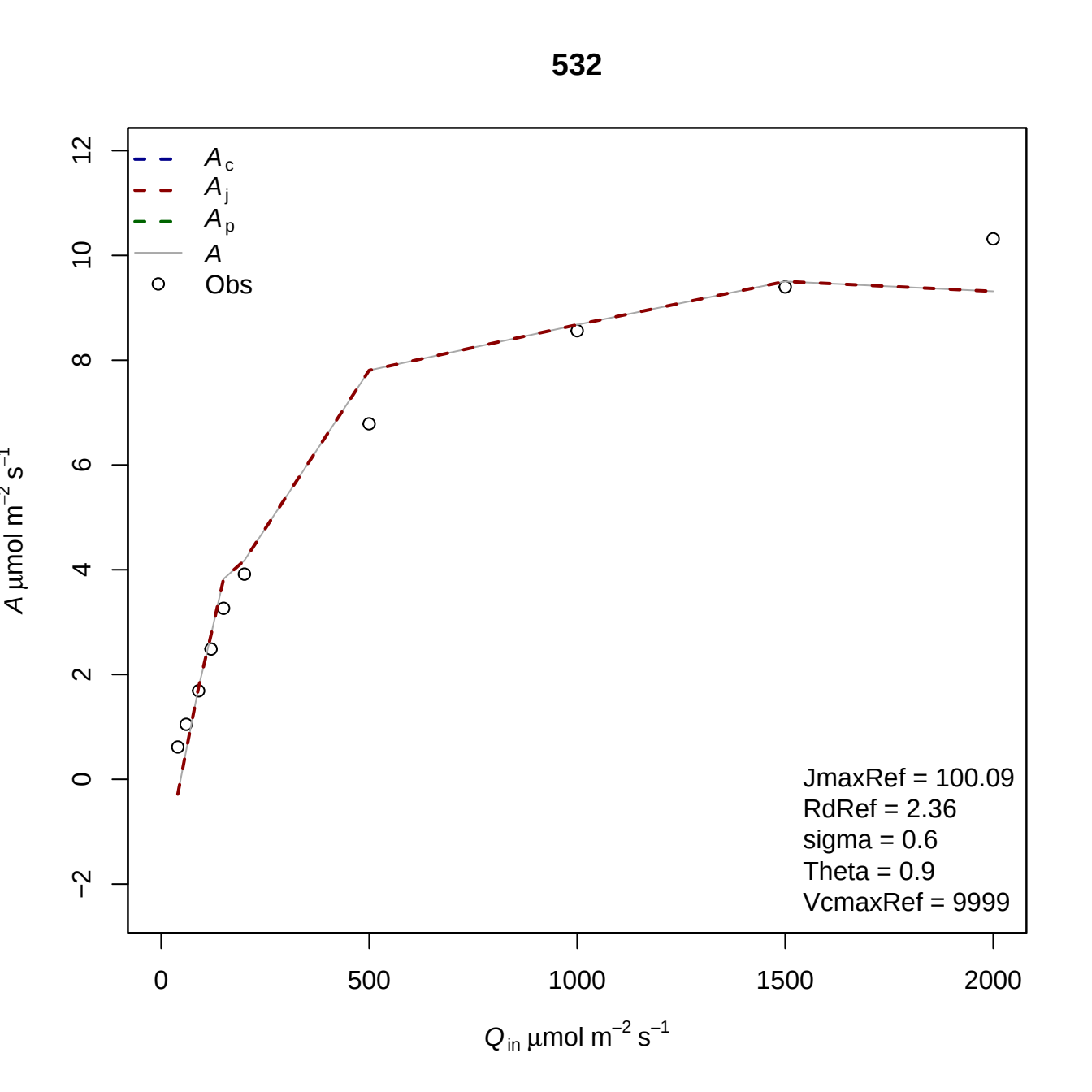

525
A_c
A_j
A_p
A
Obs
JmaxRef = 89.48
RdRef = 2.53
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

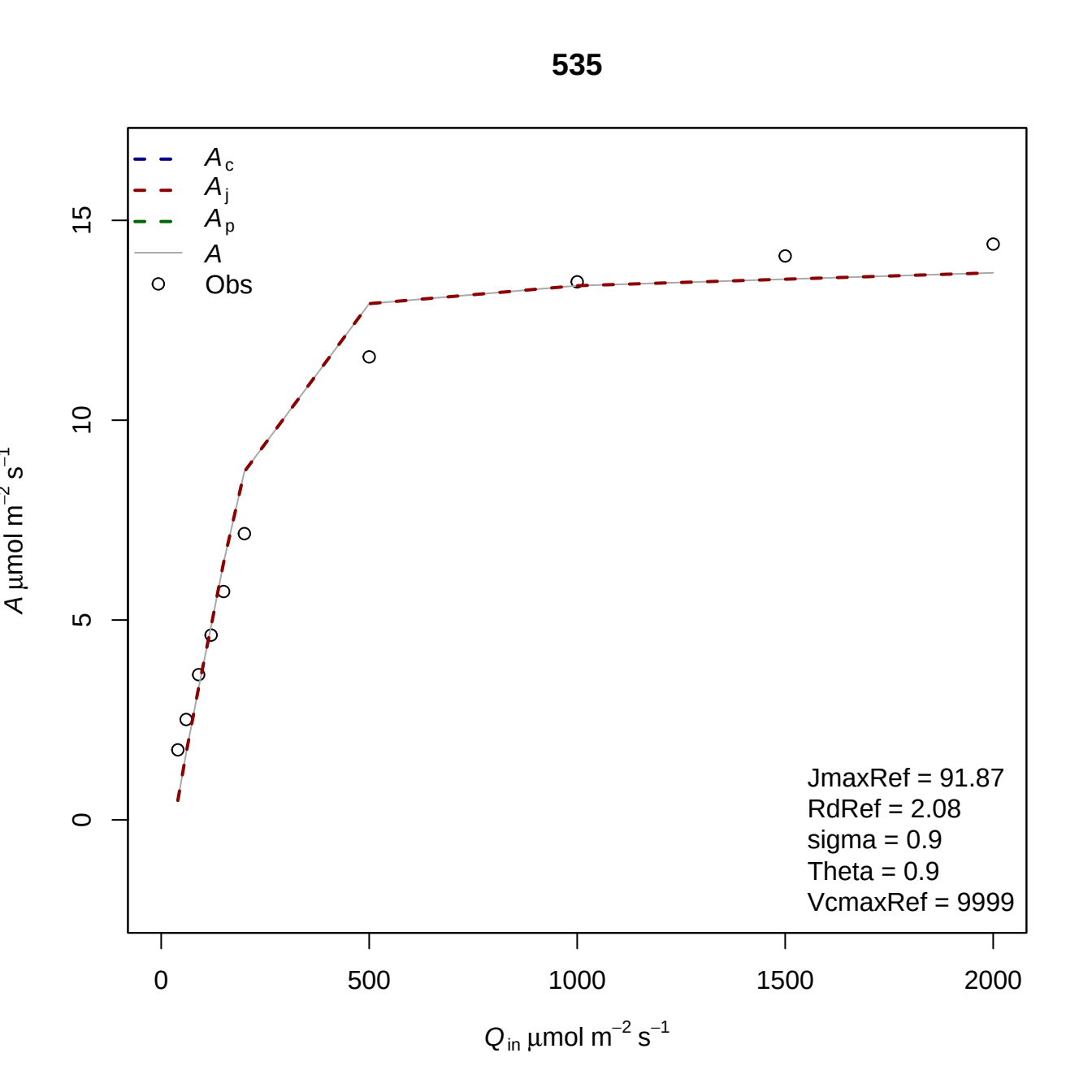

526
A_c
A_j
A_p
A
Obs
JmaxRef = 103.43
RdRef = 2.09
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

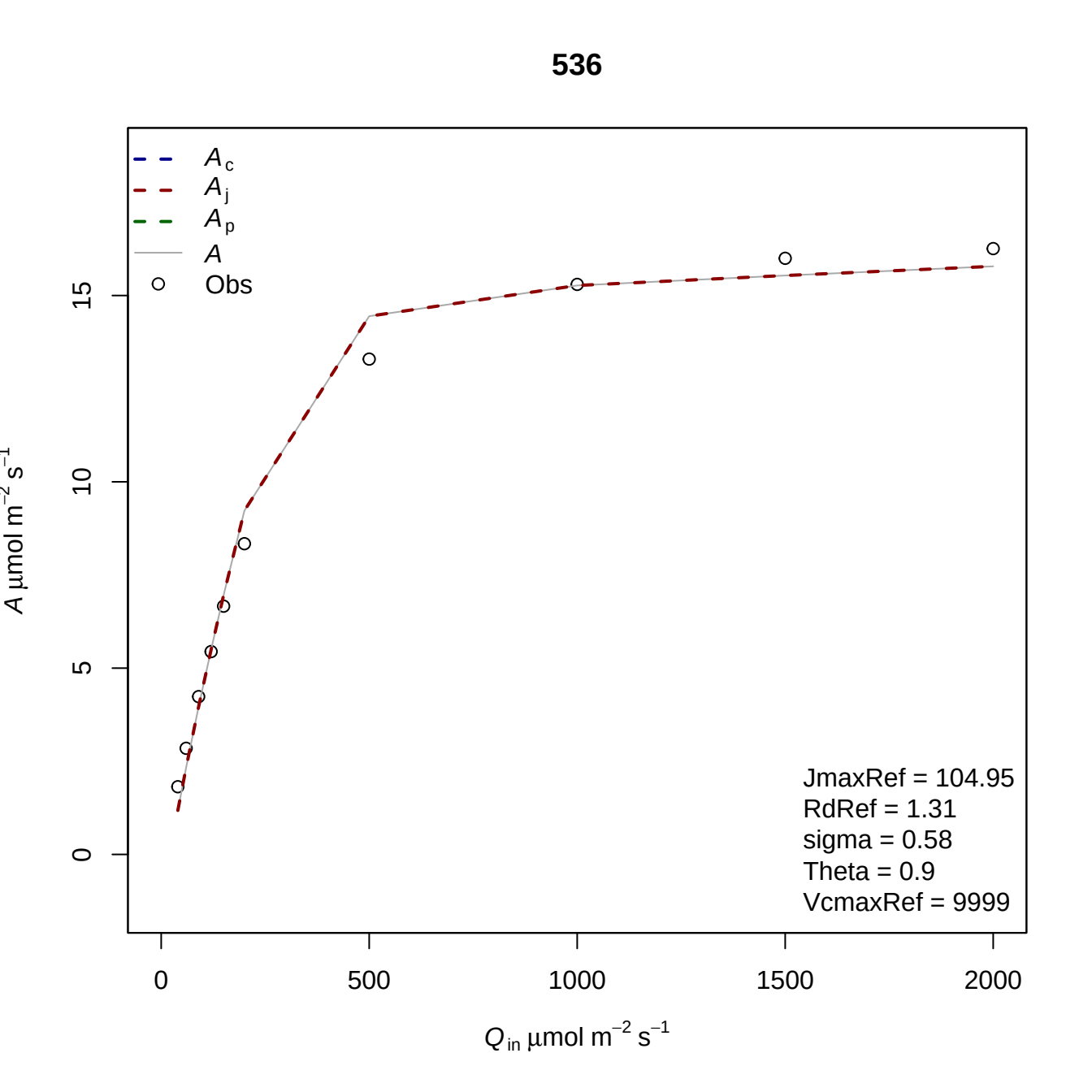

529
A_c
A_j
A_p
A
Obs
JmaxRef = 78.44
RdRef = 1.76
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

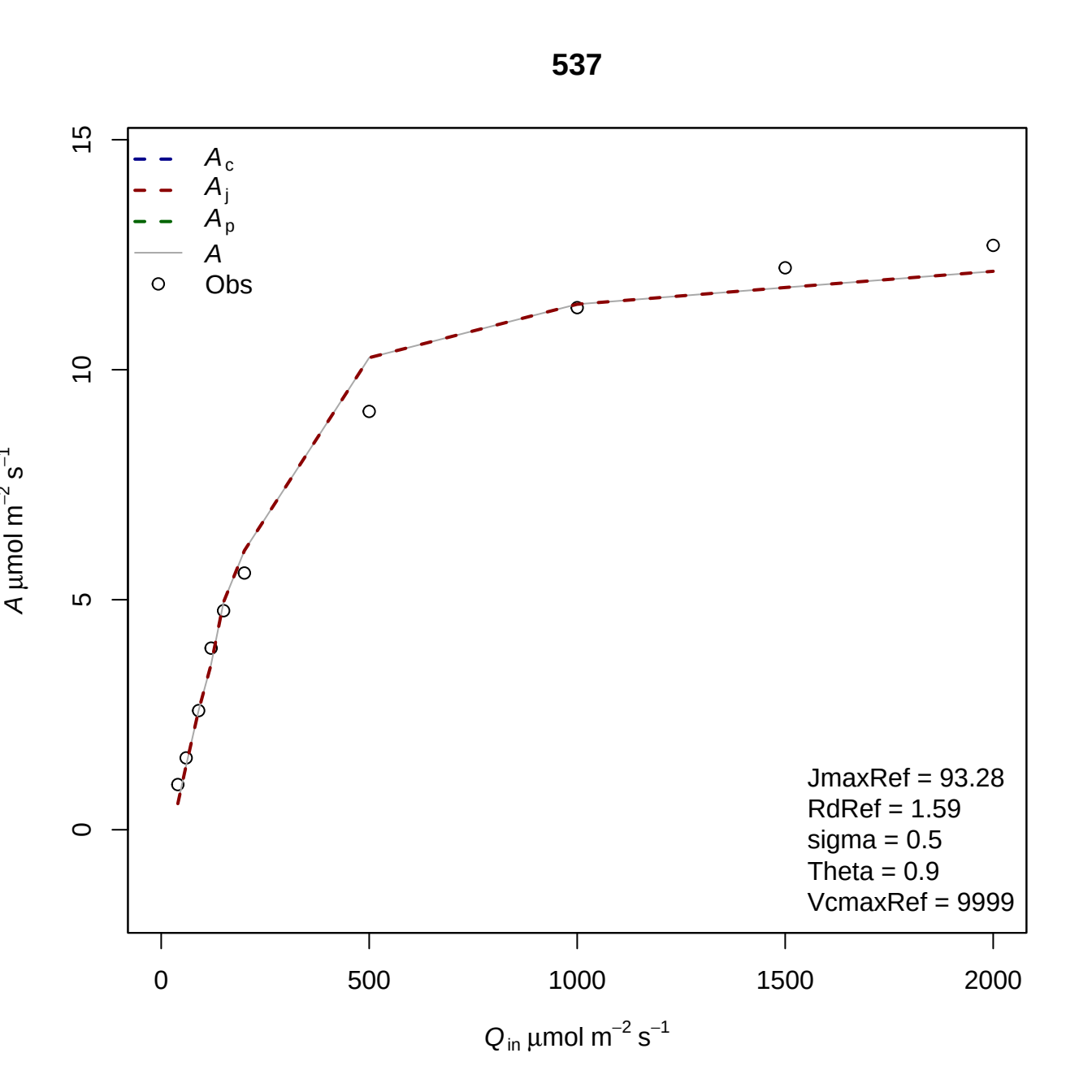

530
A_c
A_j
A_p
A
Obs
JmaxRef = 99.56
RdRef = 1.32
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

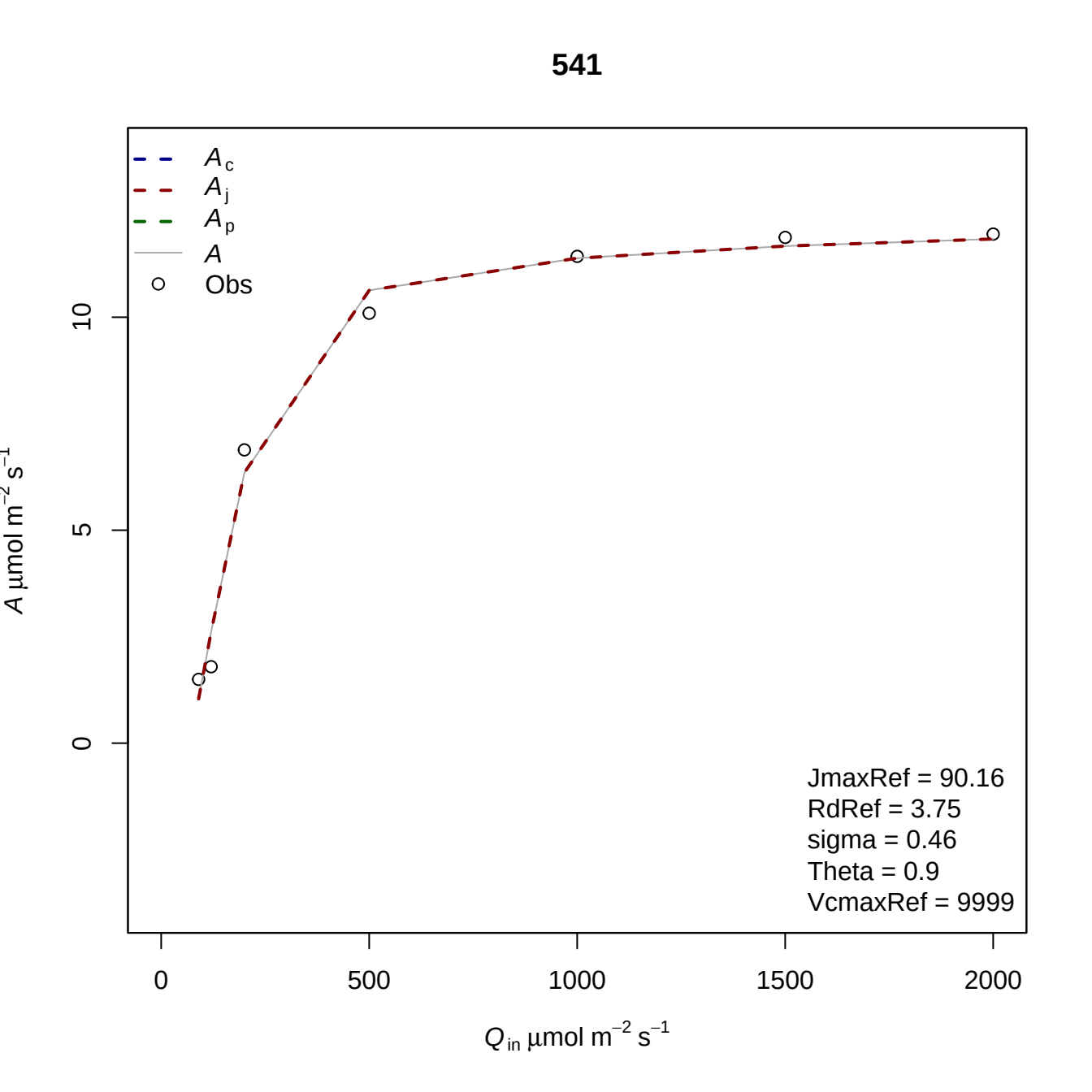

531
A_c
A_j
A_p
A
Obs
JmaxRef = 102.13
RdRef = 1.33
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

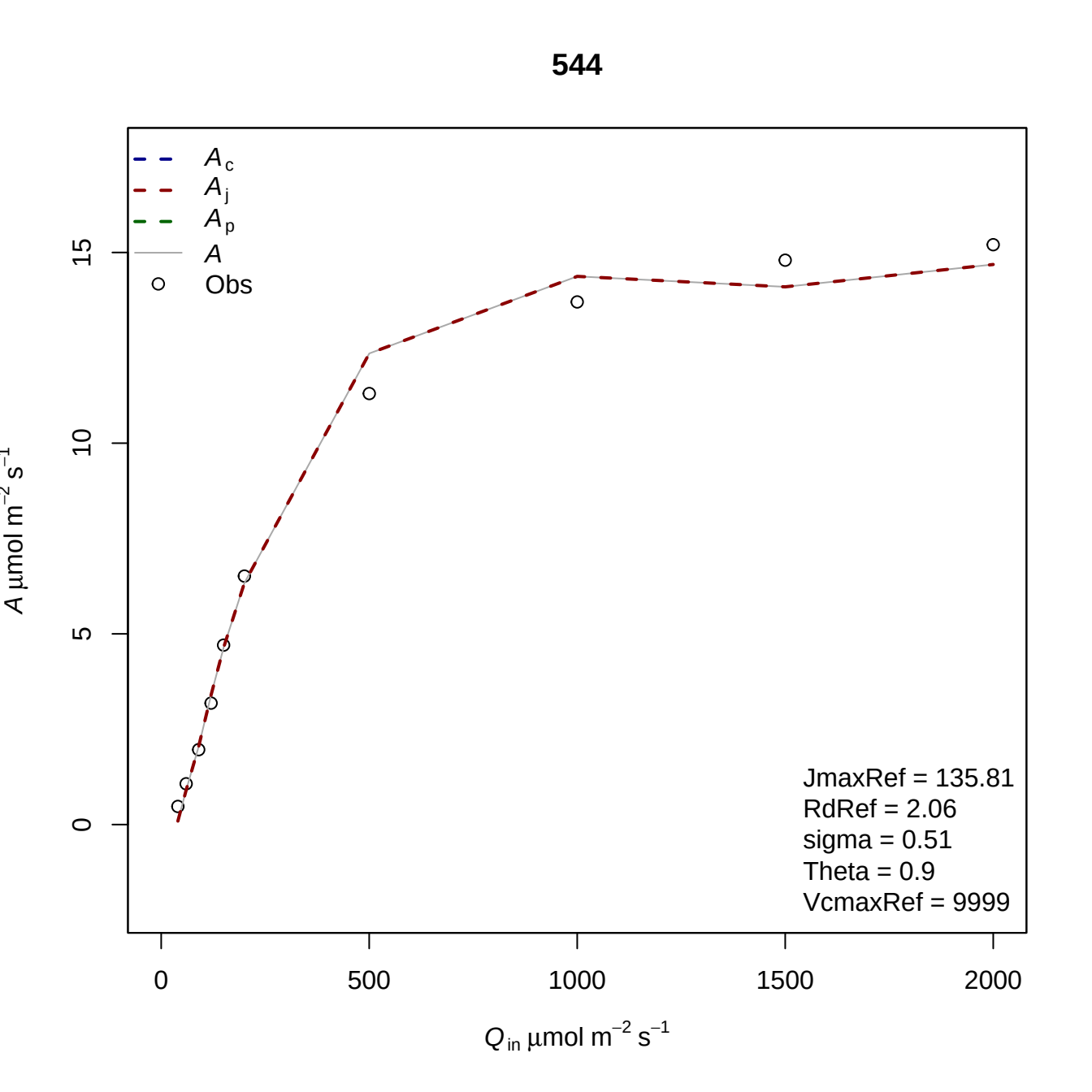

532
A_c
A_j
A_p
A
Obs
JmaxRef = 100.09
RdRef = 2.36
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

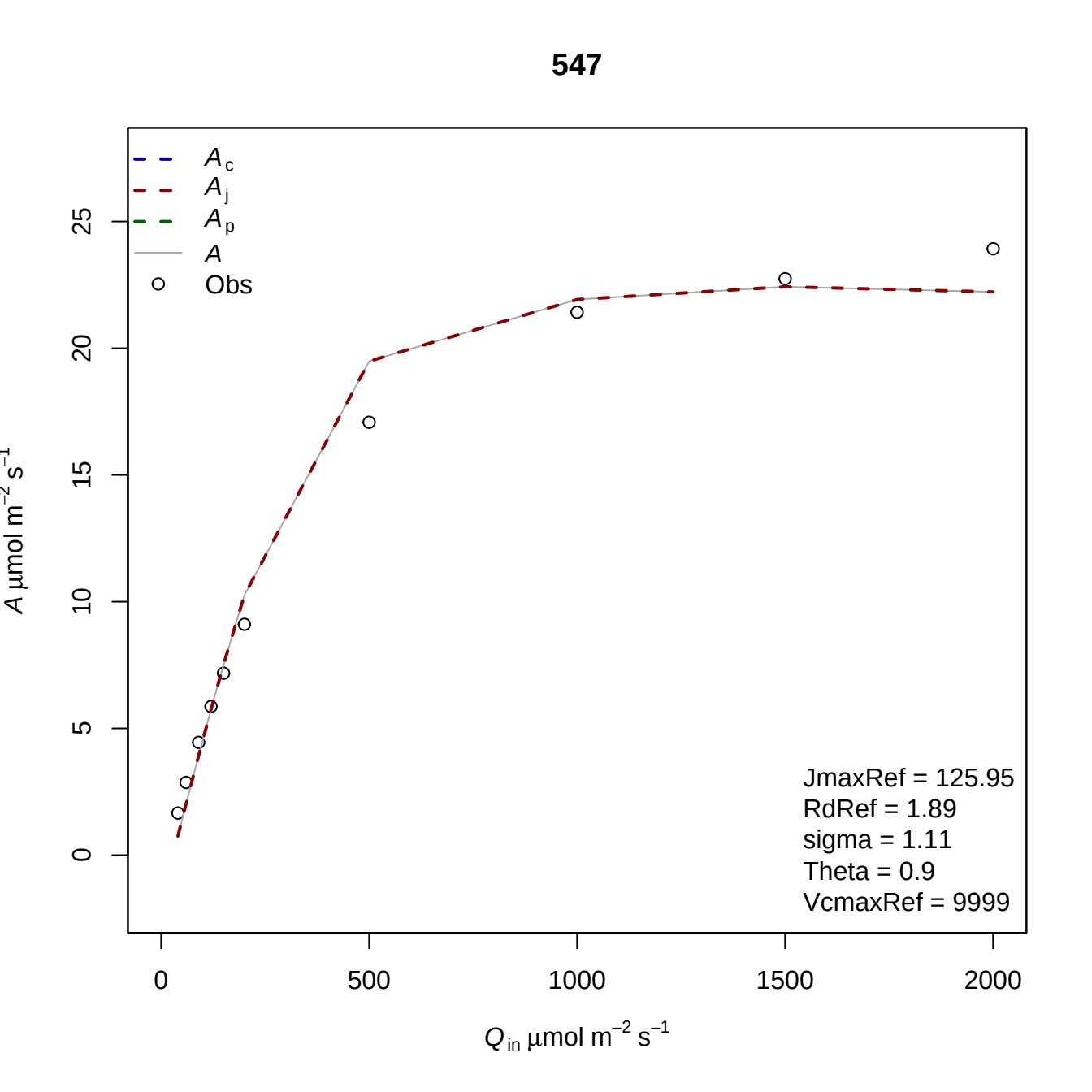

535
A_c
A_j
A_p
A
Obs
JmaxRef = 91.87
RdRef = 2.08
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

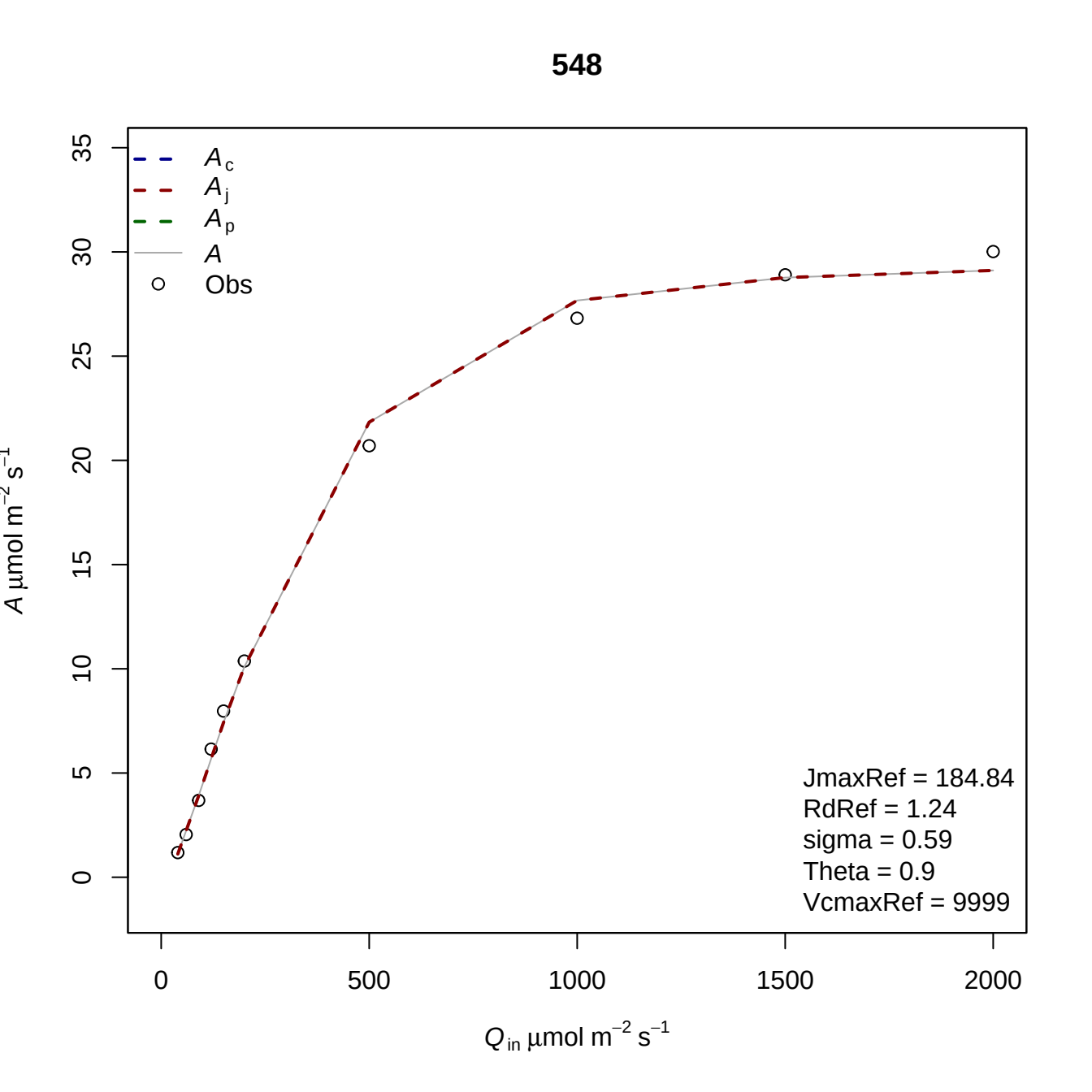

536
A_c
A_j
A_p
A
Obs
JmaxRef = 104.95
RdRef = 1.31
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

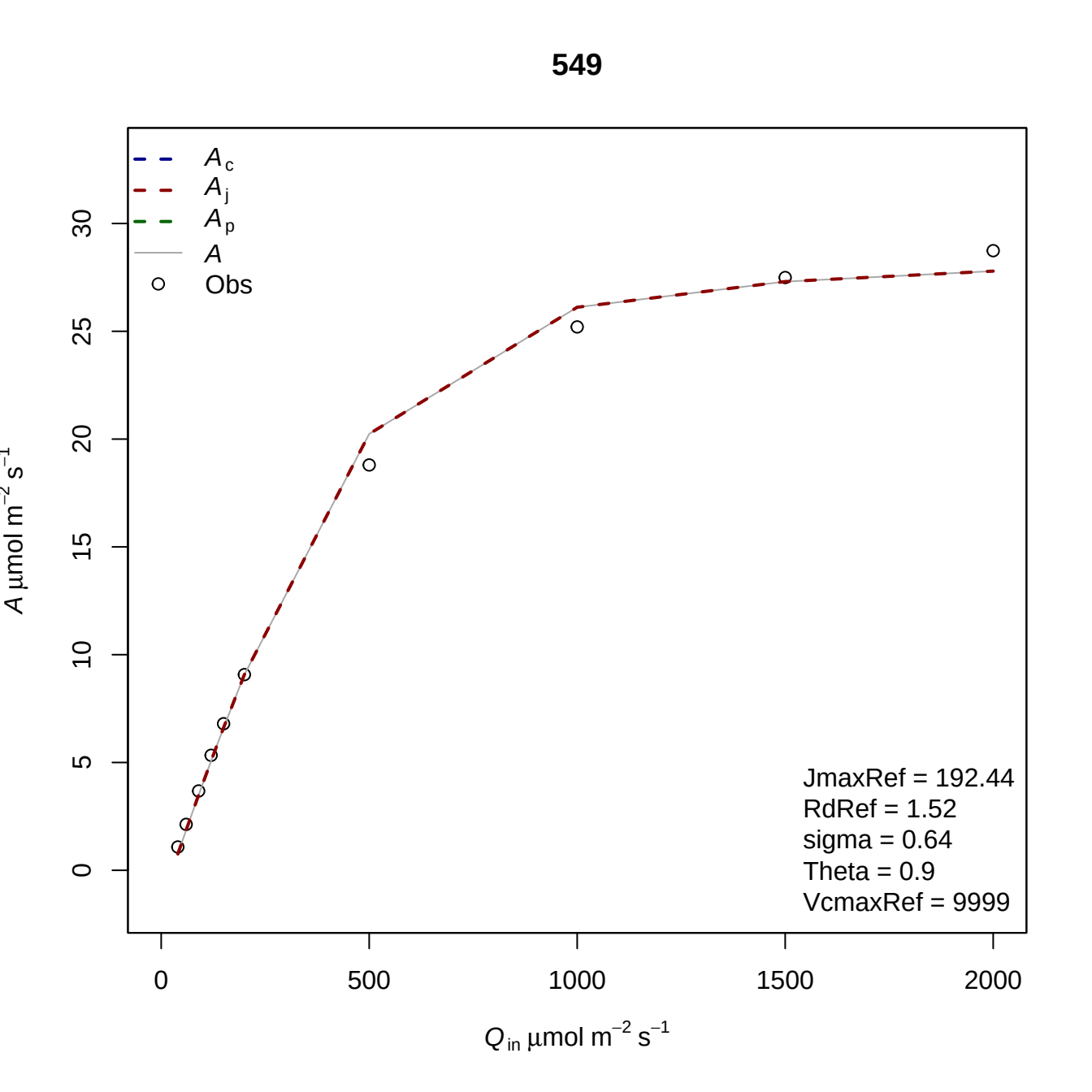

537
A_c
A_j
A_p
A
Obs
JmaxRef = 93.28
RdRef = 1.59
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

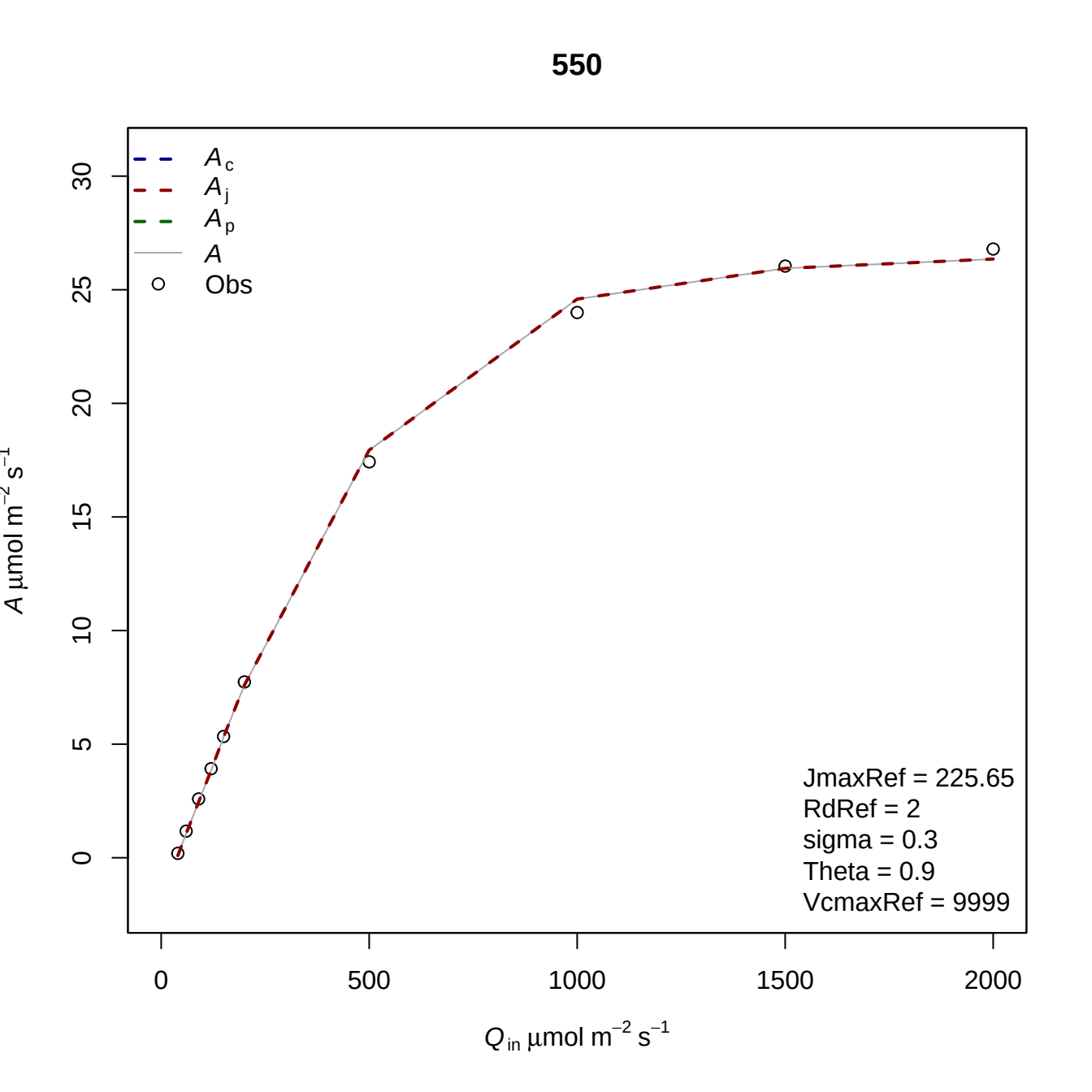

541
A_c
A_j
A_p
A
Obs
JmaxRef = 90.16
RdRef = 3.75
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

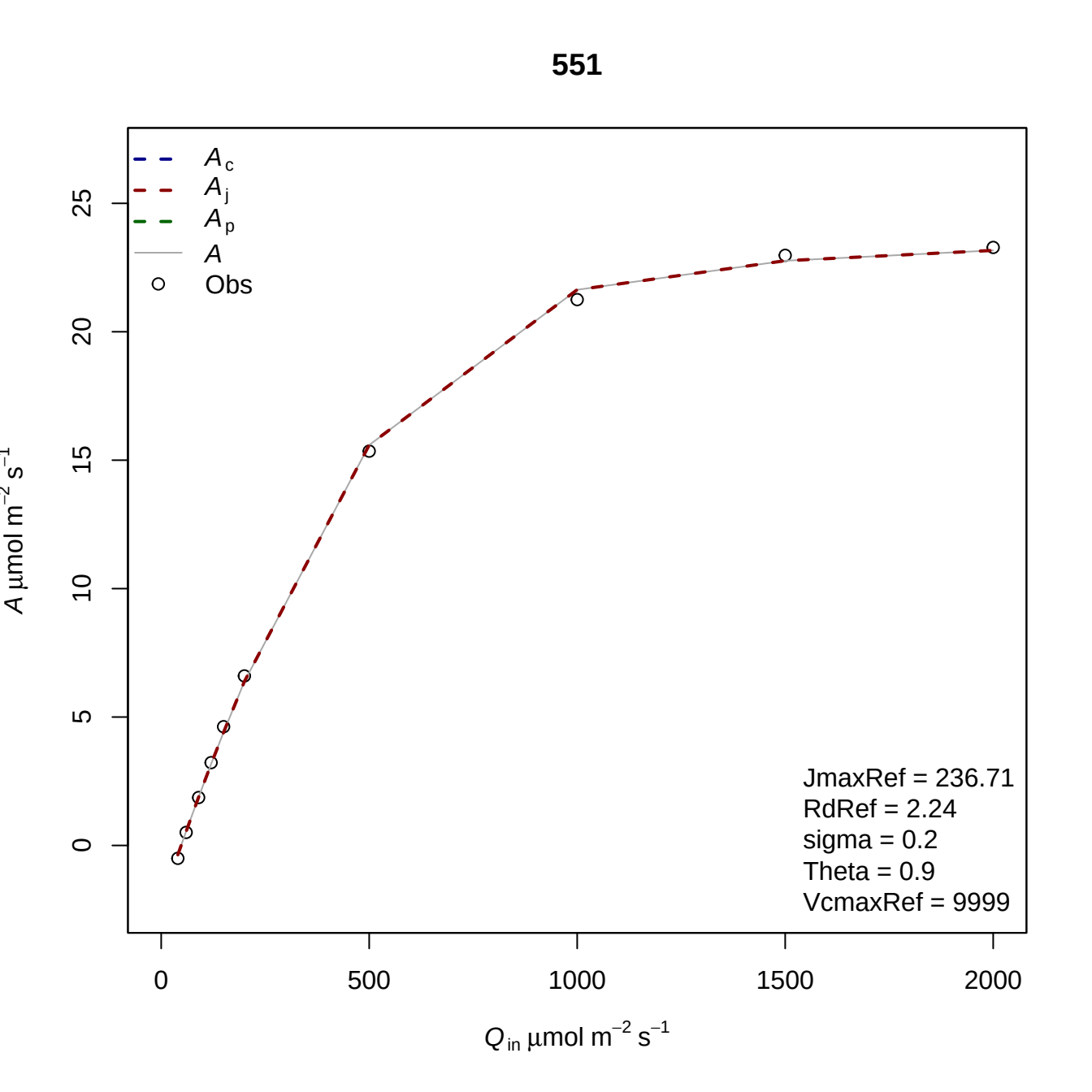

544
A_c
A_j
A_p
A
Obs
JmaxRef = 135.81
RdRef = 2.06
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

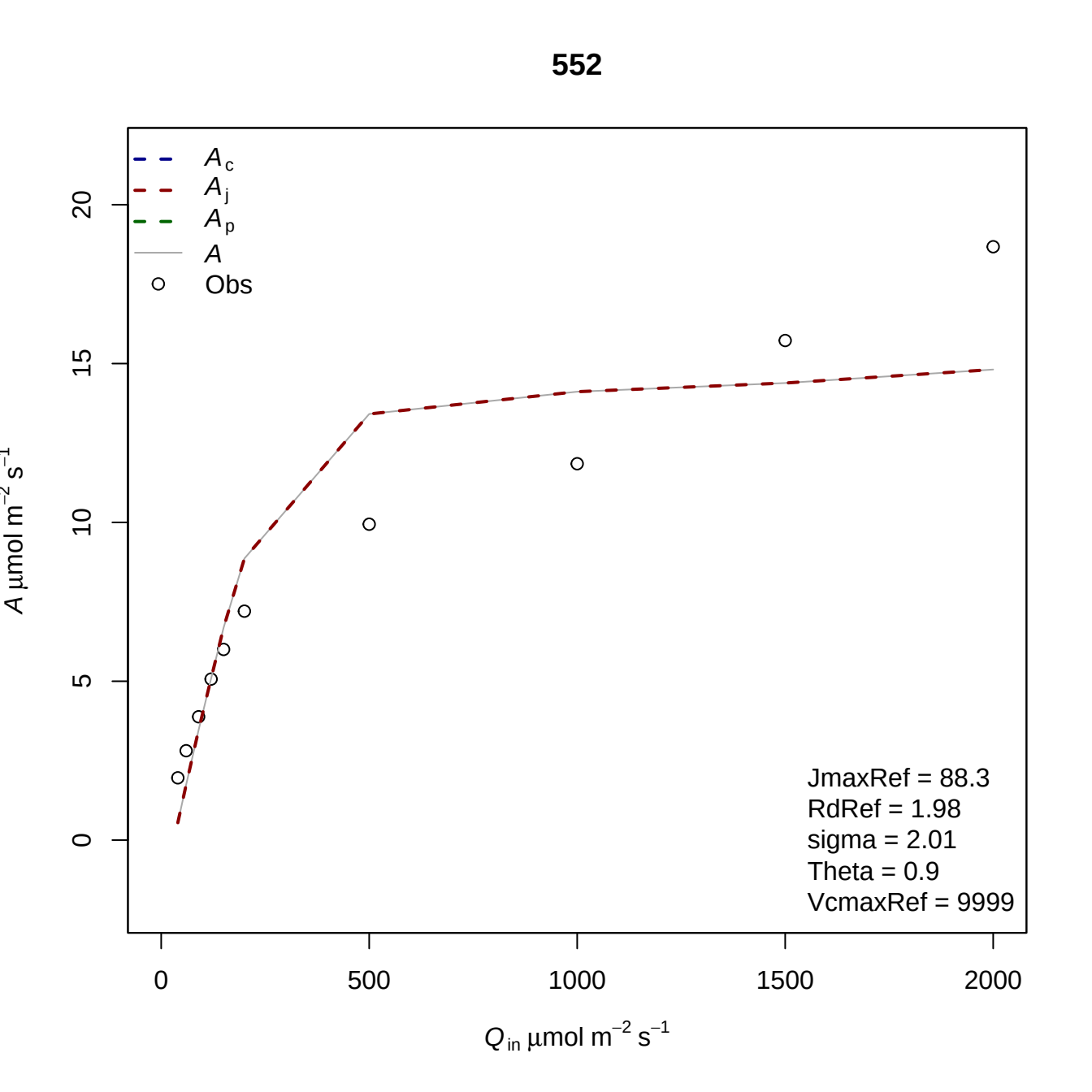

547
A_c
A_j
A_p
A
Obs
JmaxRef = 125.95
RdRef = 1.89
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

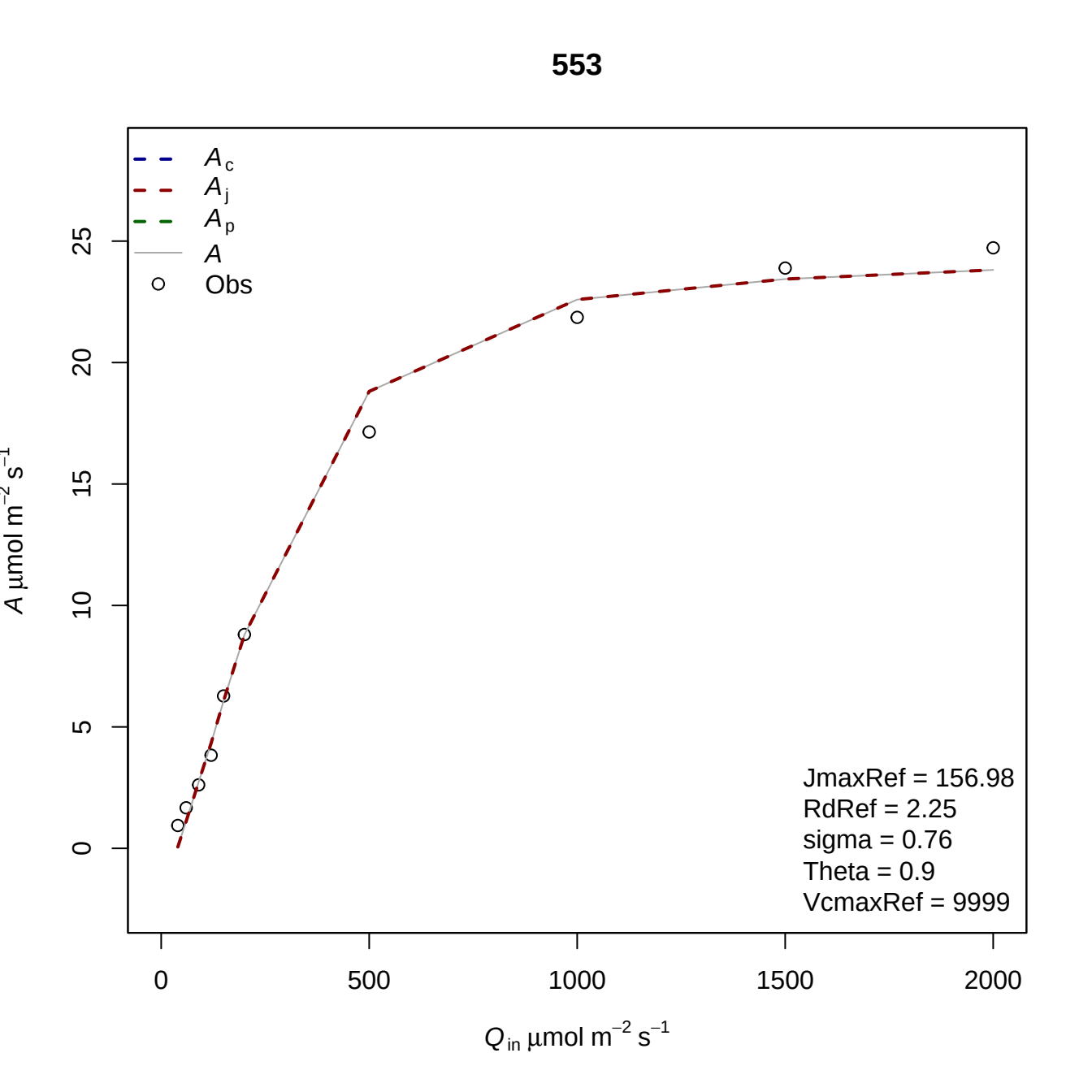

548
A_c
A_j
A_p
A
Obs
JmaxRef = 184.84
RdRef = 1.24
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

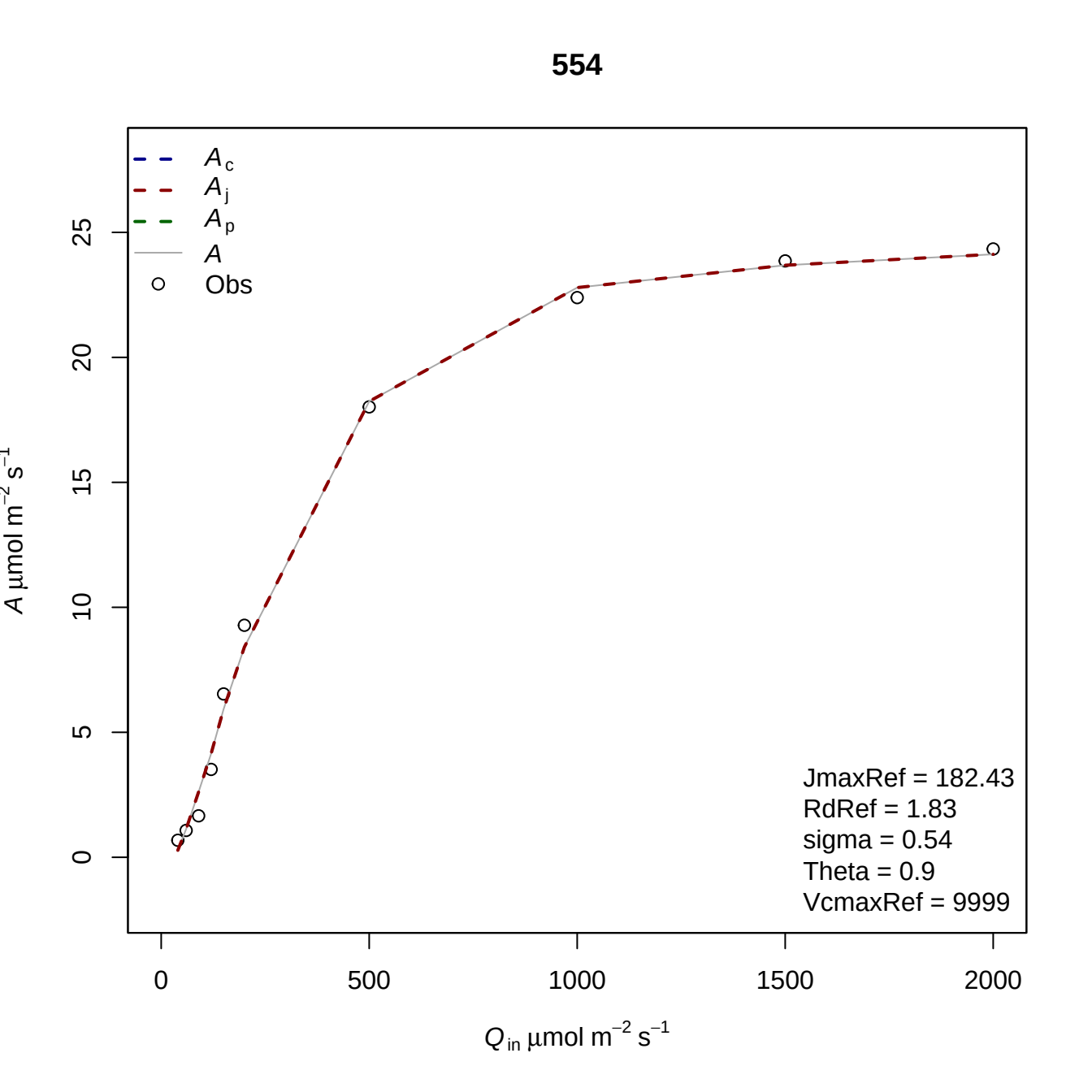

549
A_c
A_j
A_p
A
Obs
JmaxRef = 192.44
RdRef = 1.52
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

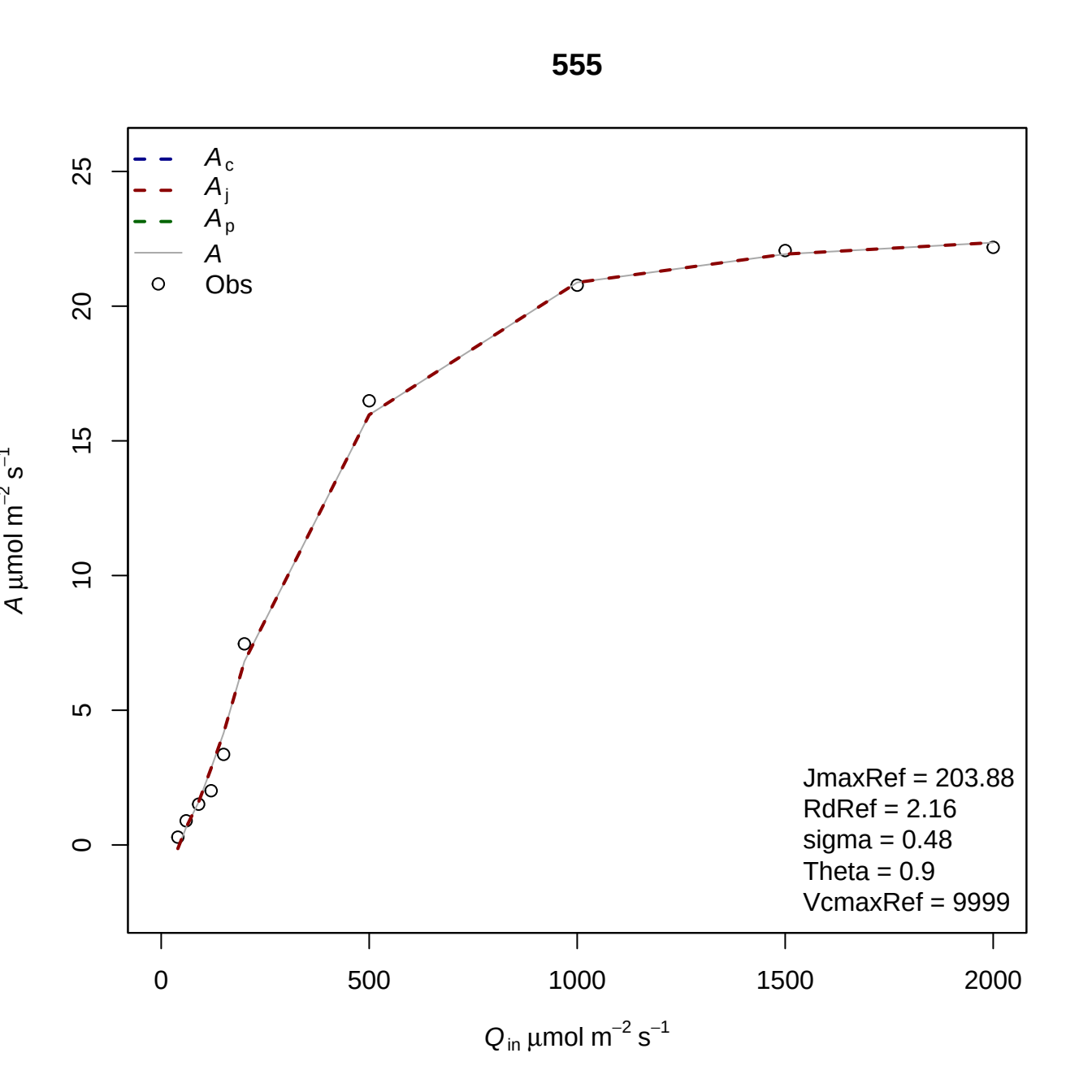

550
A_c
A_j
A_p
A
Obs
JmaxRef = 225.65
RdRef = 2
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

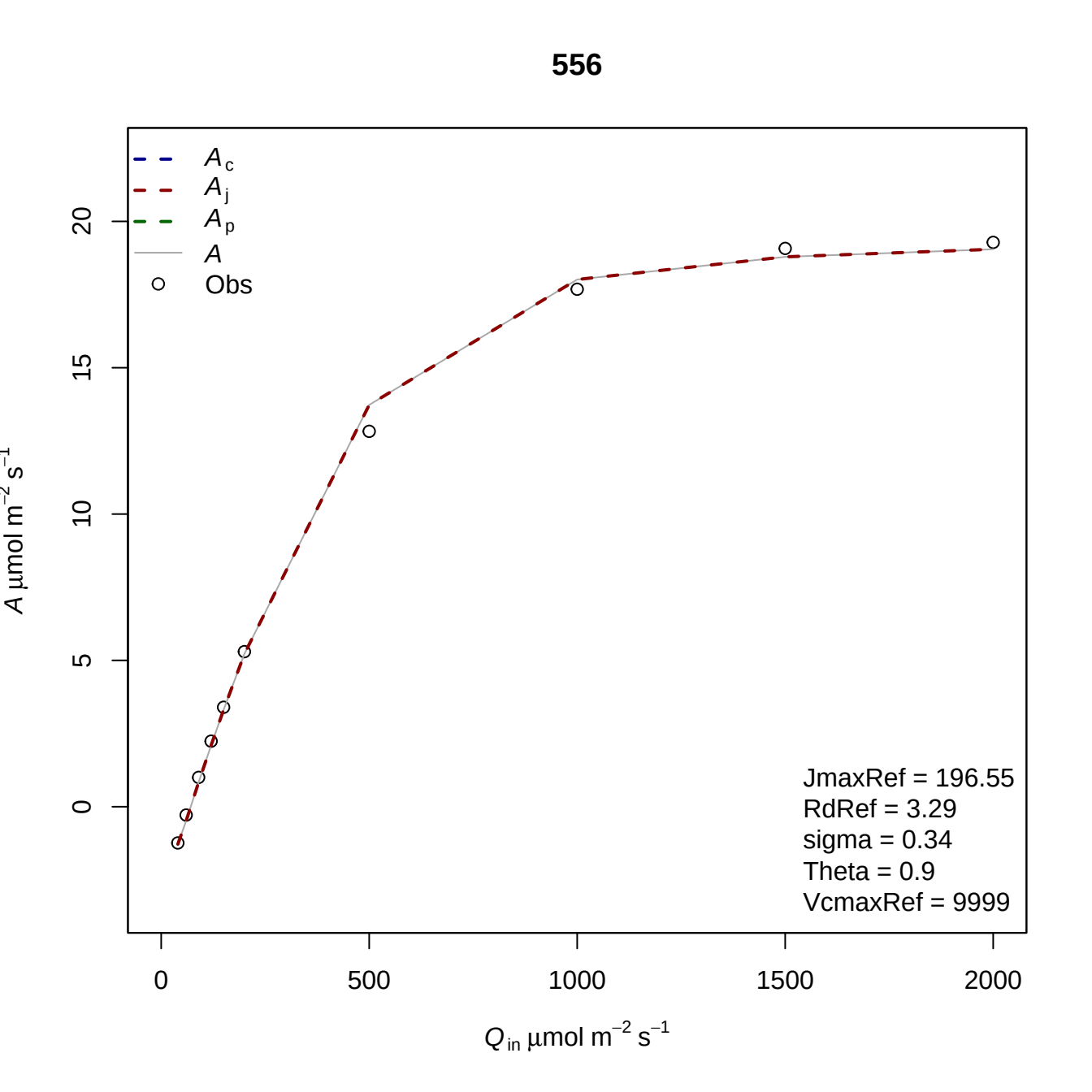

551
A_c
A_j
A_p
A
Obs
JmaxRef = 236.71
RdRef = 2.24
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

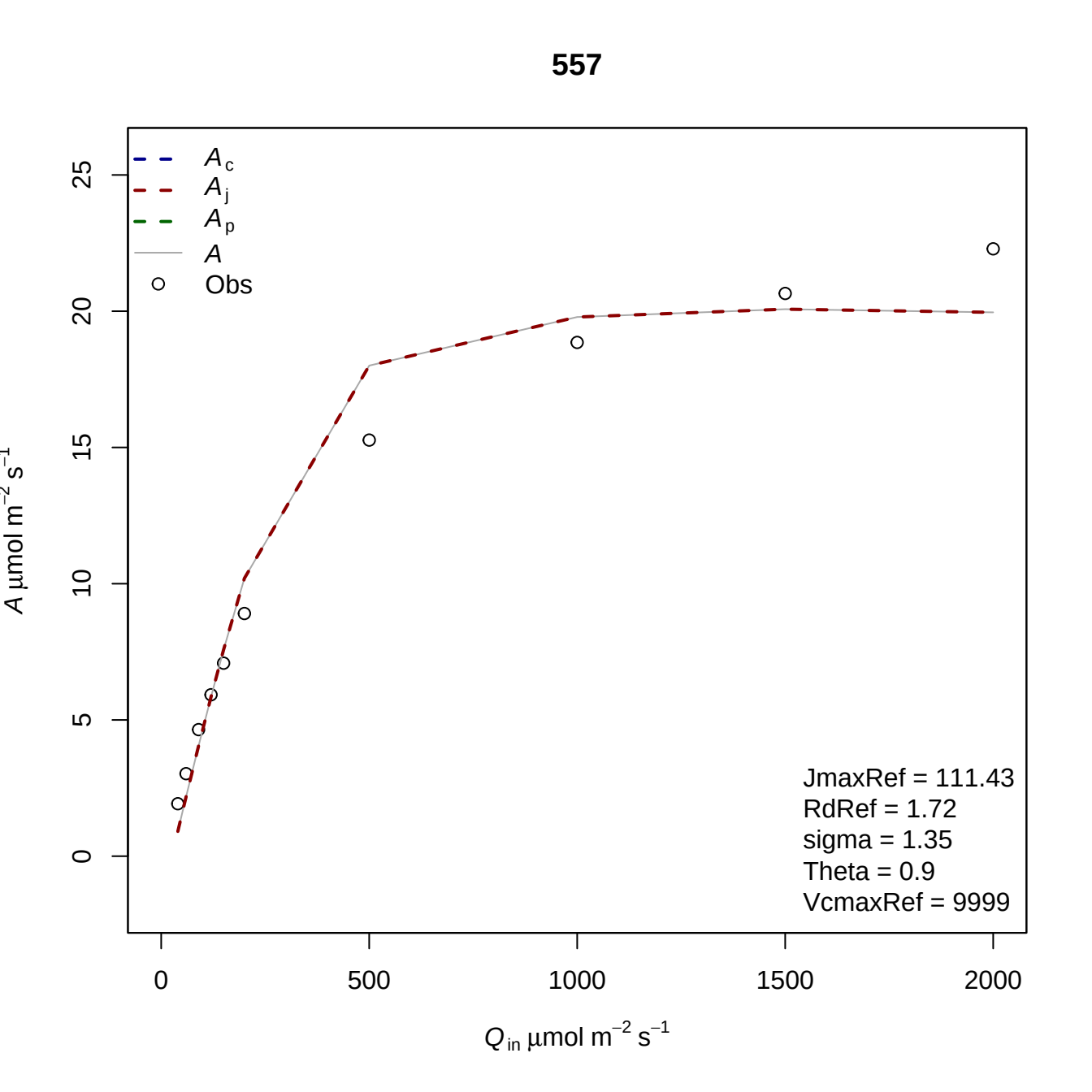

552
A_c
A_j
A_p
A
Obs
JmaxRef = 88.3
RdRef = 1.98
sigma = 2.01
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

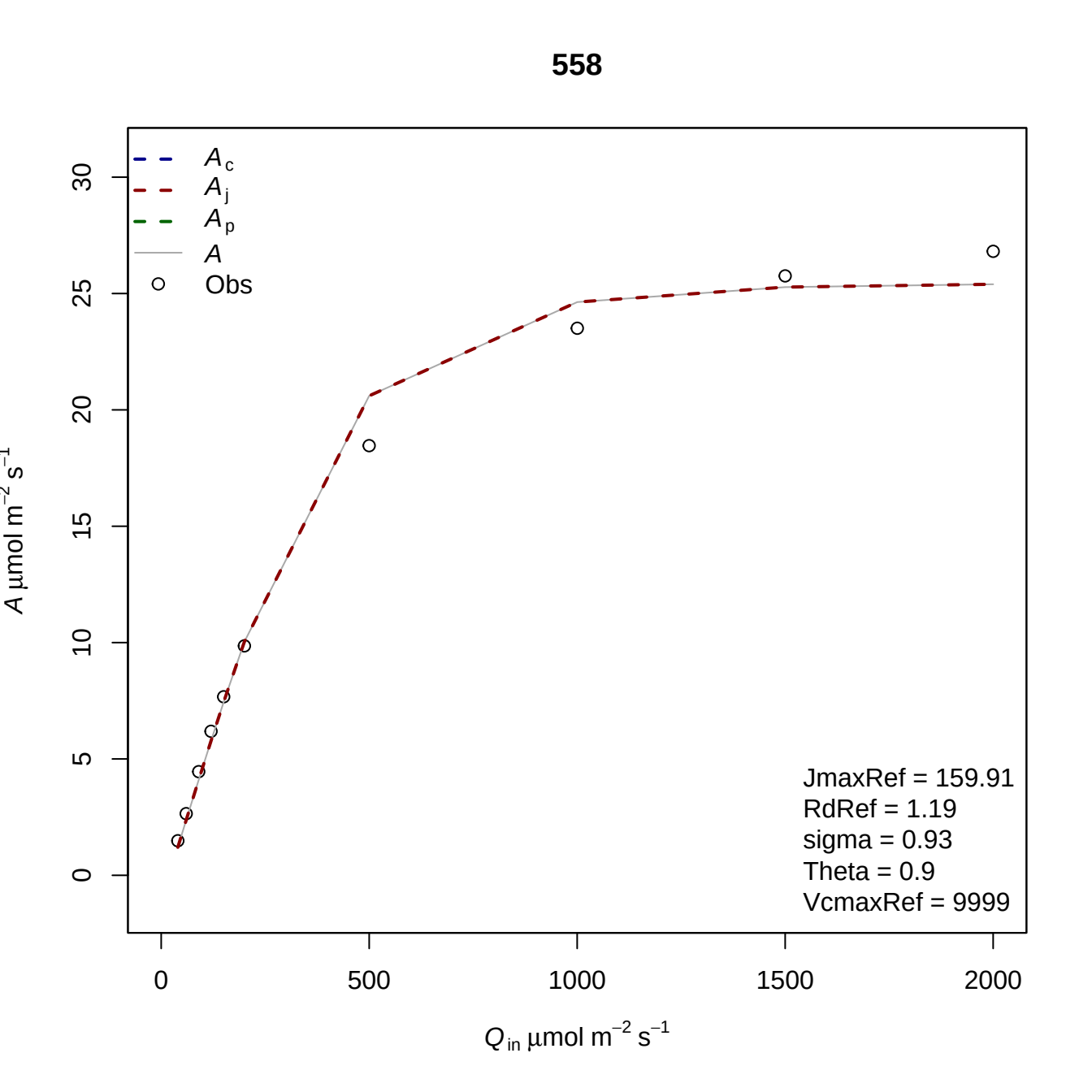

553
A_c
A_j
A_p
A
Obs
JmaxRef = 156.98
RdRef = 2.25
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

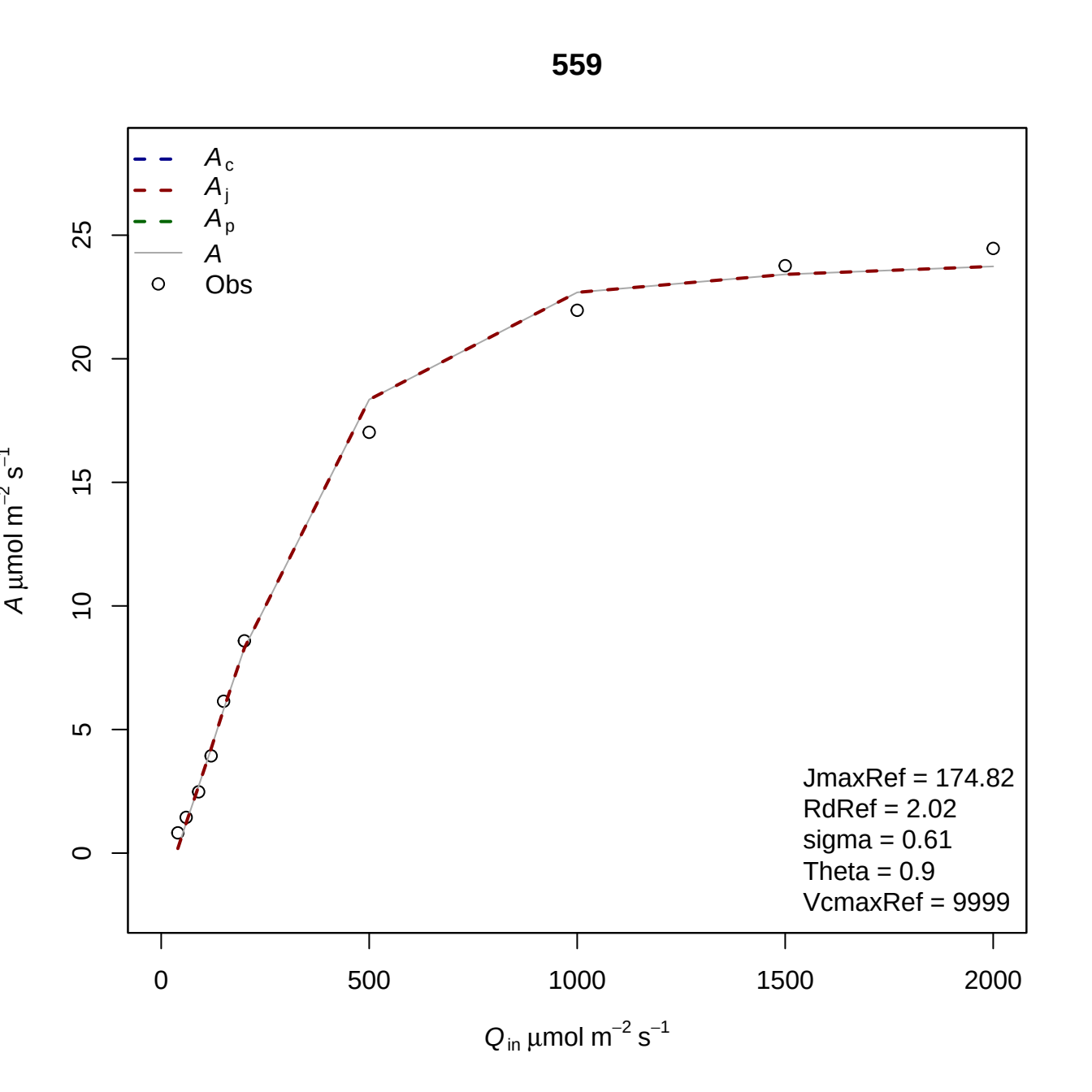

554
A_c
A_j
A_p
A
Obs
JmaxRef = 182.43
RdRef = 1.83
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

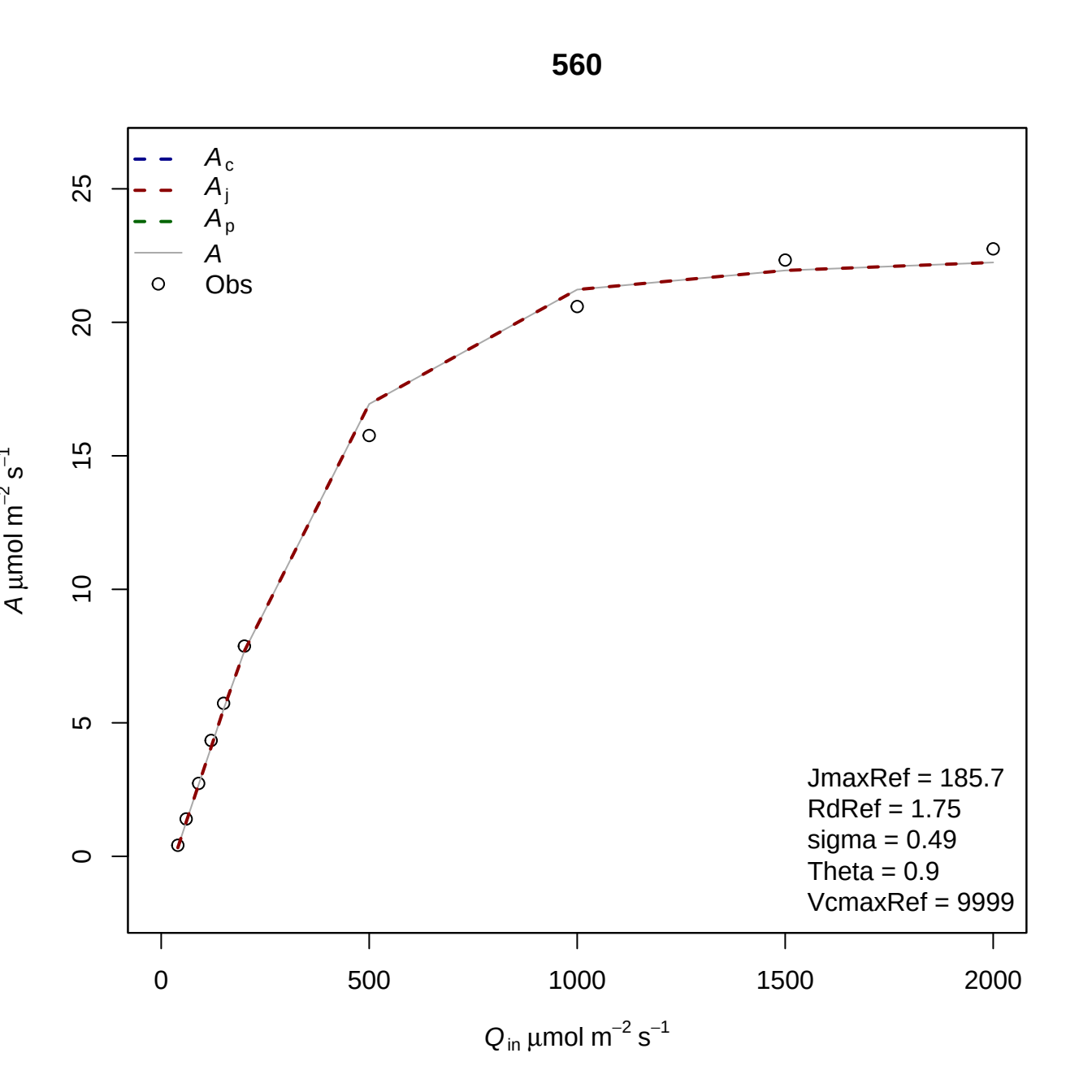

555
A_c
A_j
A_p
A
Obs
JmaxRef = 203.88
RdRef = 2.16
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

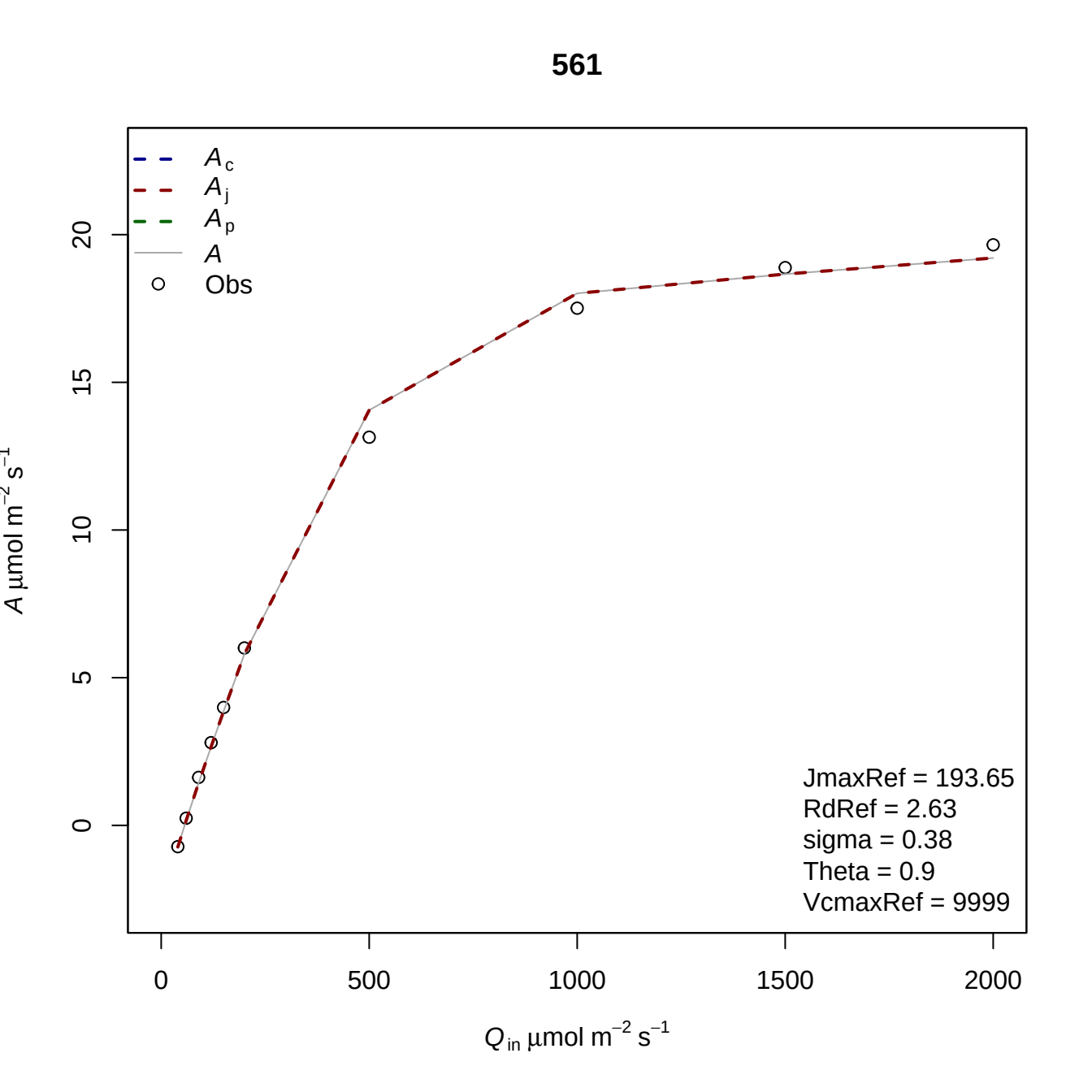

556
A_c
A_j
A_p
A
Obs
JmaxRef = 196.55
RdRef = 3.29
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

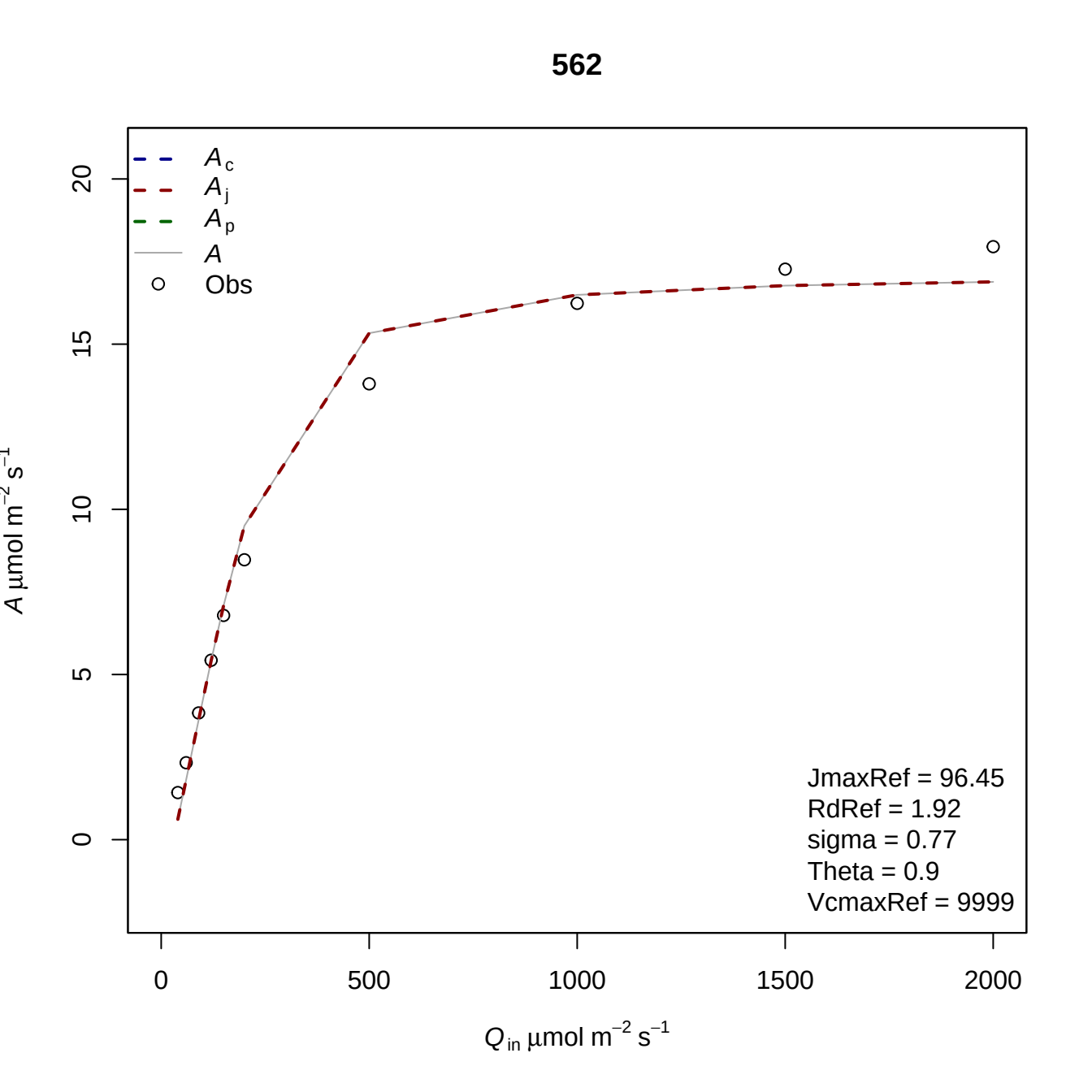

557
A_c
A_j
A_p
A
Obs
JmaxRef = 111.43
RdRef = 1.72
sigma = 1.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

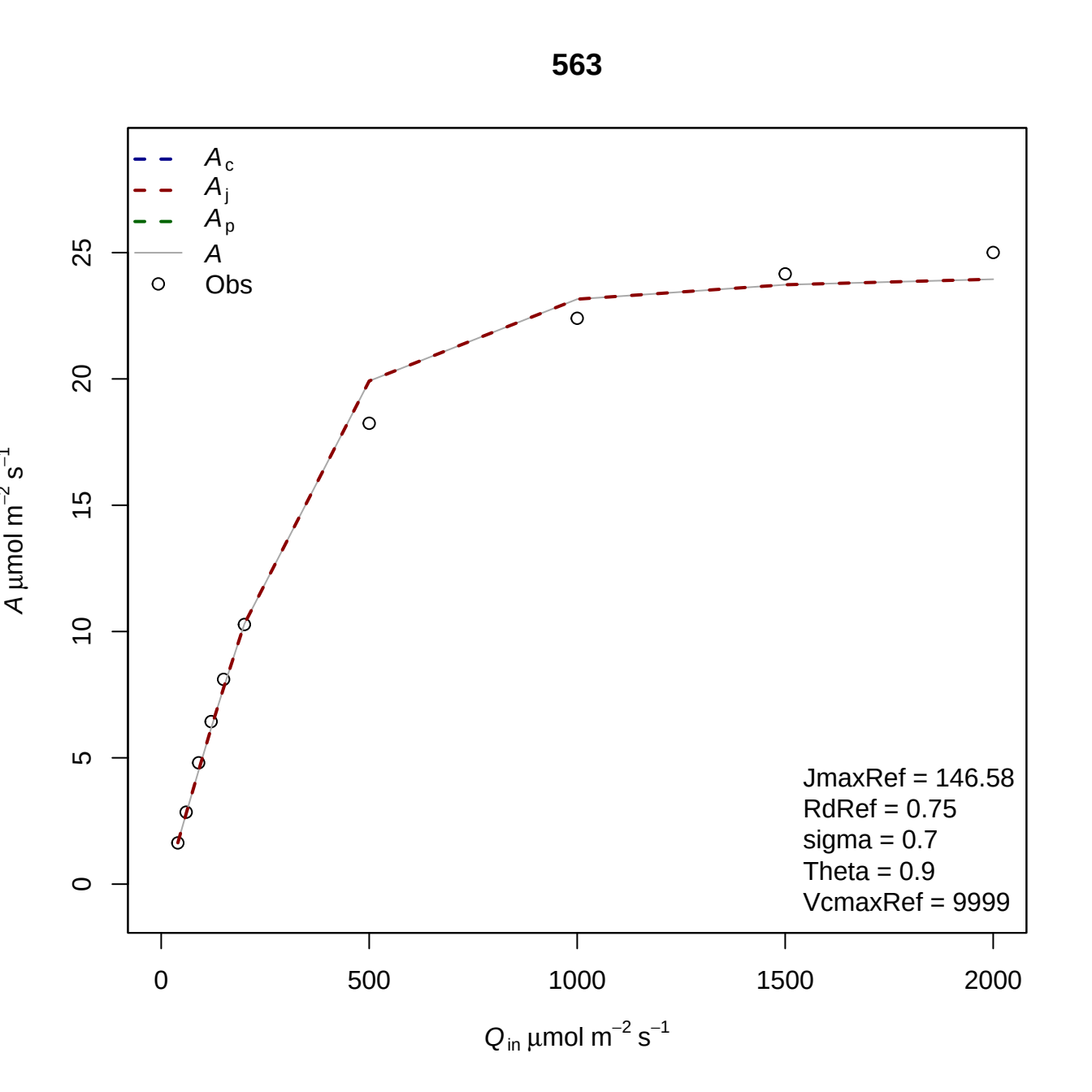

558
A_c
A_j
A_p
A
Obs
JmaxRef = 159.91
RdRef = 1.19
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

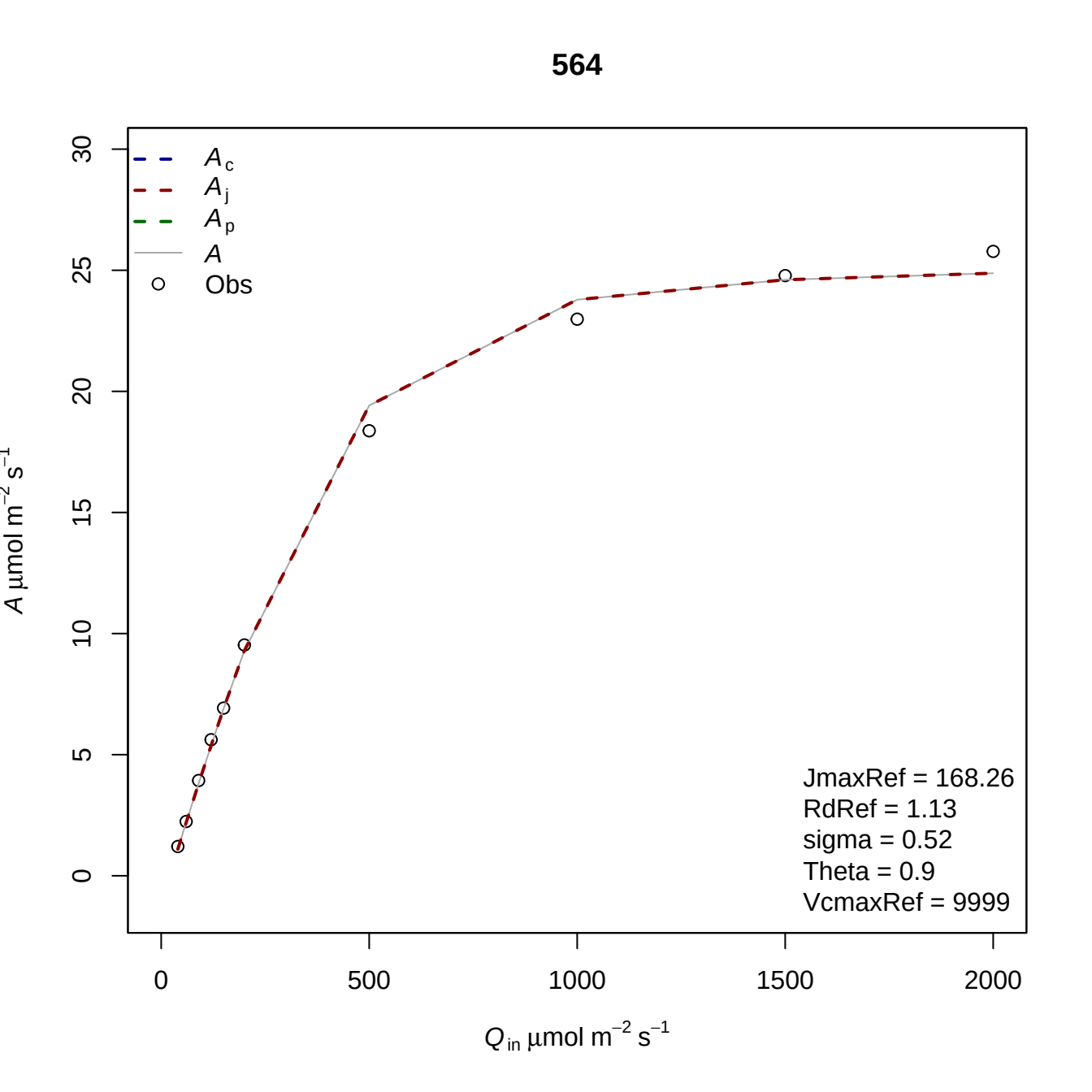

559
A_c
A_j
A_p
A
Obs
JmaxRef = 174.82
RdRef = 2.02
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

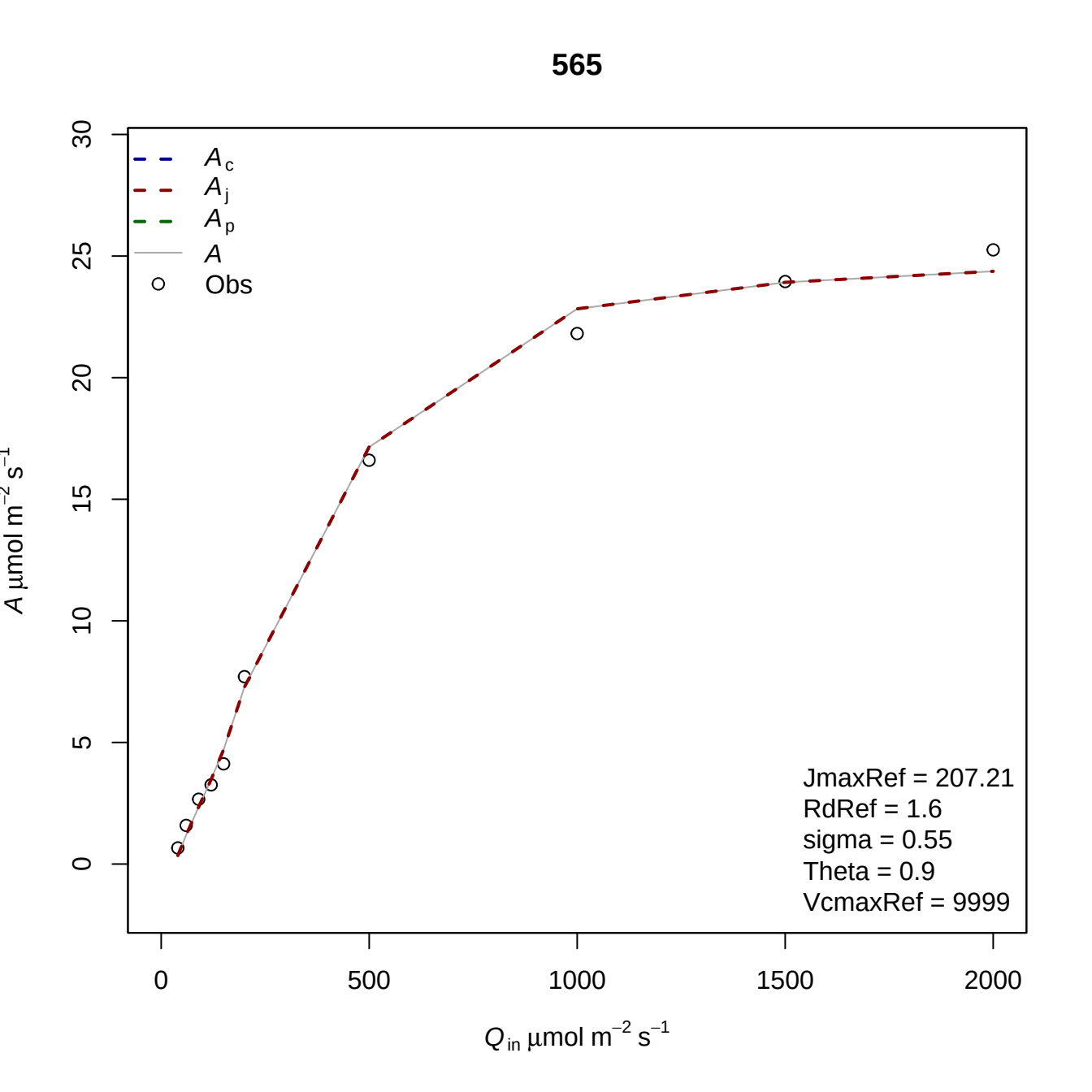

560
A_c
A_j
A_p
A
Obs
JmaxRef = 185.7
RdRef = 1.75
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

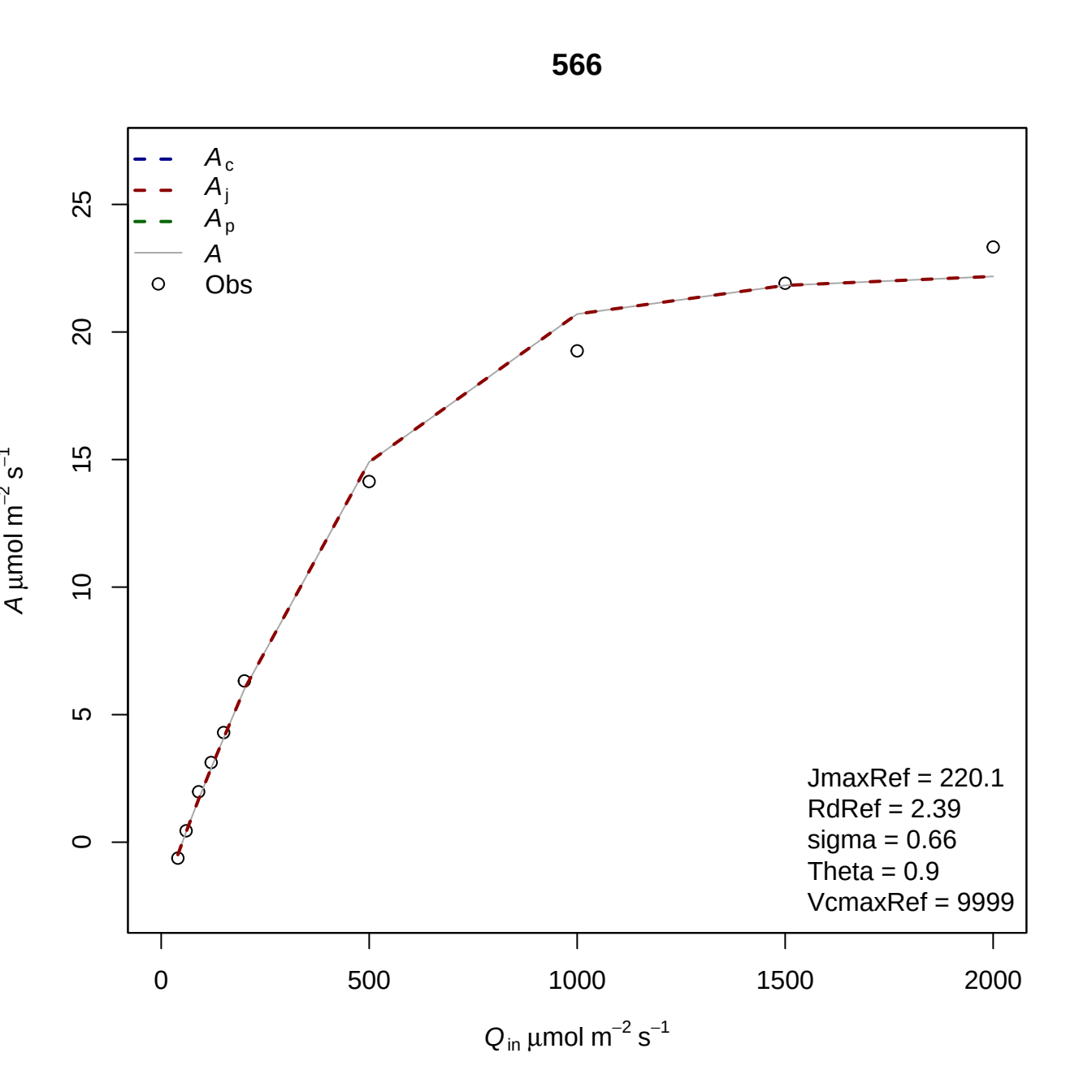

561
A_c
A_j
A_p
A
Obs
JmaxRef = 193.65
RdRef = 2.63
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

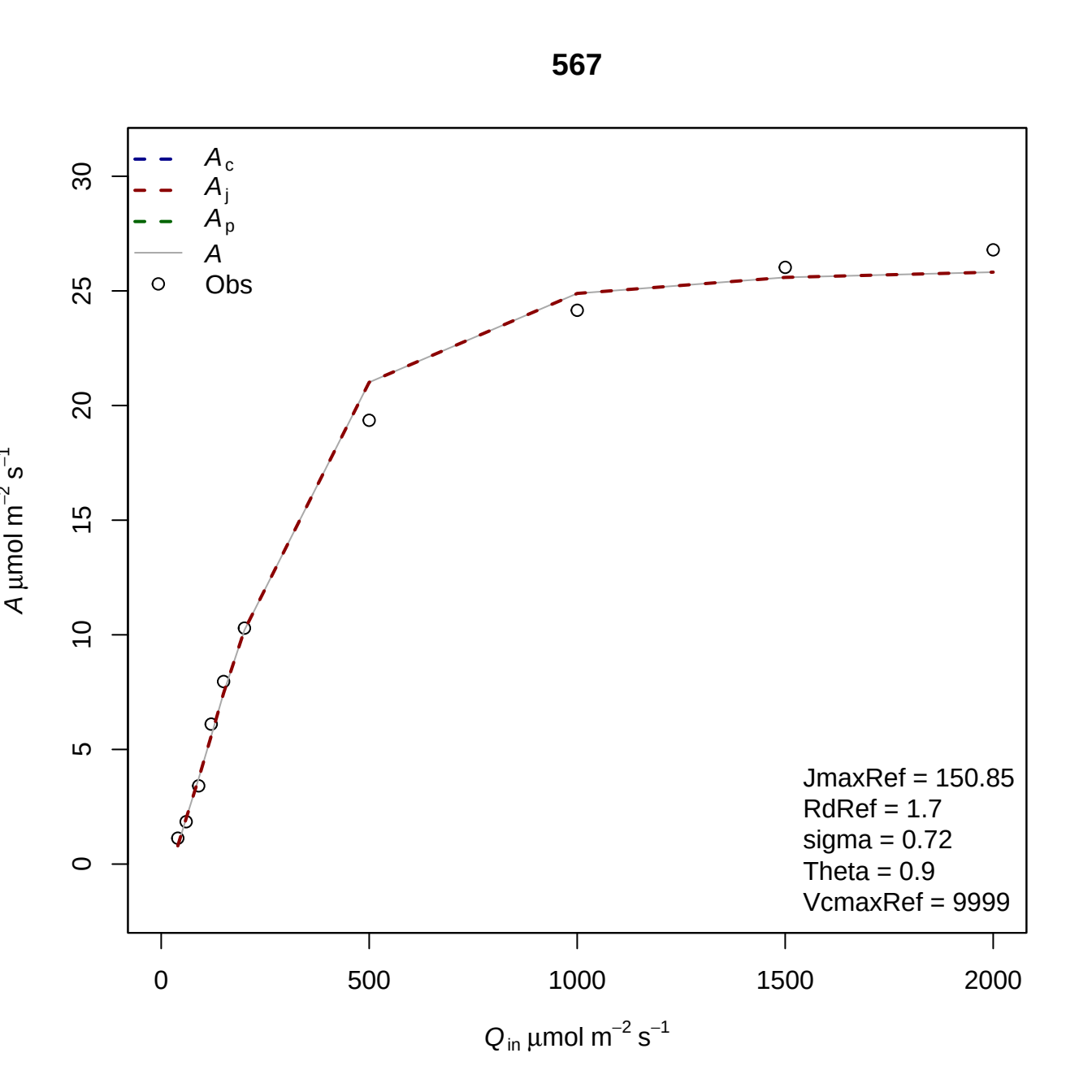

562
A_c
A_j
A_p
A
Obs
JmaxRef = 96.45
RdRef = 1.92
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

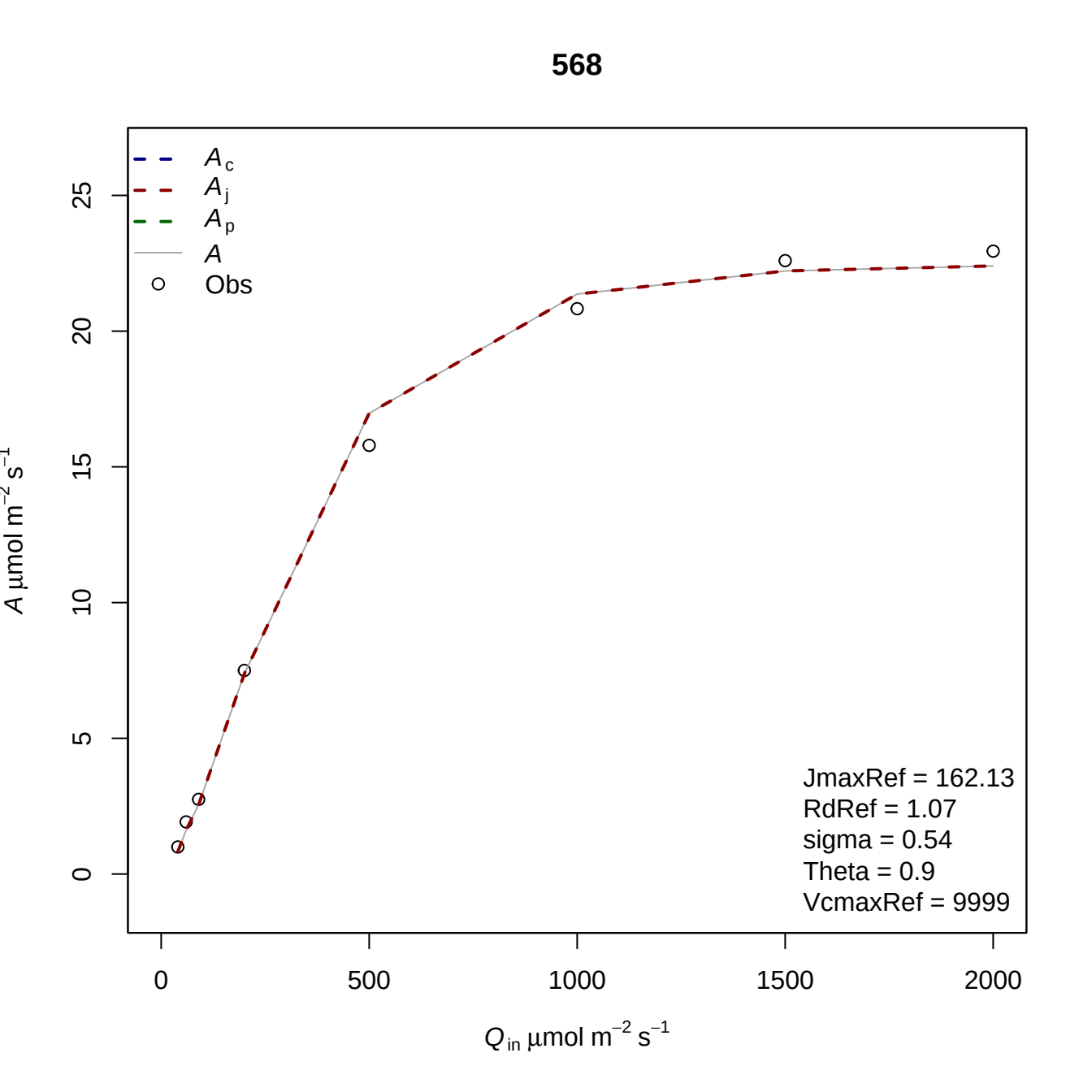

563
A_c
A_j
A_p
A
Obs
JmaxRef = 146.58
RdRef = 0.75
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

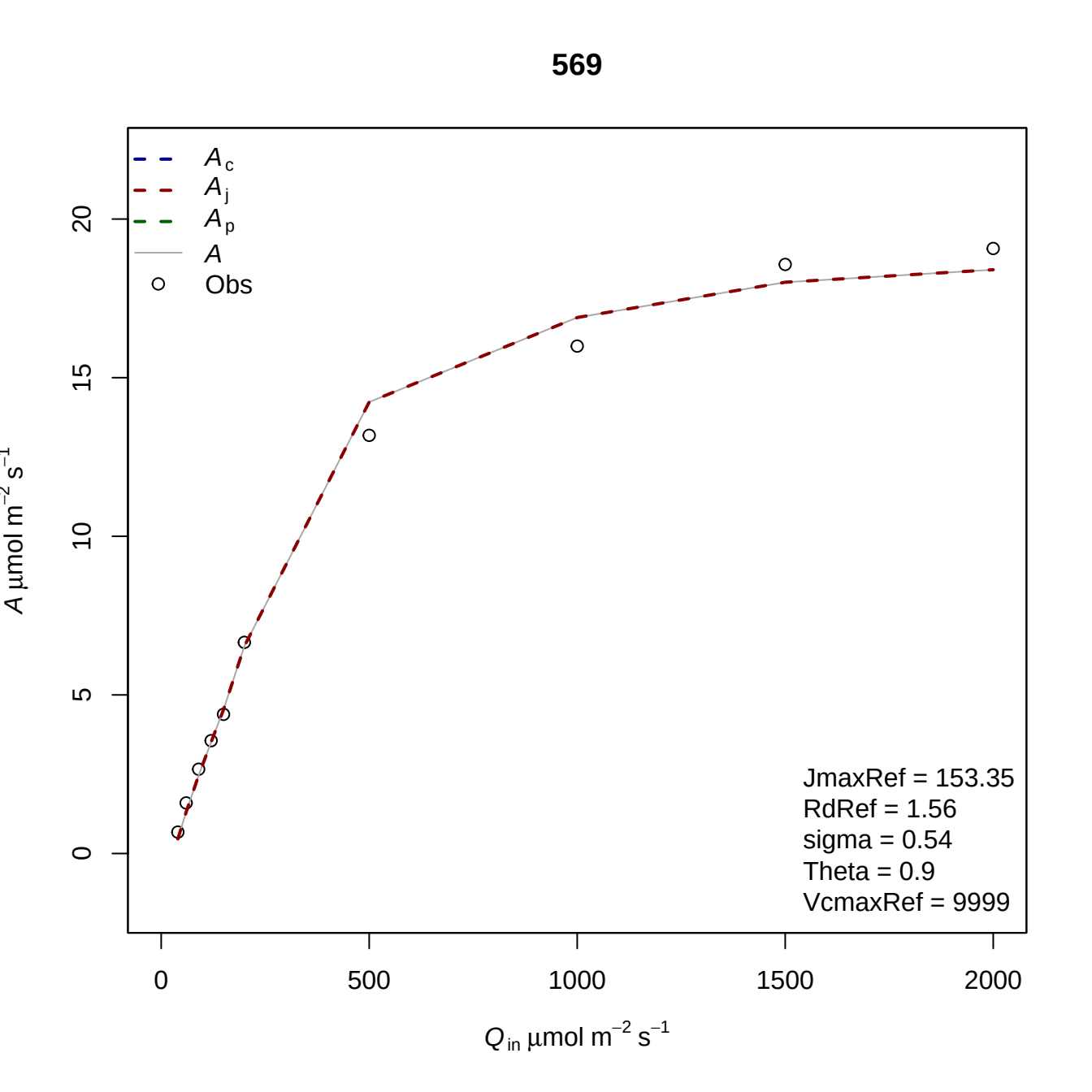

564
A_c
A_j
A_p
A
Obs
JmaxRef = 168.26
RdRef = 1.13
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

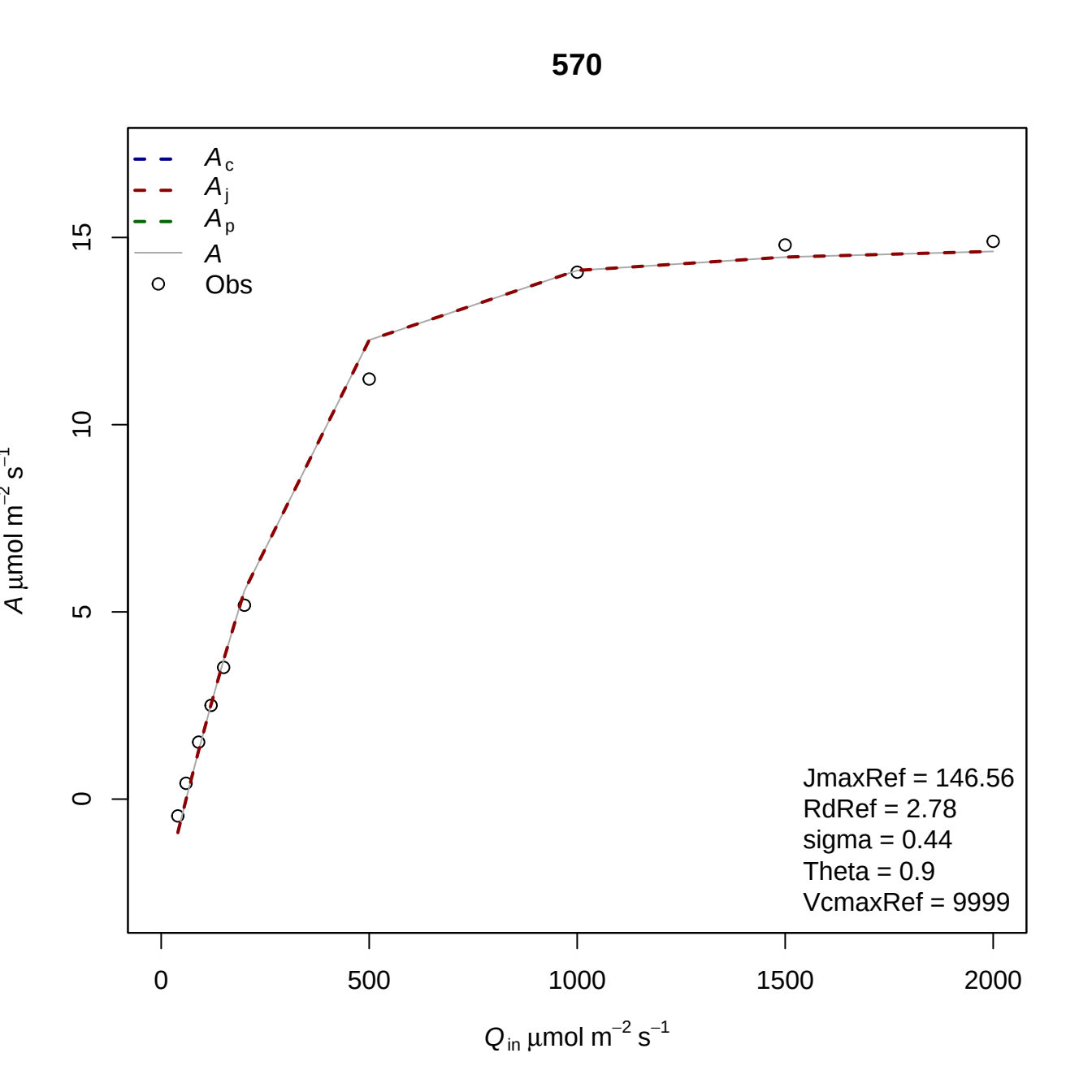

565
A_c
A_j
A_p
A
Obs
JmaxRef = 207.21
RdRef = 1.6
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

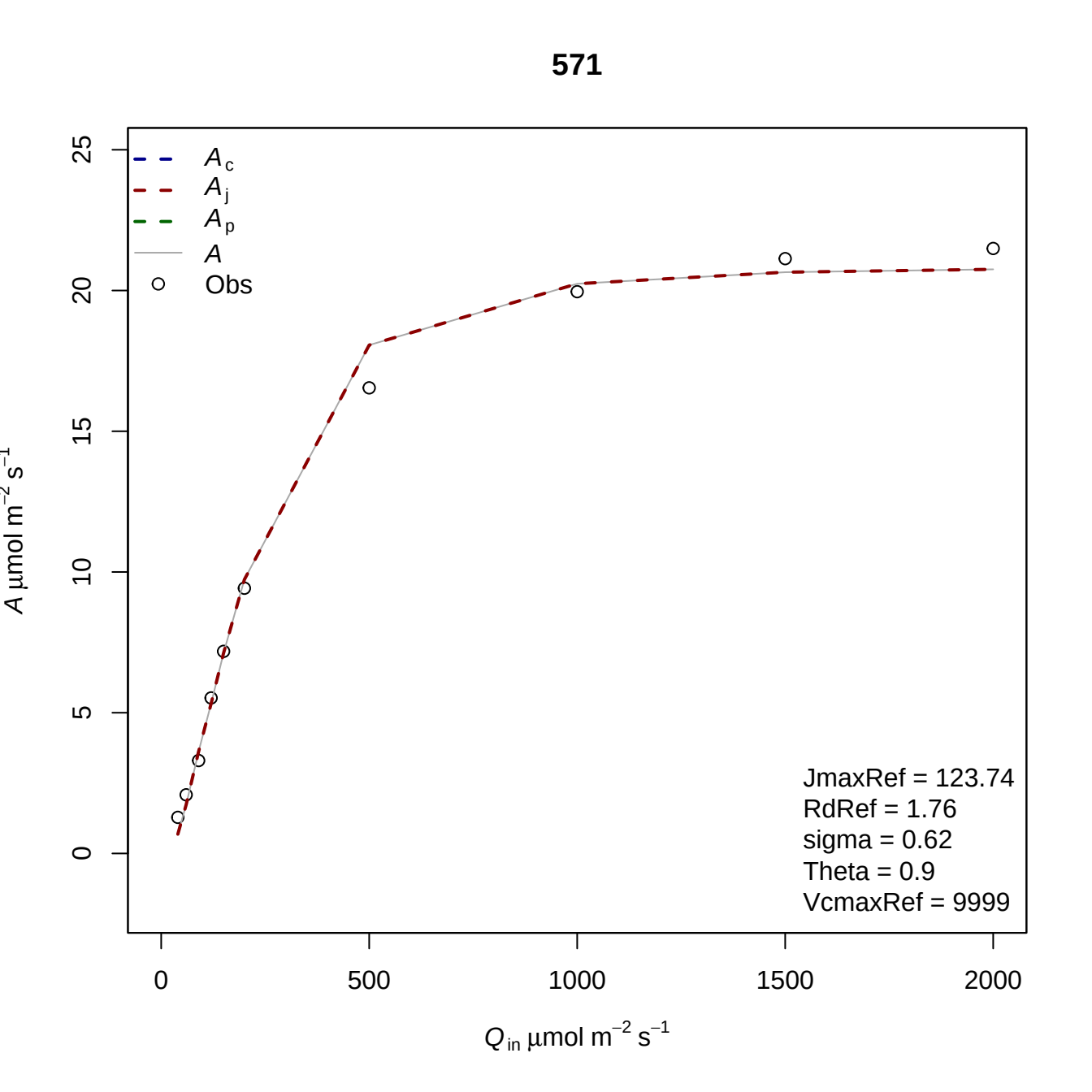

566
A_c
A_j
A_p
A
Obs
JmaxRef = 220.1
RdRef = 2.39
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

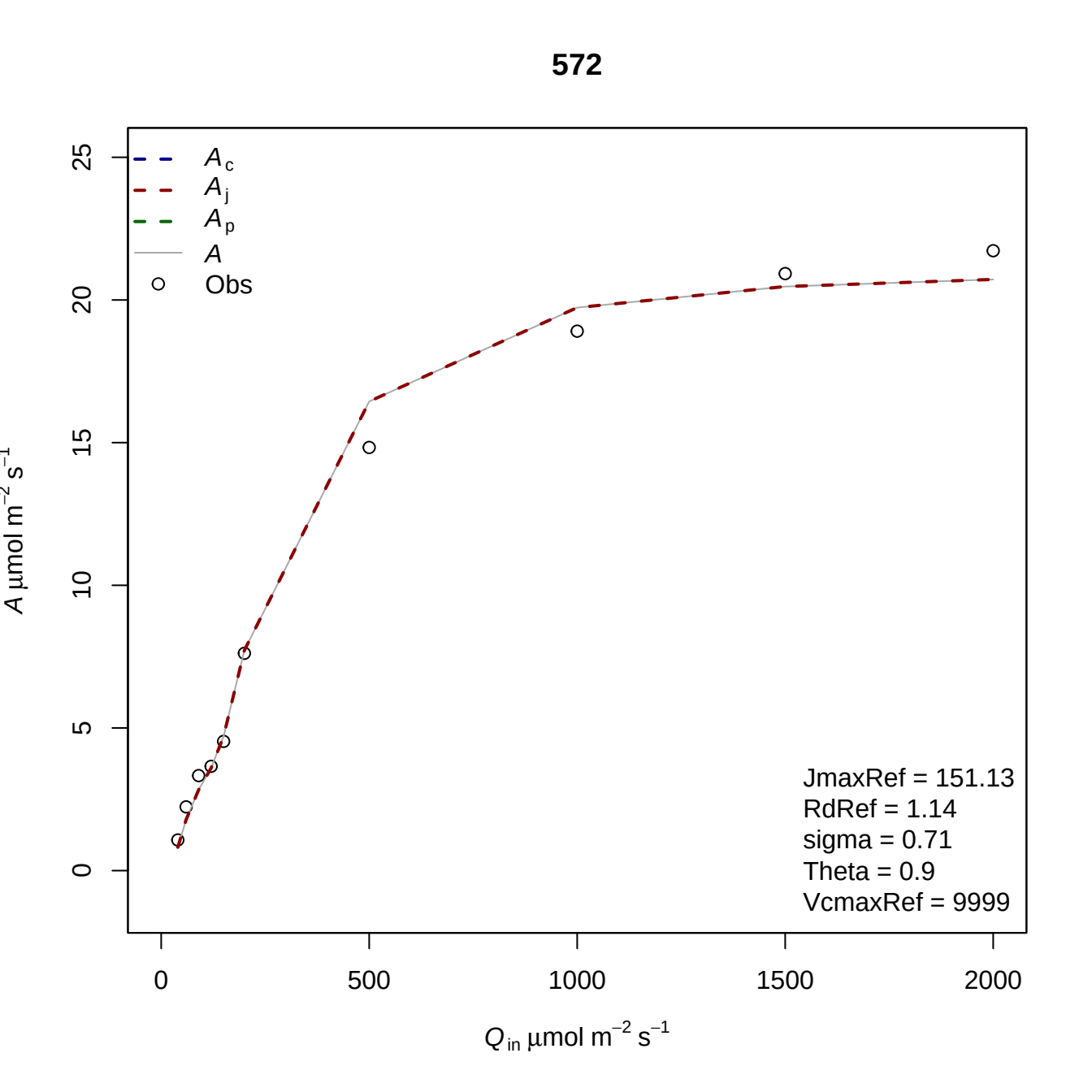

567
A_c
A_j
A_p
A
Obs
JmaxRef = 150.85
RdRef = 1.7
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

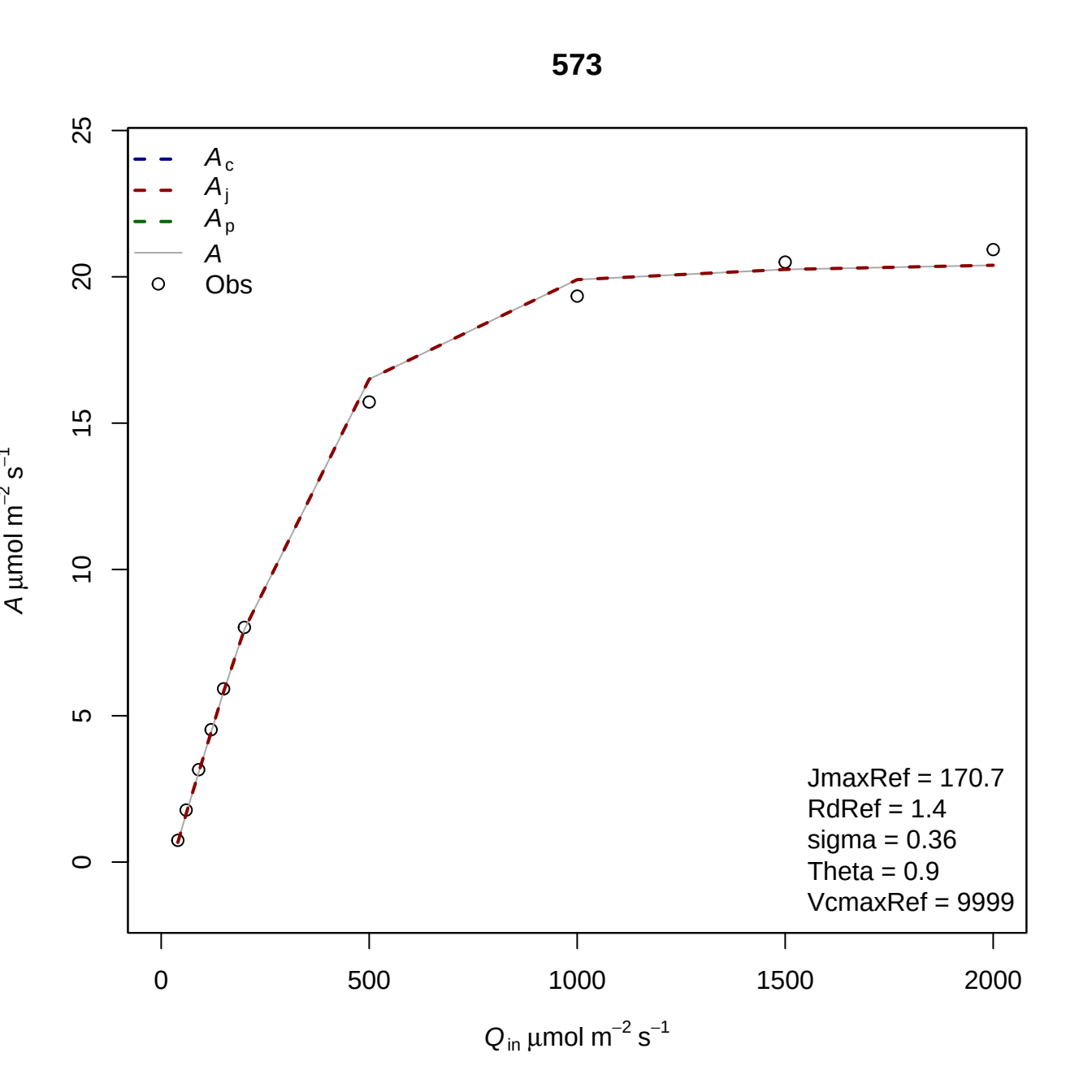

568
A_c
A_j
A_p
A
Obs
JmaxRef = 162.13
RdRef = 1.07
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

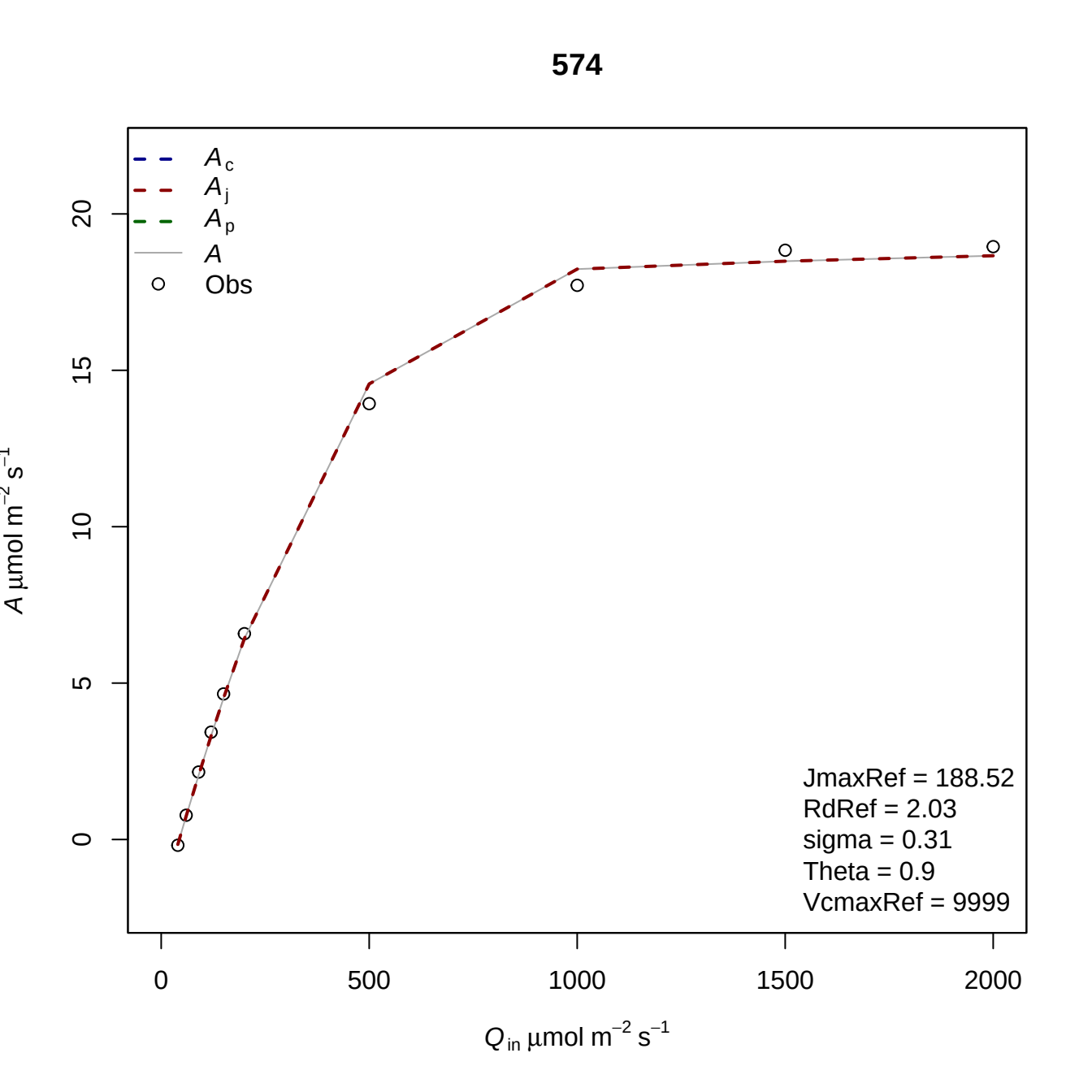

569
A_c
A_j
A_p
A
Obs
JmaxRef = 153.35
RdRef = 1.56
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

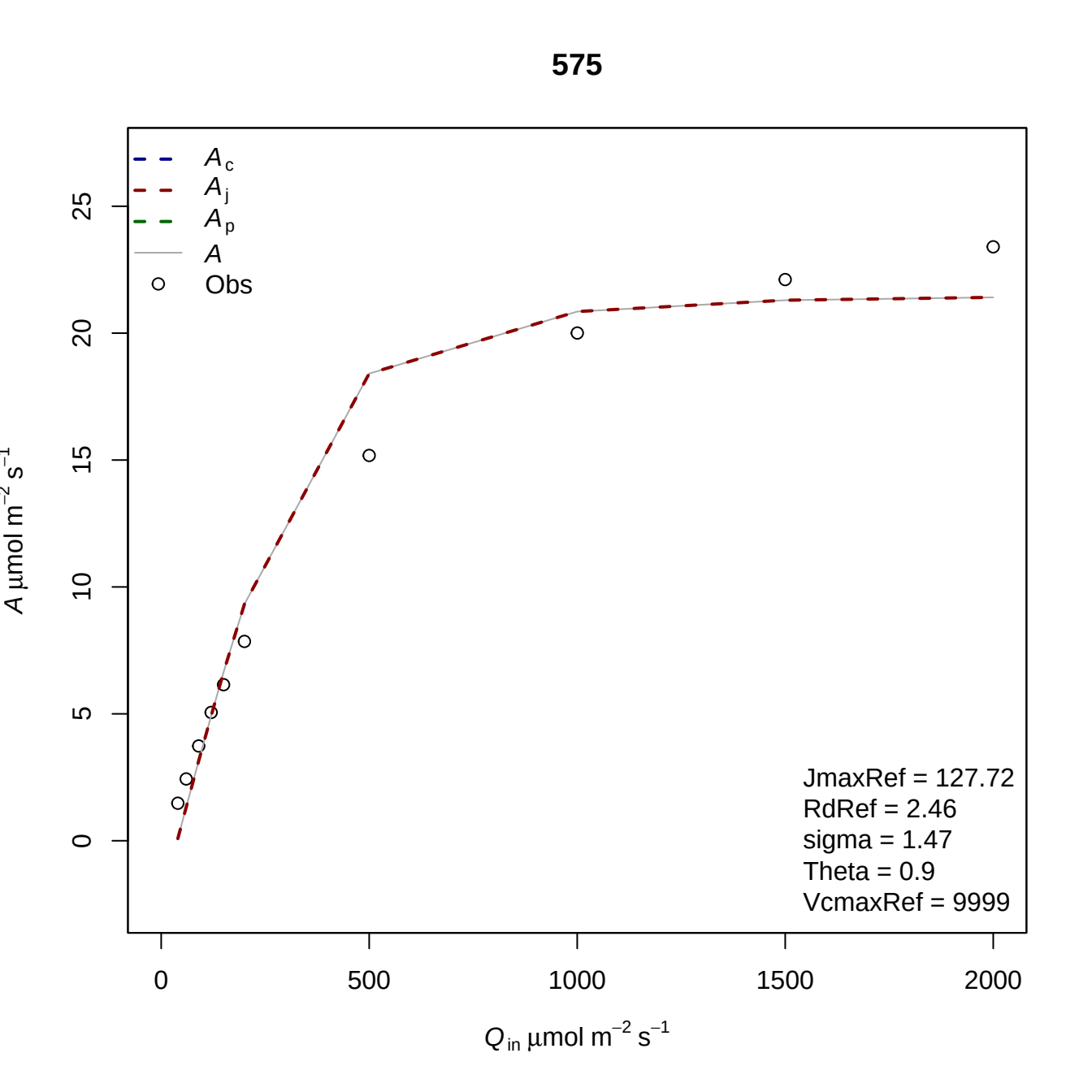

570
A_c
A_j
A_p
A
Obs
JmaxRef = 146.56
RdRef = 2.78
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

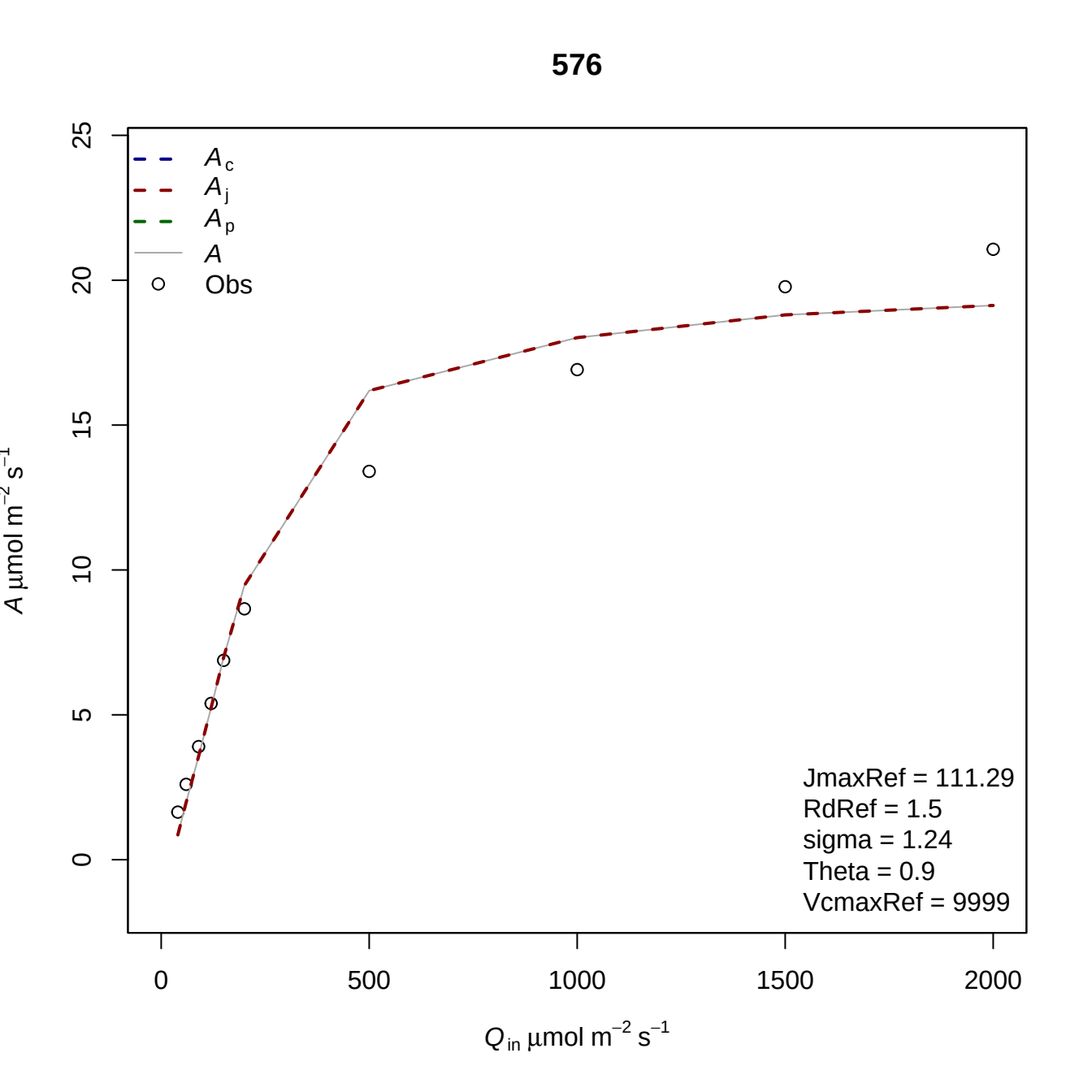

571
A_c
A_j
A_p
A
Obs
JmaxRef = 123.74
RdRef = 1.76
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

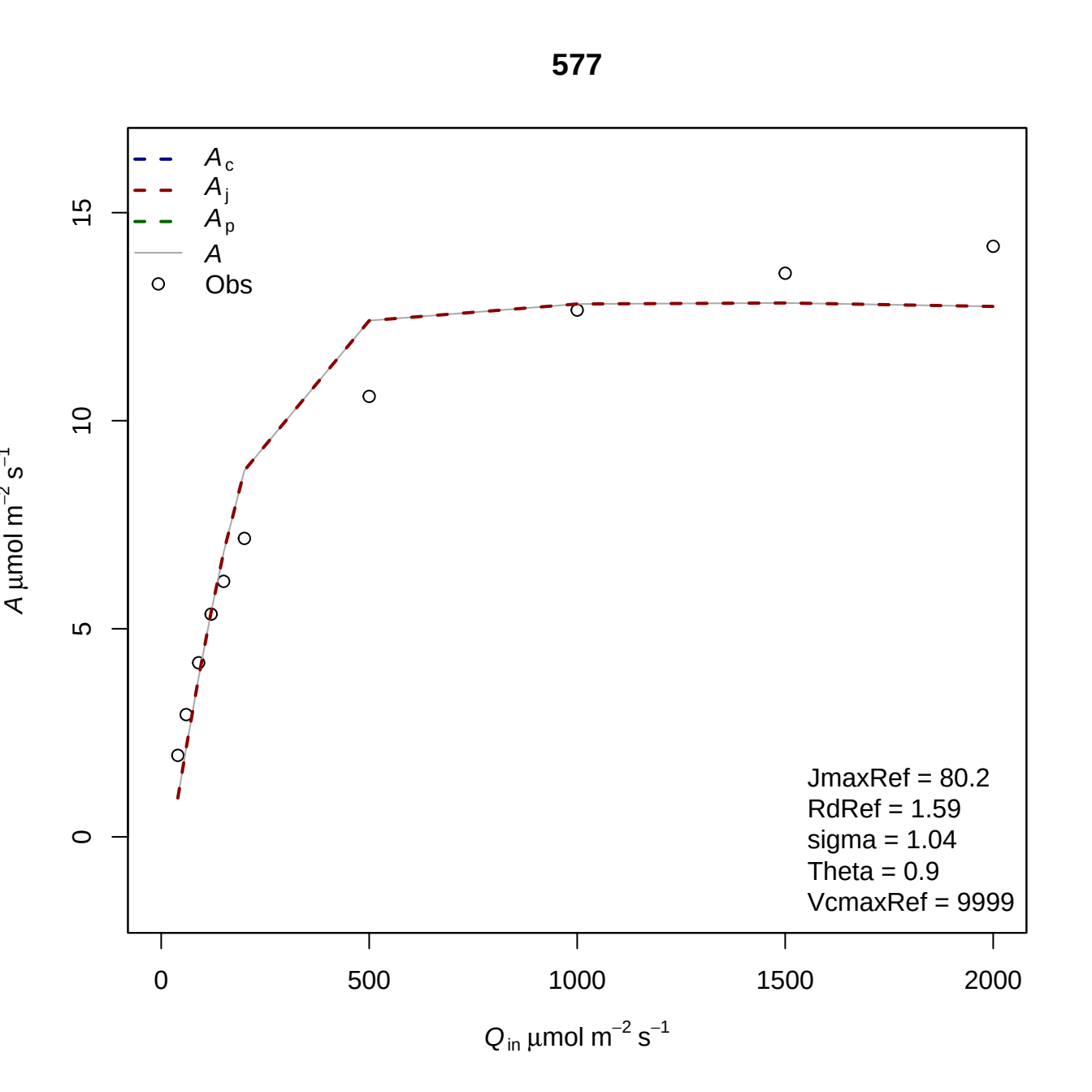

572
A_c
A_j
A_p
A
Obs
JmaxRef = 151.13
RdRef = 1.14
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

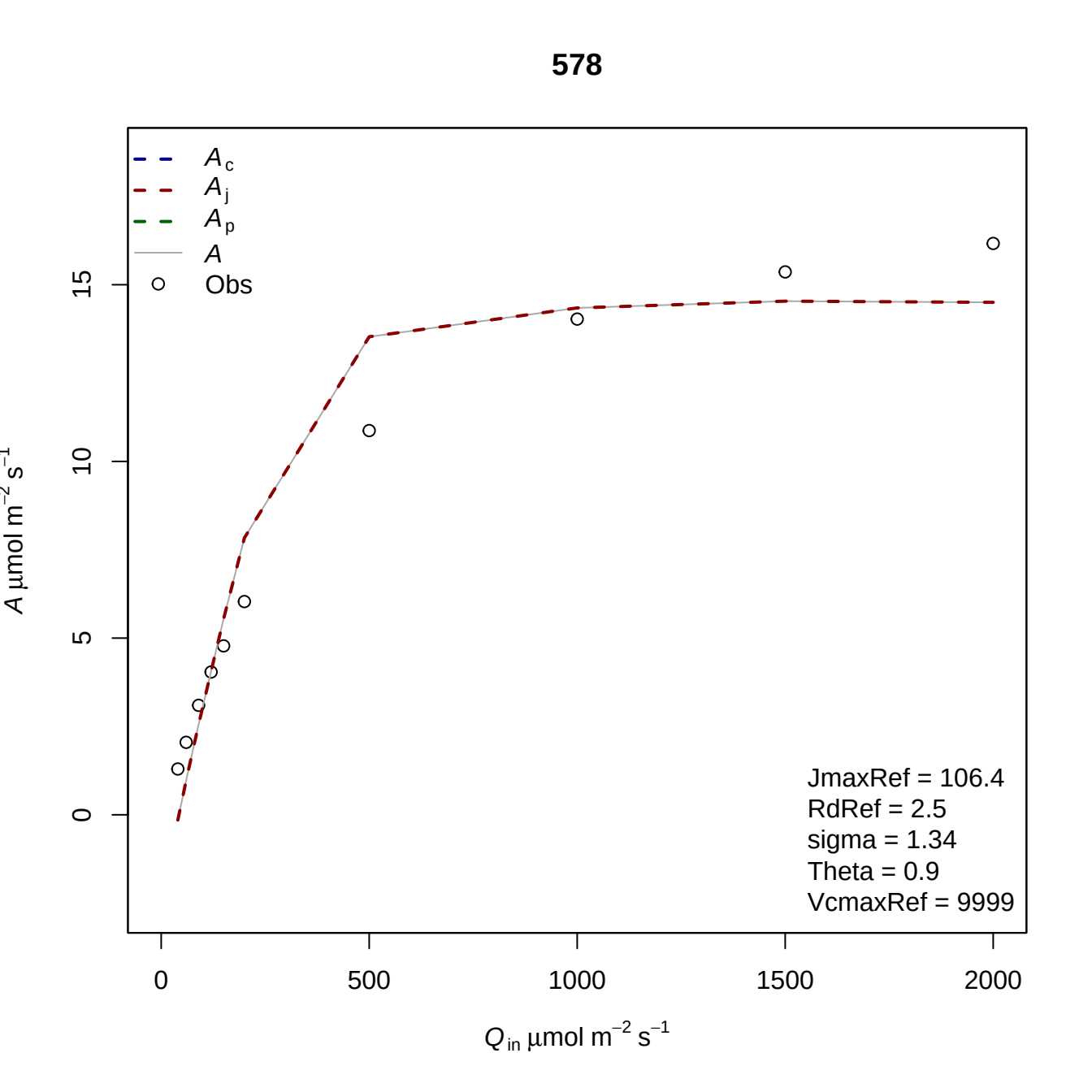

573
A_c
A_j
A_p
A
Obs
JmaxRef = 170.7
RdRef = 1.4
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

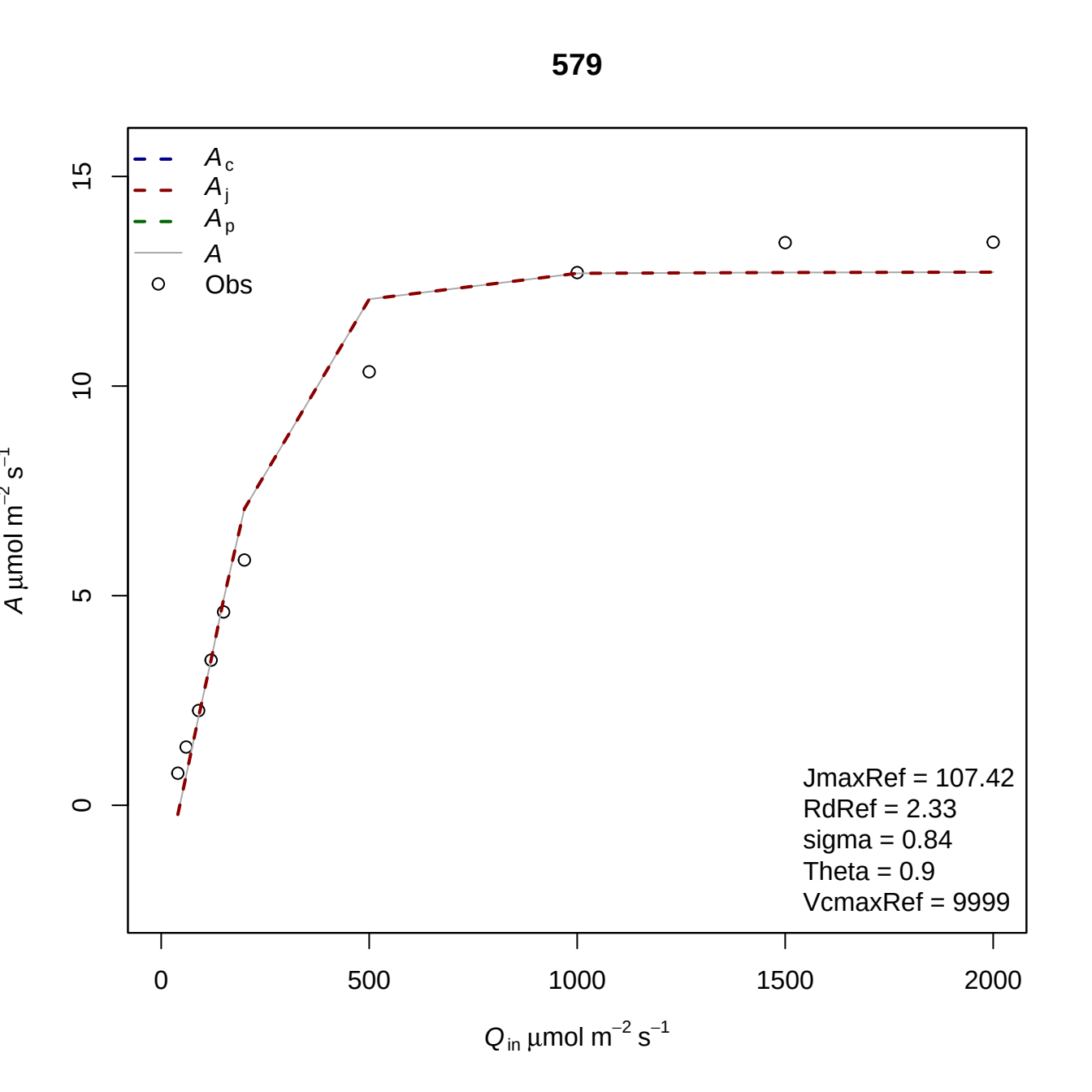

574
A_c
A_j
A_p
A
Obs
JmaxRef = 188.52
RdRef = 2.03
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

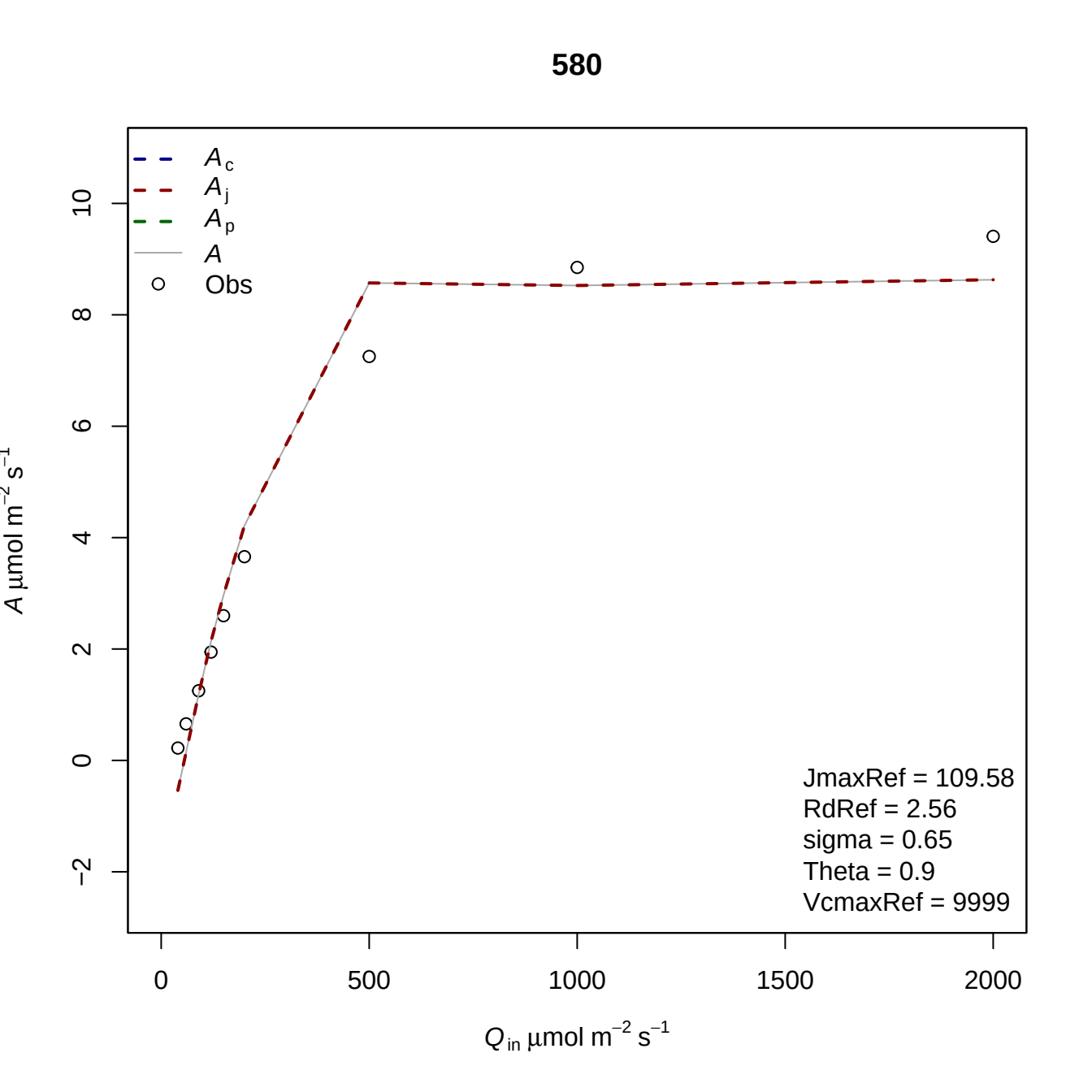

575
A_c
A_j
A_p
A
Obs
JmaxRef = 127.72
RdRef = 2.46
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

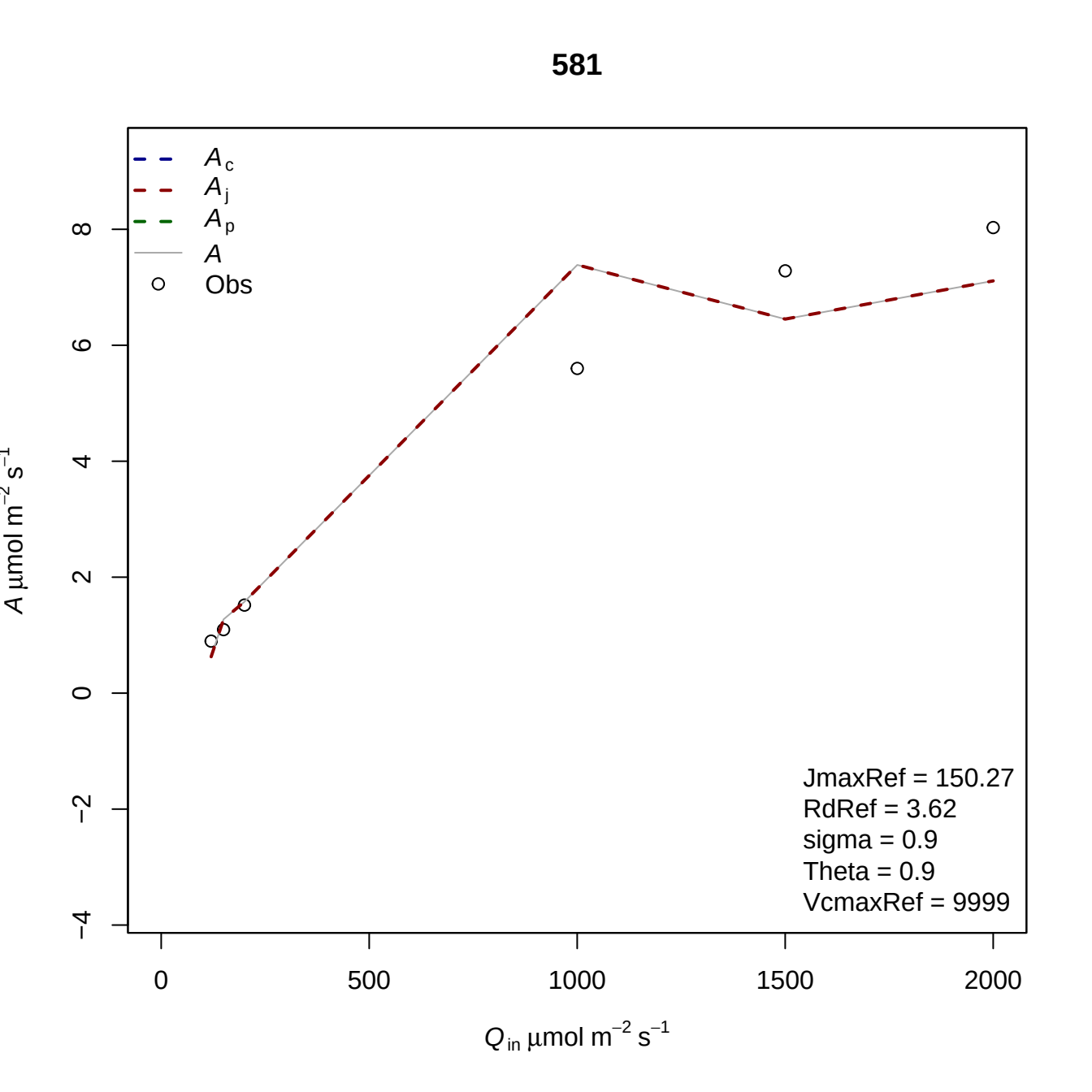

576
A_c
A_j
A_p
A
Obs
JmaxRef = 111.29
RdRef = 1.5
sigma = 1.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

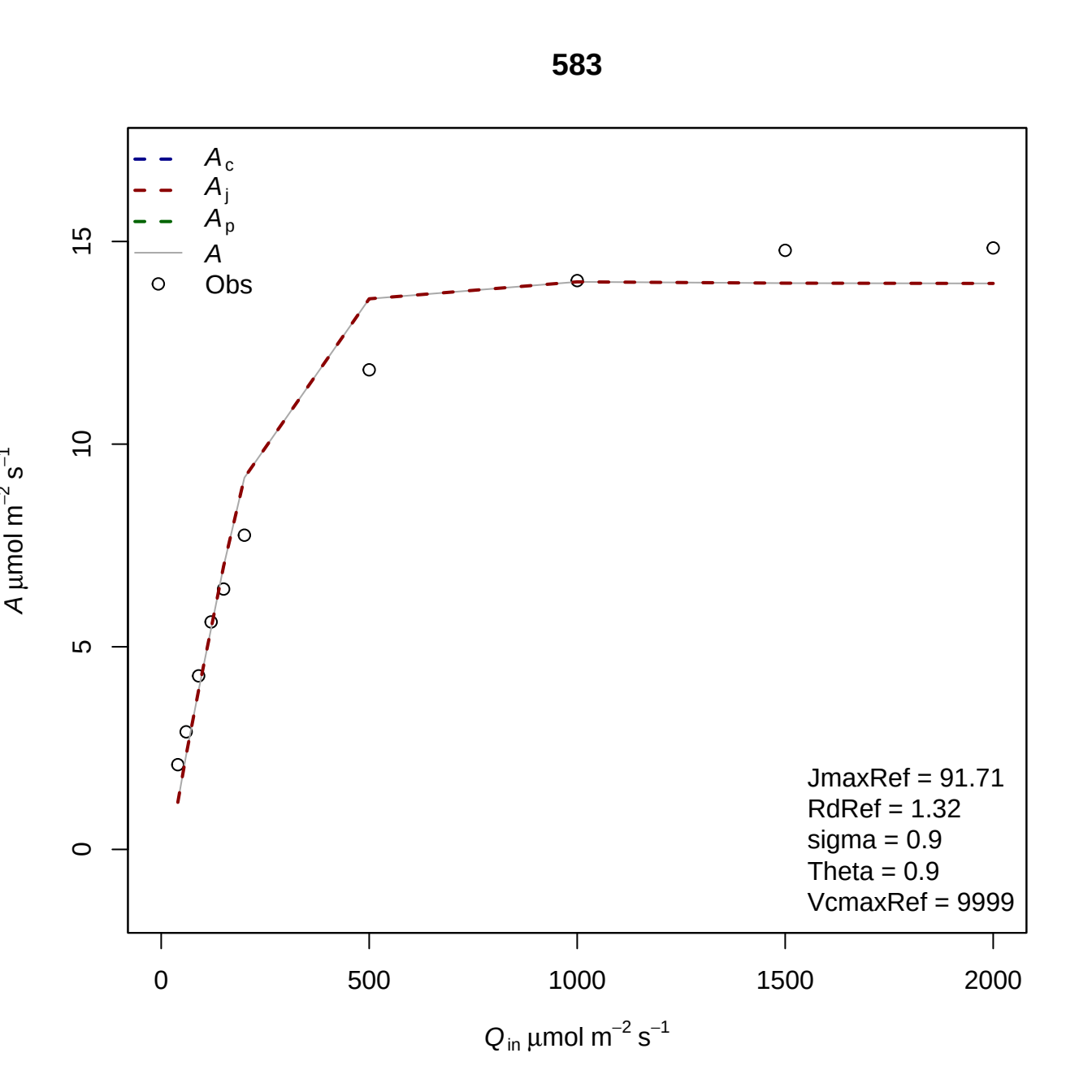

577
A_c
A_j
A_p
A
Obs
JmaxRef = 80.2
RdRef = 1.59
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

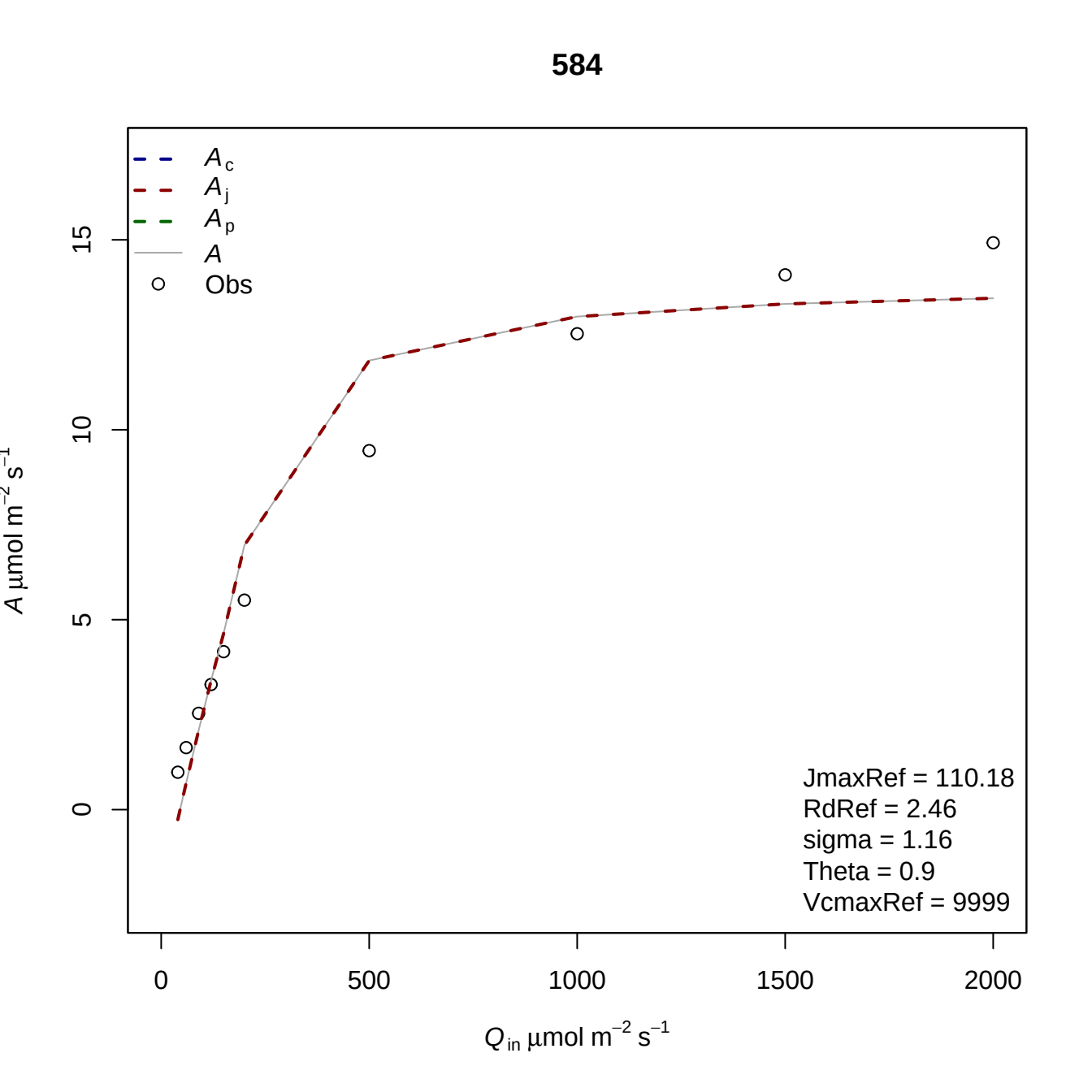

578
A_c
A_j
A_p
A
Obs
JmaxRef = 106.4
RdRef = 2.5
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

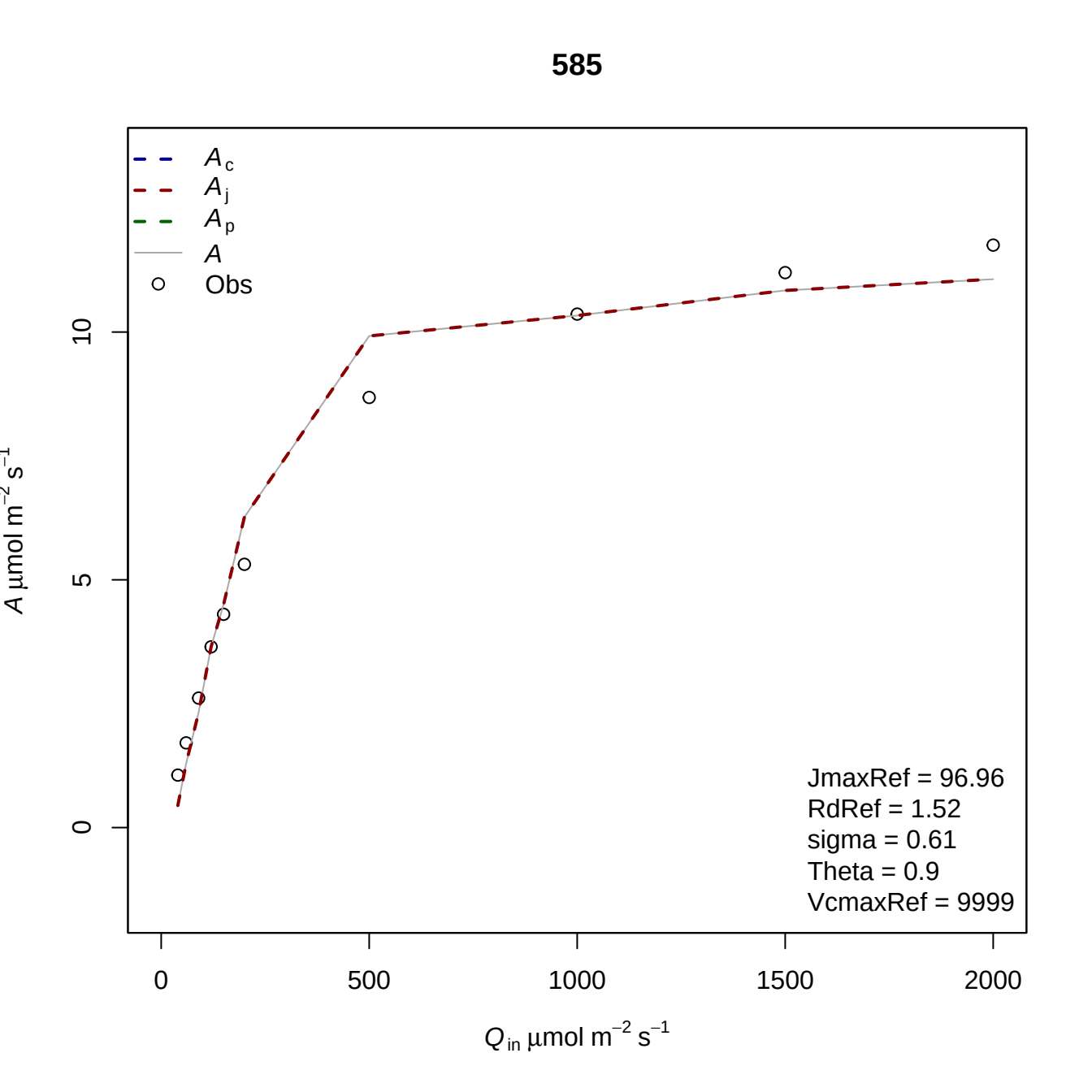

579
A_c
A_j
A_p
A
Obs
JmaxRef = 107.42
RdRef = 2.33
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

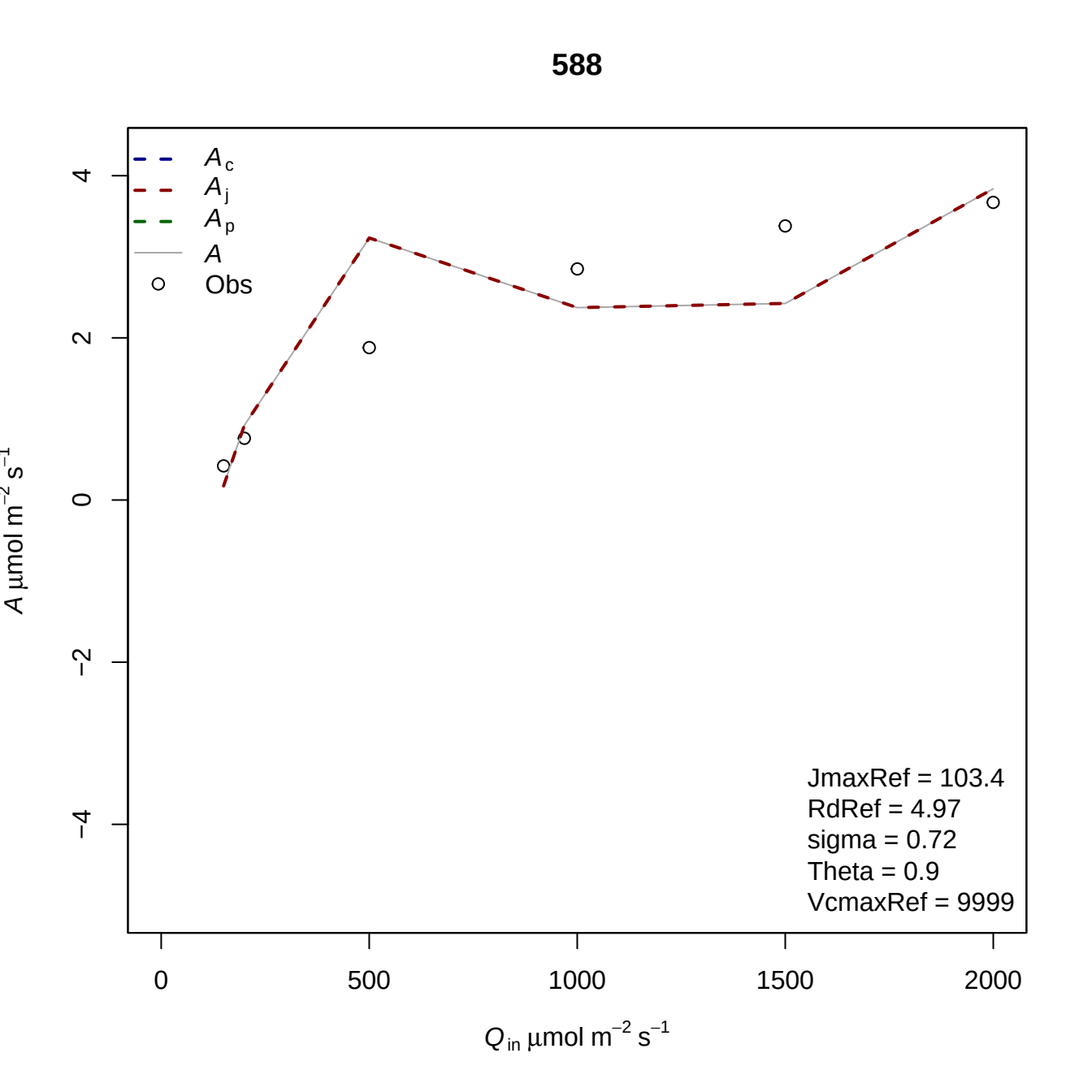

580
A_c
A_j
A_p
A
Obs
JmaxRef = 109.58
RdRef = 2.56
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

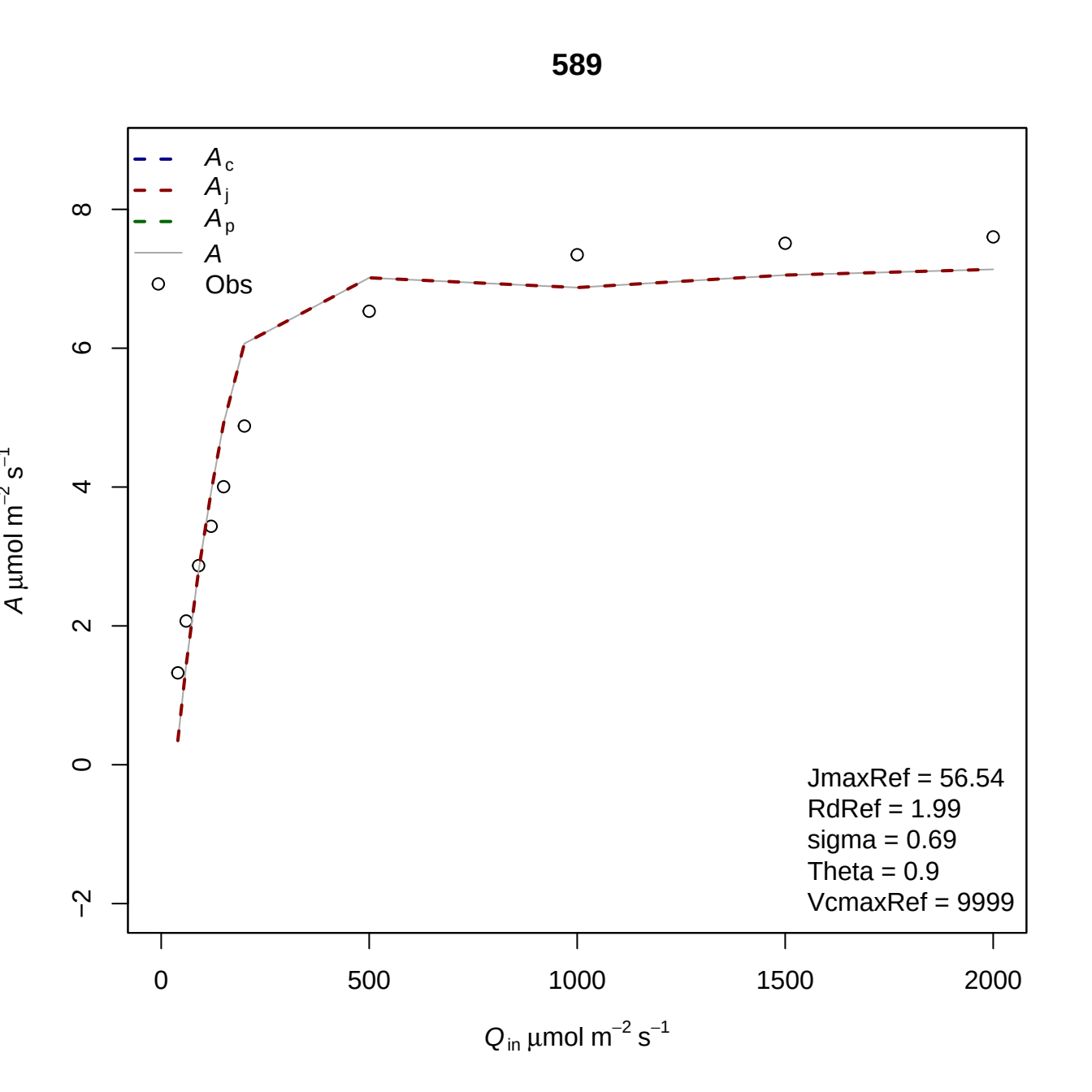

581
A_c
A_j
A_p
A
Obs
JmaxRef = 150.27
RdRef = 3.62
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

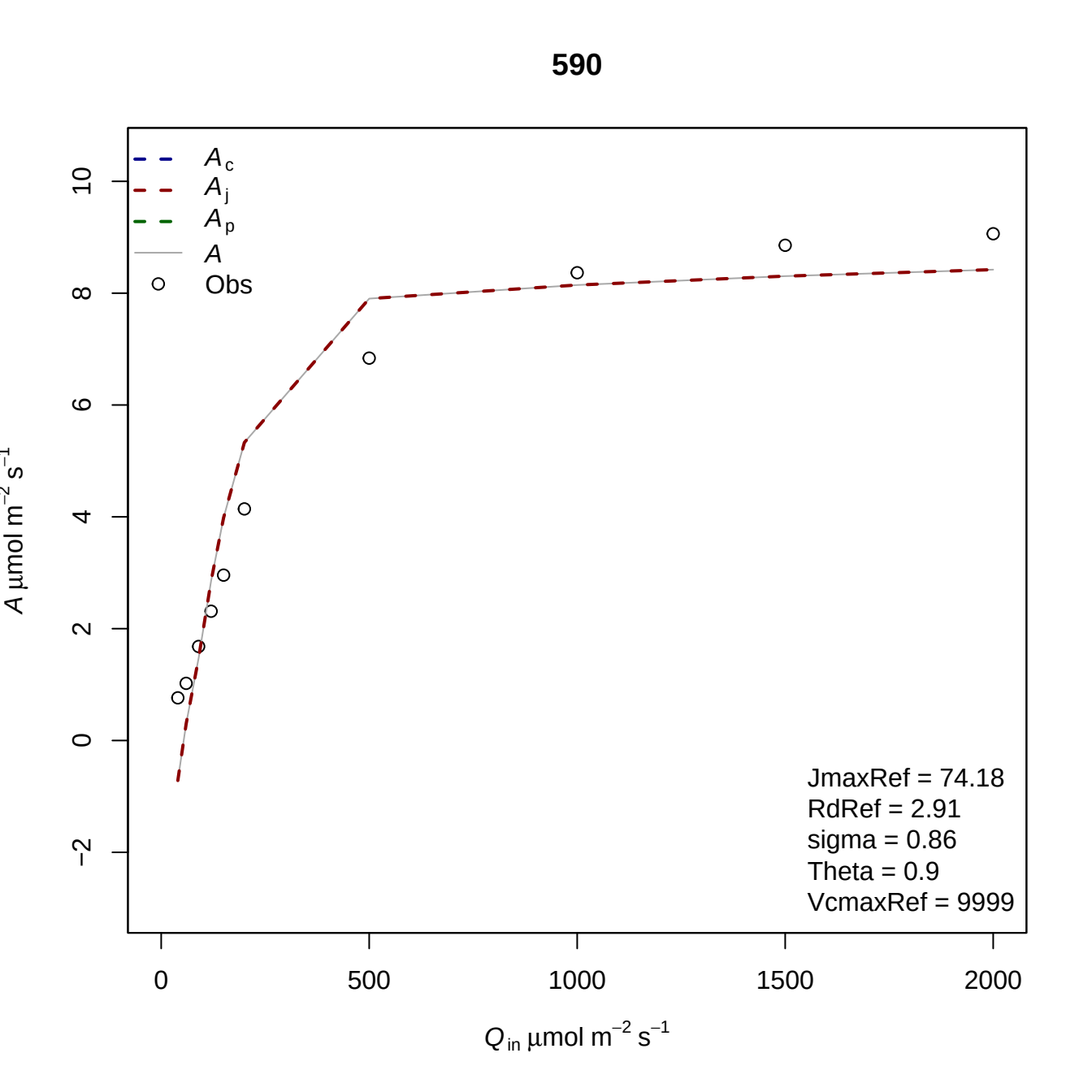

583
A_c
A_j
A_p
A
Obs
JmaxRef = 91.71
RdRef = 1.32
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

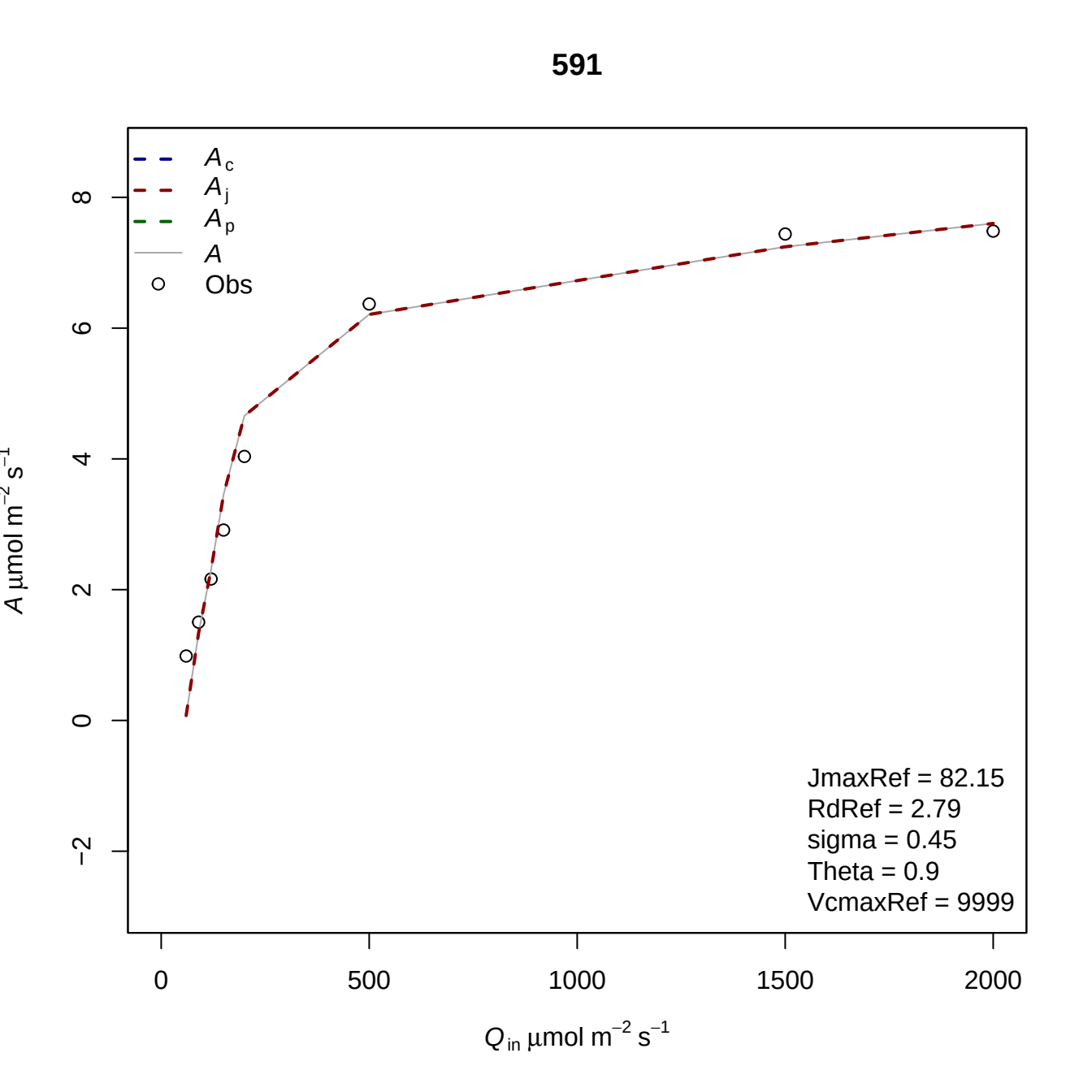

584
A_c
A_j
A_p
A
Obs
JmaxRef = 110.18
RdRef = 2.46
sigma = 1.16
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

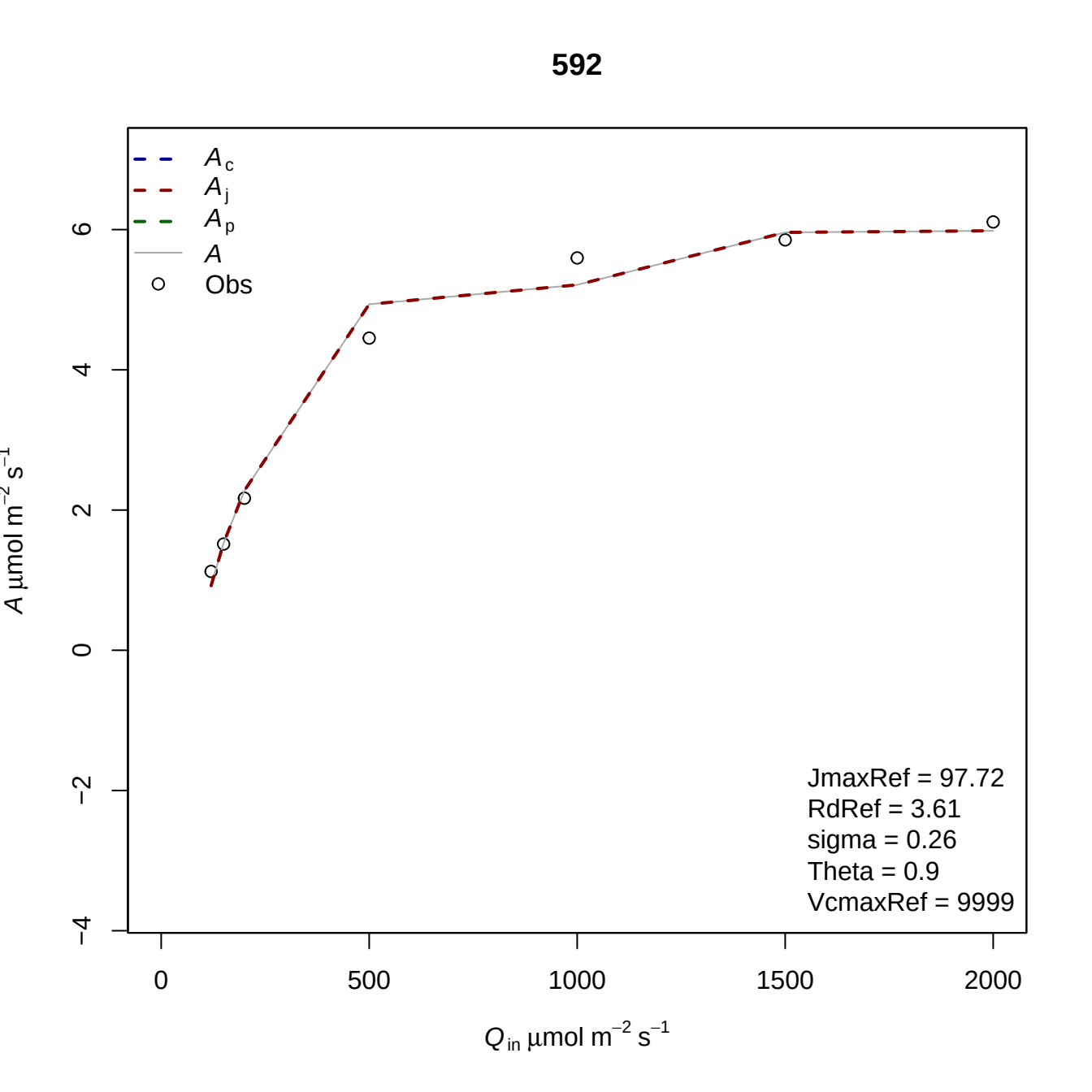

585
A_c
A_j
A_p
A
Obs
JmaxRef = 96.96
RdRef = 1.52
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

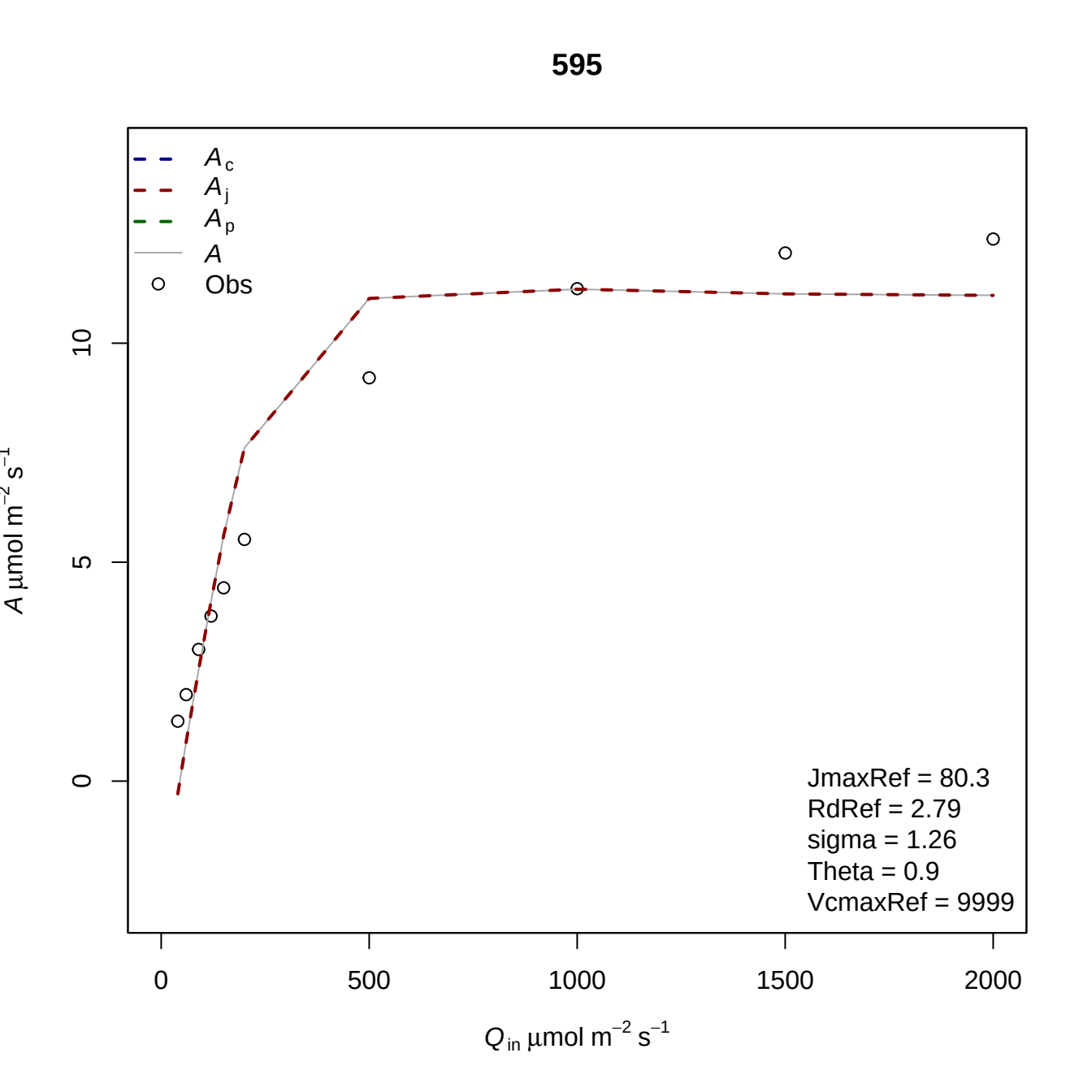

588
A_c
A_j
A_p
A
Obs
JmaxRef = 103.4
RdRef = 4.97
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

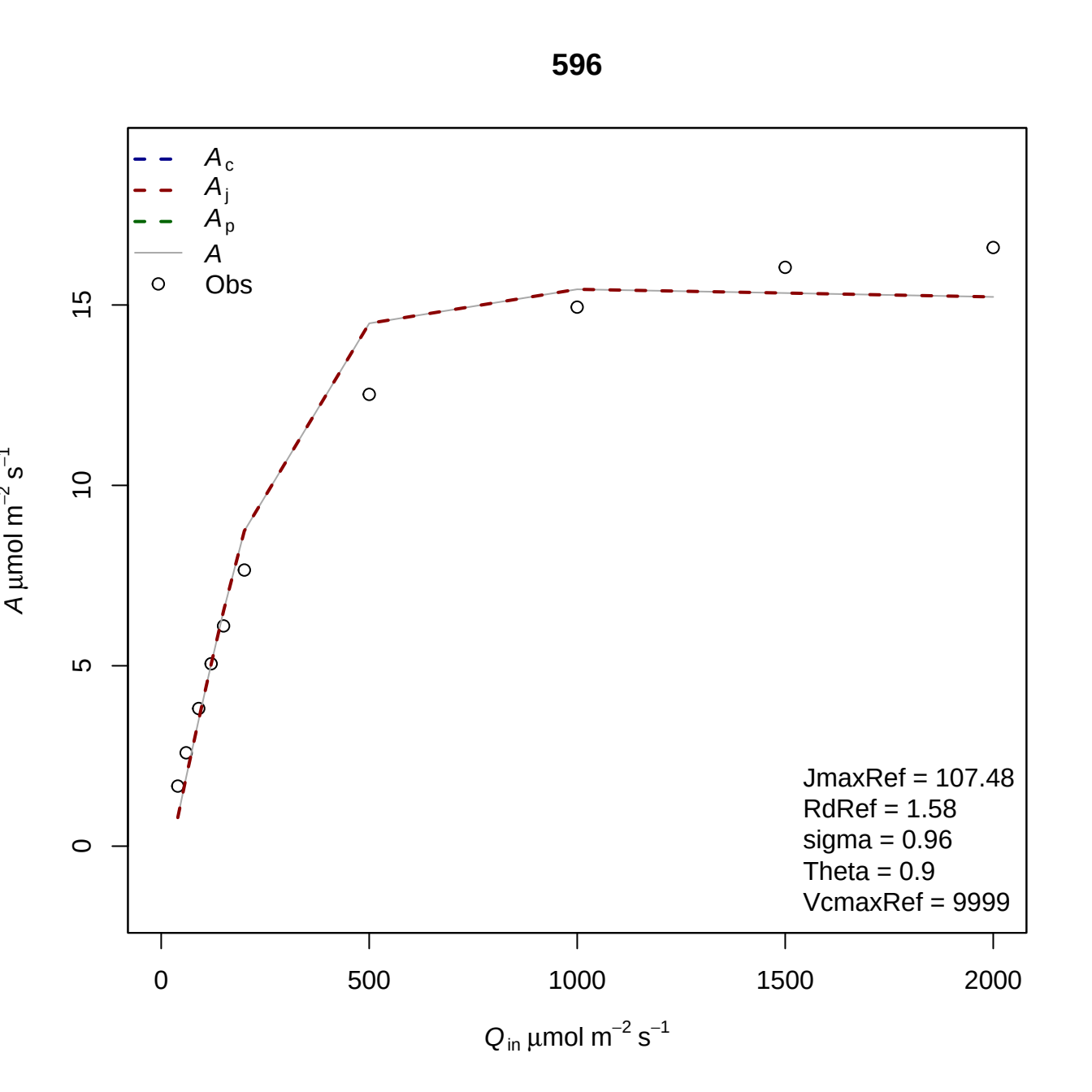

589
A_c
A_j
A_p
A
Obs
JmaxRef = 56.54
RdRef = 1.99
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

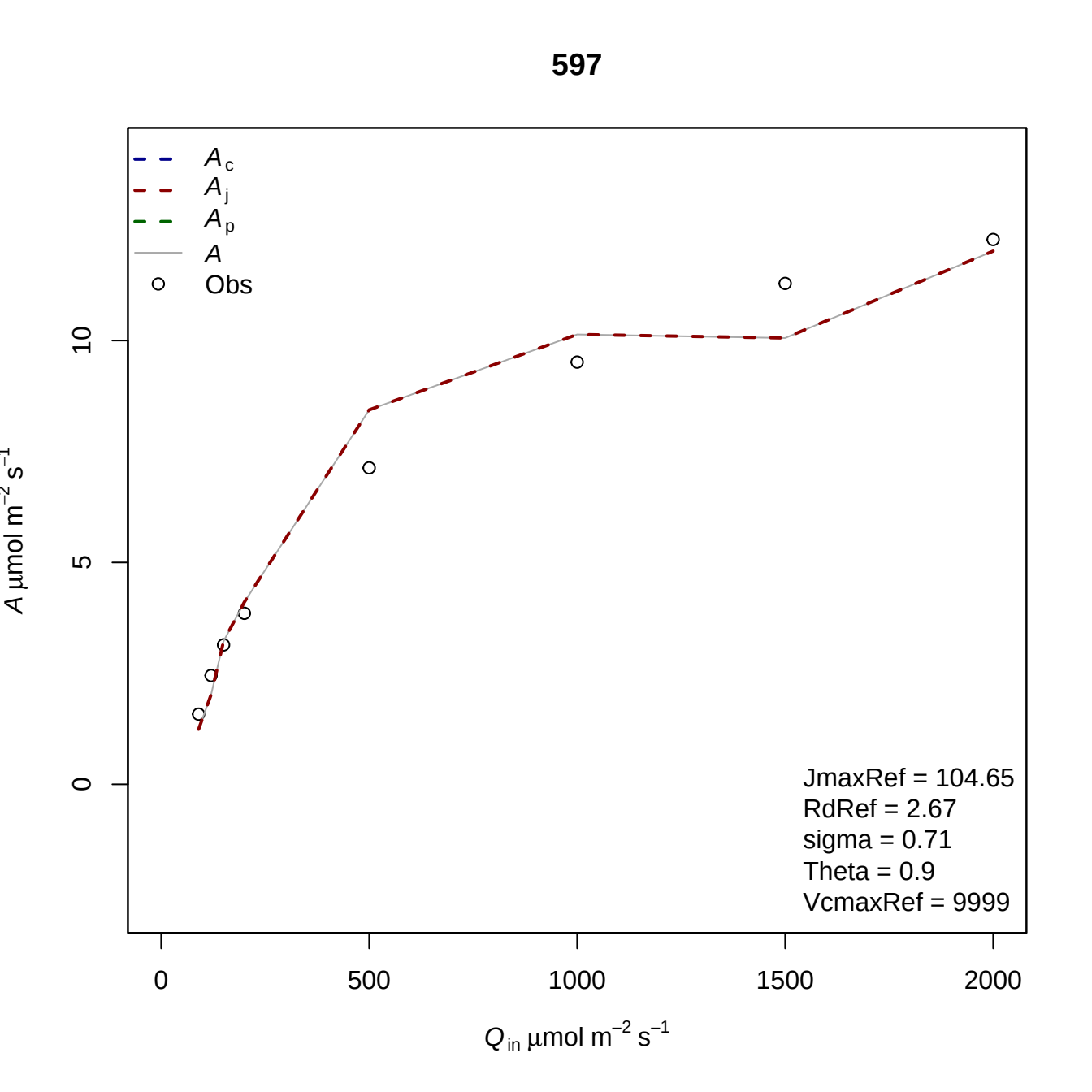

590
A_c
A_j
A_p
A
Obs
JmaxRef = 74.18
RdRef = 2.91
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

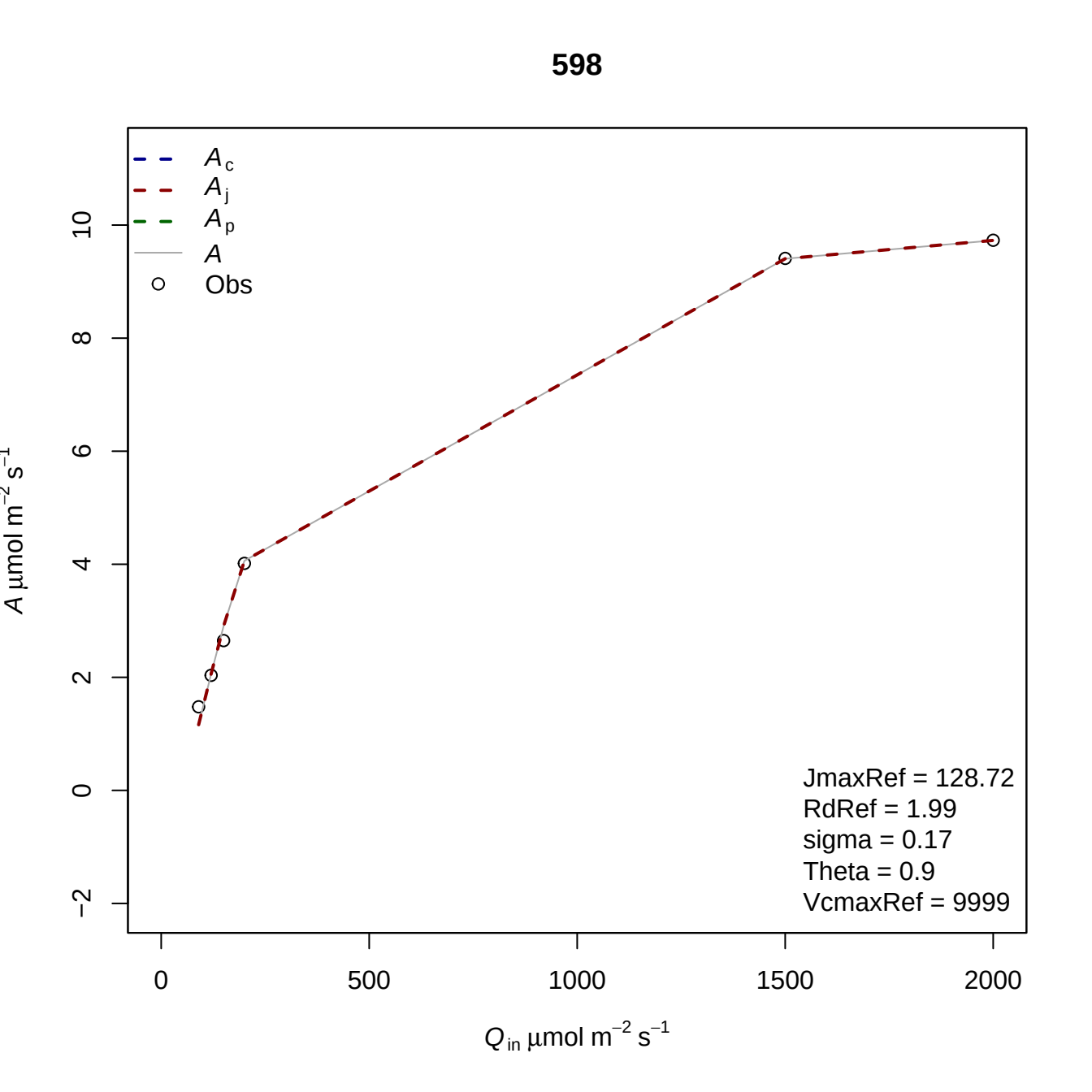

591
A_c
A_j
A_p
A
Obs
JmaxRef = 82.15
RdRef = 2.79
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

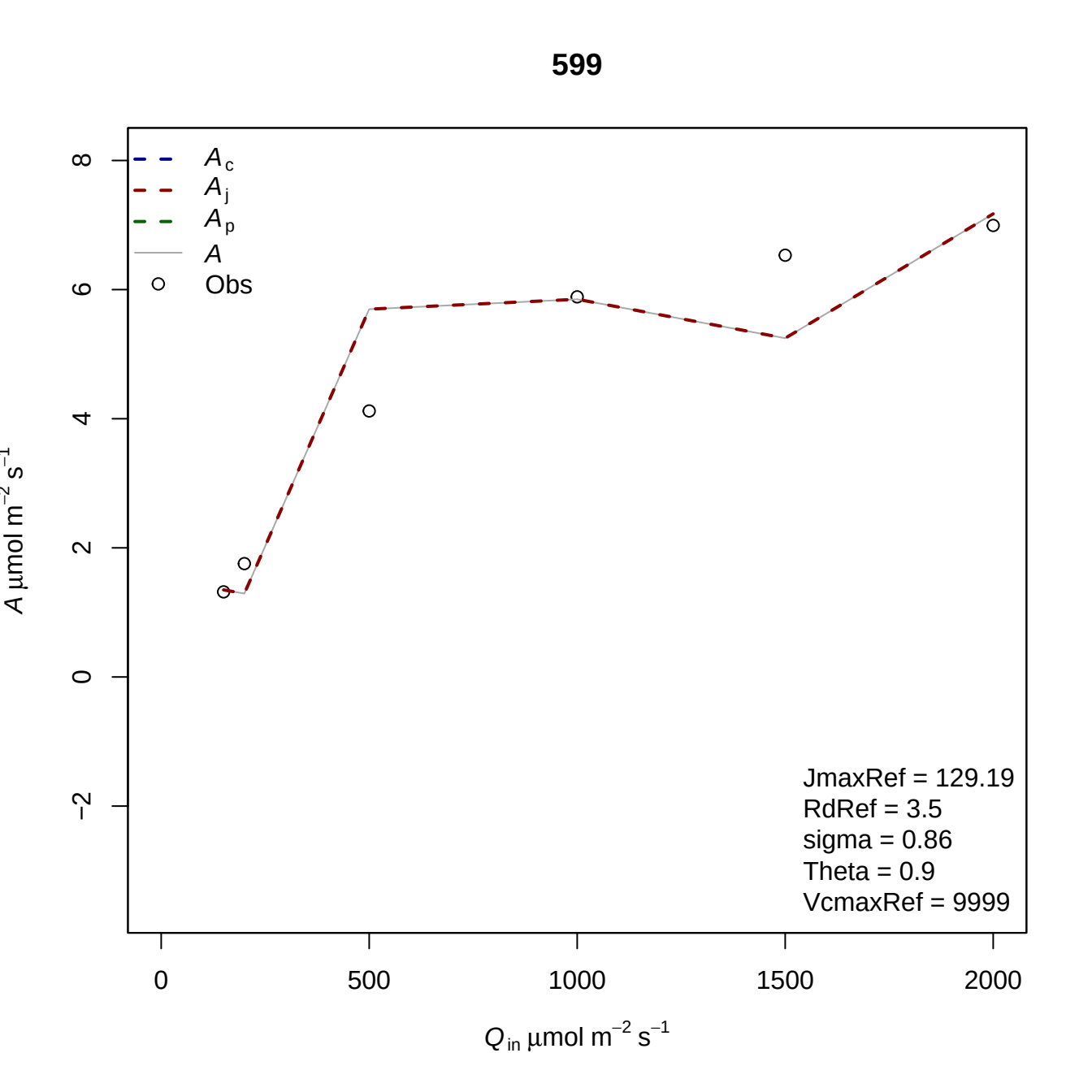

592
A_c
A_j
A_p
A
Obs
JmaxRef = 97.72
RdRef = 3.61
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

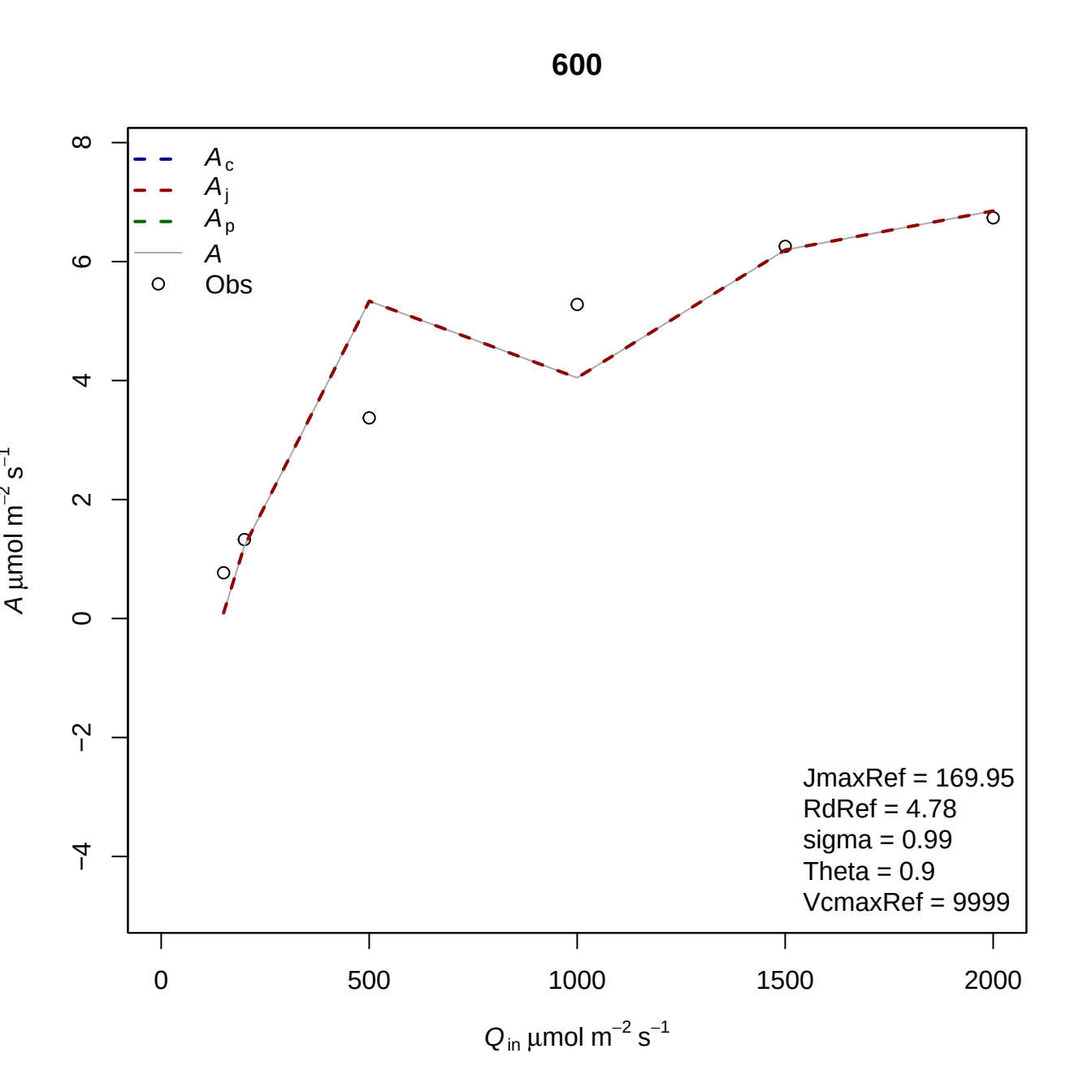

595
A_c
A_j
A_p
A
Obs
JmaxRef = 80.3
RdRef = 2.79
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

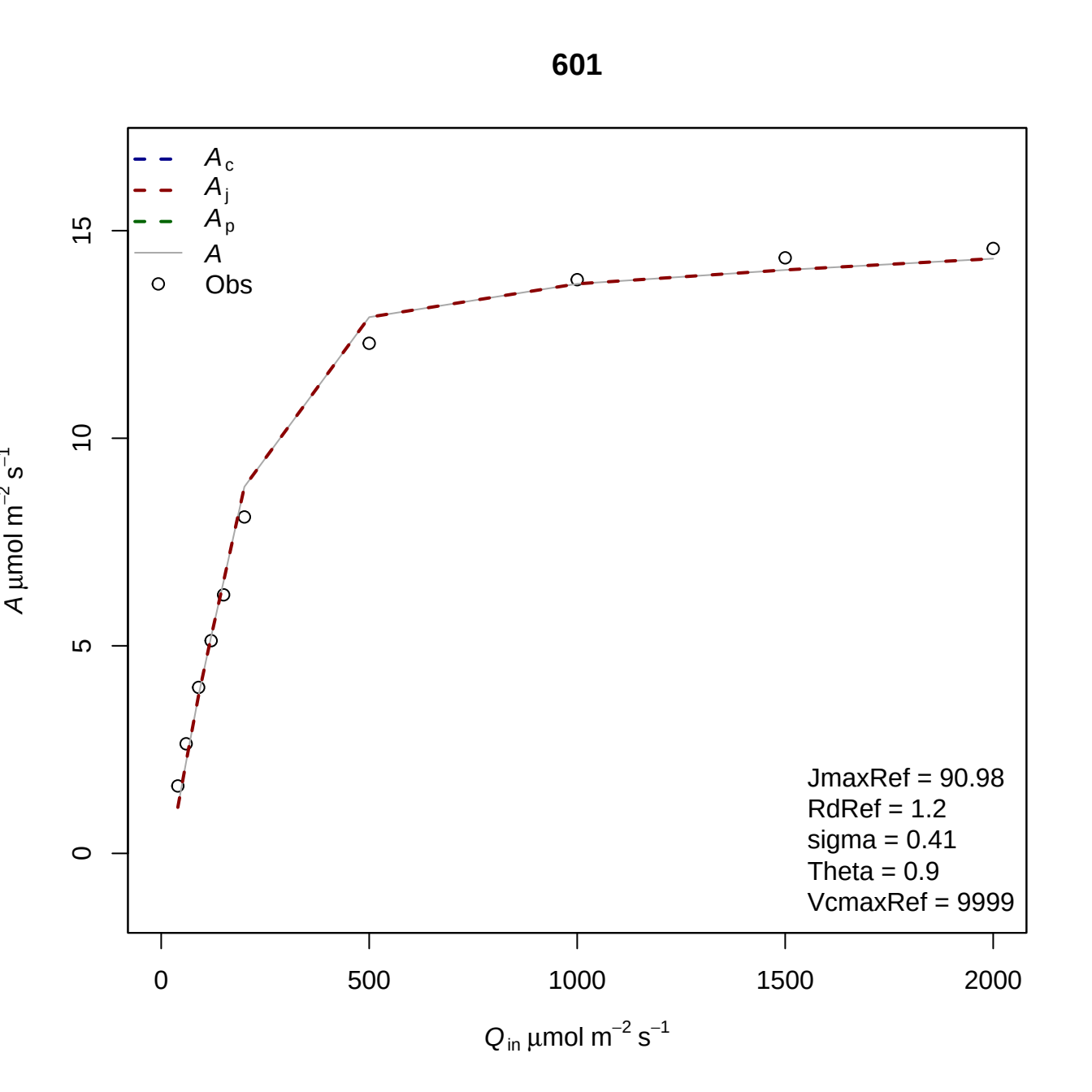

596
A_c
A_j
A_p
A
Obs
JmaxRef = 107.48
RdRef = 1.58
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

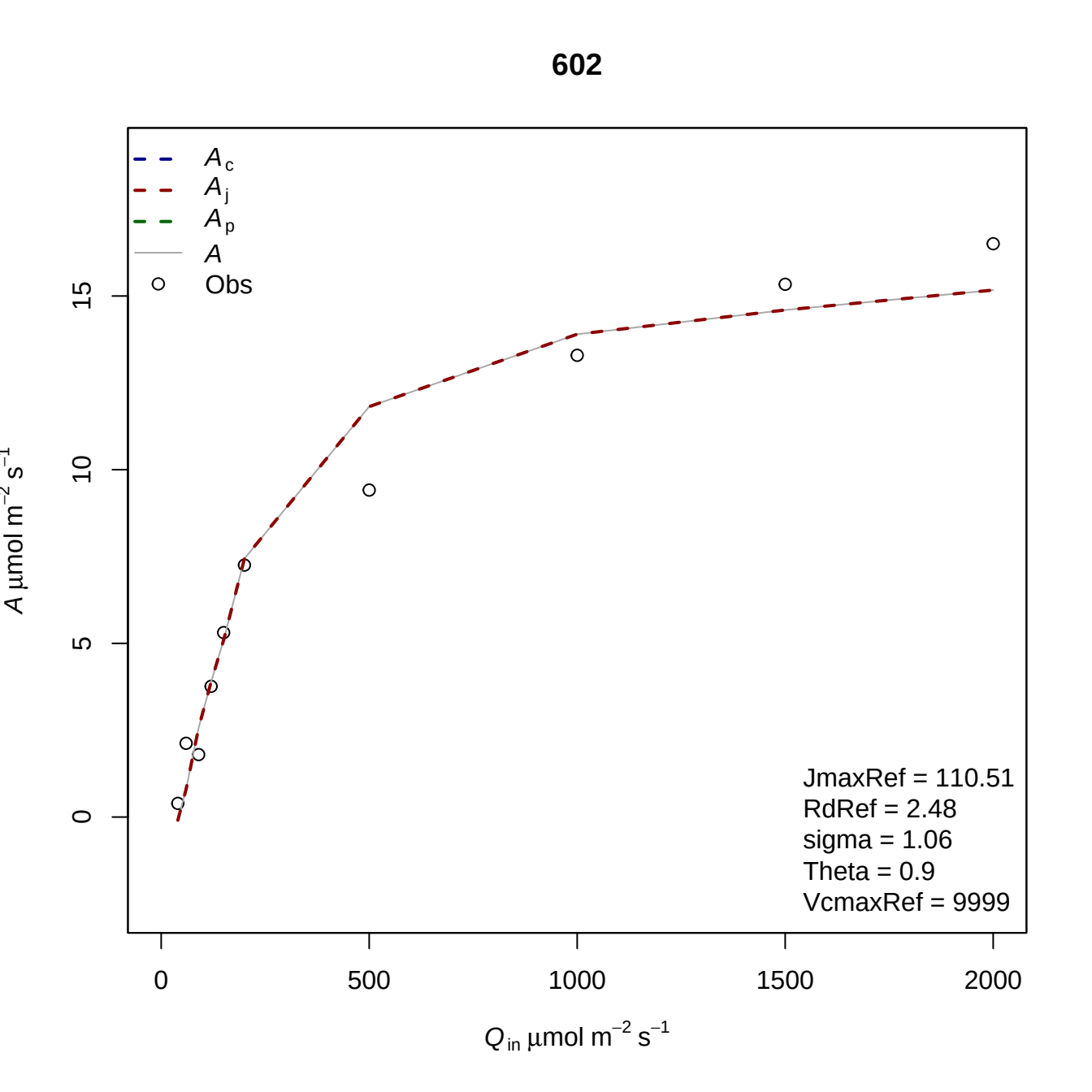

597
A_c
A_j
A_p
A
Obs
JmaxRef = 104.65
RdRef = 2.67
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

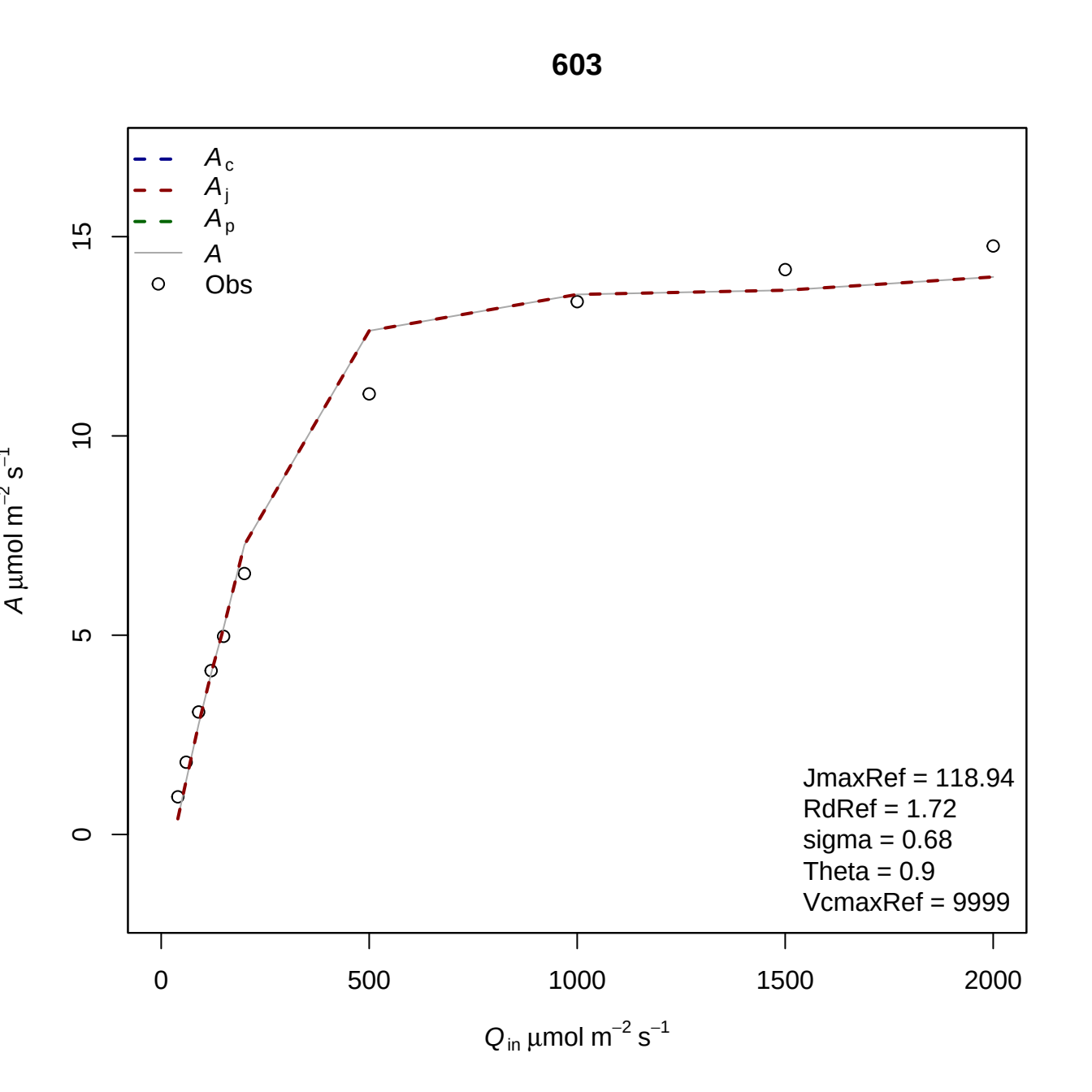

598
A_c
A_j
A_p
A
Obs
JmaxRef = 128.72
RdRef = 1.99
sigma = 0.17
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

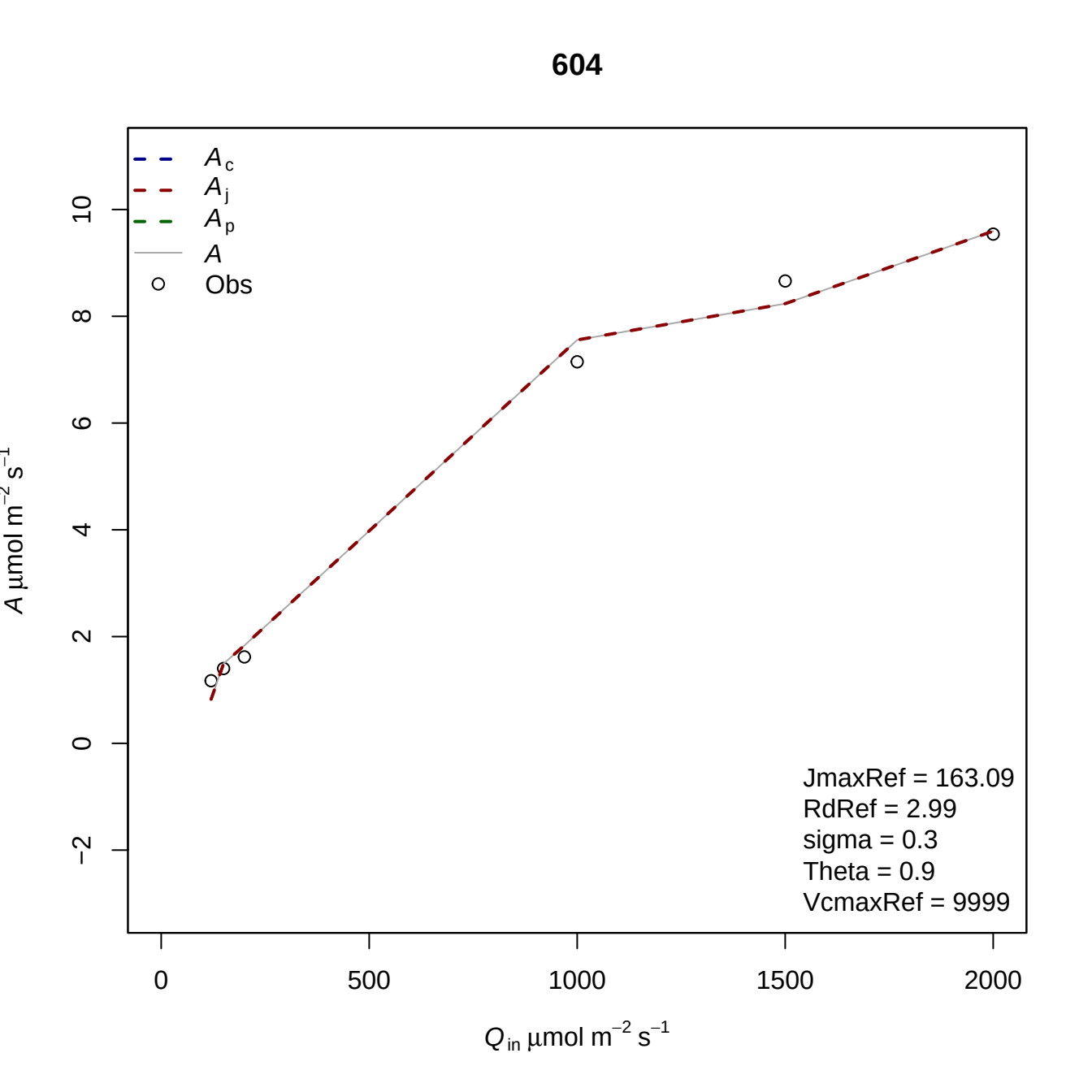

599
A_c
A_j
A_p
A
Obs
JmaxRef = 129.19
RdRef = 3.5
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

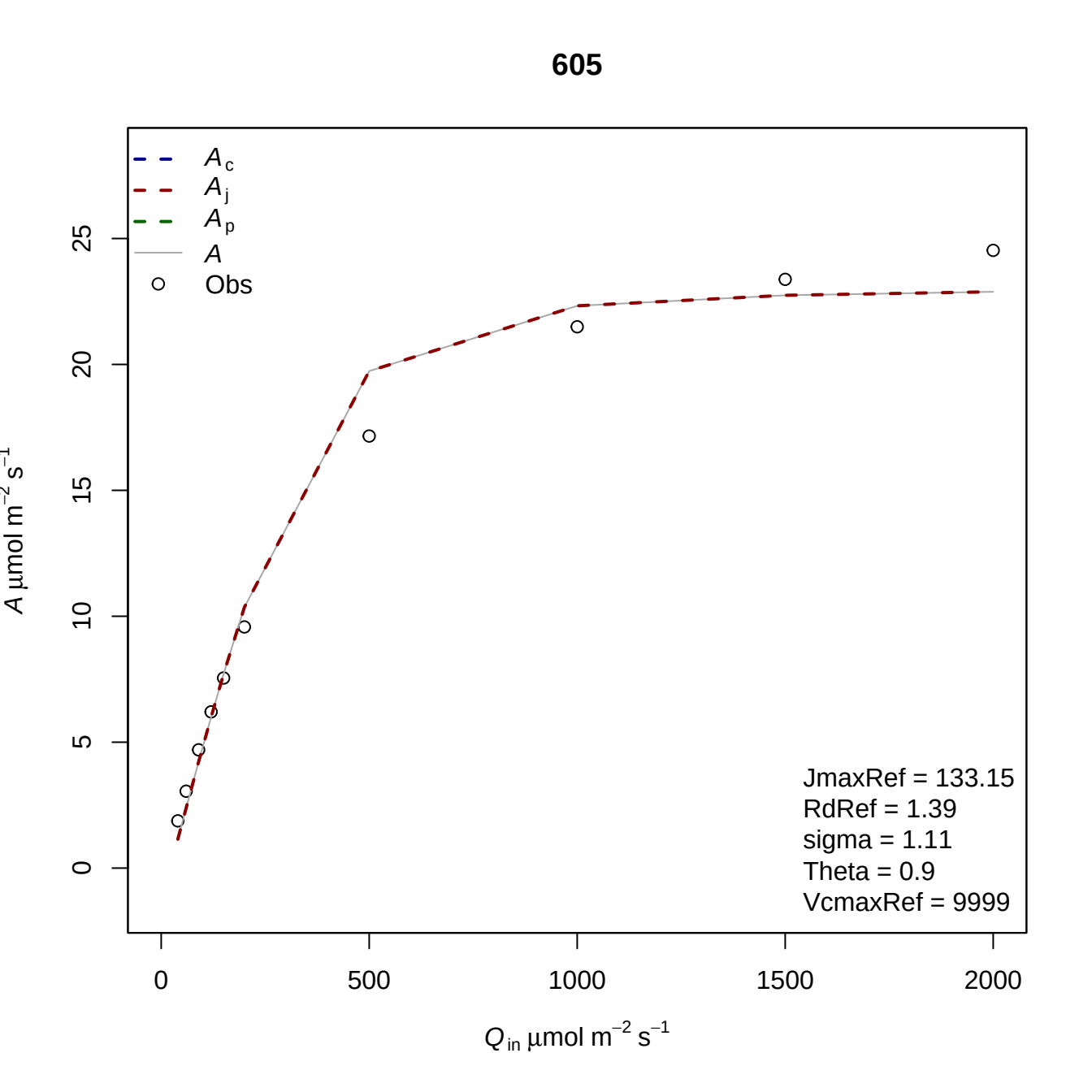

600
A_c
A_j
A_p
A
Obs
JmaxRef = 169.95
RdRef = 4.78
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

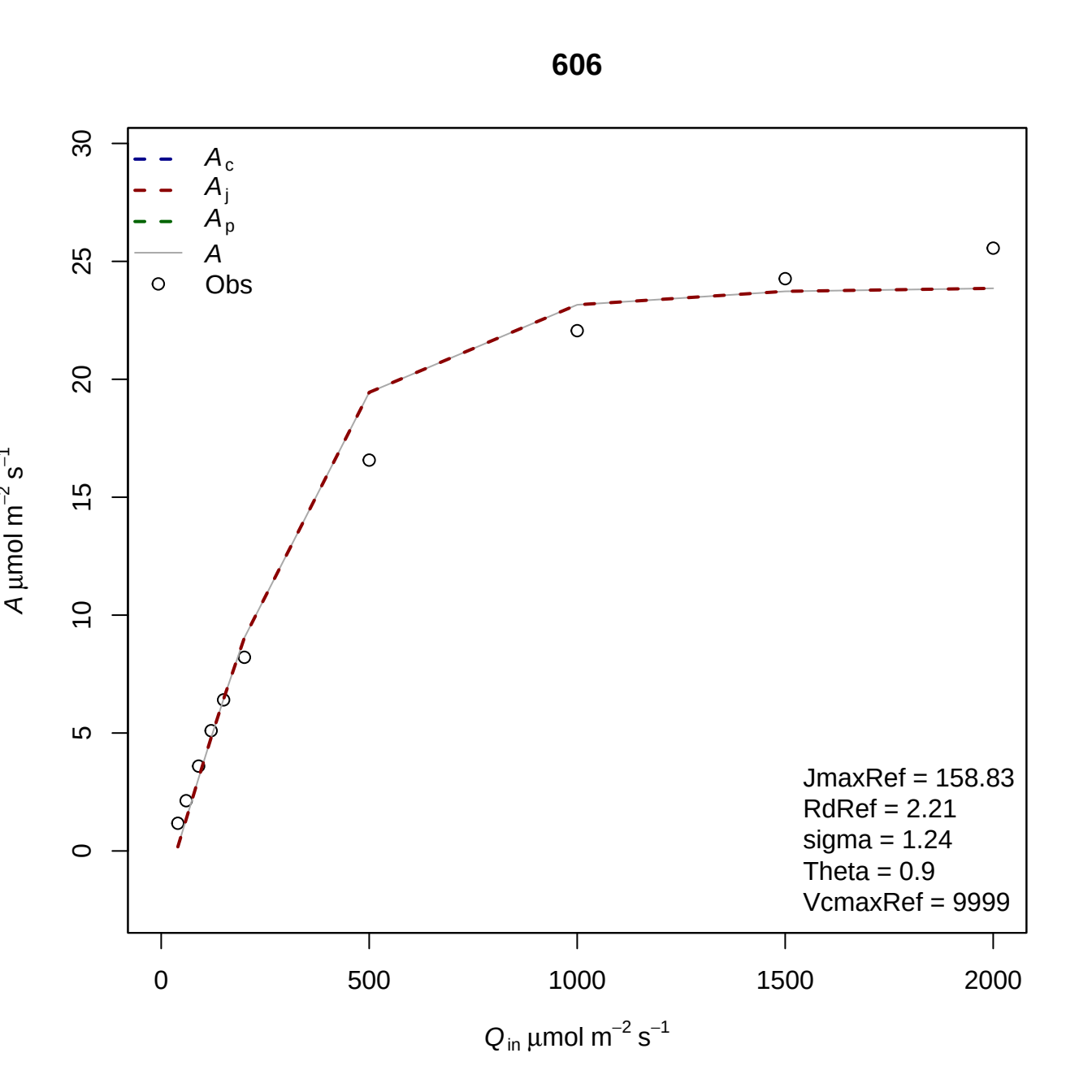

601
A_c
A_j
A_p
A
Obs
JmaxRef = 90.98
RdRef = 1.2
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

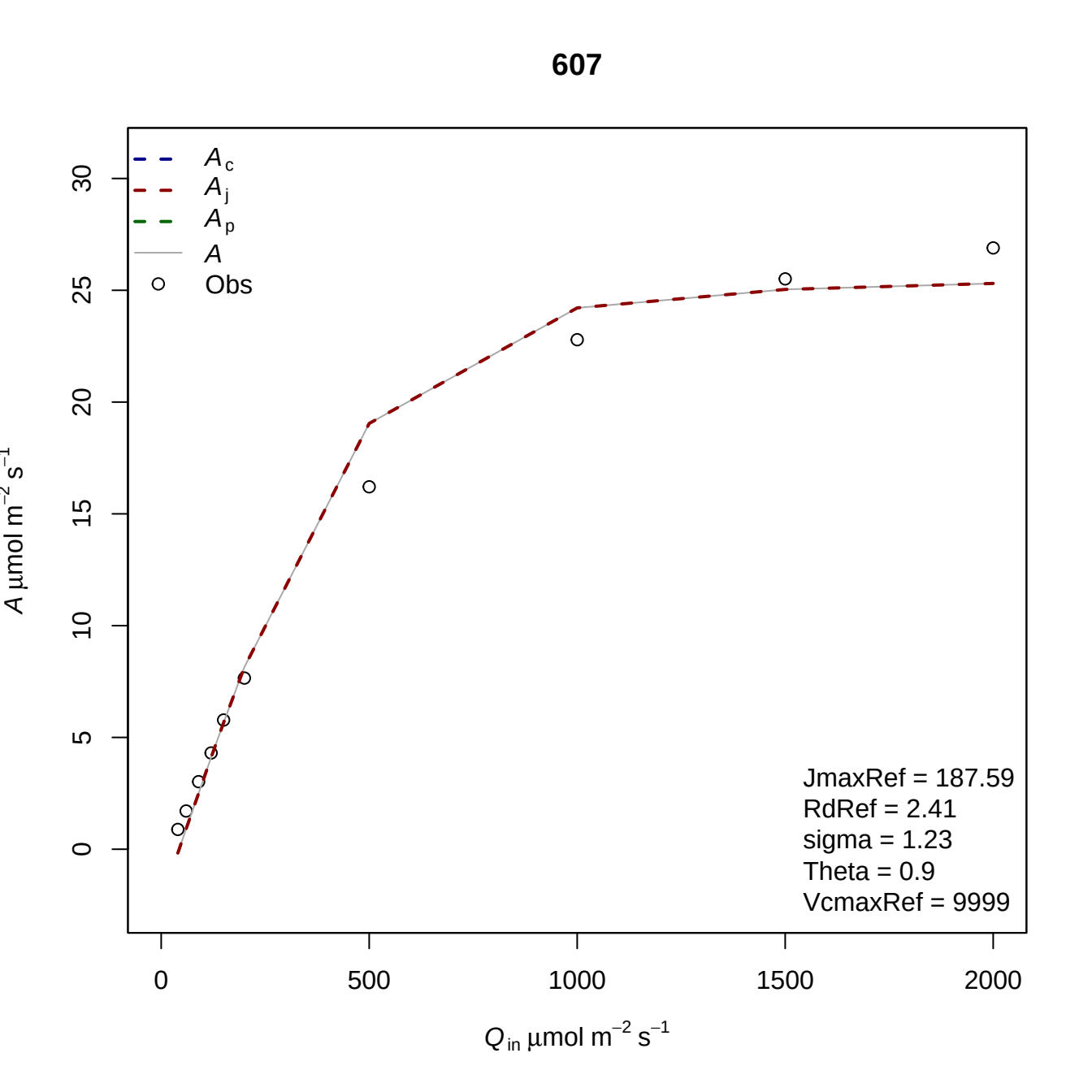

602
A_c
A_j
A_p
A
Obs
JmaxRef = 110.51
RdRef = 2.48
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

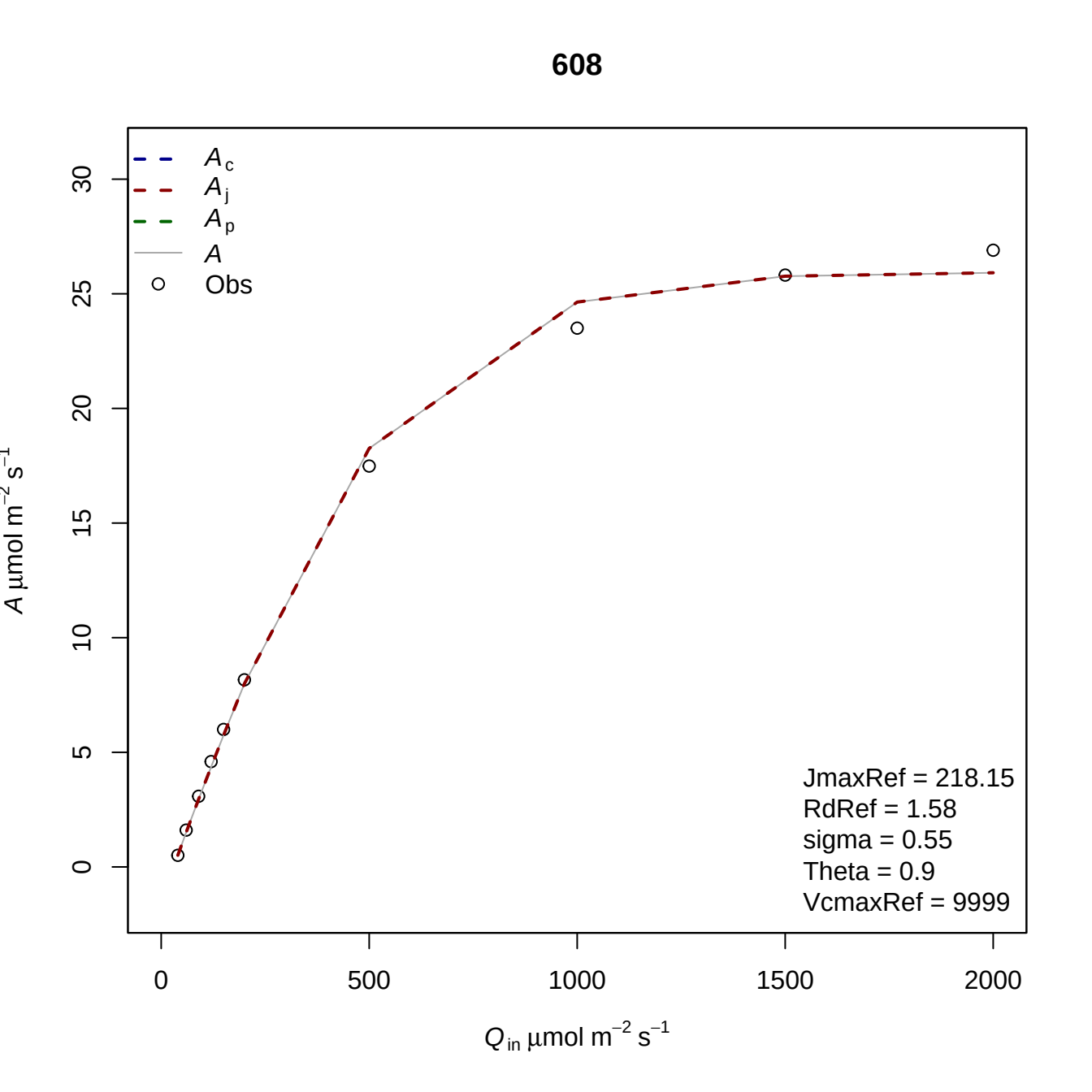

603
A_c
A_j
A_p
A
Obs
JmaxRef = 118.94
RdRef = 1.72
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

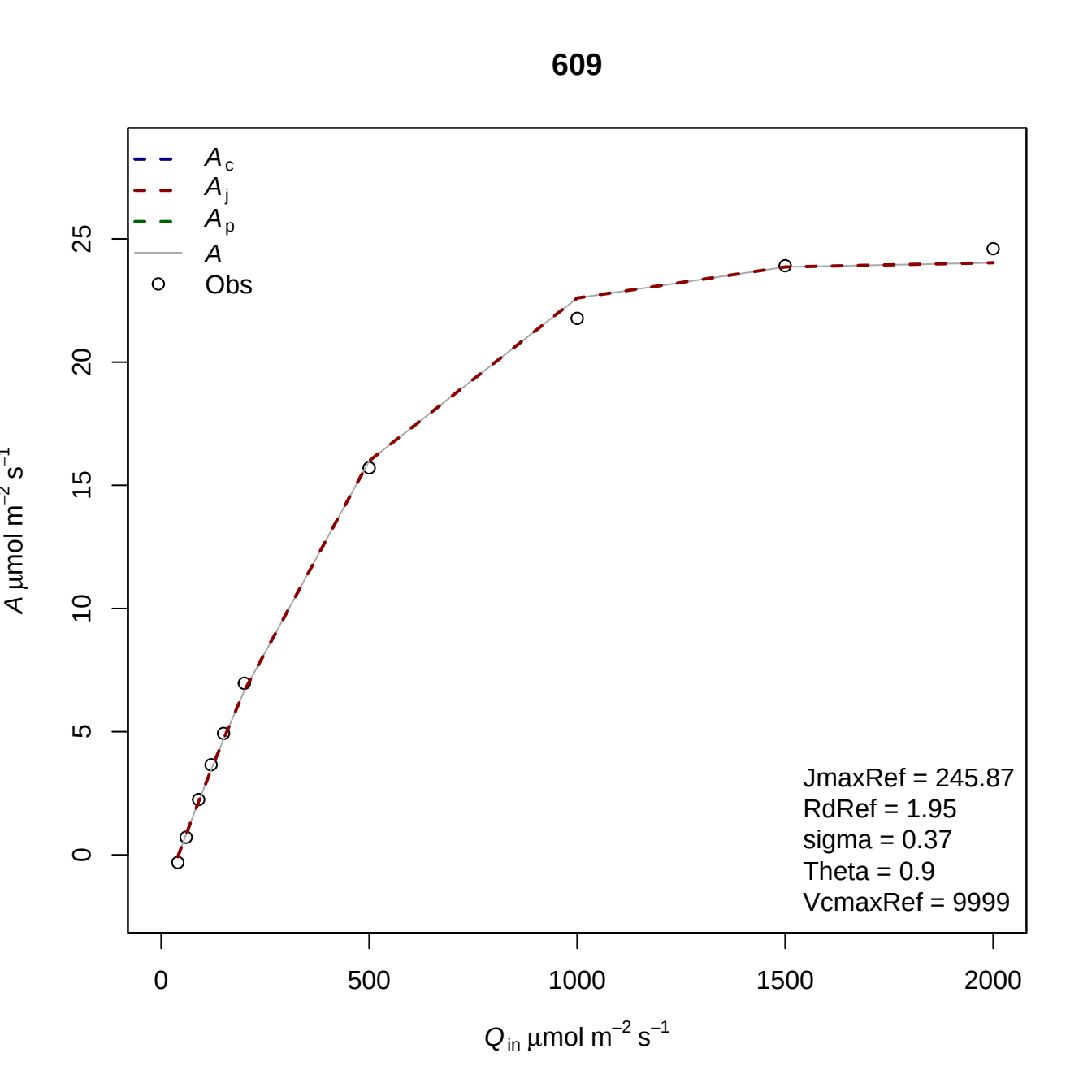

604
A_c
A_j
A_p
A
Obs
JmaxRef = 163.09
RdRef = 2.99
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

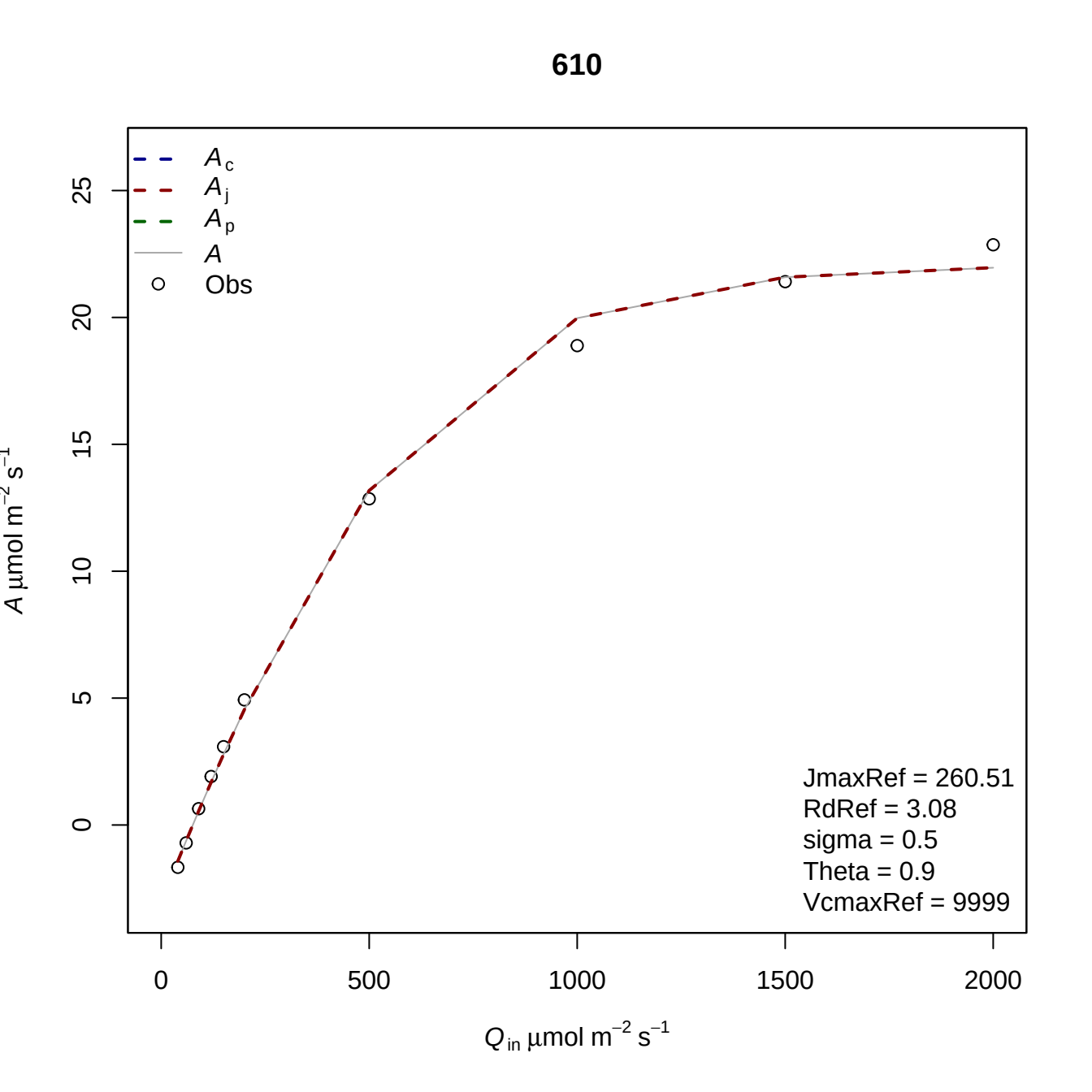

605
A_c
A_j
A_p
A
Obs
JmaxRef = 133.15
RdRef = 1.39
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

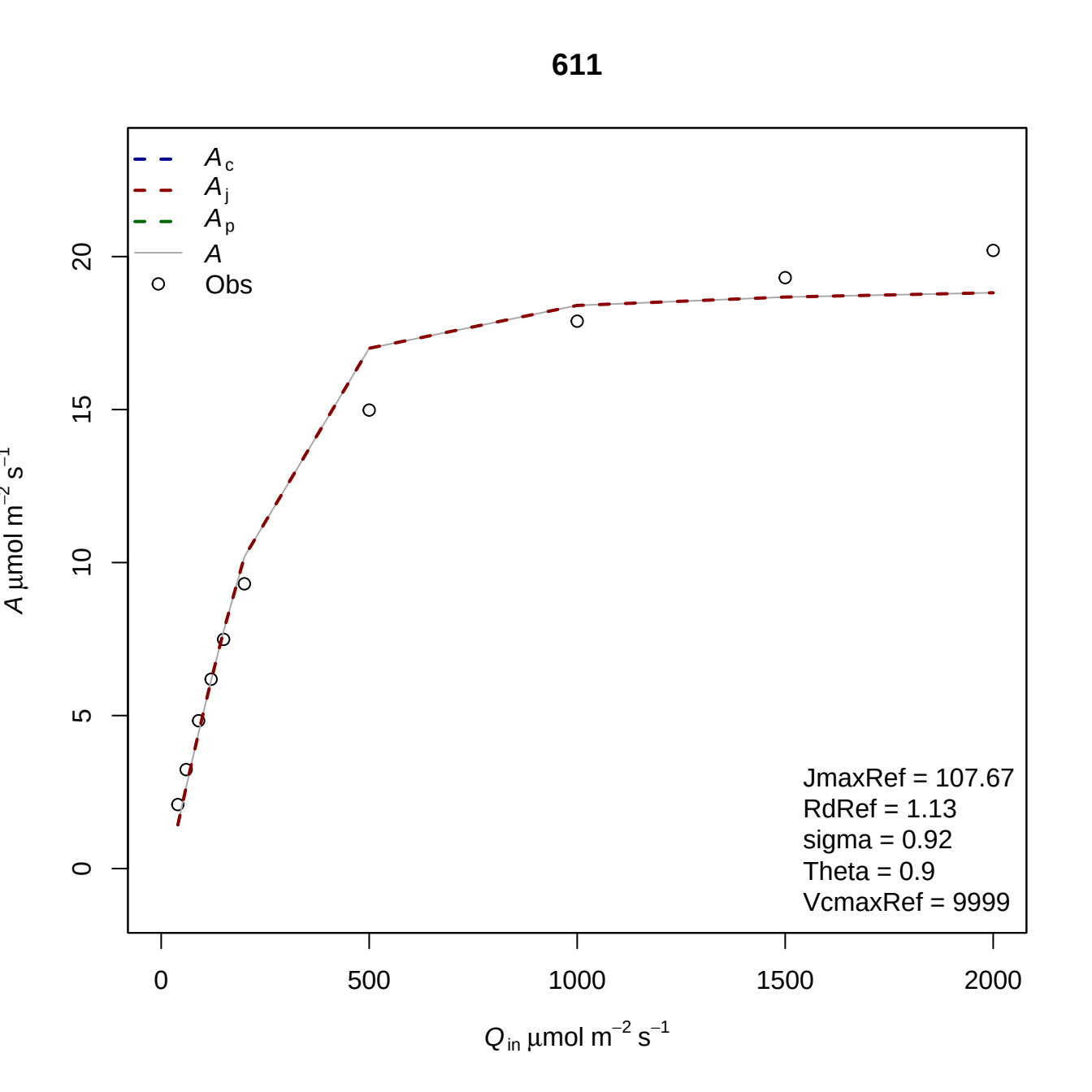

606
A_c
A_j
A_p
A
Obs
JmaxRef = 158.83
RdRef = 2.21
sigma = 1.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

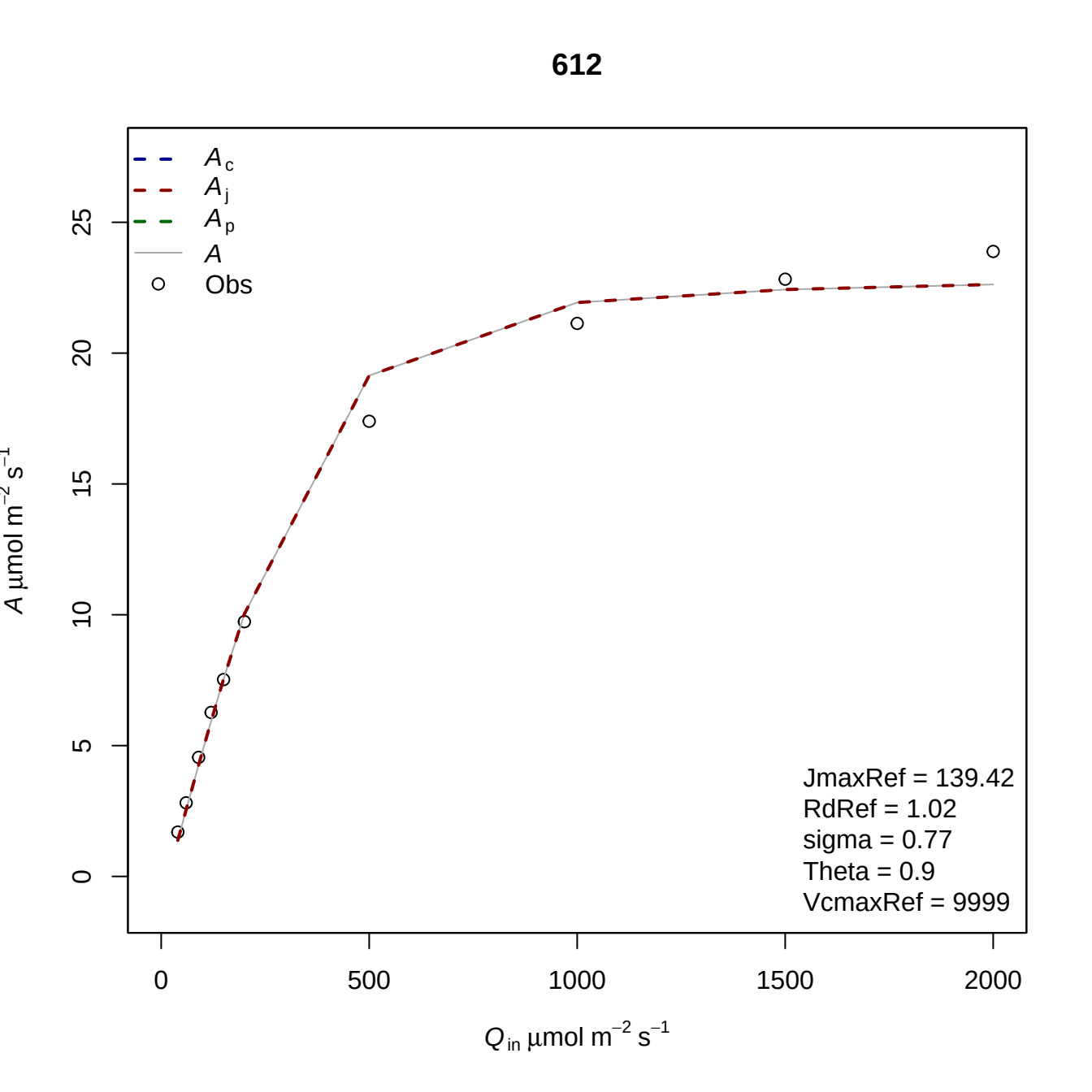

607
A_c
A_j
A_p
A
Obs
JmaxRef = 187.59
RdRef = 2.41
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

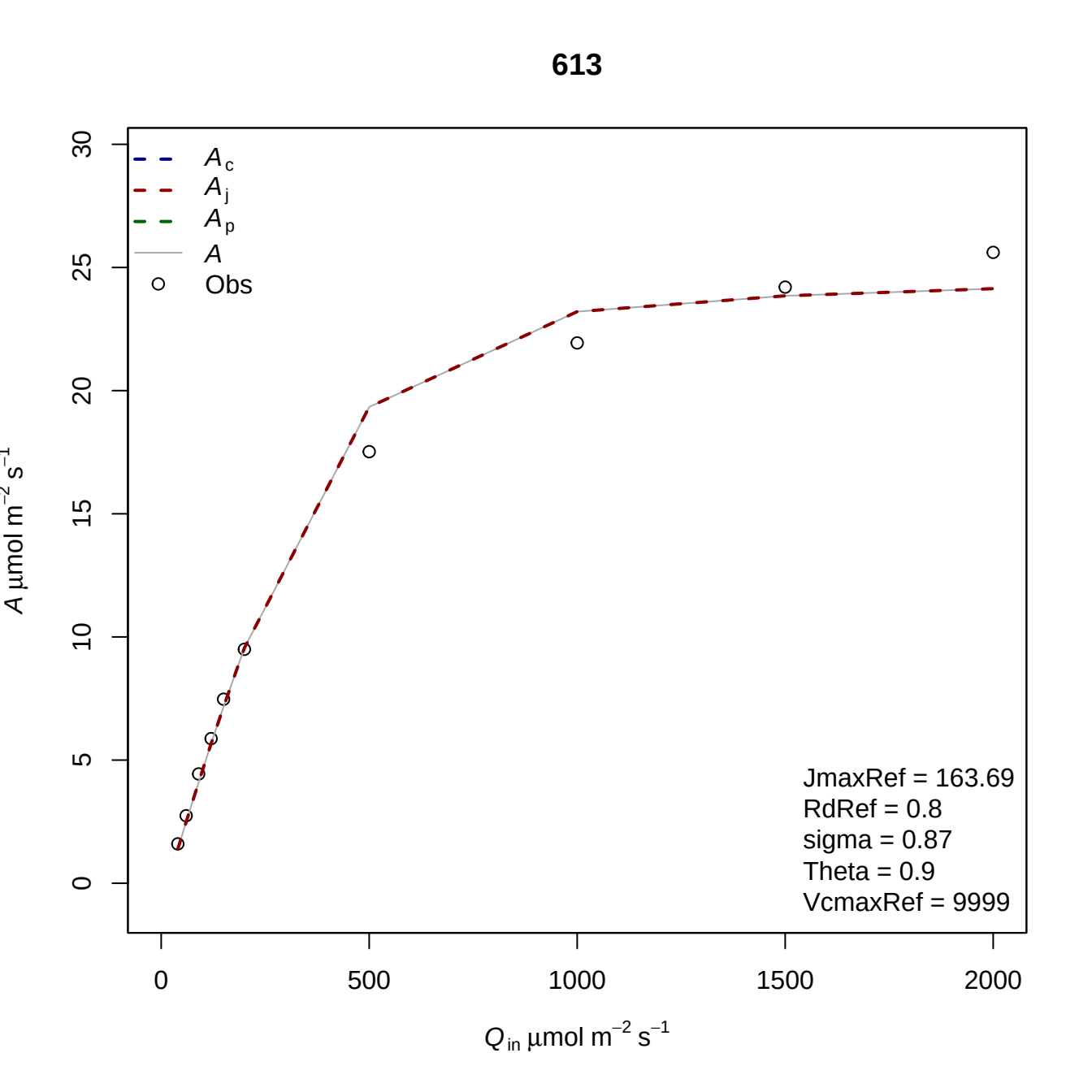

608
A_c
A_j
A_p
A
Obs
JmaxRef = 218.15
RdRef = 1.58
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

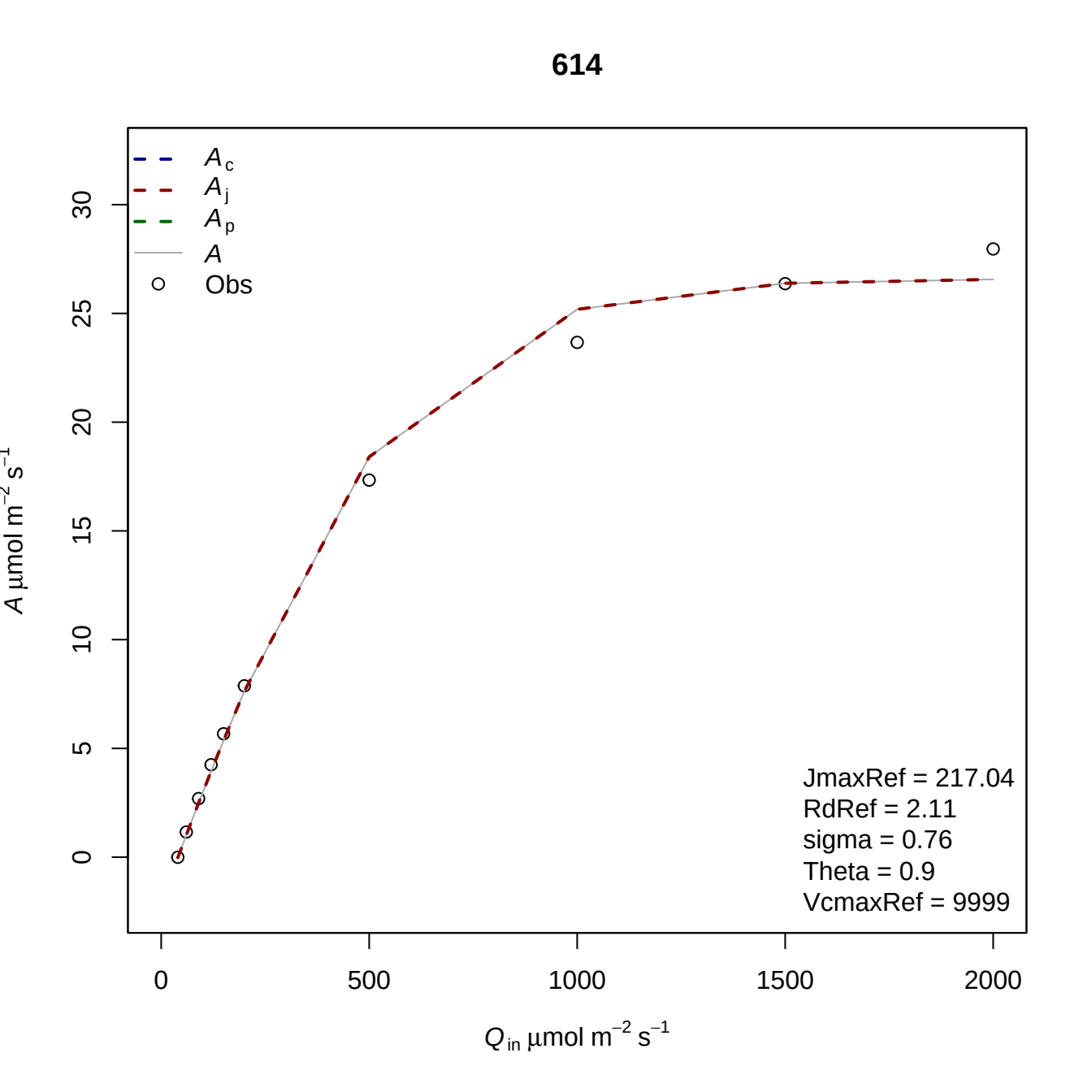

609
A_c
A_j
A_p
A
Obs
JmaxRef = 245.87
RdRef = 1.95
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

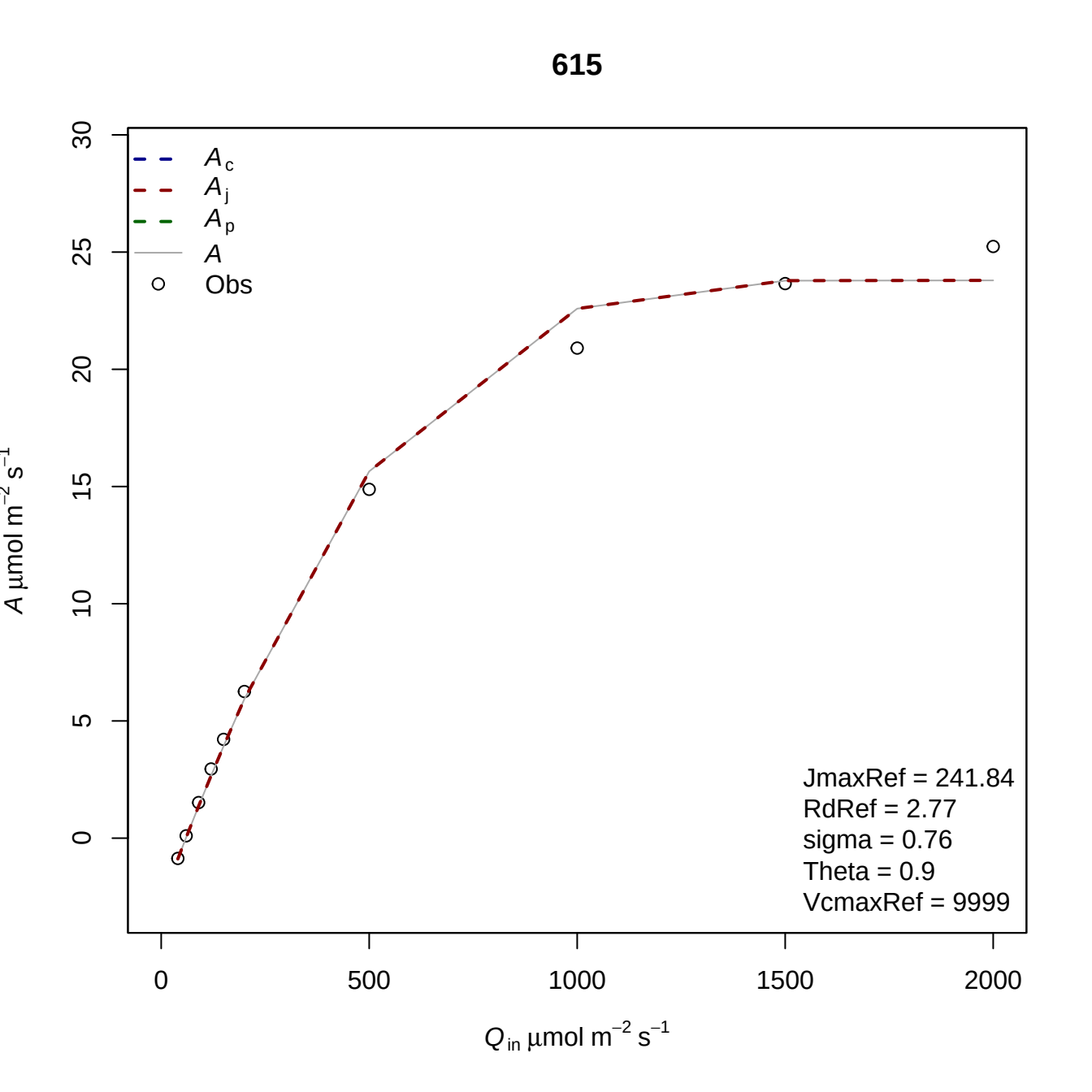

610
A_c
A_j
A_p
A
Obs
JmaxRef = 260.51
RdRef = 3.08
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

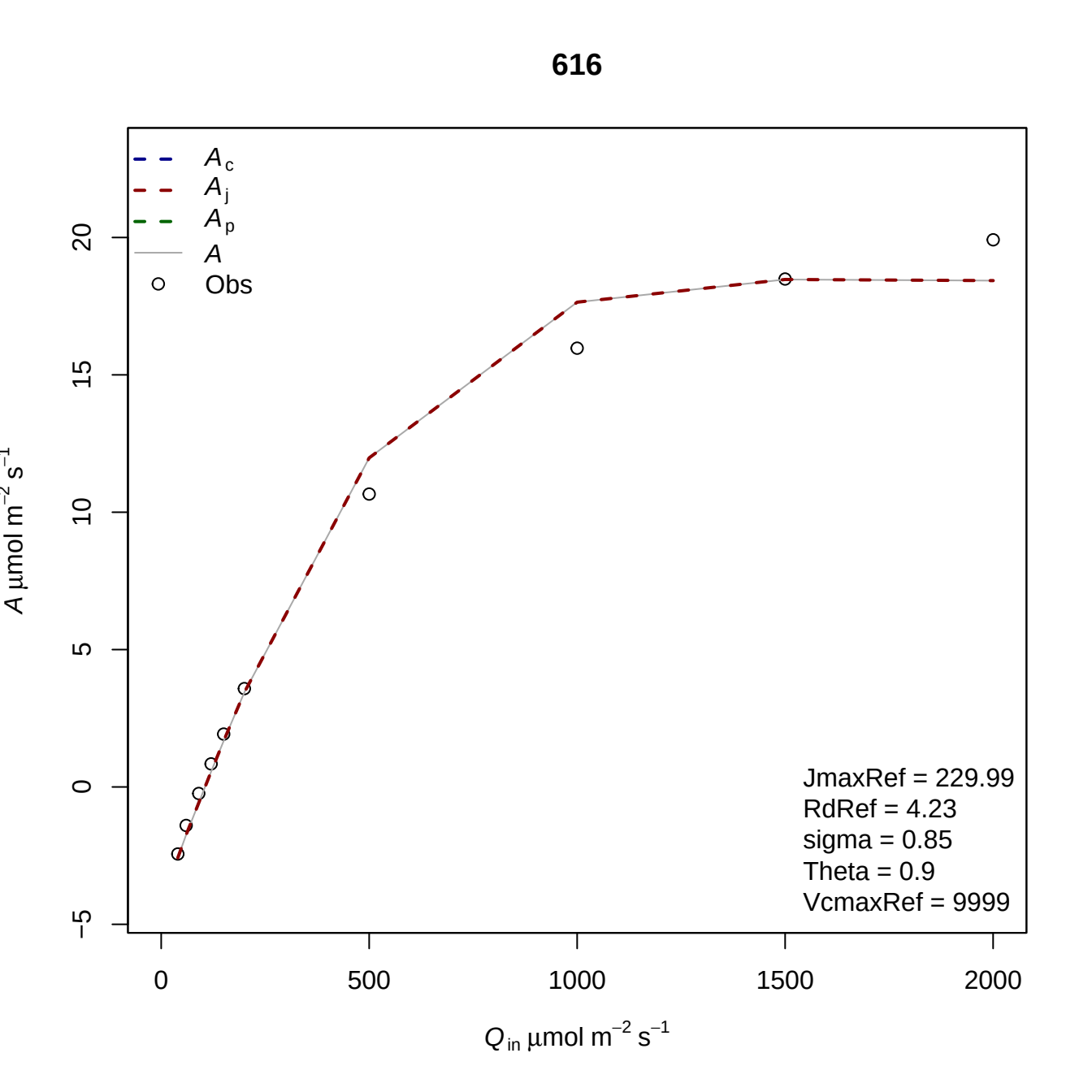

611
A_c
A_j
A_p
A
Obs
JmaxRef = 107.67
RdRef = 1.13
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

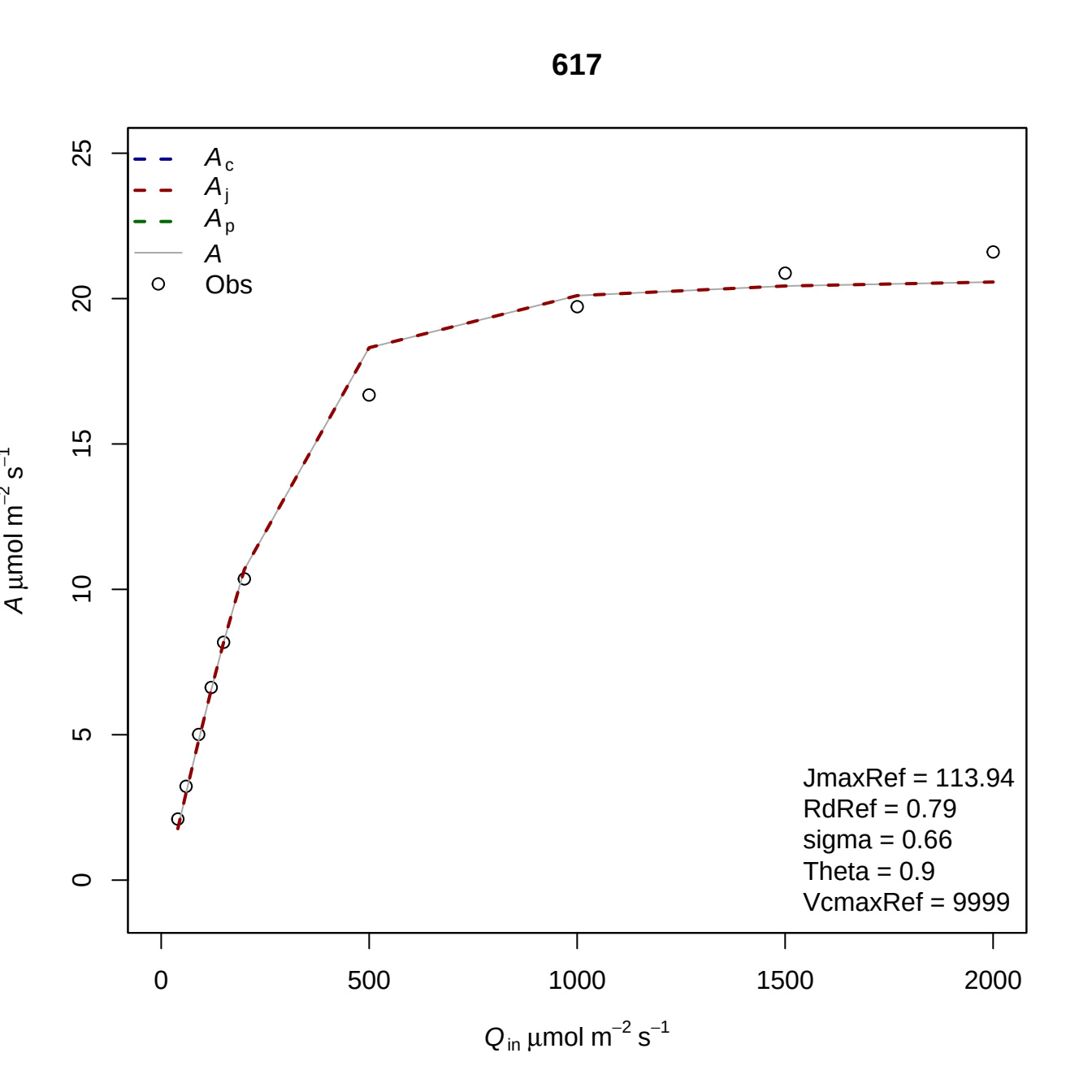

612
A_c
A_j
A_p
A
Obs
JmaxRef = 139.42
RdRef = 1.02
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

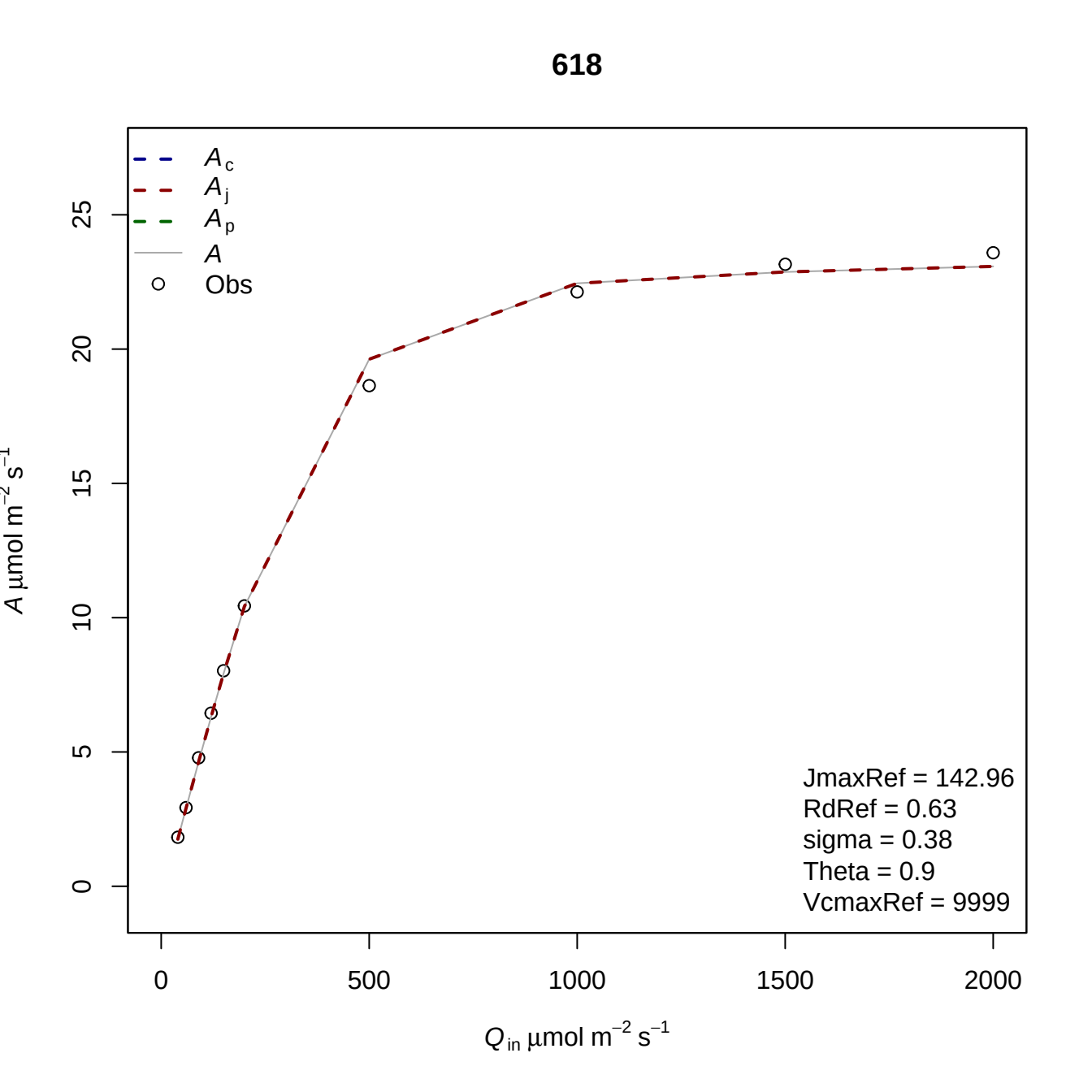

613
A_c
A_j
A_p
A
Obs
JmaxRef = 163.69
RdRef = 0.8
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

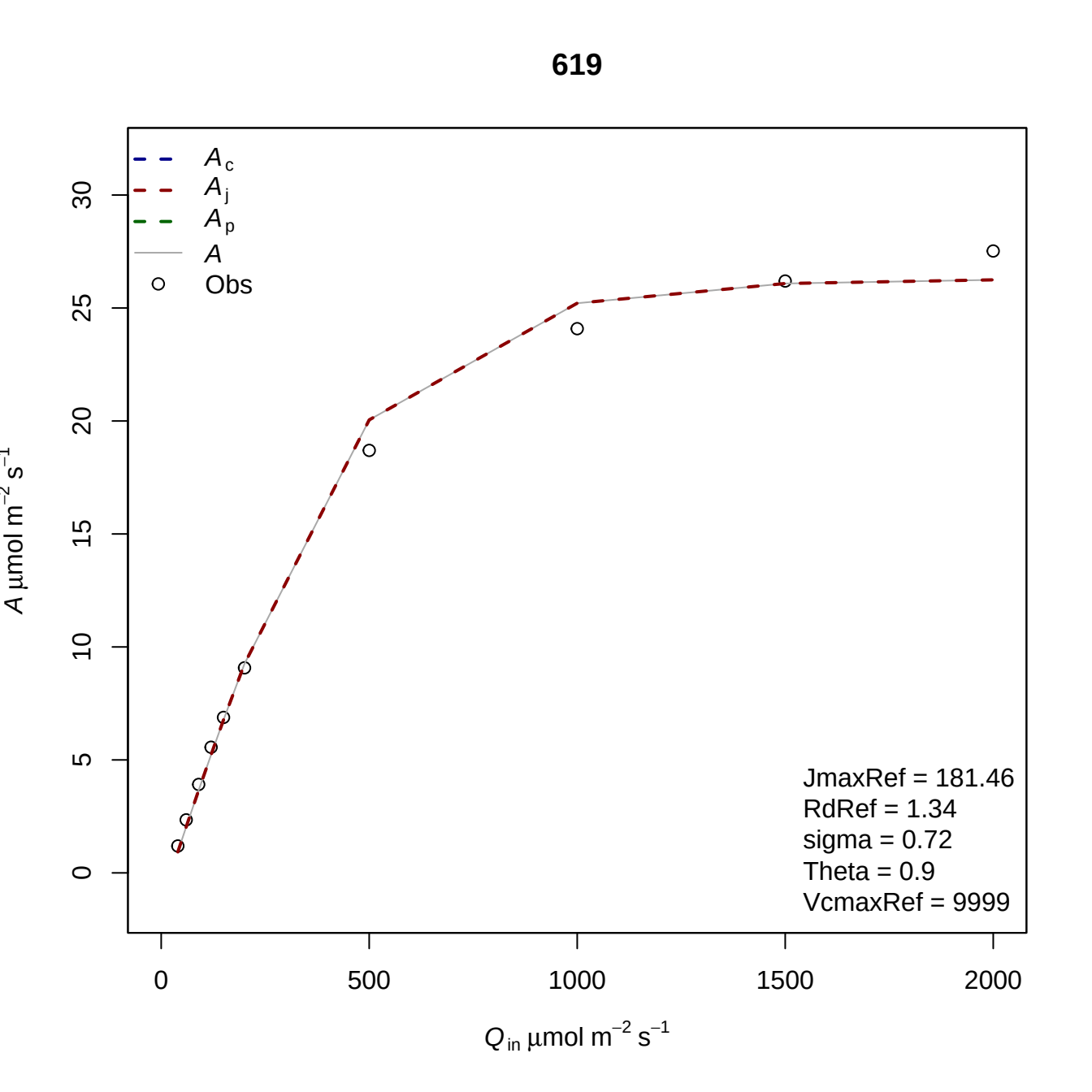

614
A_c
A_j
A_p
A
Obs
JmaxRef = 217.04
RdRef = 2.11
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

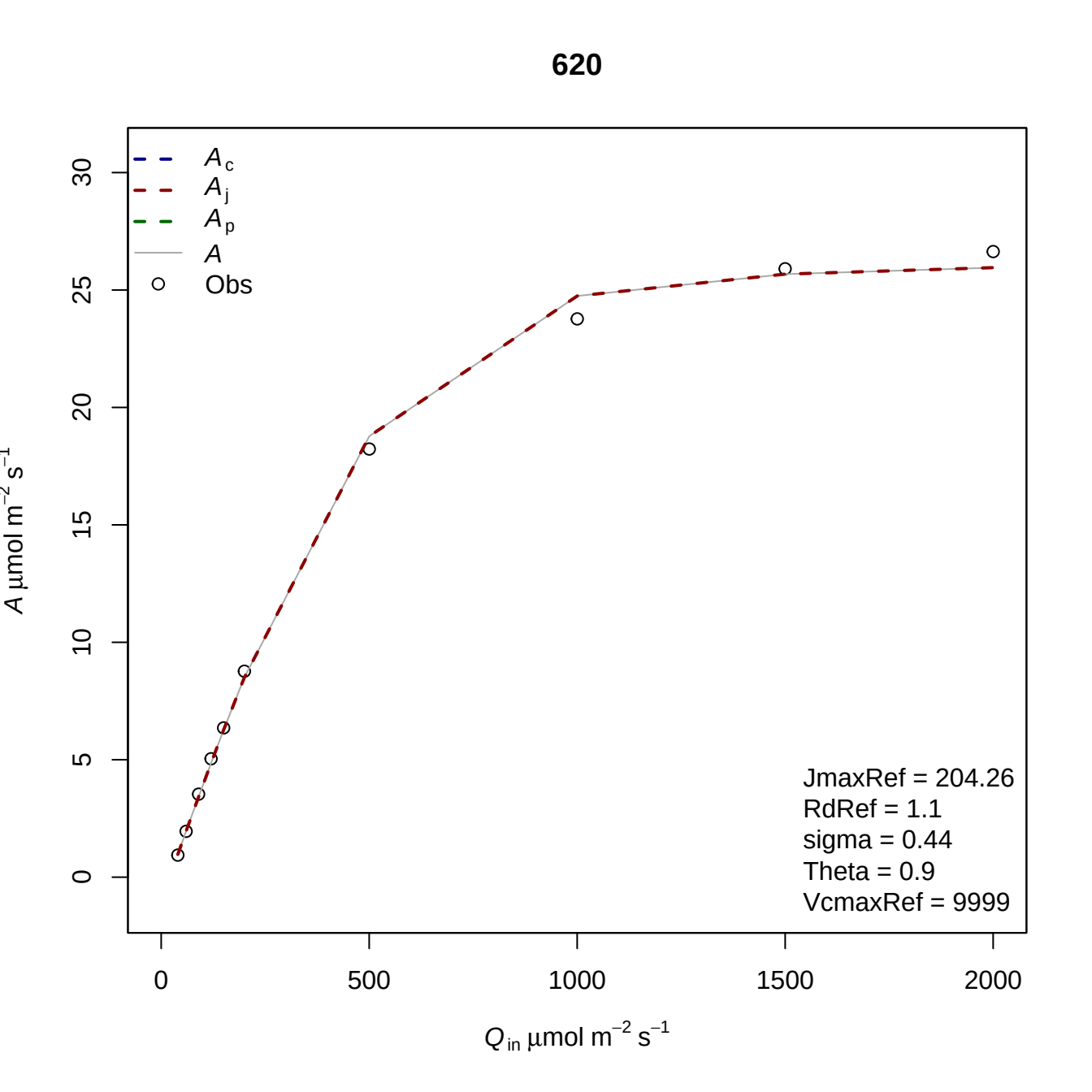

615
A_c
A_j
A_p
A
Obs
JmaxRef = 241.84
RdRef = 2.77
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

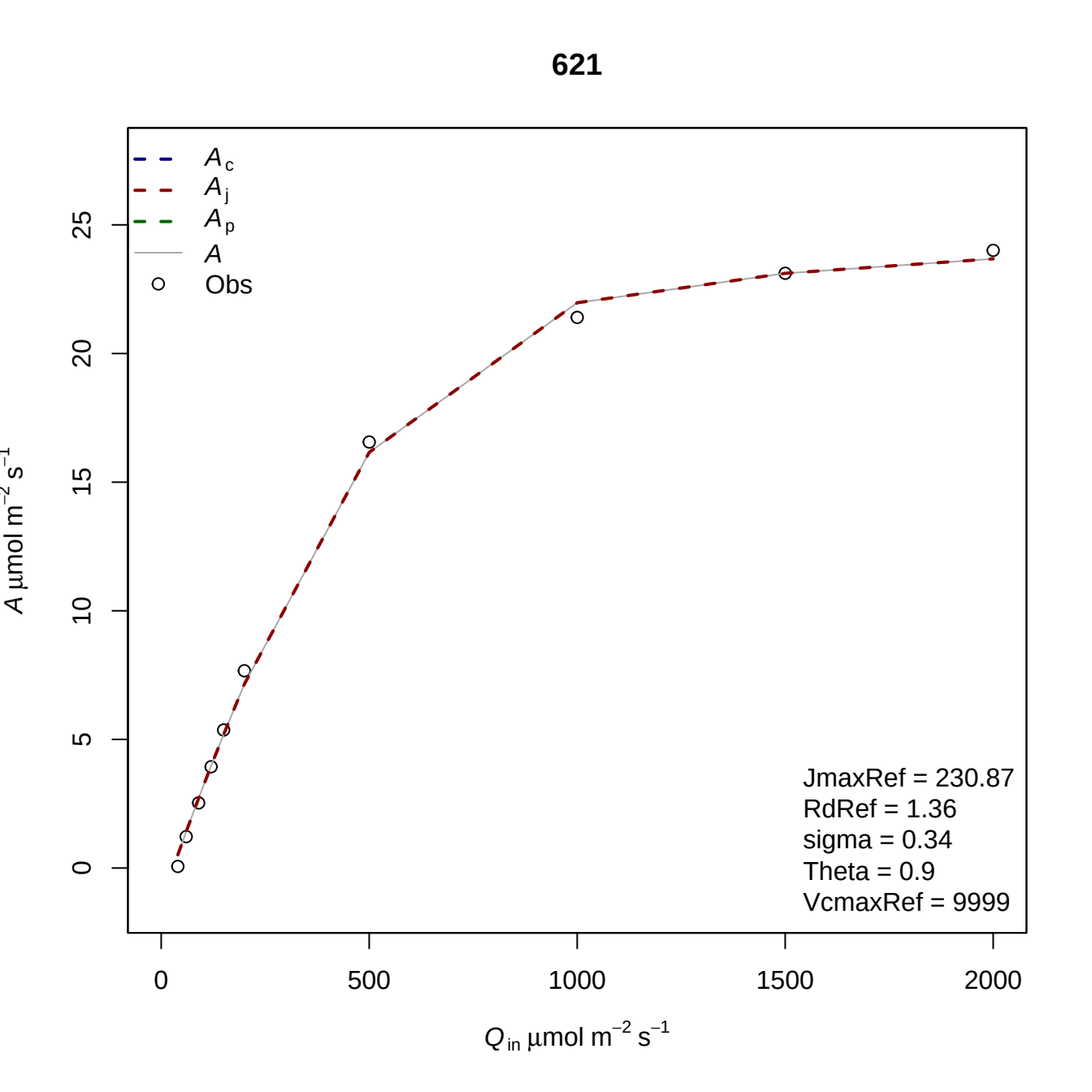

616
A_c
A_j
A_p
A
Obs
JmaxRef = 229.99
RdRef = 4.23
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

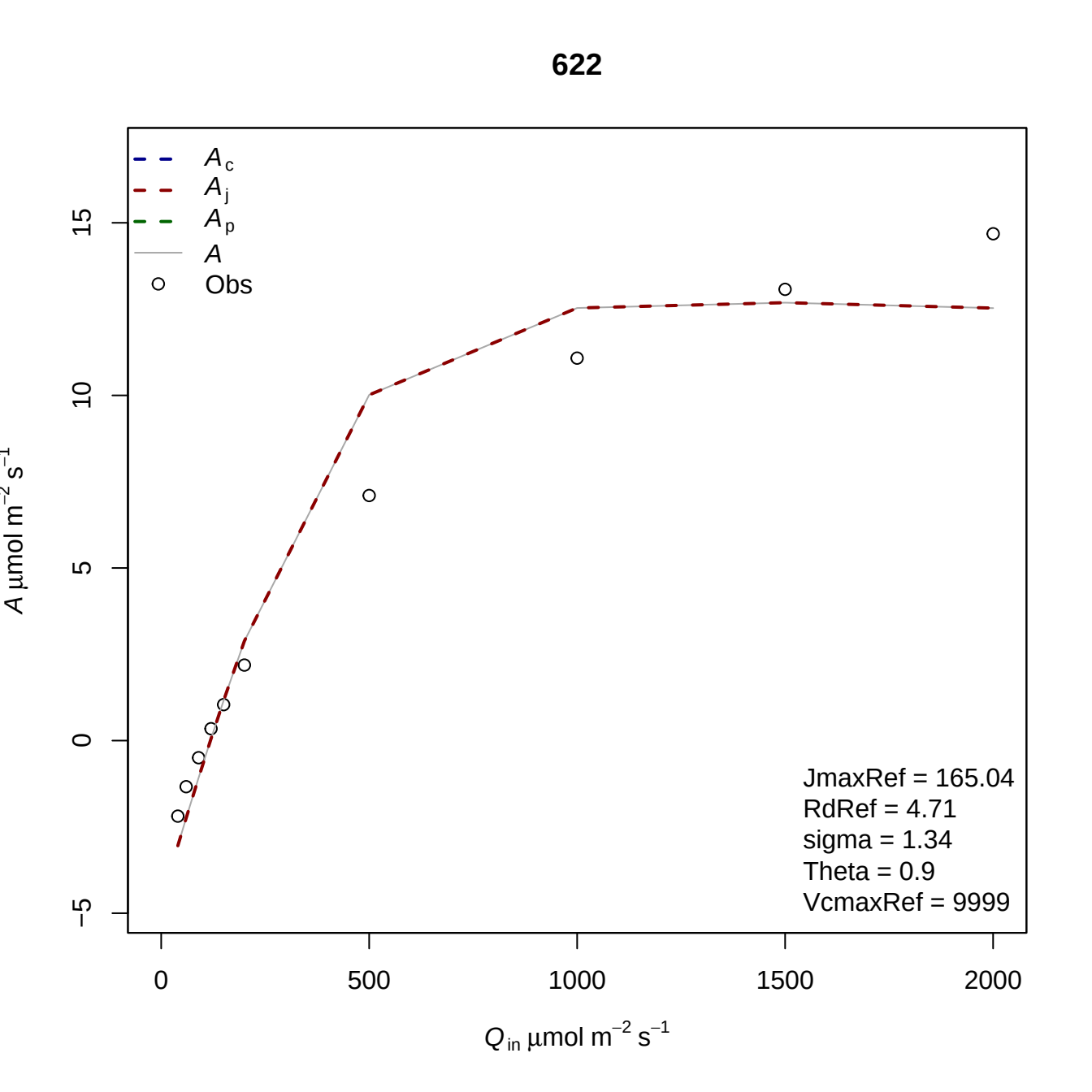

617
A_c
A_j
A_p
A
Obs
JmaxRef = 113.94
RdRef = 0.79
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

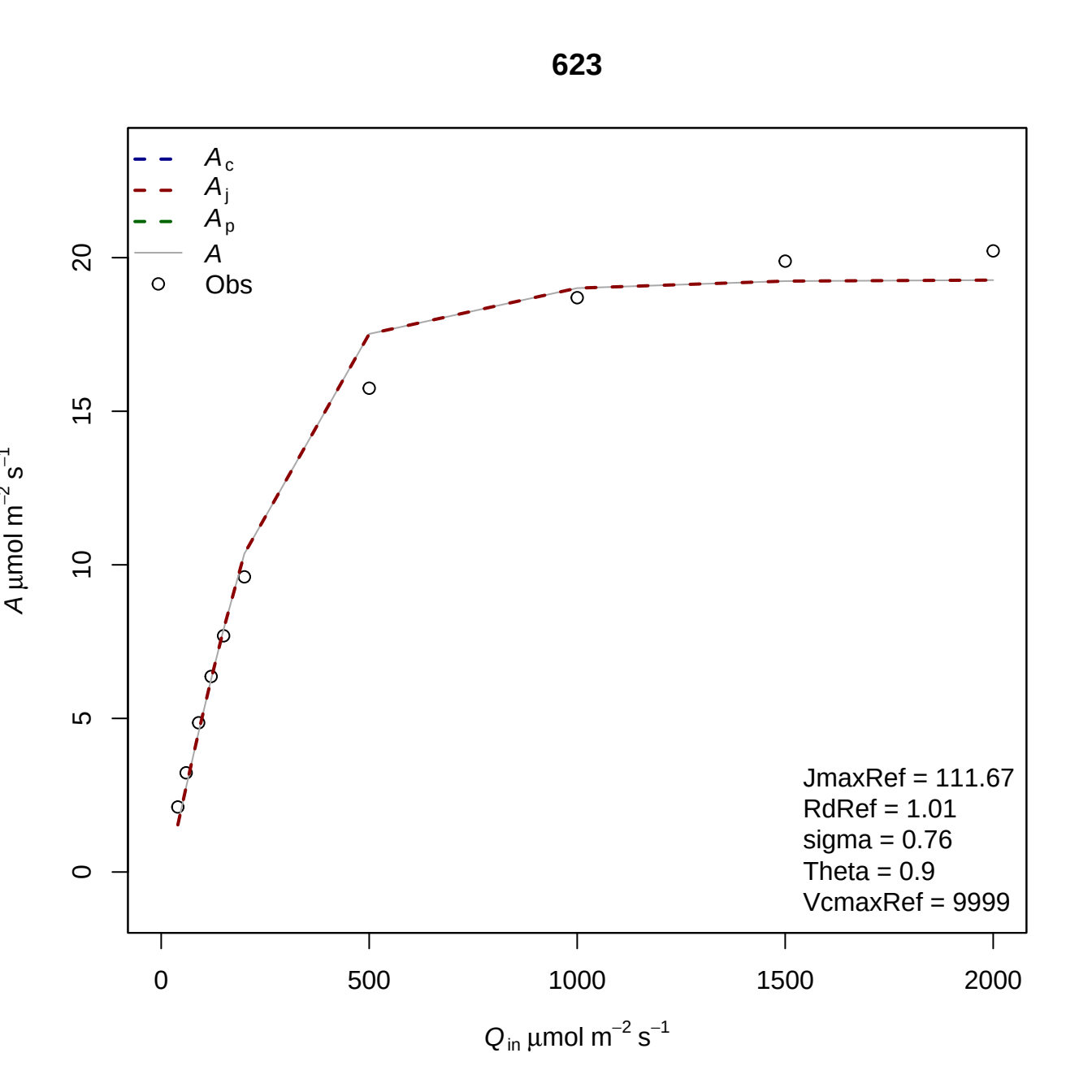

618
A_c
A_j
A_p
A
Obs
JmaxRef = 142.96
RdRef = 0.63
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

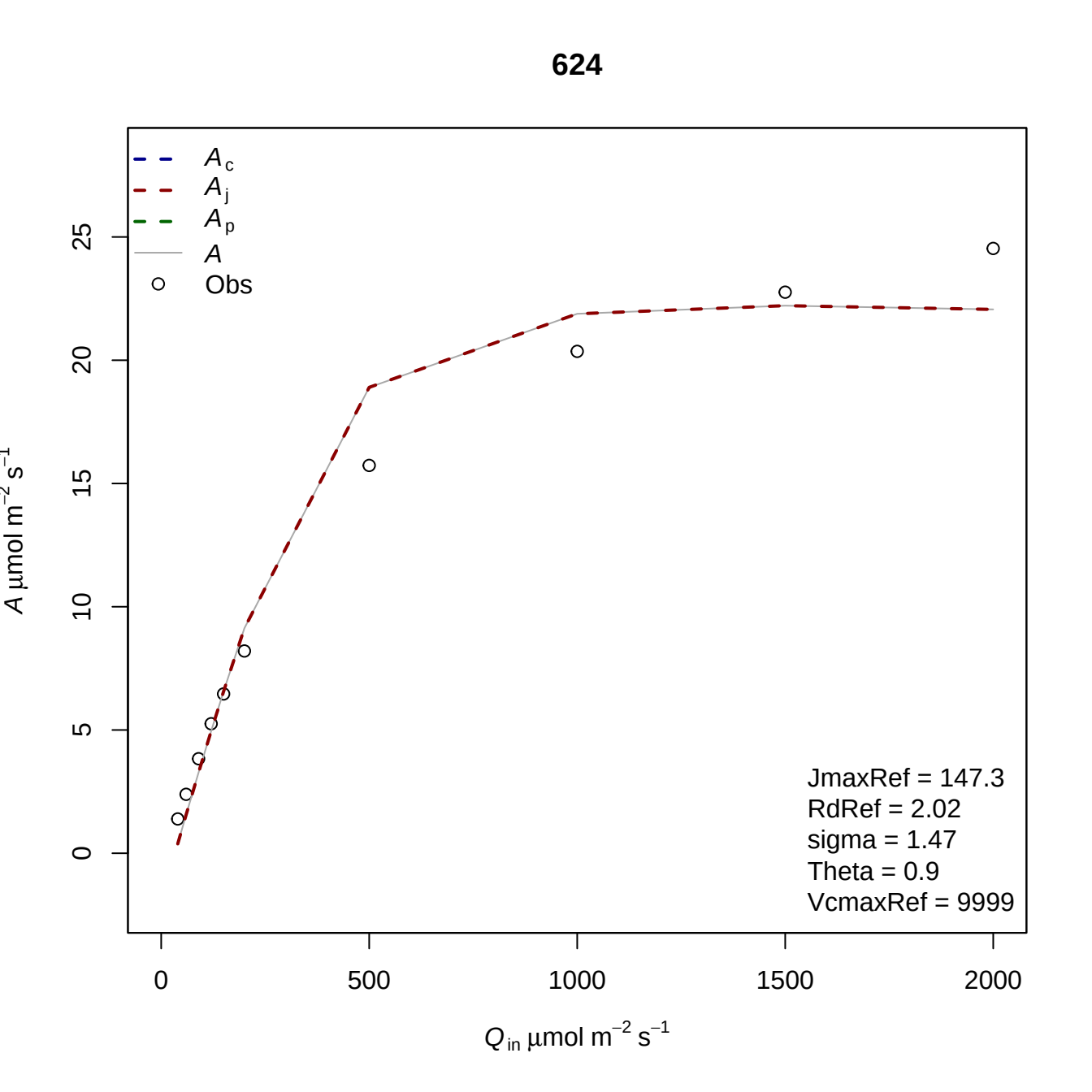

619
A_c
A_j
A_p
A
Obs
JmaxRef = 181.46
RdRef = 1.34
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

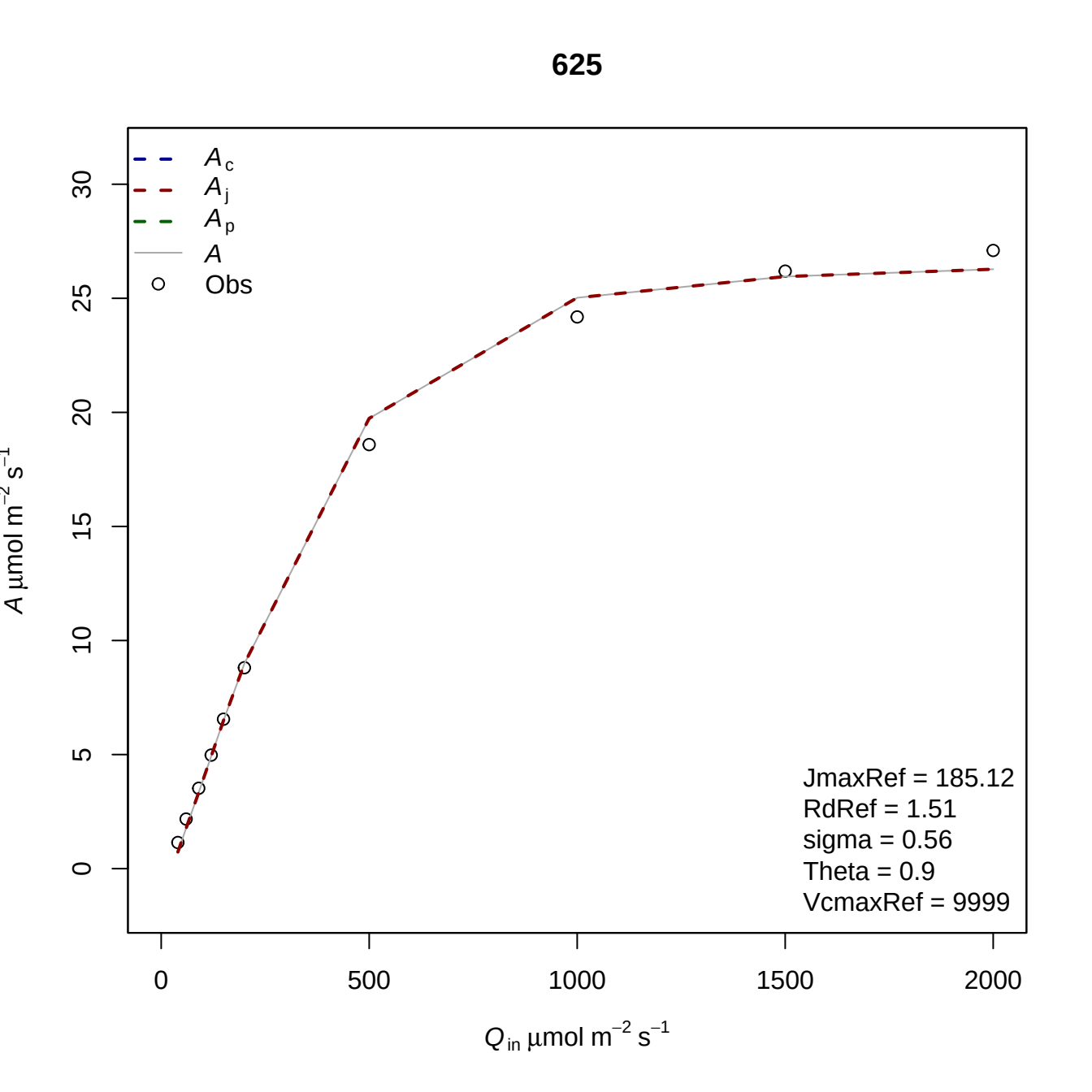

620
A_c
A_j
A_p
A
Obs
JmaxRef = 204.26
RdRef = 1.1
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

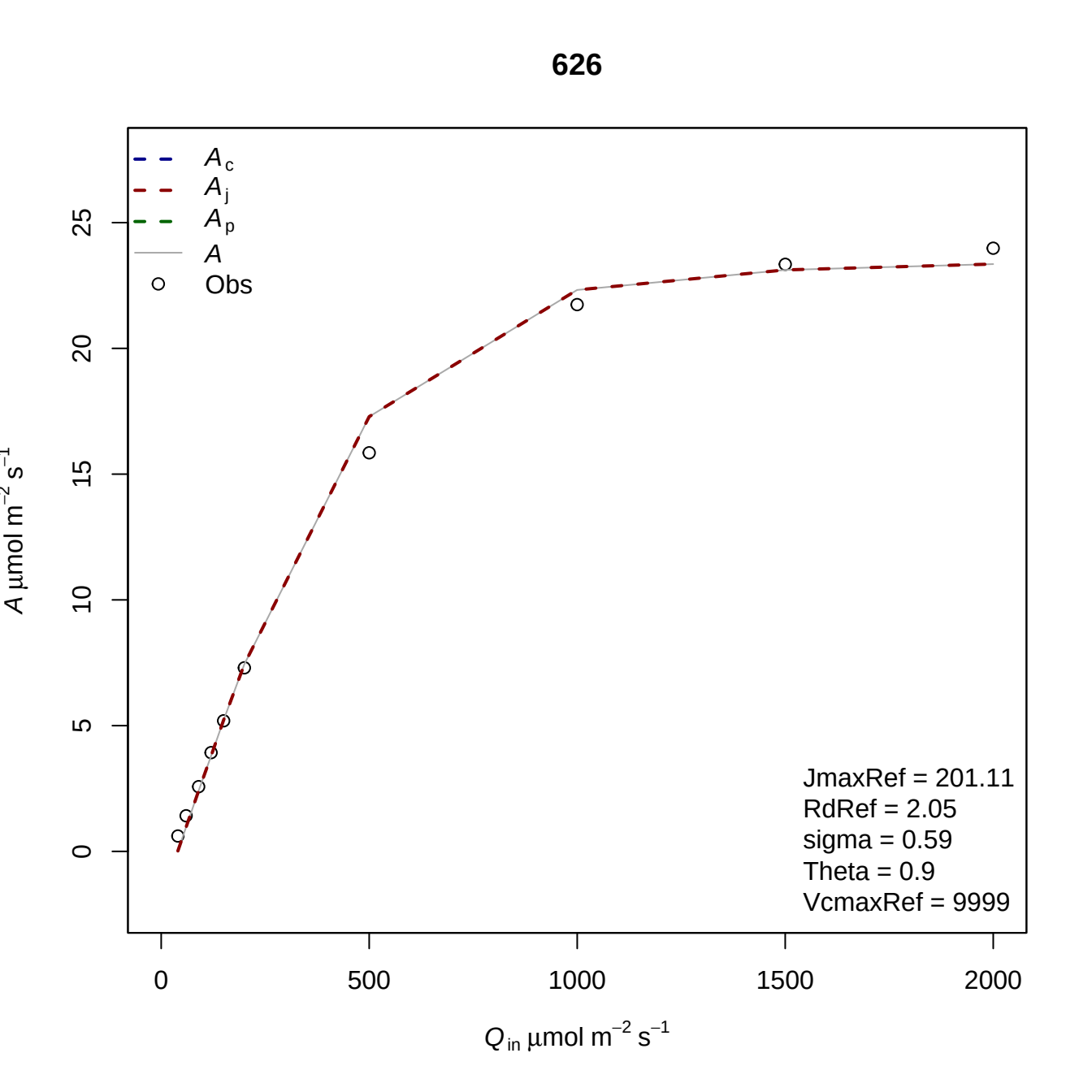

621
A_c
A_j
A_p
A
Obs
JmaxRef = 230.87
RdRef = 1.36
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

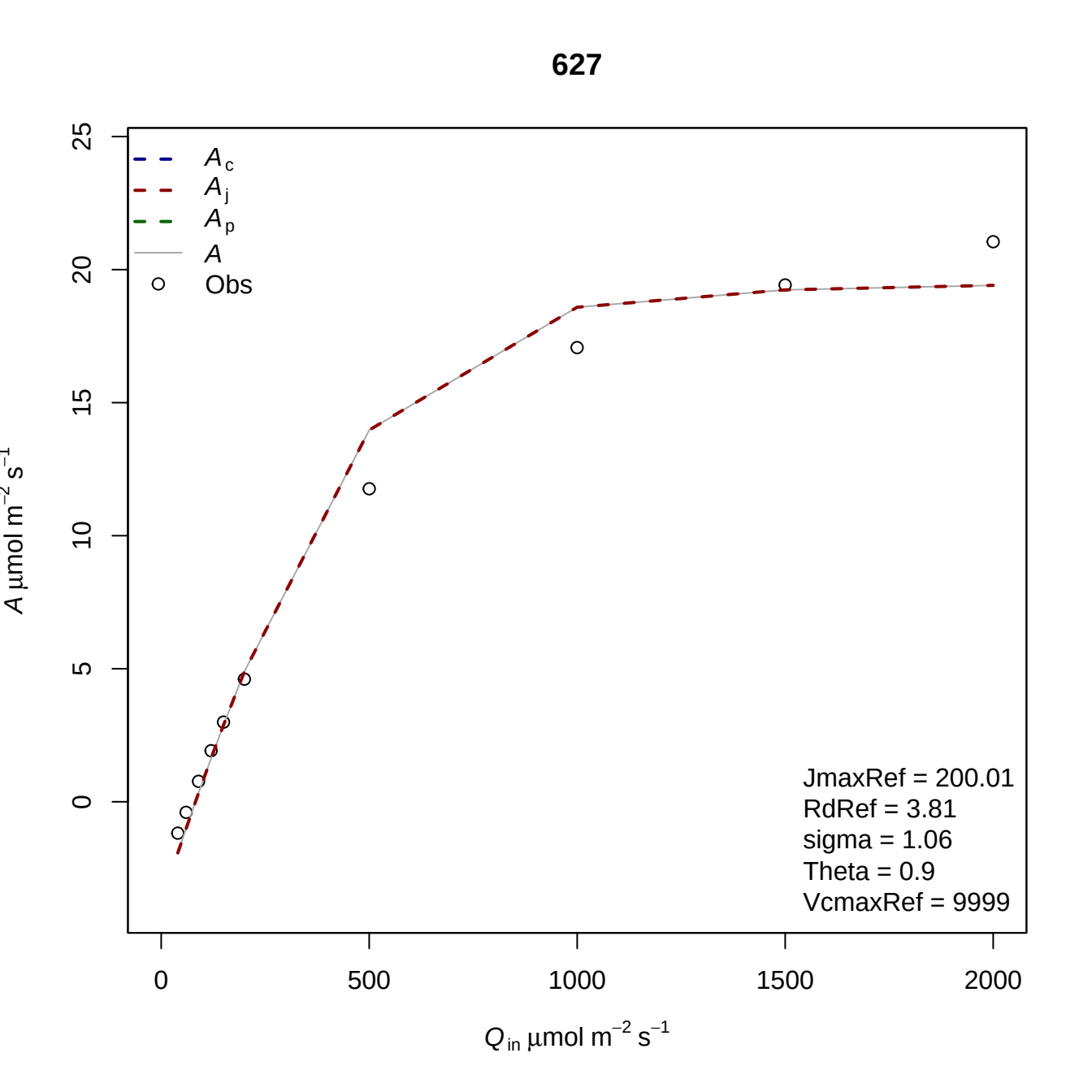

622
A_c
A_j
A_p
A
Obs
JmaxRef = 165.04
RdRef = 4.71
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

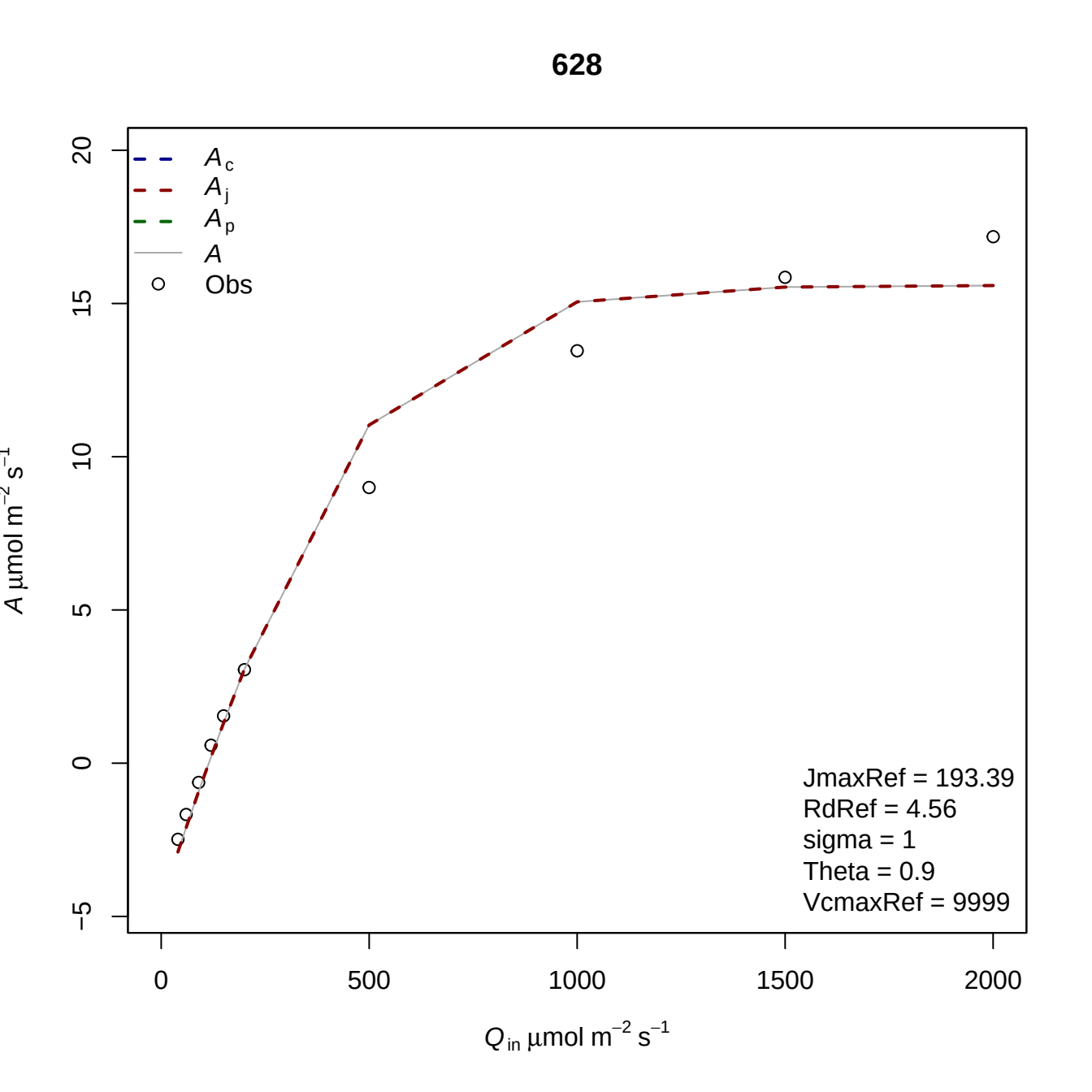

623
A_c
A_j
A_p
A
Obs
JmaxRef = 111.67
RdRef = 1.01
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

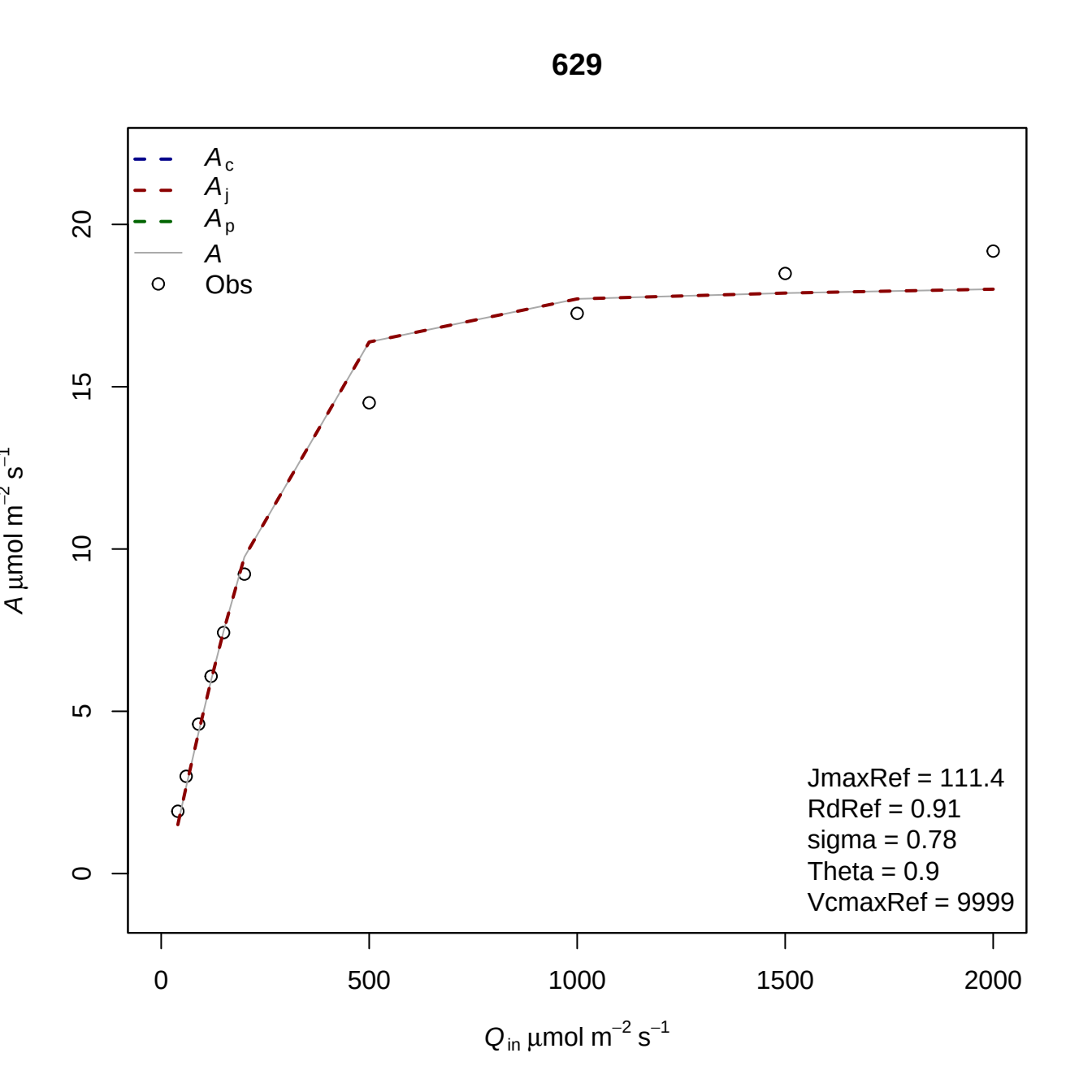

624
A_c
A_j
A_p
A
Obs
JmaxRef = 147.3
RdRef = 2.02
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

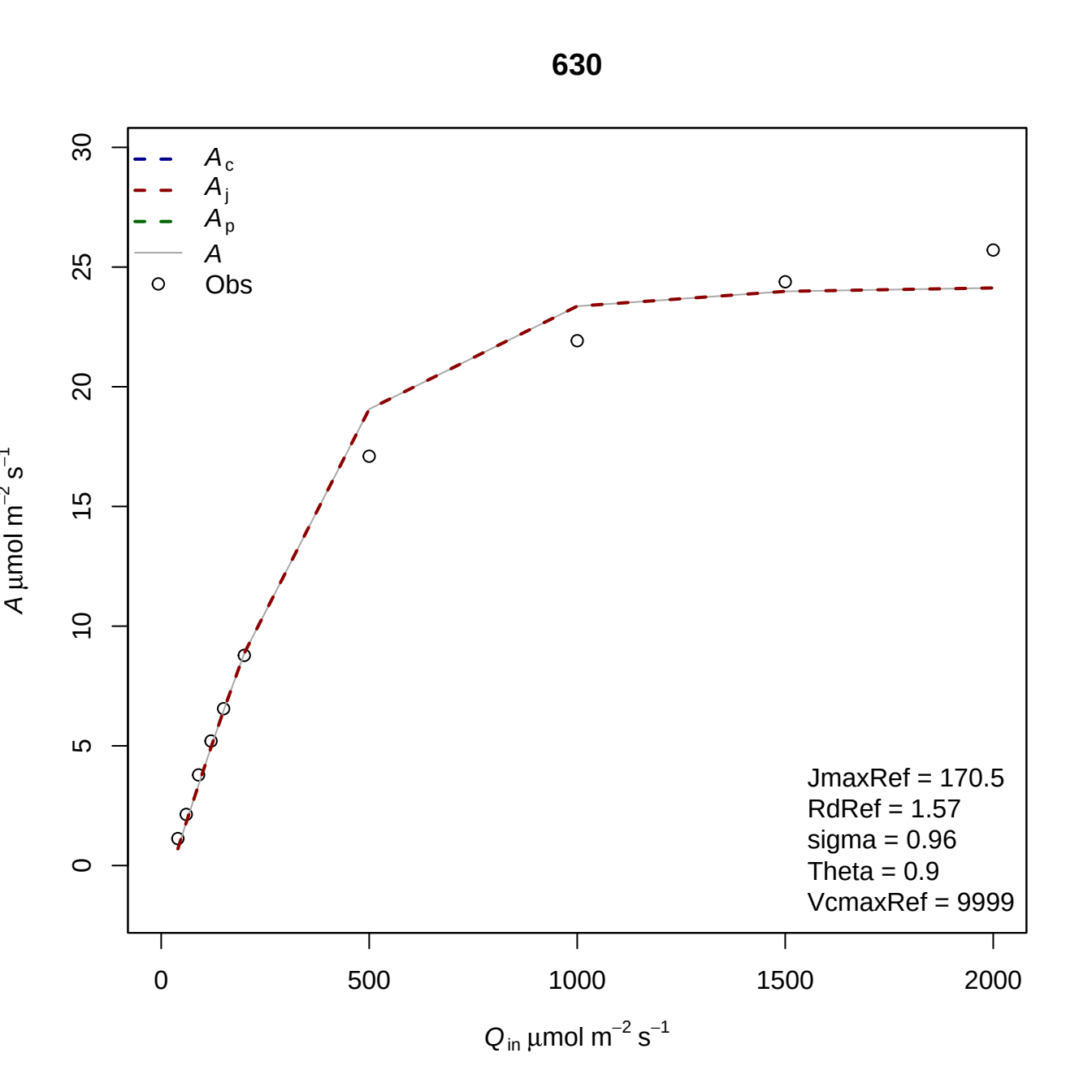

625
A_c
A_j
A_p
A
Obs
JmaxRef = 185.12
RdRef = 1.51
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

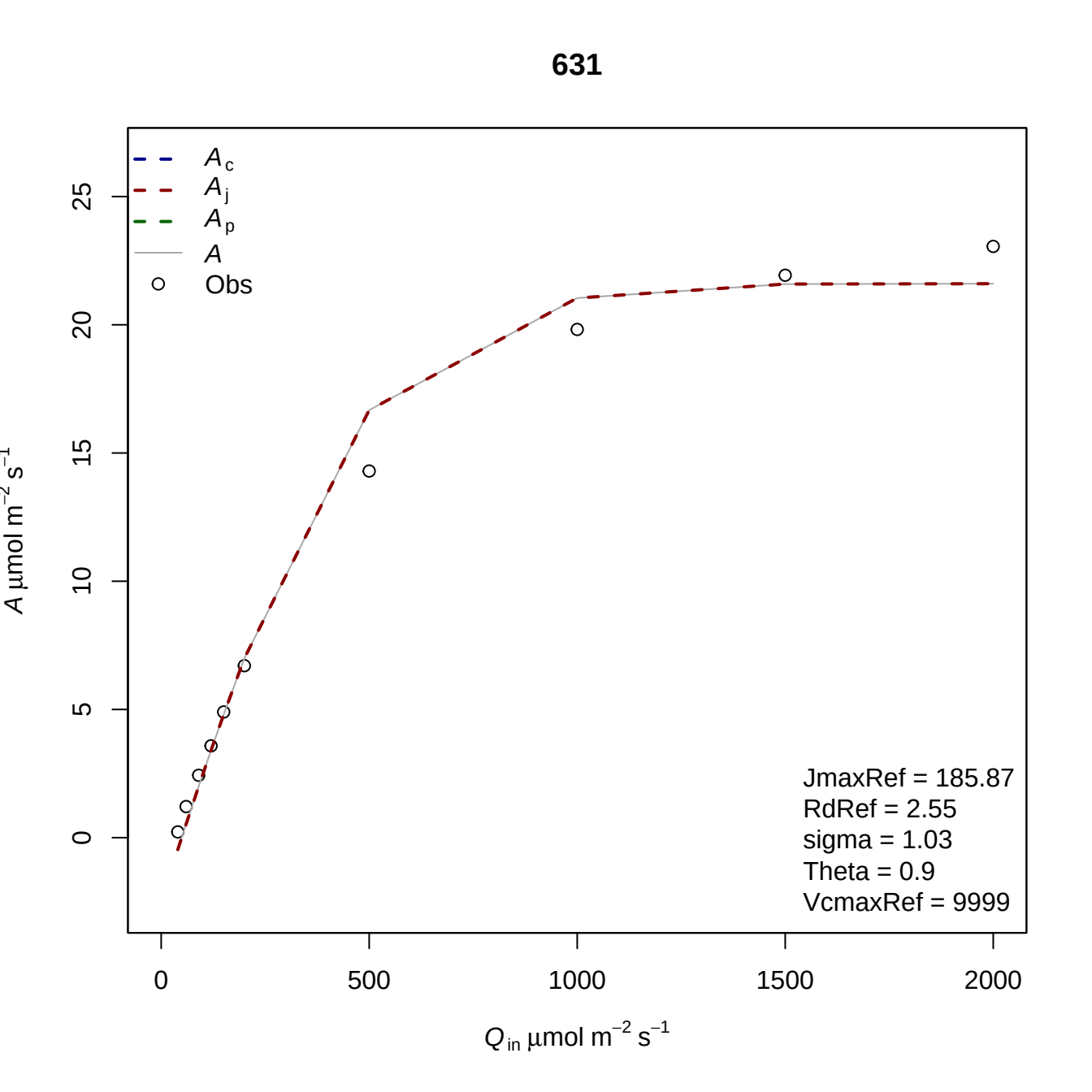

626
A_c
A_j
A_p
A
Obs
JmaxRef = 201.11
RdRef = 2.05
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

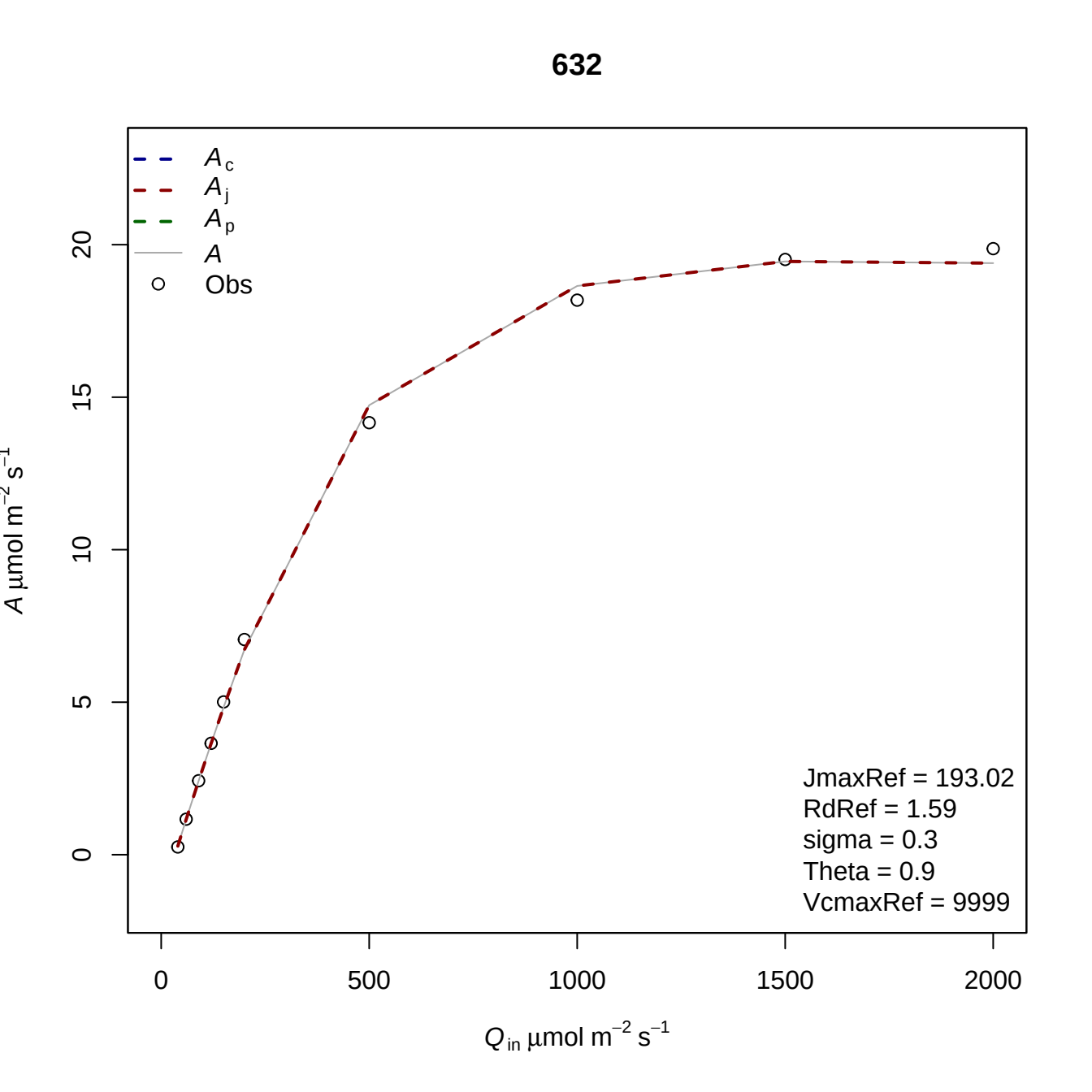

627
A_c
A_j
A_p
A
Obs
JmaxRef = 200.01
RdRef = 3.81
sigma = 1.06
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

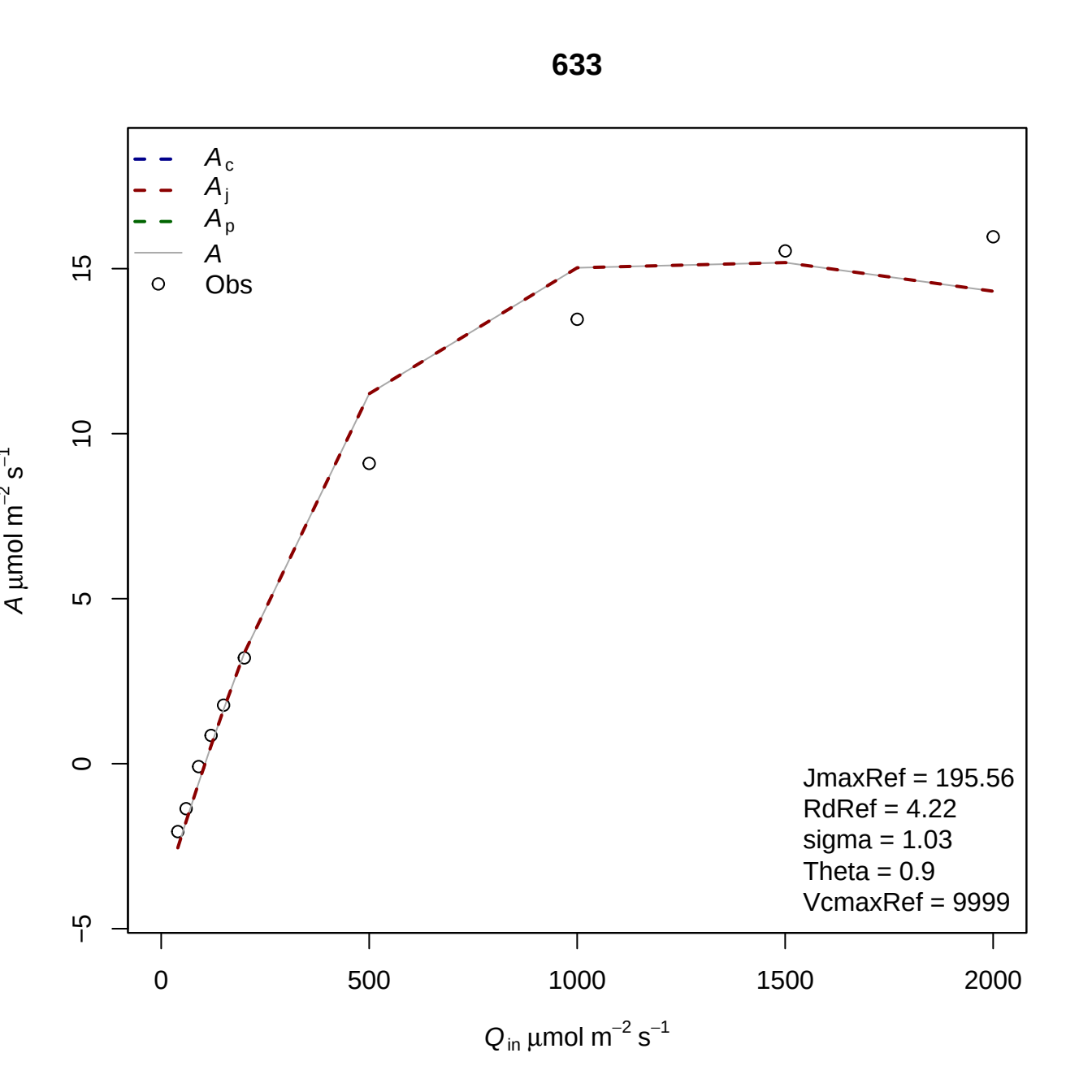

628
A_c
A_j
A_p
A
Obs
JmaxRef = 193.39
RdRef = 4.56
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

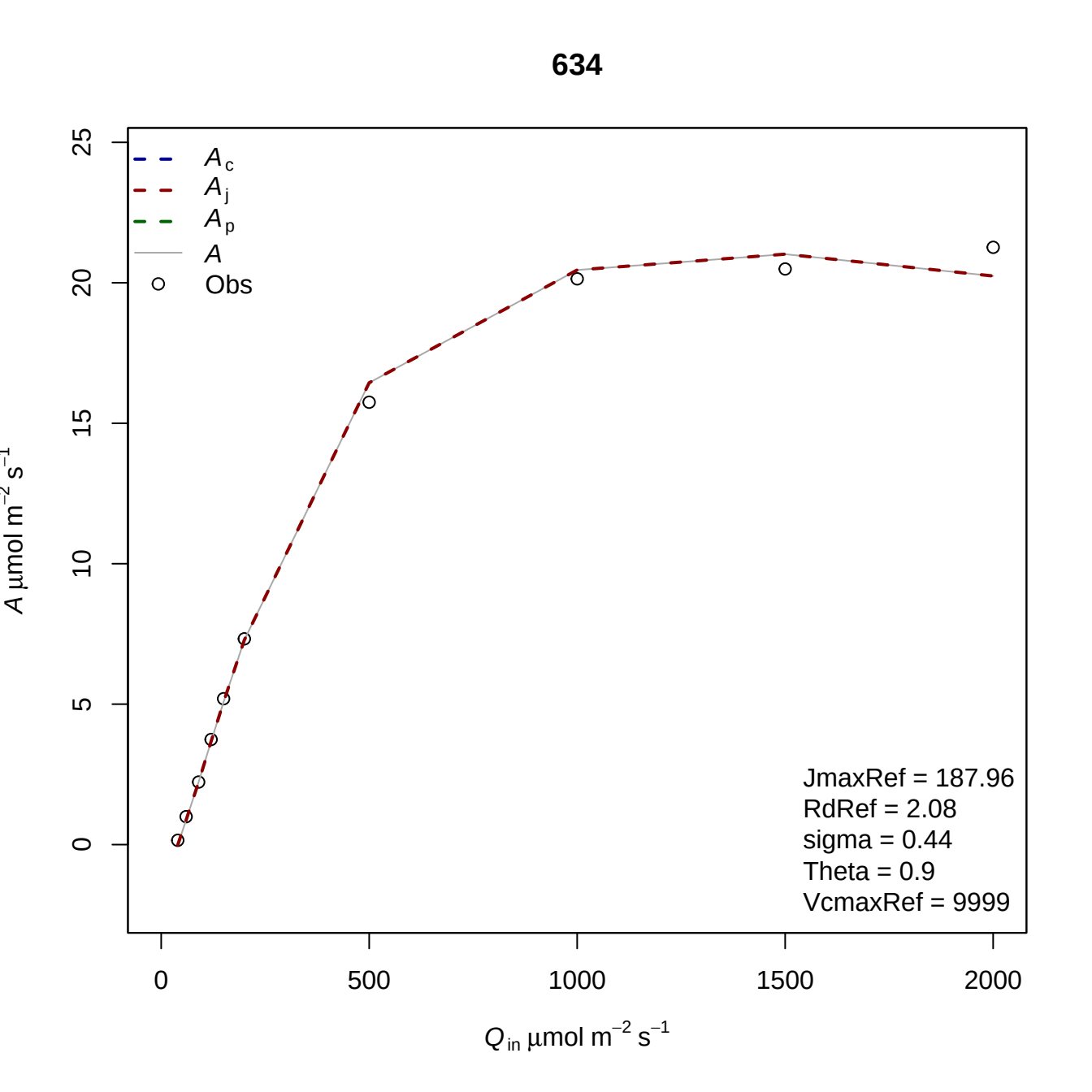

629
A_c
A_j
A_p
A
Obs
JmaxRef = 111.4
RdRef = 0.91
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

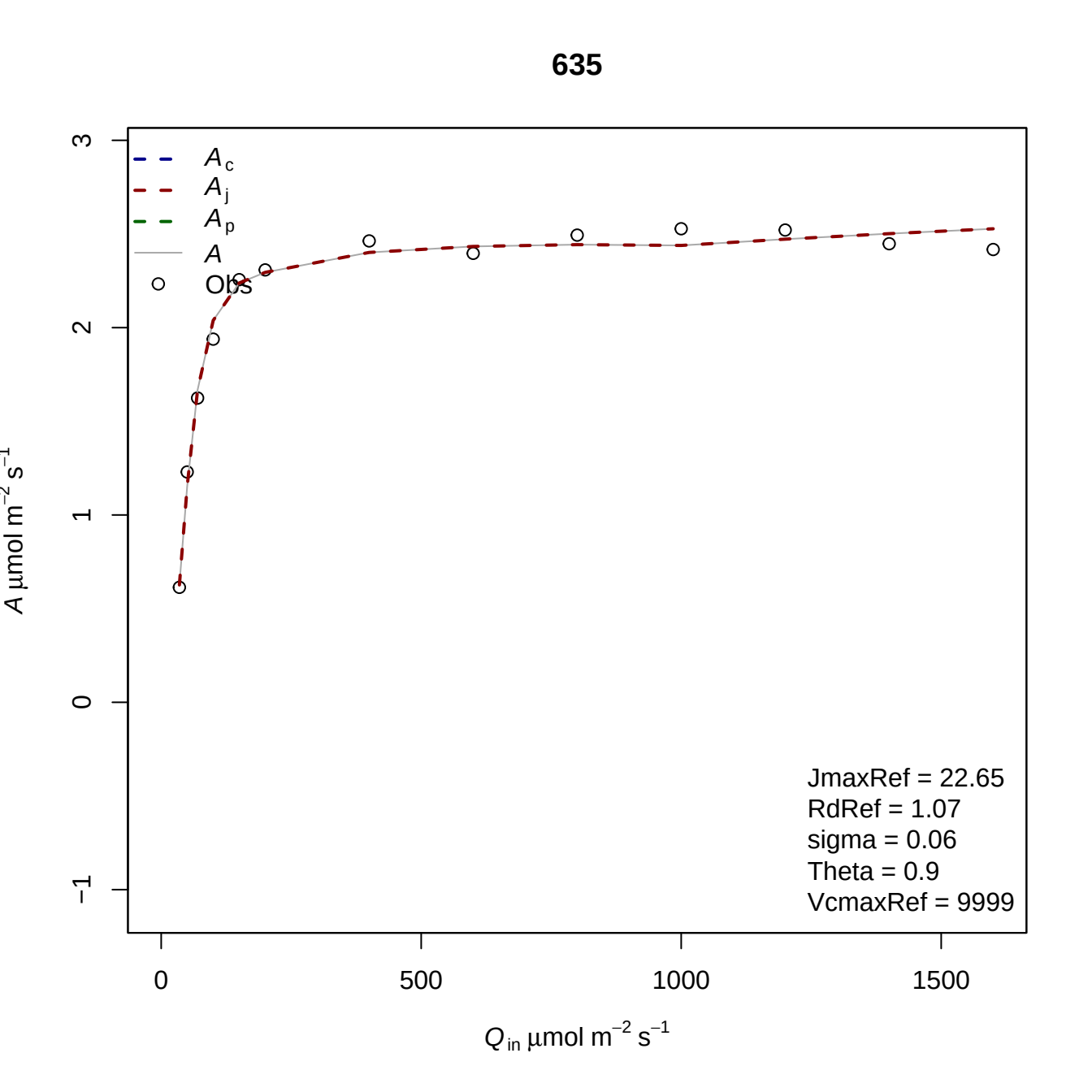

630
A_c
A_j
A_p
A
Obs
JmaxRef = 170.5
RdRef = 1.57
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

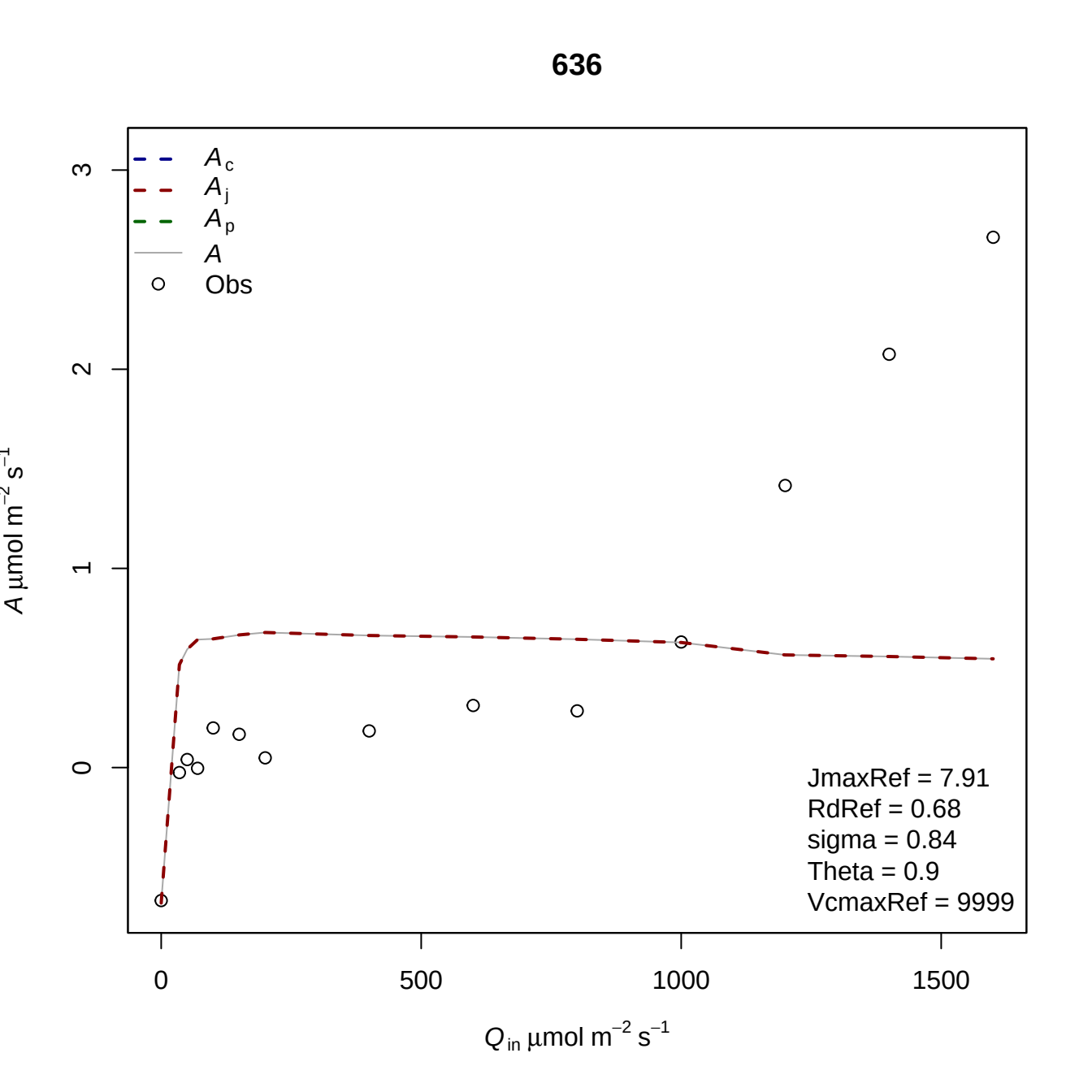

631
A_c
A_j
A_p
A
Obs
JmaxRef = 185.87
RdRef = 2.55
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

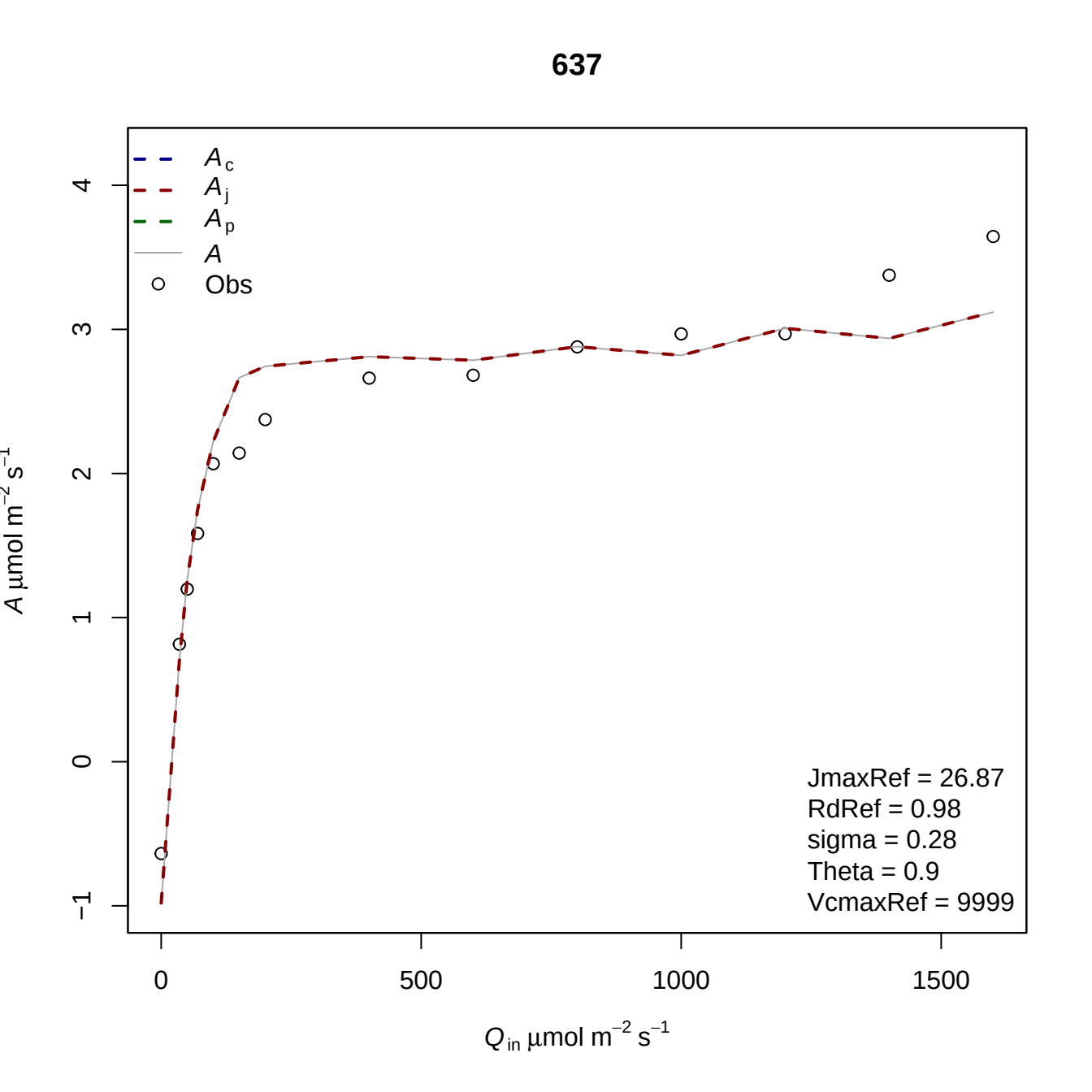

632
A_c
A_j
A_p
A
Obs
JmaxRef = 193.02
RdRef = 1.59
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

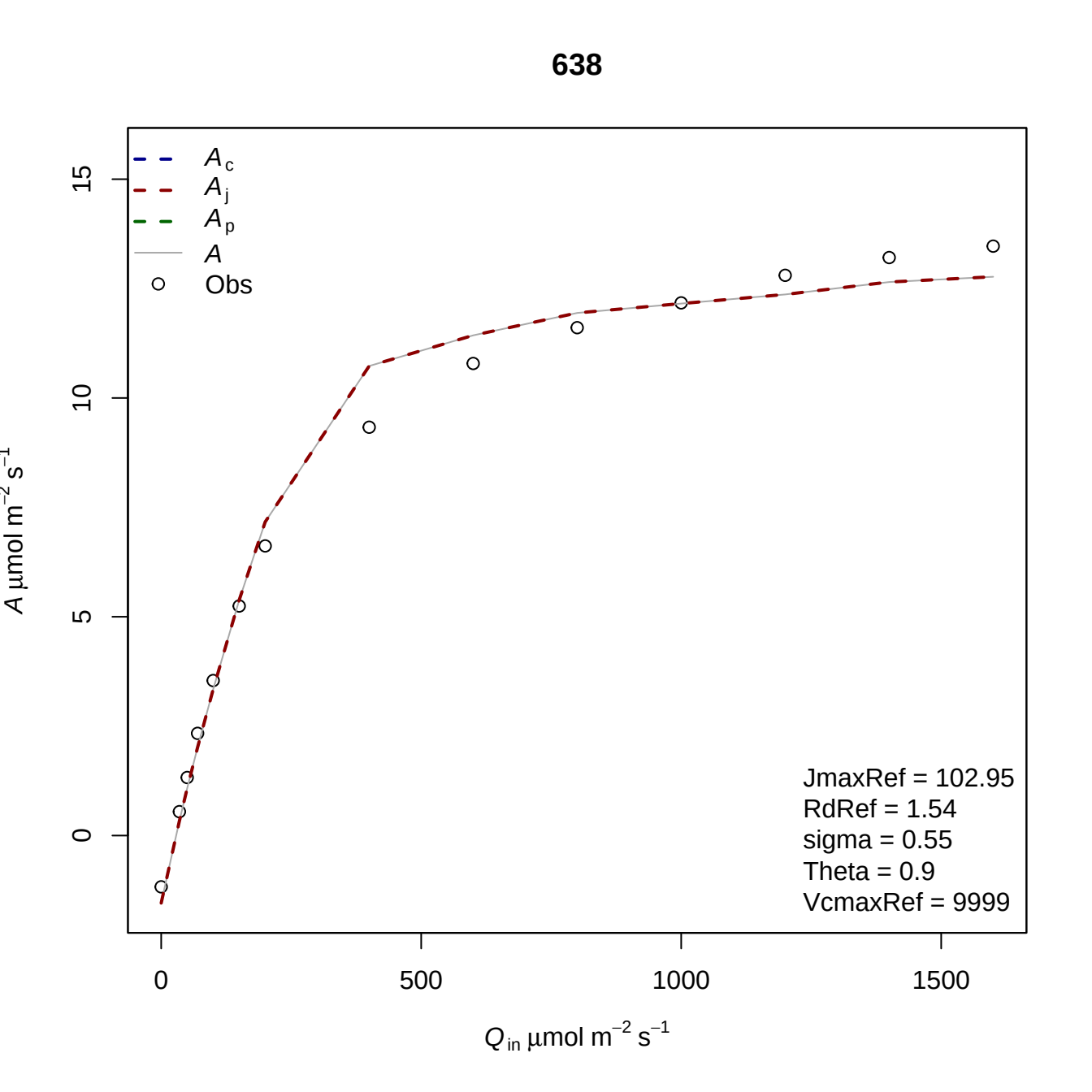

633
A_c
A_j
A_p
A
Obs
JmaxRef = 195.56
RdRef = 4.22
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

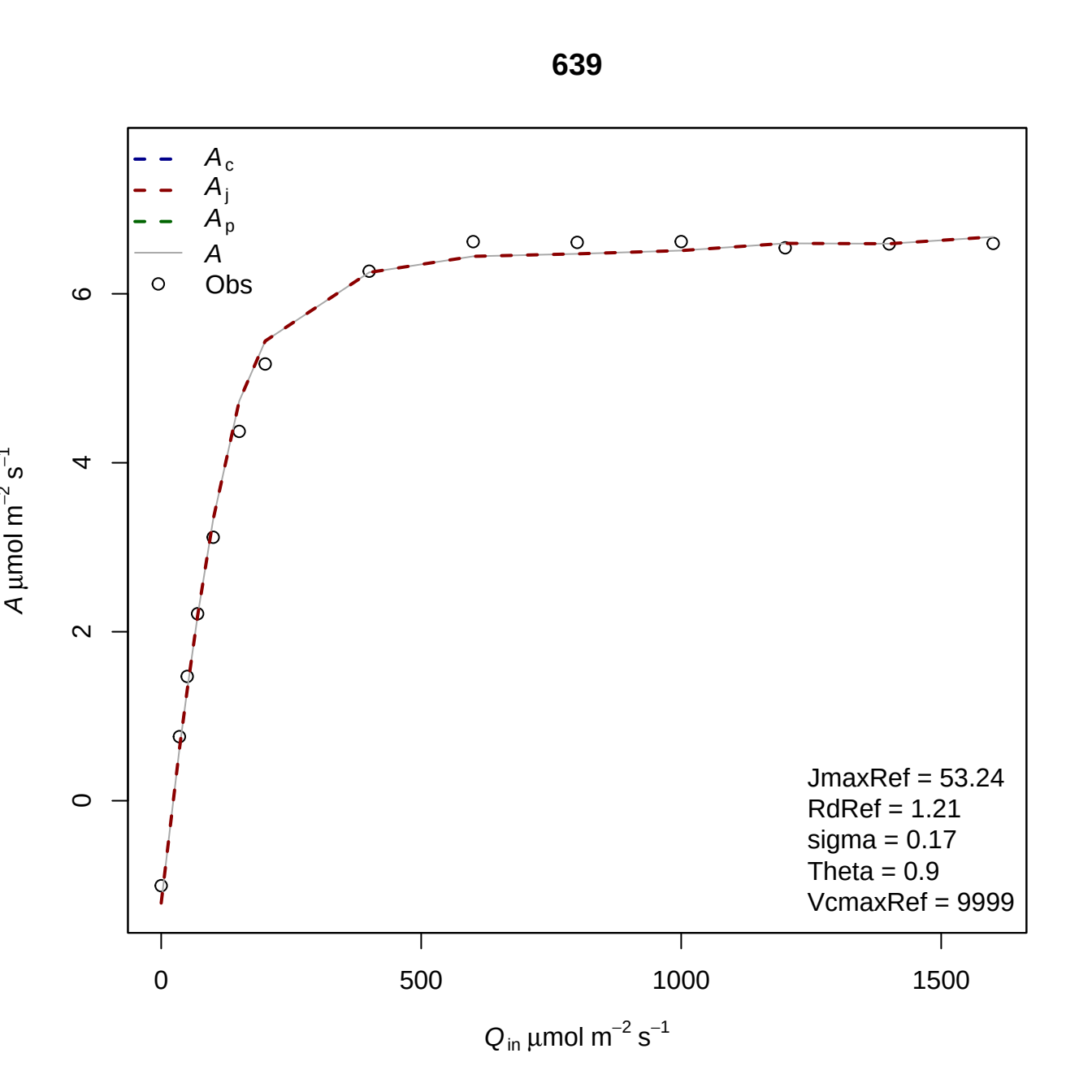

634
A_c
A_j
A_p
A
Obs
JmaxRef = 187.96
RdRef = 2.08
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

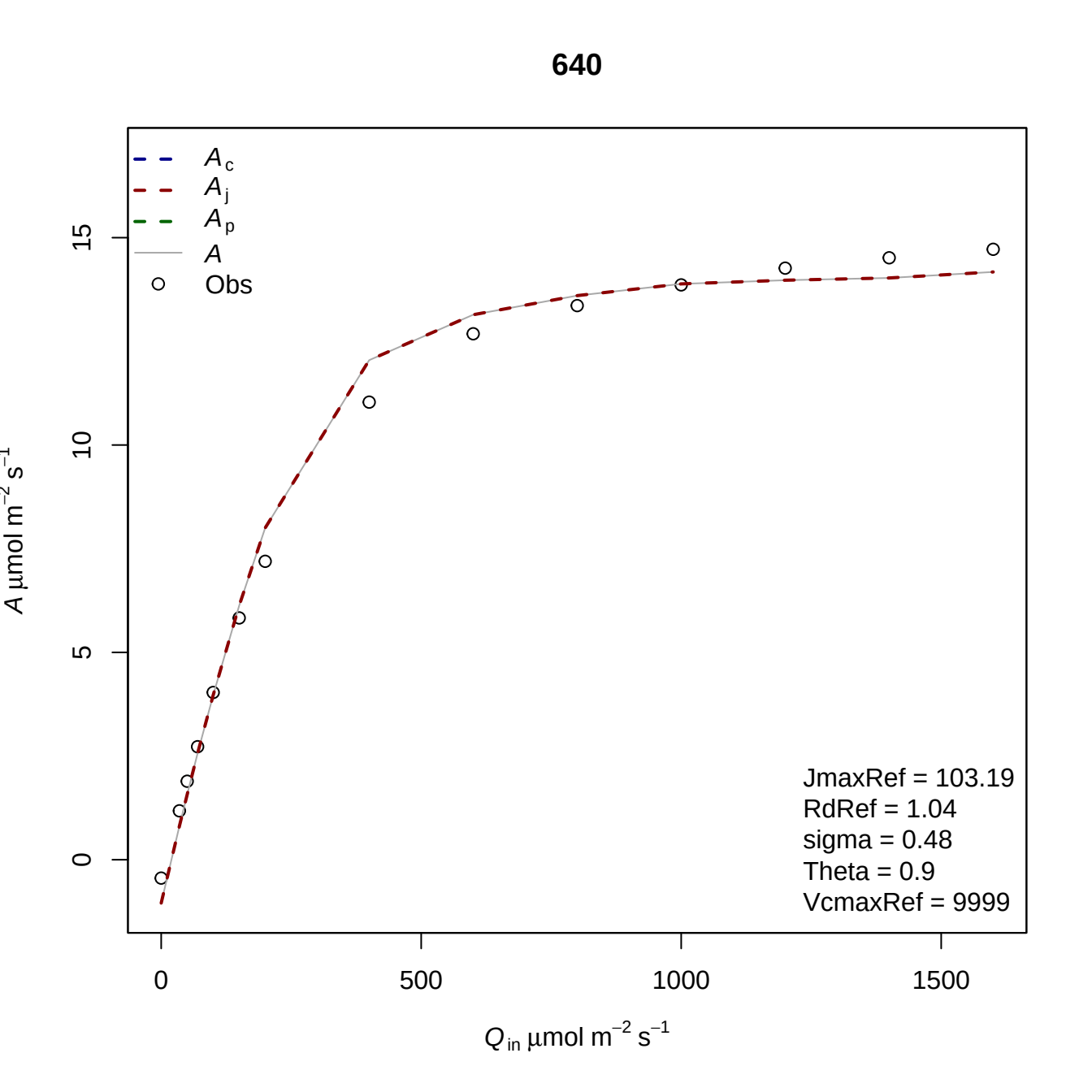

635
A_c
A_j
A_p
A
Obs
JmaxRef = 22.65
RdRef = 1.07
sigma = 0.06
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

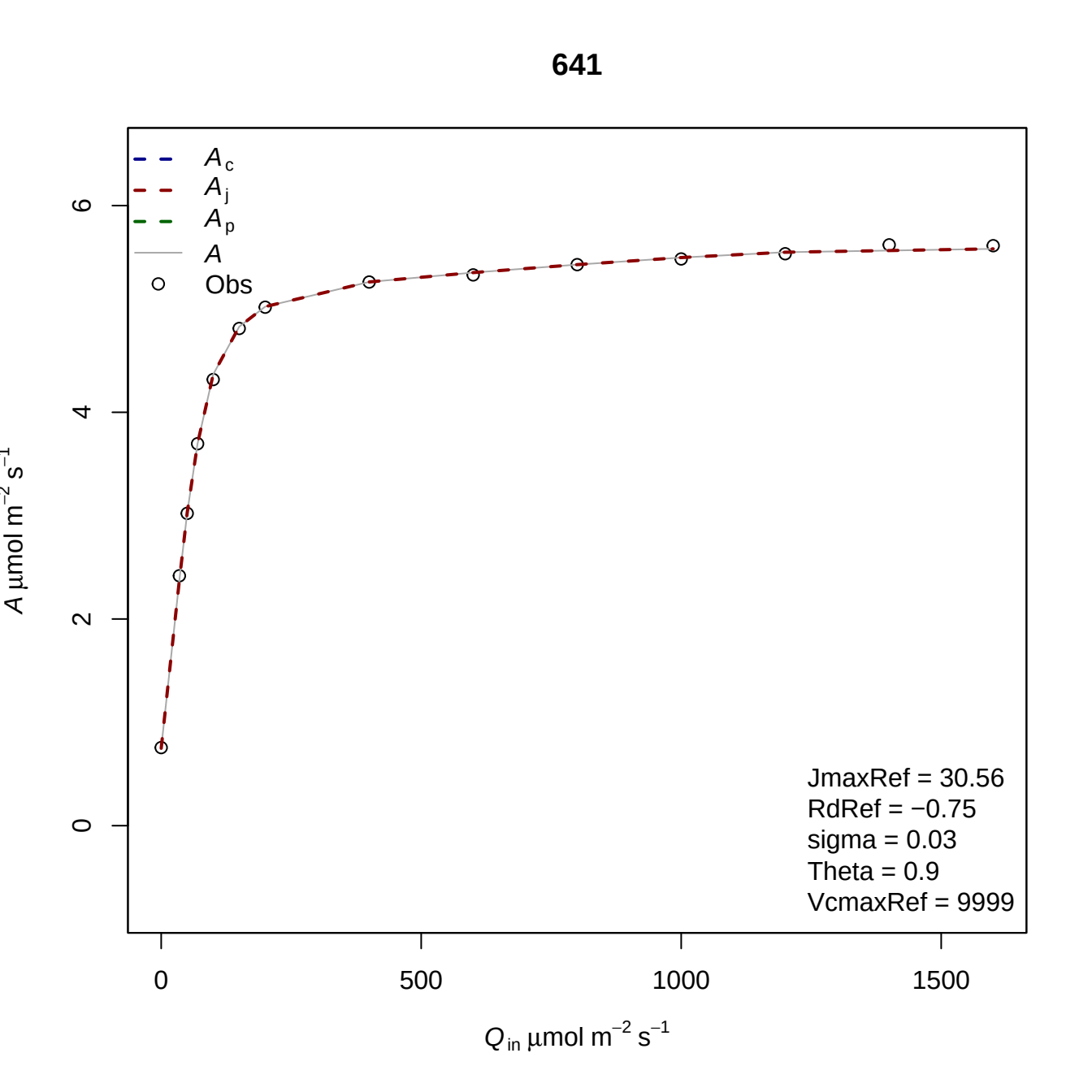

636
A_c
A_j
A_p
A
Obs
JmaxRef = 7.91
RdRef = 0.68
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

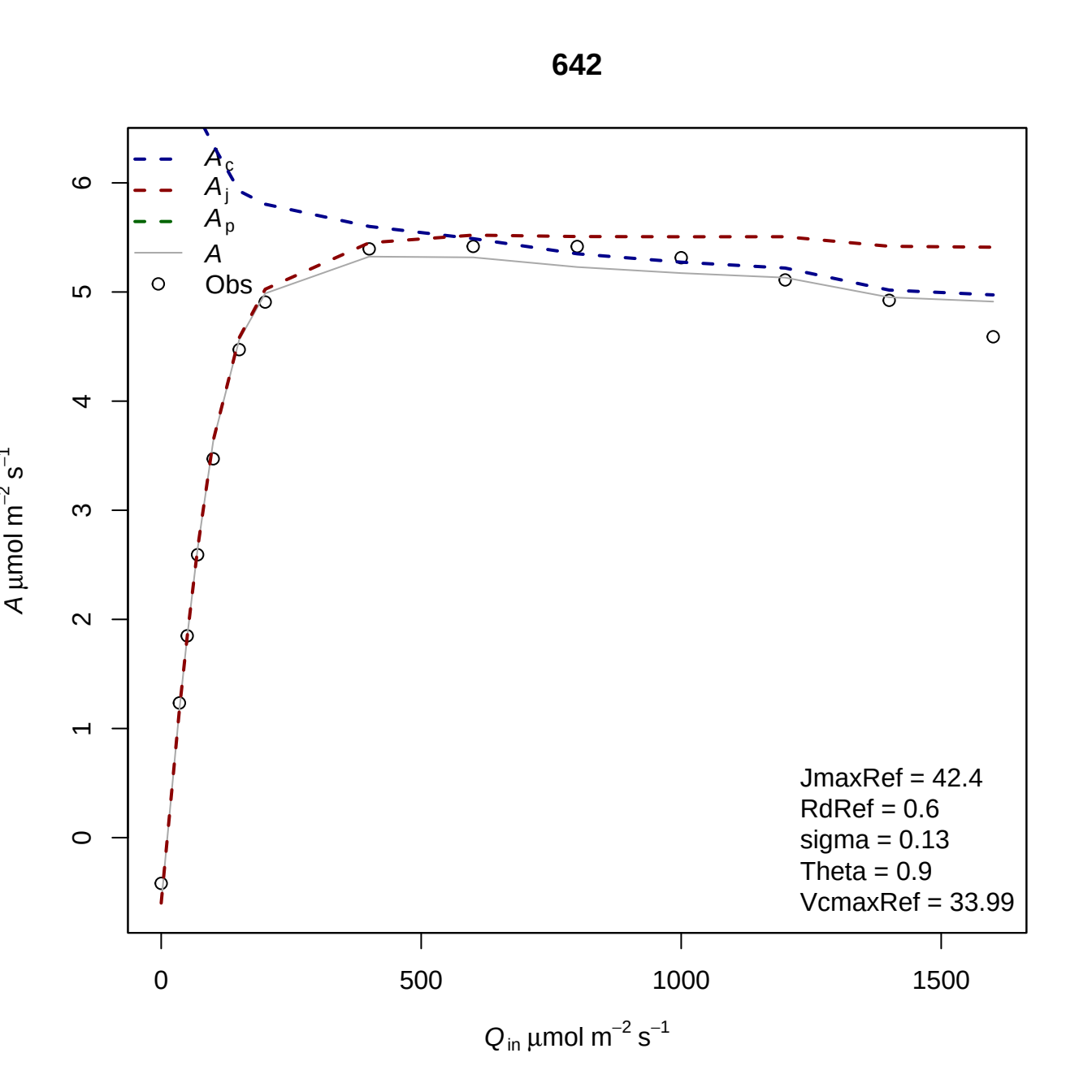

637
A_c
A_j
A_p
A
Obs
JmaxRef = 26.87
RdRef = 0.98
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

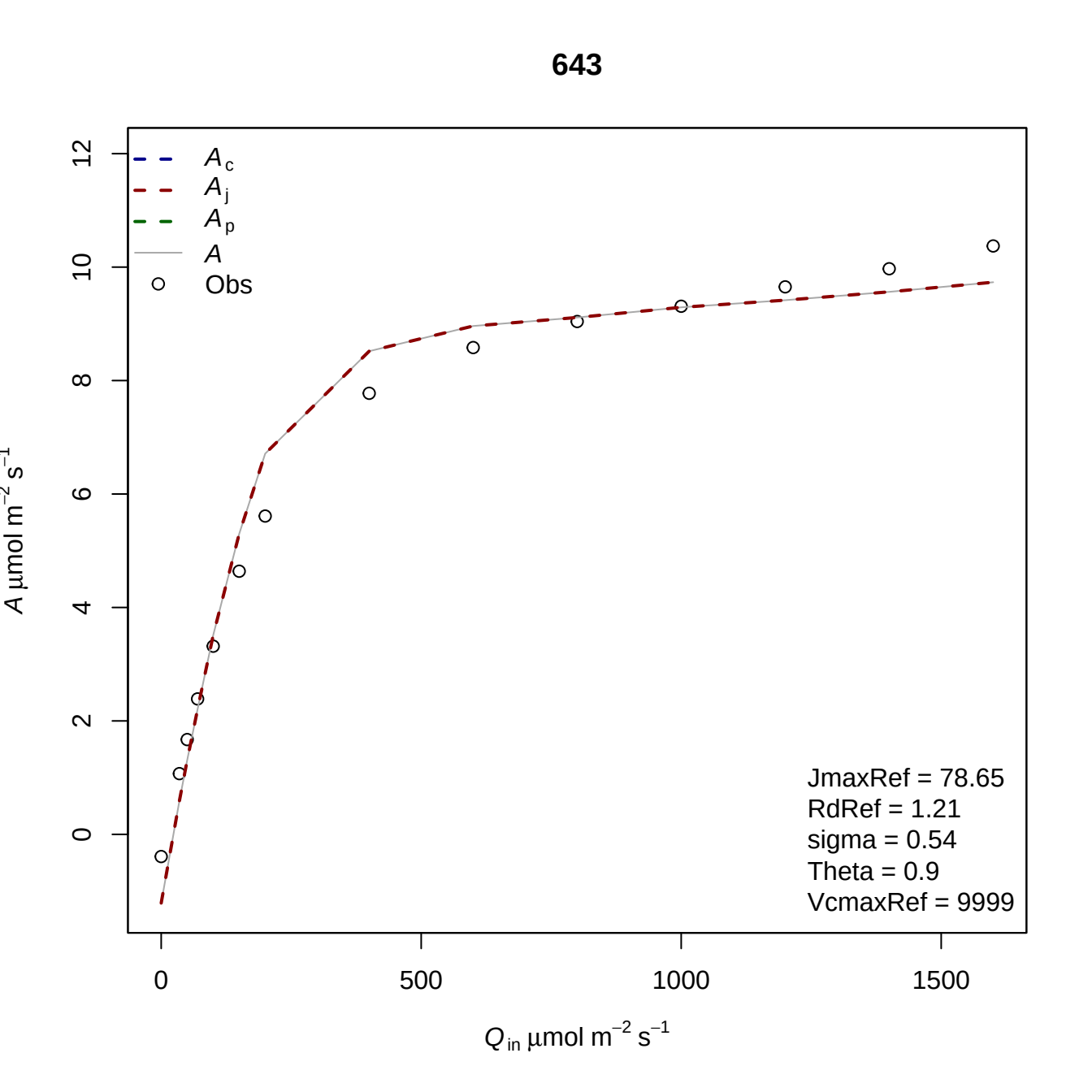

638
A_c
A_j
A_p
A
Obs
JmaxRef = 102.95
RdRef = 1.54
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

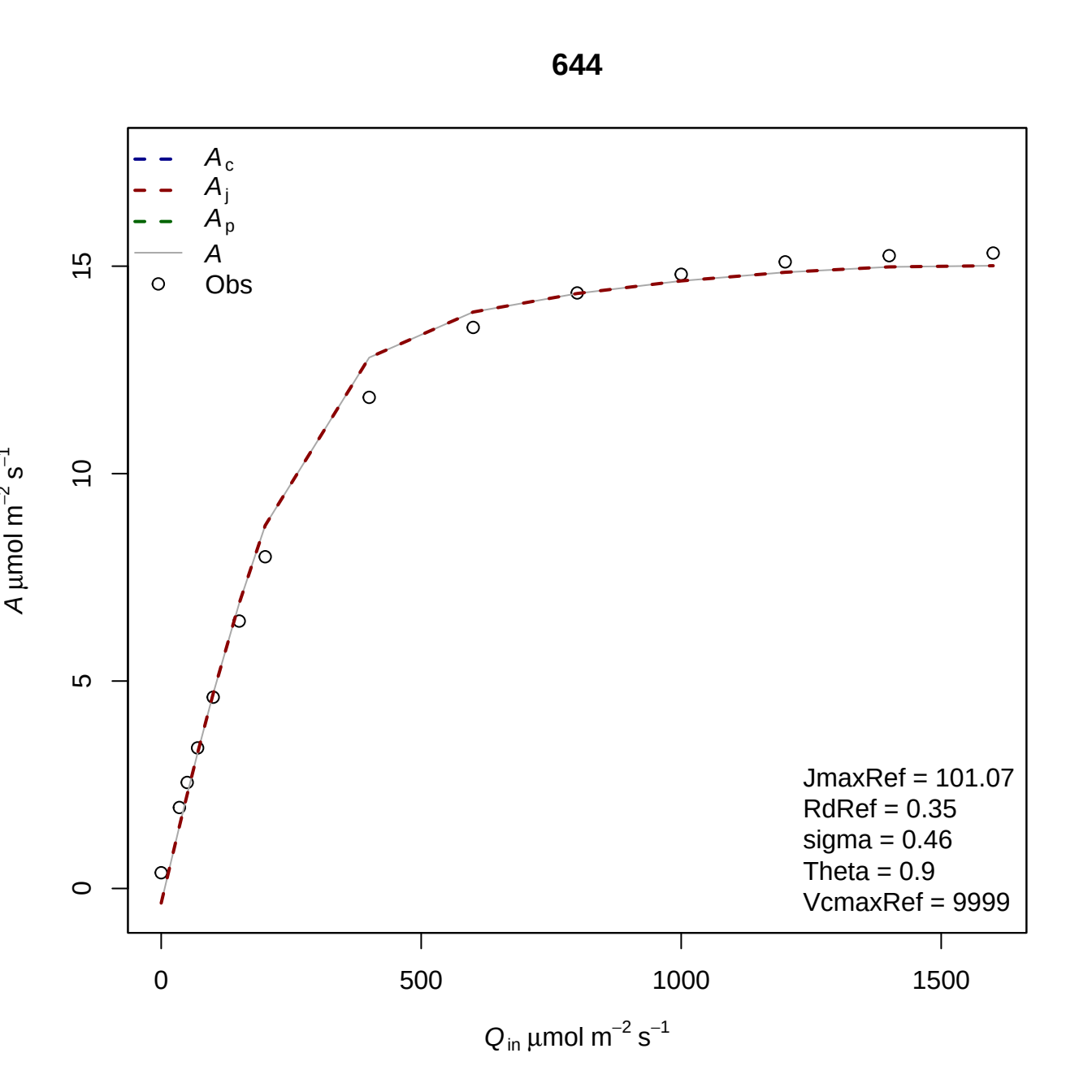

639
A_c
A_j
A_p
A
Obs
JmaxRef = 53.24
RdRef = 1.21
sigma = 0.17
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

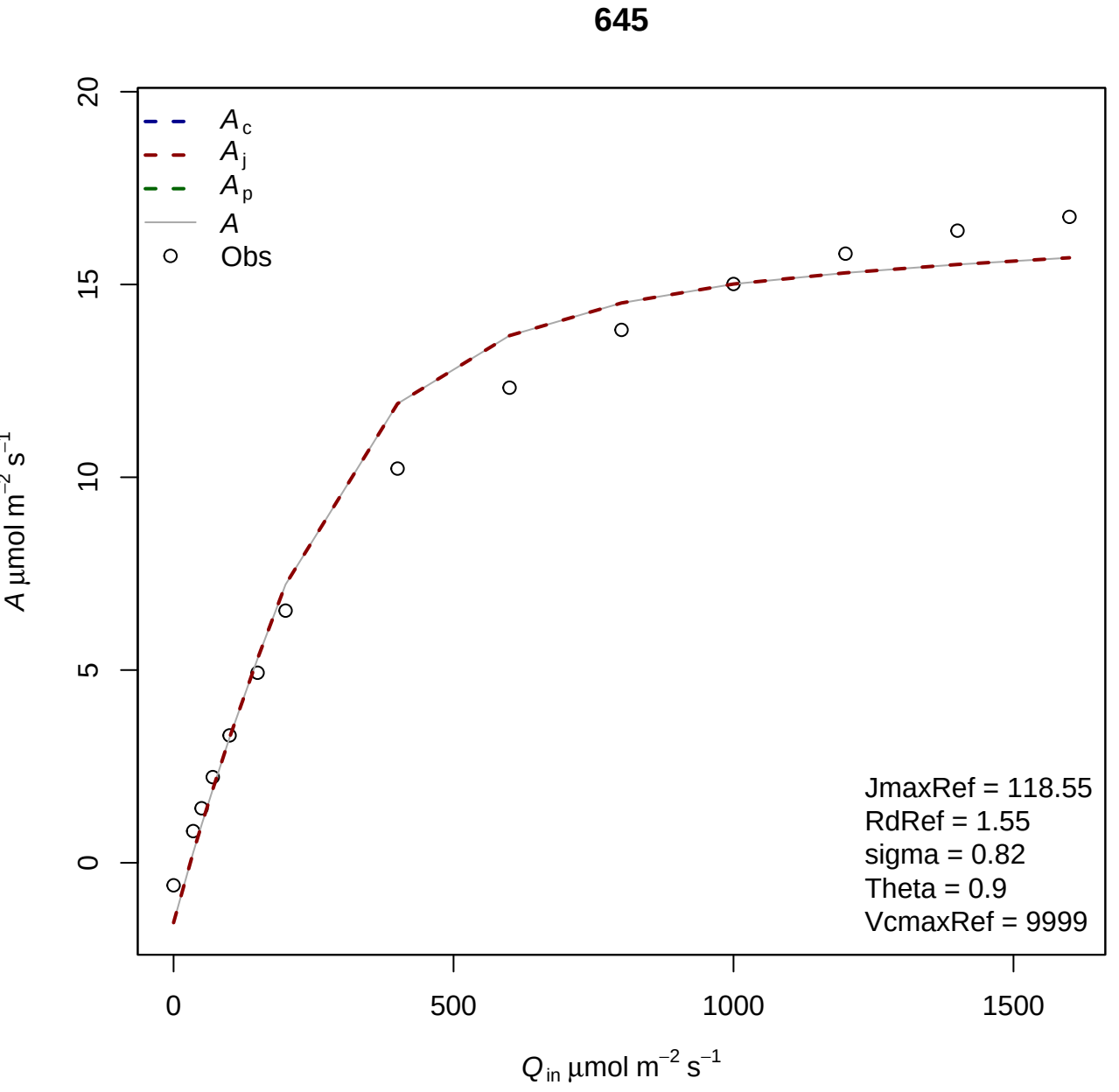

640
A_c
A_j
A_p
A
Obs
JmaxRef = 103.19
RdRef = 1.04
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

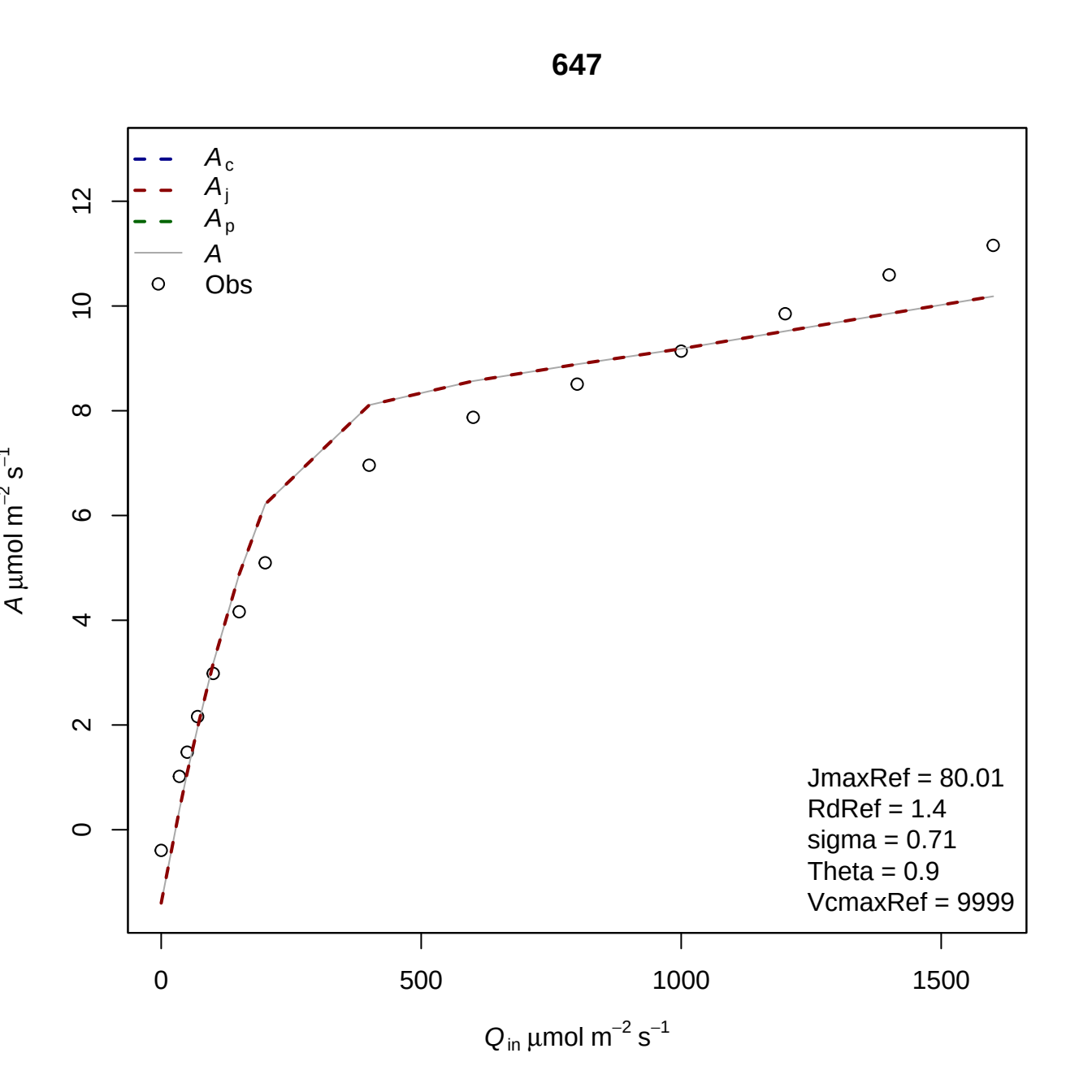

641
A_c
A_j
A_p
A
Obs
JmaxRef = 30.56
RdRef = −0.75
sigma = 0.03
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

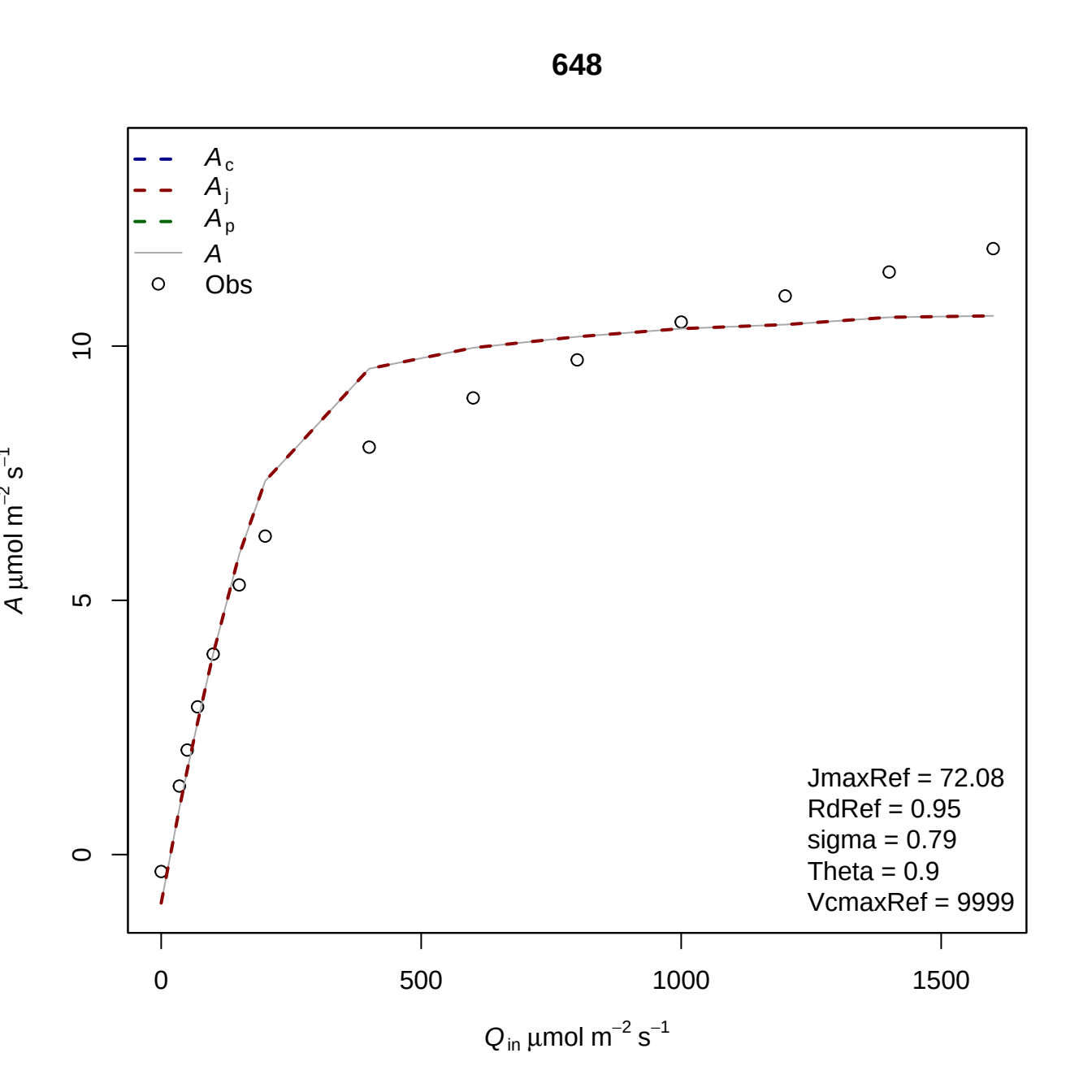

642
A_c
A_j
A_p
A
Obs
JmaxRef = 42.4
RdRef = 0.6
sigma = 0.13
Theta = 0.9
VcmaxRef = 33.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

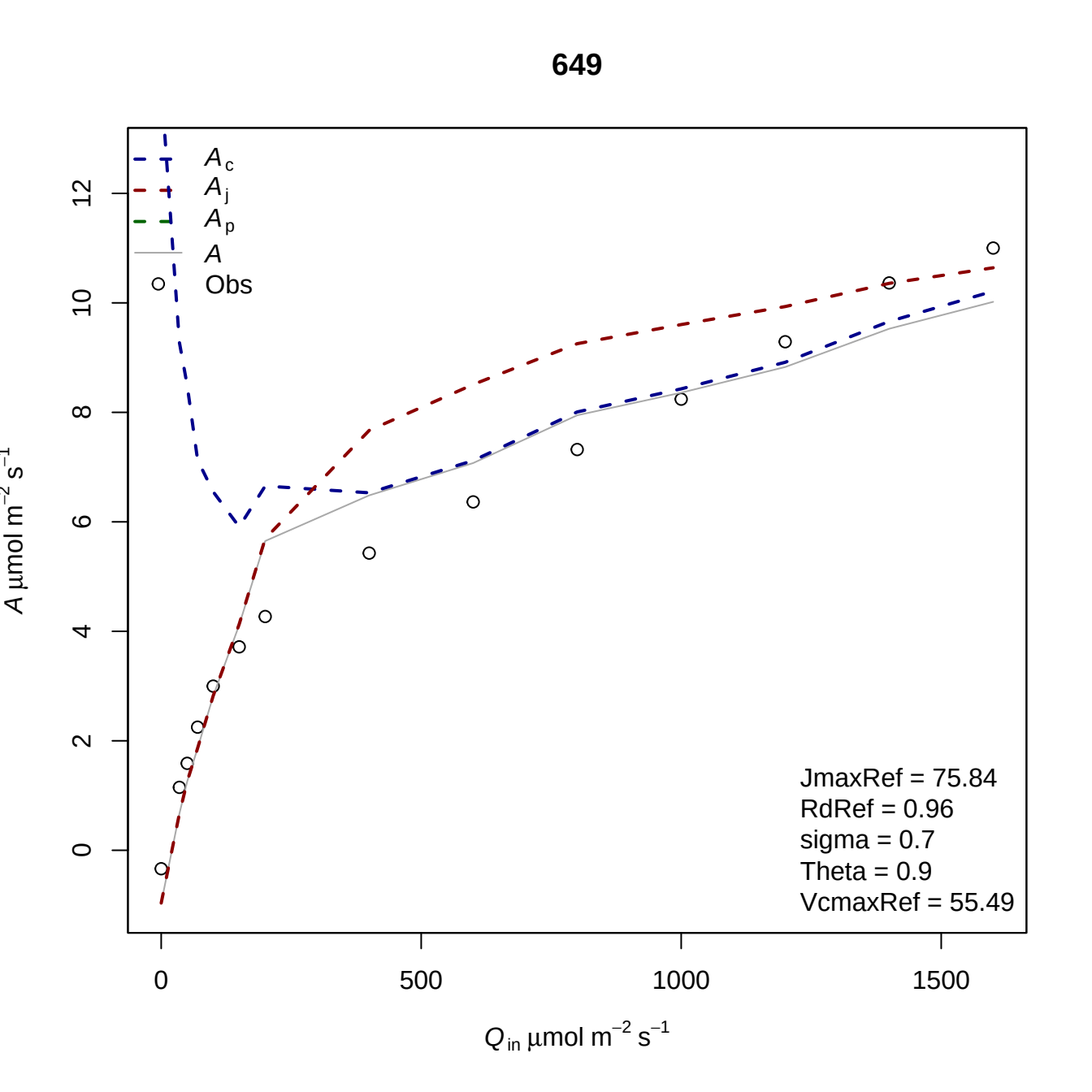

643
A_c
A_j
A_p
A
Obs
JmaxRef = 78.65
RdRef = 1.21
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

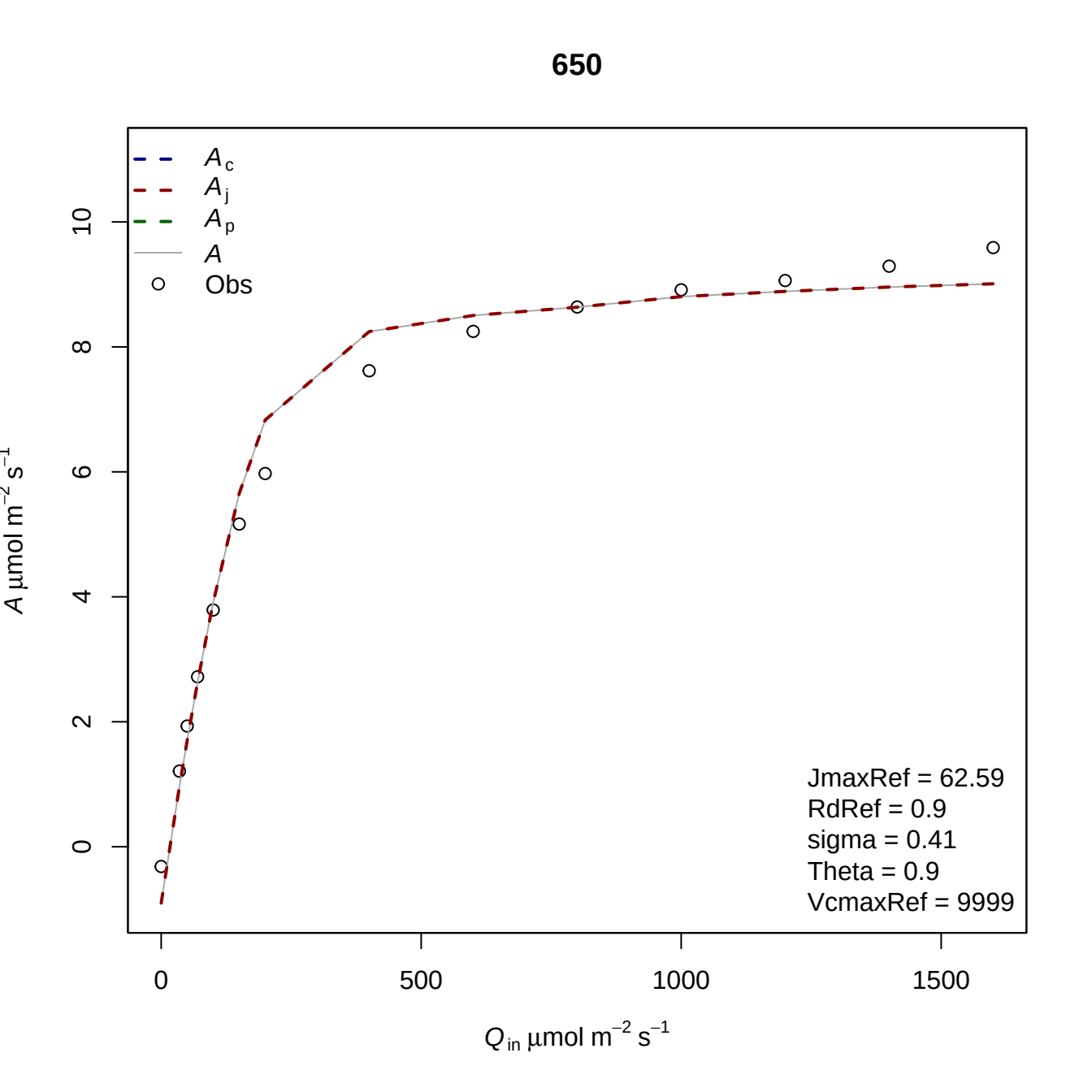

644
A_c
A_j
A_p
A
Obs
JmaxRef = 101.07
RdRef = 0.35
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

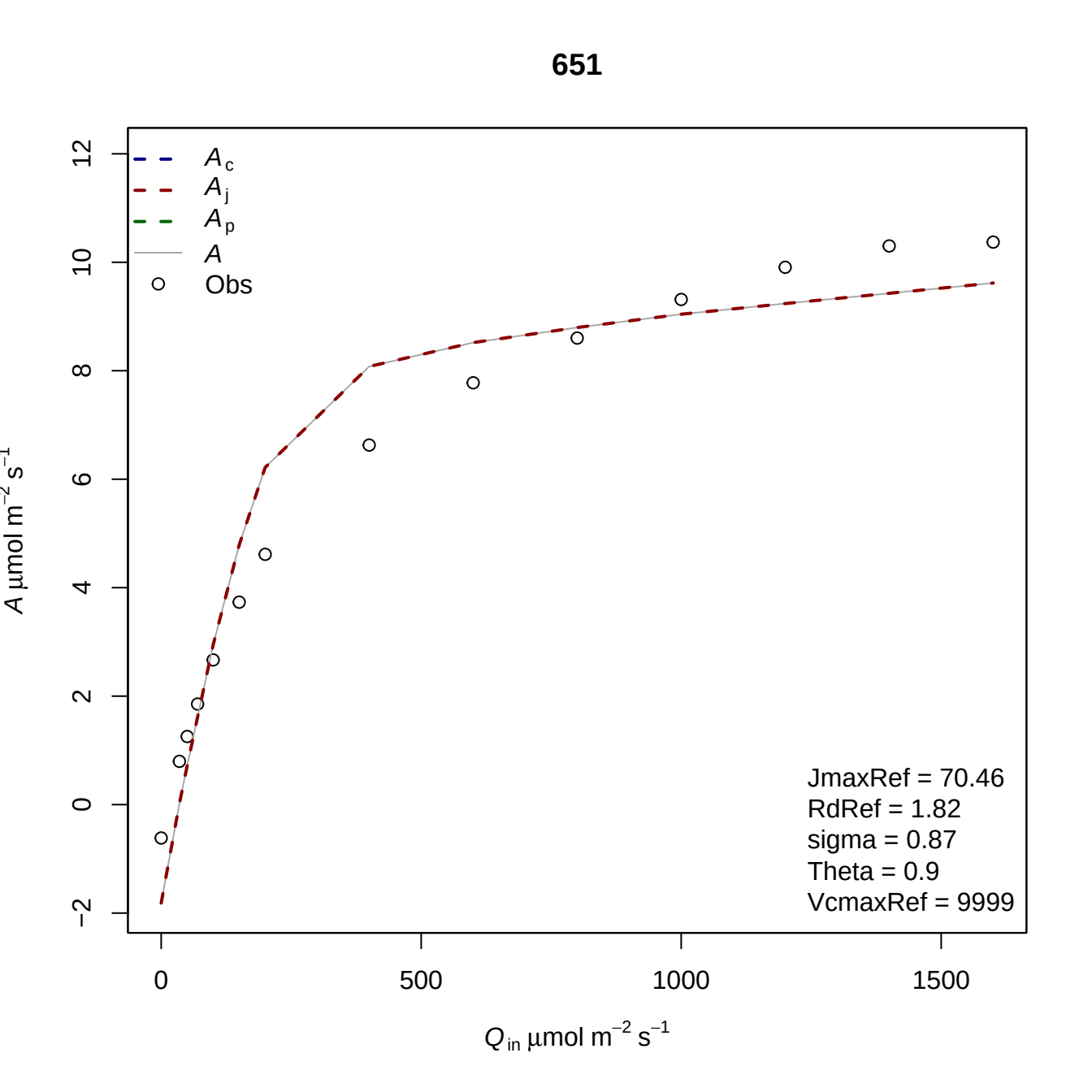

645
A_c
A_j
A_p
A
Obs
JmaxRef = 118.55
RdRef = 1.55
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

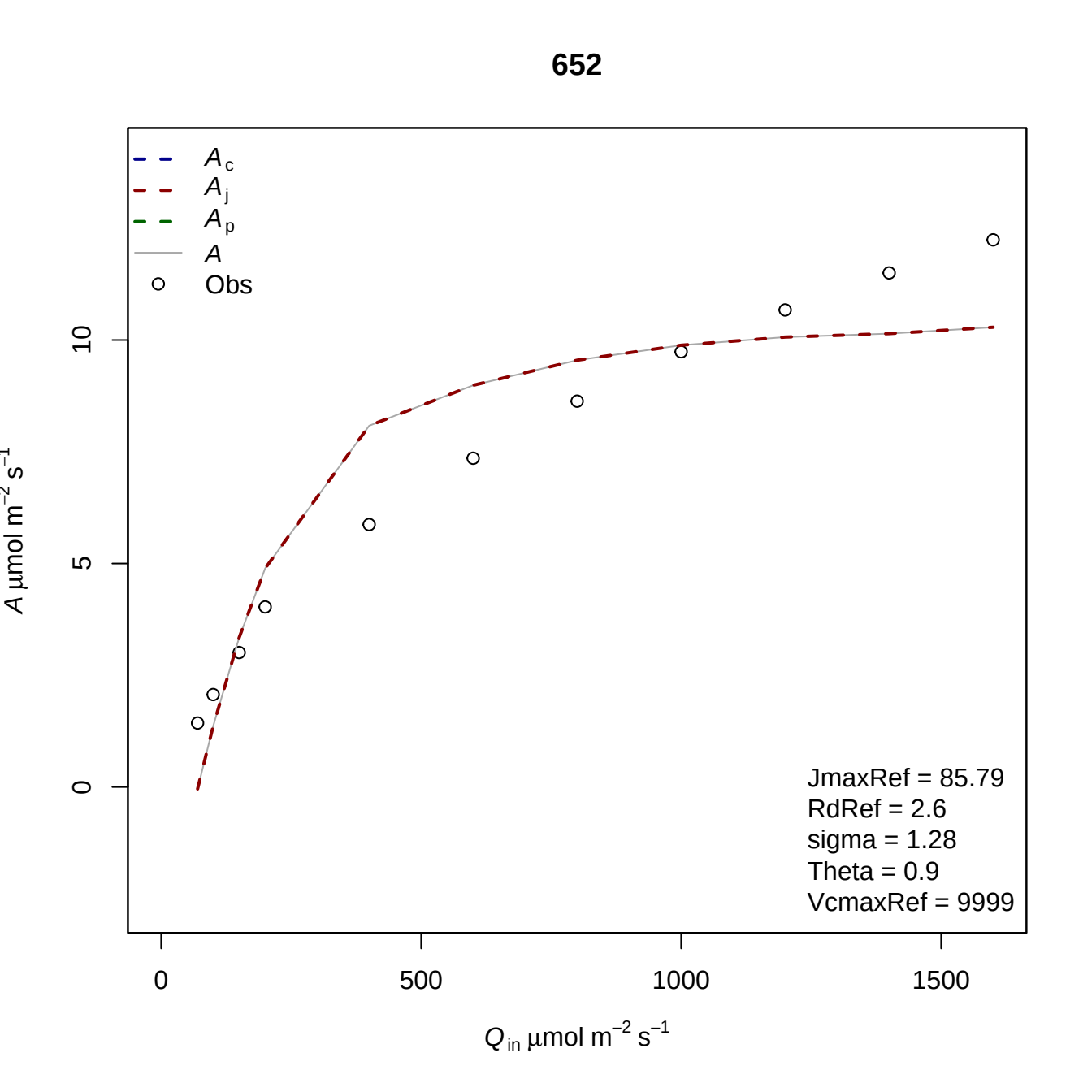

647
A_c
A_j
A_p
A
Obs
JmaxRef = 80.01
RdRef = 1.4
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

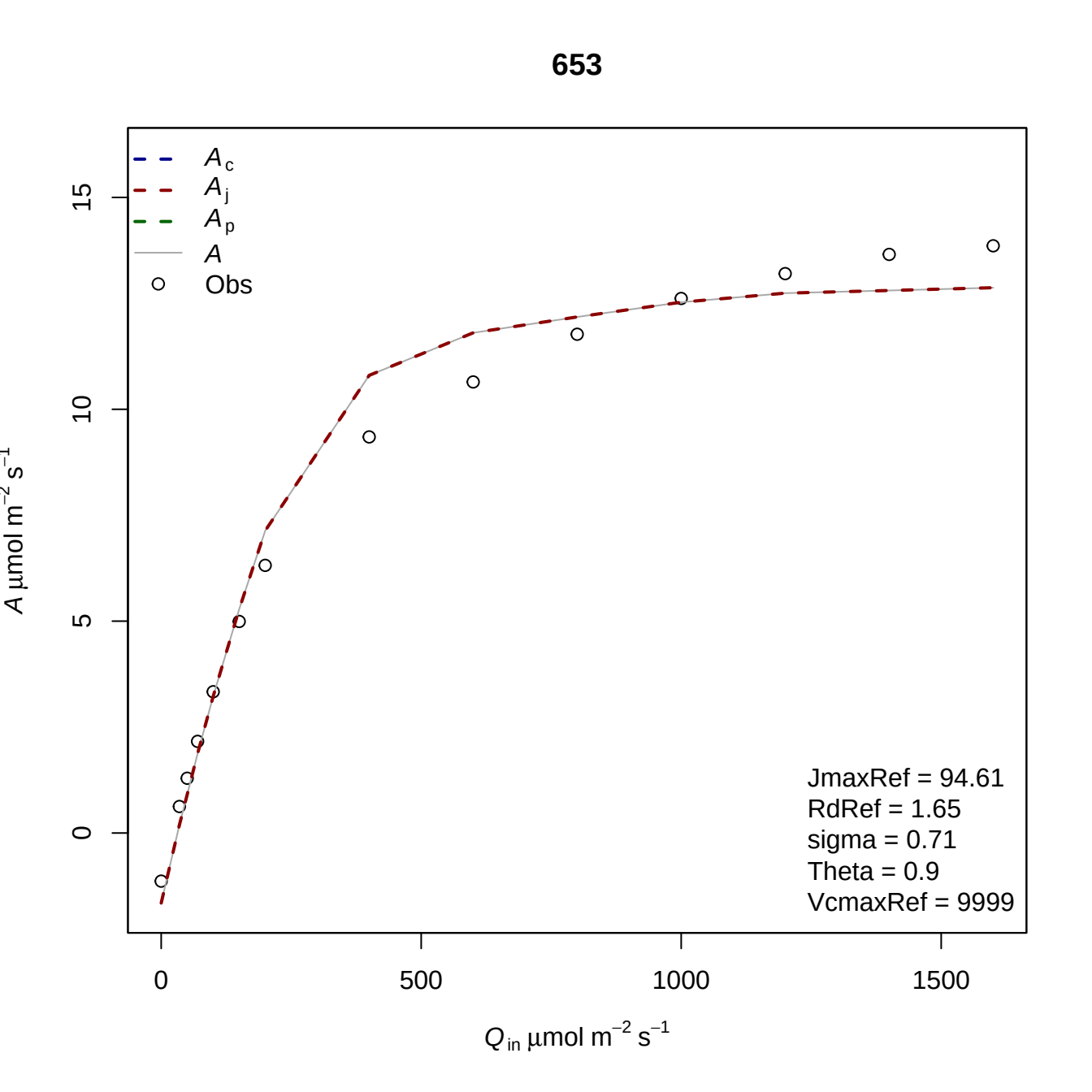

648
A_c
A_j
A_p
A
Obs
JmaxRef = 72.08
RdRef = 0.95
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

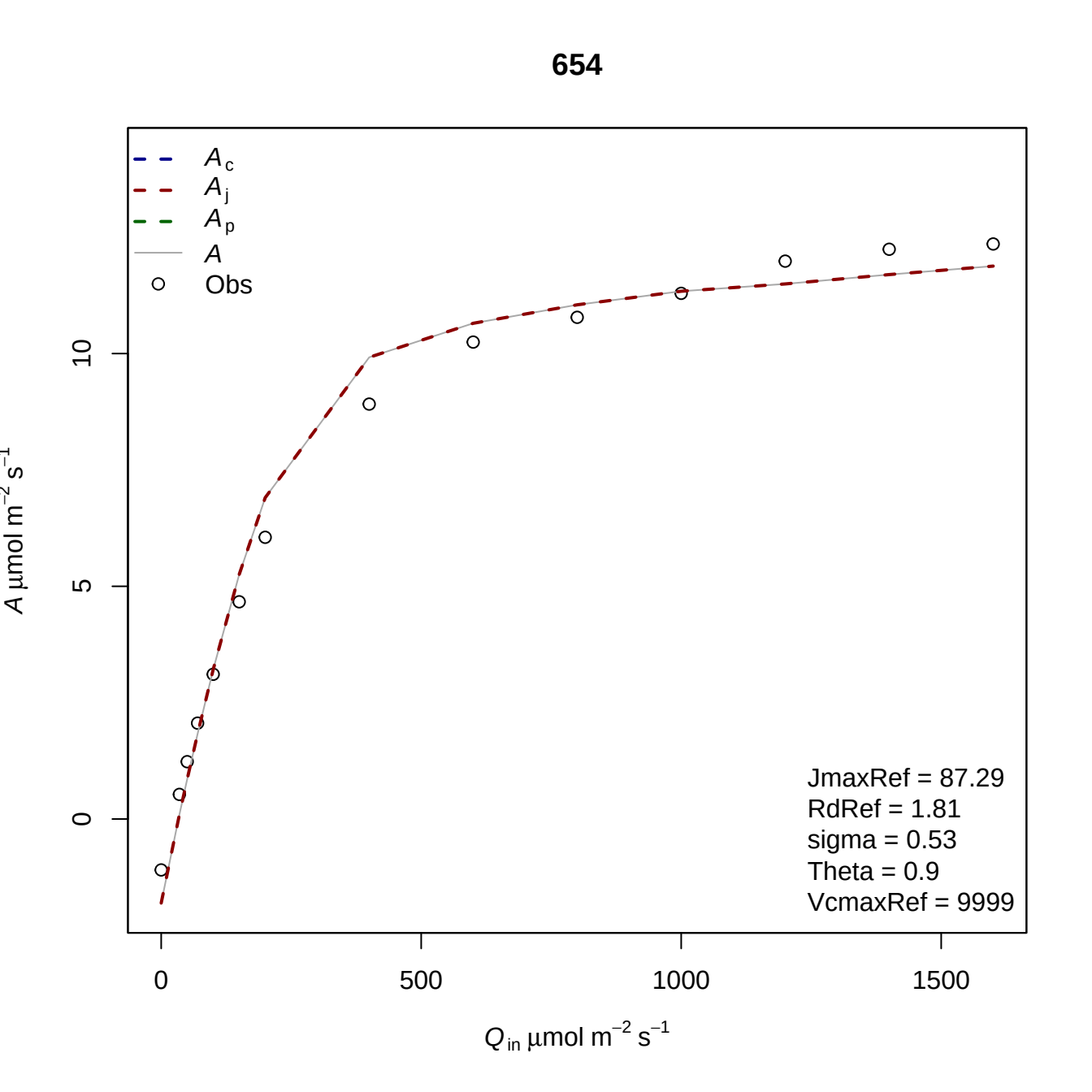

649
A_c
A_j
A_p
A
Obs
JmaxRef = 75.84
RdRef = 0.96
sigma = 0.7
Theta = 0.9
VcmaxRef = 55.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

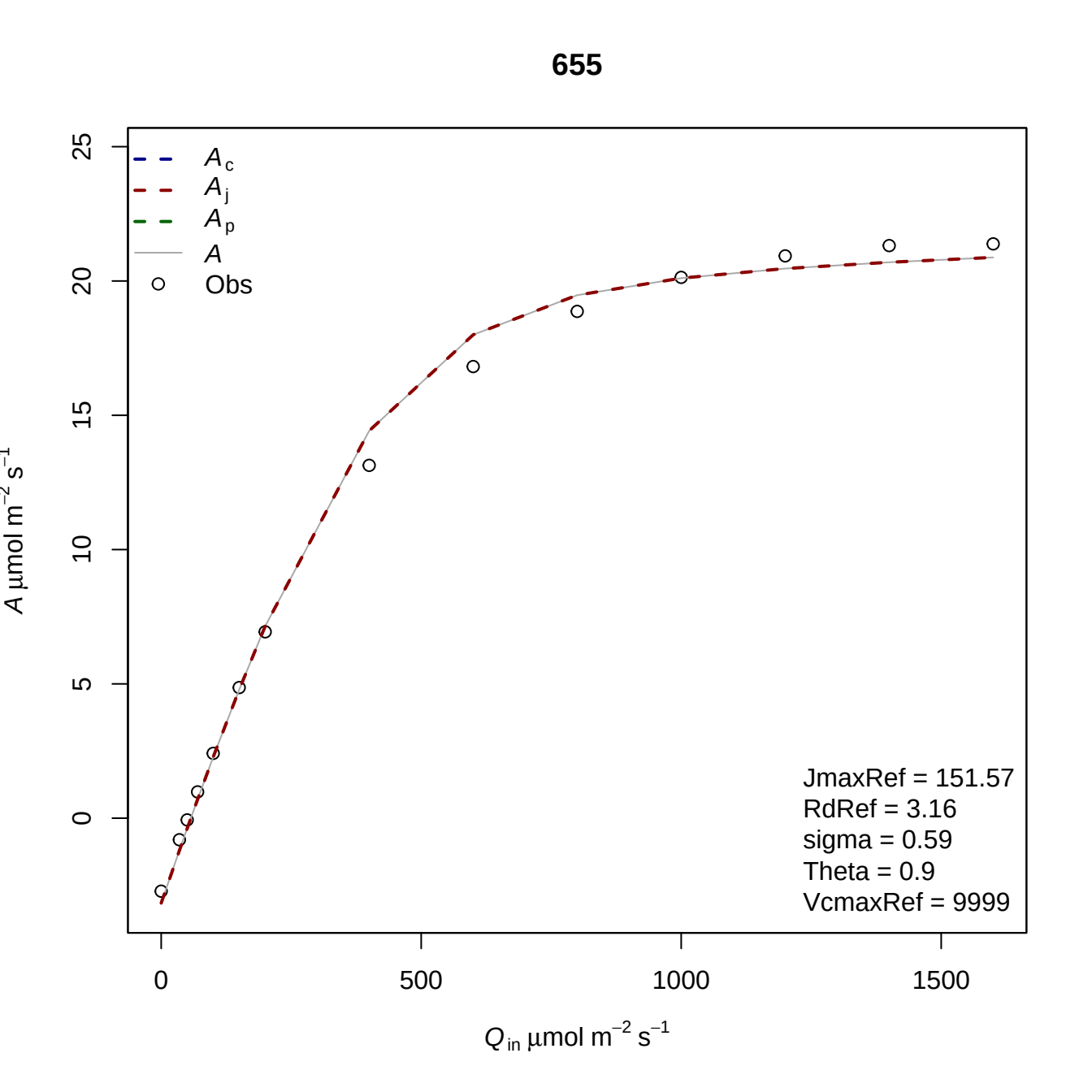

650
A_c
A_j
A_p
A
Obs
JmaxRef = 62.59
RdRef = 0.9
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

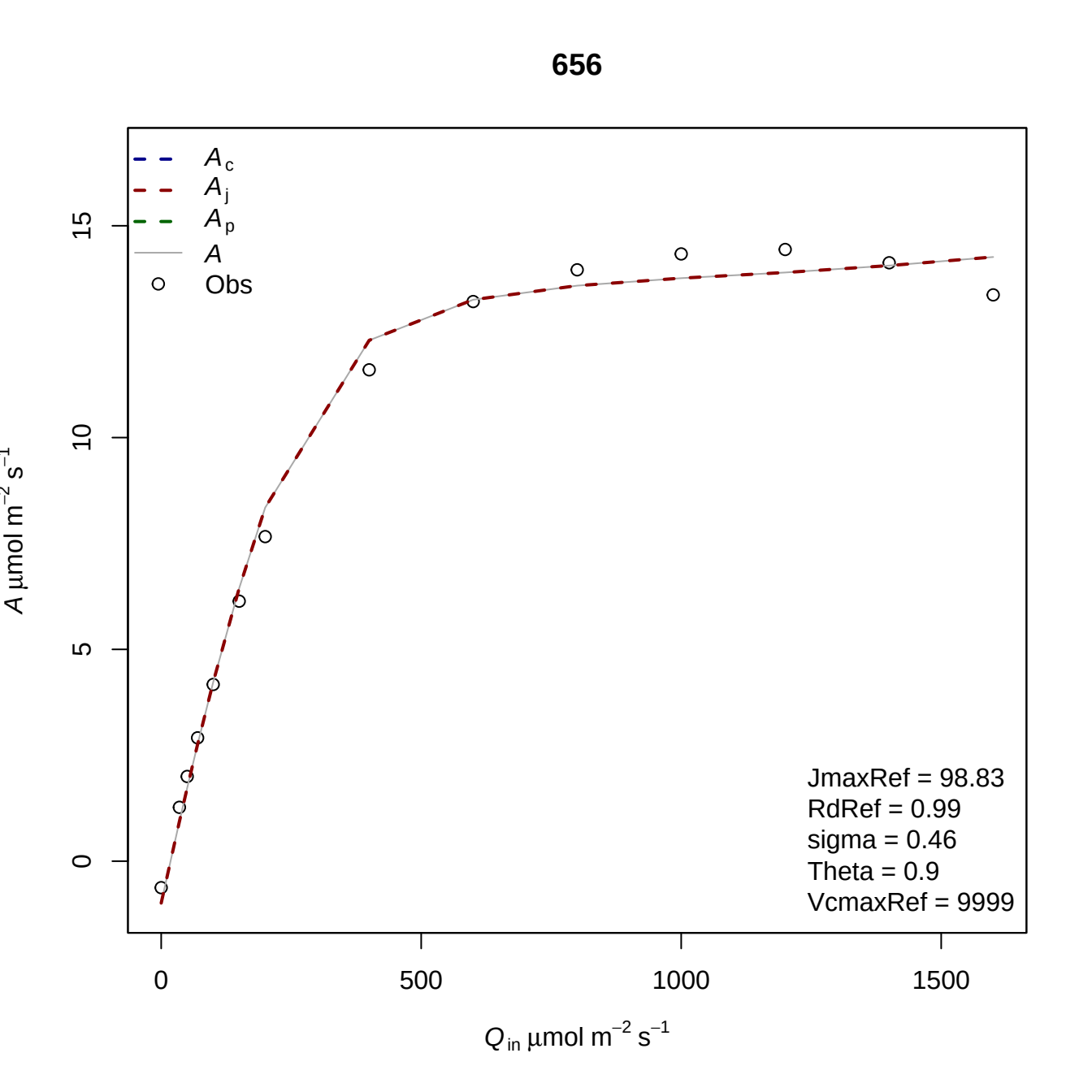

651
A_c
A_j
A_p
A
Obs
JmaxRef = 70.46
RdRef = 1.82
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

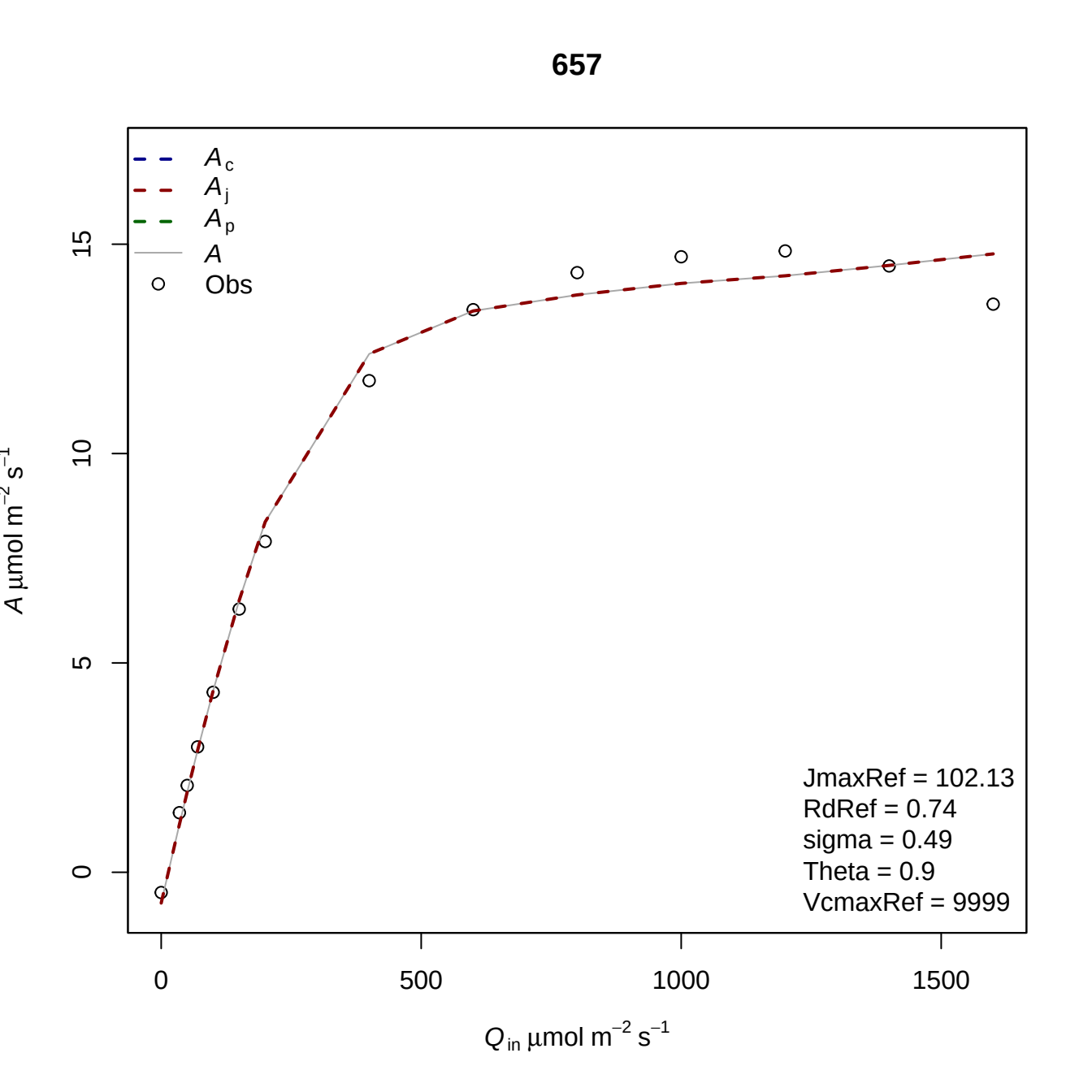

652
A_c
A_j
A_p
A
Obs
JmaxRef = 85.79
RdRef = 2.6
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

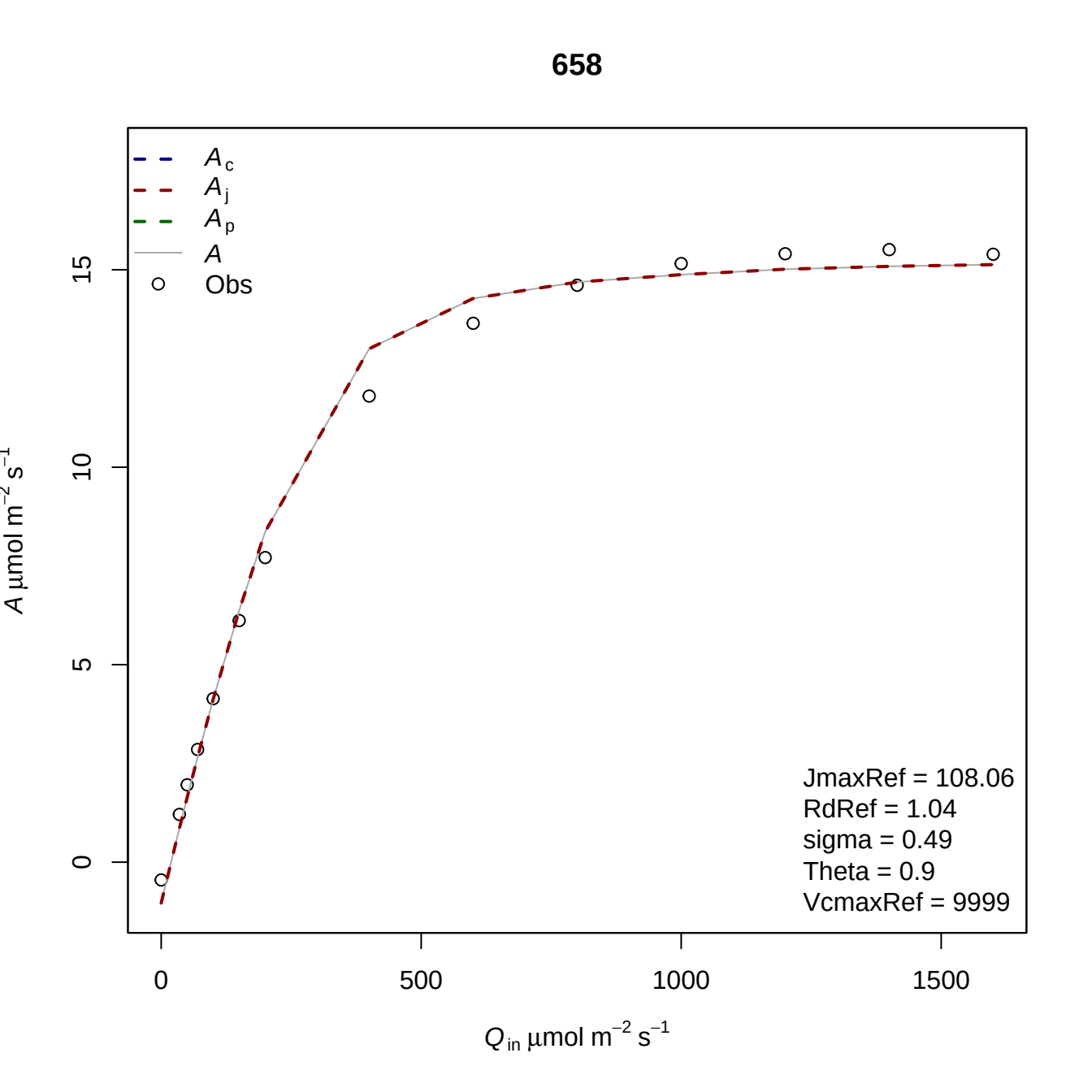

653
A_c
A_j
A_p
A
Obs
JmaxRef = 94.61
RdRef = 1.65
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

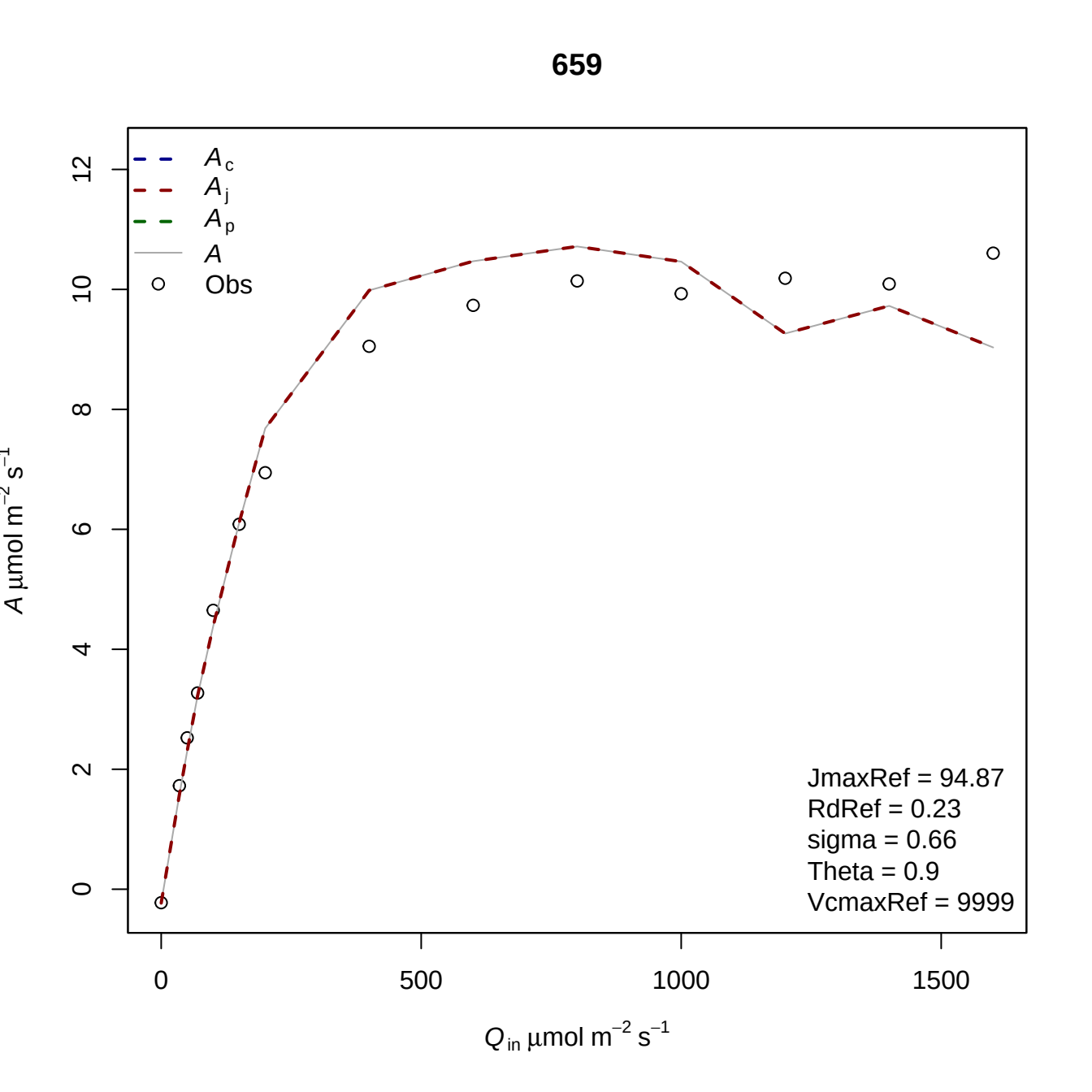

654
A_c
A_j
A_p
A
Obs
JmaxRef = 87.29
RdRef = 1.81
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

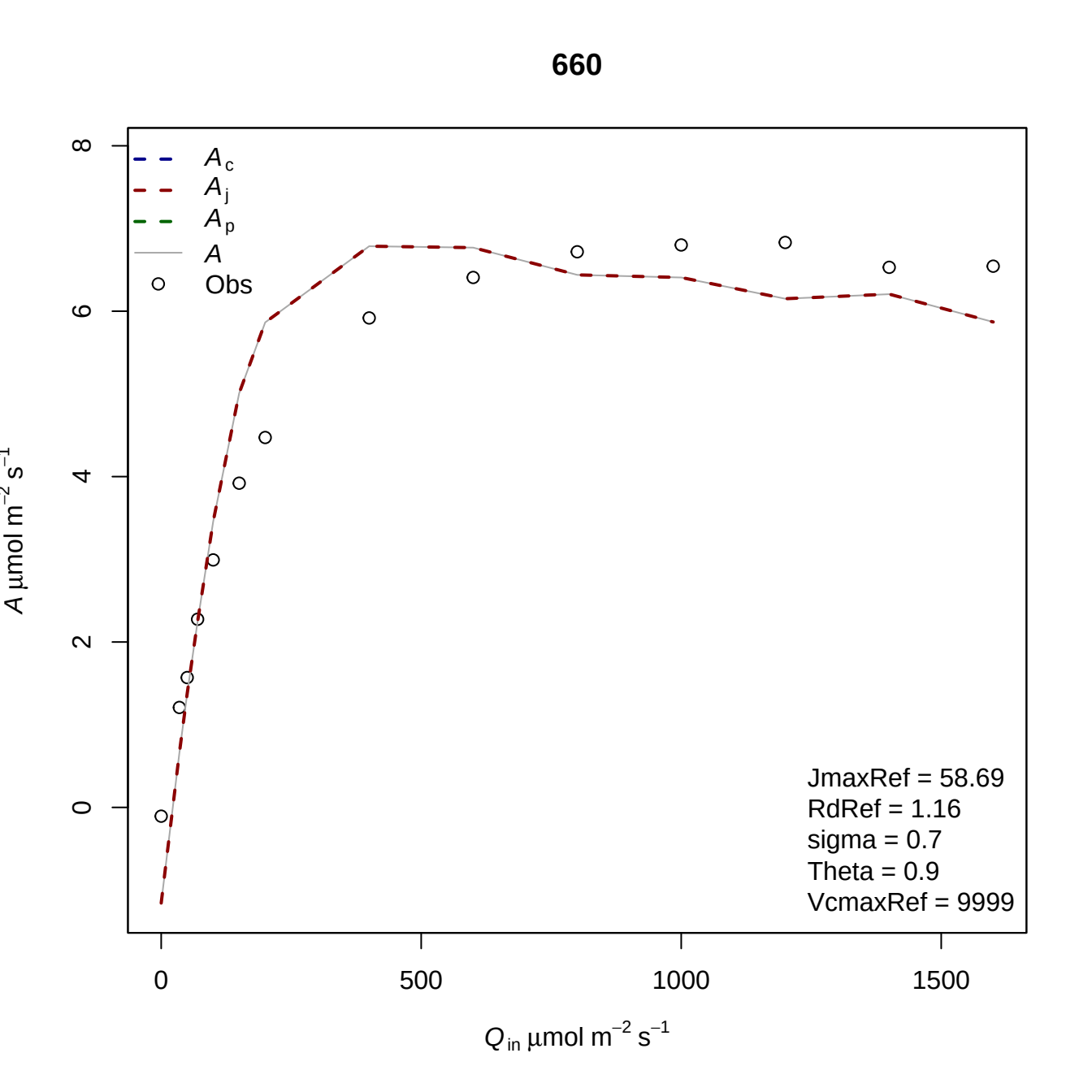

655
A_c
A_j
A_p
A
Obs
JmaxRef = 151.57
RdRef = 3.16
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

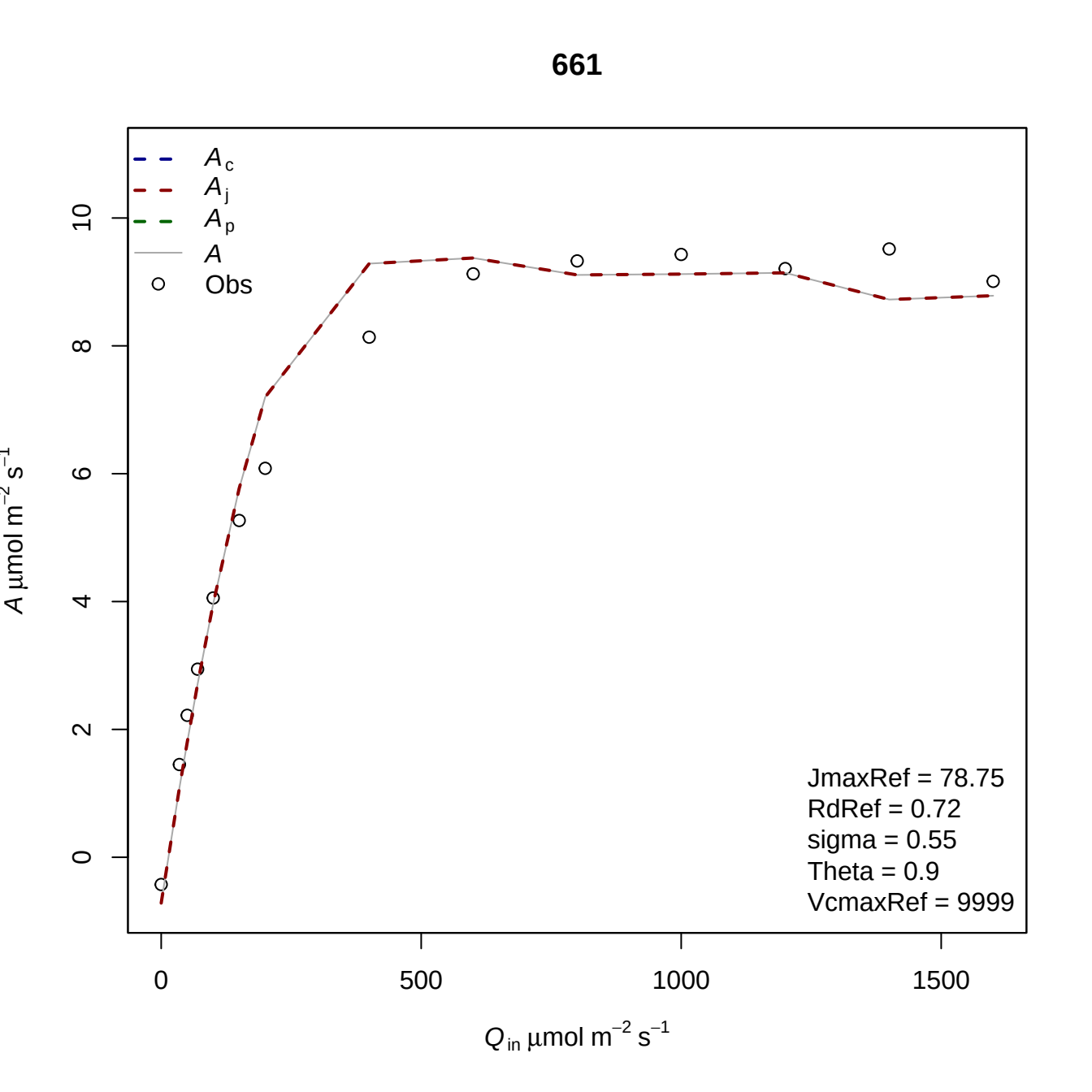

656
A_c
A_j
A_p
A
Obs
JmaxRef = 98.83
RdRef = 0.99
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

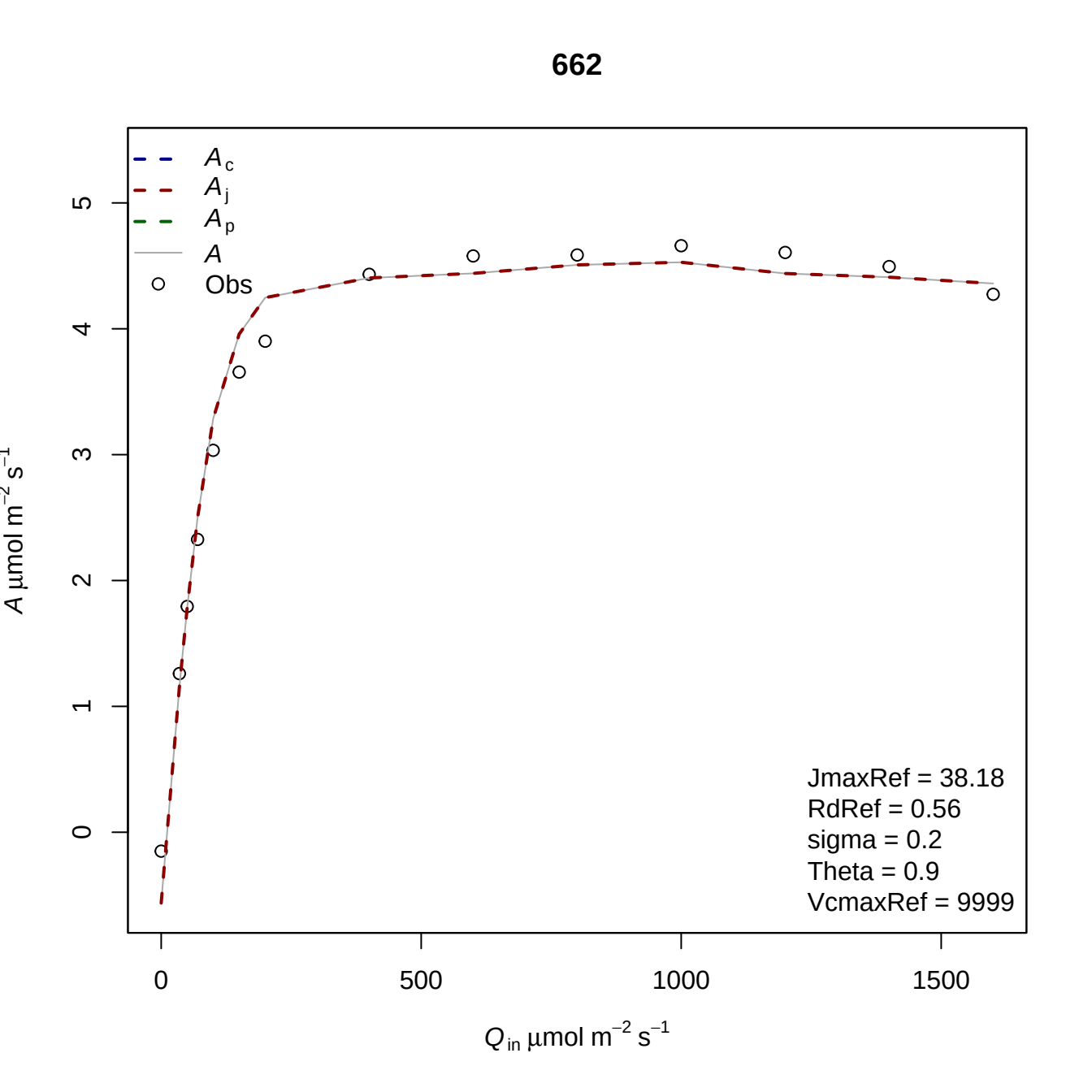

657
A_c
A_j
A_p
A
Obs
JmaxRef = 102.13
RdRef = 0.74
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

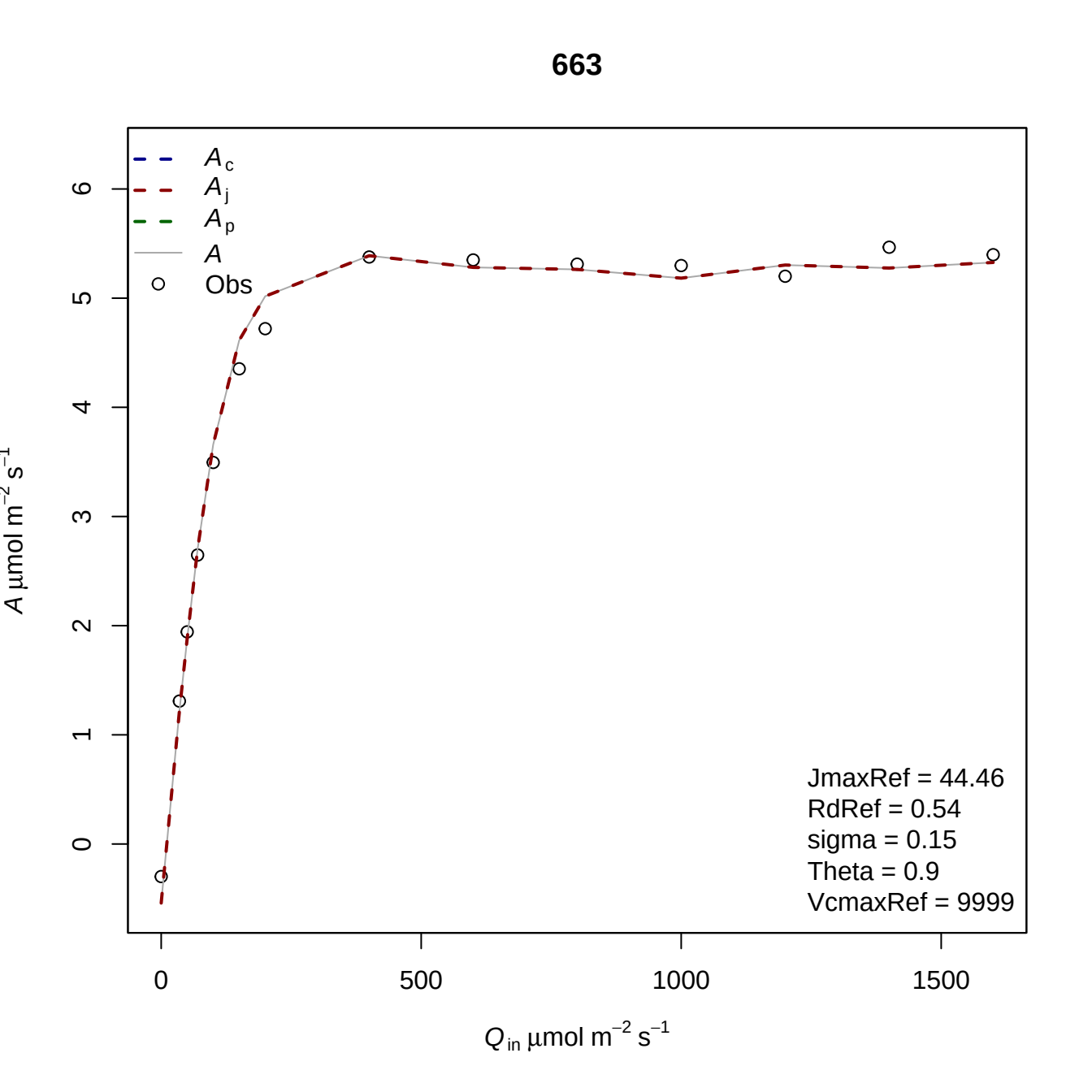

658
A_c
A_j
A_p
A
Obs
JmaxRef = 108.06
RdRef = 1.04
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

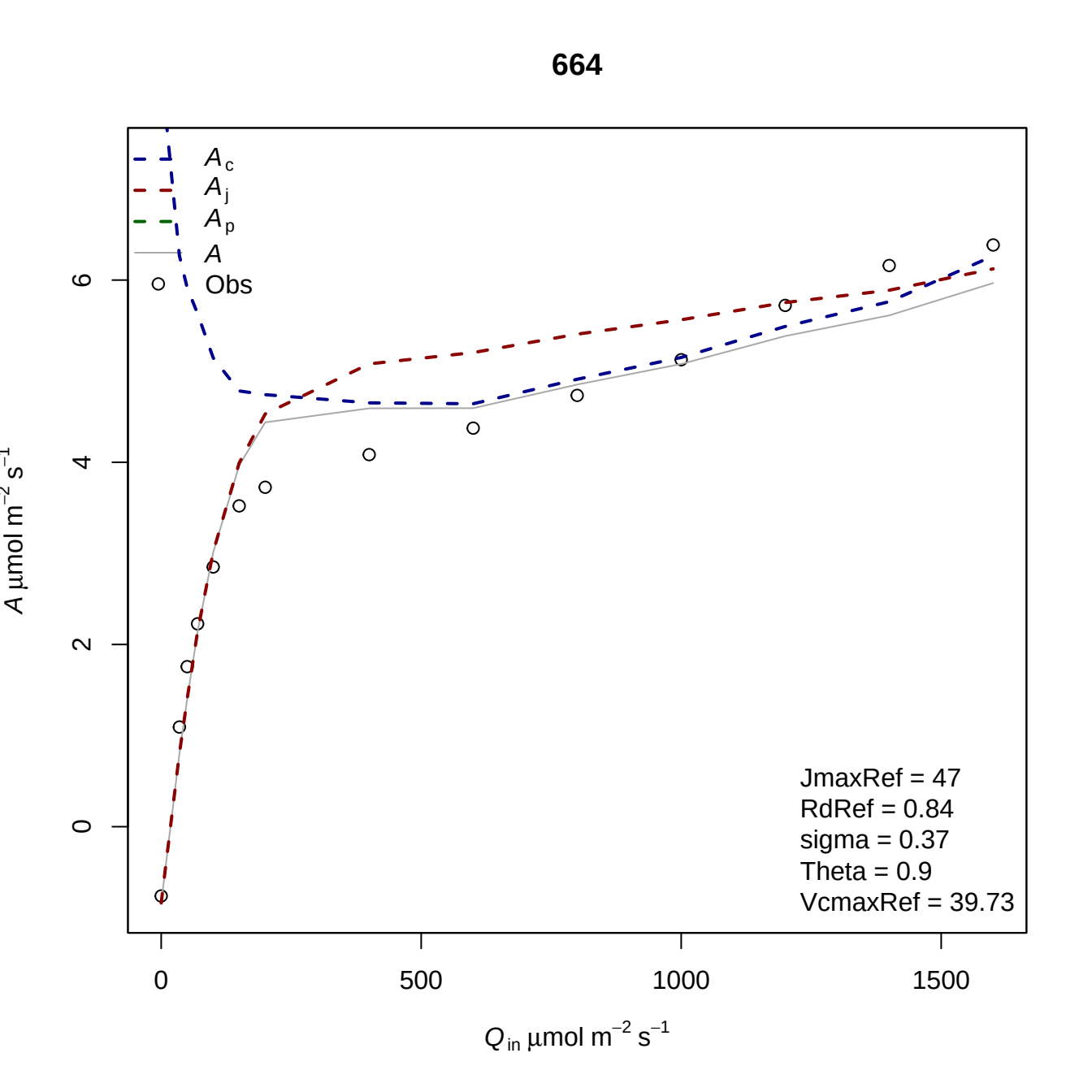

659
A_c
A_j
A_p
A
Obs
JmaxRef = 94.87
RdRef = 0.23
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

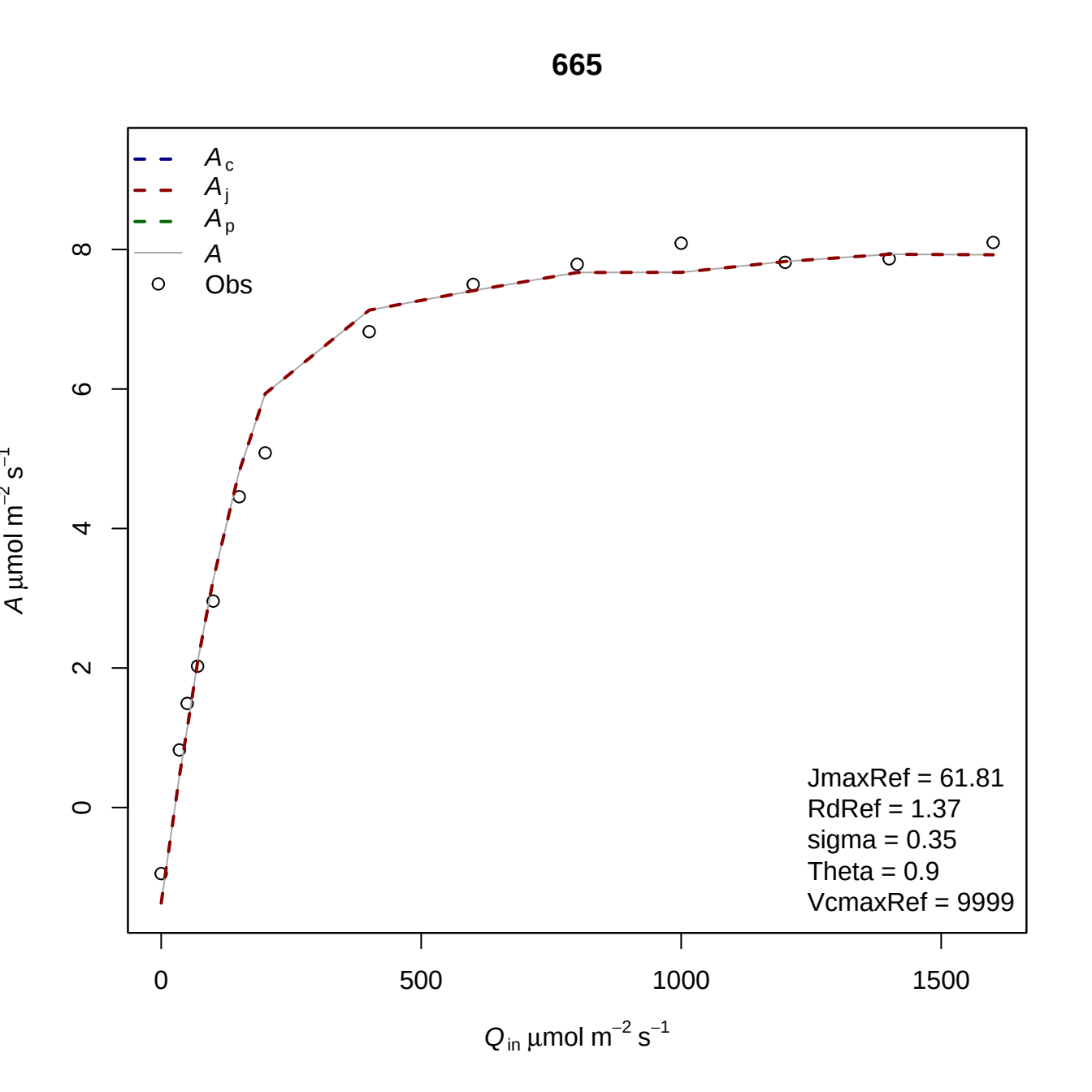

660
A_c
A_j
A_p
A
Obs
JmaxRef = 58.69
RdRef = 1.16
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

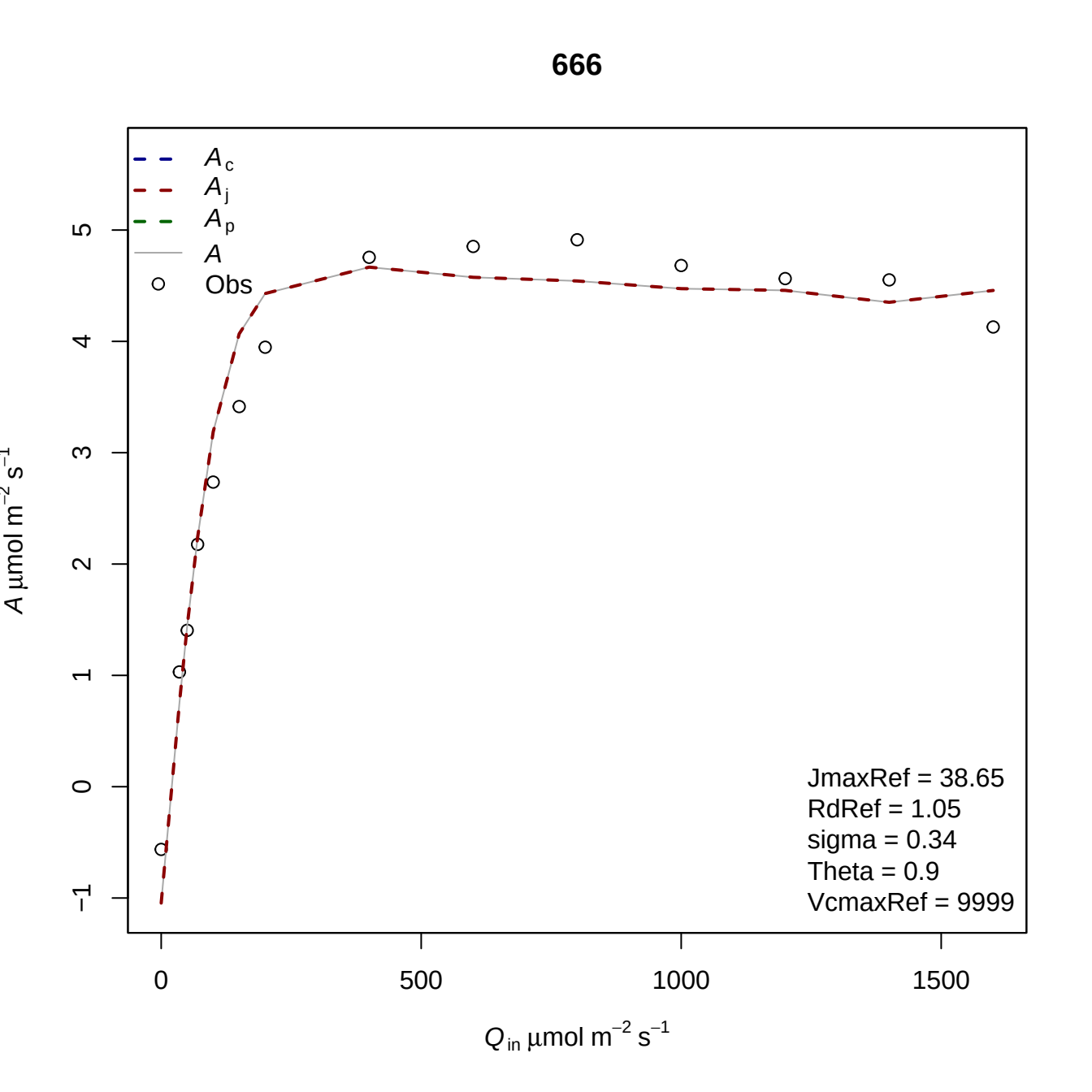

661
A_c
A_j
A_p
A
Obs
JmaxRef = 78.75
RdRef = 0.72
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

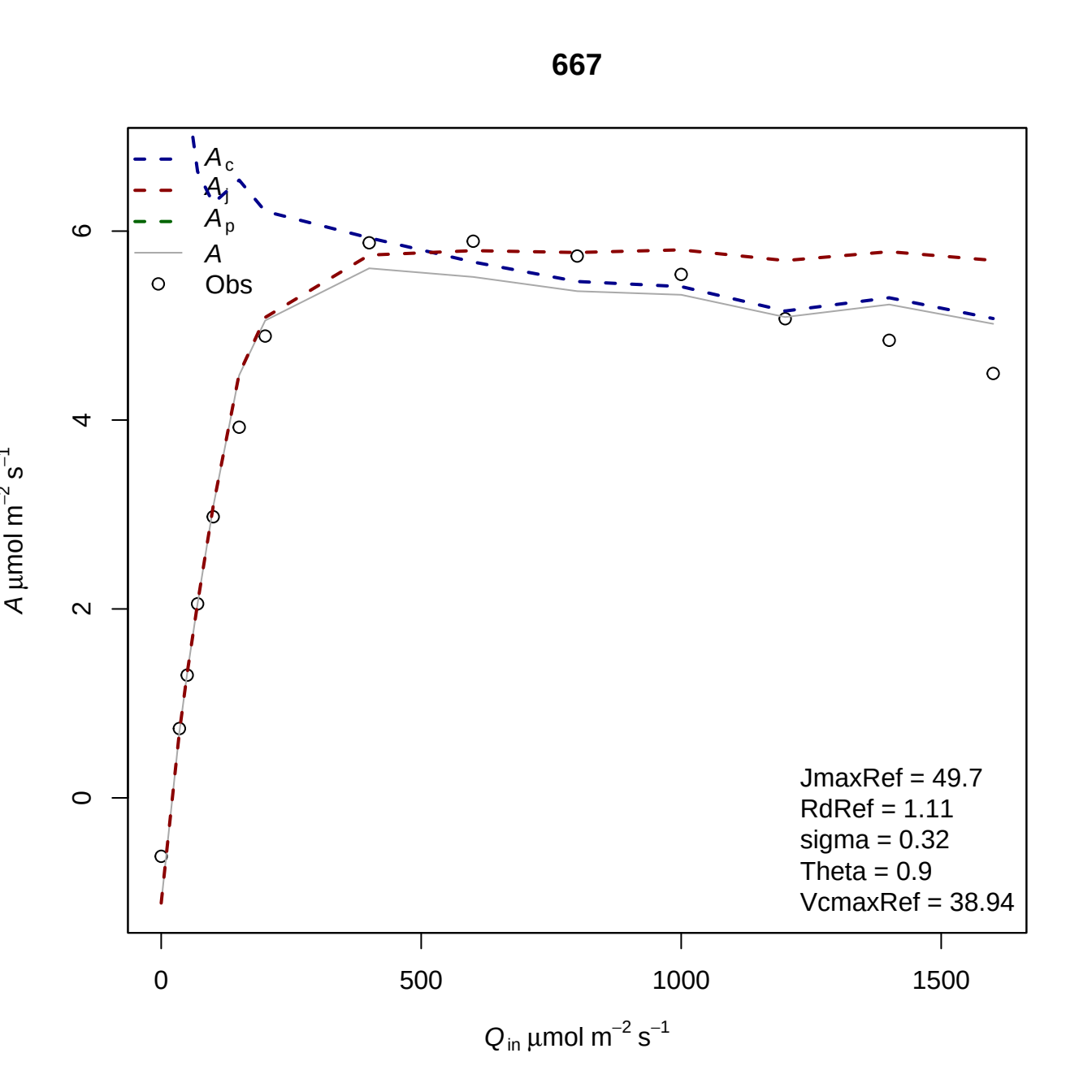

662
A_c
A_j
A_p
A
Obs
JmaxRef = 38.18
RdRef = 0.56
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

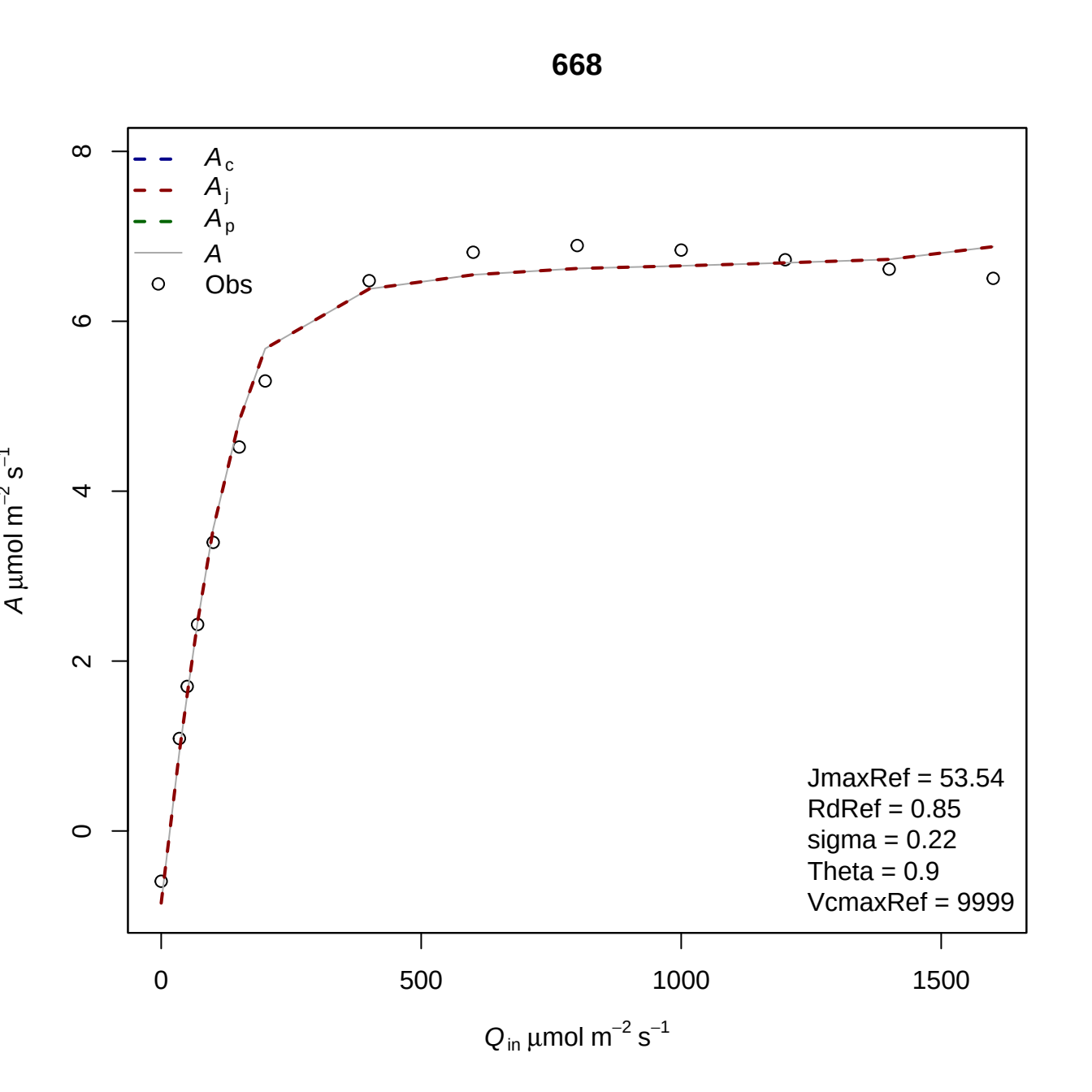

663
A_c
A_j
A_p
A
Obs
JmaxRef = 44.46
RdRef = 0.54
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

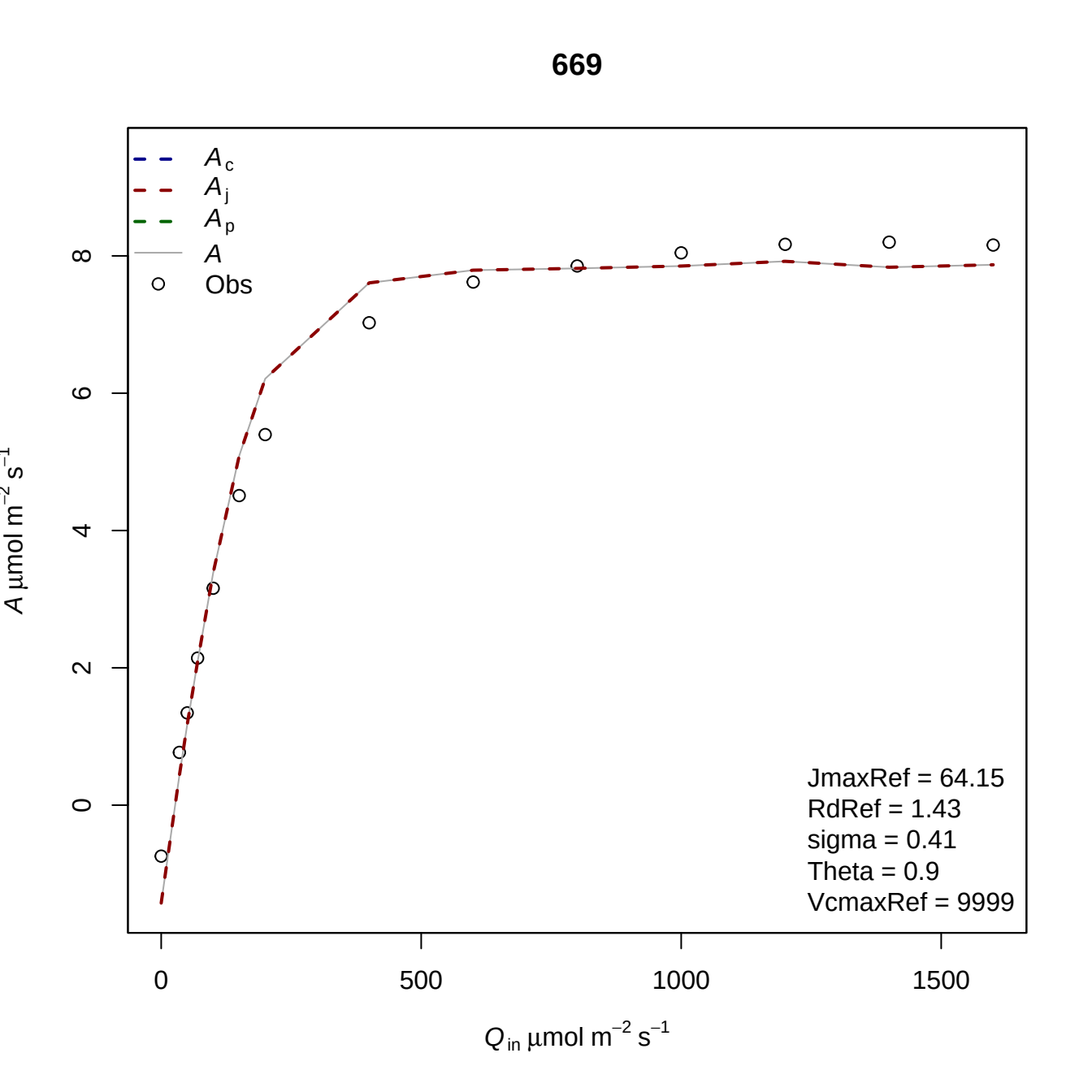

664
A_c
A_j
A_p
A
Obs
JmaxRef = 47
RdRef = 0.84
sigma = 0.37
Theta = 0.9
VcmaxRef = 39.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

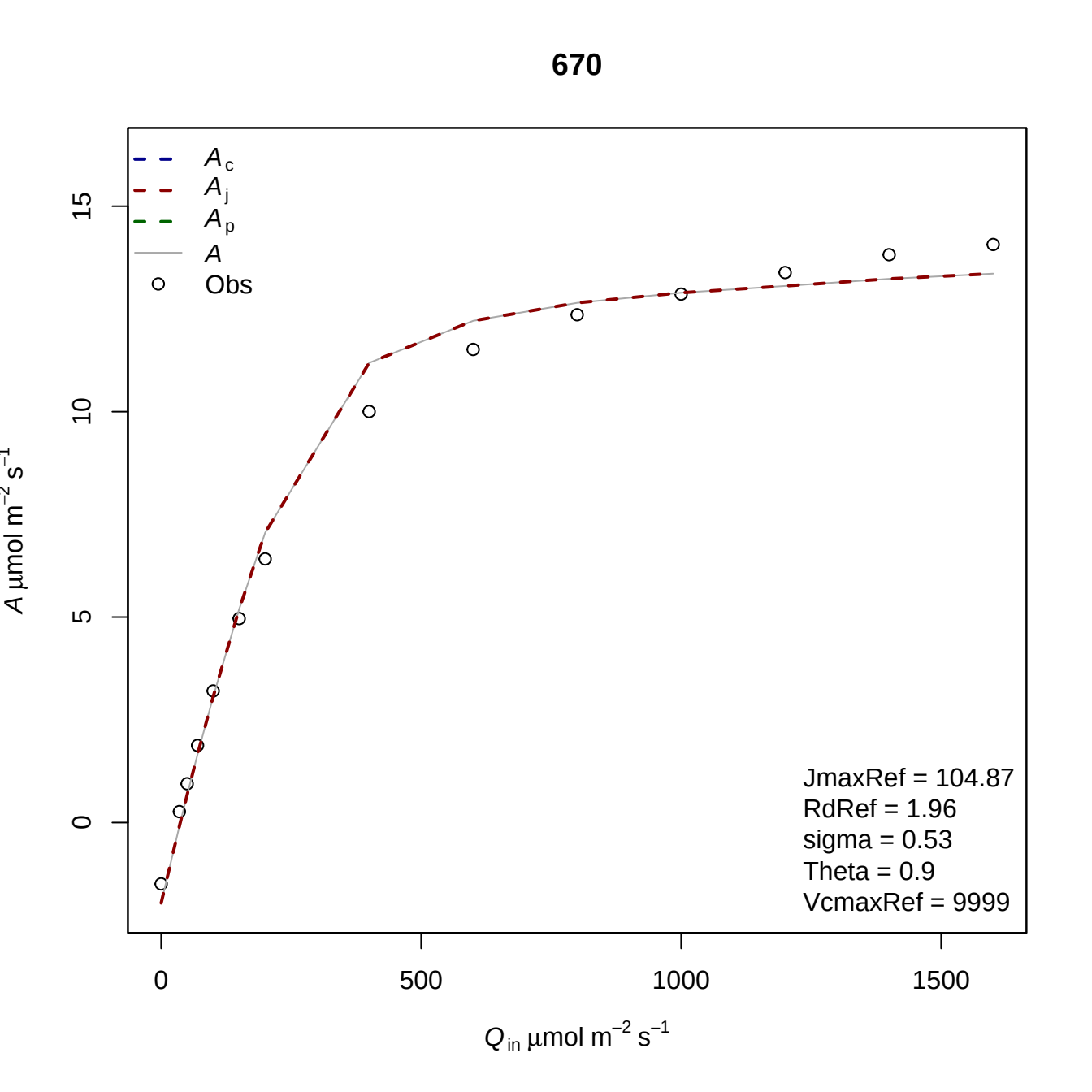

665
A_c
A_j
A_p
A
Obs
JmaxRef = 61.81
RdRef = 1.37
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

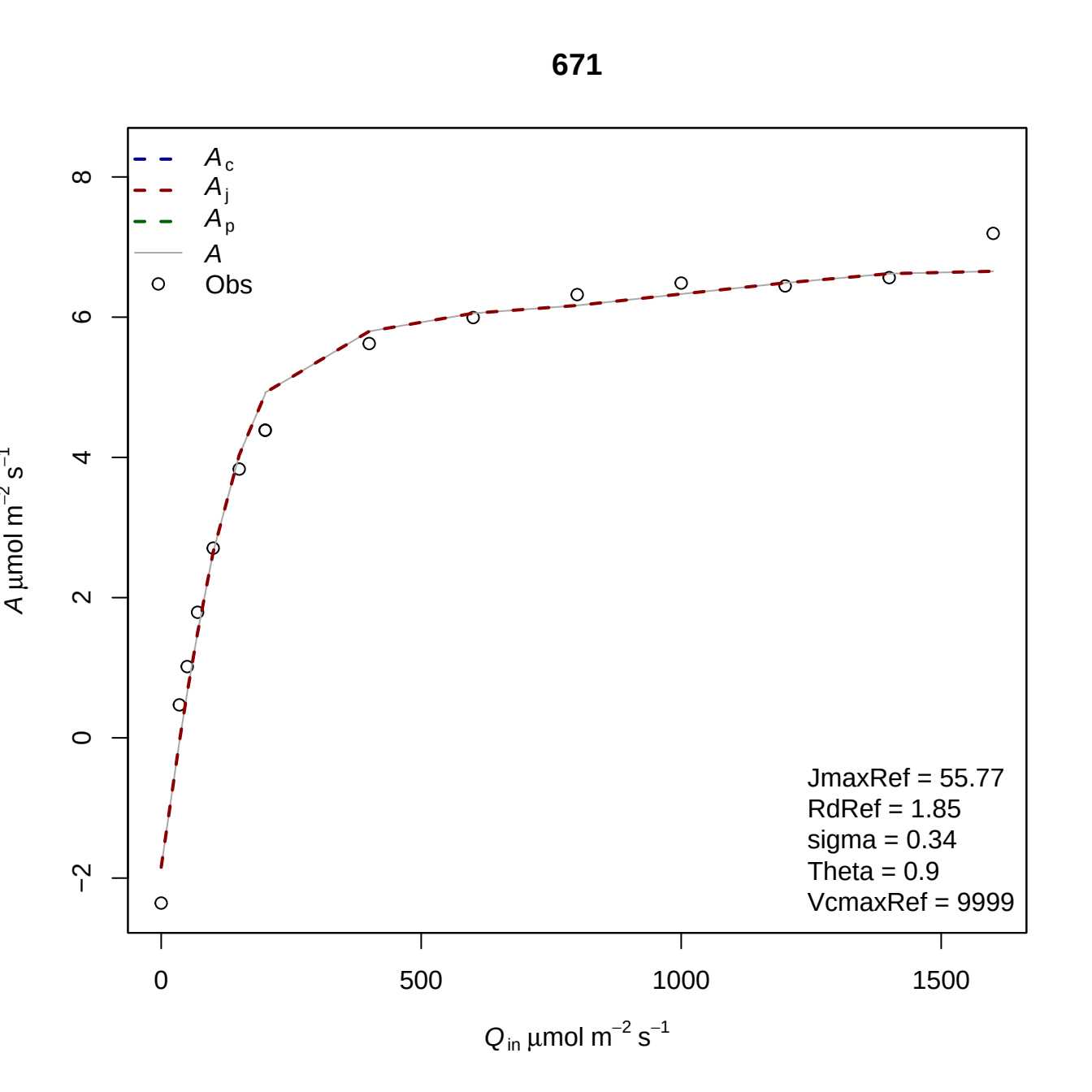

666
A_c
A_j
A_p
A
Obs
JmaxRef = 38.65
RdRef = 1.05
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

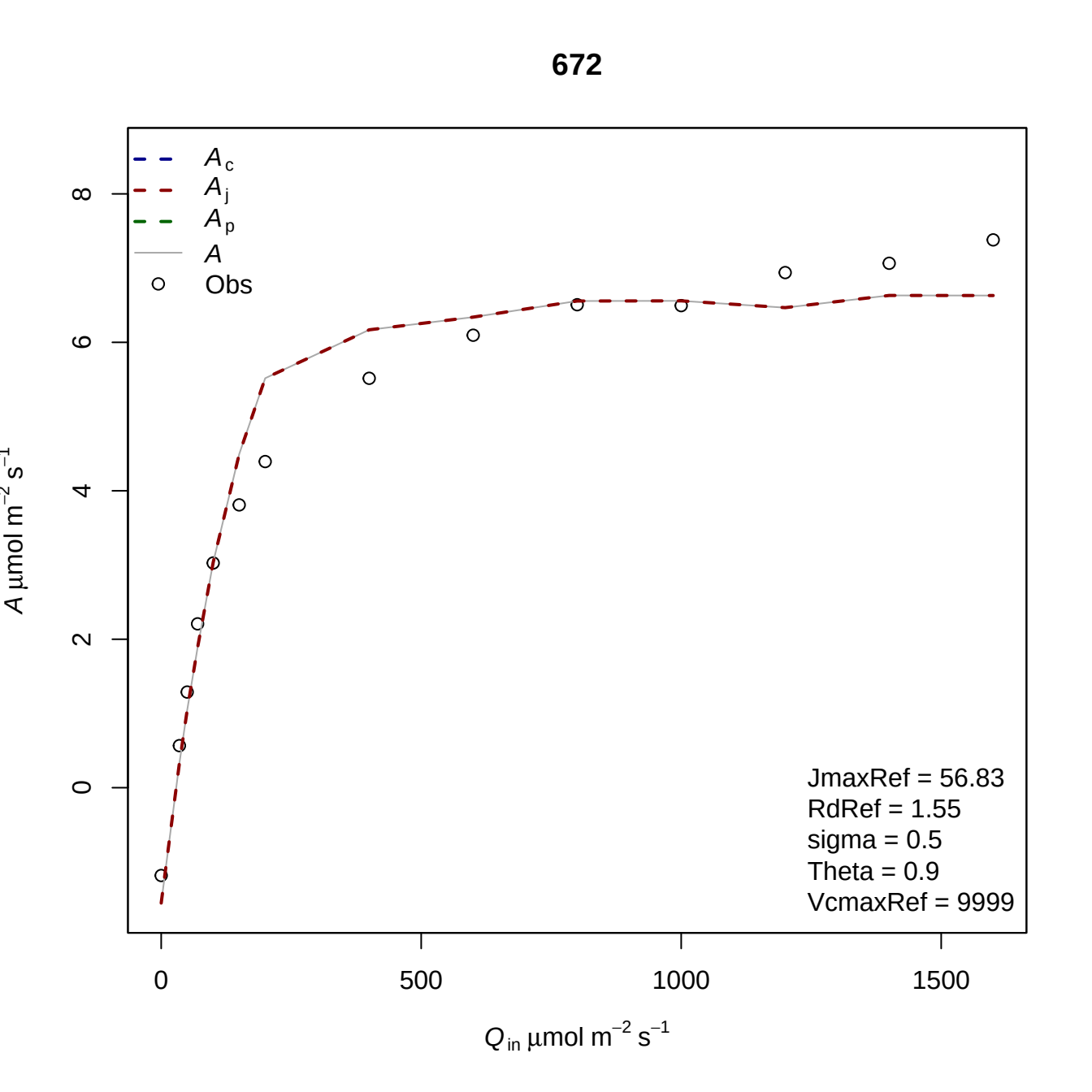

667
A_c
A_j
A_p
A
Obs
JmaxRef = 49.7
RdRef = 1.11
sigma = 0.32
Theta = 0.9
VcmaxRef = 38.94
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

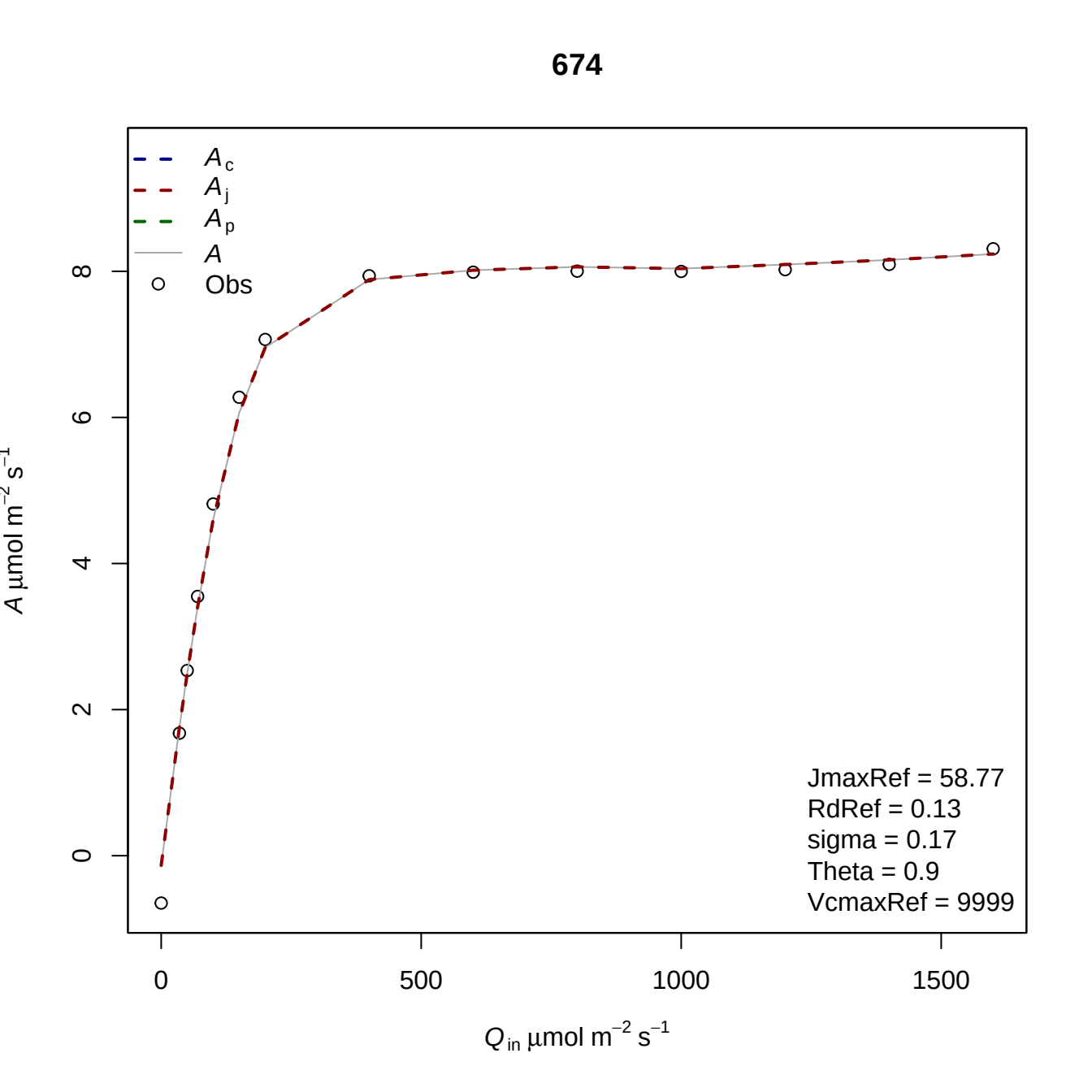

668
A_c
A_j
A_p
A
Obs
JmaxRef = 53.54
RdRef = 0.85
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

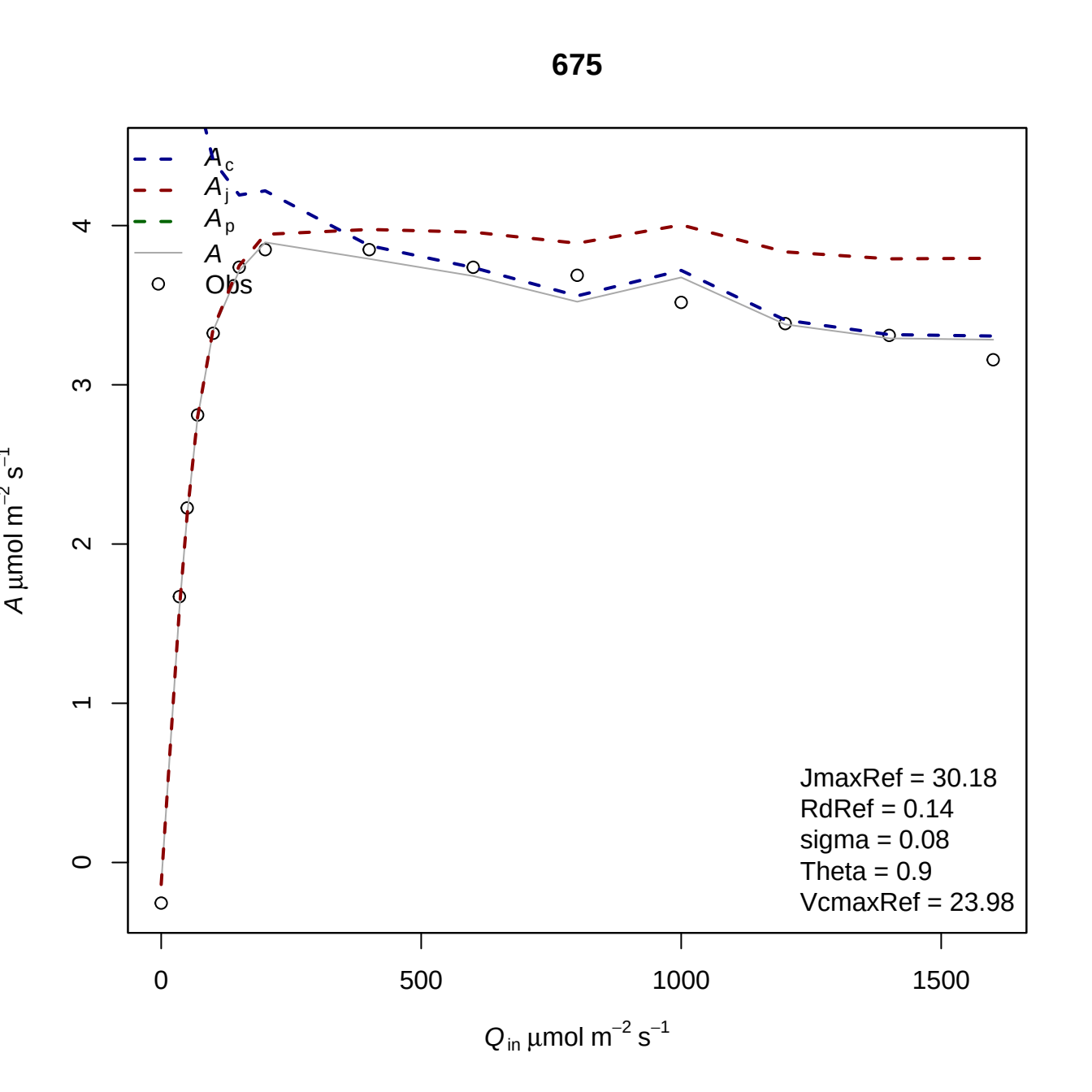

669
A_c
A_j
A_p
A
Obs
JmaxRef = 64.15
RdRef = 1.43
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

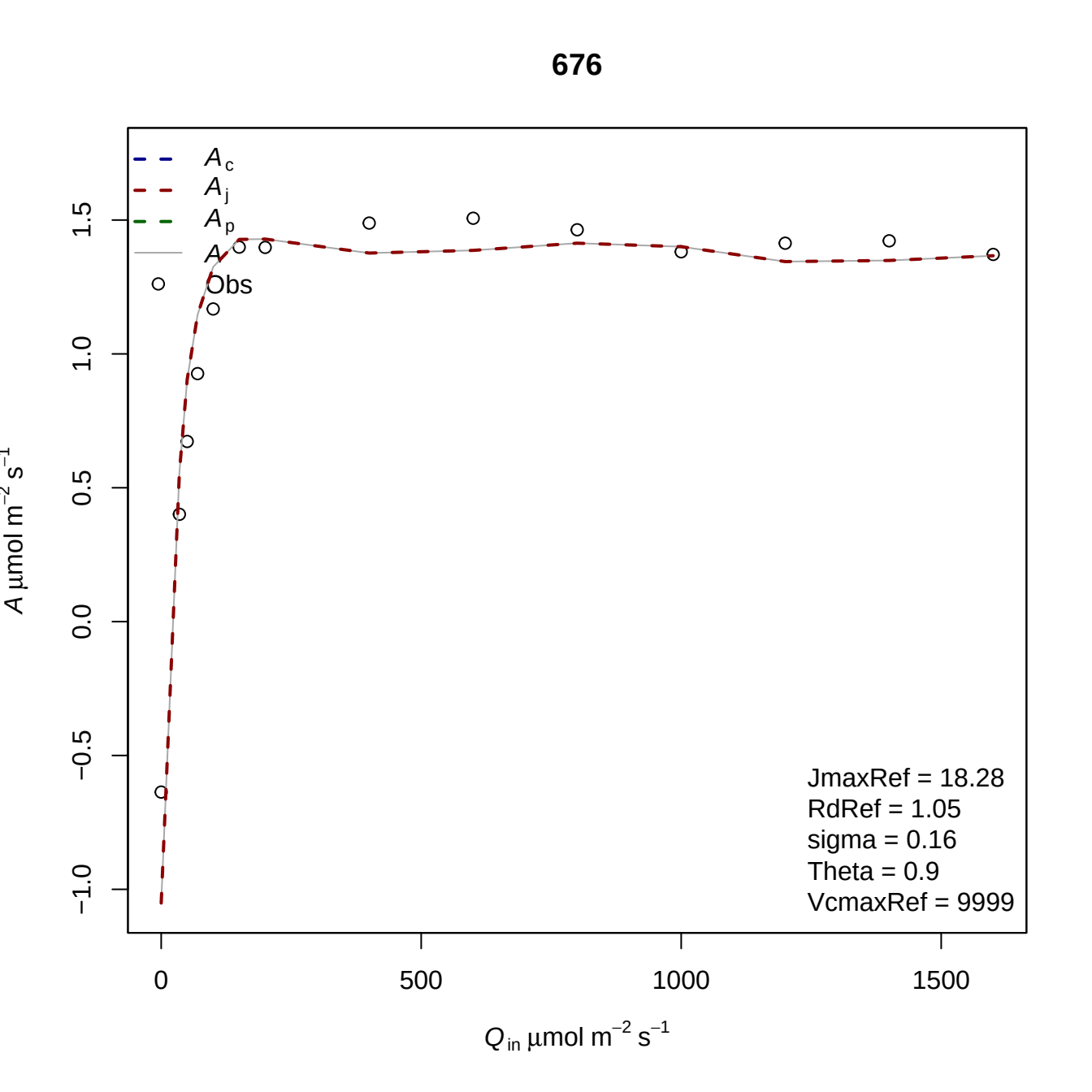

670
A_c
A_j
A_p
A
Obs
JmaxRef = 104.87
RdRef = 1.96
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

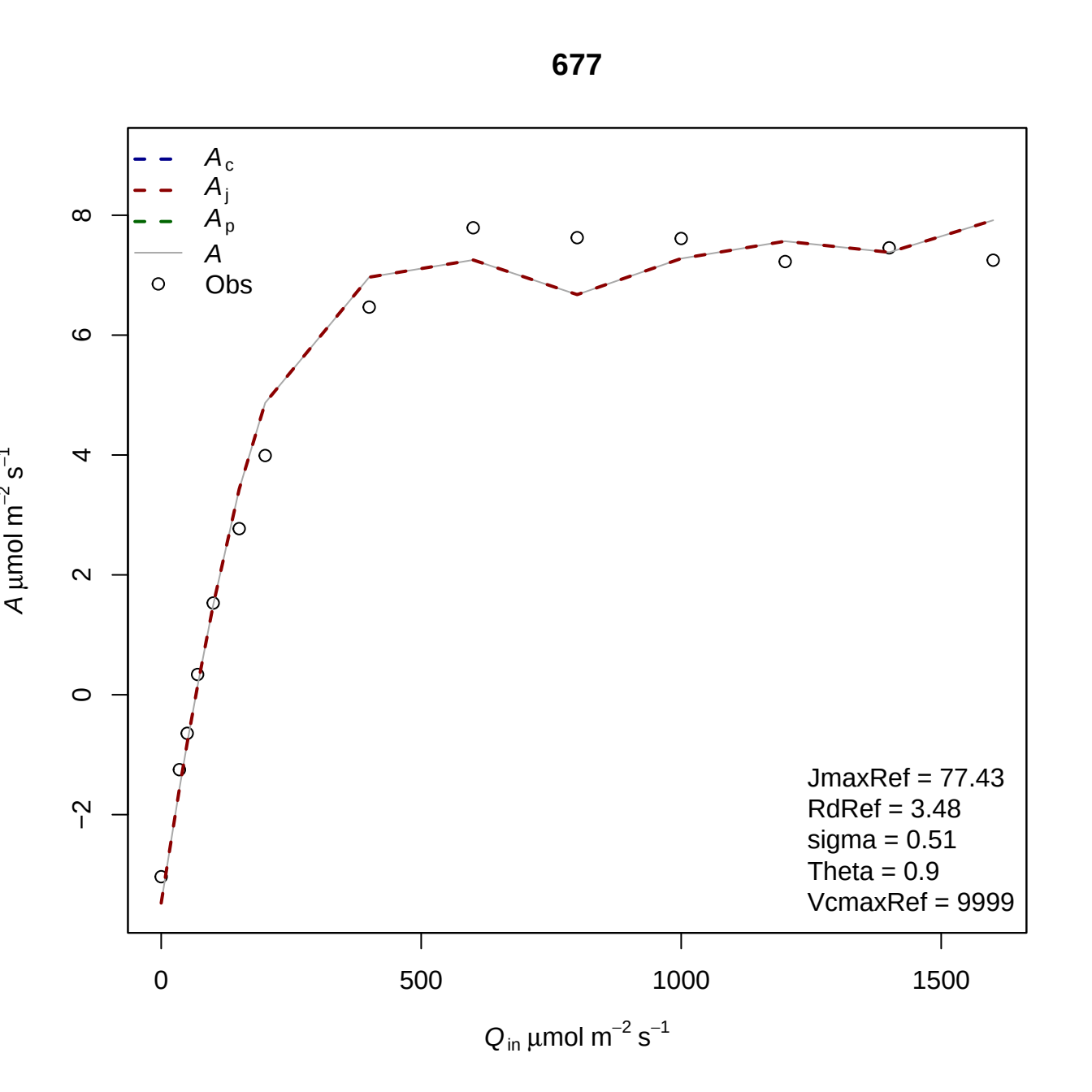

671
A_c
A_j
A_p
A
Obs
JmaxRef = 55.77
RdRef = 1.85
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

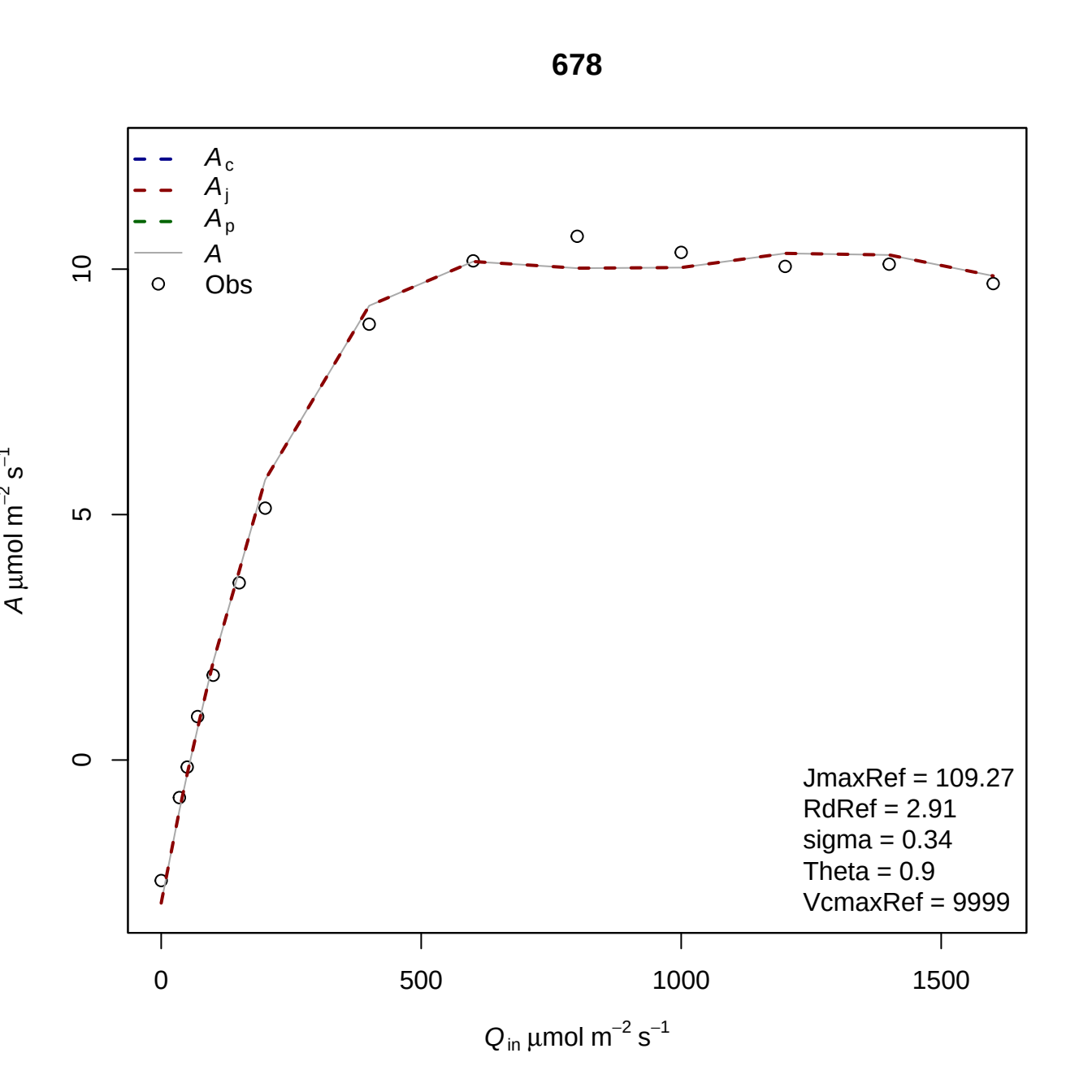

672
A_c
A_j
A_p
A
Obs
JmaxRef = 56.83
RdRef = 1.55
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

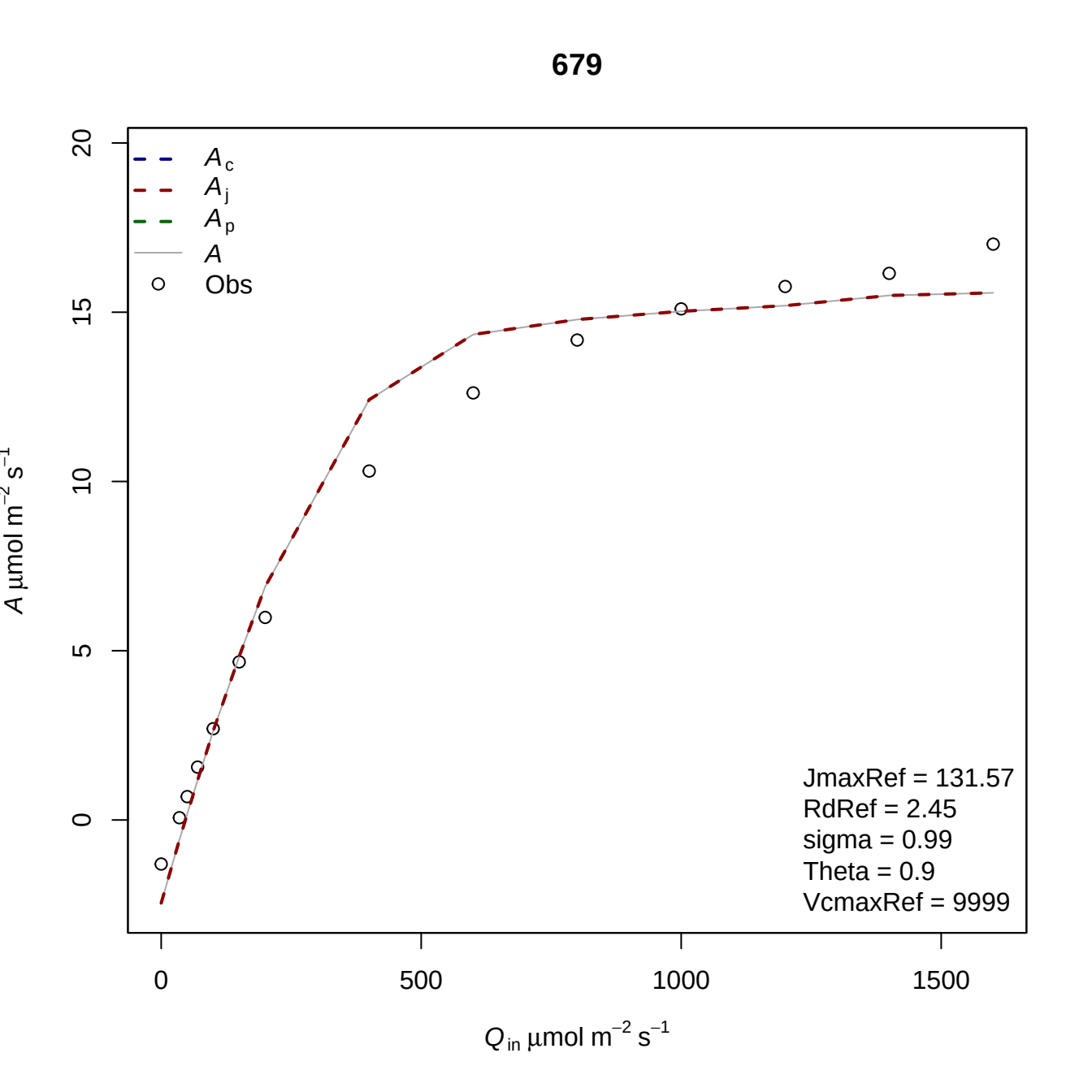

674
A_c
A_j
A_p
A
Obs
JmaxRef = 58.77
RdRef = 0.13
sigma = 0.17
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

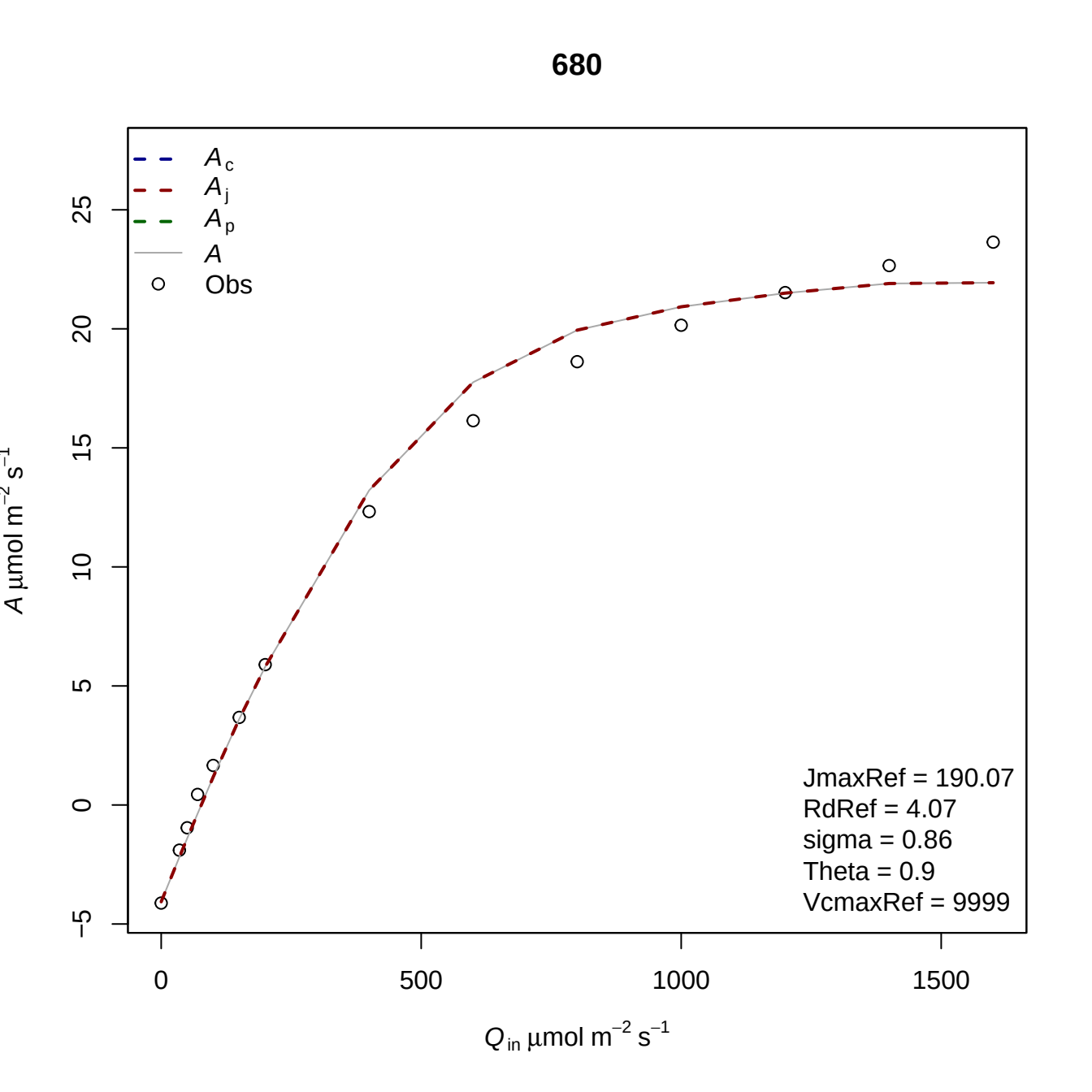

675
A_c
A_j
A_p
A
Obs
JmaxRef = 30.18
RdRef = 0.14
sigma = 0.08
Theta = 0.9
VcmaxRef = 23.98
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

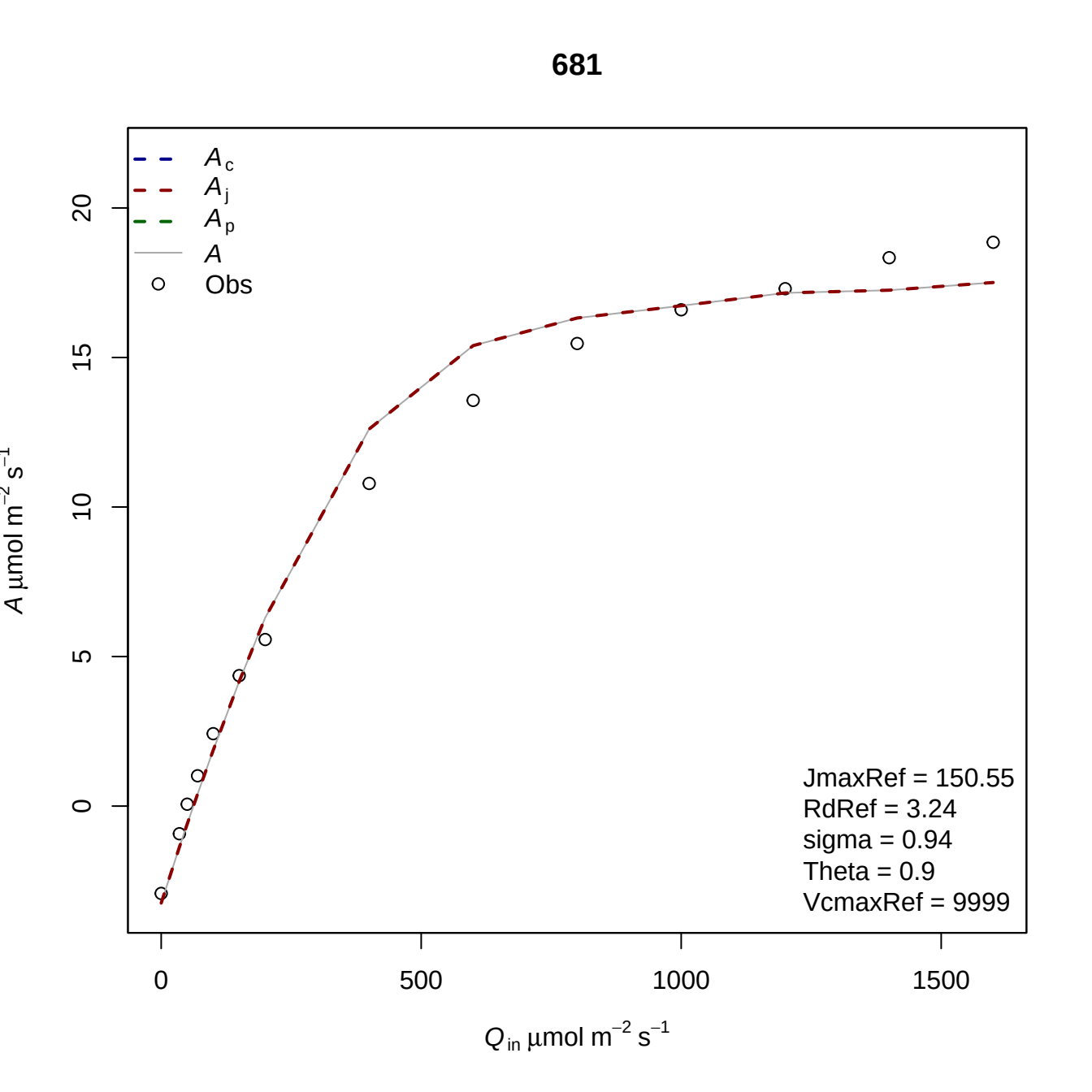

676
A_c
A_j
A_p
A
Obs
JmaxRef = 18.28
RdRef = 1.05
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

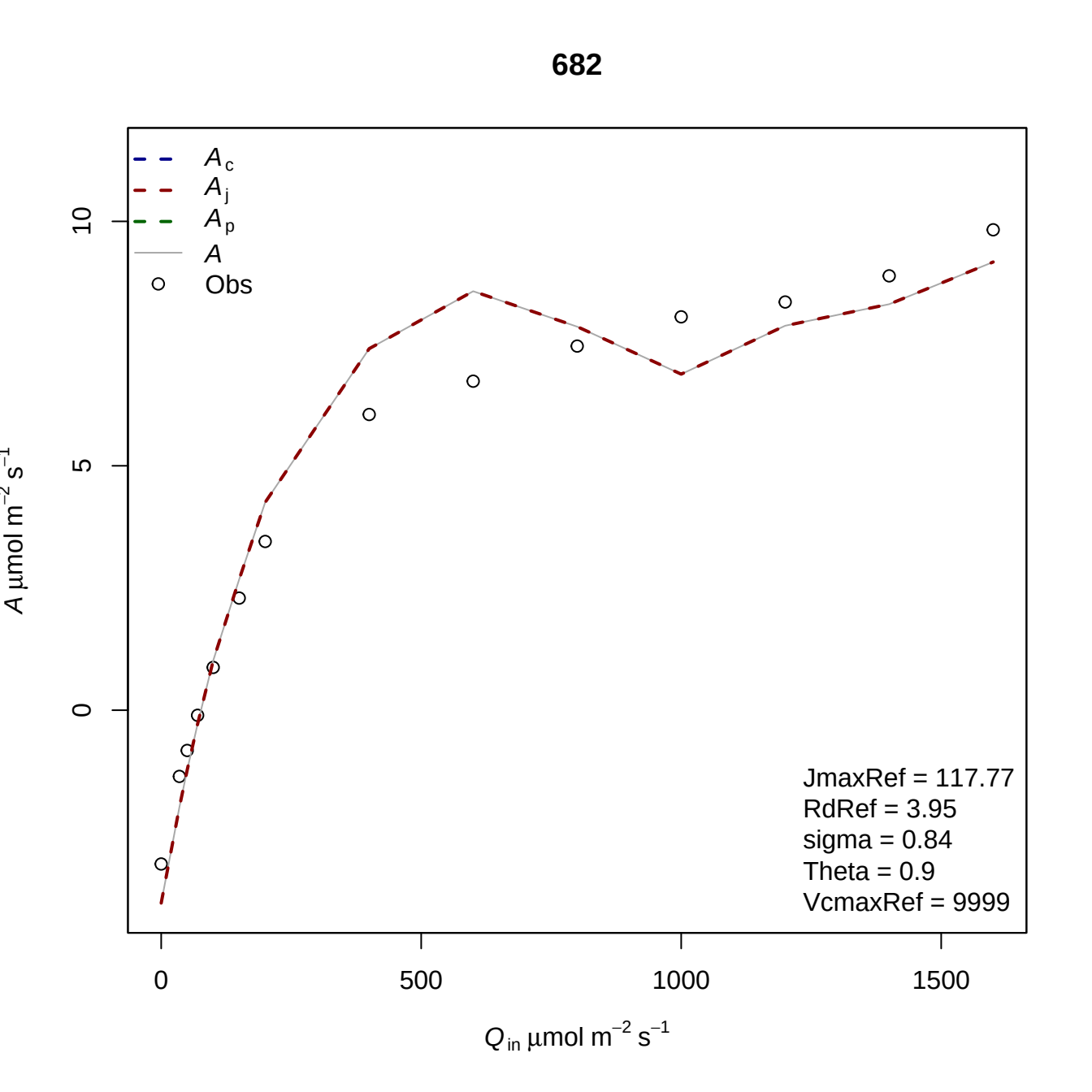

677
A_c
A_j
A_p
A
Obs
JmaxRef = 77.43
RdRef = 3.48
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

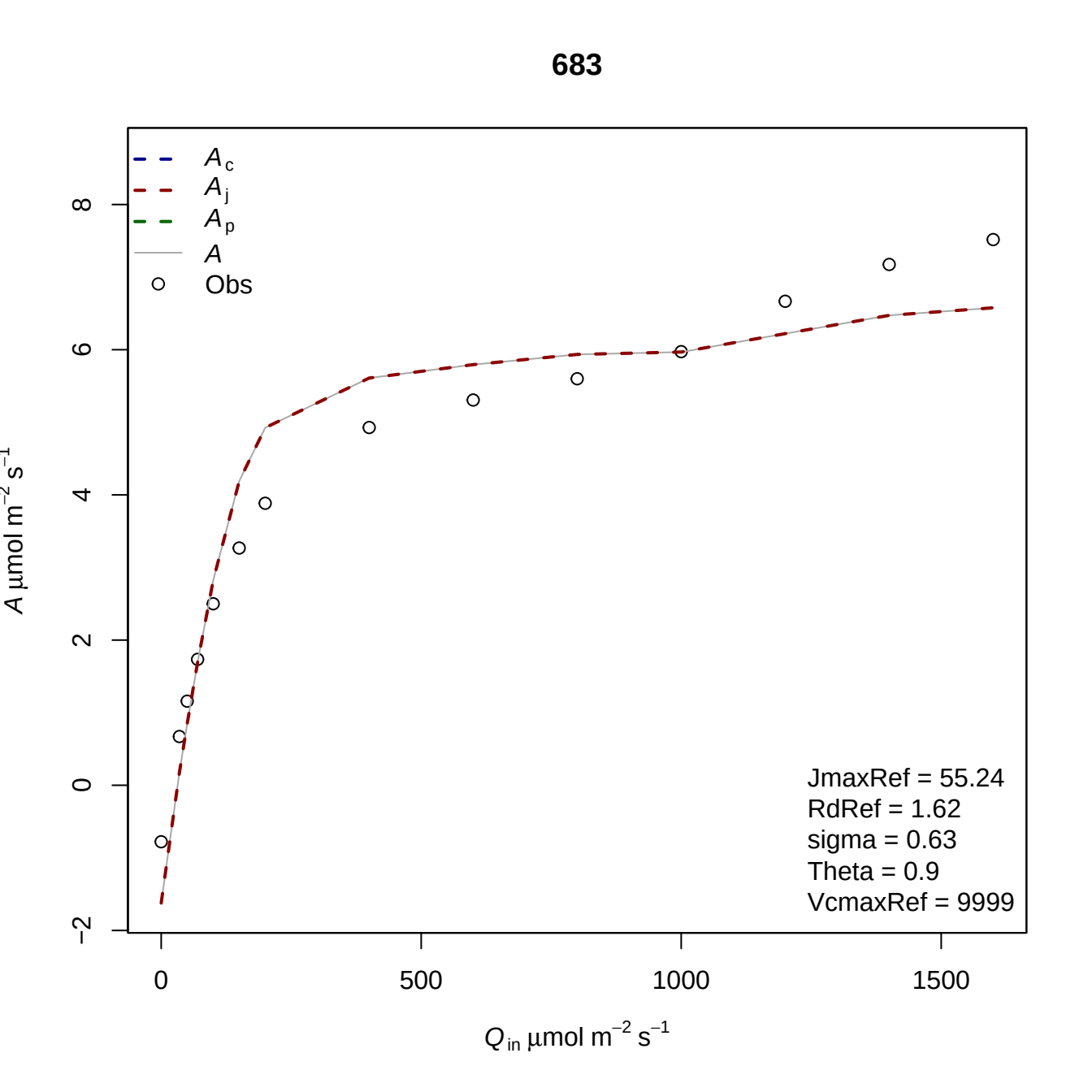

678
A_c
A_j
A_p
A
Obs
JmaxRef = 109.27
RdRef = 2.91
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

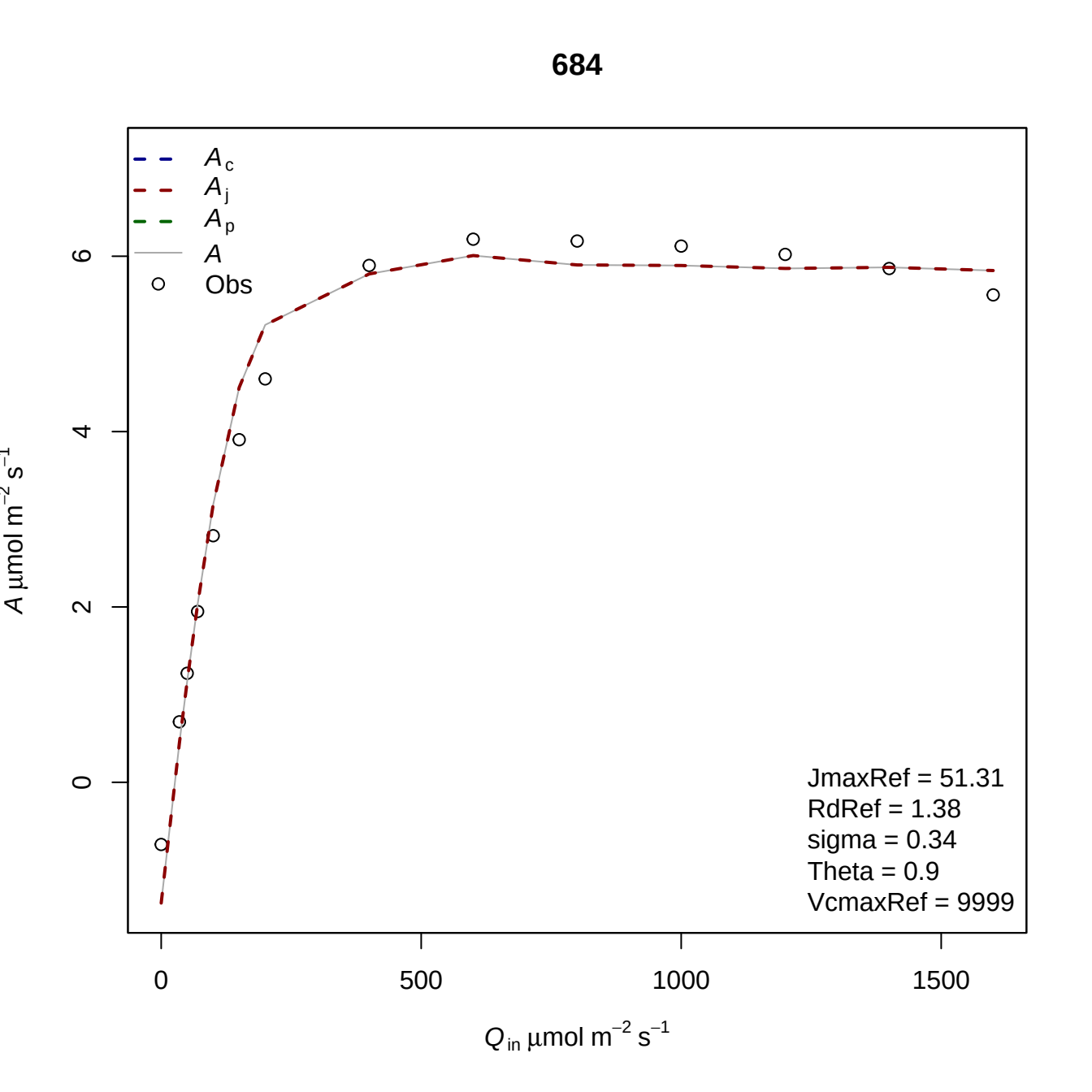

679
A_c
A_j
A_p
A
Obs
JmaxRef = 131.57
RdRef = 2.45
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

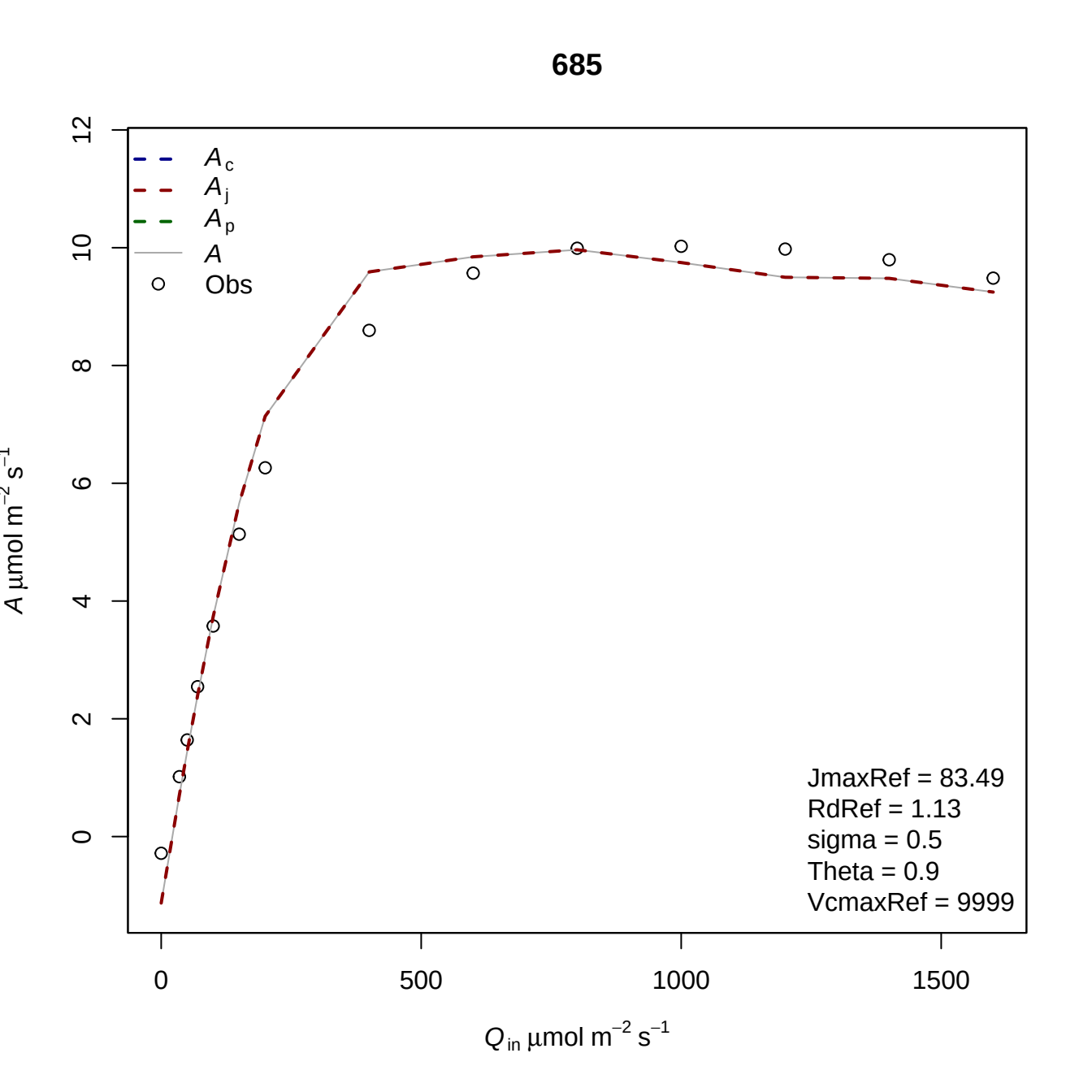

680
A_c
A_j
A_p
A
Obs
JmaxRef = 190.07
RdRef = 4.07
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

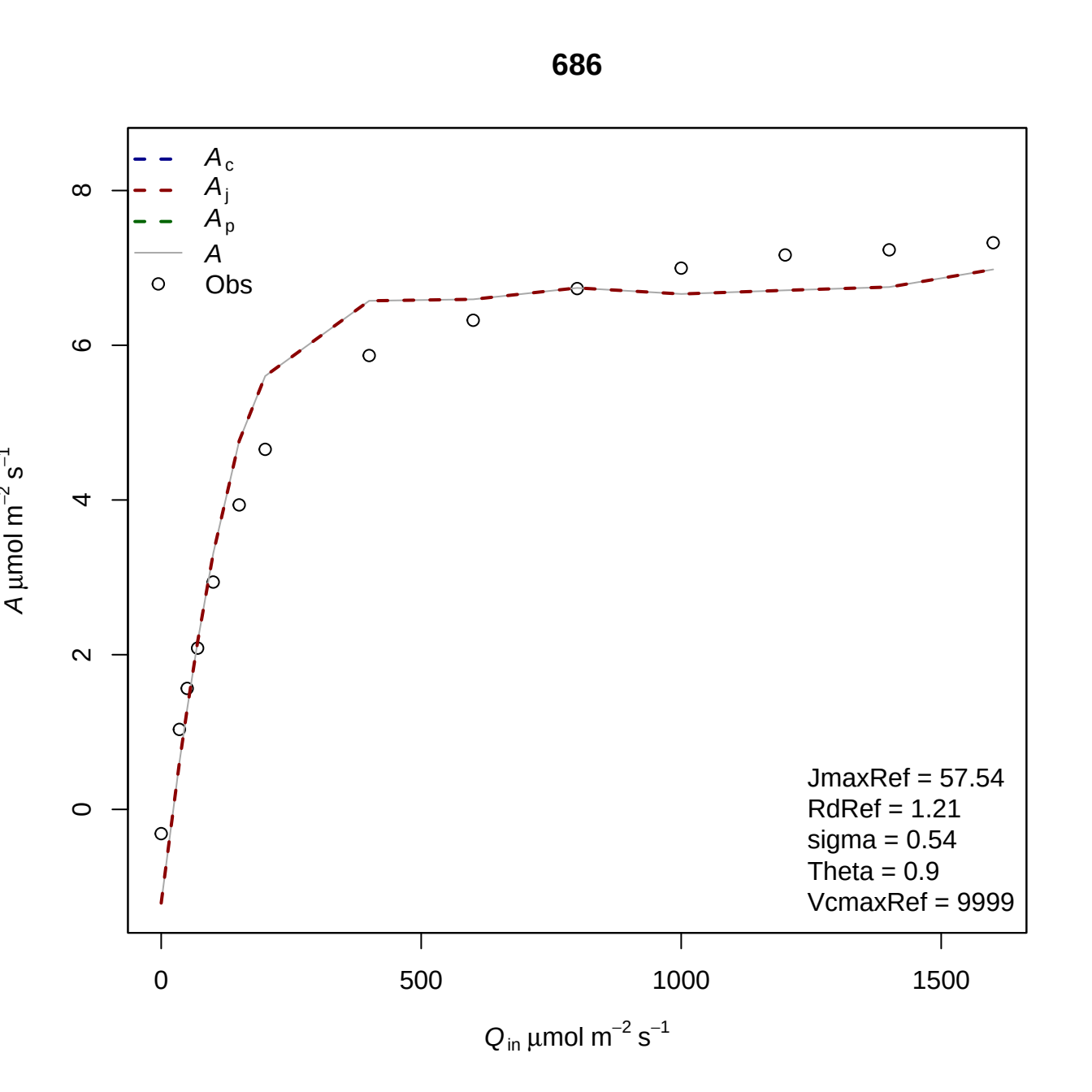

681
A_c
A_j
A_p
A
Obs
JmaxRef = 150.55
RdRef = 3.24
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

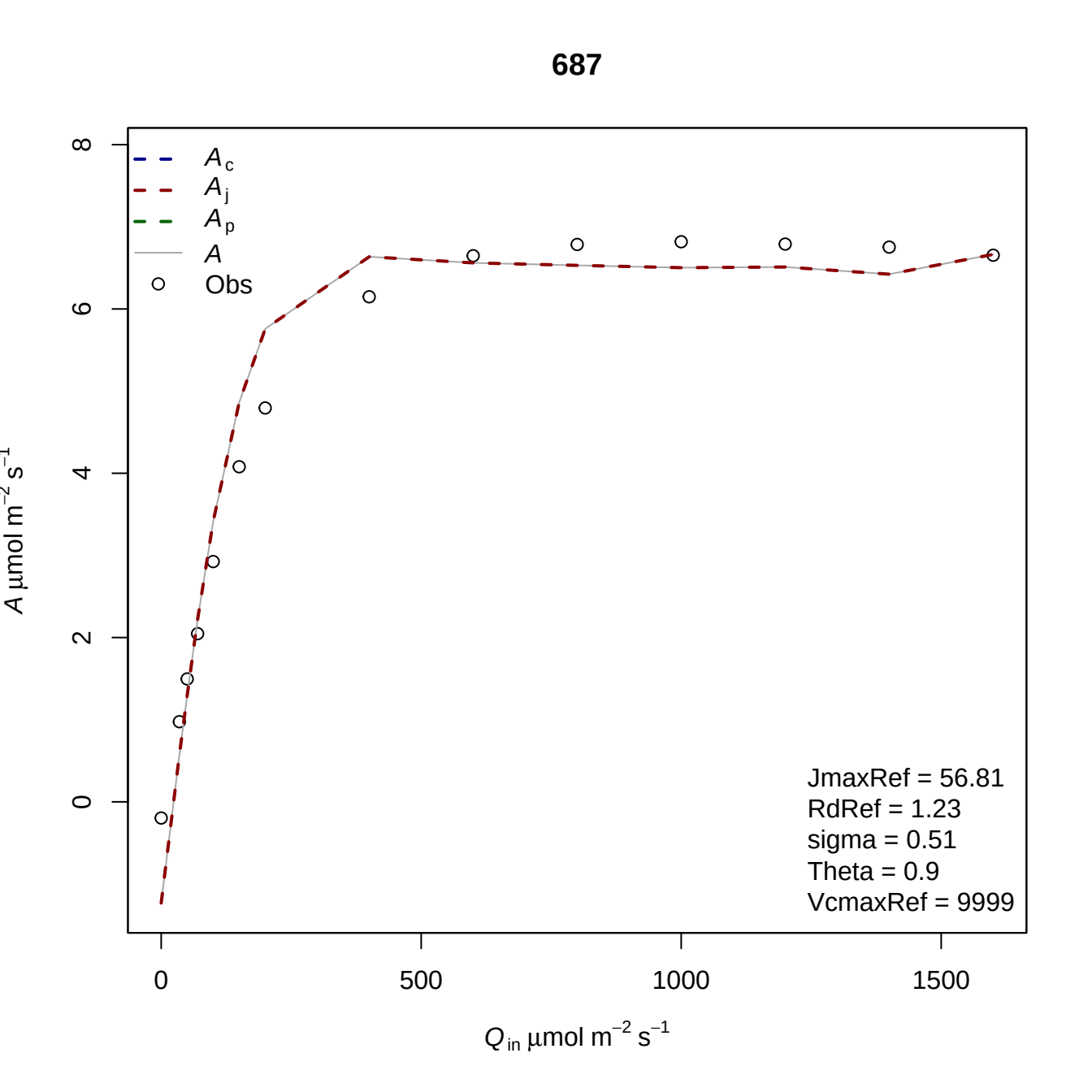

682
A_c
A_j
A_p
A
Obs
JmaxRef = 117.77
RdRef = 3.95
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

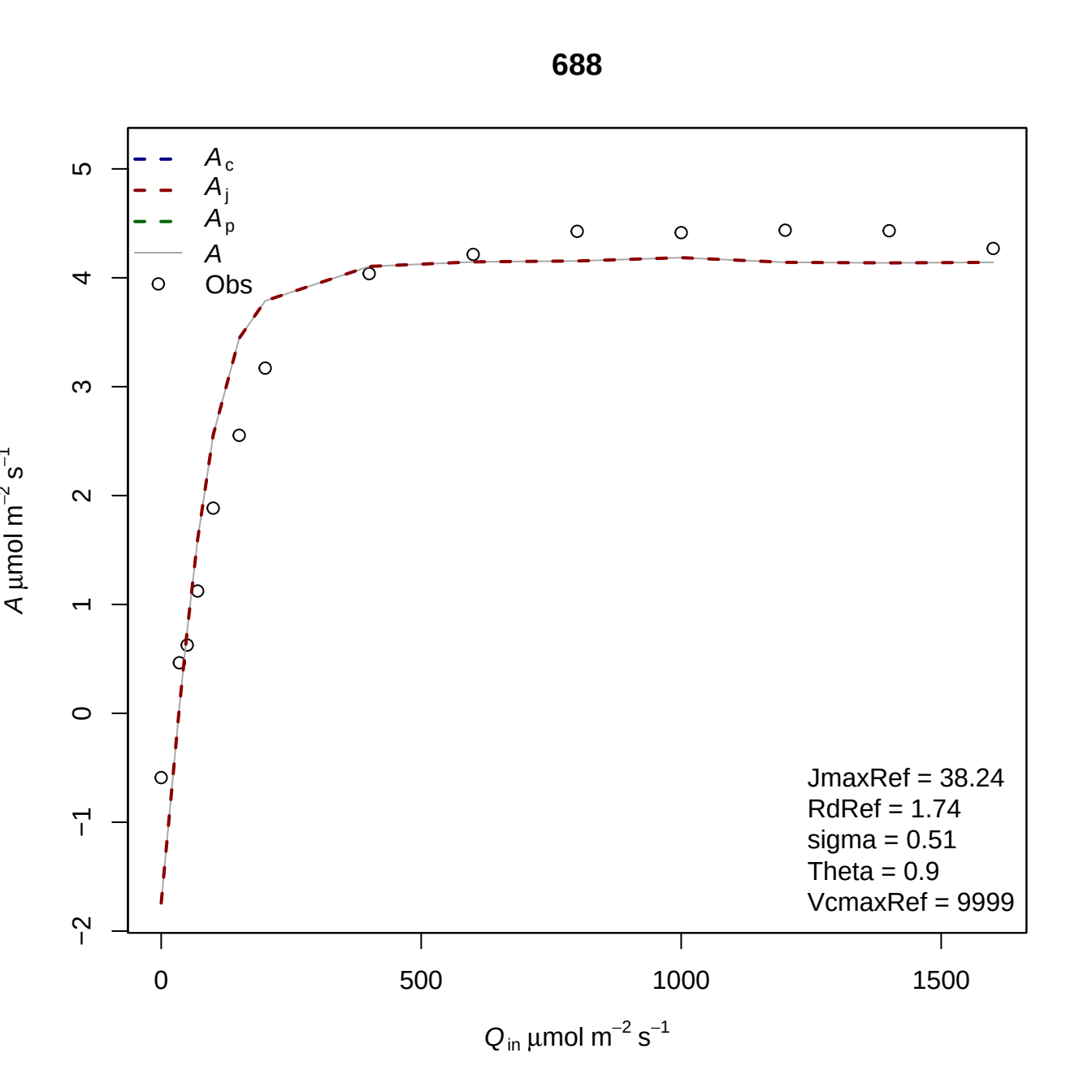

683
A_c
A_j
A_p
A
Obs
JmaxRef = 55.24
RdRef = 1.62
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

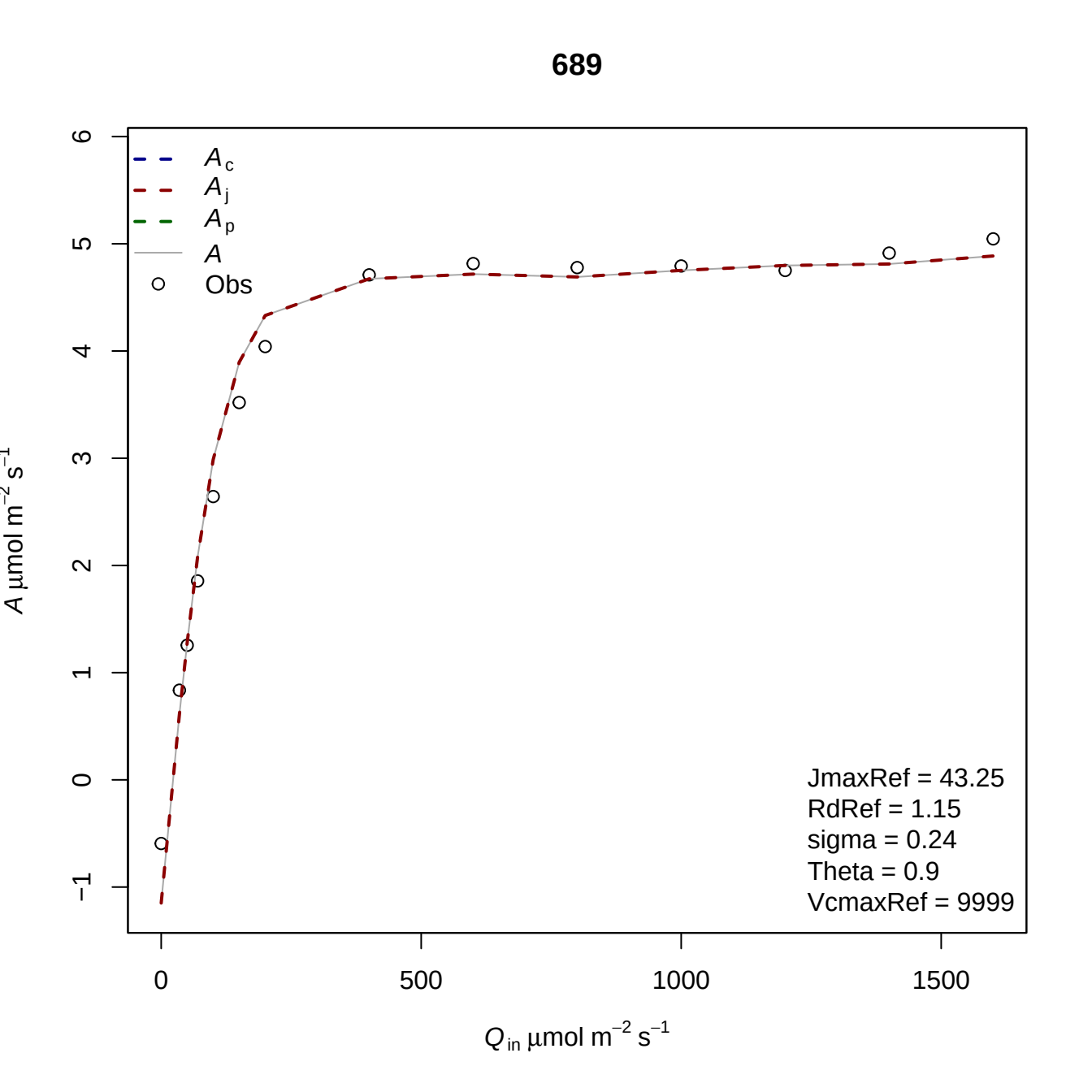

684
A_c
A_j
A_p
A
Obs
JmaxRef = 51.31
RdRef = 1.38
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

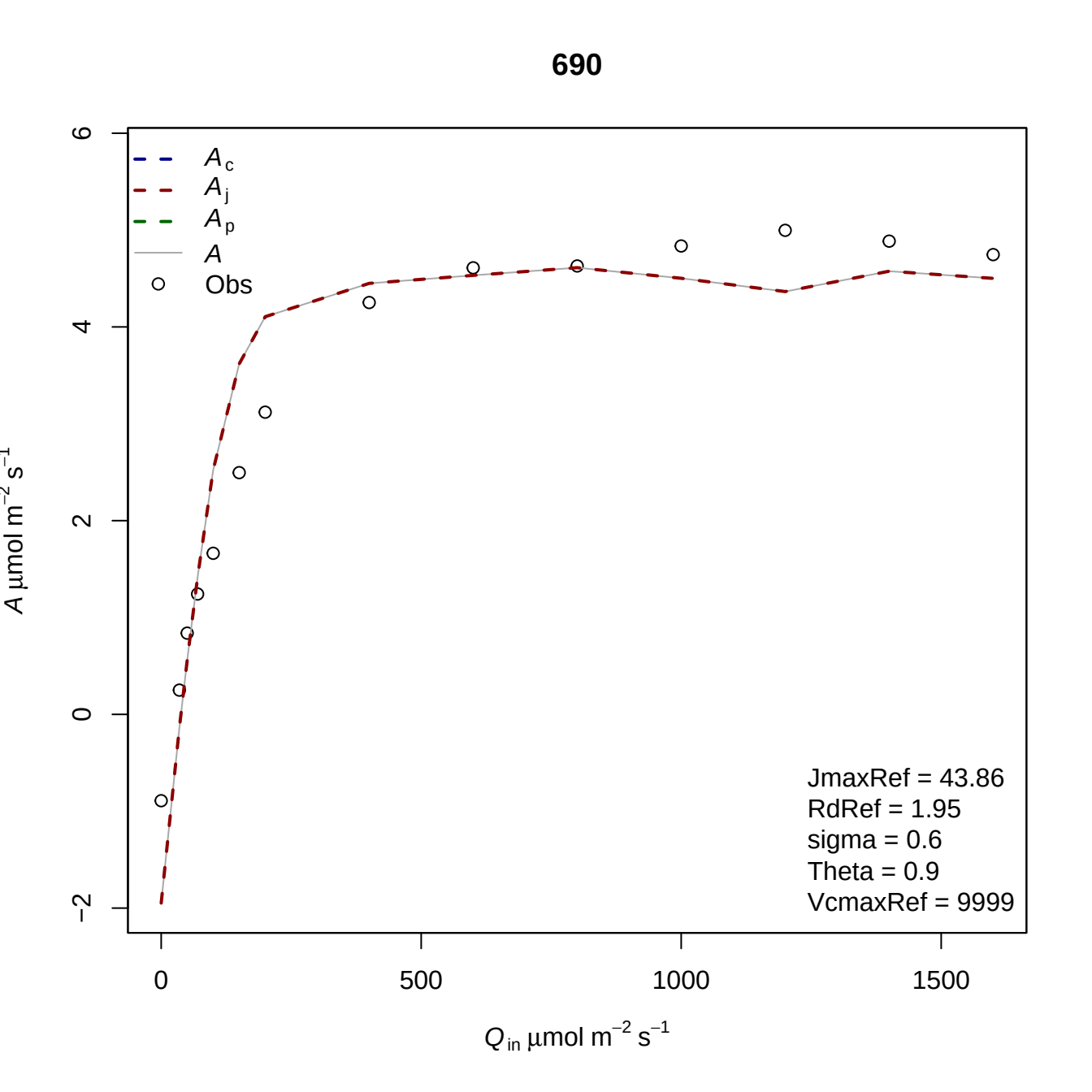

685
A_c
A_j
A_p
A
Obs
JmaxRef = 83.49
RdRef = 1.13
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

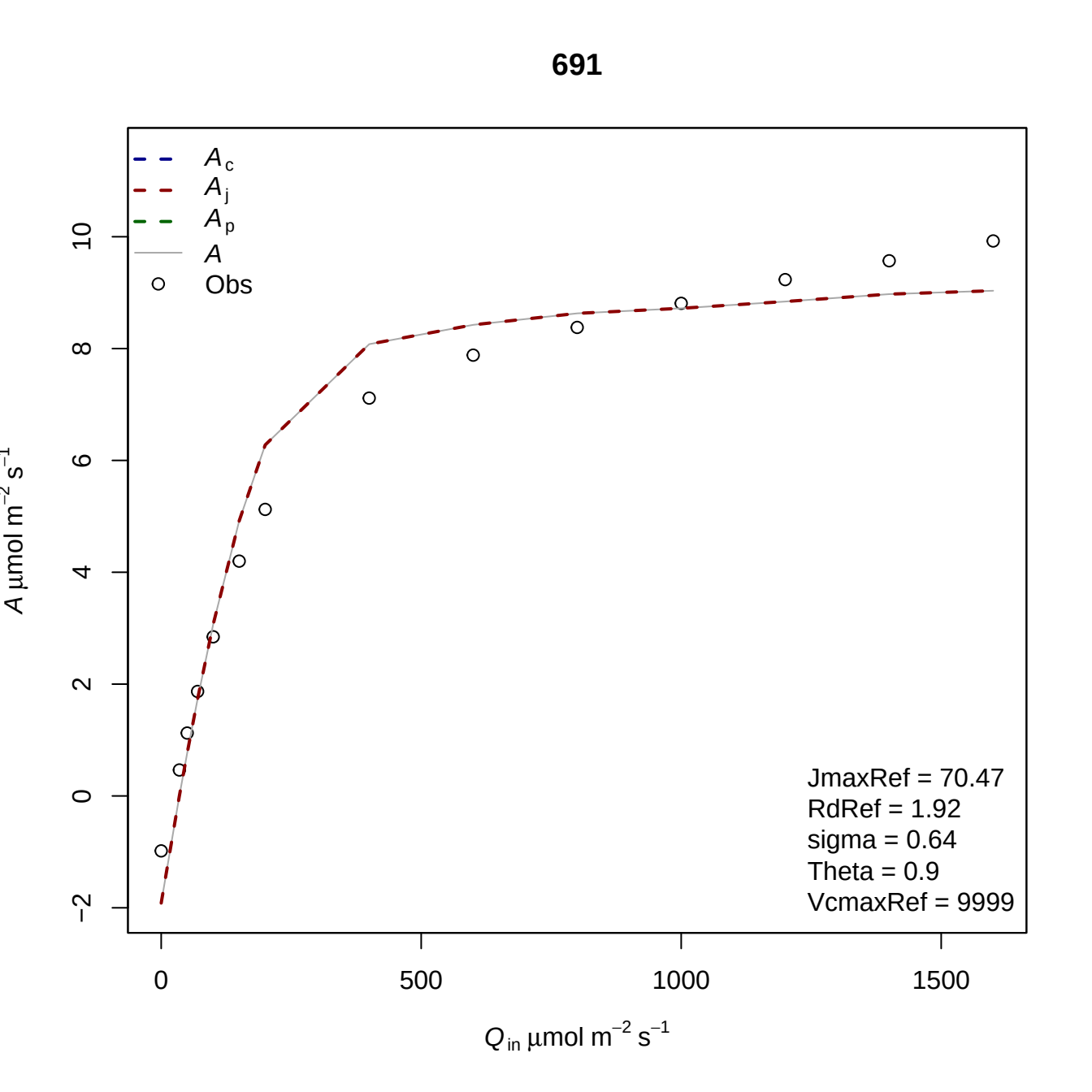

686
A_c
A_j
A_p
A
Obs
JmaxRef = 57.54
RdRef = 1.21
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

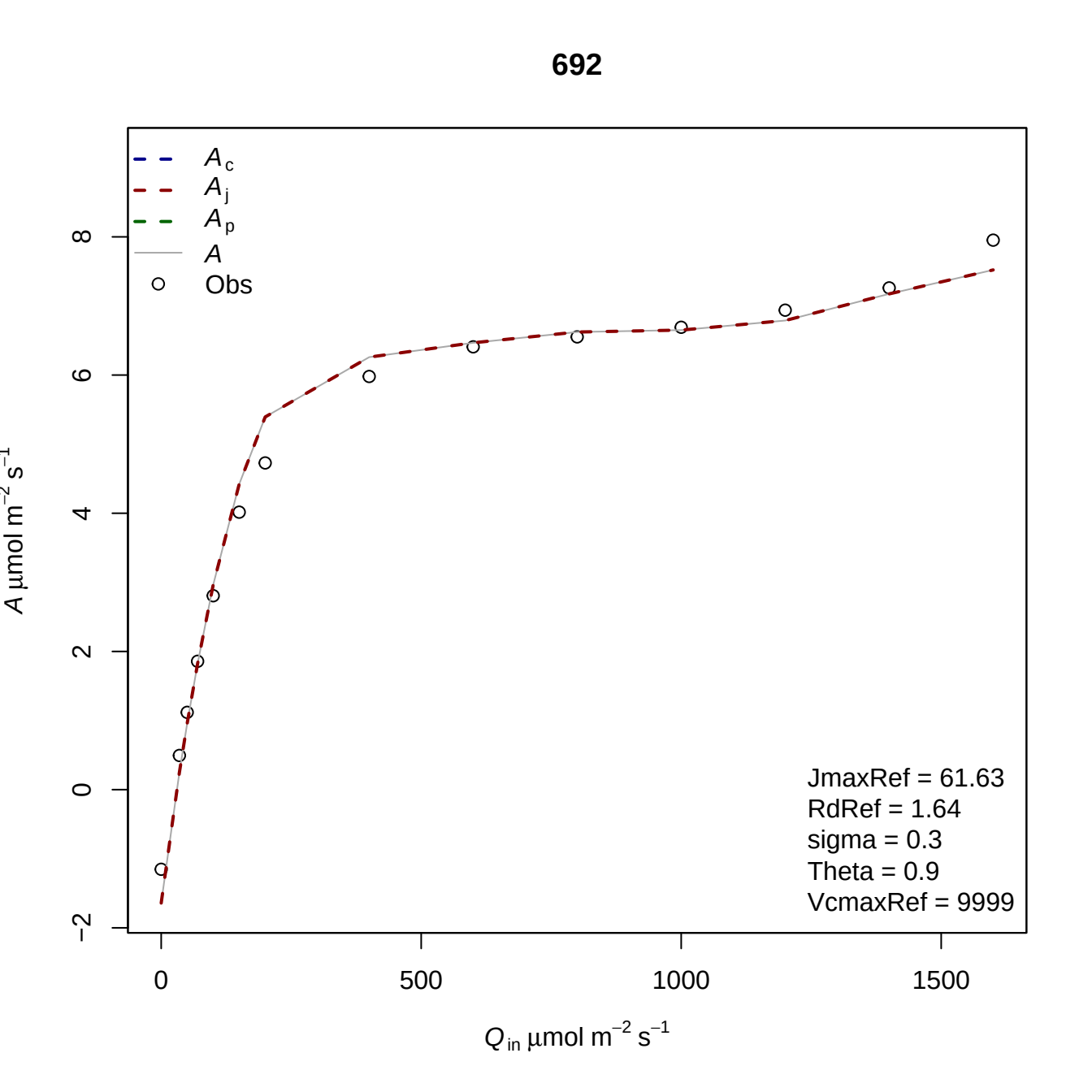

687
A_c
A_j
A_p
A
Obs
JmaxRef = 56.81
RdRef = 1.23
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

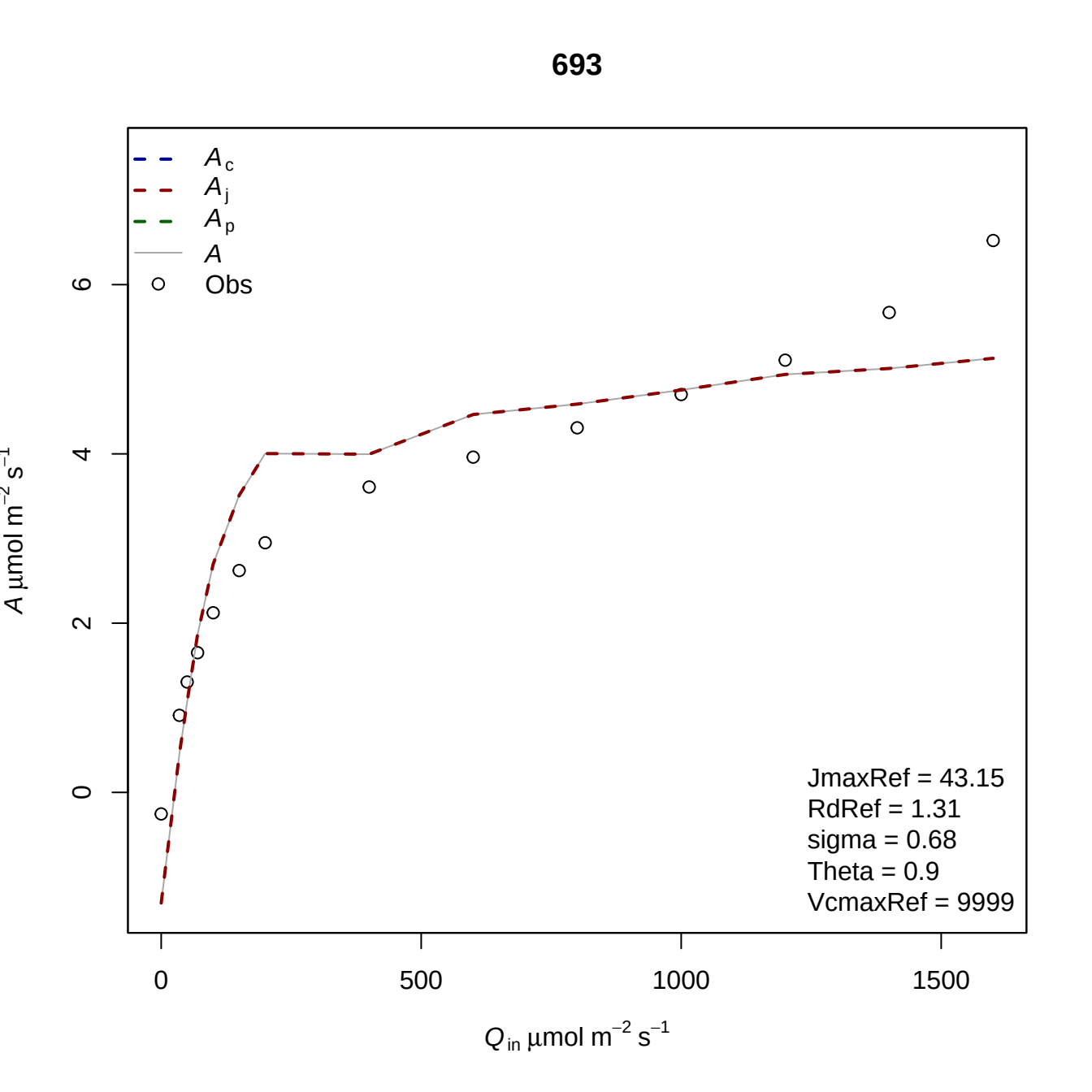

688
A_c
A_j
A_p
A
Obs
JmaxRef = 38.24
RdRef = 1.74
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
500
1000
1500

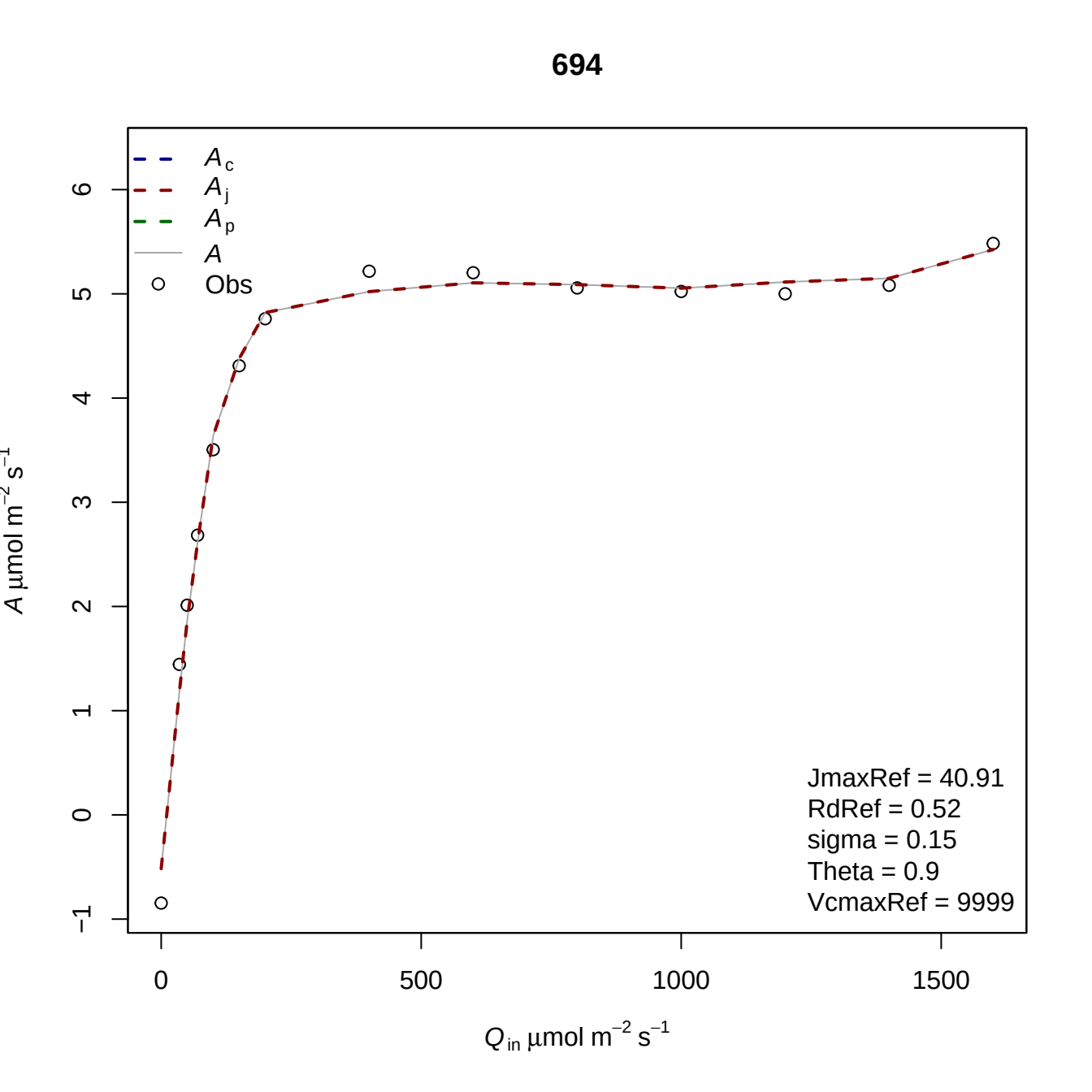

689
A_c
A_j
A_p
A
Obs
JmaxRef = 43.25
RdRef = 1.15
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

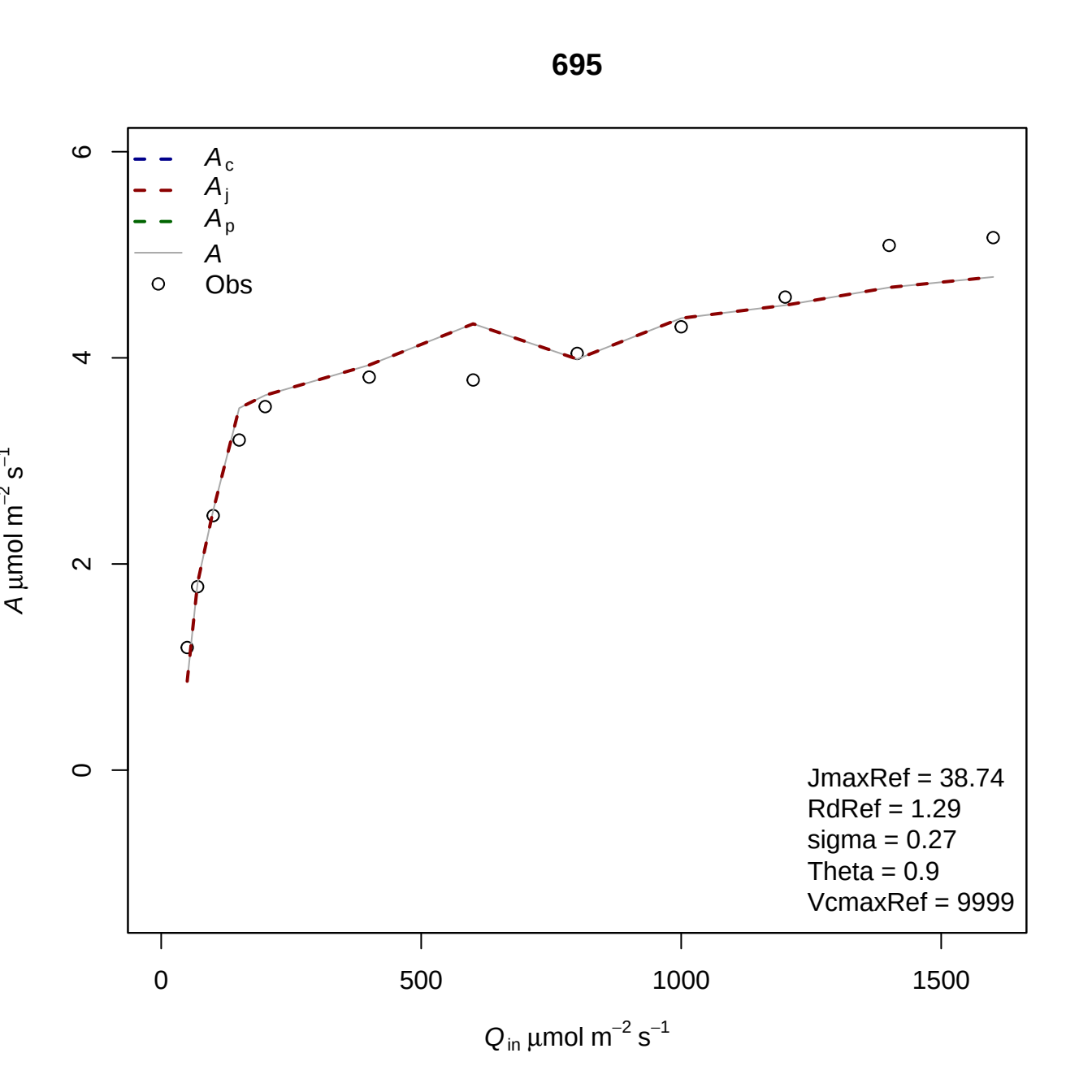

690
A_c
A_j
A_p
A
Obs
JmaxRef = 43.86
RdRef = 1.95
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

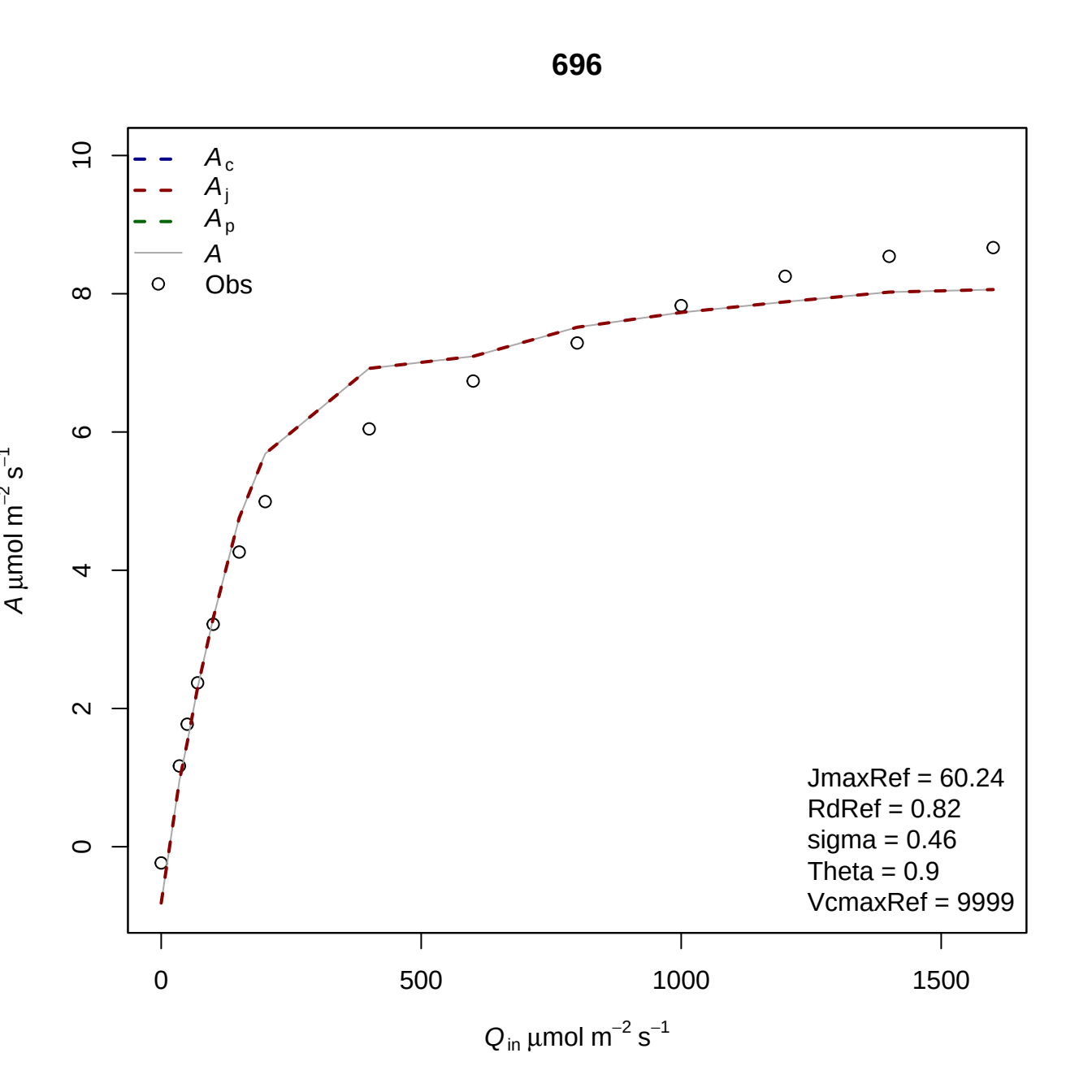

691
A_c
A_j
A_p
A
Obs
JmaxRef = 70.47
RdRef = 1.92
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

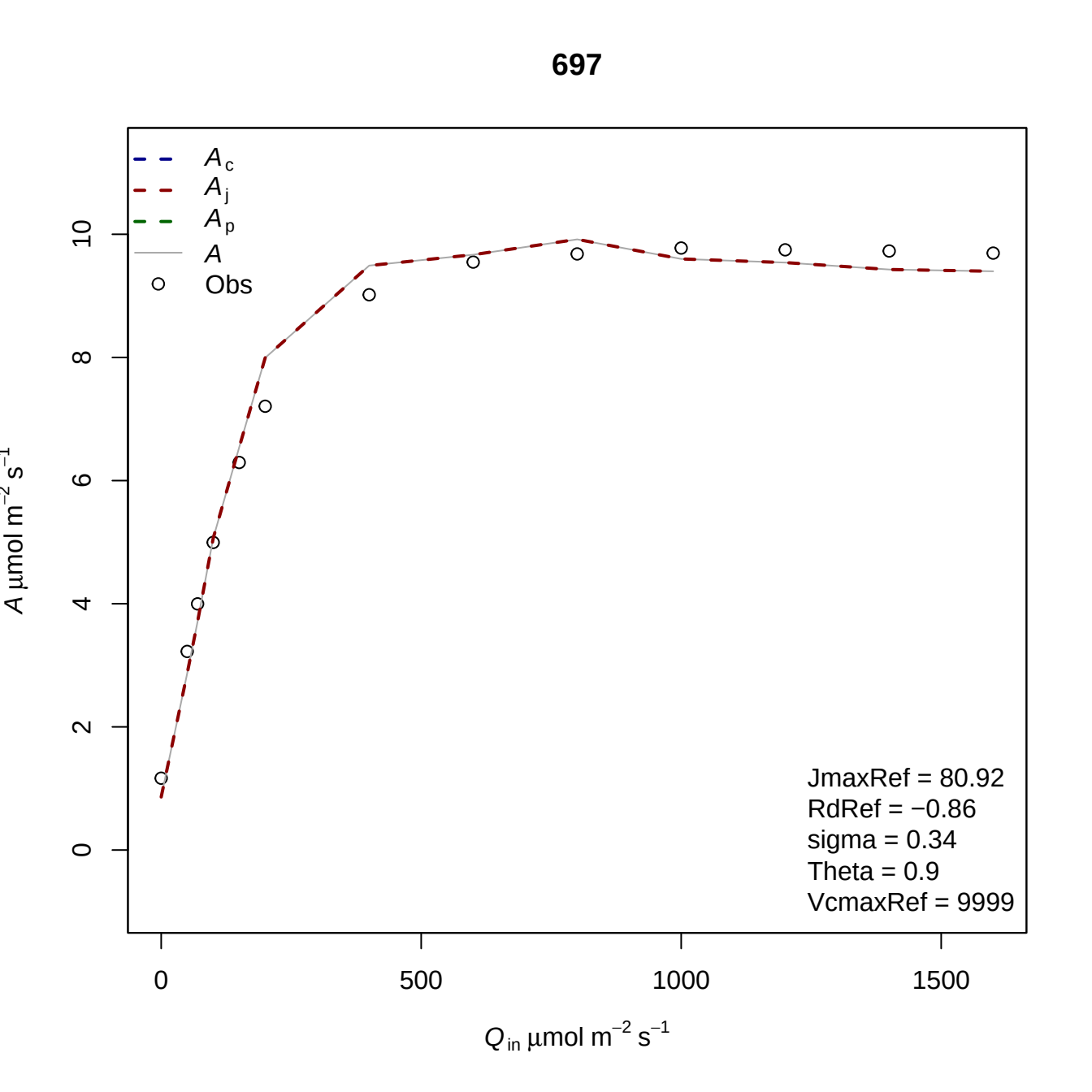

692
A_c
A_j
A_p
A
Obs
JmaxRef = 61.63
RdRef = 1.64
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

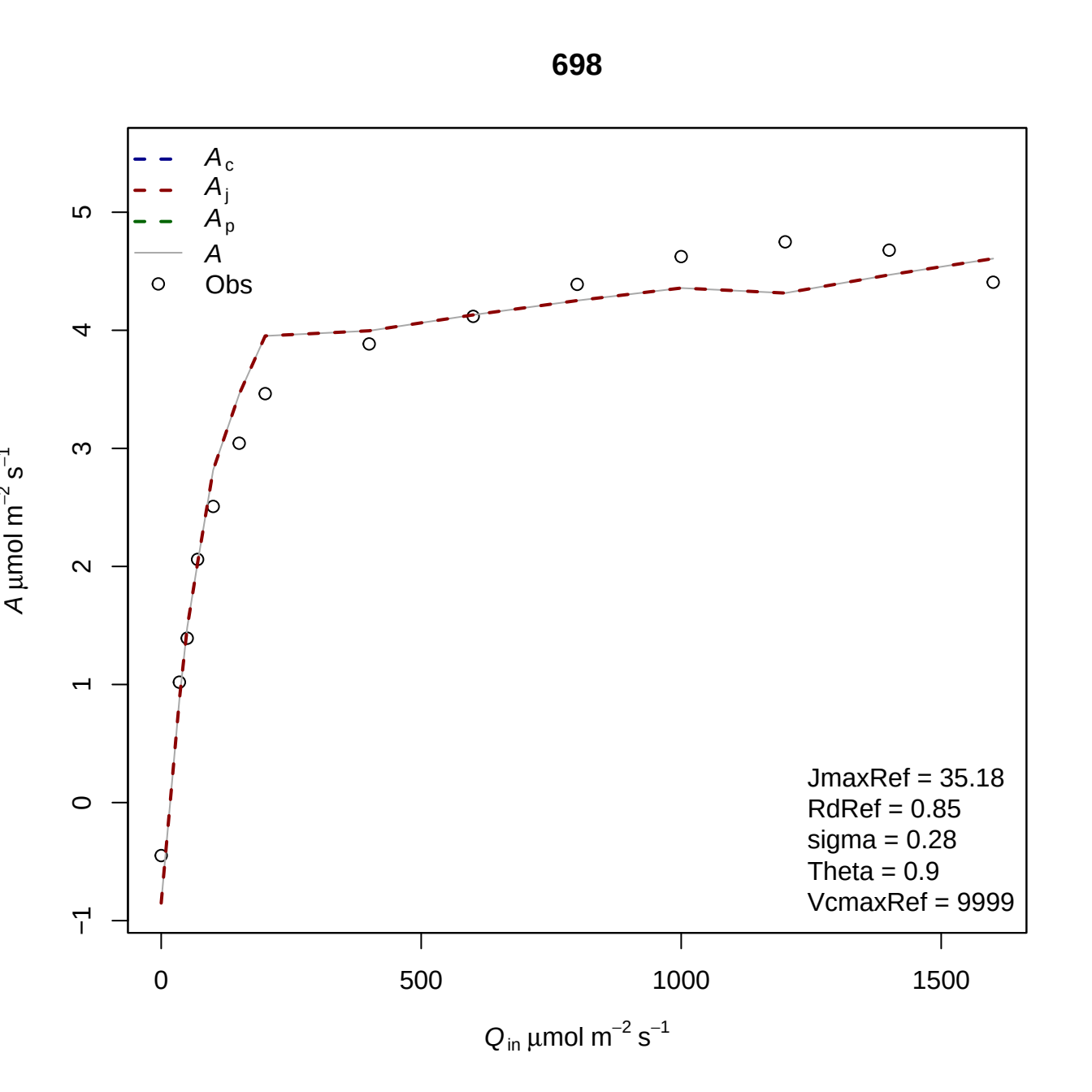

693
A_c
A_j
A_p
A
Obs
JmaxRef = 43.15
RdRef = 1.31
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

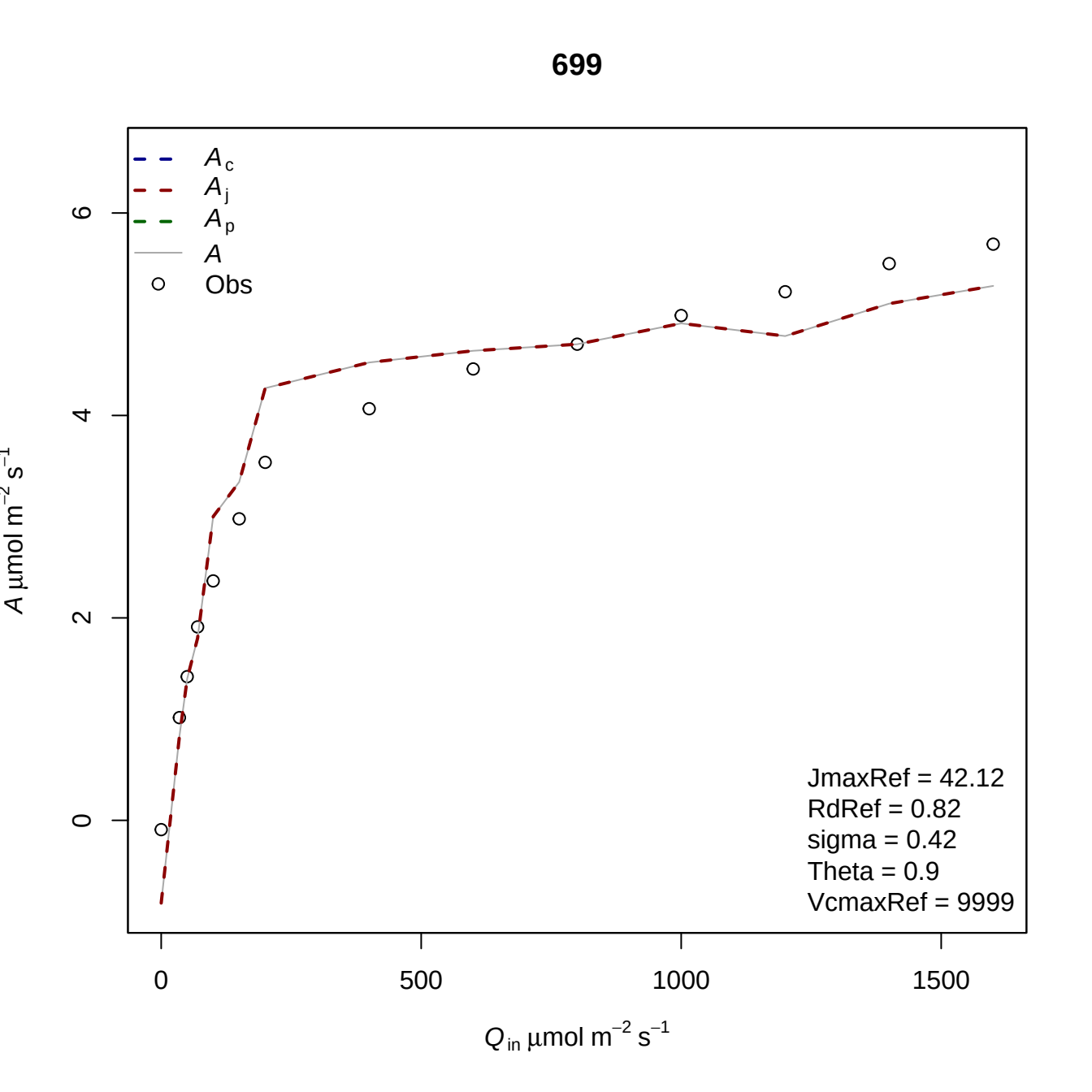

694
A_c
A_j
A_p
A
Obs
JmaxRef = 40.91
RdRef = 0.52
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

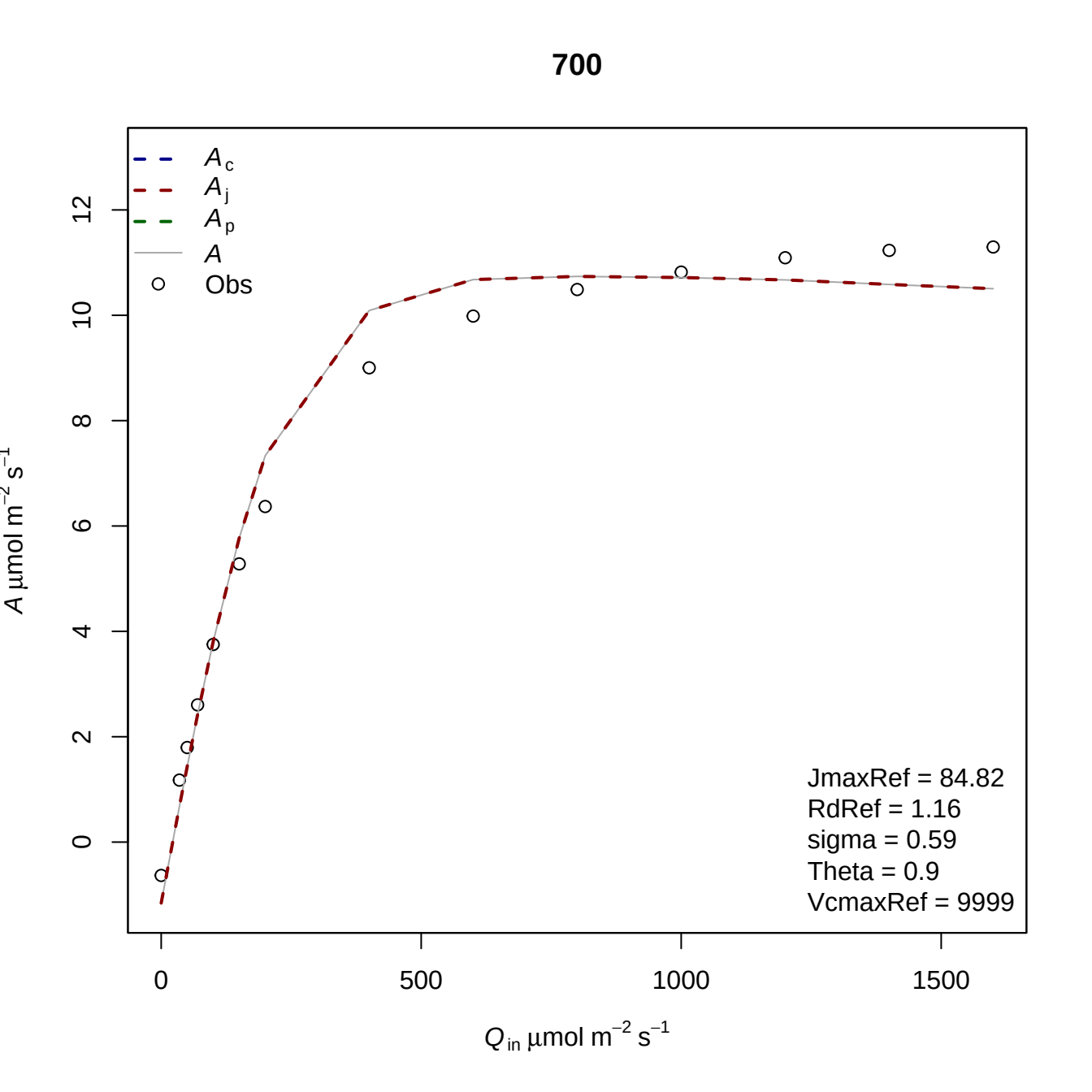

695
A_c
A_j
A_p
A
Obs
JmaxRef = 38.74
RdRef = 1.29
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

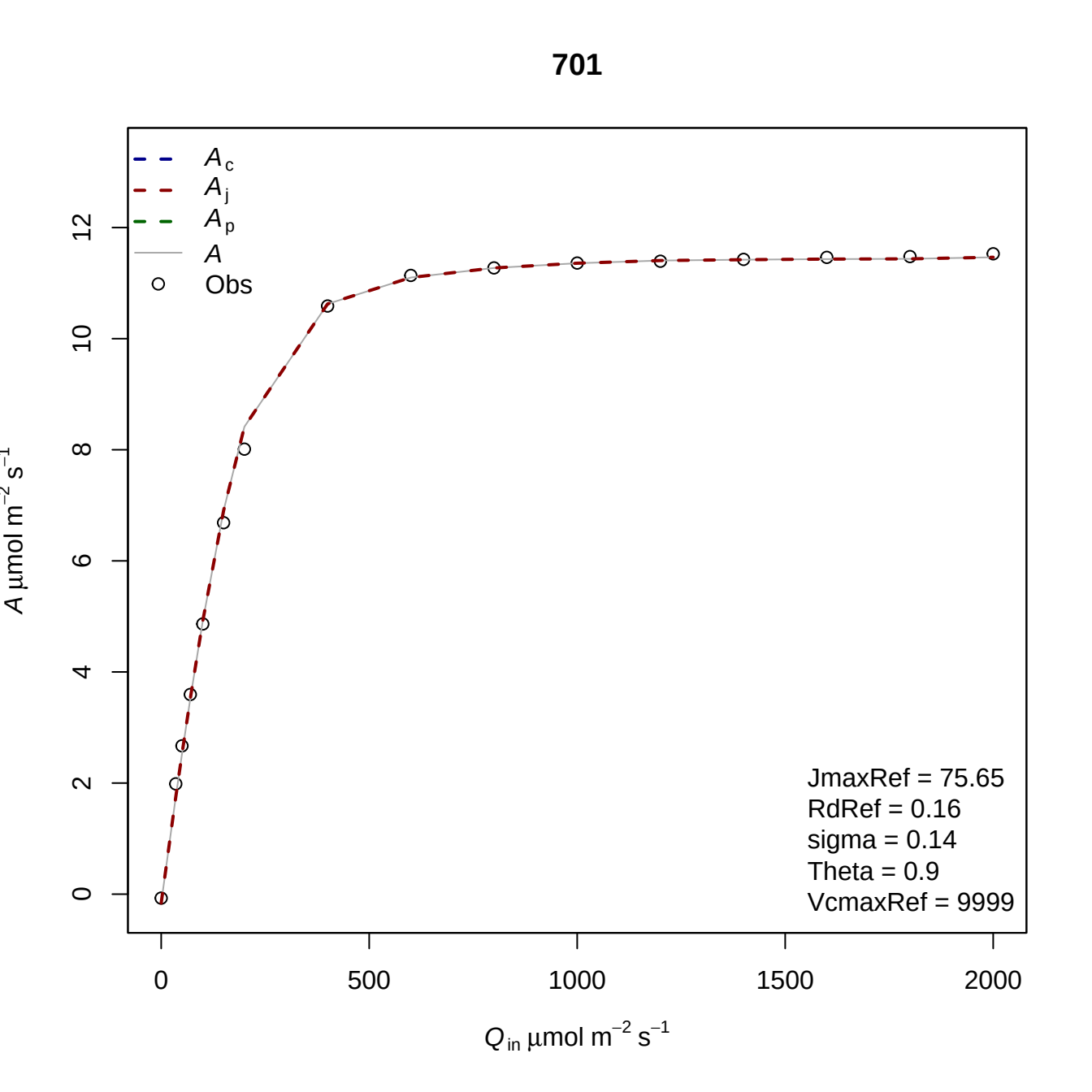

696
A_c
A_j
A_p
A
Obs
JmaxRef = 60.24
RdRef = 0.82
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

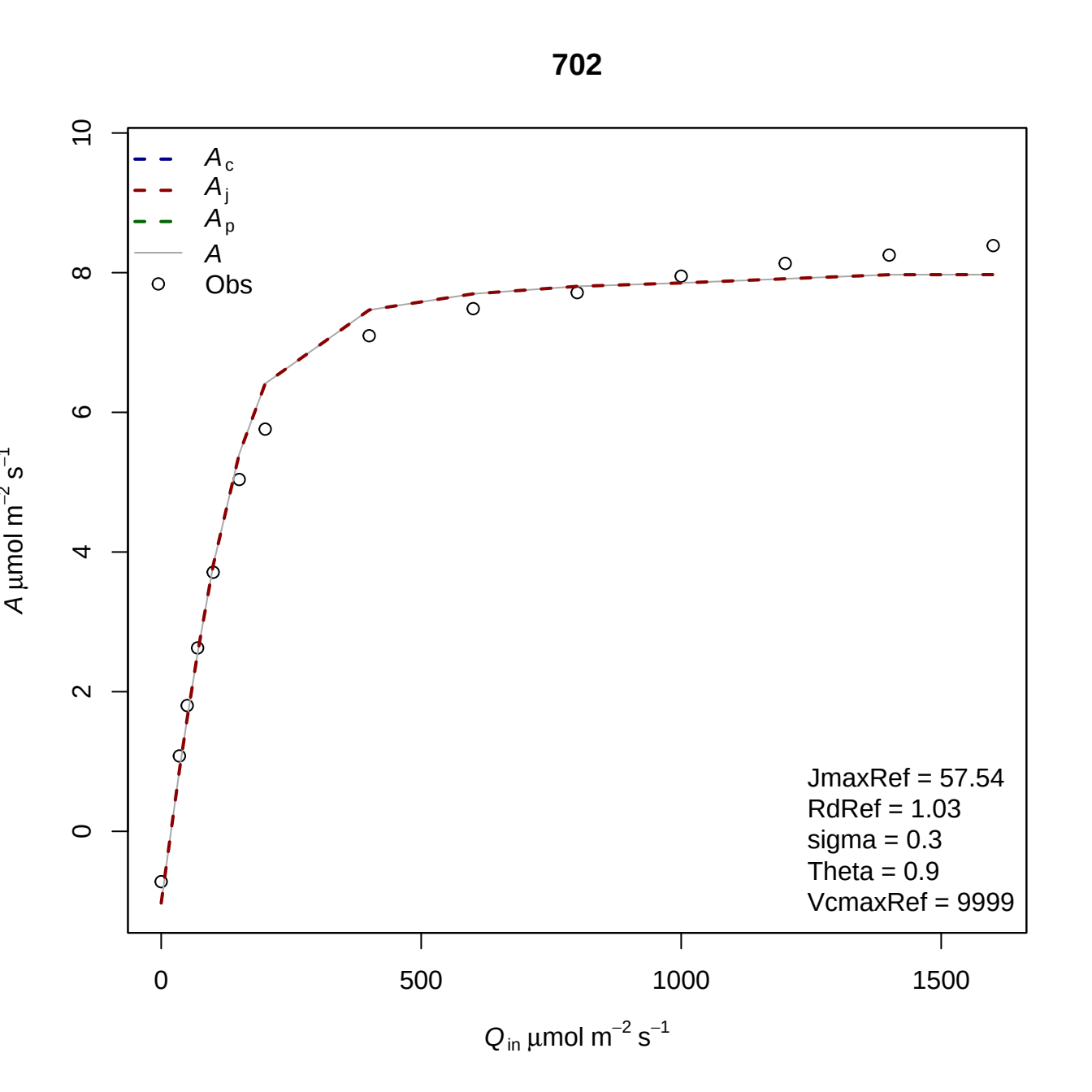

697
A_c
A_j
A_p
A
Obs
JmaxRef = 80.92
RdRef = −0.86
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

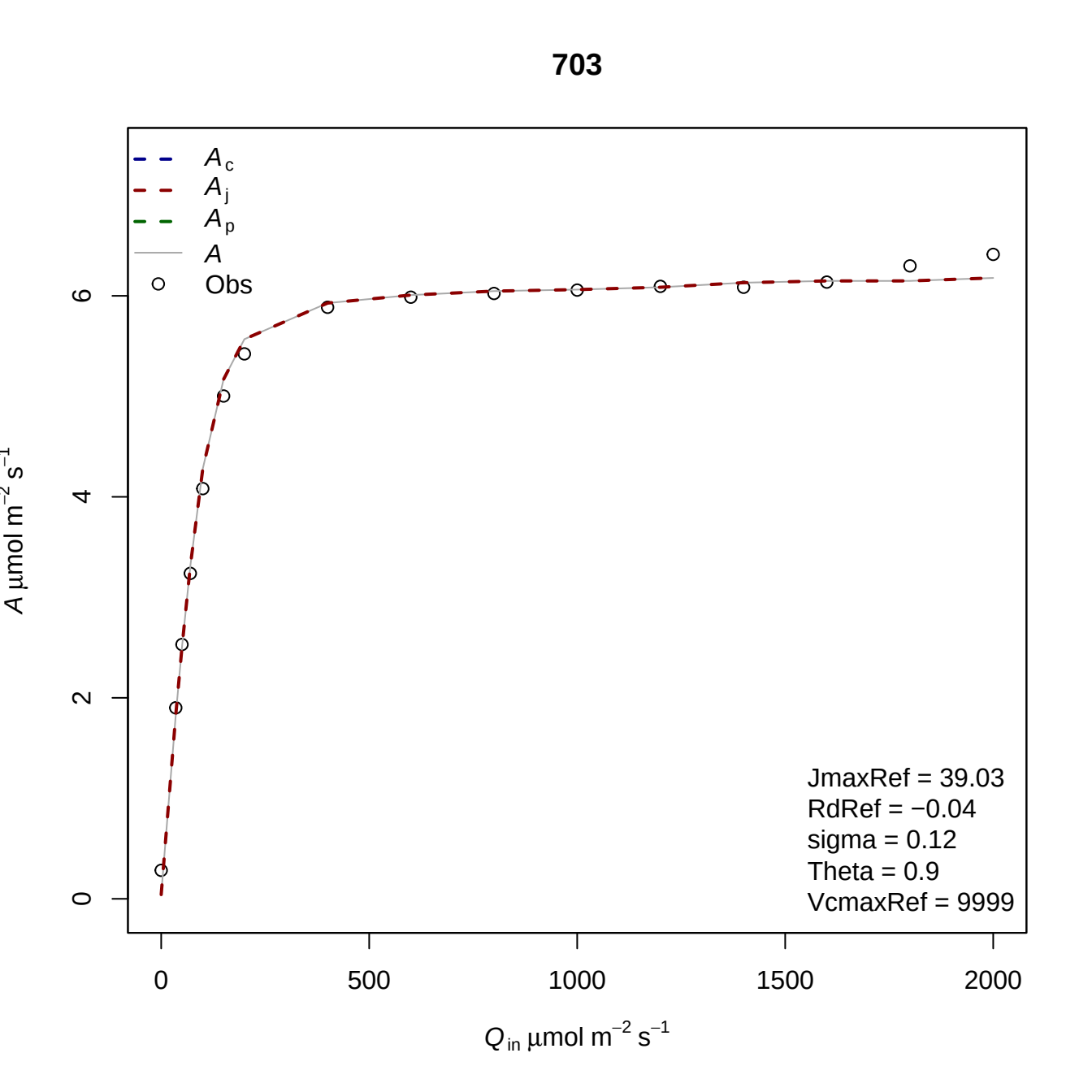

698
A_c
A_j
A_p
A
Obs
JmaxRef = 35.18
RdRef = 0.85
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

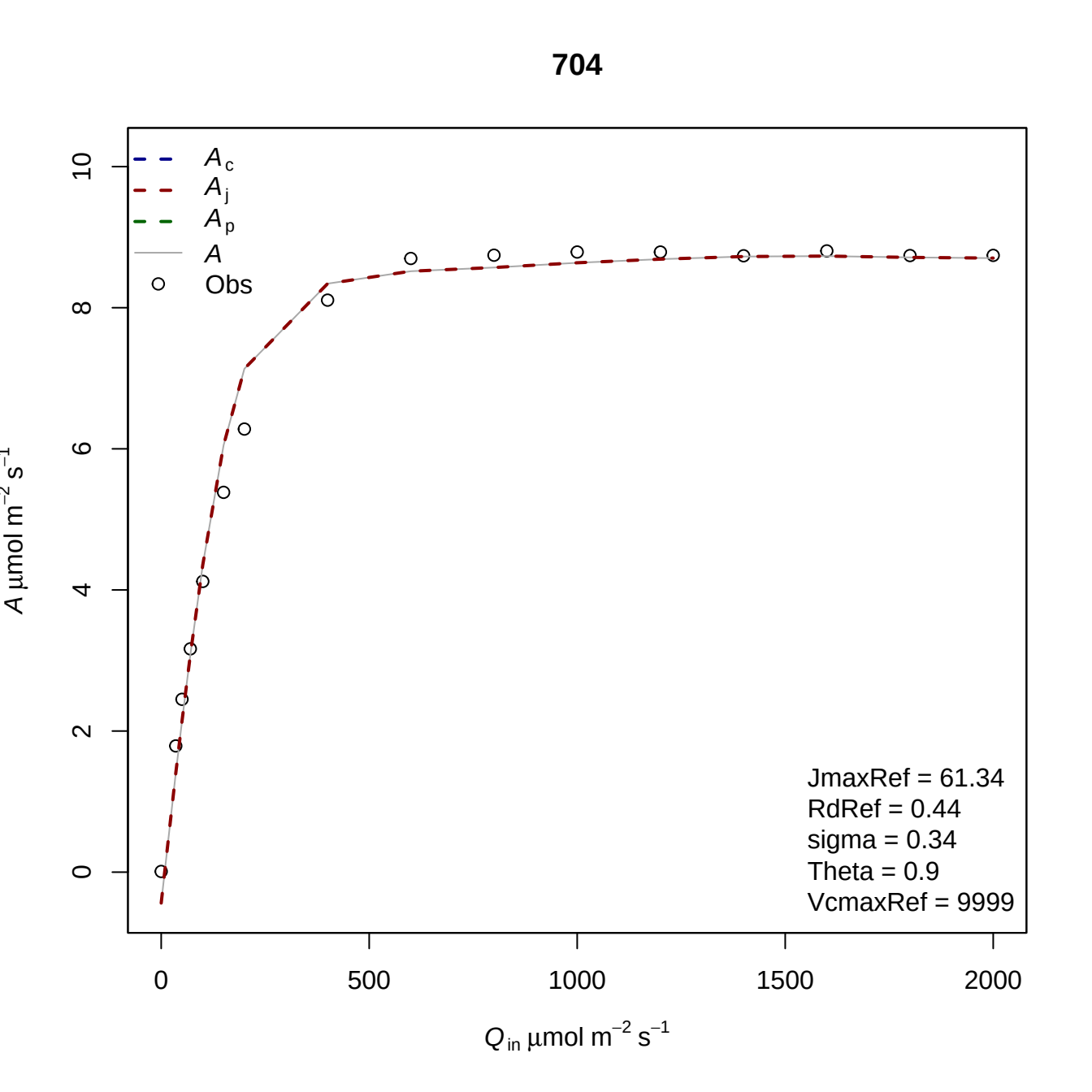

699
A_c
A_j
A_p
A
Obs
JmaxRef = 42.12
RdRef = 0.82
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

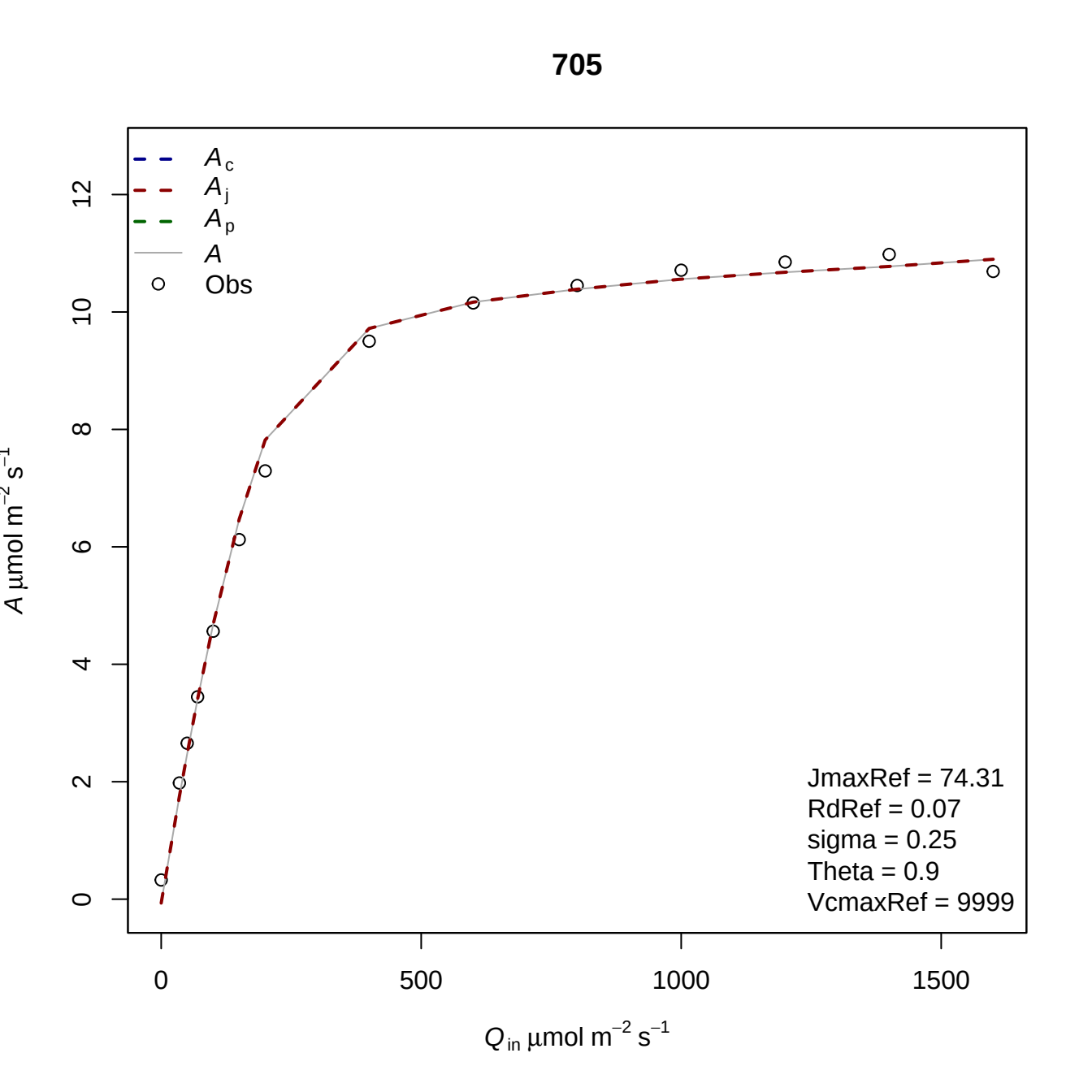

700
A_c
A_j
A_p
A
Obs
JmaxRef = 84.82
RdRef = 1.16
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

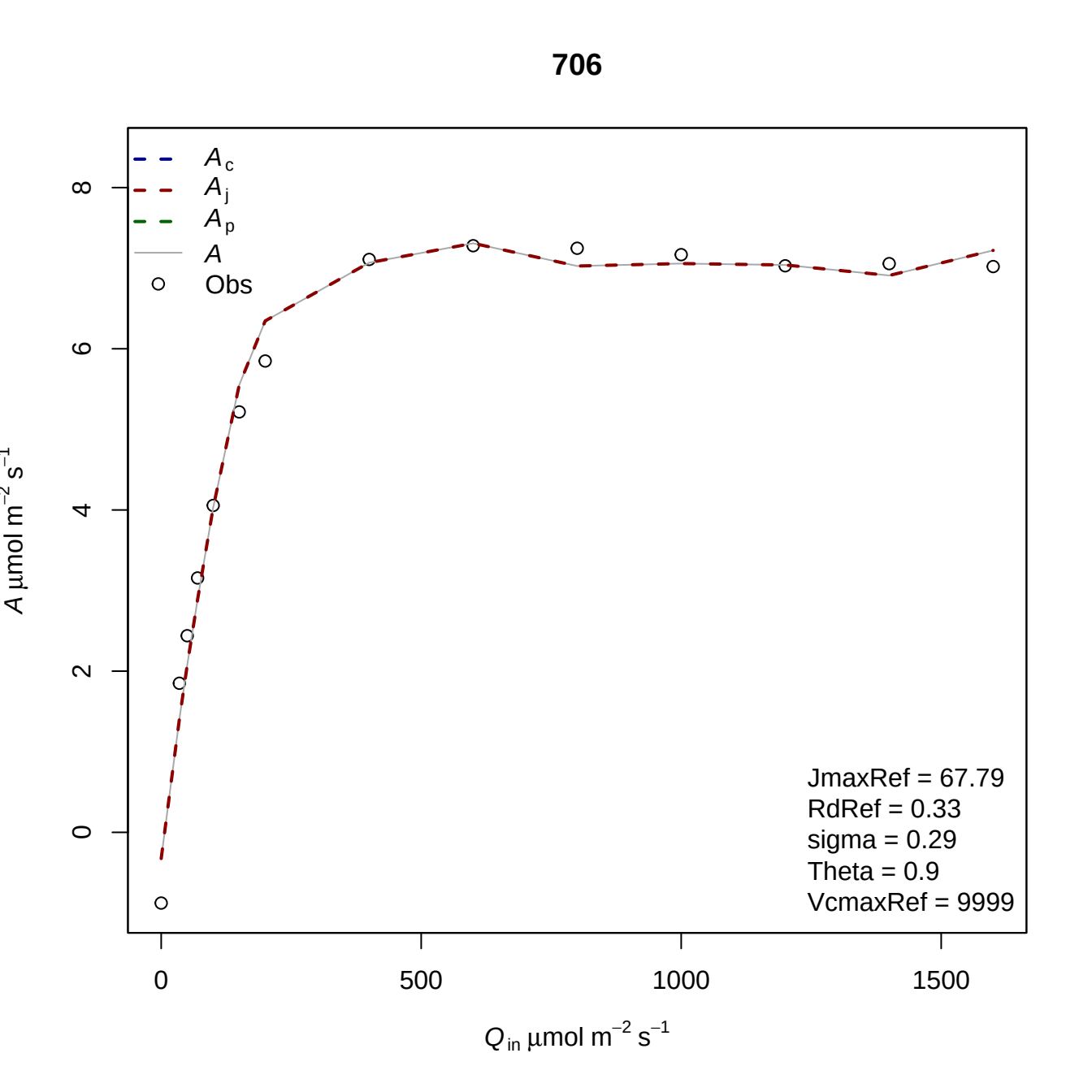

701
A_c
A_j
A_p
A
Obs
JmaxRef = 75.65
RdRef = 0.16
sigma = 0.14
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

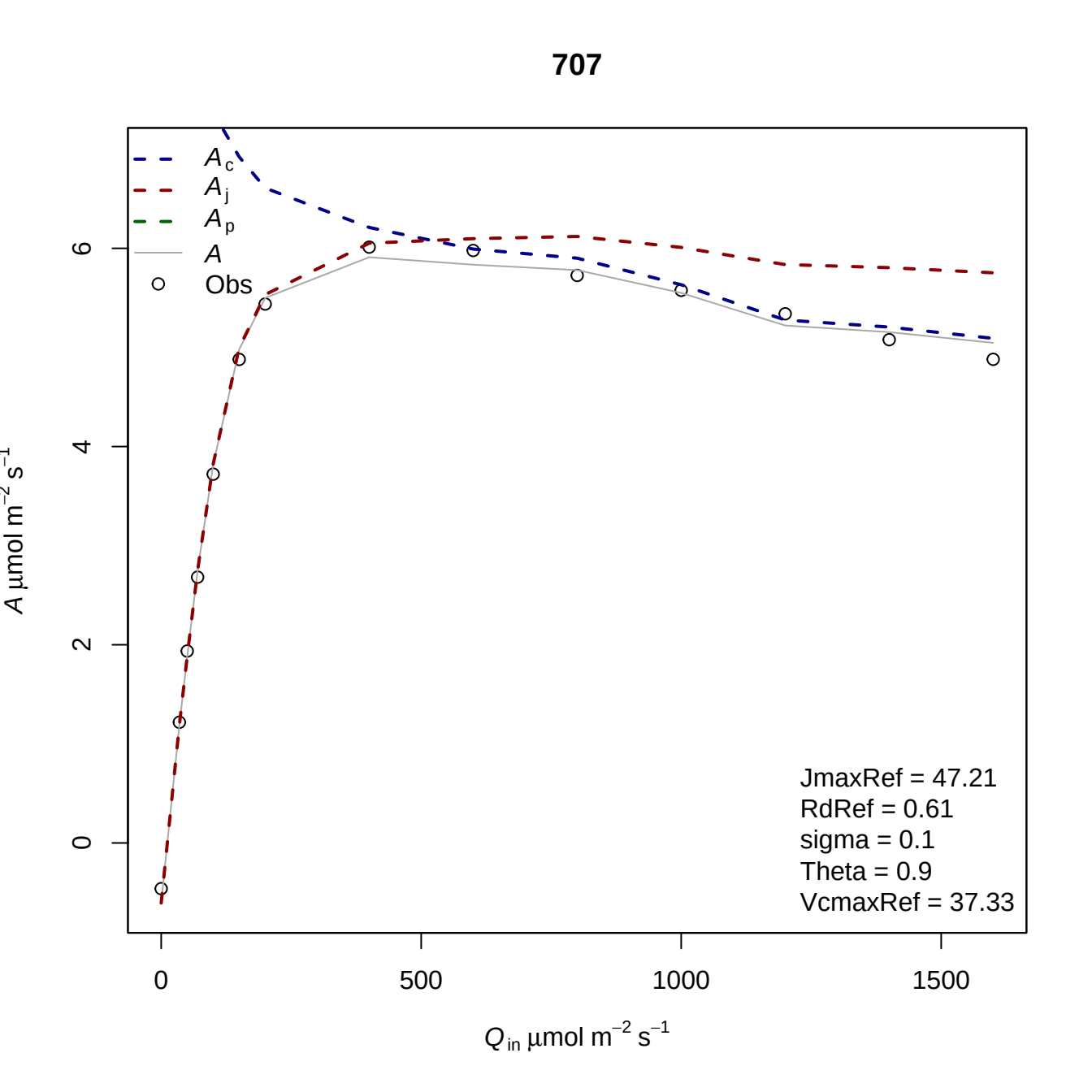

702
A_c
A_j
A_p
A
Obs
JmaxRef = 57.54
RdRef = 1.03
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

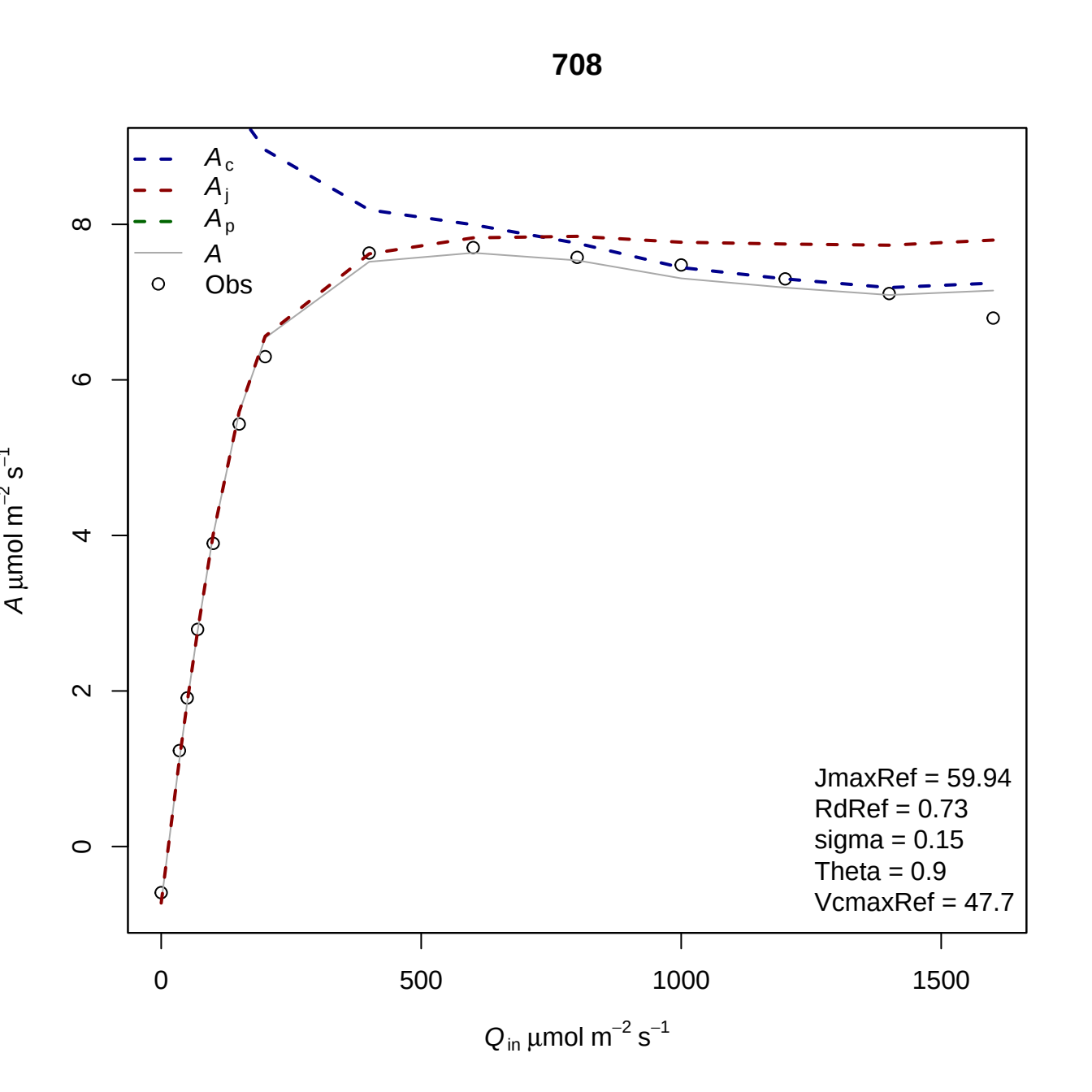

703
A_c
A_j
A_p
A
Obs
JmaxRef = 39.03
RdRef = −0.04
sigma = 0.12
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

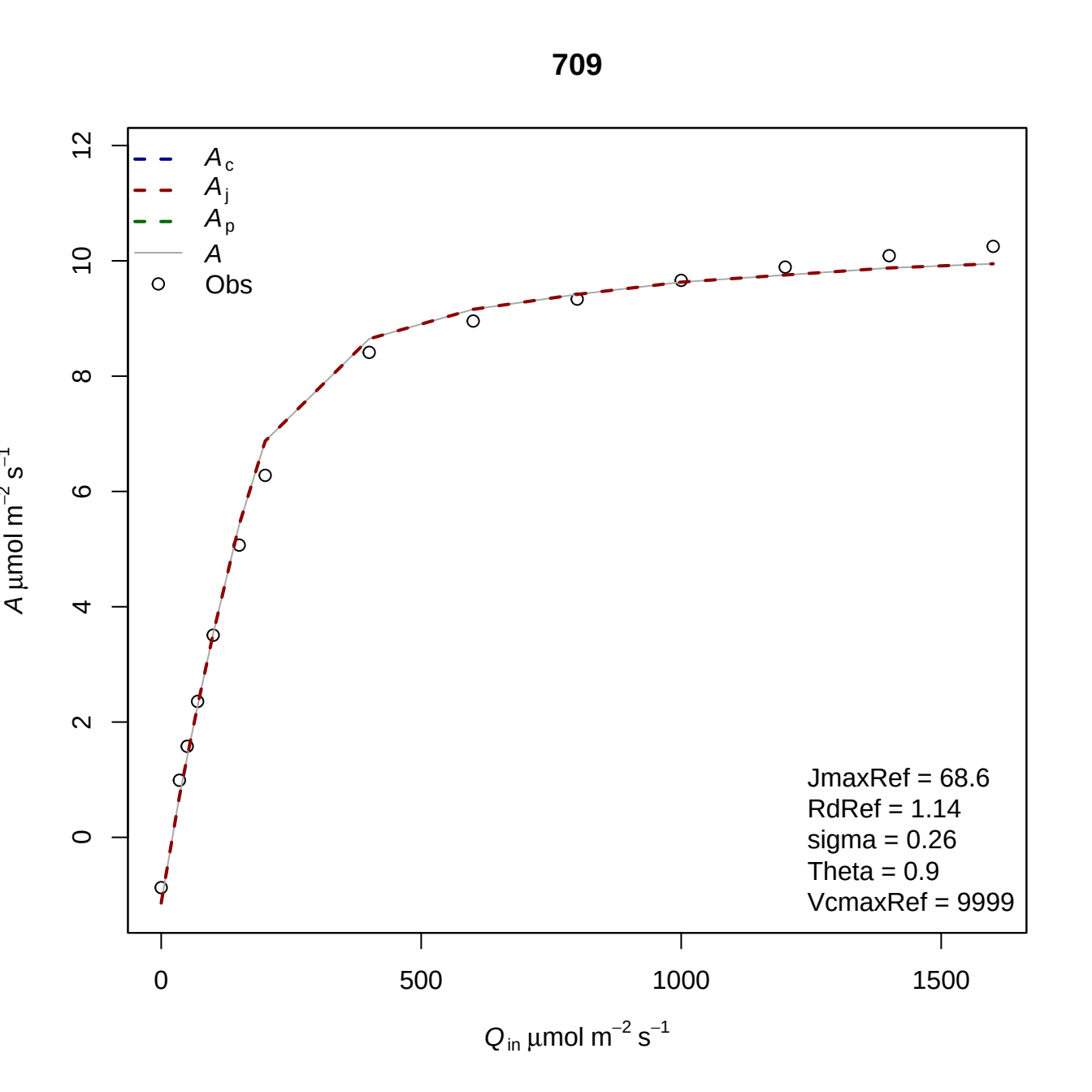

704
A_c
A_j
A_p
A
Obs
JmaxRef = 61.34
RdRef = 0.44
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

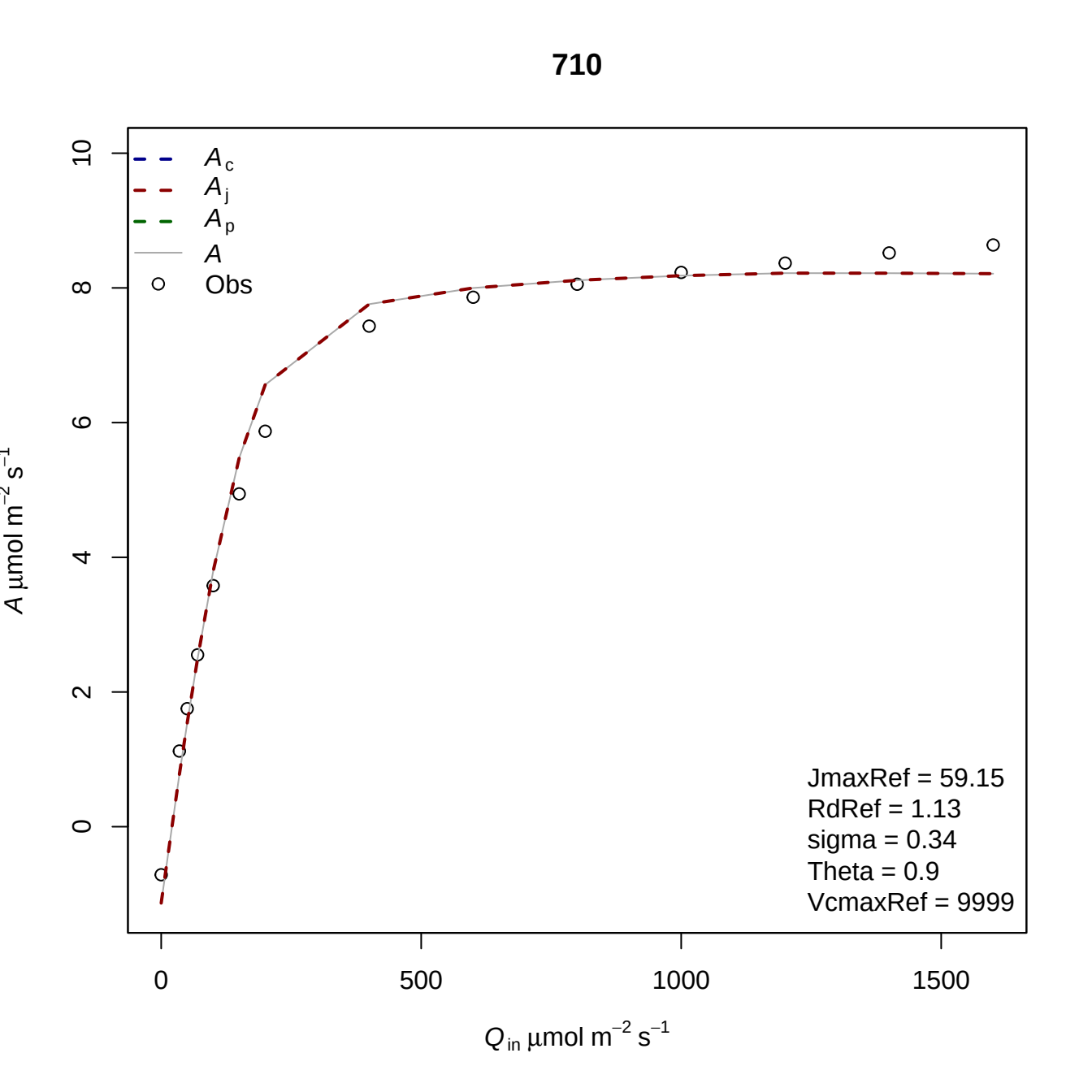

705
A_c
A_j
A_p
A
Obs
JmaxRef = 74.31
RdRef = 0.07
sigma = 0.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

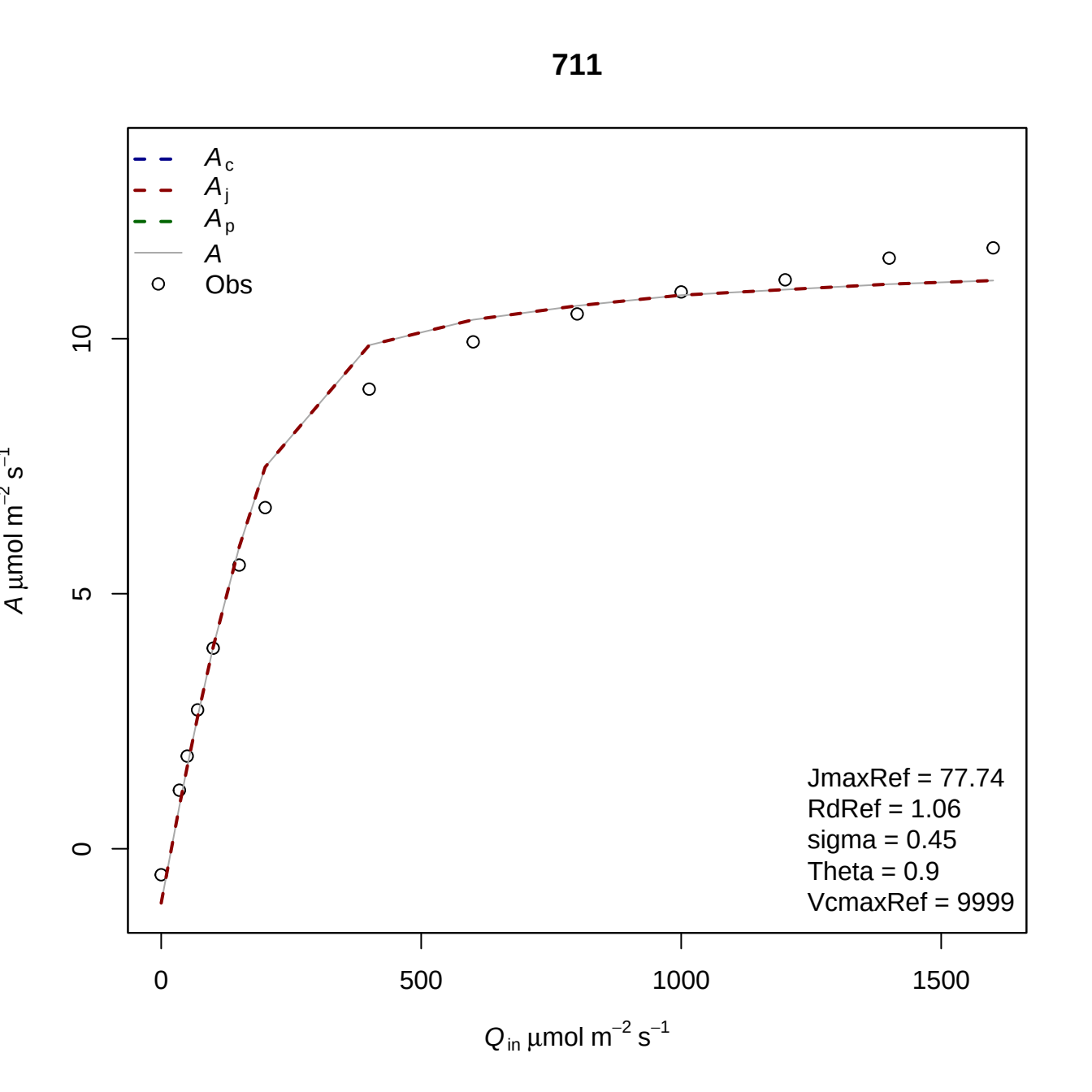

706
A_c
A_j
A_p
A
Obs
JmaxRef = 67.79
RdRef = 0.33
sigma = 0.29
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

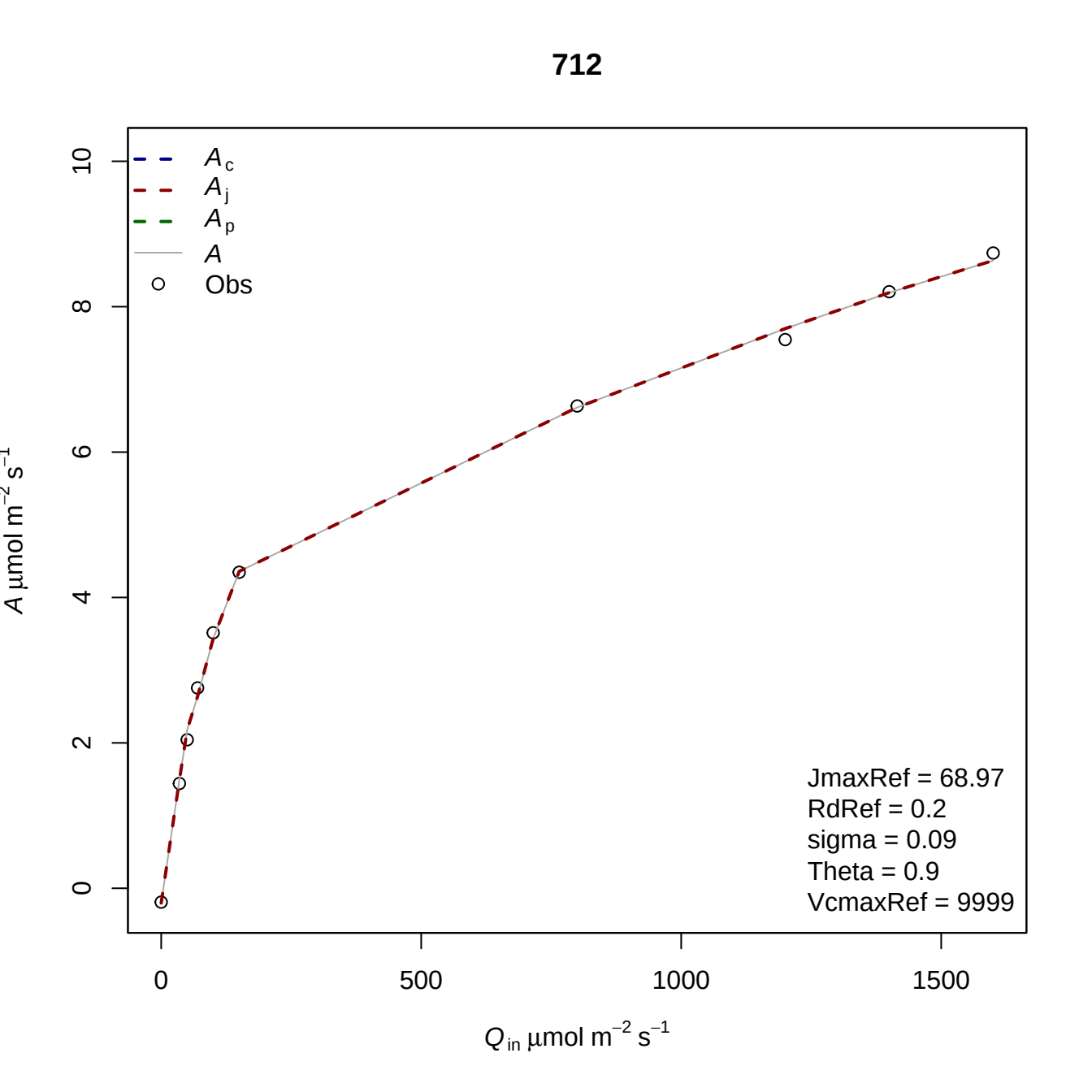

707
A_c
A_j
A_p
A
Obs
JmaxRef = 47.21
RdRef = 0.61
sigma = 0.1
Theta = 0.9
VcmaxRef = 37.33
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

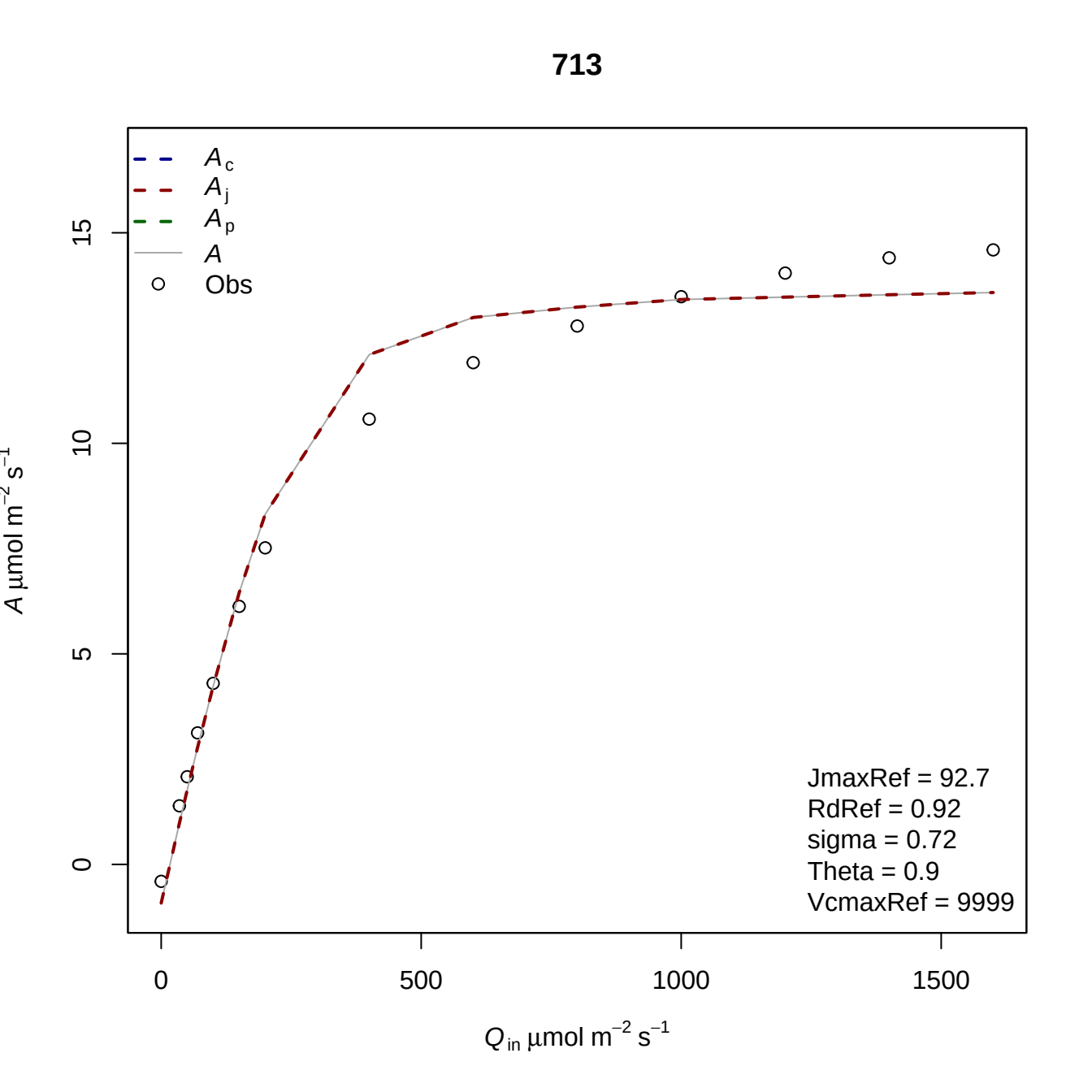

708
A_c
A_j
A_p
A
Obs
JmaxRef = 59.94
RdRef = 0.73
sigma = 0.15
Theta = 0.9
VcmaxRef = 47.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

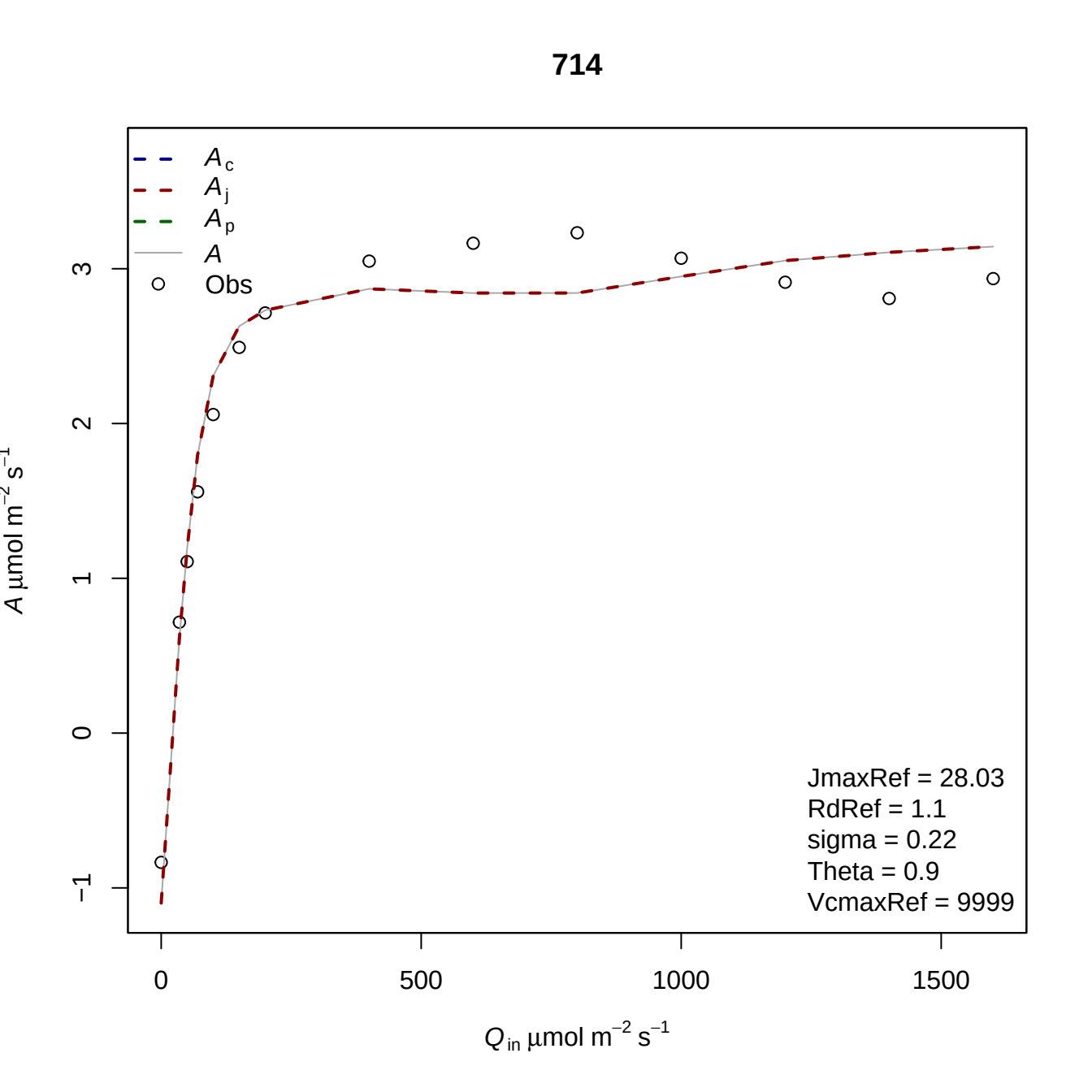

709
A_c
A_j
A_p
A
Obs
JmaxRef = 68.6
RdRef = 1.14
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

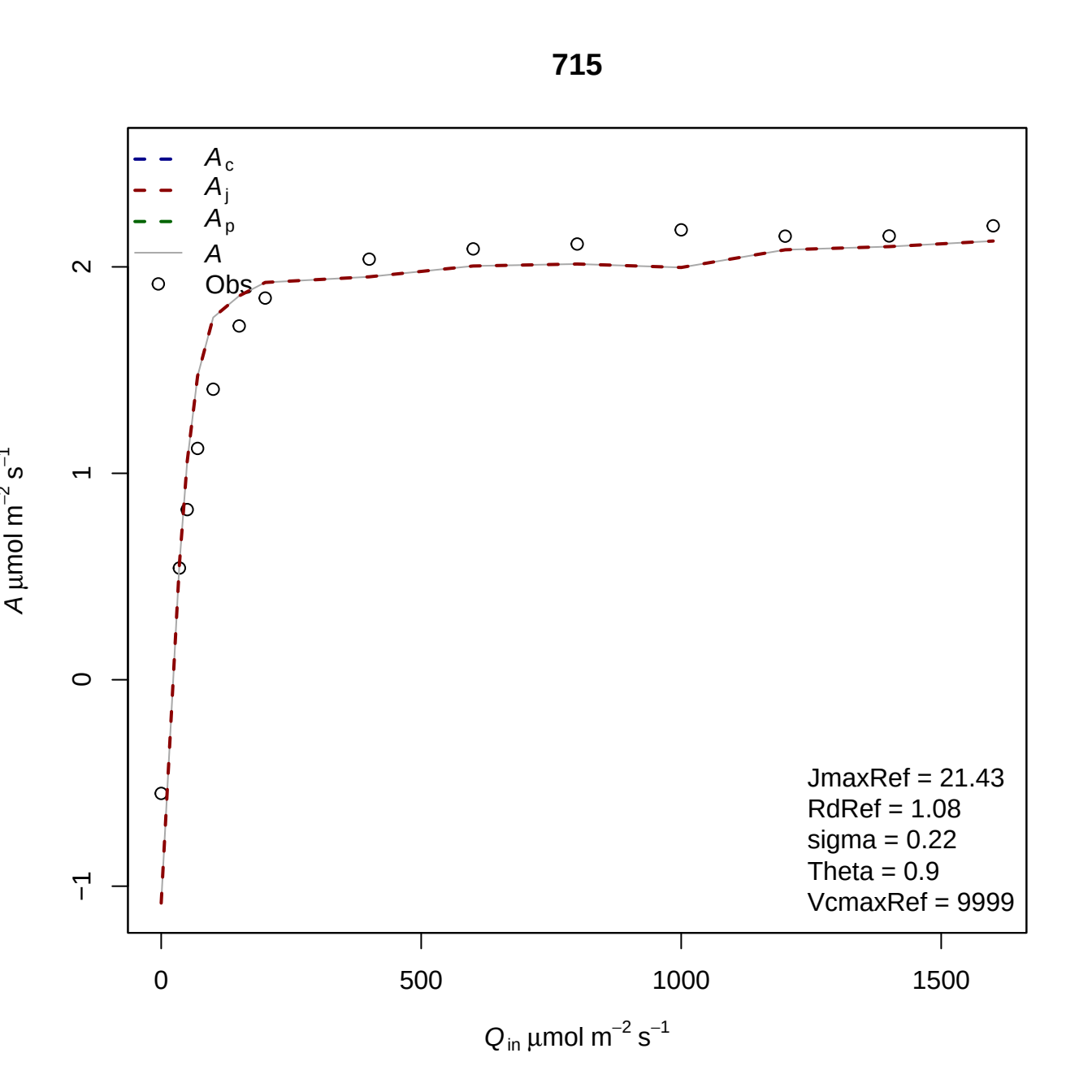

710
A_c
A_j
A_p
A
Obs
JmaxRef = 59.15
RdRef = 1.13
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

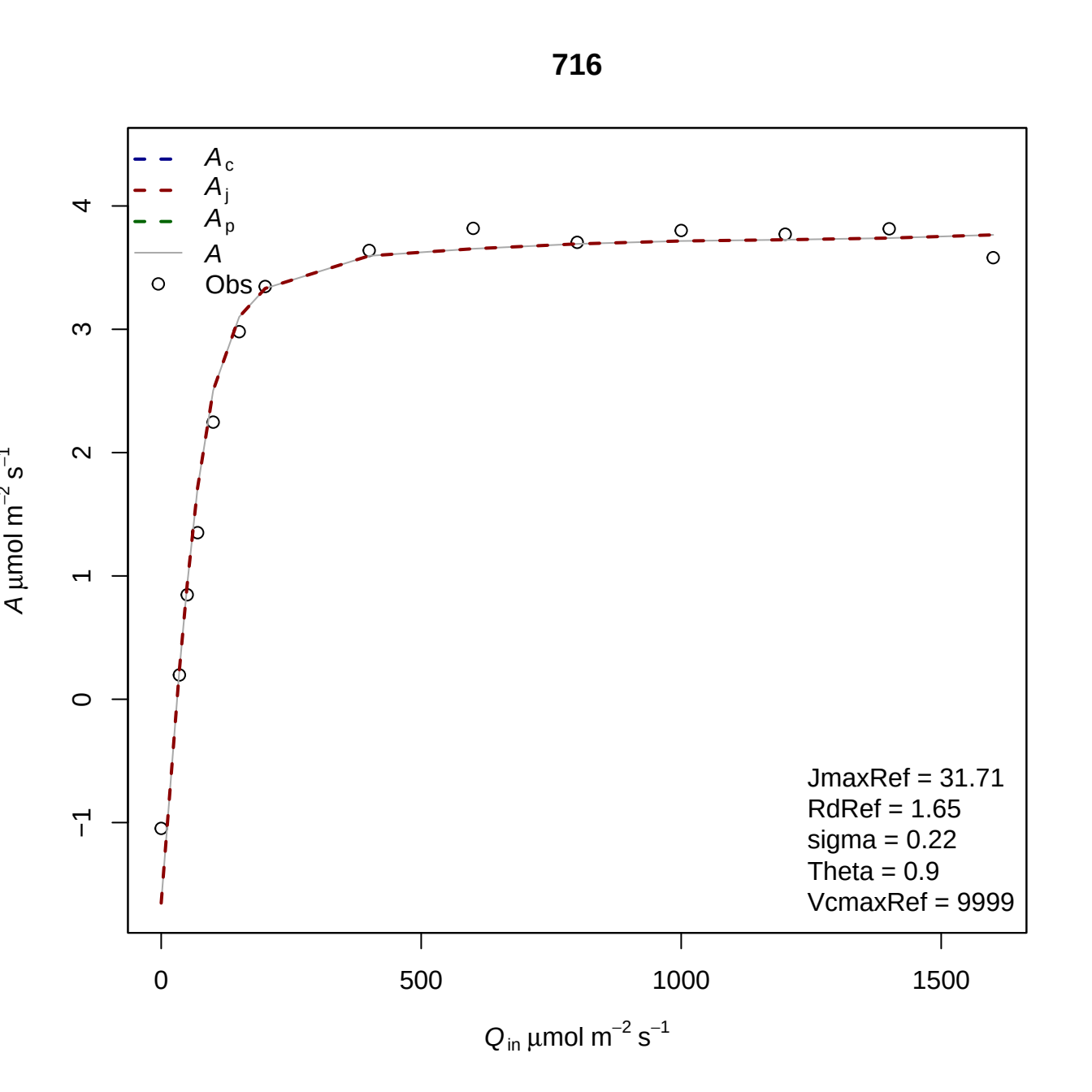

711
A_c
A_j
A_p
A
Obs
JmaxRef = 77.74
RdRef = 1.06
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

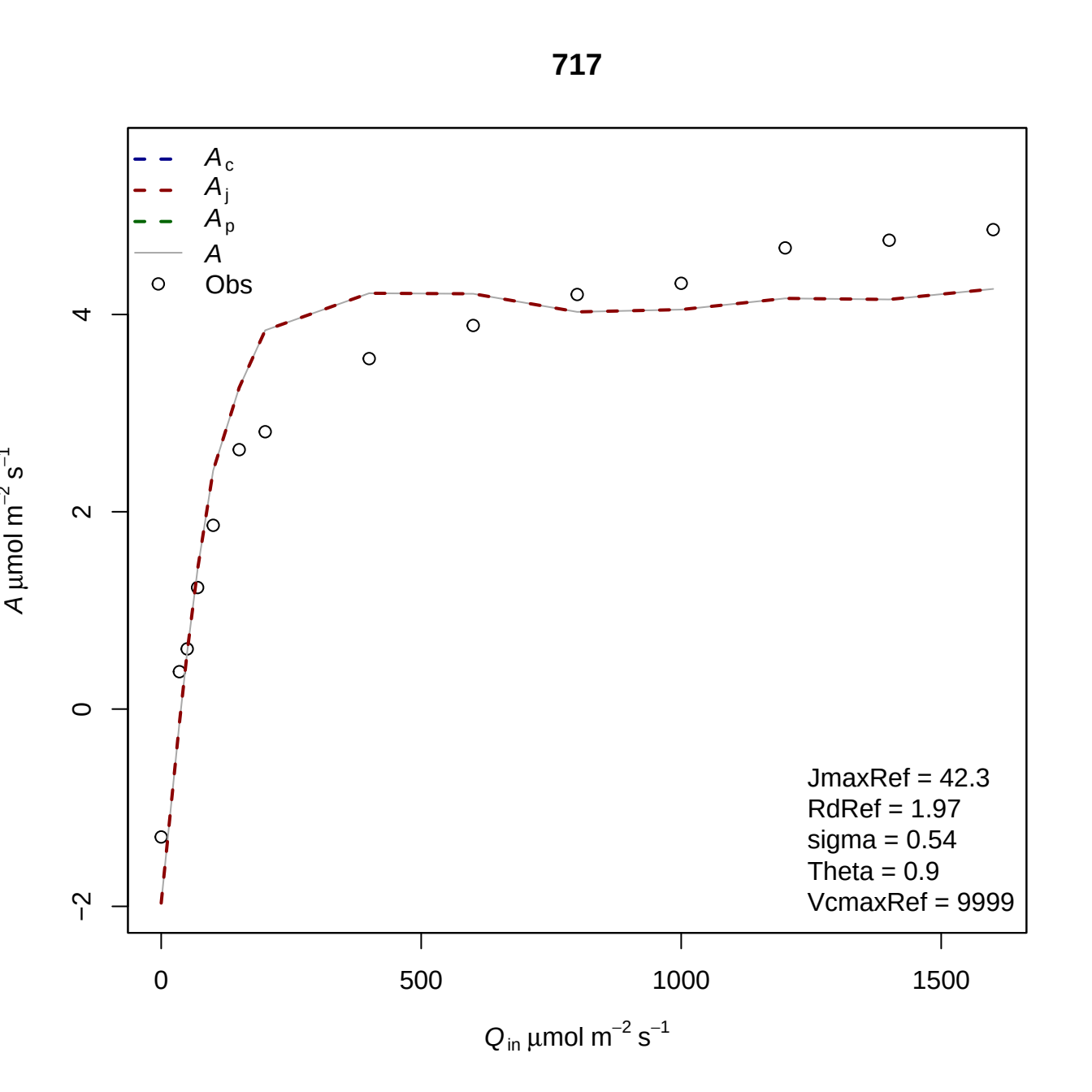

712
A_c
A_j
A_p
A
Obs
JmaxRef = 68.97
RdRef = 0.2
sigma = 0.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

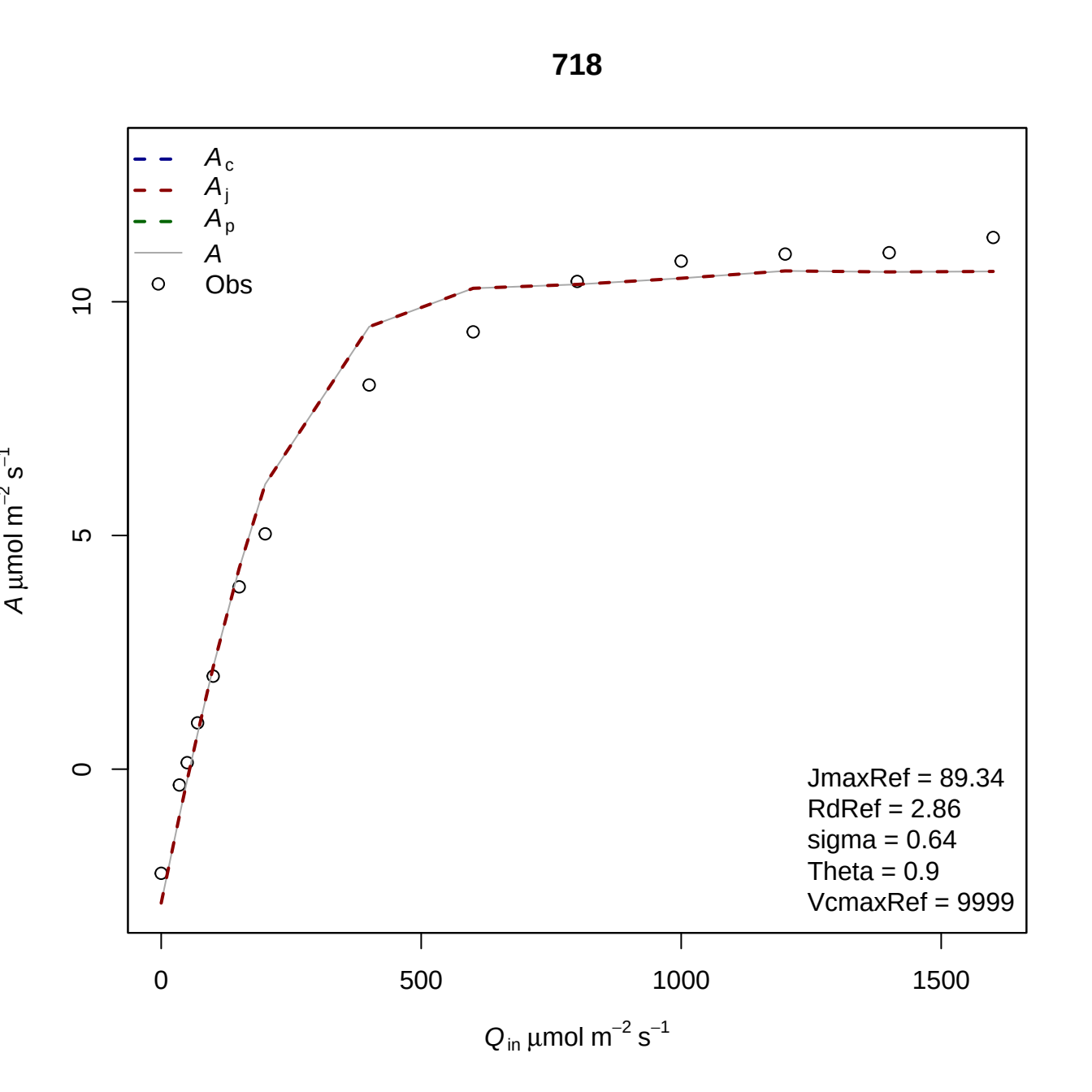

713
A_c
A_j
A_p
A
Obs
JmaxRef = 92.7
RdRef = 0.92
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

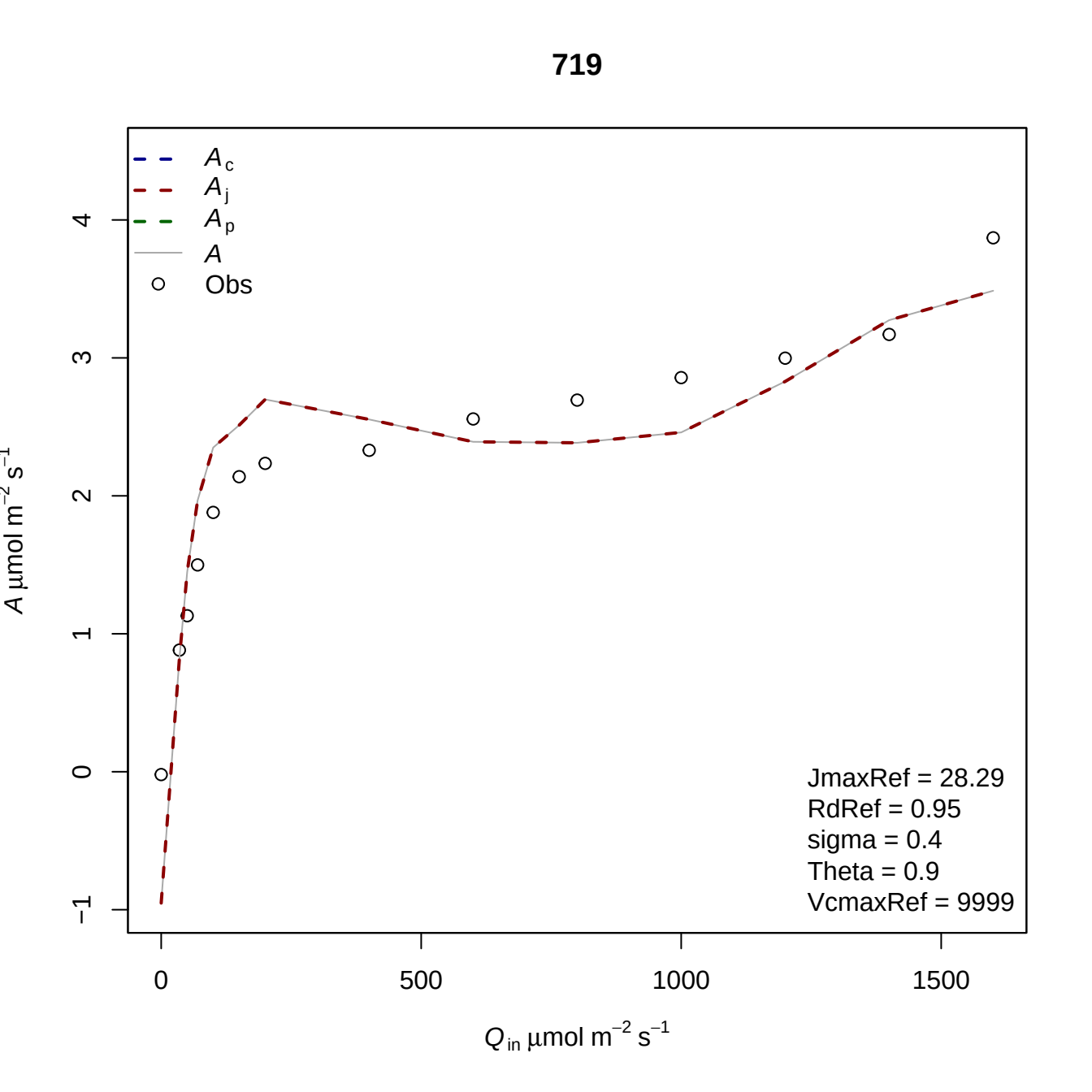

714
A_c
A_j
A_p
A
Obs
JmaxRef = 28.03
RdRef = 1.1
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

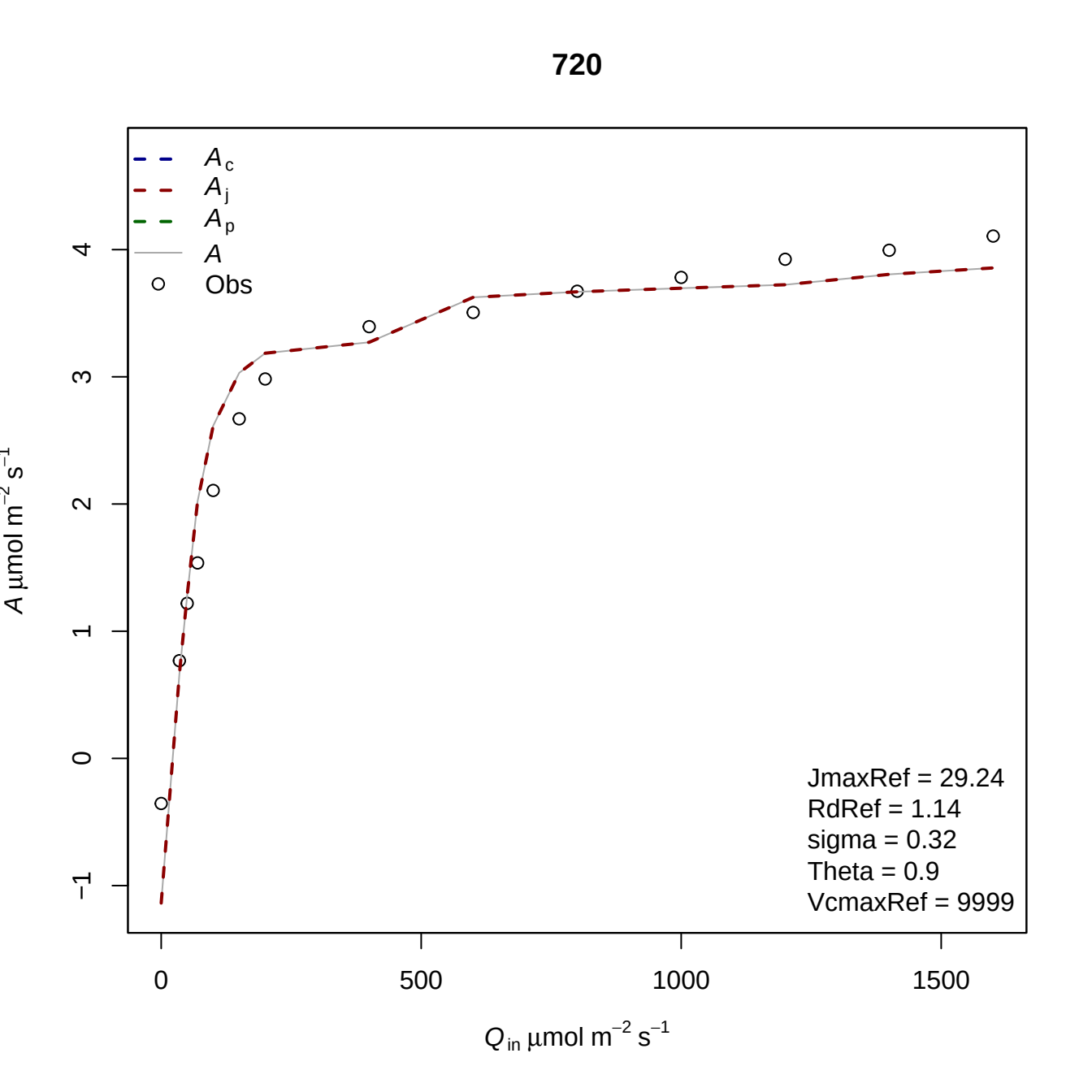

715
A_c
A_j
A_p
A
Obs
JmaxRef = 21.43
RdRef = 1.08
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

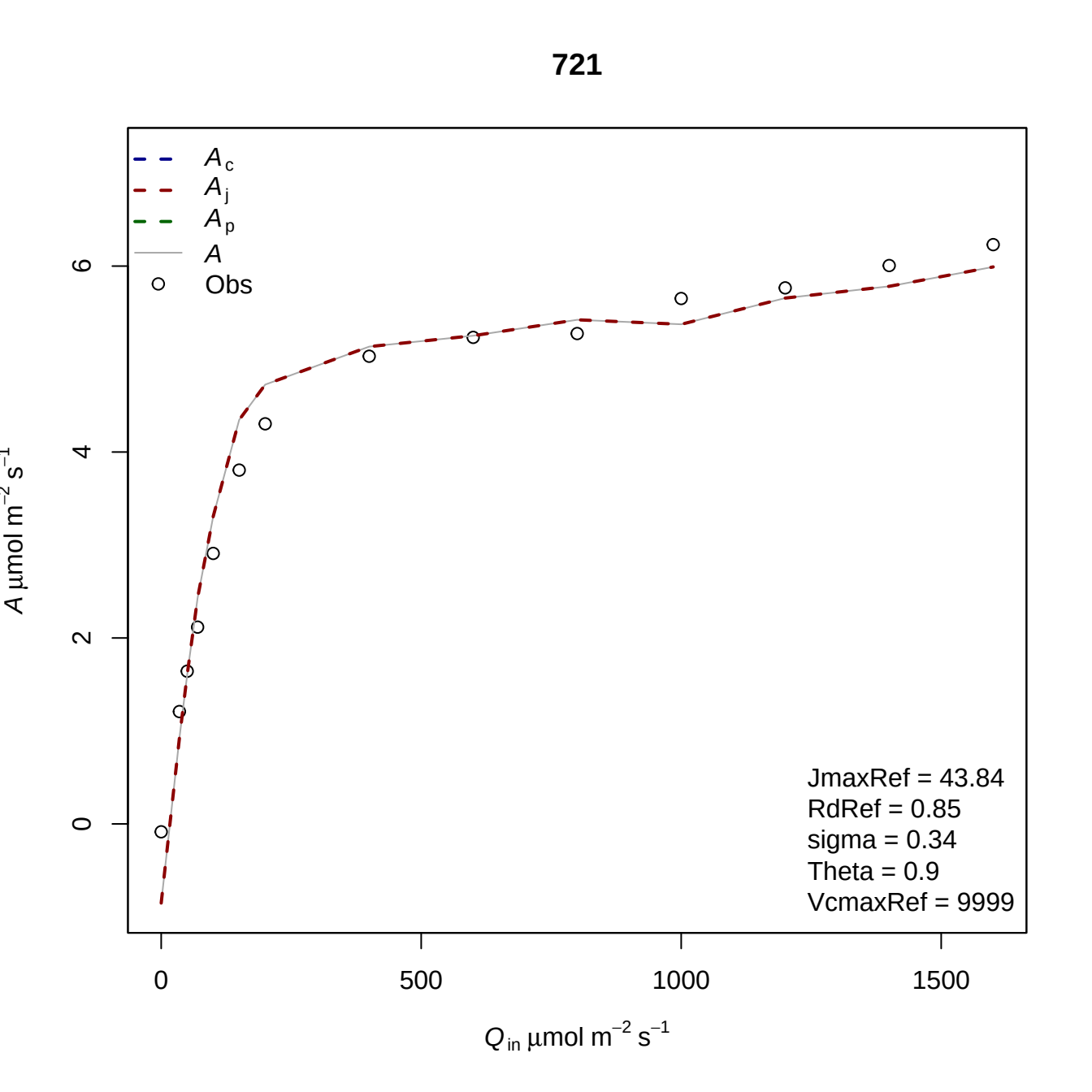

716
A_c
A_j
A_p
A
Obs
JmaxRef = 31.71
RdRef = 1.65
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

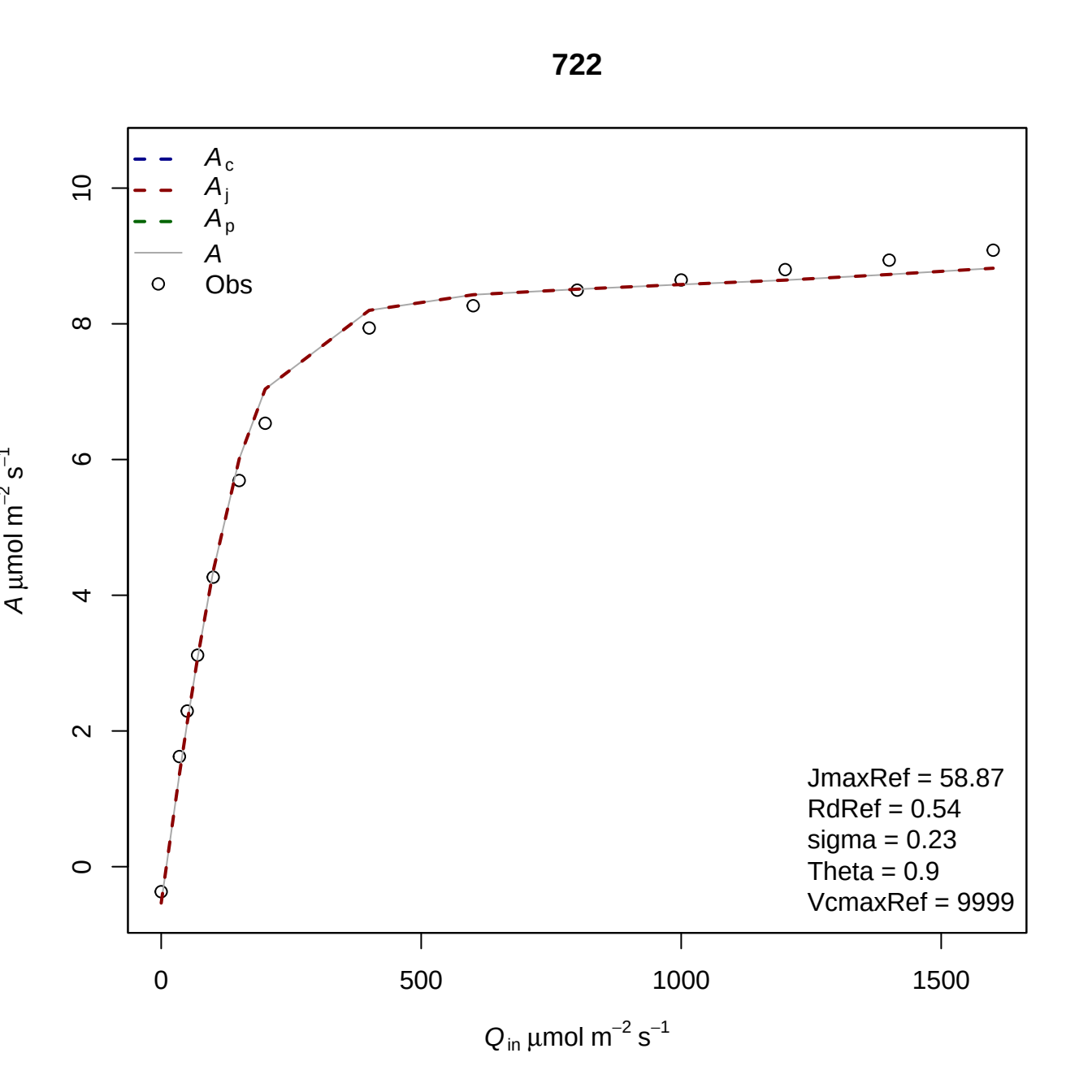

717
A_c
A_j
A_p
A
Obs
JmaxRef = 42.3
RdRef = 1.97
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

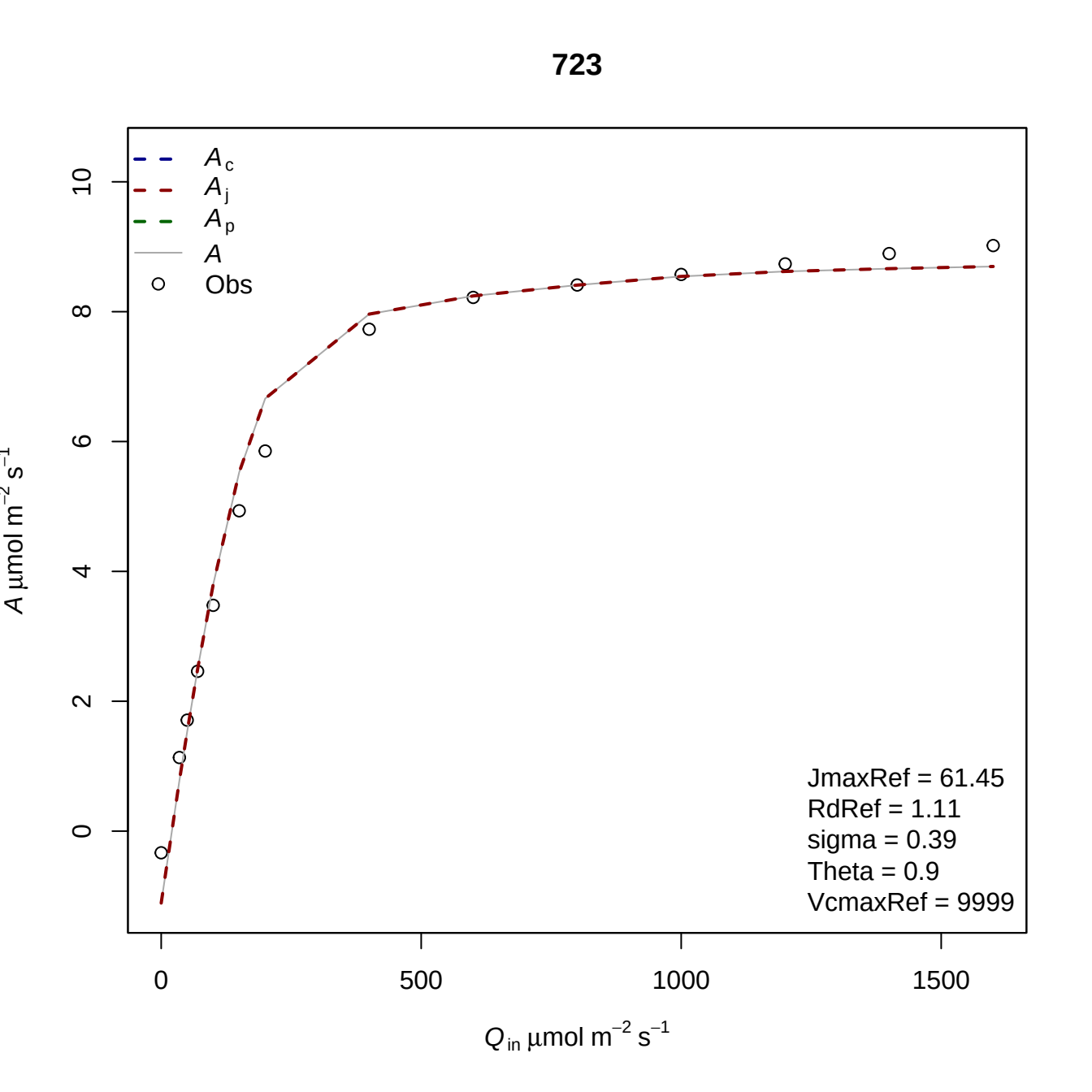

718
A_c
A_j
A_p
A
Obs
JmaxRef = 89.34
RdRef = 2.86
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

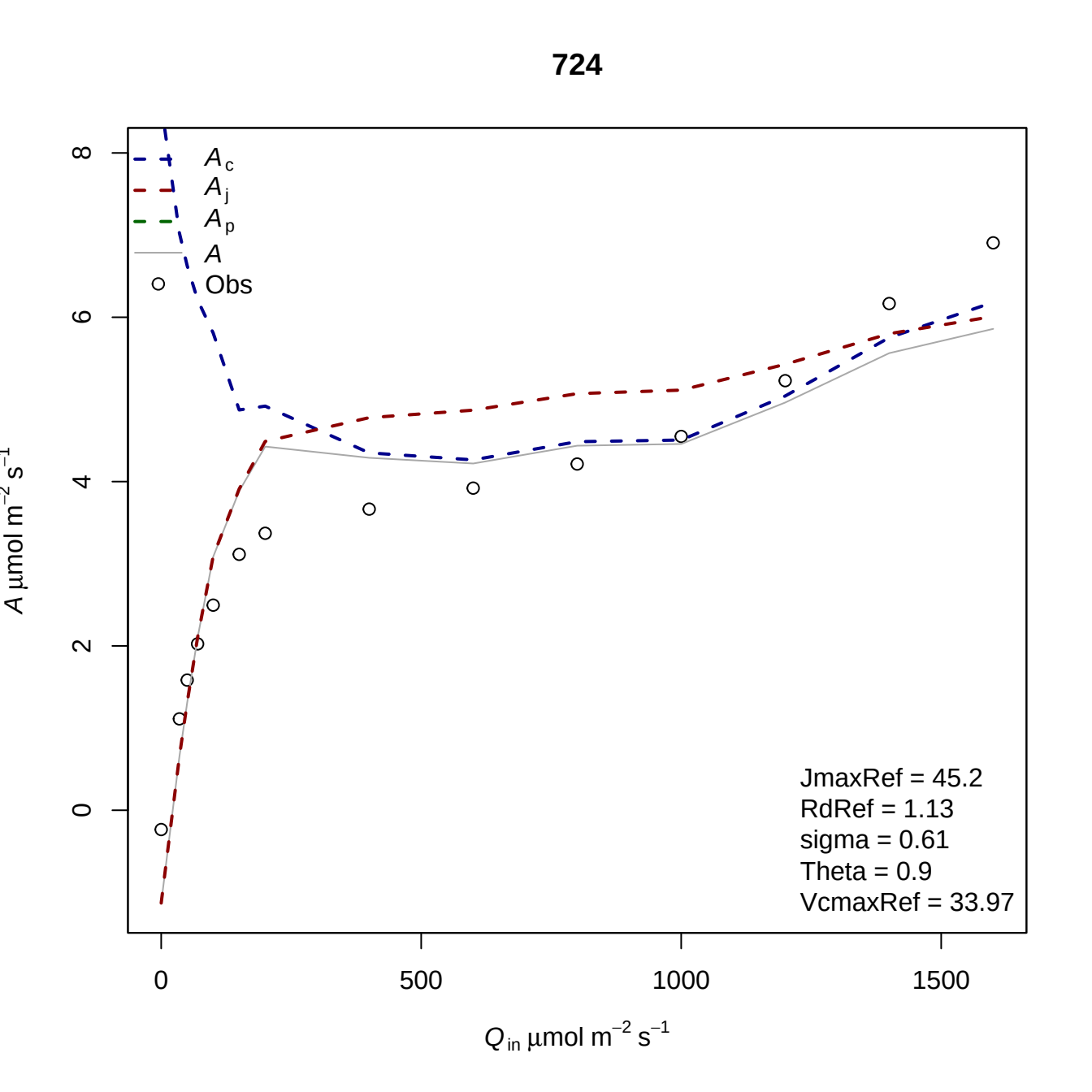

719
A_c
A_j
A_p
A
Obs
JmaxRef = 28.29
RdRef = 0.95
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

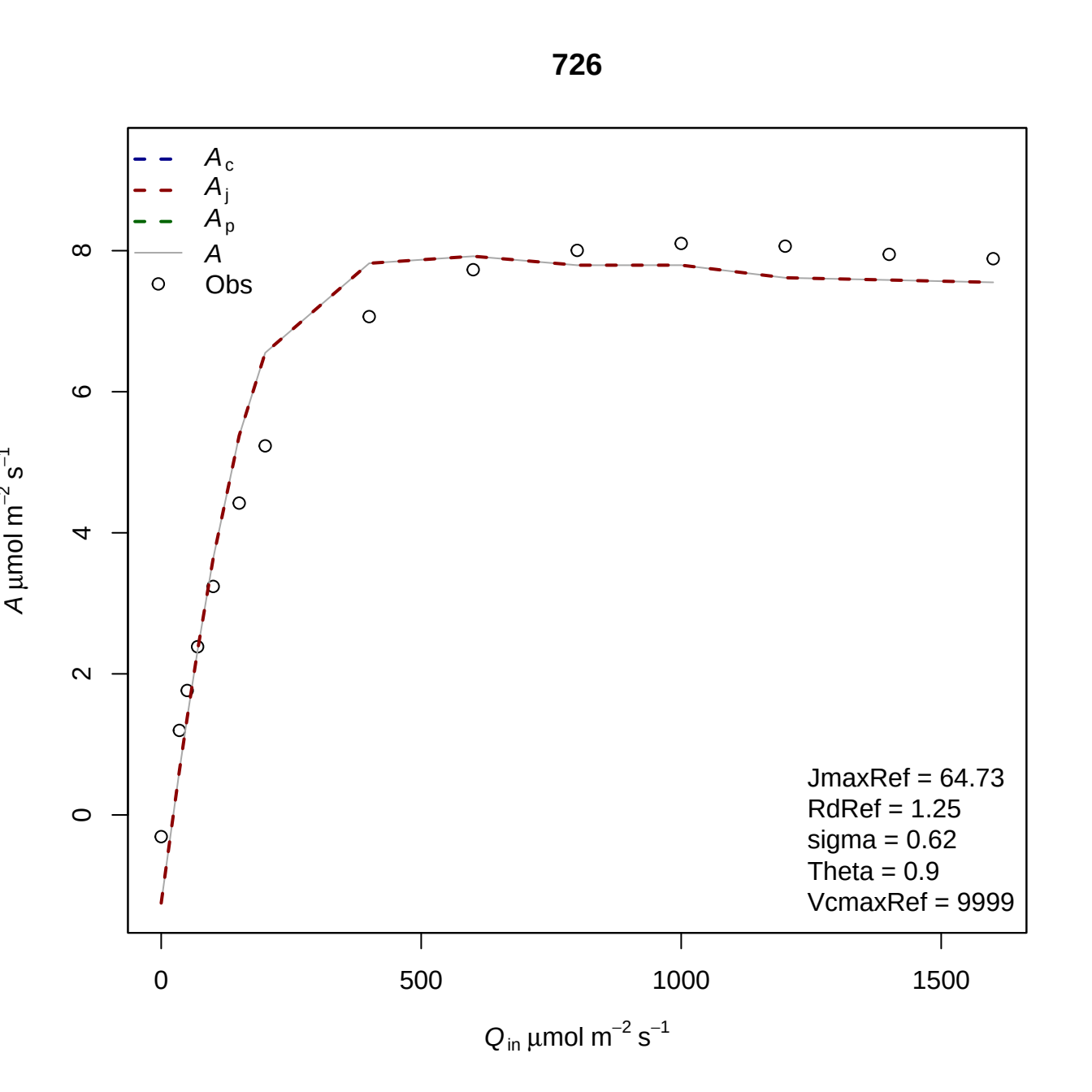

720
A_c
A_j
A_p
A
Obs
JmaxRef = 29.24
RdRef = 1.14
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

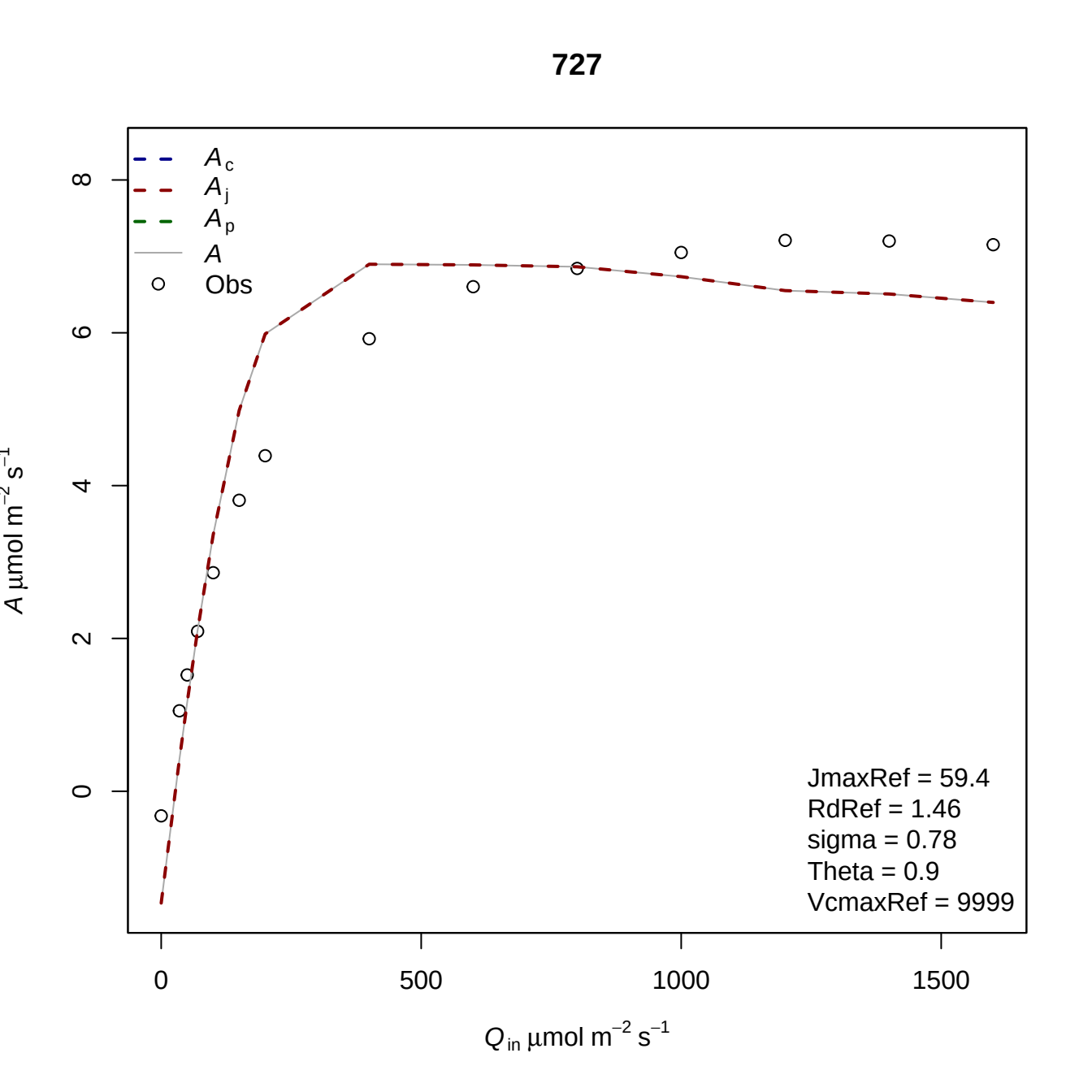

721
A_c
A_j
A_p
A
Obs
JmaxRef = 43.84
RdRef = 0.85
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

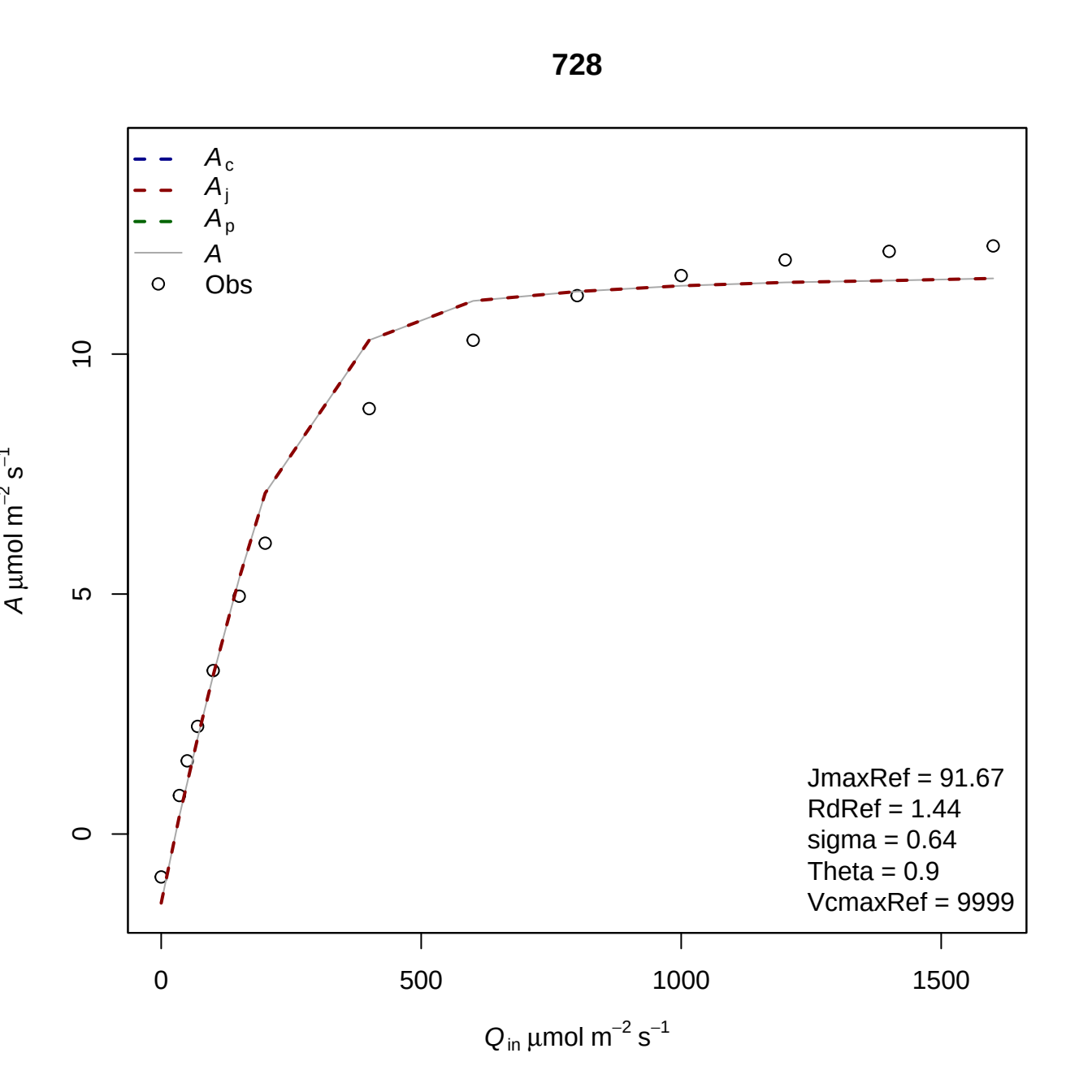

722
A_c
A_j
A_p
A
Obs
JmaxRef = 58.87
RdRef = 0.54
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

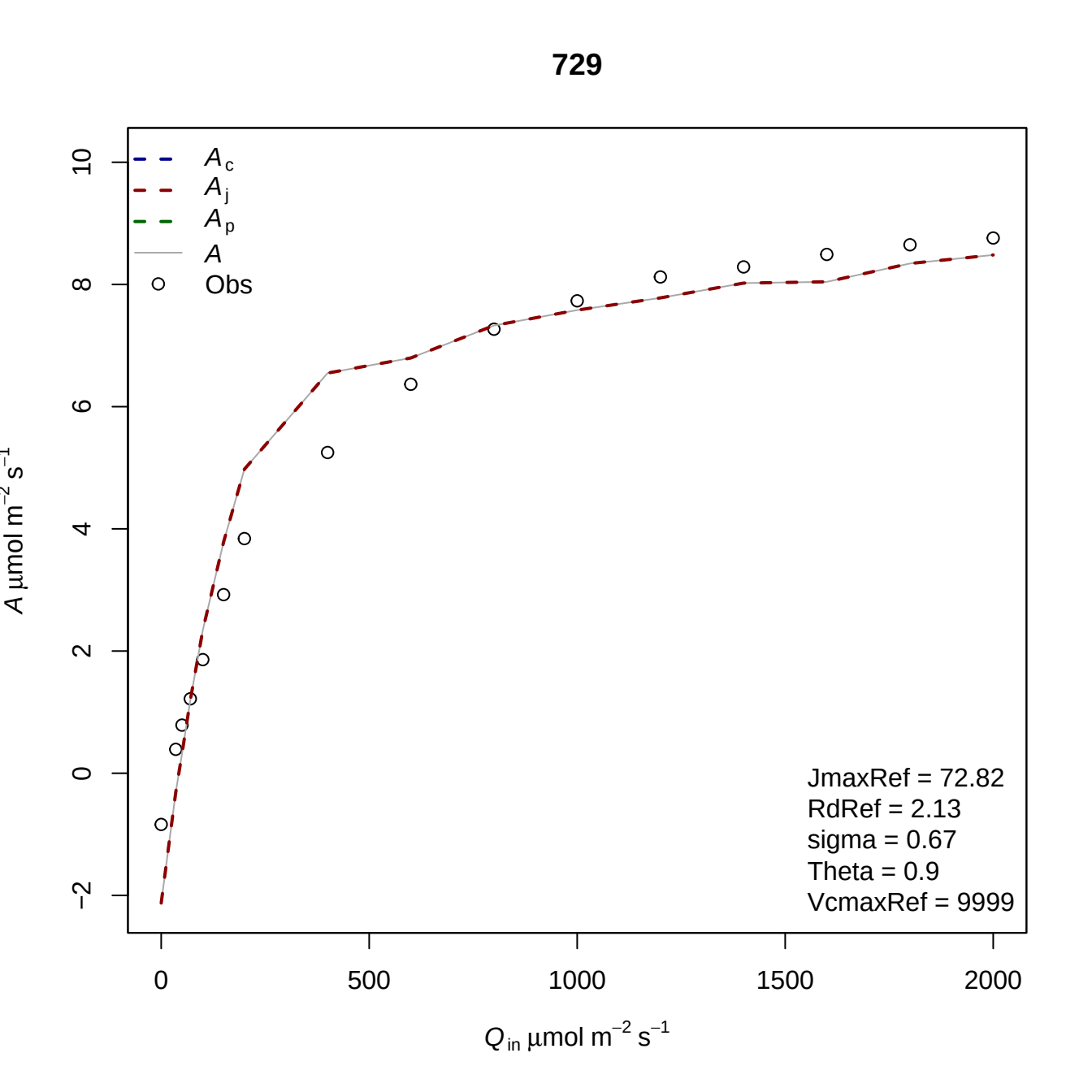

723
A_c
A_j
A_p
A
Obs
JmaxRef = 61.45
RdRef = 1.11
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

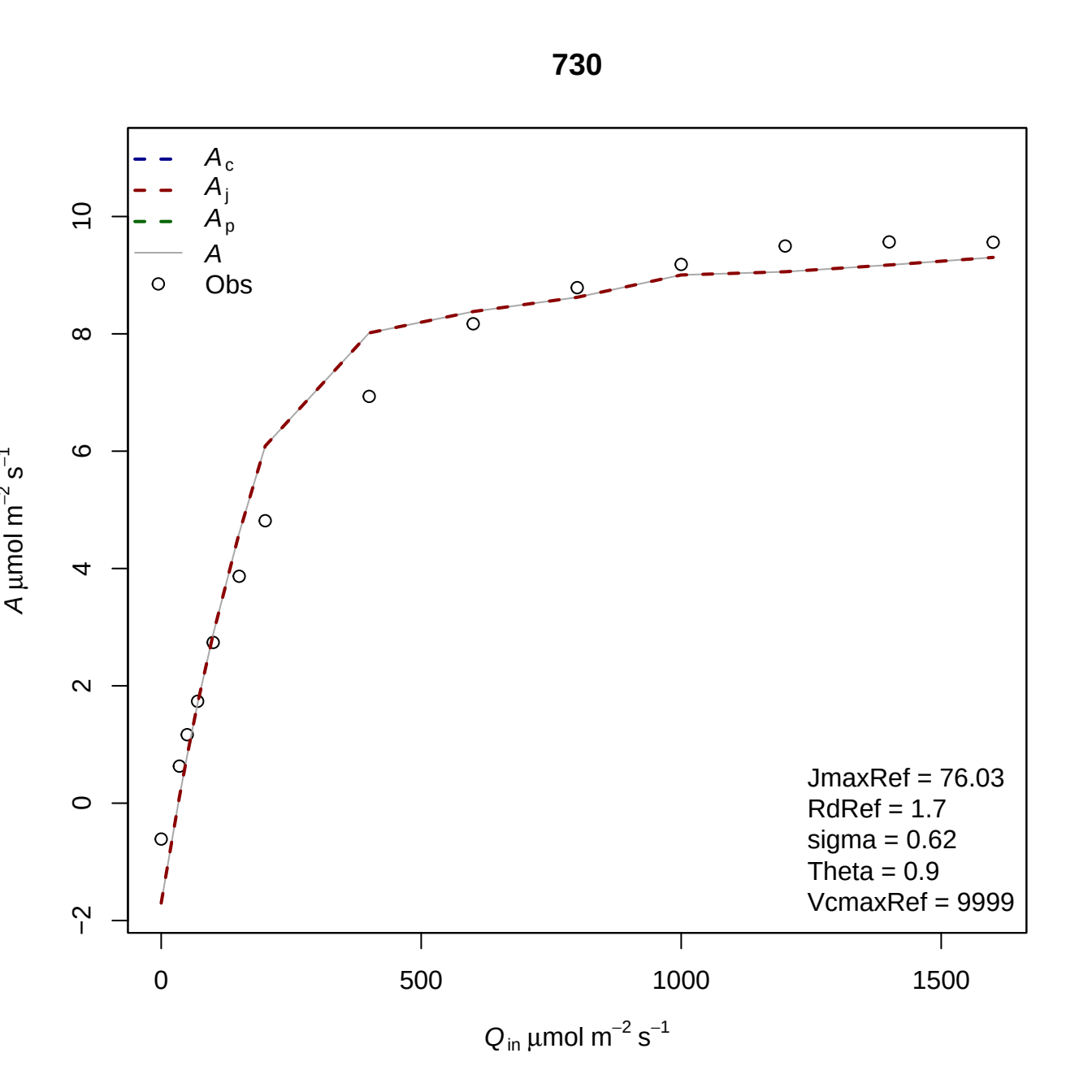

724
A_c
A_j
A_p
A
Obs
JmaxRef = 45.2
RdRef = 1.13
sigma = 0.61
Theta = 0.9
VcmaxRef = 33.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

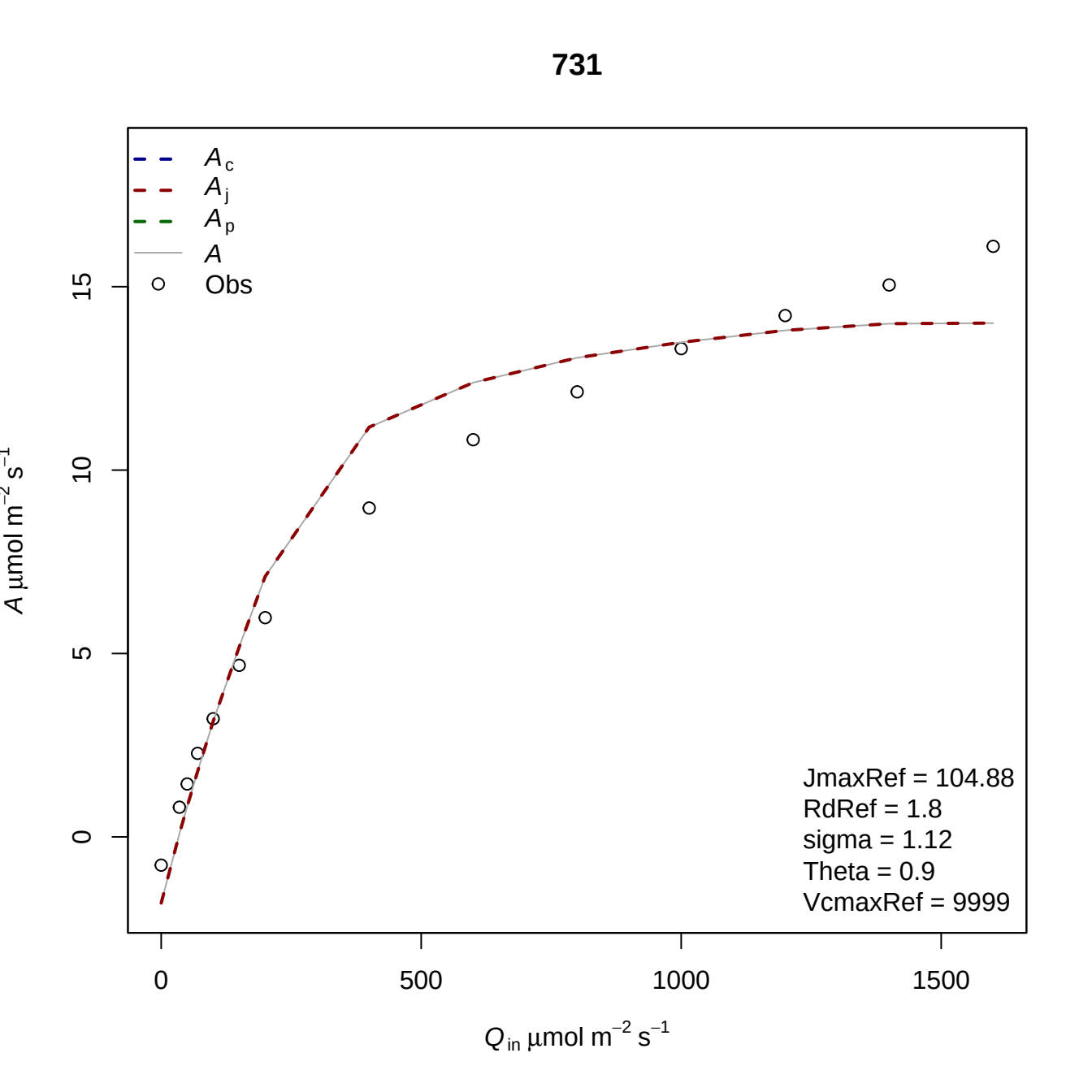

726
A_c
A_j
A_p
A
Obs
JmaxRef = 64.73
RdRef = 1.25
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

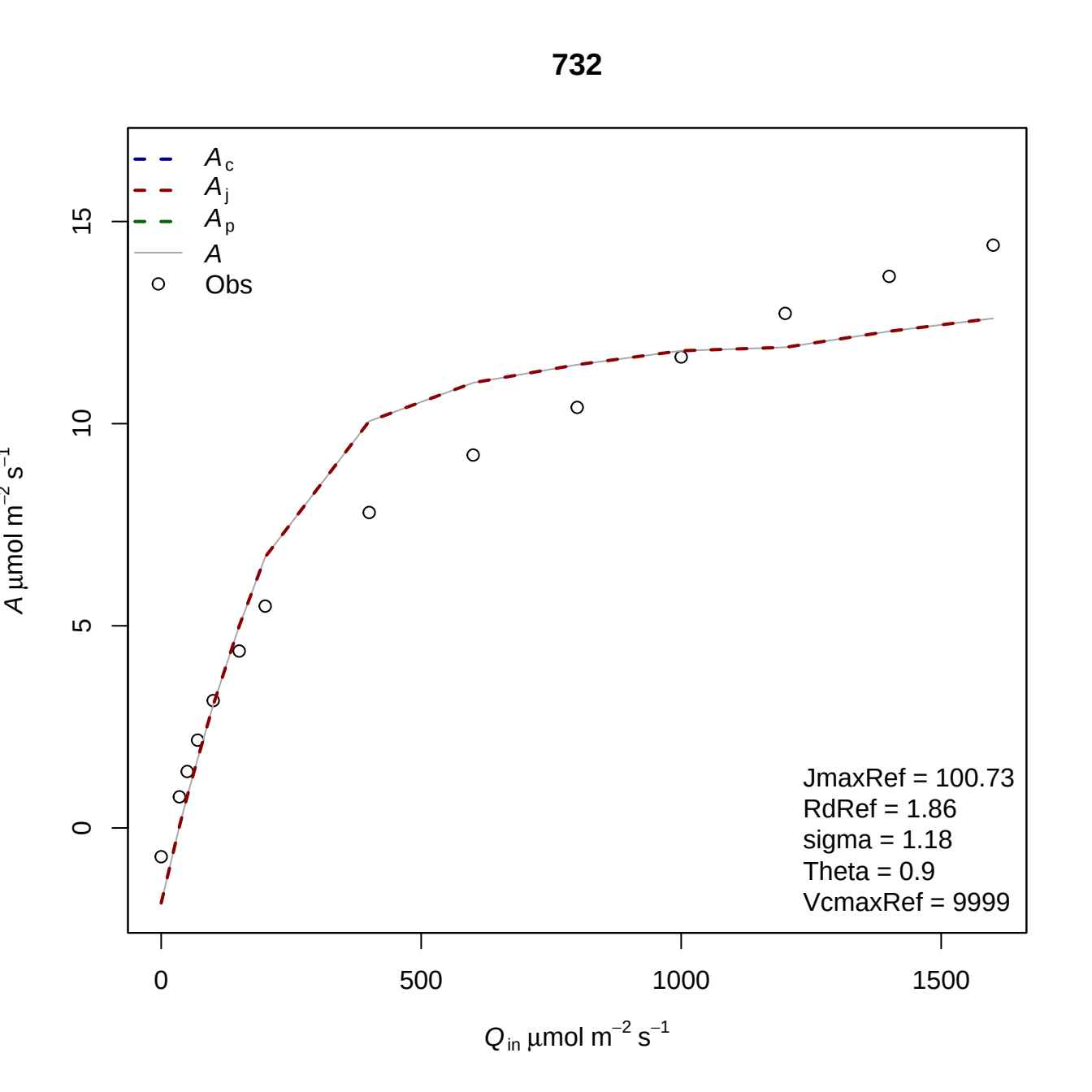

727
A_c
A_j
A_p
A
Obs
JmaxRef = 59.4
RdRef = 1.46
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

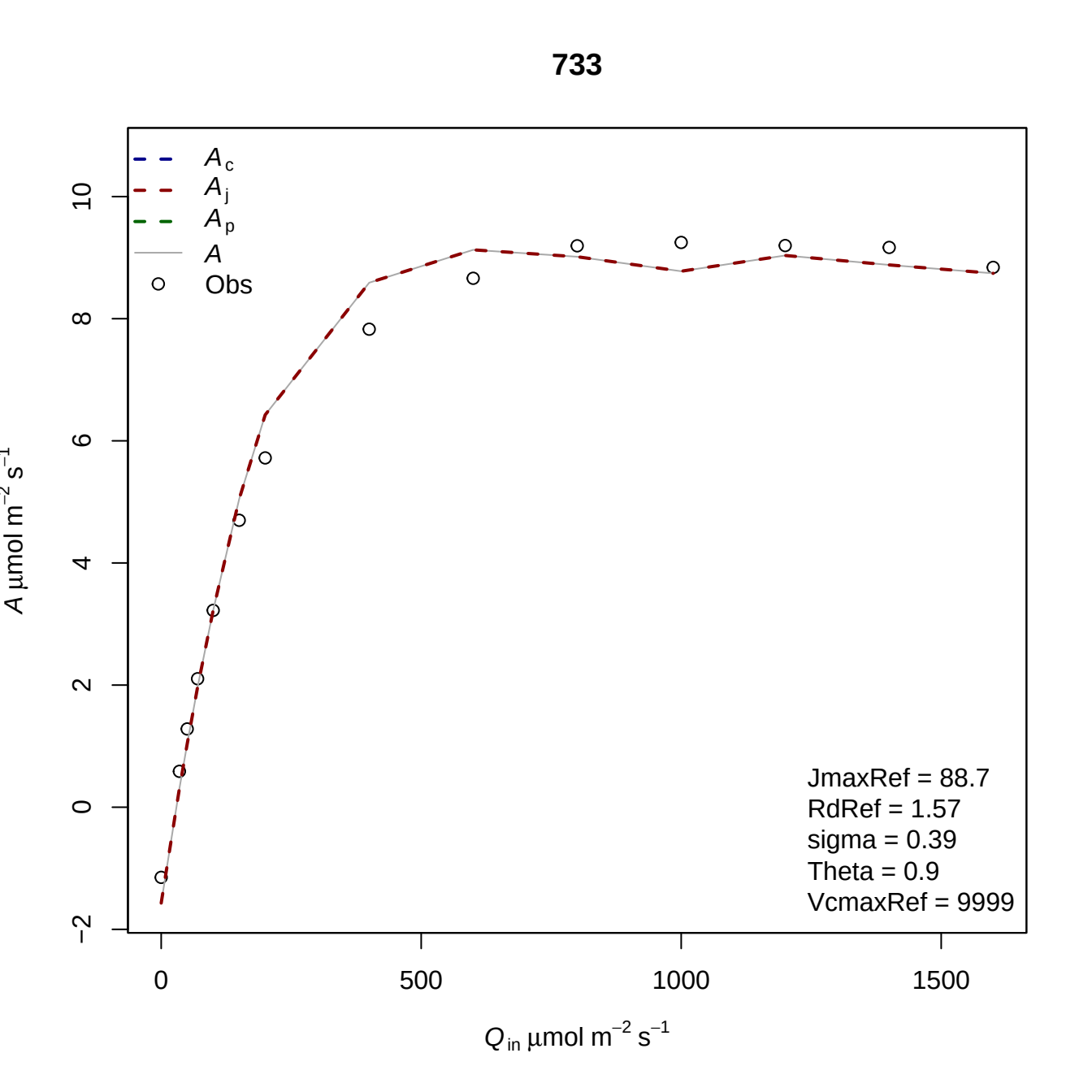

728
A_c
A_j
A_p
A
Obs
JmaxRef = 91.67
RdRef = 1.44
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

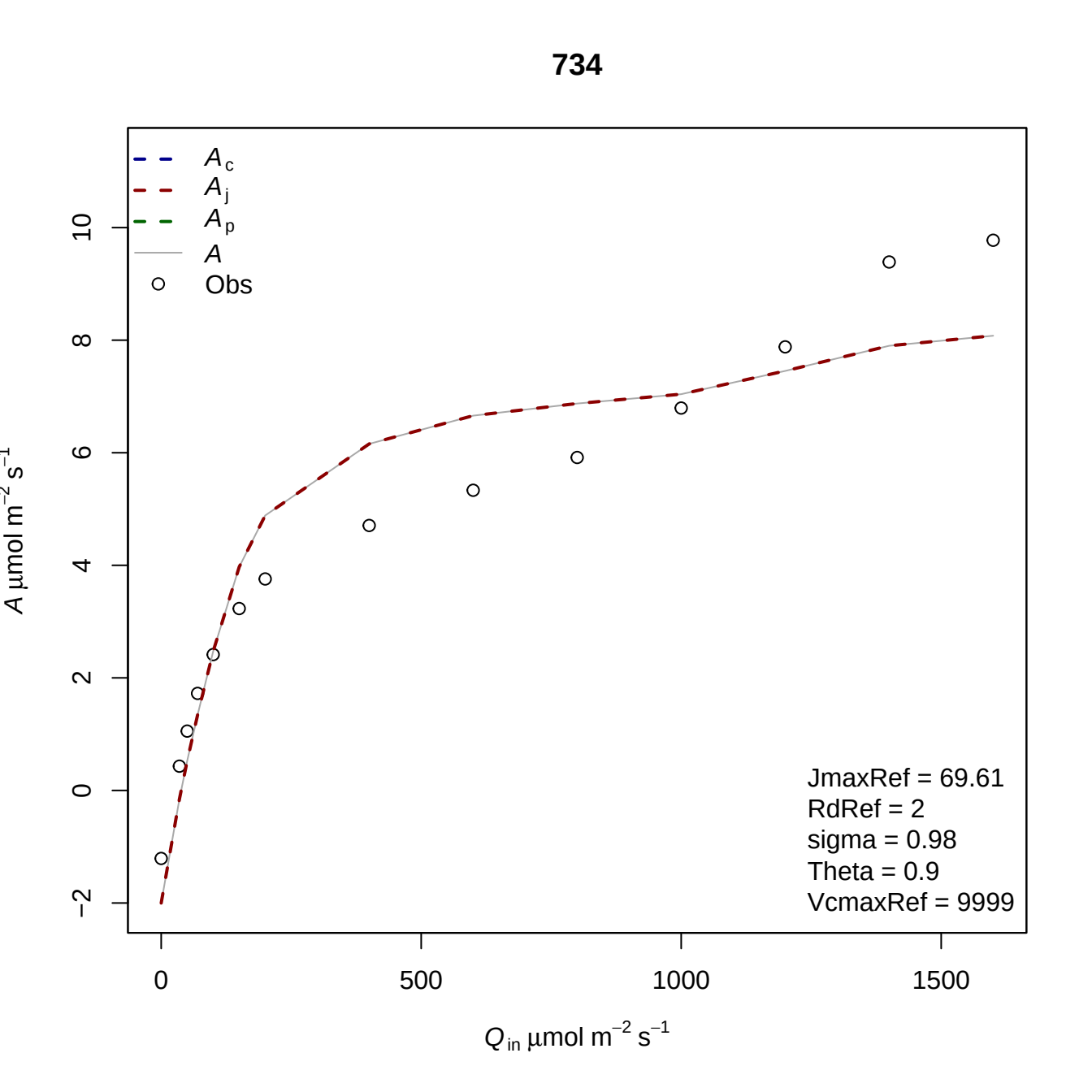

729
A_c
A_j
A_p
A
Obs
JmaxRef = 72.82
RdRef = 2.13
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

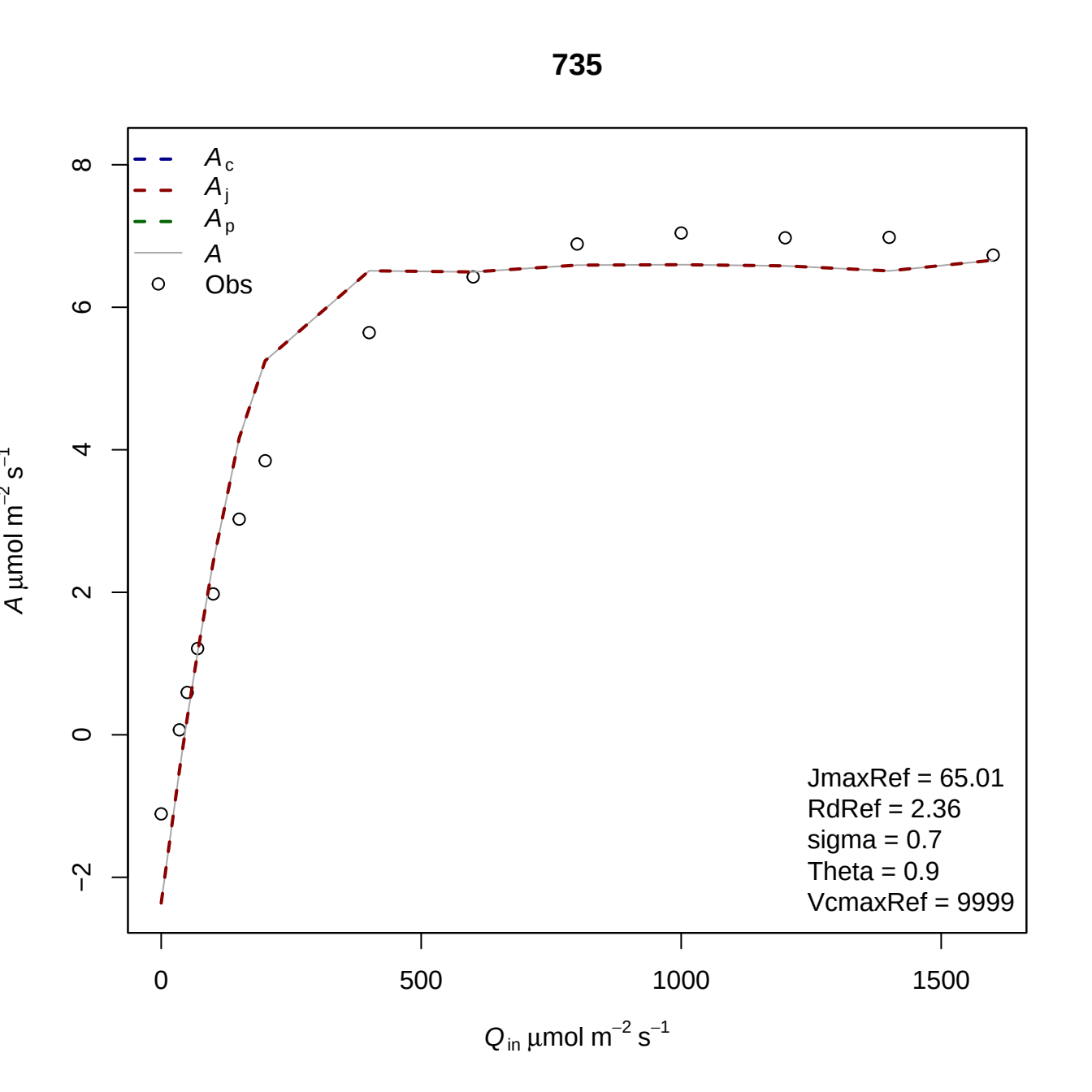

730
A_c
A_j
A_p
A
Obs
JmaxRef = 76.03
RdRef = 1.7
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

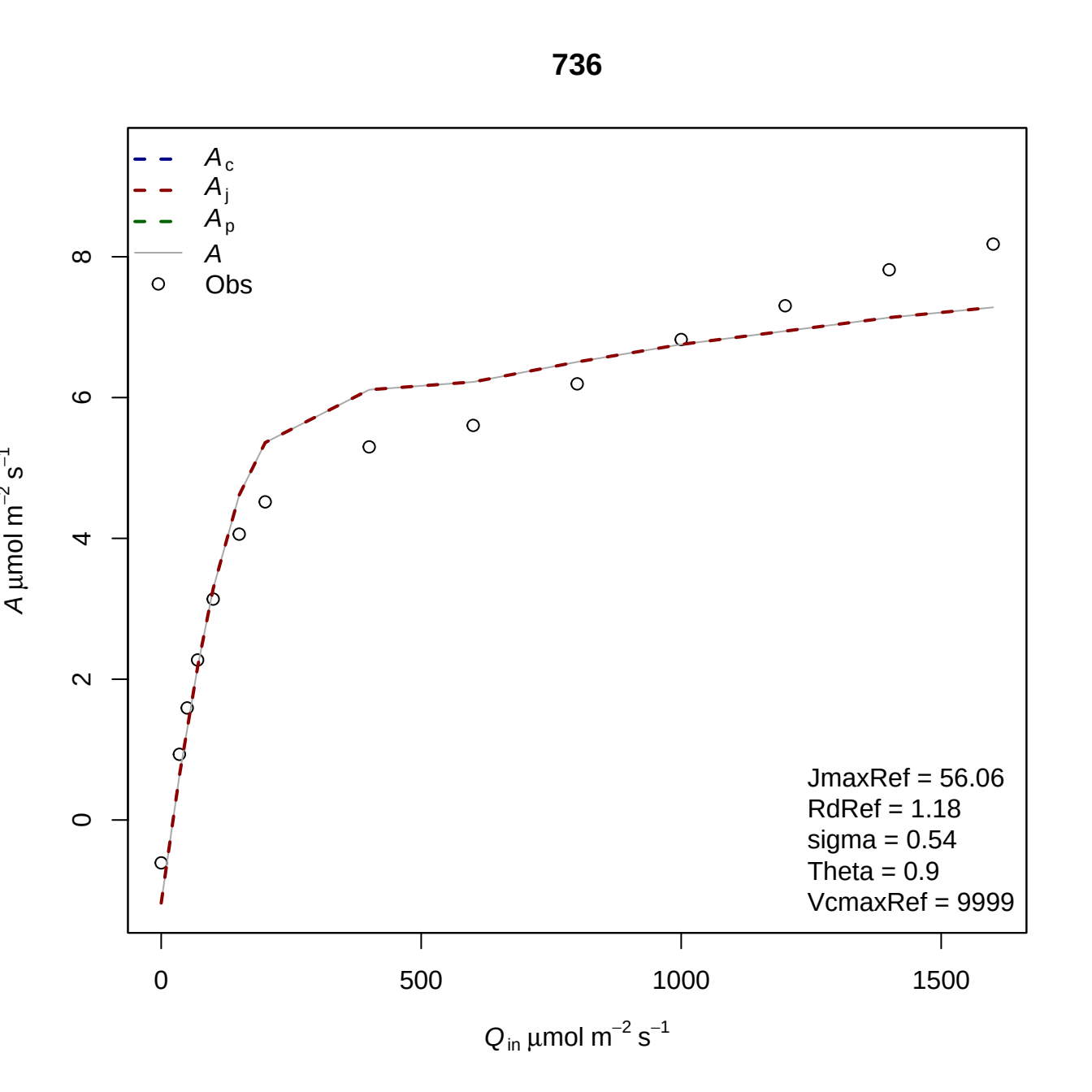

731
A_c
A_j
A_p
A
Obs
JmaxRef = 104.88
RdRef = 1.8
sigma = 1.12
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

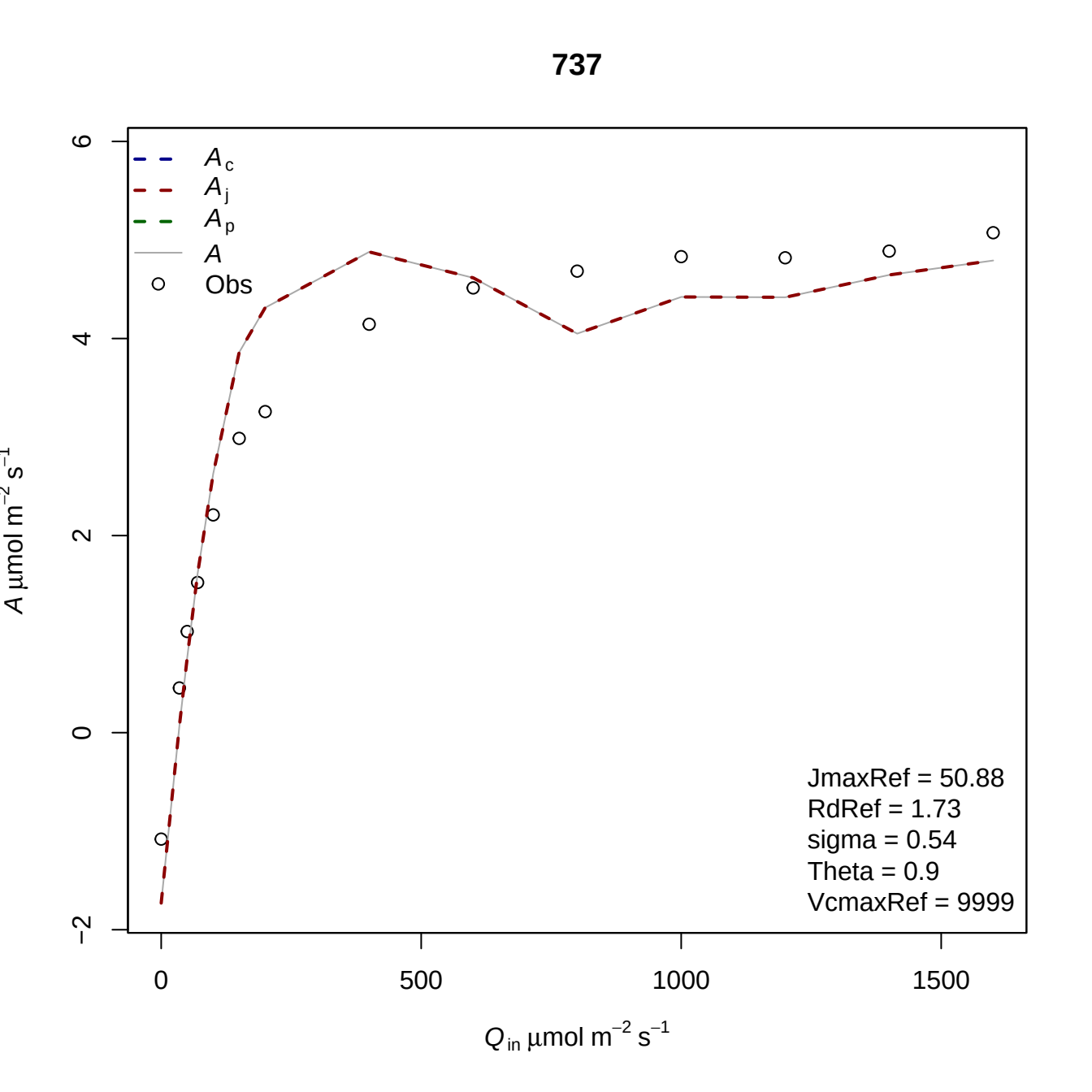

732
A_c
A_j
A_p
A
Obs
JmaxRef = 100.73
RdRef = 1.86
sigma = 1.18
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

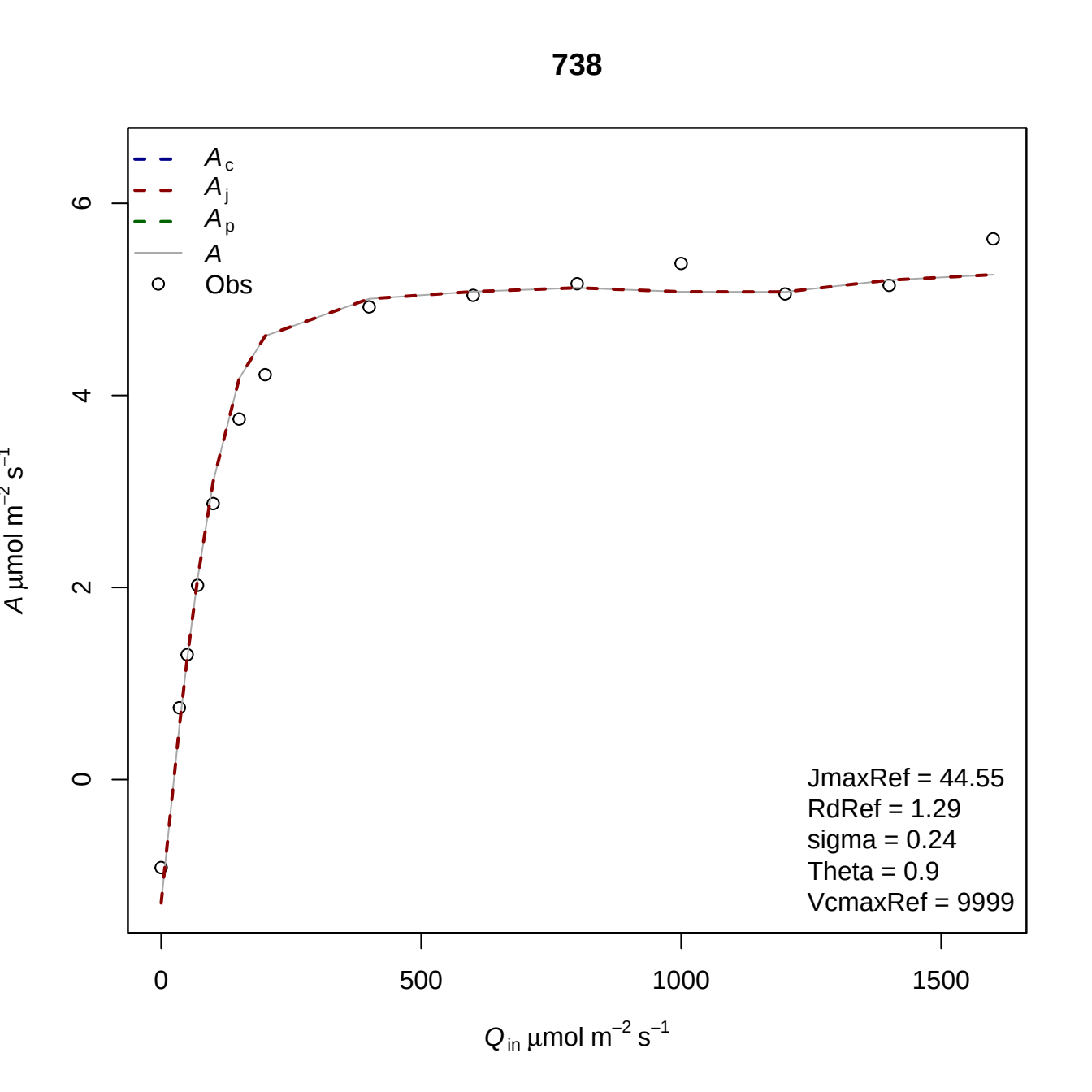

733
A_c
A_j
A_p
A
Obs
JmaxRef = 88.7
RdRef = 1.57
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

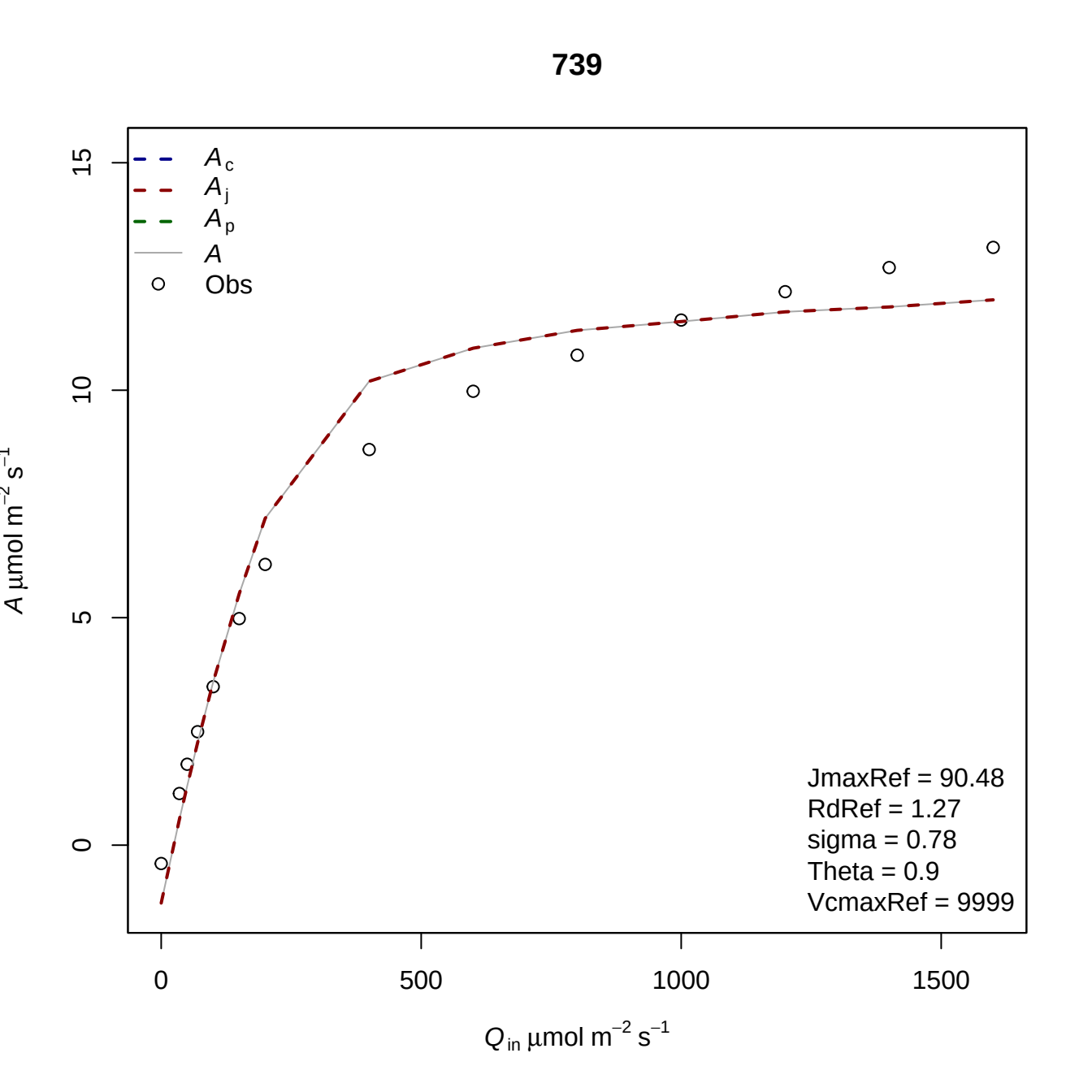

734
A_c
A_j
A_p
A
Obs
JmaxRef = 69.61
RdRef = 2
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

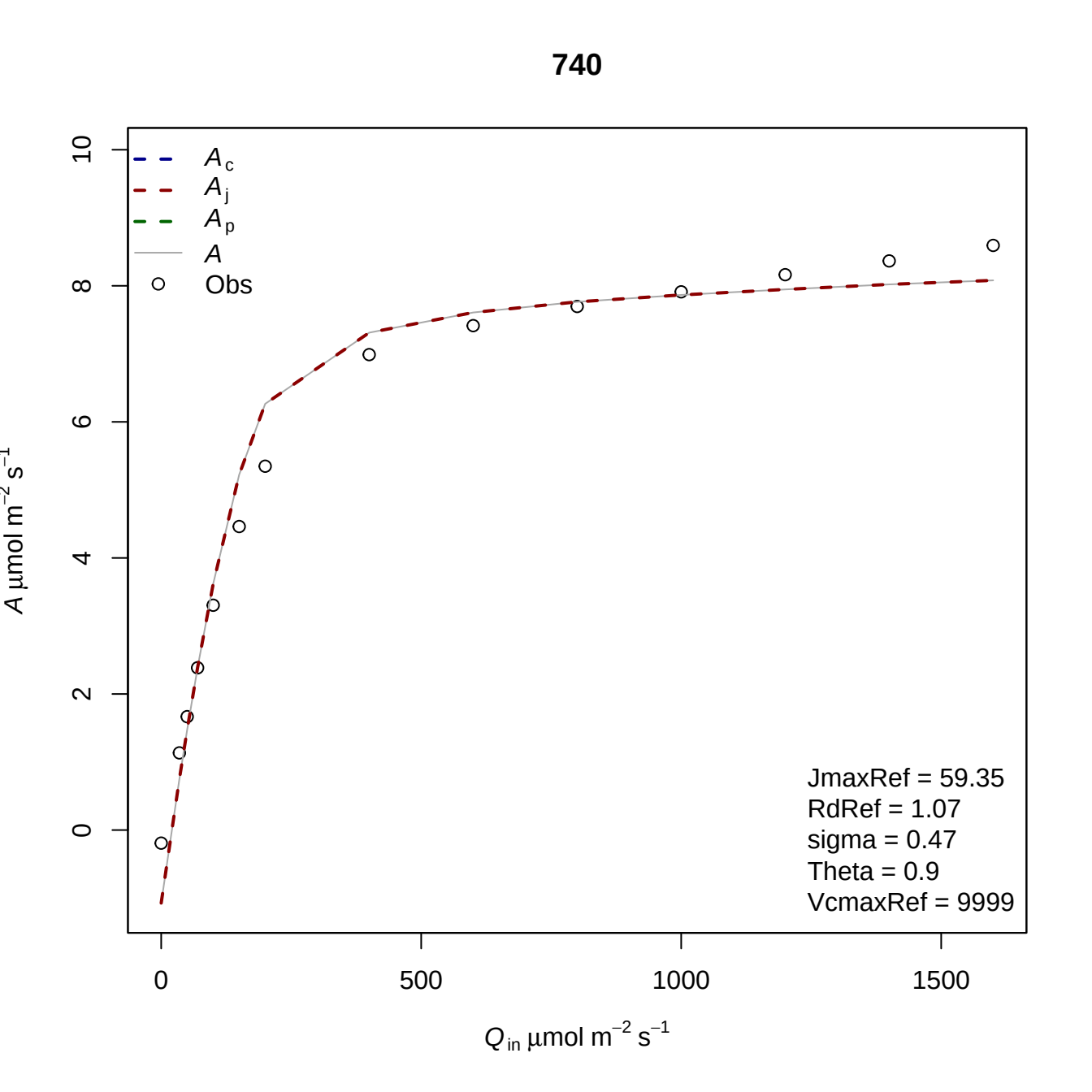

735
A_c
A_j
A_p
A
Obs
JmaxRef = 65.01
RdRef = 2.36
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

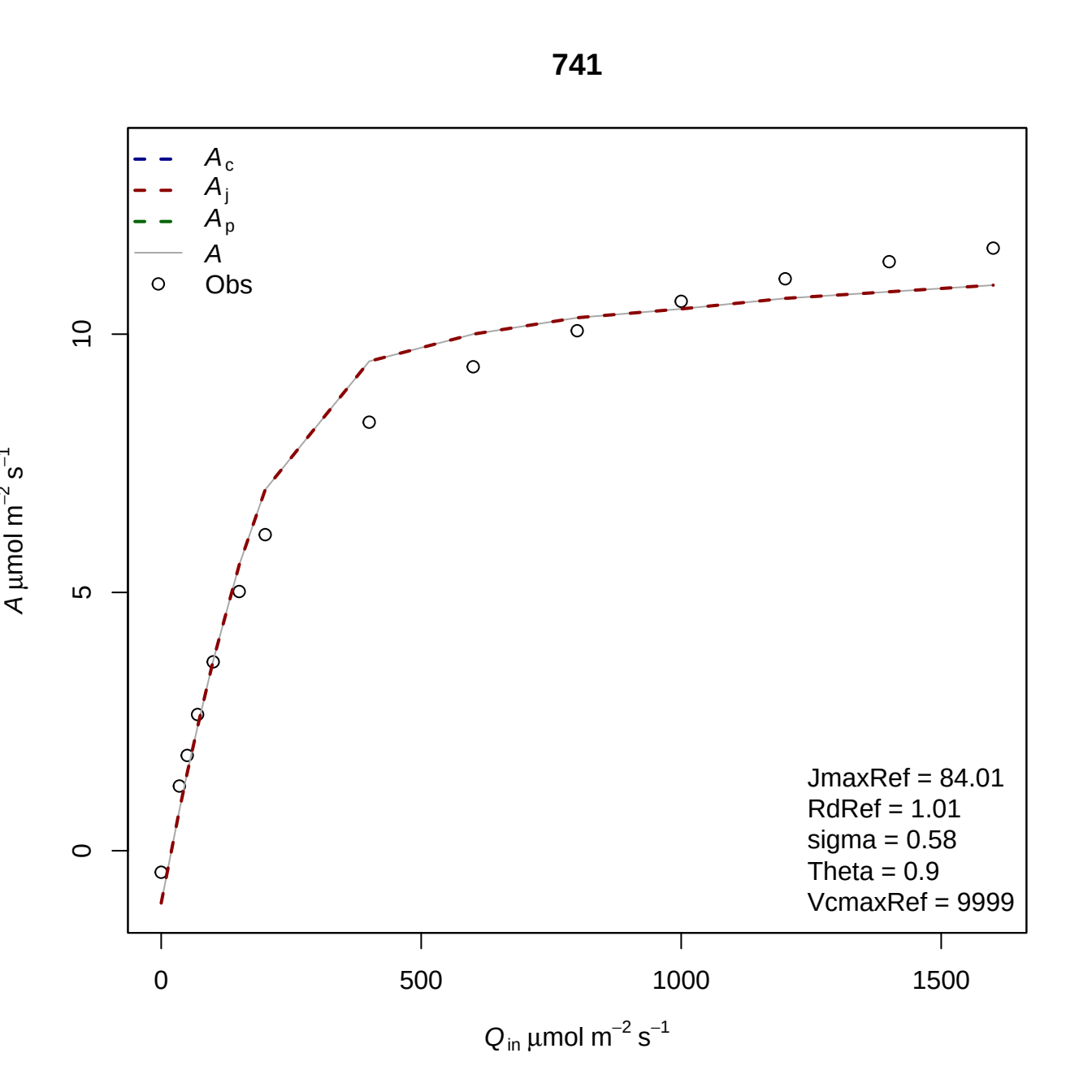

736
A_c
A_j
A_p
A
Obs
JmaxRef = 56.06
RdRef = 1.18
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

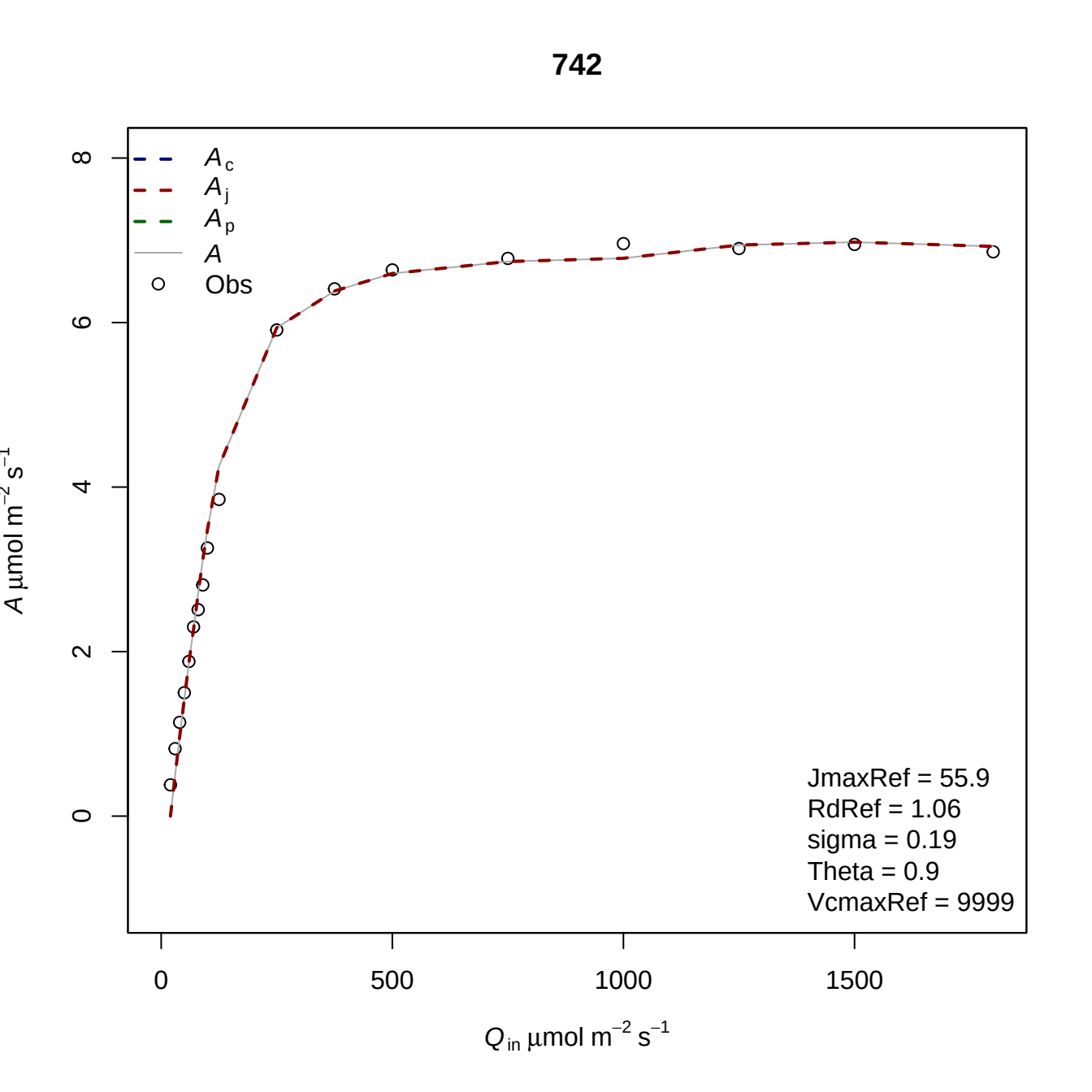

737
A_c
A_j
A_p
A
Obs
JmaxRef = 50.88
RdRef = 1.73
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

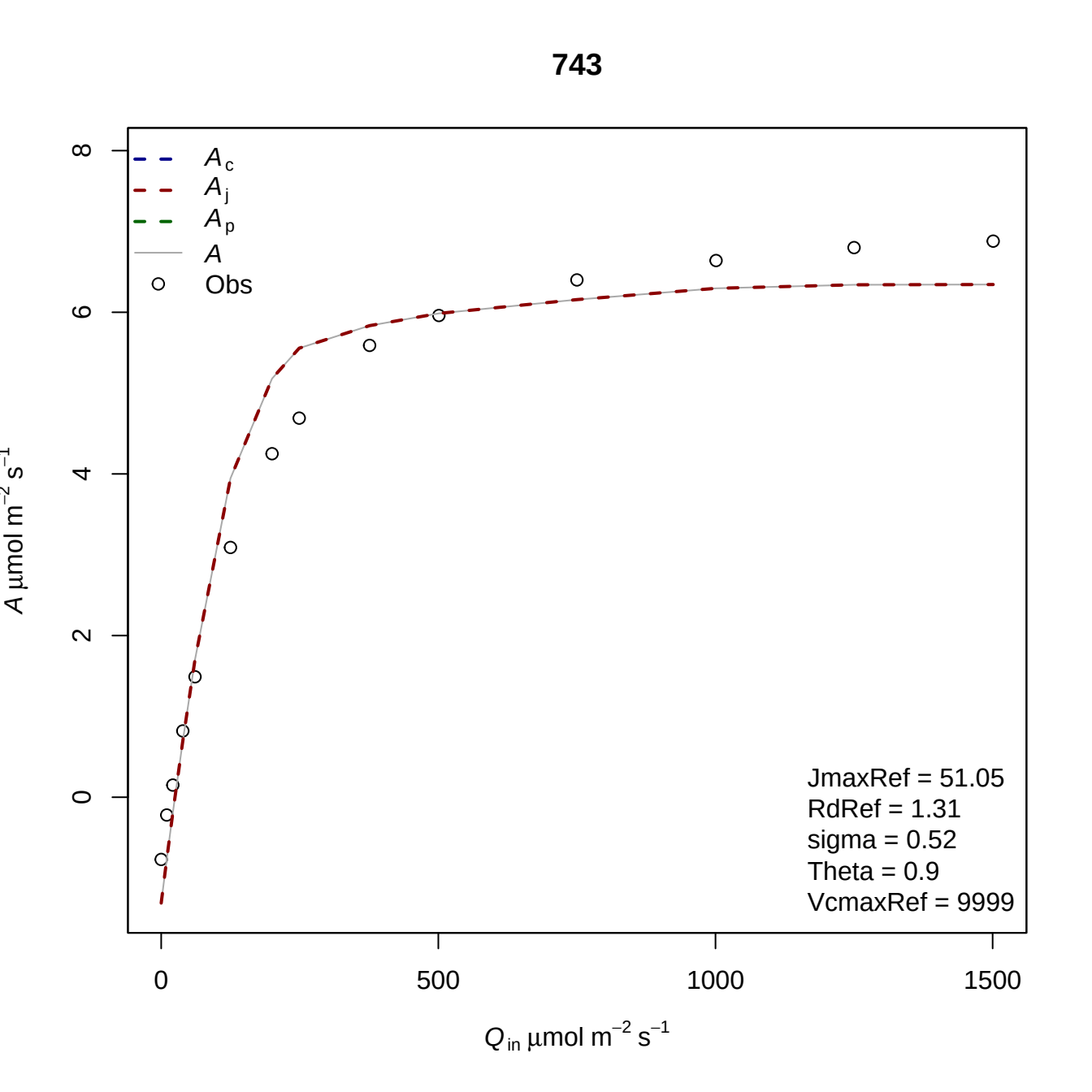

738
A_c
A_j
A_p
A
Obs
JmaxRef = 44.55
RdRef = 1.29
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

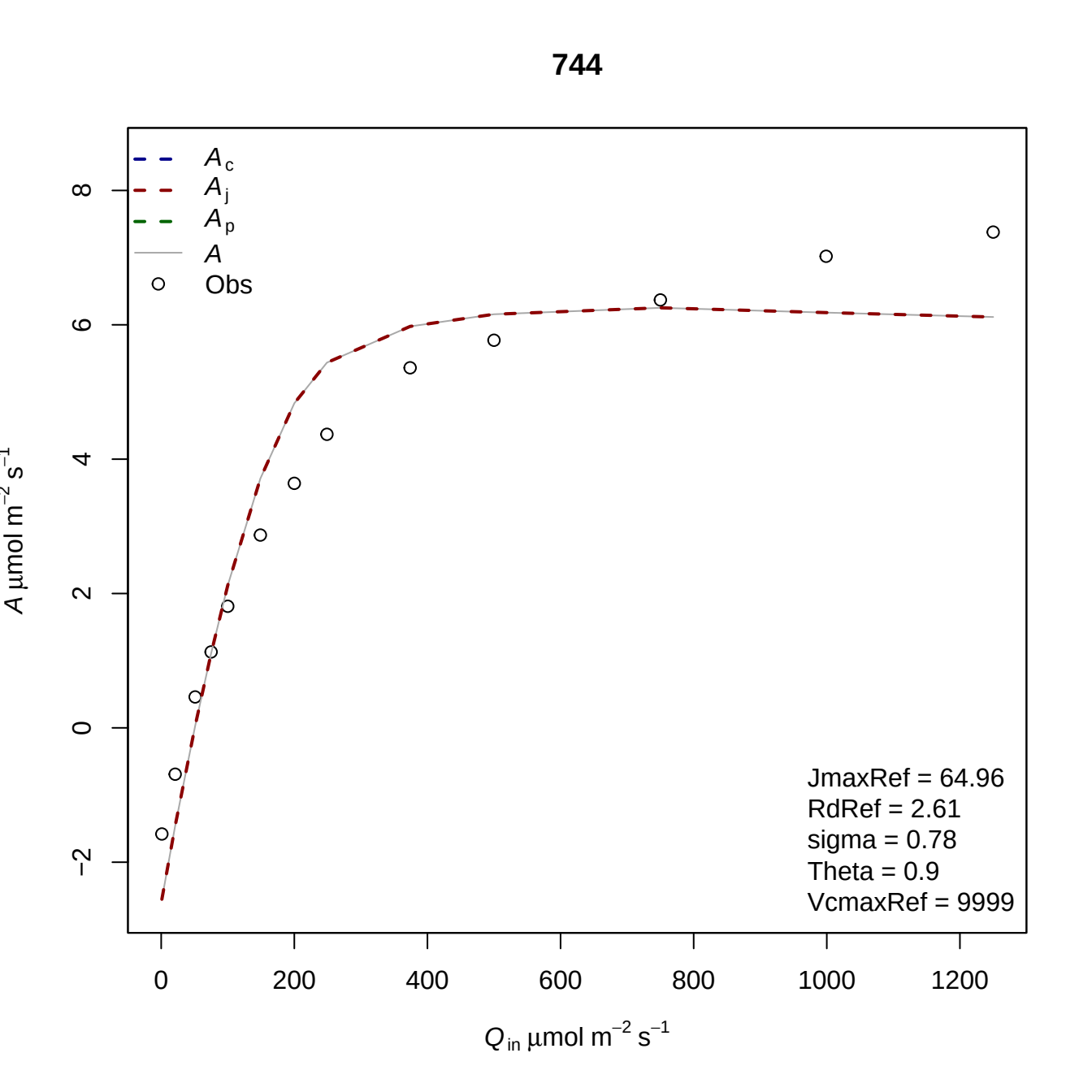

739
A_c
A_j
A_p
A
Obs
JmaxRef = 90.48
RdRef = 1.27
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

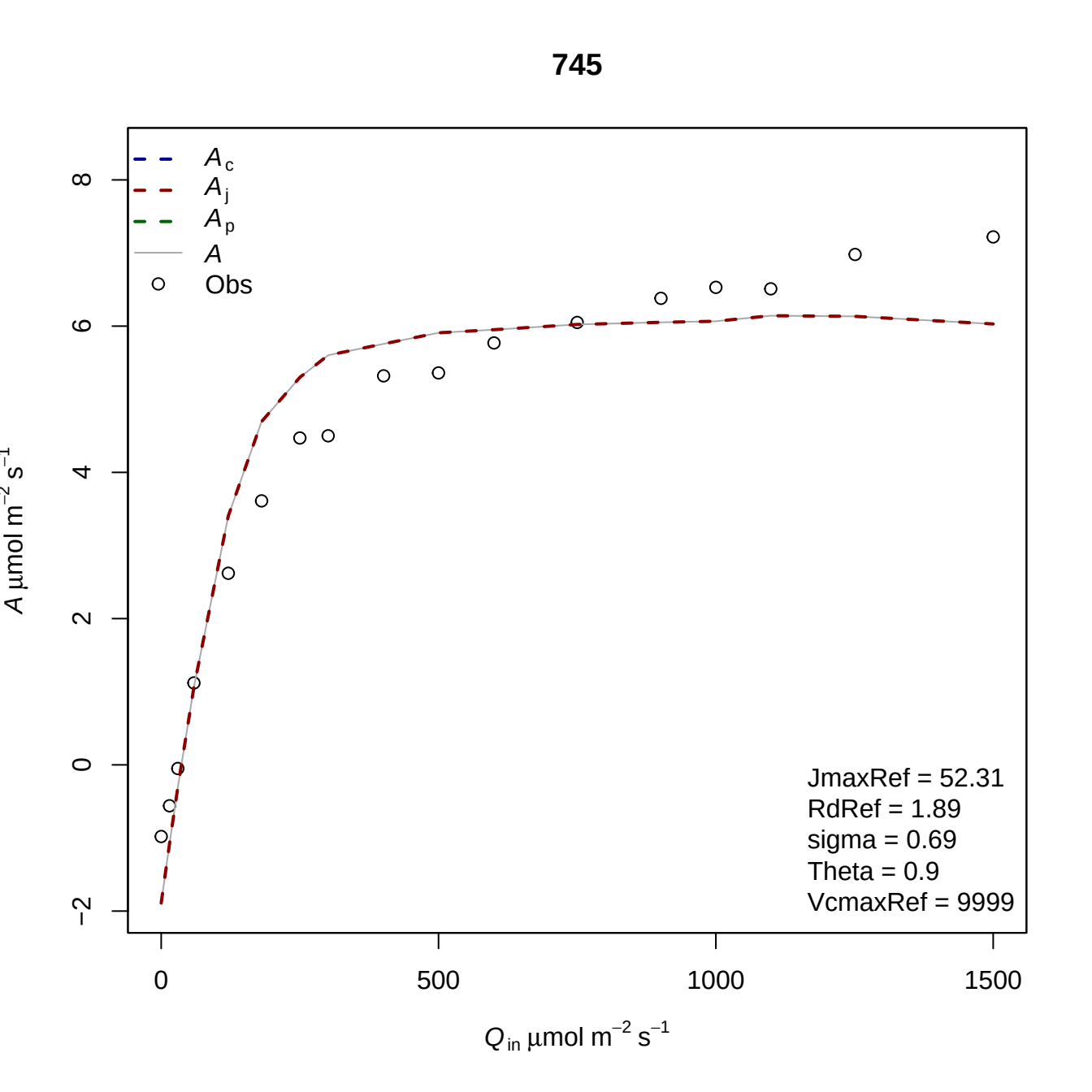

740
A_c
A_j
A_p
A
Obs
JmaxRef = 59.35
RdRef = 1.07
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

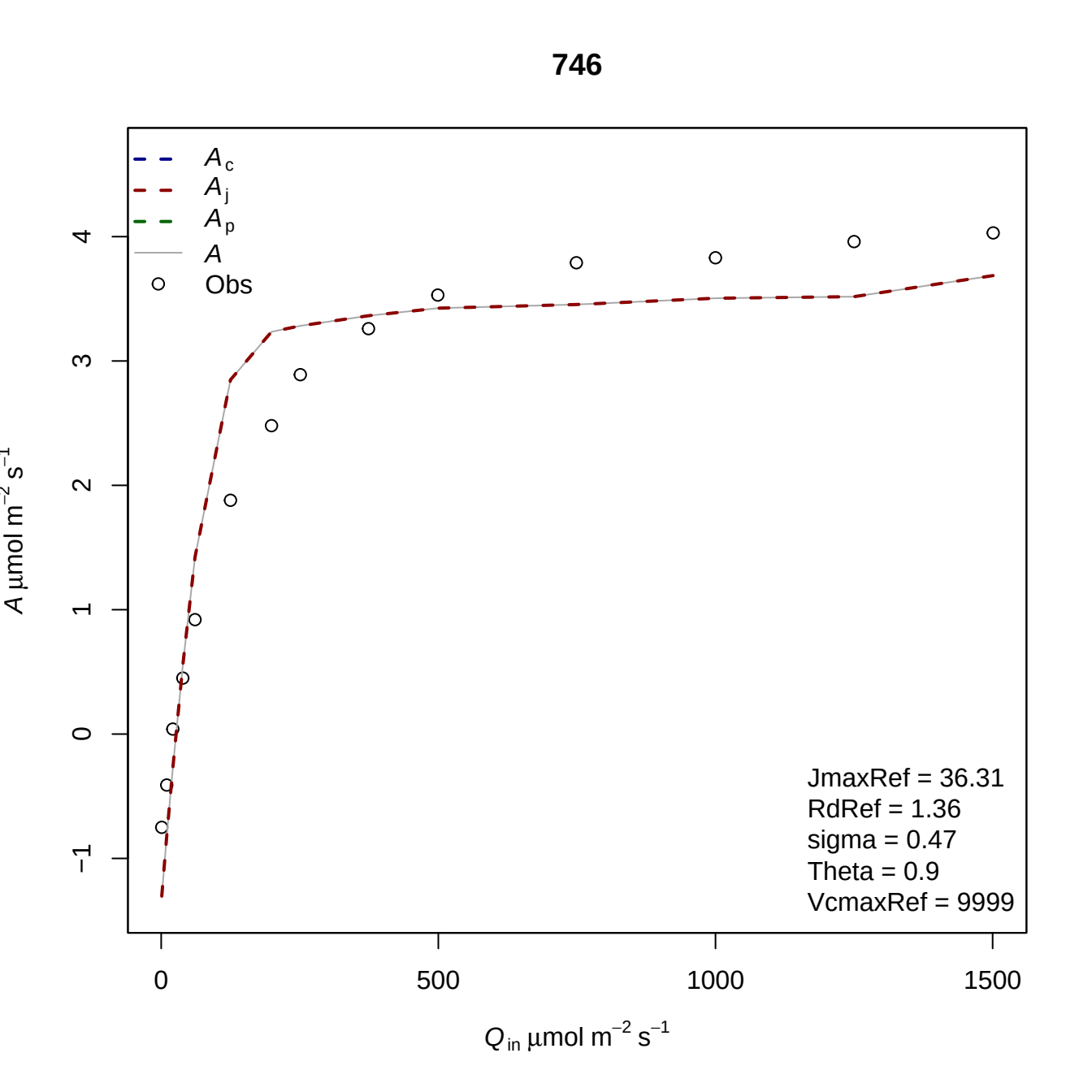

741
A_c
A_j
A_p
A
Obs
JmaxRef = 84.01
RdRef = 1.01
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

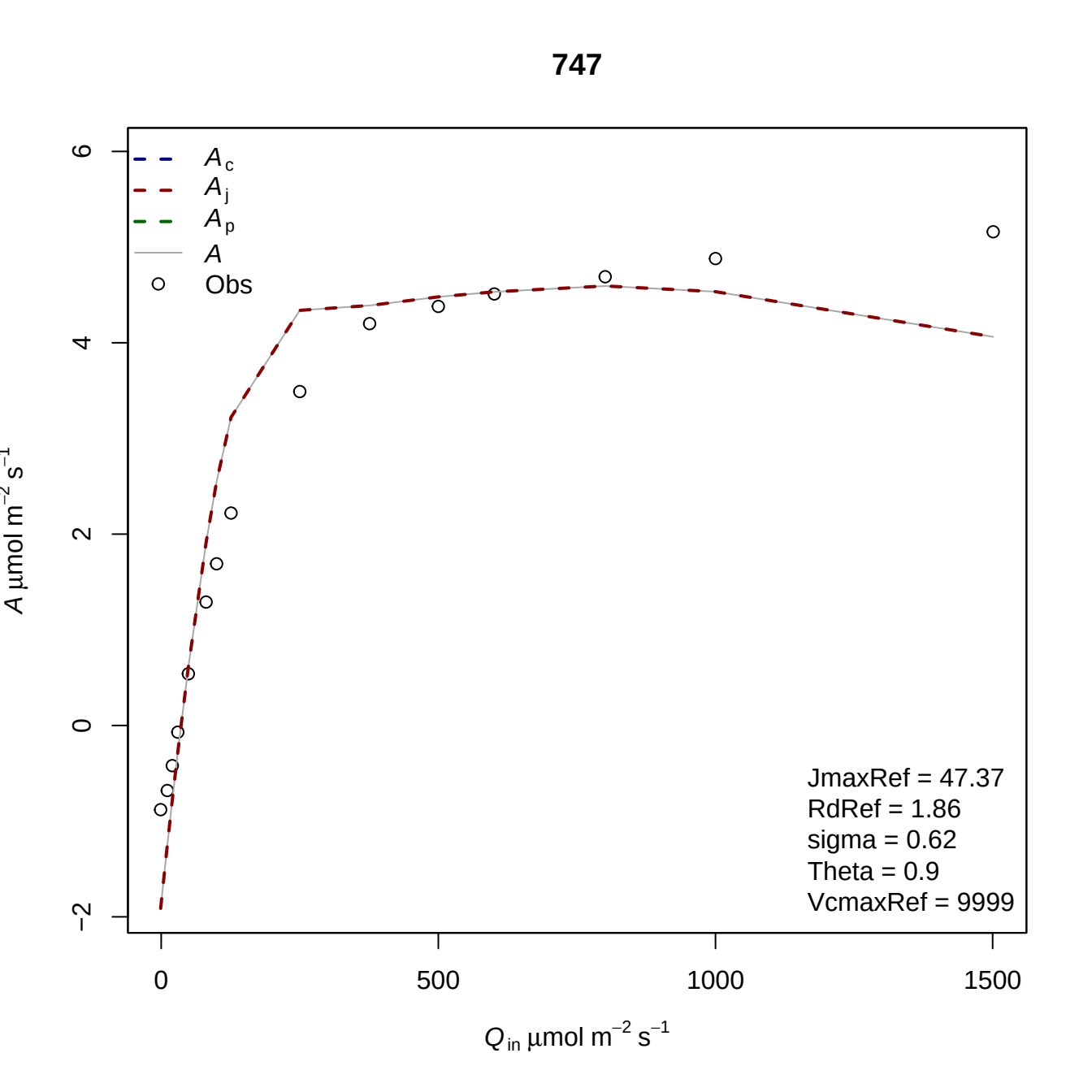

742
A_c
A_j
A_p
A
Obs
JmaxRef = 55.9
RdRef = 1.06
sigma = 0.19
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

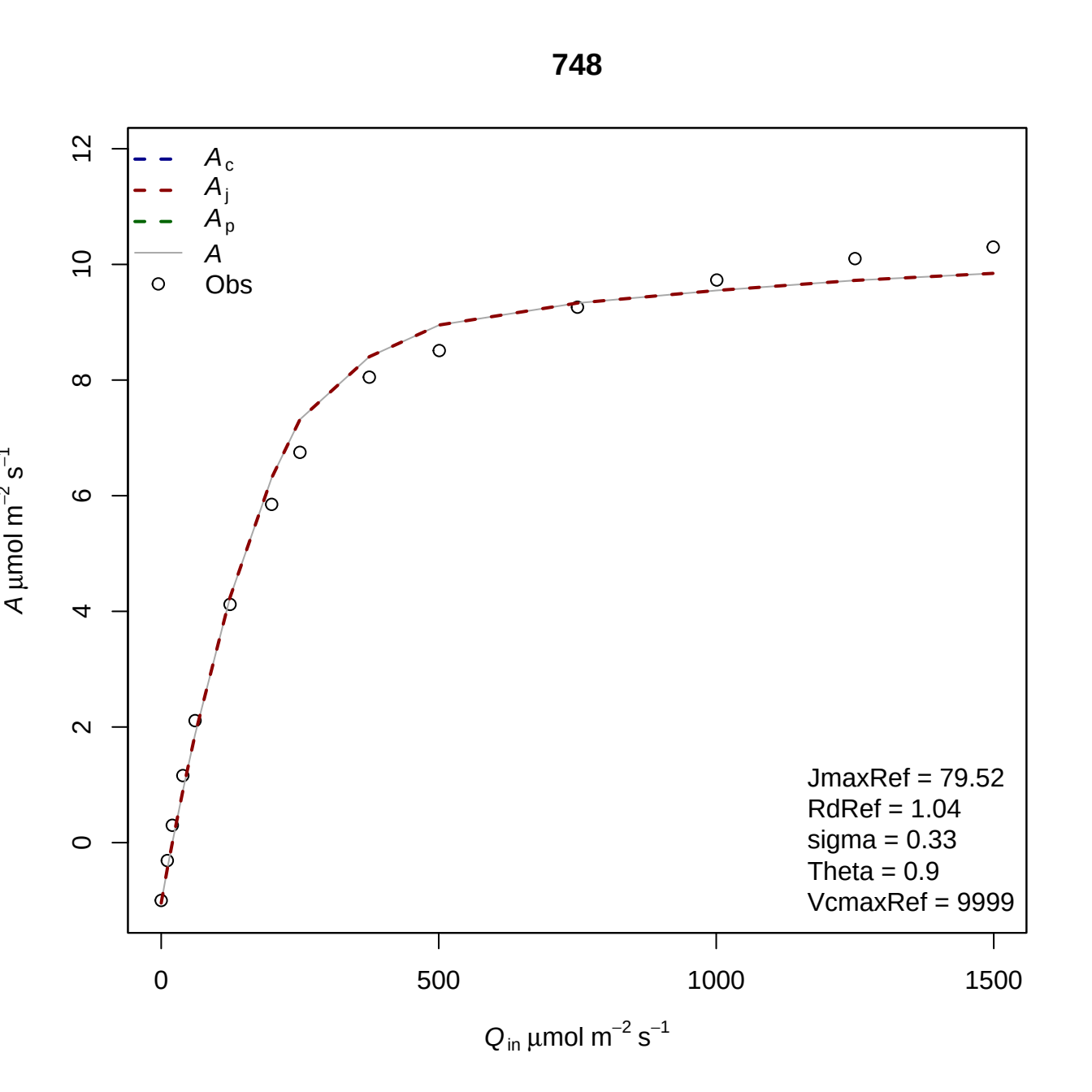

743
A_c
A_j
A_p
A
Obs
JmaxRef = 51.05
RdRef = 1.31
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

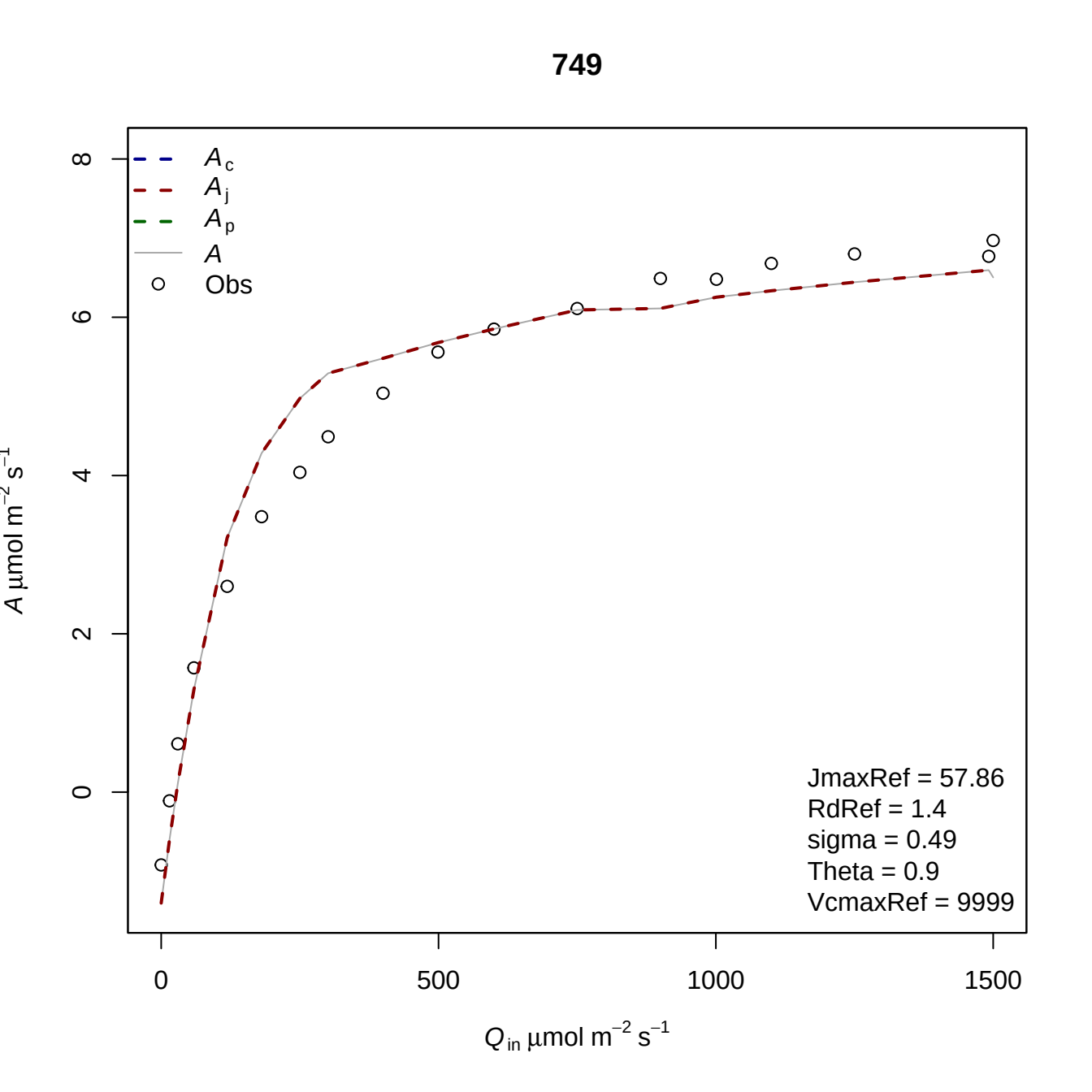

744
A_c
A_j
A_p
A
Obs
JmaxRef = 64.96
RdRef = 2.61
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

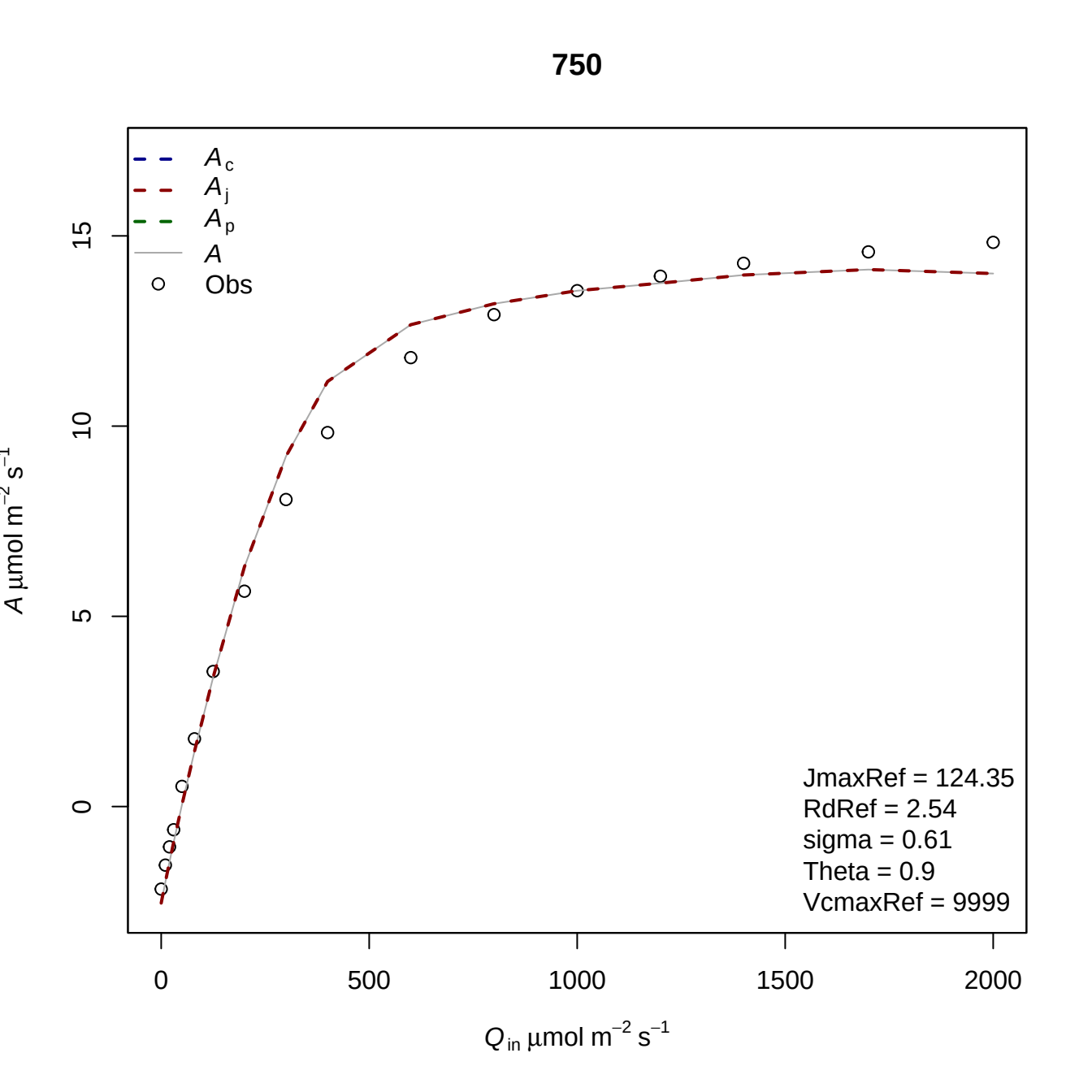

745
A_c
A_j
A_p
A
Obs
JmaxRef = 52.31
RdRef = 1.89
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

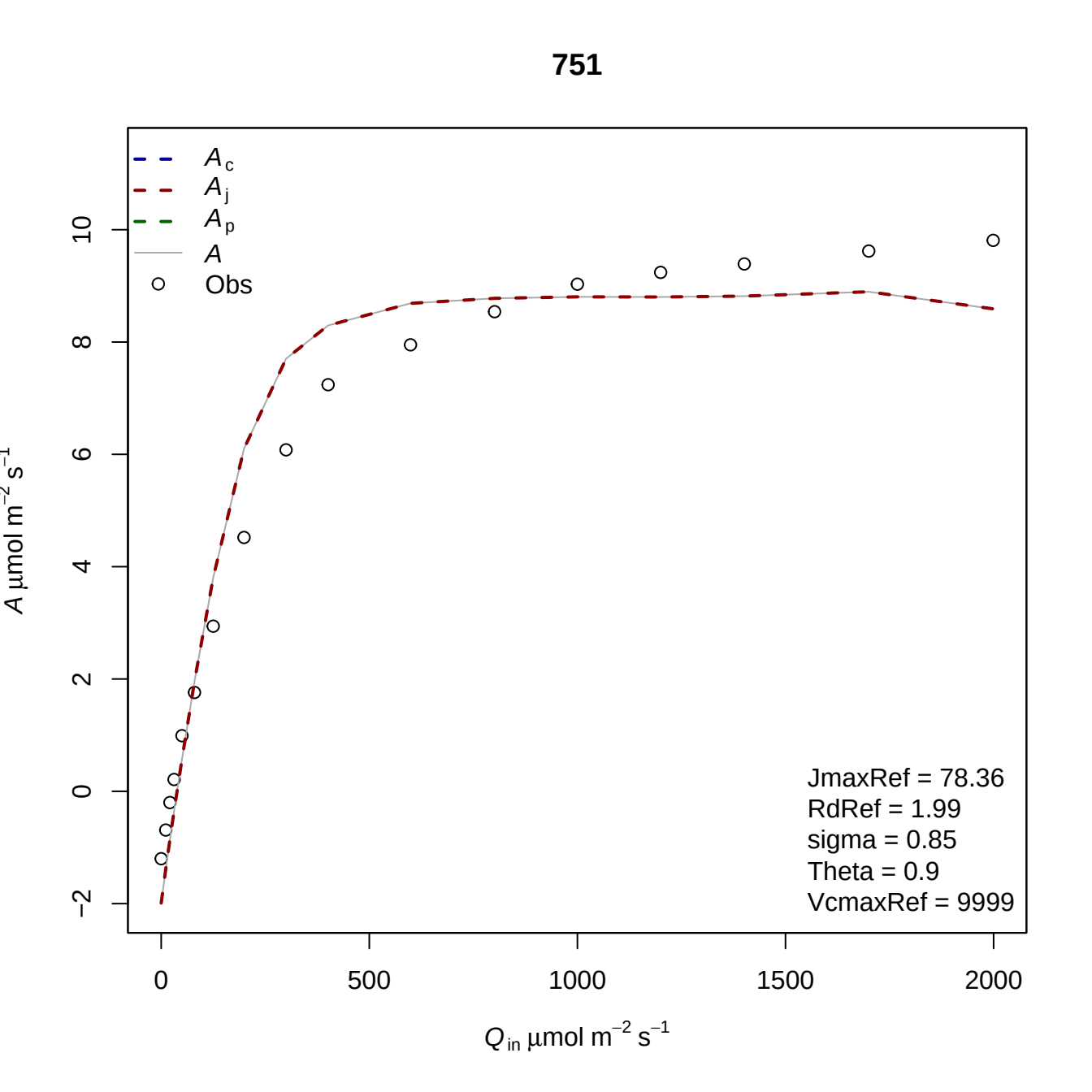

746
A_c
A_j
A_p
A
Obs
JmaxRef = 36.31
RdRef = 1.36
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

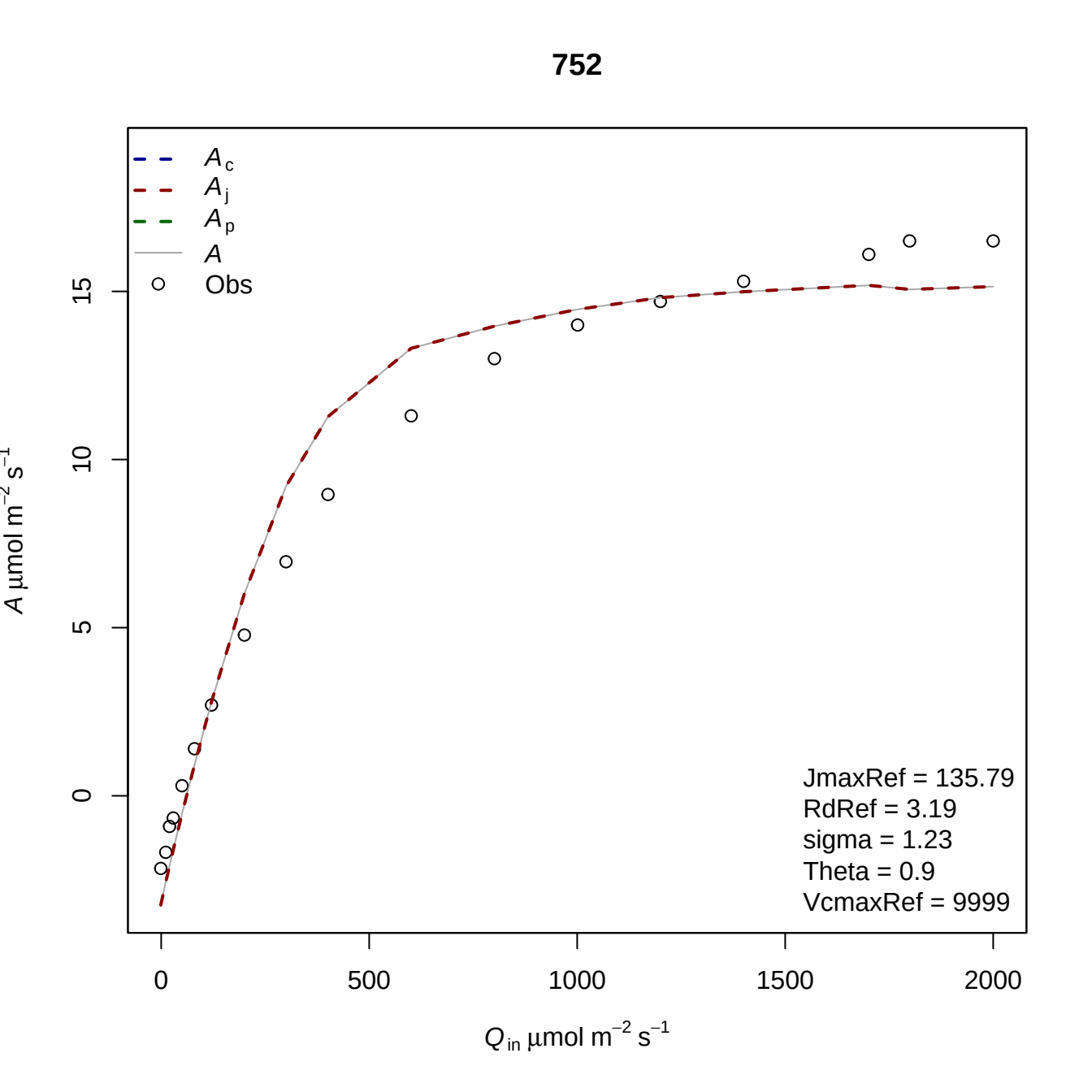

747
A_c
A_j
A_p
A
Obs
JmaxRef = 47.37
RdRef = 1.86
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

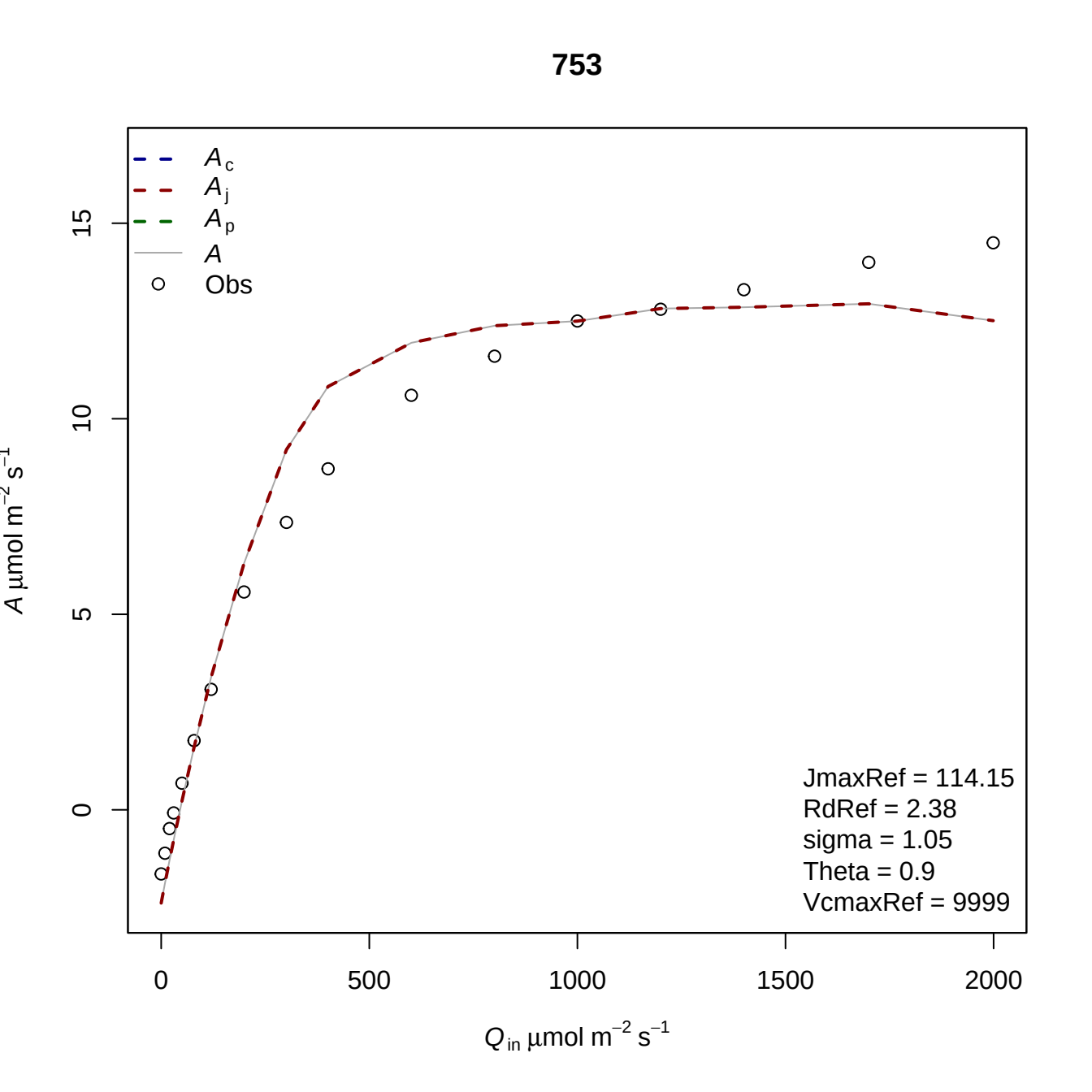

748
A_c
A_j
A_p
A
Obs
JmaxRef = 79.52
RdRef = 1.04
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

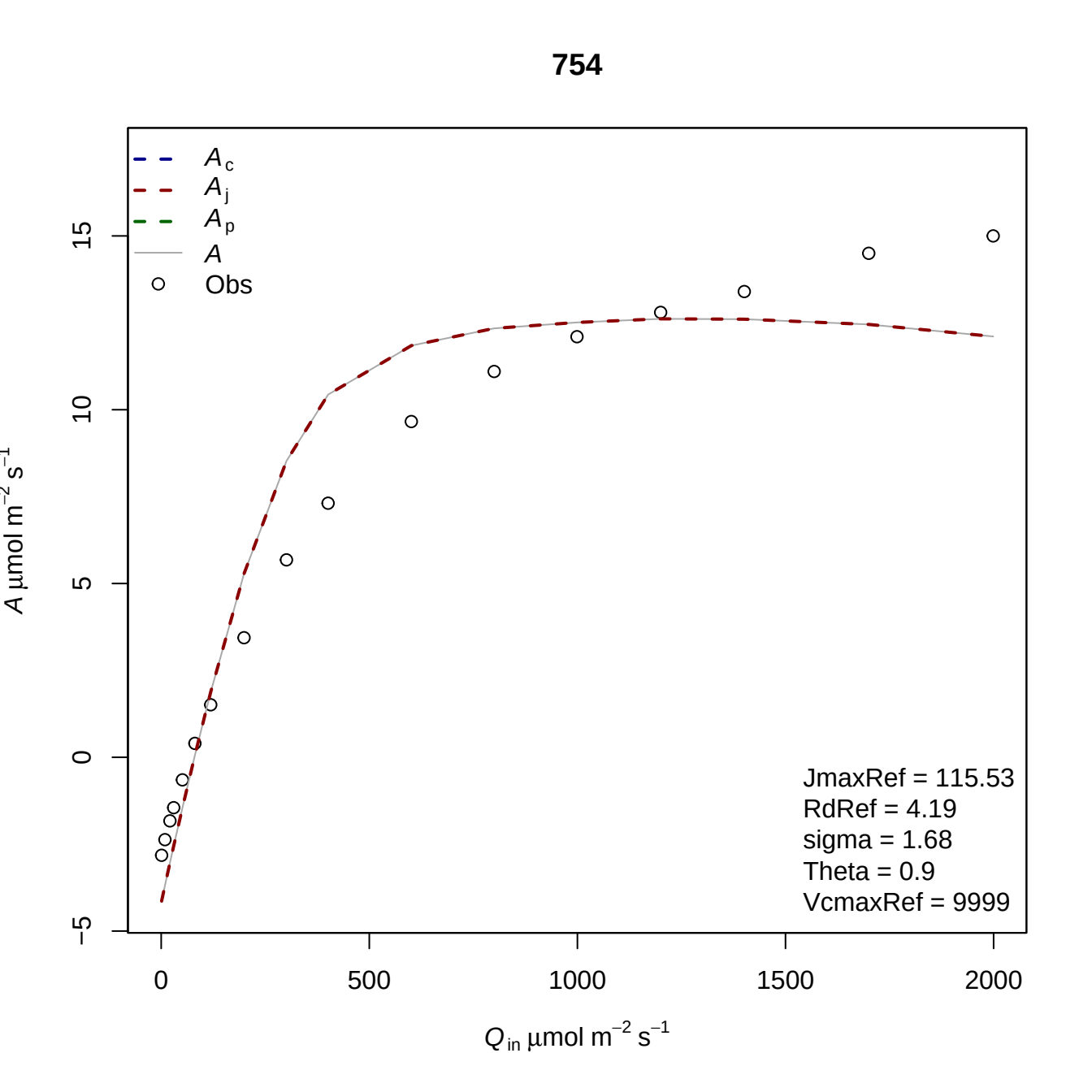

749
A_c
A_j
A_p
A
Obs
JmaxRef = 57.86
RdRef = 1.4
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

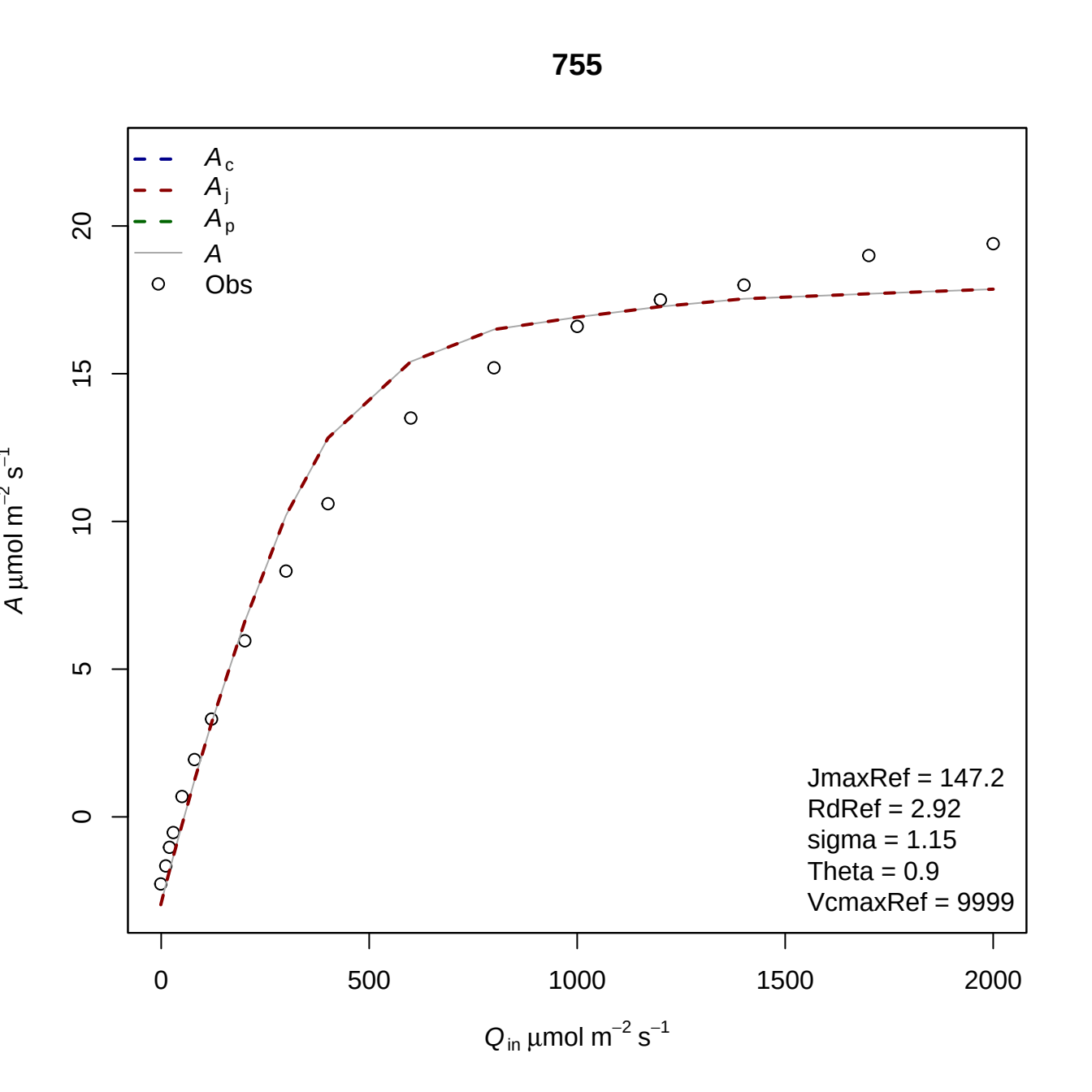

750
A_c
A_j
A_p
A
Obs
JmaxRef = 124.35
RdRef = 2.54
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

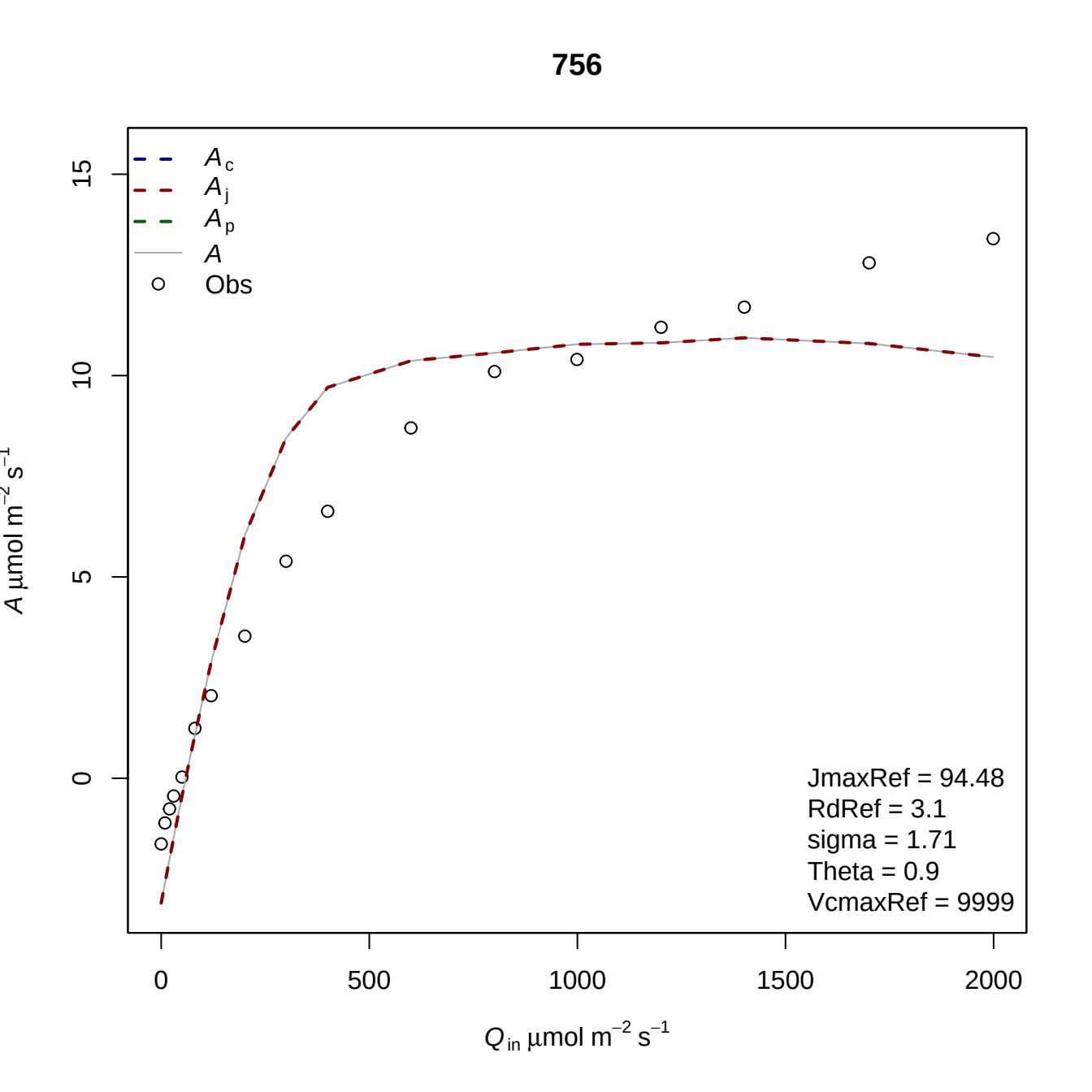

751
A_c
A_j
A_p
A
Obs
JmaxRef = 78.36
RdRef = 1.99
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

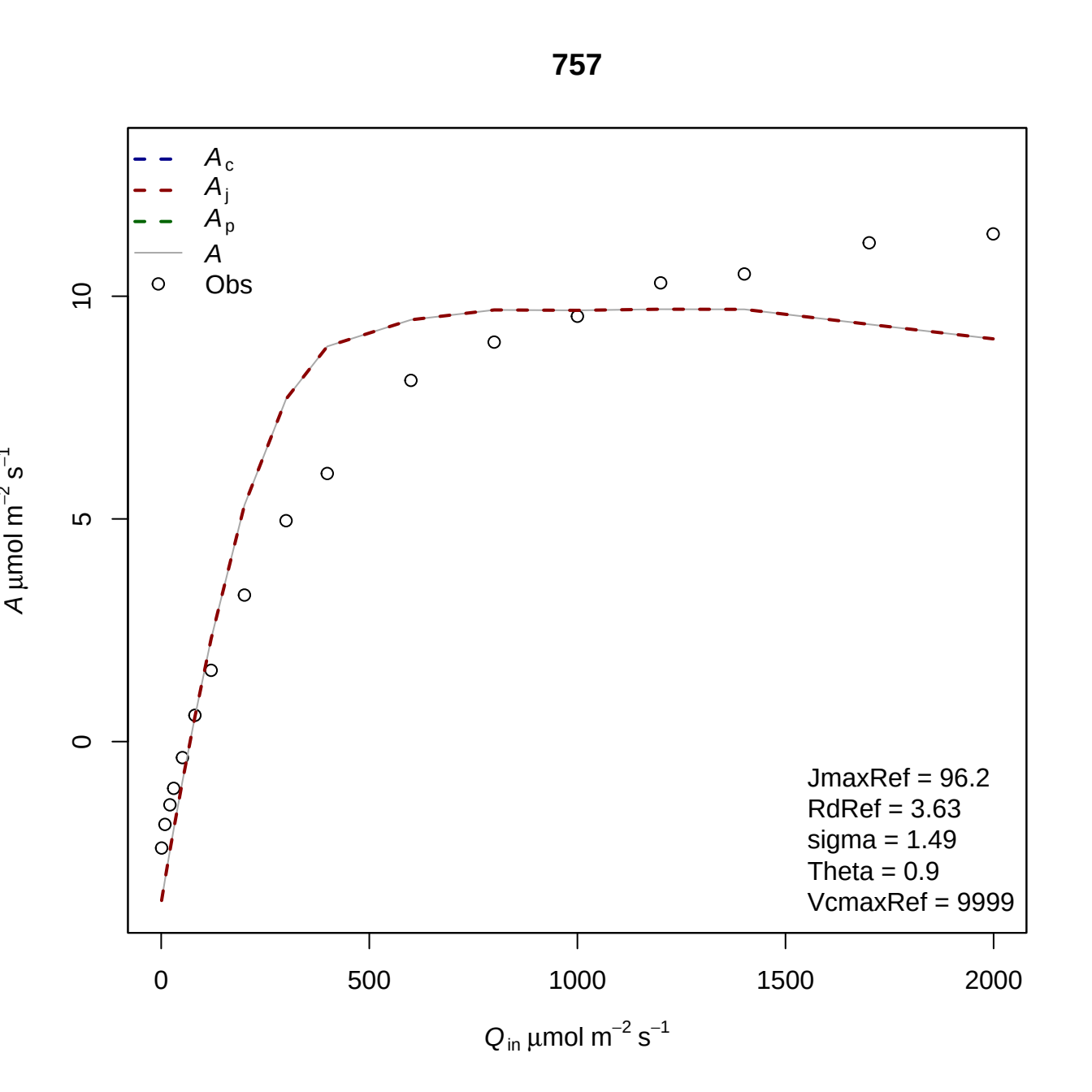

752
A_c
A_j
A_p
A
Obs
JmaxRef = 135.79
RdRef = 3.19
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

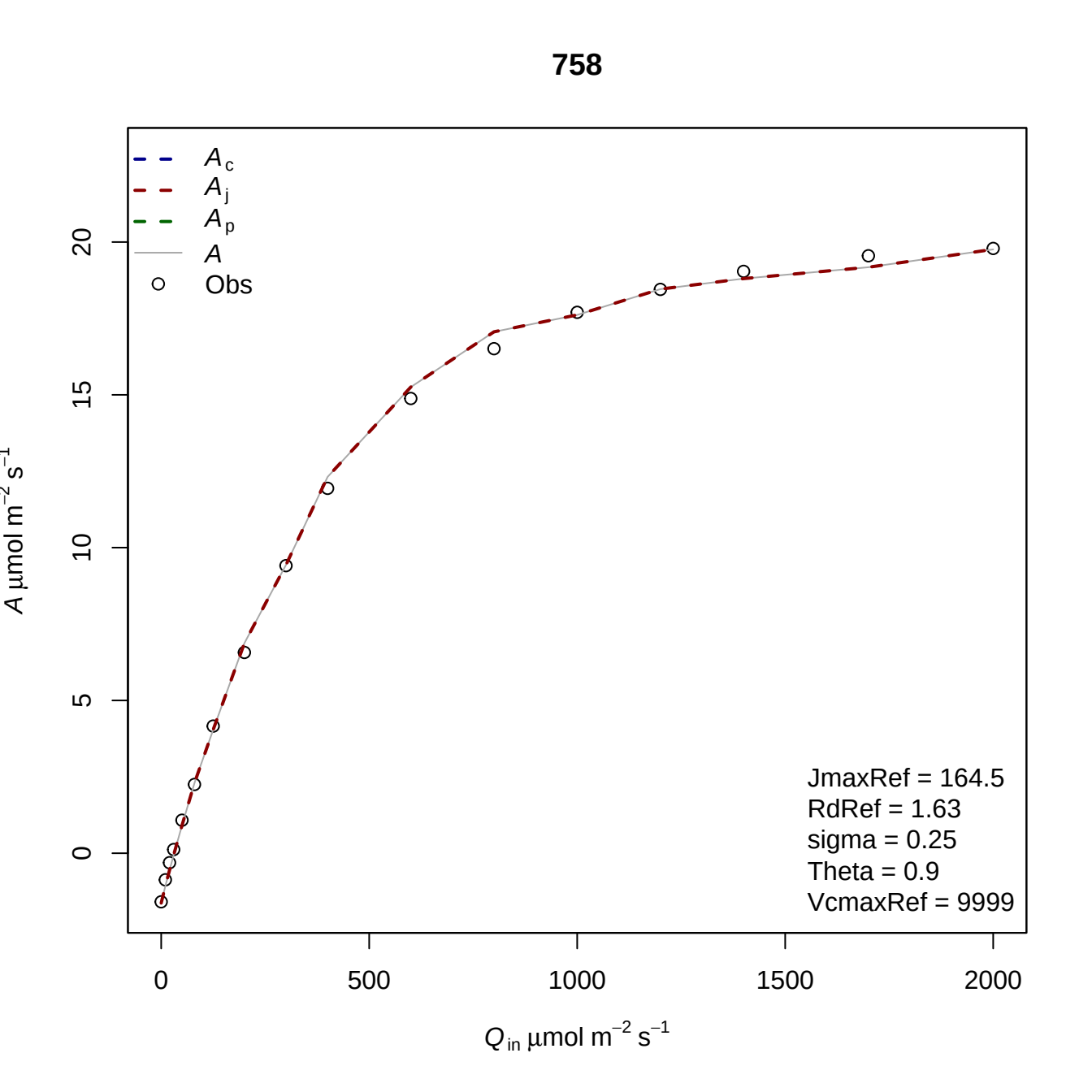

753
A_c
A_j
A_p
A
Obs
JmaxRef = 114.15
RdRef = 2.38
sigma = 1.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

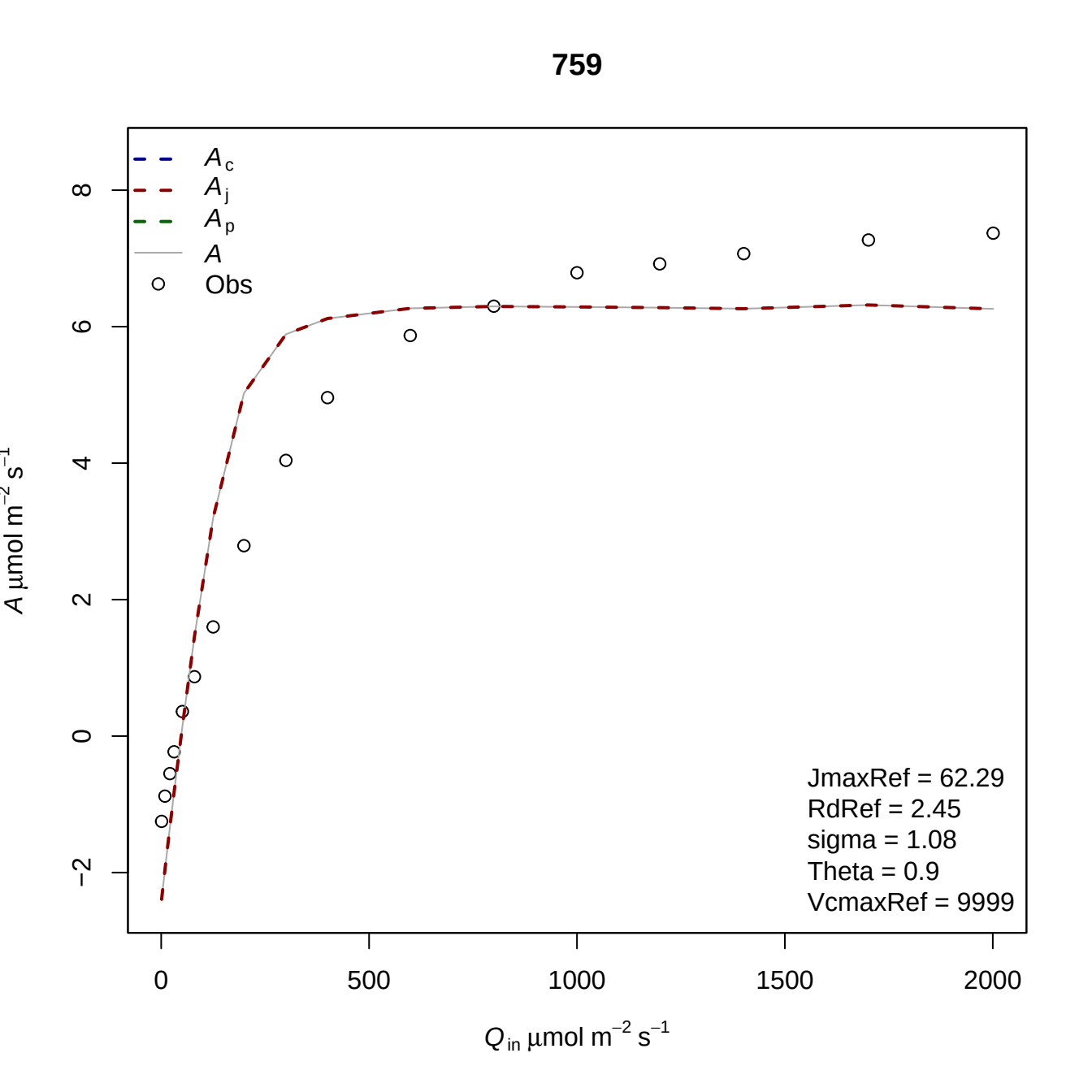

754
A_c
A_j
A_p
A
Obs
JmaxRef = 115.53
RdRef = 4.19
sigma = 1.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

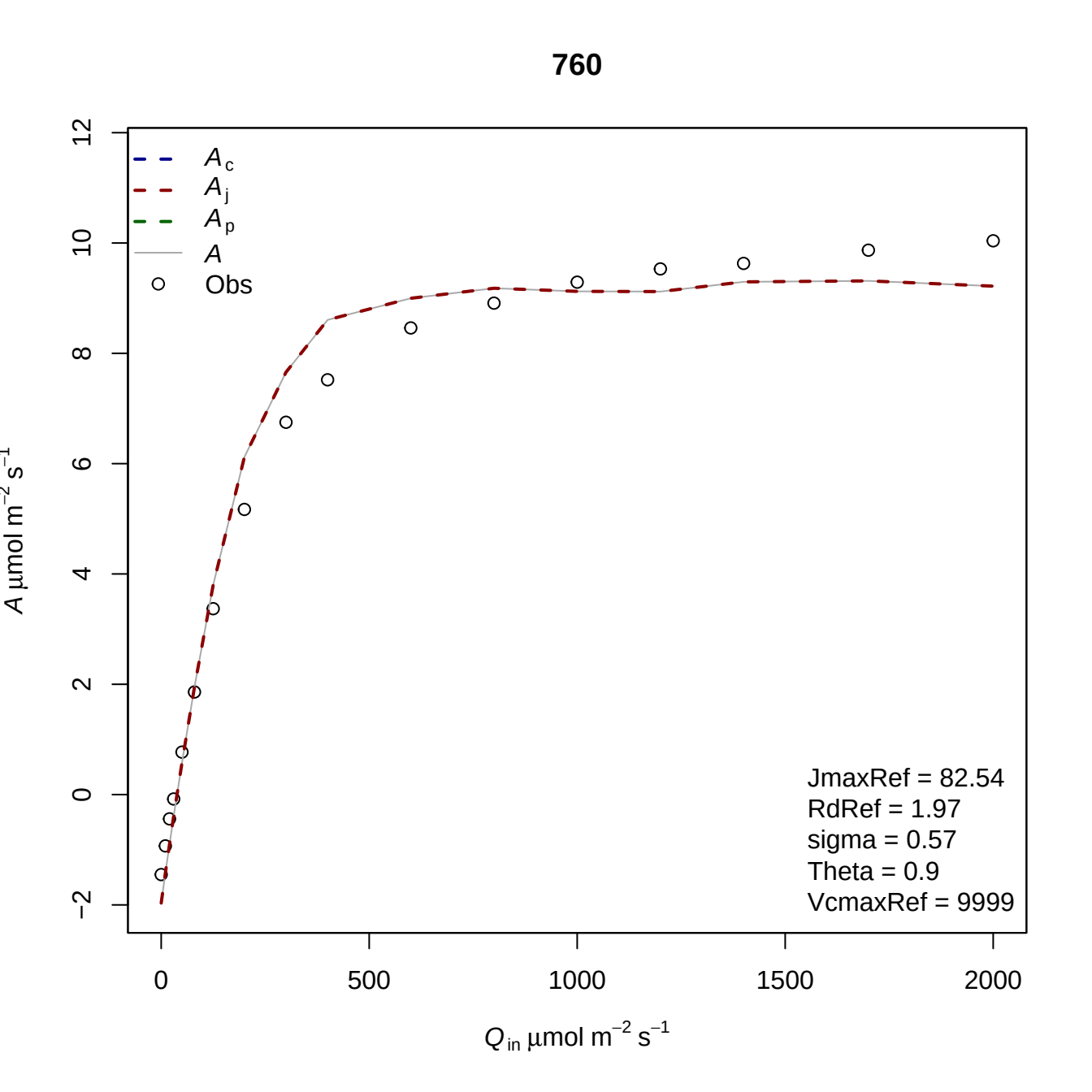

755
A_c
A_j
A_p
A
Obs
JmaxRef = 147.2
RdRef = 2.92
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

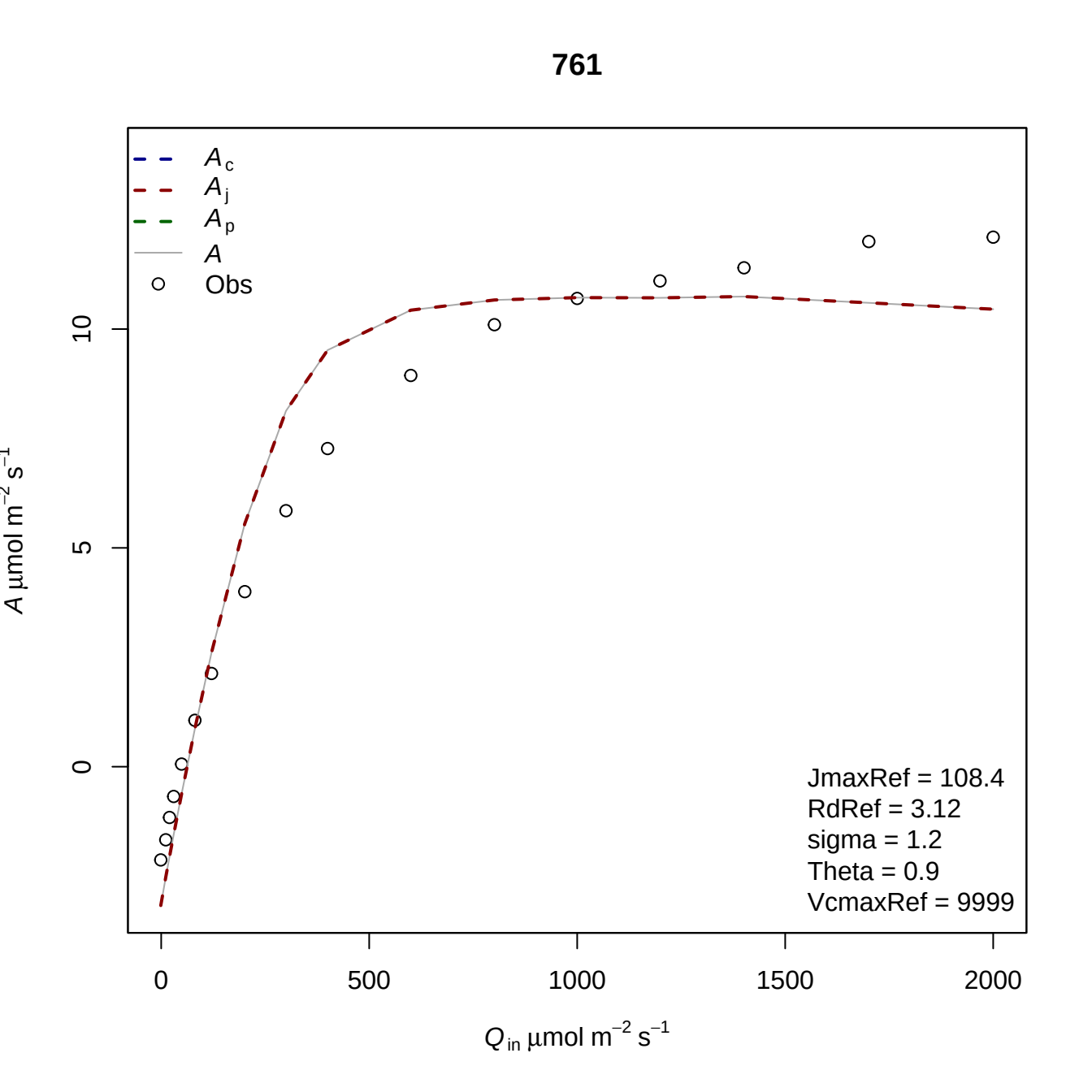

756
A_c
A_j
A_p
A
Obs
JmaxRef = 94.48
RdRef = 3.1
sigma = 1.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

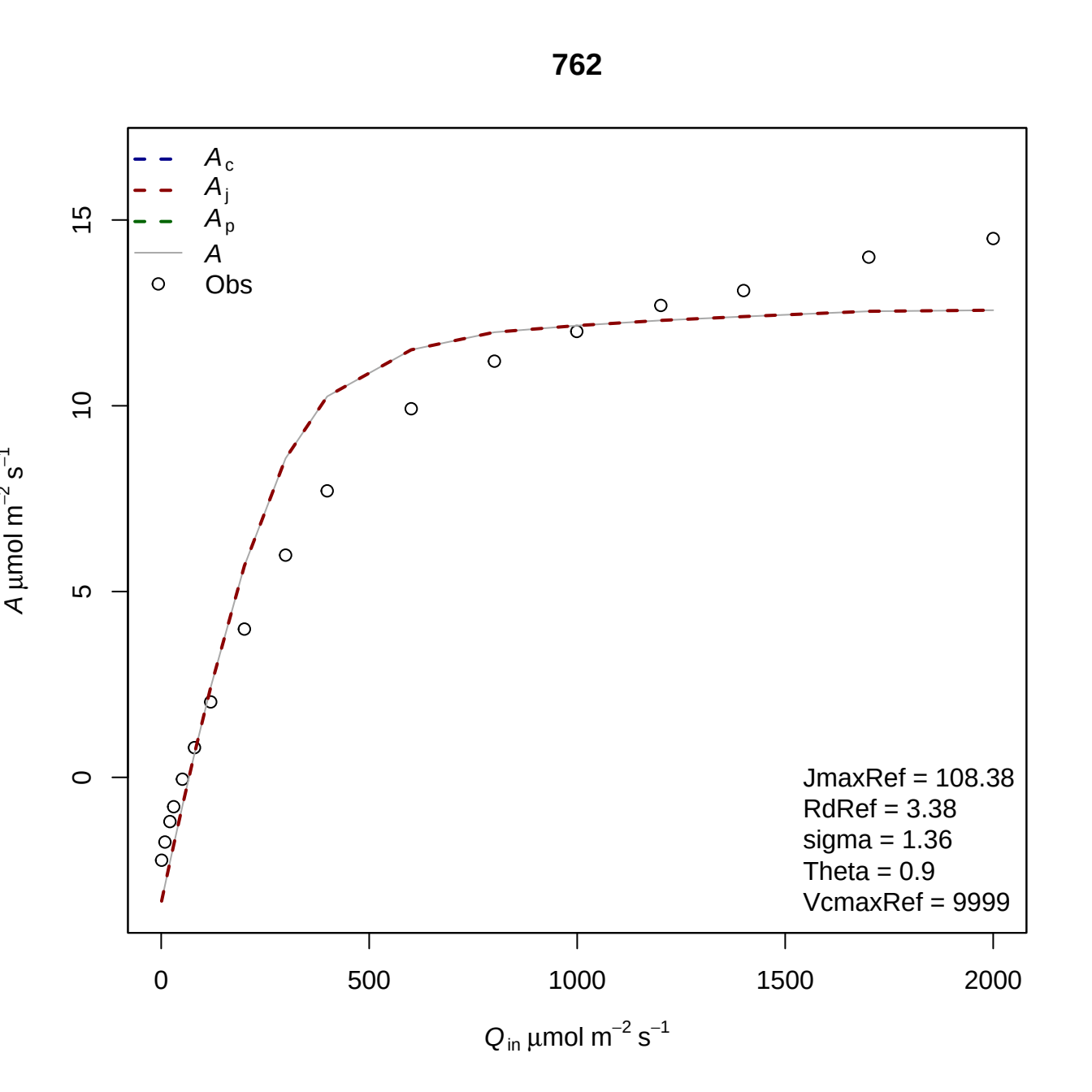

757
A_c
A_j
A_p
A
Obs
JmaxRef = 96.2
RdRef = 3.63
sigma = 1.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

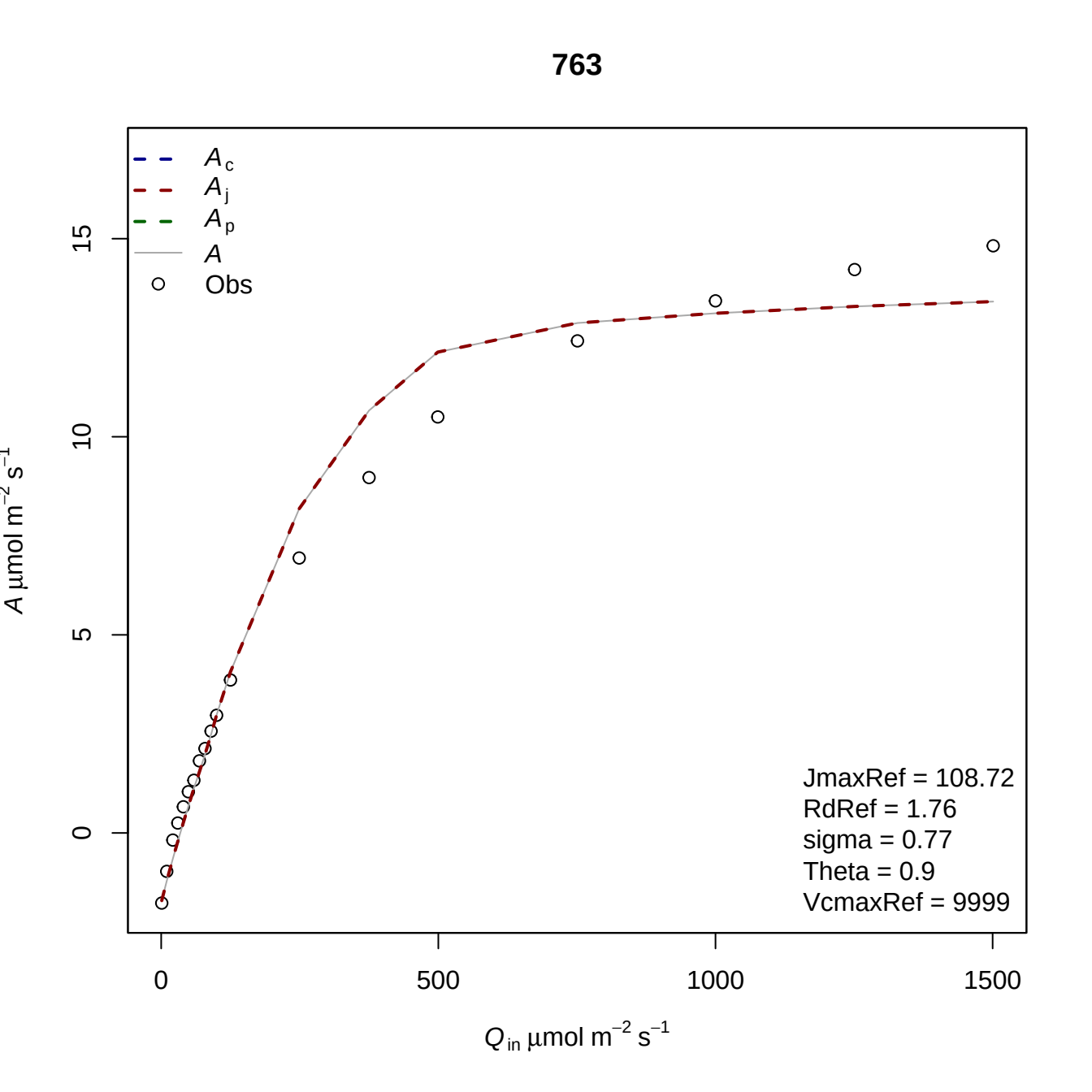

758
A_c
A_j
A_p
A
Obs
JmaxRef = 164.5
RdRef = 1.63
sigma = 0.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

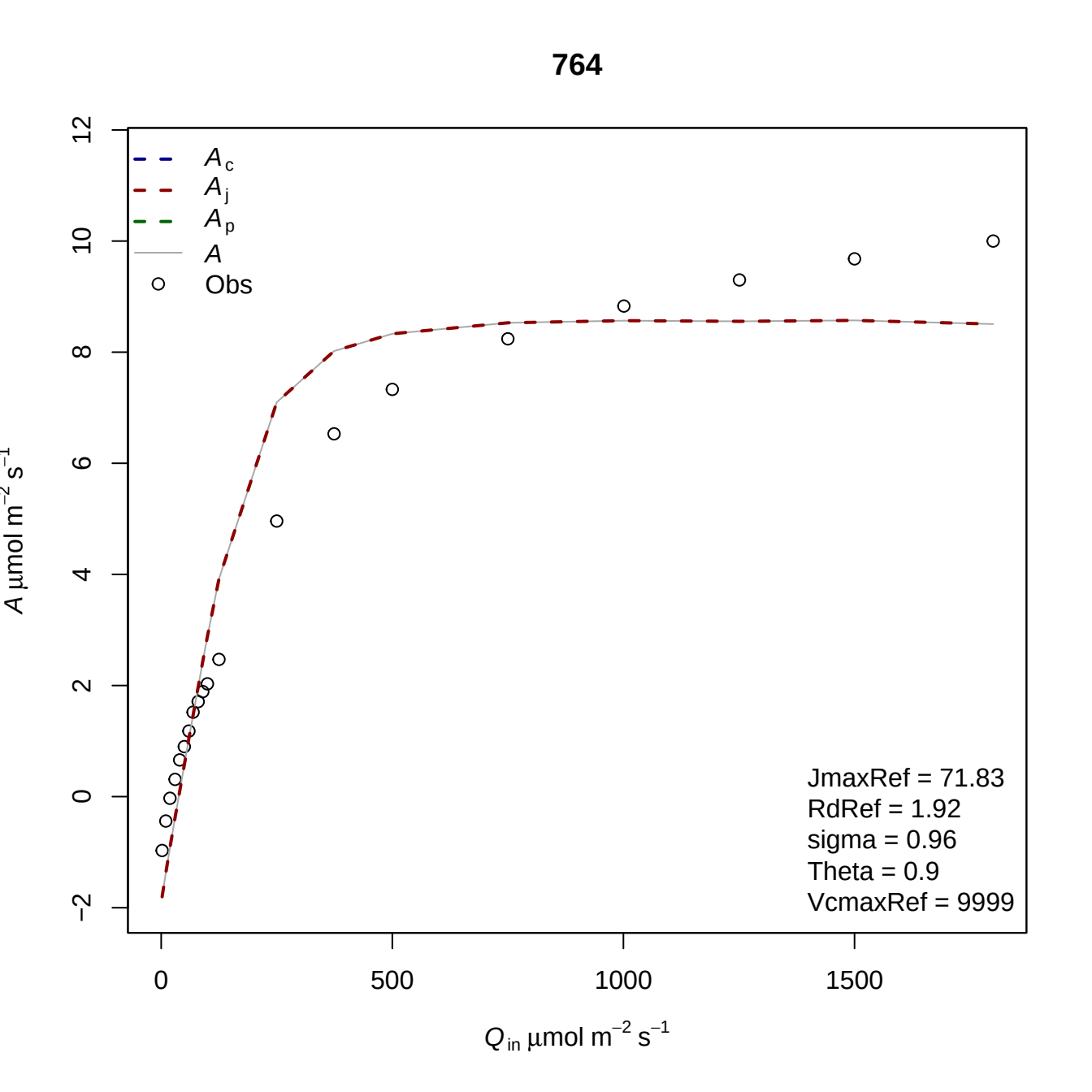

759
A_c
A_j
A_p
A
Obs
JmaxRef = 62.29
RdRef = 2.45
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

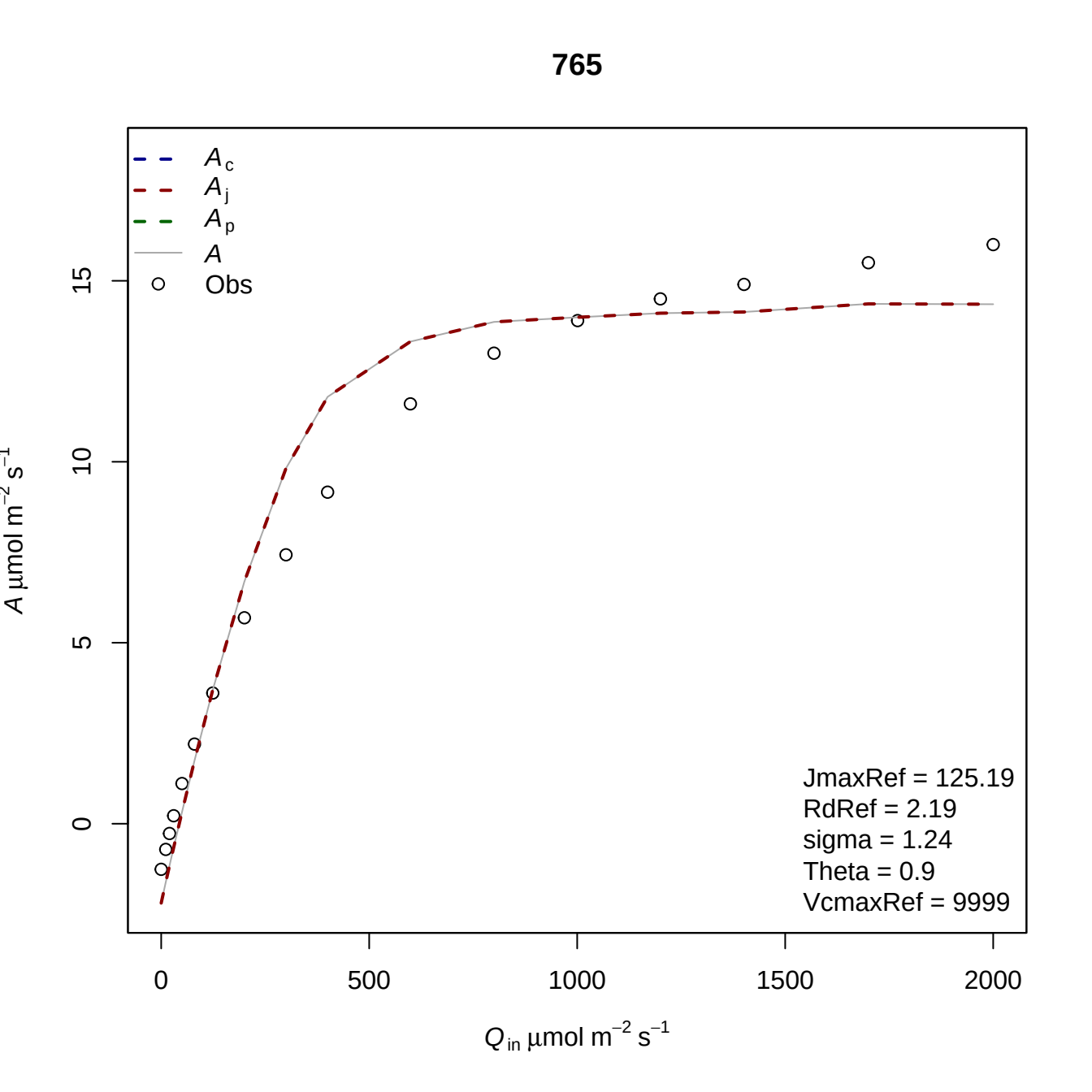

760
A_c
A_j
A_p
A
Obs
JmaxRef = 82.54
RdRef = 1.97
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

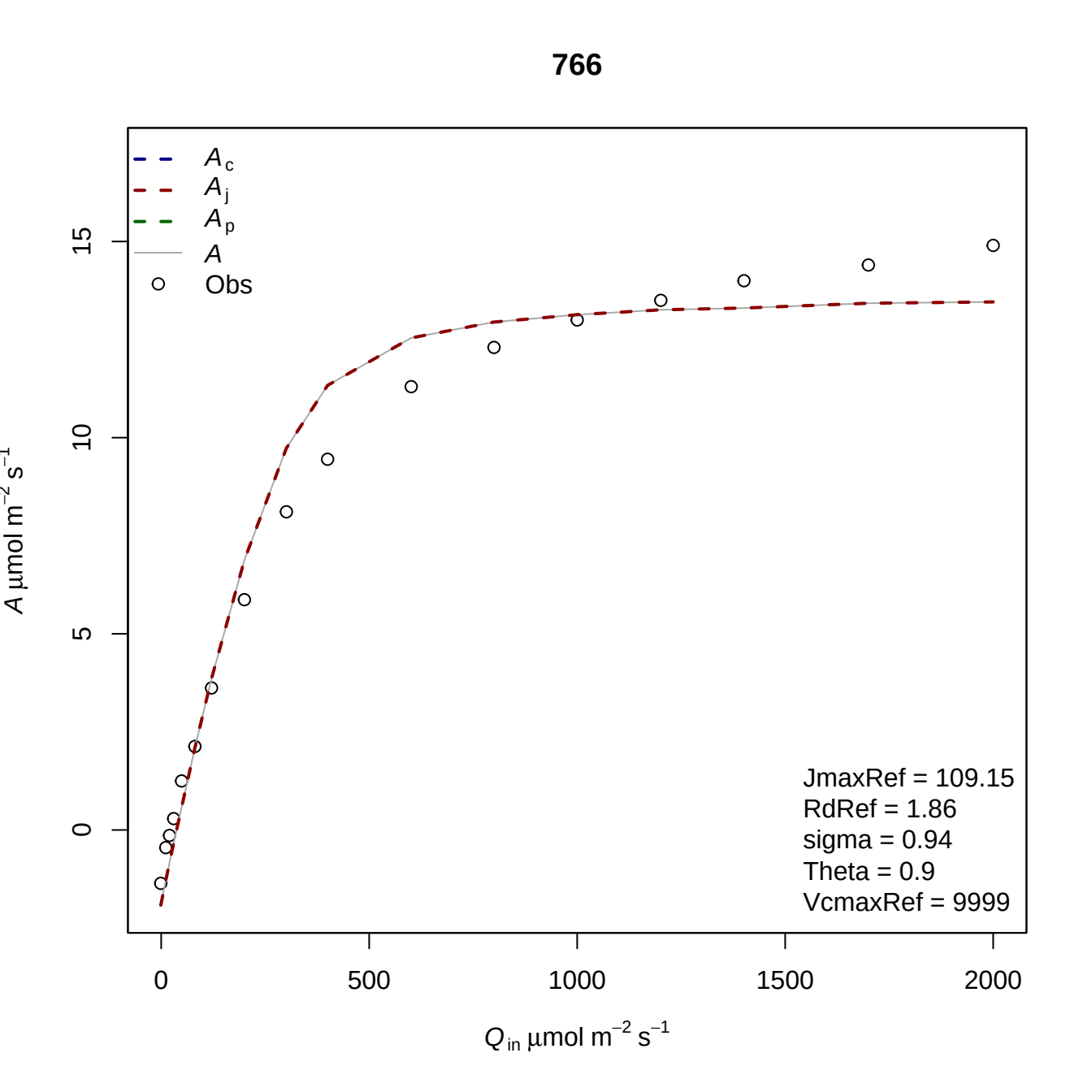

761
A_c
A_j
A_p
A
Obs
JmaxRef = 108.4
RdRef = 3.12
sigma = 1.2
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

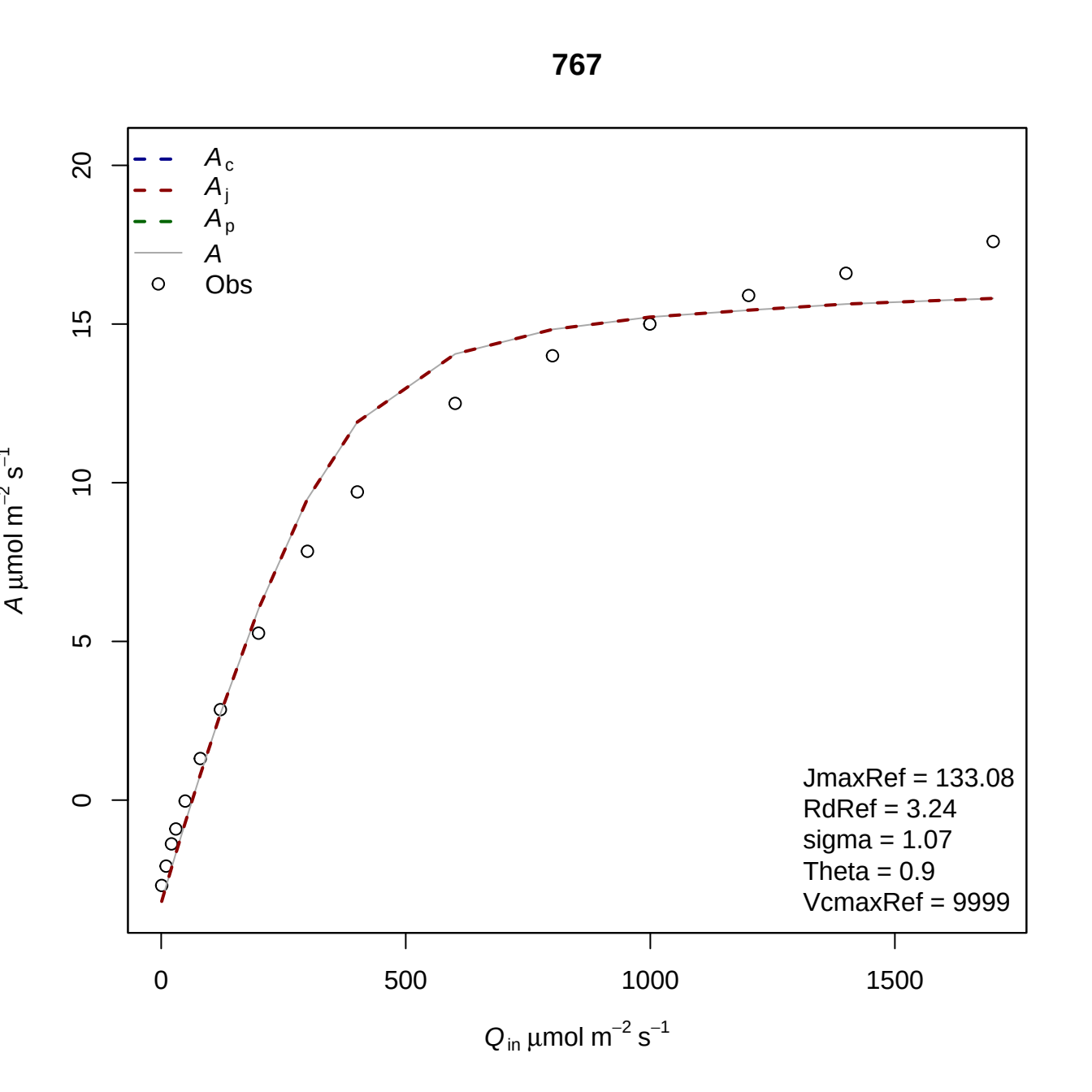

762
A_c
A_j
A_p
A
Obs
JmaxRef = 108.38
RdRef = 3.38
sigma = 1.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

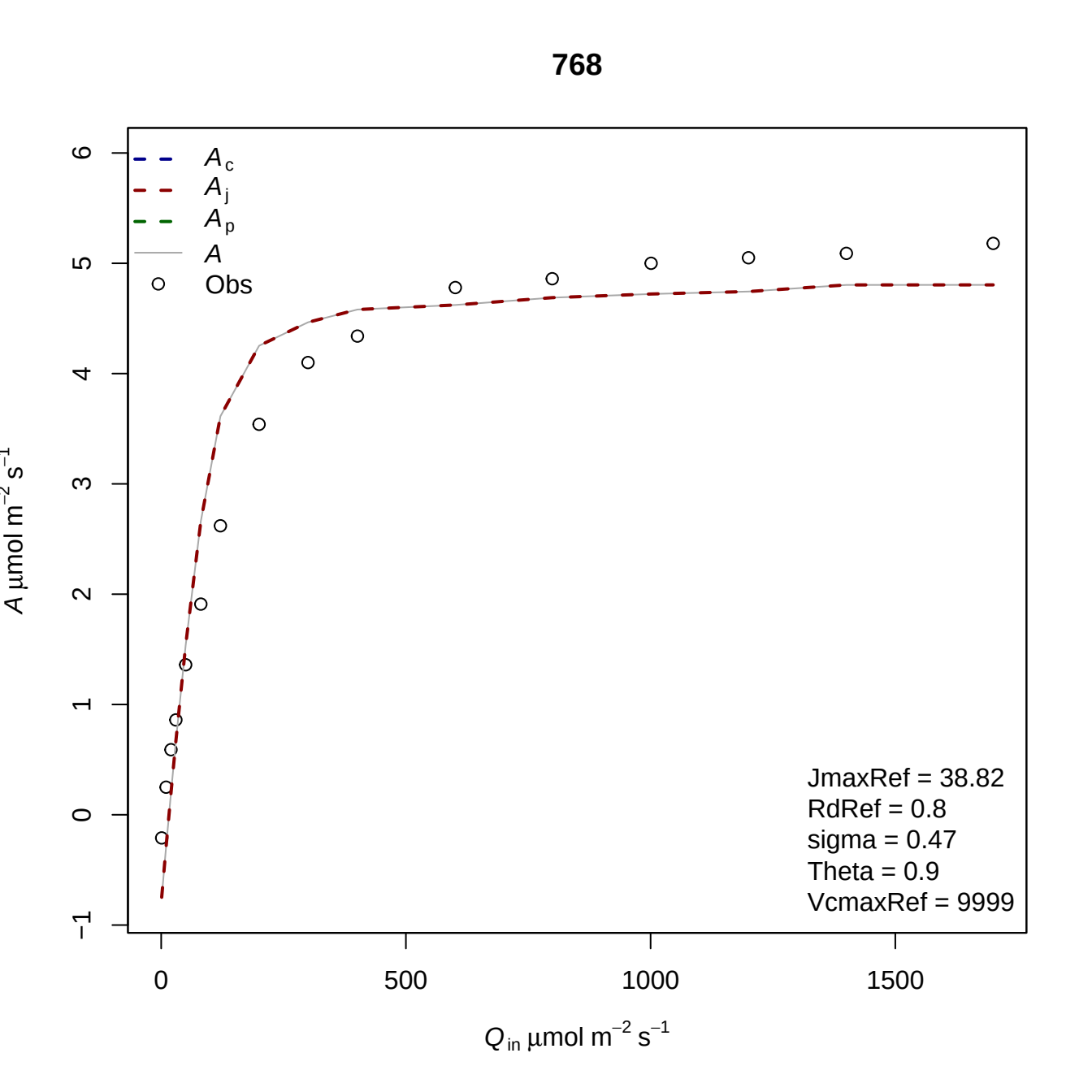

763
A_c
A_j
A_p
A
Obs
JmaxRef = 108.72
RdRef = 1.76
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

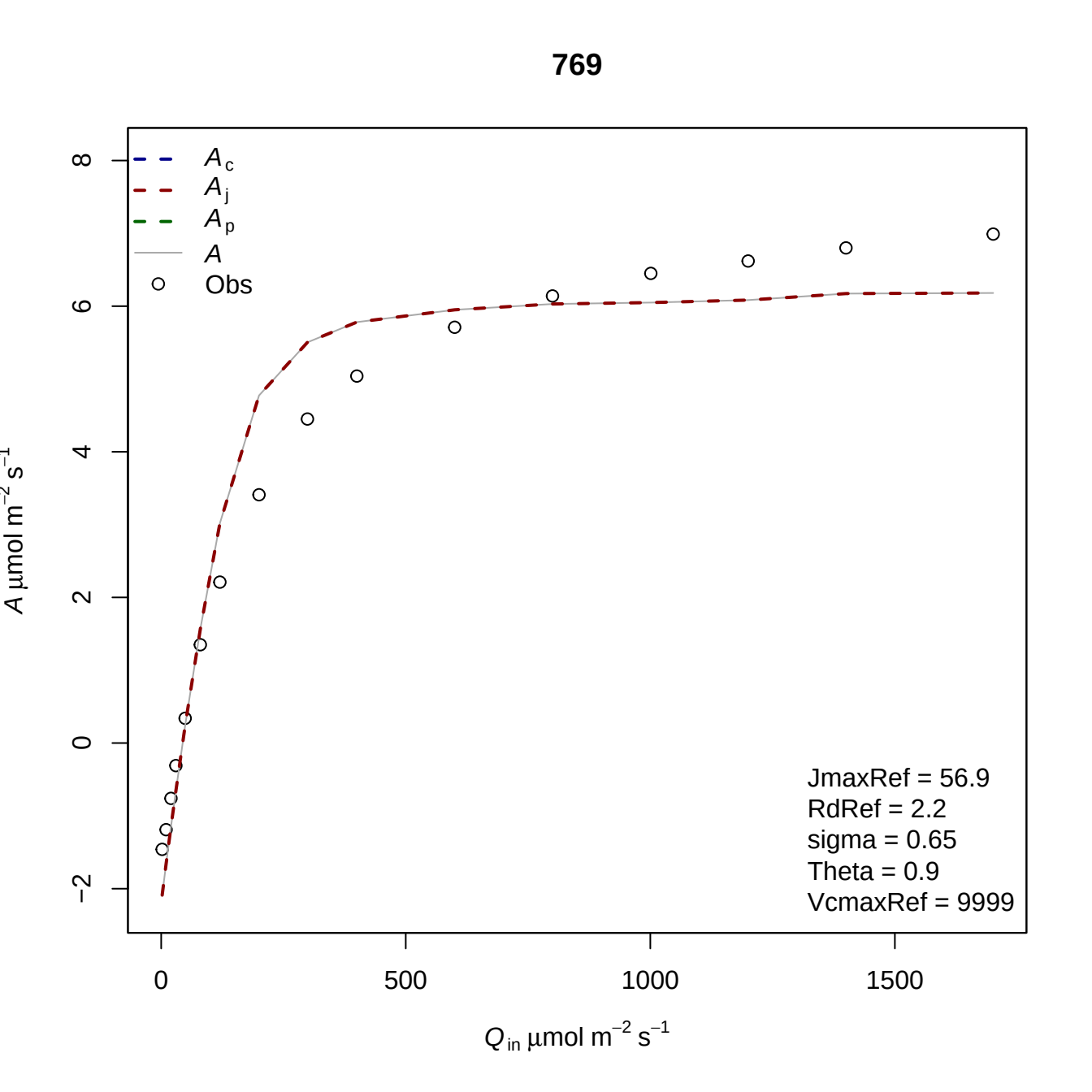

764
A_c
A_j
A_p
A
Obs
JmaxRef = 71.83
RdRef = 1.92
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

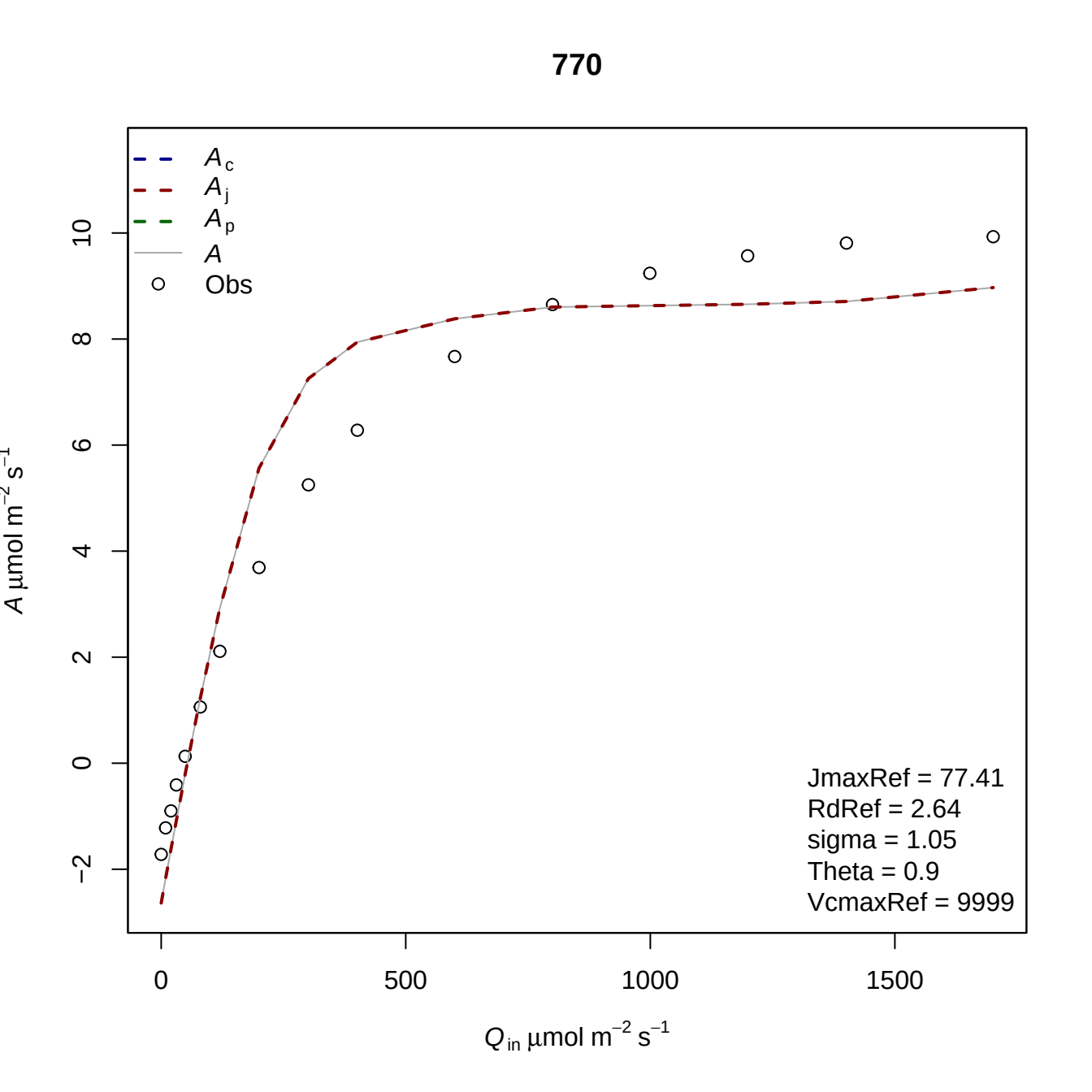

765
A_c
A_j
A_p
A
Obs
JmaxRef = 125.19
RdRef = 2.19
sigma = 1.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

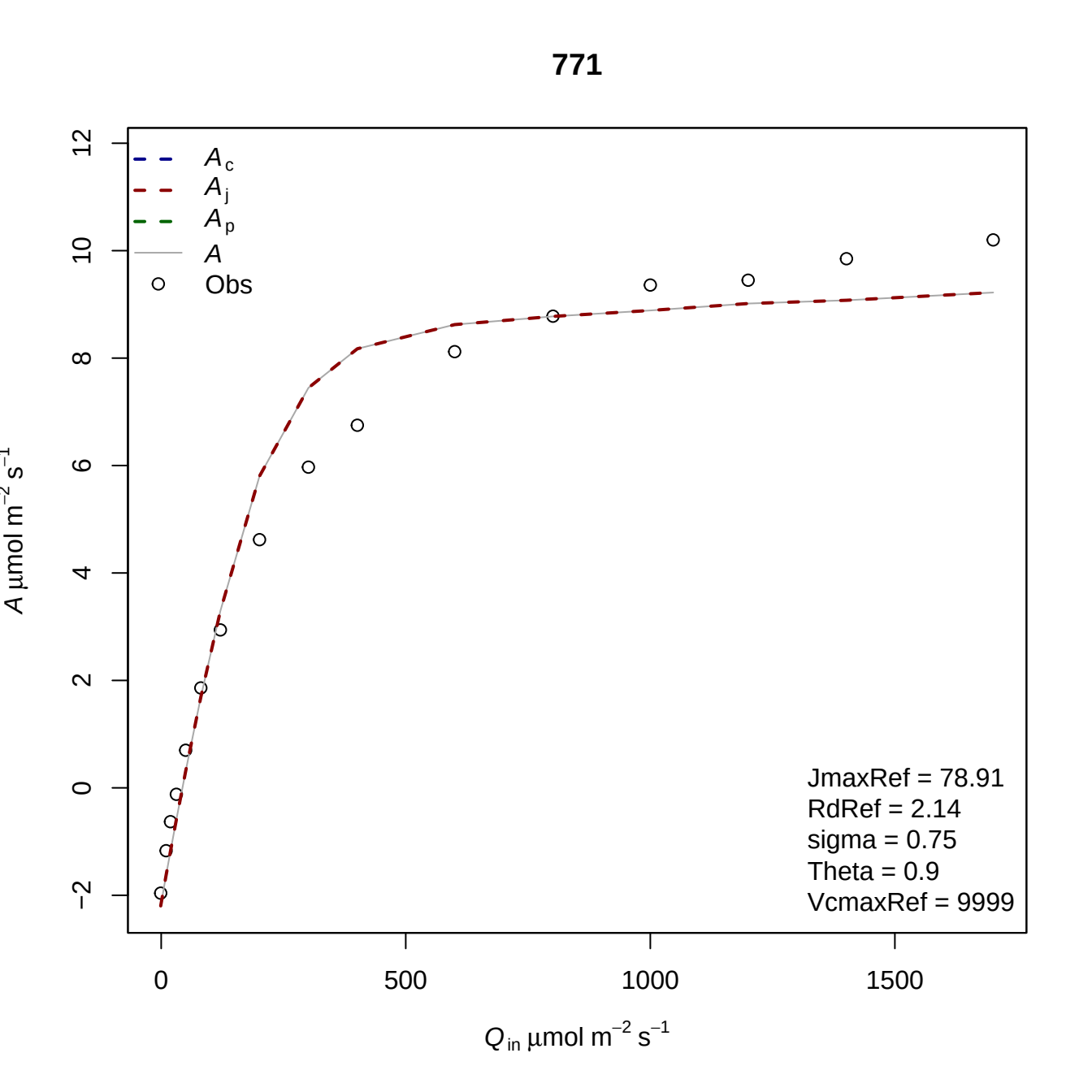

766
A_c
A_j
A_p
A
Obs
JmaxRef = 109.15
RdRef = 1.86
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

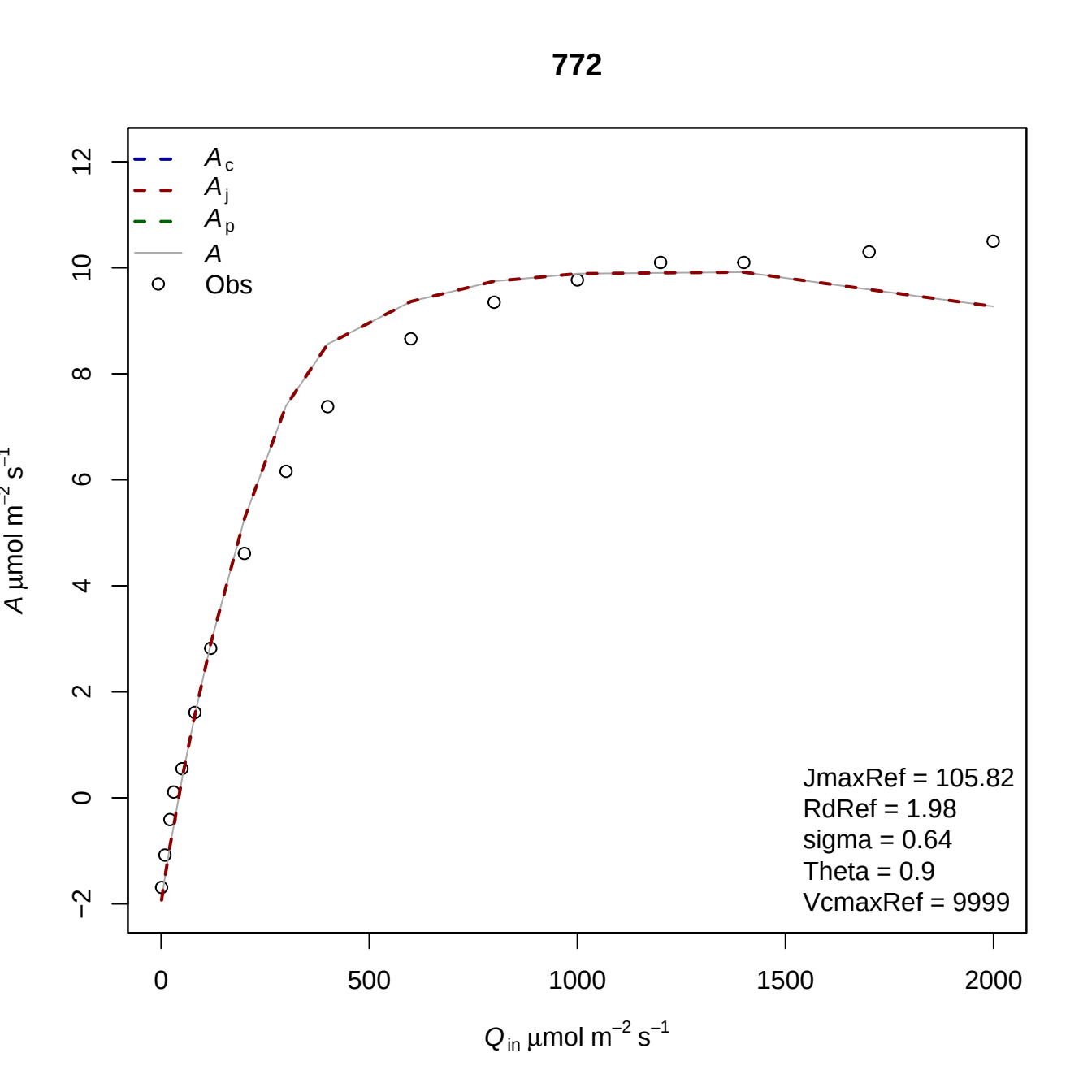

767
A_c
A_j
A_p
A
Obs
JmaxRef = 133.08
RdRef = 3.24
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

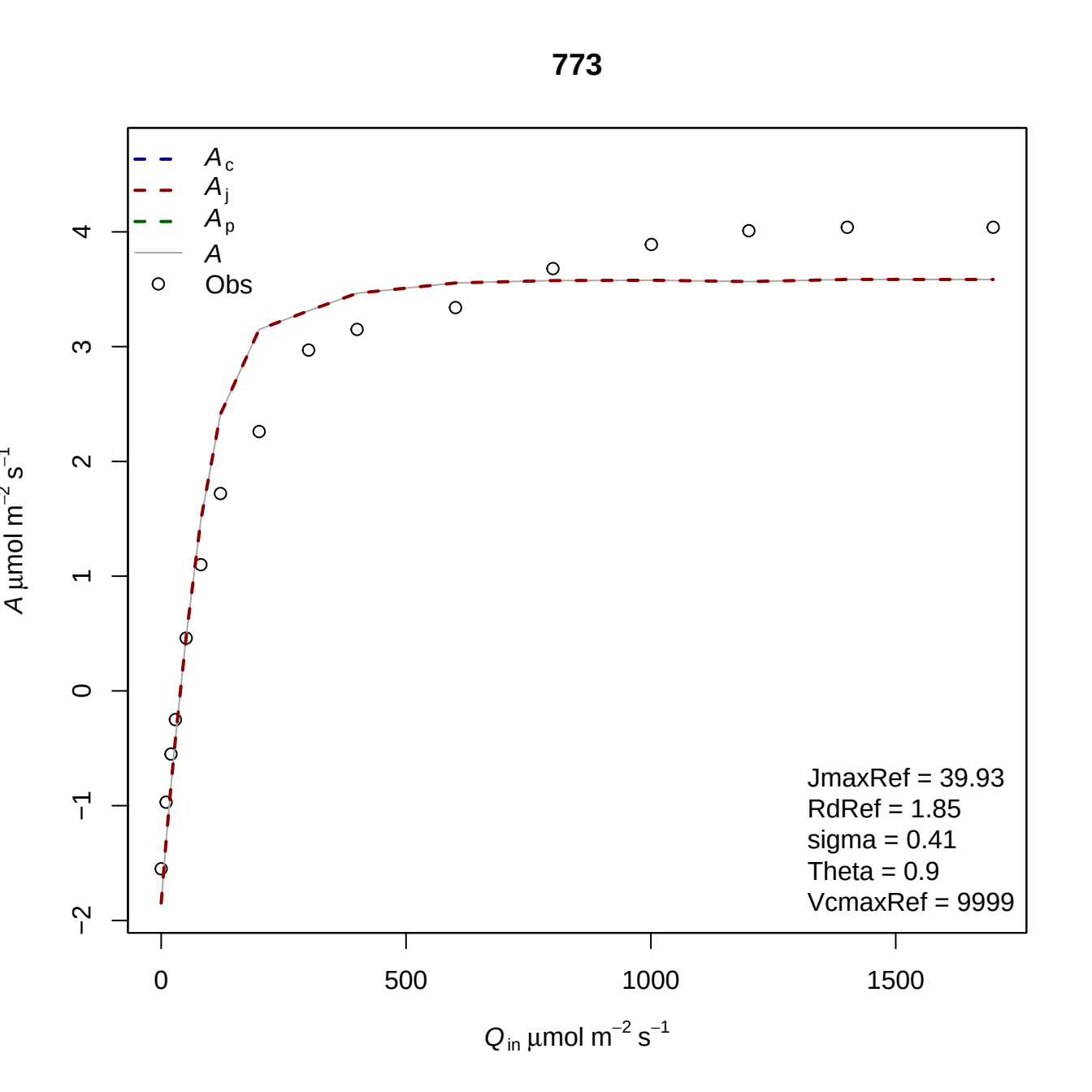

768
A_c
A_j
A_p
A
Obs
JmaxRef = 38.82
RdRef = 0.8
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

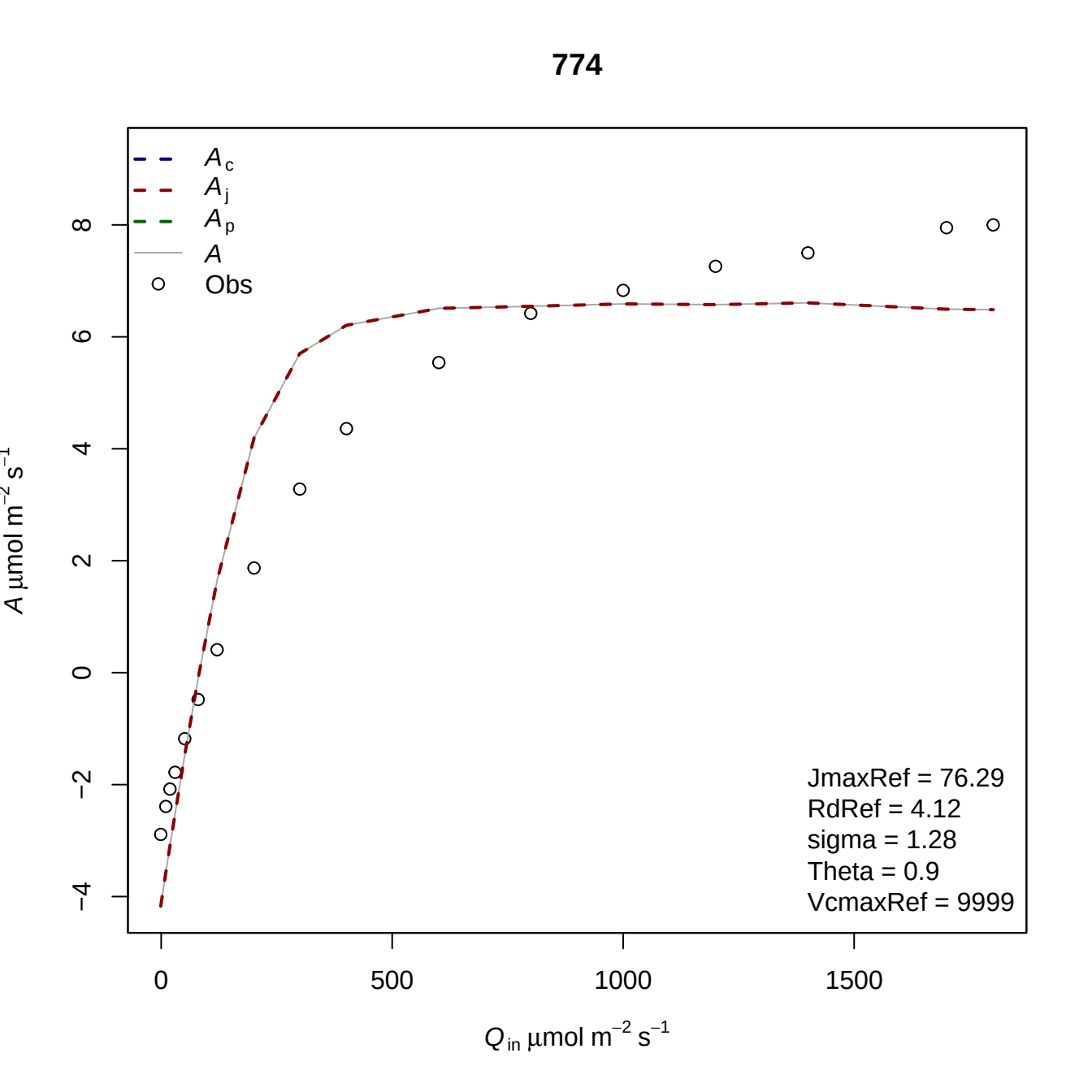

769
A_c
A_j
A_p
A
Obs
JmaxRef = 56.9
RdRef = 2.2
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

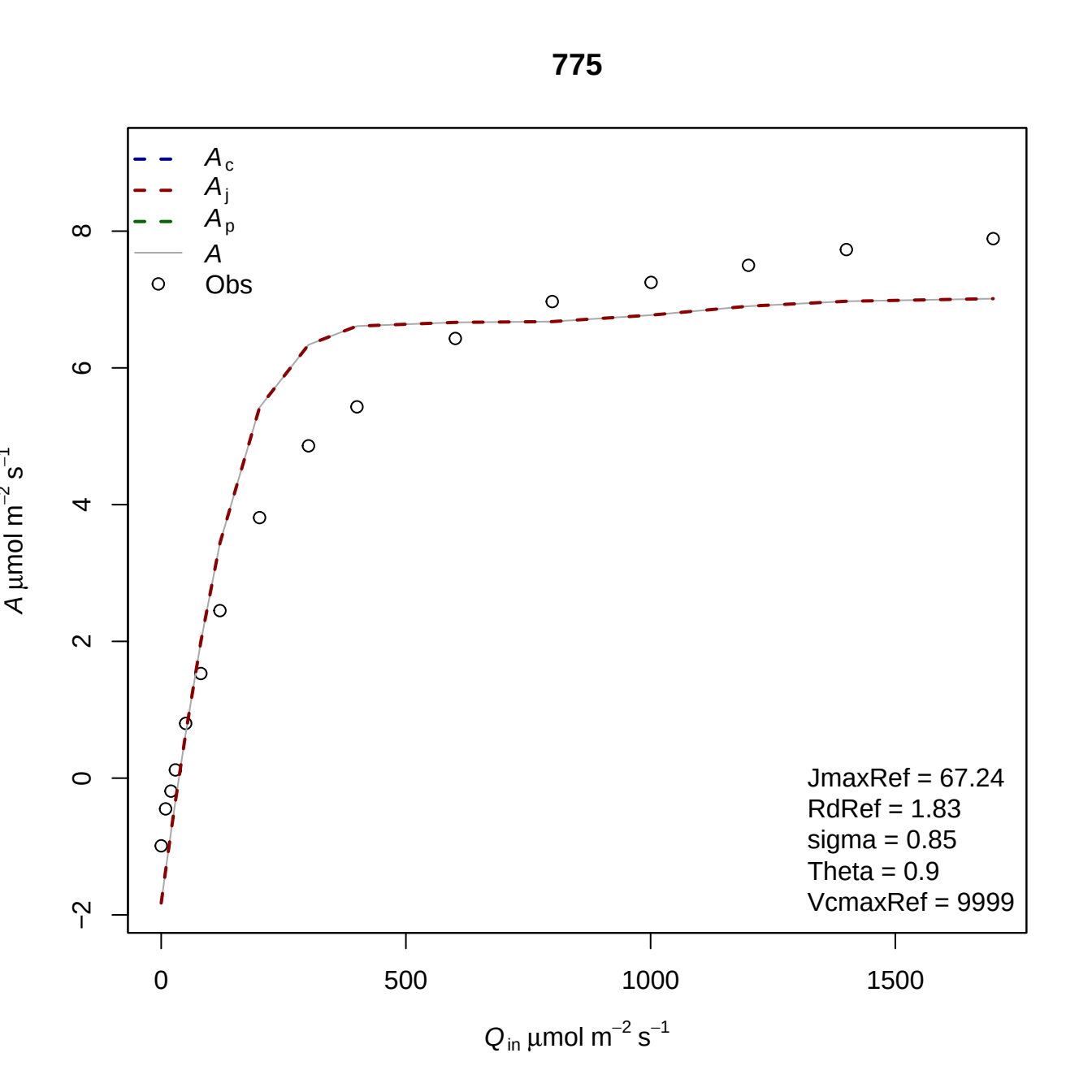

770
A_c
A_j
A_p
A
Obs
JmaxRef = 77.41
RdRef = 2.64
sigma = 1.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

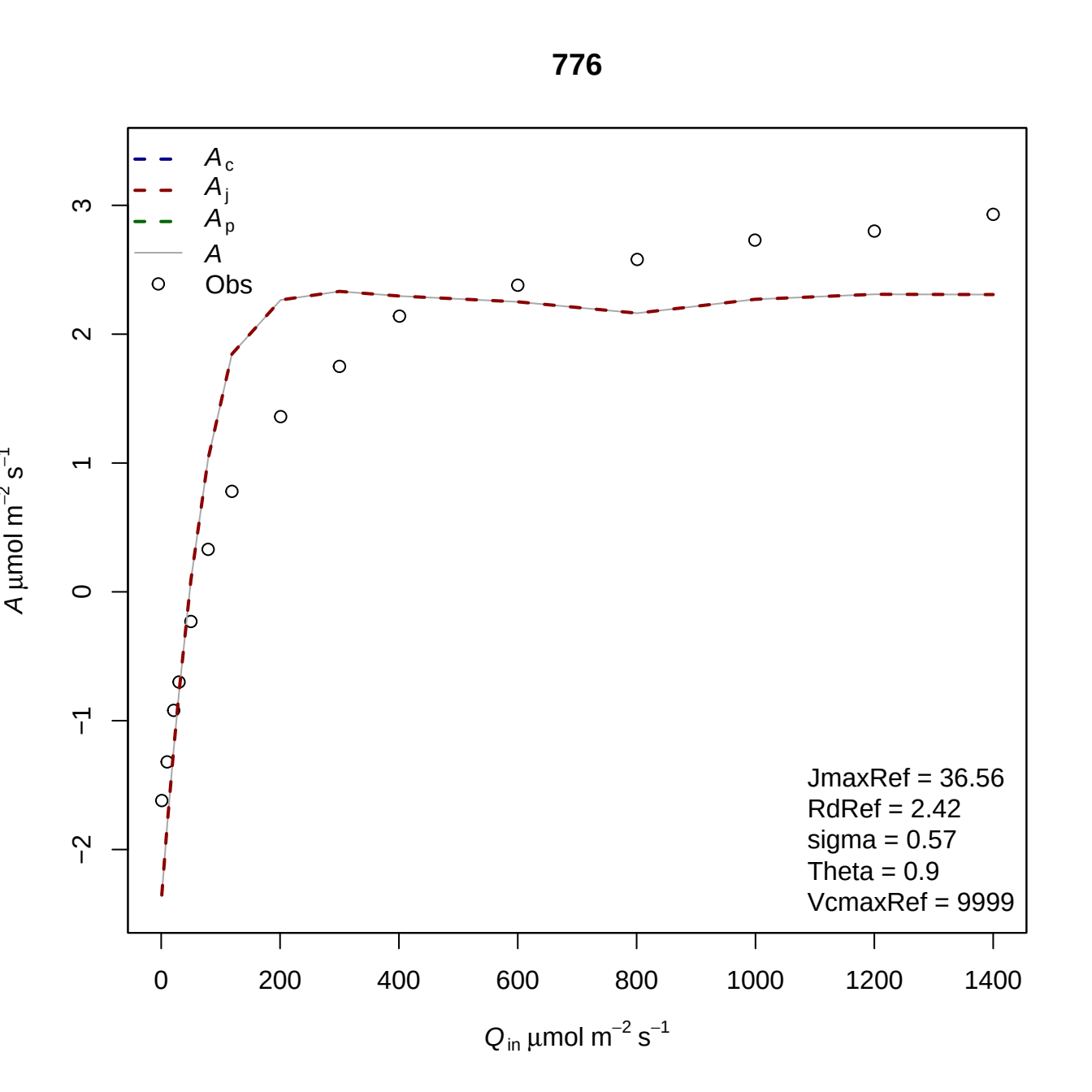

771
A_c
A_j
A_p
A
Obs
JmaxRef = 78.91
RdRef = 2.14
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

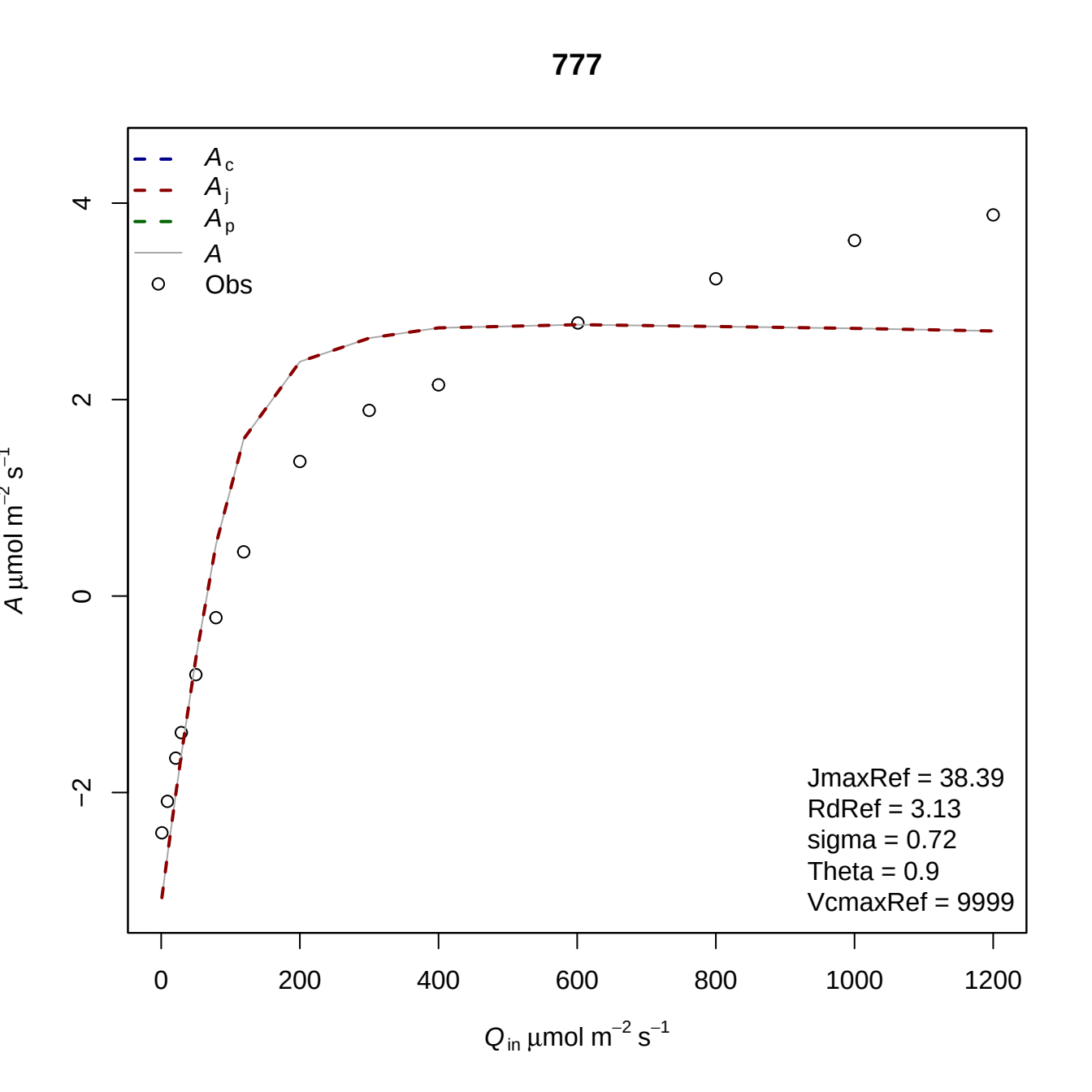

772
A_c
A_j
A_p
A
Obs
JmaxRef = 105.82
RdRef = 1.98
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

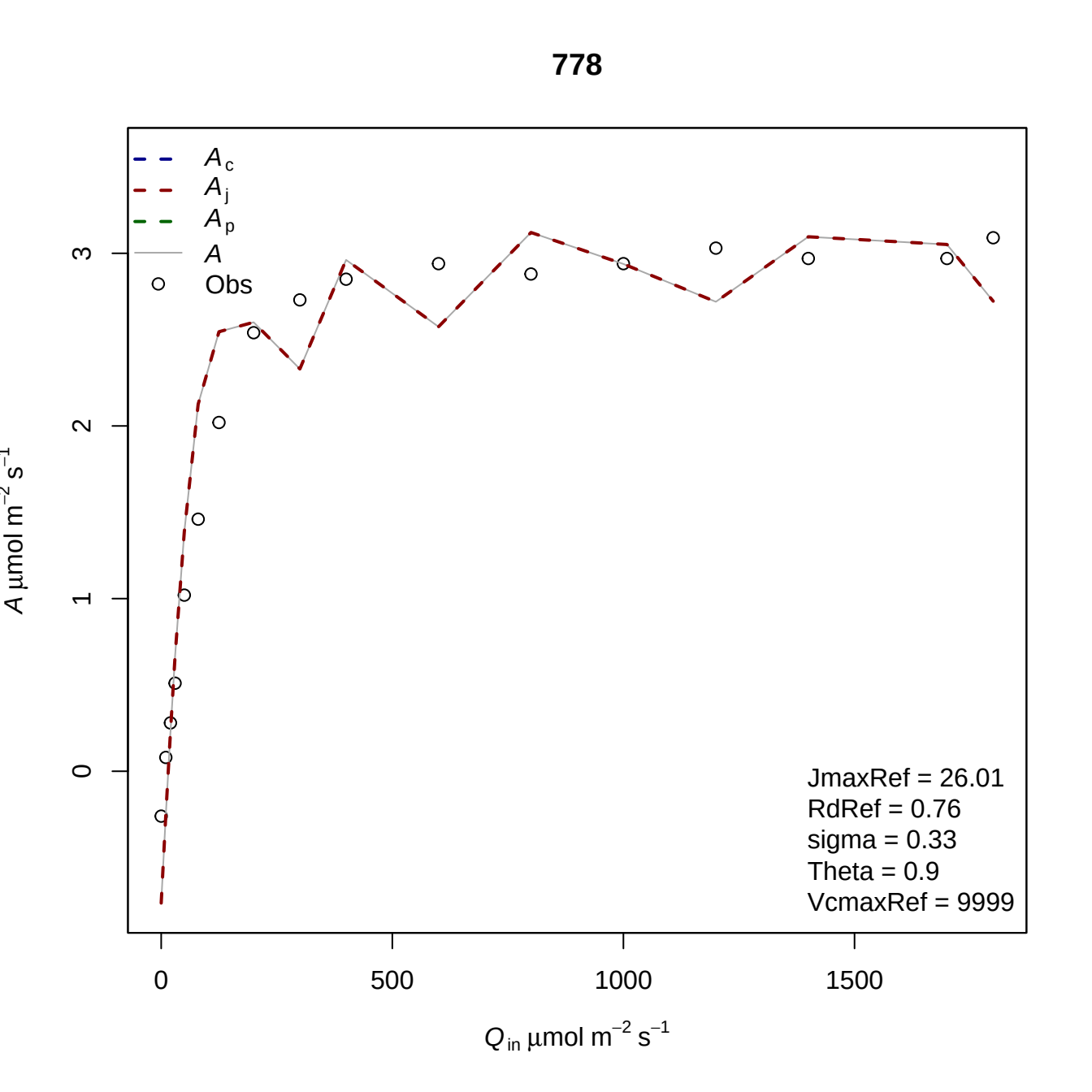

773
A_c
A_j
A_p
A
Obs
JmaxRef = 39.93
RdRef = 1.85
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

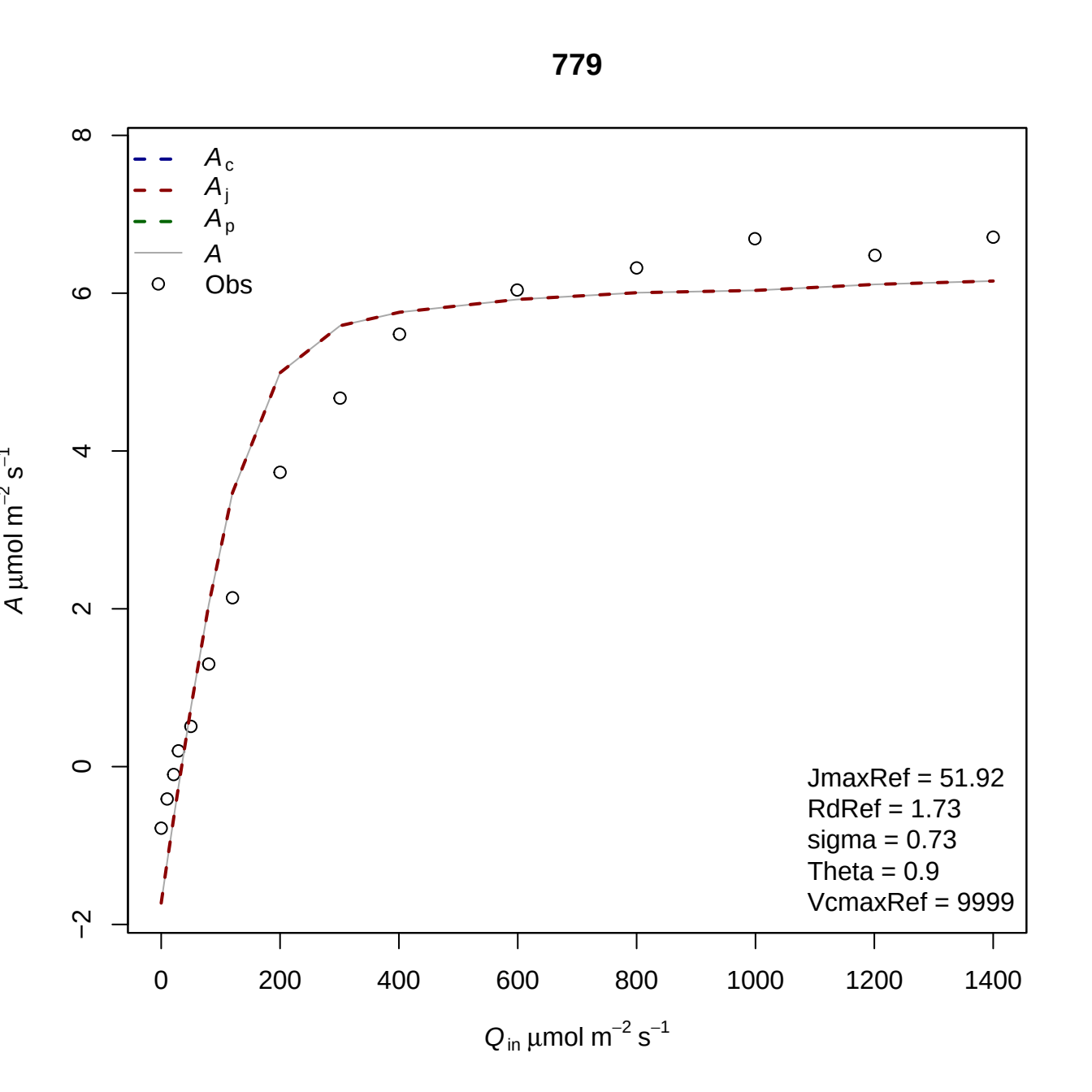

774
A_c
A_j
A_p
A
Obs
JmaxRef = 76.29
RdRef = 4.12
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

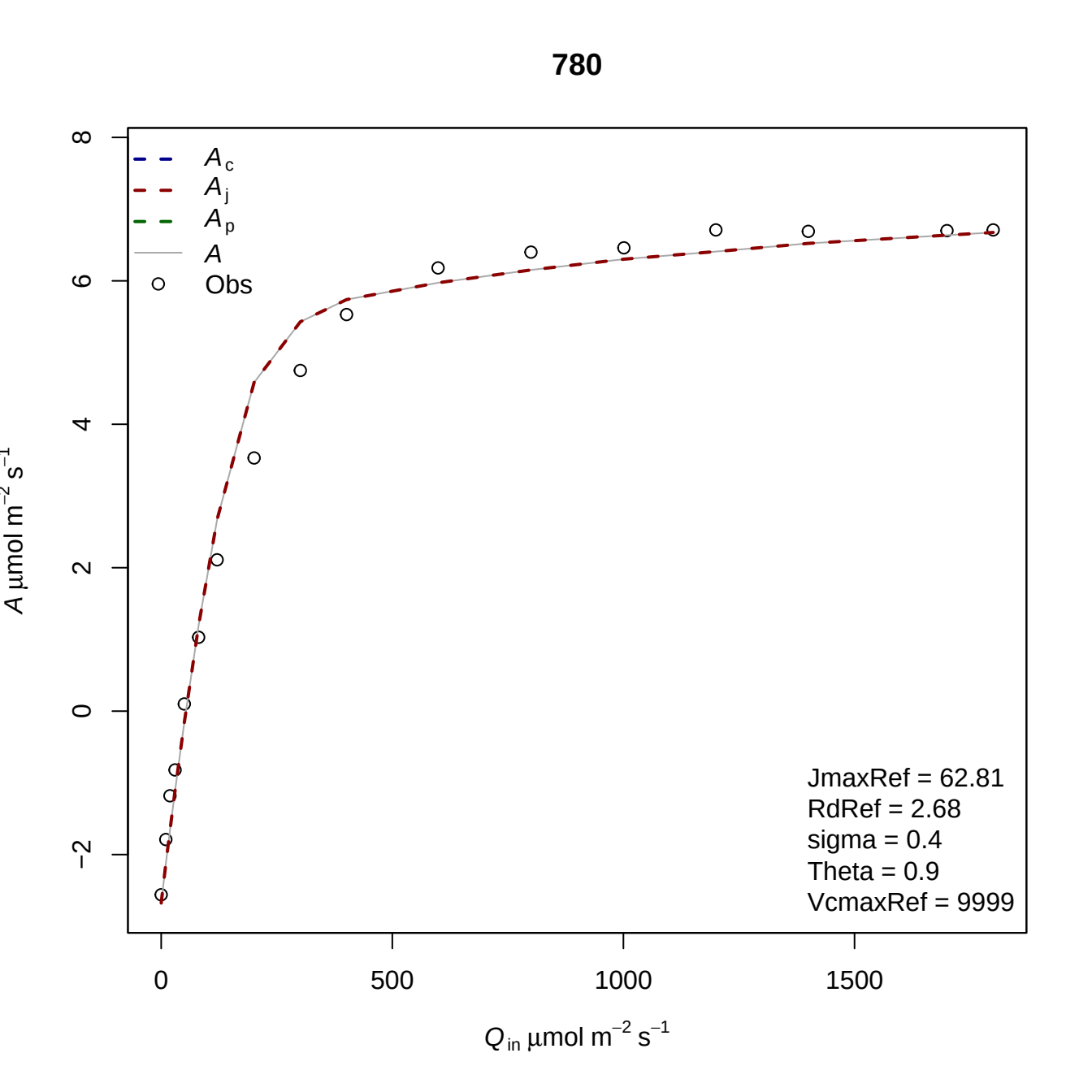

775
A_c
A_j
A_p
A
Obs
JmaxRef = 67.24
RdRef = 1.83
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

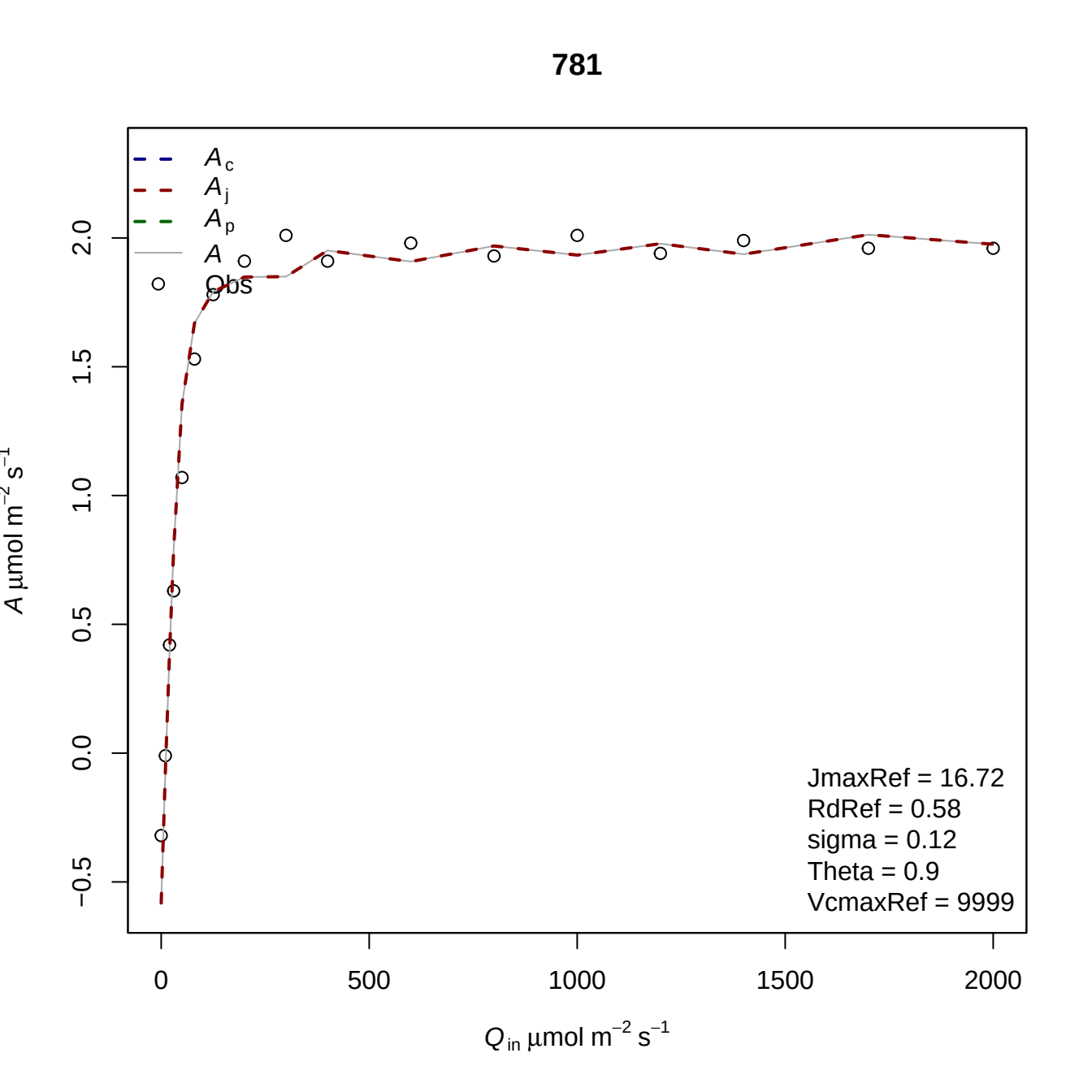

776
A_c
A_j
A_p
A
Obs
JmaxRef = 36.56
RdRef = 2.42
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

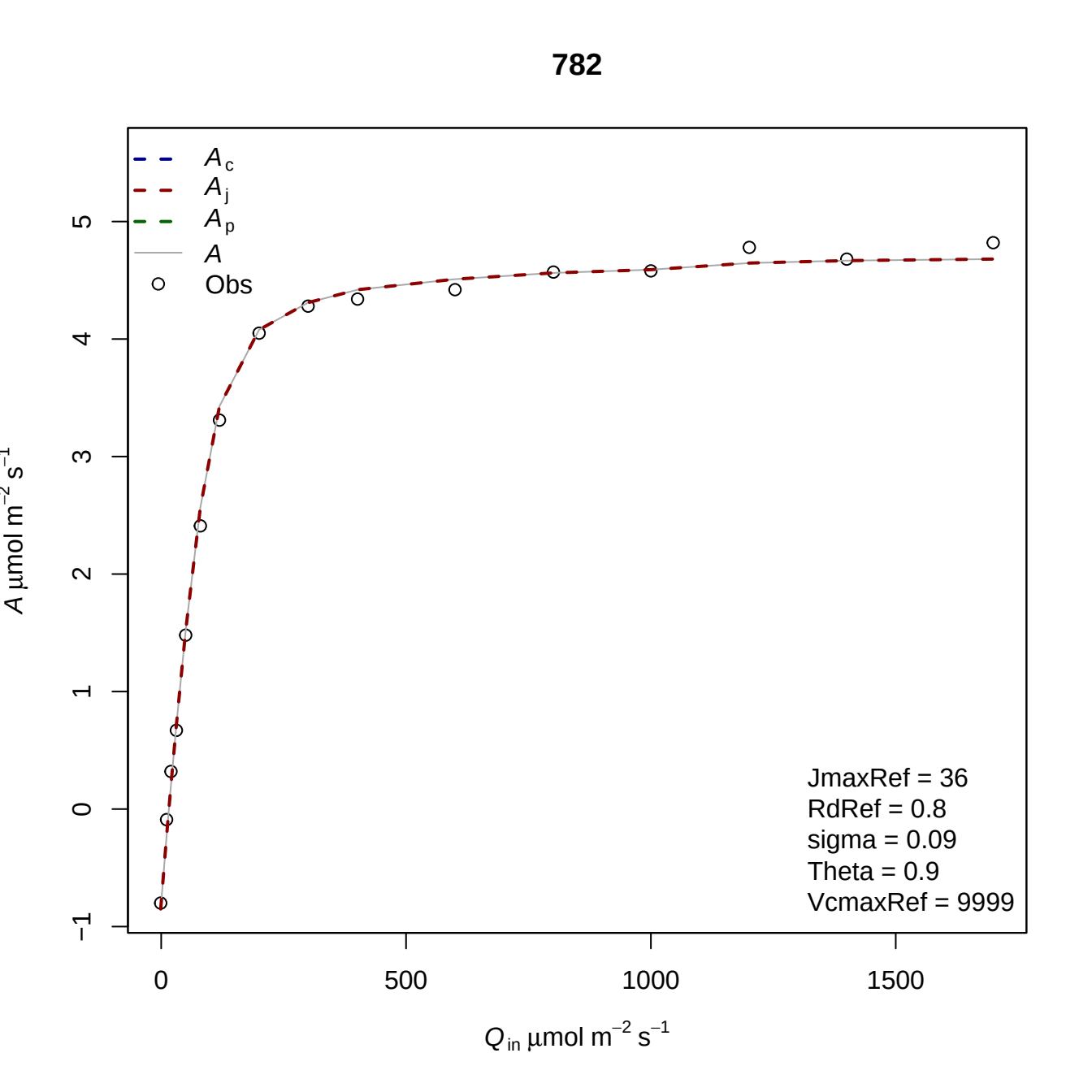

777
A_c
A_j
A_p
A
Obs
JmaxRef = 38.39
RdRef = 3.13
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000
1200

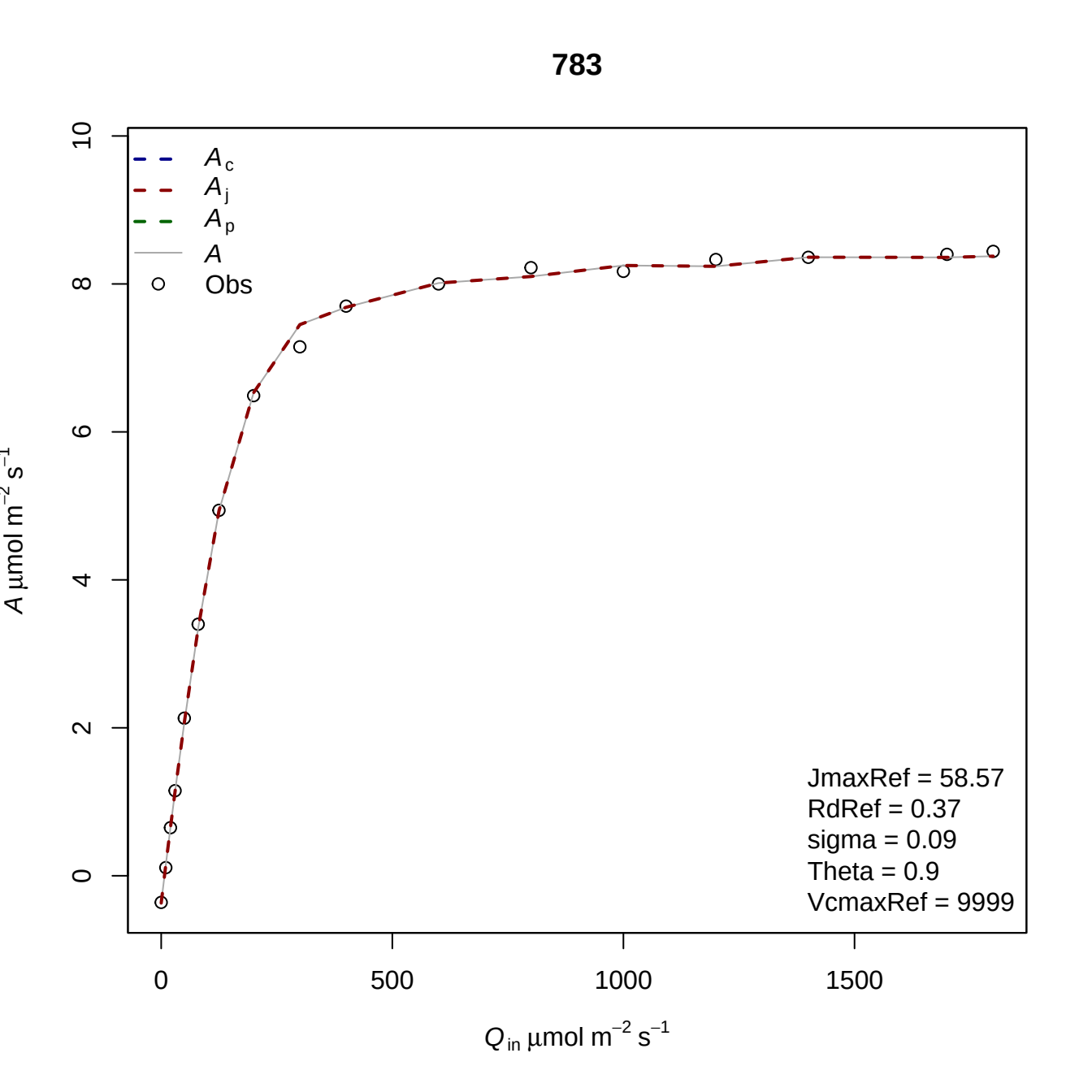

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.01
RdRef = 0.76
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

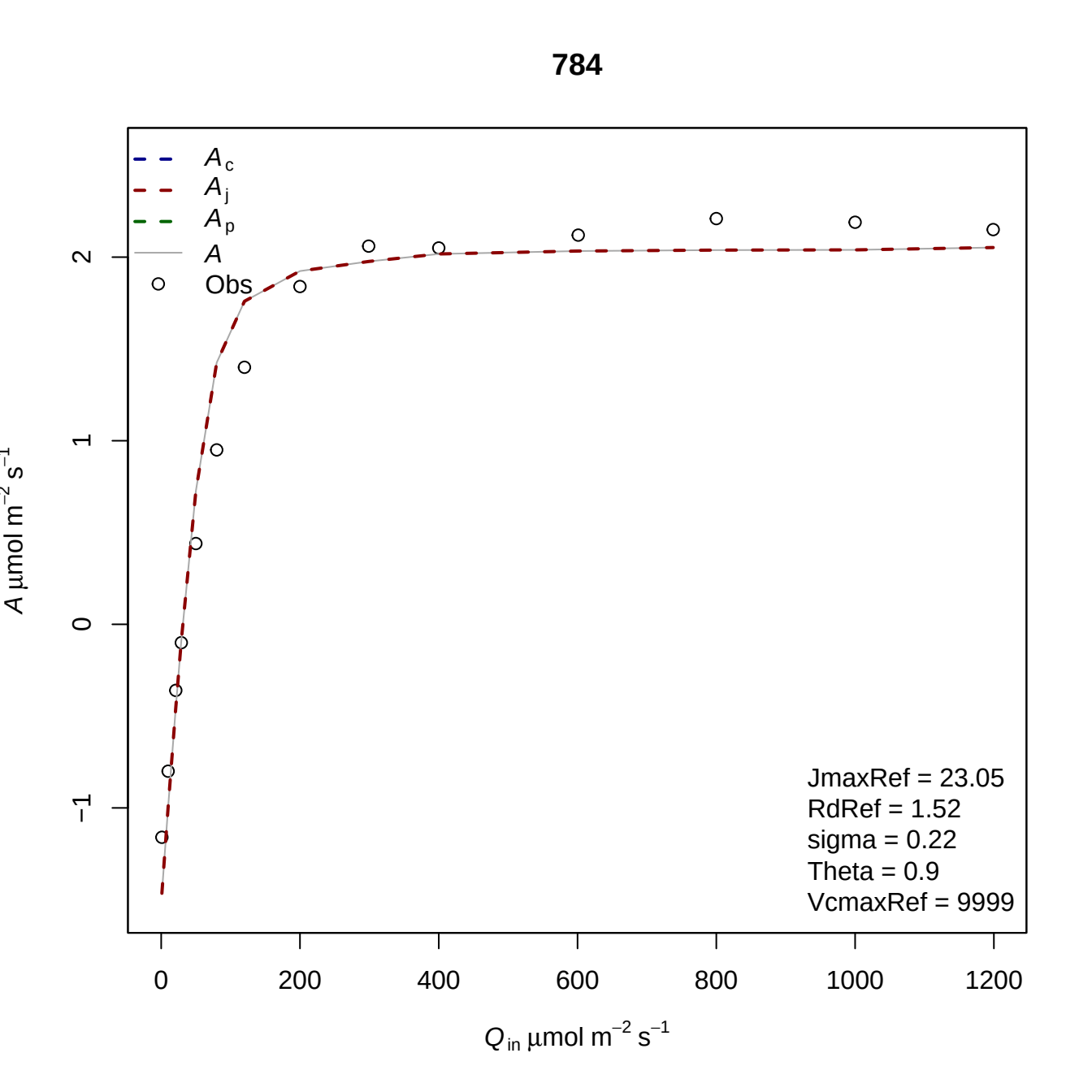

779
A_c
A_j
A_p
A
Obs
JmaxRef = 51.92
RdRef = 1.73
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

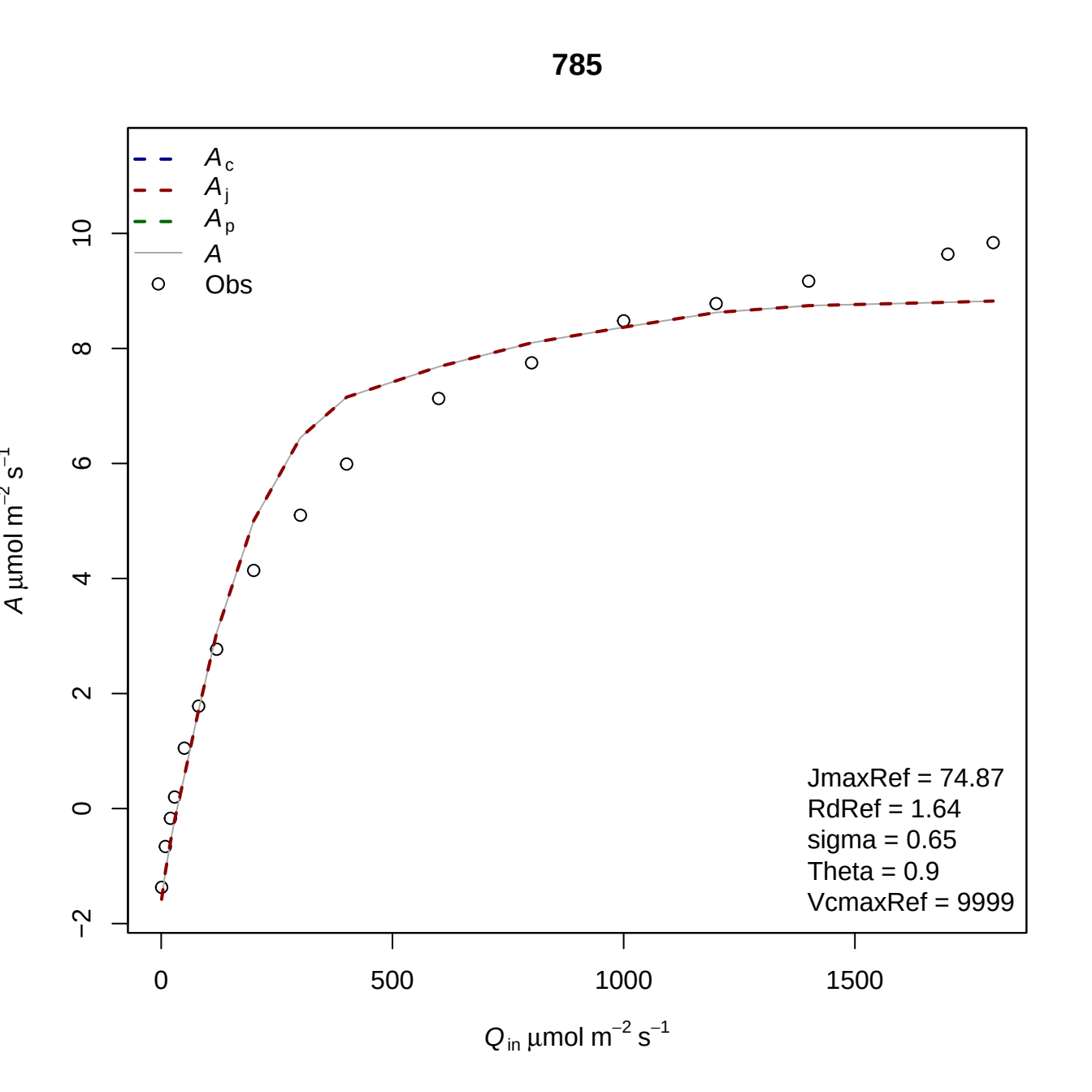

780
A_c
A_j
A_p
A
Obs
JmaxRef = 62.81
RdRef = 2.68
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

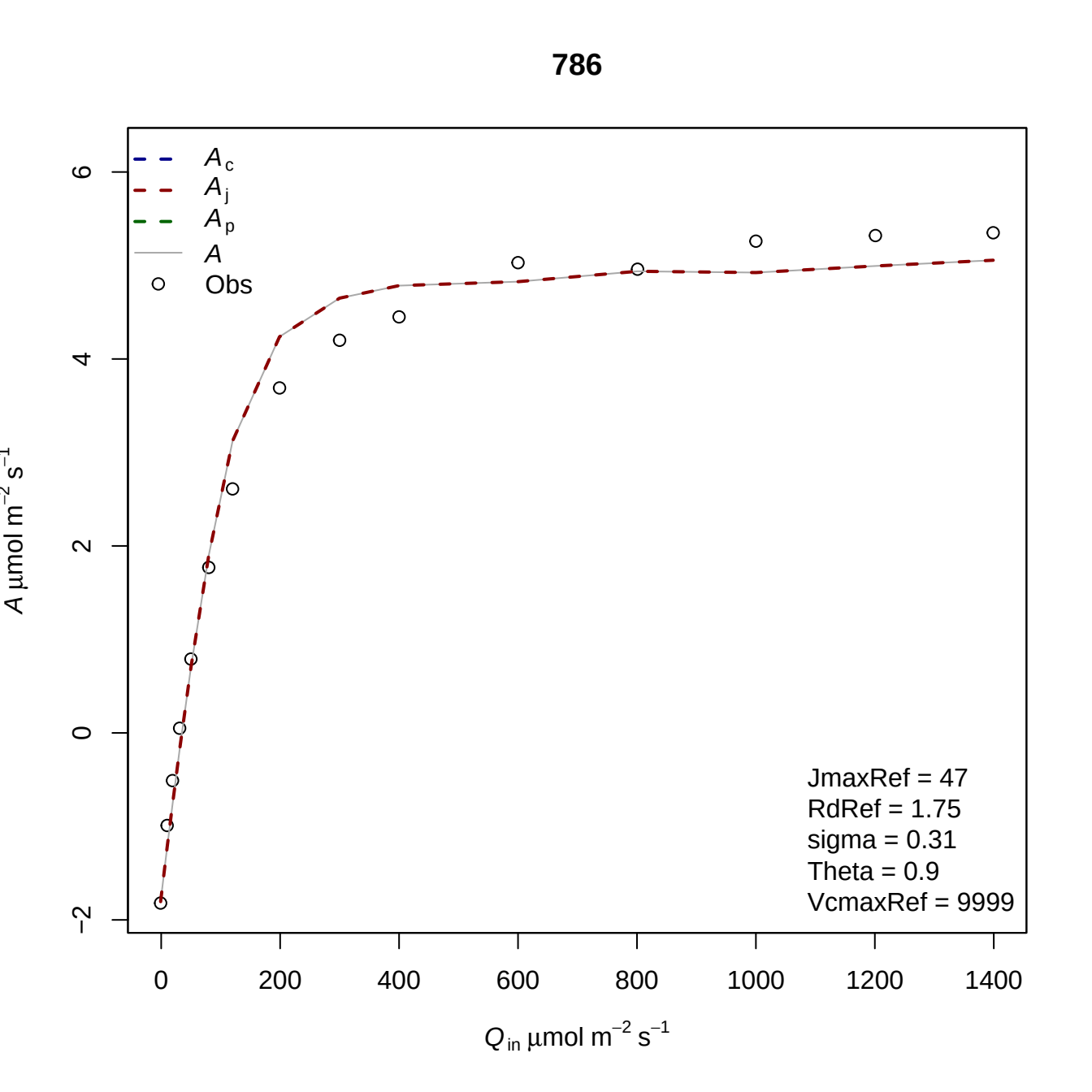

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.72
RdRef = 0.58
sigma = 0.12
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

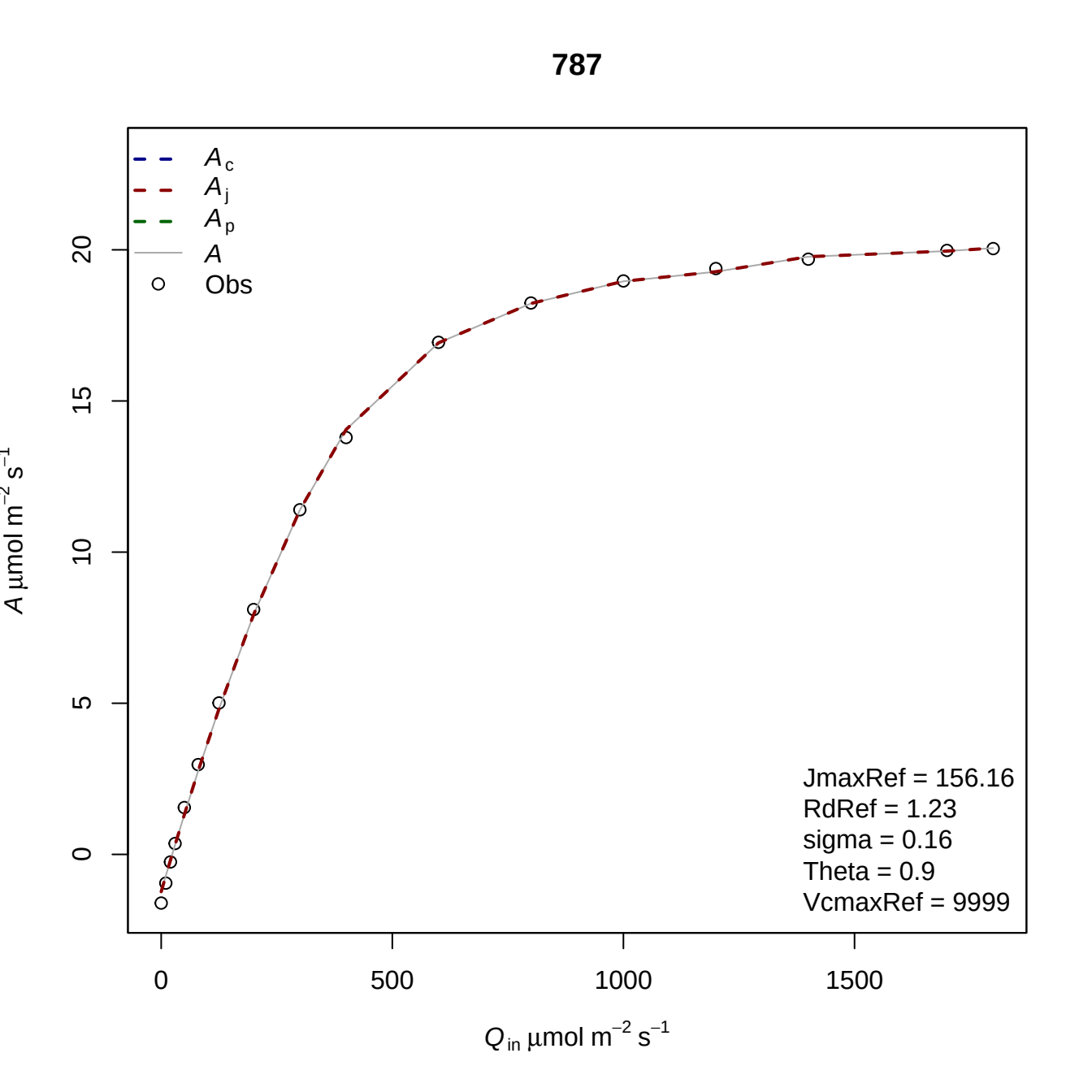

782
A_c
A_j
A_p
A
Obs
JmaxRef = 36
RdRef = 0.8
sigma = 0.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

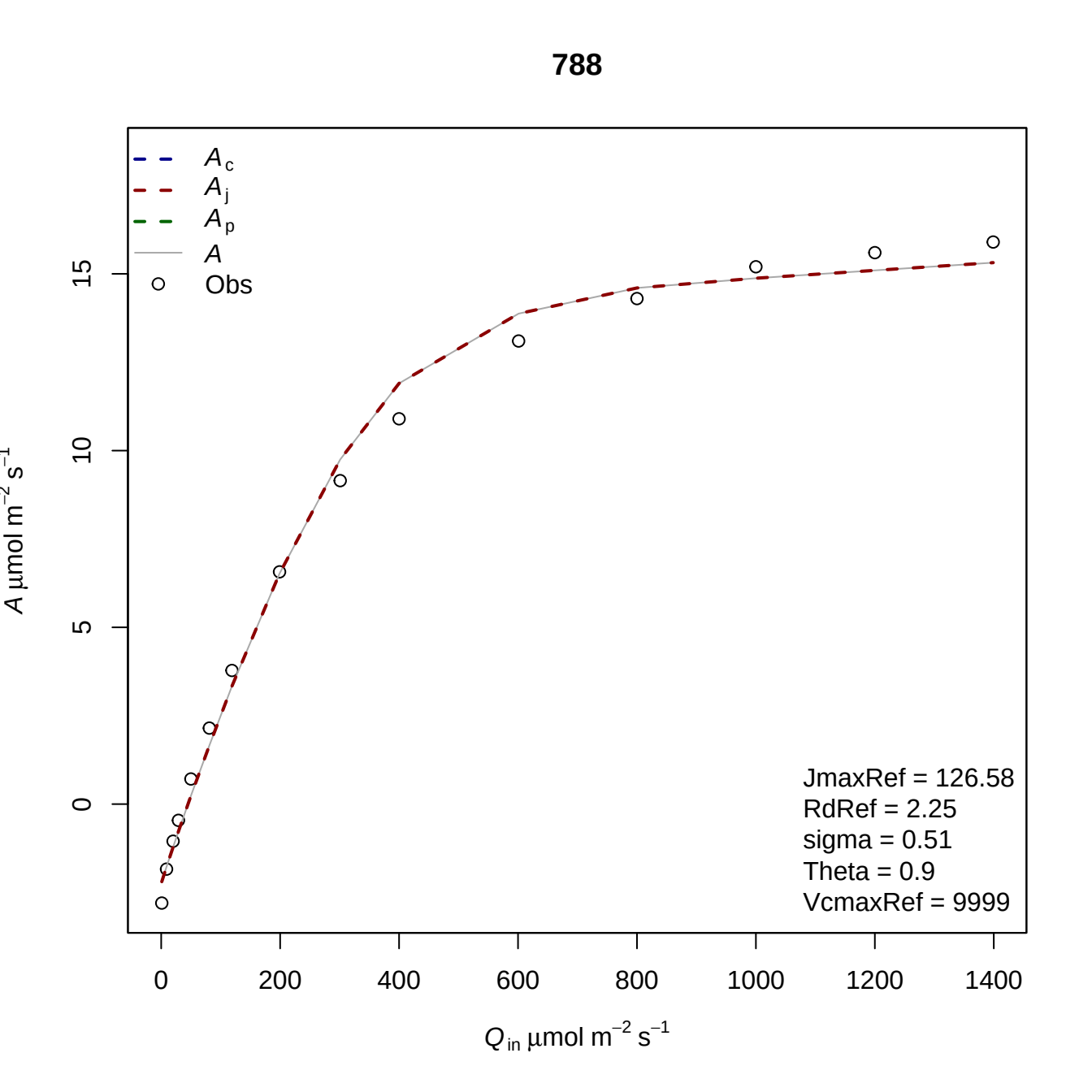

783
A_c
A_j
A_p
A
Obs
JmaxRef = 58.57
RdRef = 0.37
sigma = 0.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

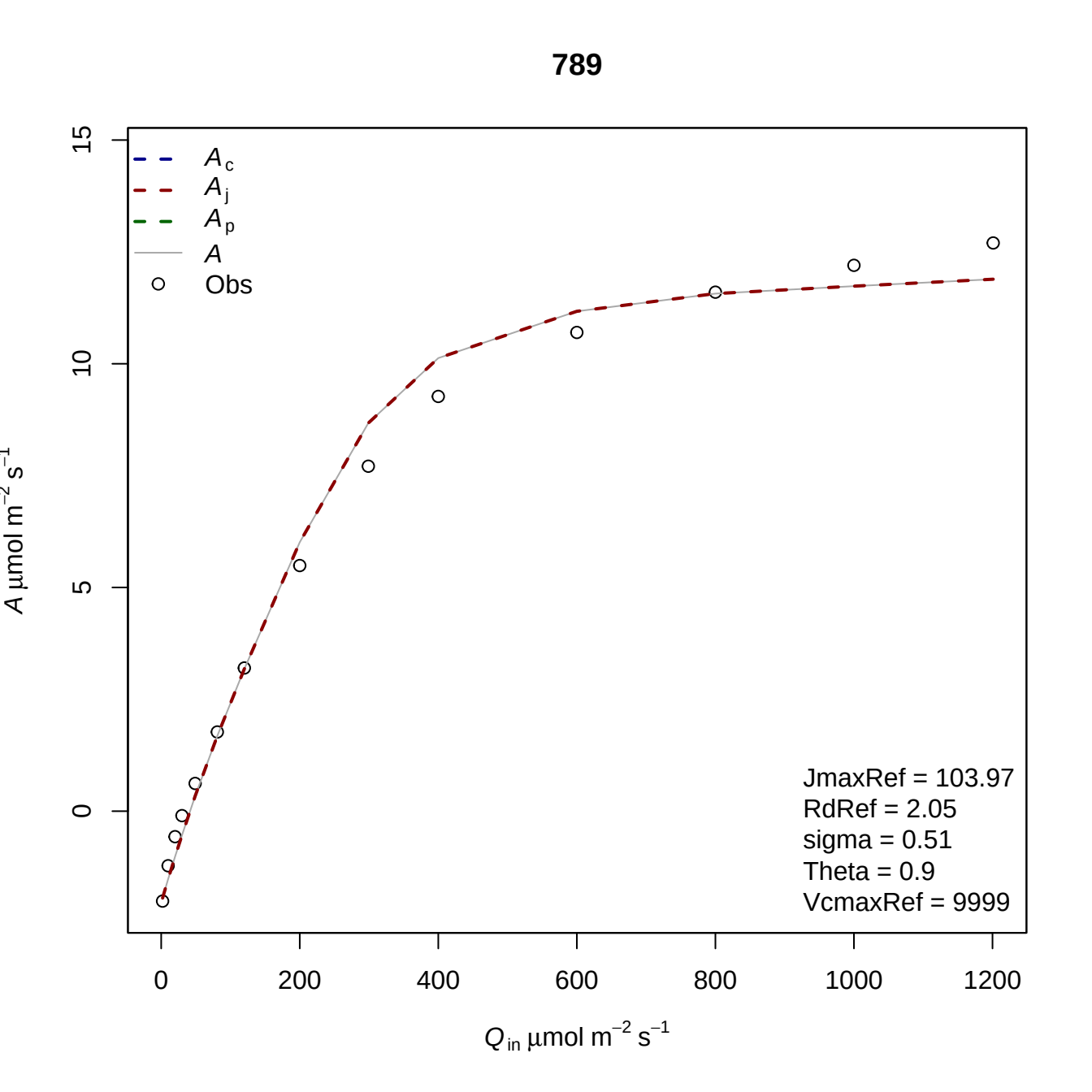

784
A_c
A_j
A_p
A
Obs
JmaxRef = 23.05
RdRef = 1.52
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

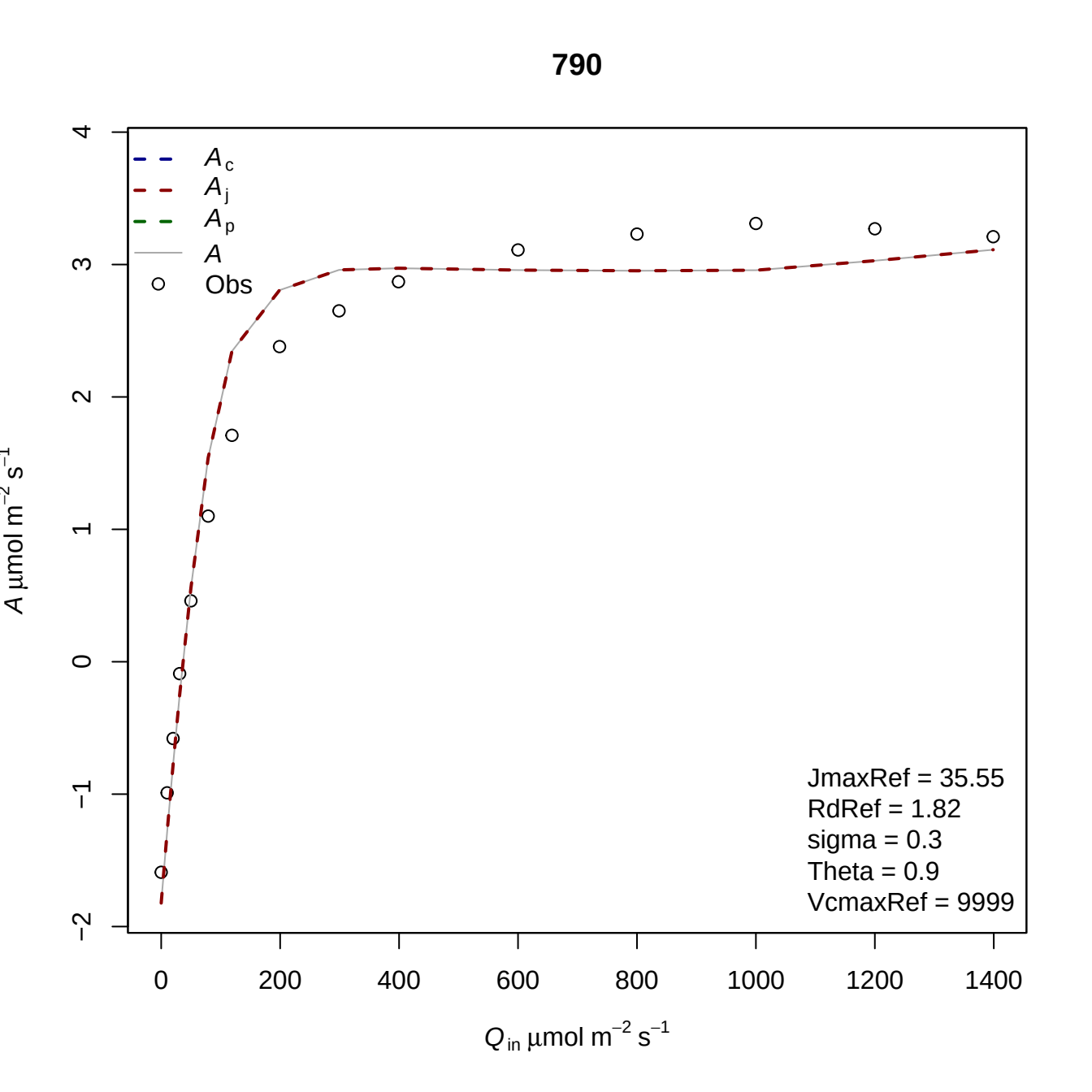

785
A_c
A_j
A_p
A
Obs
JmaxRef = 74.87
RdRef = 1.64
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

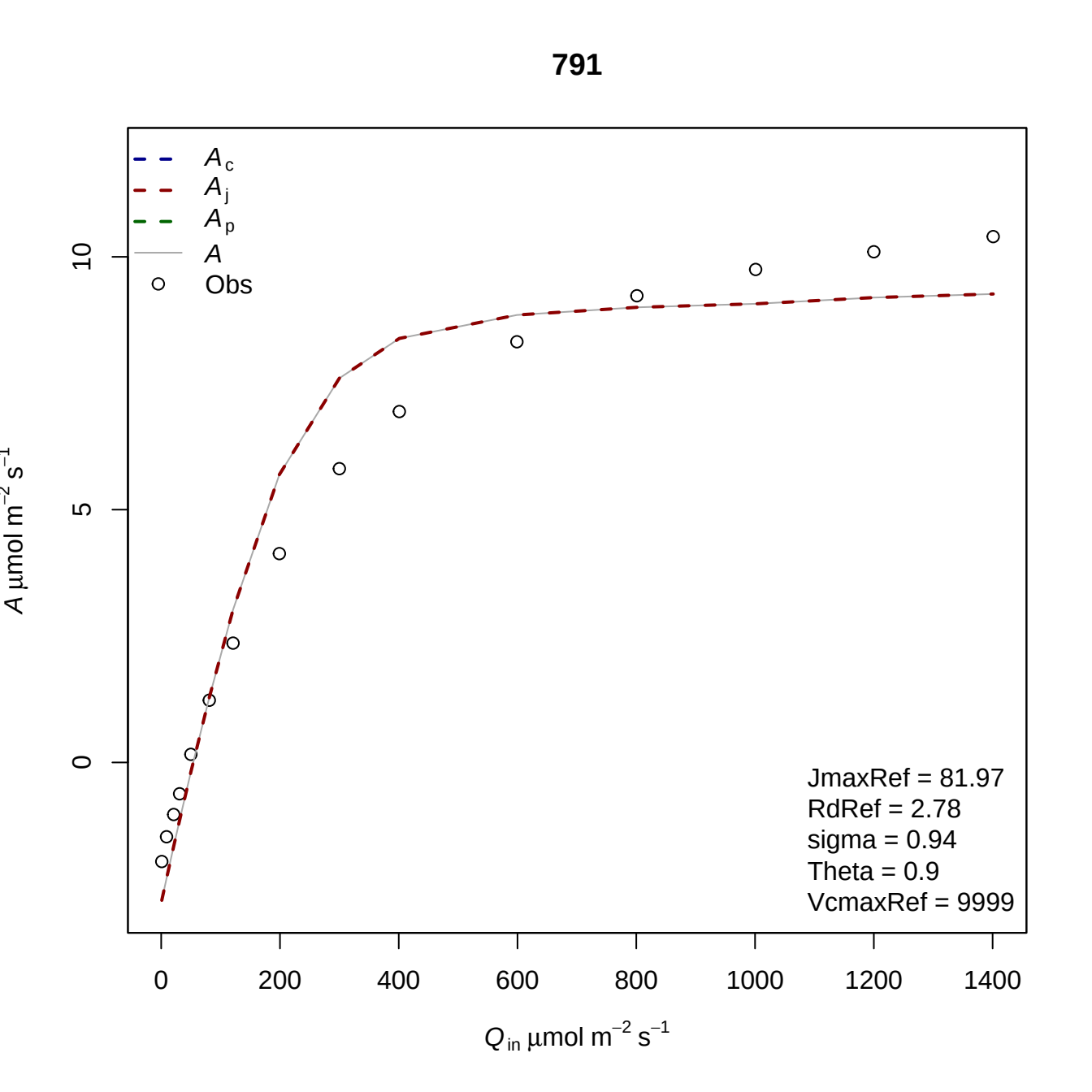

786
A_c
A_j
A_p
A
Obs
JmaxRef = 47
RdRef = 1.75
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

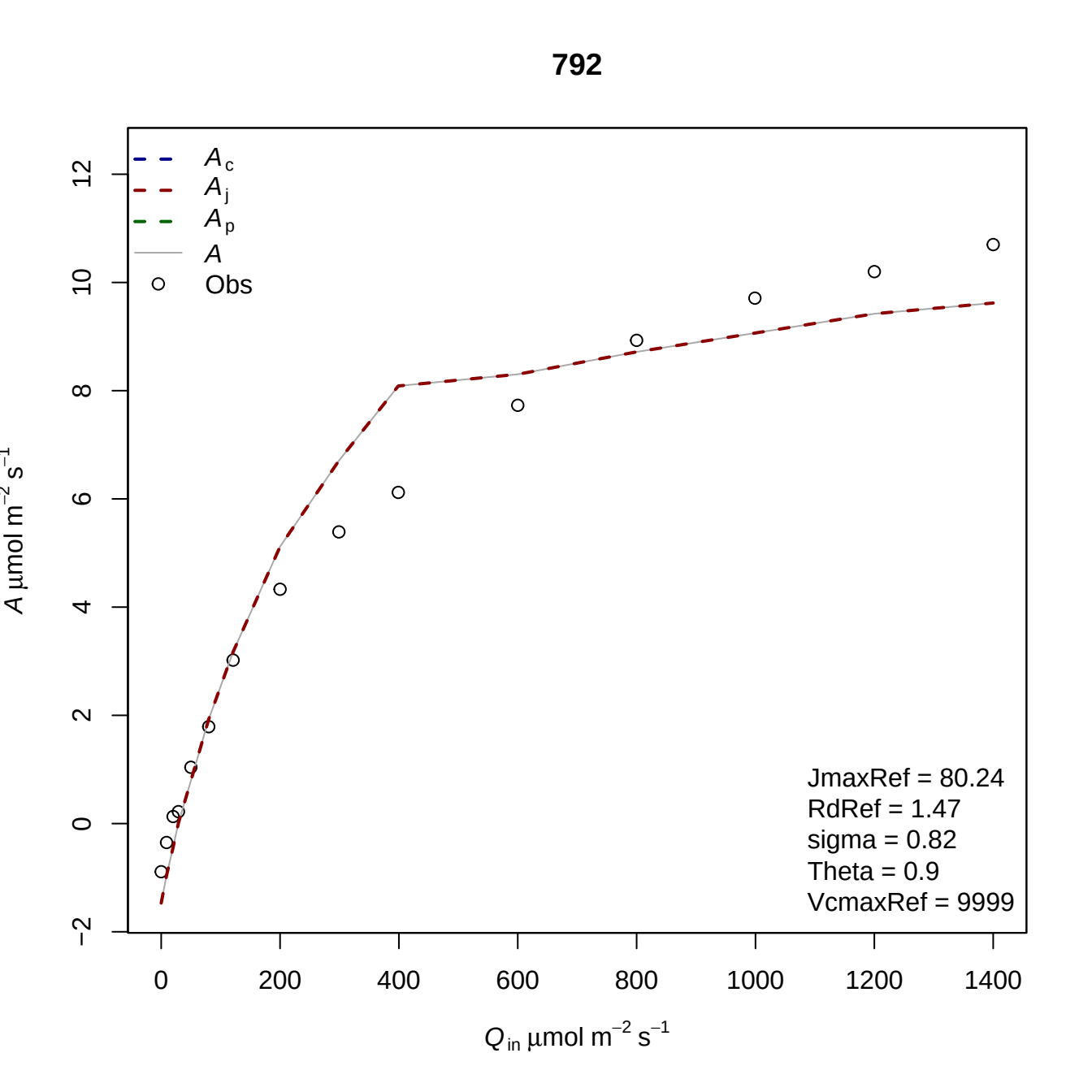

787
A_c
A_j
A_p
A
Obs
JmaxRef = 156.16
RdRef = 1.23
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

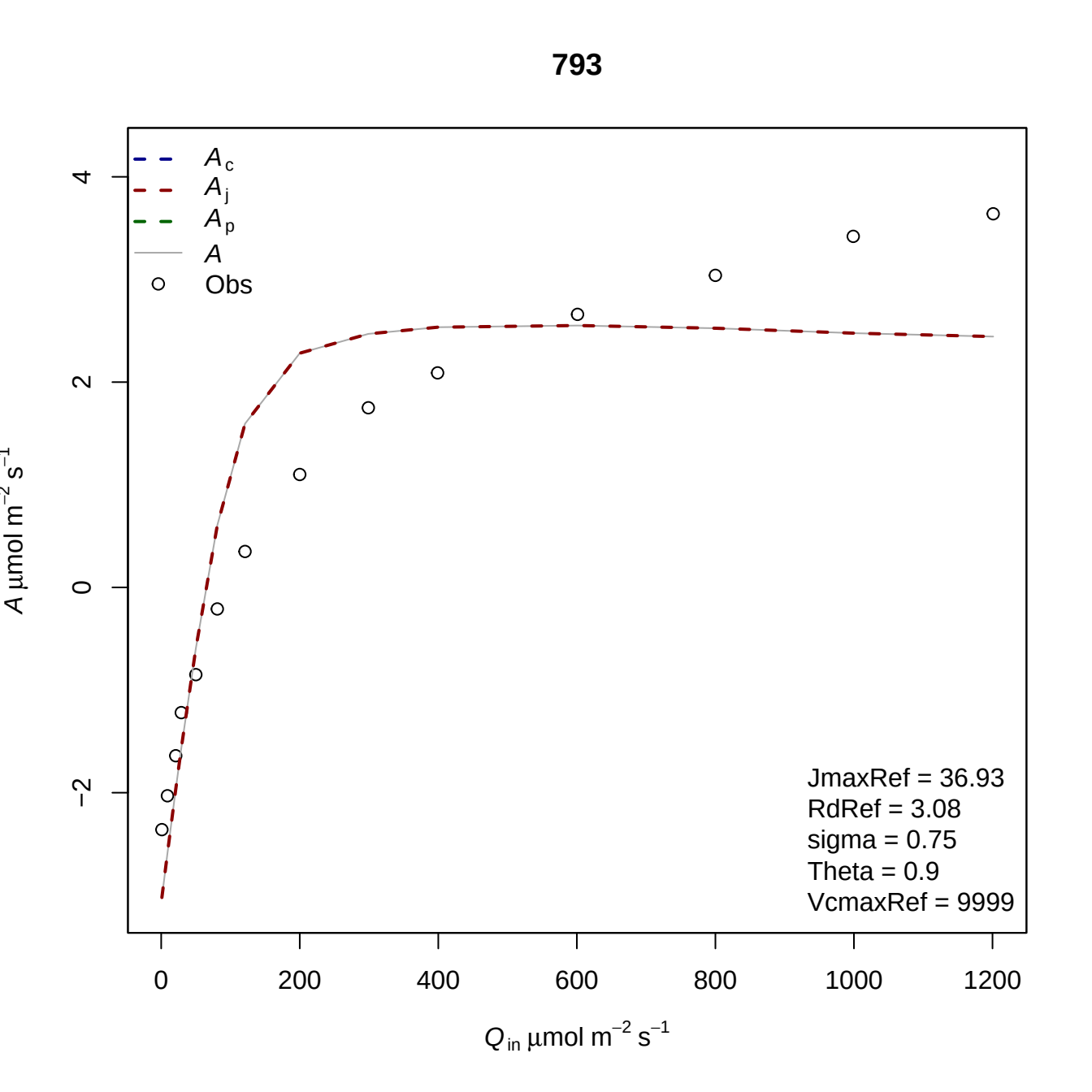

788
A_c
A_j
A_p
A
Obs
JmaxRef = 126.58
RdRef = 2.25
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

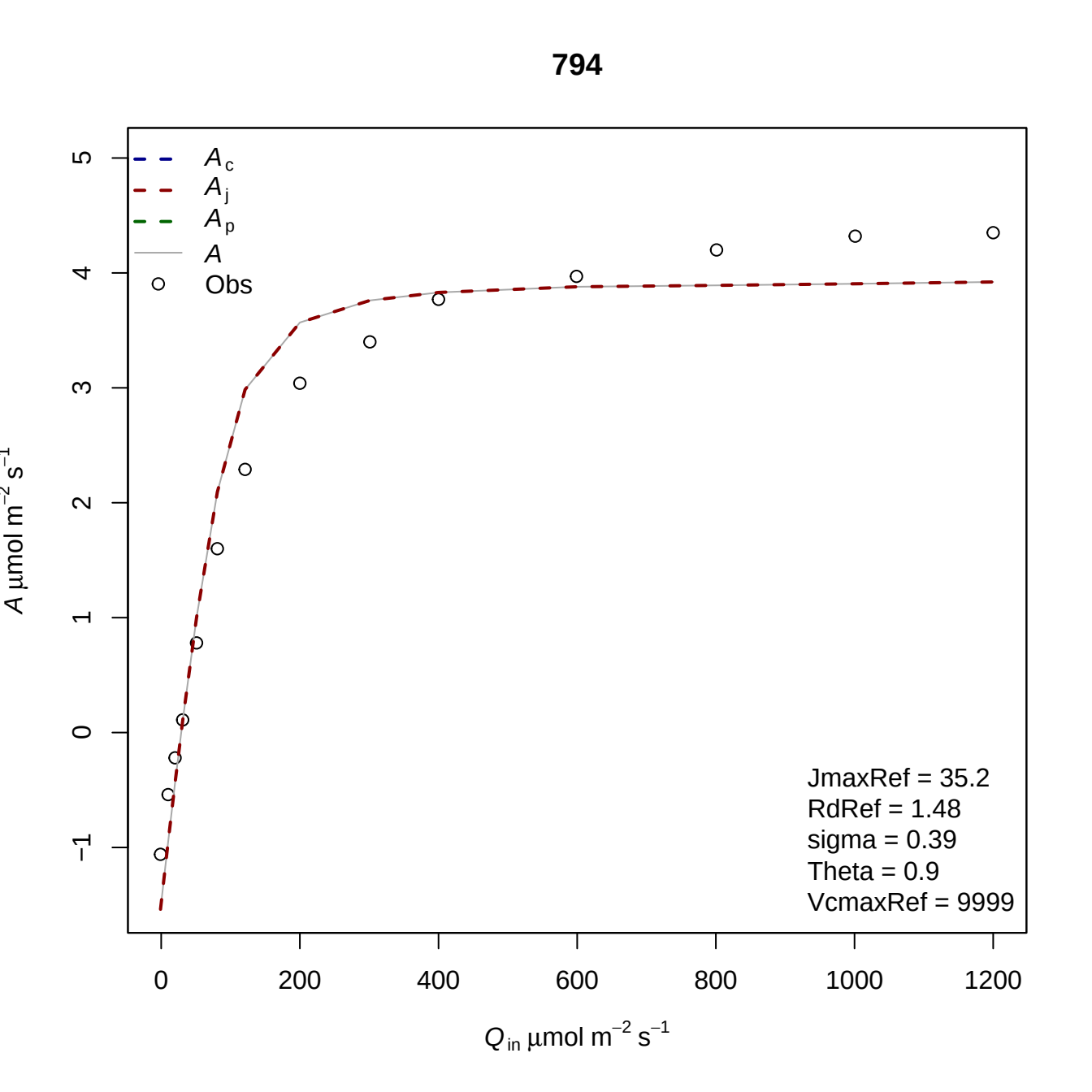

789
A_c
A_j
A_p
A
Obs
JmaxRef = 103.97
RdRef = 2.05
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

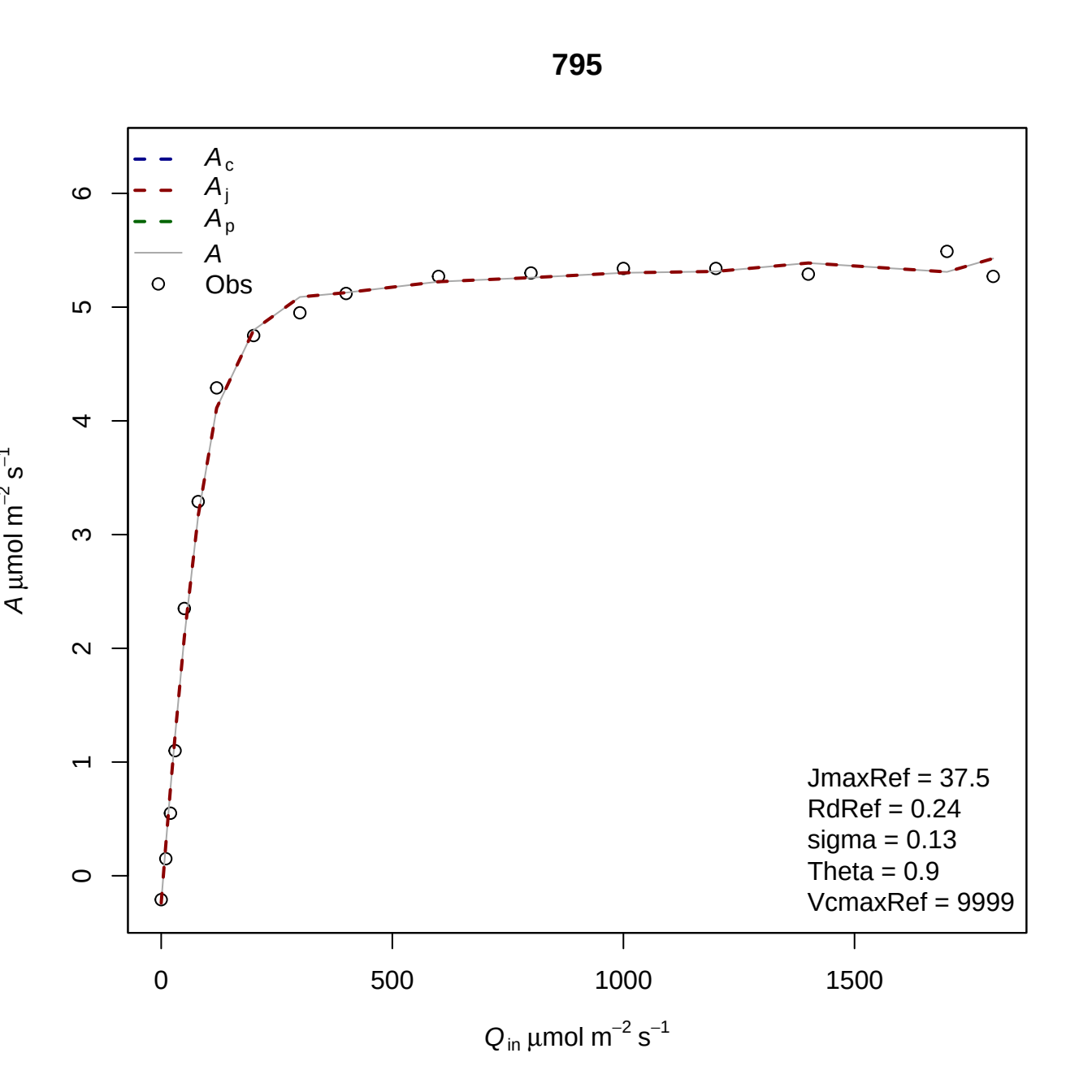

790
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 1.82
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

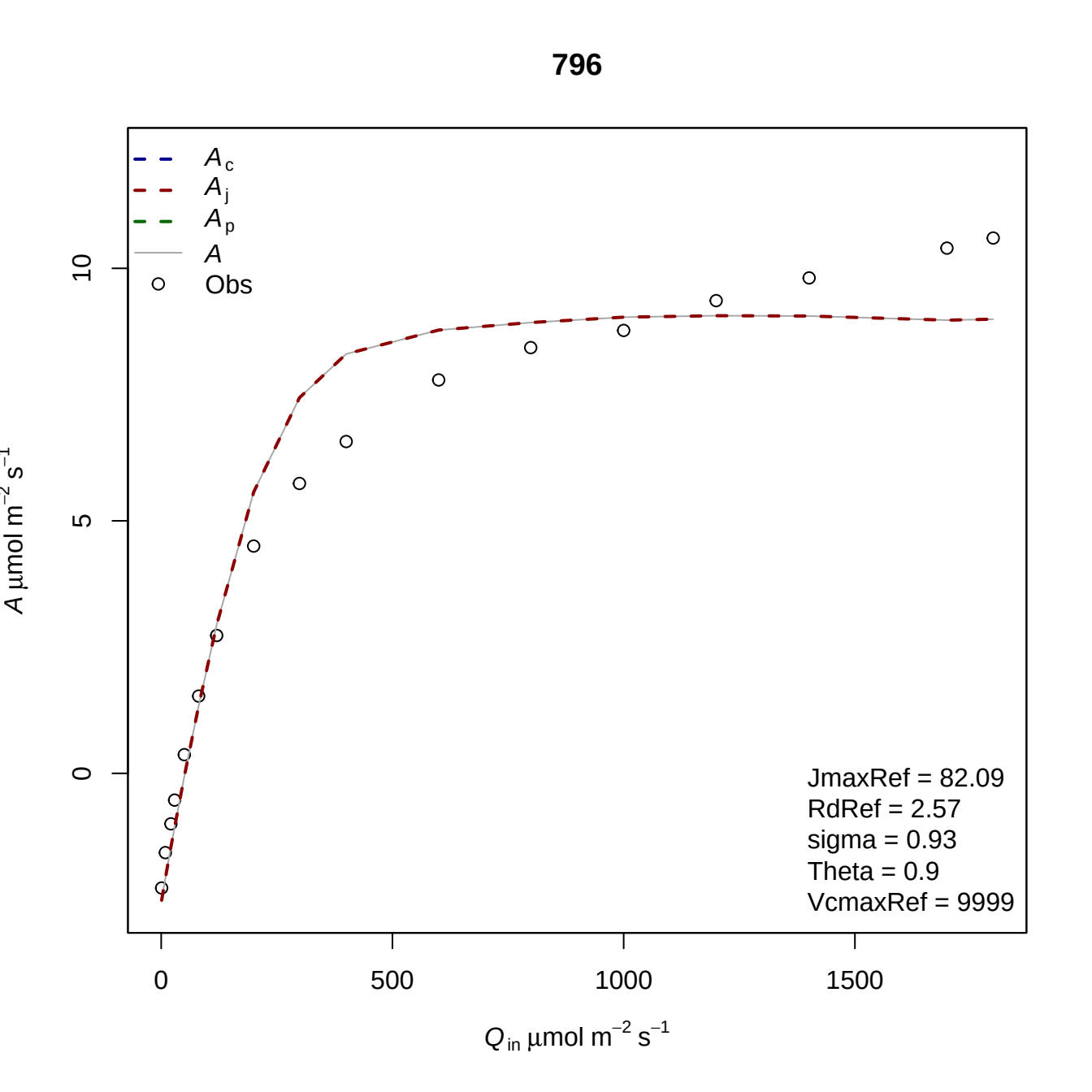

791
A_c
A_j
A_p
A
Obs
JmaxRef = 81.97
RdRef = 2.78
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
1400

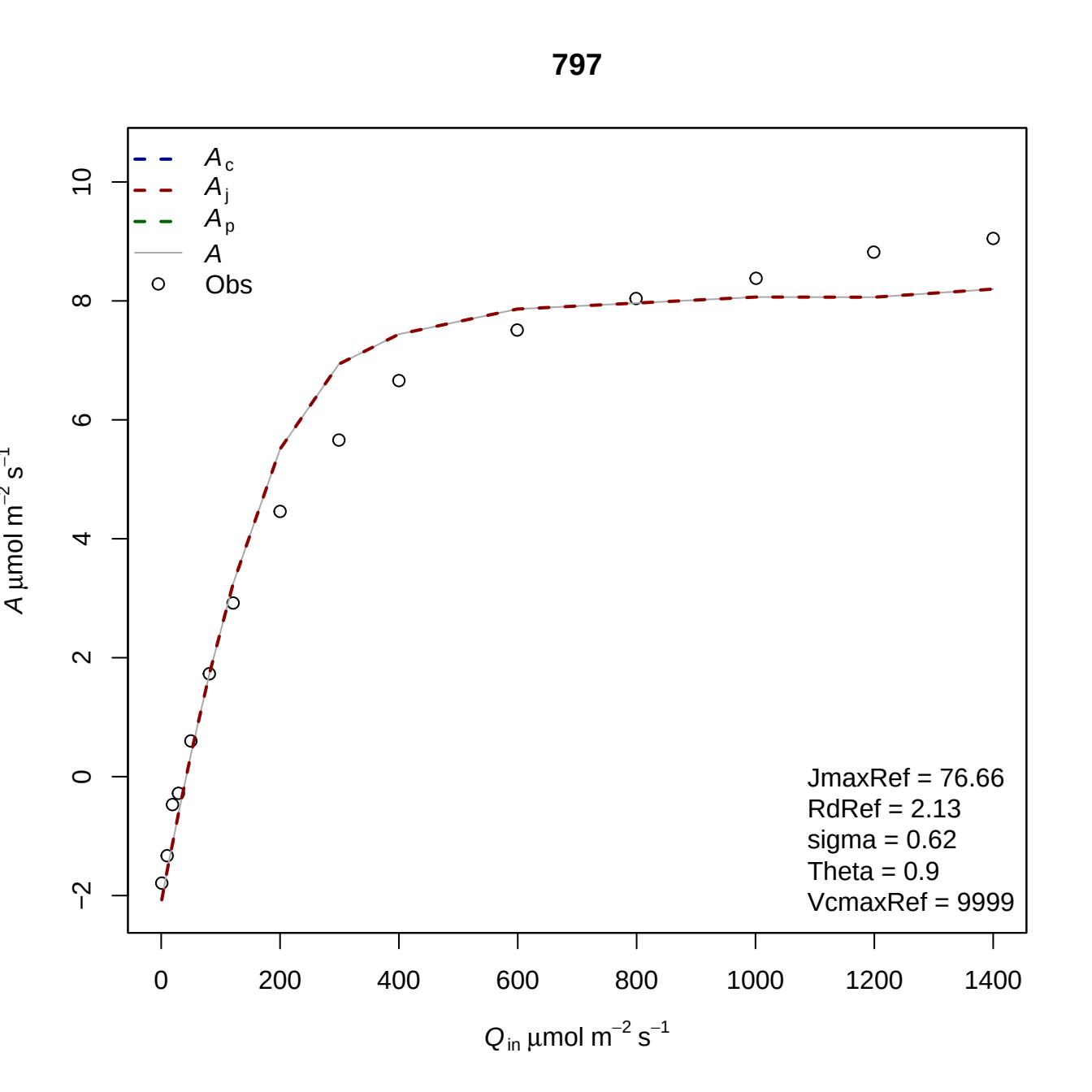

792
A_c
A_j
A_p
A
Obs
JmaxRef = 80.24
RdRef = 1.47
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

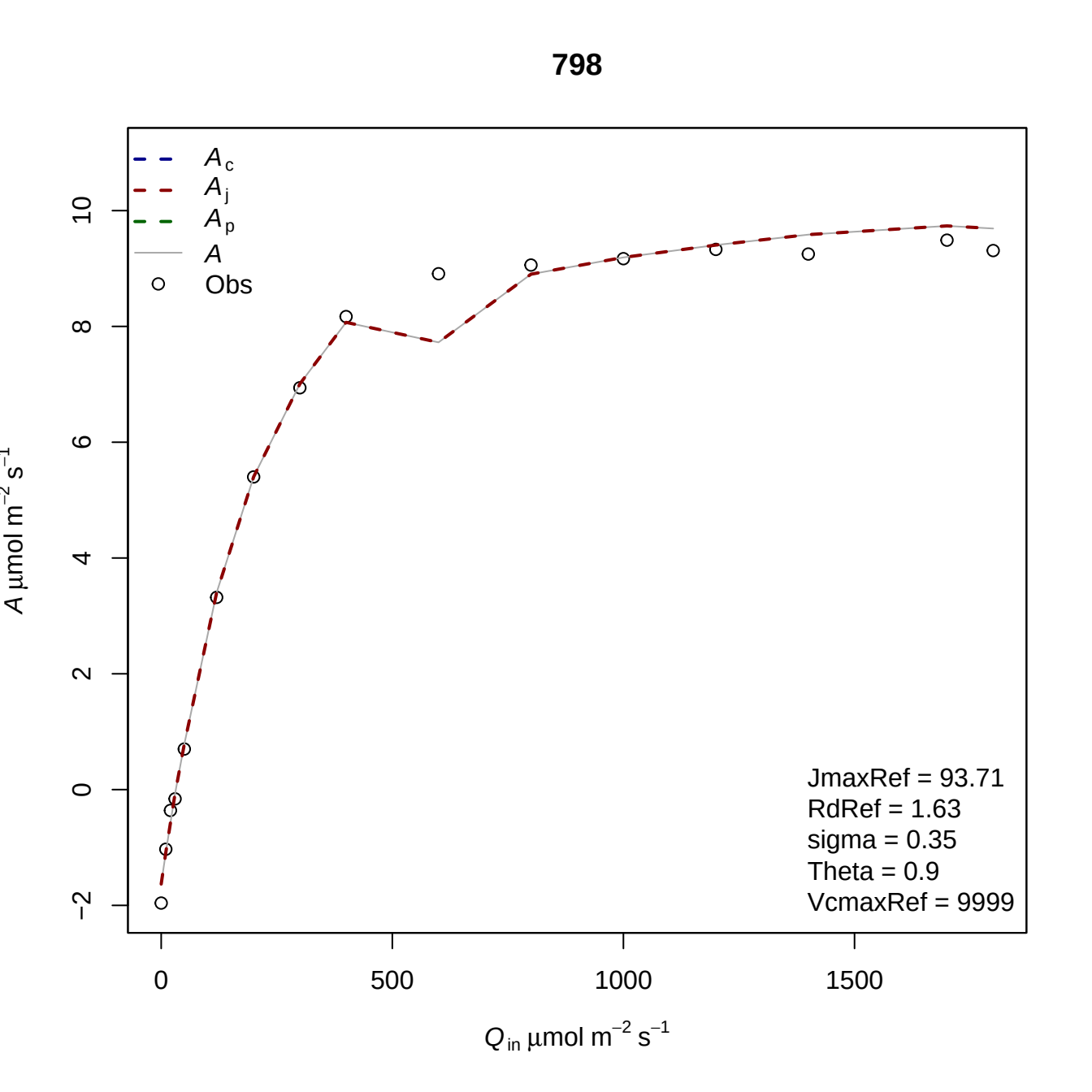

793
A_c
A_j
A_p
A
Obs
JmaxRef = 36.93
RdRef = 3.08
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000
1200

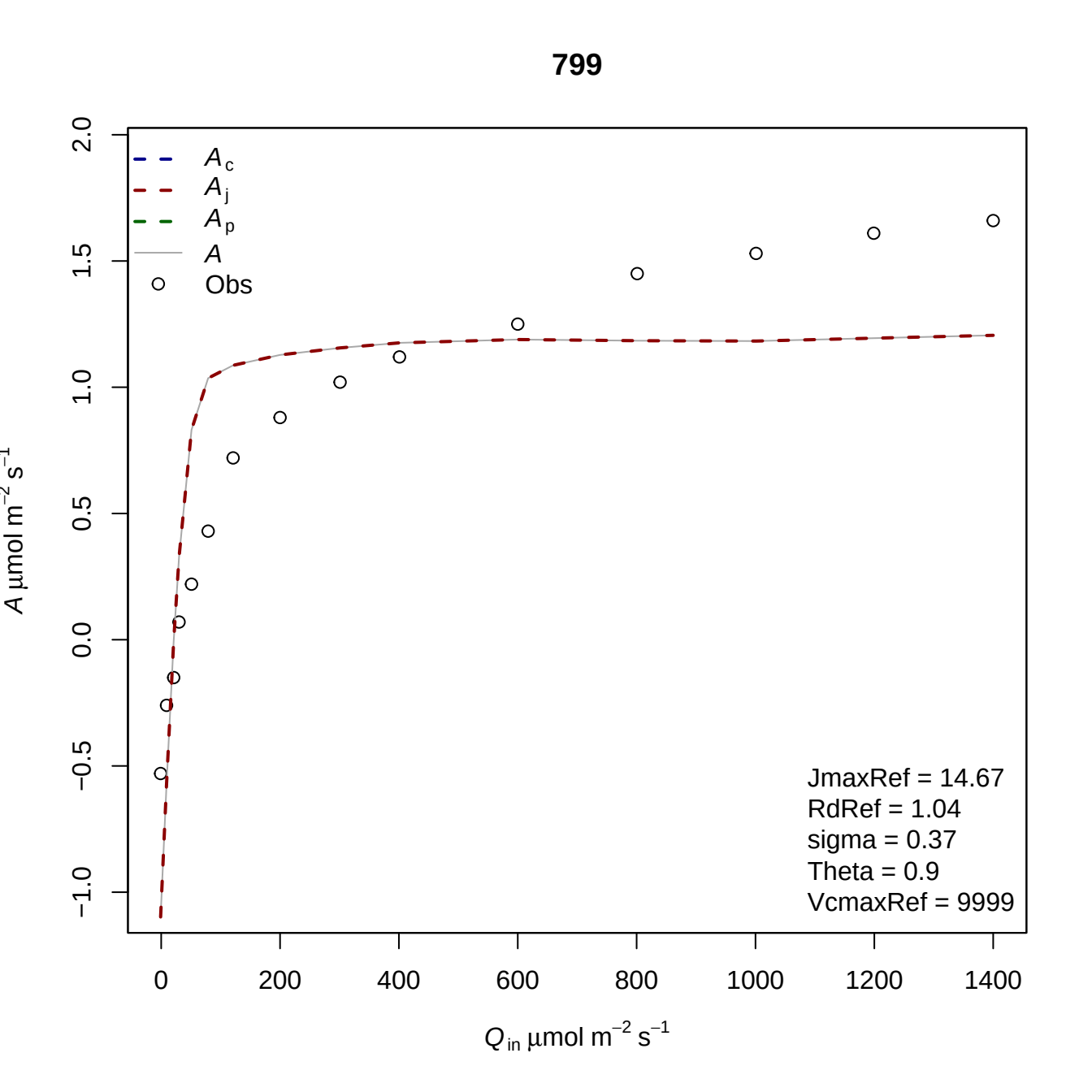

794
A_c
A_j
A_p
A
Obs
JmaxRef = 35.2
RdRef = 1.48
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

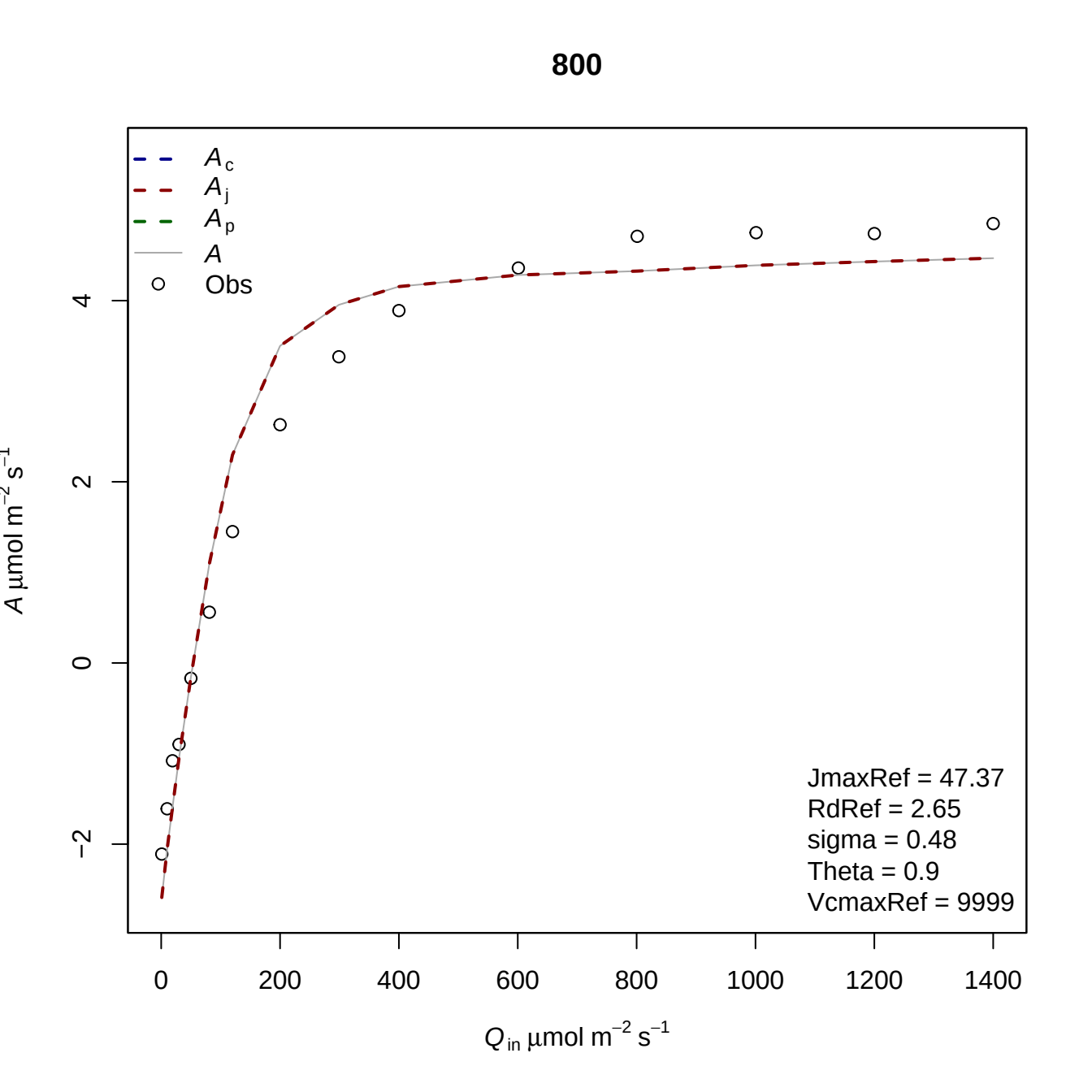

795
A_c
A_j
A_p
A
Obs
JmaxRef = 37.5
RdRef = 0.24
sigma = 0.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

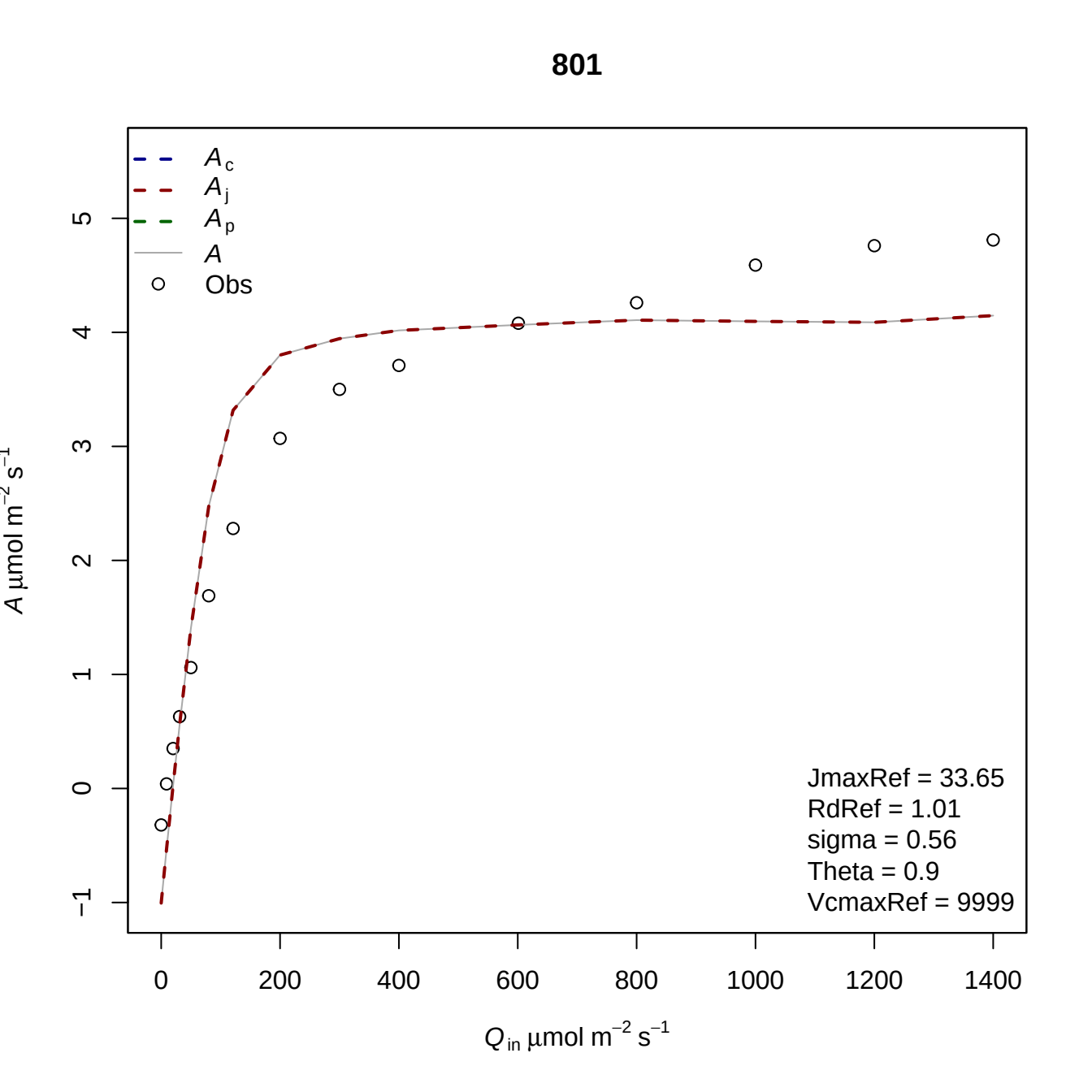

796
A_c
A_j
A_p
A
Obs
JmaxRef = 82.09
RdRef = 2.57
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

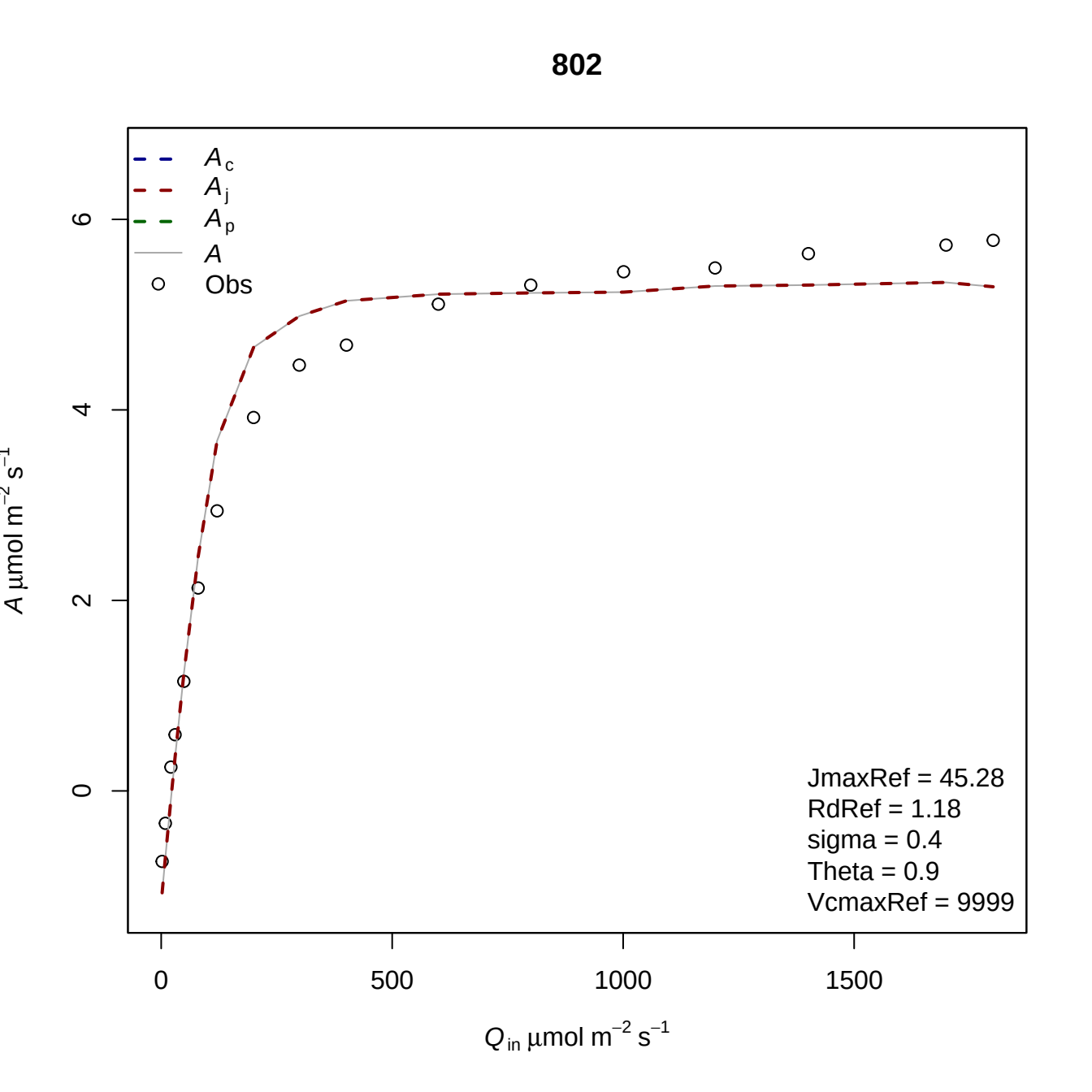

797
A_c
A_j
A_p
A
Obs
JmaxRef = 76.66
RdRef = 2.13
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

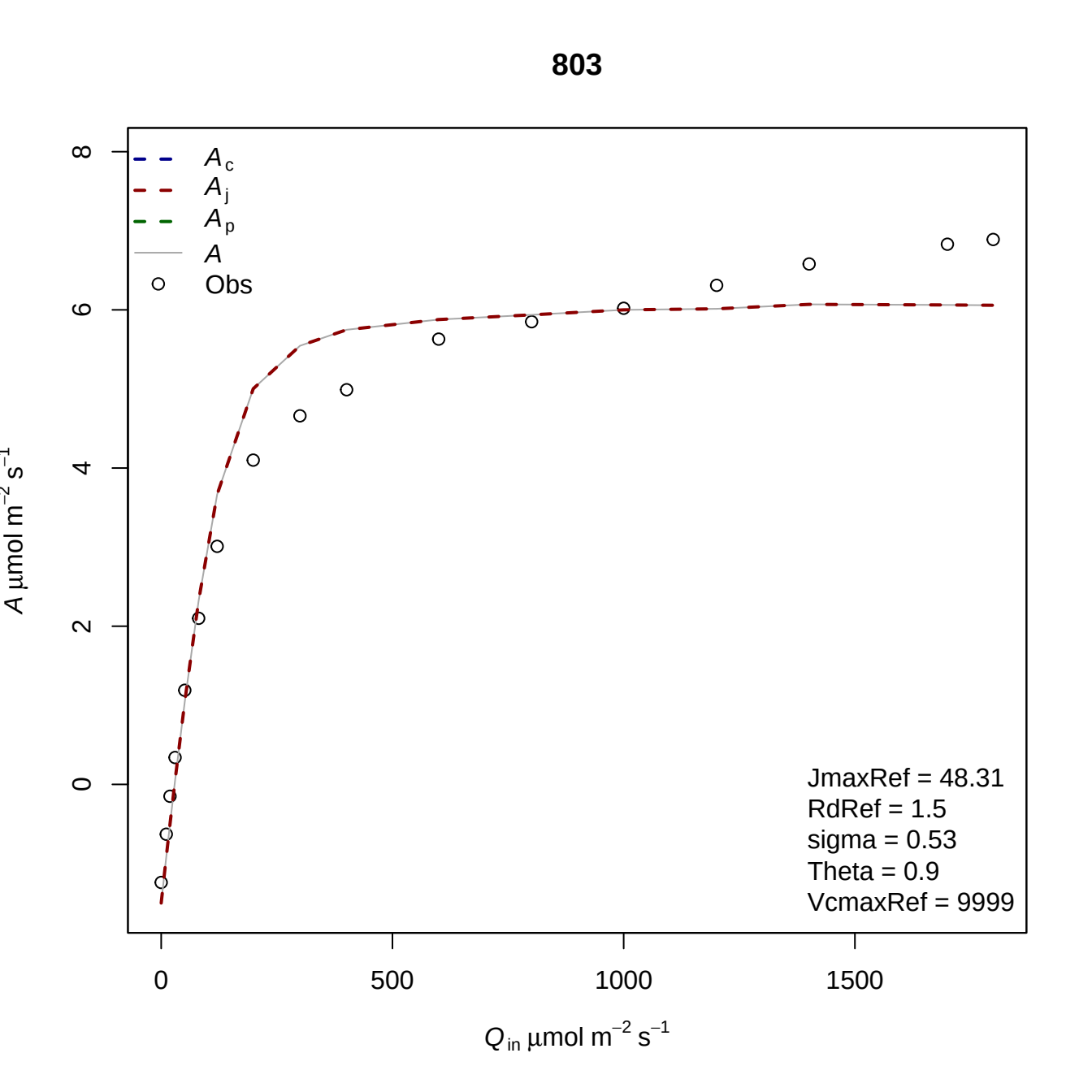

798
A_c
A_j
A_p
A
Obs
JmaxRef = 93.71
RdRef = 1.63
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

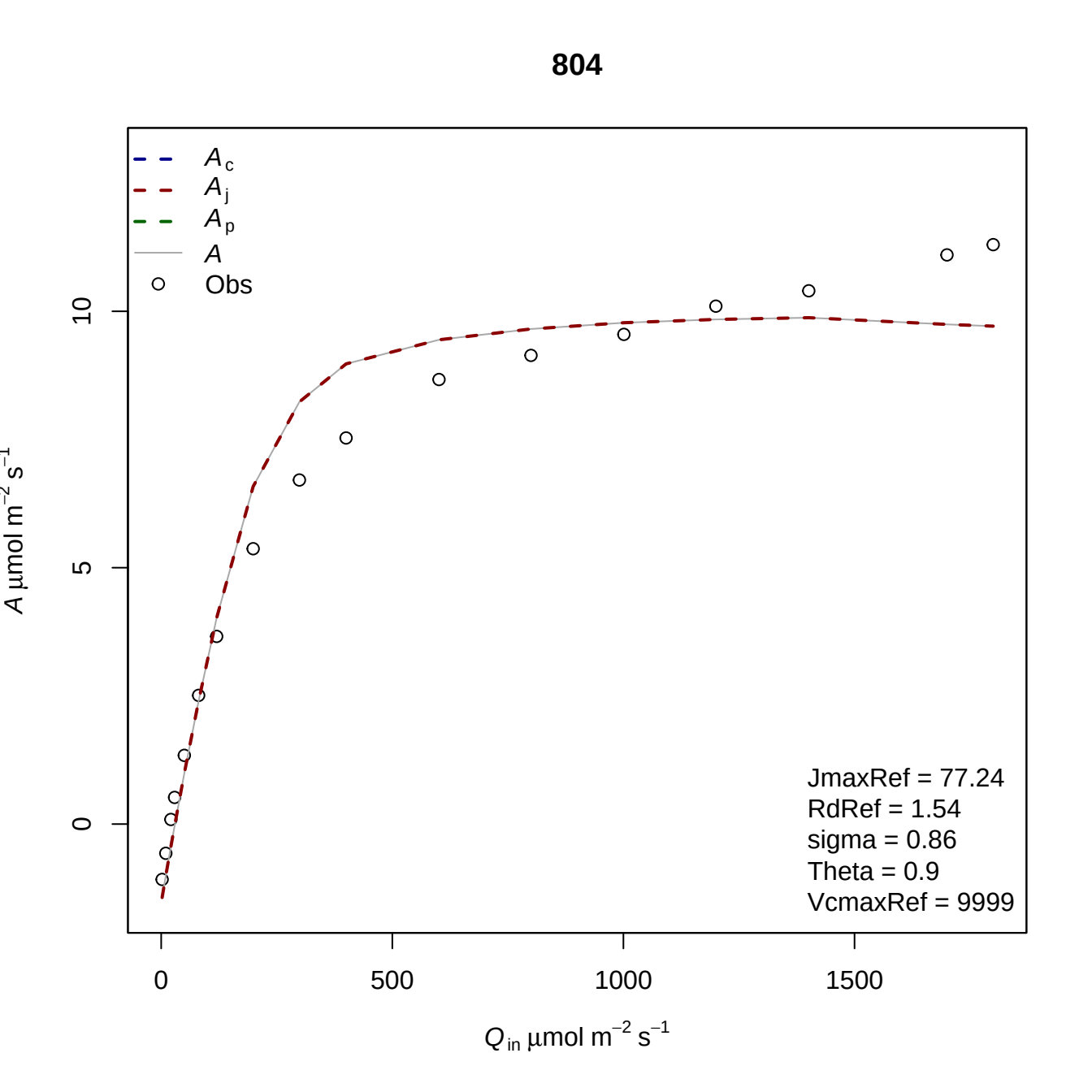

799
A_c
A_j
A_p
A
Obs
JmaxRef = 14.67
RdRef = 1.04
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

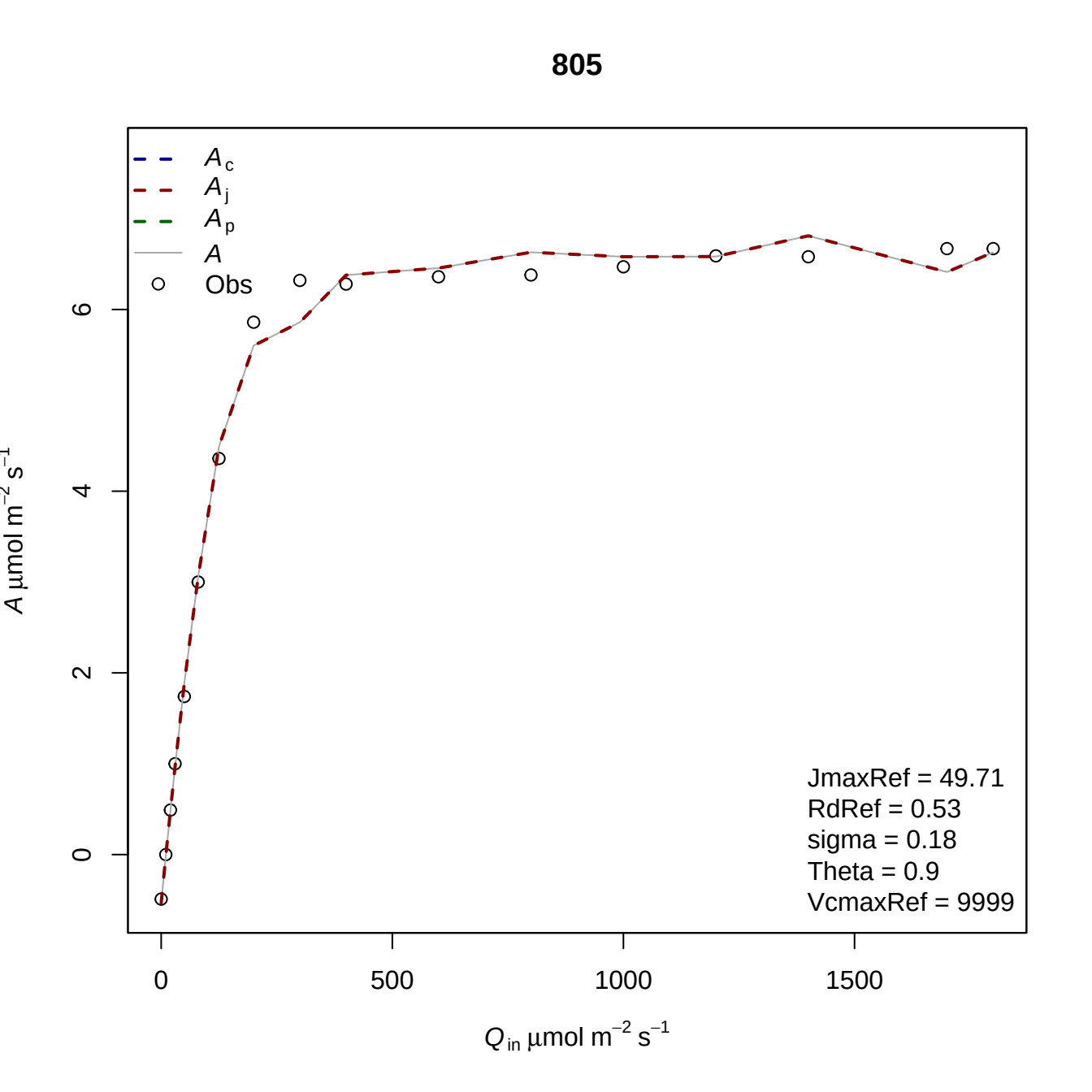

800
A_c
A_j
A_p
A
Obs
JmaxRef = 47.37
RdRef = 2.65
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

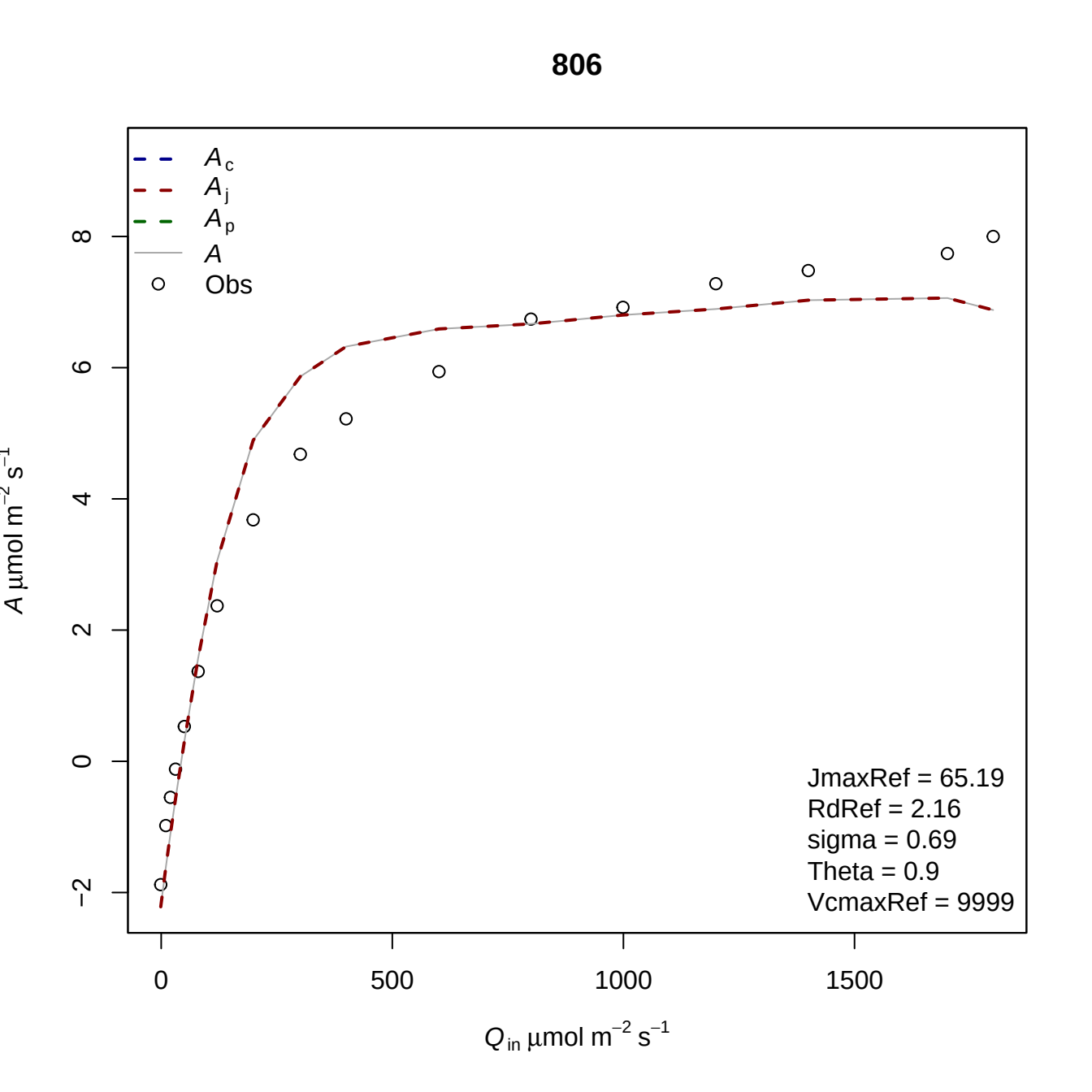

801
A_c
A_j
A_p
A
Obs
JmaxRef = 33.65
RdRef = 1.01
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

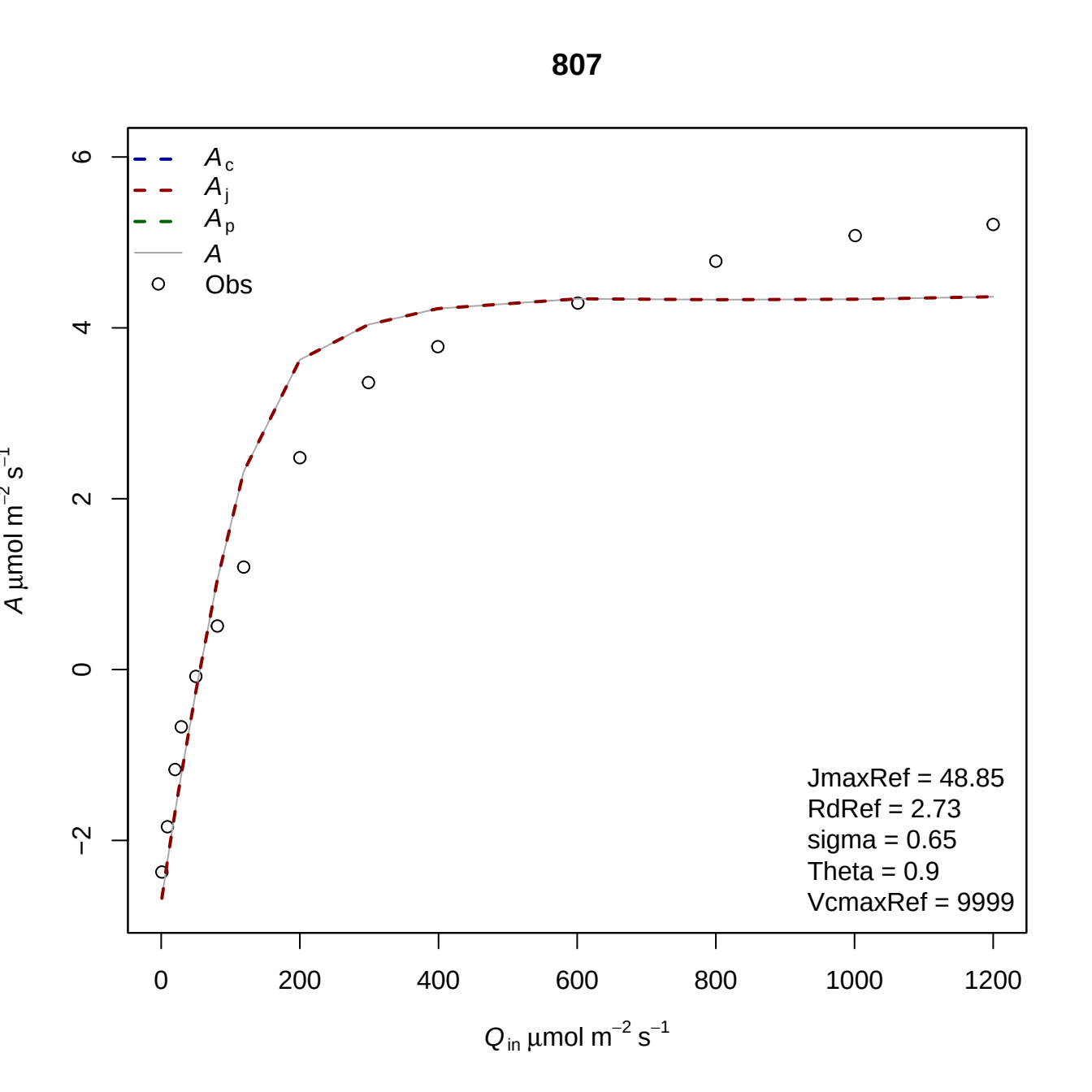

802
A_c
A_j
A_p
A
Obs
JmaxRef = 45.28
RdRef = 1.18
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

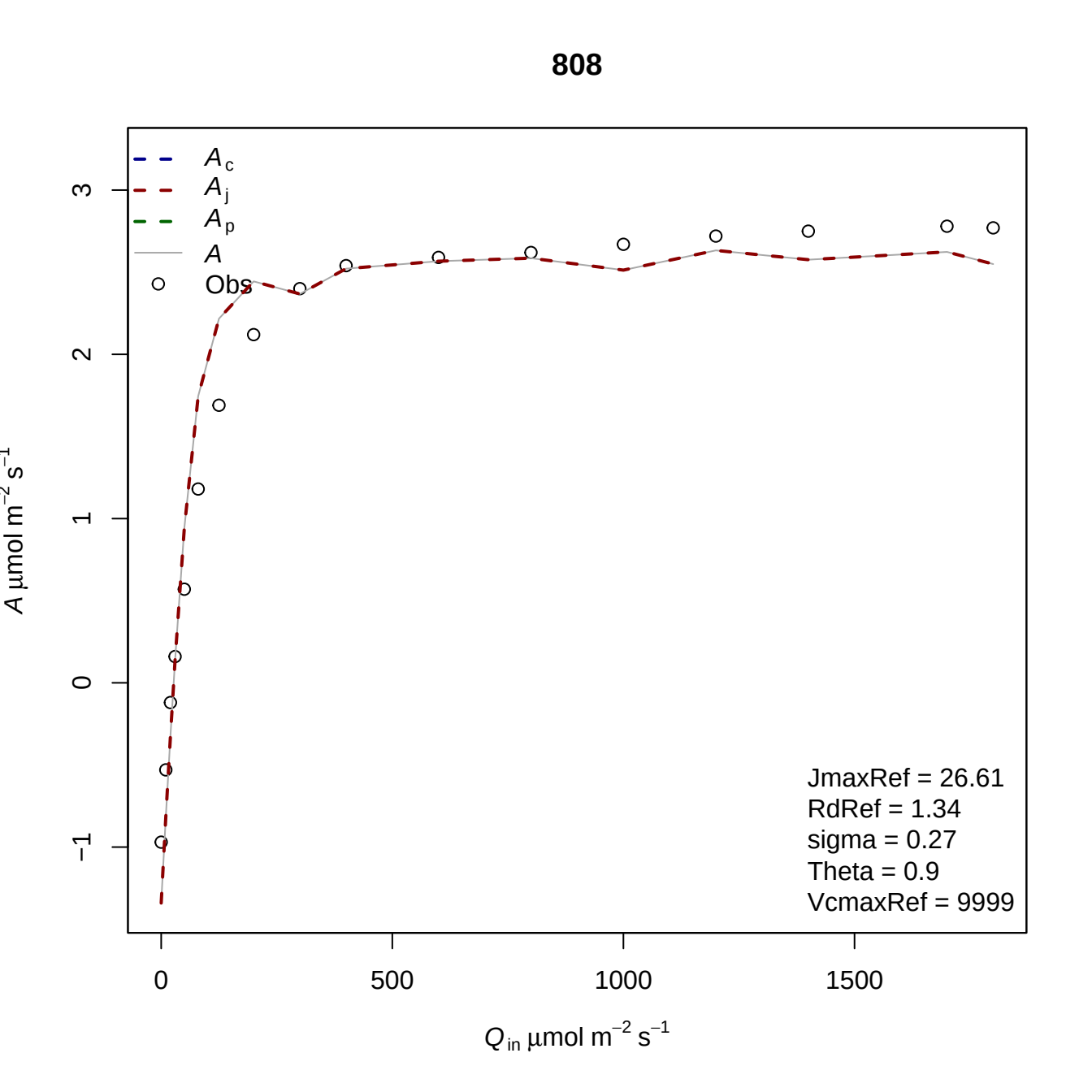

803
A_c
A_j
A_p
A
Obs
JmaxRef = 48.31
RdRef = 1.5
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

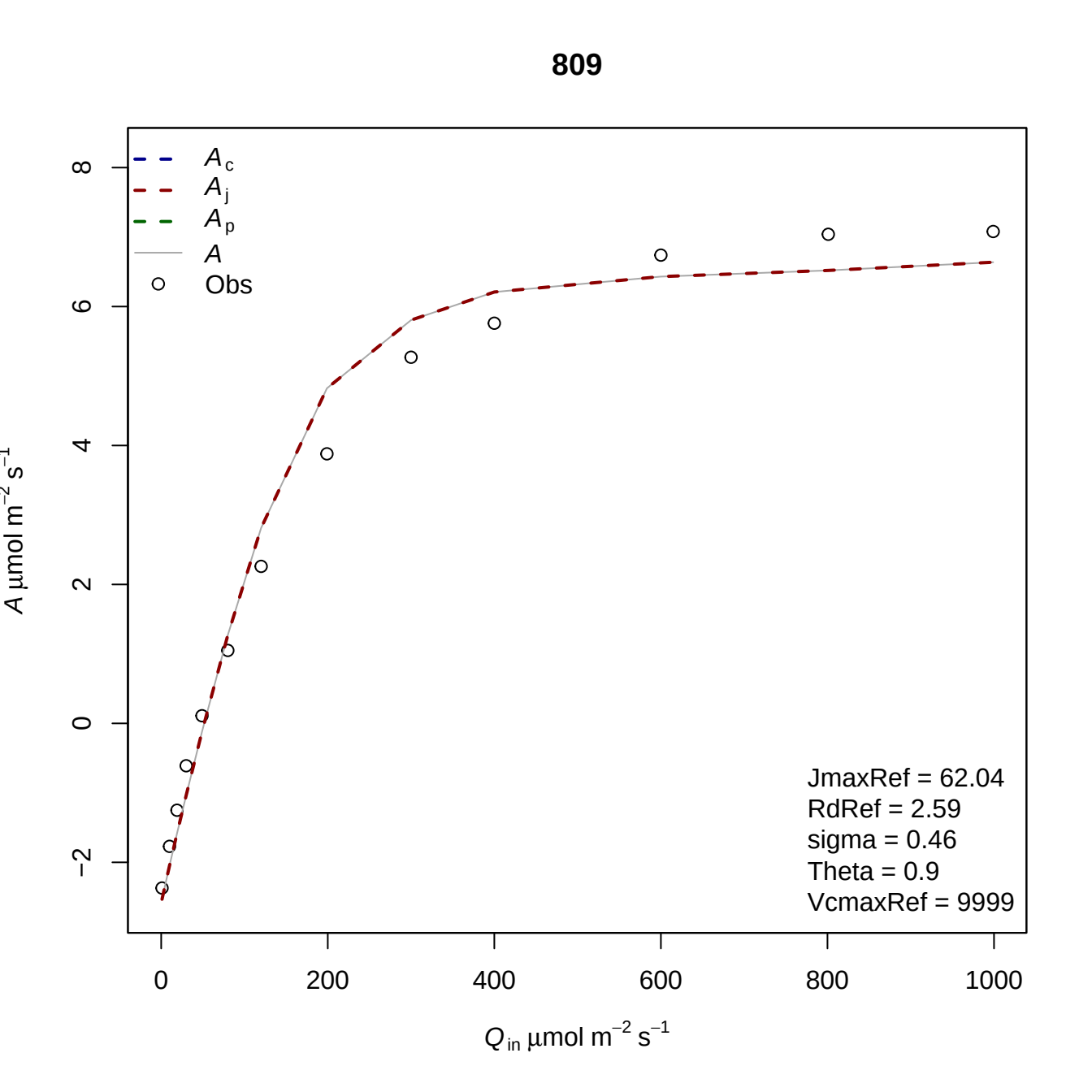

804
A_c
A_j
A_p
A
Obs
JmaxRef = 77.24
RdRef = 1.54
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

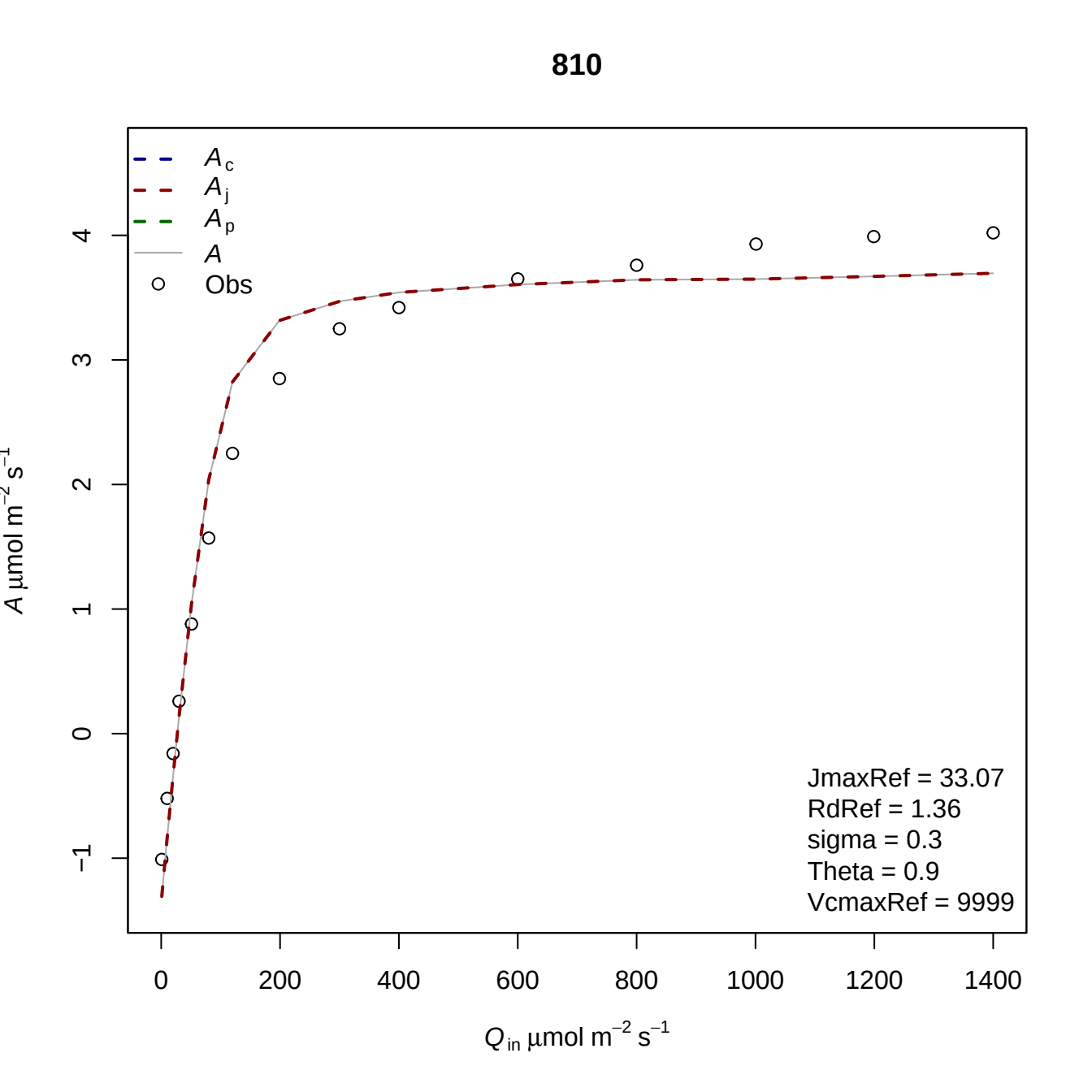

805
A_c
A_j
A_p
A
Obs
JmaxRef = 49.71
RdRef = 0.53
sigma = 0.18
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

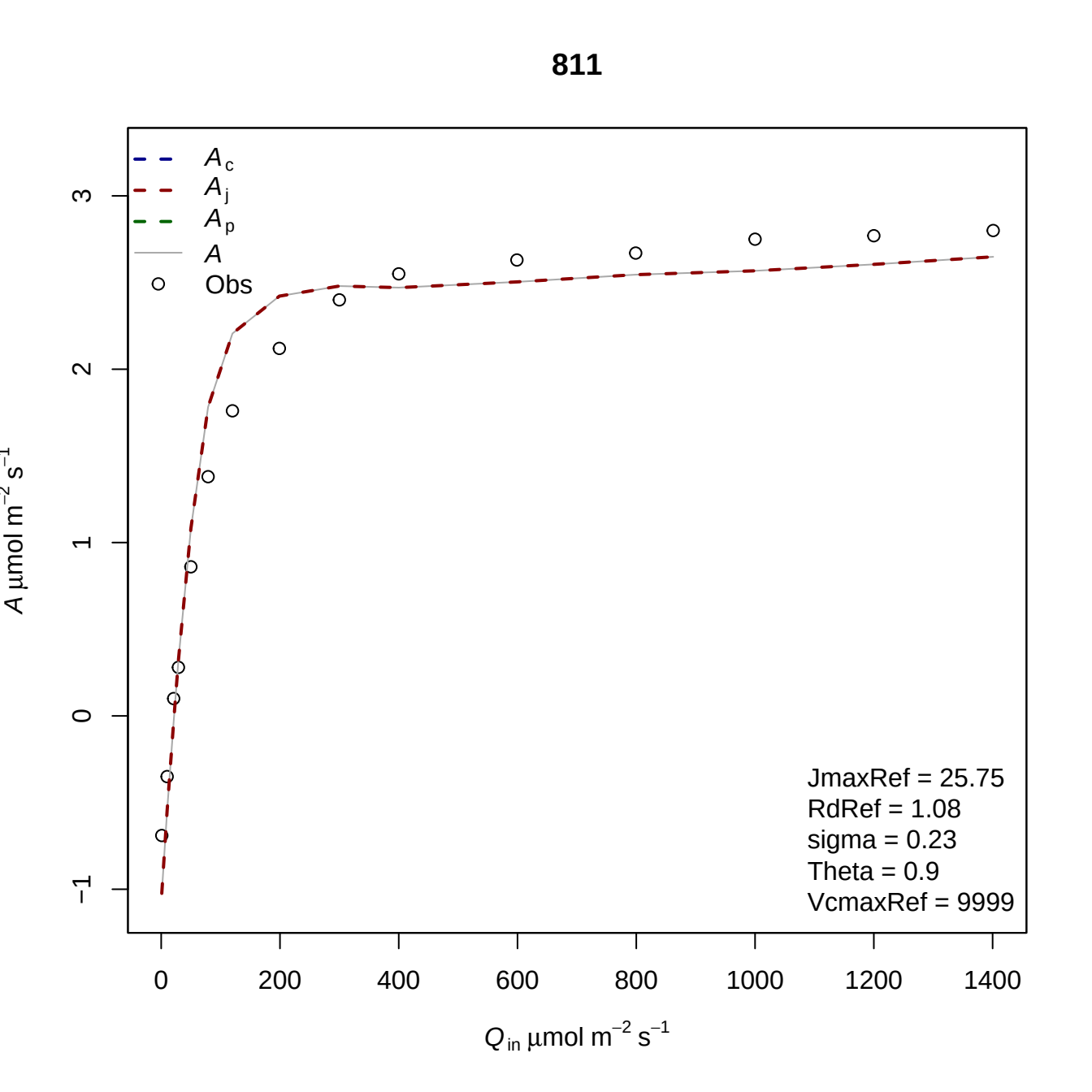

806
A_c
A_j
A_p
A
Obs
JmaxRef = 65.19
RdRef = 2.16
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

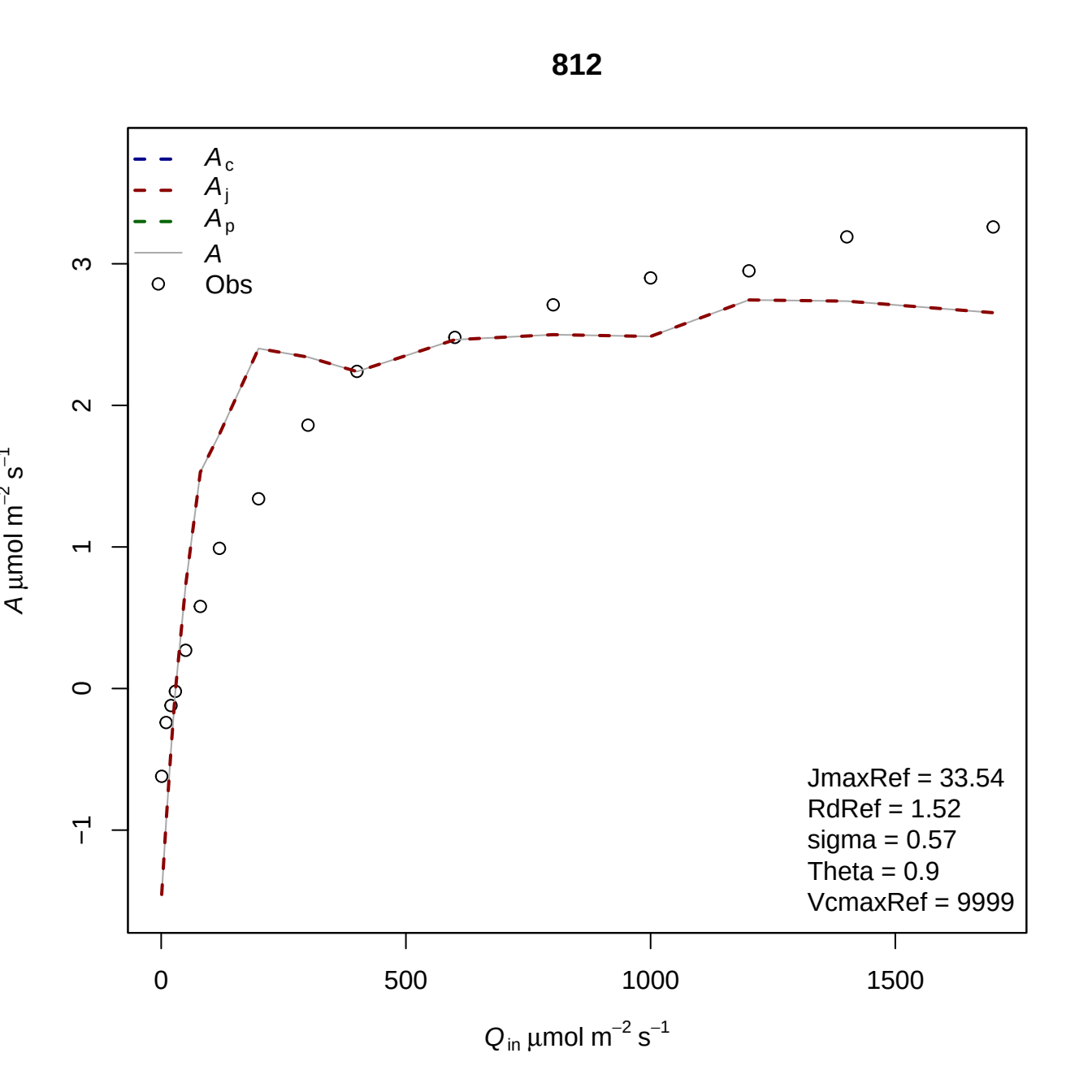

807
A_c
A_j
A_p
A
Obs
JmaxRef = 48.85
RdRef = 2.73
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

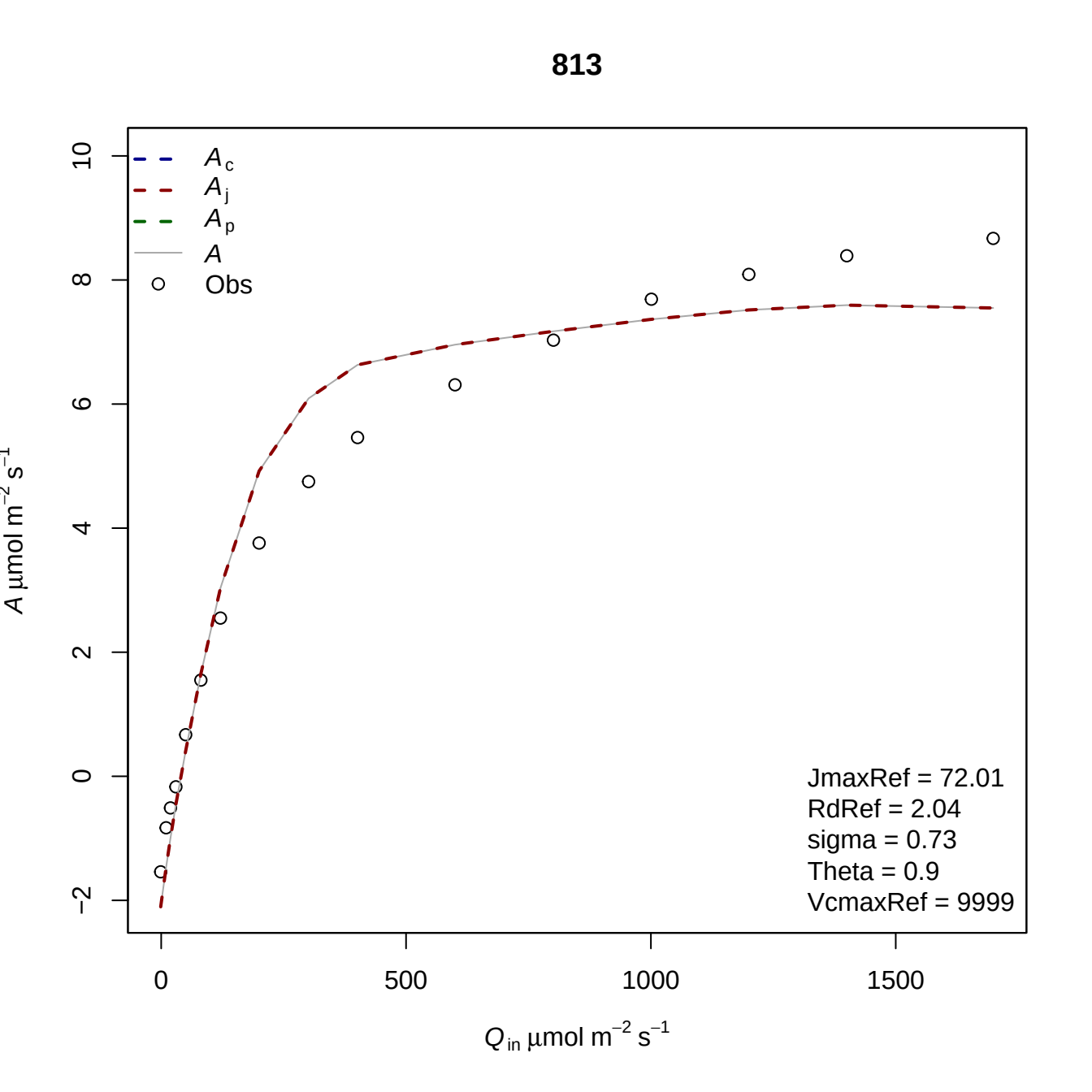

808
A_c
A_j
A_p
A
Obs
JmaxRef = 26.61
RdRef = 1.34
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

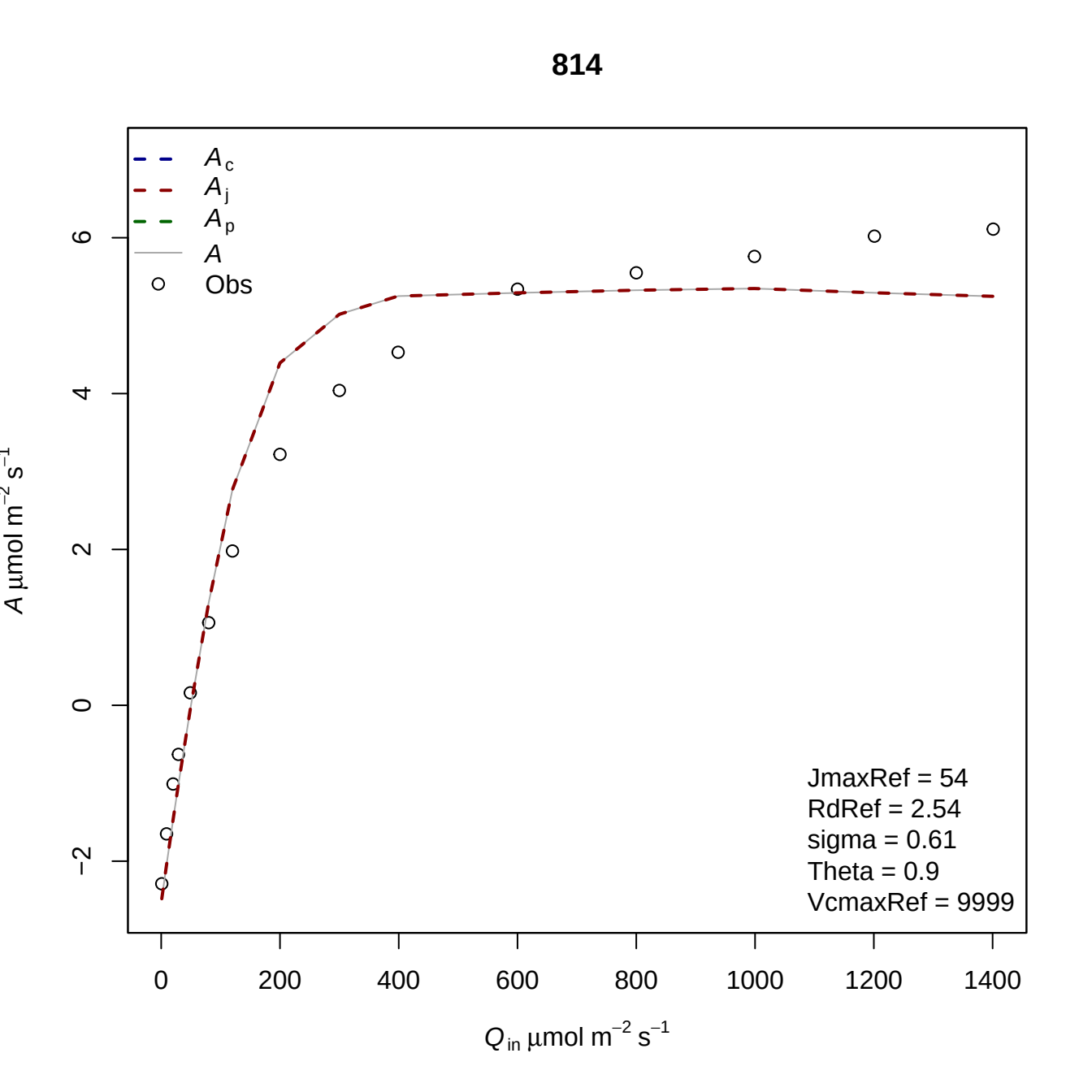

809
A_c
A_j
A_p
A
Obs
JmaxRef = 62.04
RdRef = 2.59
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

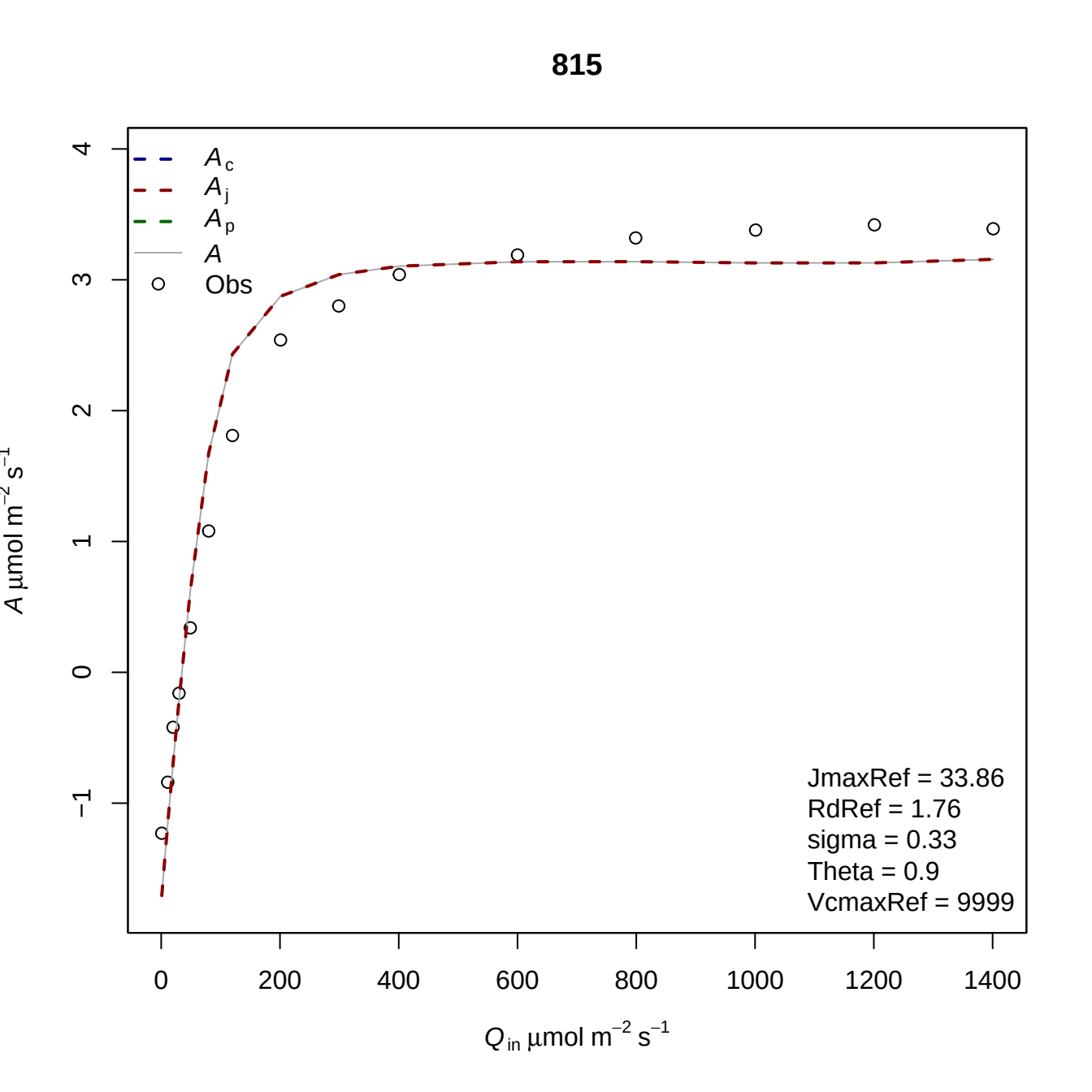

810
A_c
A_j
A_p
A
Obs
JmaxRef = 33.07
RdRef = 1.36
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

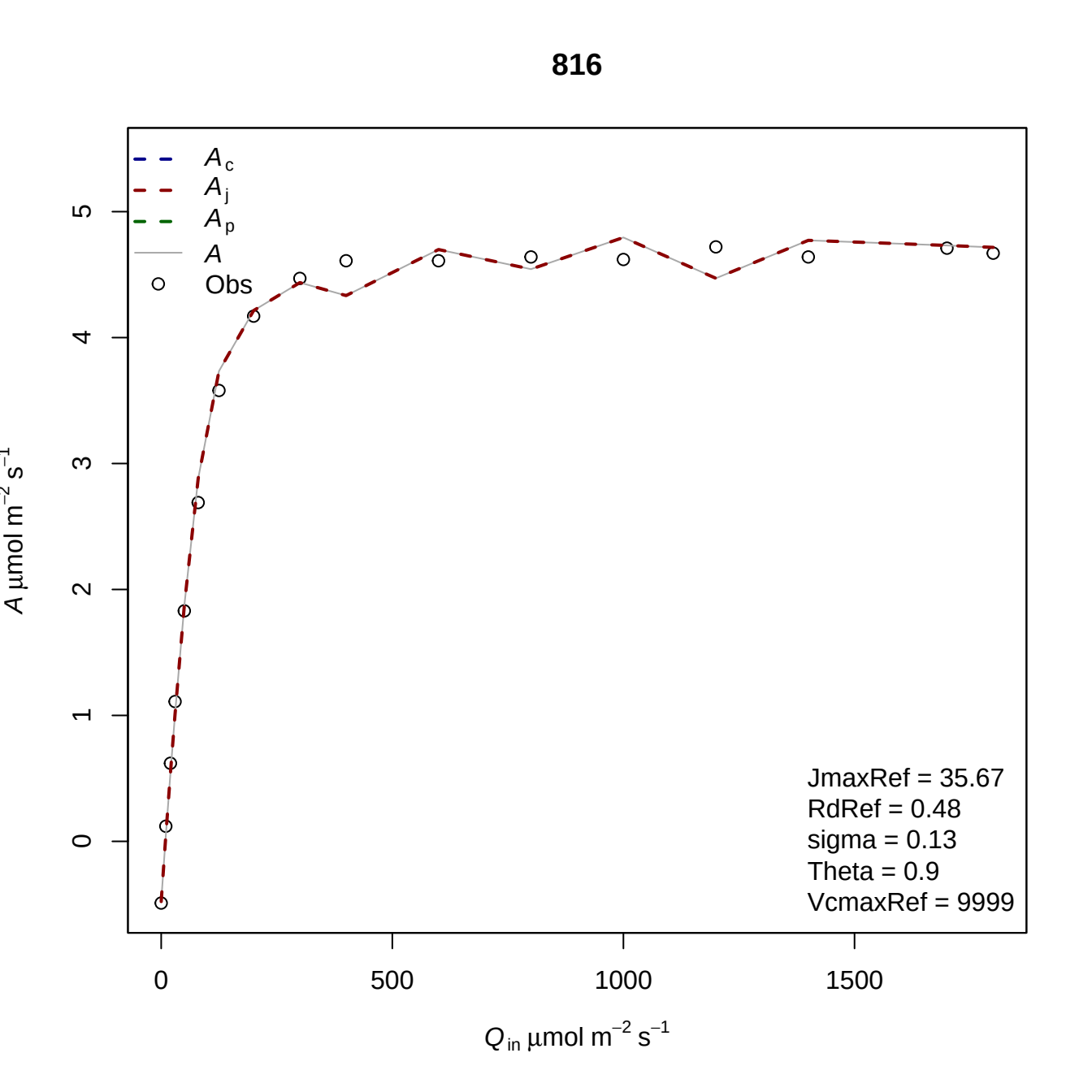

811
A_c
A_j
A_p
A
Obs
JmaxRef = 25.75
RdRef = 1.08
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

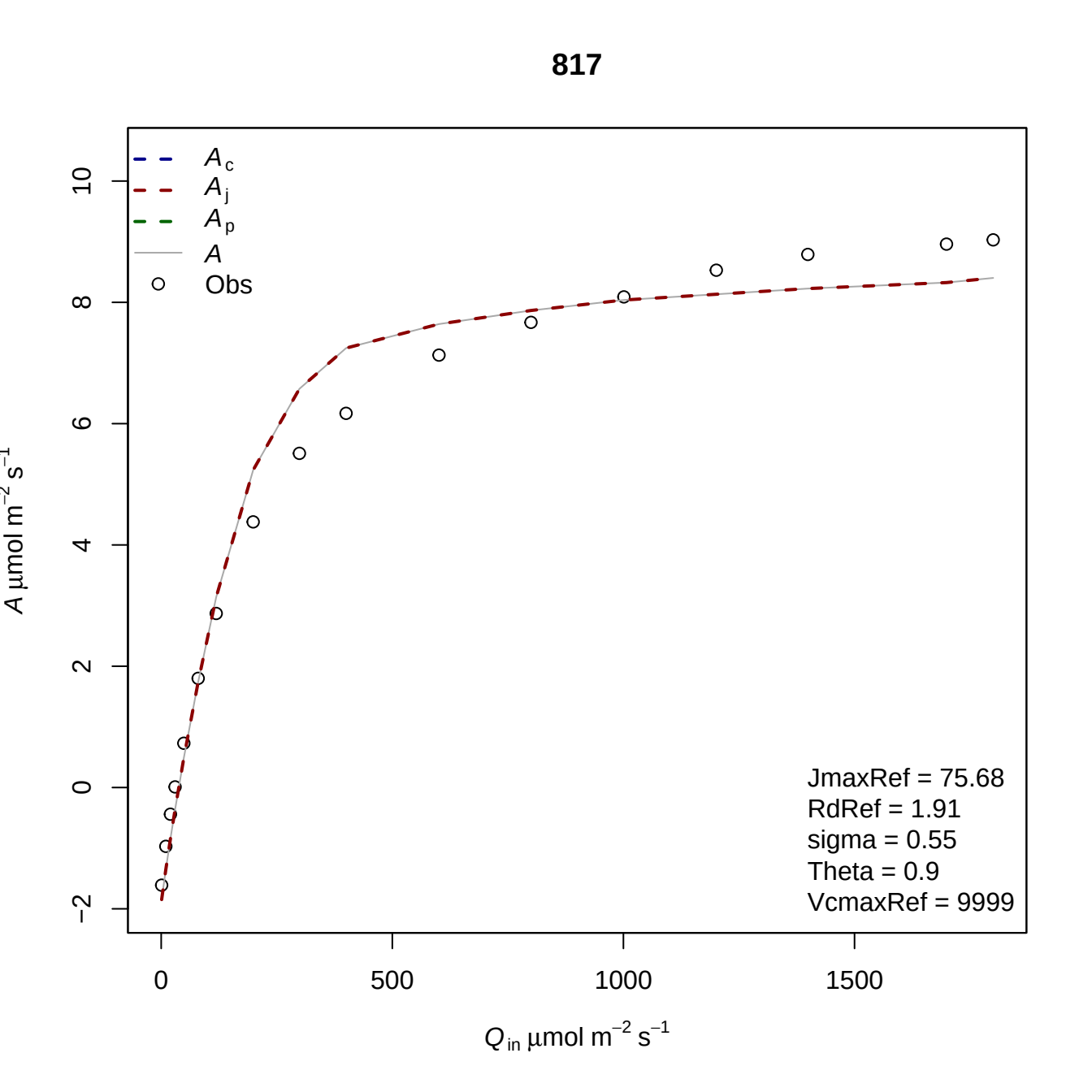

812
A_c
A_j
A_p
A
Obs
JmaxRef = 33.54
RdRef = 1.52
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

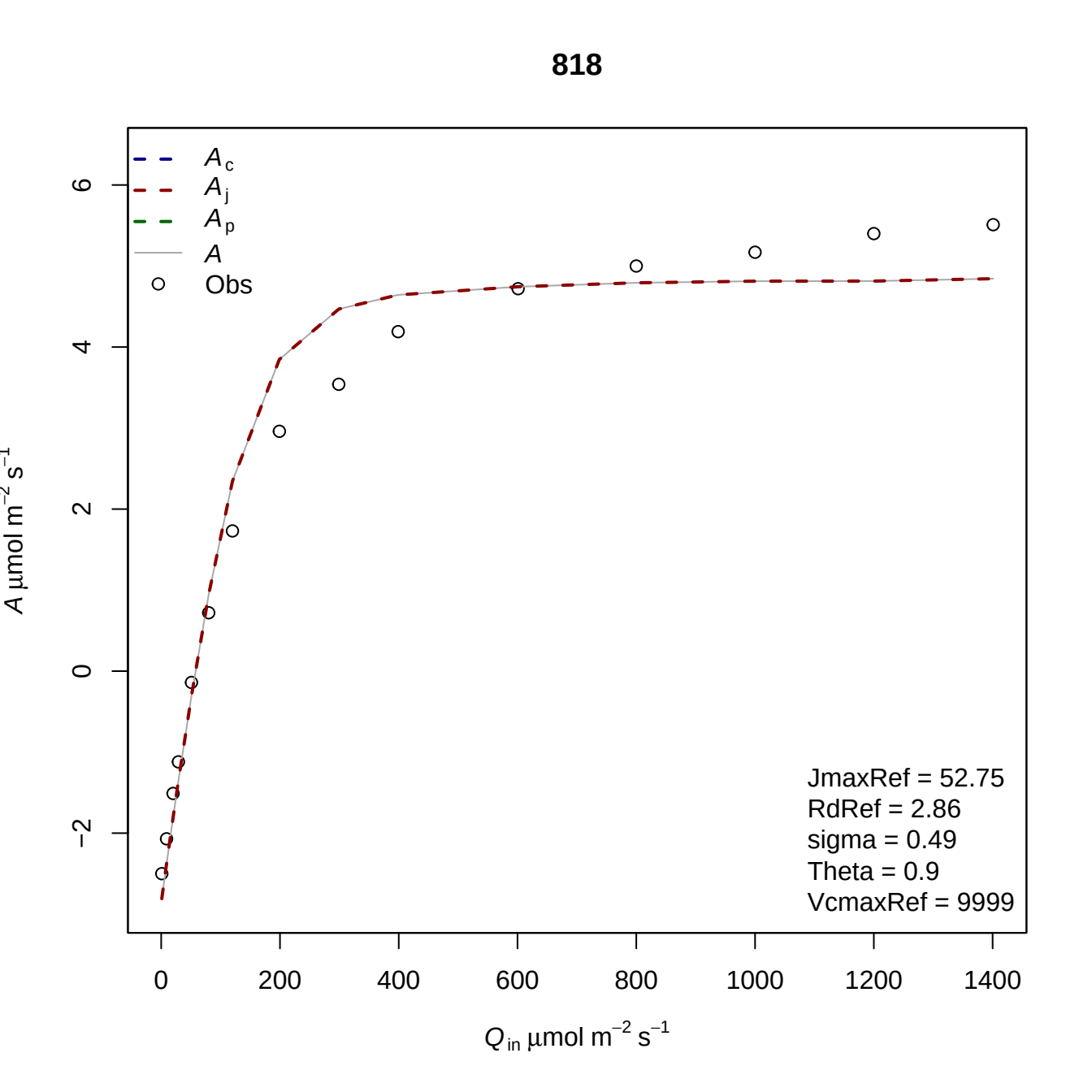

813
A_c
A_j
A_p
A
Obs
JmaxRef = 72.01
RdRef = 2.04
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

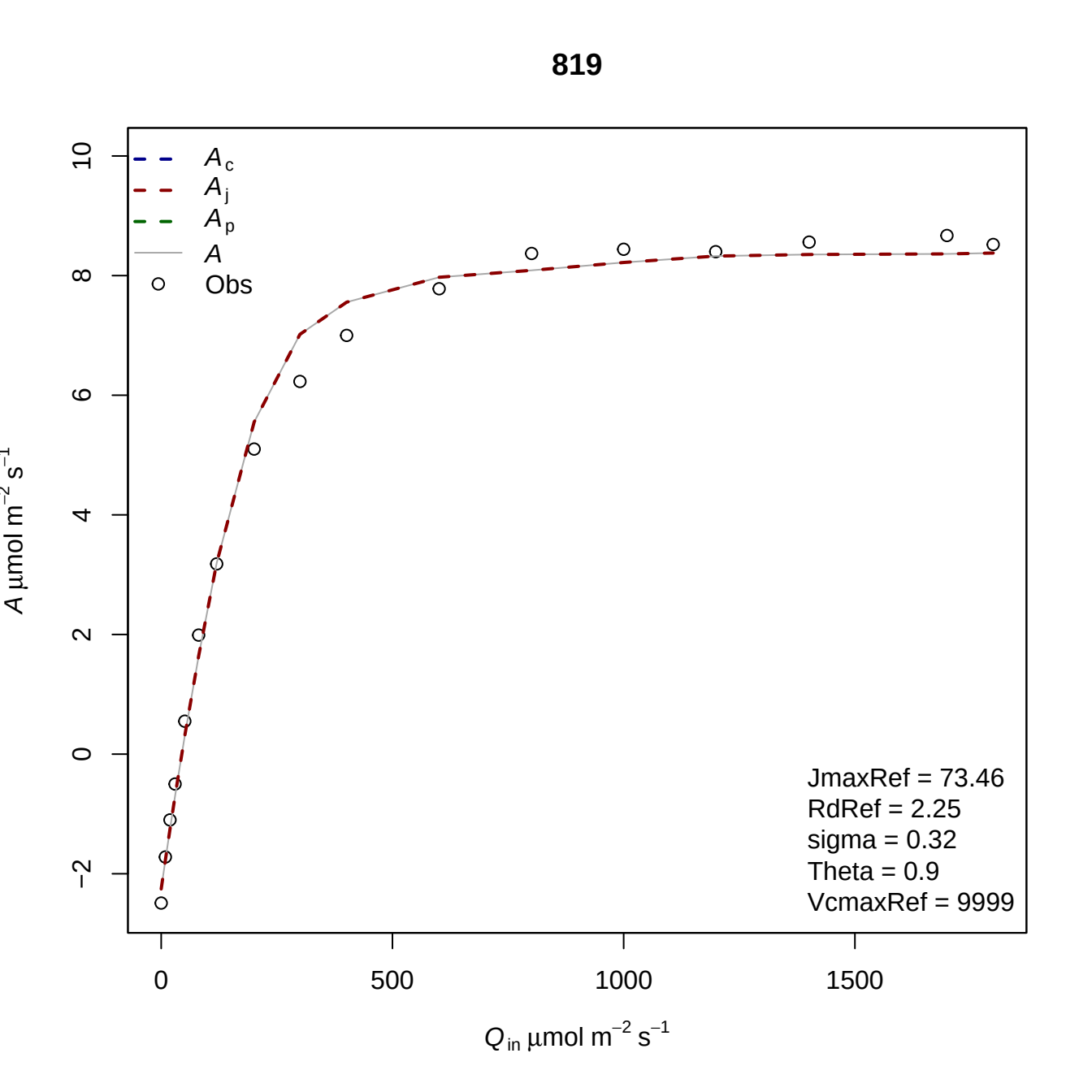

814
A_c
A_j
A_p
A
Obs
JmaxRef = 54
RdRef = 2.54
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

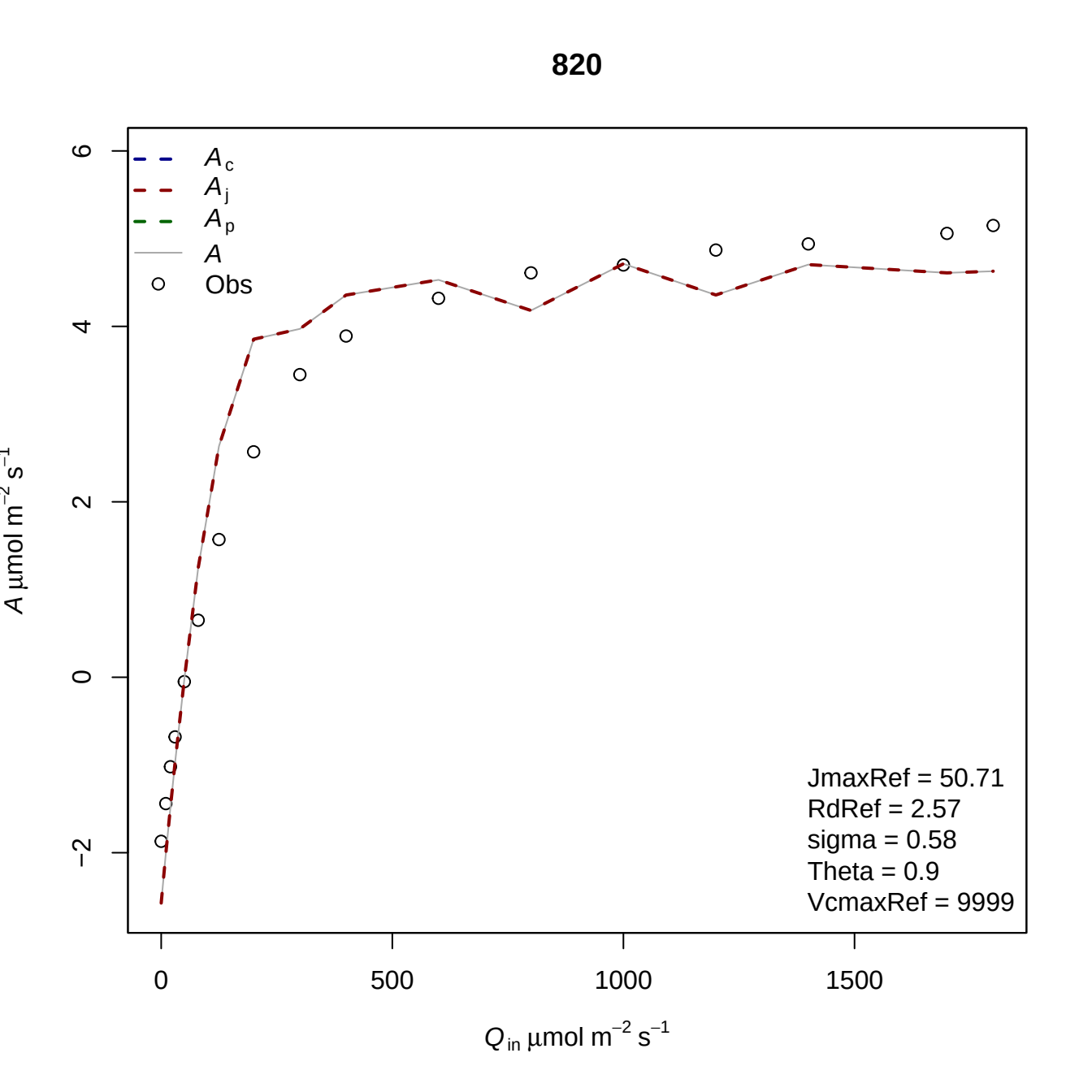

815
A_c
A_j
A_p
A
Obs
JmaxRef = 33.86
RdRef = 1.76
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

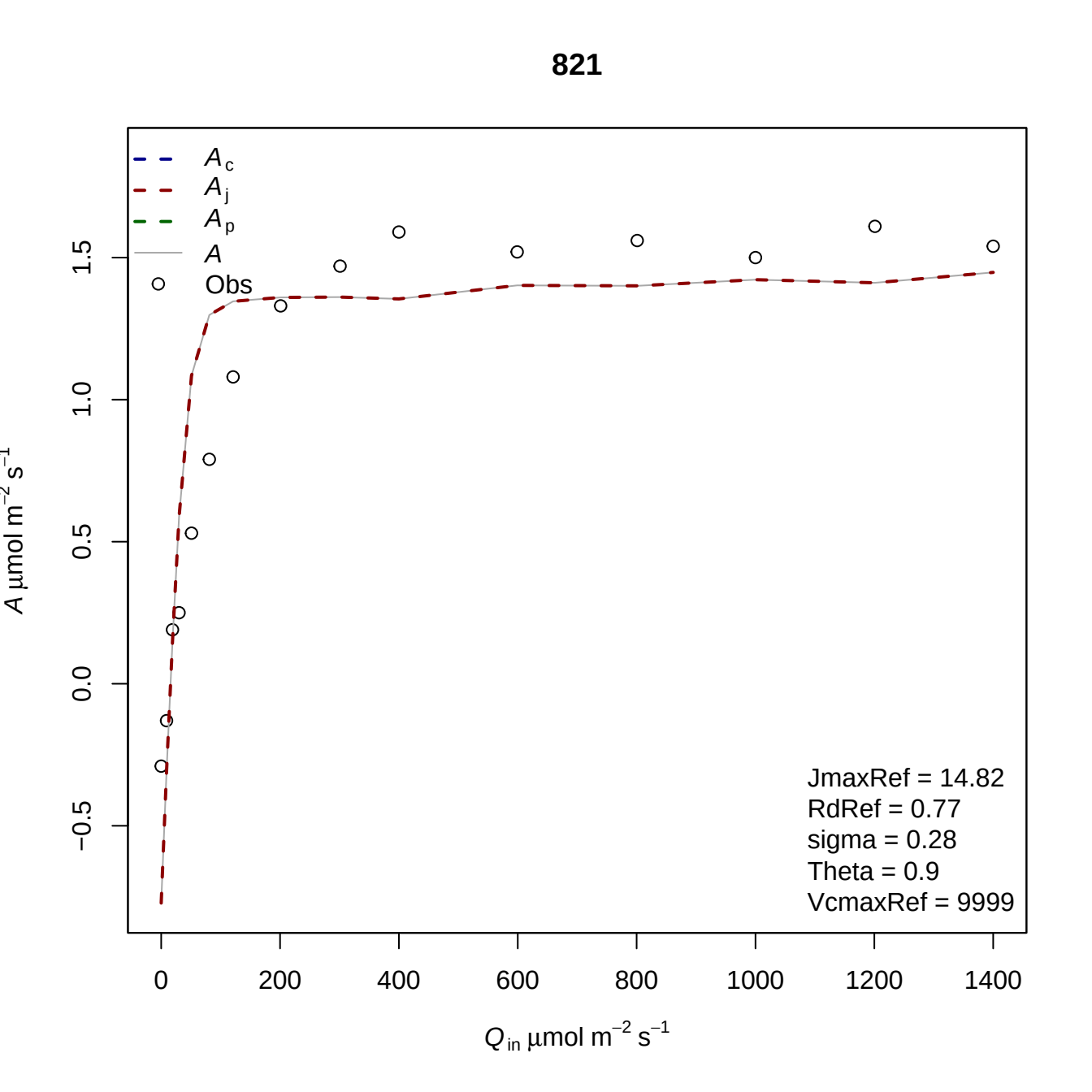

816
A_c
A_j
A_p
A
Obs
JmaxRef = 35.67
RdRef = 0.48
sigma = 0.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

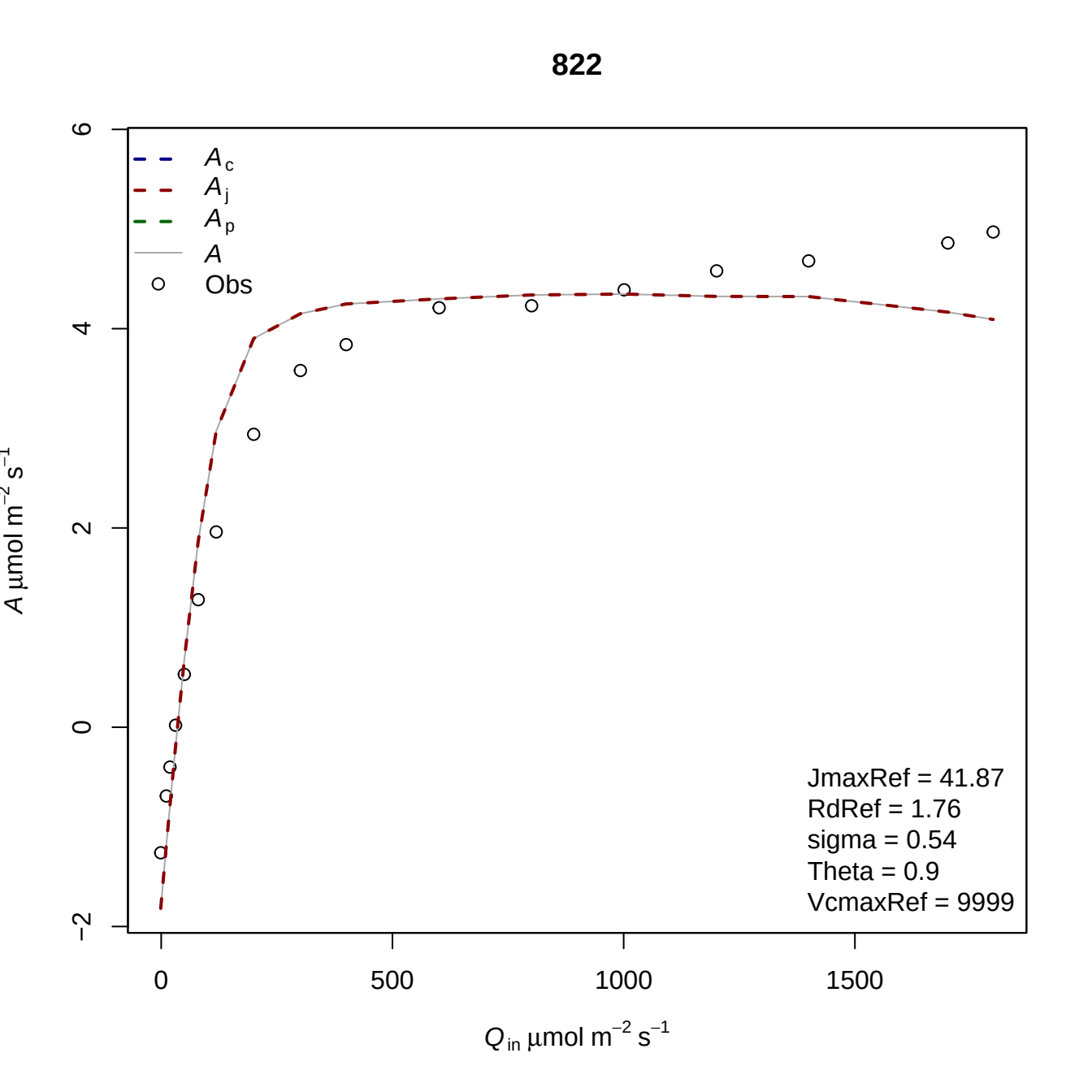

817
A_c
A_j
A_p
A
Obs
JmaxRef = 75.68
RdRef = 1.91
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

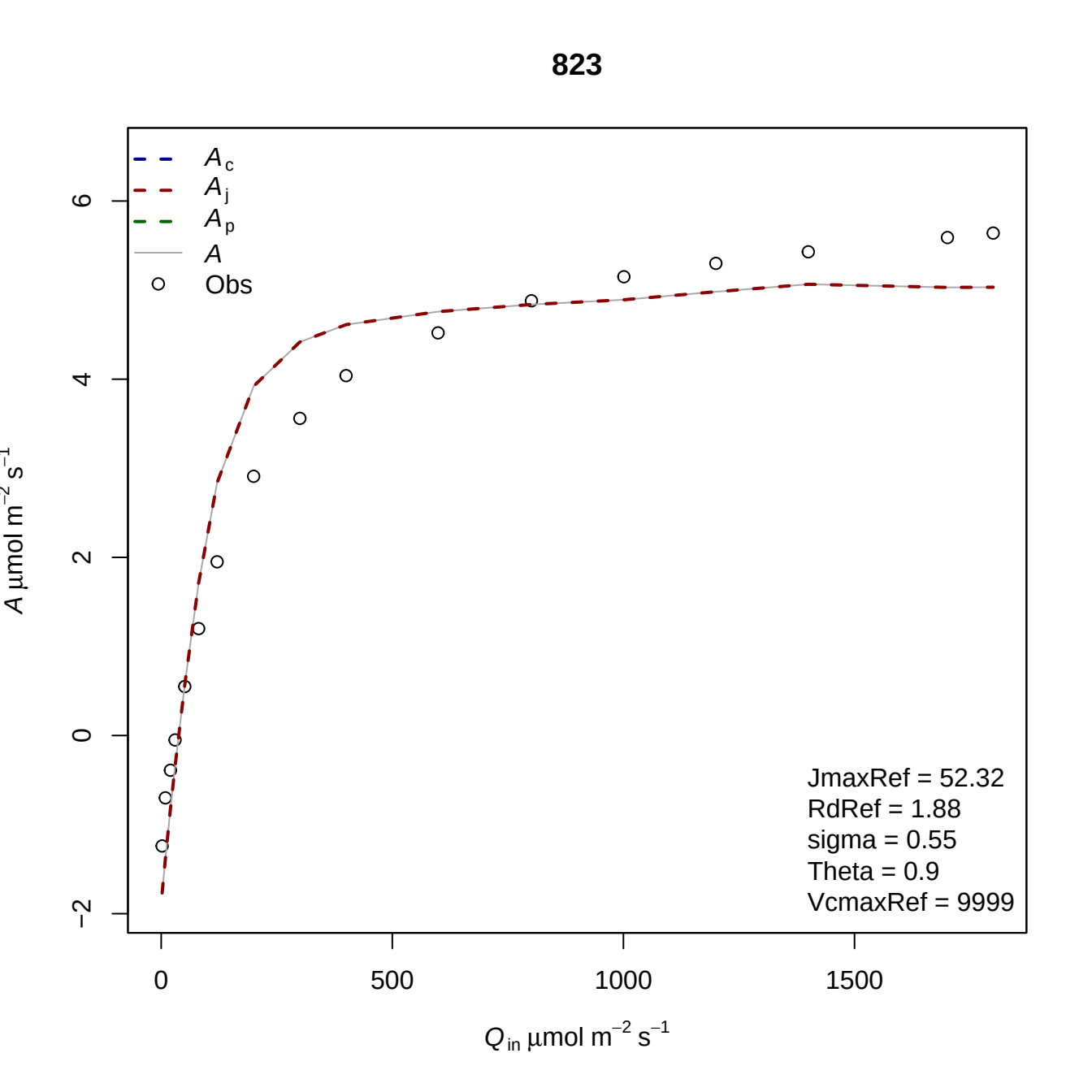

818
A_c
A_j
A_p
A
Obs
JmaxRef = 52.75
RdRef = 2.86
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

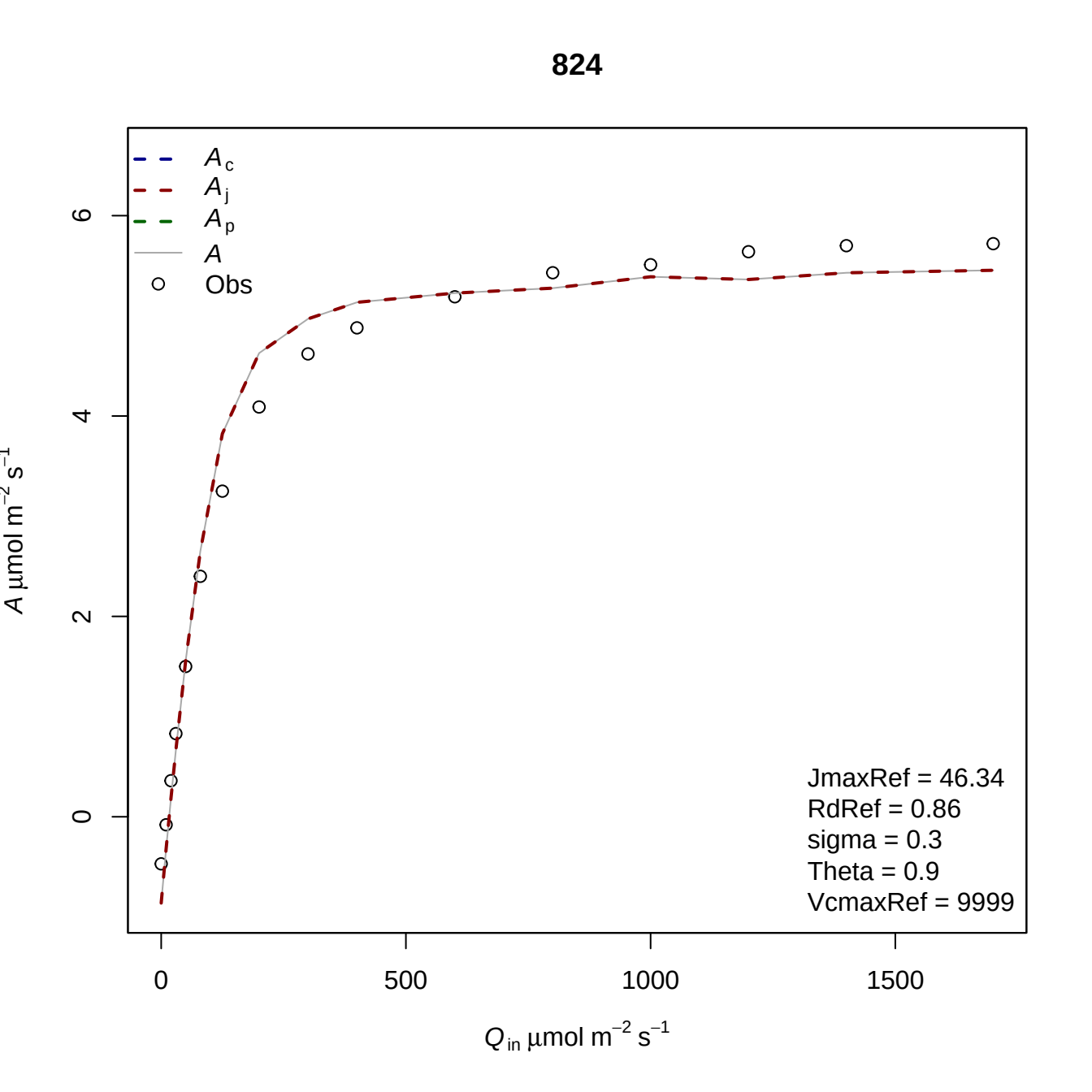

819
A_c
A_j
A_p
A
Obs
JmaxRef = 73.46
RdRef = 2.25
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

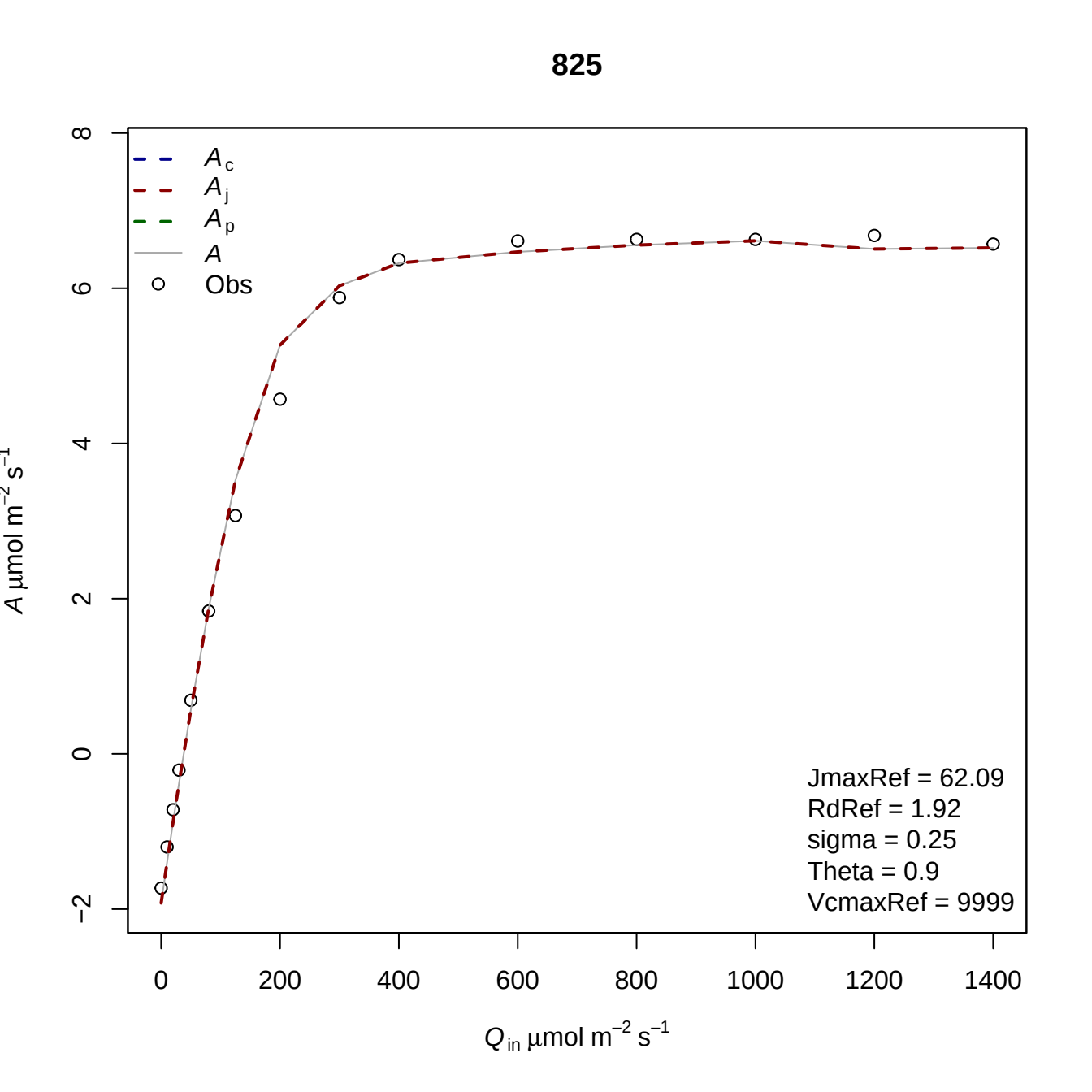

820
A_c
A_j
A_p
A
Obs
JmaxRef = 50.71
RdRef = 2.57
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

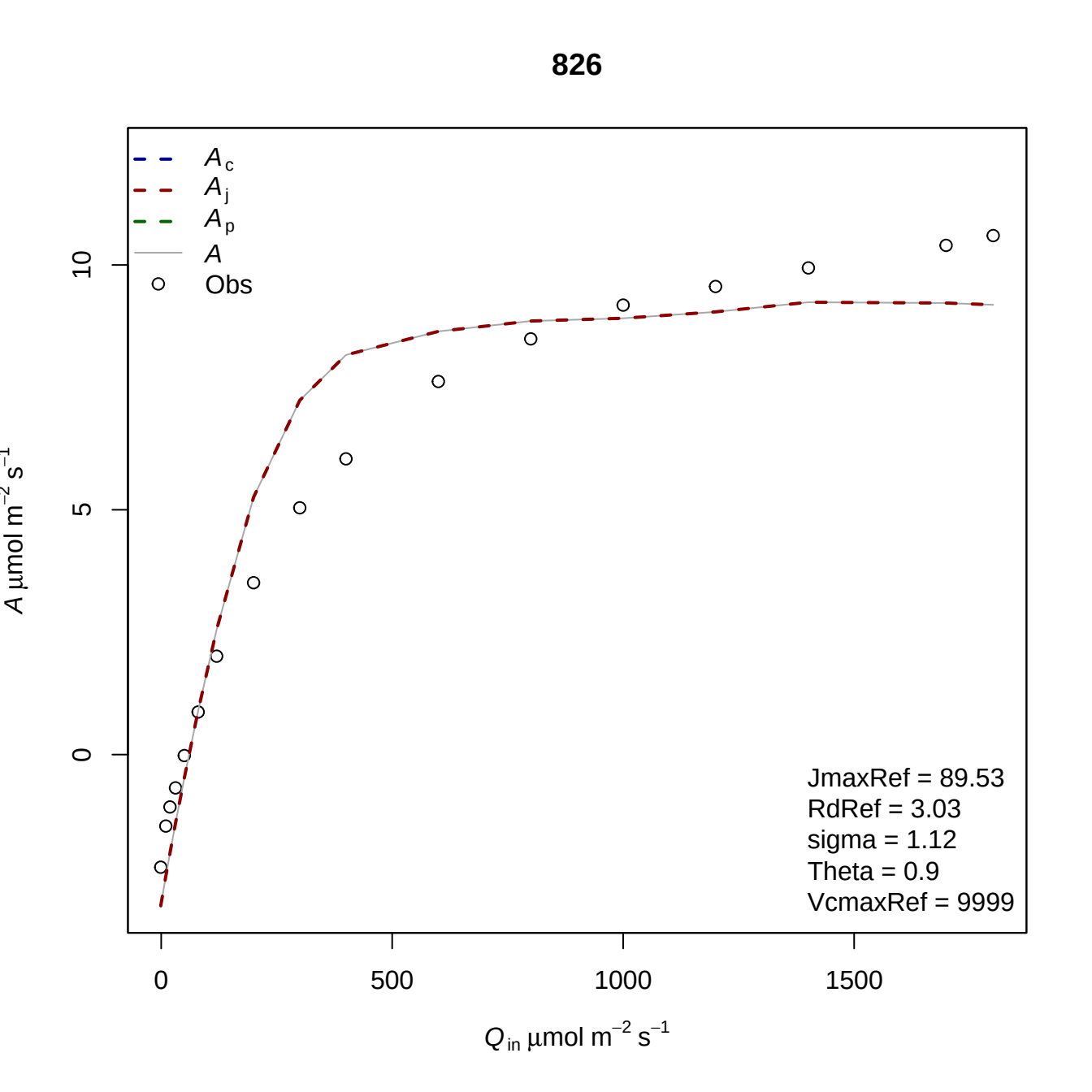

821
A_c
A_j
A_p
A
Obs
JmaxRef = 14.82
RdRef = 0.77
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

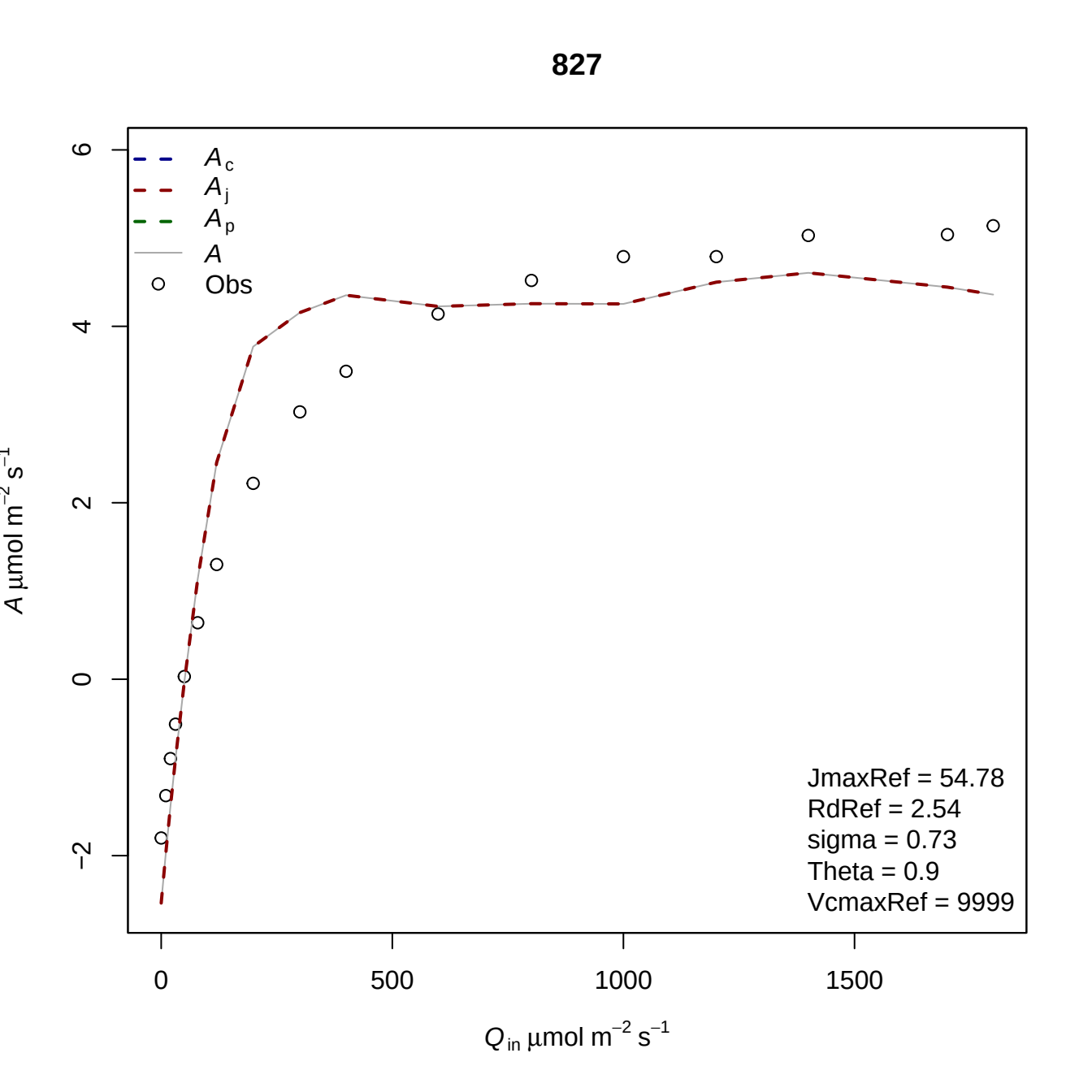

822
A_c
A_j
A_p
A
Obs
JmaxRef = 41.87
RdRef = 1.76
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

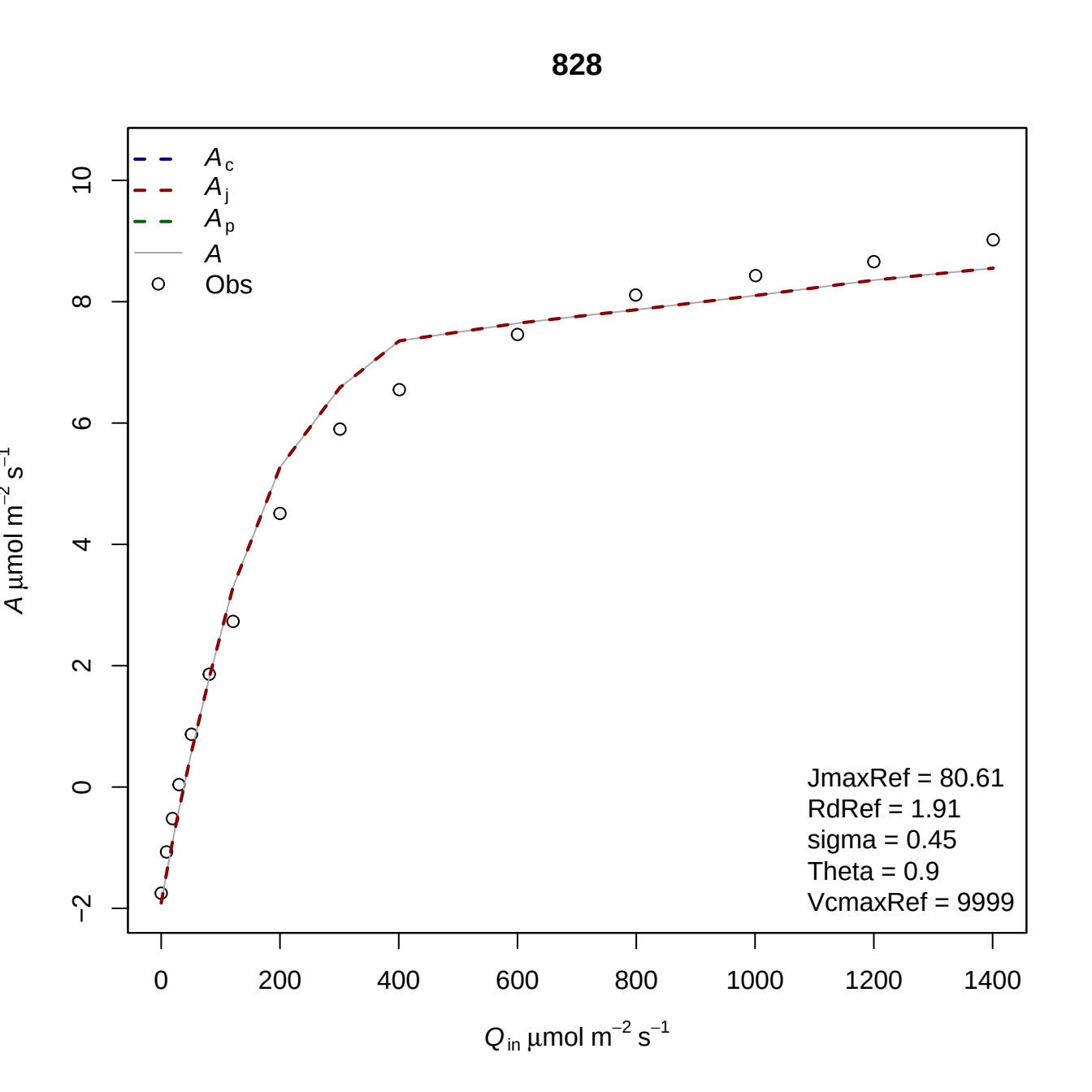

823
A_c
A_j
A_p
A
Obs
JmaxRef = 52.32
RdRef = 1.88
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

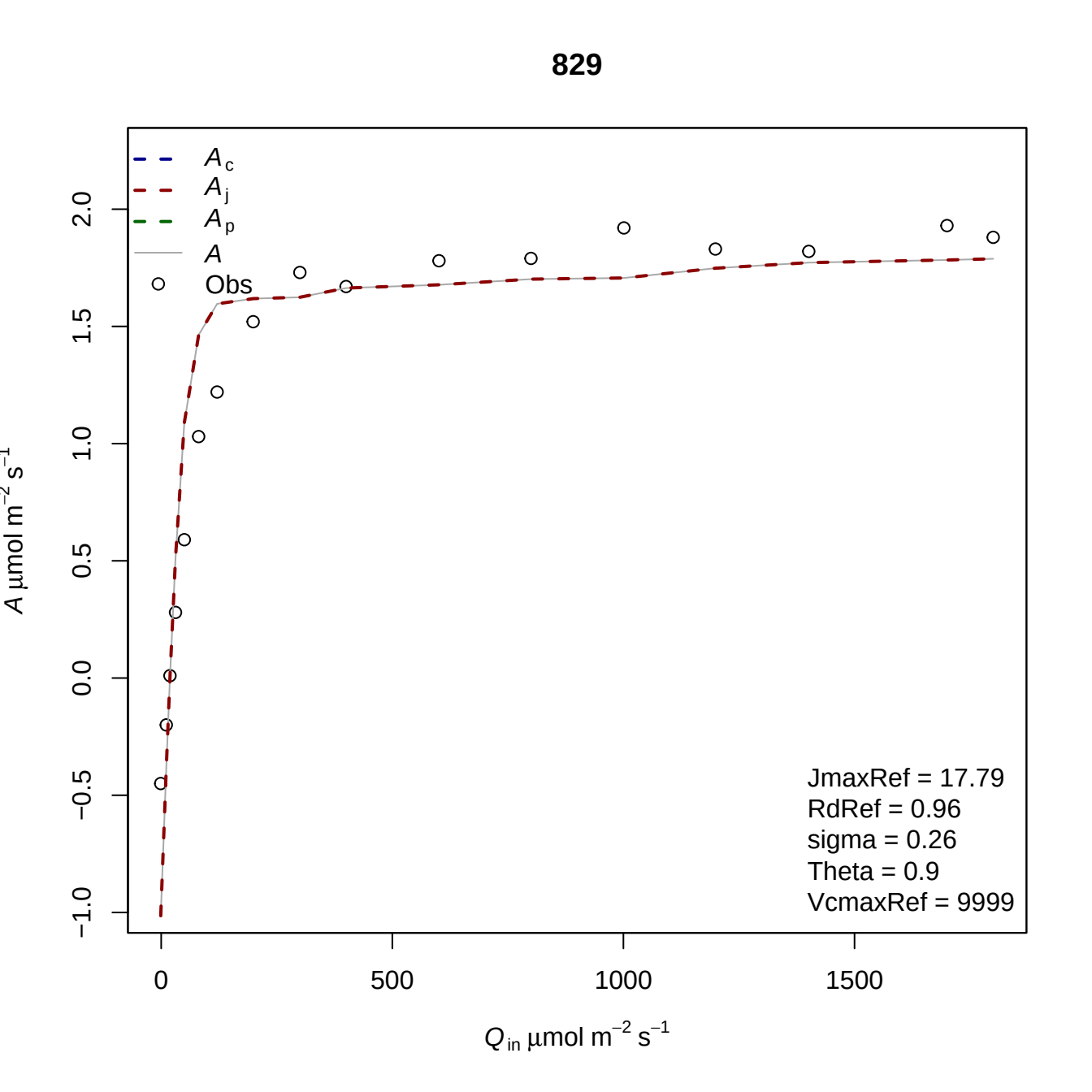

824
A_c
A_j
A_p
A
Obs
JmaxRef = 46.34
RdRef = 0.86
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

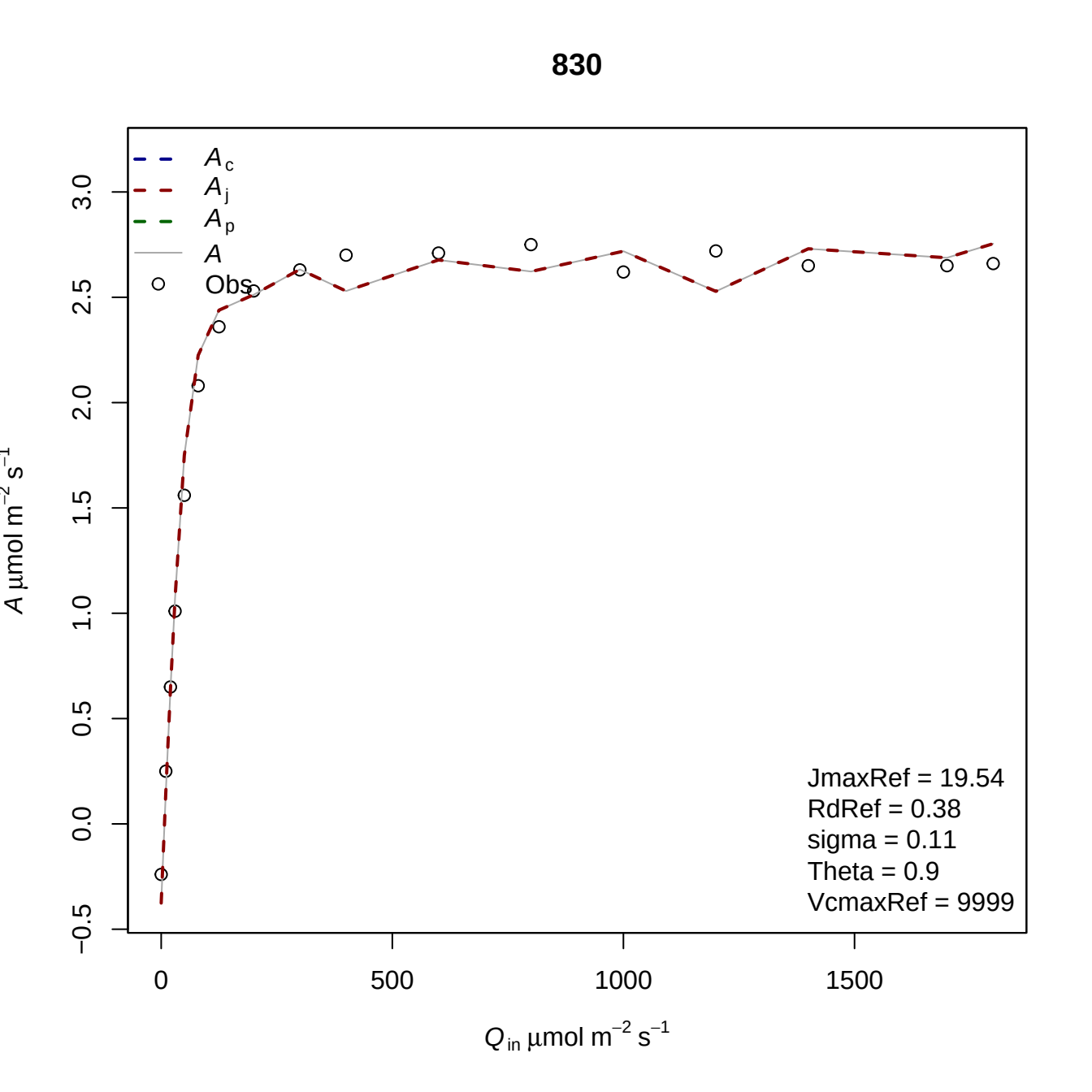

825
A_c
A_j
A_p
A
Obs
JmaxRef = 62.09
RdRef = 1.92
sigma = 0.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

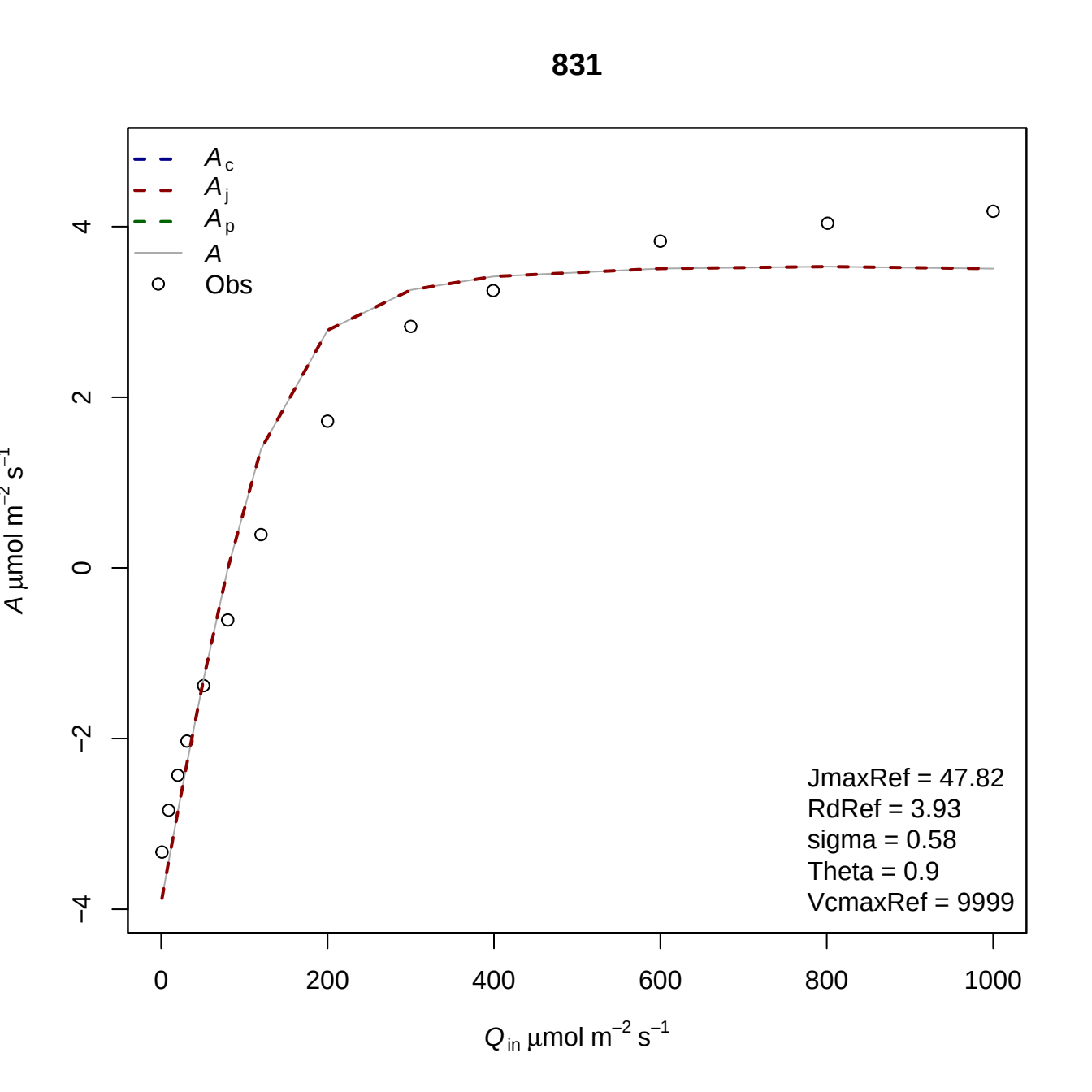

826
A_c
A_j
A_p
A
Obs
JmaxRef = 89.53
RdRef = 3.03
sigma = 1.12
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

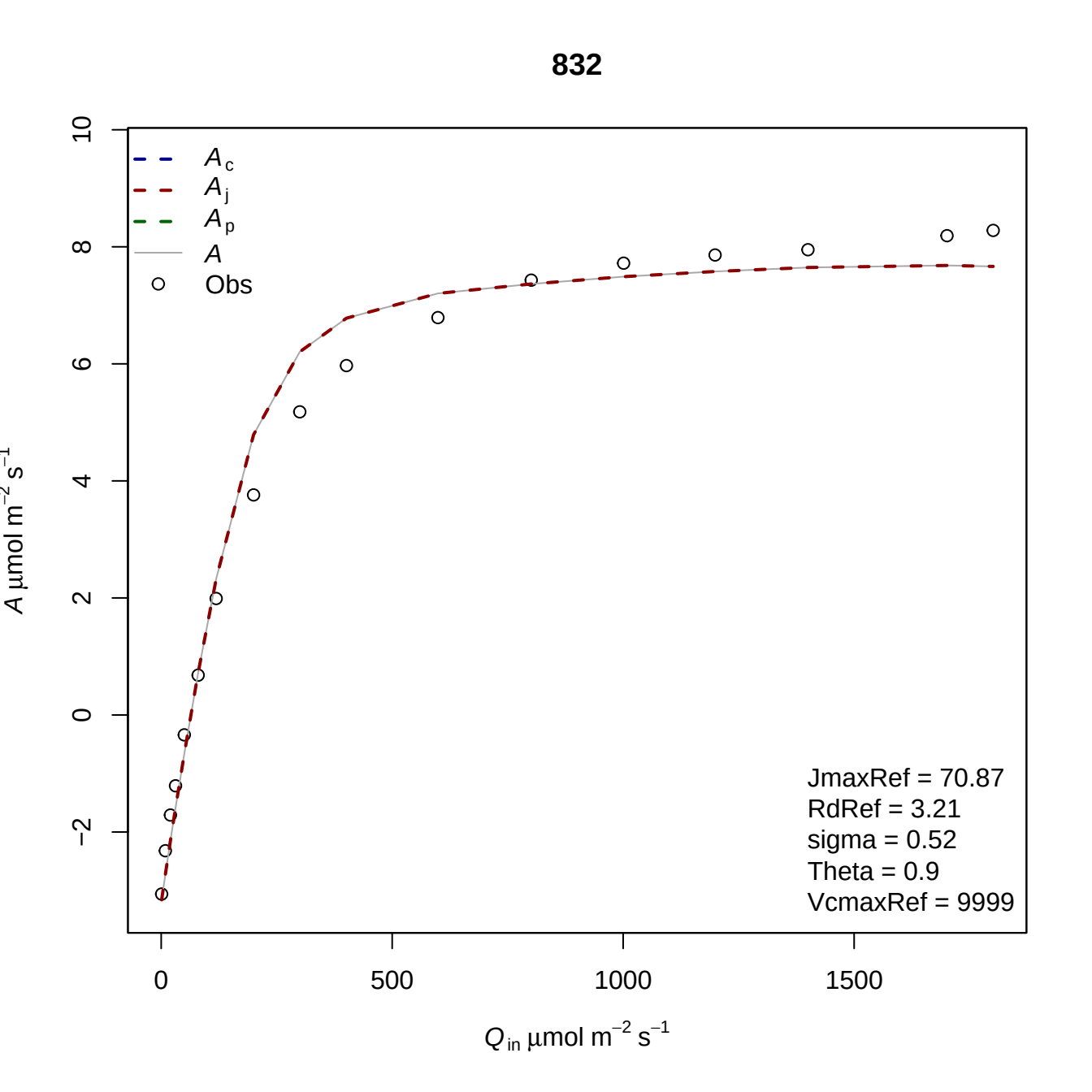

827
A_c
A_j
A_p
A
Obs
JmaxRef = 54.78
RdRef = 2.54
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

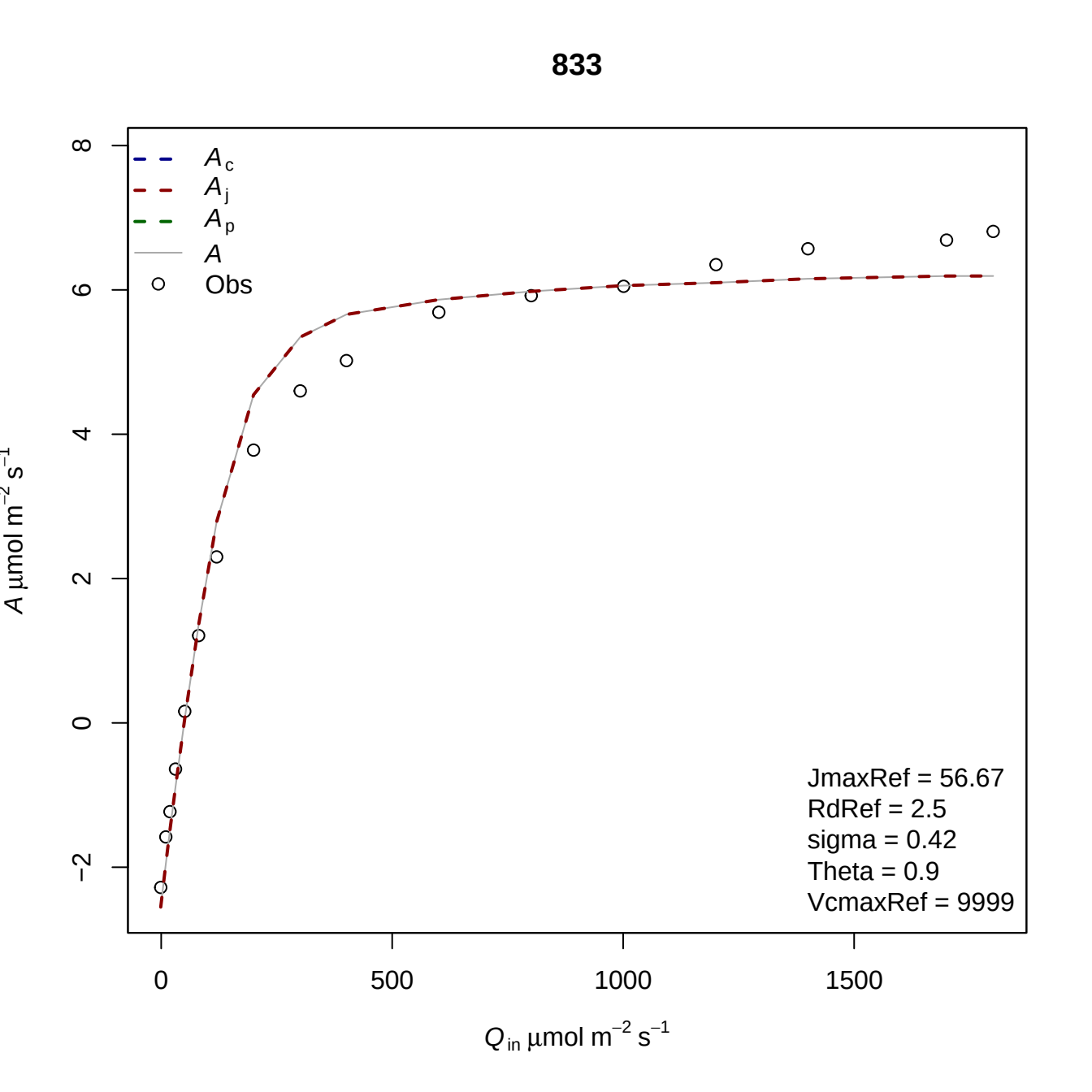

828
A_c
A_j
A_p
A
Obs
JmaxRef = 80.61
RdRef = 1.91
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

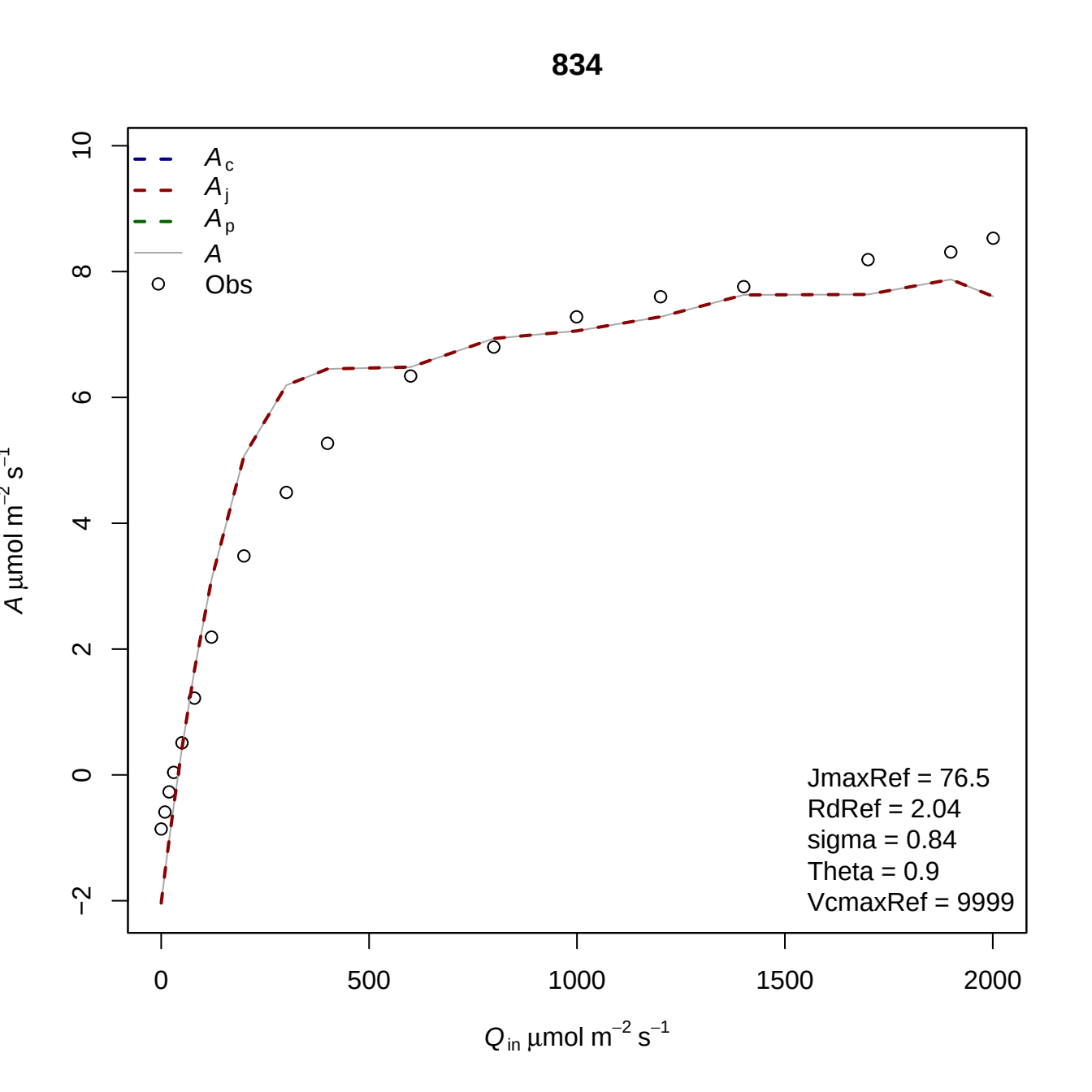

829
A_c
A_j
A_p
A
Obs
JmaxRef = 17.79
RdRef = 0.96
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

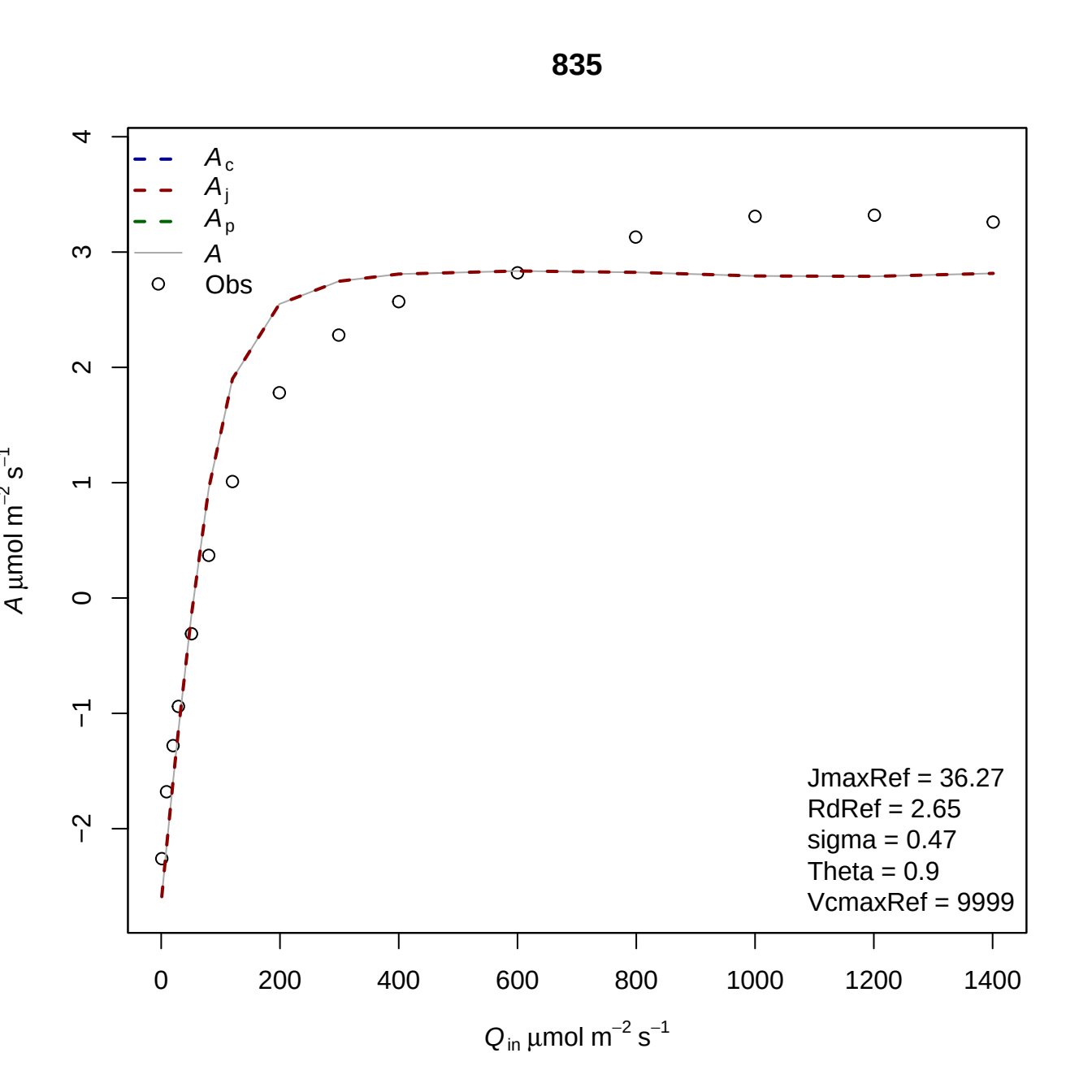

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.54
RdRef = 0.38
sigma = 0.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

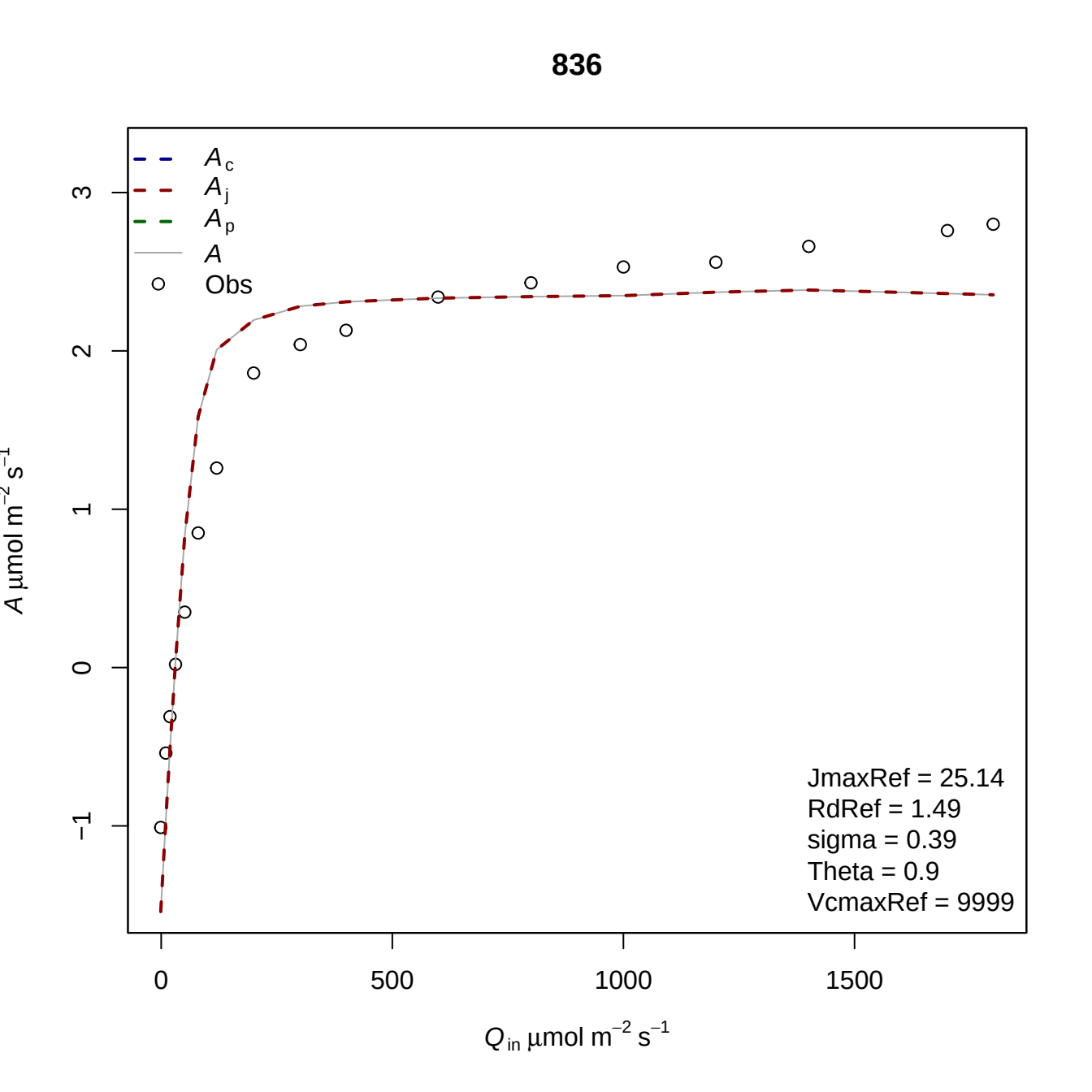

831
A_c
A_j
A_p
A
Obs
JmaxRef = 47.82
RdRef = 3.93
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

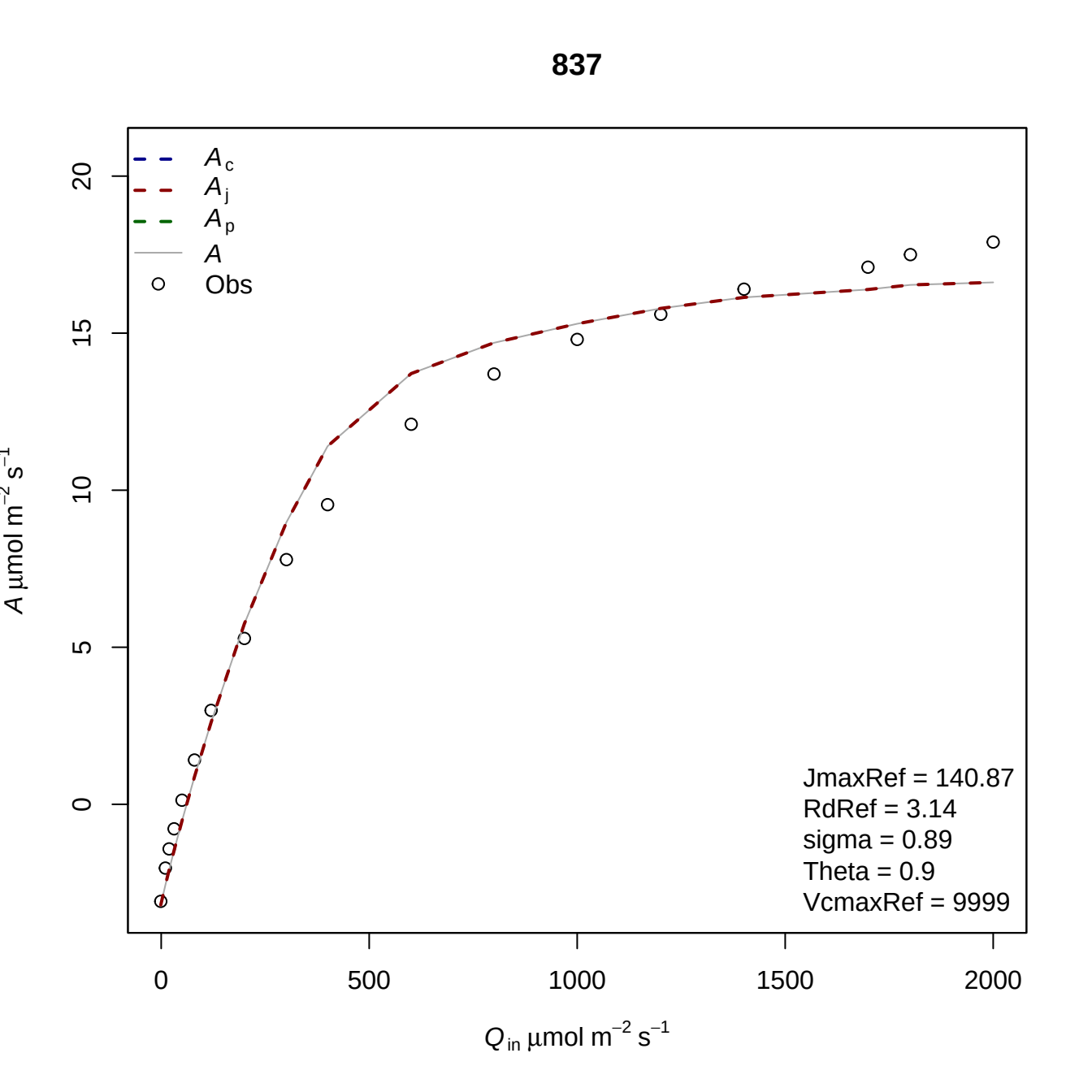

832
A_c
A_j
A_p
A
Obs
JmaxRef = 70.87
RdRef = 3.21
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

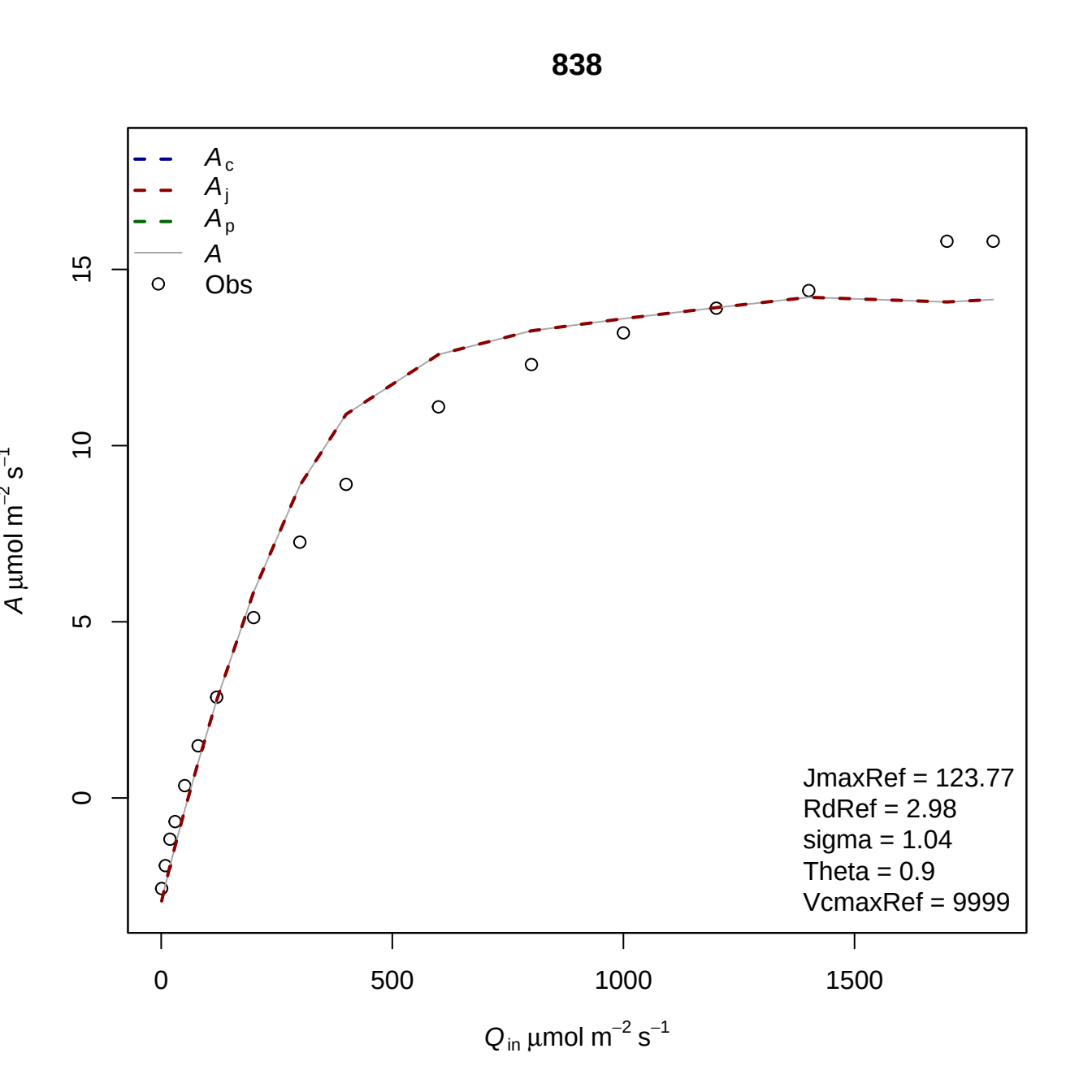

833
A_c
A_j
A_p
A
Obs
JmaxRef = 56.67
RdRef = 2.5
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

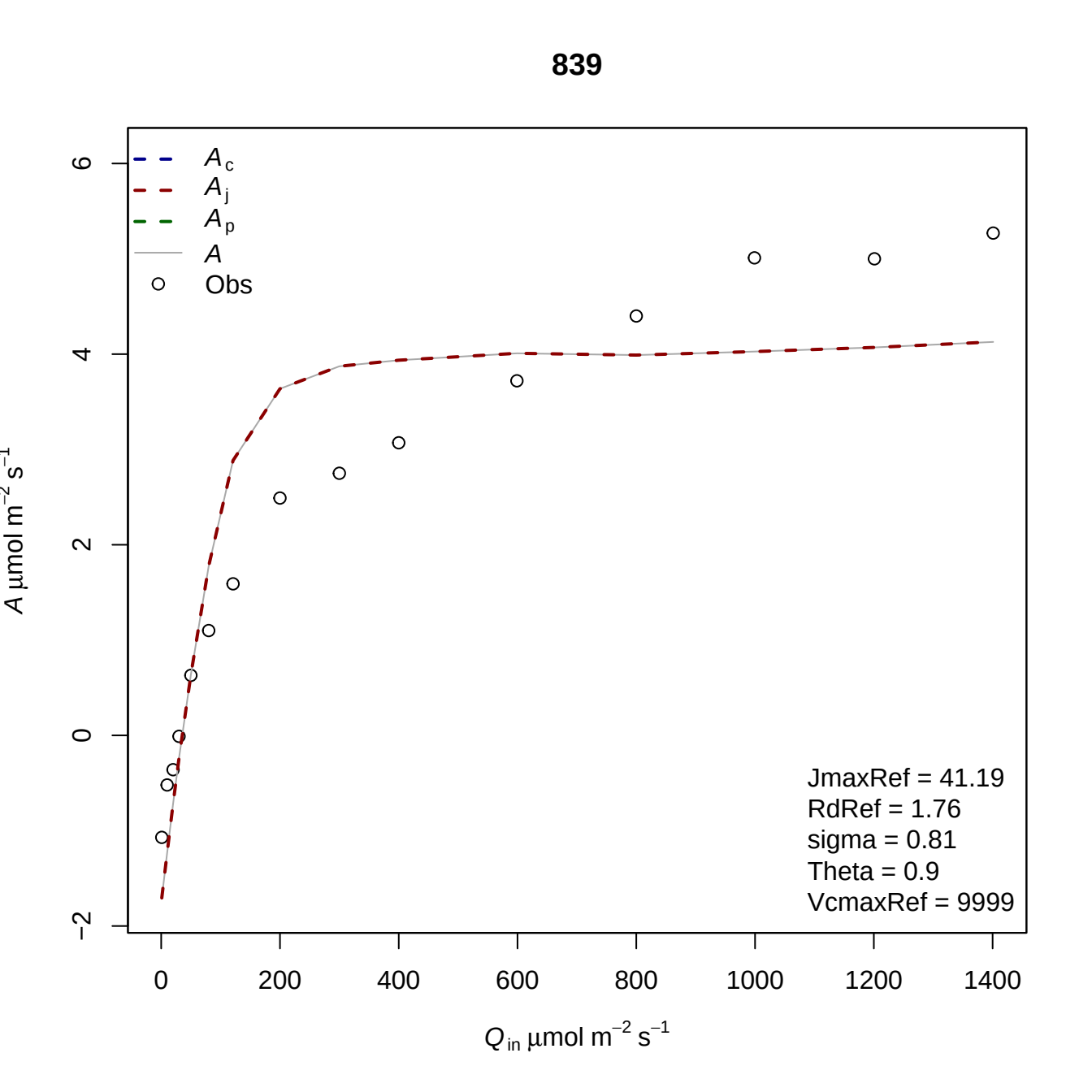

834
A_c
A_j
A_p
A
Obs
JmaxRef = 76.5
RdRef = 2.04
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

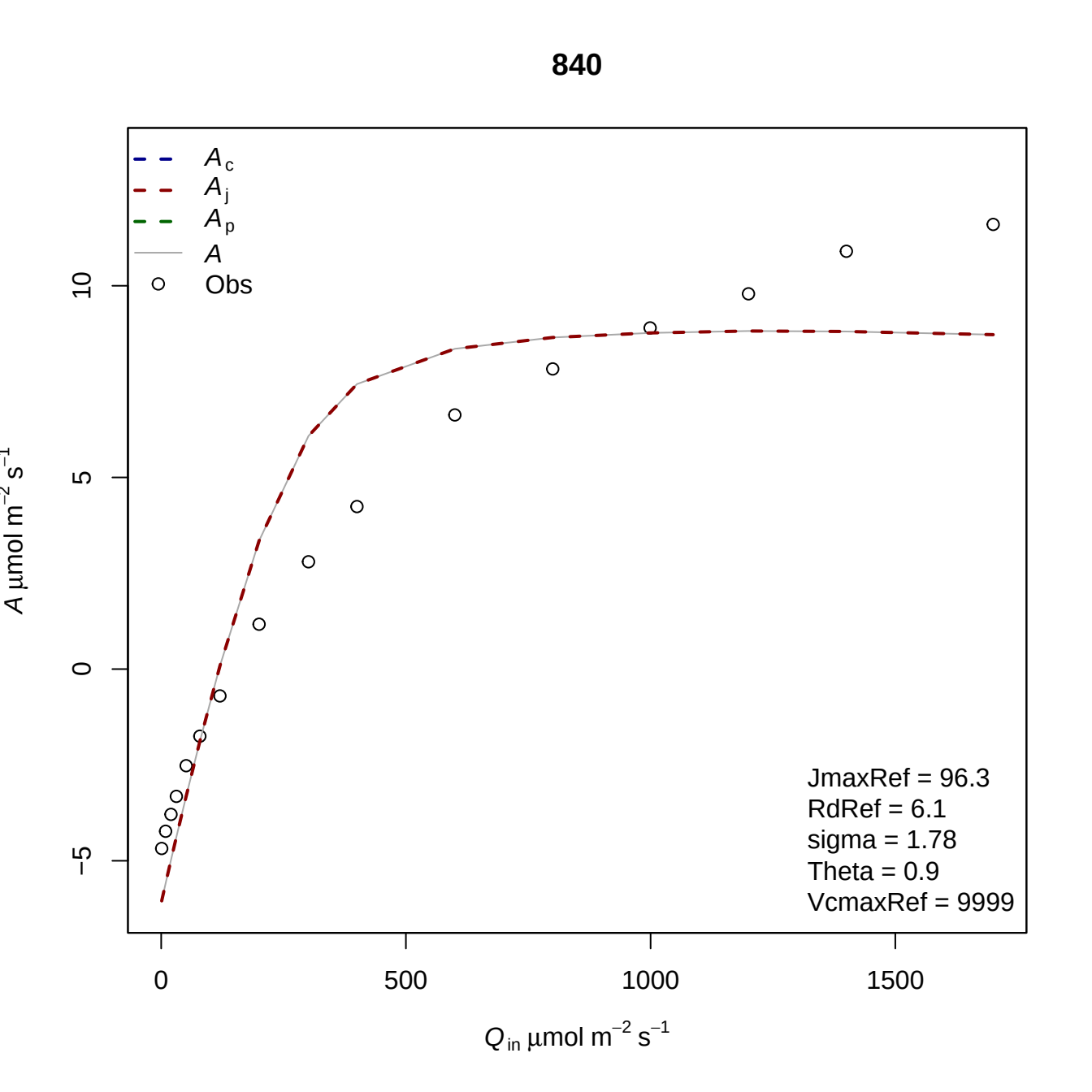

835
A_c
A_j
A_p
A
Obs
JmaxRef = 36.27
RdRef = 2.65
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

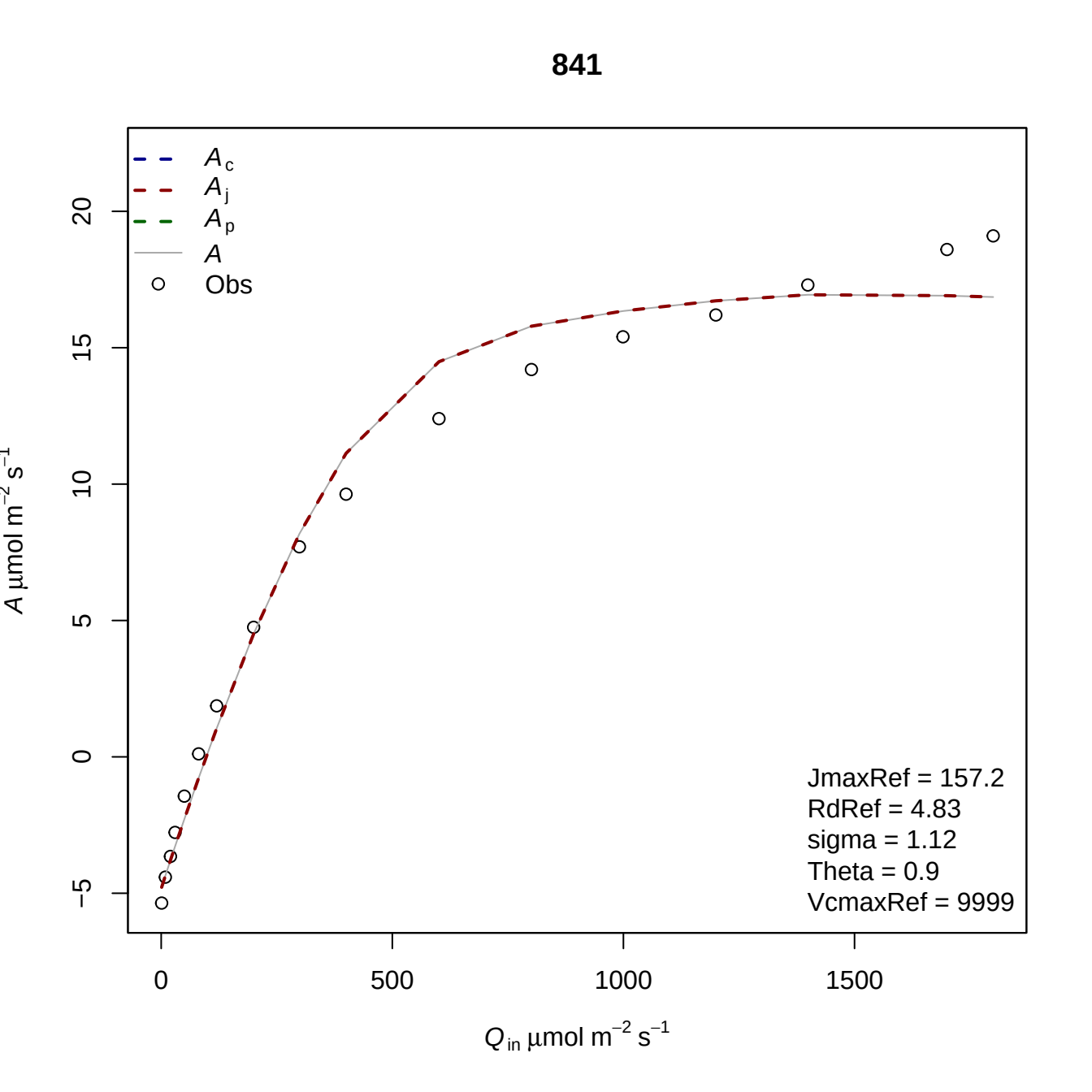

836
A_c
A_j
A_p
A
Obs
JmaxRef = 25.14
RdRef = 1.49
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

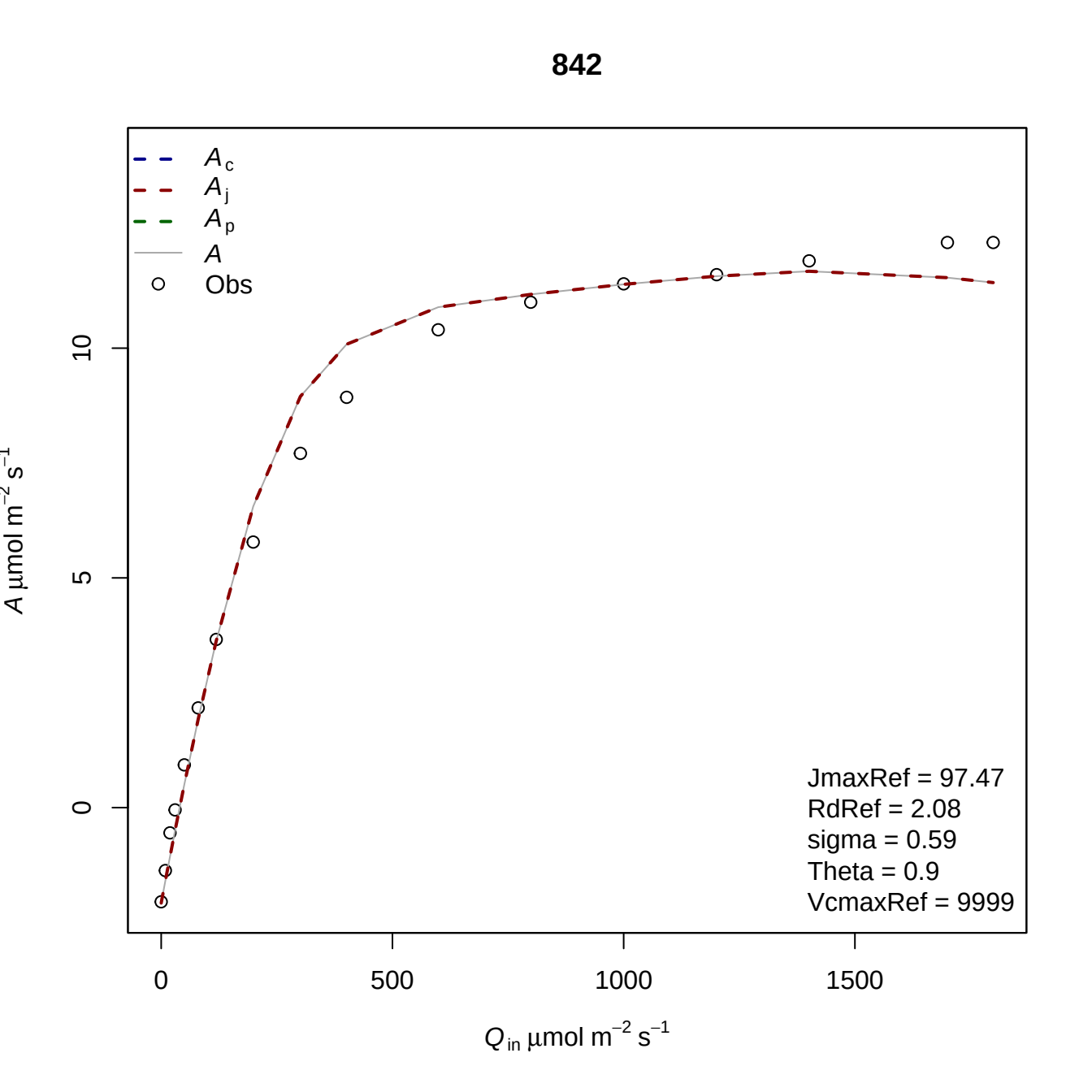

837
A_c
A_j
A_p
A
Obs
JmaxRef = 140.87
RdRef = 3.14
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

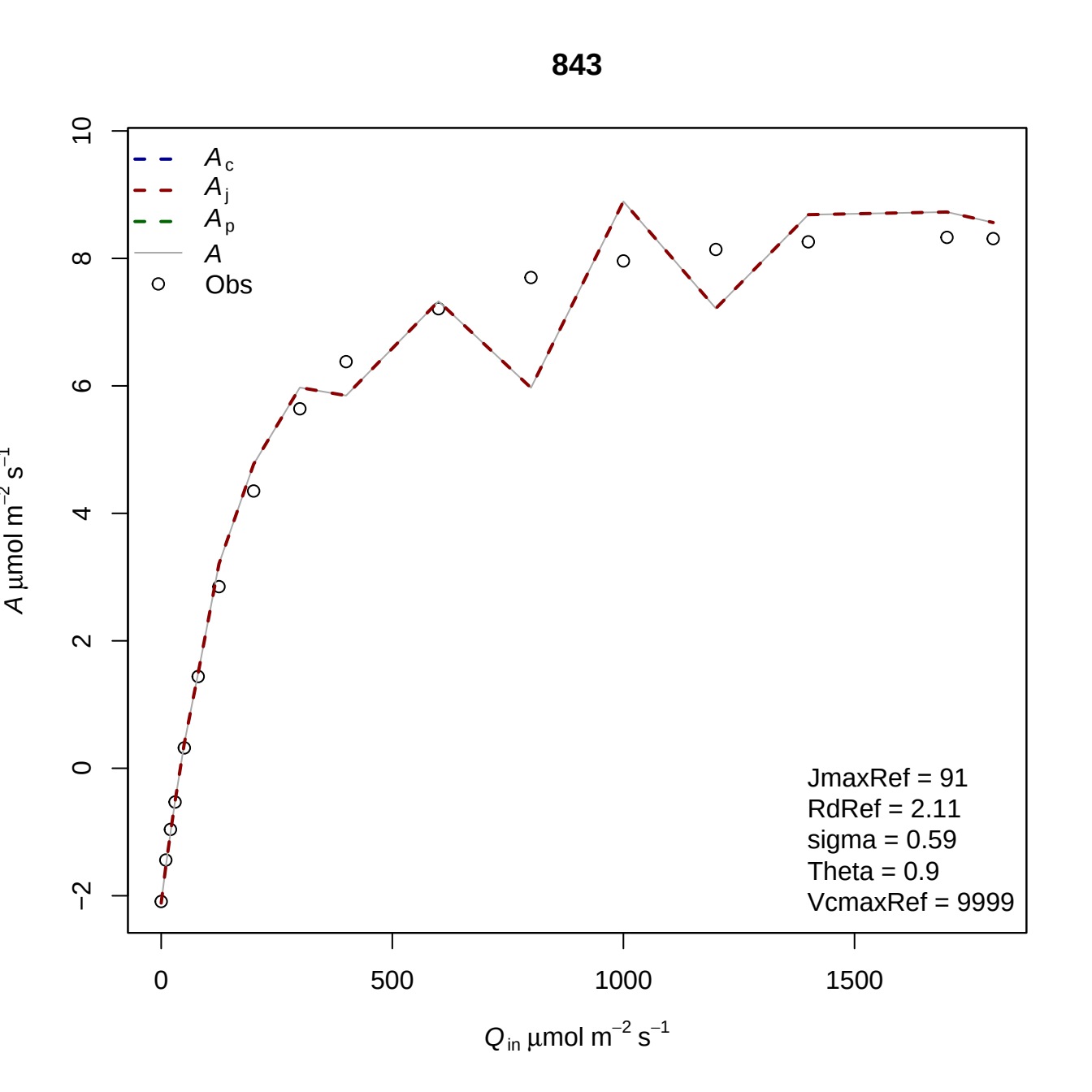

838
A_c
A_j
A_p
A
Obs
JmaxRef = 123.77
RdRef = 2.98
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

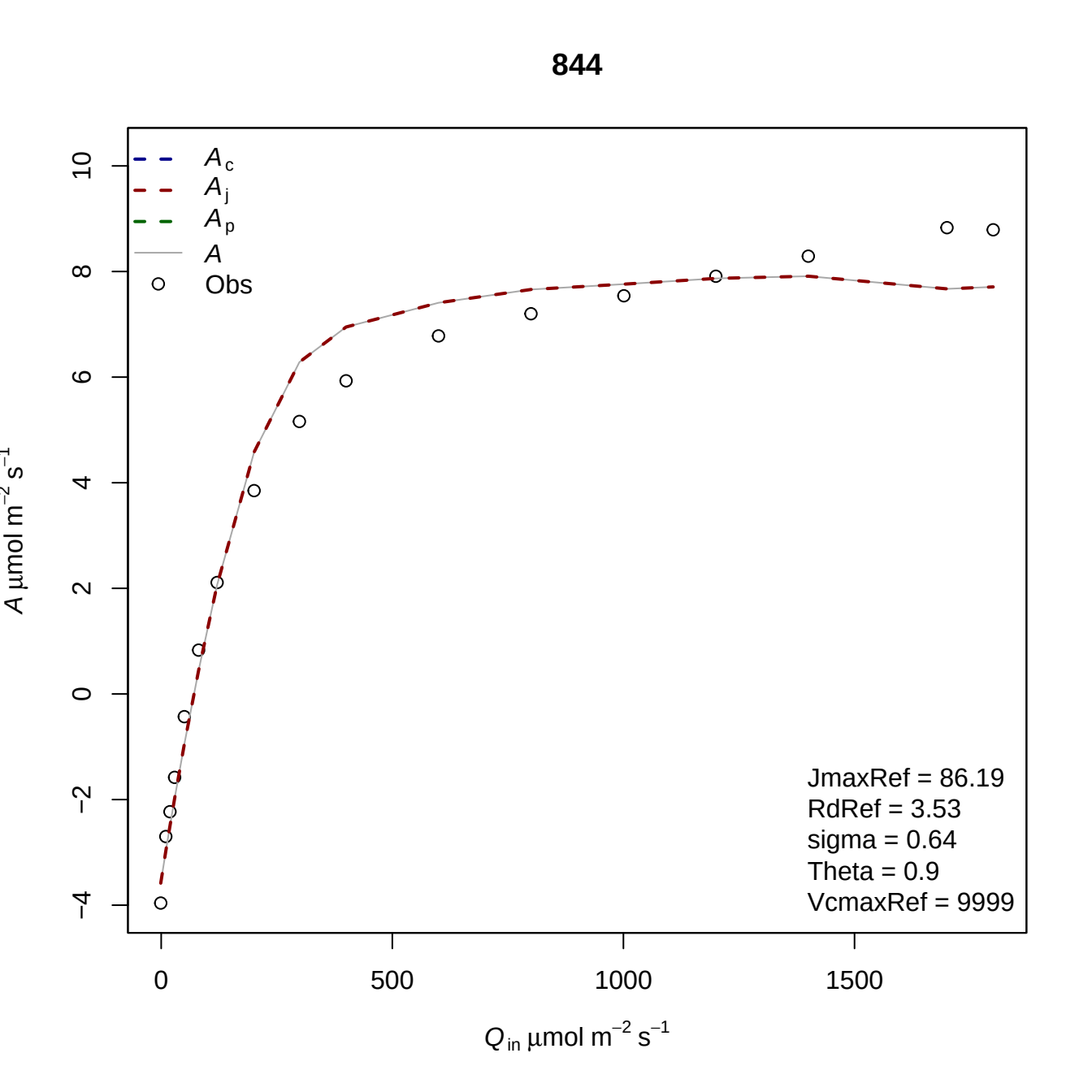

839
A_c
A_j
A_p
A
Obs
JmaxRef = 41.19
RdRef = 1.76
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

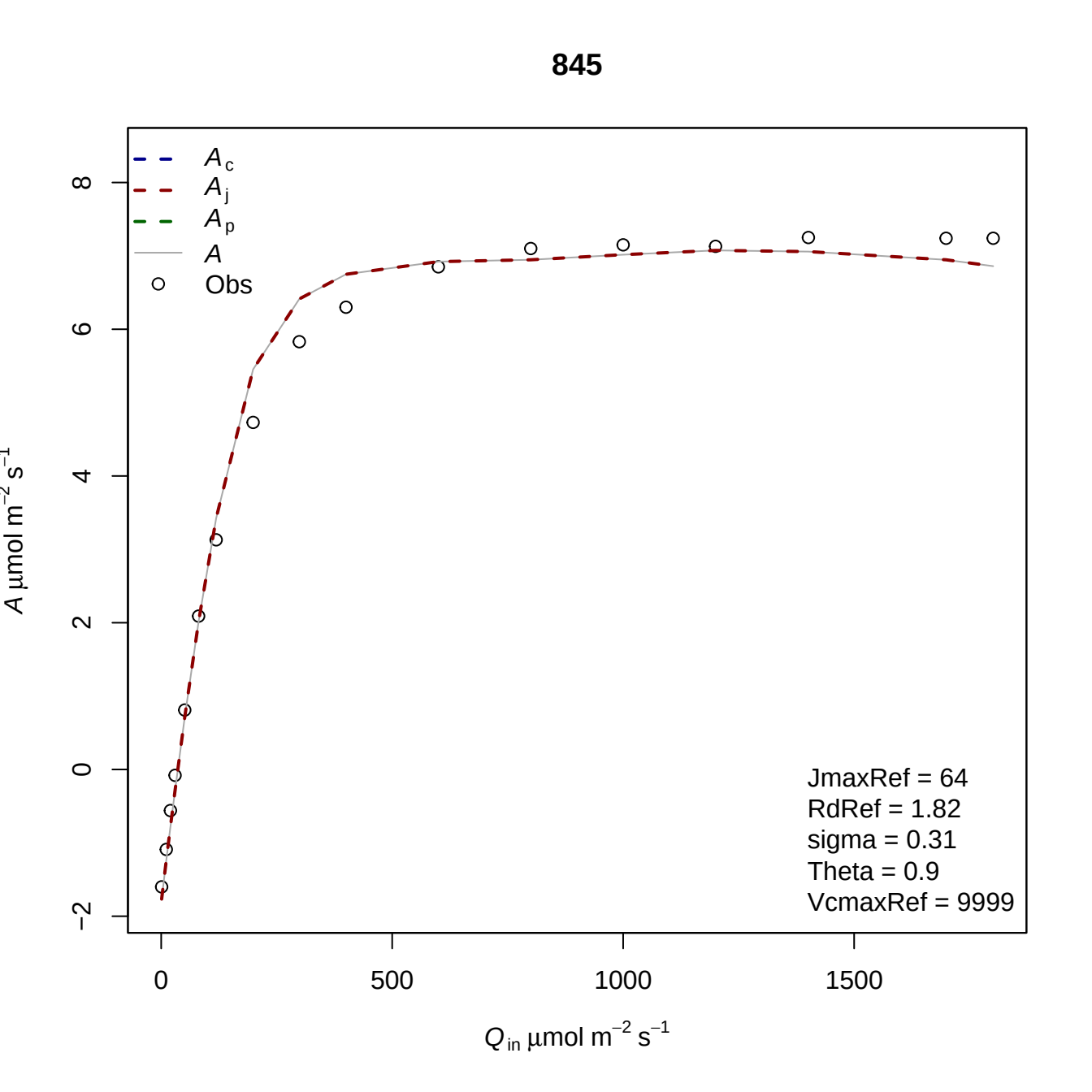

840
A_c
A_j
A_p
A
Obs
JmaxRef = 96.3
RdRef = 6.1
sigma = 1.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

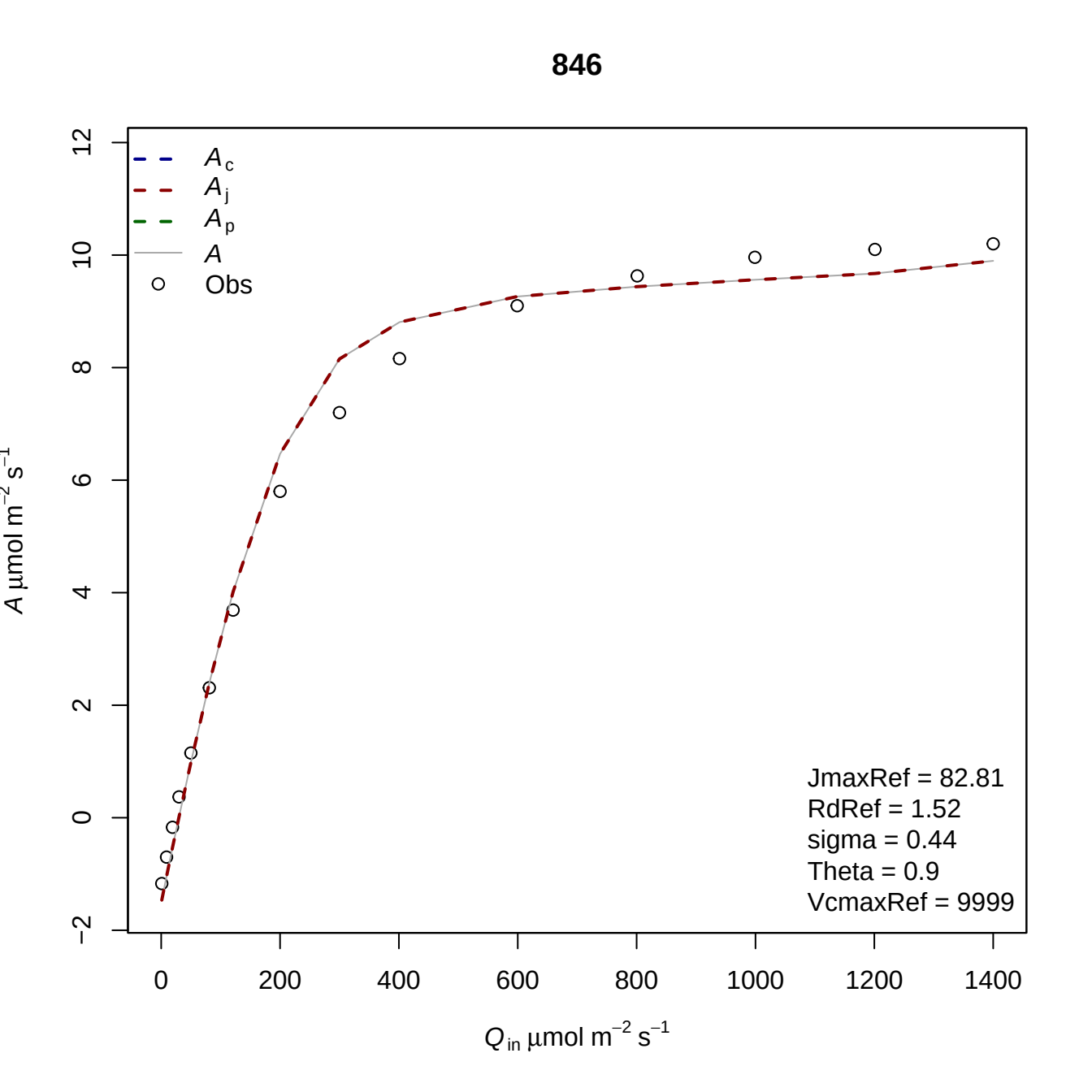

841
A_c
A_j
A_p
A
Obs
JmaxRef = 157.2
RdRef = 4.83
sigma = 1.12
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

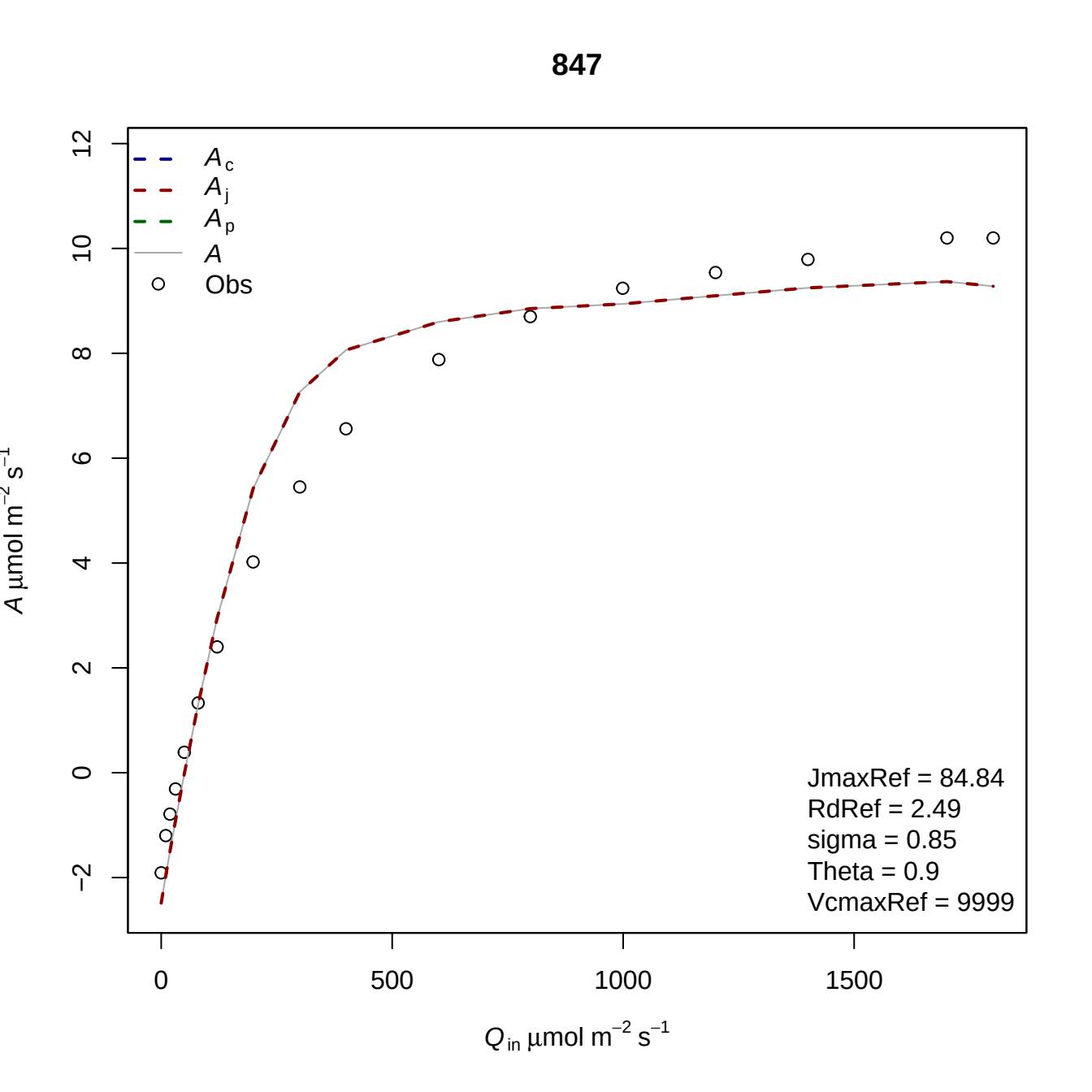

842
A_c
A_j
A_p
A
Obs
JmaxRef = 97.47
RdRef = 2.08
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

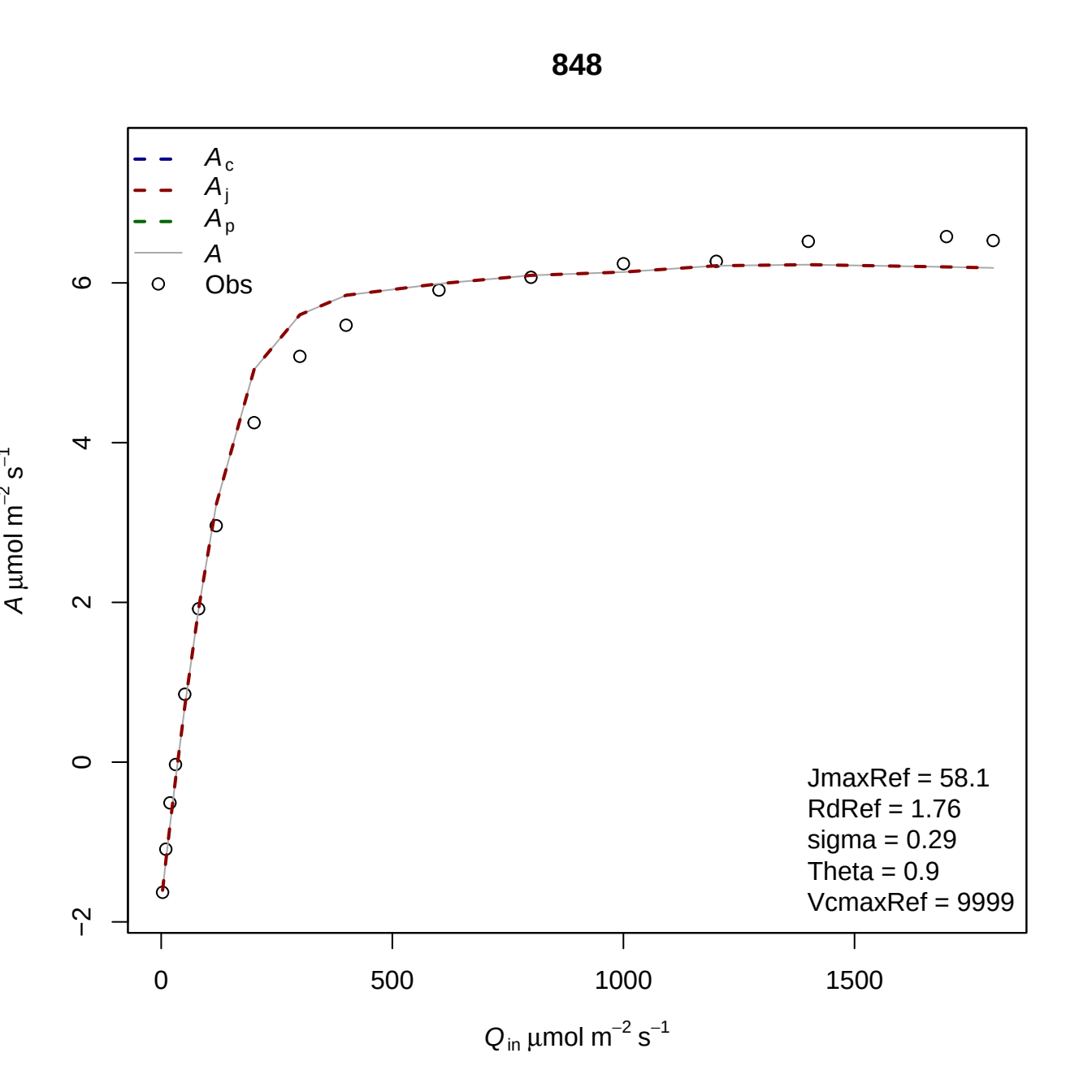

843
A_c
A_j
A_p
A
Obs
JmaxRef = 91
RdRef = 2.11
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

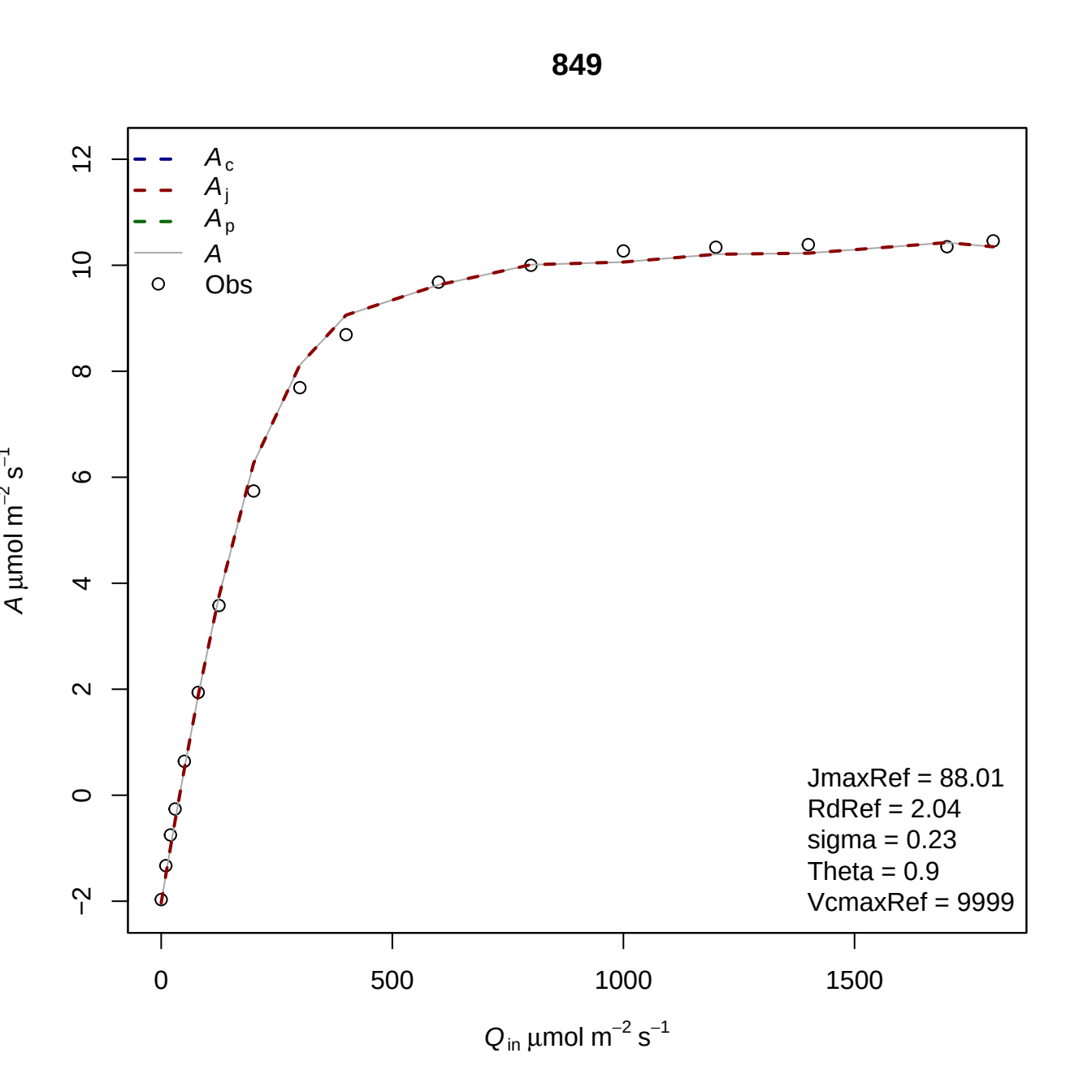

844
A_c
A_j
A_p
A
Obs
JmaxRef = 86.19
RdRef = 3.53
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

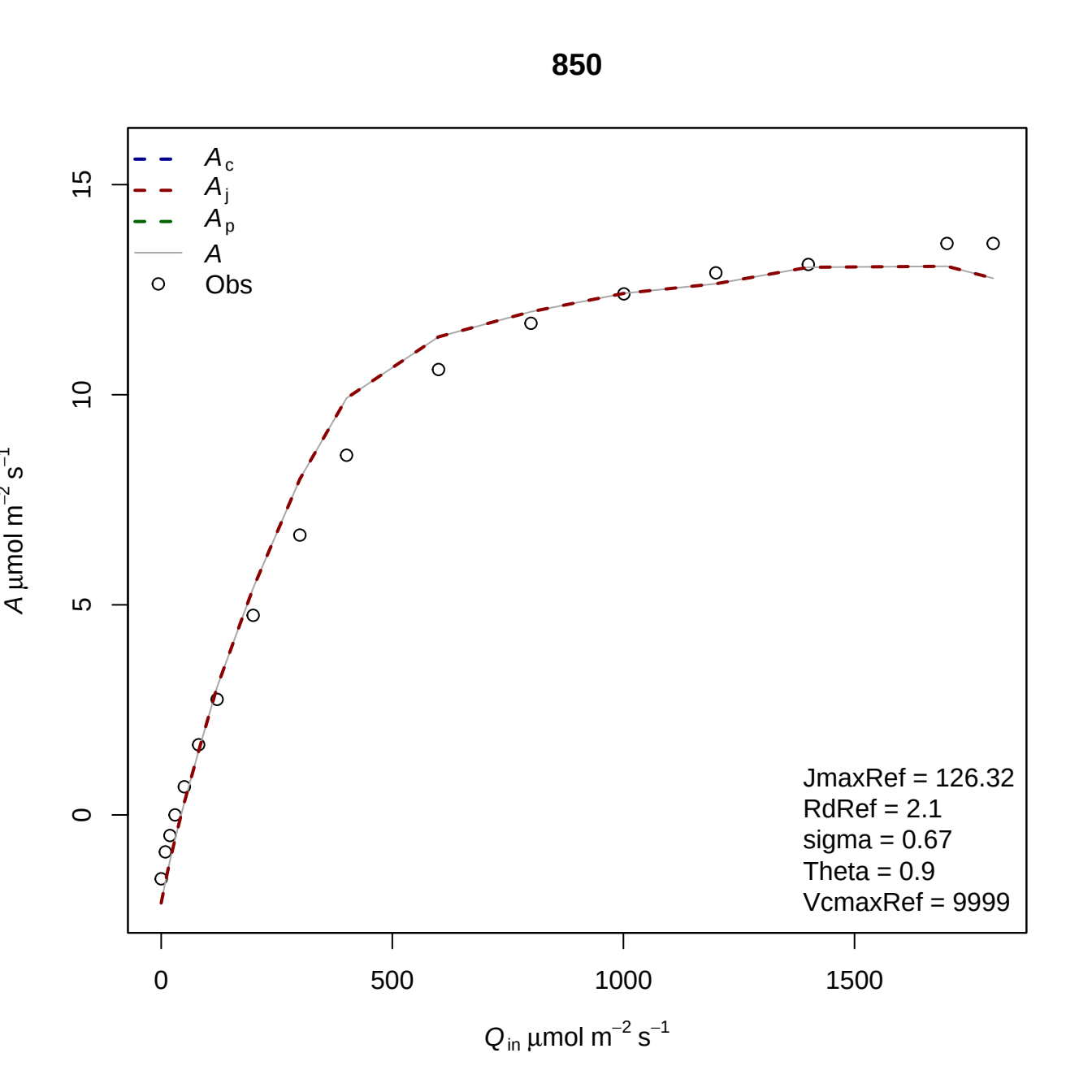

845
A_c
A_j
A_p
A
Obs
JmaxRef = 64
RdRef = 1.82
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

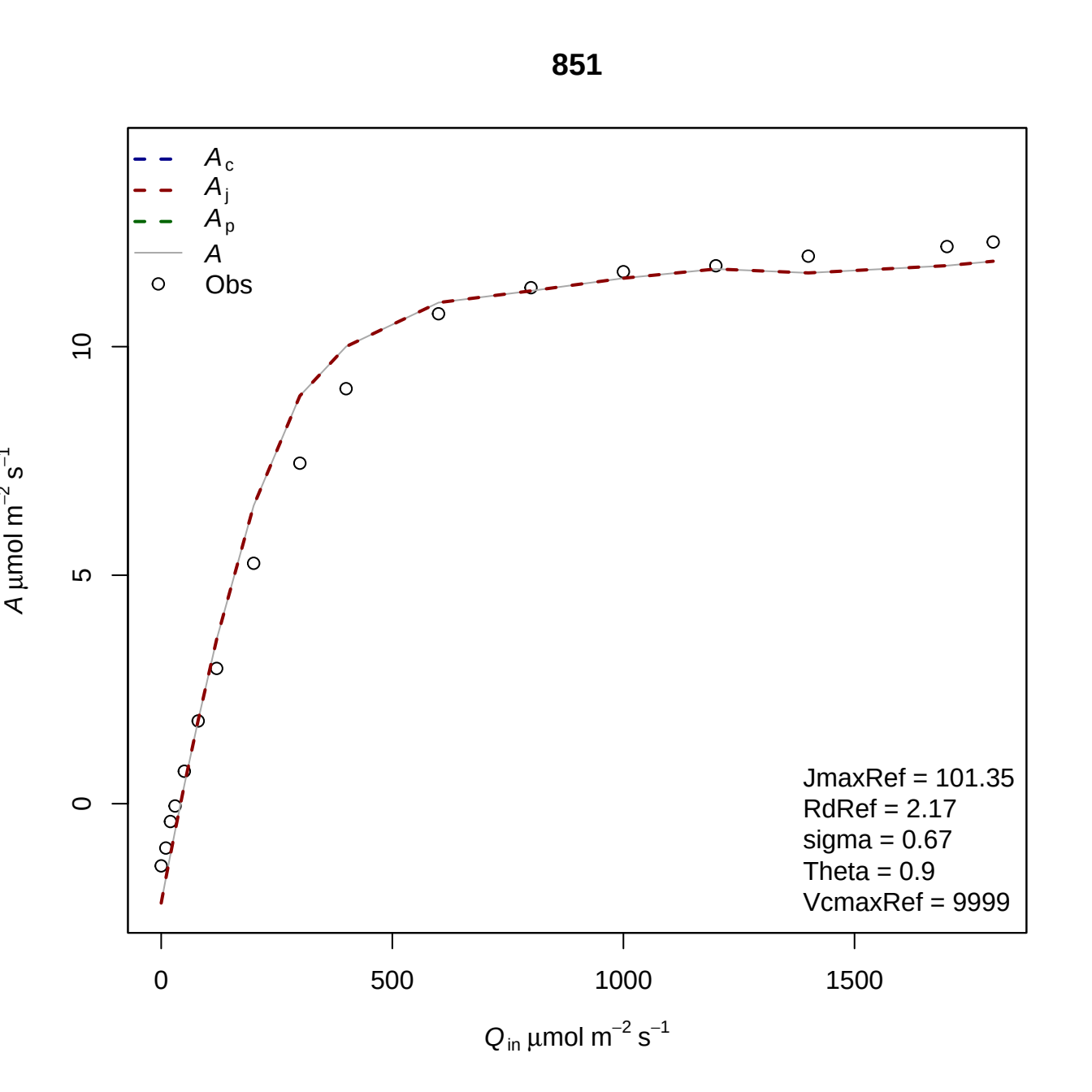

846
A_c
A_j
A_p
A
Obs
JmaxRef = 82.81
RdRef = 1.52
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

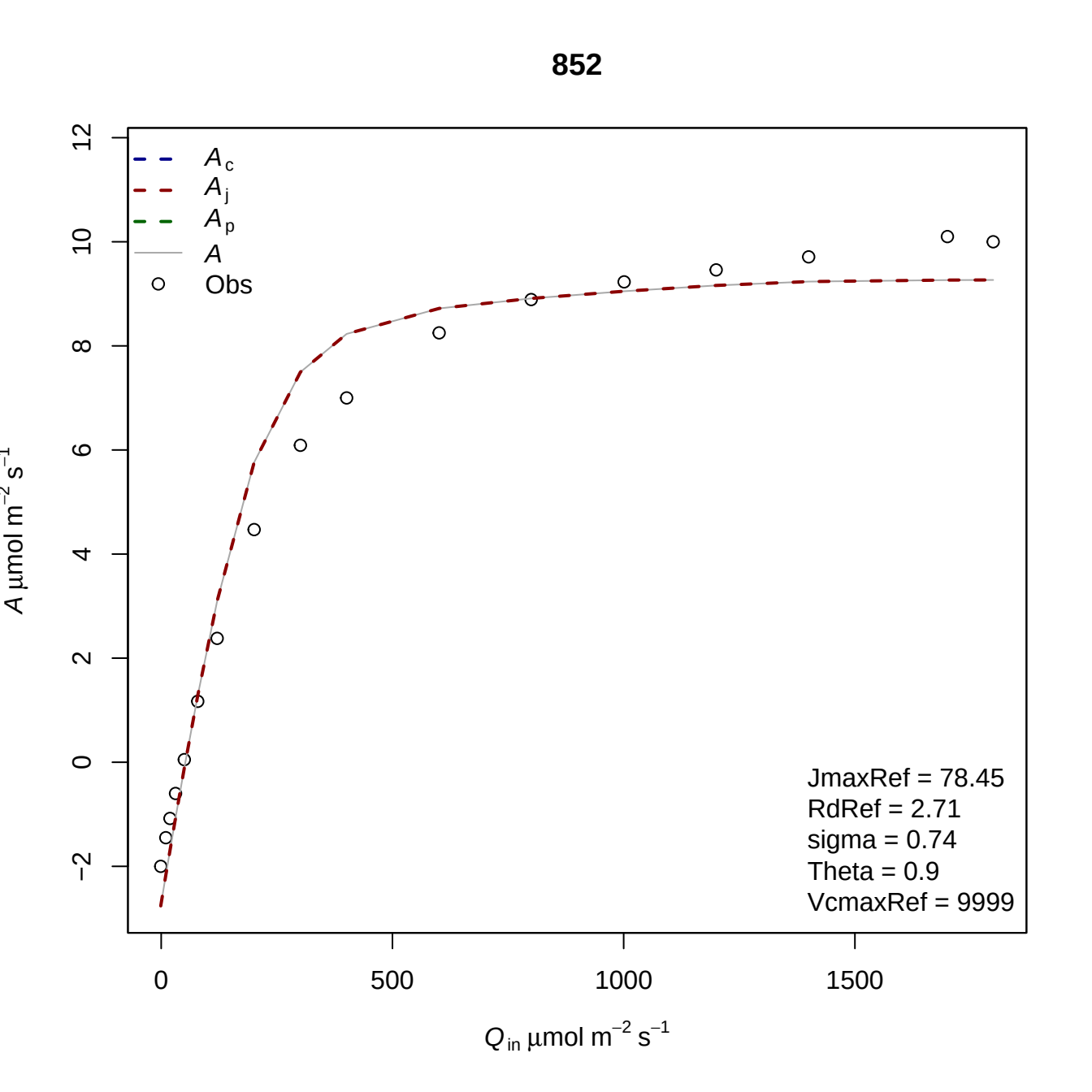

847
A_c
A_j
A_p
A
Obs
JmaxRef = 84.84
RdRef = 2.49
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

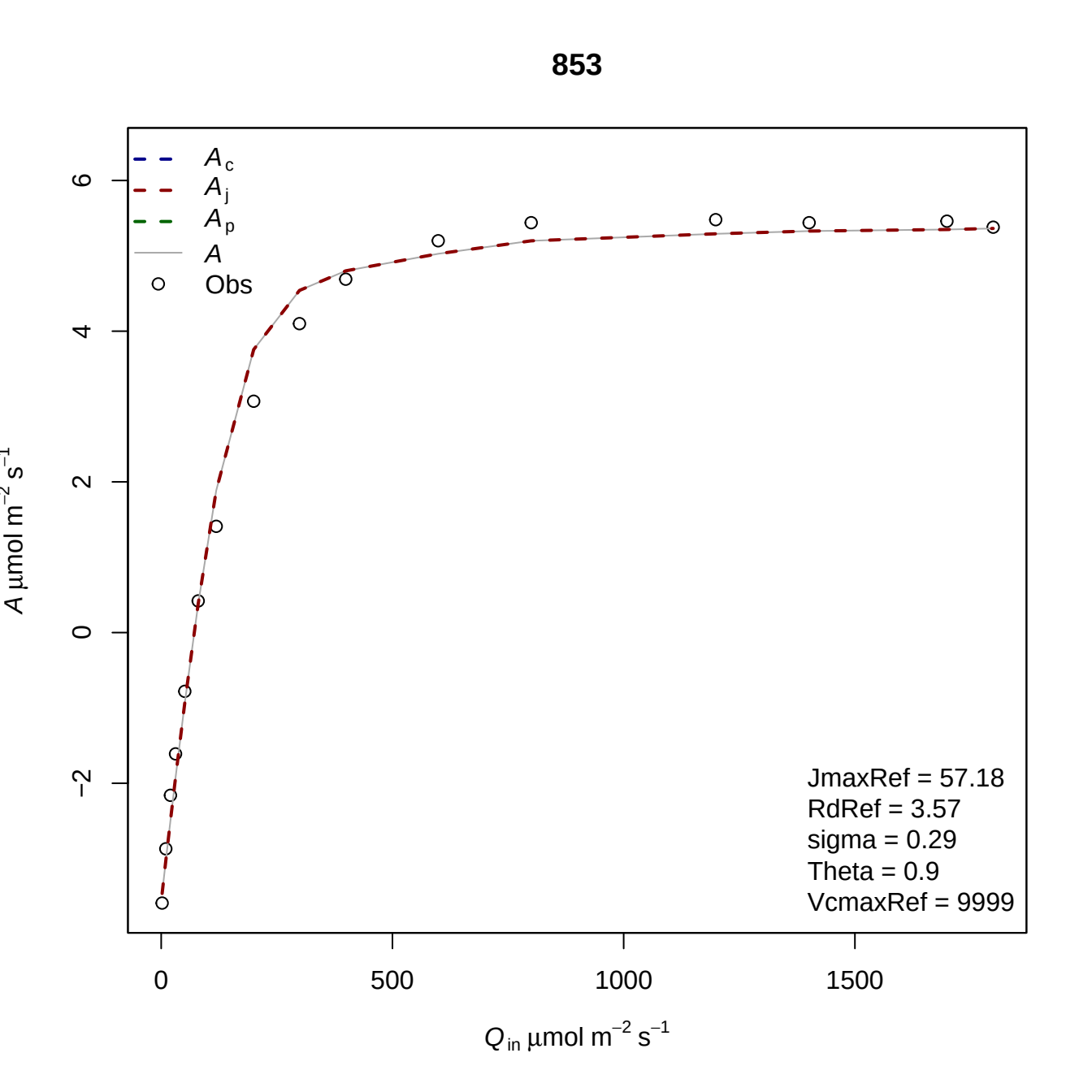

848
A_c
A_j
A_p
A
Obs
JmaxRef = 58.1
RdRef = 1.76
sigma = 0.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

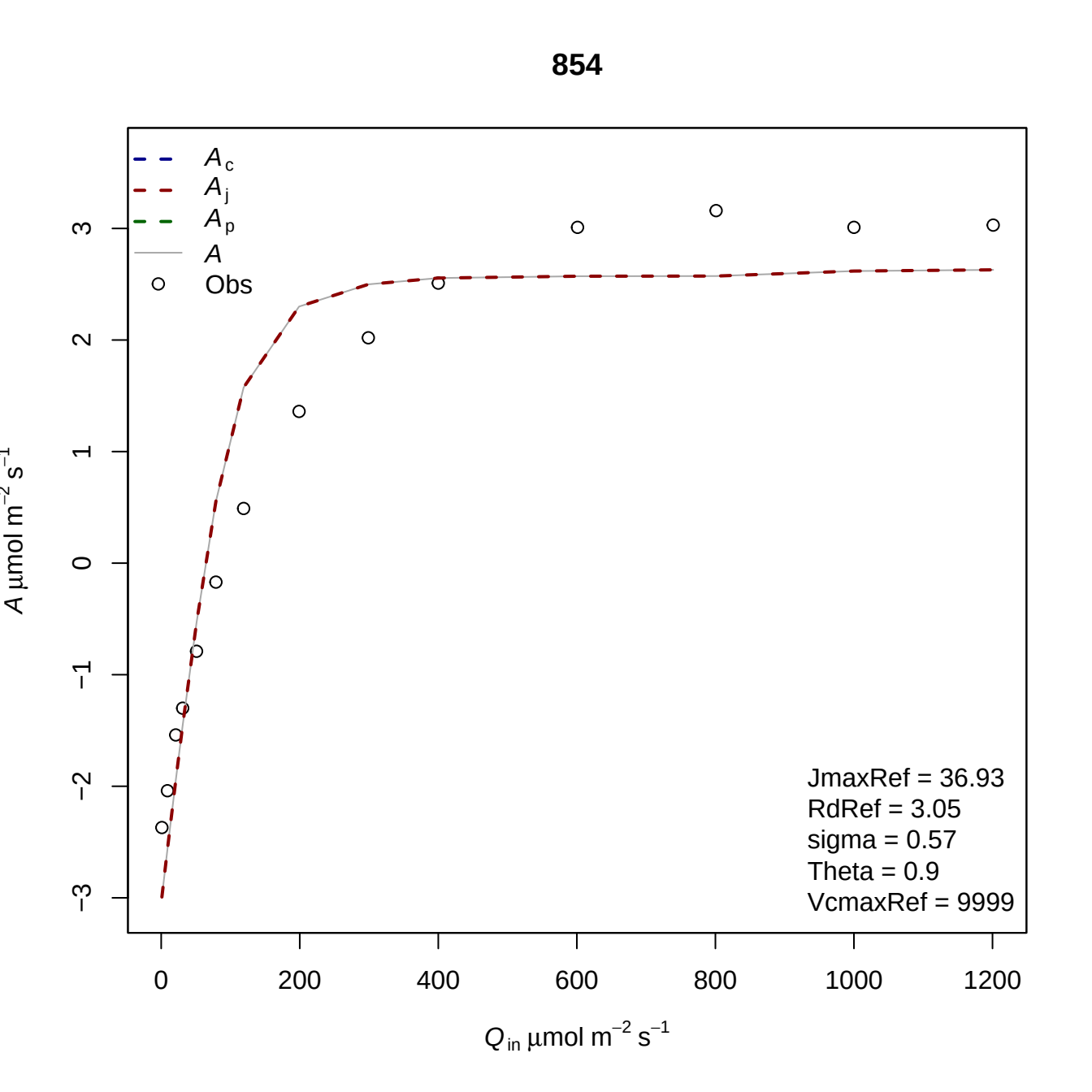

849
A_c
A_j
A_p
A
Obs
JmaxRef = 88.01
RdRef = 2.04
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

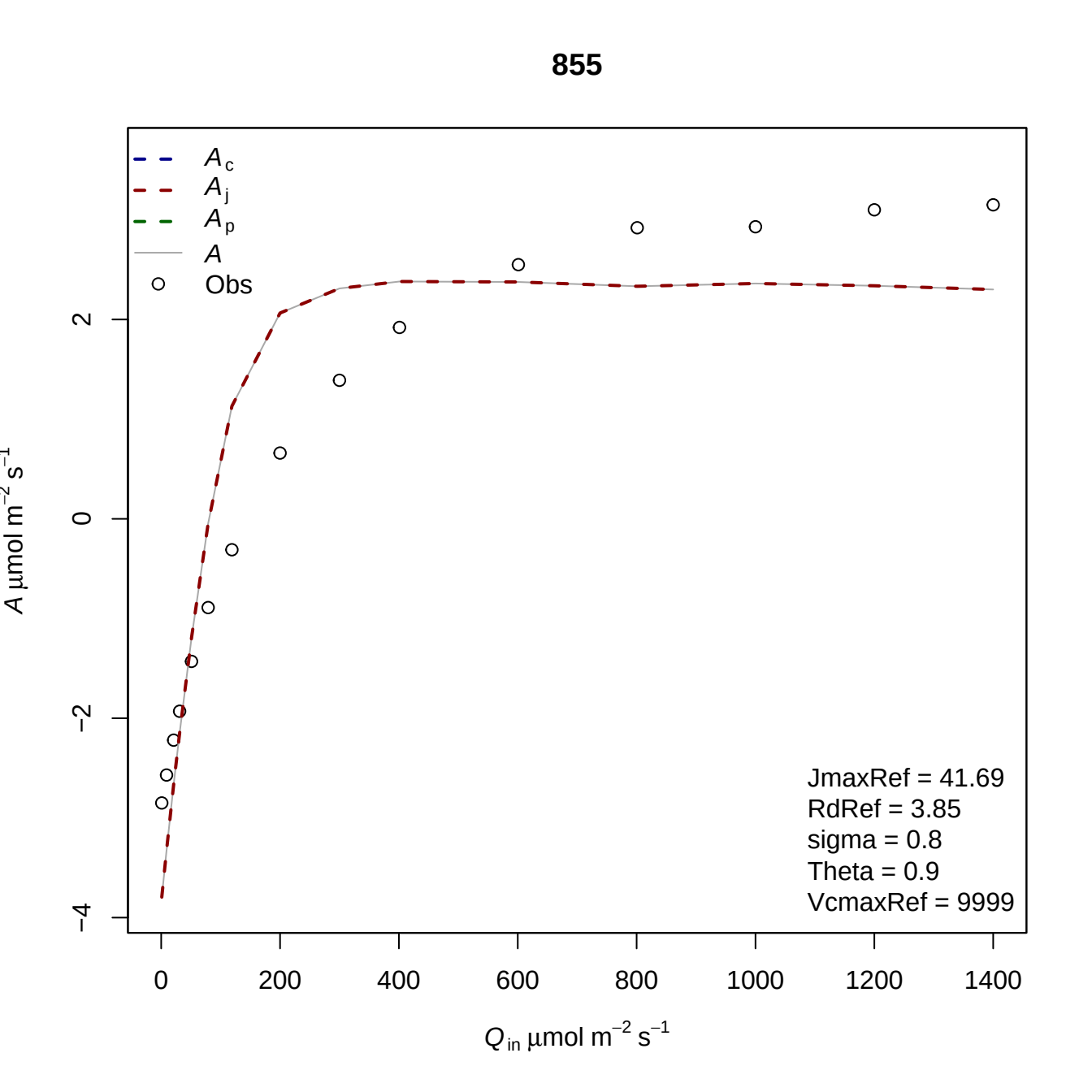

850
A_c
A_j
A_p
A
Obs
JmaxRef = 126.32
RdRef = 2.1
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

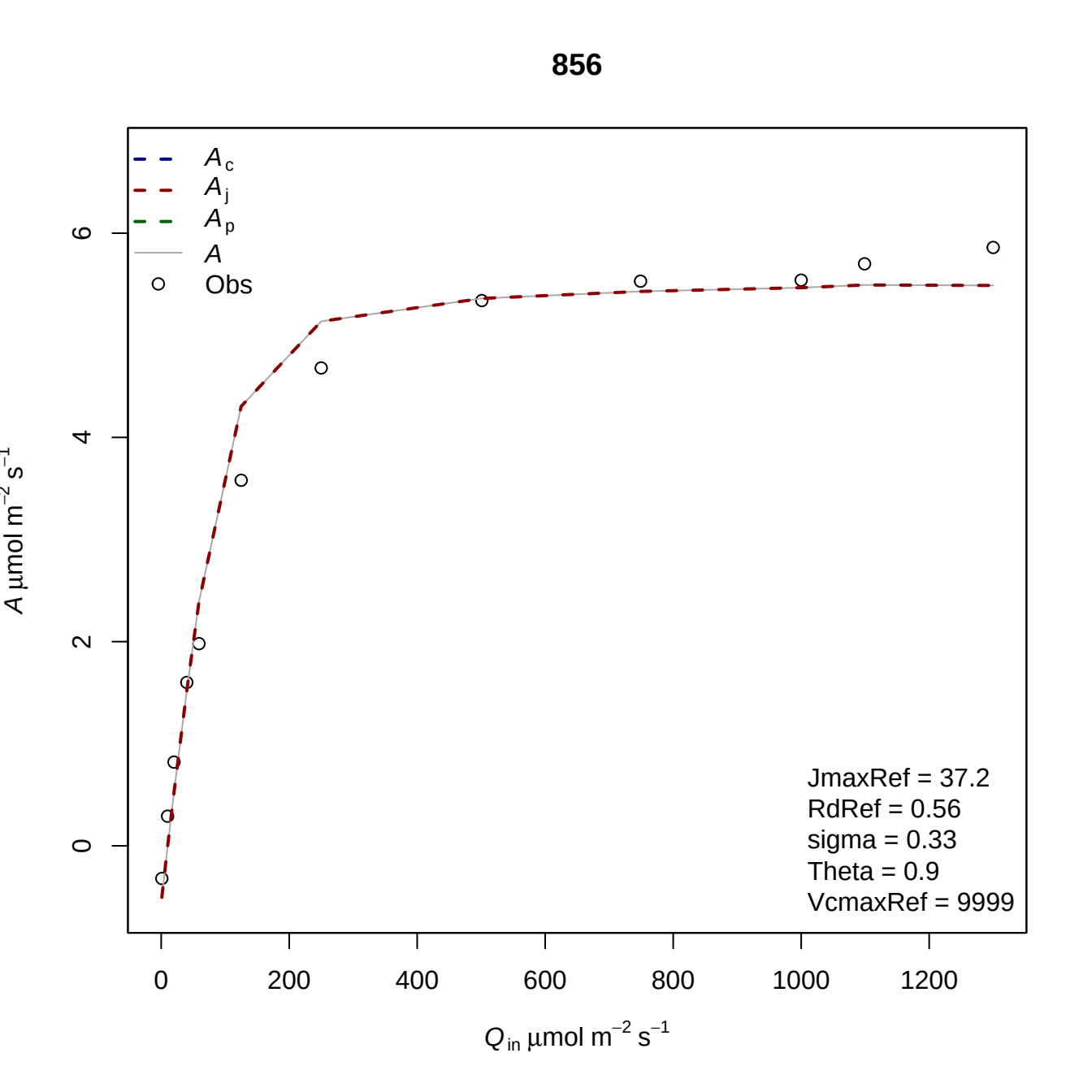

851
A_c
A_j
A_p
A
Obs
JmaxRef = 101.35
RdRef = 2.17
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

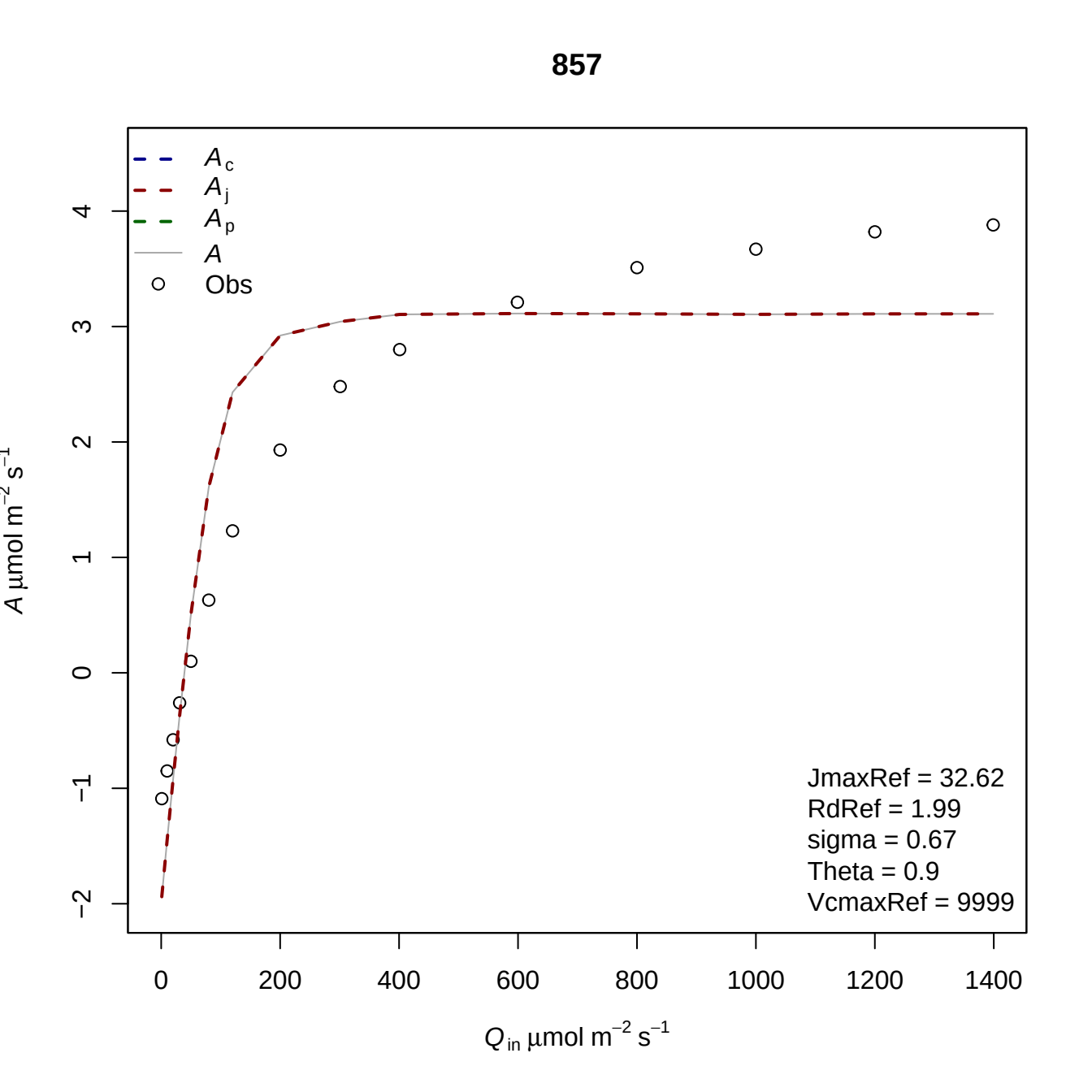

852
A_c
A_j
A_p
A
Obs
JmaxRef = 78.45
RdRef = 2.71
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

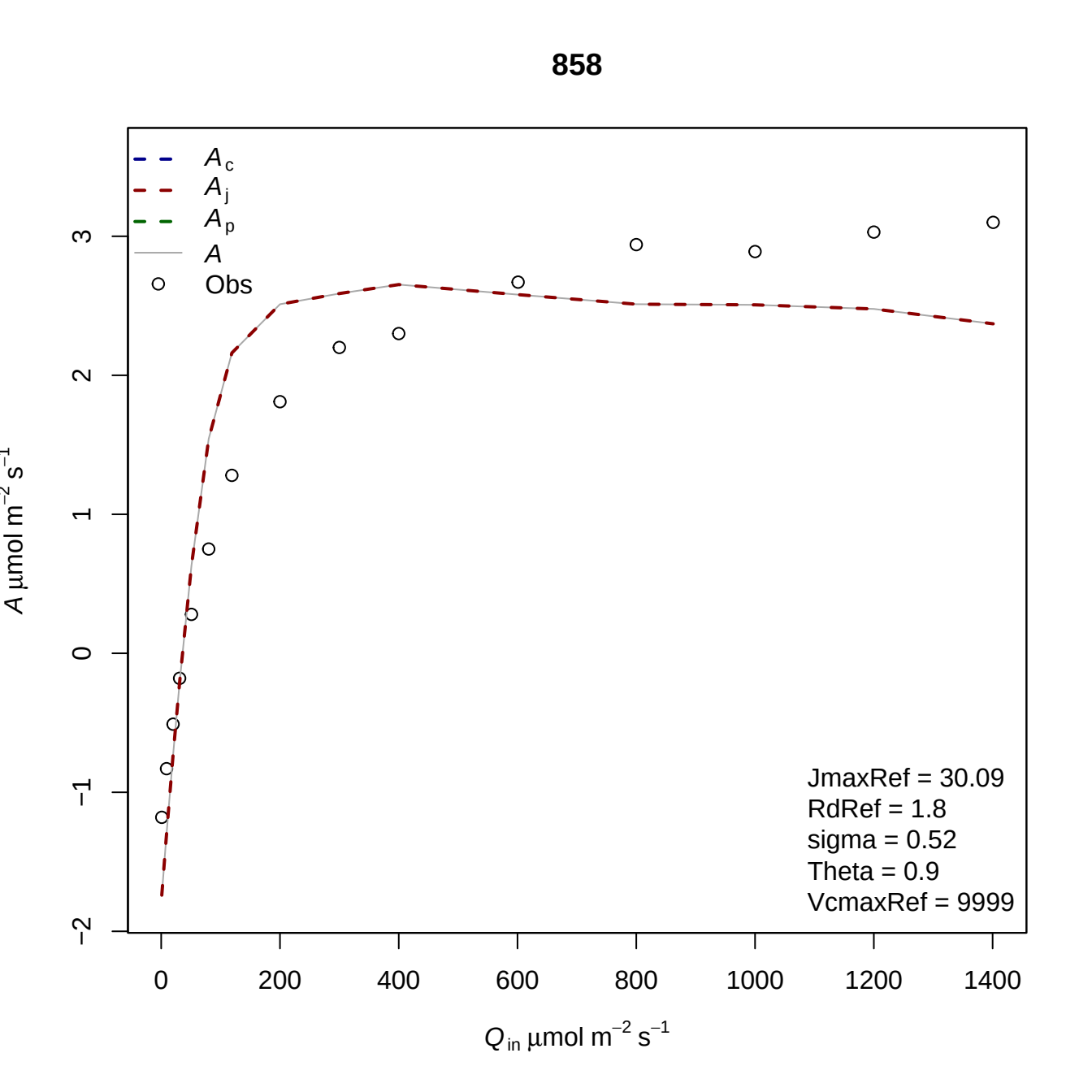

853
A_c
A_j
A_p
A
Obs
JmaxRef = 57.18
RdRef = 3.57
sigma = 0.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

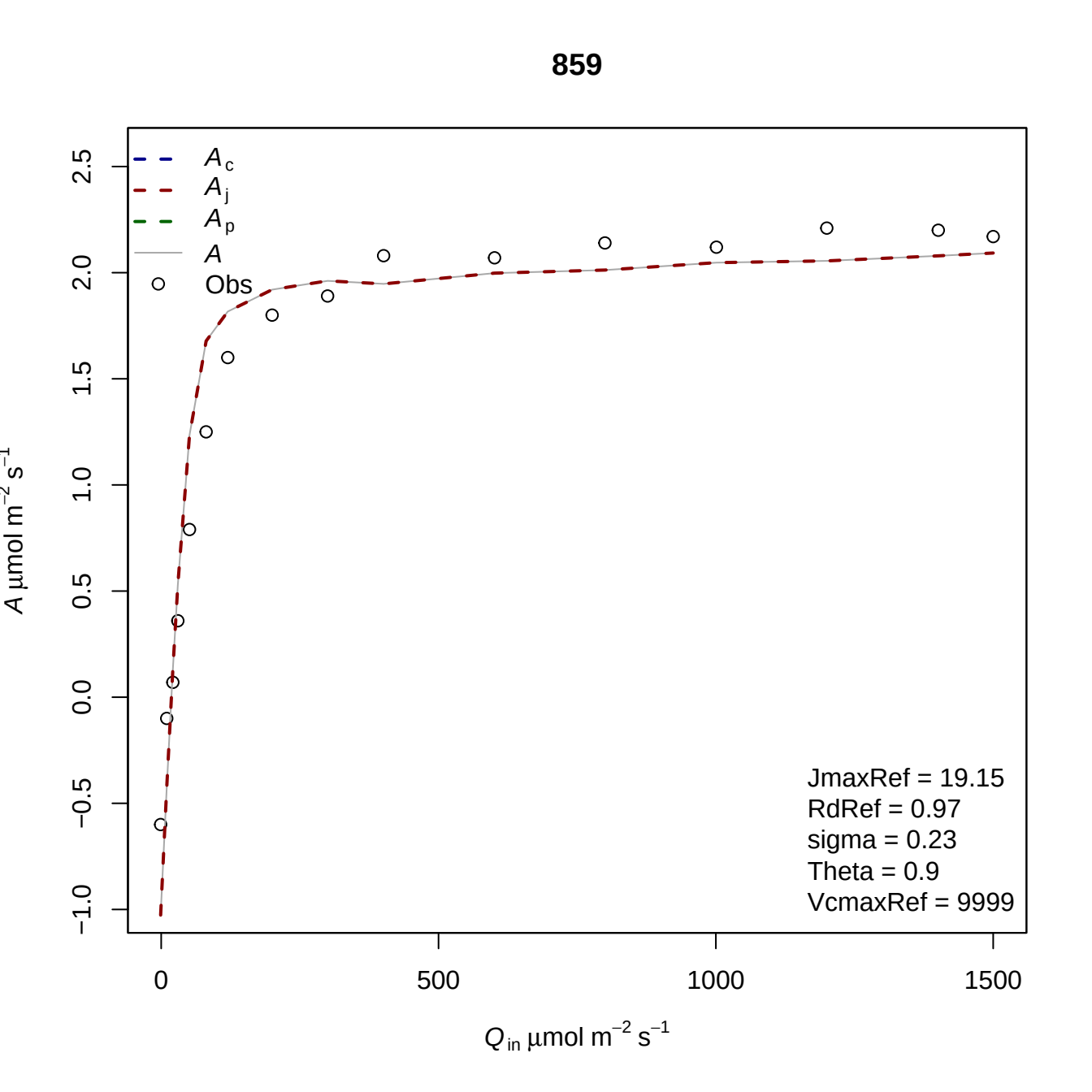

854
A_c
A_j
A_p
A
Obs
JmaxRef = 36.93
RdRef = 3.05
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

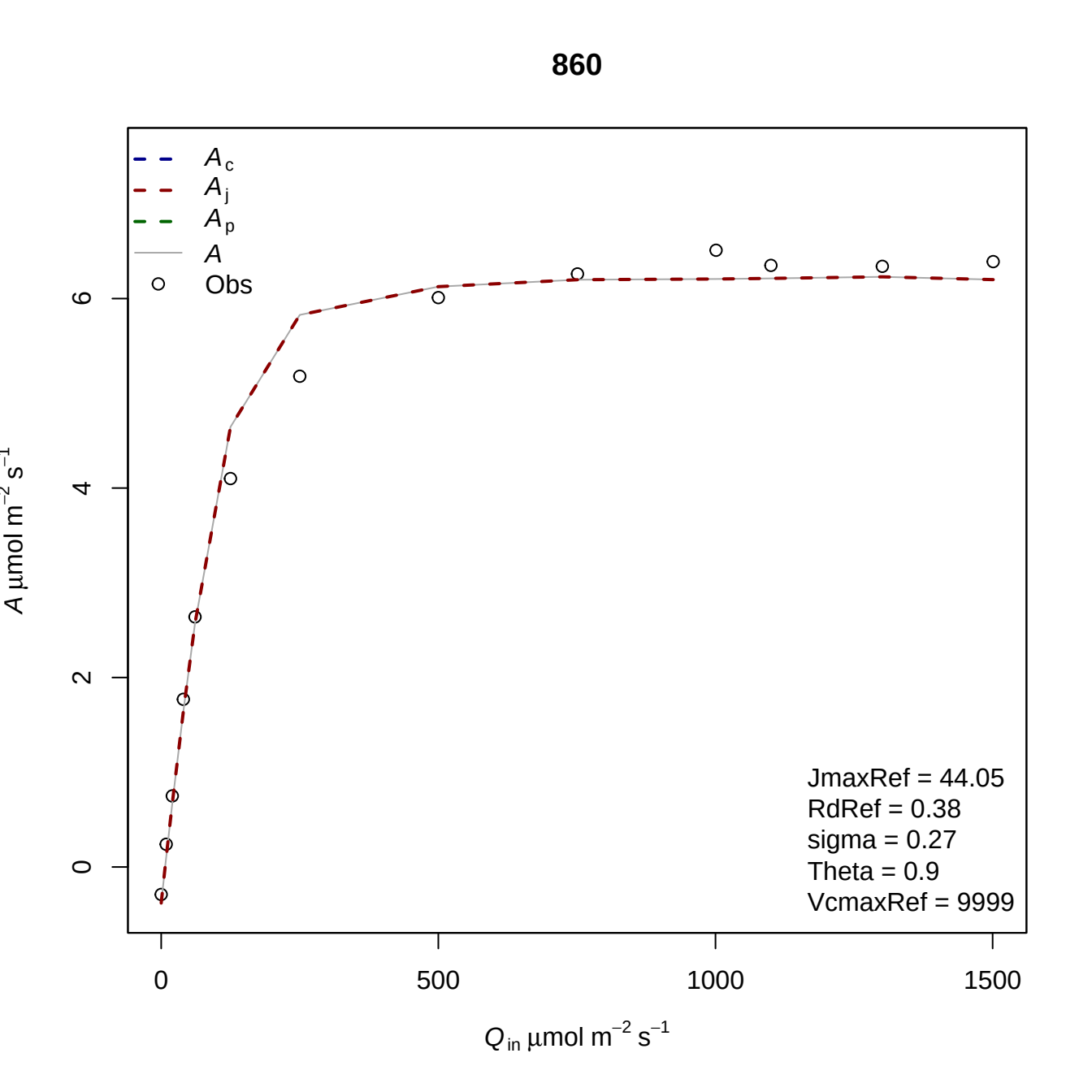

855
A_c
A_j
A_p
A
Obs
JmaxRef = 41.69
RdRef = 3.85
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
0
200
400
600
800
1000
1200
1400

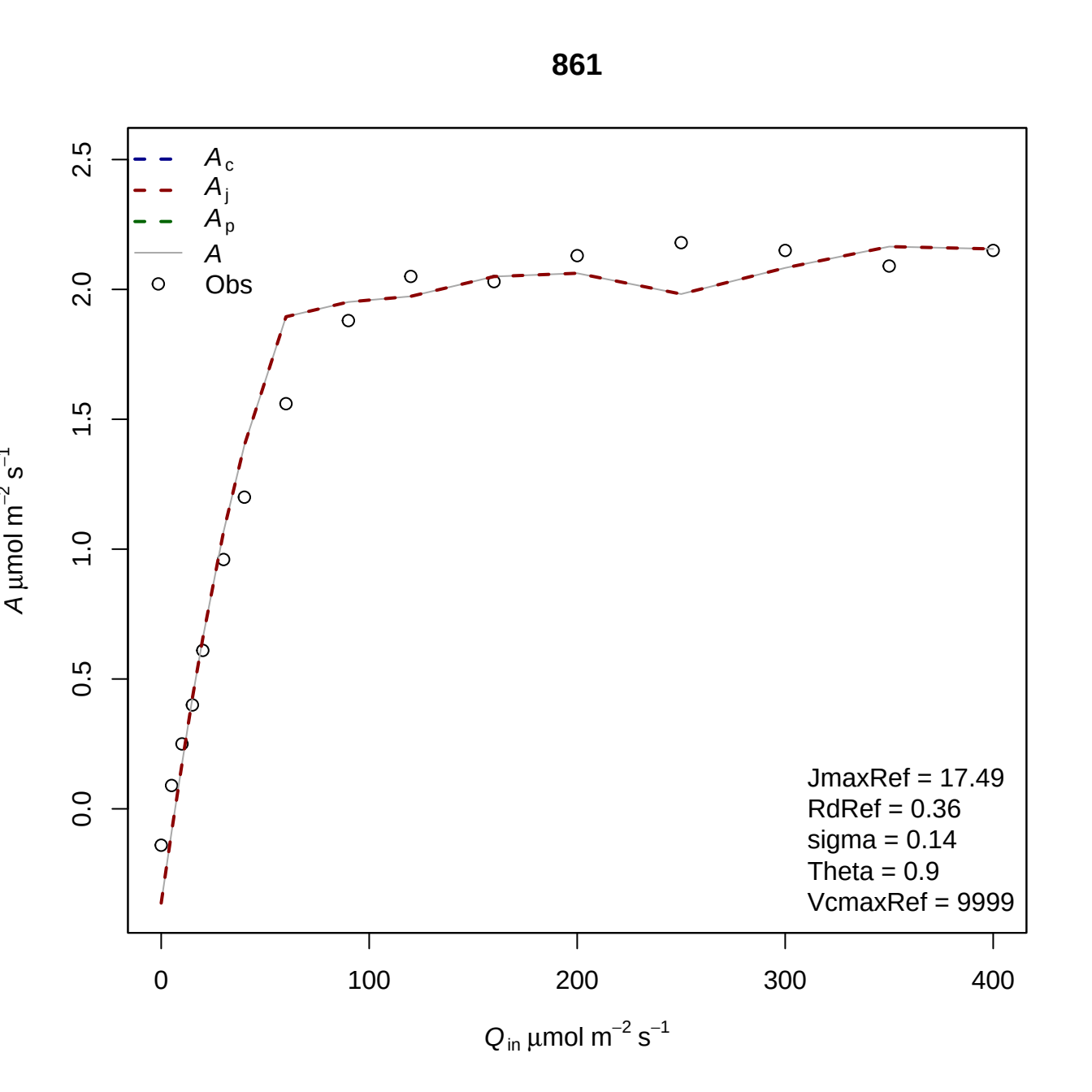

856
A_c
A_j
A_p
A
Obs
JmaxRef = 37.2
RdRef = 0.56
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200

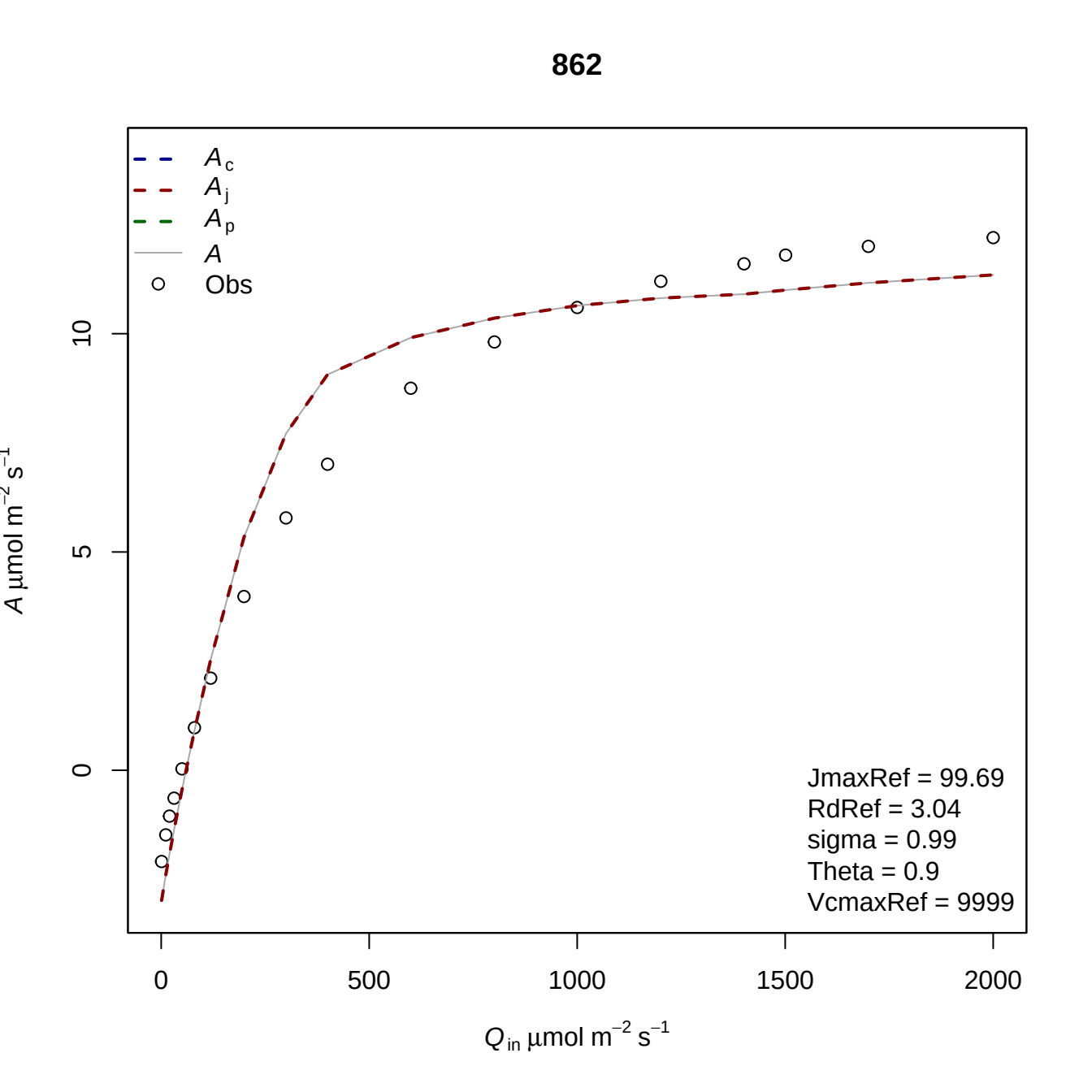

857
A_c
A_j
A_p
A
Obs
JmaxRef = 32.62
RdRef = 1.99
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

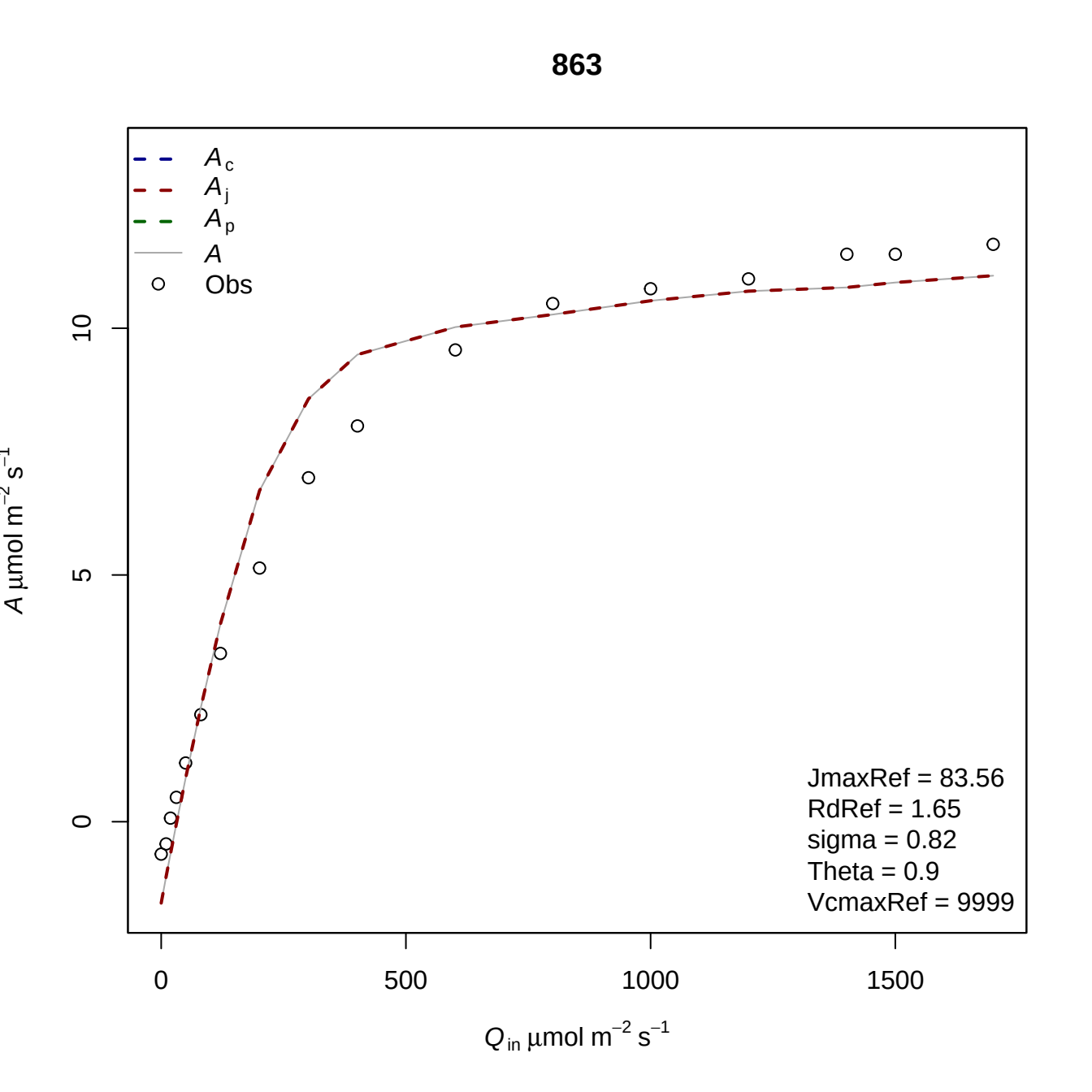

858
A_c
A_j
A_p
A
Obs
JmaxRef = 30.09
RdRef = 1.8
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

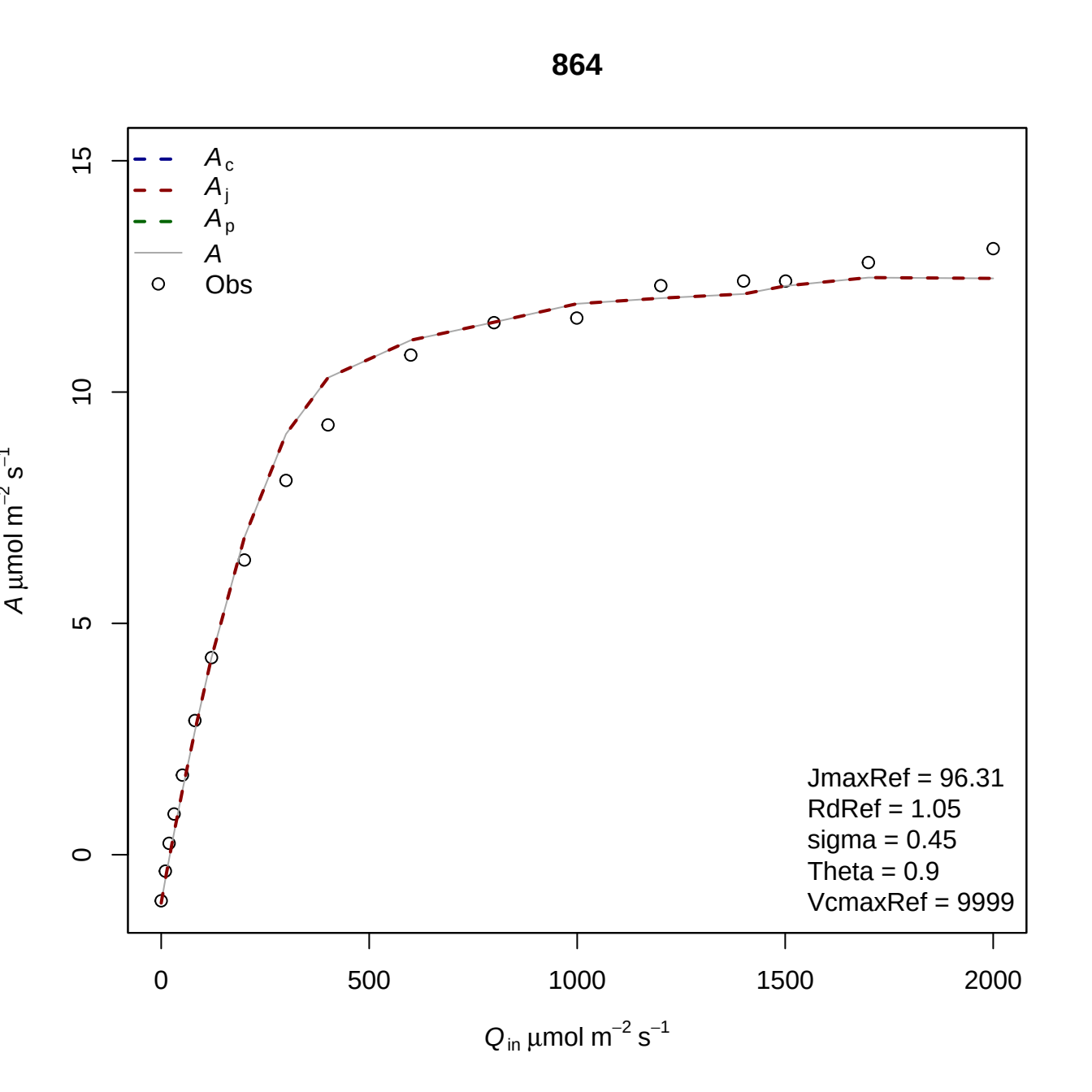

859
A_c
A_j
A_p
A
Obs
JmaxRef = 19.15
RdRef = 0.97
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

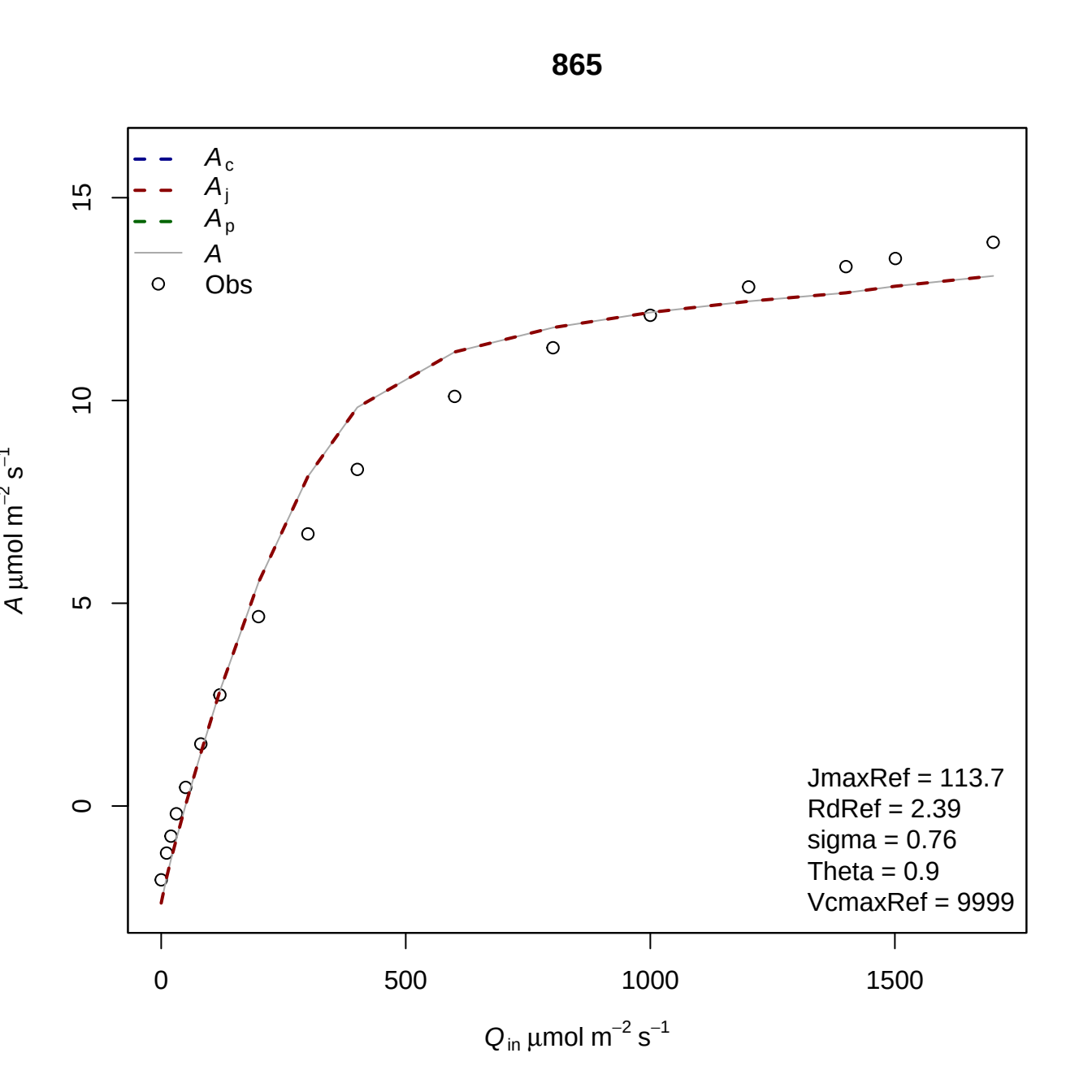

860
A_c
A_j
A_p
A
Obs
JmaxRef = 44.05
RdRef = 0.38
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

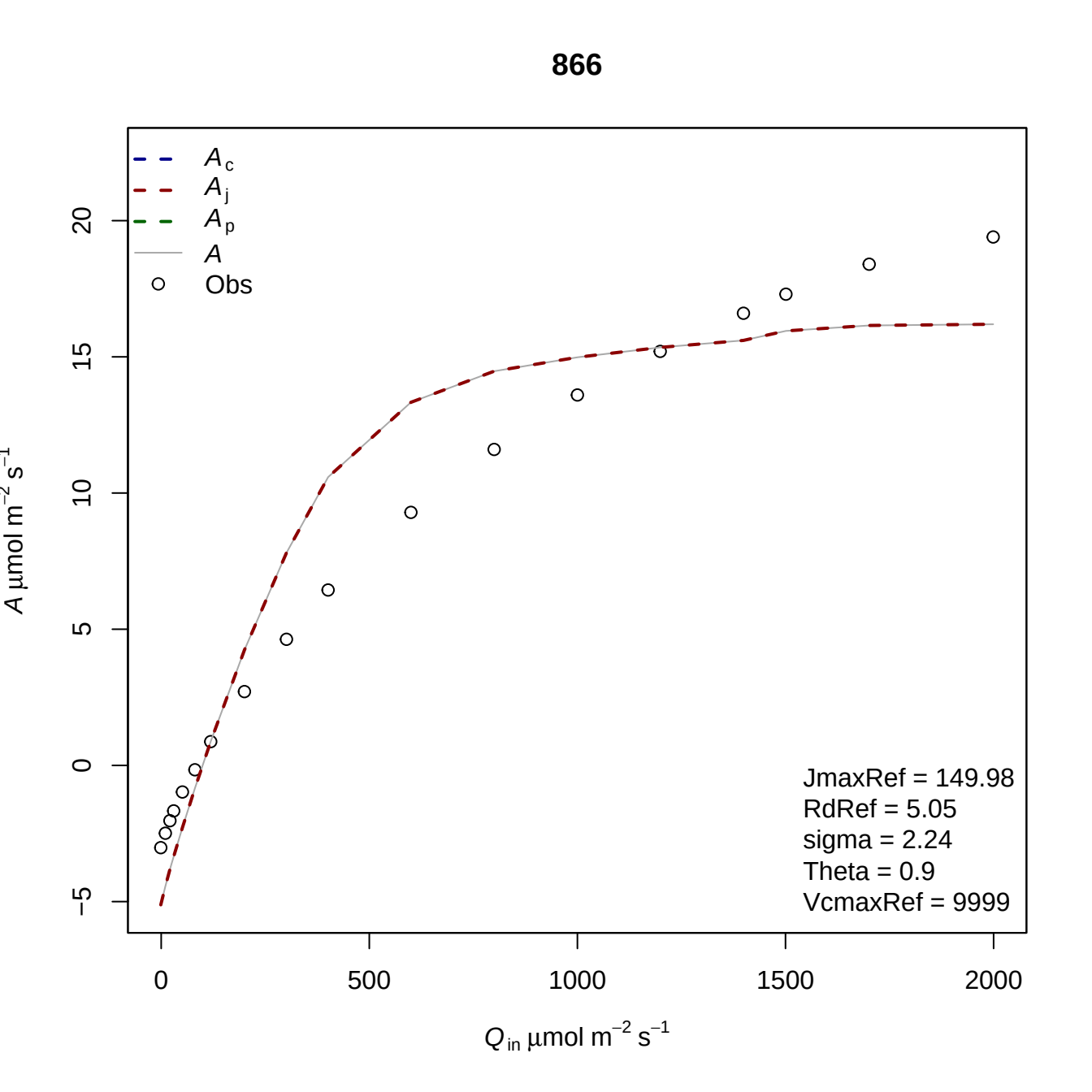

861
A_c
A_j
A_p
A
Obs
JmaxRef = 17.49
RdRef = 0.36
sigma = 0.14
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

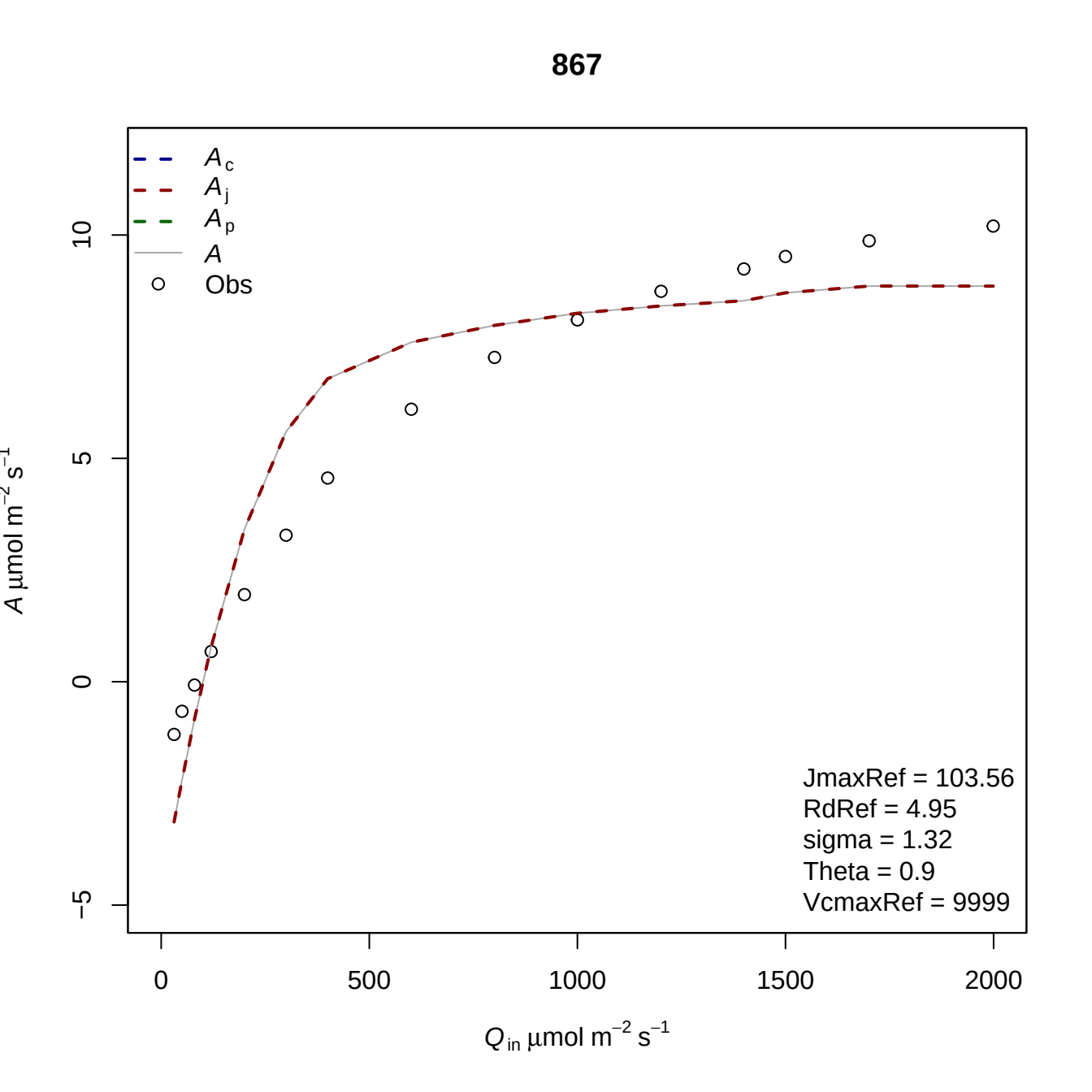

862
A_c
A_j
A_p
A
Obs
JmaxRef = 99.69
RdRef = 3.04
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

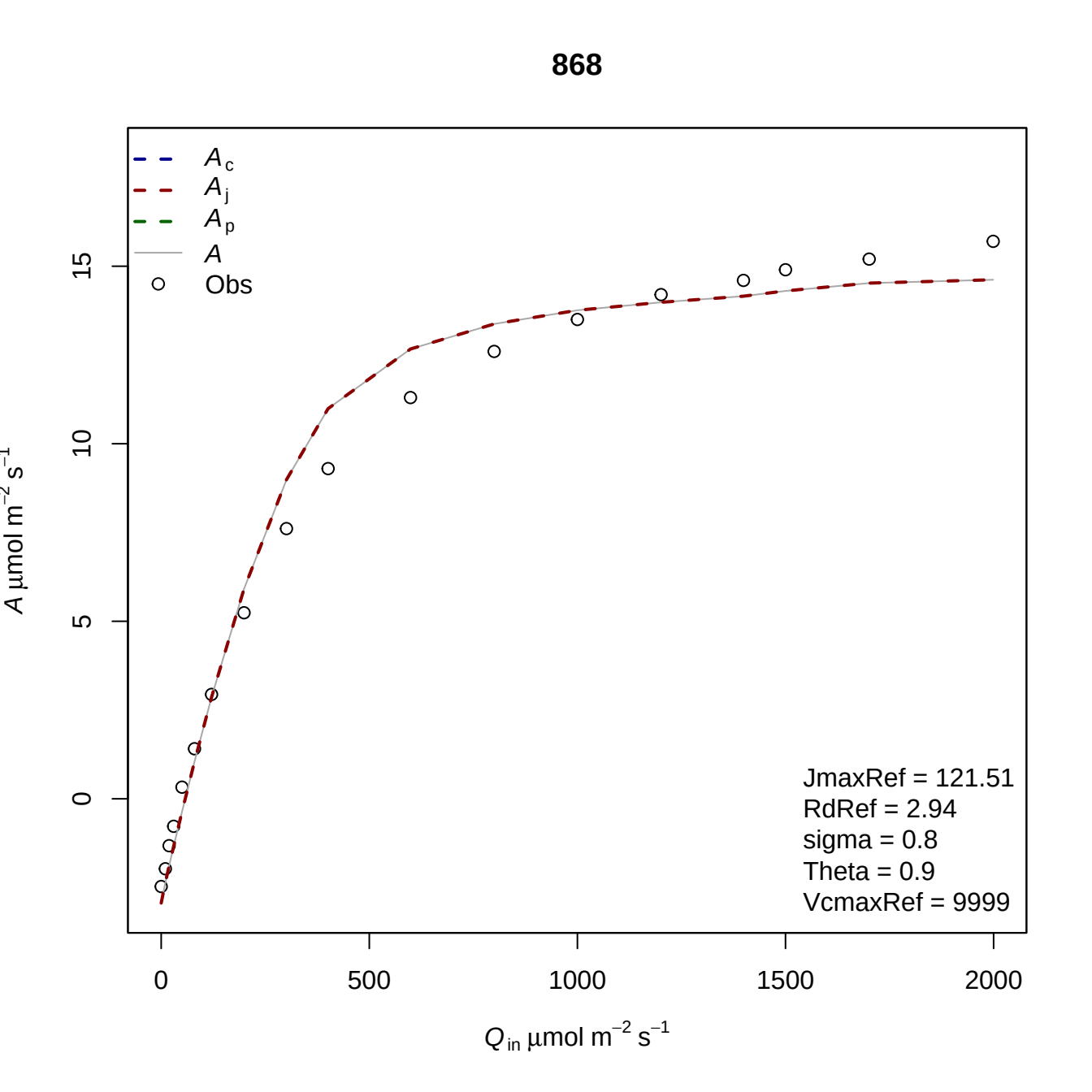

863
A_c
A_j
A_p
A
Obs
JmaxRef = 83.56
RdRef = 1.65
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

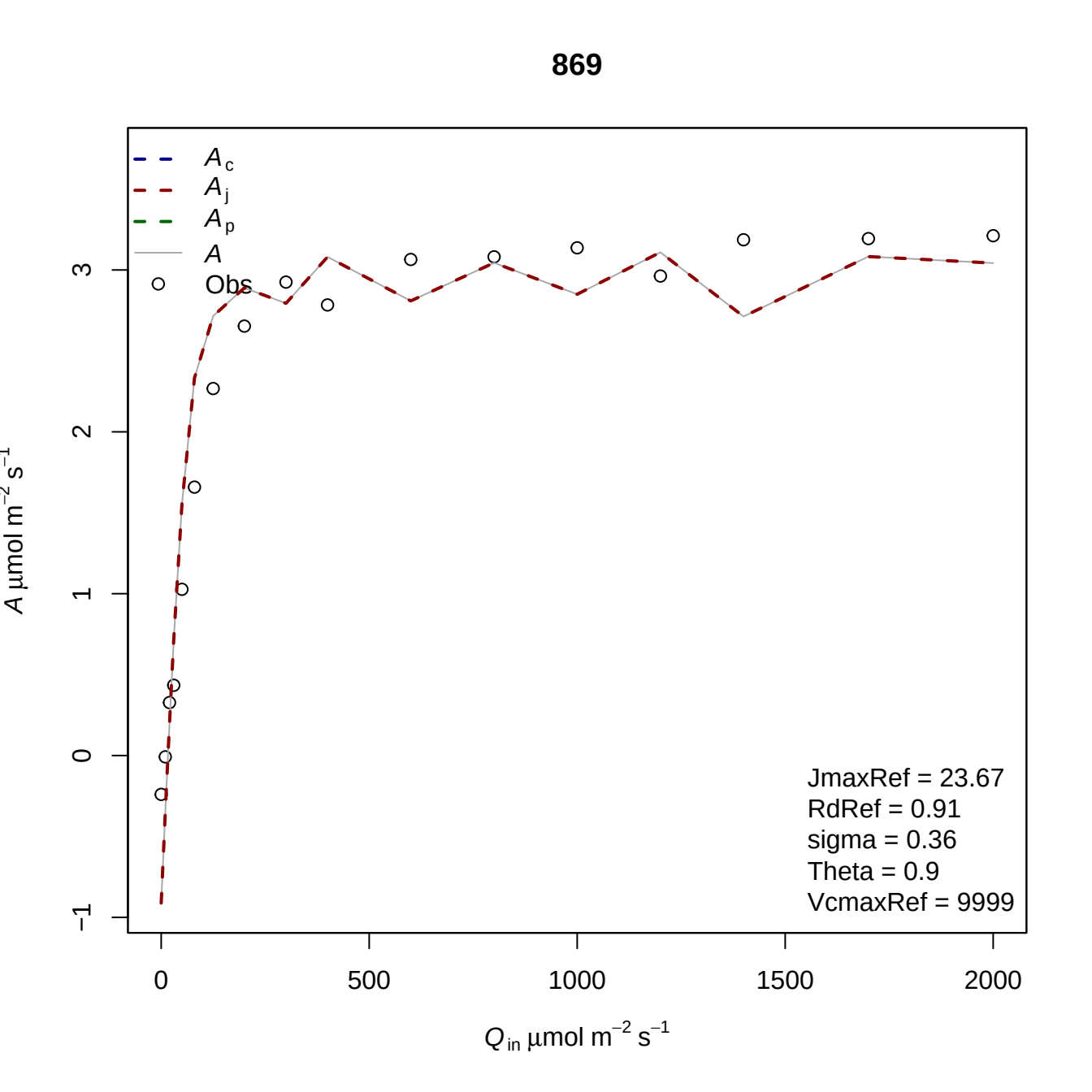

864
A_c
A_j
A_p
A
Obs
JmaxRef = 96.31
RdRef = 1.05
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

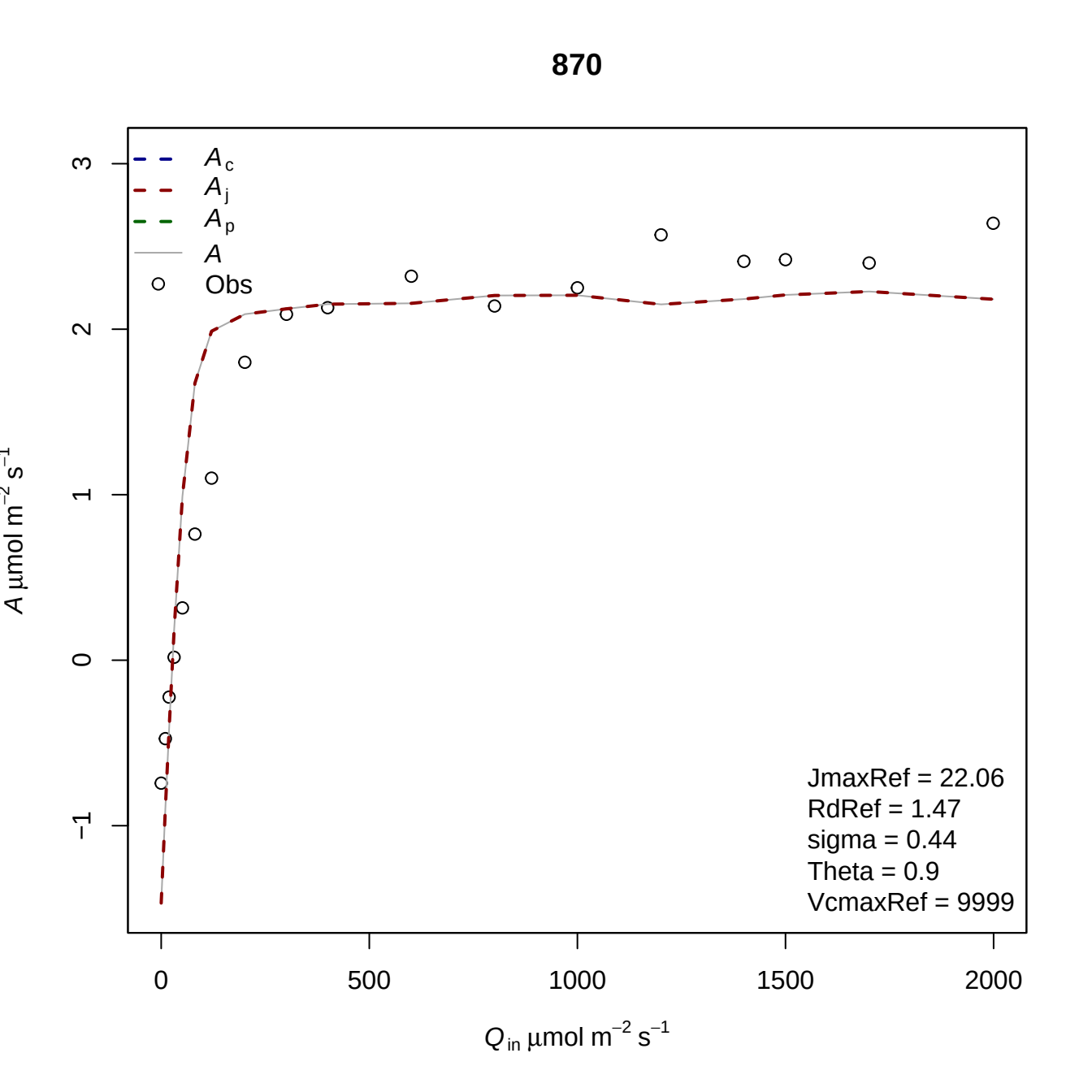

865
A_c
A_j
A_p
A
Obs
JmaxRef = 113.7
RdRef = 2.39
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

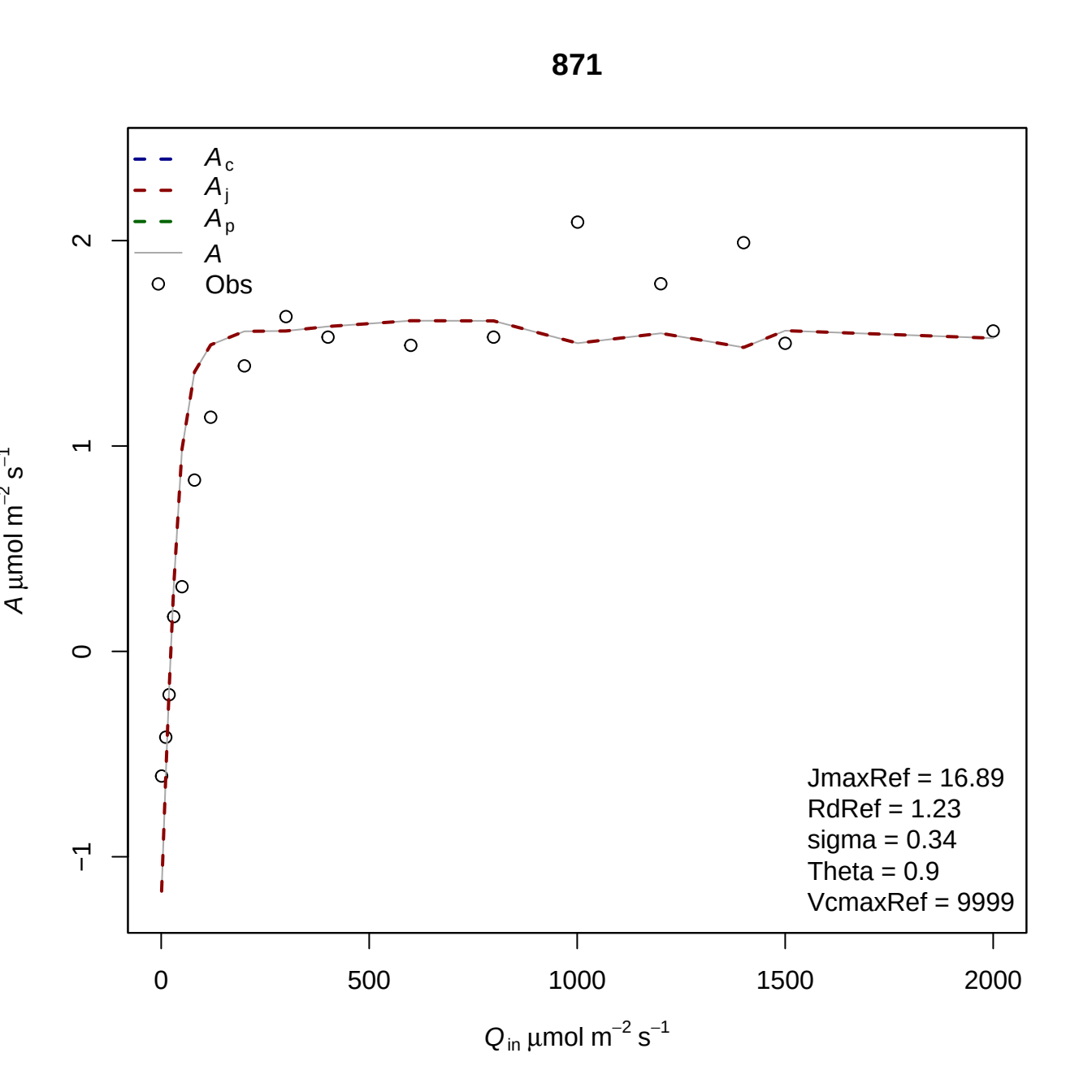

866
A_c
A_j
A_p
A
Obs
JmaxRef = 149.98
RdRef = 5.05
sigma = 2.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

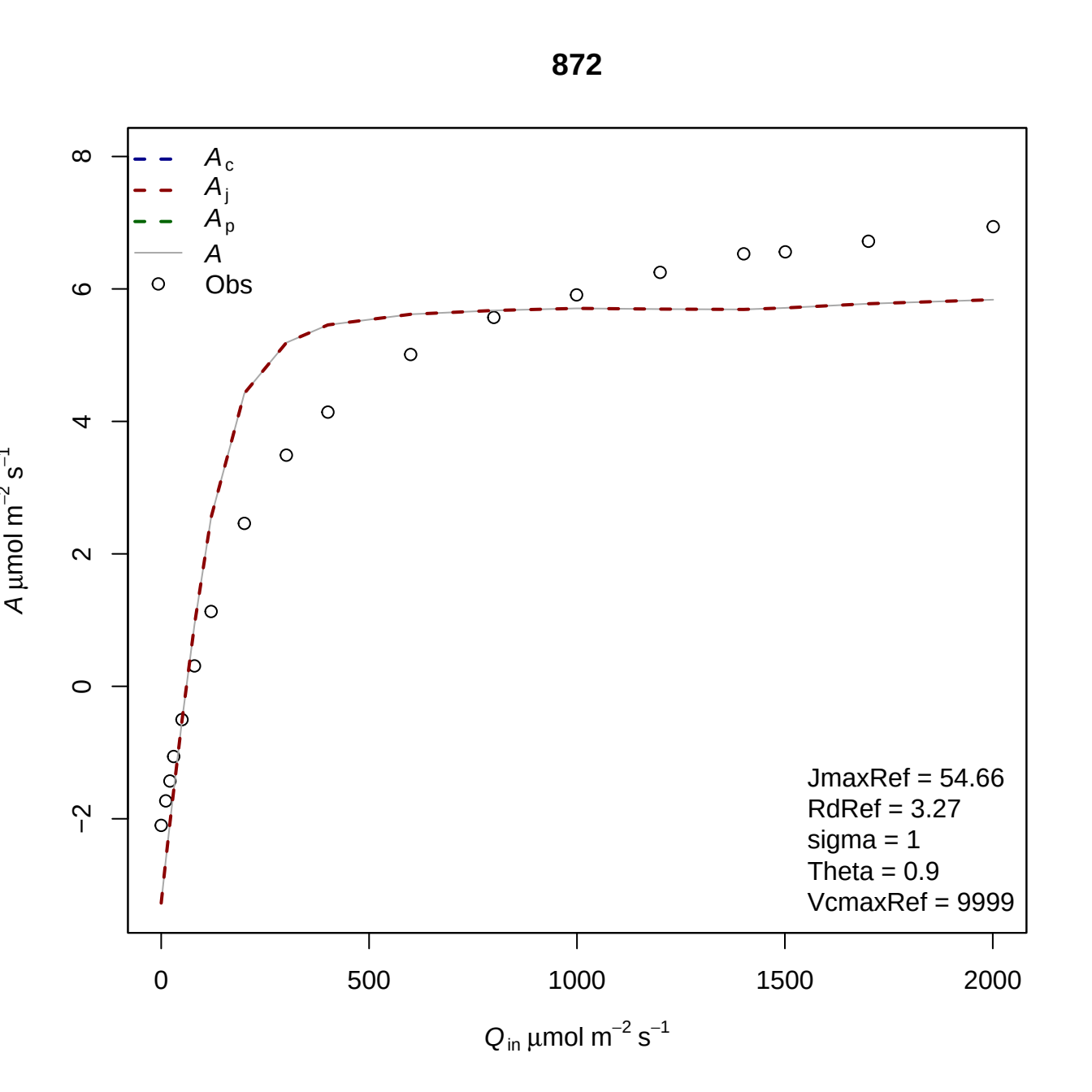

867
A_c
A_j
A_p
A
Obs
JmaxRef = 103.56
RdRef = 4.95
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

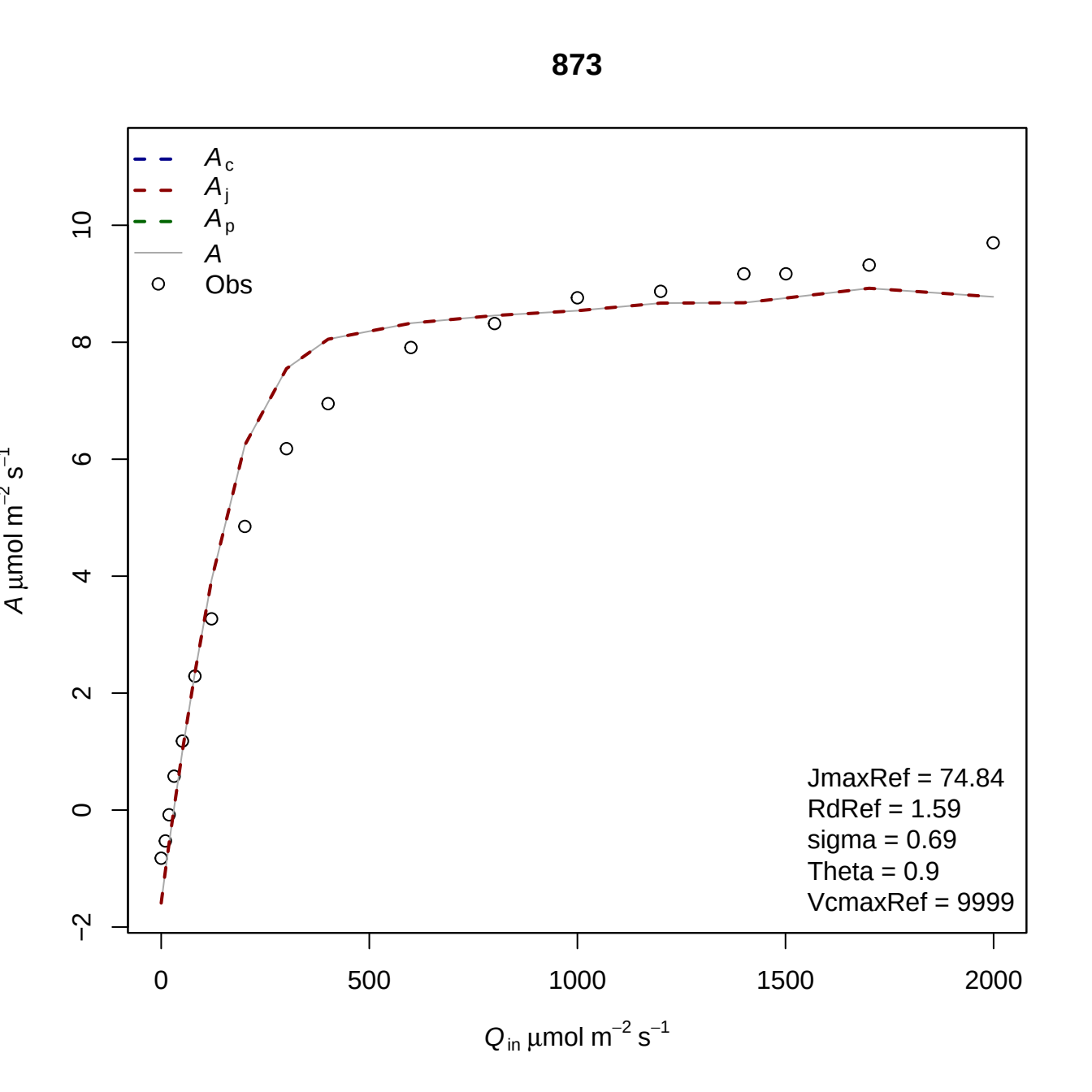

868
A_c
A_j
A_p
A
Obs
JmaxRef = 121.51
RdRef = 2.94
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

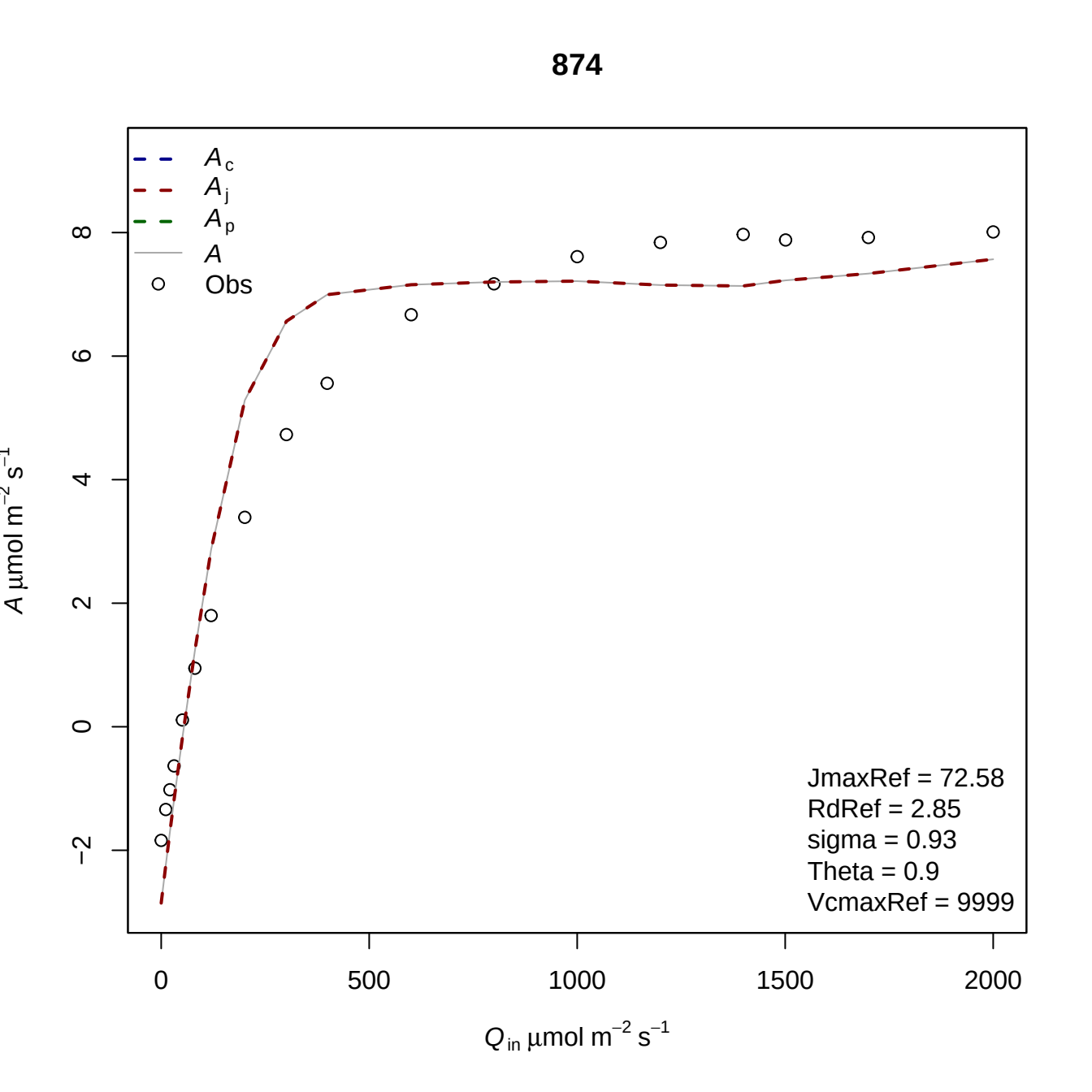

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.67
RdRef = 0.91
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

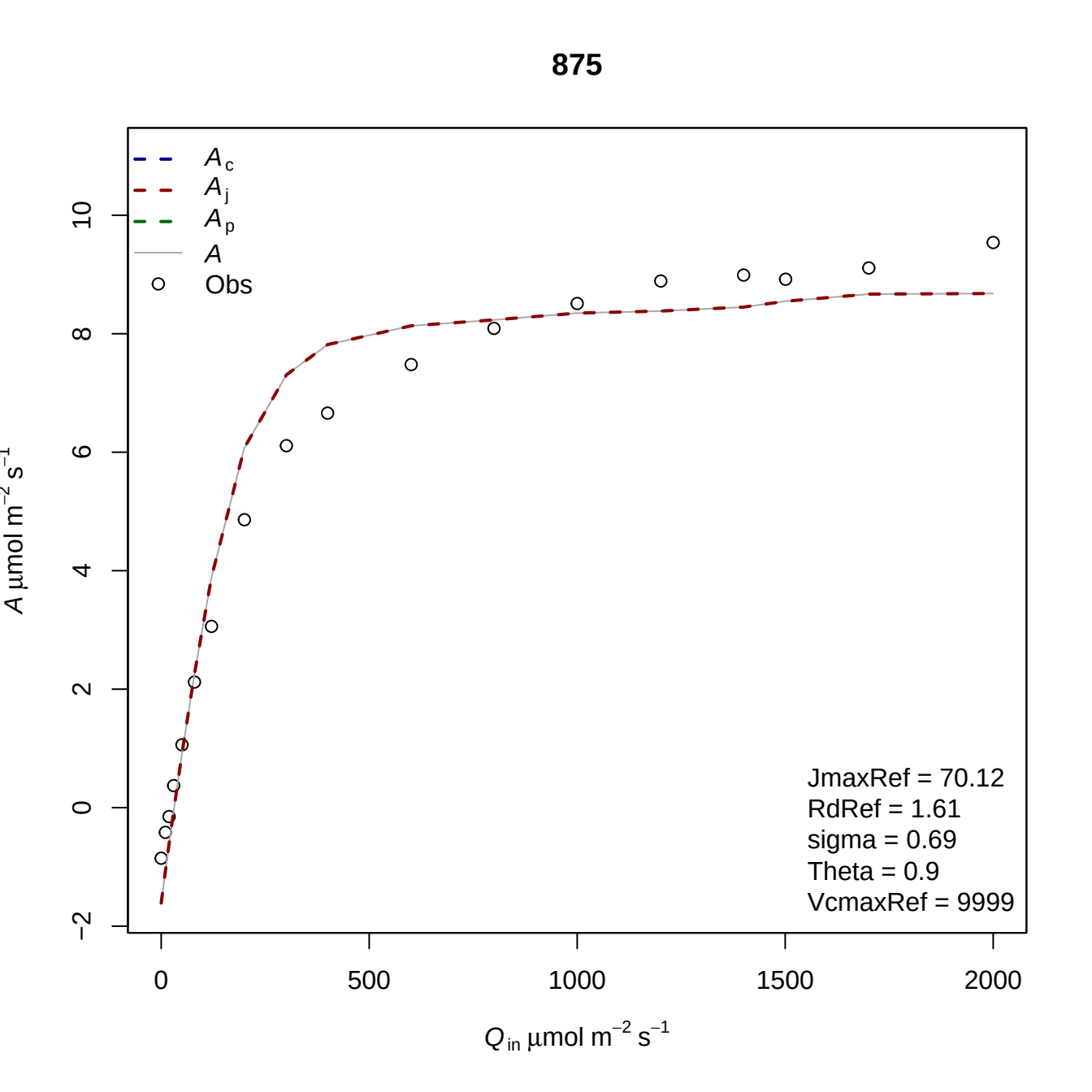

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.06
RdRef = 1.47
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

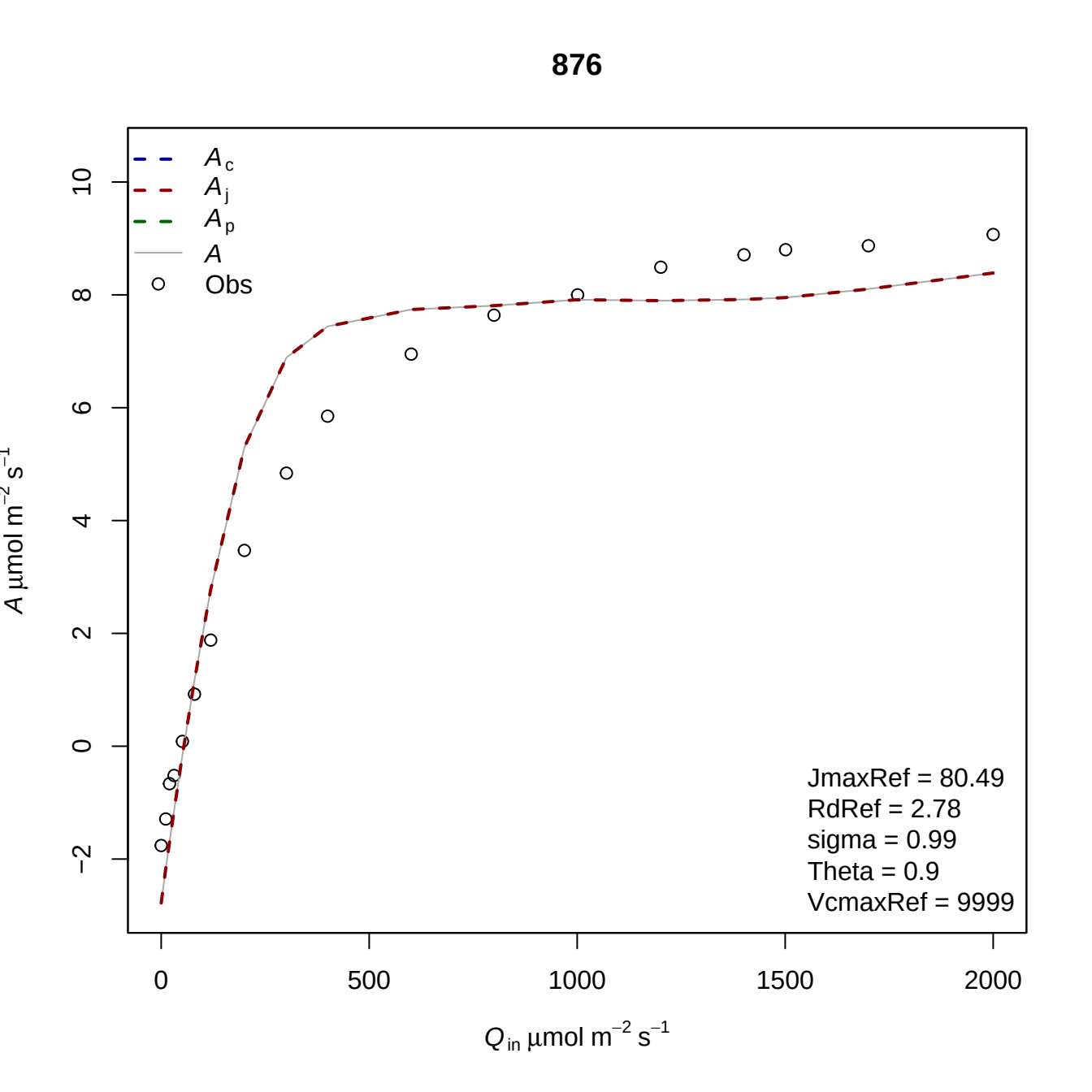

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.89
RdRef = 1.23
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

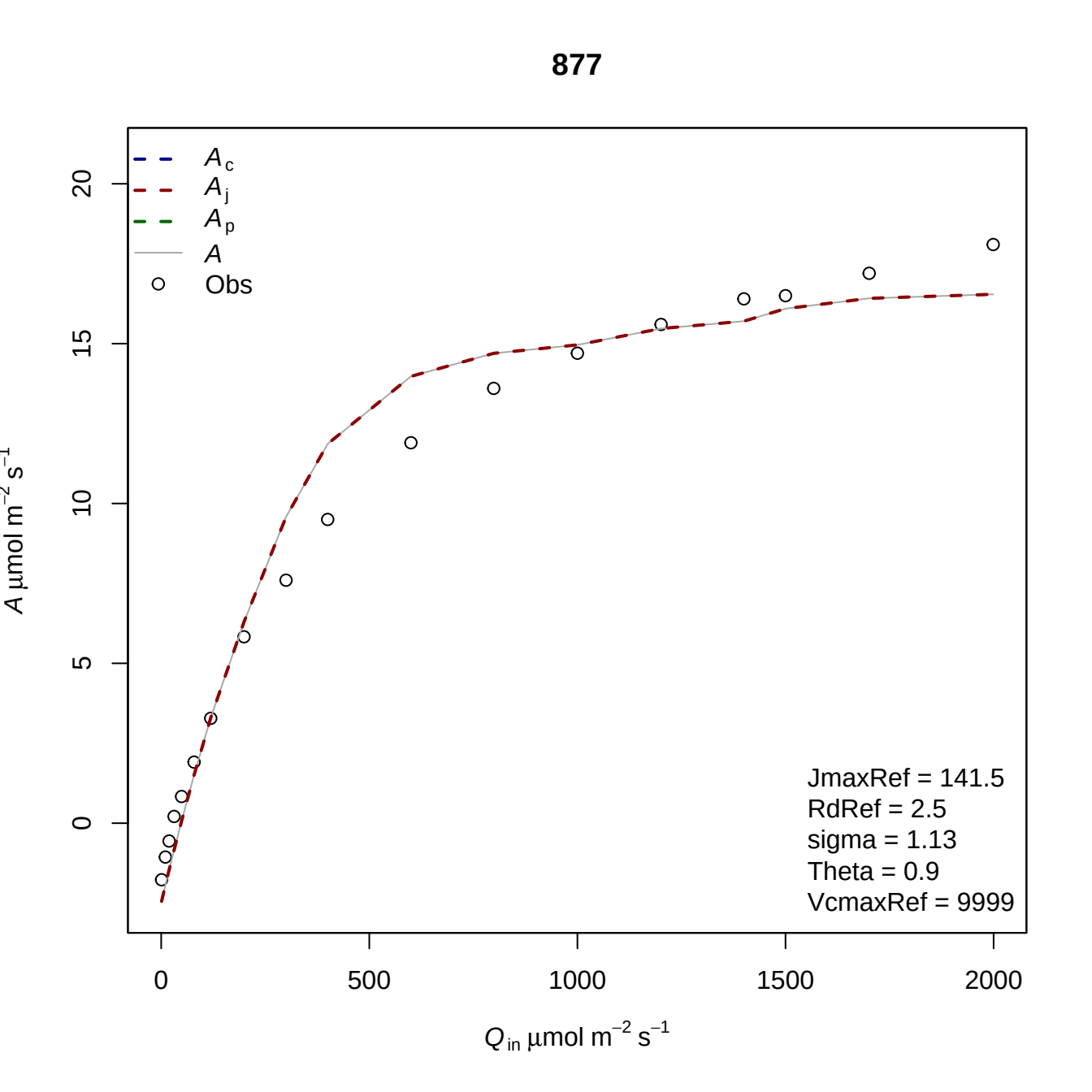

872
A_c
A_j
A_p
A
Obs
JmaxRef = 54.66
RdRef = 3.27
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

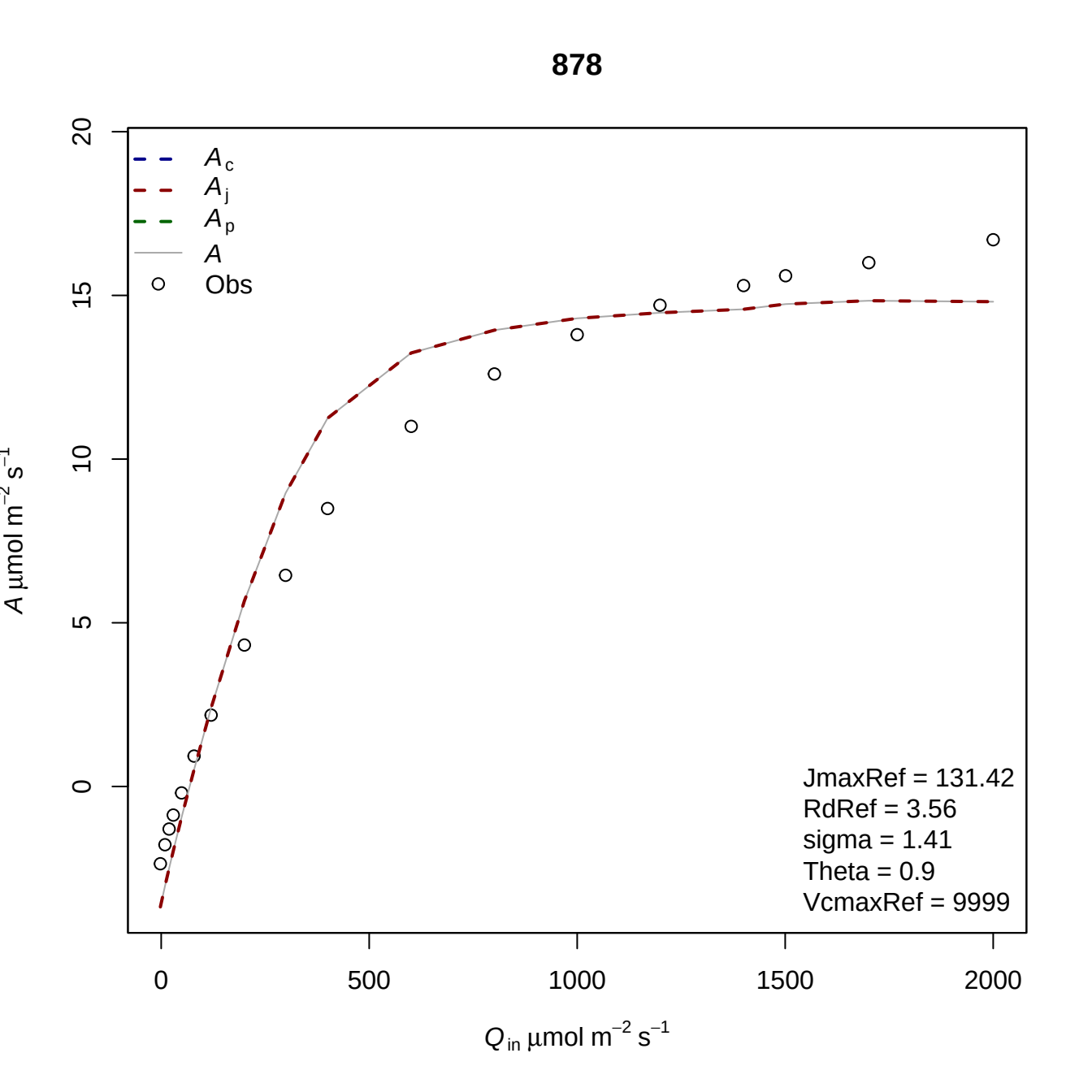

873
A_c
A_j
A_p
A
Obs
JmaxRef = 74.84
RdRef = 1.59
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

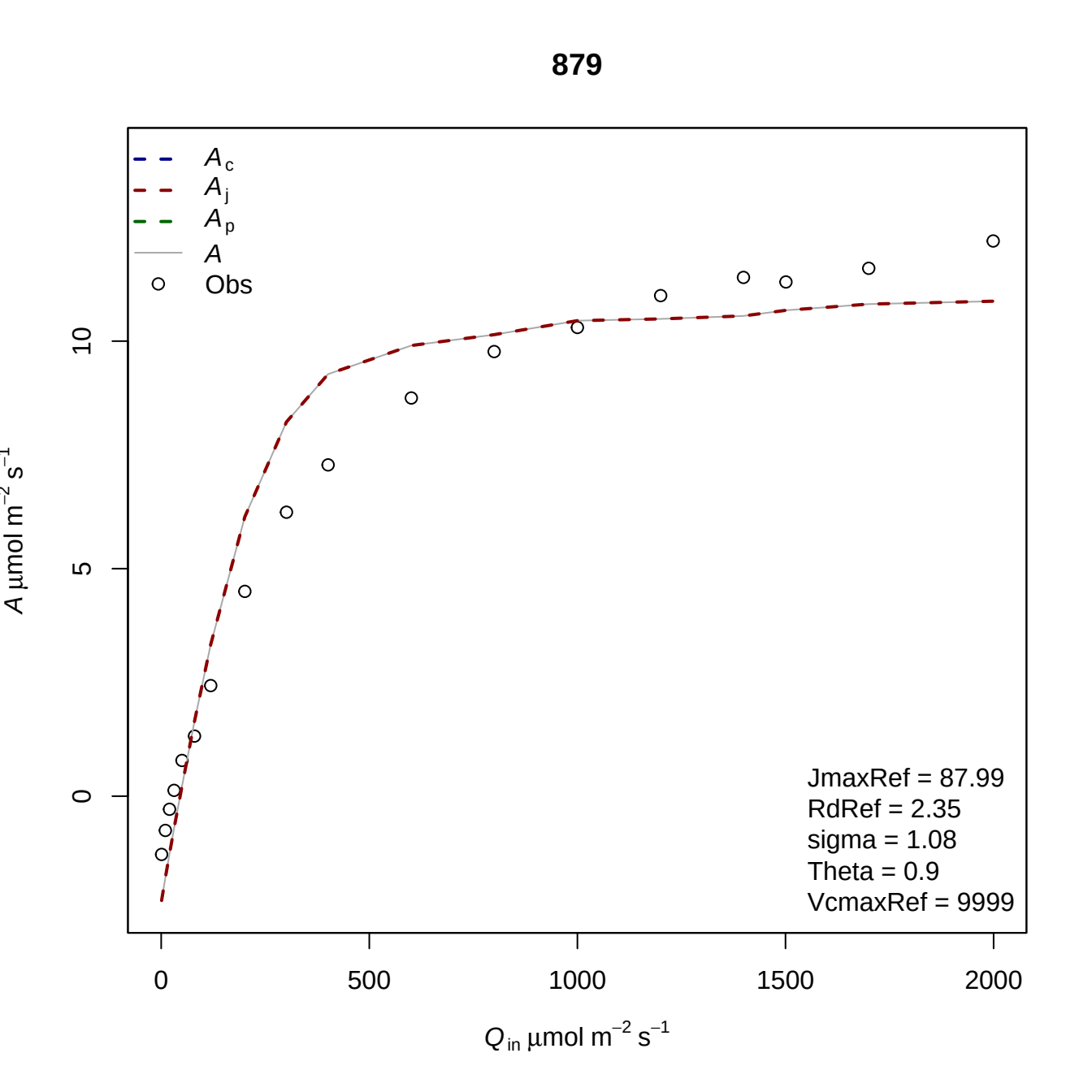

874
A_c
A_j
A_p
A
Obs
JmaxRef = 72.58
RdRef = 2.85
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

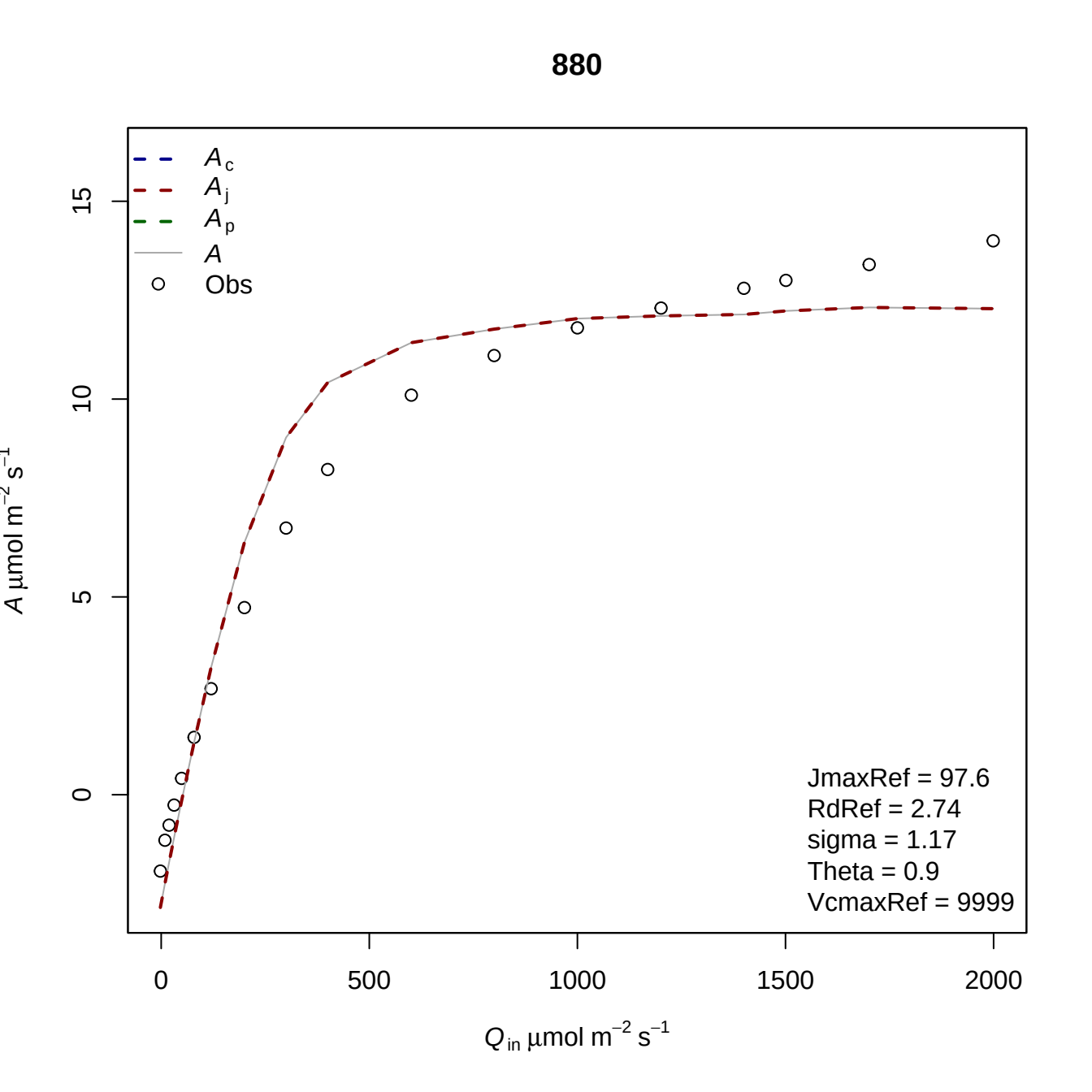

875
A_c
A_j
A_p
A
Obs
JmaxRef = 70.12
RdRef = 1.61
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

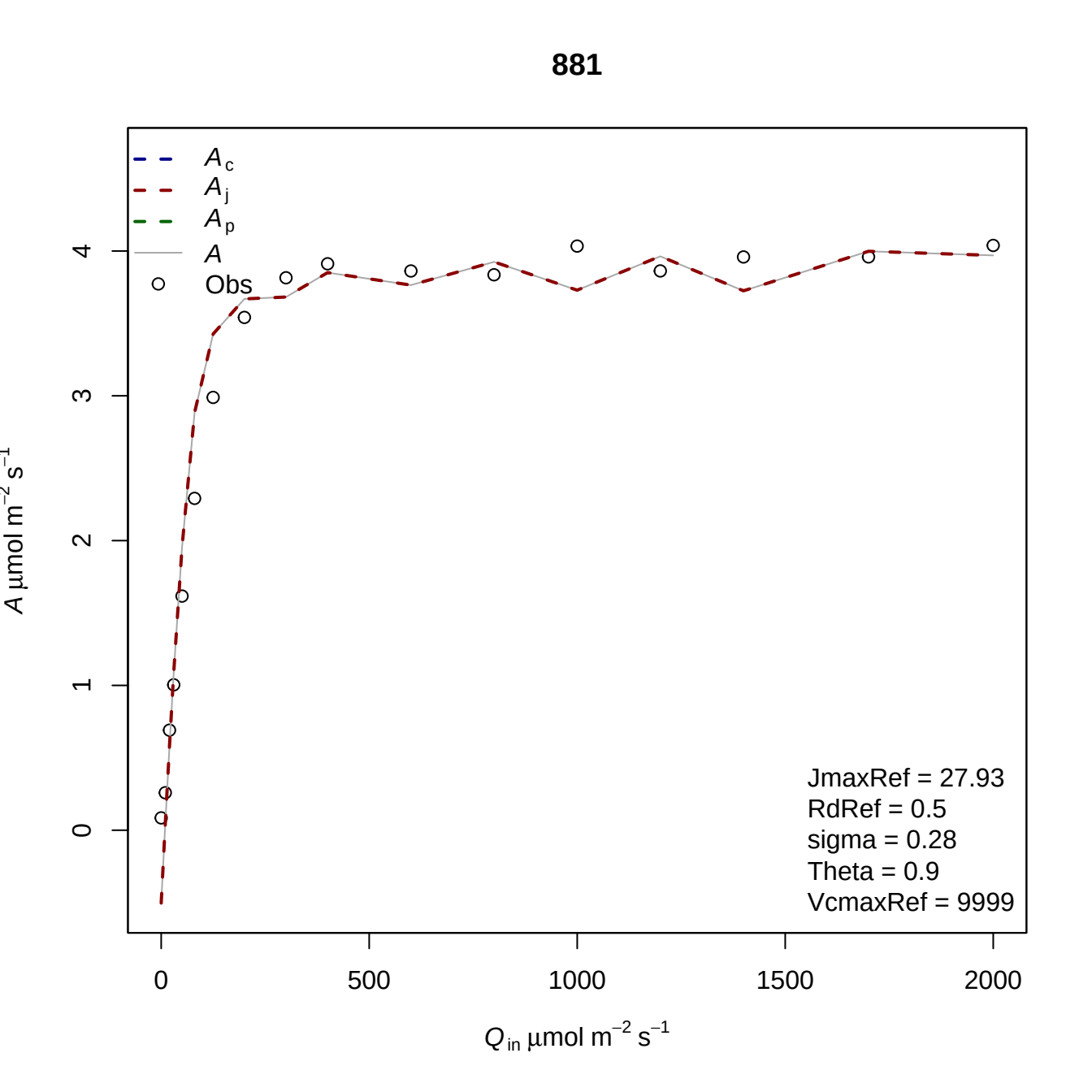

876
A_c
A_j
A_p
A
Obs
JmaxRef = 80.49
RdRef = 2.78
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

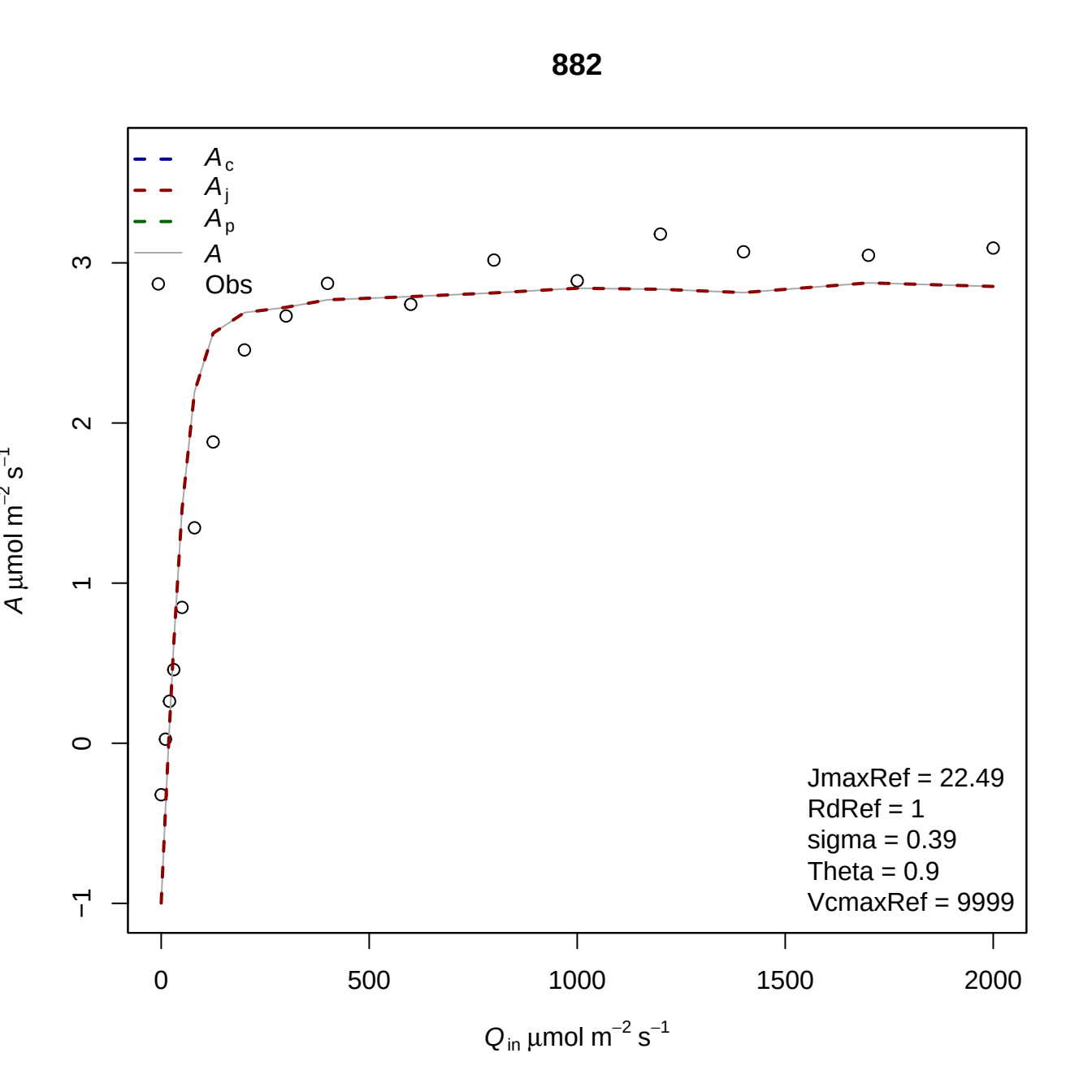

877
A_c
A_j
A_p
A
Obs
JmaxRef = 141.5
RdRef = 2.5
sigma = 1.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

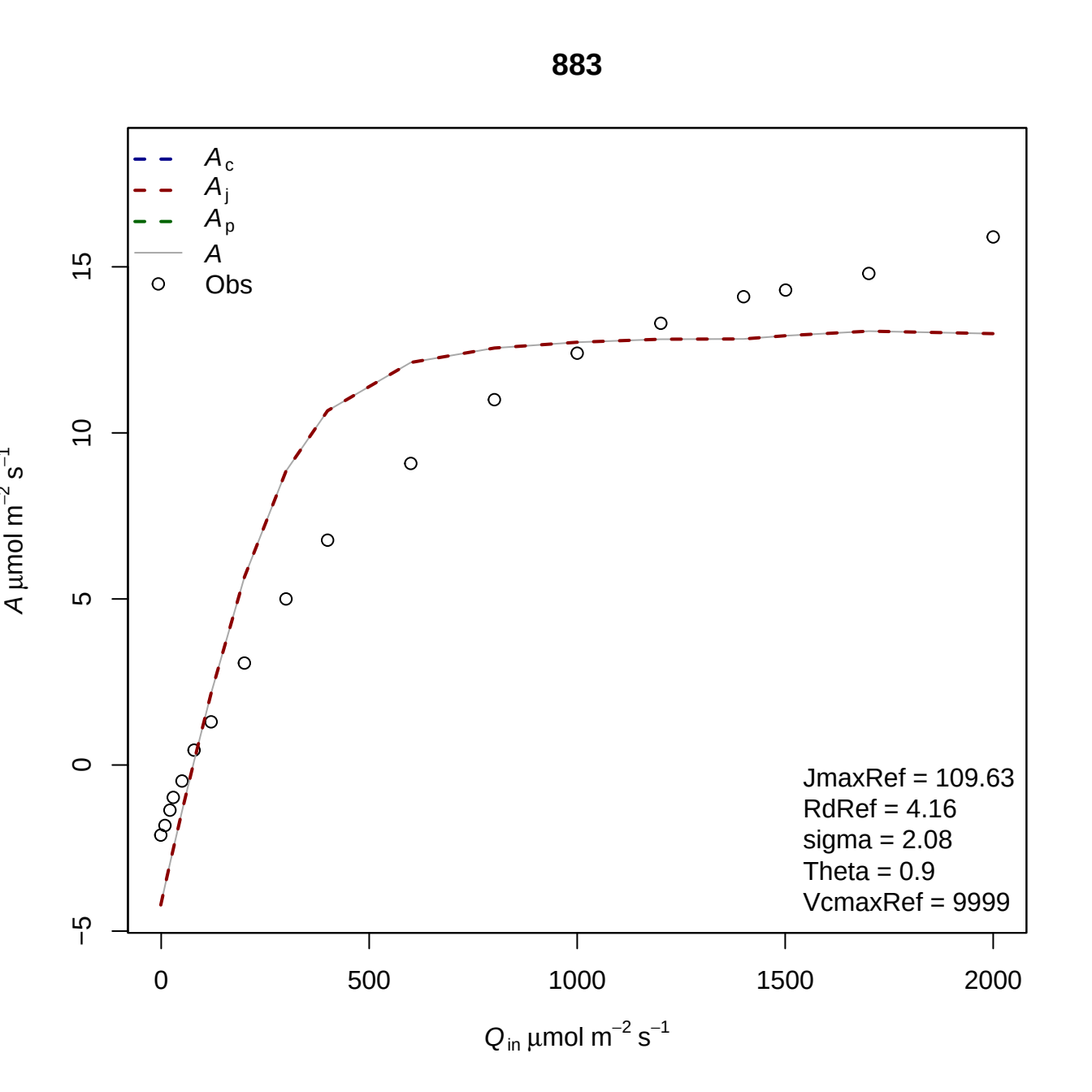

878
A_c
A_j
A_p
A
Obs
JmaxRef = 131.42
RdRef = 3.56
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

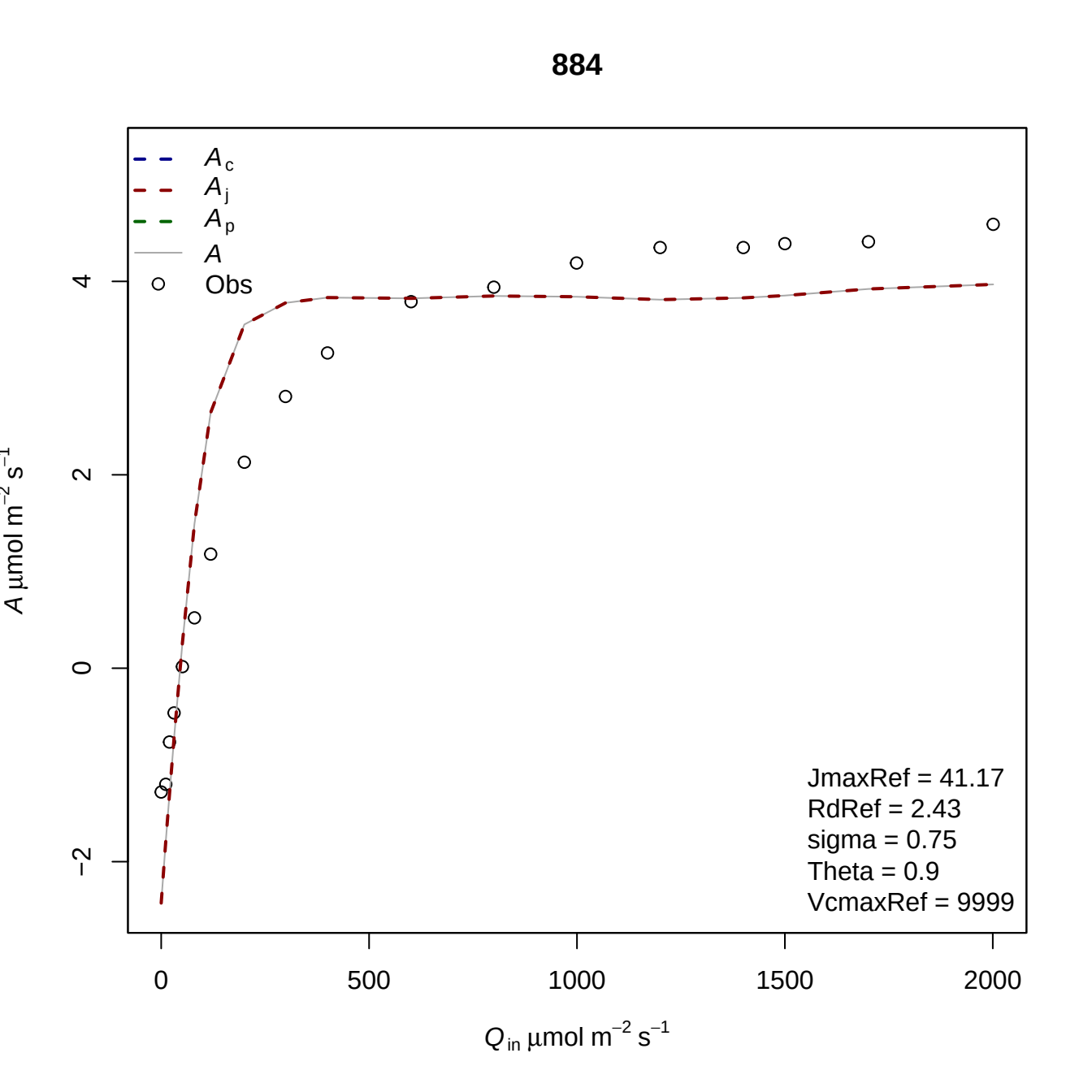

879
A_c
A_j
A_p
A
Obs
JmaxRef = 87.99
RdRef = 2.35
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

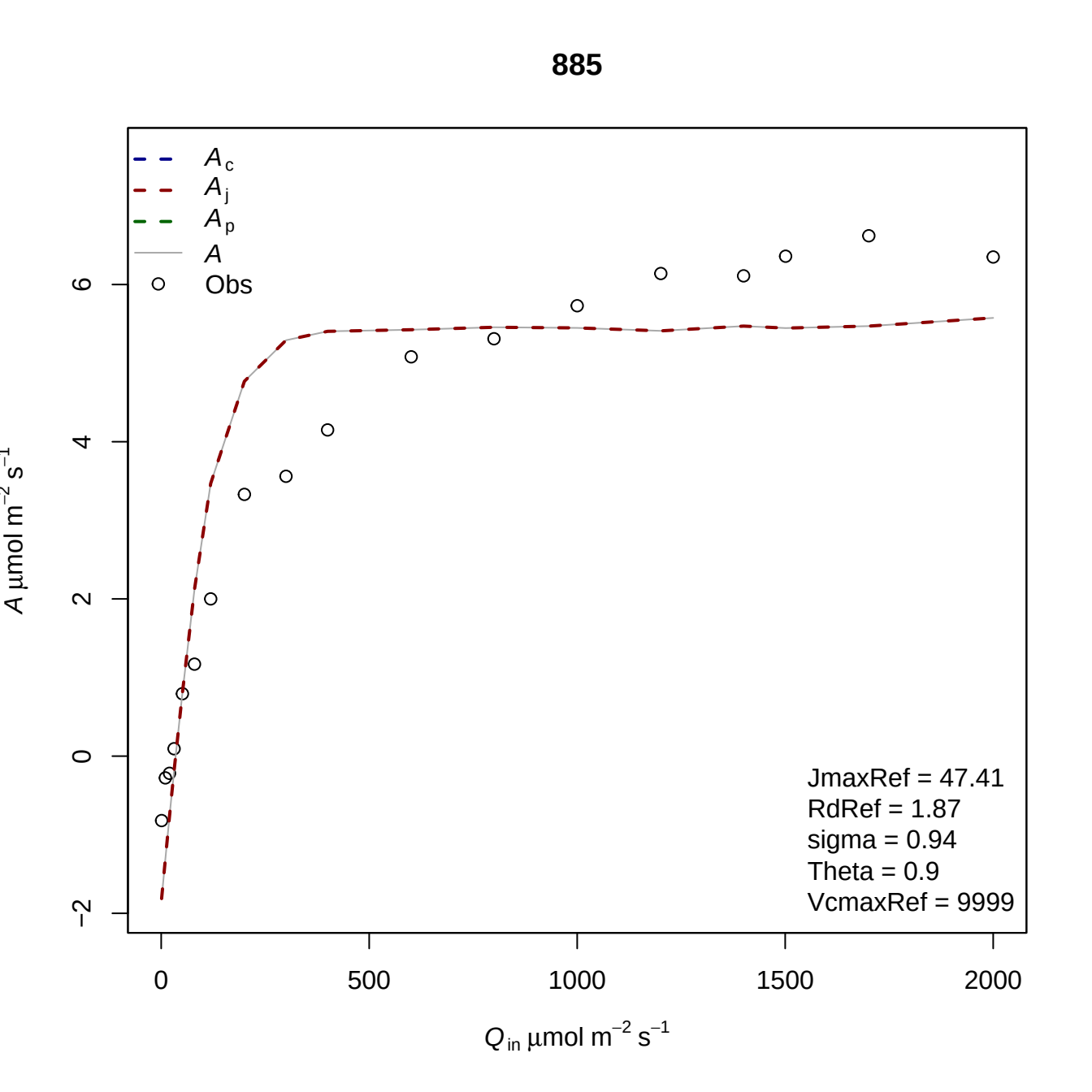

880
A_c
A_j
A_p
A
Obs
JmaxRef = 97.6
RdRef = 2.74
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

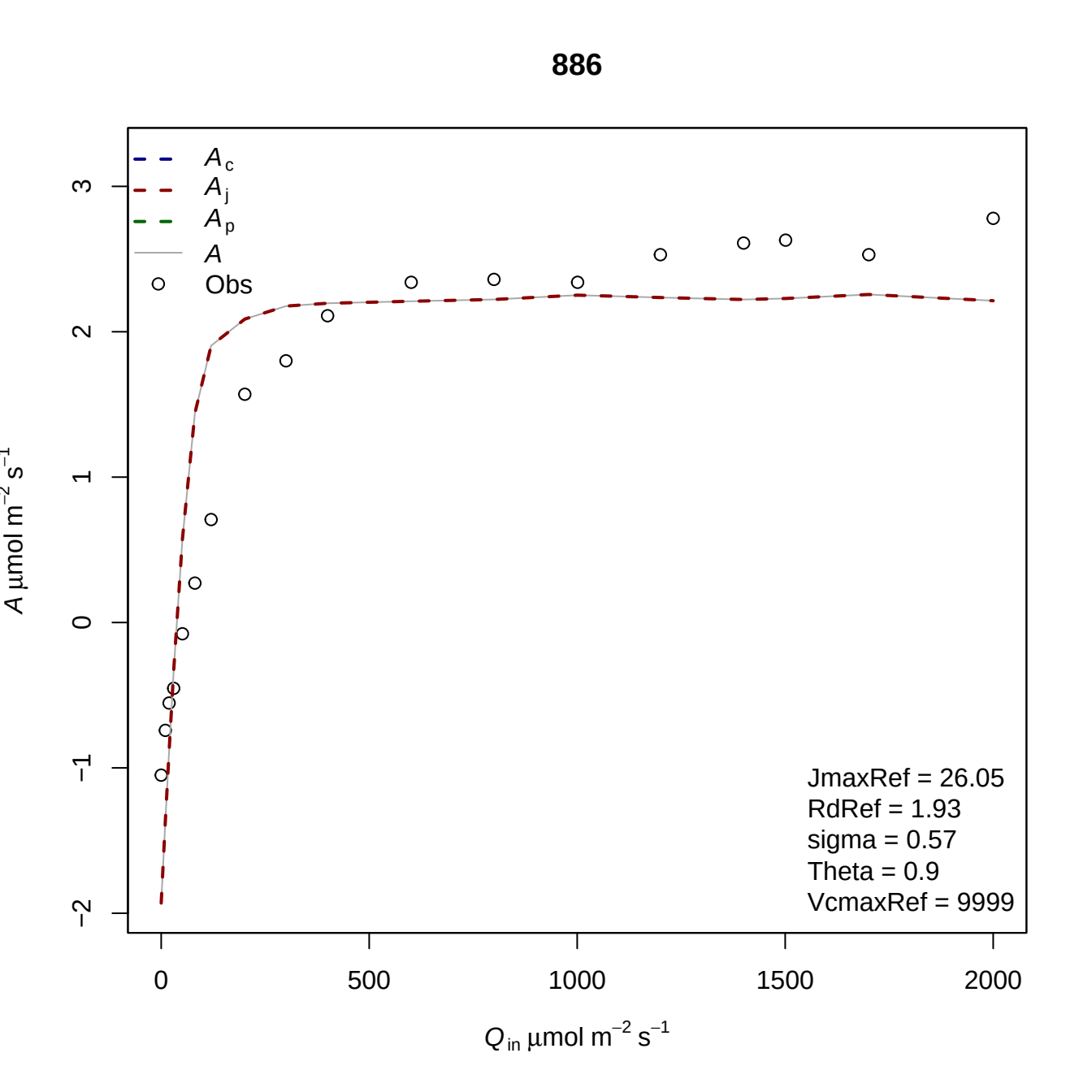

881
A_c
A_j
A_p
A
Obs
JmaxRef = 27.93
RdRef = 0.5
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

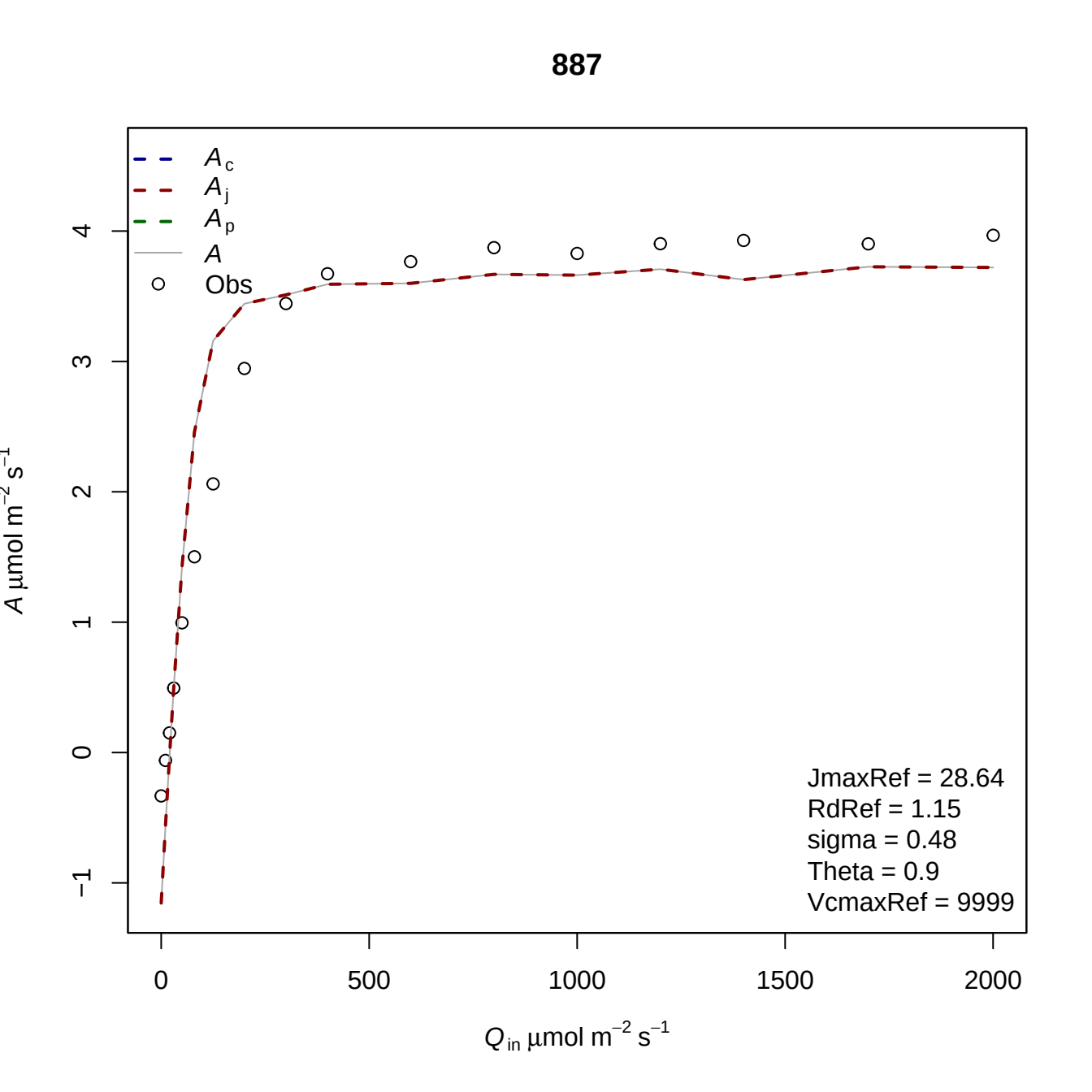

882
A_c
A_j
A_p
A
Obs
JmaxRef = 22.49
RdRef = 1
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

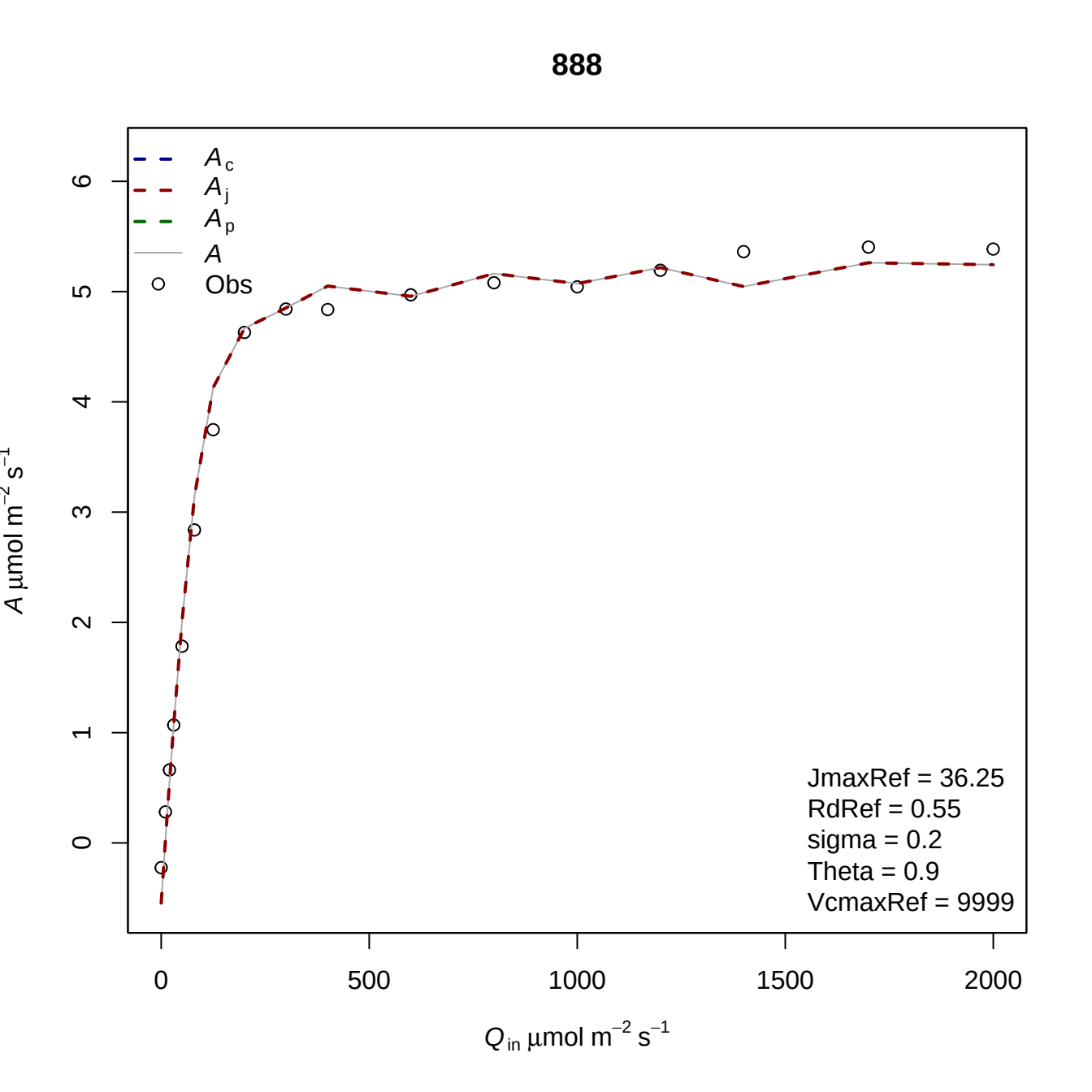

883
A_c
A_j
A_p
A
Obs
JmaxRef = 109.63
RdRef = 4.16
sigma = 2.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

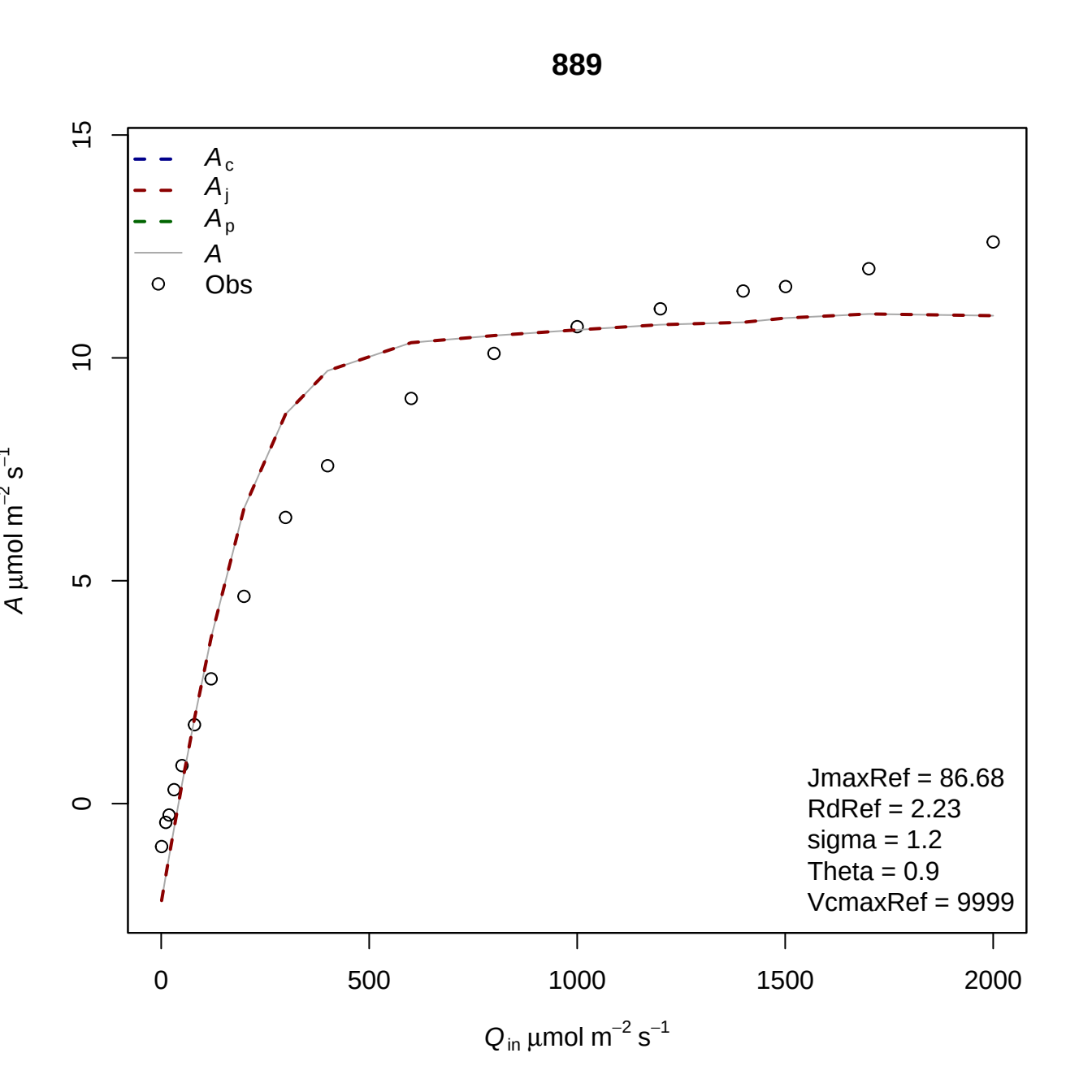

884
A_c
A_j
A_p
A
Obs
JmaxRef = 41.17
RdRef = 2.43
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

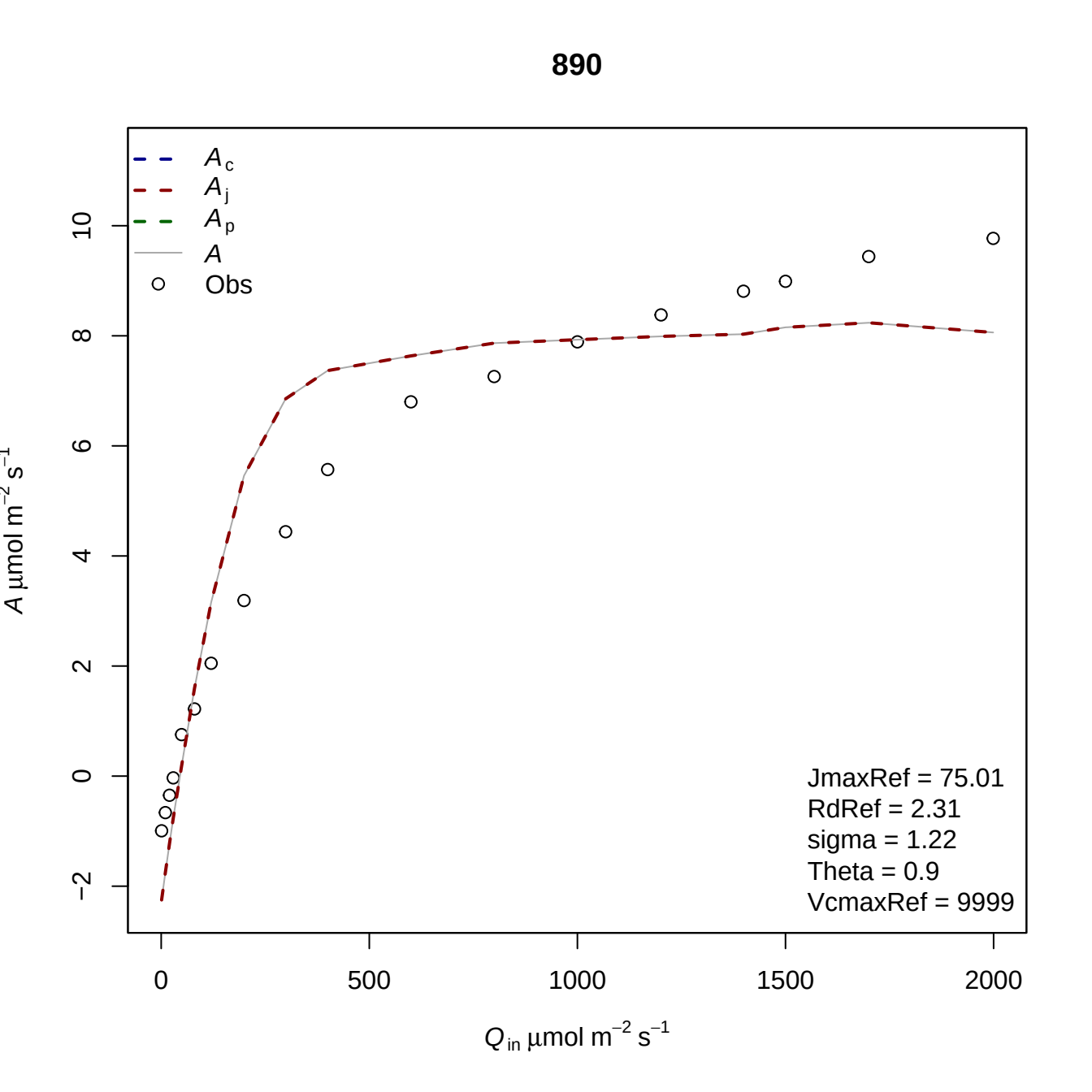

885
A_c
A_j
A_p
A
Obs
JmaxRef = 47.41
RdRef = 1.87
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

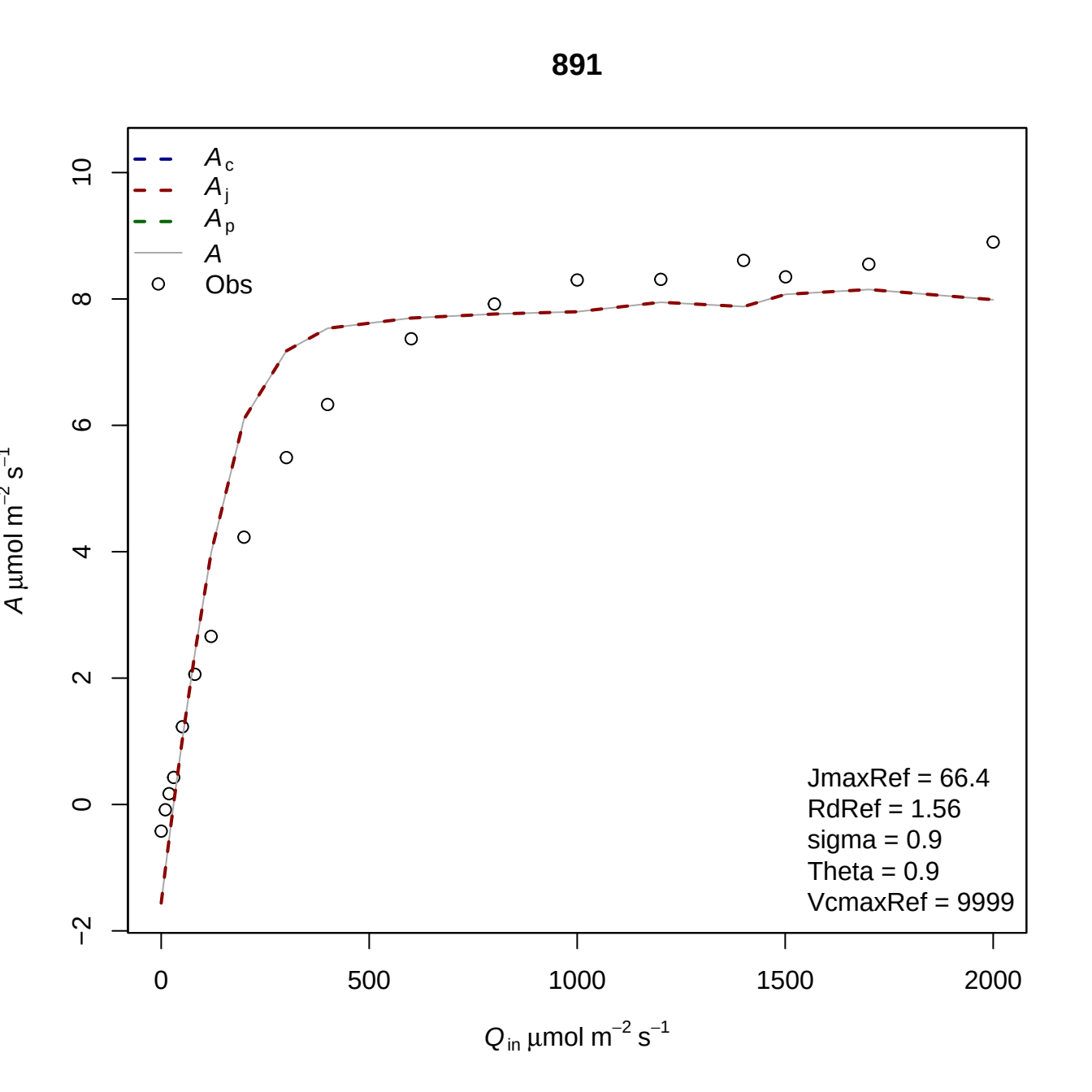

886
A_c
A_j
A_p
A
Obs
JmaxRef = 26.05
RdRef = 1.93
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500
2000

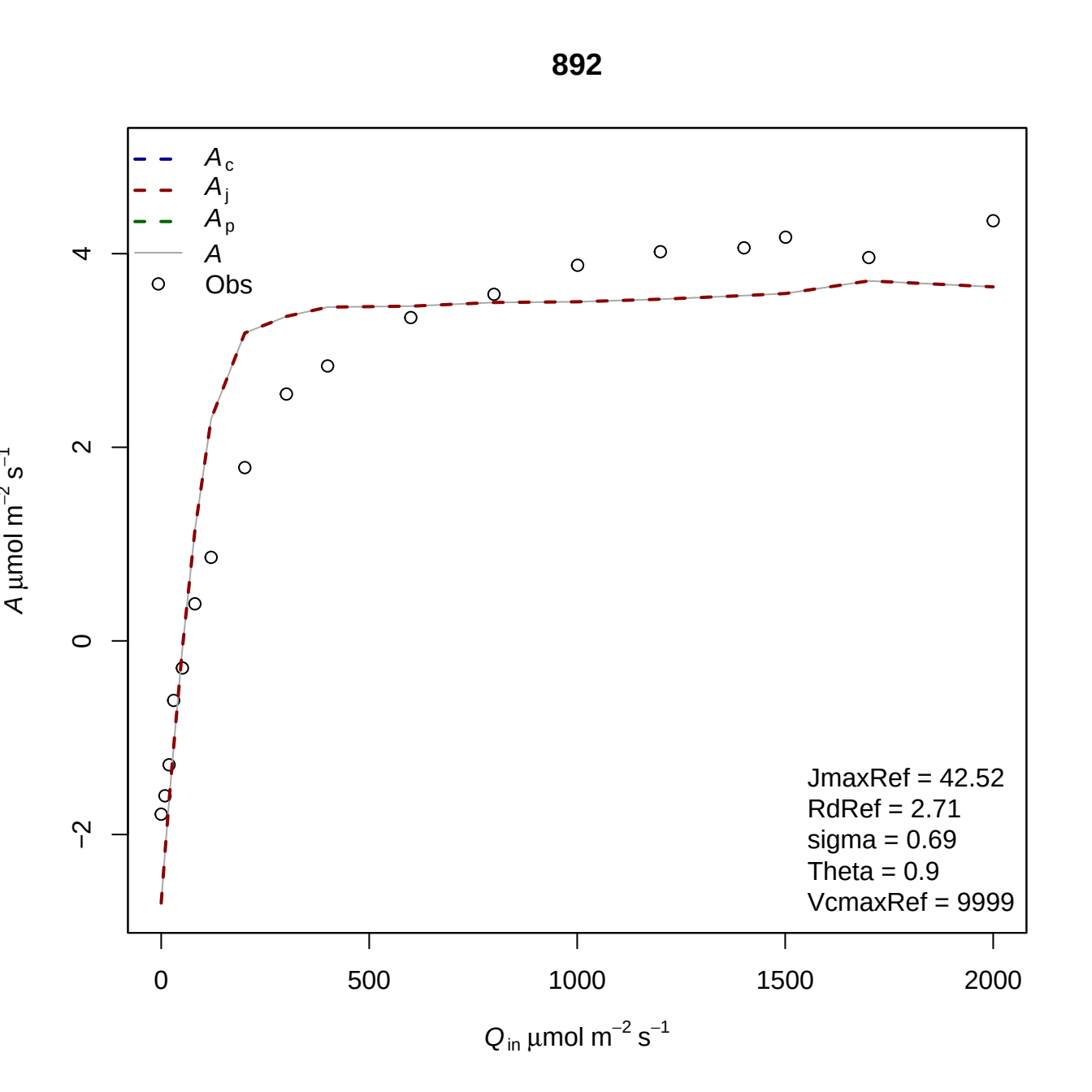

887
A_c
A_j
A_p
A
Obs
JmaxRef = 28.64
RdRef = 1.15
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

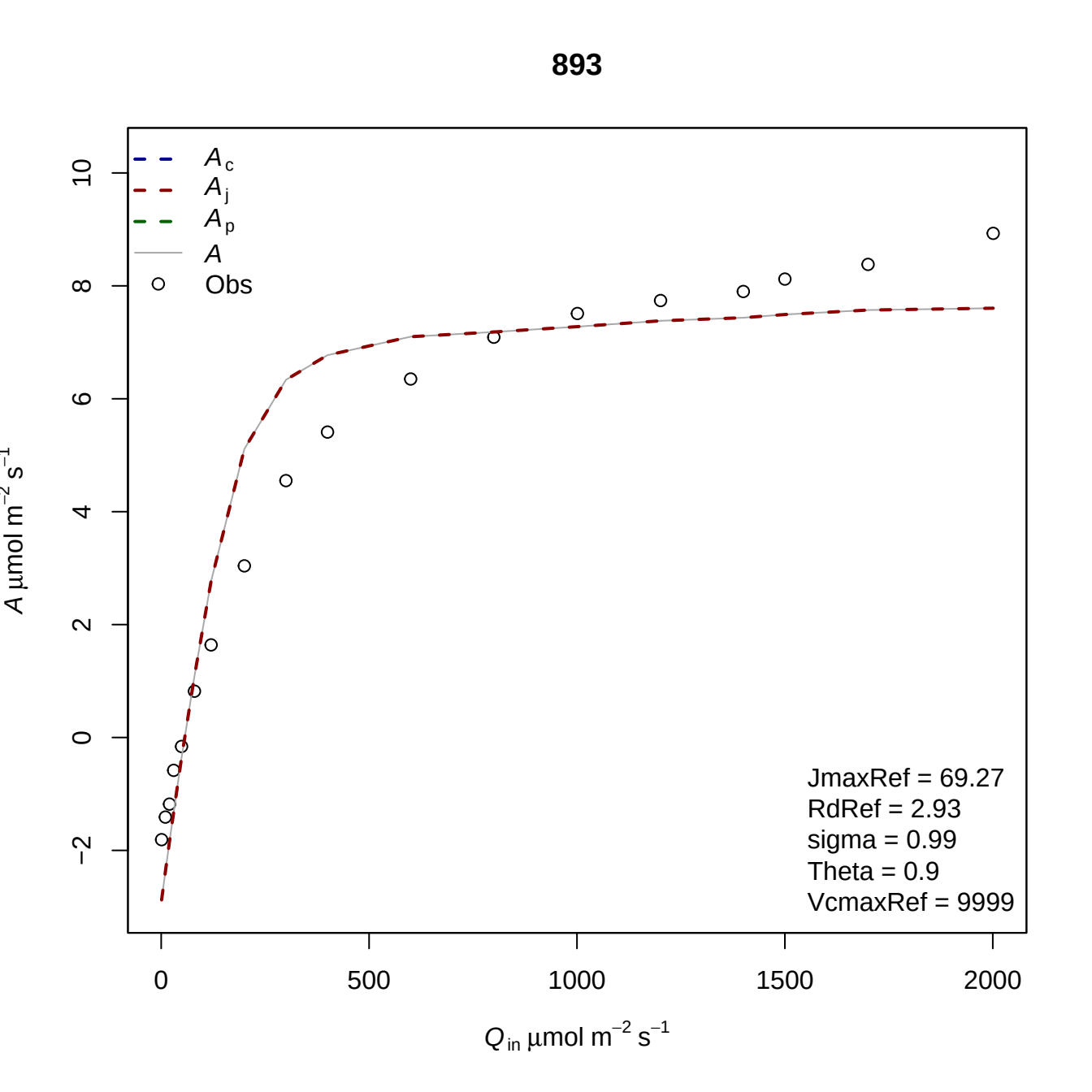

888
A_c
A_j
A_p
A
Obs
JmaxRef = 36.25
RdRef = 0.55
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

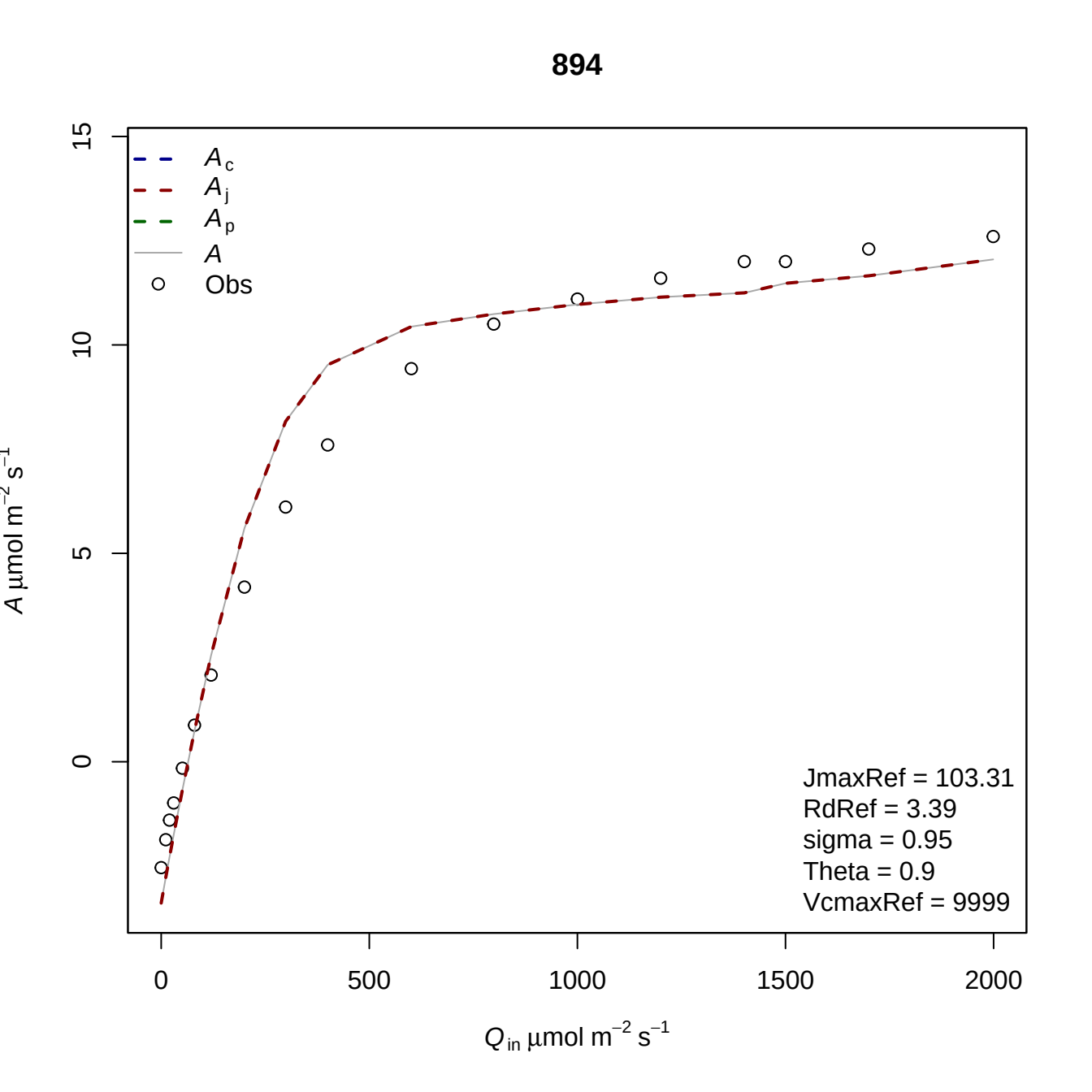

889
A_c
A_j
A_p
A
Obs
JmaxRef = 86.68
RdRef = 2.23
sigma = 1.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

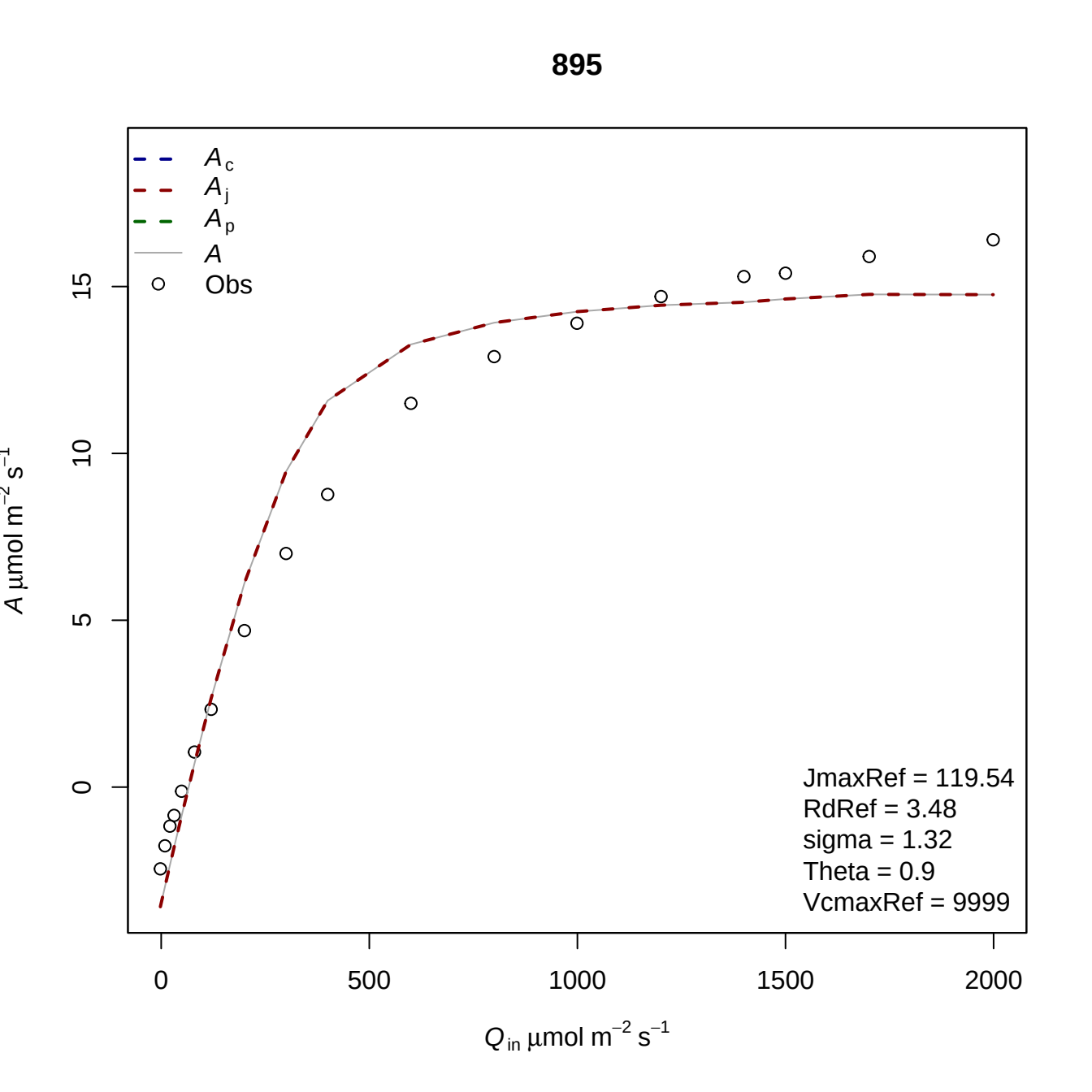

890
A_c
A_j
A_p
A
Obs
JmaxRef = 75.01
RdRef = 2.31
sigma = 1.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

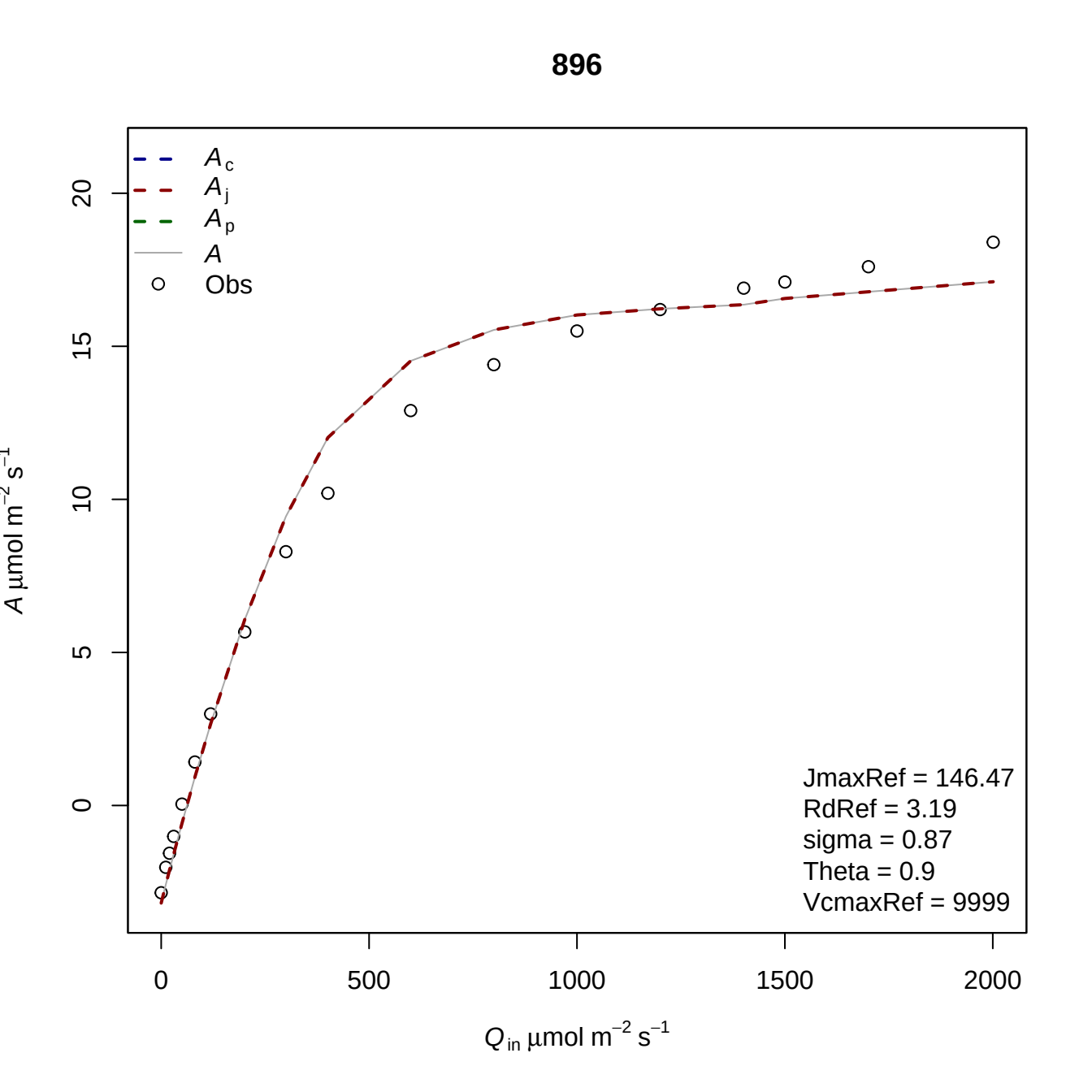

891
A_c
A_j
A_p
A
Obs
JmaxRef = 66.4
RdRef = 1.56
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

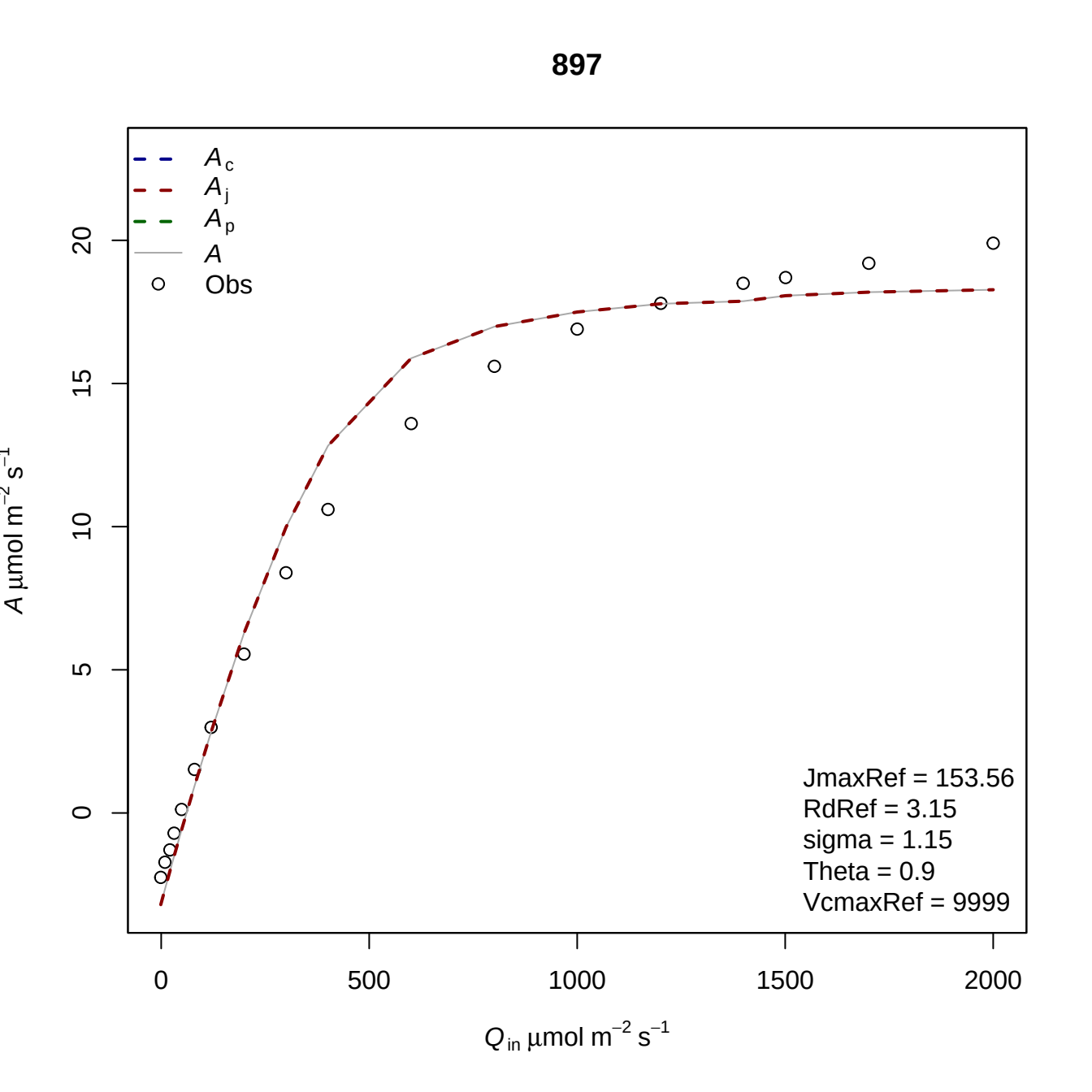

892
A_c
A_j
A_p
A
Obs
JmaxRef = 42.52
RdRef = 2.71
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

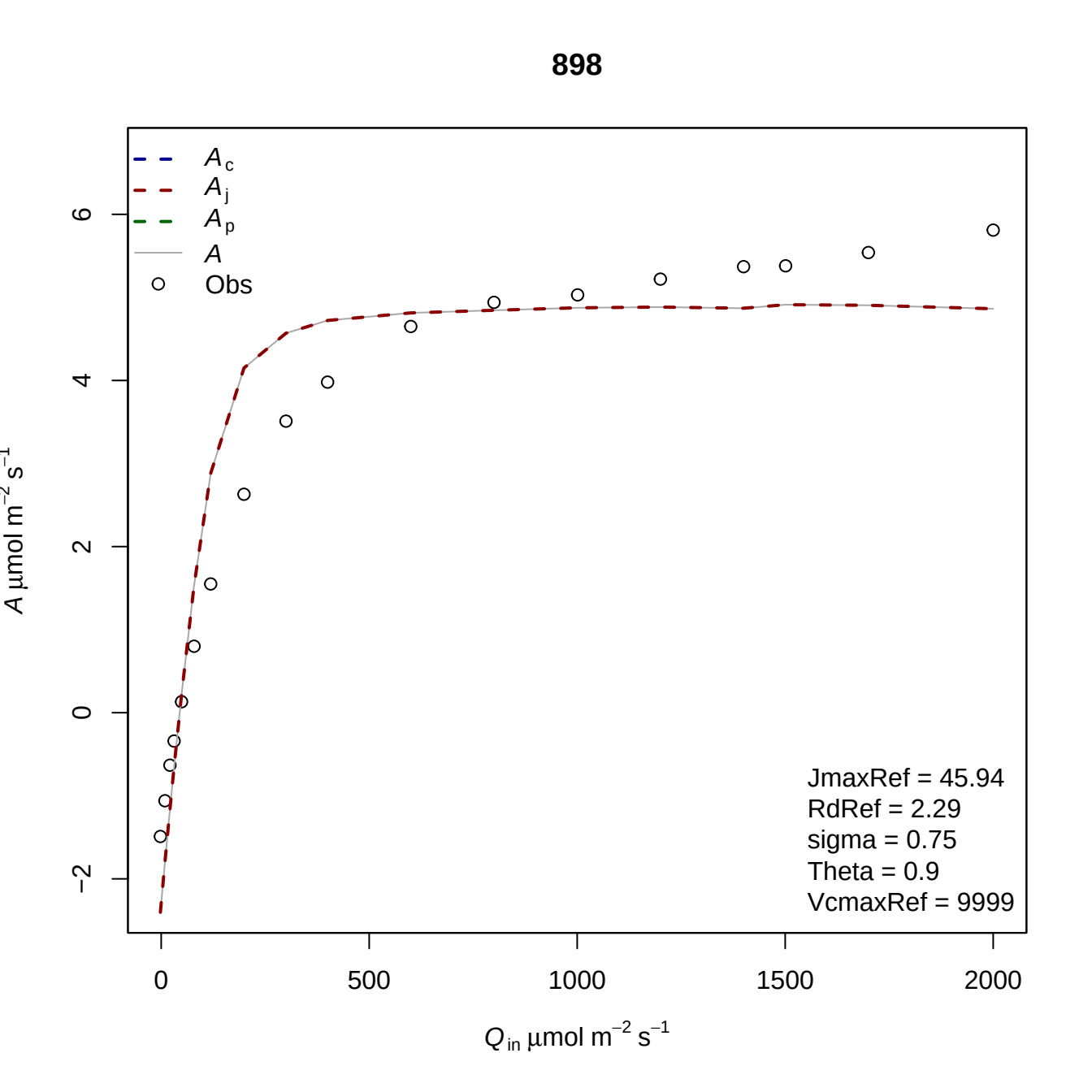

893
A_c
A_j
A_p
A
Obs
JmaxRef = 69.27
RdRef = 2.93
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

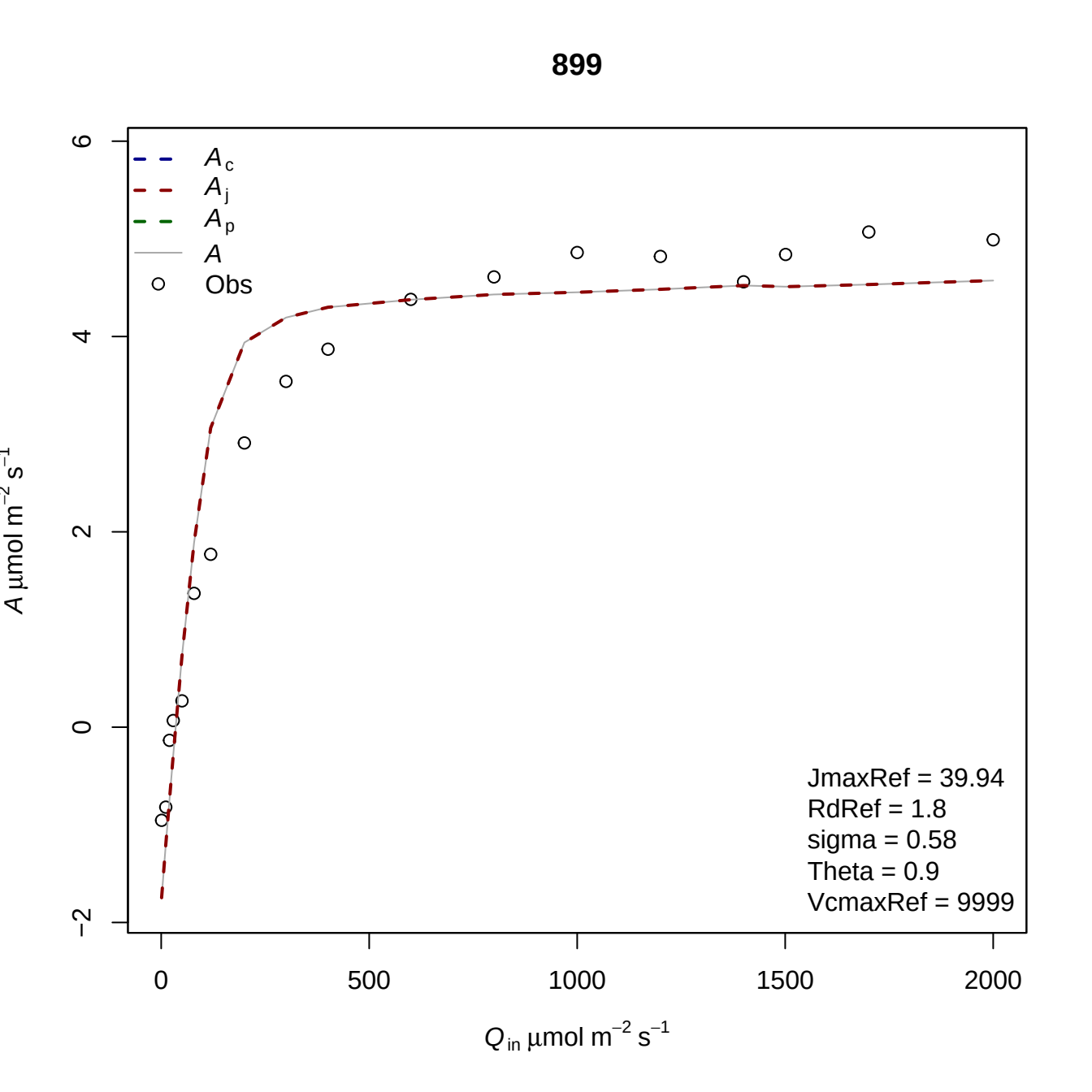

894
A_c
A_j
A_p
A
Obs
JmaxRef = 103.31
RdRef = 3.39
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

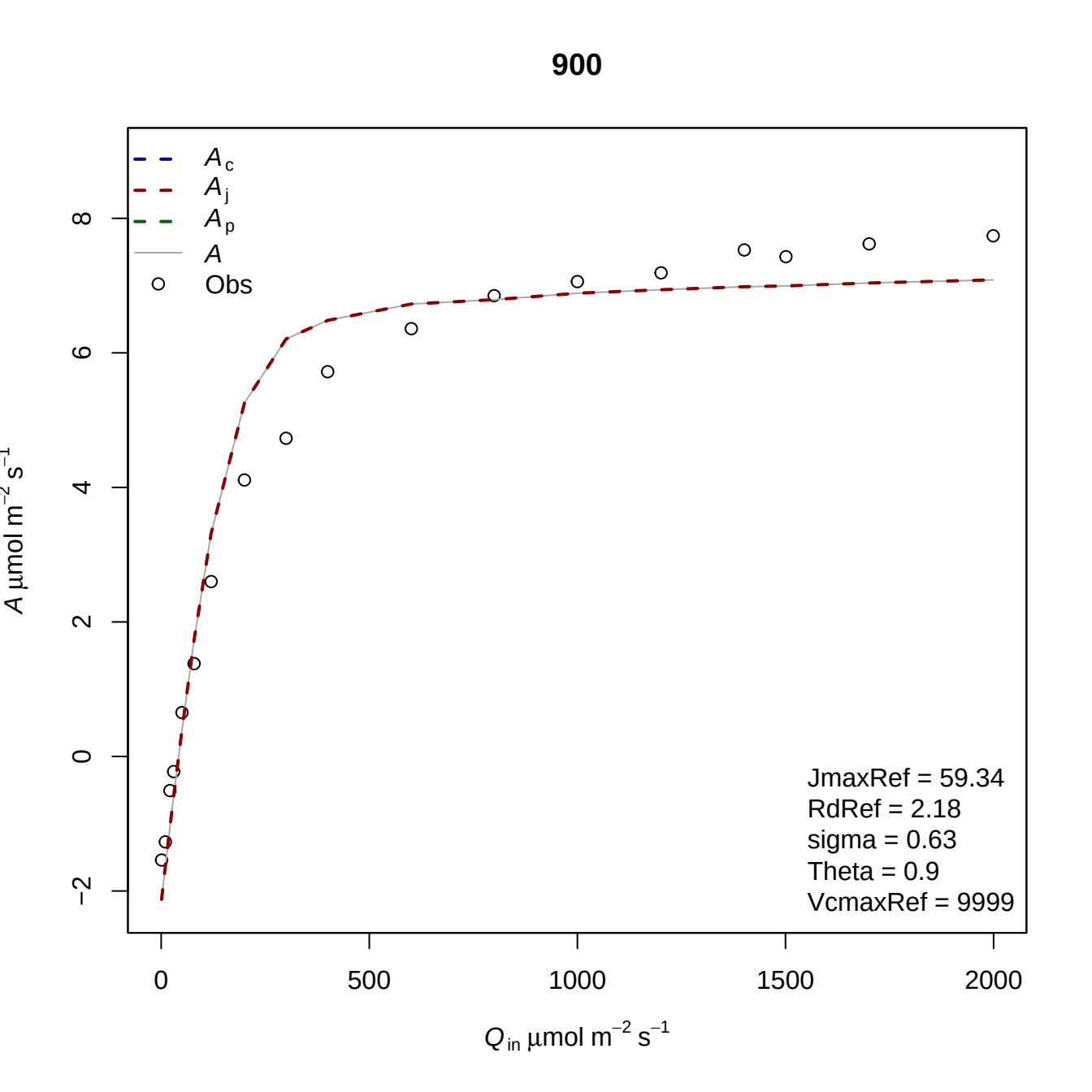

895
A_c
A_j
A_p
A
Obs
JmaxRef = 119.54
RdRef = 3.48
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

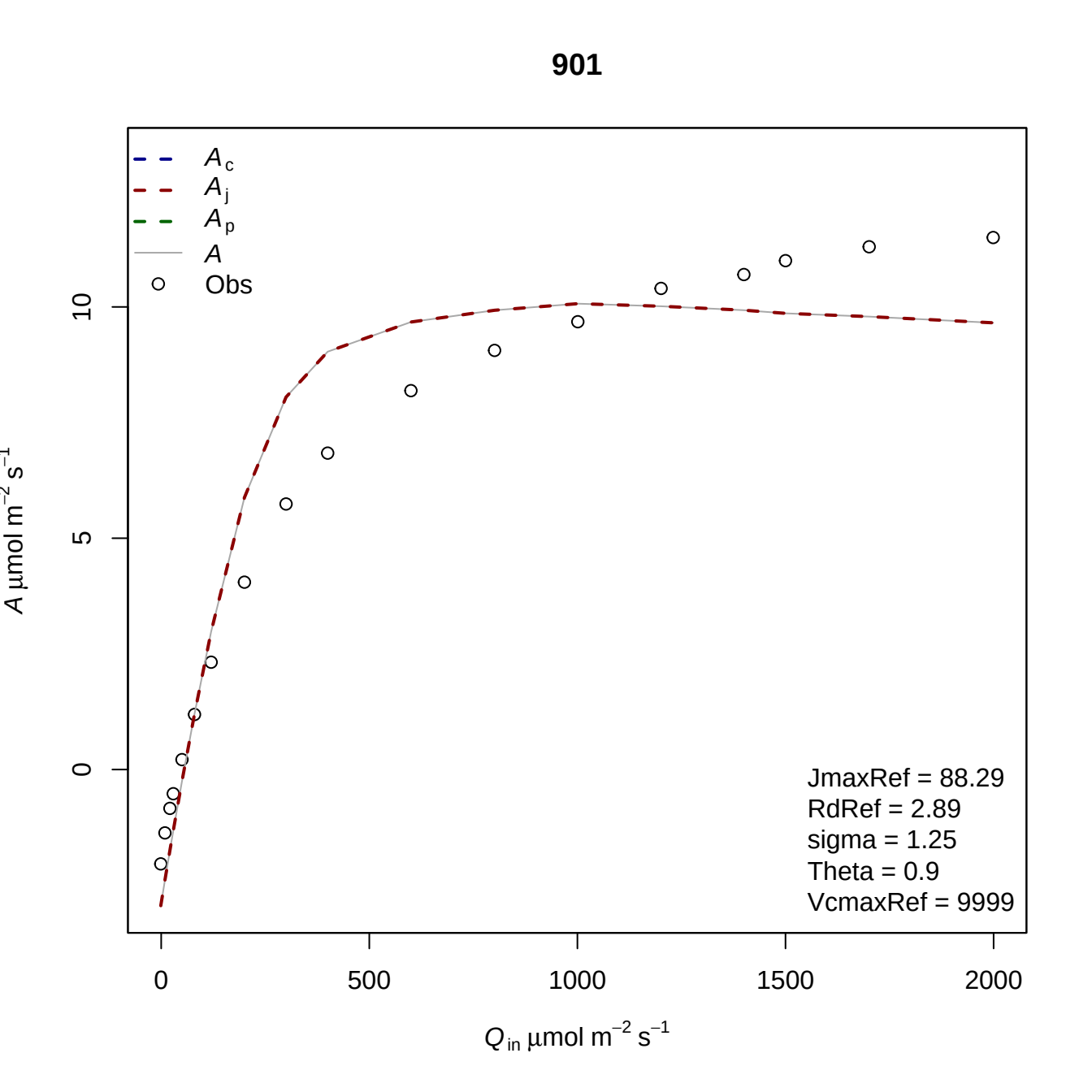

896
A_c
A_j
A_p
A
Obs
JmaxRef = 146.47
RdRef = 3.19
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

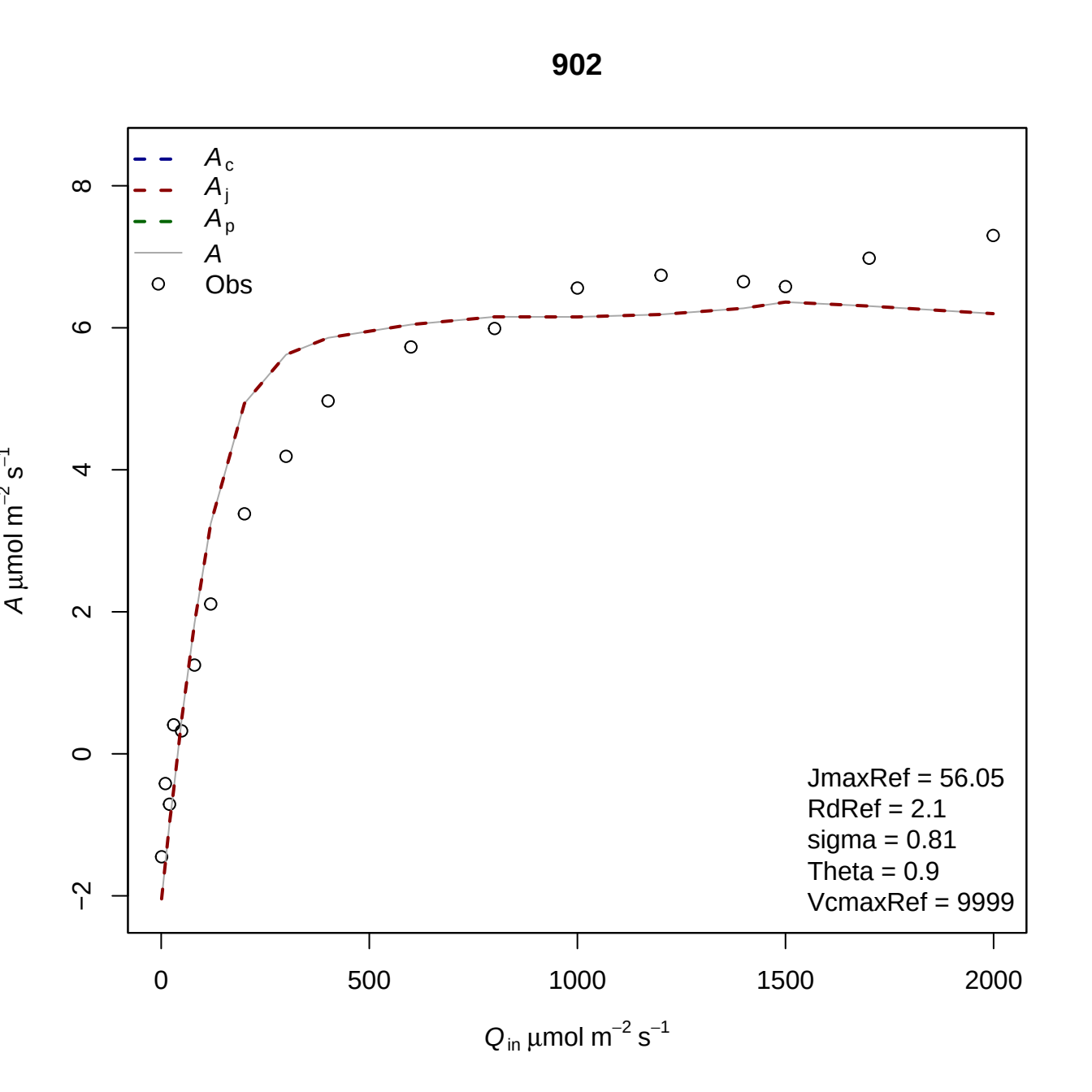

897
A_c
A_j
A_p
A
Obs
JmaxRef = 153.56
RdRef = 3.15
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

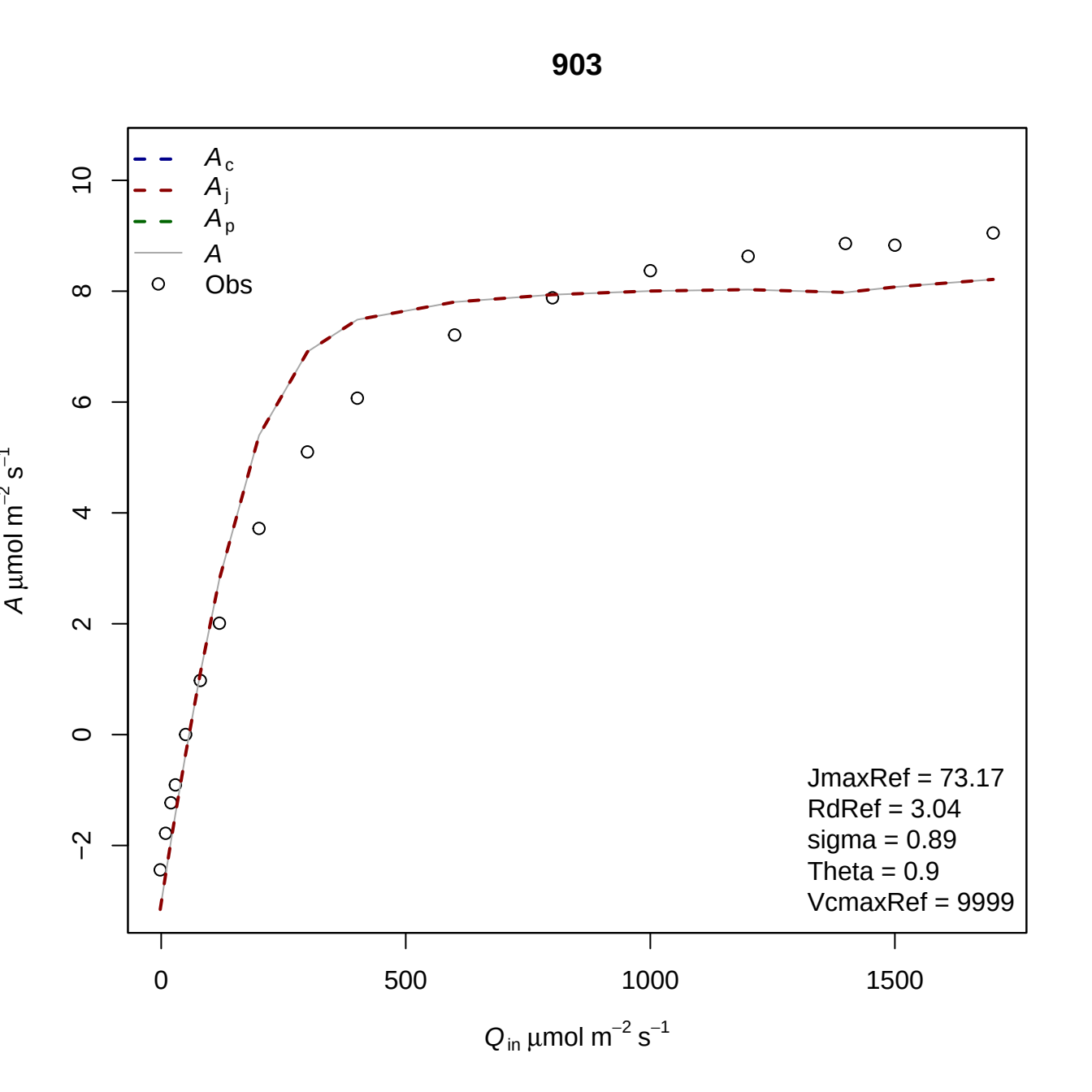

898
A_c
A_j
A_p
A
Obs
JmaxRef = 45.94
RdRef = 2.29
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

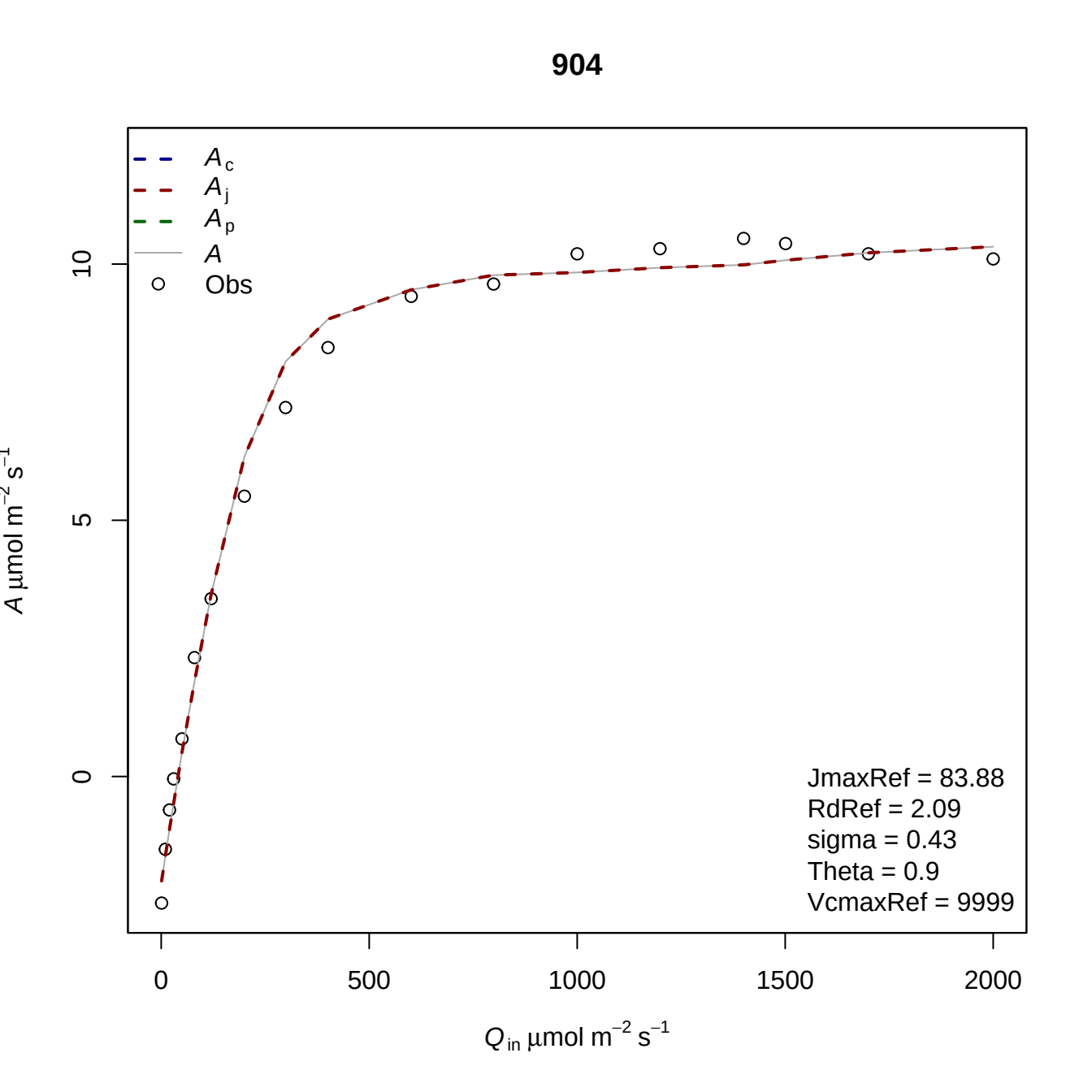

899
A_c
A_j
A_p
A
Obs
JmaxRef = 39.94
RdRef = 1.8
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

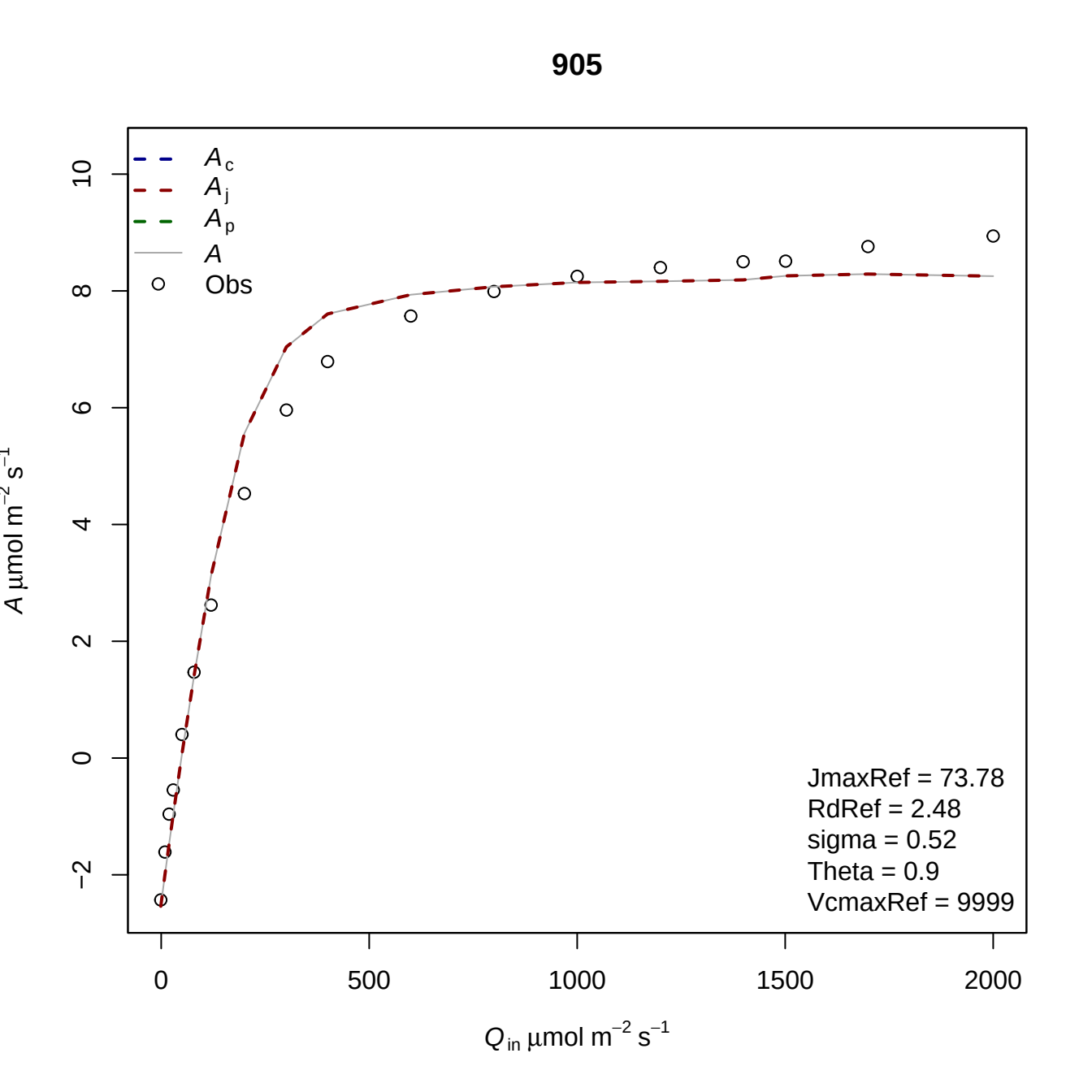

900
A_c
A_j
A_p
A
Obs
JmaxRef = 59.34
RdRef = 2.18
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

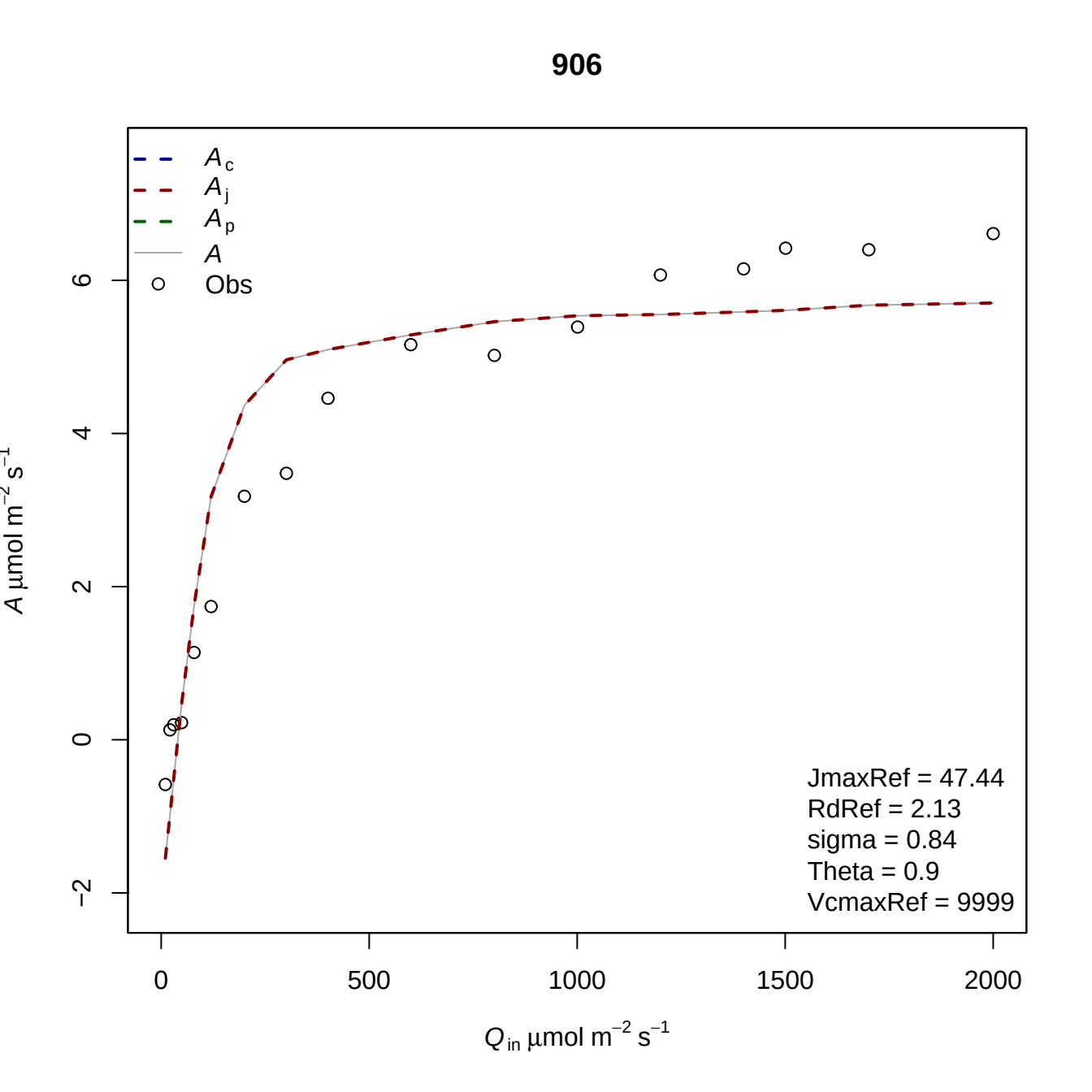

901
A_c
A_j
A_p
A
Obs
JmaxRef = 88.29
RdRef = 2.89
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

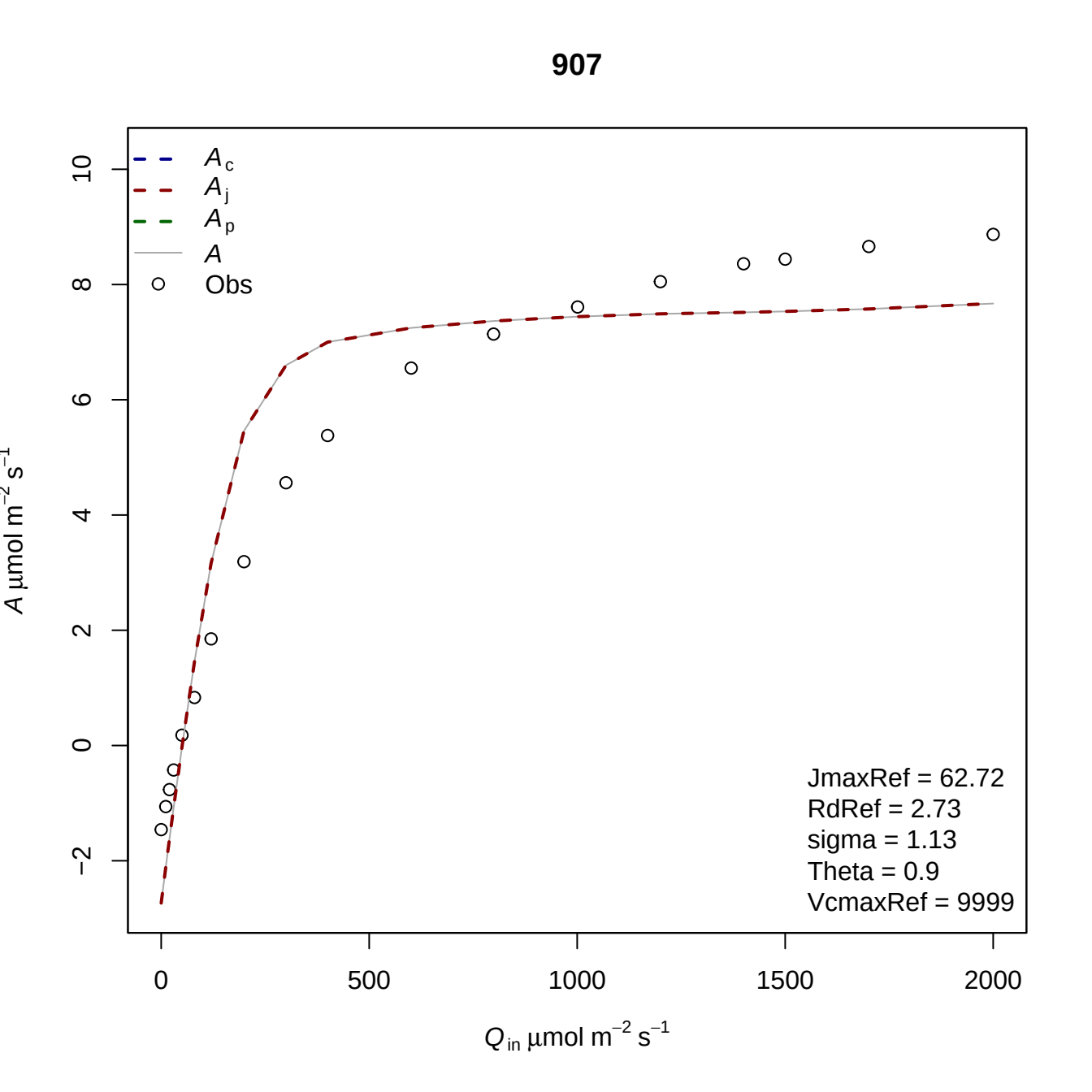

902
A_c
A_j
A_p
A
Obs
JmaxRef = 56.05
RdRef = 2.1
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

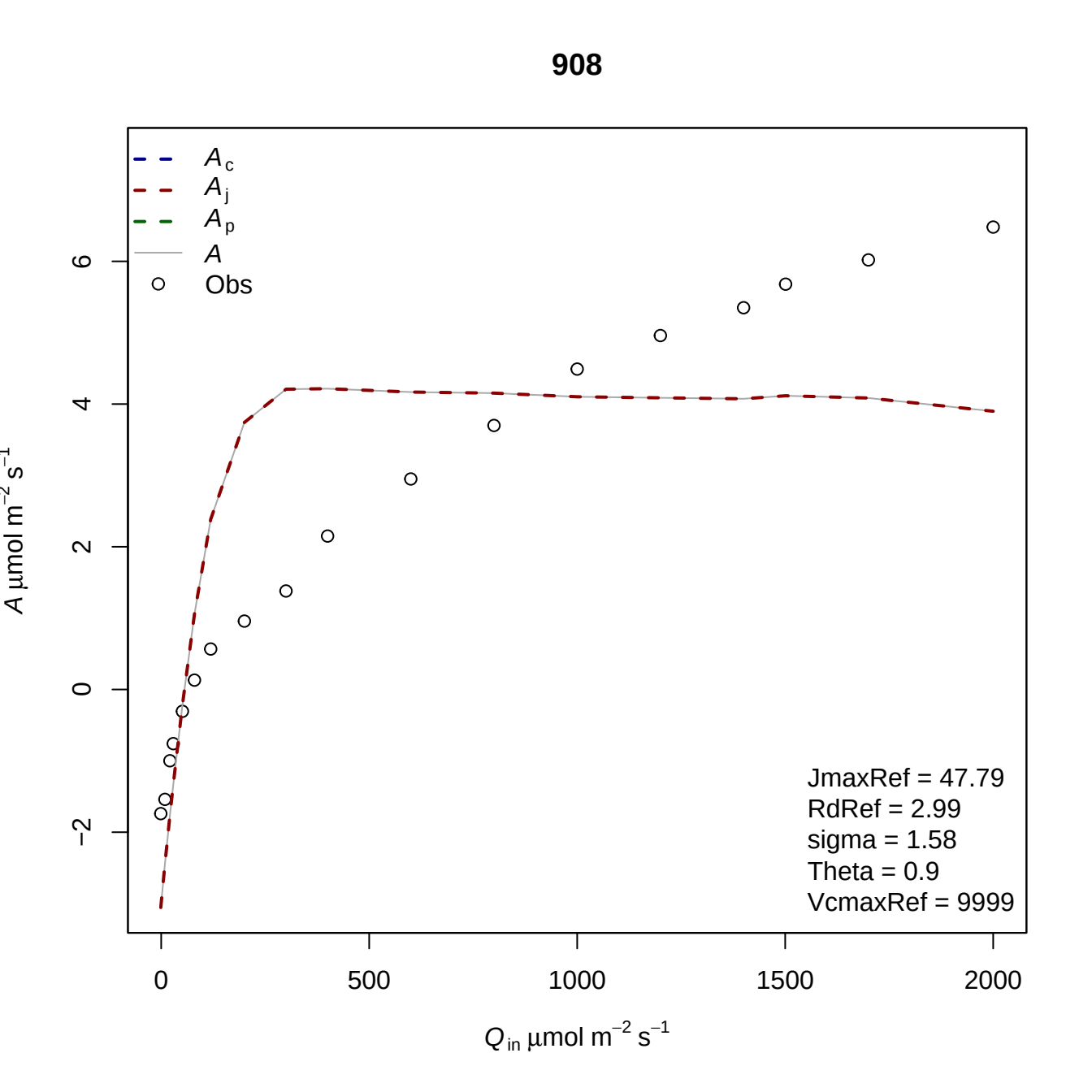

903
A_c
A_j
A_p
A
Obs
JmaxRef = 73.17
RdRef = 3.04
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

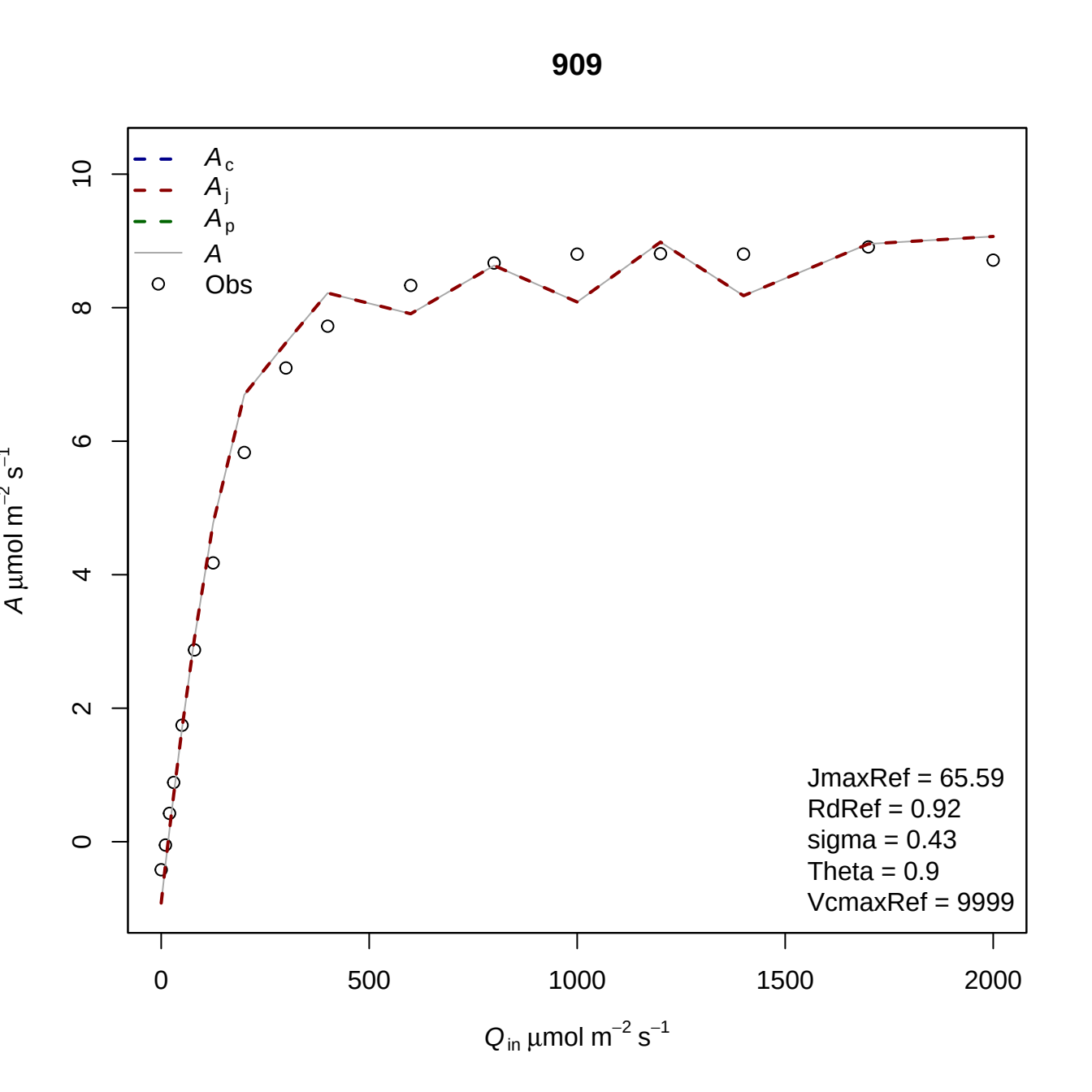

904
A_c
A_j
A_p
A
Obs
JmaxRef = 83.88
RdRef = 2.09
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

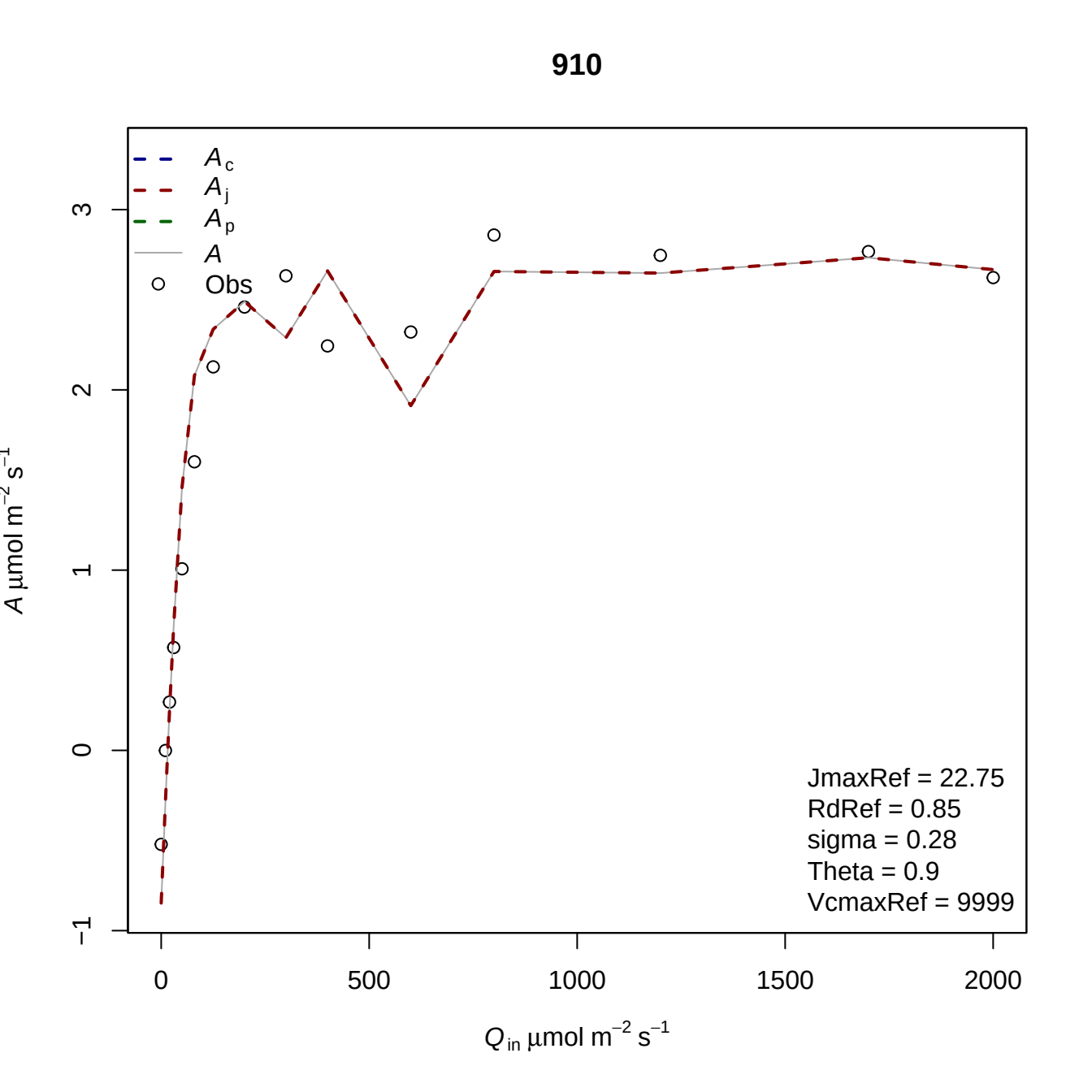

905
A_c
A_j
A_p
A
Obs
JmaxRef = 73.78
RdRef = 2.48
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

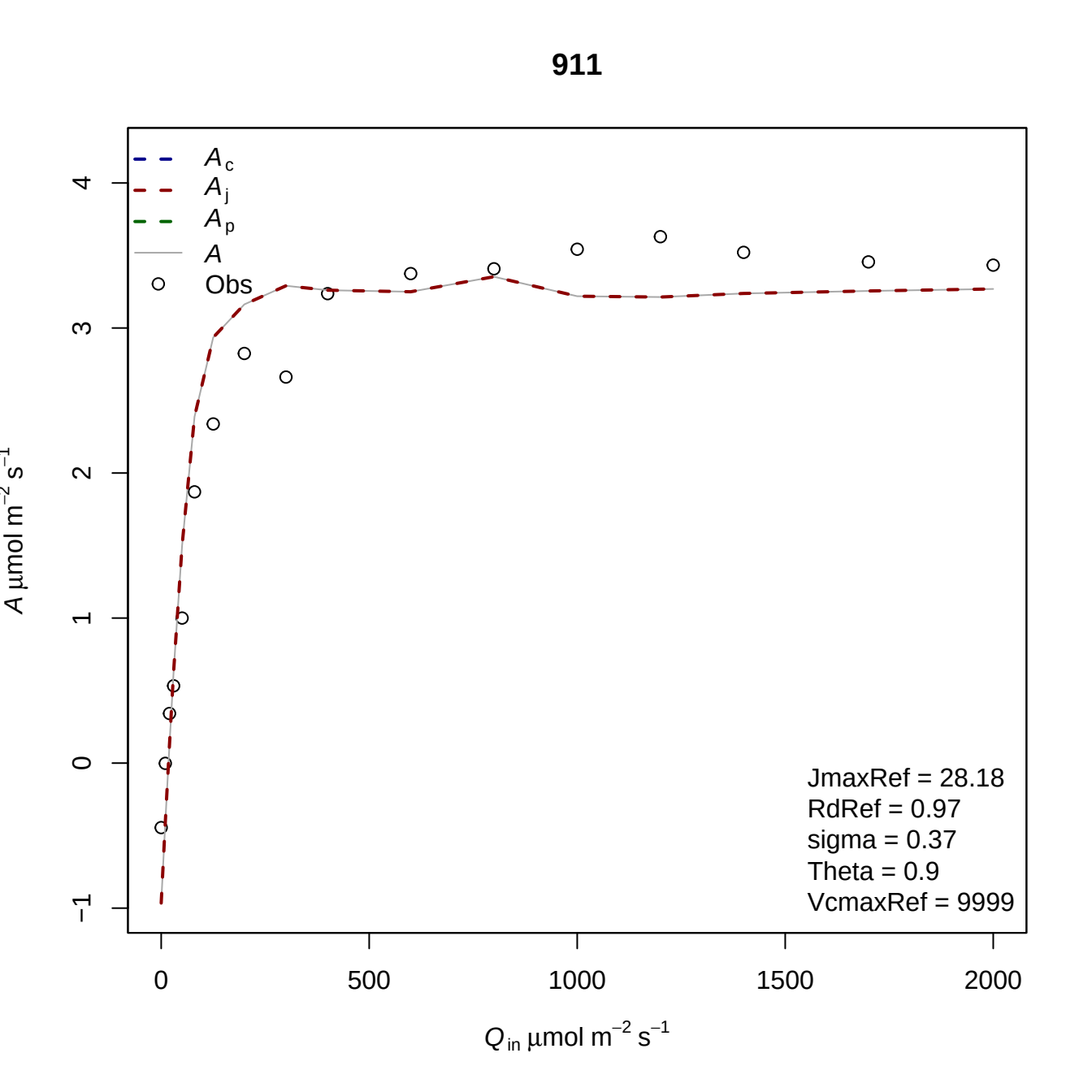

906
A_c
A_j
A_p
A
Obs
JmaxRef = 47.44
RdRef = 2.13
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

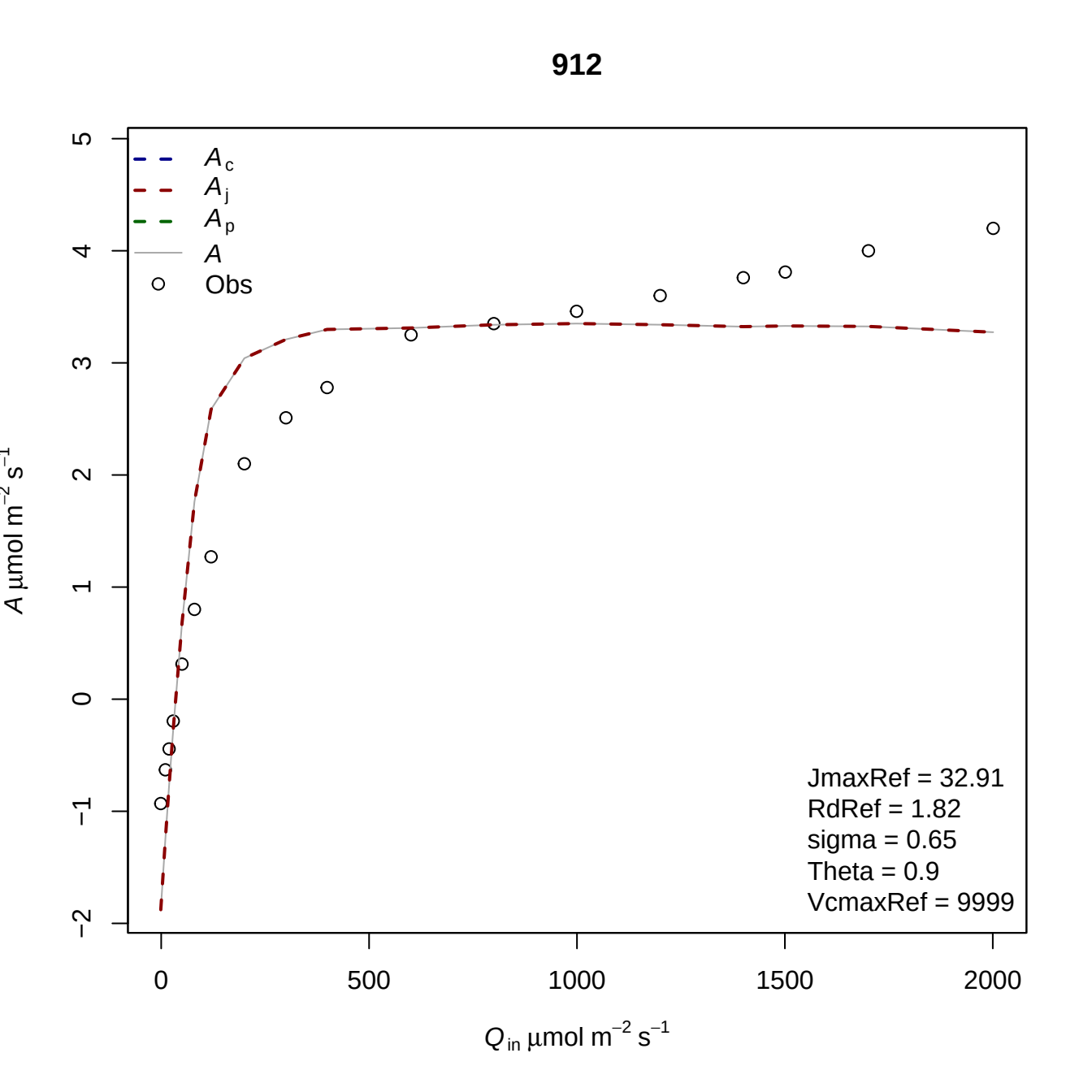

907
A_c
A_j
A_p
A
Obs
JmaxRef = 62.72
RdRef = 2.73
sigma = 1.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

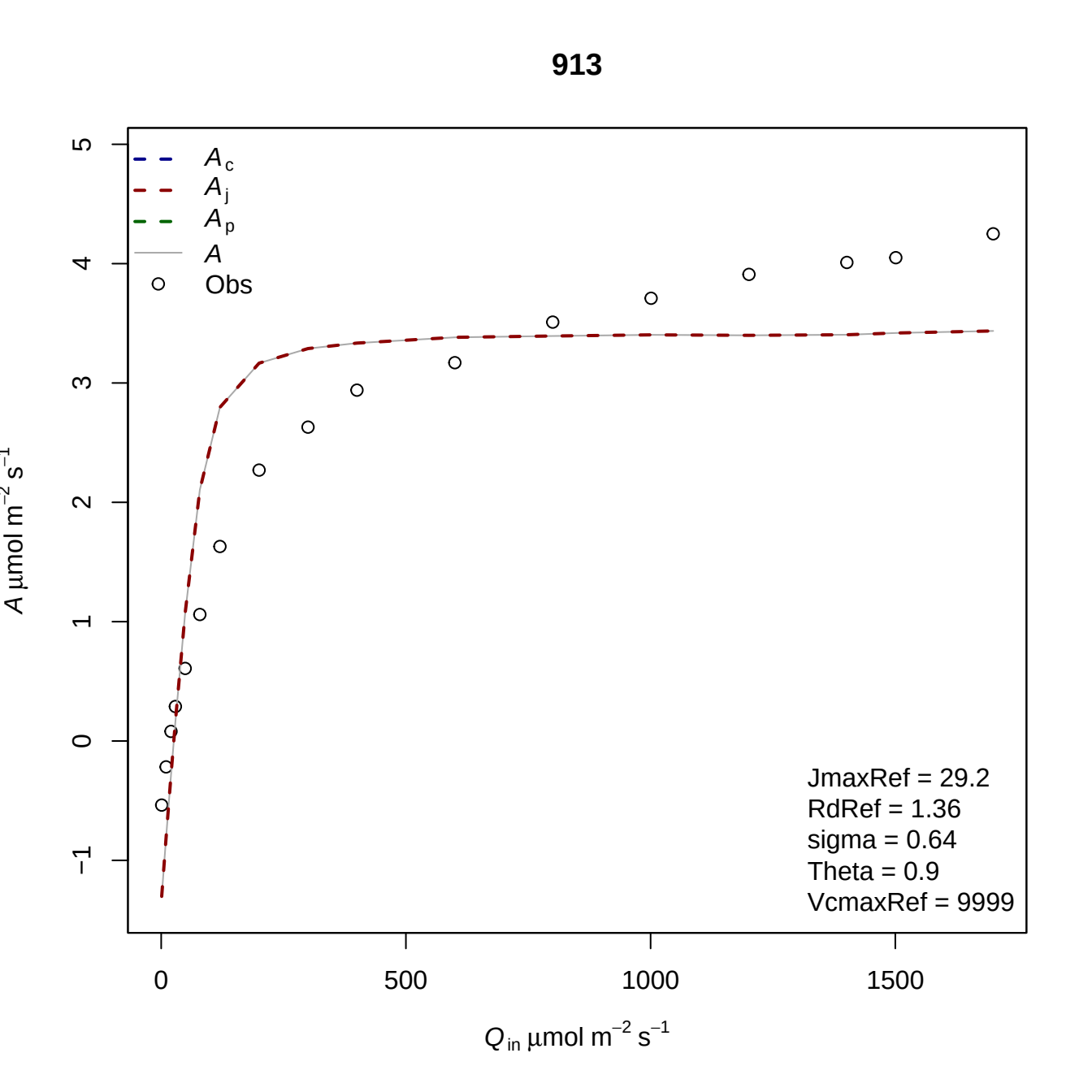

908
A_c
A_j
A_p
A
Obs
JmaxRef = 47.79
RdRef = 2.99
sigma = 1.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

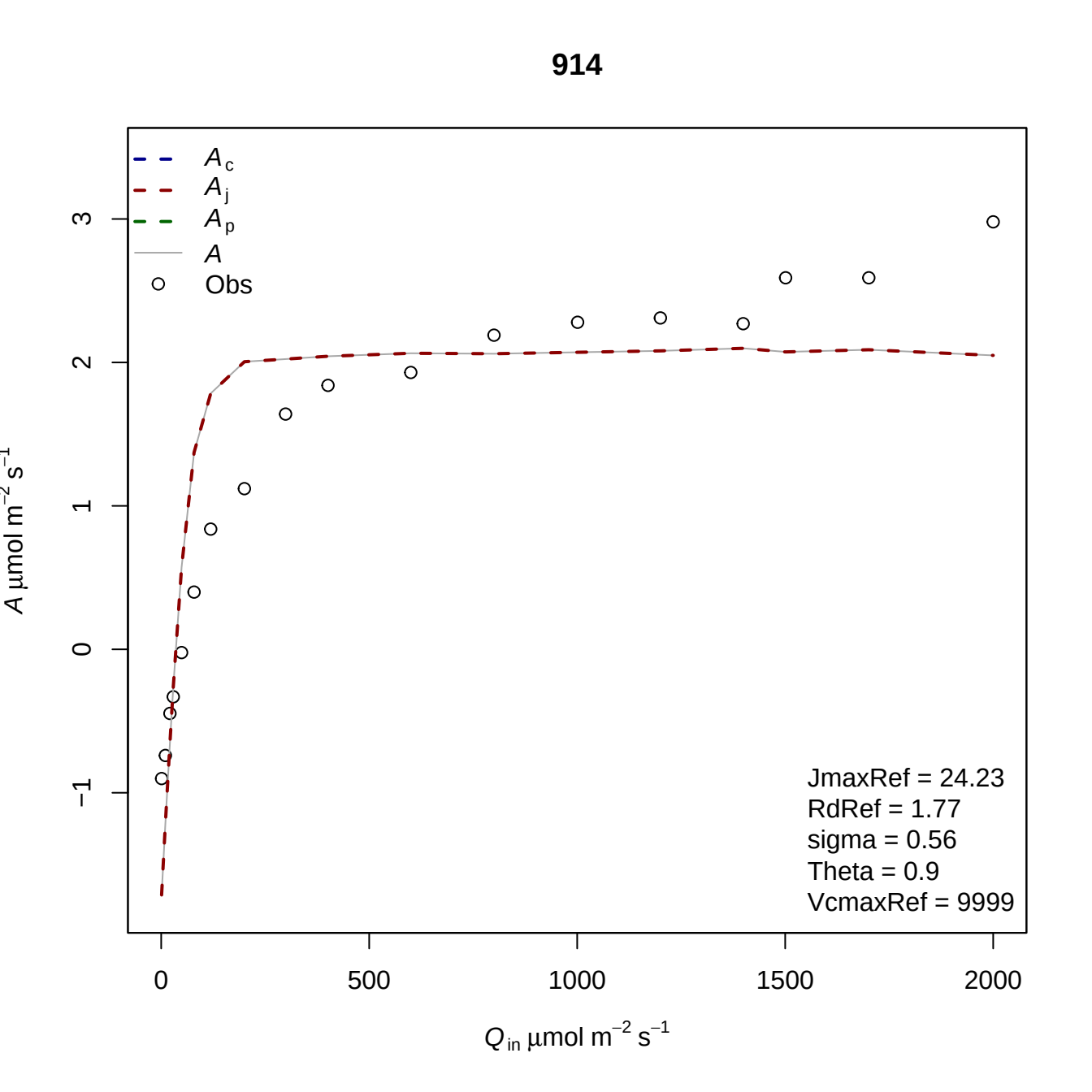

909
A_c
A_j
A_p
A
Obs
JmaxRef = 65.59
RdRef = 0.92
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

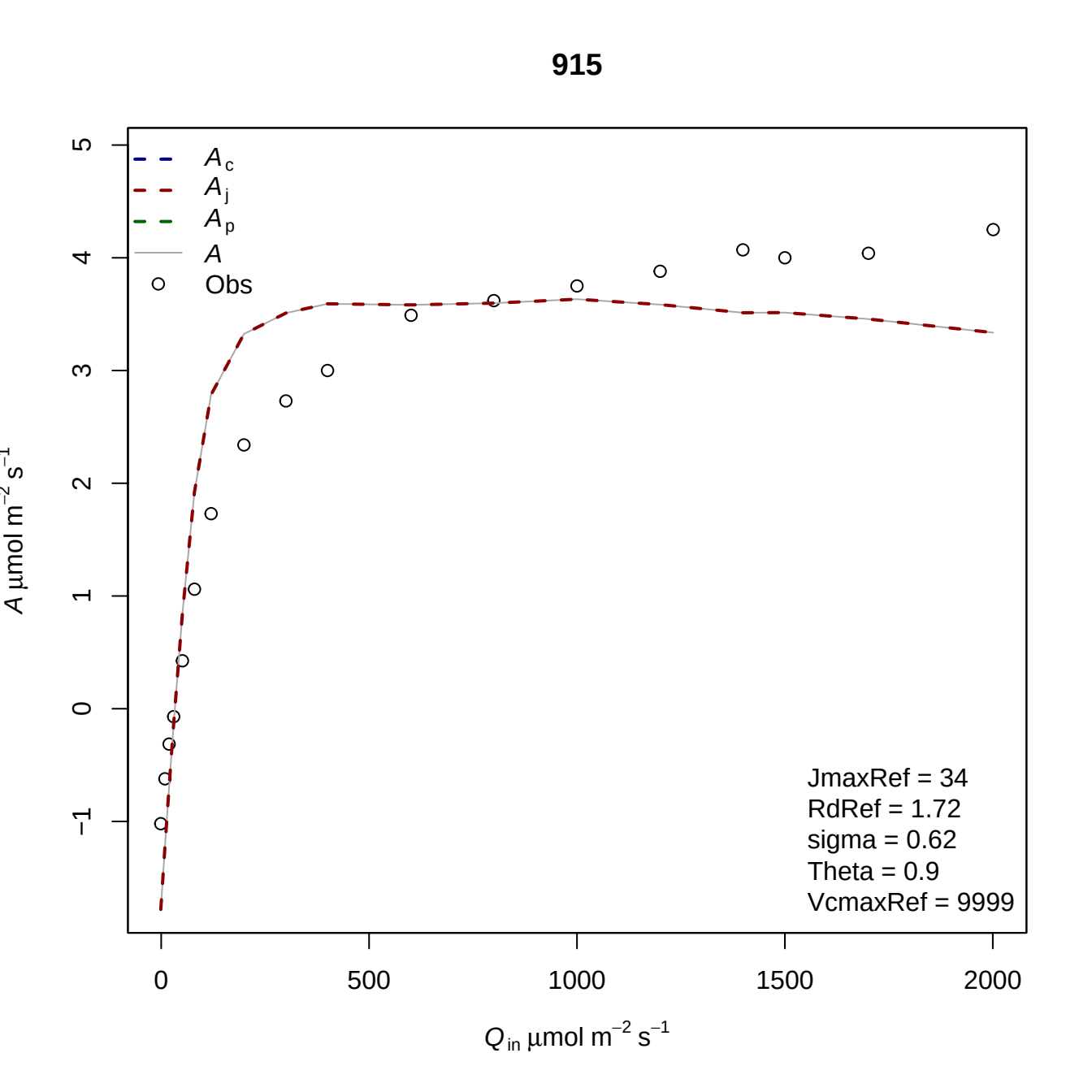

910
A_c
A_j
A_p
A
Obs
JmaxRef = 22.75
RdRef = 0.85
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

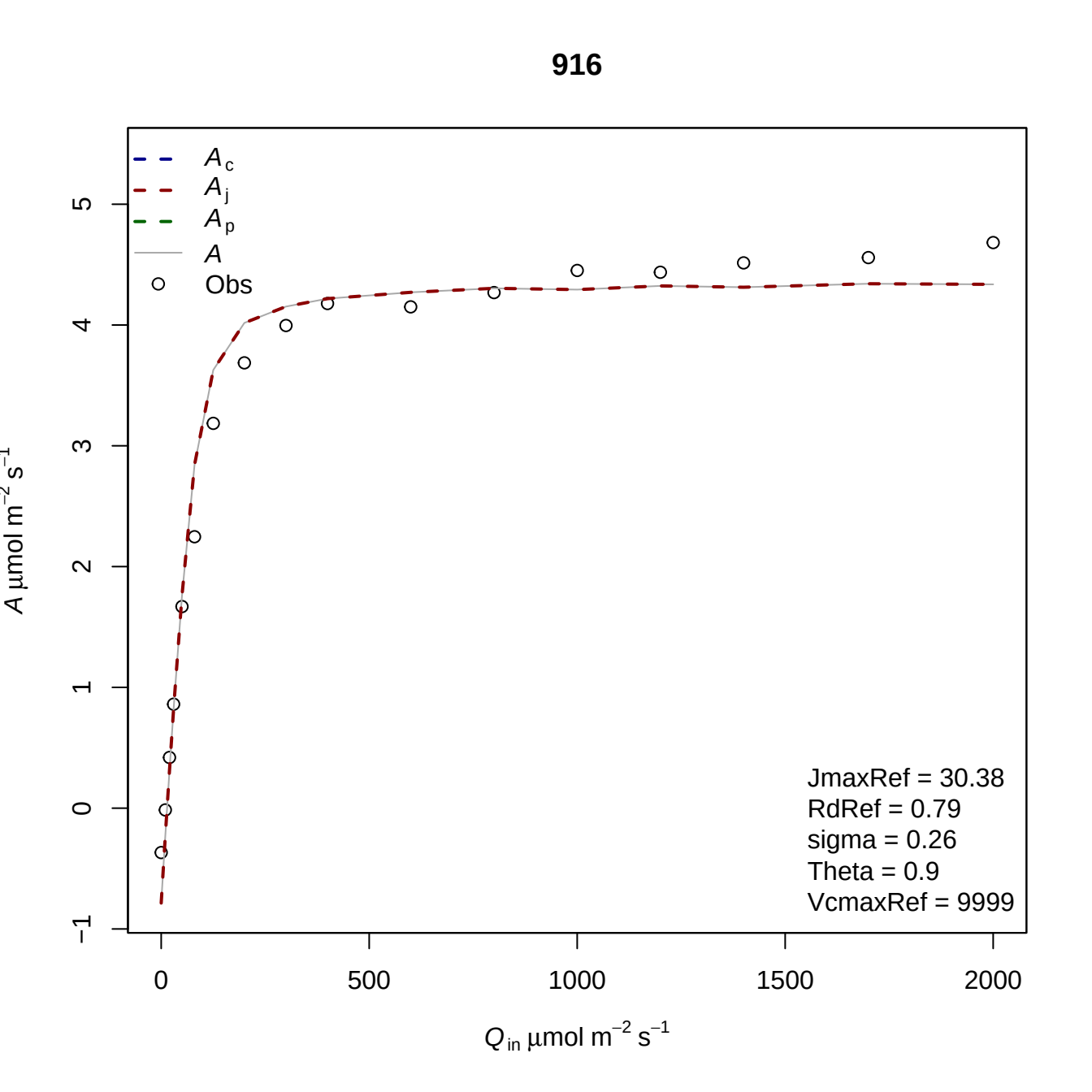

911
A_c
A_j
A_p
A
Obs
JmaxRef = 28.18
RdRef = 0.97
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

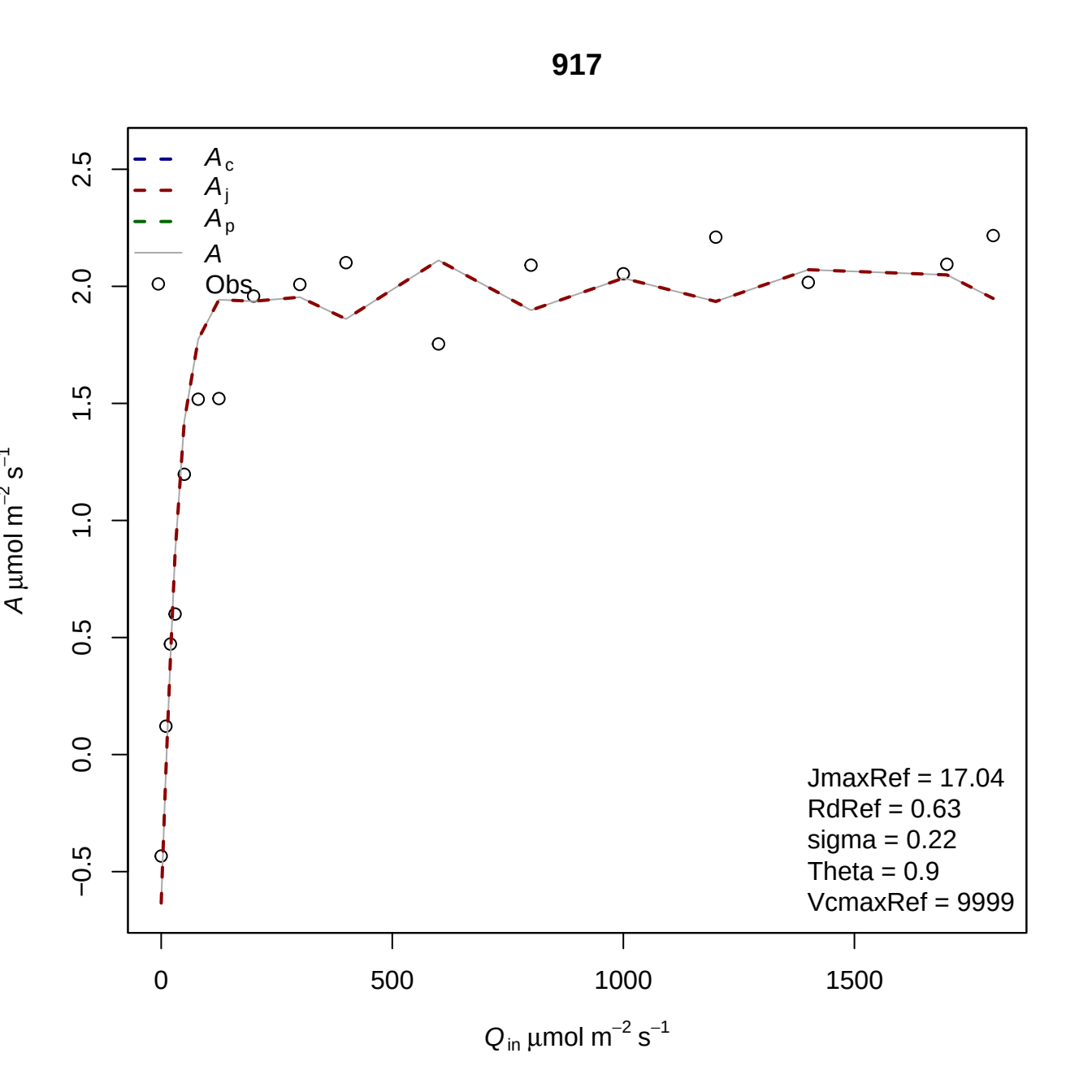

912
A_c
A_j
A_p
A
Obs
JmaxRef = 32.91
RdRef = 1.82
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

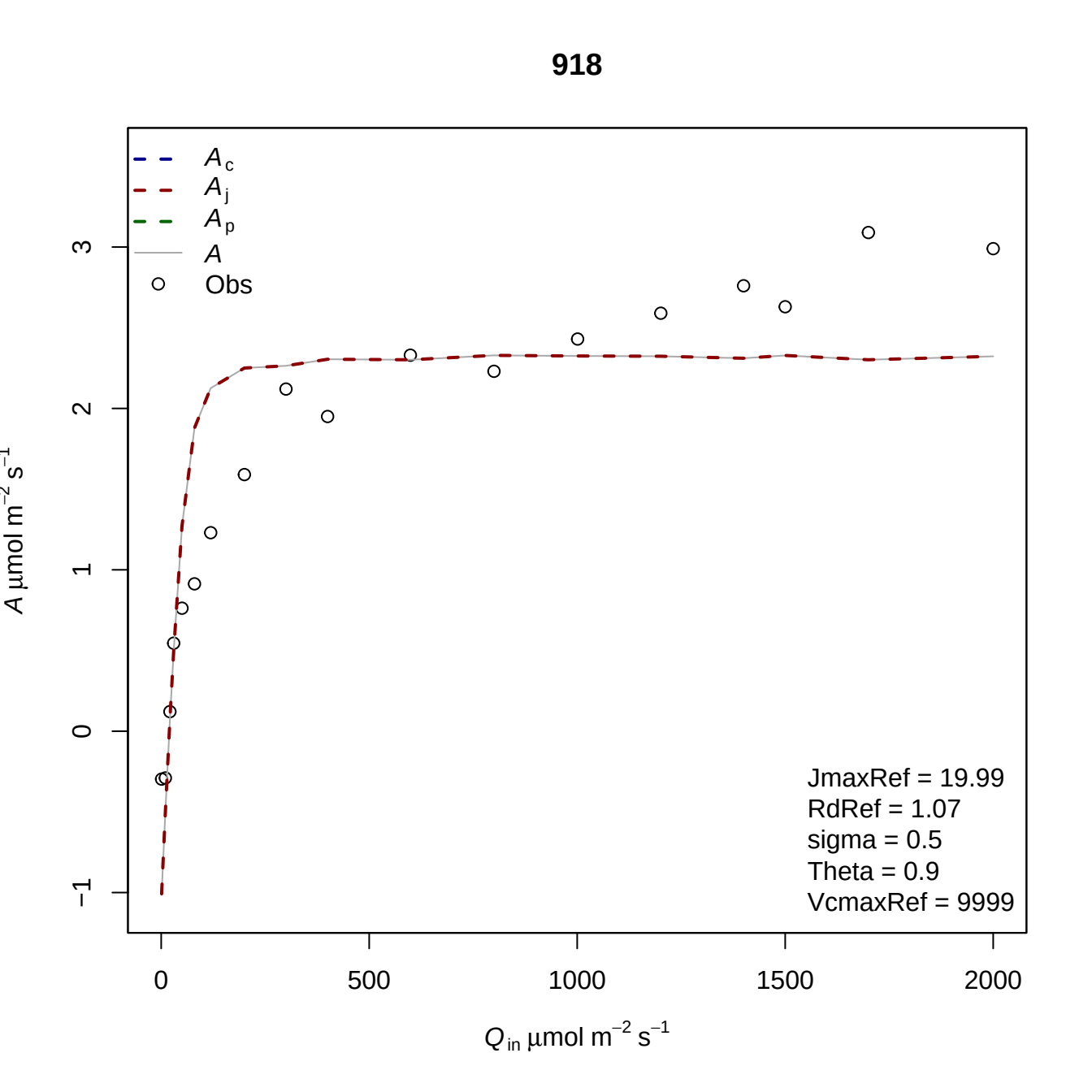

913
A_c
A_j
A_p
A
Obs
JmaxRef = 29.2
RdRef = 1.36
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

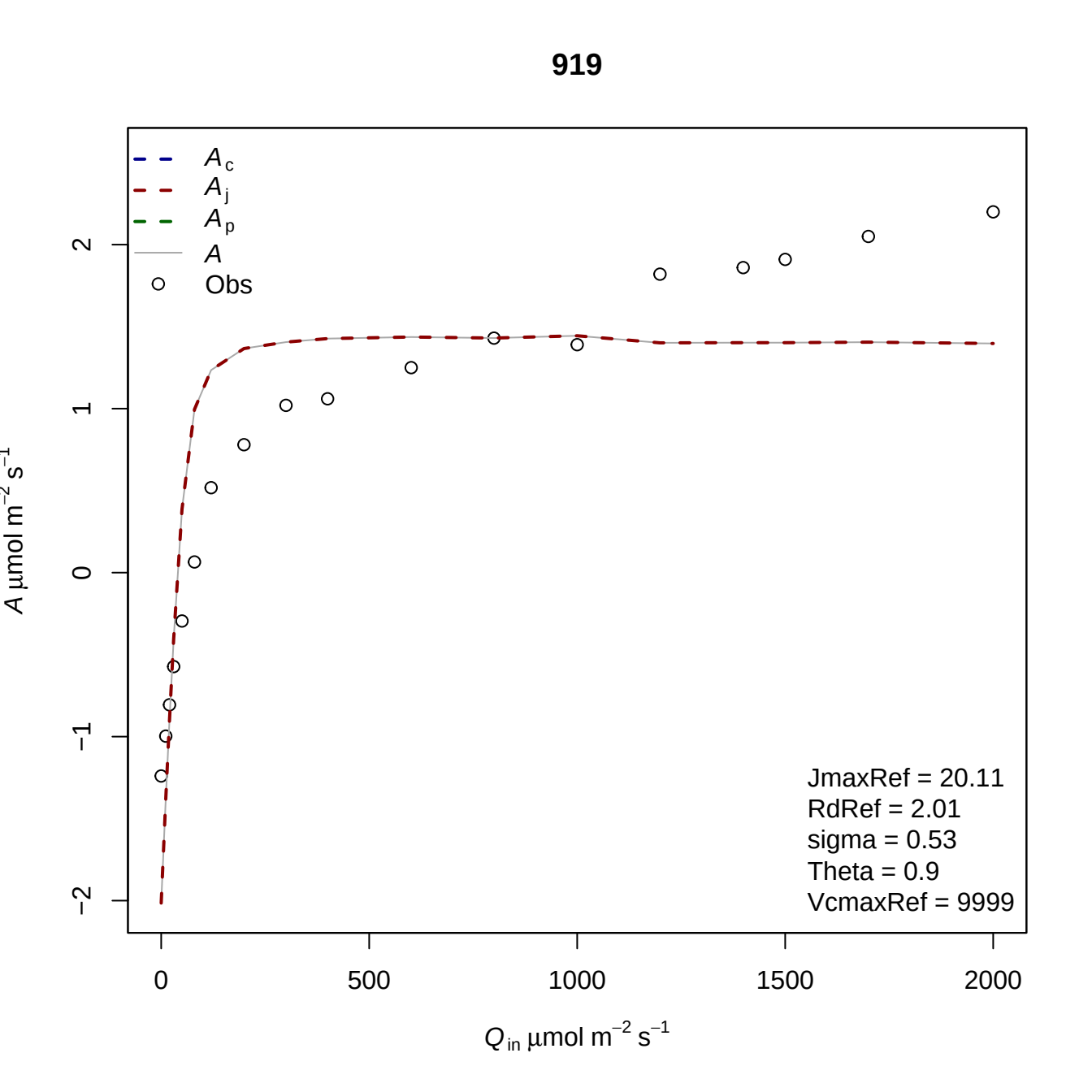

914
A_c
A_j
A_p
A
Obs
JmaxRef = 24.23
RdRef = 1.77
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

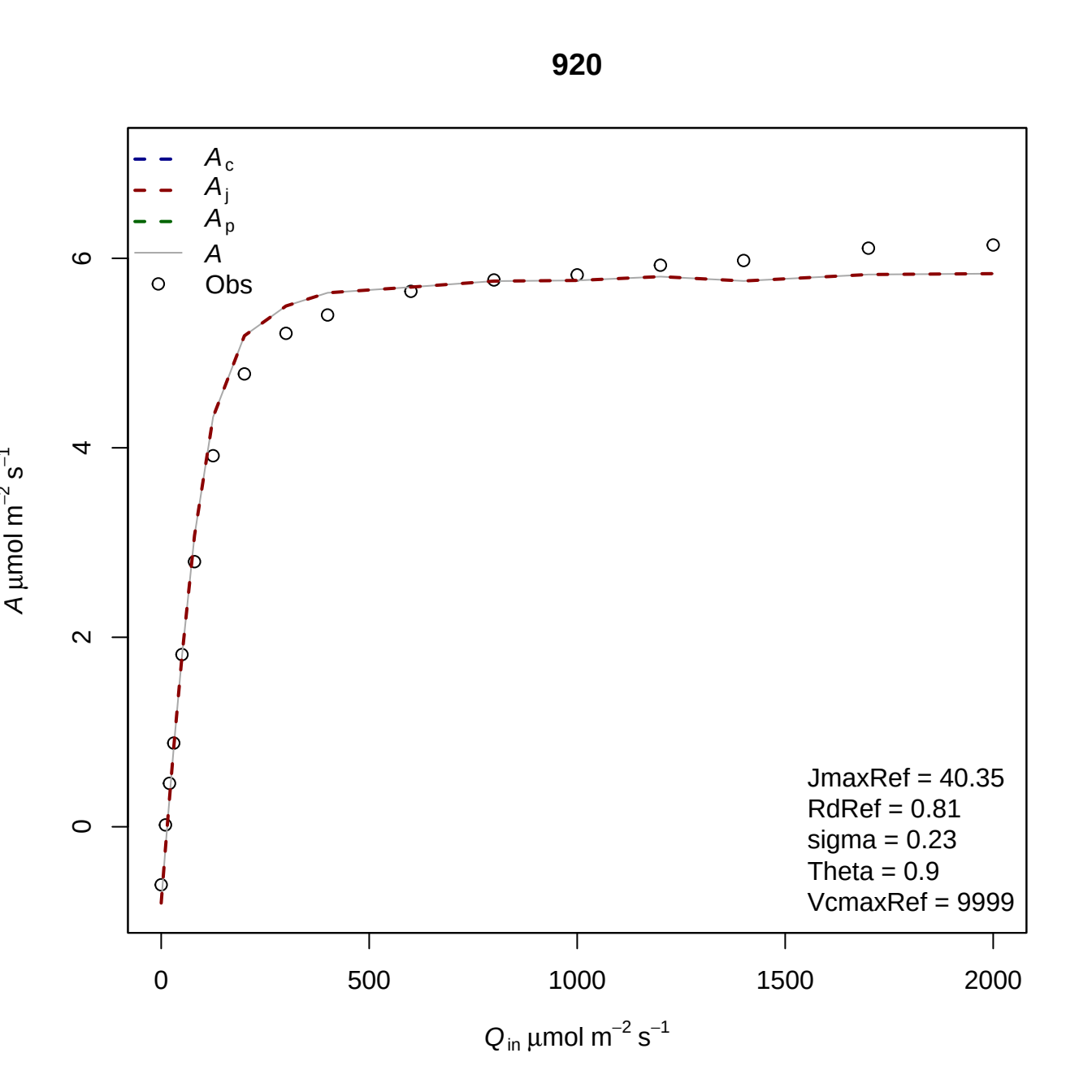

915
A_c
A_j
A_p
A
Obs
JmaxRef = 34
RdRef = 1.72
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

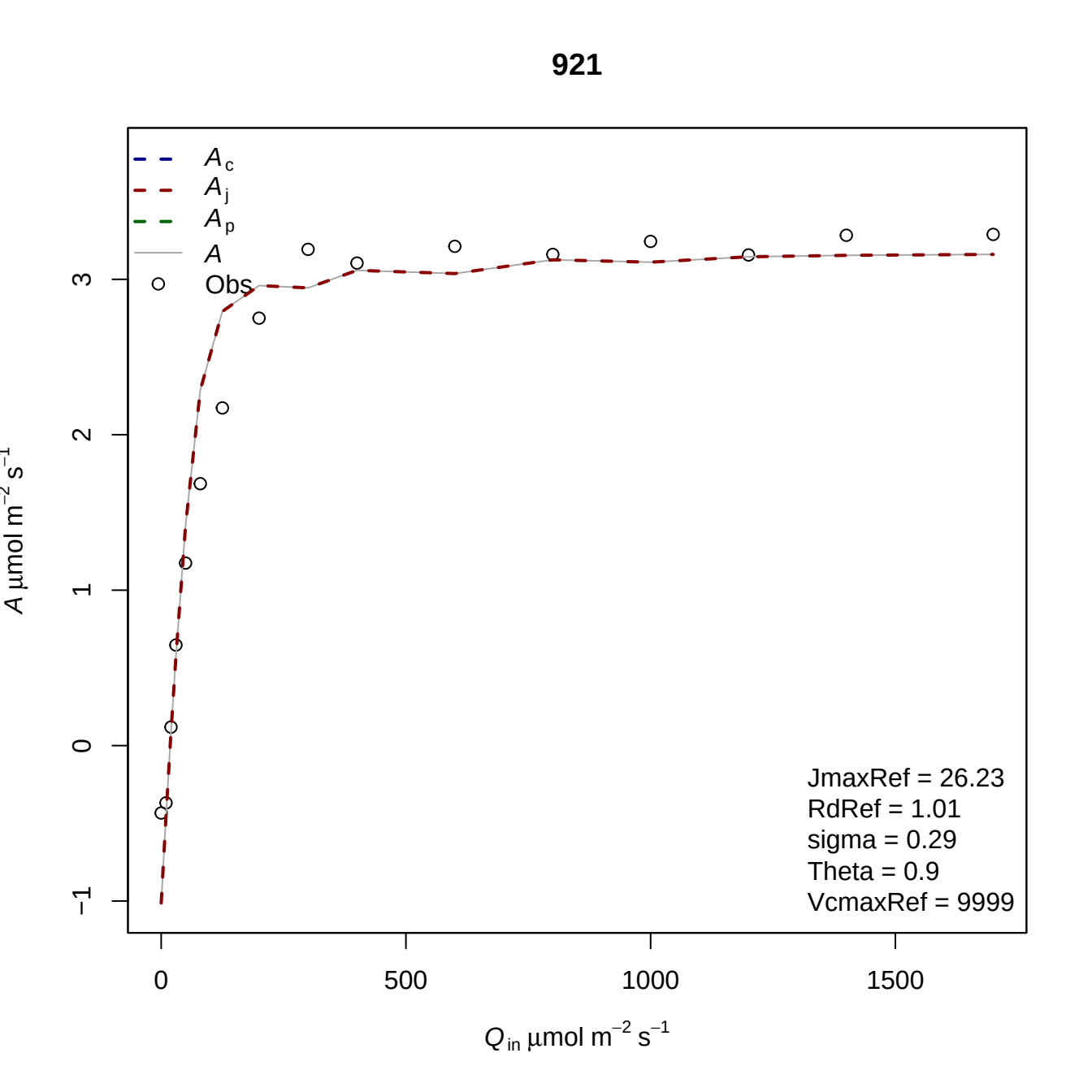

916
A_c
A_j
A_p
A
Obs
JmaxRef = 30.38
RdRef = 0.79
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

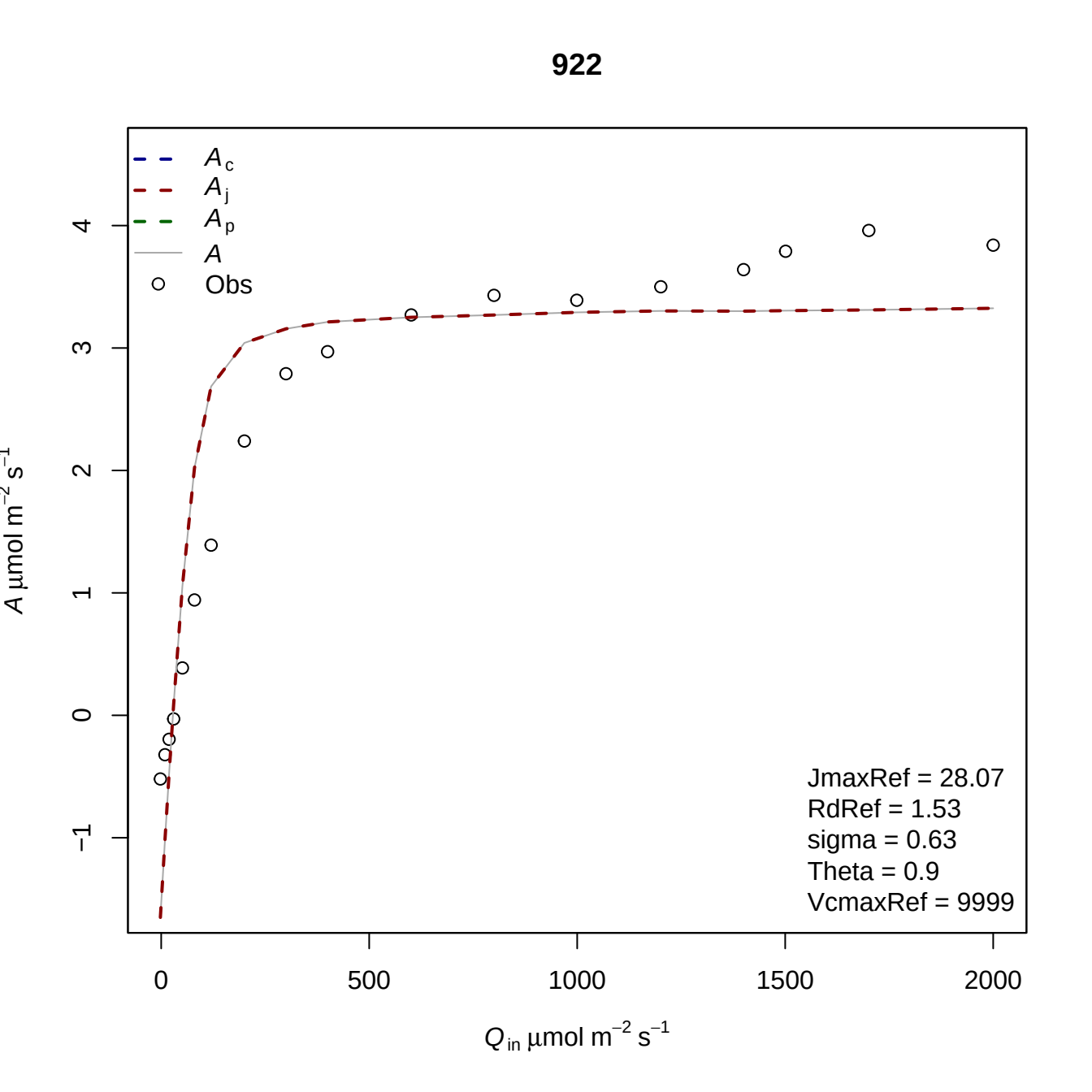

917
A_c
A_j
A_p
A
Obs
JmaxRef = 17.04
RdRef = 0.63
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

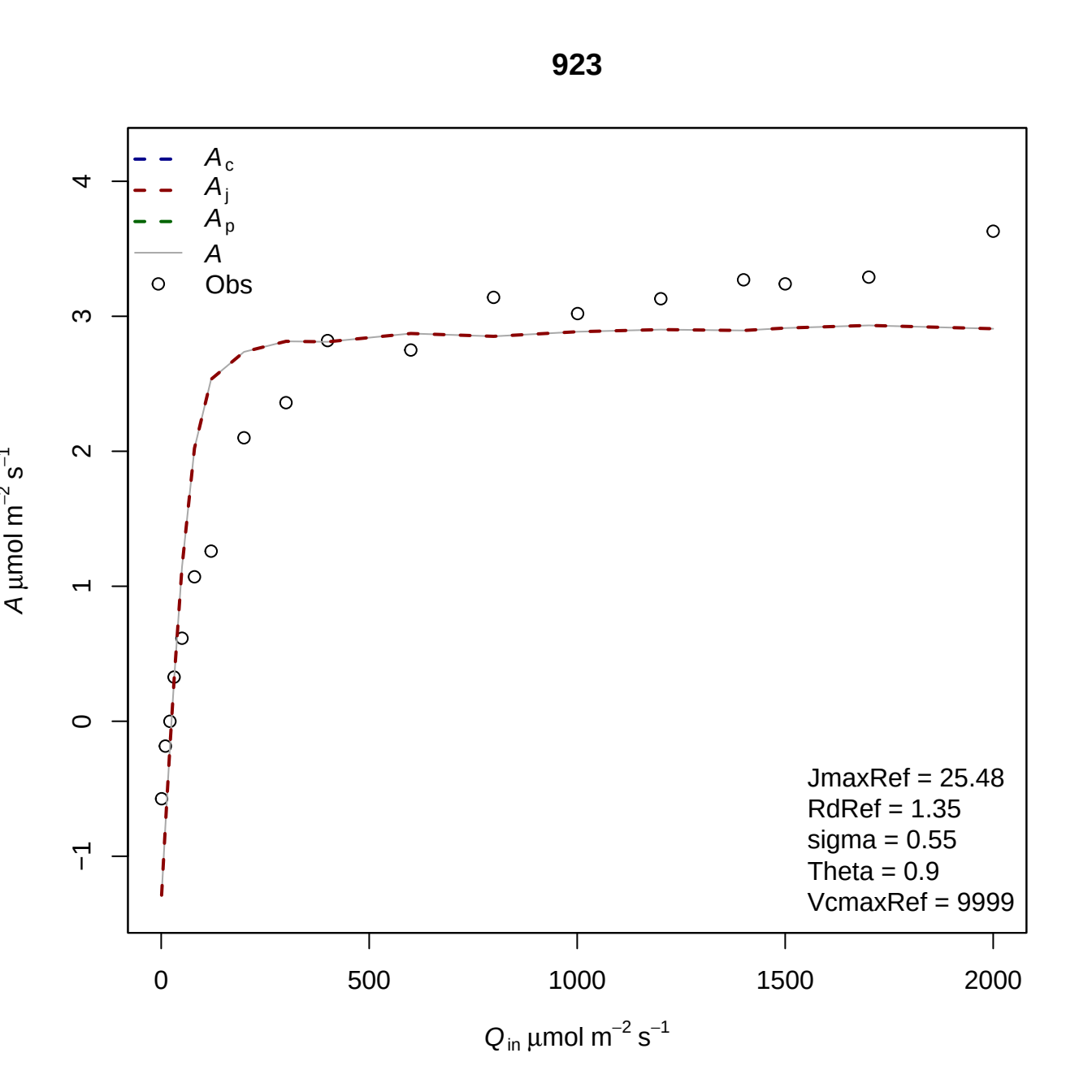

918
A_c
A_j
A_p
A
Obs
JmaxRef = 19.99
RdRef = 1.07
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

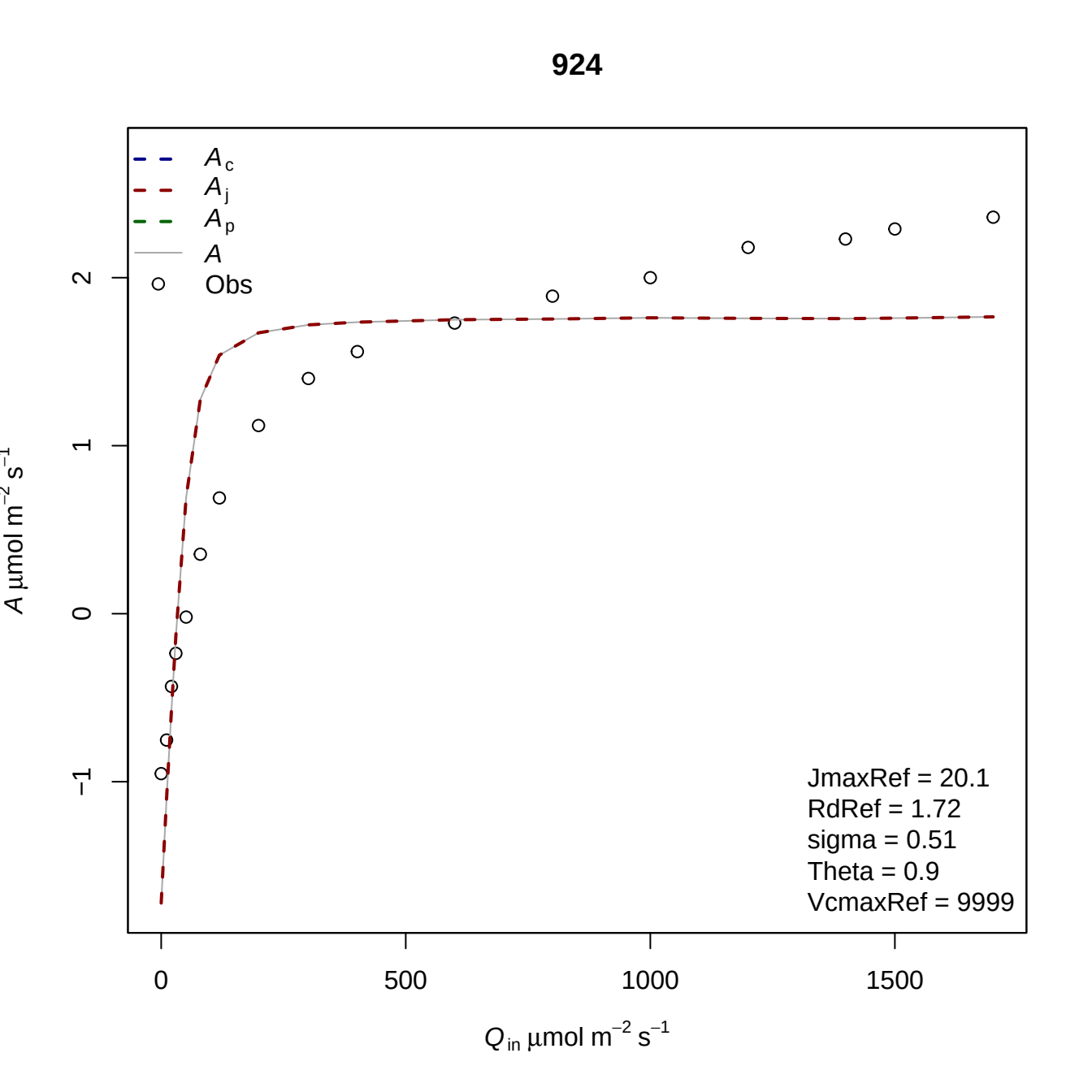

919
A_c
A_j
A_p
A
Obs
JmaxRef = 20.11
RdRef = 2.01
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500
2000

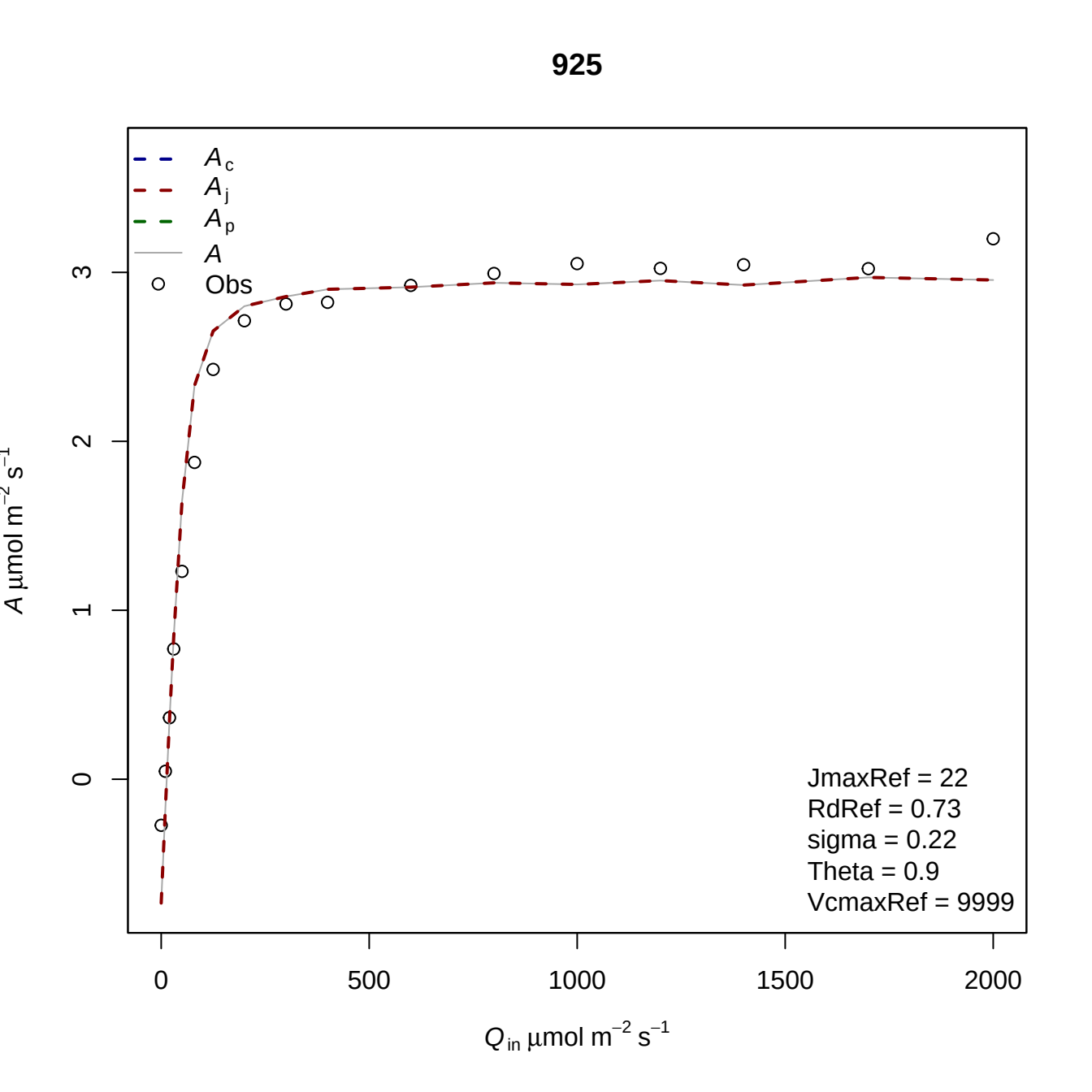

920
A_c
A_j
A_p
A
Obs
JmaxRef = 40.35
RdRef = 0.81
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

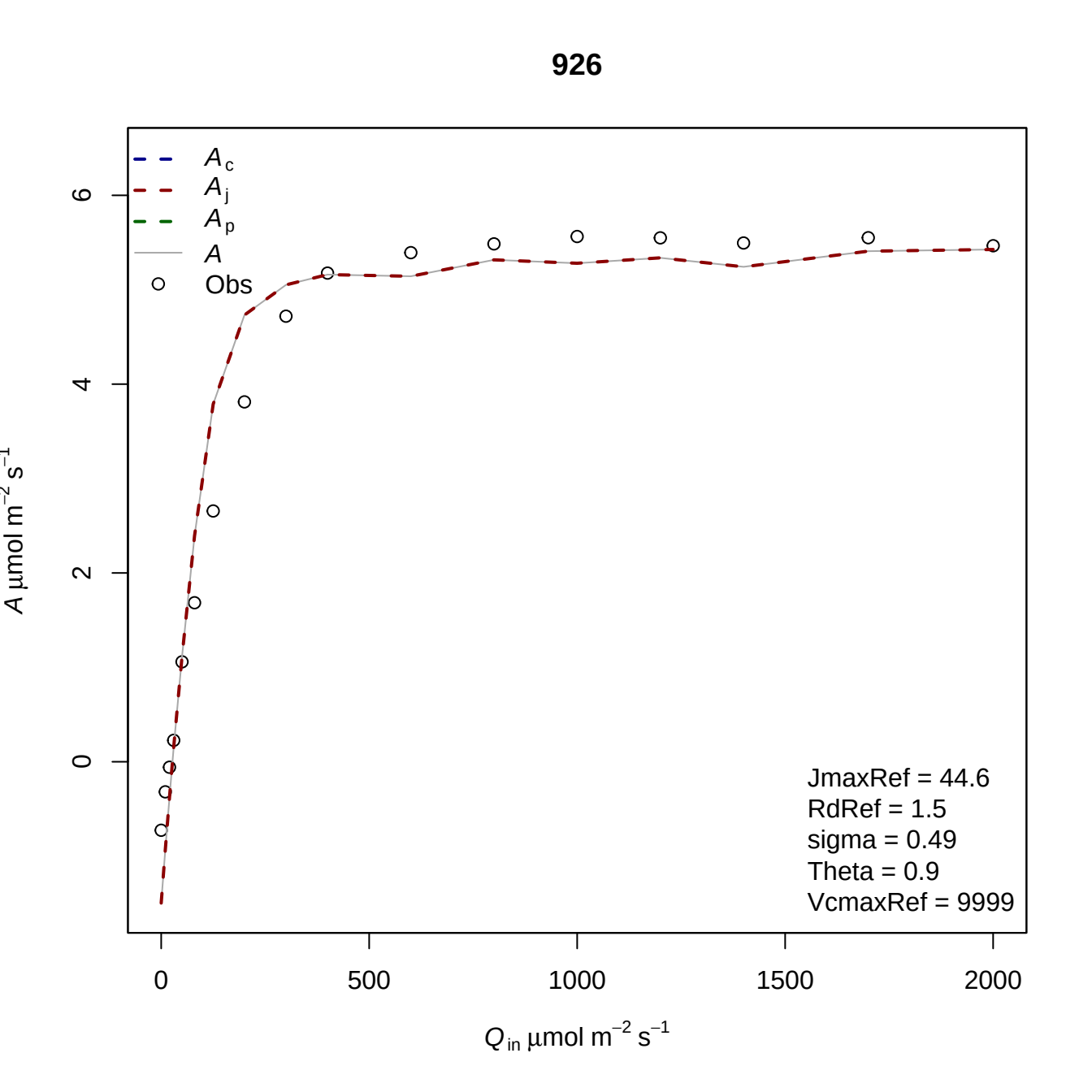

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.23
RdRef = 1.01
sigma = 0.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

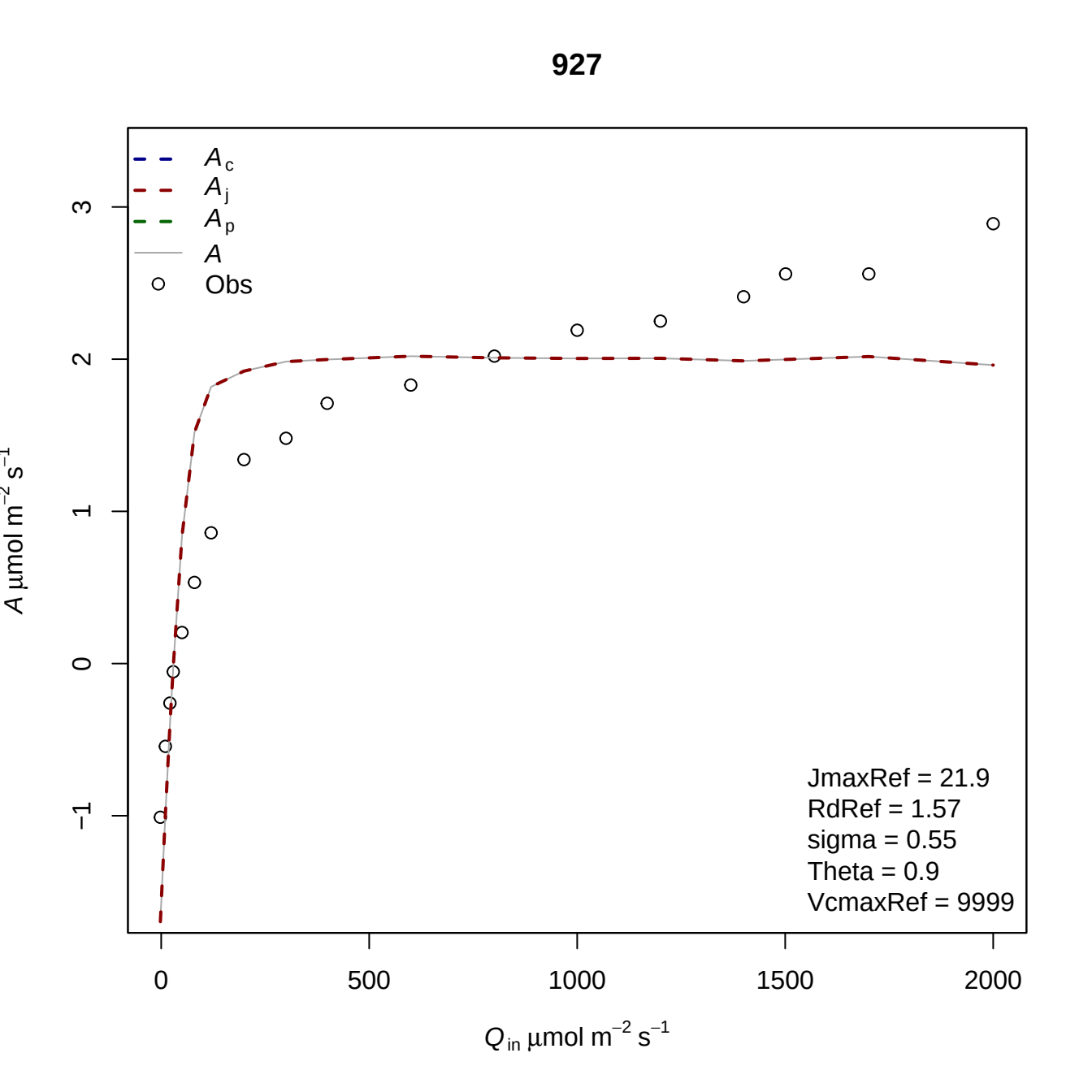

922
A_c
A_j
A_p
A
Obs
JmaxRef = 28.07
RdRef = 1.53
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

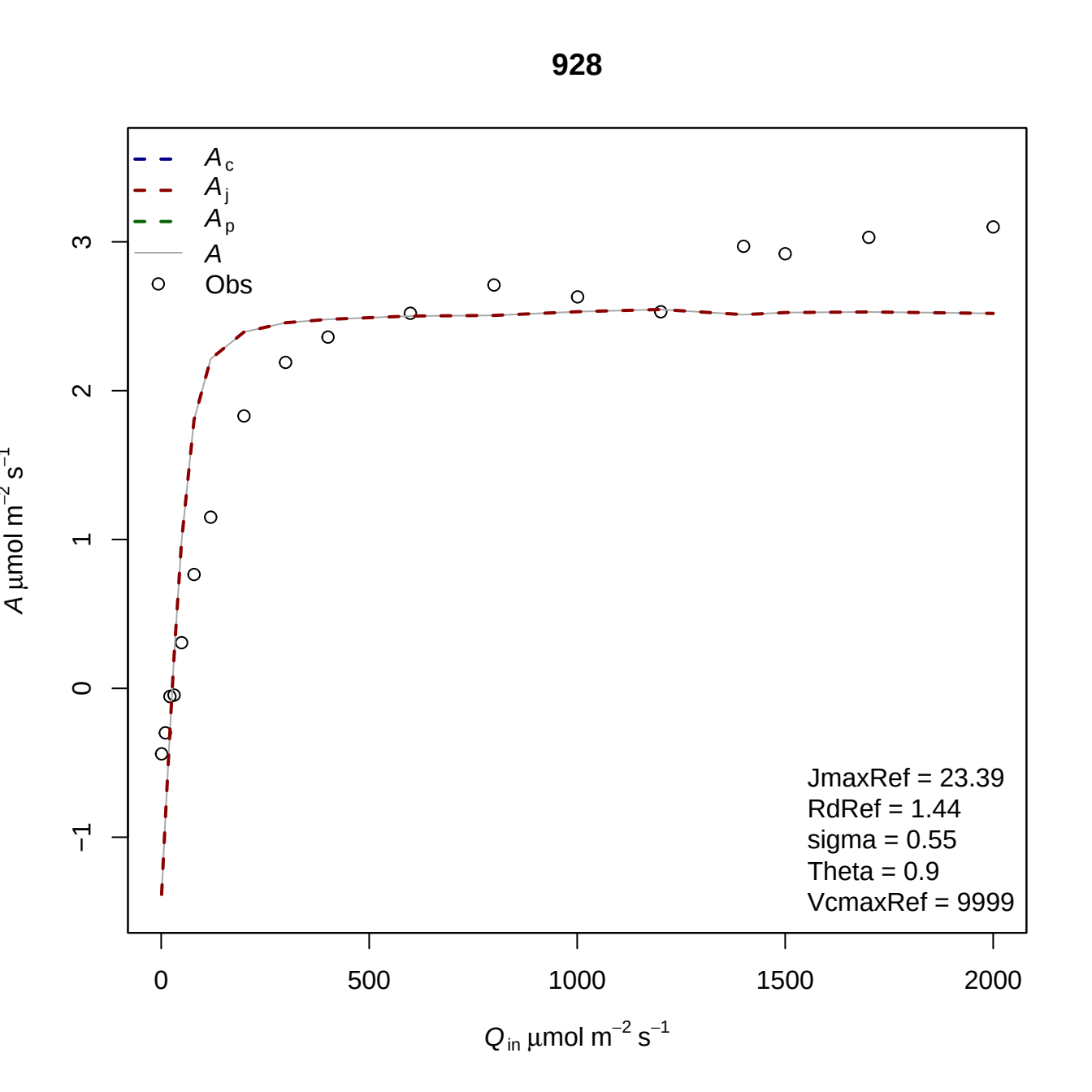

923
A_c
A_j
A_p
A
Obs
JmaxRef = 25.48
RdRef = 1.35
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

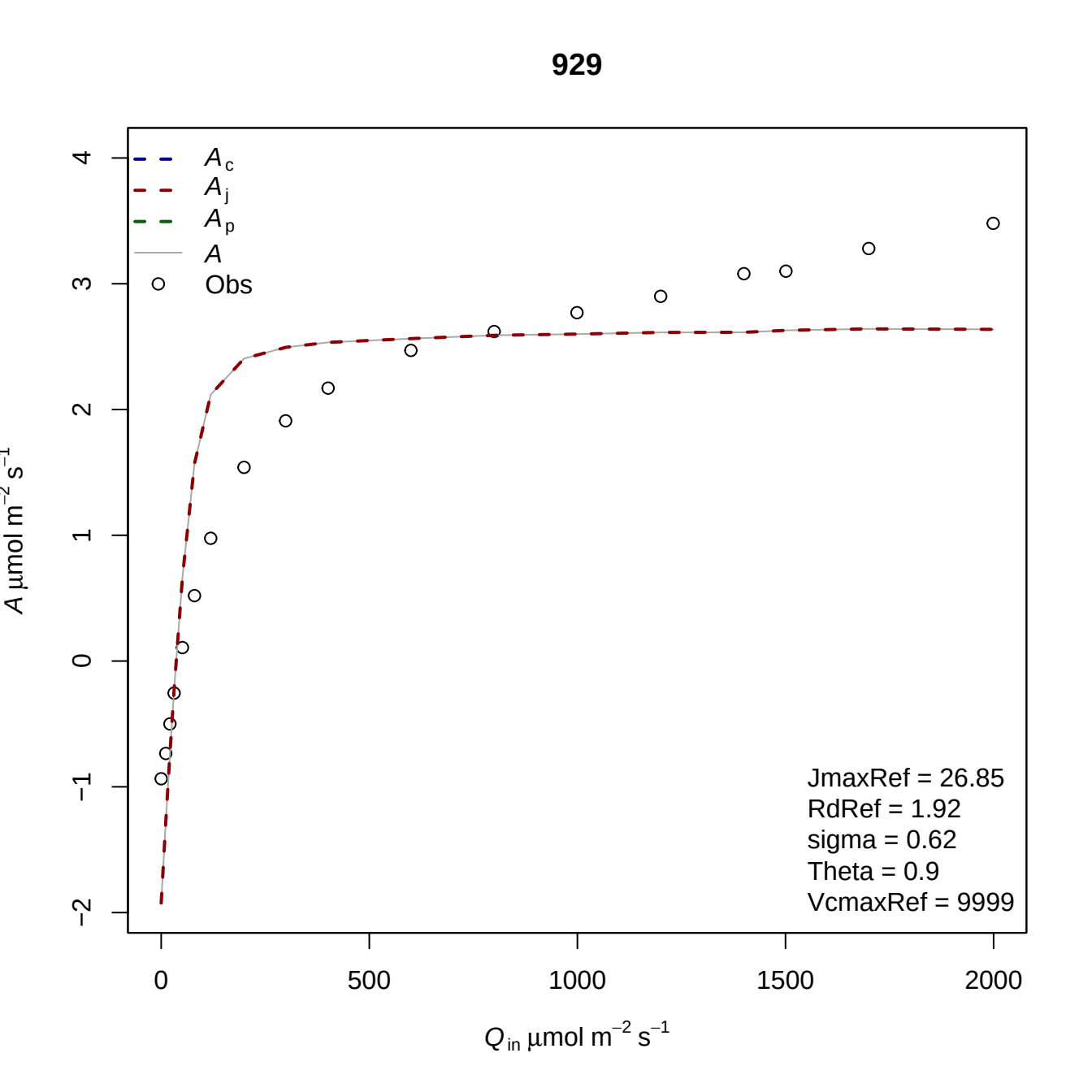

924
A_c
A_j
A_p
A
Obs
JmaxRef = 20.1
RdRef = 1.72
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

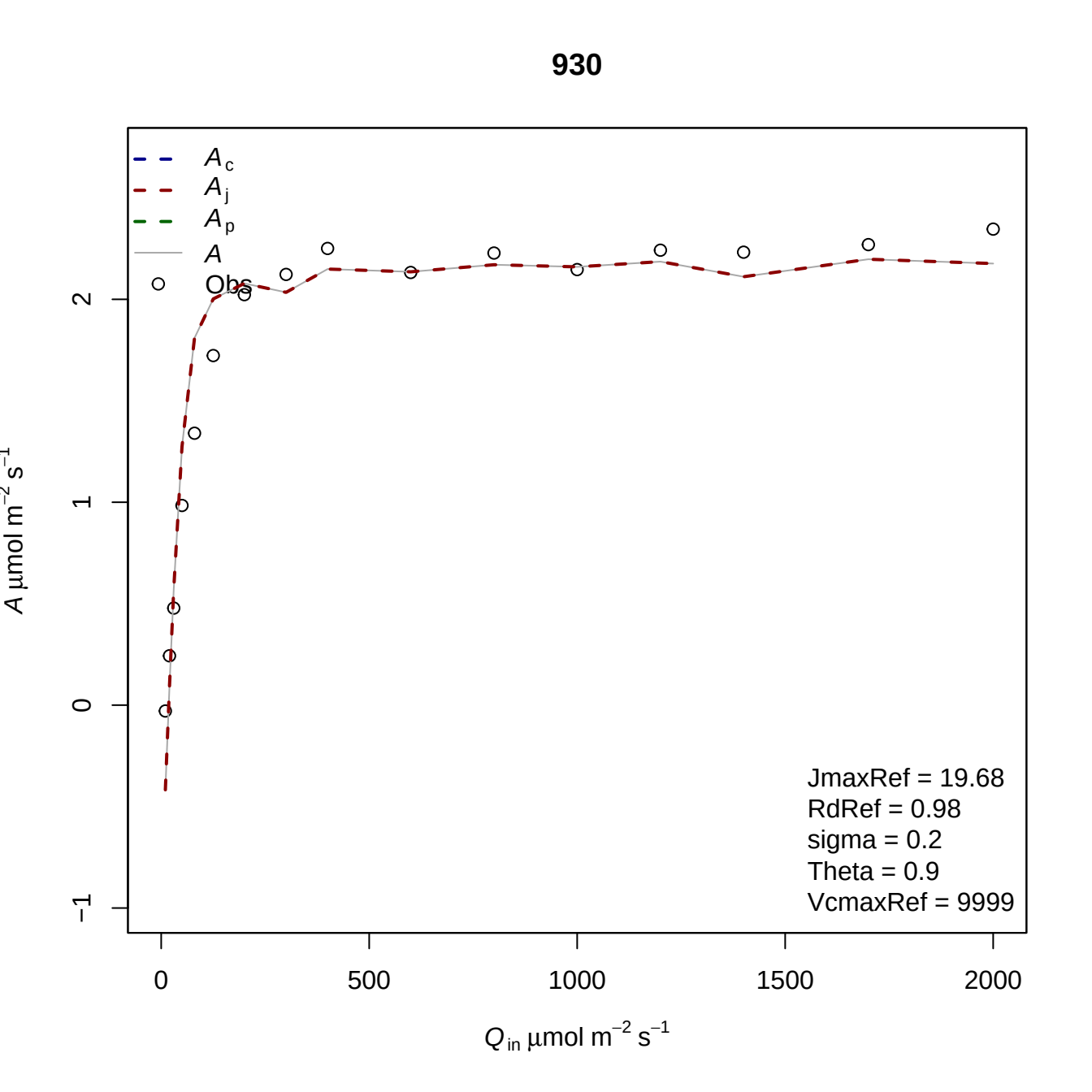

925
A_c
A_j
A_p
A
Obs
JmaxRef = 22
RdRef = 0.73
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

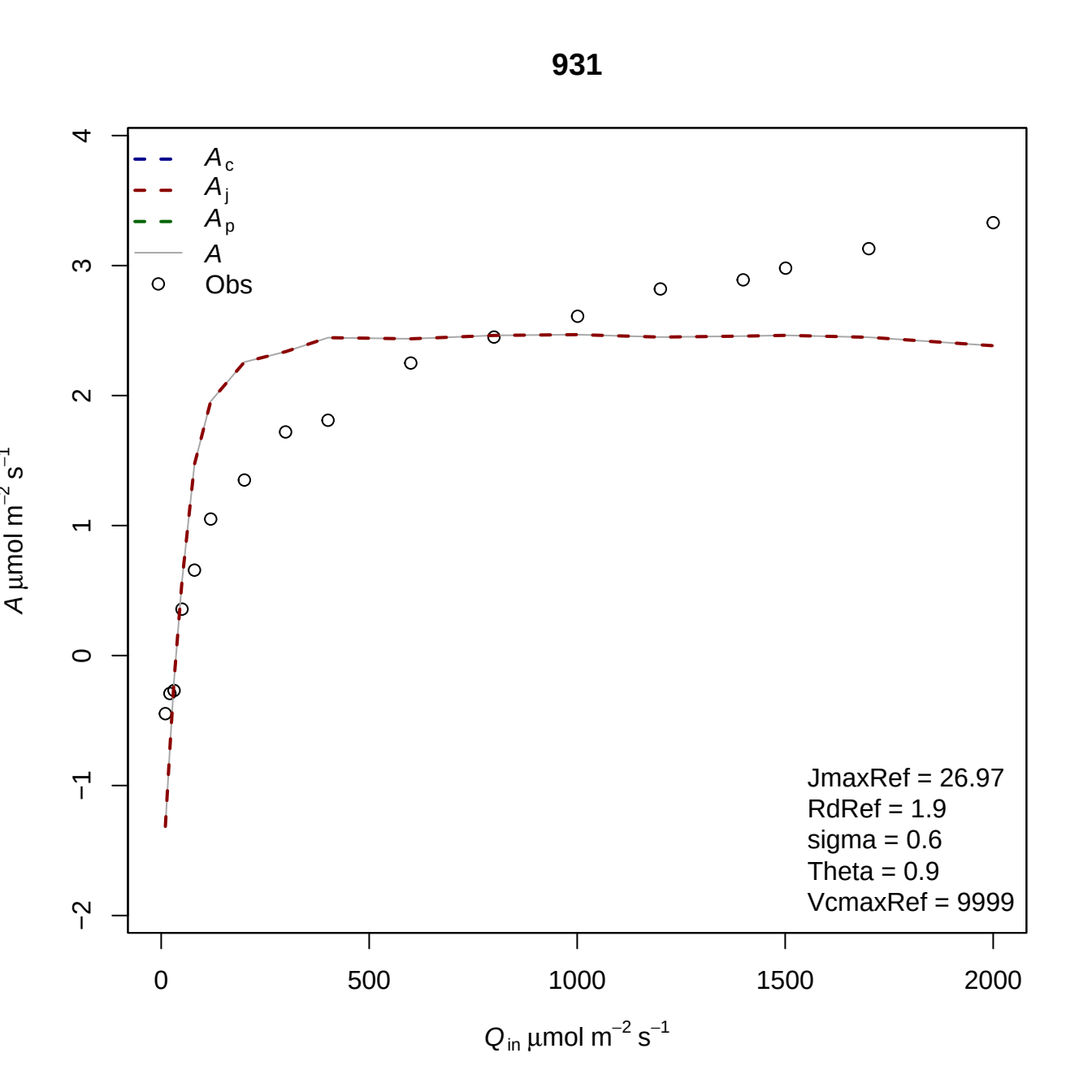

926
A_c
A_j
A_p
A
Obs
JmaxRef = 44.6
RdRef = 1.5
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

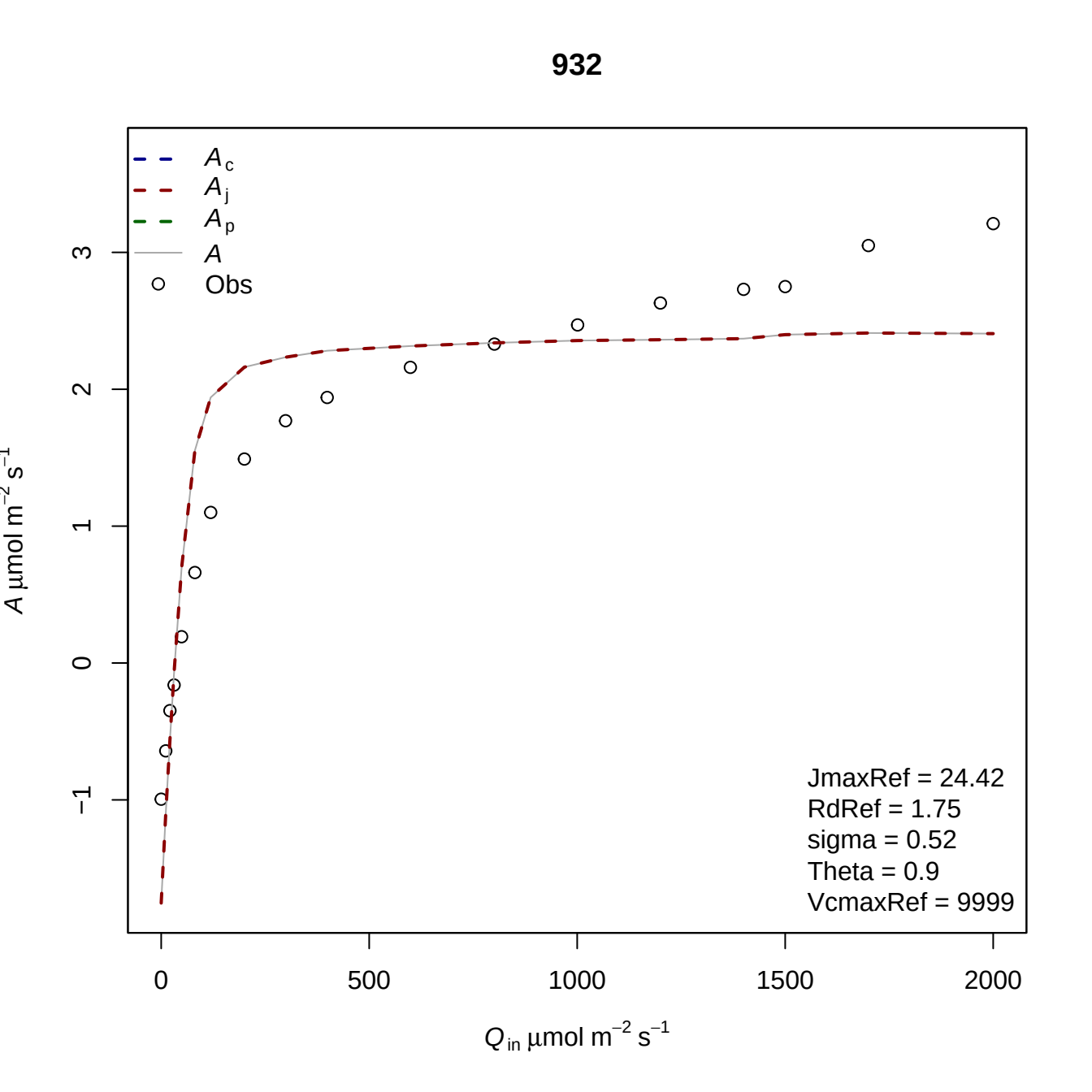

927
A_c
A_j
A_p
A
Obs
JmaxRef = 21.9
RdRef = 1.57
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

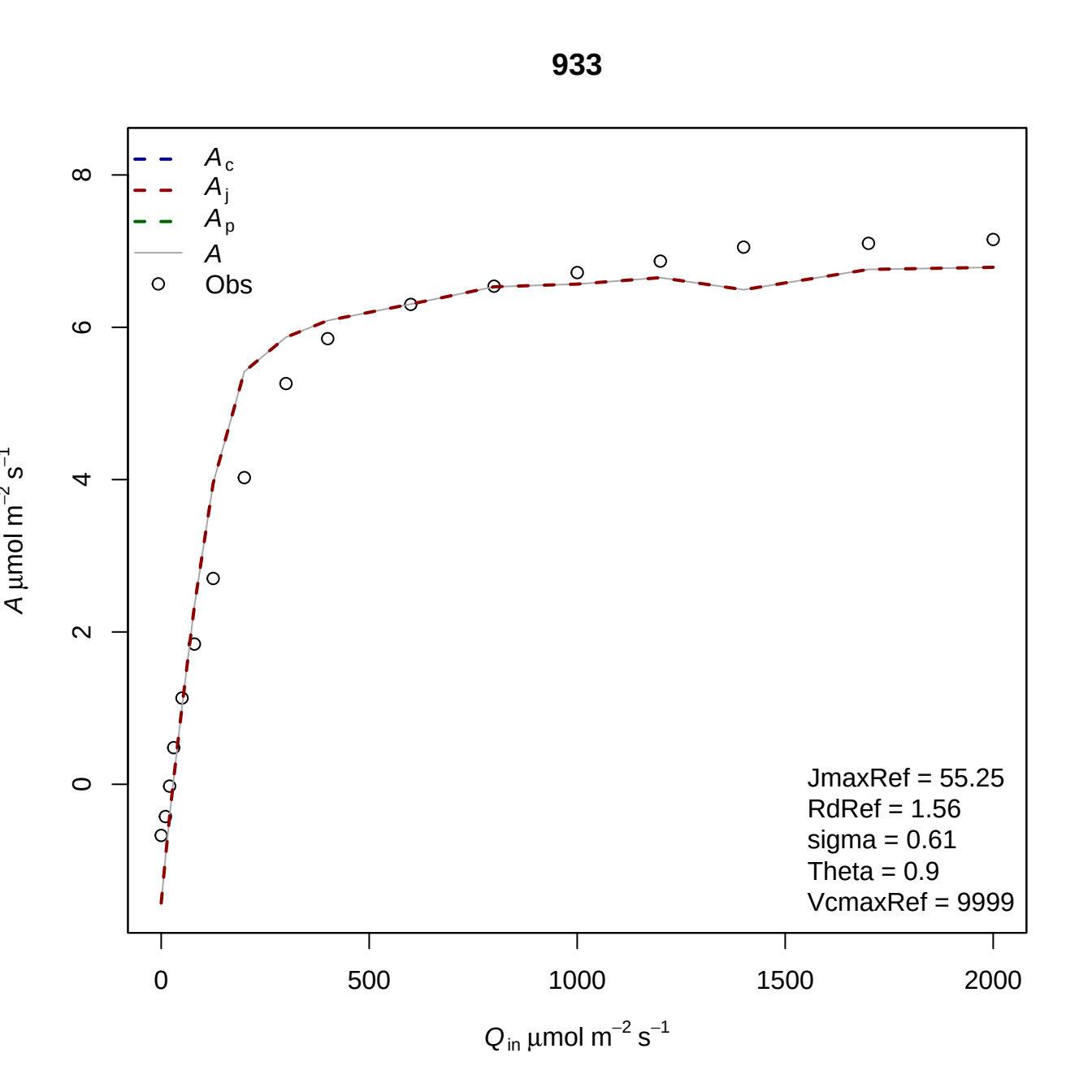

928
A_c
A_j
A_p
A
Obs
JmaxRef = 23.39
RdRef = 1.44
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

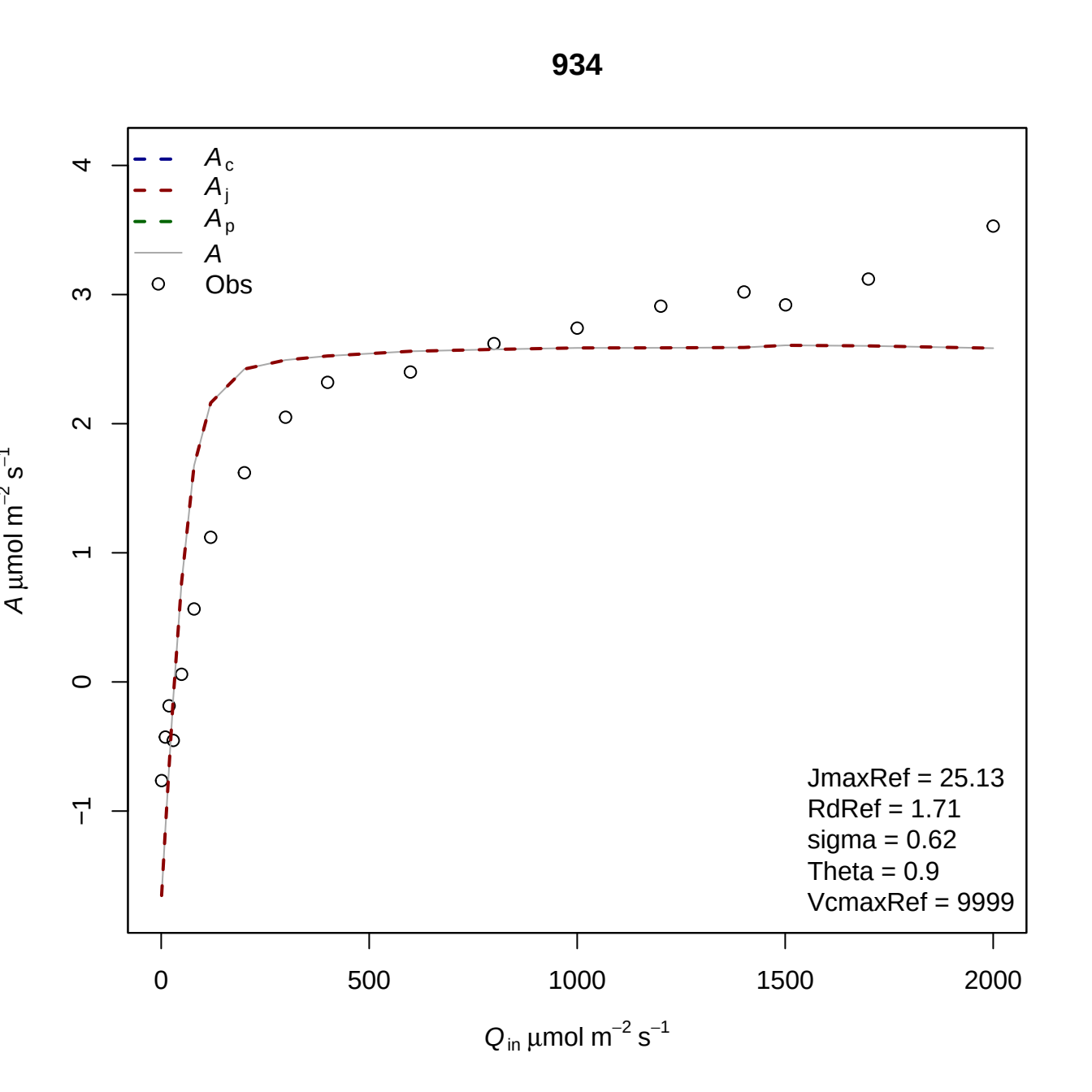

929
A_c
A_j
A_p
A
Obs
JmaxRef = 26.85
RdRef = 1.92
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

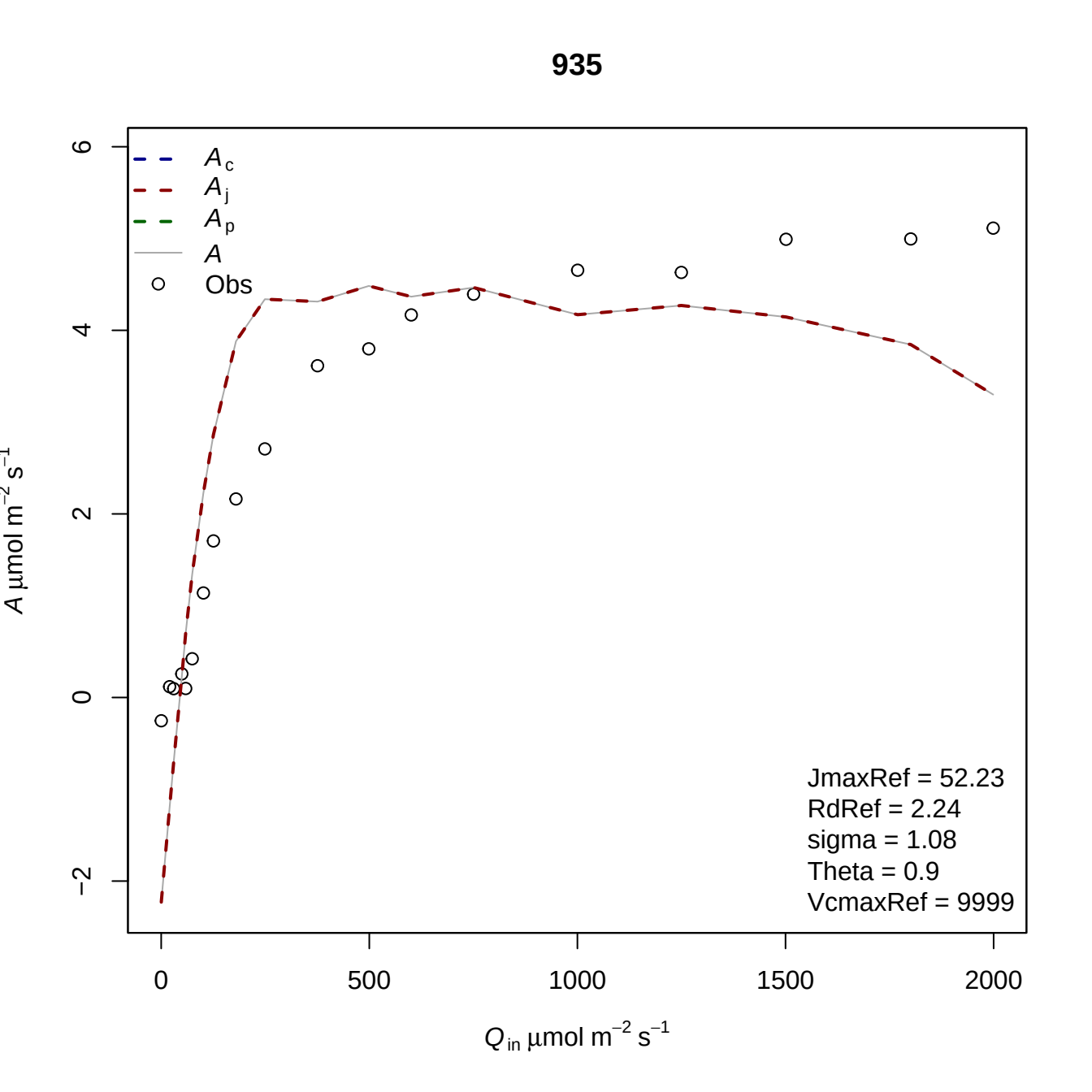

930
A_c
A_j
A_p
A
Obs
JmaxRef = 19.68
RdRef = 0.98
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

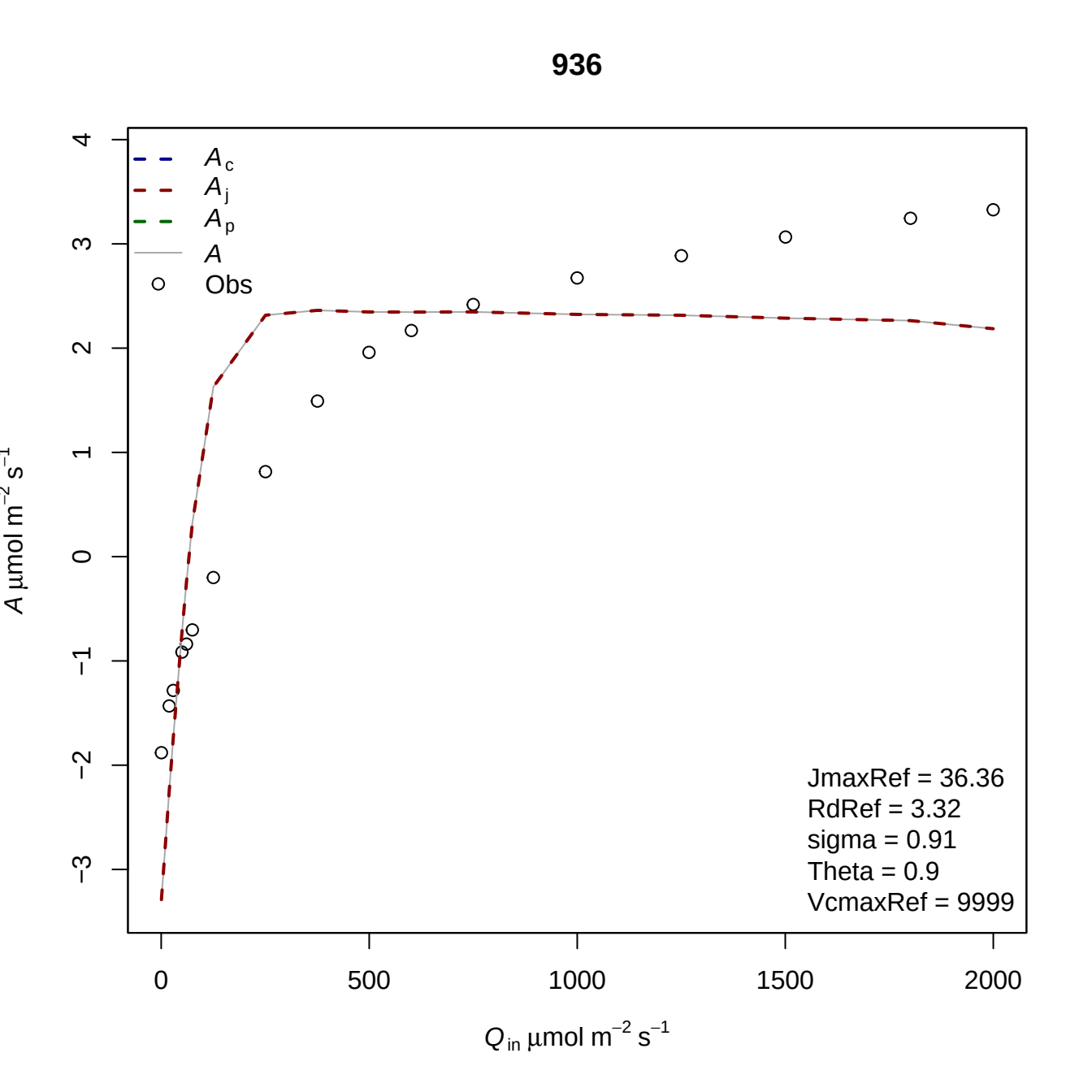

931
A_c
A_j
A_p
A
Obs
JmaxRef = 26.97
RdRef = 1.9
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

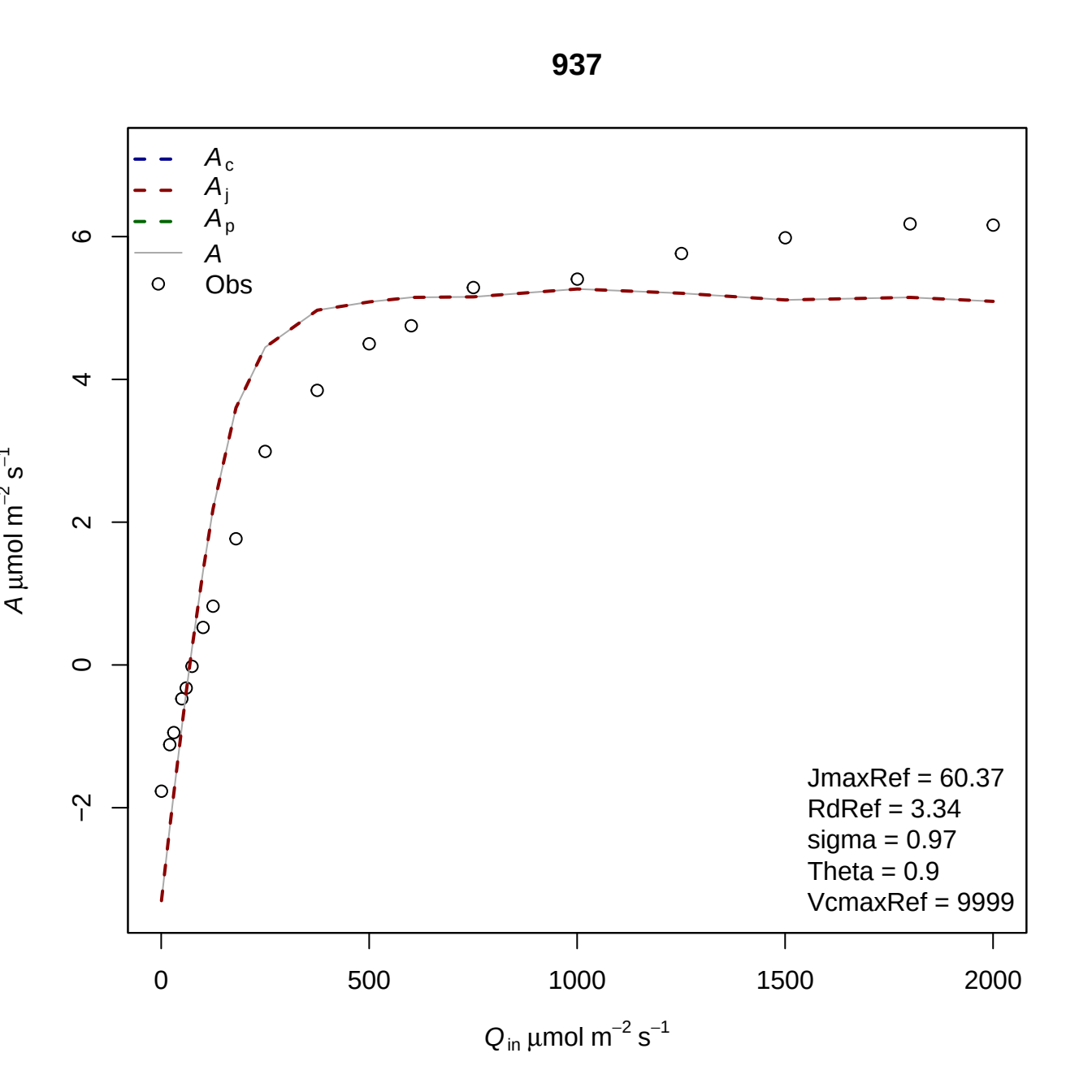

932
A_c
A_j
A_p
A
Obs
JmaxRef = 24.42
RdRef = 1.75
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

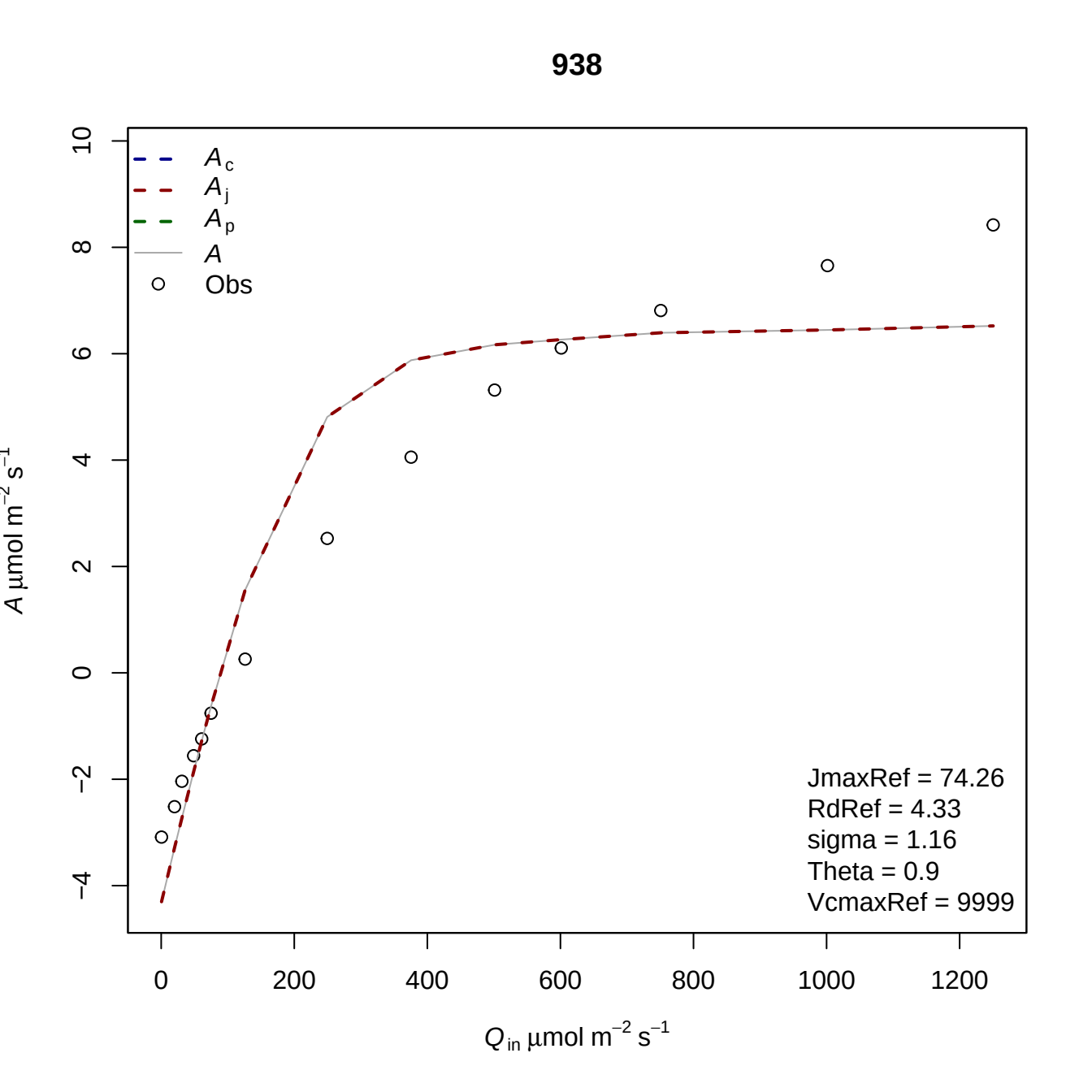

933
A_c
A_j
A_p
A
Obs
JmaxRef = 55.25
RdRef = 1.56
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

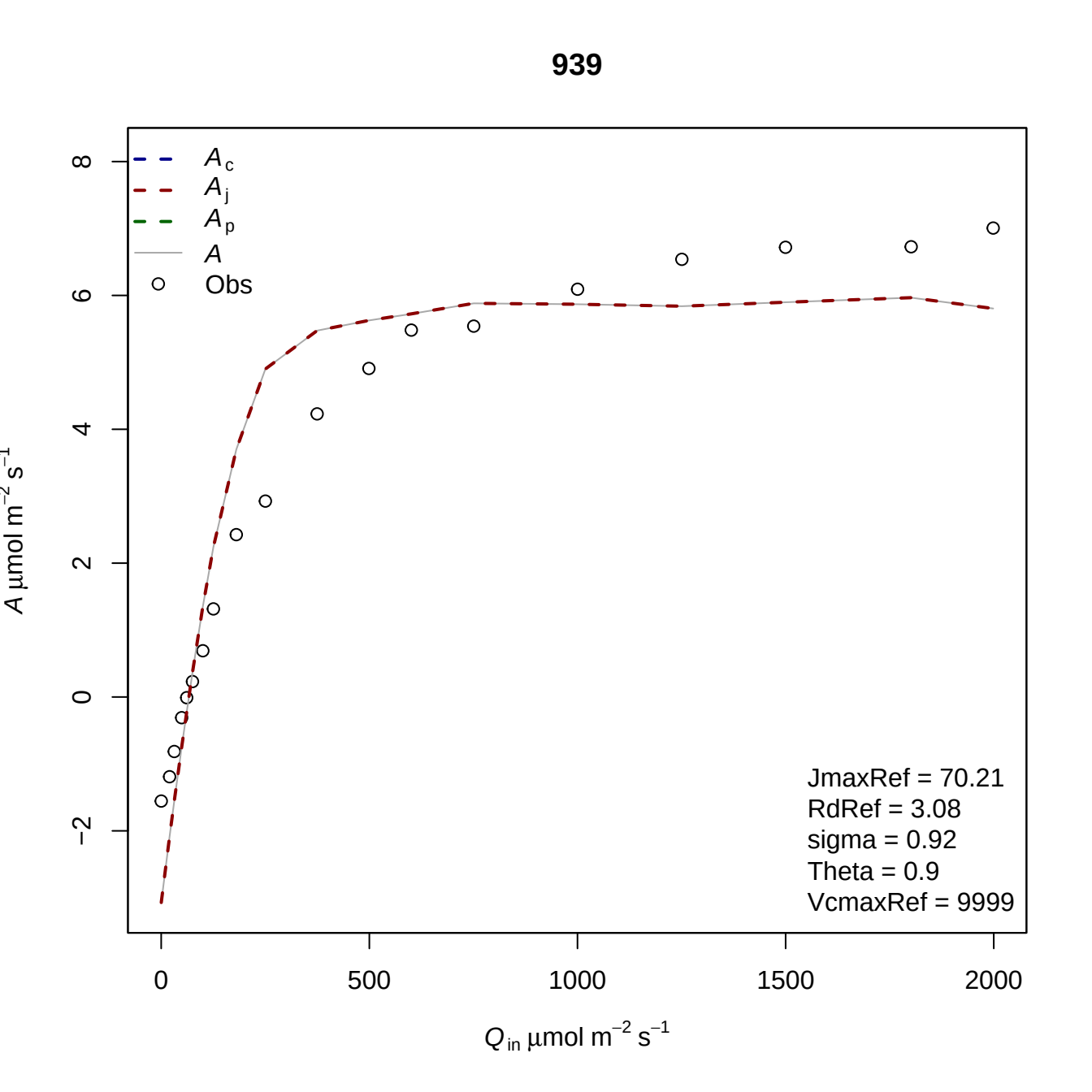

934
A_c
A_j
A_p
A
Obs
JmaxRef = 25.13
RdRef = 1.71
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

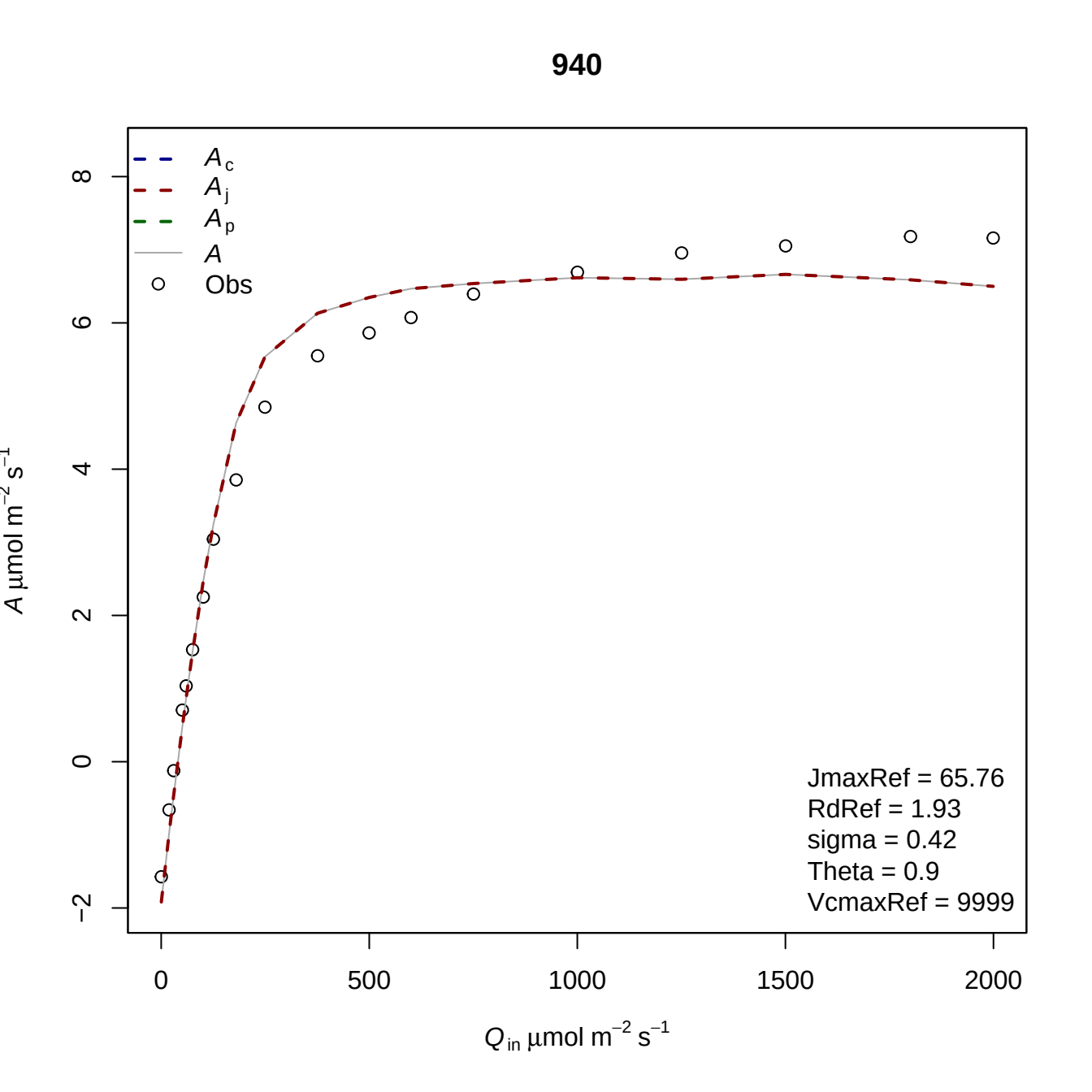

935
A_c
A_j
A_p
A
Obs
JmaxRef = 52.23
RdRef = 2.24
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

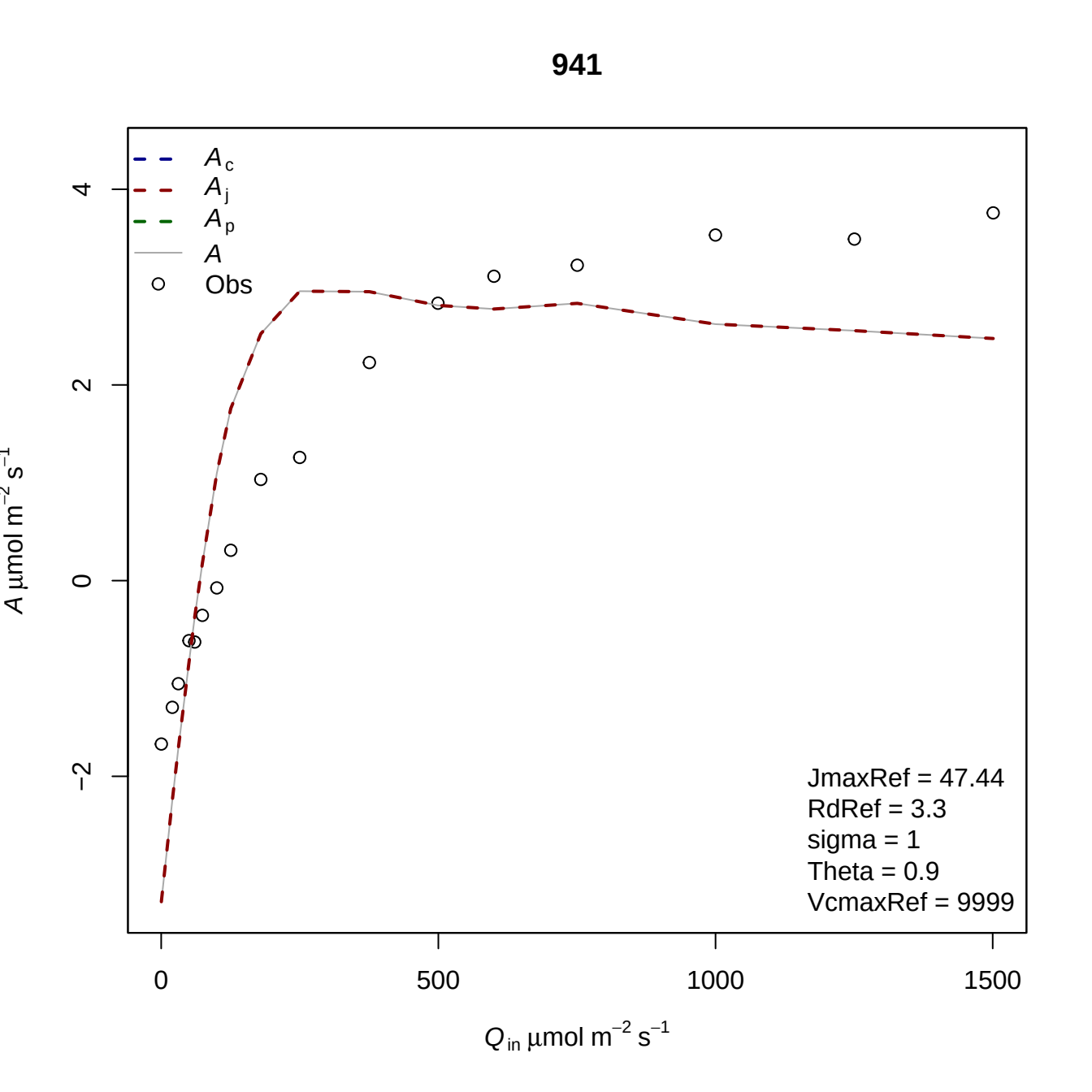

936
A_c
A_j
A_p
A
Obs
JmaxRef = 36.36
RdRef = 3.32
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

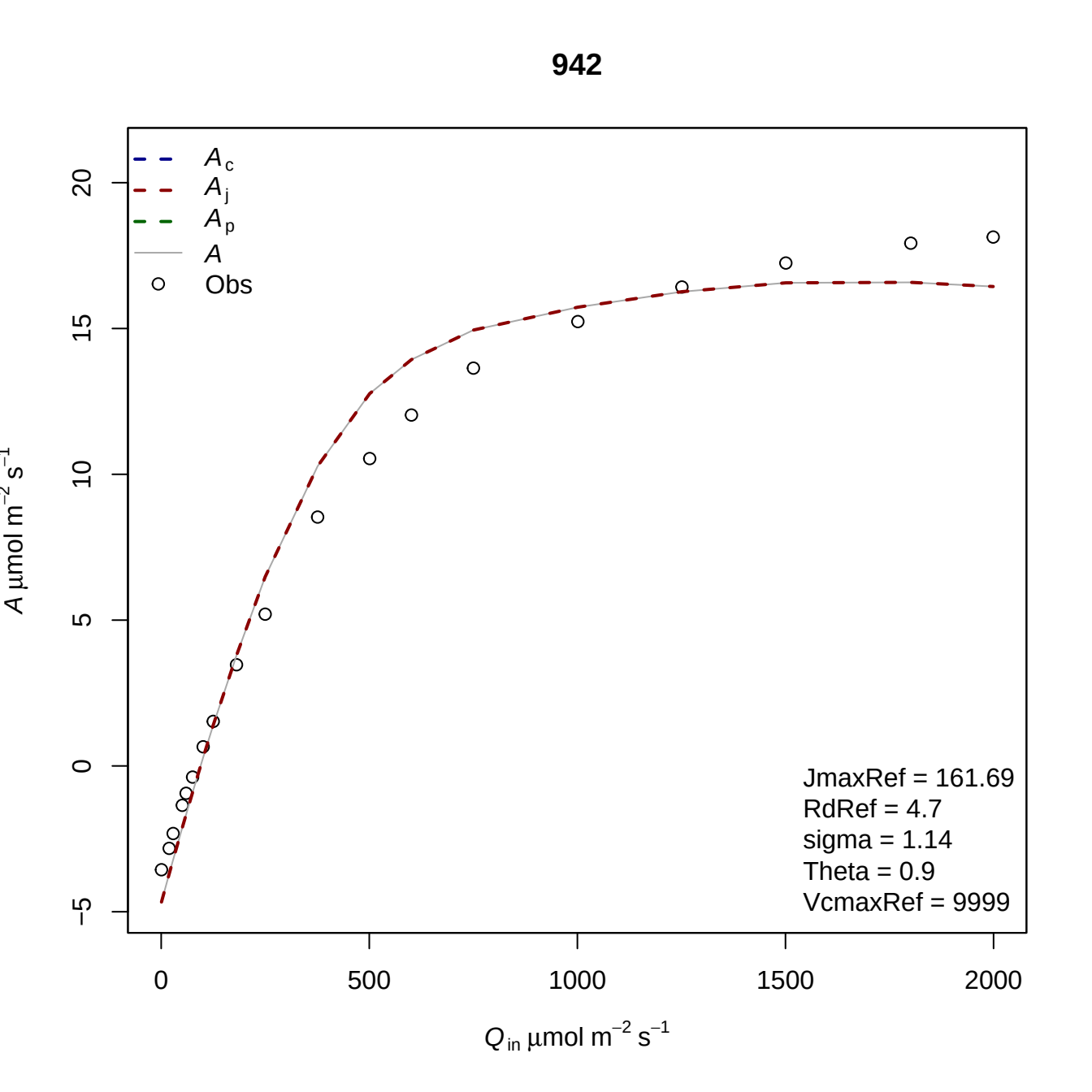

937
A_c
A_j
A_p
A
Obs
JmaxRef = 60.37
RdRef = 3.34
sigma = 0.97
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

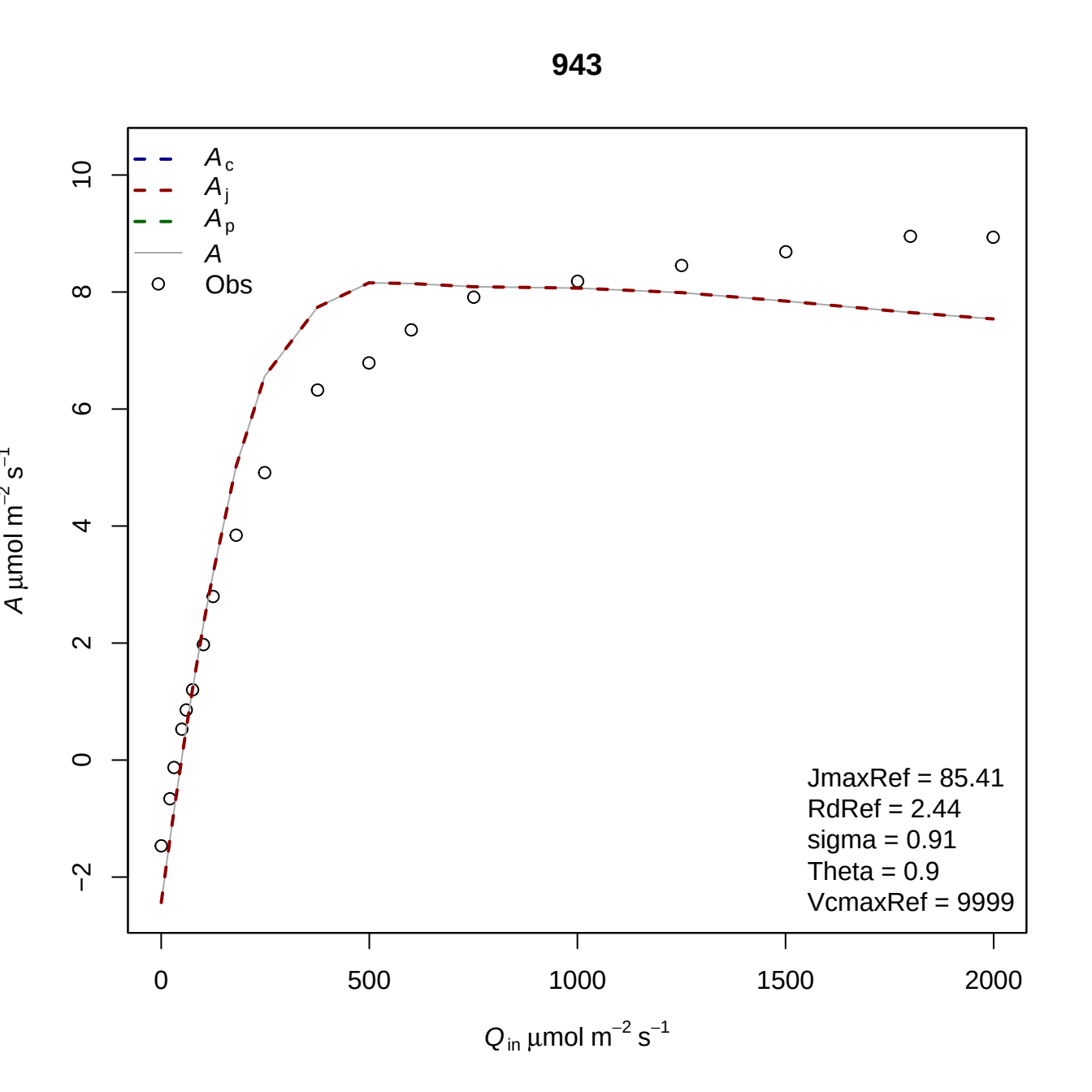

938
A_c
A_j
A_p
A
Obs
JmaxRef = 74.26
RdRef = 4.33
sigma = 1.16
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

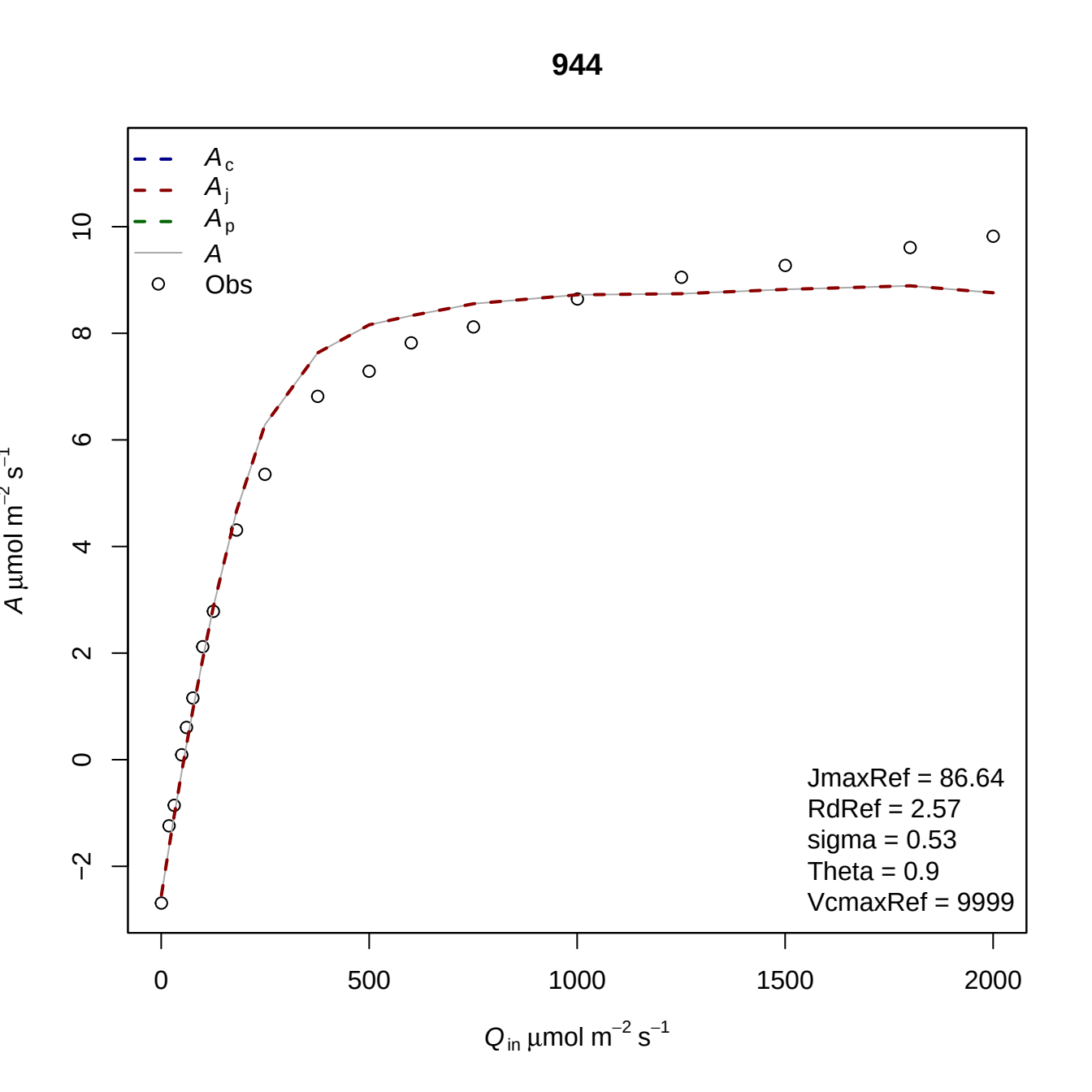

939
A_c
A_j
A_p
A
Obs
JmaxRef = 70.21
RdRef = 3.08
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

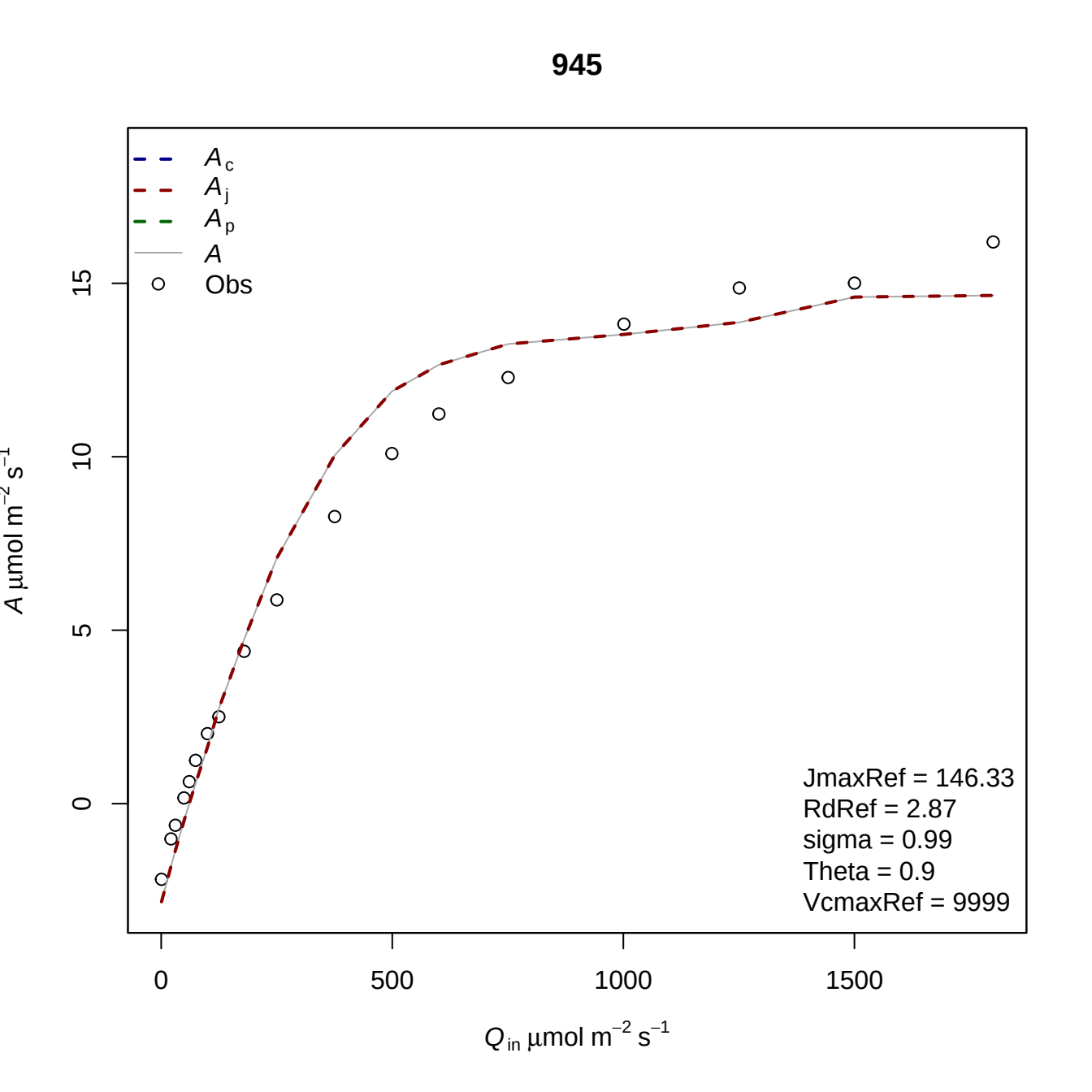

940
A_c
A_j
A_p
A
Obs
JmaxRef = 65.76
RdRef = 1.93
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

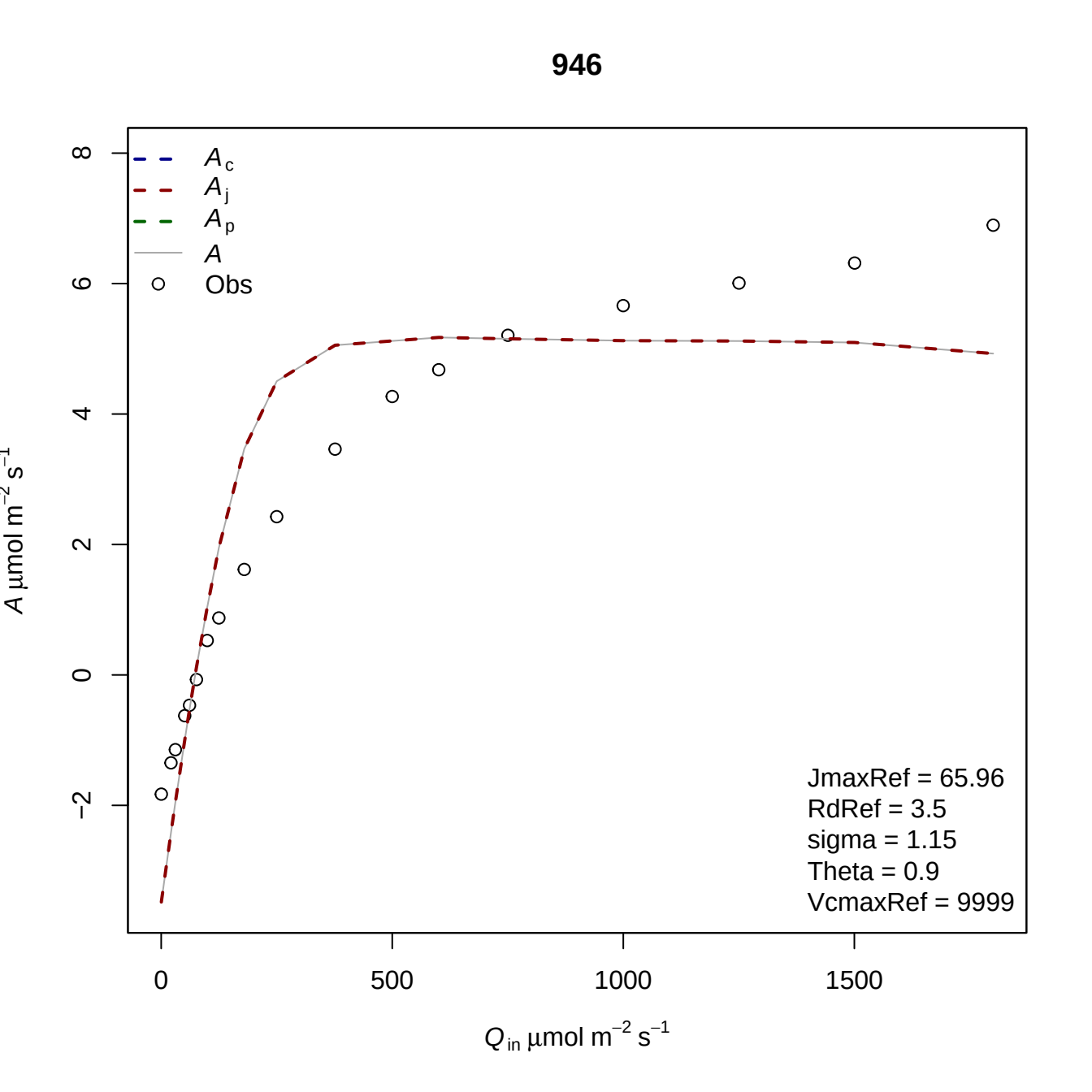

941
A_c
A_j
A_p
A
Obs
JmaxRef = 47.44
RdRef = 3.3
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500

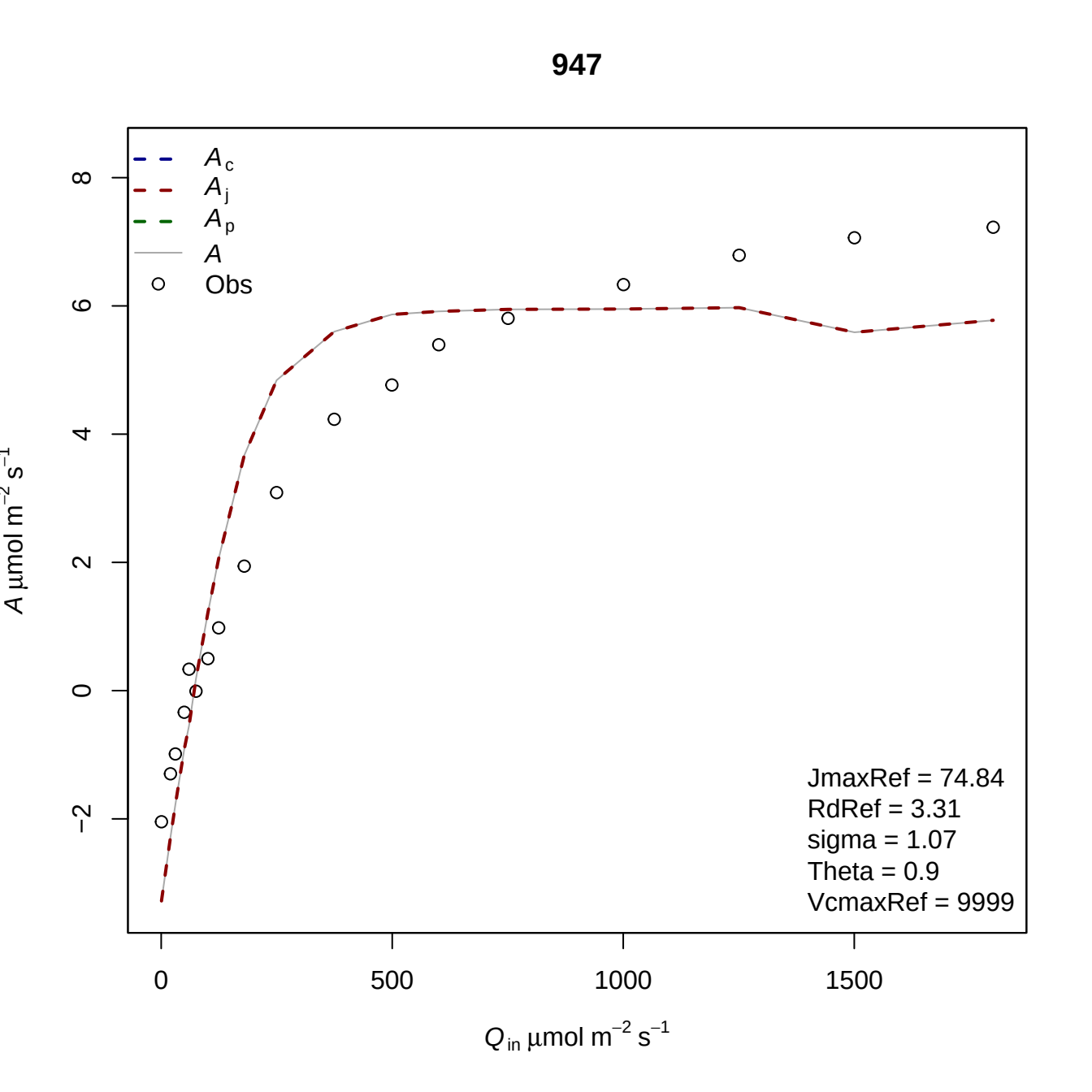

942
A_c
A_j
A_p
A
Obs
JmaxRef = 161.69
RdRef = 4.7
sigma = 1.14
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

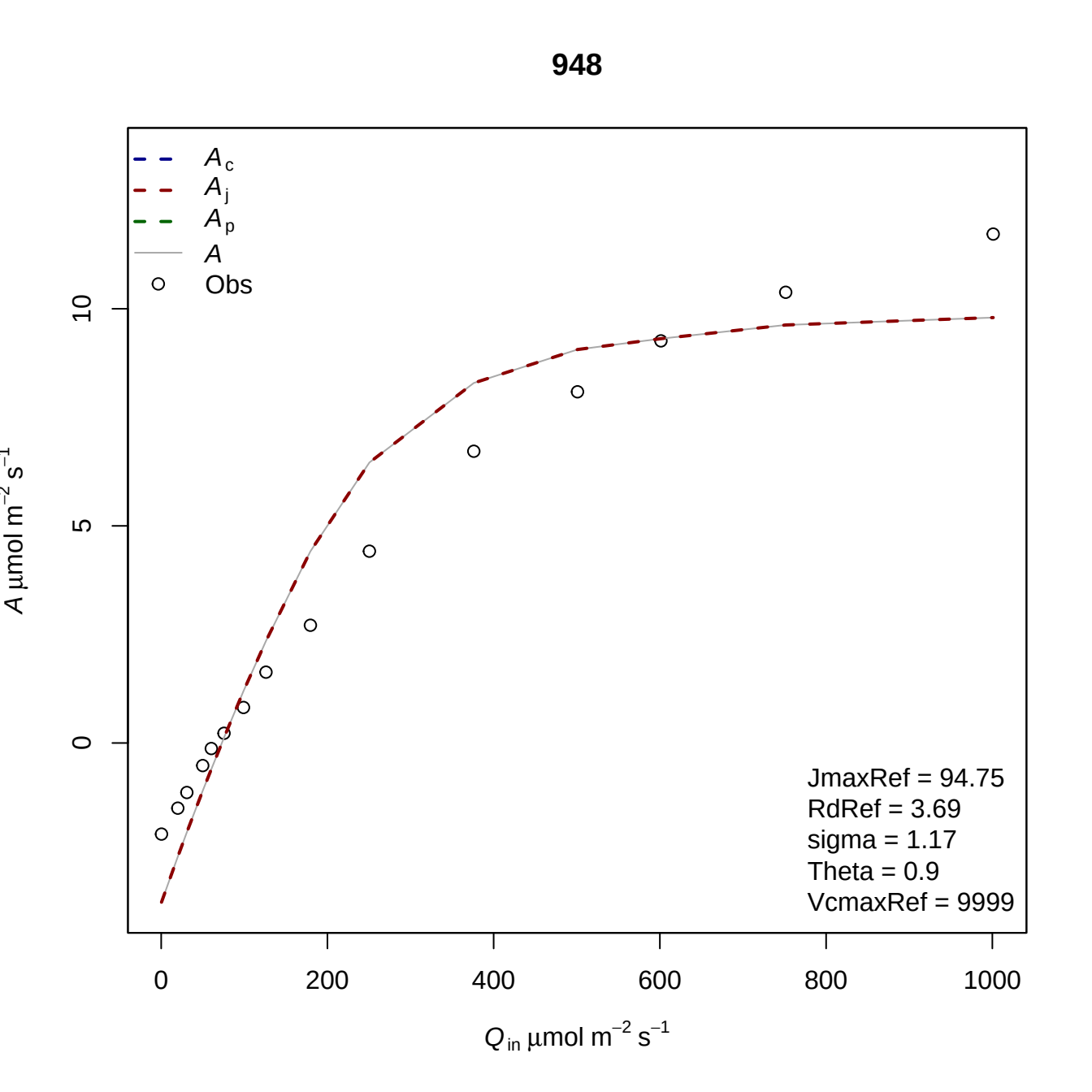

943
A_c
A_j
A_p
A
Obs
JmaxRef = 85.41
RdRef = 2.44
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

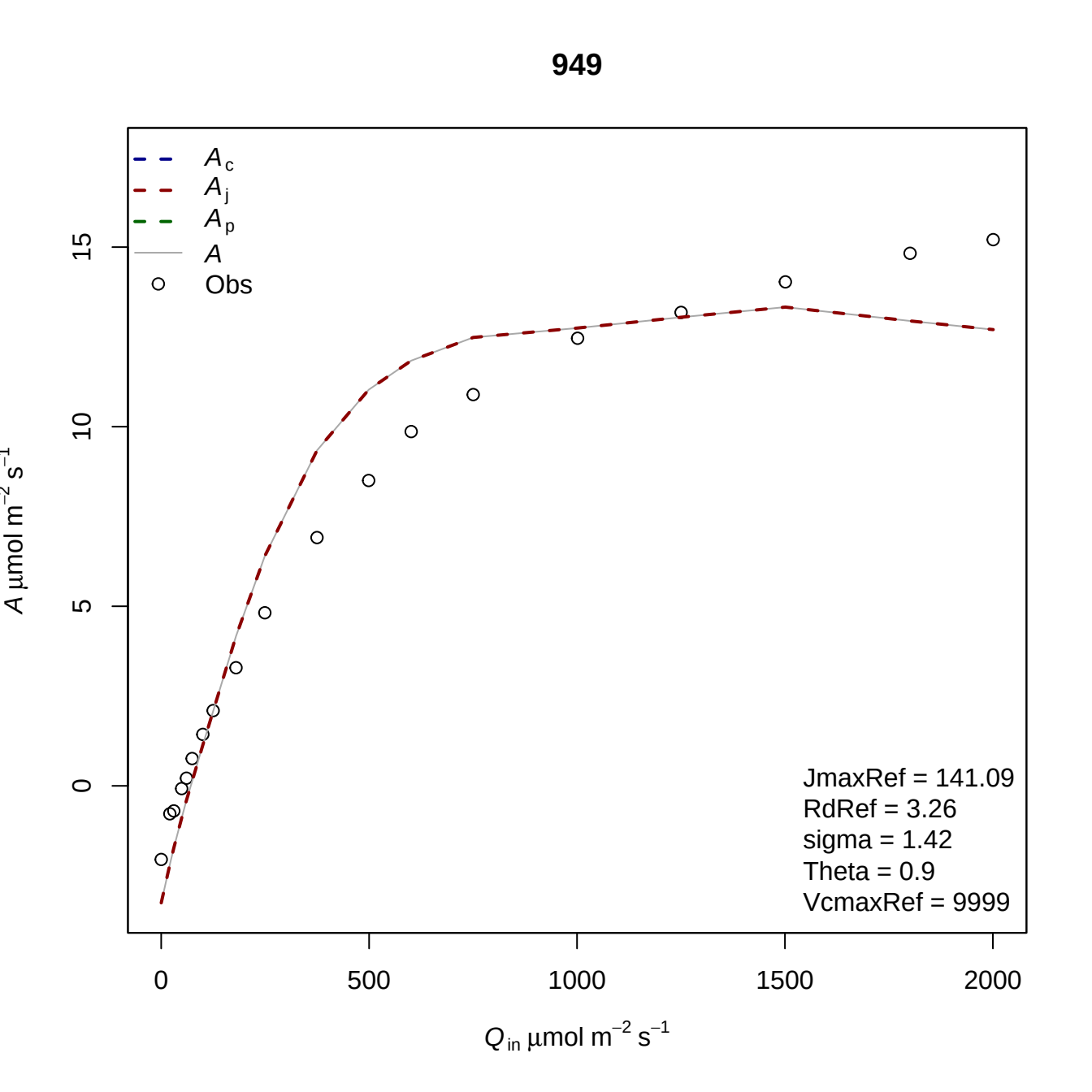

944
A_c
A_j
A_p
A
Obs
JmaxRef = 86.64
RdRef = 2.57
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

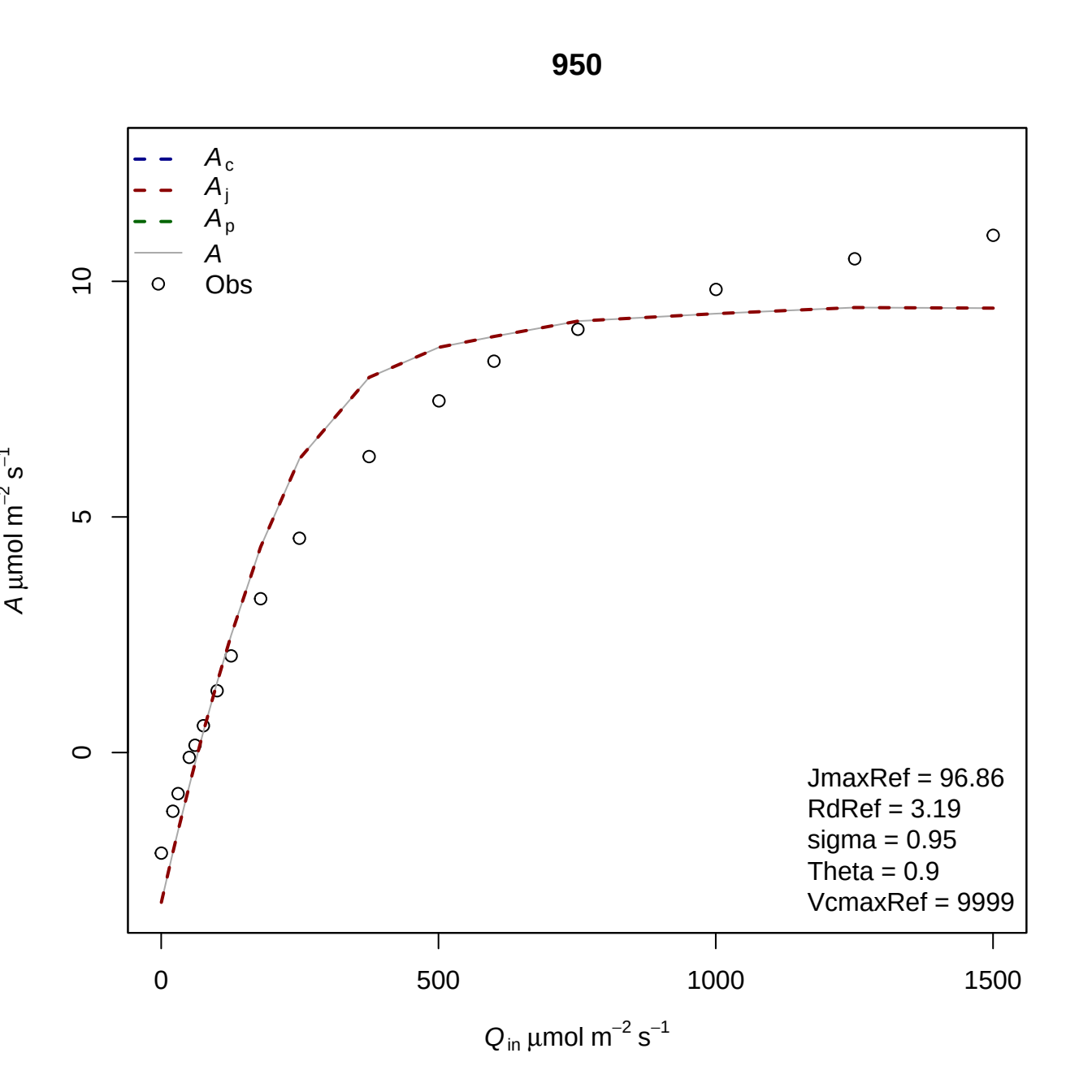

945
A_c
A_j
A_p
A
Obs
JmaxRef = 146.33
RdRef = 2.87
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

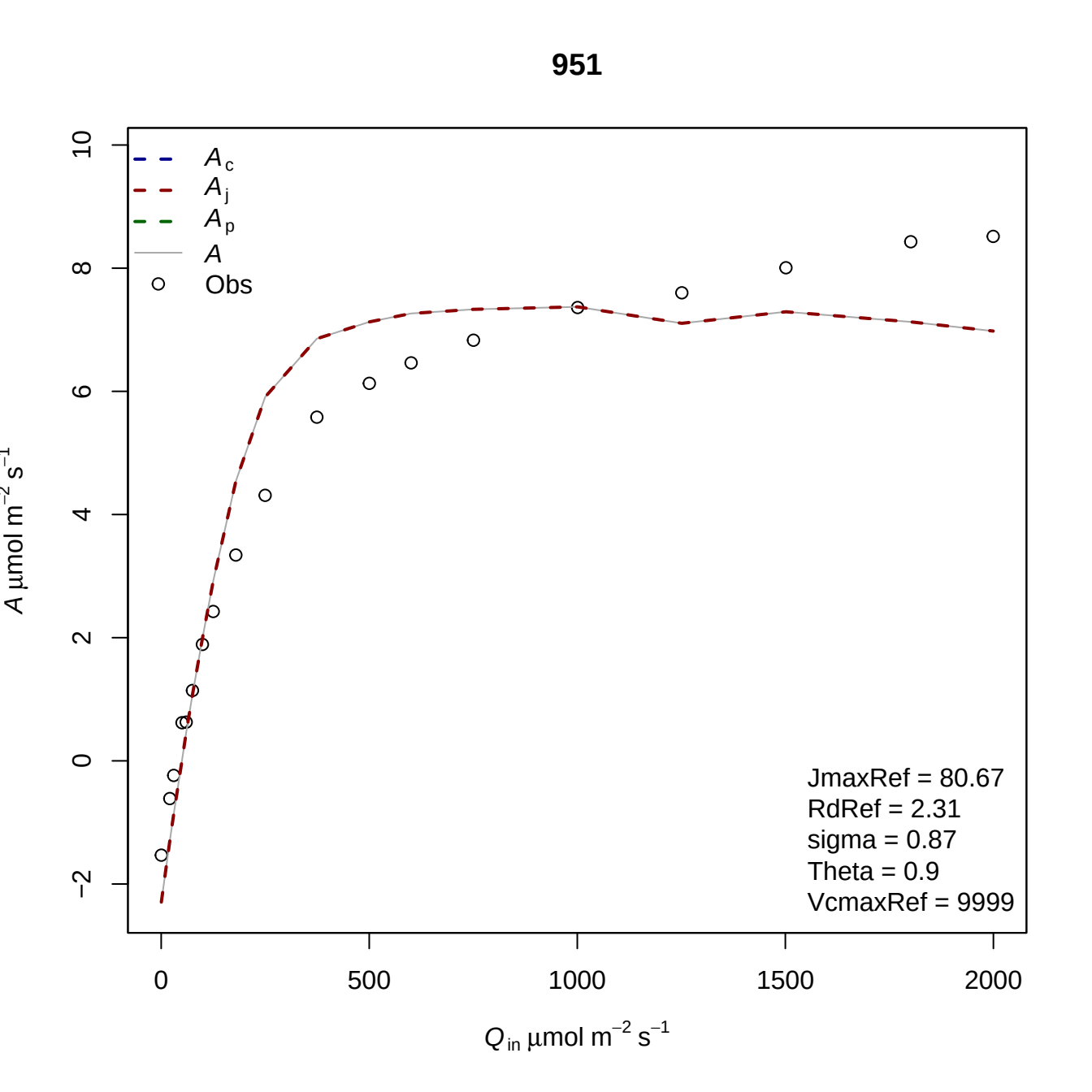

946
A_c
A_j
A_p
A
Obs
JmaxRef = 65.96
RdRef = 3.5
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

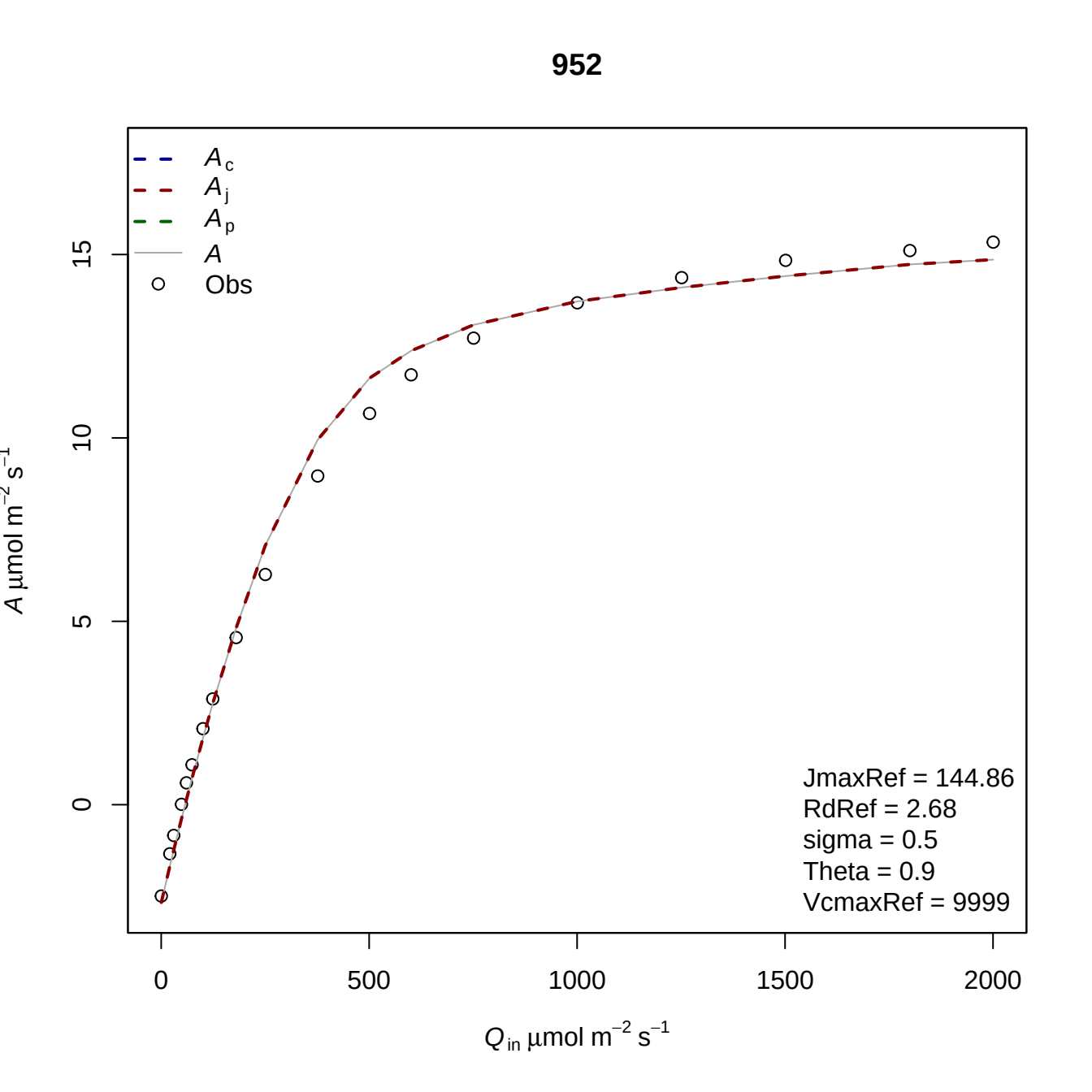

947
A_c
A_j
A_p
A
Obs
JmaxRef = 74.84
RdRef = 3.31
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

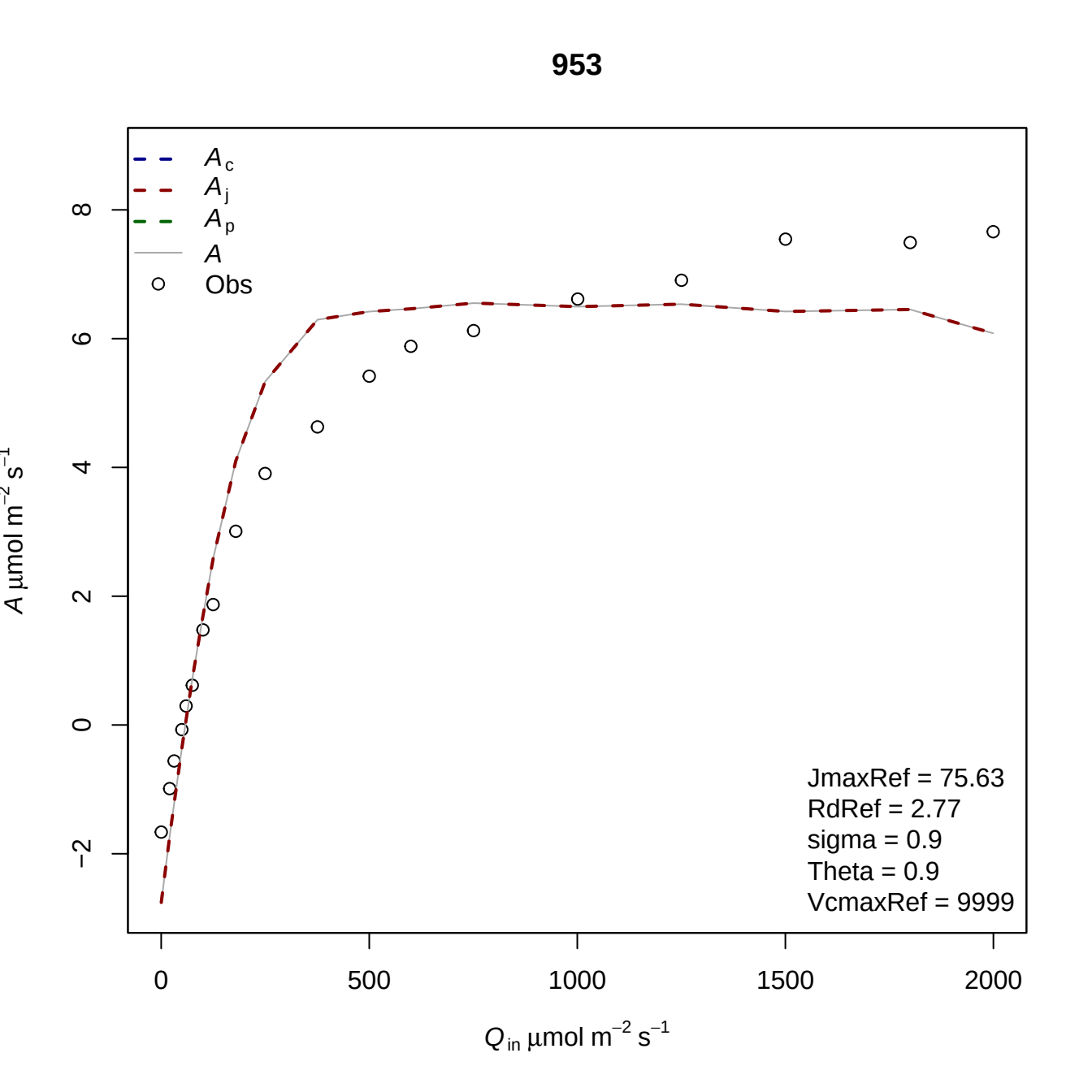

948
A_c
A_j
A_p
A
Obs
JmaxRef = 94.75
RdRef = 3.69
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

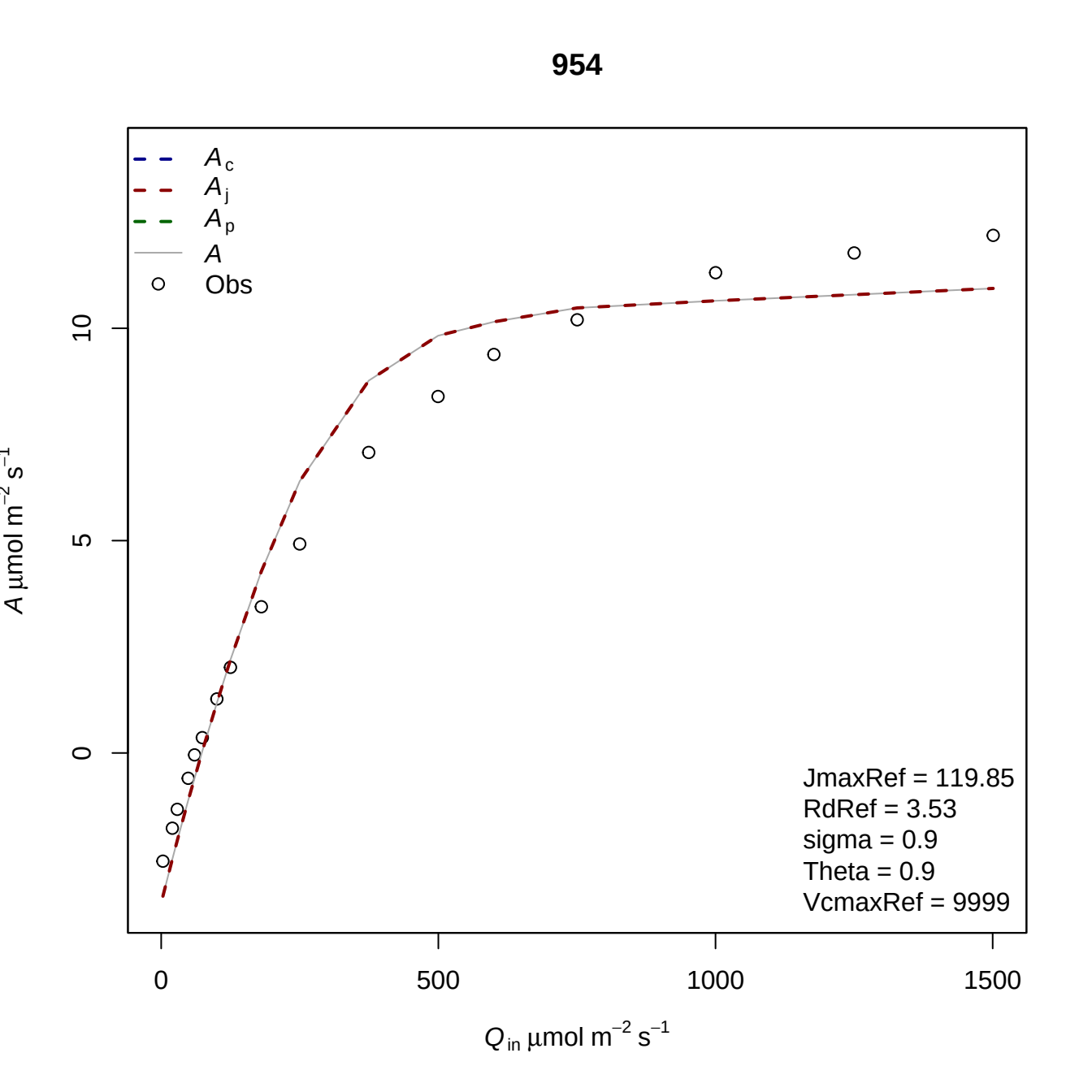

949
A_c
A_j
A_p
A
Obs
JmaxRef = 141.09
RdRef = 3.26
sigma = 1.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

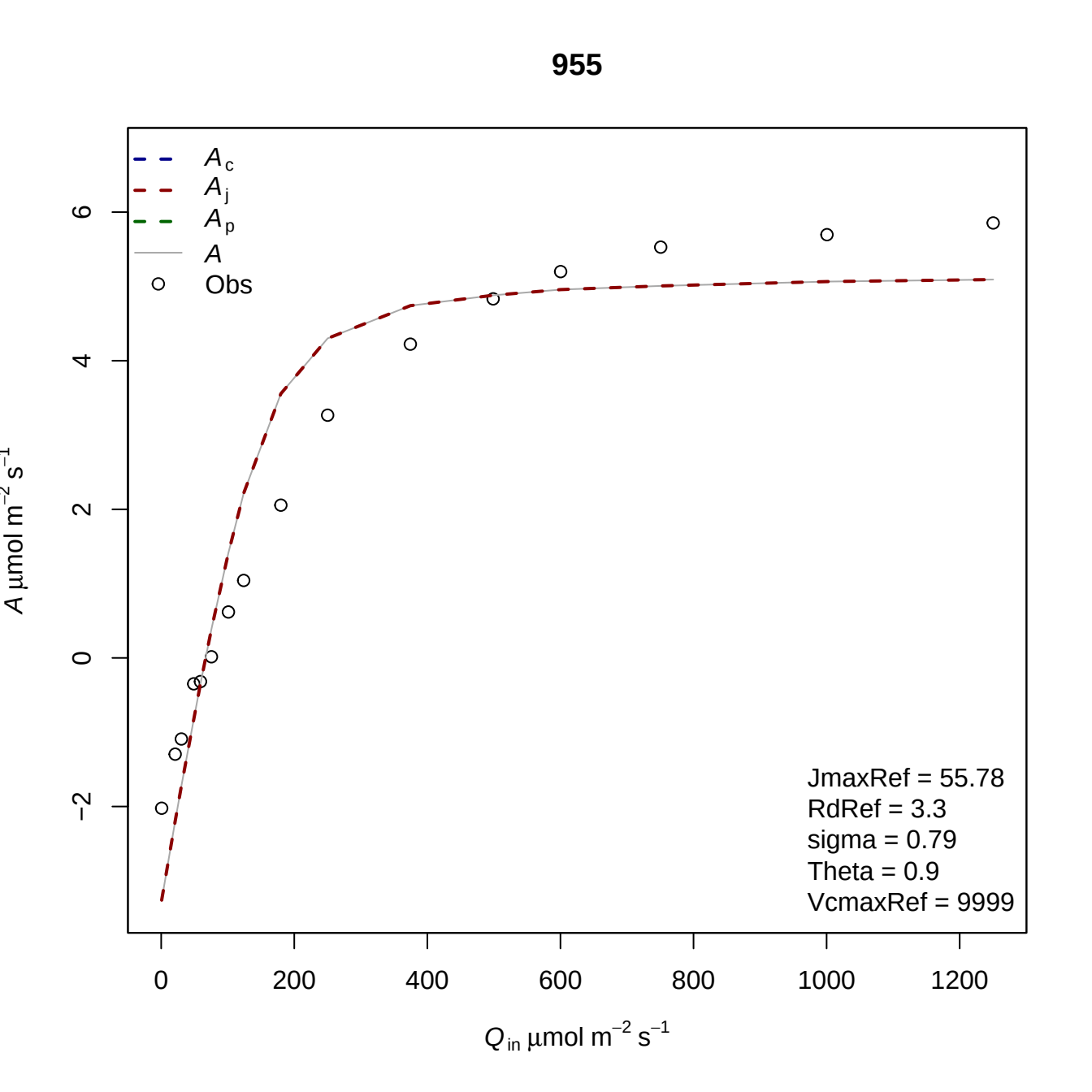

950
A_c
A_j
A_p
A
Obs
JmaxRef = 96.86
RdRef = 3.19
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

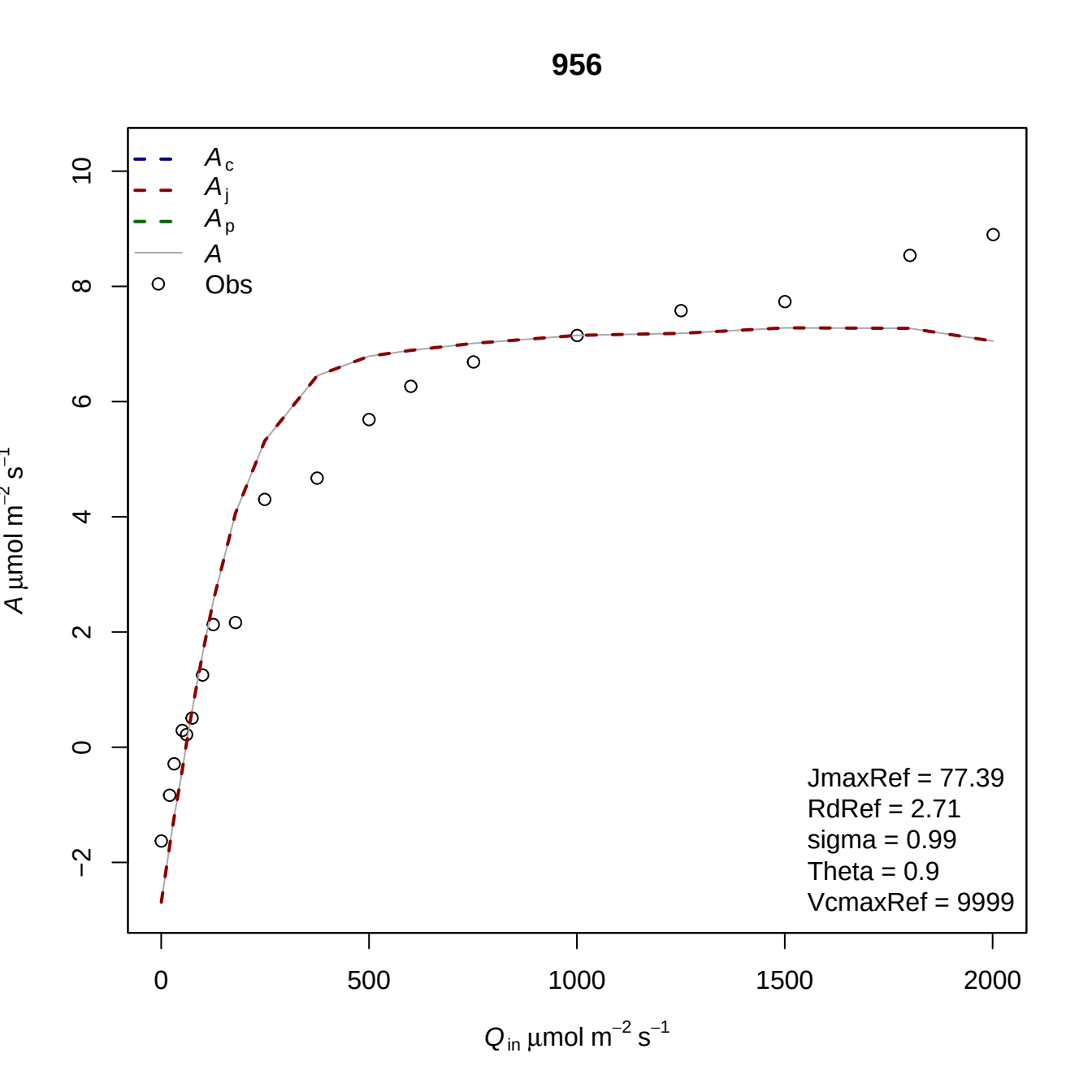

951
A_c
A_j
A_p
A
Obs
JmaxRef = 80.67
RdRef = 2.31
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

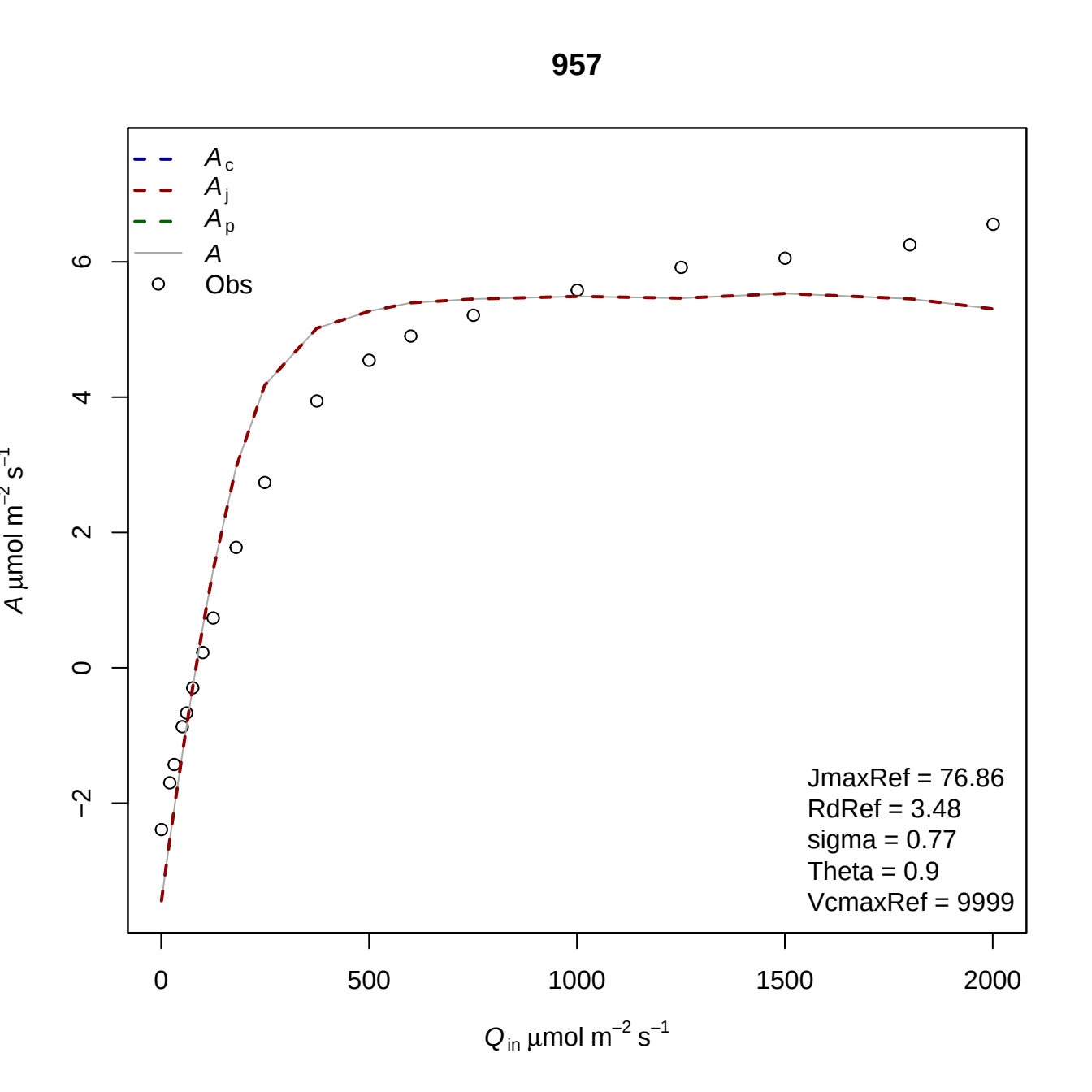

952
A_c
A_j
A_p
A
Obs
JmaxRef = 144.86
RdRef = 2.68
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

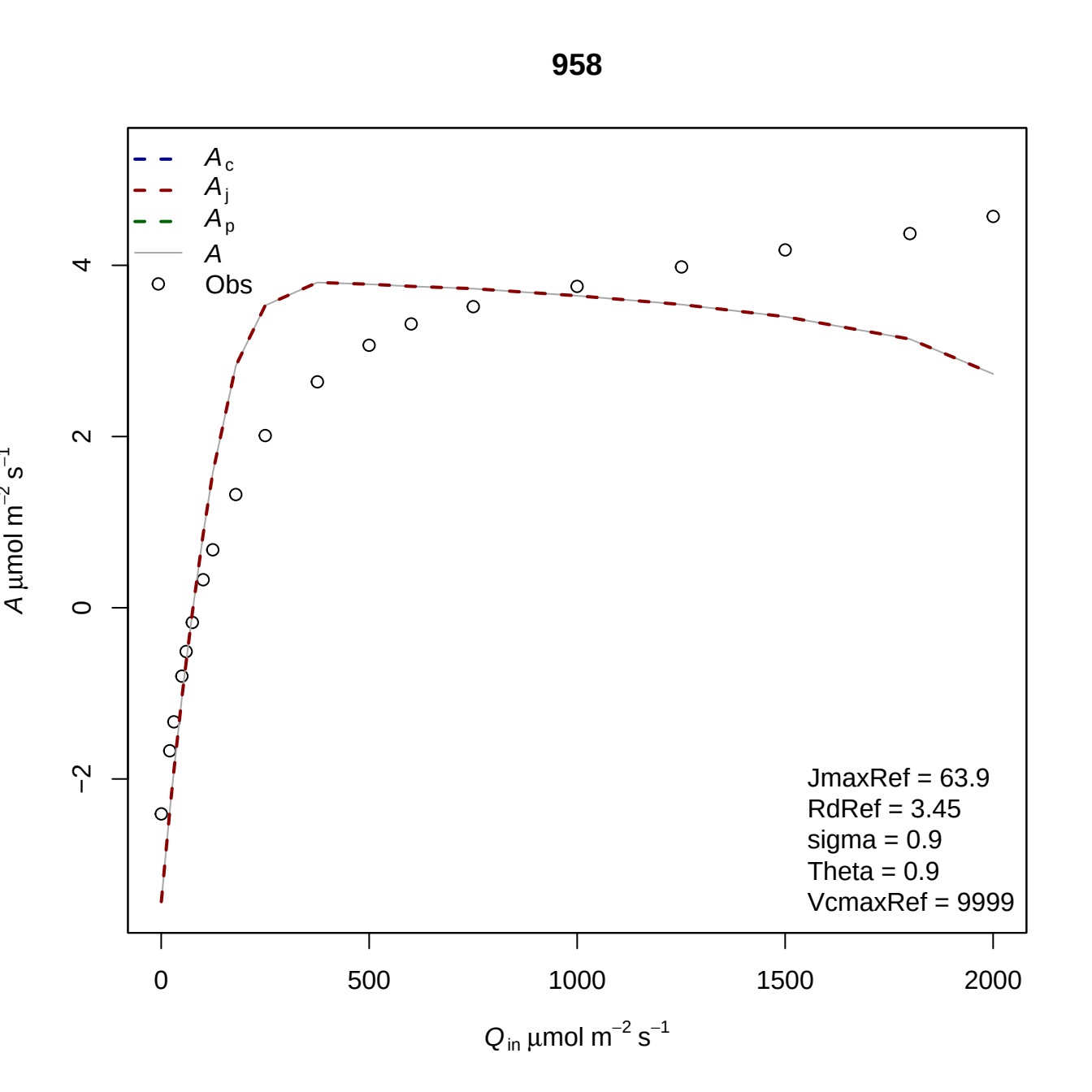

953
A_c
A_j
A_p
A
Obs
JmaxRef = 75.63
RdRef = 2.77
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

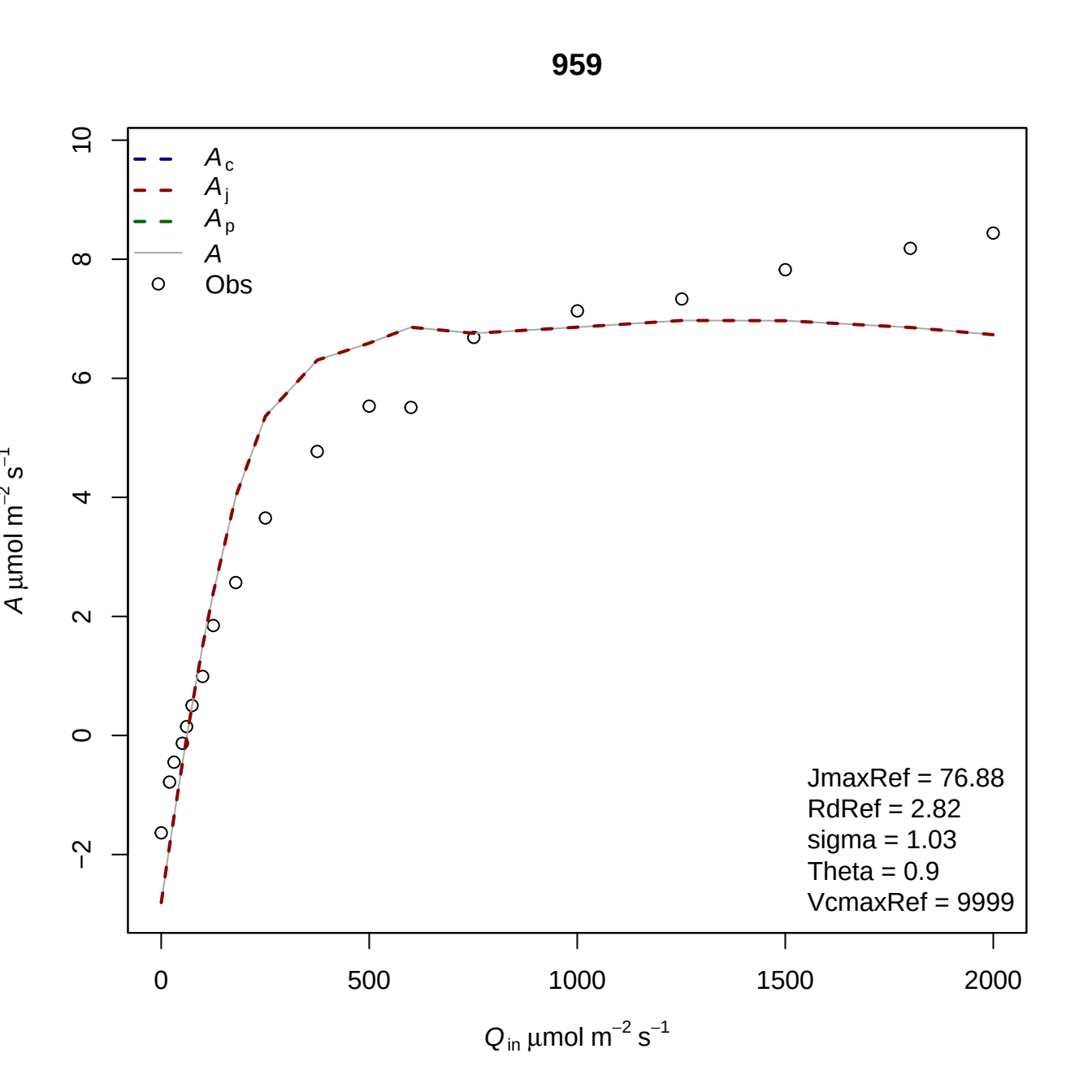

954
A_c
A_j
A_p
A
Obs
JmaxRef = 119.85
RdRef = 3.53
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

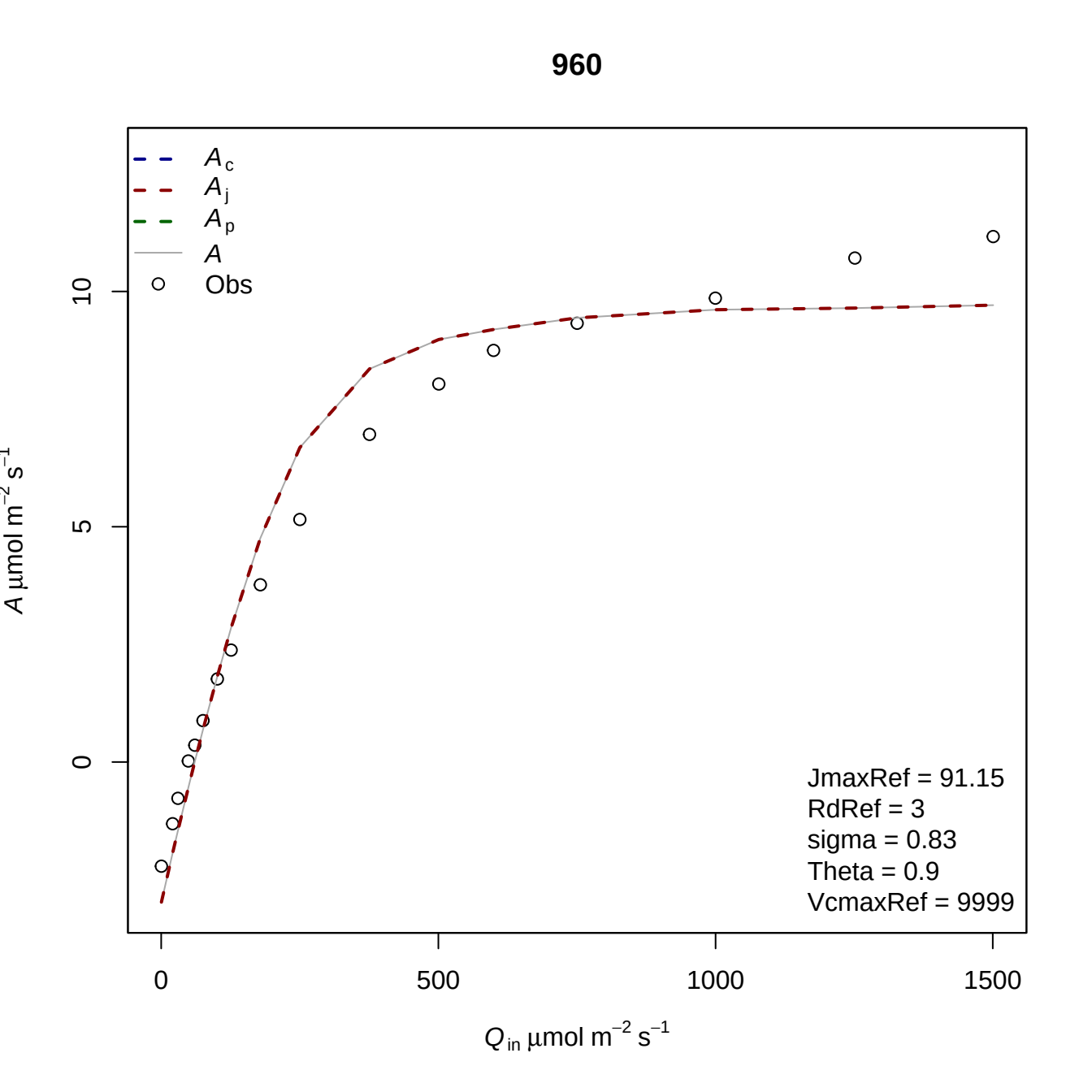

955
A_c
A_j
A_p
A
Obs
JmaxRef = 55.78
RdRef = 3.3
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

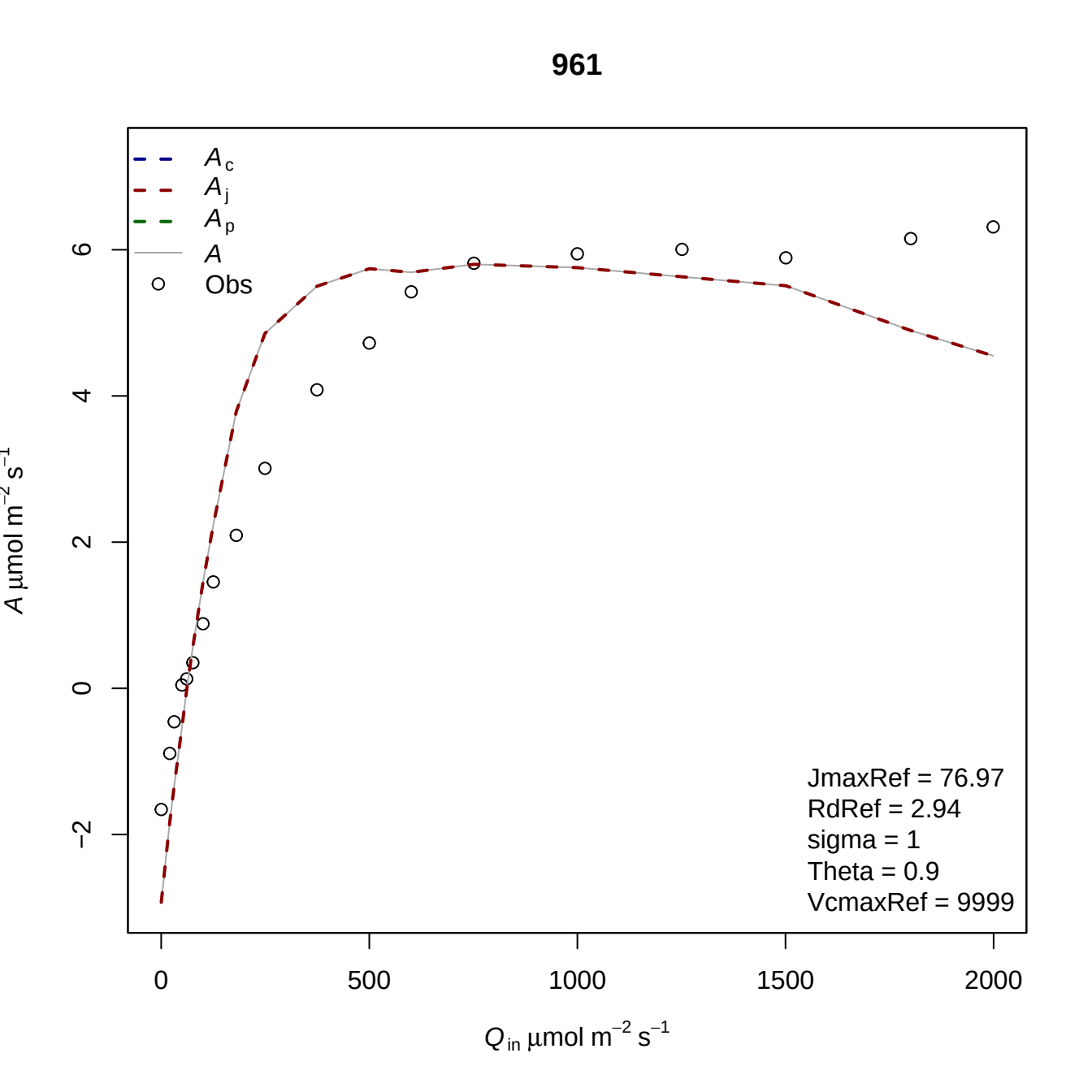

956
A_c
A_j
A_p
A
Obs
JmaxRef = 77.39
RdRef = 2.71
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

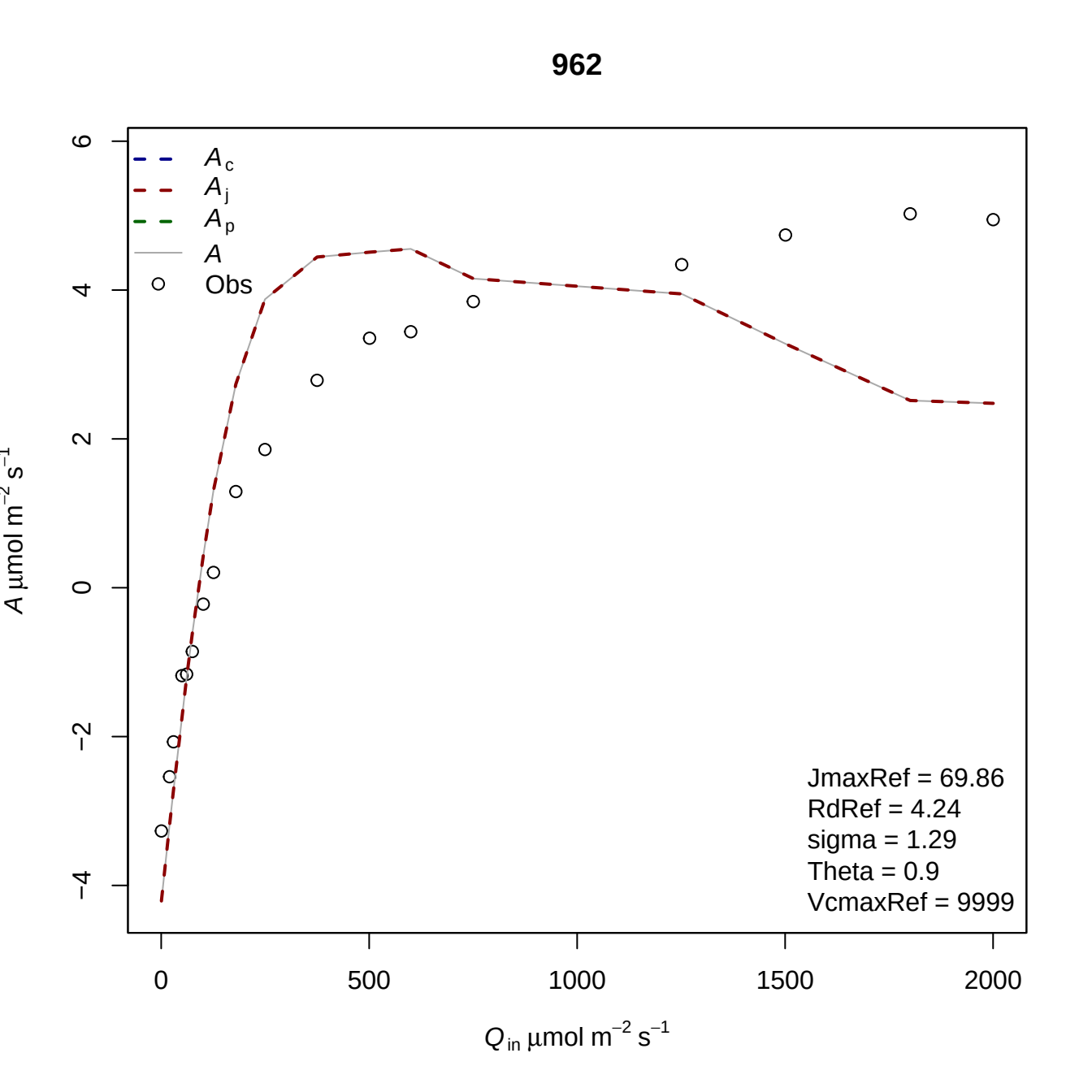

957
A_c
A_j
A_p
A
Obs
JmaxRef = 76.86
RdRef = 3.48
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

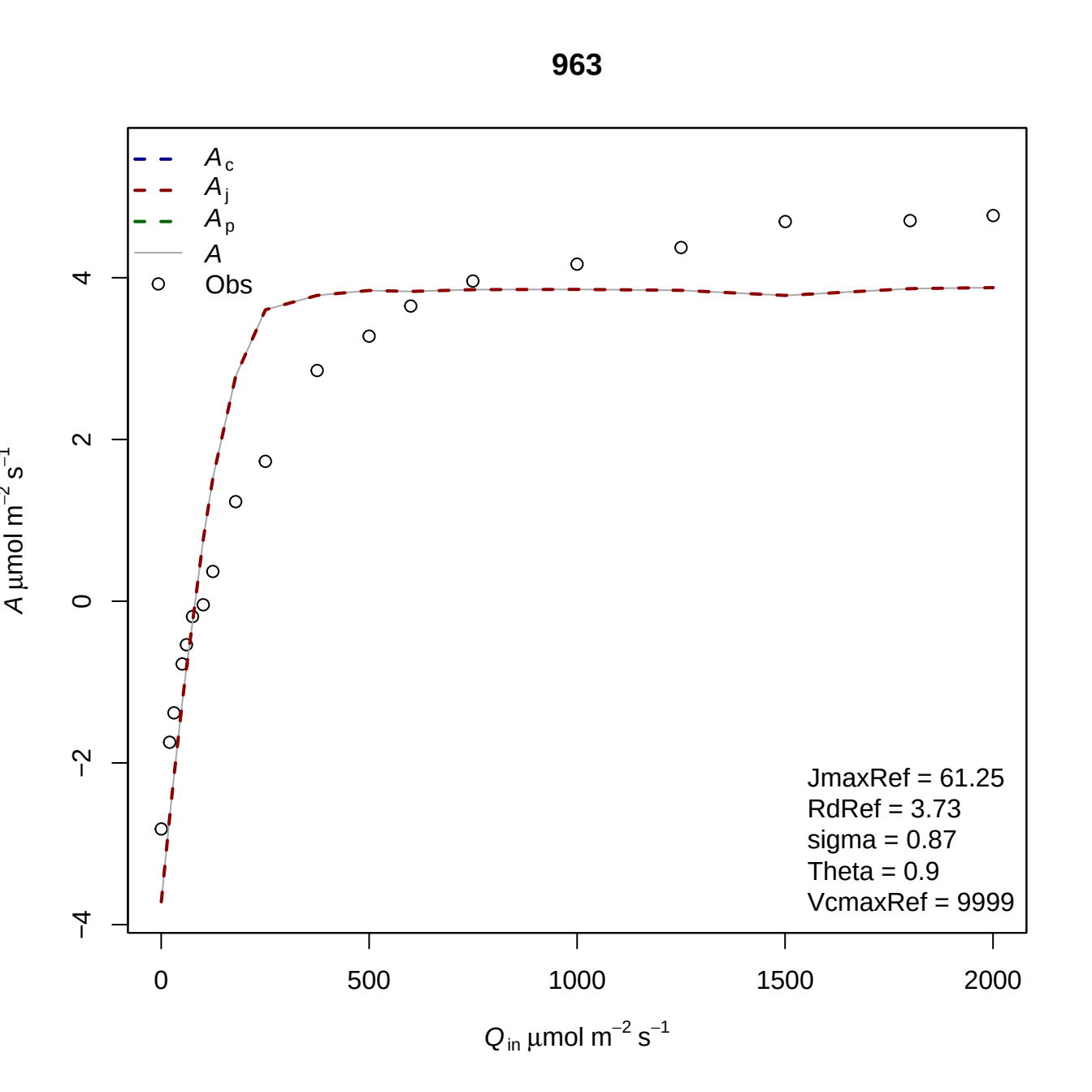

958
A_c
A_j
A_p
A
Obs
JmaxRef = 63.9
RdRef = 3.45
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

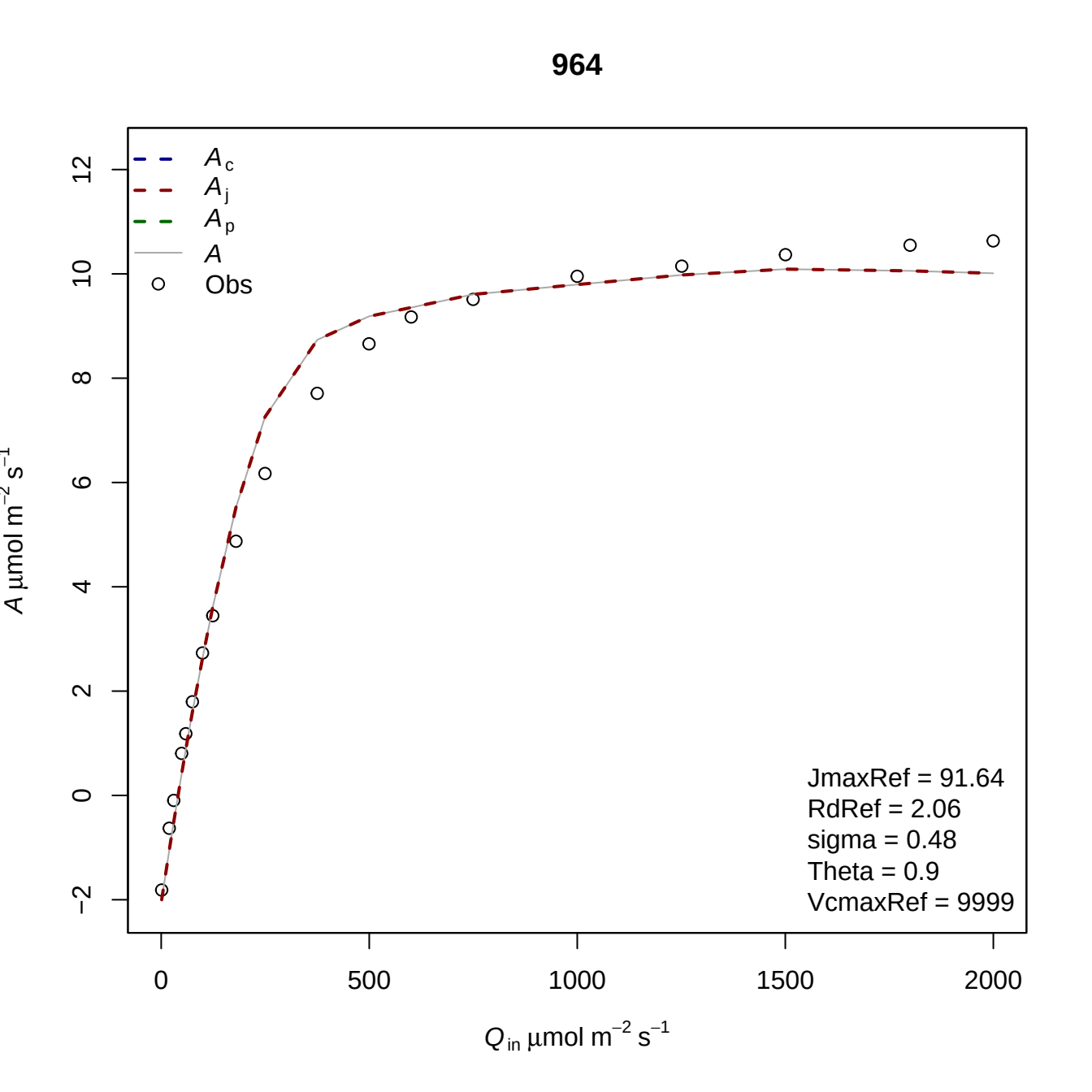

959
A_c
A_j
A_p
A
Obs
JmaxRef = 76.88
RdRef = 2.82
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

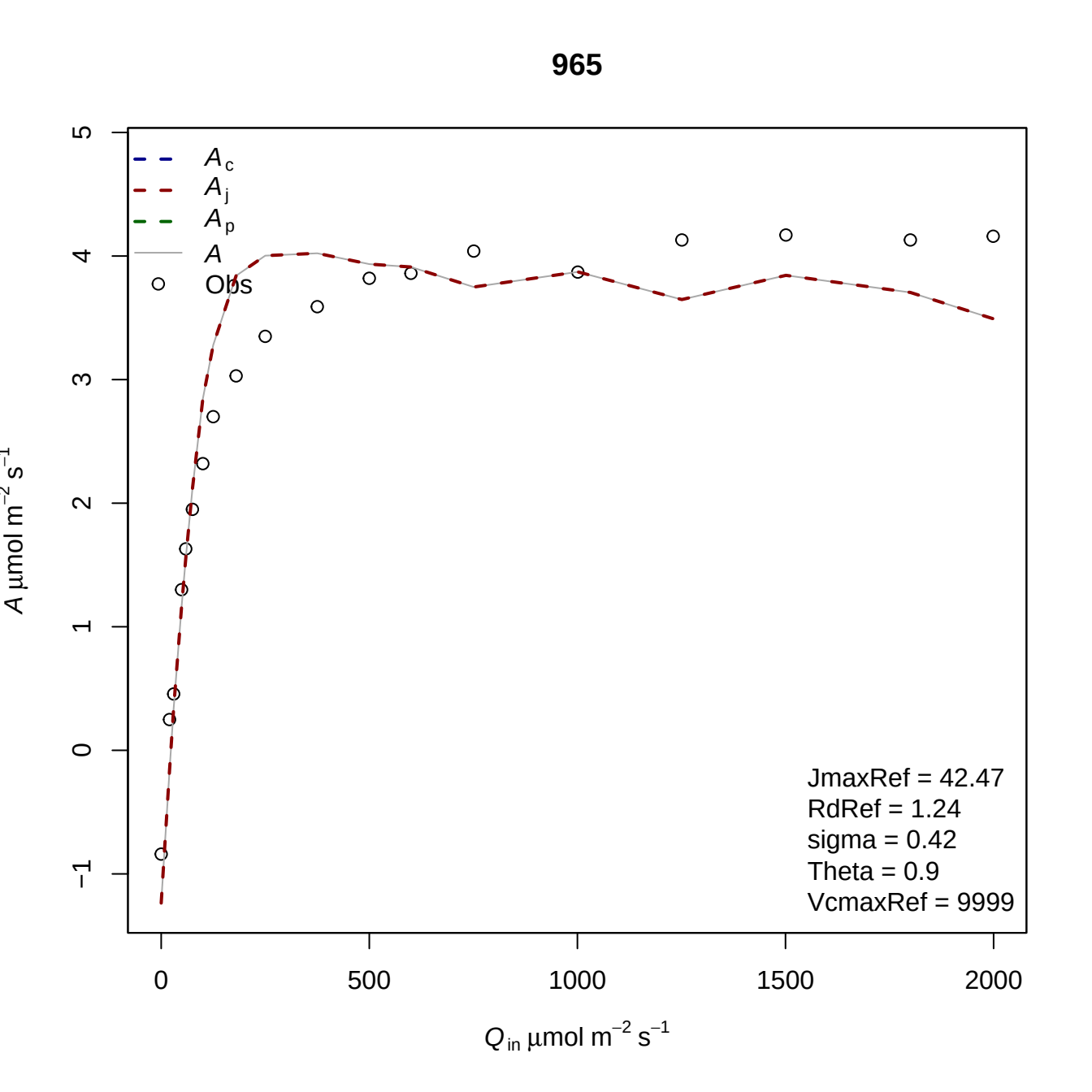

960
A_c
A_j
A_p
A
Obs
JmaxRef = 91.15
RdRef = 3
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

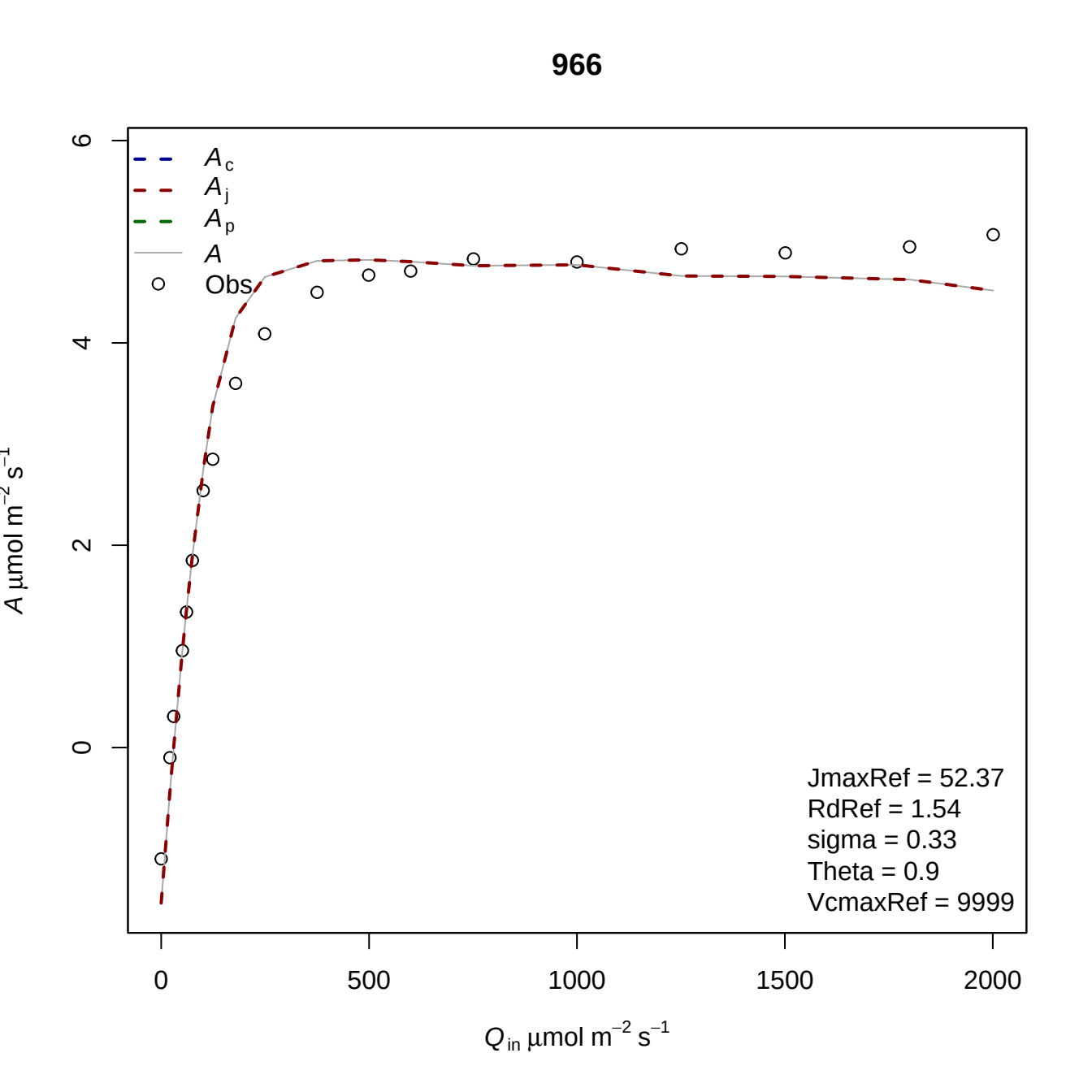

961
A_c
A_j
A_p
A
Obs
JmaxRef = 76.97
RdRef = 2.94
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

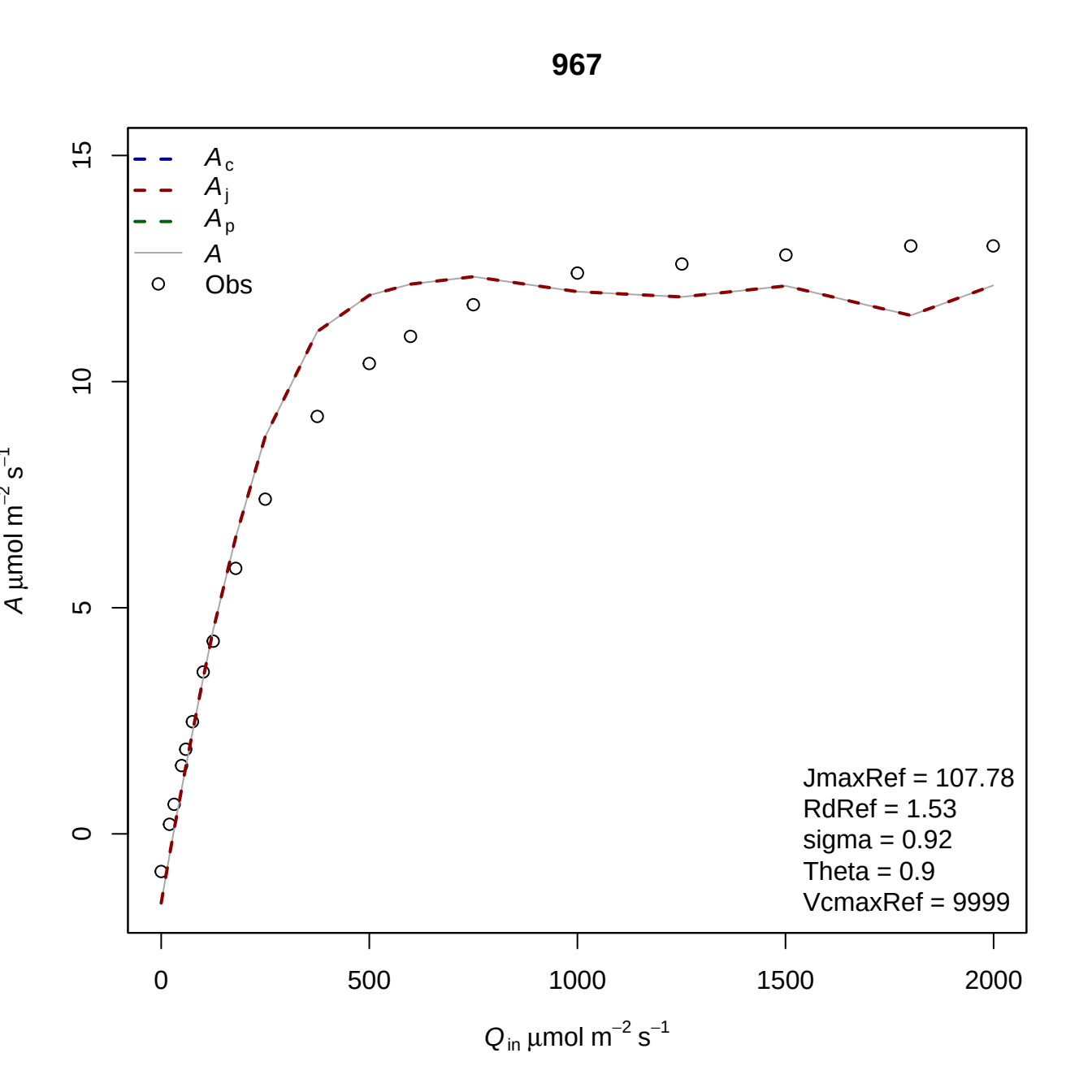

962
A_c
A_j
A_p
A
Obs
JmaxRef = 69.86
RdRef = 4.24
sigma = 1.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

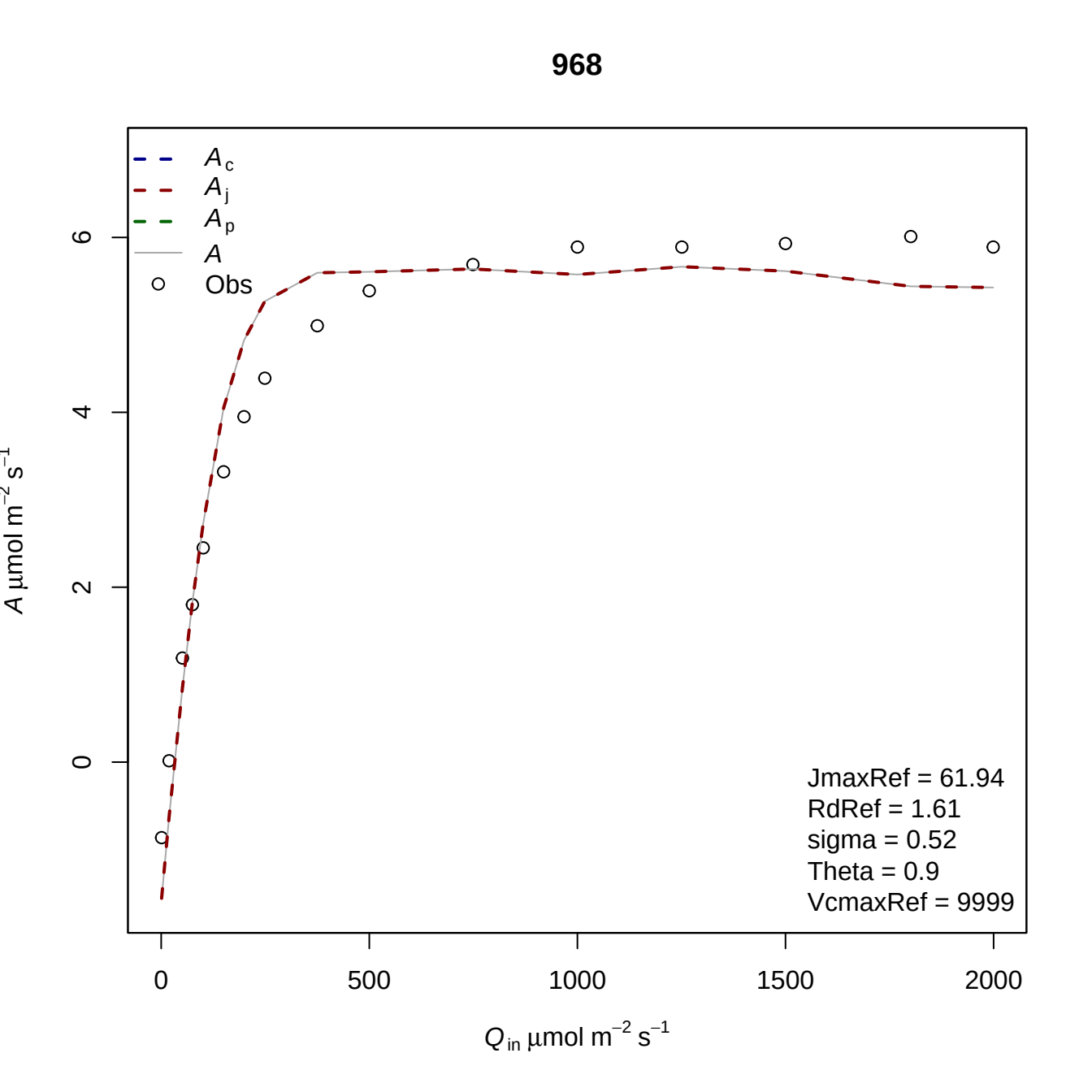

963
A_c
A_j
A_p
A
Obs
JmaxRef = 61.25
RdRef = 3.73
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500
2000

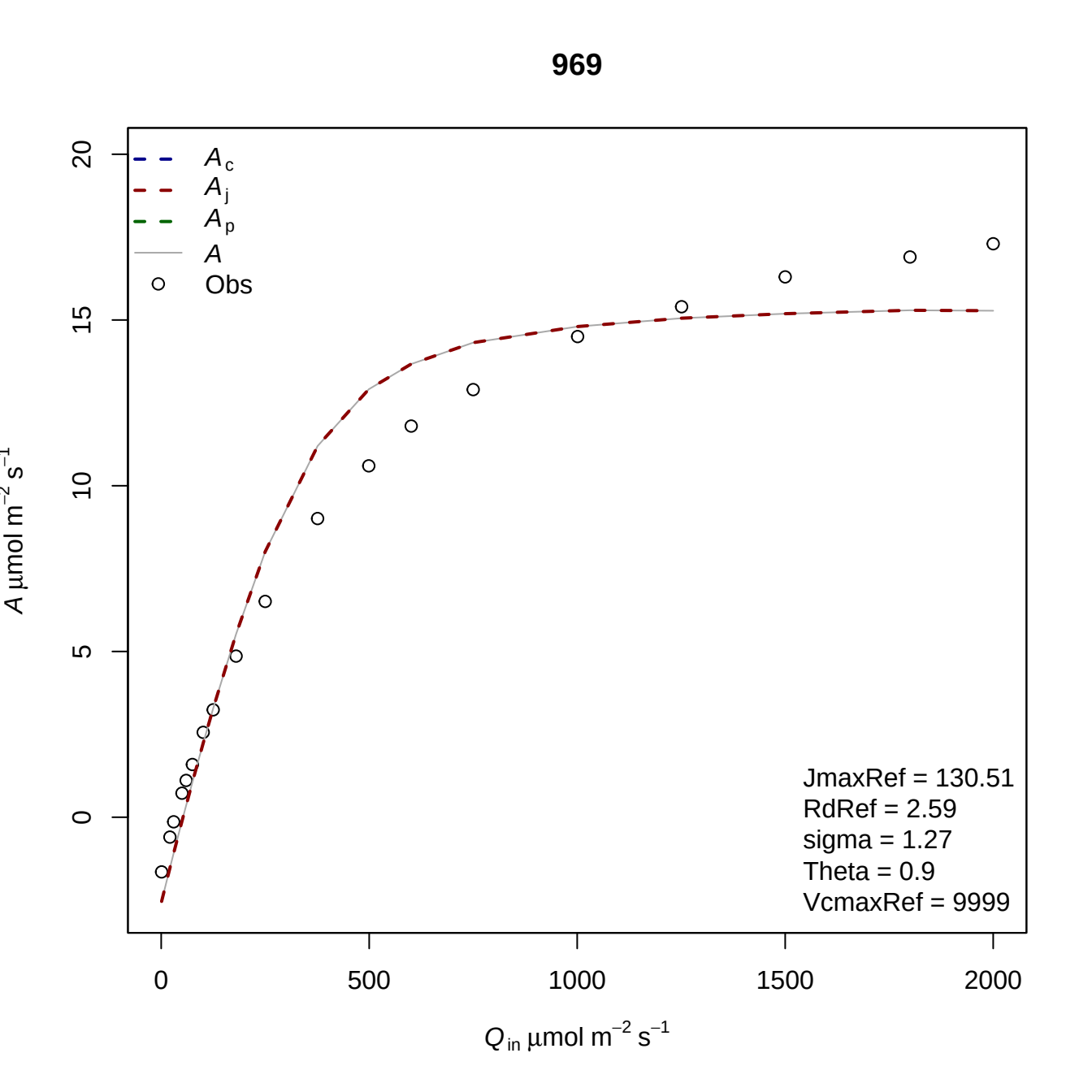

964
A_c
A_j
A_p
A
Obs
JmaxRef = 91.64
RdRef = 2.06
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

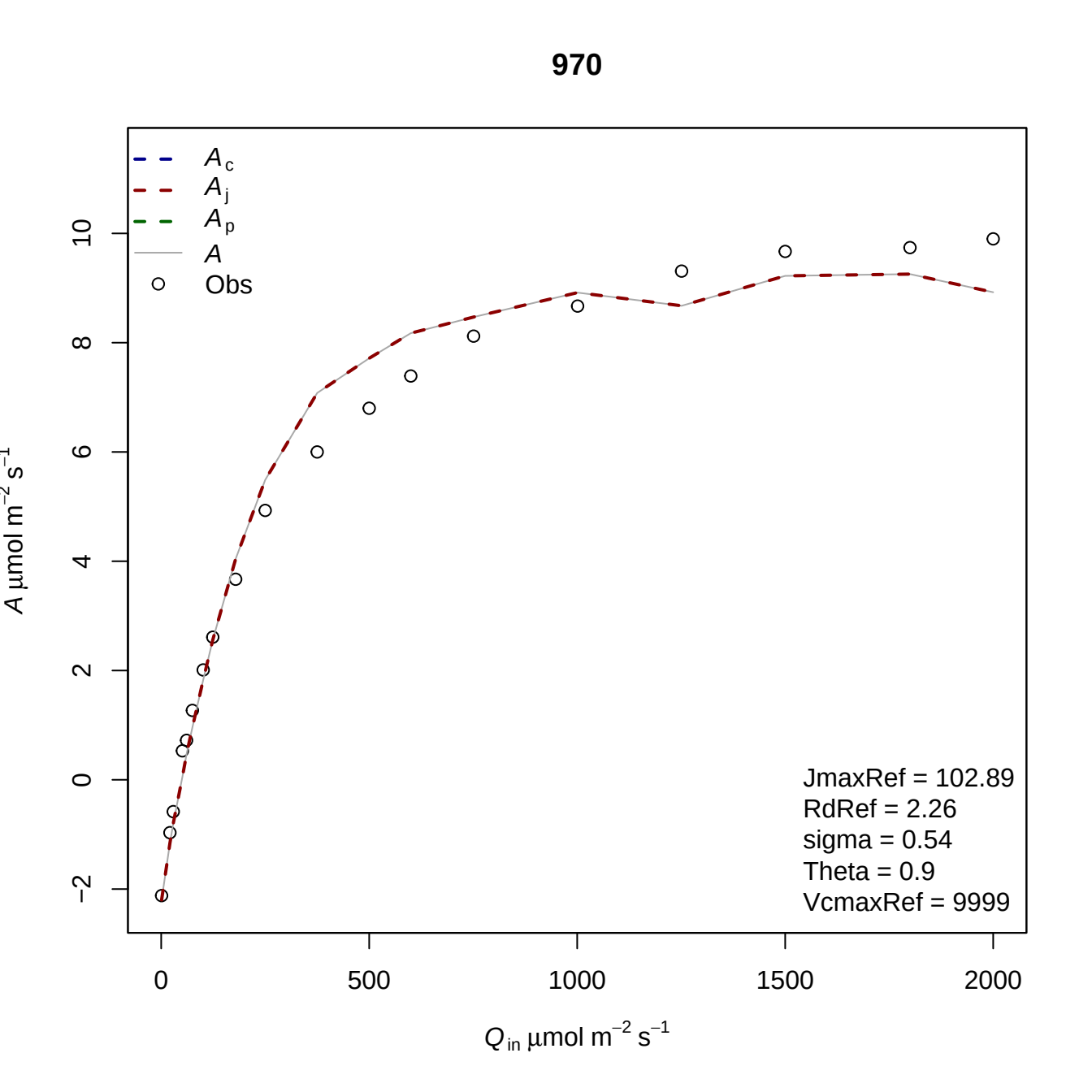

965
A_c
A_j
A_p
A
Obs
JmaxRef = 42.47
RdRef = 1.24
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

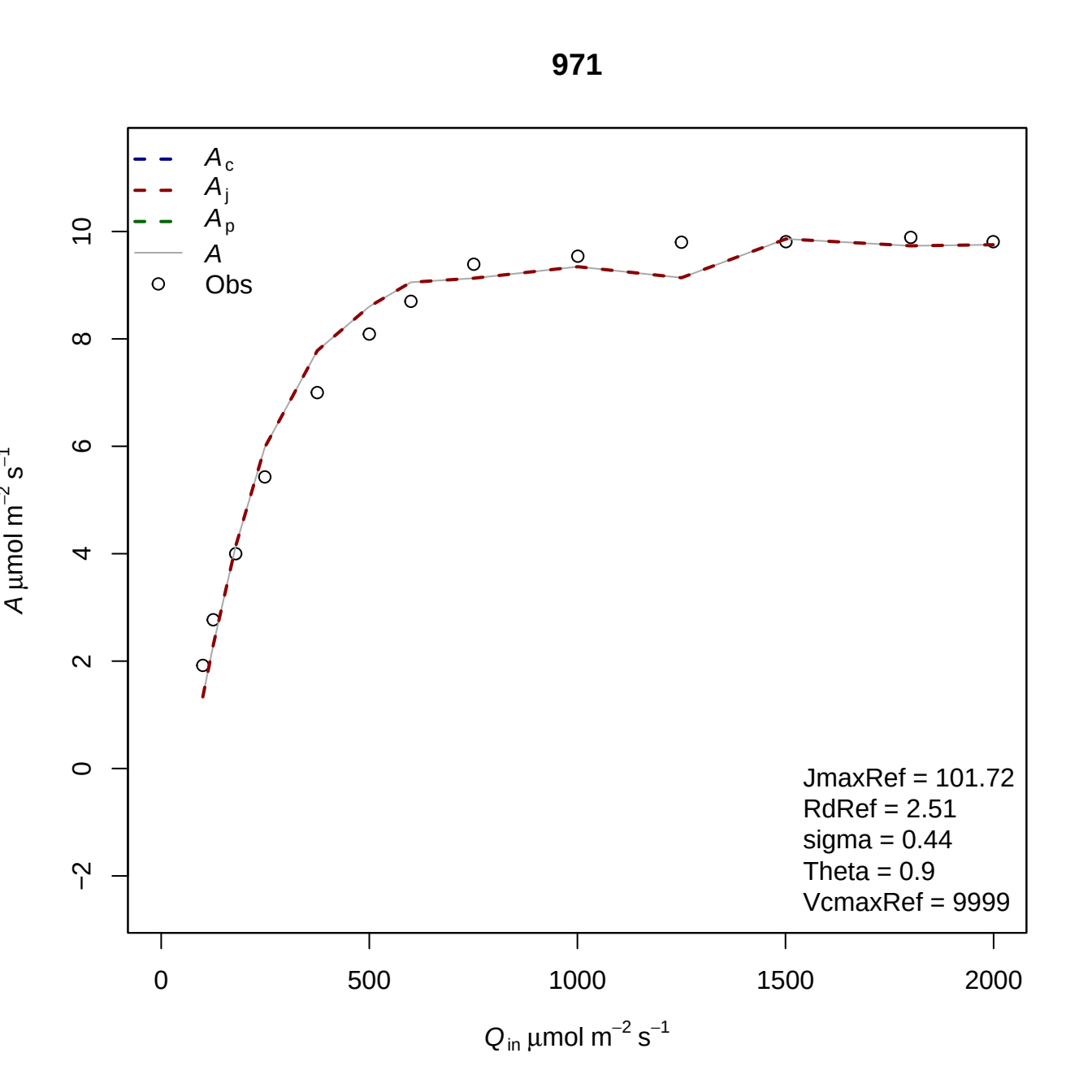

966
A_c
A_j
A_p
A
Obs
JmaxRef = 52.37
RdRef = 1.54
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

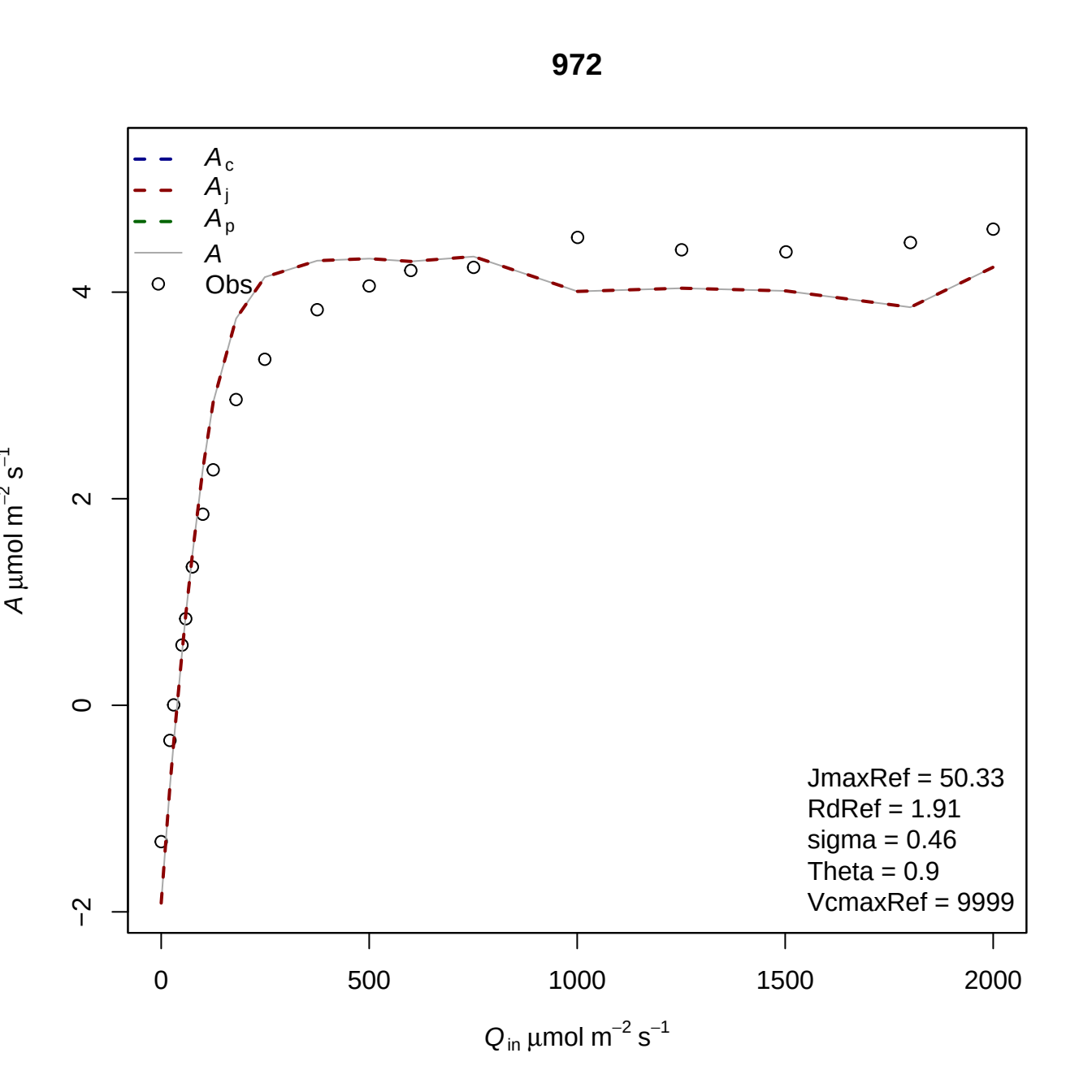

967
A_c
A_j
A_p
A
Obs
JmaxRef = 107.78
RdRef = 1.53
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

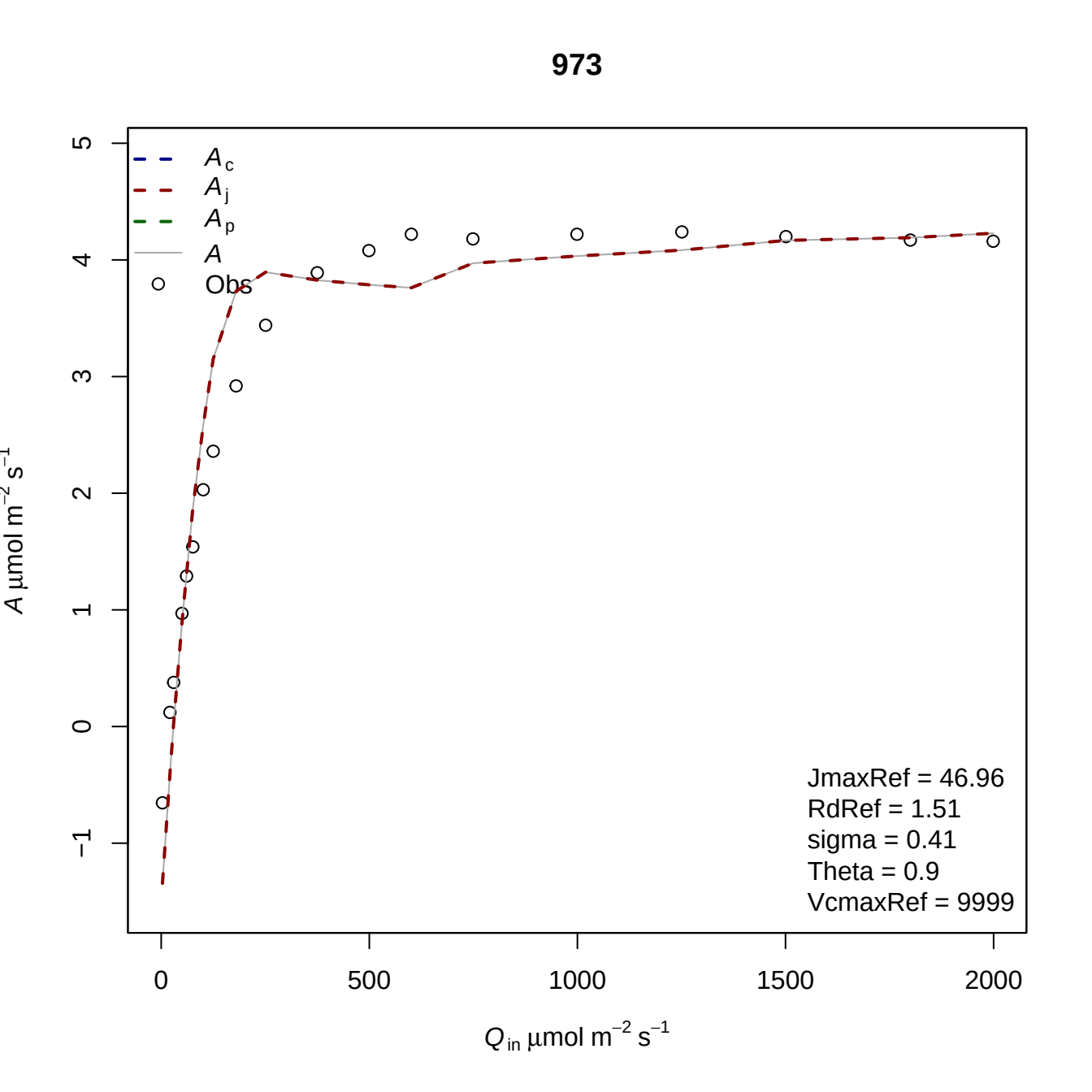

968
A_c
A_j
A_p
A
Obs
JmaxRef = 61.94
RdRef = 1.61
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

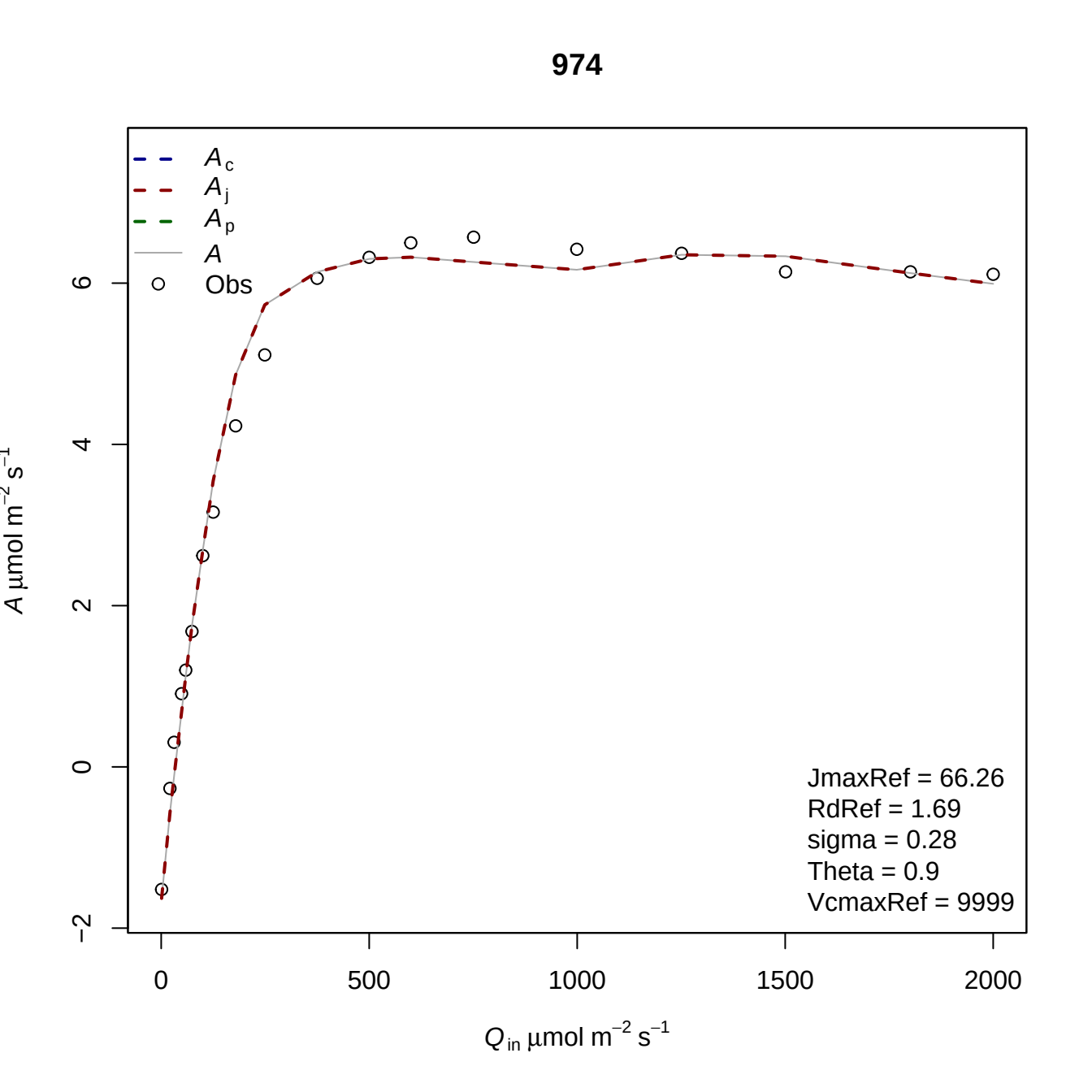

969
A_c
A_j
A_p
A
Obs
JmaxRef = 130.51
RdRef = 2.59
sigma = 1.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

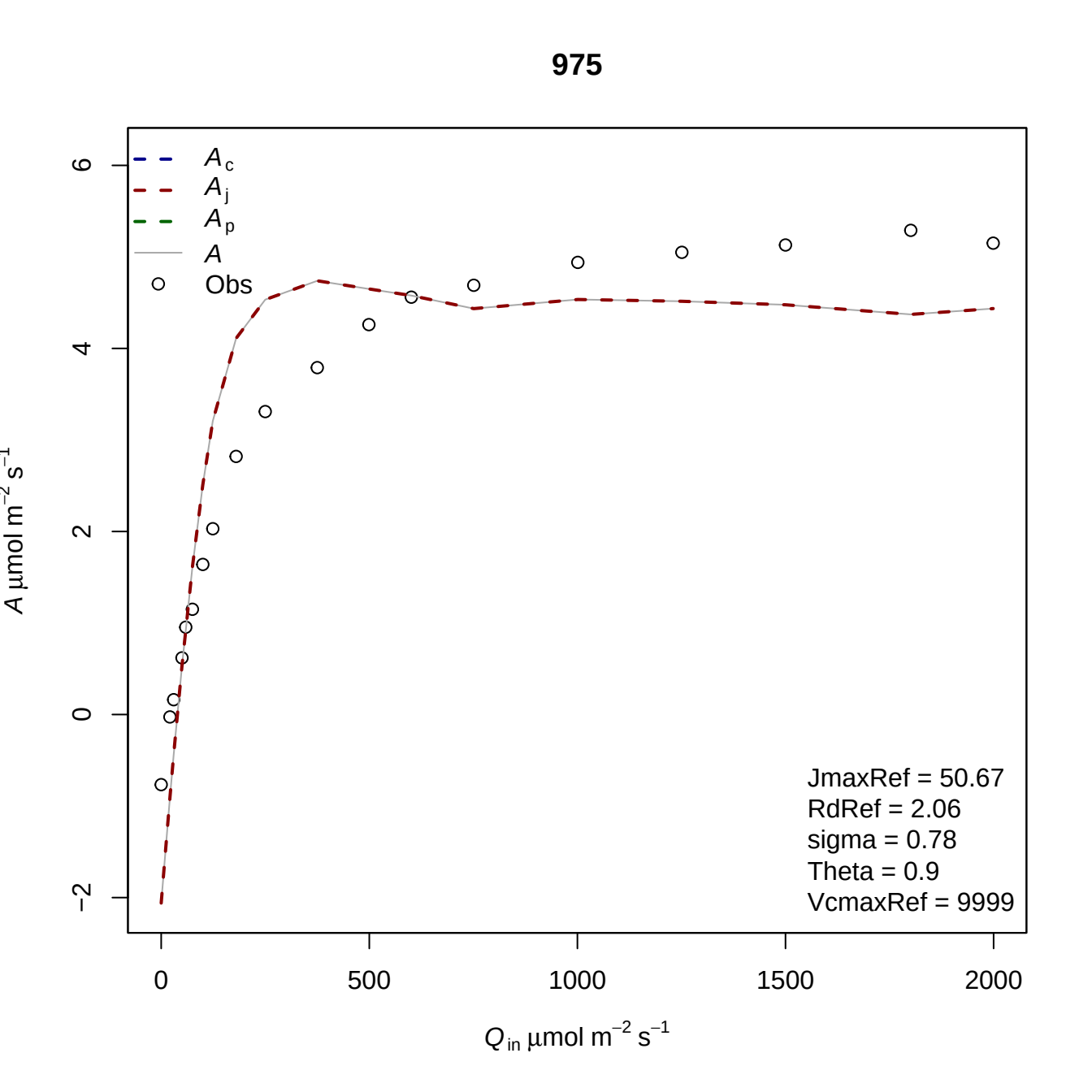

970
A_c
A_j
A_p
A
Obs
JmaxRef = 102.89
RdRef = 2.26
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

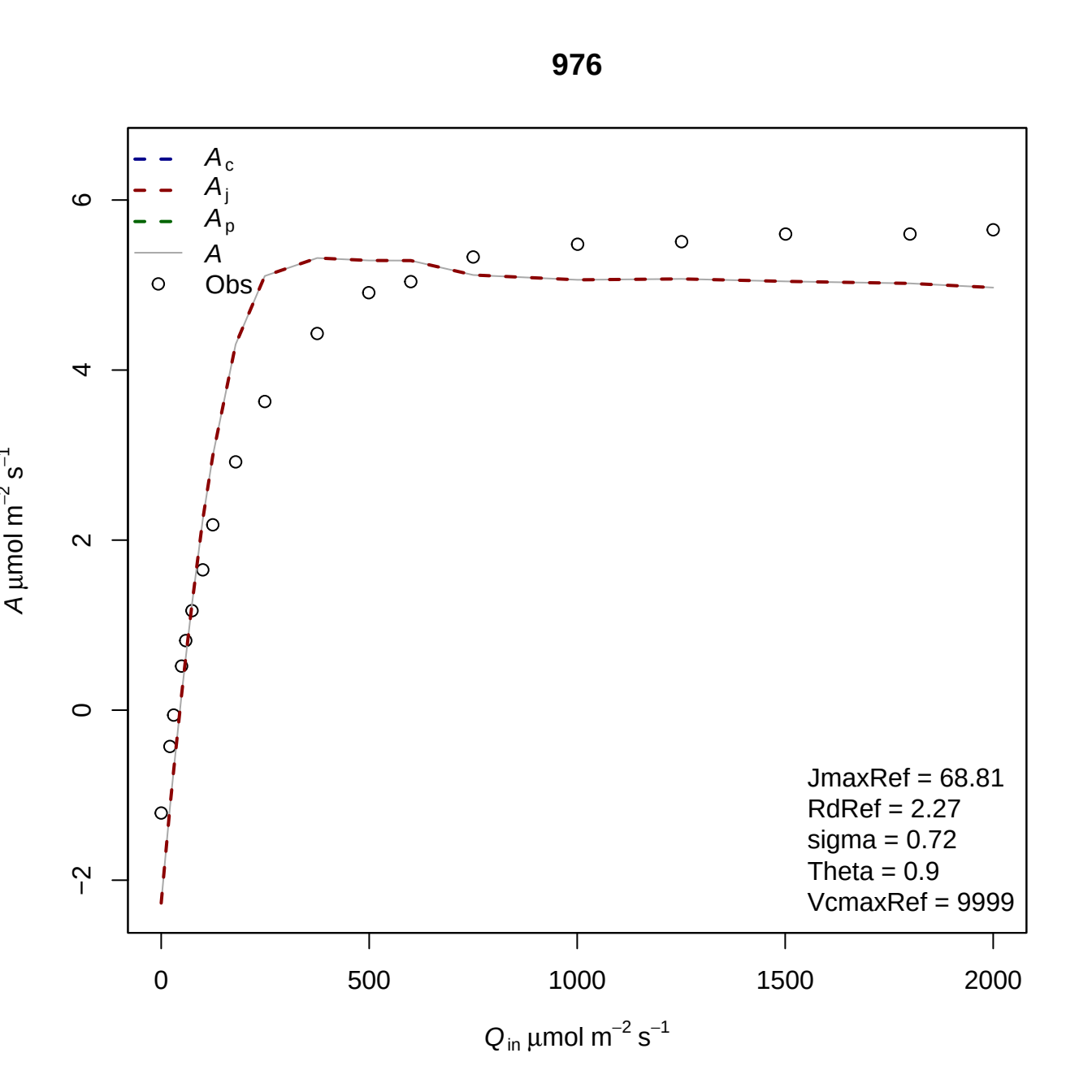

971
A_c
A_j
A_p
A
Obs
JmaxRef = 101.72
RdRef = 2.51
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

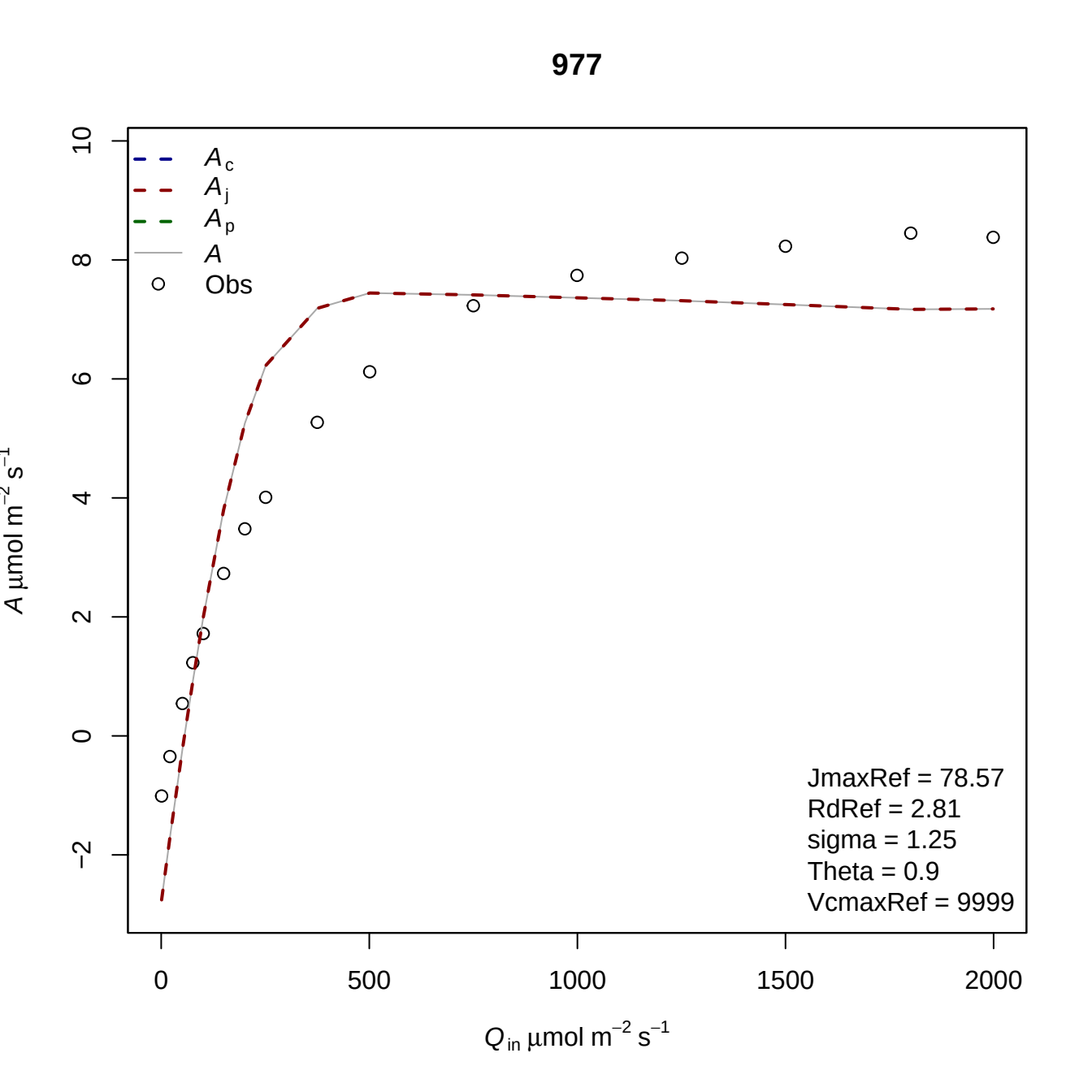

972
A_c
A_j
A_p
A
Obs
JmaxRef = 50.33
RdRef = 1.91
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

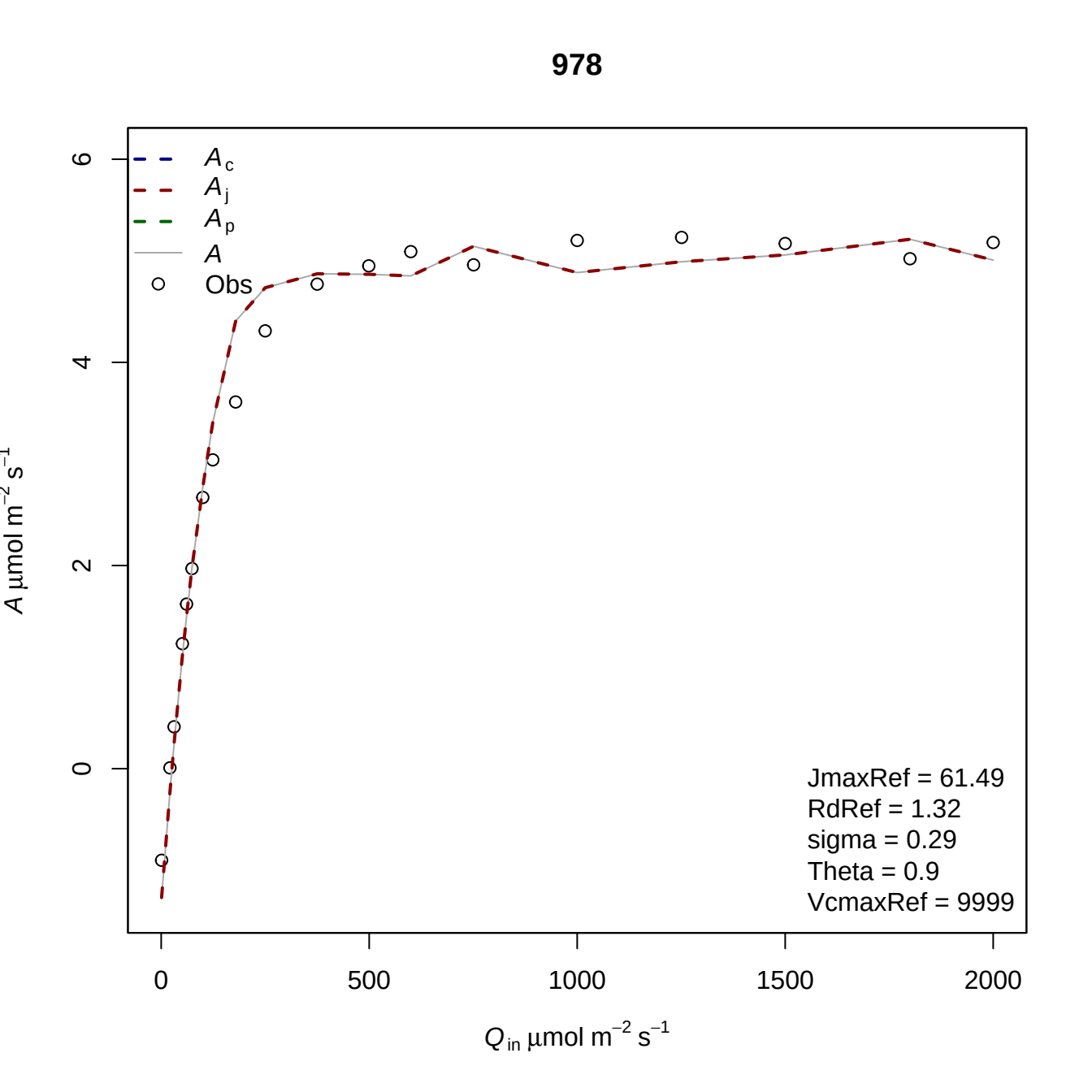

973
A_c
A_j
A_p
A
Obs
JmaxRef = 46.96
RdRef = 1.51
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

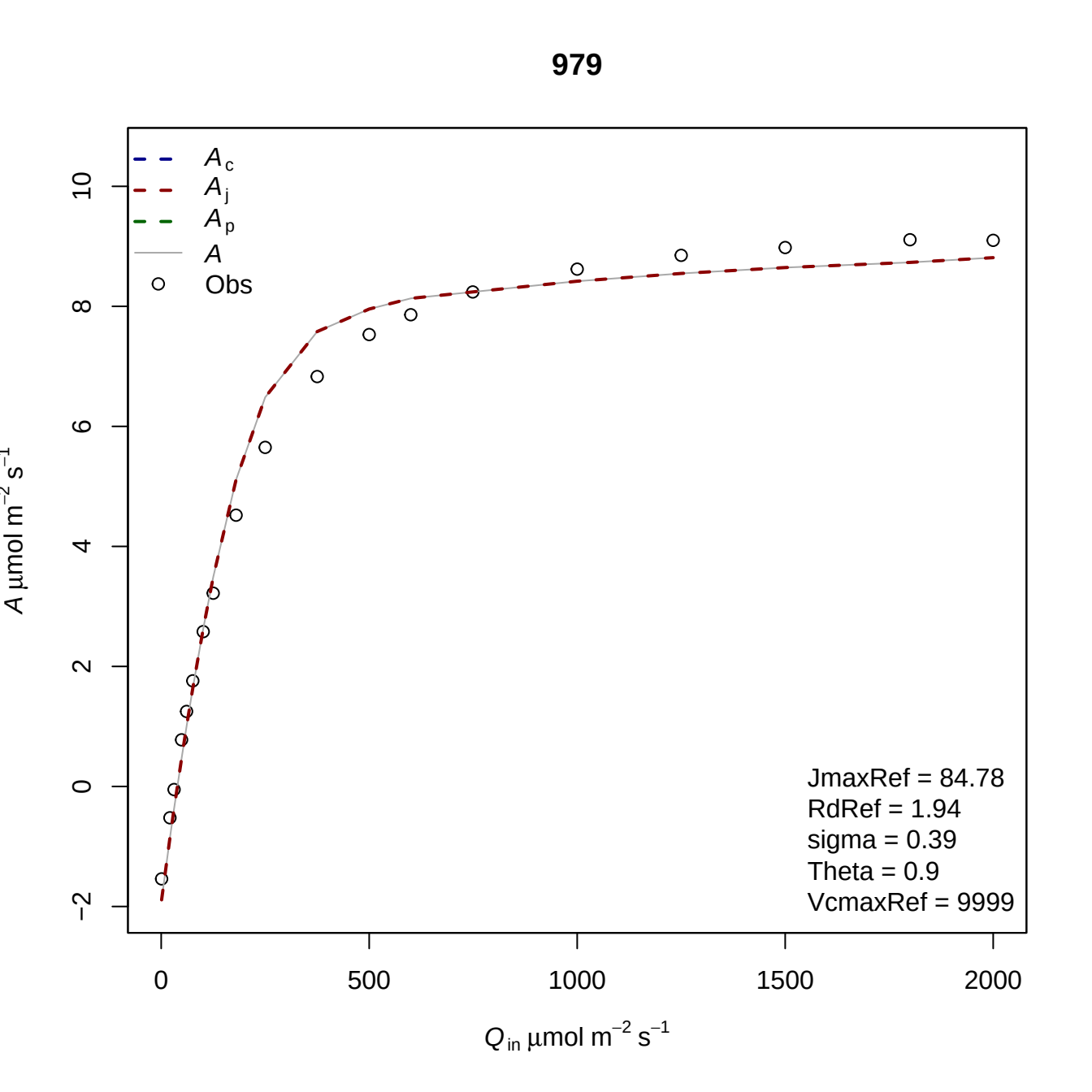

974
A_c
A_j
A_p
A
Obs
JmaxRef = 66.26
RdRef = 1.69
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

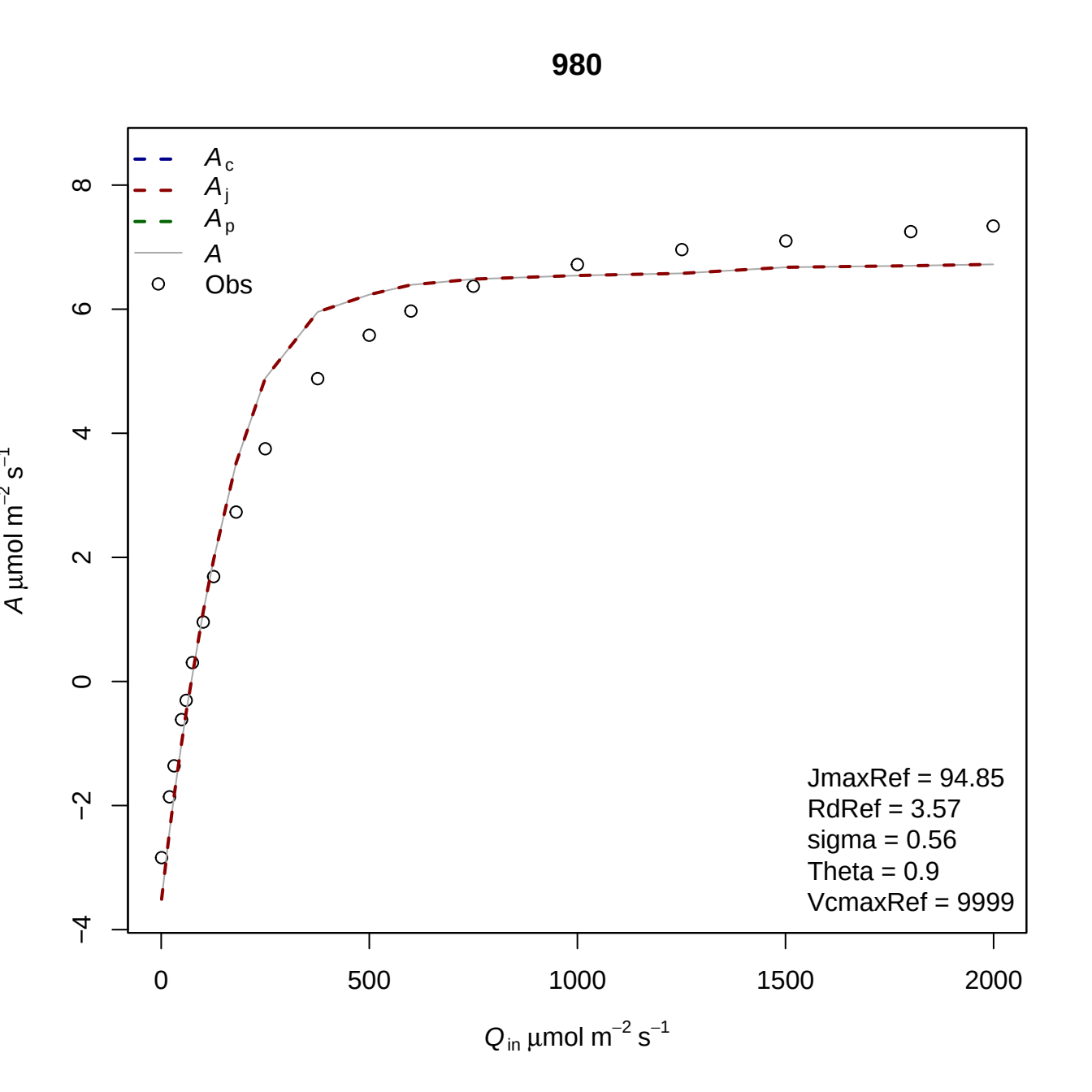

975
A_c
A_j
A_p
A
Obs
JmaxRef = 50.67
RdRef = 2.06
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

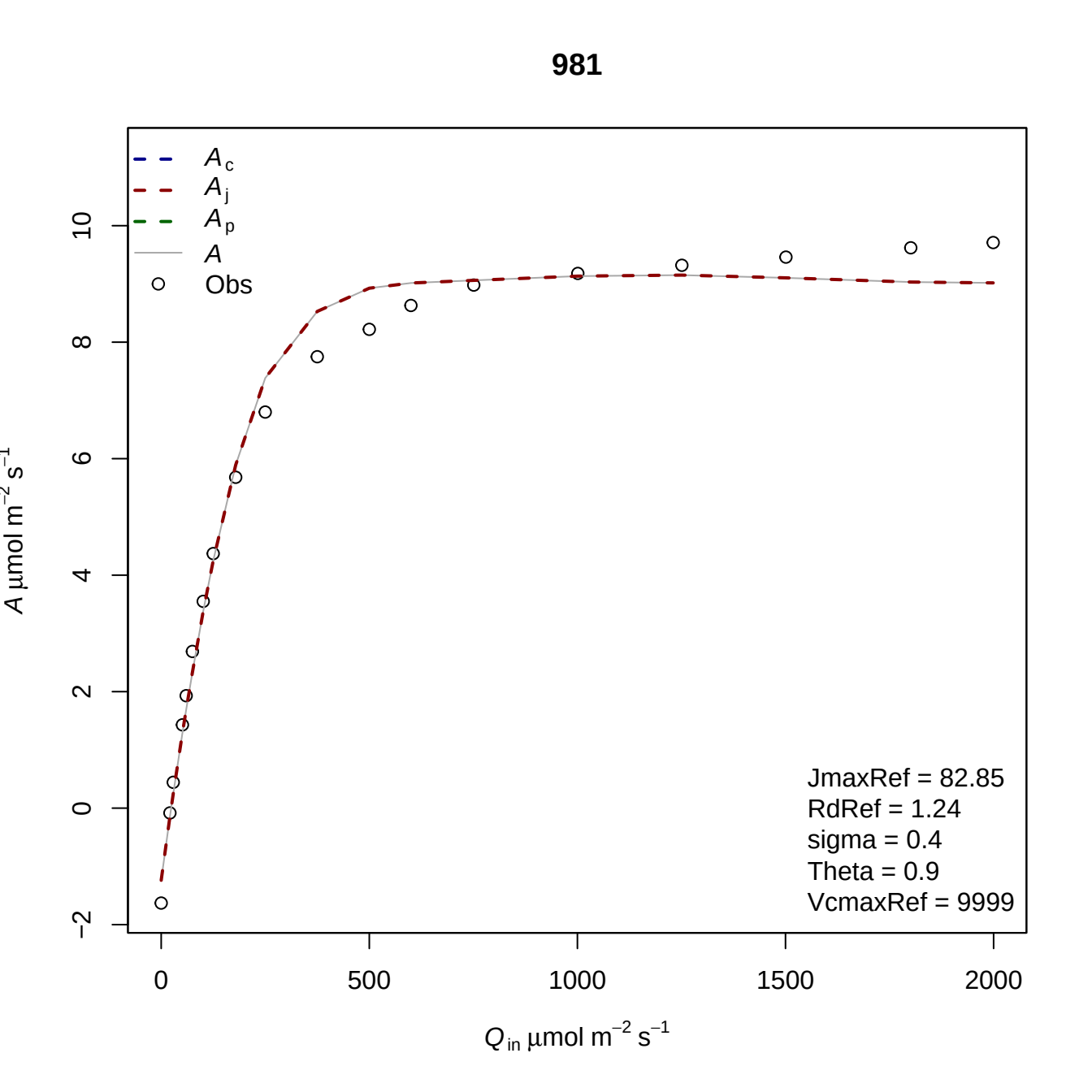

976
A_c
A_j
A_p
A
Obs
JmaxRef = 68.81
RdRef = 2.27
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

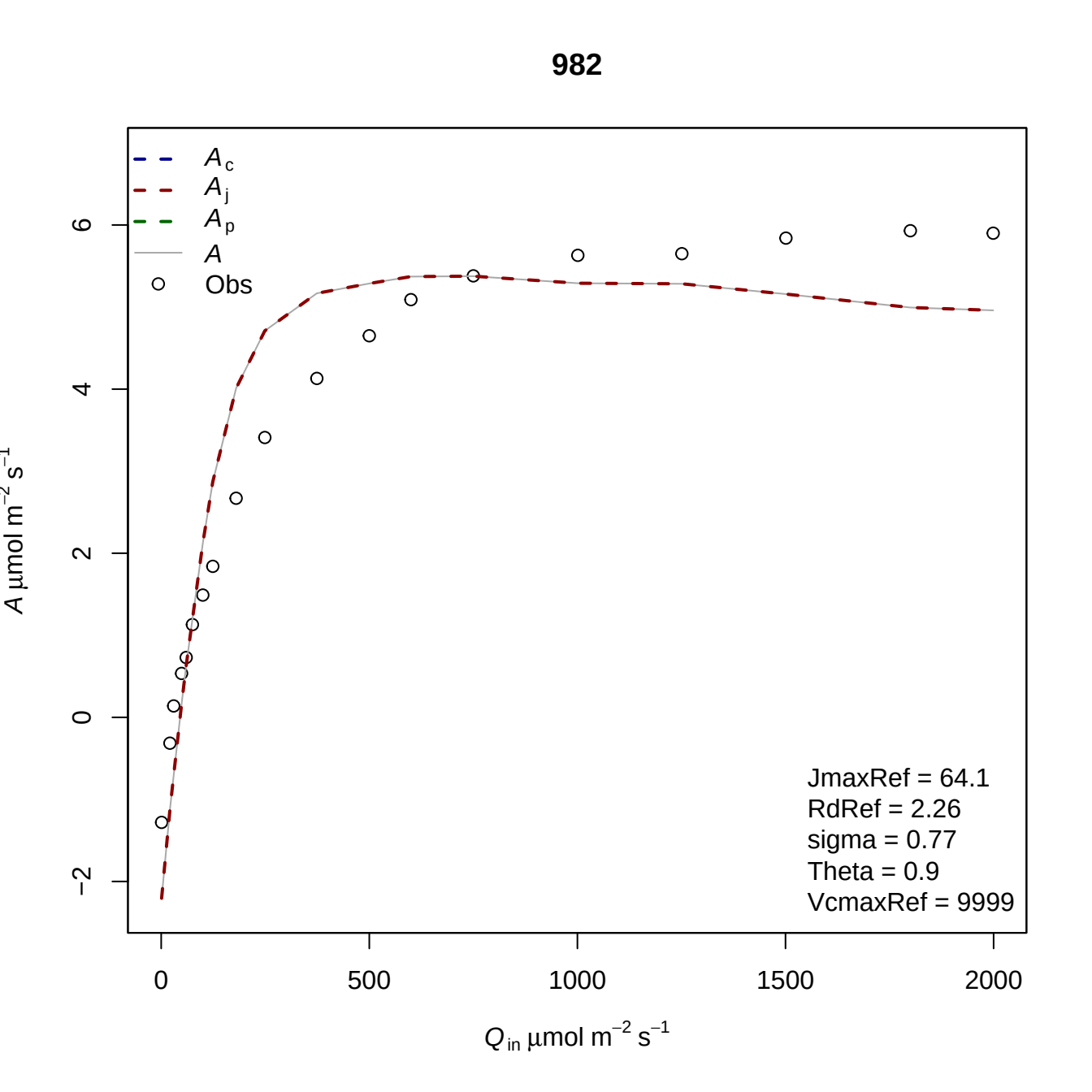

977
A_c
A_j
A_p
A
Obs
JmaxRef = 78.57
RdRef = 2.81
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

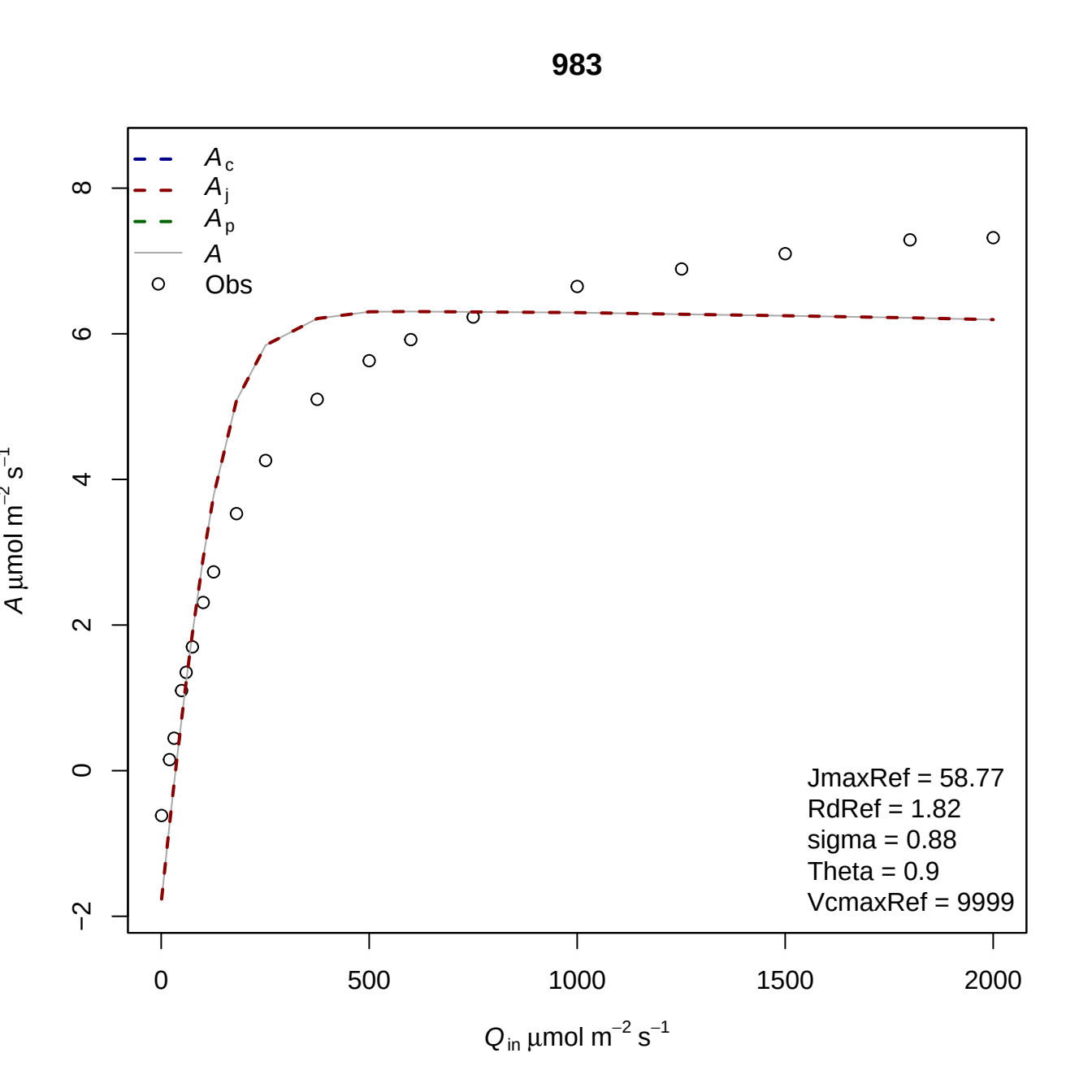

978
A_c
A_j
A_p
A
Obs
JmaxRef = 61.49
RdRef = 1.32
sigma = 0.29
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

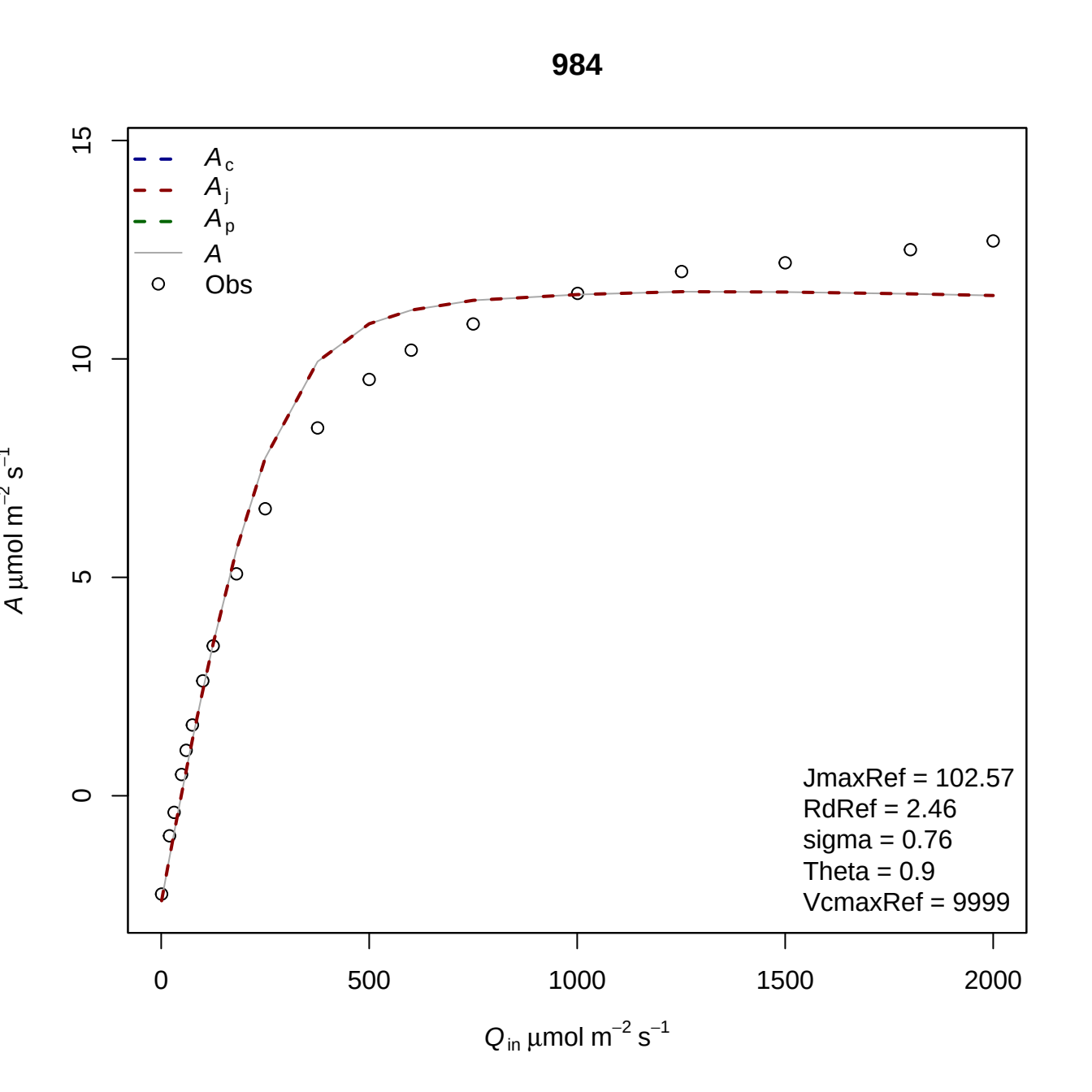

979
A_c
A_j
A_p
A
Obs
JmaxRef = 84.78
RdRef = 1.94
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

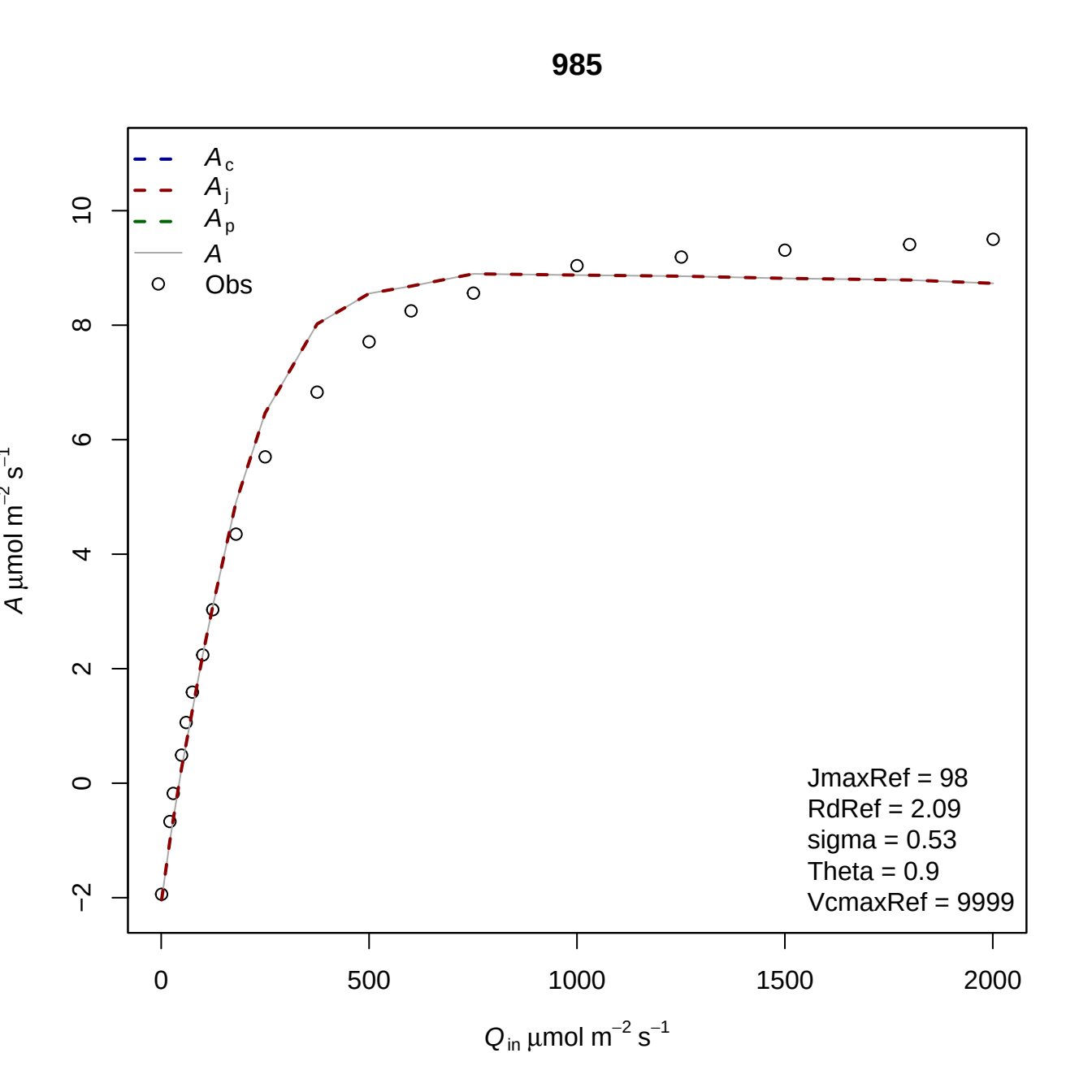

980
A_c
A_j
A_p
A
Obs
JmaxRef = 94.85
RdRef = 3.57
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

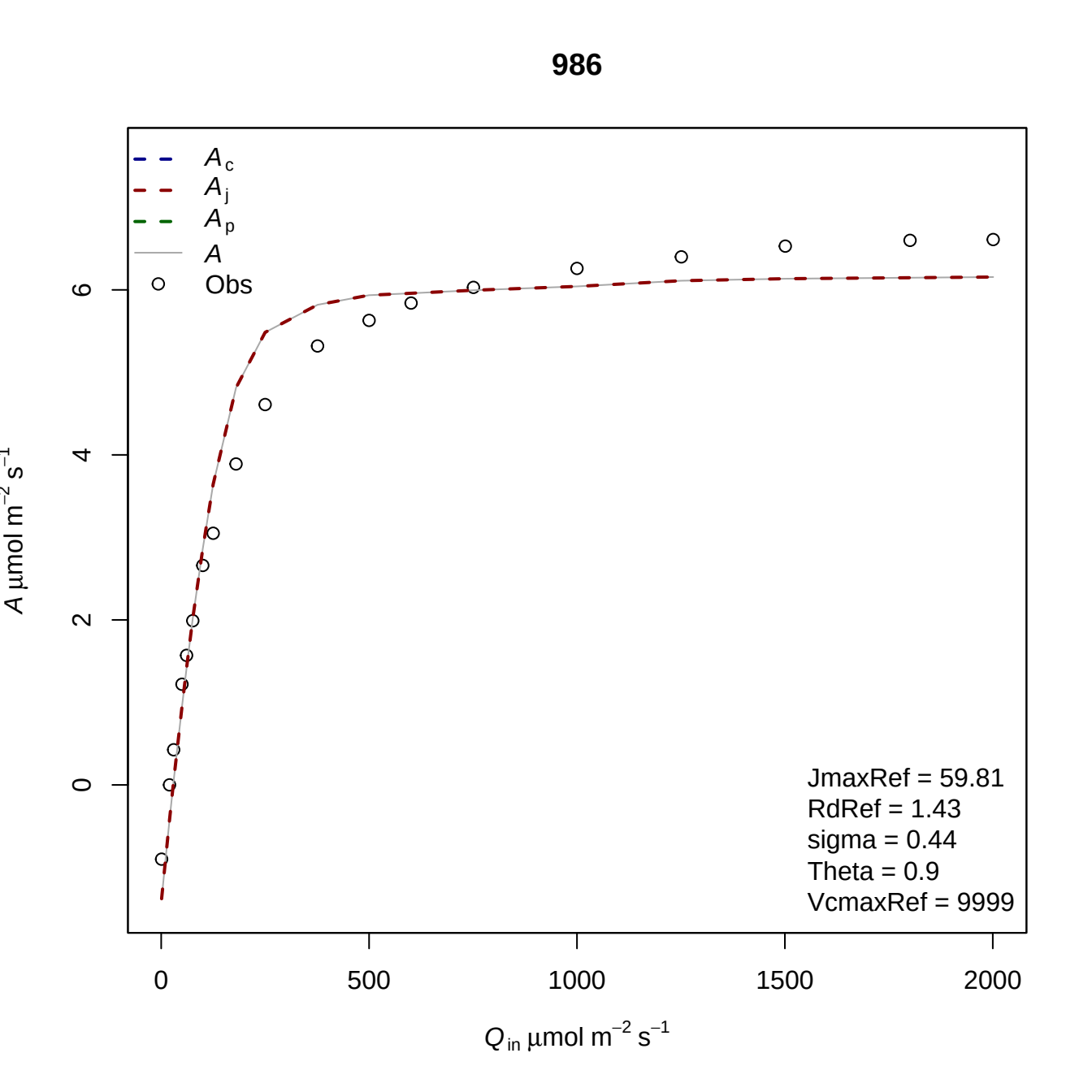

981
A_c
A_j
A_p
A
Obs
JmaxRef = 82.85
RdRef = 1.24
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

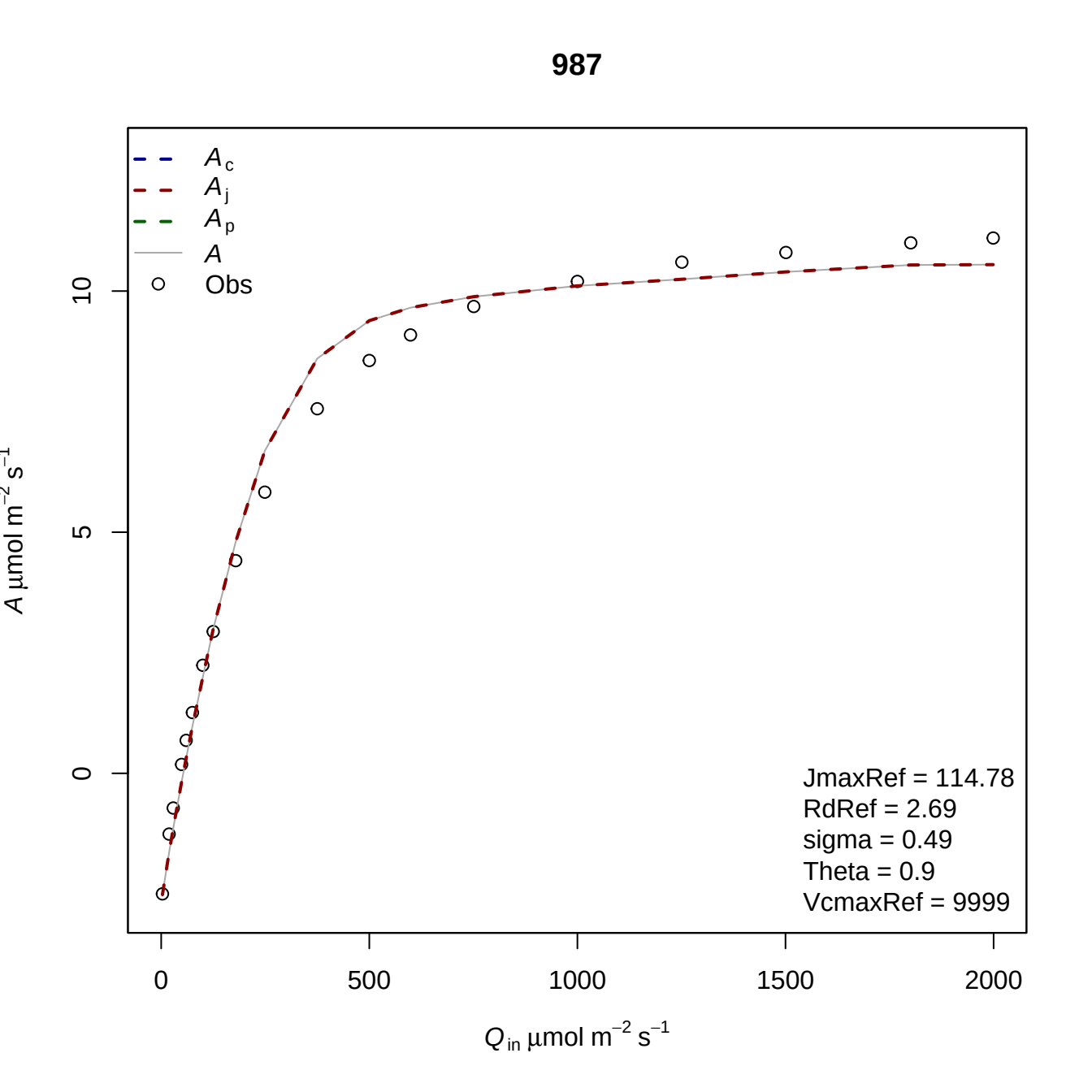

982
A_c
A_j
A_p
A
Obs
JmaxRef = 64.1
RdRef = 2.26
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

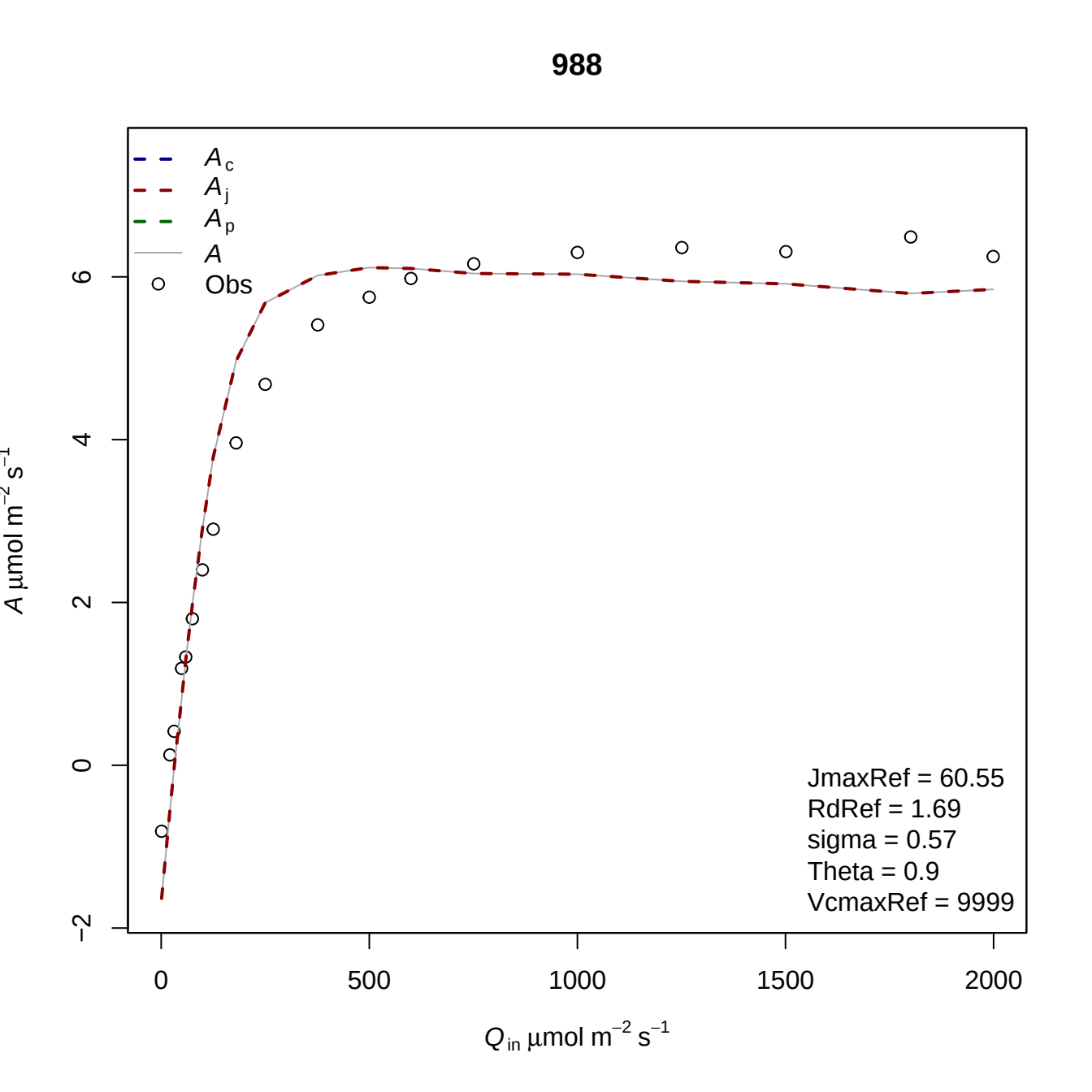

983
A_c
A_j
A_p
A
Obs
JmaxRef = 58.77
RdRef = 1.82
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

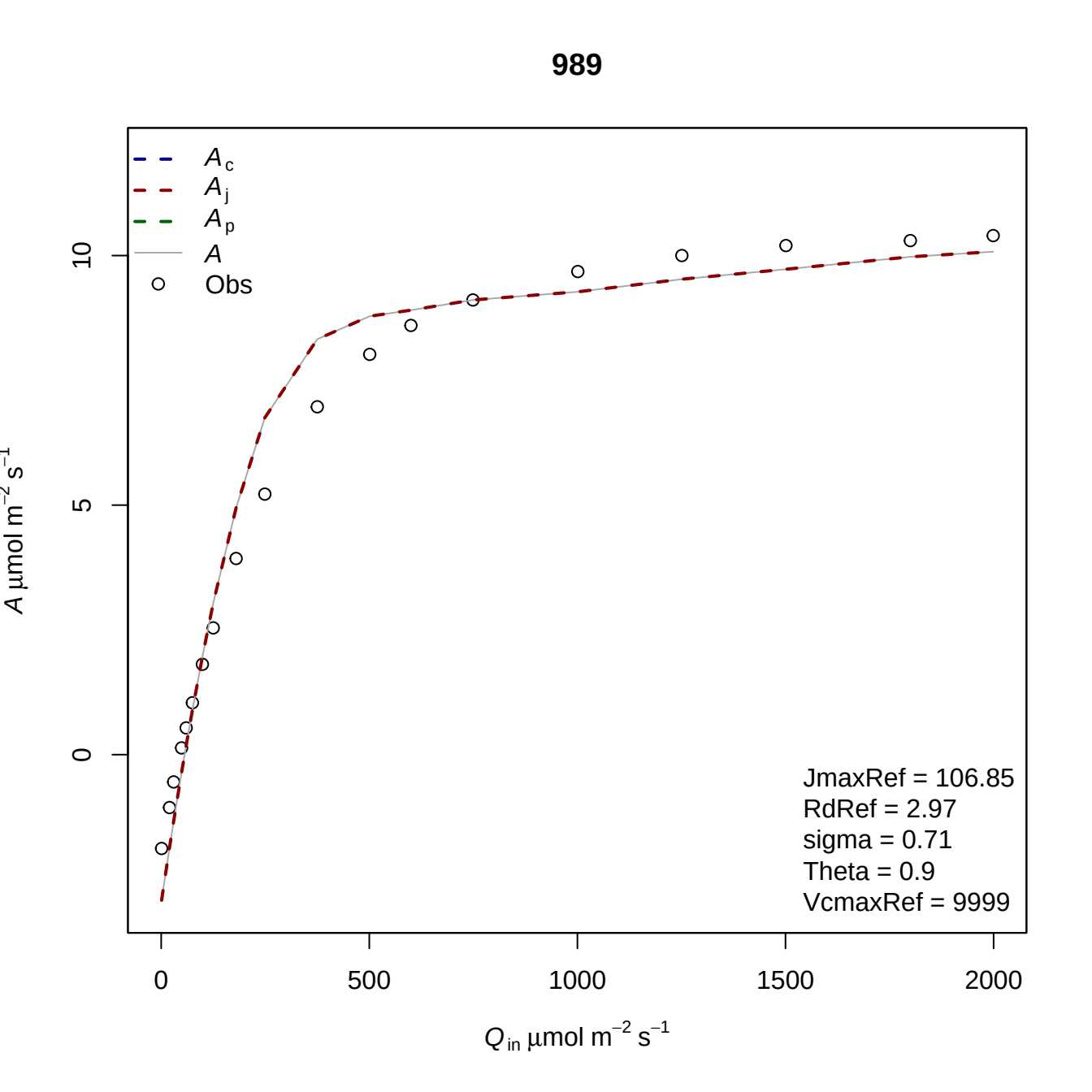

984
A_c
A_j
A_p
A
Obs
JmaxRef = 102.57
RdRef = 2.46
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

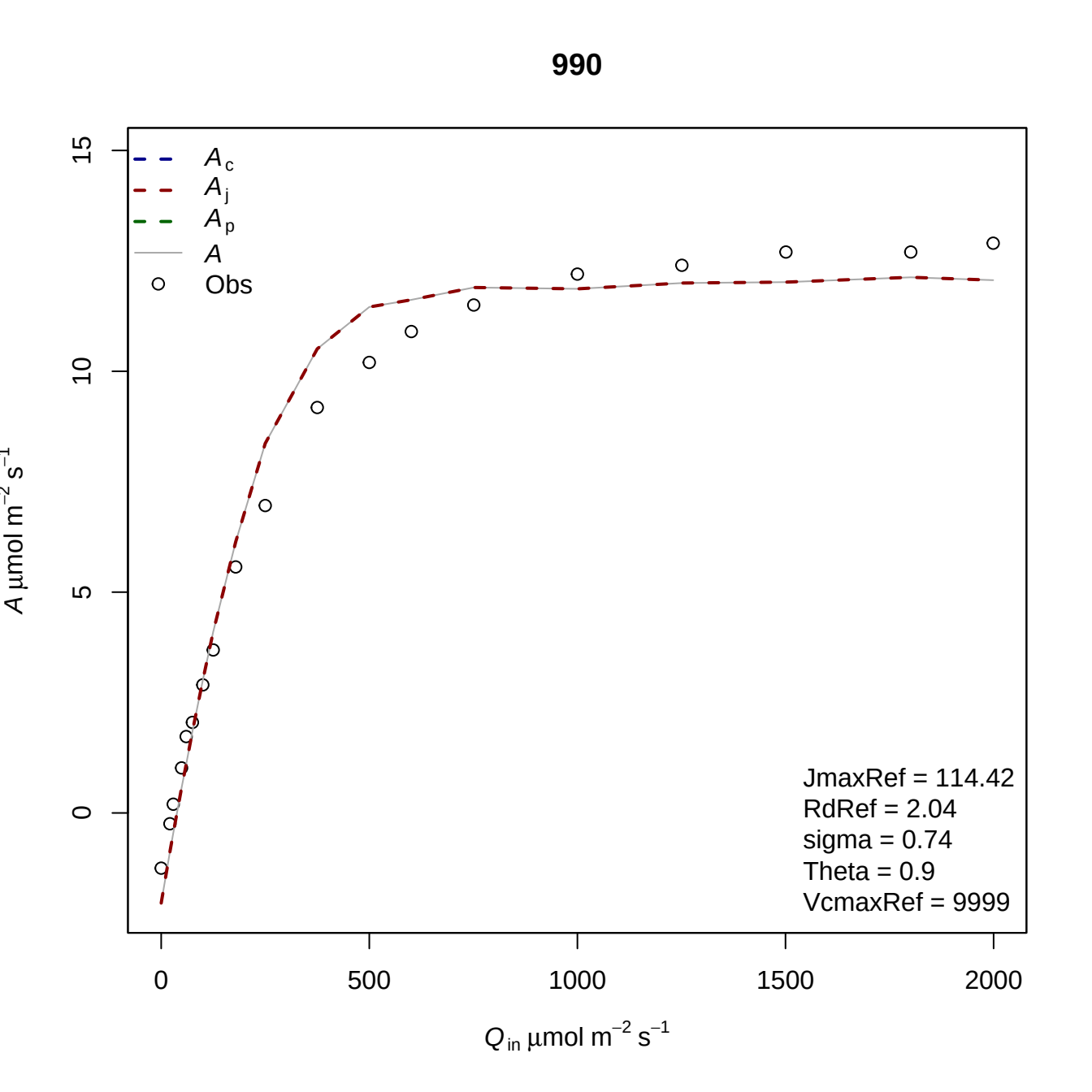

985
A_c
A_j
A_p
A
Obs
JmaxRef = 98
RdRef = 2.09
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

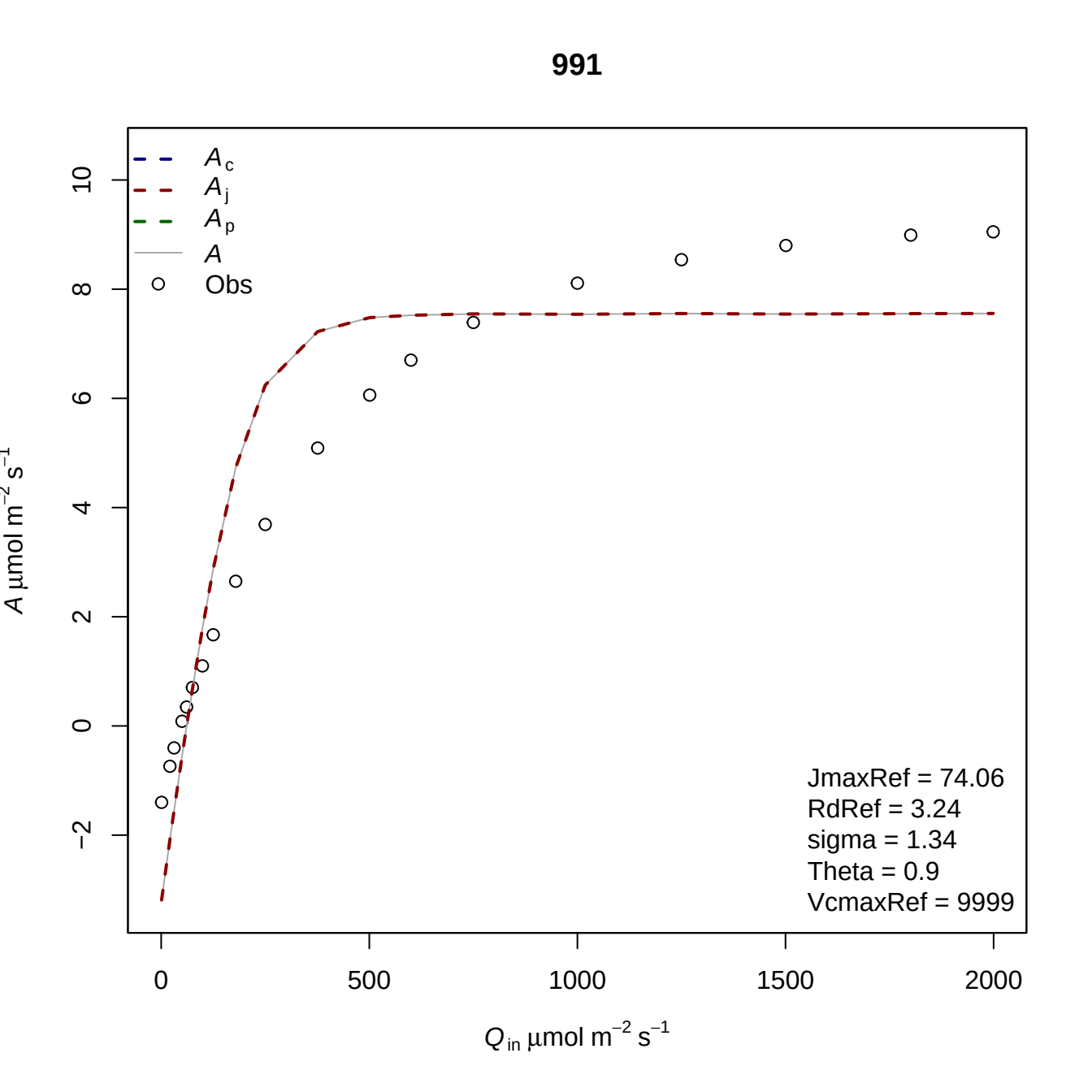

986
A_c
A_j
A_p
A
Obs
JmaxRef = 59.81
RdRef = 1.43
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

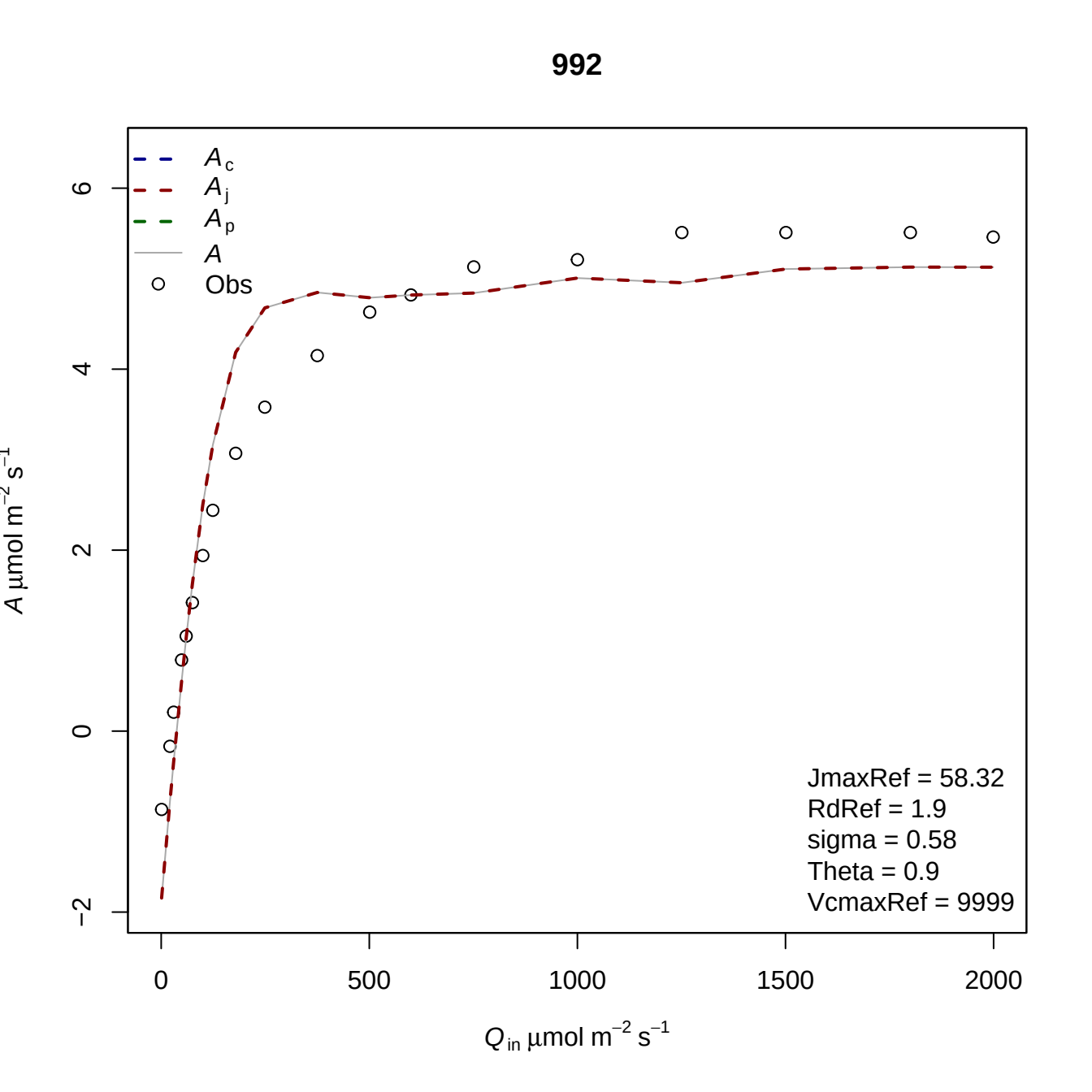

987
A_c
A_j
A_p
A
Obs
JmaxRef = 114.78
RdRef = 2.69
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

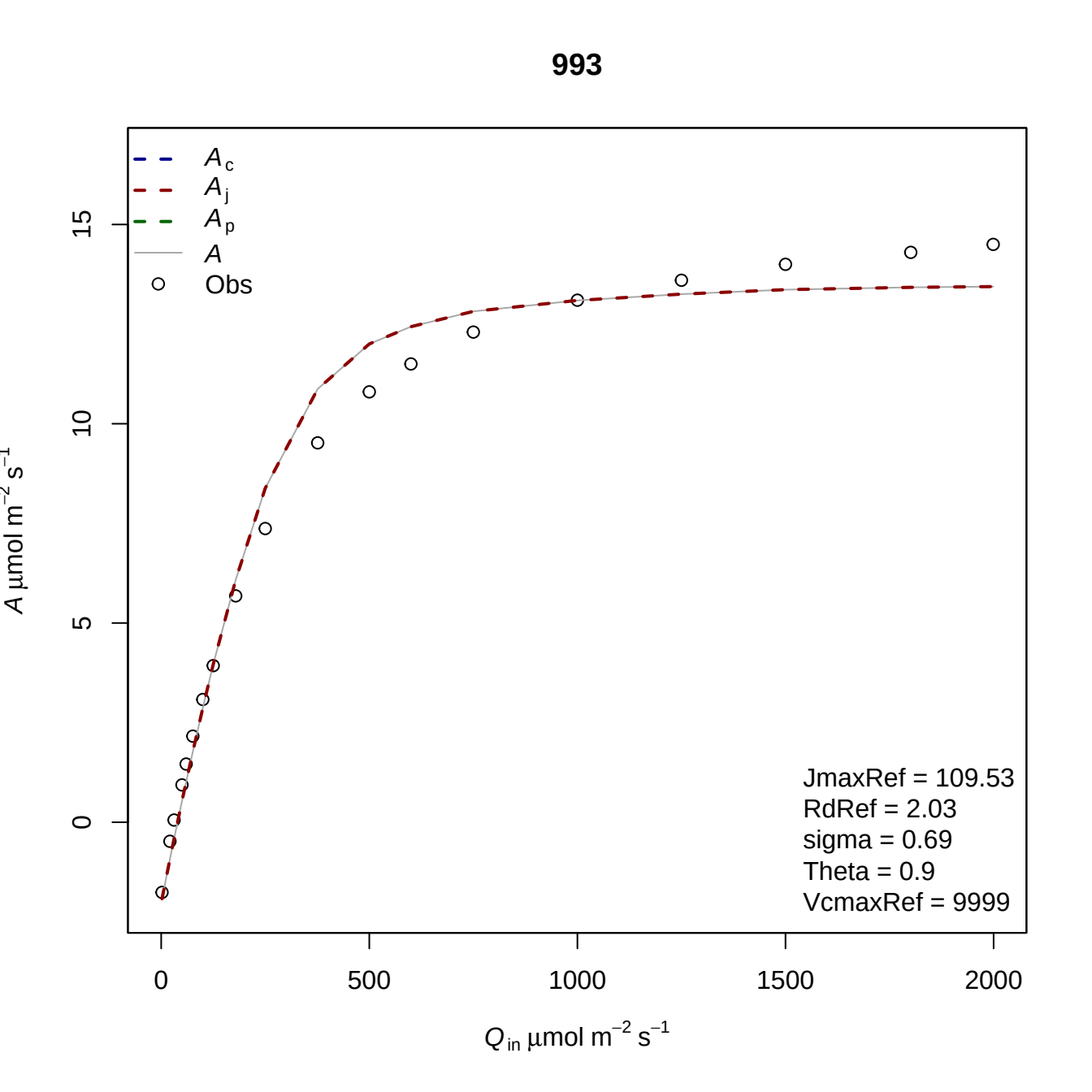

988
A_c
A_j
A_p
A
Obs
JmaxRef = 60.55
RdRef = 1.69
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

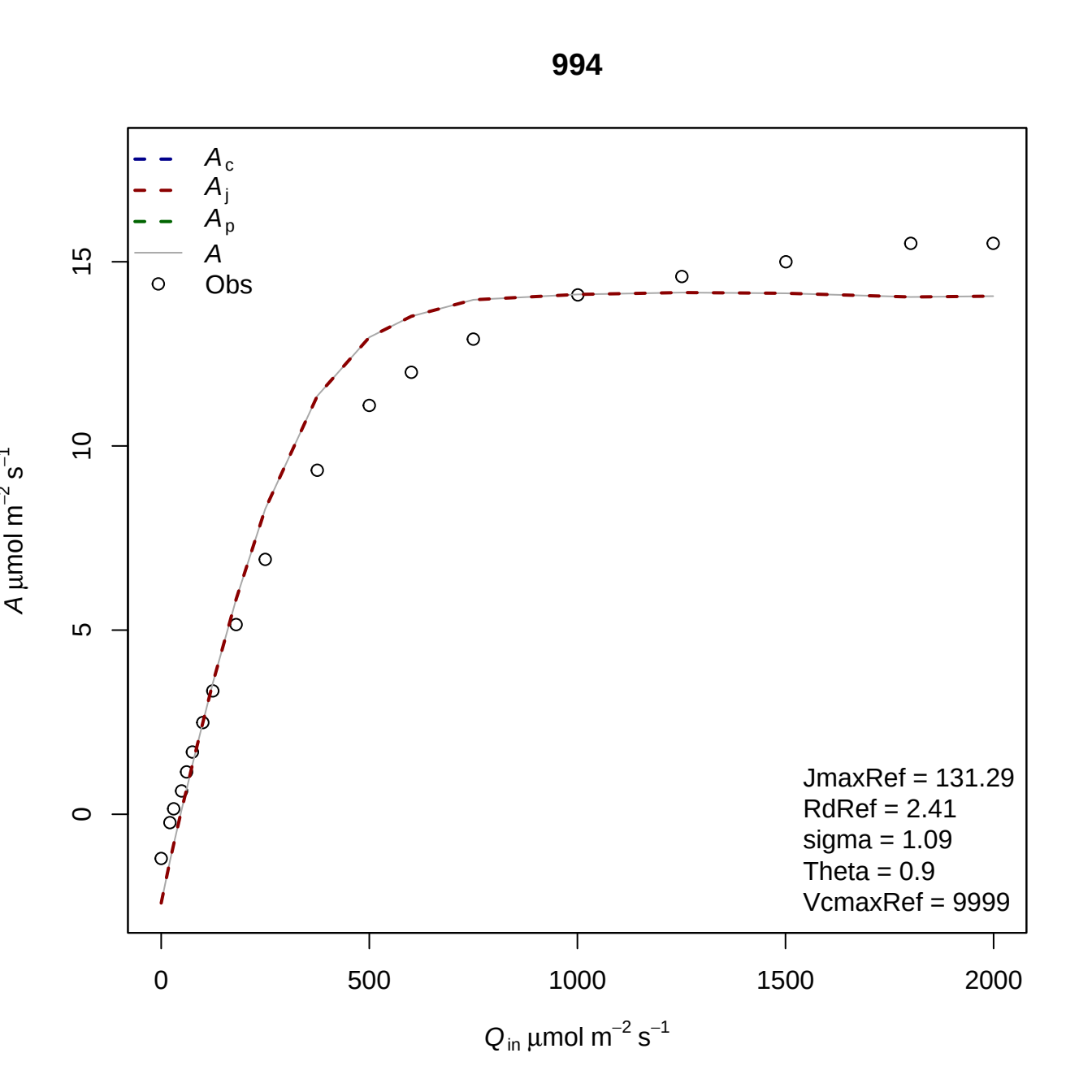

989
A_c
A_j
A_p
A
Obs
JmaxRef = 106.85
RdRef = 2.97
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

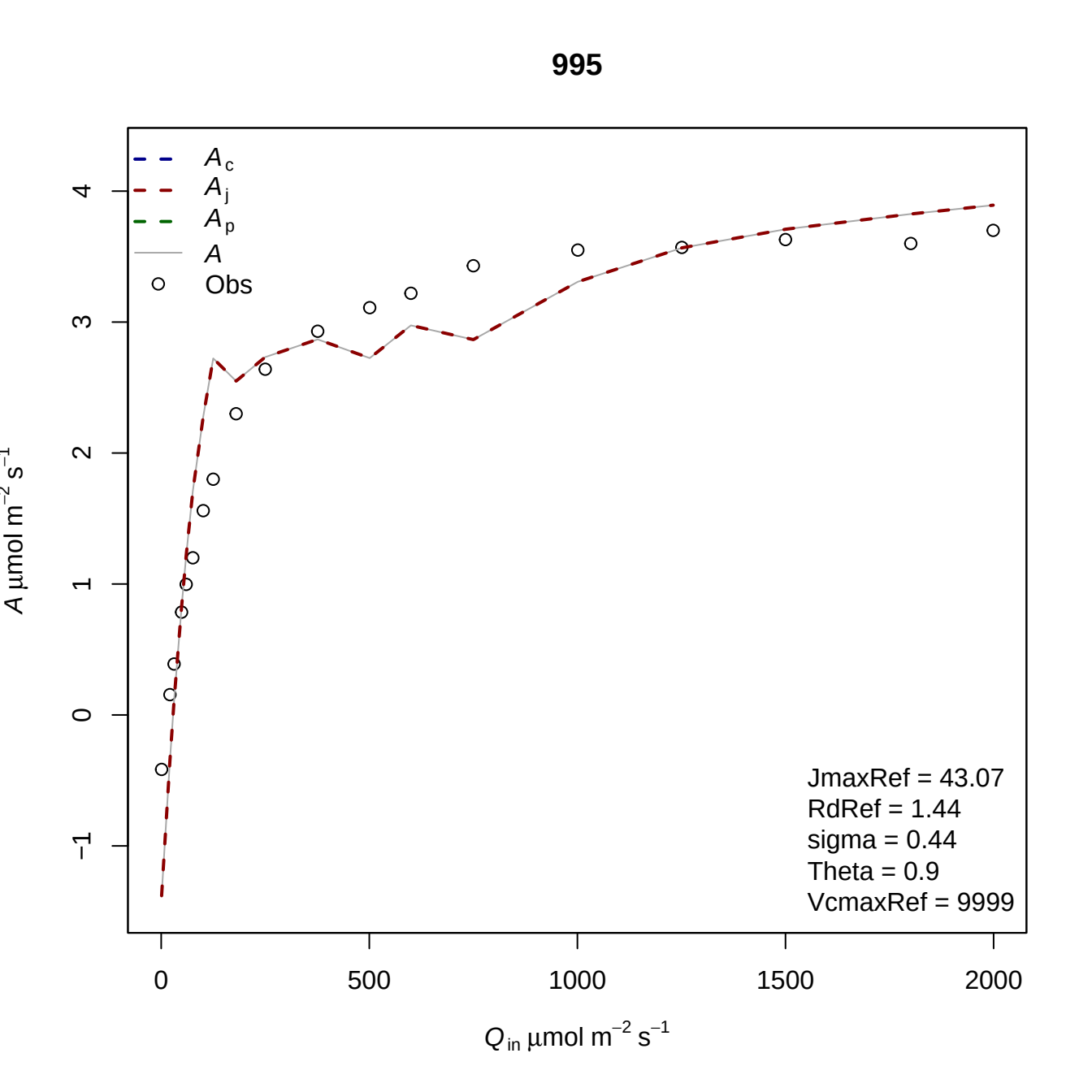

990
A_c
A_j
A_p
A
Obs
JmaxRef = 114.42
RdRef = 2.04
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

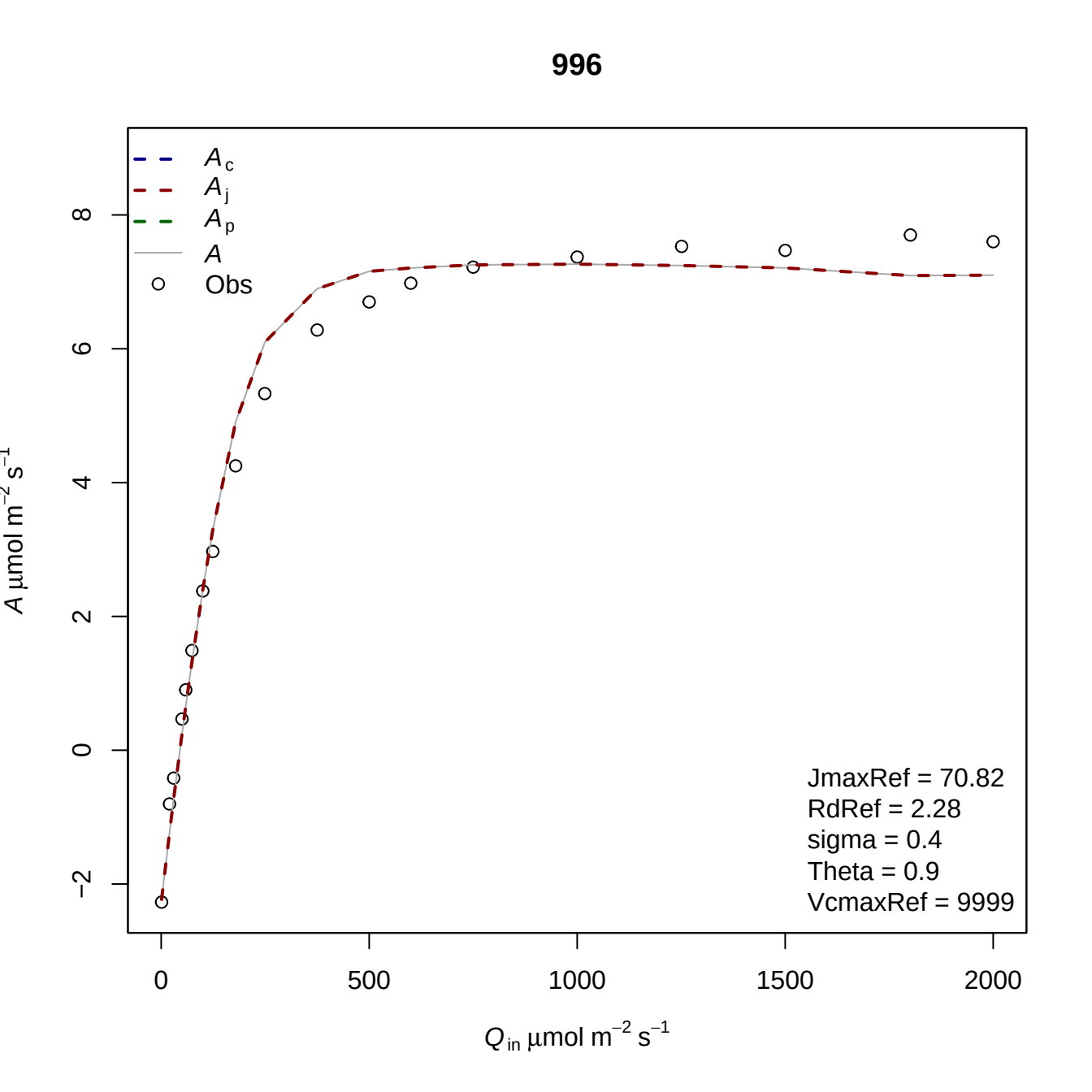

991
A_c
A_j
A_p
A
Obs
JmaxRef = 74.06
RdRef = 3.24
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

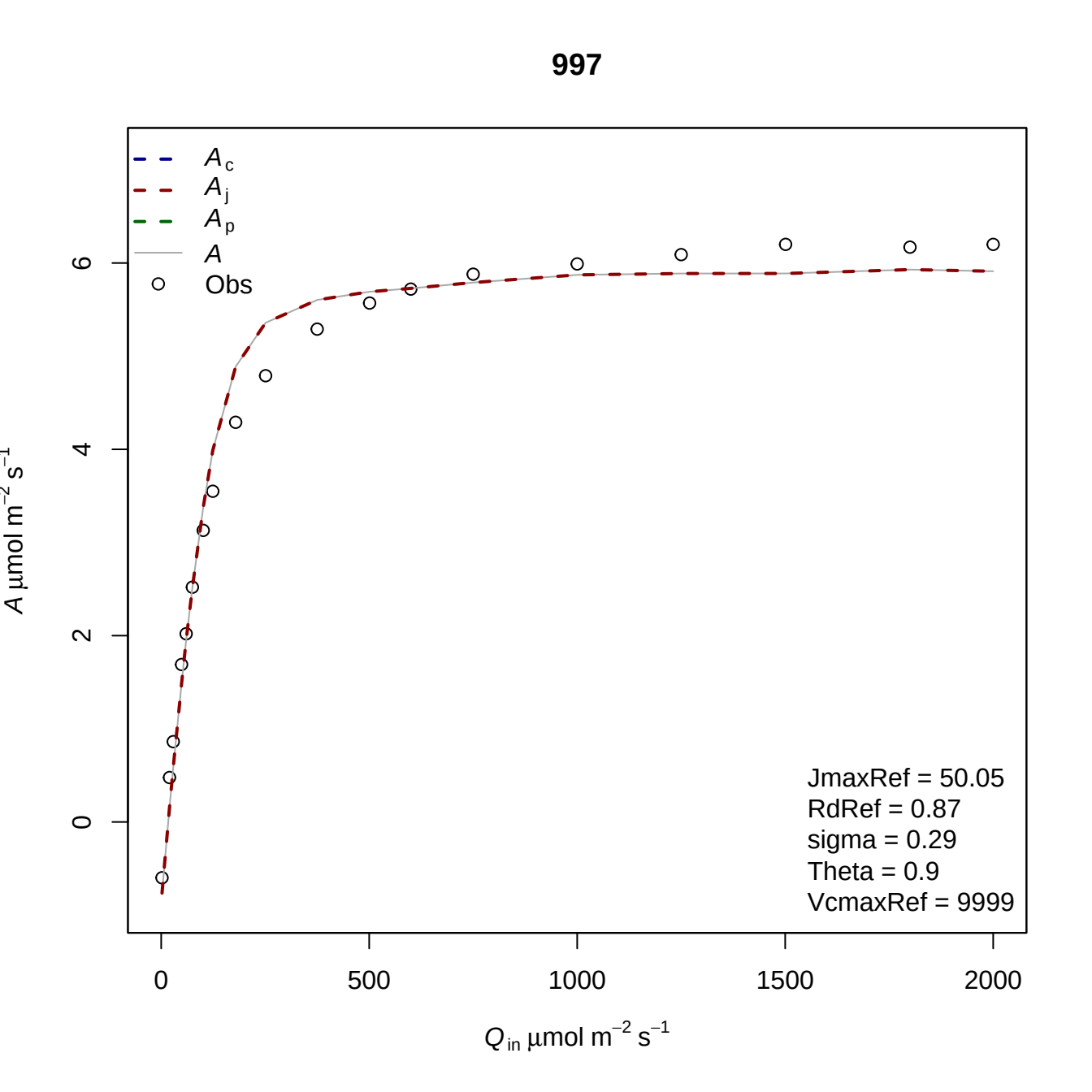

992
A_c
A_j
A_p
A
Obs
JmaxRef = 58.32
RdRef = 1.9
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

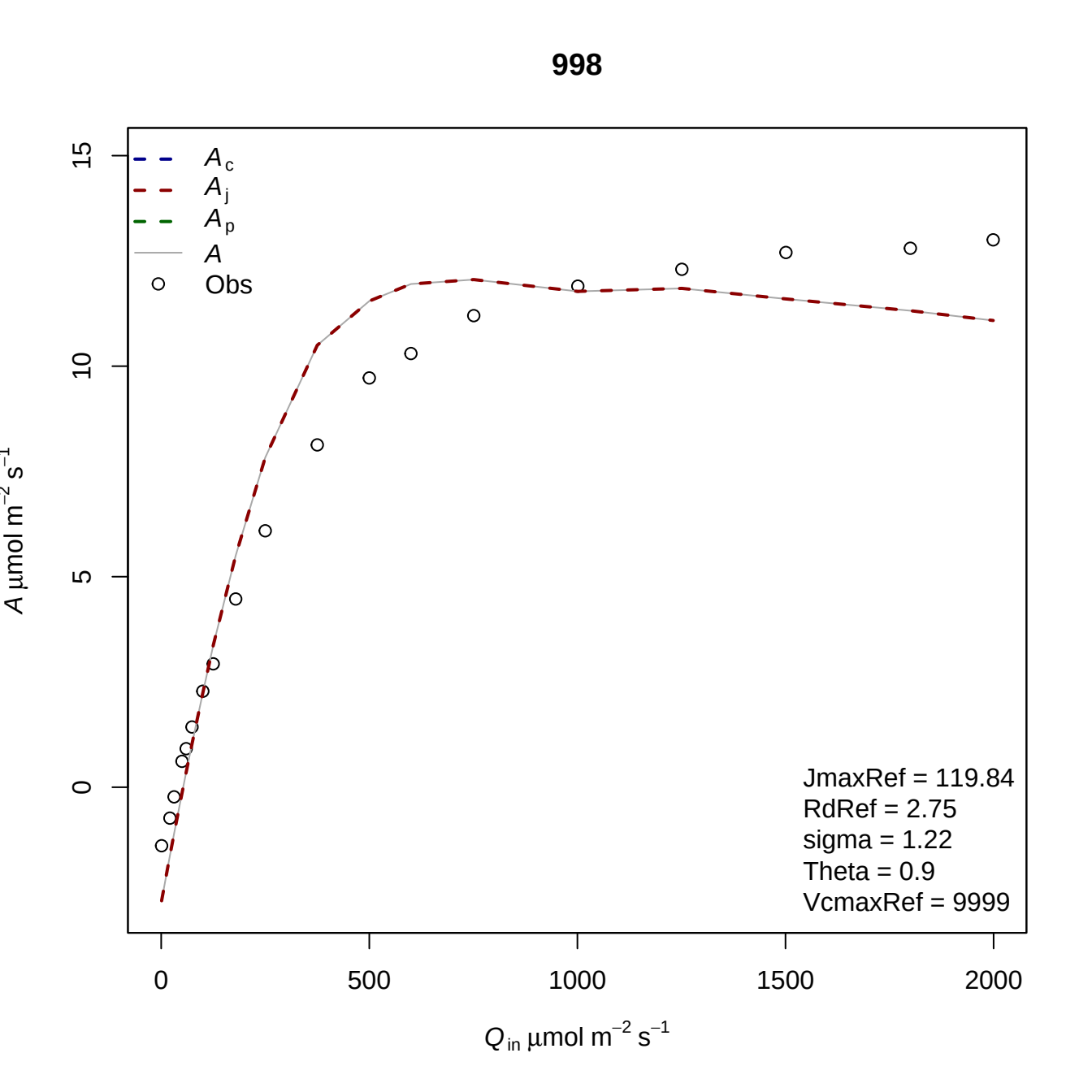

993
A_c
A_j
A_p
A
Obs
JmaxRef = 109.53
RdRef = 2.03
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

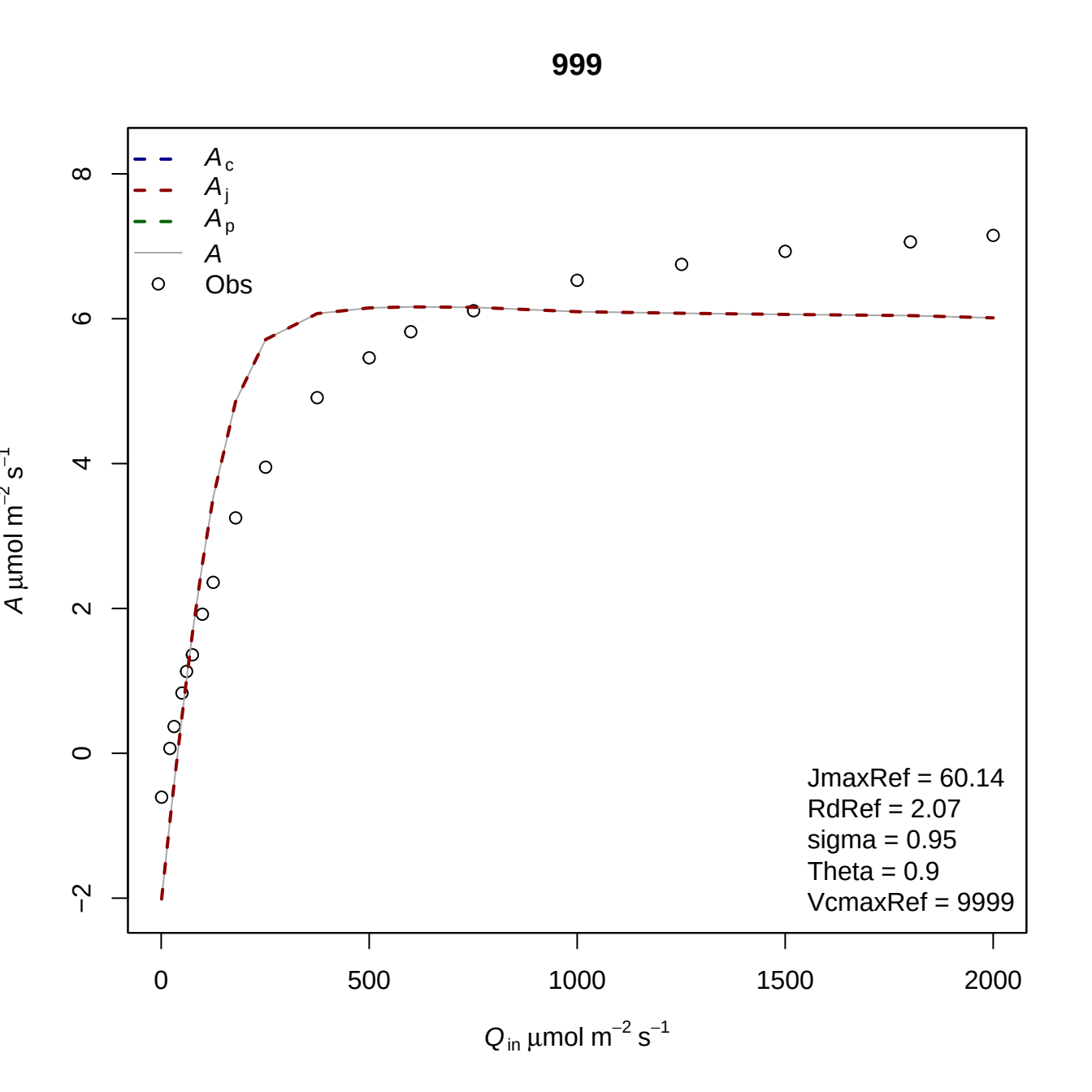

994
A_c
A_j
A_p
A
Obs
JmaxRef = 131.29
RdRef = 2.41
sigma = 1.09
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

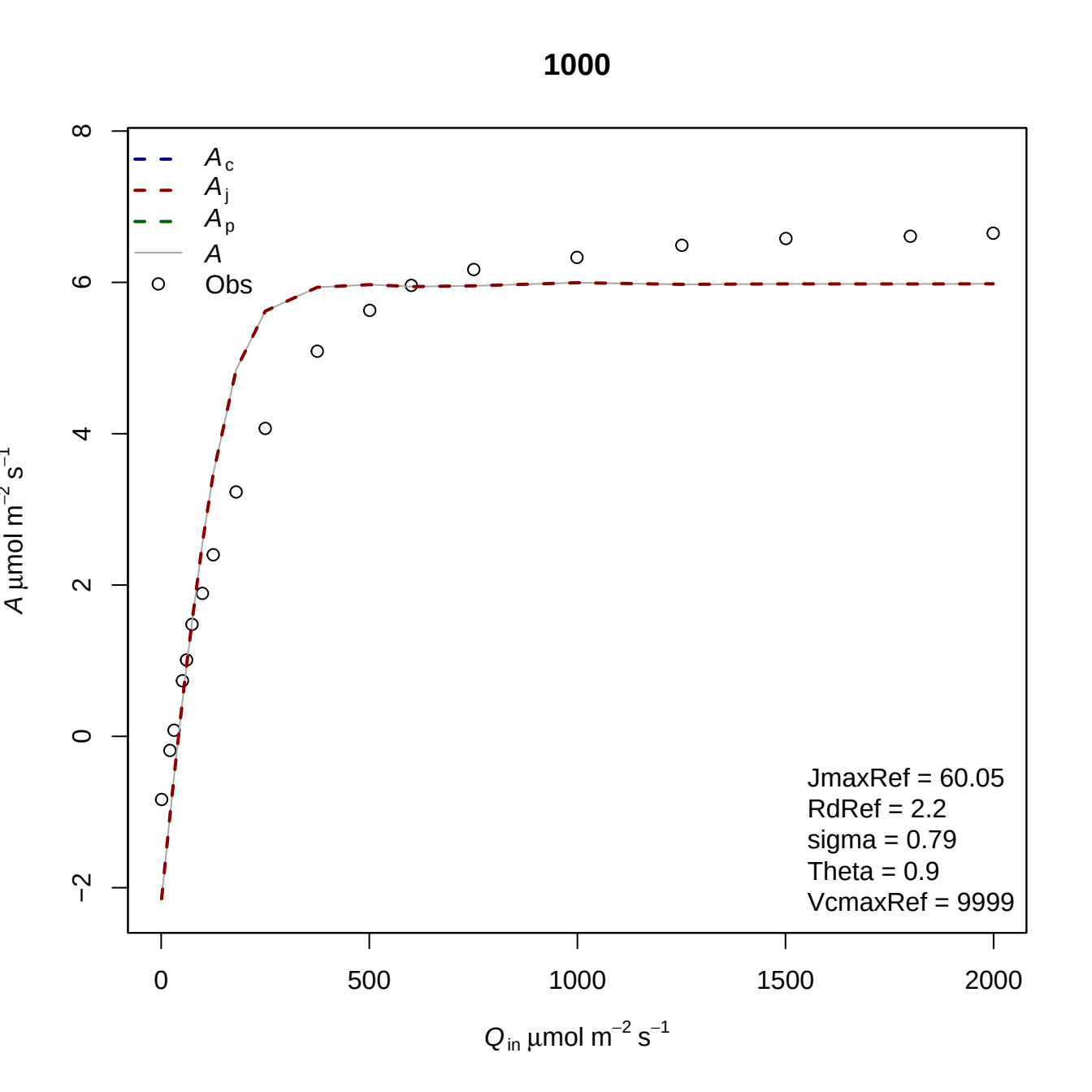

995
A_c
A_j
A_p
A
Obs
JmaxRef = 43.07
RdRef = 1.44
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

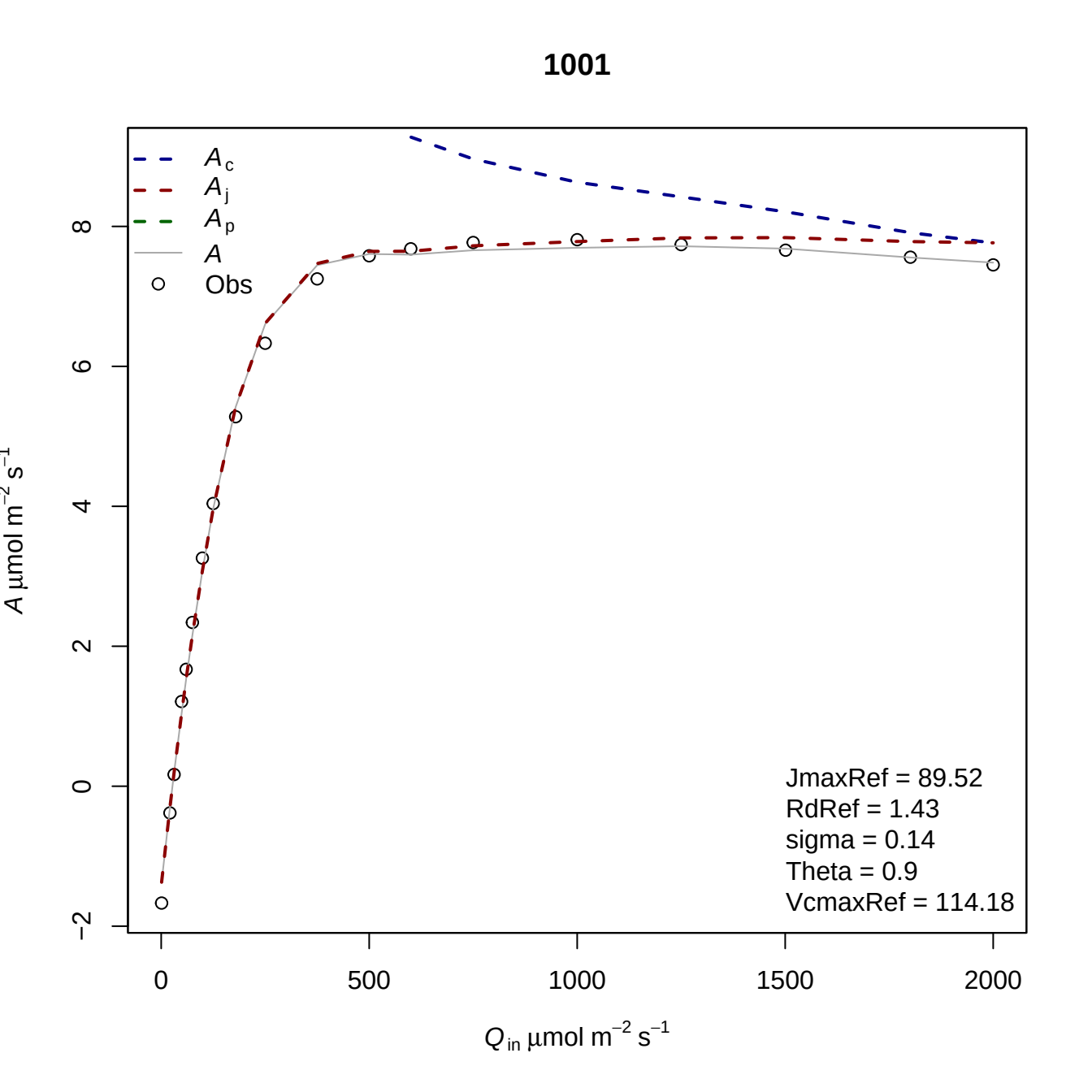

996
A_c
A_j
A_p
A
Obs
JmaxRef = 70.82
RdRef = 2.28
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

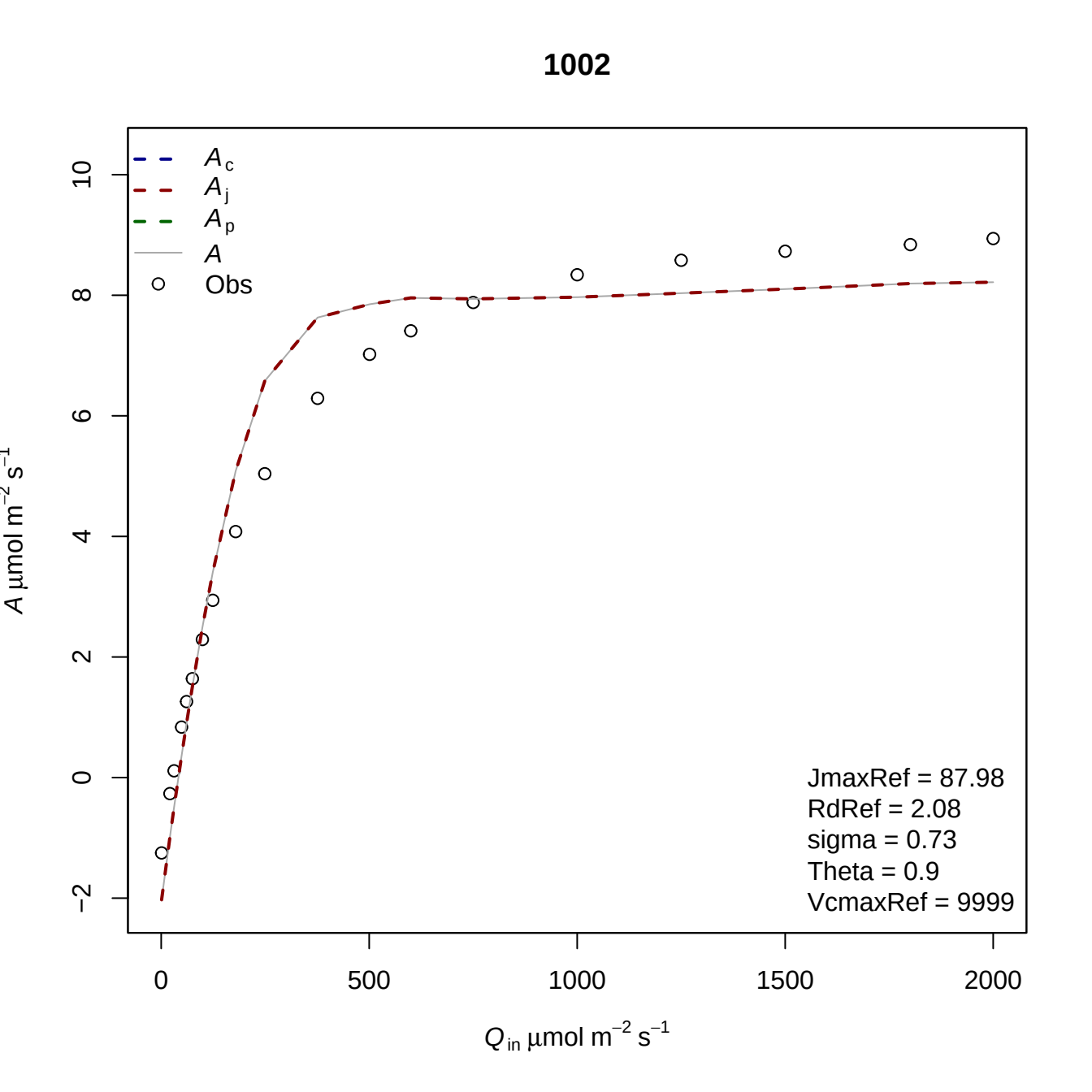

997
A_c
A_j
A_p
A
Obs
JmaxRef = 50.05
RdRef = 0.87
sigma = 0.29
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

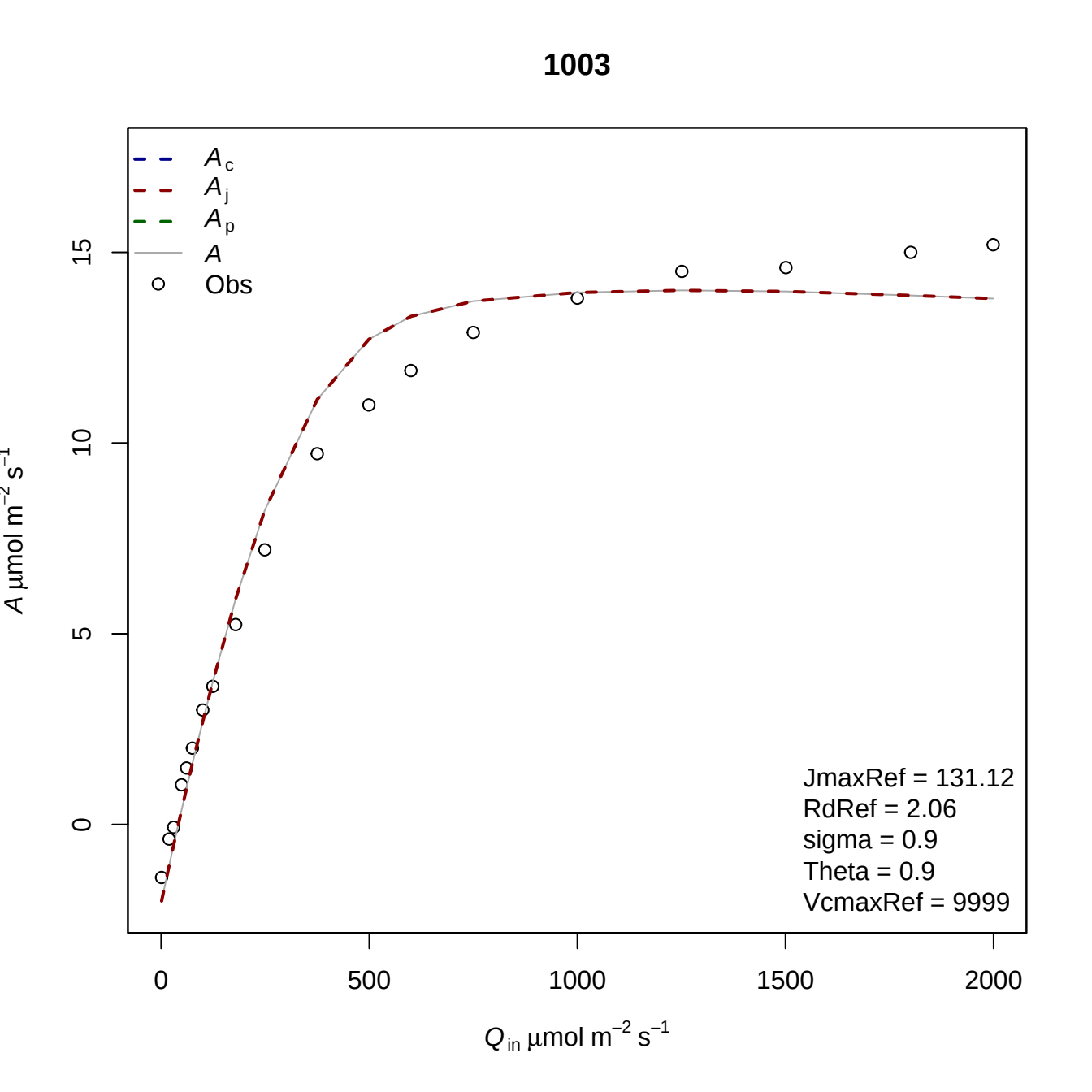

998
A_c
A_j
A_p
A
Obs
JmaxRef = 119.84
RdRef = 2.75
sigma = 1.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

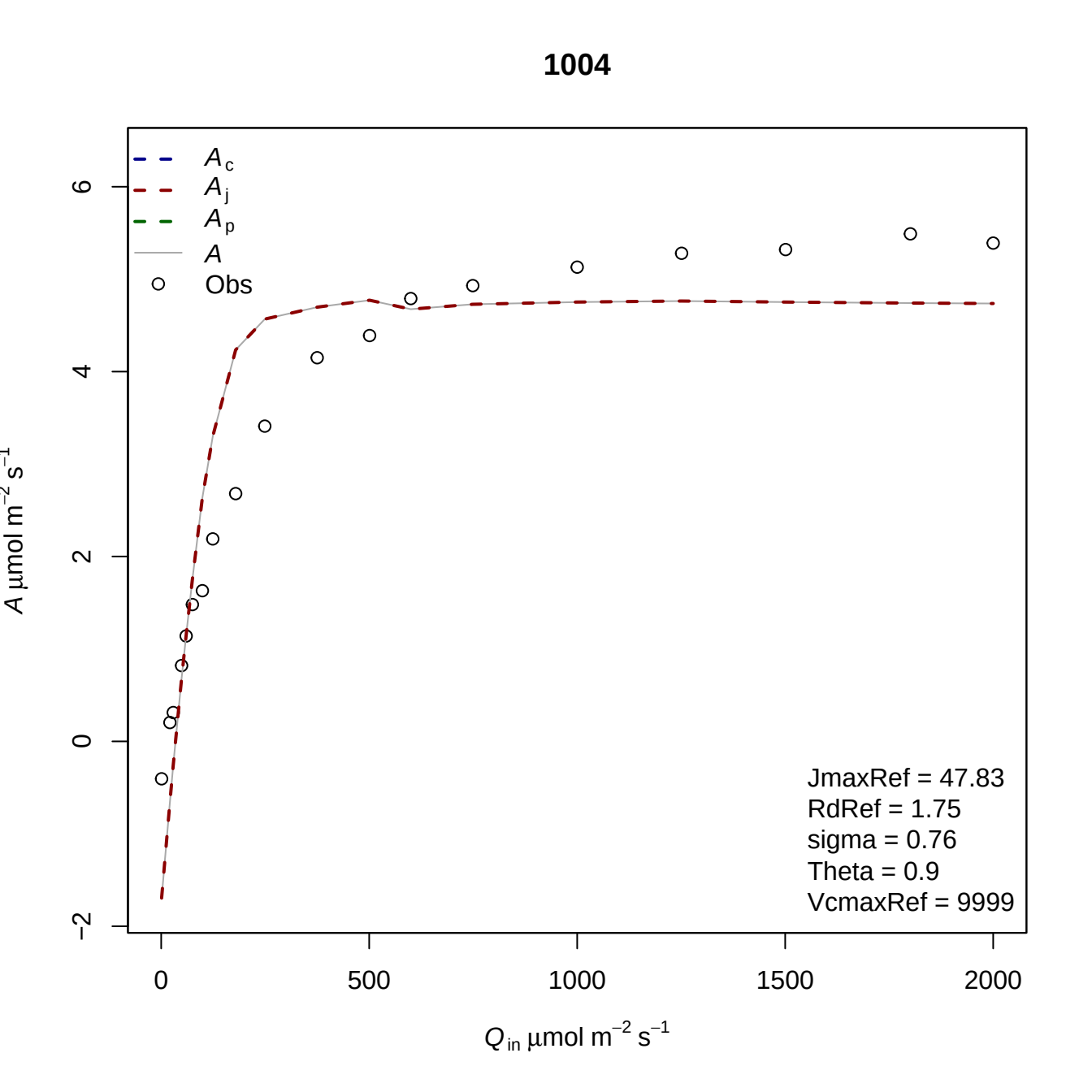

999
A_c
A_j
A_p
A
Obs
JmaxRef = 60.14
RdRef = 2.07
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

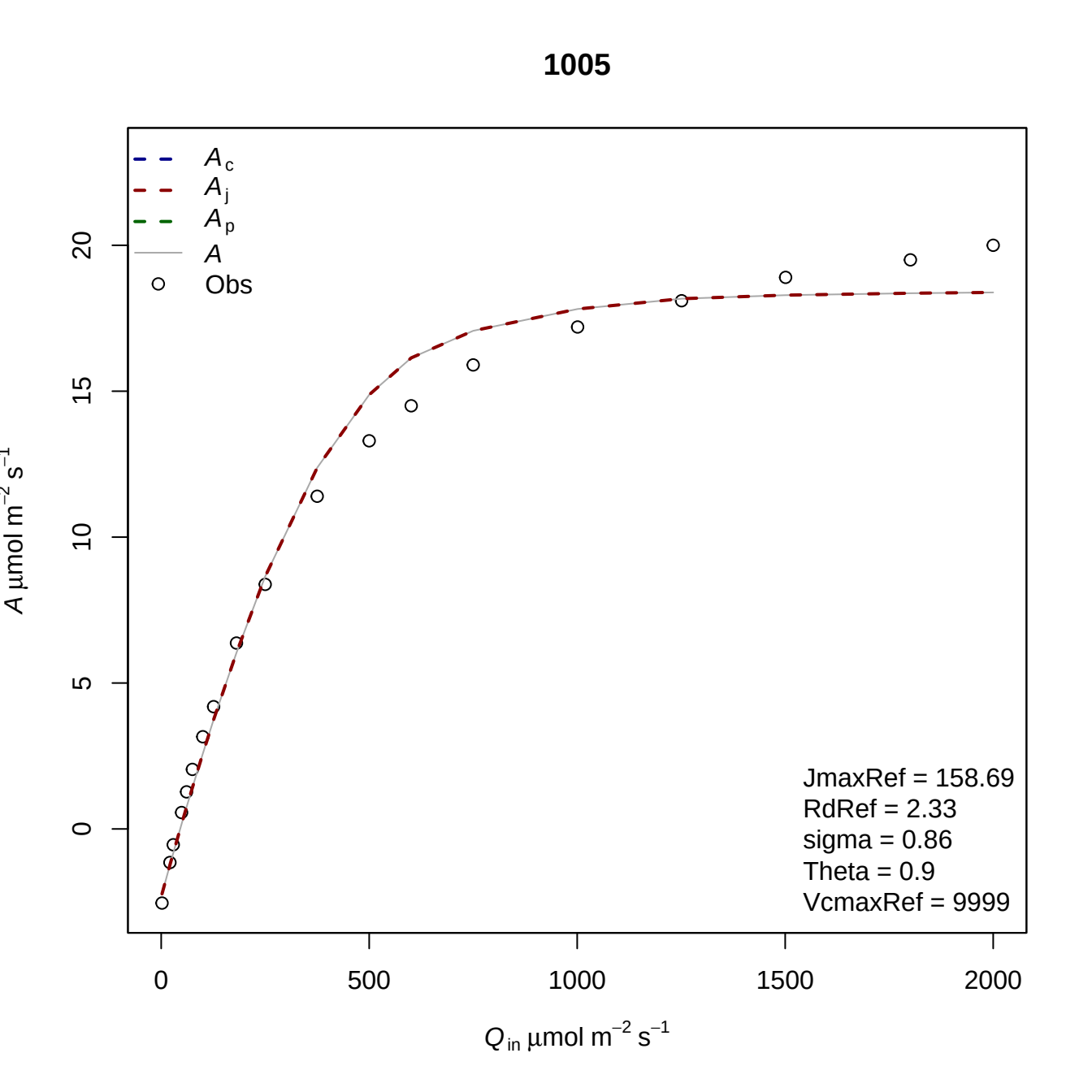

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 60.05
RdRef = 2.2
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

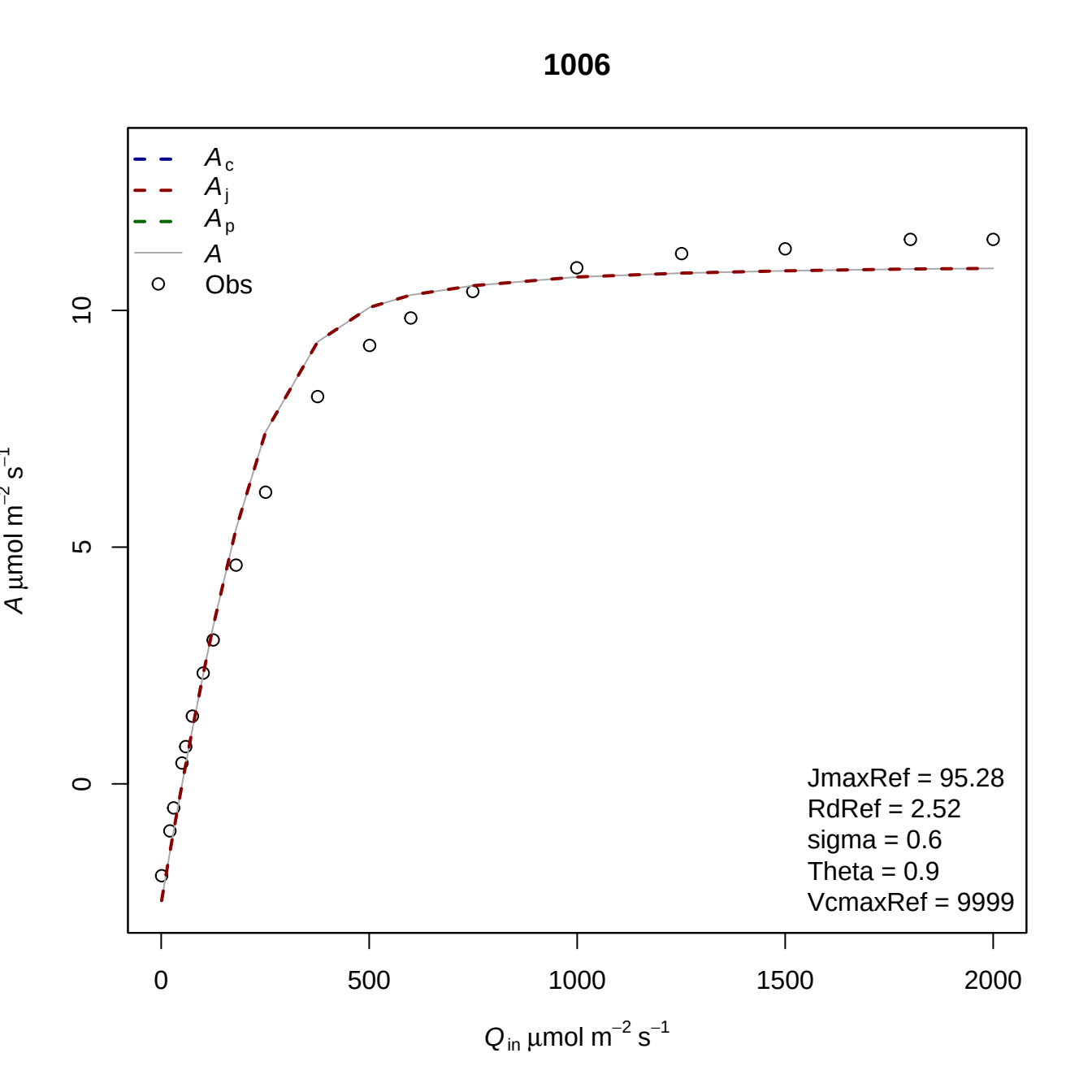

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 89.52
RdRef = 1.43
sigma = 0.14
Theta = 0.9
VcmaxRef = 114.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

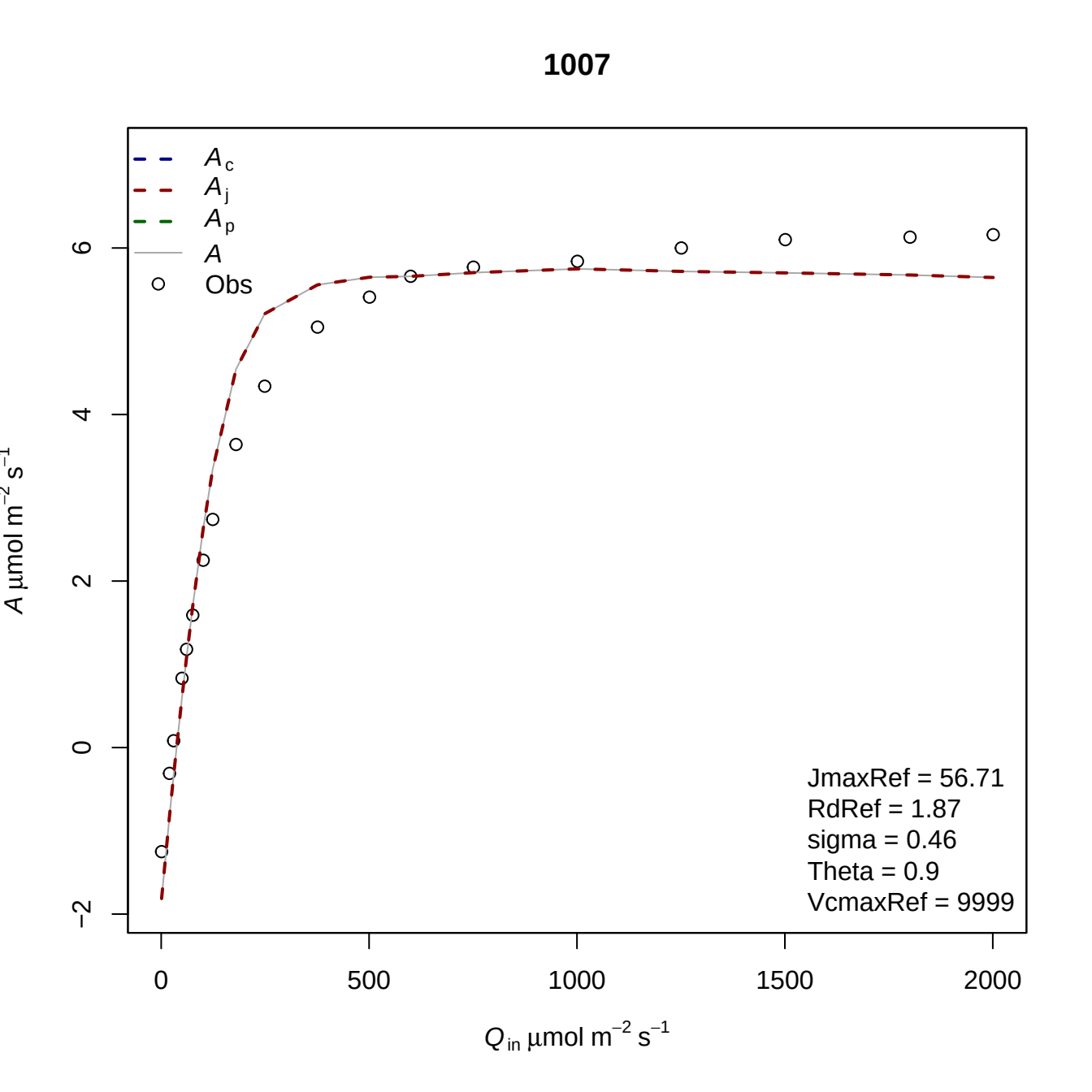

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 87.98
RdRef = 2.08
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

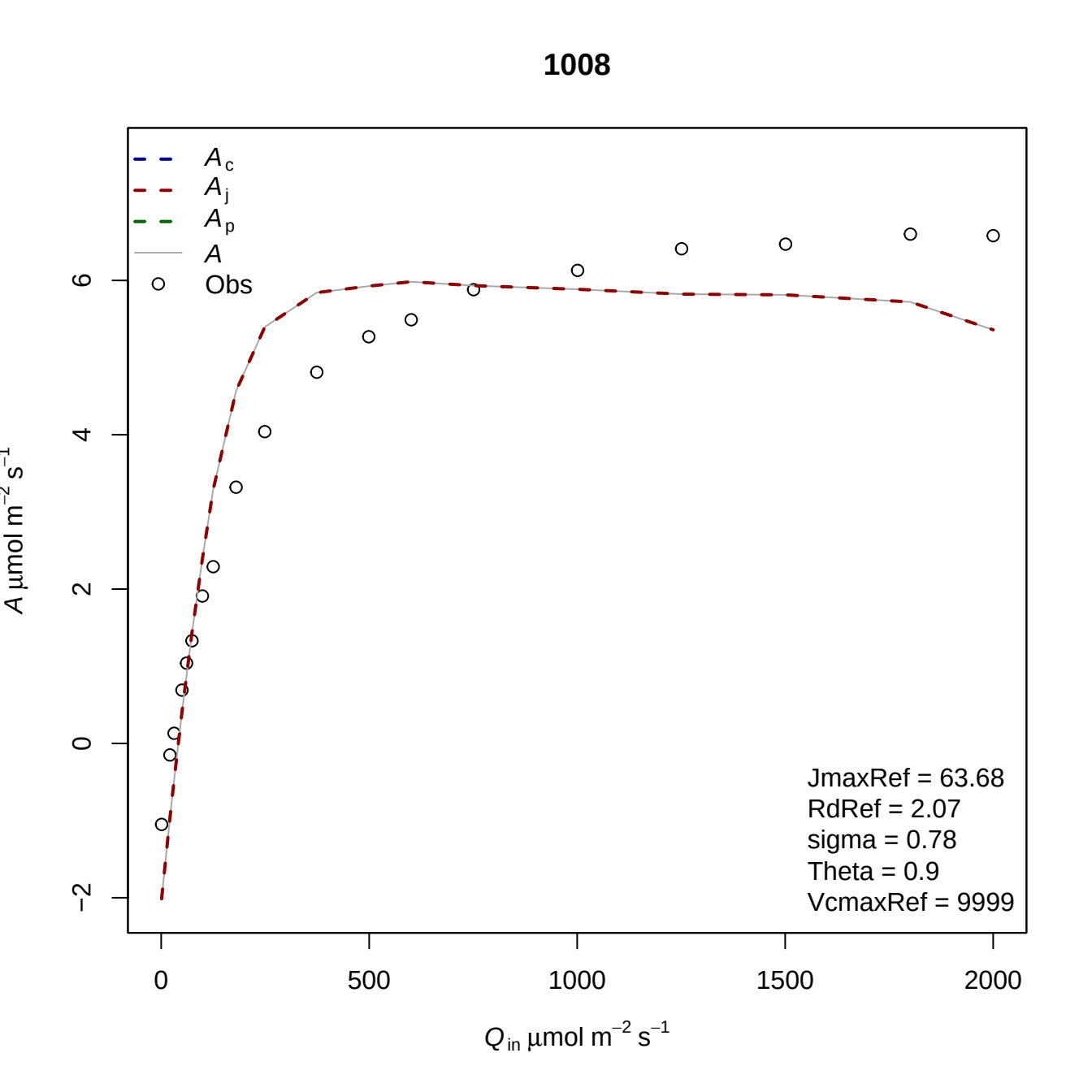

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 131.12
RdRef = 2.06
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

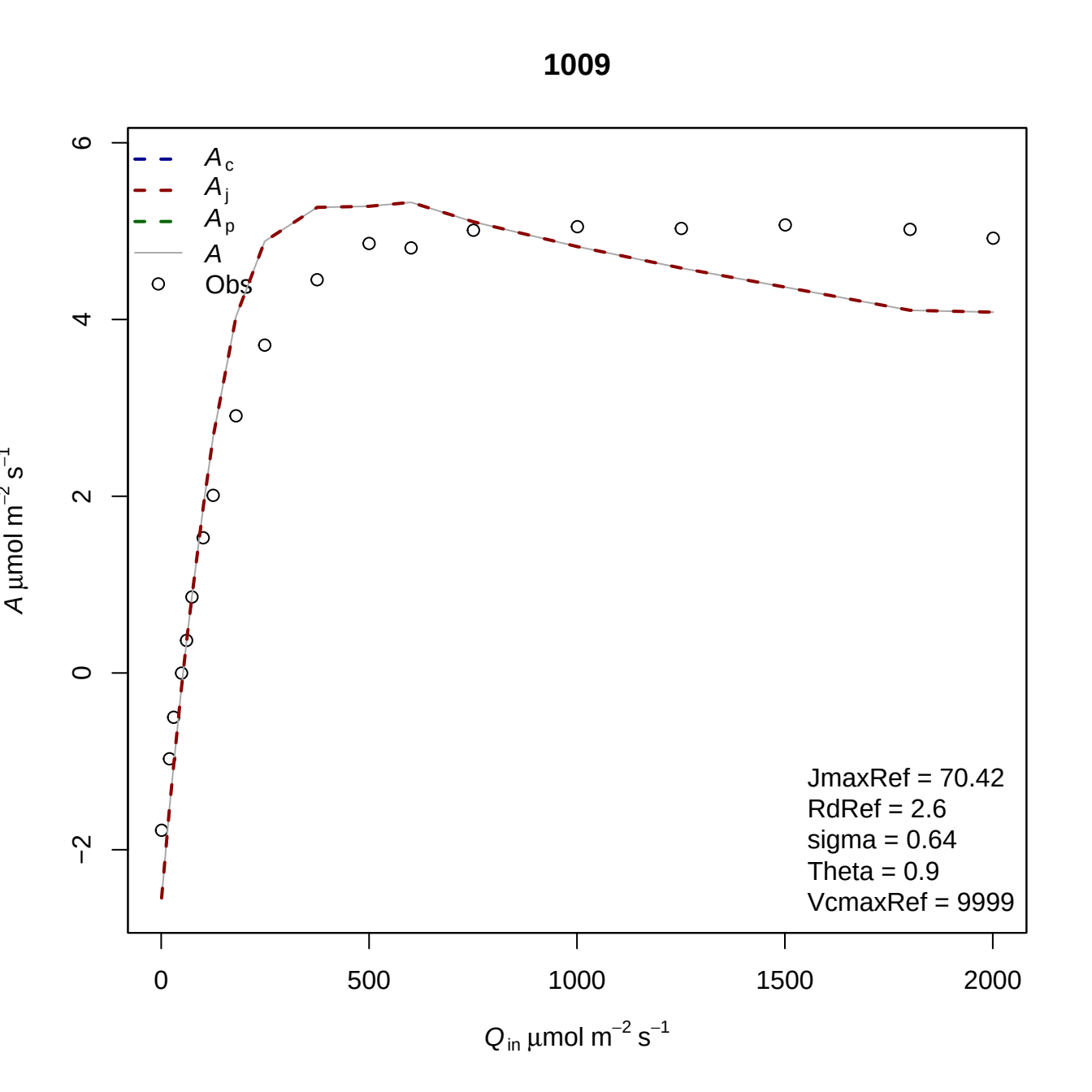

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 47.83
RdRef = 1.75
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

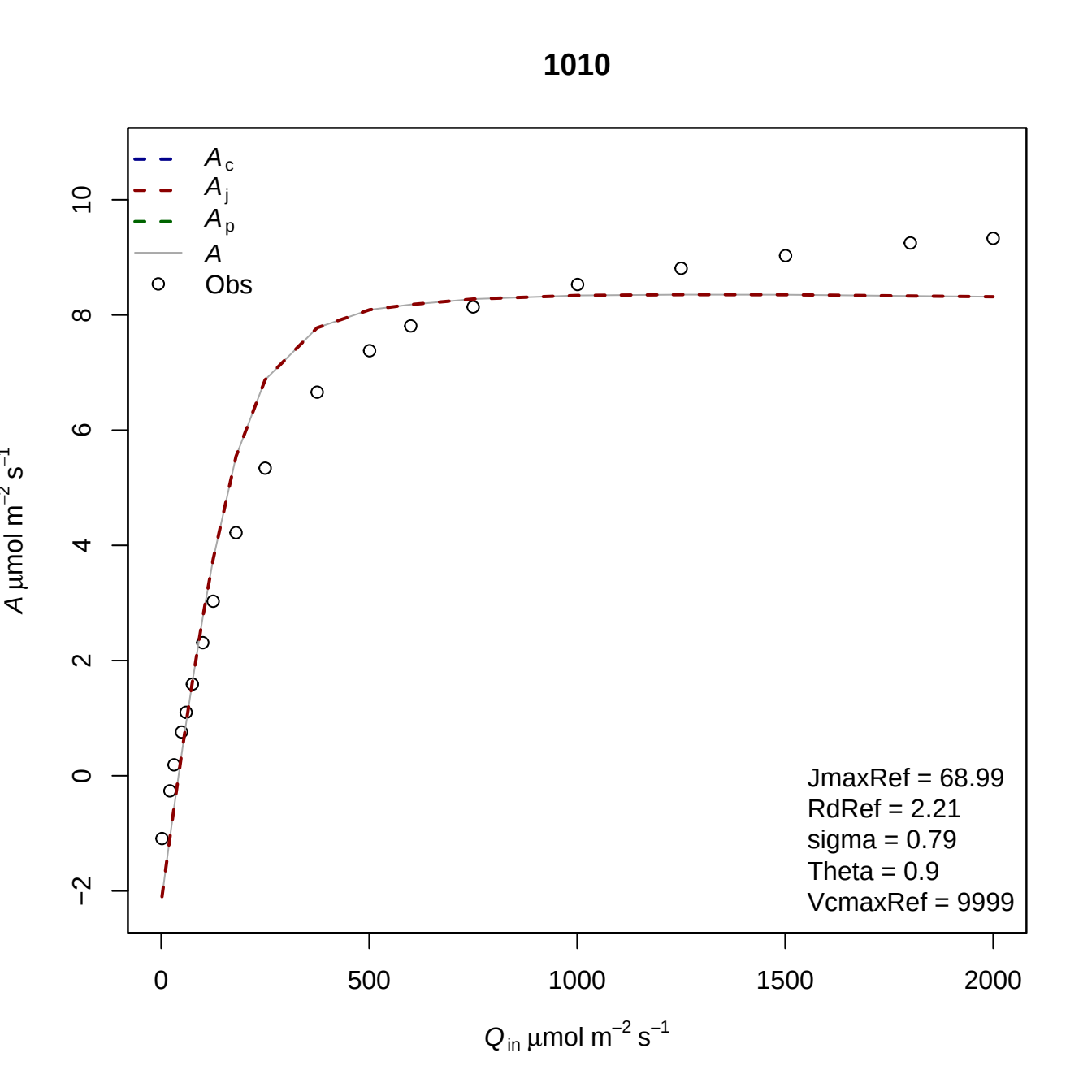

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 158.69
RdRef = 2.33
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

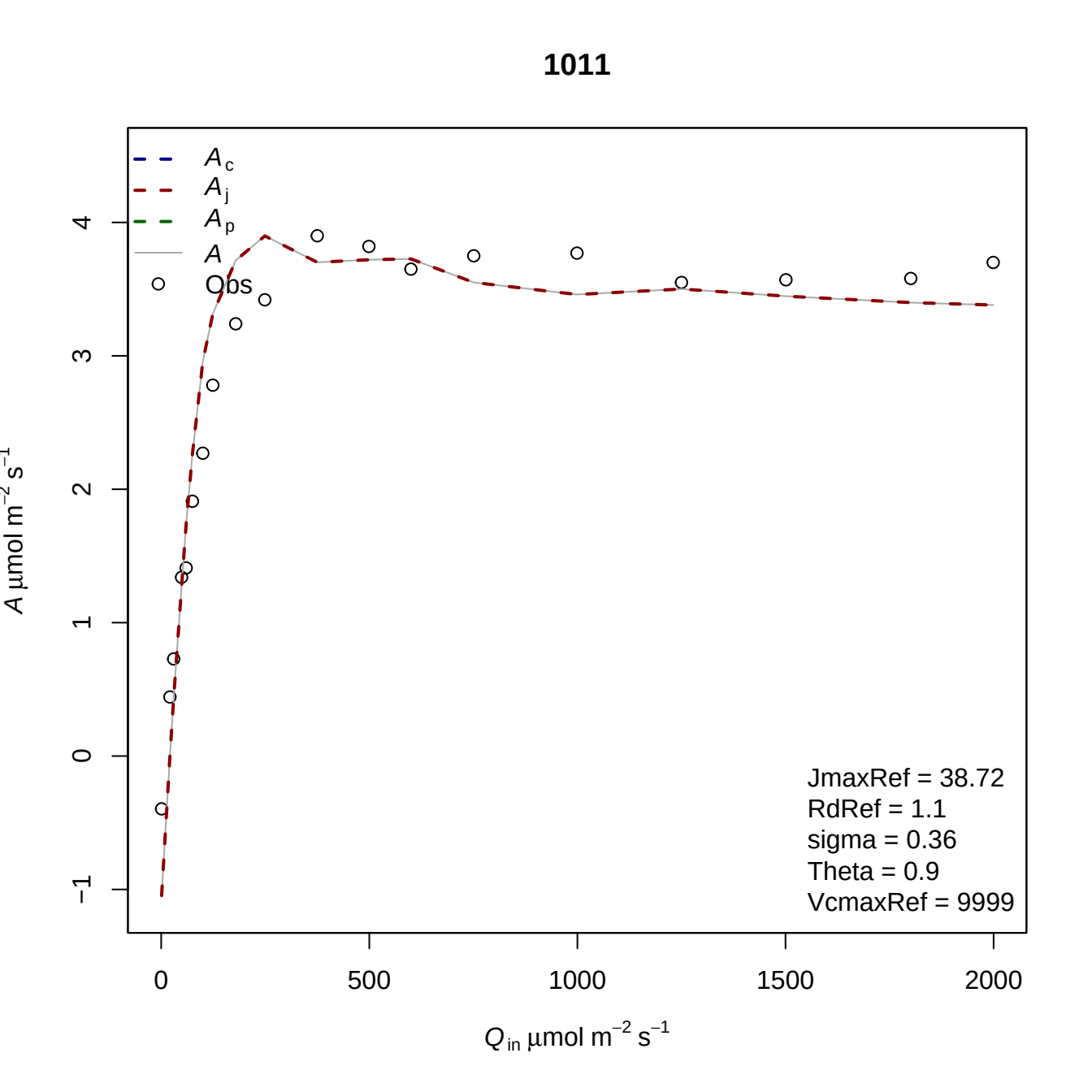

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 95.28
RdRef = 2.52
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

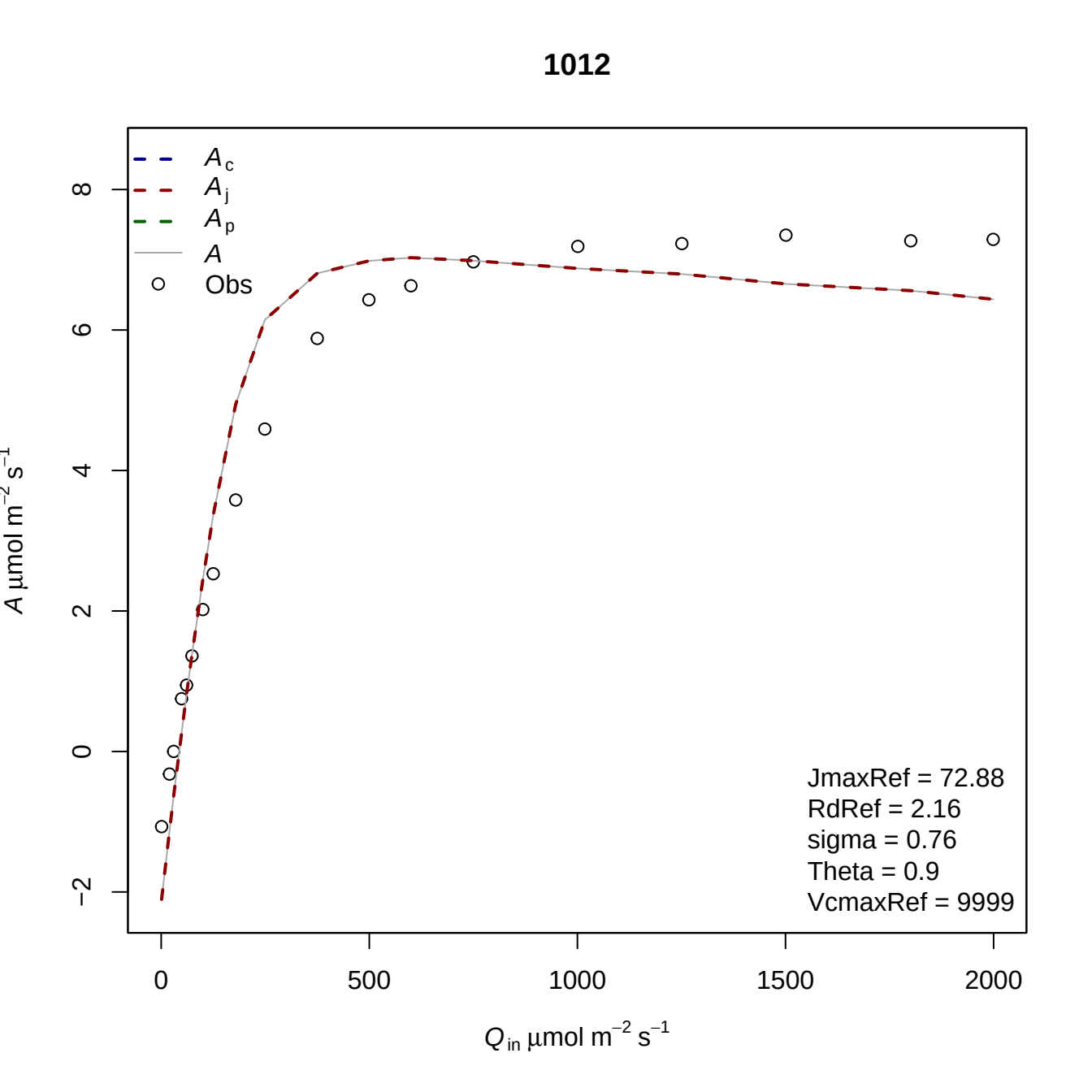

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 56.71
RdRef = 1.87
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

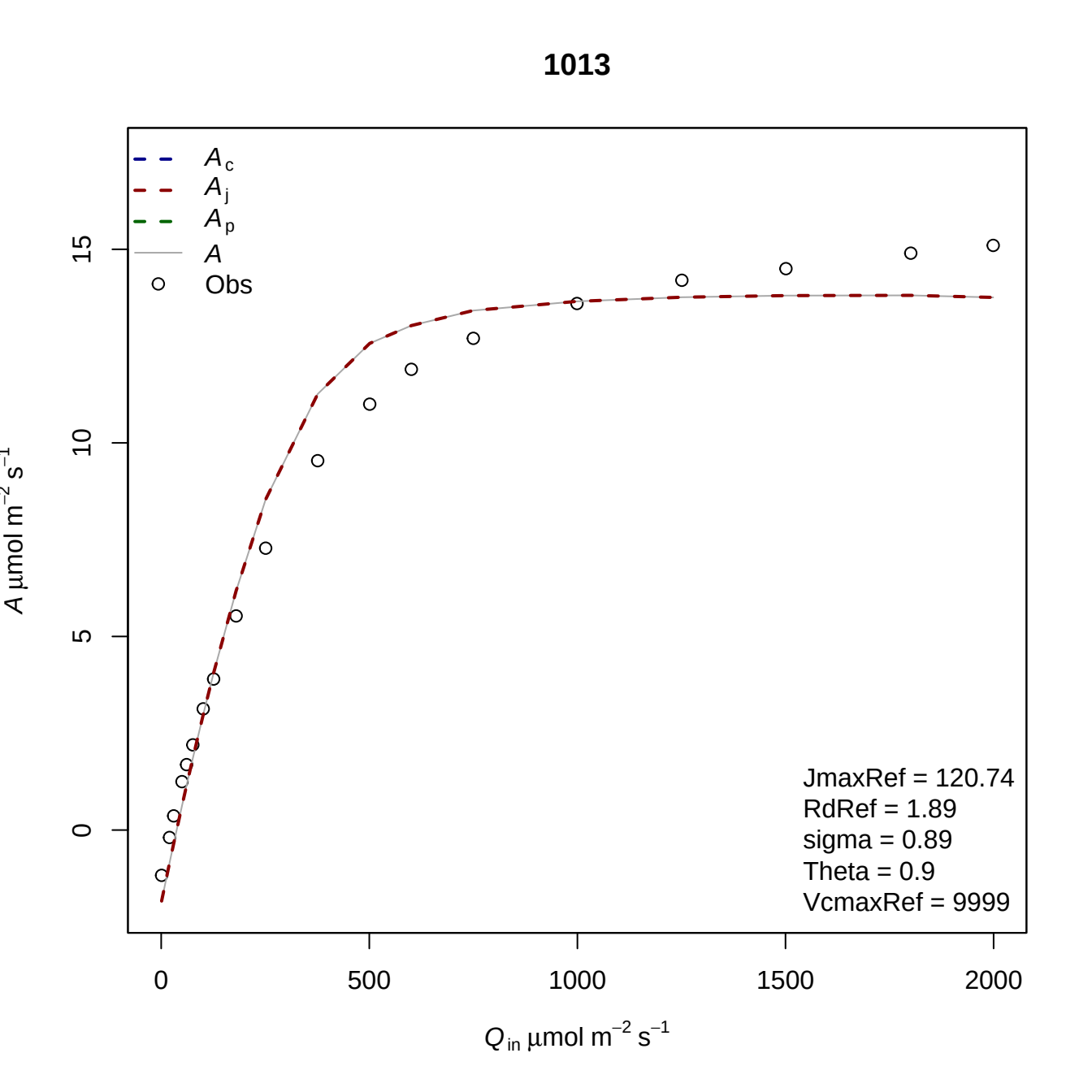

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 63.68
RdRef = 2.07
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

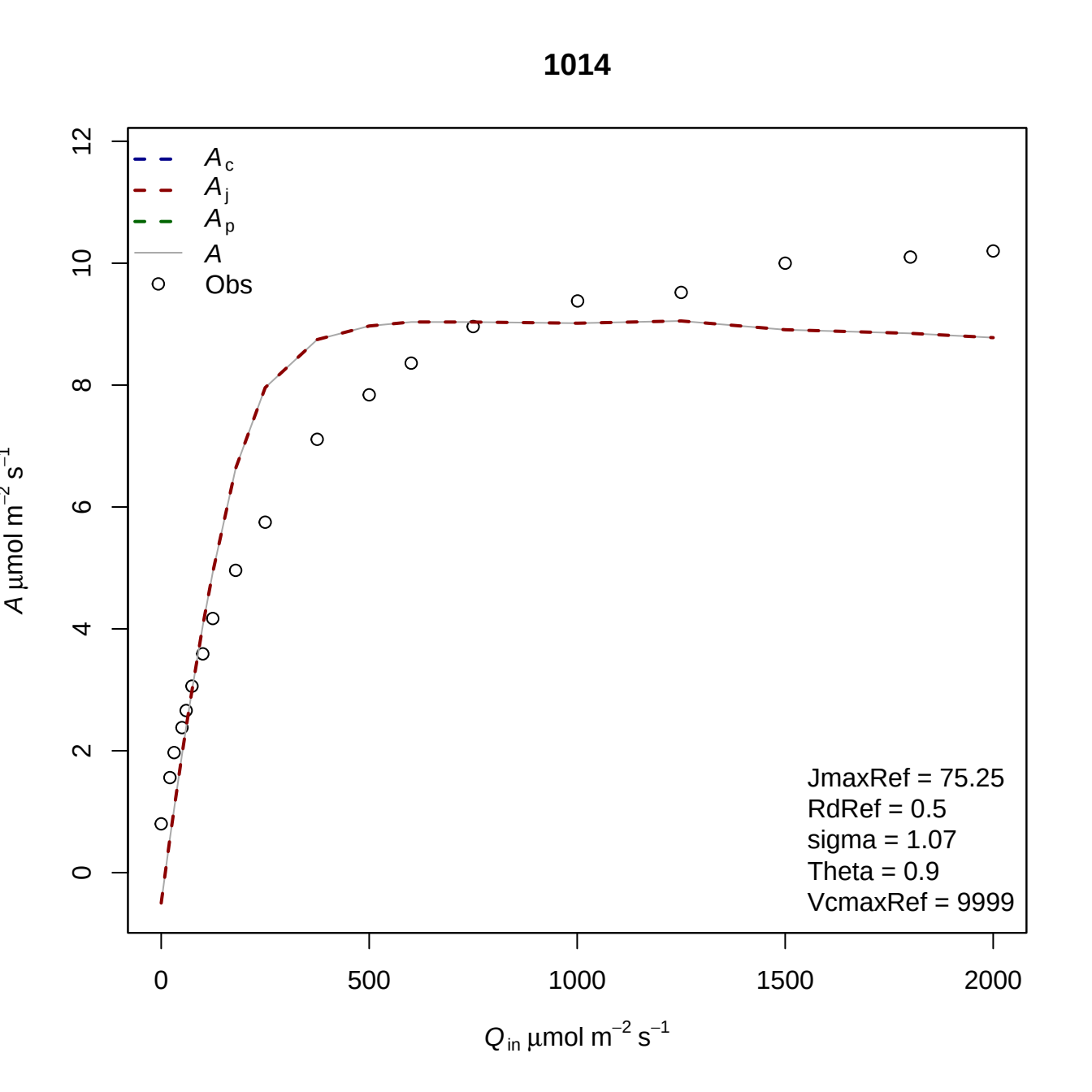

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 70.42
RdRef = 2.6
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

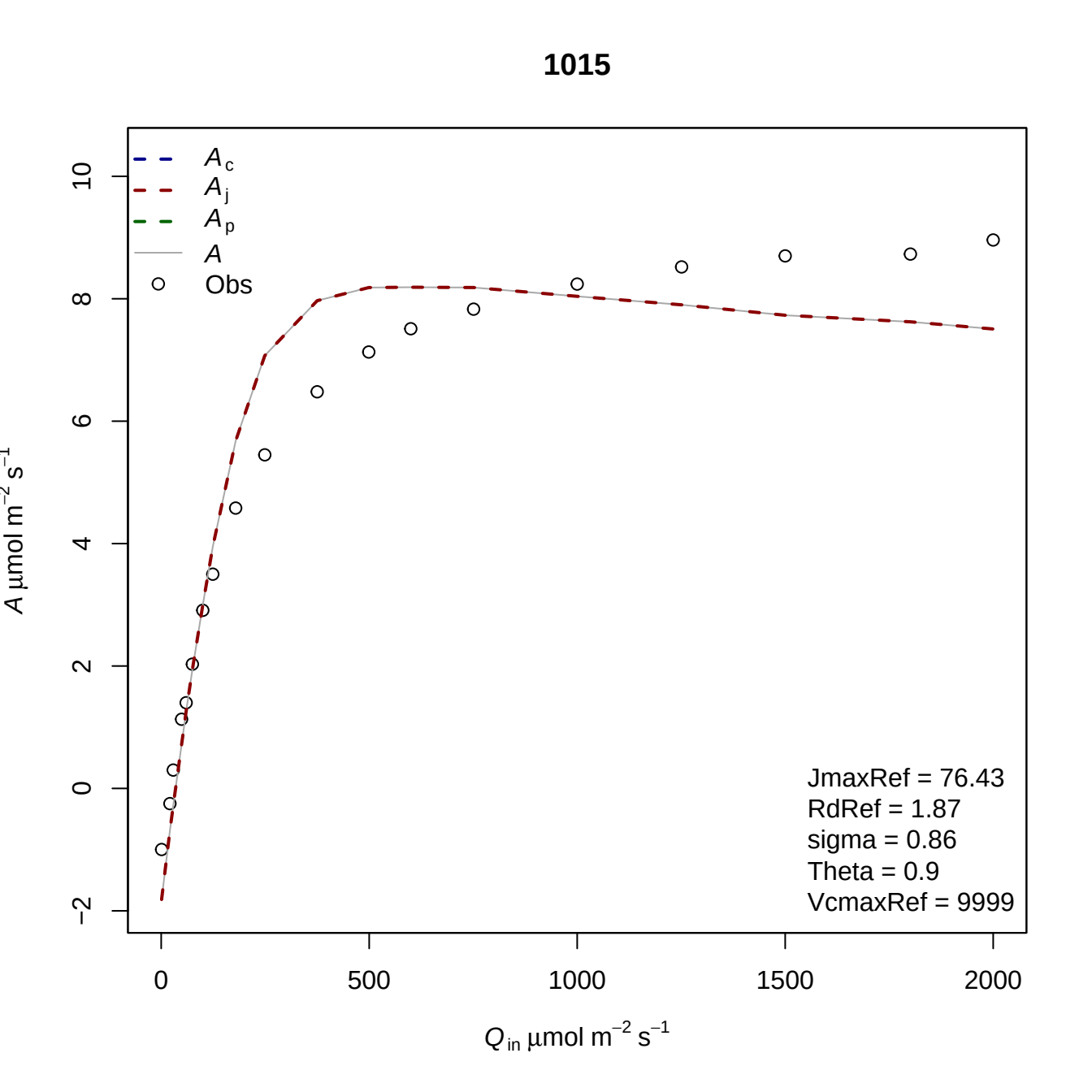

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 68.99
RdRef = 2.21
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

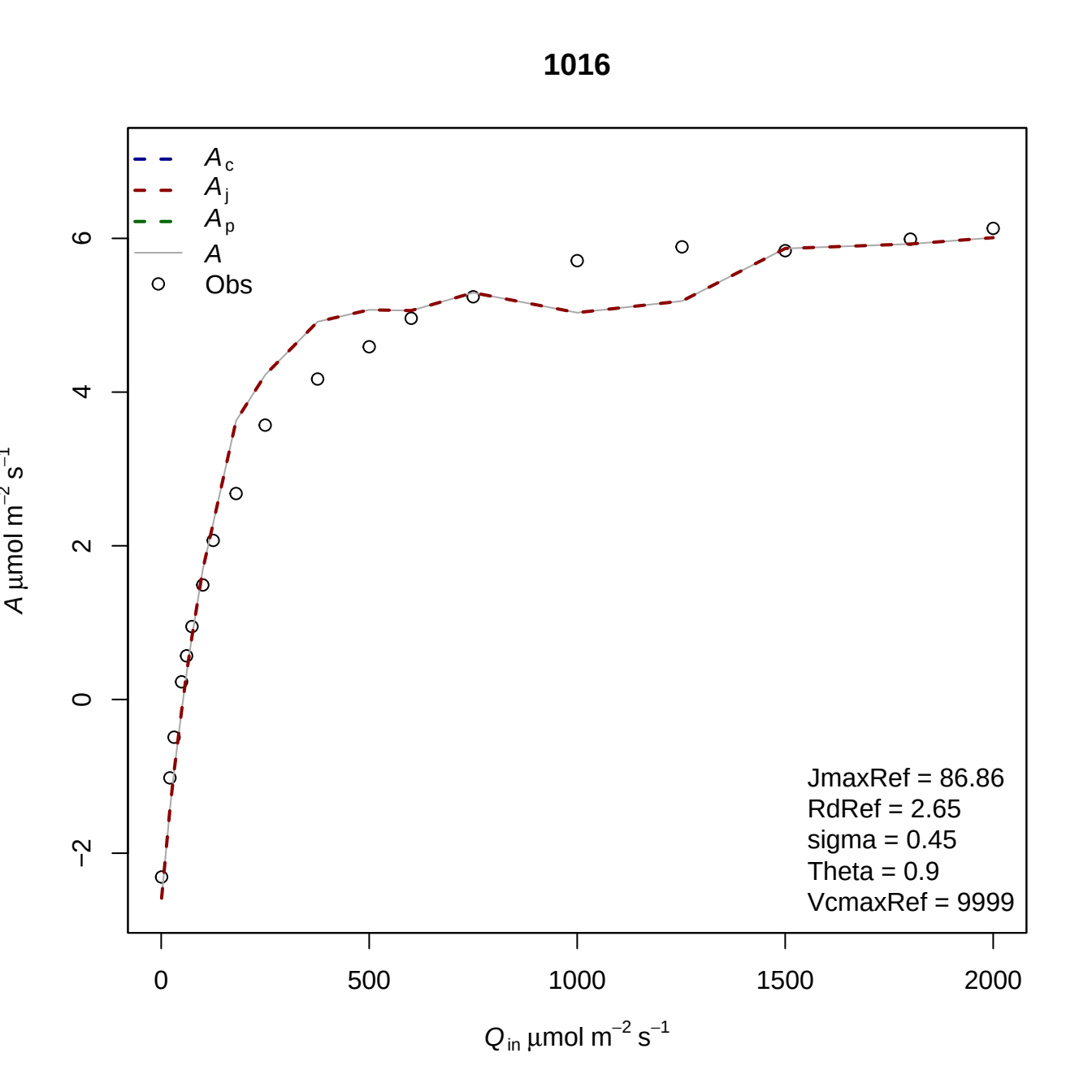

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 38.72
RdRef = 1.1
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

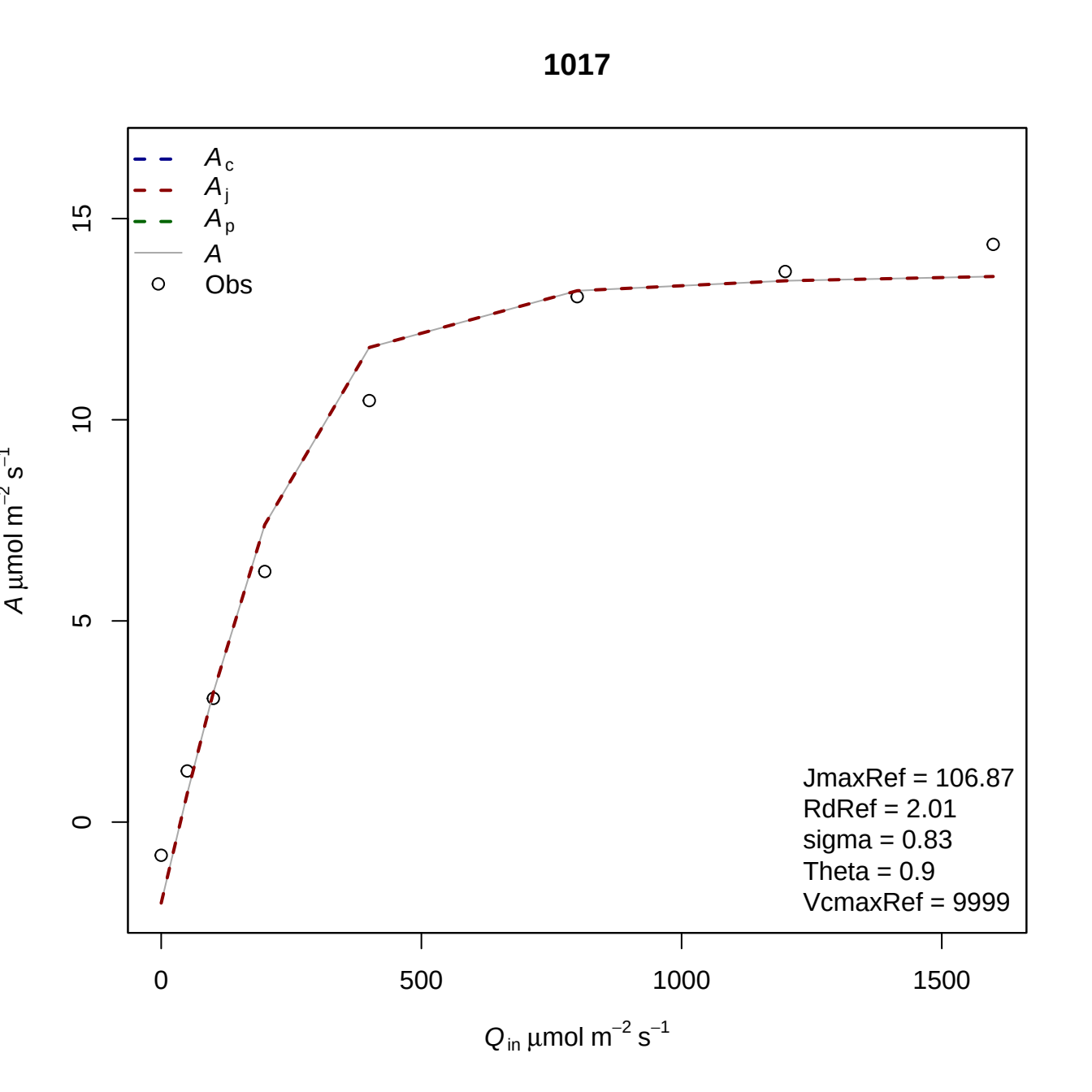

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 72.88
RdRef = 2.16
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

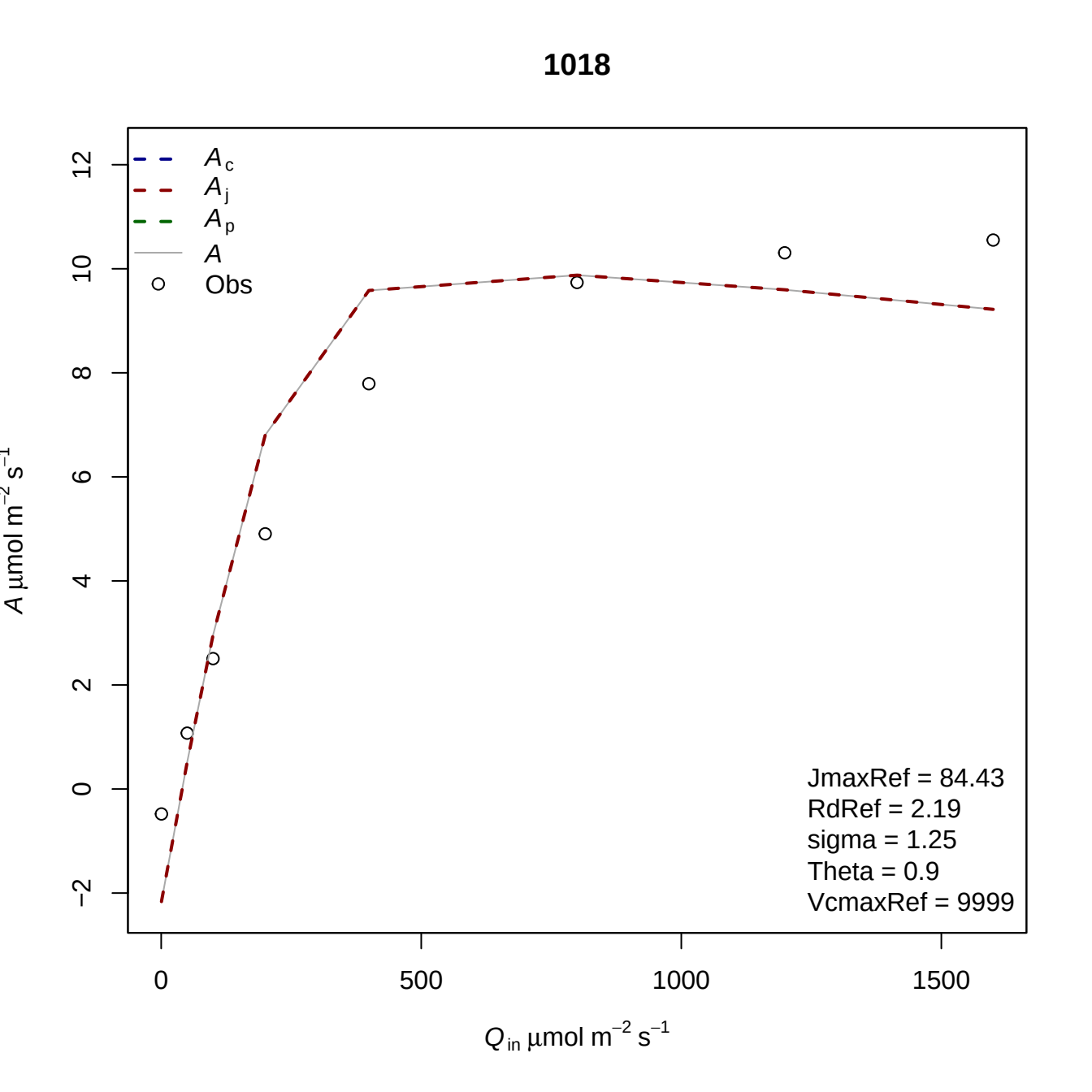

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 120.74
RdRef = 1.89
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

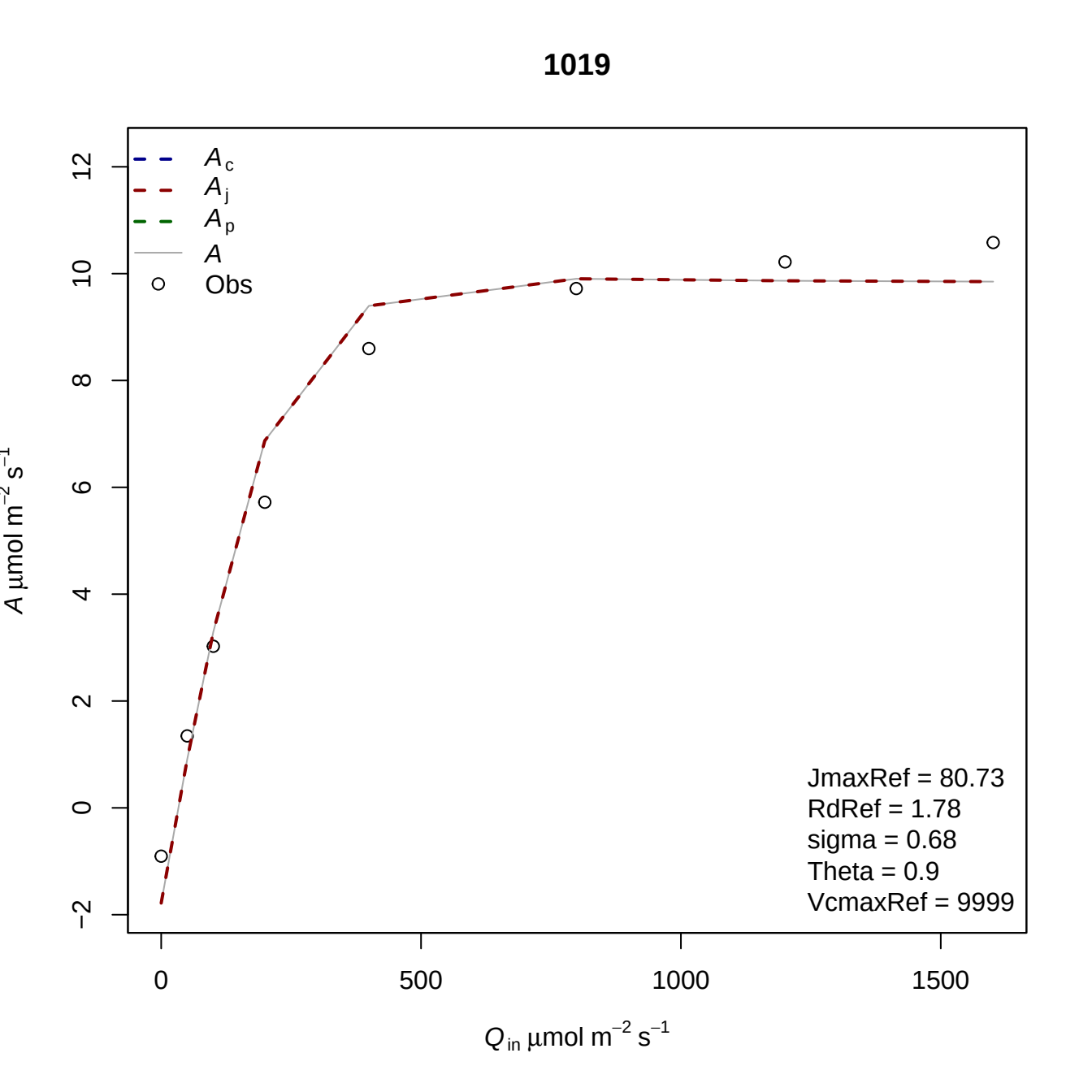

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 75.25
RdRef = 0.5
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

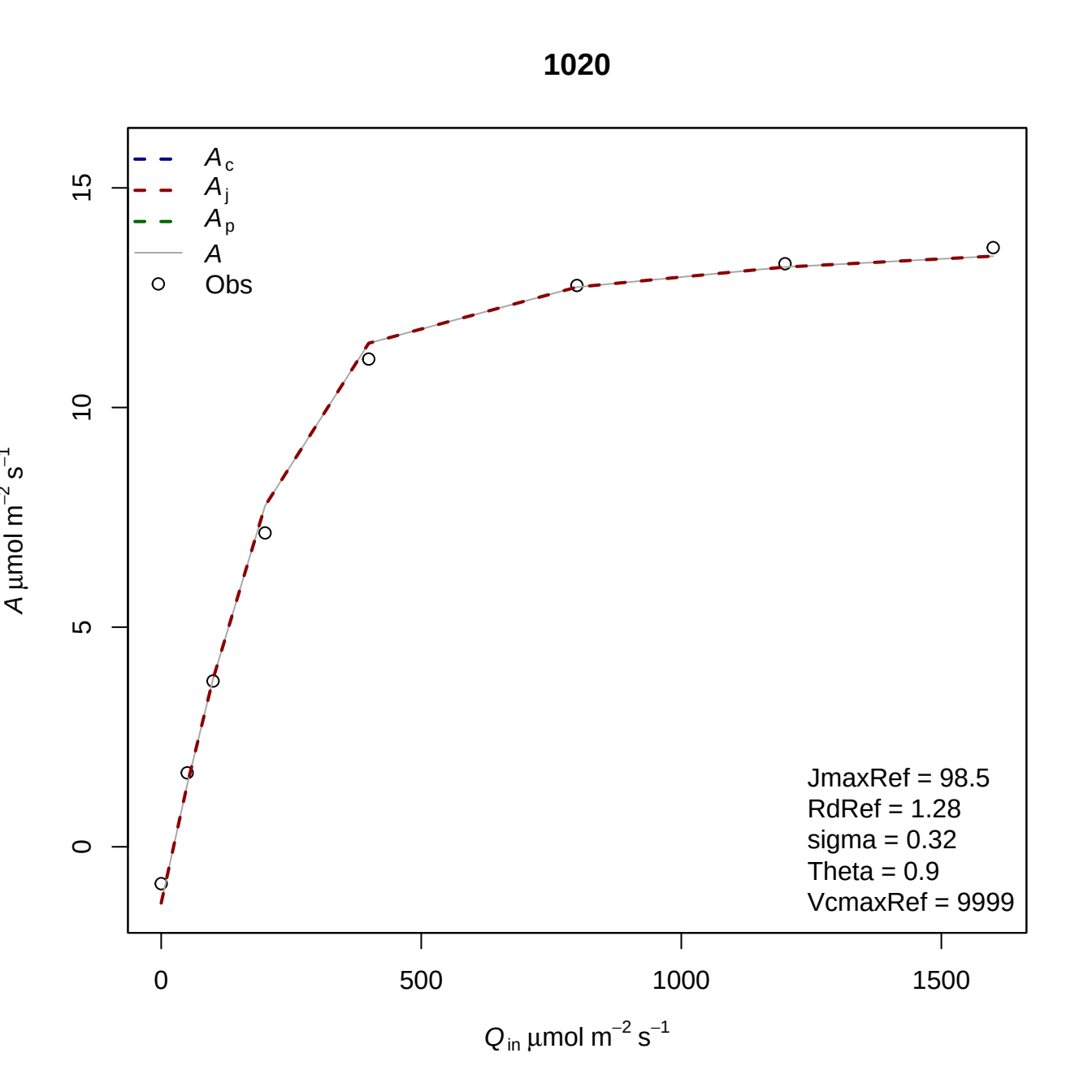

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 76.43
RdRef = 1.87
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

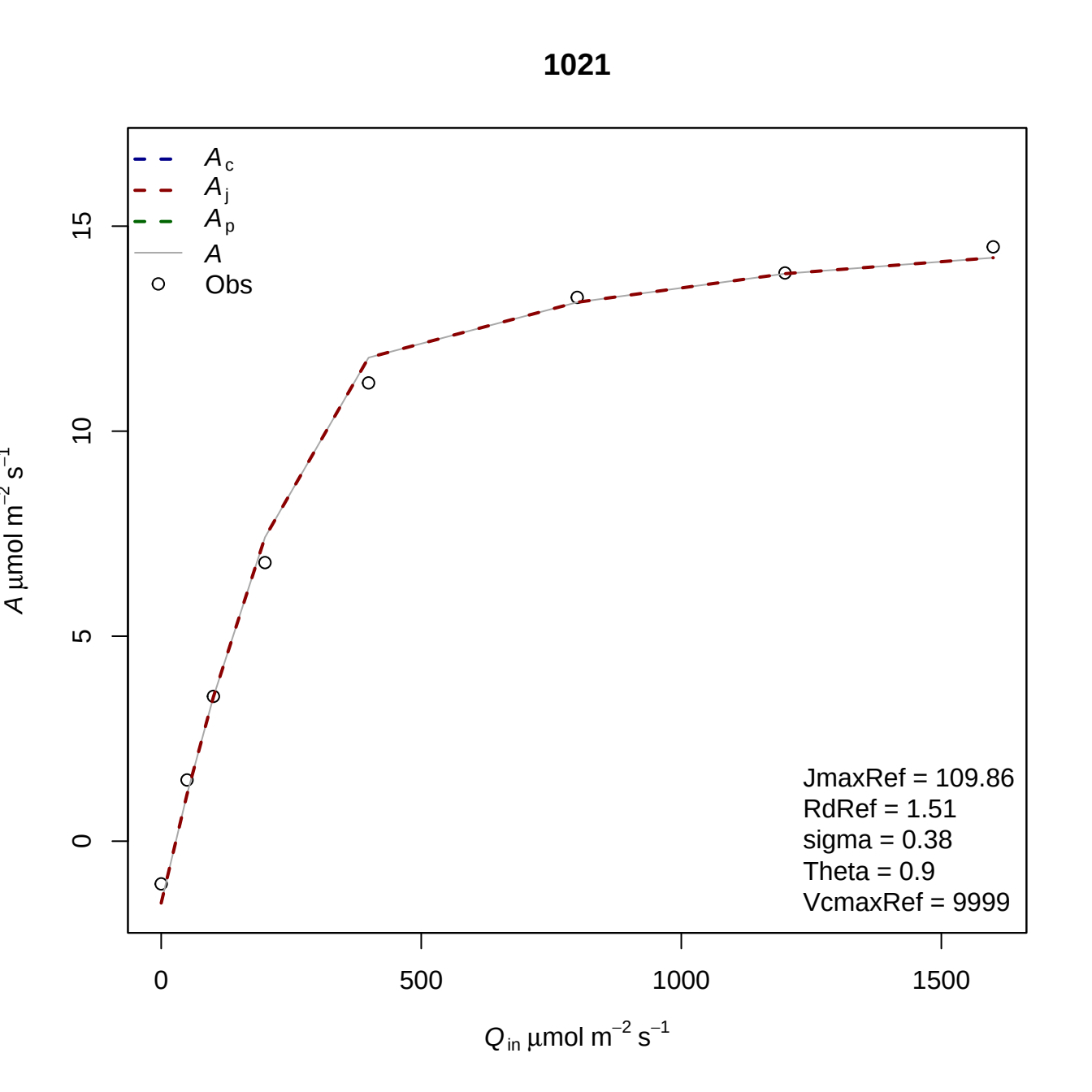

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 86.86
RdRef = 2.65
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

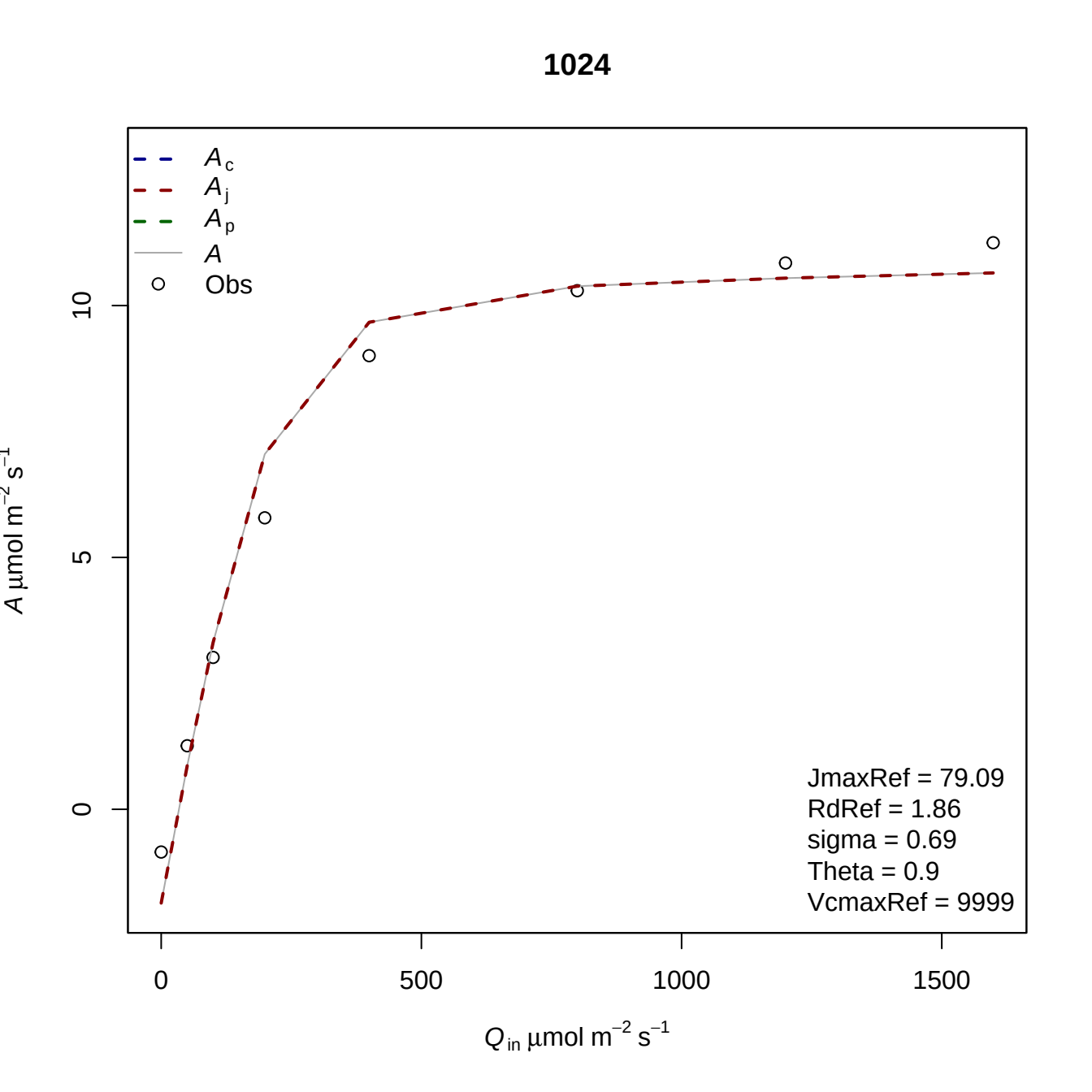

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 106.87
RdRef = 2.01
sigma = 0.83
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

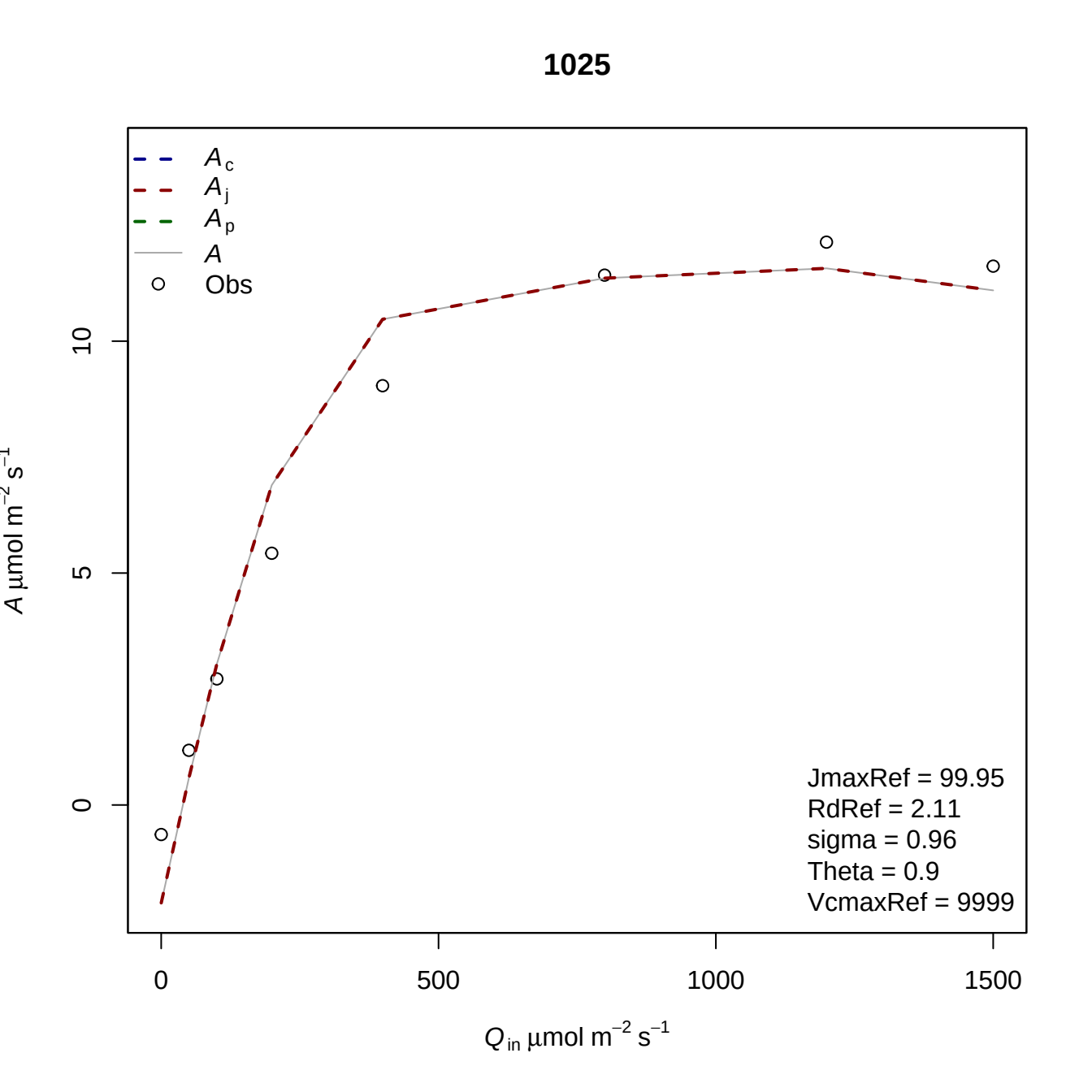

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 84.43
RdRef = 2.19
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

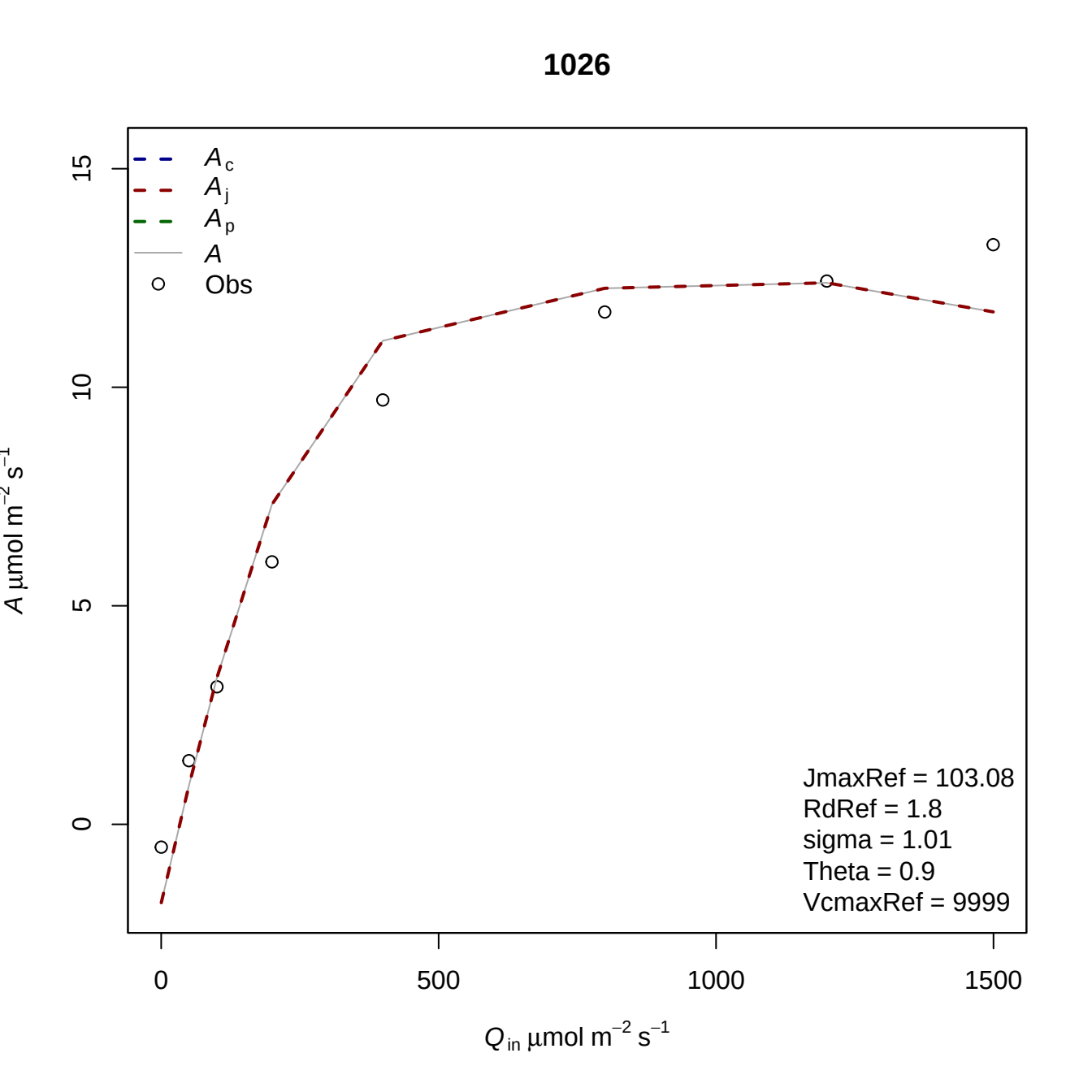

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 80.73
RdRef = 1.78
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

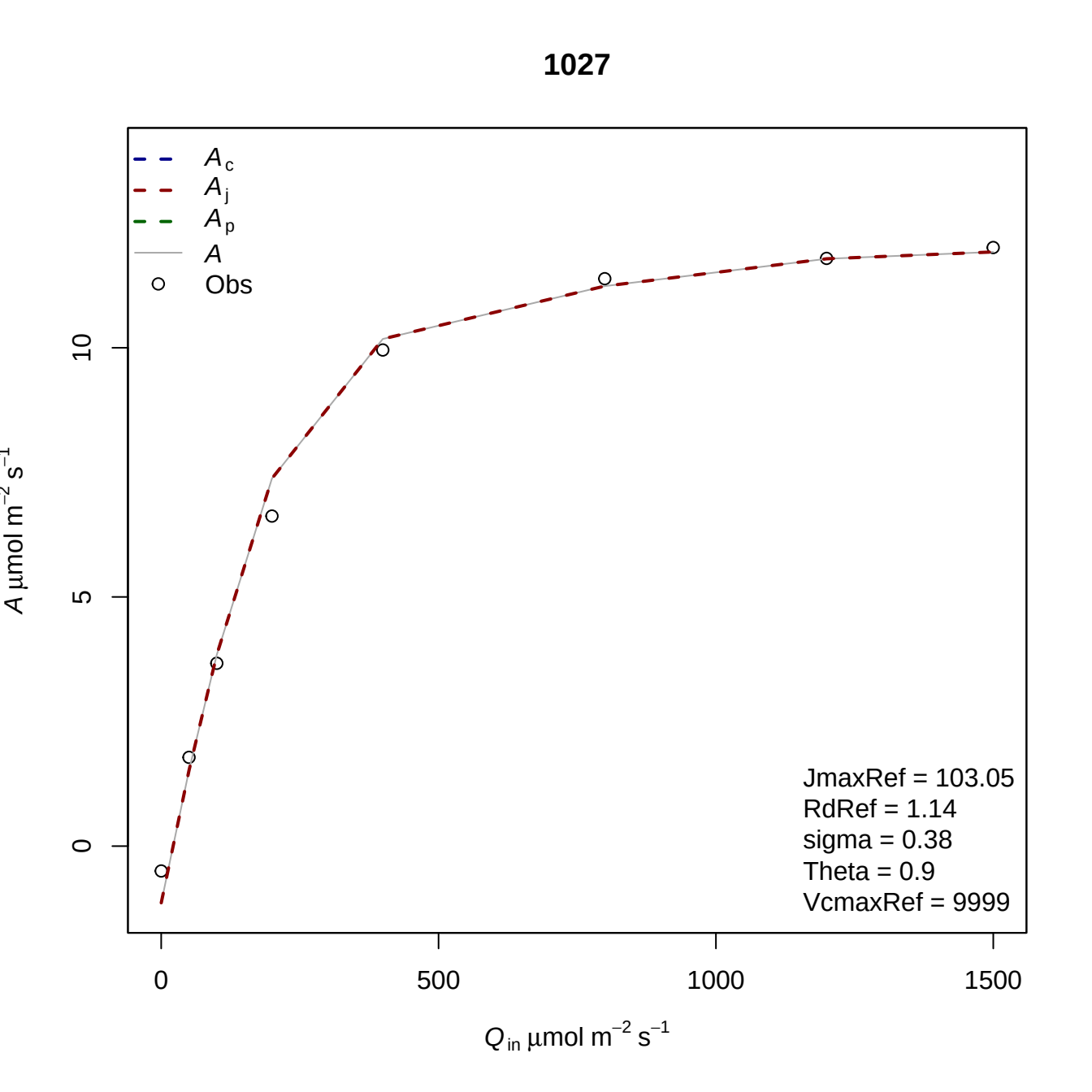

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 98.5
RdRef = 1.28
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

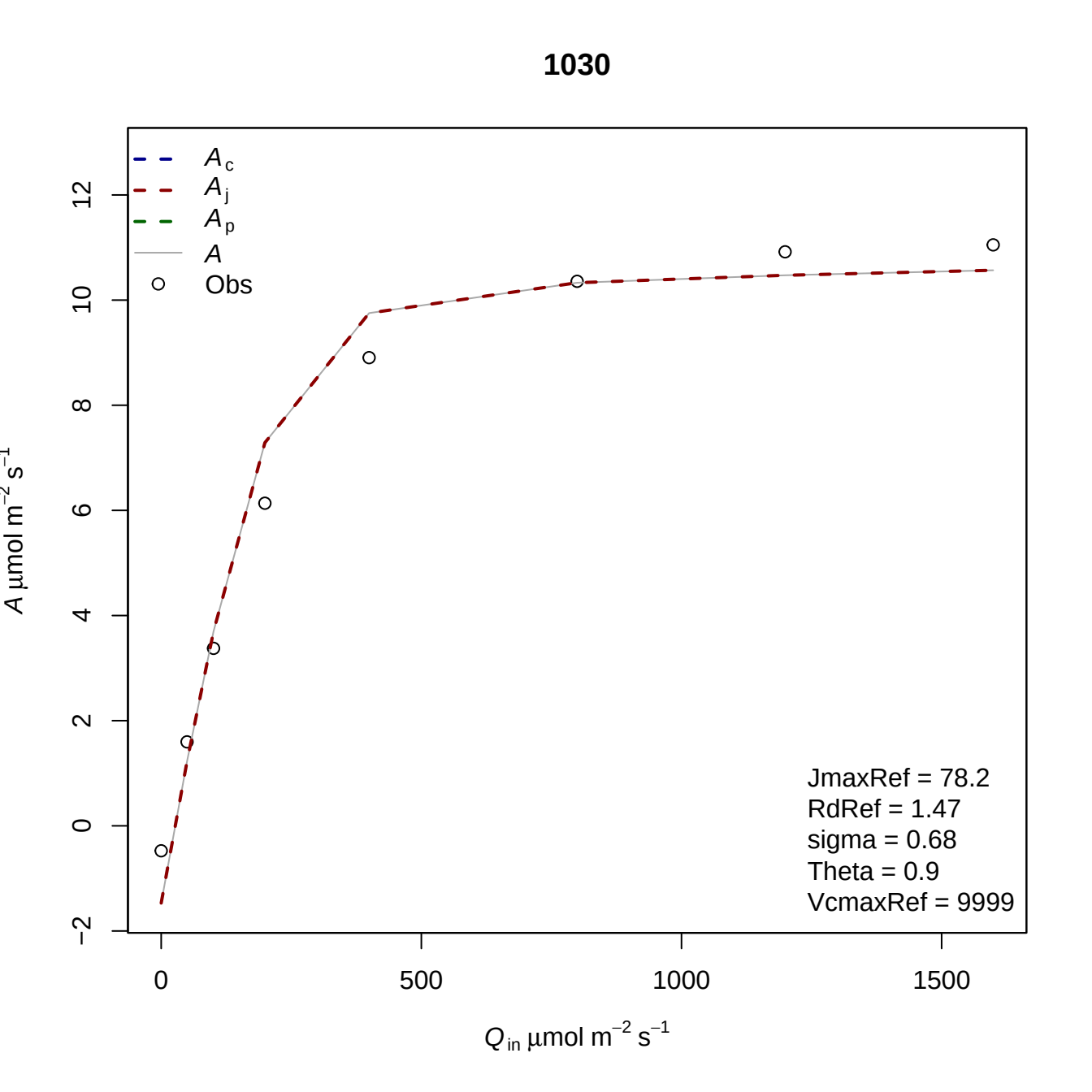

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 109.86
RdRef = 1.51
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

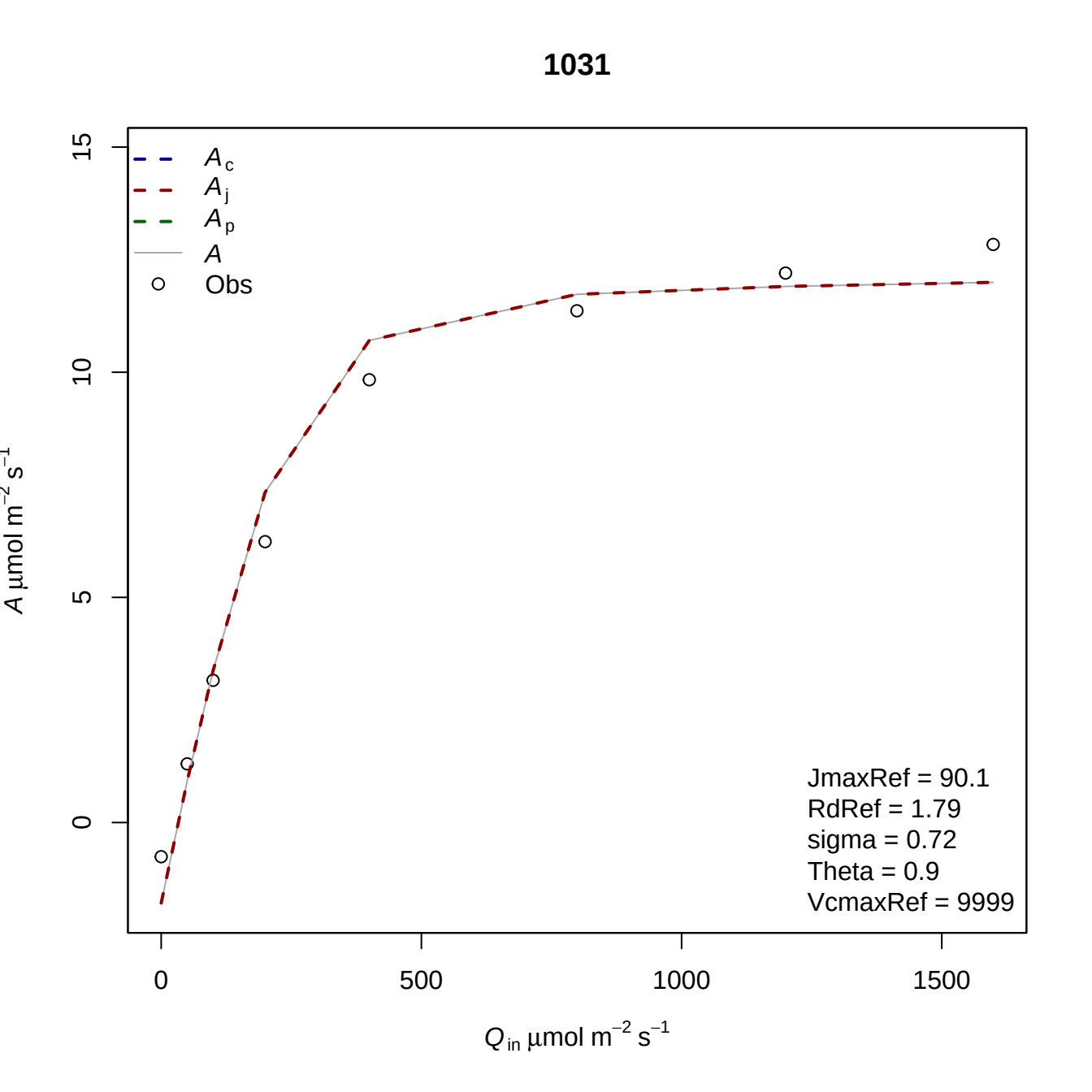

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 79.09
RdRef = 1.86
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

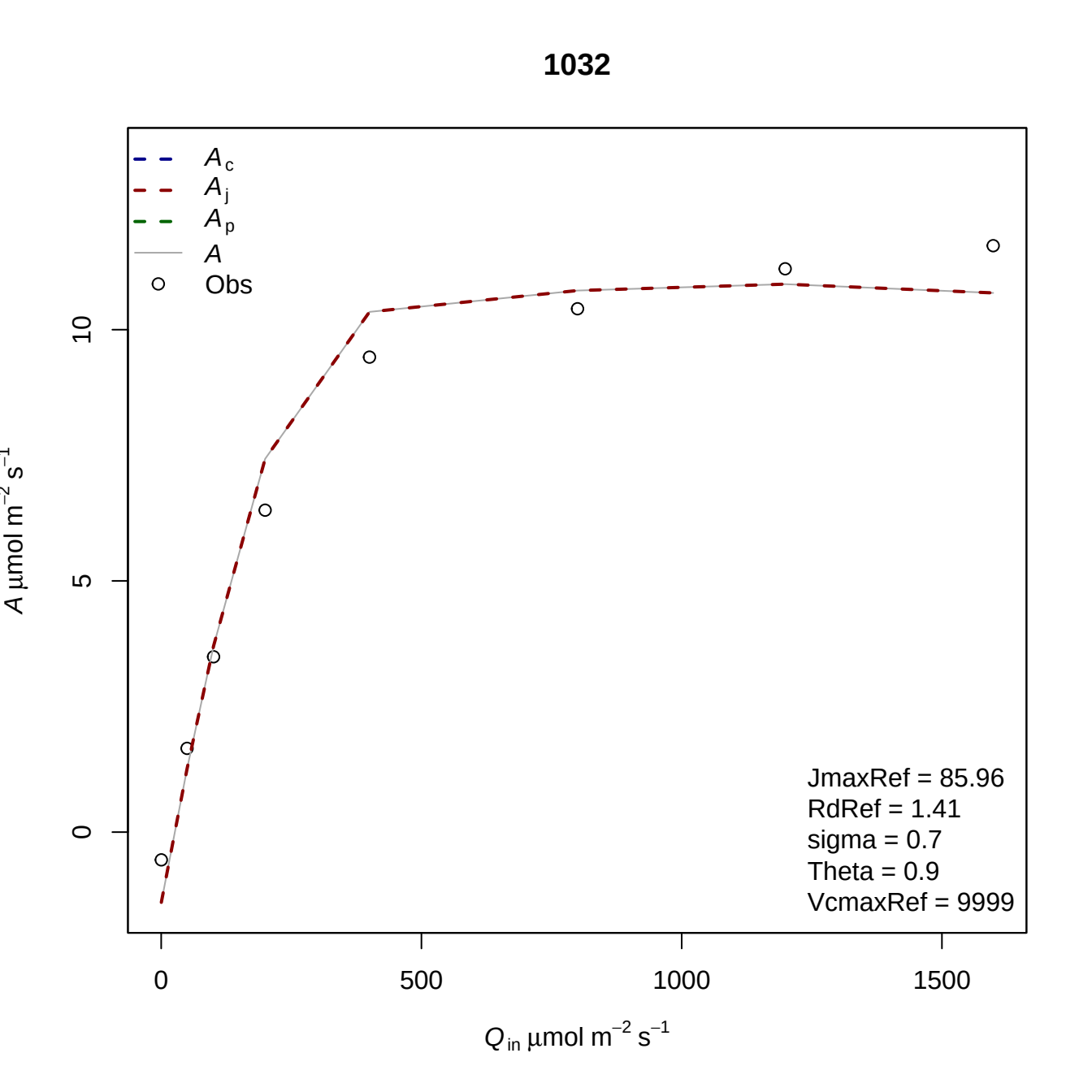

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 99.95
RdRef = 2.11
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

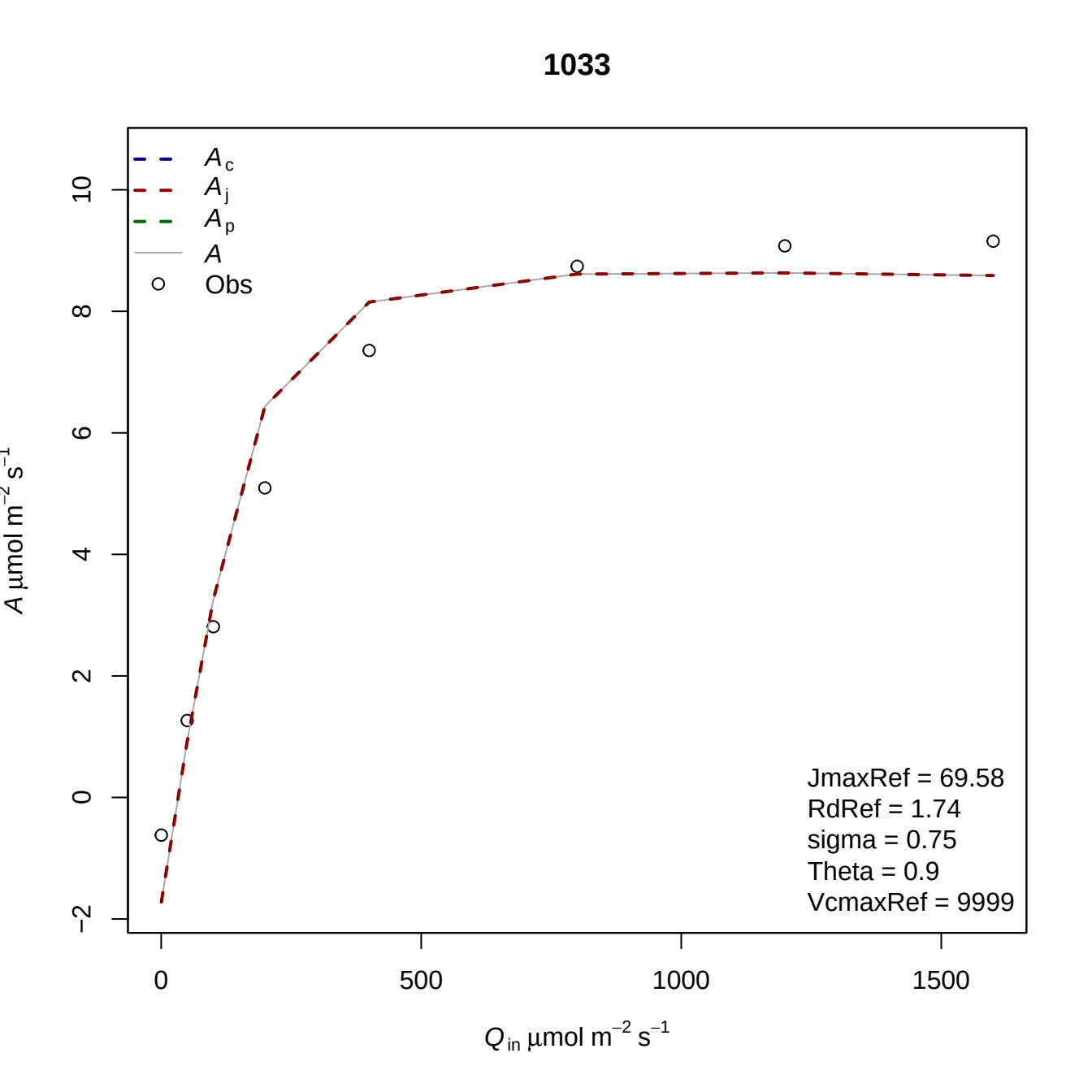

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 103.08
RdRef = 1.8
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

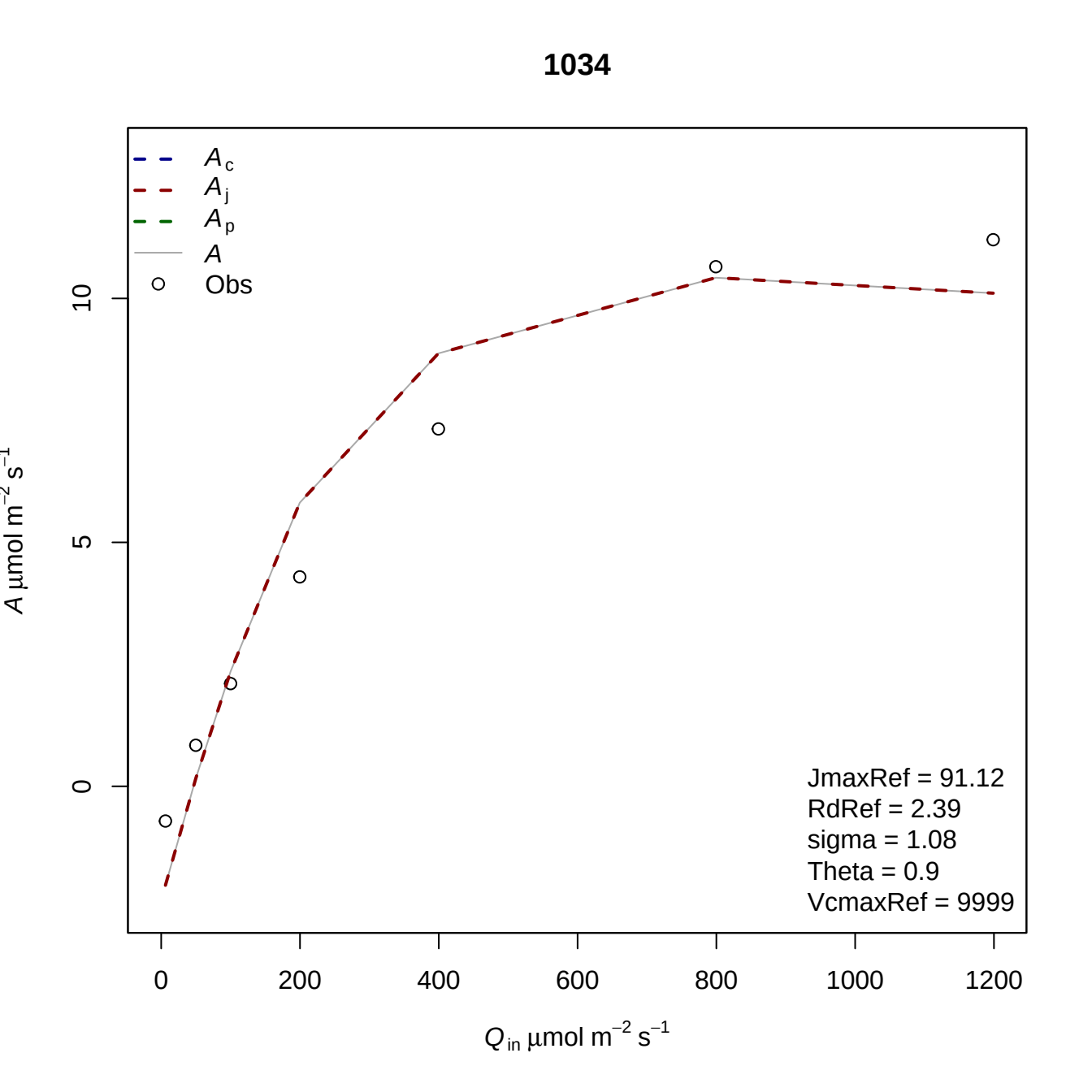

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 103.05
RdRef = 1.14
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

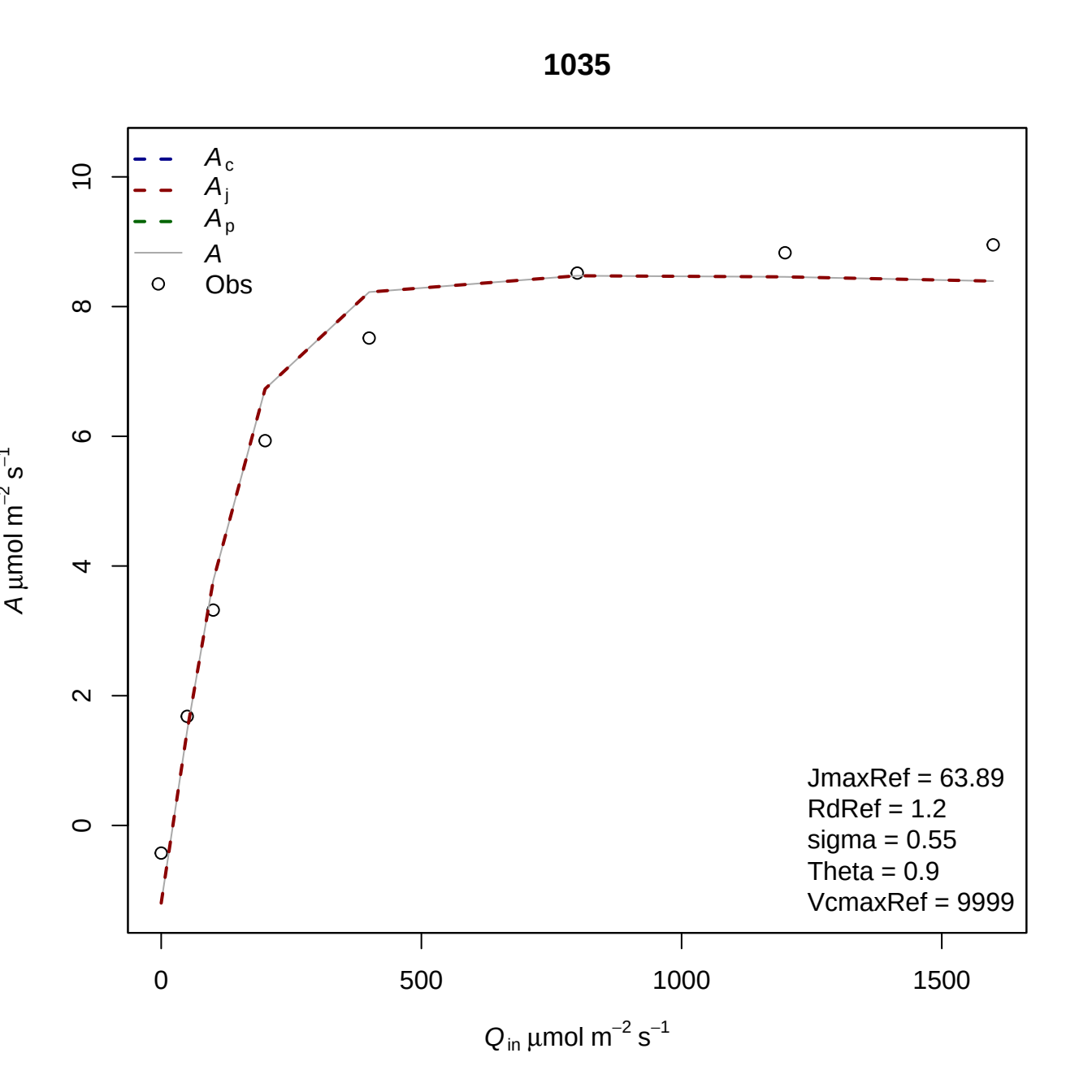

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 78.2
RdRef = 1.47
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

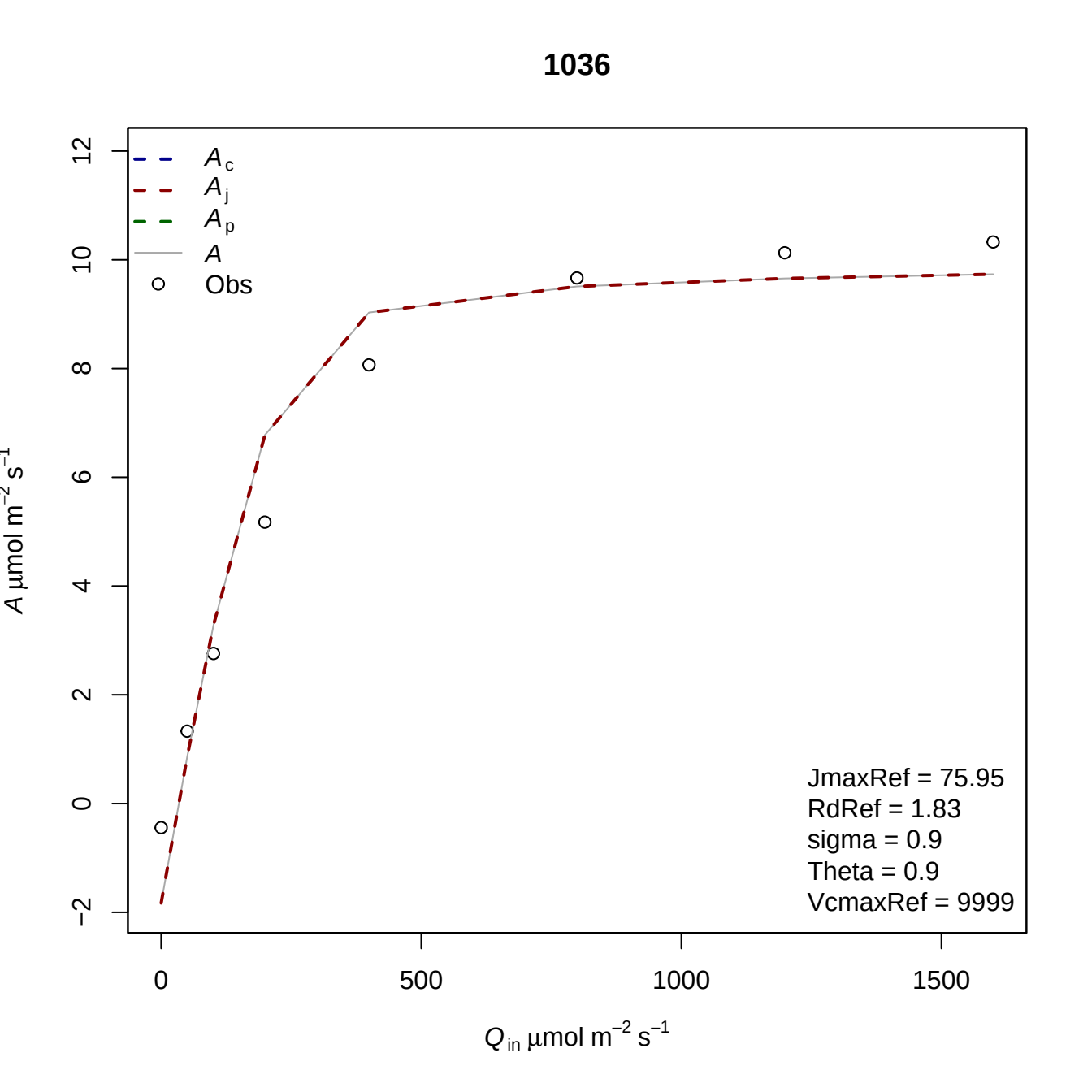

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 90.1
RdRef = 1.79
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

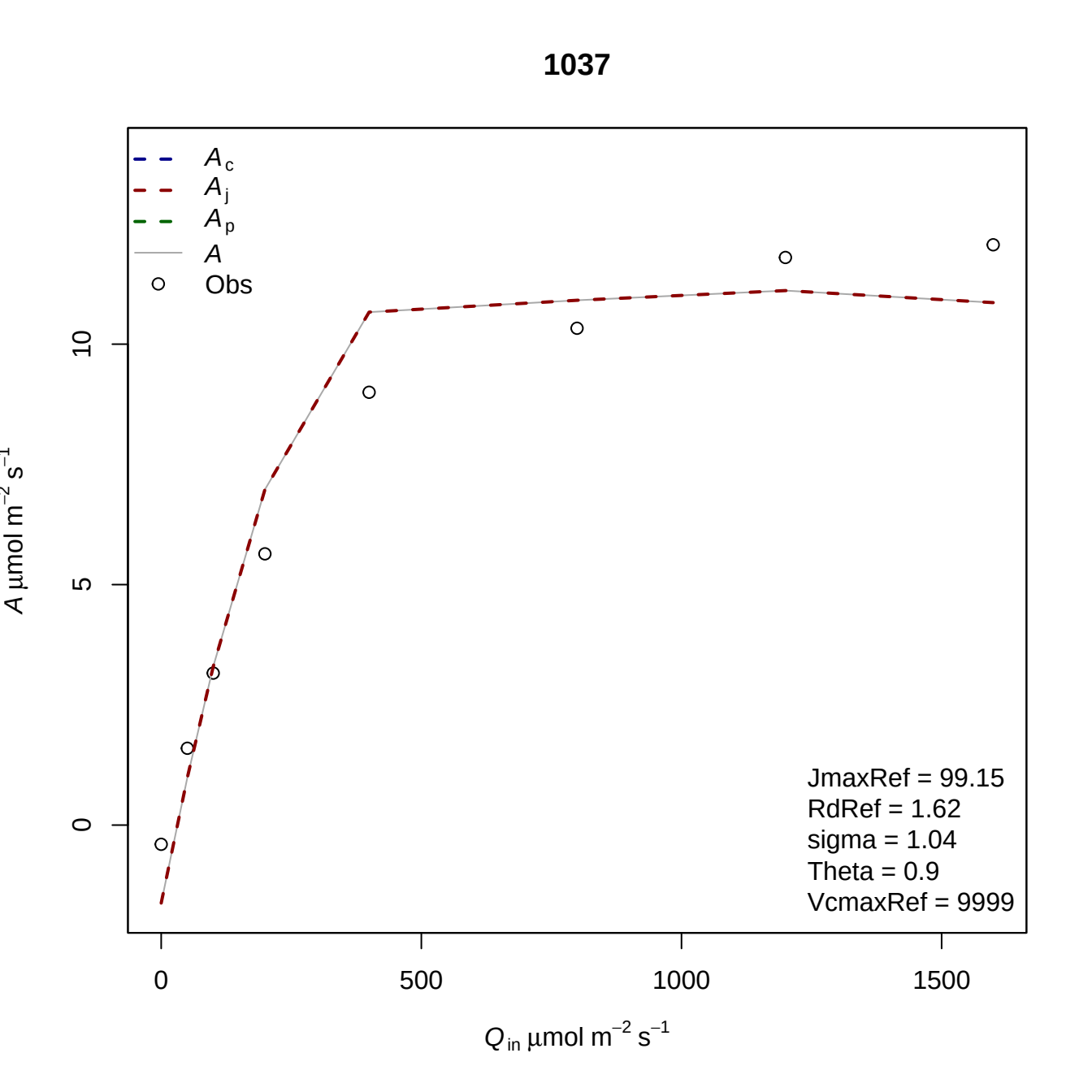

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 85.96
RdRef = 1.41
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

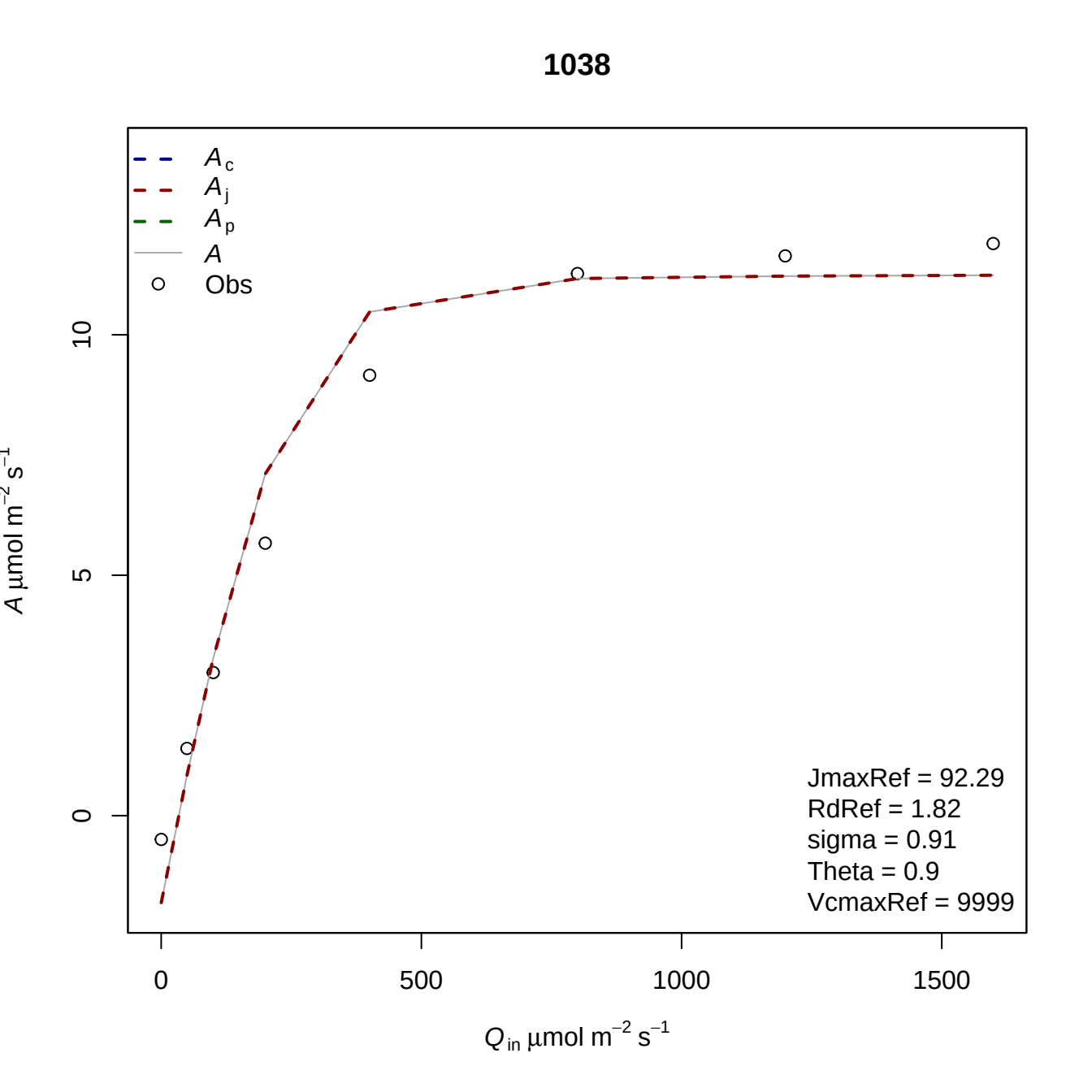

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 69.58
RdRef = 1.74
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

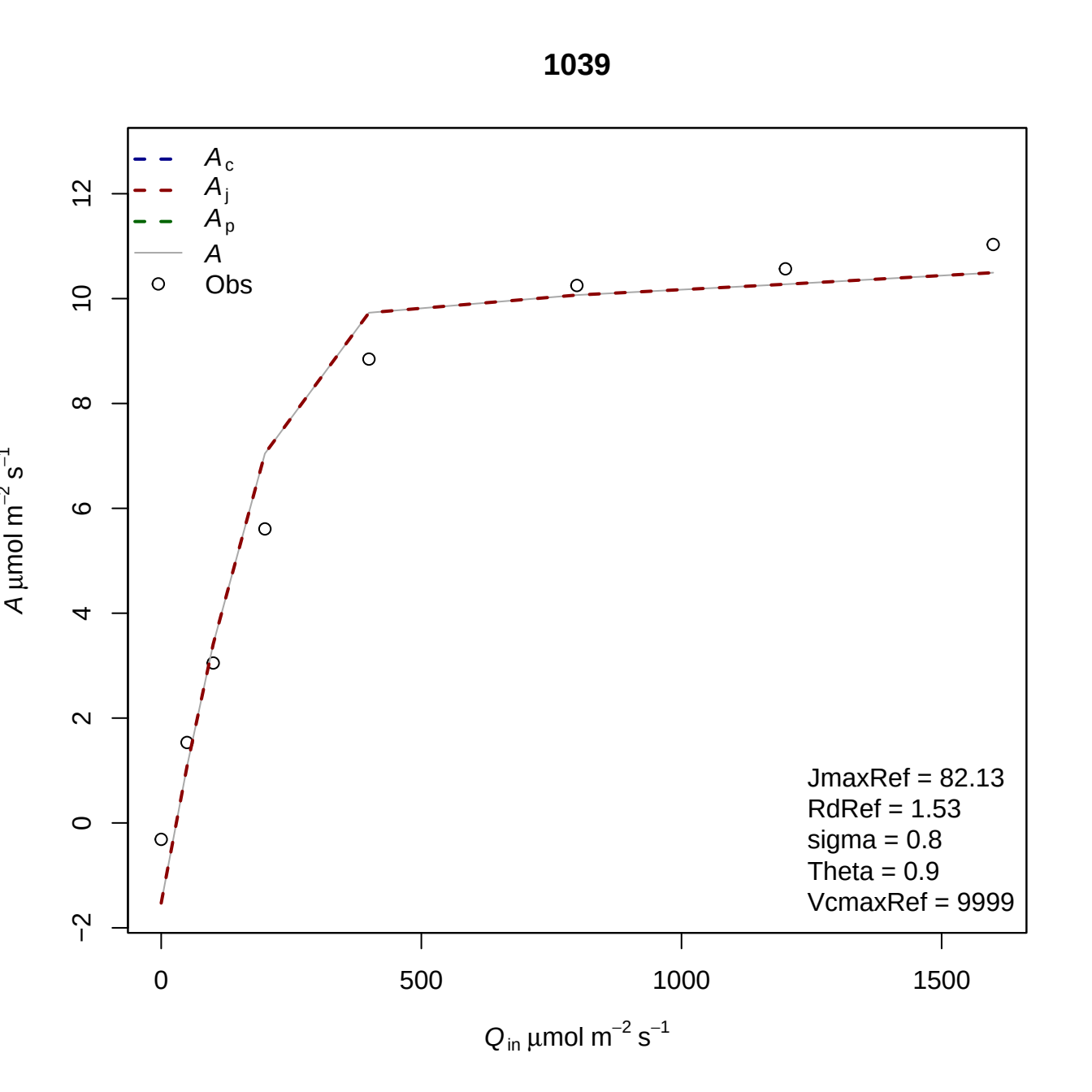

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 91.12
RdRef = 2.39
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

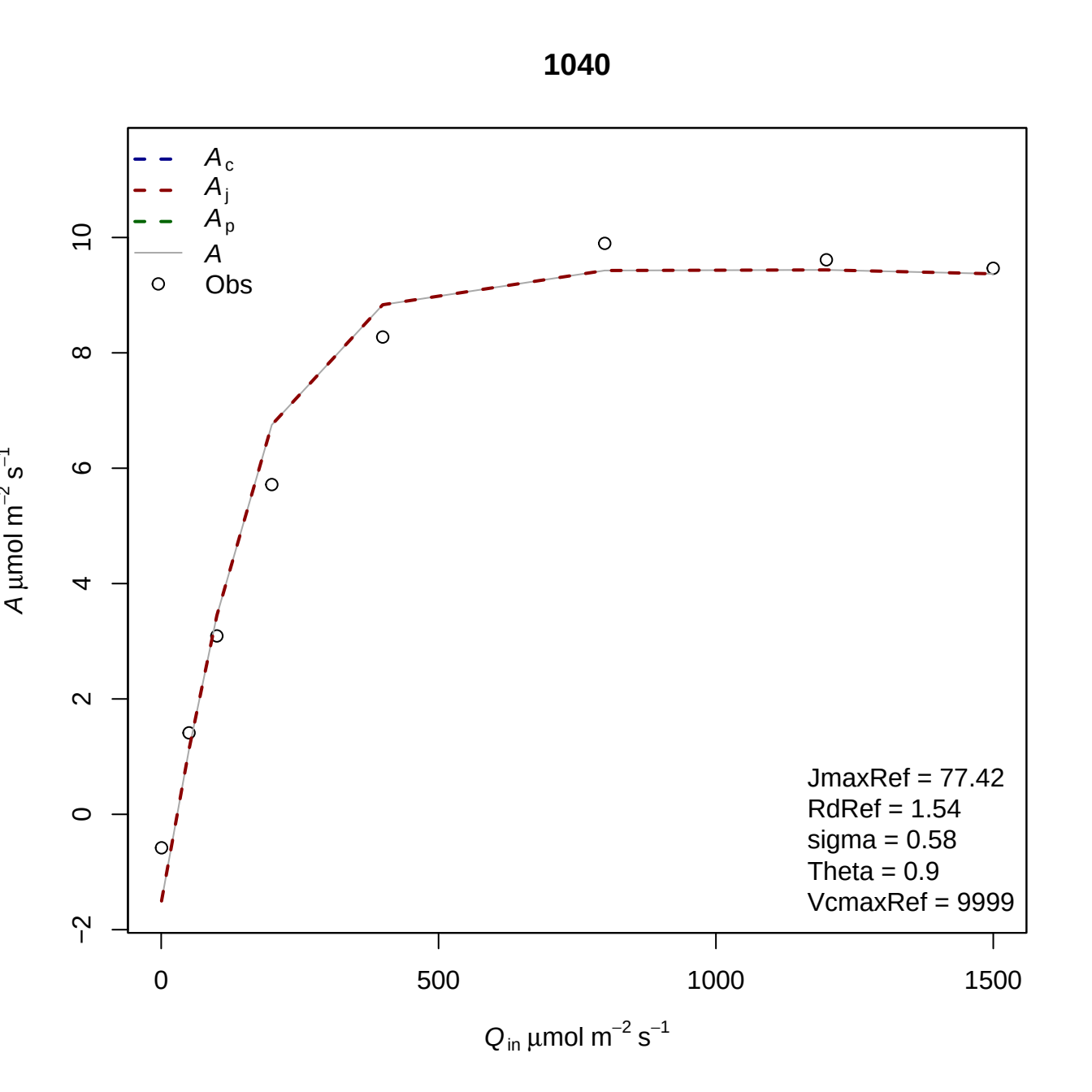

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 63.89
RdRef = 1.2
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

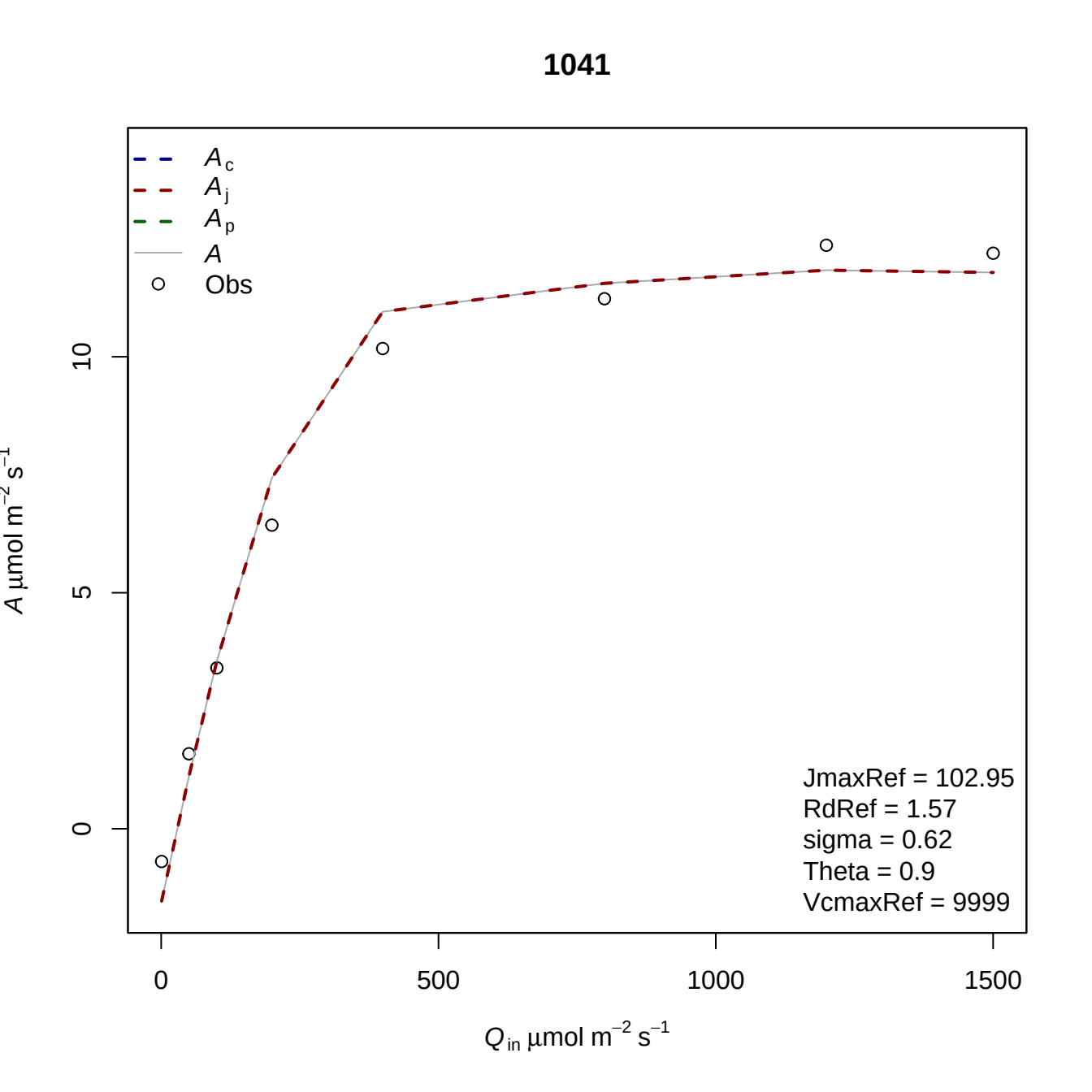

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 75.95
RdRef = 1.83
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

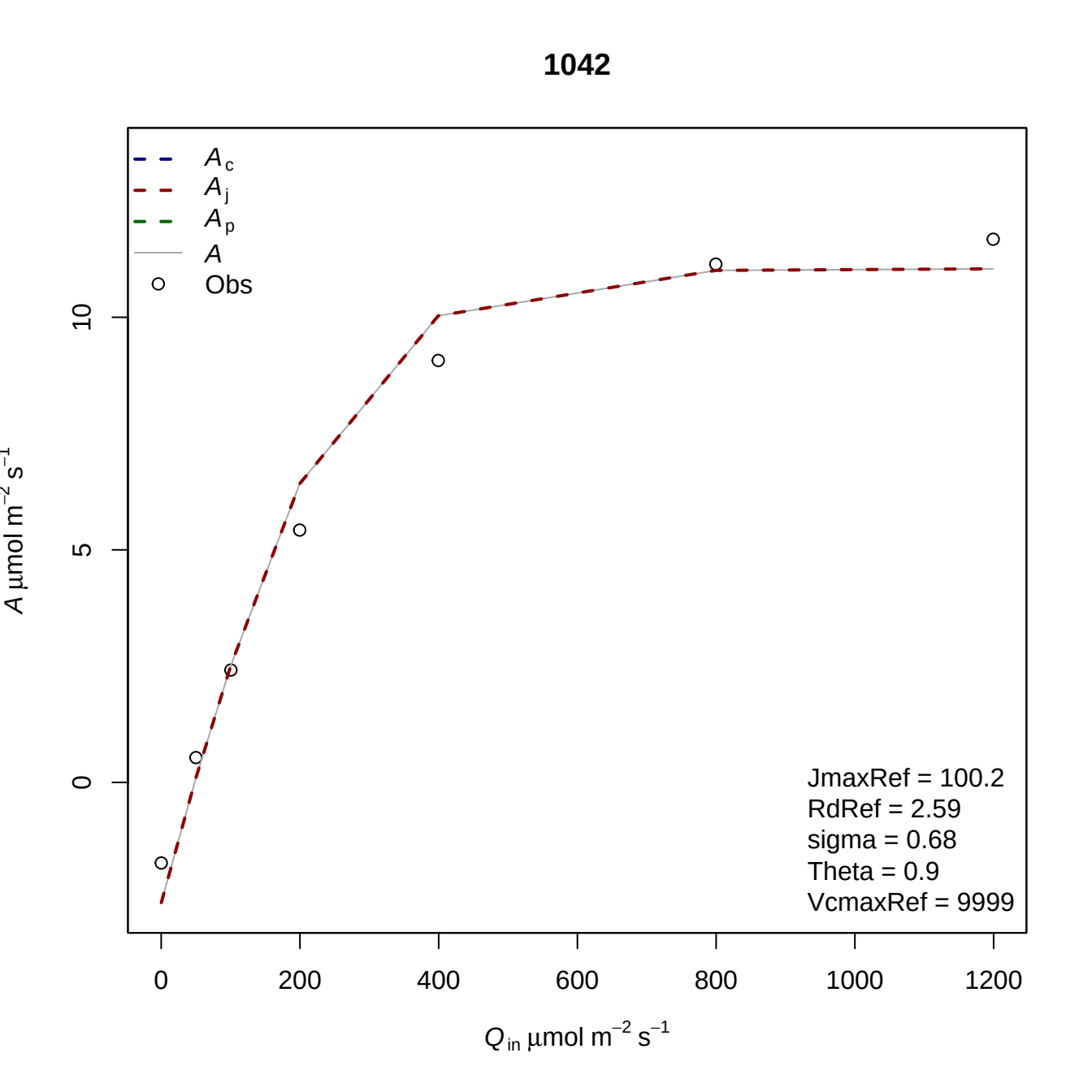

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 99.15
RdRef = 1.62
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

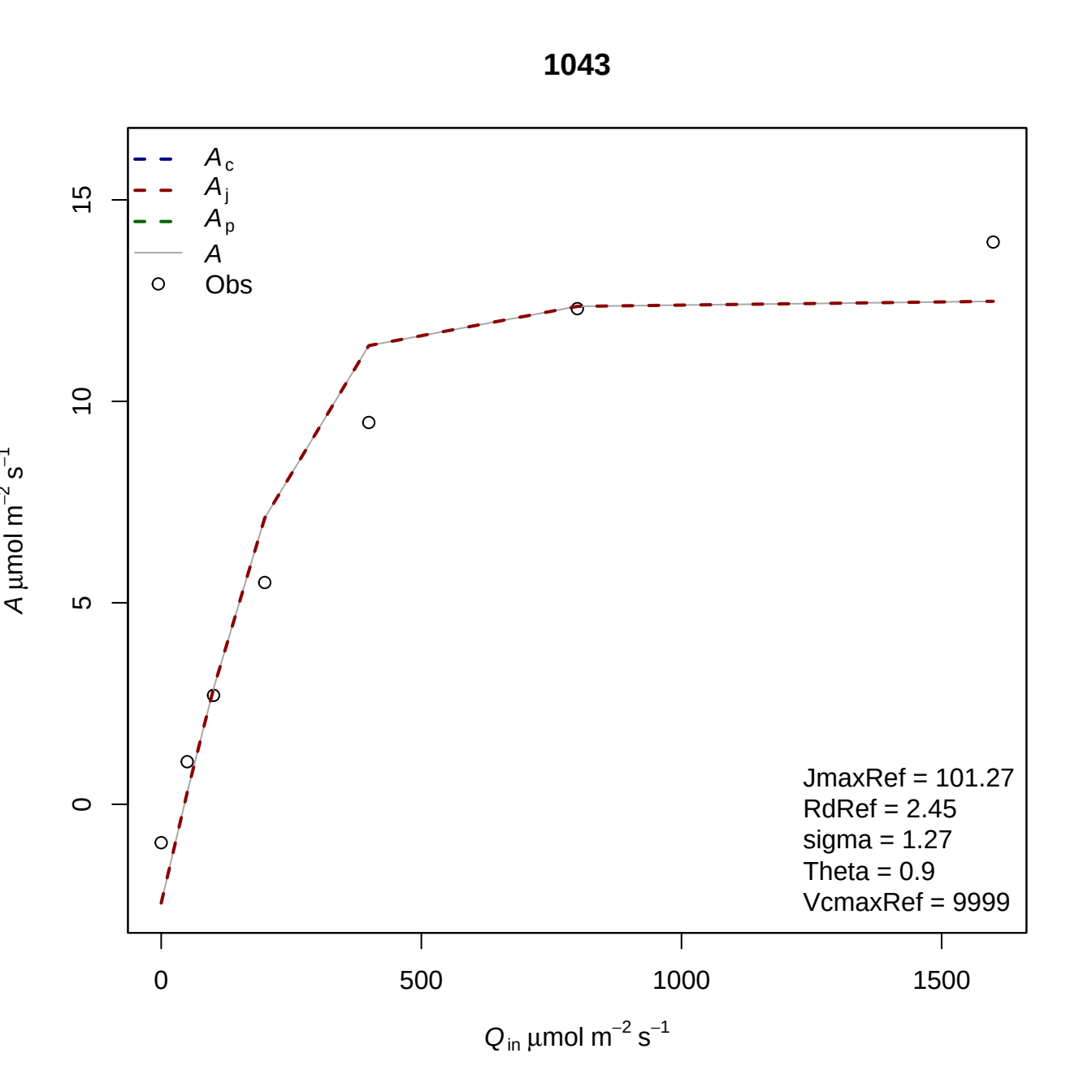

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 92.29
RdRef = 1.82
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

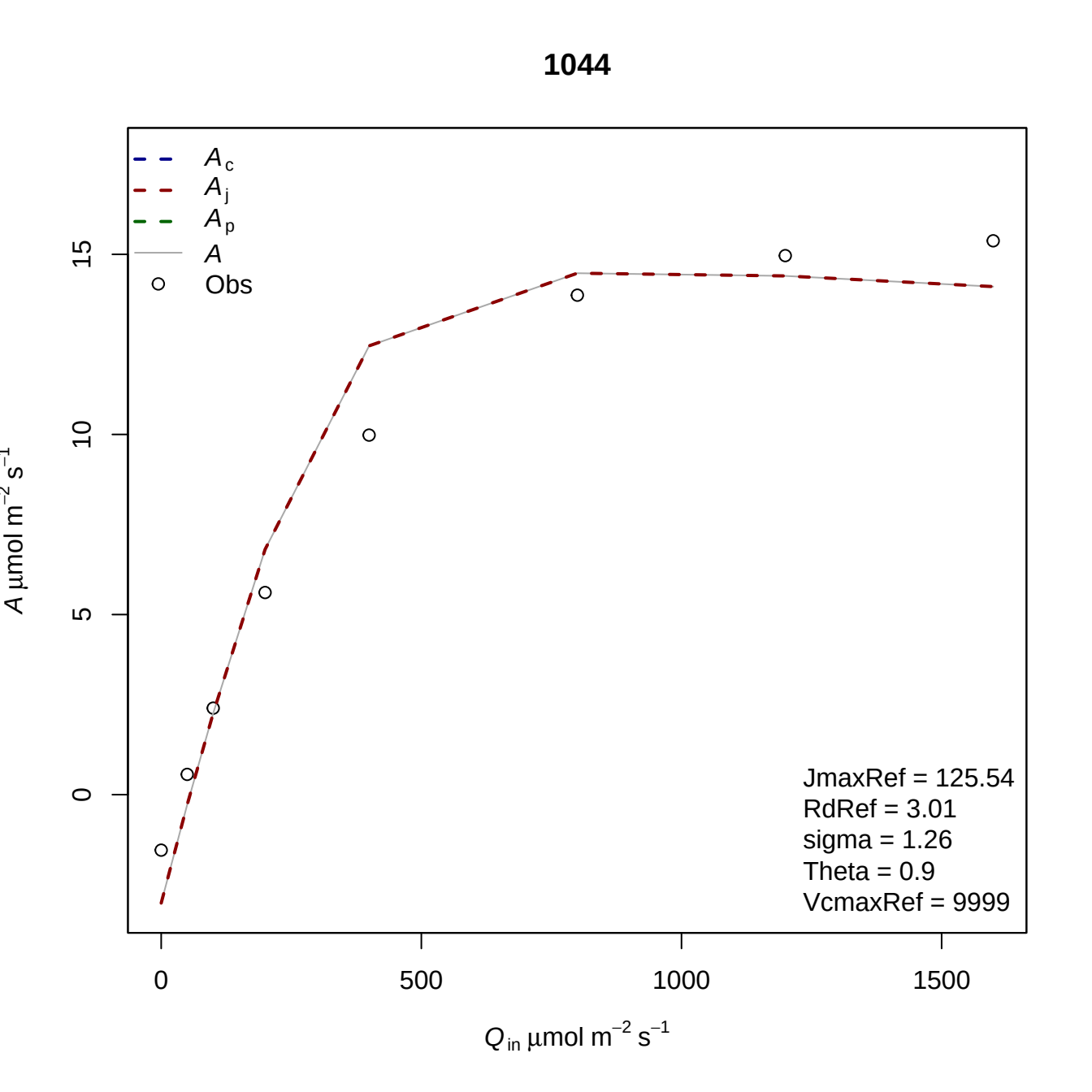

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 82.13
RdRef = 1.53
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

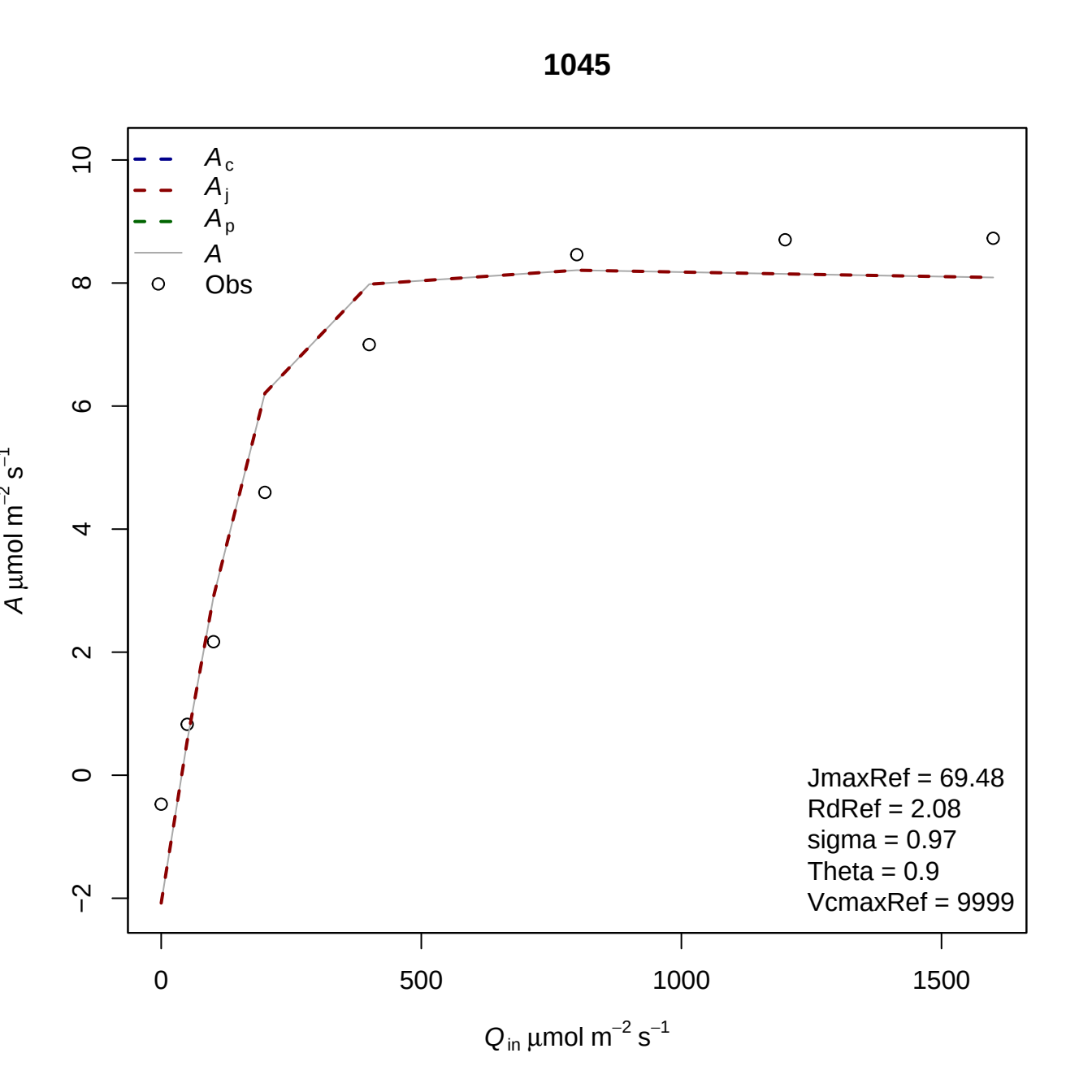

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 77.42
RdRef = 1.54
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

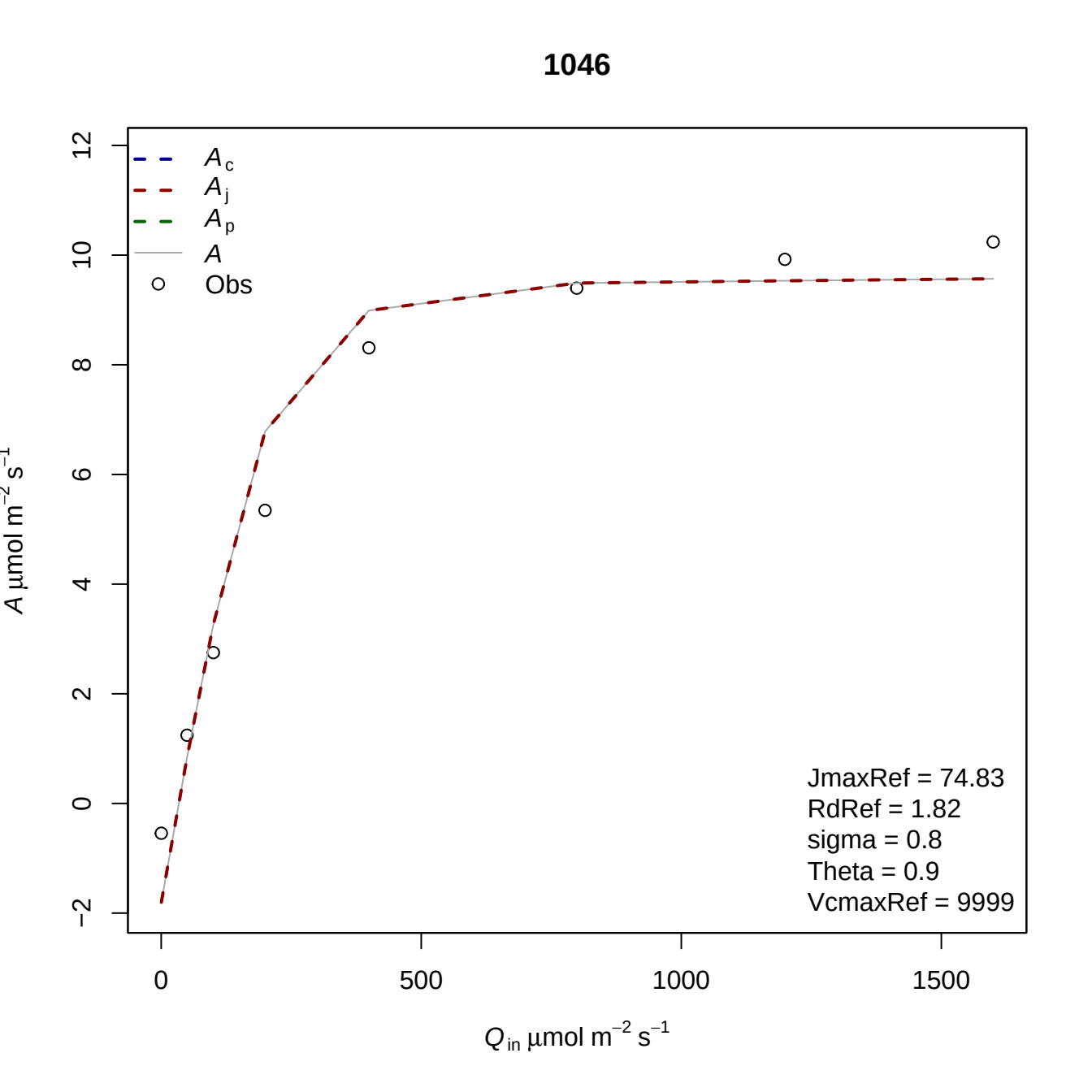

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 102.95
RdRef = 1.57
sigma = 0.62
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

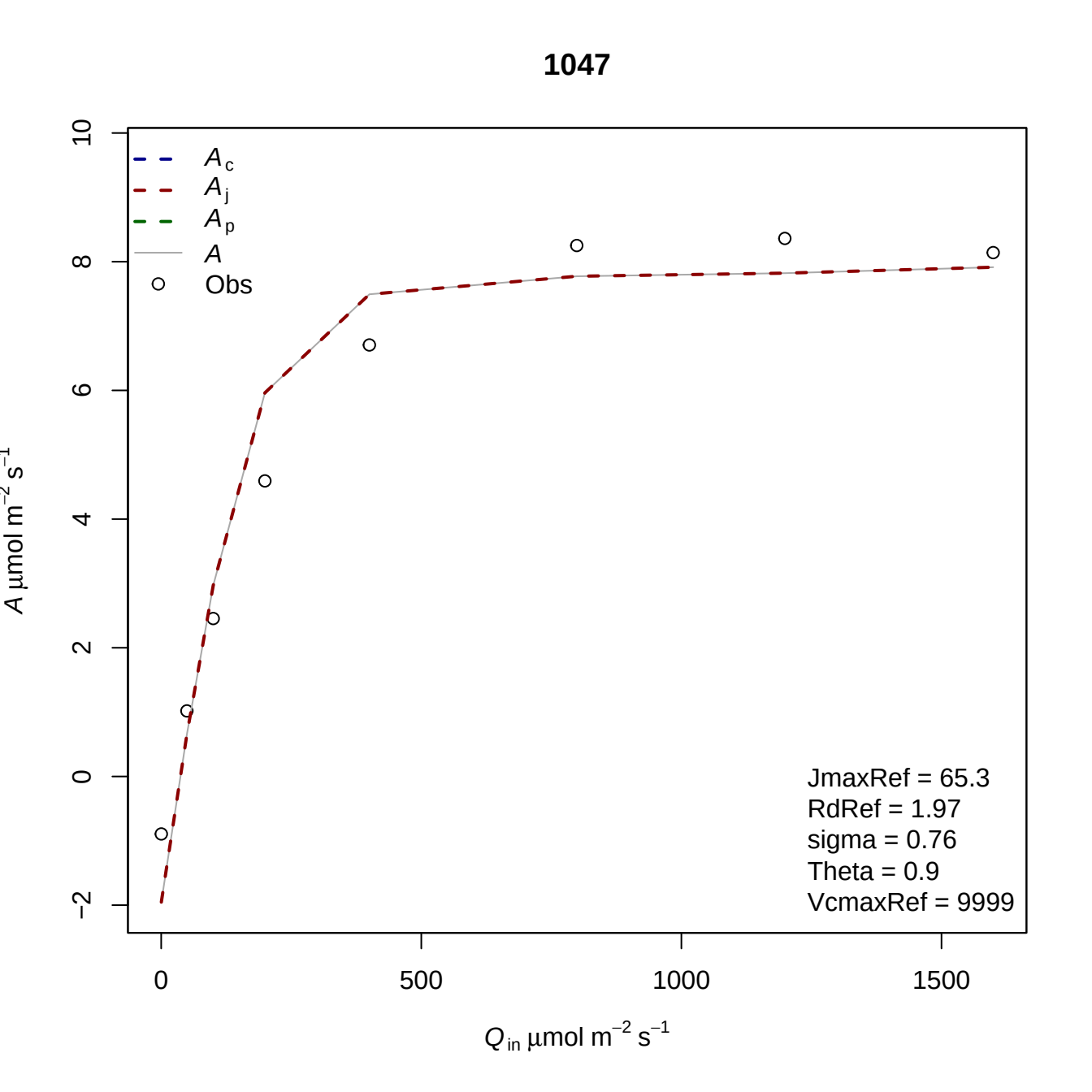

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 100.2
RdRef = 2.59
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

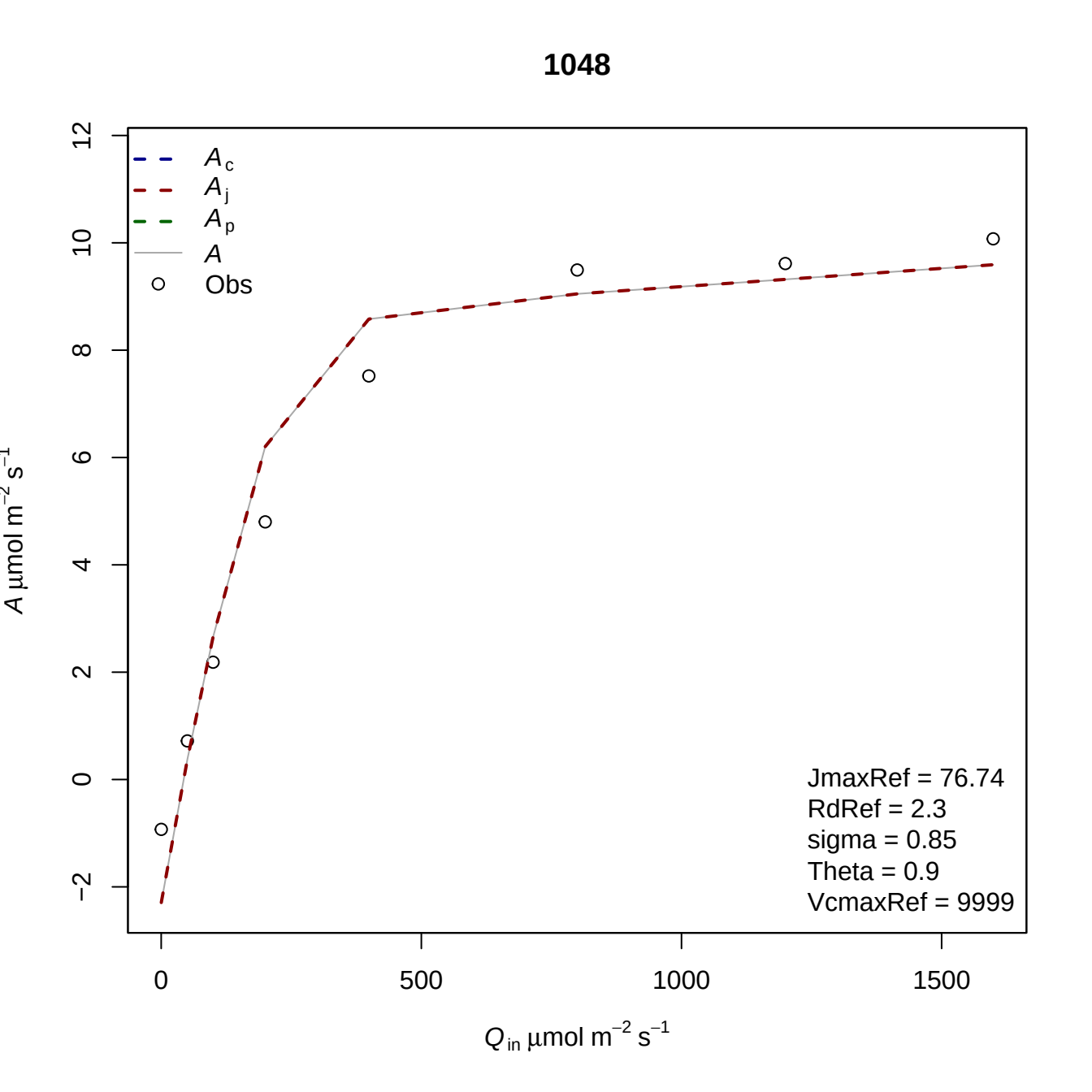

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 101.27
RdRef = 2.45
sigma = 1.27
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

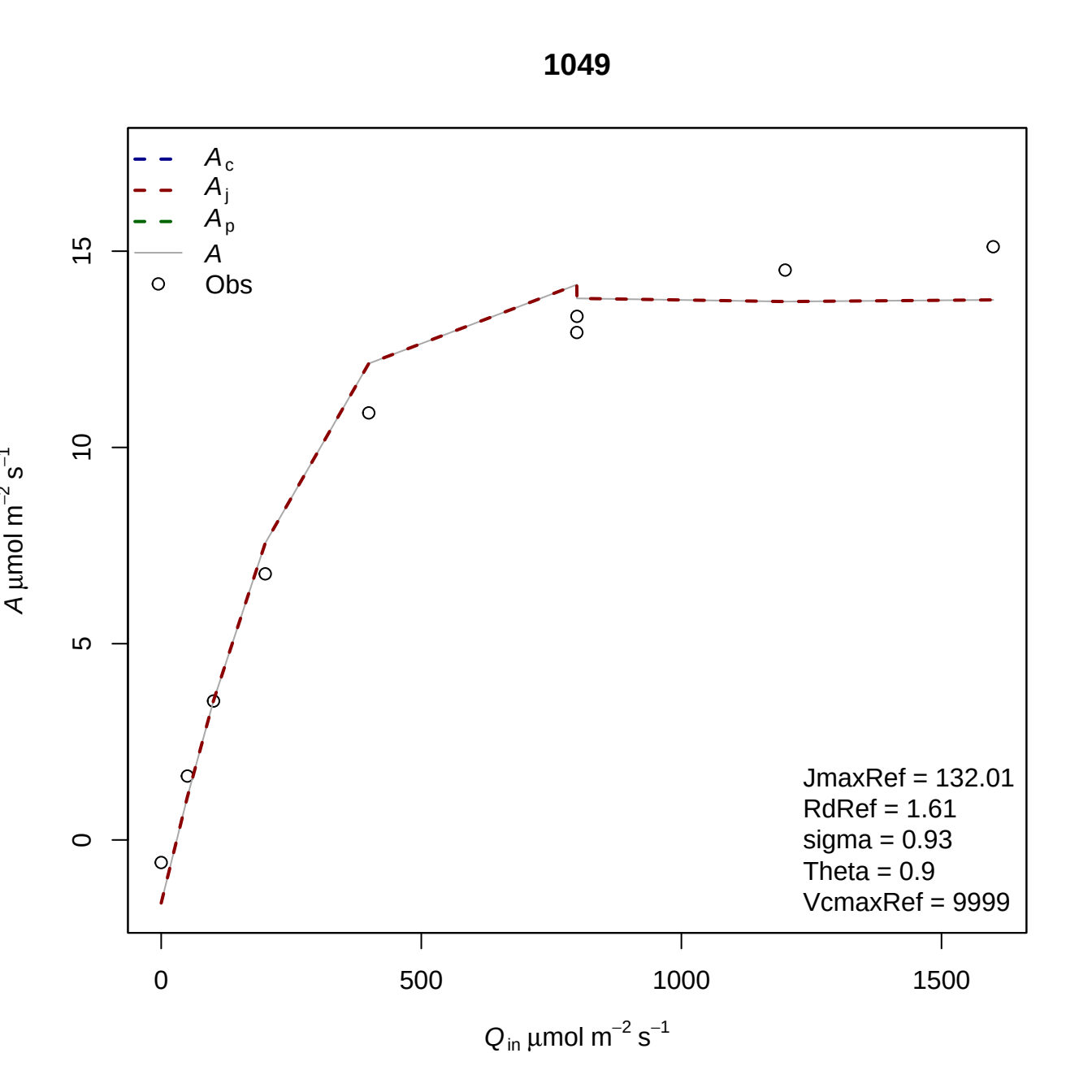

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 125.54
RdRef = 3.01
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

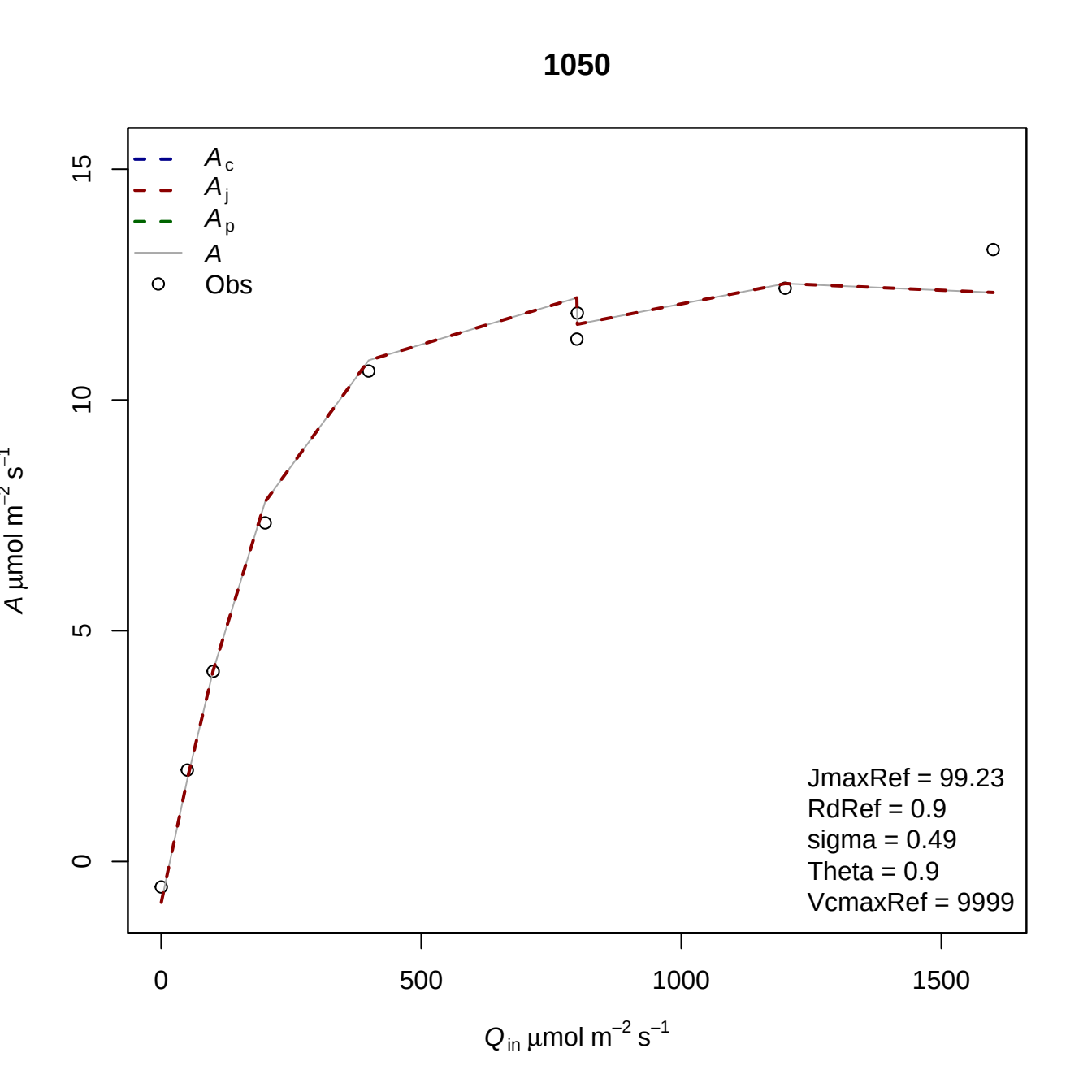

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 69.48
RdRef = 2.08
sigma = 0.97
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

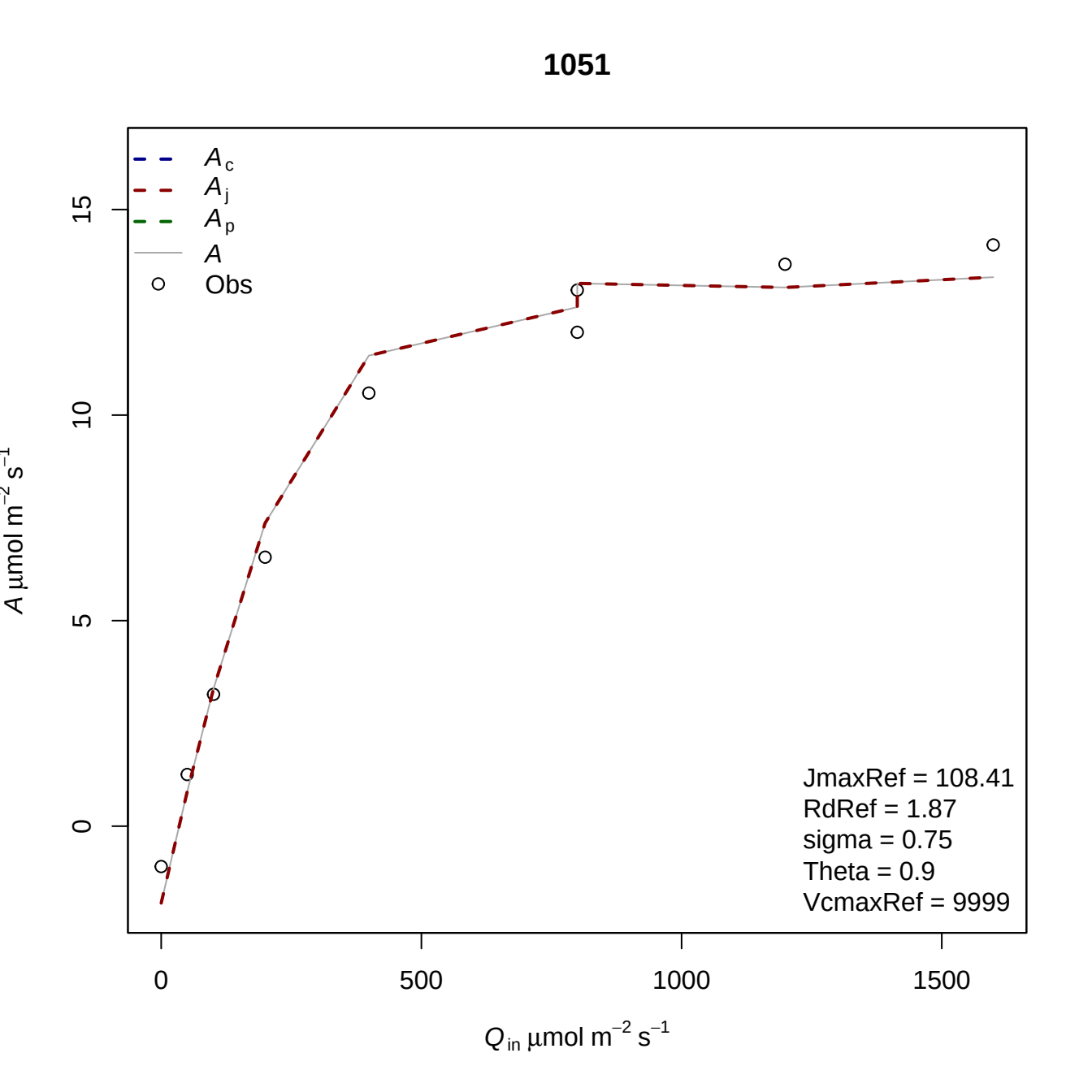

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 74.83
RdRef = 1.82
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

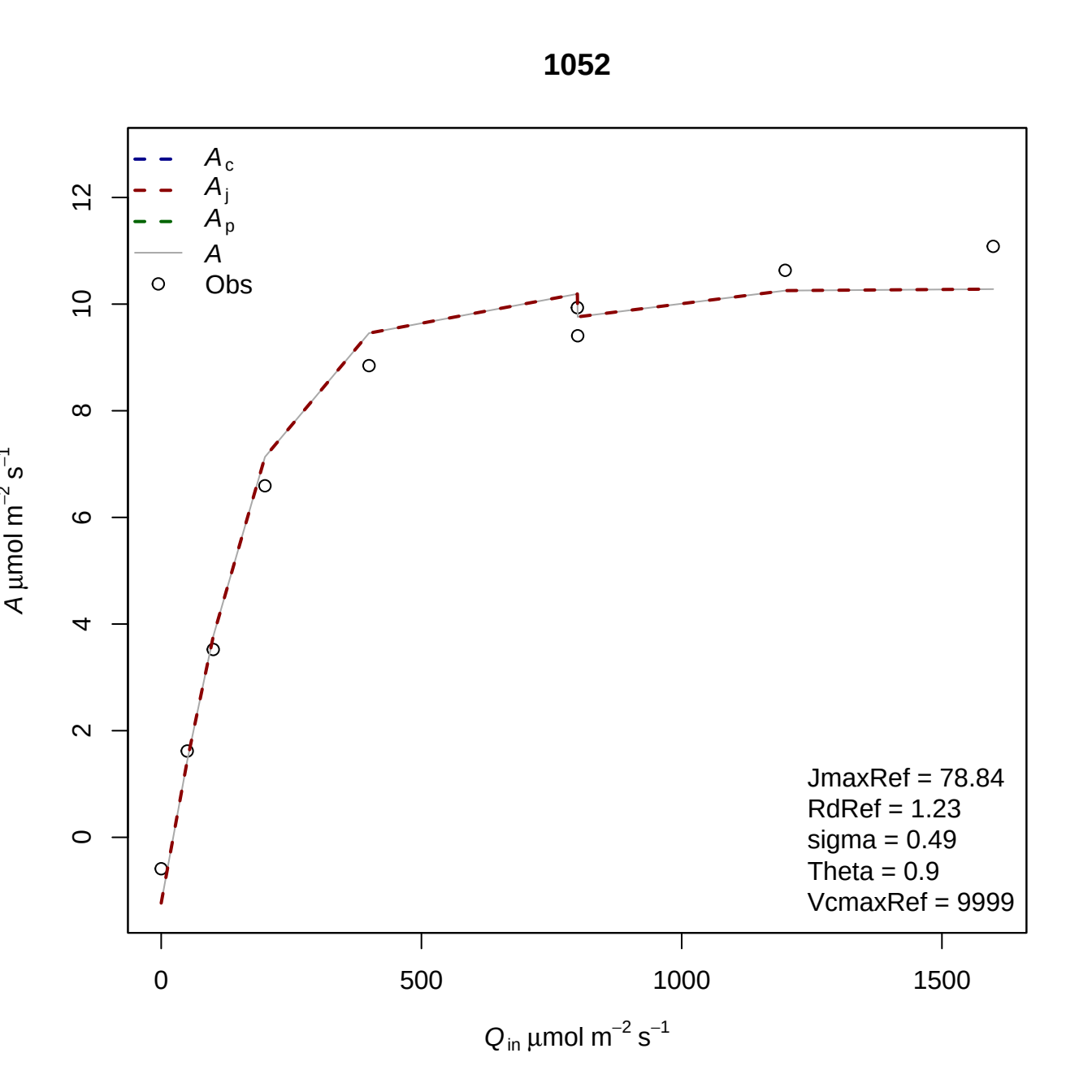

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 65.3
RdRef = 1.97
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

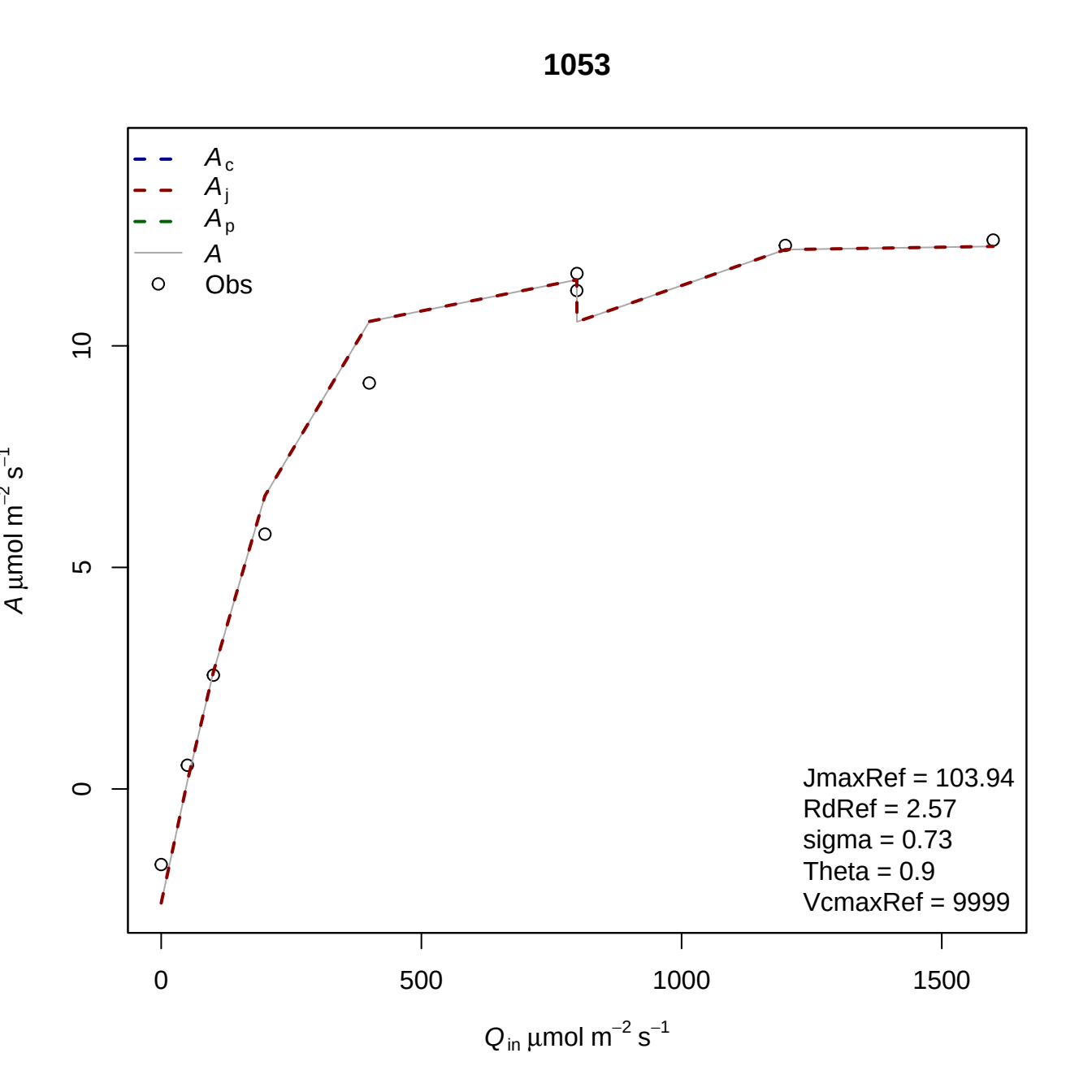

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 76.74
RdRef = 2.3
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

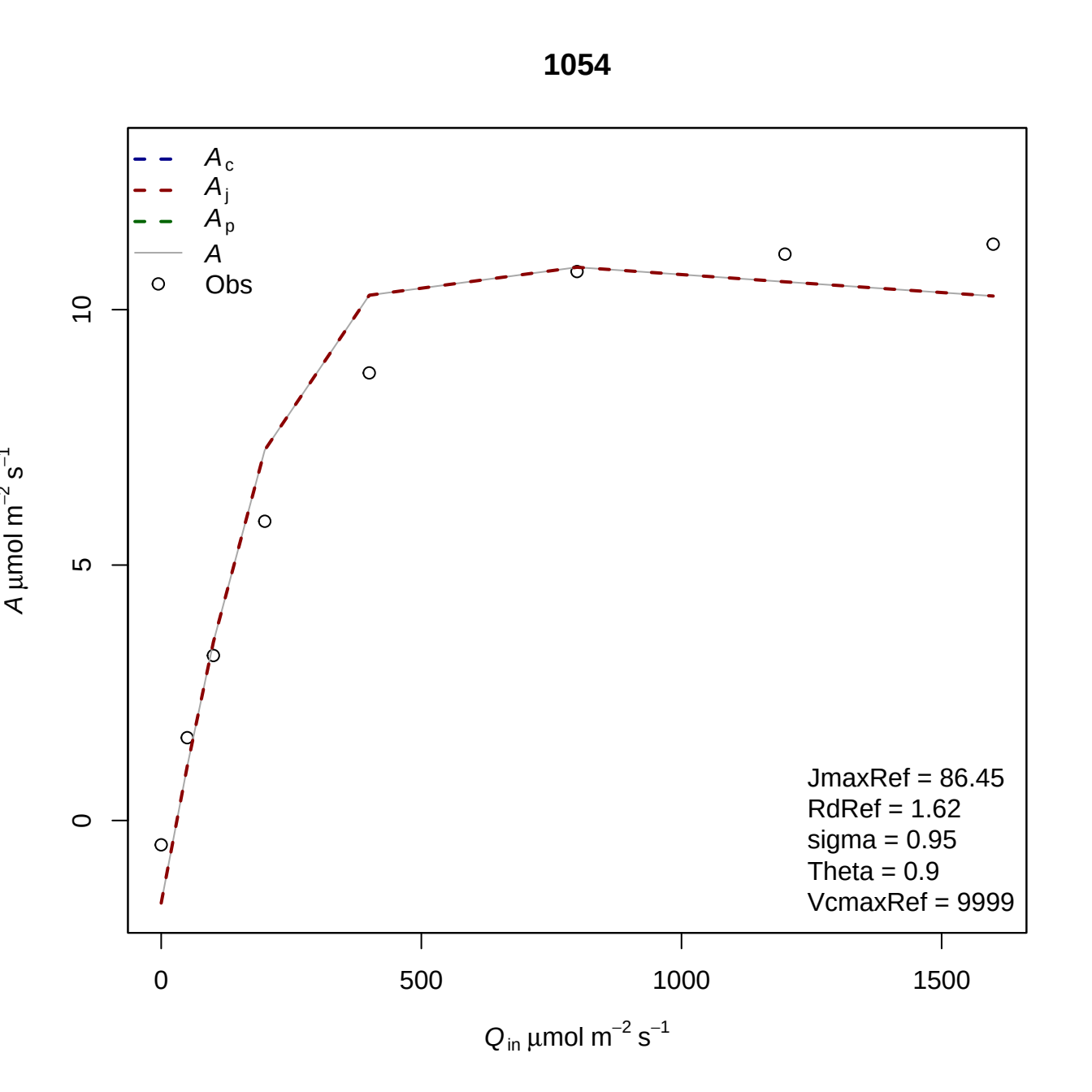

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 132.01
RdRef = 1.61
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

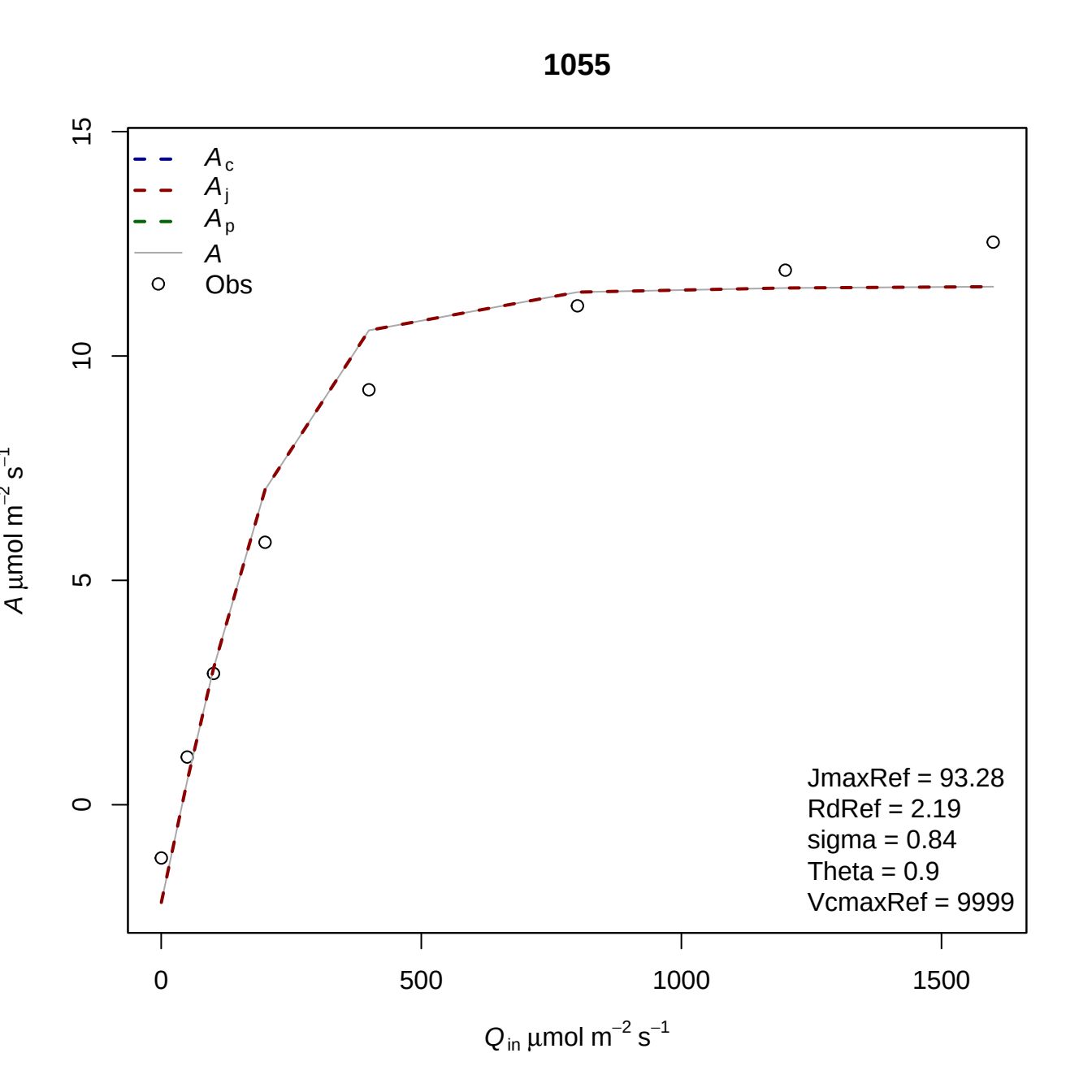

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 99.23
RdRef = 0.9
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

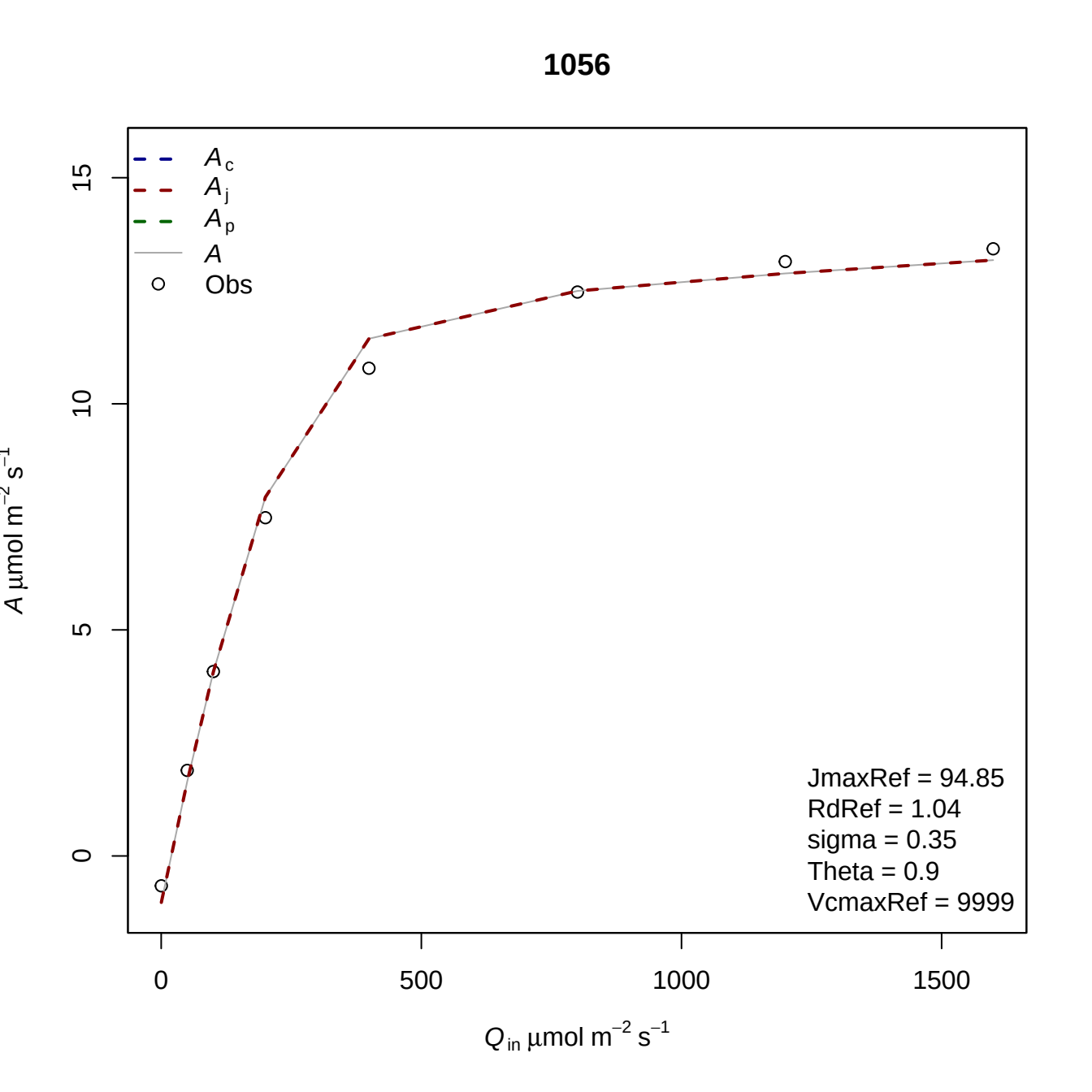

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 108.41
RdRef = 1.87
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

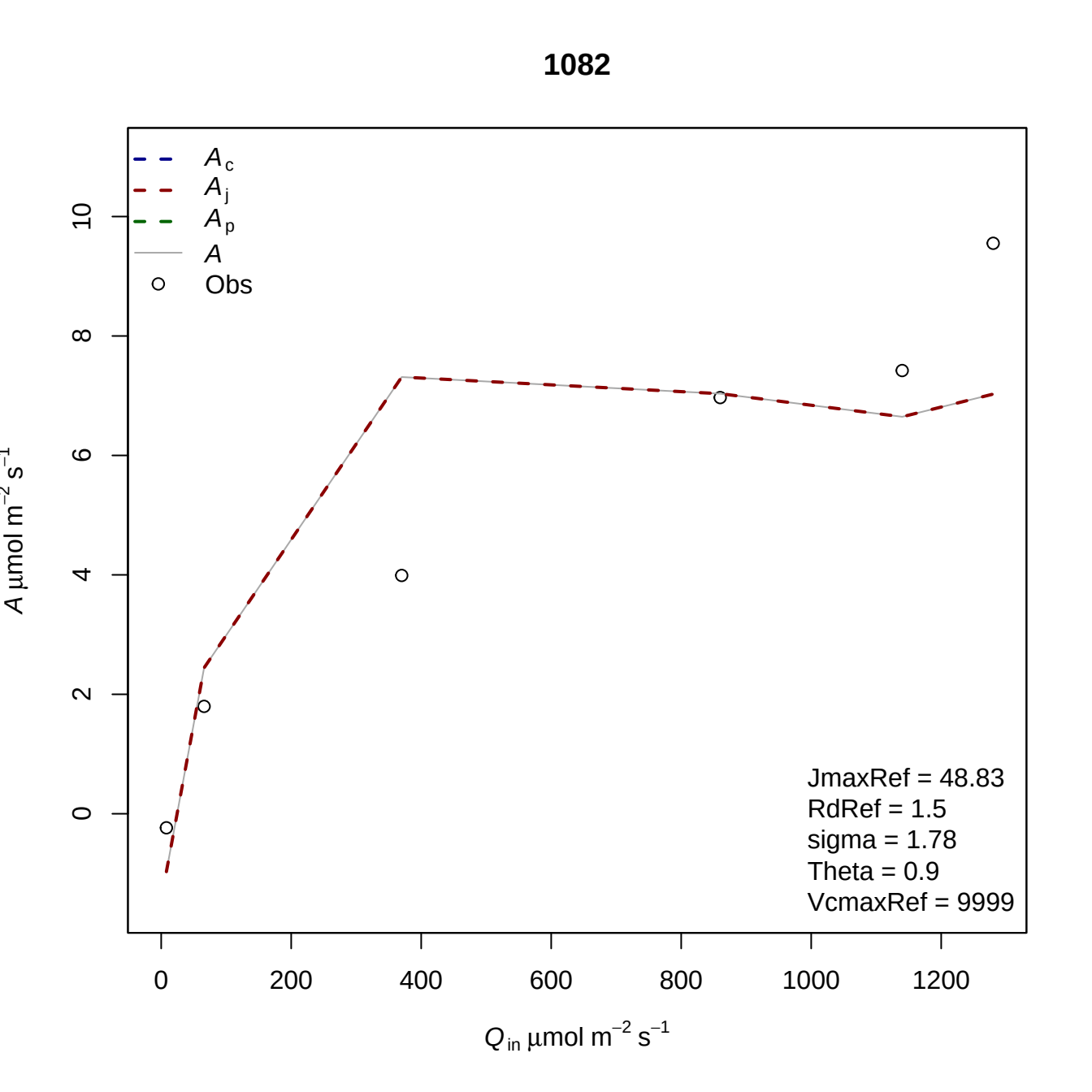

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 78.84
RdRef = 1.23
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

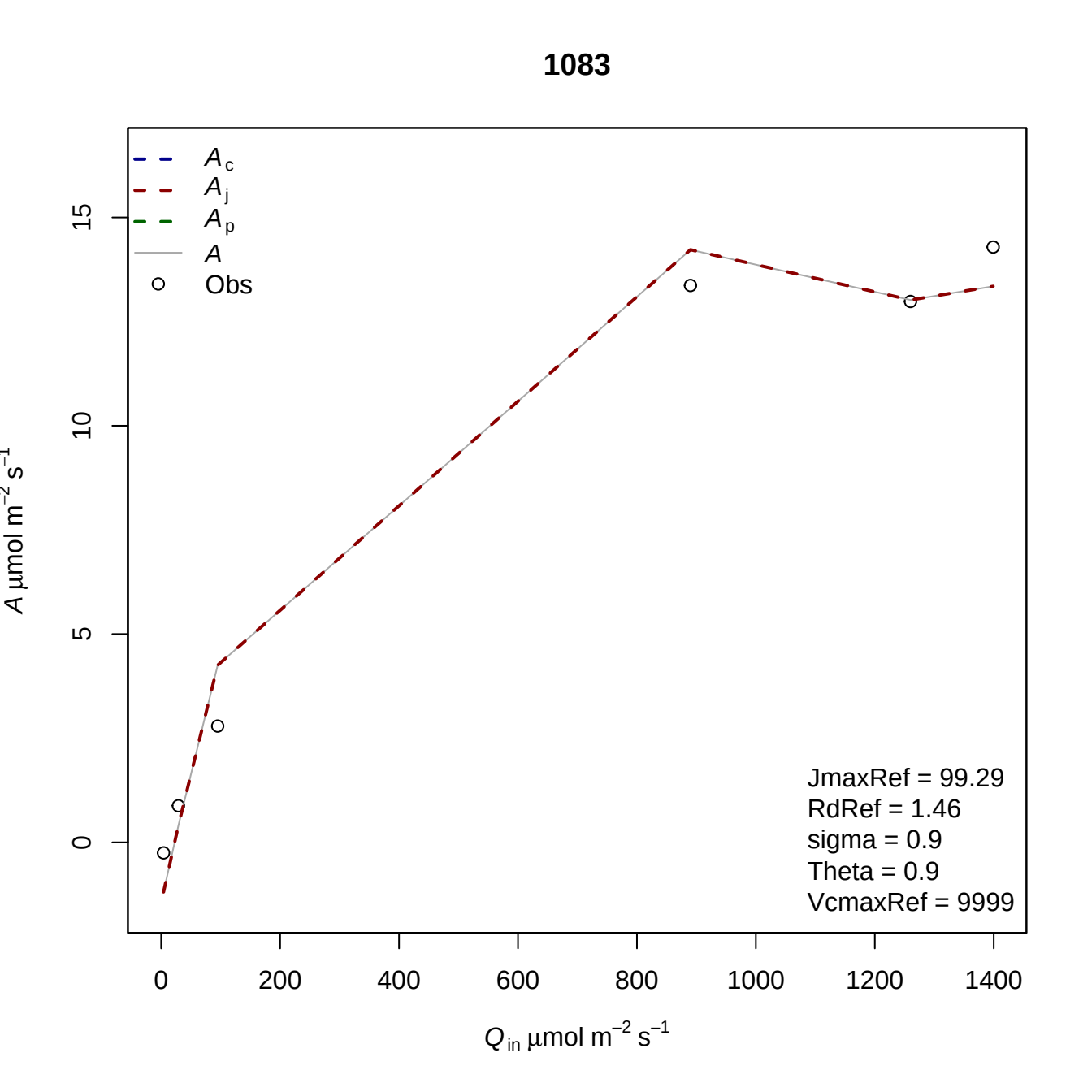

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 103.94
RdRef = 2.57
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

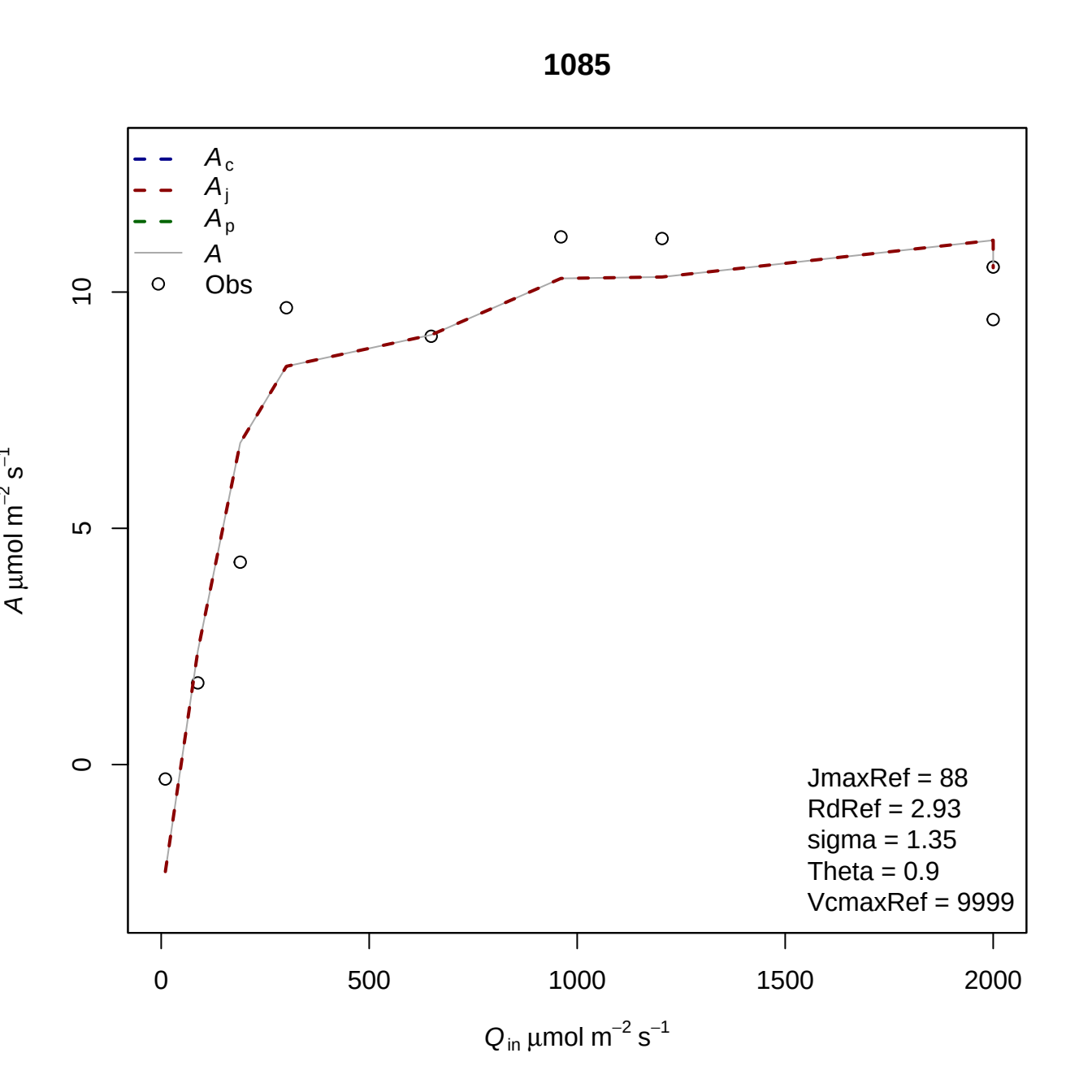

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 86.45
RdRef = 1.62
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

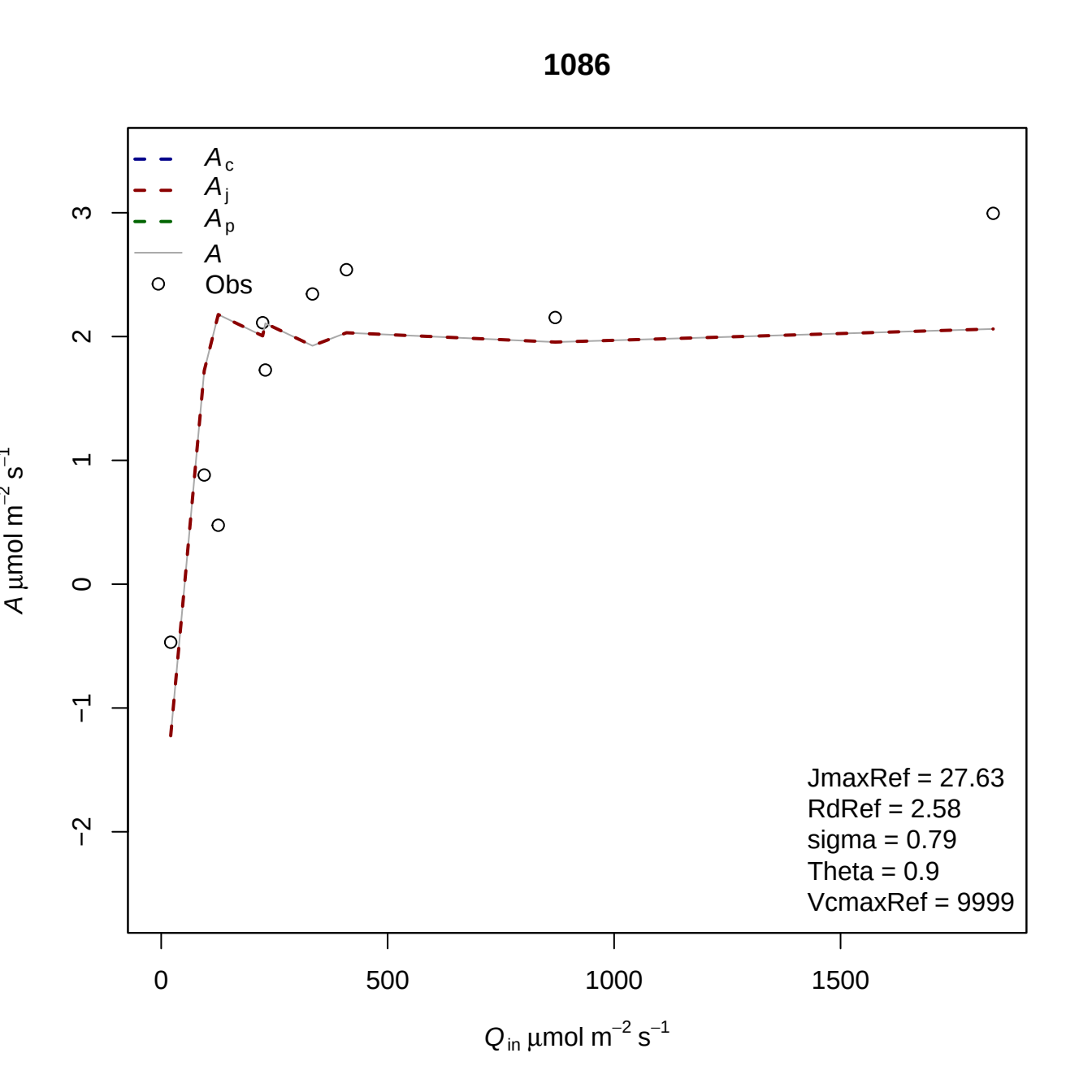

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 93.28
RdRef = 2.19
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

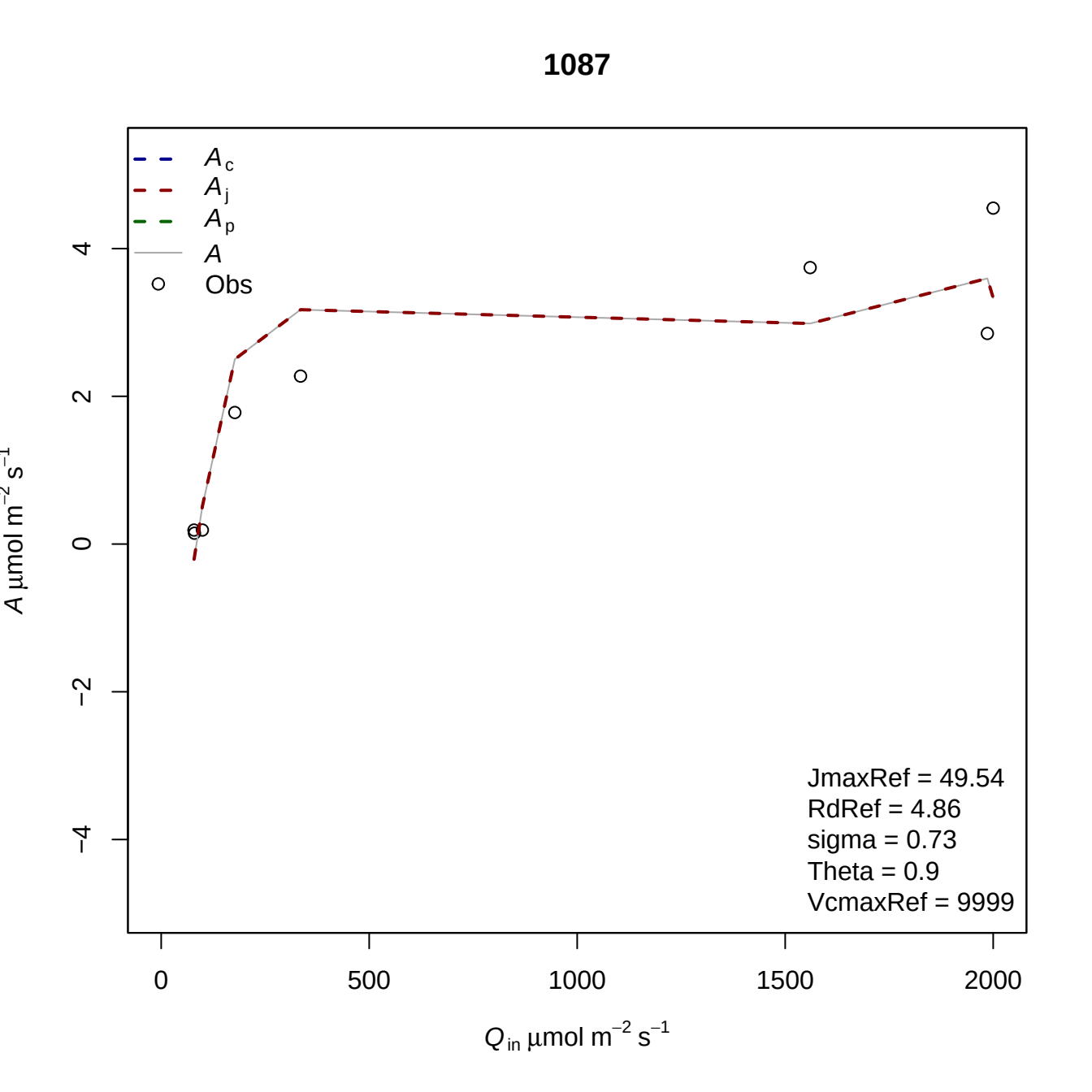

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 94.85
RdRef = 1.04
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

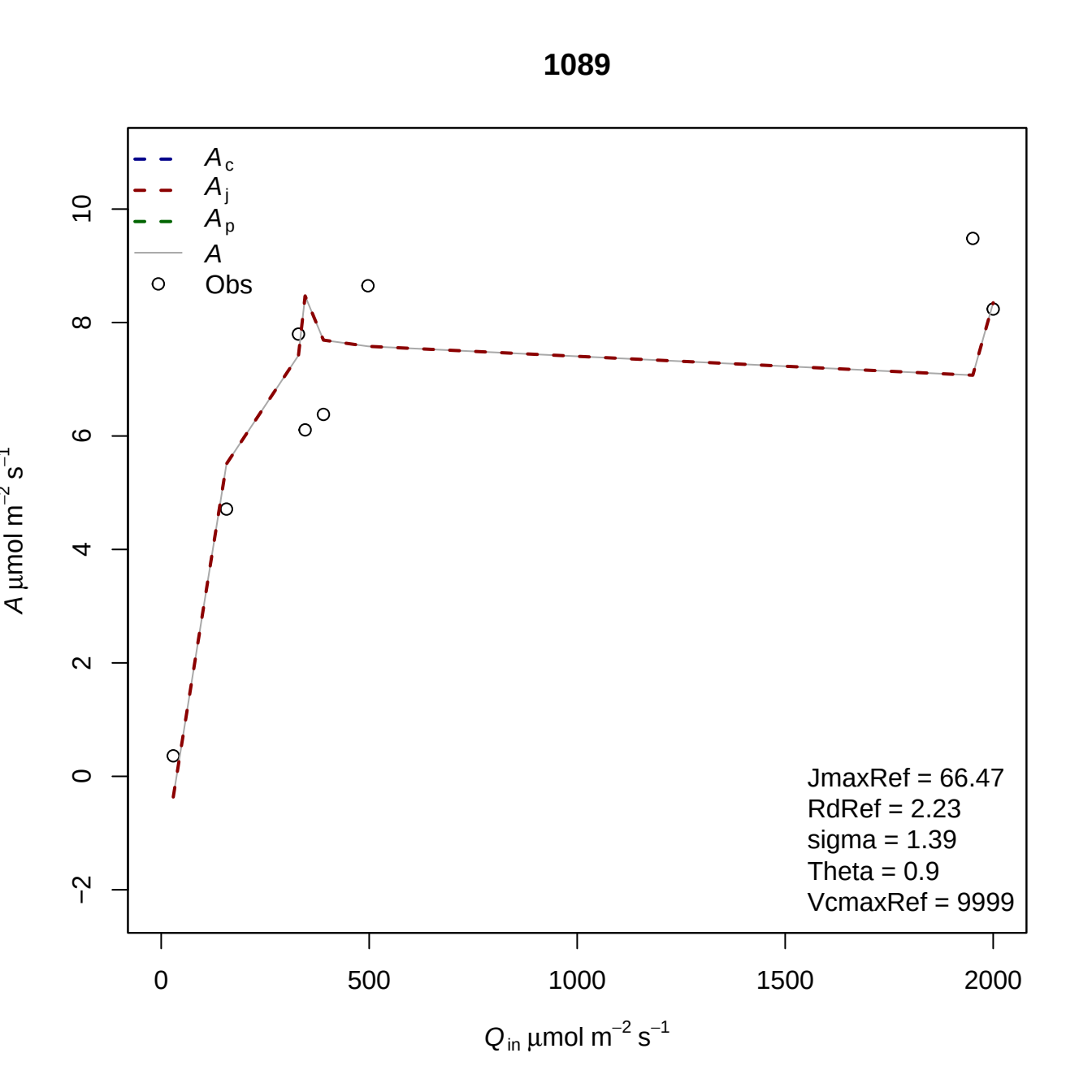

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 48.83
RdRef = 1.5
sigma = 1.78
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

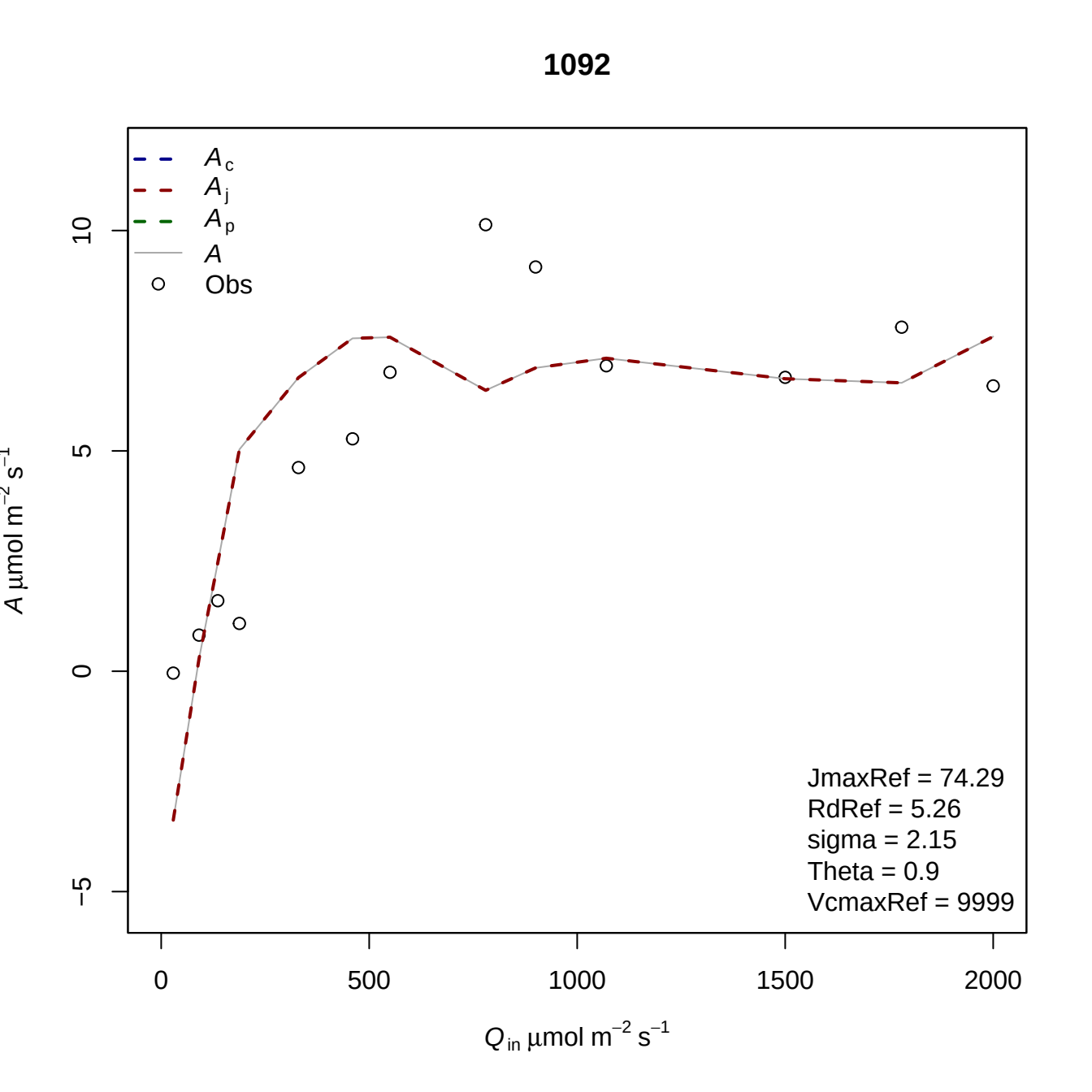

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 99.29
RdRef = 1.46
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

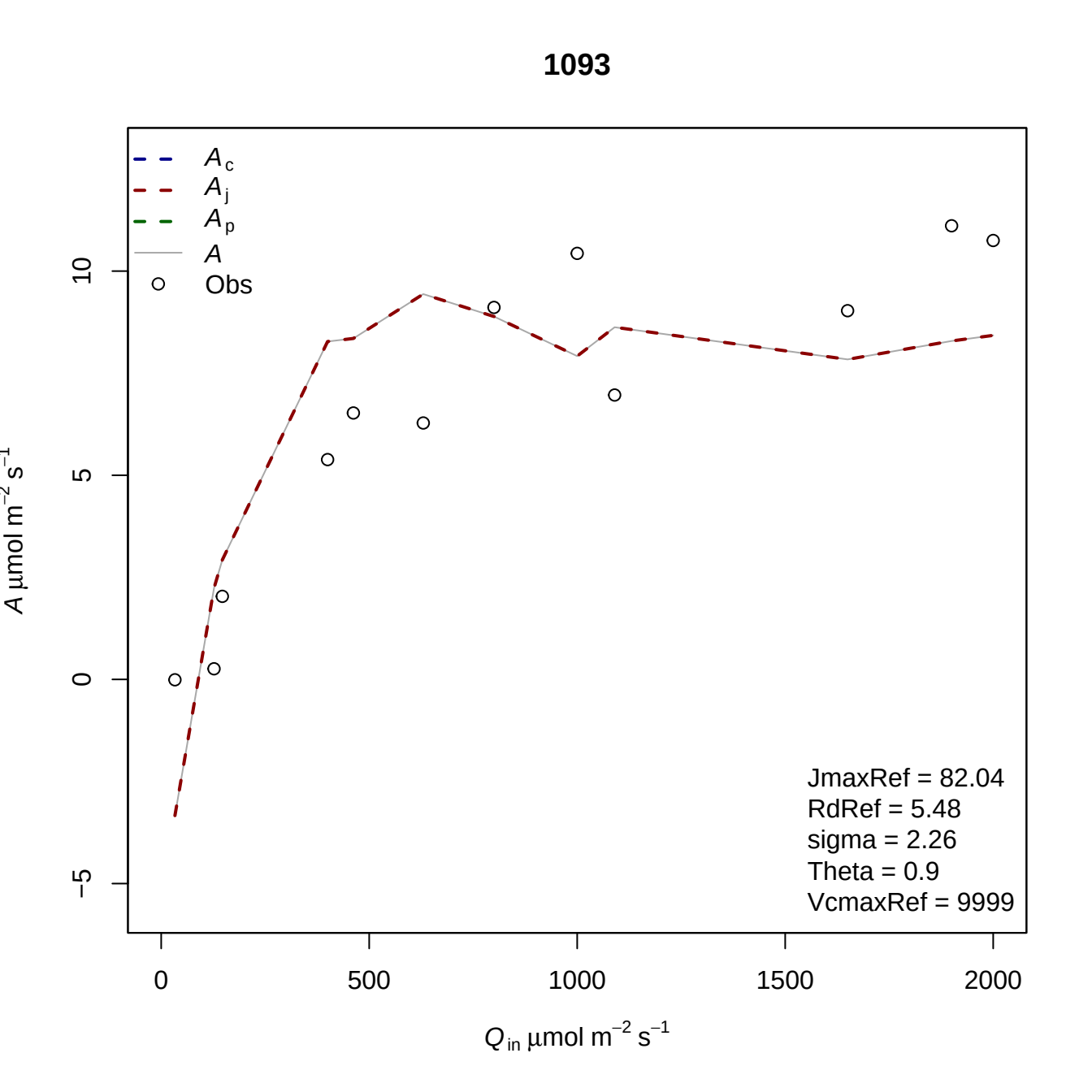

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 88
RdRef = 2.93
sigma = 1.35
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

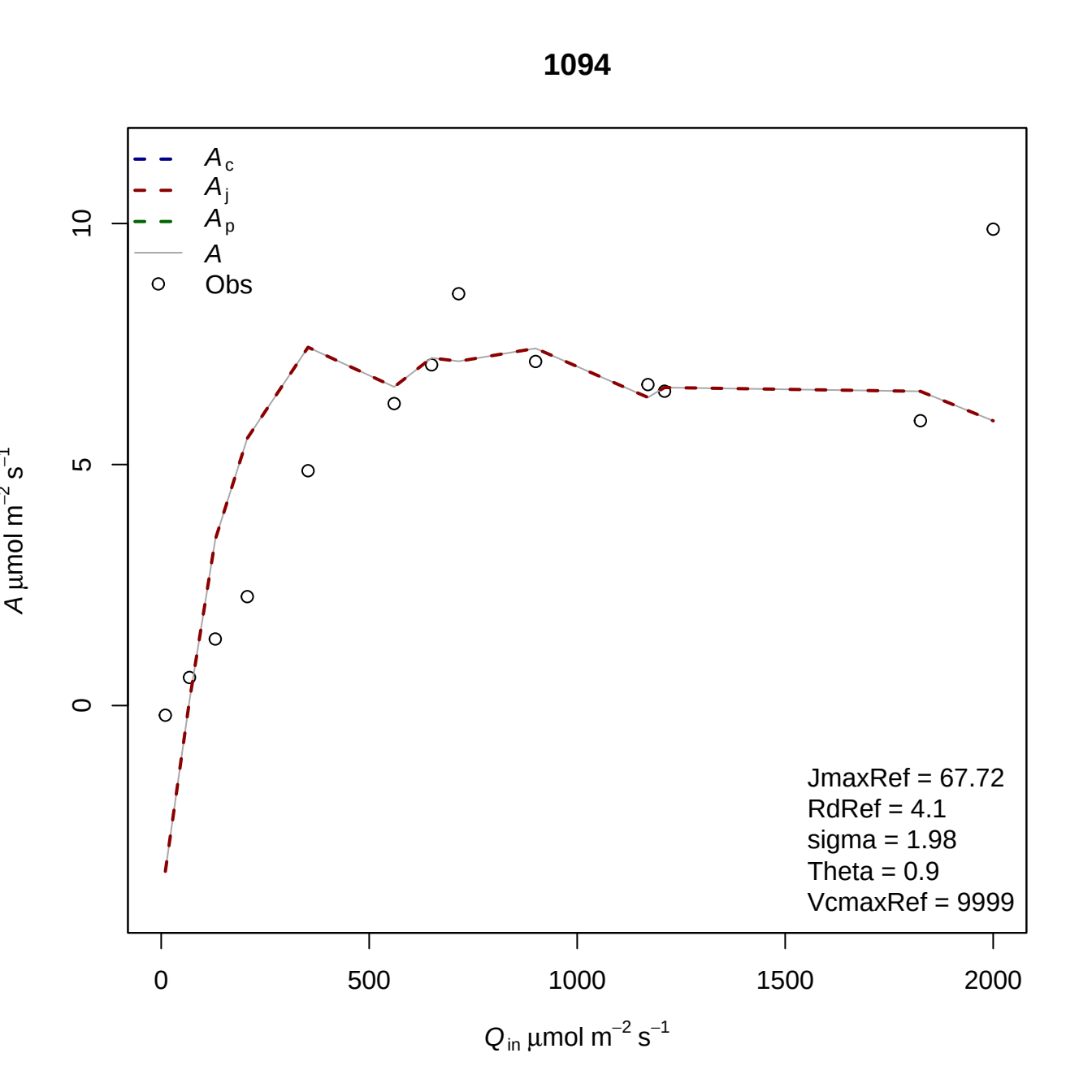

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 27.63
RdRef = 2.58
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
3
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

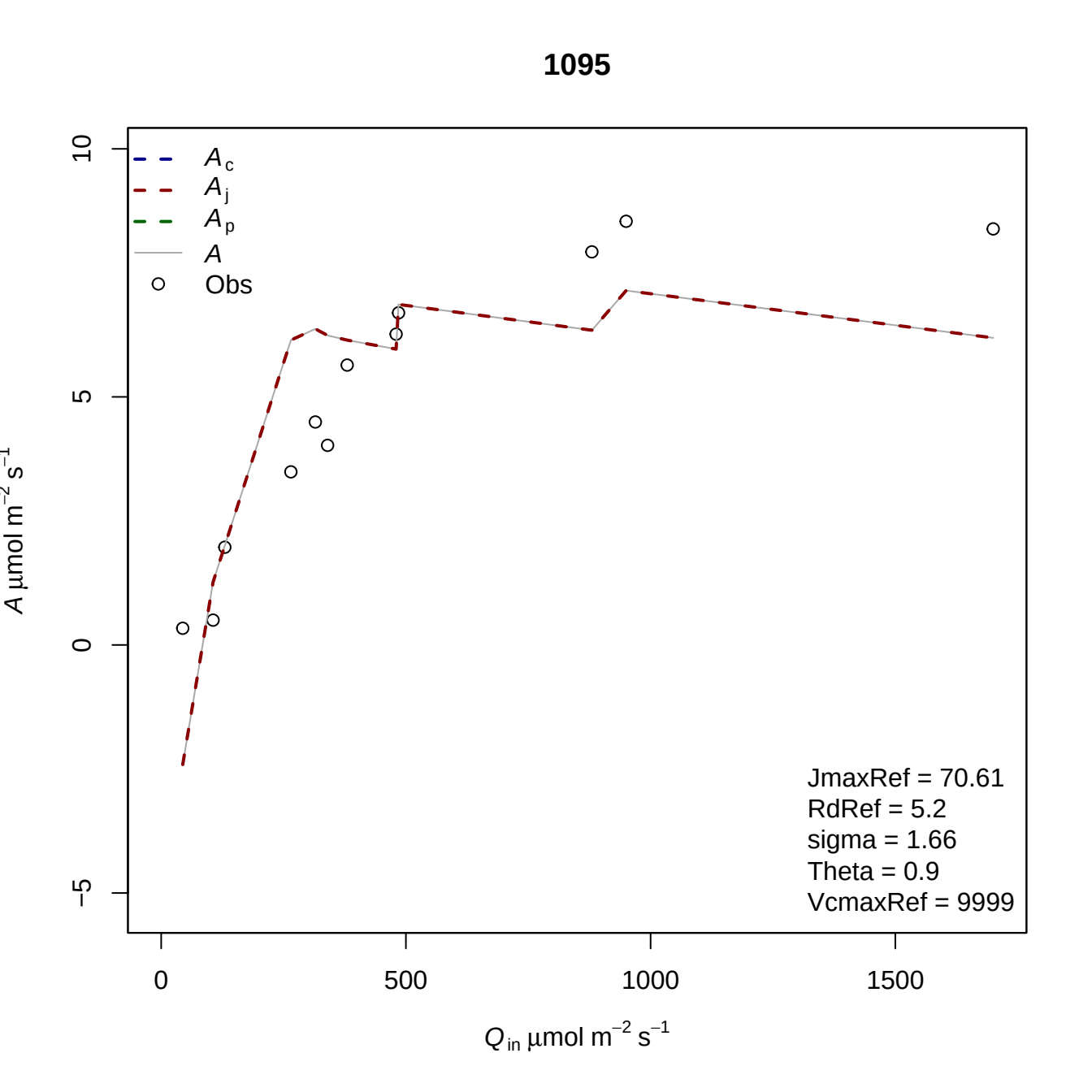

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 49.54
RdRef = 4.86
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500
2000

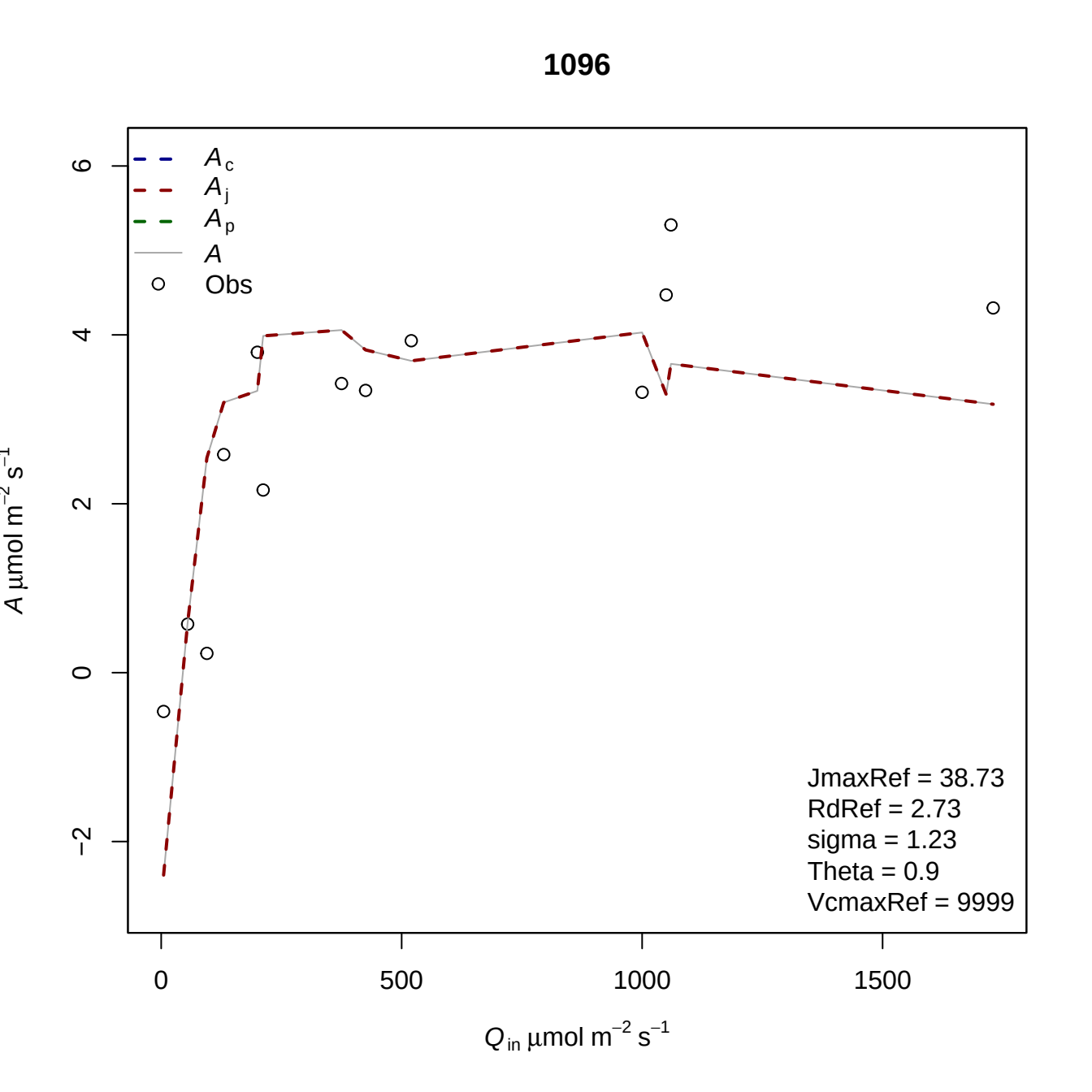

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 66.47
RdRef = 2.23
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

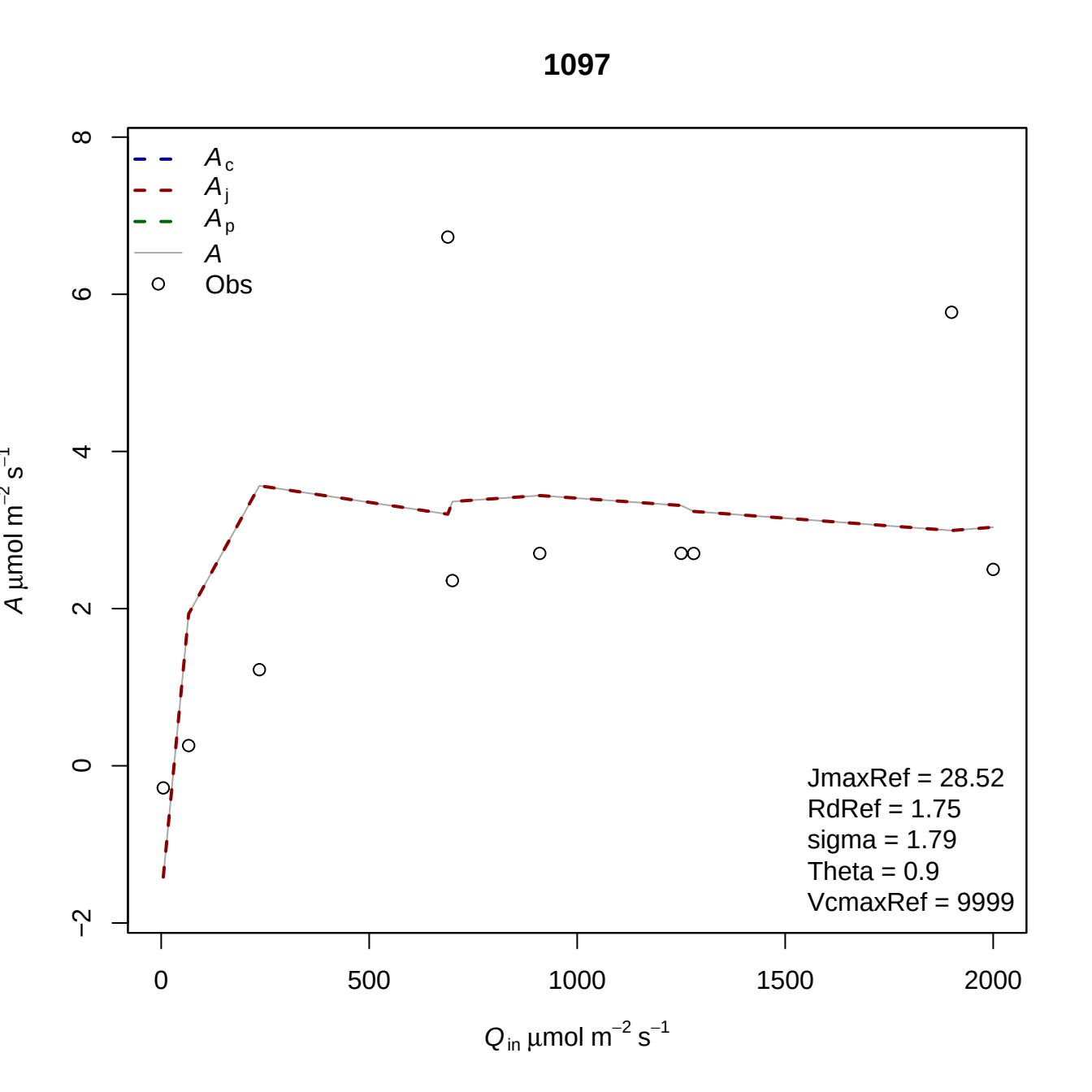

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 74.29
RdRef = 5.26
sigma = 2.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

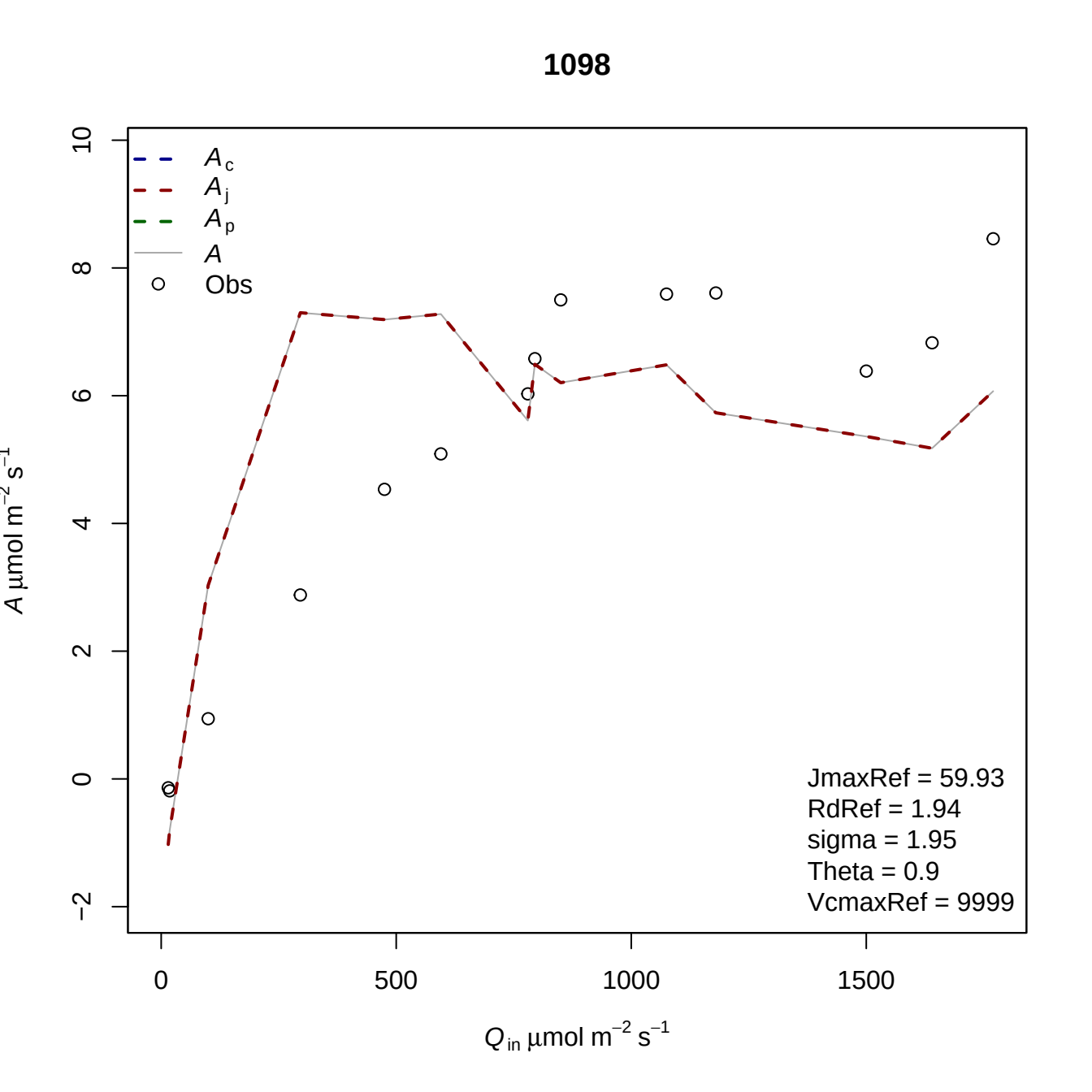

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 82.04
RdRef = 5.48
sigma = 2.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

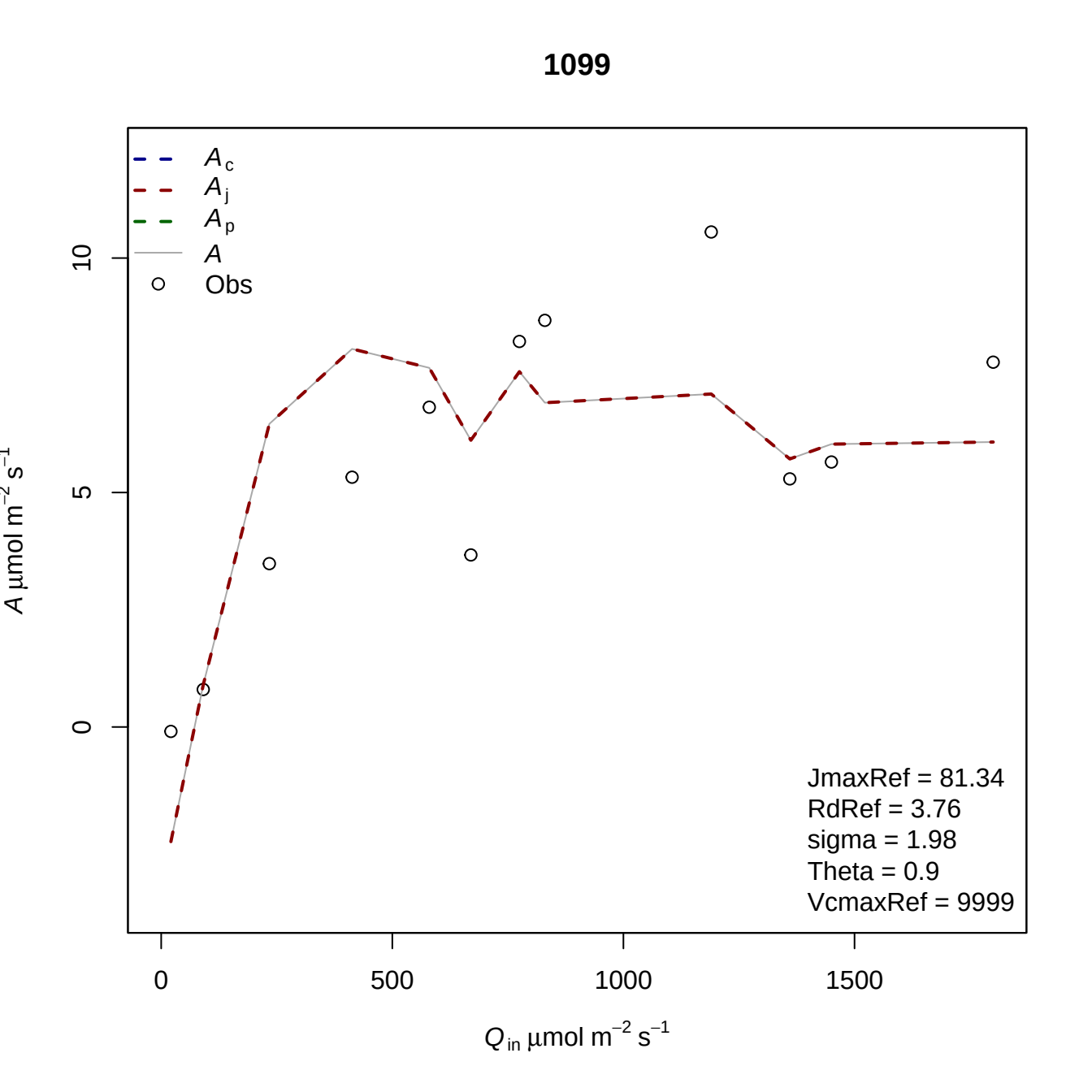

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 67.72
RdRef = 4.1
sigma = 1.98
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

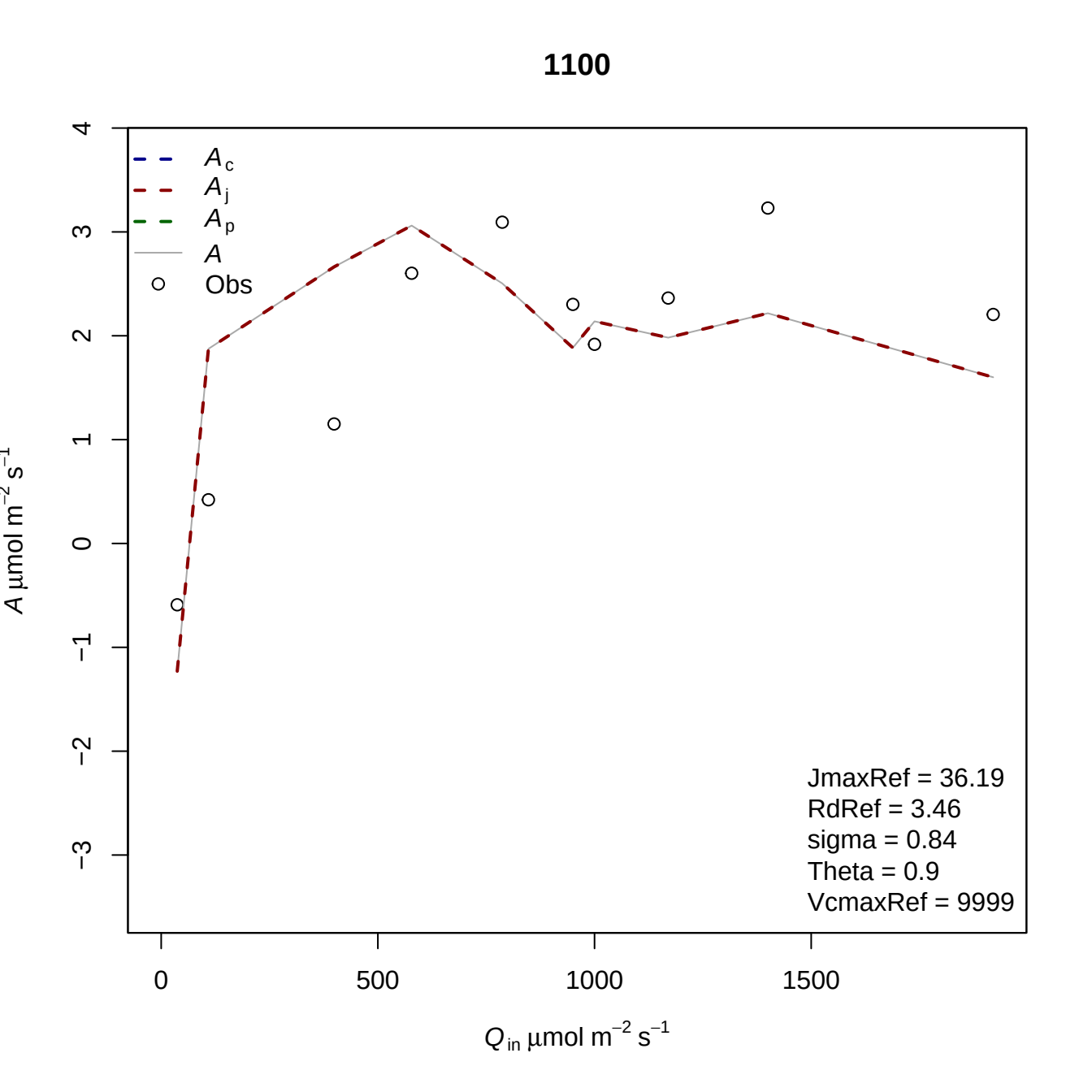

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 70.61
RdRef = 5.2
sigma = 1.66
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

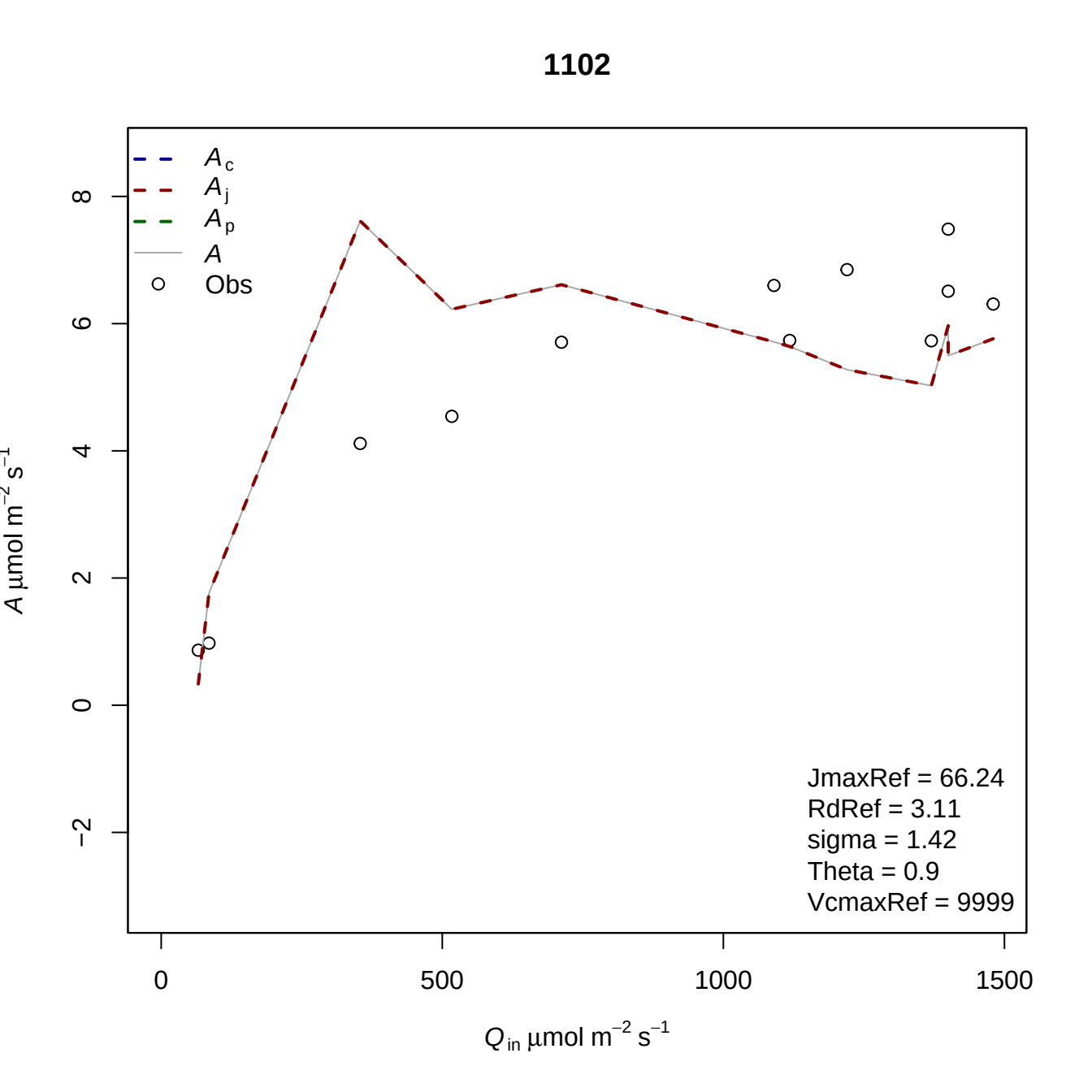

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 38.73
RdRef = 2.73
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

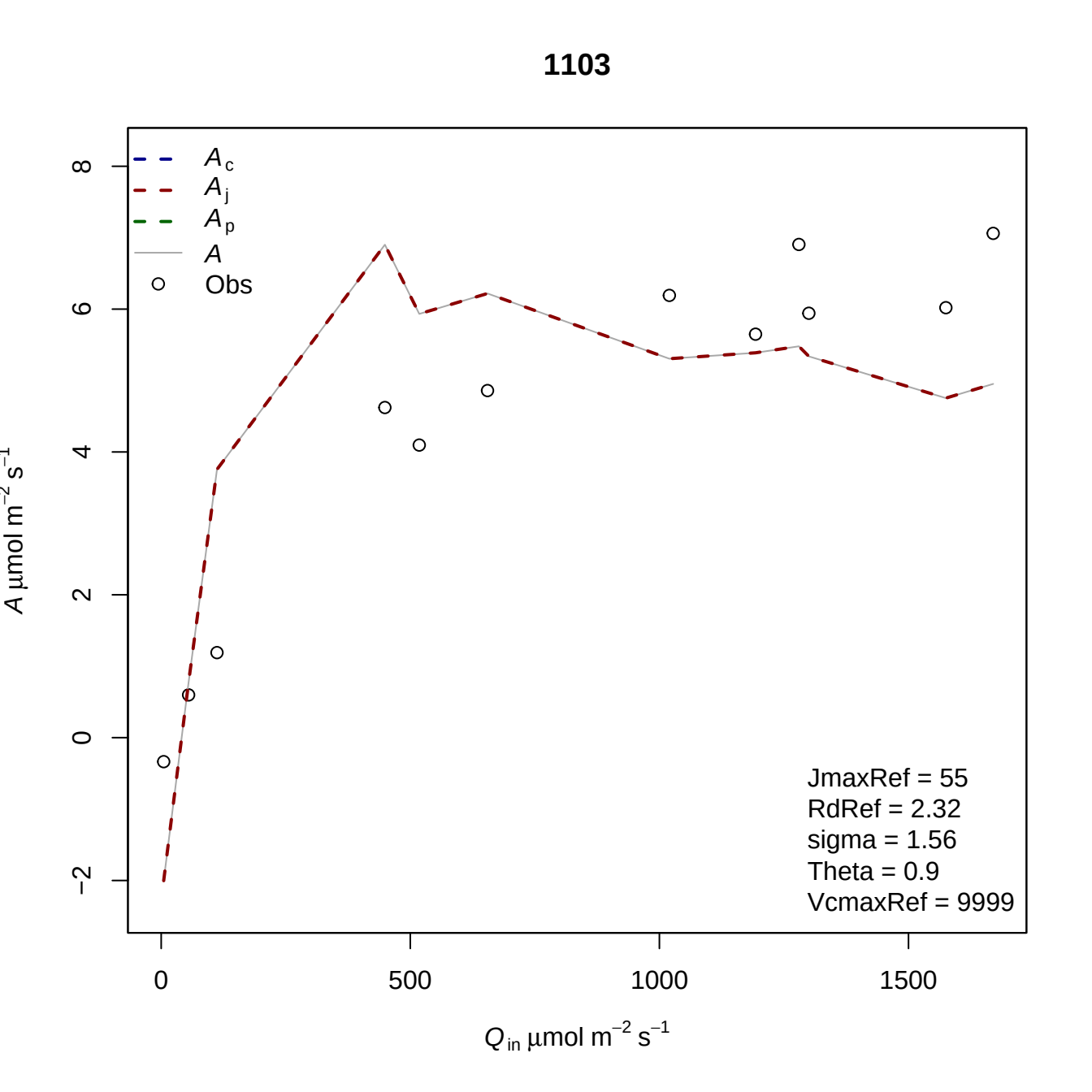

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 28.52
RdRef = 1.75
sigma = 1.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

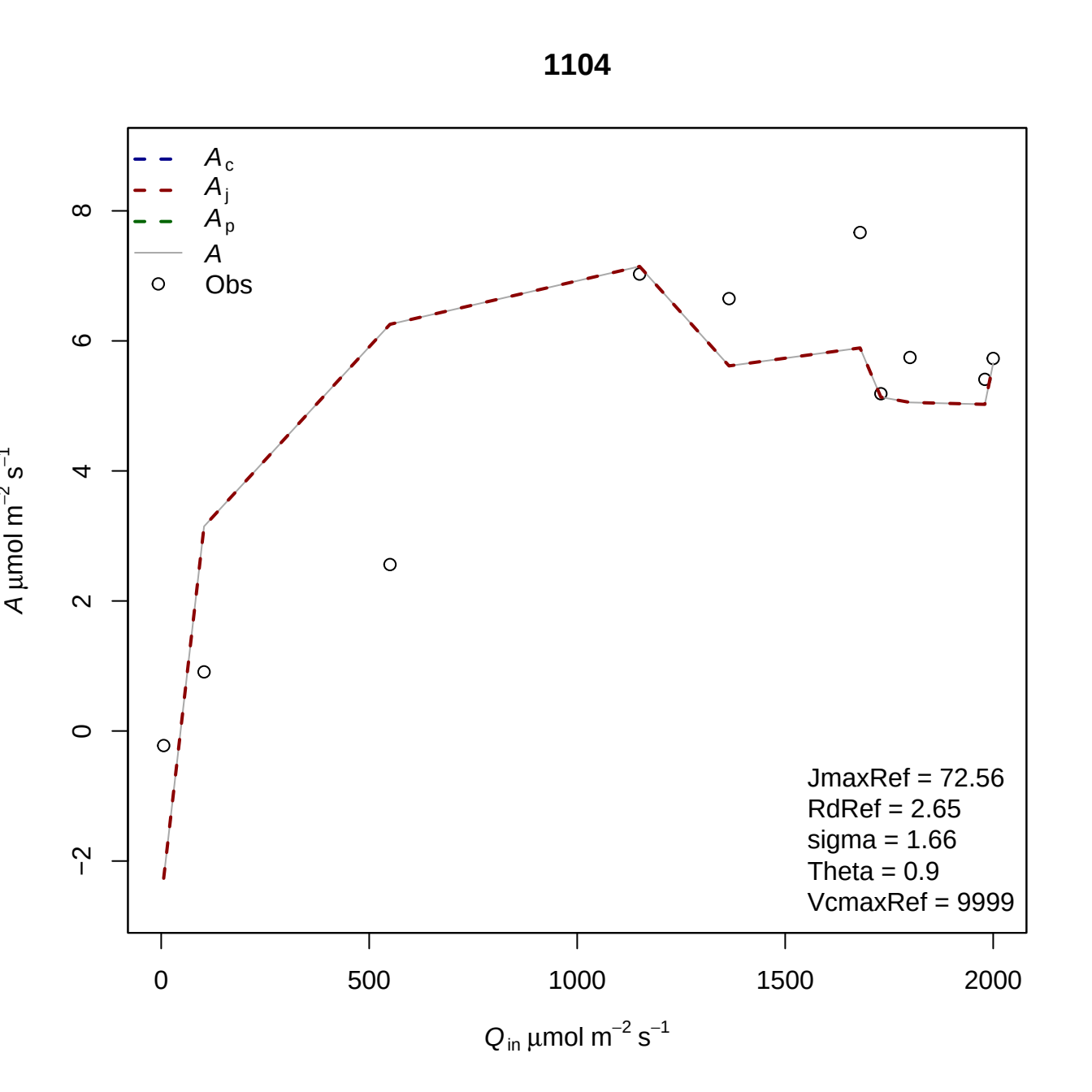

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 59.93
RdRef = 1.94
sigma = 1.95
Theta = 0.9
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

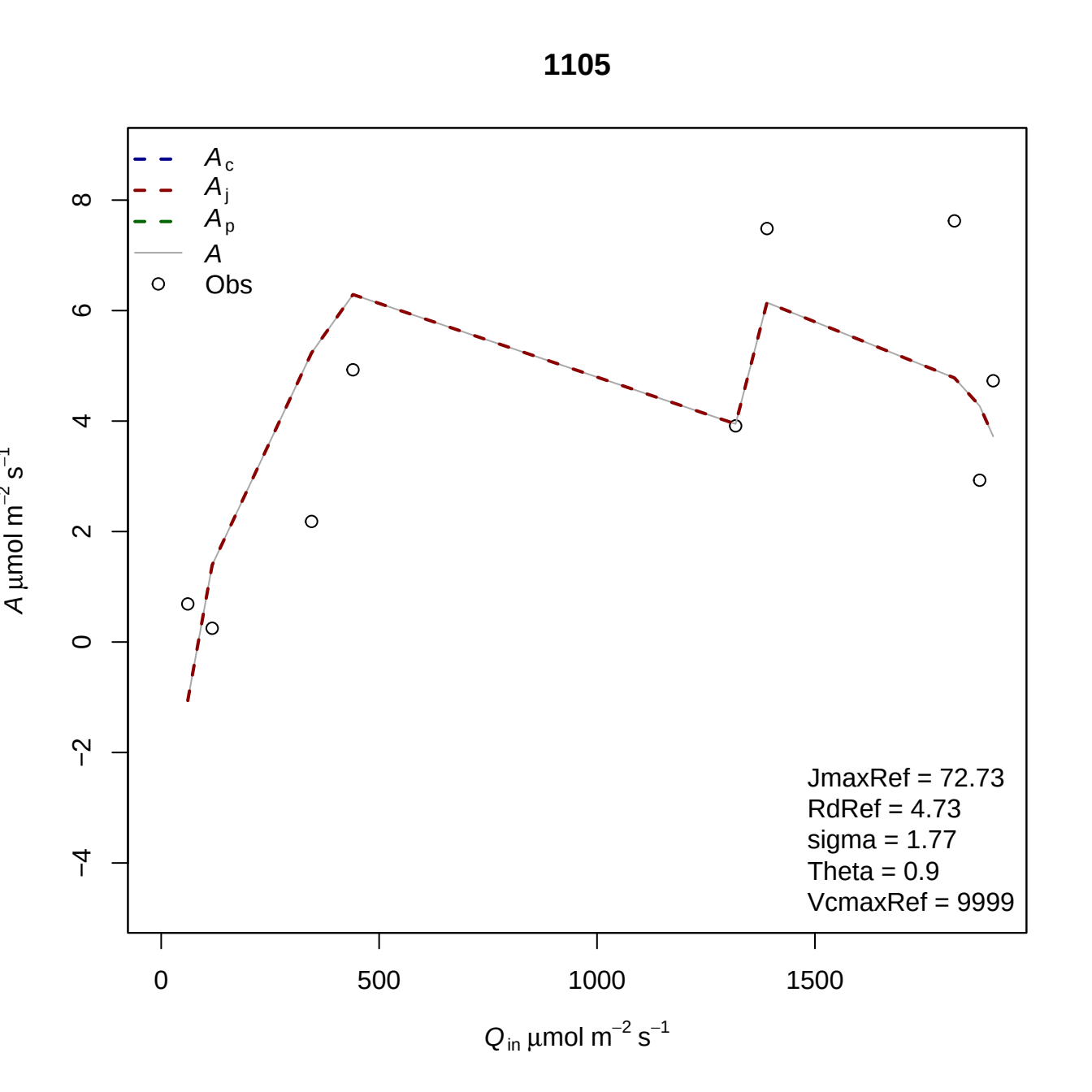

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 81.34
RdRef = 3.76
sigma = 1.98
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

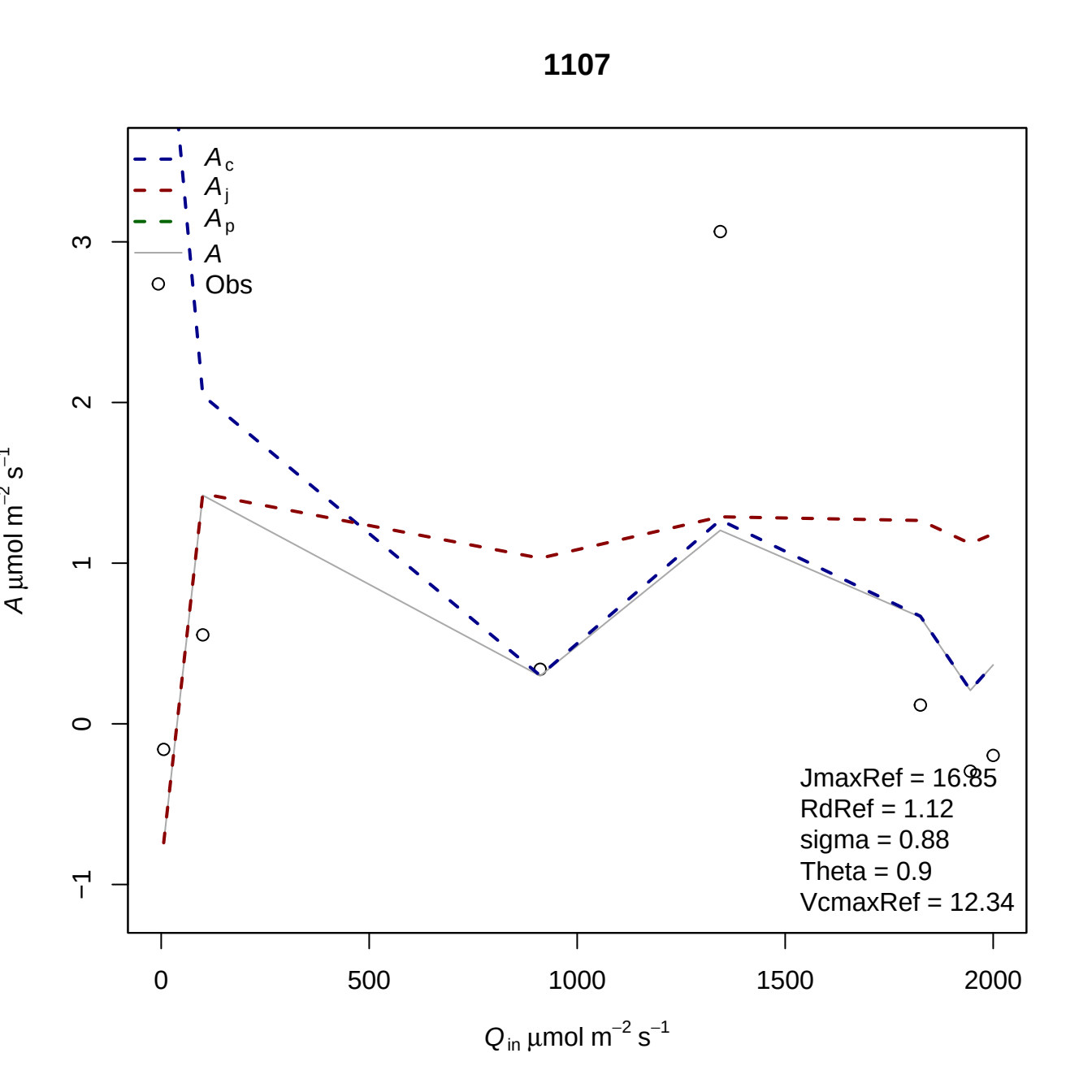

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 36.19
RdRef = 3.46
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

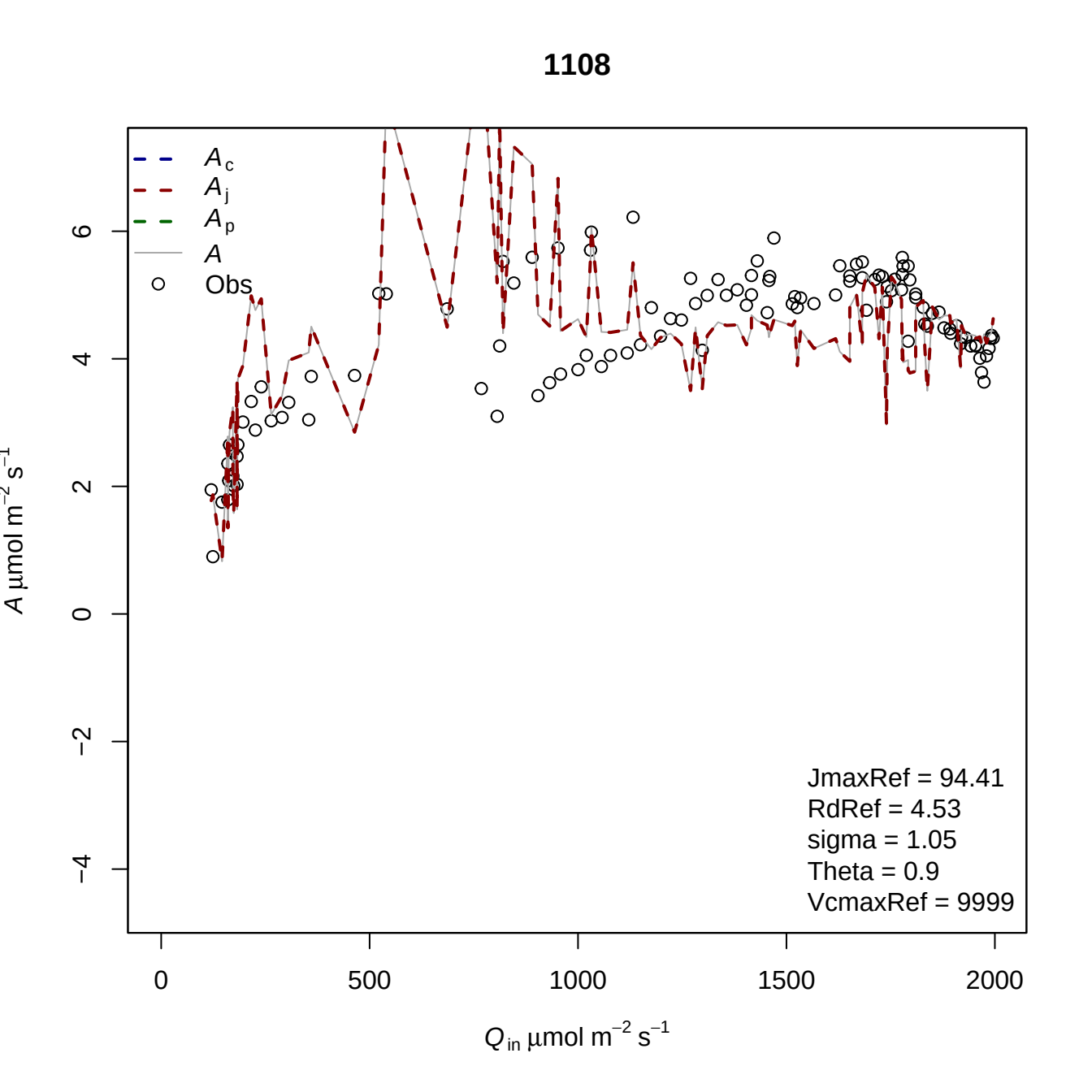

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 66.24
RdRef = 3.11
sigma = 1.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

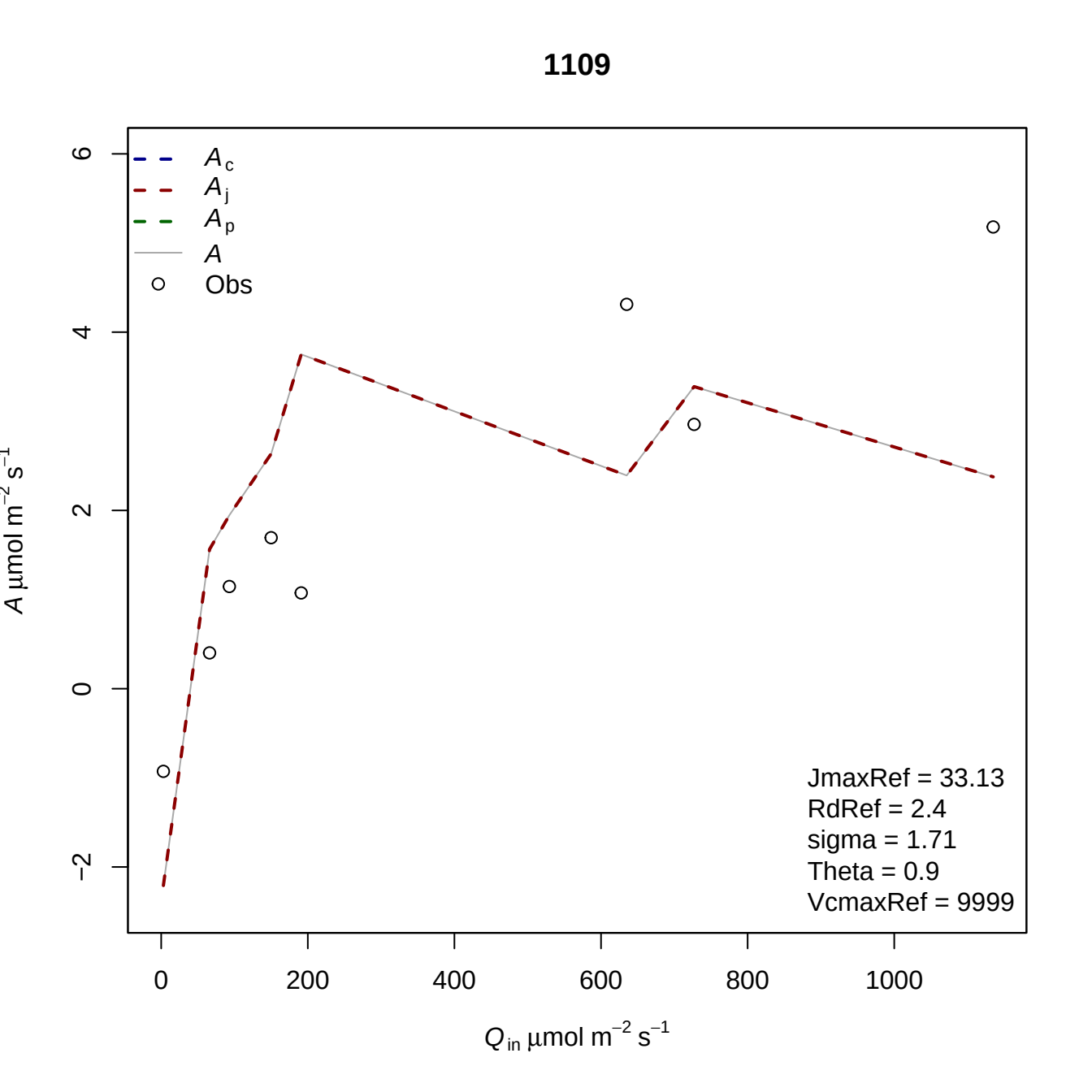

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 55
RdRef = 2.32
sigma = 1.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

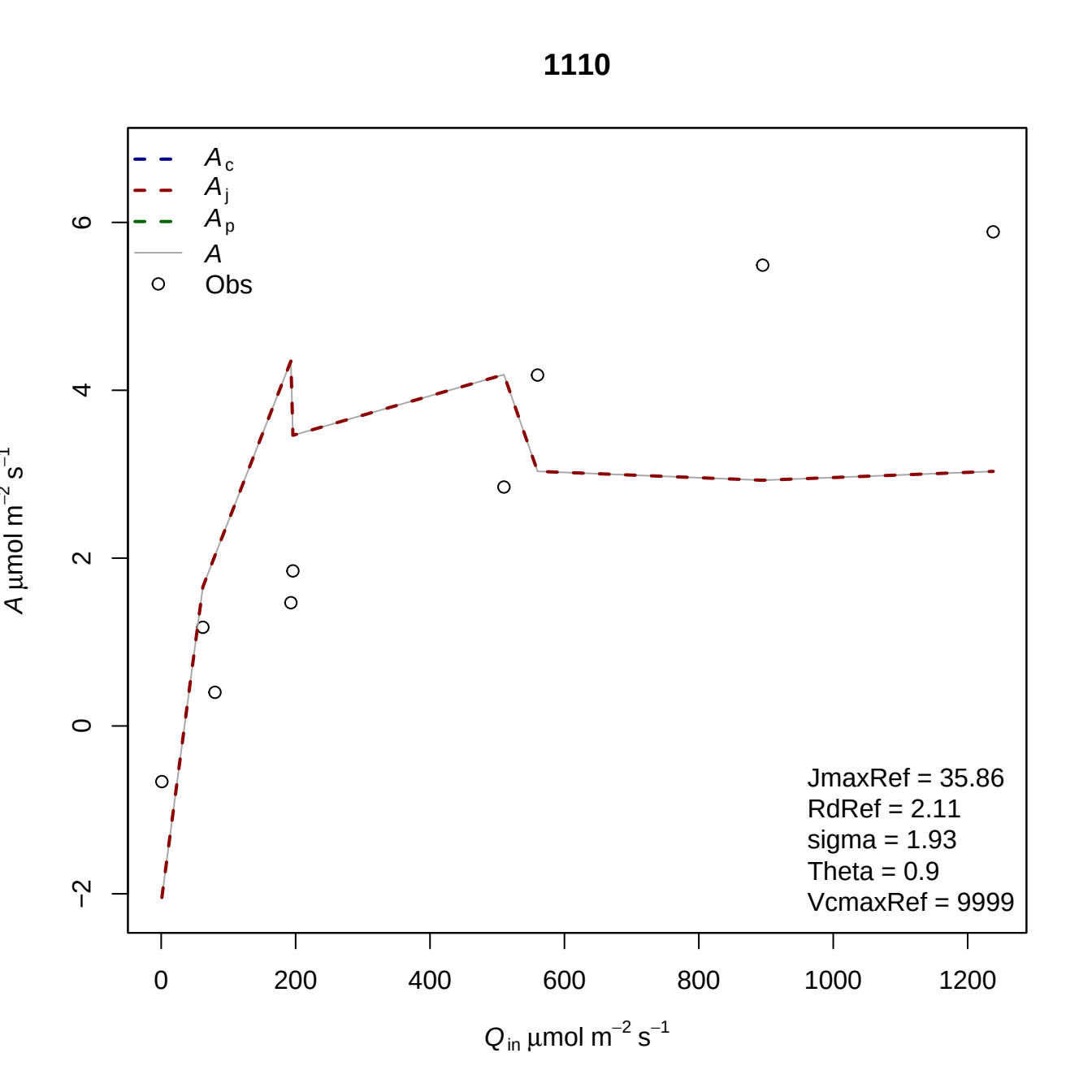

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 72.56
RdRef = 2.65
sigma = 1.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

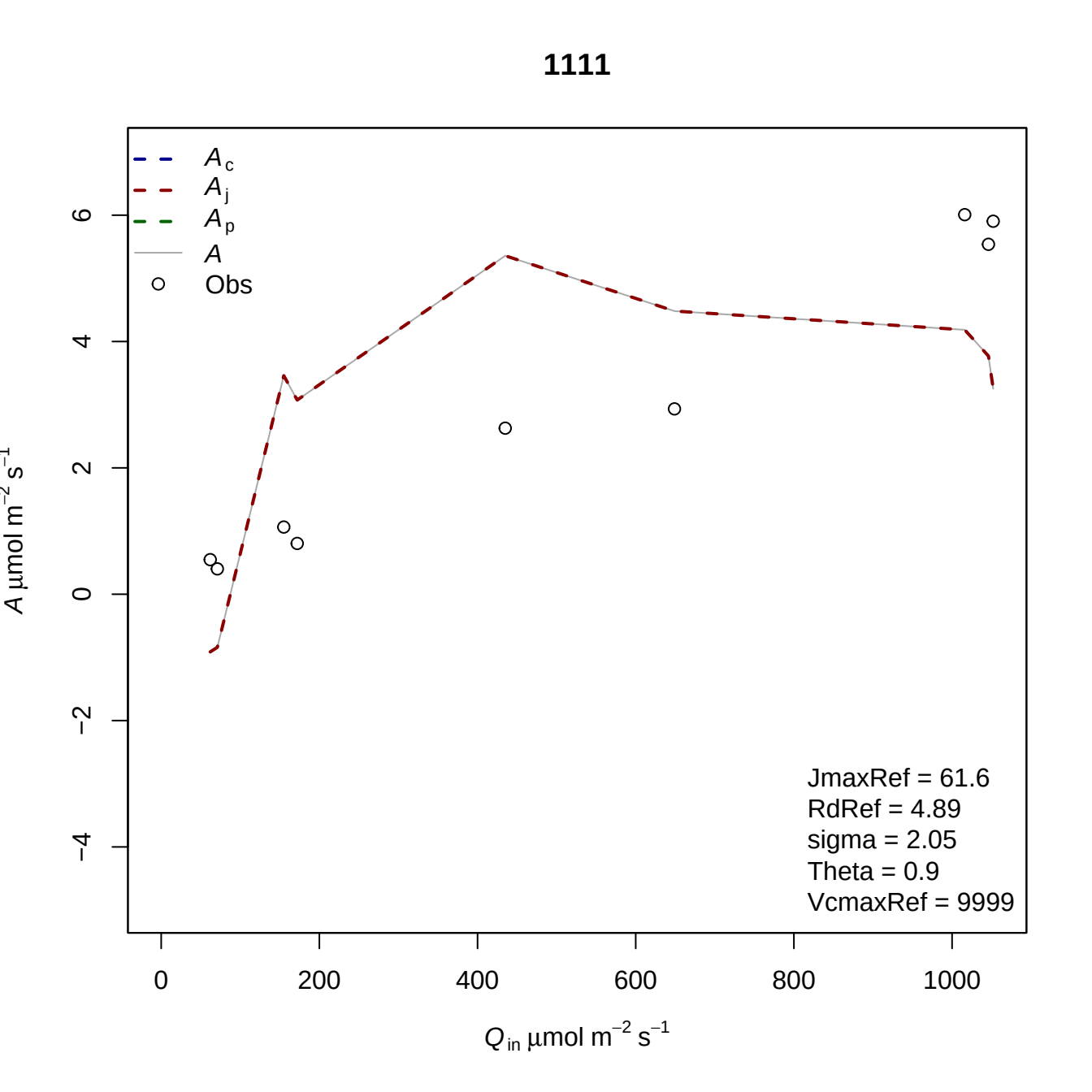

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 72.73
RdRef = 4.73
sigma = 1.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

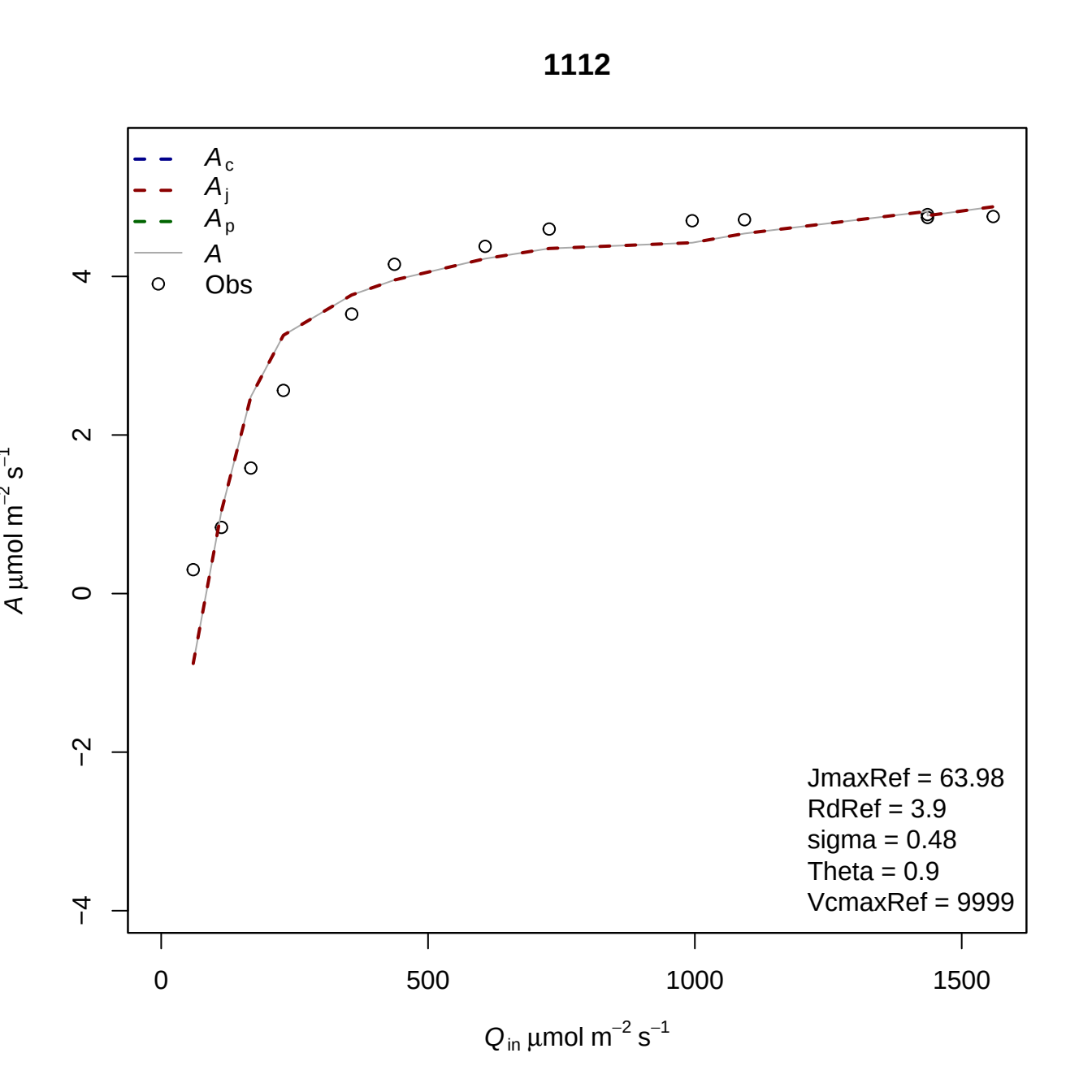

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 16.85
RdRef = 1.12
sigma = 0.88
Theta = 0.9
VcmaxRef = 12.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

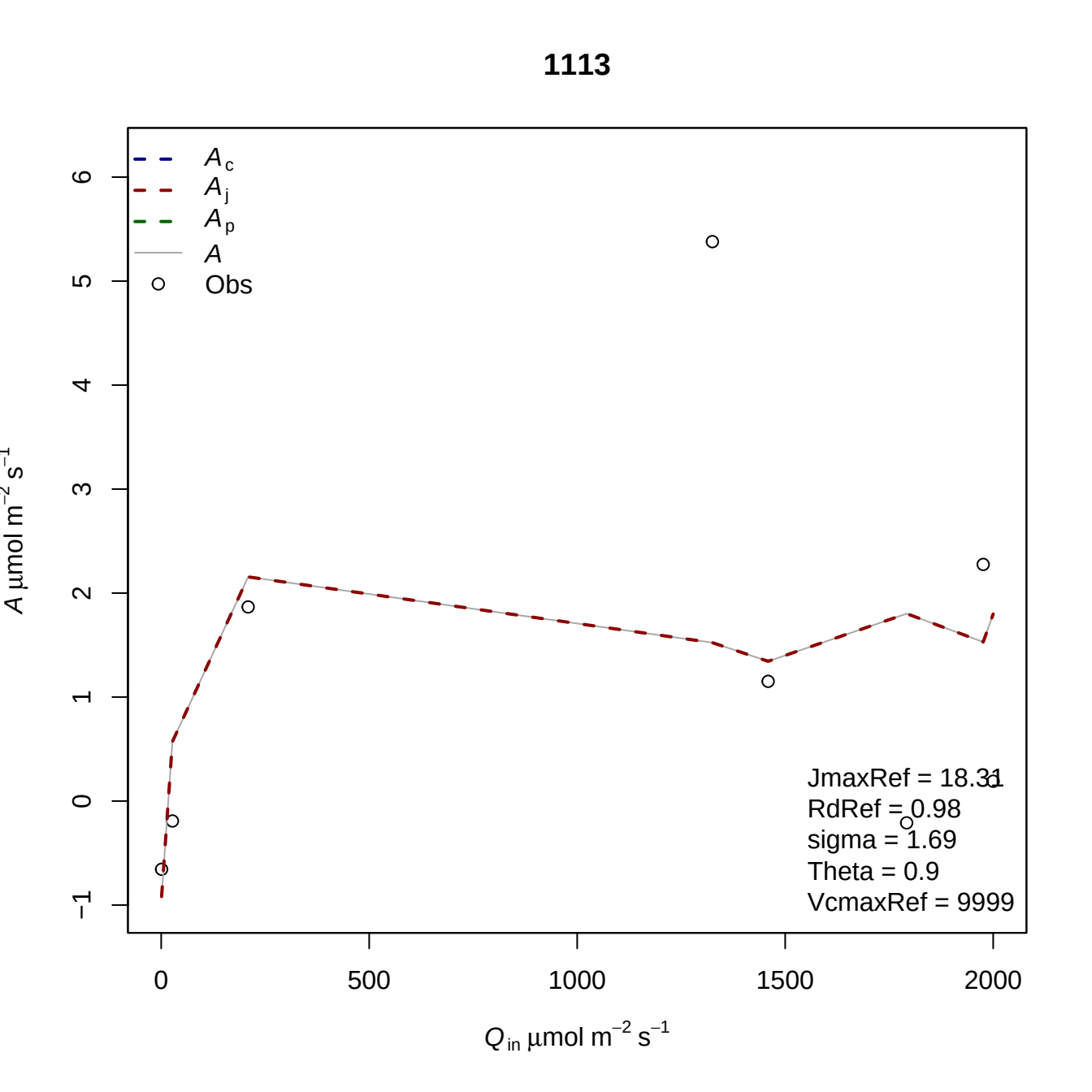

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 94.41
RdRef = 4.53
sigma = 1.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

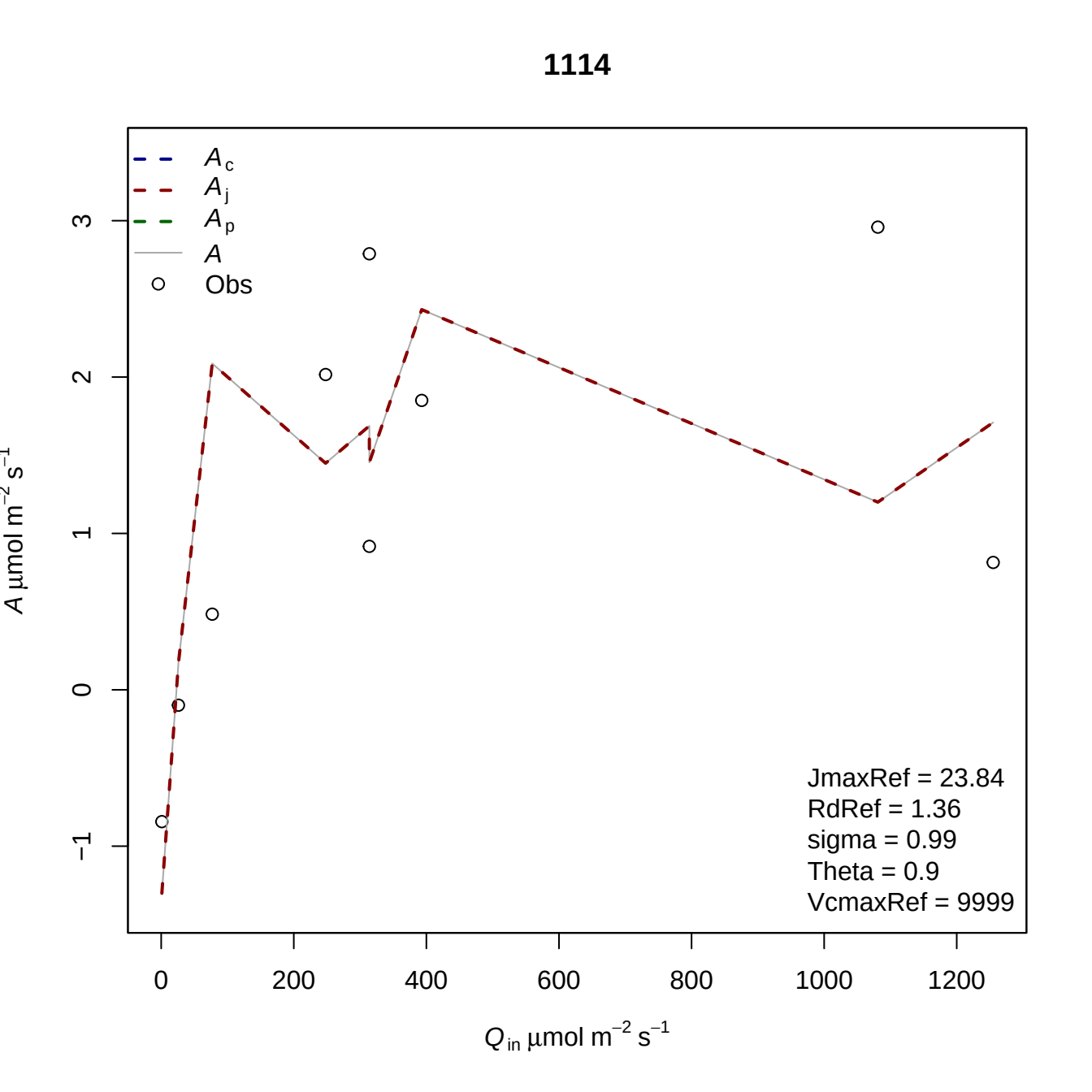

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 33.13
RdRef = 2.4
sigma = 1.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

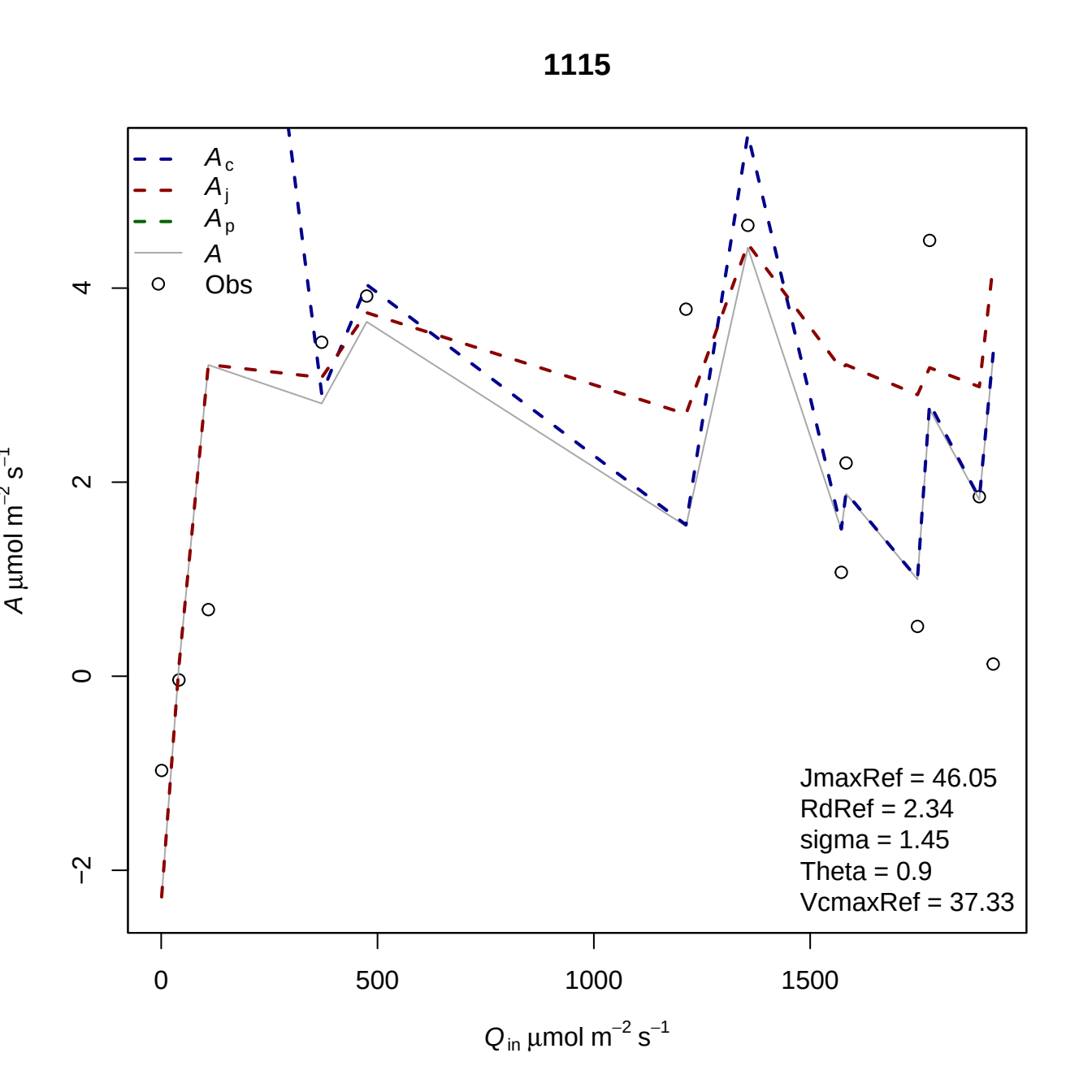

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 35.86
RdRef = 2.11
sigma = 1.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

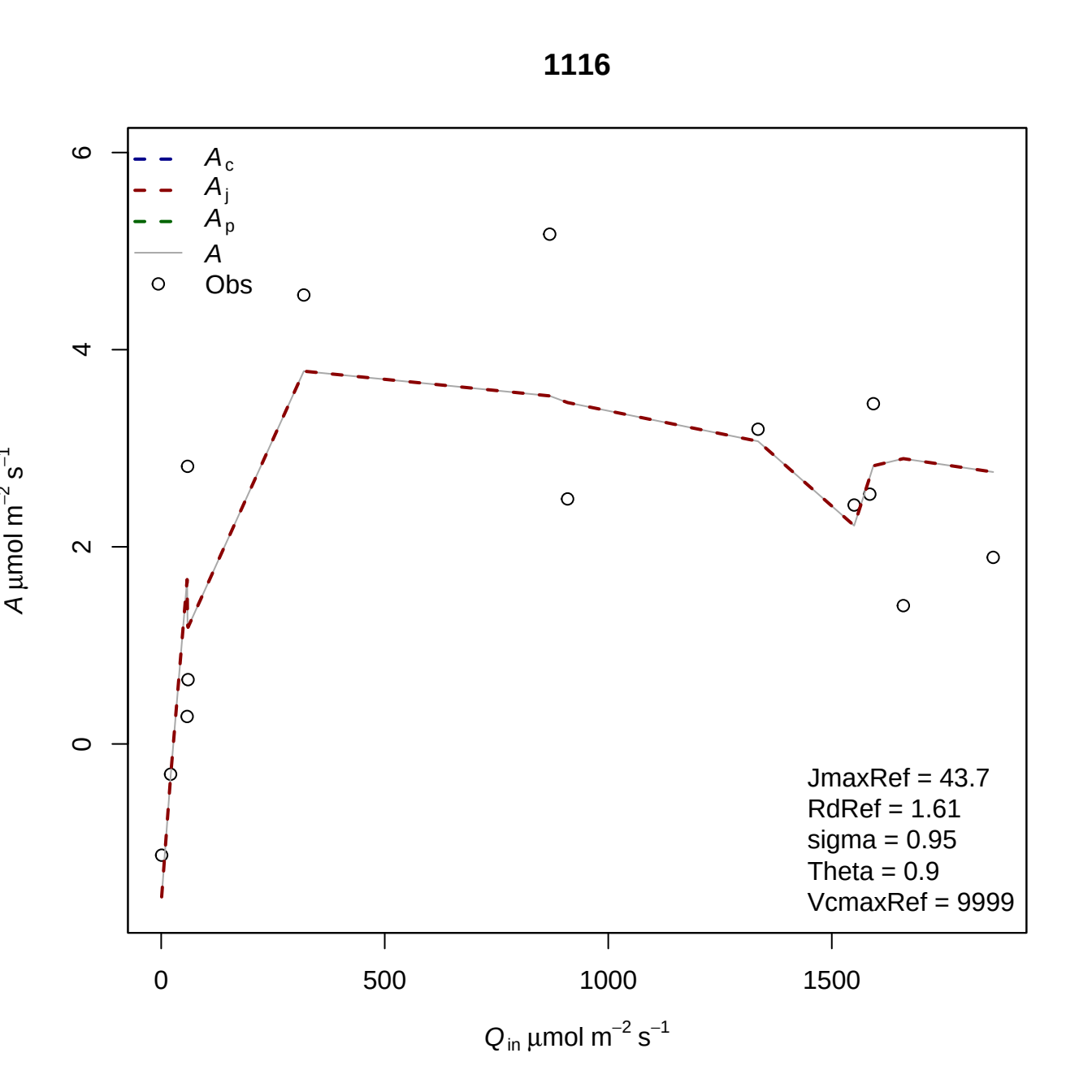

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 61.6
RdRef = 4.89
sigma = 2.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

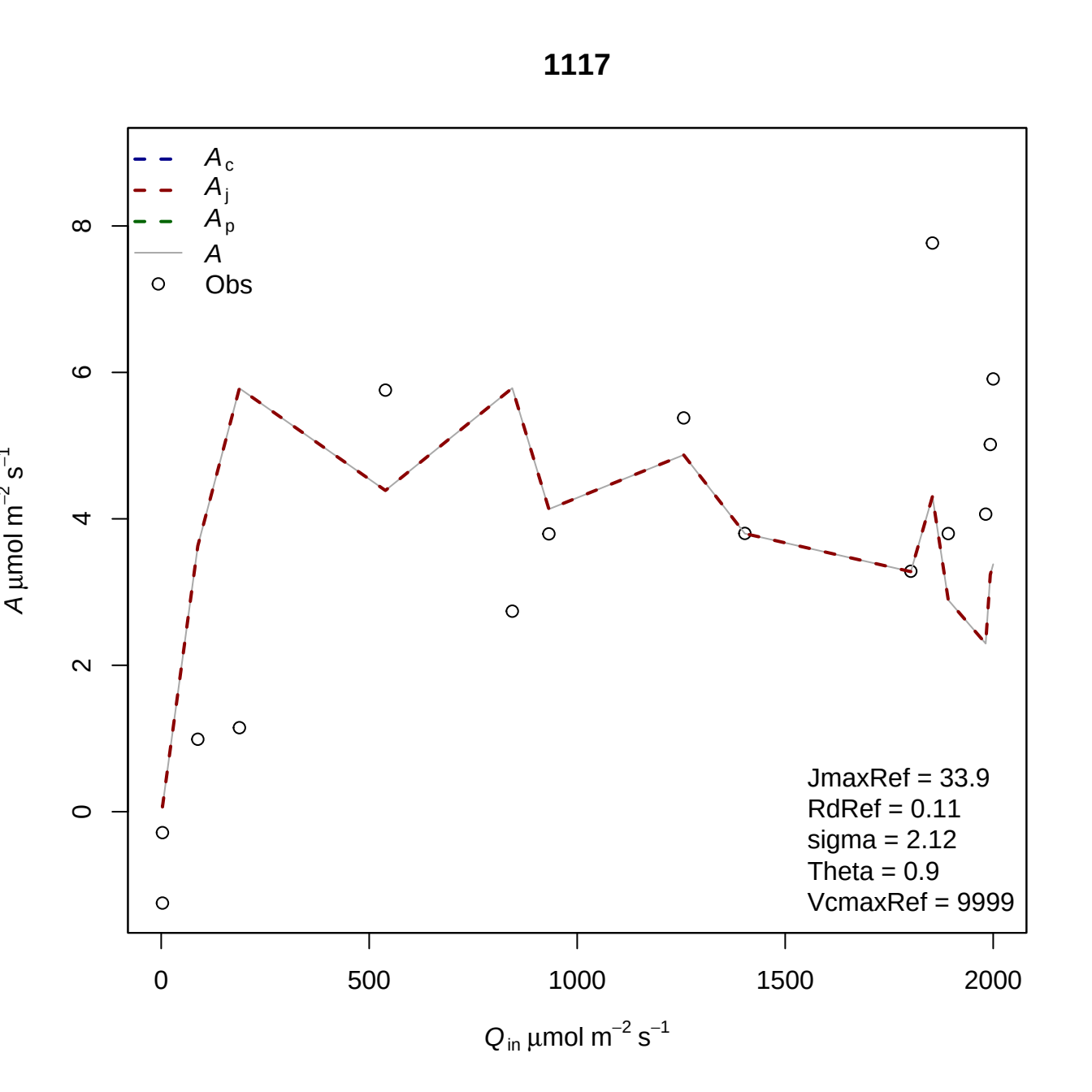

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 63.98
RdRef = 3.9
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

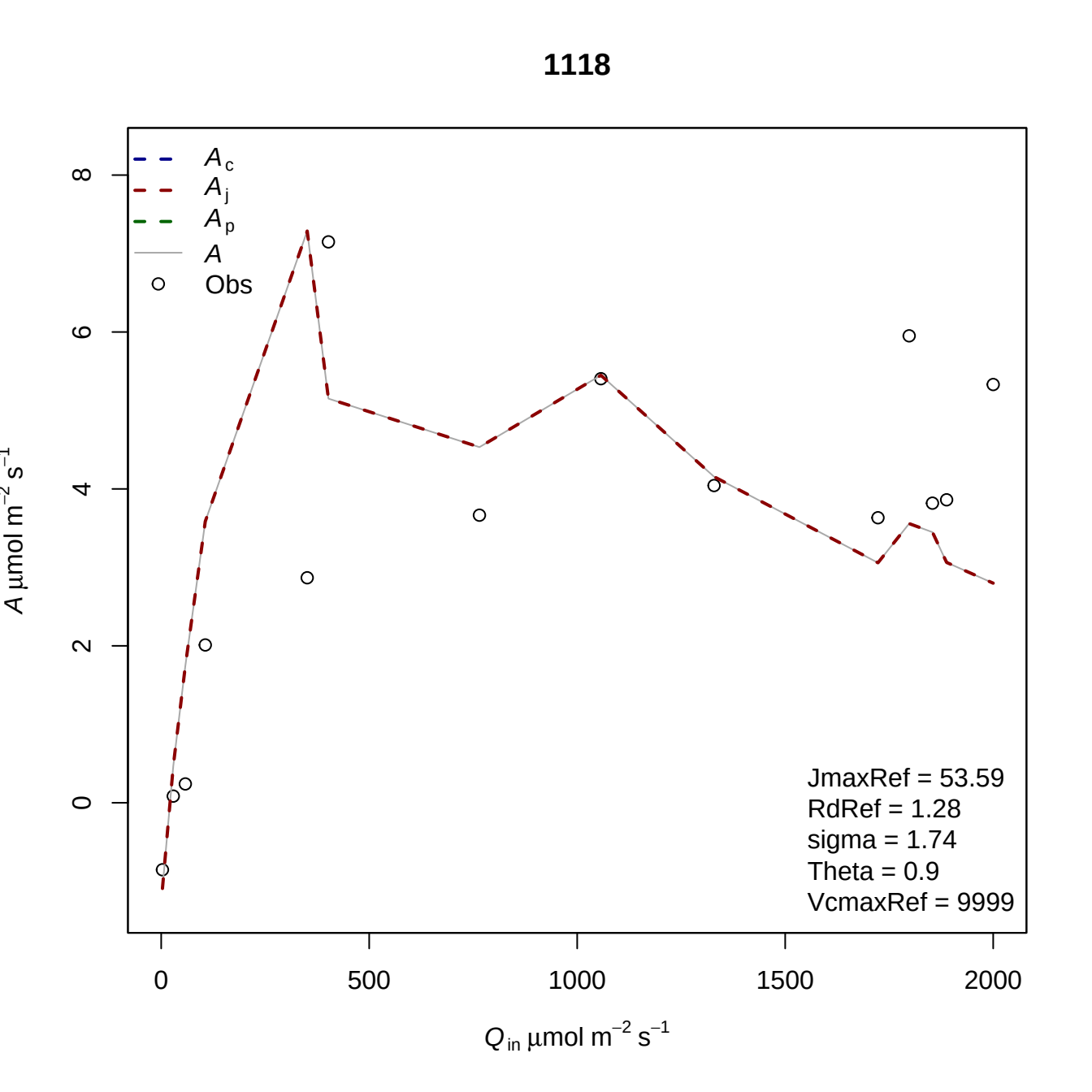

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 18.31
RdRef = 0.98
sigma = 1.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

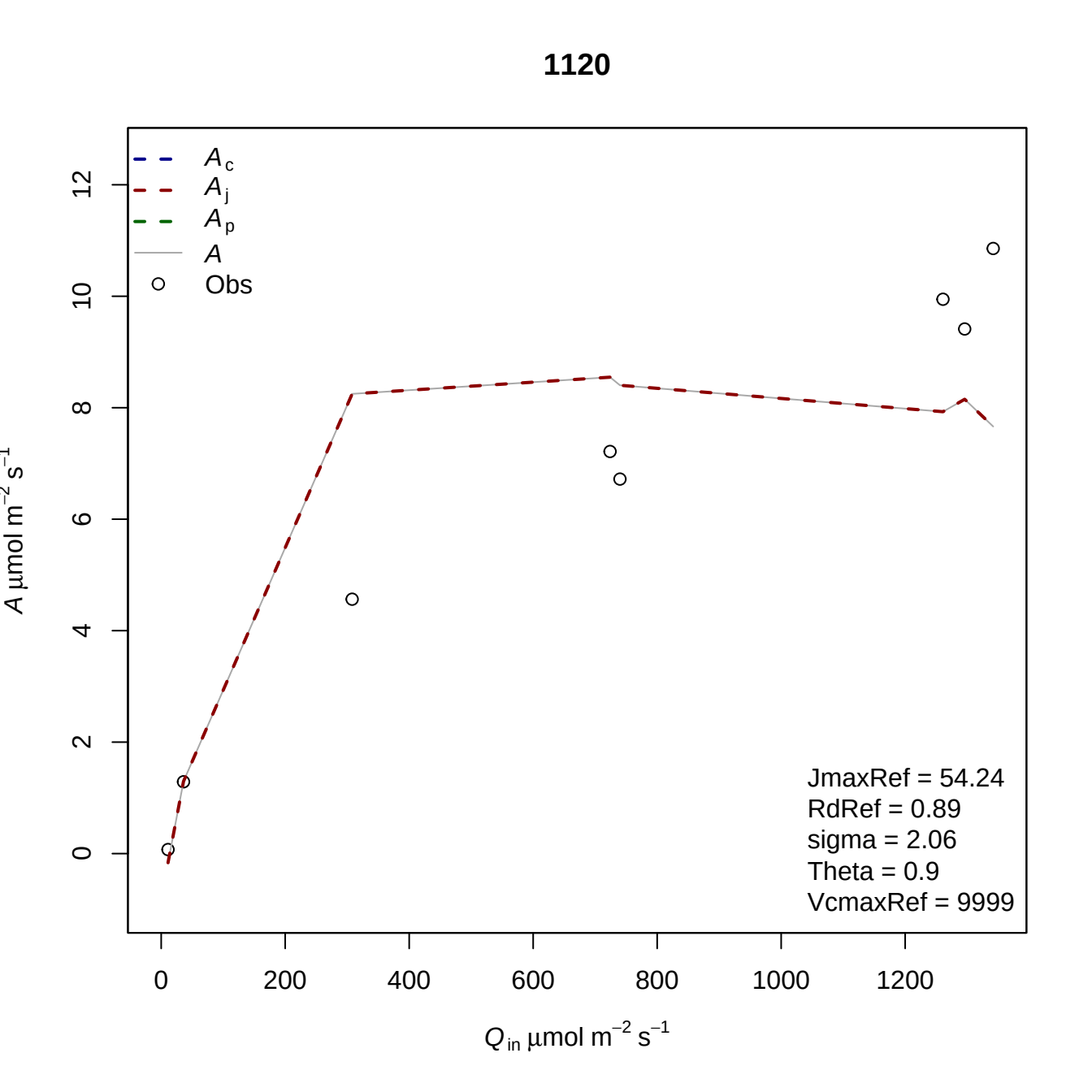

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 23.84
RdRef = 1.36
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
200
400
600
800
1000
1200

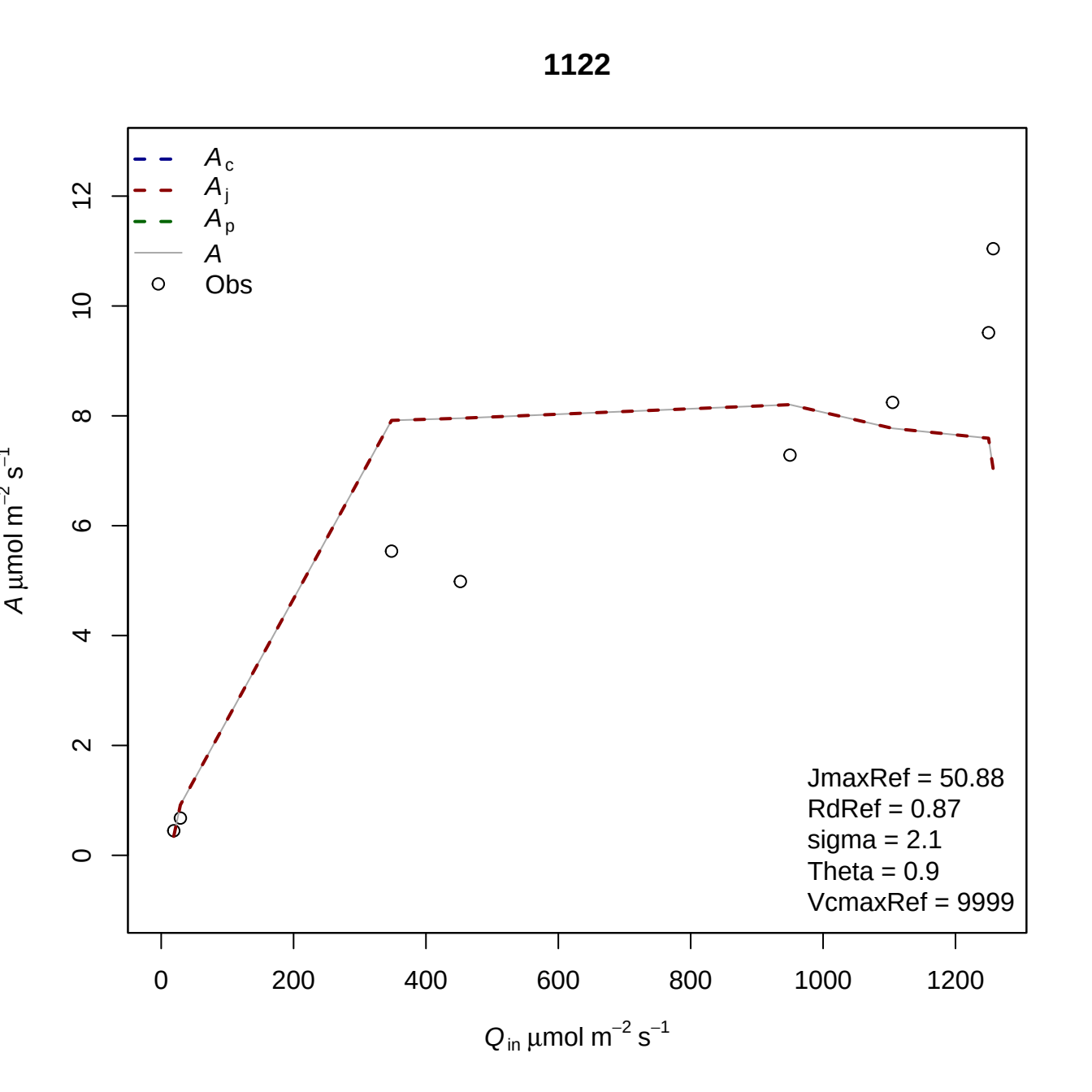

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 46.05
RdRef = 2.34
sigma = 1.45
Theta = 0.9
VcmaxRef = 37.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

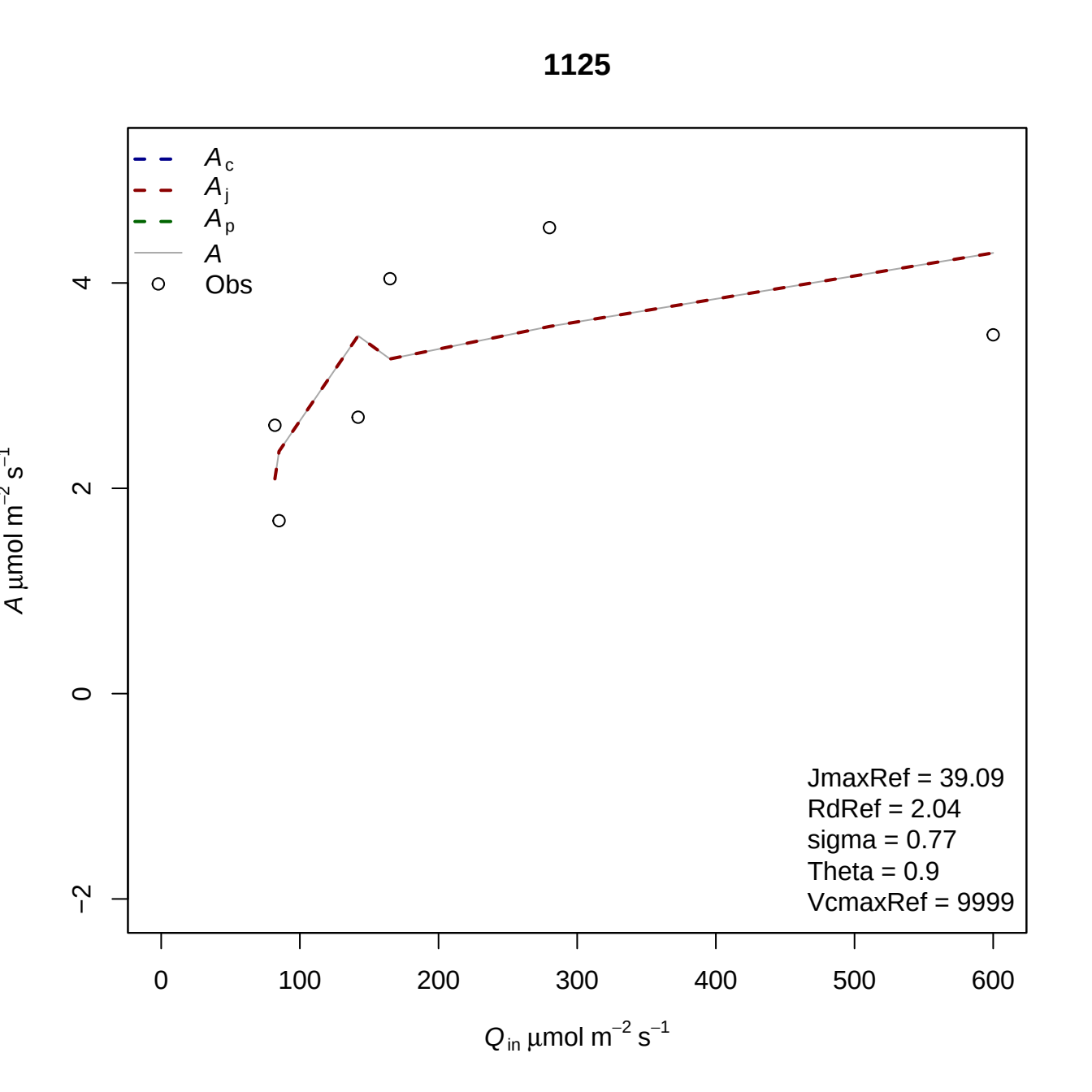

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 43.7
RdRef = 1.61
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

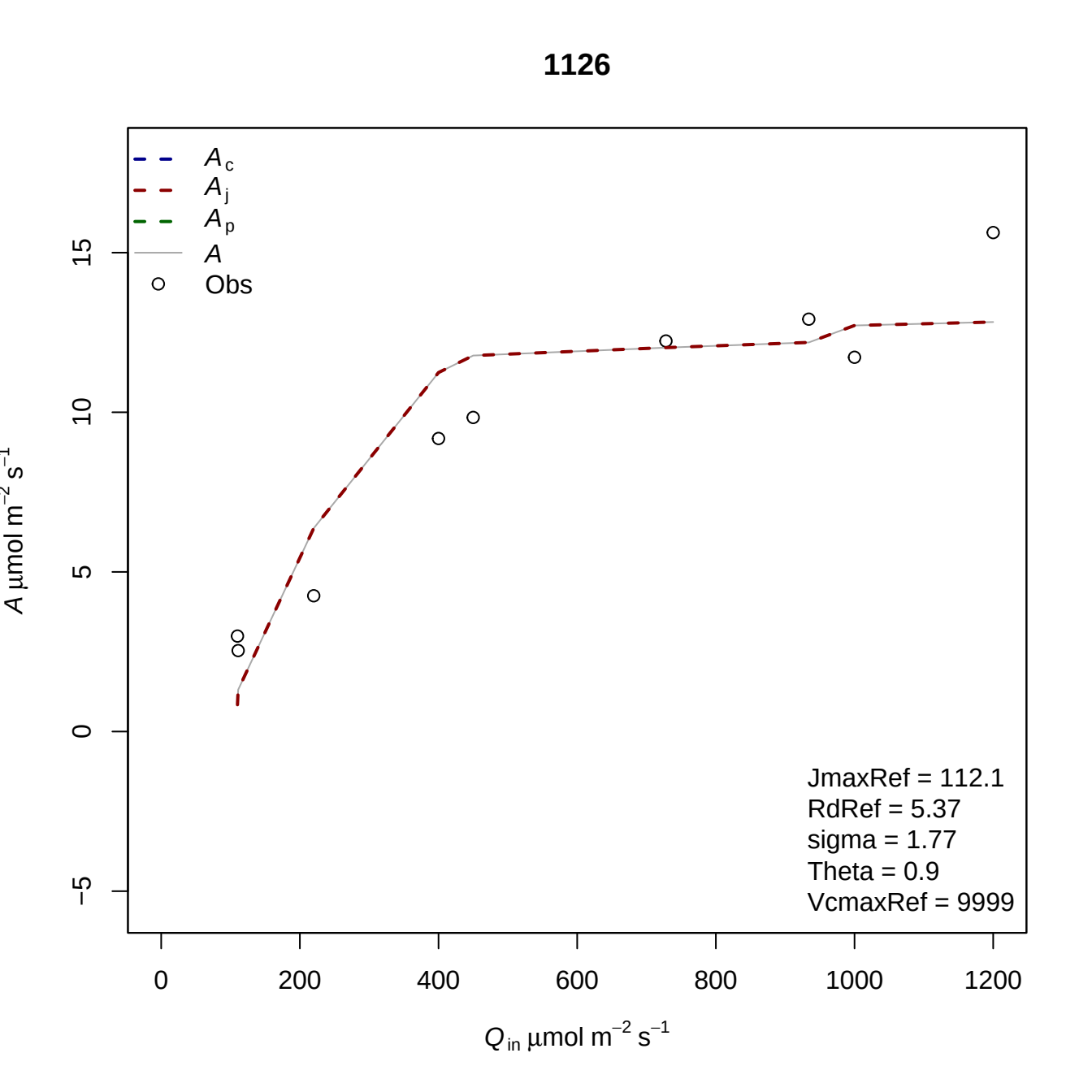

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 33.9
RdRef = 0.11
sigma = 2.12
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

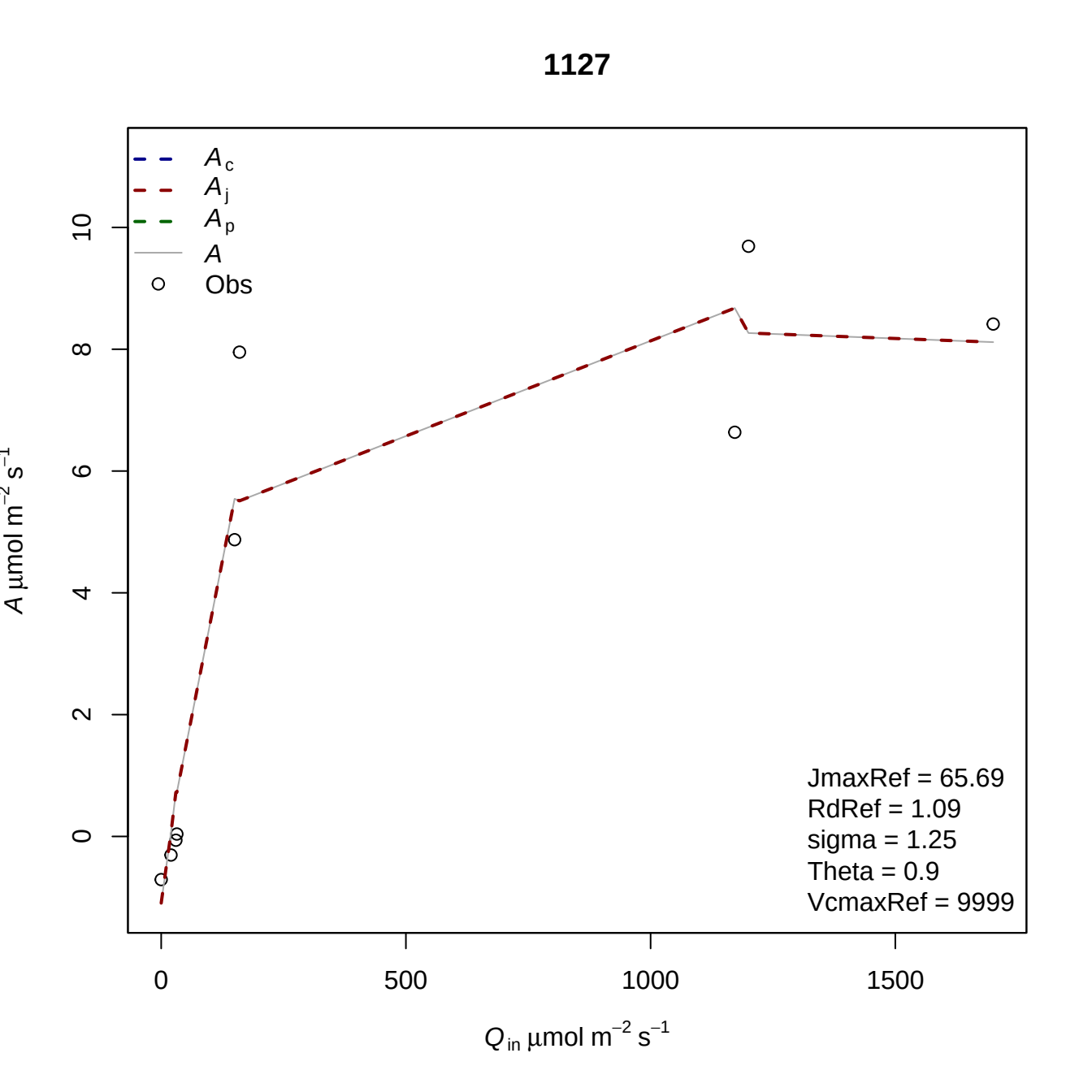

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 53.59
RdRef = 1.28
sigma = 1.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

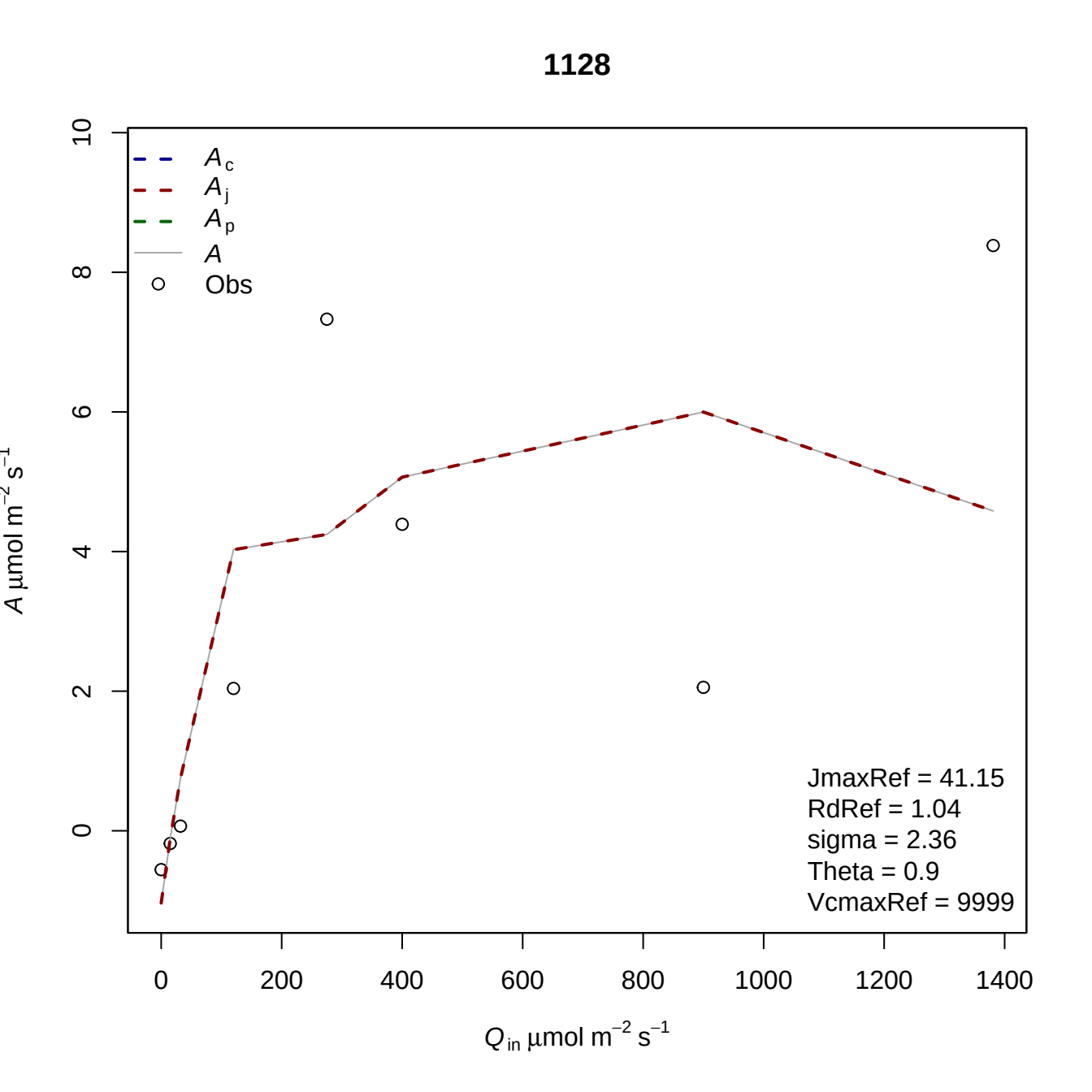

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 54.24
RdRef = 0.89
sigma = 2.06
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

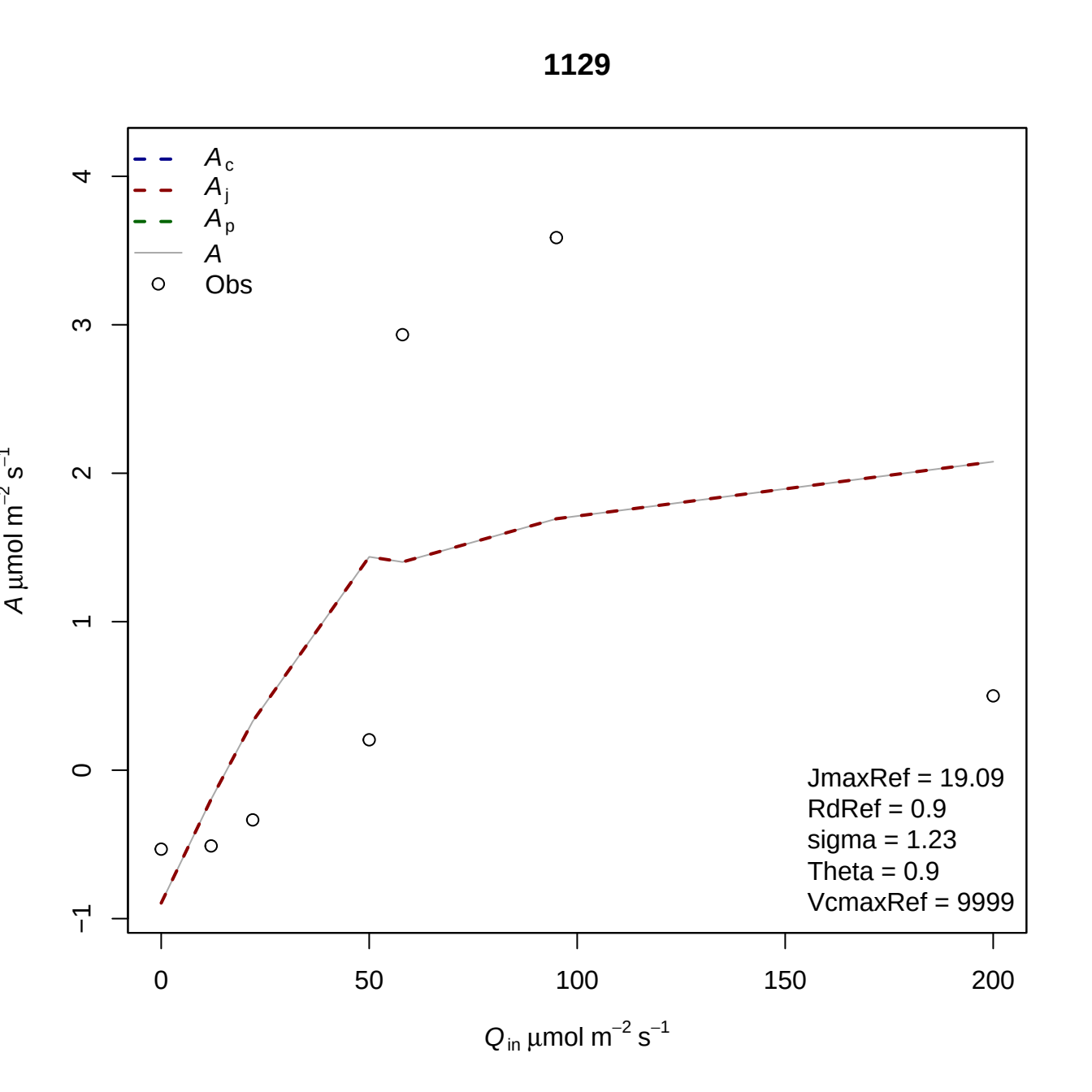

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 50.88
RdRef = 0.87
sigma = 2.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

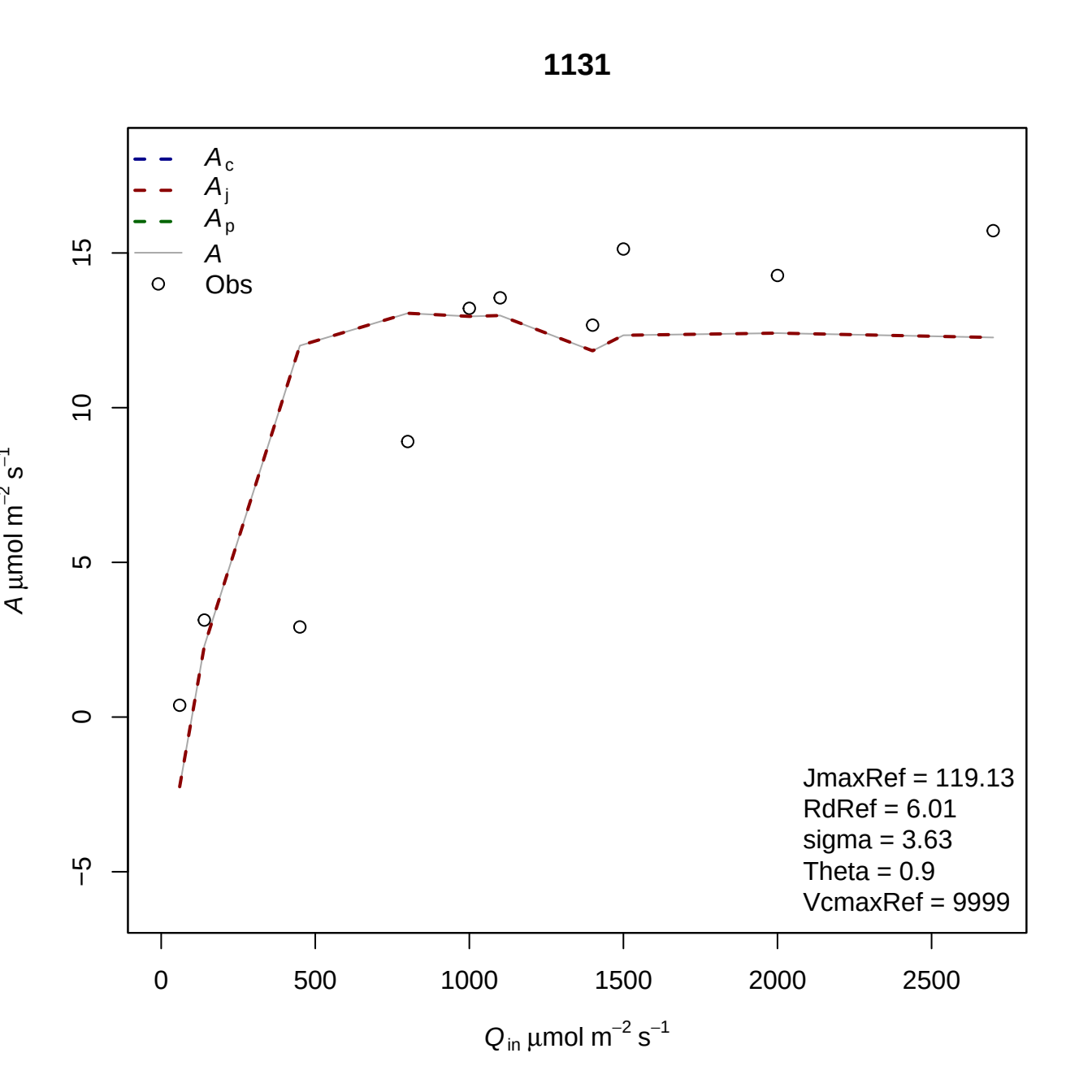

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 39.09
RdRef = 2.04
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
100
200
300
400
500
600

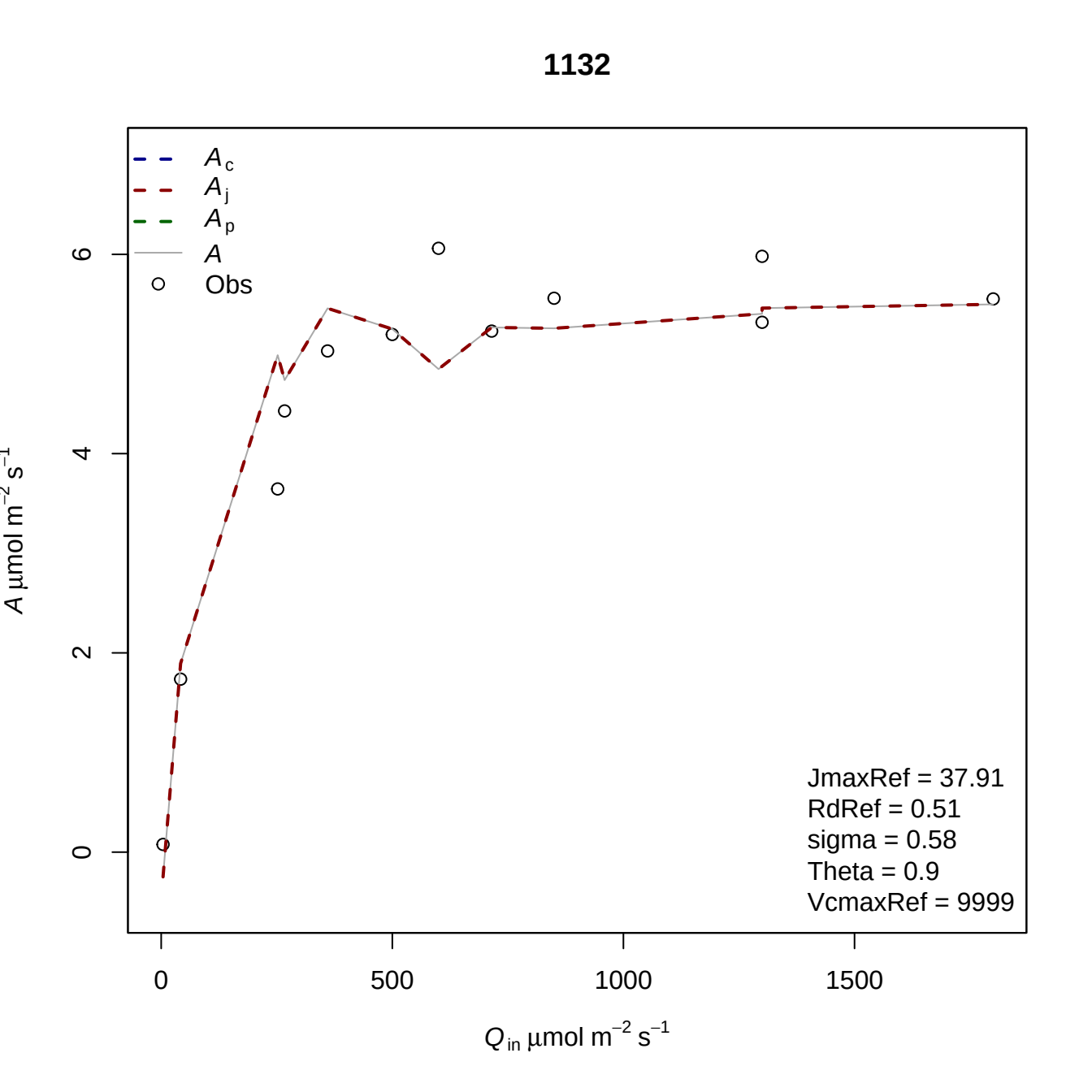

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 112.1
RdRef = 5.37
sigma = 1.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
200
400
600
800
1000
1200

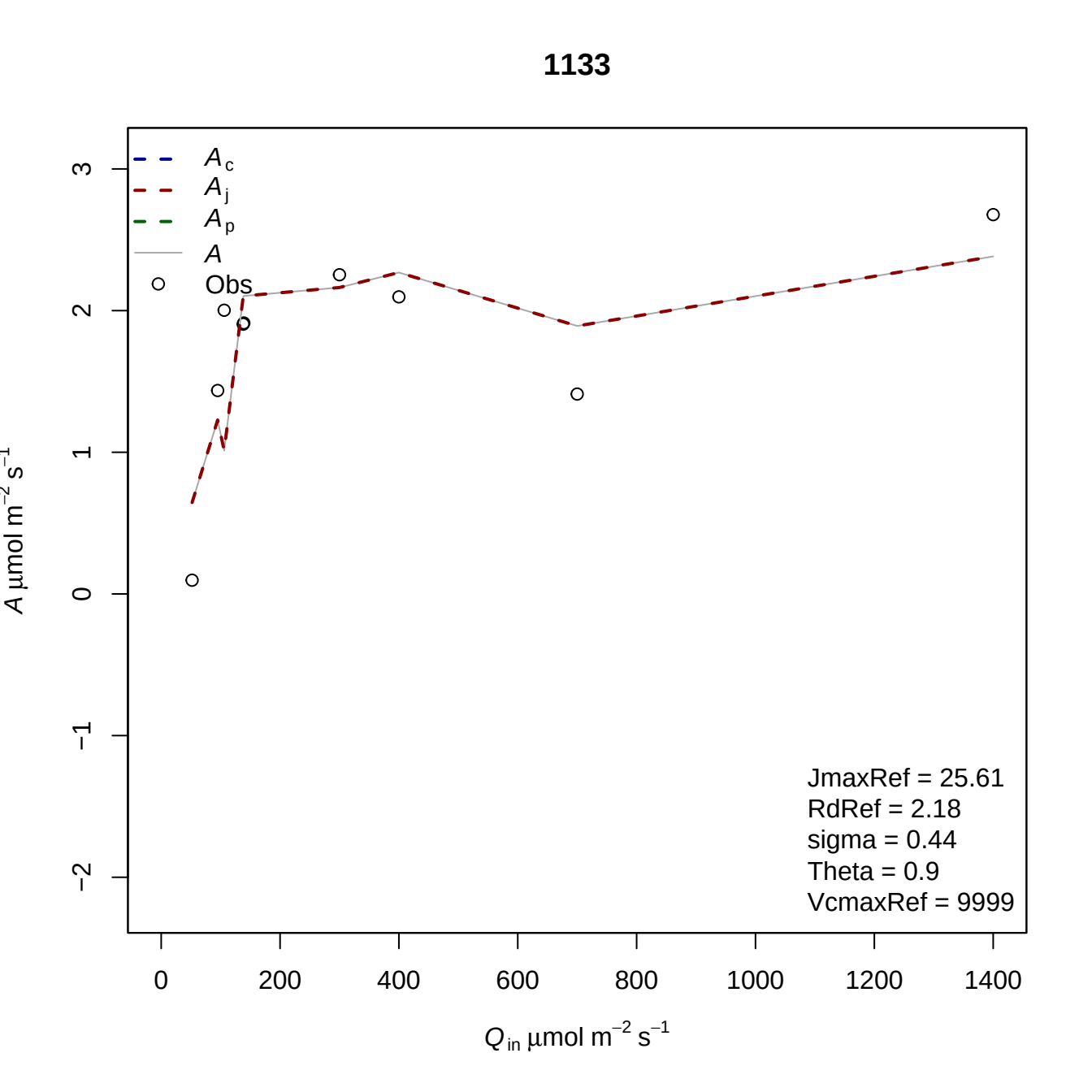

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 65.69
RdRef = 1.09
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

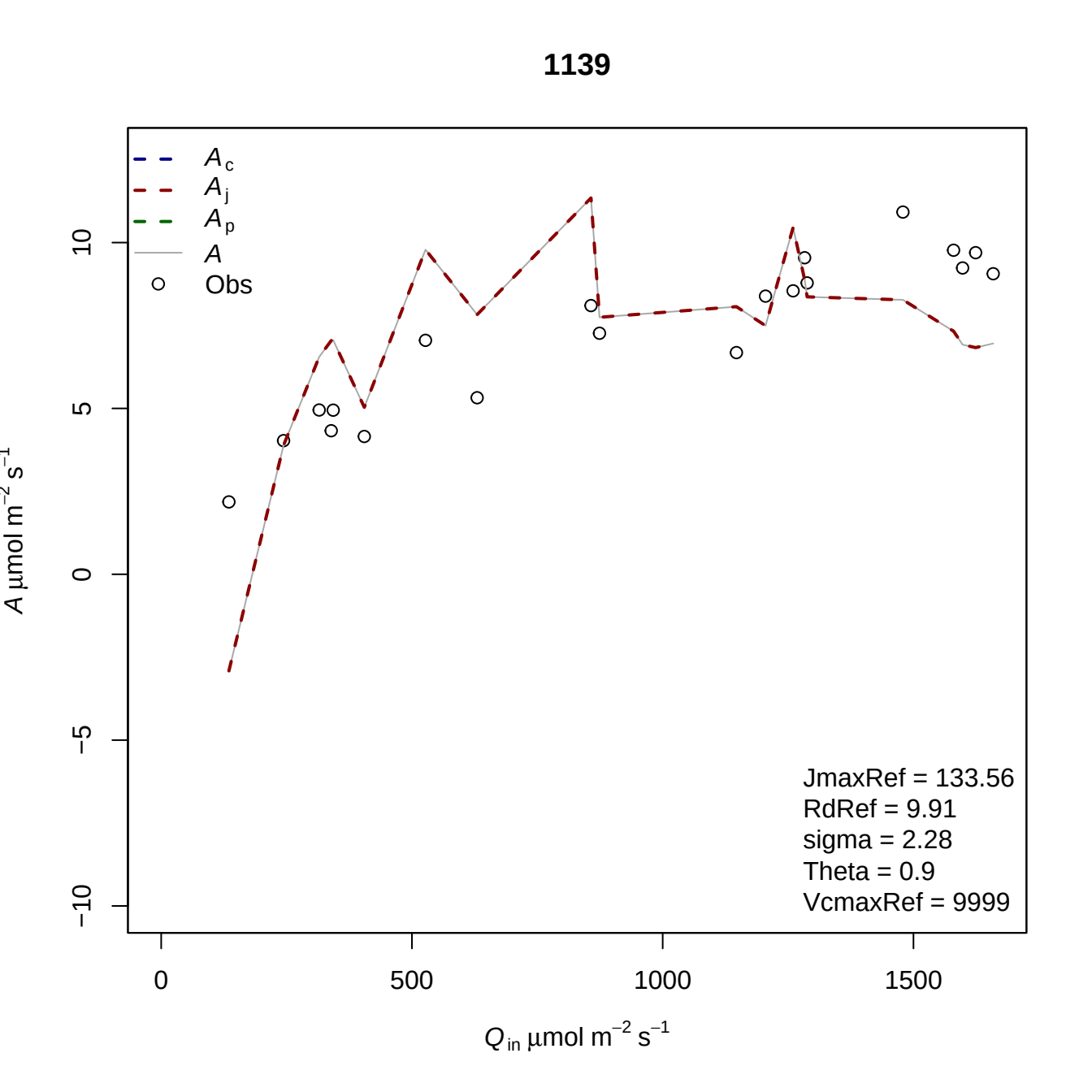

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 41.15
RdRef = 1.04
sigma = 2.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

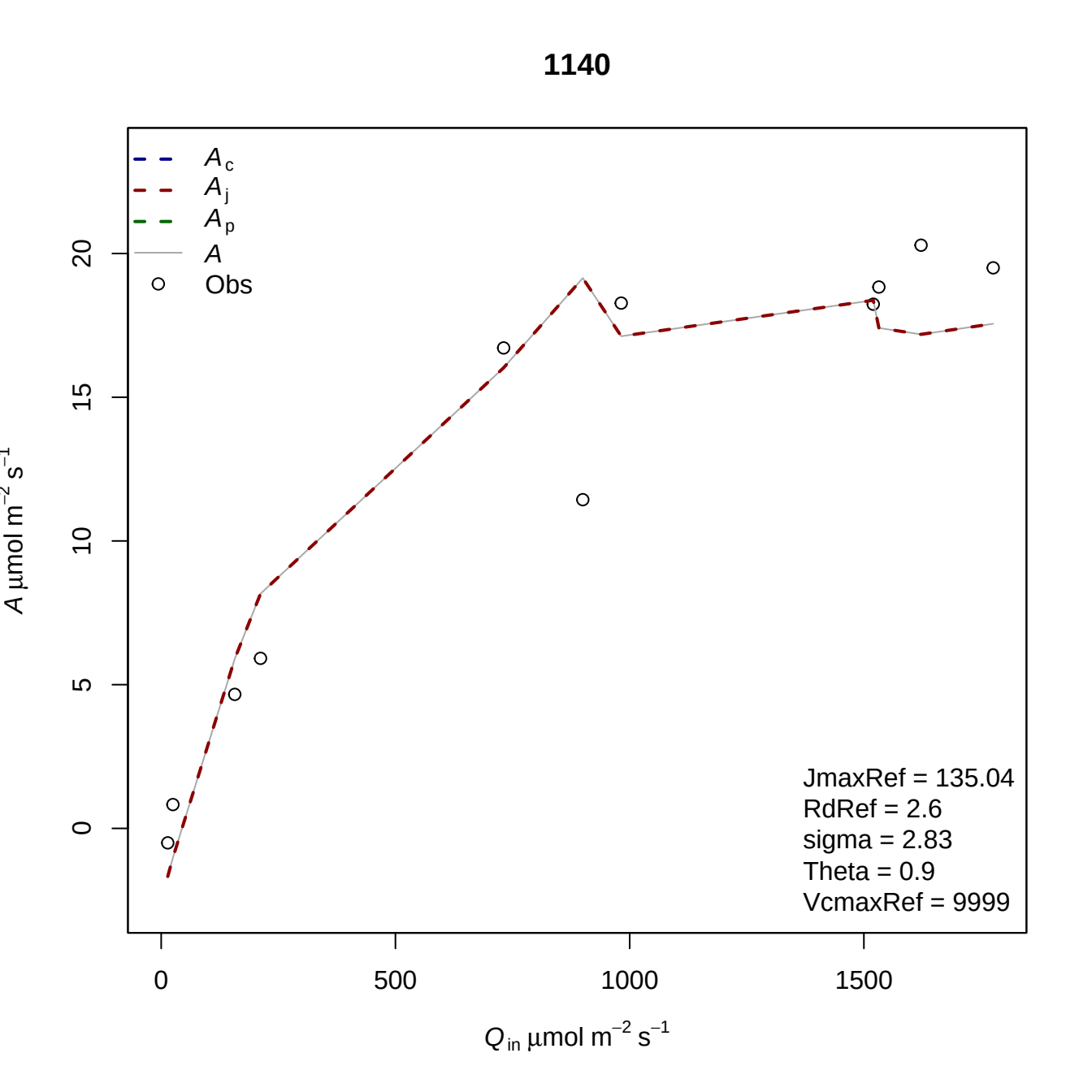

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 19.09
RdRef = 0.9
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

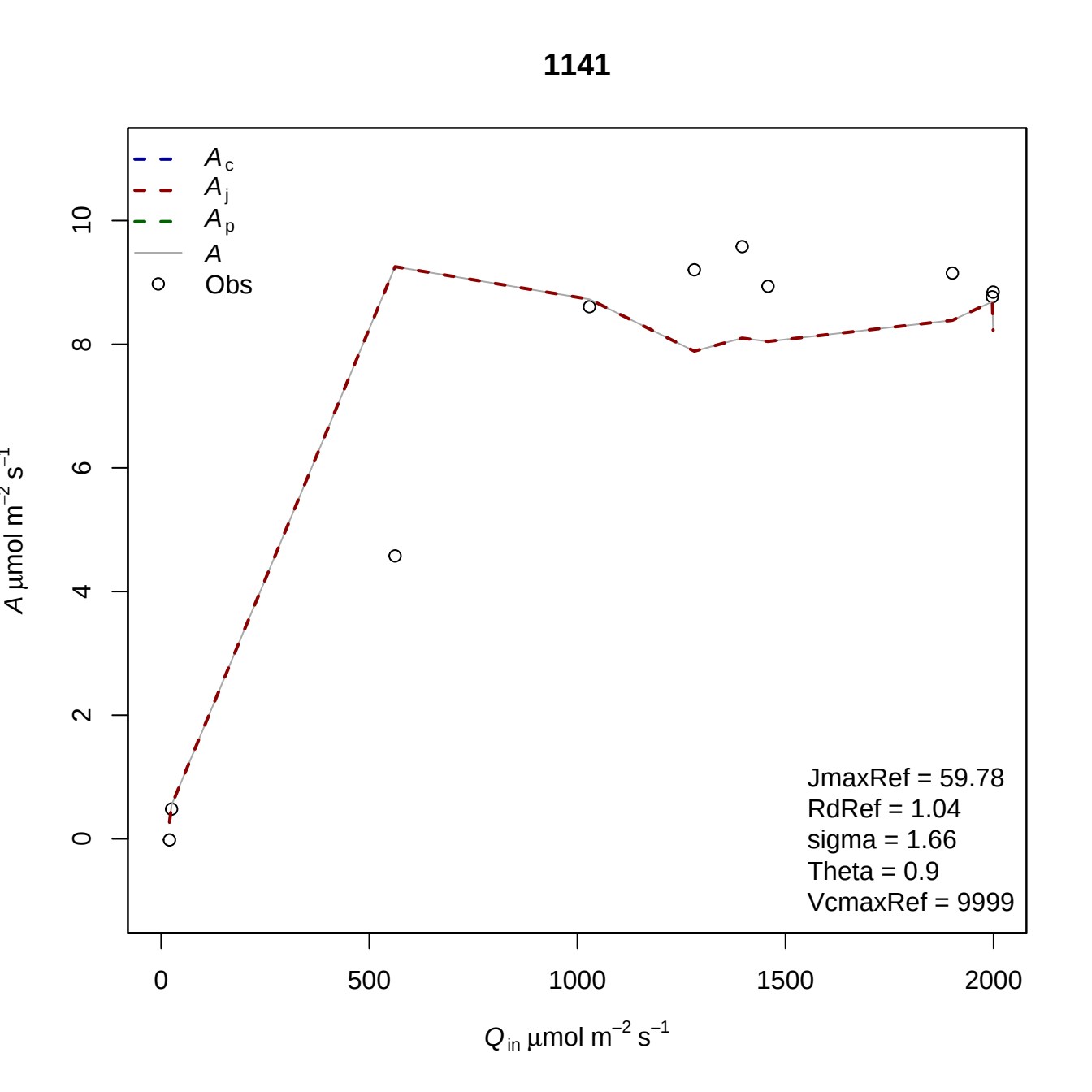

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 119.13
RdRef = 6.01
sigma = 3.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000
2500

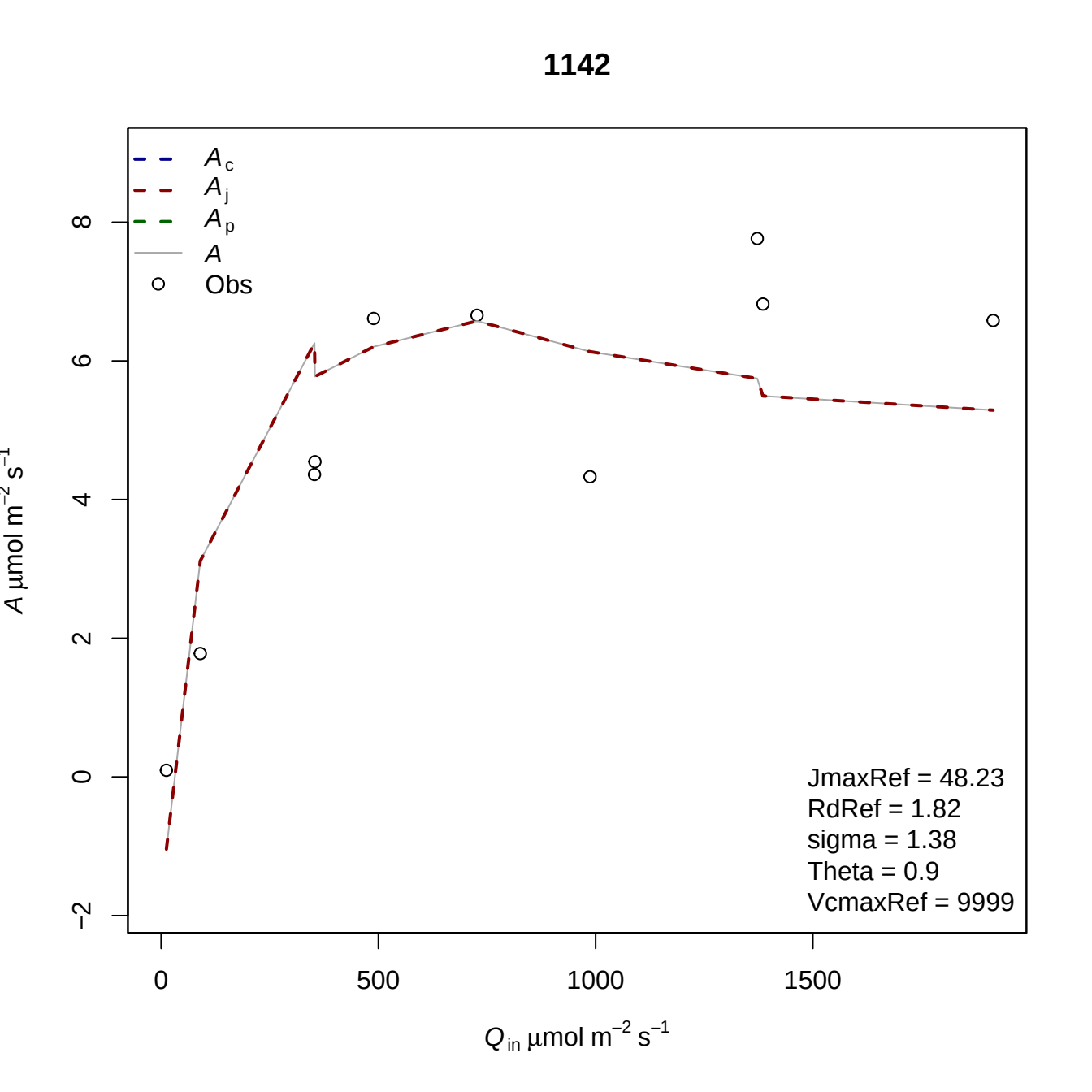

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 37.91
RdRef = 0.51
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

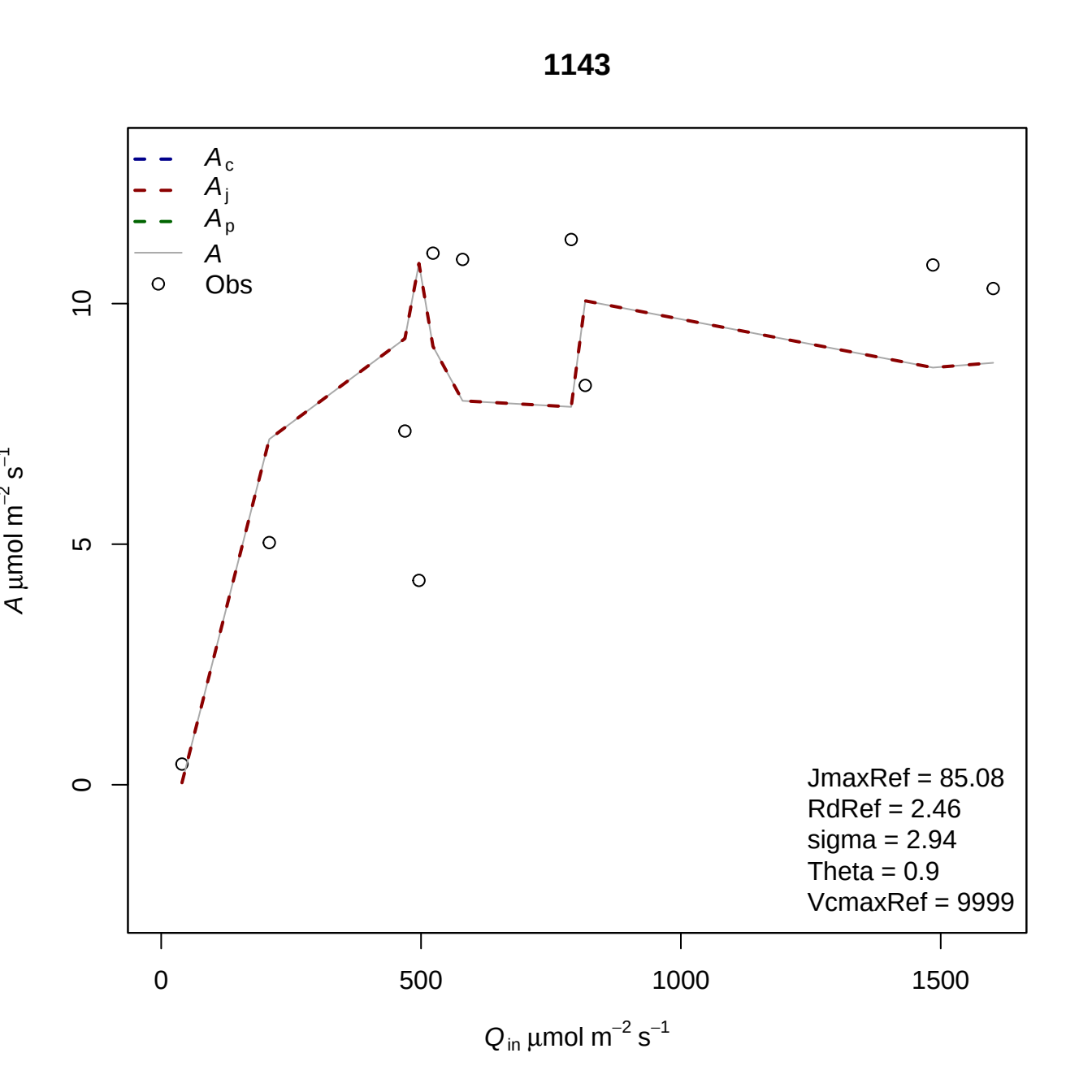

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 25.61
RdRef = 2.18
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

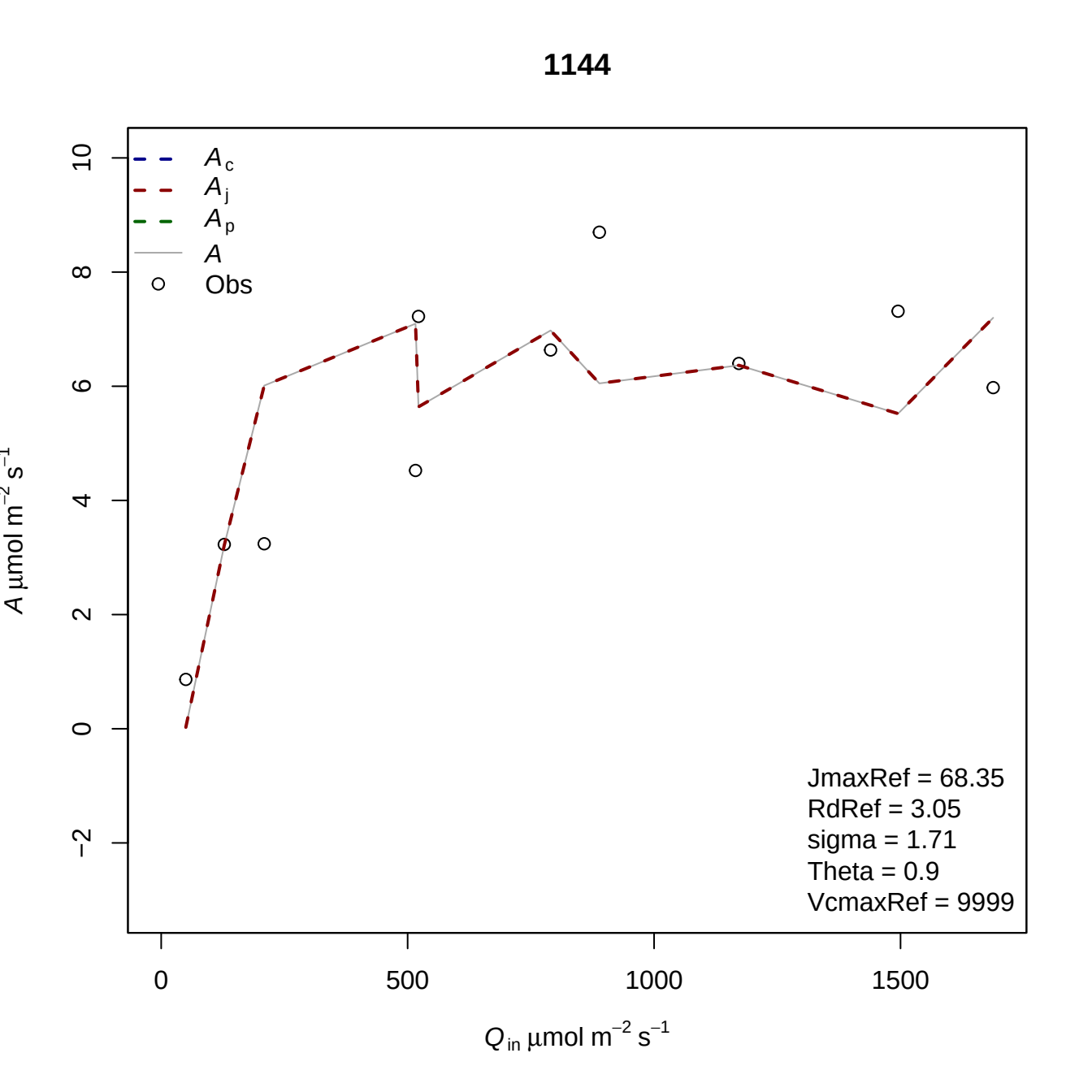

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 133.56
RdRef = 9.91
sigma = 2.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
10
5
0
−5
−10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

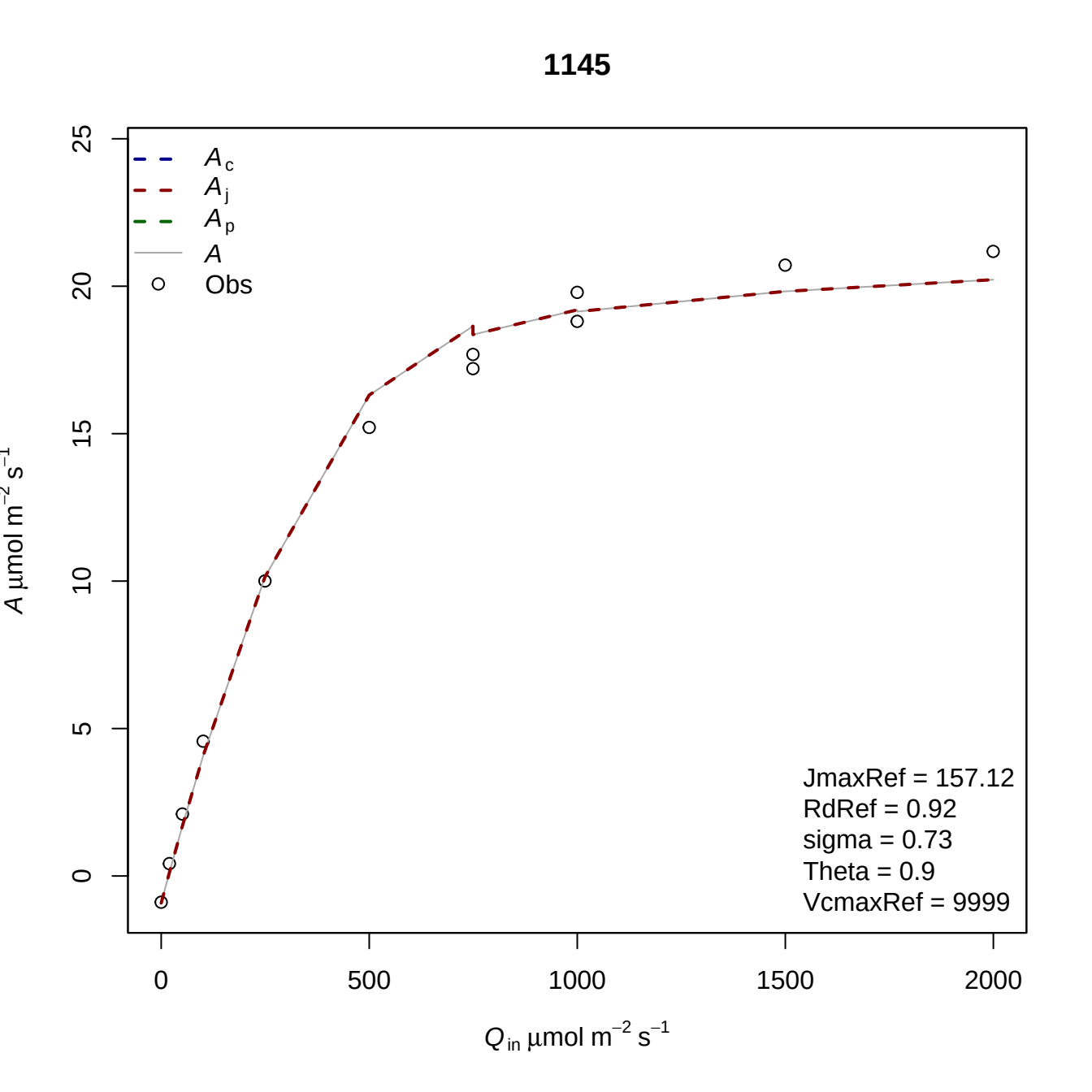

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 135.04
RdRef = 2.6
sigma = 2.83
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

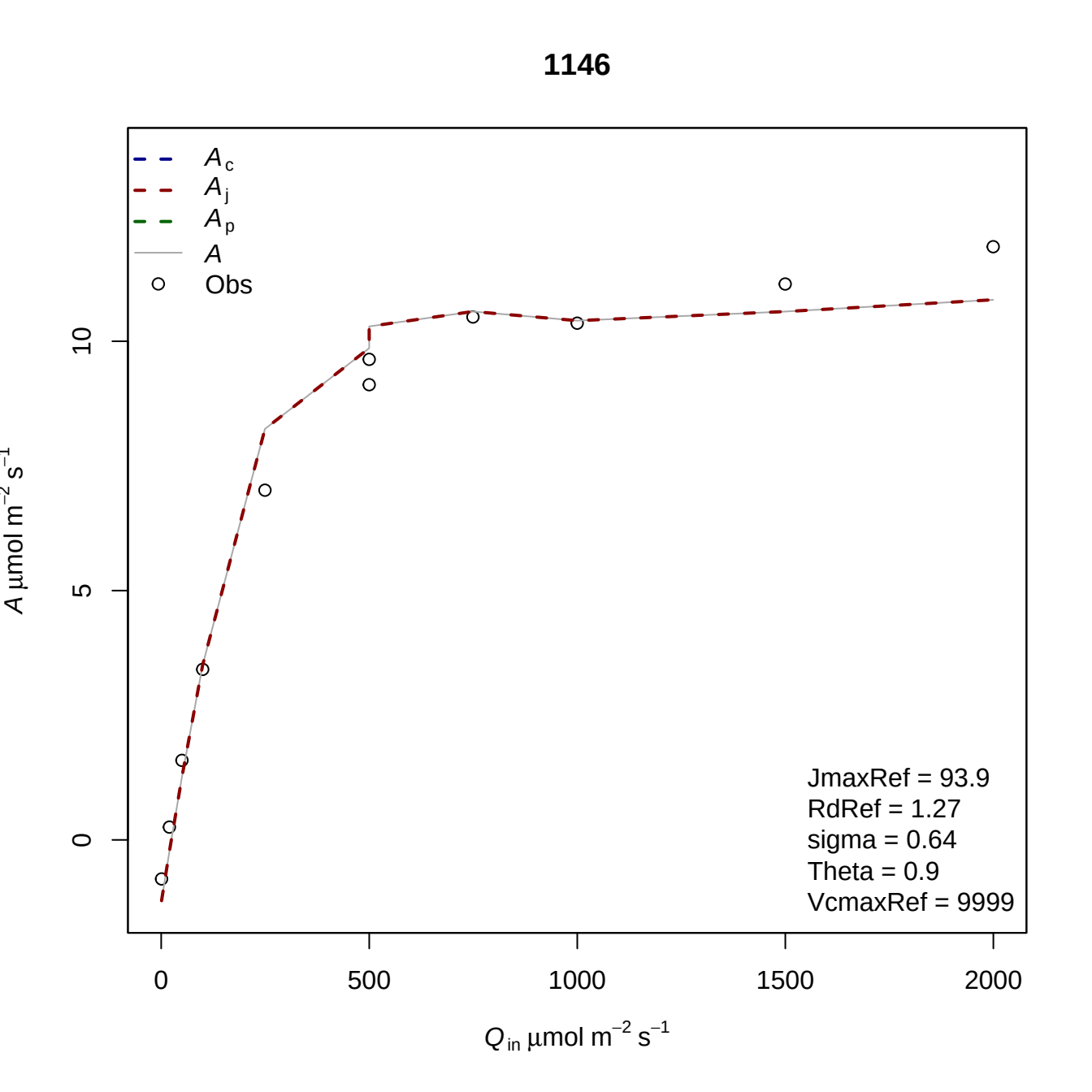

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 59.78
RdRef = 1.04
sigma = 1.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

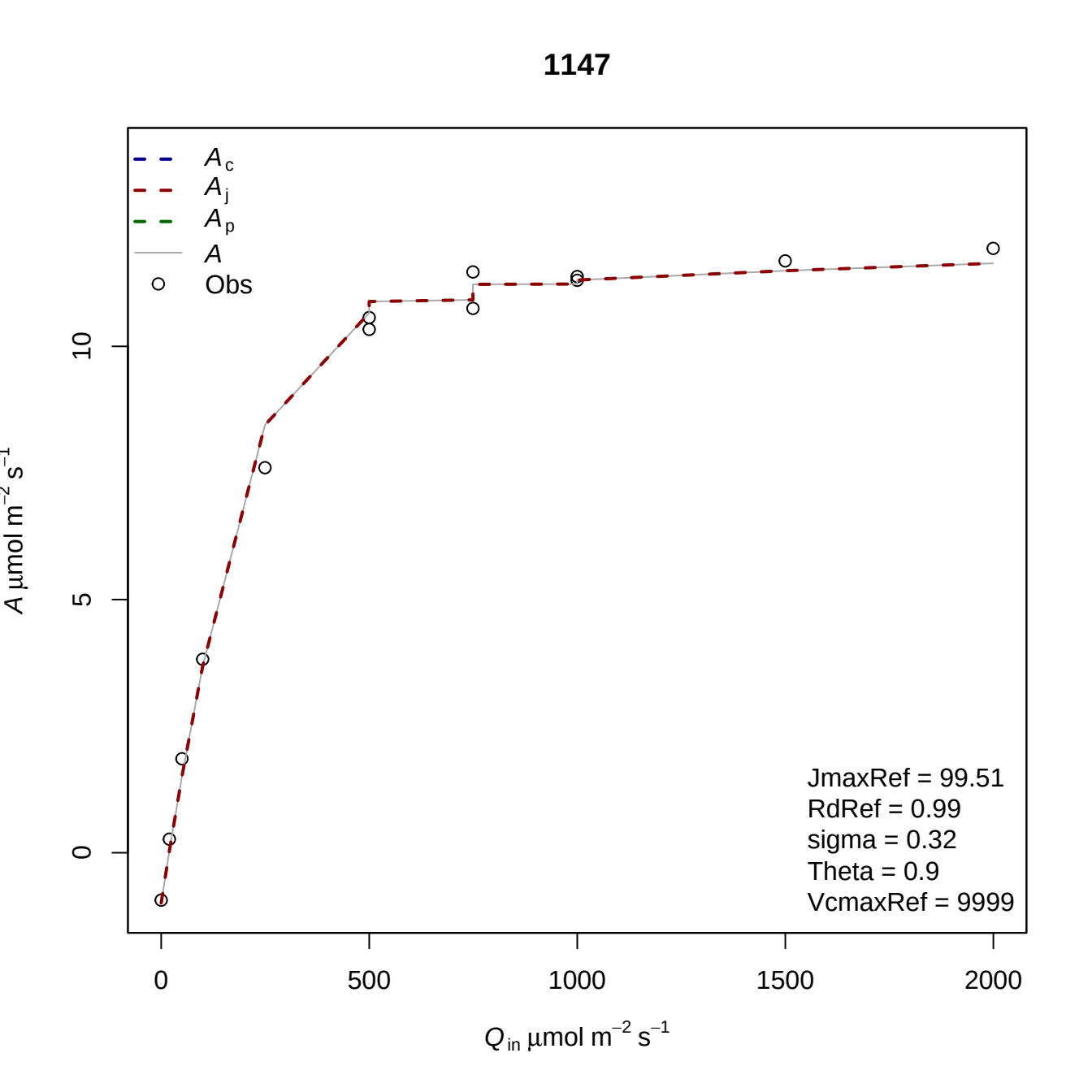

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 48.23
RdRef = 1.82
sigma = 1.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

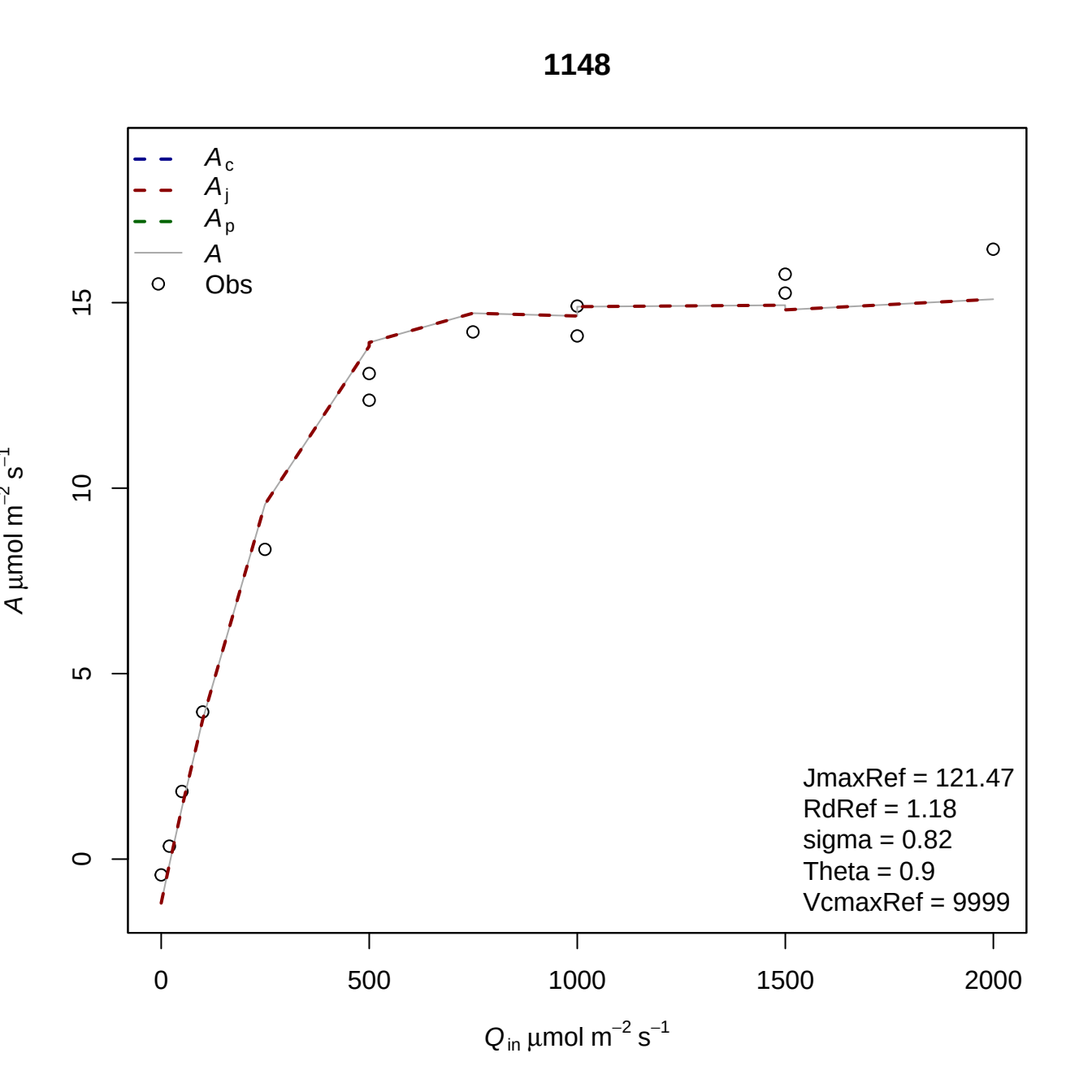

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 85.08
RdRef = 2.46
sigma = 2.94
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

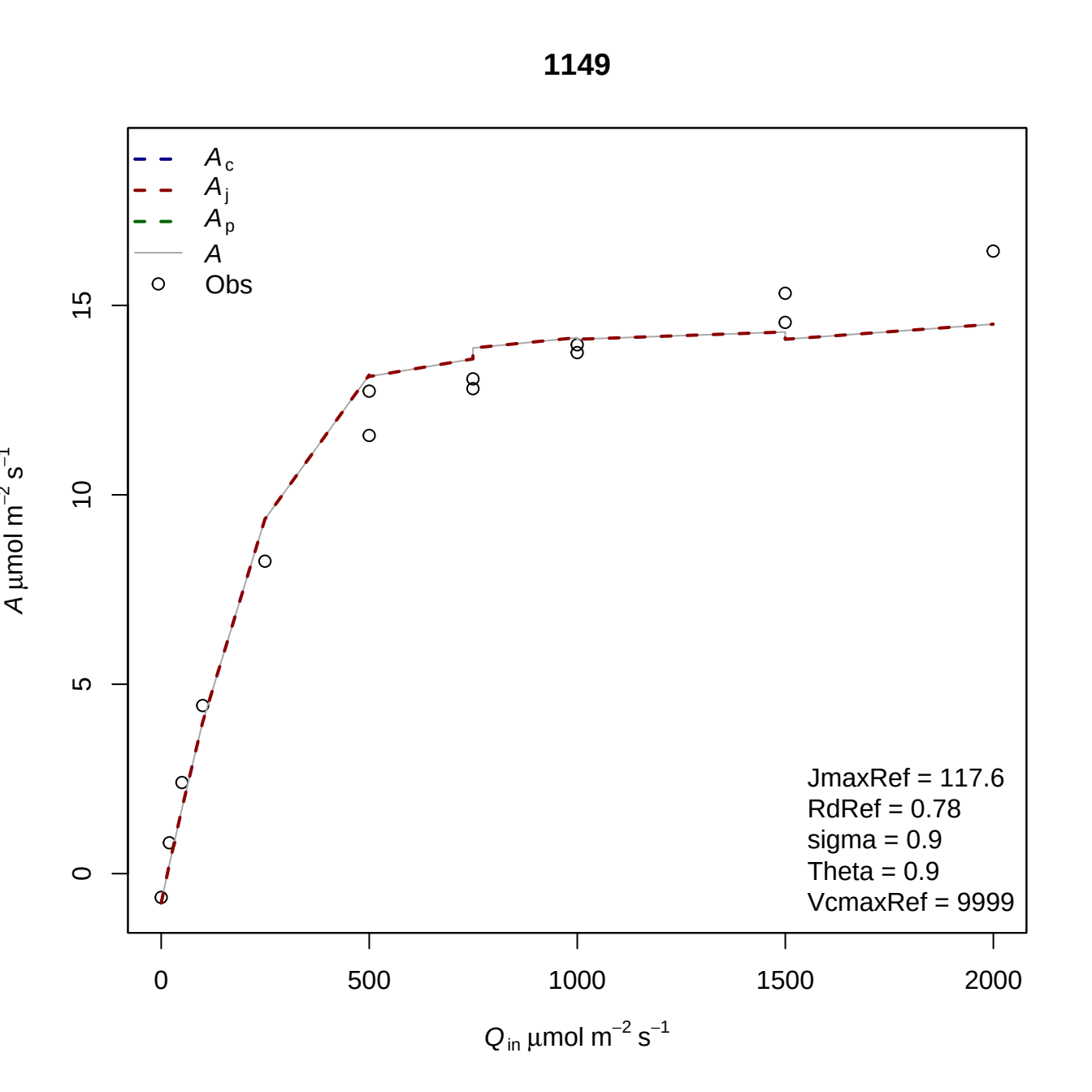

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 68.35
RdRef = 3.05
sigma = 1.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

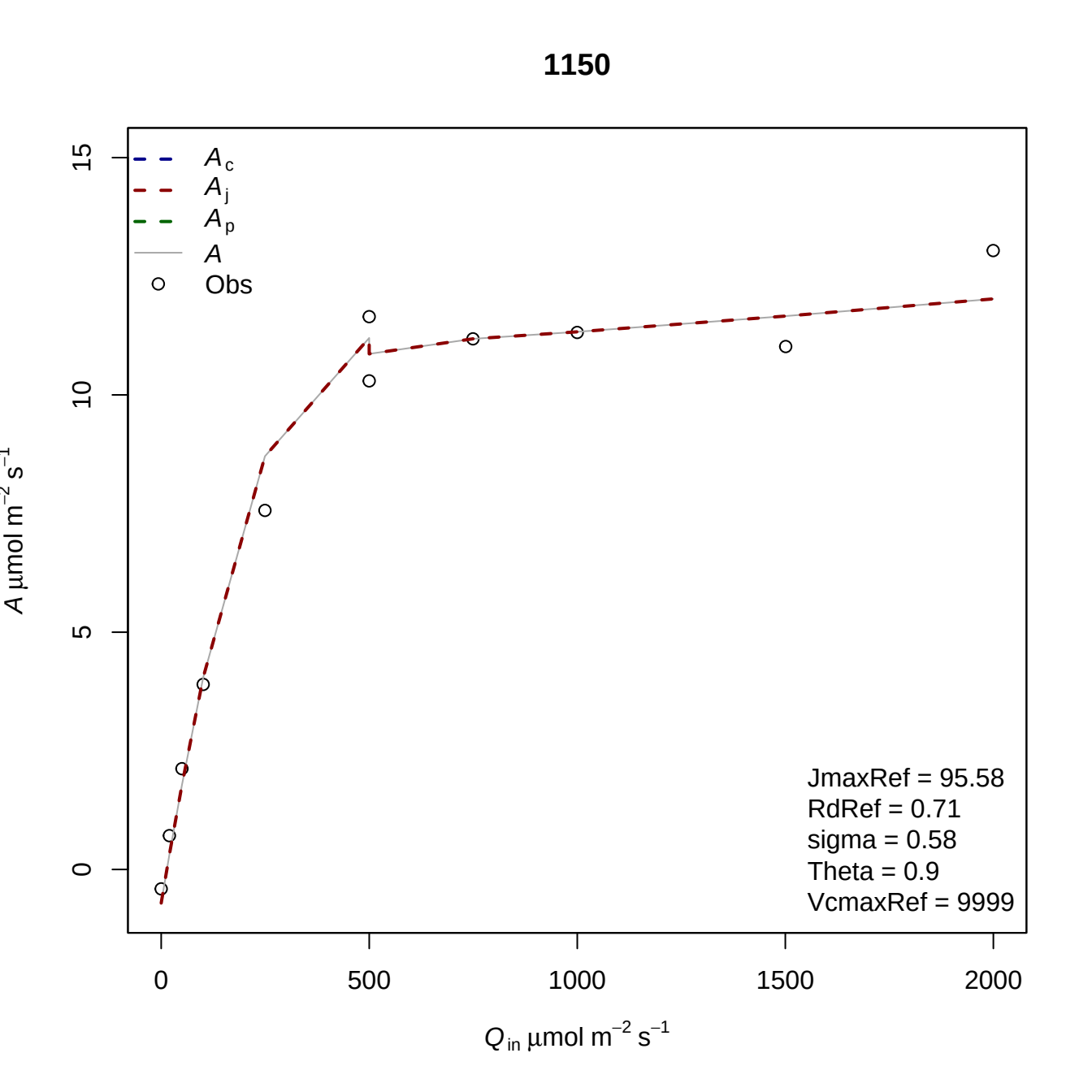

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 157.12
RdRef = 0.92
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

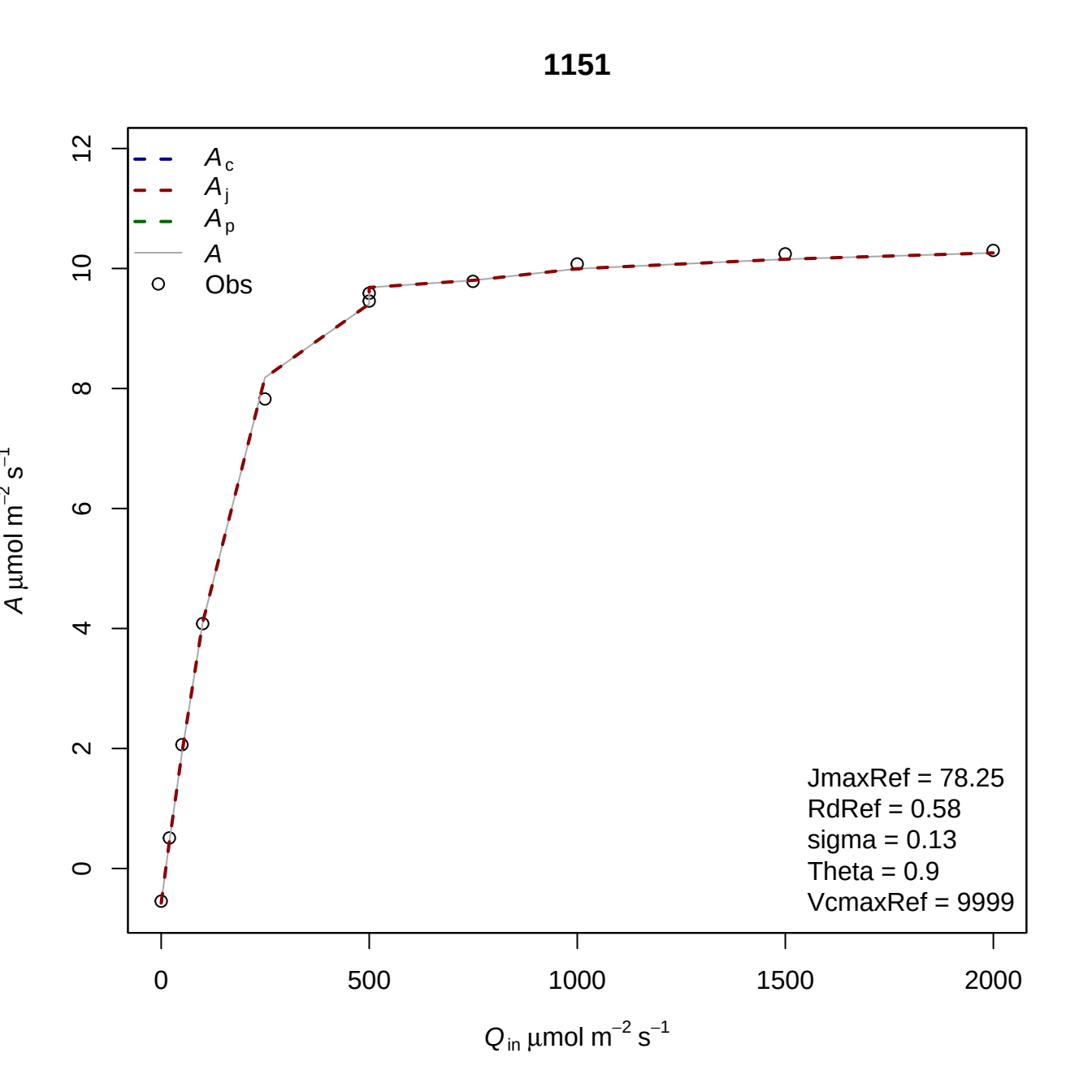

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 93.9
RdRef = 1.27
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

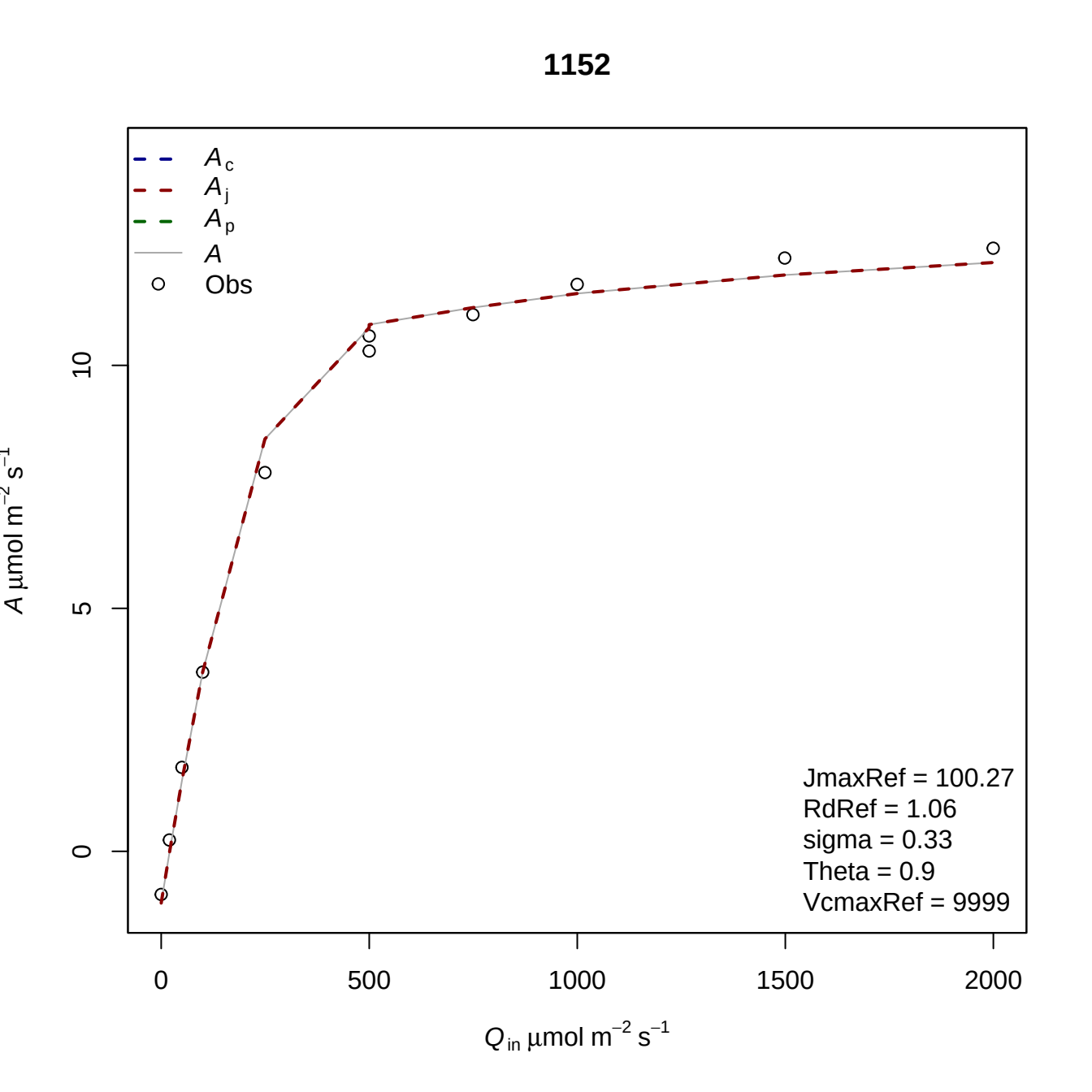

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 99.51
RdRef = 0.99
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

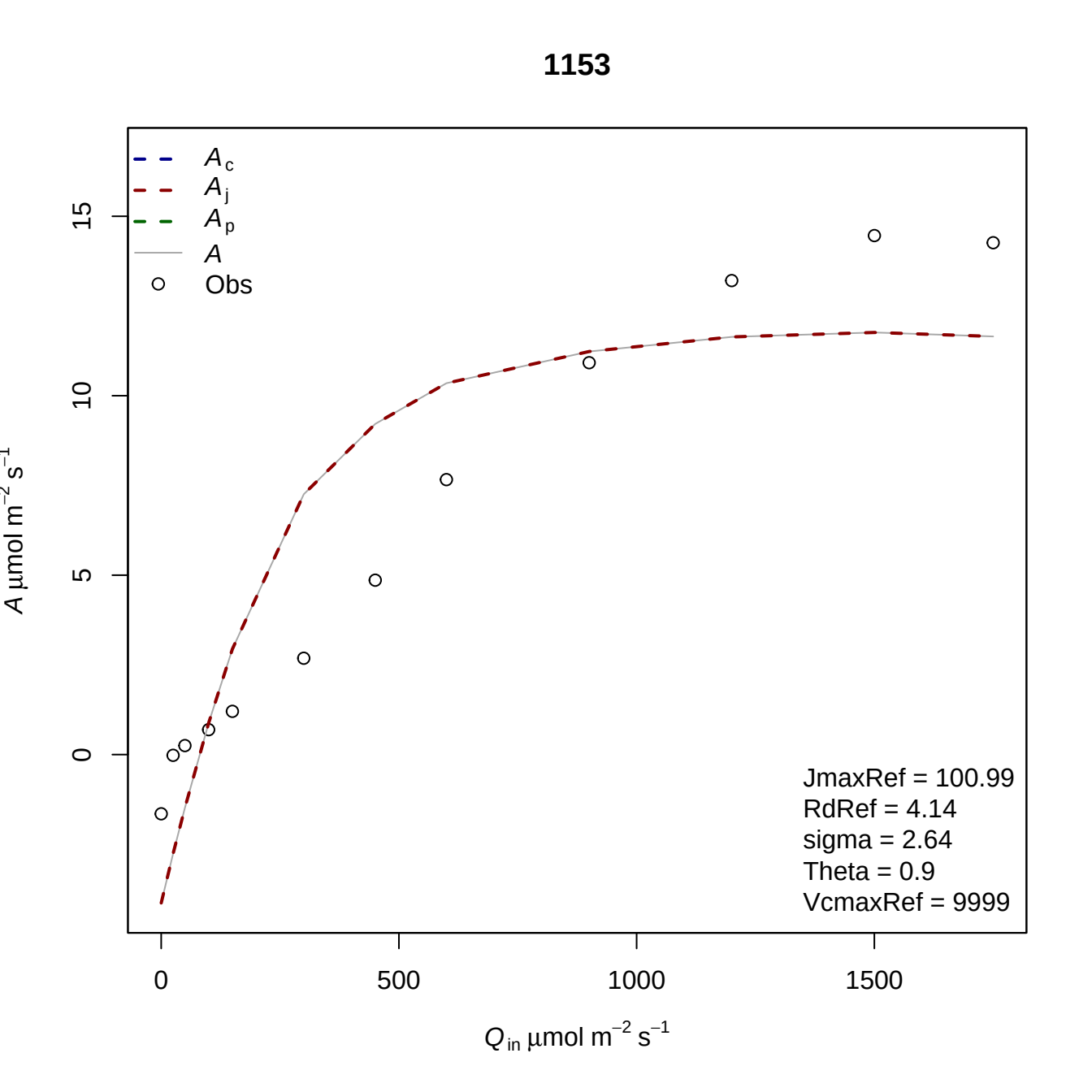

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 121.47
RdRef = 1.18
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

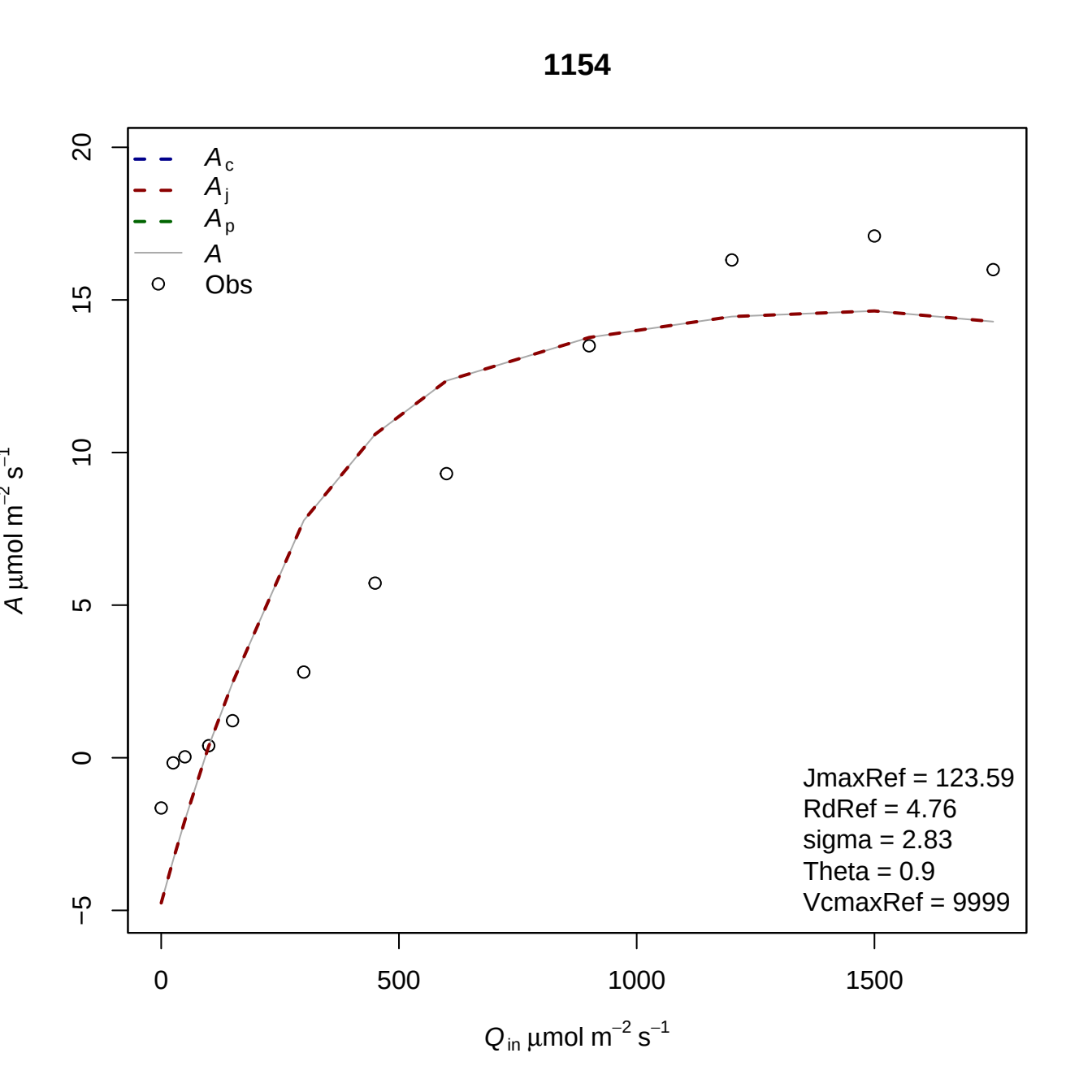

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 117.6
RdRef = 0.78
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

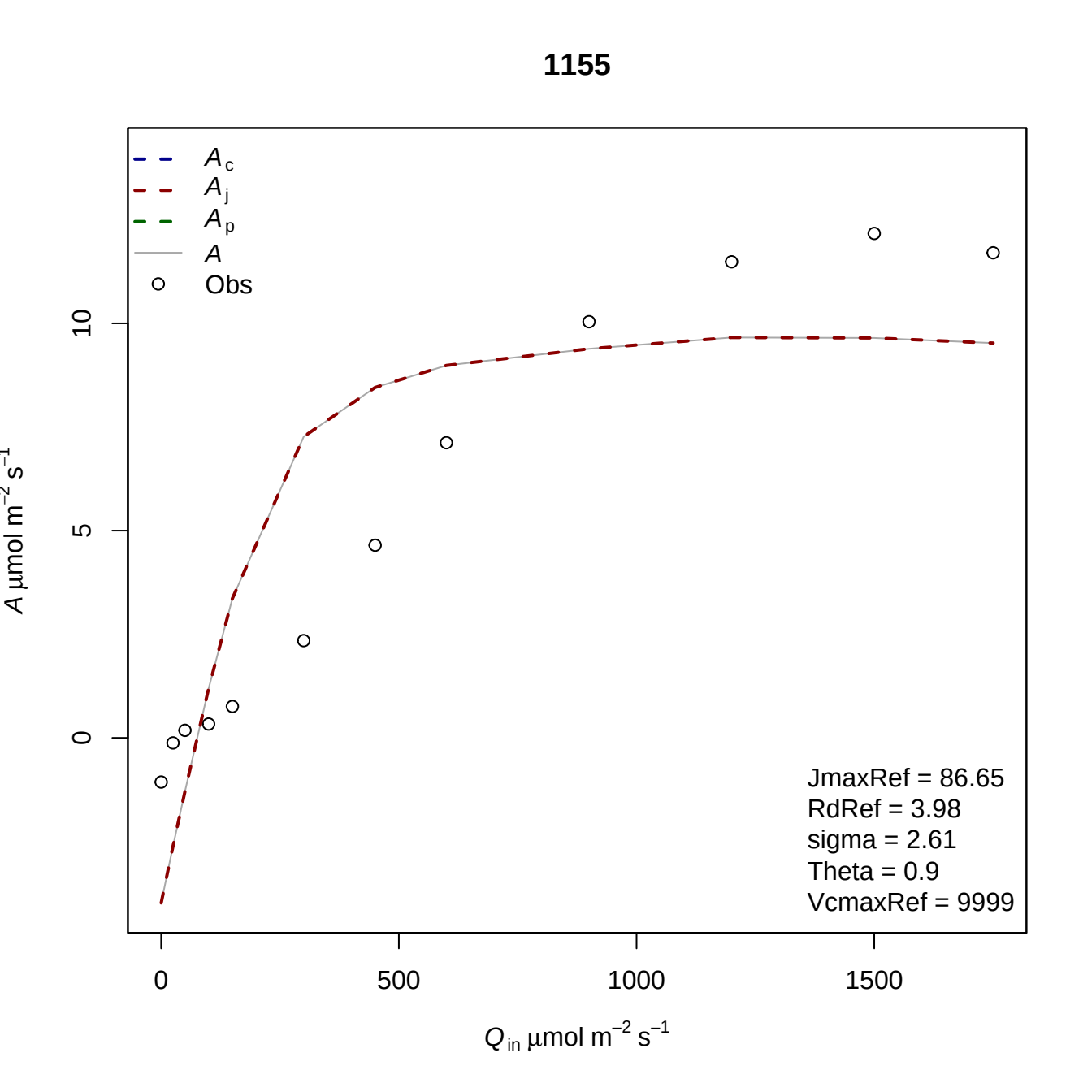

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 95.58
RdRef = 0.71
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

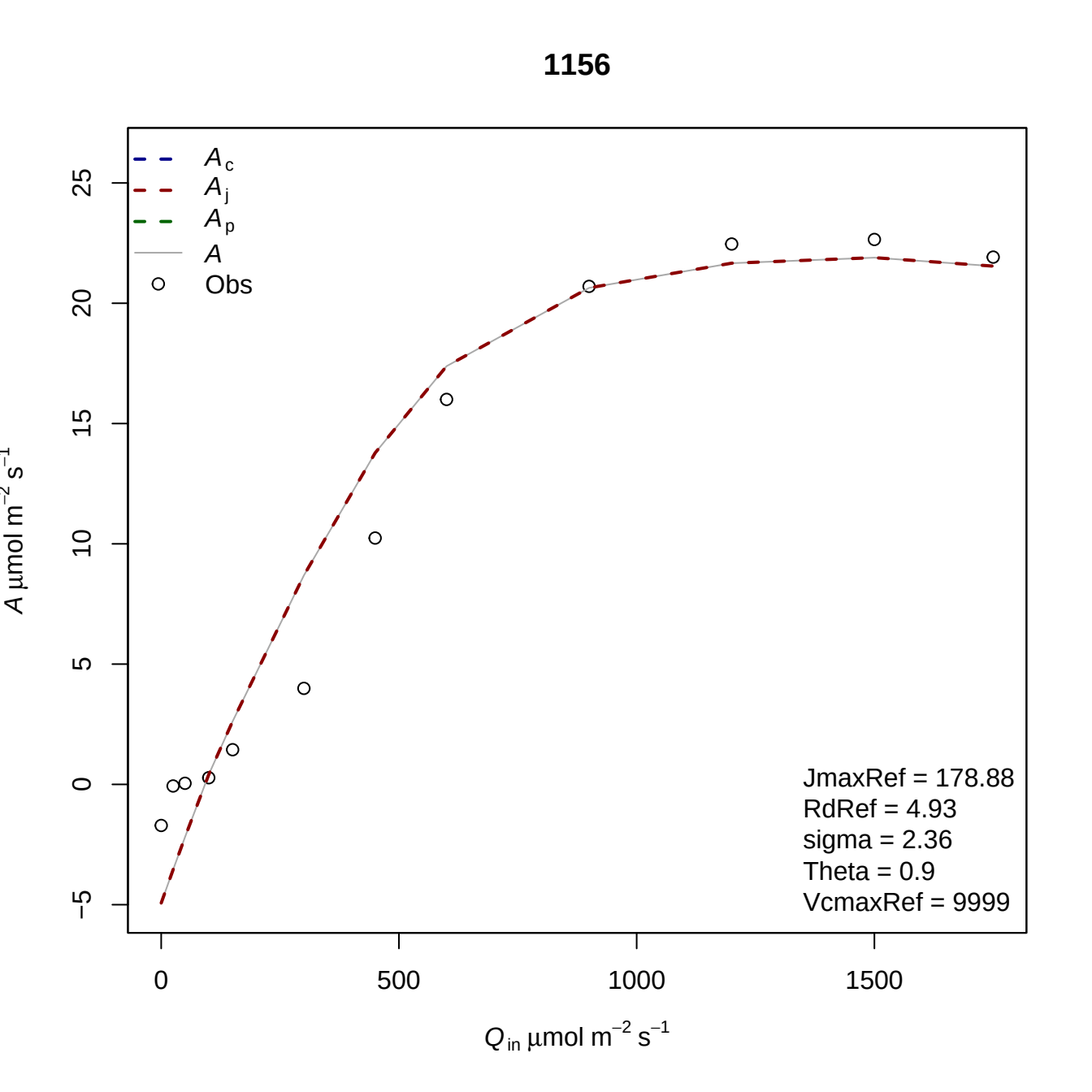

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 78.25
RdRef = 0.58
sigma = 0.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

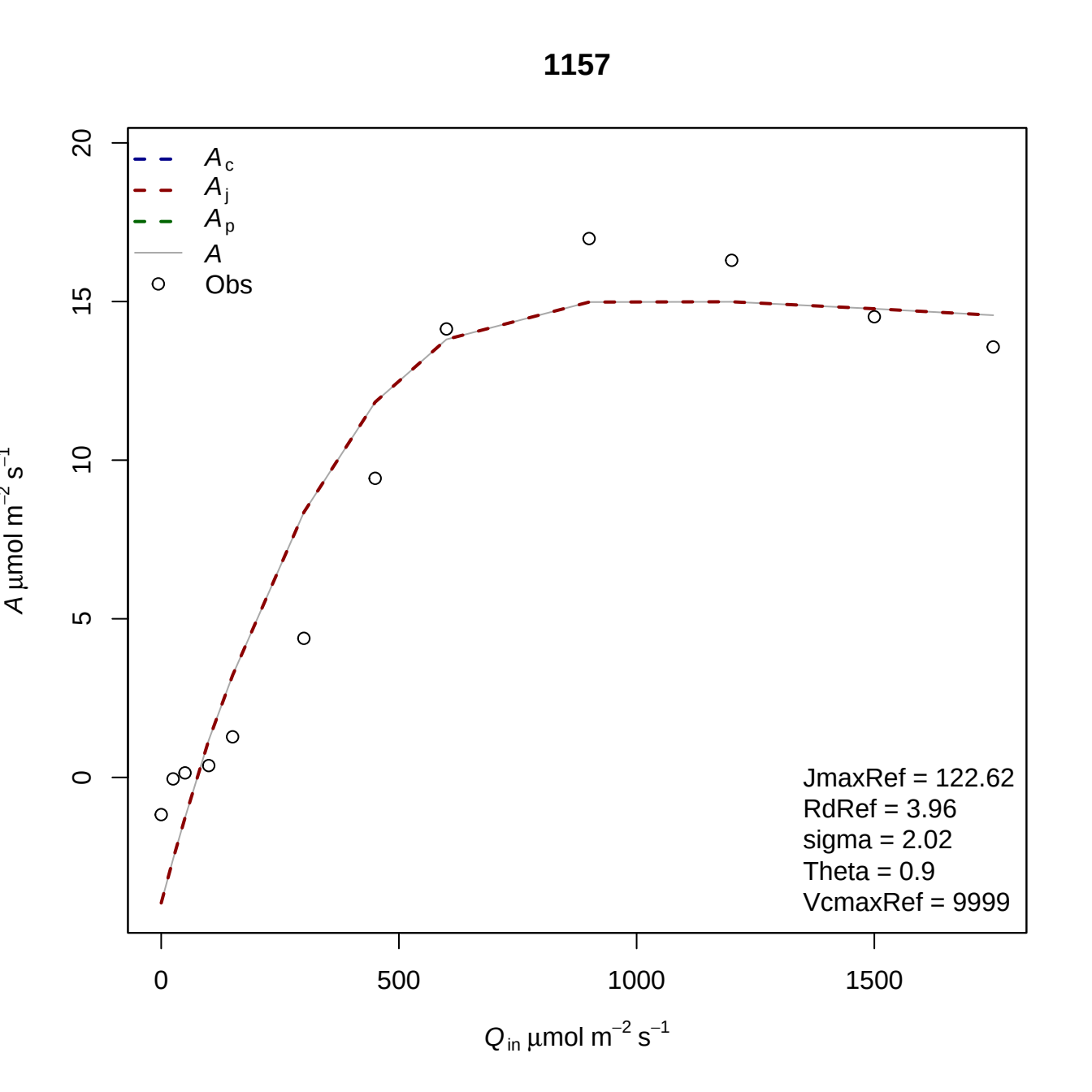

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 100.27
RdRef = 1.06
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

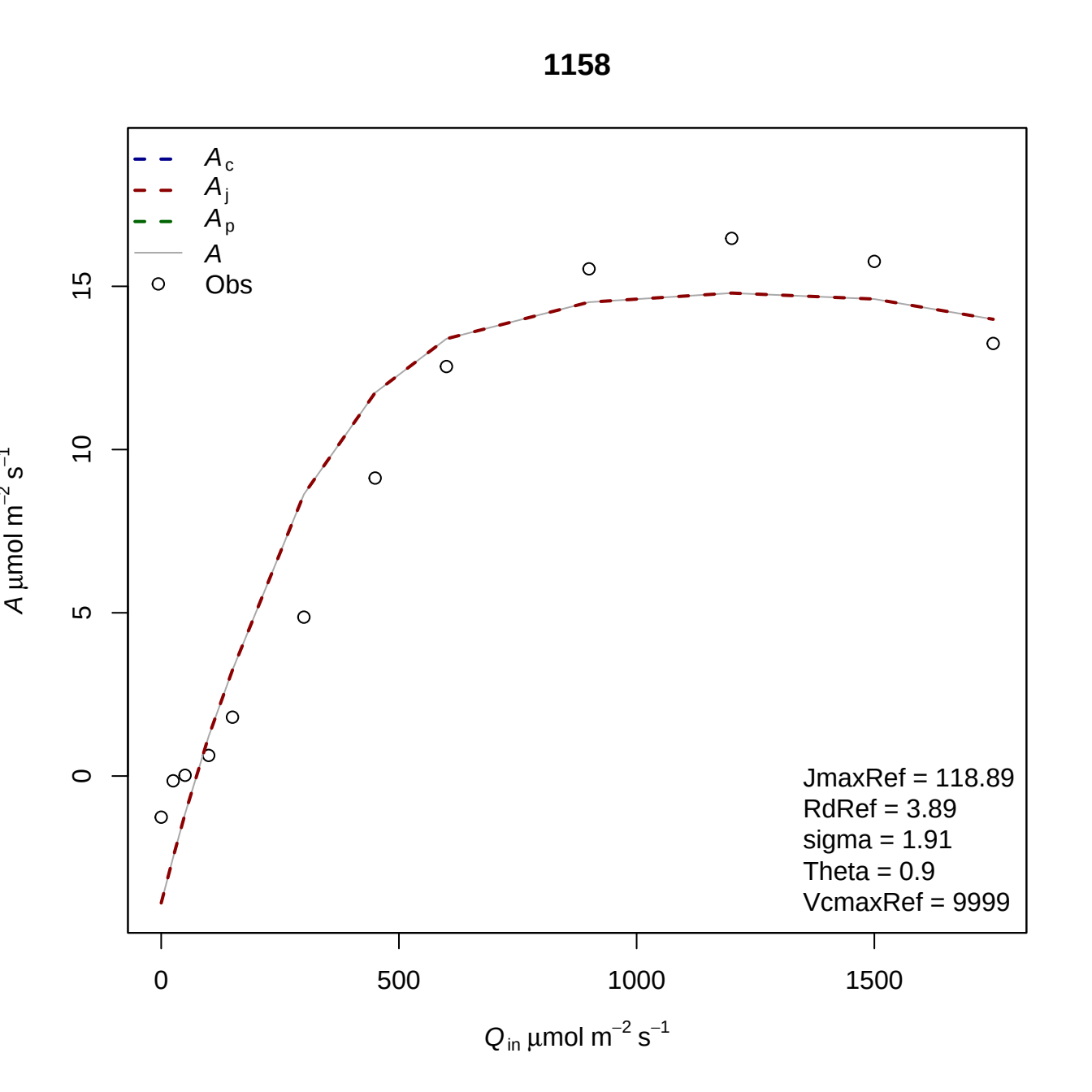

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 100.99
RdRef = 4.14
sigma = 2.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

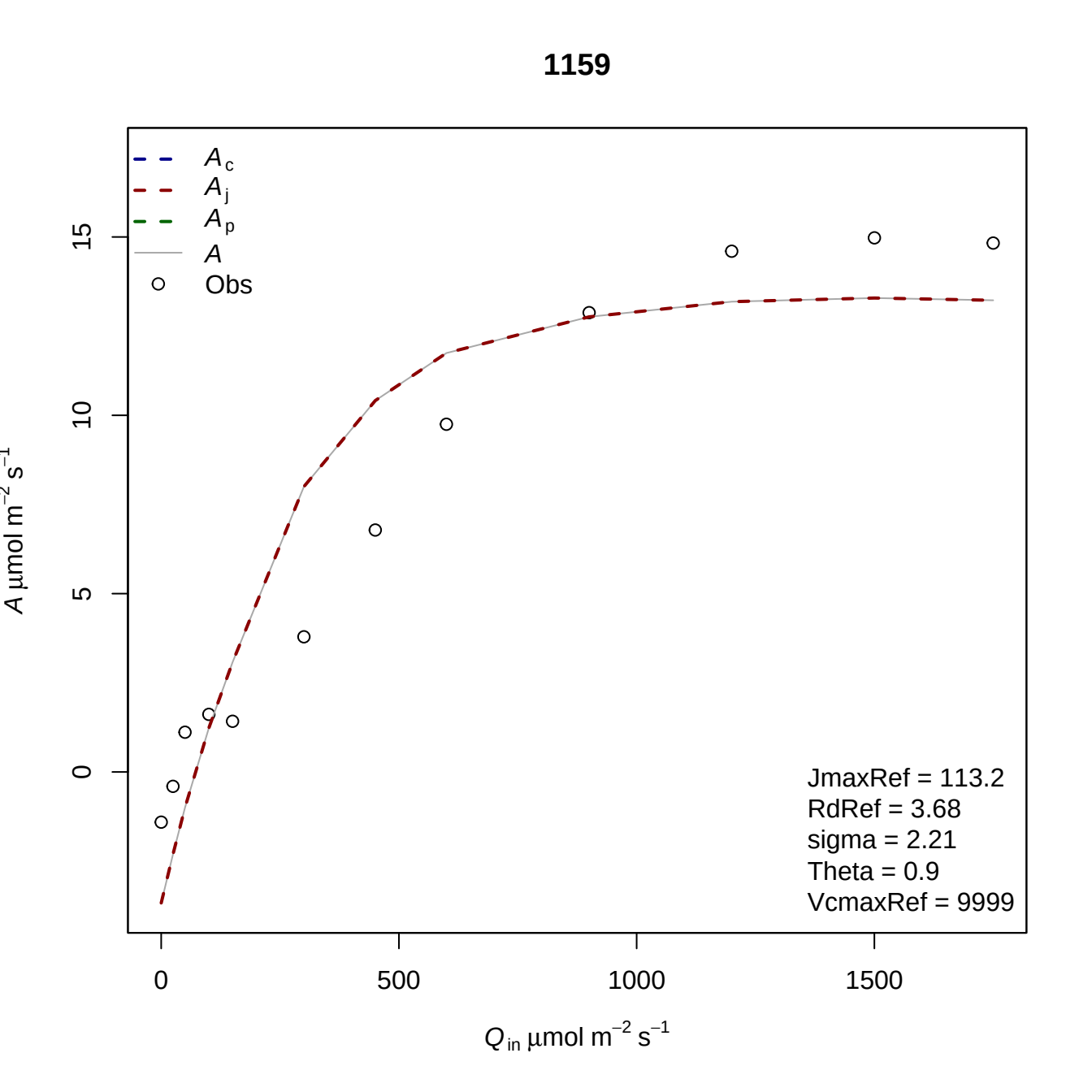

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 123.59
RdRef = 4.76
sigma = 2.83
Theta = 0.9
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

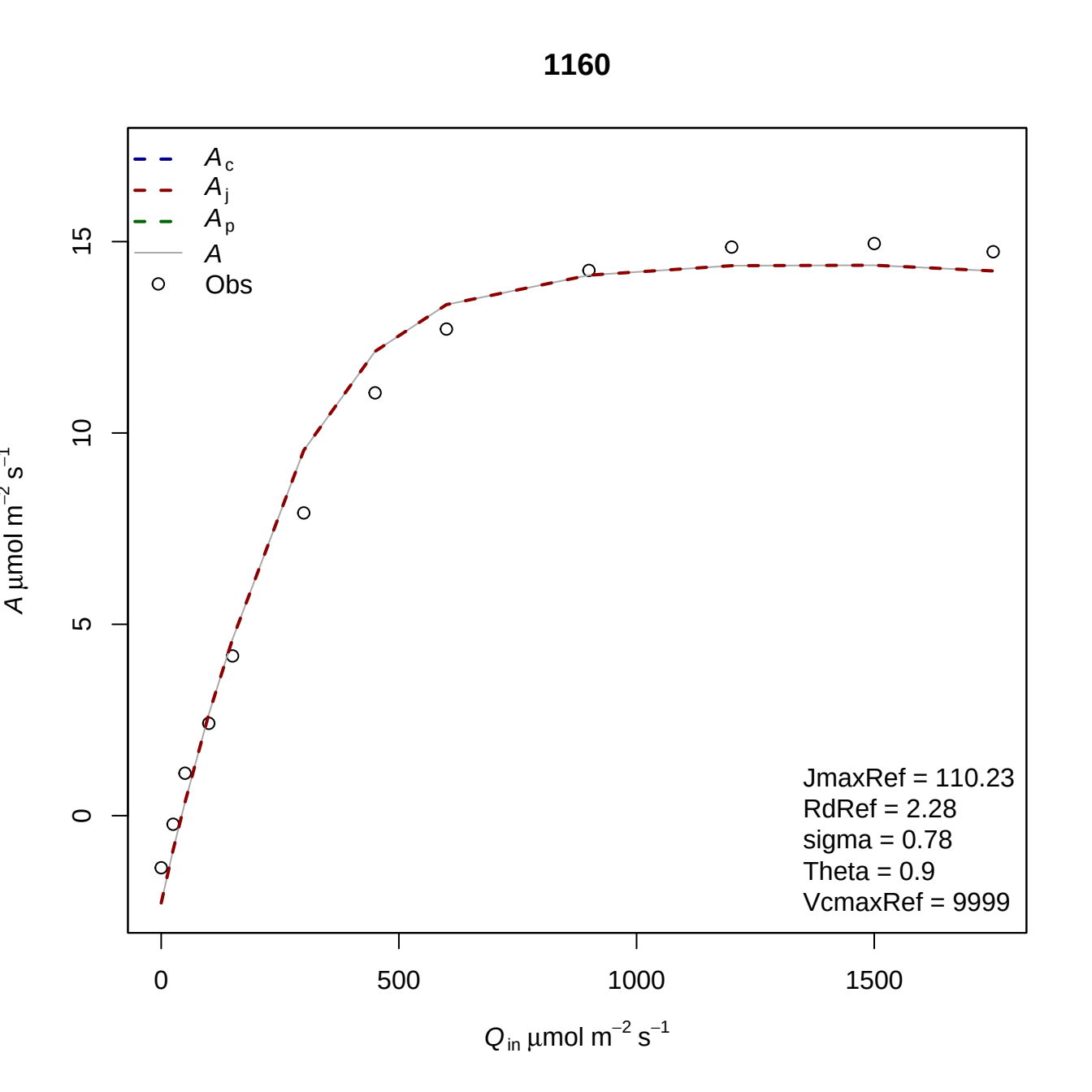

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 86.65
RdRef = 3.98
sigma = 2.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

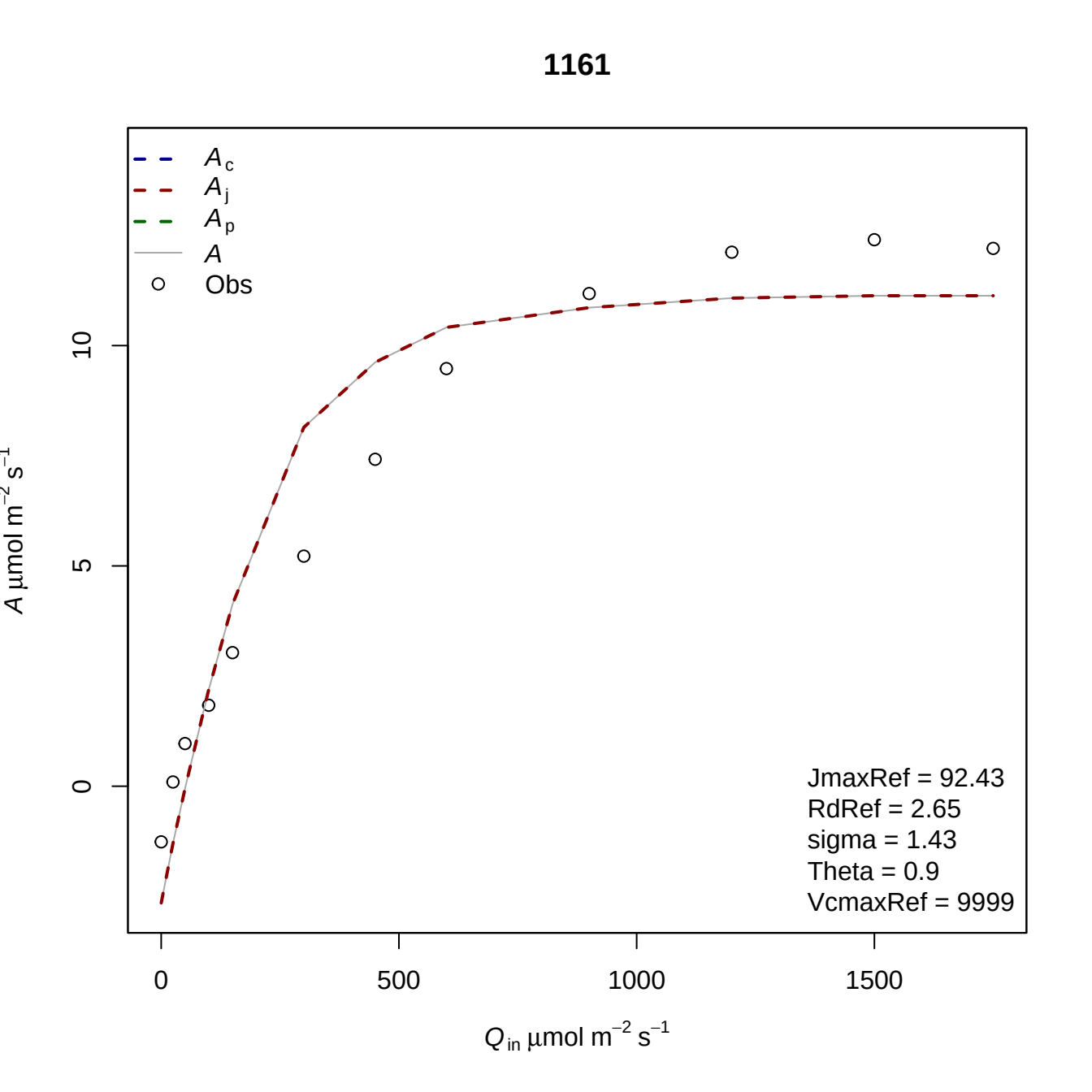

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 178.88
RdRef = 4.93
sigma = 2.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500

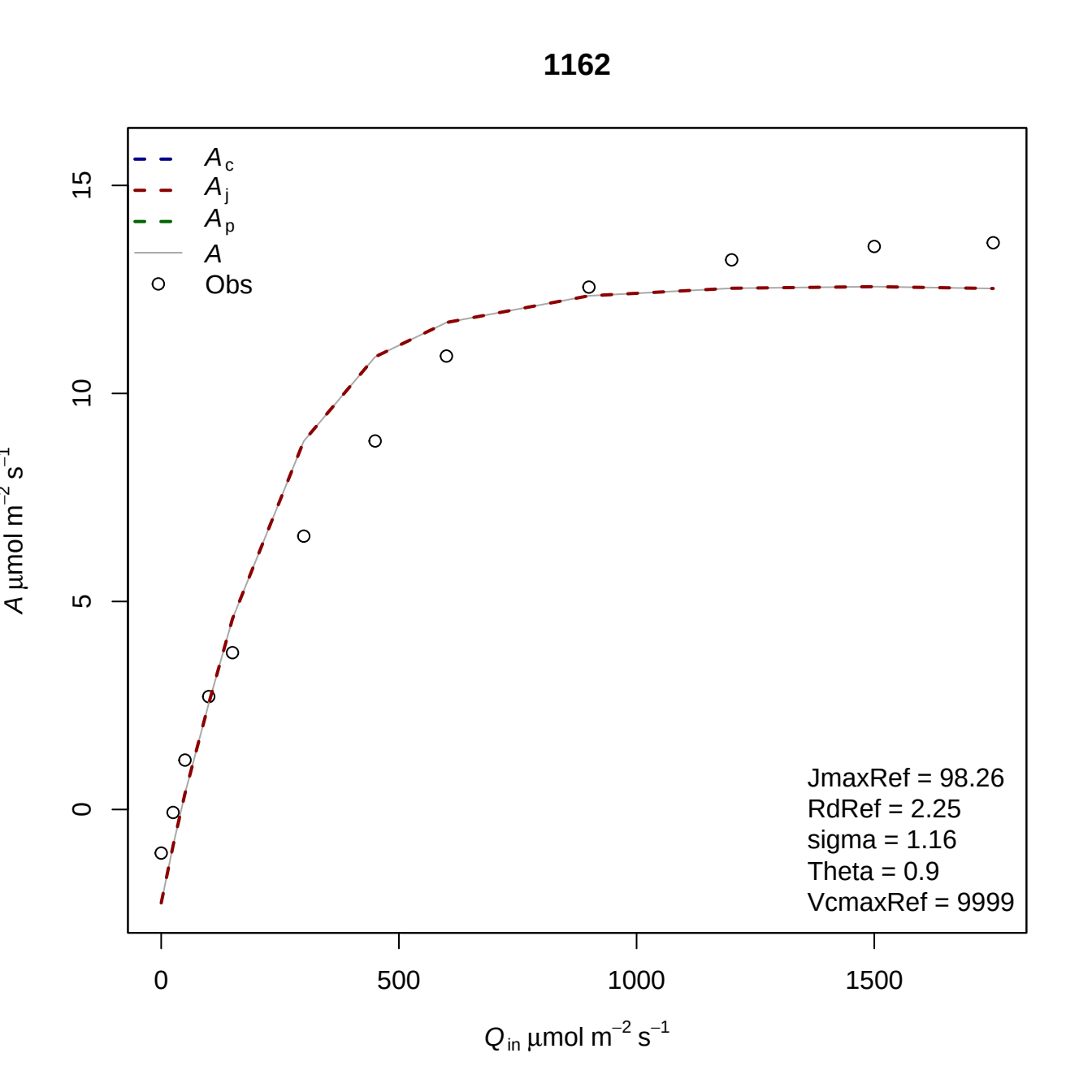

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 122.62
RdRef = 3.96
sigma = 2.02
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

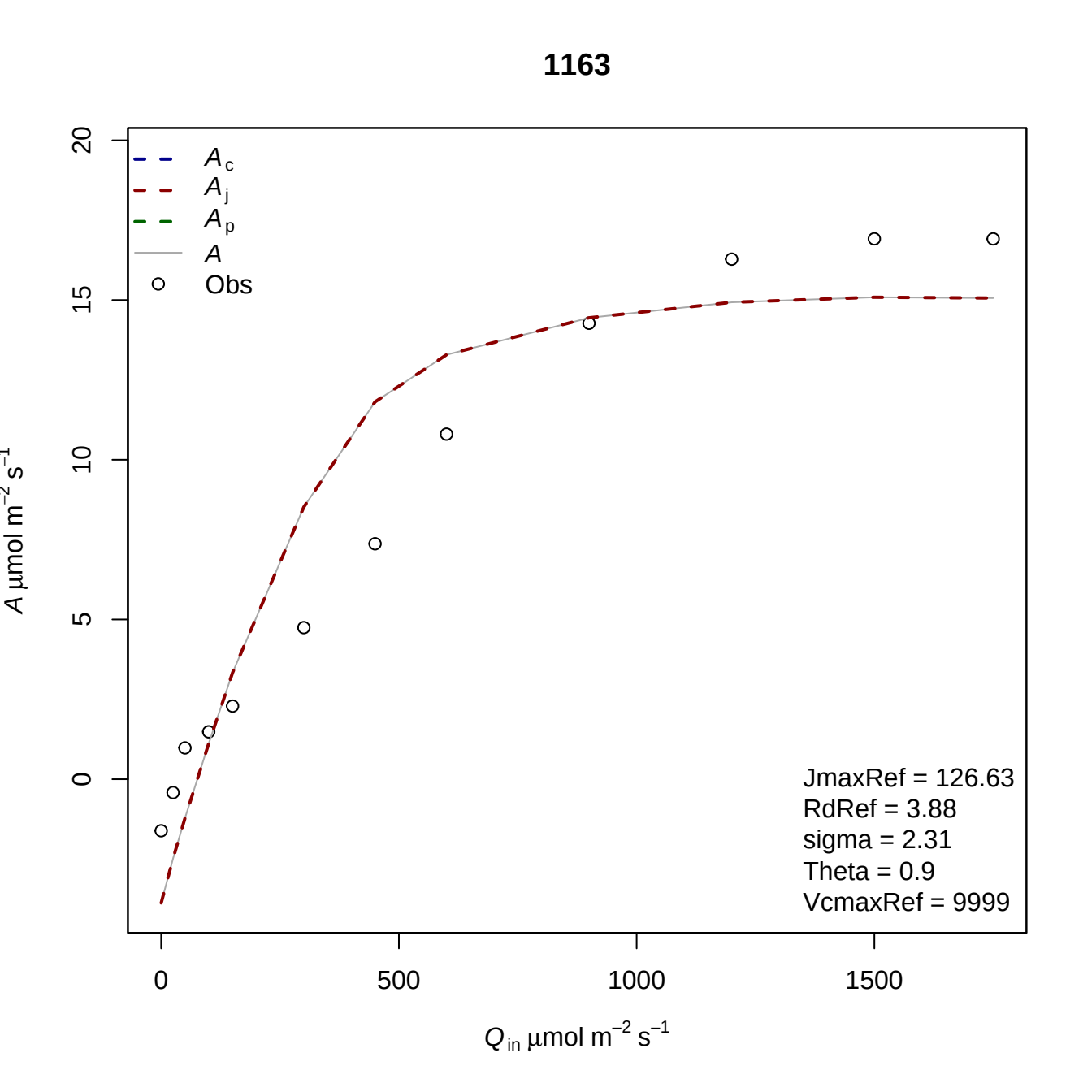

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 118.89
RdRef = 3.89
sigma = 1.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

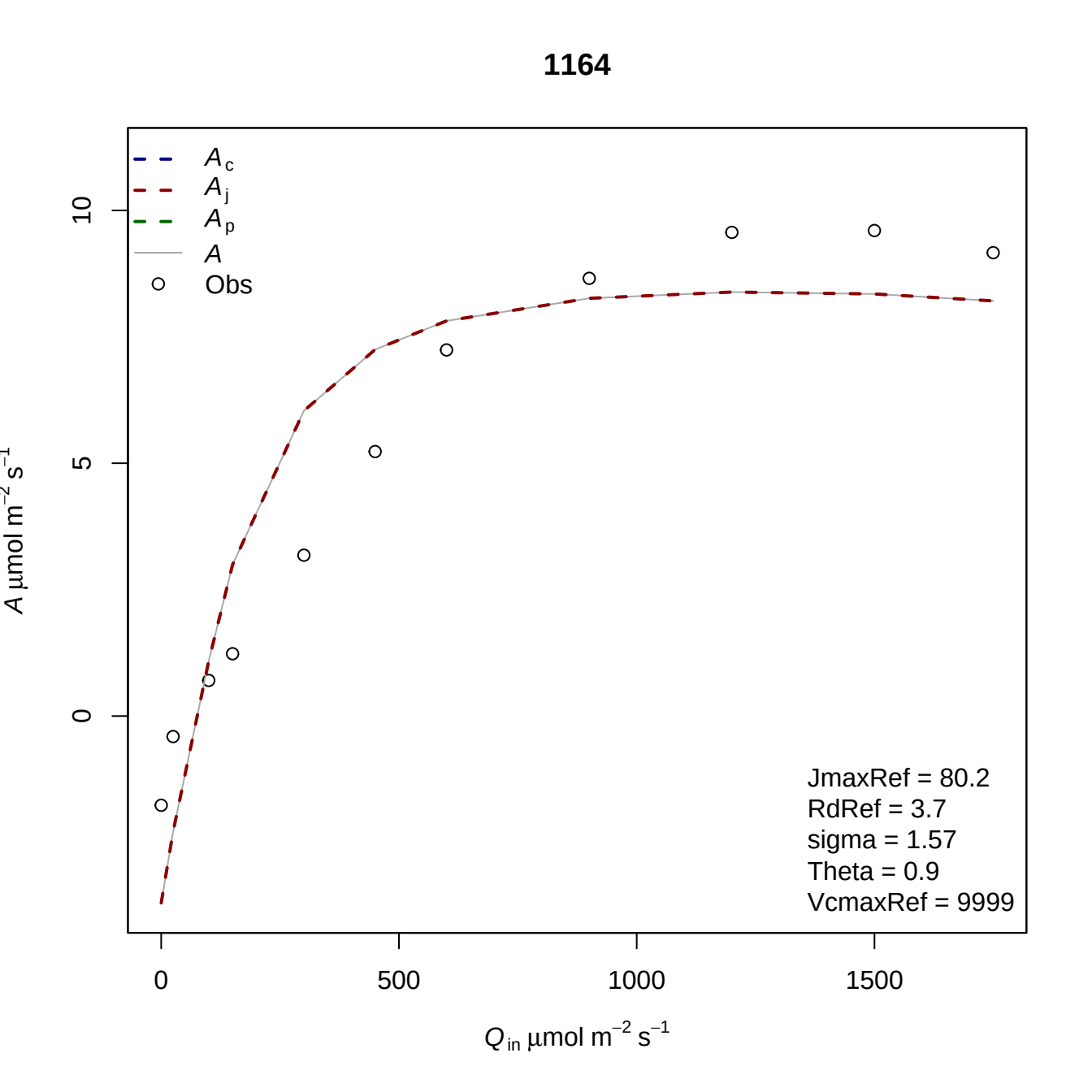

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 113.2
RdRef = 3.68
sigma = 2.21
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

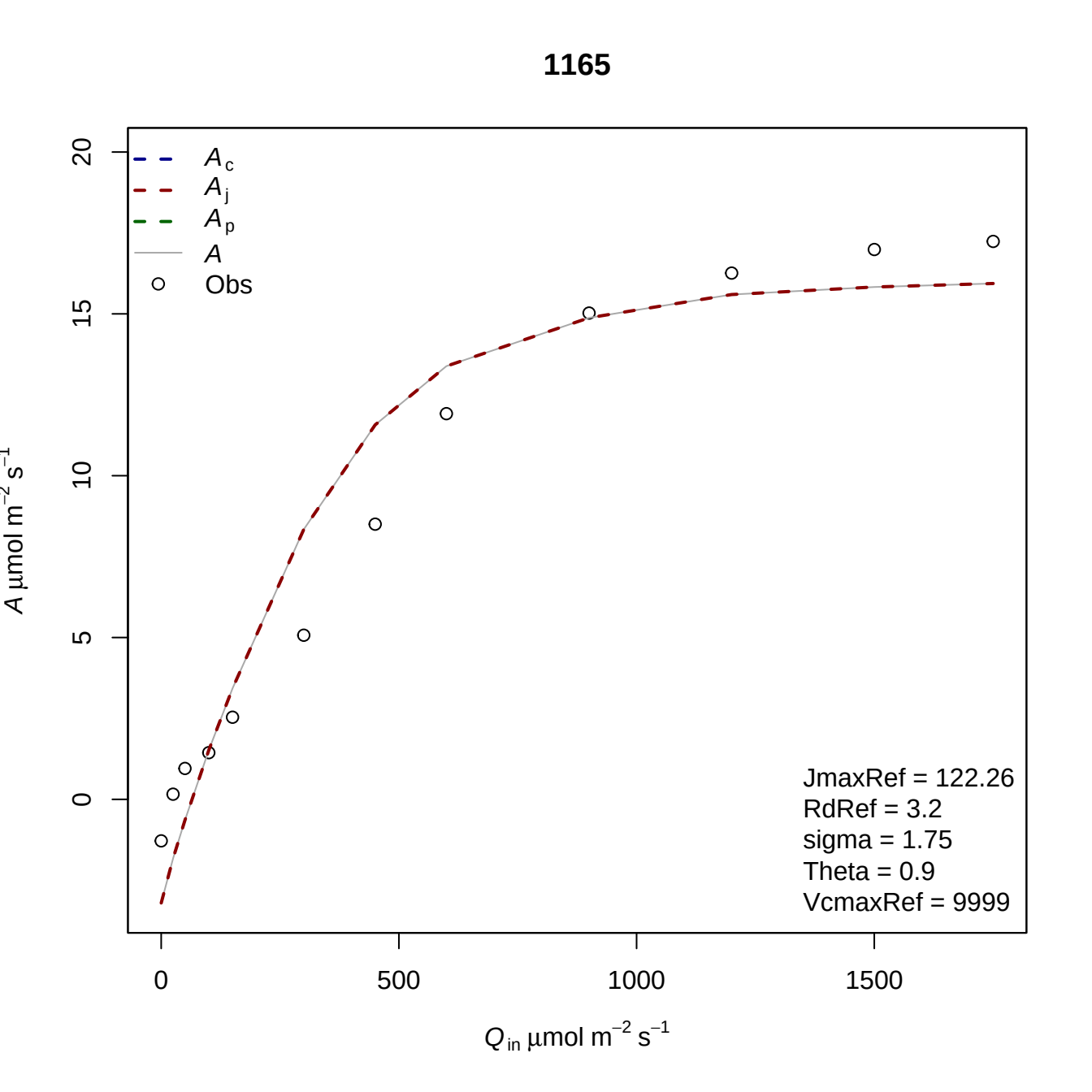

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 110.23
RdRef = 2.28
sigma = 0.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

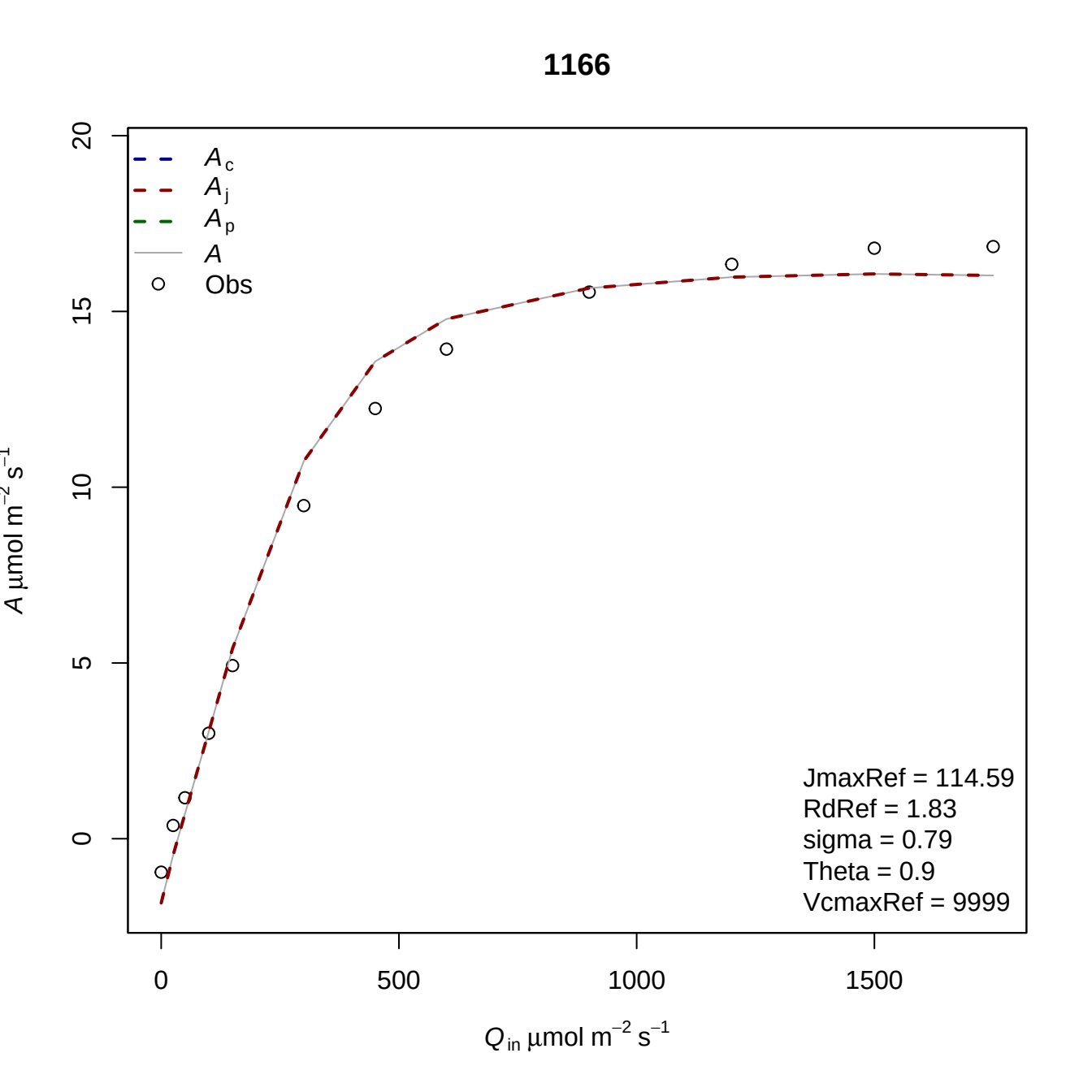

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 92.43
RdRef = 2.65
sigma = 1.43
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

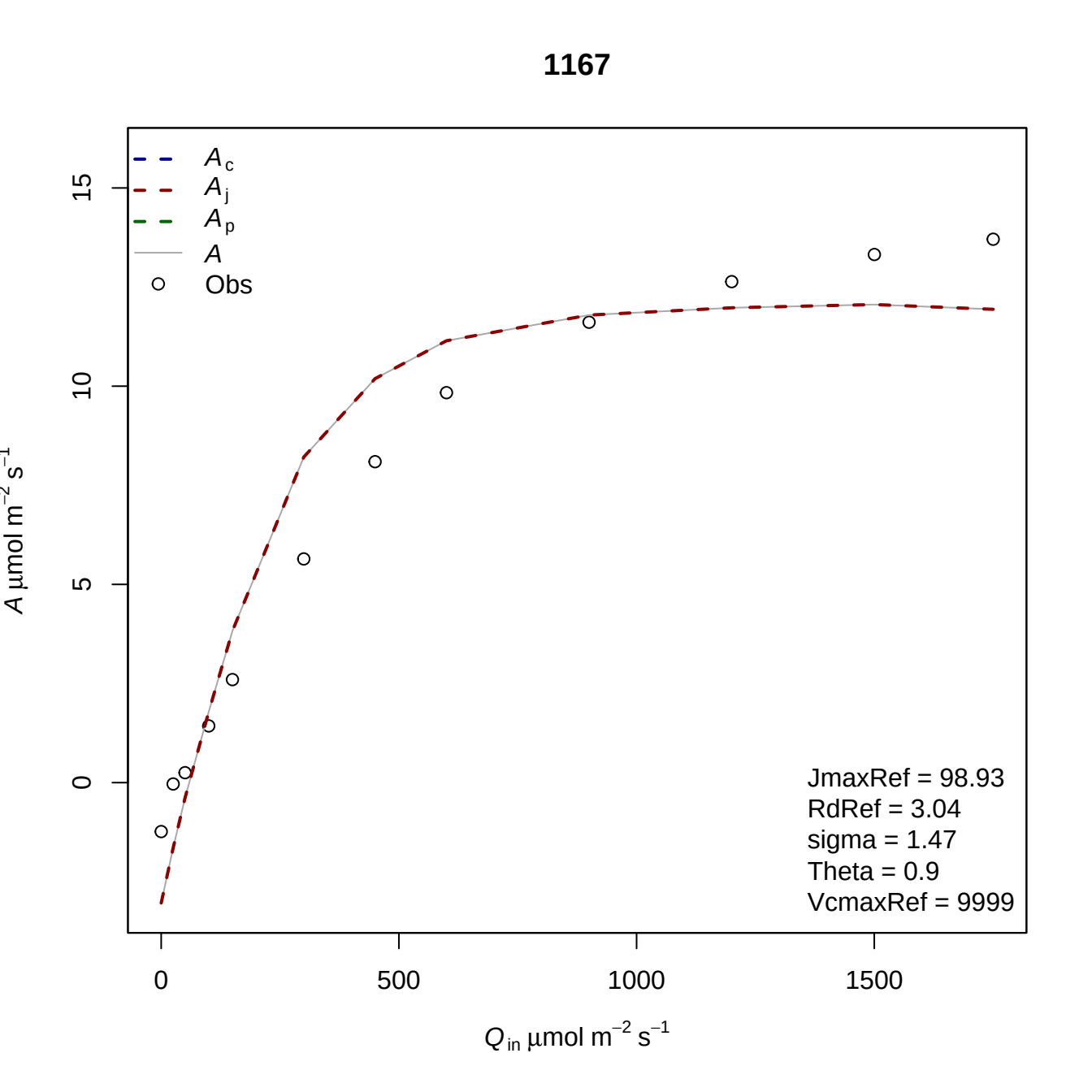

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 98.26
RdRef = 2.25
sigma = 1.16
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

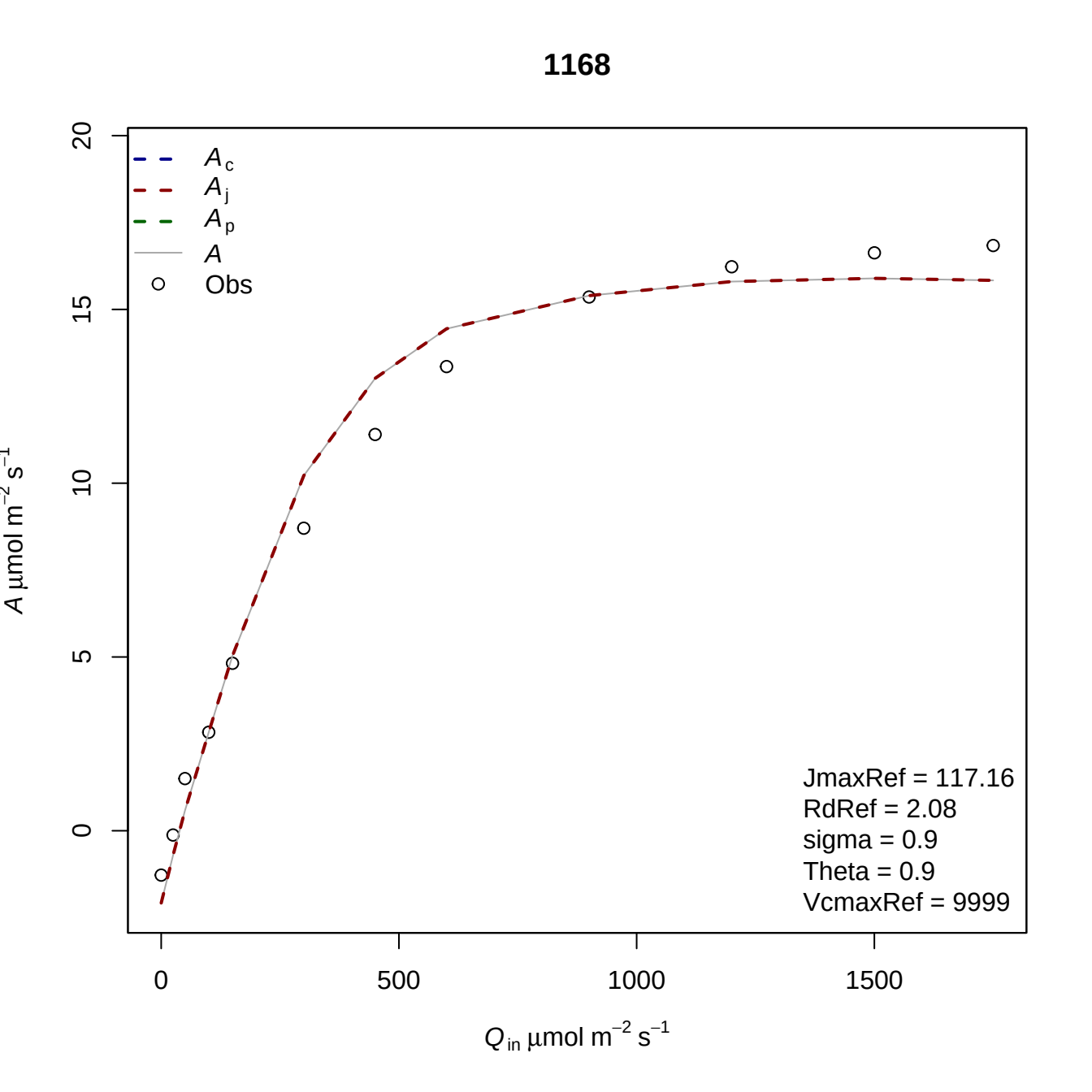

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 126.63
RdRef = 3.88
sigma = 2.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

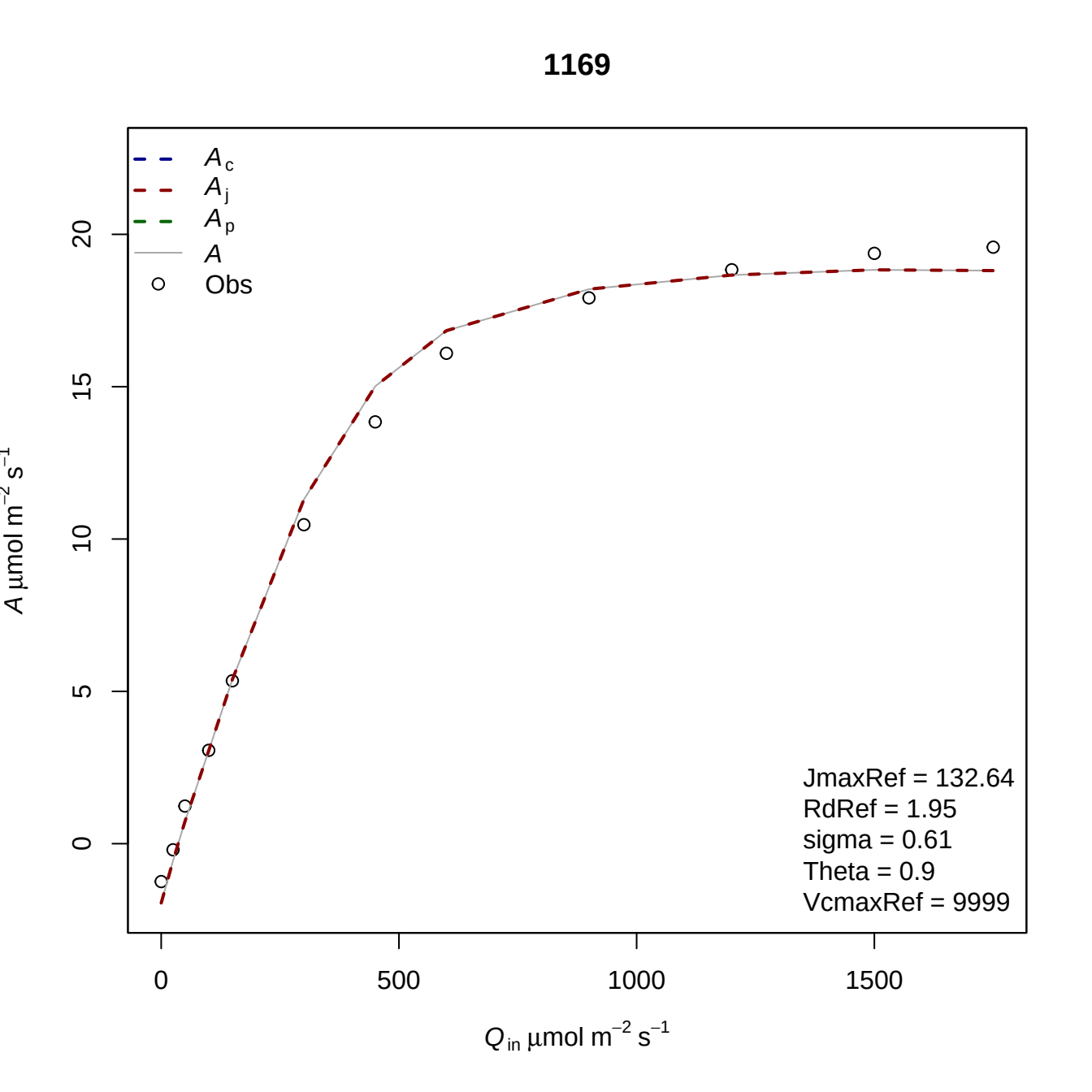

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 80.2
RdRef = 3.7
sigma = 1.57
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

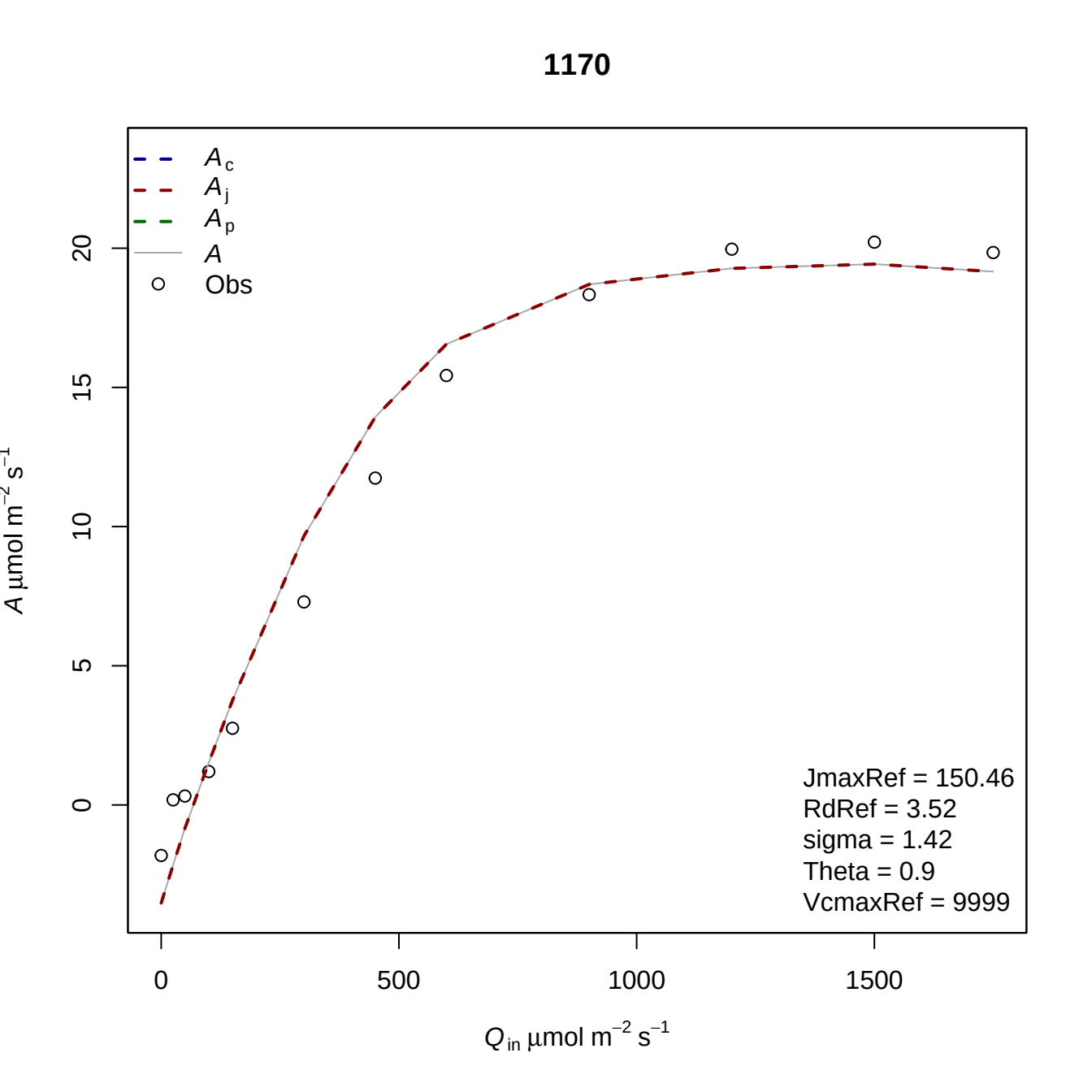

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 122.26
RdRef = 3.2
sigma = 1.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

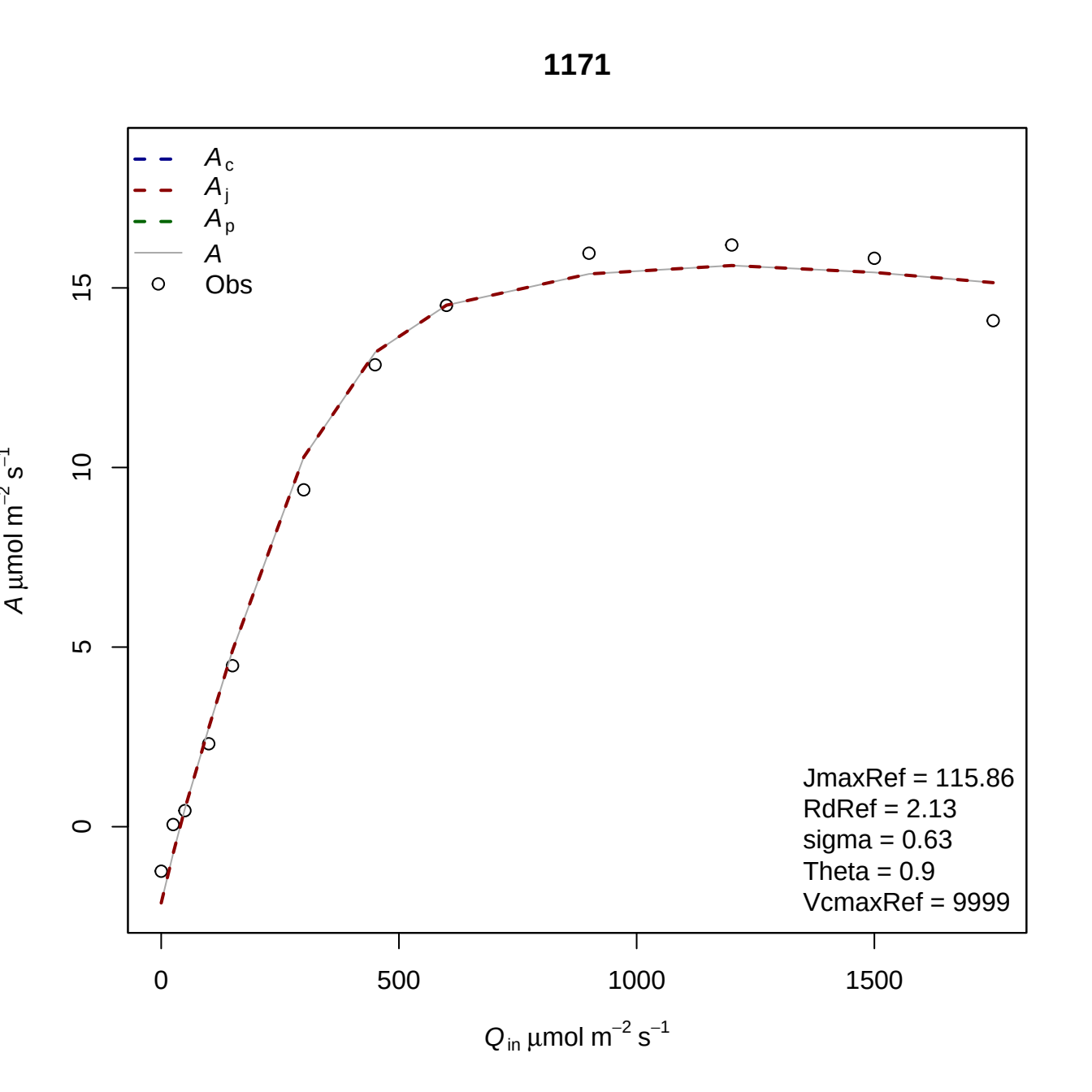

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 114.59
RdRef = 1.83
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

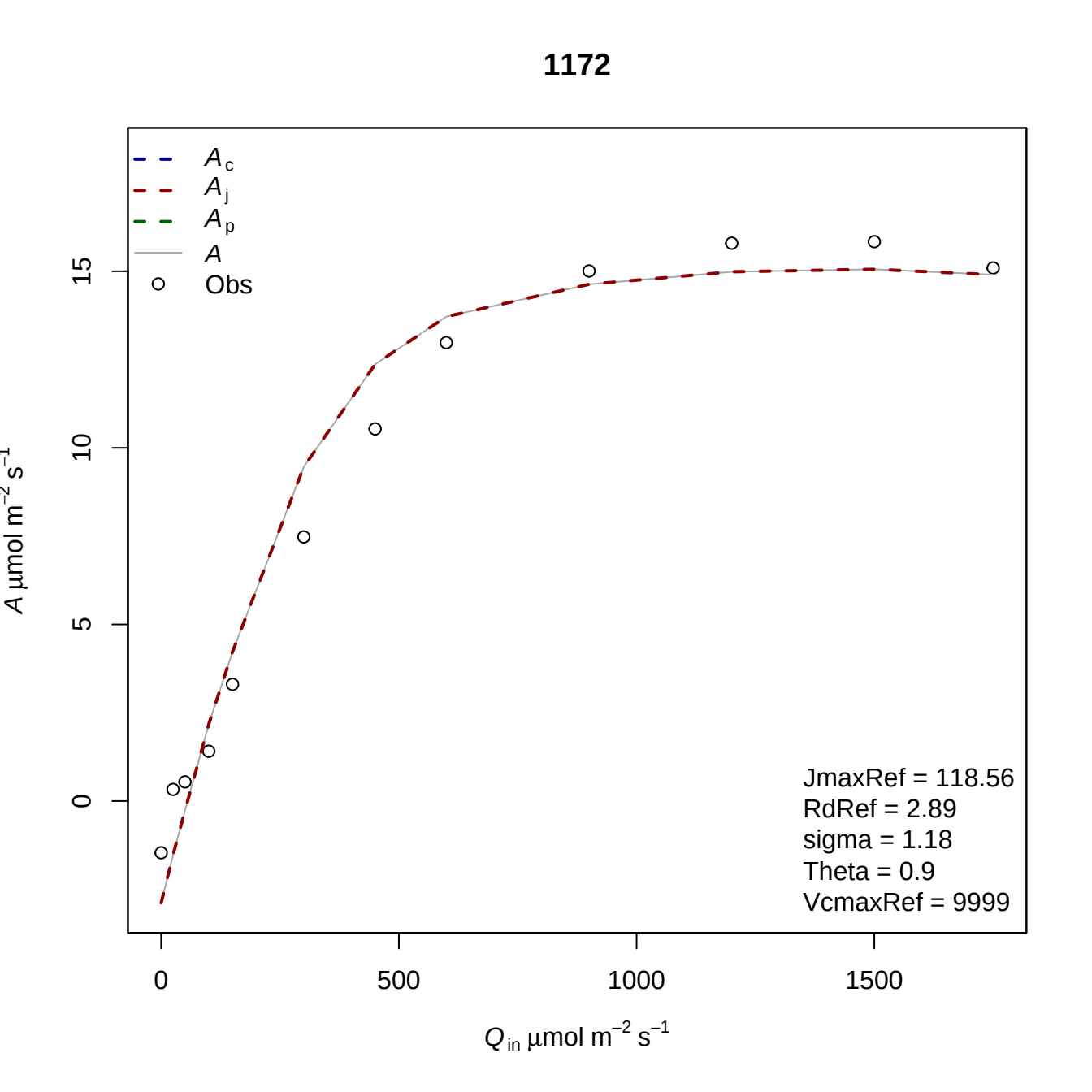

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 98.93
RdRef = 3.04
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

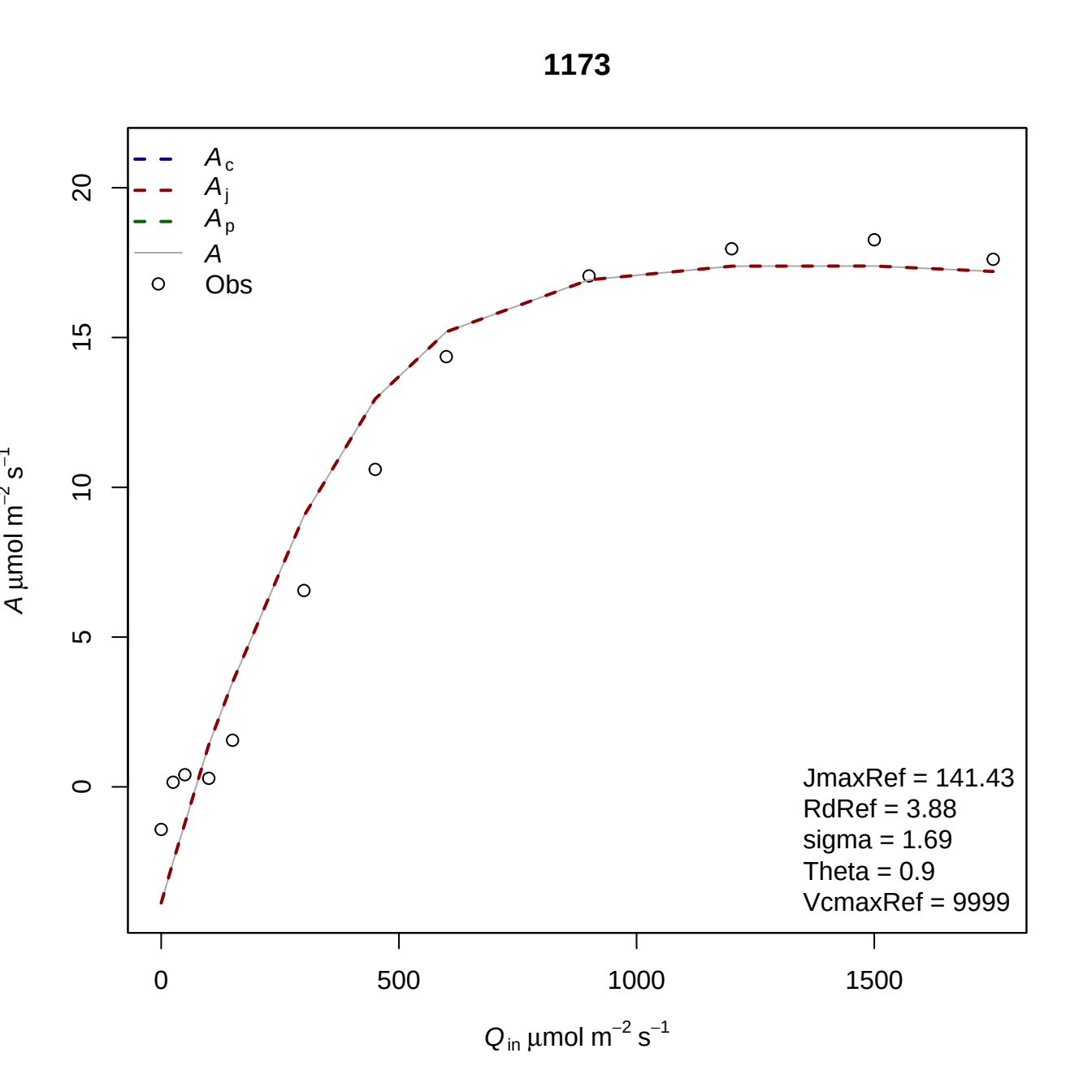

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 117.16
RdRef = 2.08
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

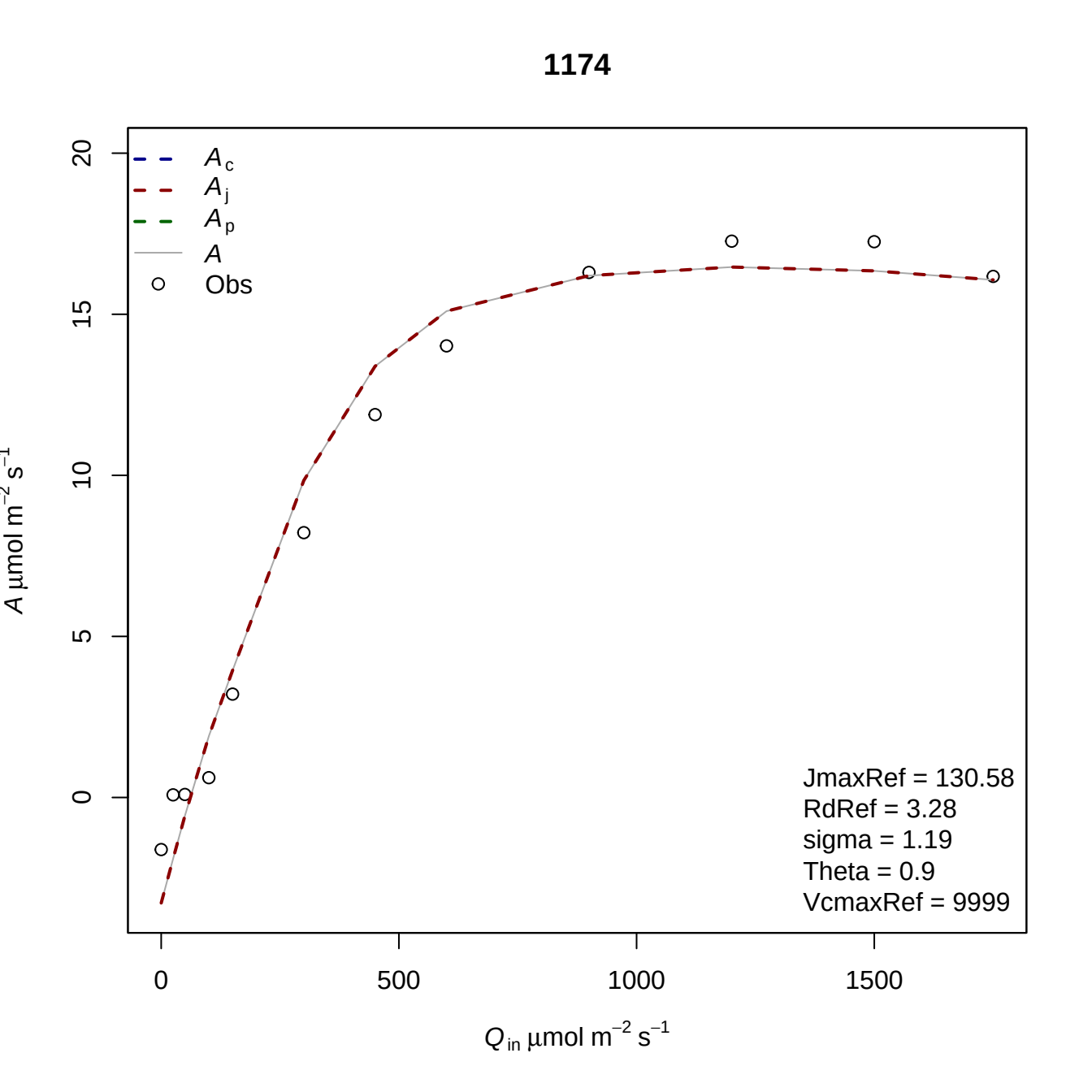

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 132.64
RdRef = 1.95
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

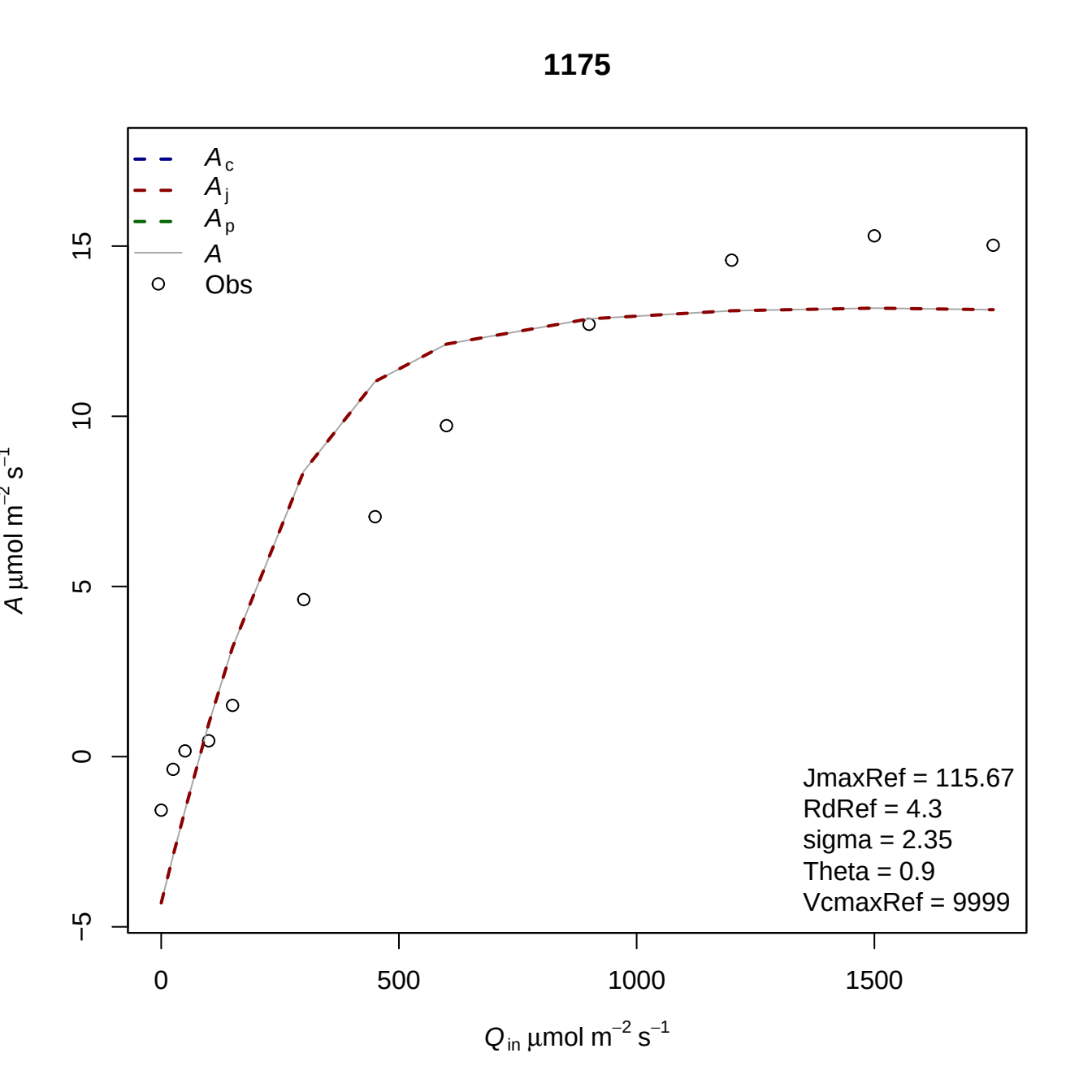

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 150.46
RdRef = 3.52
sigma = 1.42
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

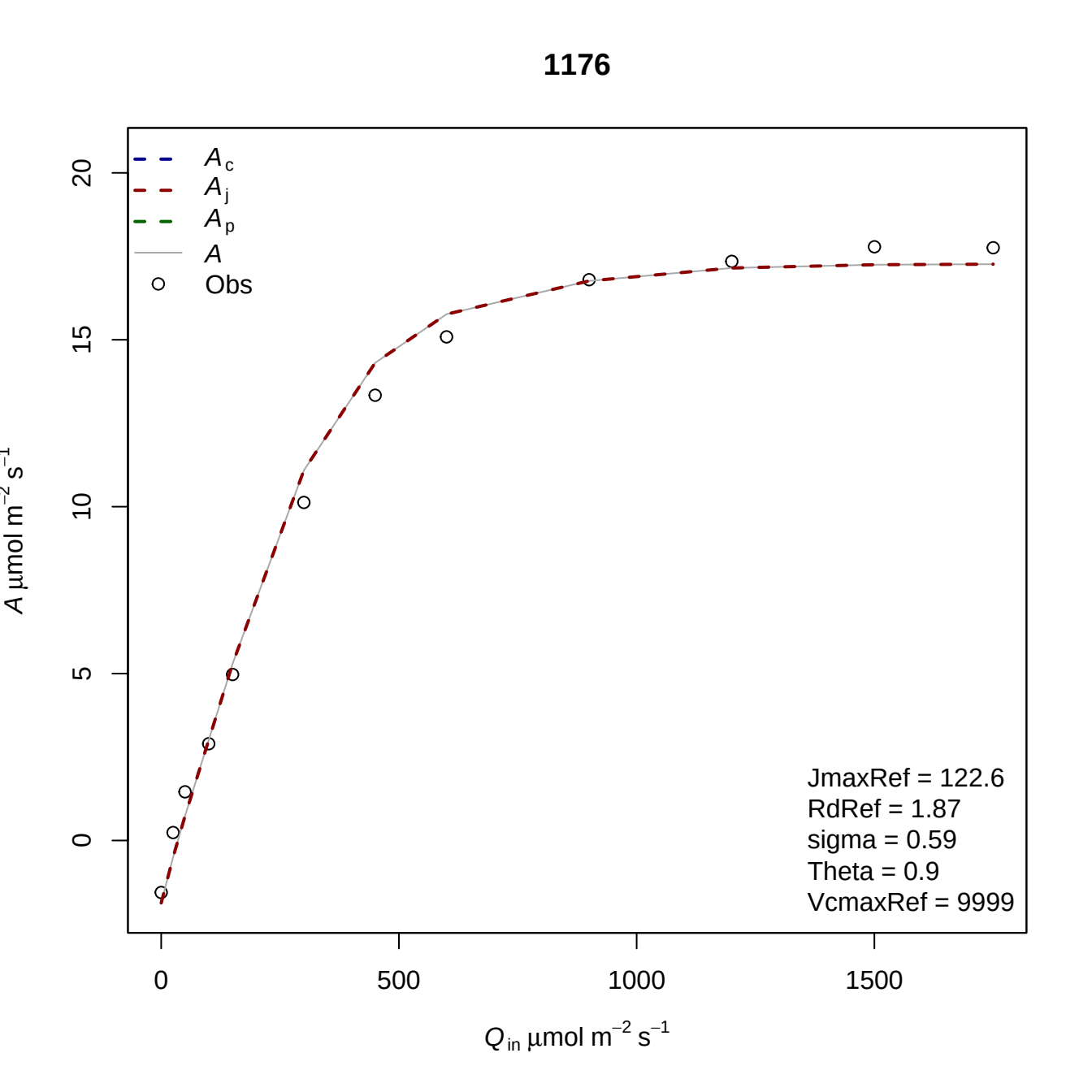

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 2.13
sigma = 0.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

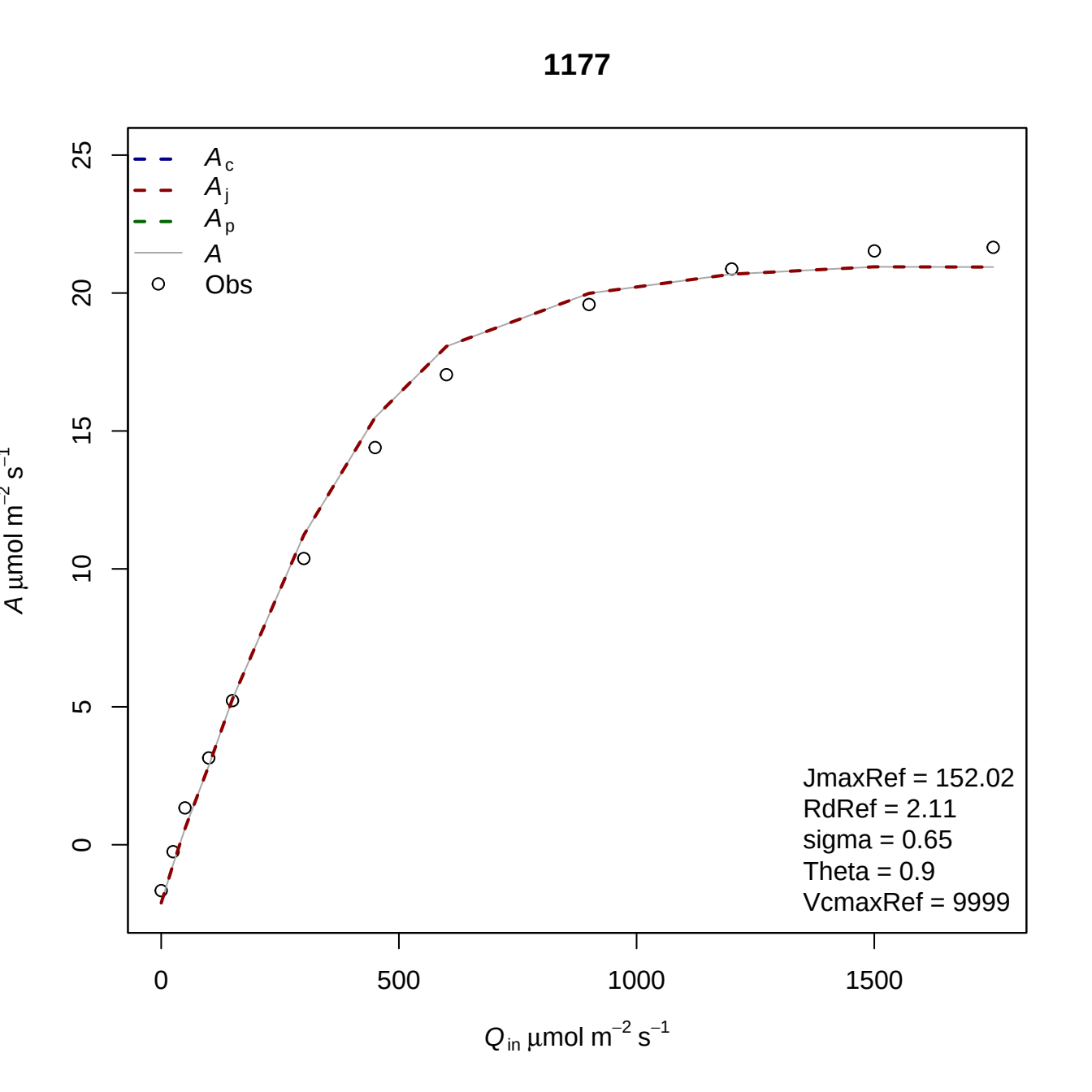

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 118.56
RdRef = 2.89
sigma = 1.18
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

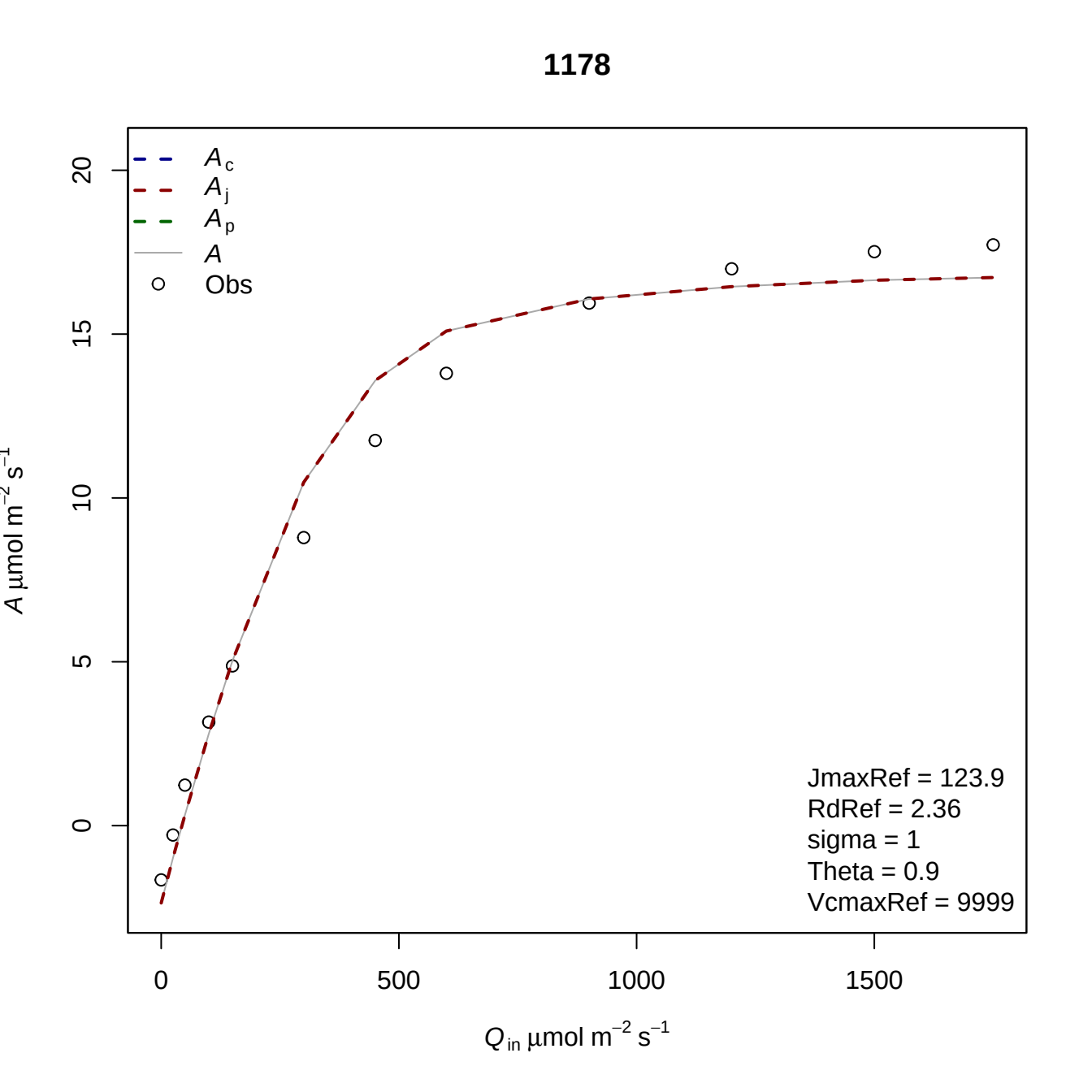

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 141.43
RdRef = 3.88
sigma = 1.69
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

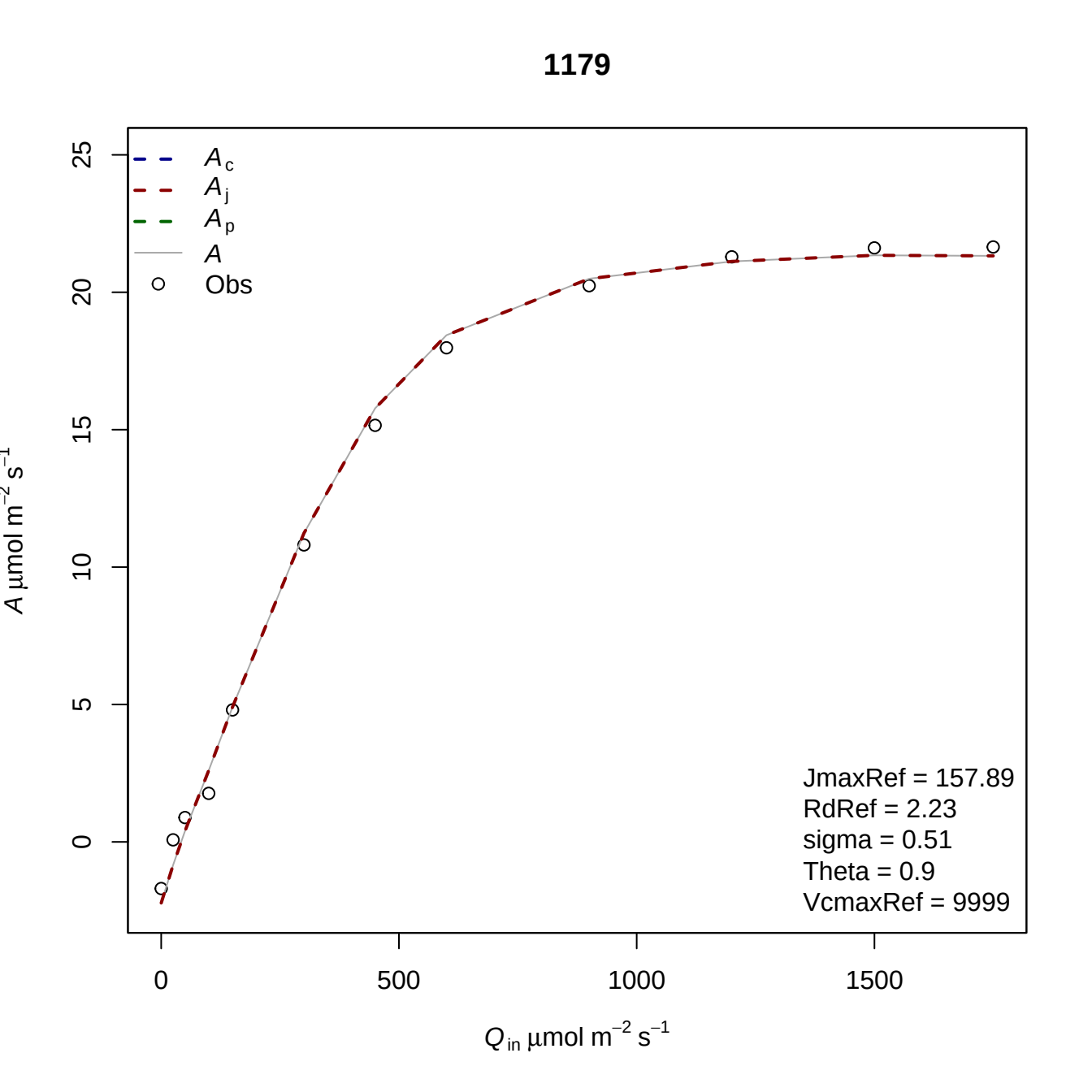

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 130.58
RdRef = 3.28
sigma = 1.19
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

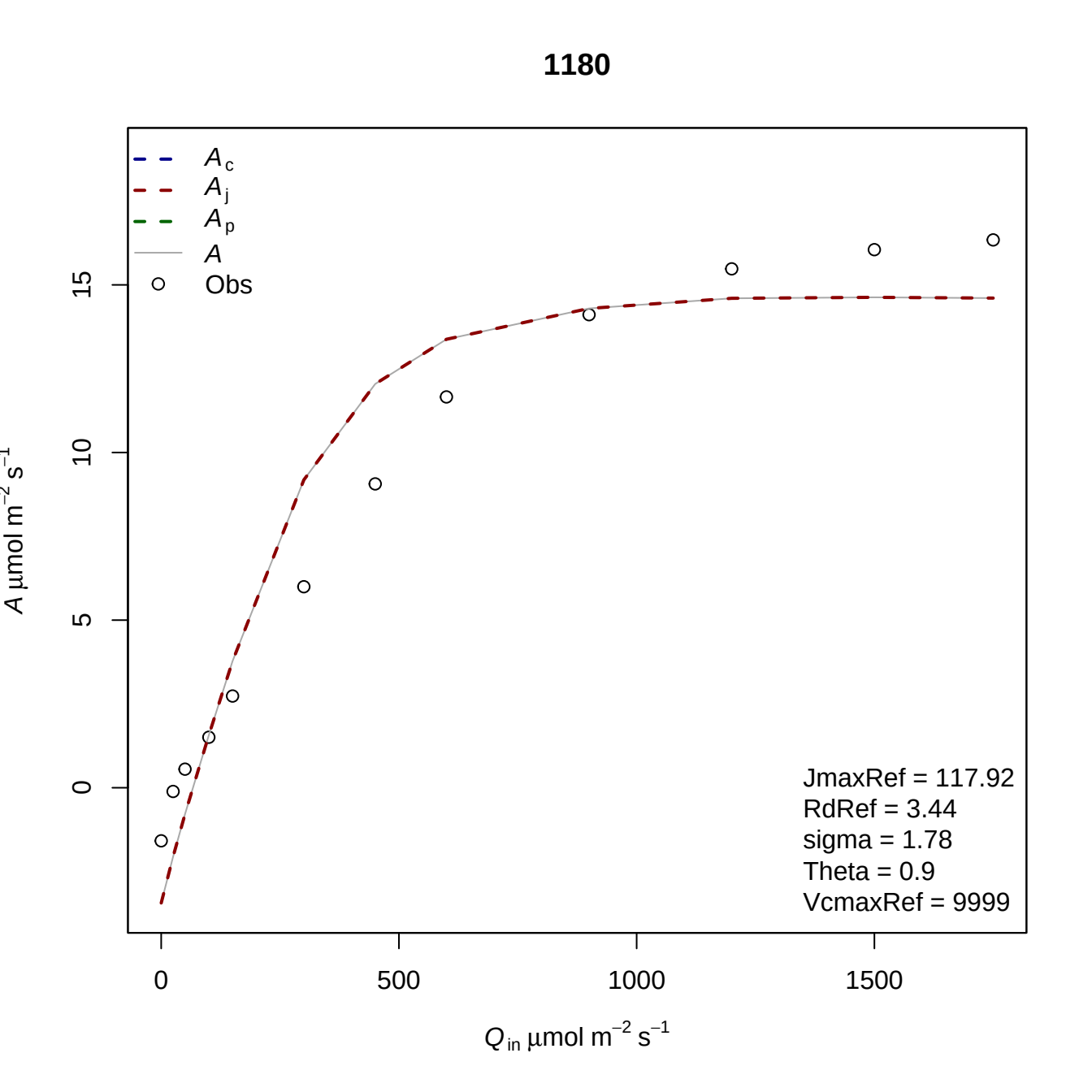

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 115.67
RdRef = 4.3
sigma = 2.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

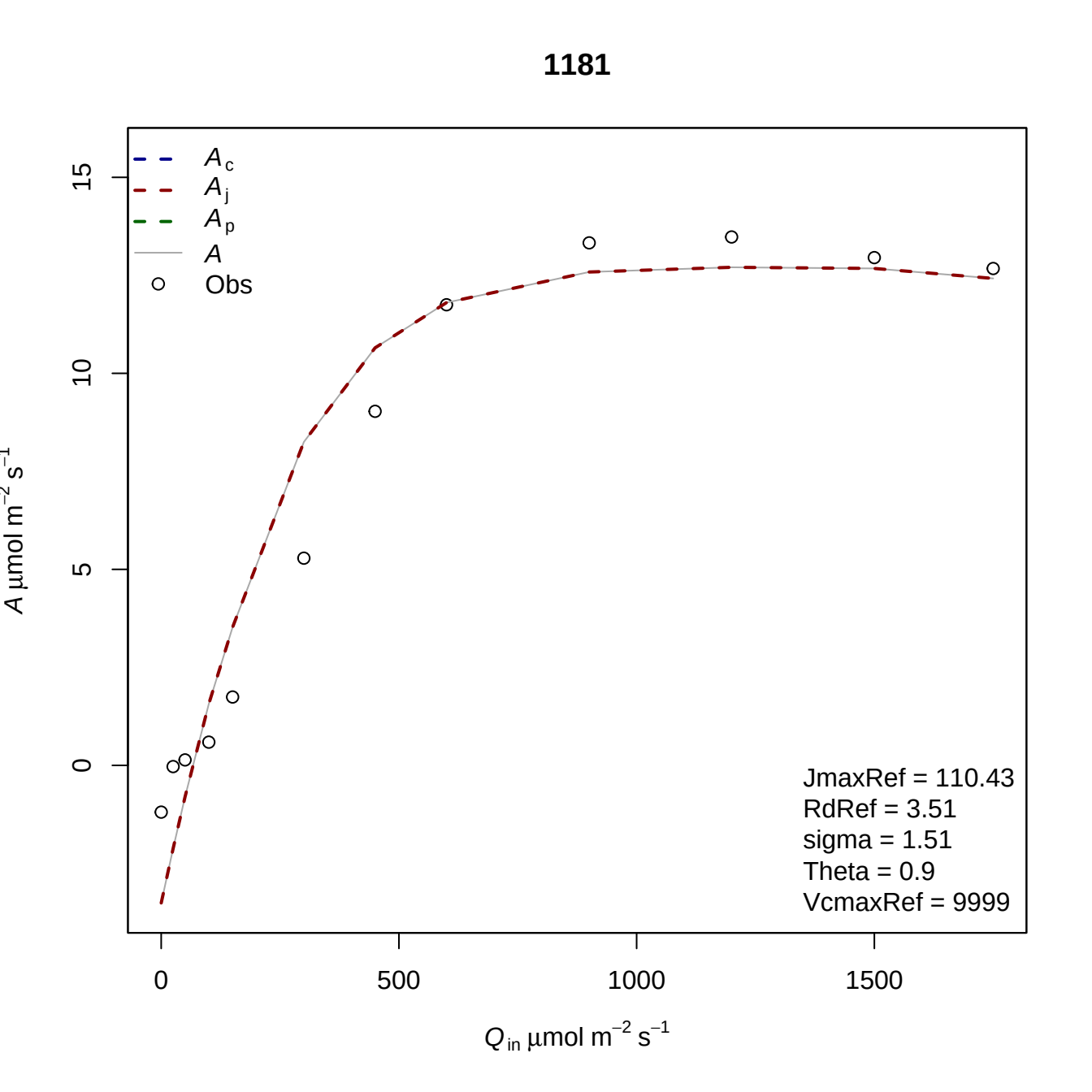

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 122.6
RdRef = 1.87
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

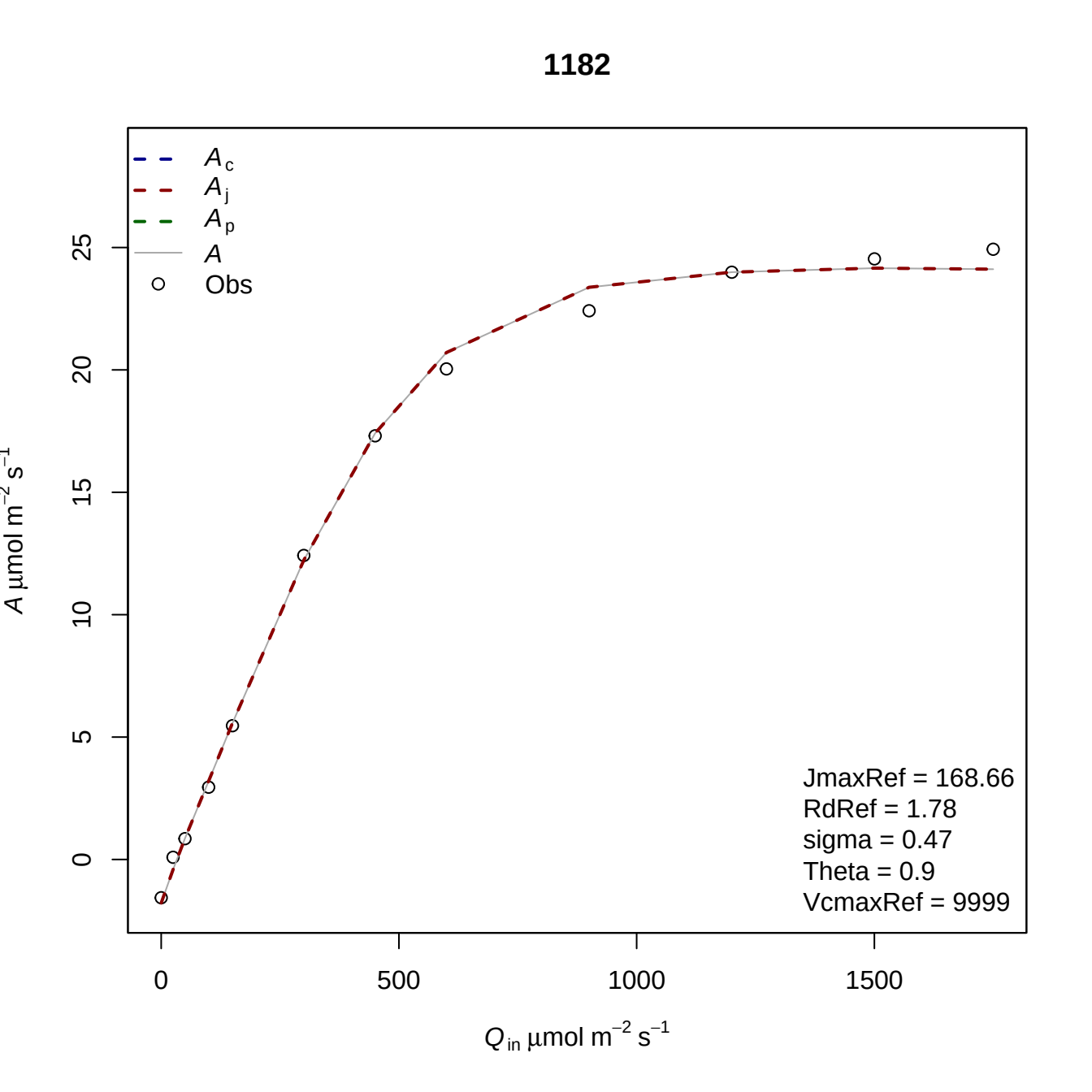

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 152.02
RdRef = 2.11
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

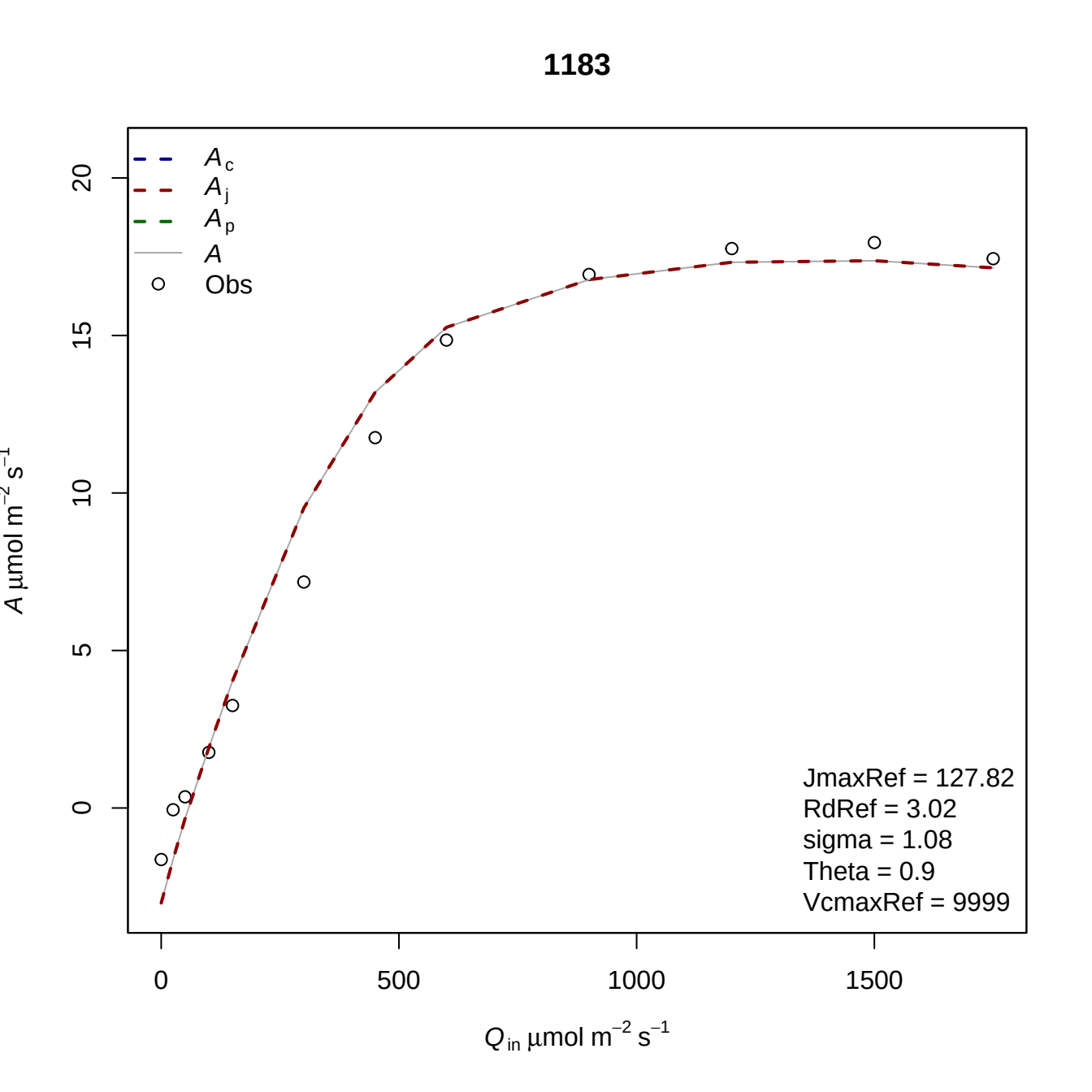

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 123.9
RdRef = 2.36
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

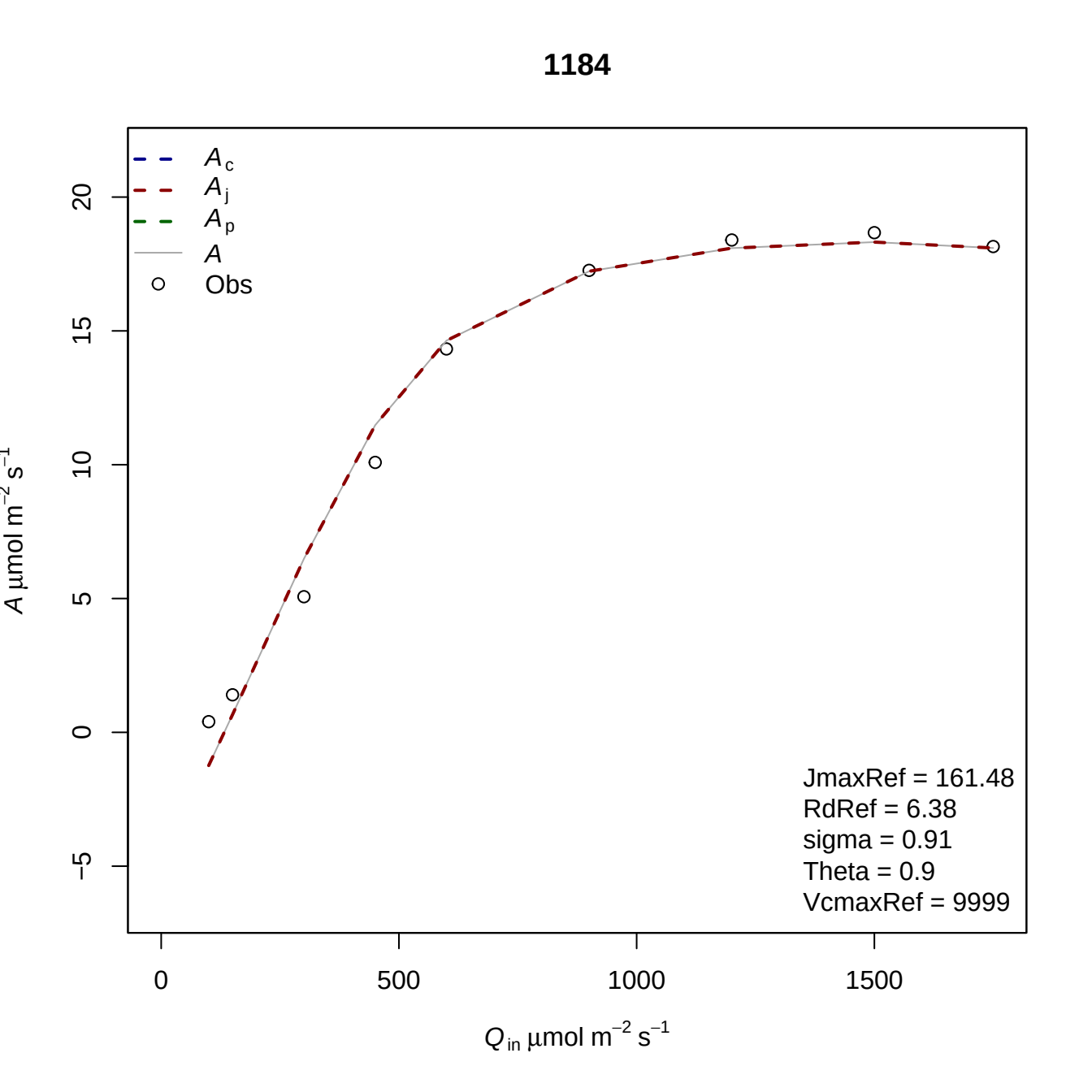

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 157.89
RdRef = 2.23
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

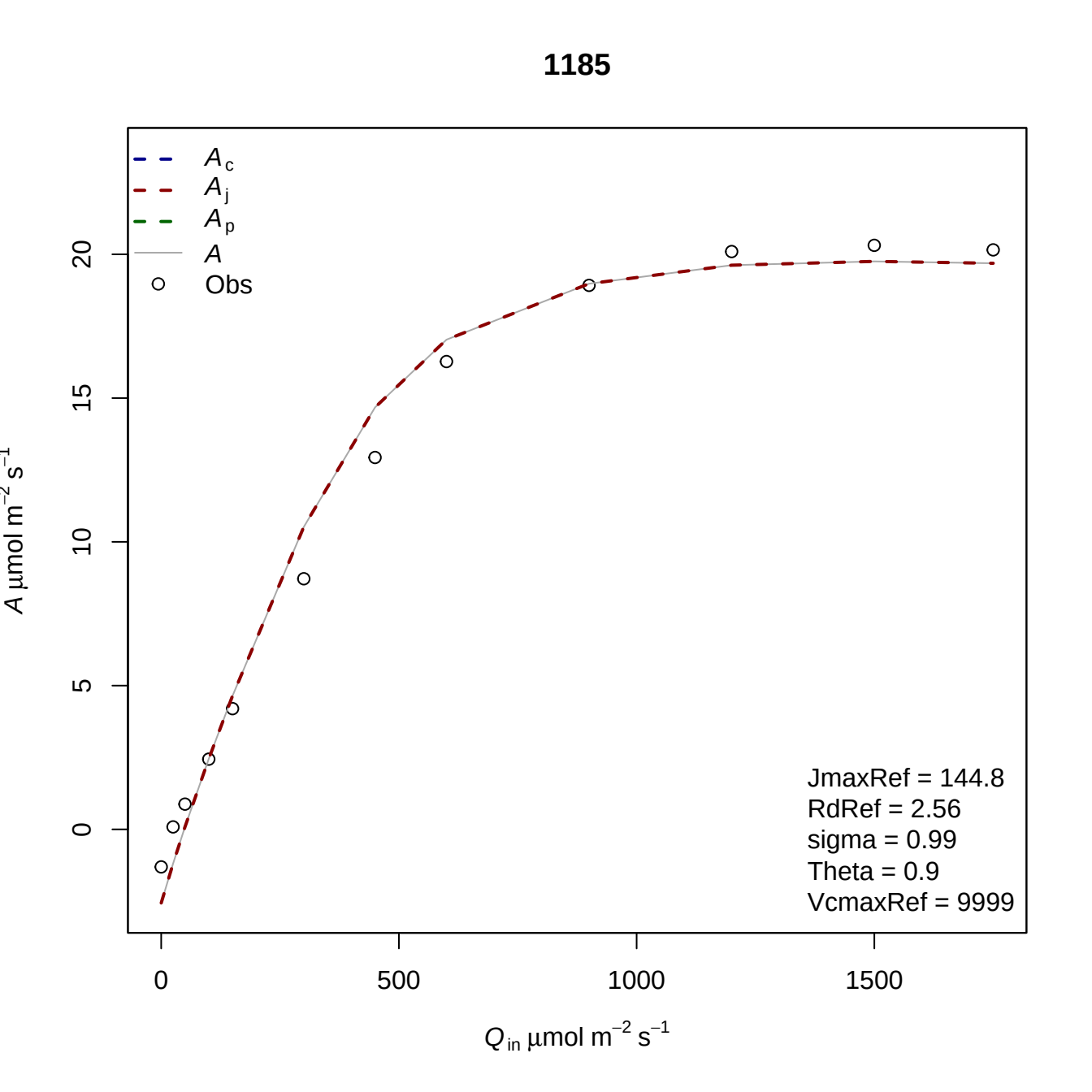

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 117.92
RdRef = 3.44
sigma = 1.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

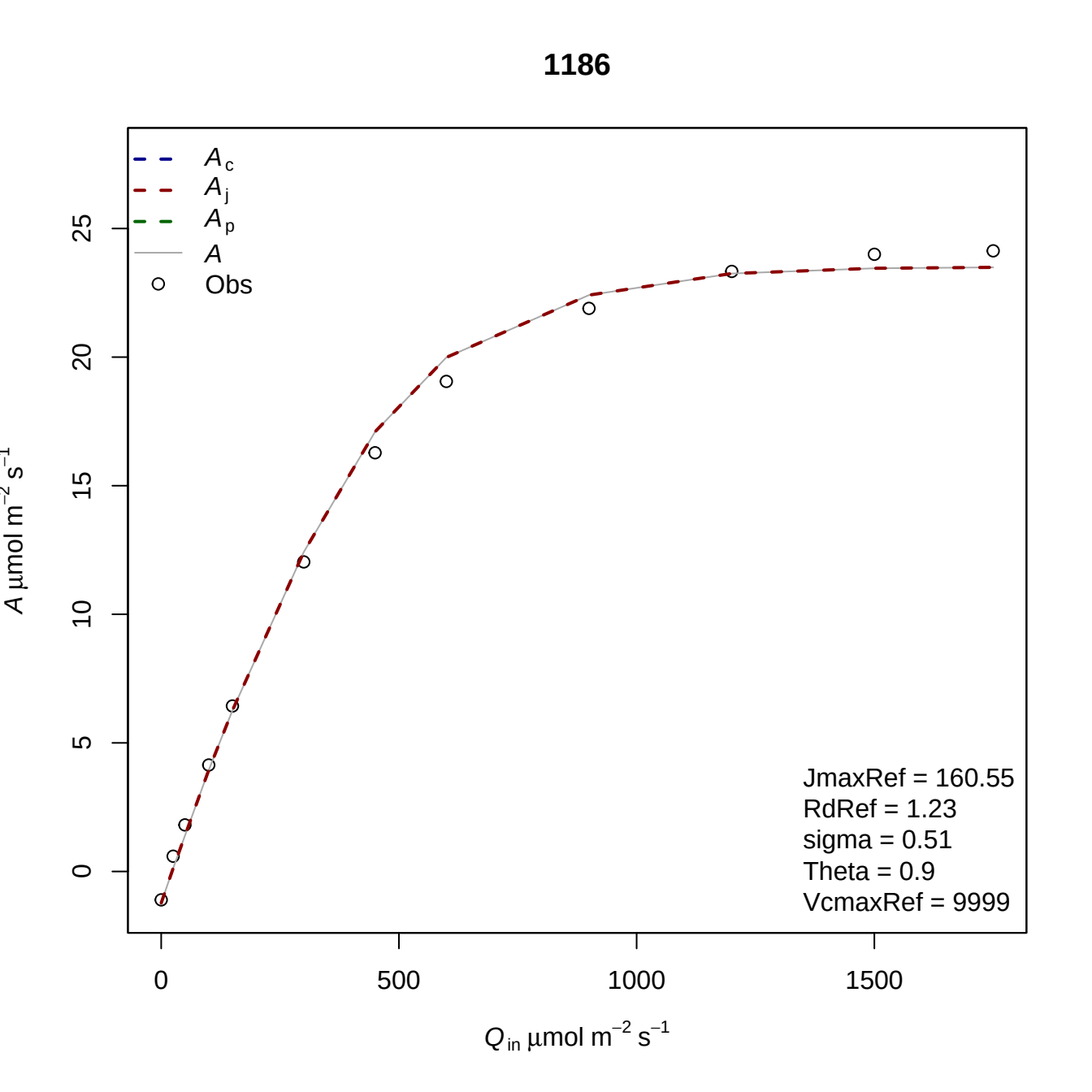

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 110.43
RdRef = 3.51
sigma = 1.51
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

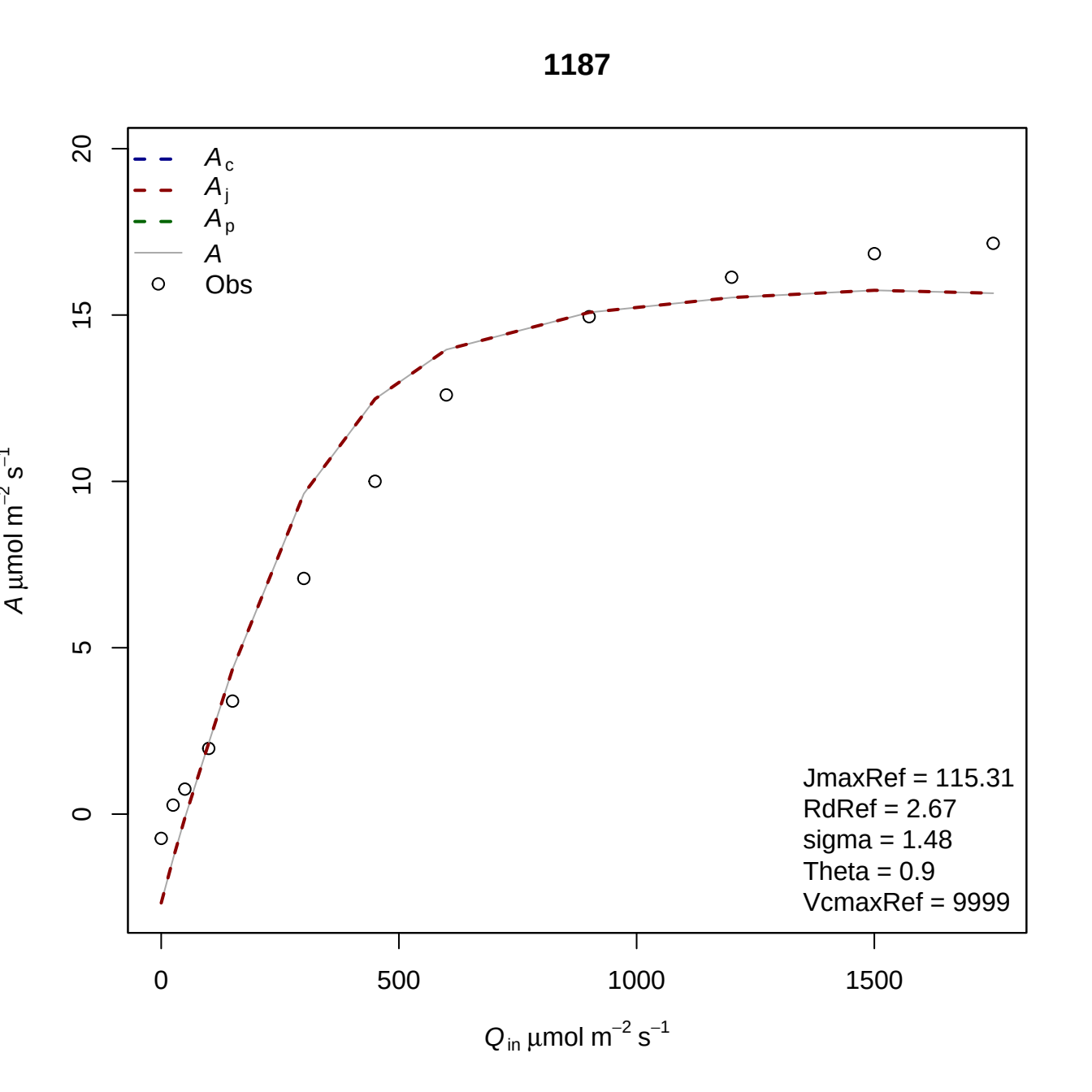

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 168.66
RdRef = 1.78
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

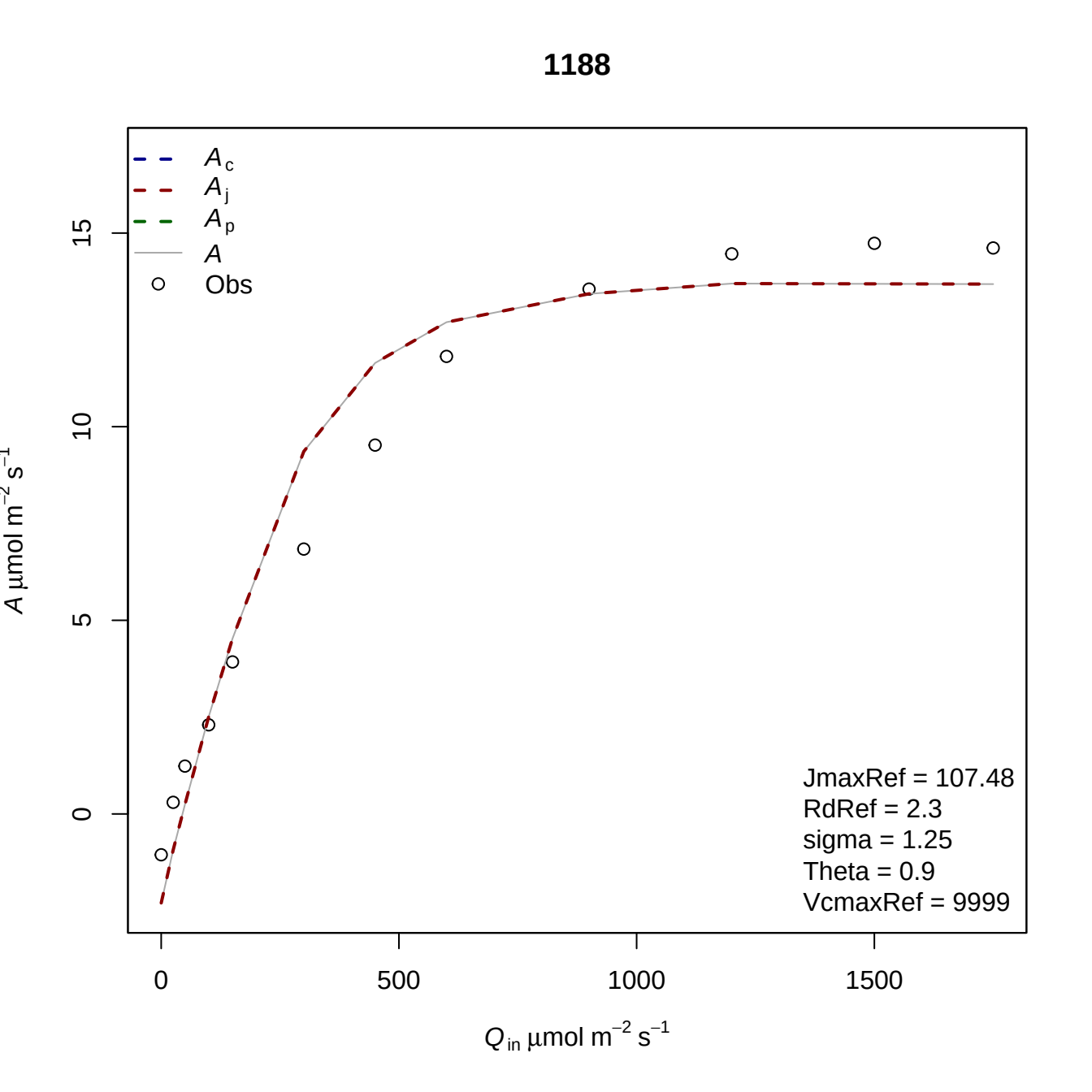

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 127.82
RdRef = 3.02
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

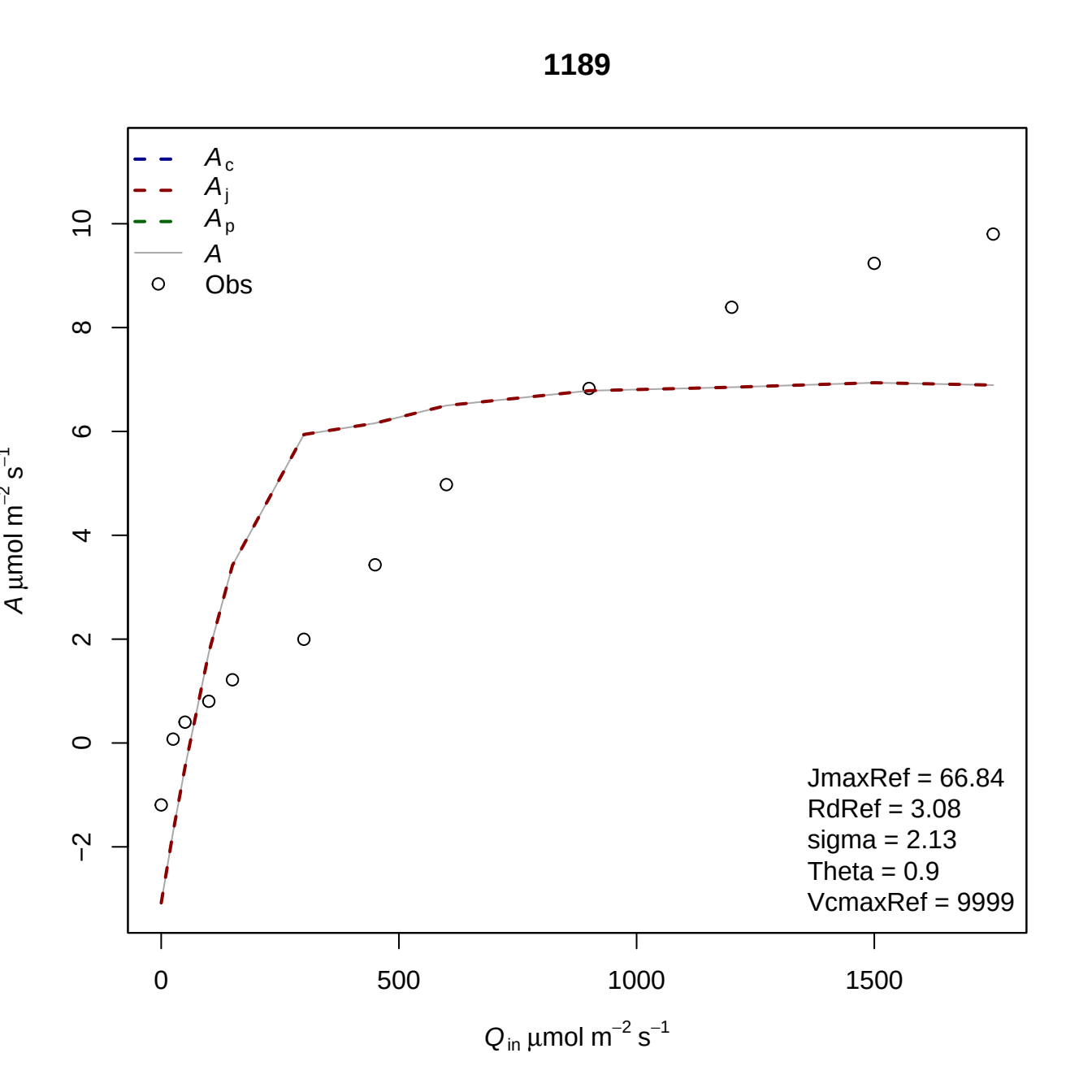

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 161.48
RdRef = 6.38
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

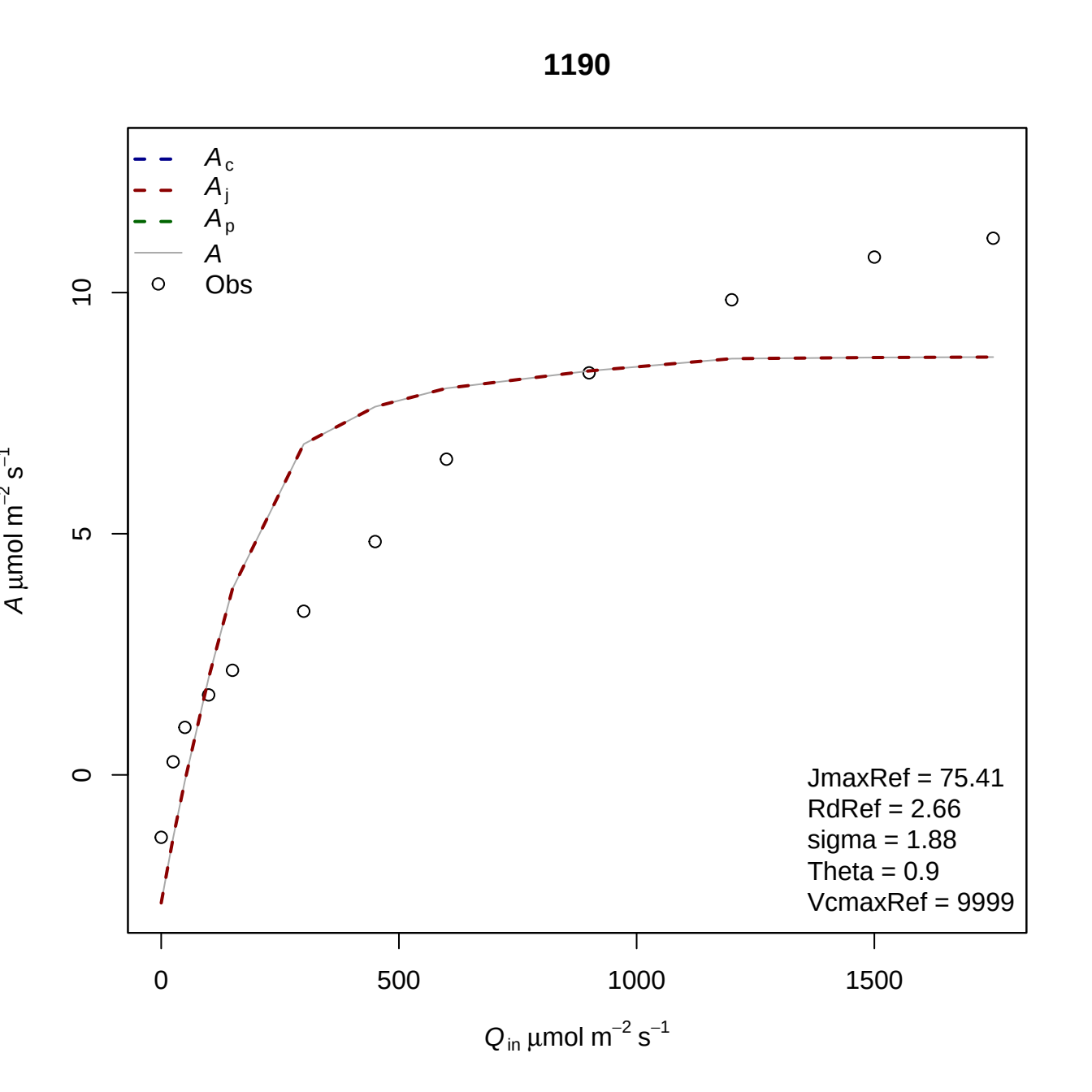

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 144.8
RdRef = 2.56
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

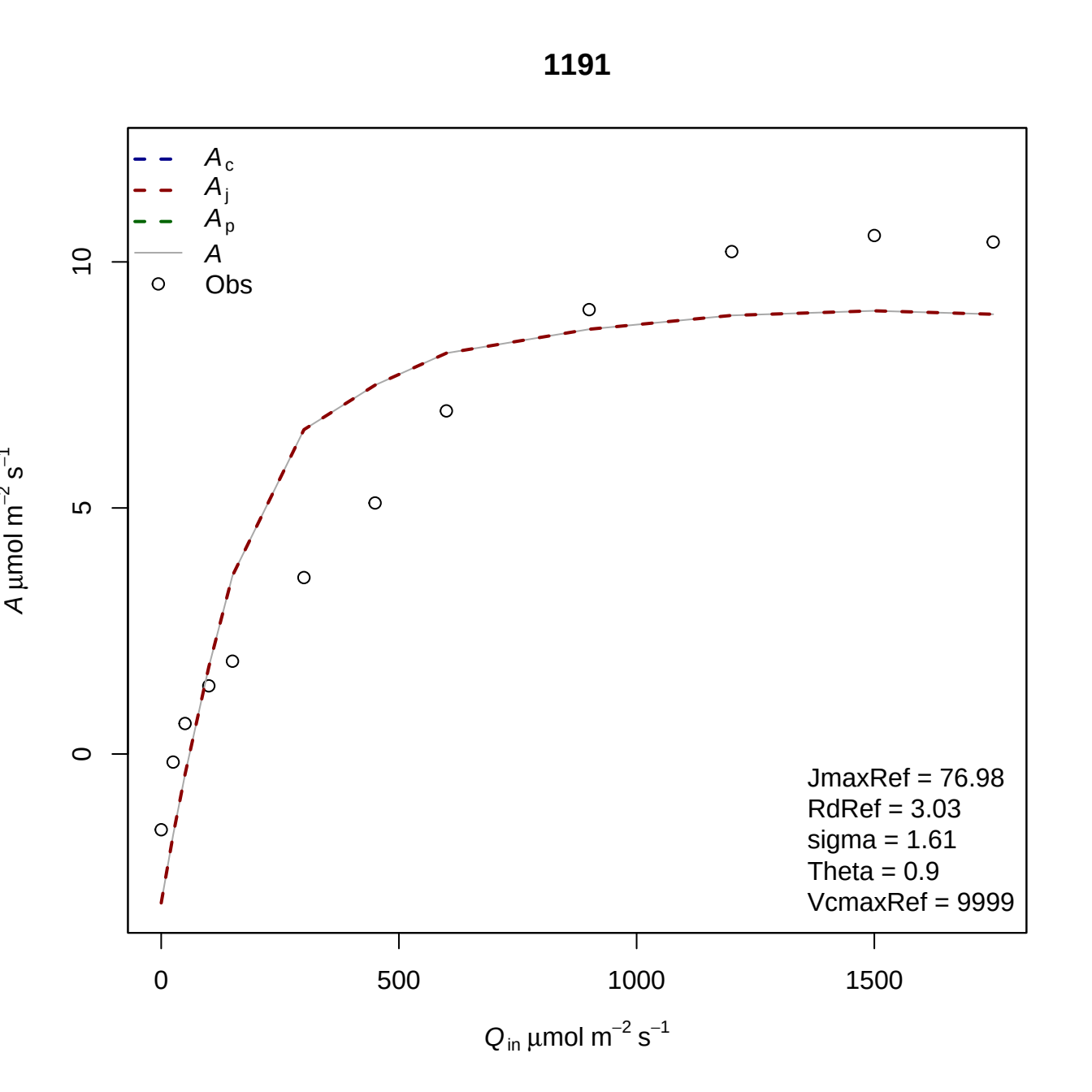

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 160.55
RdRef = 1.23
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

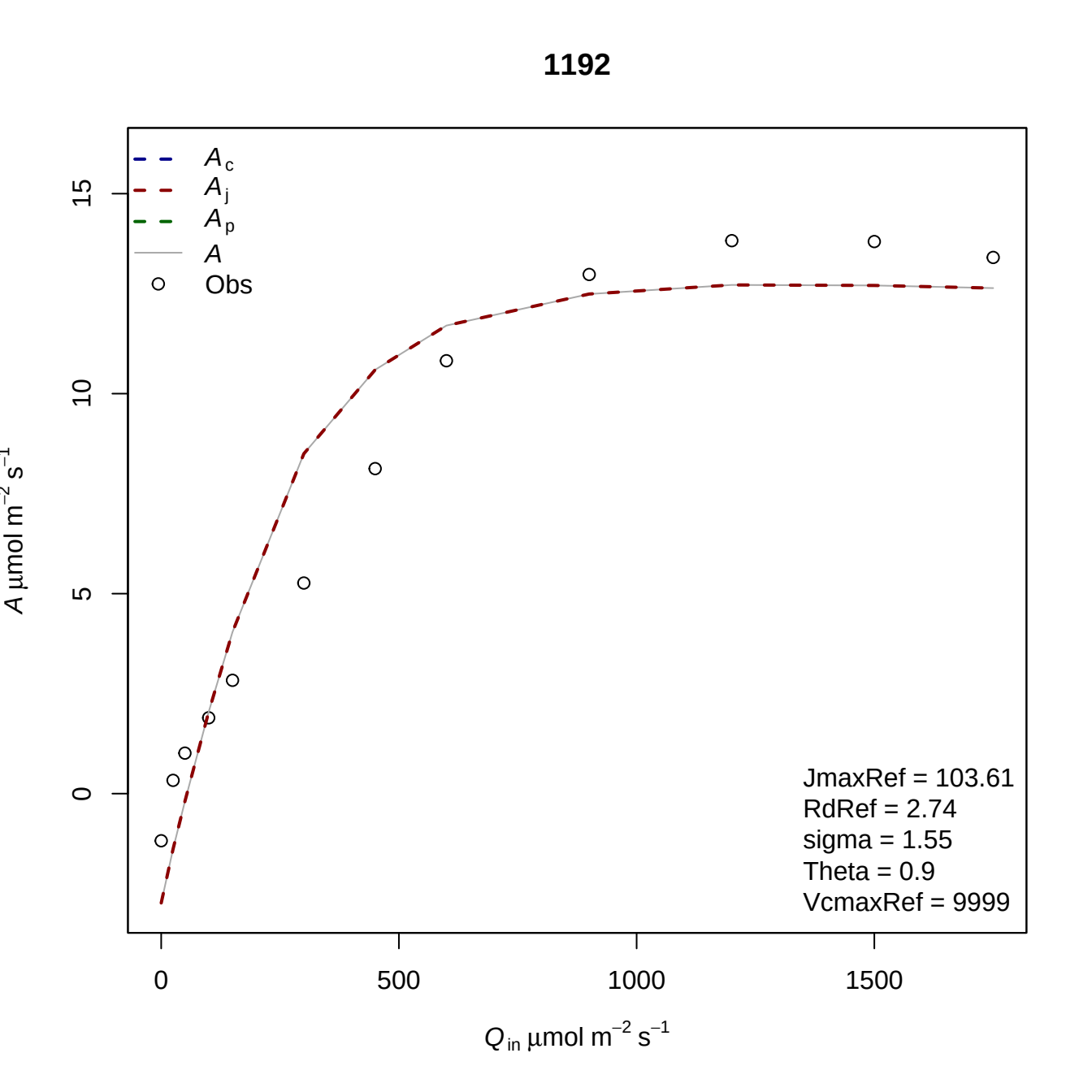

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 115.31
RdRef = 2.67
sigma = 1.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

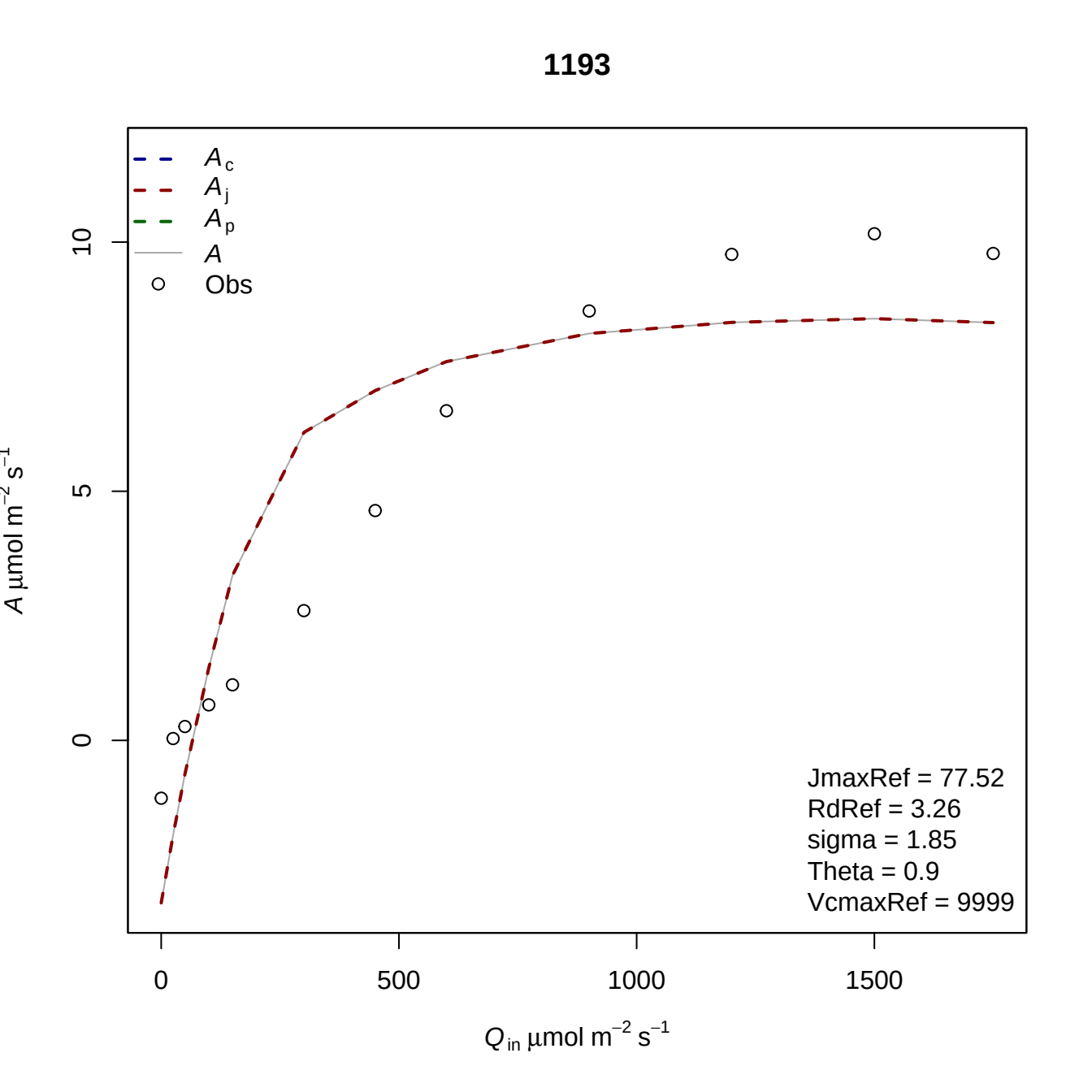

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 107.48
RdRef = 2.3
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

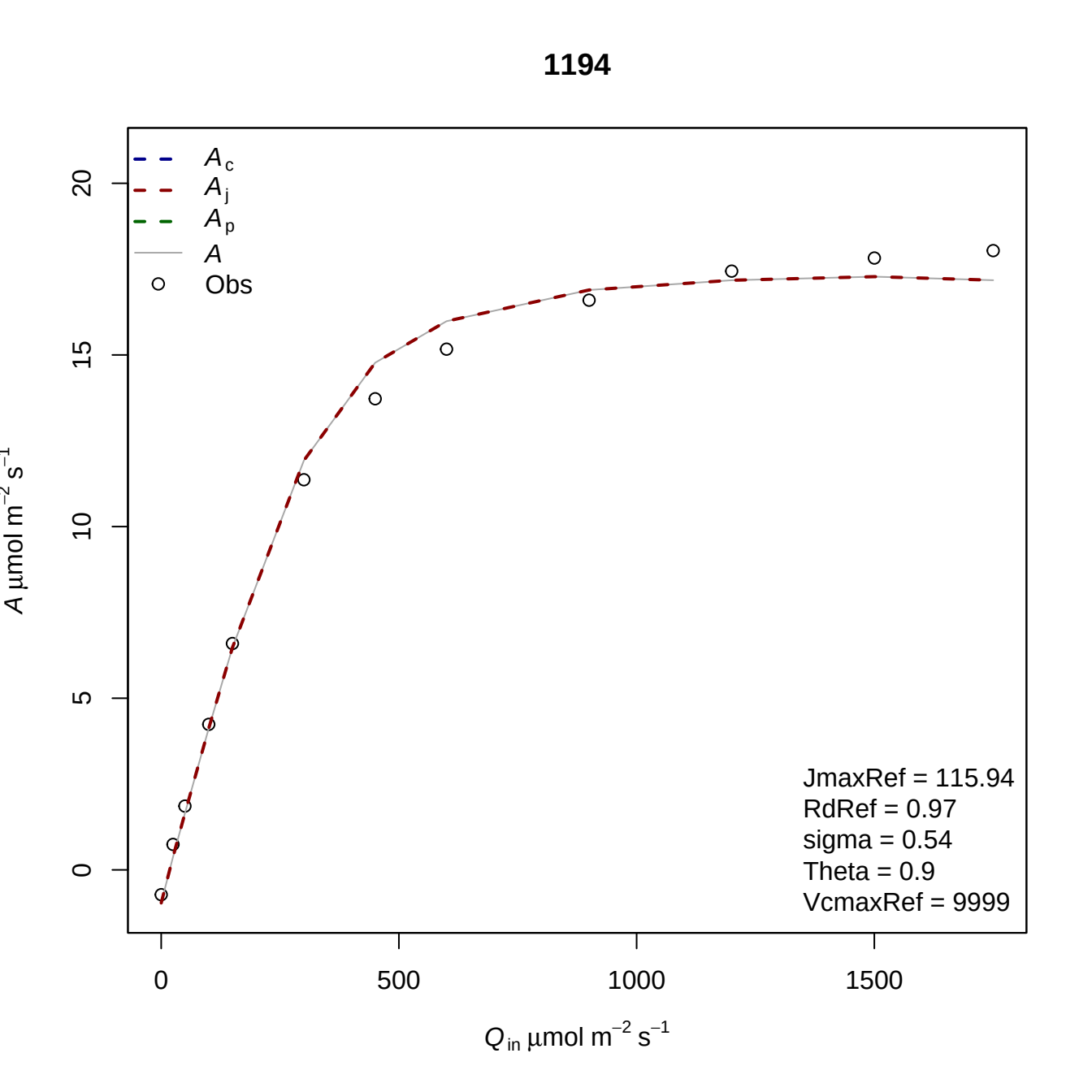

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 66.84
RdRef = 3.08
sigma = 2.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

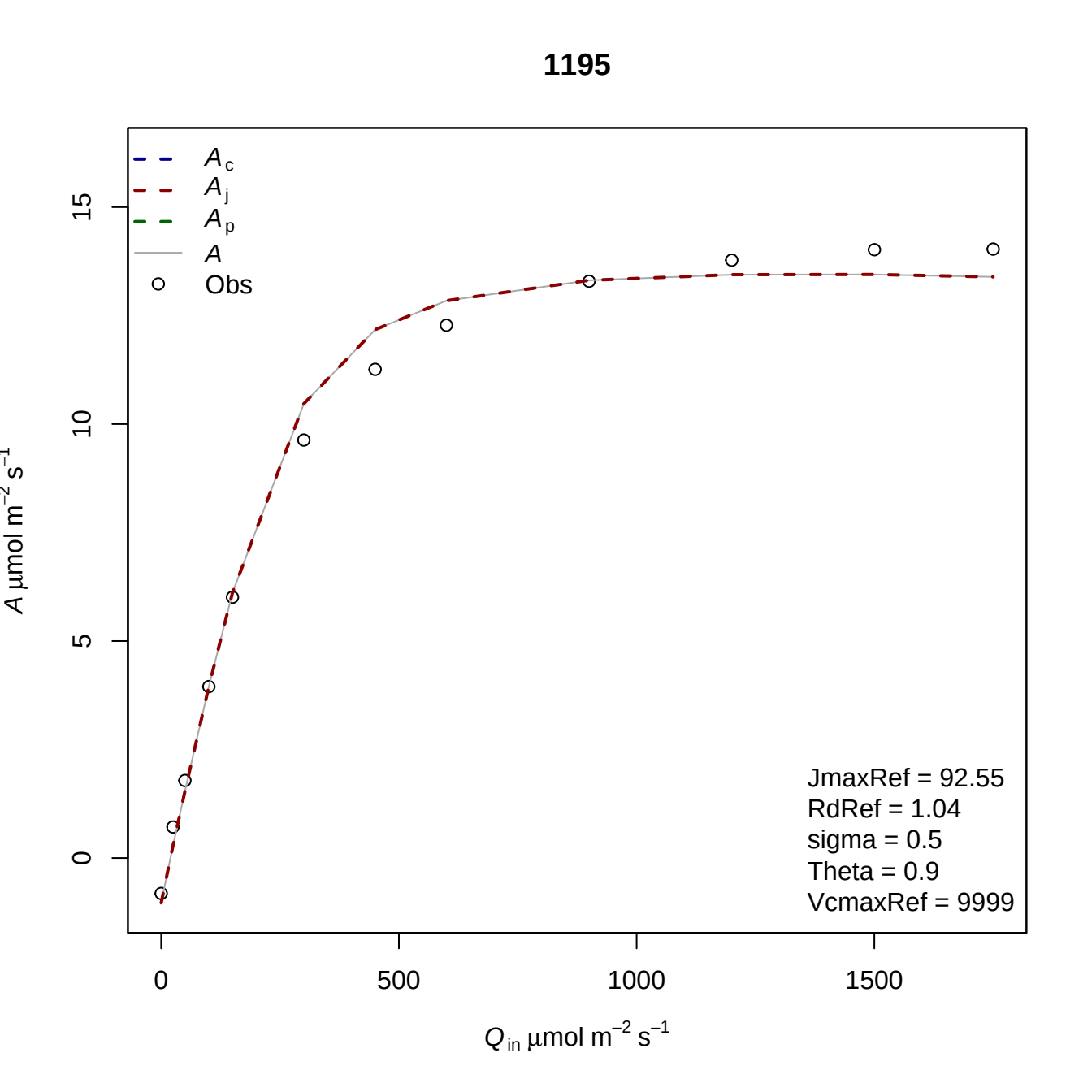

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 75.41
RdRef = 2.66
sigma = 1.88
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

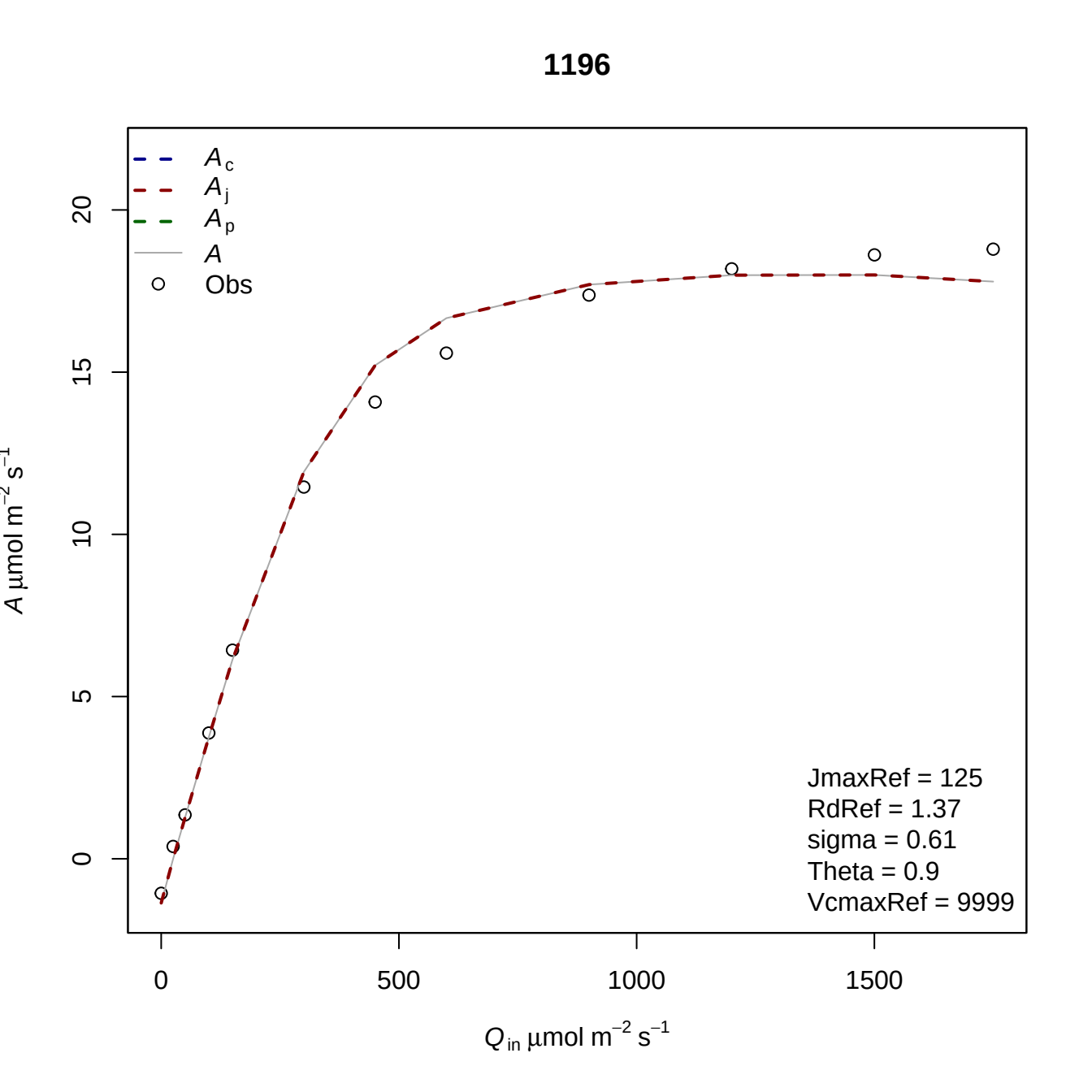

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 76.98
RdRef = 3.03
sigma = 1.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

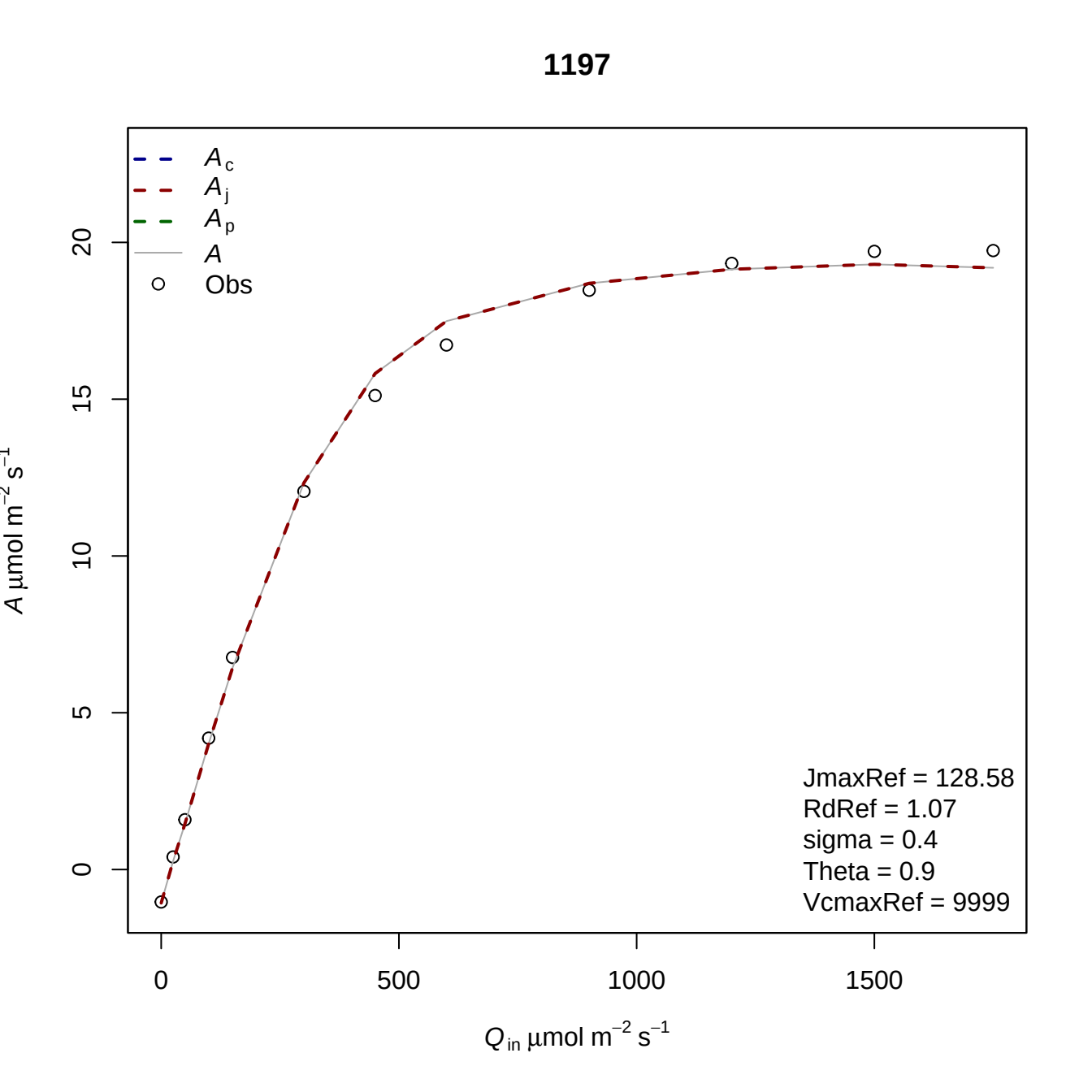

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 103.61
RdRef = 2.74
sigma = 1.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

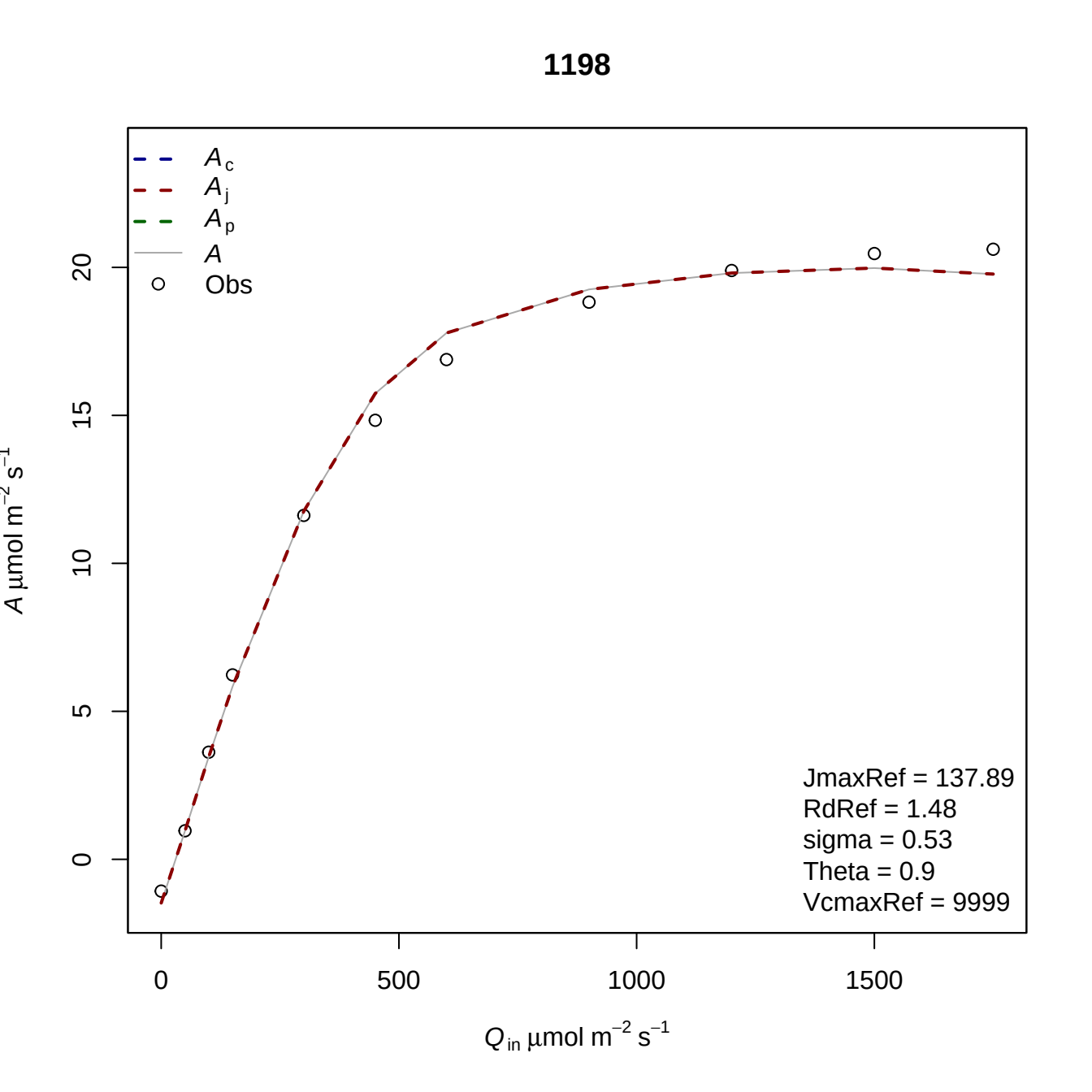

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 77.52
RdRef = 3.26
sigma = 1.85
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

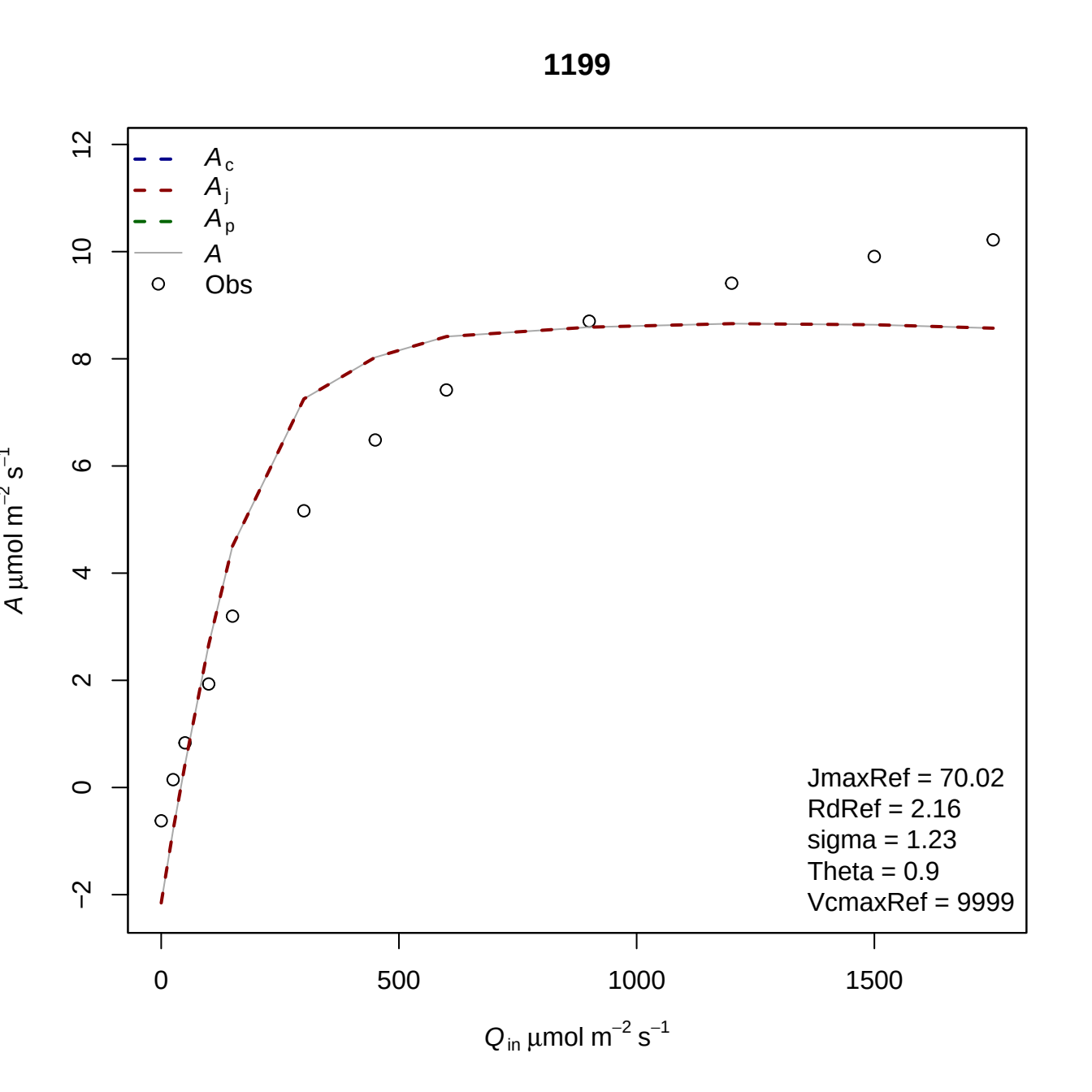

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 115.94
RdRef = 0.97
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

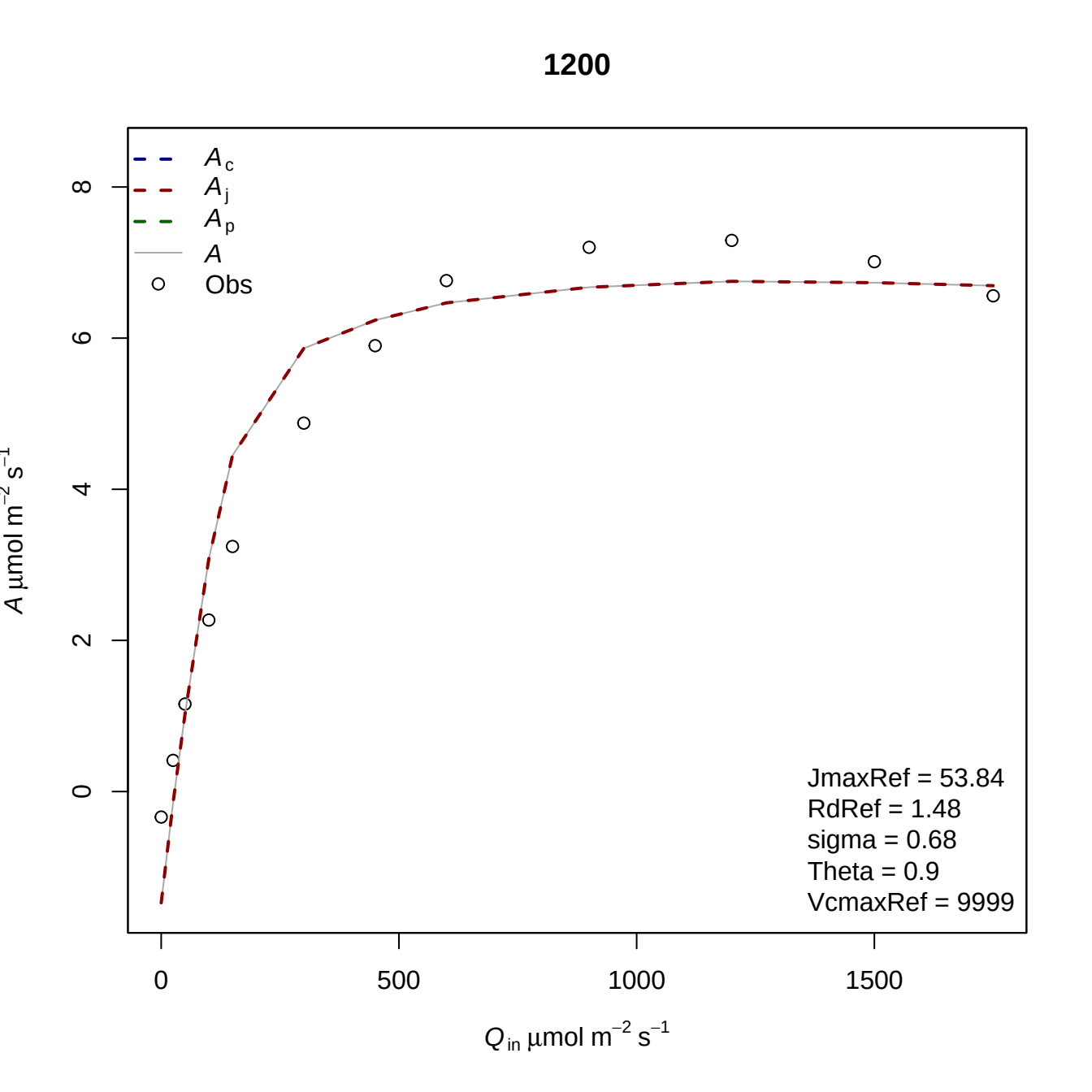

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 92.55
RdRef = 1.04
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

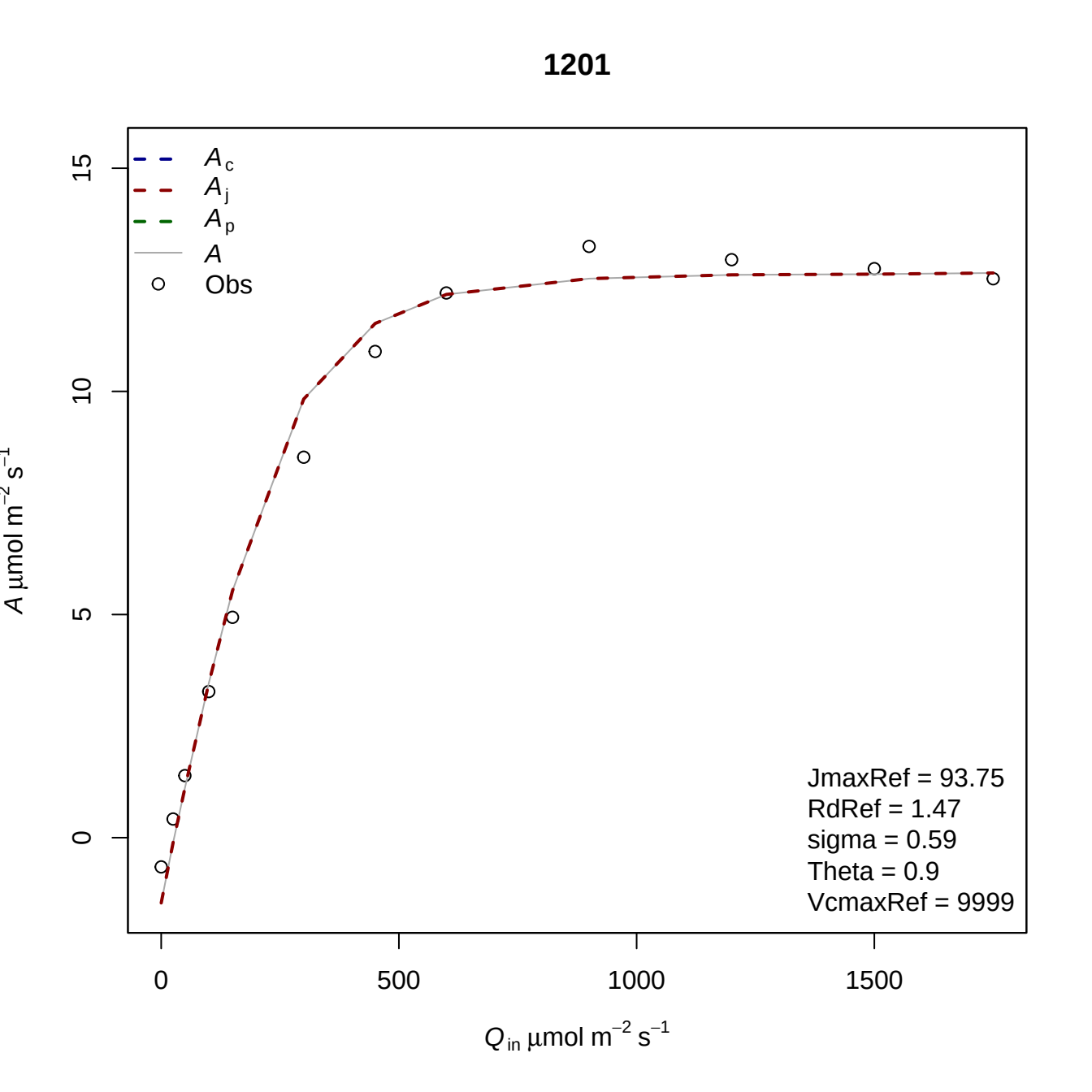

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 125
RdRef = 1.37
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

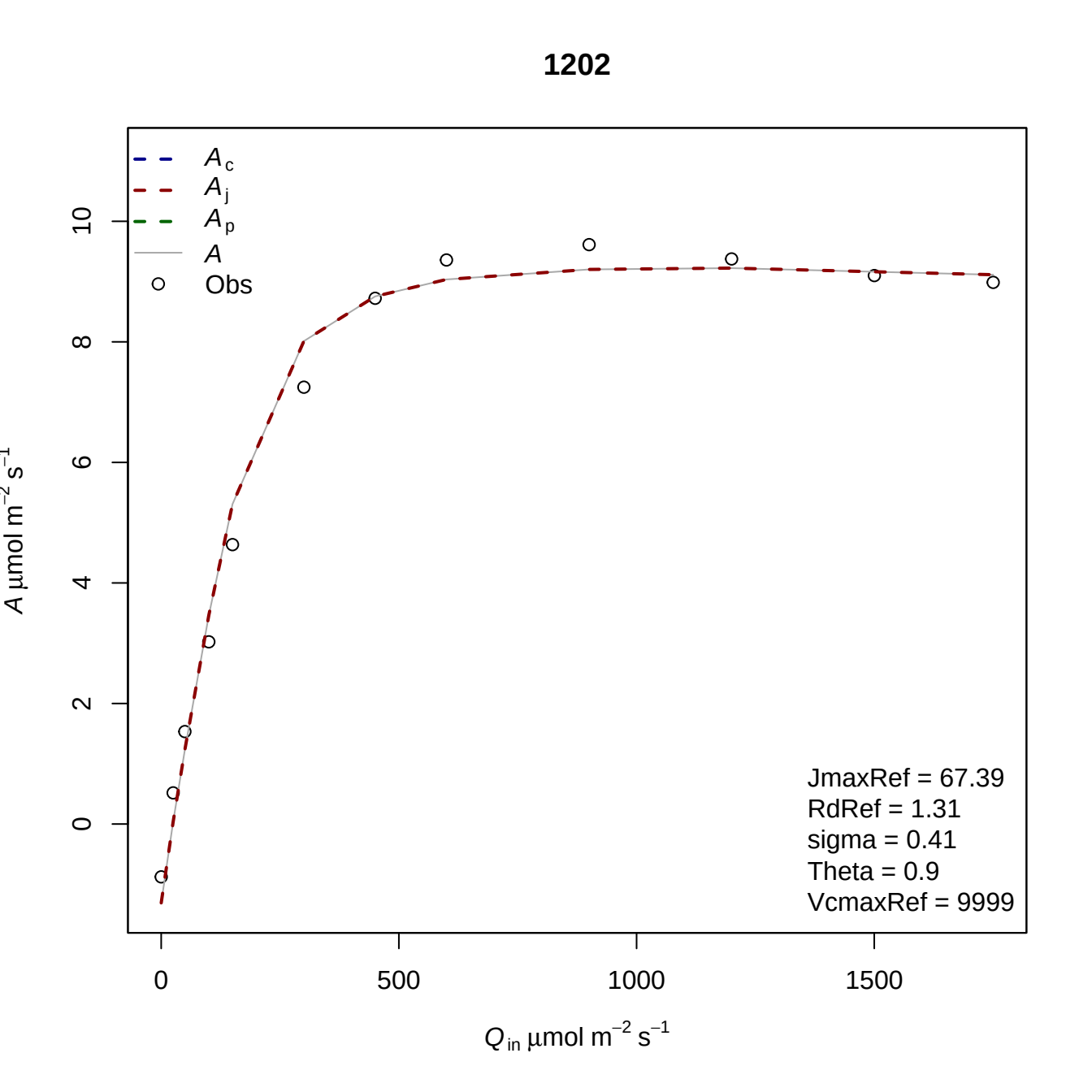

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 128.58
RdRef = 1.07
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

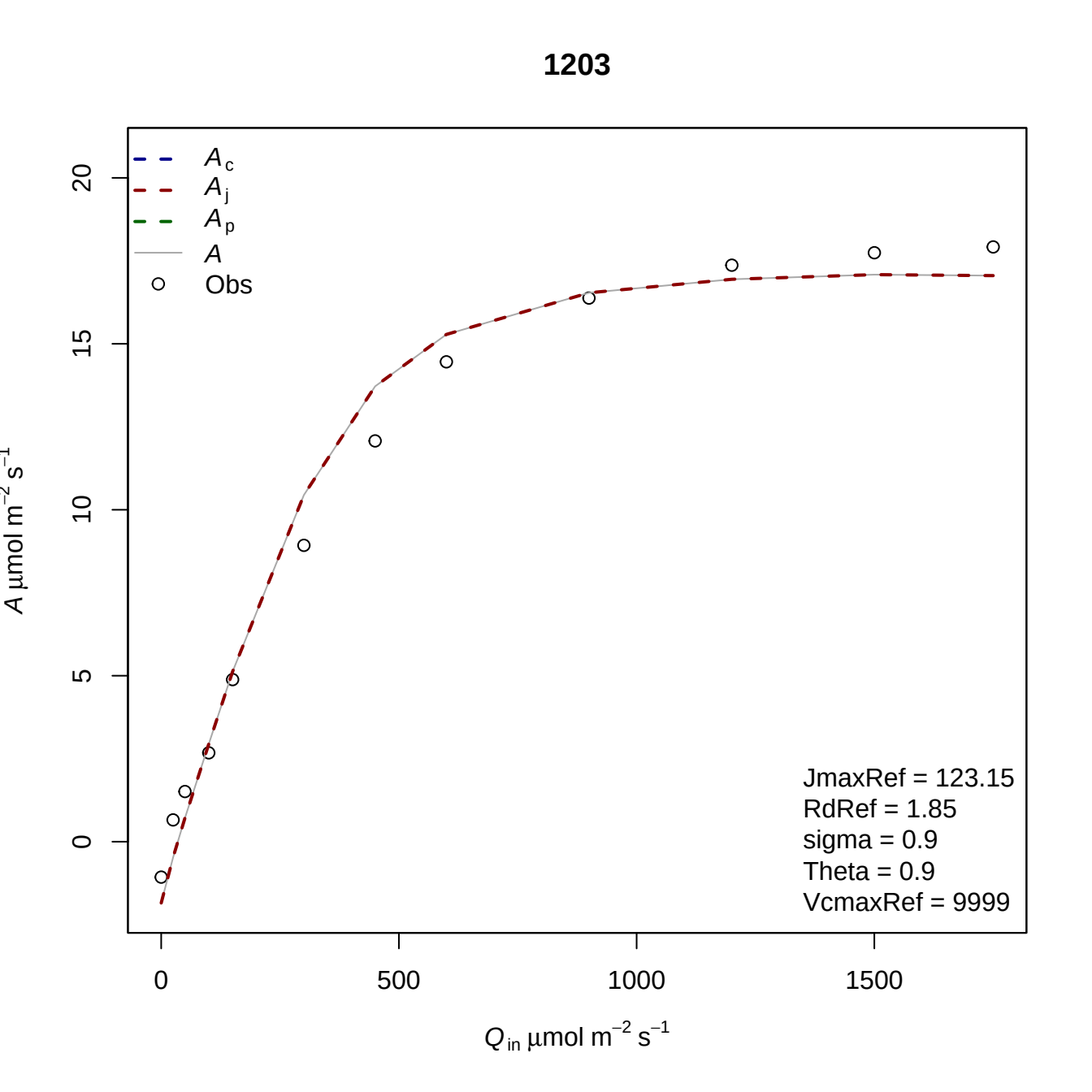

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 137.89
RdRef = 1.48
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

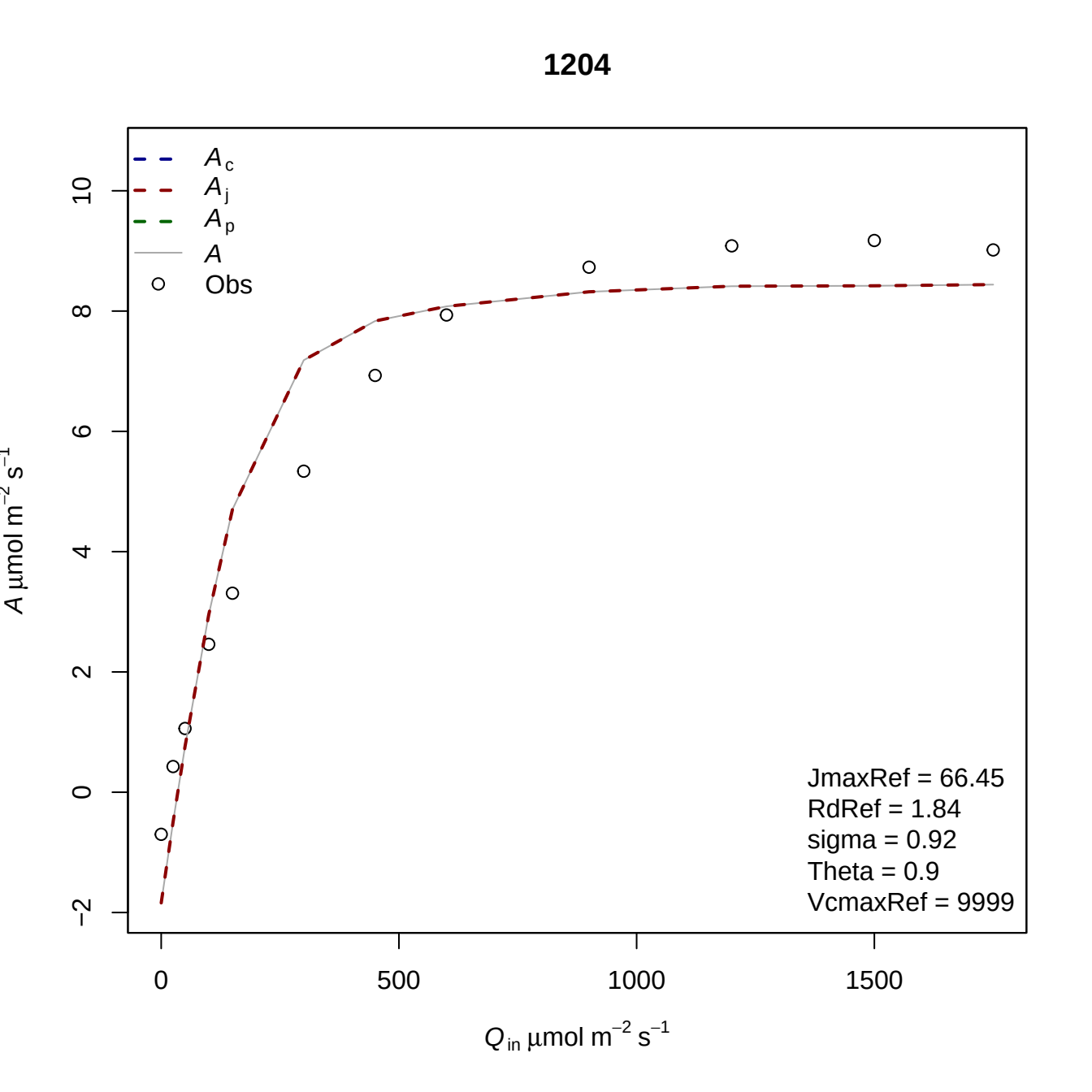

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 70.02
RdRef = 2.16
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

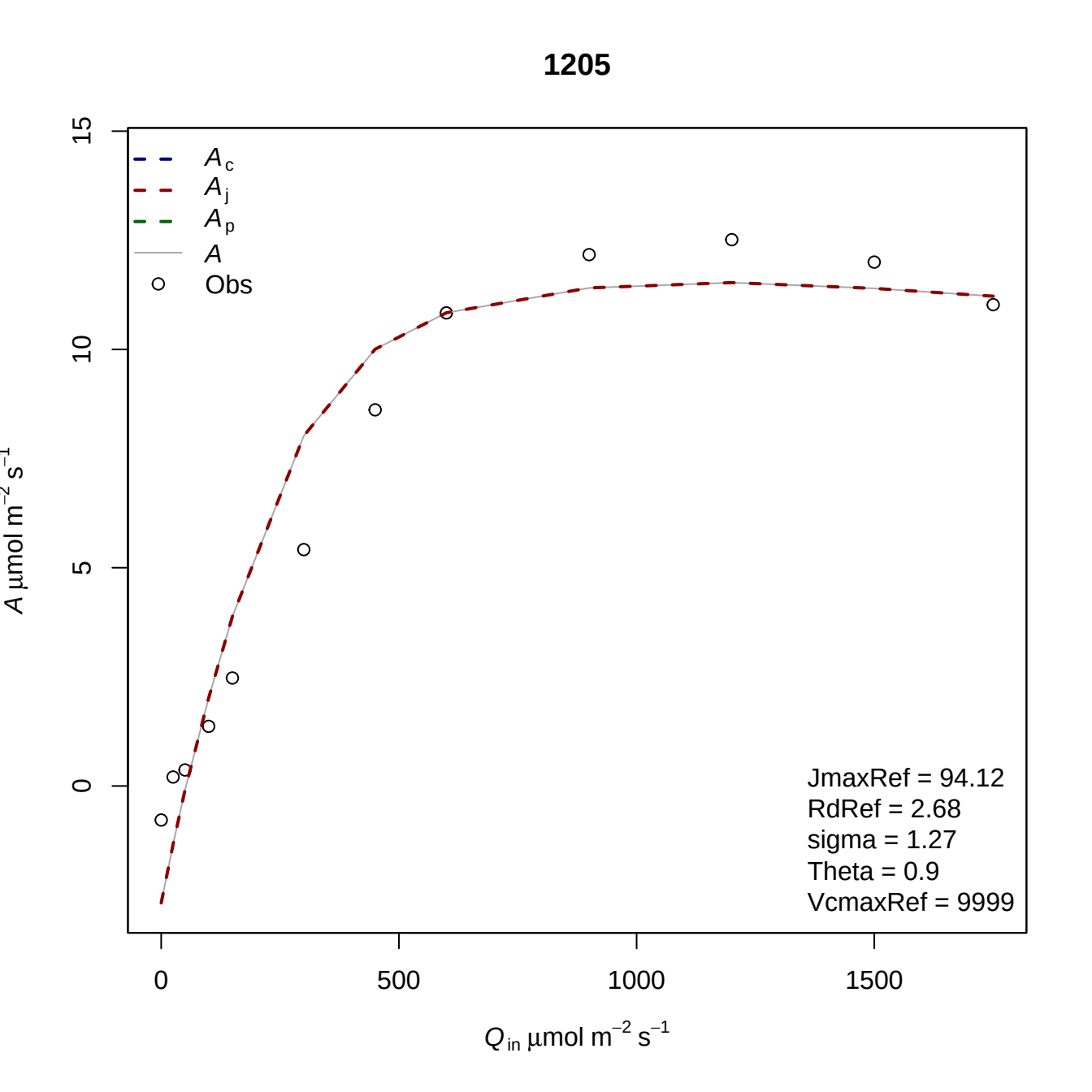

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 53.84
RdRef = 1.48
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

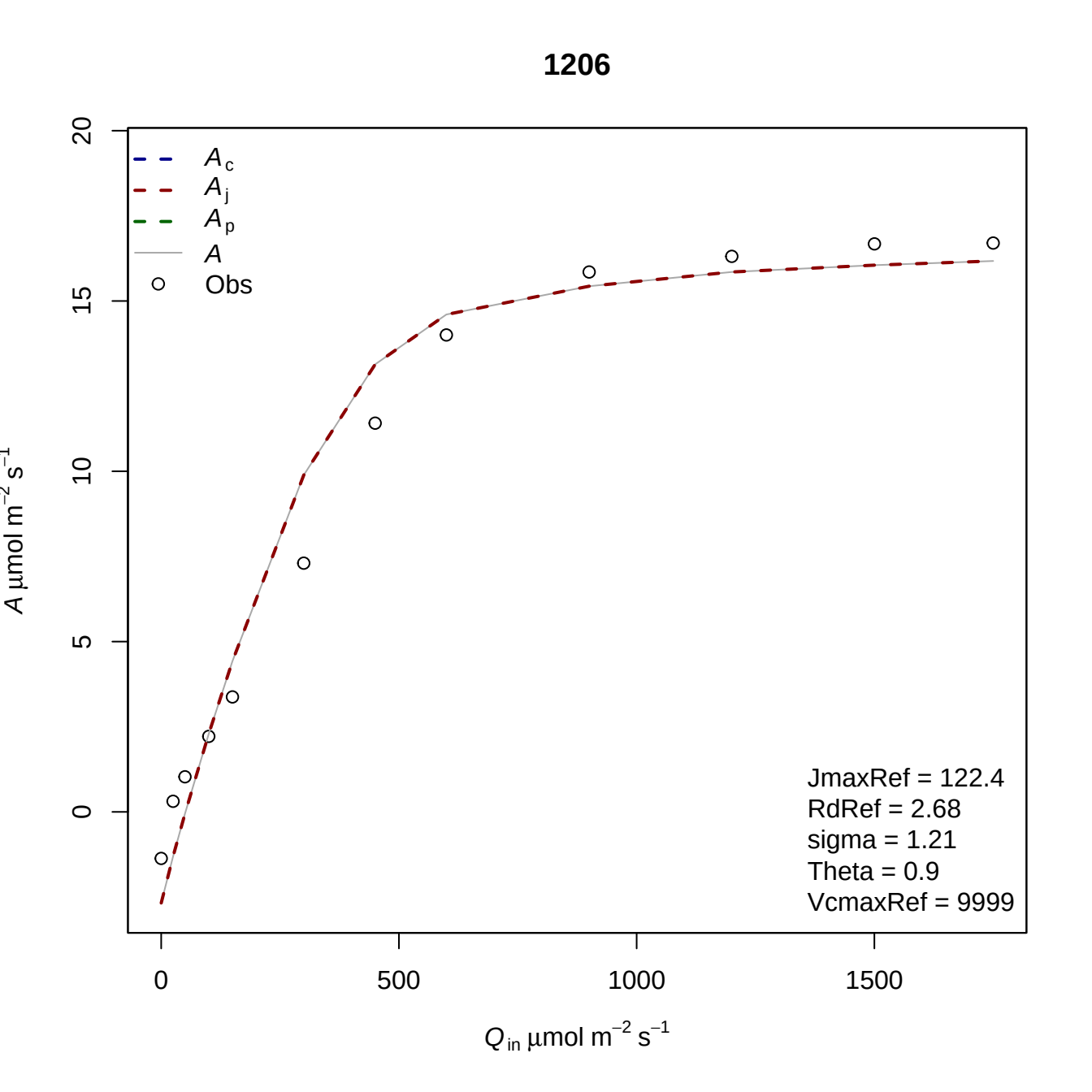

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 93.75
RdRef = 1.47
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

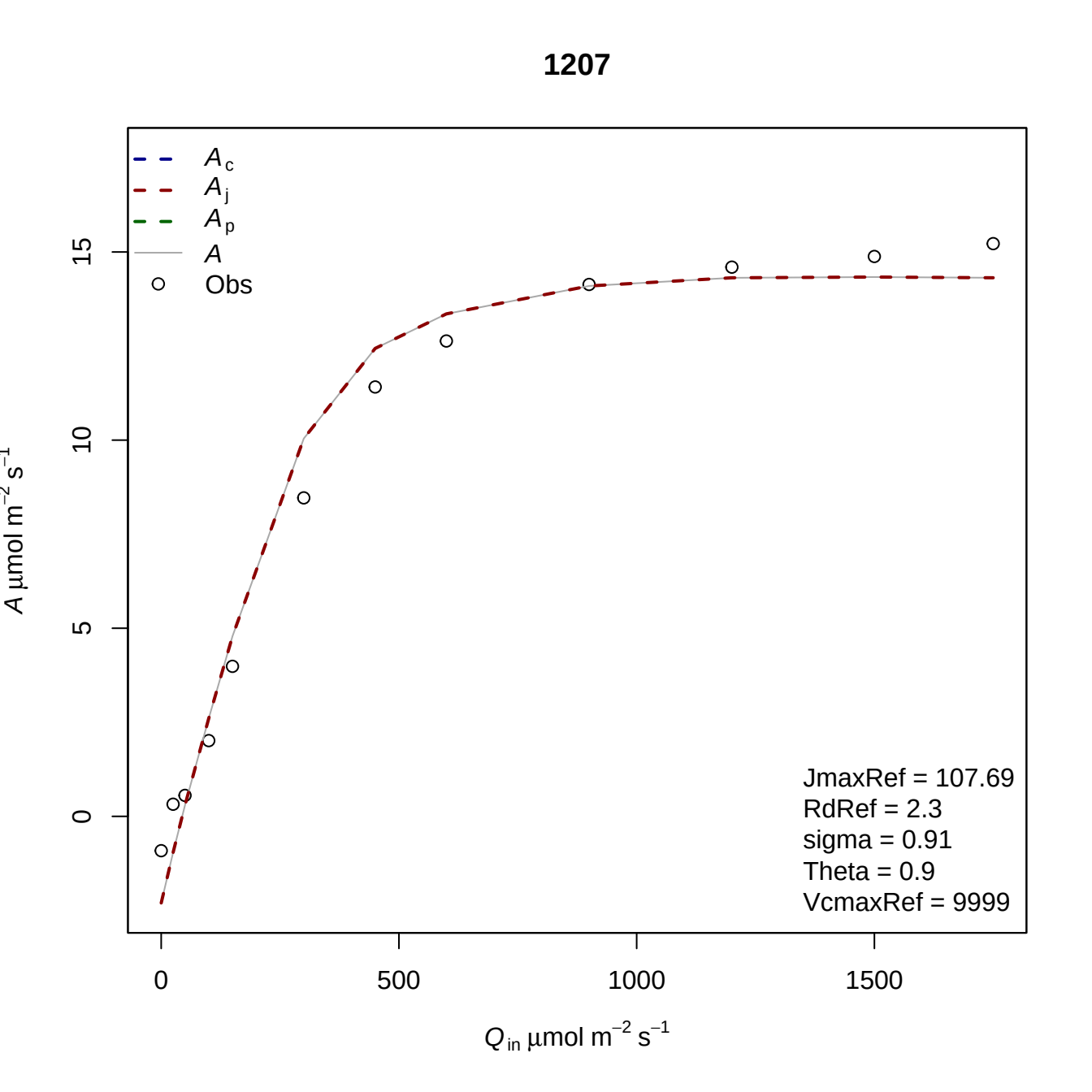

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 67.39
RdRef = 1.31
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

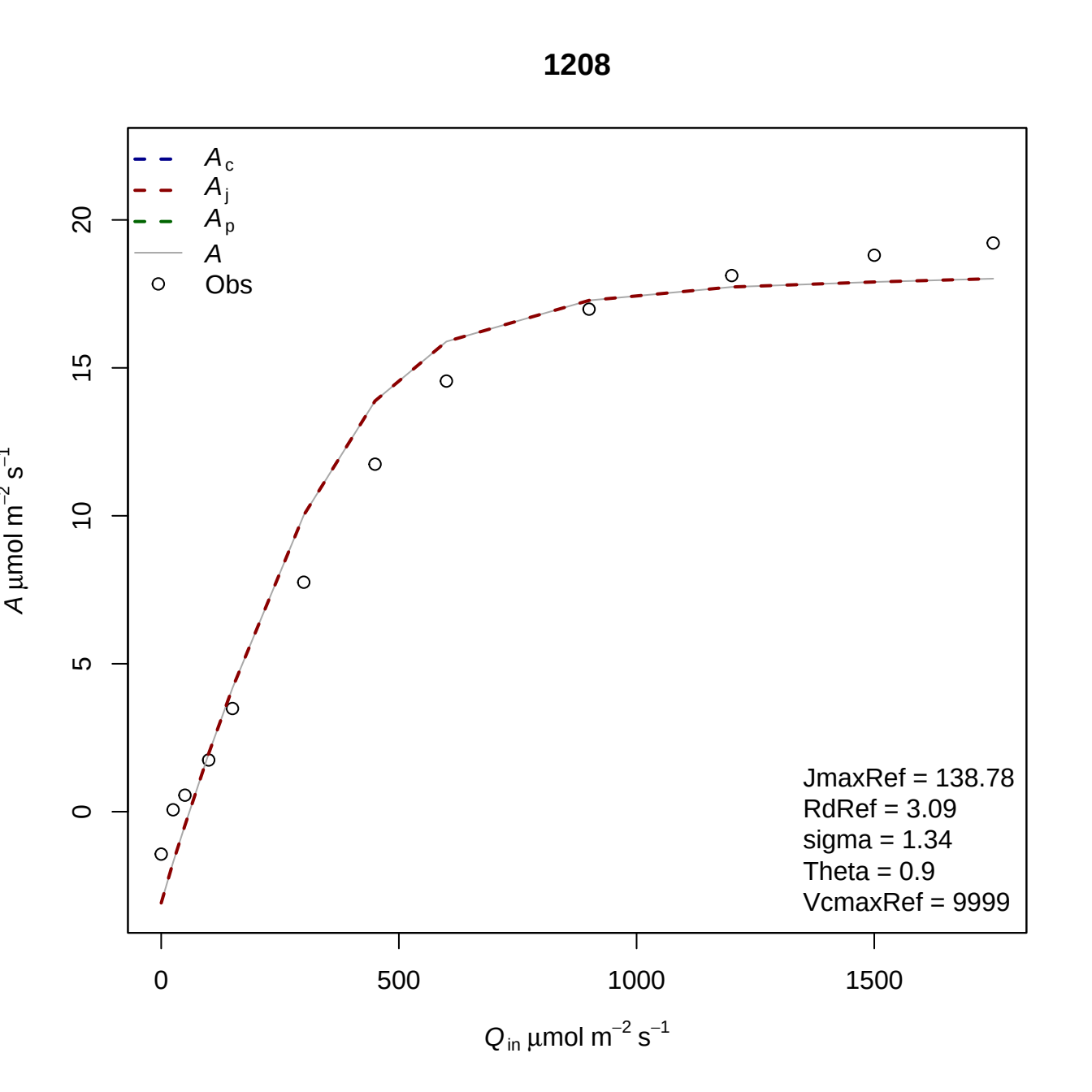

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 123.15
RdRef = 1.85
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

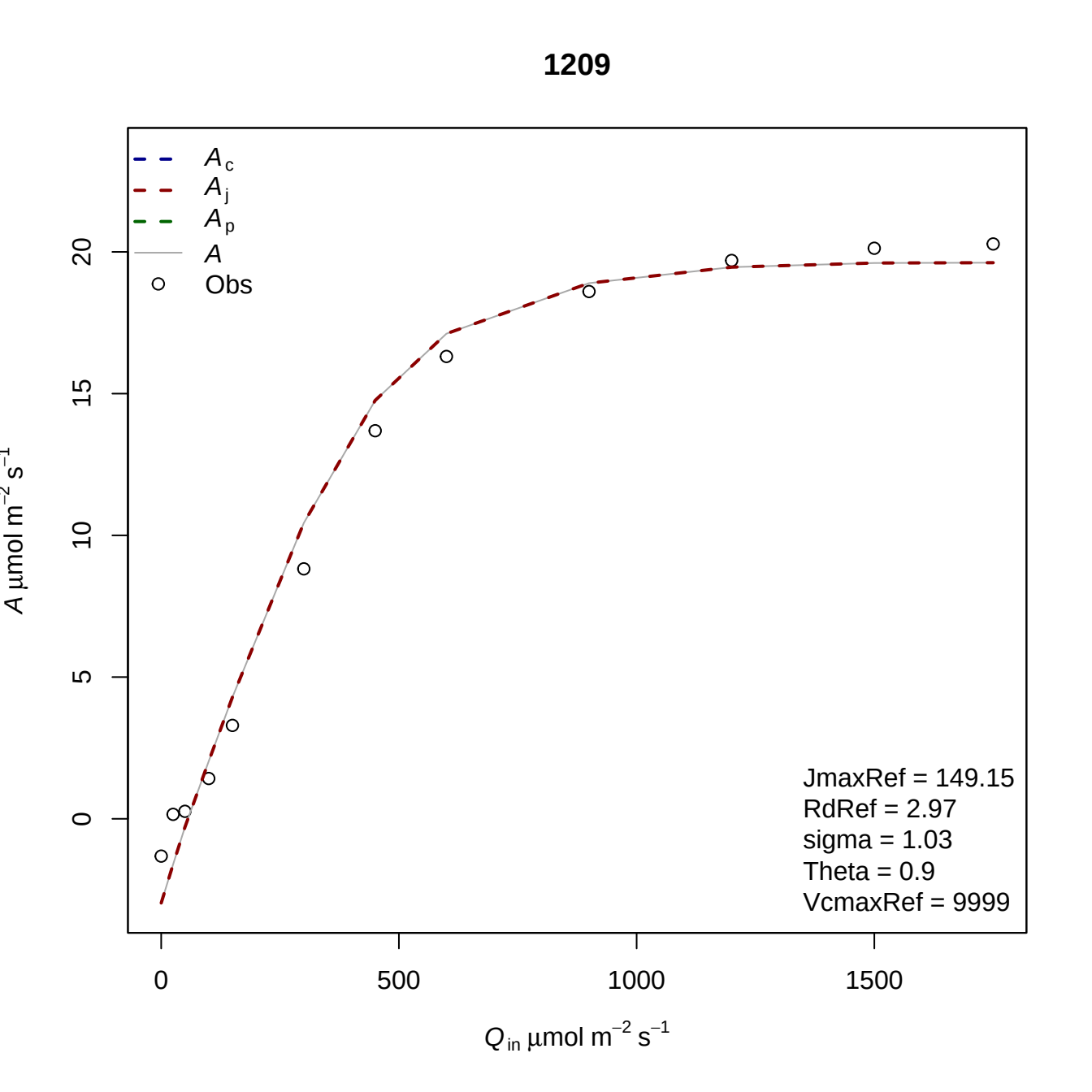

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 66.45
RdRef = 1.84
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

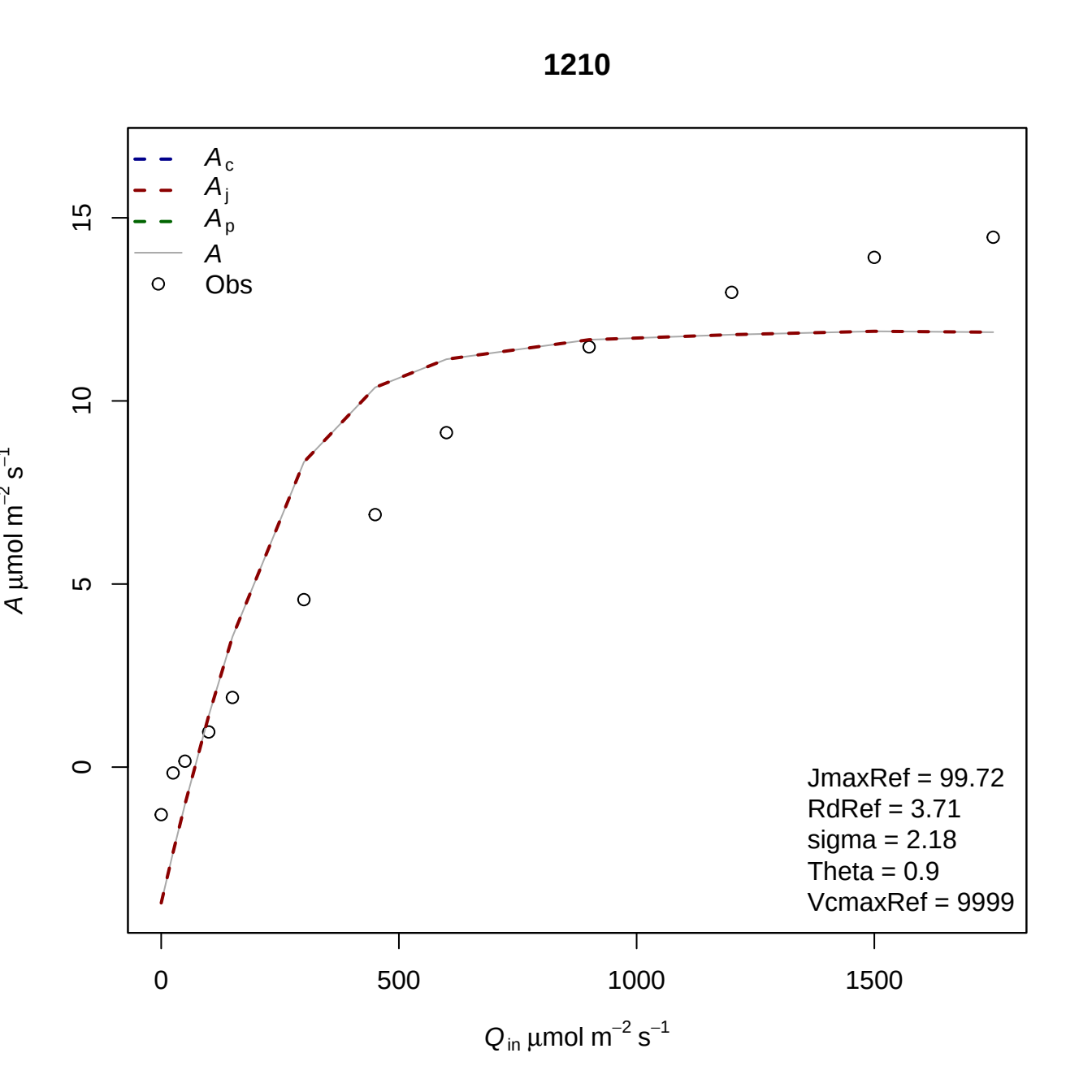

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 94.12
RdRef = 2.68
sigma = 1.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

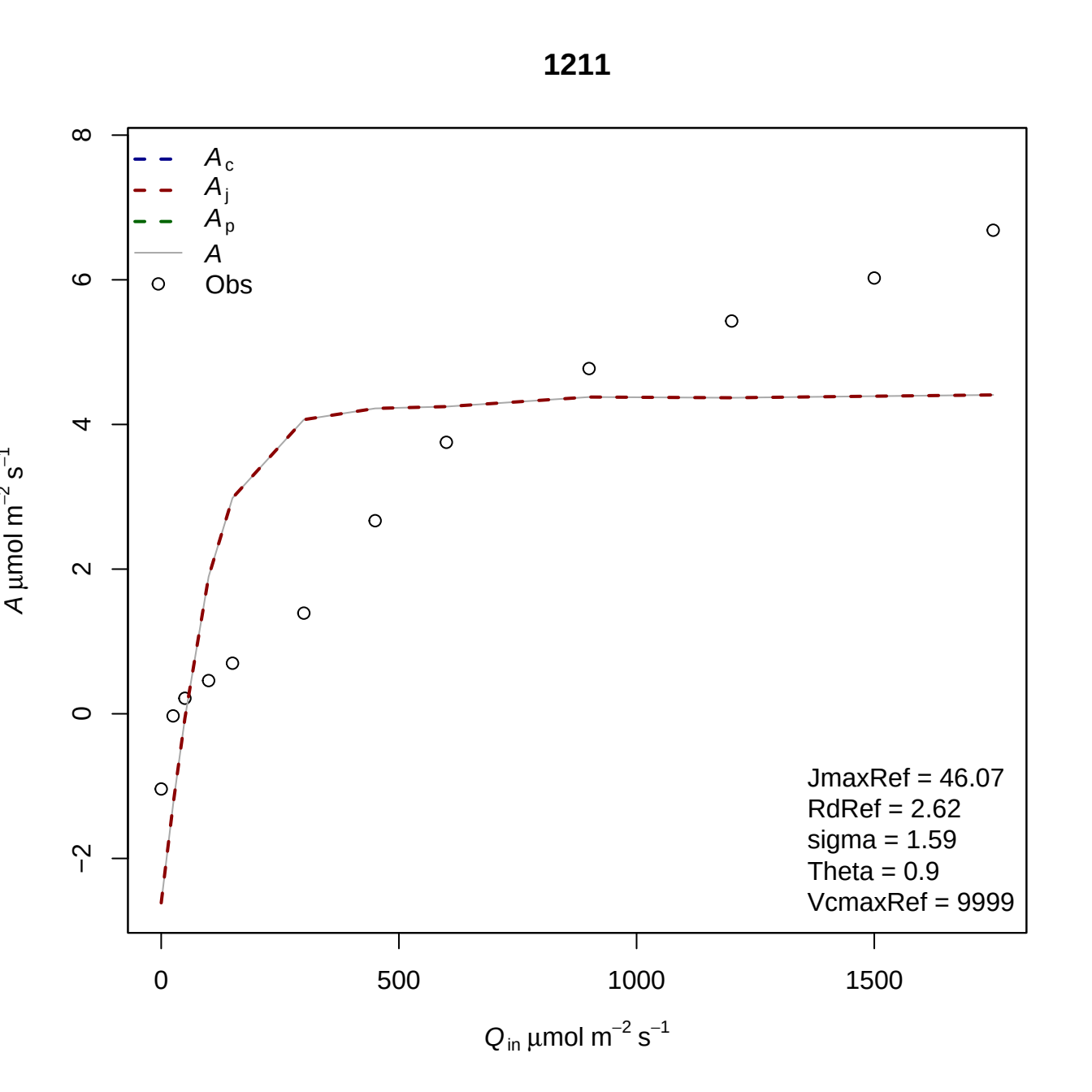

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 122.4
RdRef = 2.68
sigma = 1.21
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

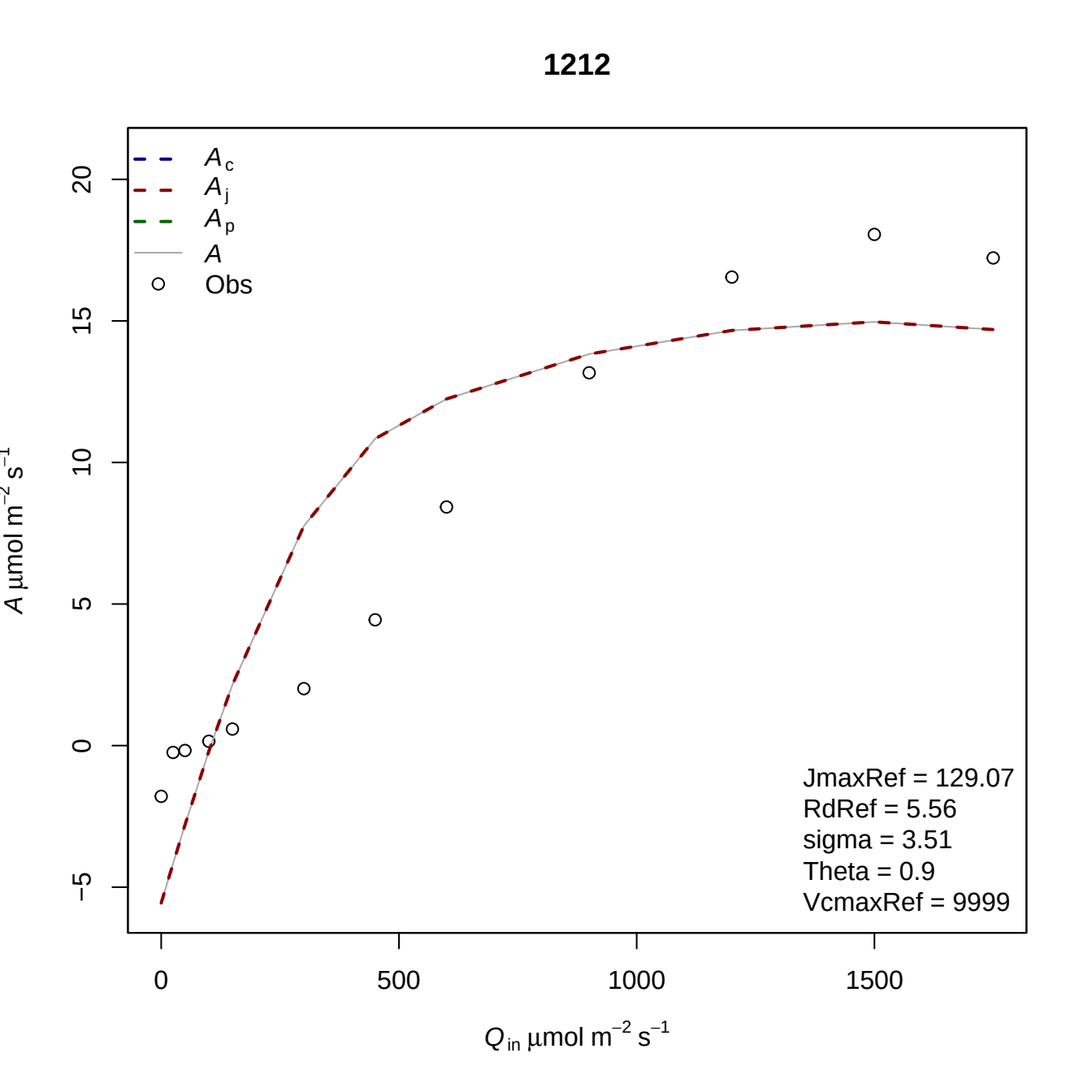

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 107.69
RdRef = 2.3
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

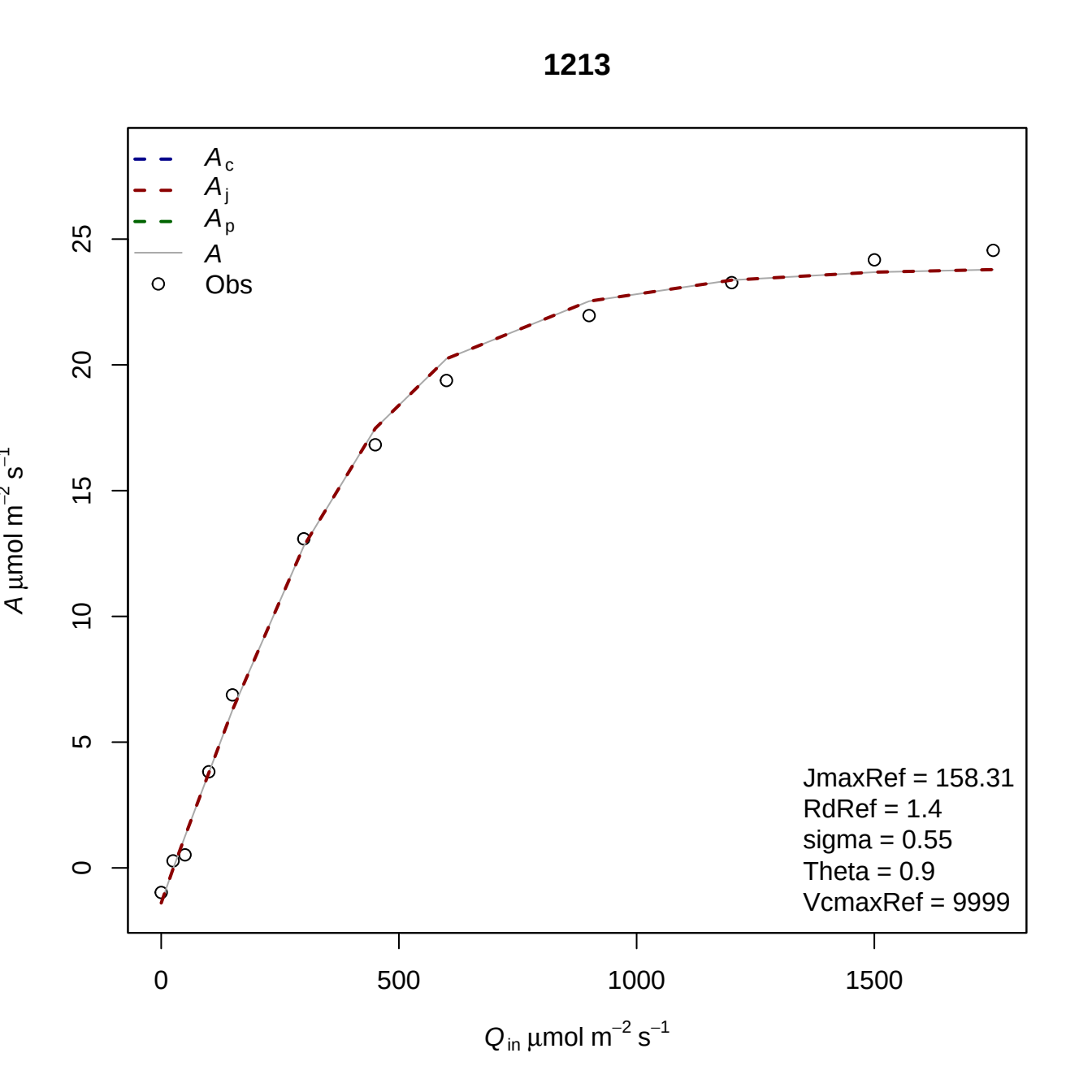

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 138.78
RdRef = 3.09
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

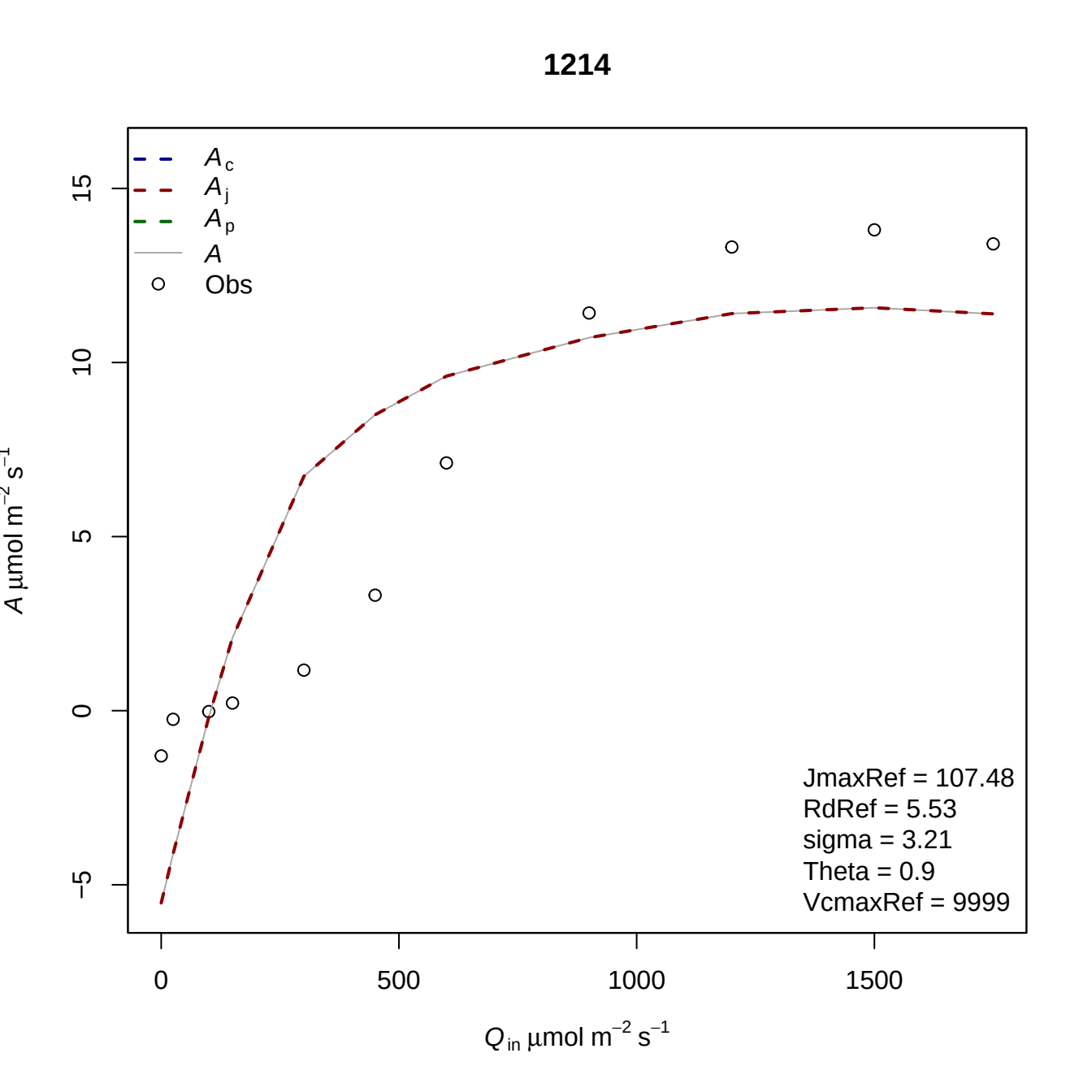

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 149.15
RdRef = 2.97
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

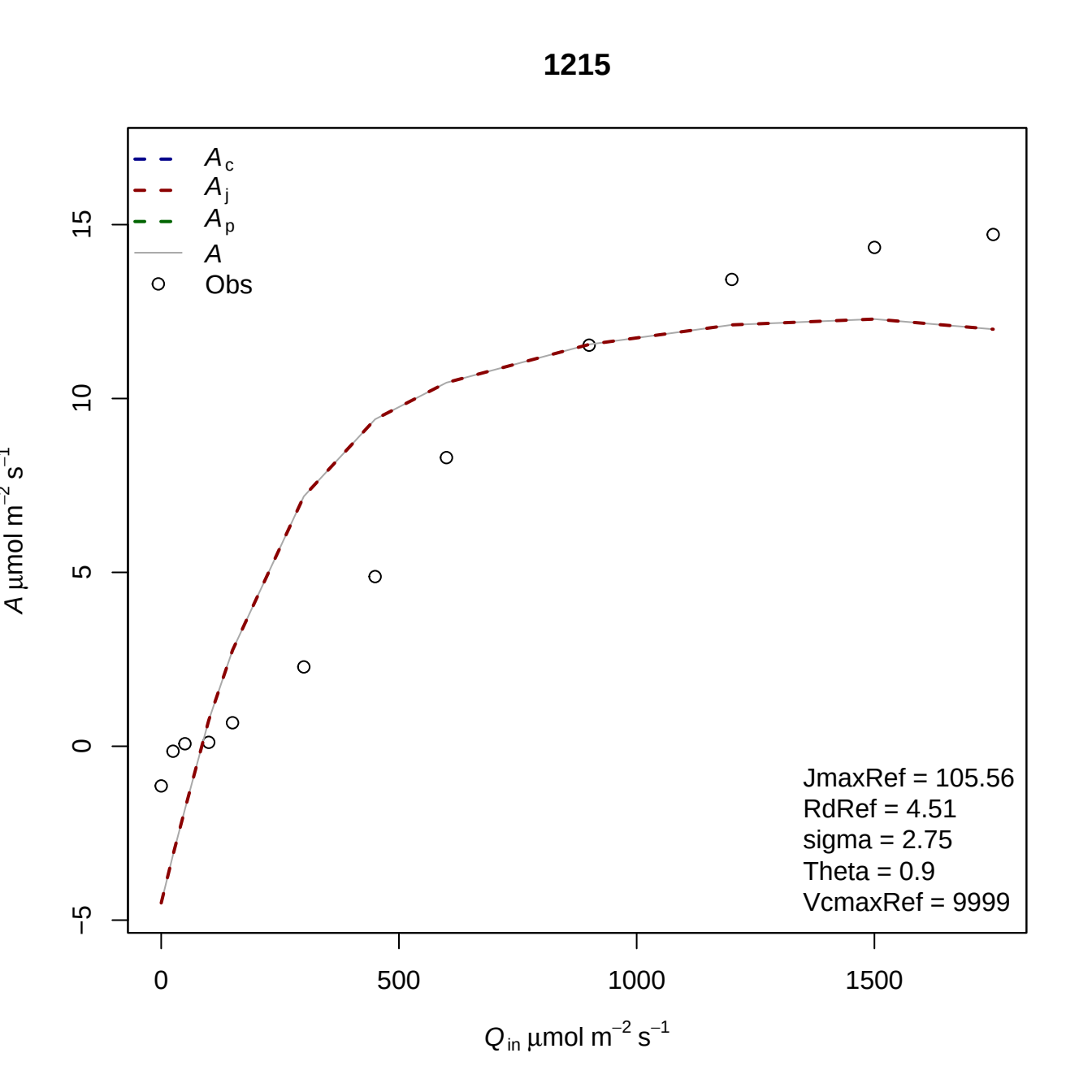

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 99.72
RdRef = 3.71
sigma = 2.18
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

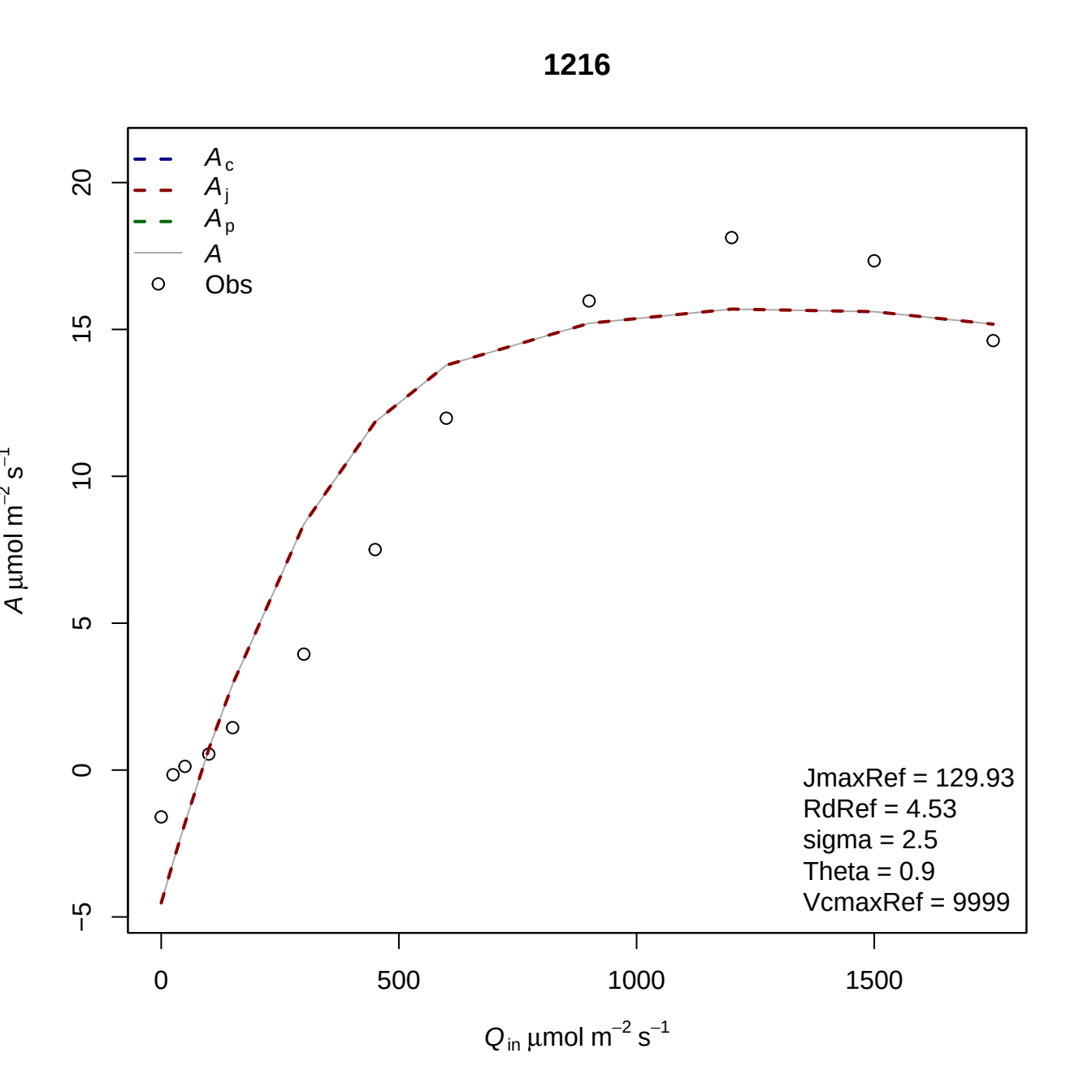

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 46.07
RdRef = 2.62
sigma = 1.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

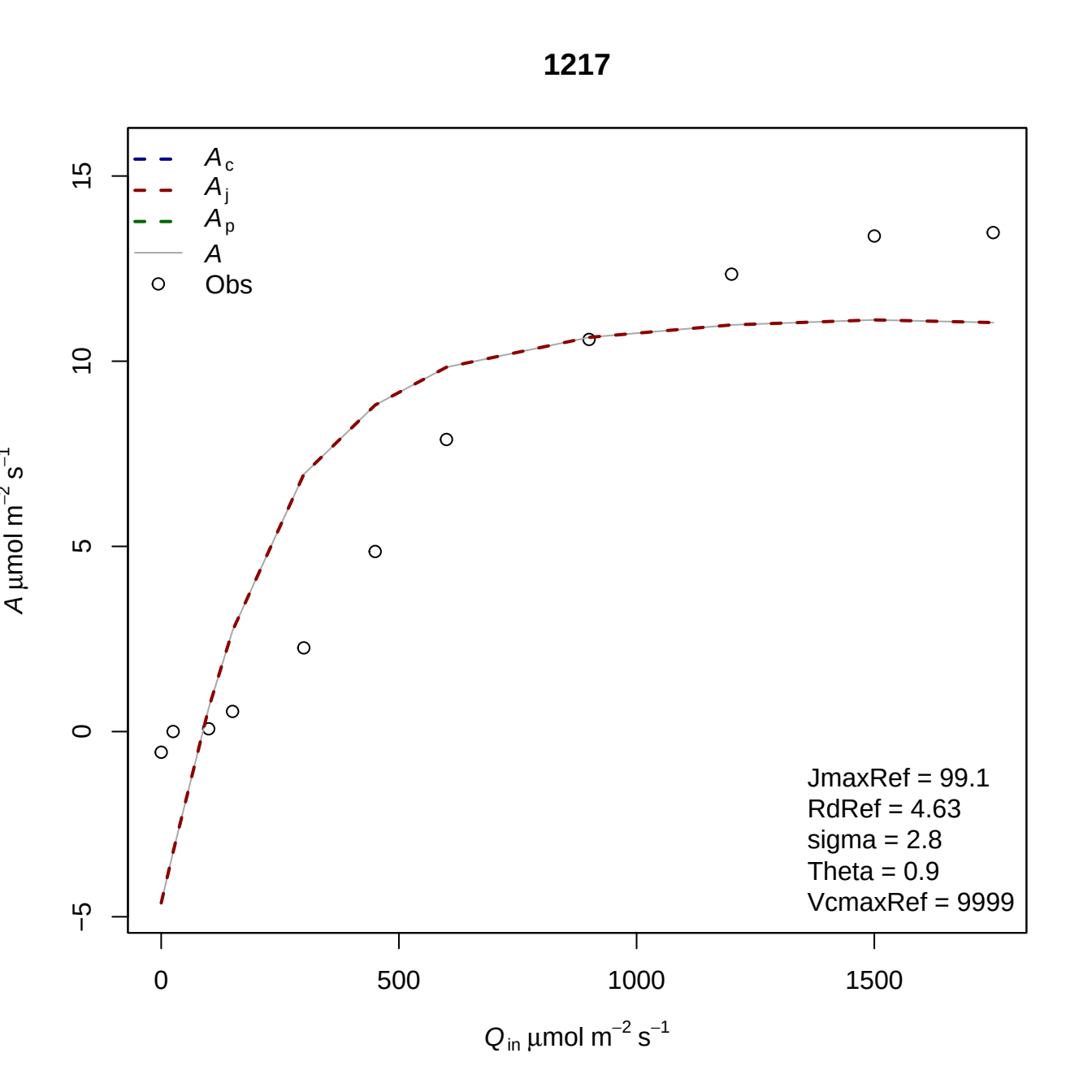

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 129.07
RdRef = 5.56
sigma = 3.51
Theta = 0.9
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

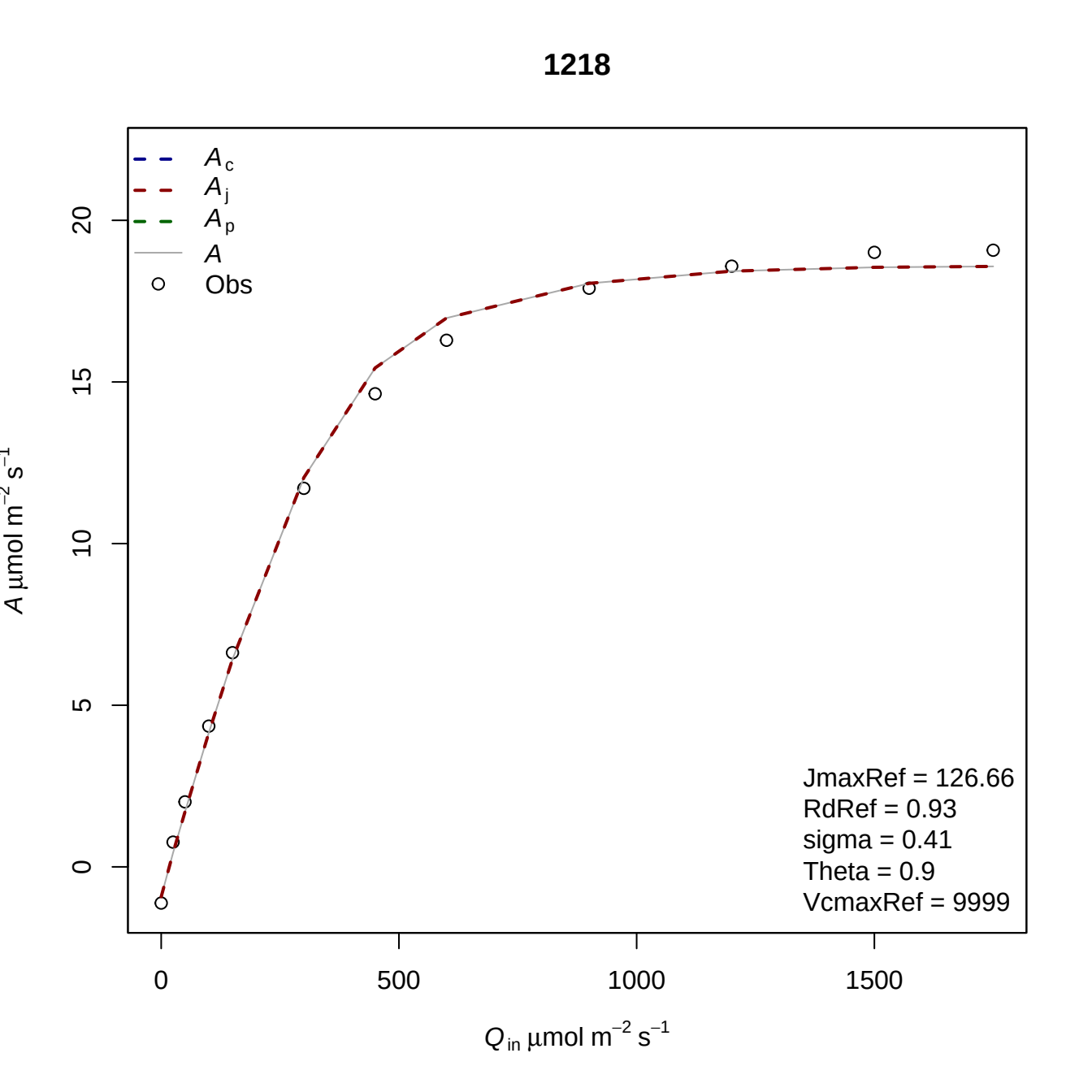

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 158.31
RdRef = 1.4
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

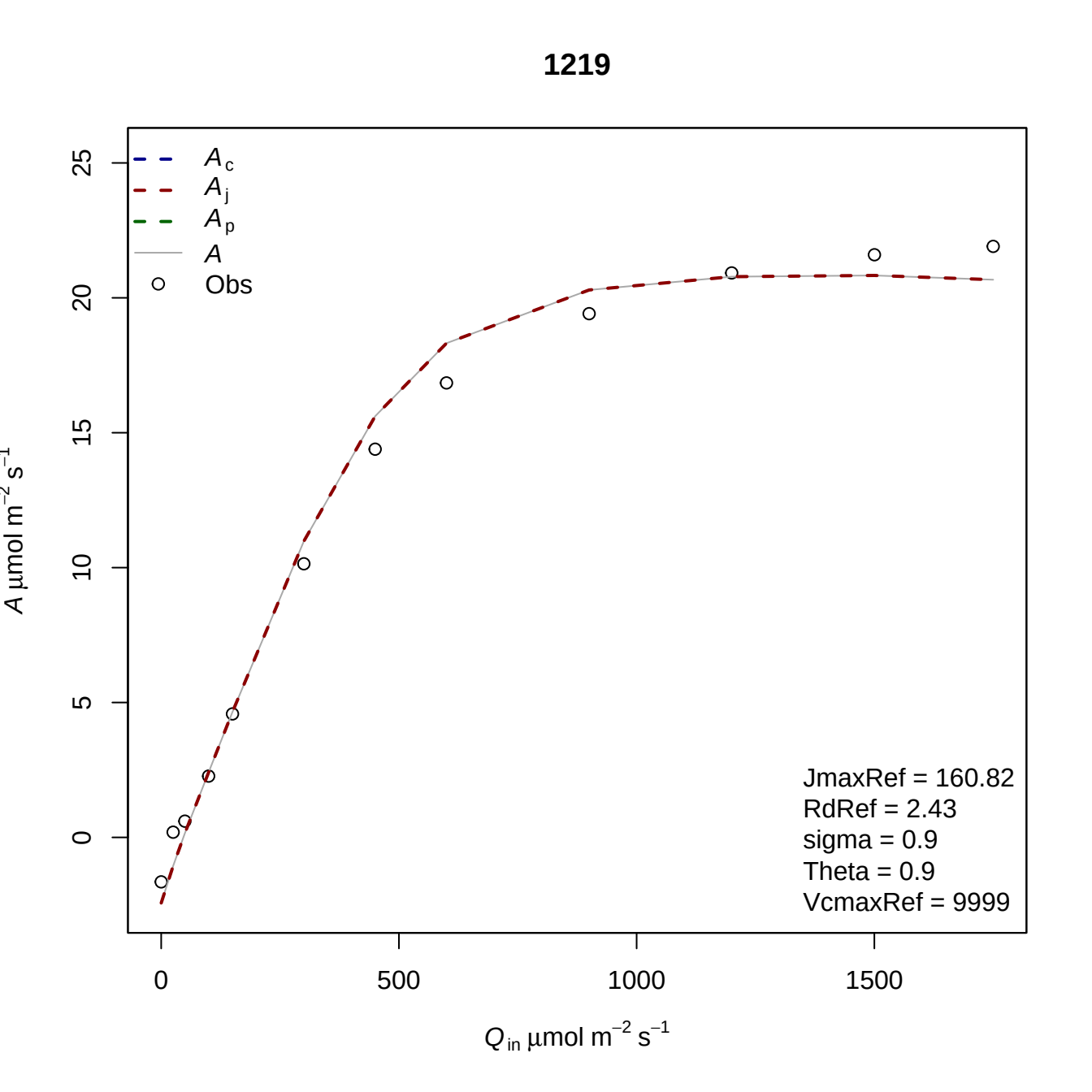

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 107.48
RdRef = 5.53
sigma = 3.21
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

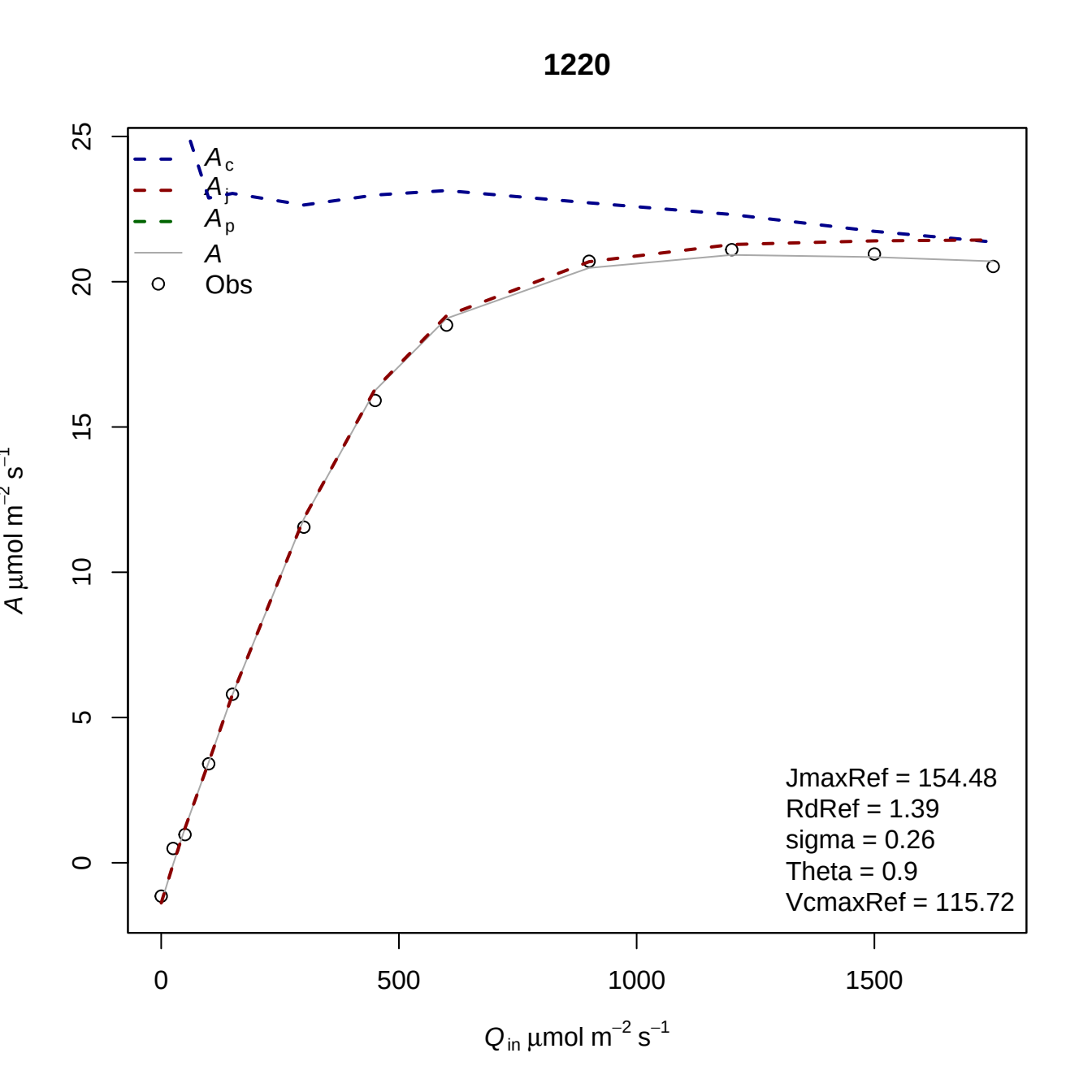

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 105.56
RdRef = 4.51
sigma = 2.75
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

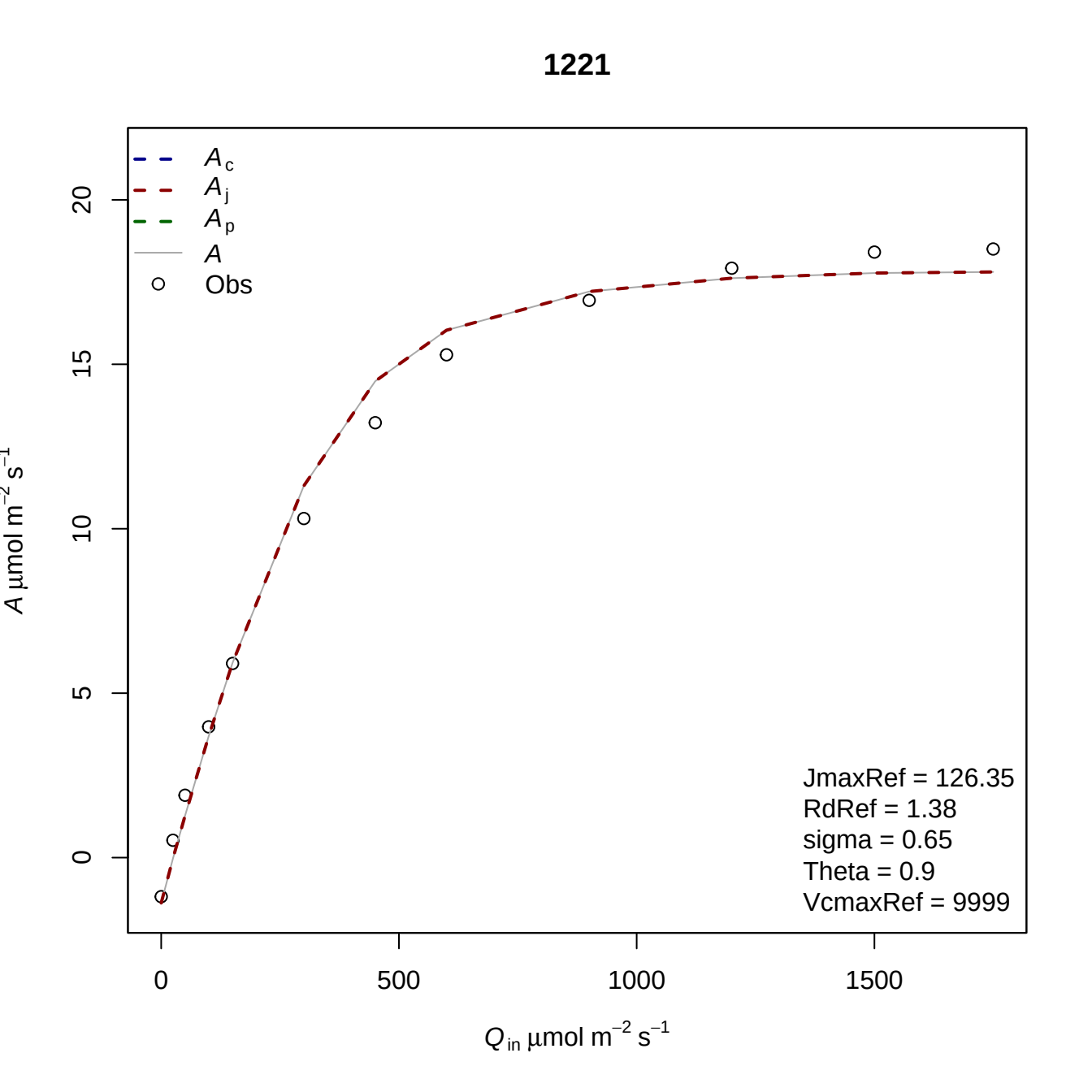

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 129.93
RdRef = 4.53
sigma = 2.5
Theta = 0.9
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

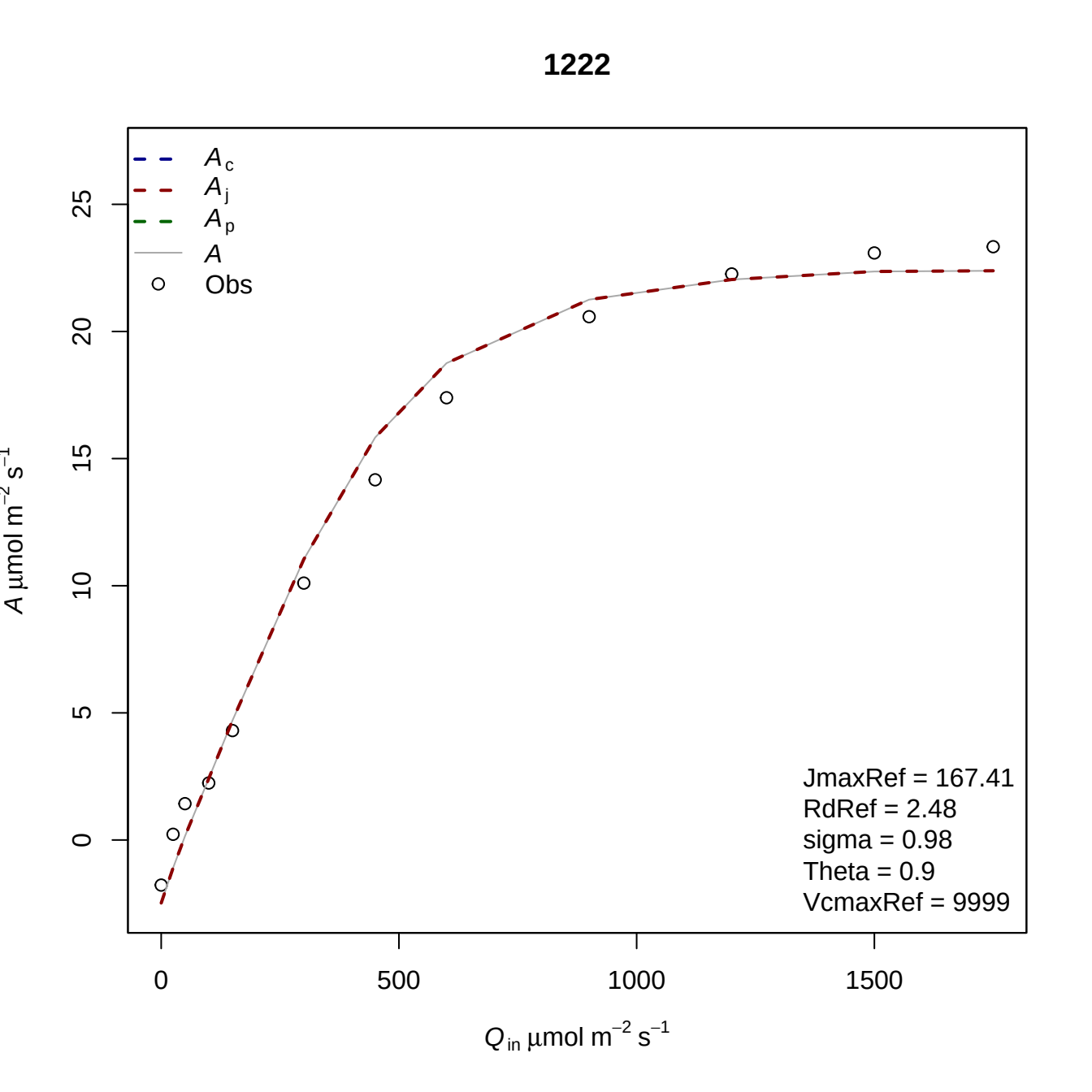

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 99.1
RdRef = 4.63
sigma = 2.8
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

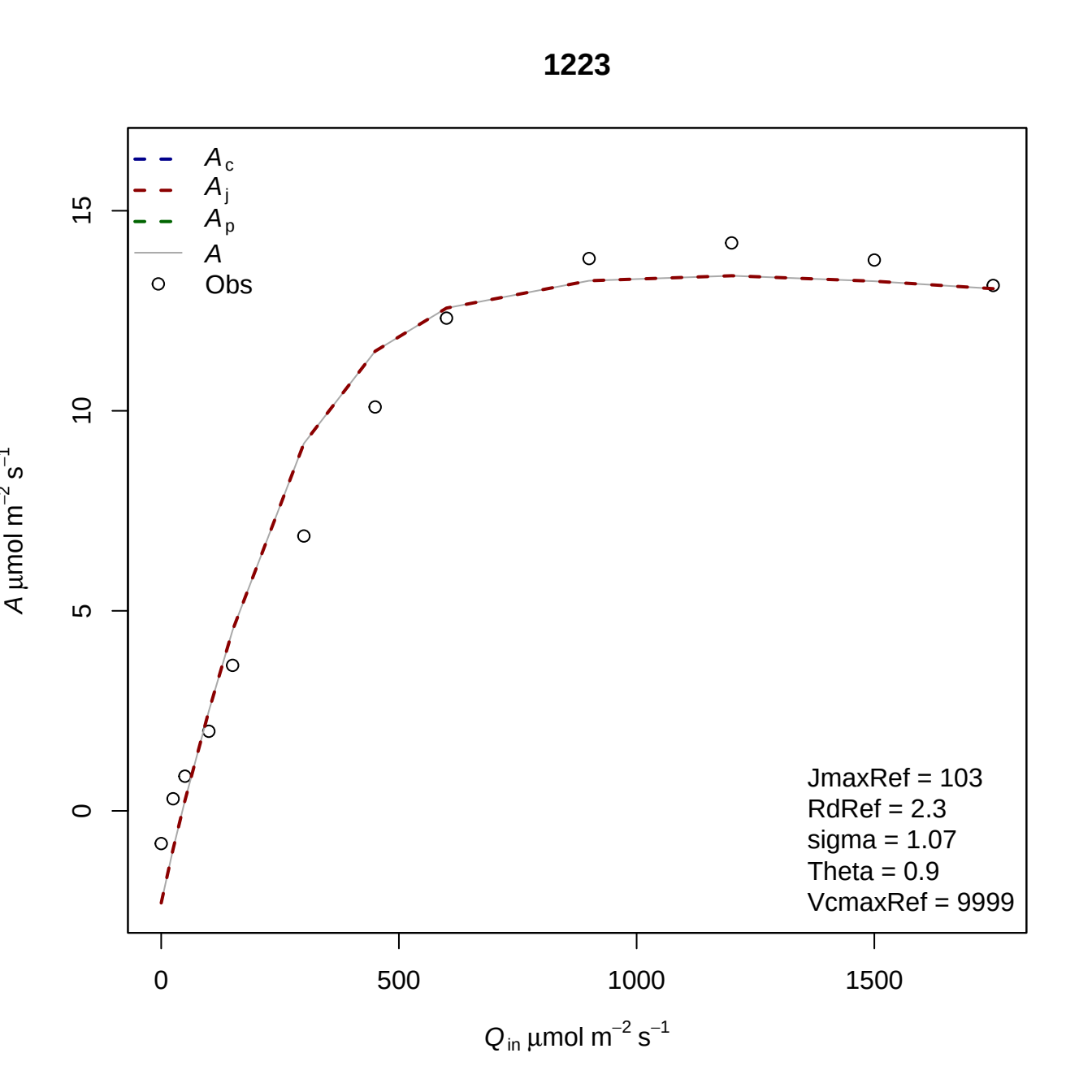

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 126.66
RdRef = 0.93
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

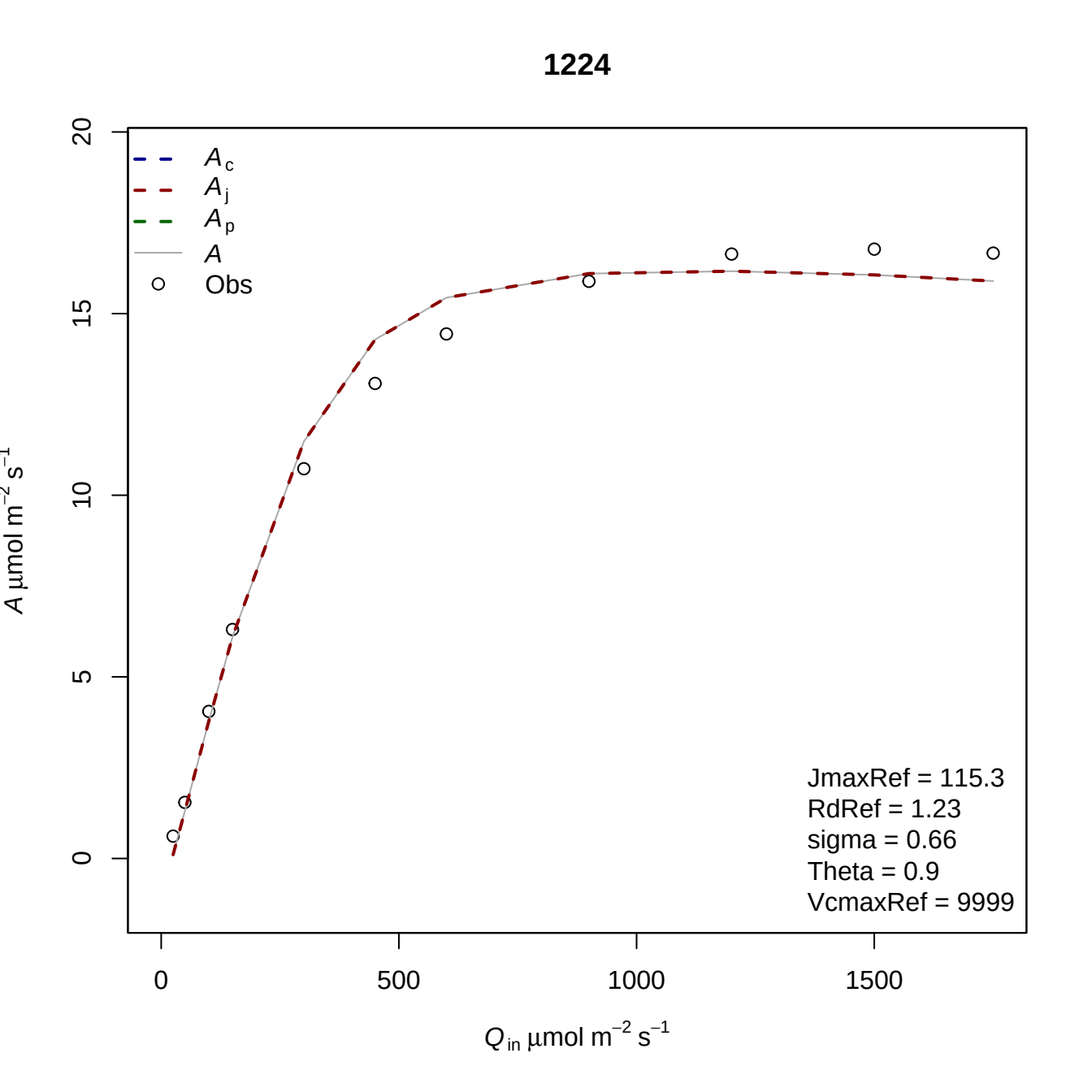

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 160.82
RdRef = 2.43
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

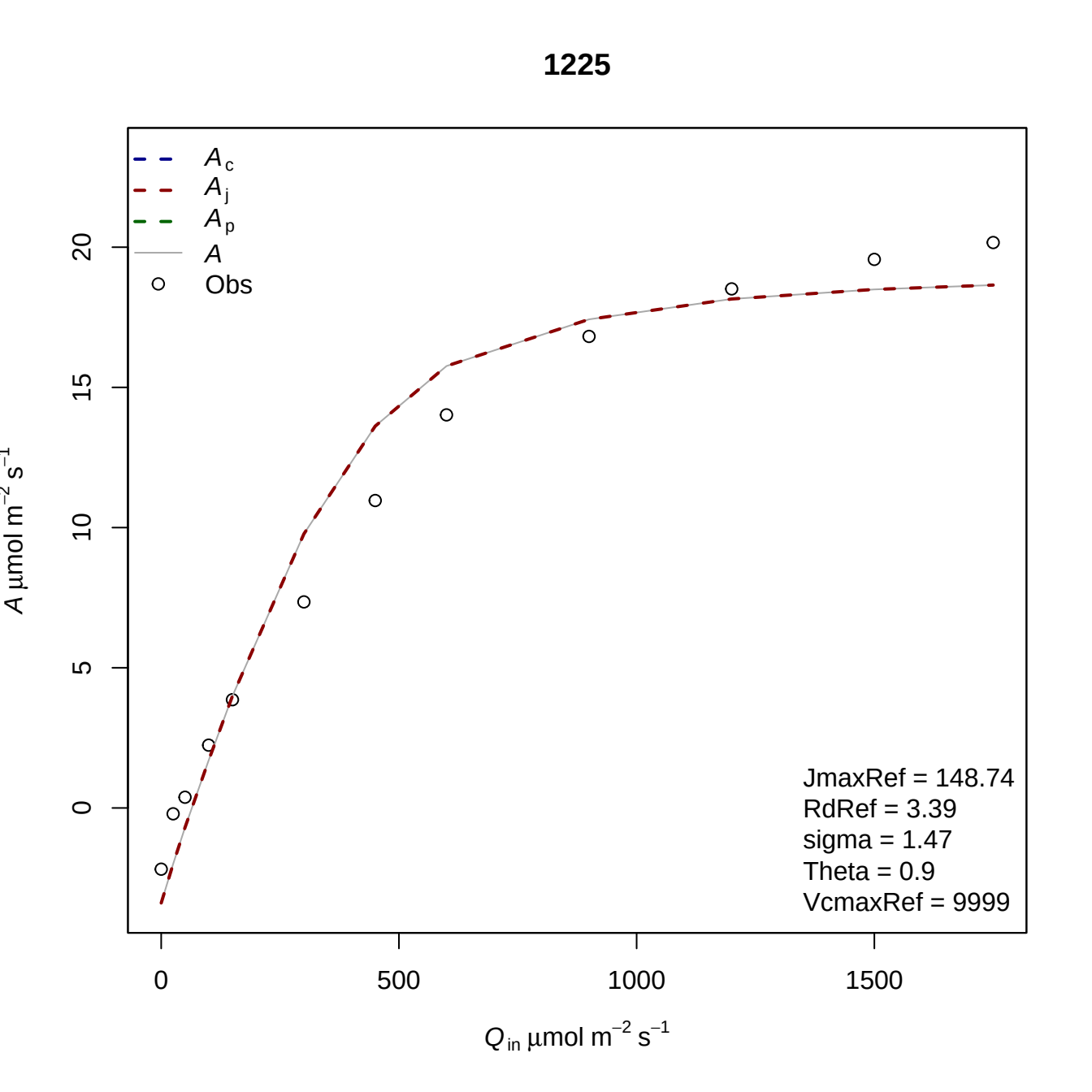

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 154.48
RdRef = 1.39
sigma = 0.26
Theta = 0.9
VcmaxRef = 115.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

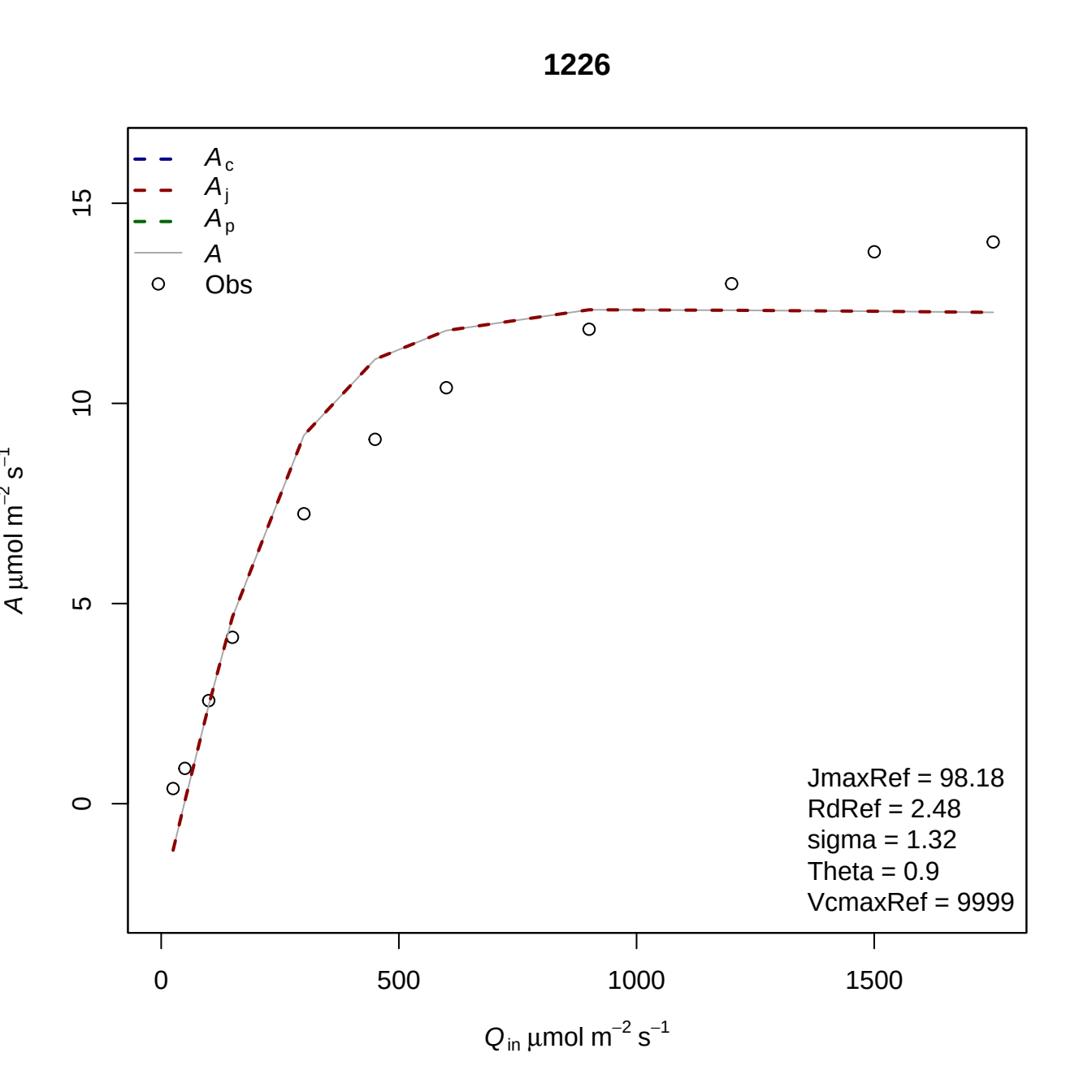

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 126.35
RdRef = 1.38
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

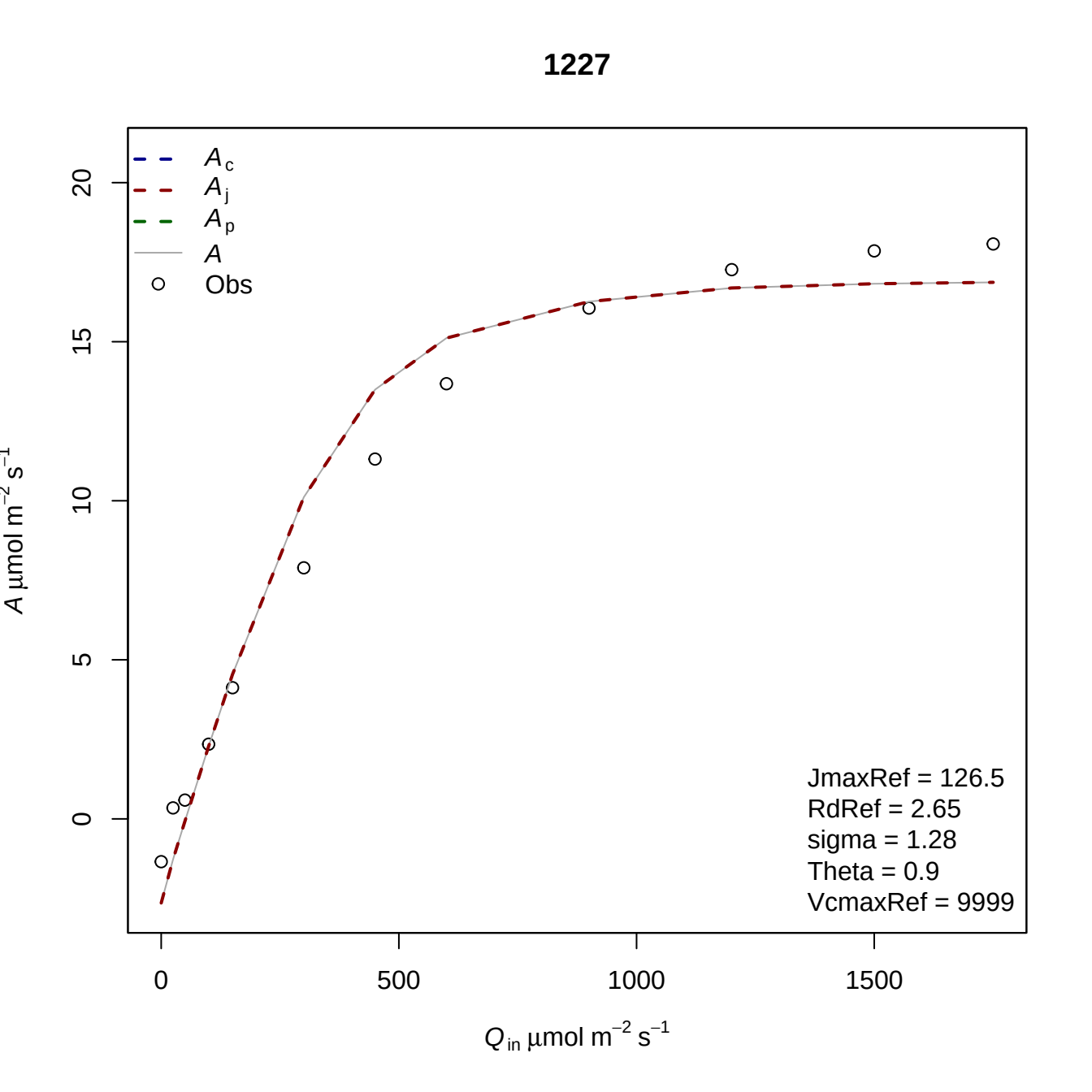

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 167.41
RdRef = 2.48
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

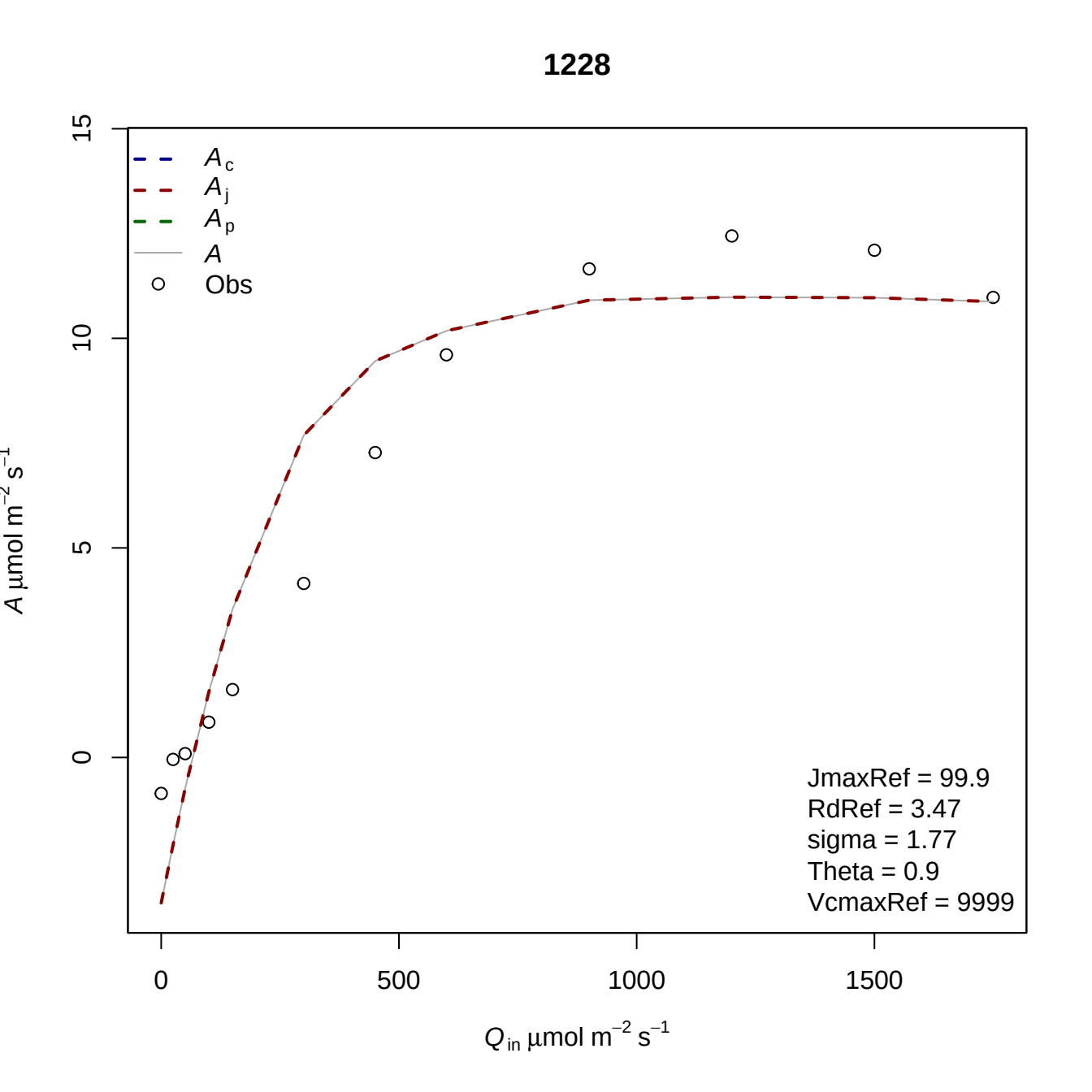

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 103
RdRef = 2.3
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

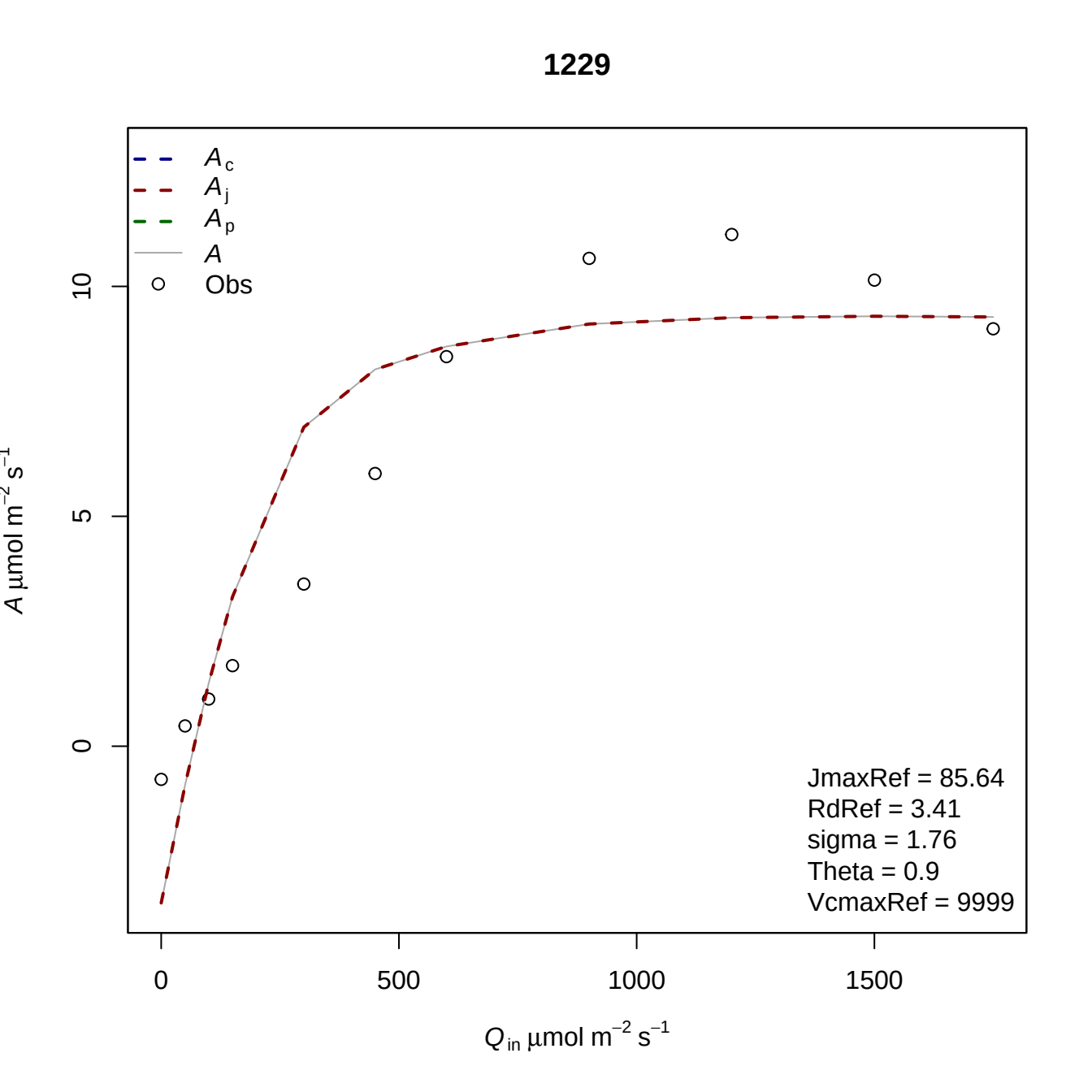

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 115.3
RdRef = 1.23
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

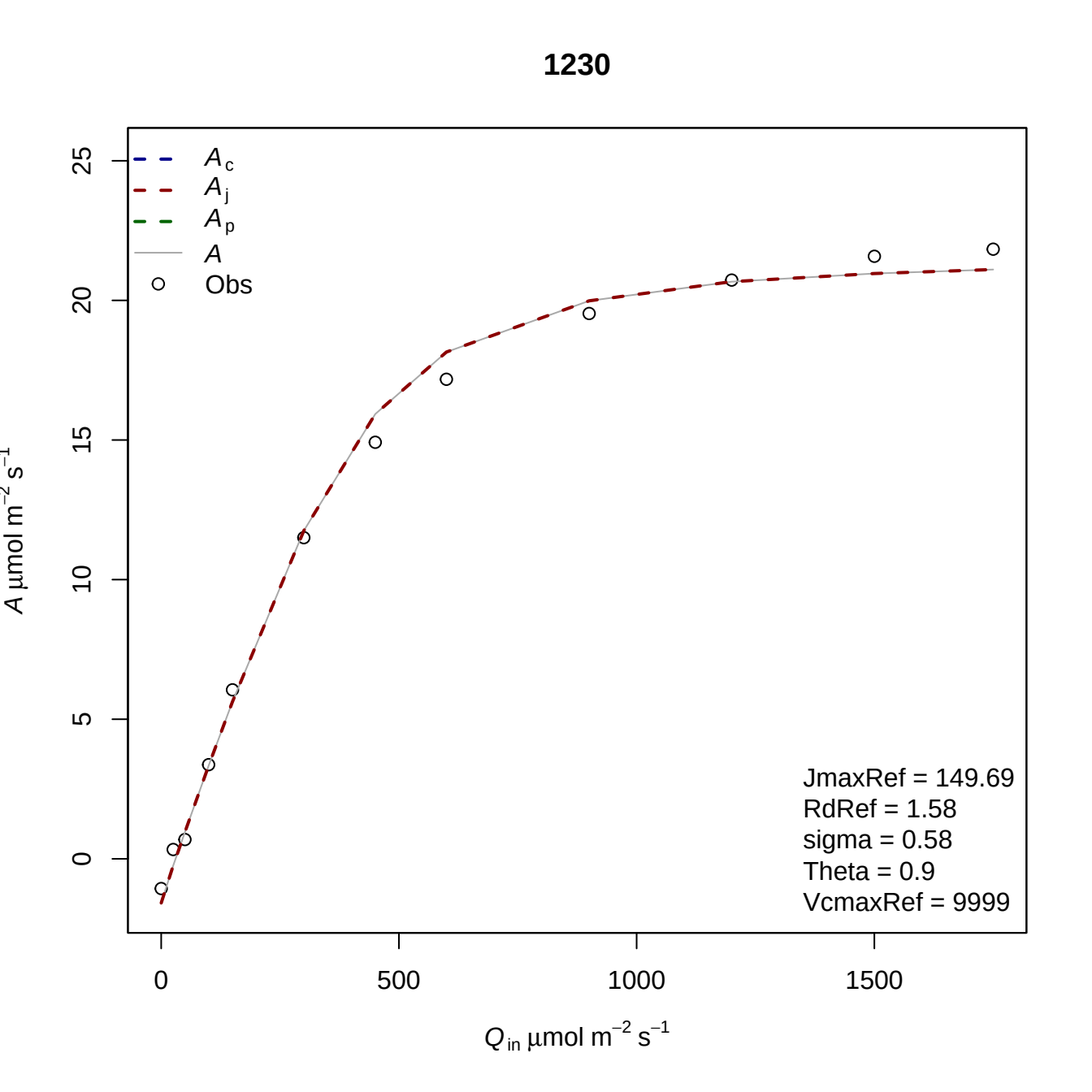

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 148.74
RdRef = 3.39
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

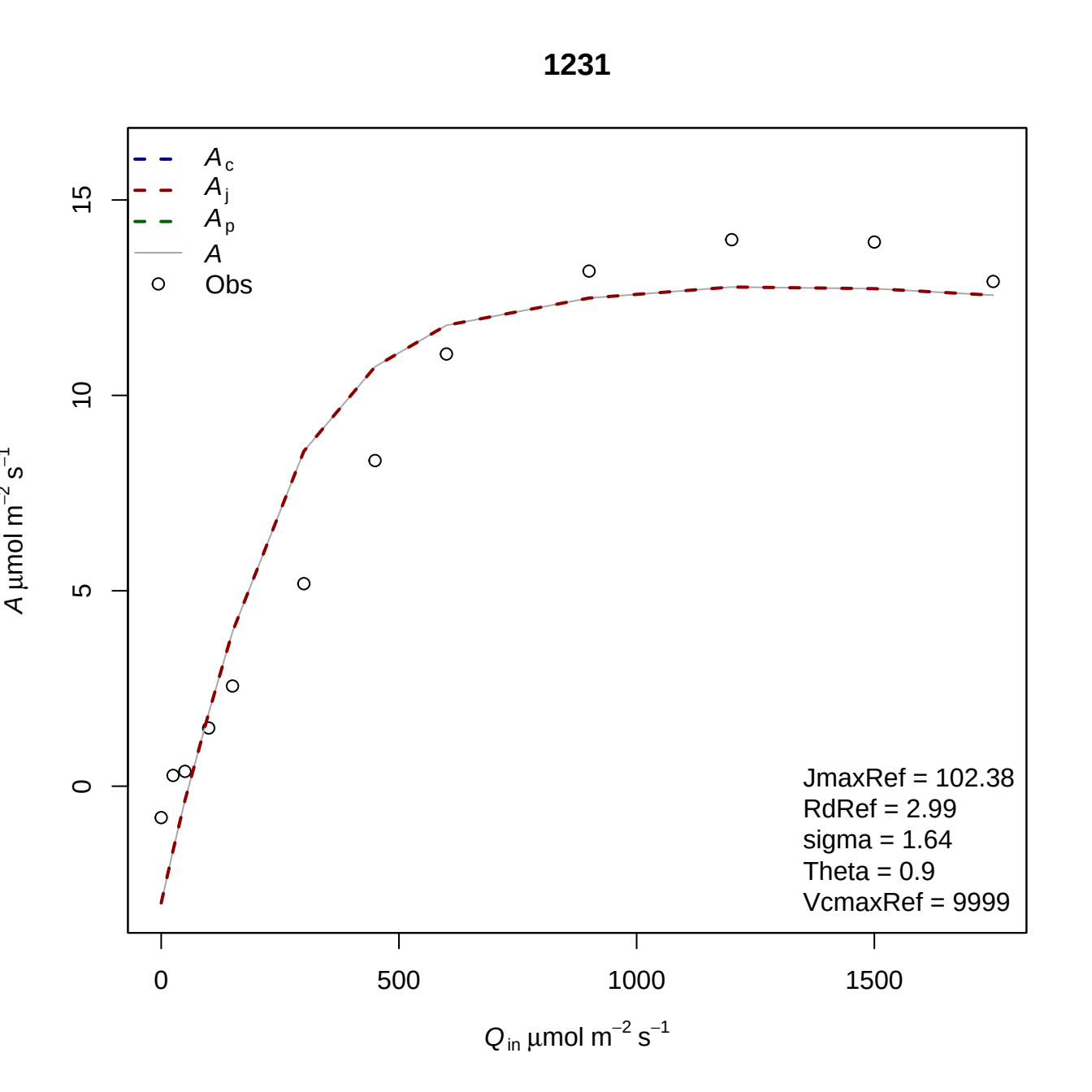

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 98.18
RdRef = 2.48
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

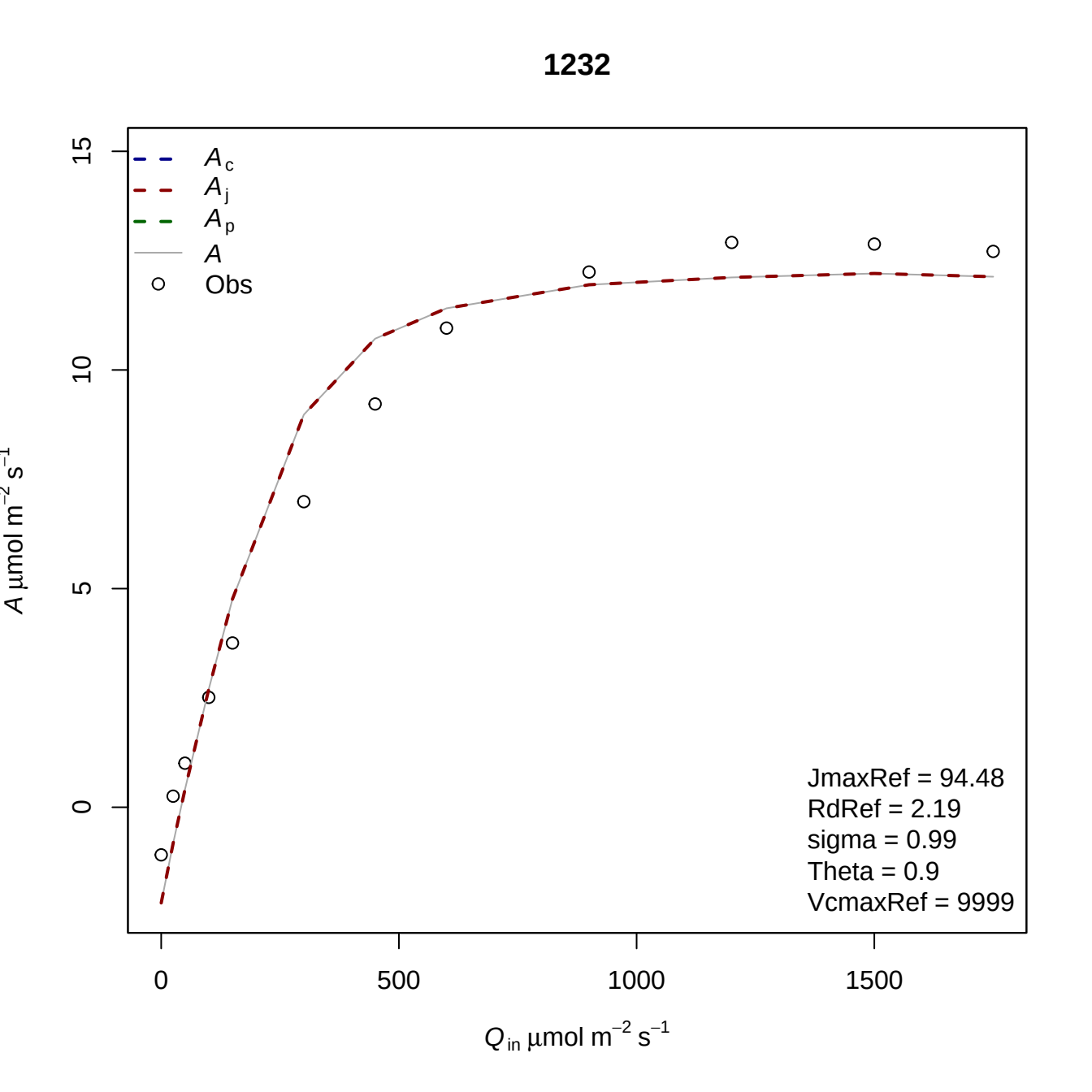

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 126.5
RdRef = 2.65
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

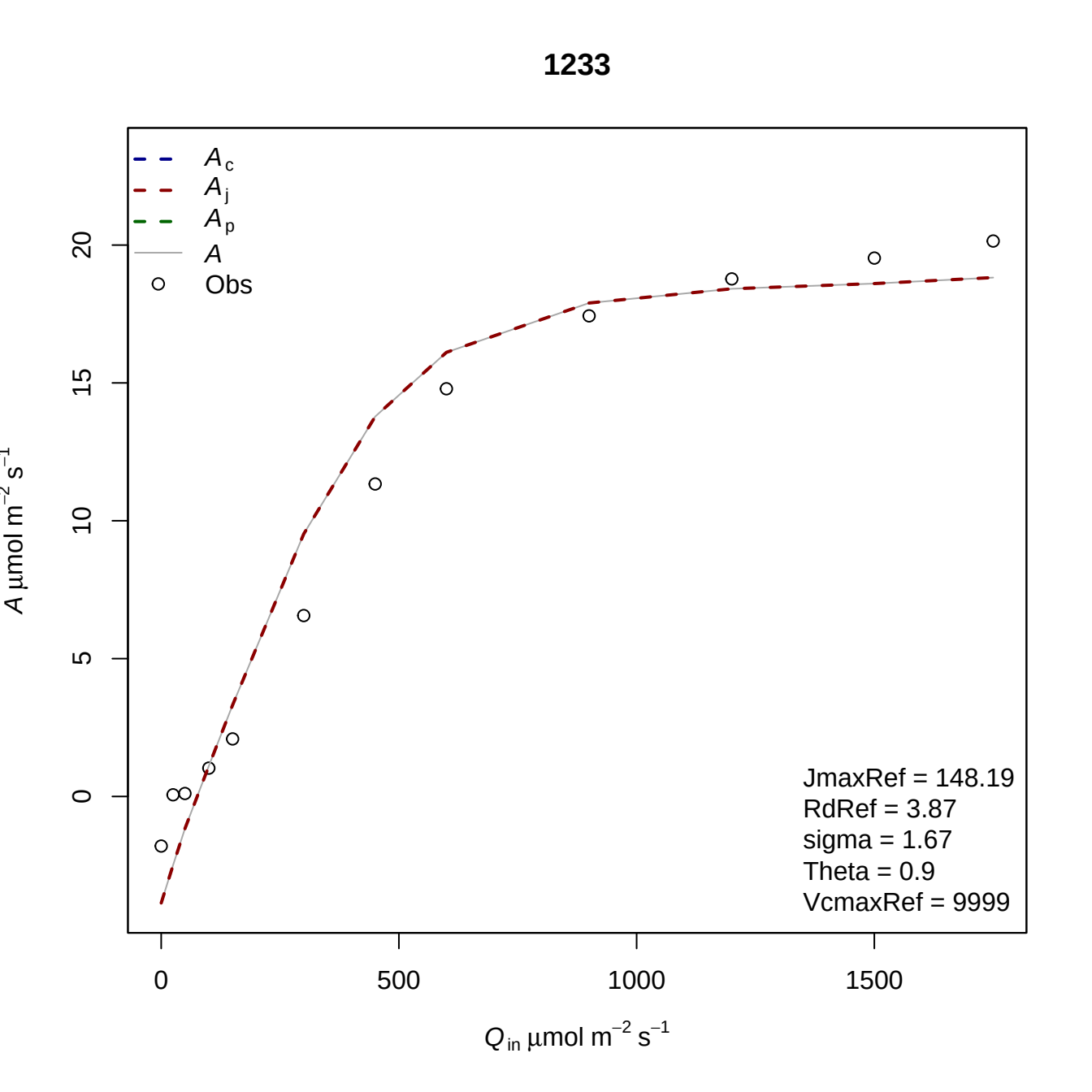

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 99.9
RdRef = 3.47
sigma = 1.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

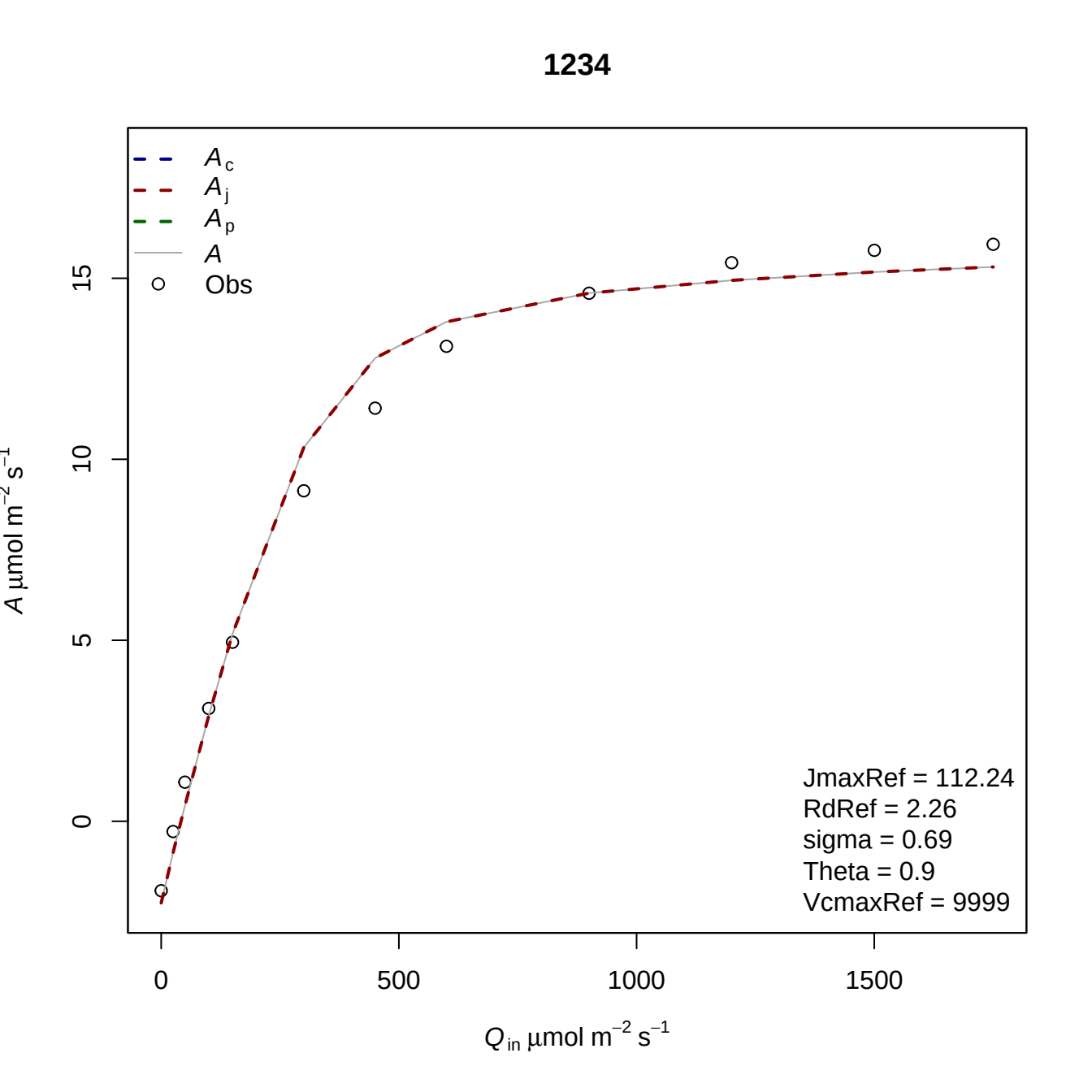

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 85.64
RdRef = 3.41
sigma = 1.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

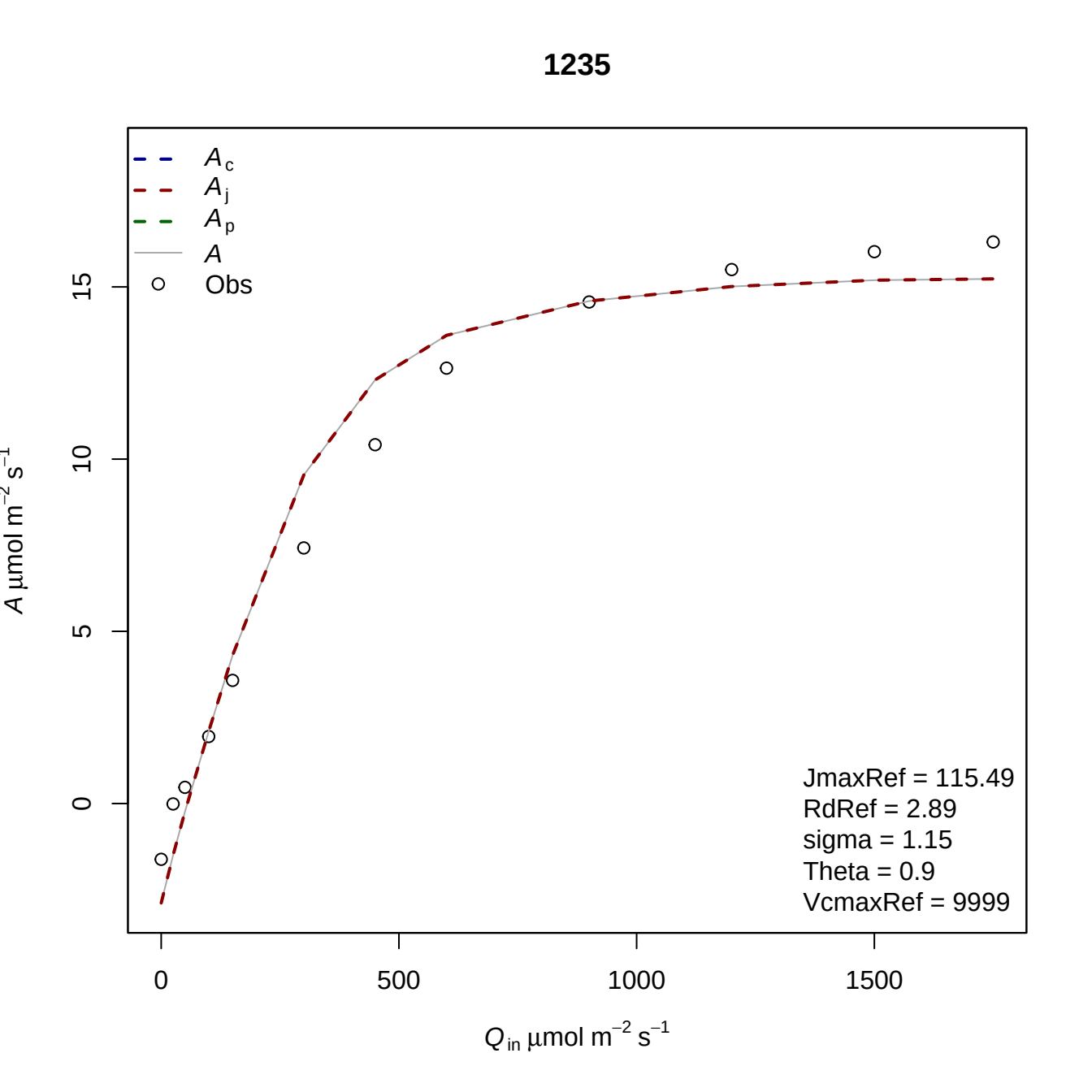

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 149.69
RdRef = 1.58
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

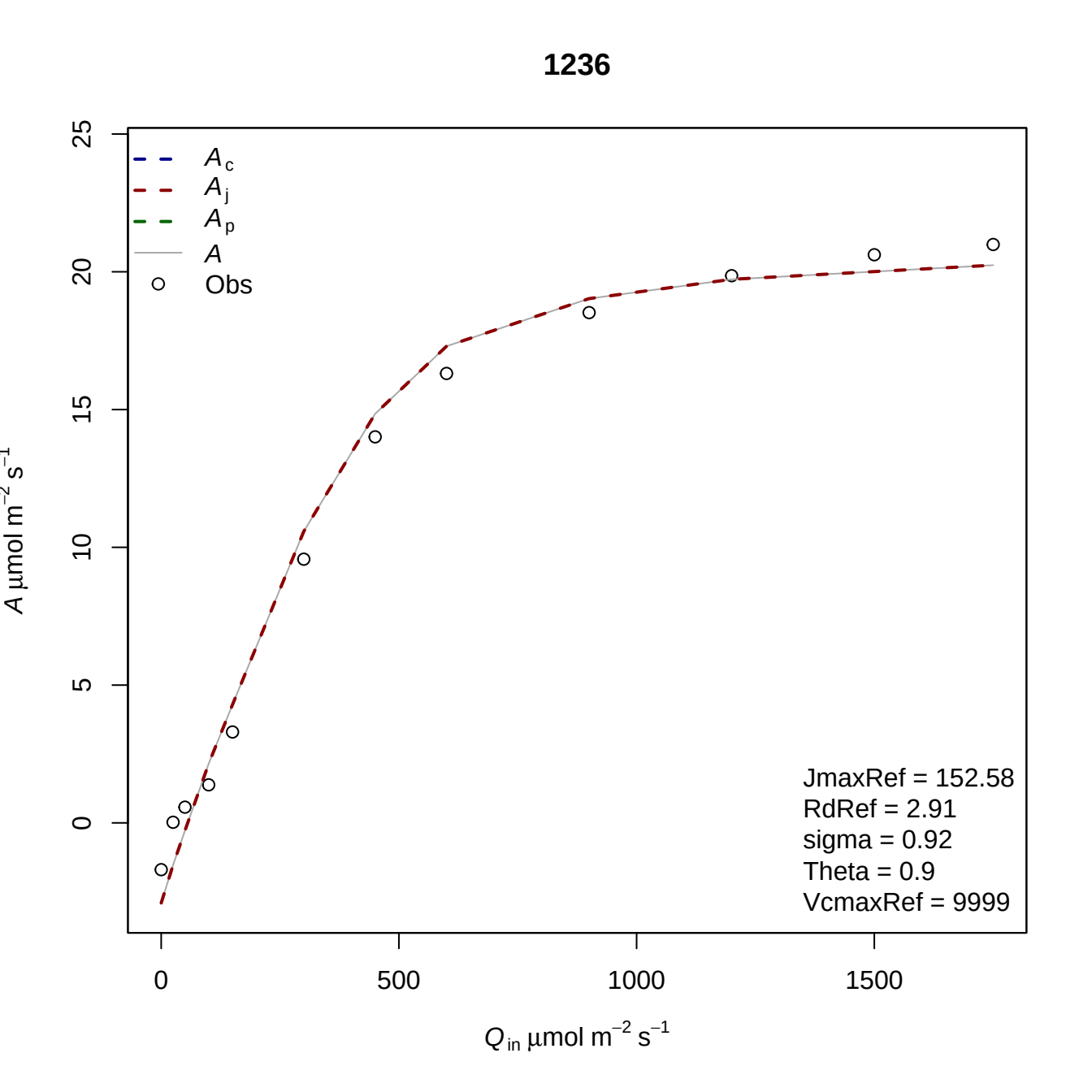

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 102.38
RdRef = 2.99
sigma = 1.64
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

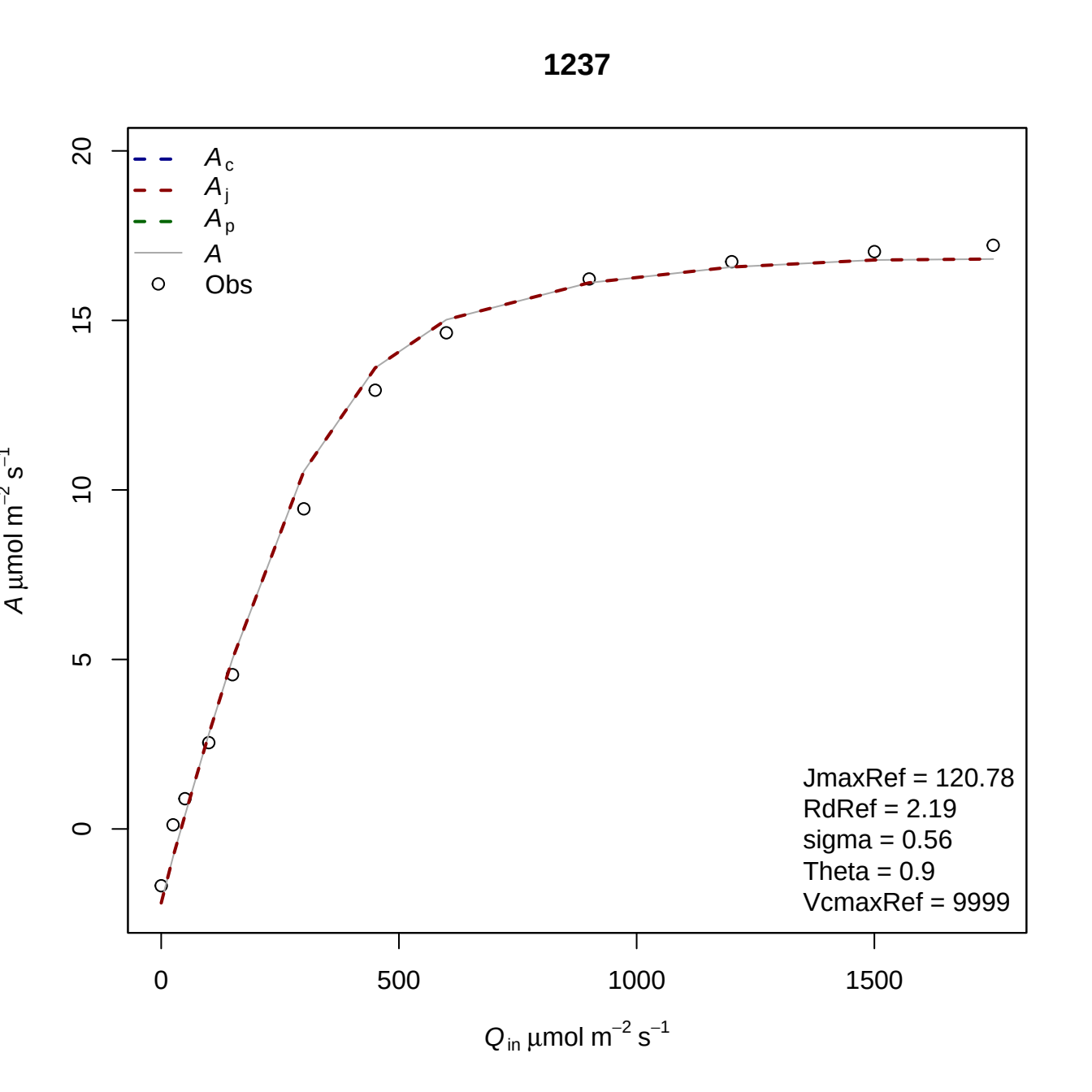

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 94.48
RdRef = 2.19
sigma = 0.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

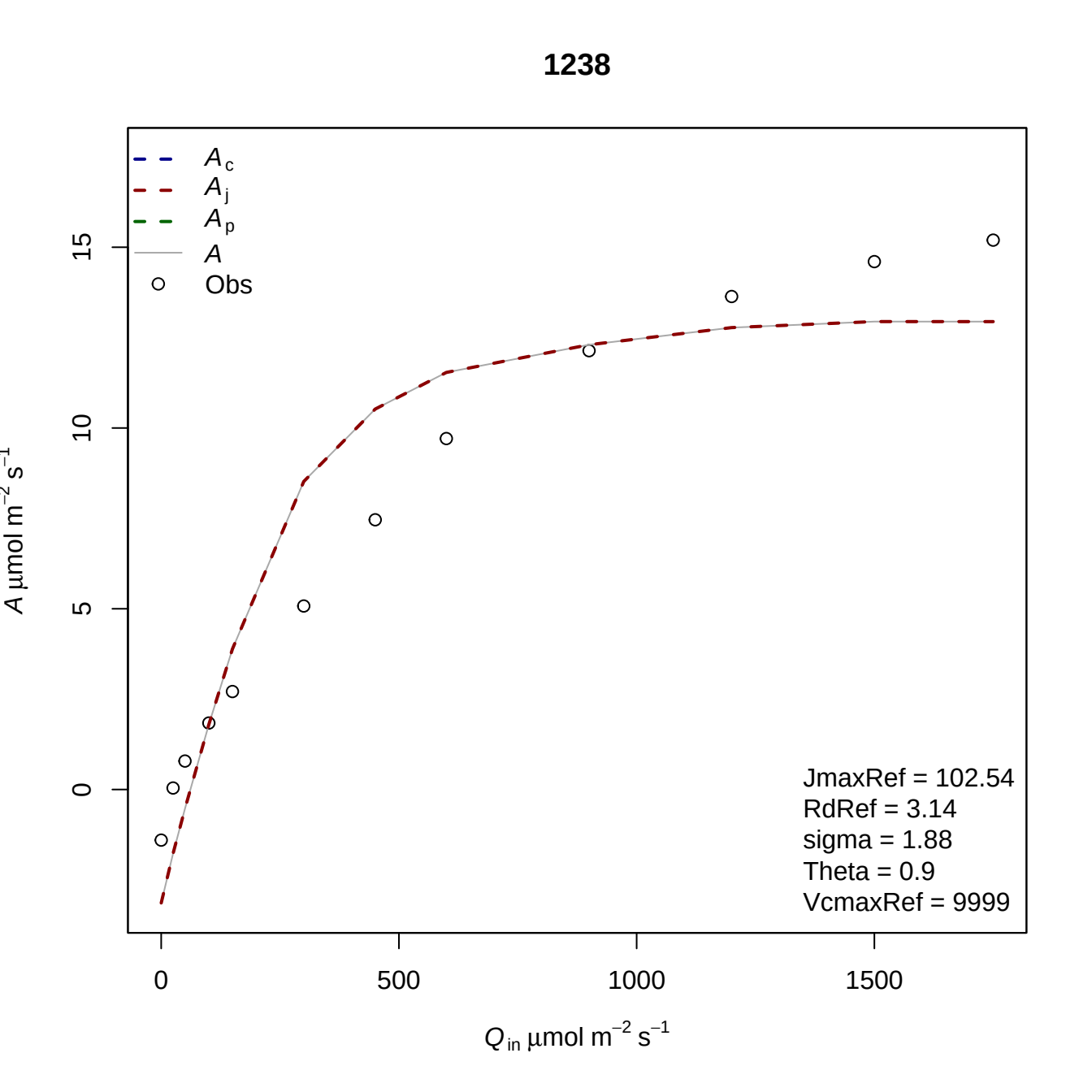

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 148.19
RdRef = 3.87
sigma = 1.67
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

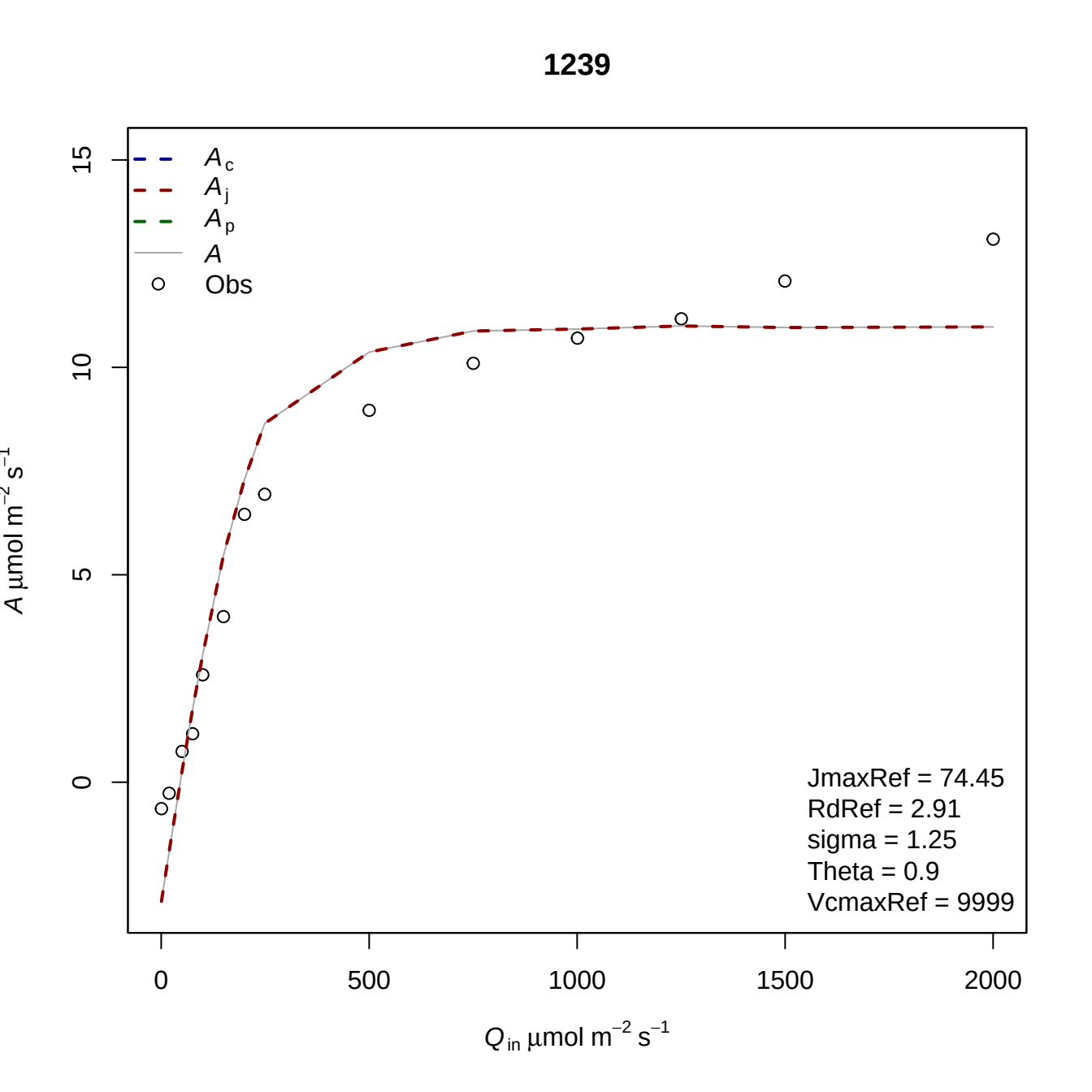

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 112.24
RdRef = 2.26
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

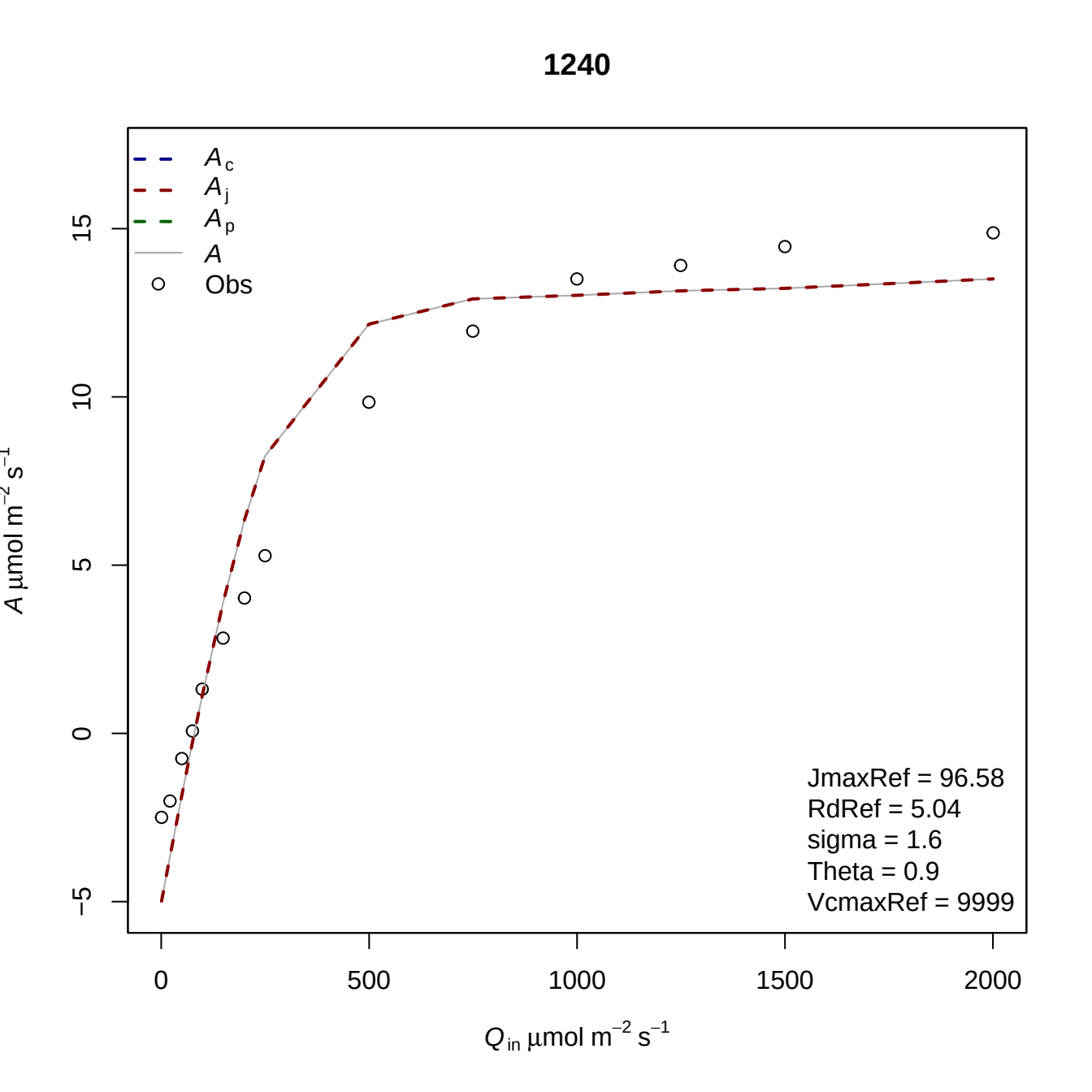

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 115.49
RdRef = 2.89
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

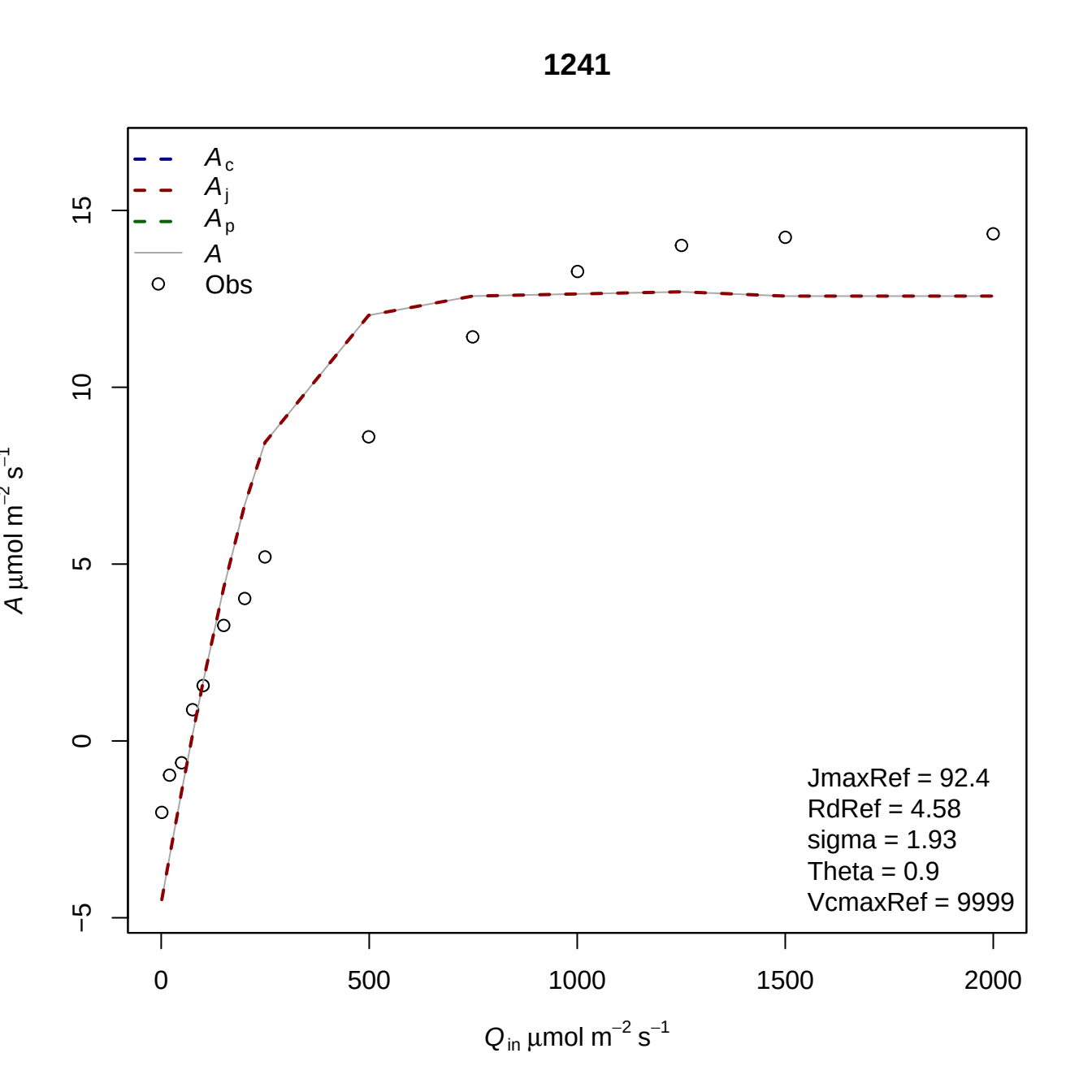

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 152.58
RdRef = 2.91
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

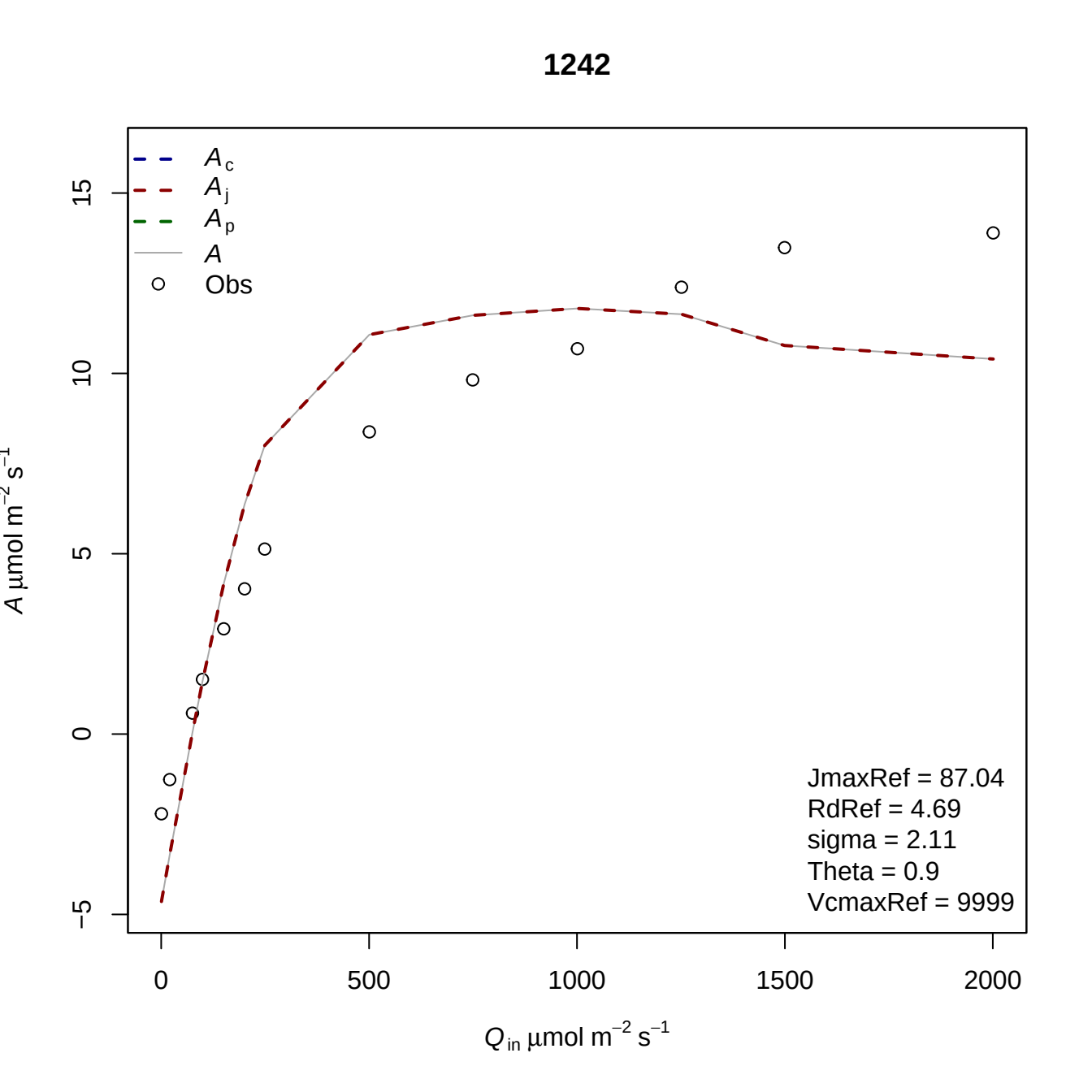

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 120.78
RdRef = 2.19
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

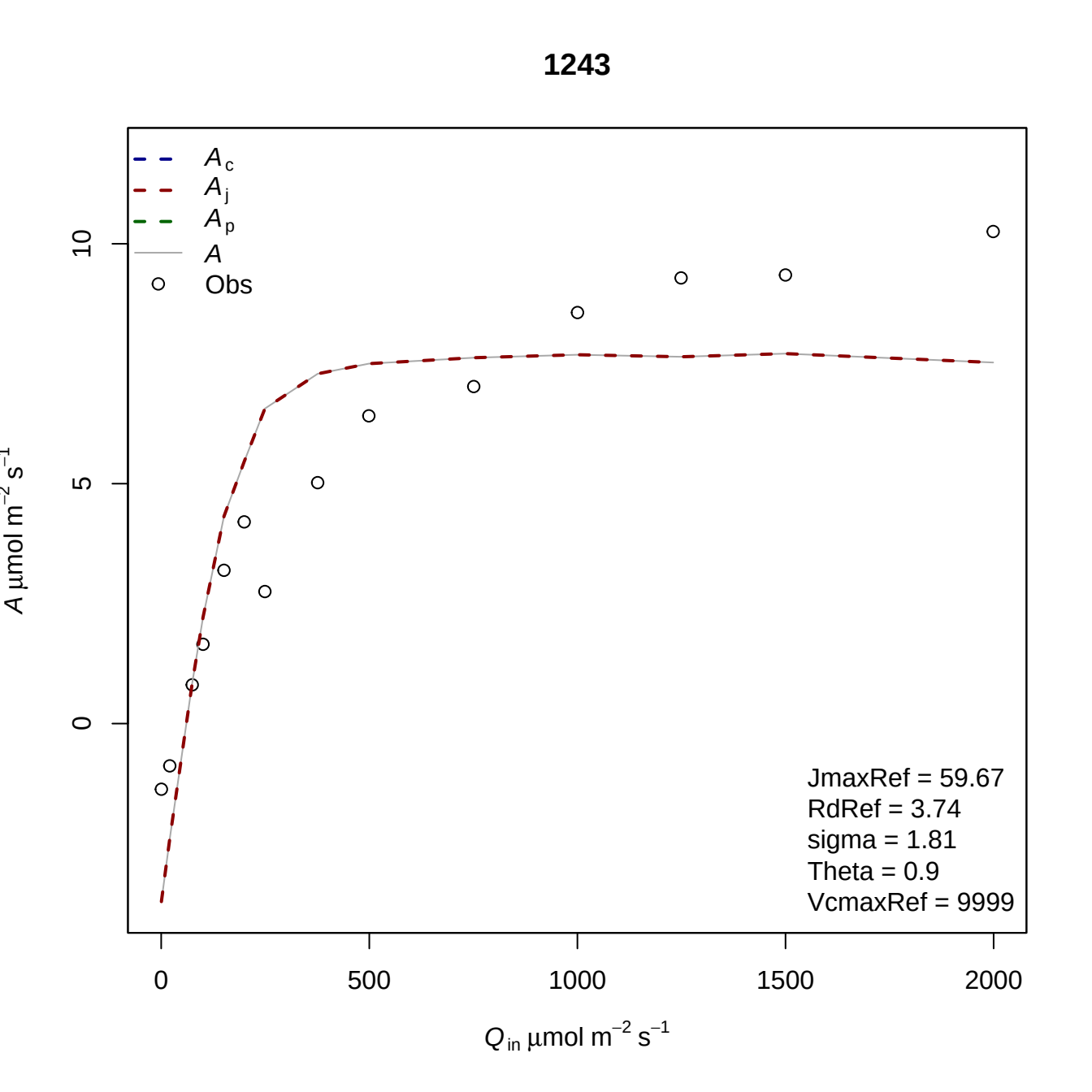

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 102.54
RdRef = 3.14
sigma = 1.88
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

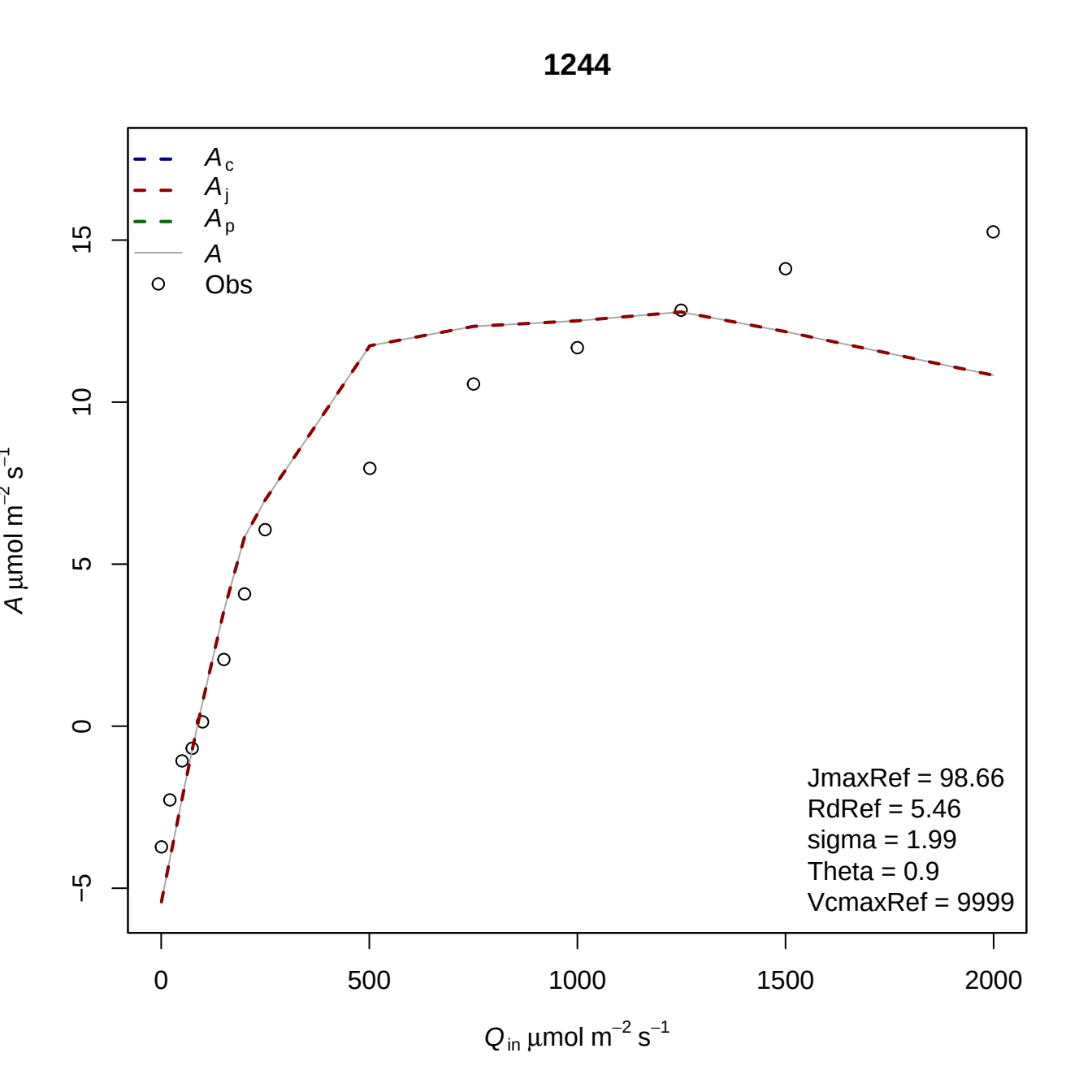

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 74.45
RdRef = 2.91
sigma = 1.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

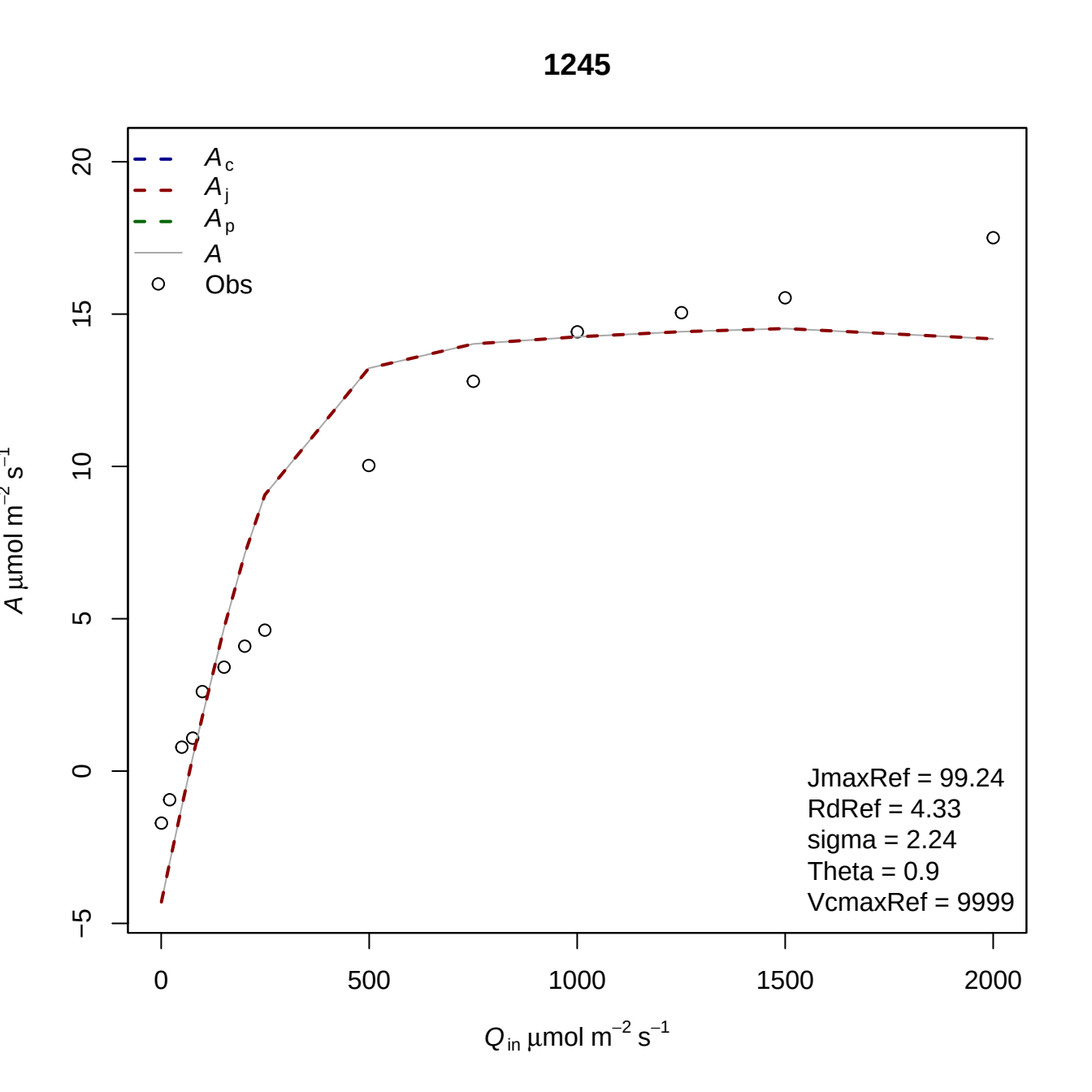

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 96.58
RdRef = 5.04
sigma = 1.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

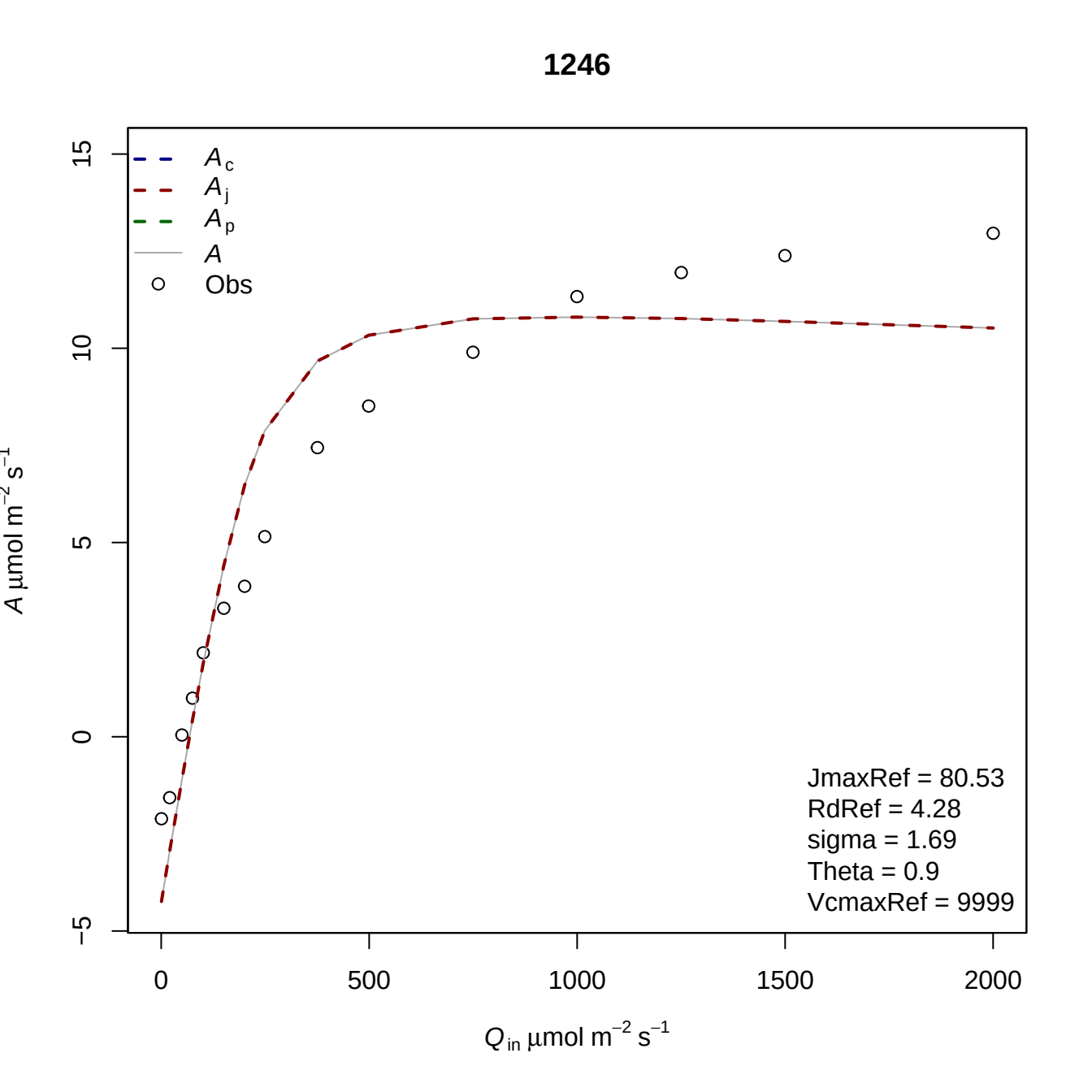

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 92.4
RdRef = 4.58
sigma = 1.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

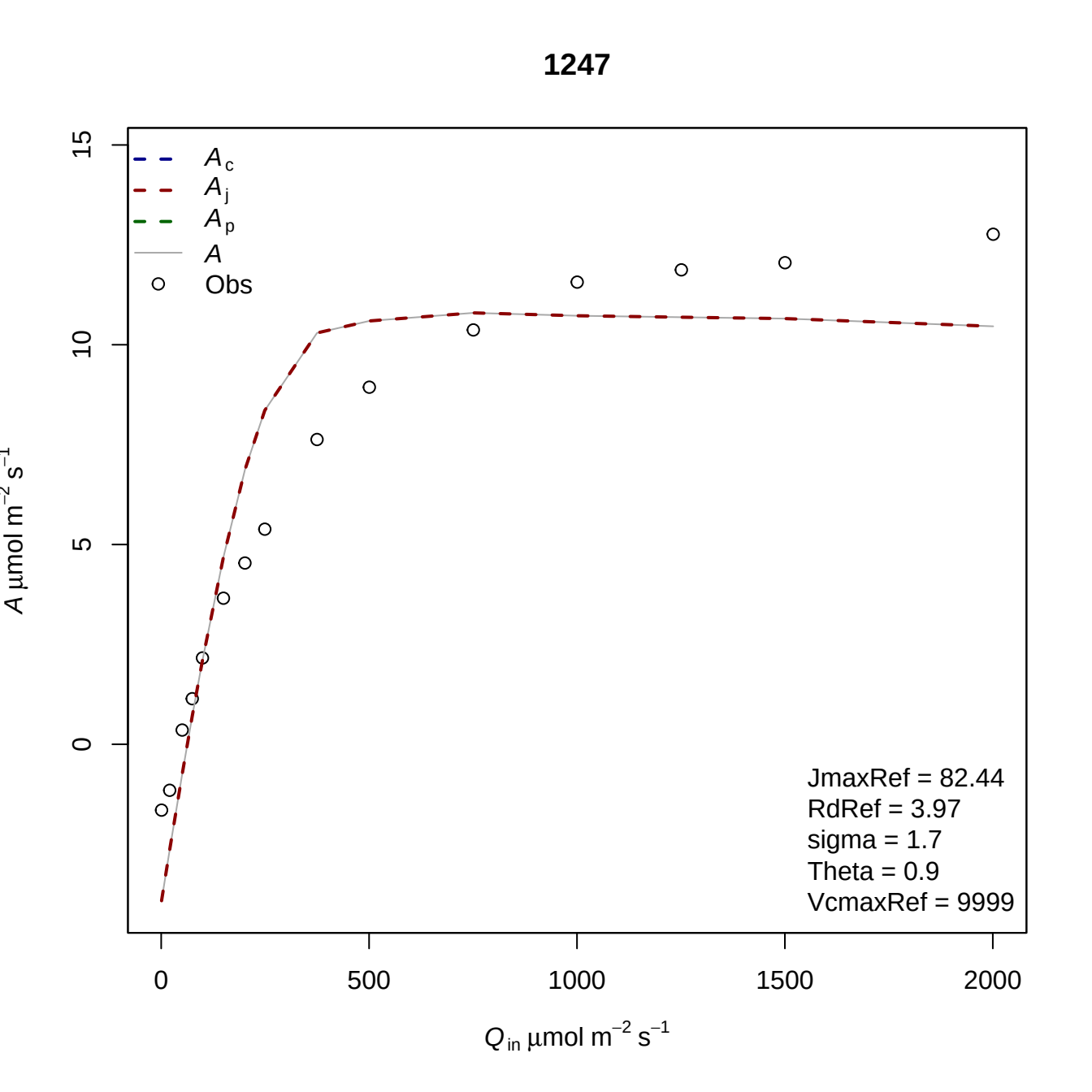

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 87.04
RdRef = 4.69
sigma = 2.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

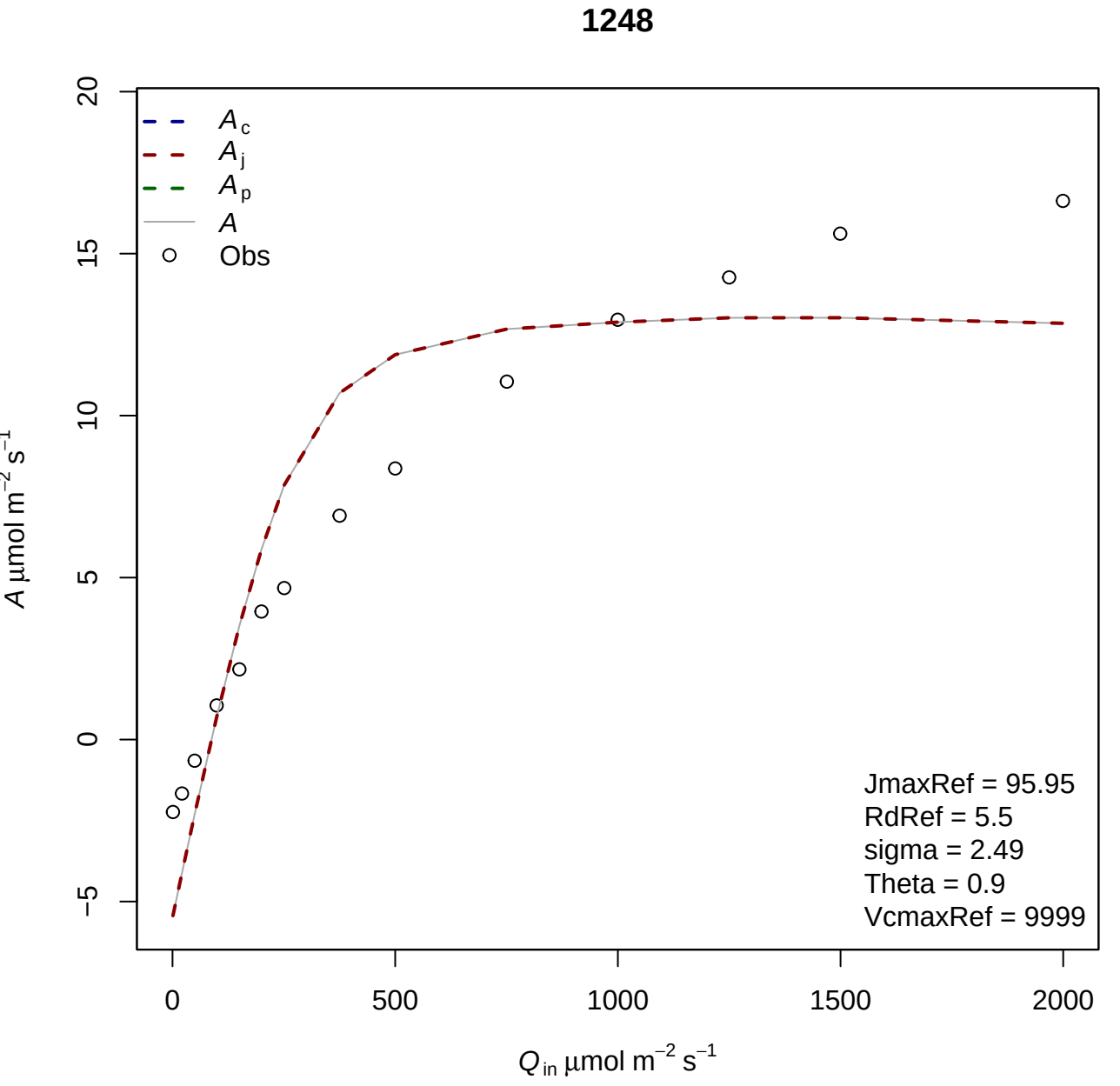

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 59.67
RdRef = 3.74
sigma = 1.81
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

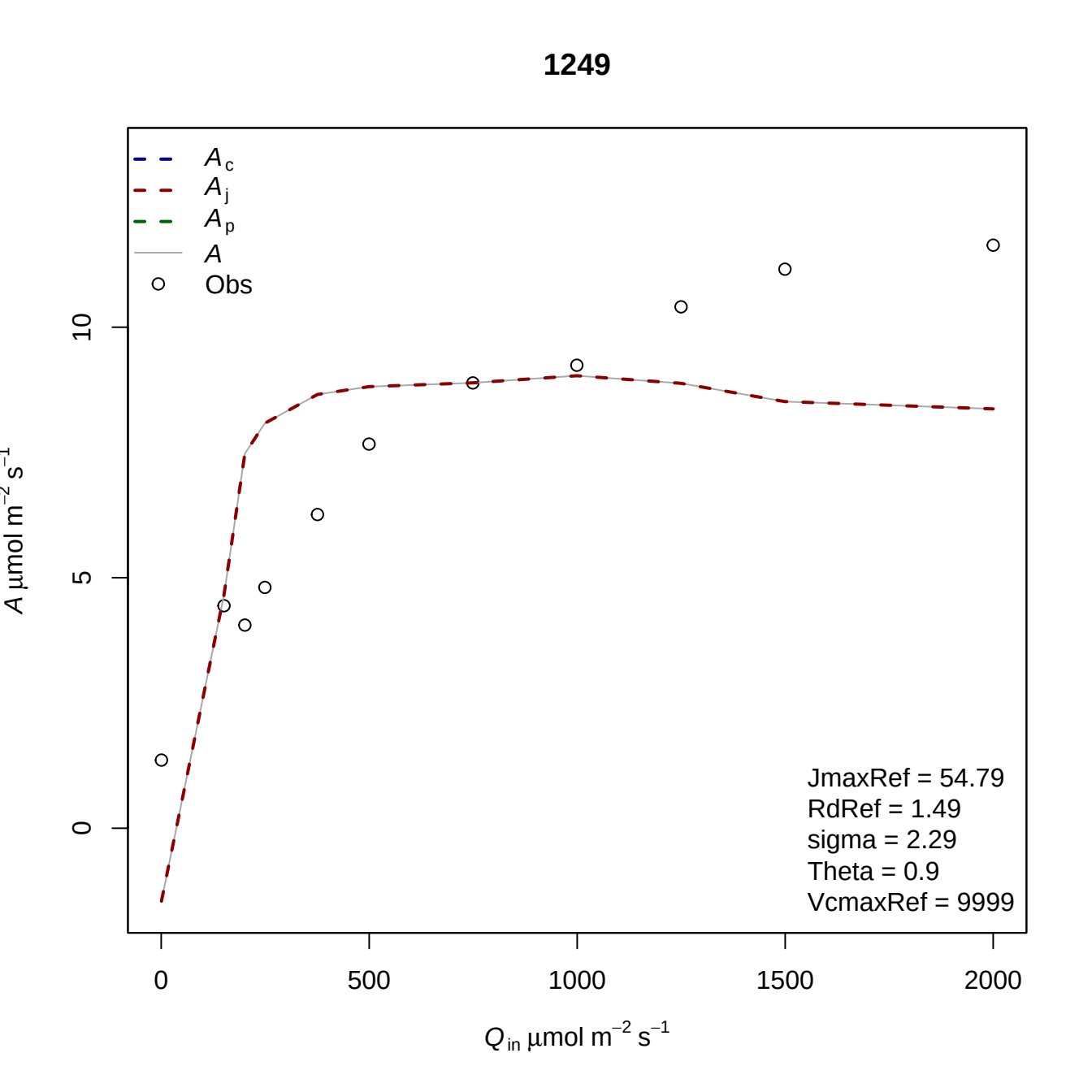

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 98.66
RdRef = 5.46
sigma = 1.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

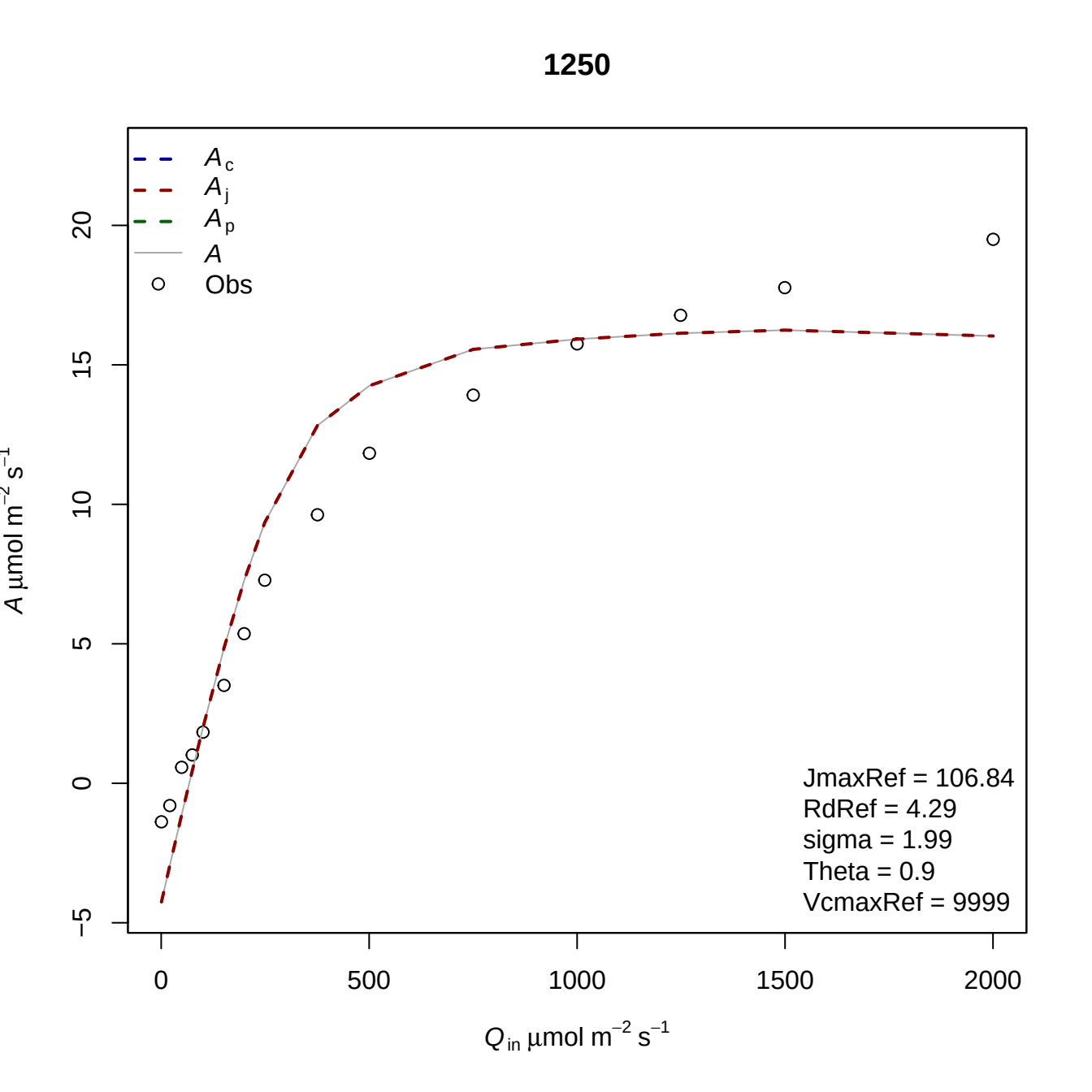

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 99.24
RdRef = 4.33
sigma = 2.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

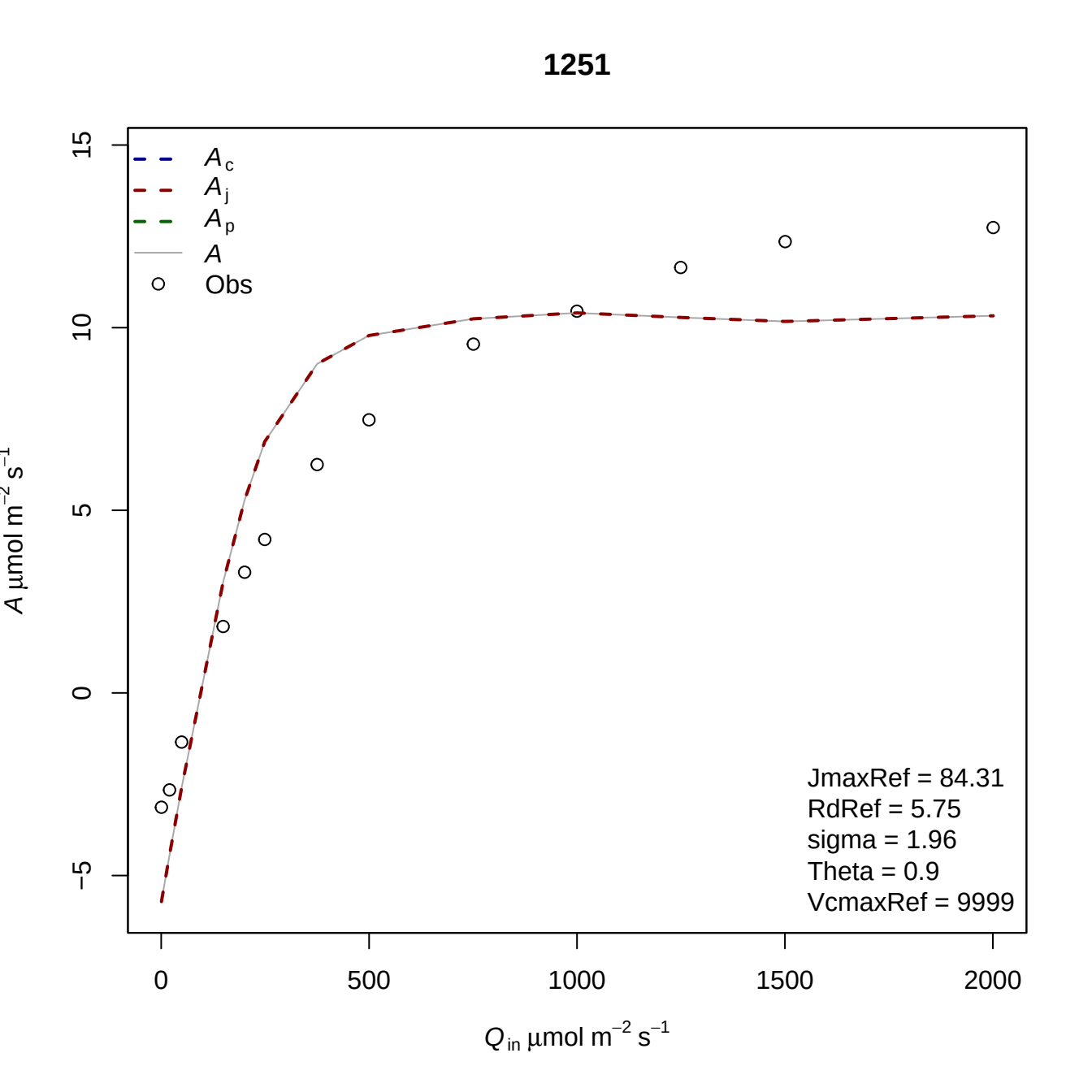

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 80.53
RdRef = 4.28
sigma = 1.69
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

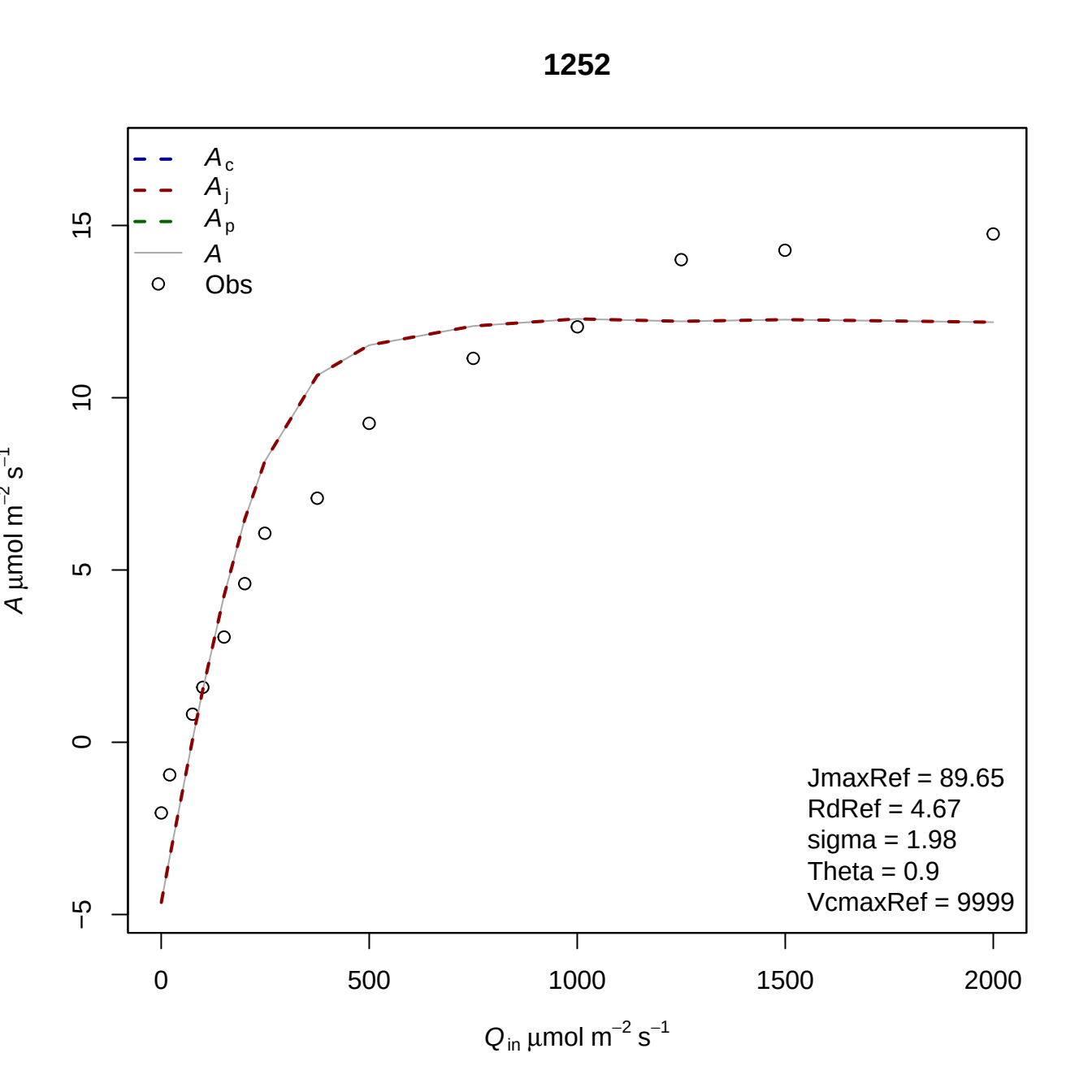

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 82.44
RdRef = 3.97
sigma = 1.7
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

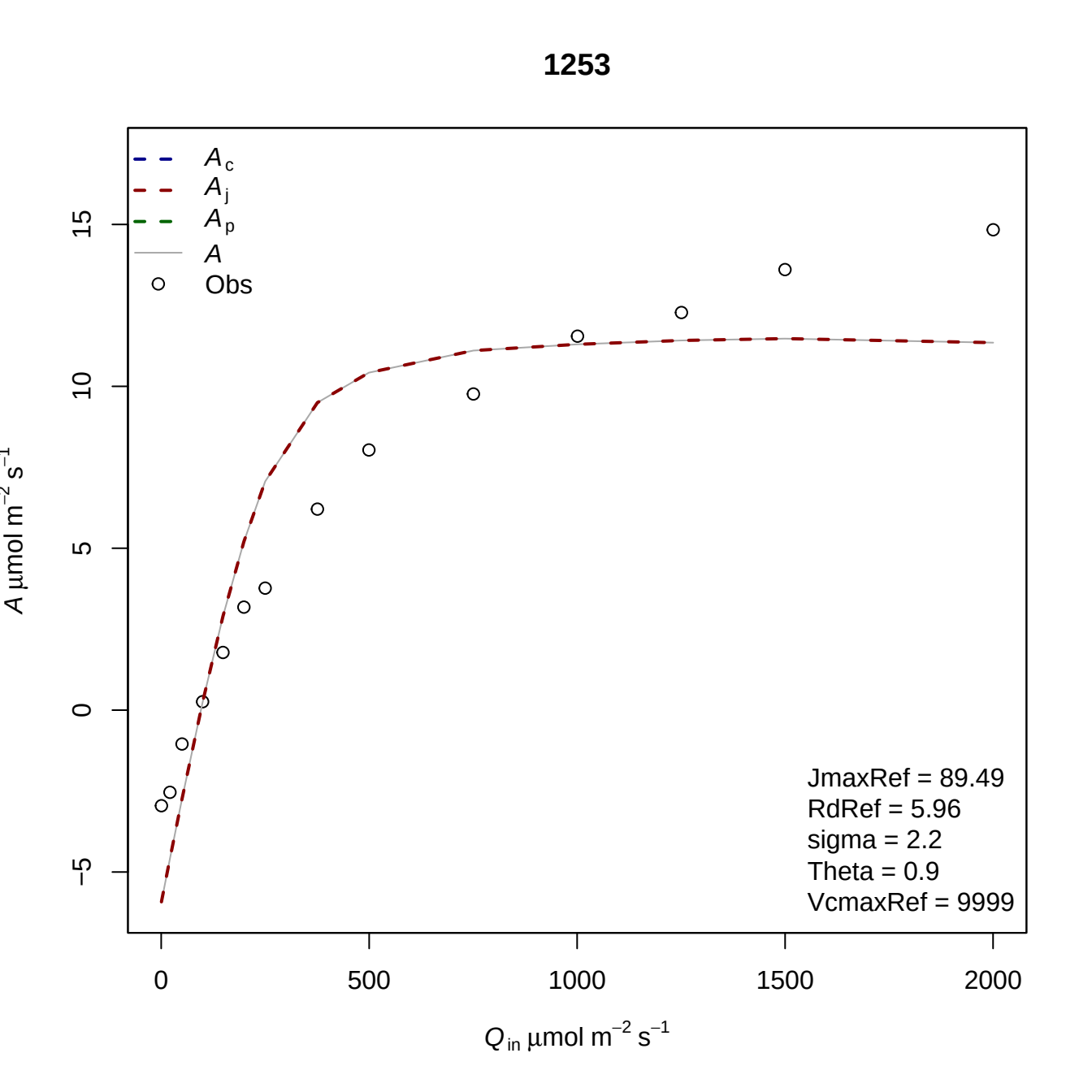

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 95.95
RdRef = 5.5
sigma = 2.49
Theta = 0.9
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

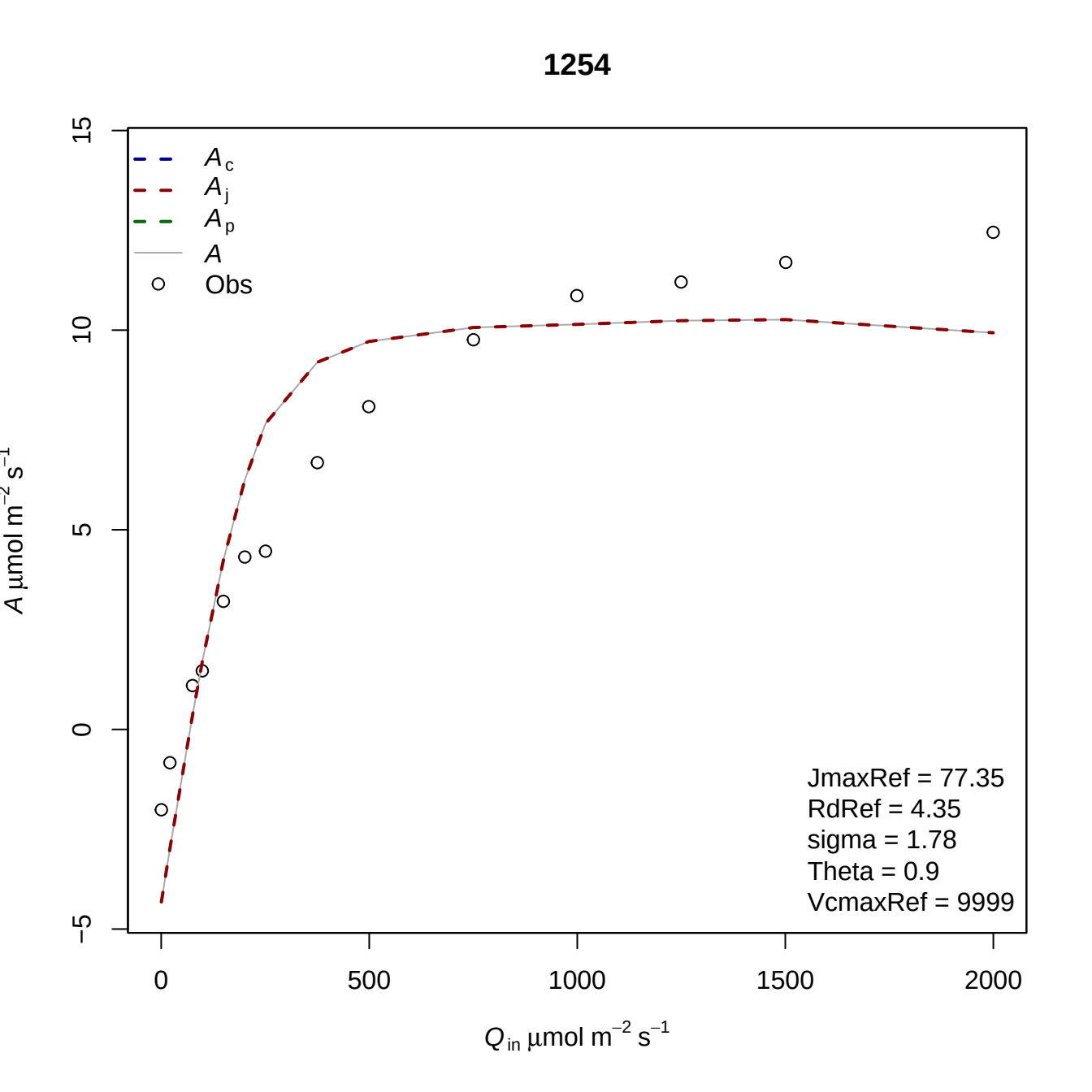

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 54.79
RdRef = 1.49
sigma = 2.29
Theta = 0.9
VcmaxRef = 9999
0
5
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

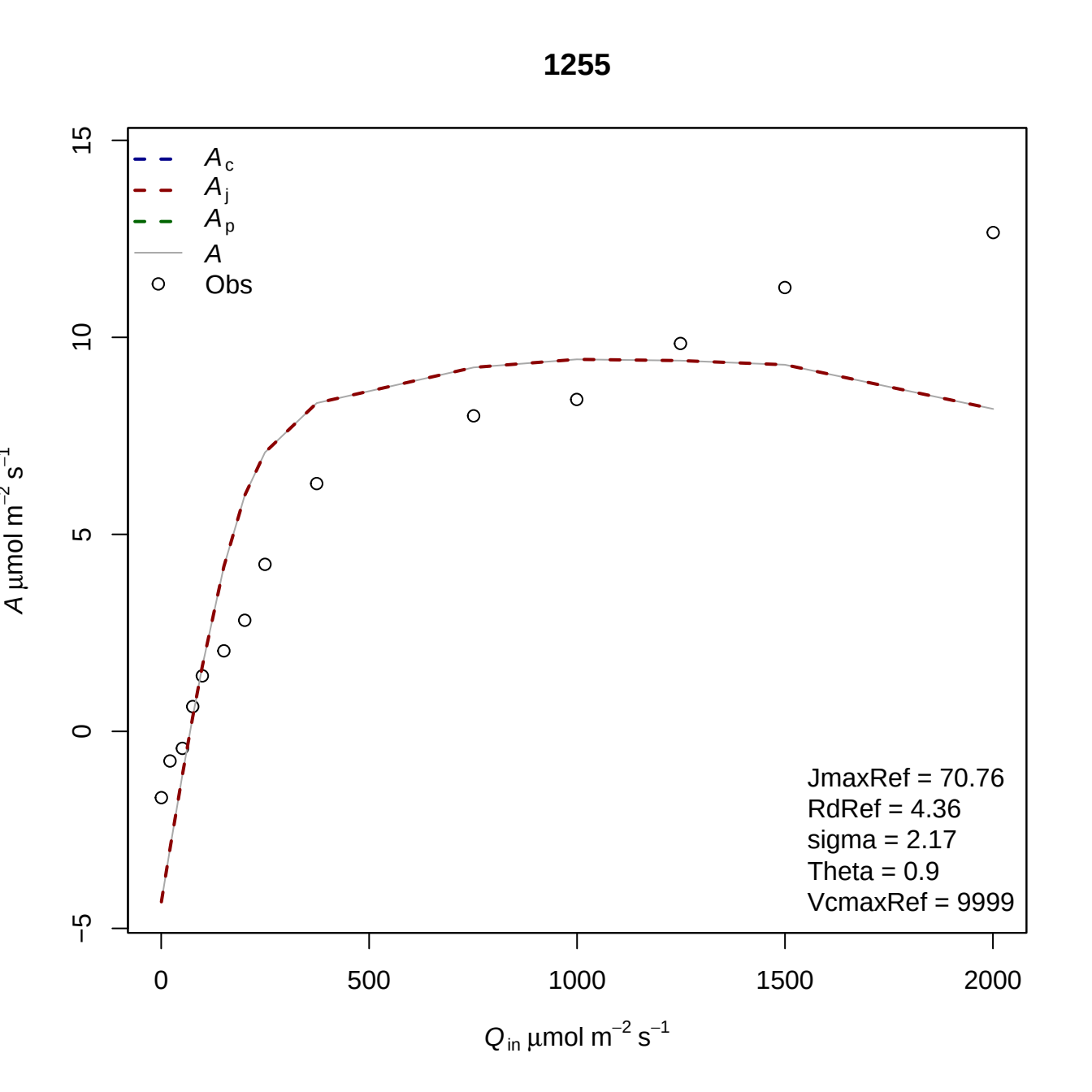

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 106.84
RdRef = 4.29
sigma = 1.99
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

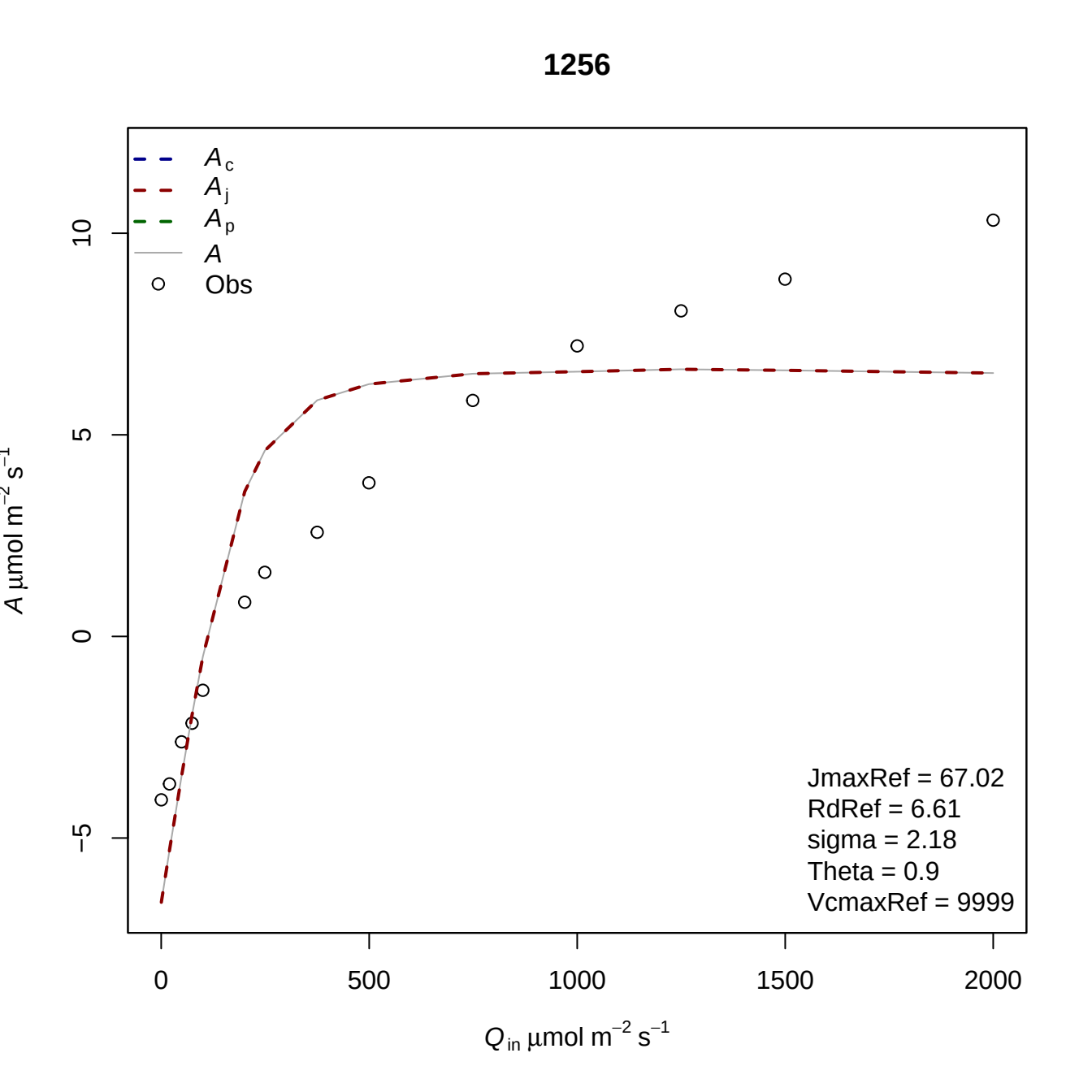

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 84.31
RdRef = 5.75
sigma = 1.96
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

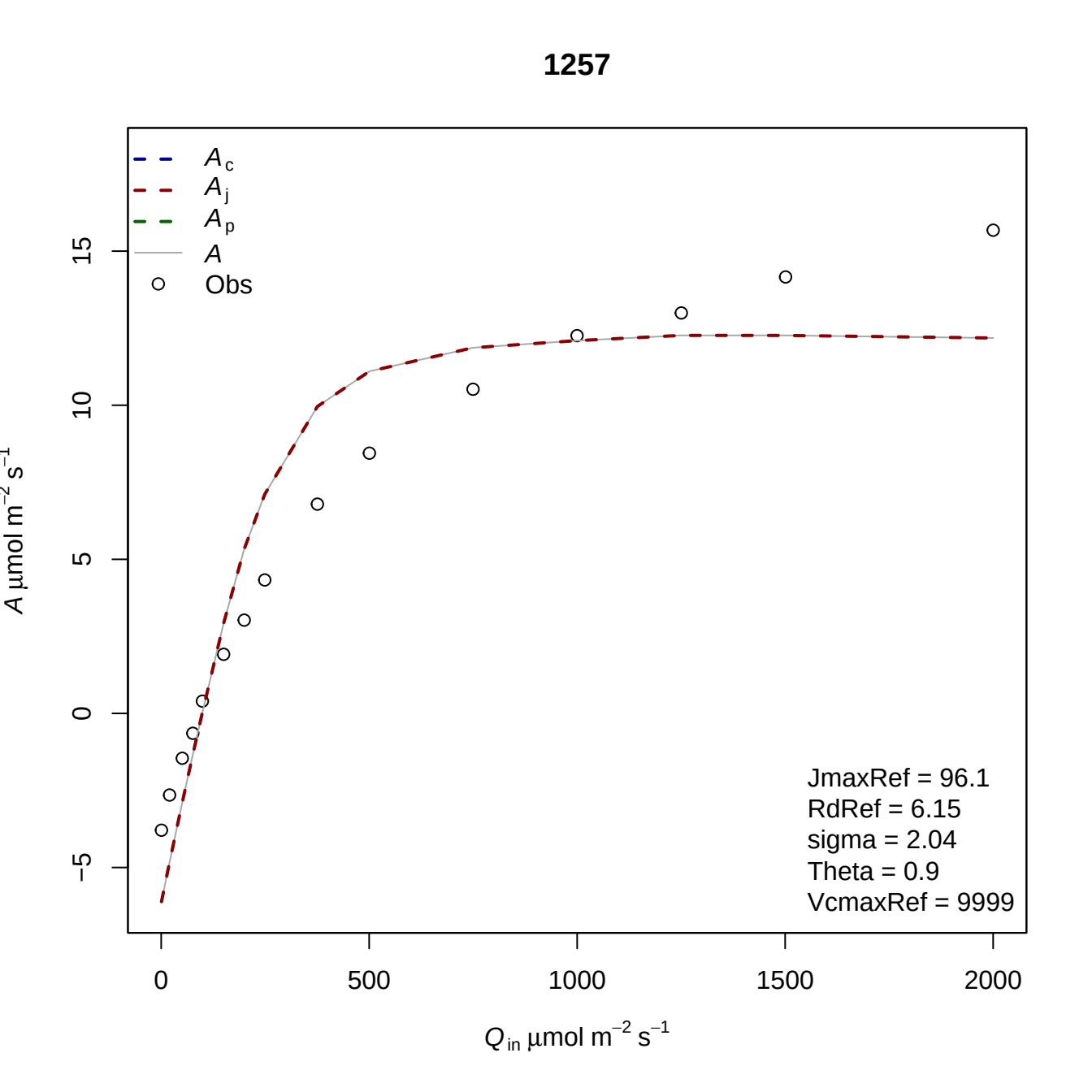

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 89.65
RdRef = 4.67
sigma = 1.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

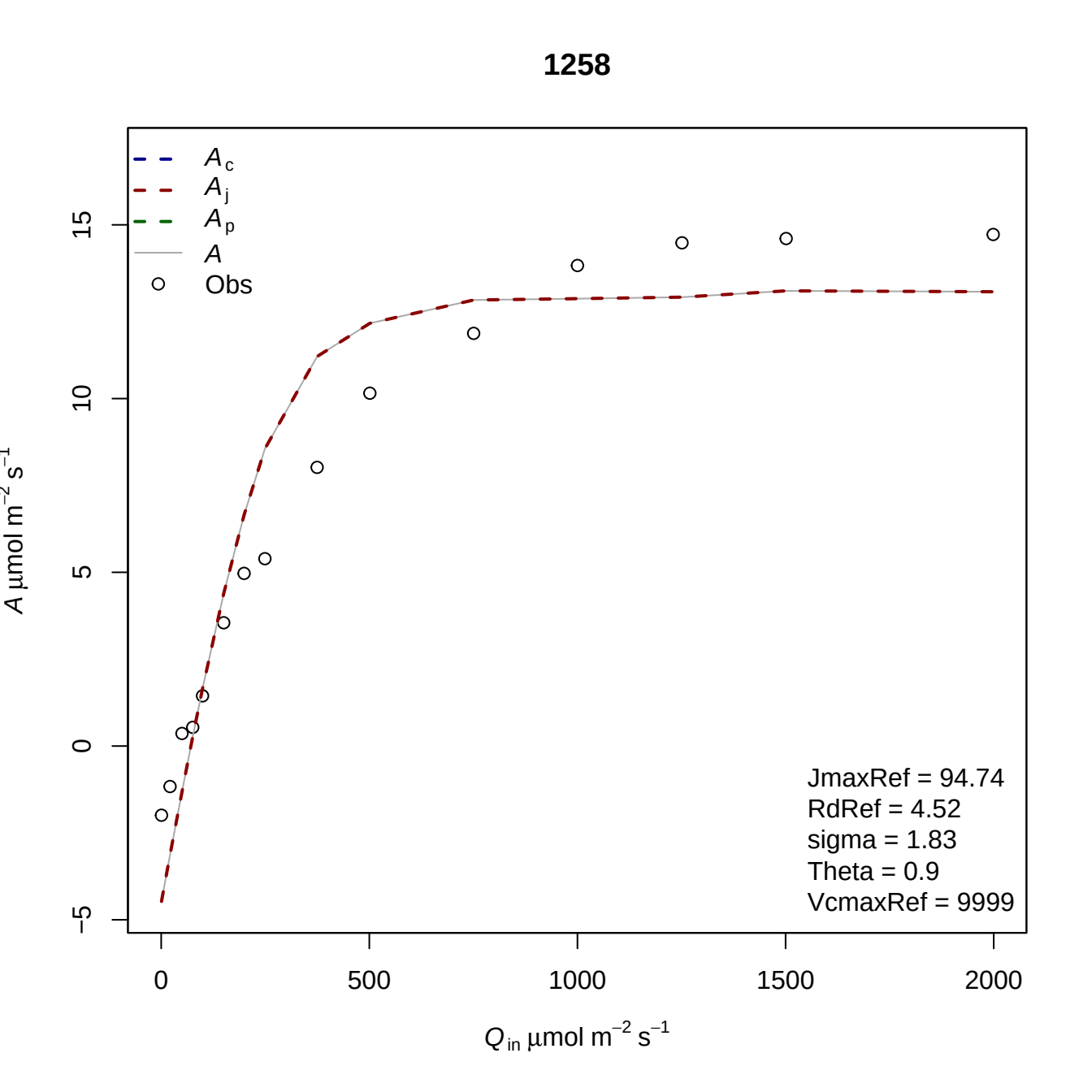

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 89.49
RdRef = 5.96
sigma = 2.2
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

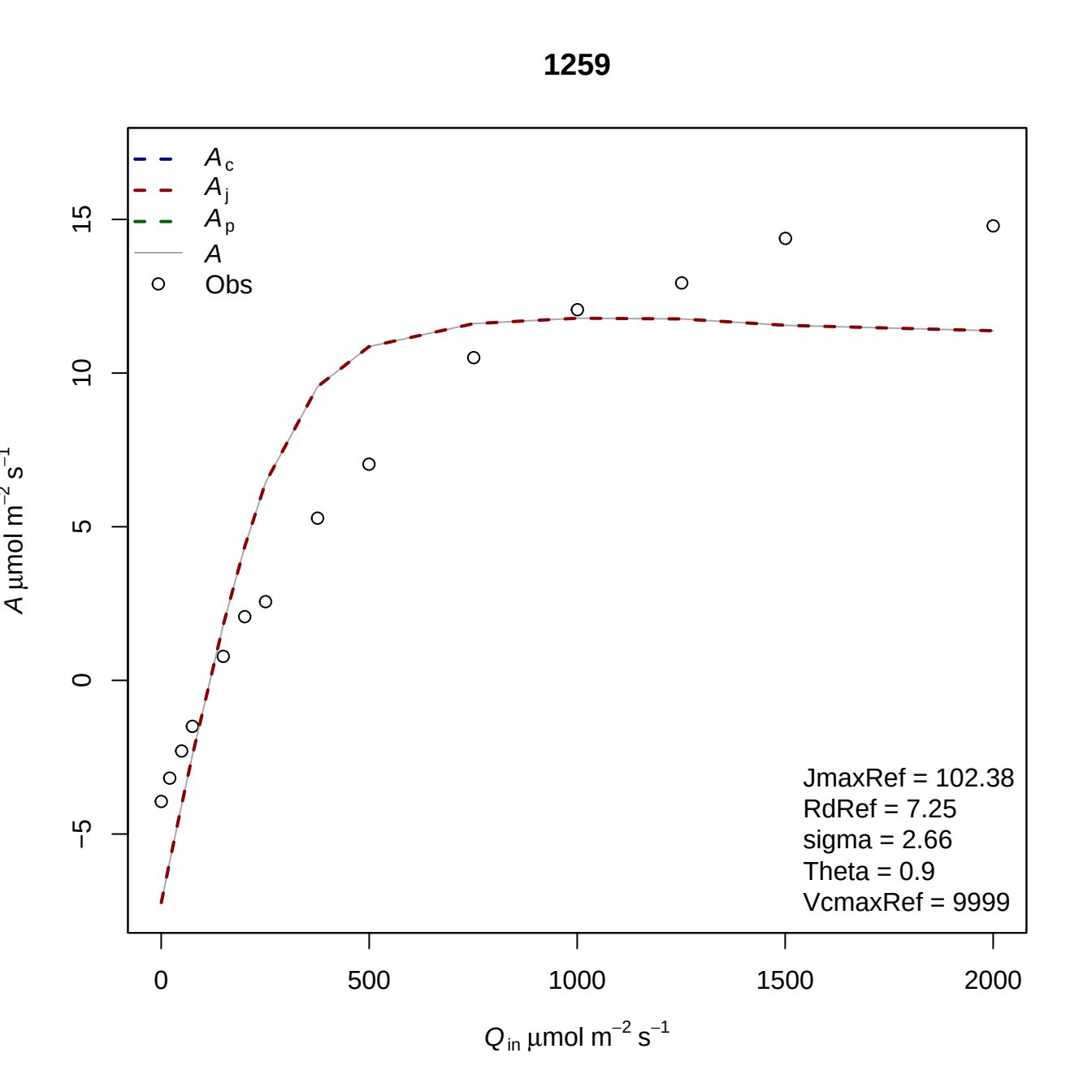

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 77.35
RdRef = 4.35
sigma = 1.78
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

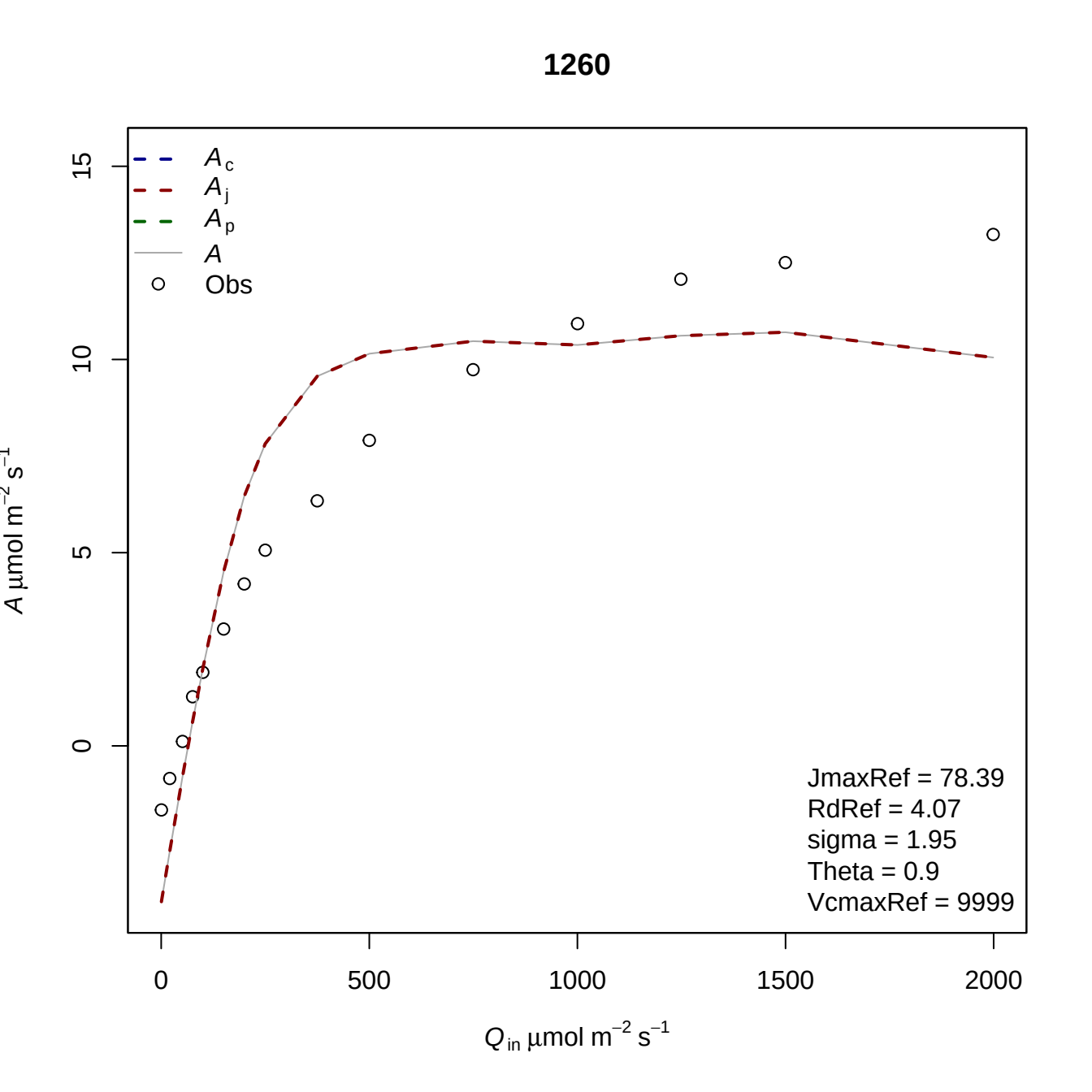

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 70.76
RdRef = 4.36
sigma = 2.17
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

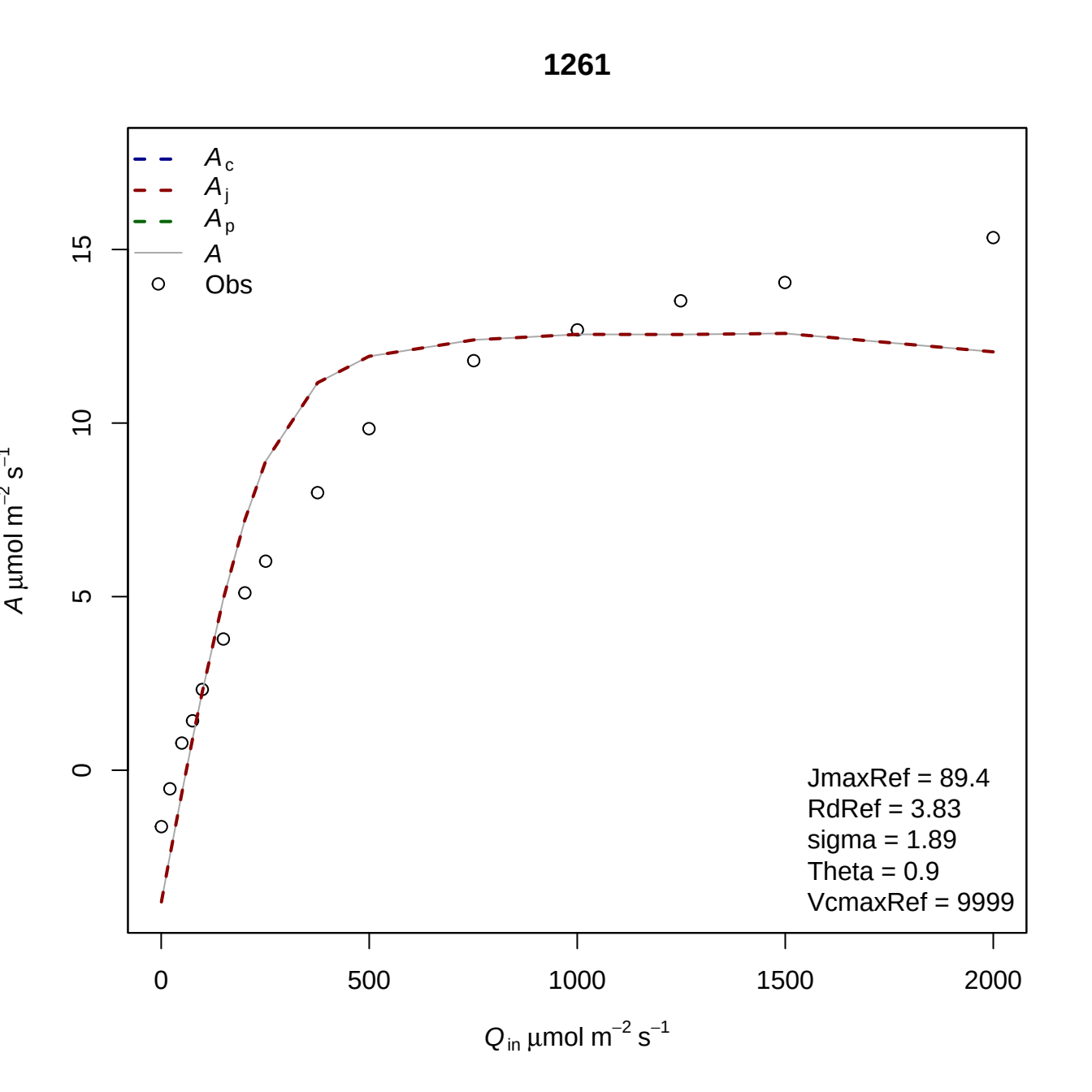

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 67.02
RdRef = 6.61
sigma = 2.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

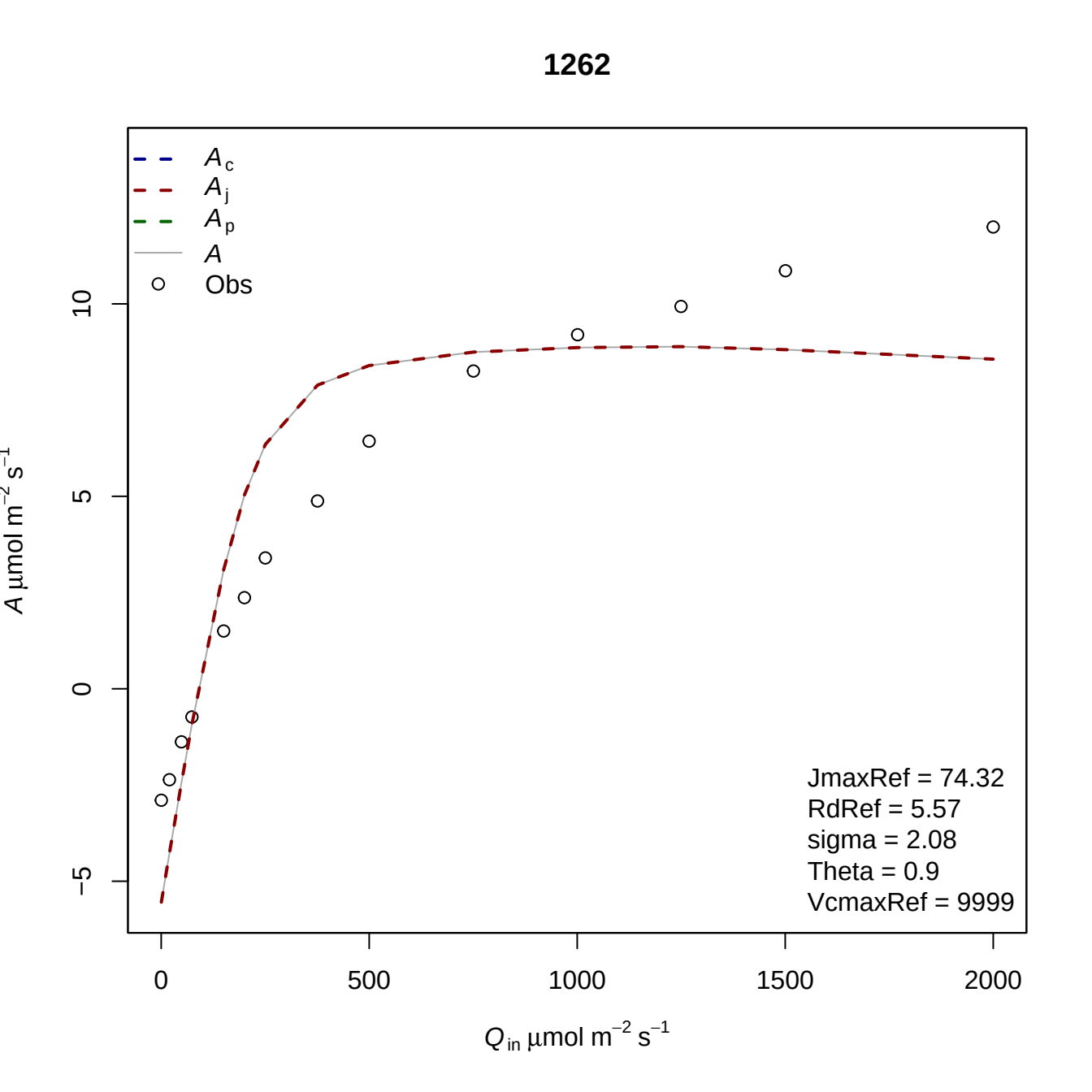

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 96.1
RdRef = 6.15
sigma = 2.04
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

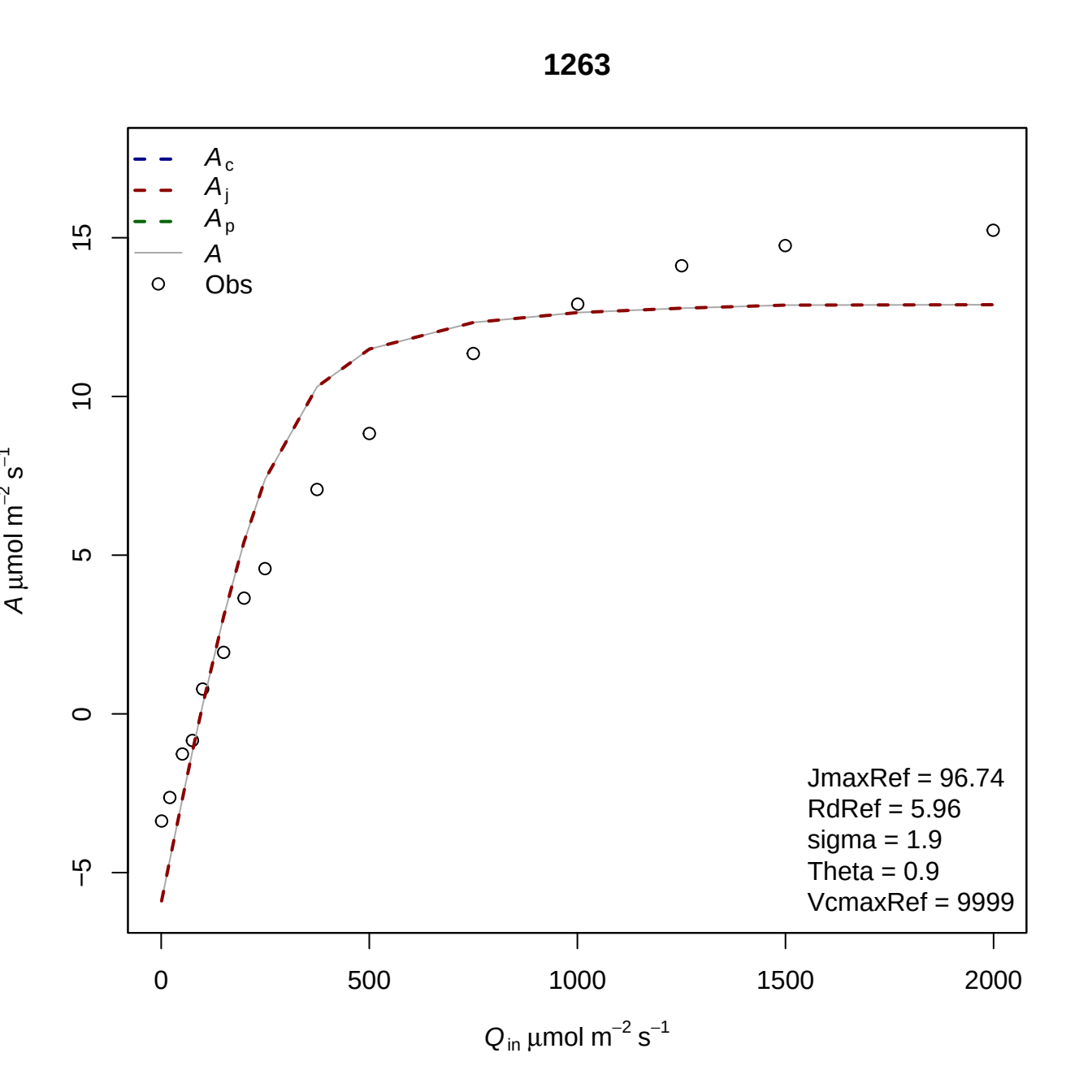

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 94.74
RdRef = 4.52
sigma = 1.83
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

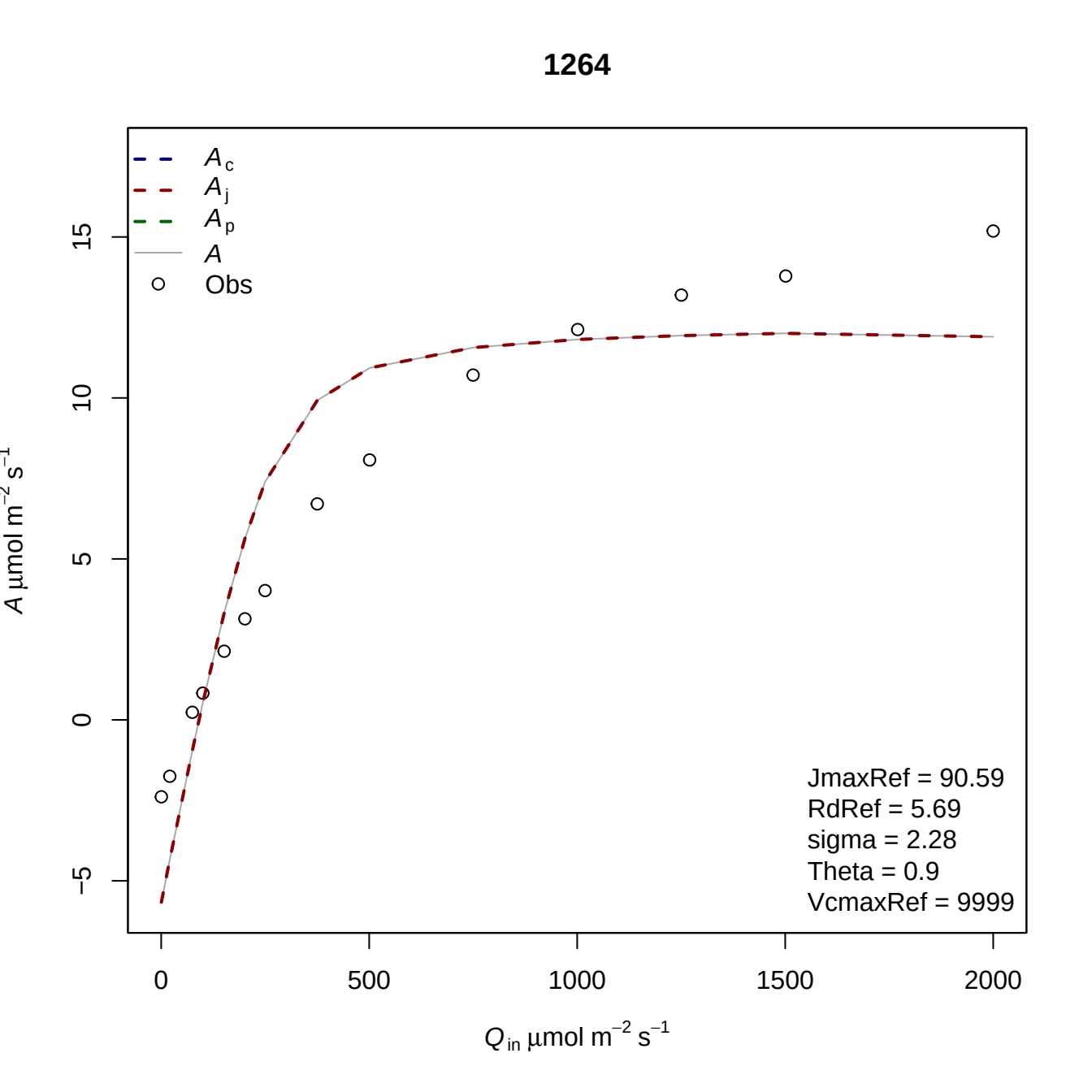

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 102.38
RdRef = 7.25
sigma = 2.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

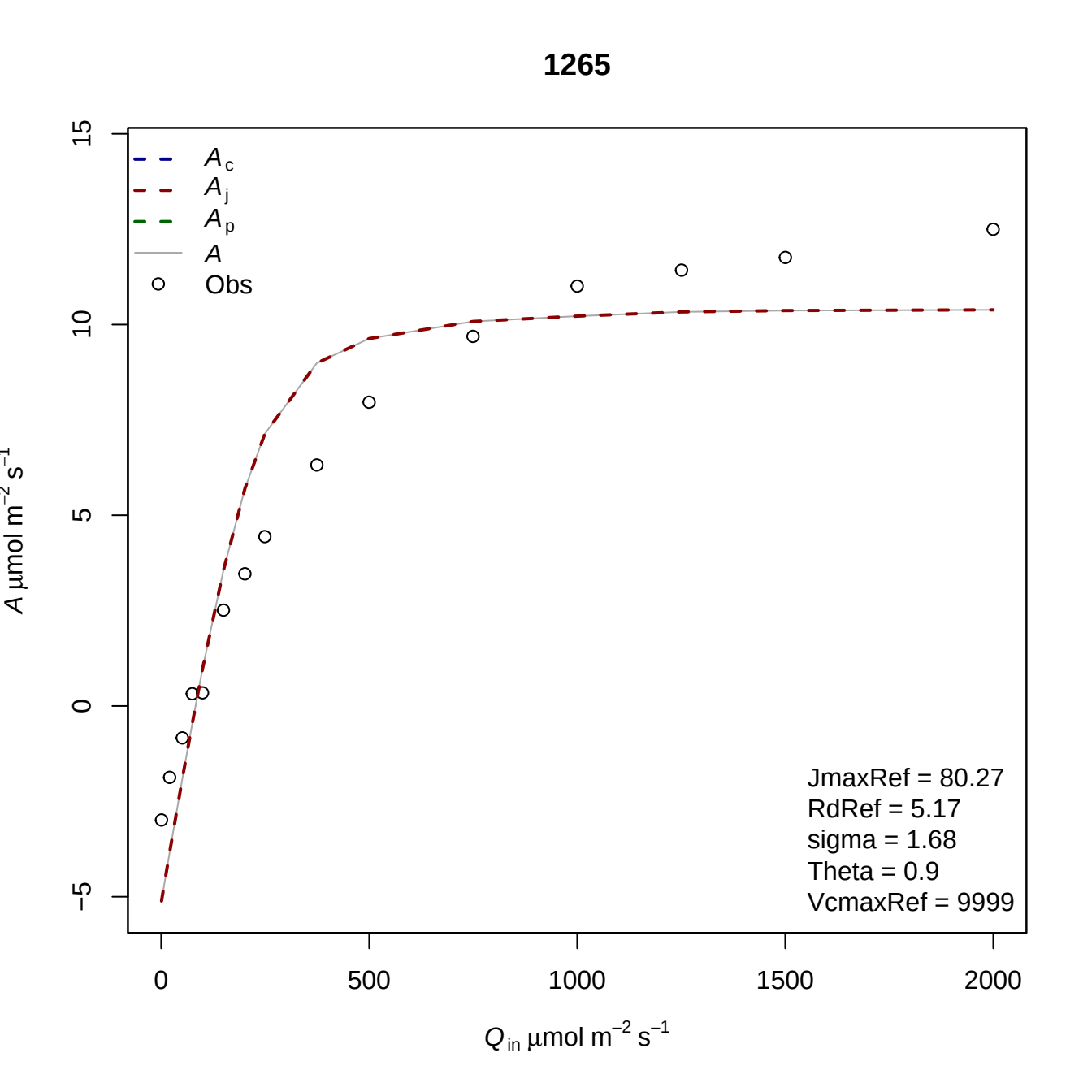

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 78.39
RdRef = 4.07
sigma = 1.95
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

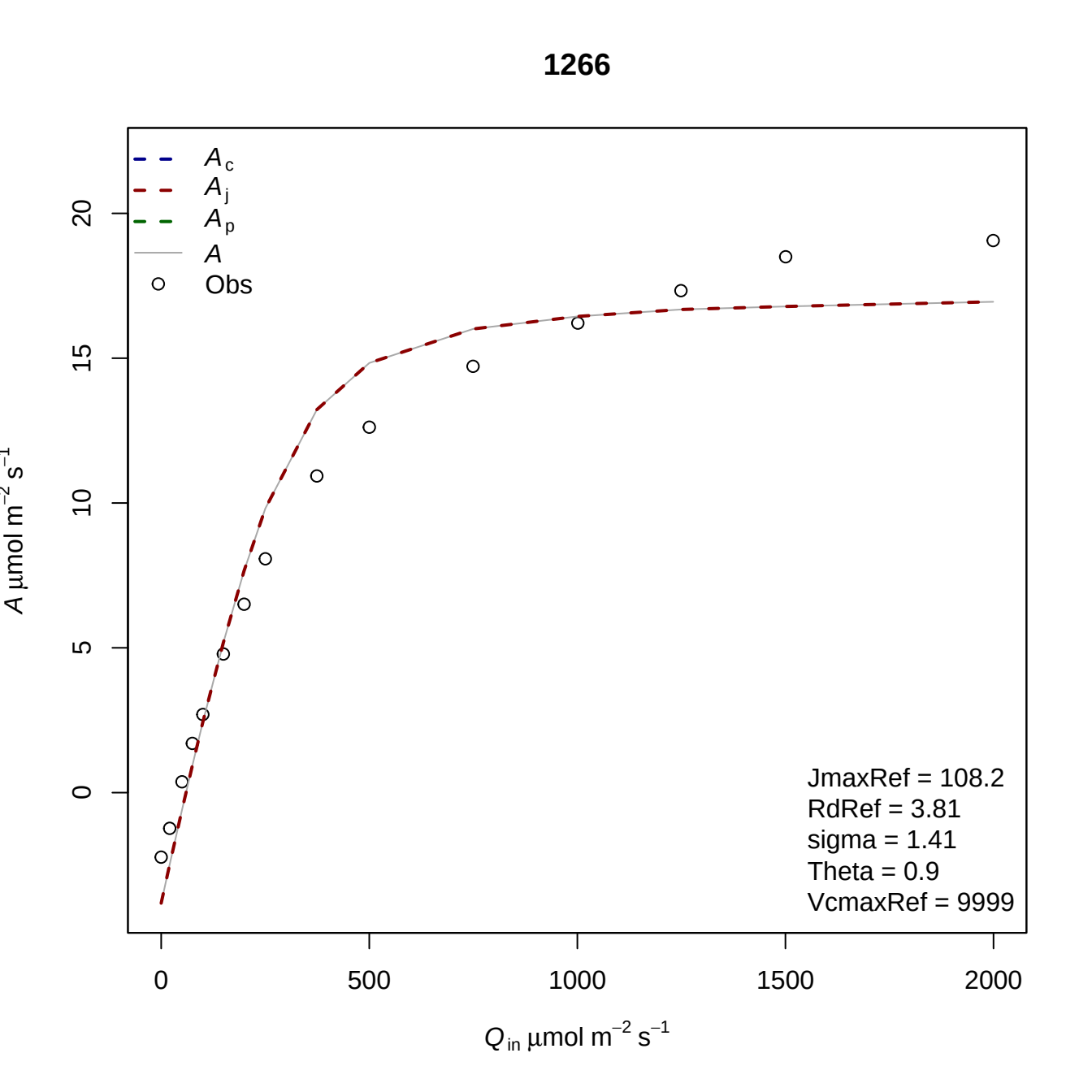

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 89.4
RdRef = 3.83
sigma = 1.89
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

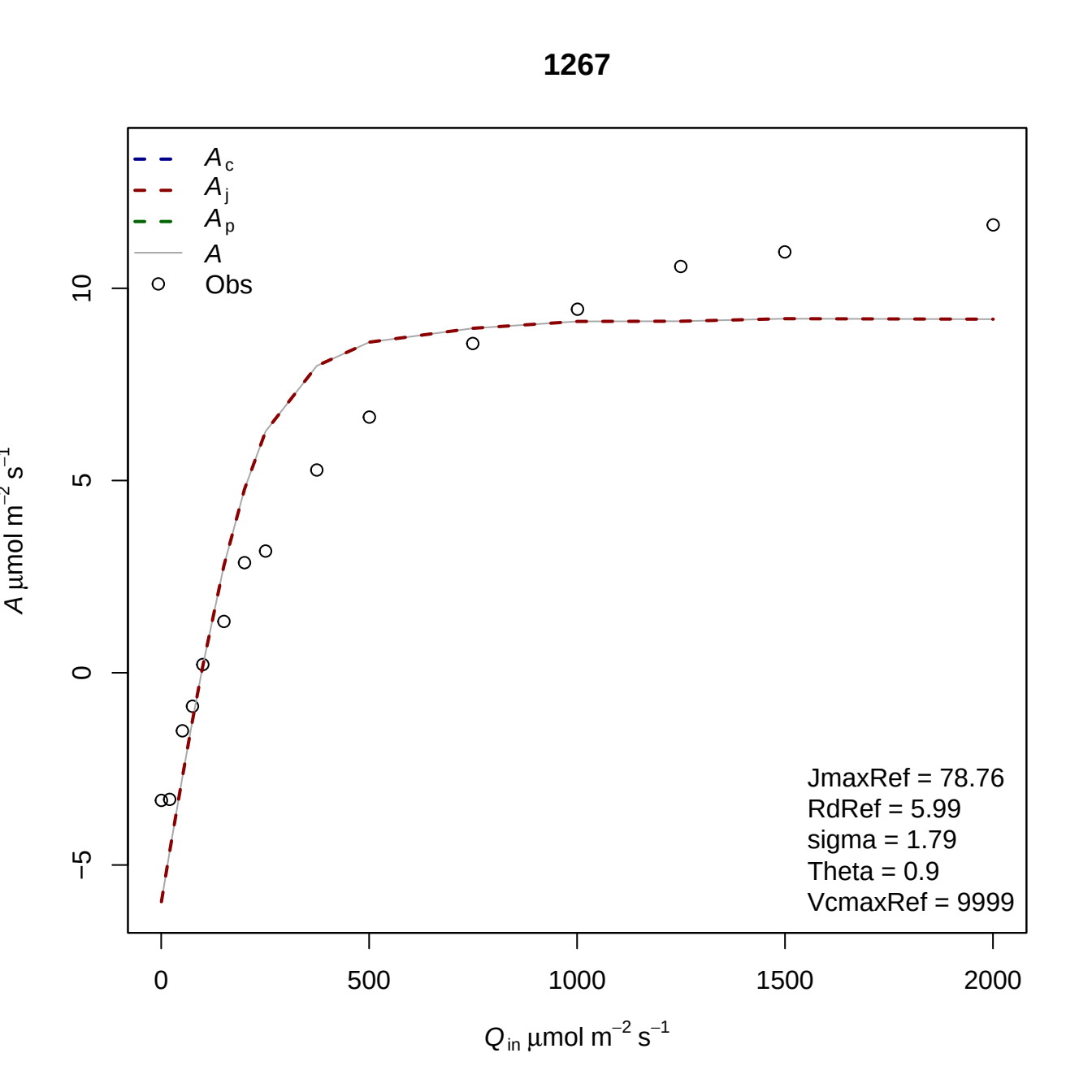

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 74.32
RdRef = 5.57
sigma = 2.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

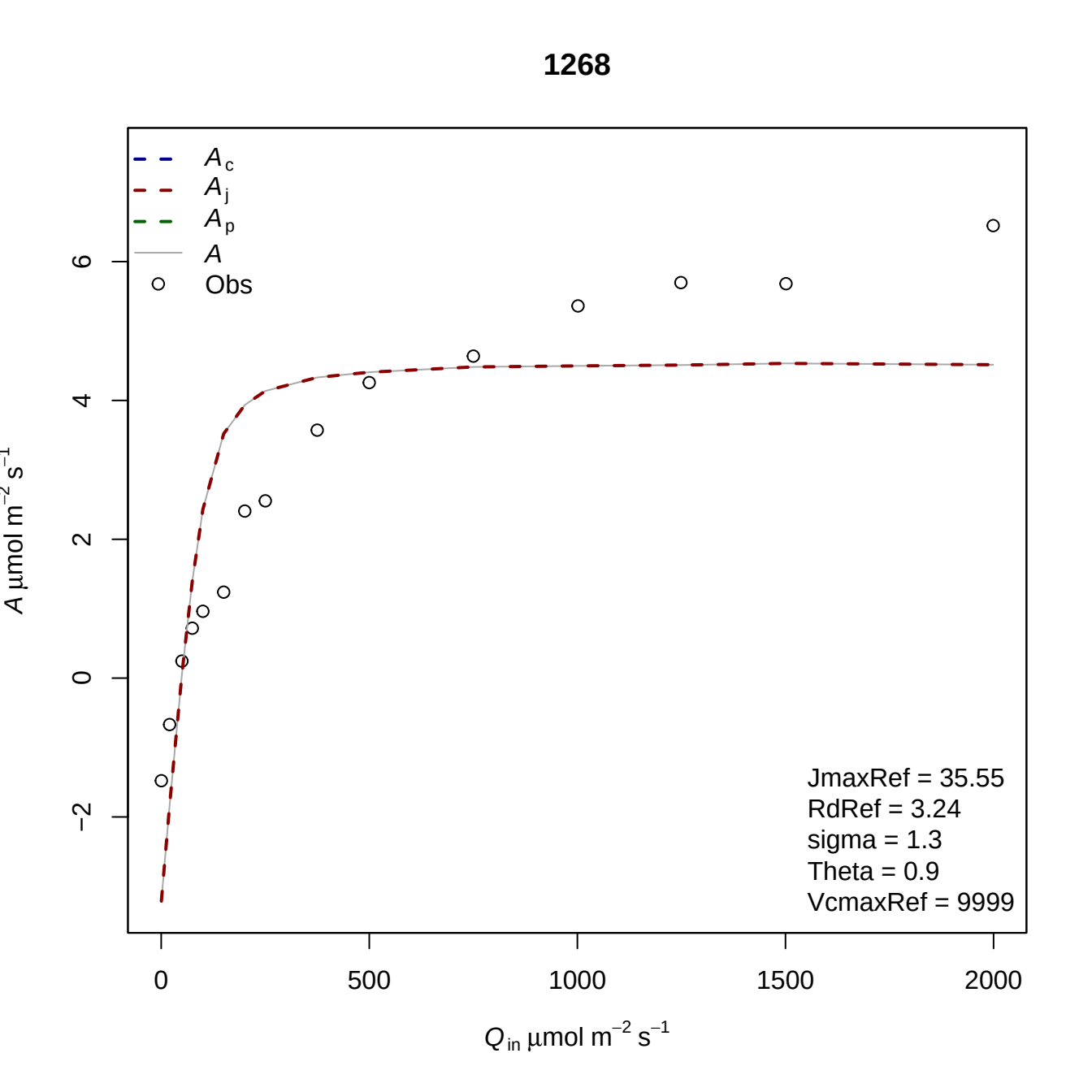

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 96.74
RdRef = 5.96
sigma = 1.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

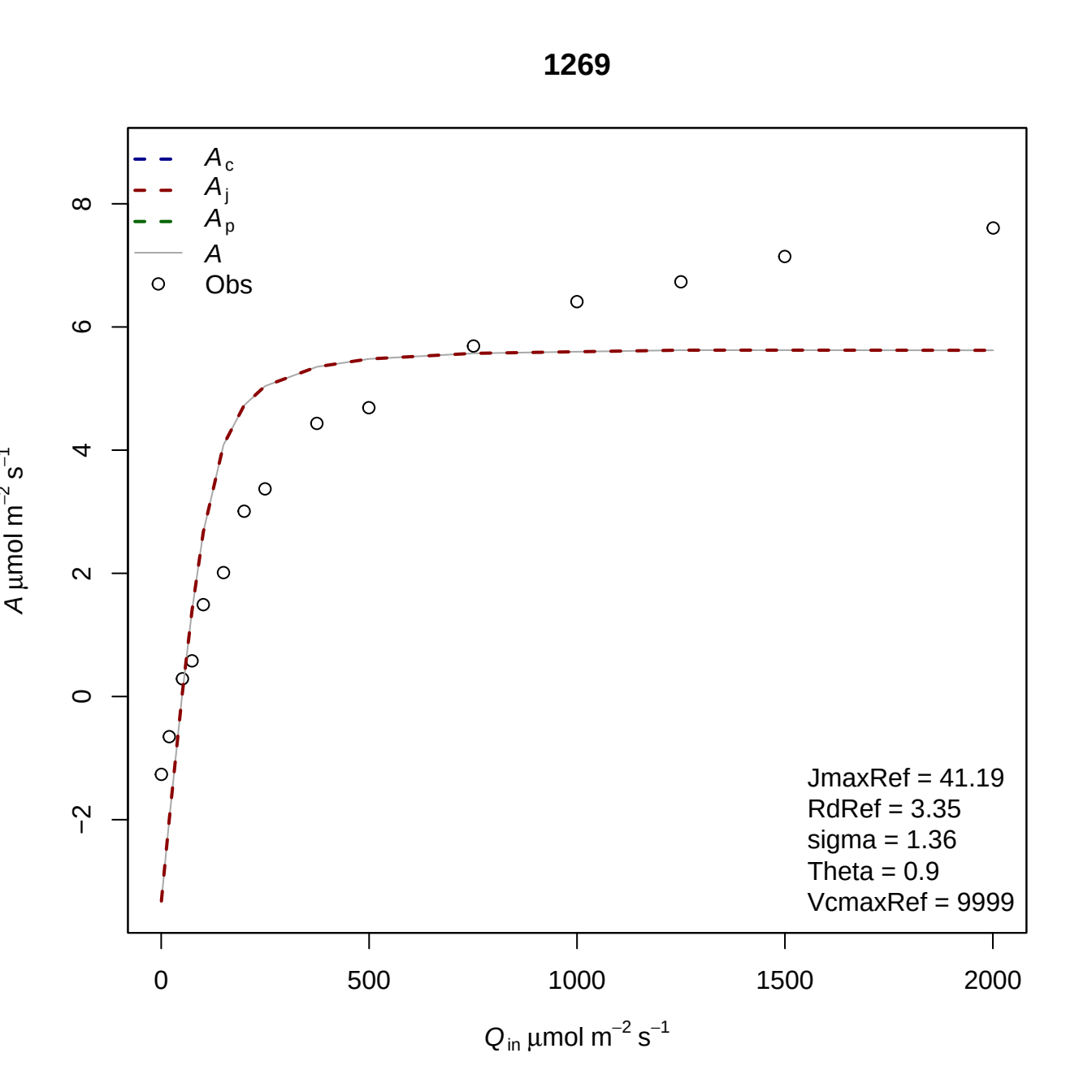

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 90.59
RdRef = 5.69
sigma = 2.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

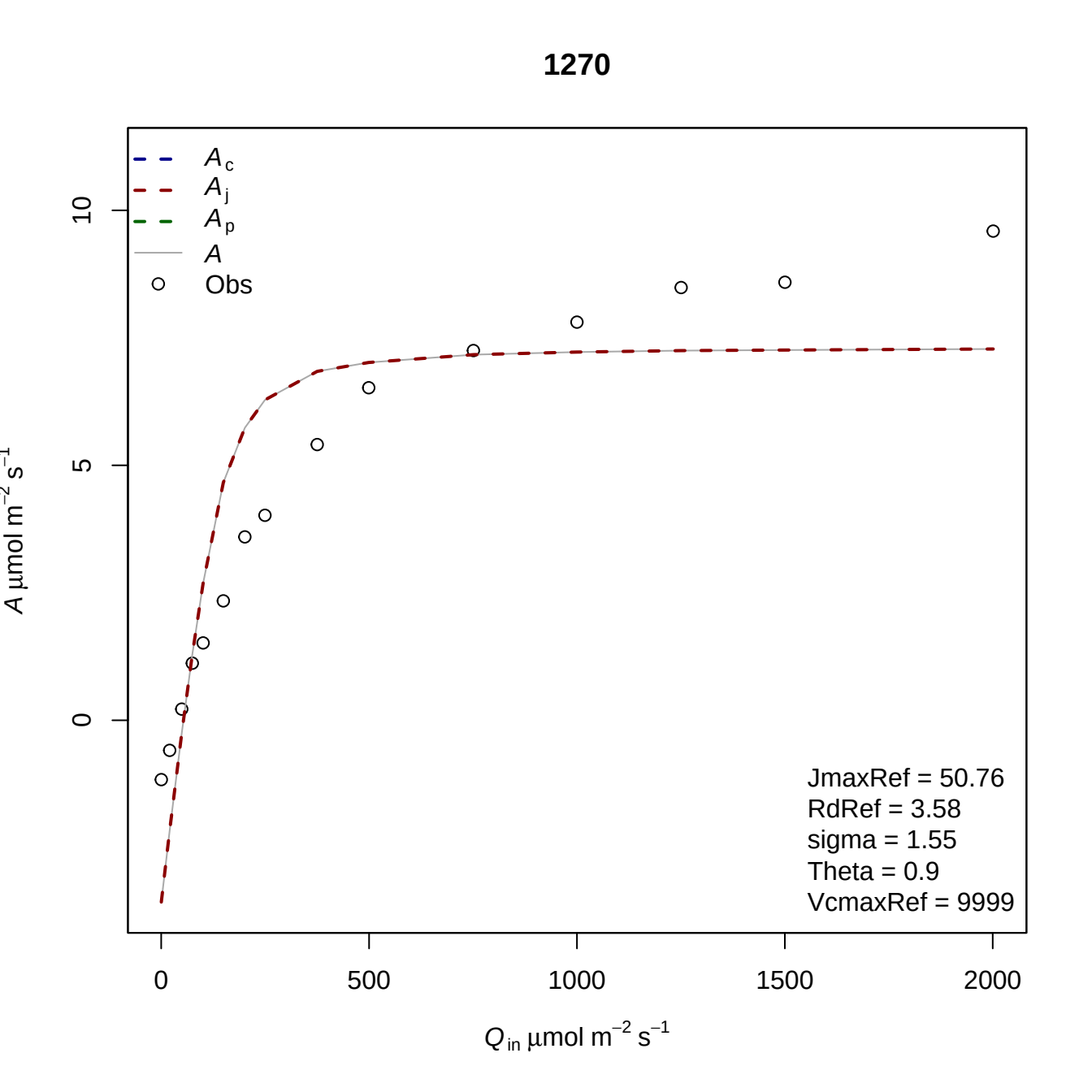

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 80.27
RdRef = 5.17
sigma = 1.68
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

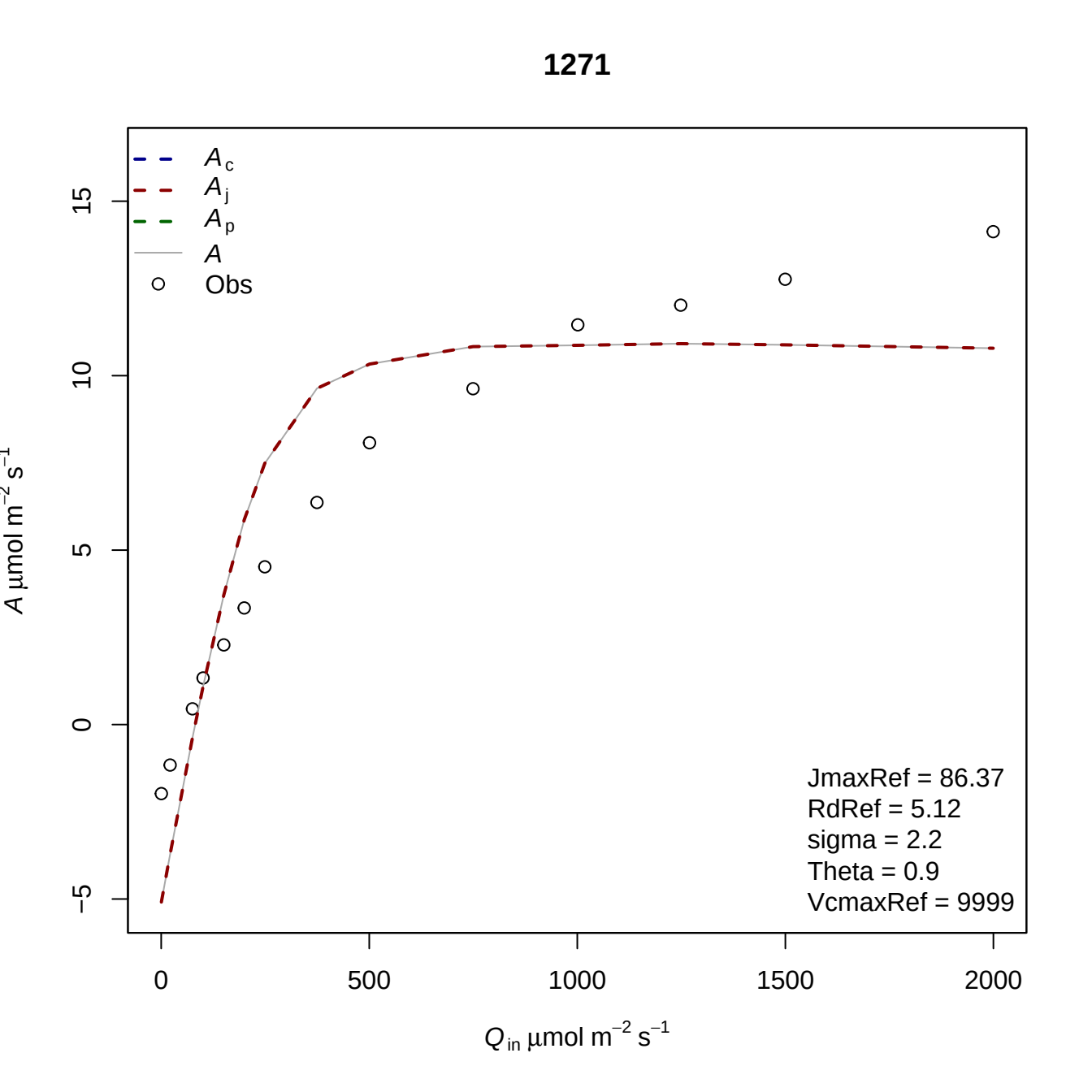

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 108.2
RdRef = 3.81
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

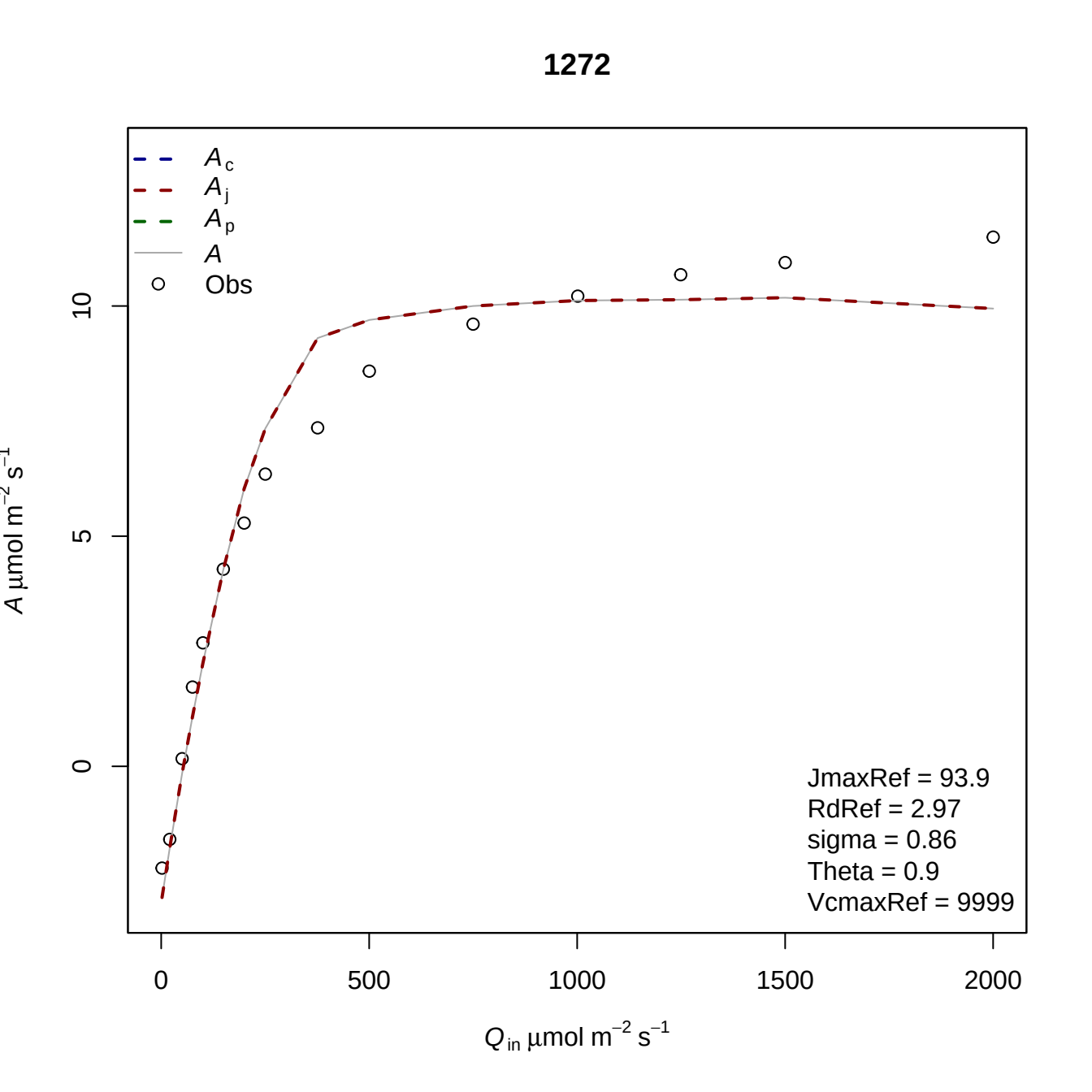

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 78.76
RdRef = 5.99
sigma = 1.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

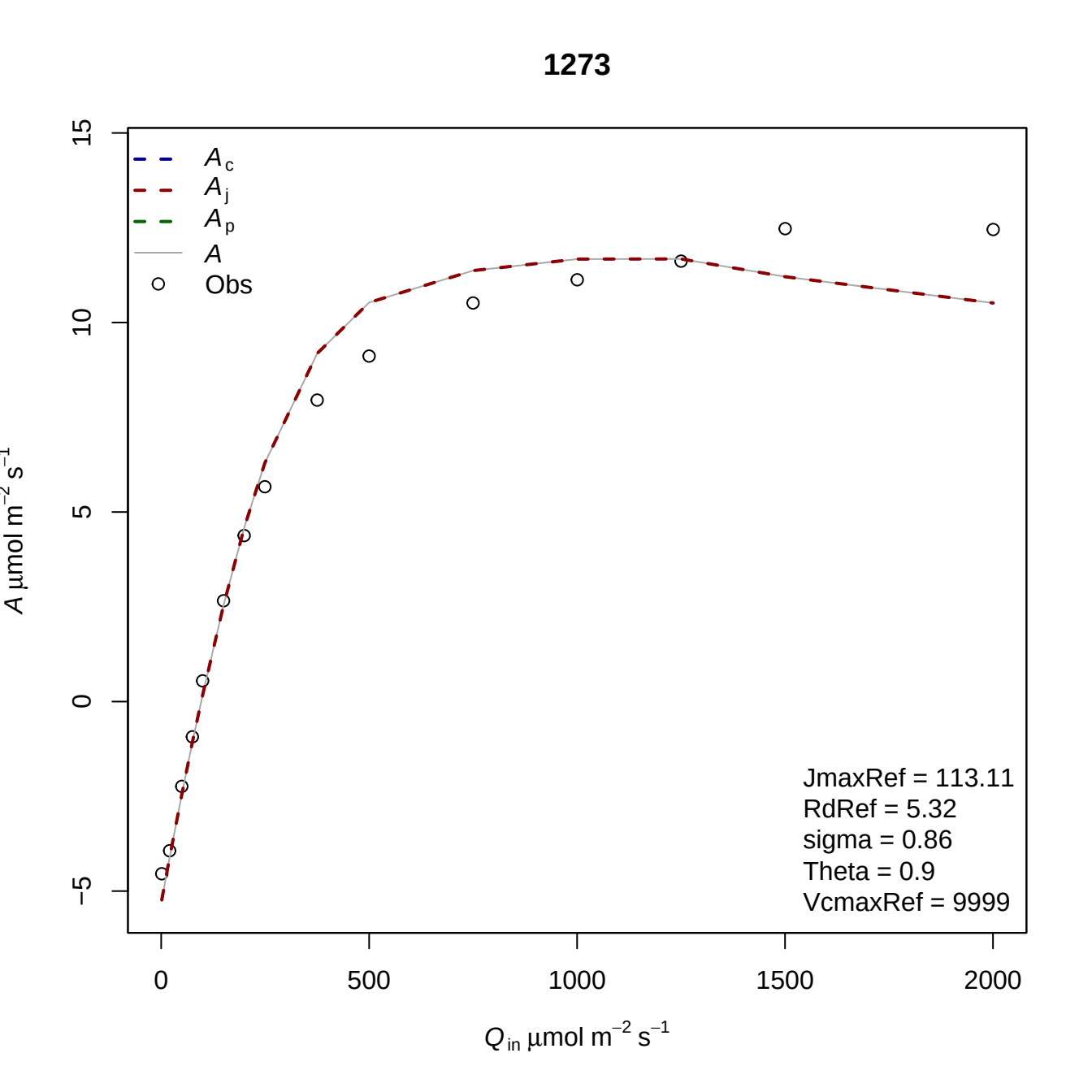

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 3.24
sigma = 1.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

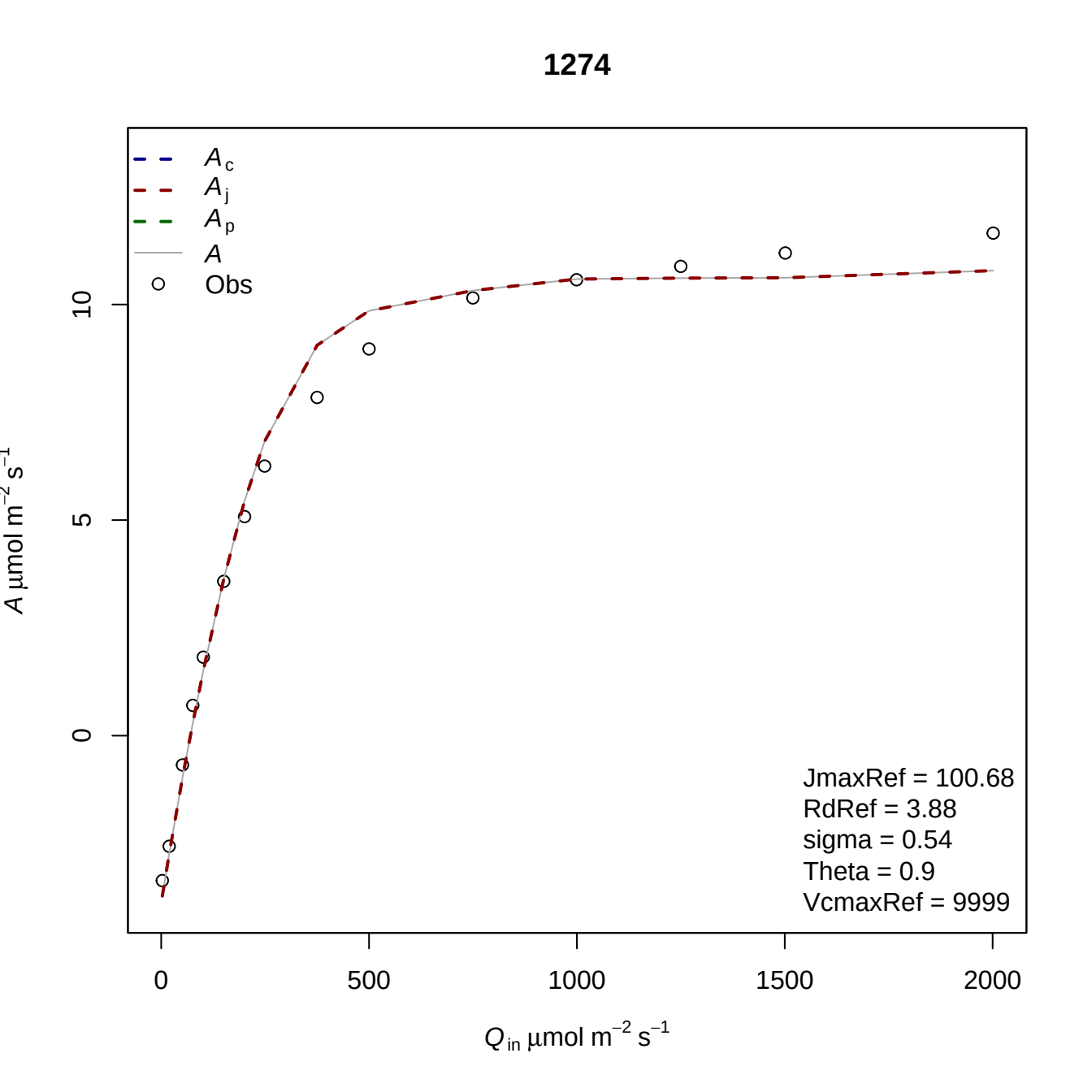

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 41.19
RdRef = 3.35
sigma = 1.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

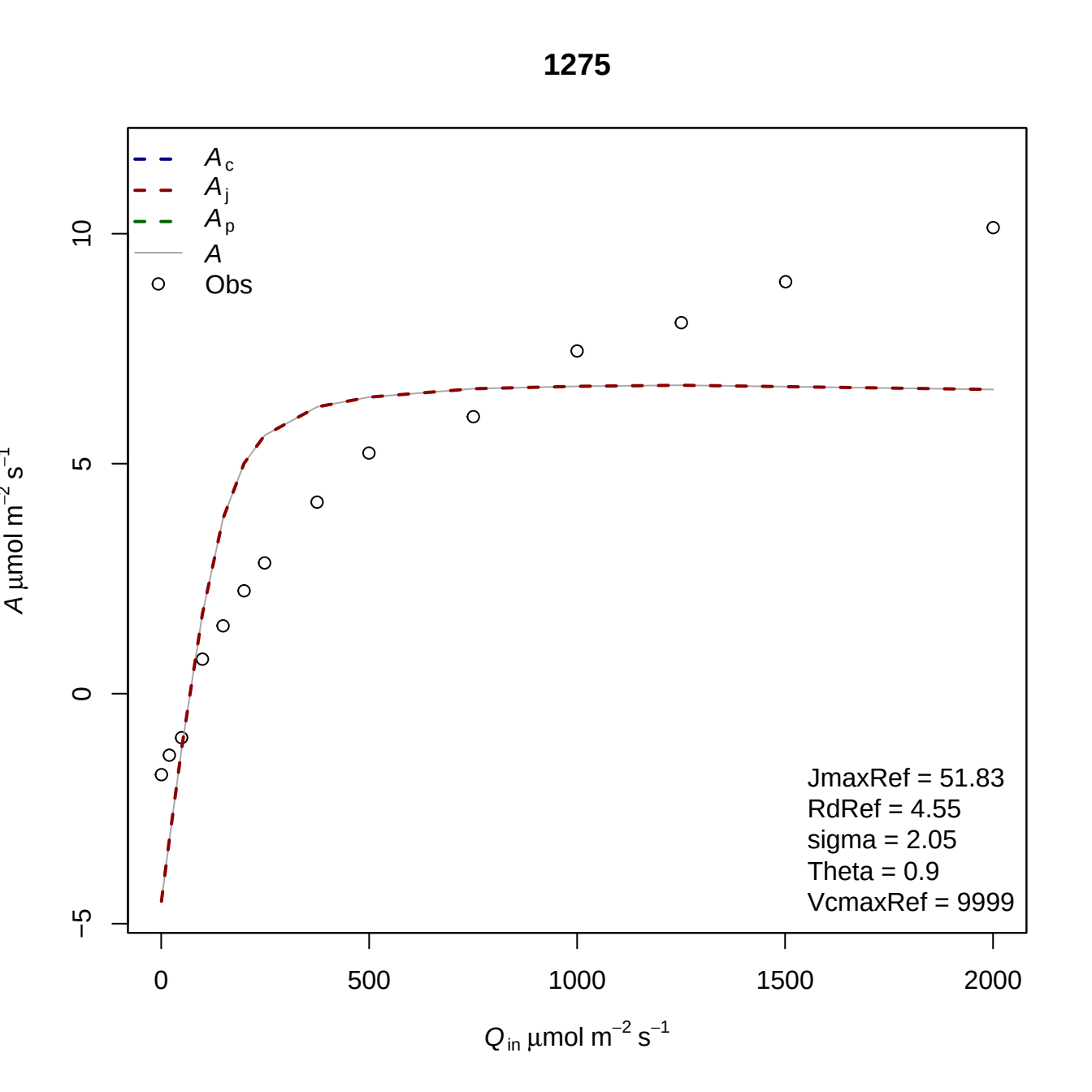

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 50.76
RdRef = 3.58
sigma = 1.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

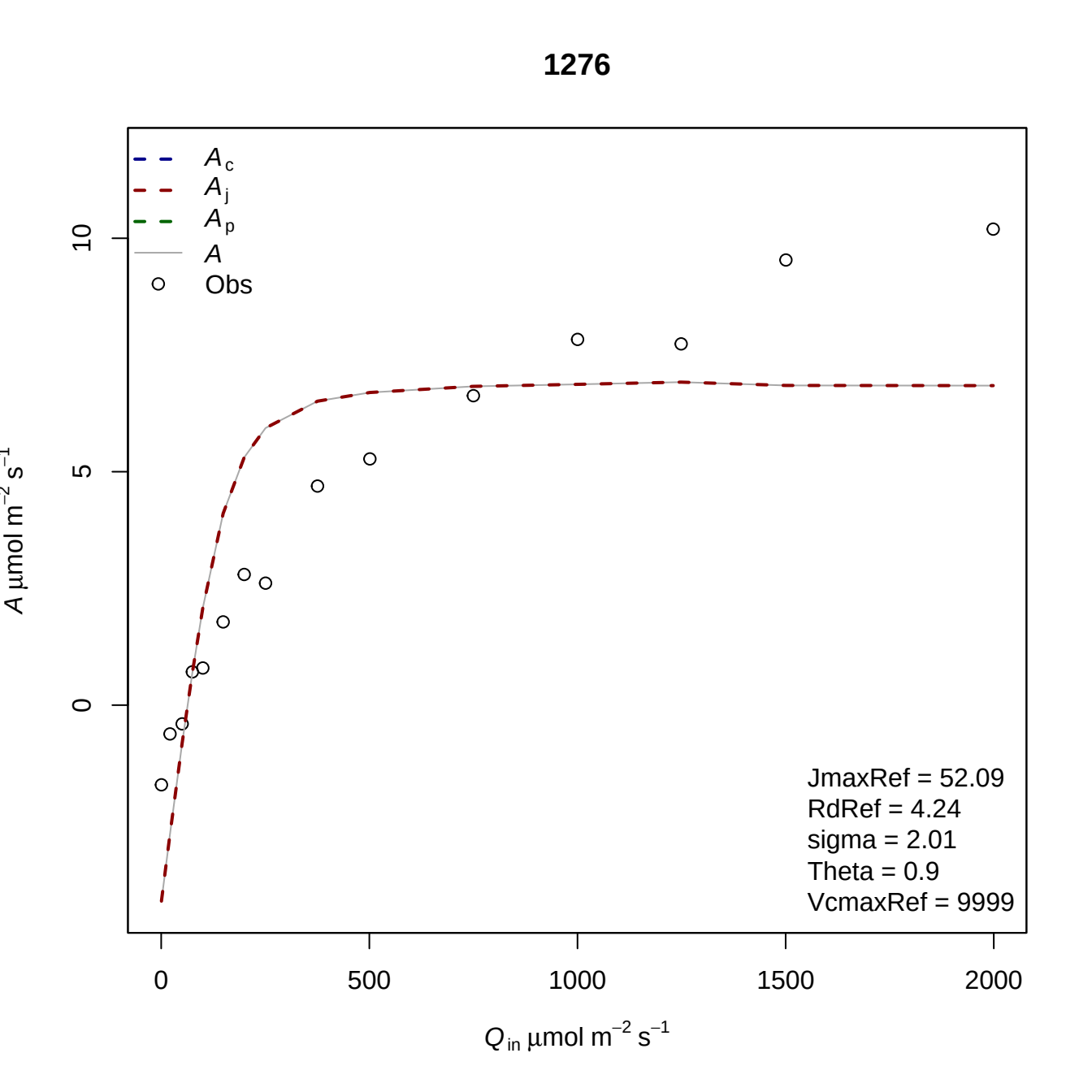

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 86.37
RdRef = 5.12
sigma = 2.2
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

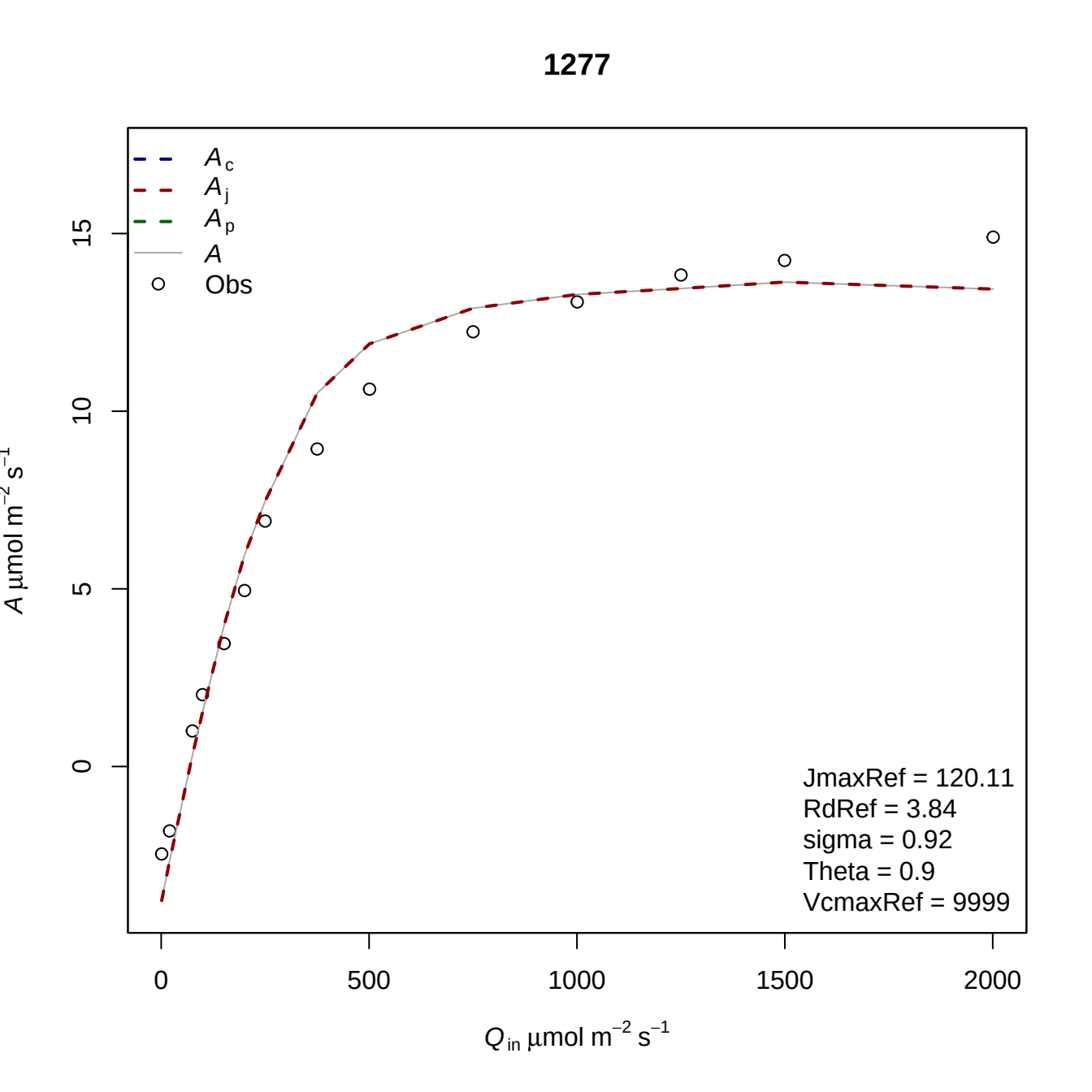

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 93.9
RdRef = 2.97
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

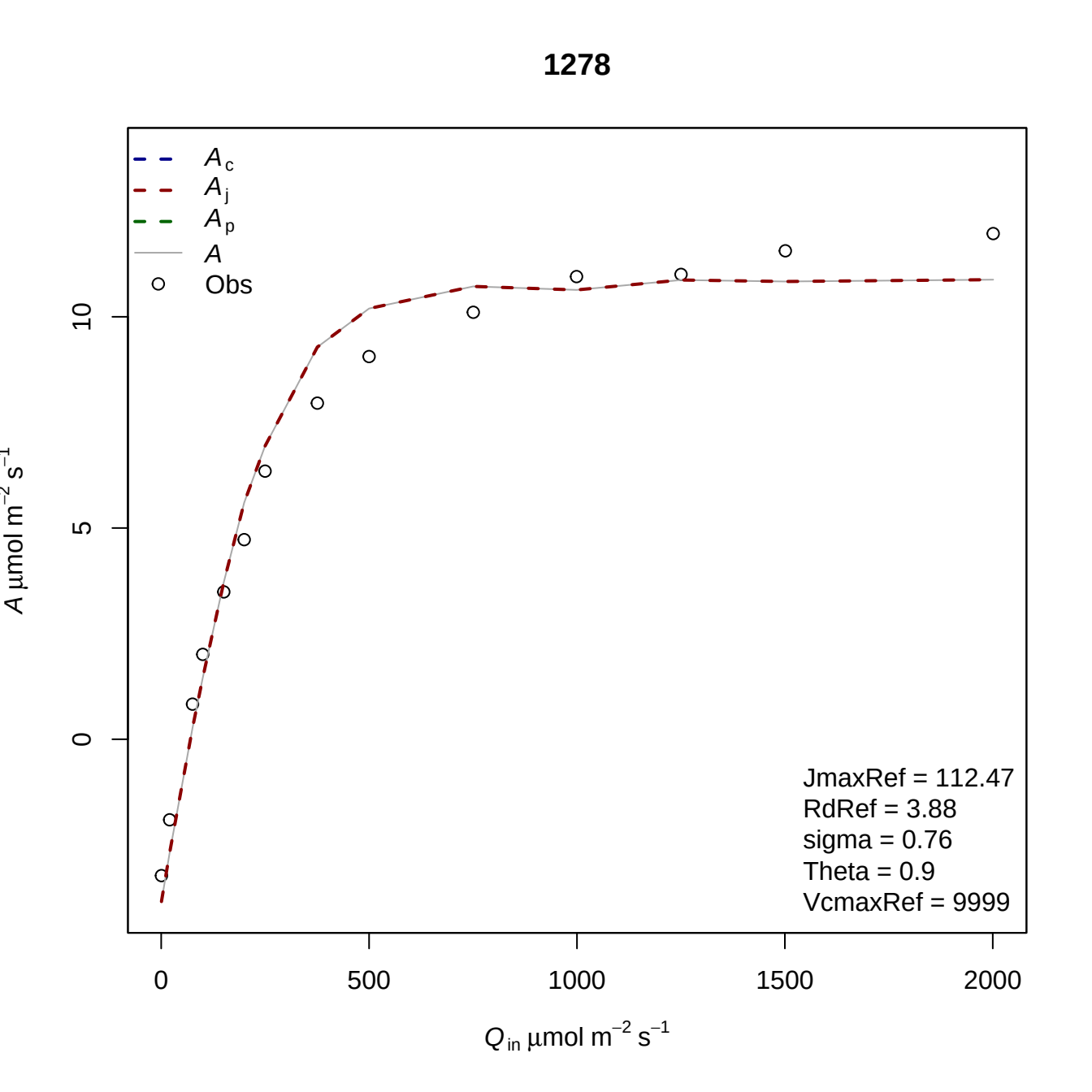

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 113.11
RdRef = 5.32
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

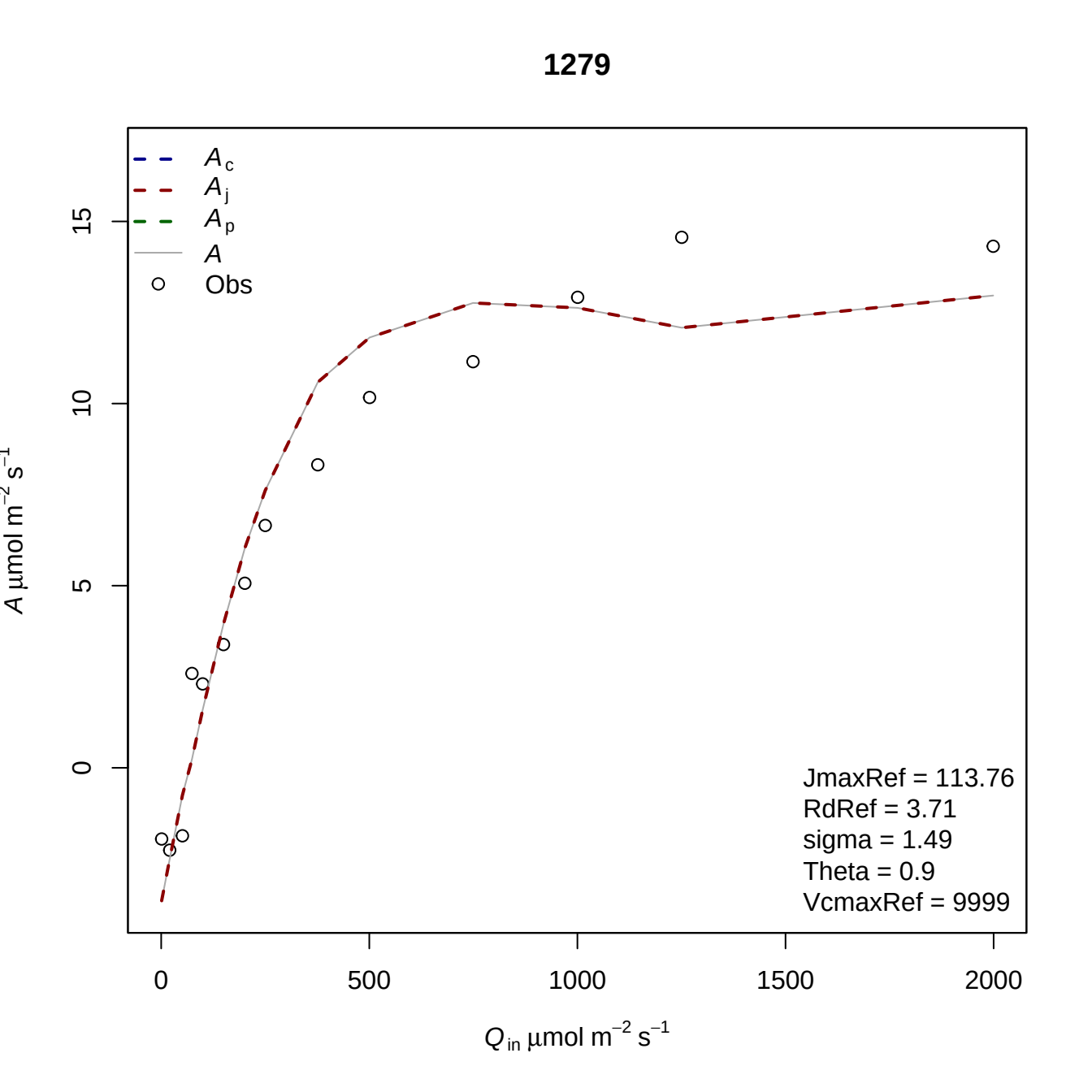

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 100.68
RdRef = 3.88
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

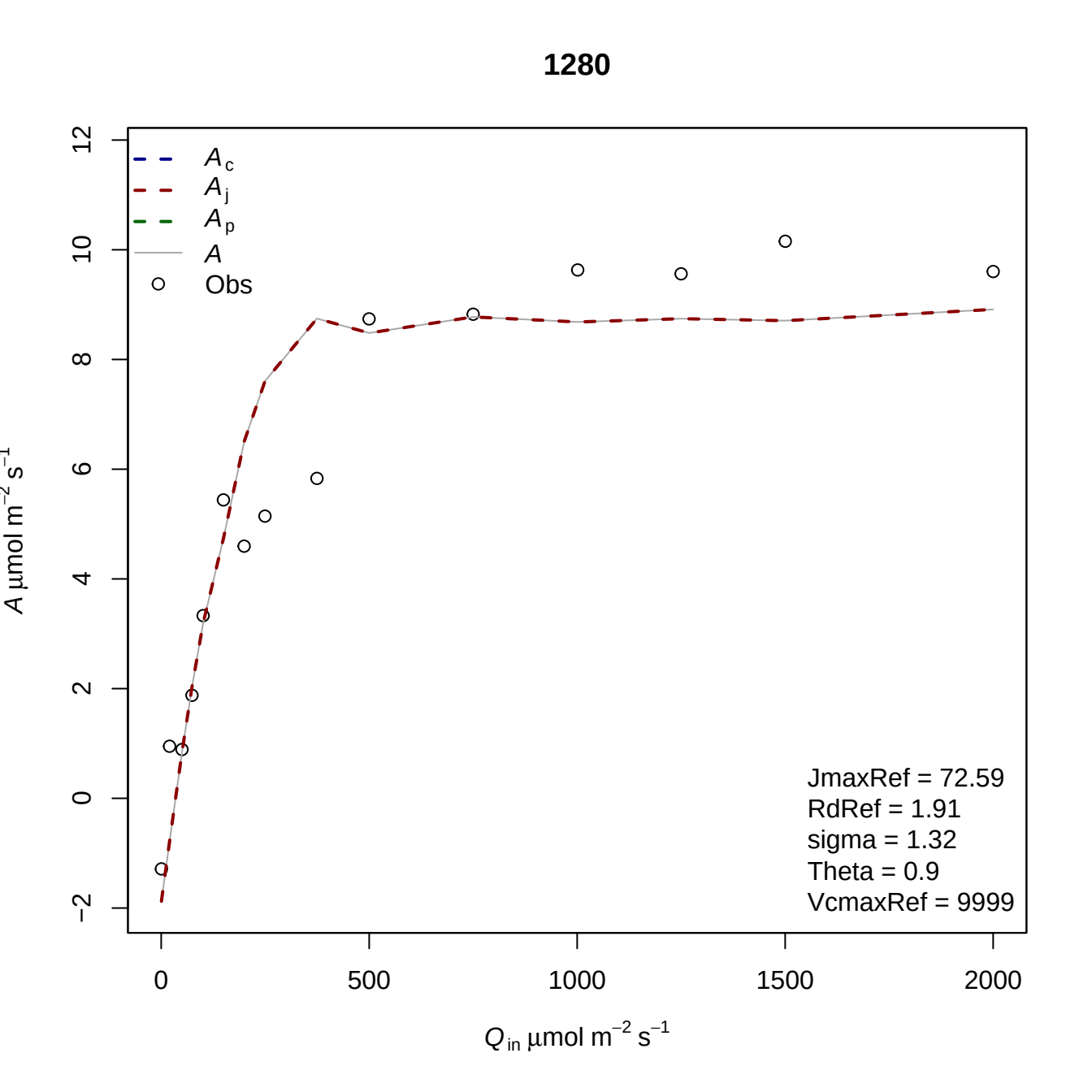

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 51.83
RdRef = 4.55
sigma = 2.05
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

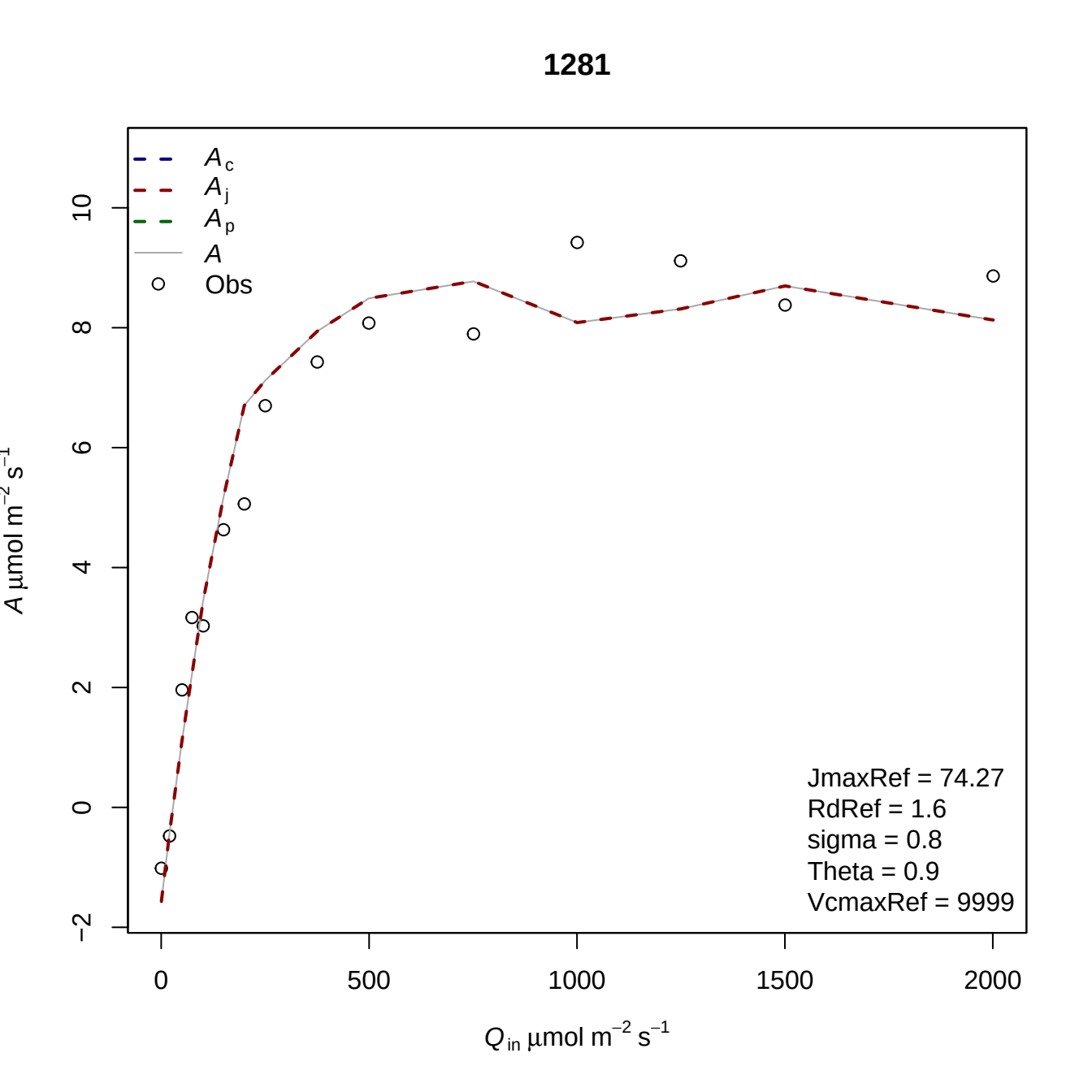

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 52.09
RdRef = 4.24
sigma = 2.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

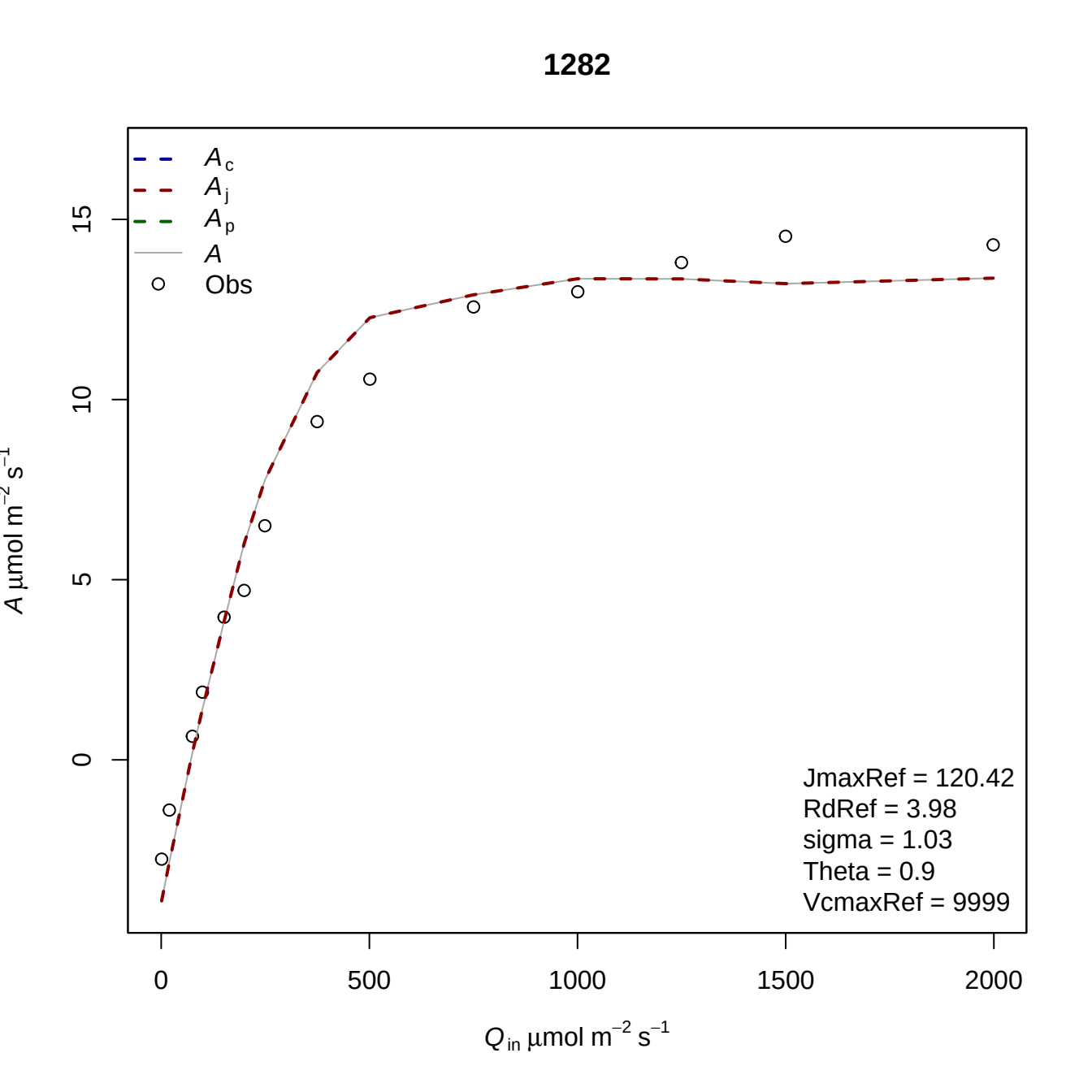

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 120.11
RdRef = 3.84
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

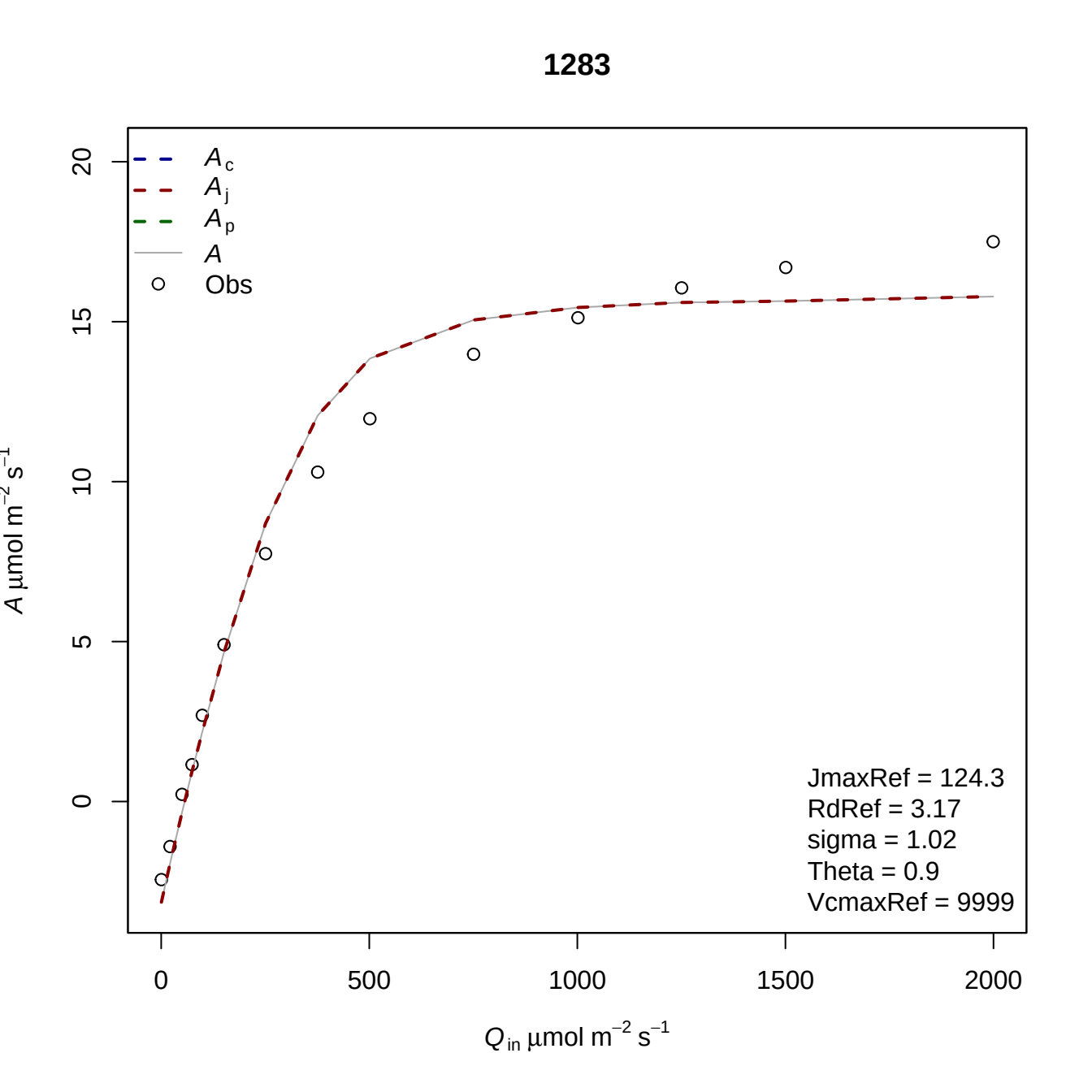

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 112.47
RdRef = 3.88
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

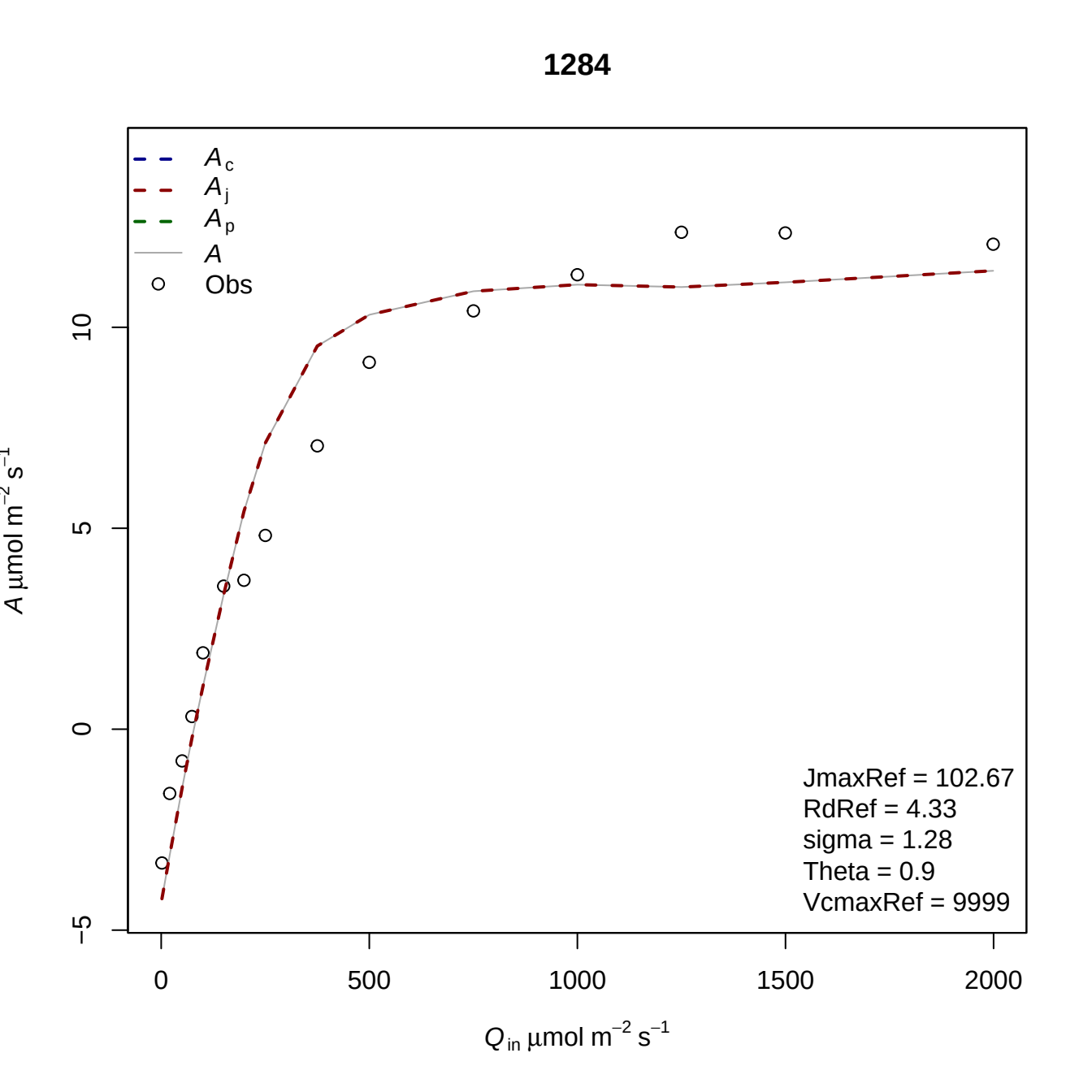

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 113.76
RdRef = 3.71
sigma = 1.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

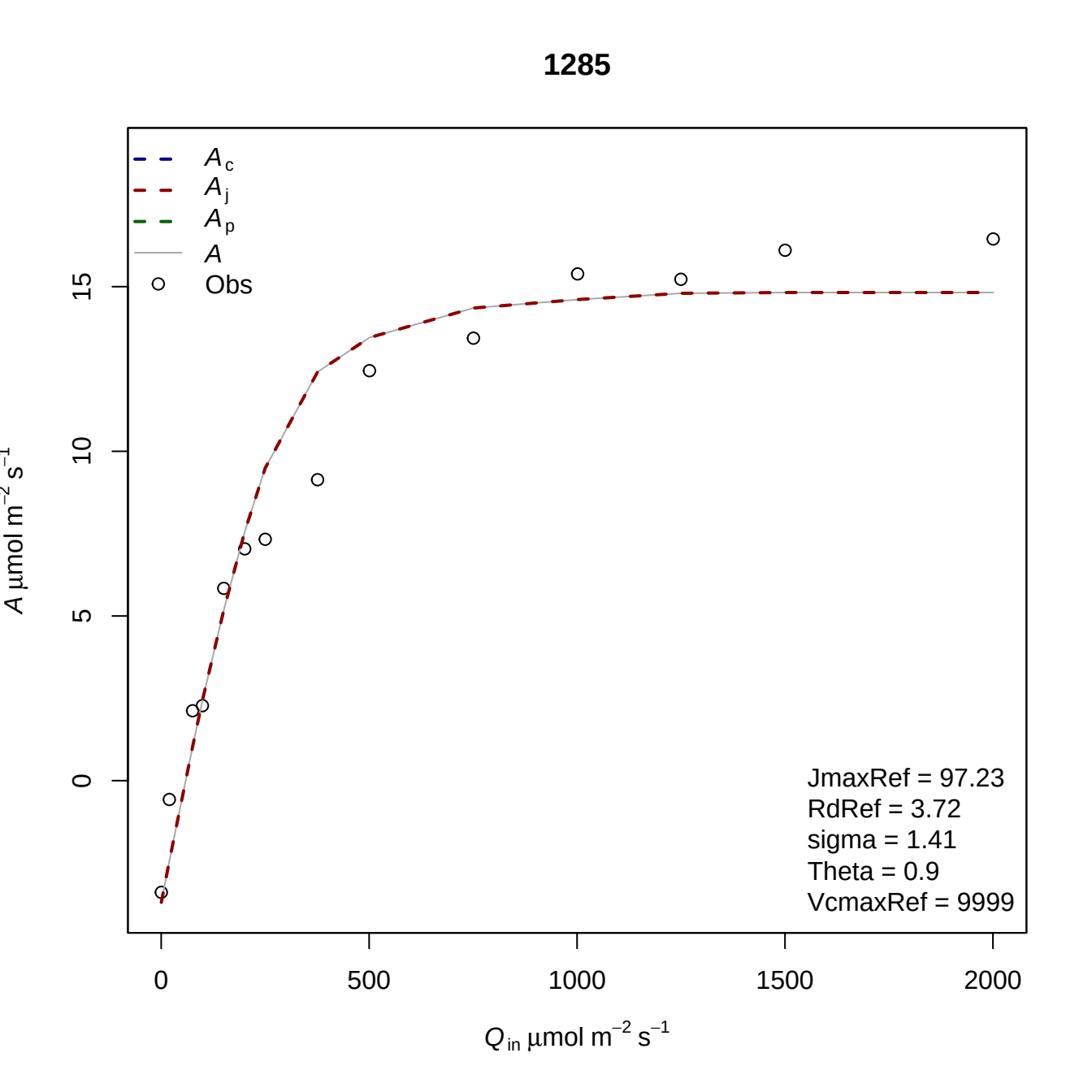

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 72.59
RdRef = 1.91
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

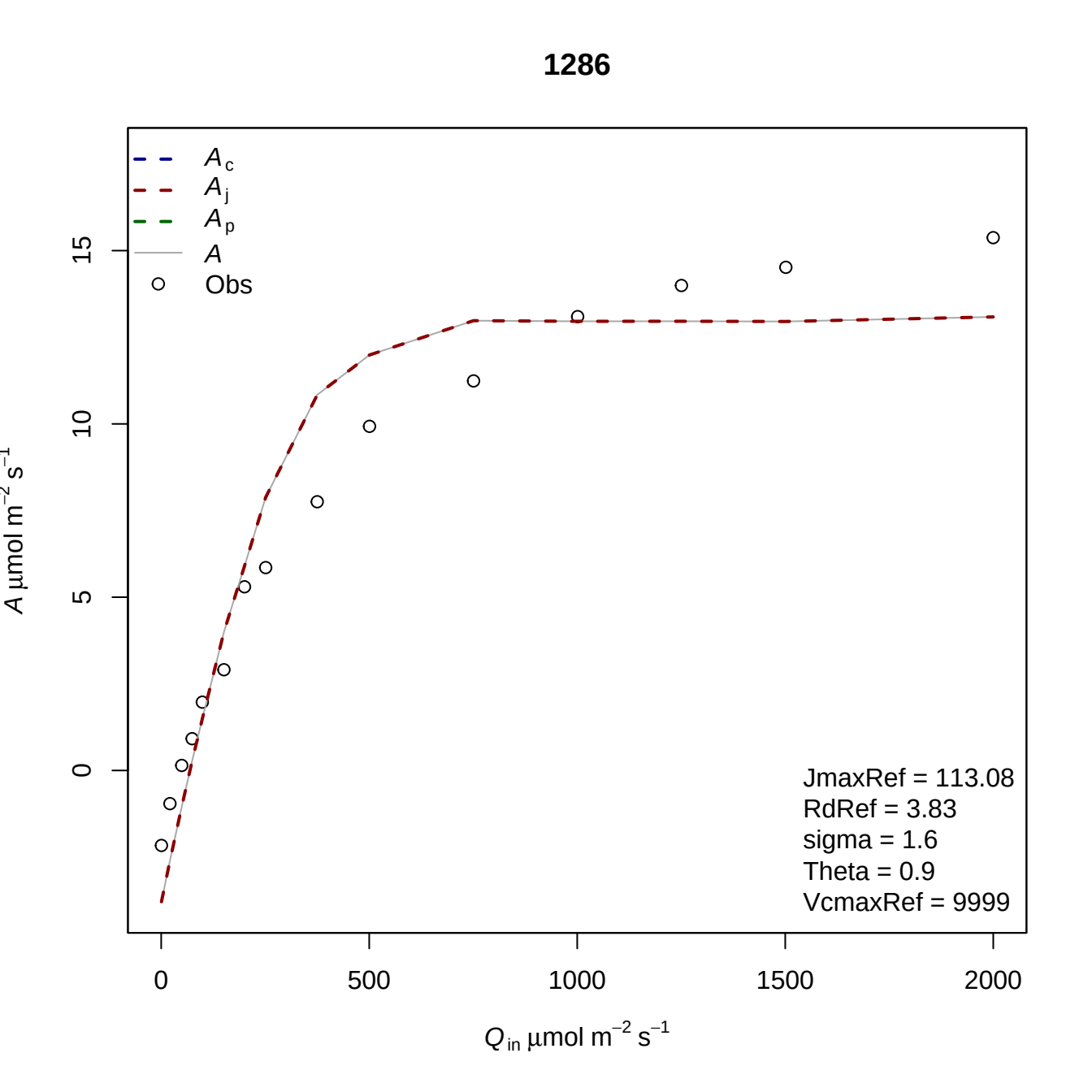

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 74.27
RdRef = 1.6
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

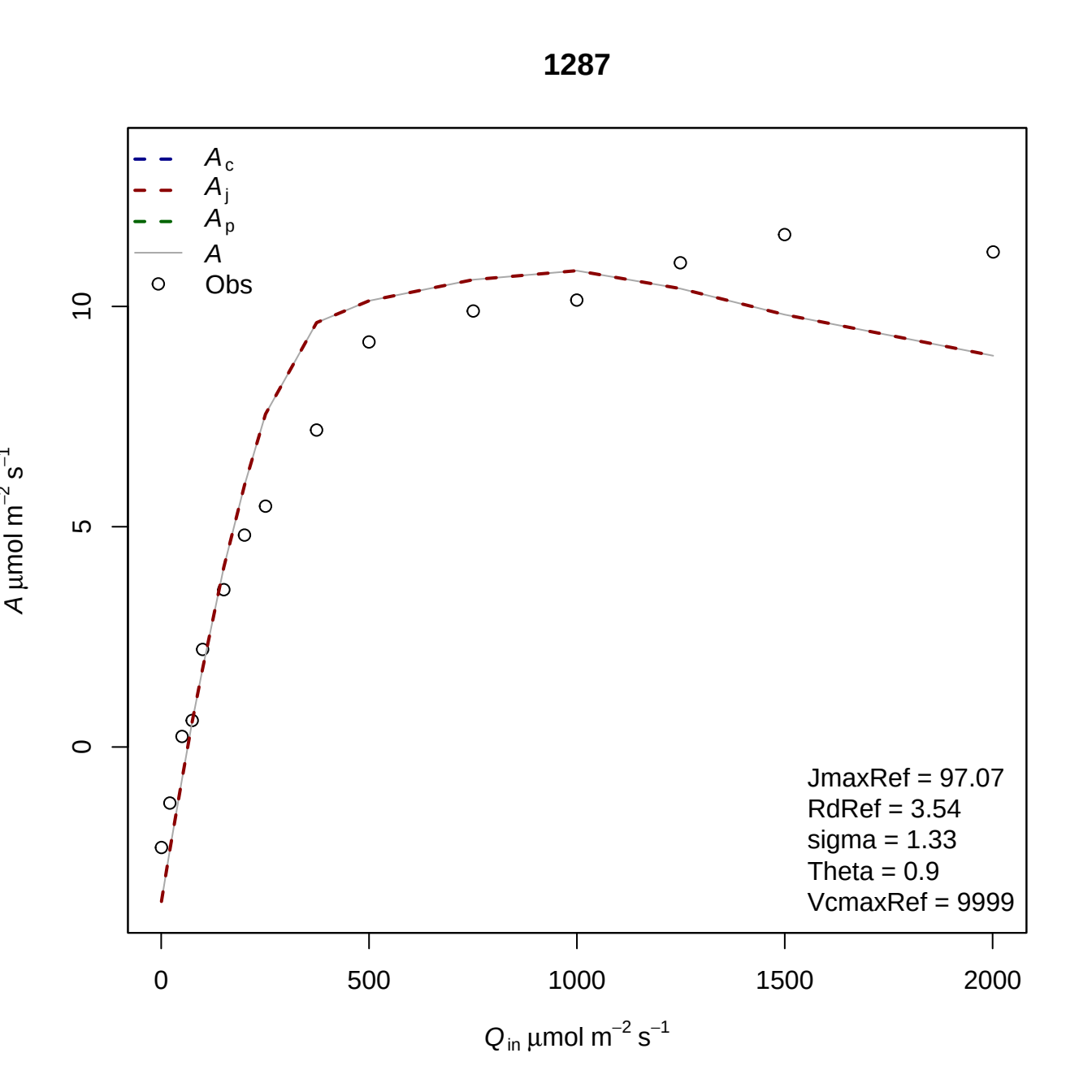

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 120.42
RdRef = 3.98
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

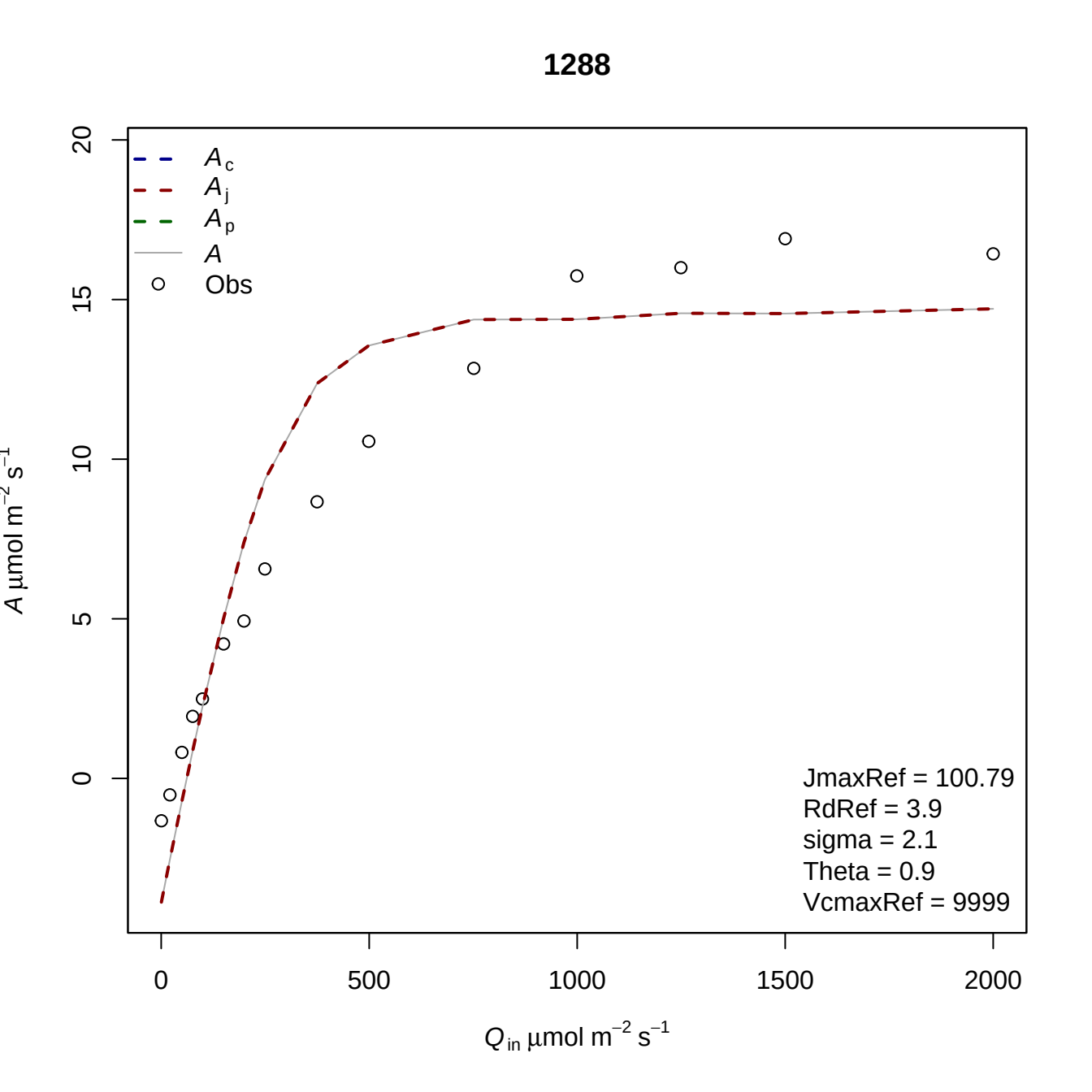

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 124.3
RdRef = 3.17
sigma = 1.02
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

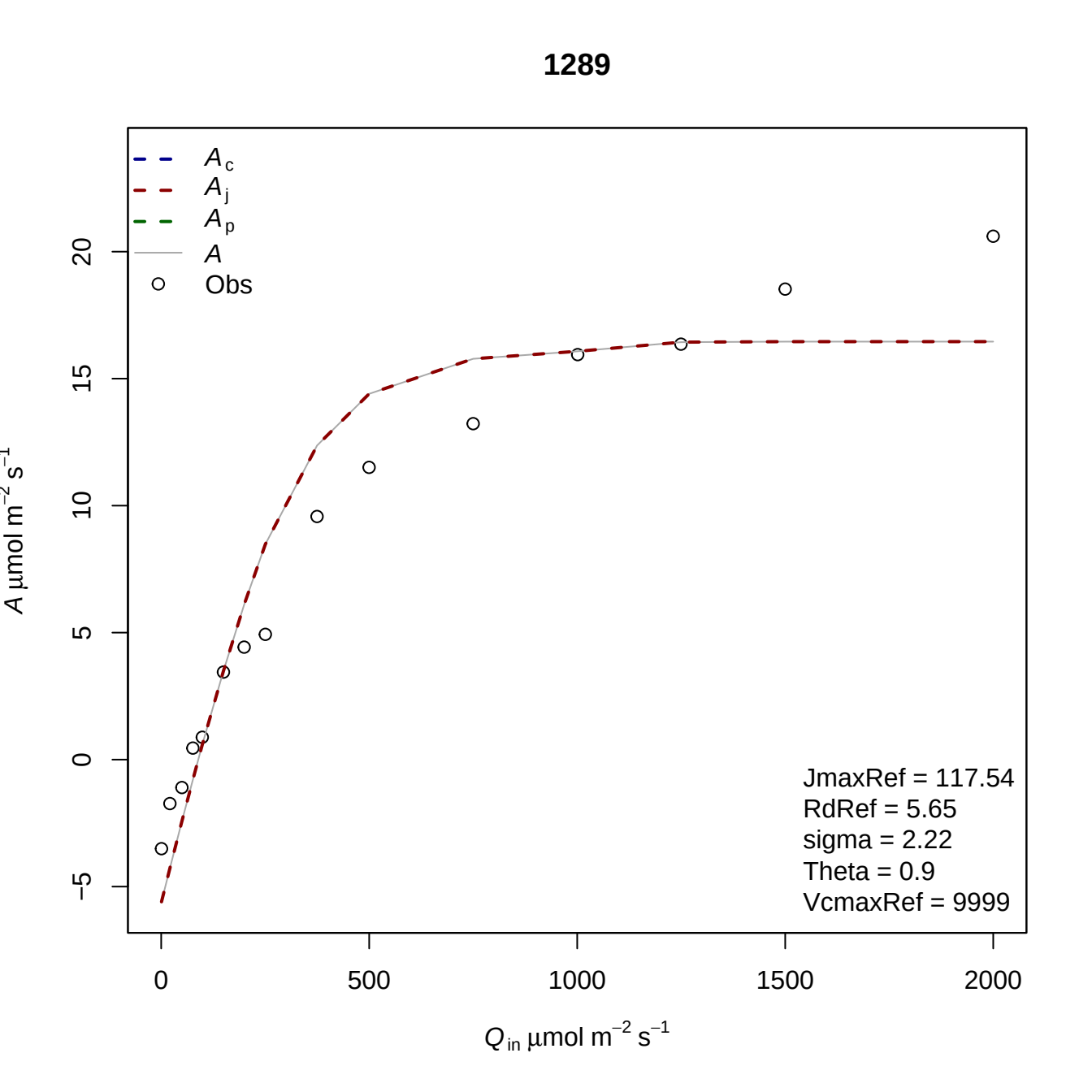

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 102.67
RdRef = 4.33
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

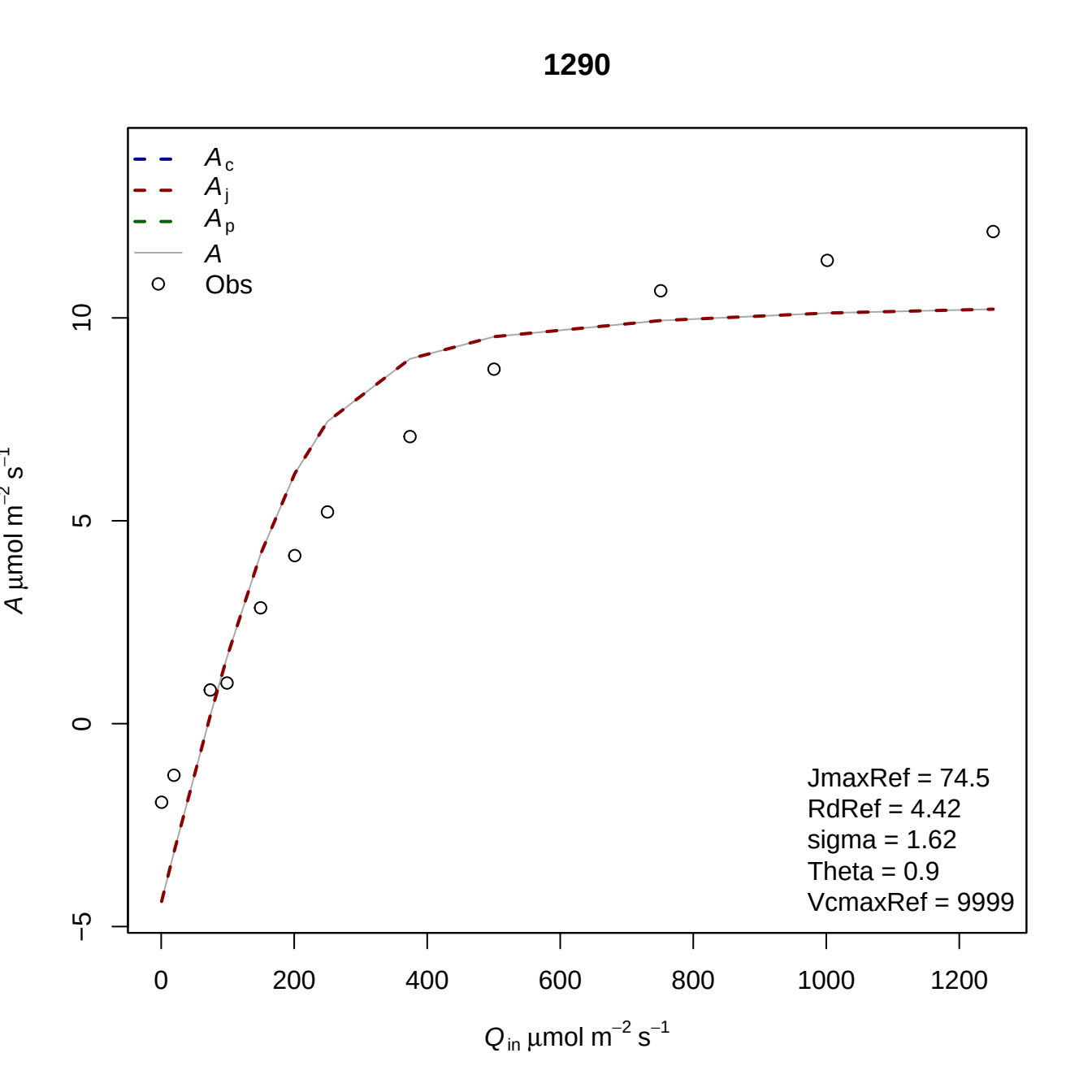

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 97.23
RdRef = 3.72
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

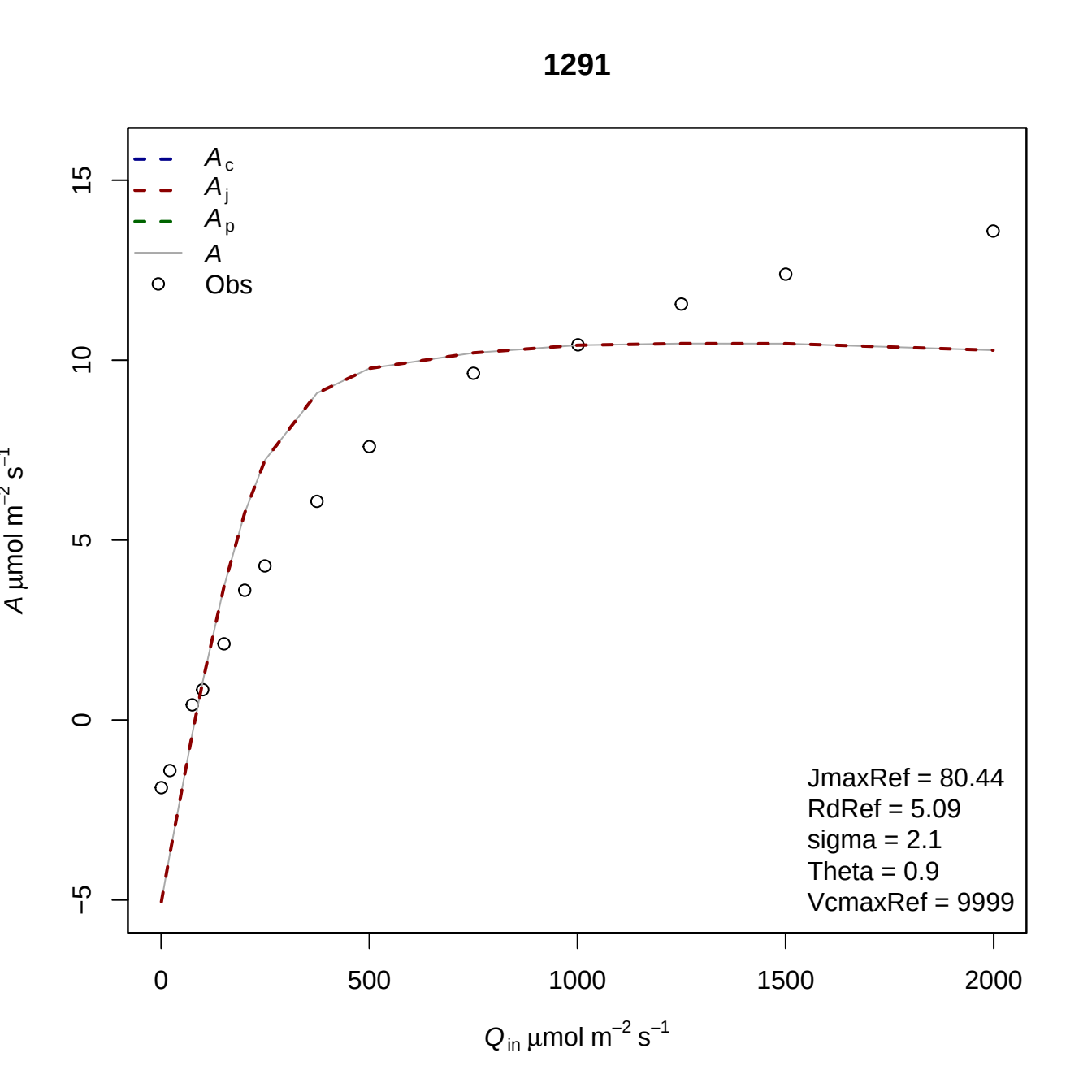

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 113.08
RdRef = 3.83
sigma = 1.6
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

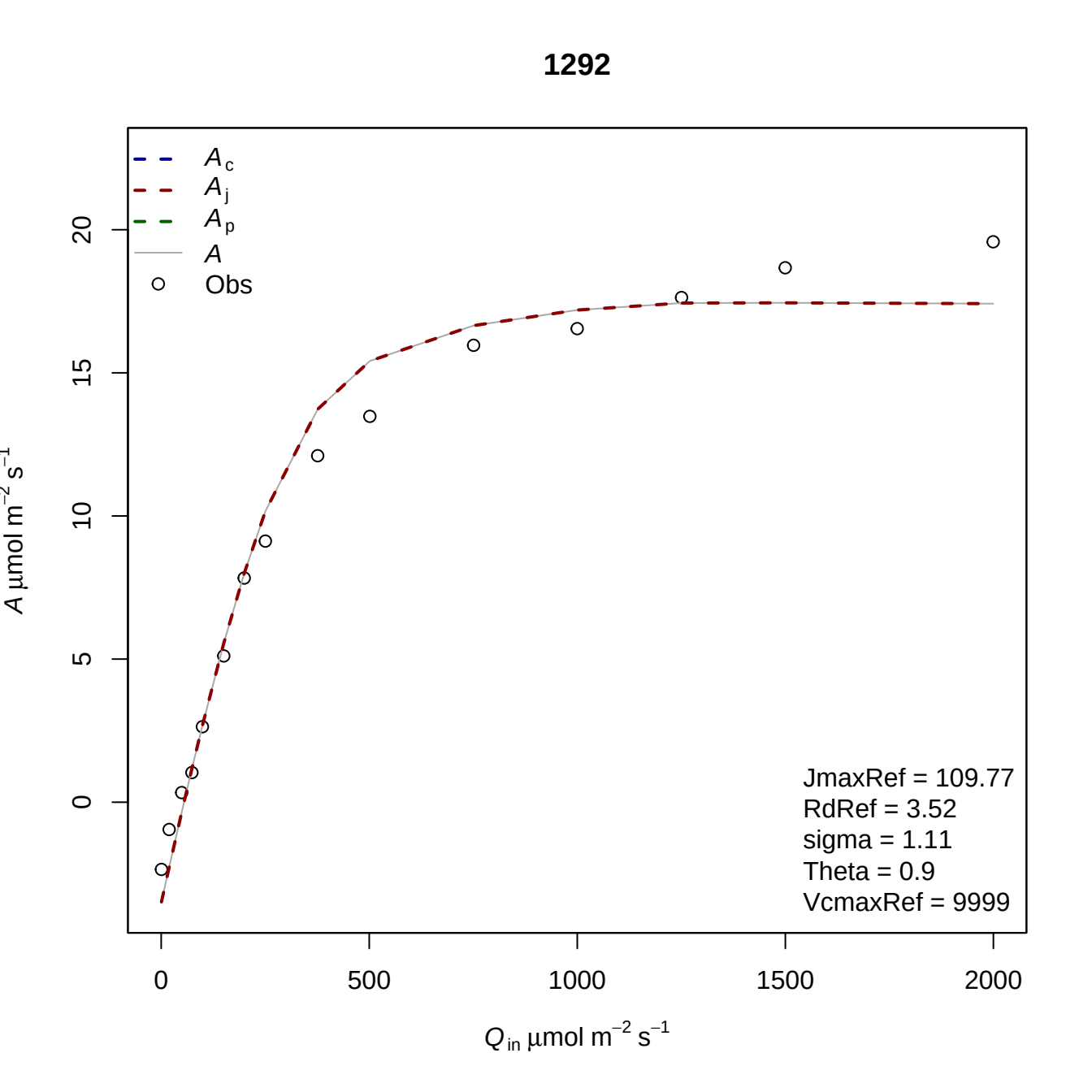

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 97.07
RdRef = 3.54
sigma = 1.33
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

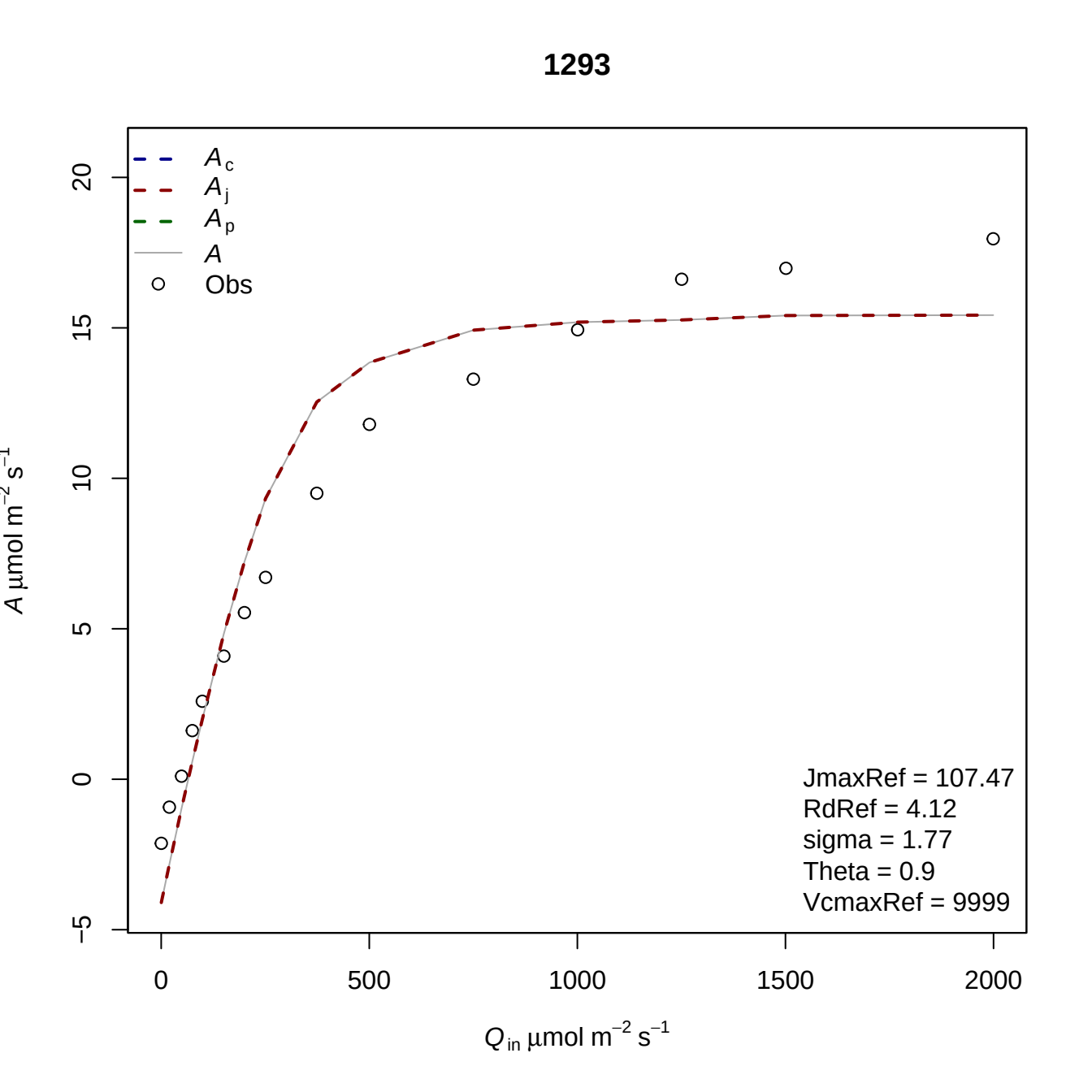

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 100.79
RdRef = 3.9
sigma = 2.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

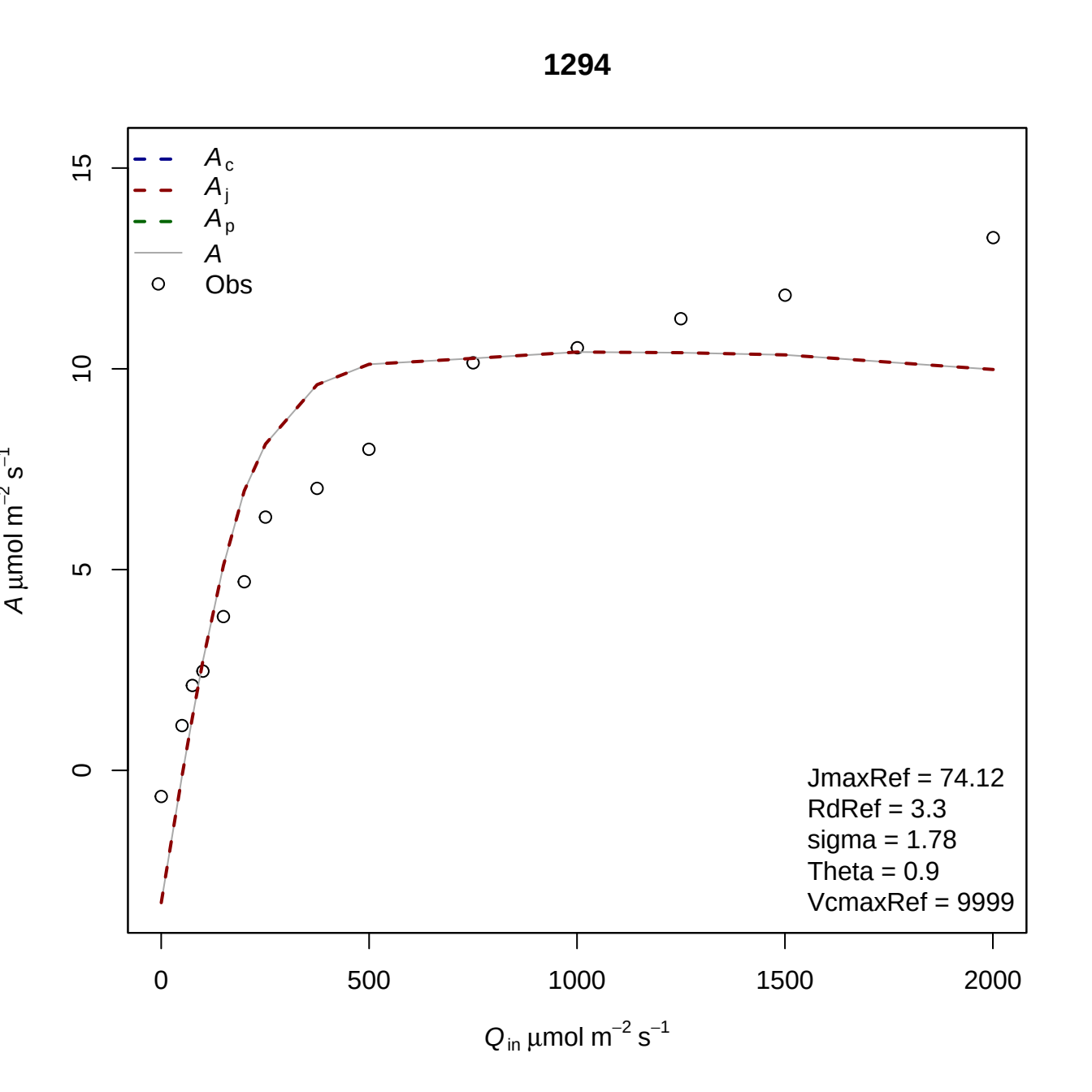

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 117.54
RdRef = 5.65
sigma = 2.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

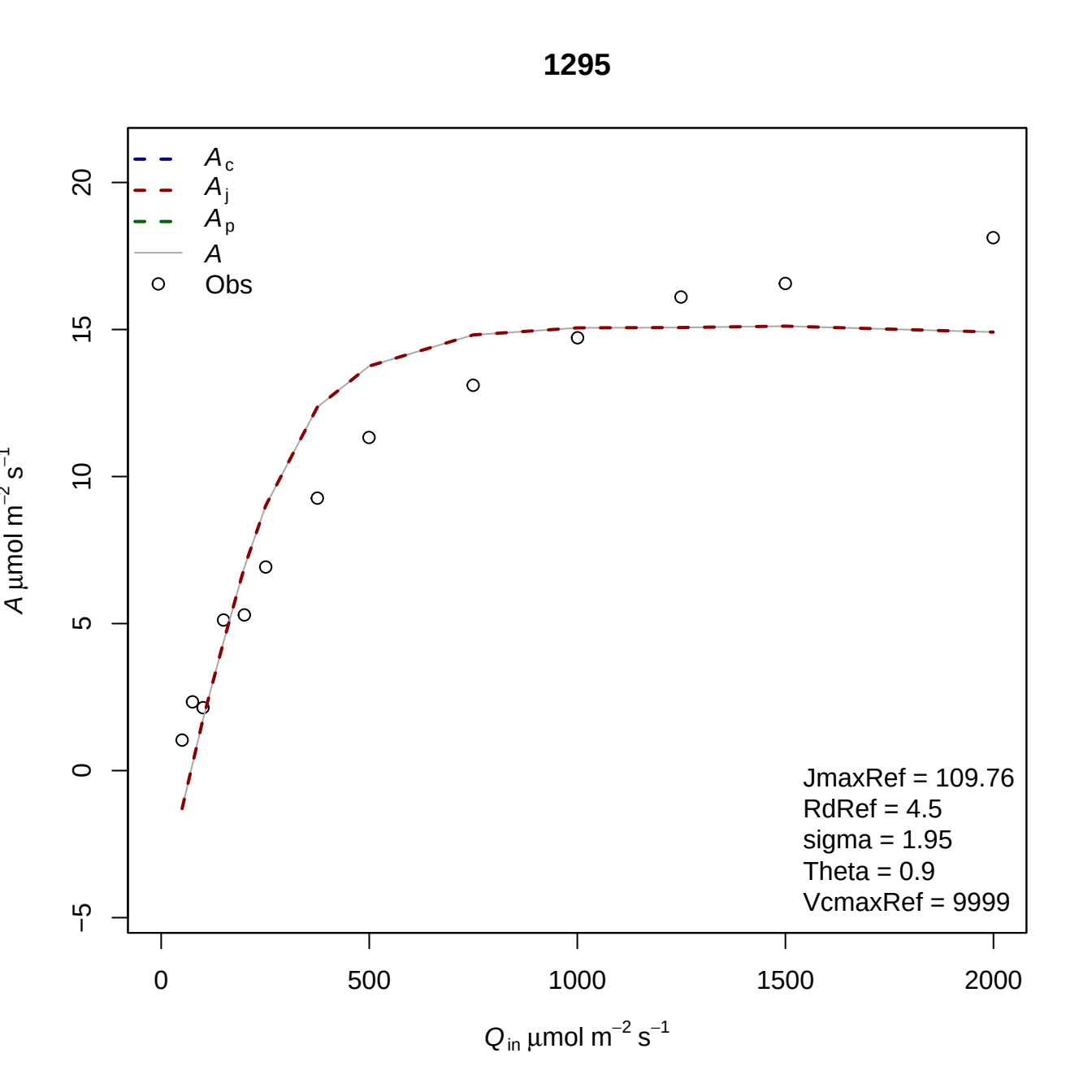

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 74.5
RdRef = 4.42
sigma = 1.62
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
200
400
600
800
1000
1200

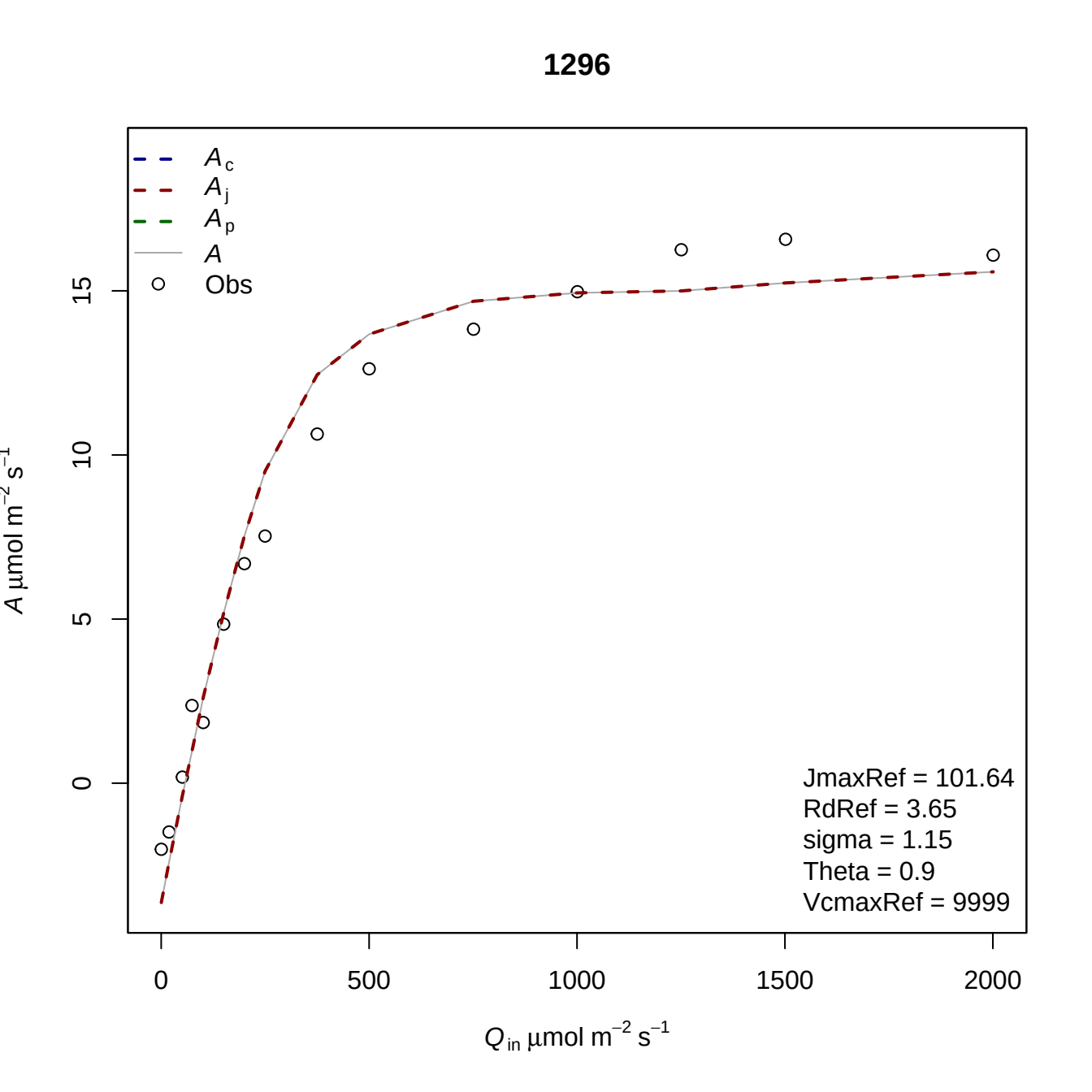

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 80.44
RdRef = 5.09
sigma = 2.1
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

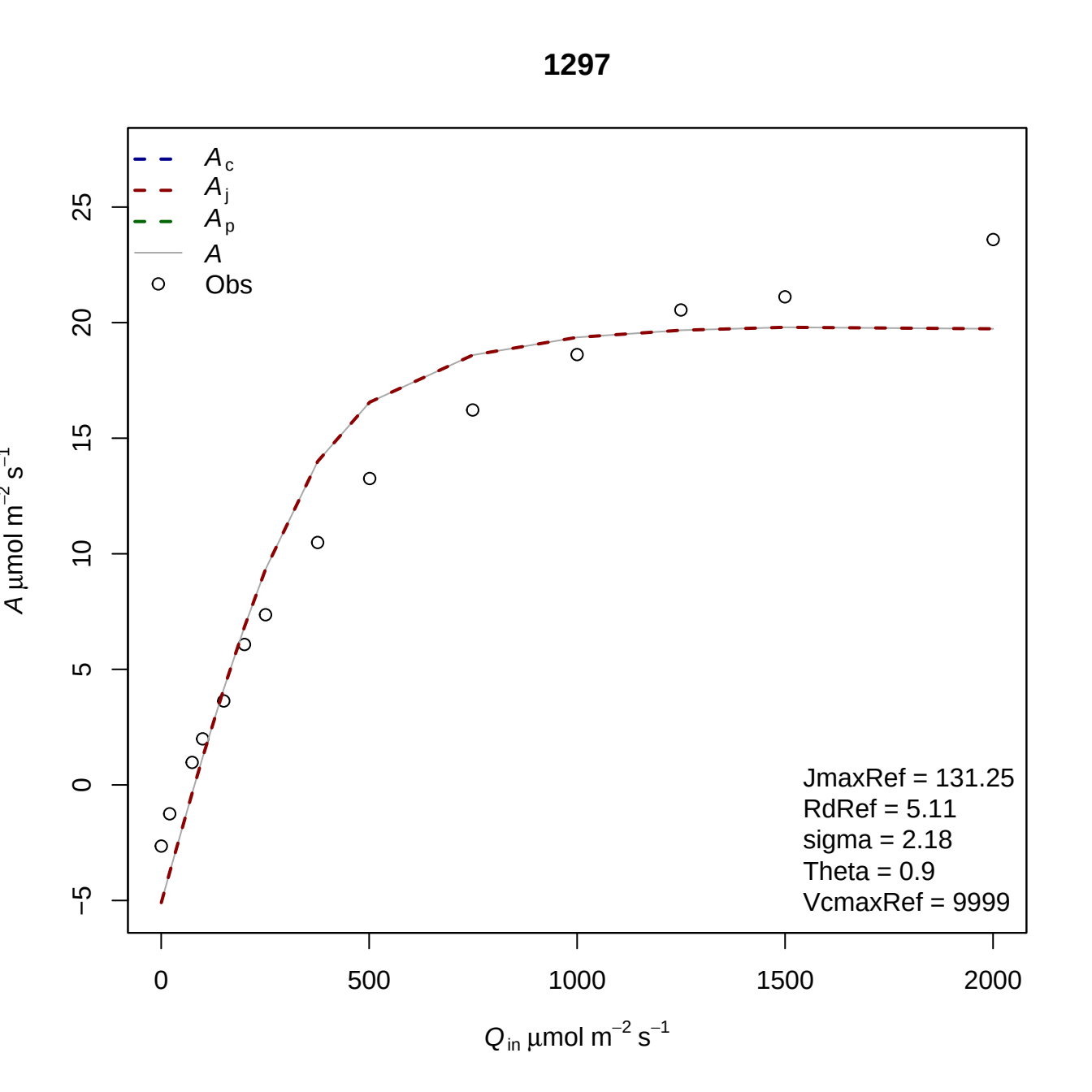

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 109.77
RdRef = 3.52
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

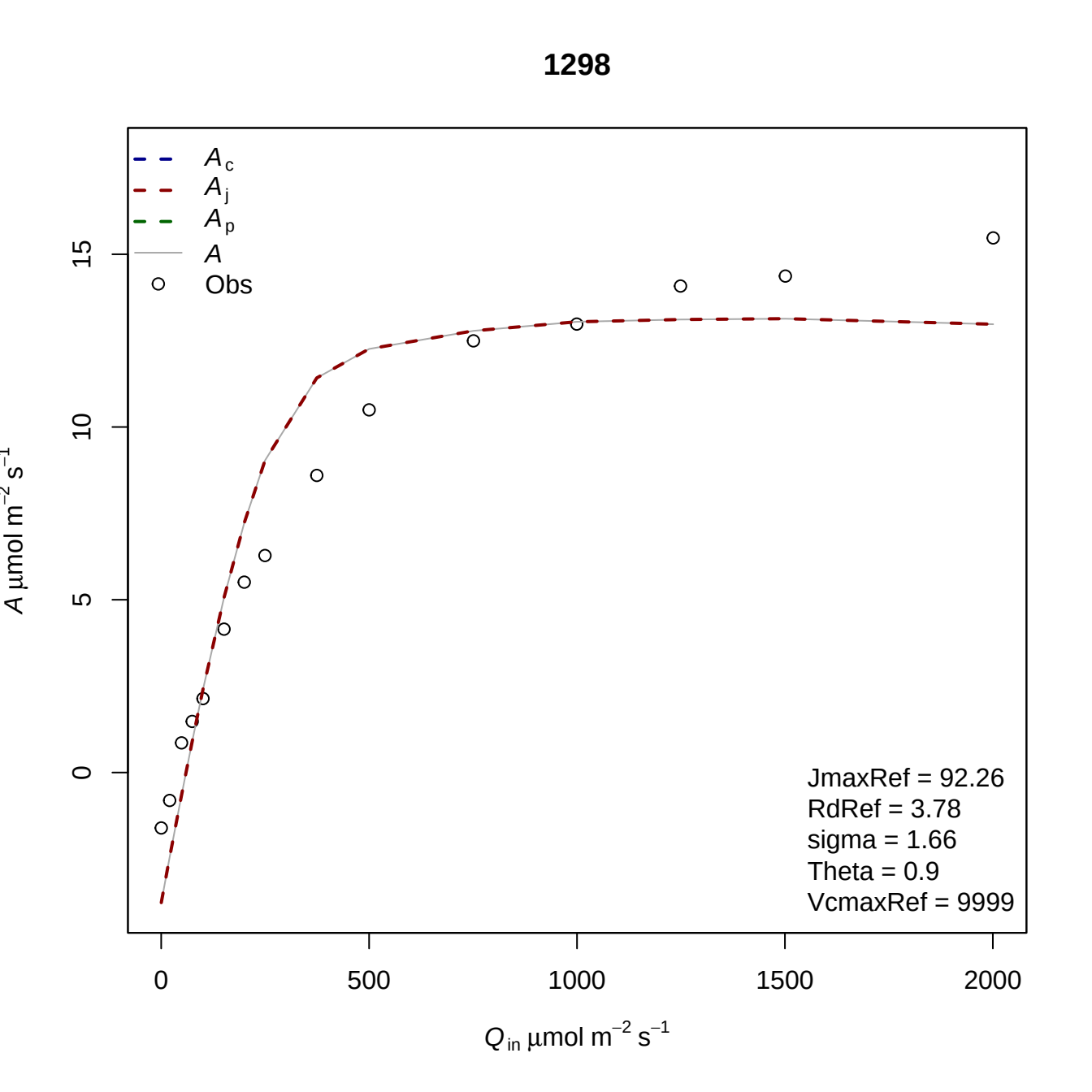

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 107.47
RdRef = 4.12
sigma = 1.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

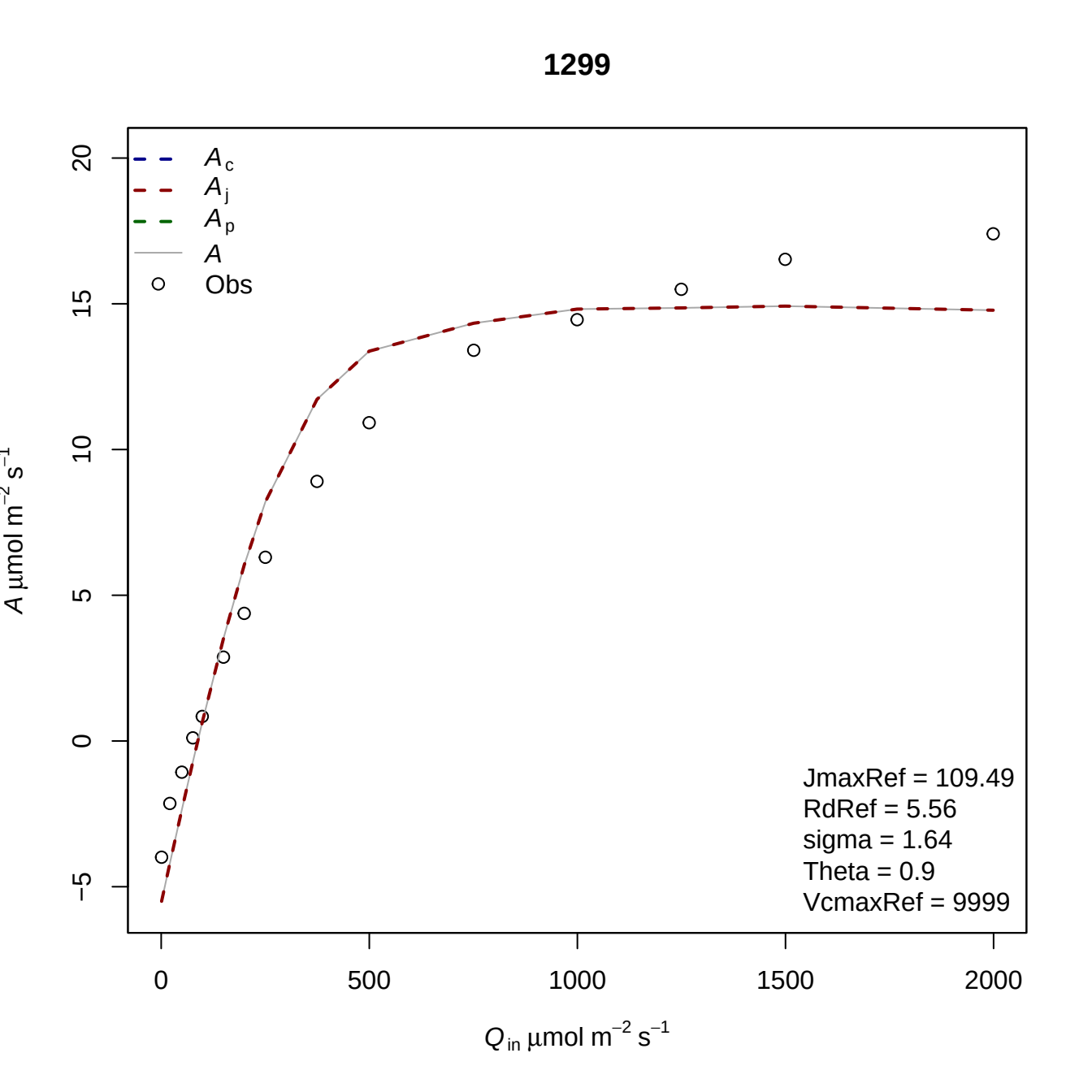

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 74.12
RdRef = 3.3
sigma = 1.78
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

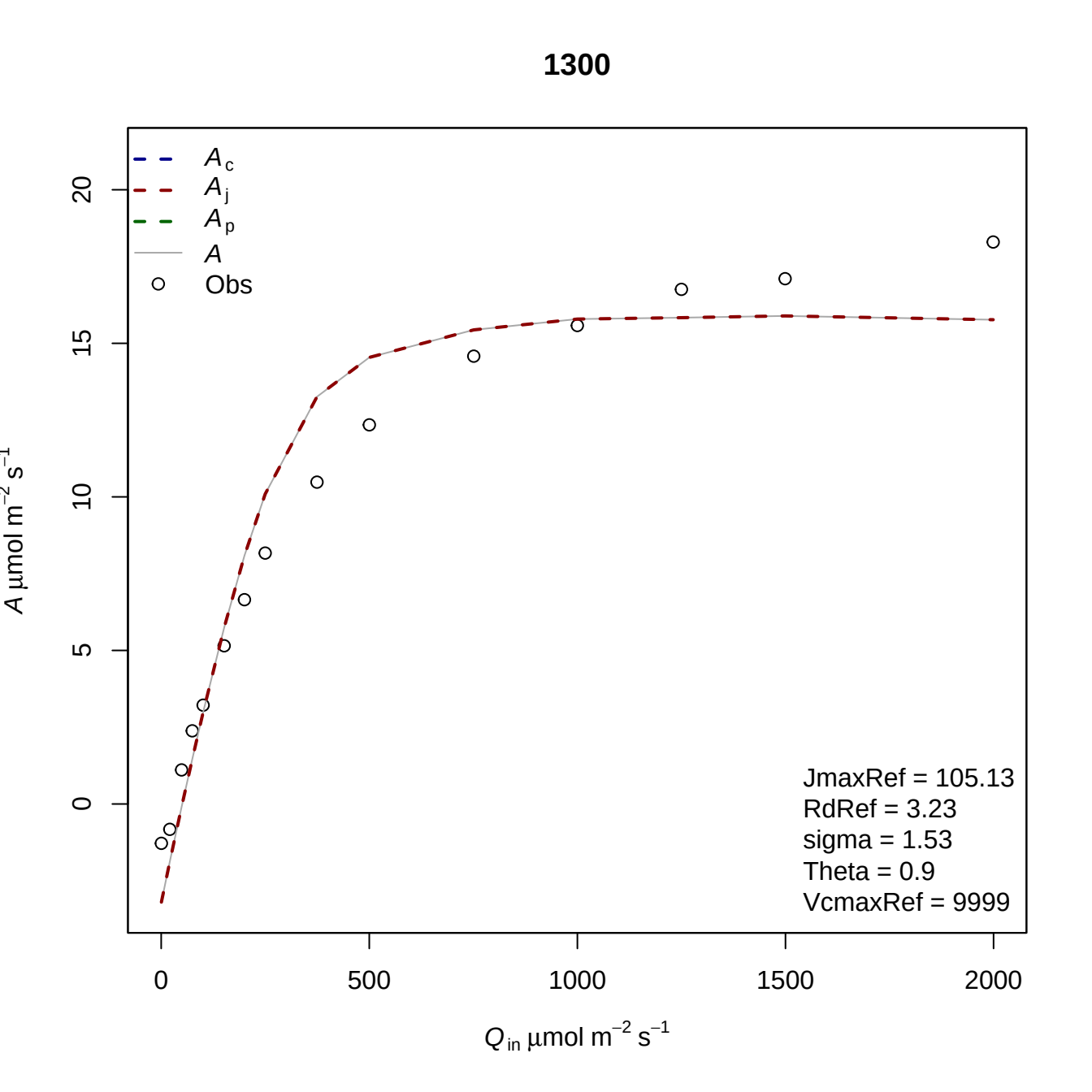

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 109.76
RdRef = 4.5
sigma = 1.95
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

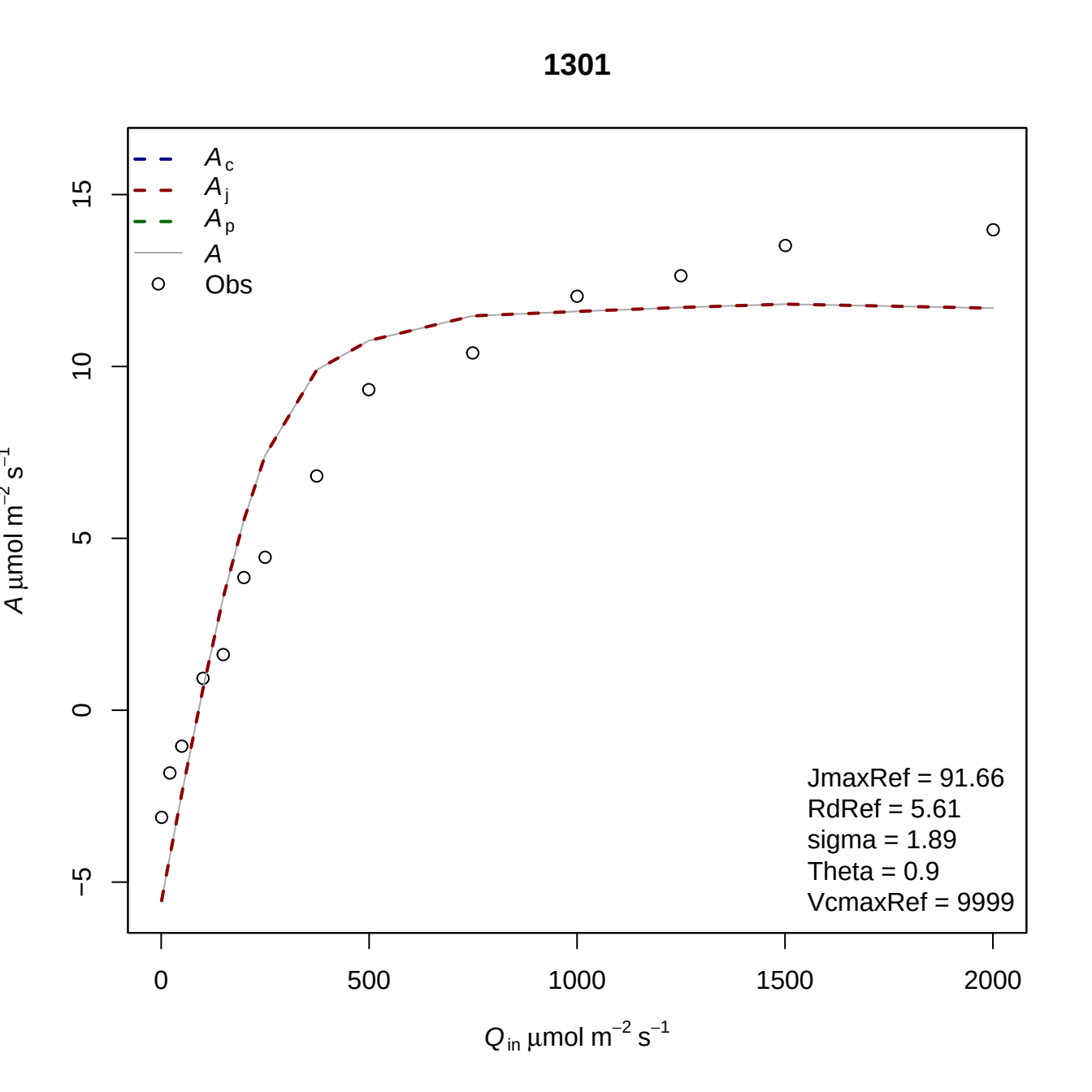

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 101.64
RdRef = 3.65
sigma = 1.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

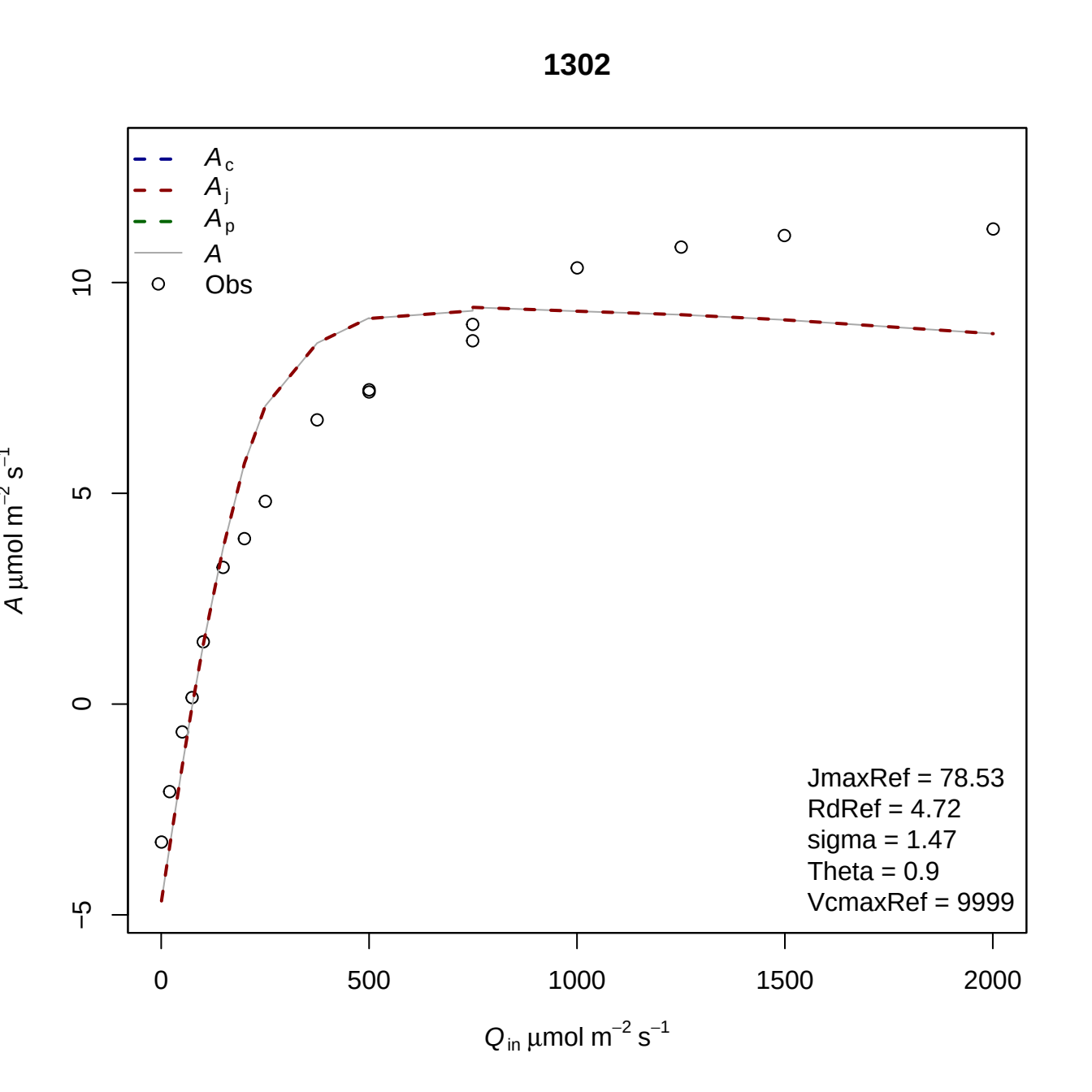

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 131.25
RdRef = 5.11
sigma = 2.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

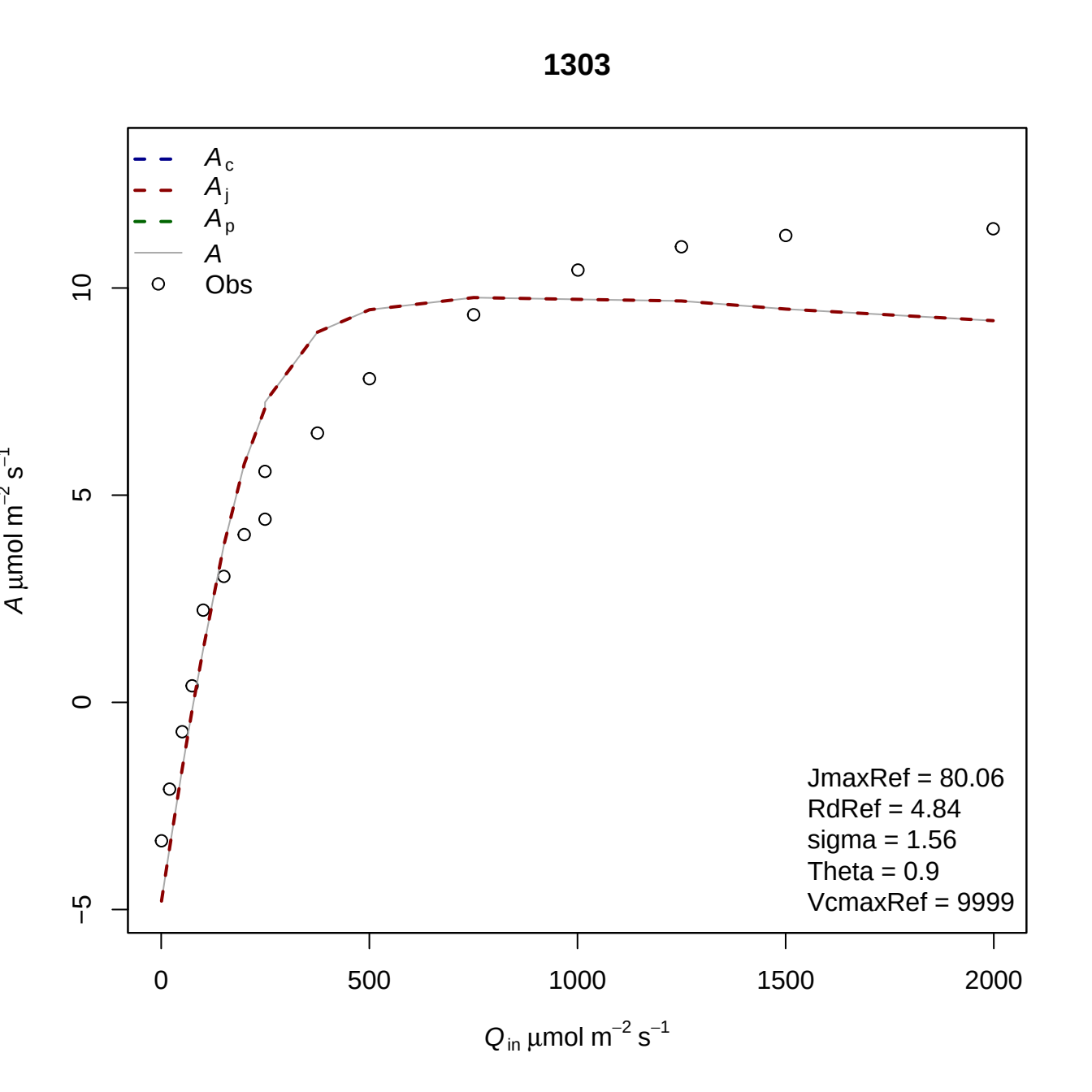

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 92.26
RdRef = 3.78
sigma = 1.66
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

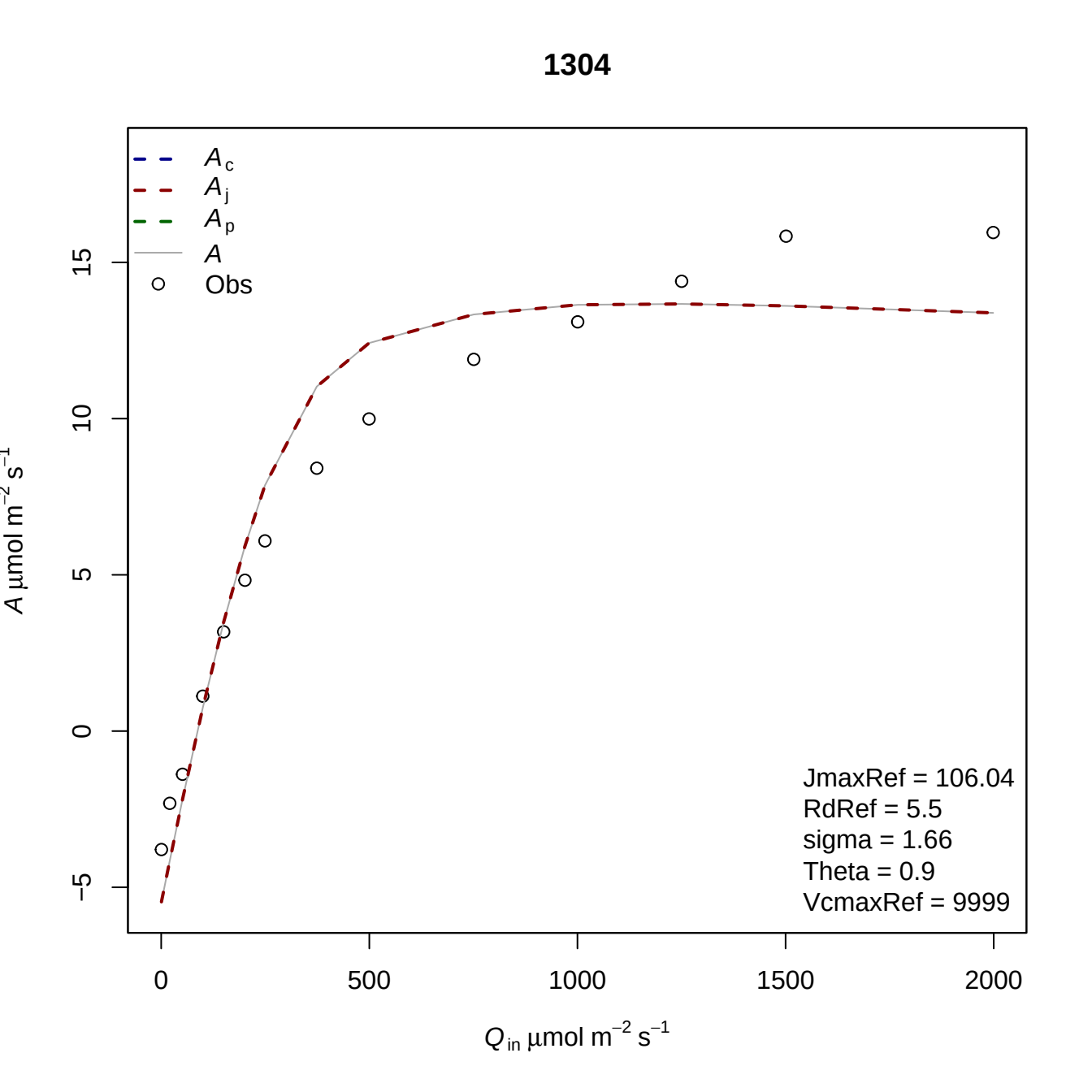

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 109.49
RdRef = 5.56
sigma = 1.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

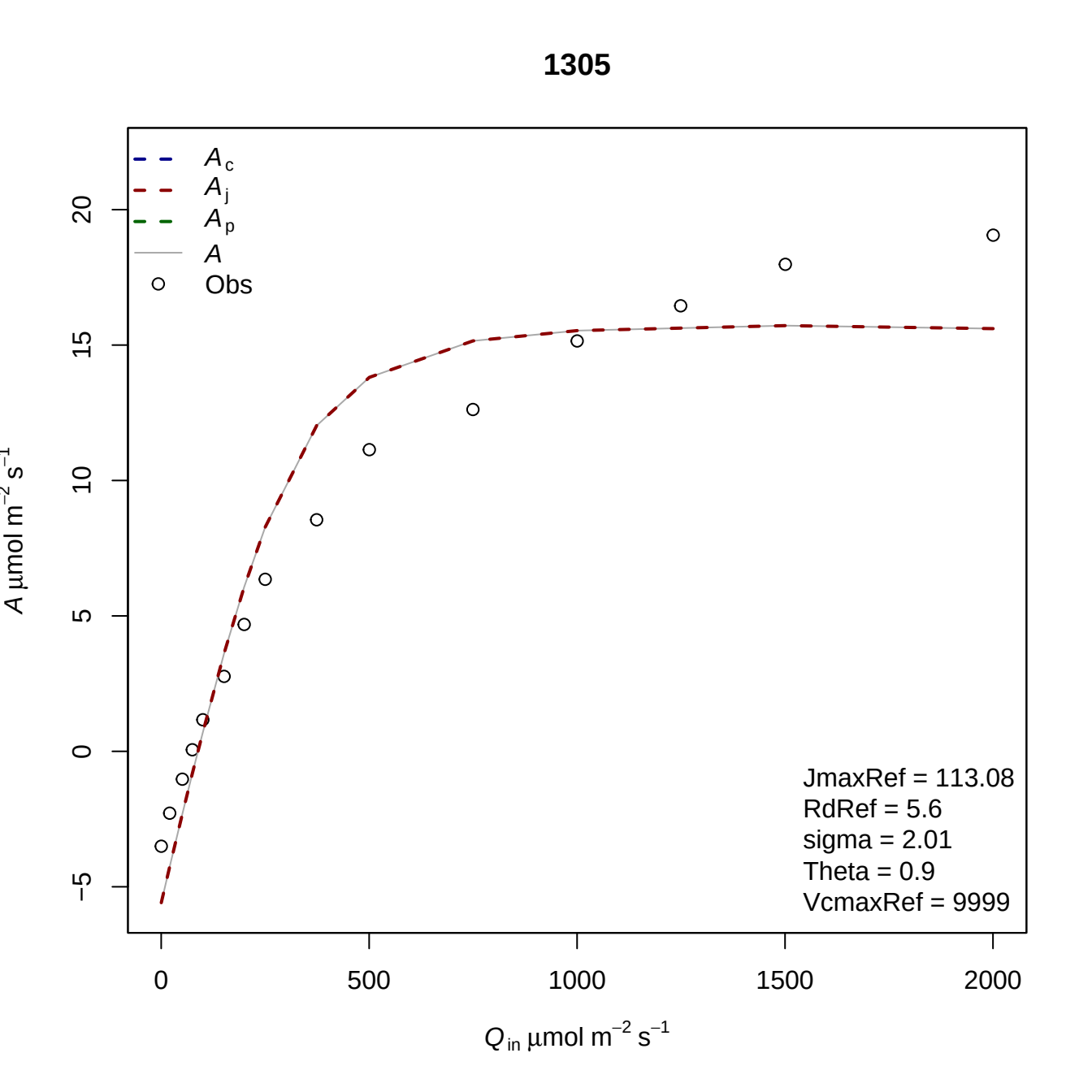

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 105.13
RdRef = 3.23
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

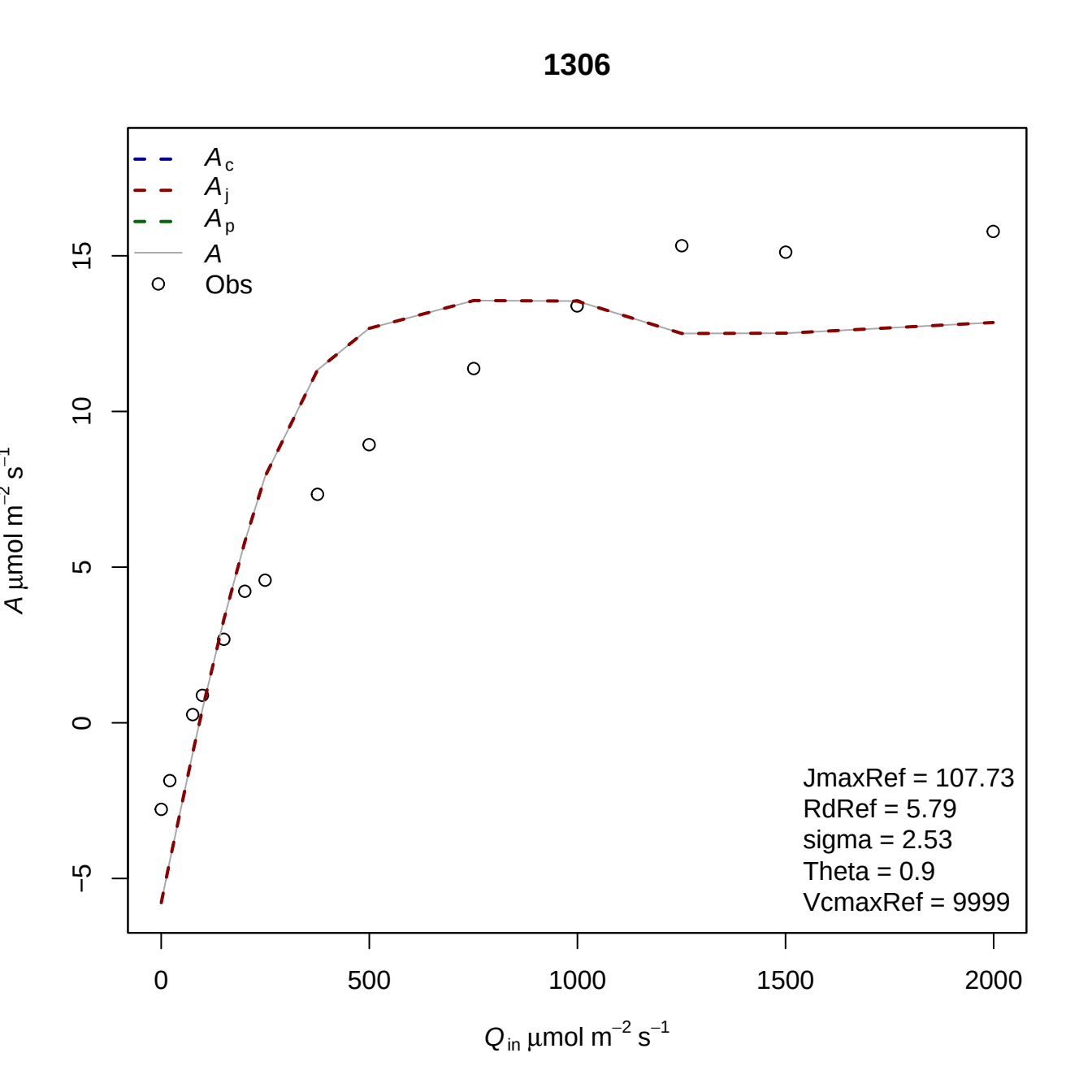

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 91.66
RdRef = 5.61
sigma = 1.89
Theta = 0.9
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

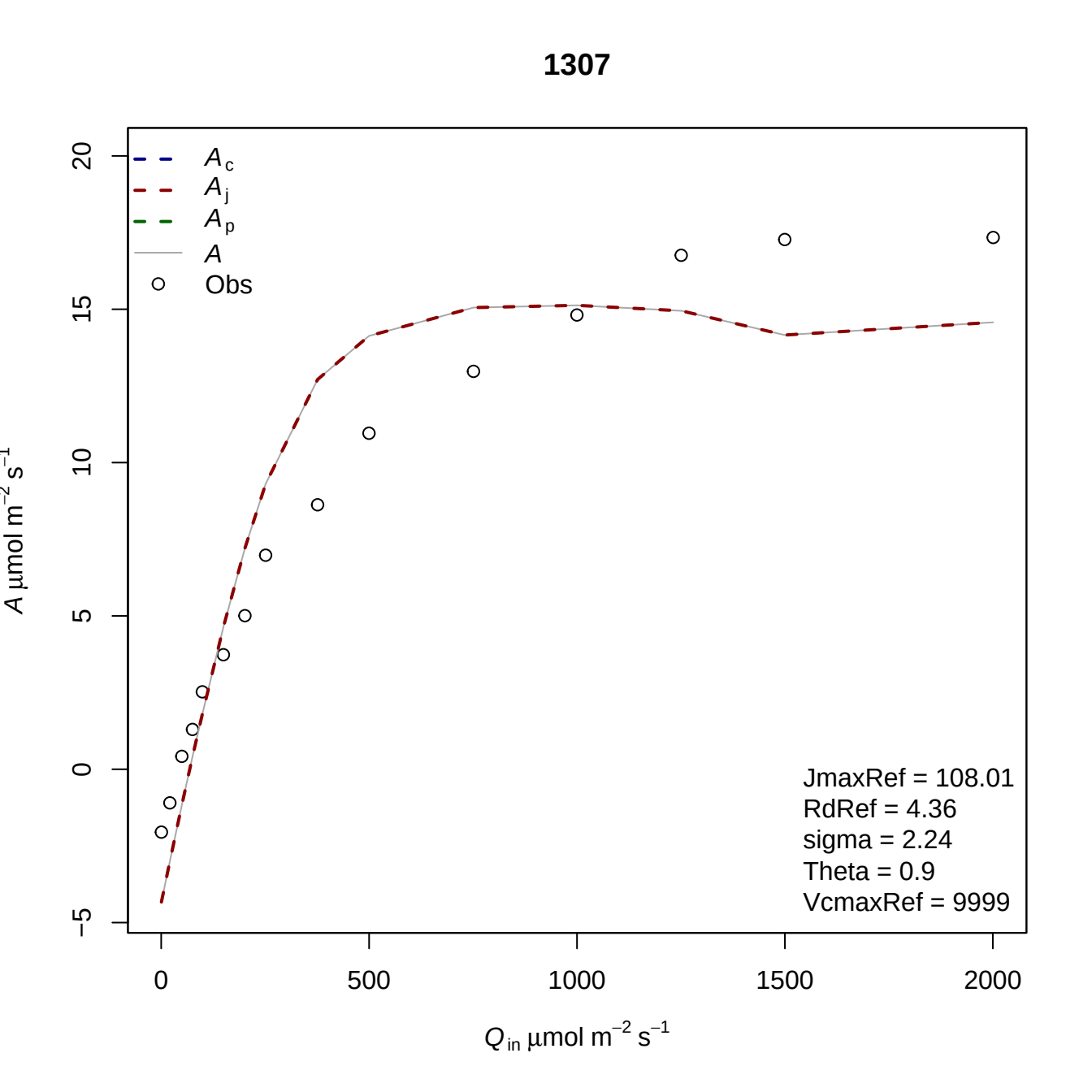

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 78.53
RdRef = 4.72
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

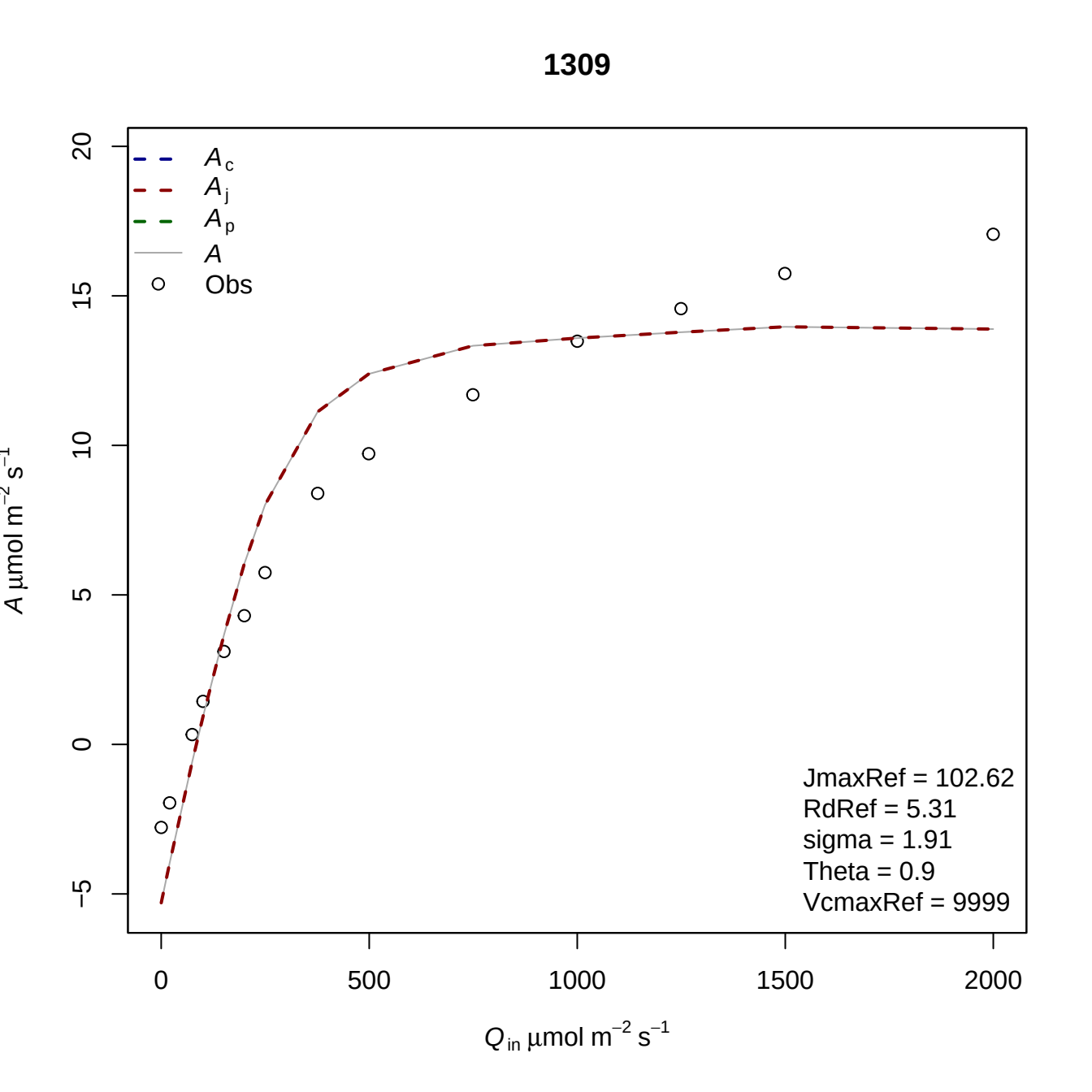

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 80.06
RdRef = 4.84
sigma = 1.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

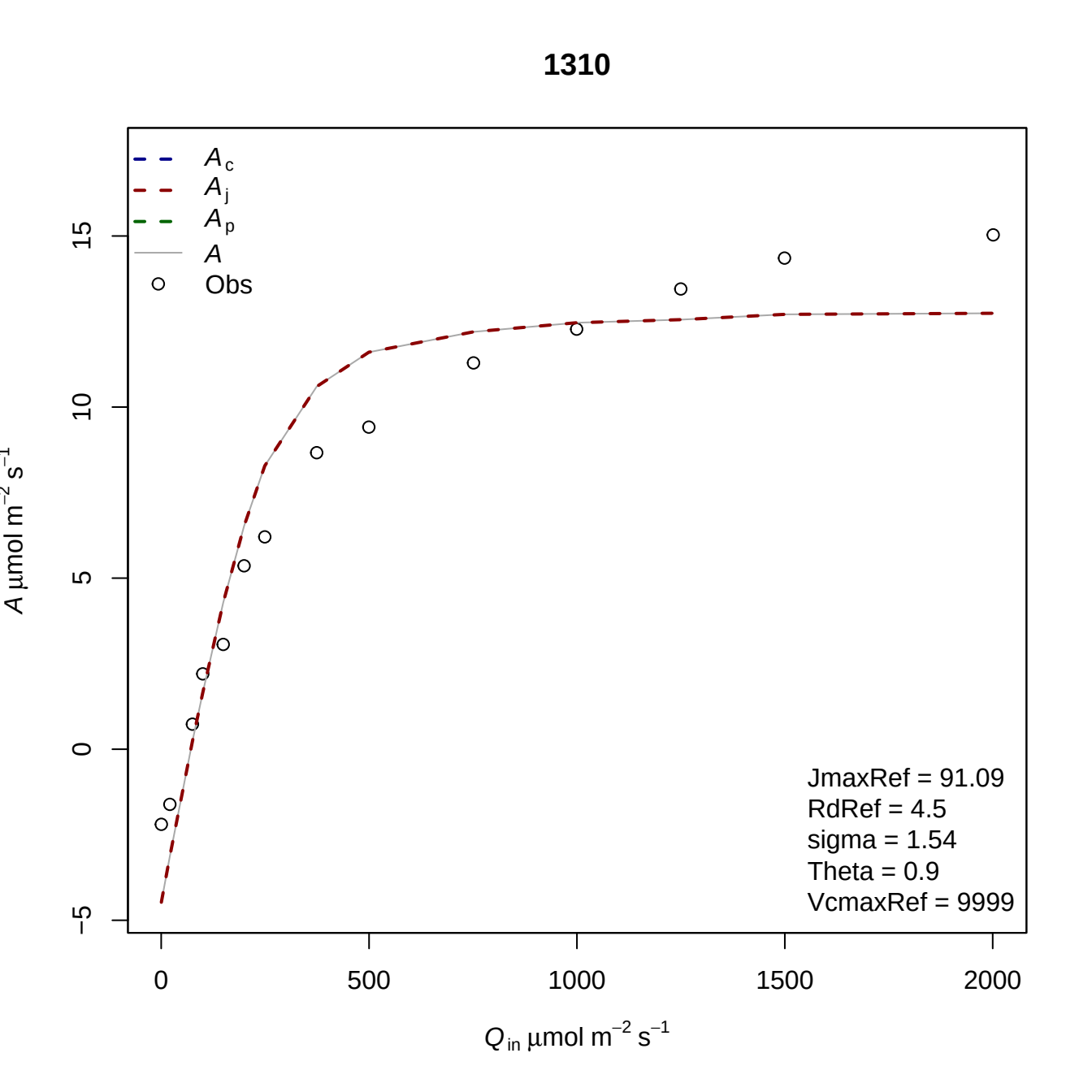

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 106.04
RdRef = 5.5
sigma = 1.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

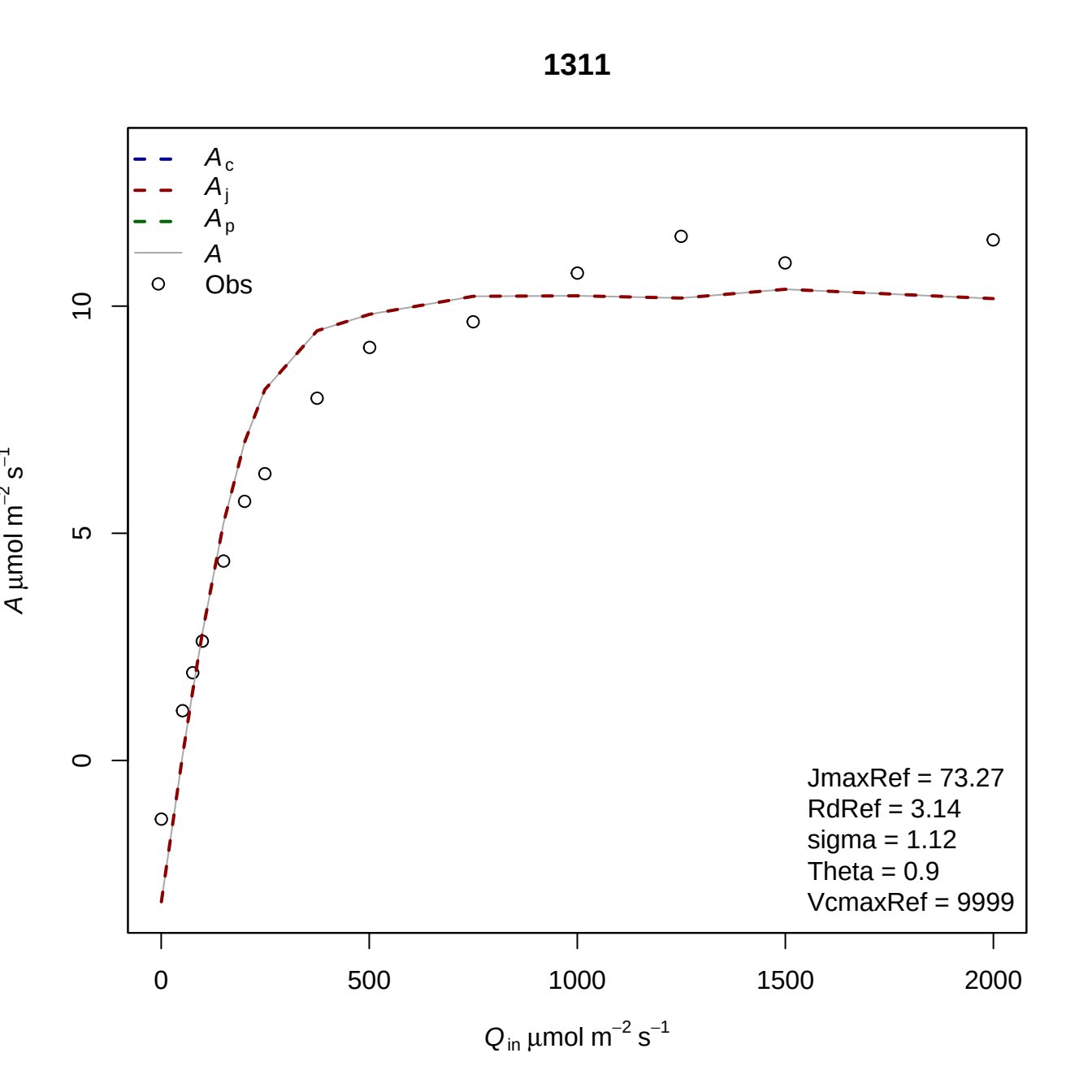

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 113.08
RdRef = 5.6
sigma = 2.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

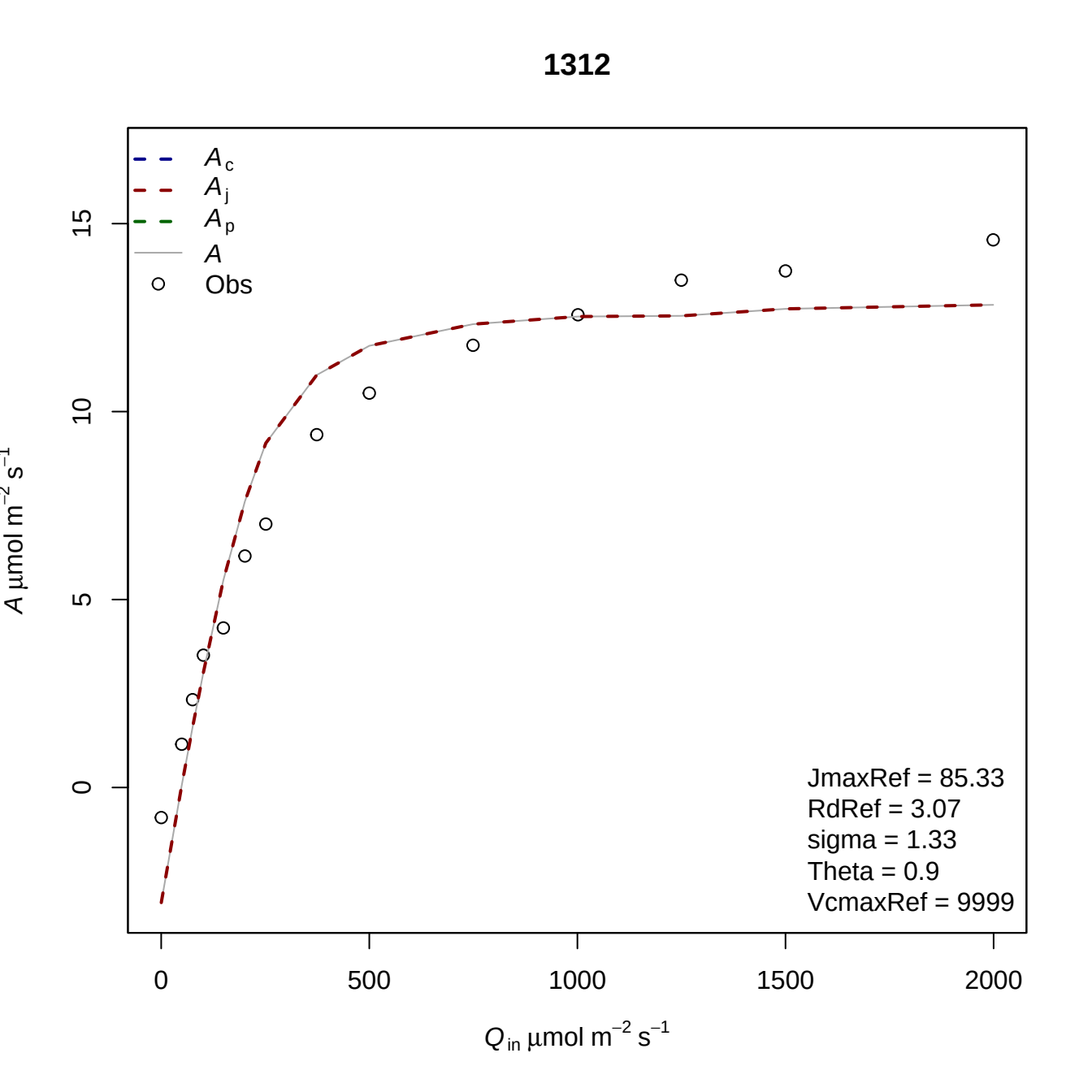

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 107.73
RdRef = 5.79
sigma = 2.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

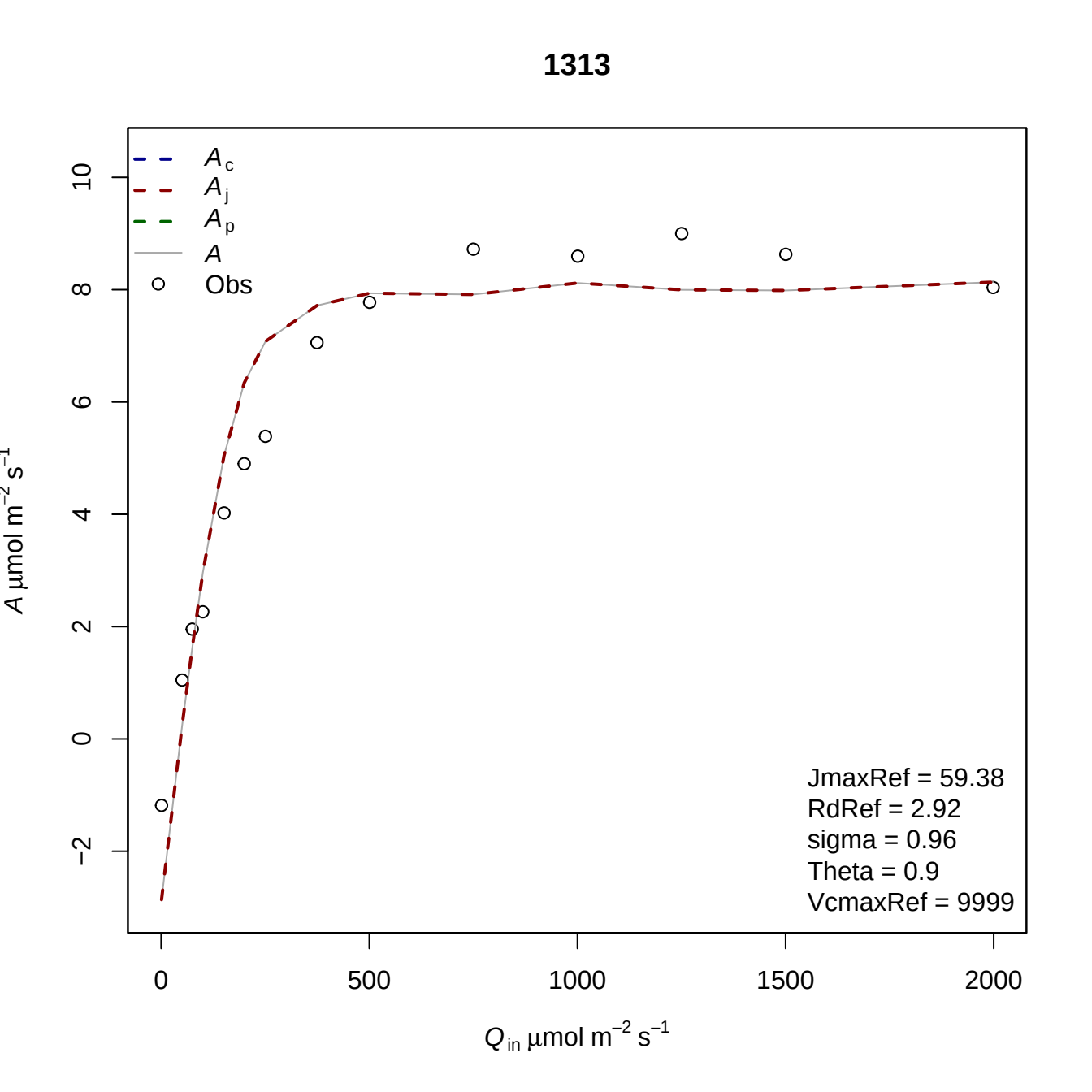

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 108.01
RdRef = 4.36
sigma = 2.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

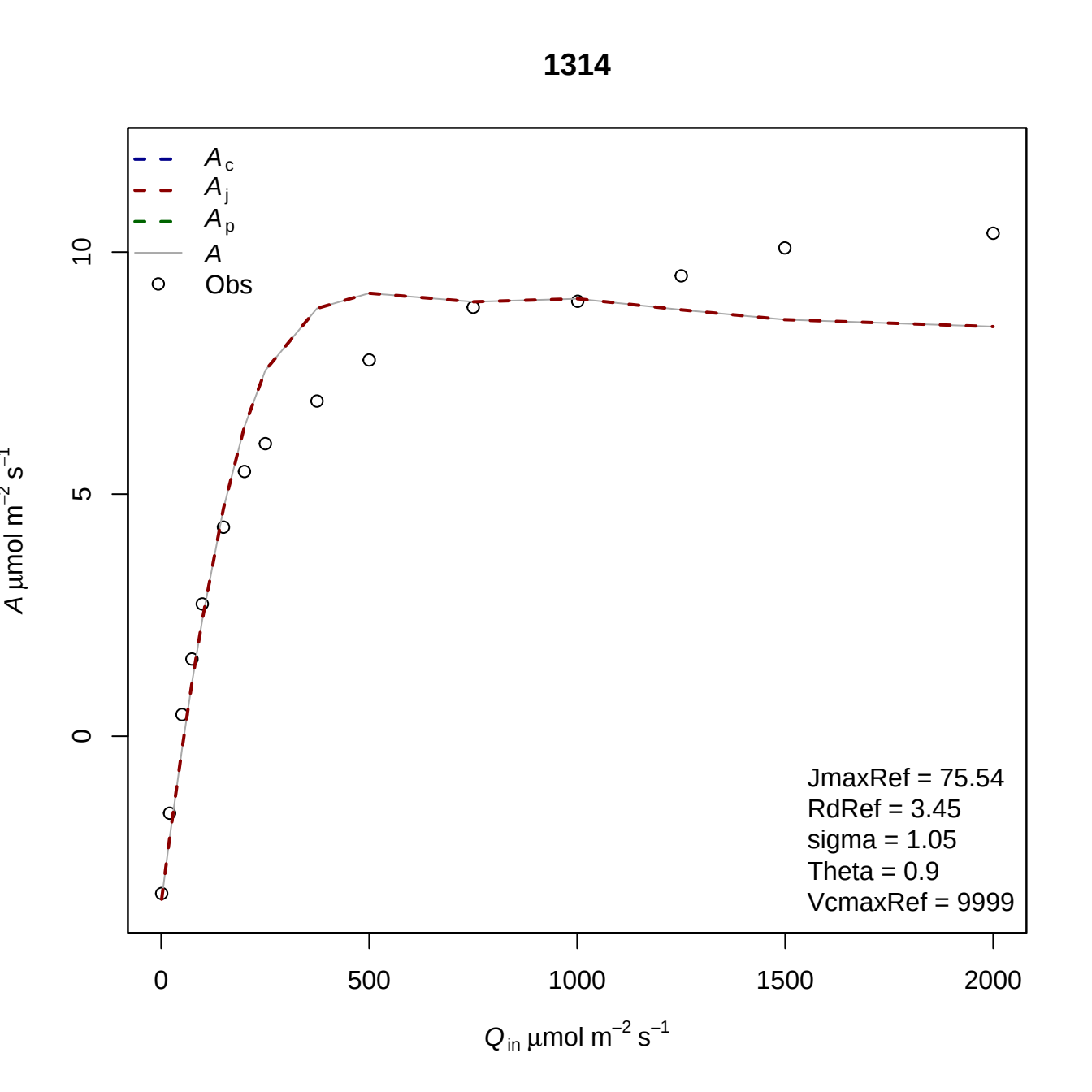

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 102.62
RdRef = 5.31
sigma = 1.91
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

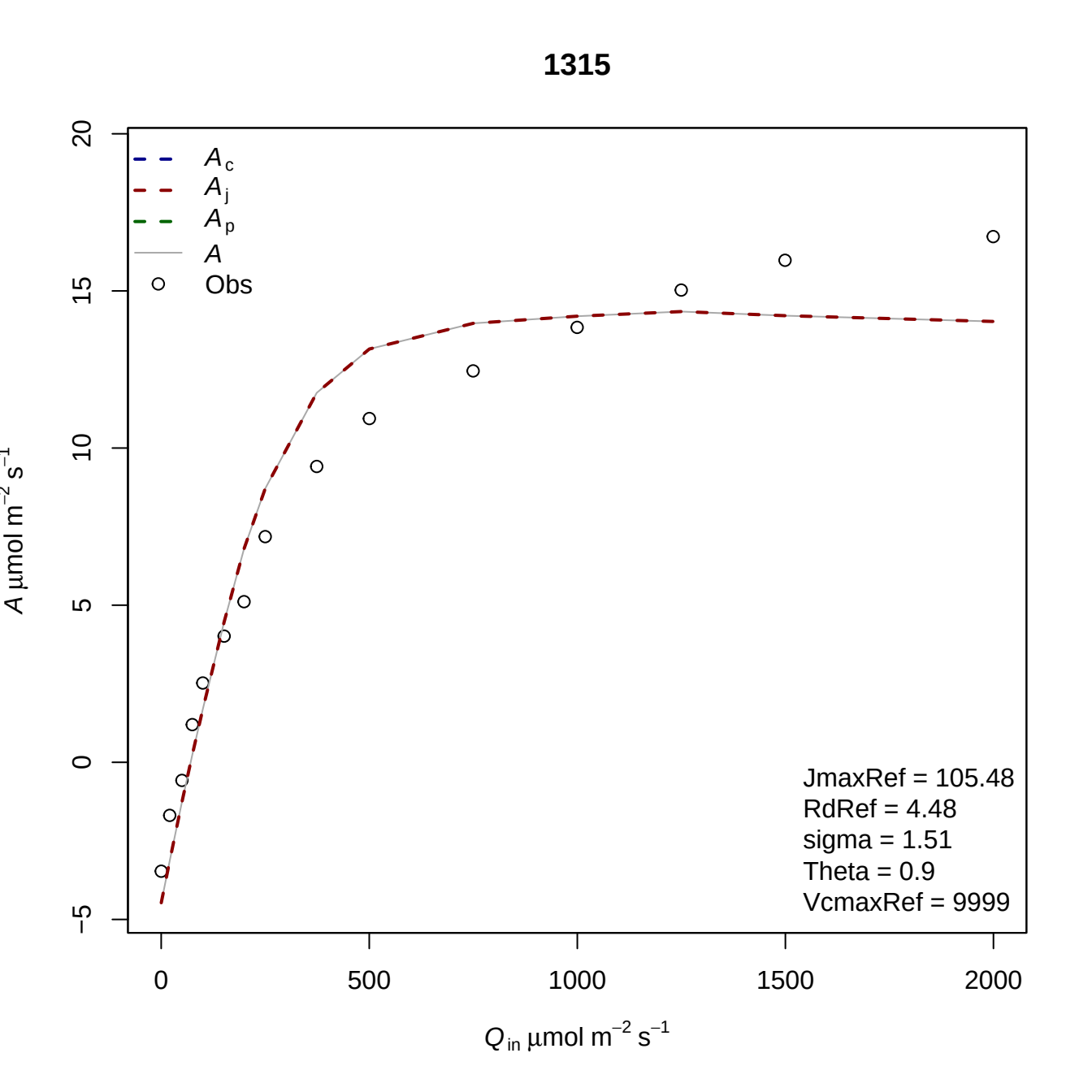

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 91.09
RdRef = 4.5
sigma = 1.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

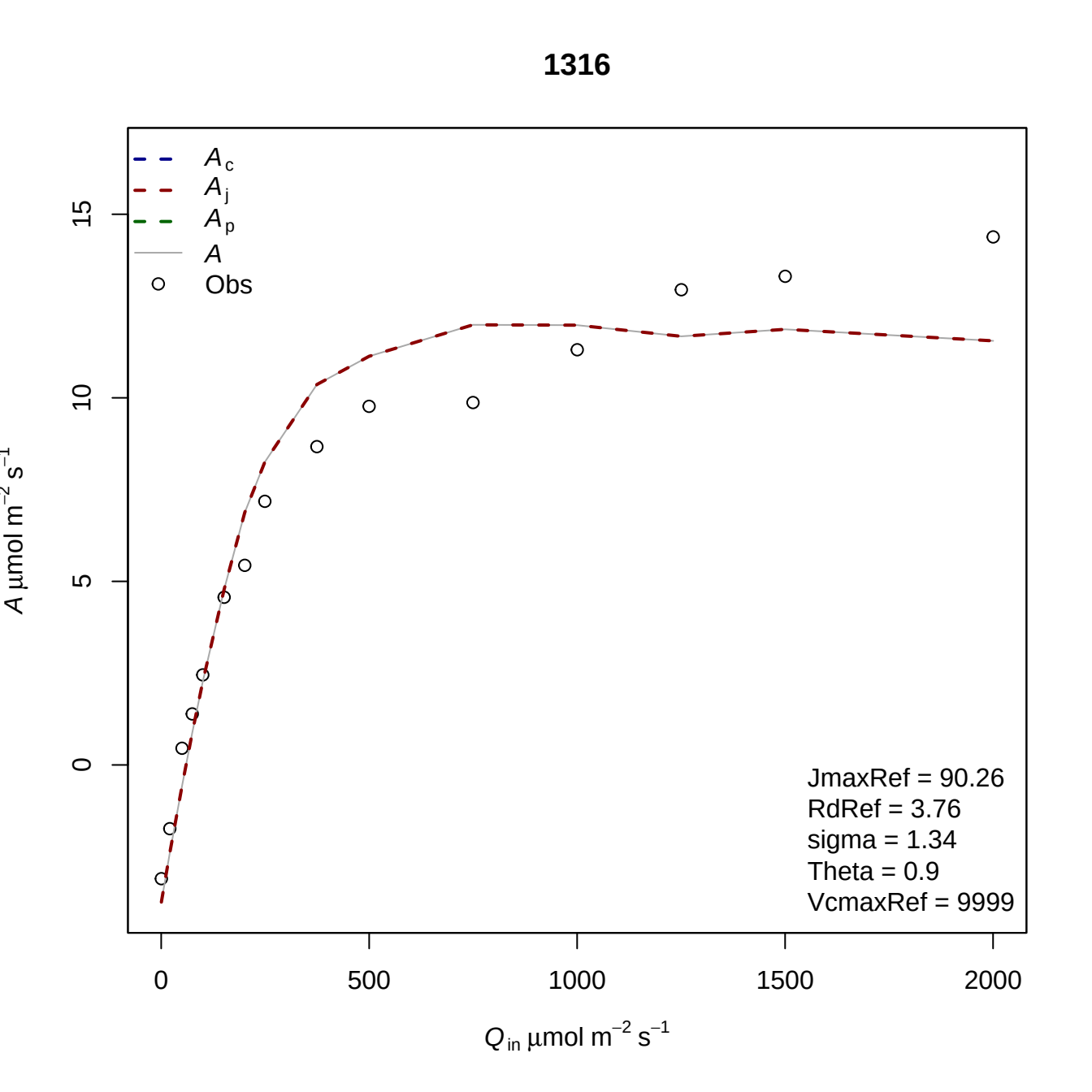

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 73.27
RdRef = 3.14
sigma = 1.12
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

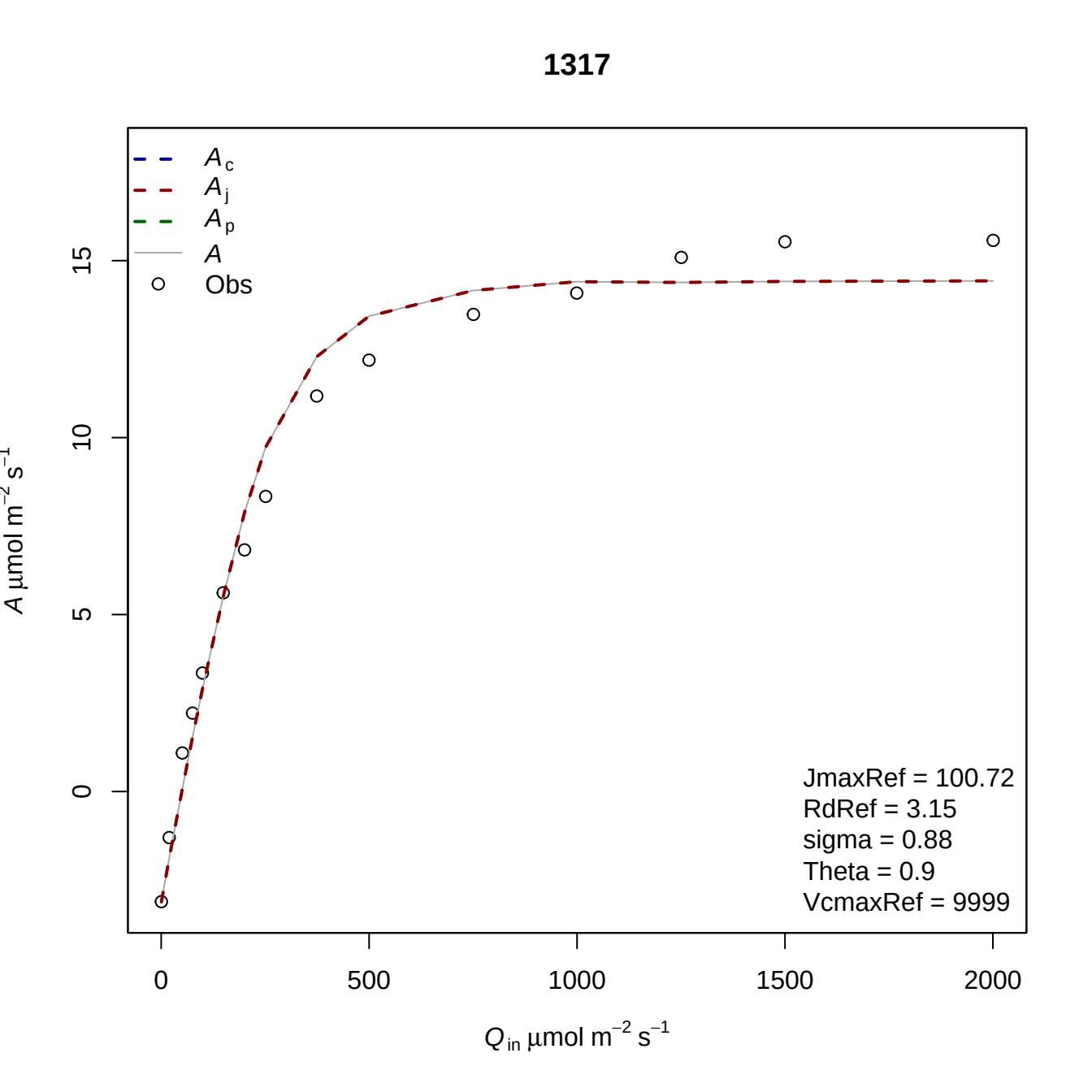

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 85.33
RdRef = 3.07
sigma = 1.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

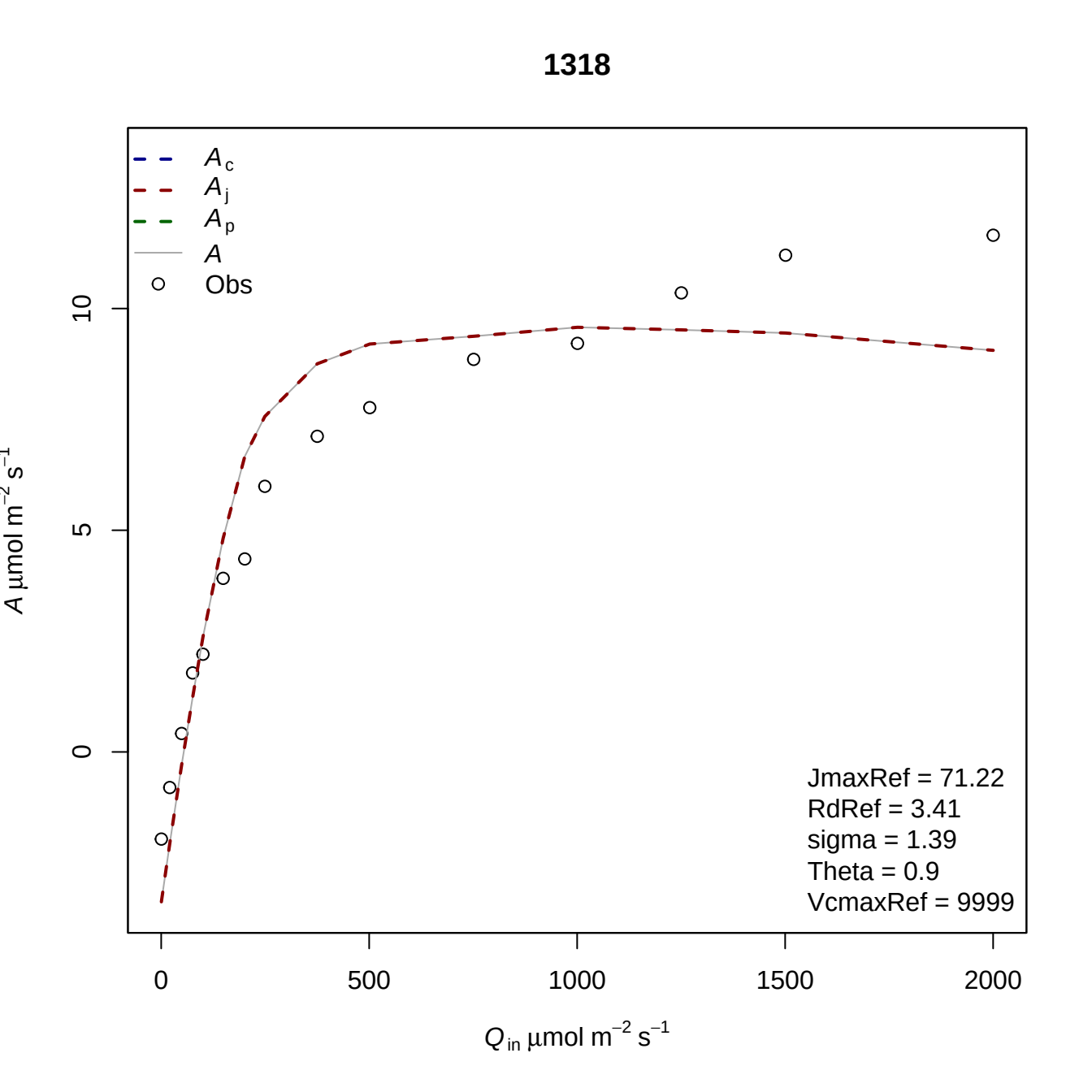

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 59.38
RdRef = 2.92
sigma = 0.96
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

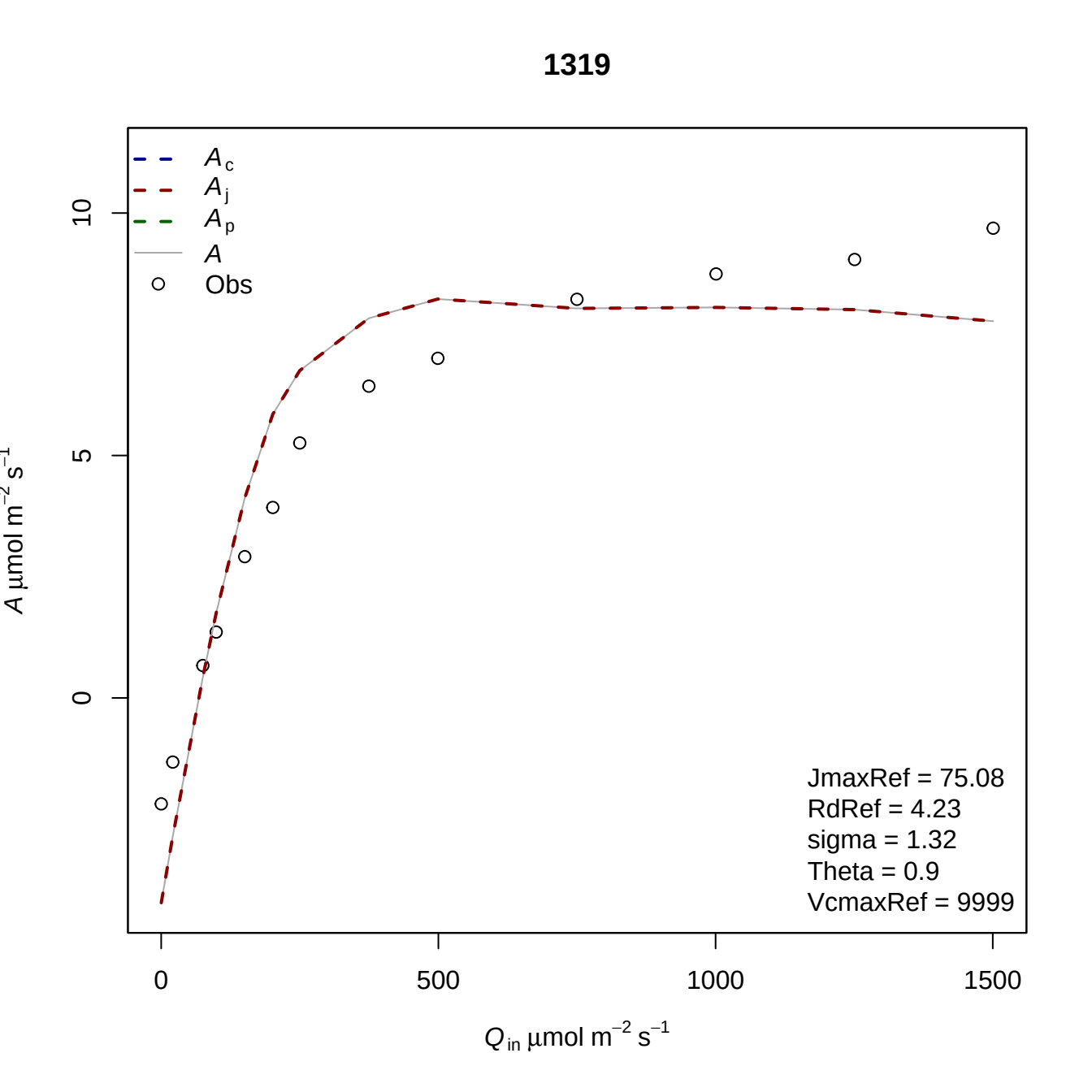

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 75.54
RdRef = 3.45
sigma = 1.05
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

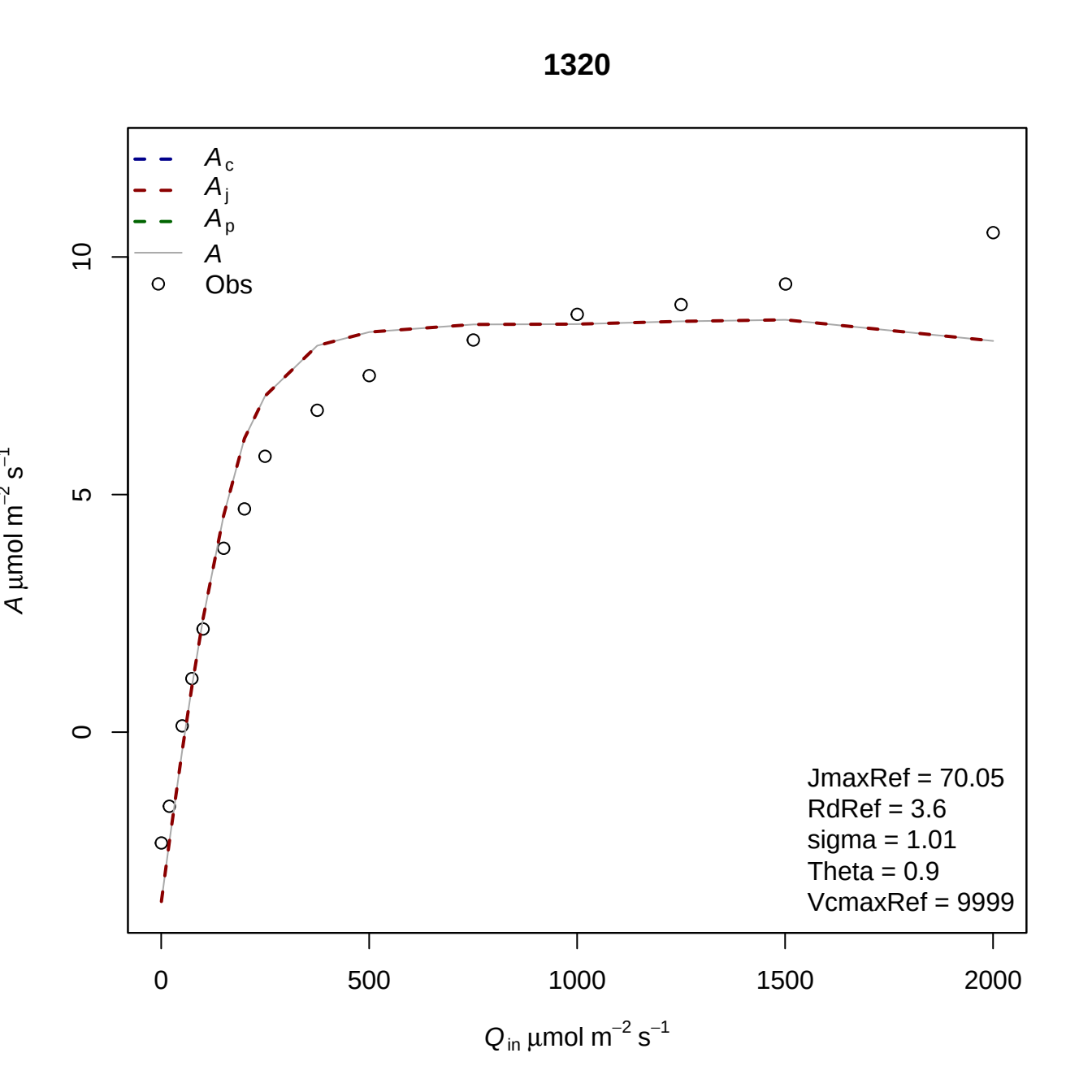

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 105.48
RdRef = 4.48
sigma = 1.51
Theta = 0.9
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

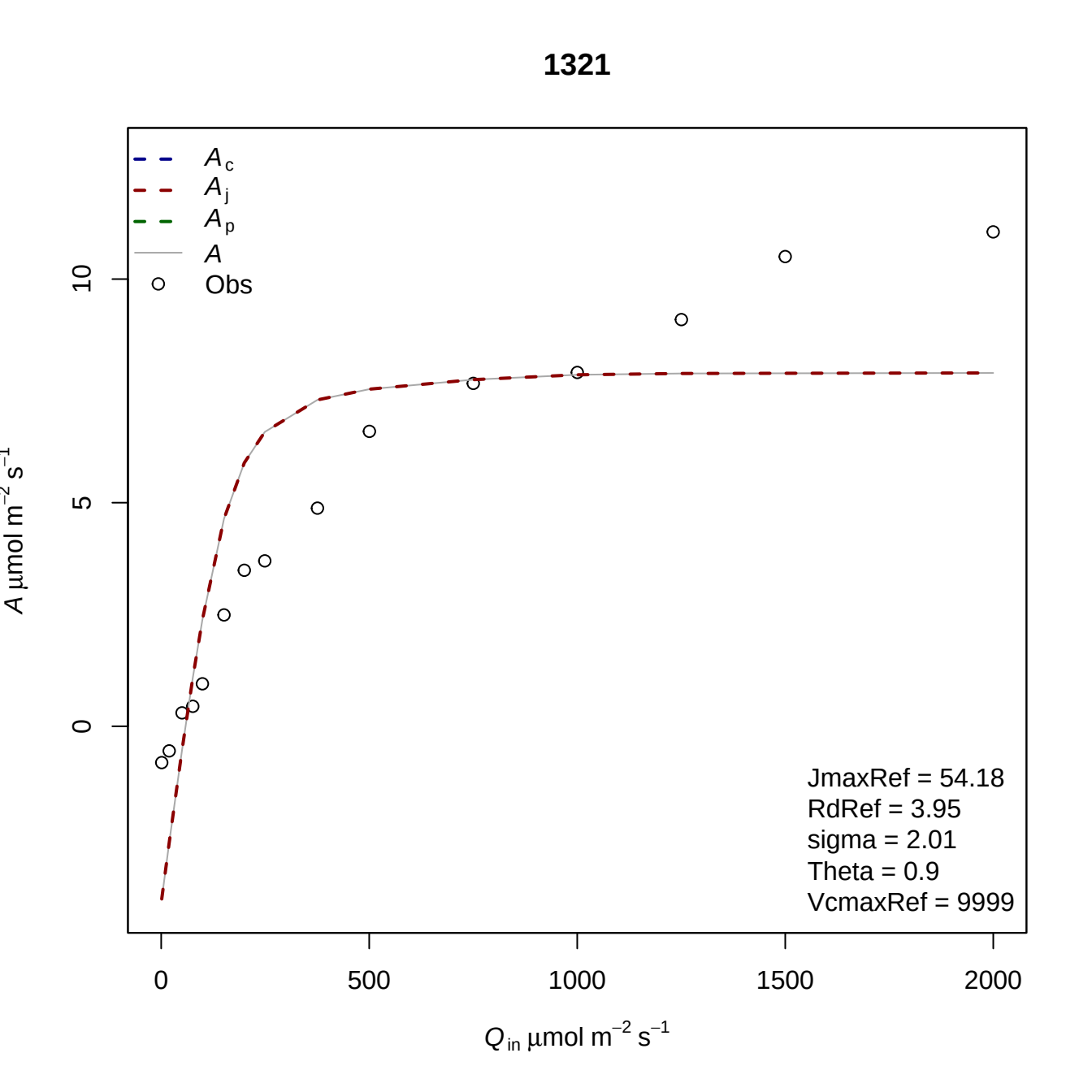

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 90.26
RdRef = 3.76
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

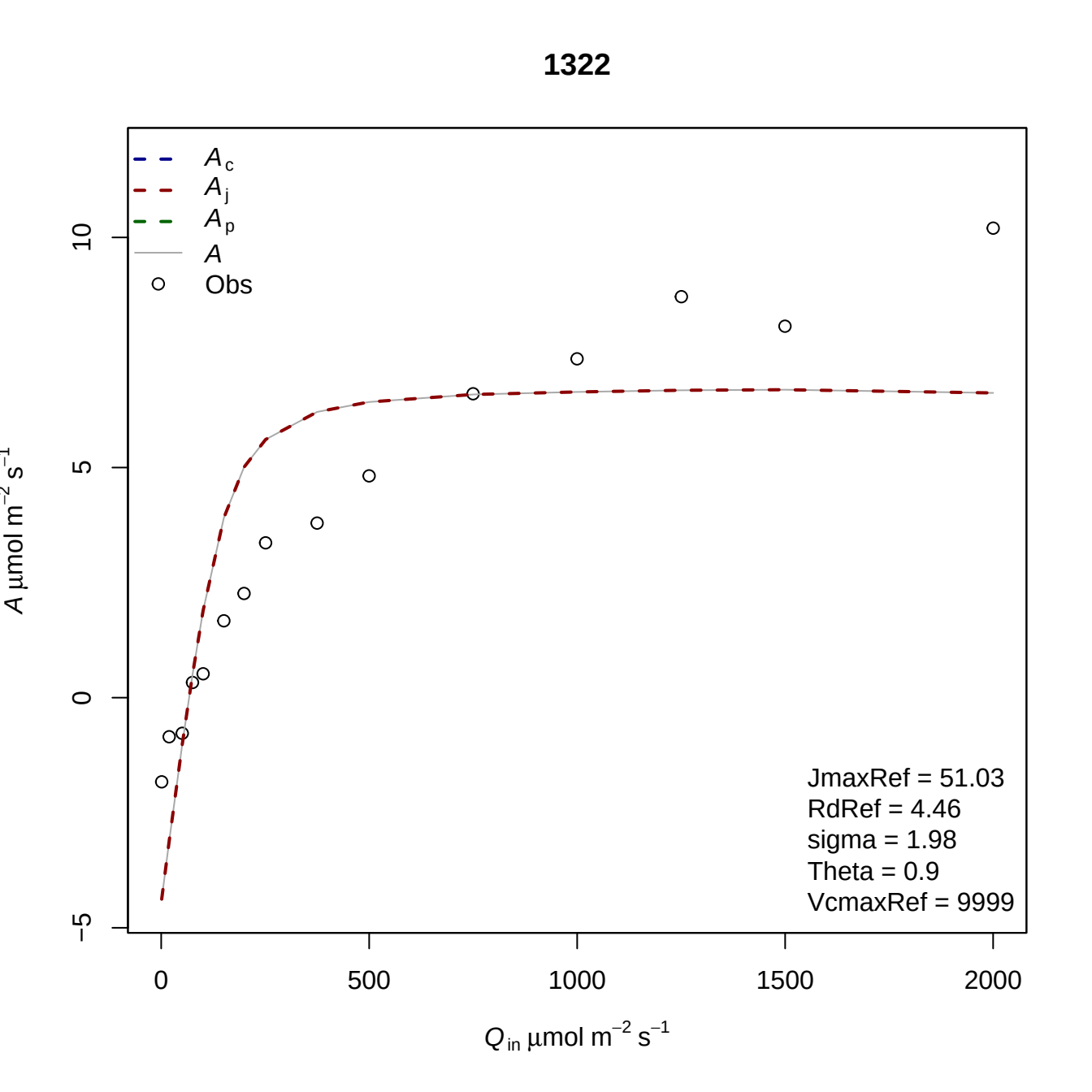

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 100.72
RdRef = 3.15
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

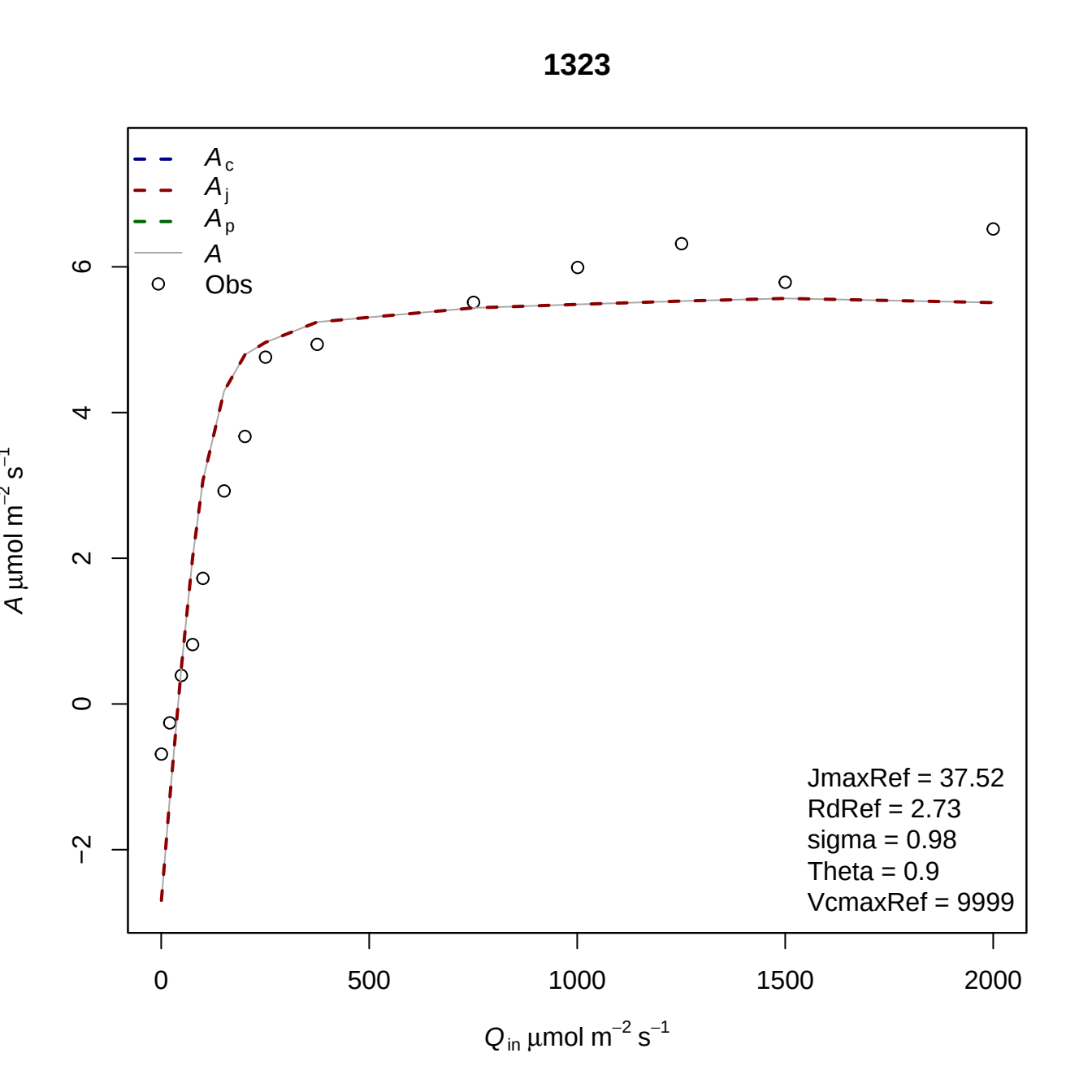

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 71.22
RdRef = 3.41
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

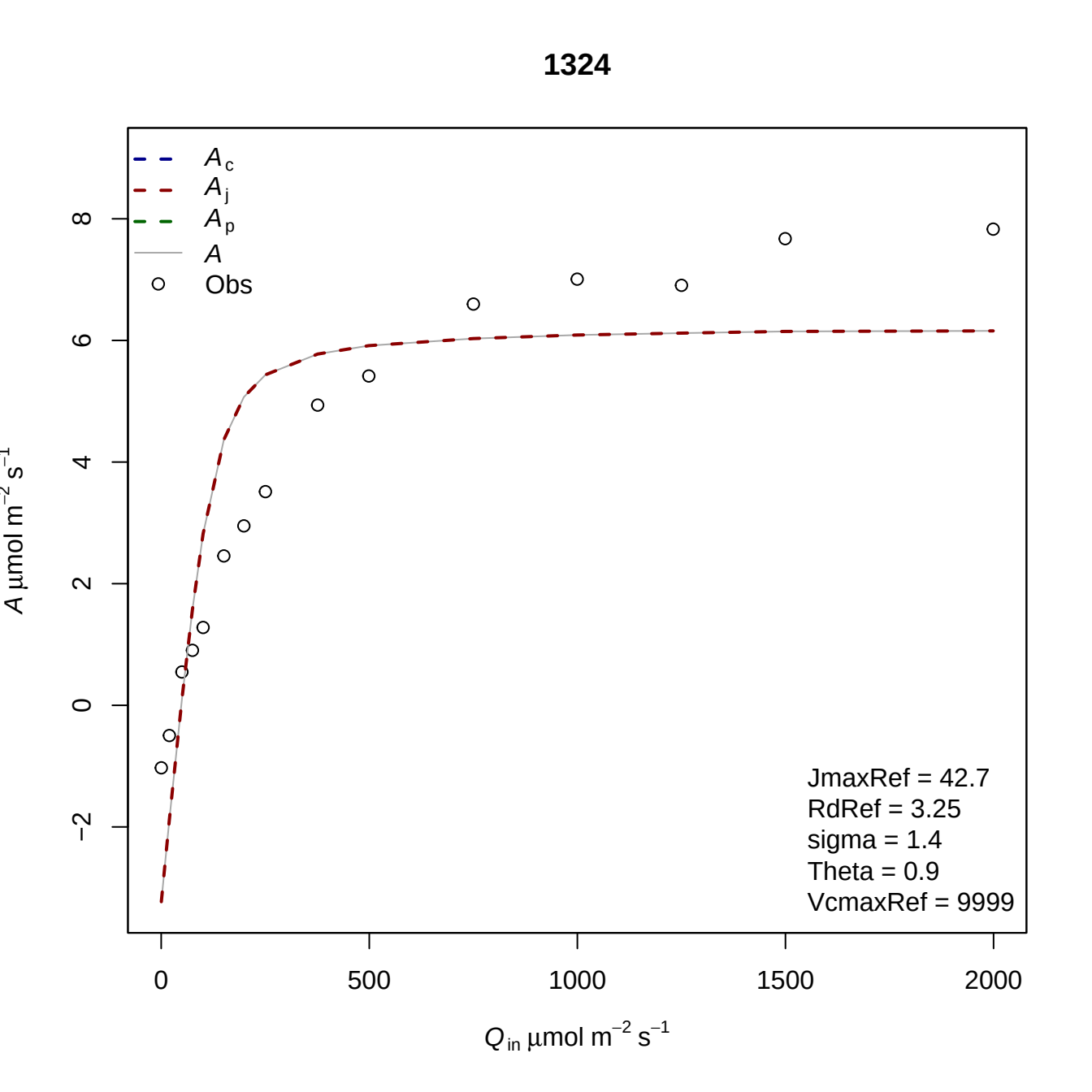

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 75.08
RdRef = 4.23
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

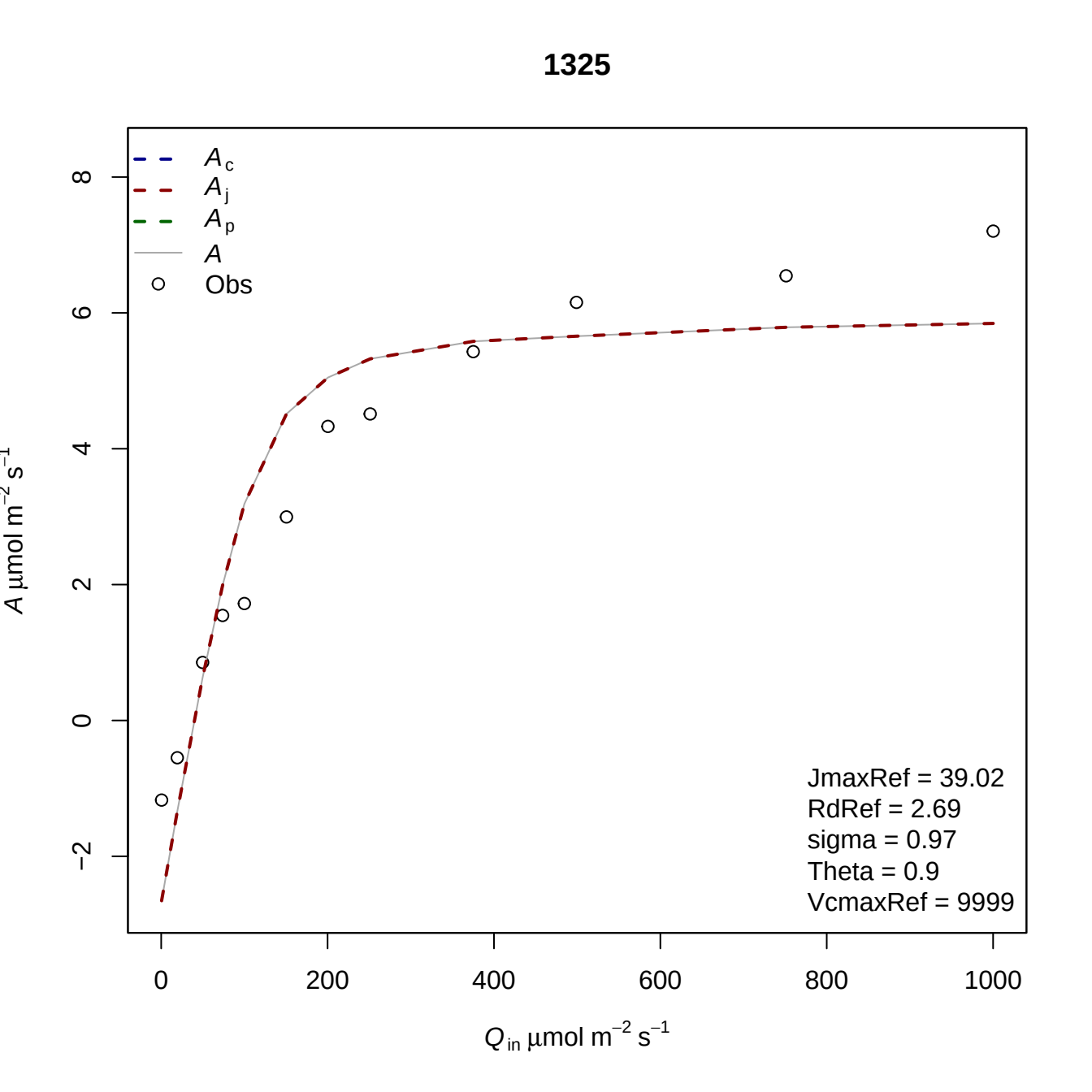

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 70.05
RdRef = 3.6
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

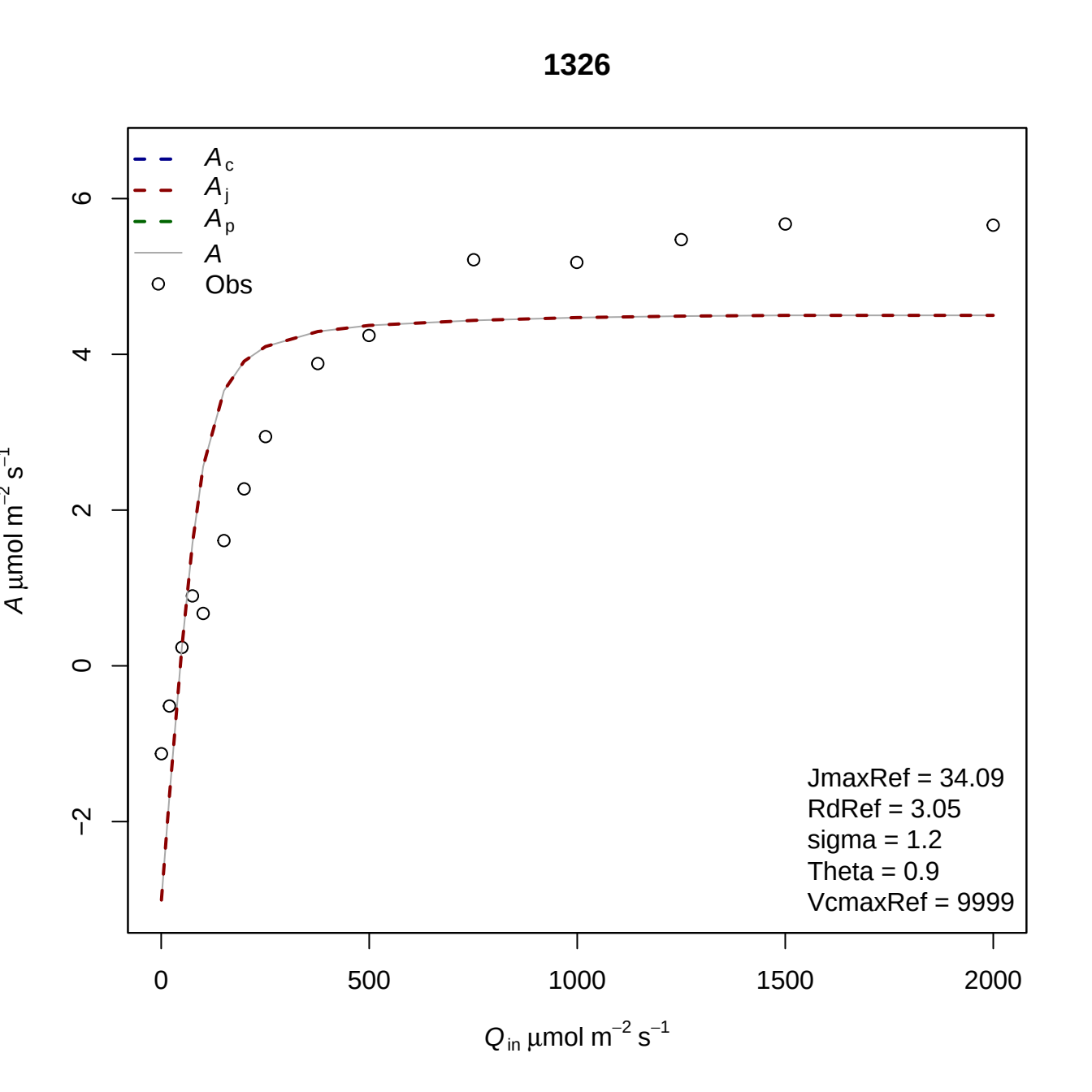

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 54.18
RdRef = 3.95
sigma = 2.01
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

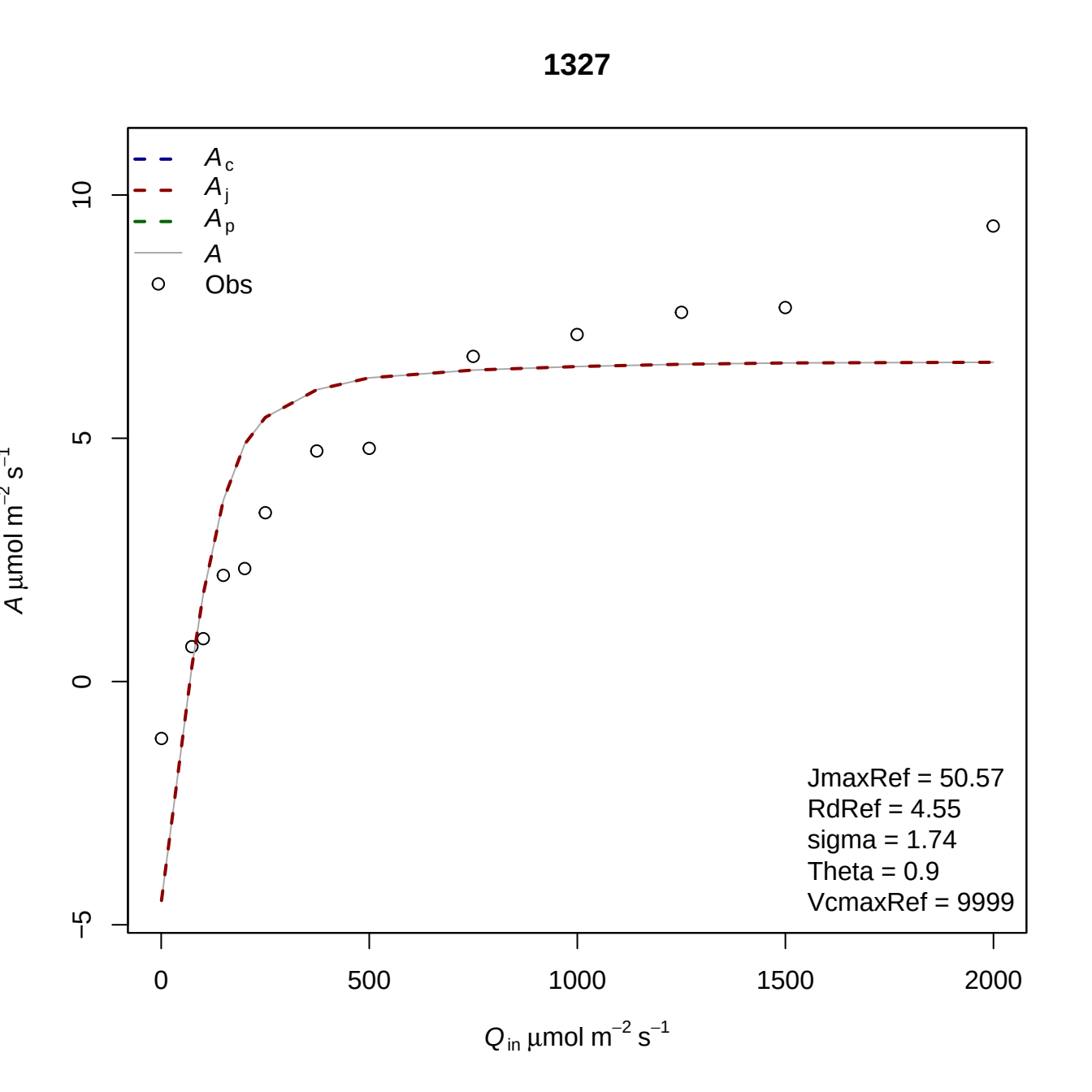

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 51.03
RdRef = 4.46
sigma = 1.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

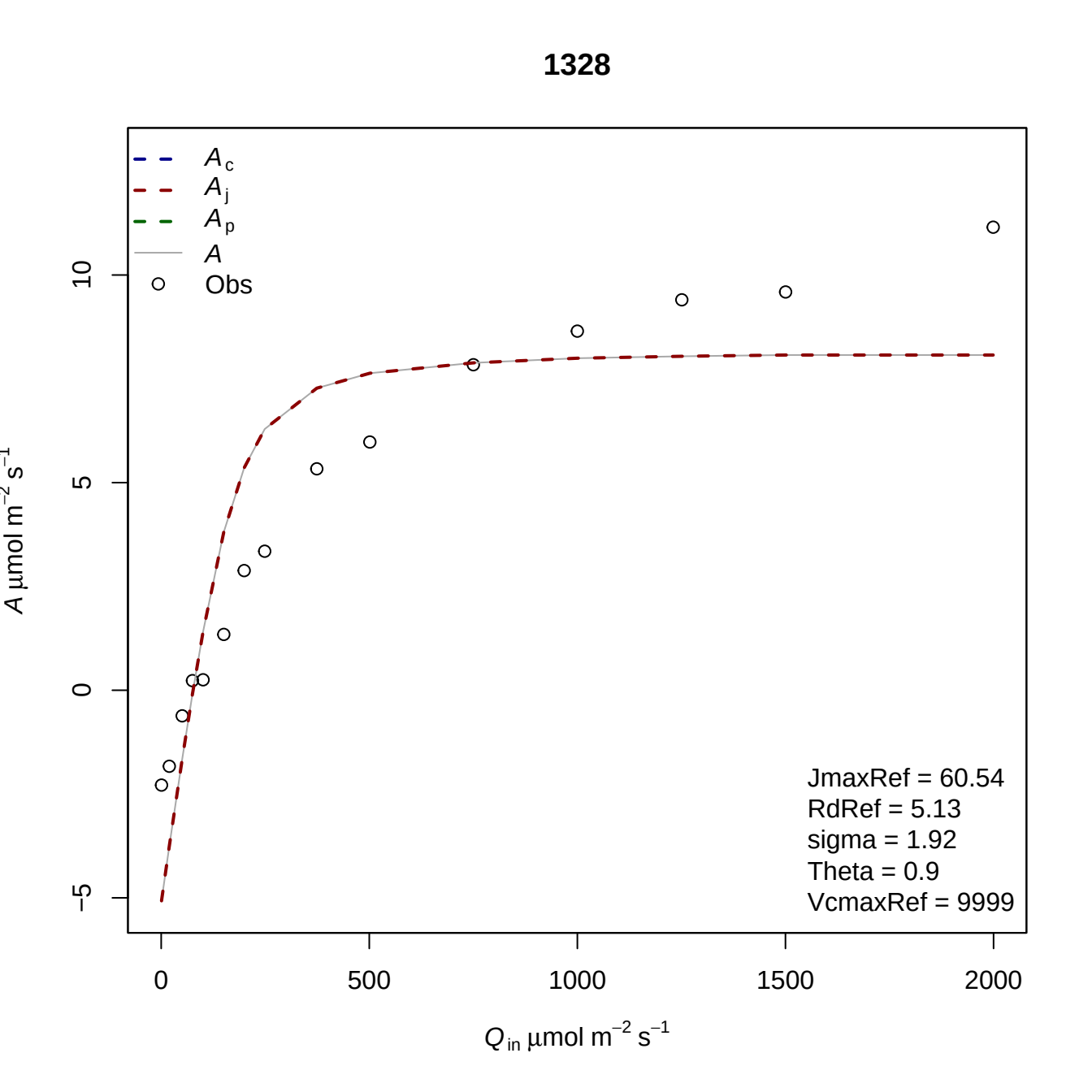

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 37.52
RdRef = 2.73
sigma = 0.98
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

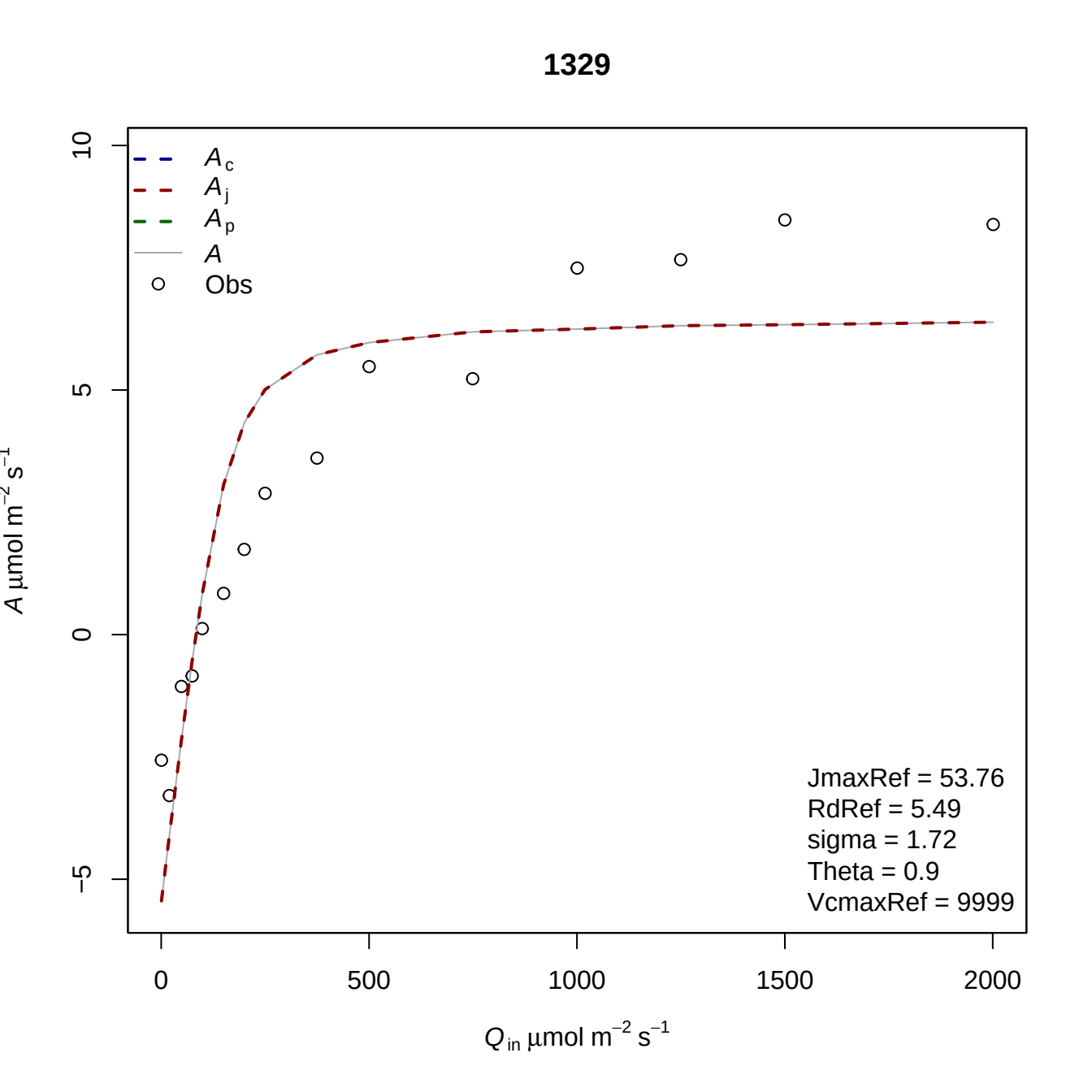

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 42.7
RdRef = 3.25
sigma = 1.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

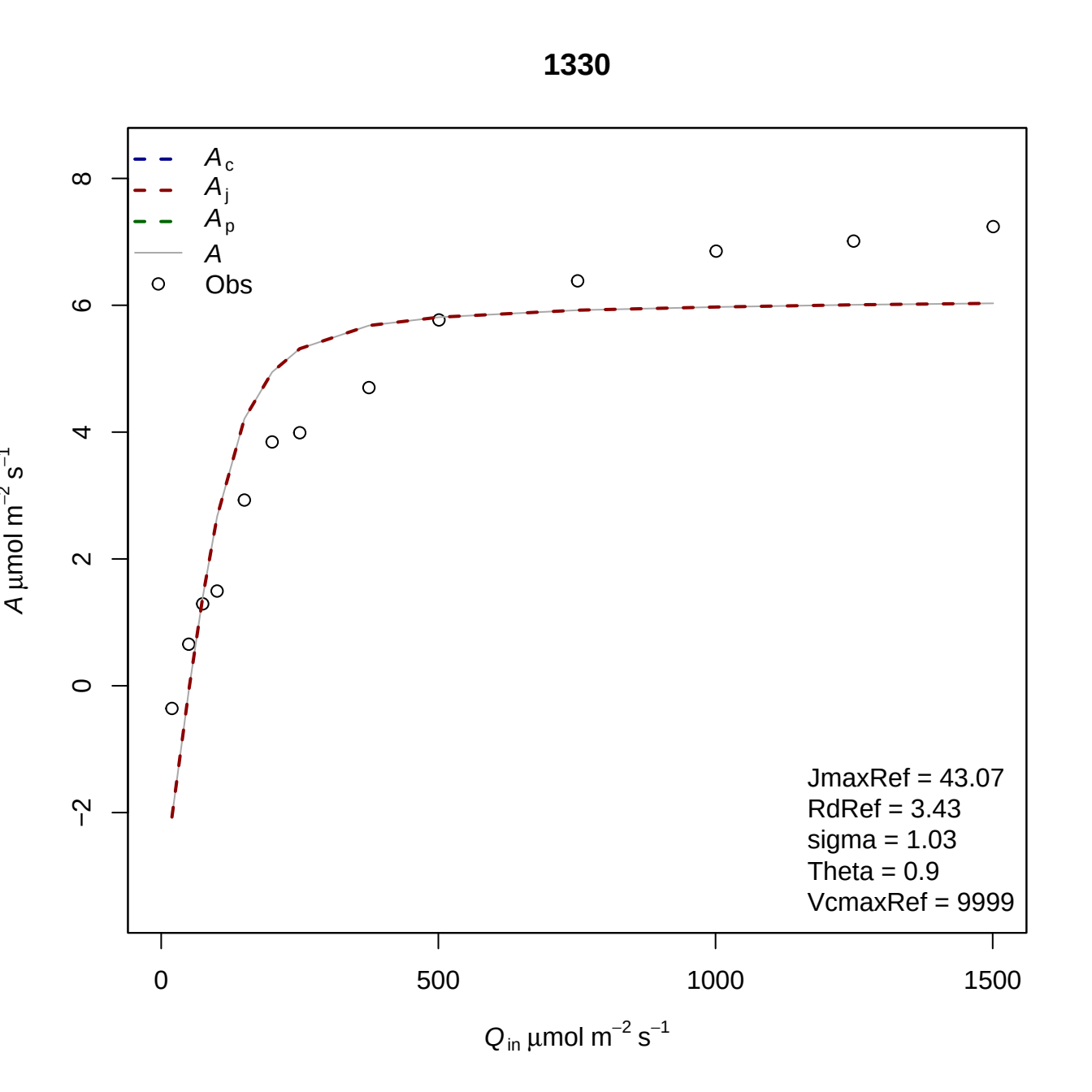

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 39.02
RdRef = 2.69
sigma = 0.97
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

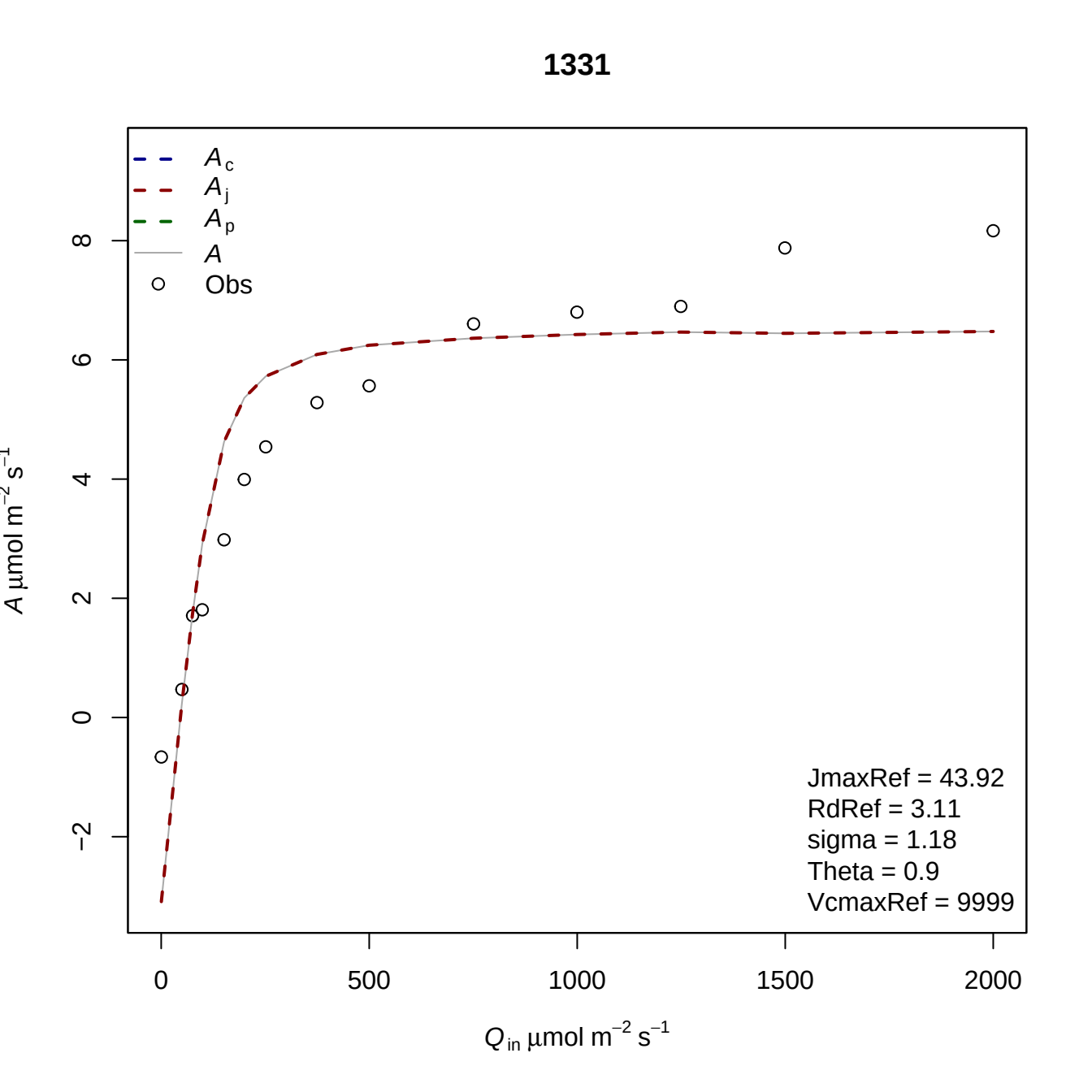

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 34.09
RdRef = 3.05
sigma = 1.2
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

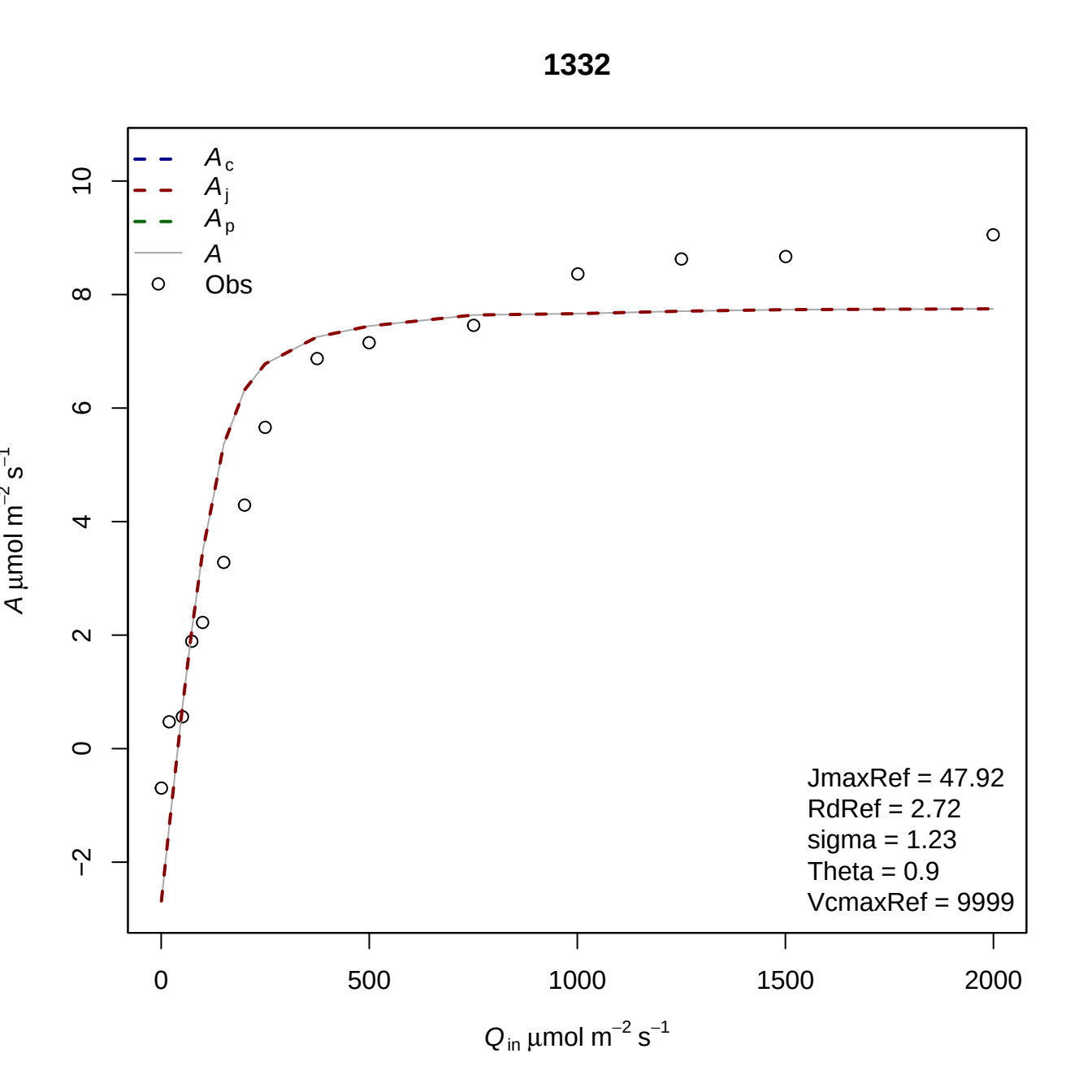

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 50.57
RdRef = 4.55
sigma = 1.74
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

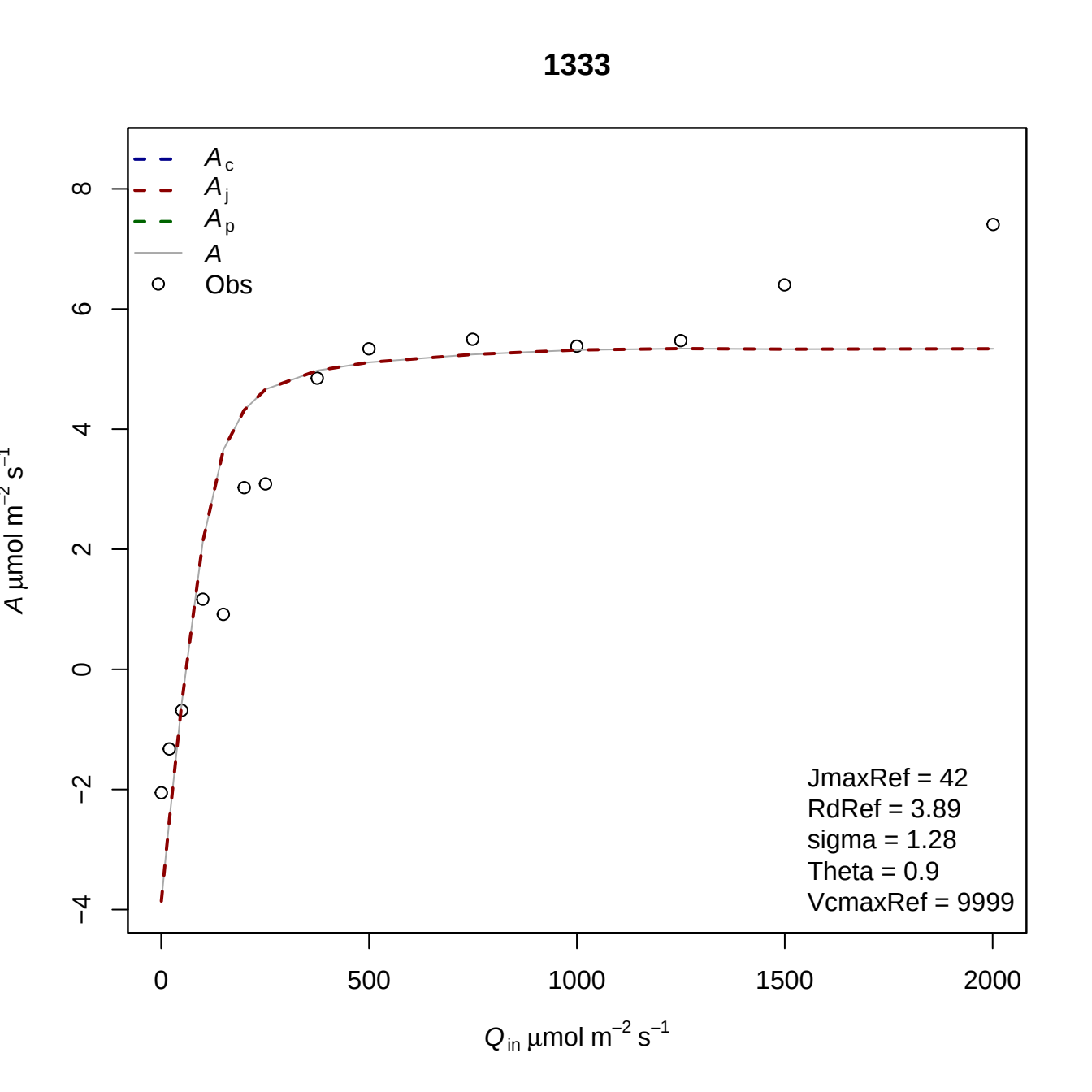

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 60.54
RdRef = 5.13
sigma = 1.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

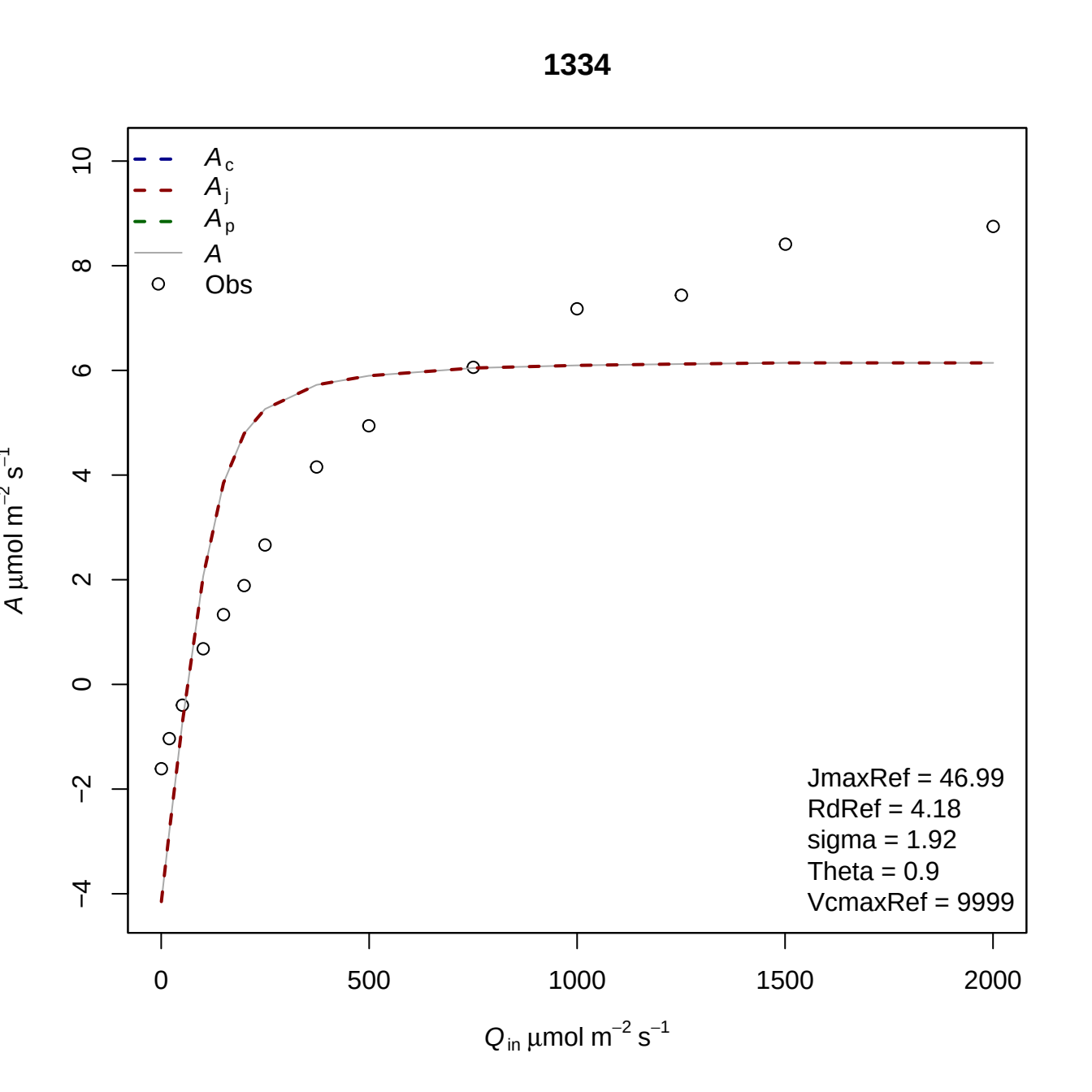

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 53.76
RdRef = 5.49
sigma = 1.72
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

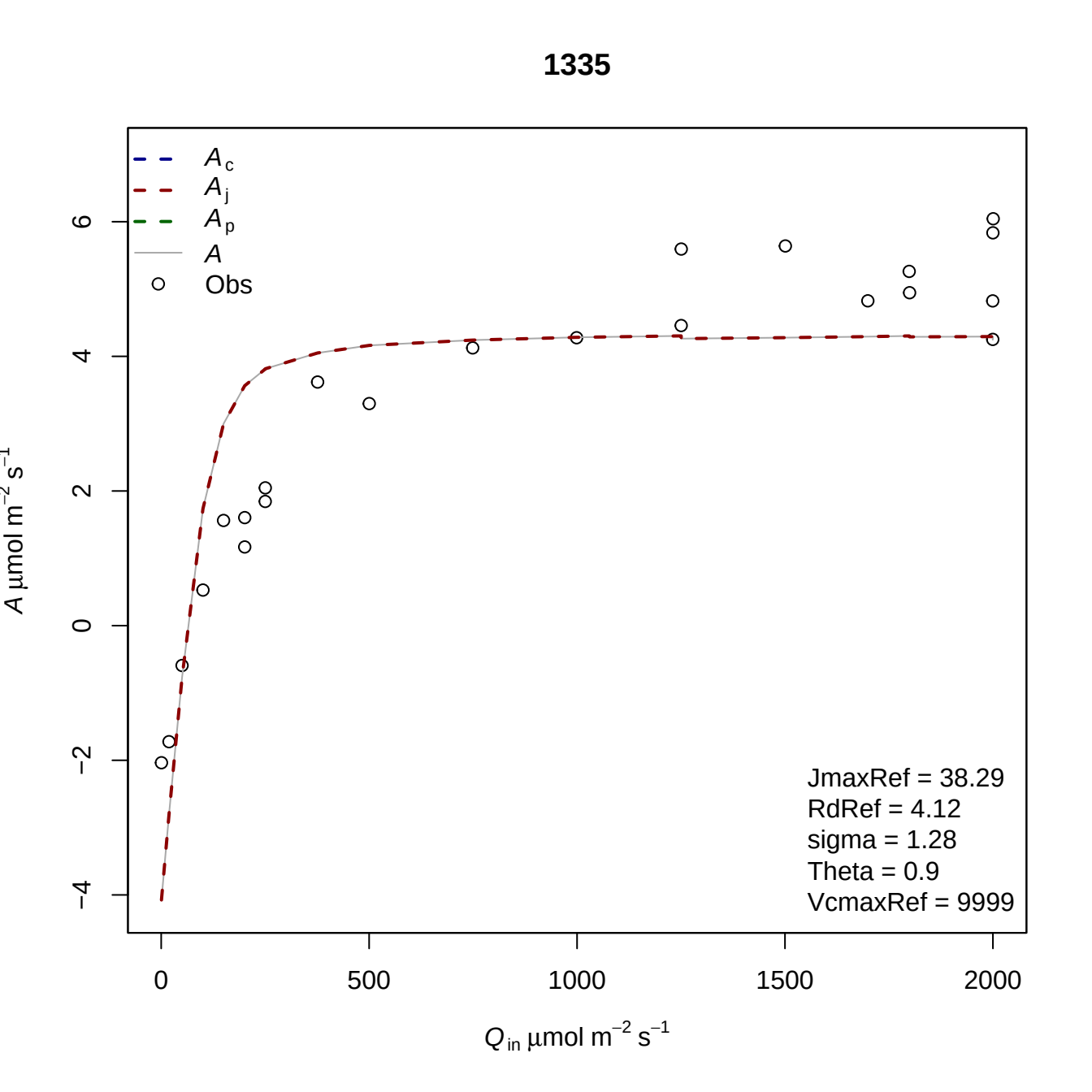

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 43.07
RdRef = 3.43
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

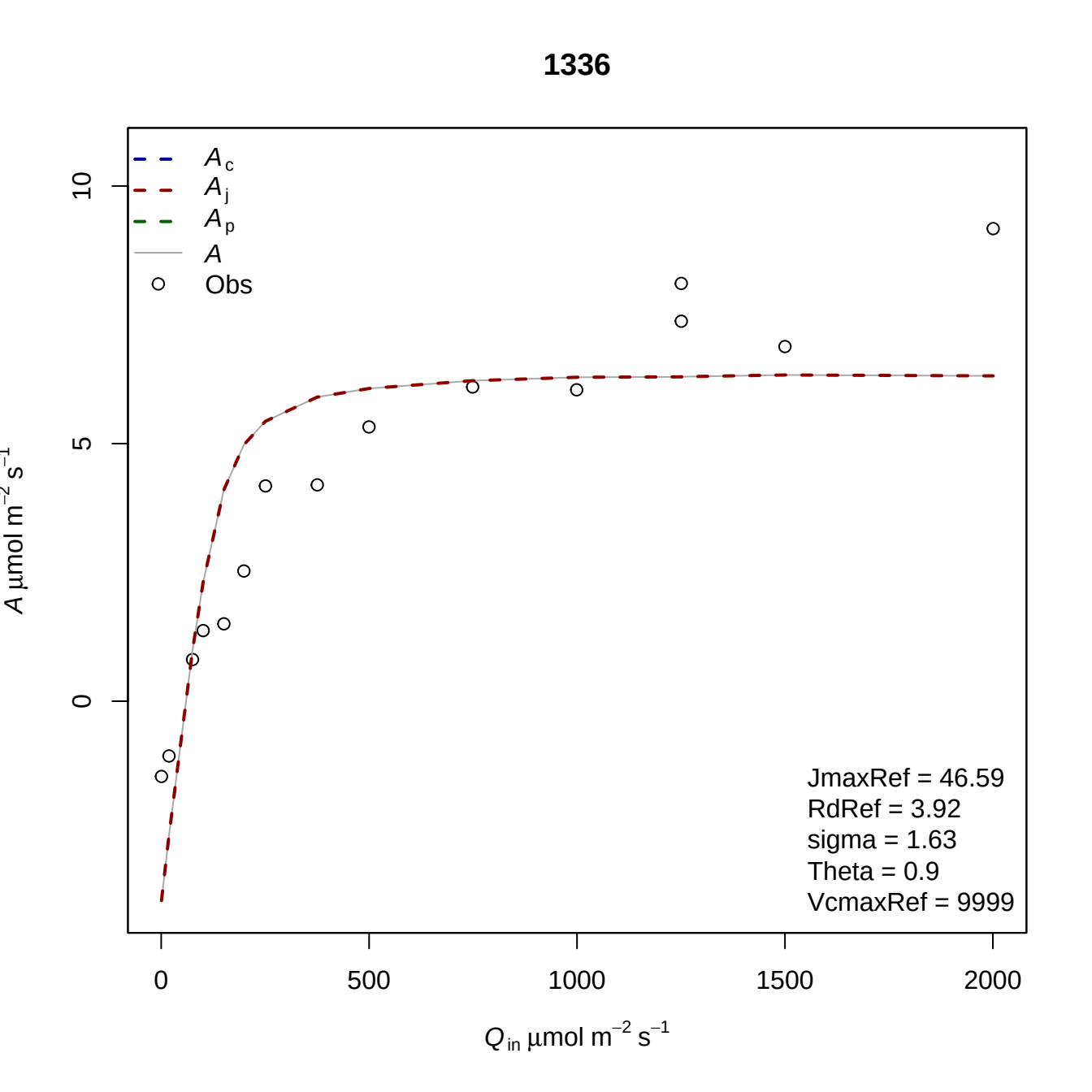

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 43.92
RdRef = 3.11
sigma = 1.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

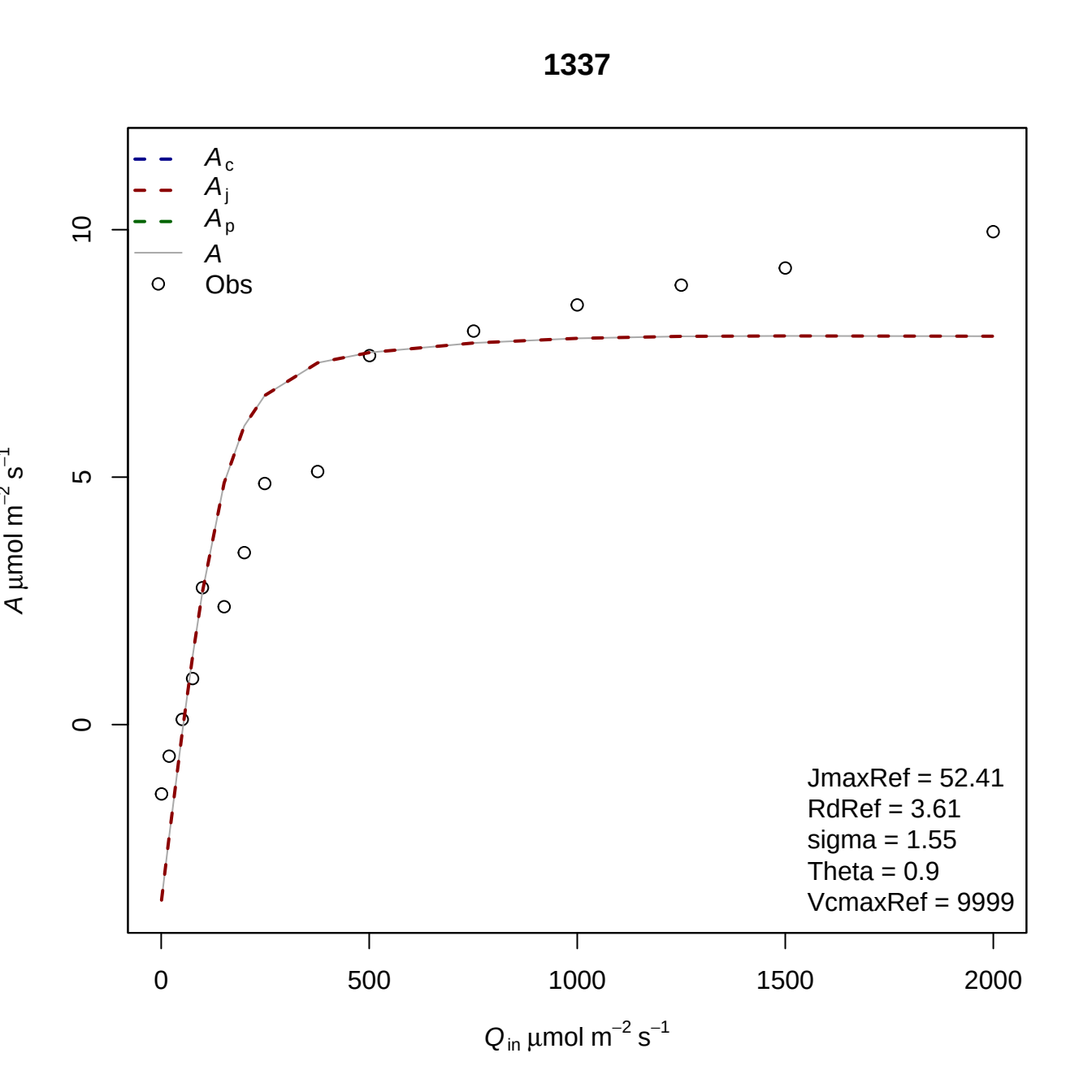

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 47.92
RdRef = 2.72
sigma = 1.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

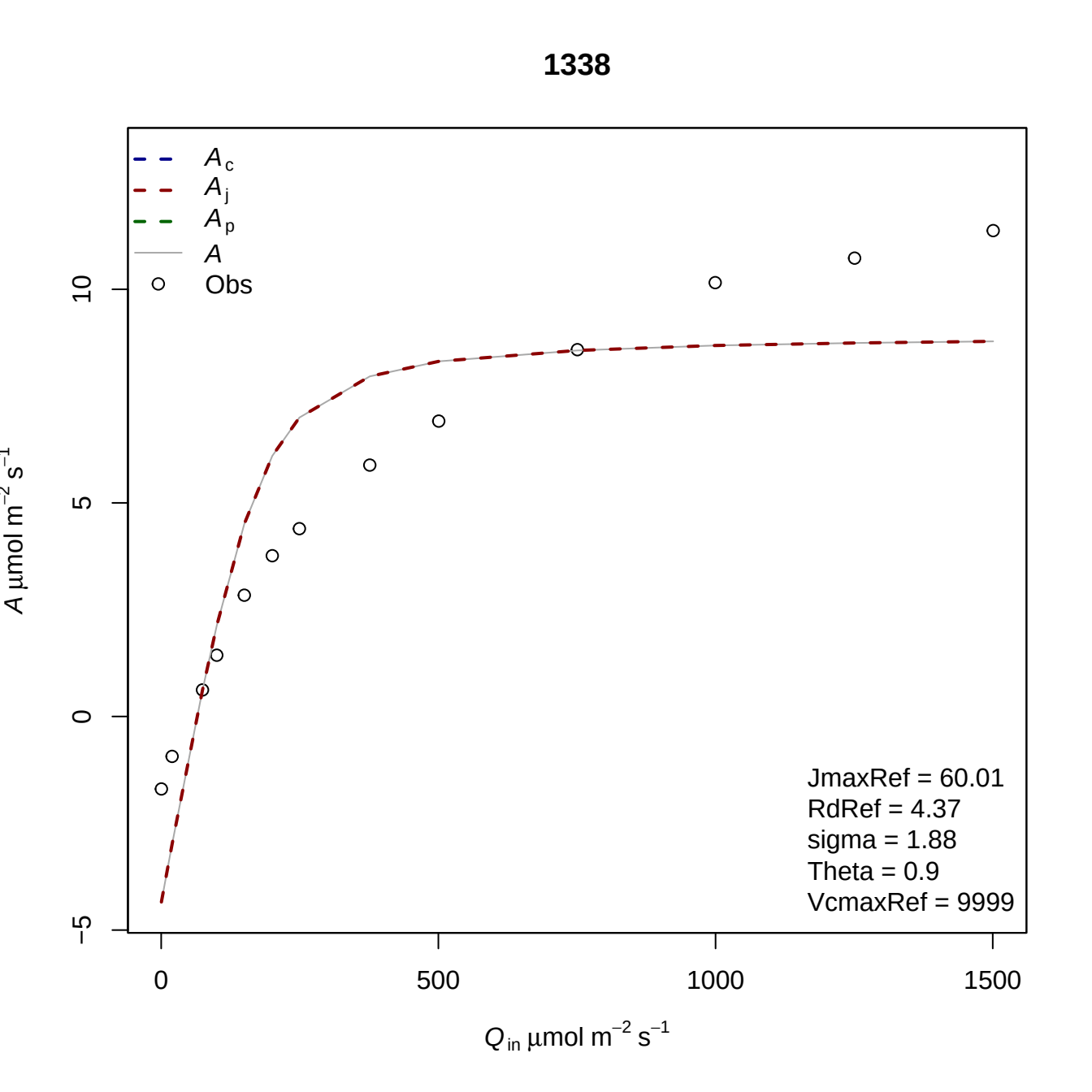

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 42
RdRef = 3.89
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

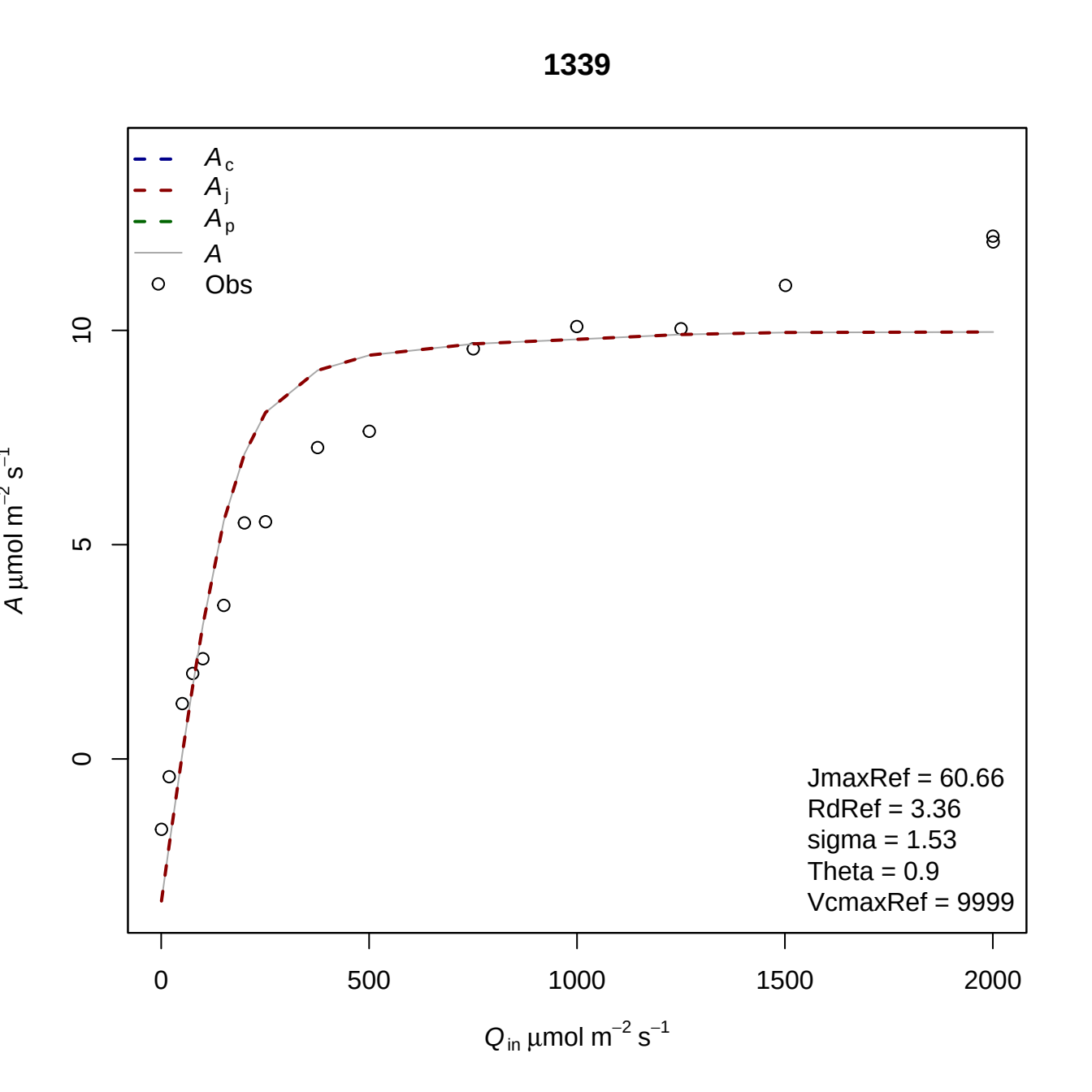

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 46.99
RdRef = 4.18
sigma = 1.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

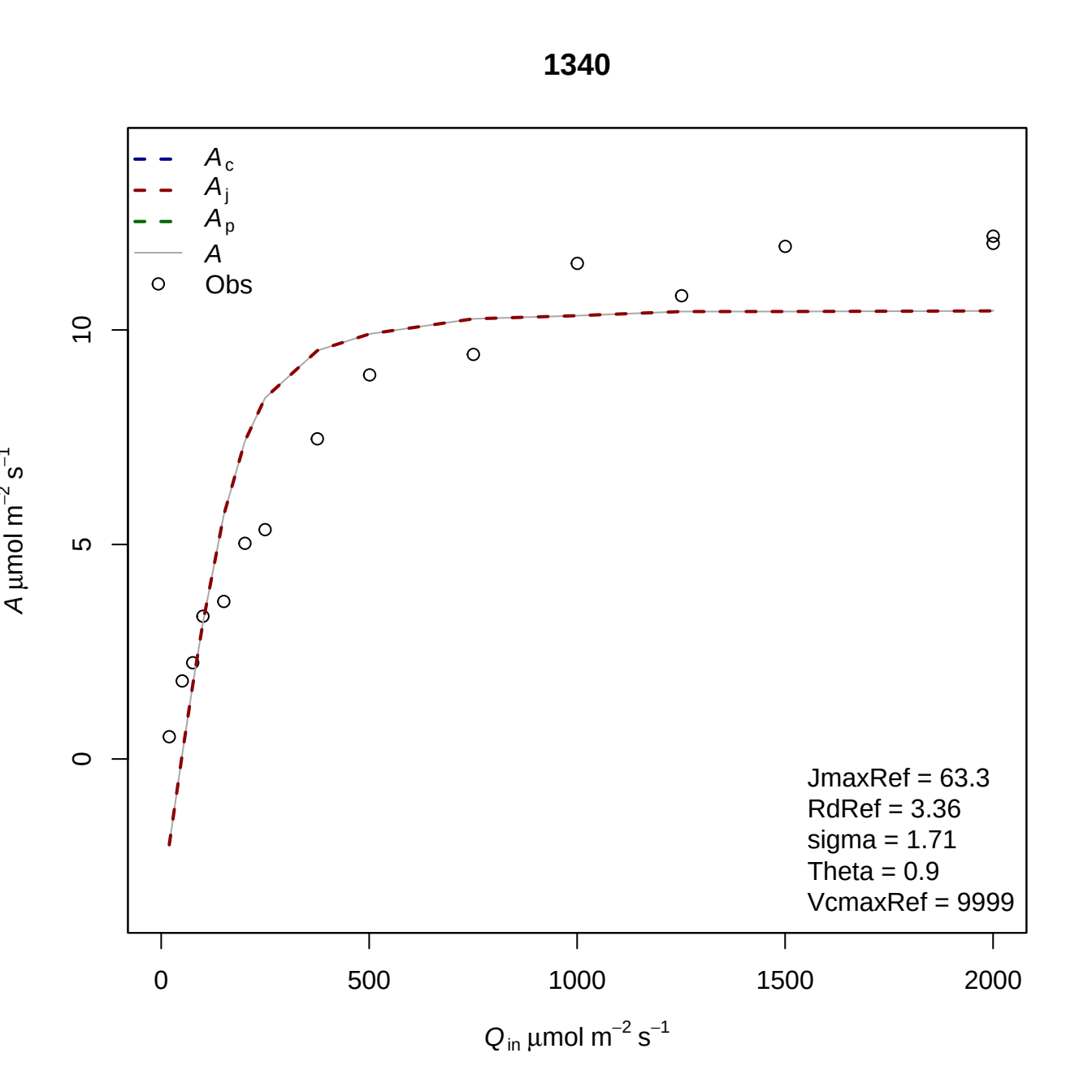

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 38.29
RdRef = 4.12
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

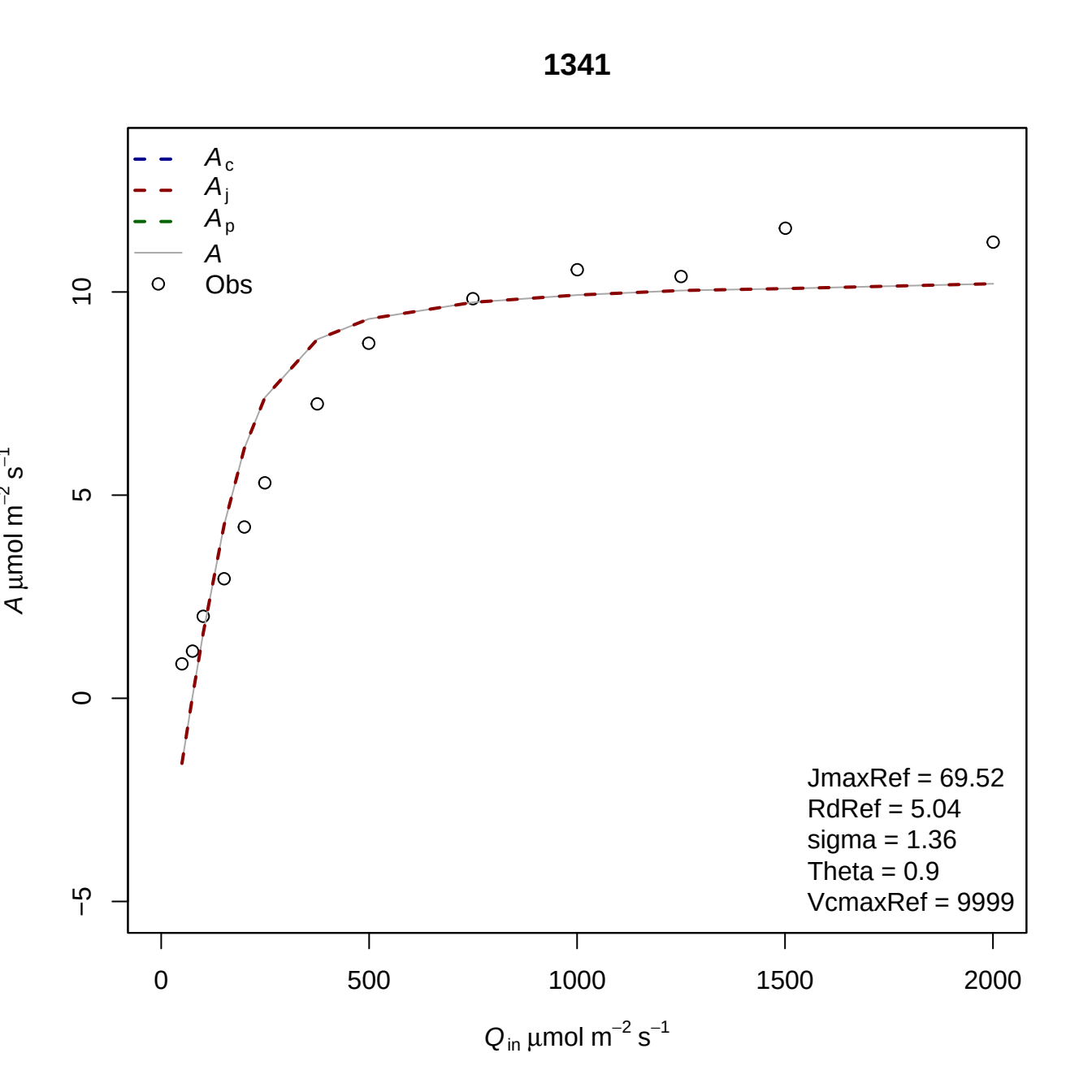

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 46.59
RdRef = 3.92
sigma = 1.63
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

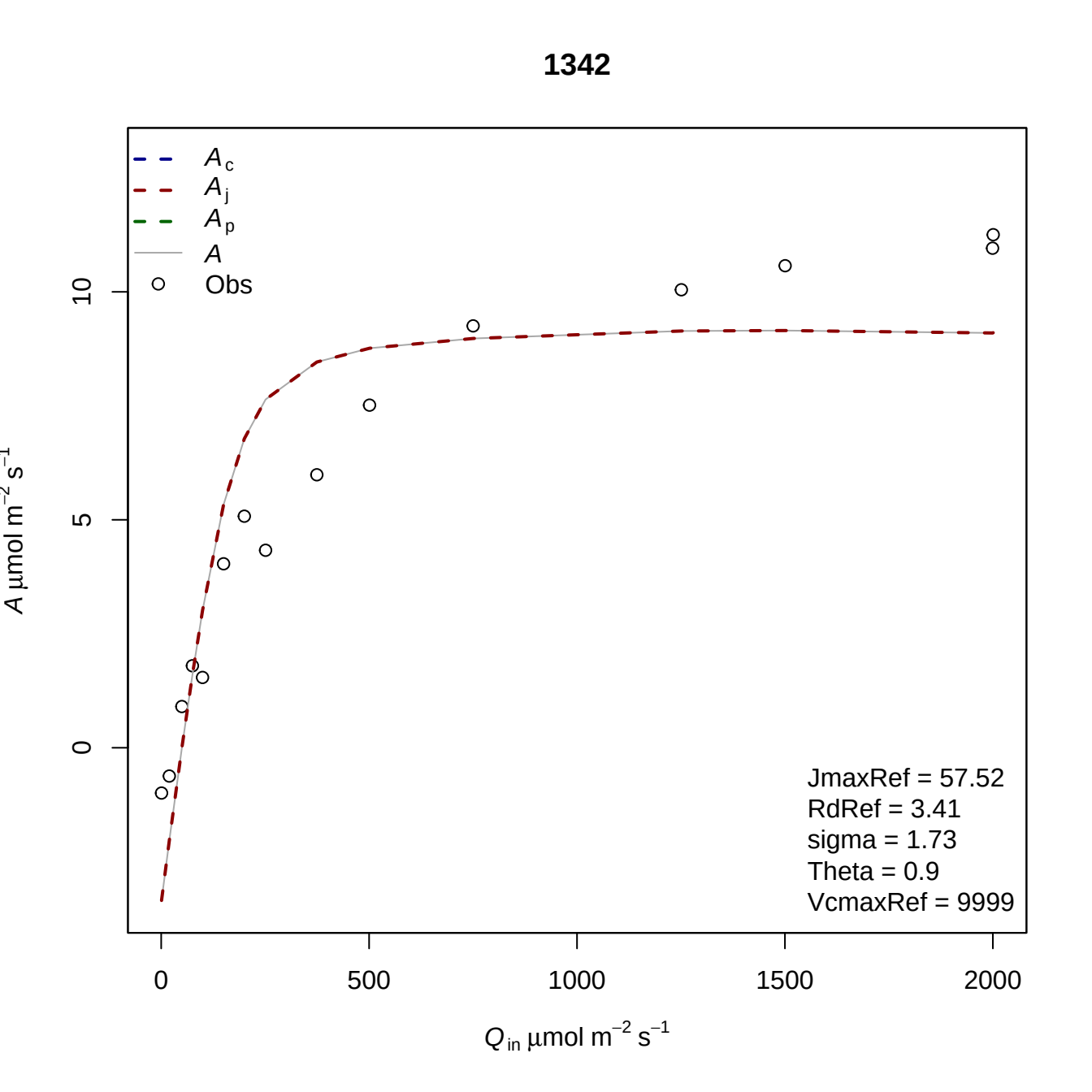

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 52.41
RdRef = 3.61
sigma = 1.55
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

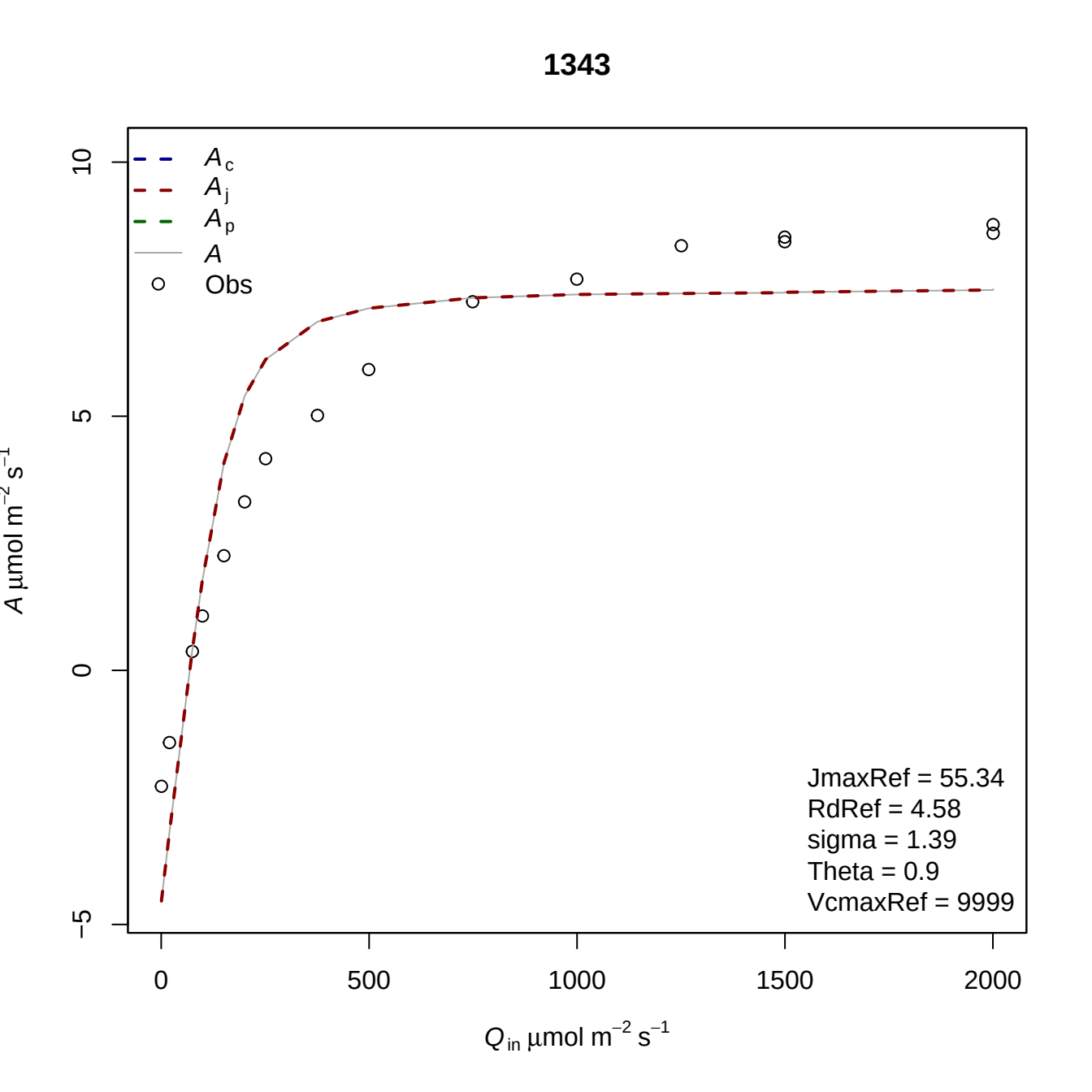

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 60.01
RdRef = 4.37
sigma = 1.88
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

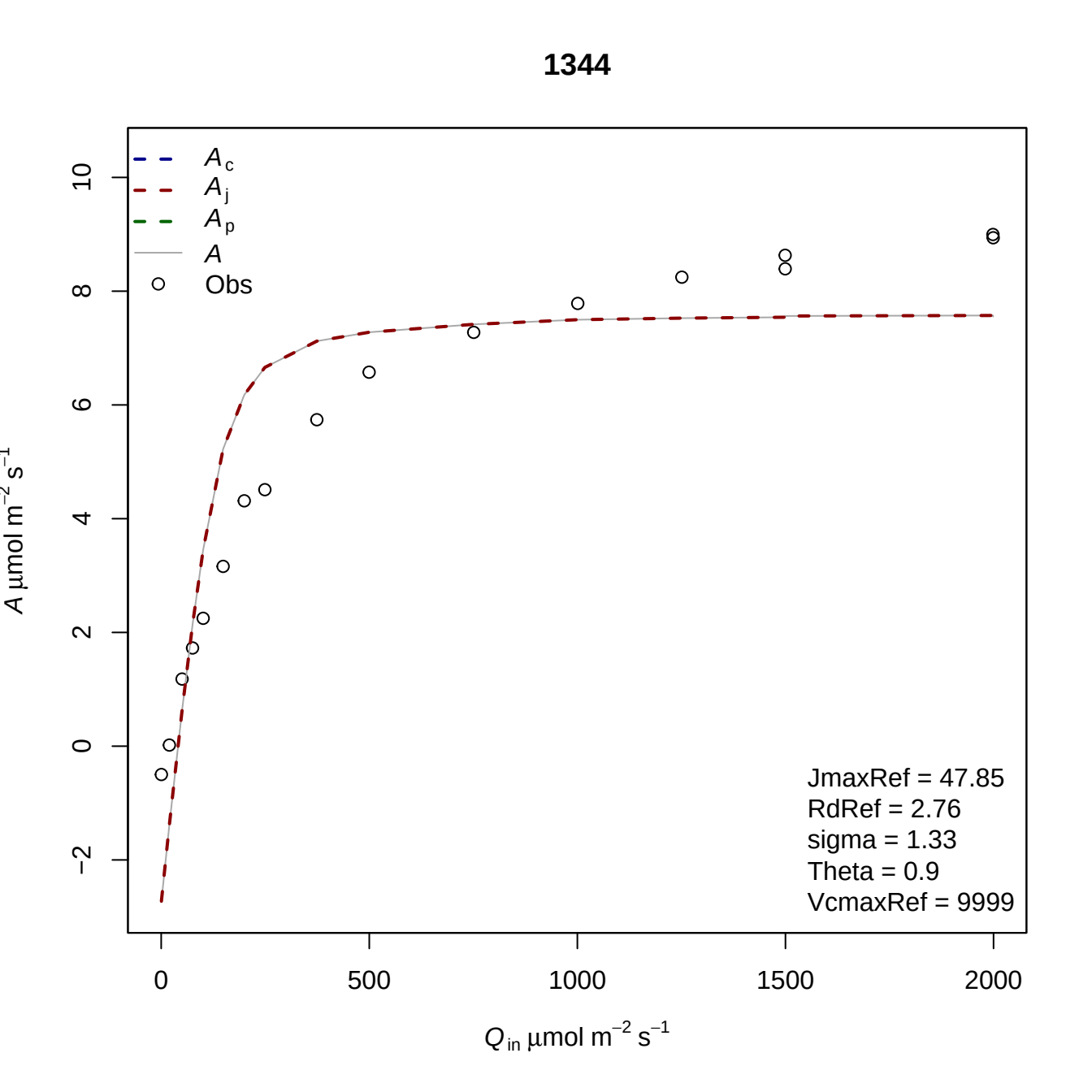

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 60.66
RdRef = 3.36
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

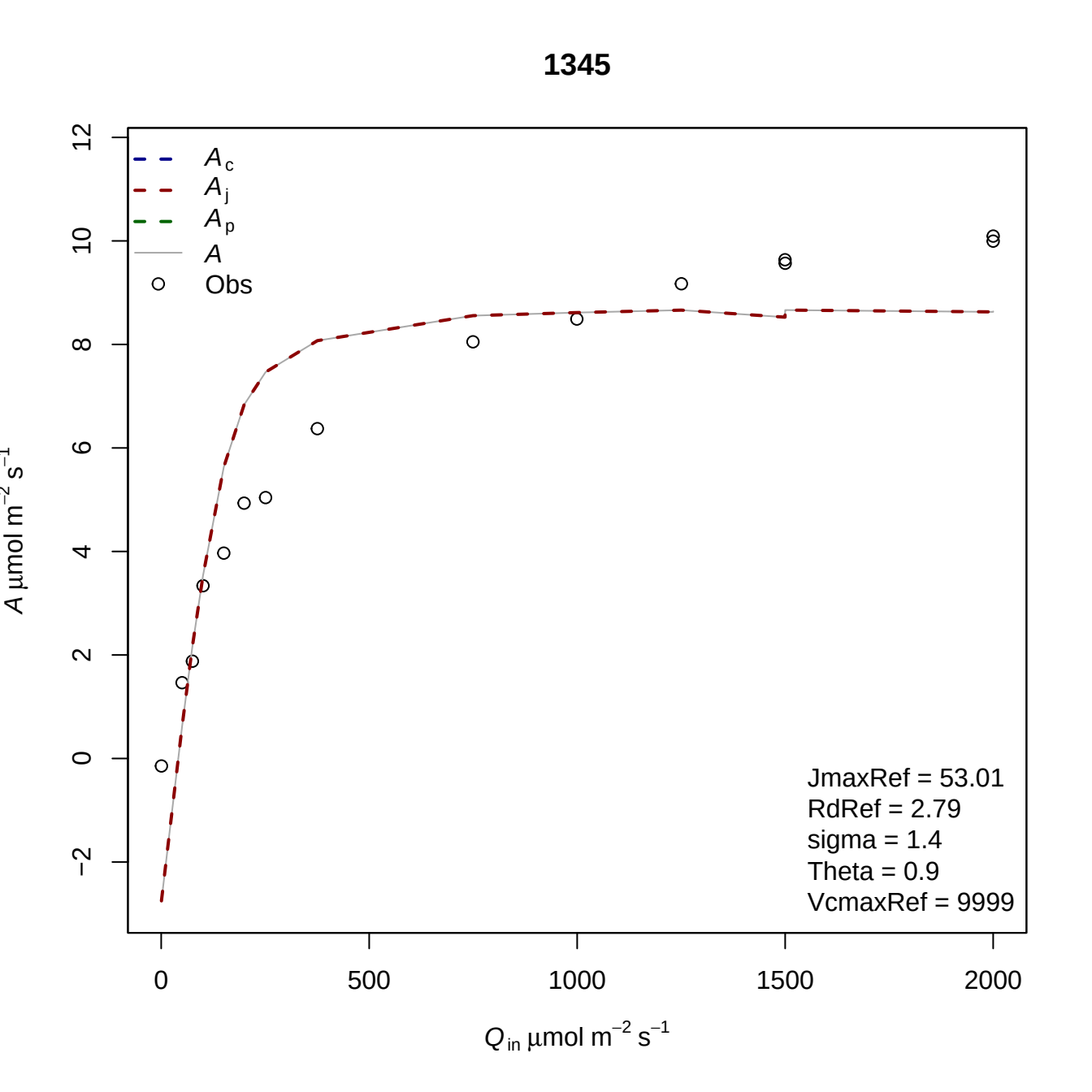

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 63.3
RdRef = 3.36
sigma = 1.71
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

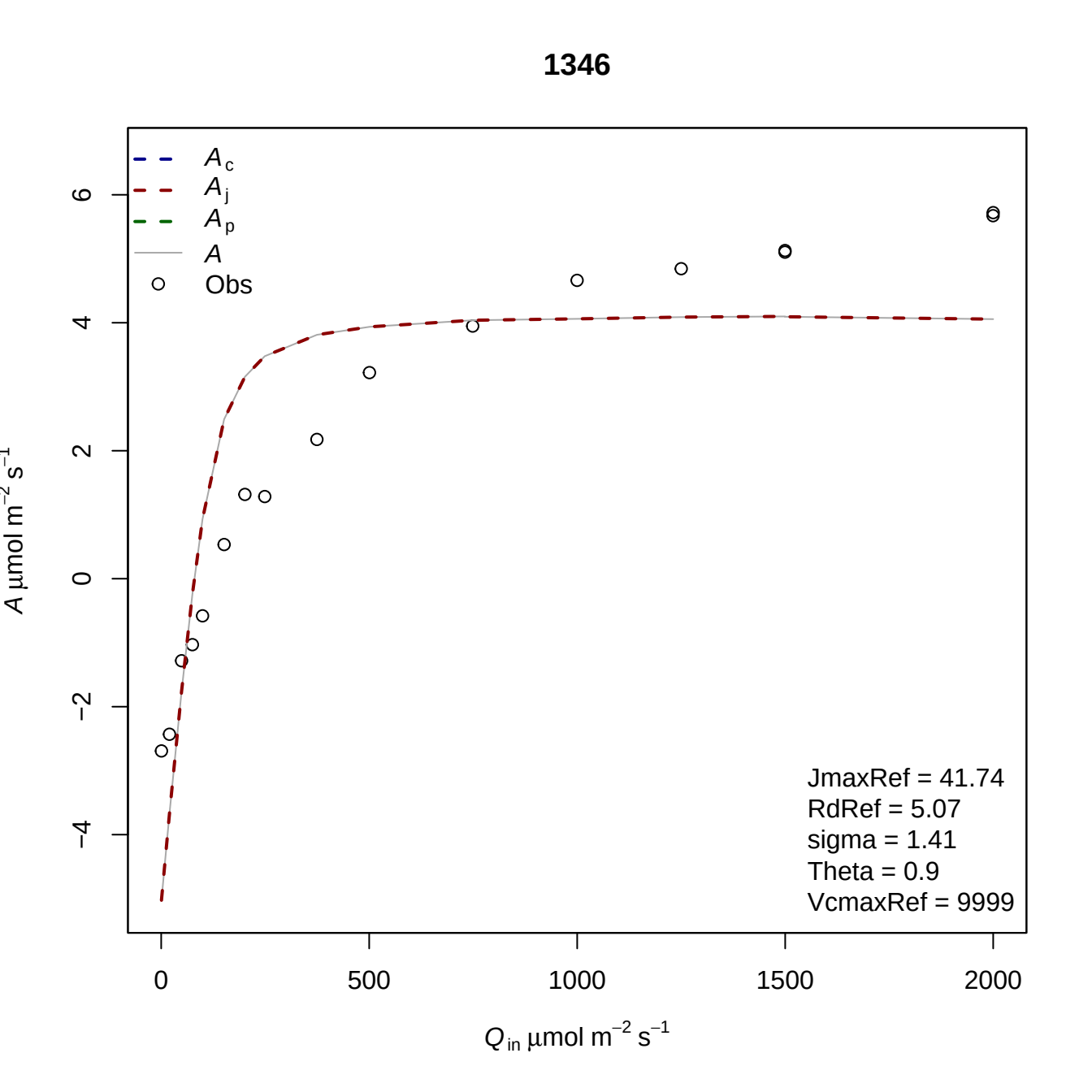

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 69.52
RdRef = 5.04
sigma = 1.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

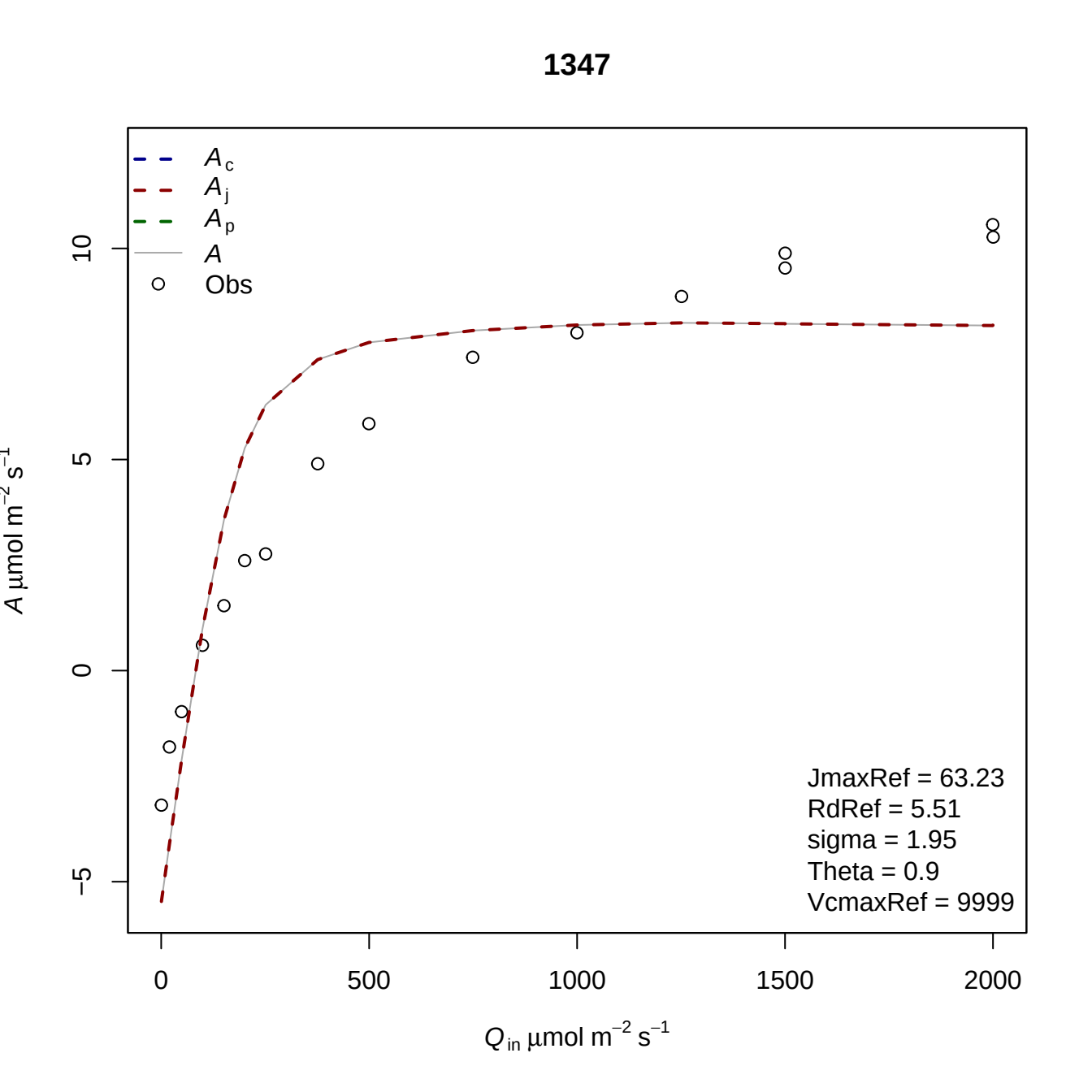

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 57.52
RdRef = 3.41
sigma = 1.73
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

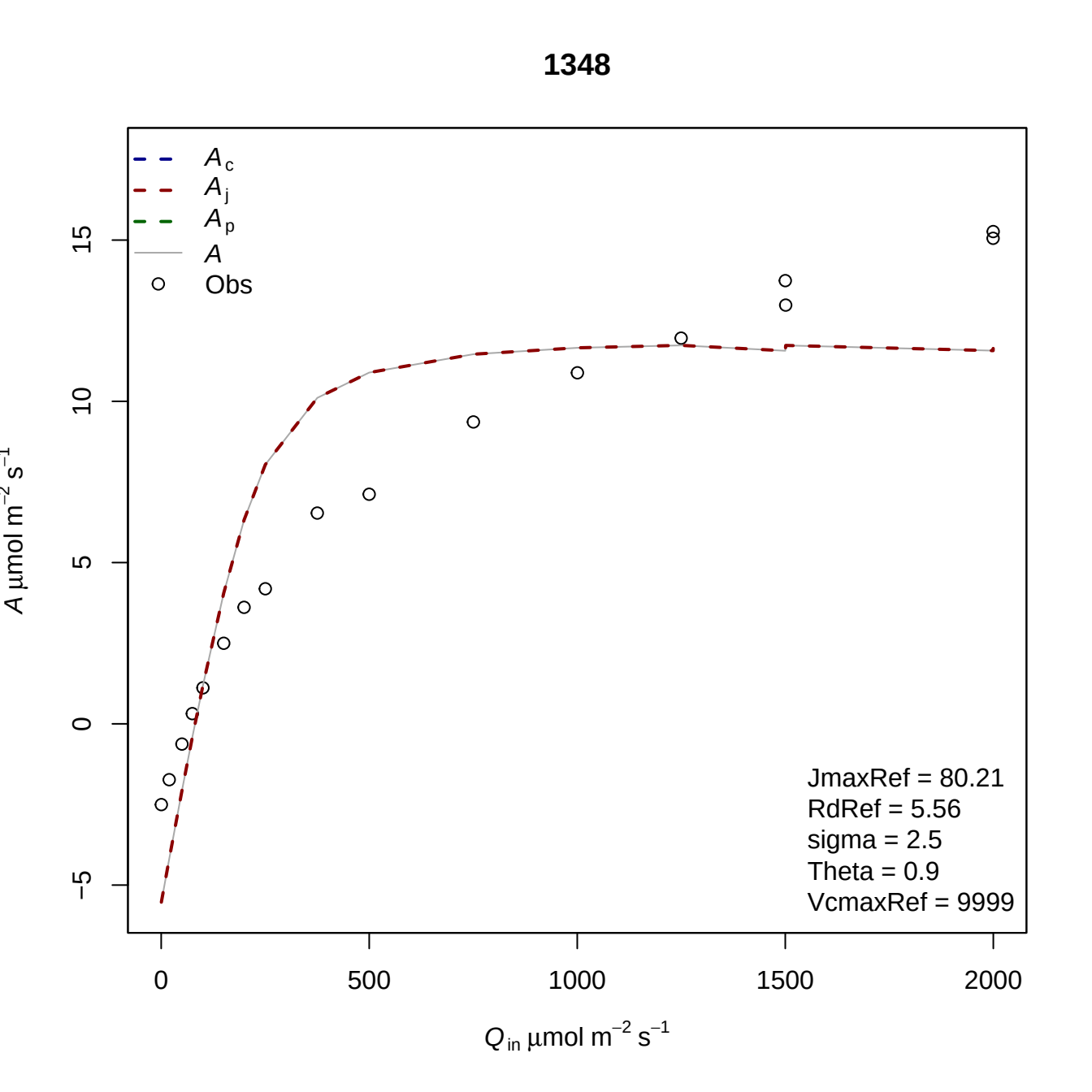

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 55.34
RdRef = 4.58
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

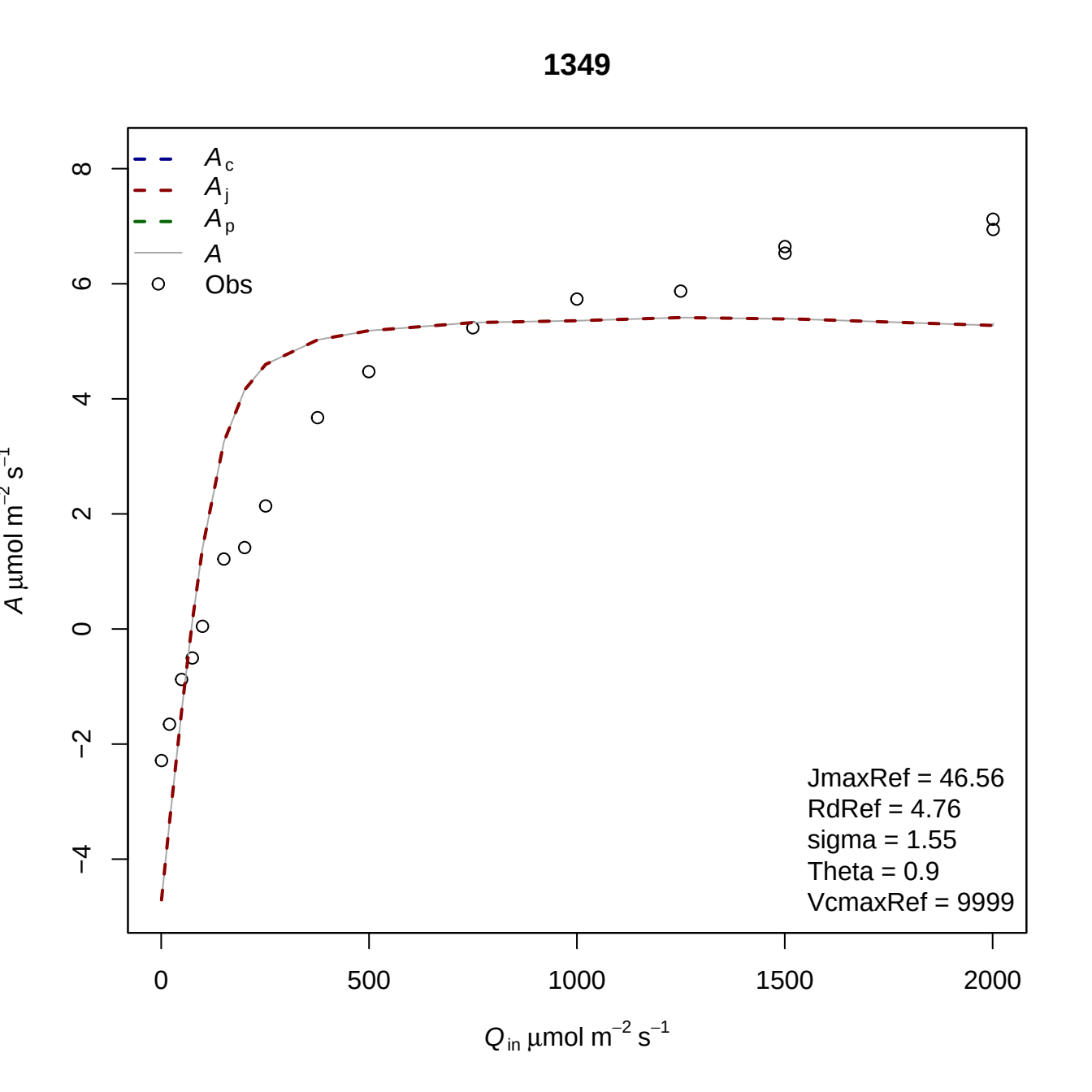

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 47.85
RdRef = 2.76
sigma = 1.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

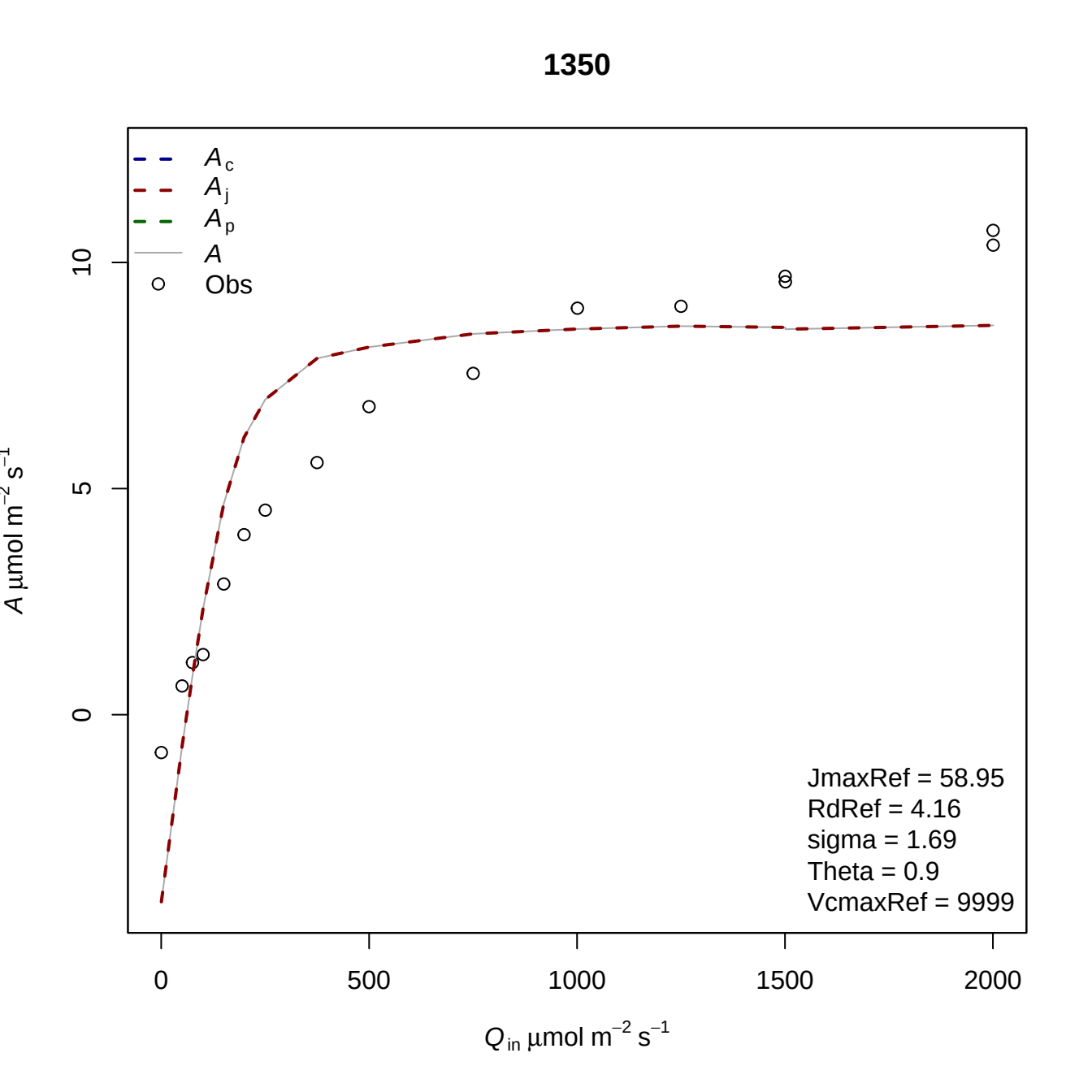

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 53.01
RdRef = 2.79
sigma = 1.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

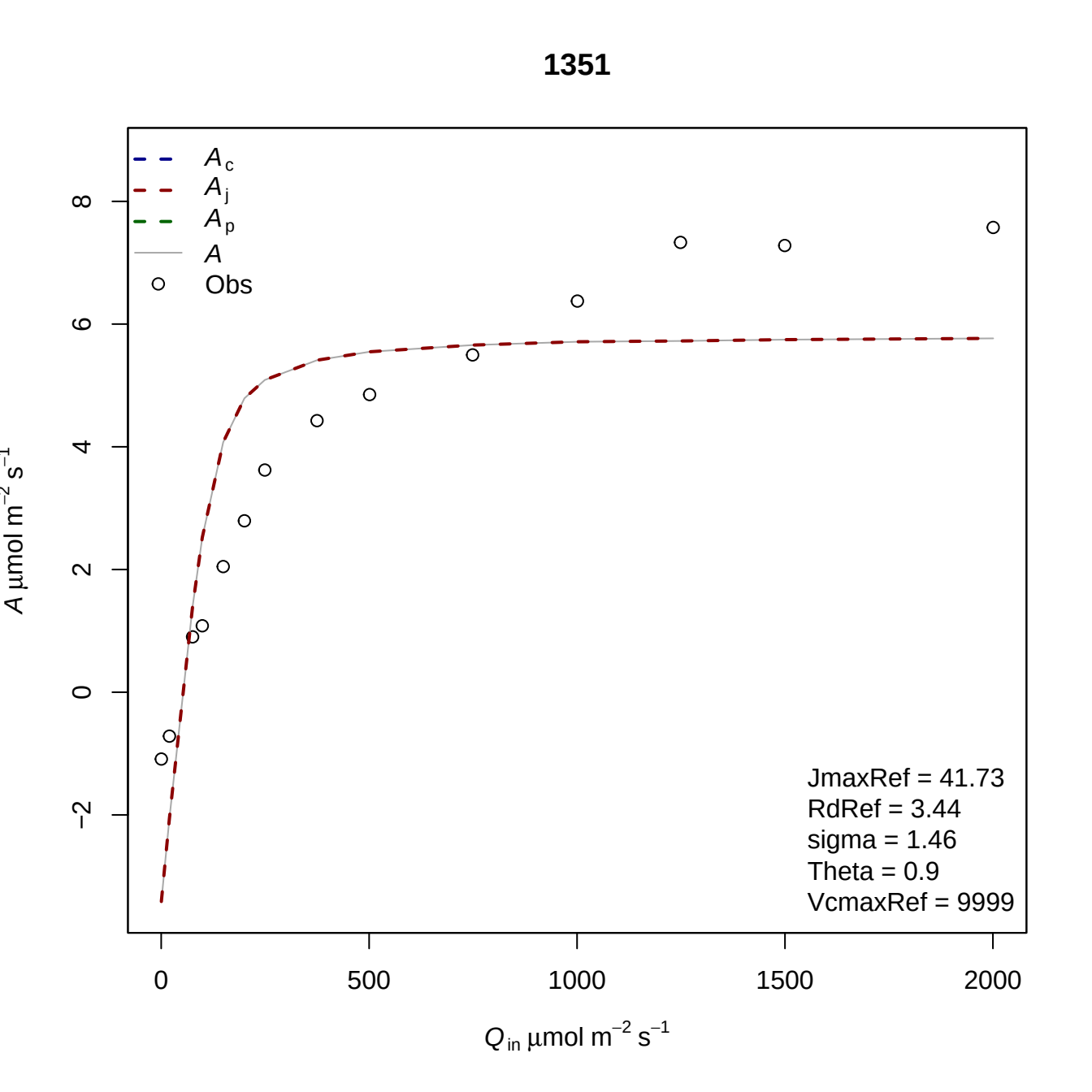

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 41.74
RdRef = 5.07
sigma = 1.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

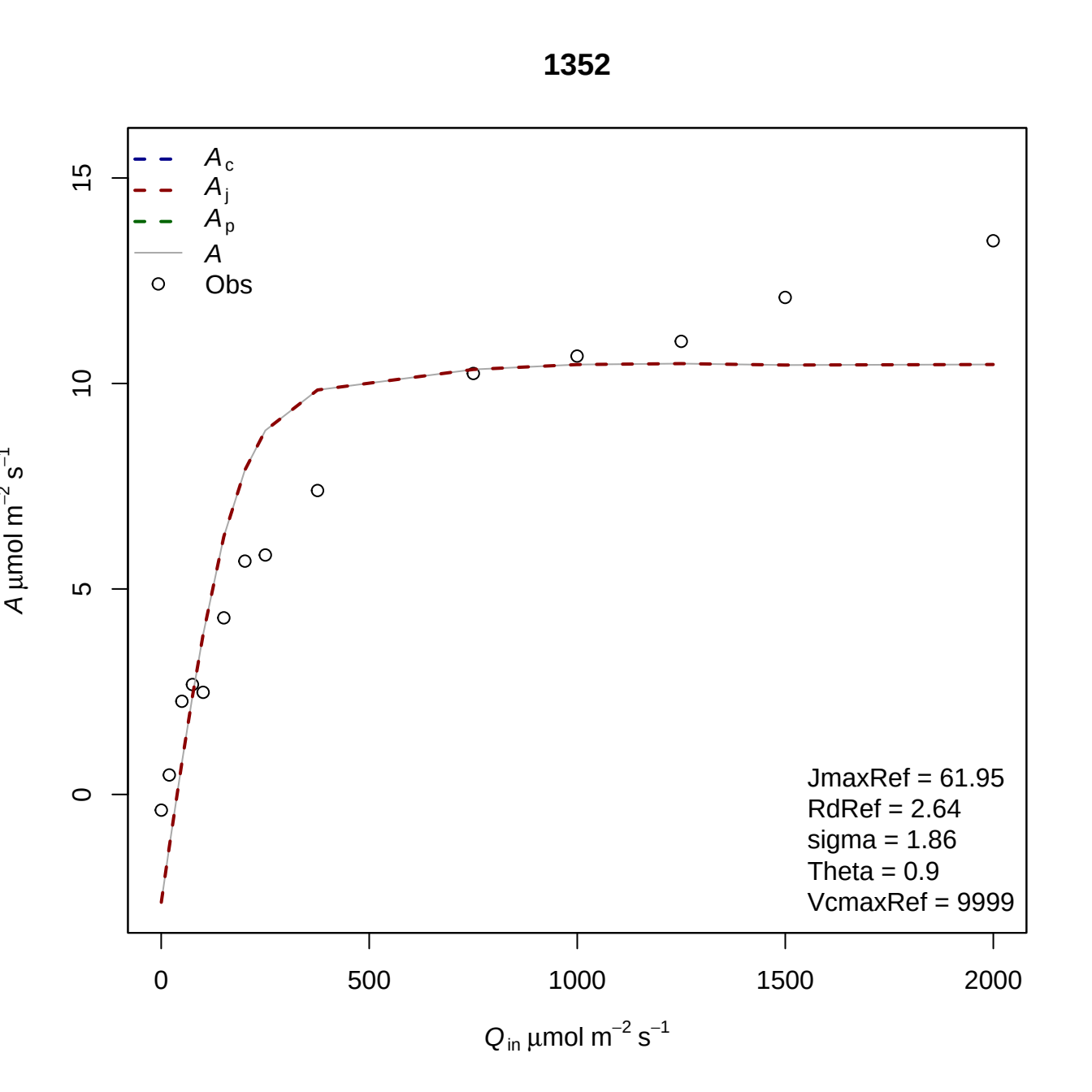

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 63.23
RdRef = 5.51
sigma = 1.95
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

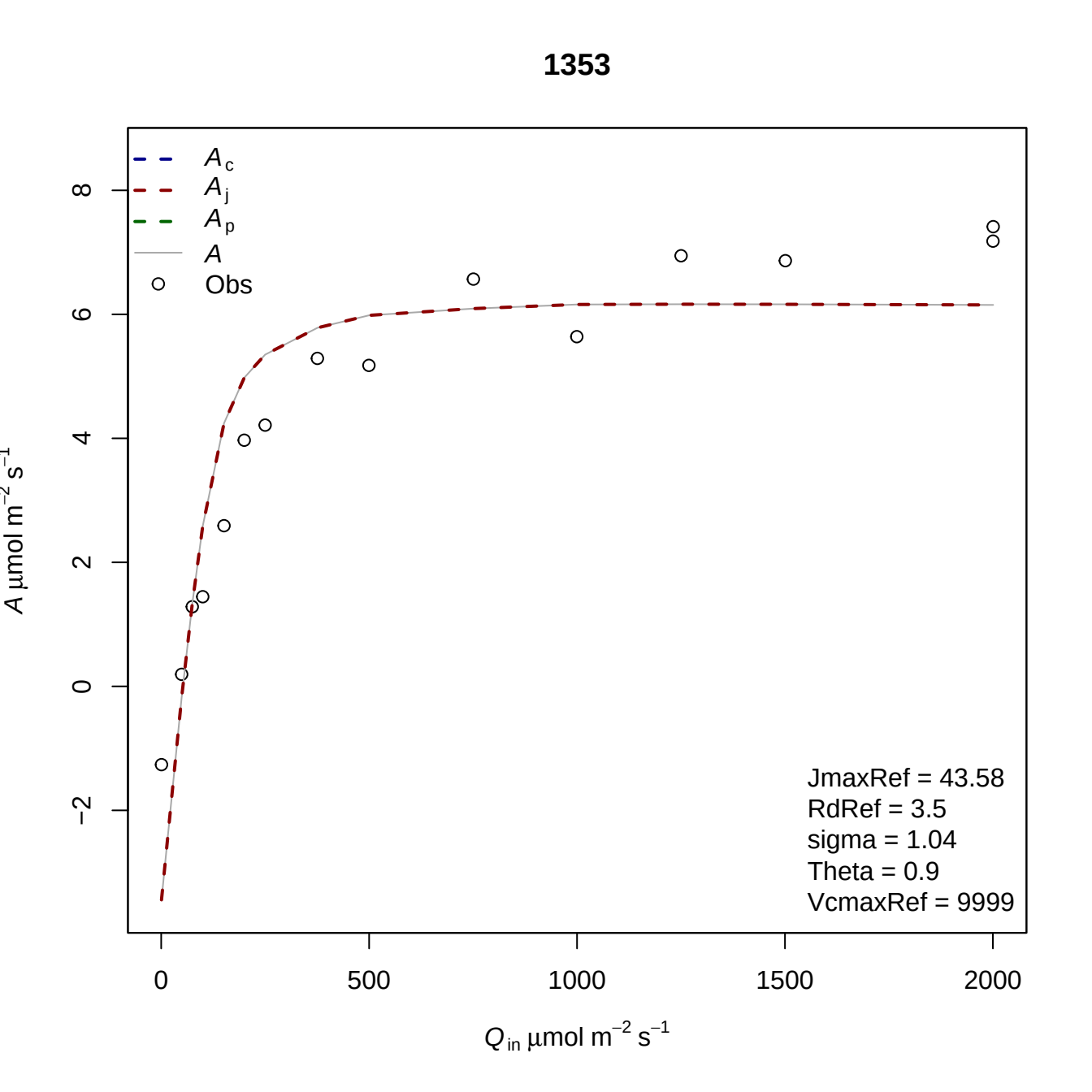

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 80.21
RdRef = 5.56
sigma = 2.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

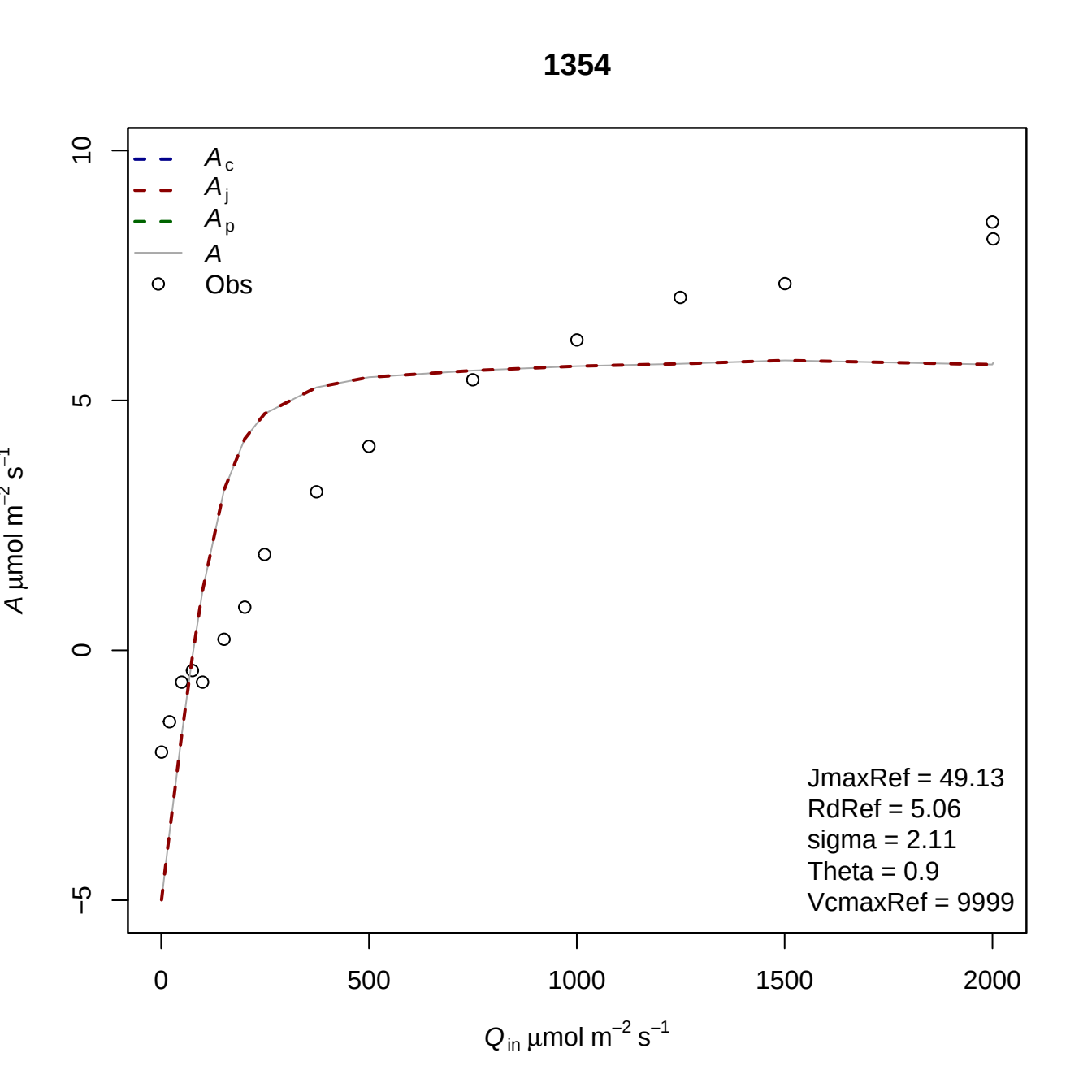

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 46.56
RdRef = 4.76
sigma = 1.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

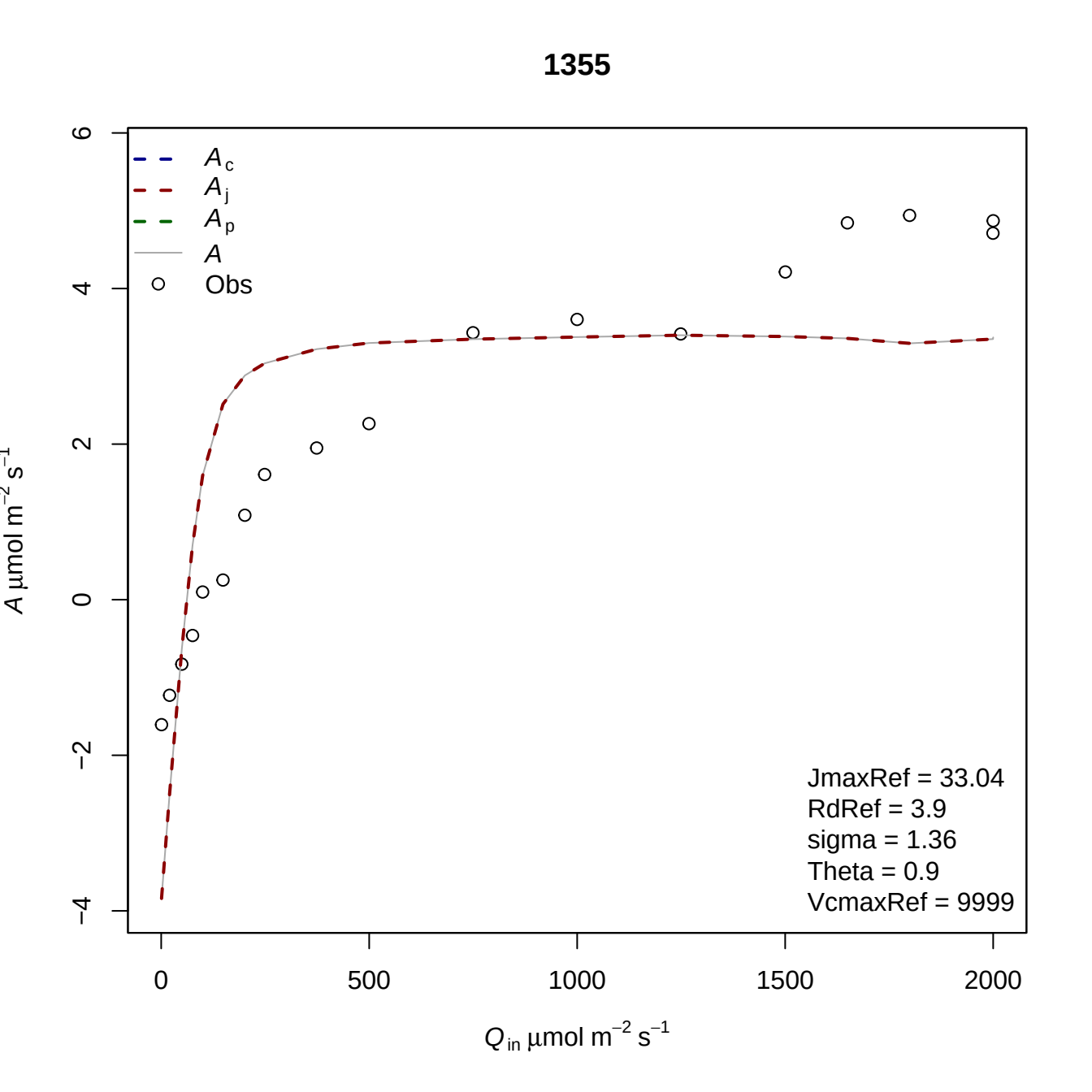

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 58.95
RdRef = 4.16
sigma = 1.69
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

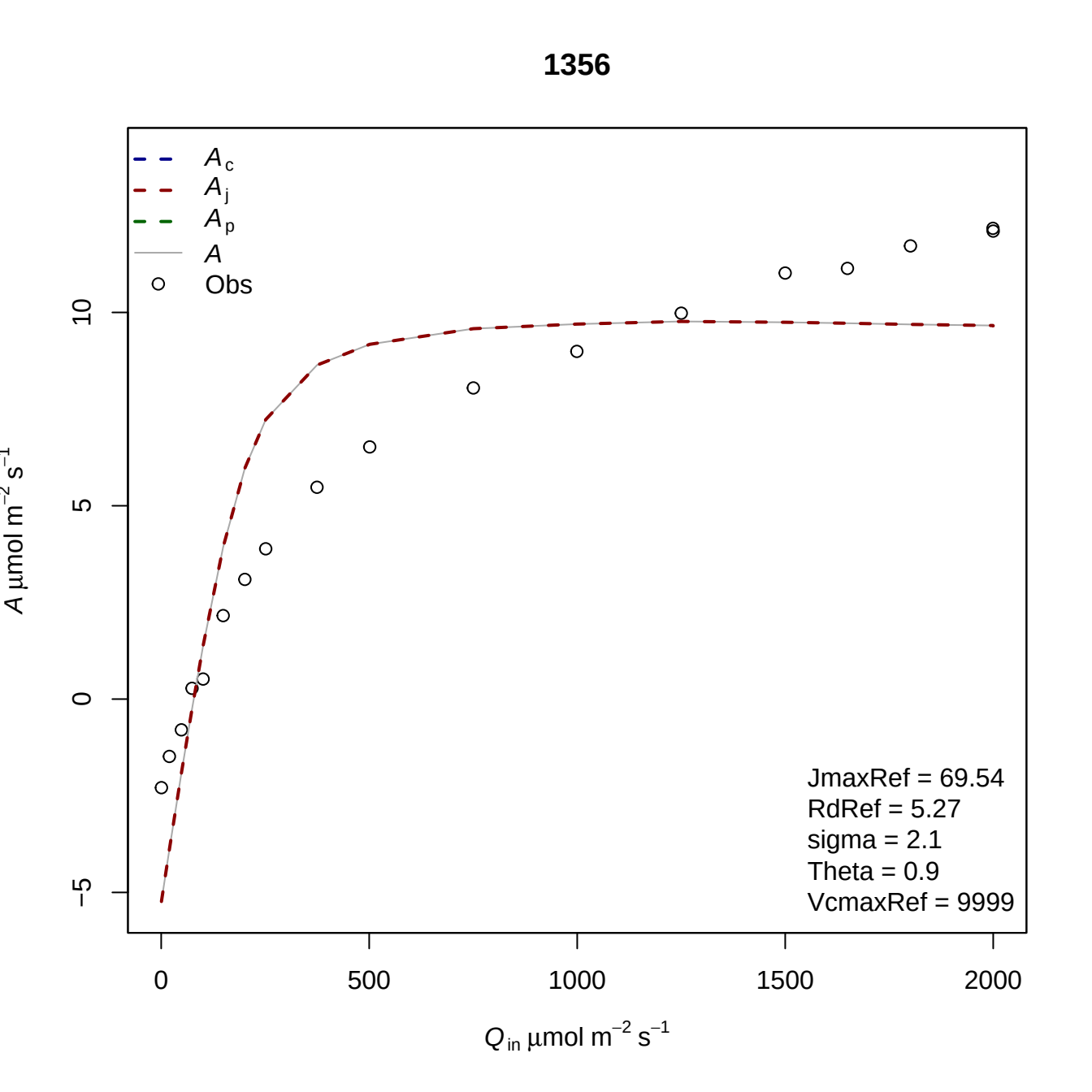

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 41.73
RdRef = 3.44
sigma = 1.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

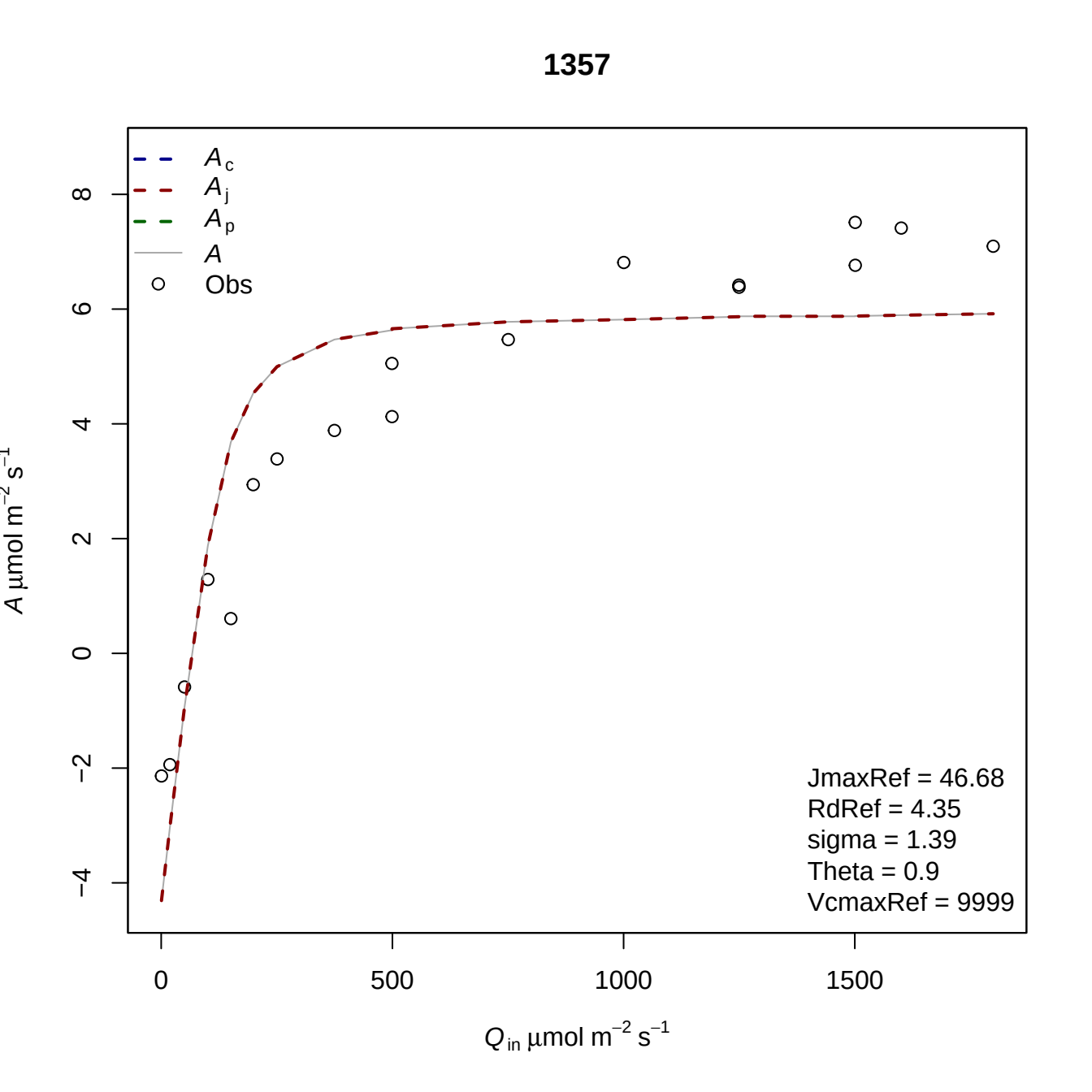

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 61.95
RdRef = 2.64
sigma = 1.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

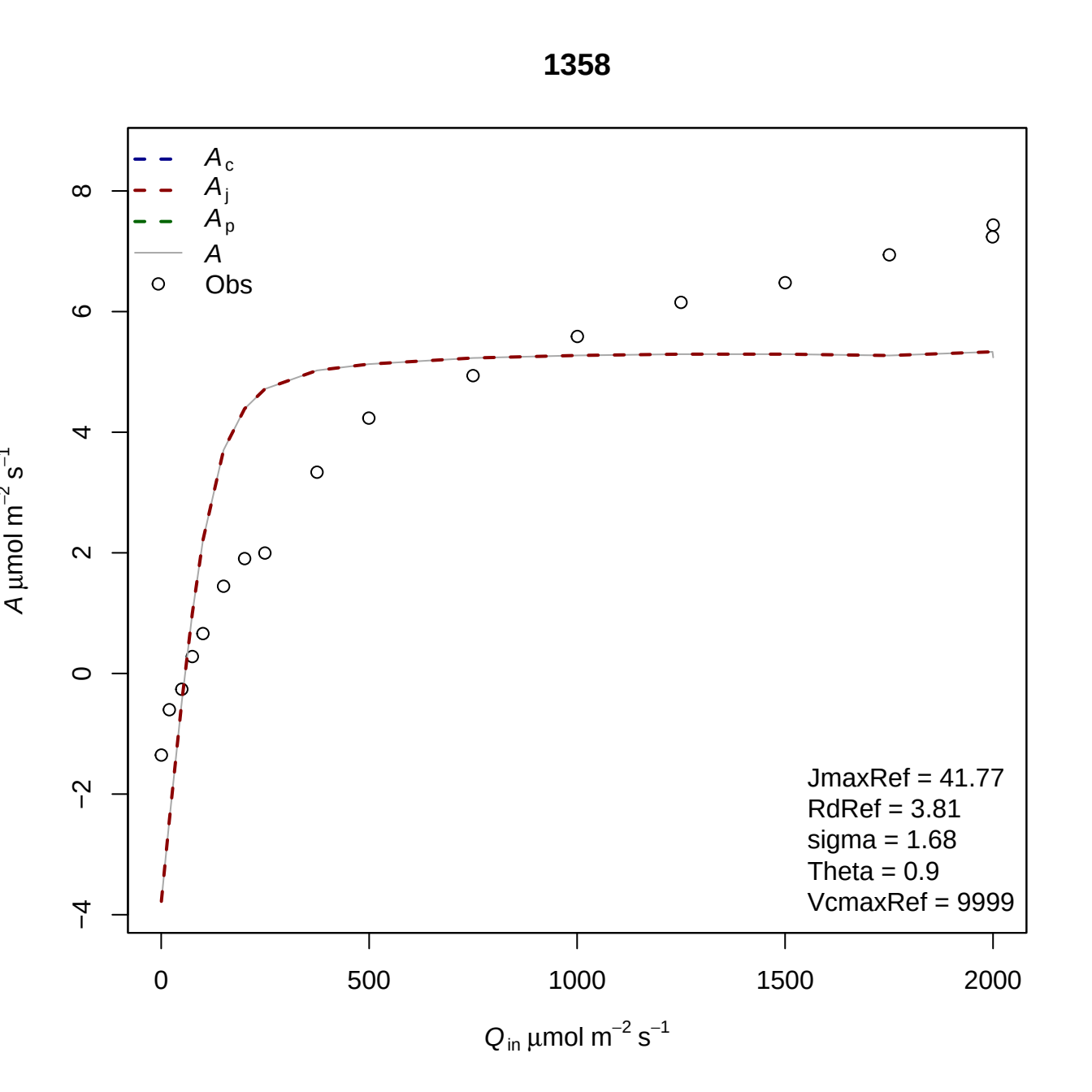

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 43.58
RdRef = 3.5
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

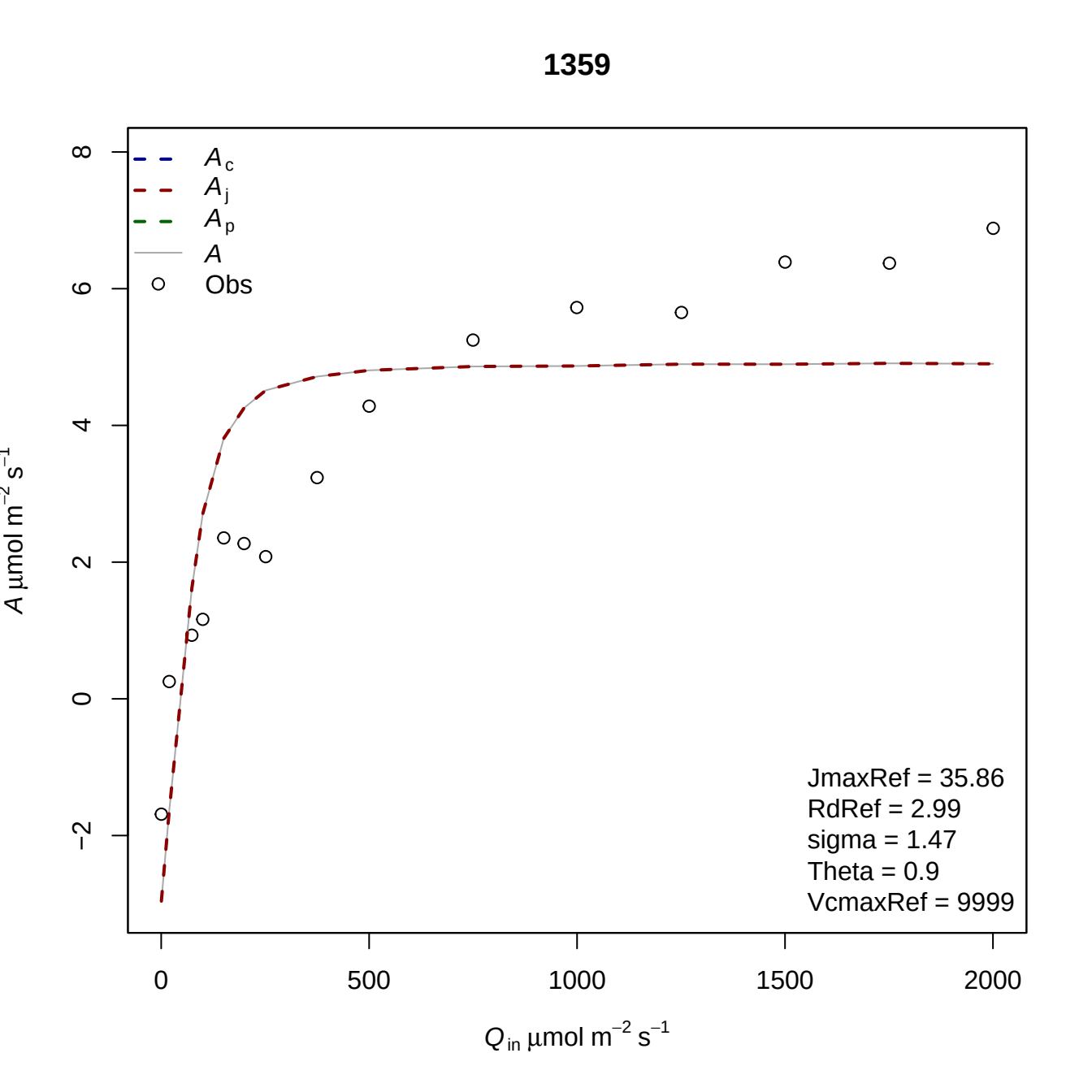

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 49.13
RdRef = 5.06
sigma = 2.11
Theta = 0.9
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

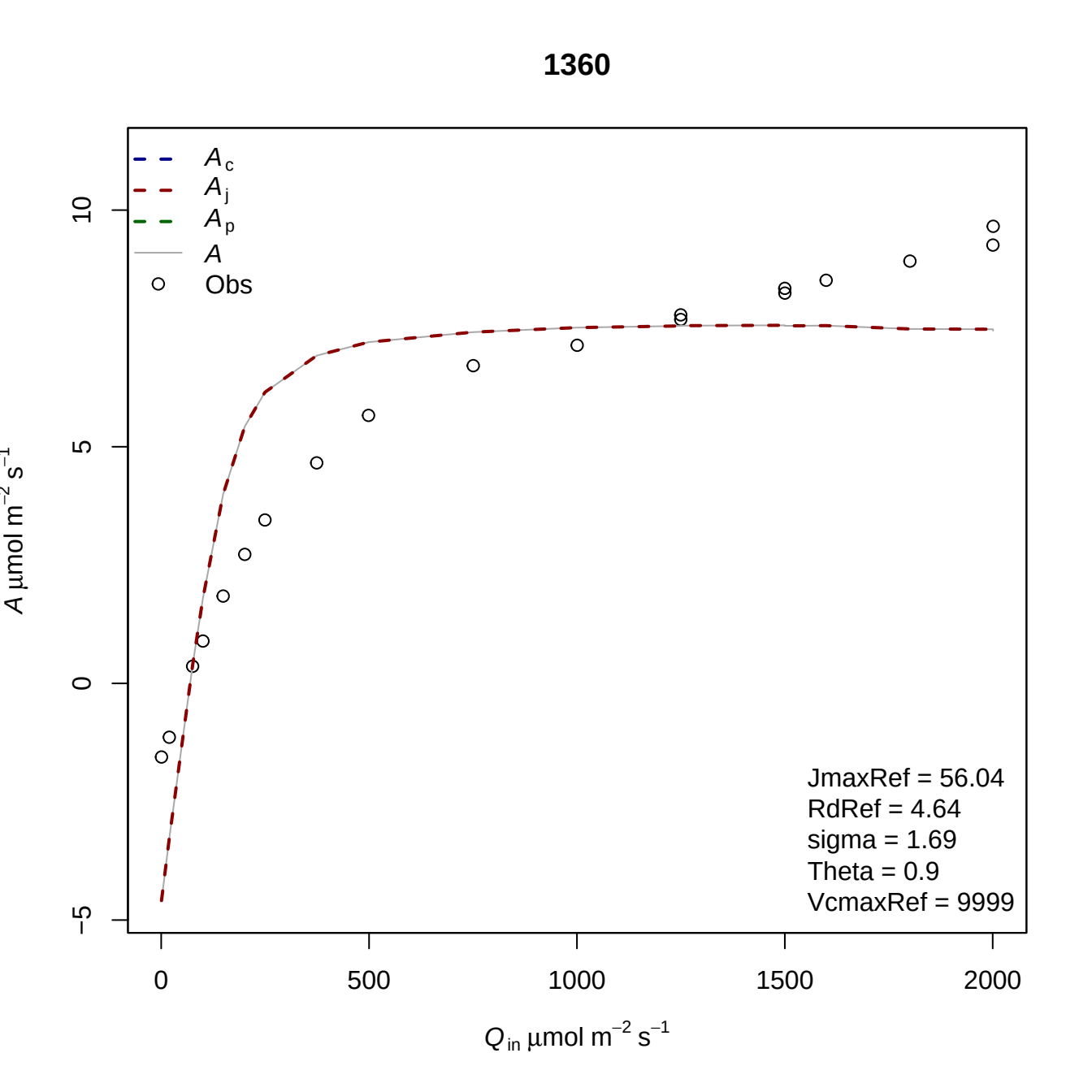

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 33.04
RdRef = 3.9
sigma = 1.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

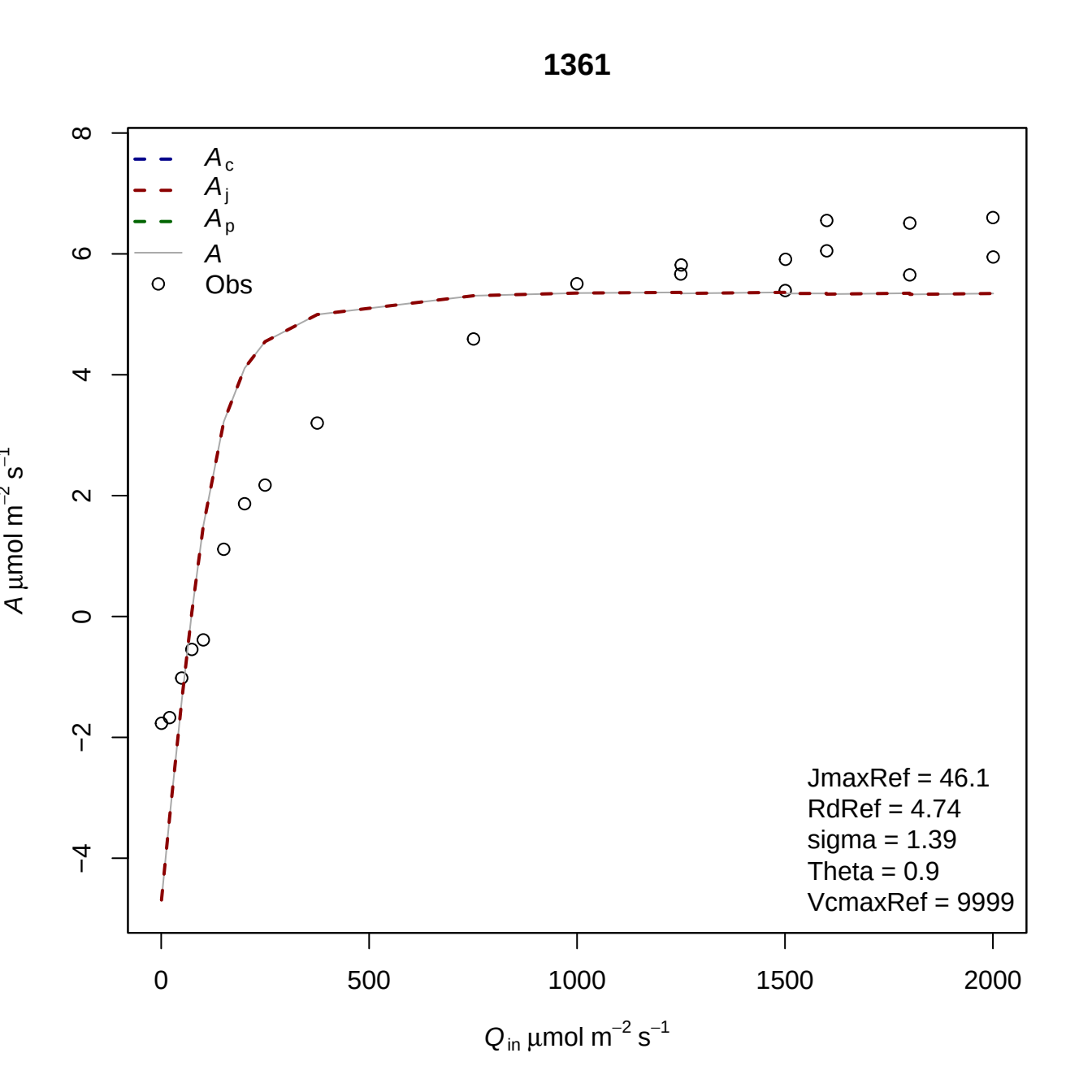

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 69.54
RdRef = 5.27
sigma = 2.1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

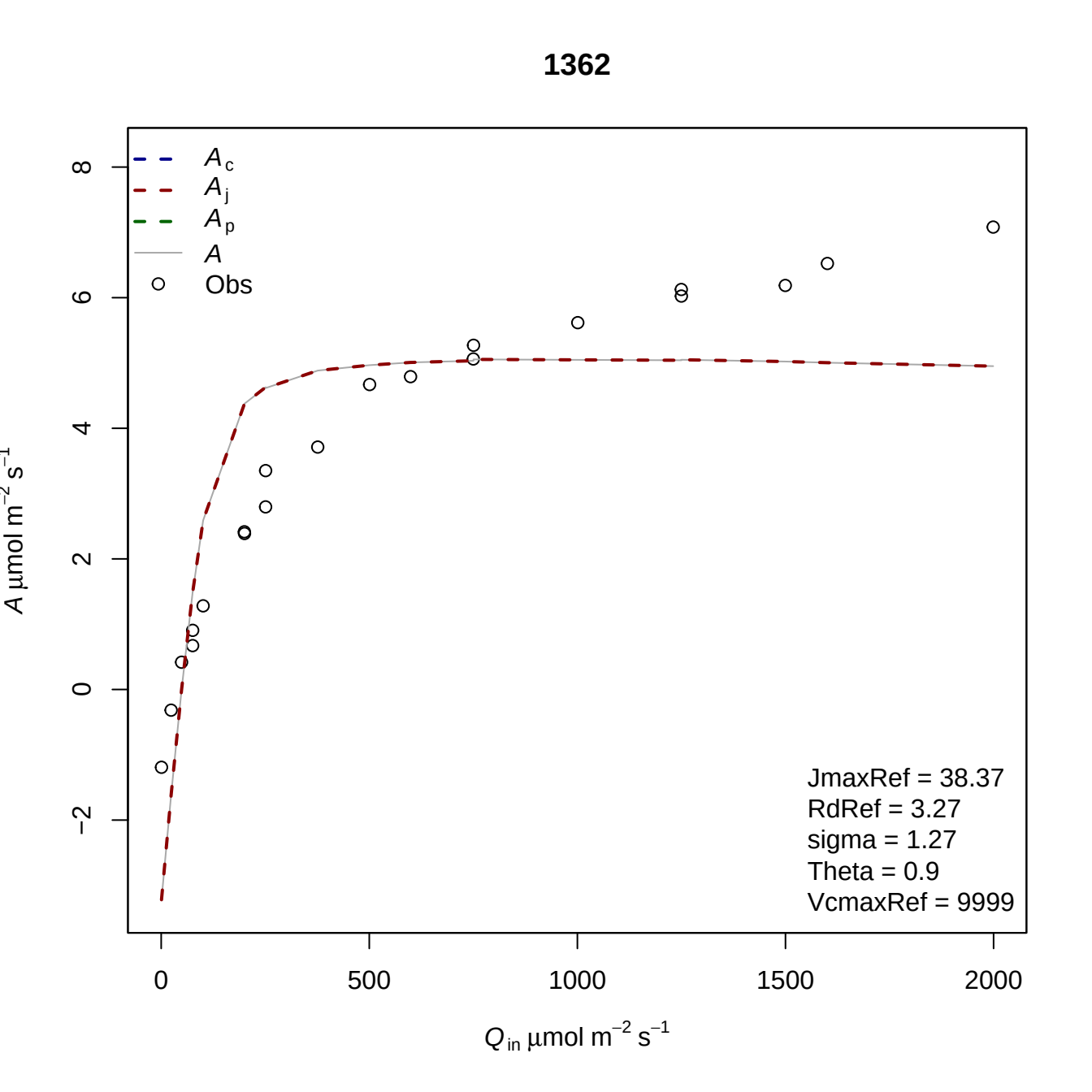

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 46.68
RdRef = 4.35
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

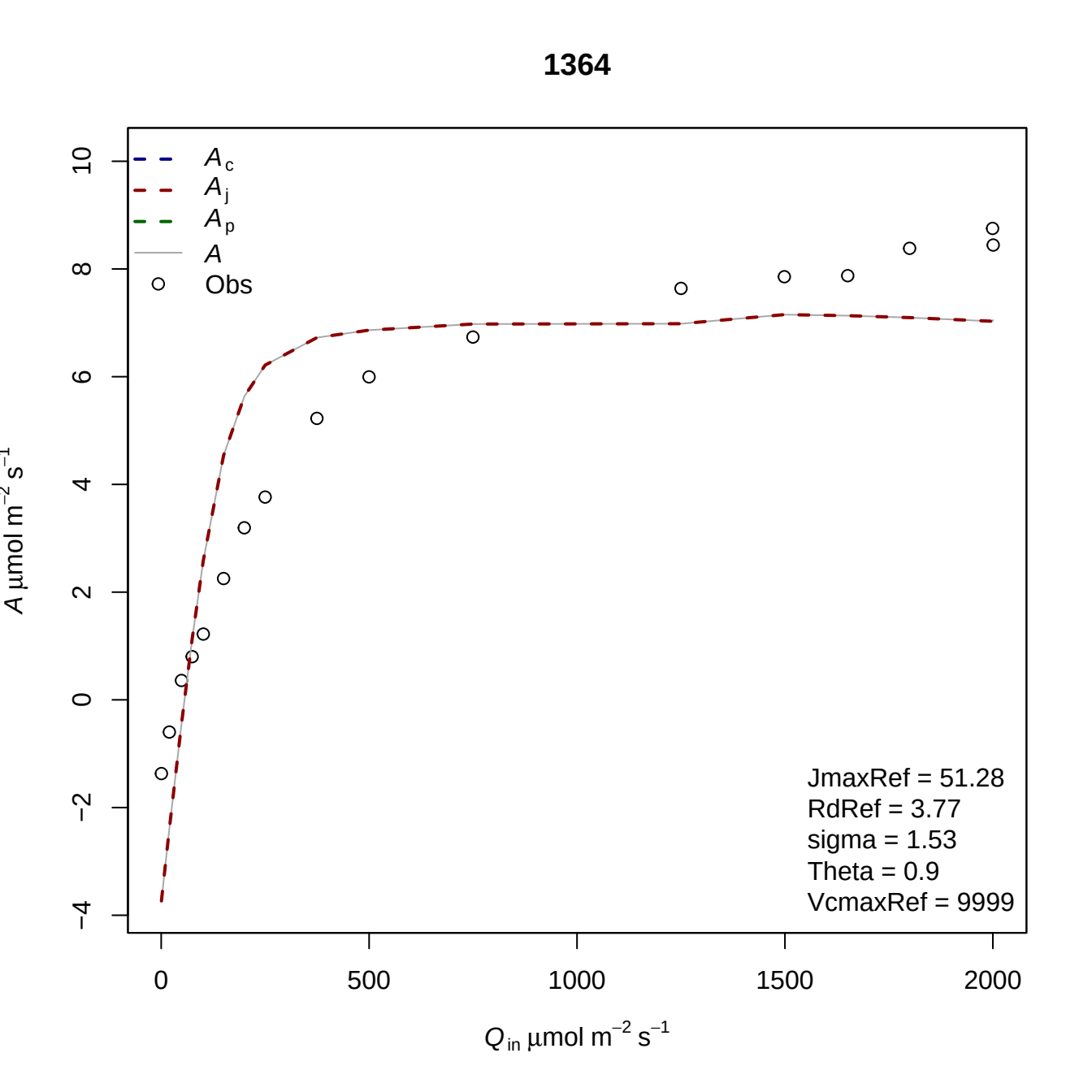

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 41.77
RdRef = 3.81
sigma = 1.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

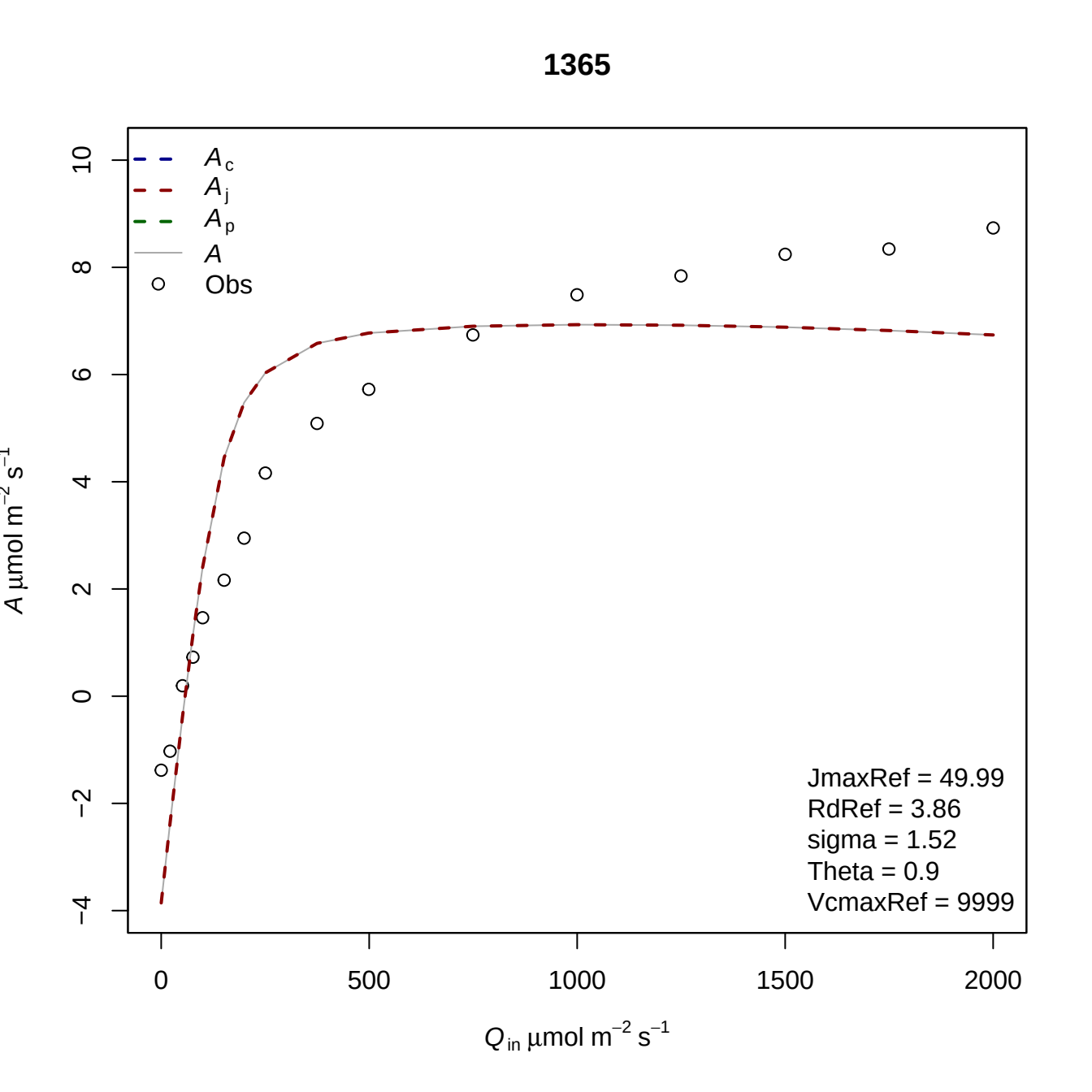

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 35.86
RdRef = 2.99
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

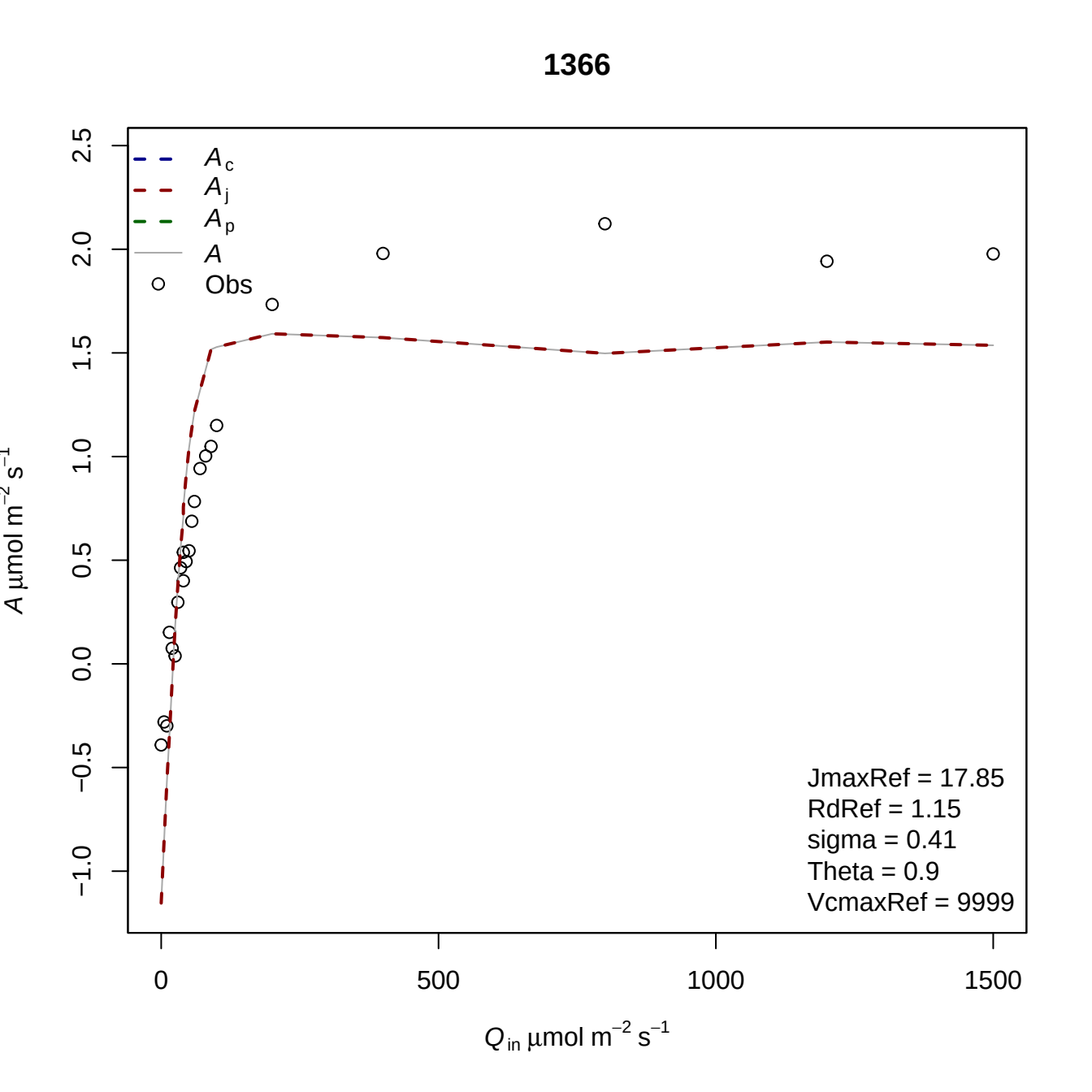

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 56.04
RdRef = 4.64
sigma = 1.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

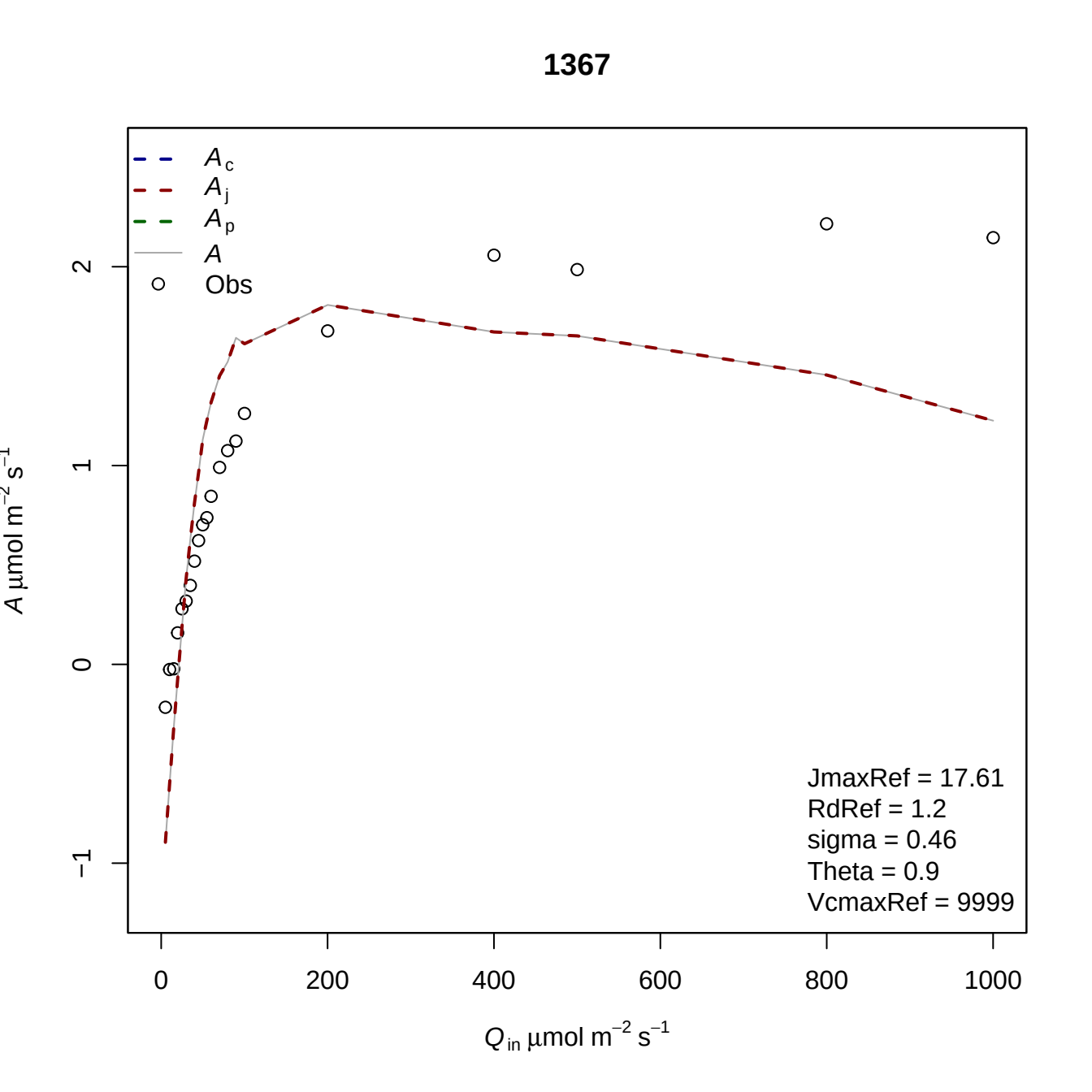

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 46.1
RdRef = 4.74
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

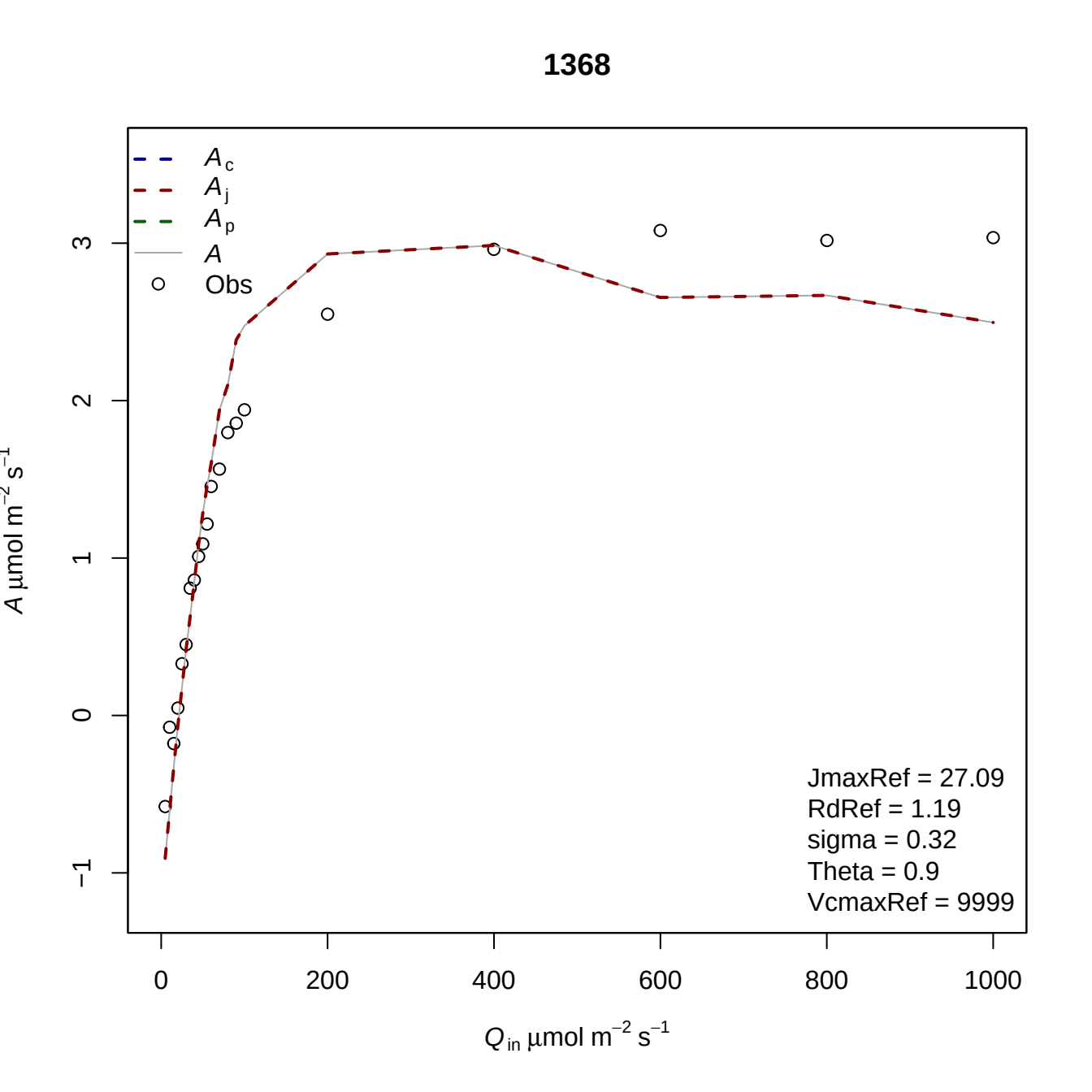

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 38.37
RdRef = 3.27
sigma = 1.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

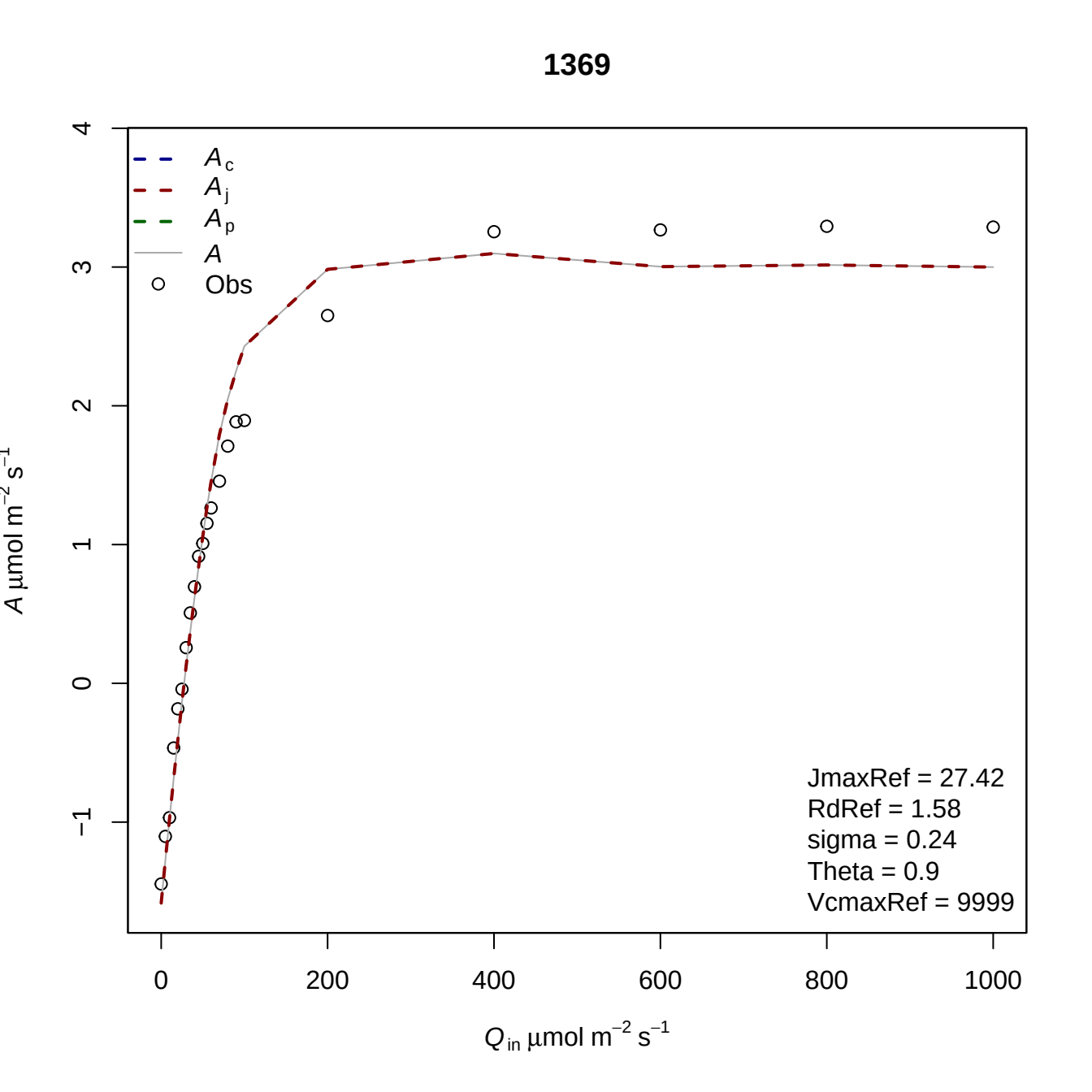

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 51.28
RdRef = 3.77
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

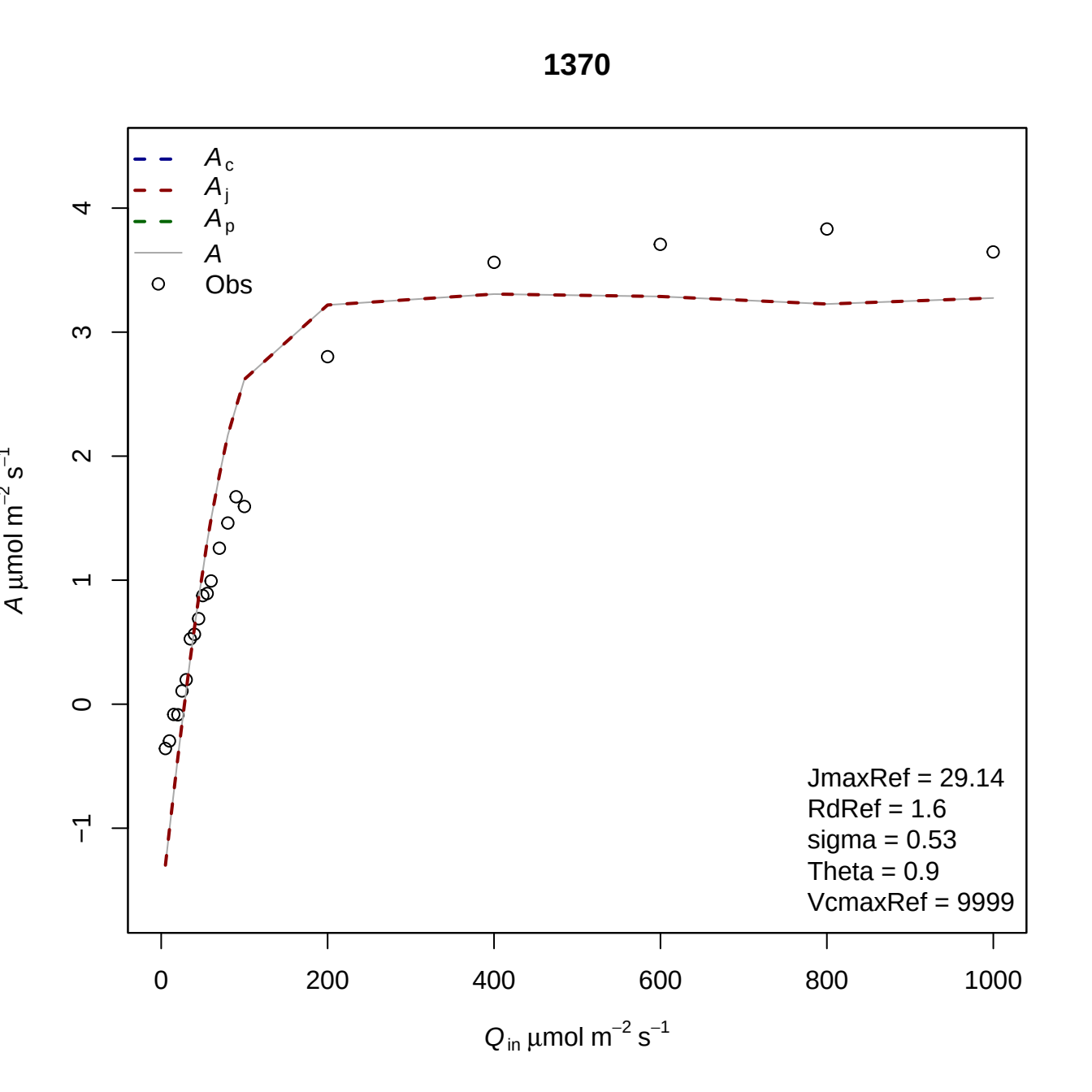

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 49.99
RdRef = 3.86
sigma = 1.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

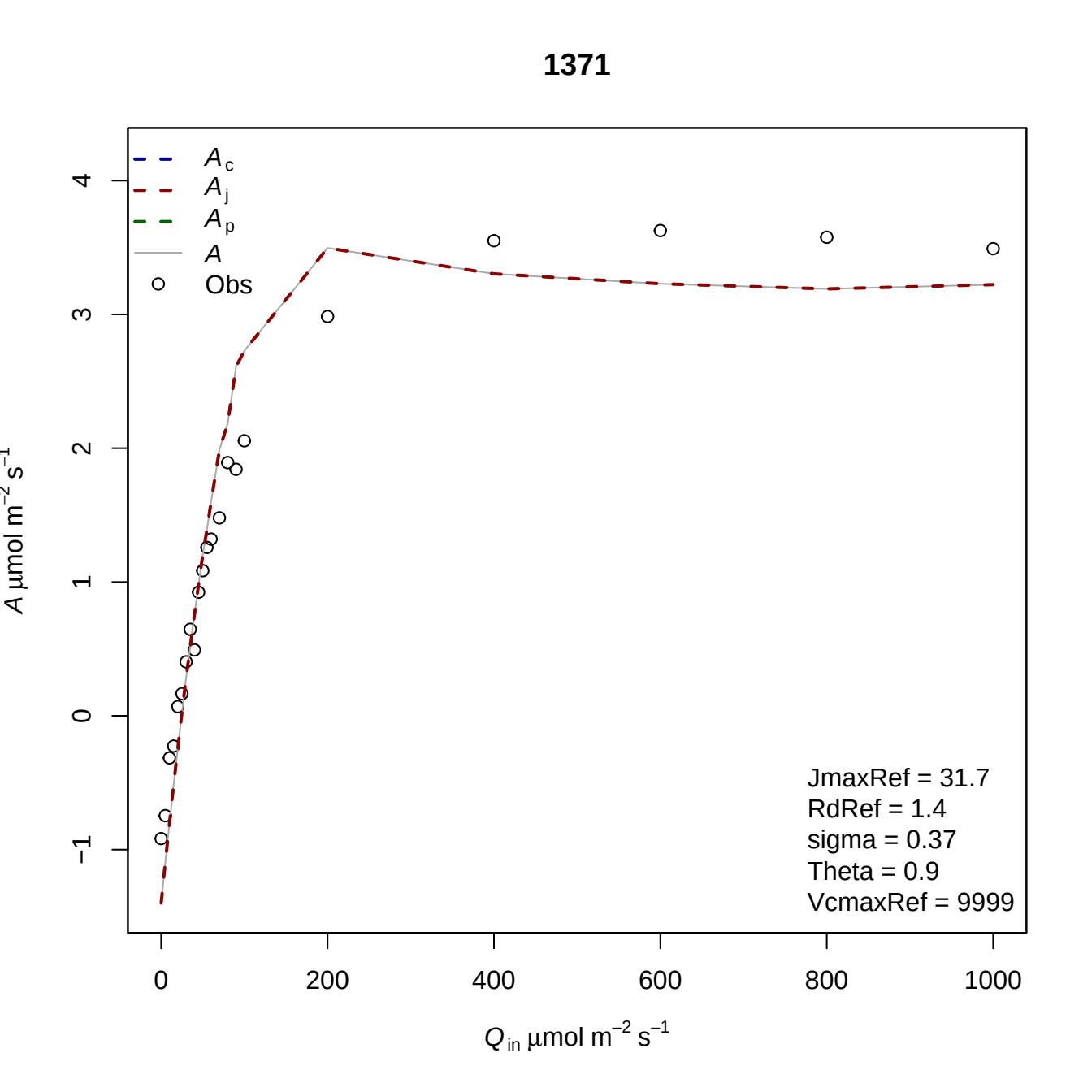

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 17.85
RdRef = 1.15
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

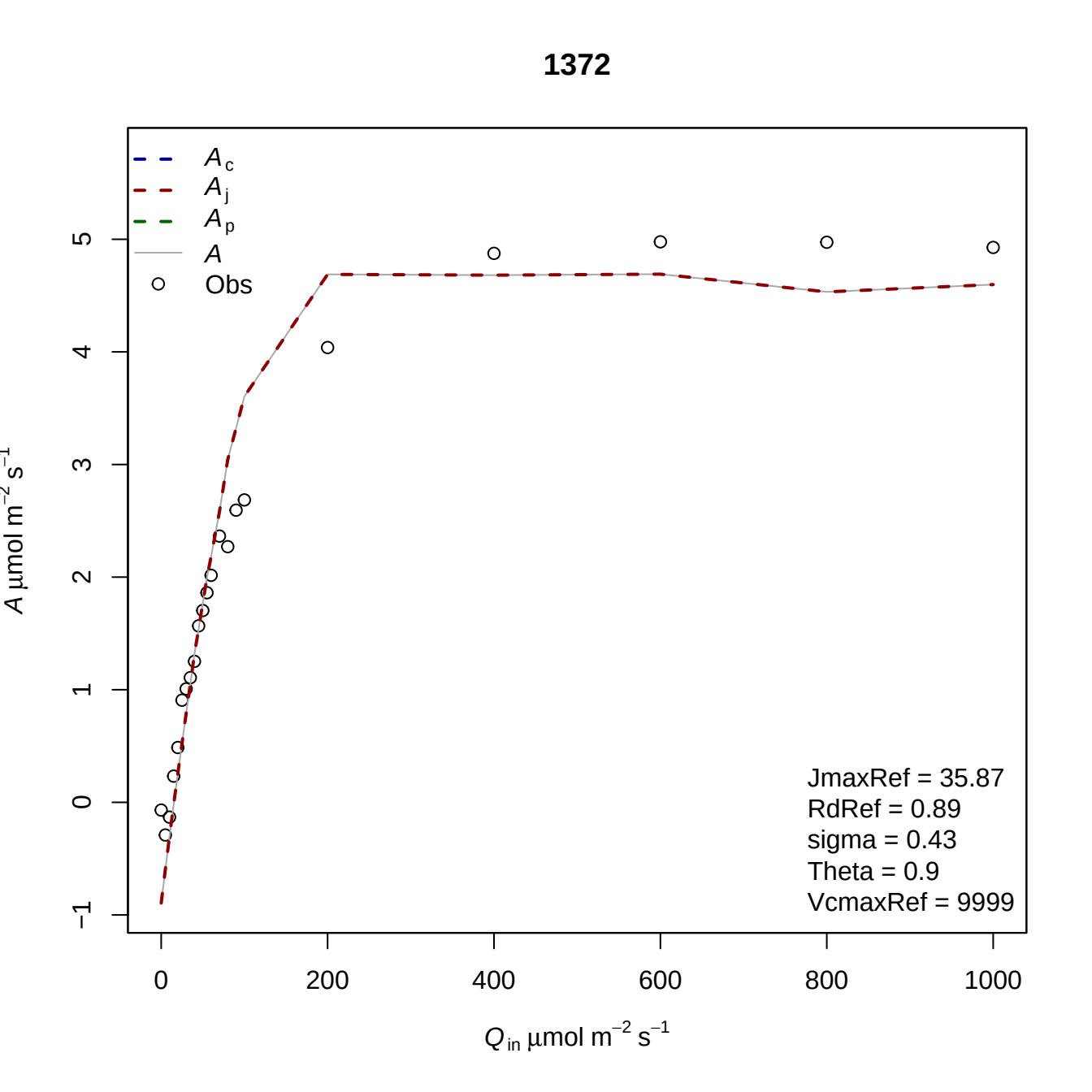

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 17.61
RdRef = 1.2
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000

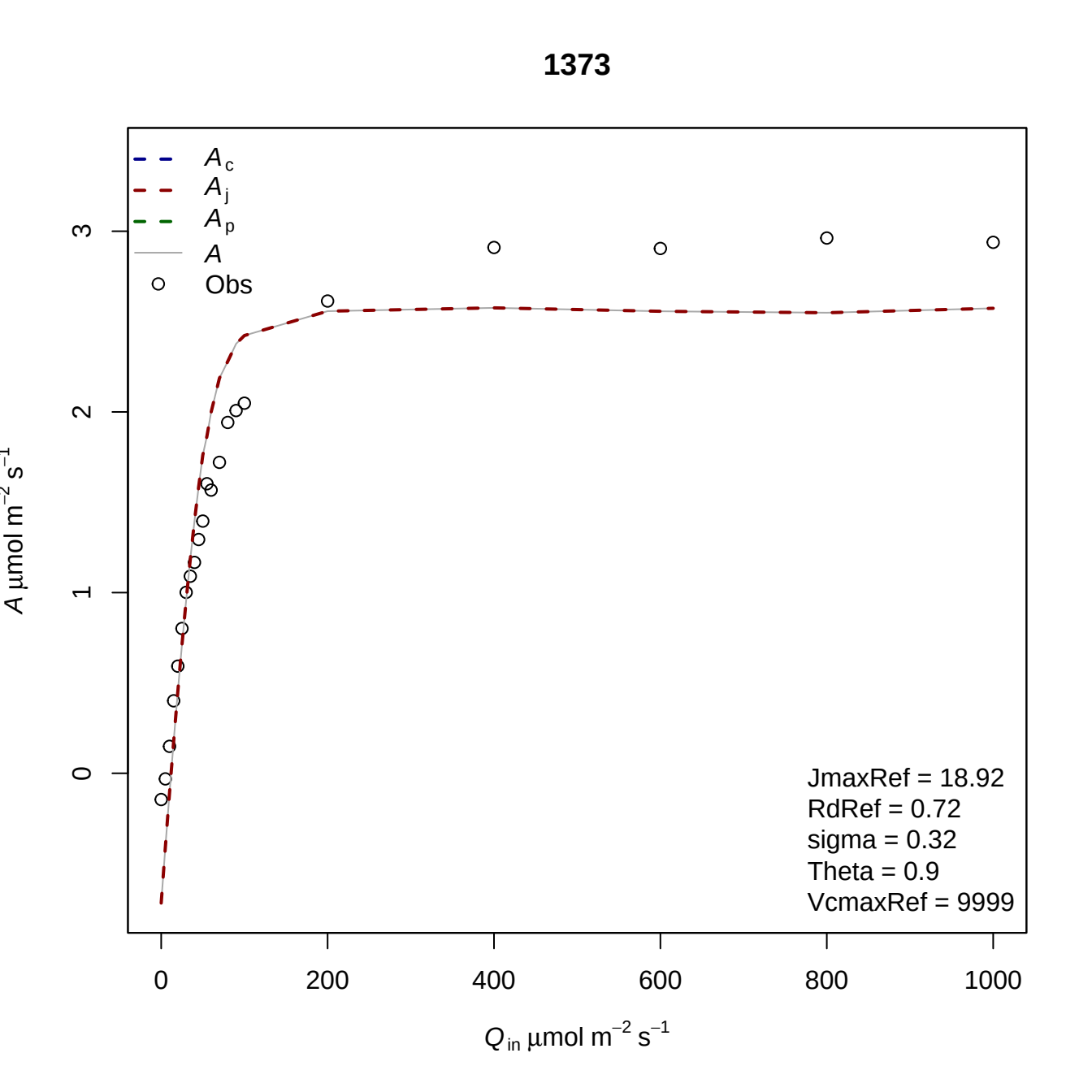

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 27.09
RdRef = 1.19
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

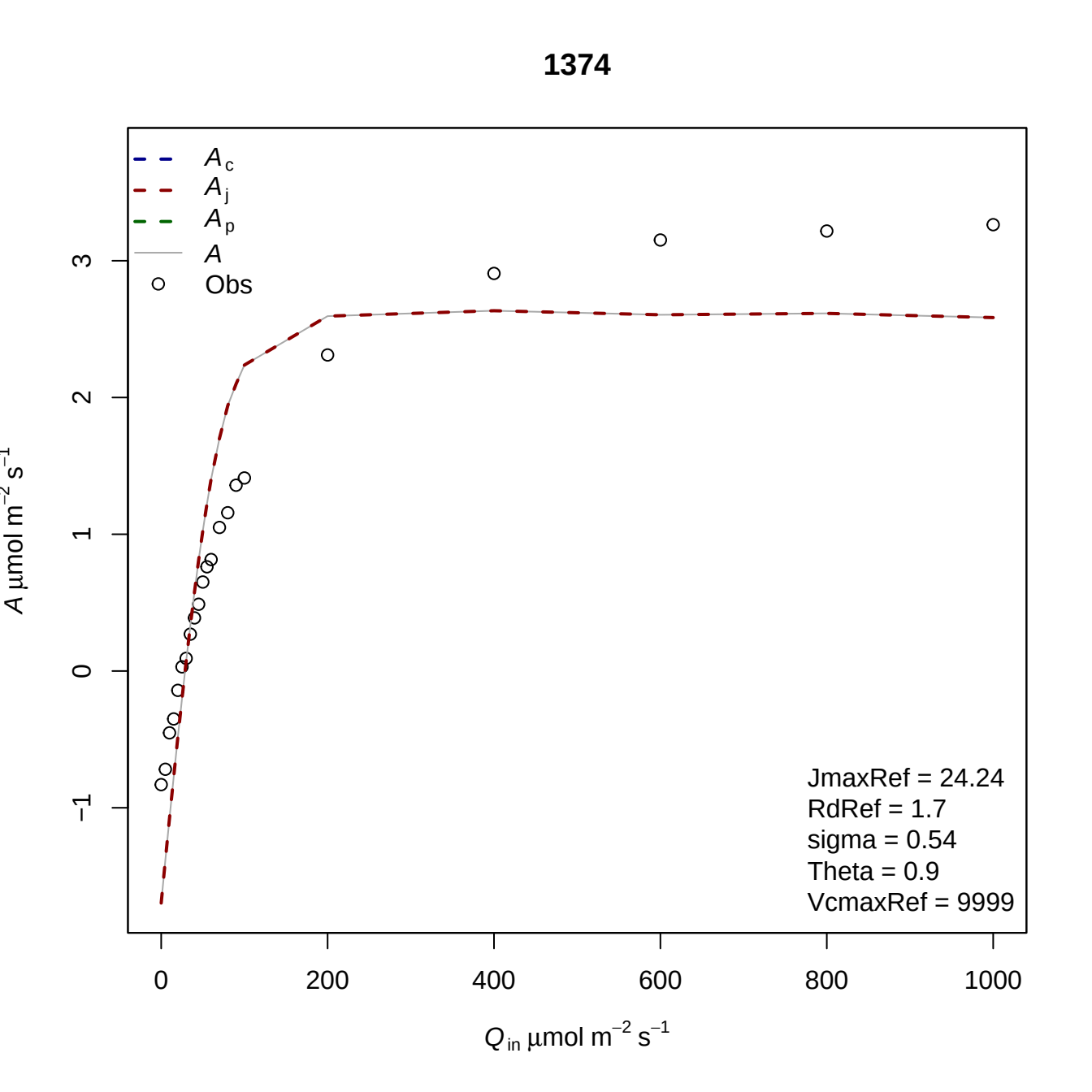

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 27.42
RdRef = 1.58
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000

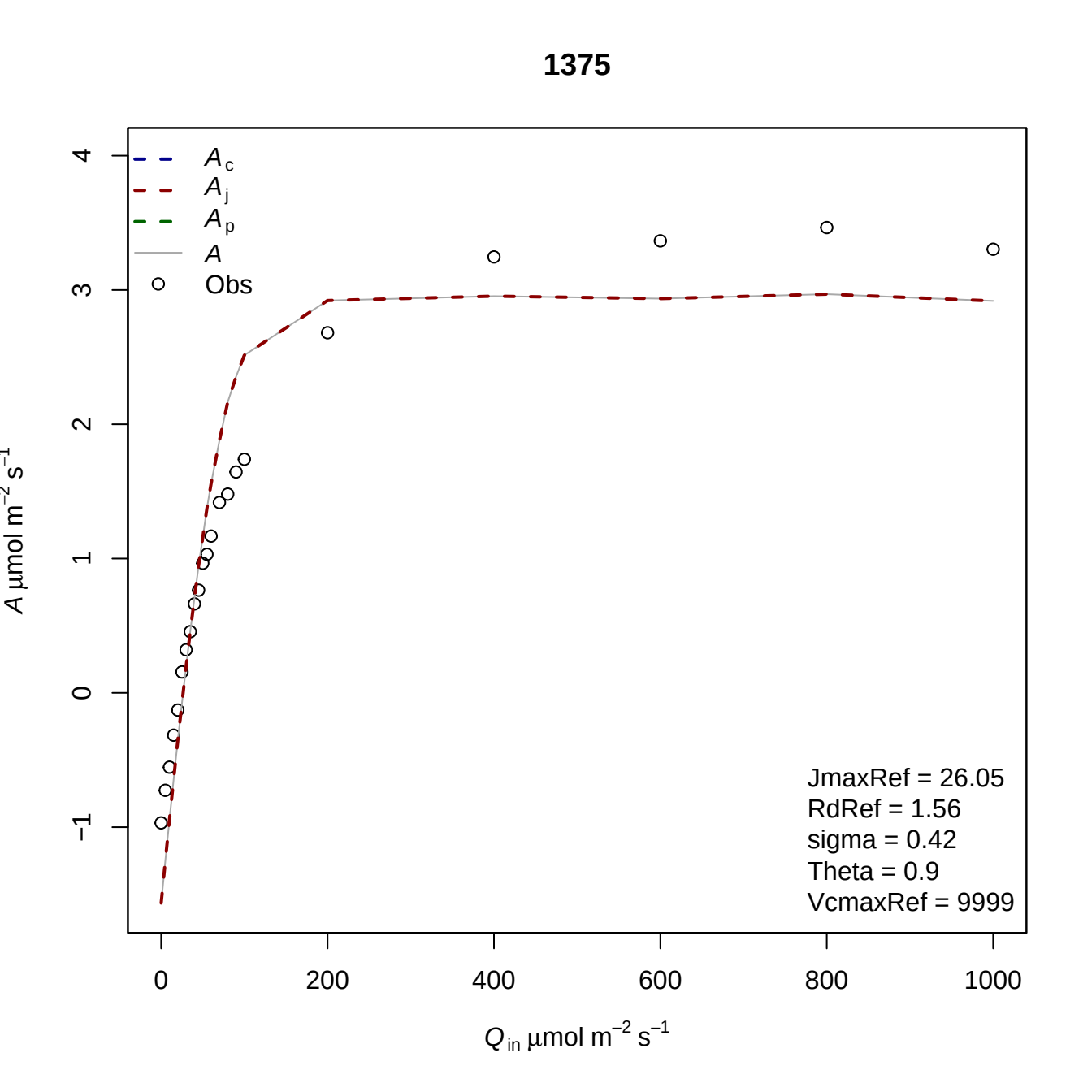

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 29.14
RdRef = 1.6
sigma = 0.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

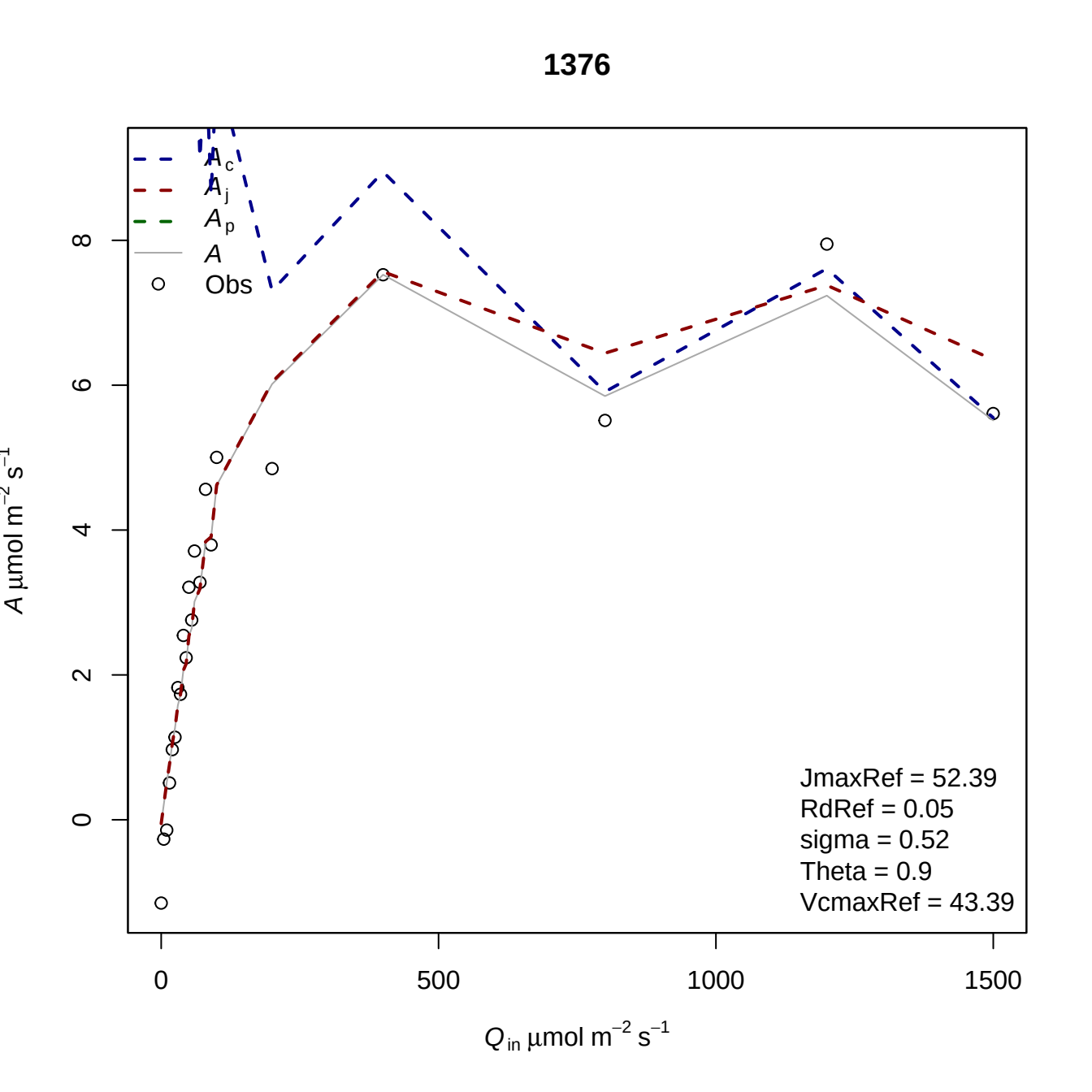

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 31.7
RdRef = 1.4
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

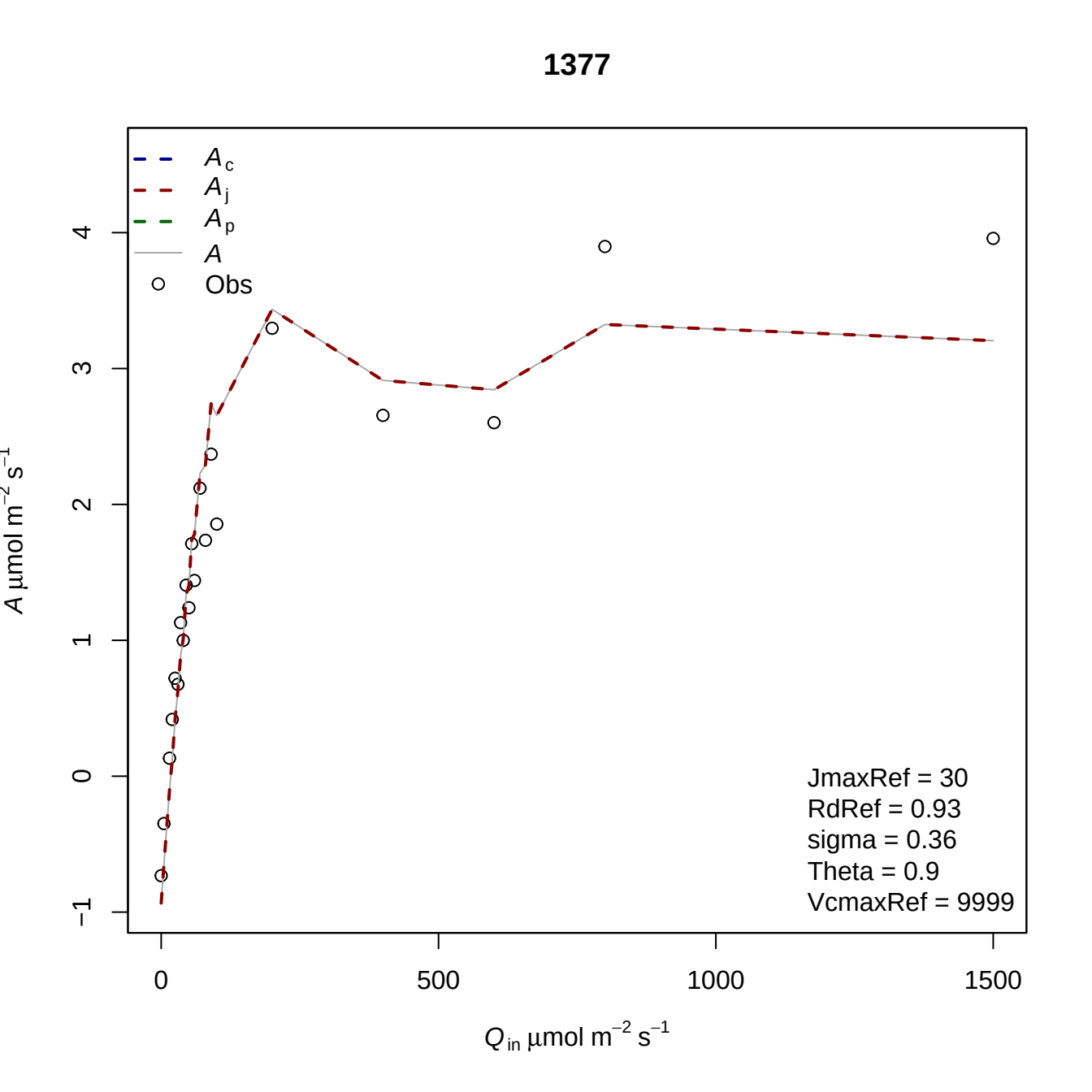

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 35.87
RdRef = 0.89
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

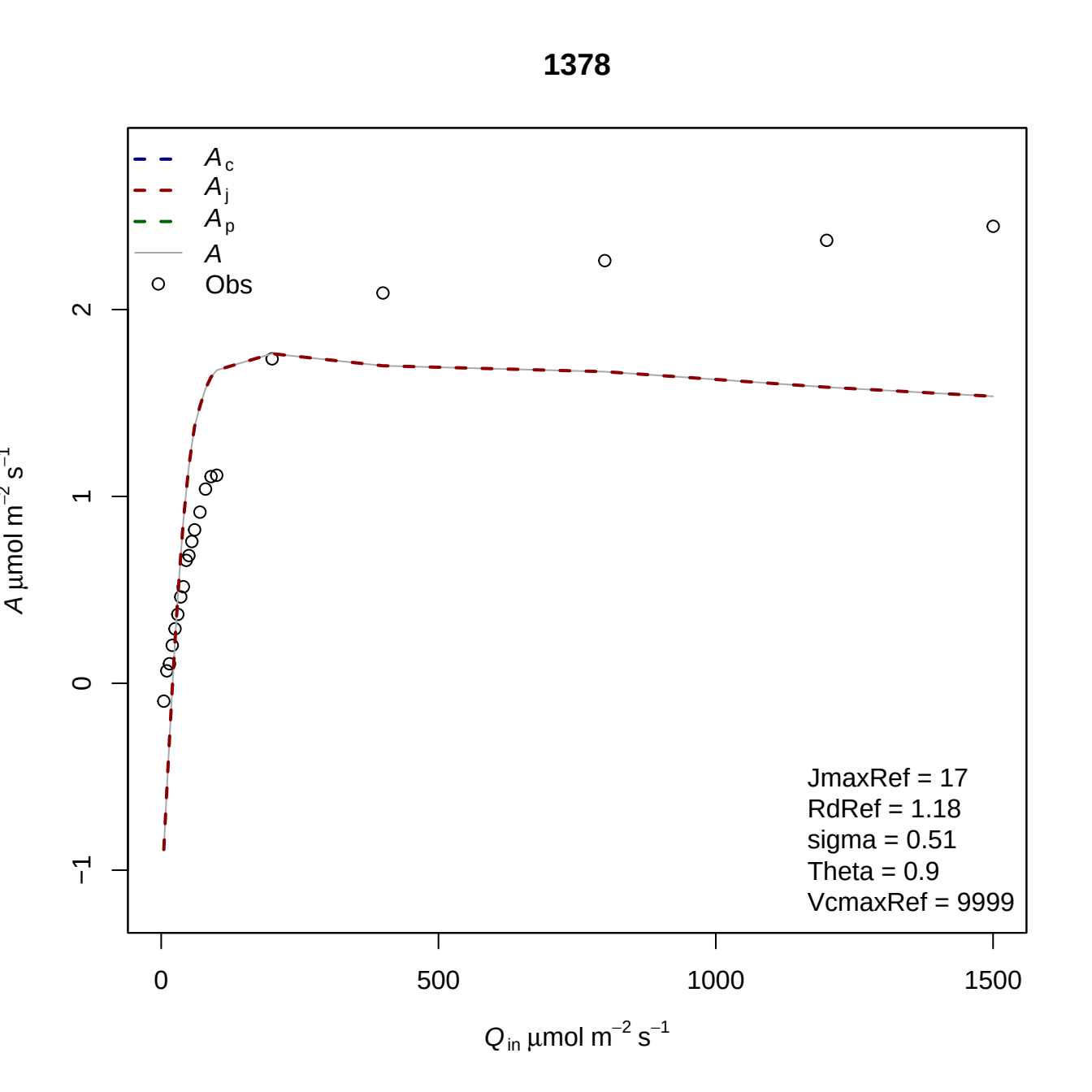

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 18.92
RdRef = 0.72
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

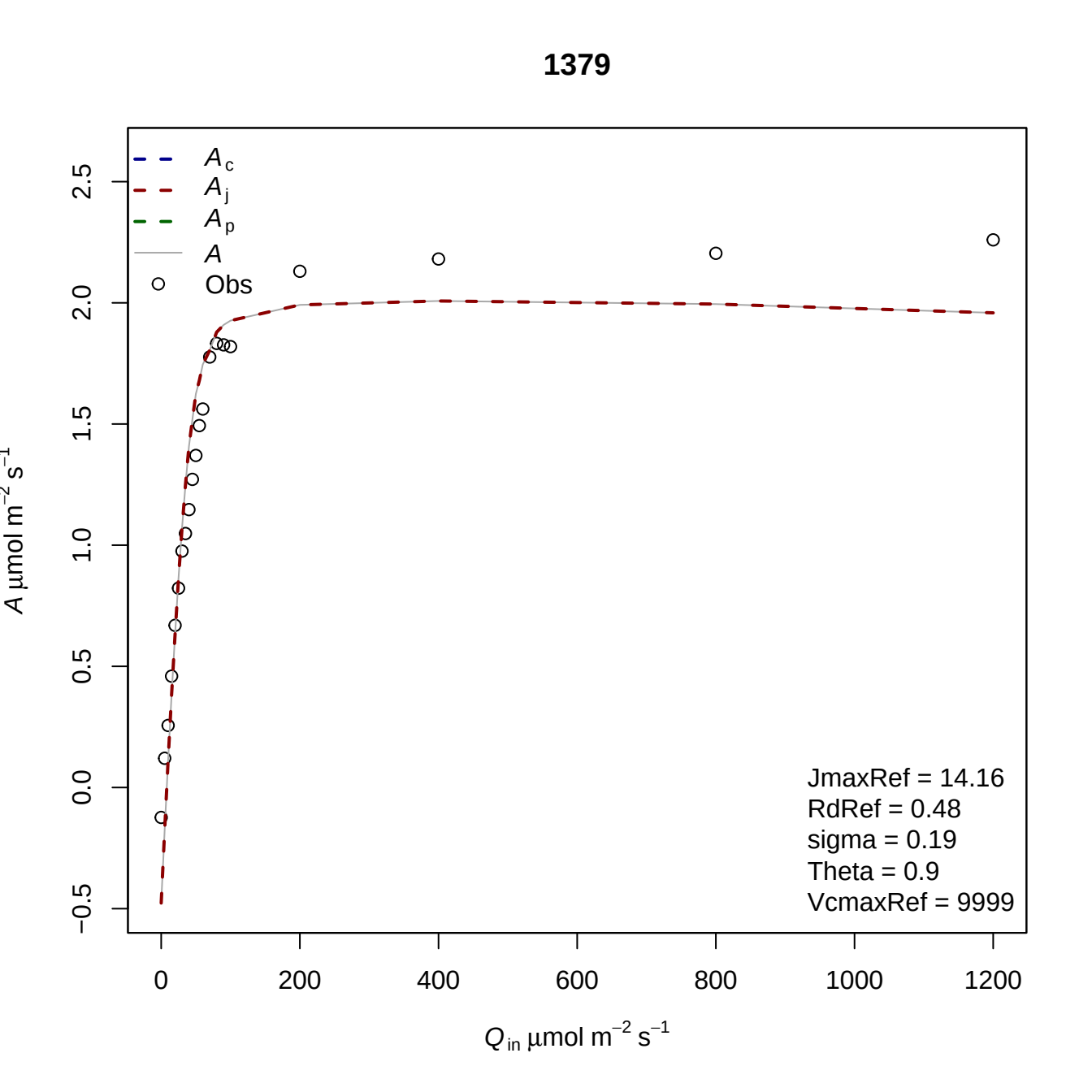

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 24.24
RdRef = 1.7
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

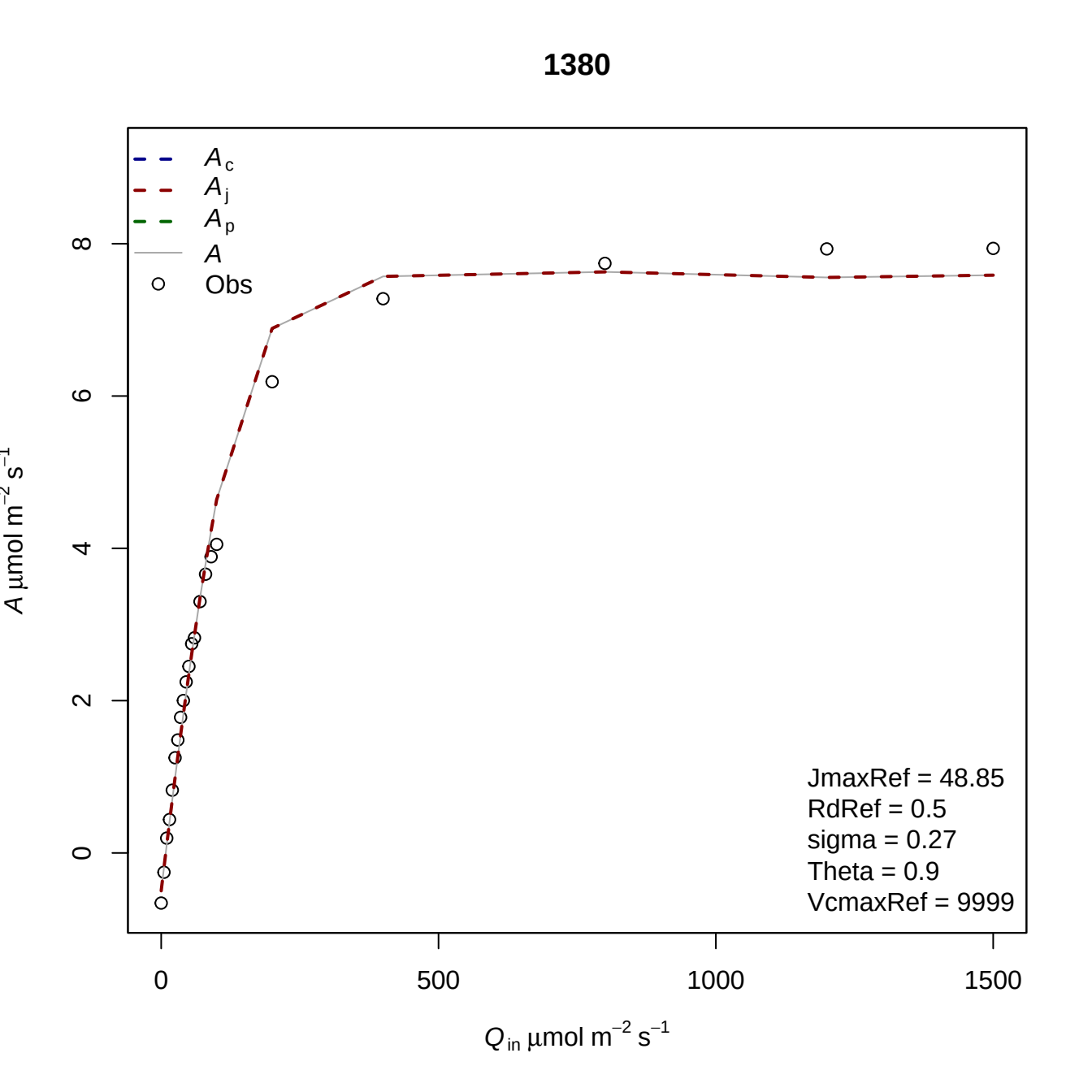

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 26.05
RdRef = 1.56
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

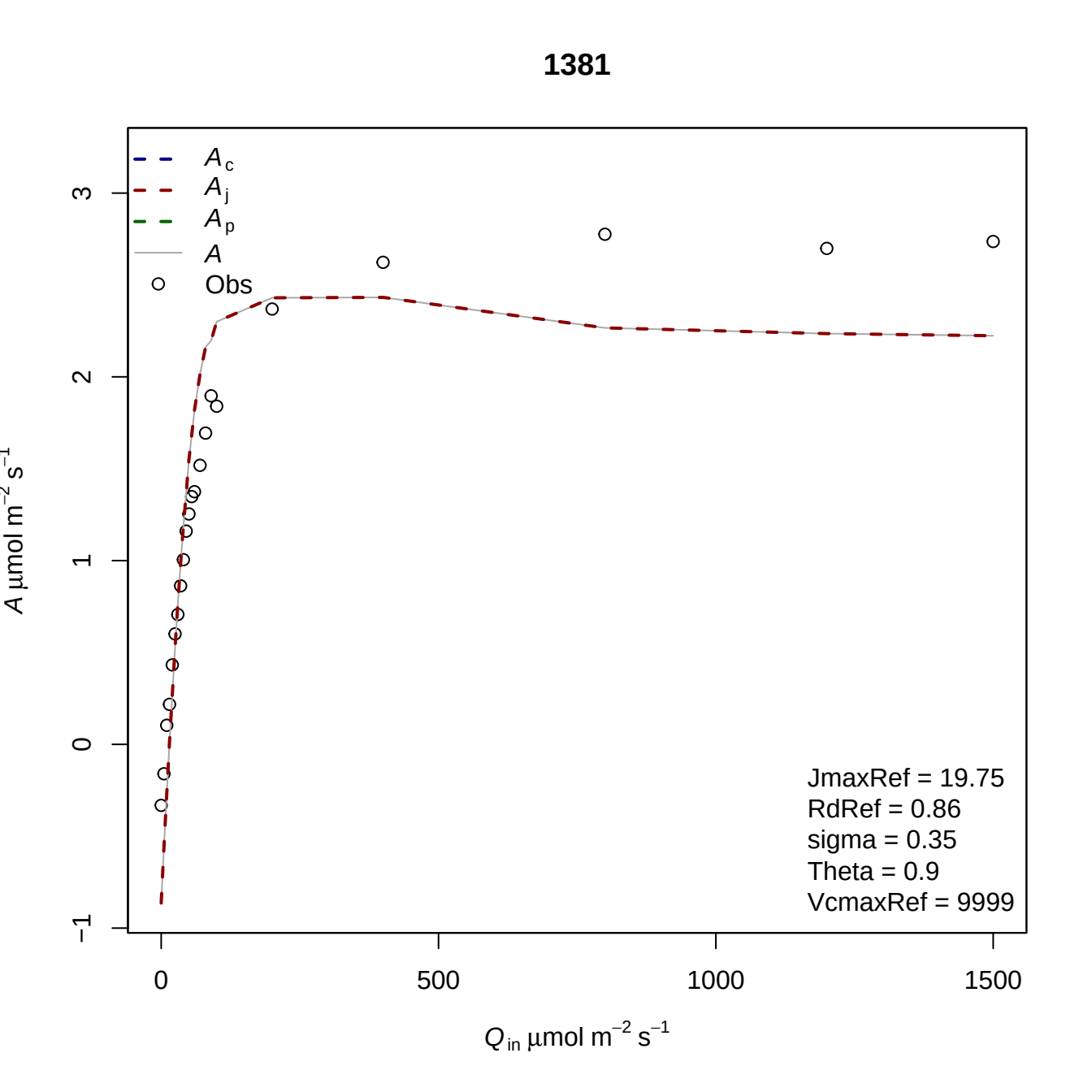

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 52.39
RdRef = 0.05
sigma = 0.52
Theta = 0.9
VcmaxRef = 43.39
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

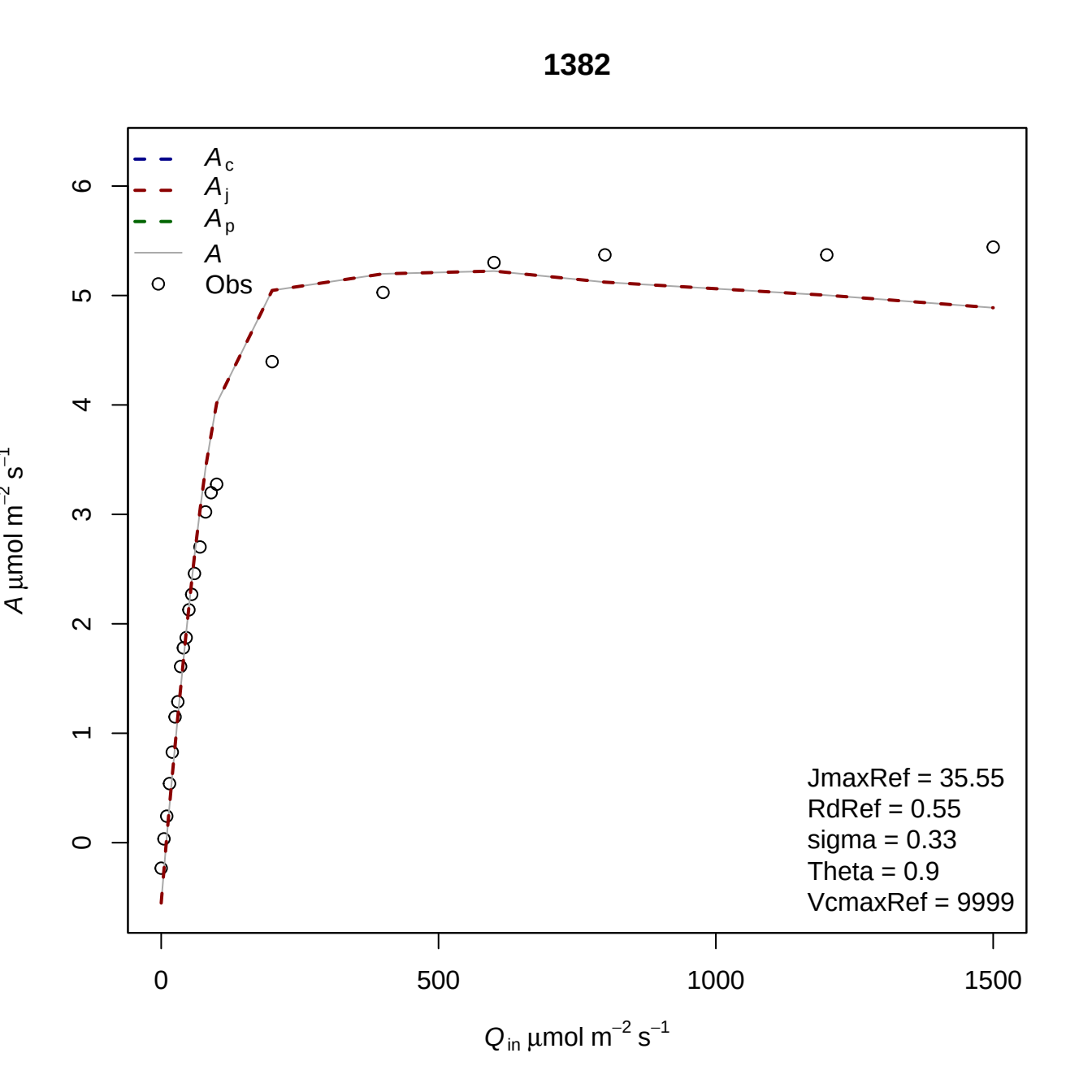

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 30
RdRef = 0.93
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

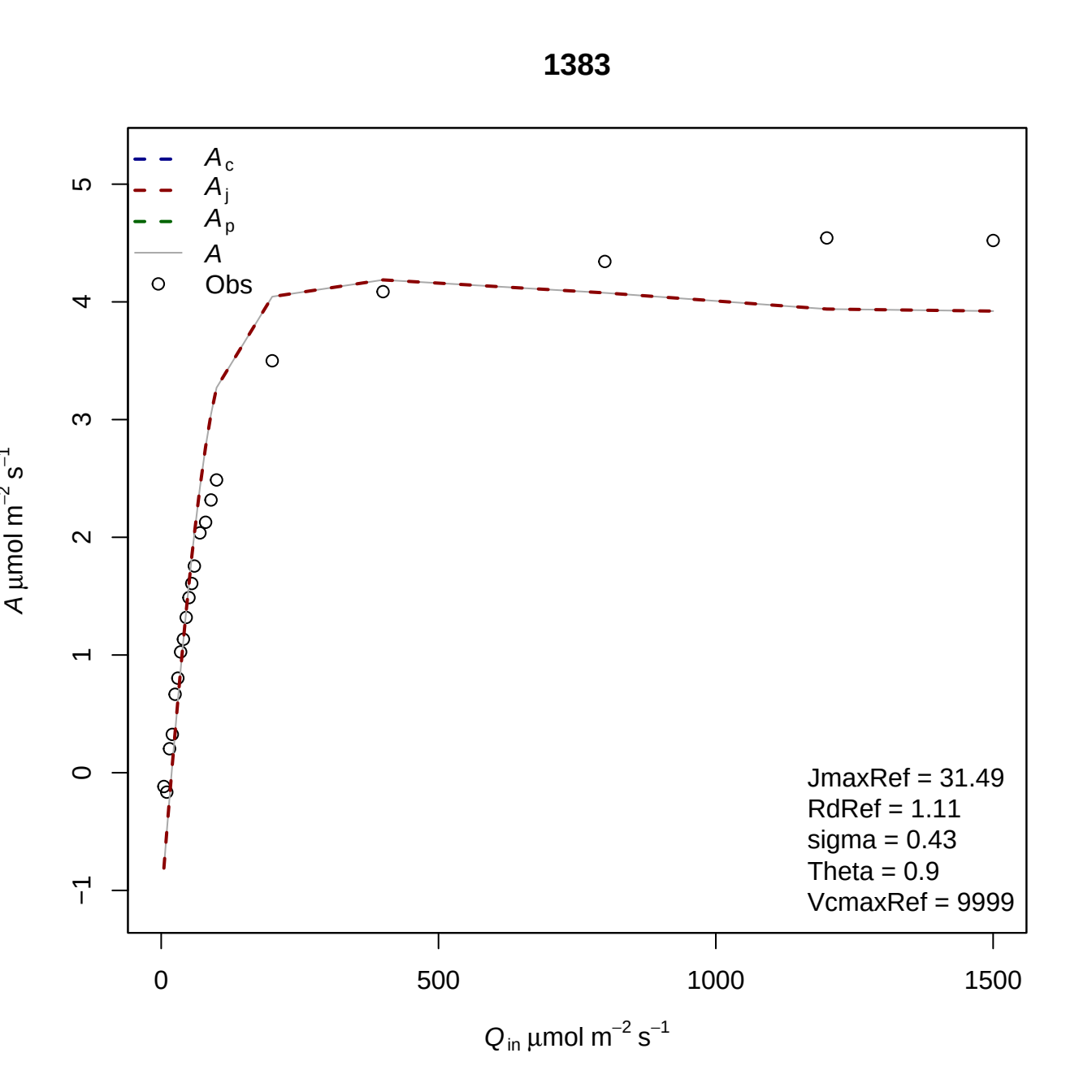

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 17
RdRef = 1.18
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

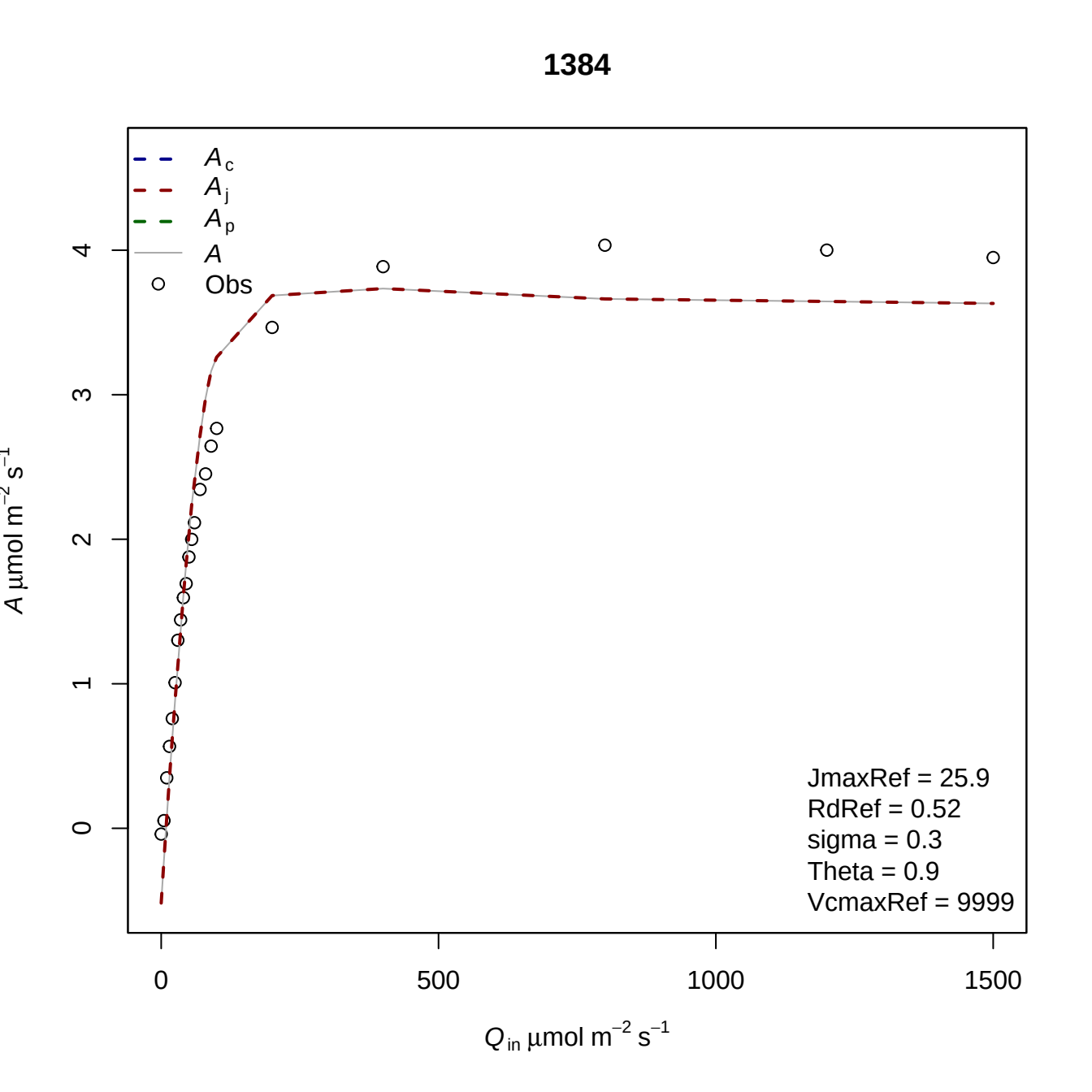

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 14.16
RdRef = 0.48
sigma = 0.19
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

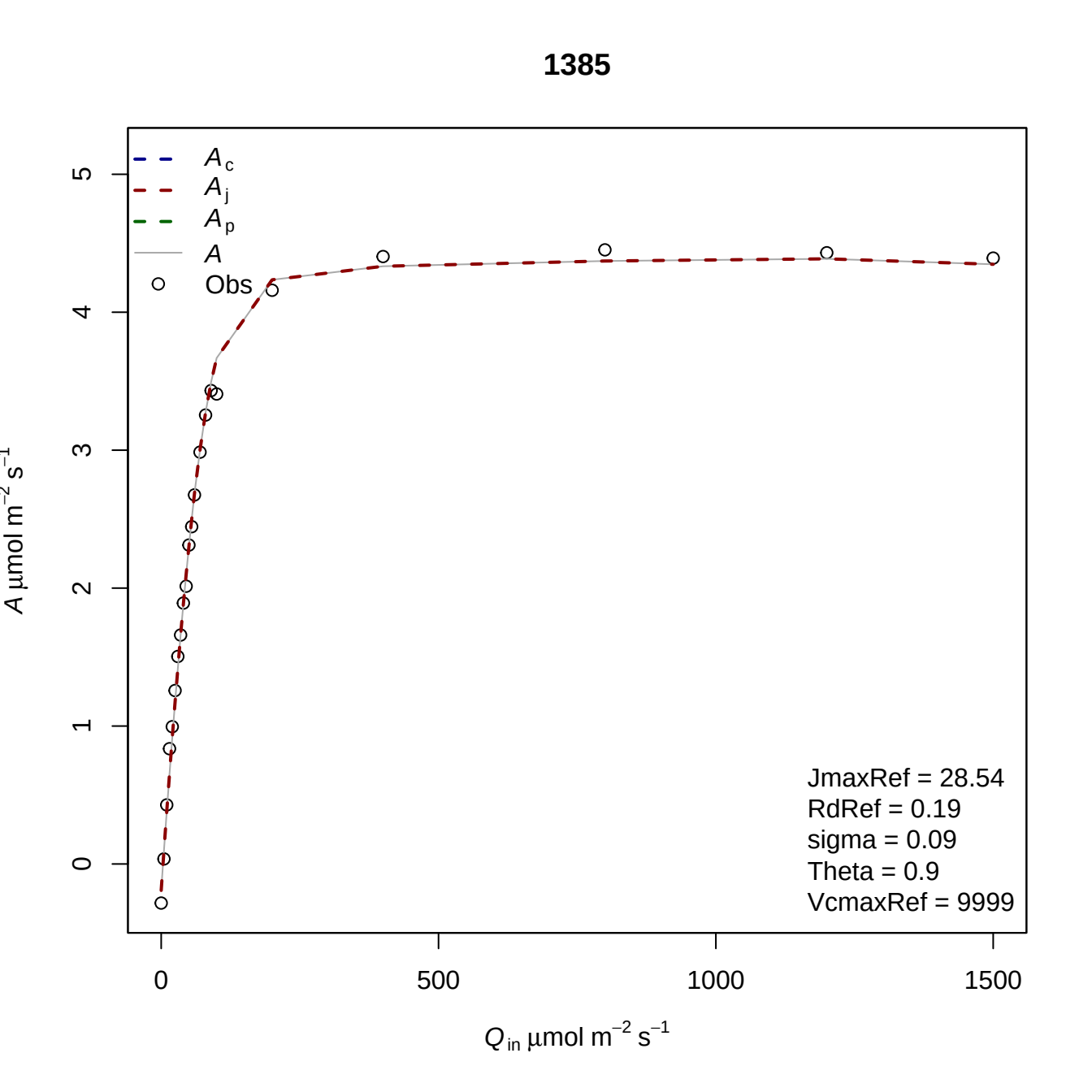

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 48.85
RdRef = 0.5
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

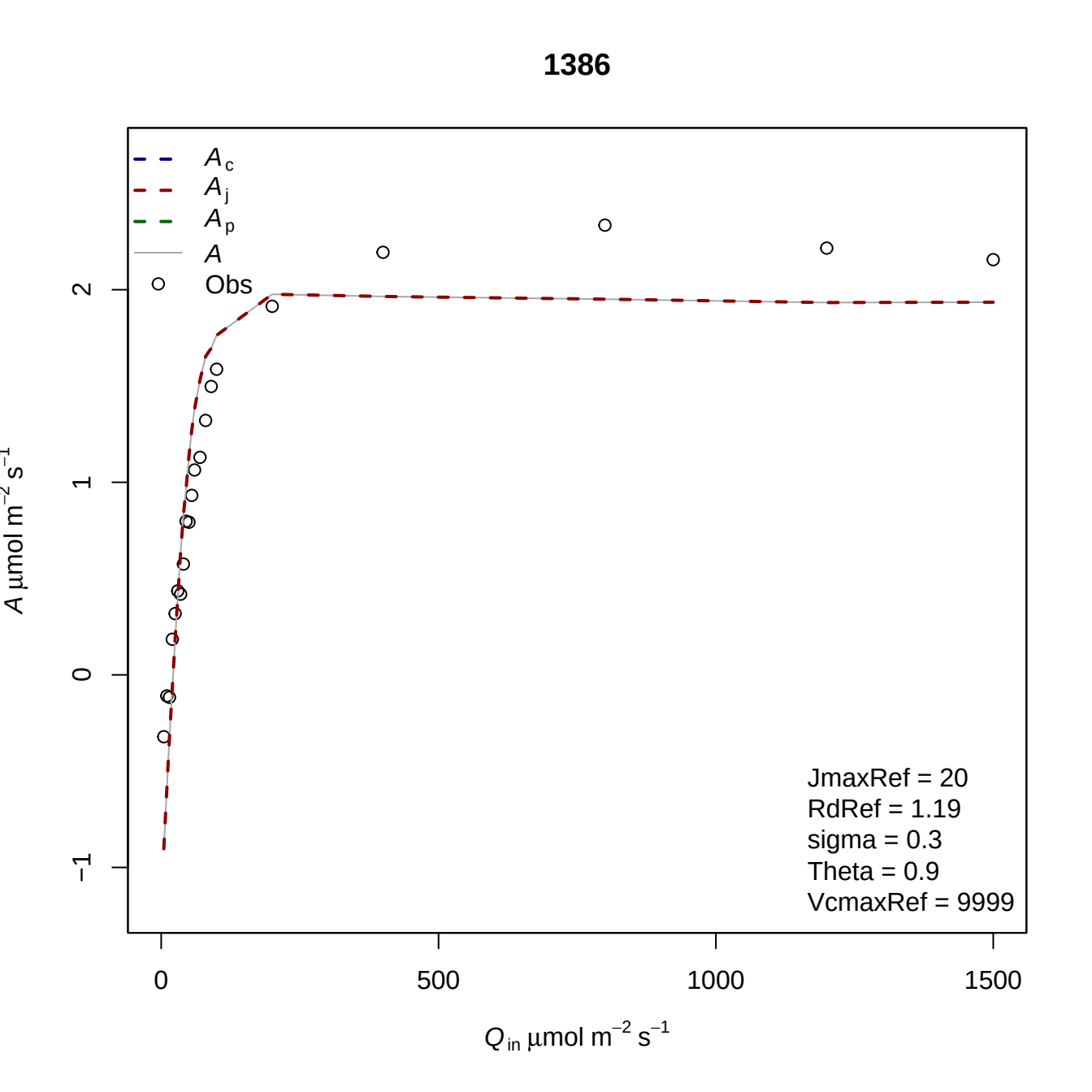

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 19.75
RdRef = 0.86
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

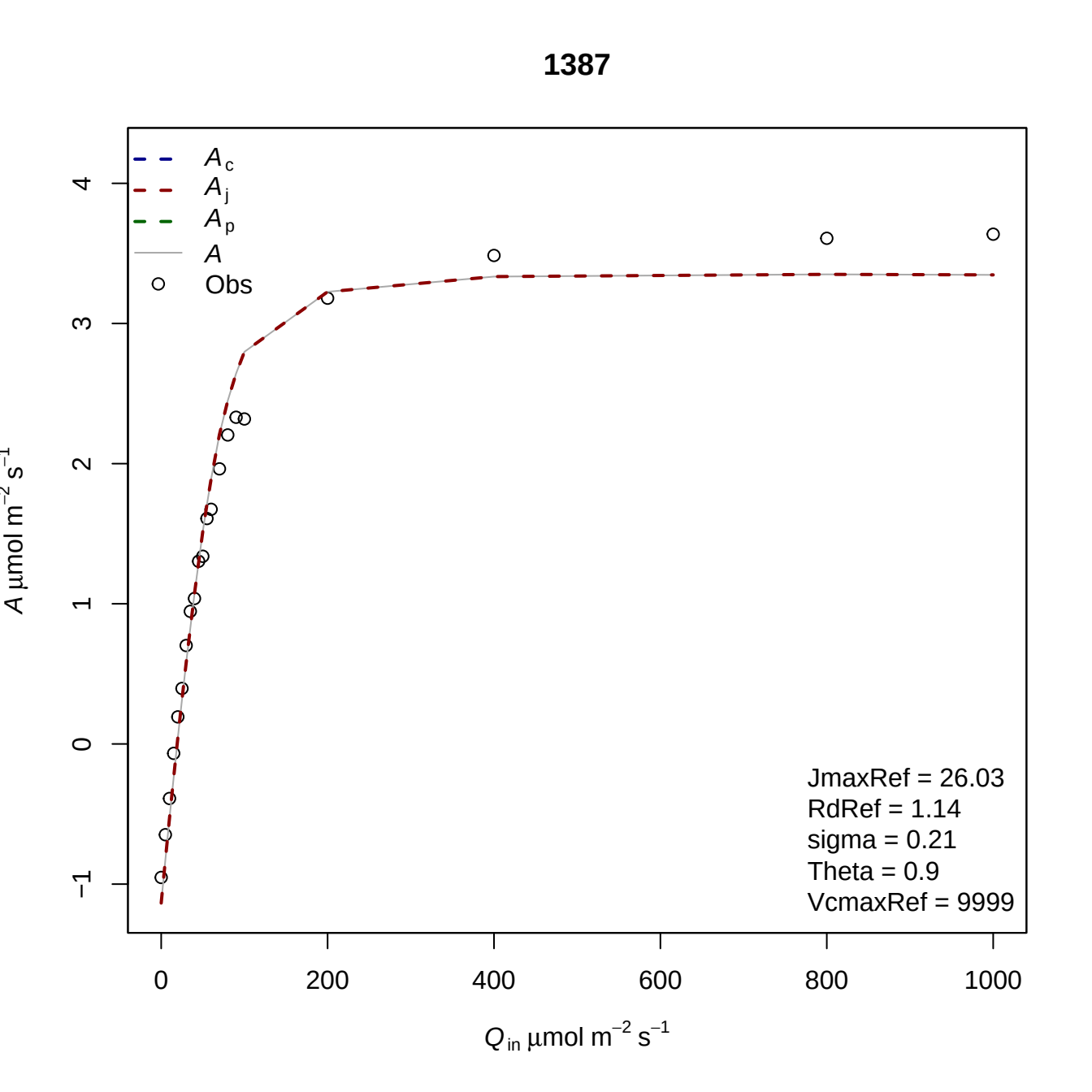

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 0.55
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

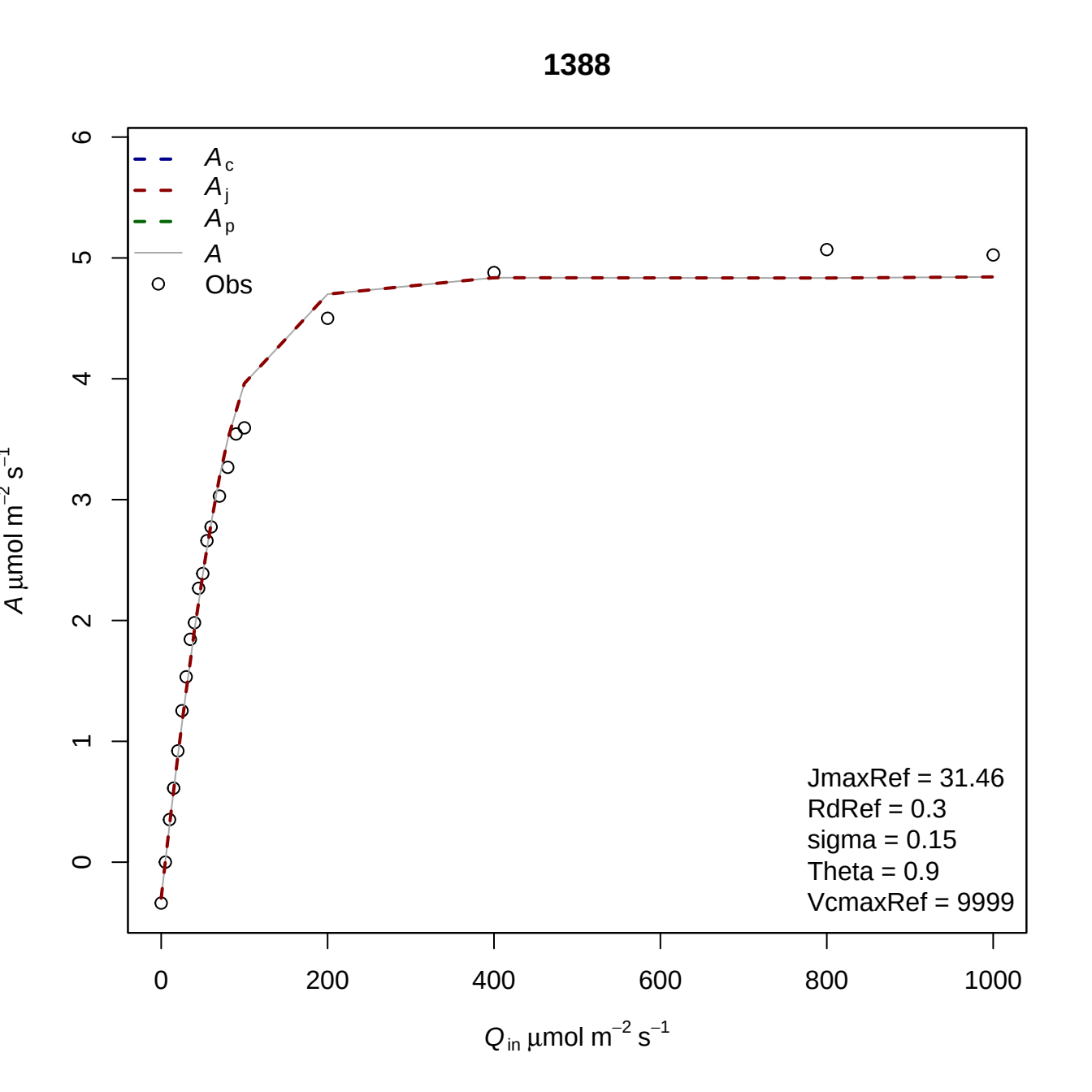

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 31.49
RdRef = 1.11
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

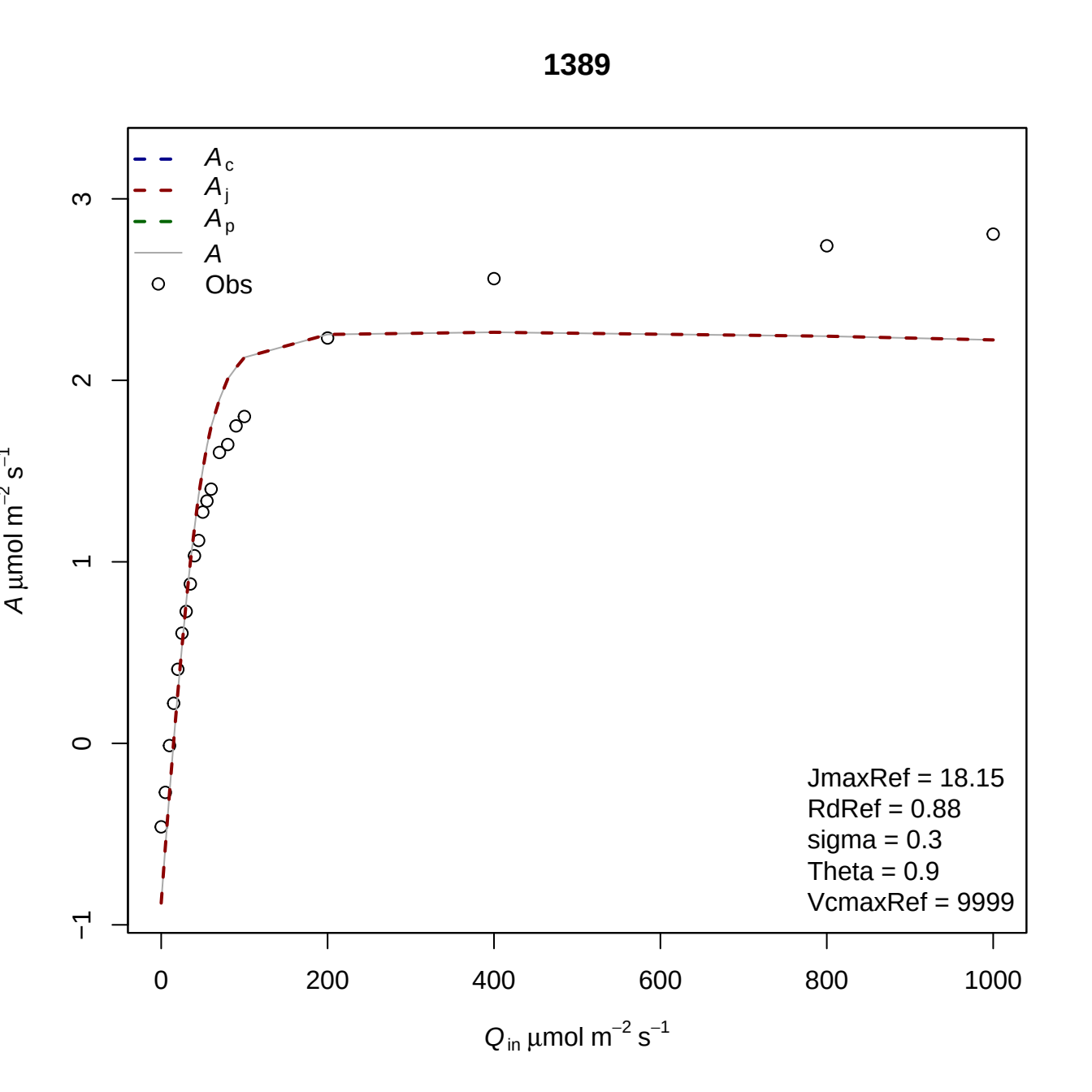

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 25.9
RdRef = 0.52
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

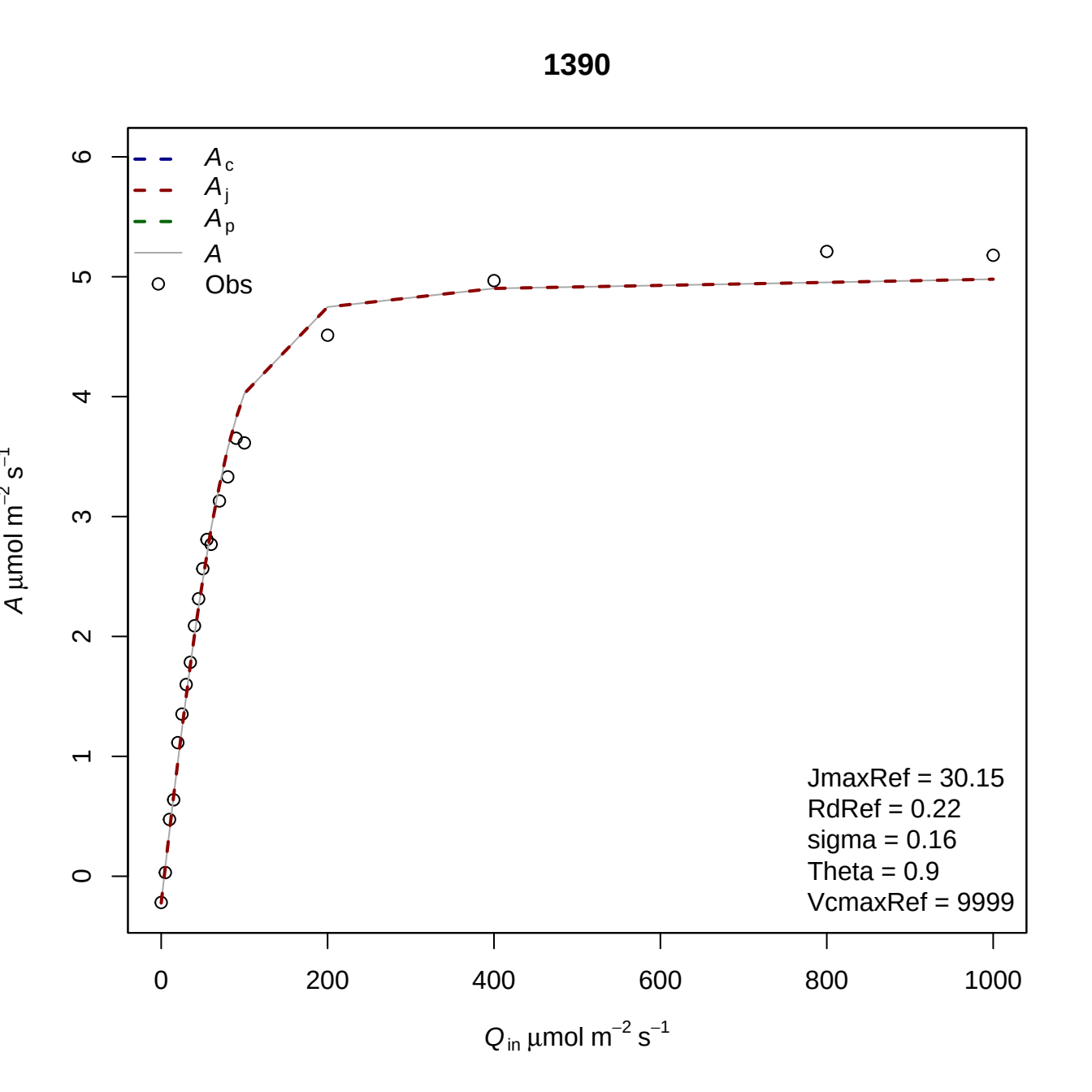

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 28.54
RdRef = 0.19
sigma = 0.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

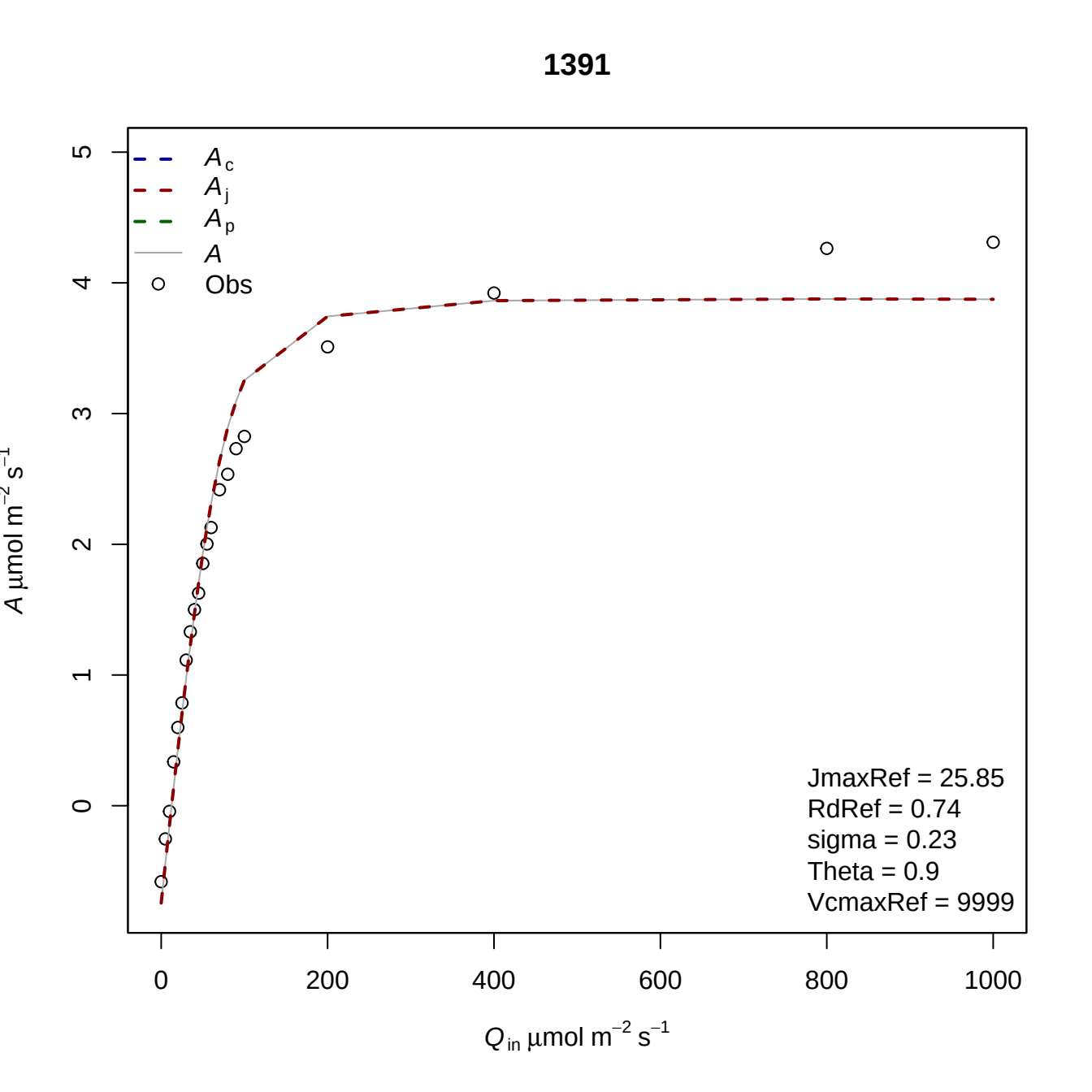

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 20
RdRef = 1.19
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

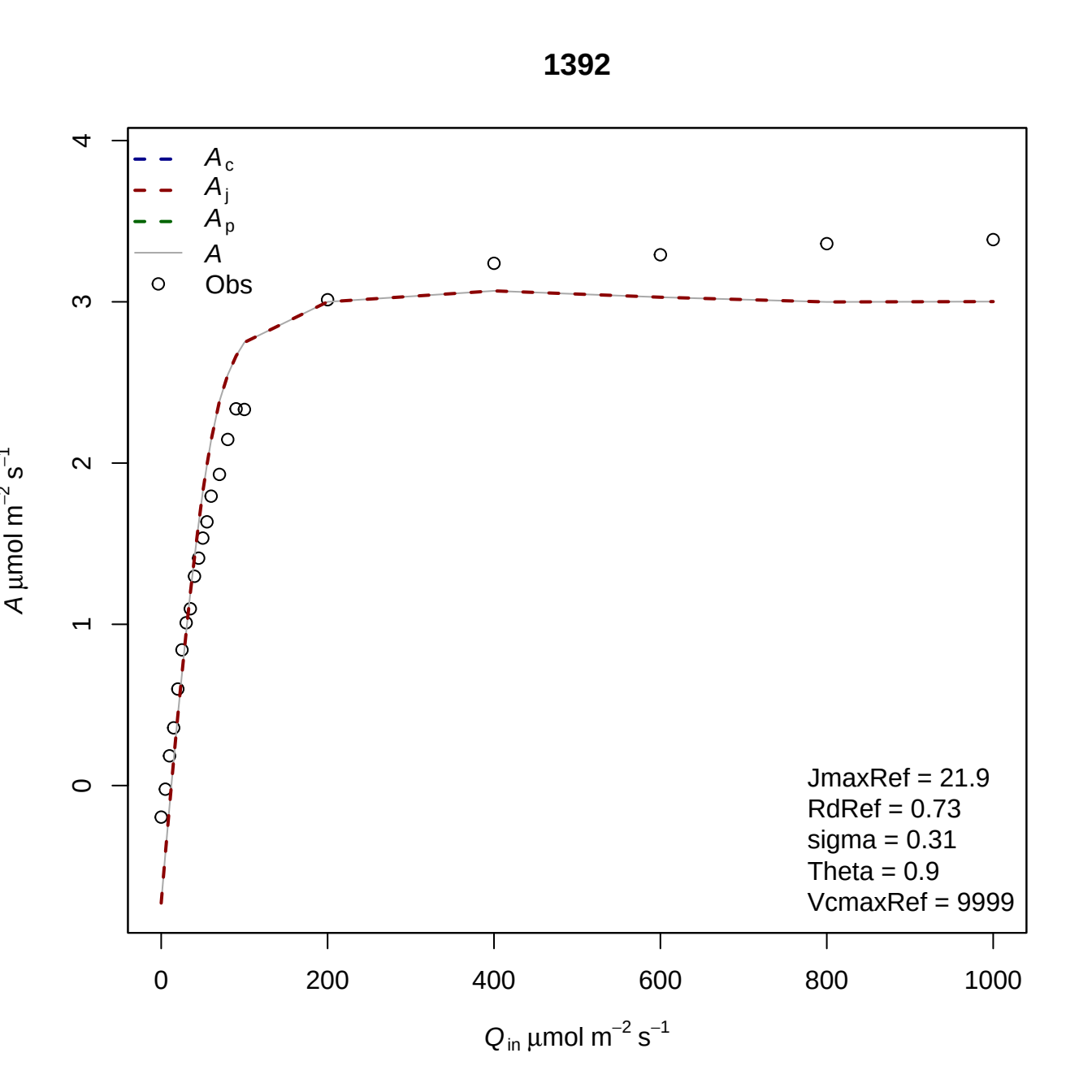

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 26.03
RdRef = 1.14
sigma = 0.21
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

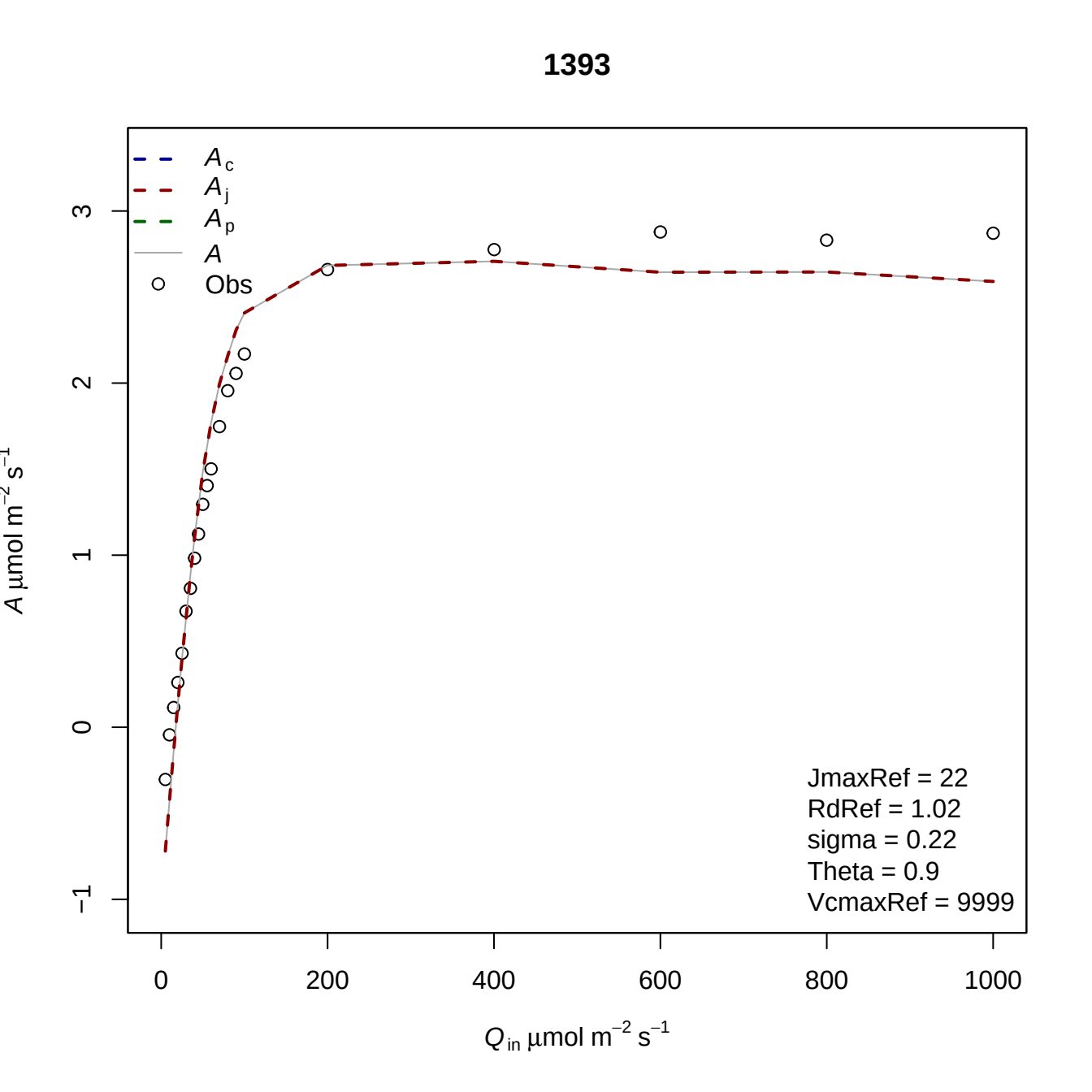

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 31.46
RdRef = 0.3
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

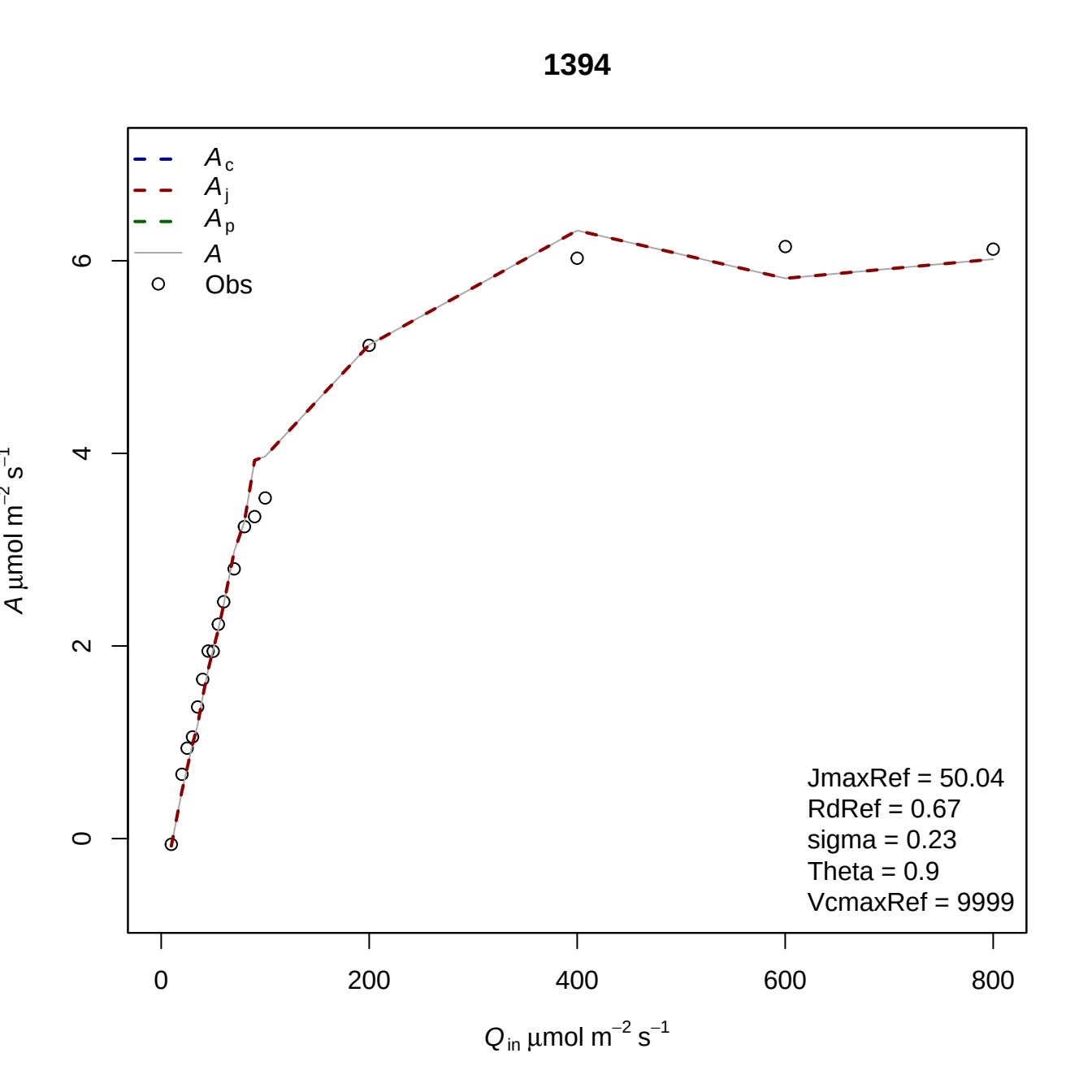

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 18.15
RdRef = 0.88
sigma = 0.3
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

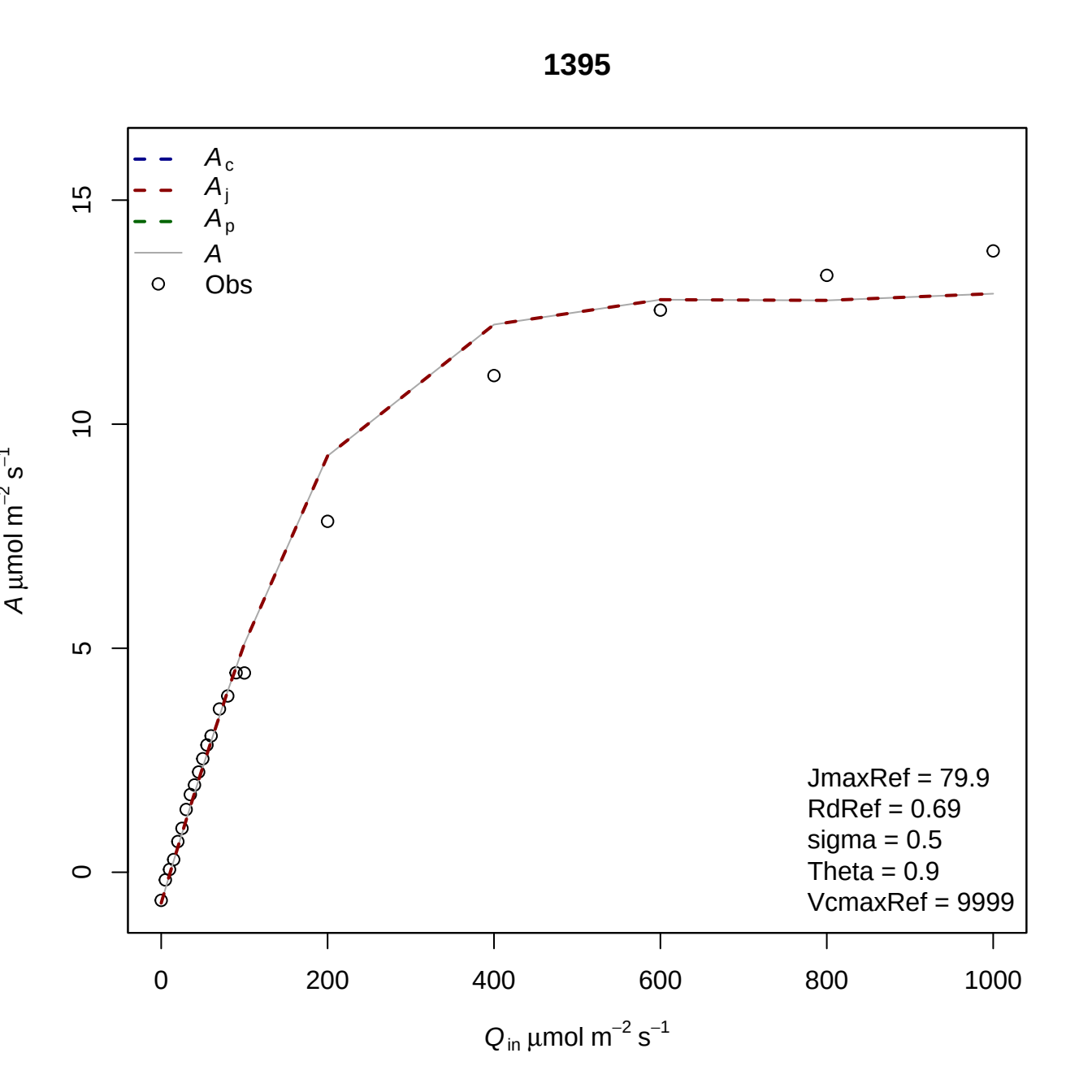

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 30.15
RdRef = 0.22
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

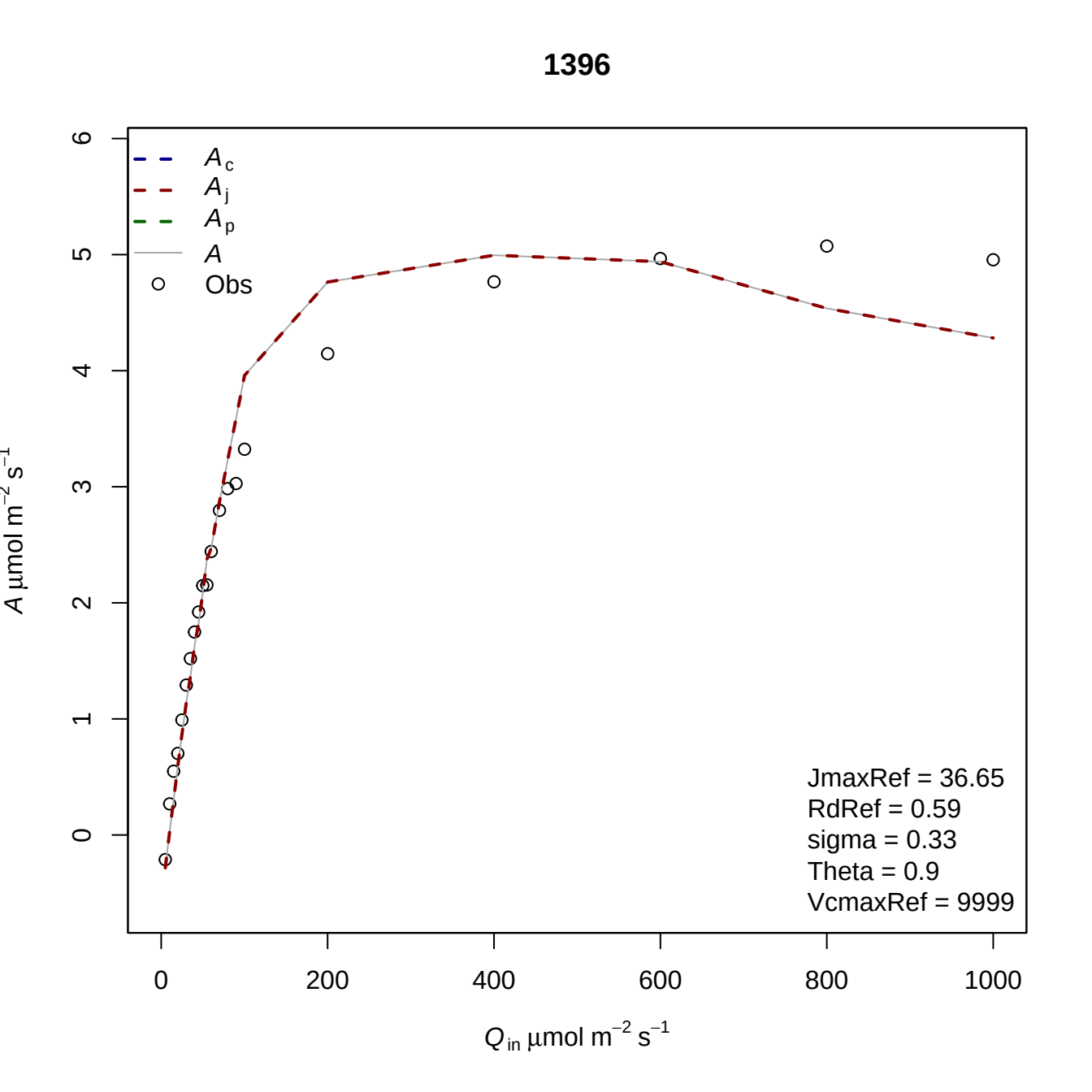

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 25.85
RdRef = 0.74
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

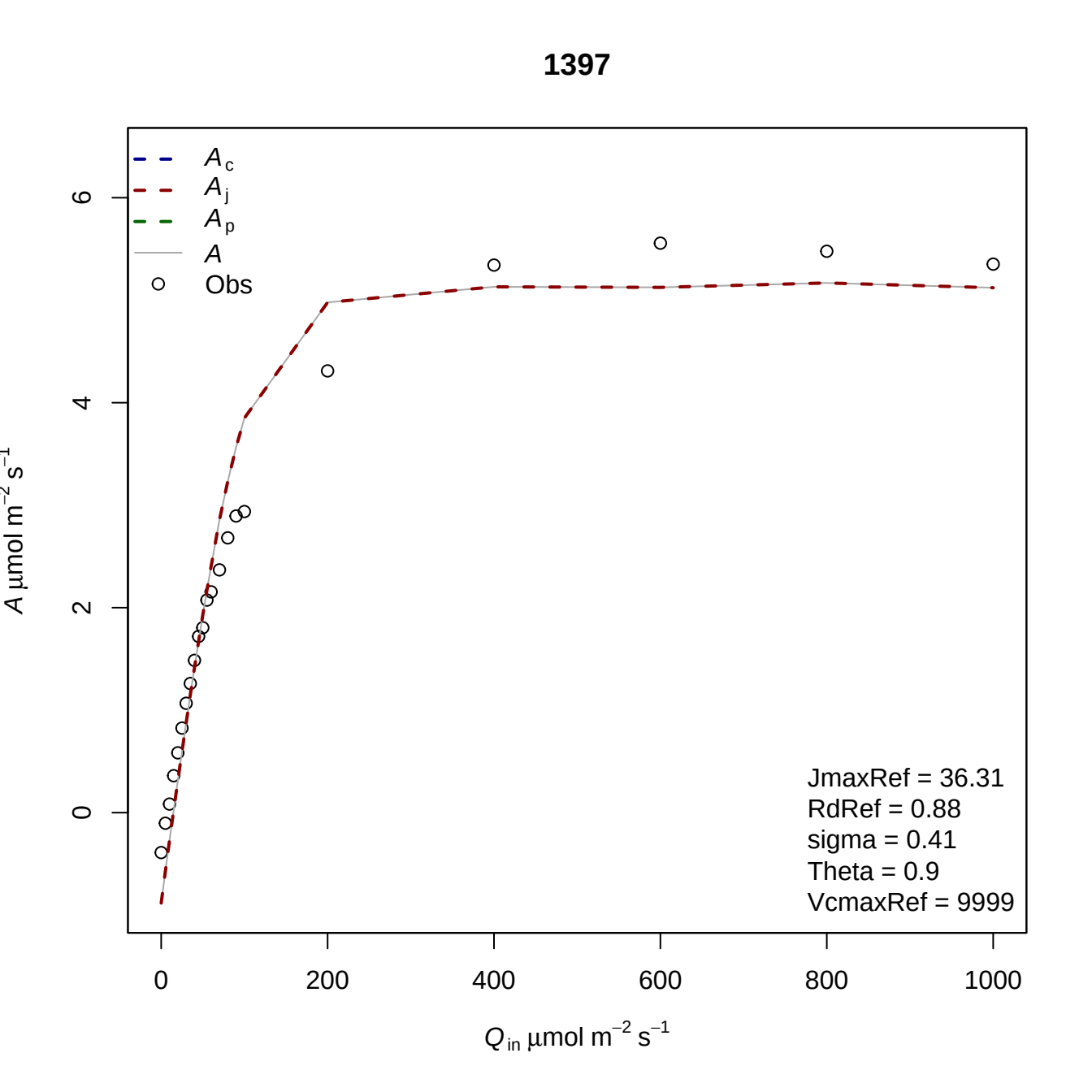

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 21.9
RdRef = 0.73
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

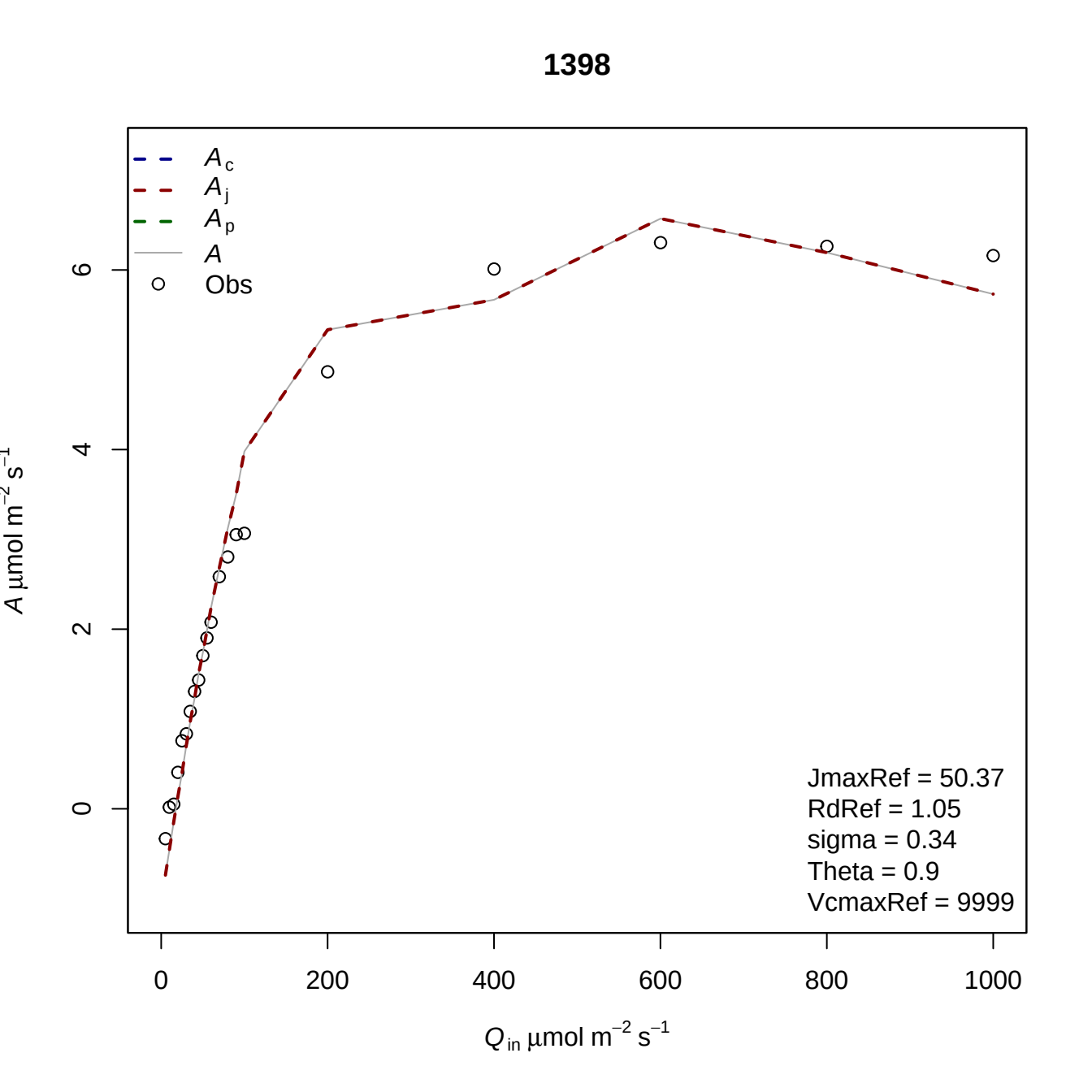

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 22
RdRef = 1.02
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

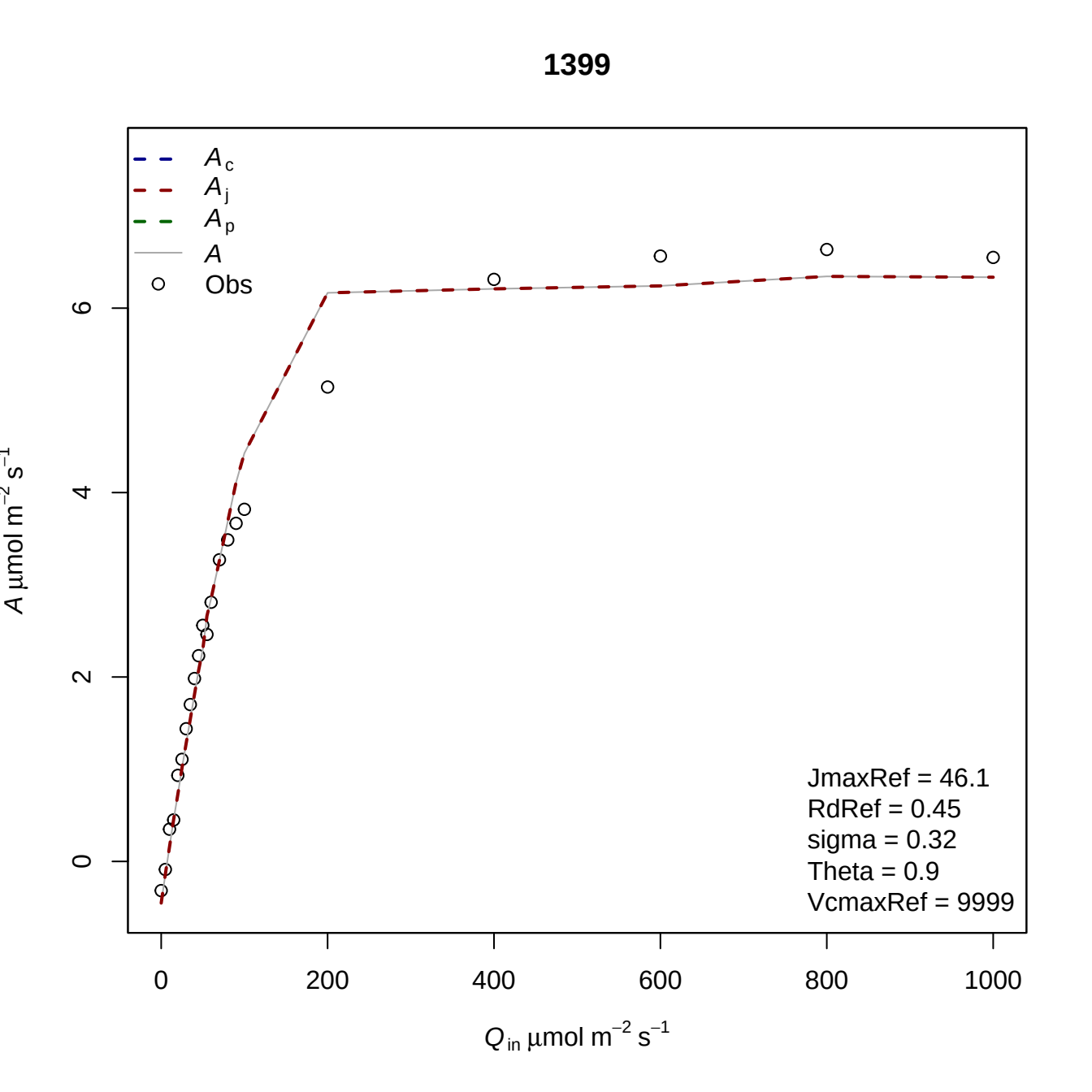

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 50.04
RdRef = 0.67
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800

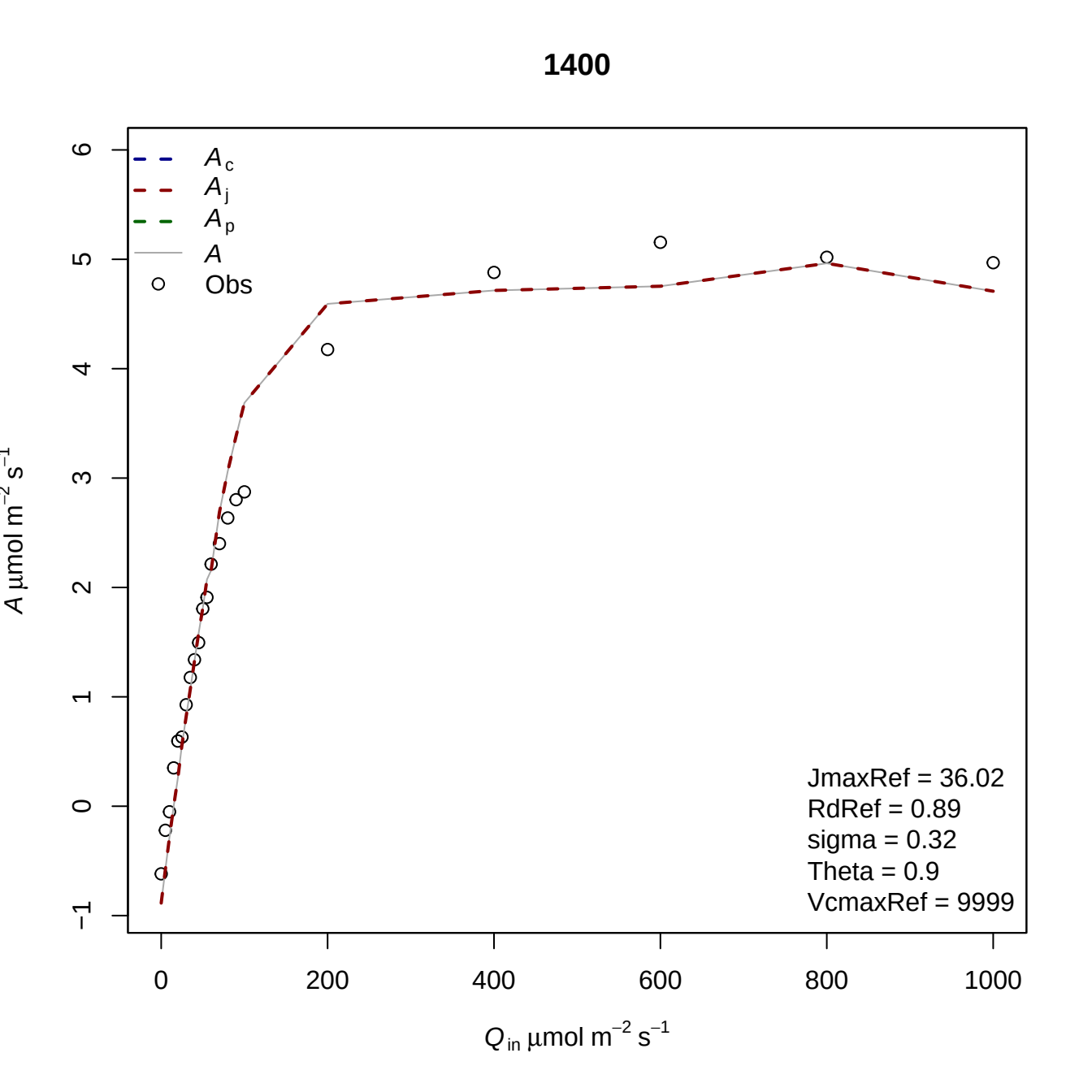

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 79.9
RdRef = 0.69
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

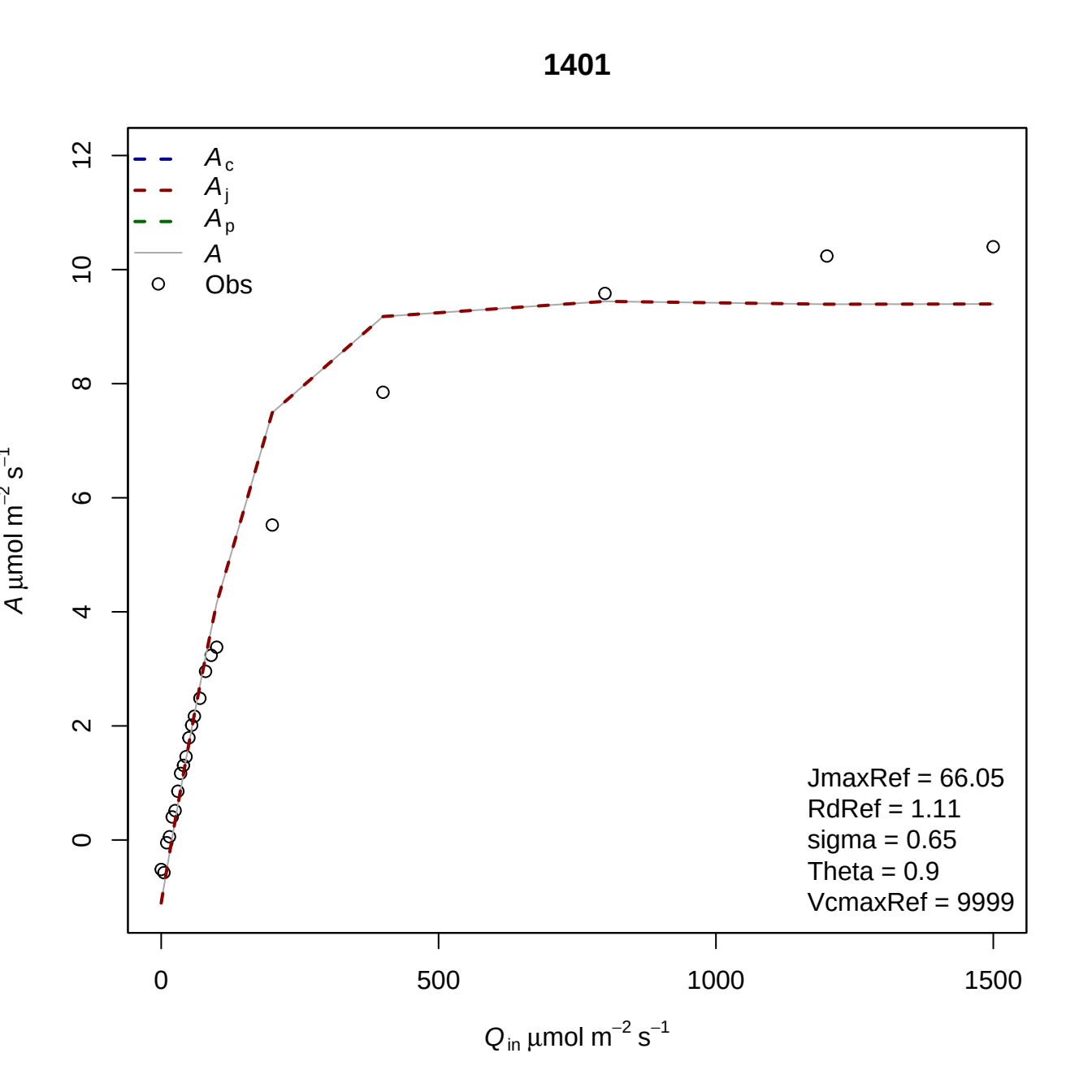

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 36.65
RdRef = 0.59
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

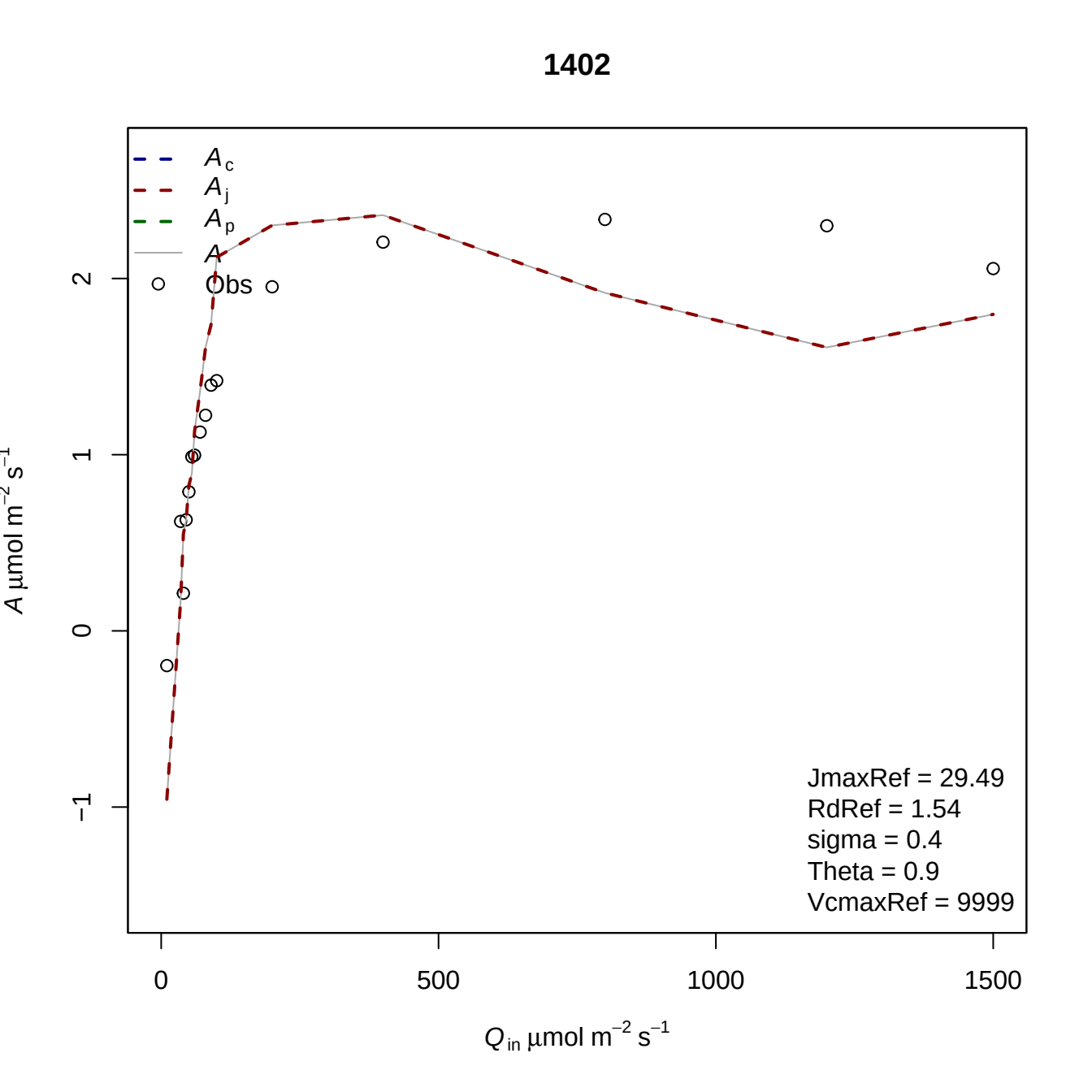

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 36.31
RdRef = 0.88
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

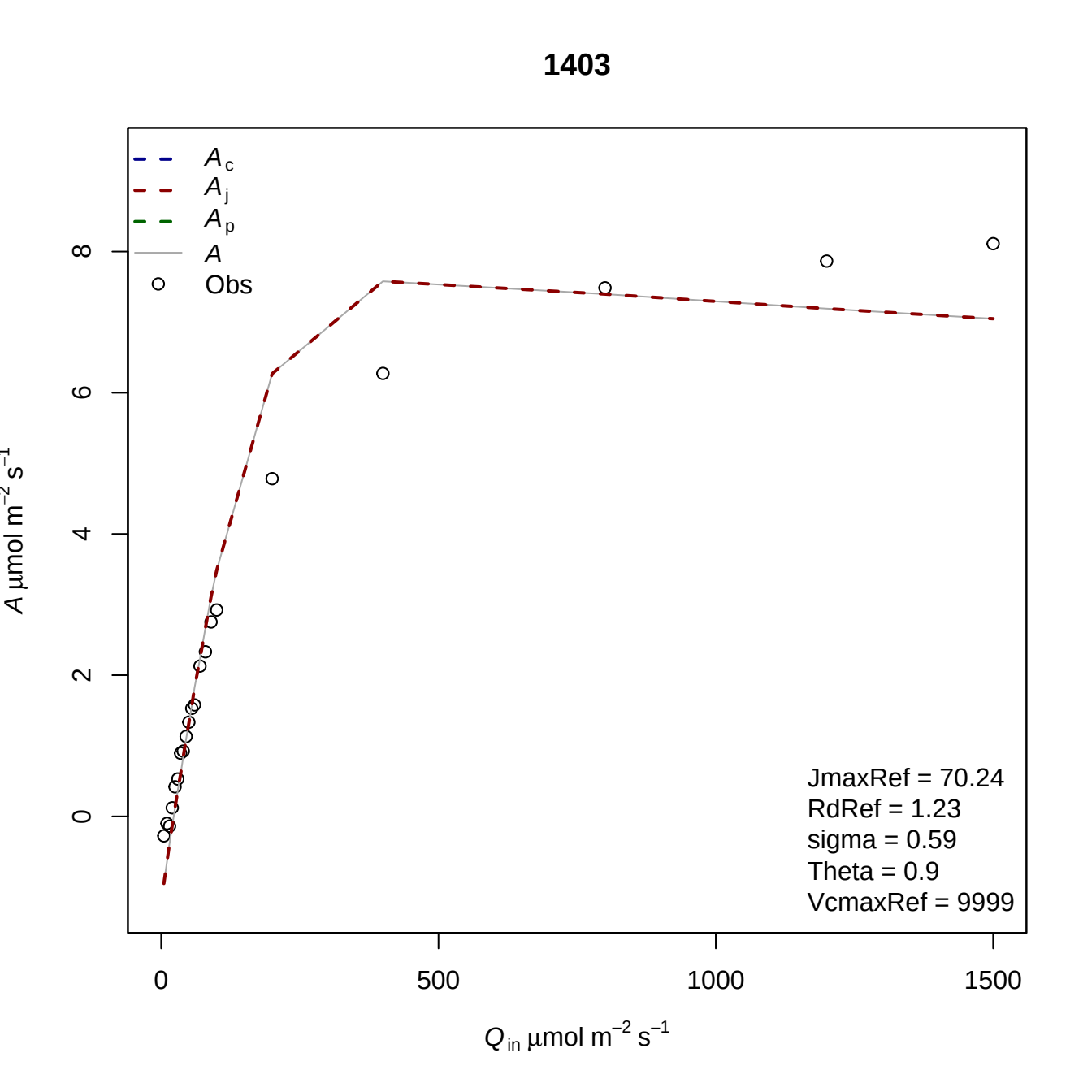

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 50.37
RdRef = 1.05
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

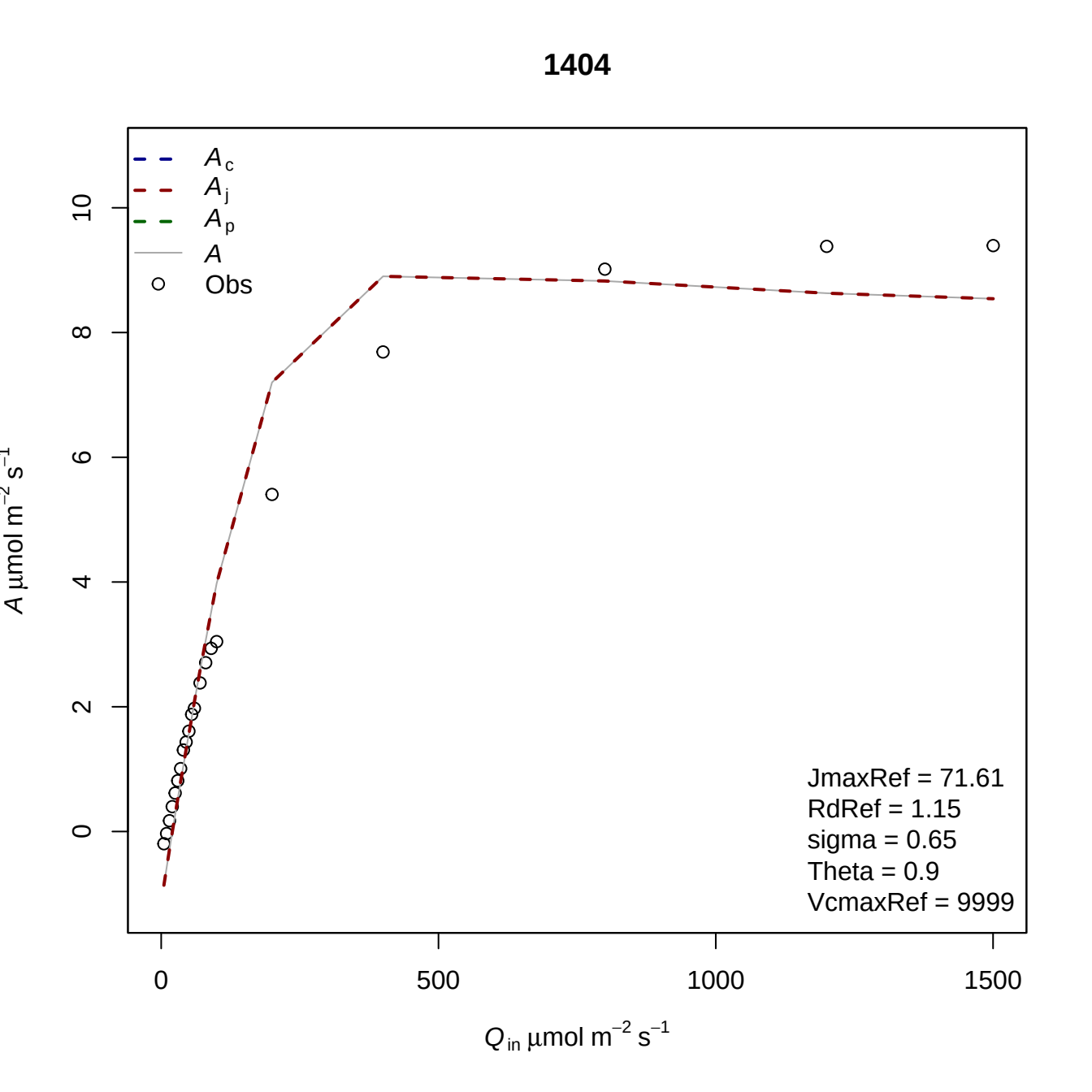

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 46.1
RdRef = 0.45
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

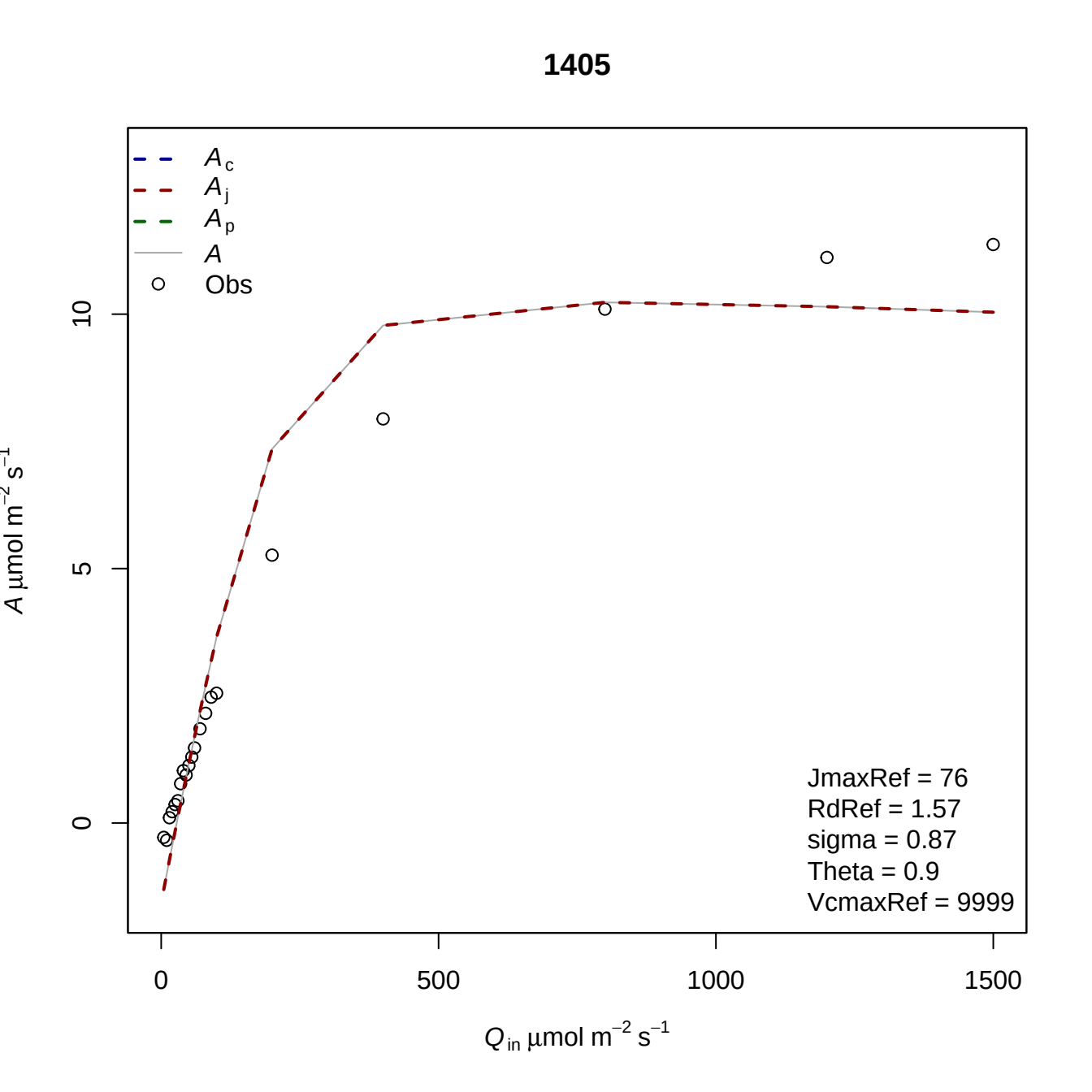

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 36.02
RdRef = 0.89
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

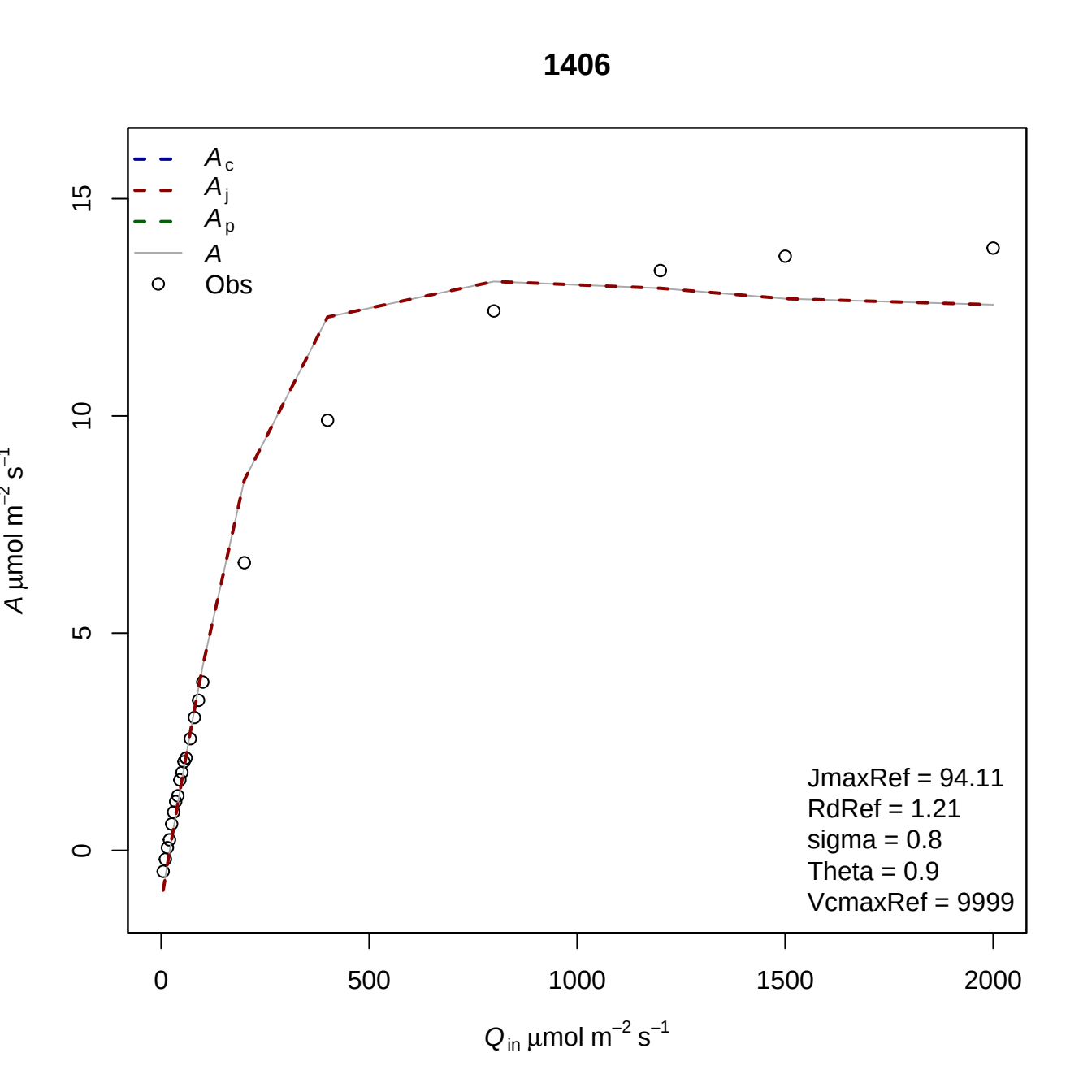

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 66.05
RdRef = 1.11
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

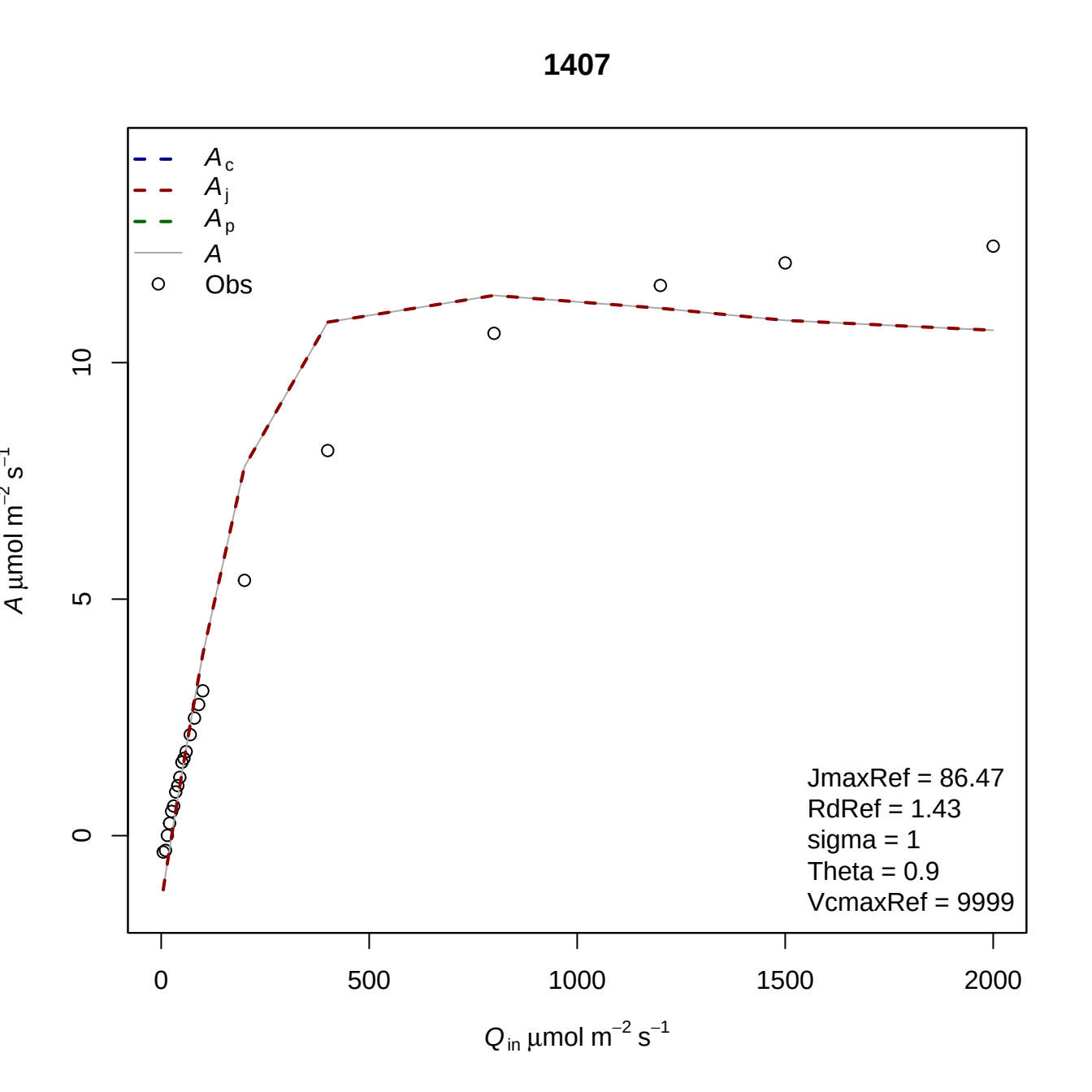

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 29.49
RdRef = 1.54
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

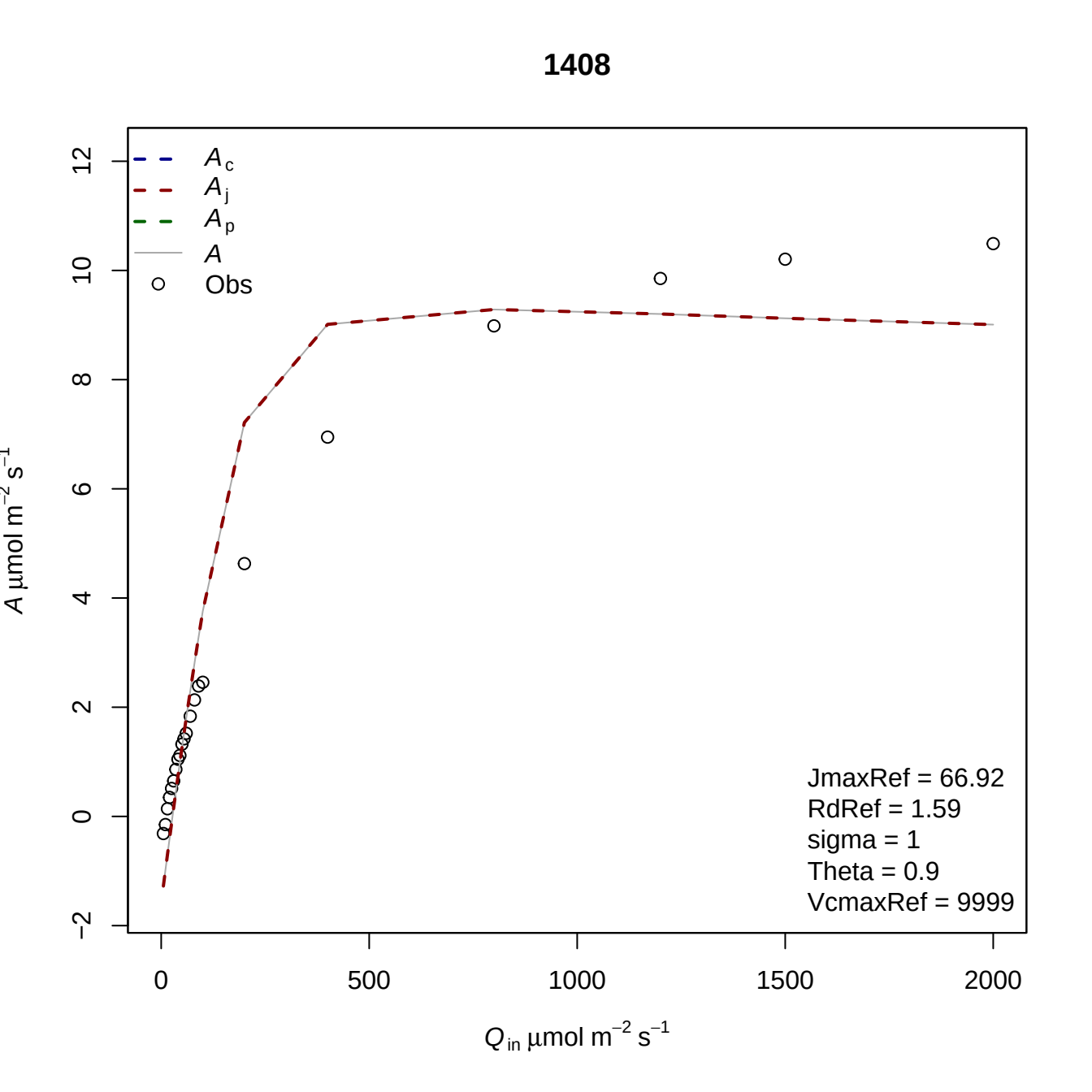

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 70.24
RdRef = 1.23
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

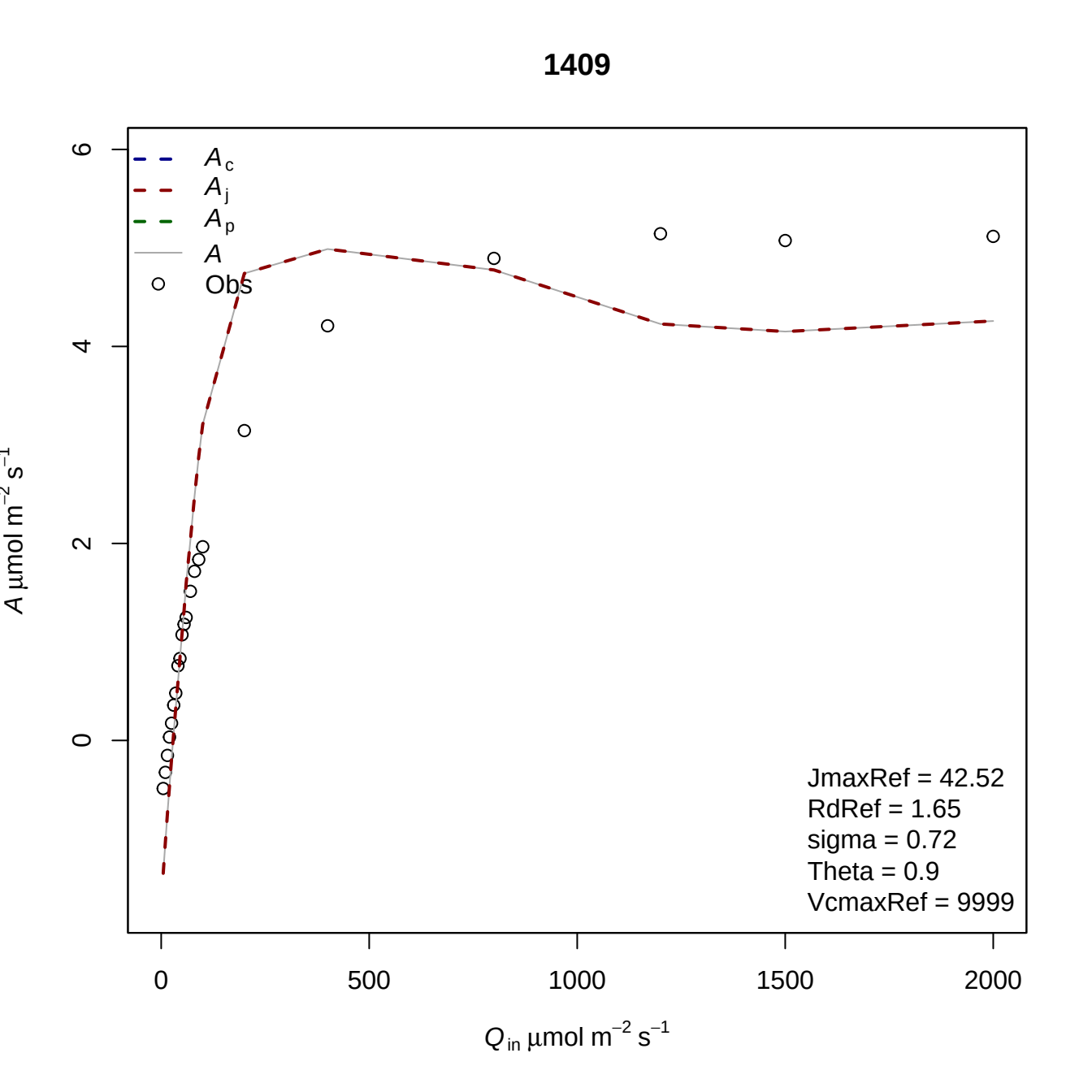

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 71.61
RdRef = 1.15
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

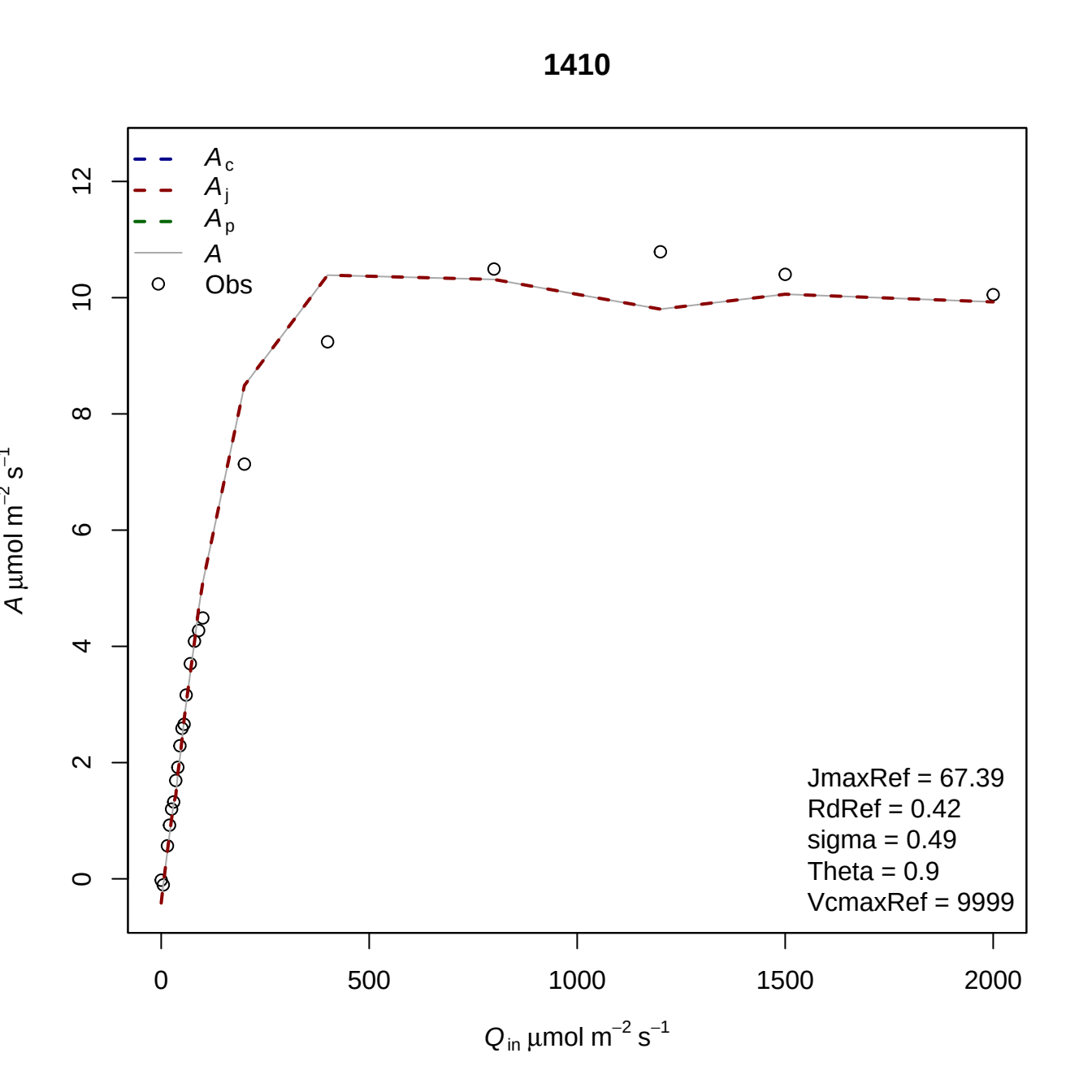

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 76
RdRef = 1.57
sigma = 0.87
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

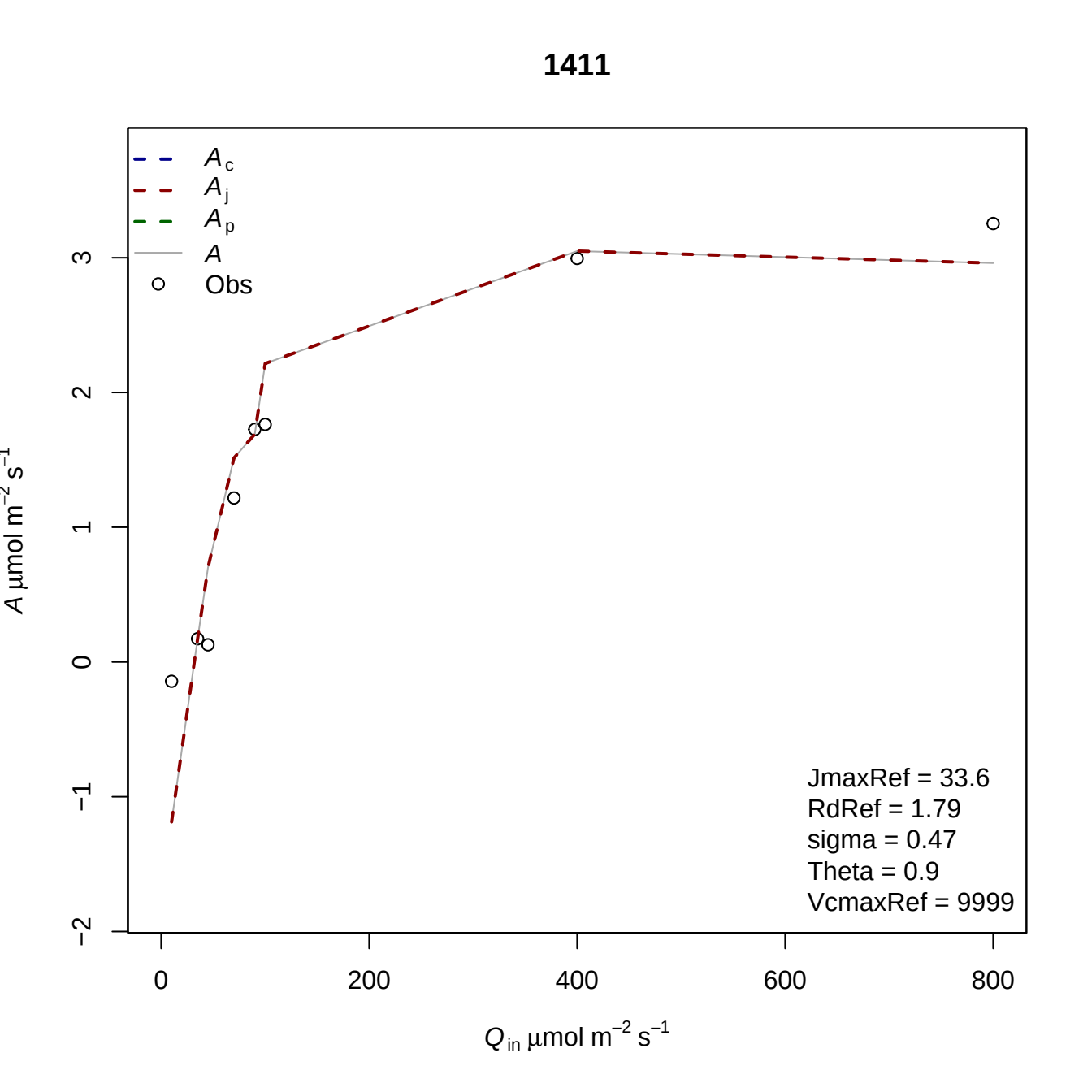

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 94.11
RdRef = 1.21
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

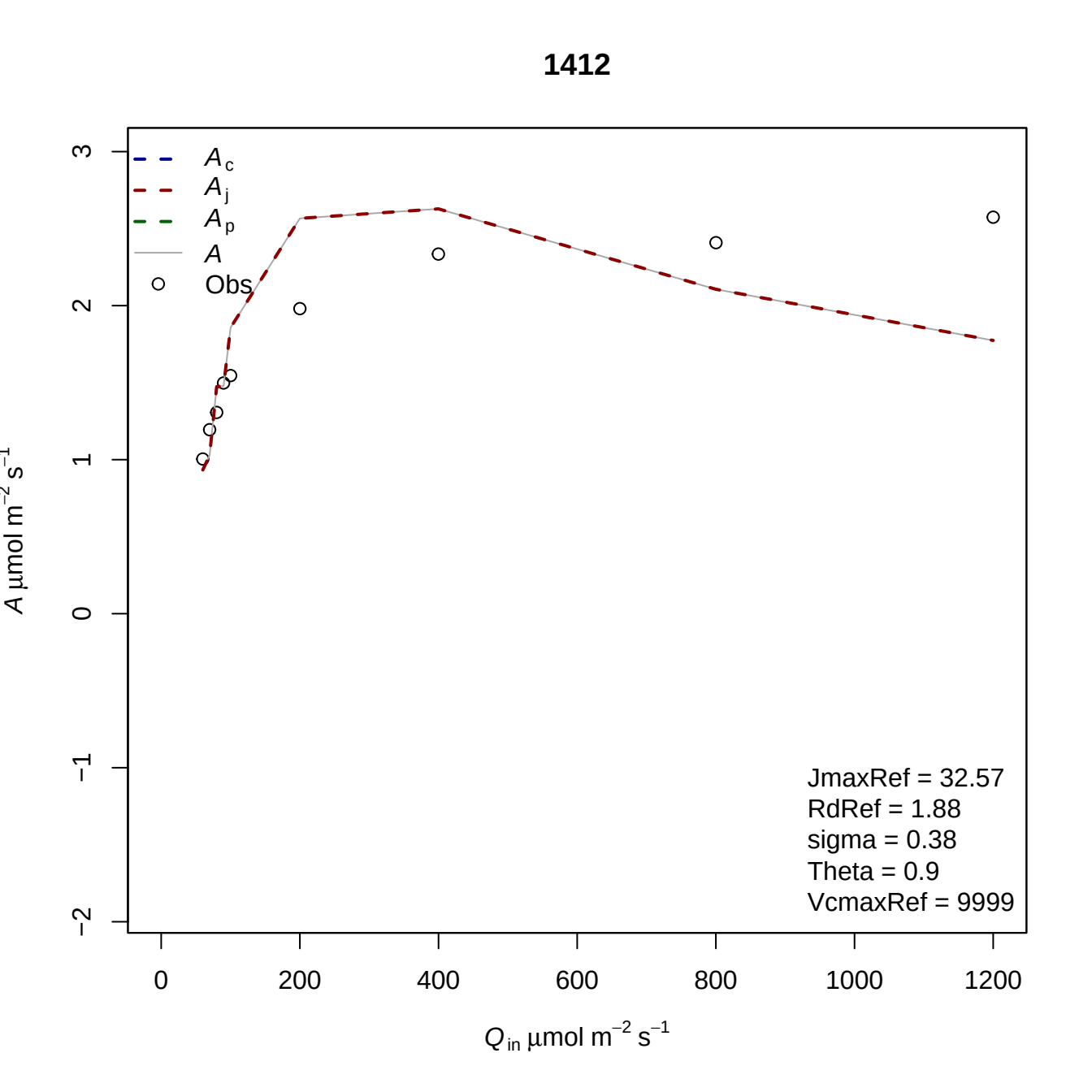

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 86.47
RdRef = 1.43
sigma = 1
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

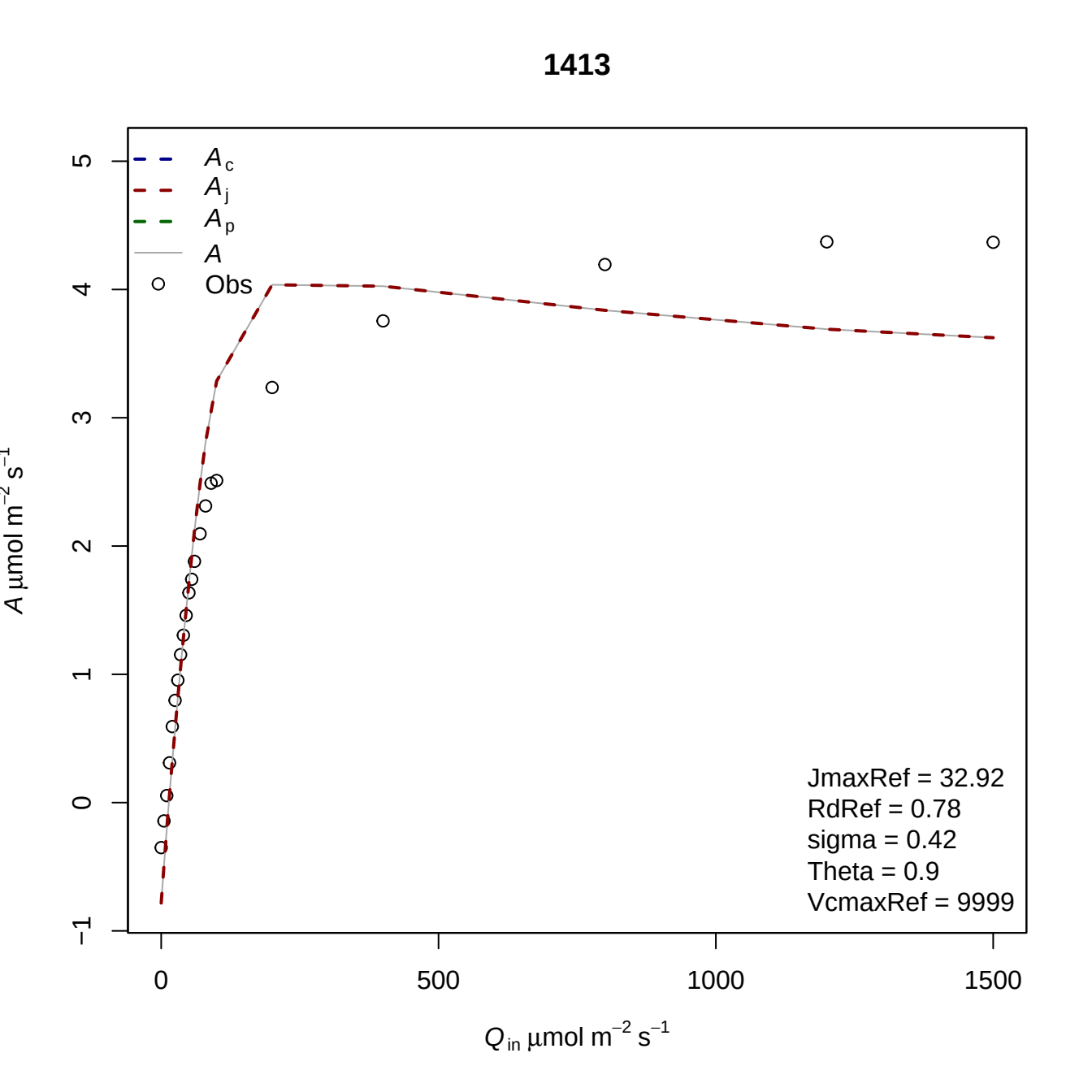

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 66.92
RdRef = 1.59
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

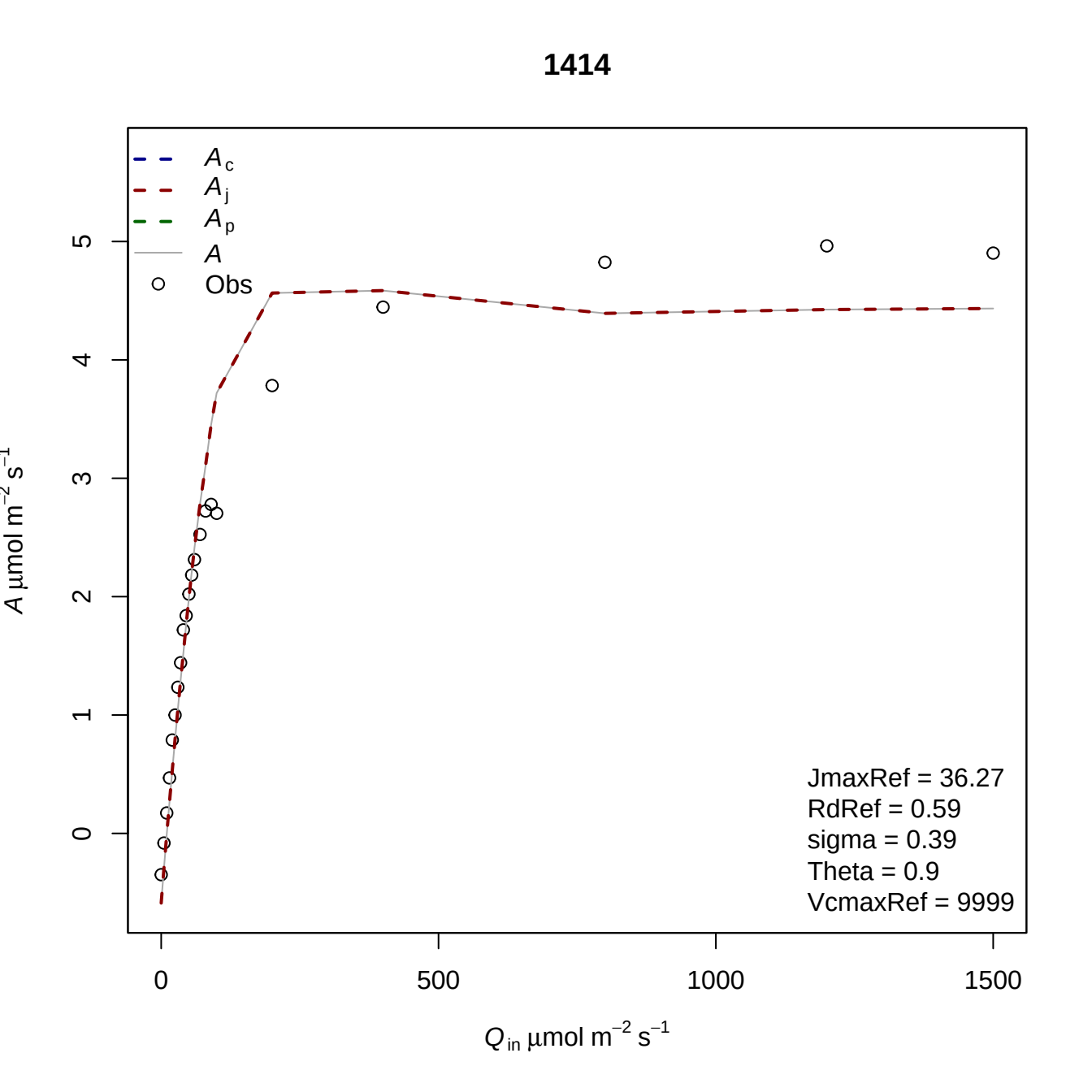

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 42.52
RdRef = 1.65
sigma = 0.72
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

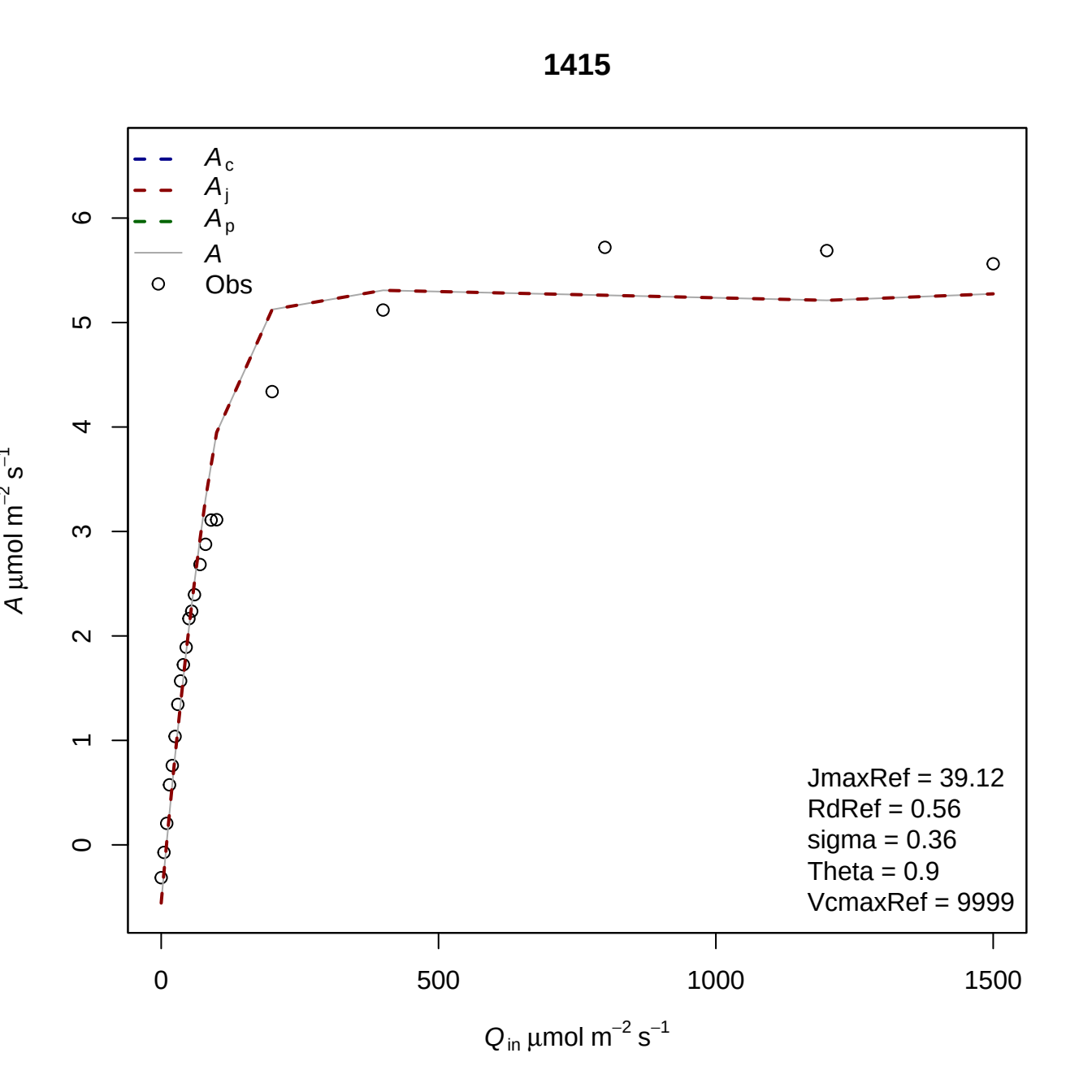

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 67.39
RdRef = 0.42
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

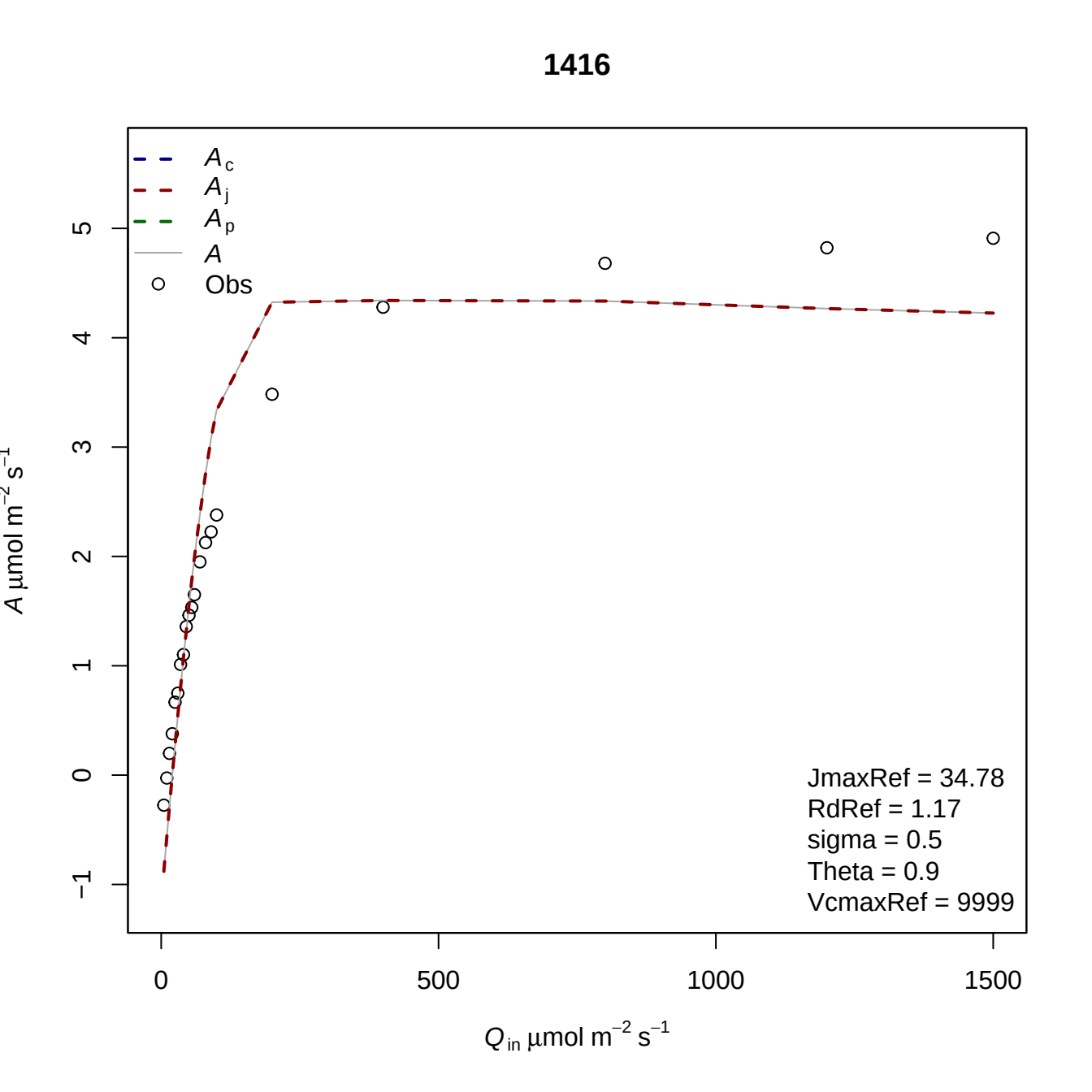

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 33.6
RdRef = 1.79
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800

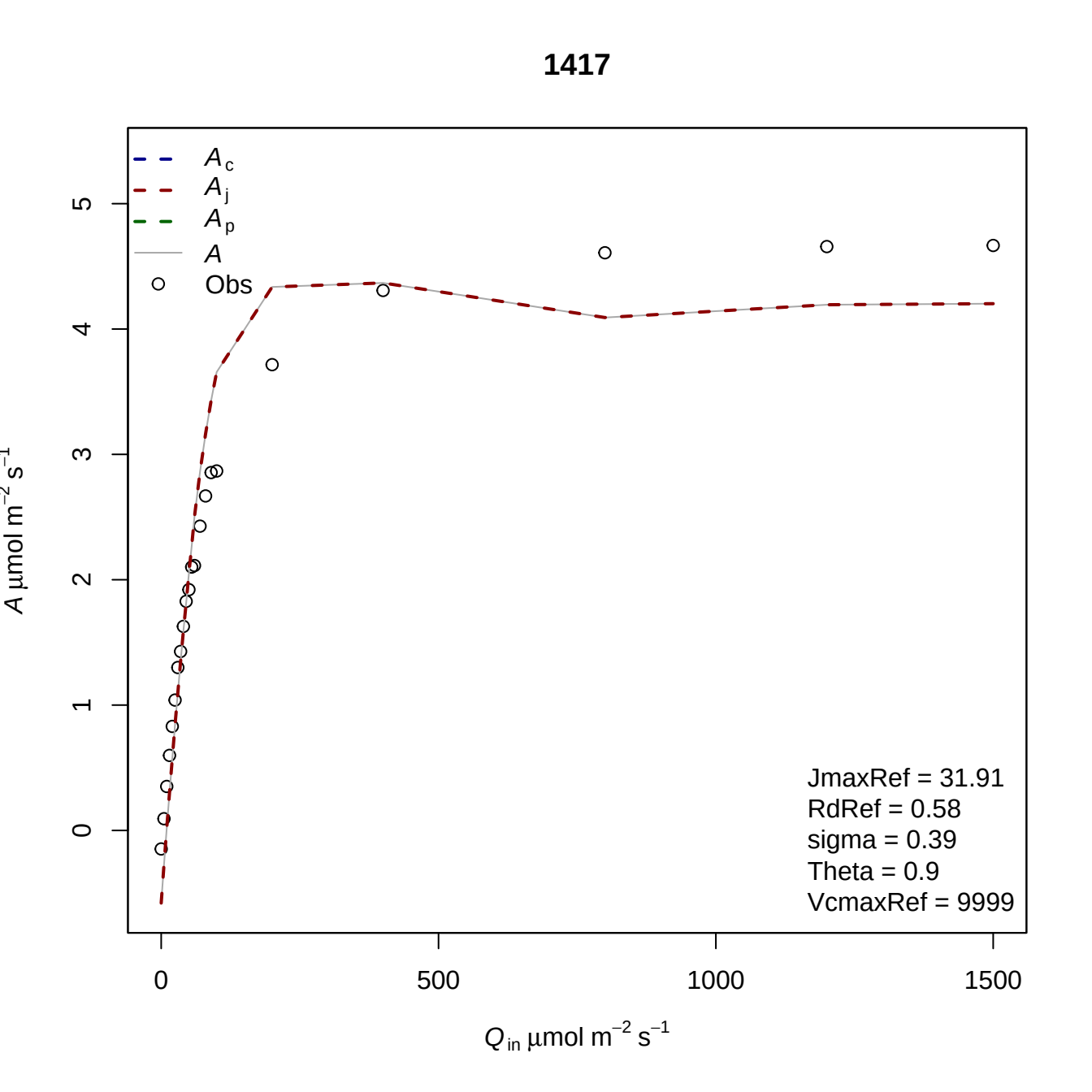

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 32.57
RdRef = 1.88
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200

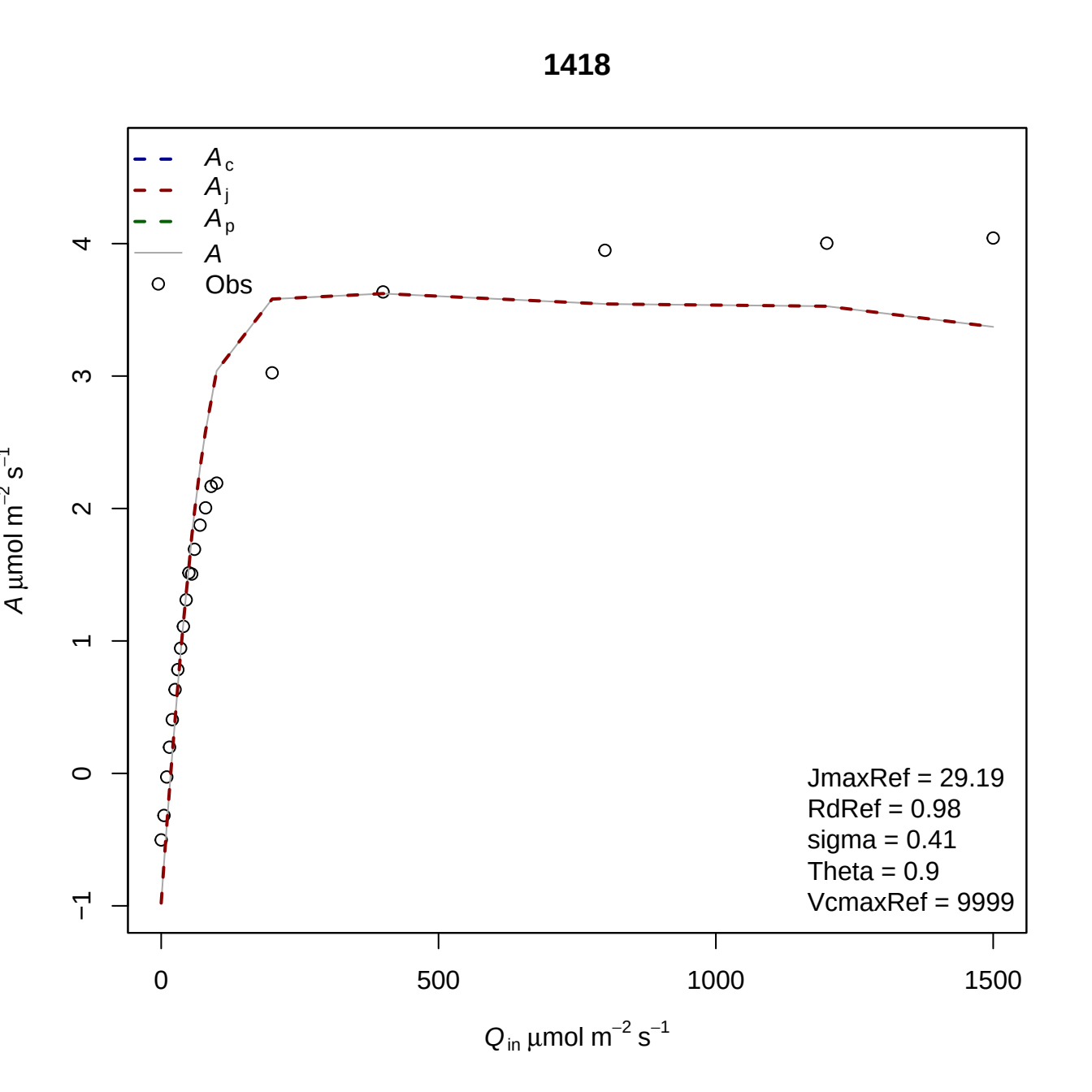

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 32.92
RdRef = 0.78
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

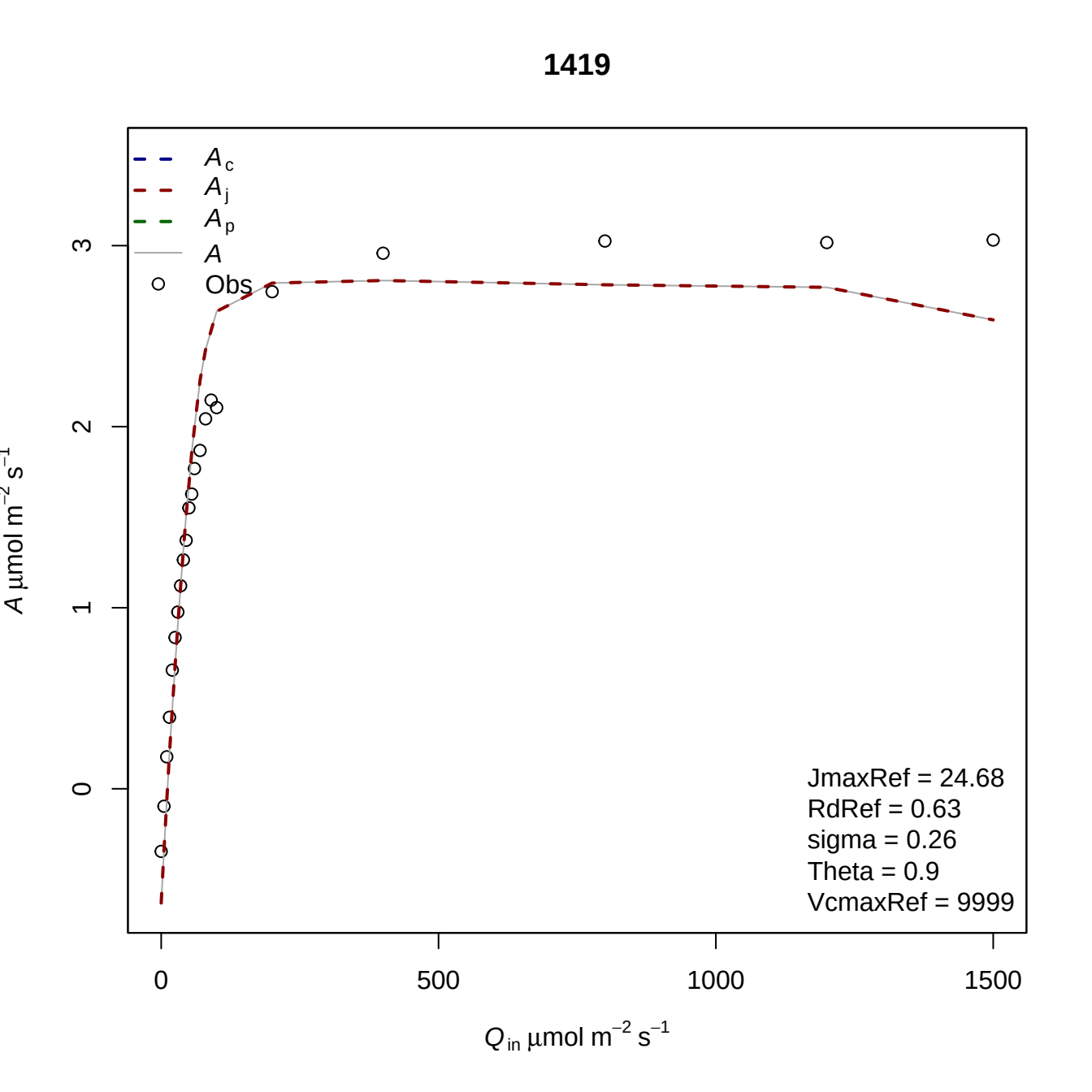

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 36.27
RdRef = 0.59
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

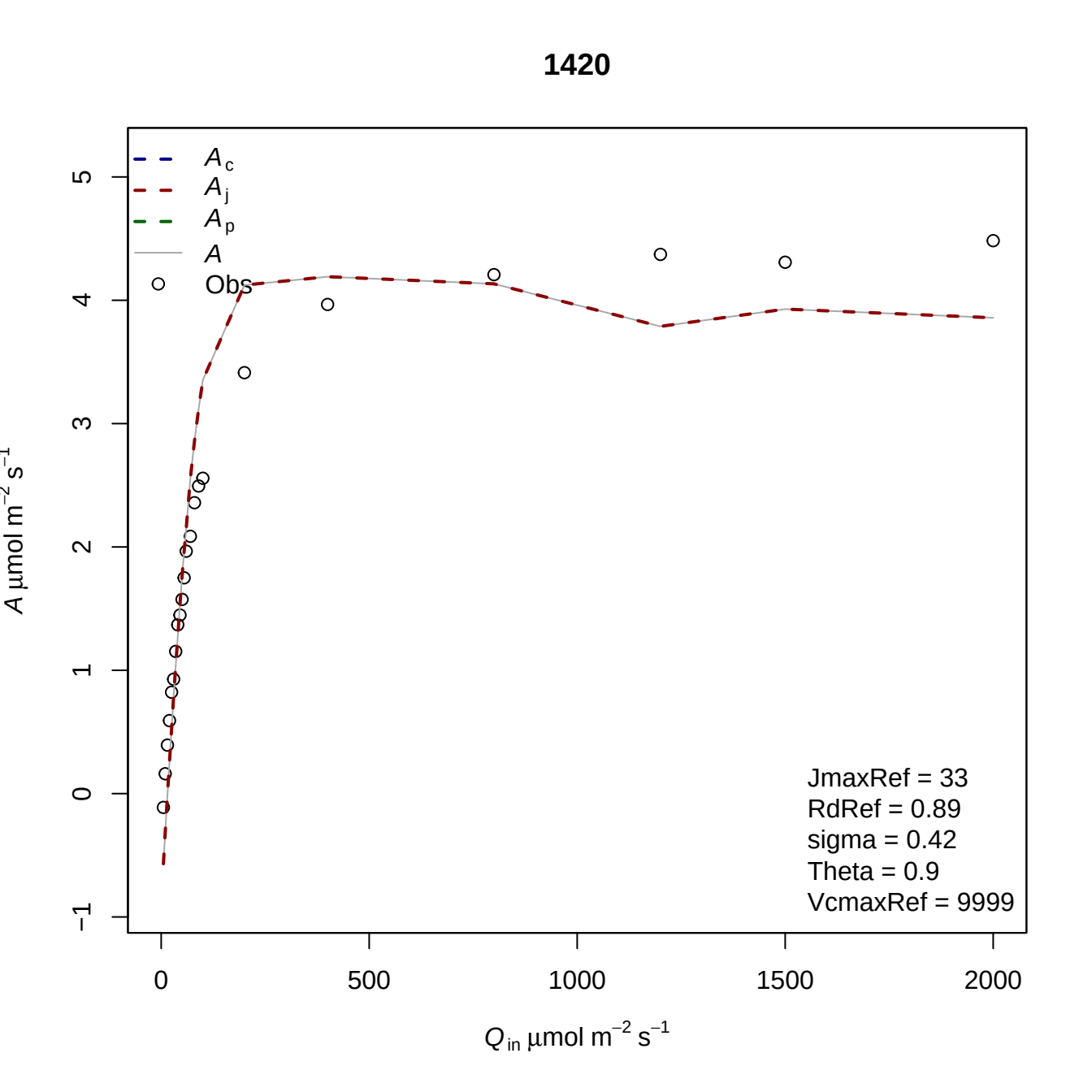

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 39.12
RdRef = 0.56
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

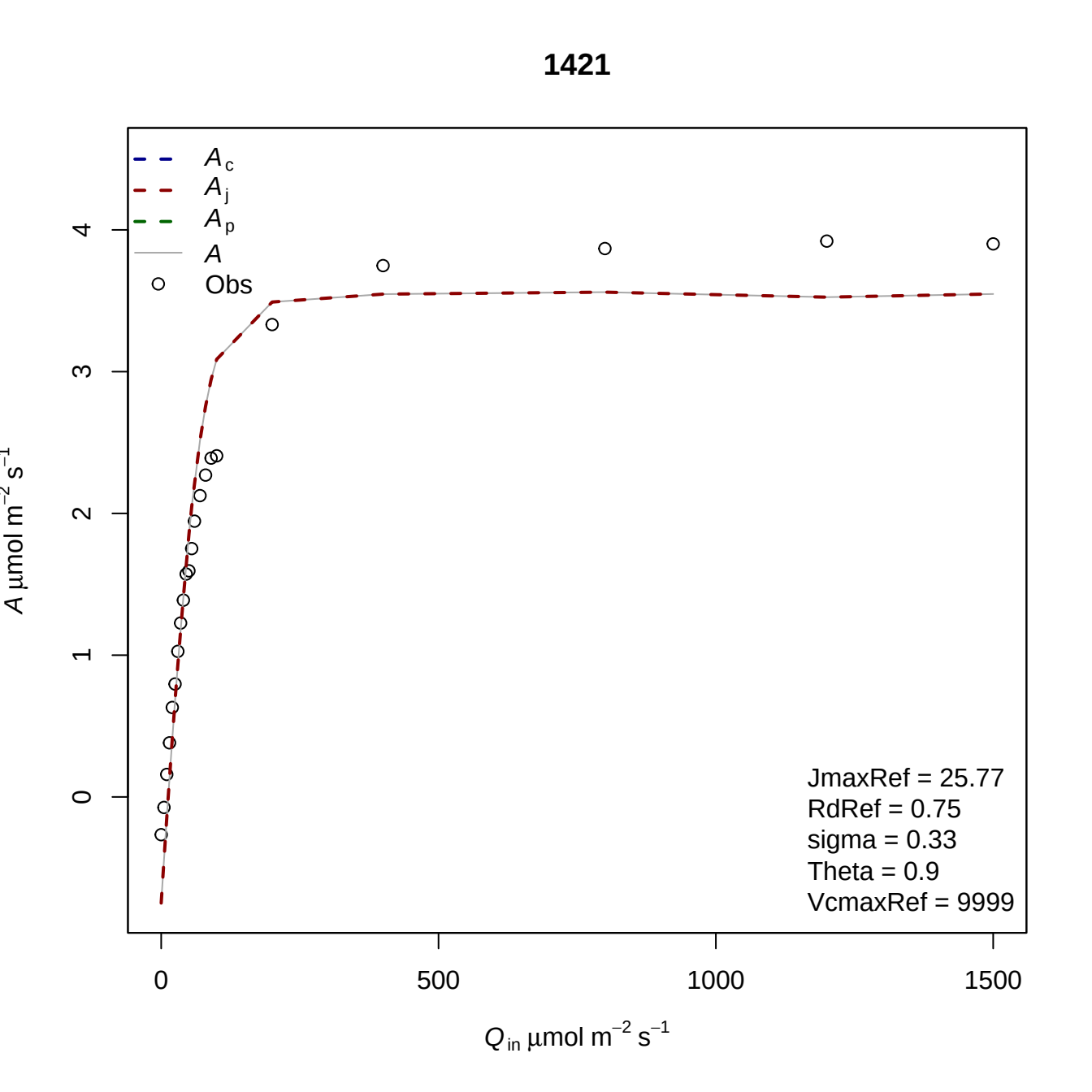

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 34.78
RdRef = 1.17
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

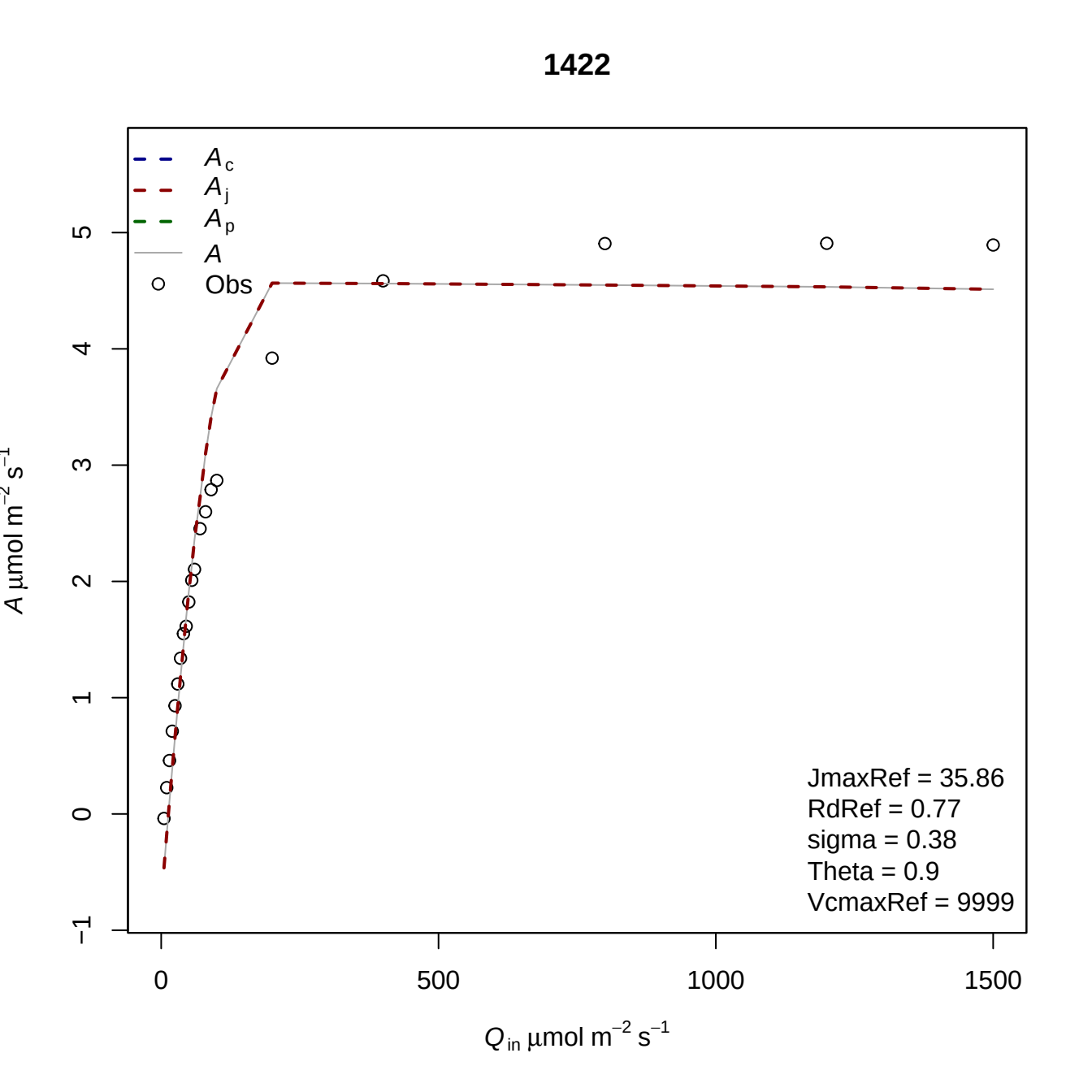

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 31.91
RdRef = 0.58
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

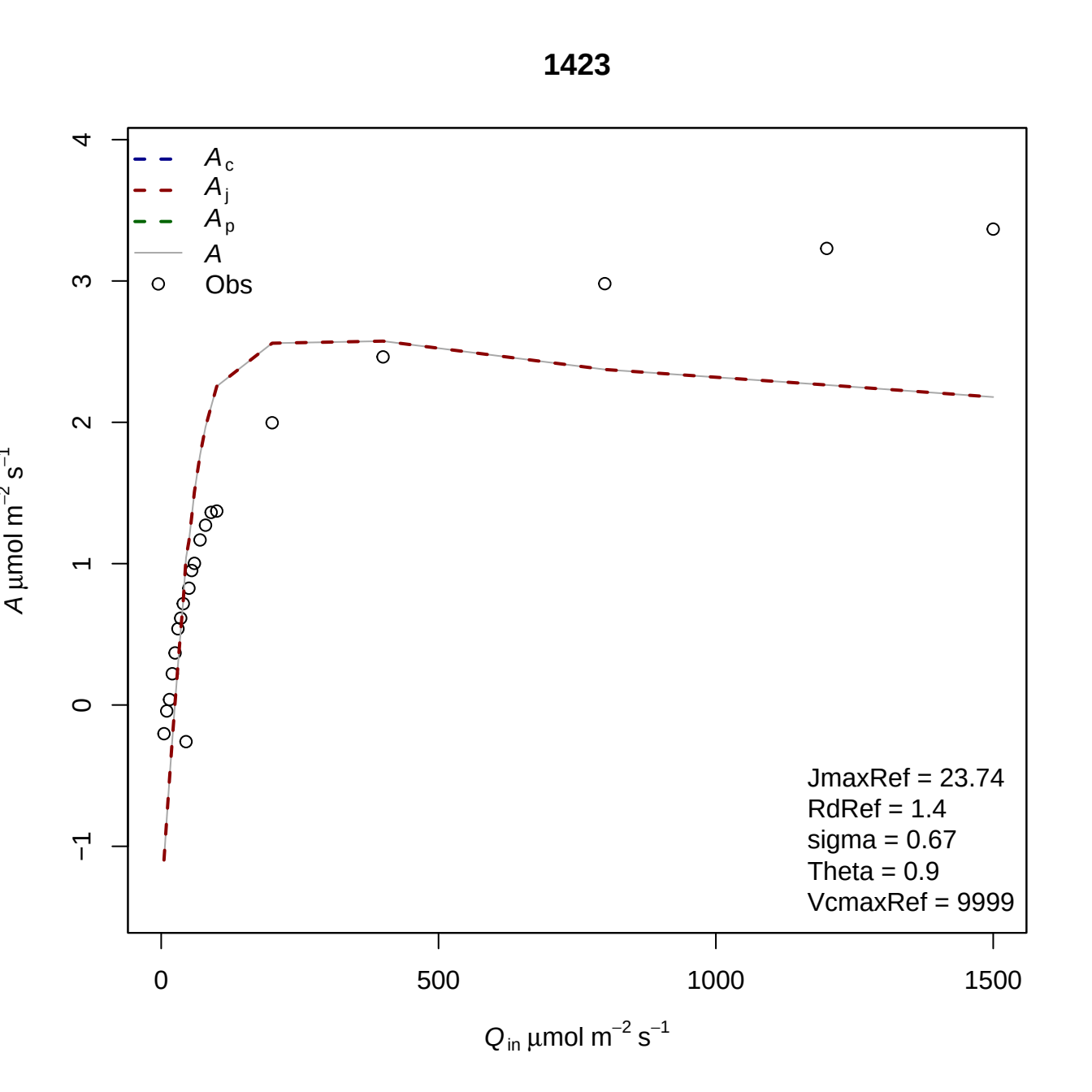

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 29.19
RdRef = 0.98
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

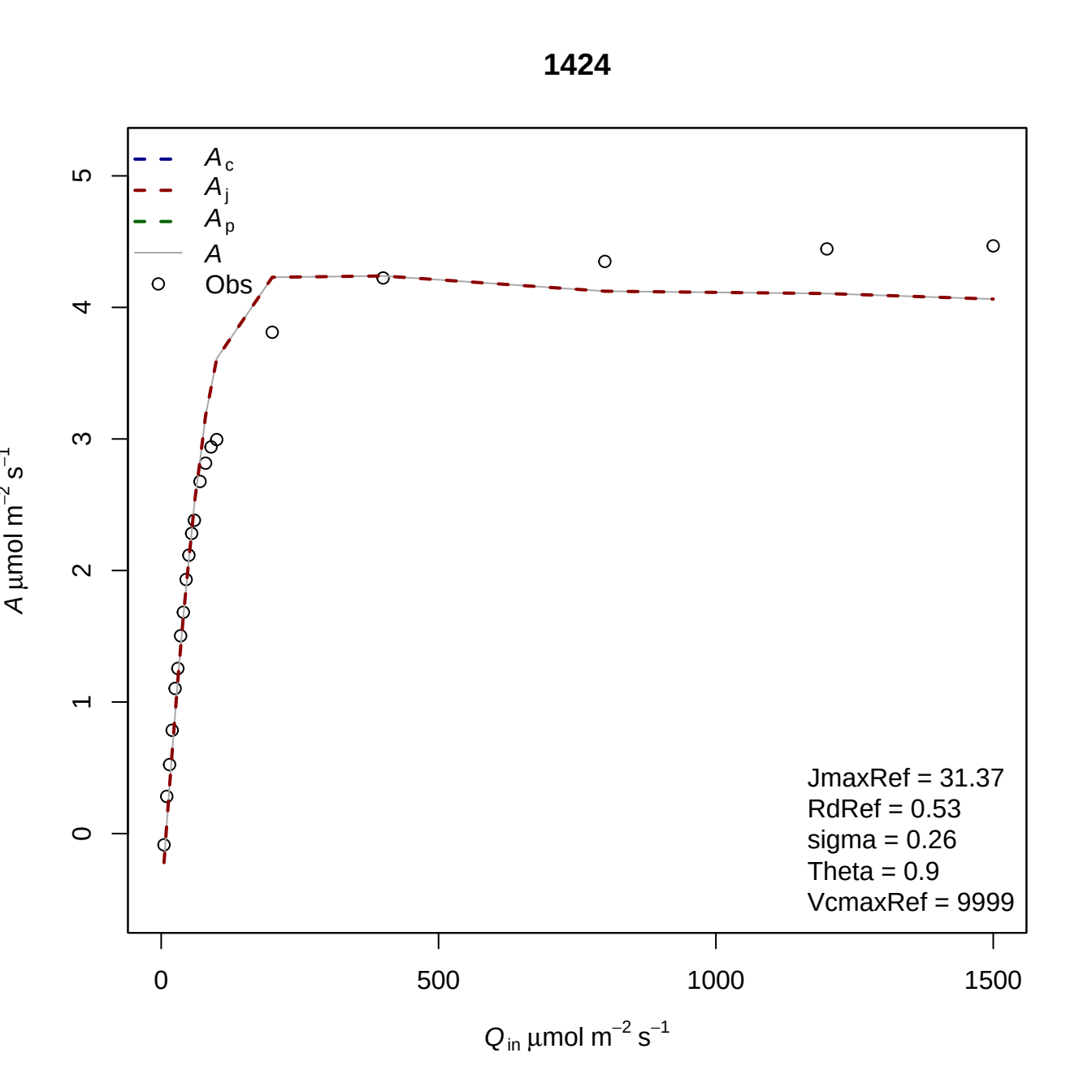

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 24.68
RdRef = 0.63
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

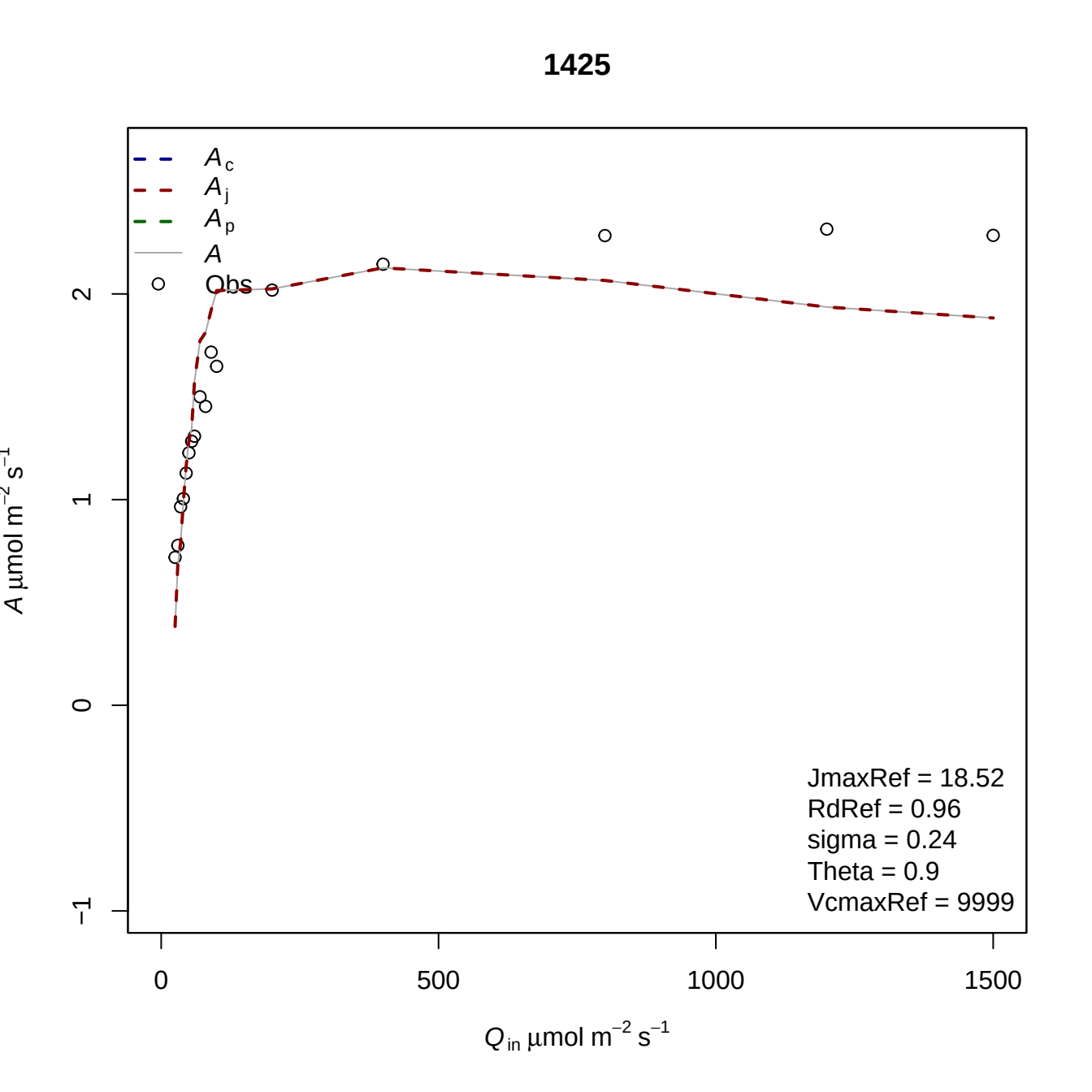

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 33
RdRef = 0.89
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

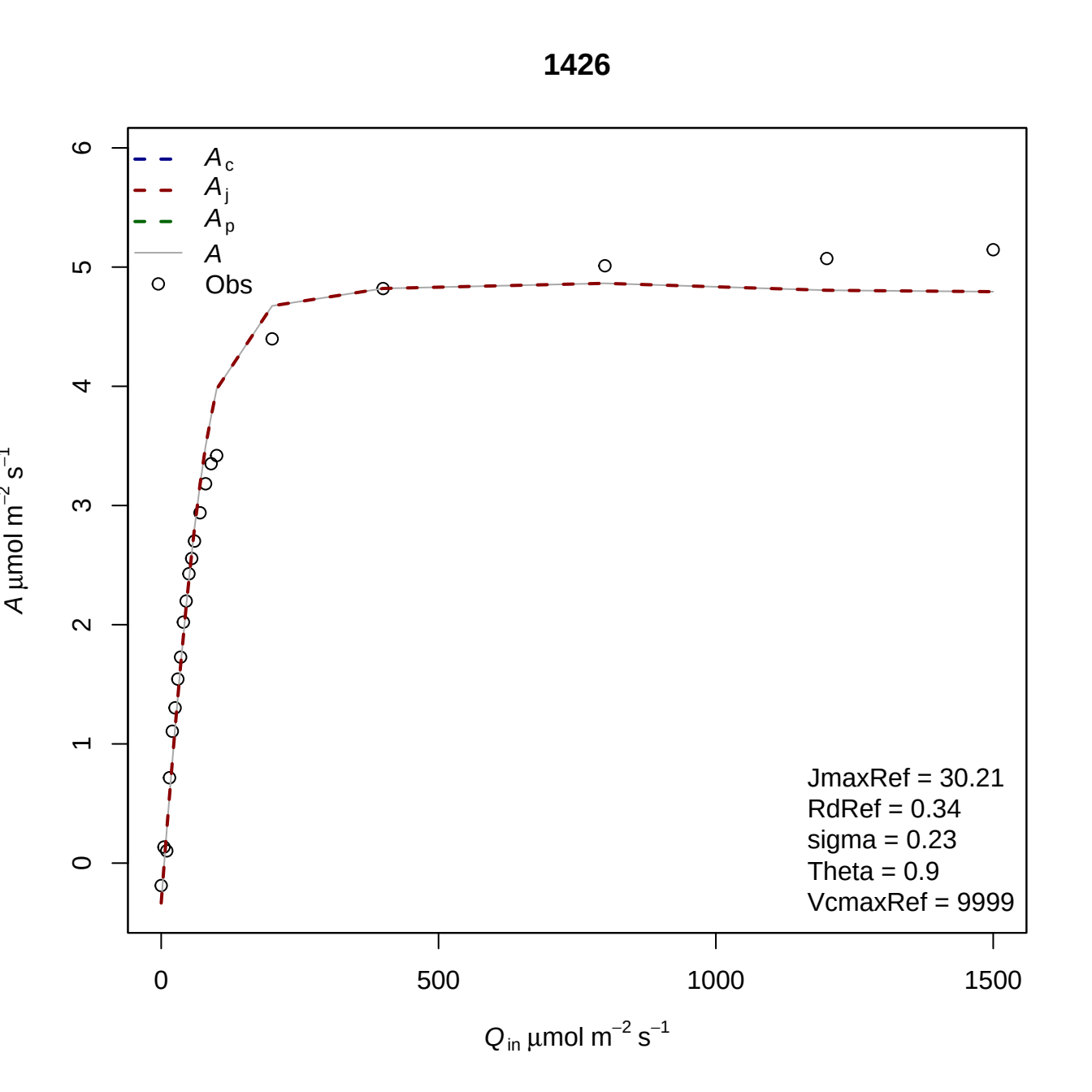

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 25.77
RdRef = 0.75
sigma = 0.33
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

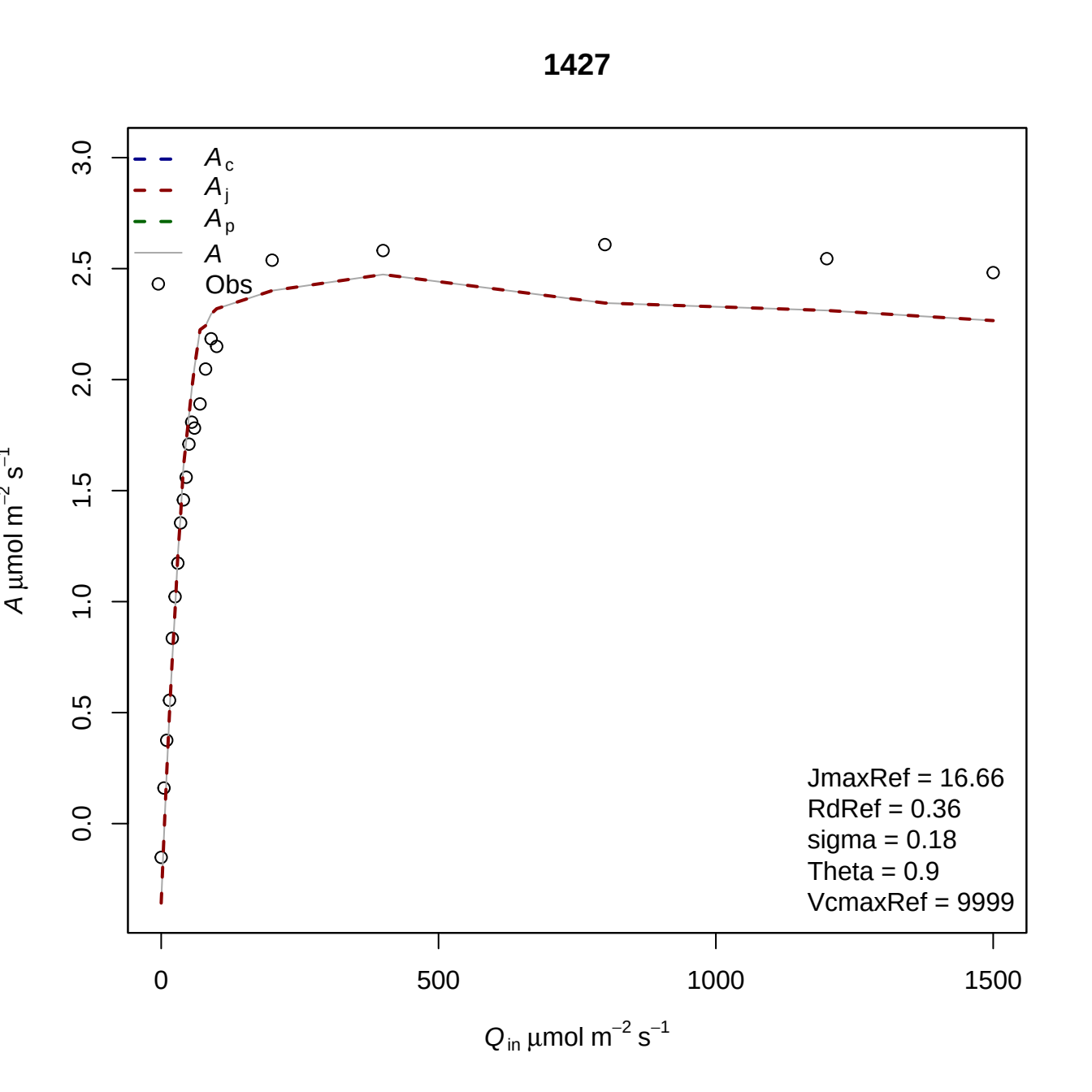

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 35.86
RdRef = 0.77
sigma = 0.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

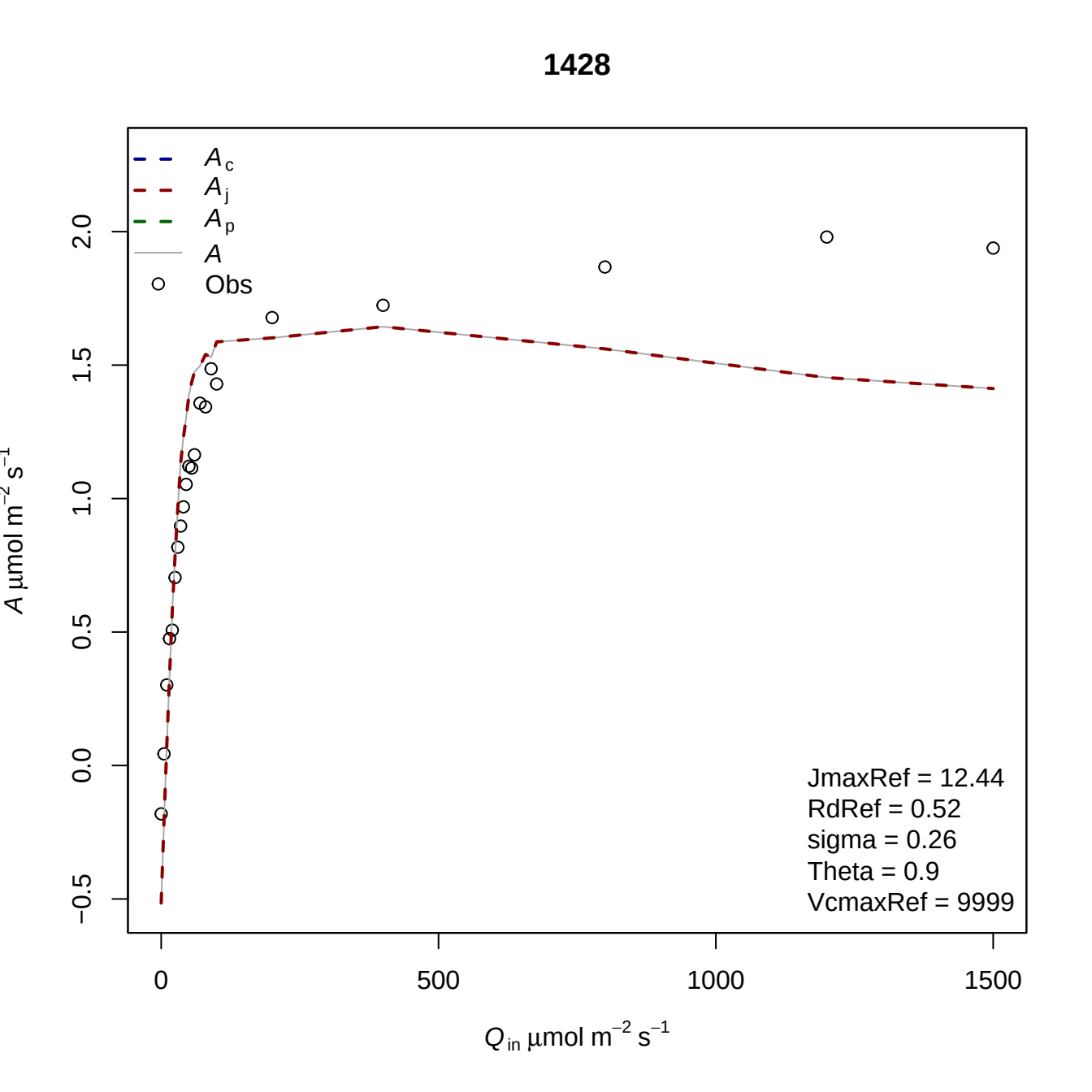

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 23.74
RdRef = 1.4
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

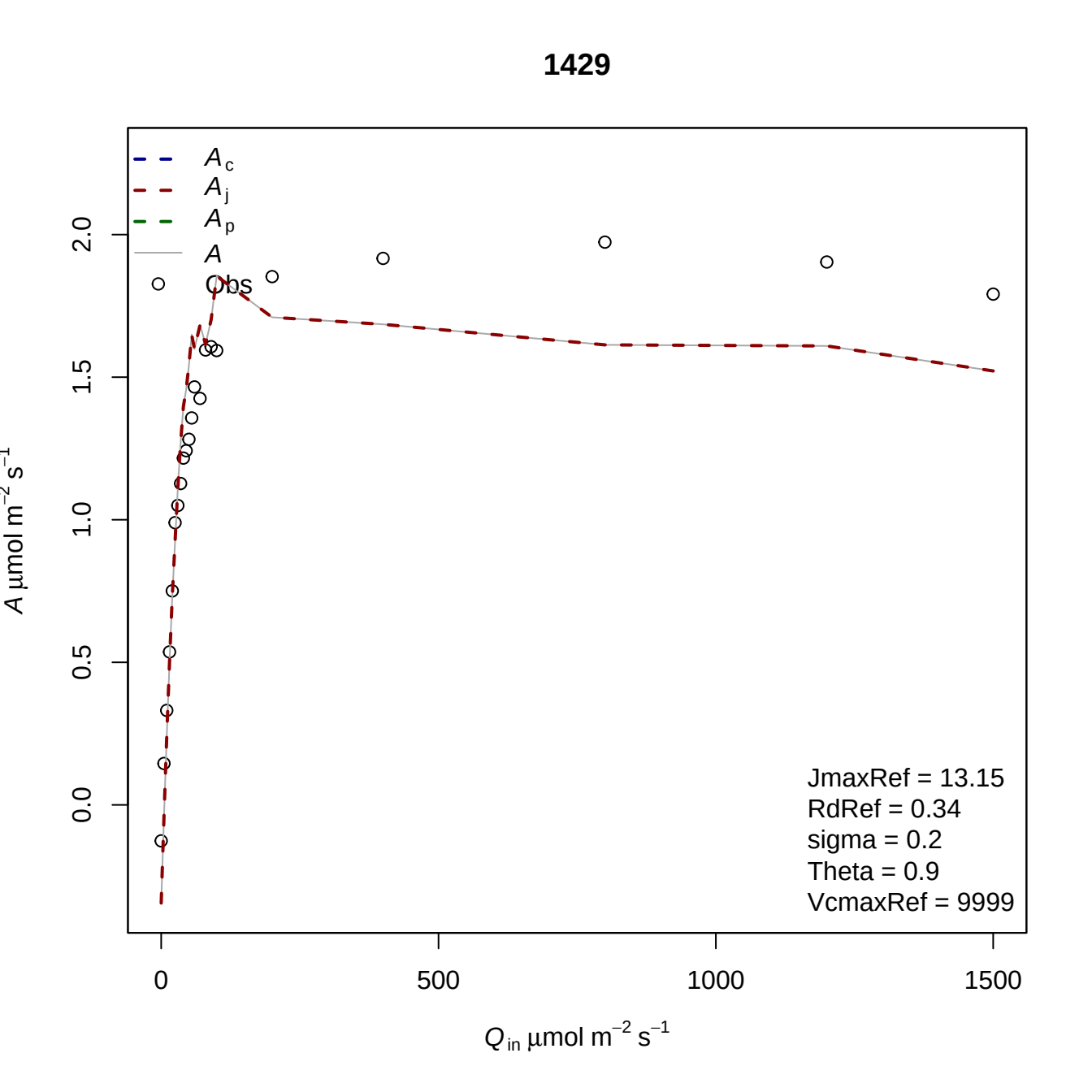

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 31.37
RdRef = 0.53
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

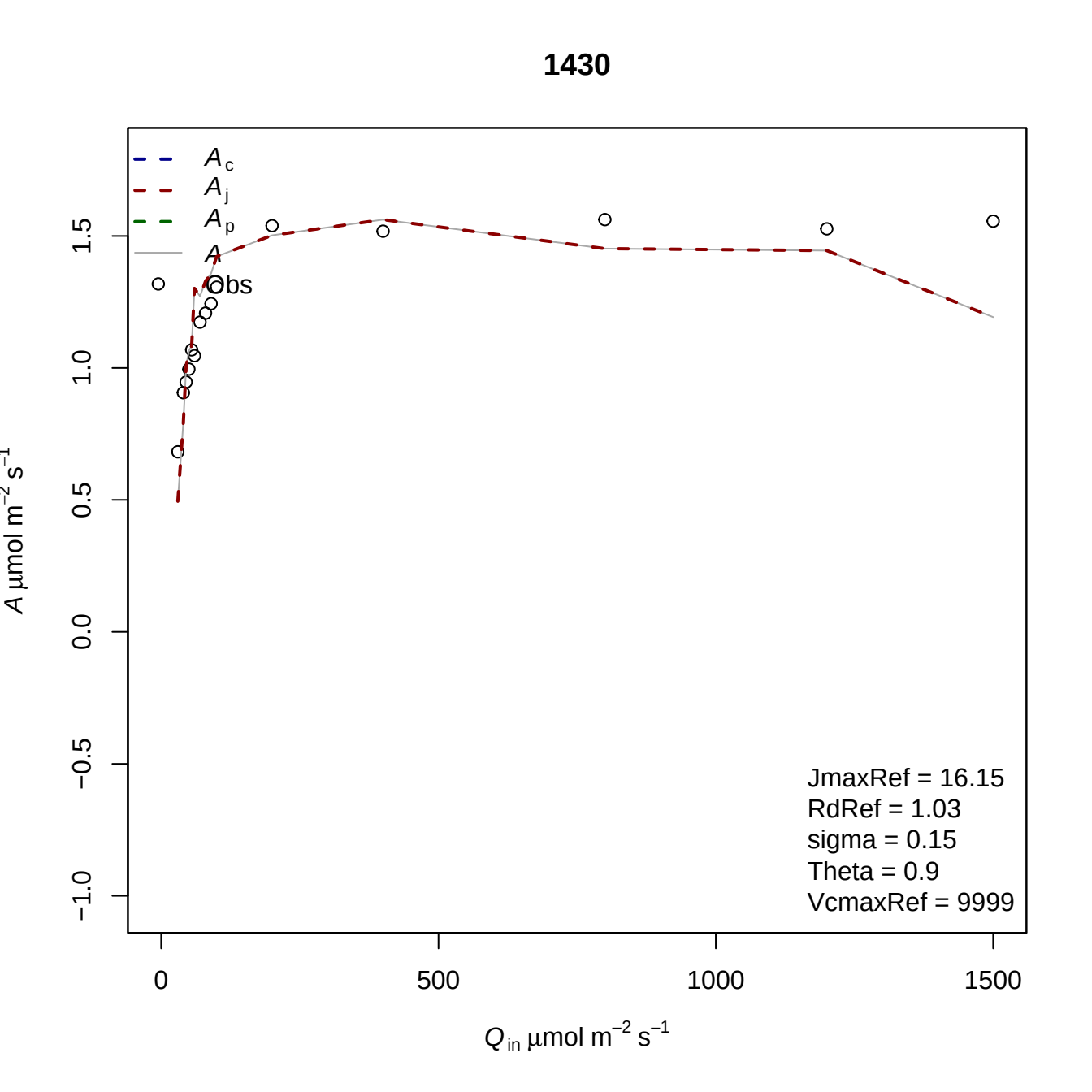

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 18.52
RdRef = 0.96
sigma = 0.24
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

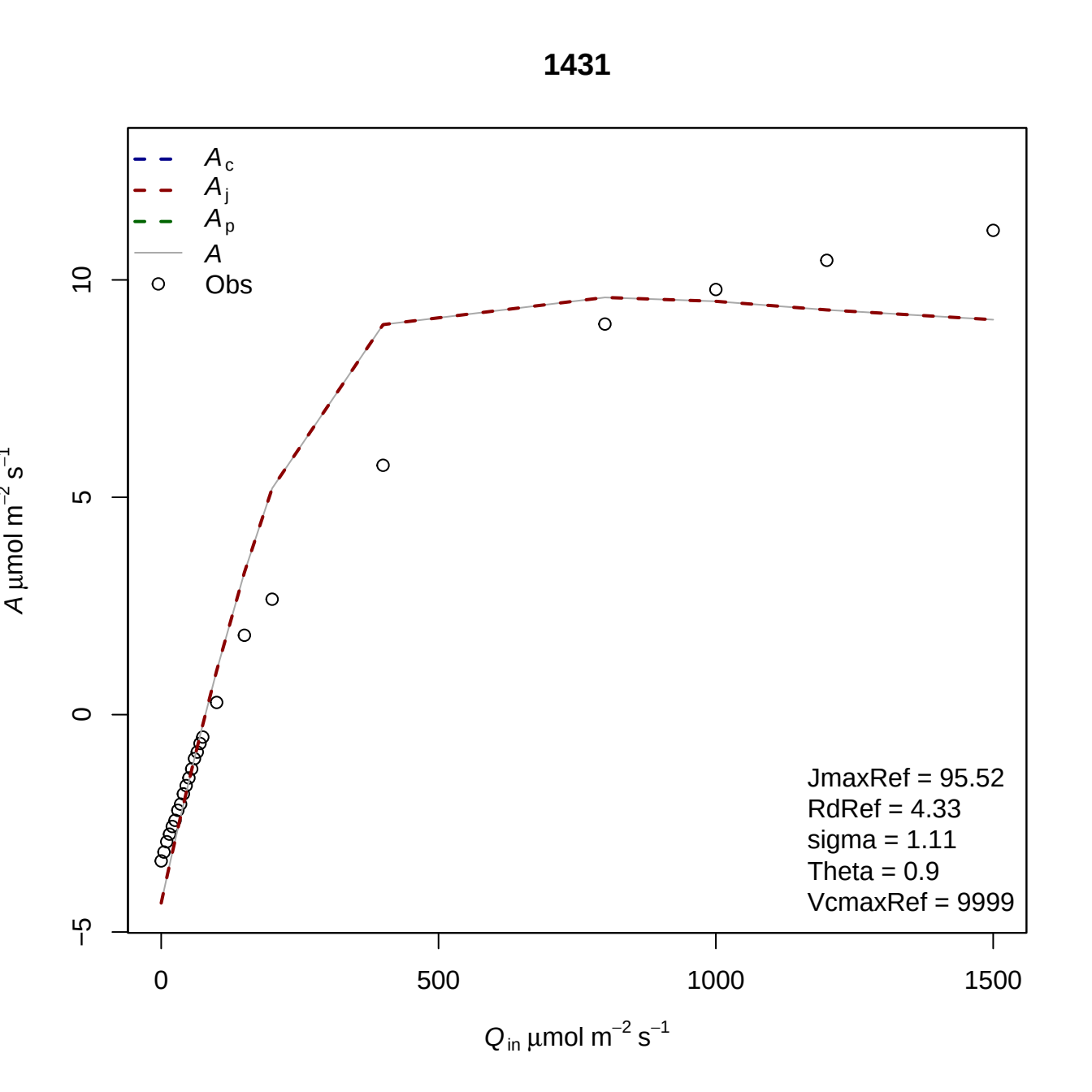

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 30.21
RdRef = 0.34
sigma = 0.23
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

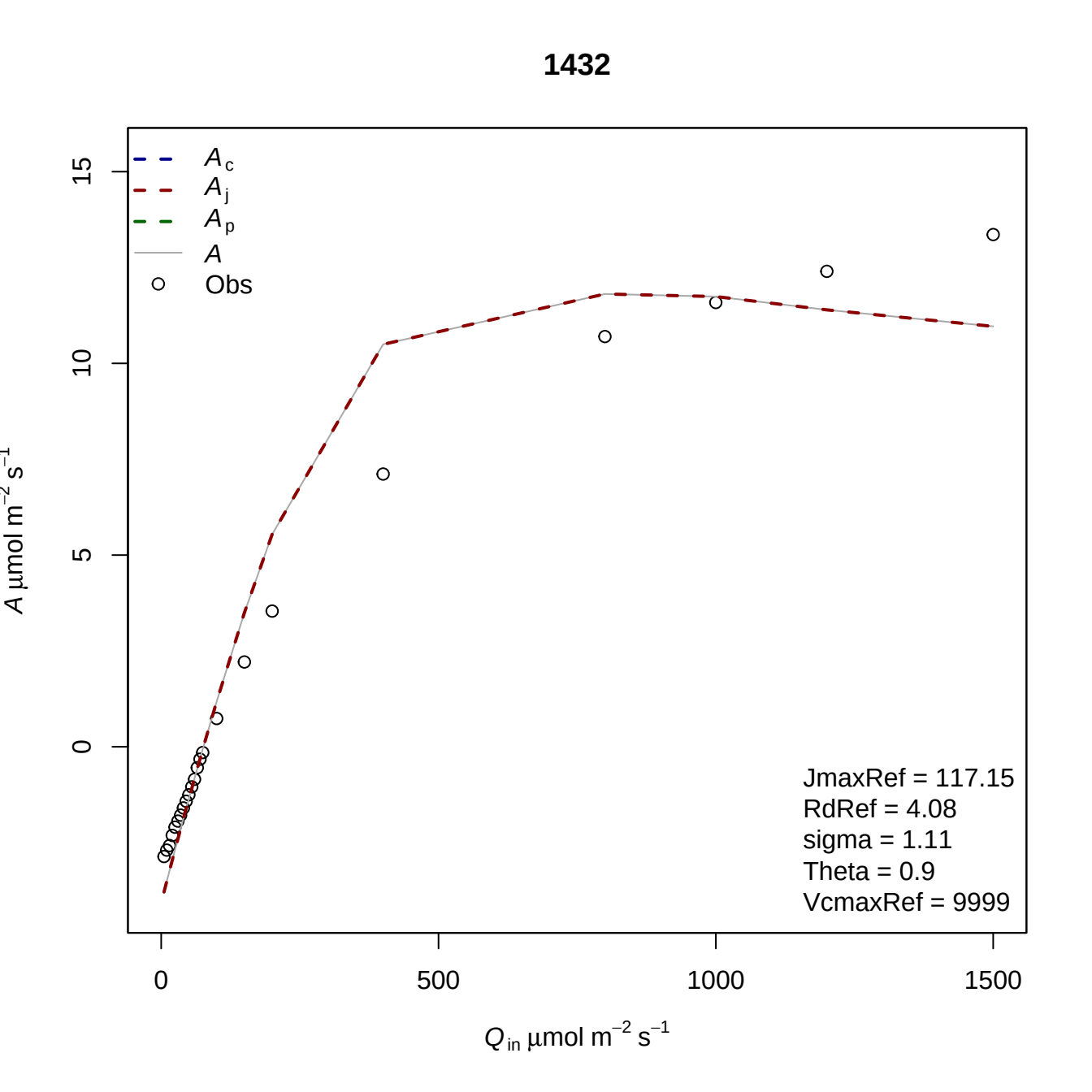

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 16.66
RdRef = 0.36
sigma = 0.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

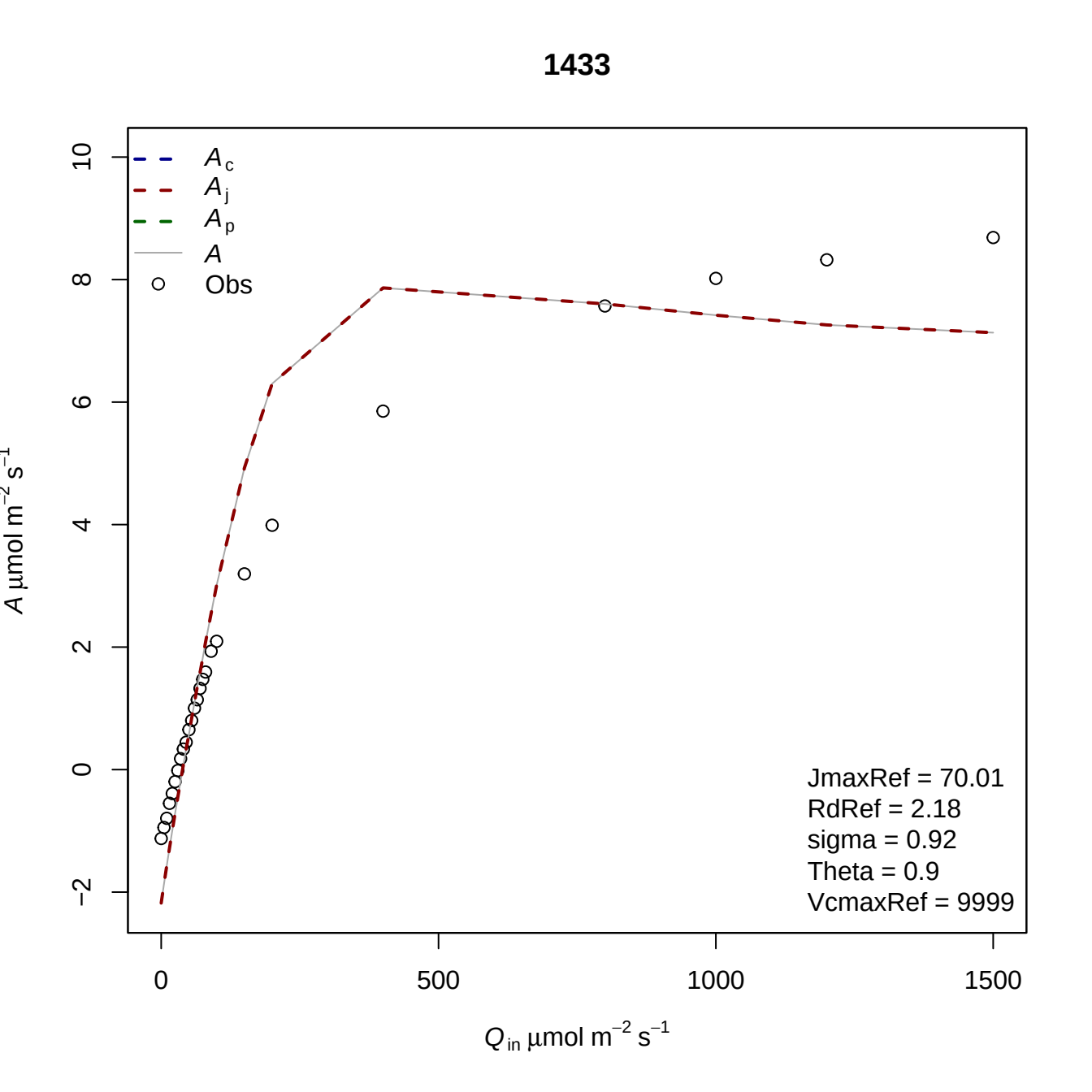

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 12.44
RdRef = 0.52
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

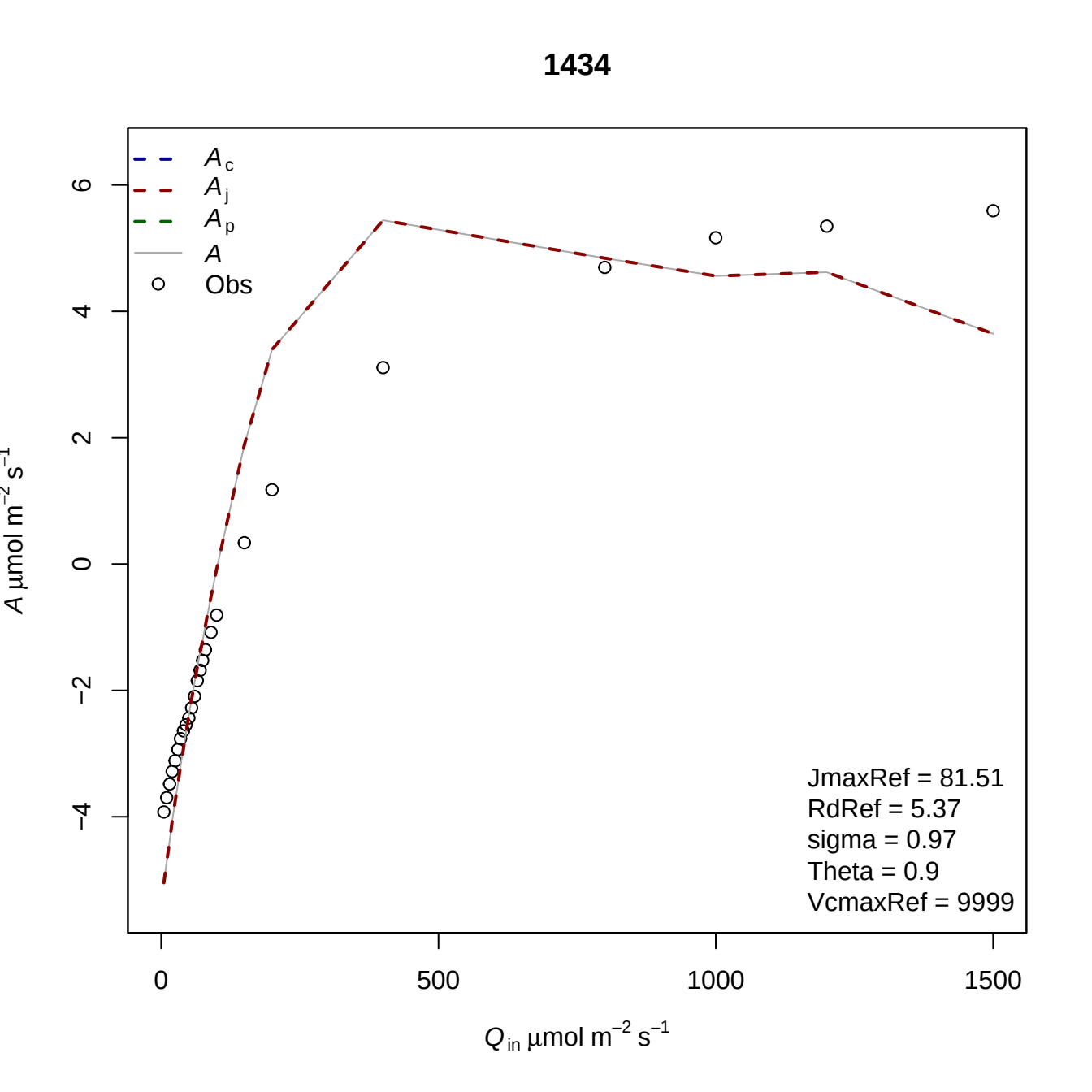

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 13.15
RdRef = 0.34
sigma = 0.2
Theta = 0.9
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

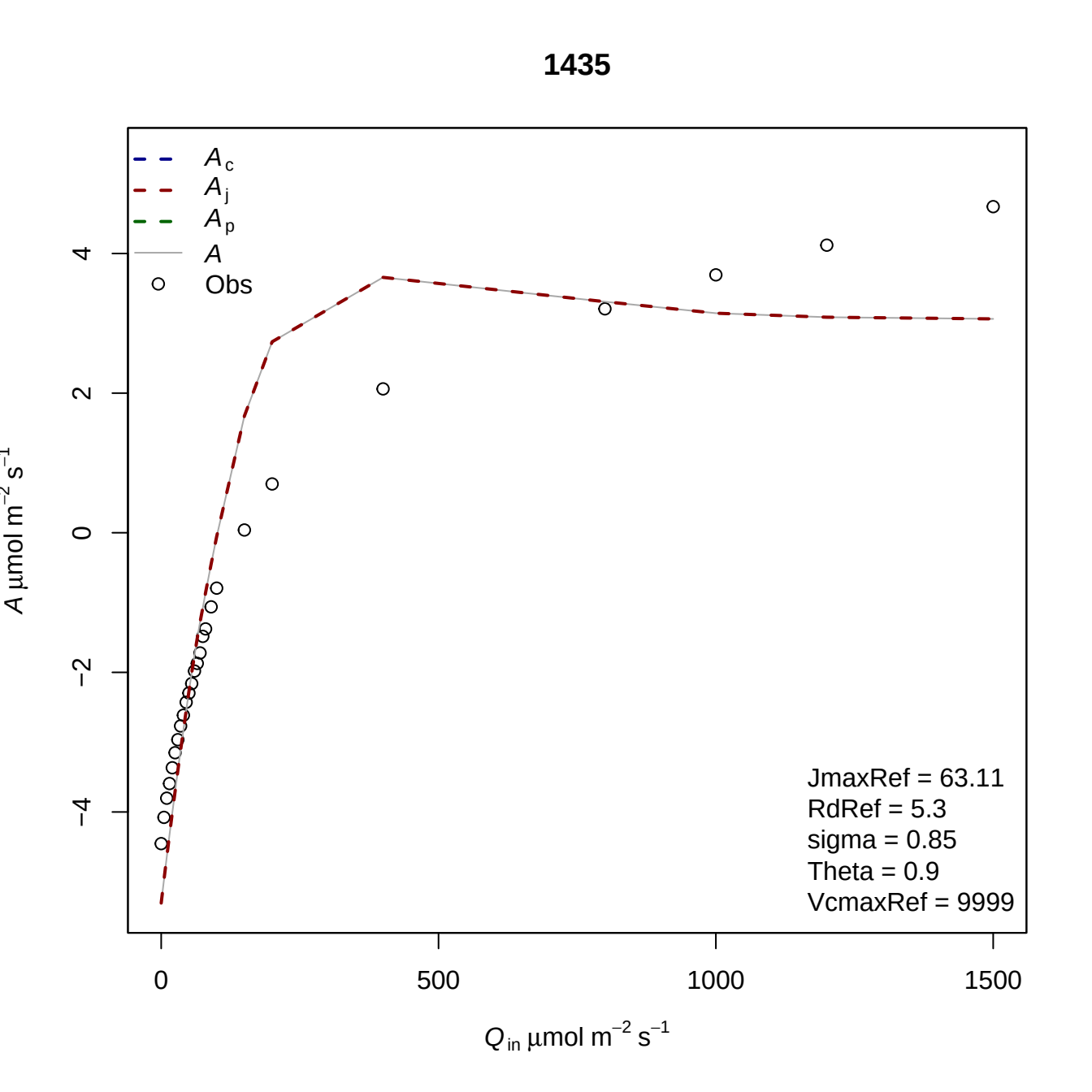

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 16.15
RdRef = 1.03
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

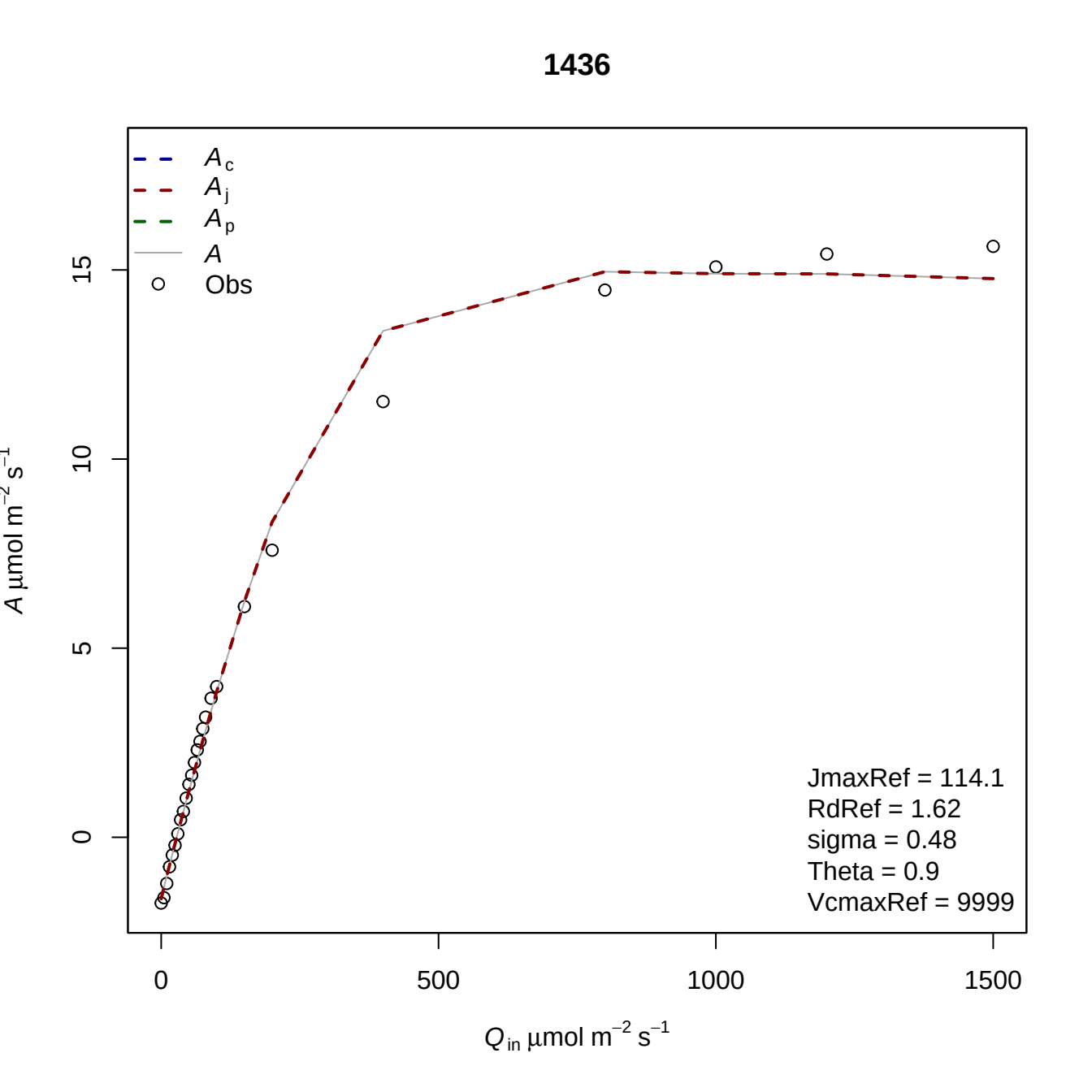

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 95.52
RdRef = 4.33
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

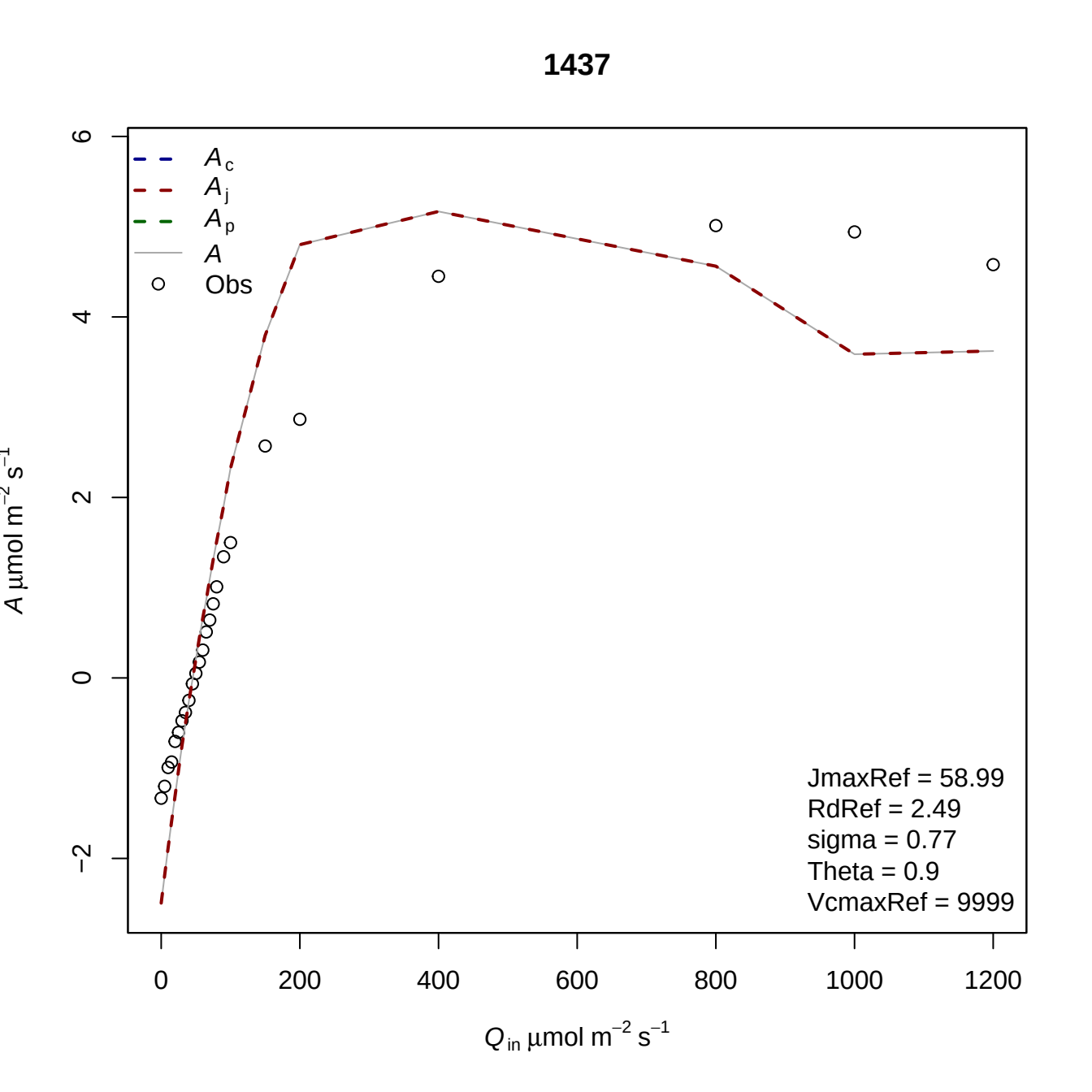

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 117.15
RdRef = 4.08
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

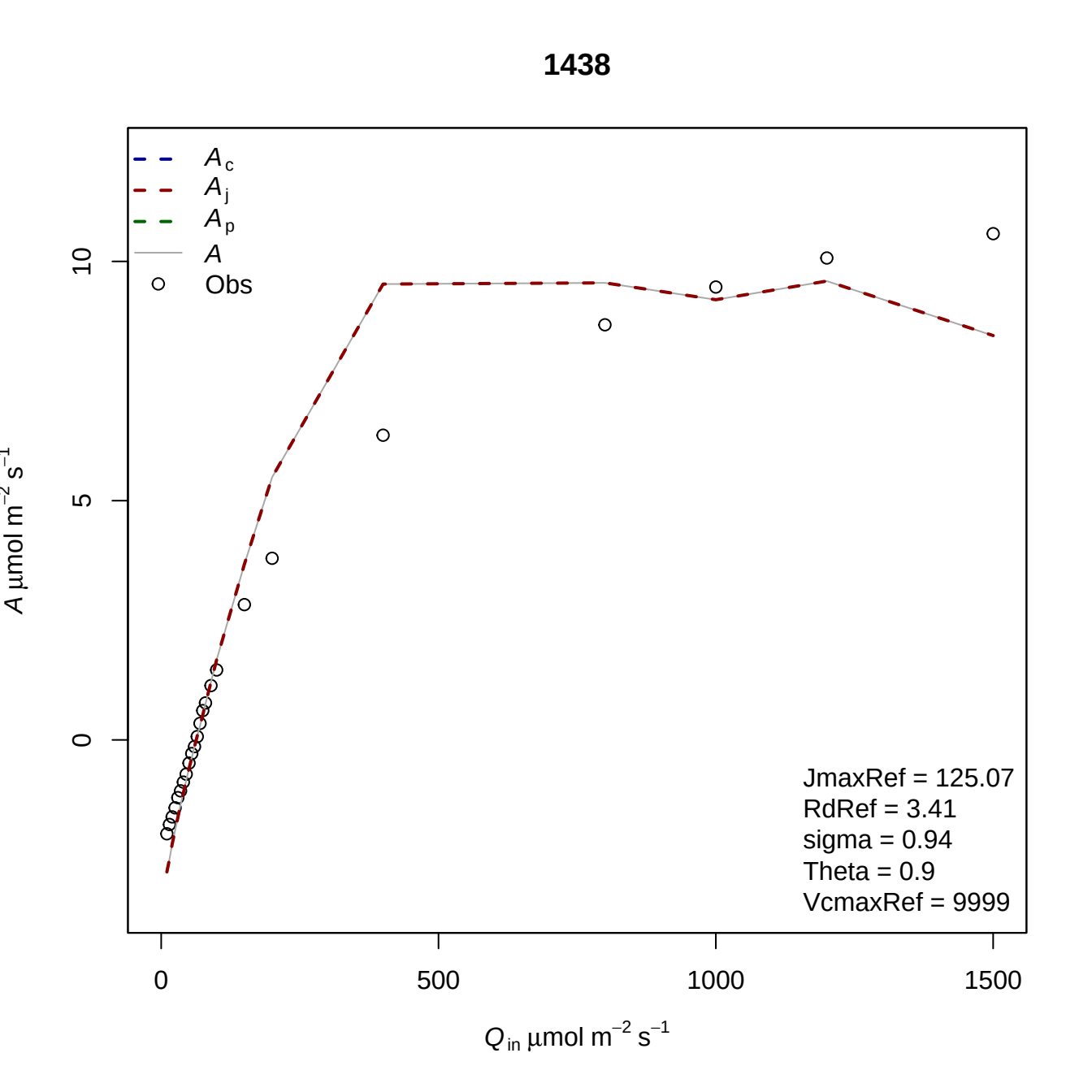

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 70.01
RdRef = 2.18
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

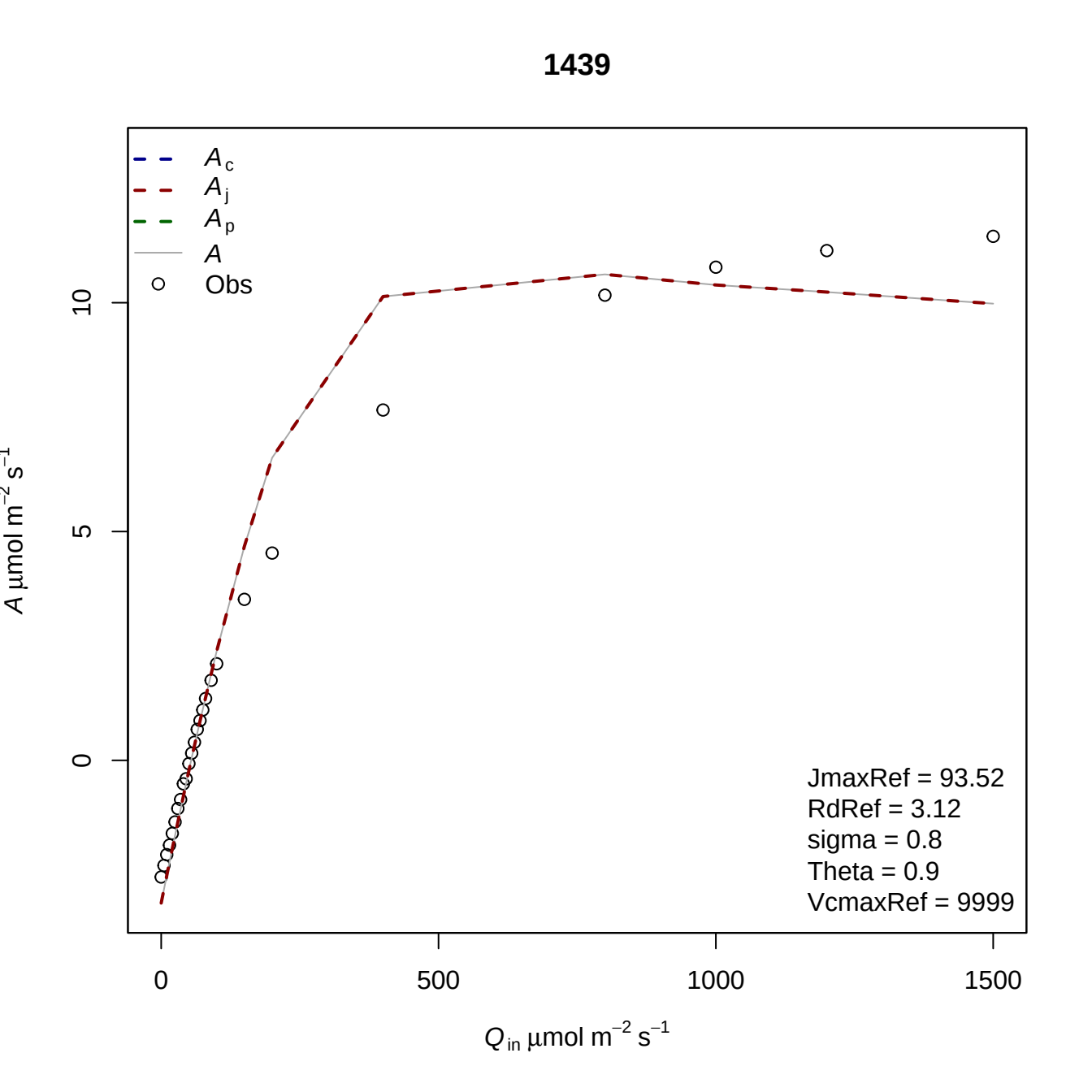

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 81.51
RdRef = 5.37
sigma = 0.97
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

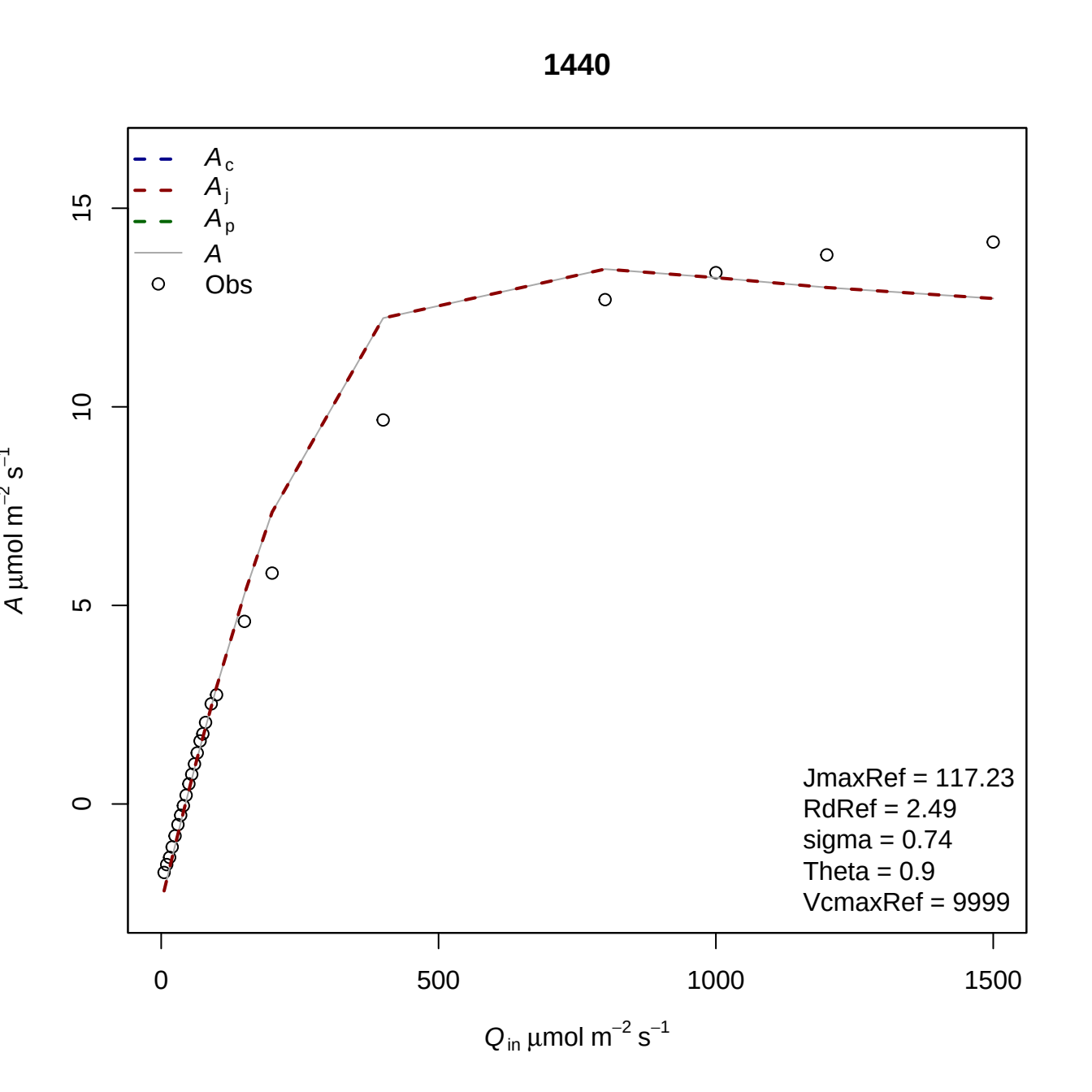

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 63.11
RdRef = 5.3
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

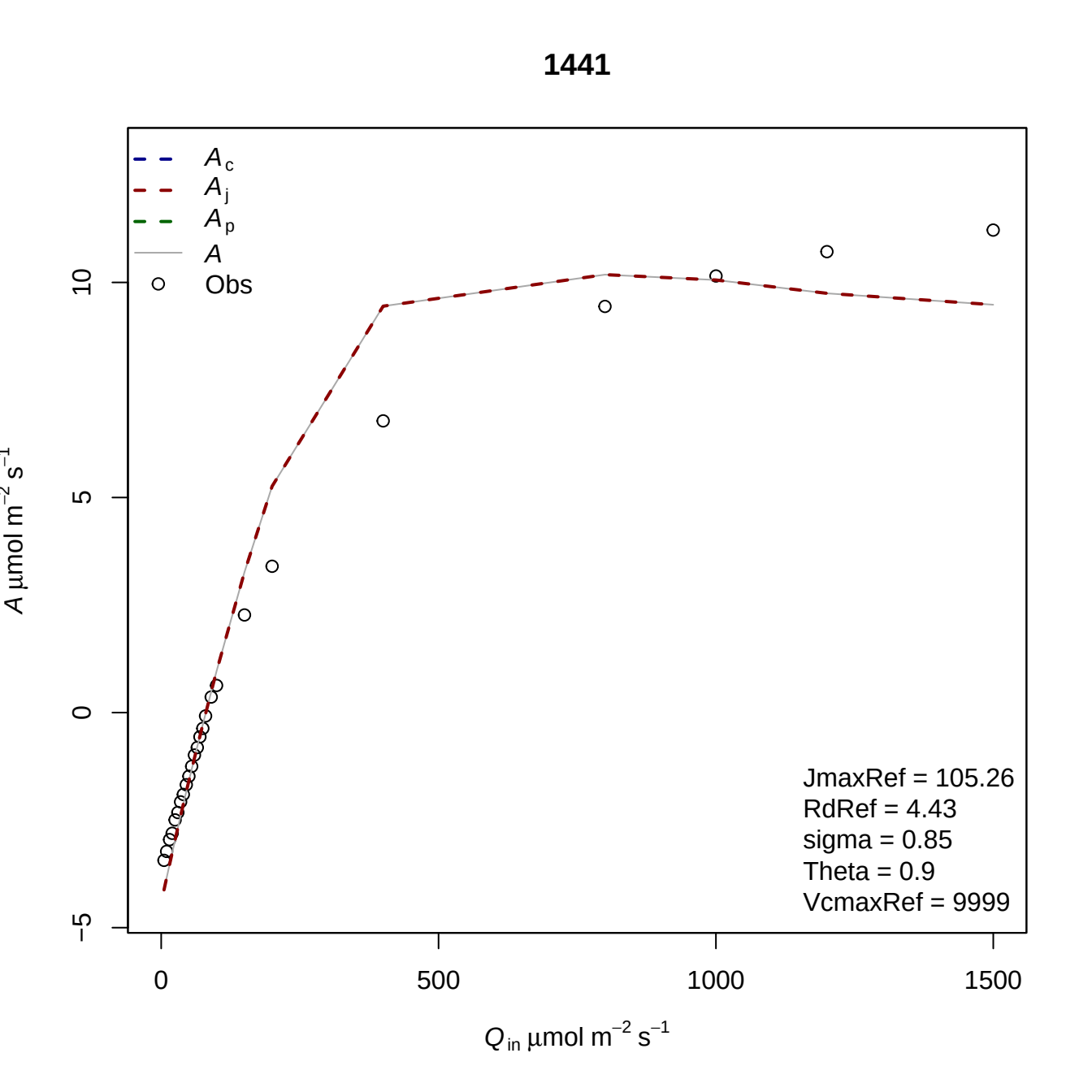

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 114.1
RdRef = 1.62
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

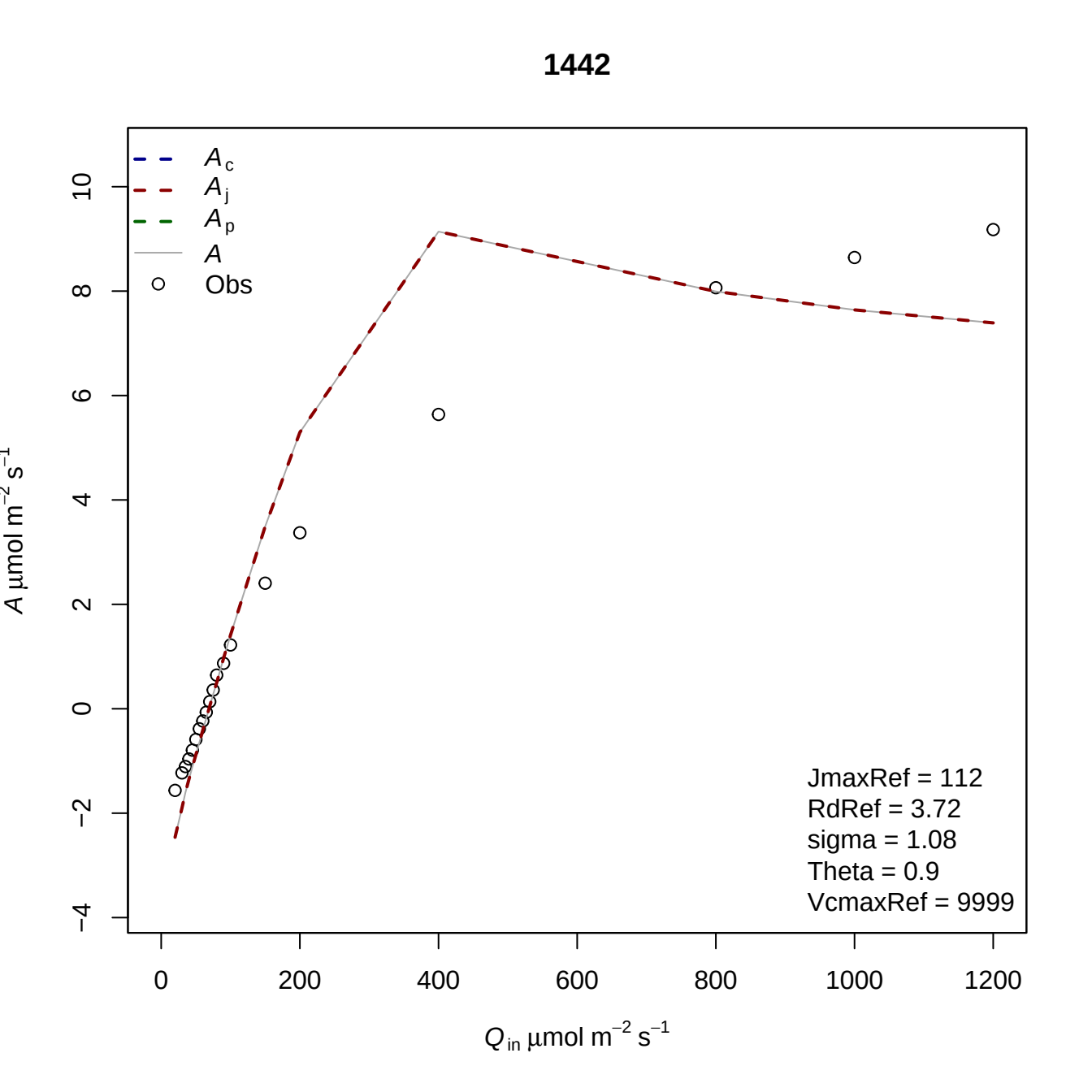

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 58.99
RdRef = 2.49
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

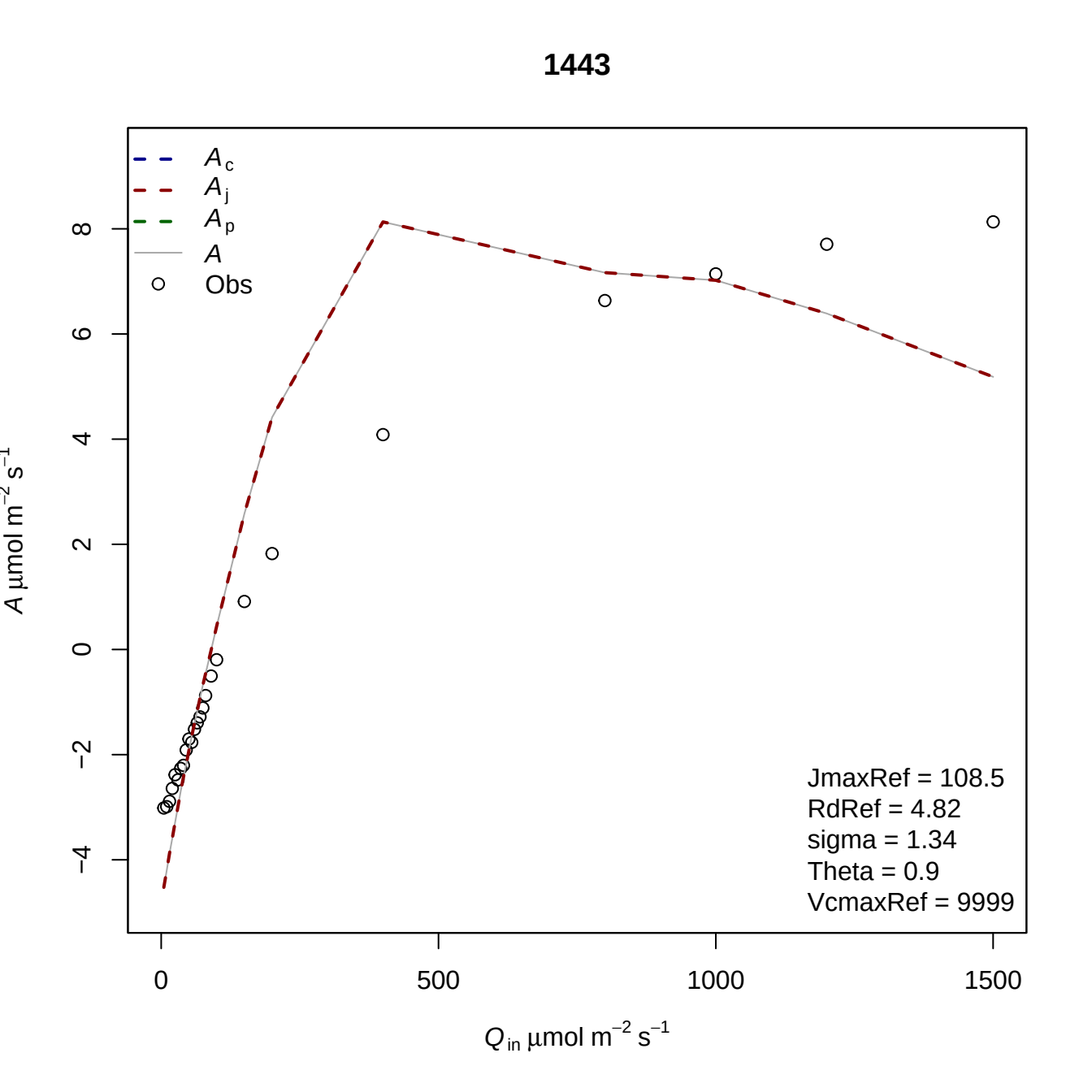

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 125.07
RdRef = 3.41
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

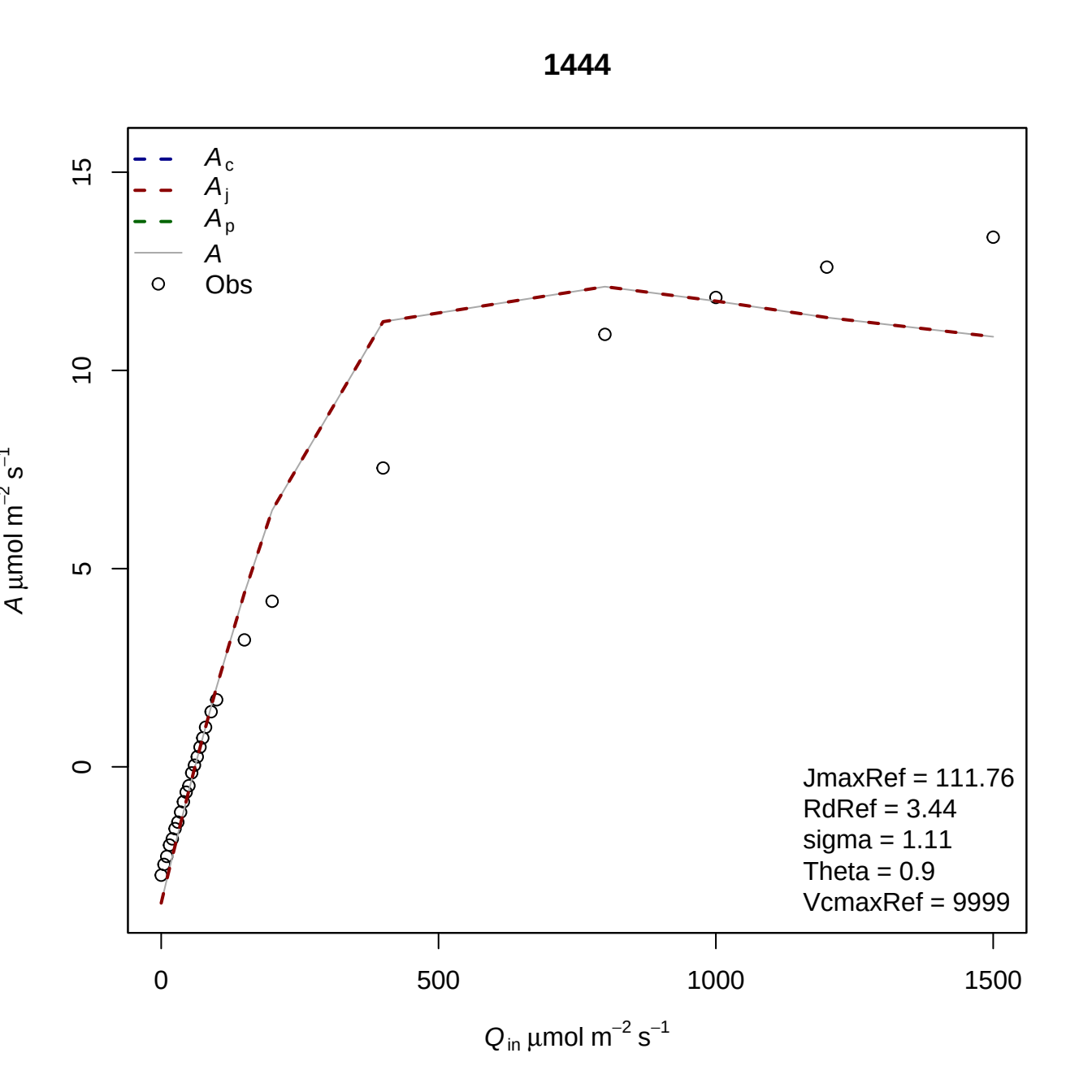

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 93.52
RdRef = 3.12
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

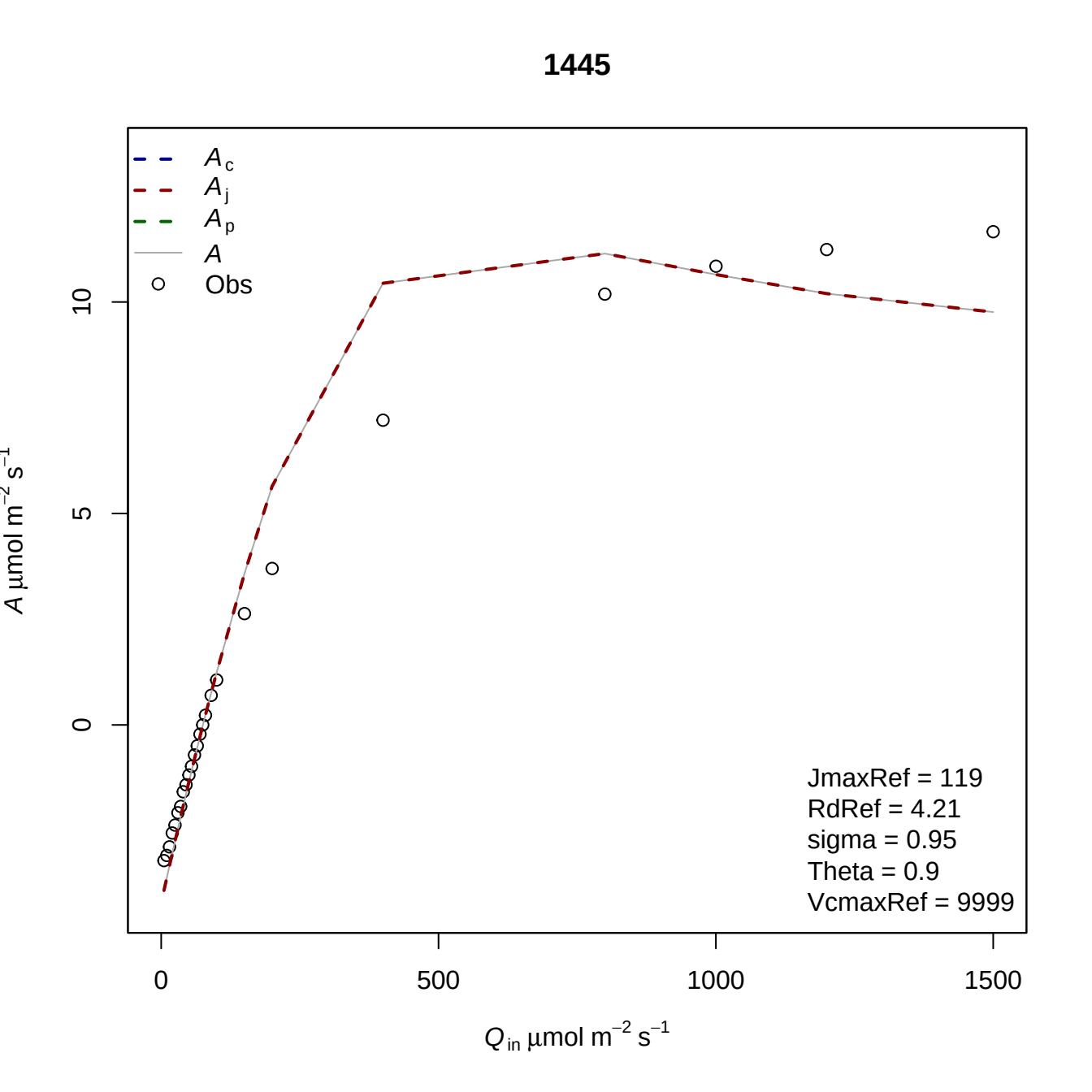

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 117.23
RdRef = 2.49
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

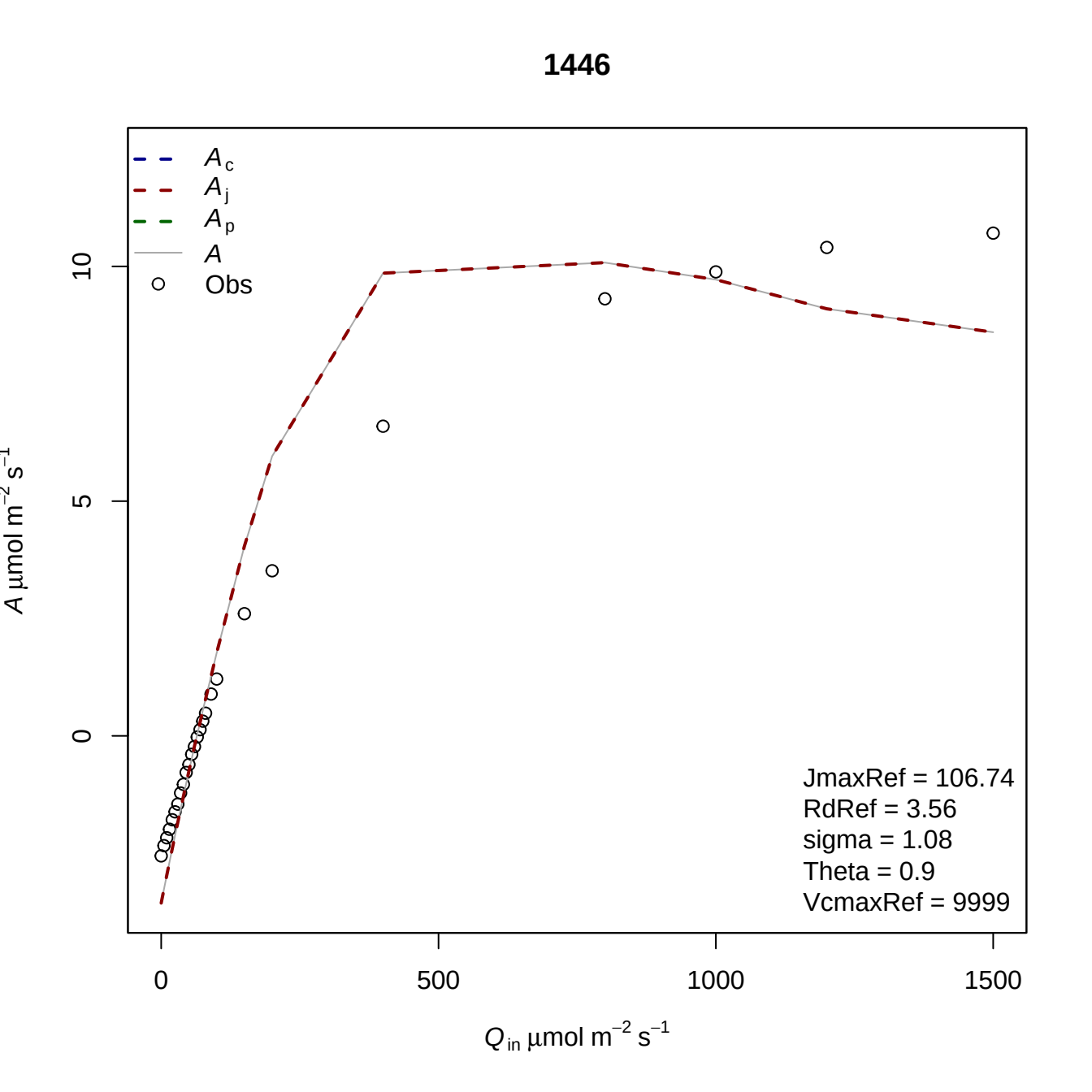

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 105.26
RdRef = 4.43
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

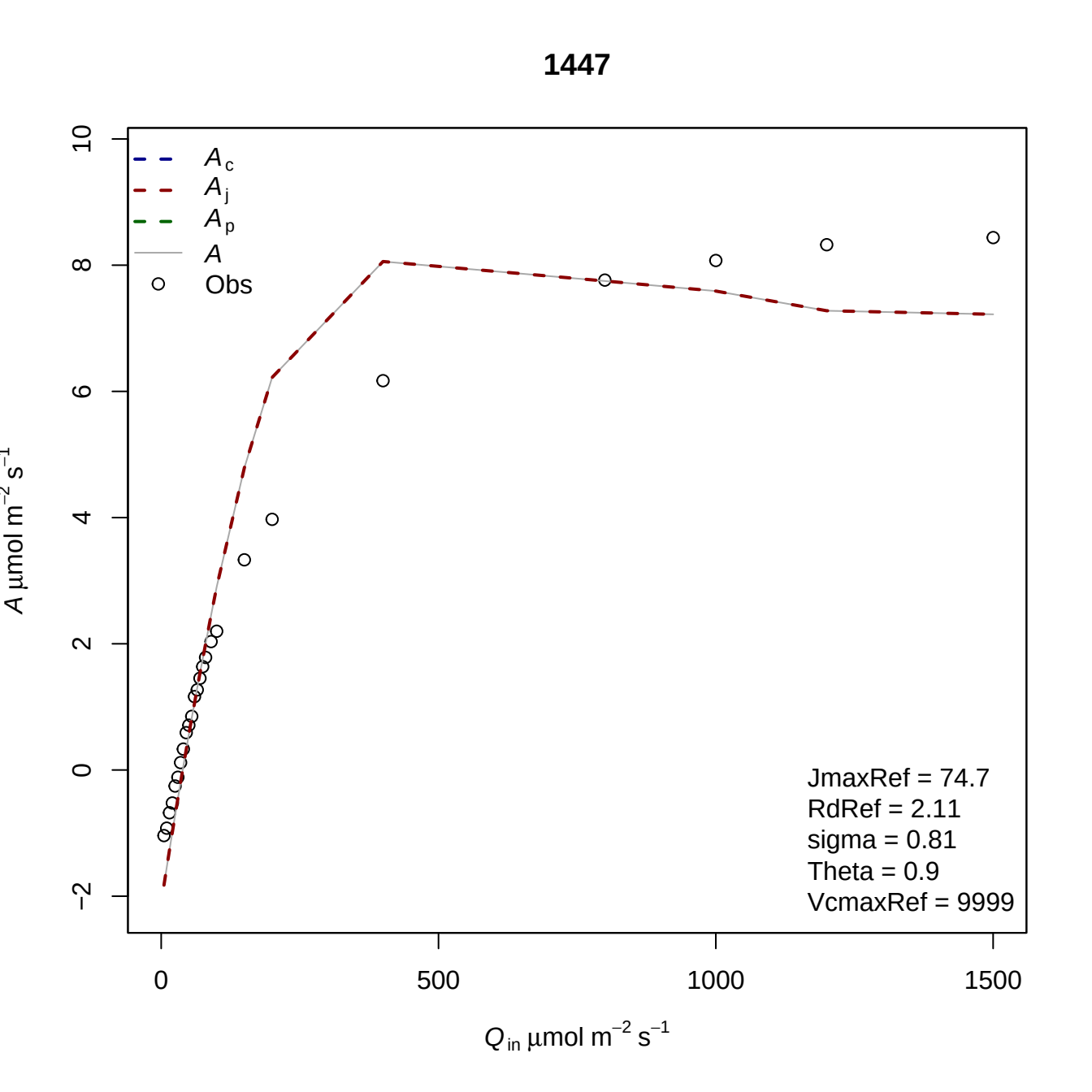

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 112
RdRef = 3.72
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

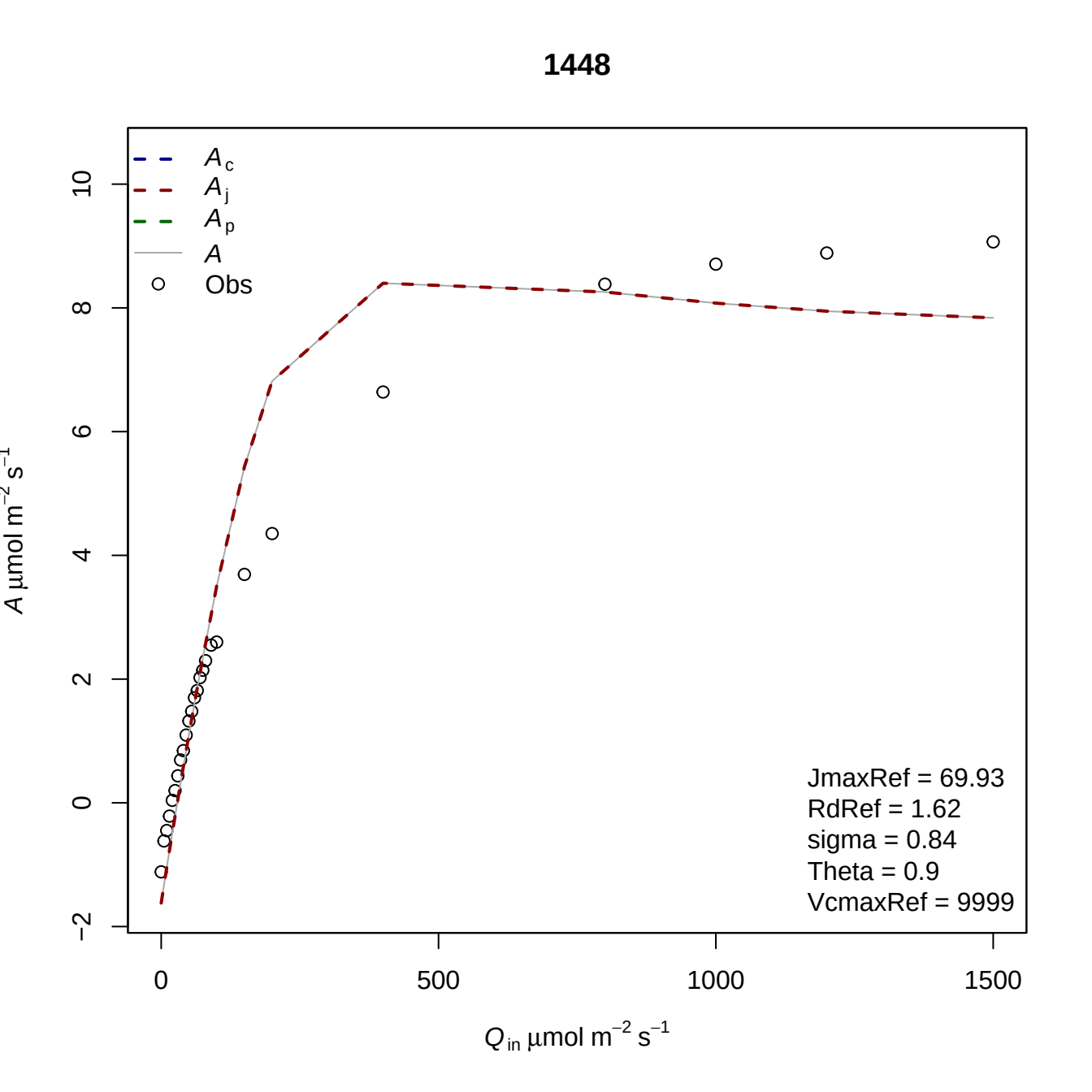

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 108.5
RdRef = 4.82
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

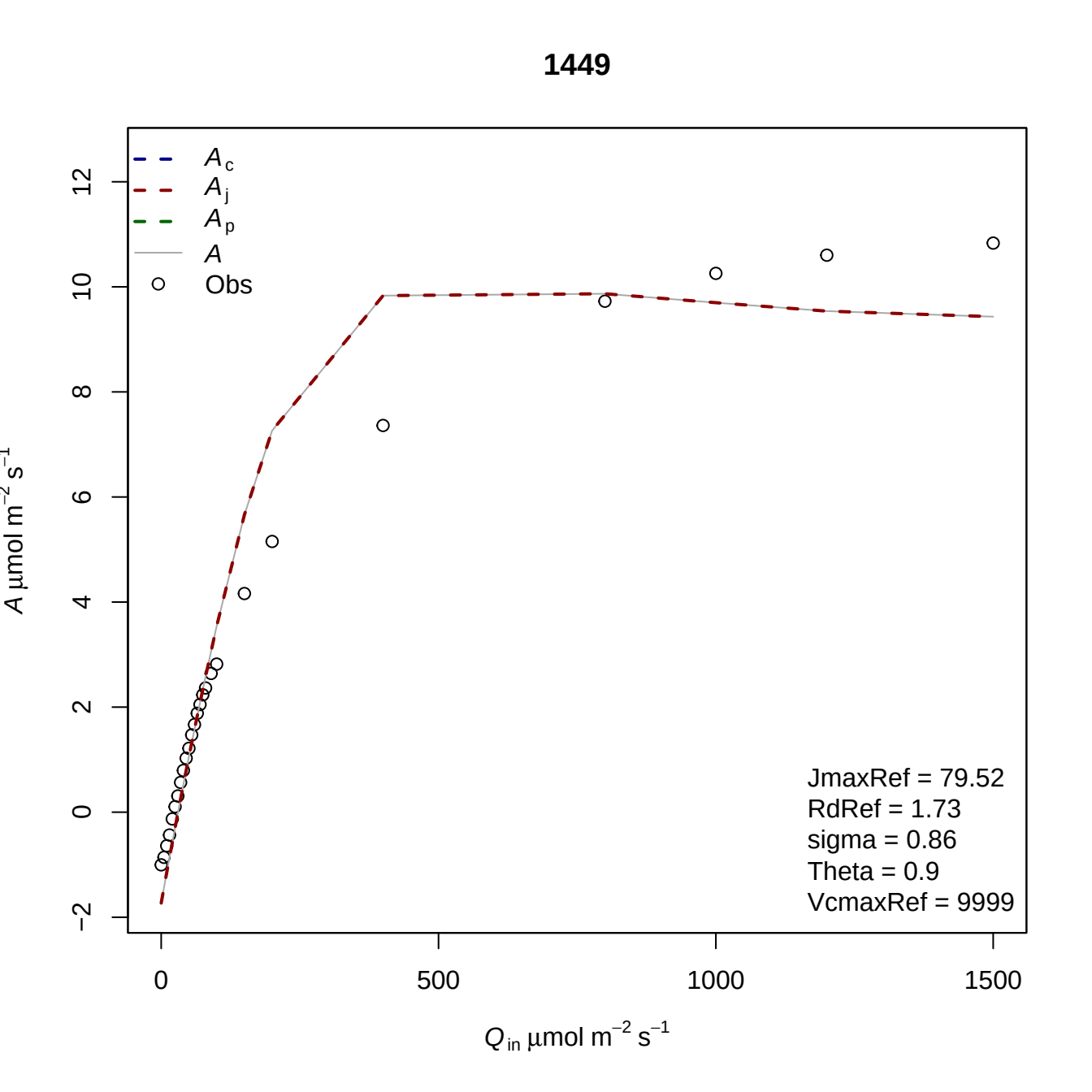

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 111.76
RdRef = 3.44
sigma = 1.11
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

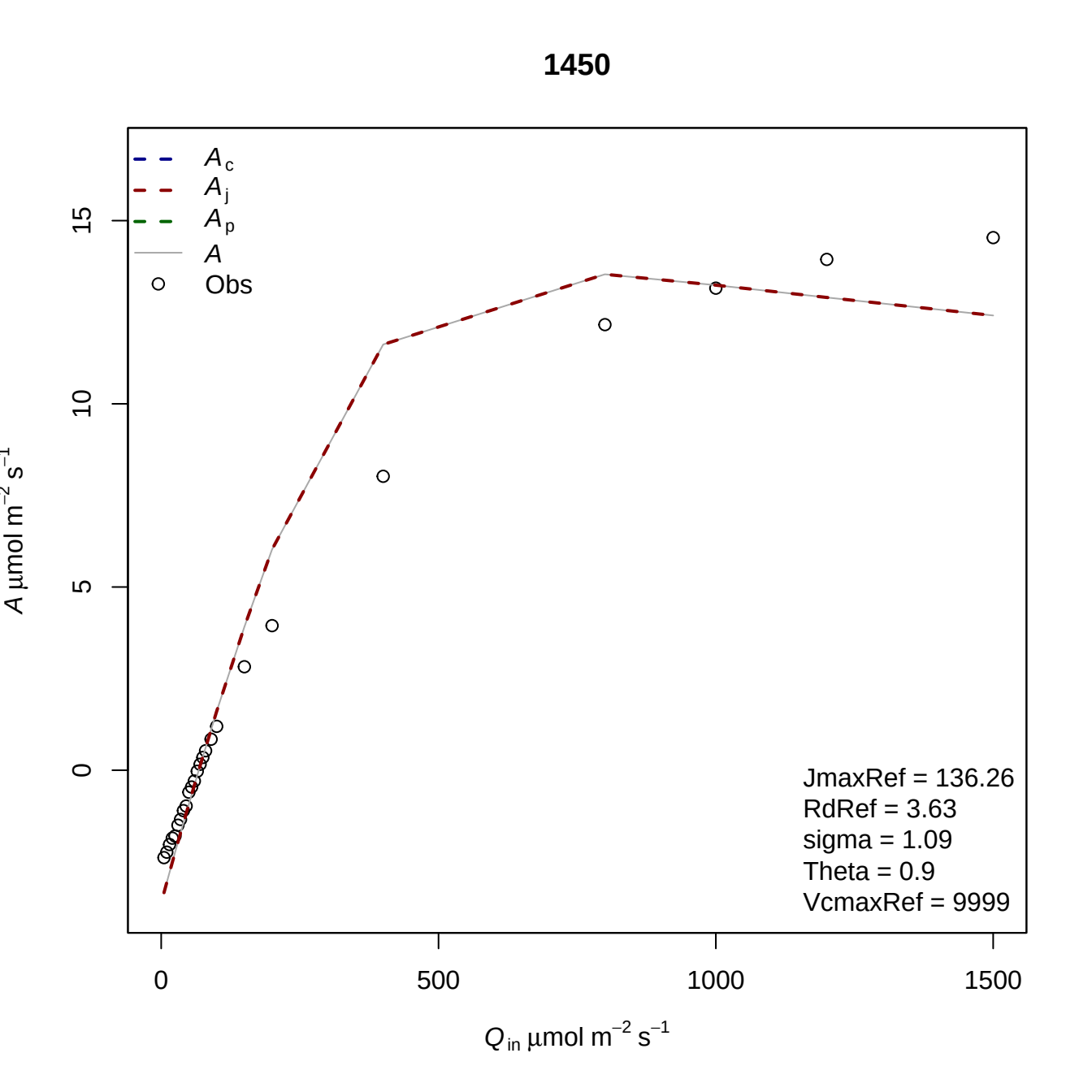

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 119
RdRef = 4.21
sigma = 0.95
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

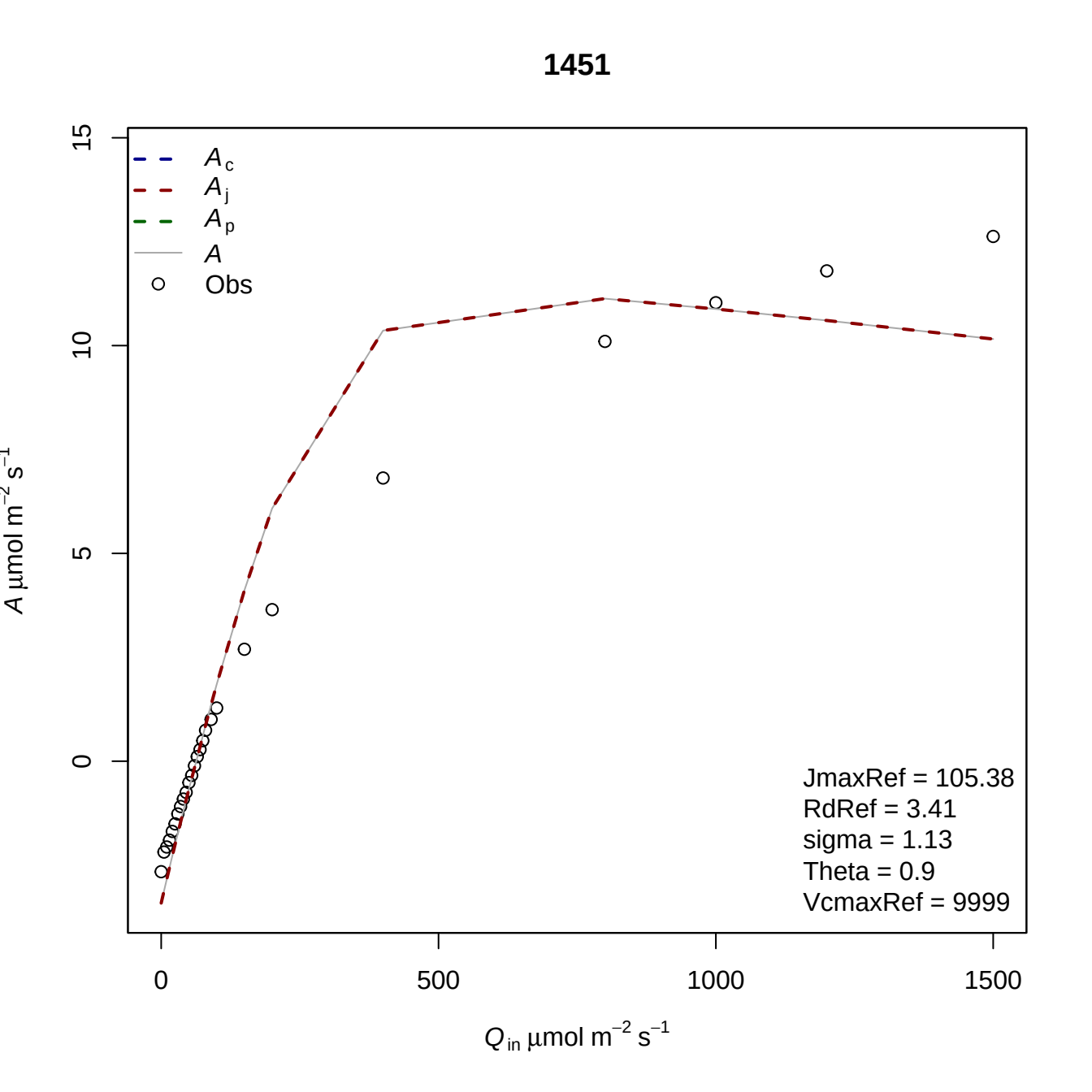

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 106.74
RdRef = 3.56
sigma = 1.08
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

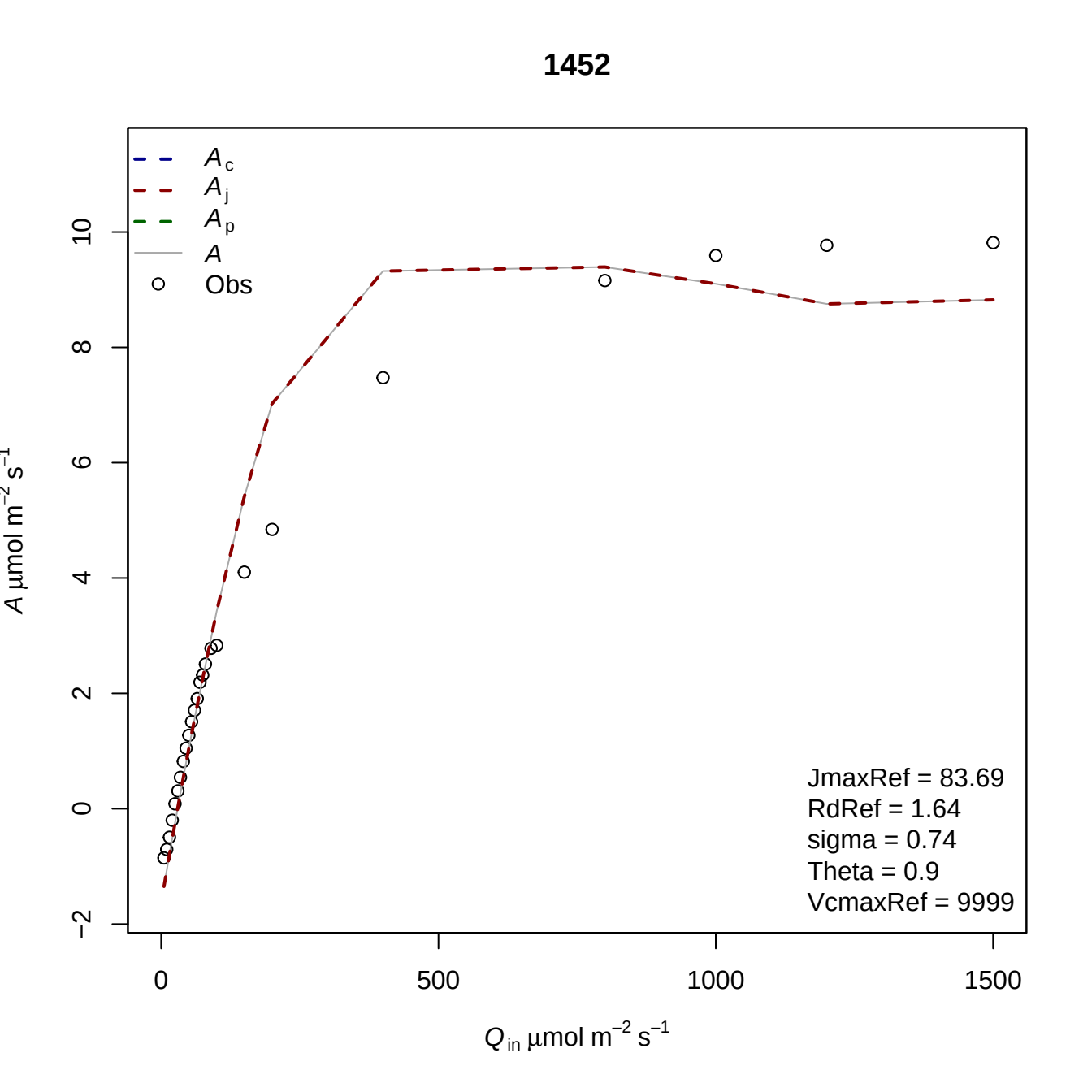

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 74.7
RdRef = 2.11
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

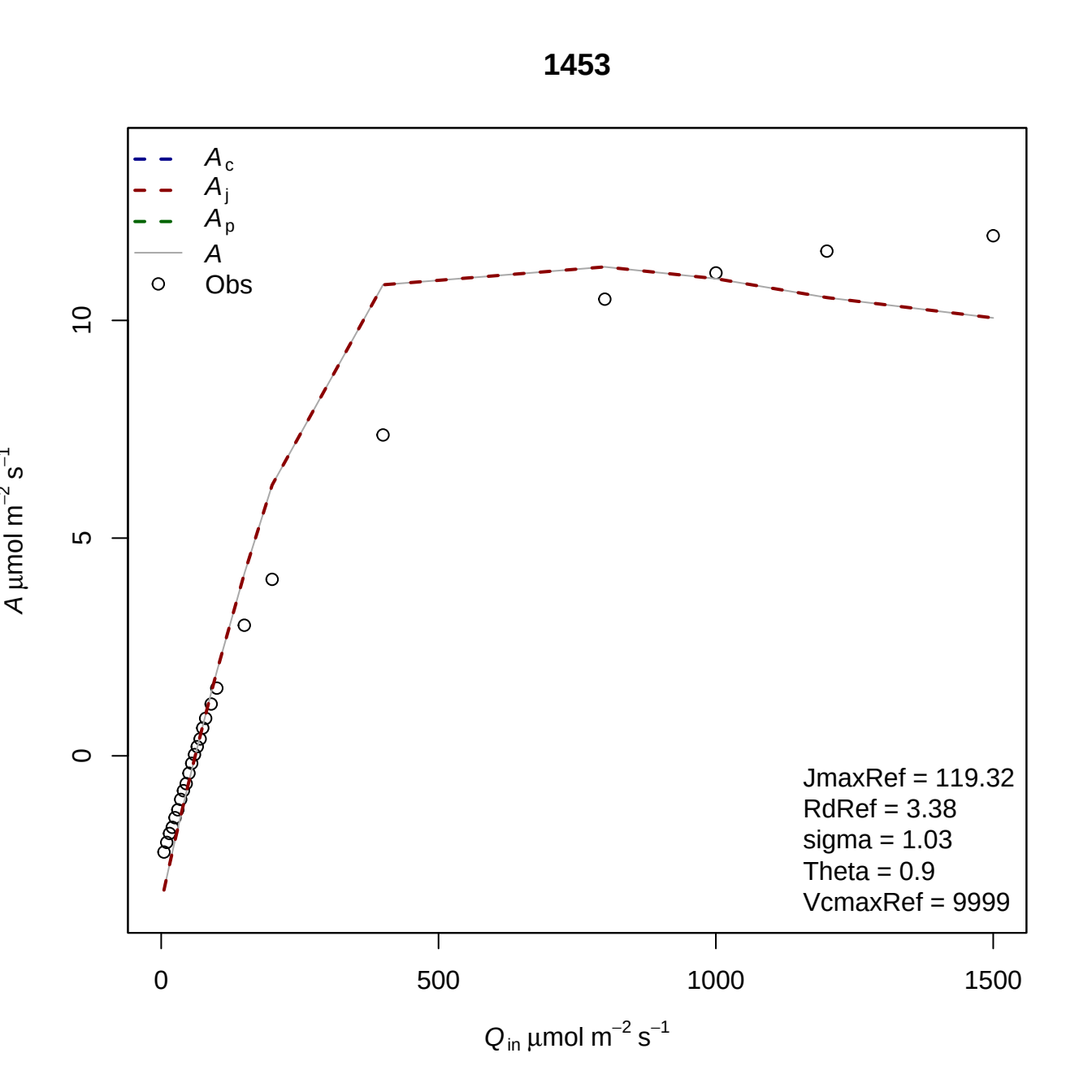

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 69.93
RdRef = 1.62
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

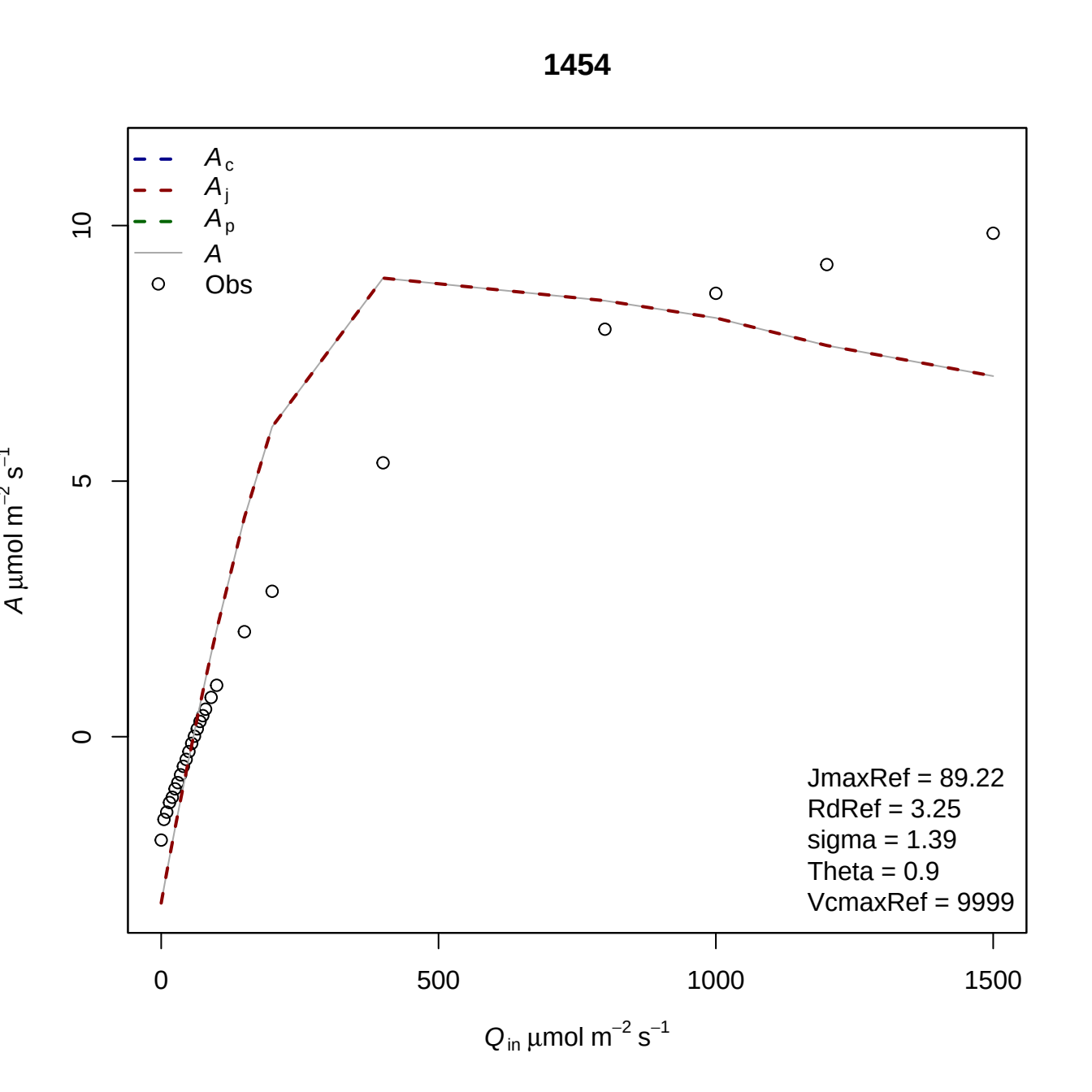

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 79.52
RdRef = 1.73
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

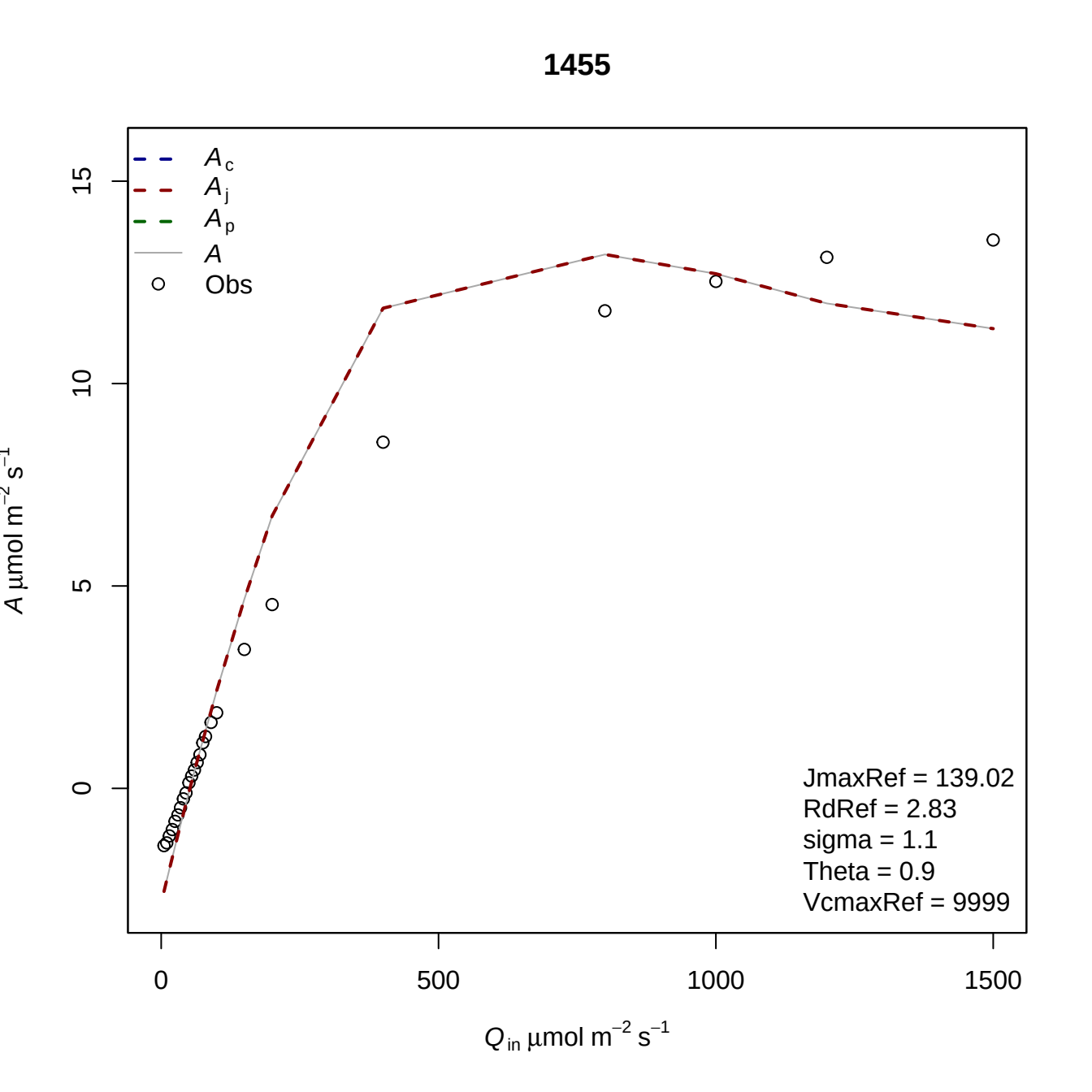

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 136.26
RdRef = 3.63
sigma = 1.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

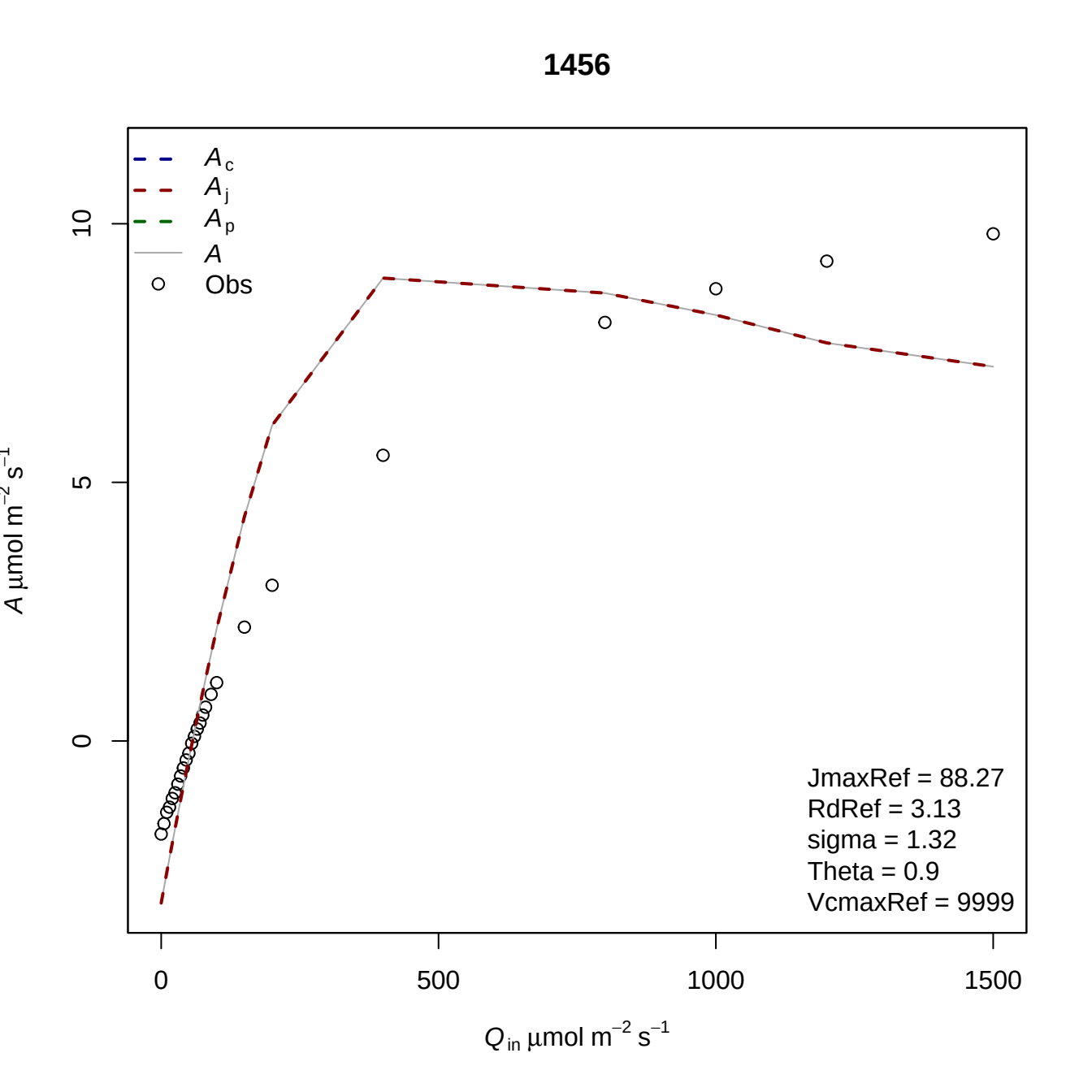

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 105.38
RdRef = 3.41
sigma = 1.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

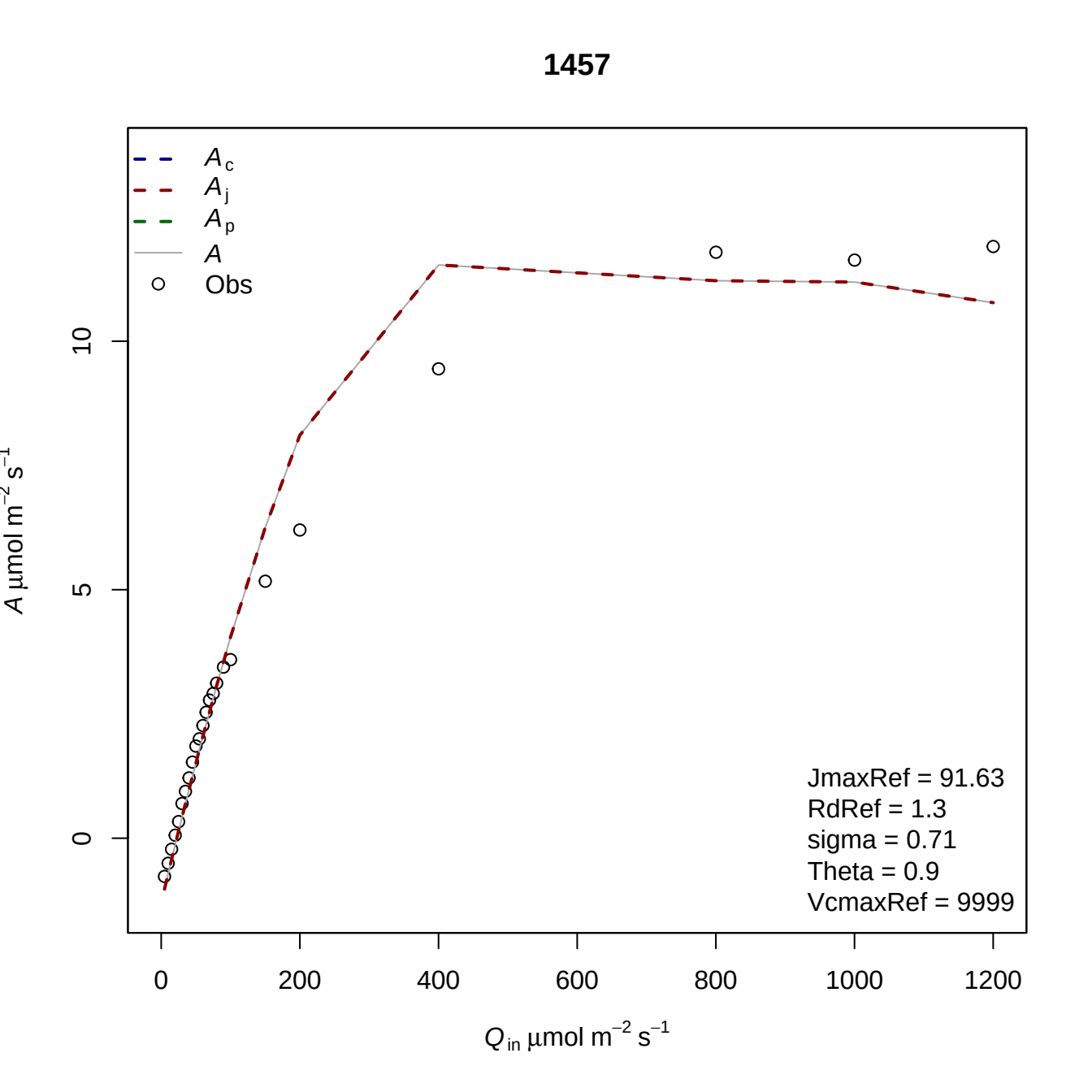

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 83.69
RdRef = 1.64
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

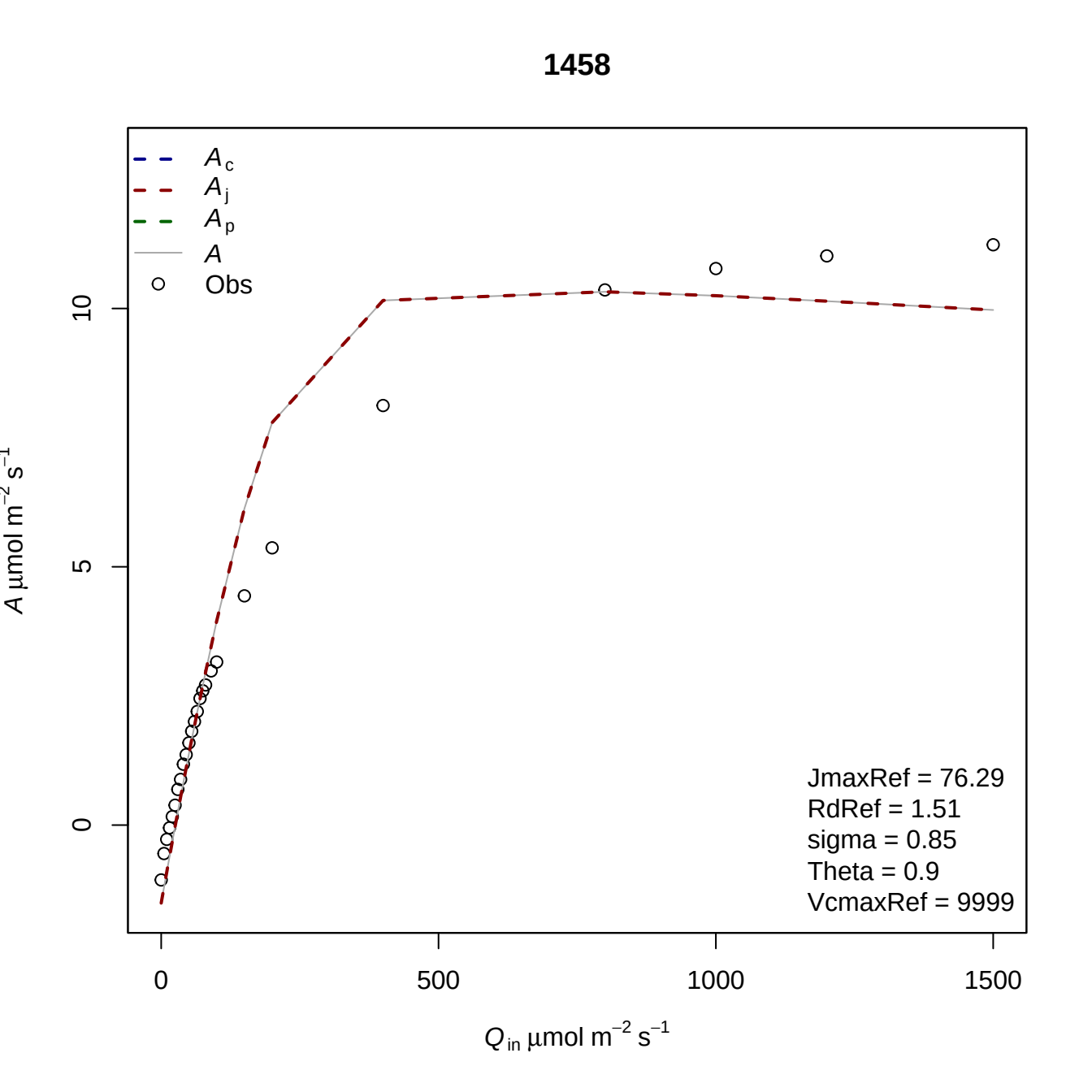

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 119.32
RdRef = 3.38
sigma = 1.03
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

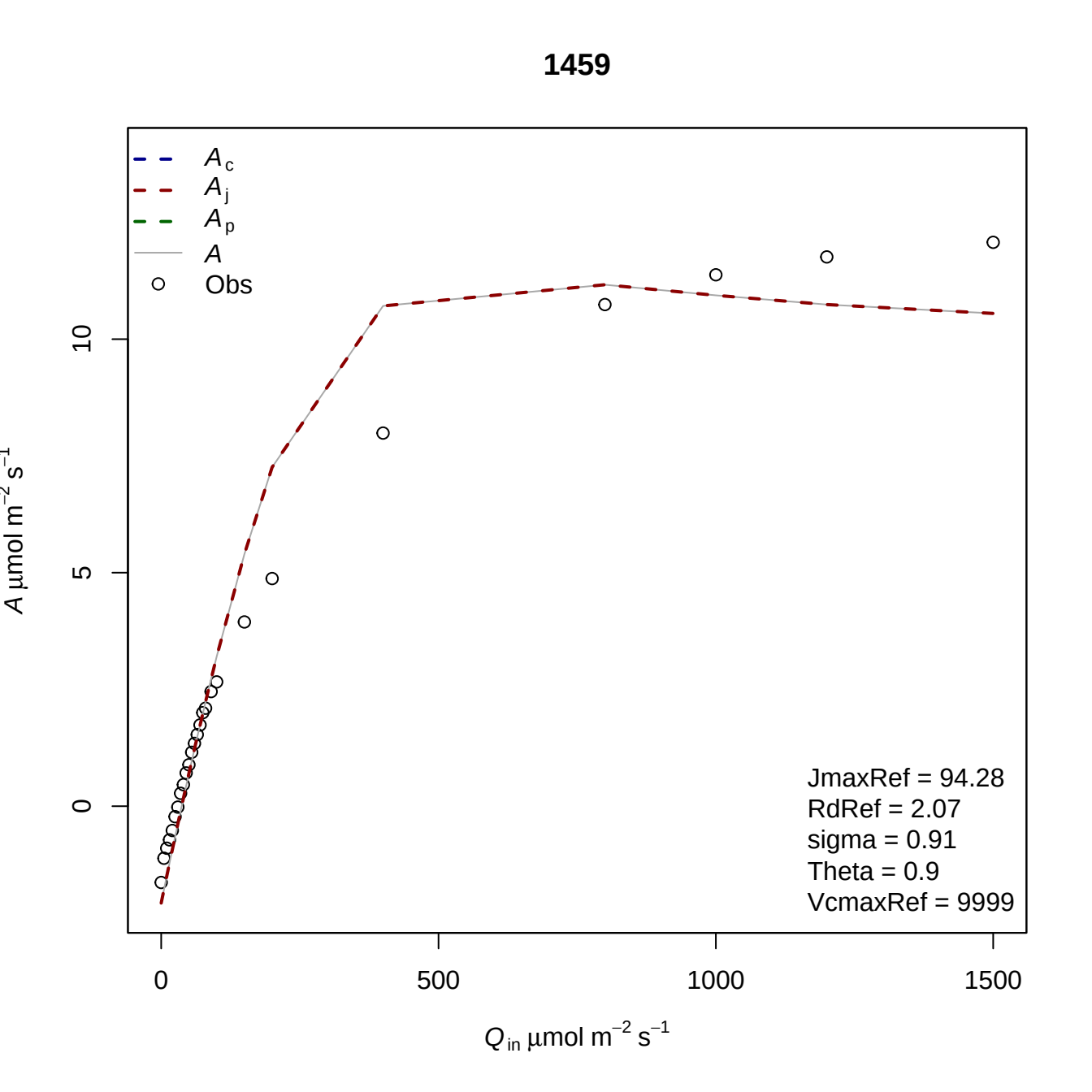

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 89.22
RdRef = 3.25
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

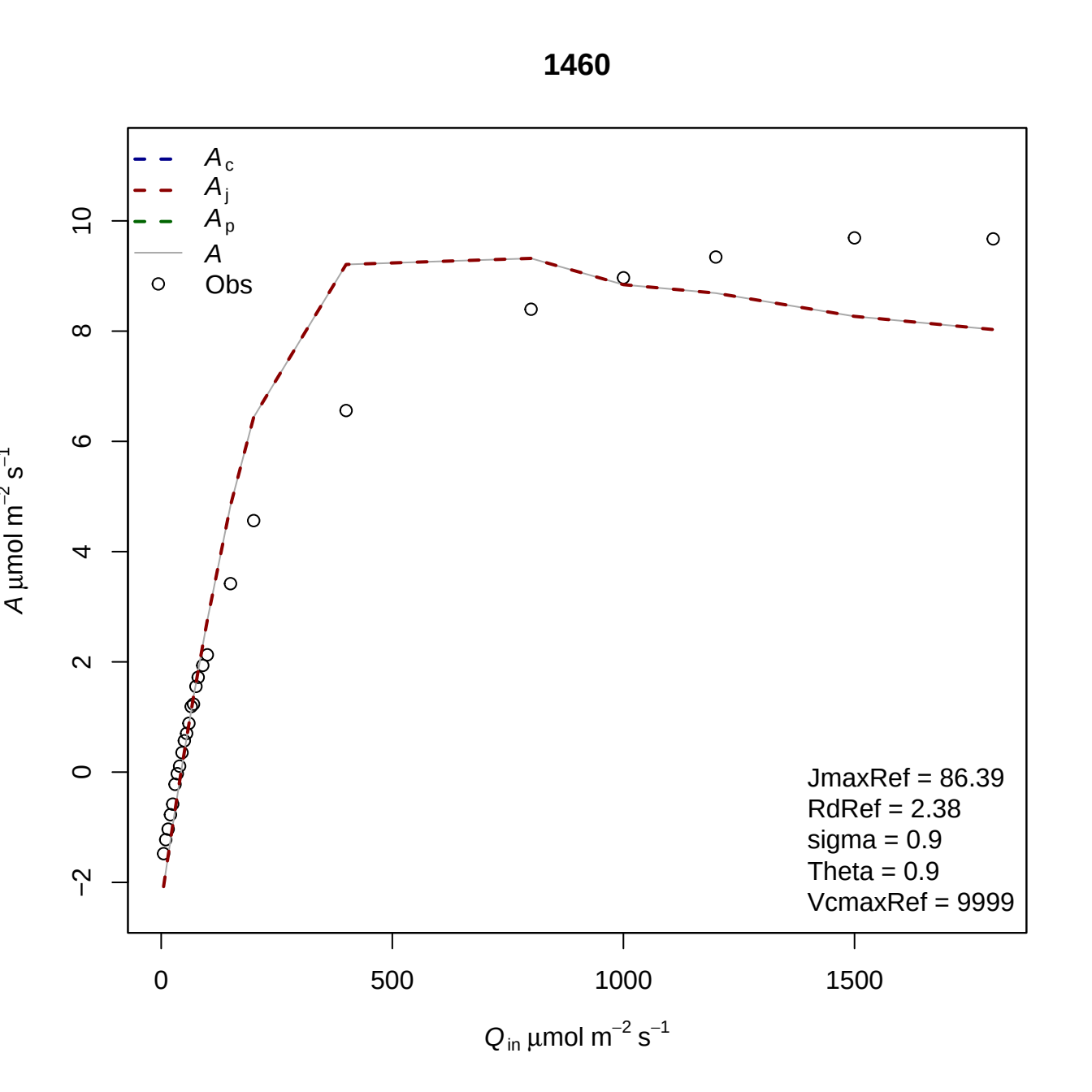

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 139.02
RdRef = 2.83
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

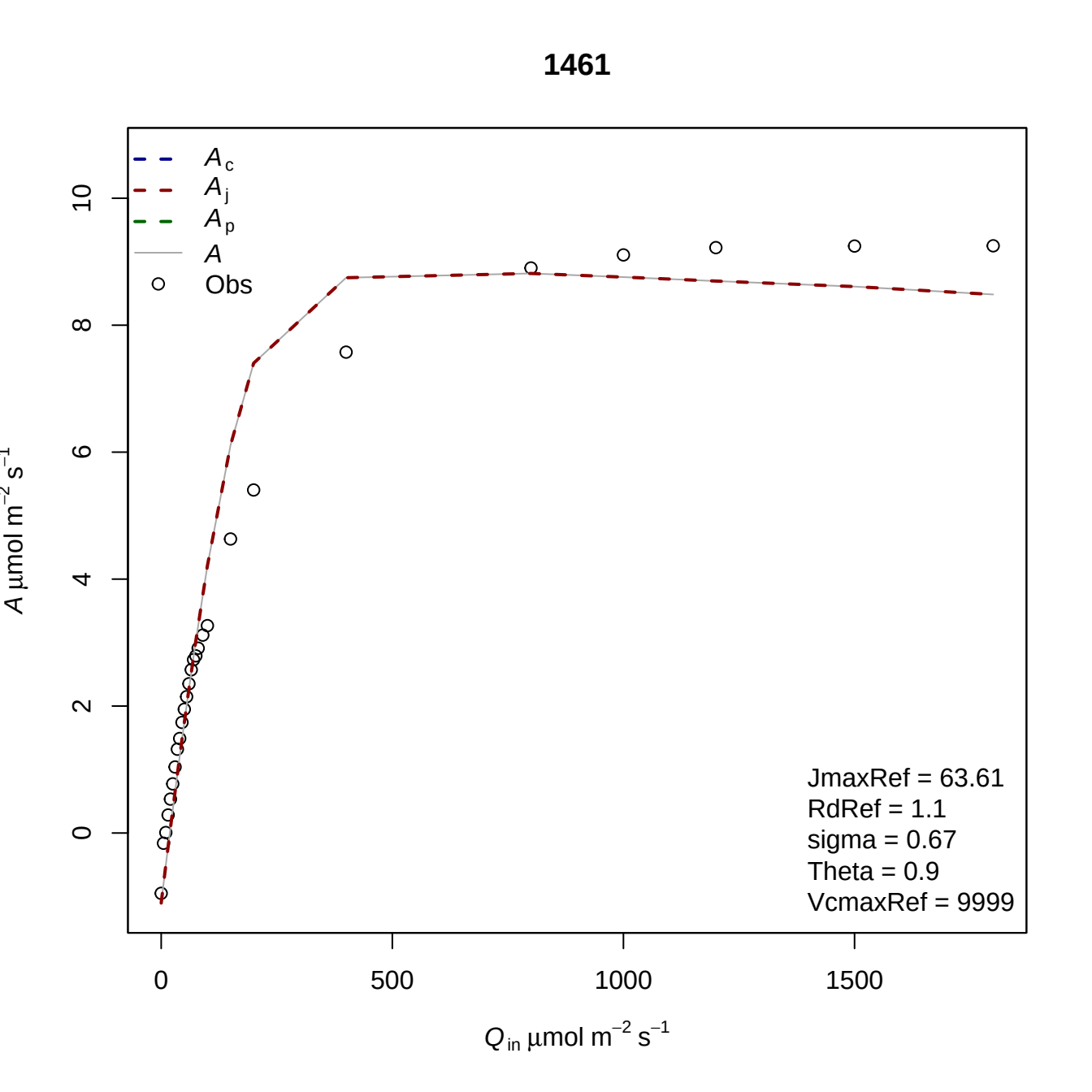

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 88.27
RdRef = 3.13
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

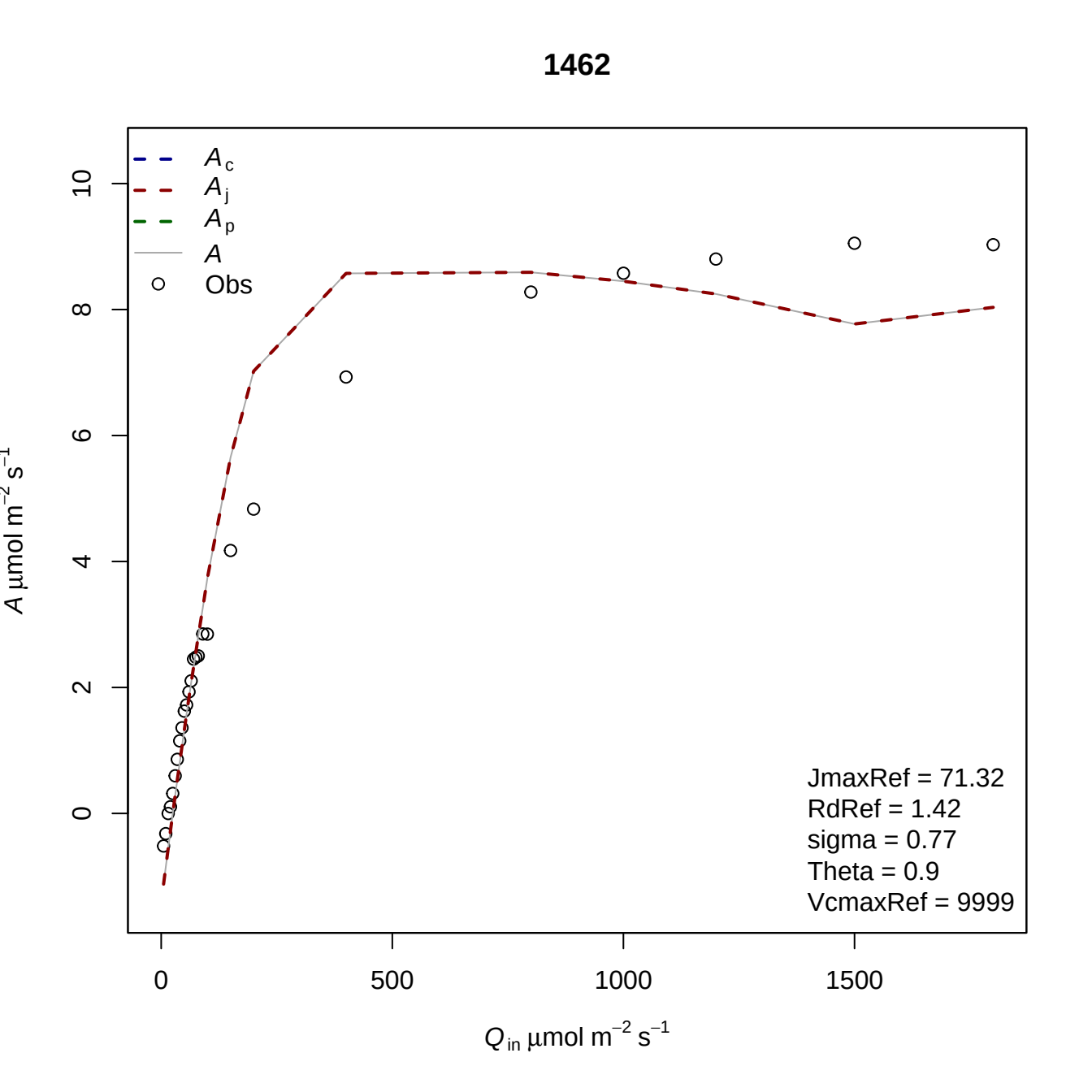

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 91.63
RdRef = 1.3
sigma = 0.71
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

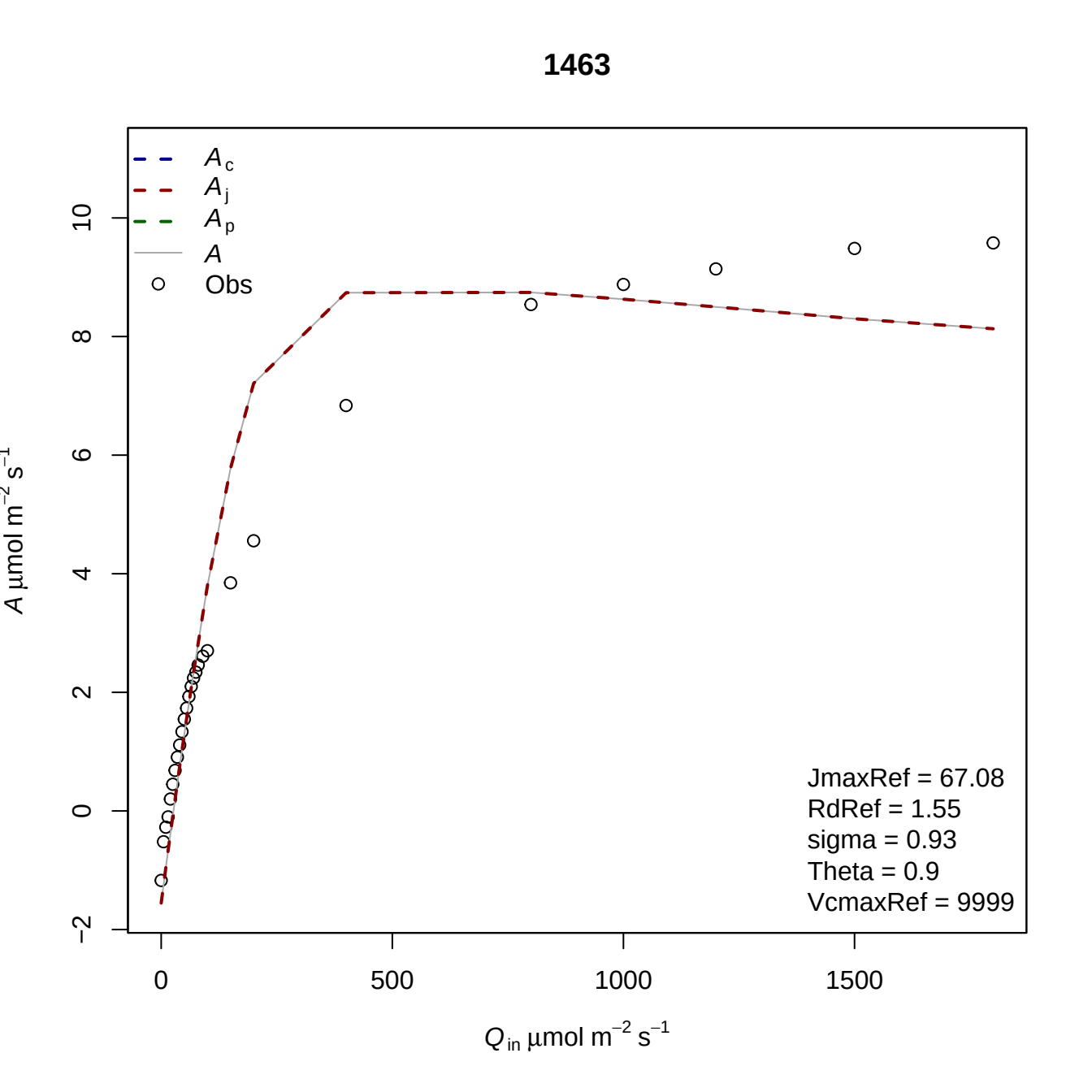

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 76.29
RdRef = 1.51
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

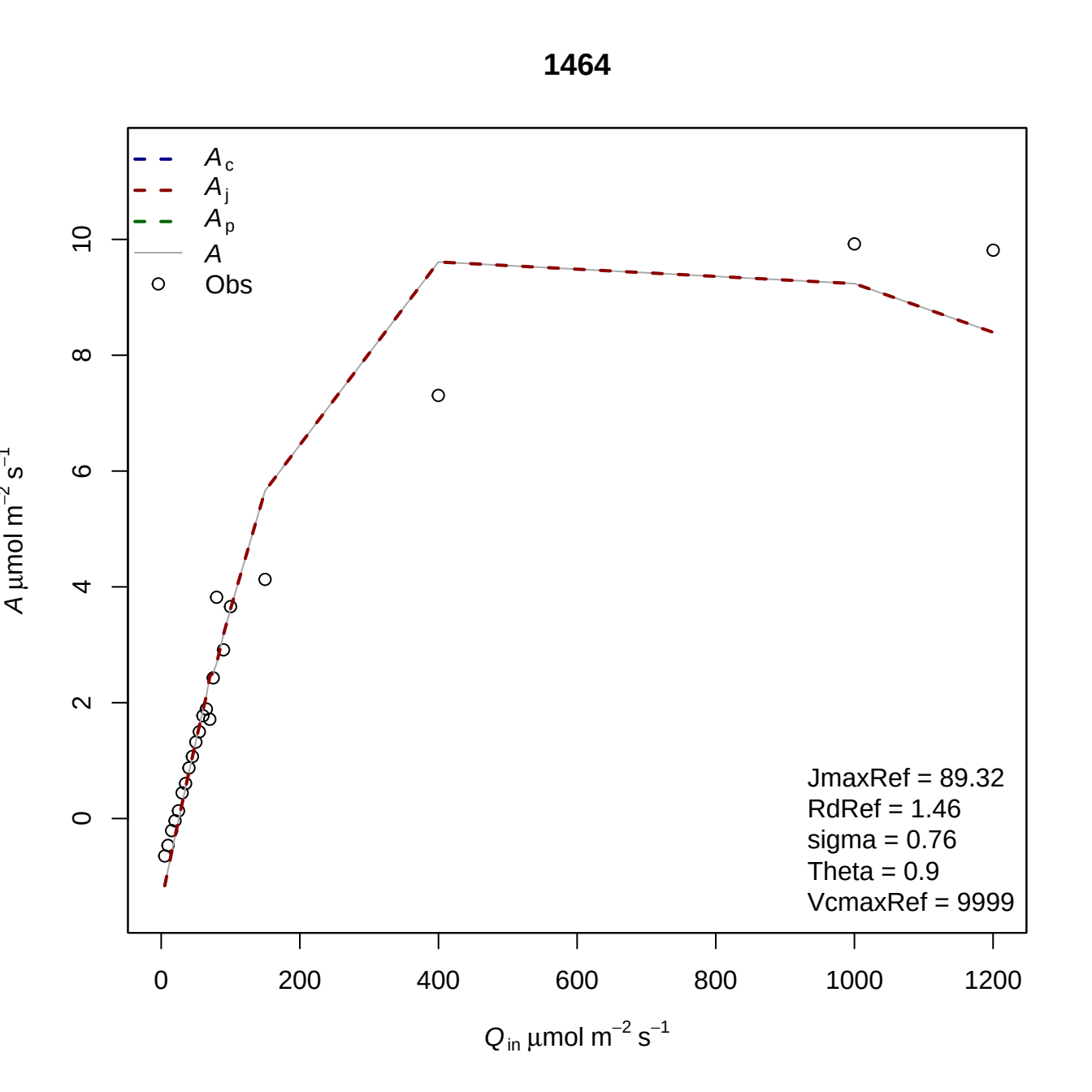

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 94.28
RdRef = 2.07
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

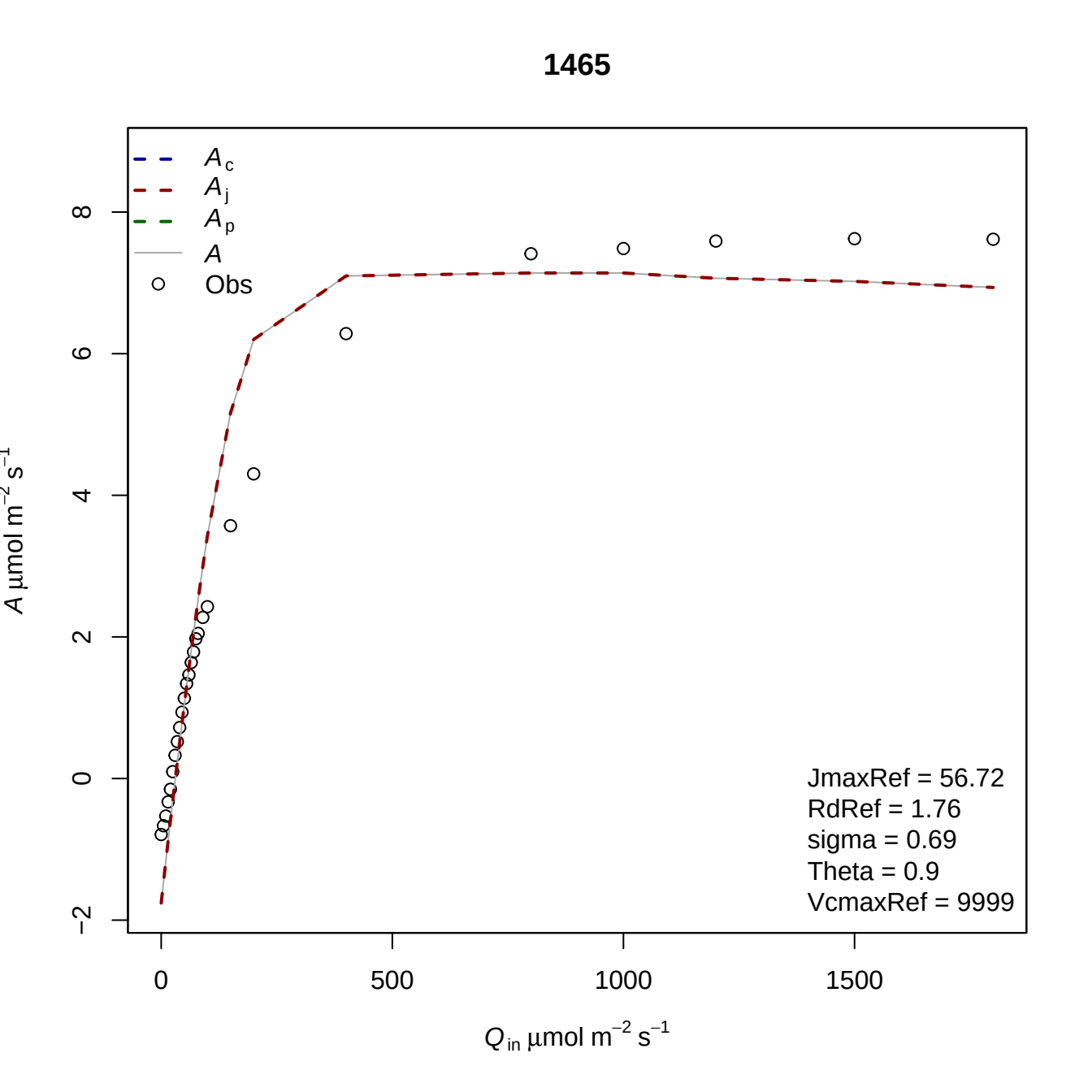

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 86.39
RdRef = 2.38
sigma = 0.9
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

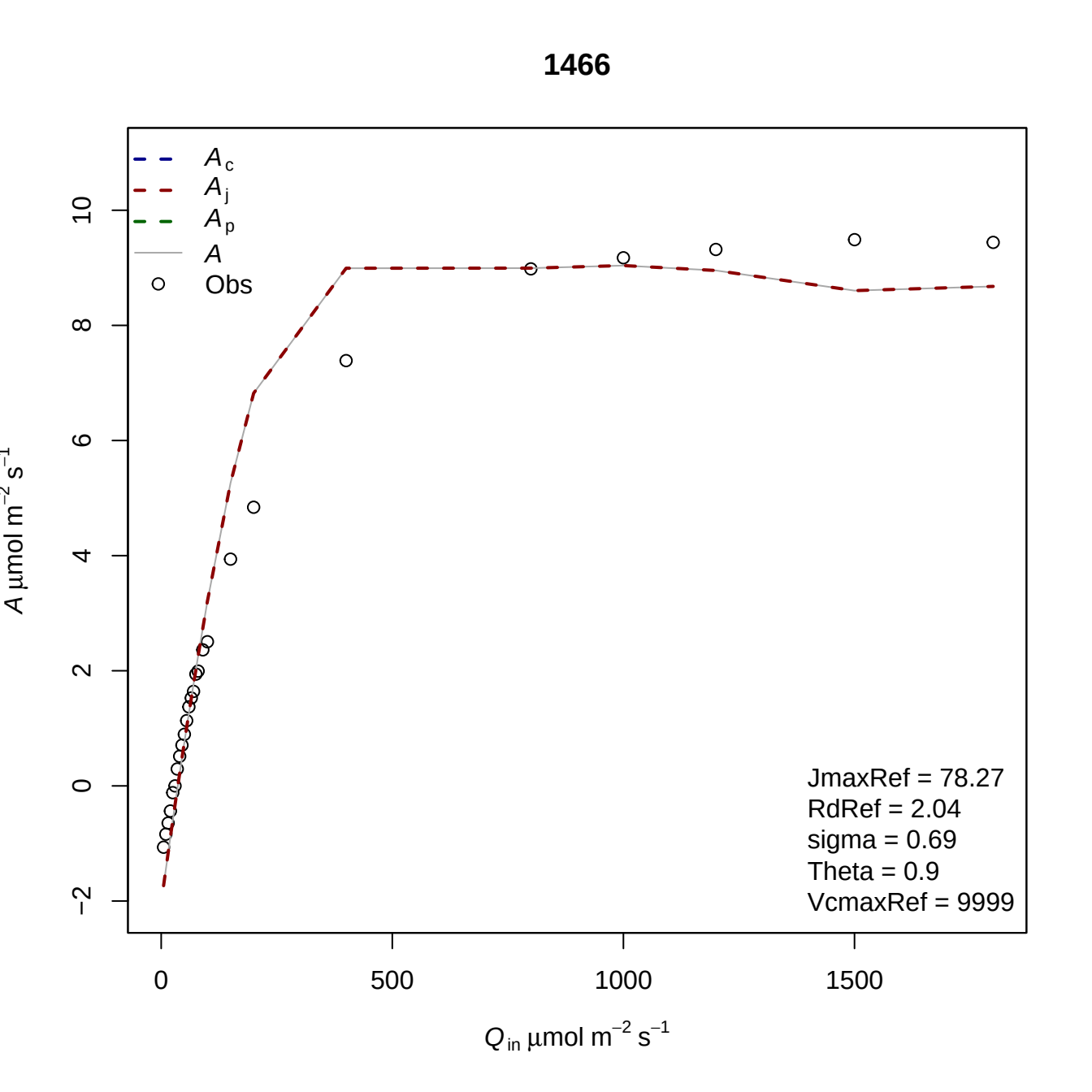

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 63.61
RdRef = 1.1
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

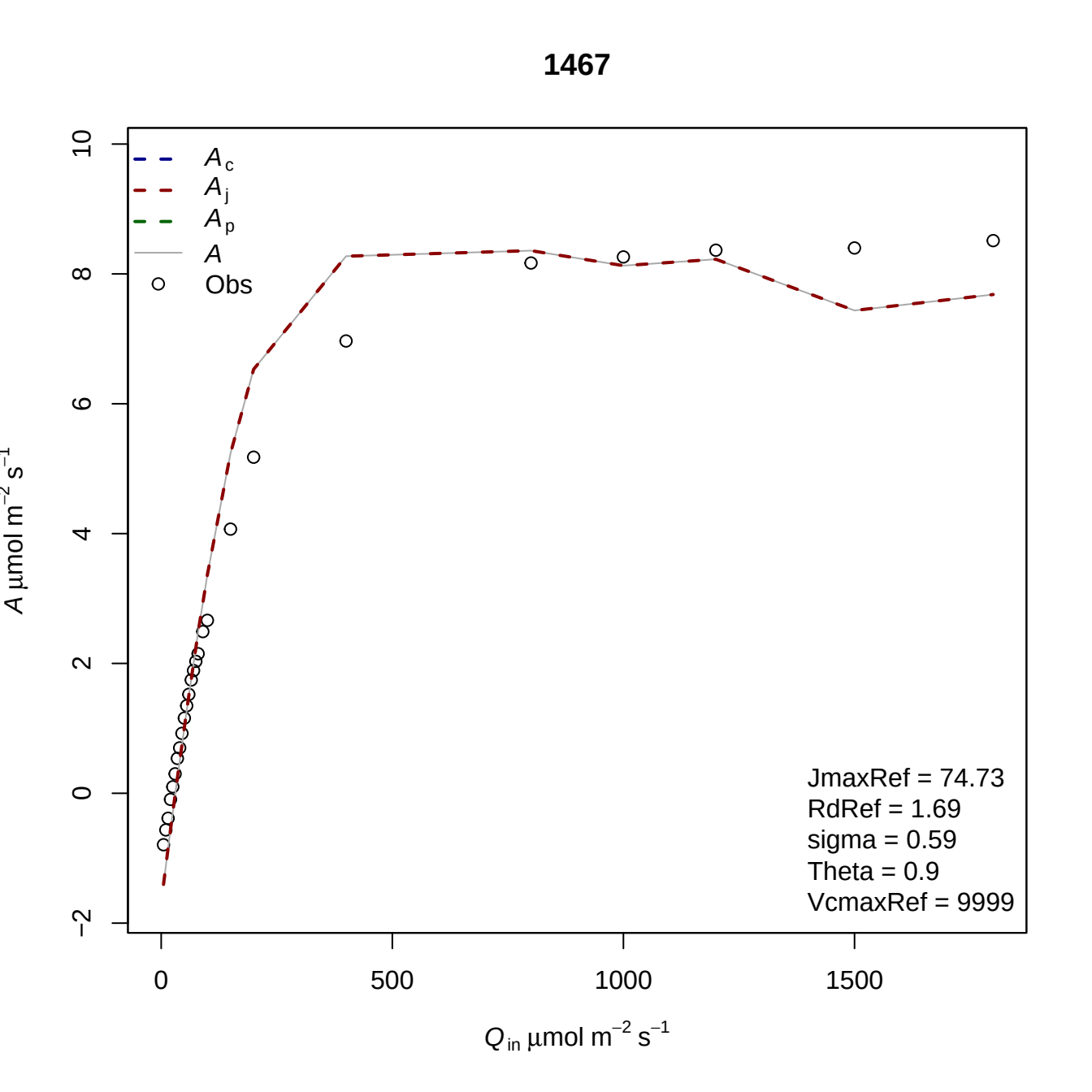

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 71.32
RdRef = 1.42
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

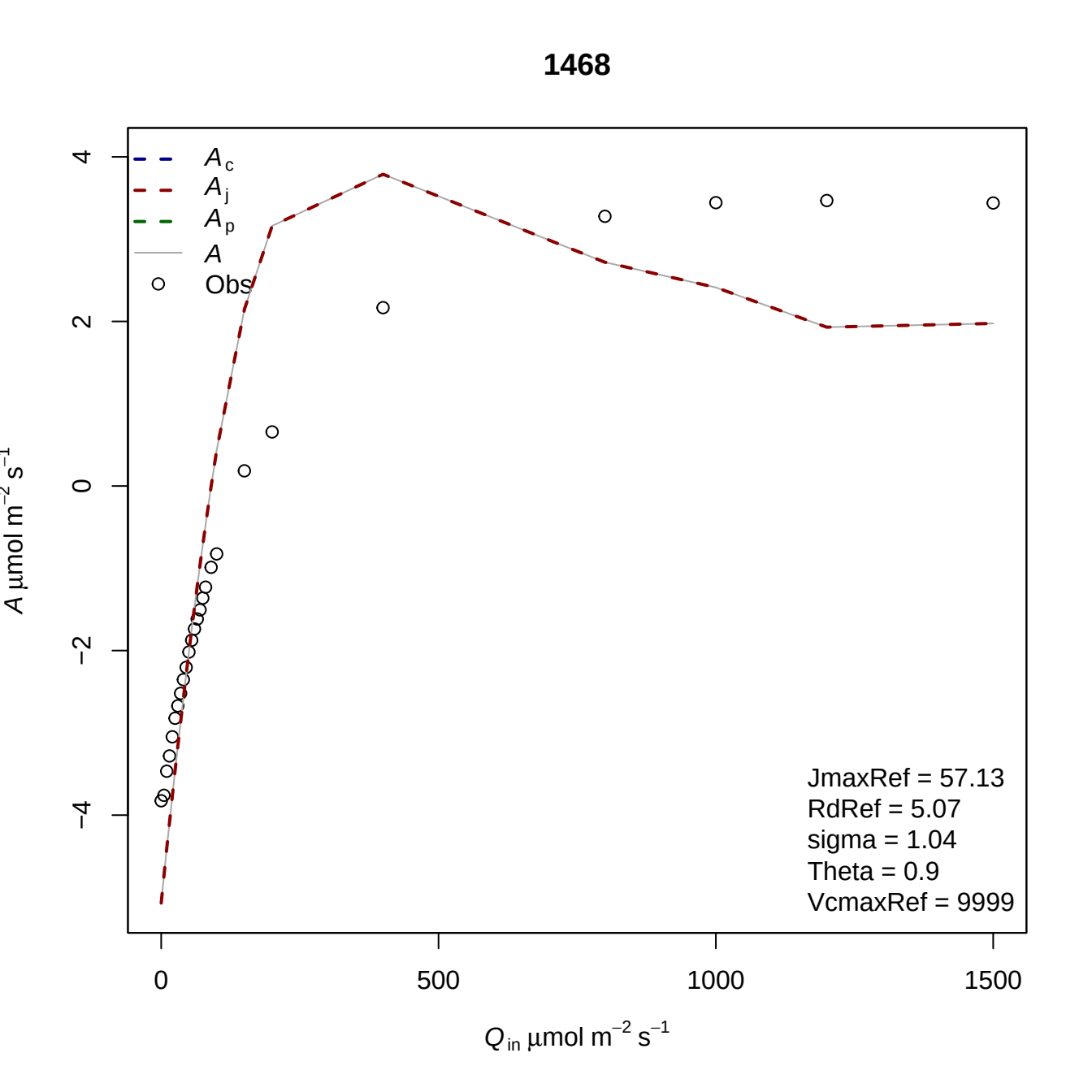

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 67.08
RdRef = 1.55
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

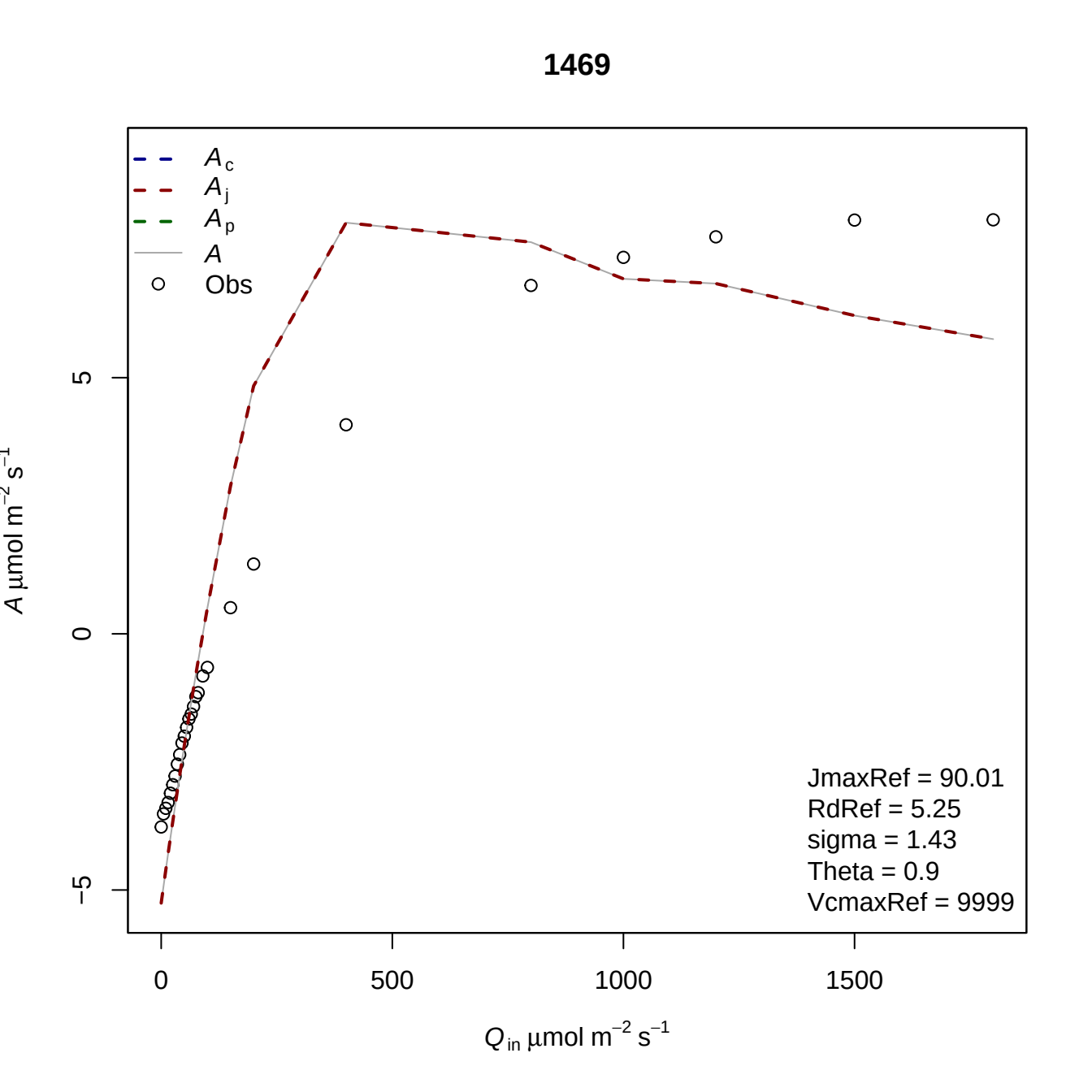

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 89.32
RdRef = 1.46
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

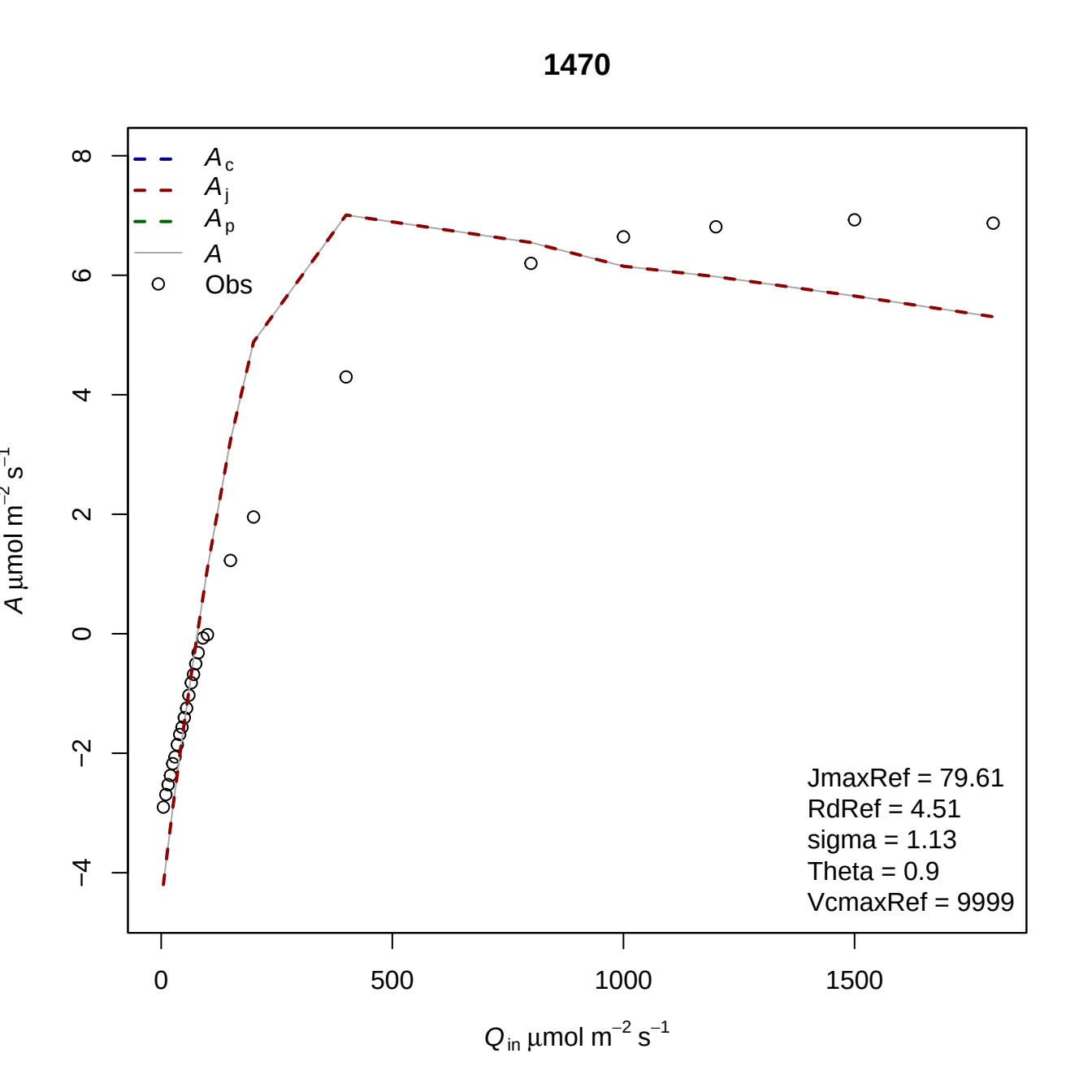

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 56.72
RdRef = 1.76
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

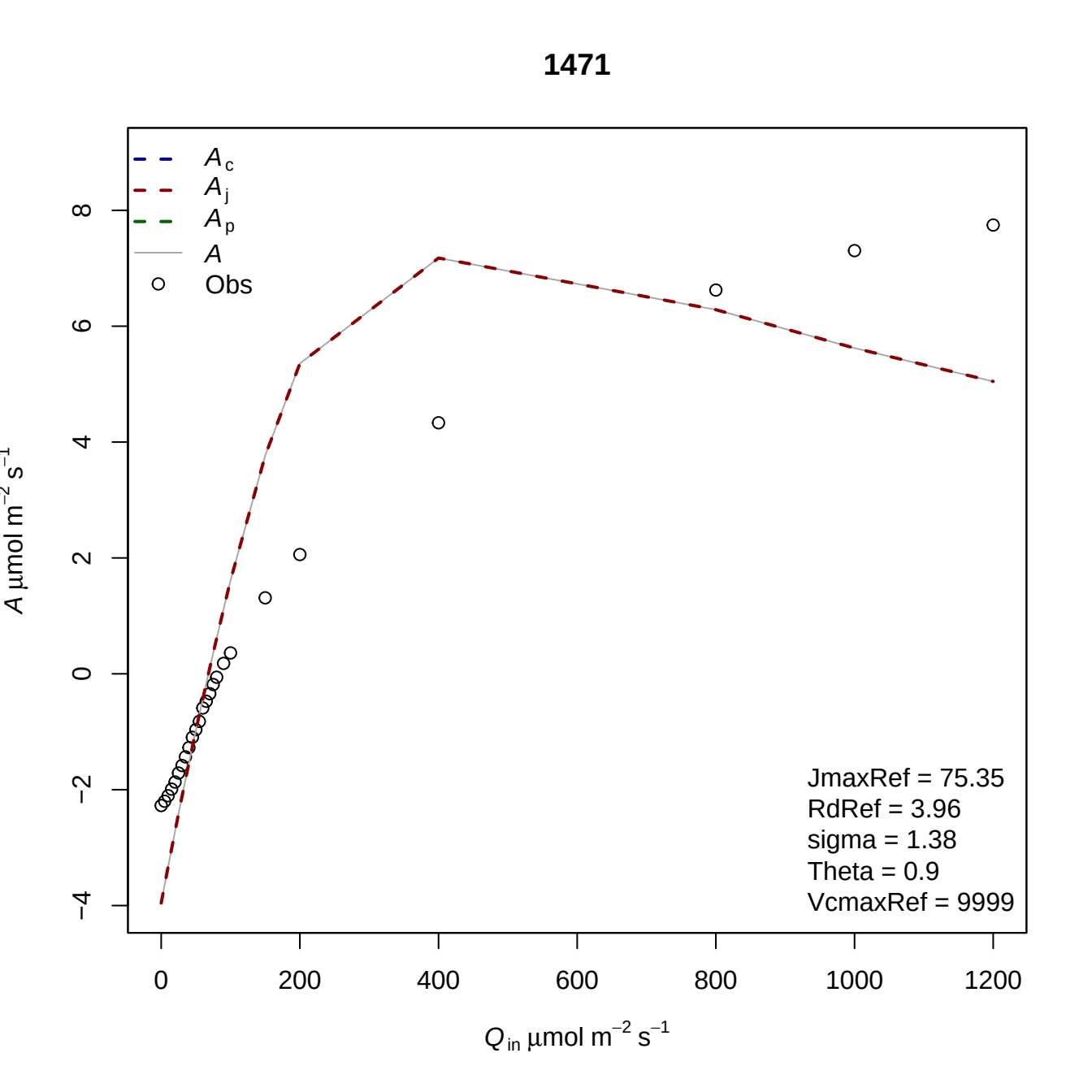

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 78.27
RdRef = 2.04
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

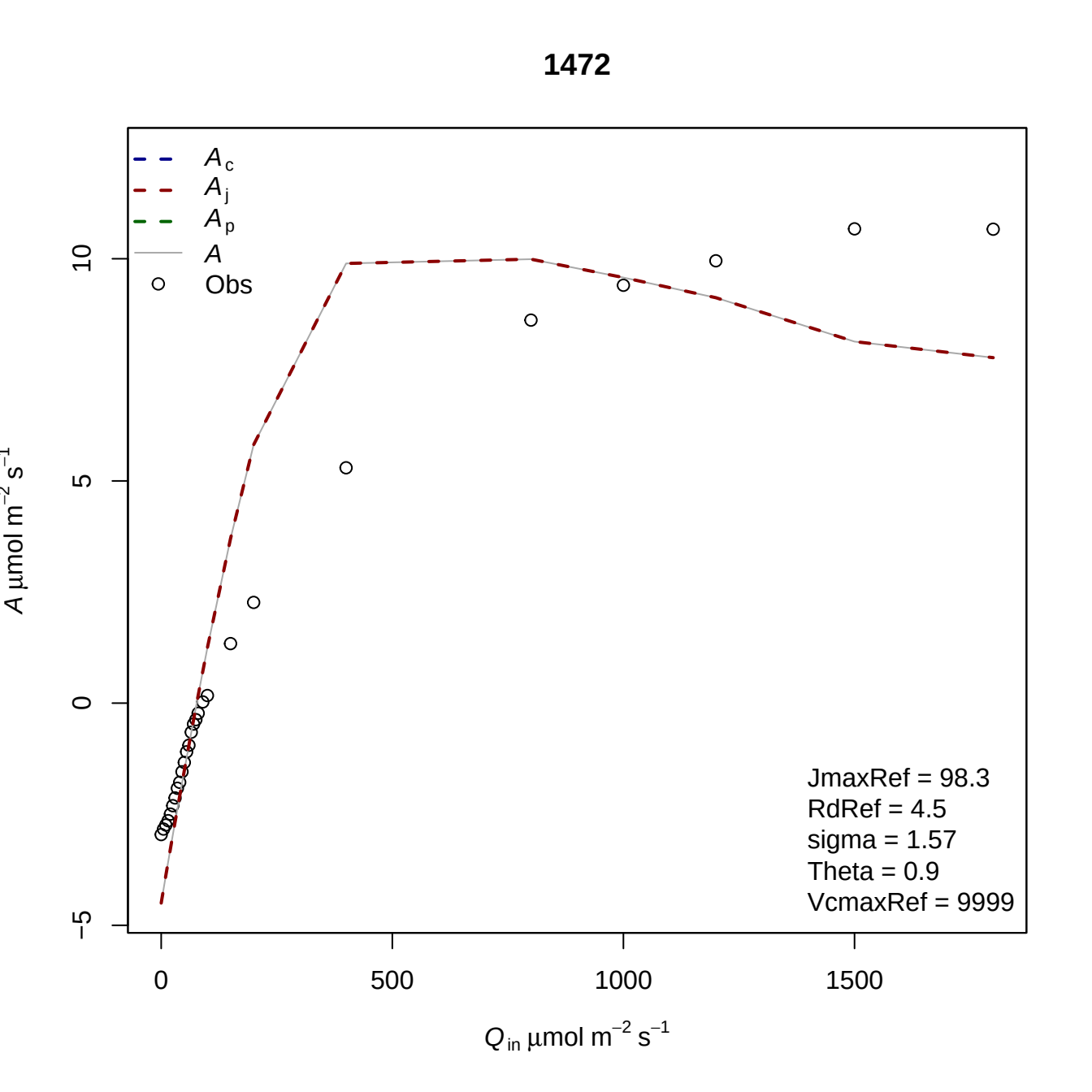

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 74.73
RdRef = 1.69
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

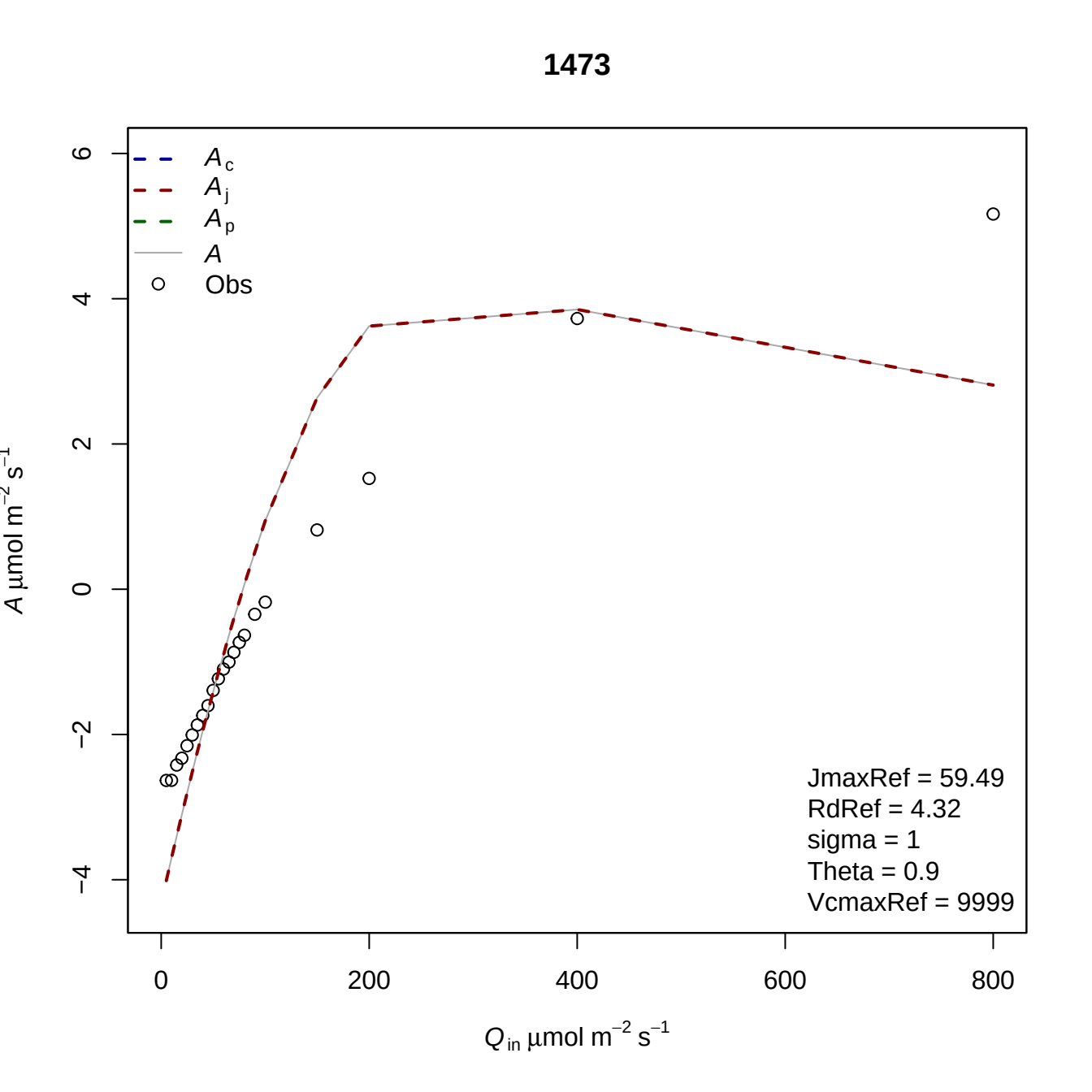

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 57.13
RdRef = 5.07
sigma = 1.04
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

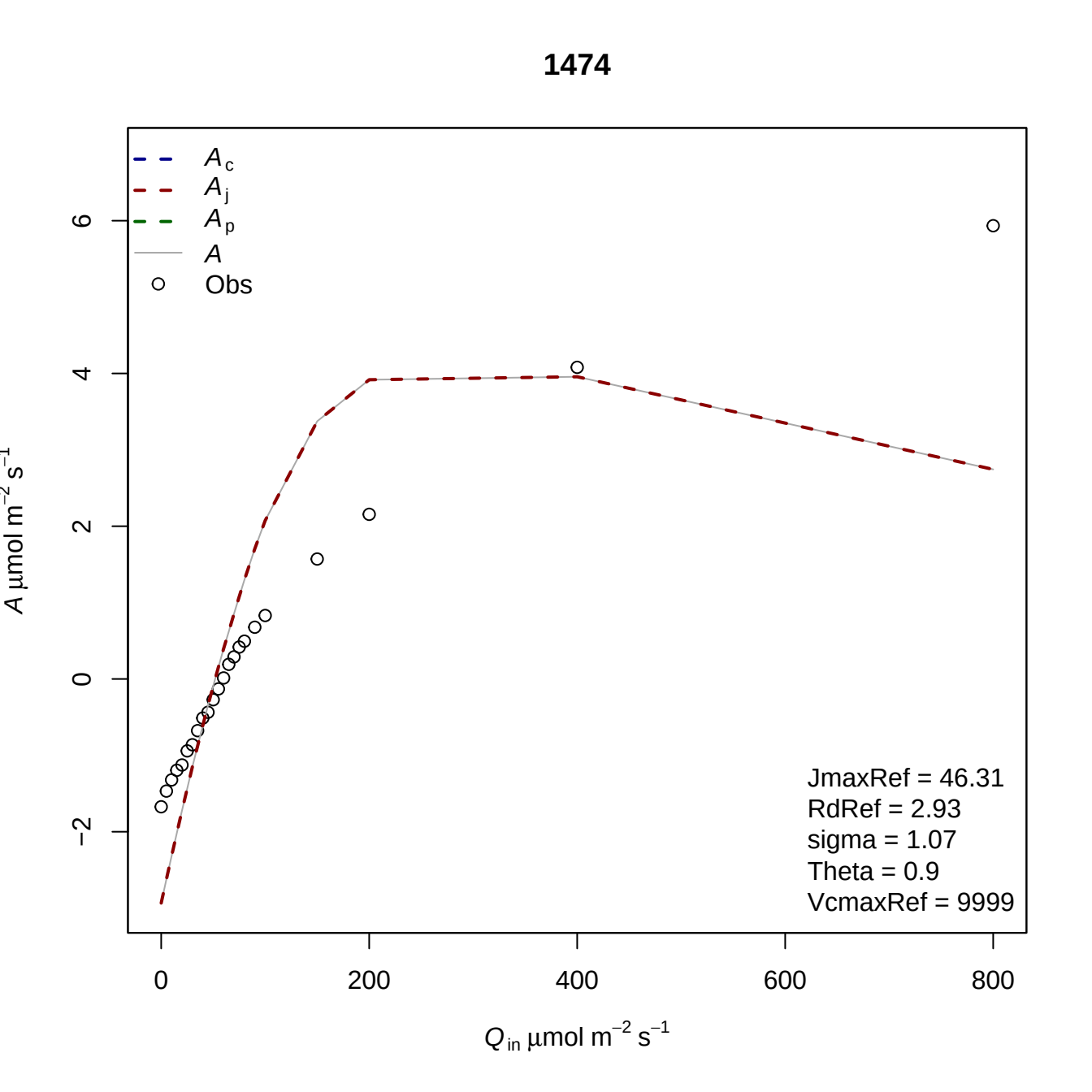

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 90.01
RdRef = 5.25
sigma = 1.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−5
0
5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

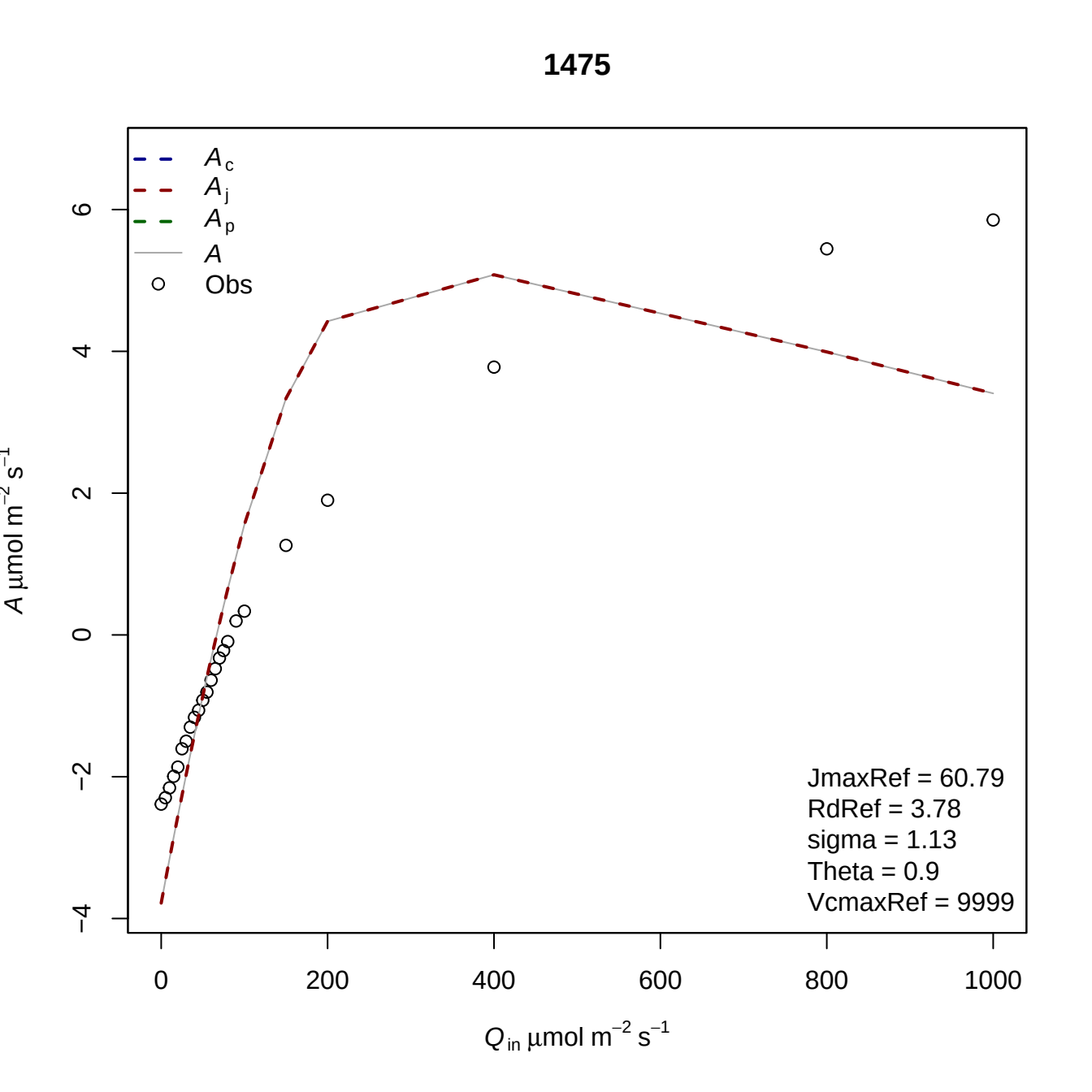

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 79.61
RdRef = 4.51
sigma = 1.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

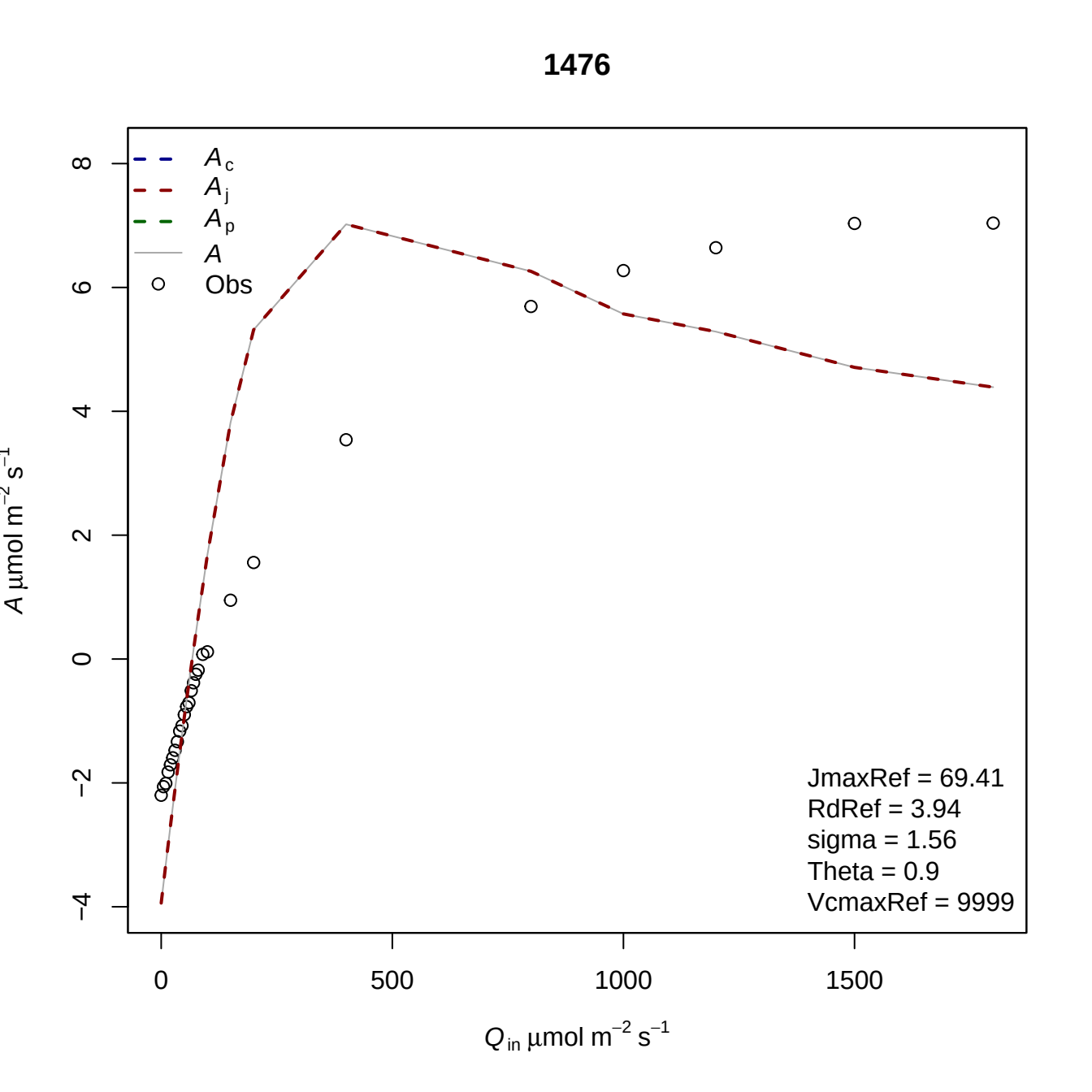

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 75.35
RdRef = 3.96
sigma = 1.38
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

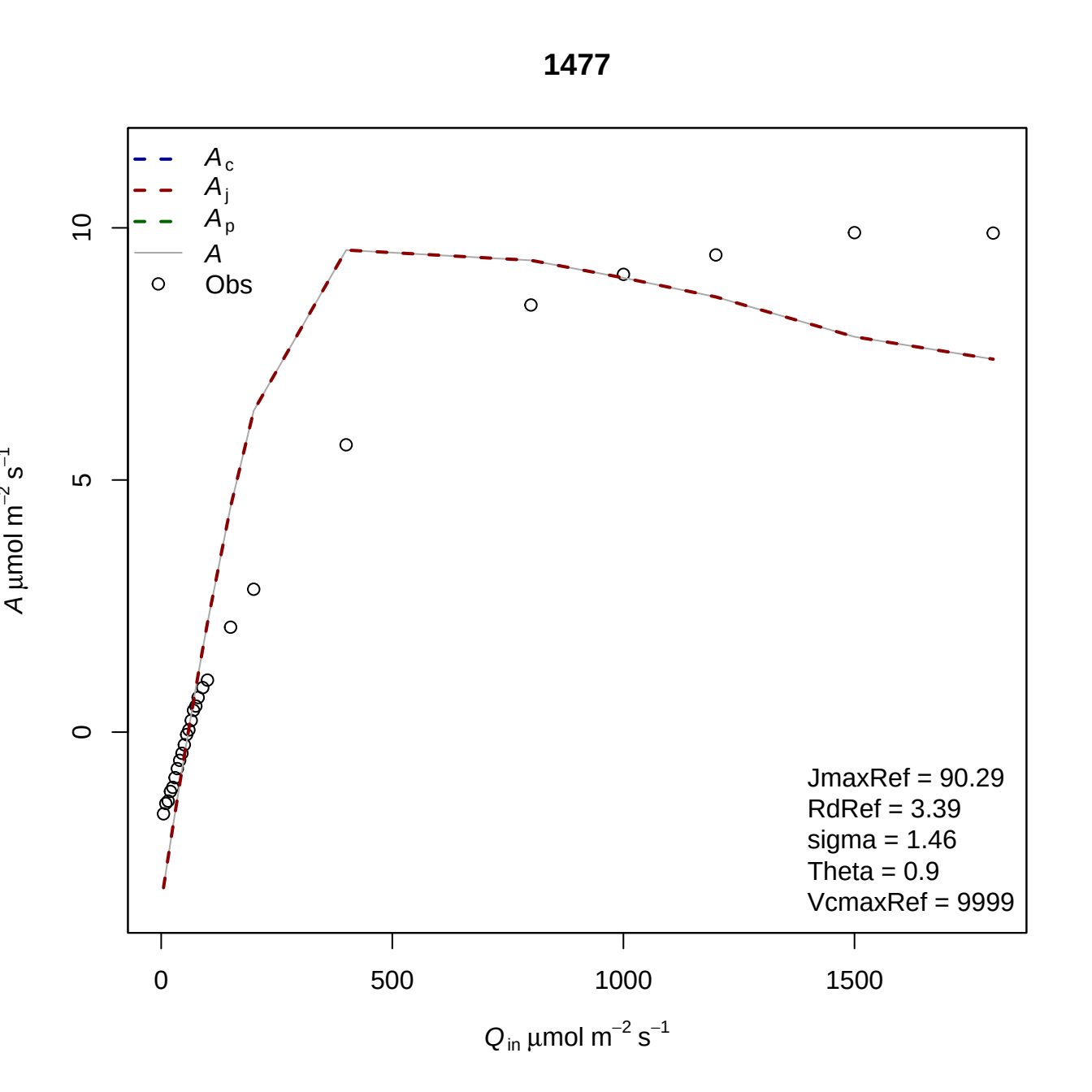

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 98.3
RdRef = 4.5
sigma = 1.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

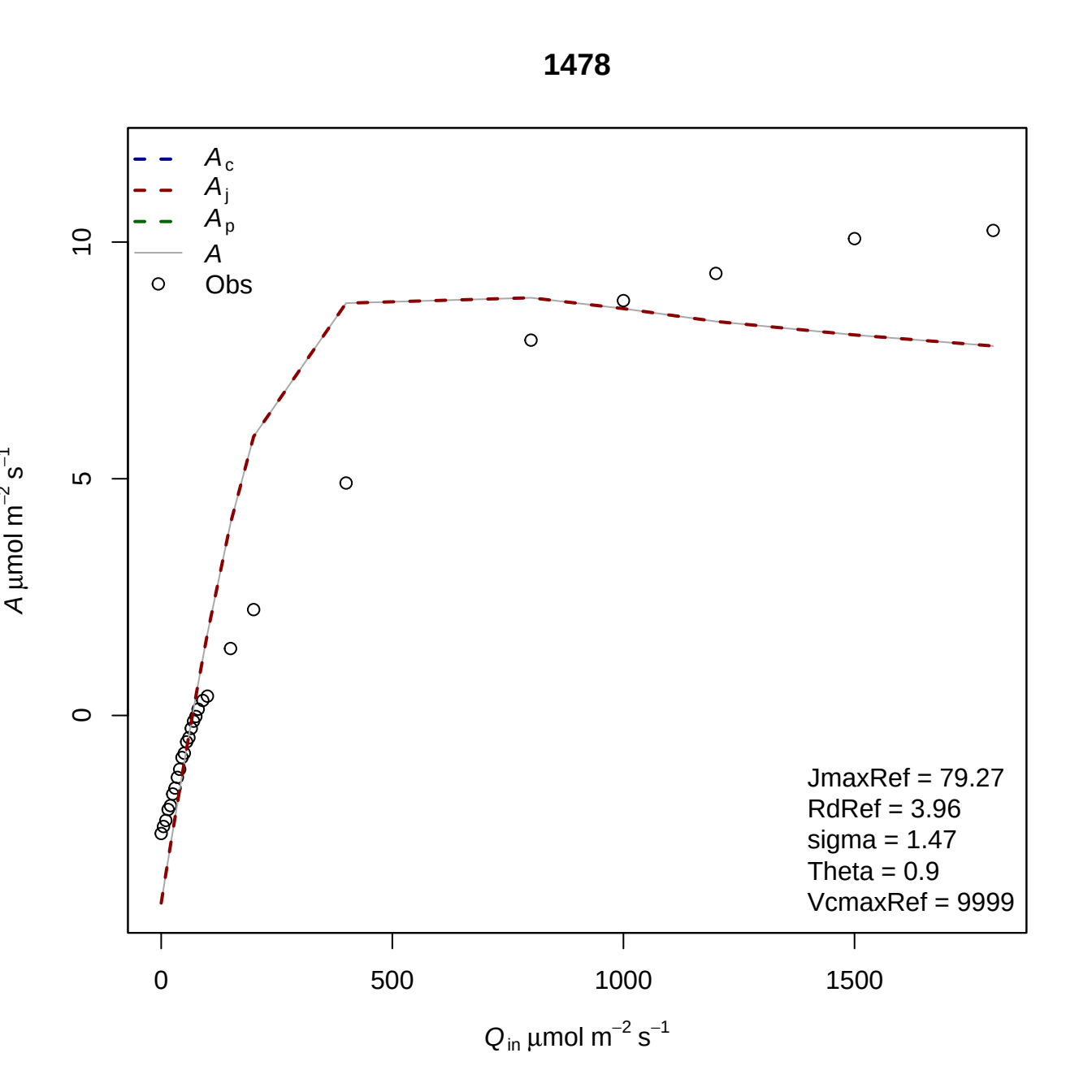

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 59.49
RdRef = 4.32
sigma = 1
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800

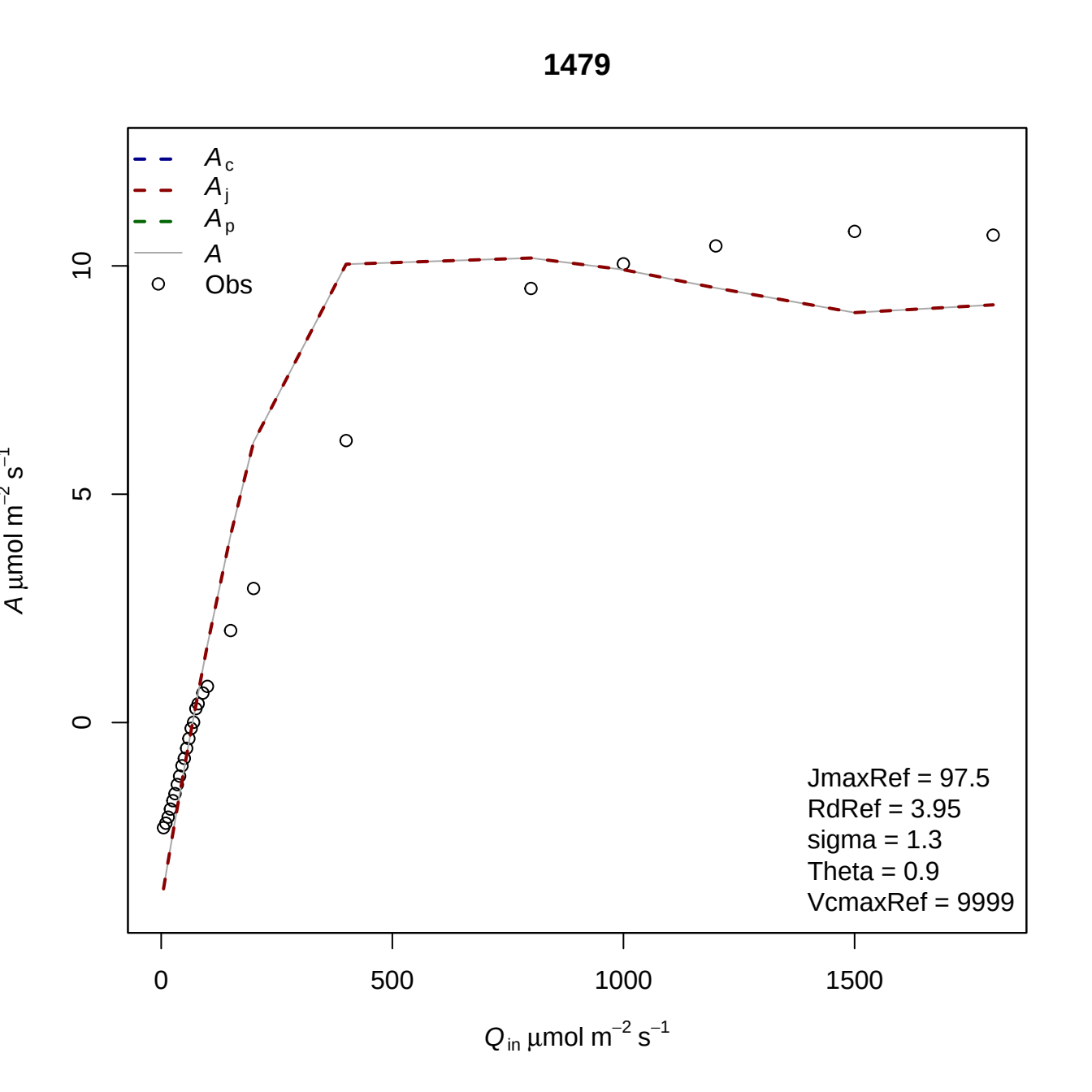

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 46.31
RdRef = 2.93
sigma = 1.07
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800

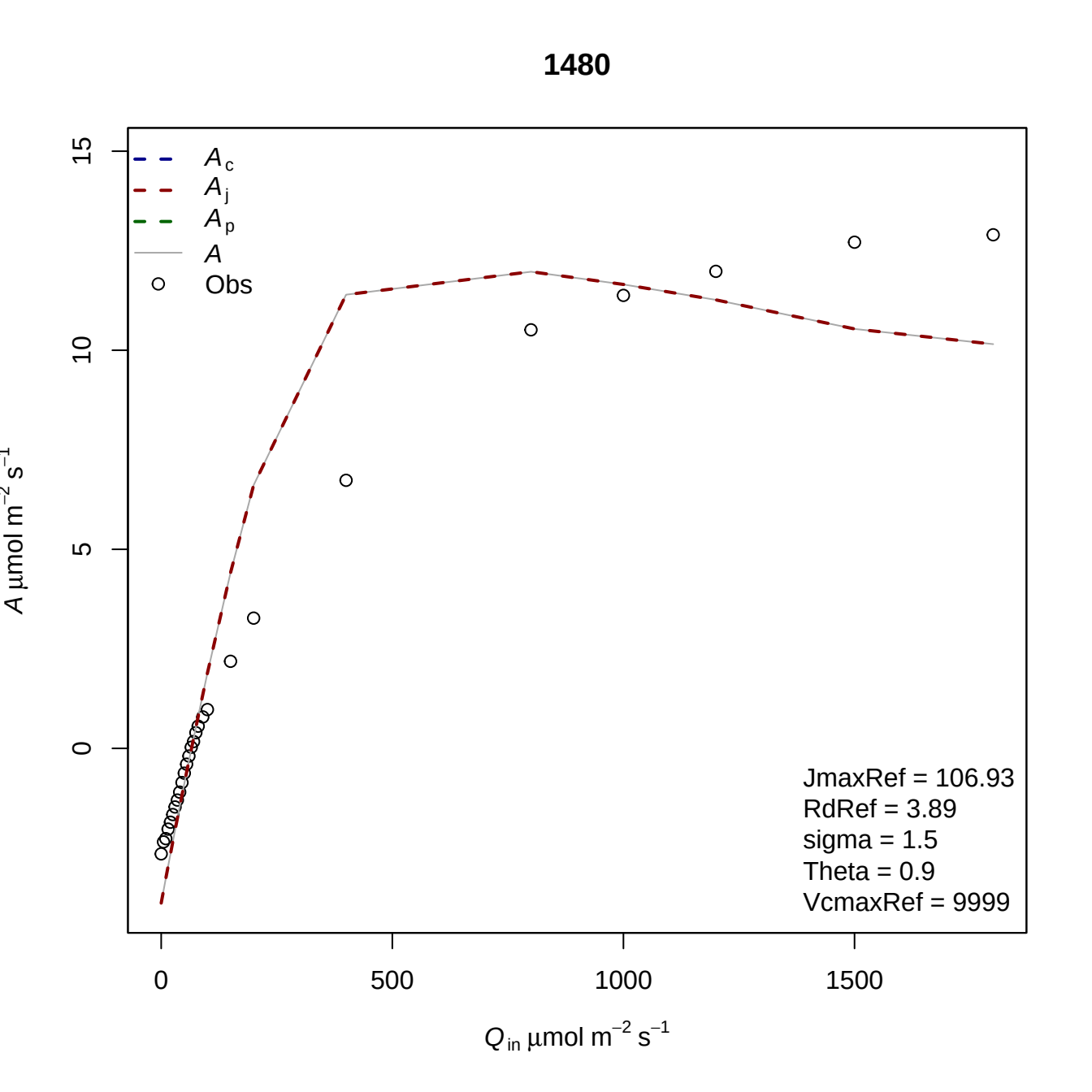

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 60.79
RdRef = 3.78
sigma = 1.13
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

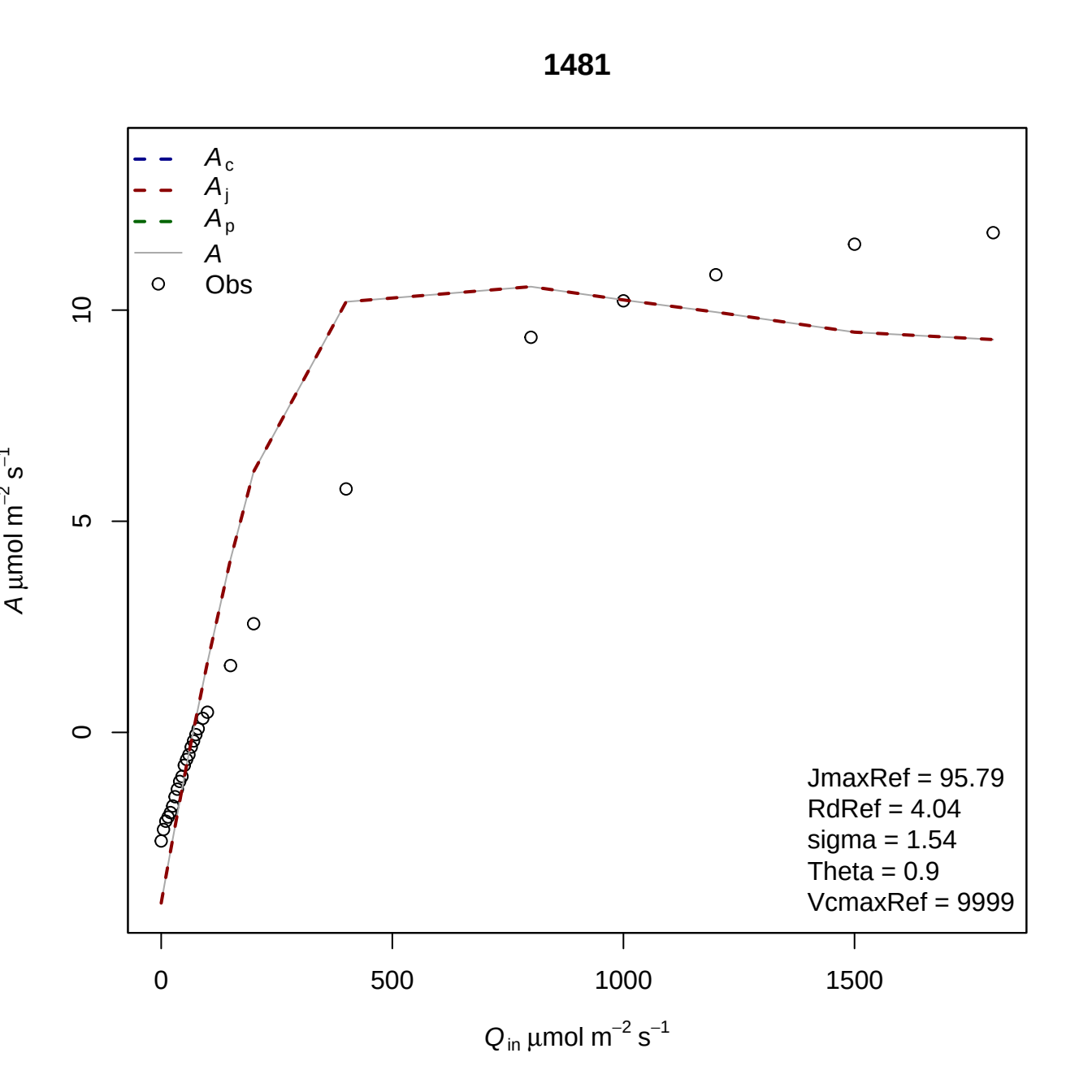

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 69.41
RdRef = 3.94
sigma = 1.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

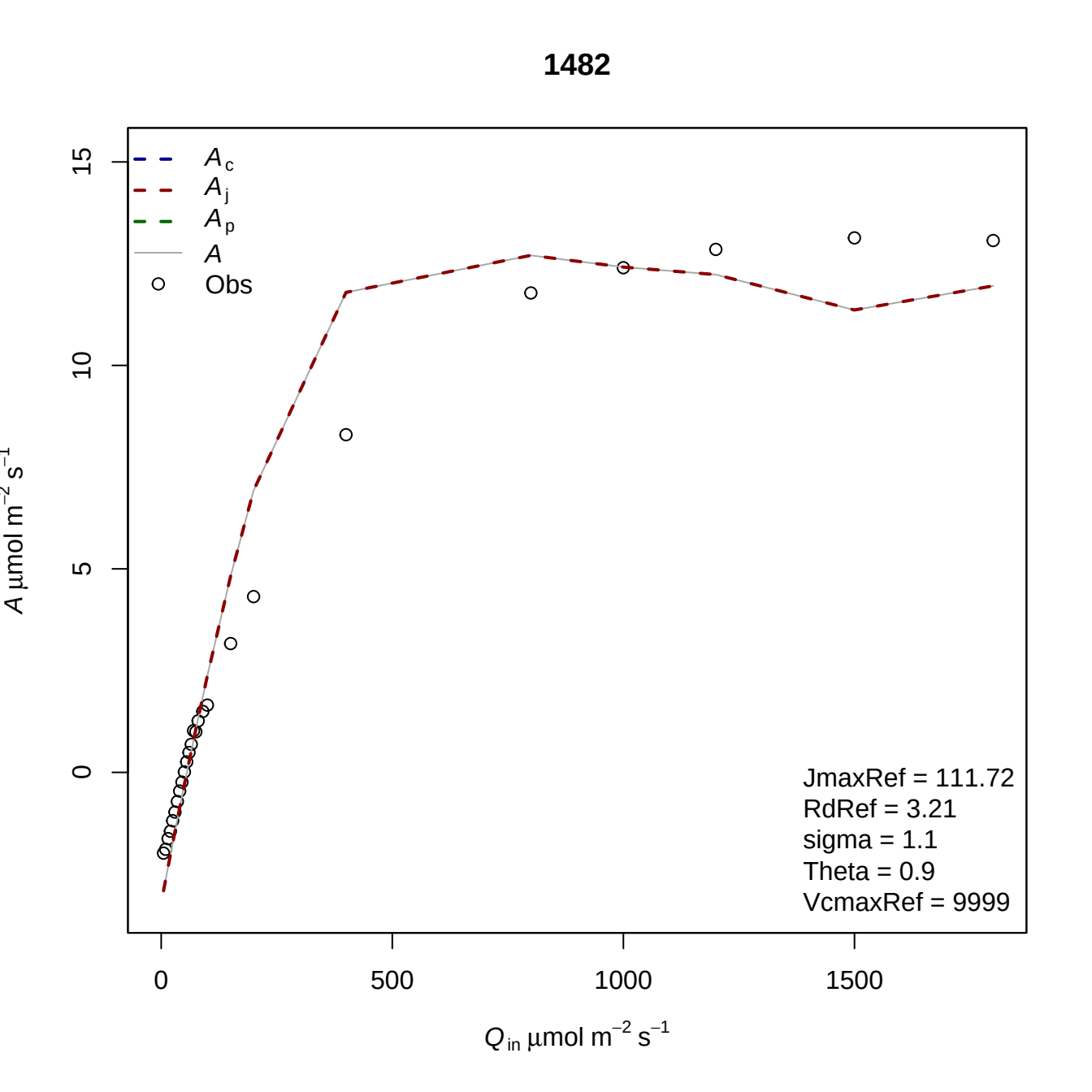

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 90.29
RdRef = 3.39
sigma = 1.46
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

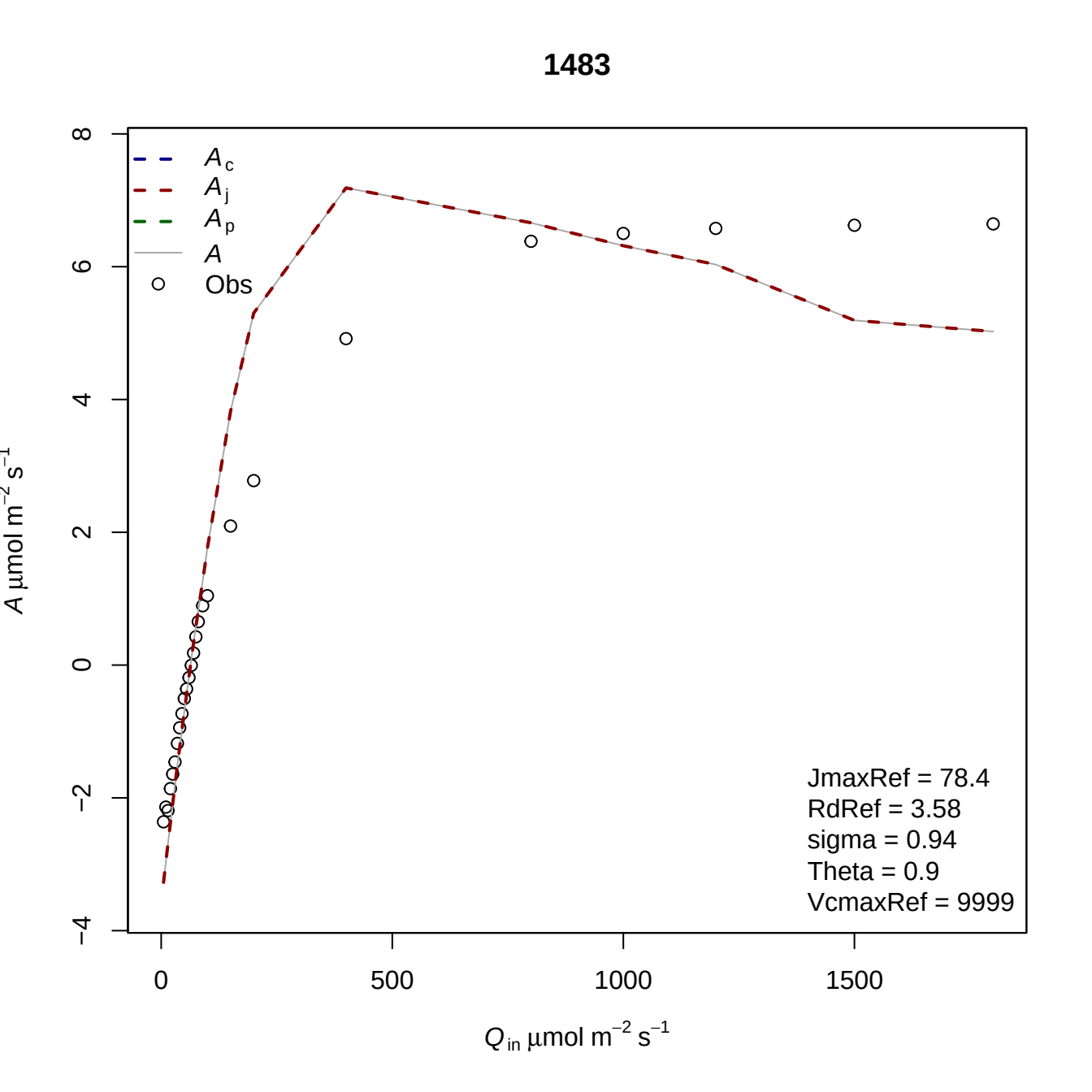

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 79.27
RdRef = 3.96
sigma = 1.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

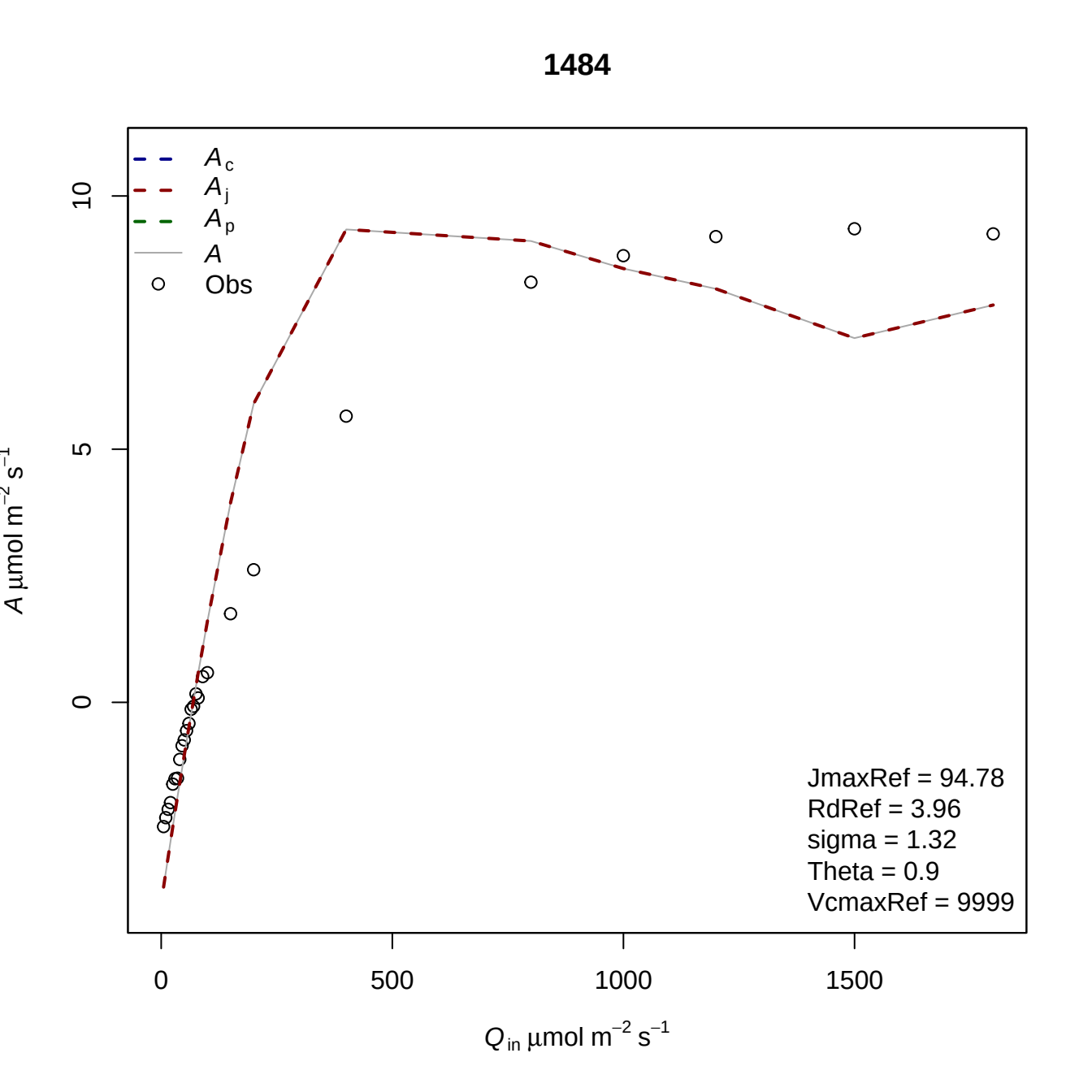

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 97.5
RdRef = 3.95
sigma = 1.3
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

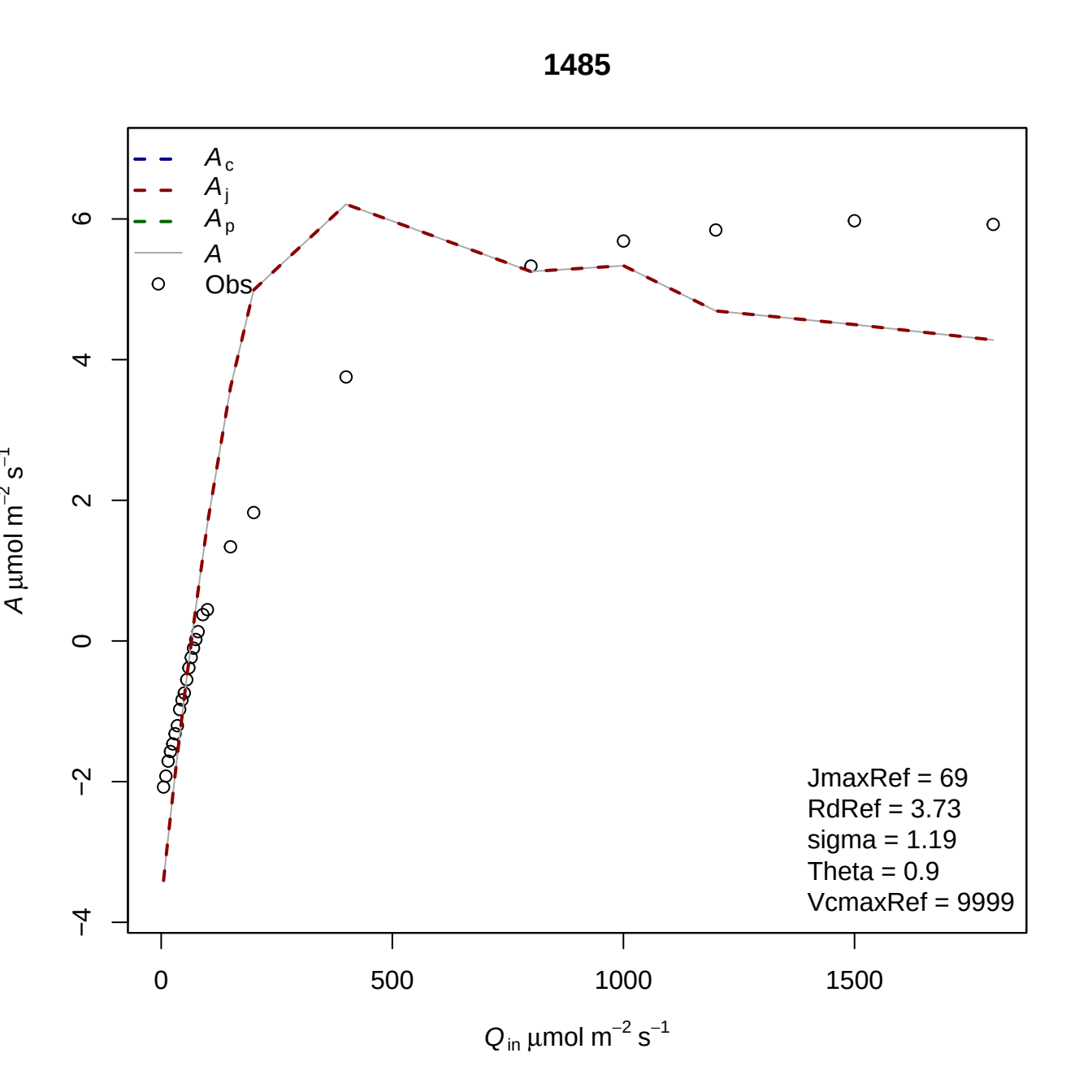

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 106.93
RdRef = 3.89
sigma = 1.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

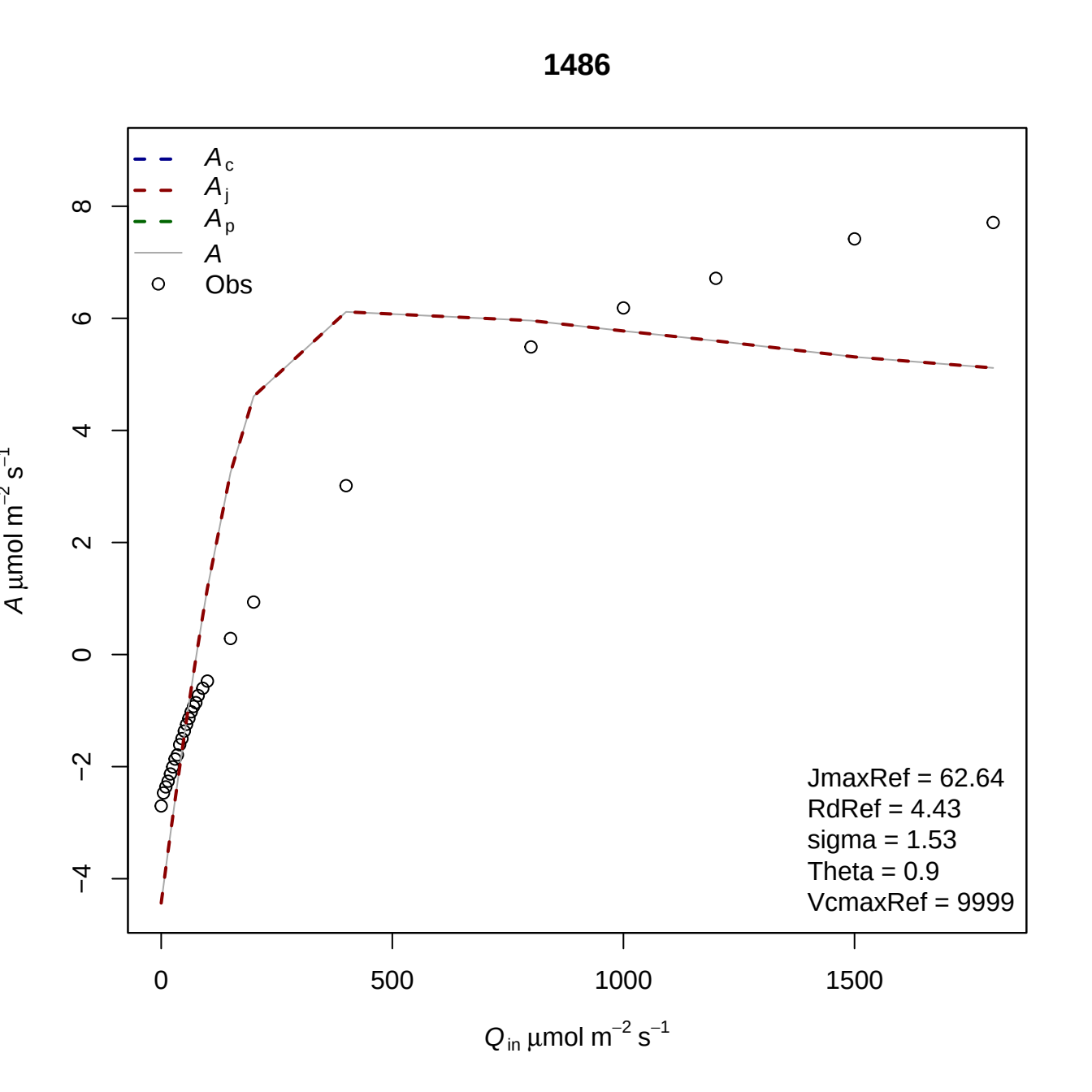

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 95.79
RdRef = 4.04
sigma = 1.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

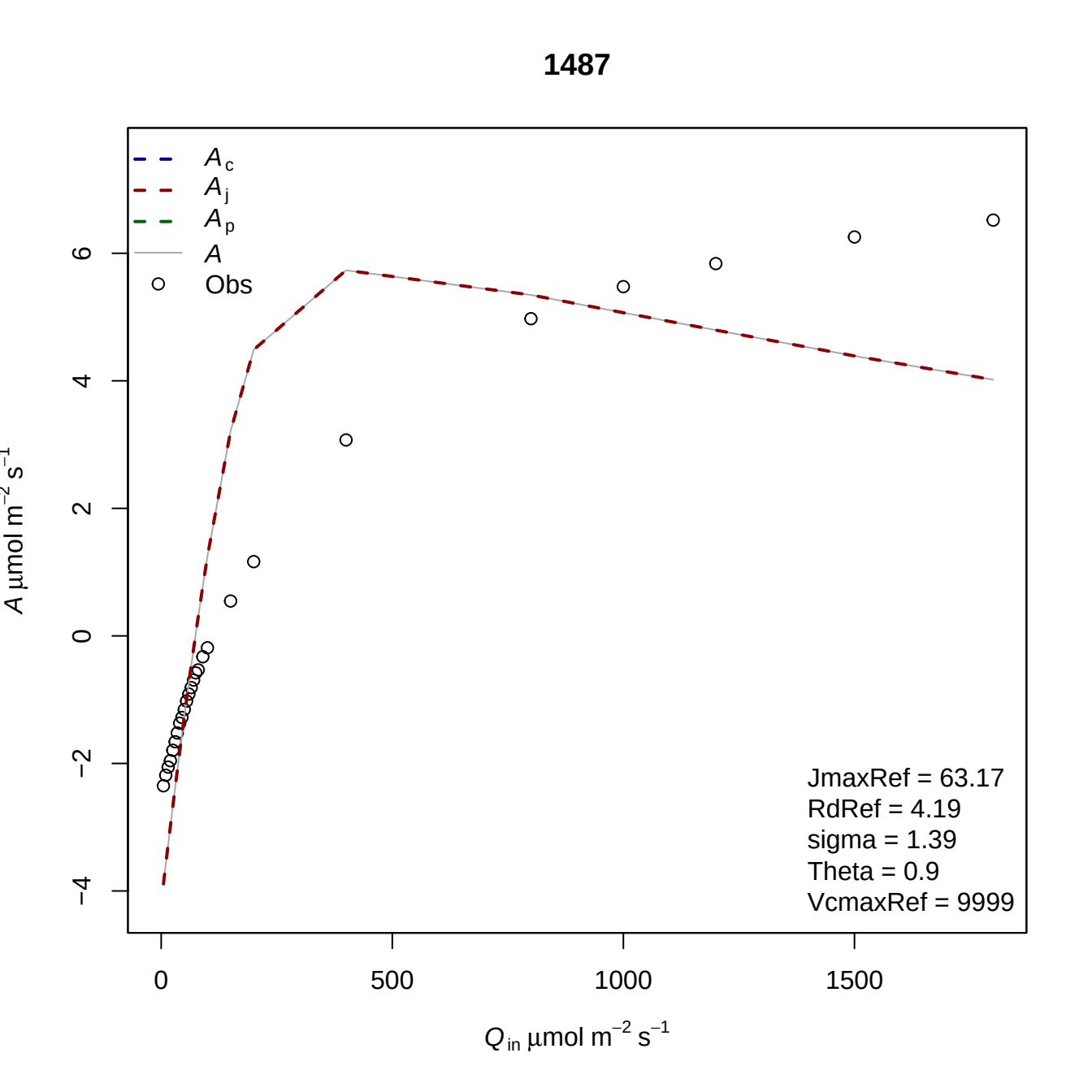

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 111.72
RdRef = 3.21
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

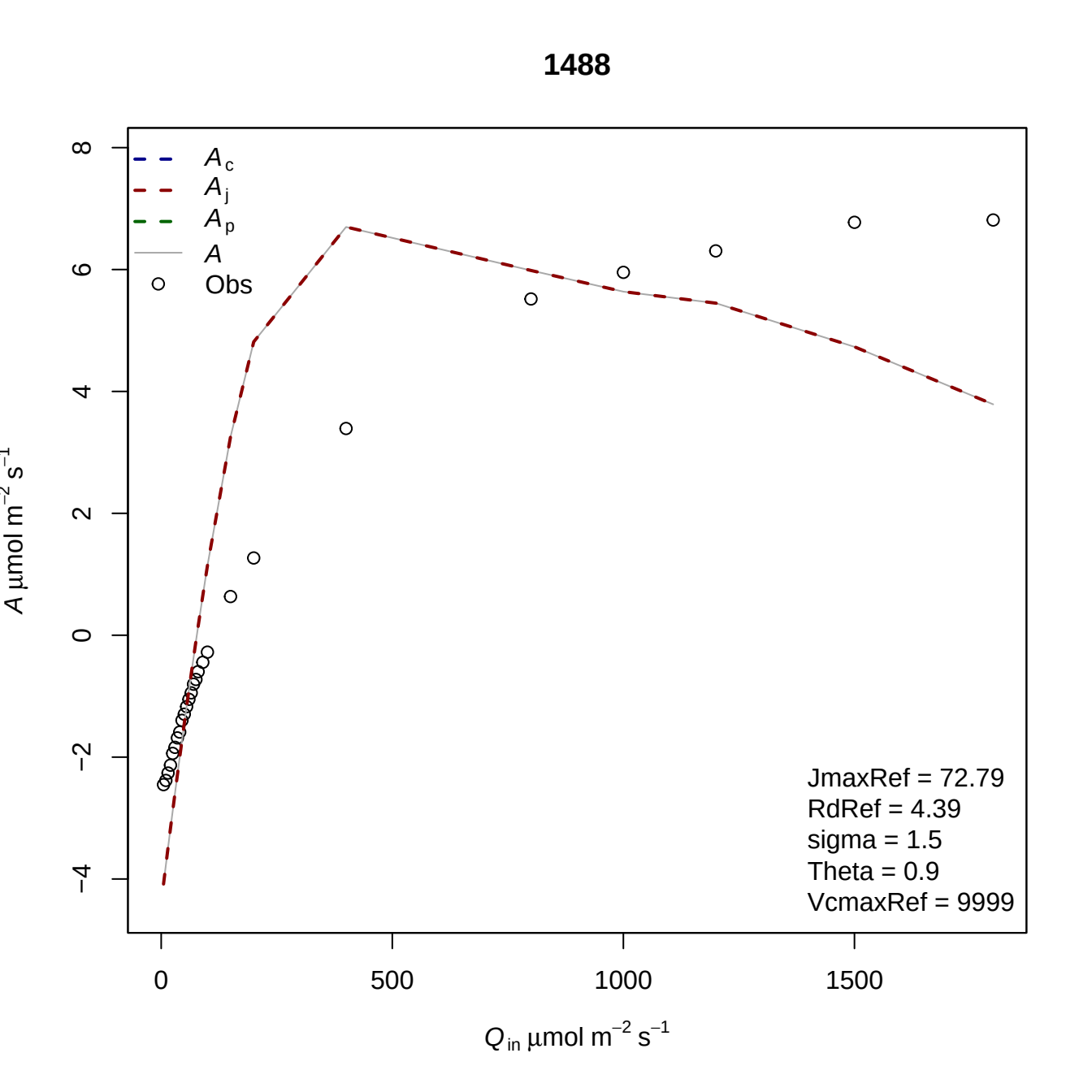

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 78.4
RdRef = 3.58
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

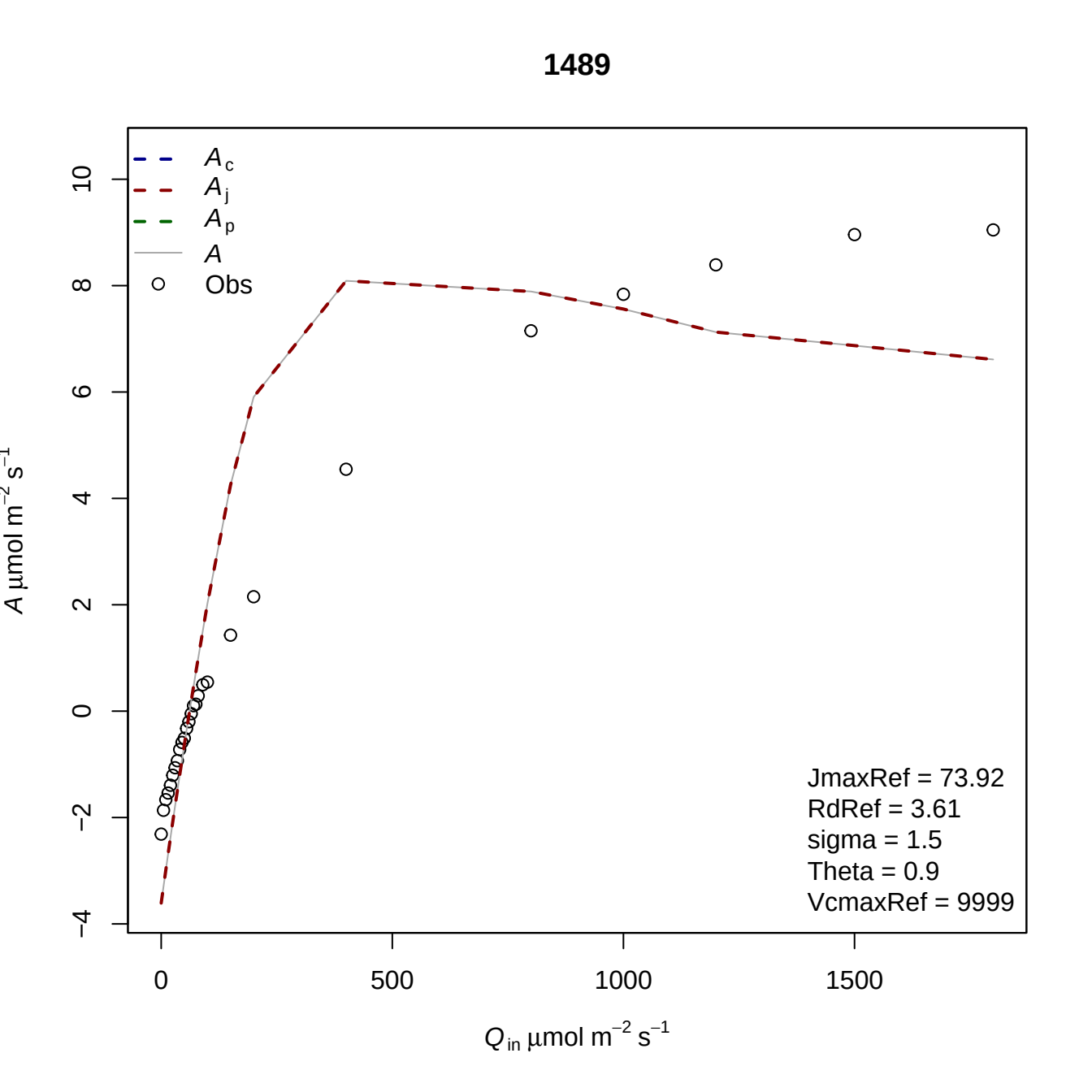

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 94.78
RdRef = 3.96
sigma = 1.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

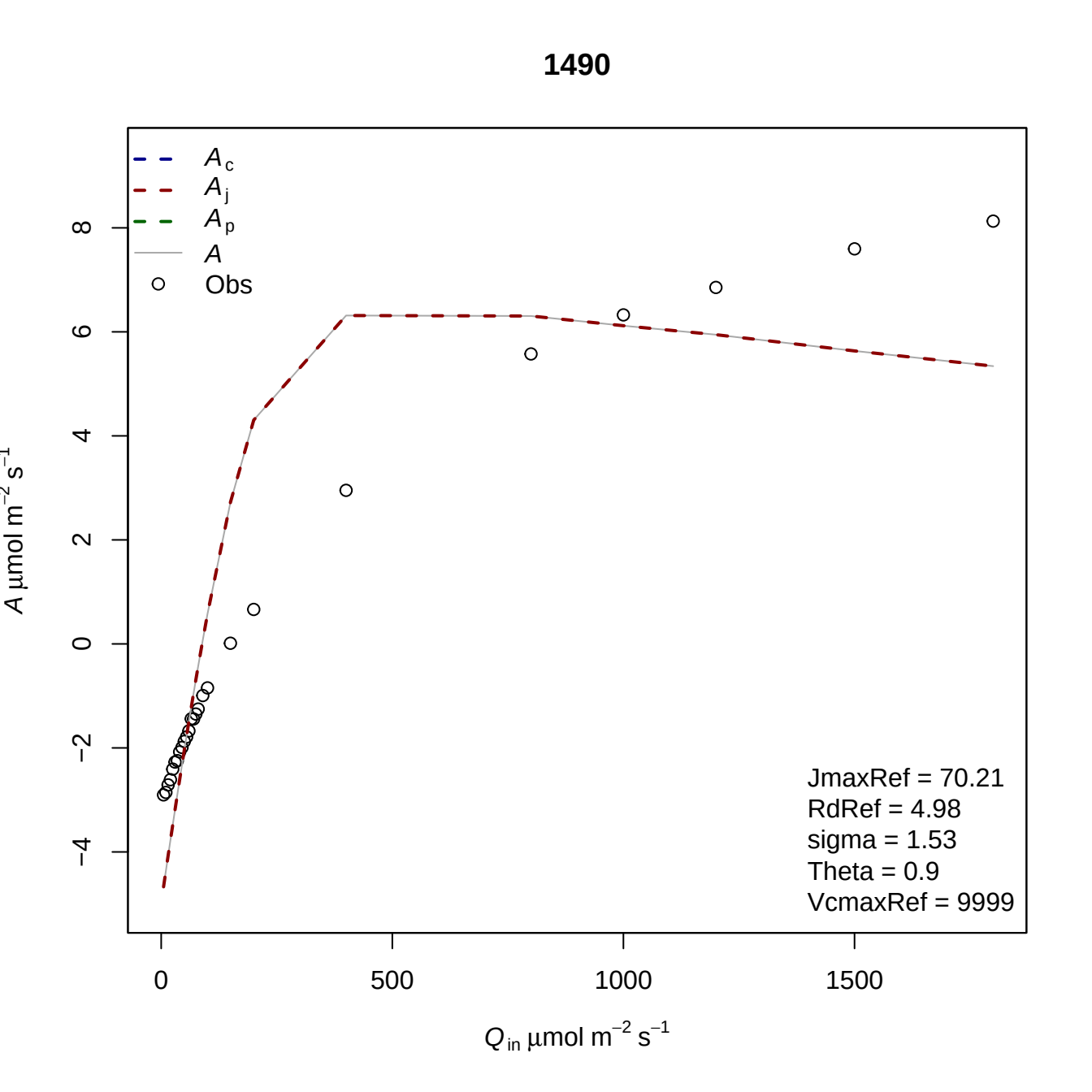

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 69
RdRef = 3.73
sigma = 1.19
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

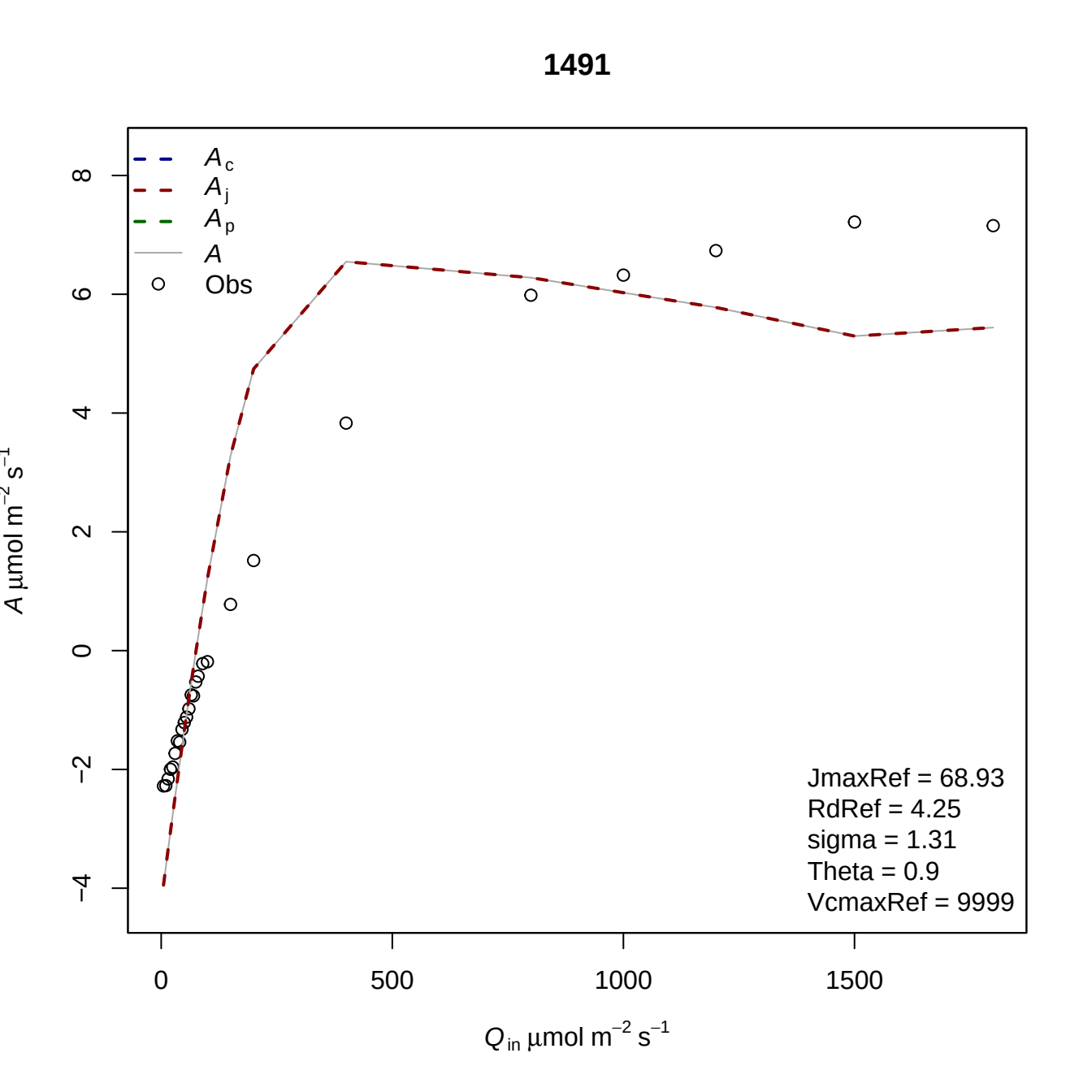

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 62.64
RdRef = 4.43
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

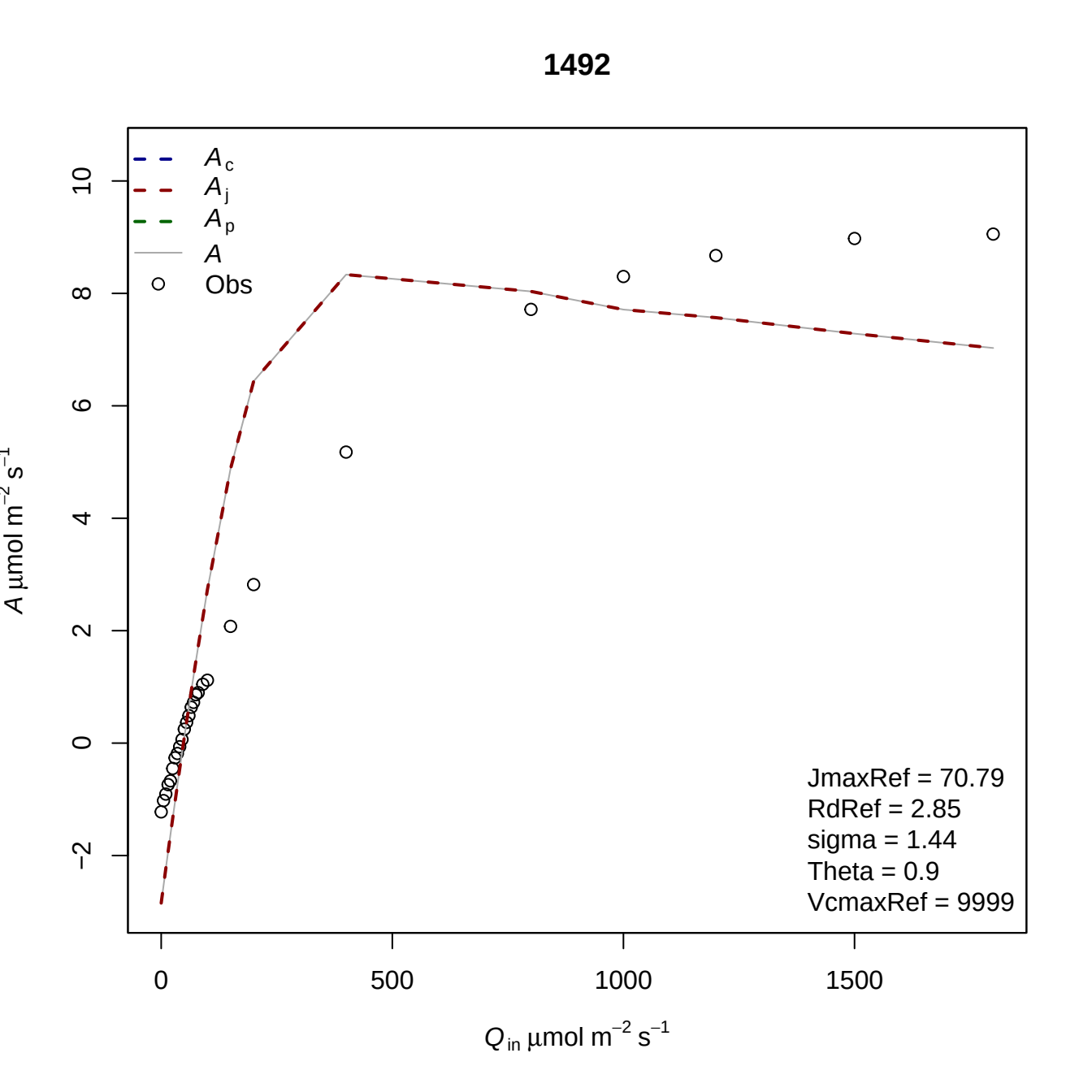

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 63.17
RdRef = 4.19
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

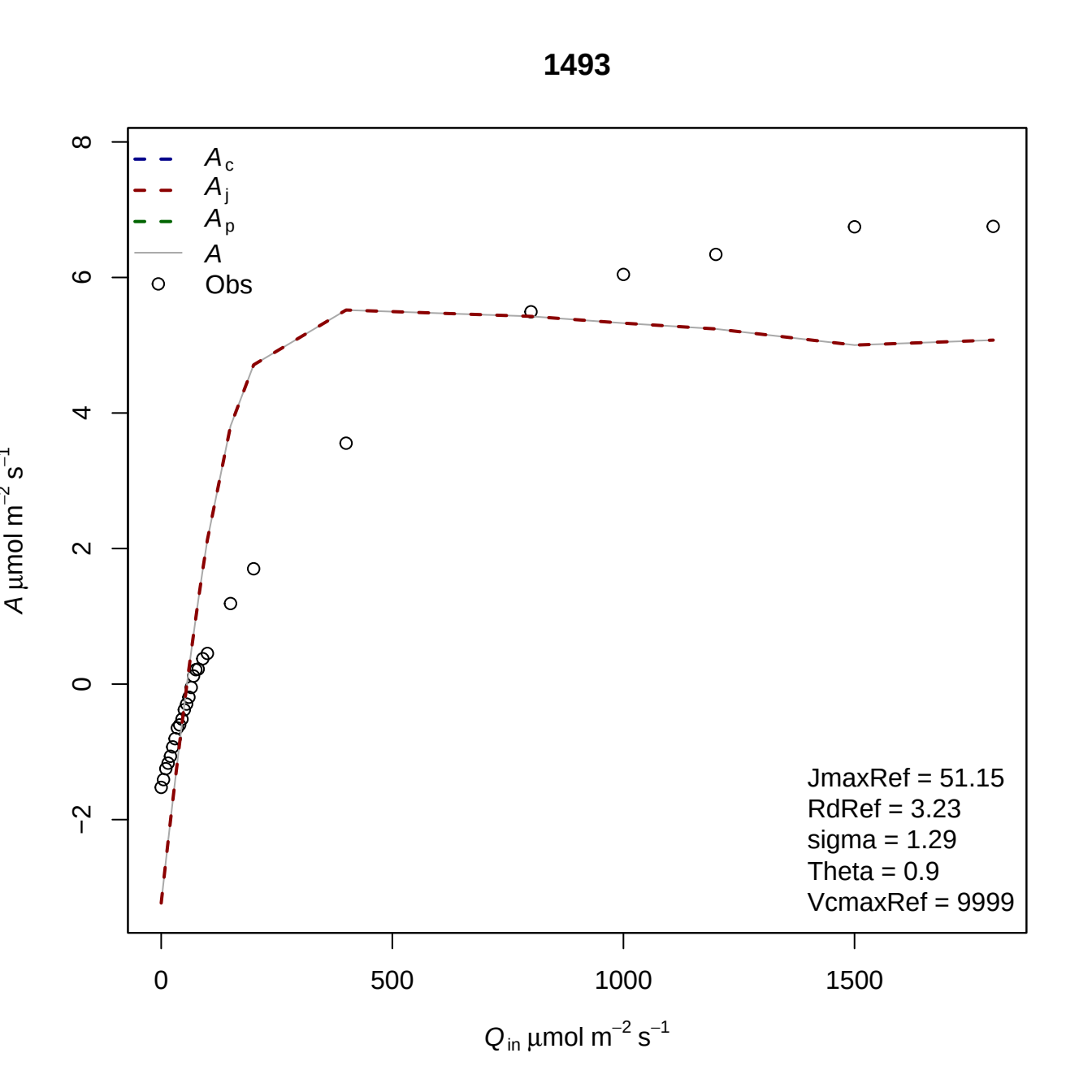

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 72.79
RdRef = 4.39
sigma = 1.5
Theta = 0.9
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

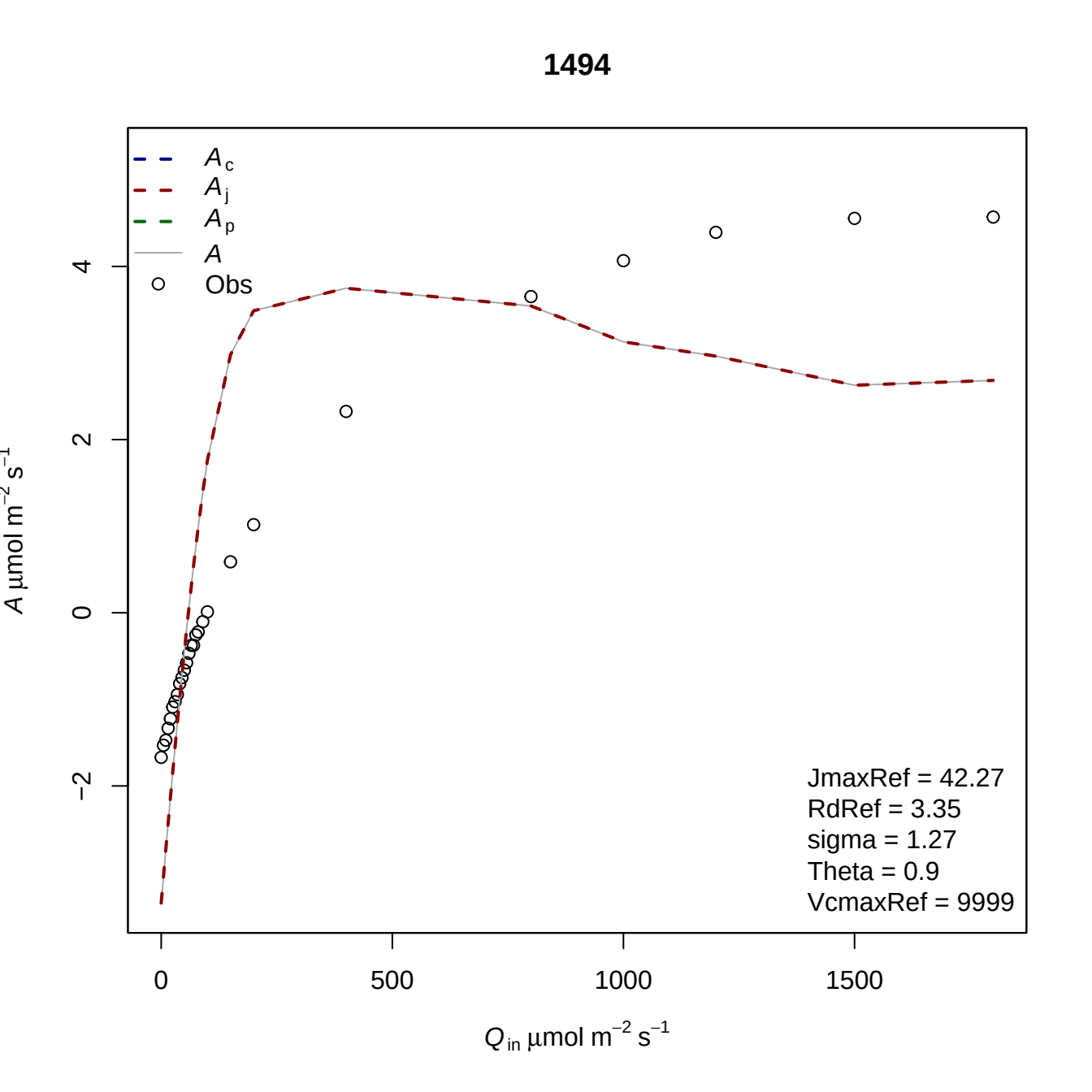

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 73.92
RdRef = 3.61
sigma = 1.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

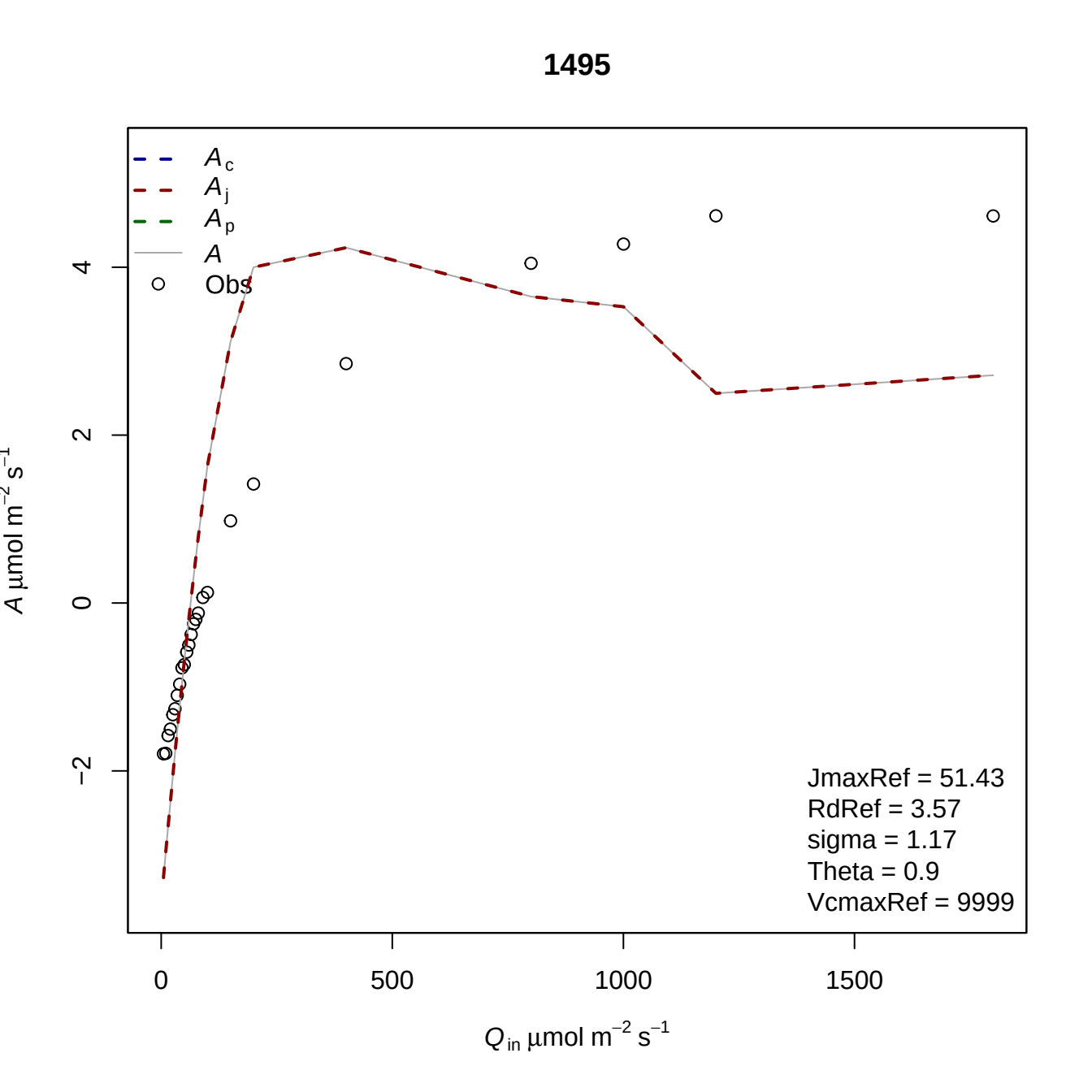

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 70.21
RdRef = 4.98
sigma = 1.53
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

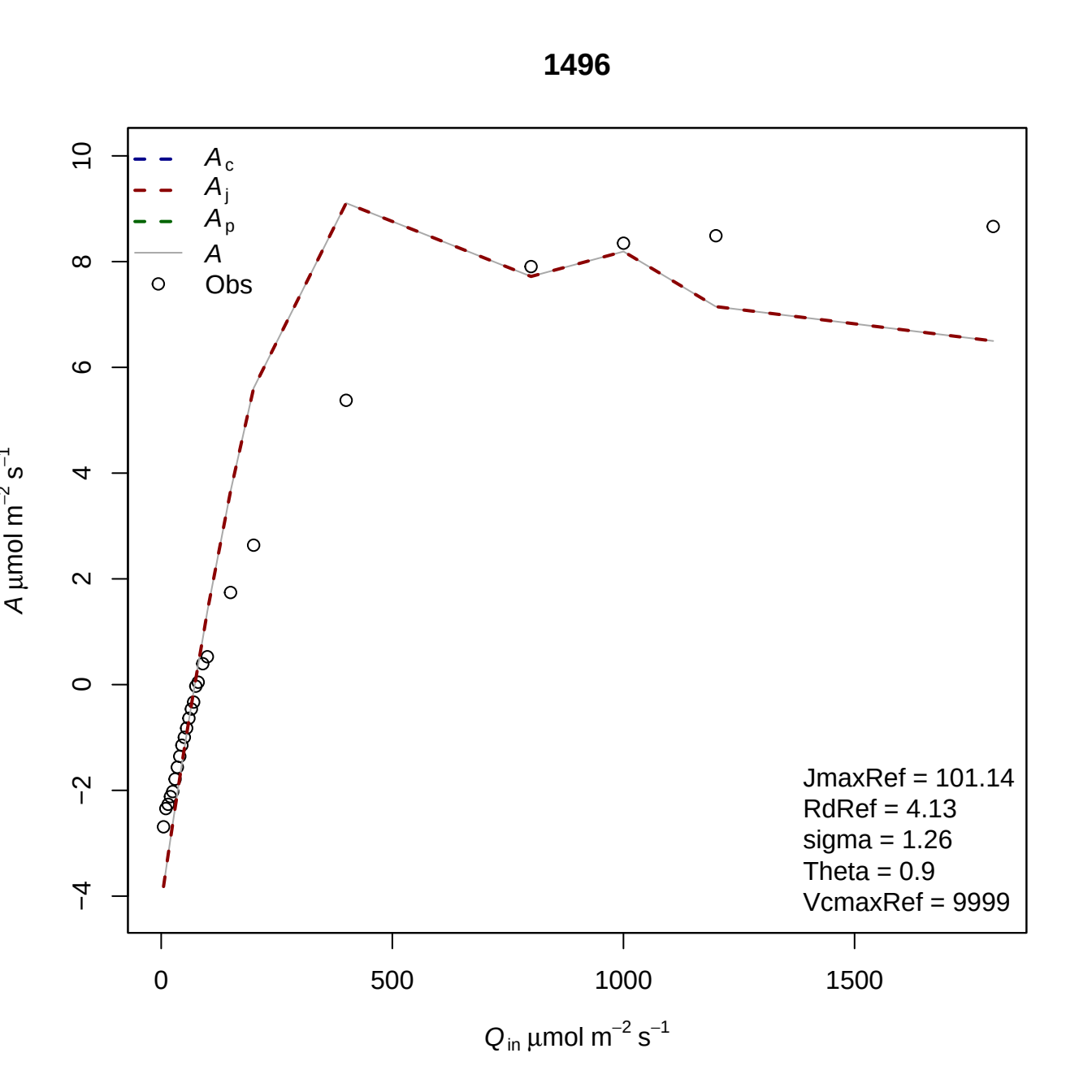

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 68.93
RdRef = 4.25
sigma = 1.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

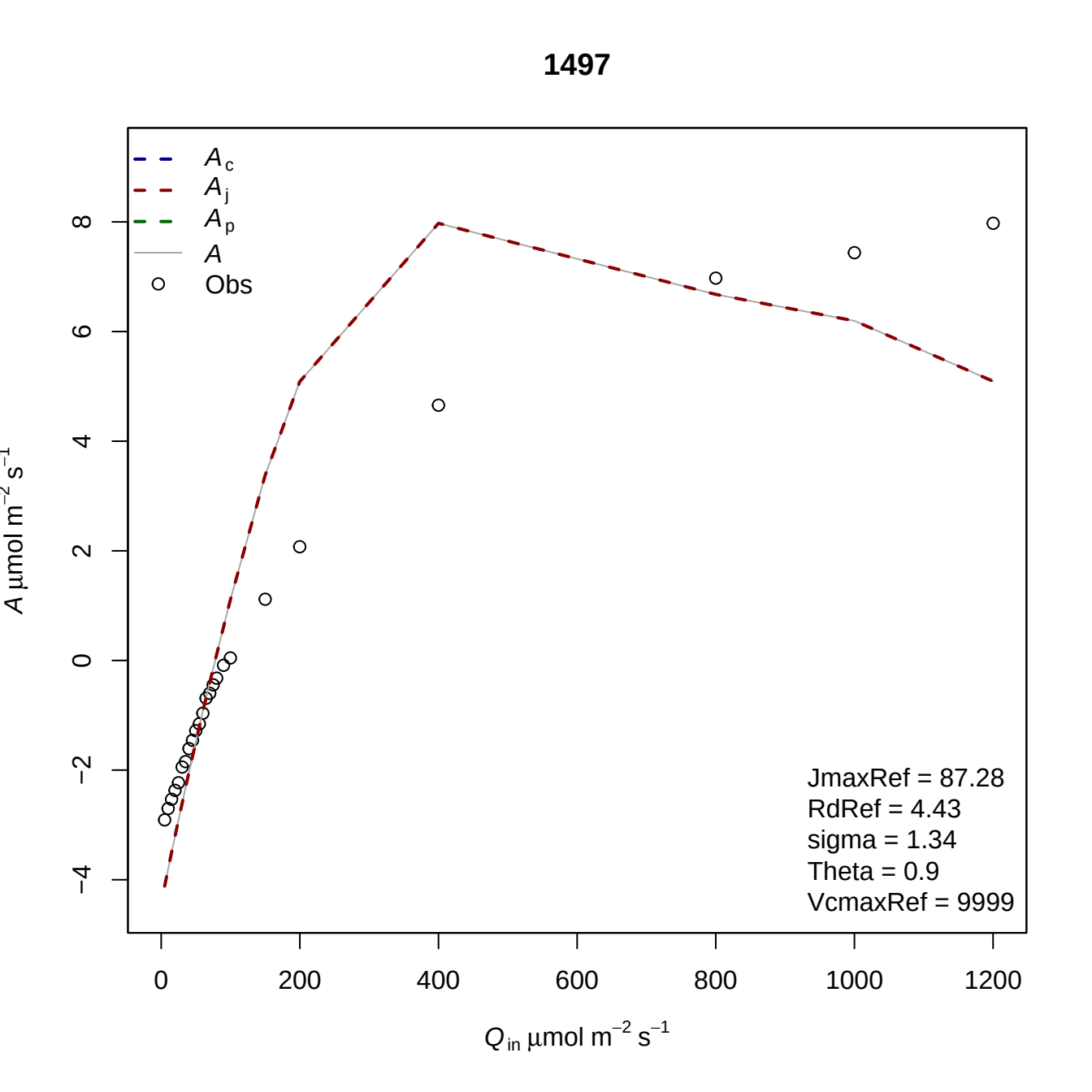

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 70.79
RdRef = 2.85
sigma = 1.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

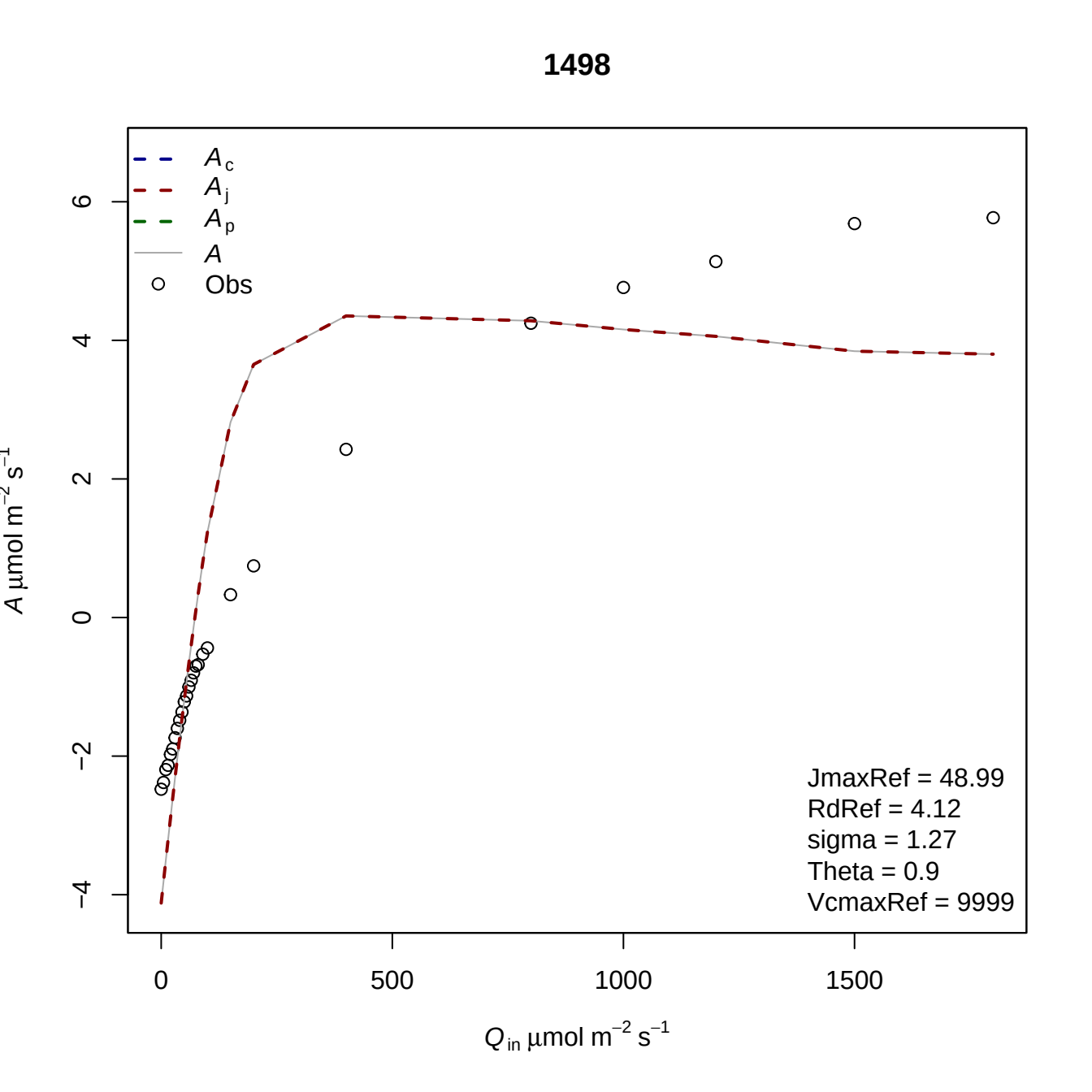

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 51.15
RdRef = 3.23
sigma = 1.29
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

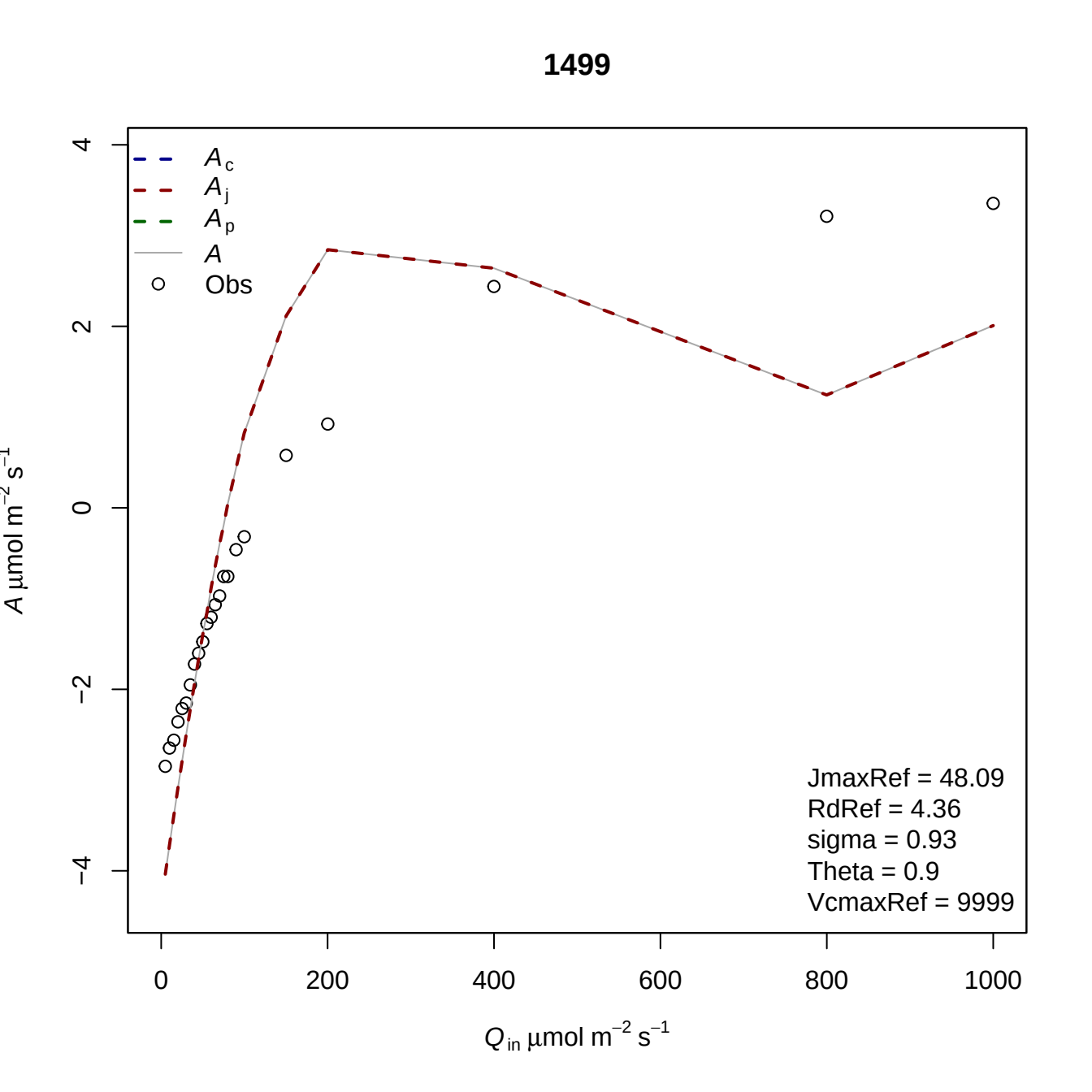

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 42.27
RdRef = 3.35
sigma = 1.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

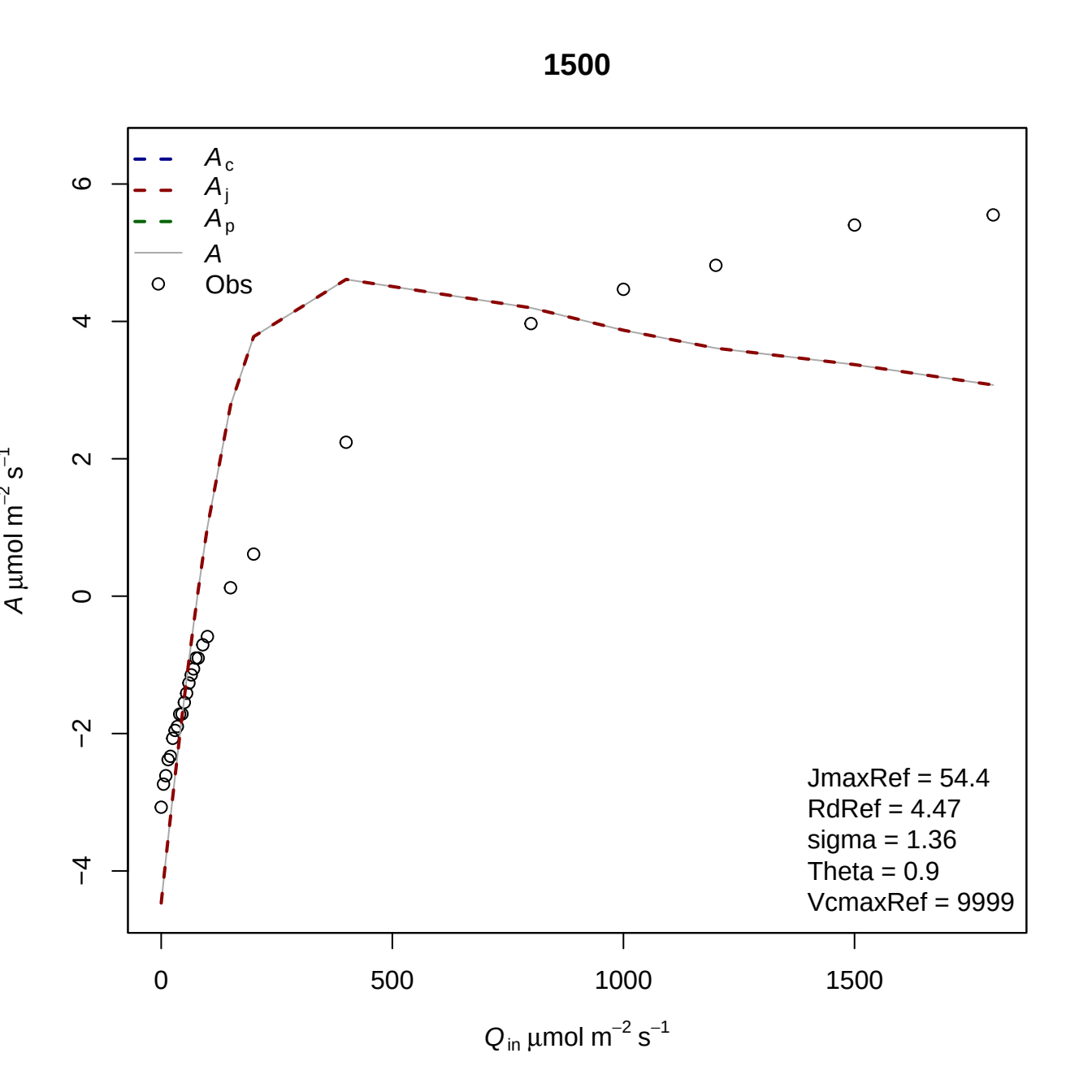

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 51.43
RdRef = 3.57
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

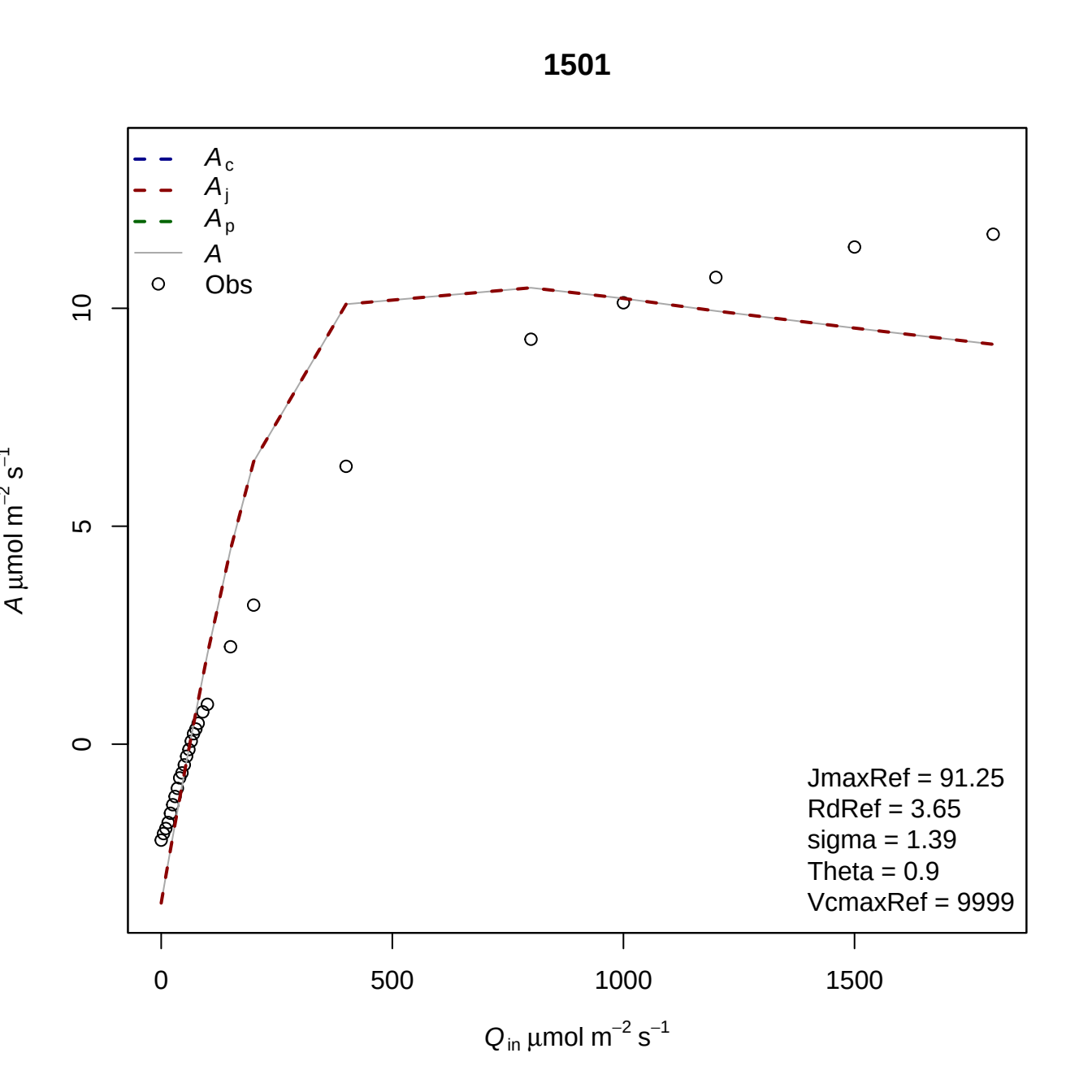

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 101.14
RdRef = 4.13
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

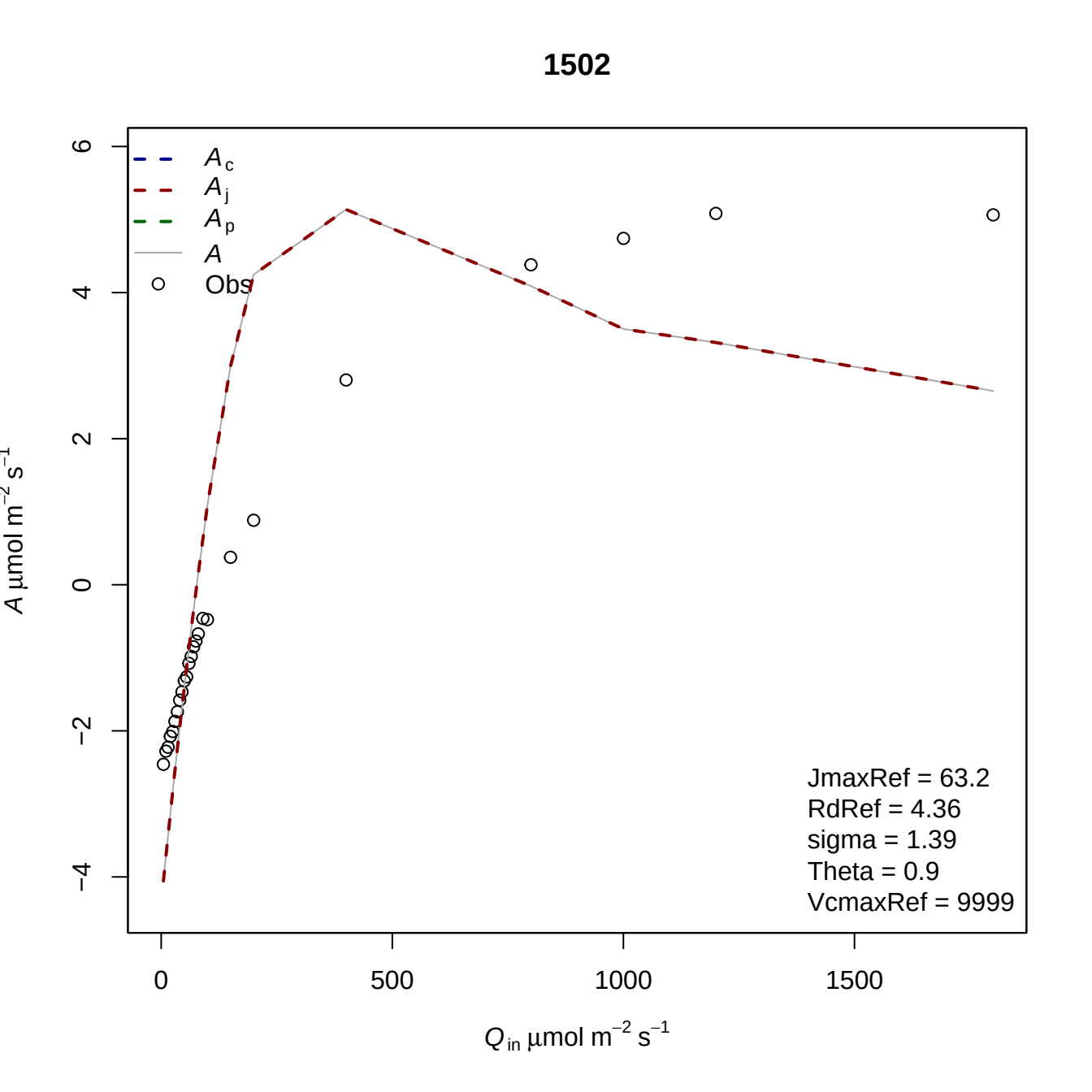

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 87.28
RdRef = 4.43
sigma = 1.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

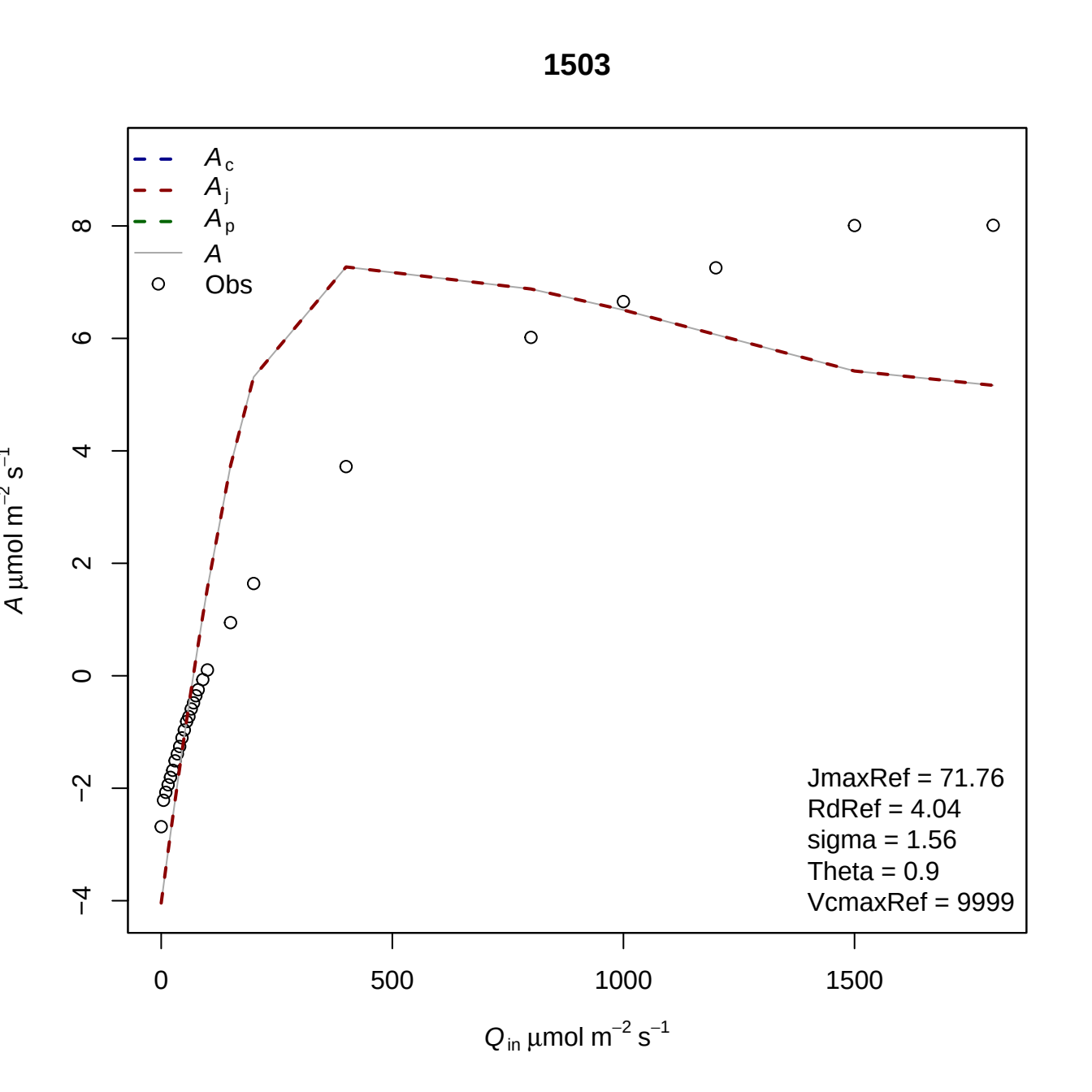

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 48.99
RdRef = 4.12
sigma = 1.27
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

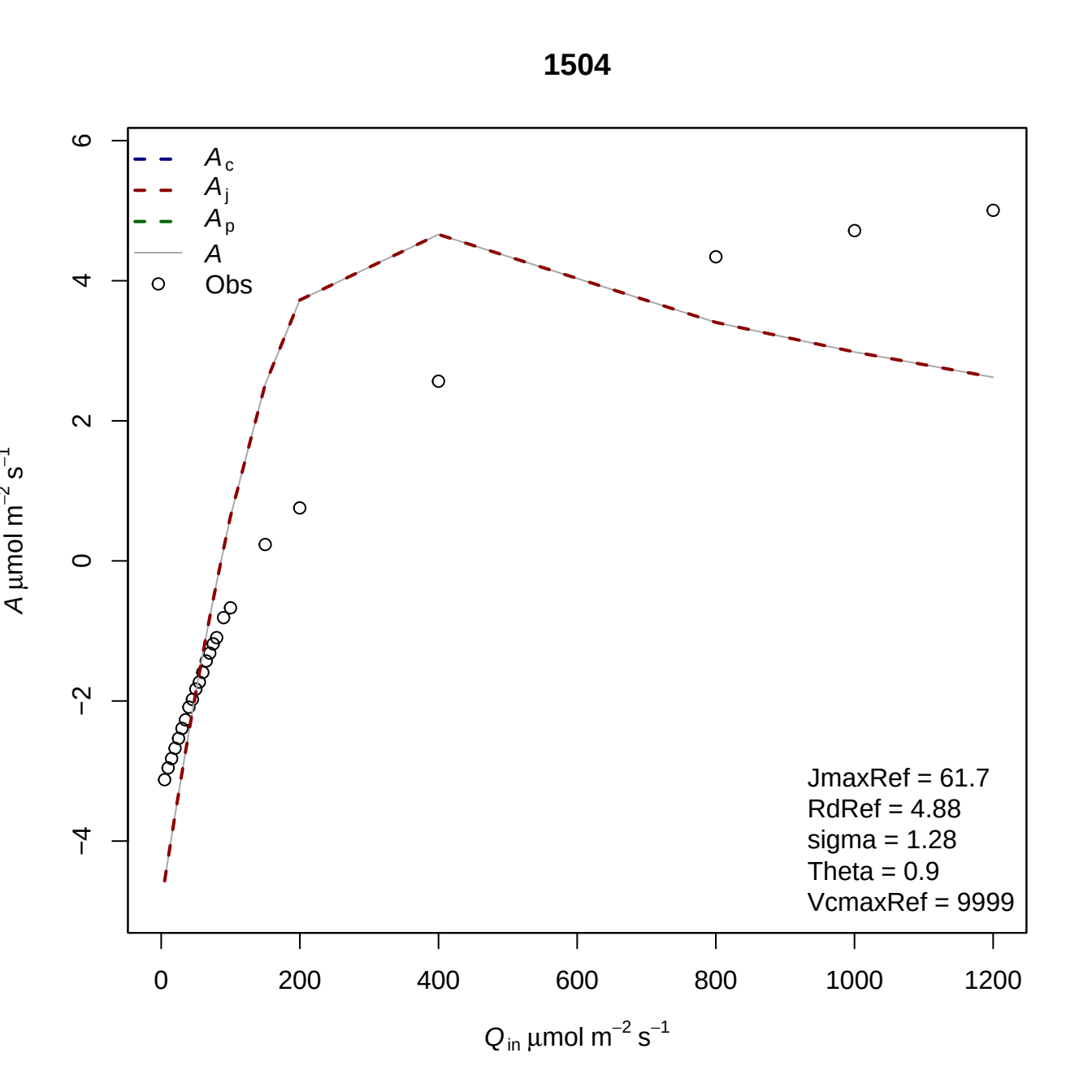

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 48.09
RdRef = 4.36
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

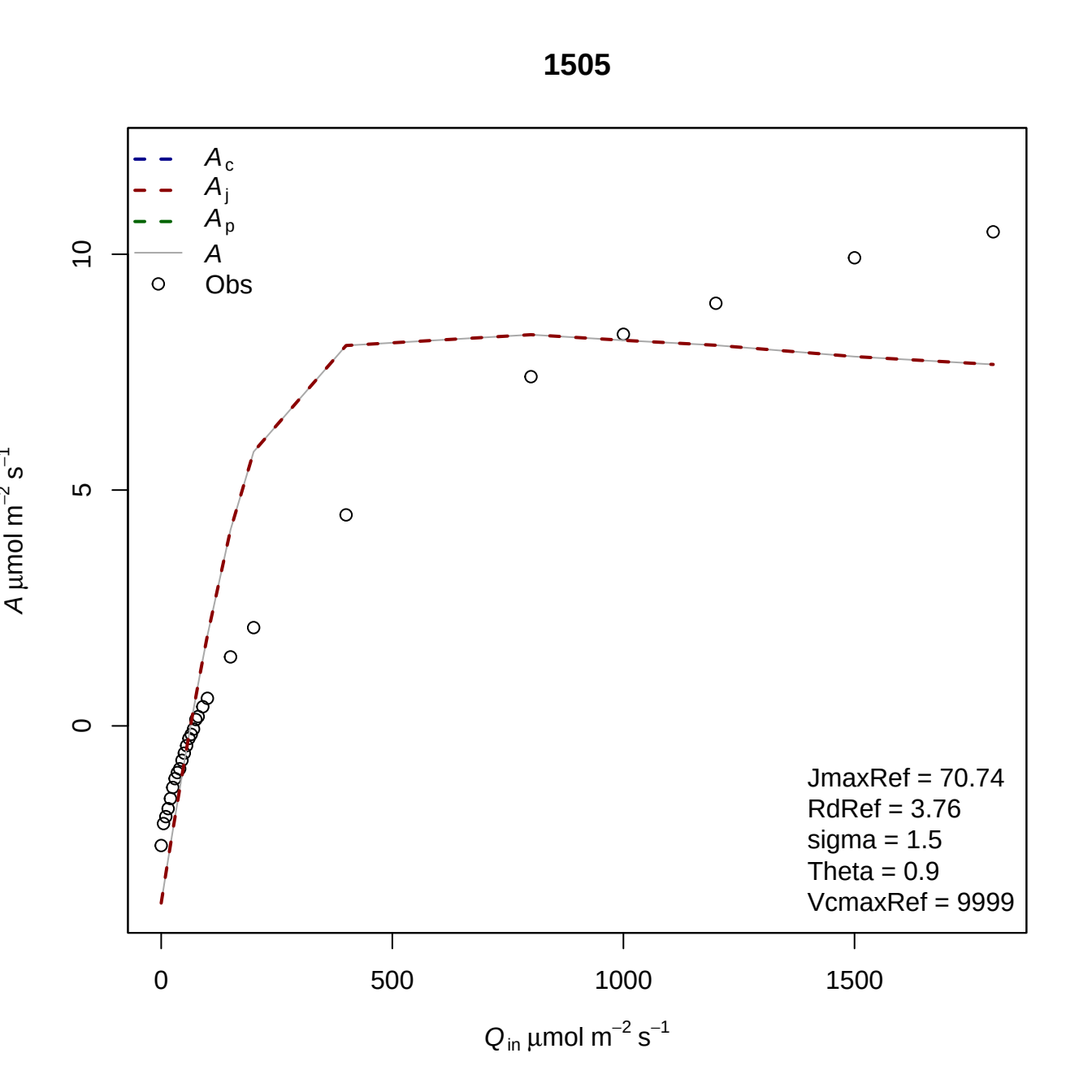

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 54.4
RdRef = 4.47
sigma = 1.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

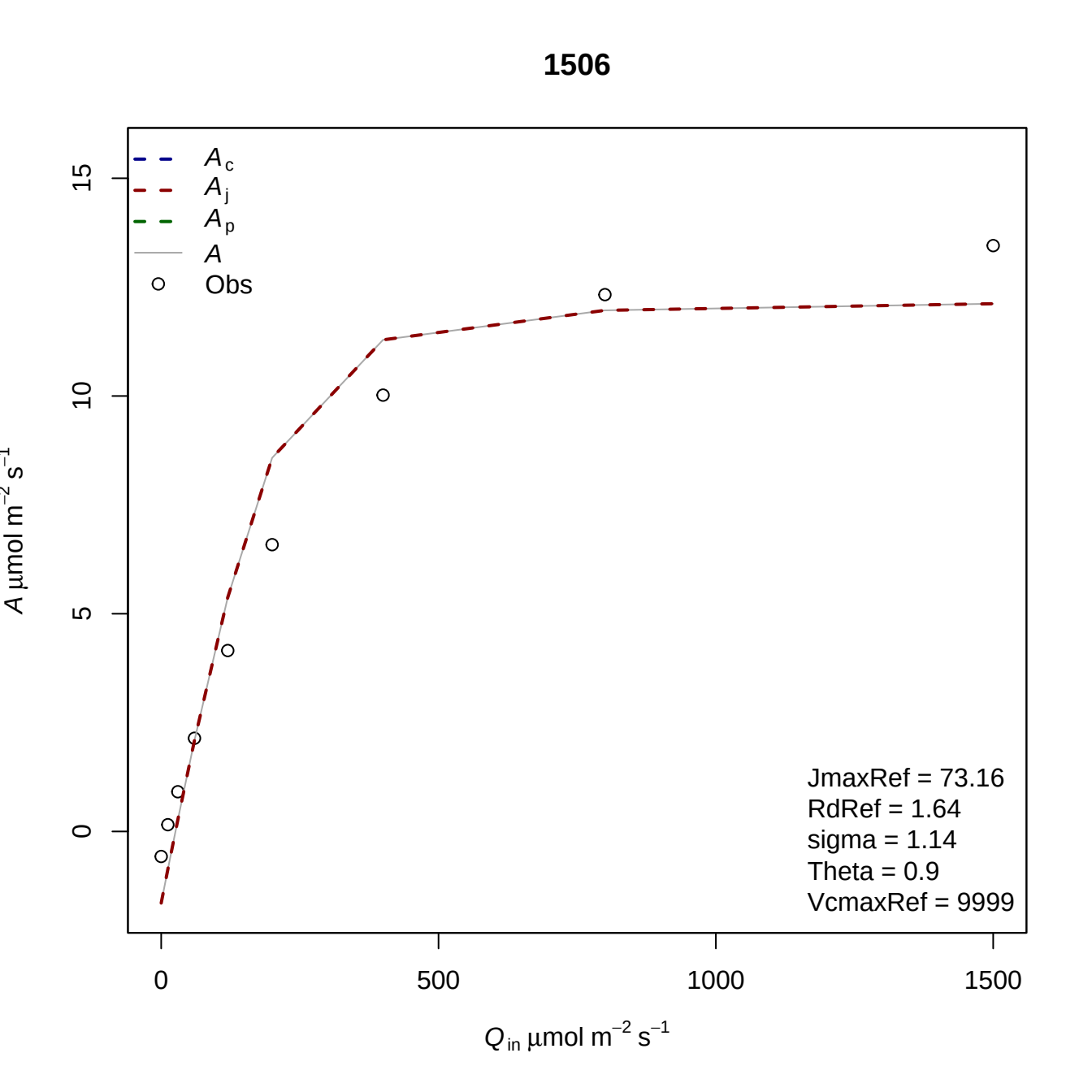

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 91.25
RdRef = 3.65
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

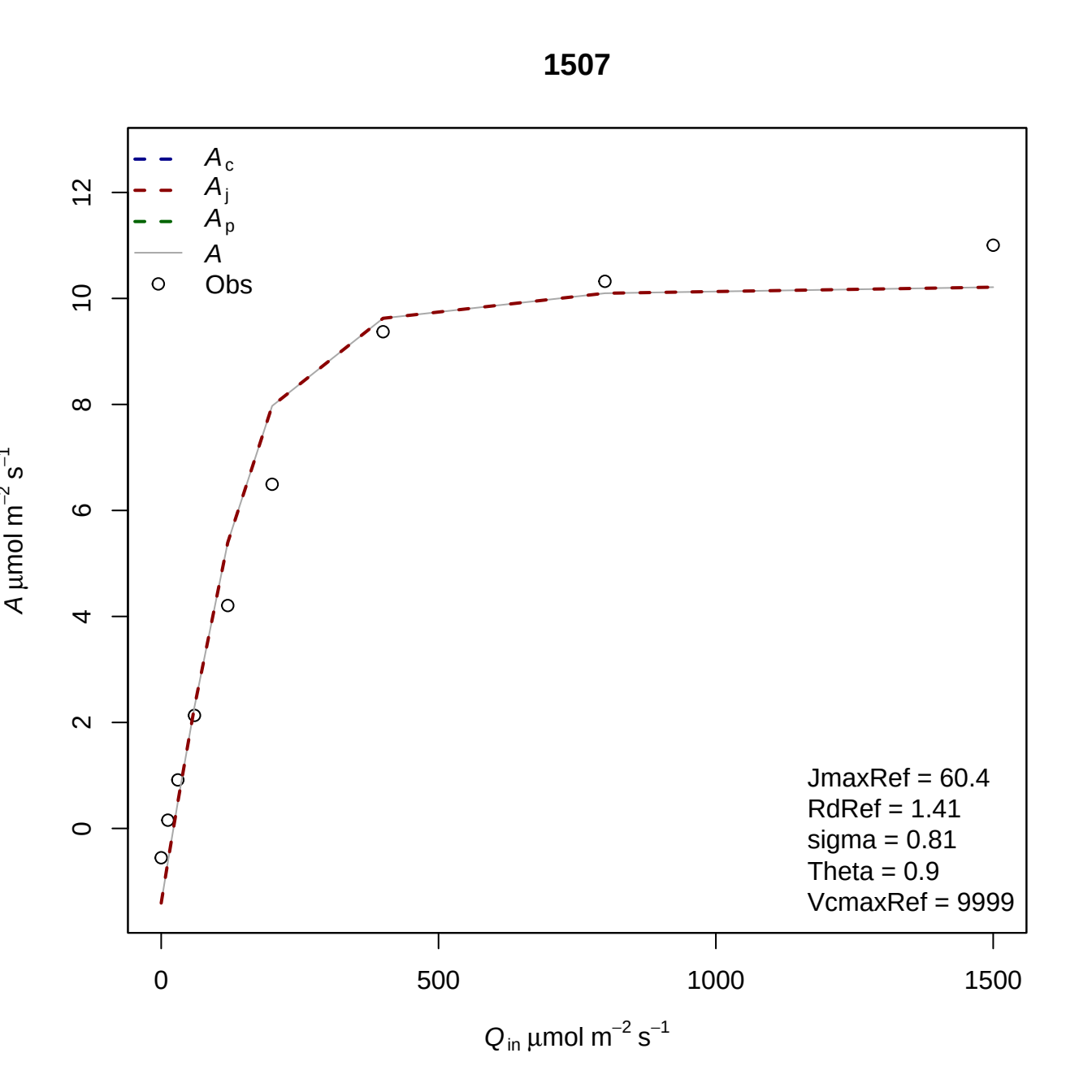

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 63.2
RdRef = 4.36
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

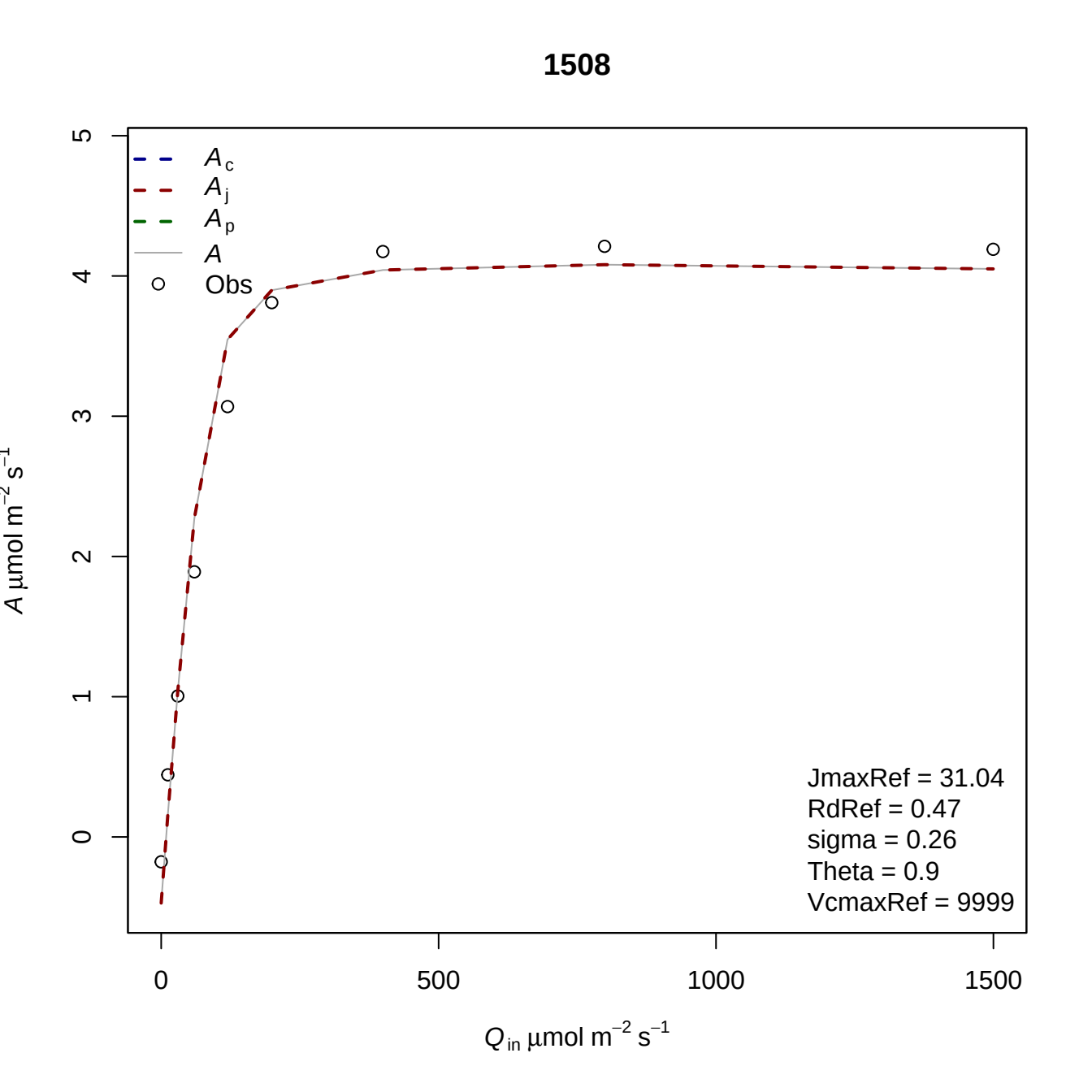

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 71.76
RdRef = 4.04
sigma = 1.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

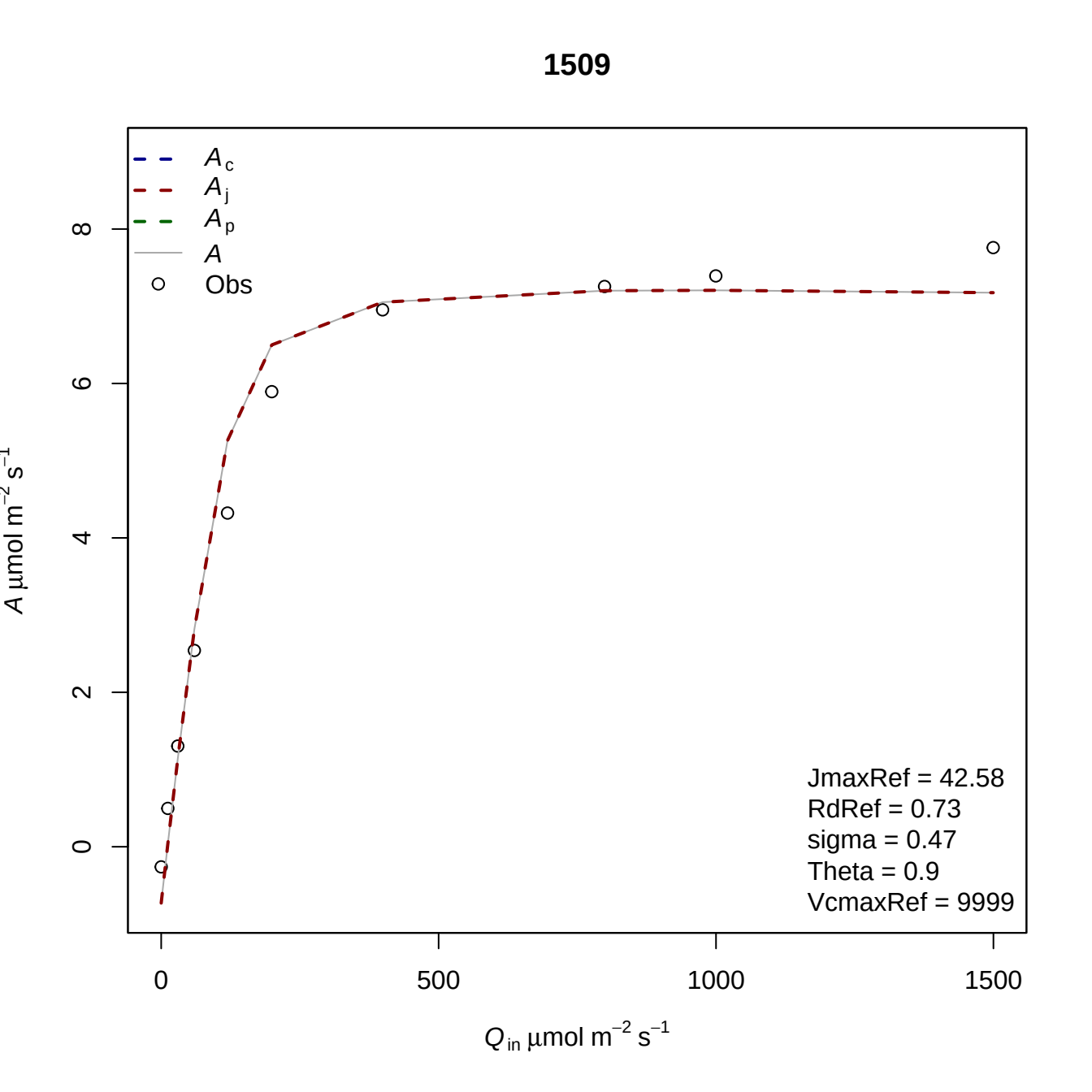

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 61.7
RdRef = 4.88
sigma = 1.28
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

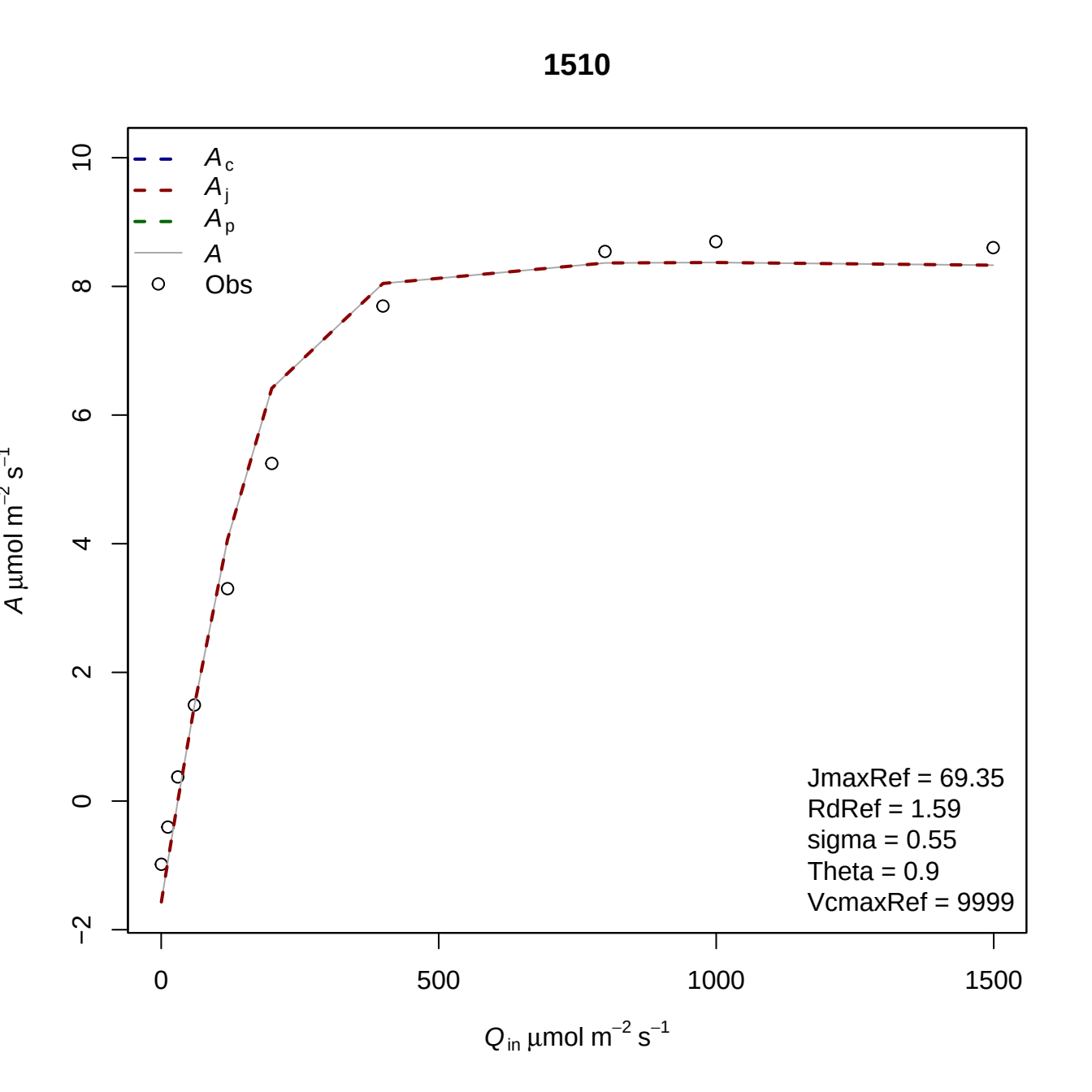

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 70.74
RdRef = 3.76
sigma = 1.5
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

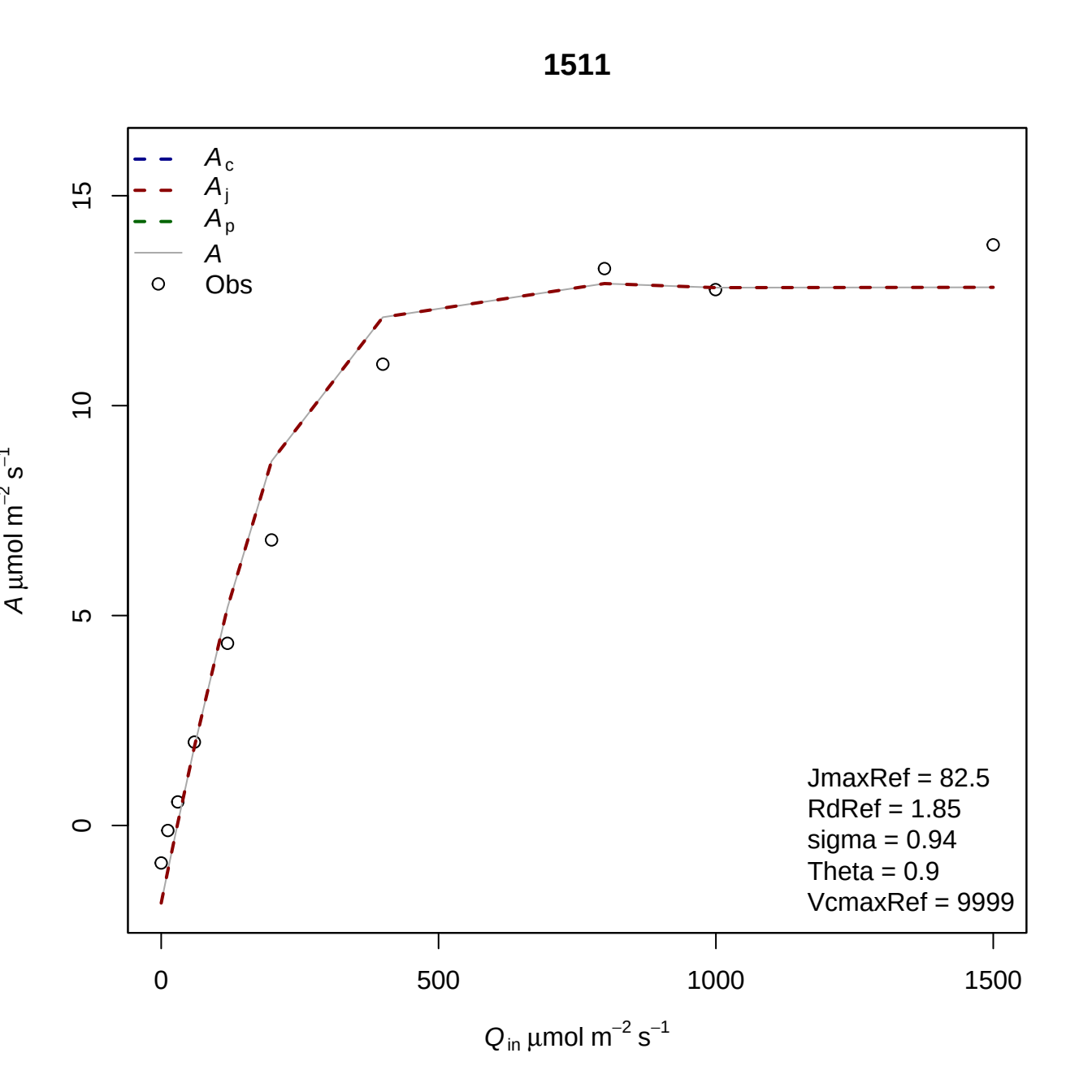

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 73.16
RdRef = 1.64
sigma = 1.14
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

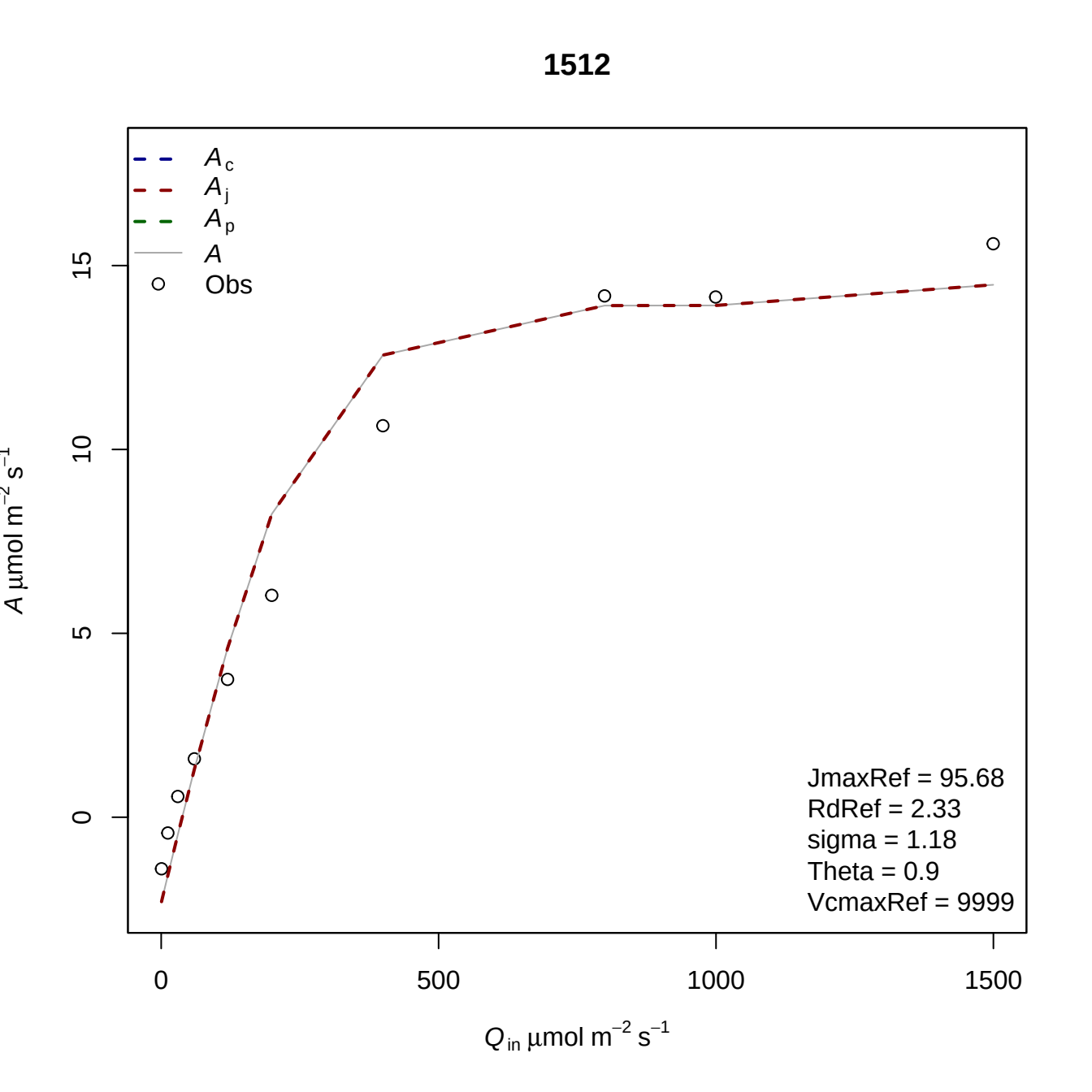

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 60.4
RdRef = 1.41
sigma = 0.81
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

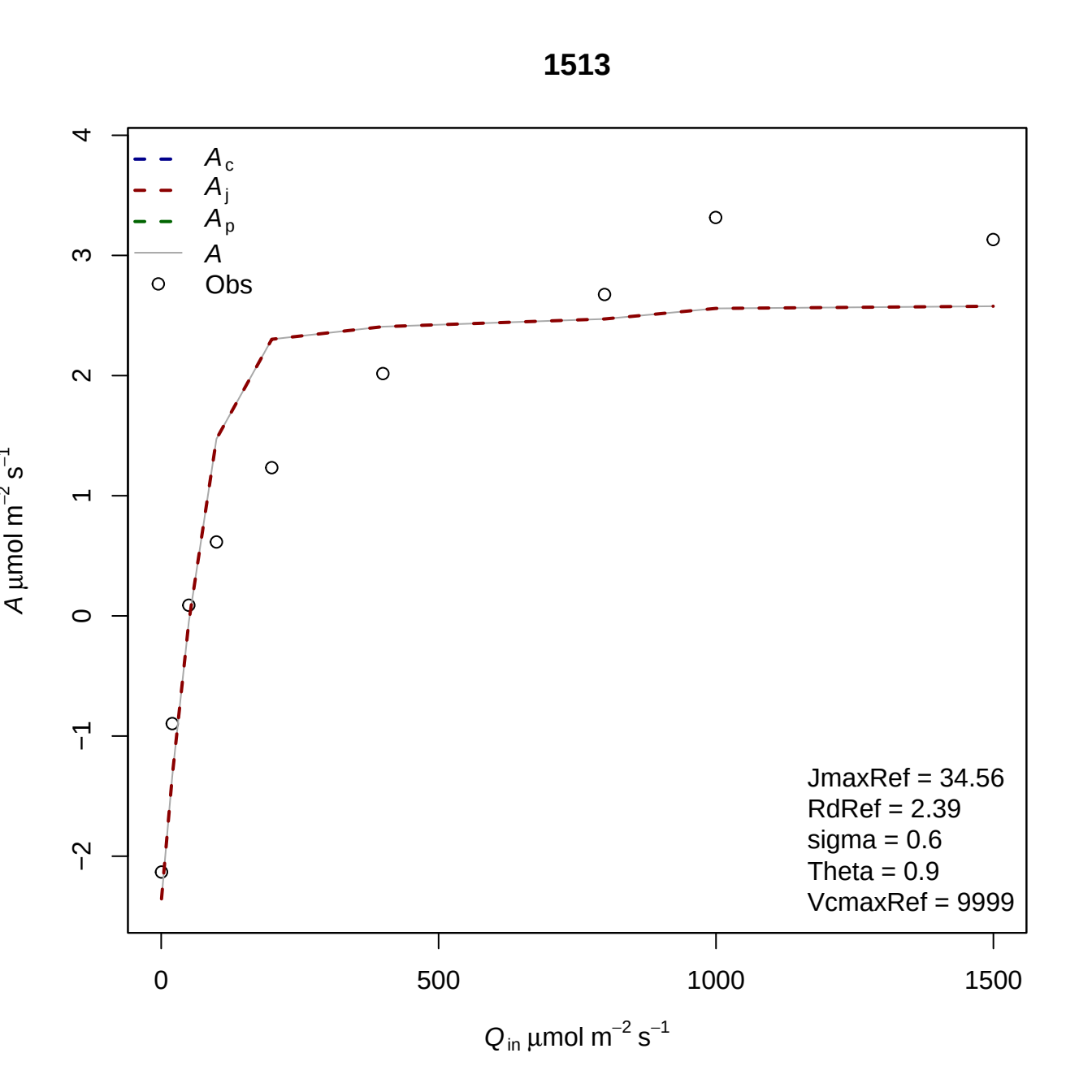

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 31.04
RdRef = 0.47
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

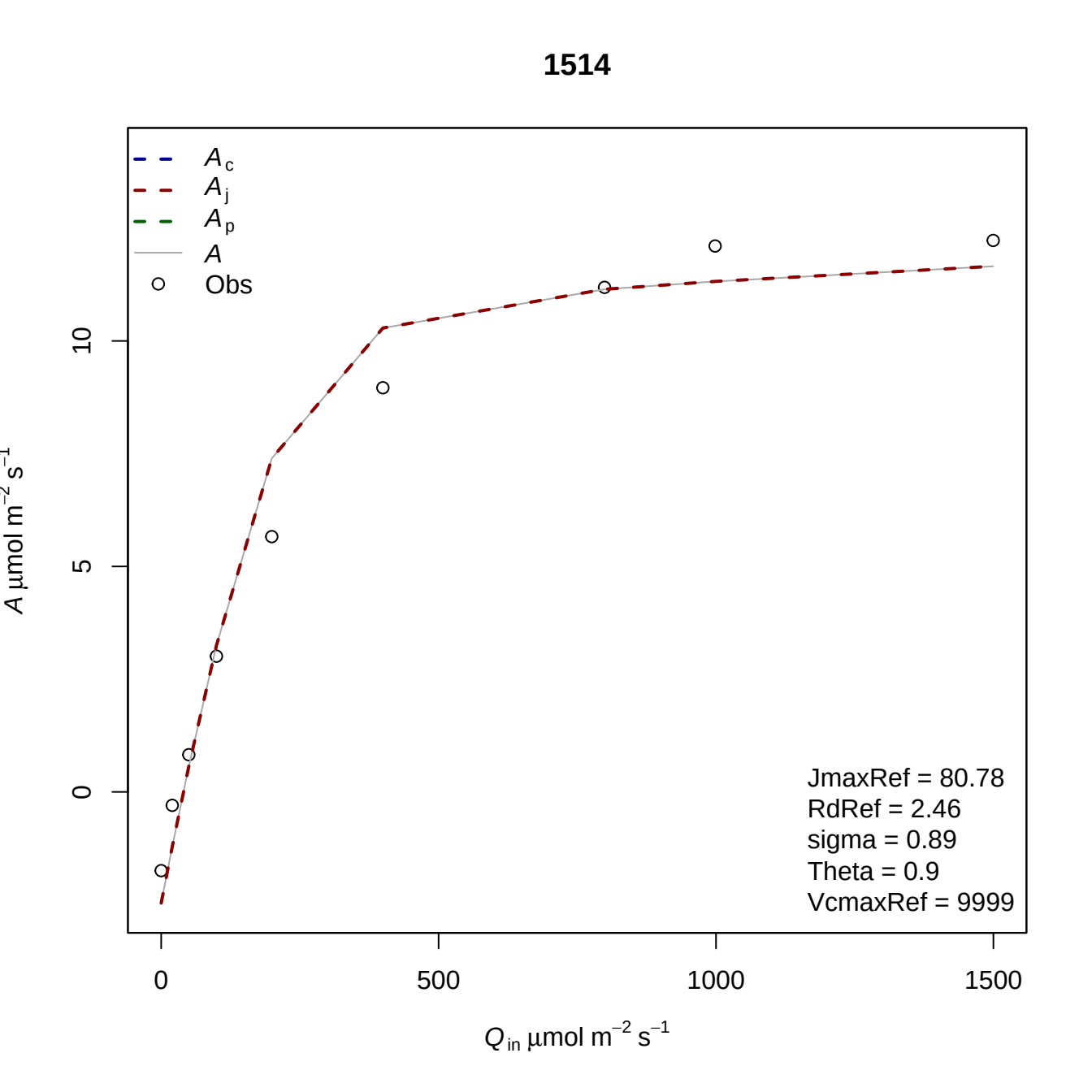

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 42.58
RdRef = 0.73
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

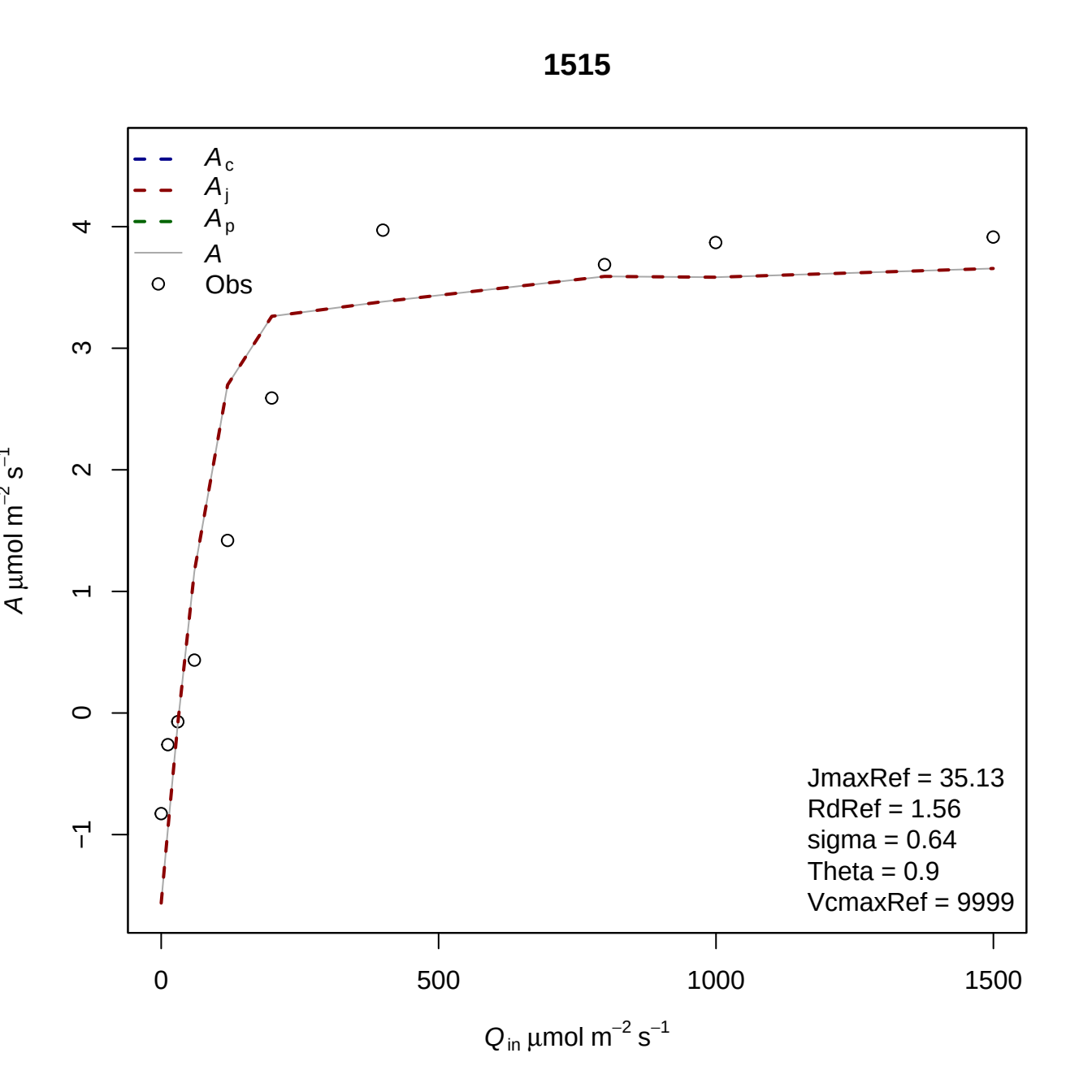

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 69.35
RdRef = 1.59
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

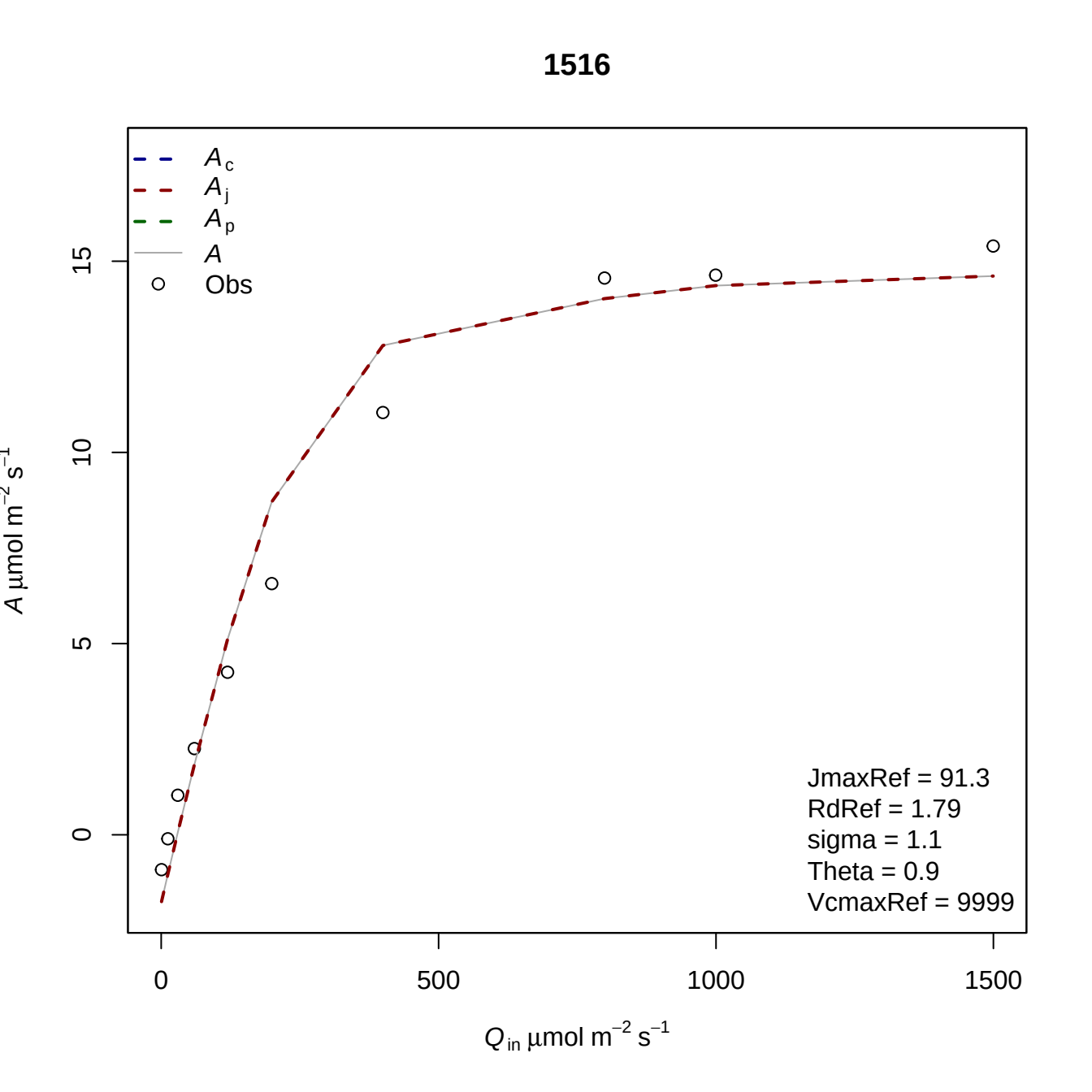

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 82.5
RdRef = 1.85
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

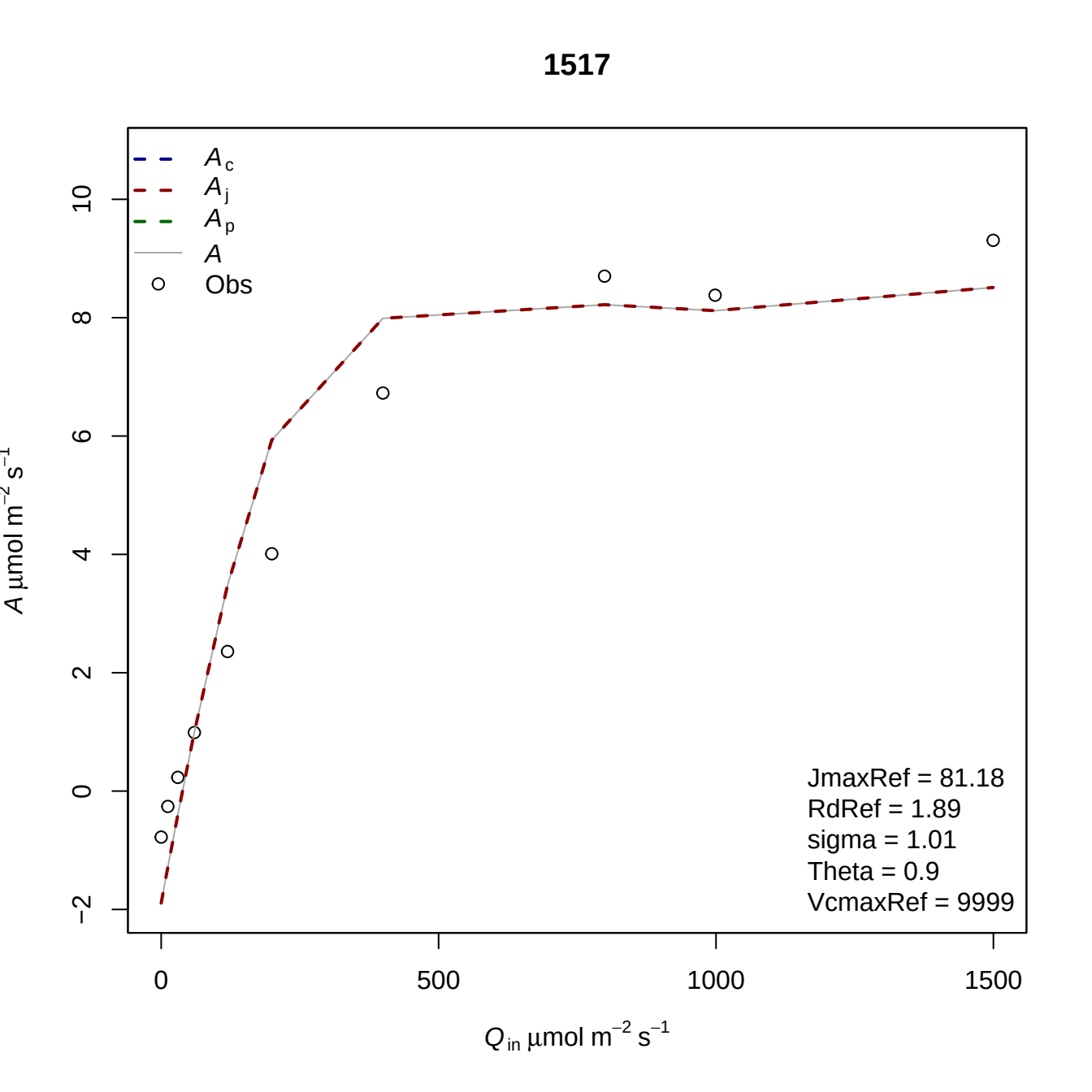

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 95.68
RdRef = 2.33
sigma = 1.18
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

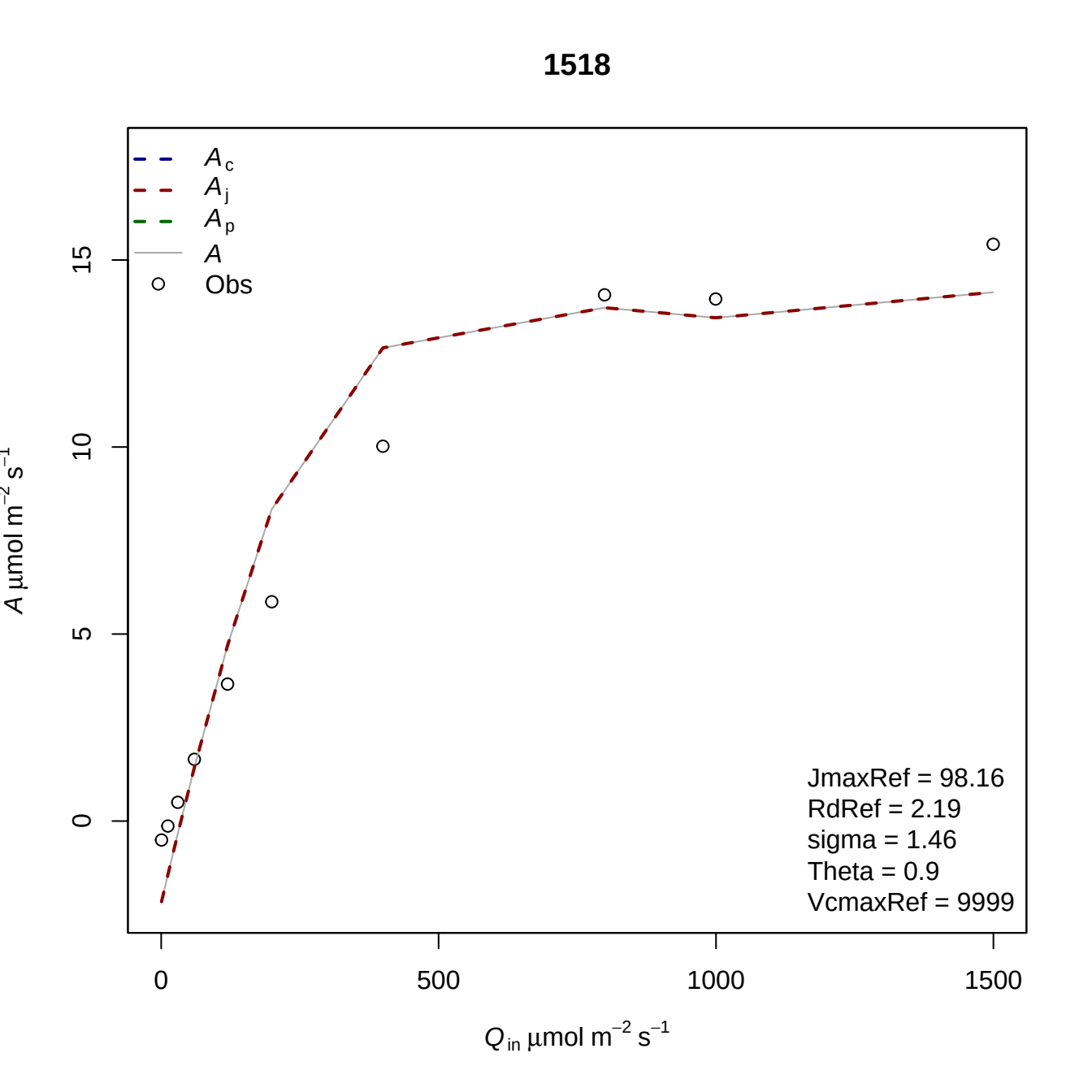

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 34.56
RdRef = 2.39
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

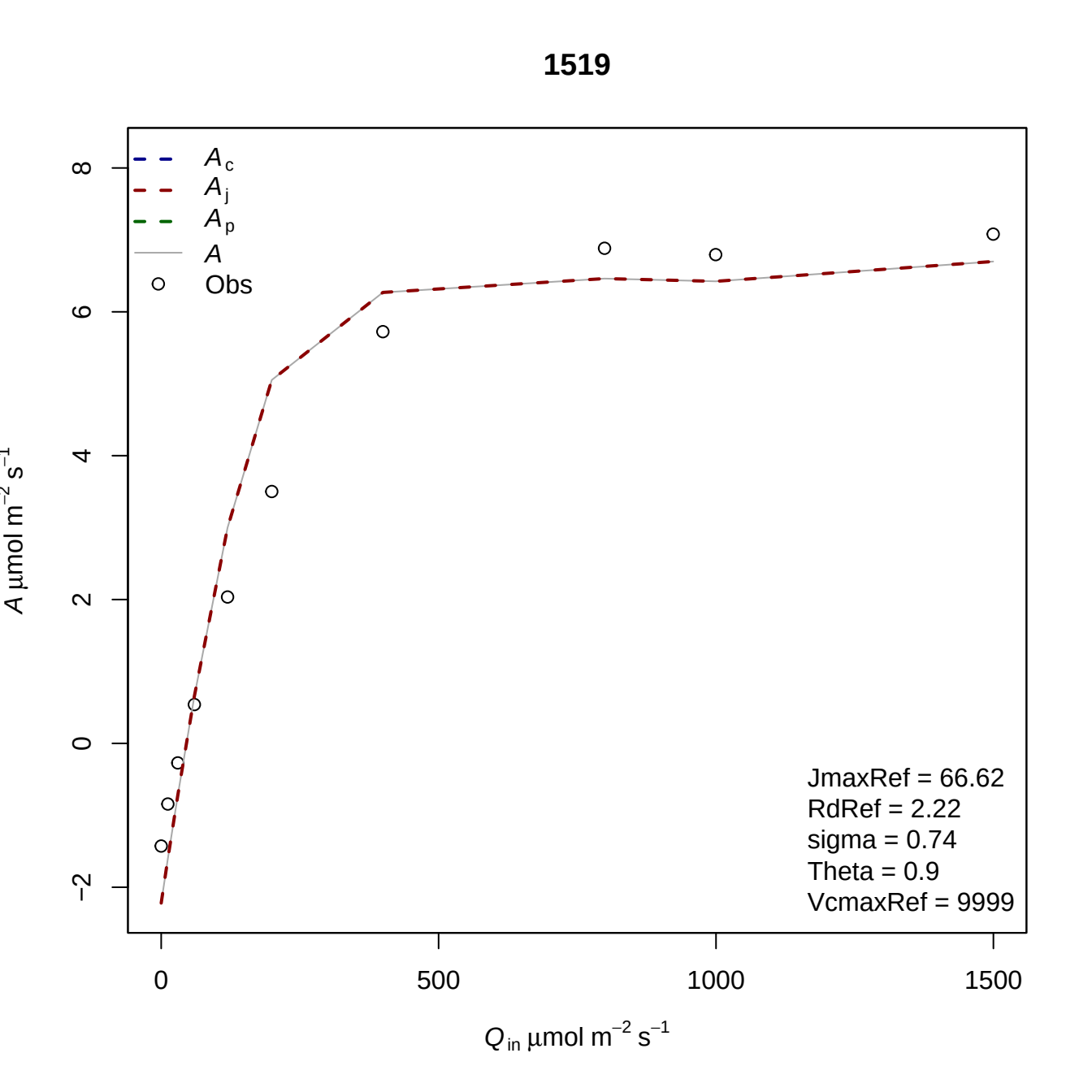

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 80.78
RdRef = 2.46
sigma = 0.89
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

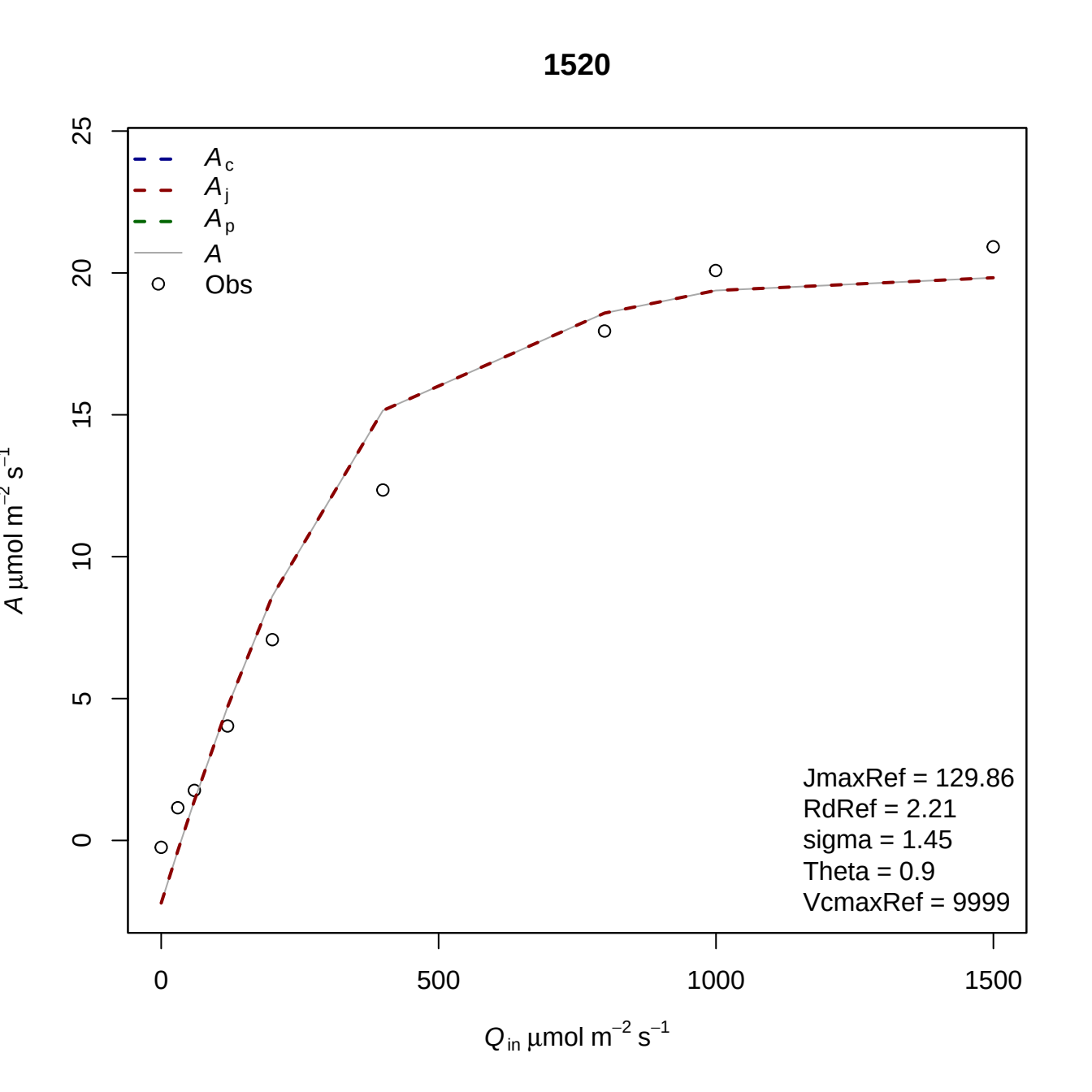

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 35.13
RdRef = 1.56
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

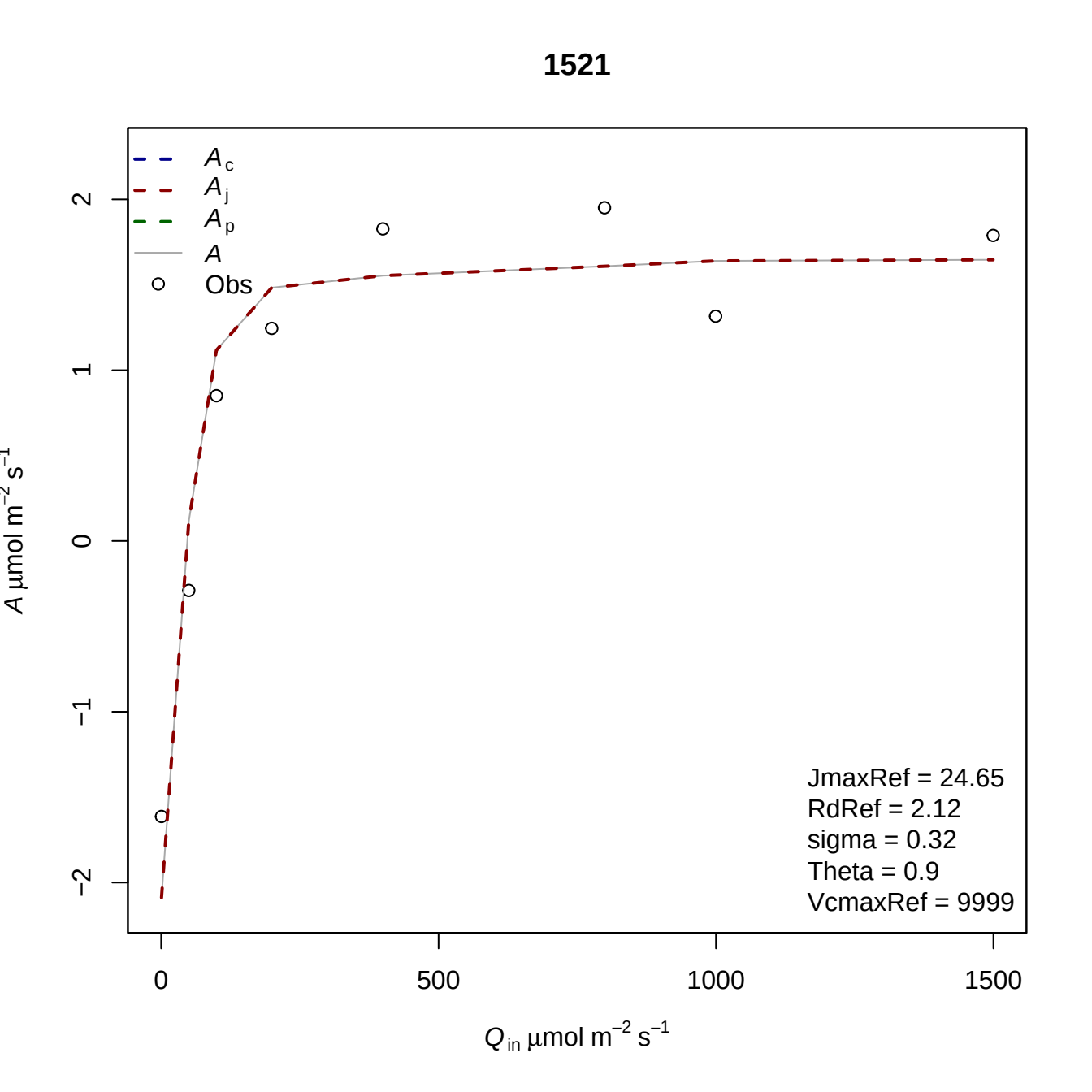

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 91.3
RdRef = 1.79
sigma = 1.1
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

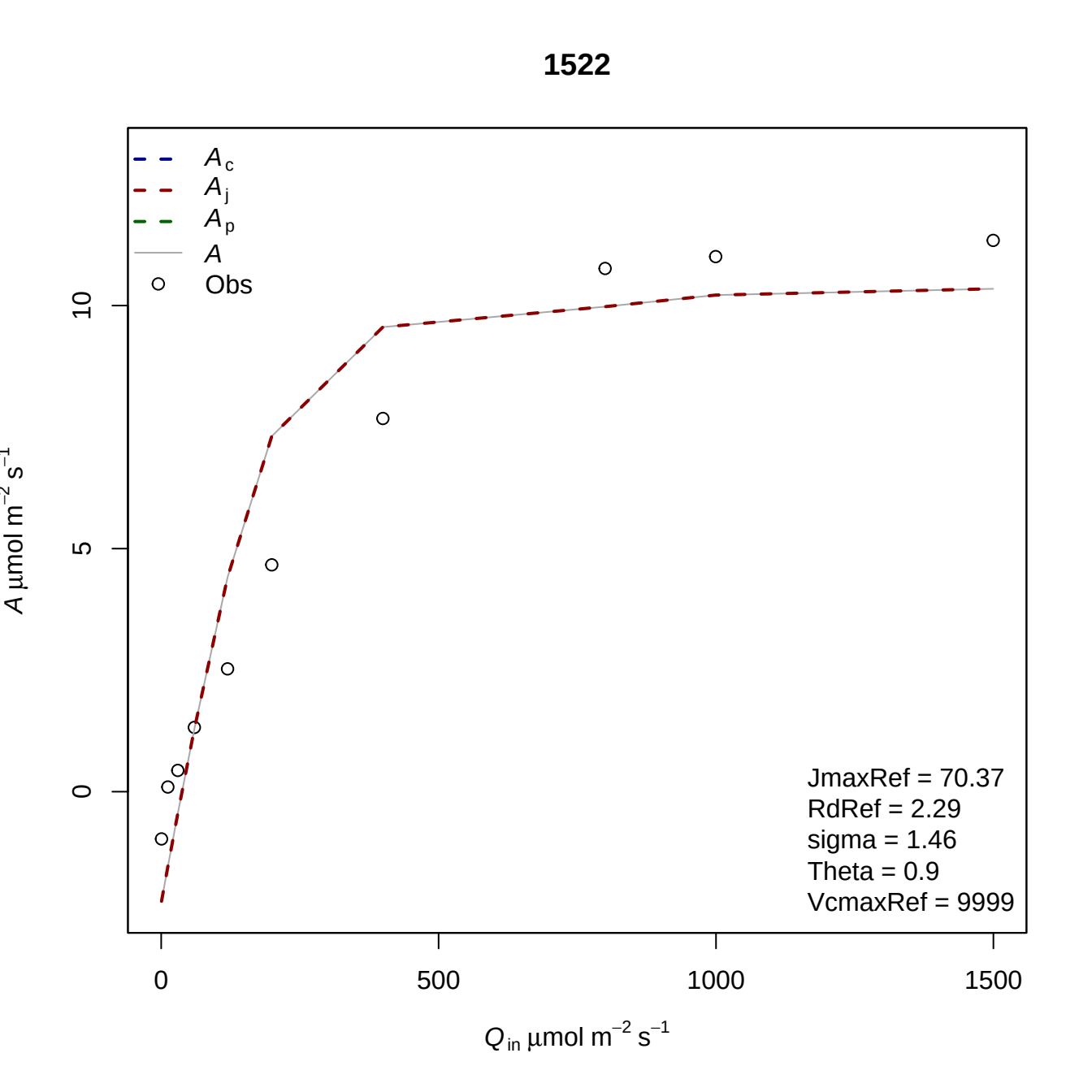

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 81.18
RdRef = 1.89
sigma = 1.01
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

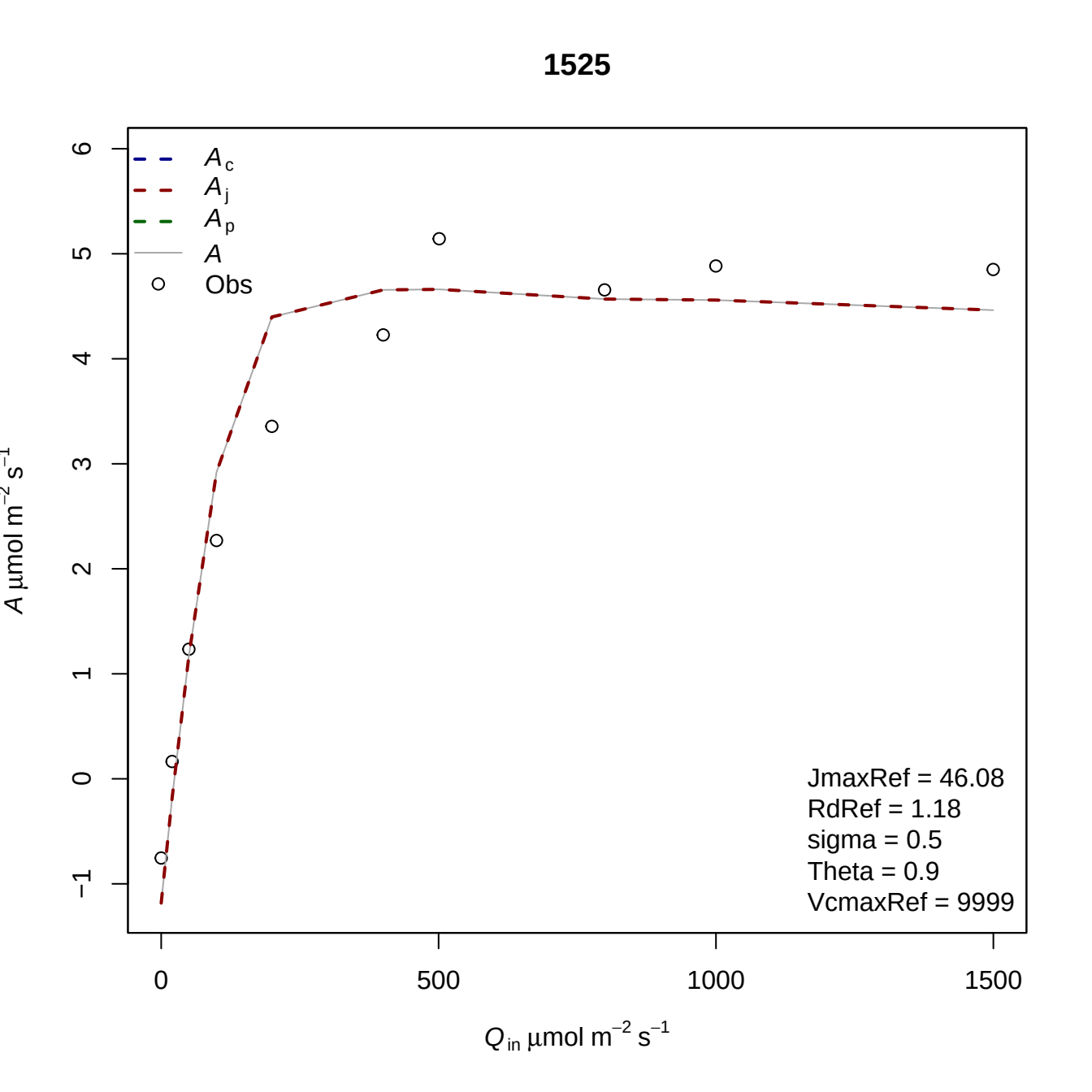

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 98.16
RdRef = 2.19
sigma = 1.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

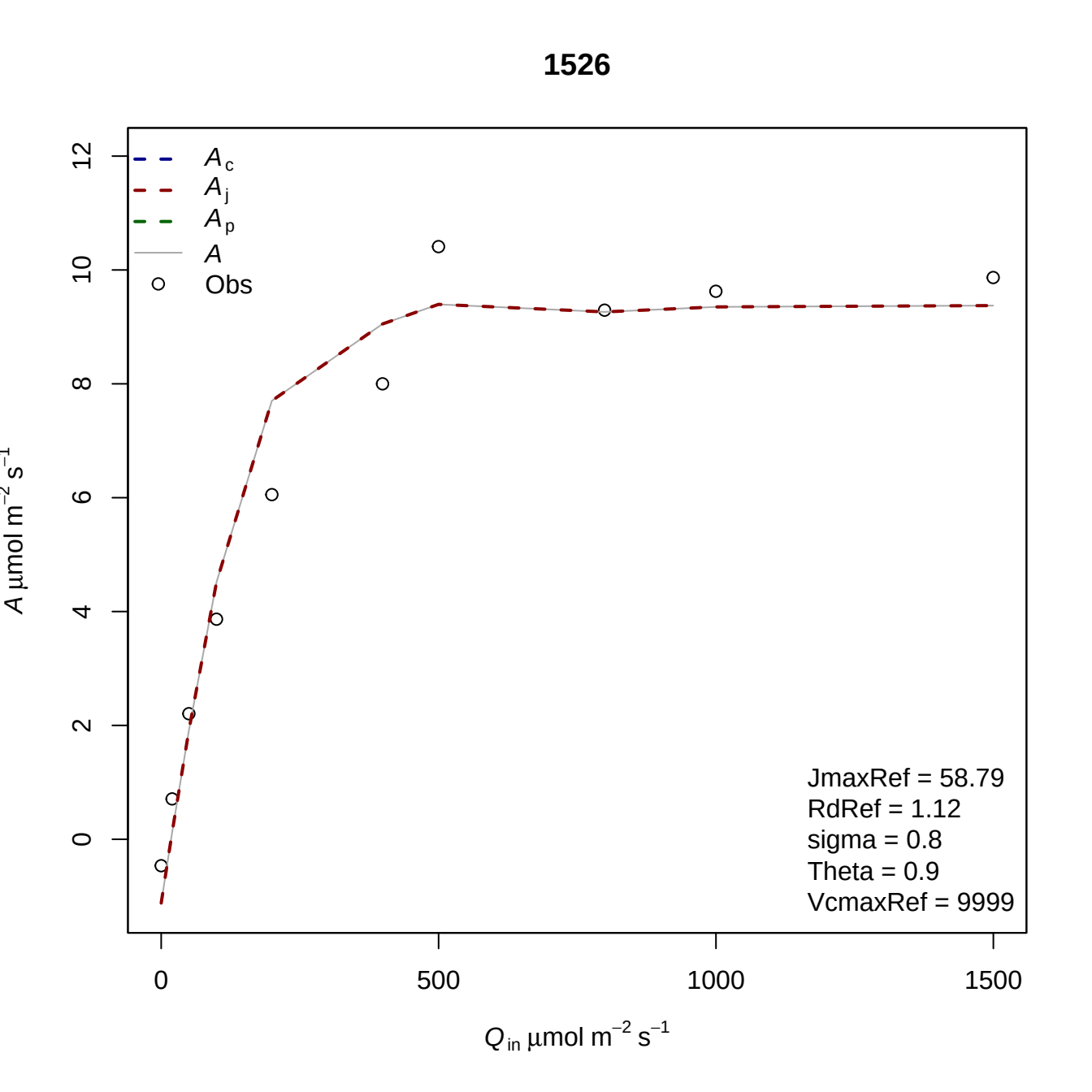

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 66.62
RdRef = 2.22
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

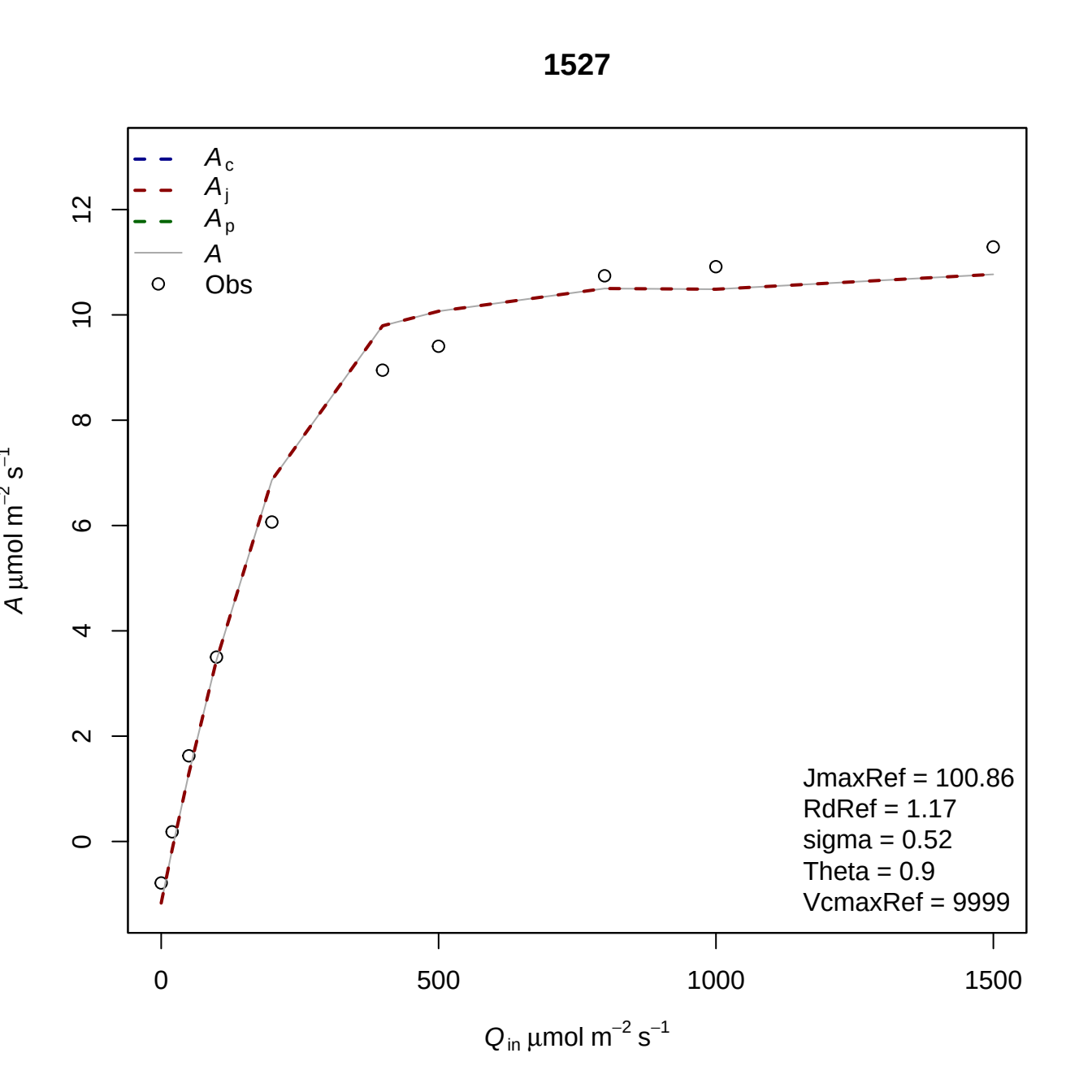

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 129.86
RdRef = 2.21
sigma = 1.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

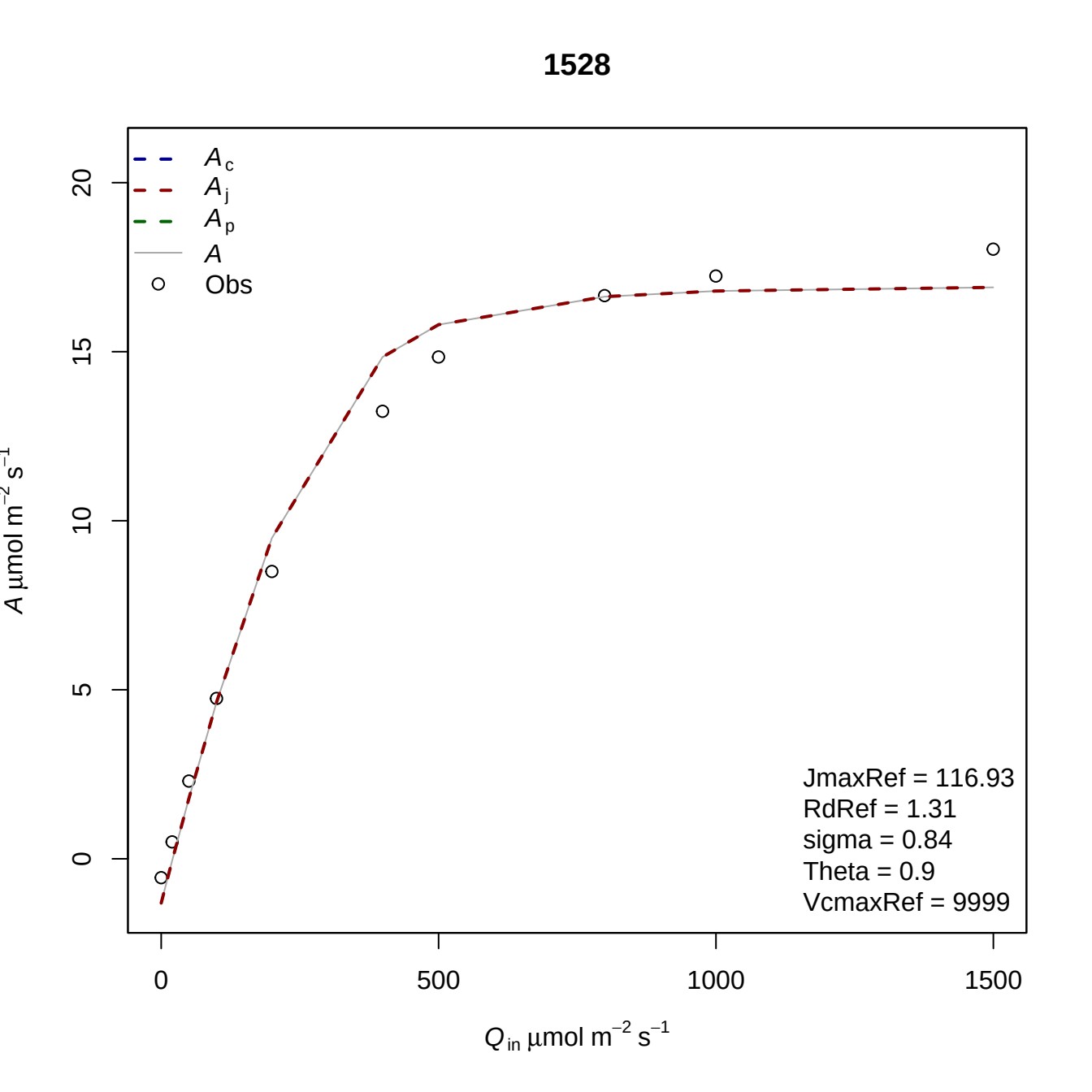

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 24.65
RdRef = 2.12
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

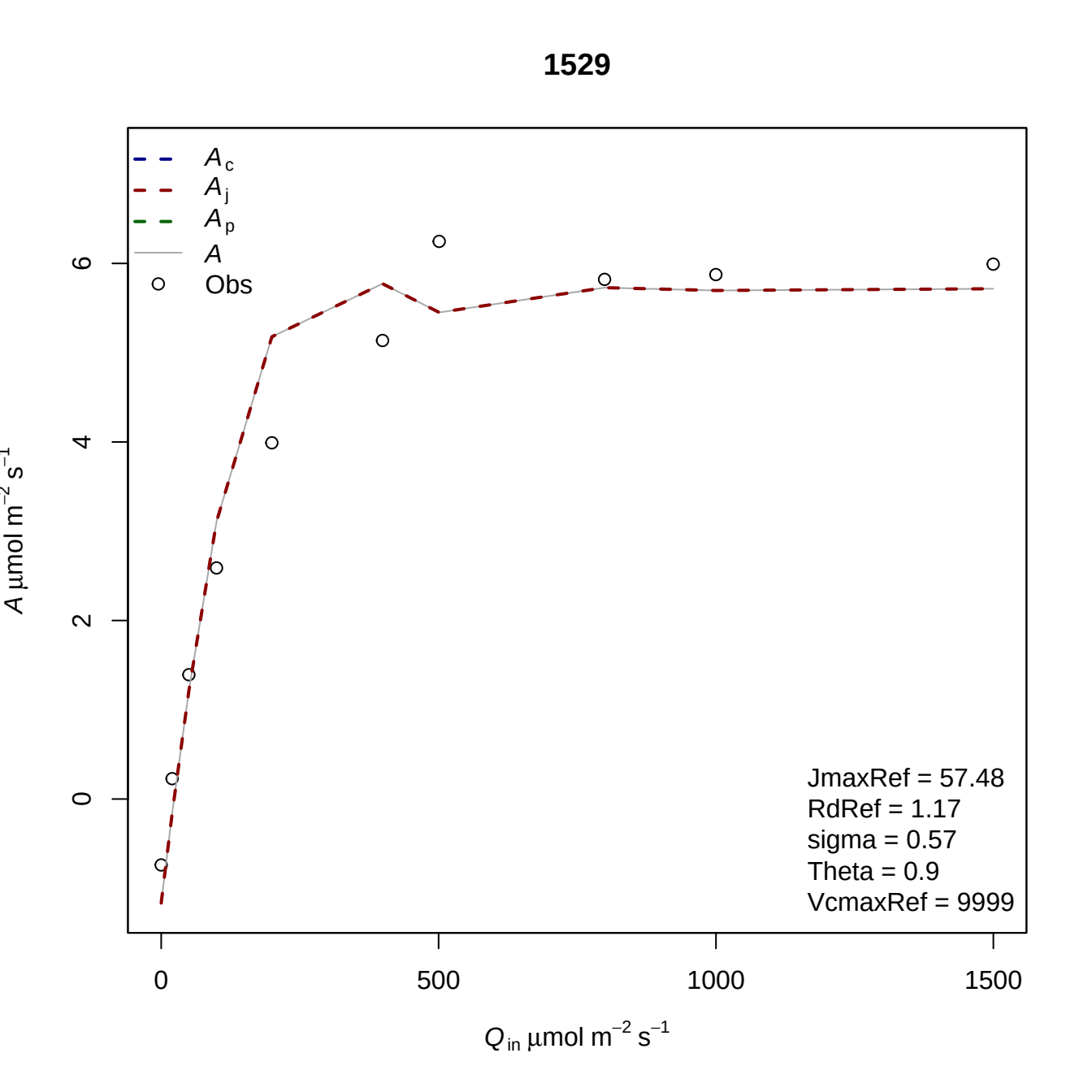

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 70.37
RdRef = 2.29
sigma = 1.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

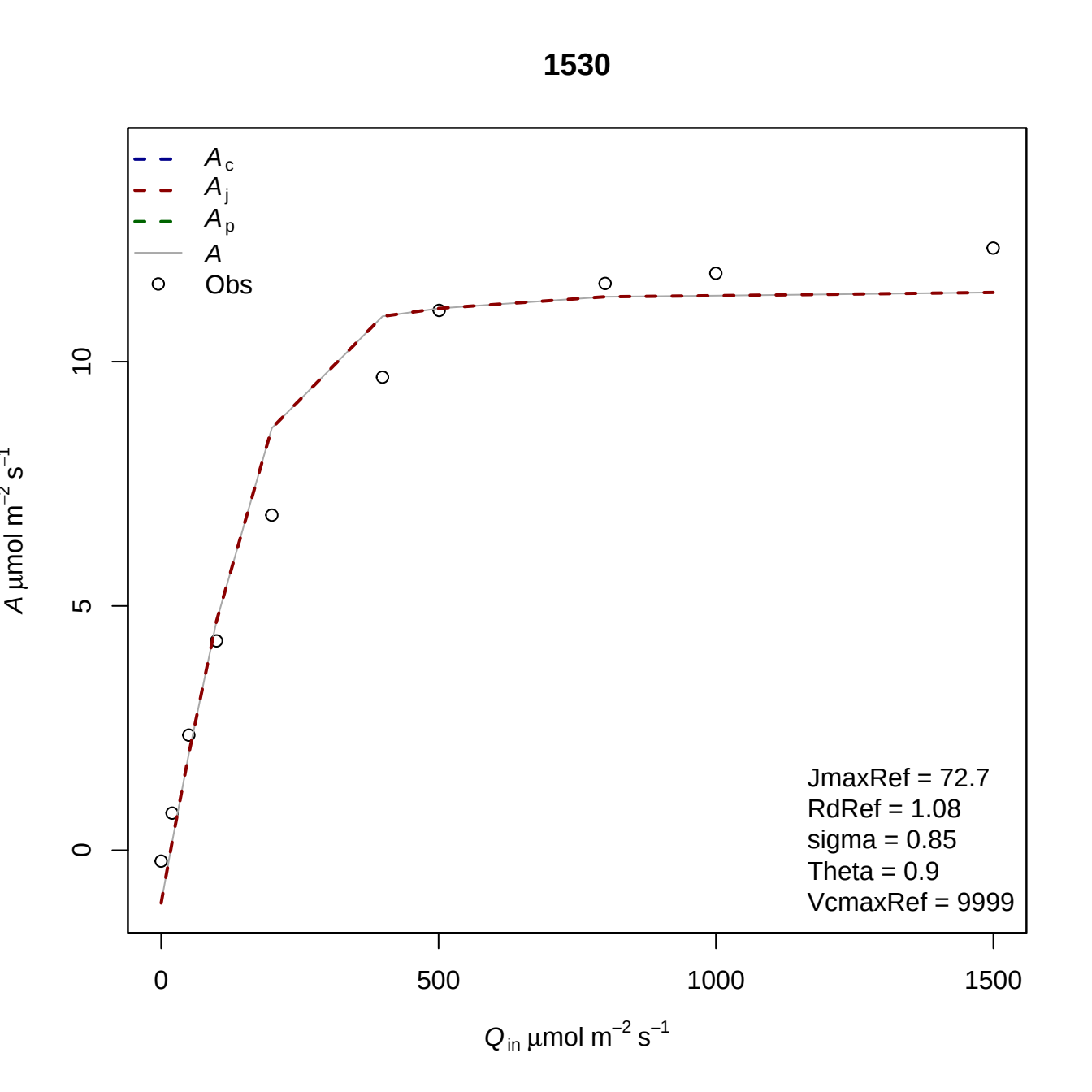

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 46.08
RdRef = 1.18
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

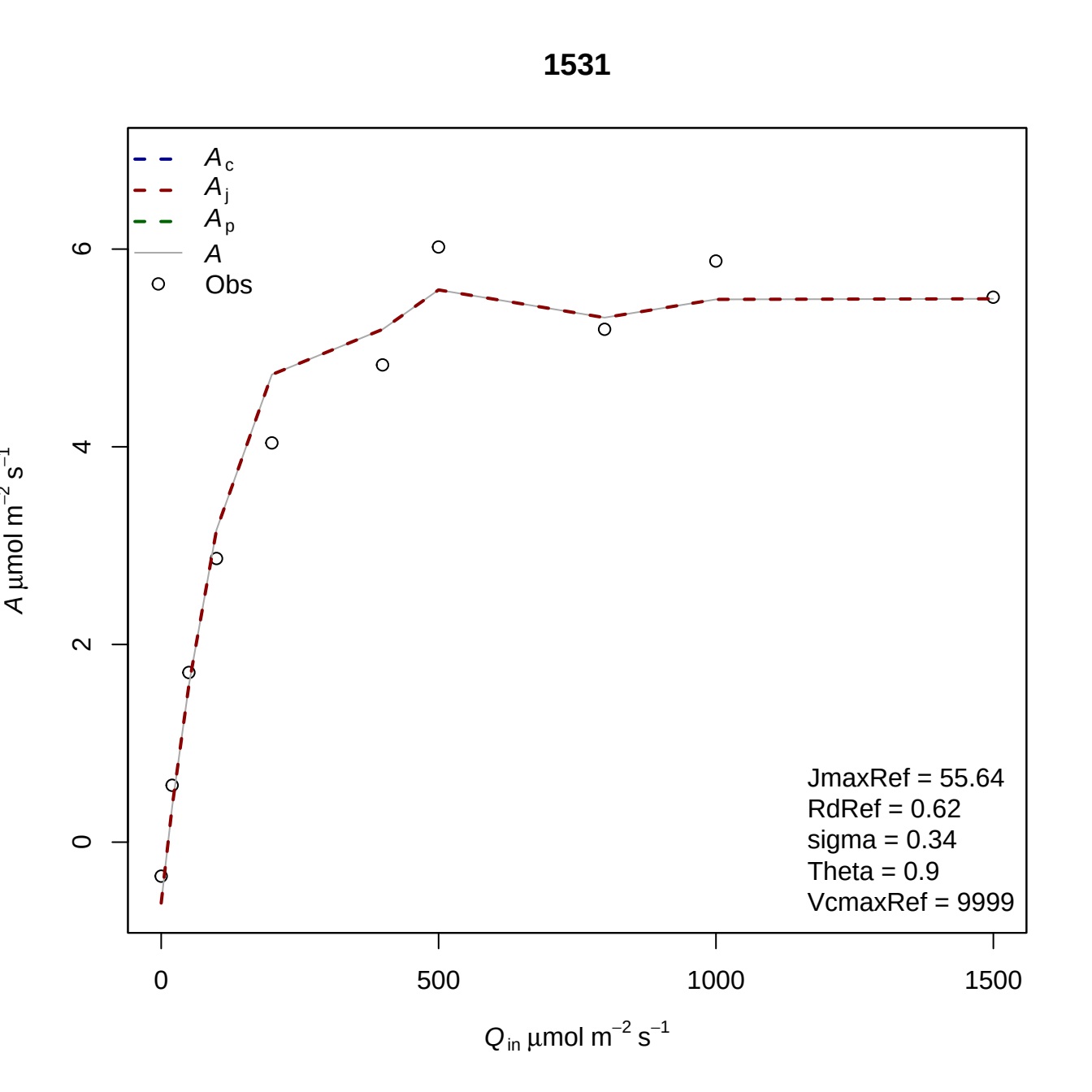

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 58.79
RdRef = 1.12
sigma = 0.8
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

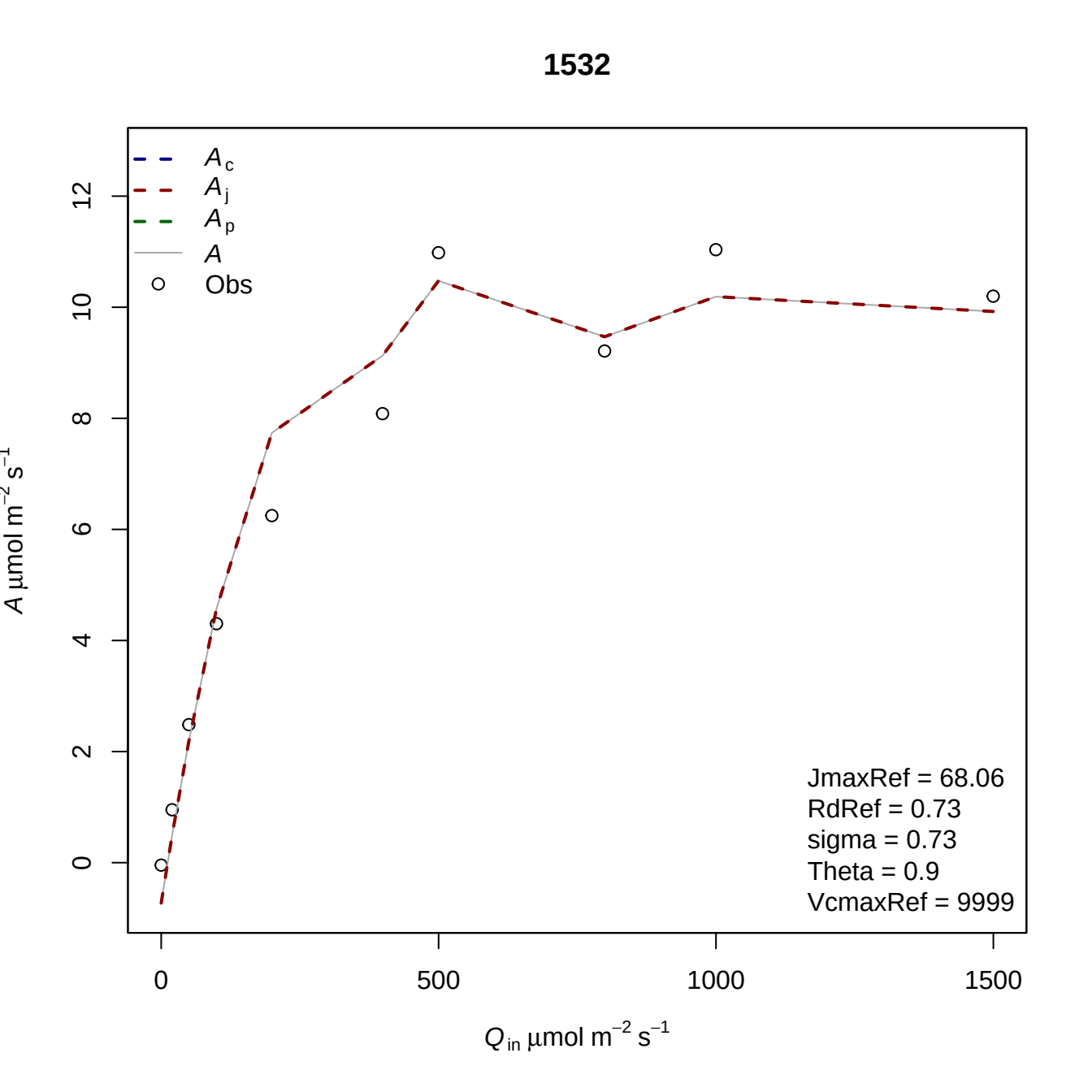

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 100.86
RdRef = 1.17
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

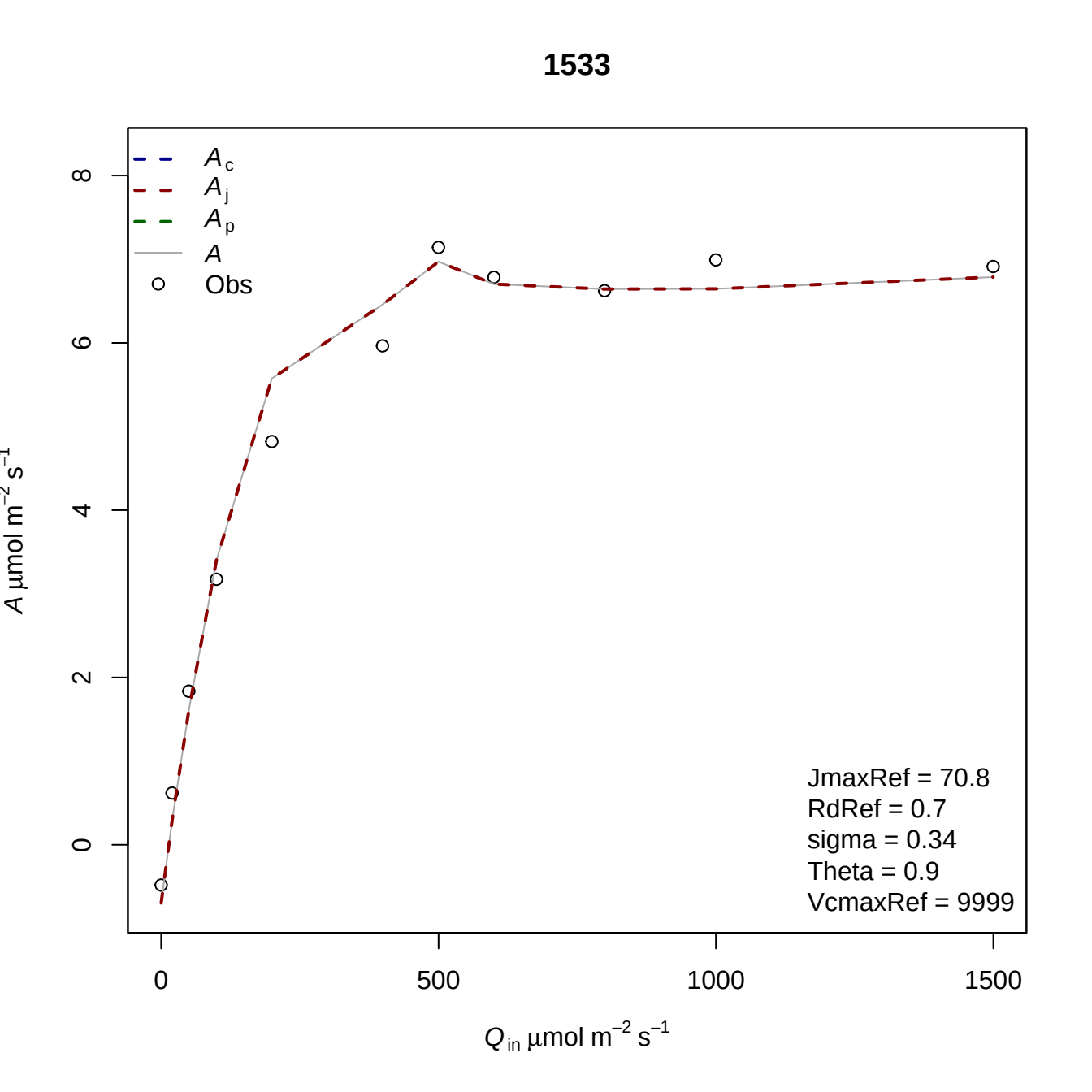

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 116.93
RdRef = 1.31
sigma = 0.84
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

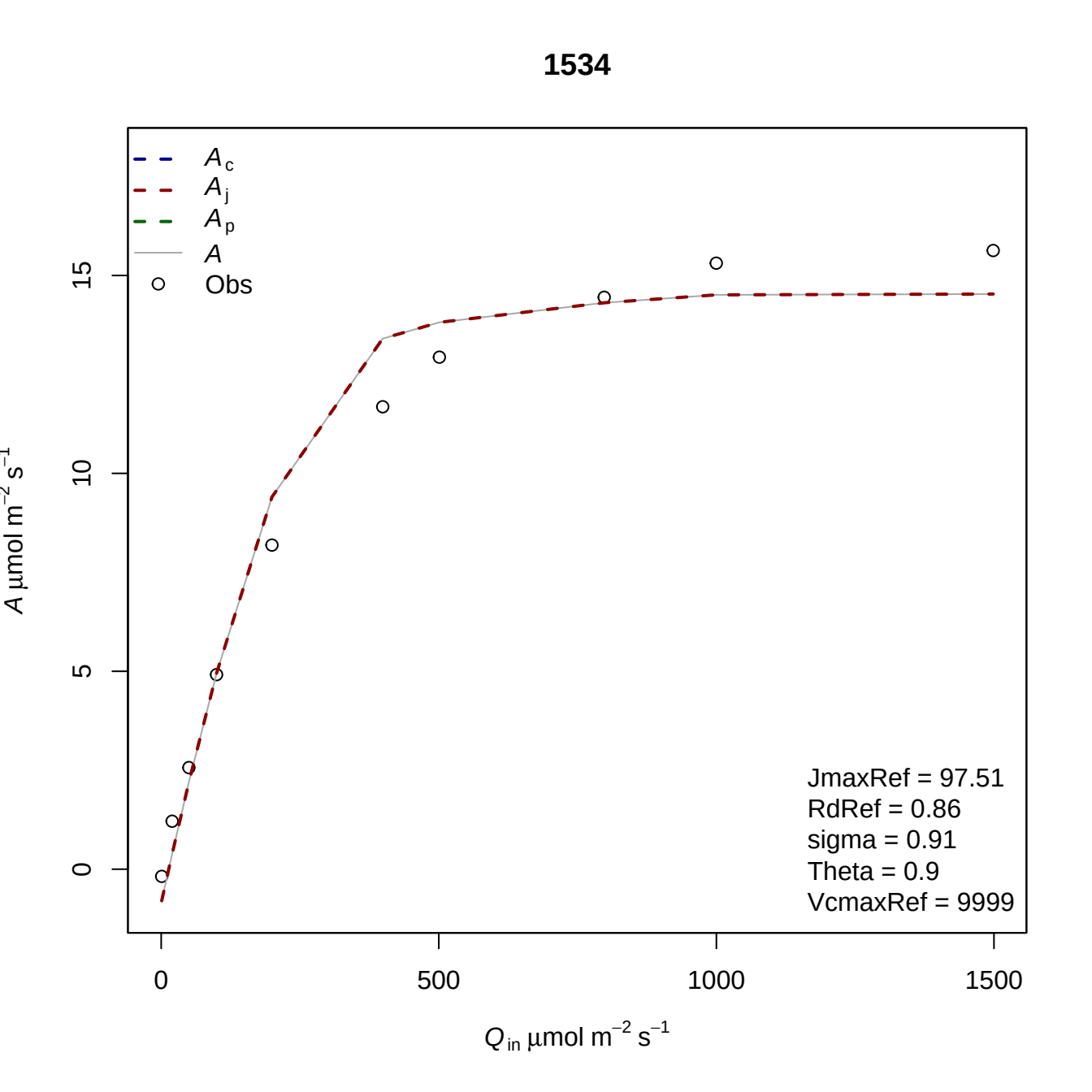

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 57.48
RdRef = 1.17
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

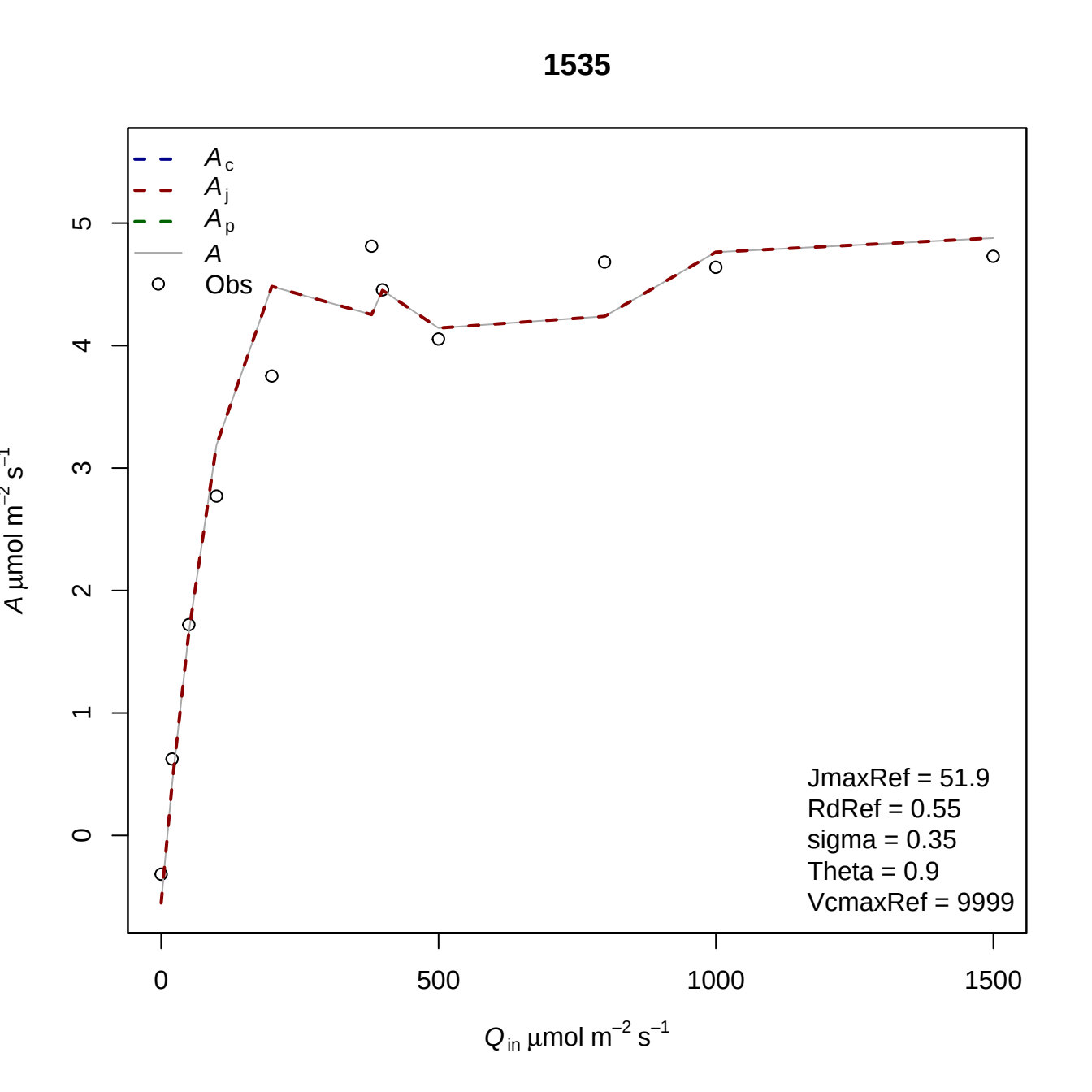

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 72.7
RdRef = 1.08
sigma = 0.85
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

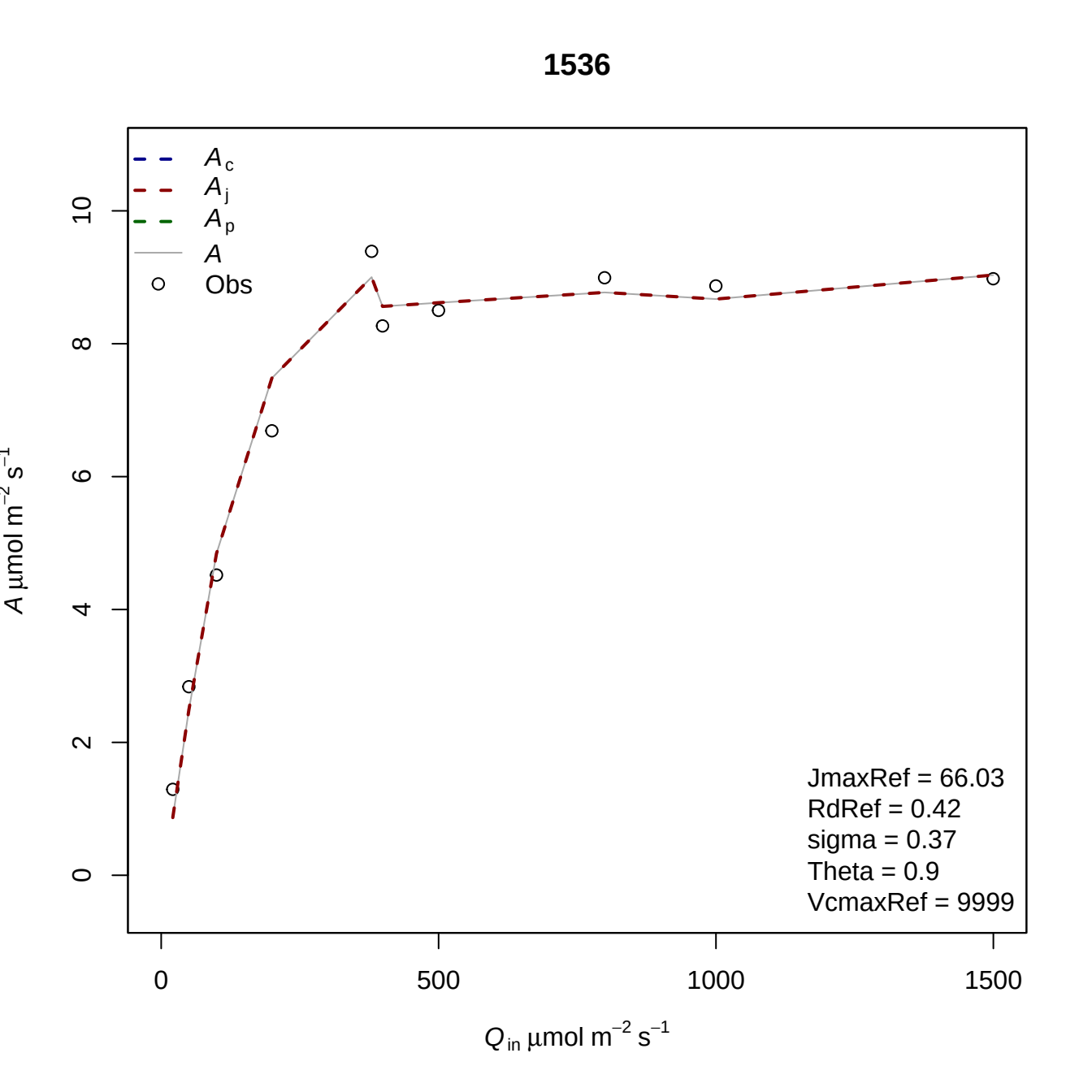

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 55.64
RdRef = 0.62
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

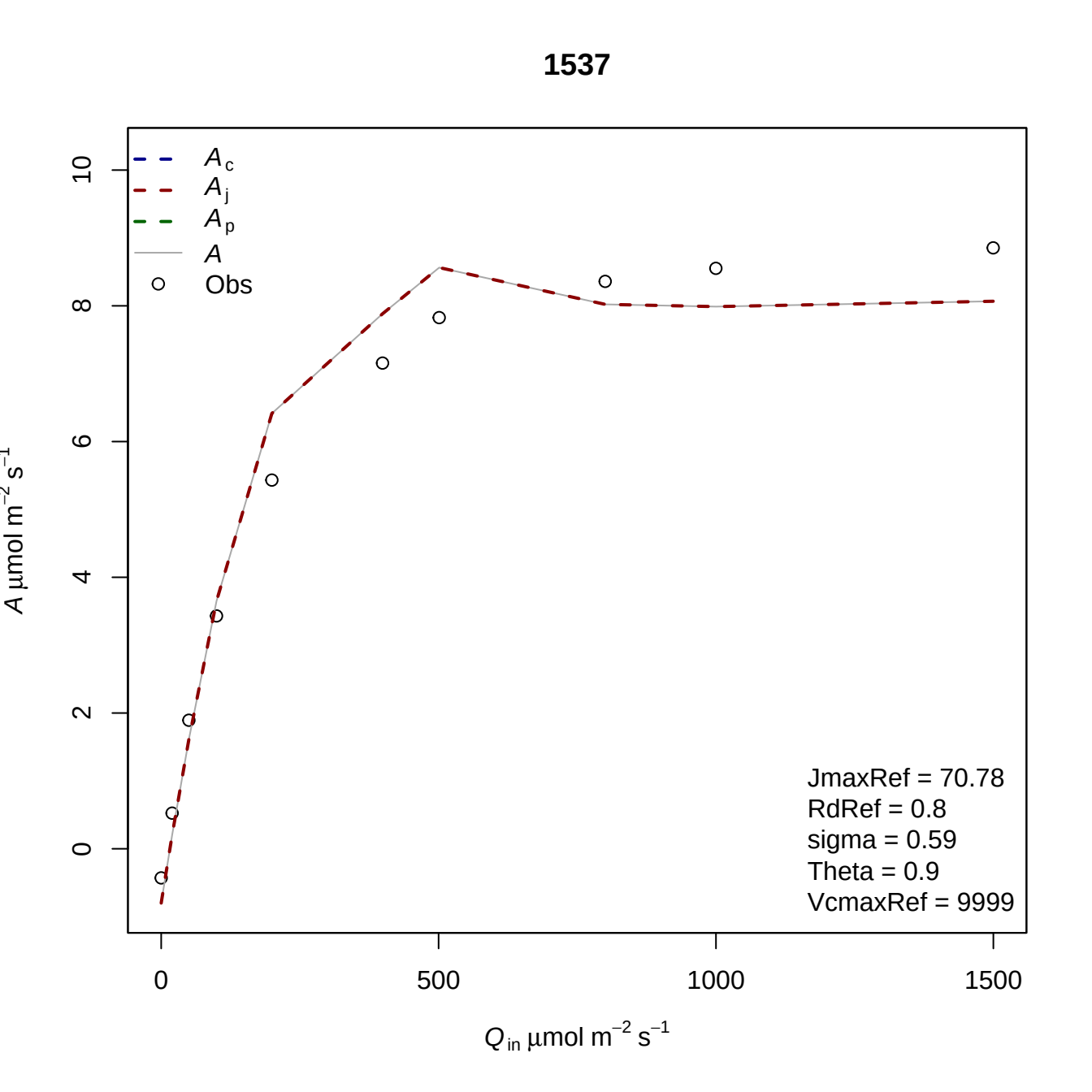

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 68.06
RdRef = 0.73
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

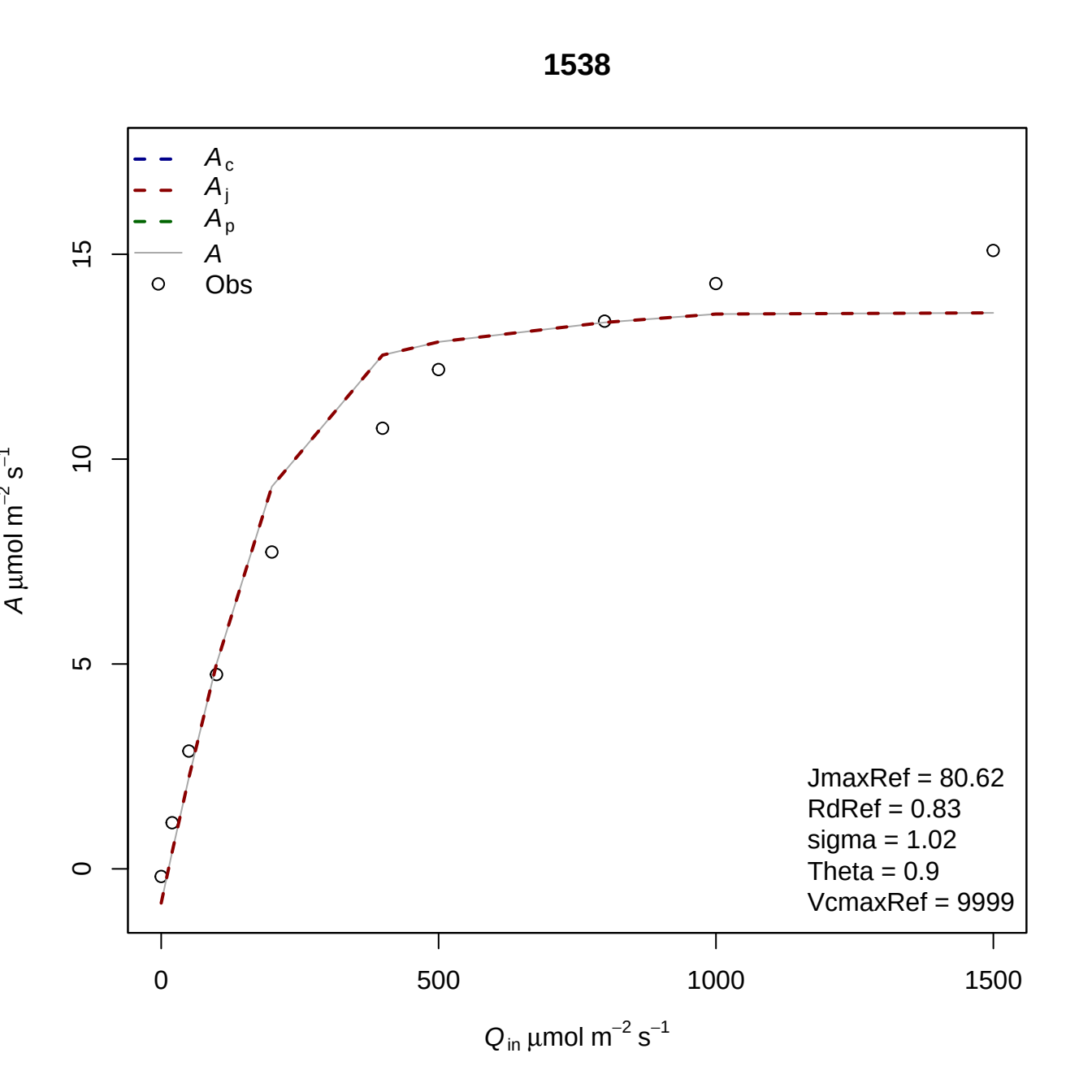

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 70.8
RdRef = 0.7
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

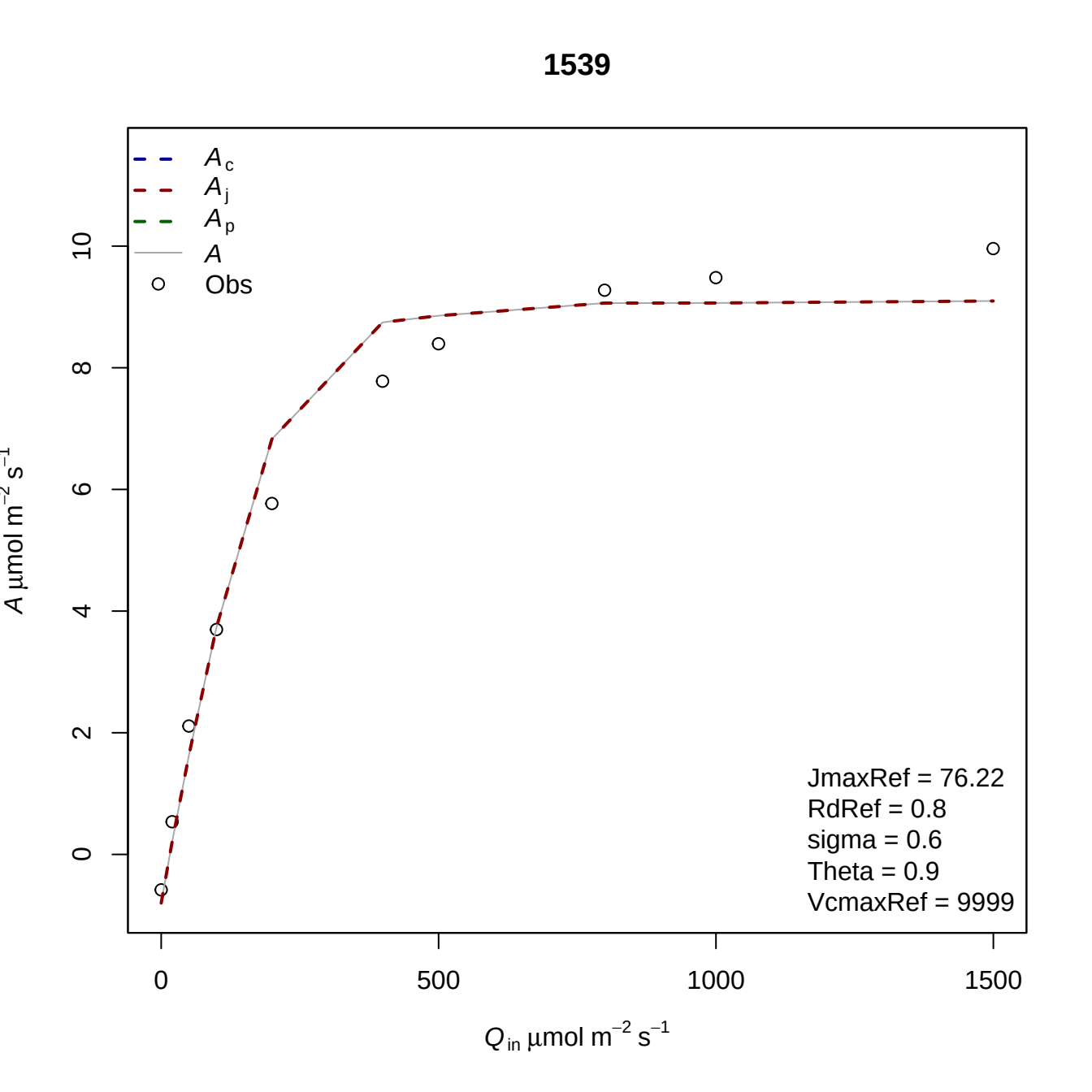

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 97.51
RdRef = 0.86
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

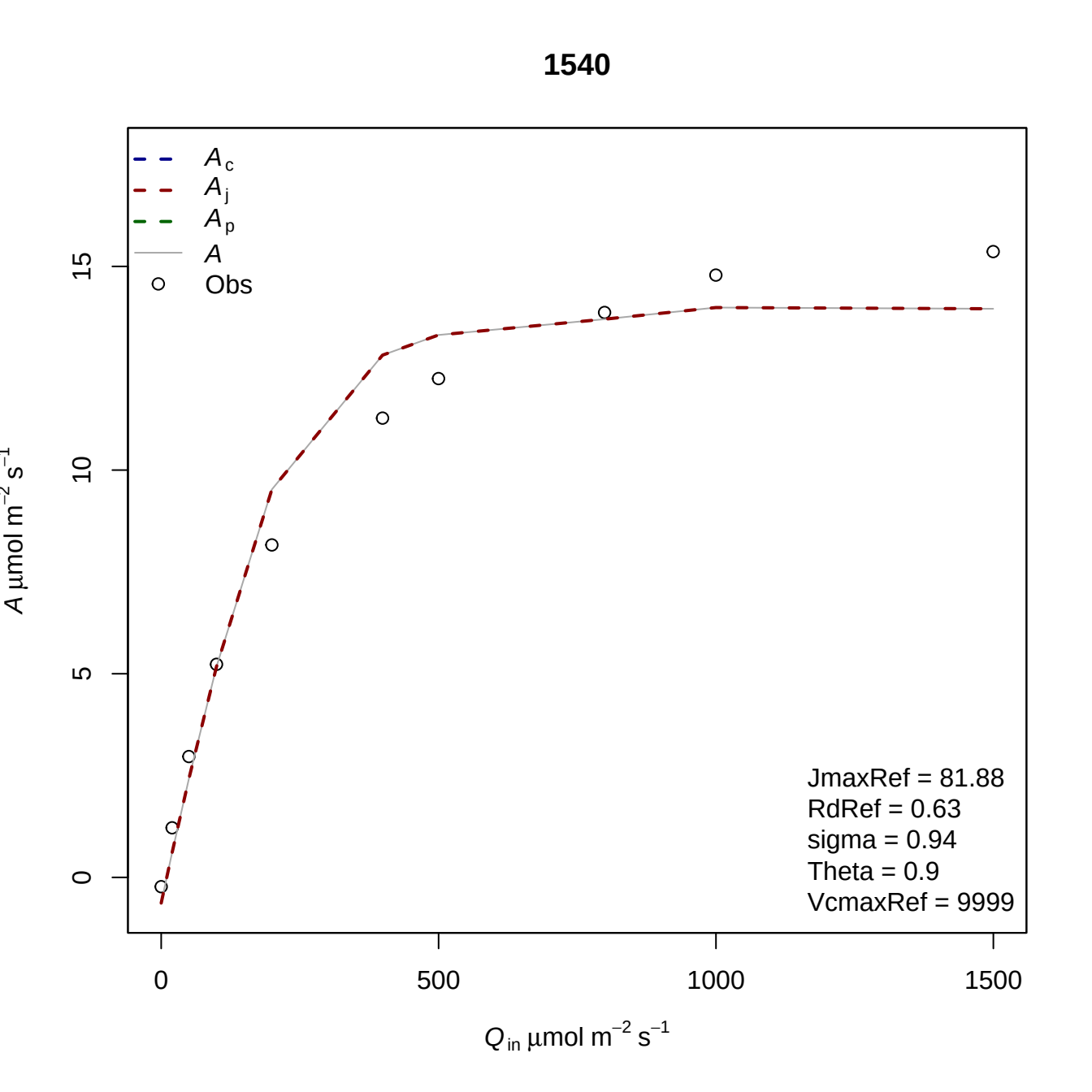

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 51.9
RdRef = 0.55
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

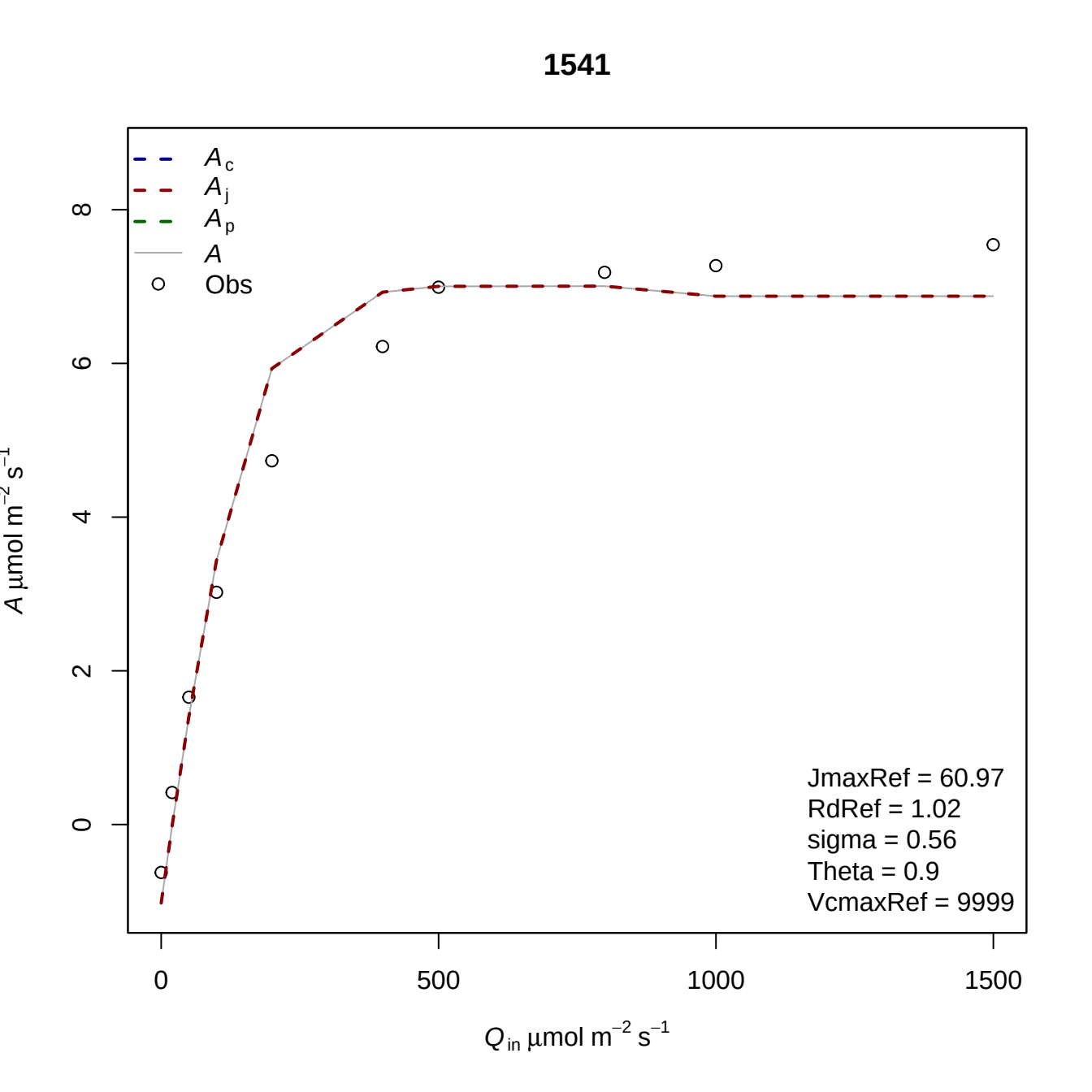

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 66.03
RdRef = 0.42
sigma = 0.37
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

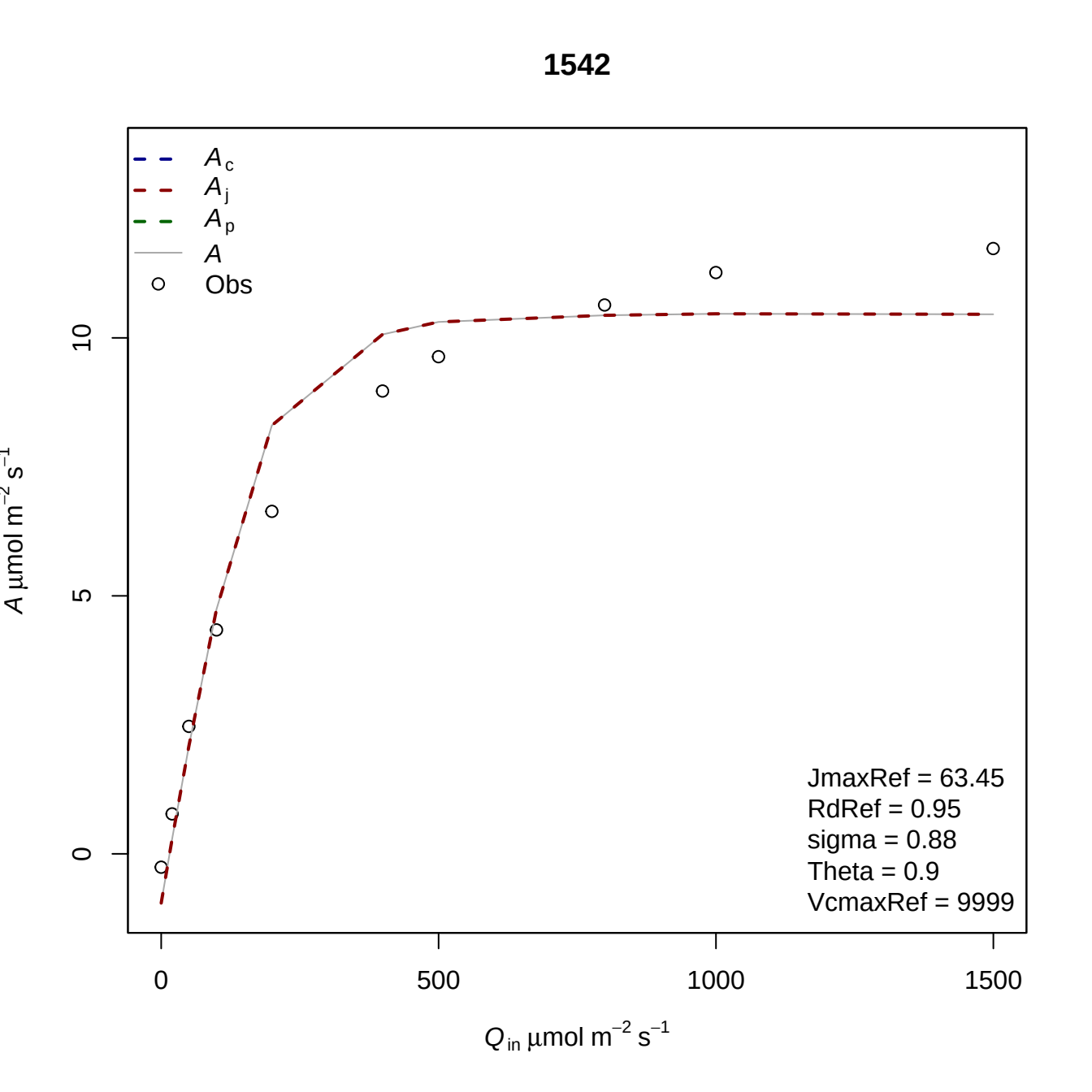

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 70.78
RdRef = 0.8
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

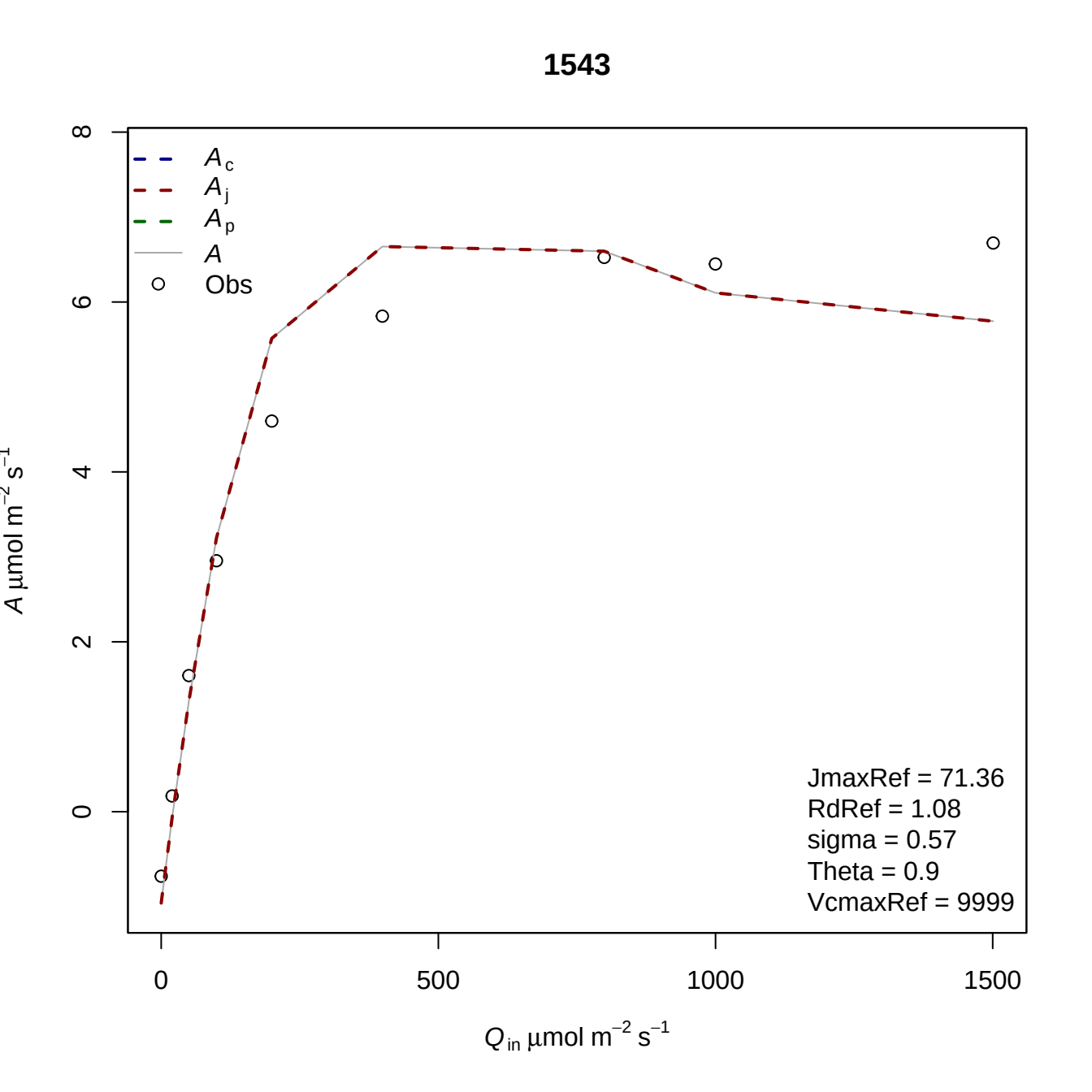

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 80.62
RdRef = 0.83
sigma = 1.02
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

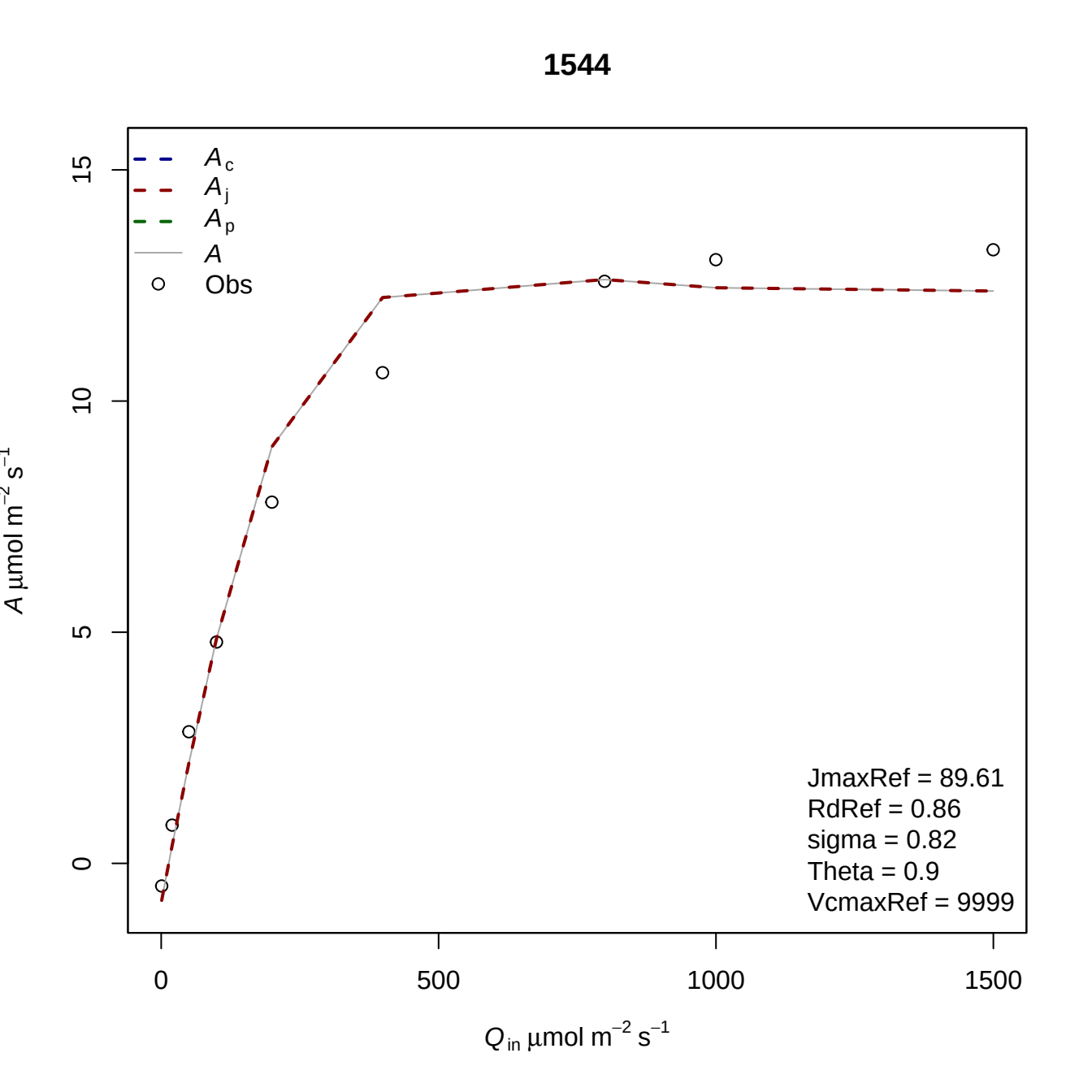

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 76.22
RdRef = 0.8
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

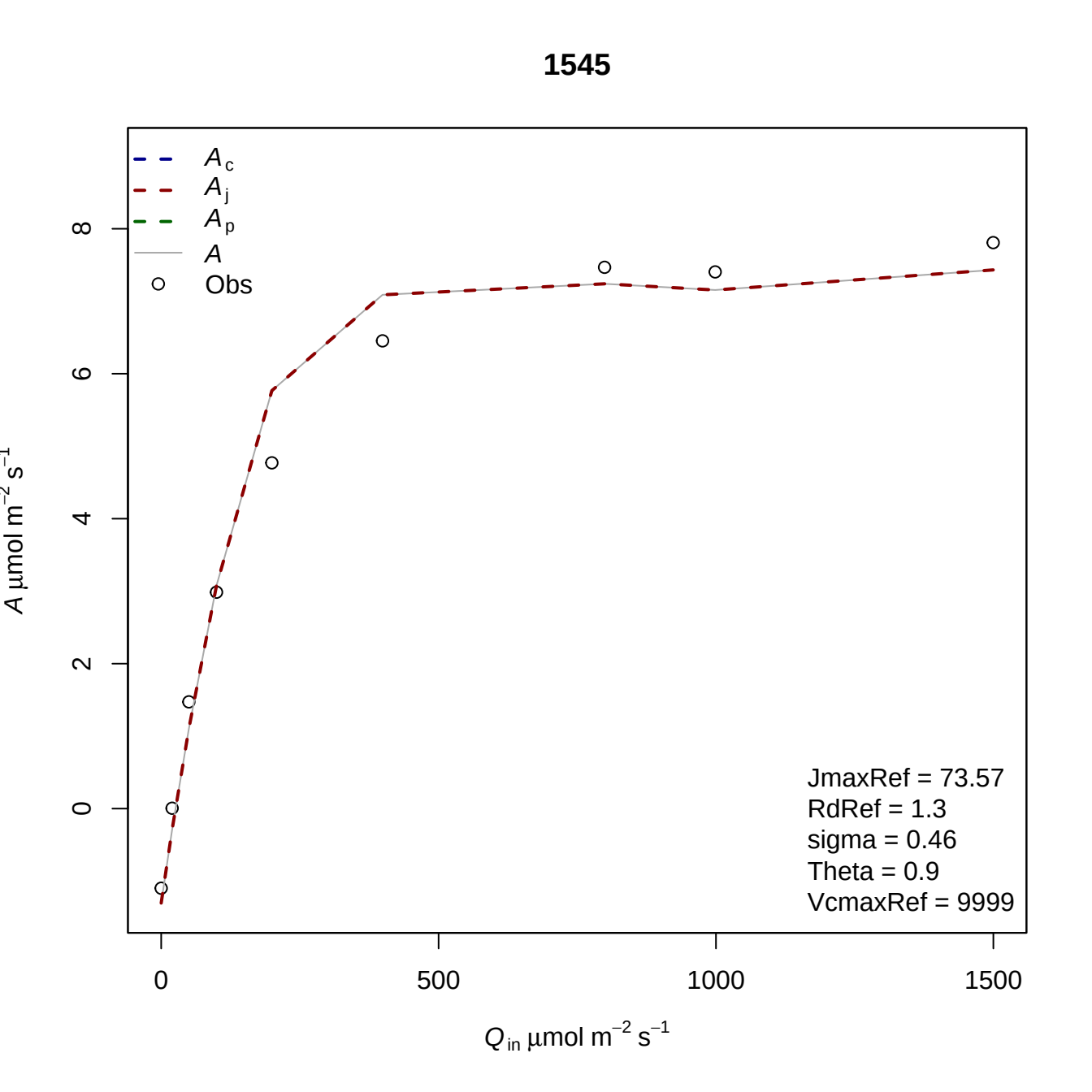

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 81.88
RdRef = 0.63
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

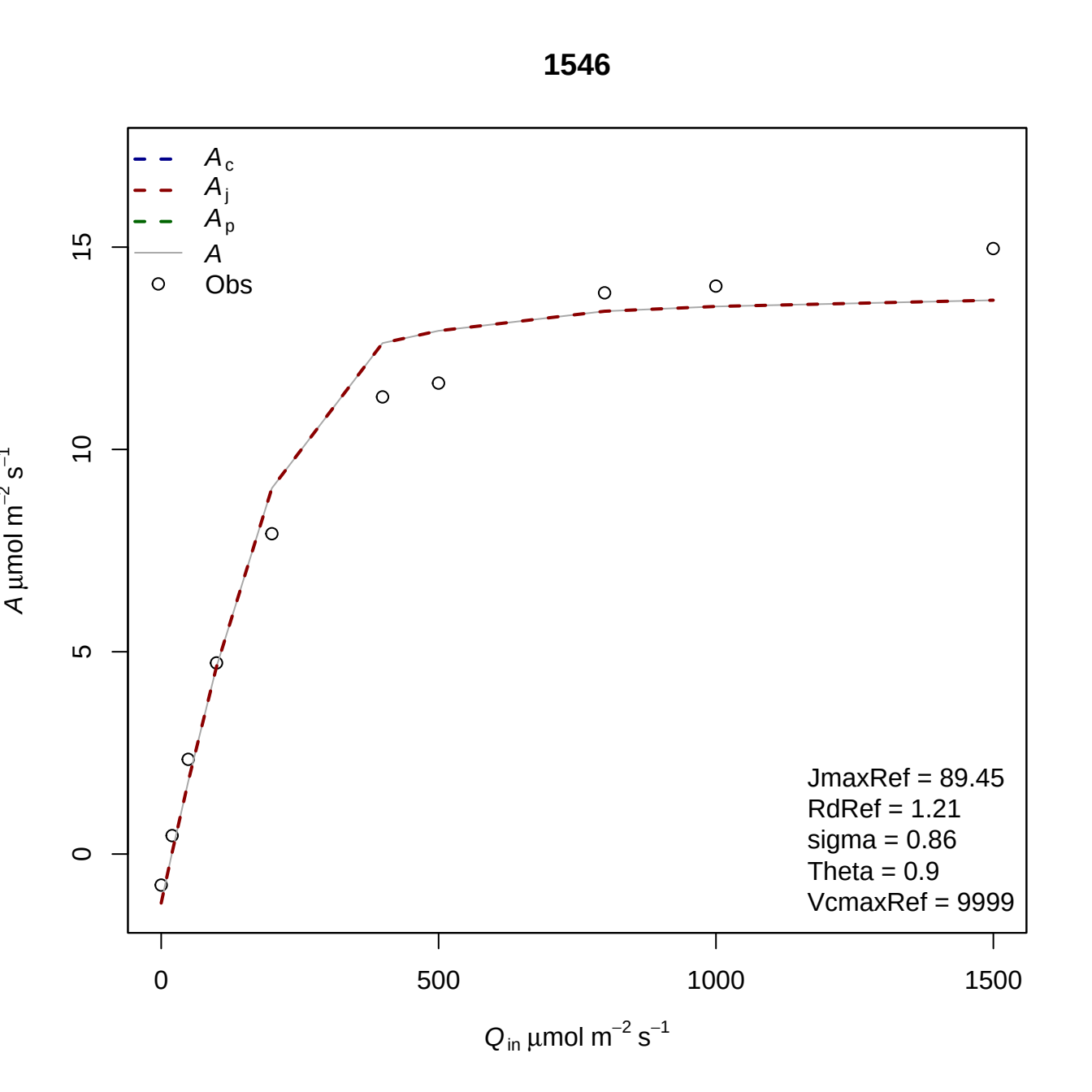

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 60.97
RdRef = 1.02
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

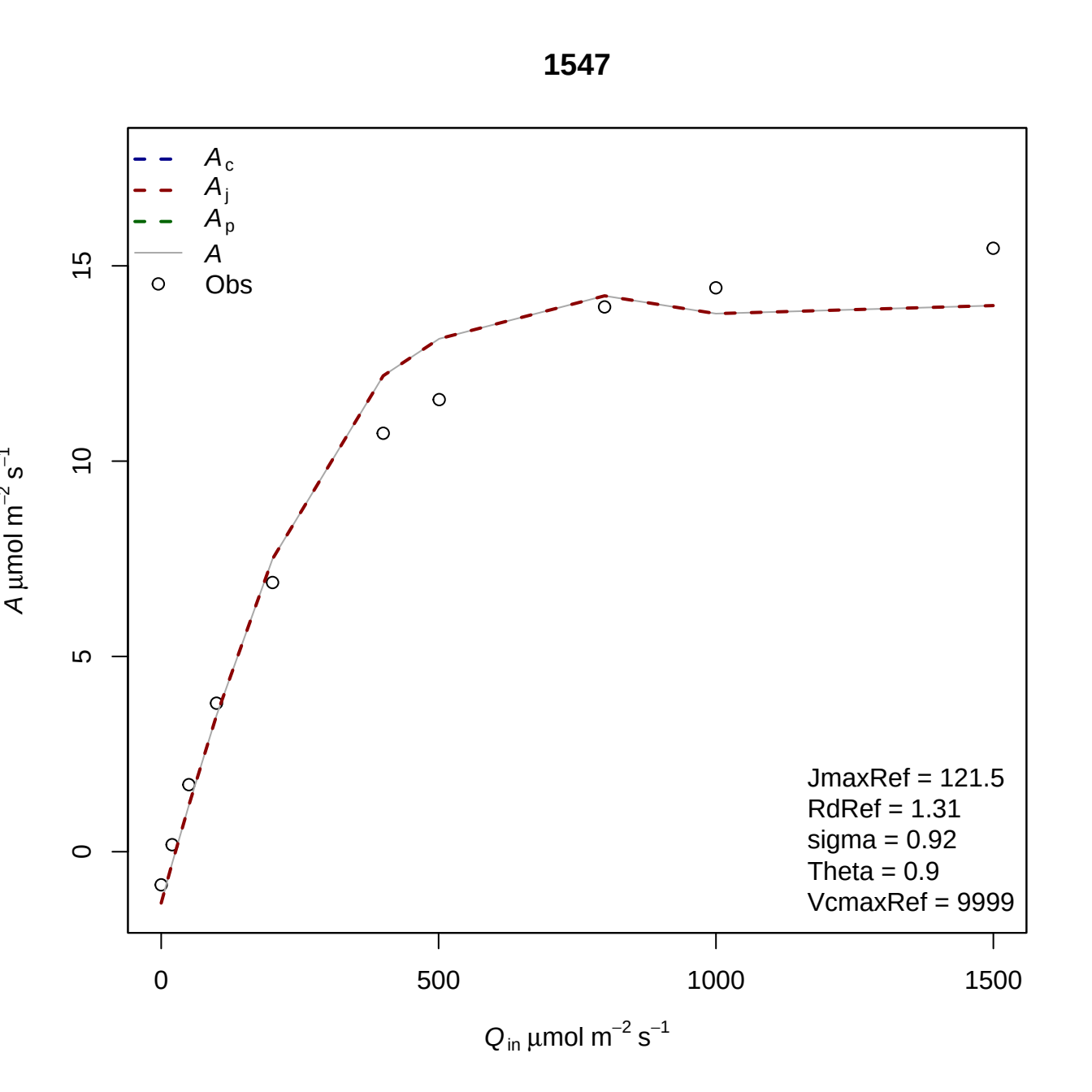

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 63.45
RdRef = 0.95
sigma = 0.88
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

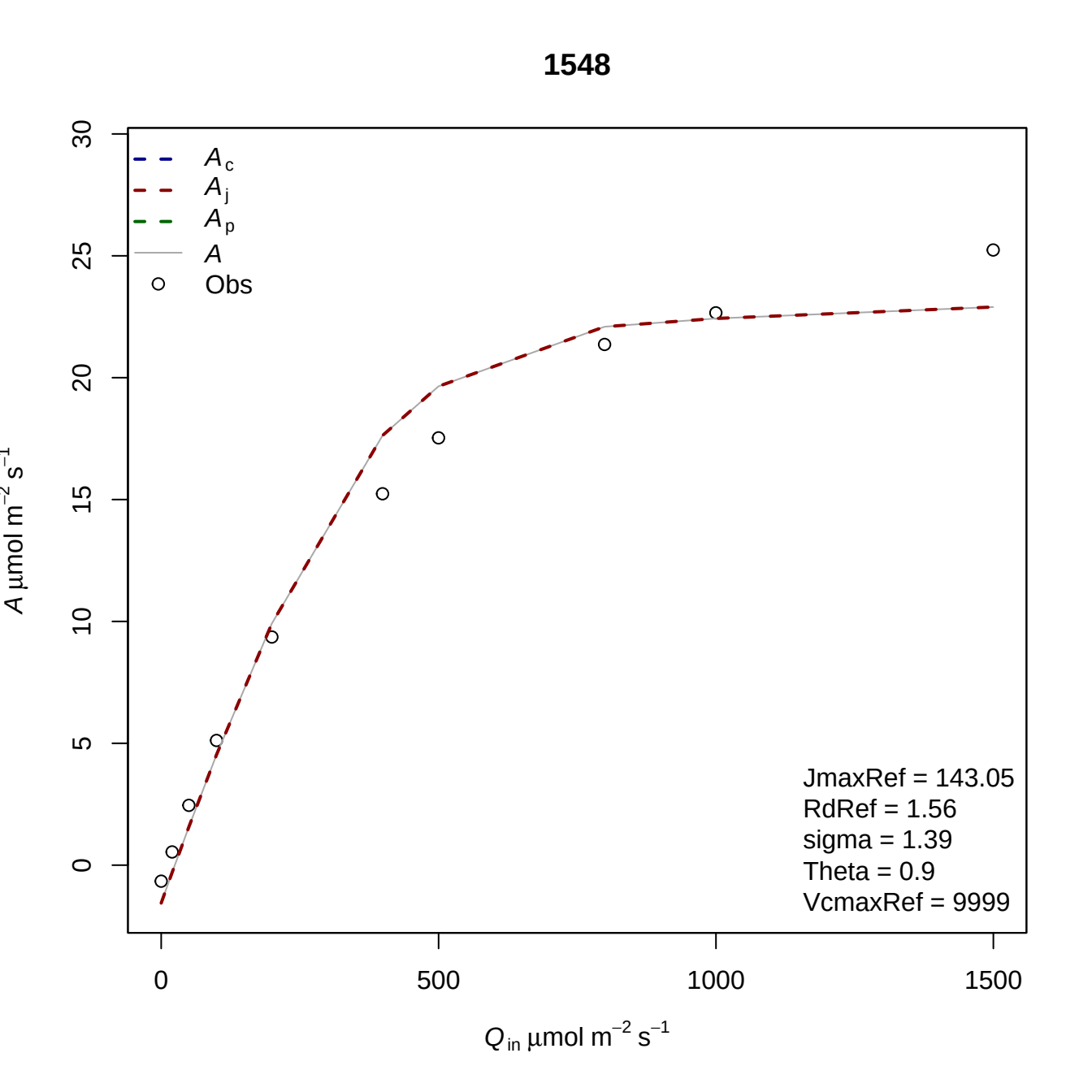

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 71.36
RdRef = 1.08
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

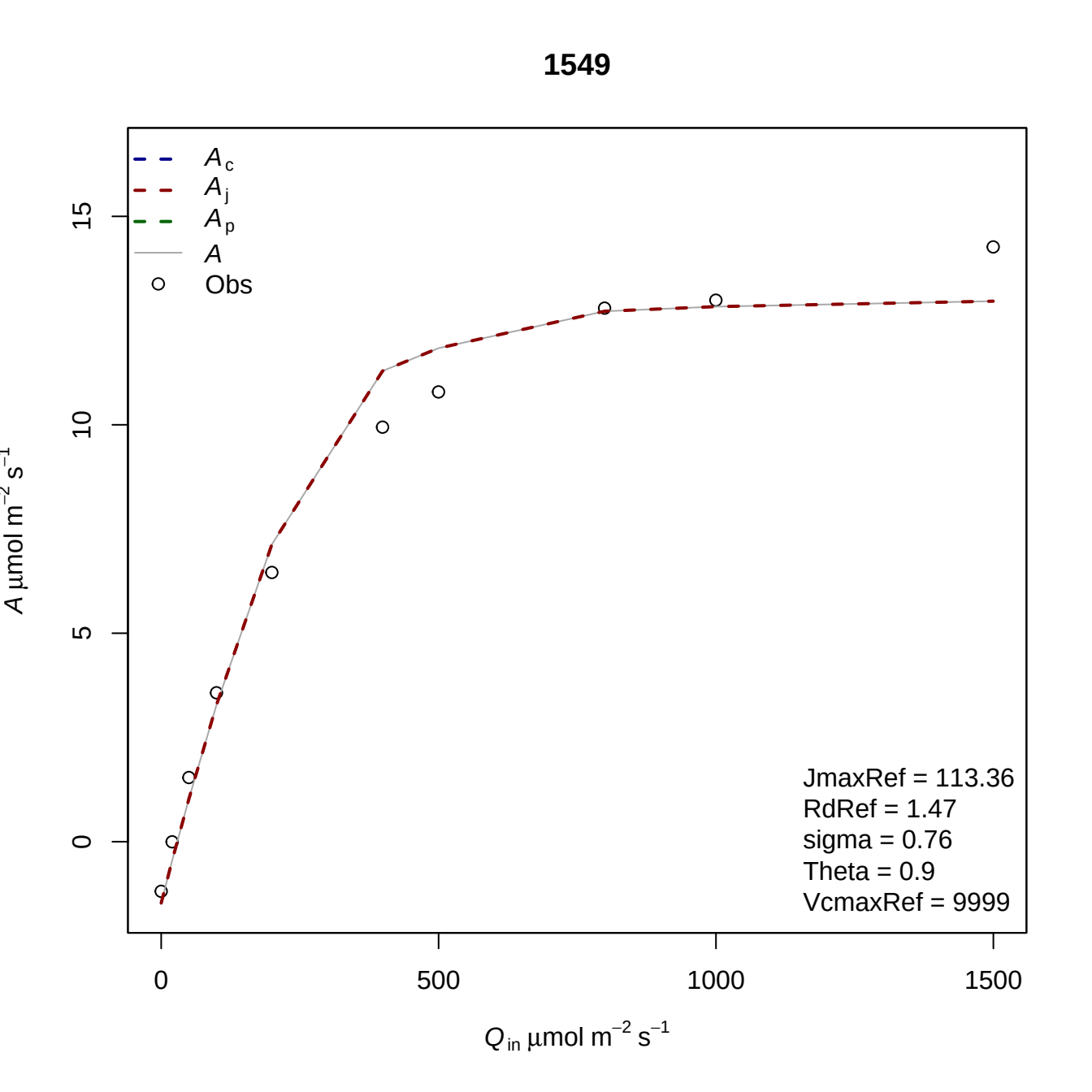

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 89.61
RdRef = 0.86
sigma = 0.82
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

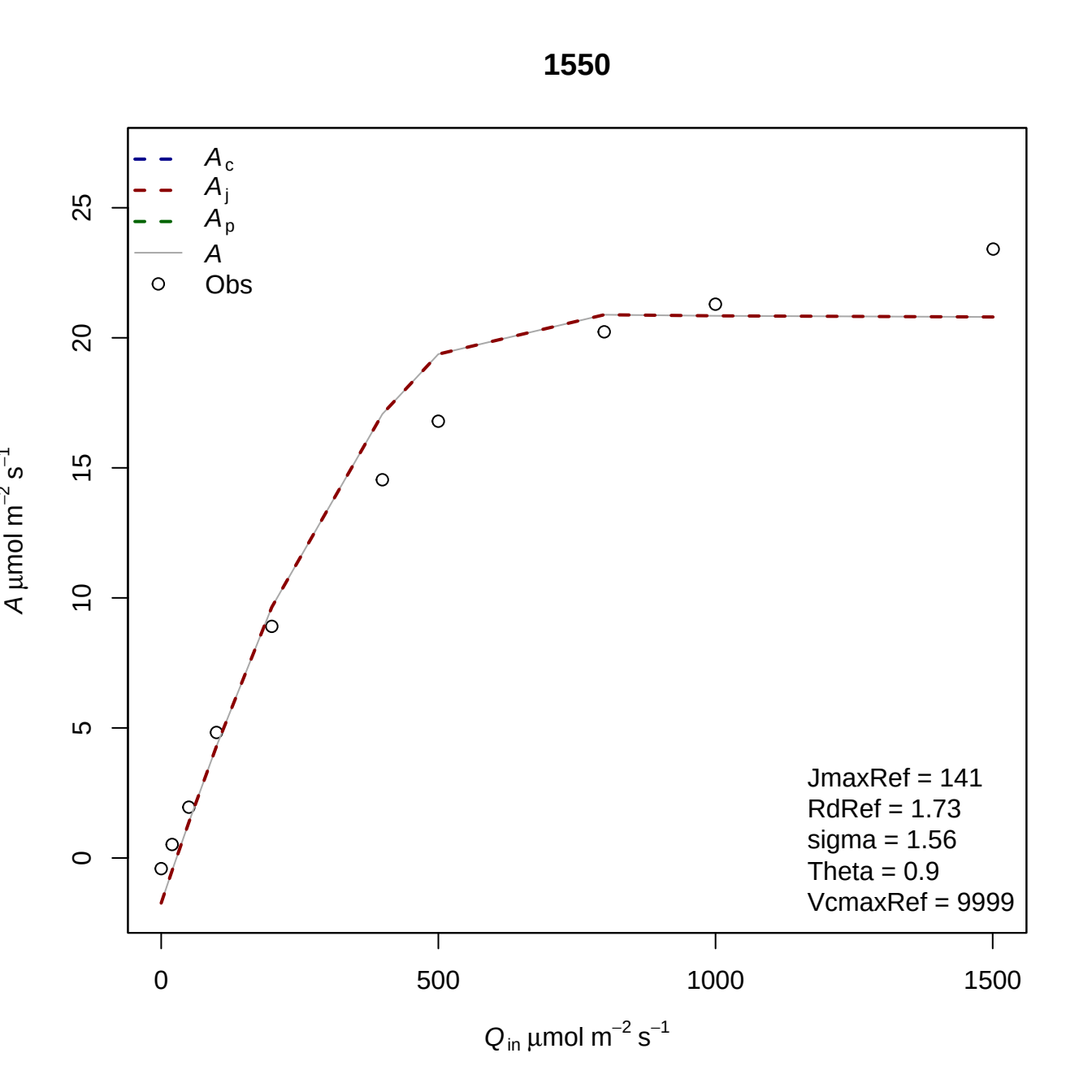

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 73.57
RdRef = 1.3
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

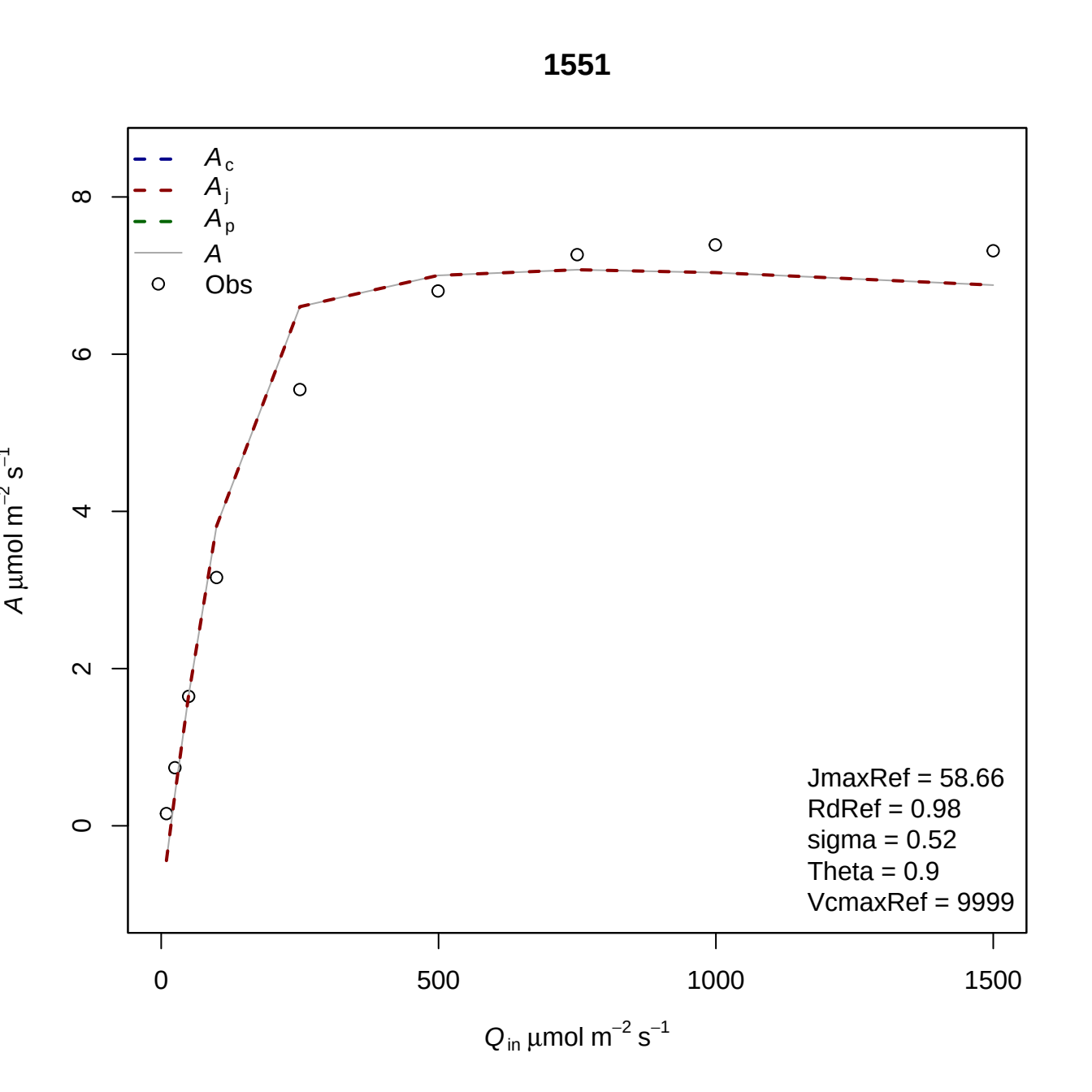

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 89.45
RdRef = 1.21
sigma = 0.86
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

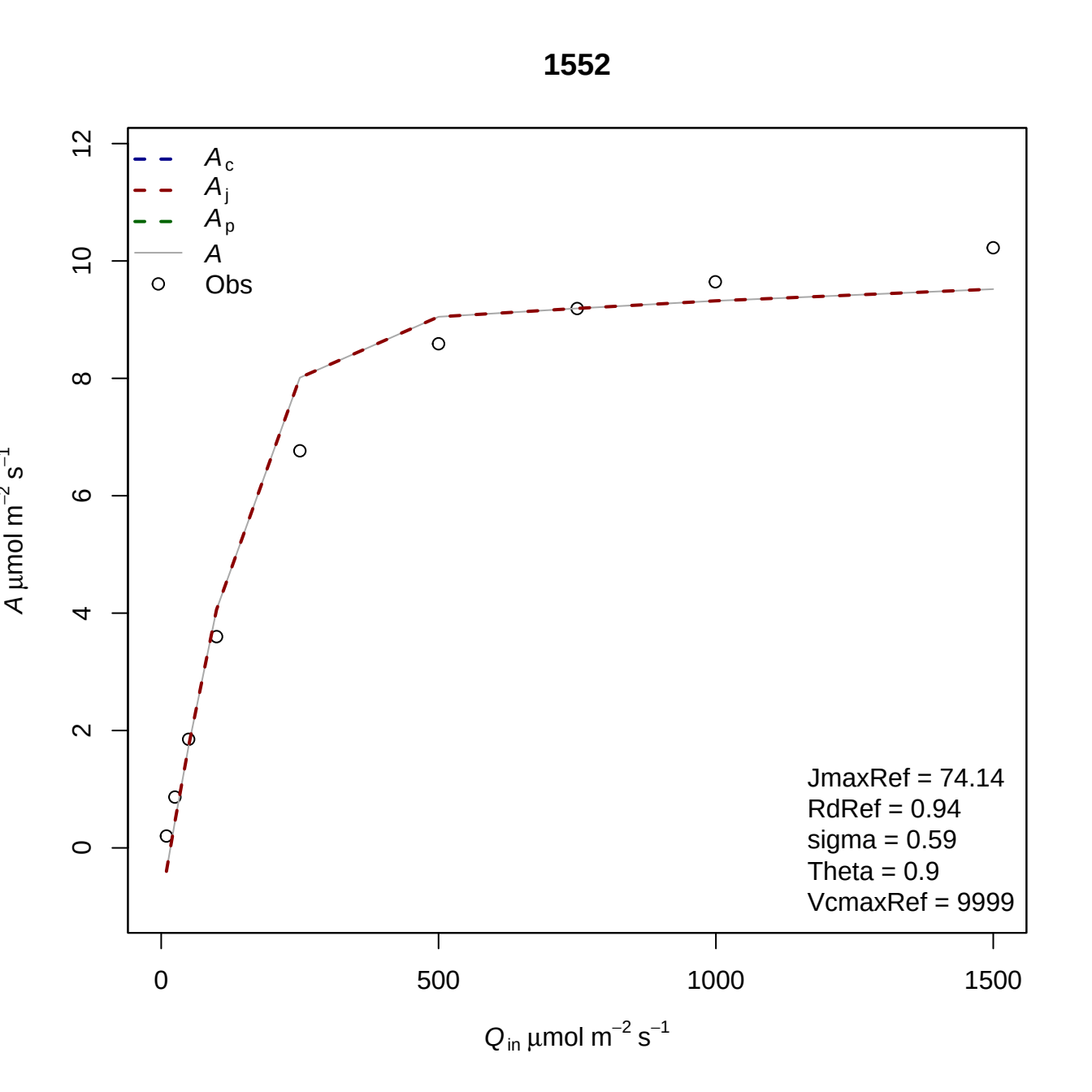

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 121.5
RdRef = 1.31
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

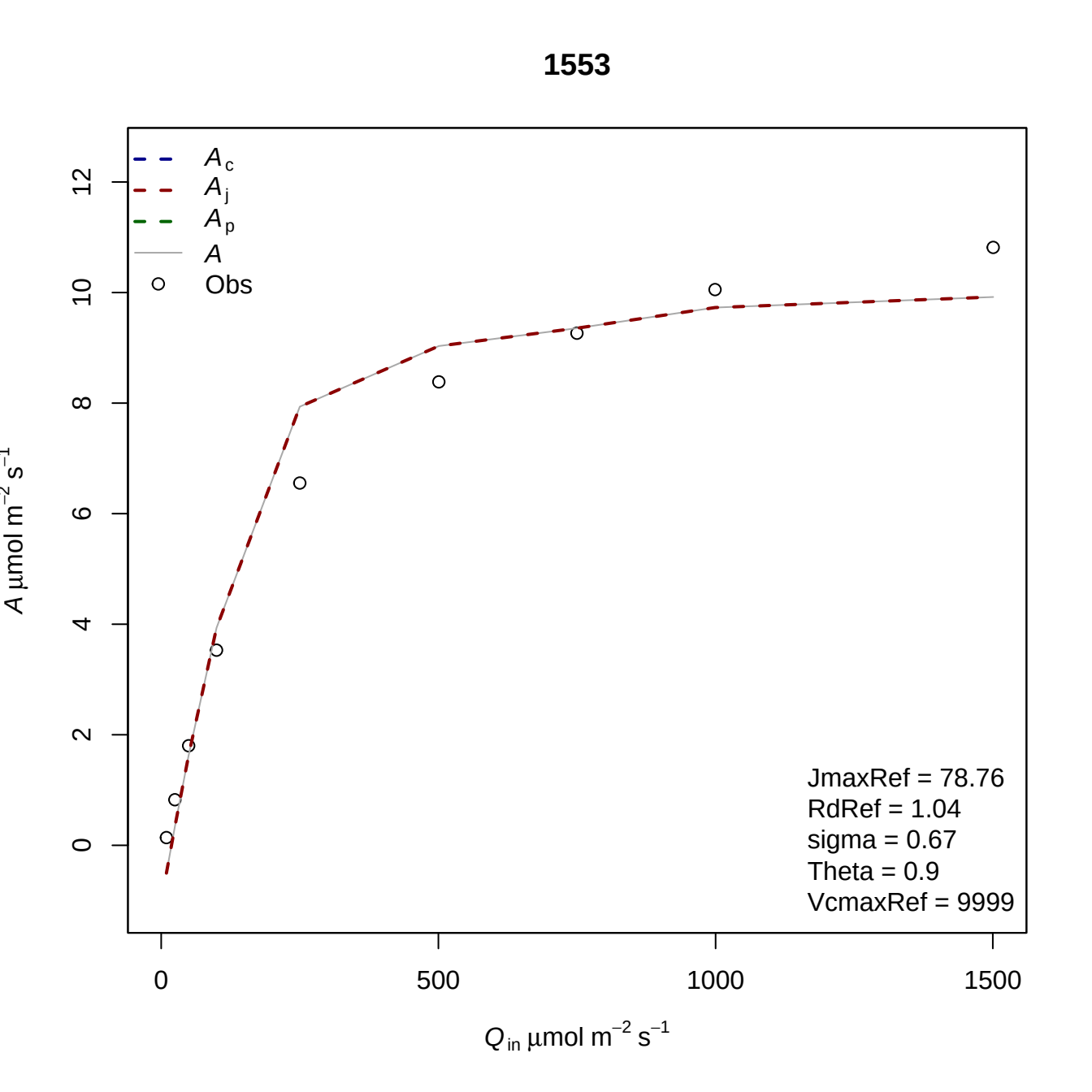

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 143.05
RdRef = 1.56
sigma = 1.39
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

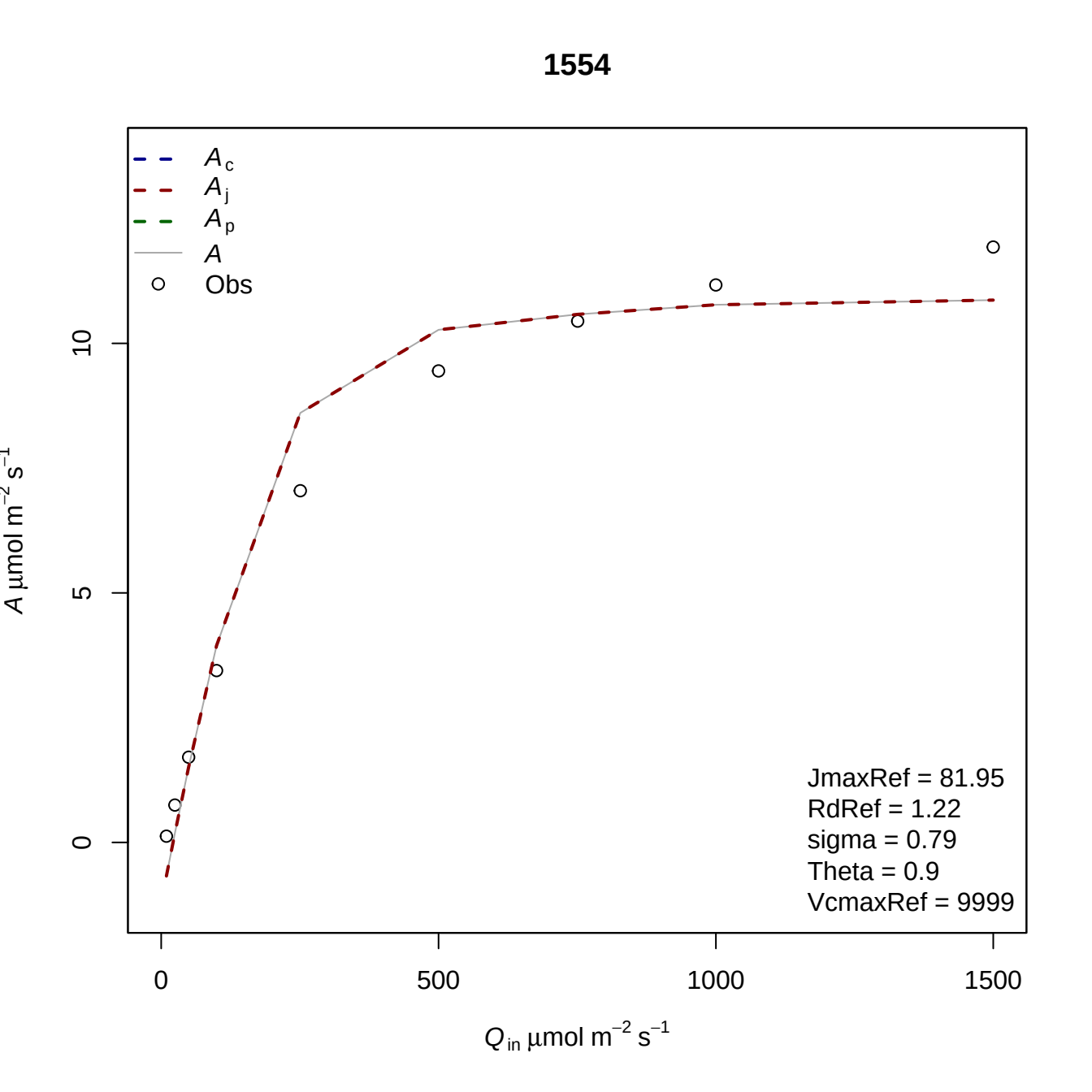

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 113.36
RdRef = 1.47
sigma = 0.76
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

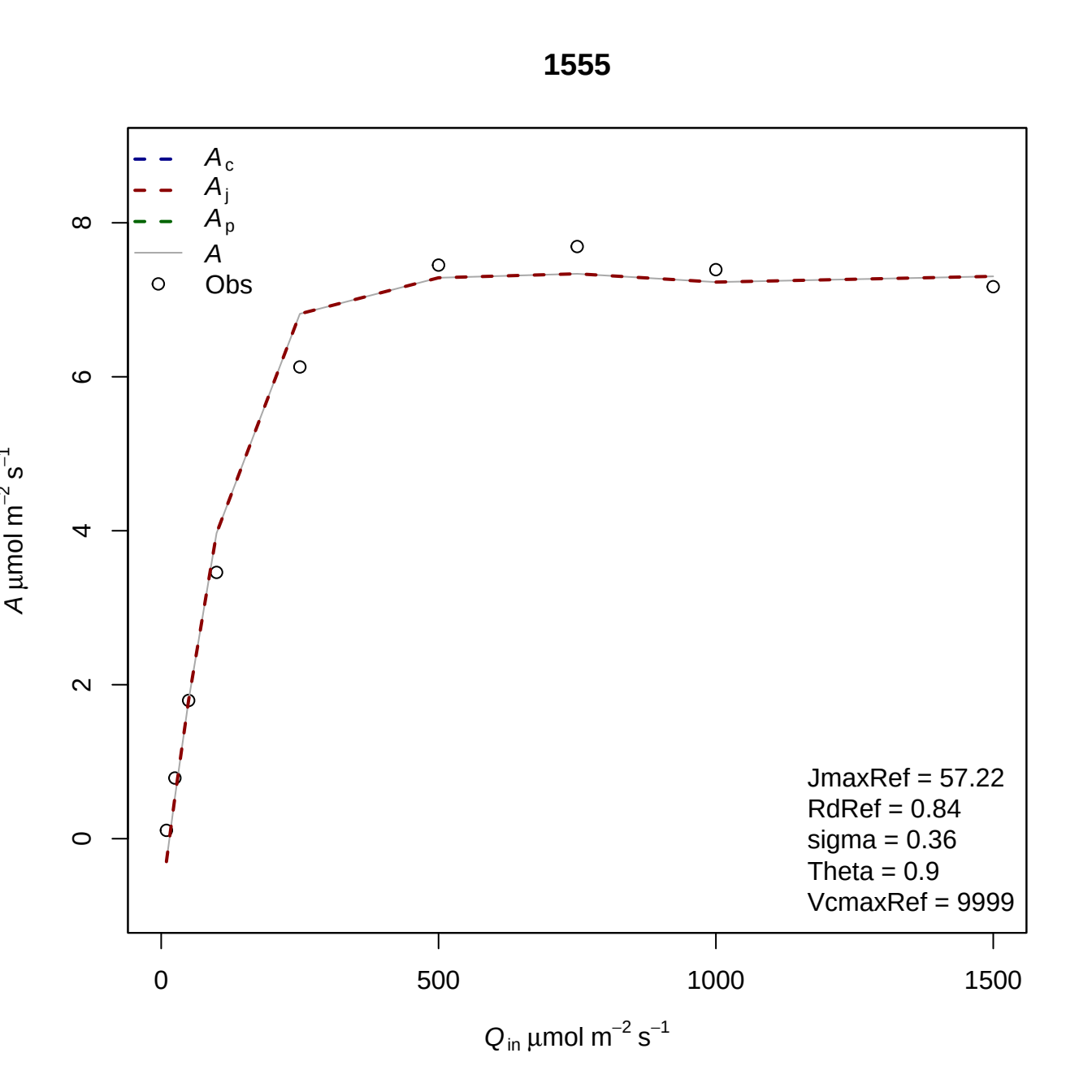

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 141
RdRef = 1.73
sigma = 1.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

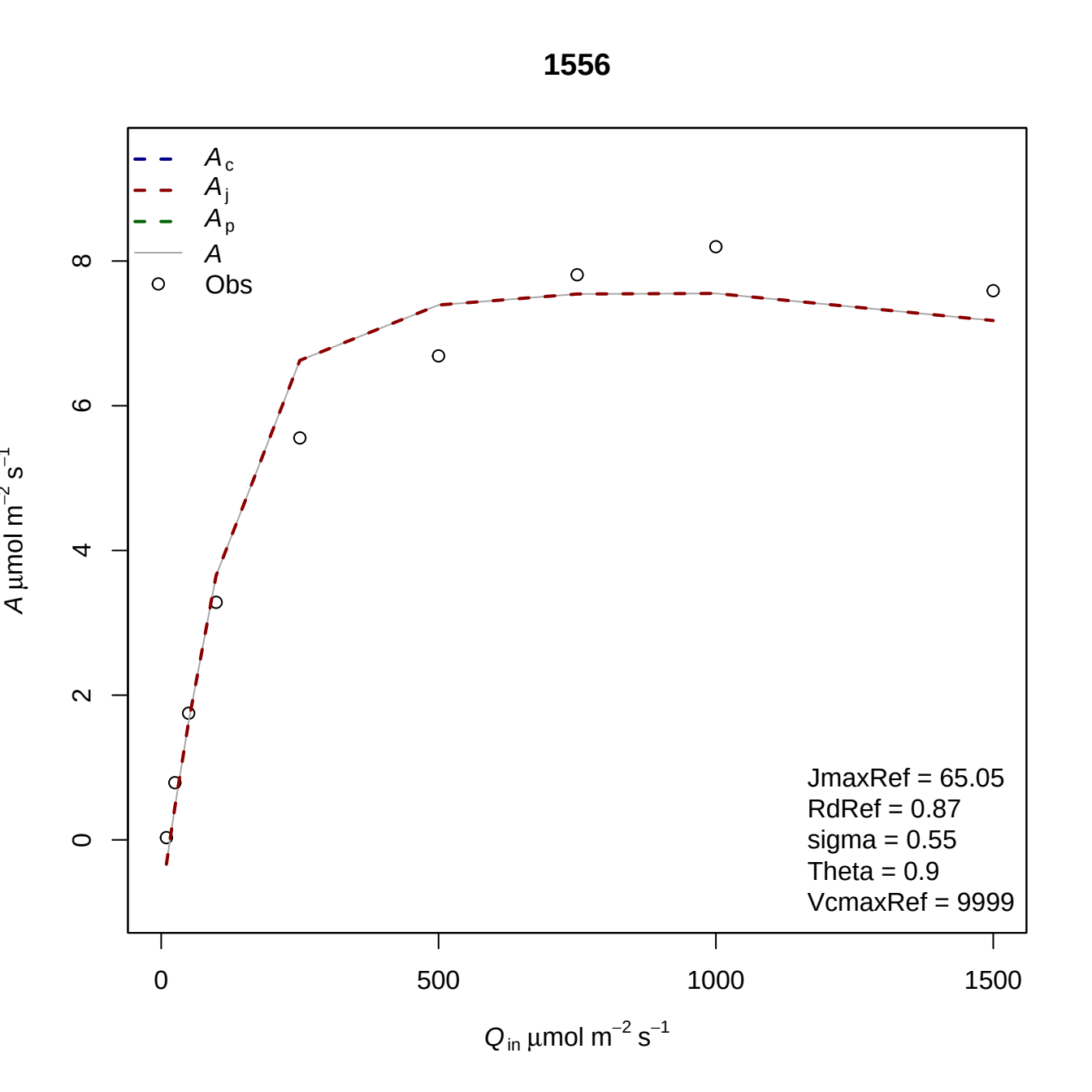

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 58.66
RdRef = 0.98
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

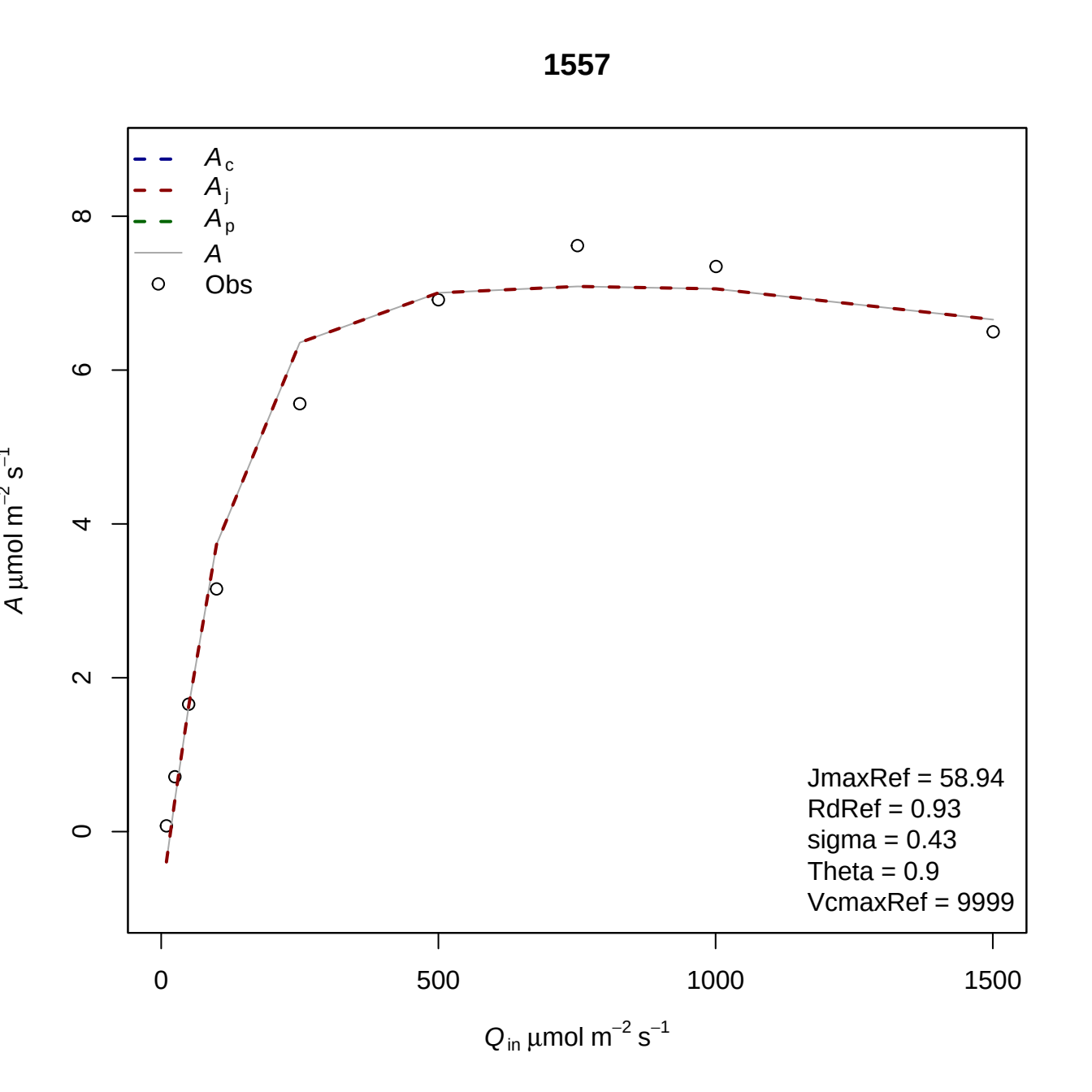

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 74.14
RdRef = 0.94
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

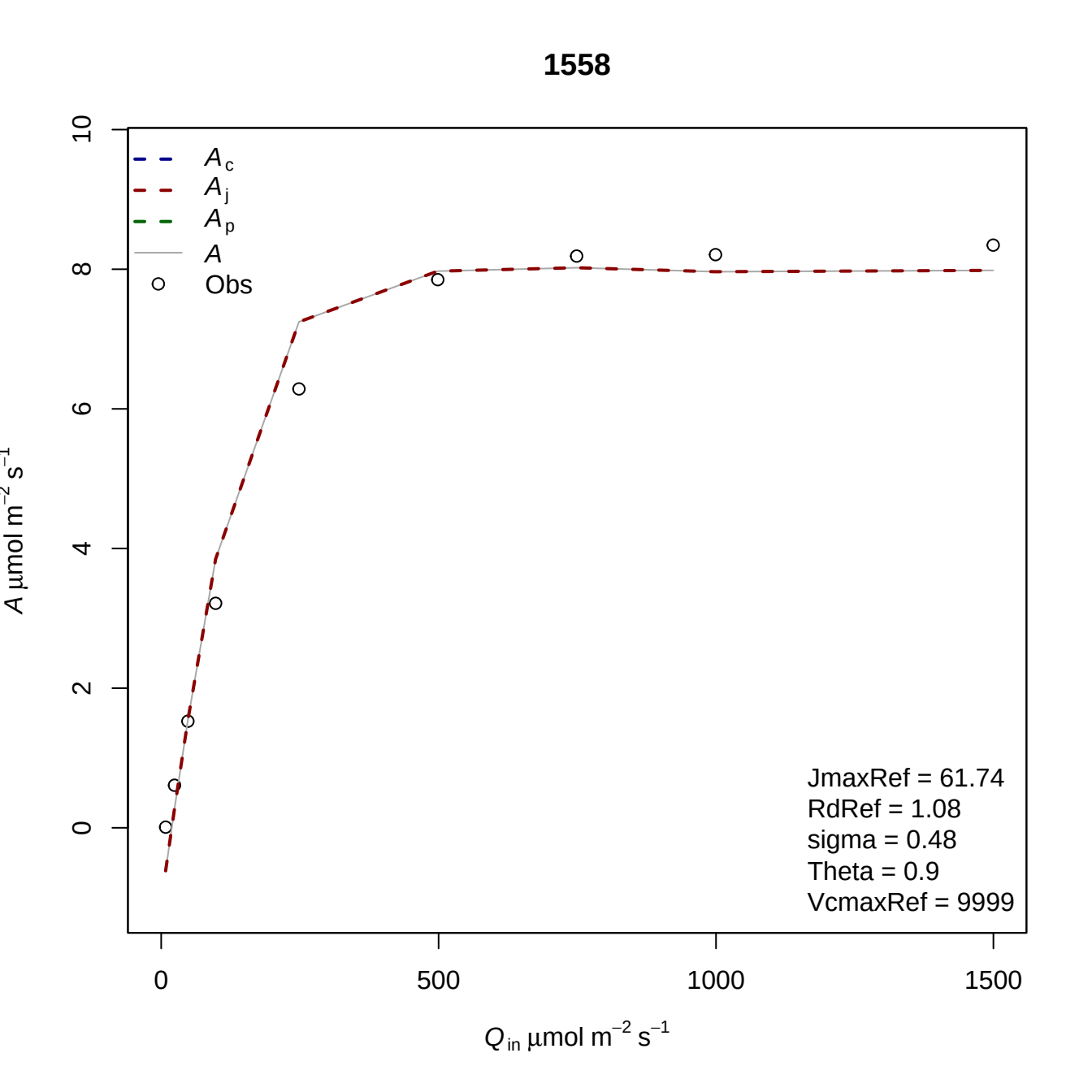

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 78.76
RdRef = 1.04
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

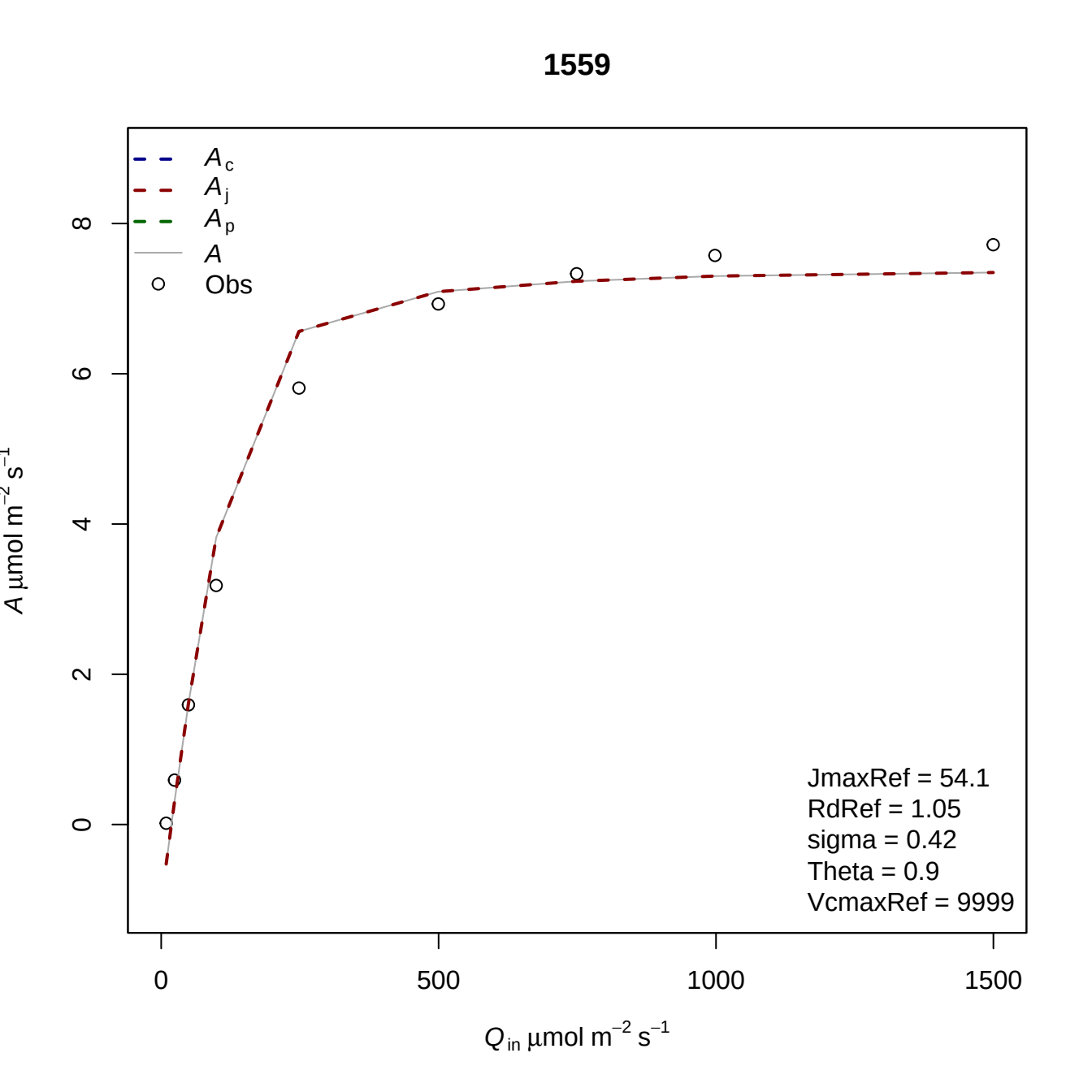

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 81.95
RdRef = 1.22
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

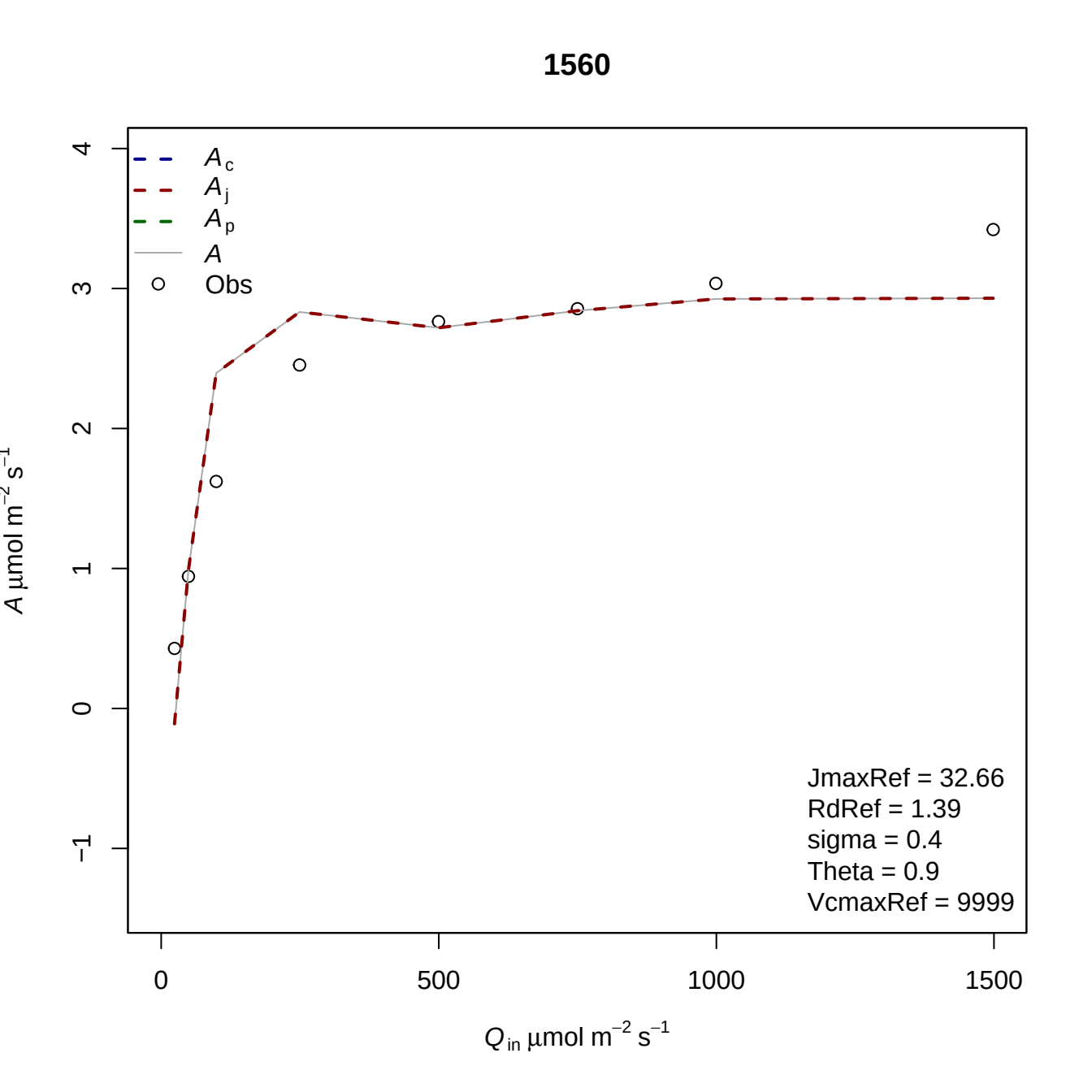

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 57.22
RdRef = 0.84
sigma = 0.36
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

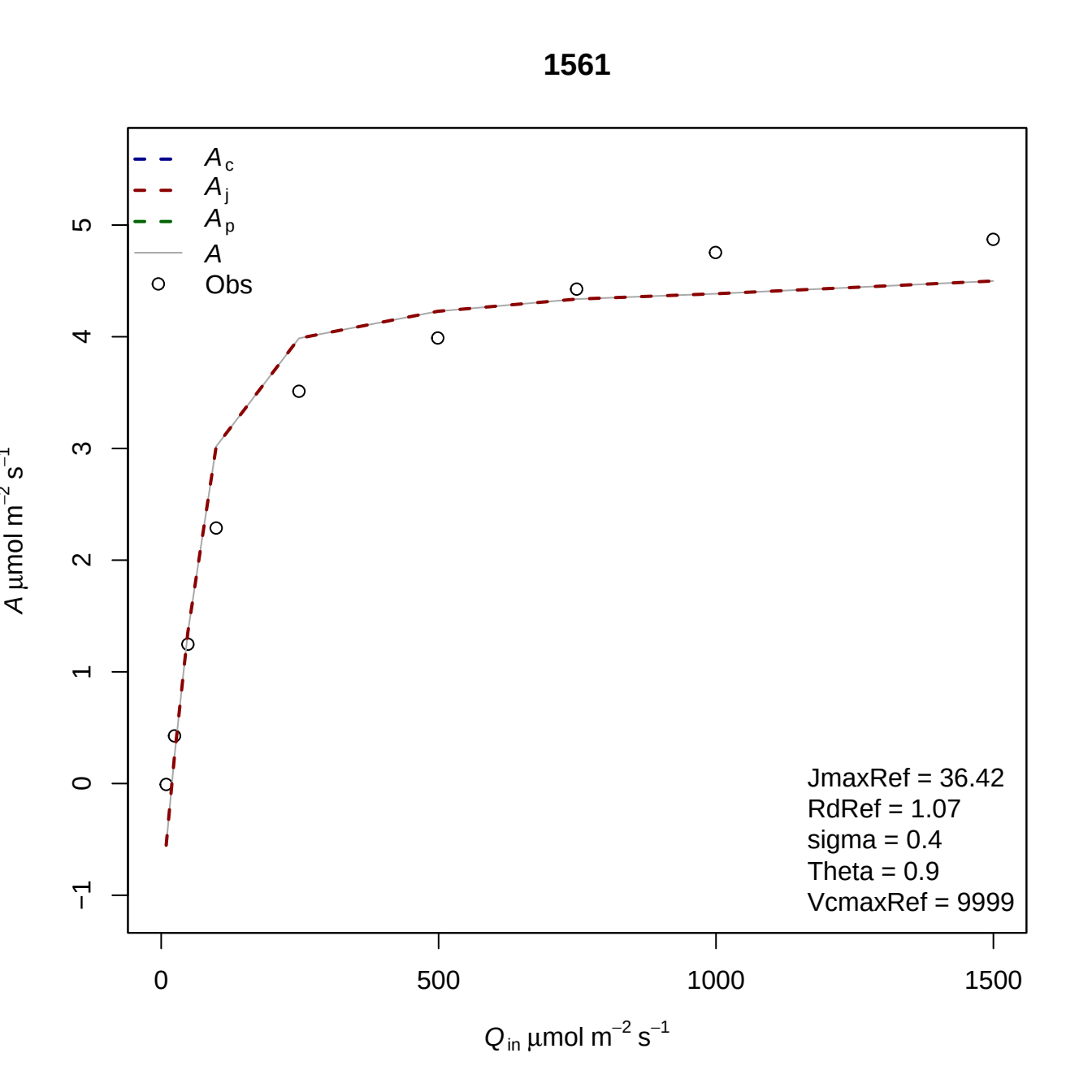

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 65.05
RdRef = 0.87
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

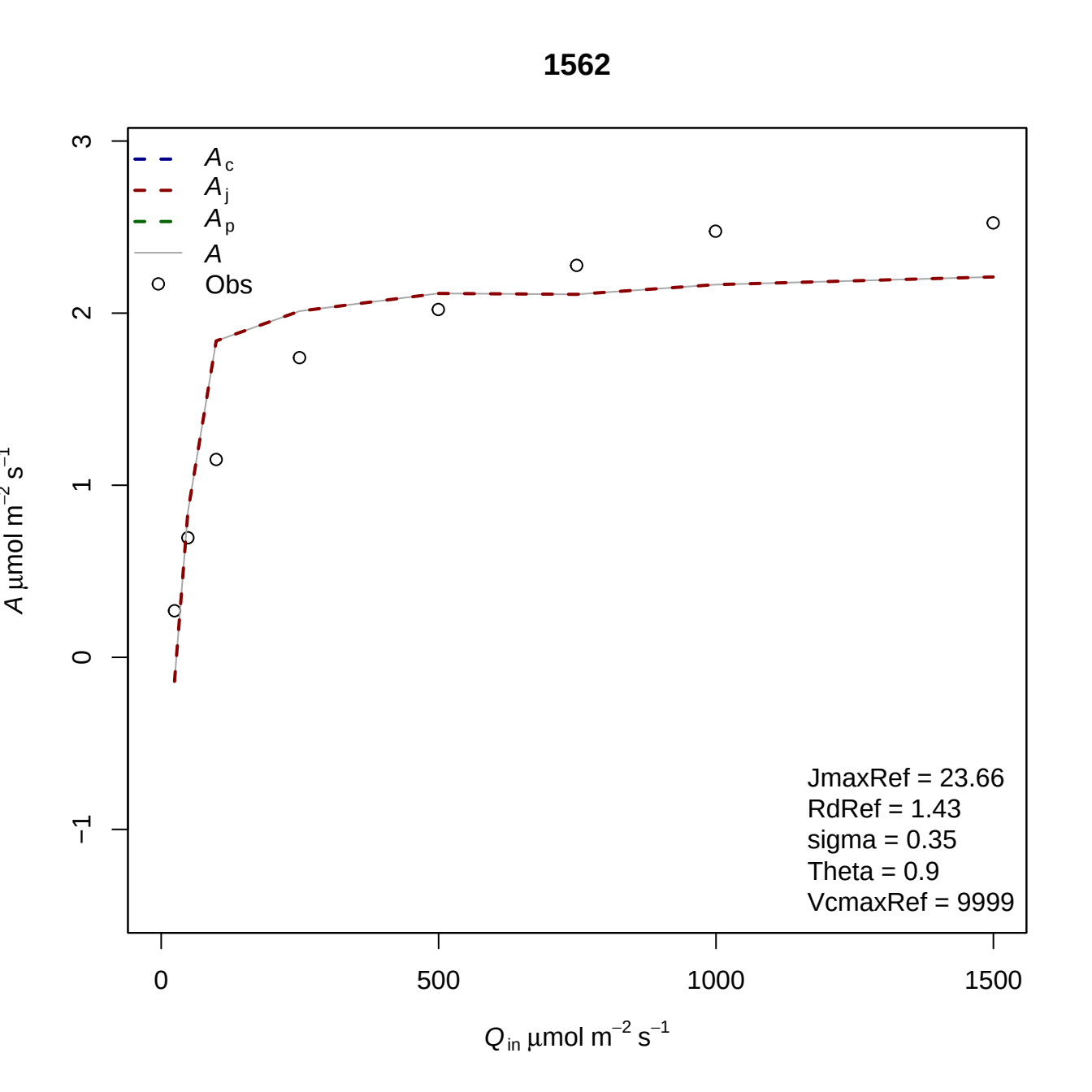

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 58.94
RdRef = 0.93
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

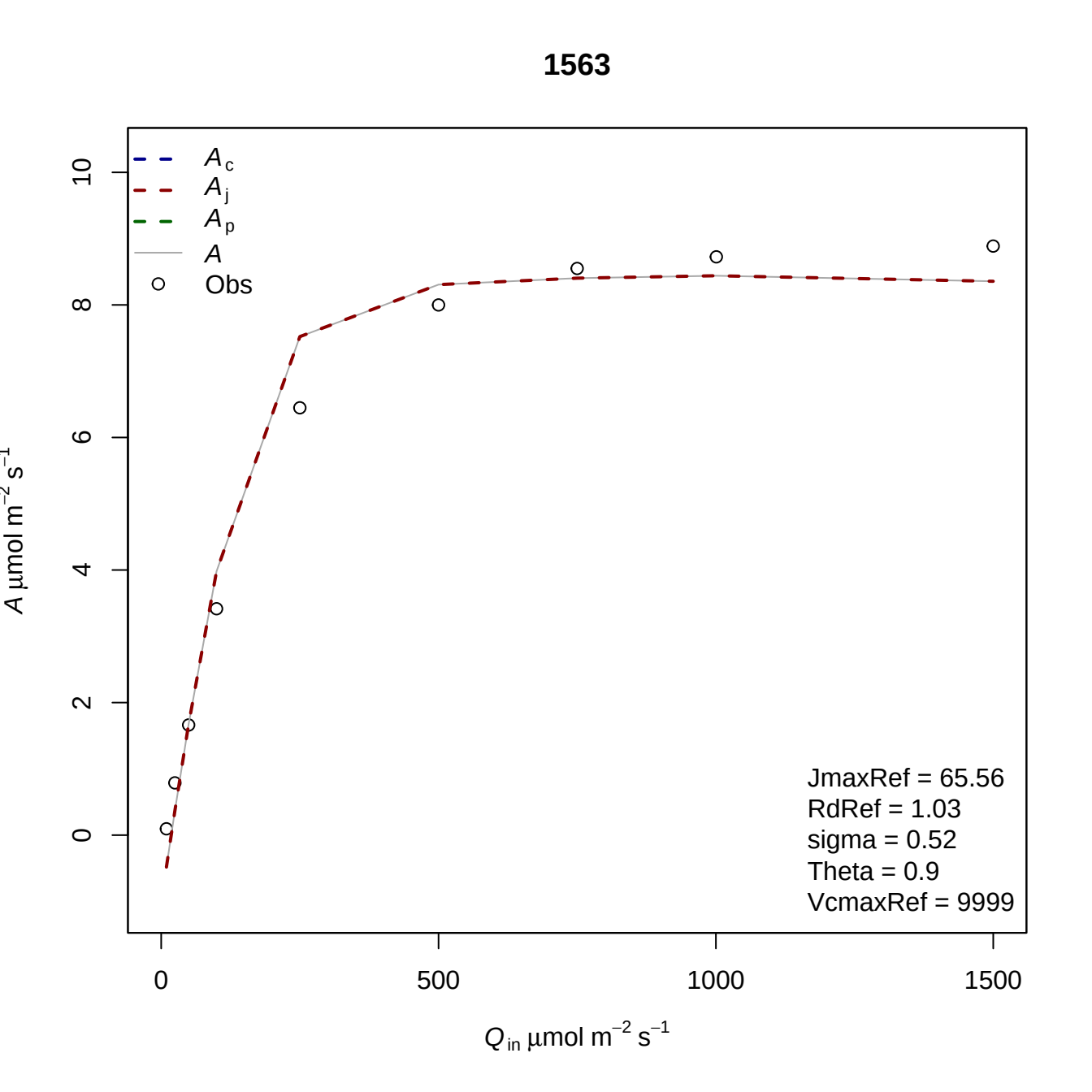

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 61.74
RdRef = 1.08
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

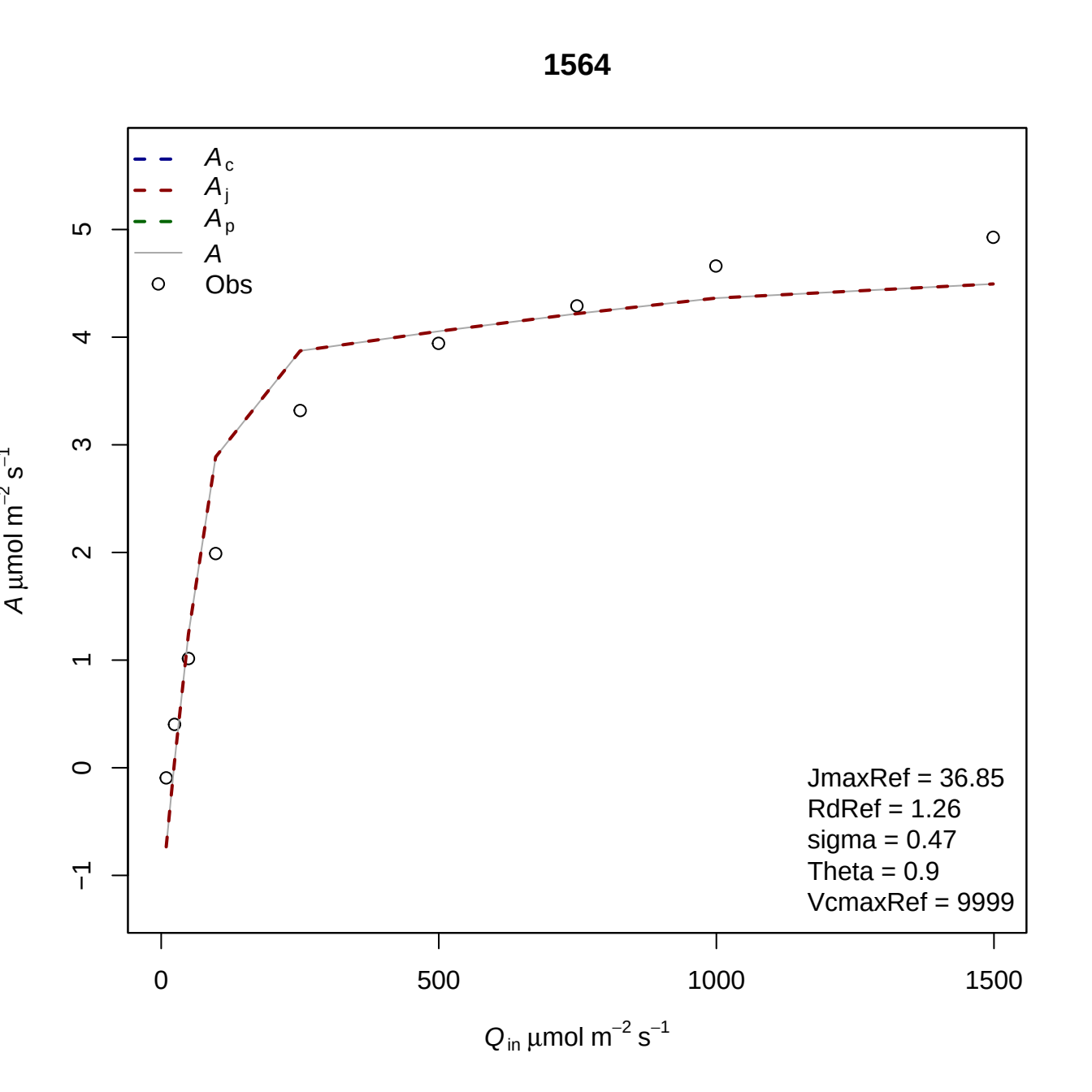

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 54.1
RdRef = 1.05
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

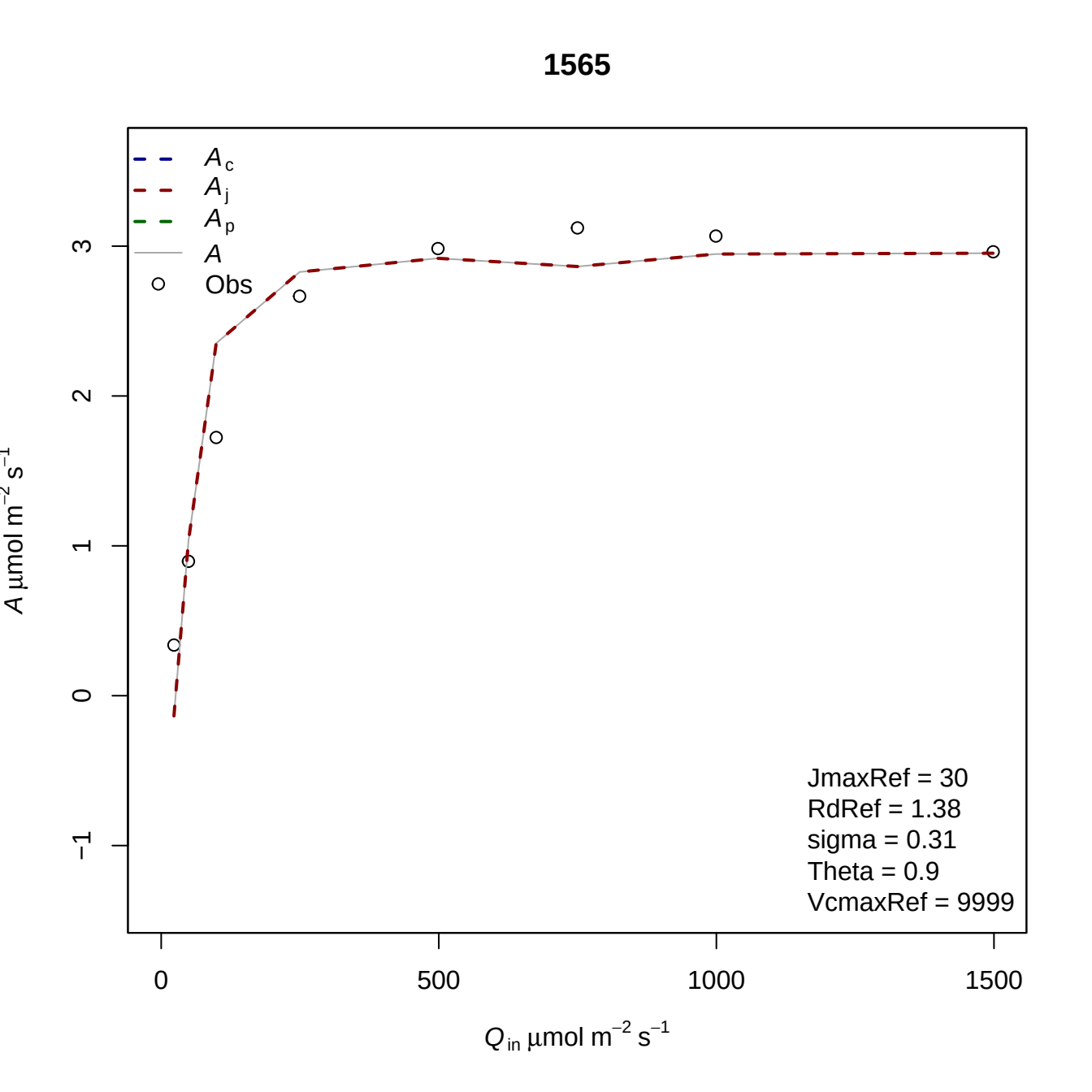

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 32.66
RdRef = 1.39
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
4
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

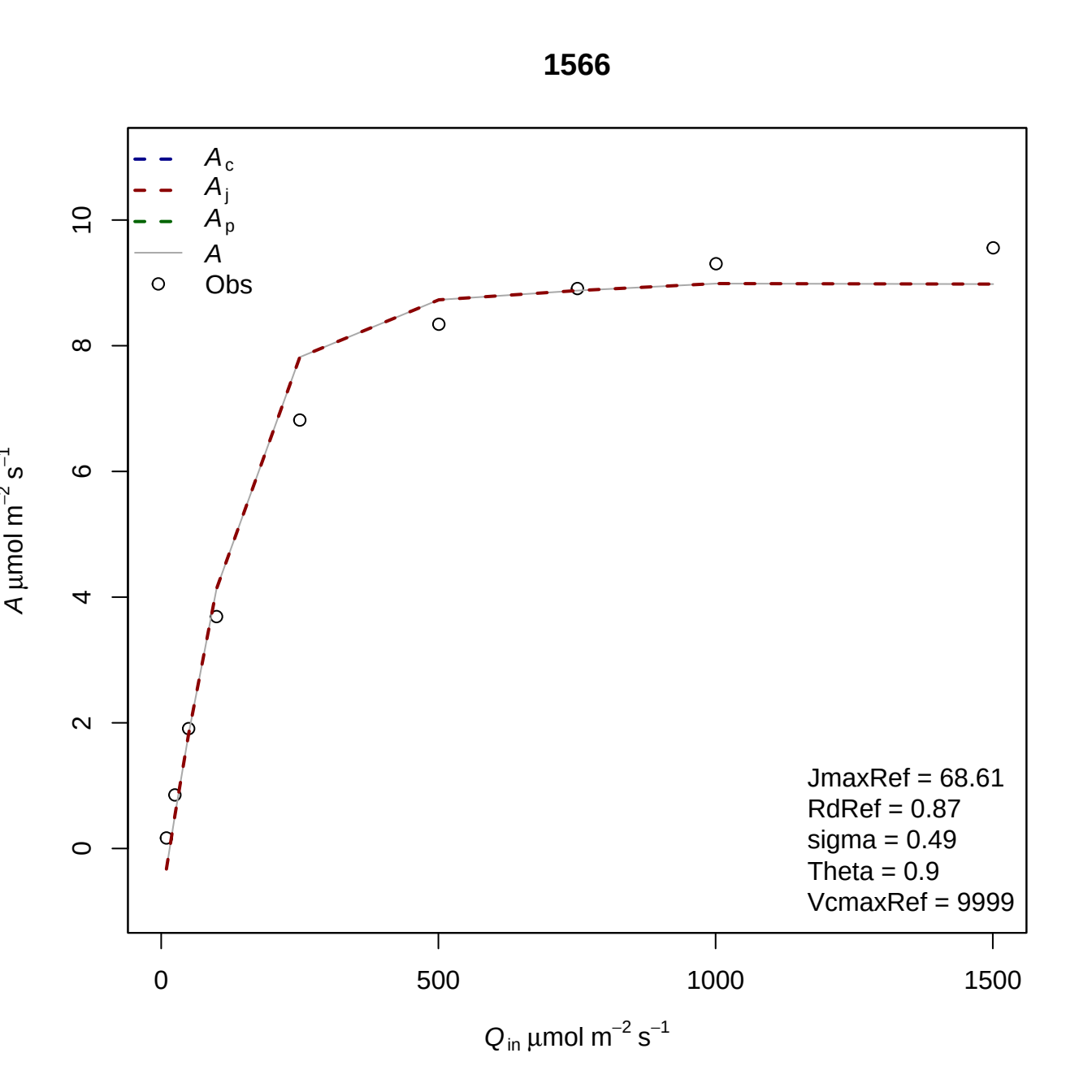

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 36.42
RdRef = 1.07
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

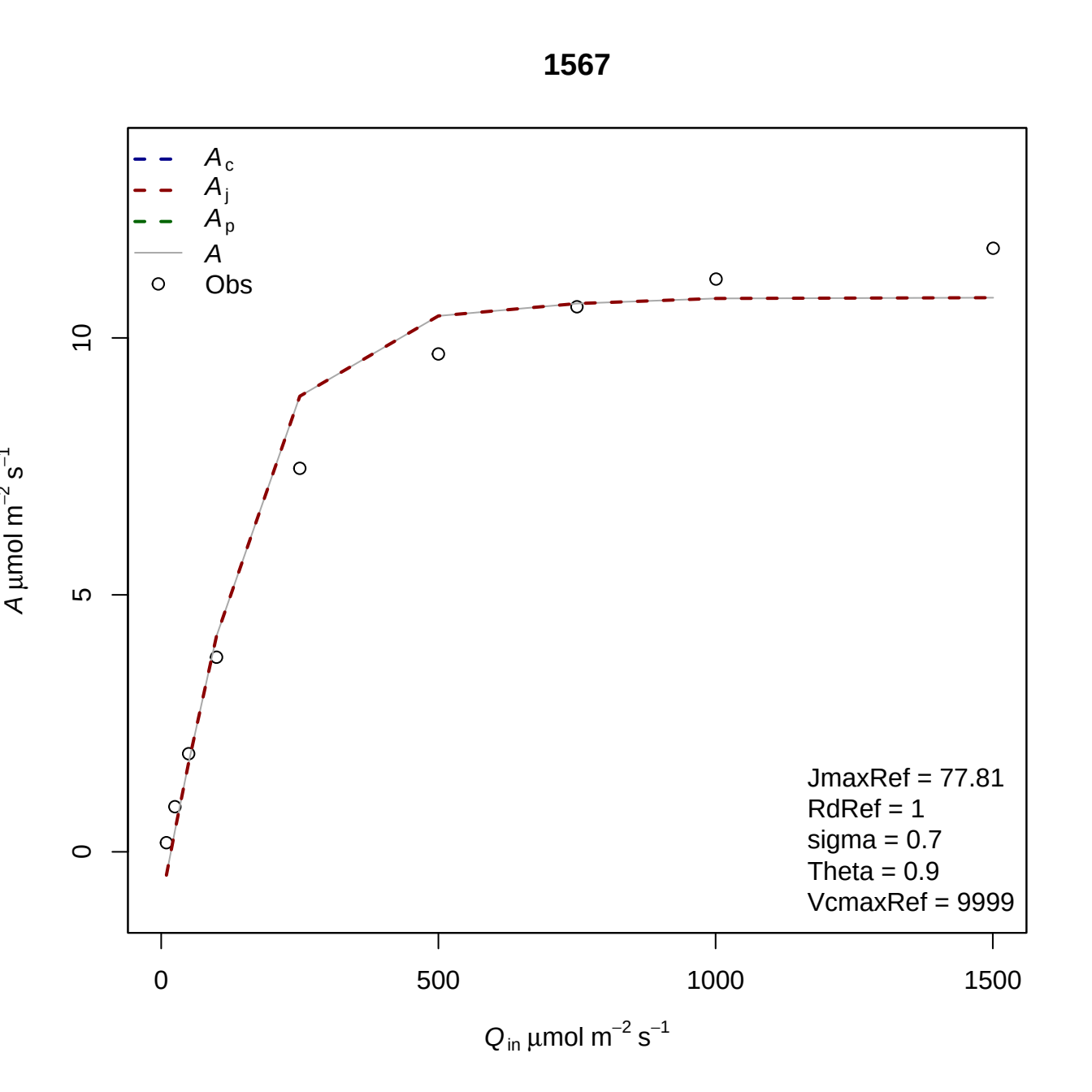

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 23.66
RdRef = 1.43
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

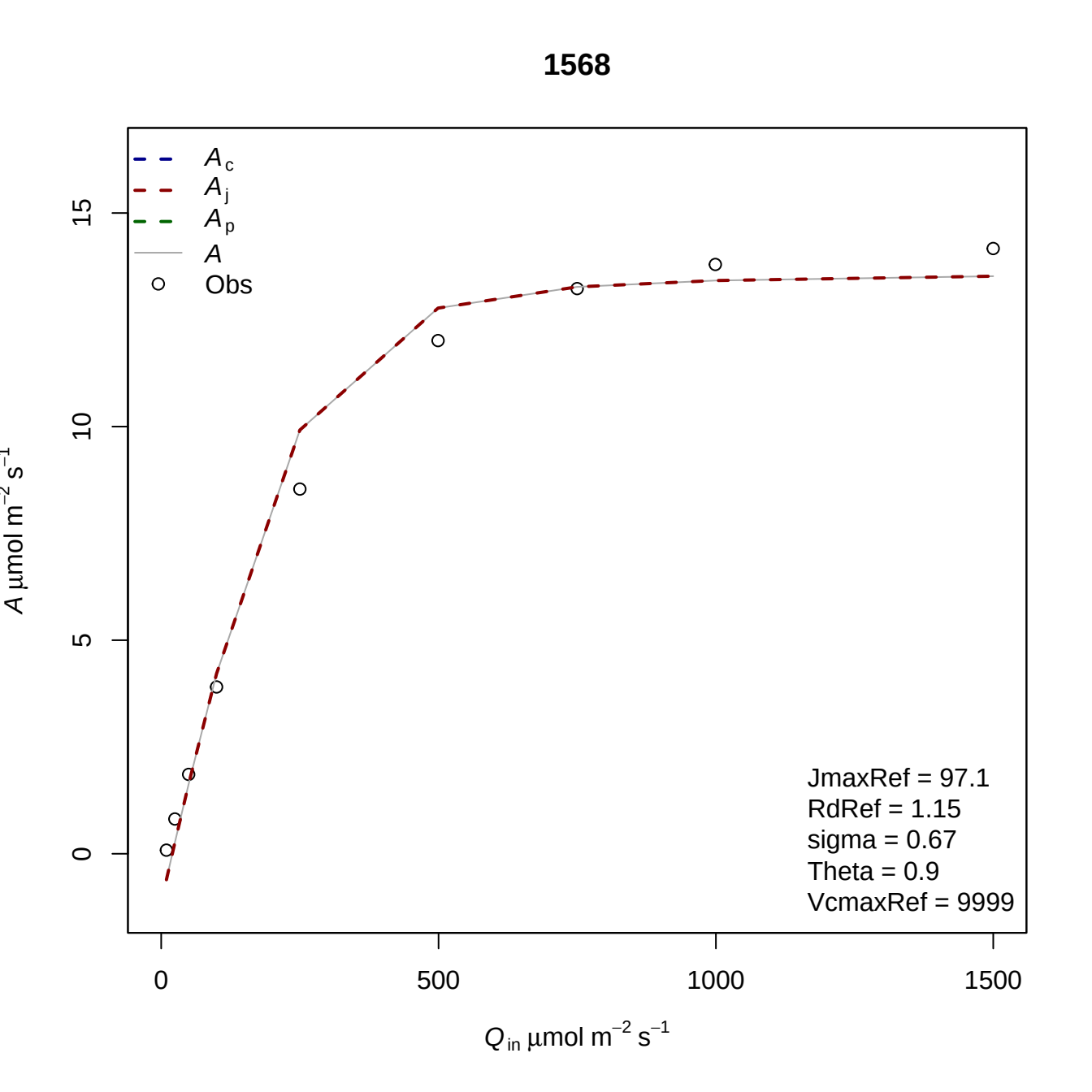

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 65.56
RdRef = 1.03
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

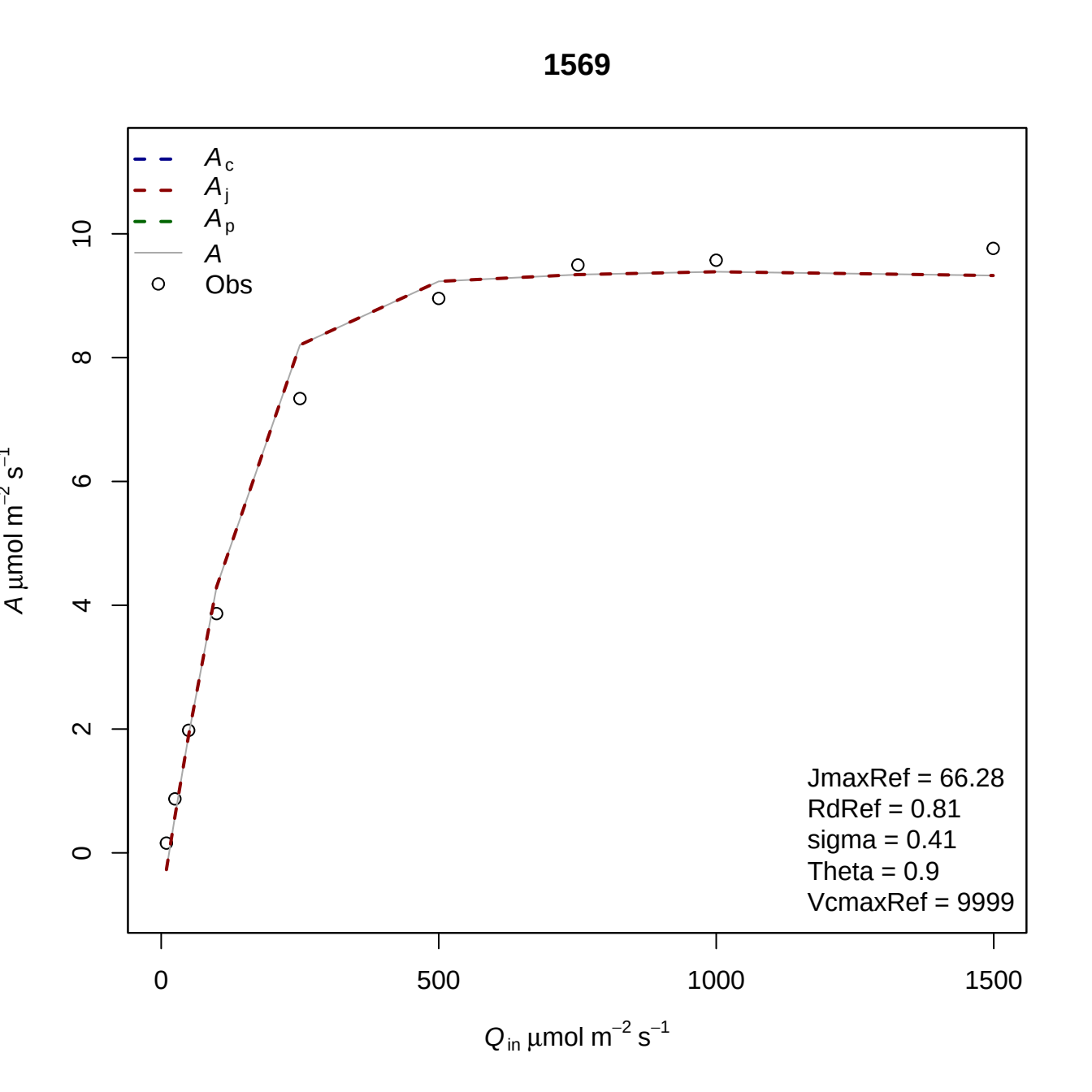

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 36.85
RdRef = 1.26
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

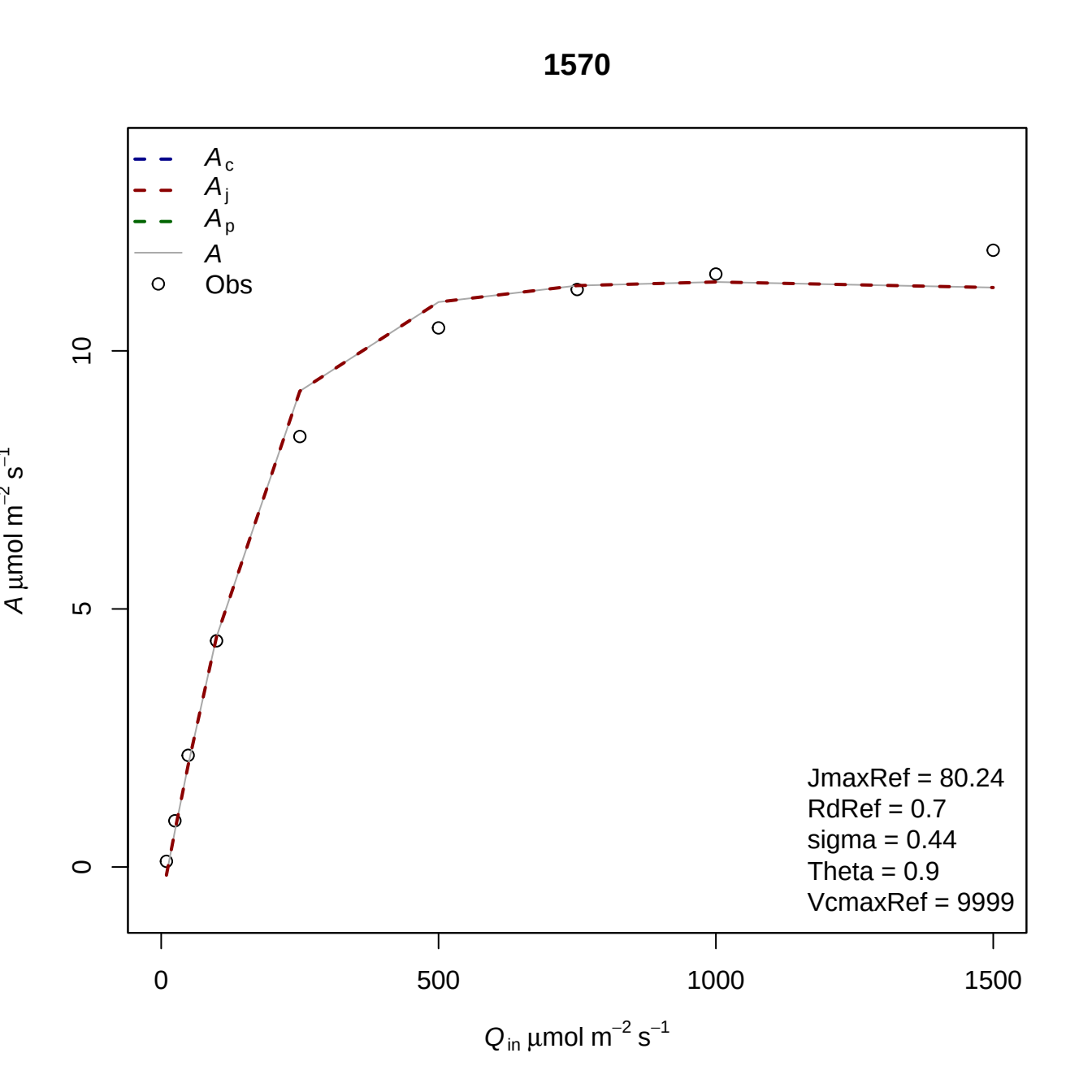

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 30
RdRef = 1.38
sigma = 0.31
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

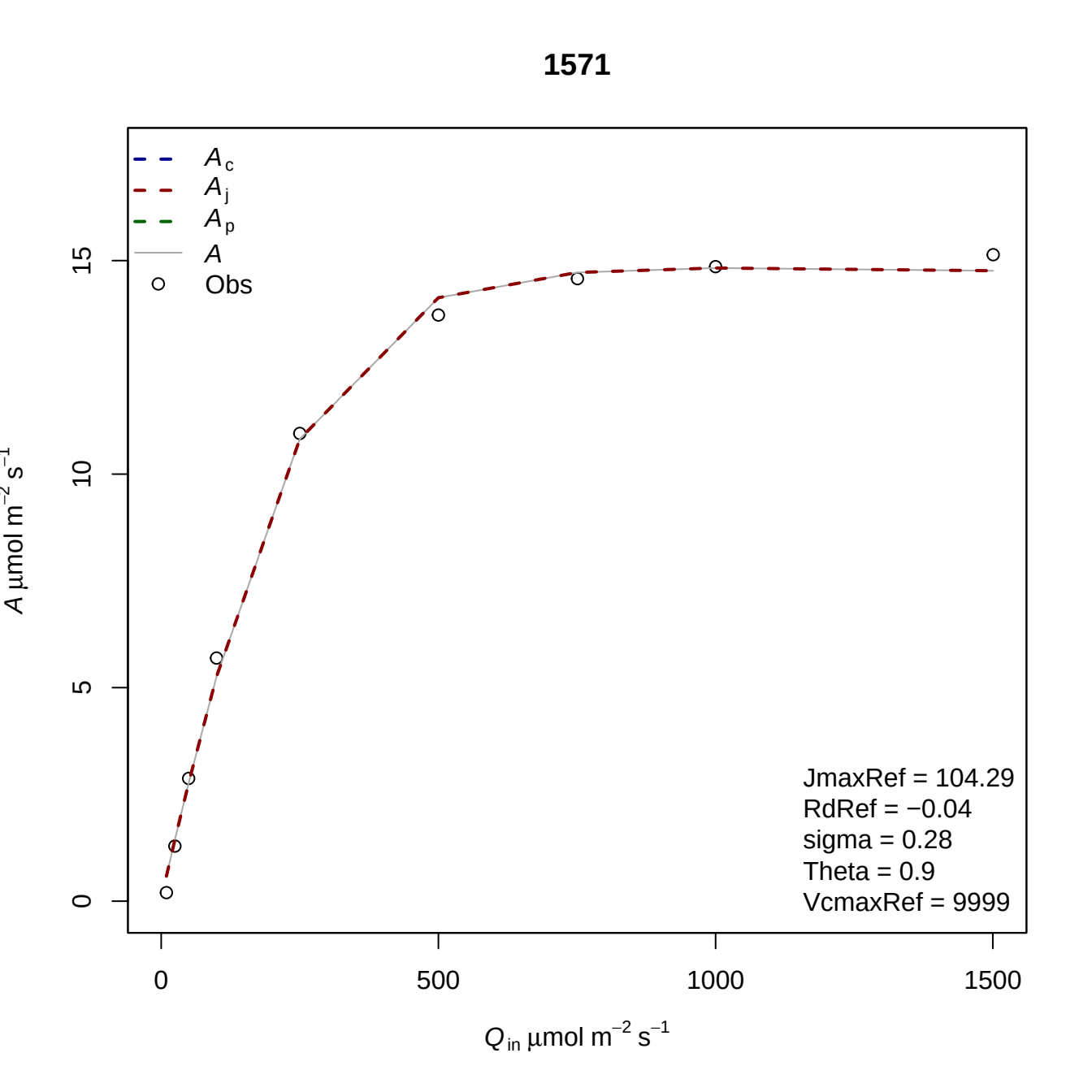

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 68.61
RdRef = 0.87
sigma = 0.49
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

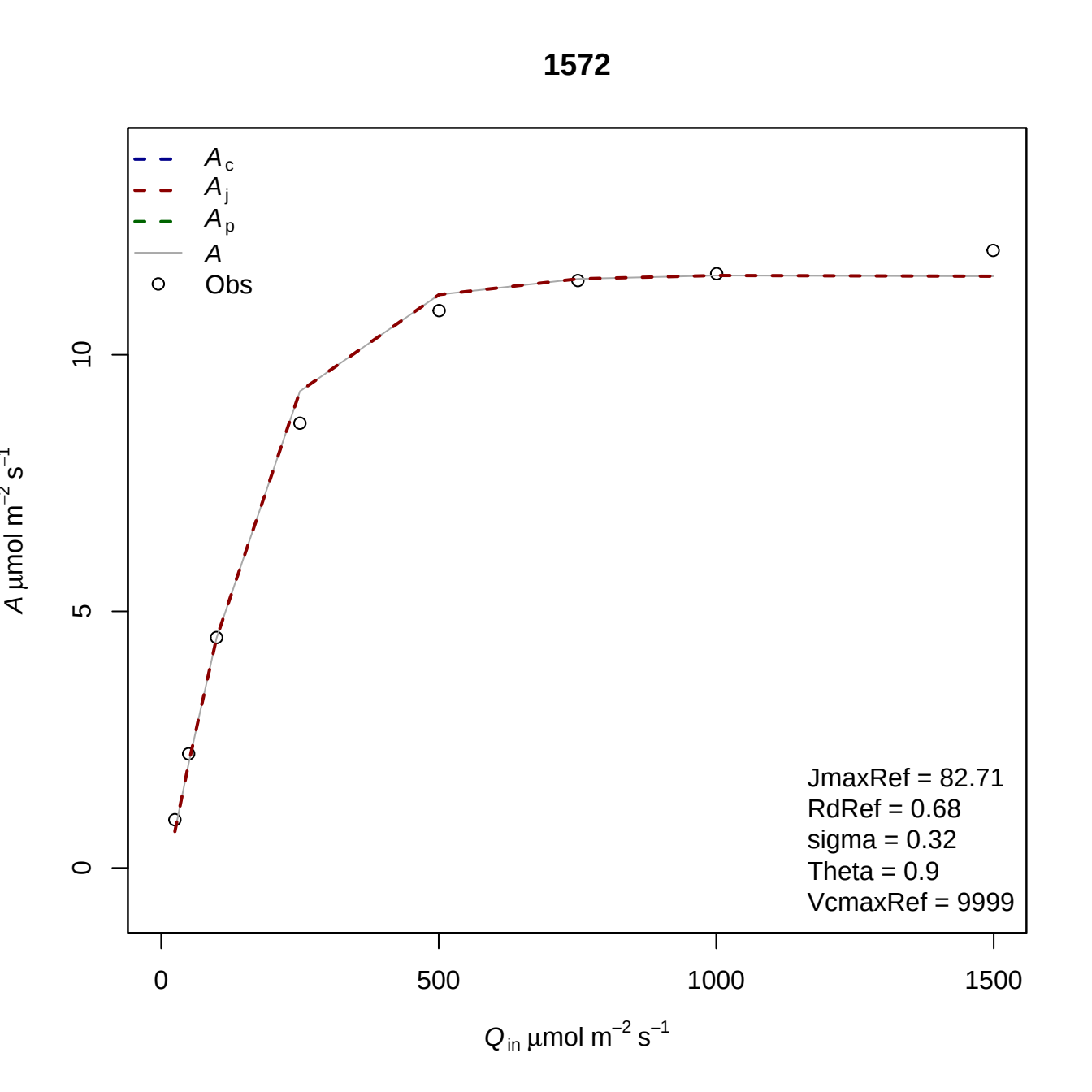

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 77.81
RdRef = 1
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

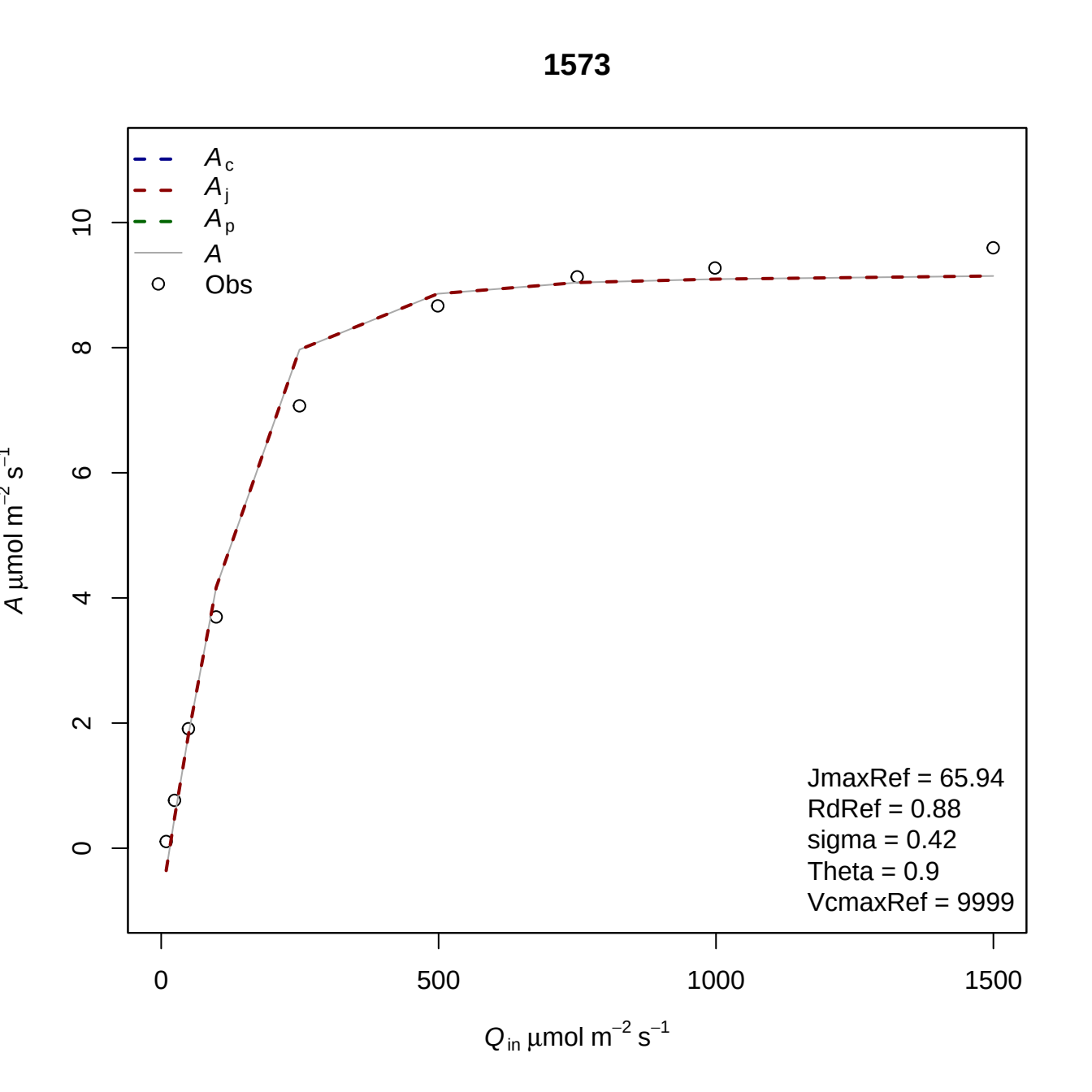

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 97.1
RdRef = 1.15
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

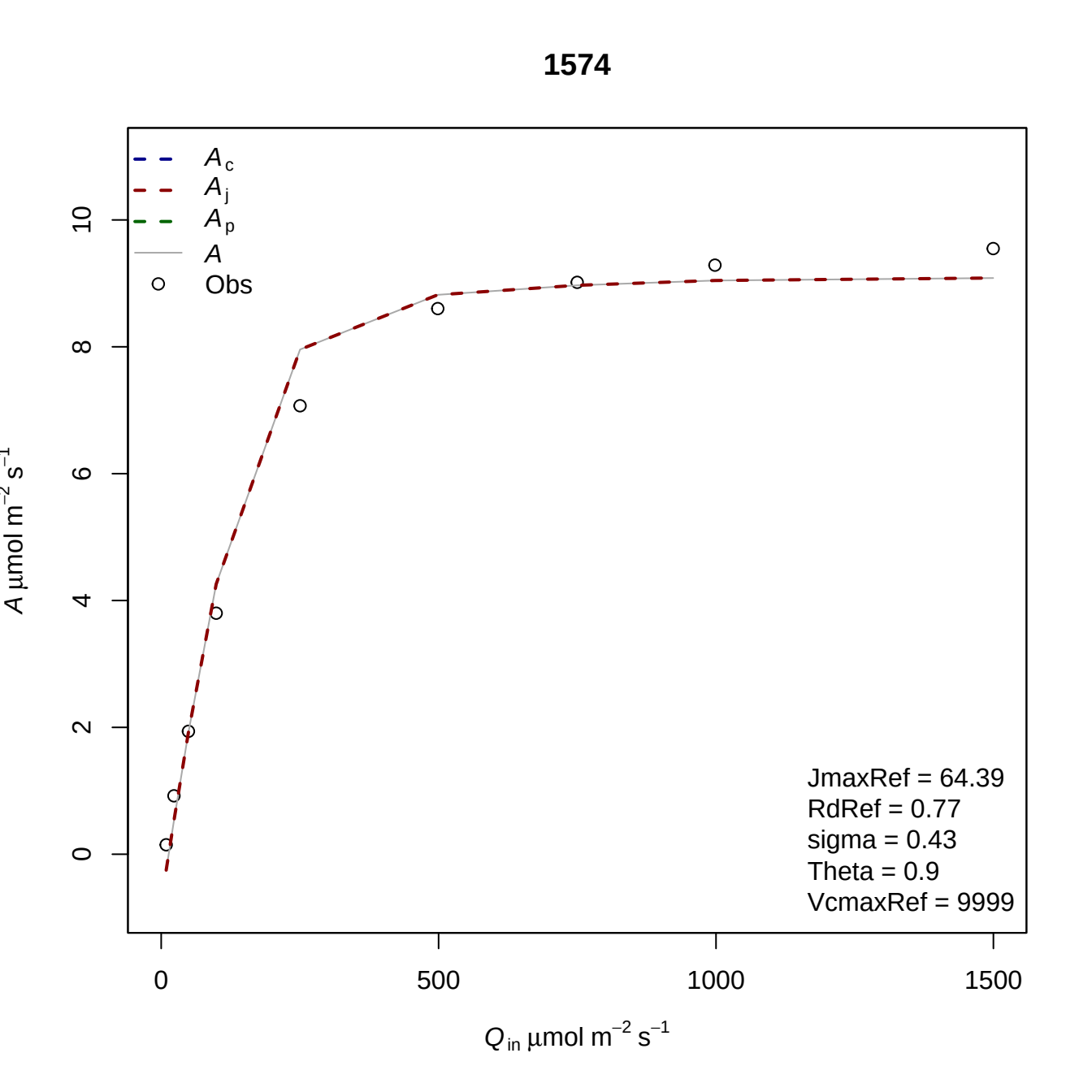

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 66.28
RdRef = 0.81
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

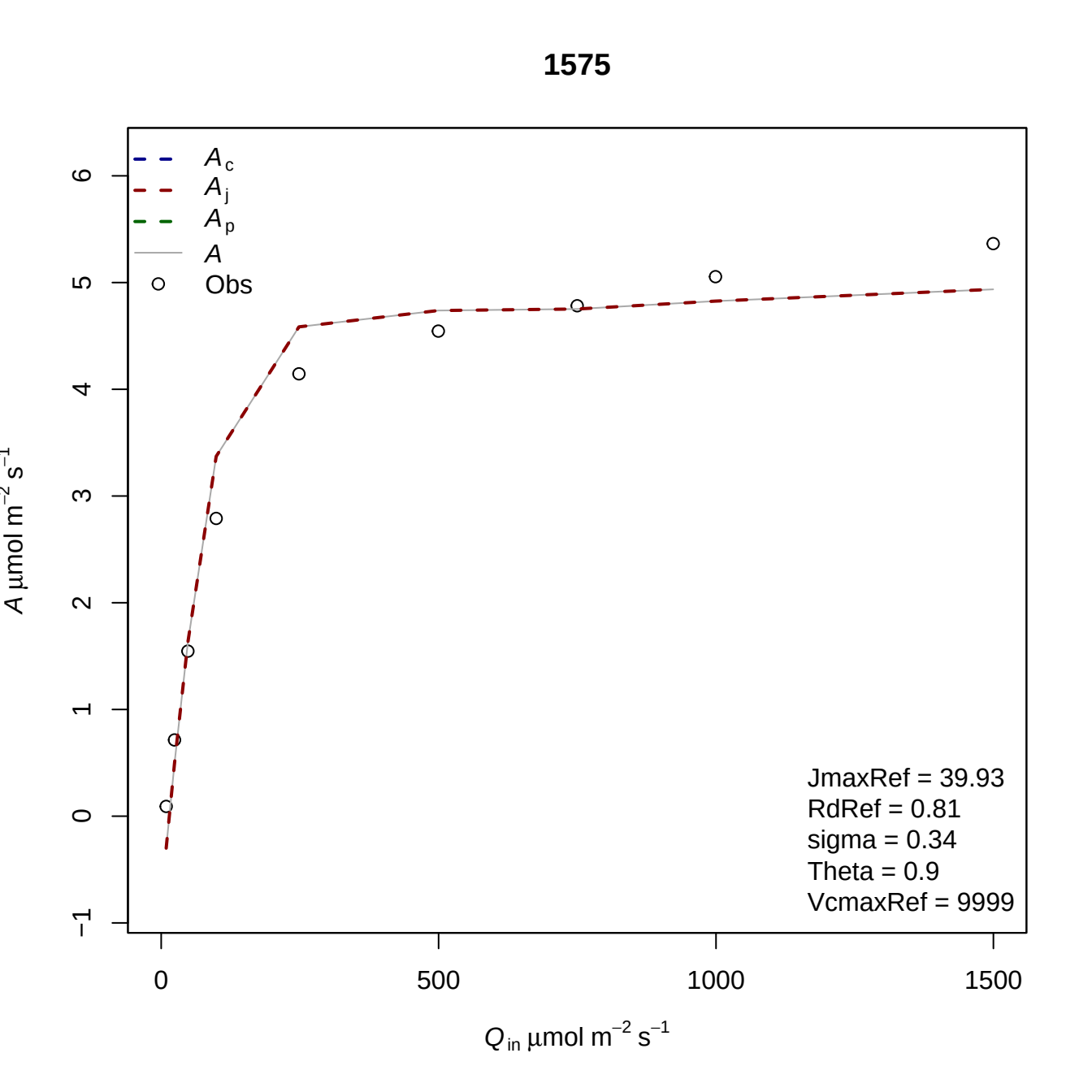

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 80.24
RdRef = 0.7
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

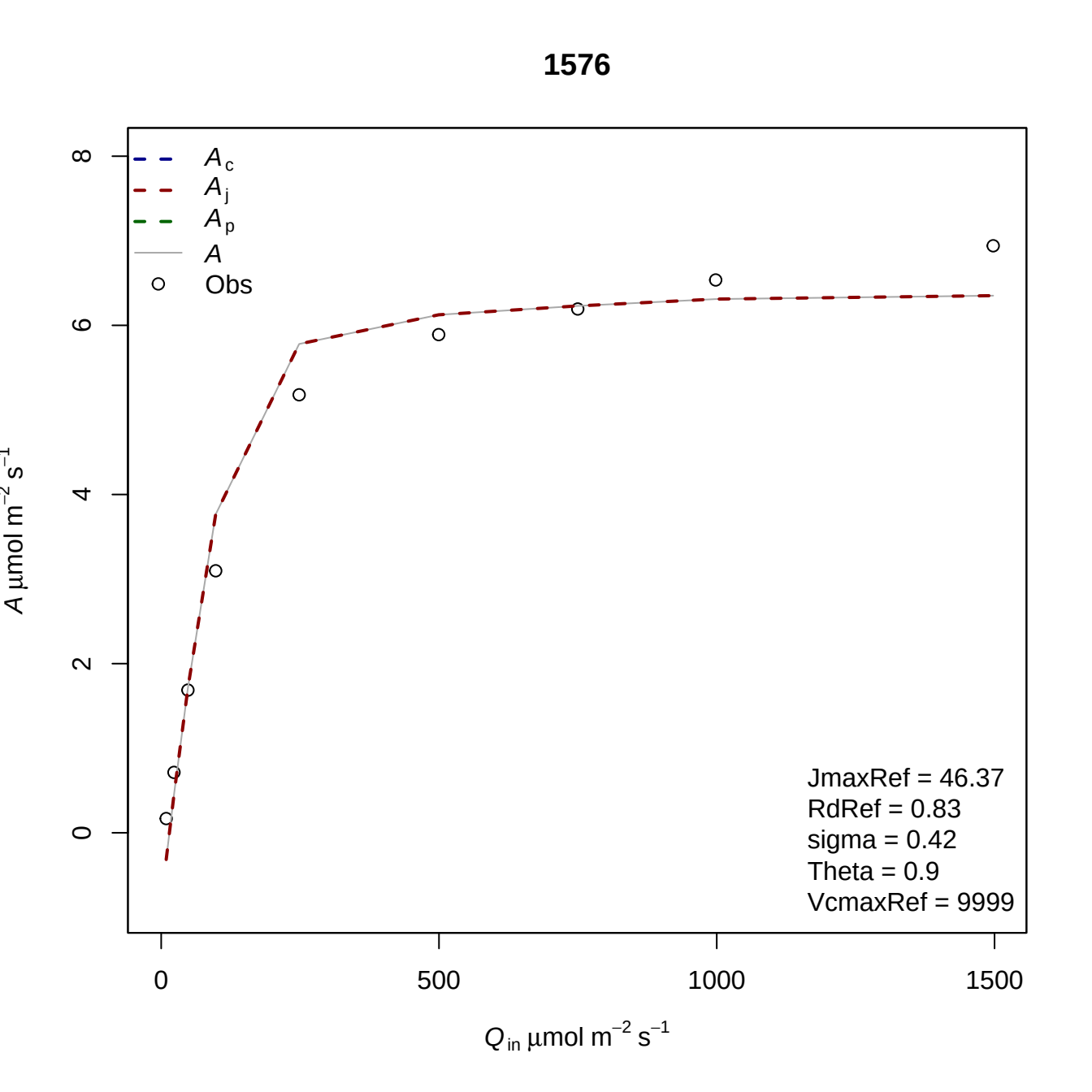

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 104.29
RdRef = −0.04
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

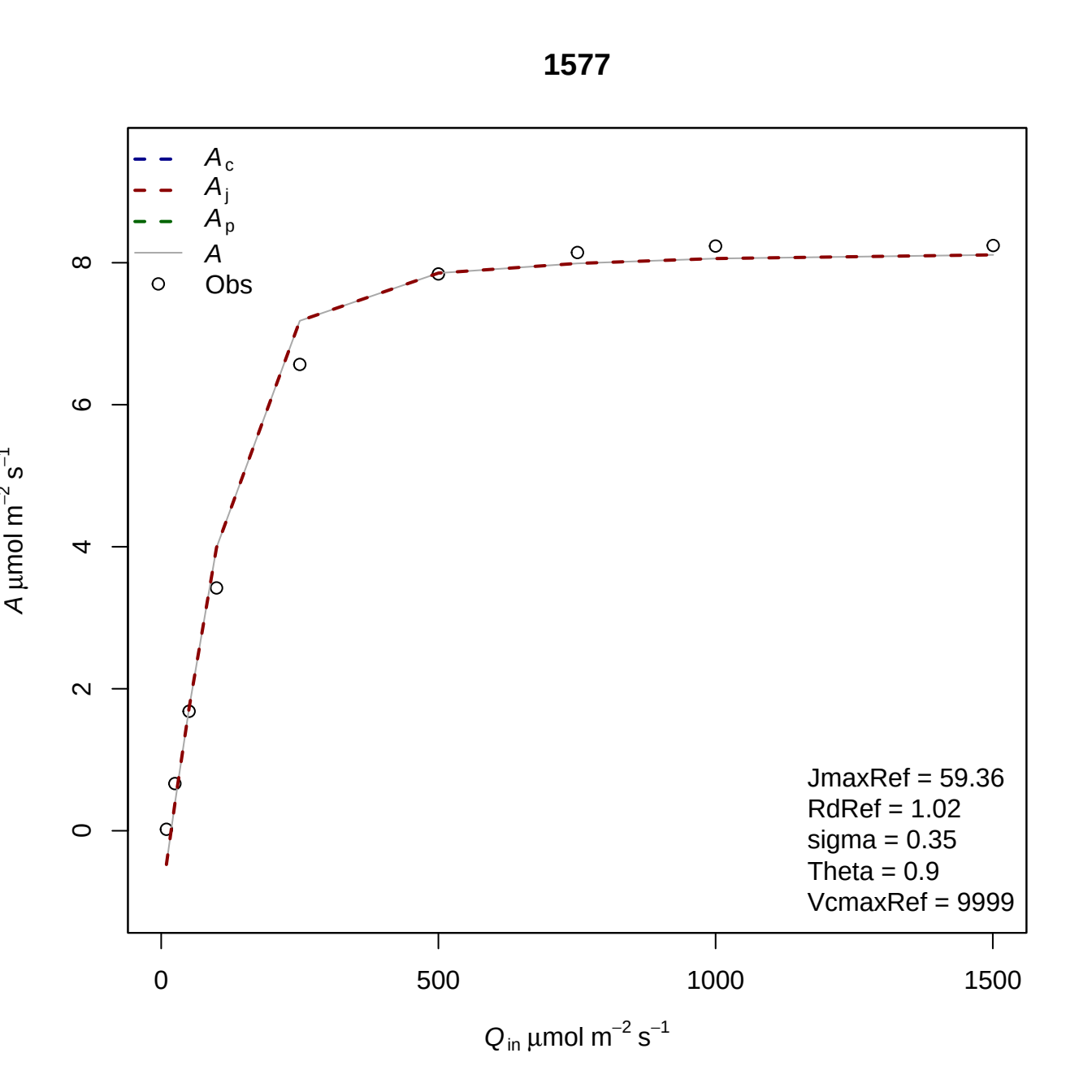

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 82.71
RdRef = 0.68
sigma = 0.32
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

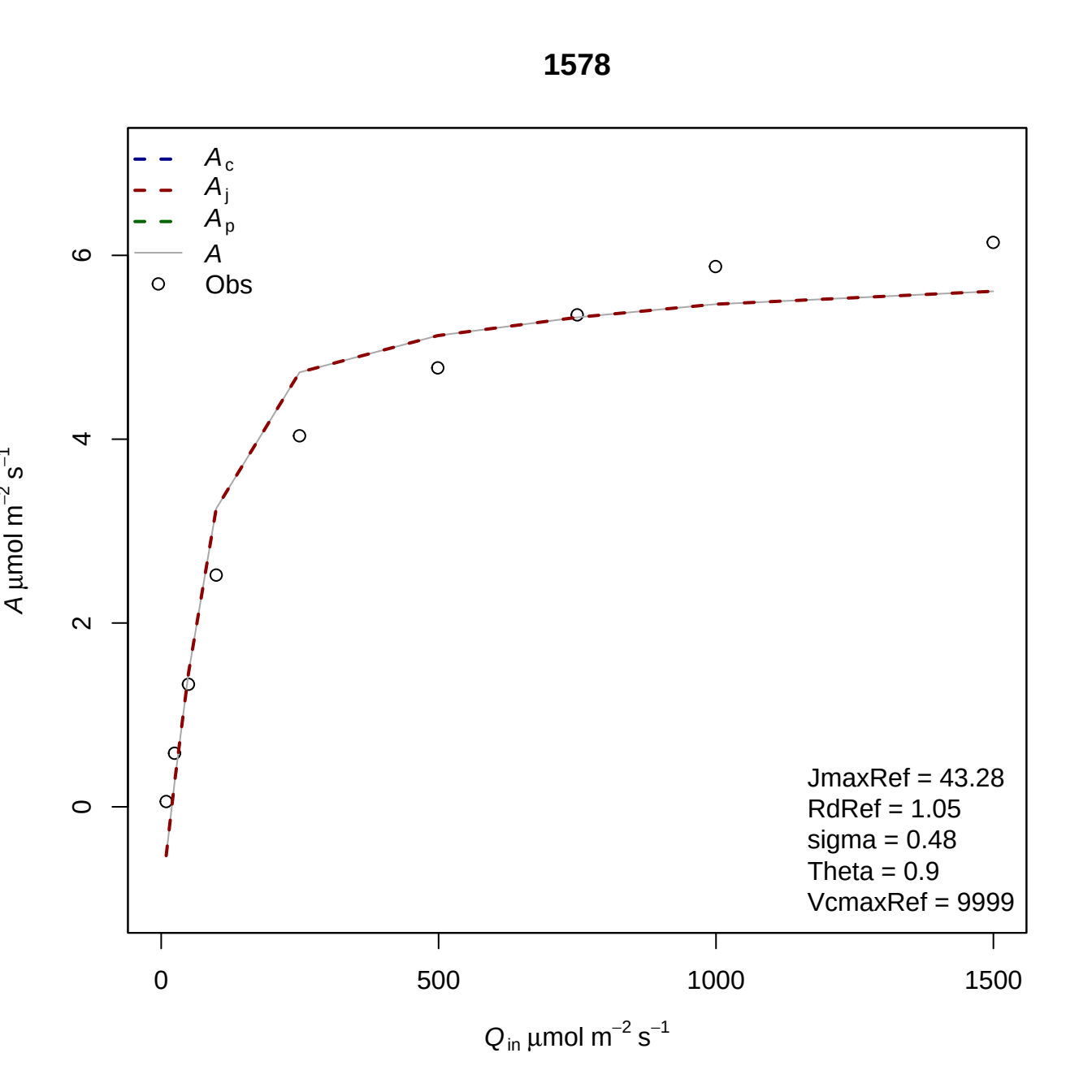

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 65.94
RdRef = 0.88
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

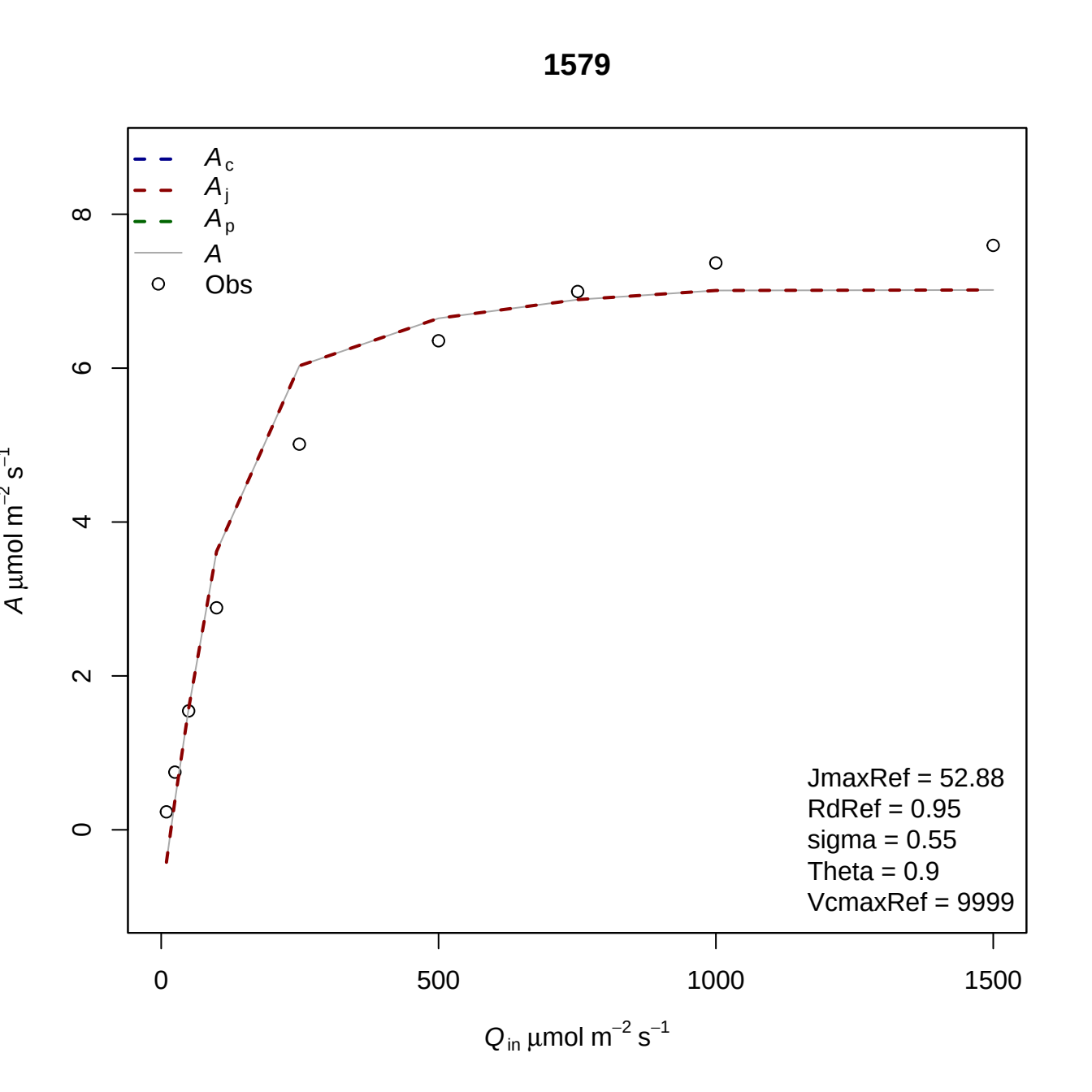

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 64.39
RdRef = 0.77
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

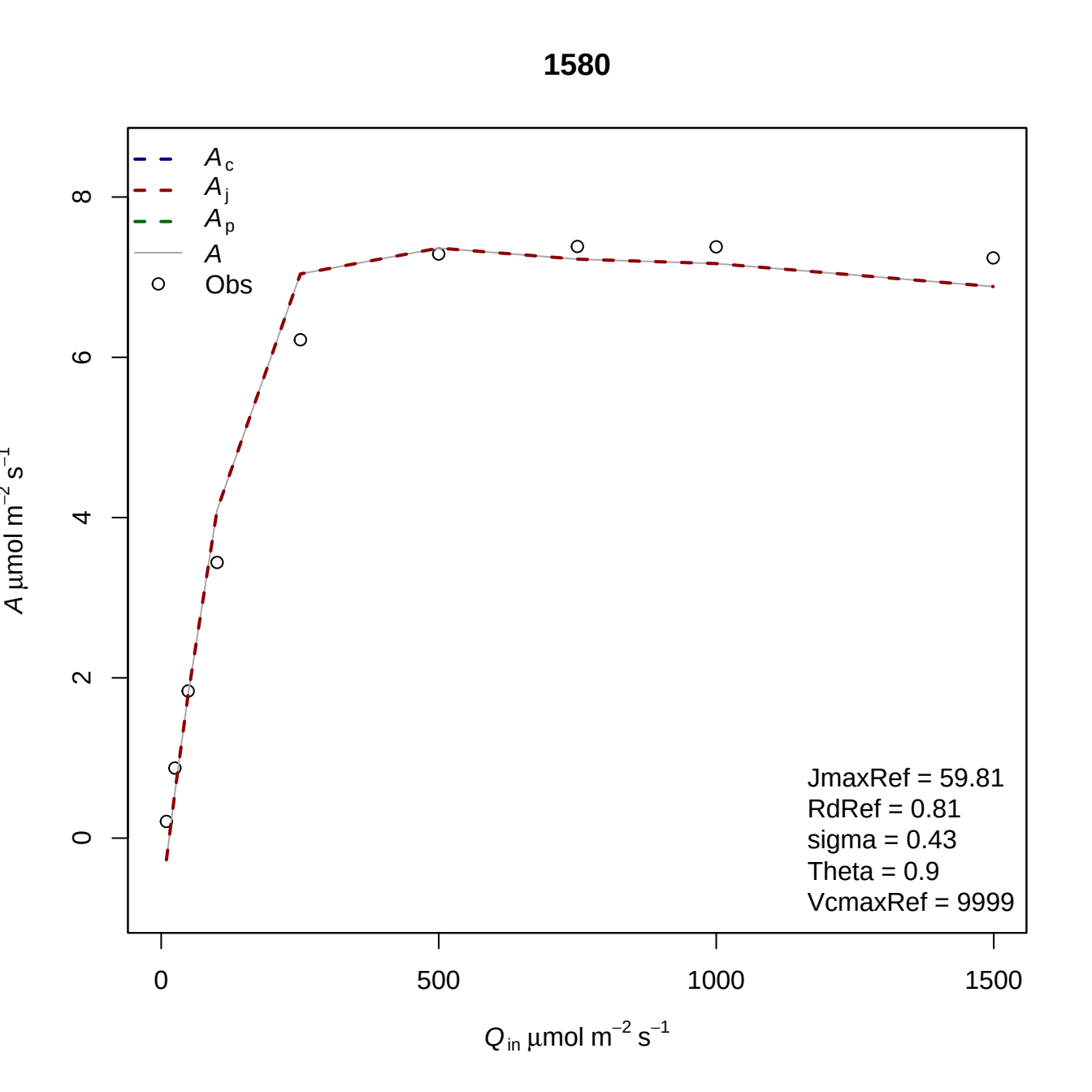

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 39.93
RdRef = 0.81
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

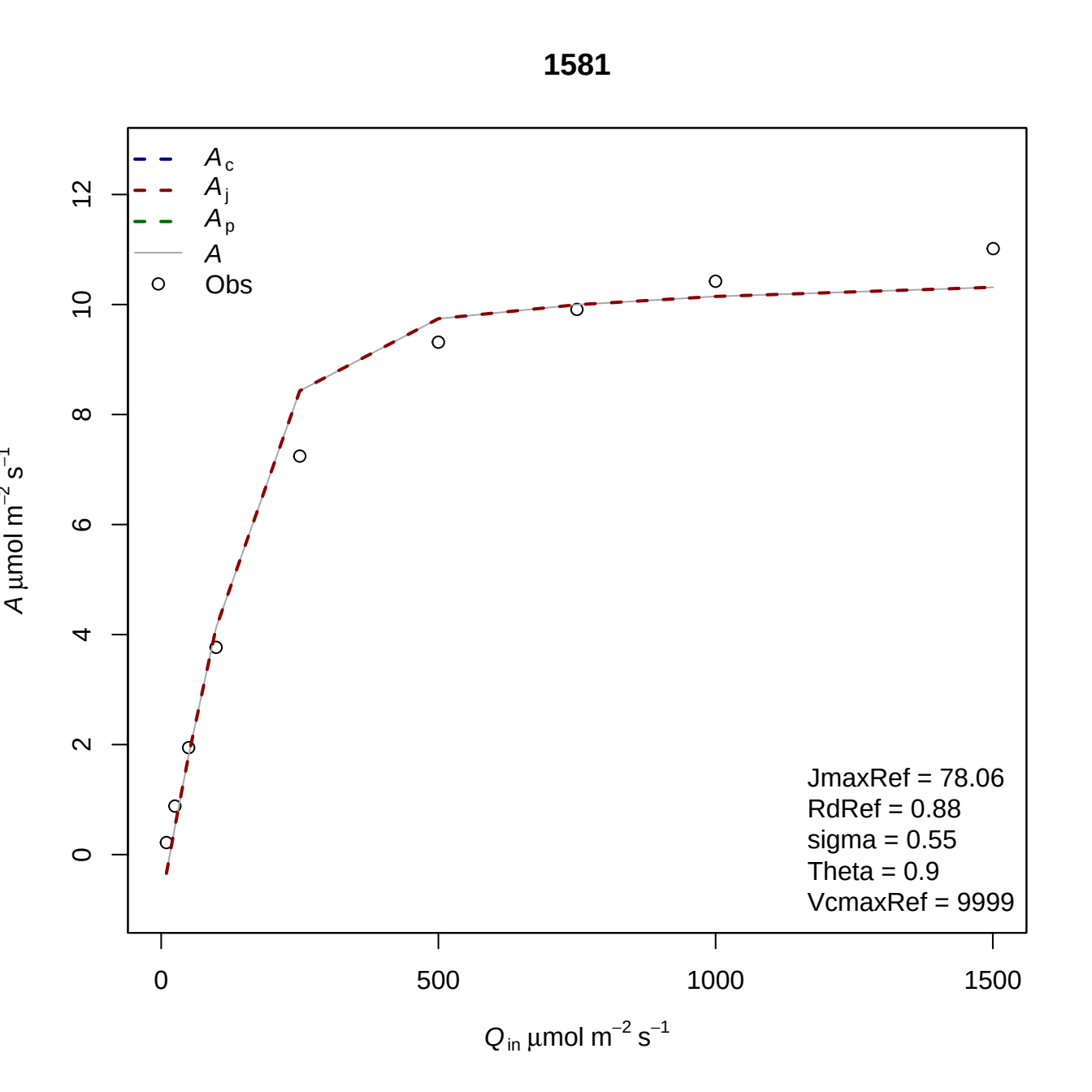

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 46.37
RdRef = 0.83
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

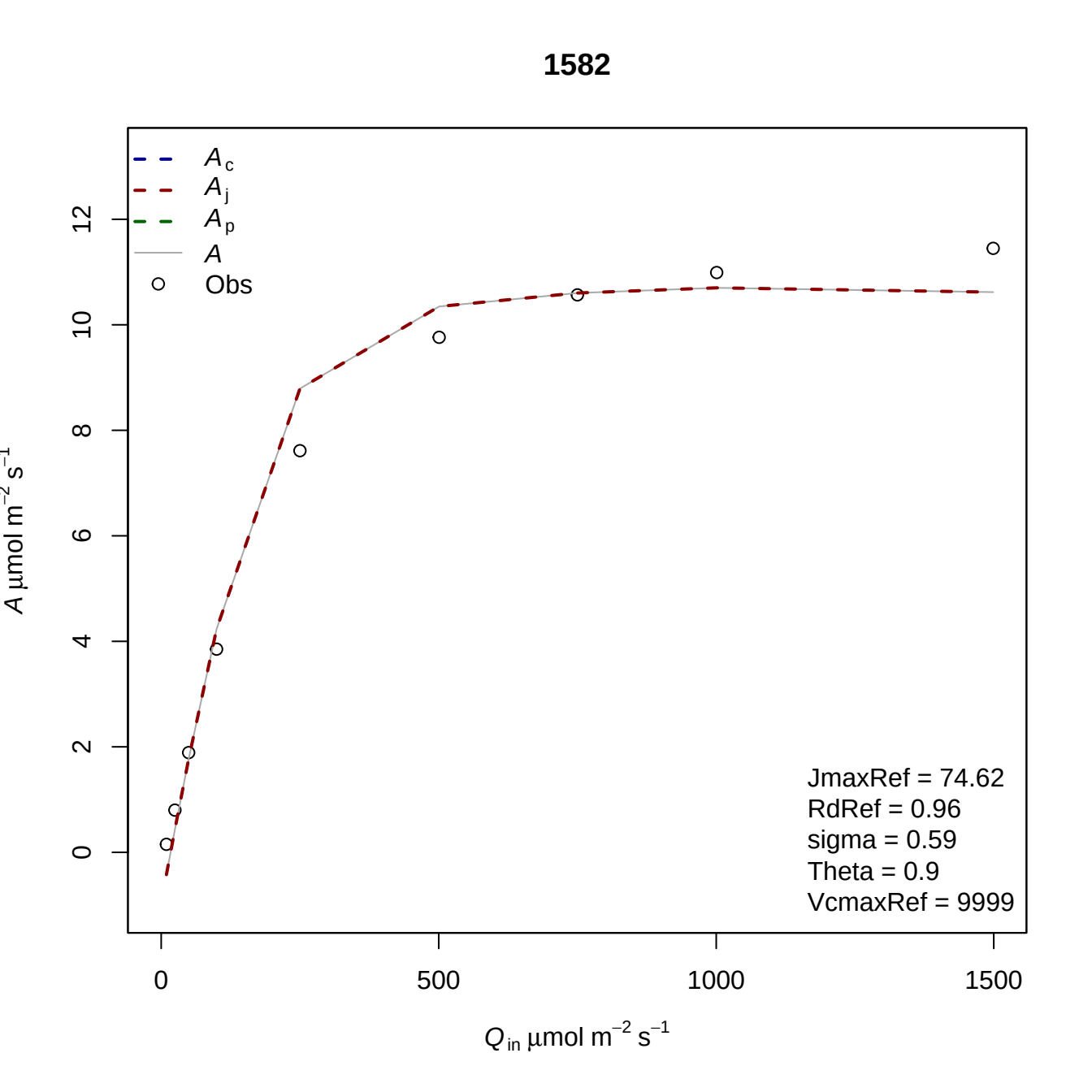

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 59.36
RdRef = 1.02
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

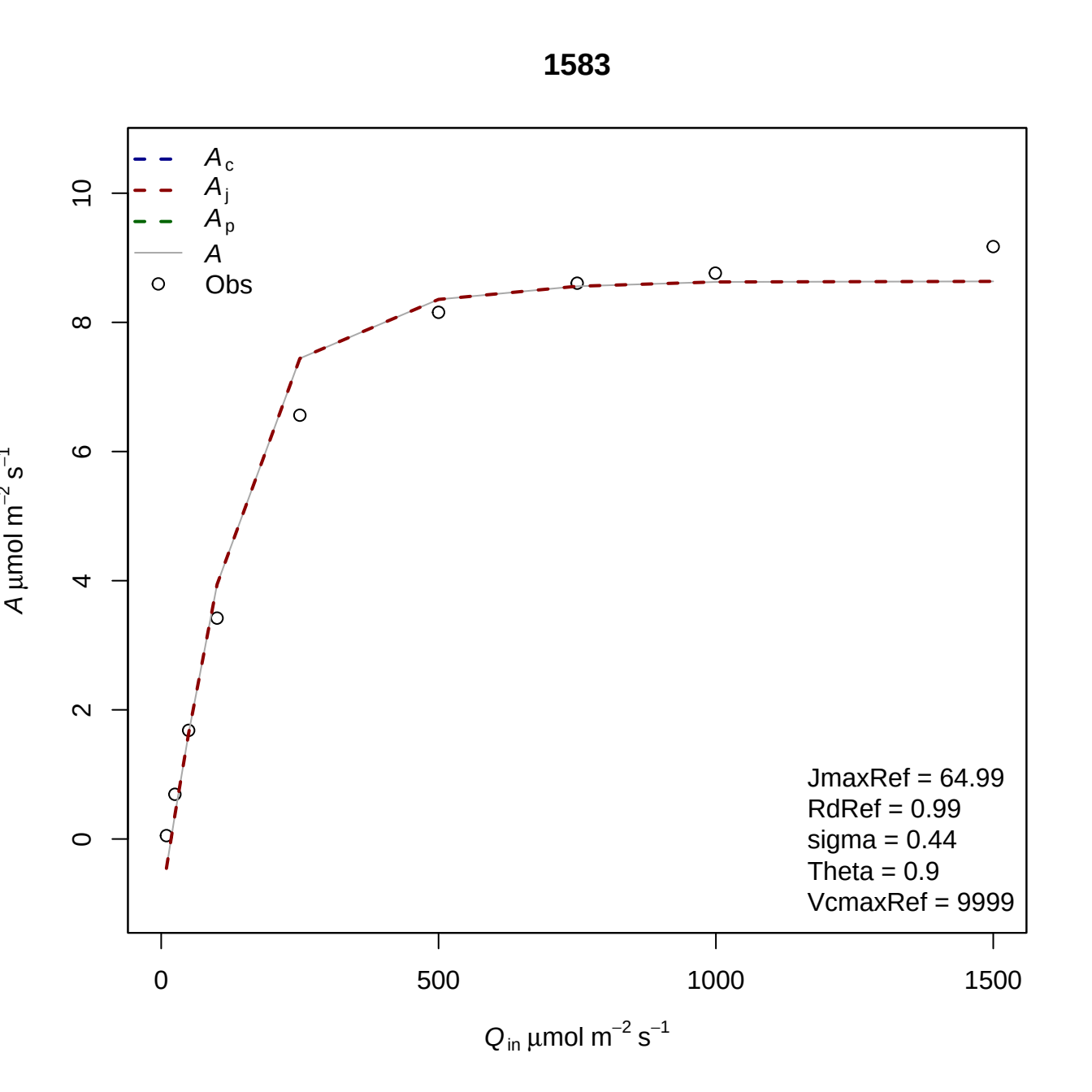

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 43.28
RdRef = 1.05
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

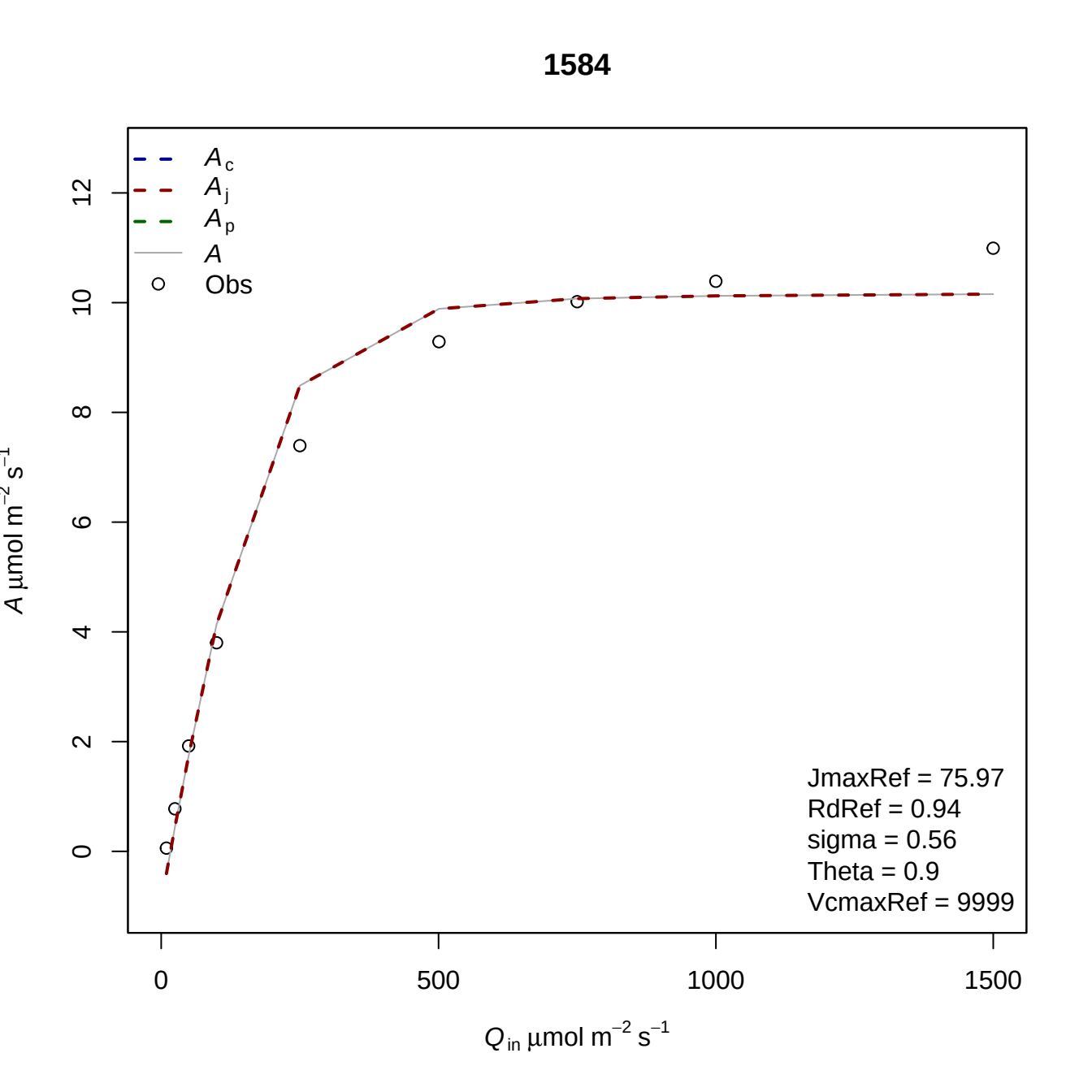

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 52.88
RdRef = 0.95
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

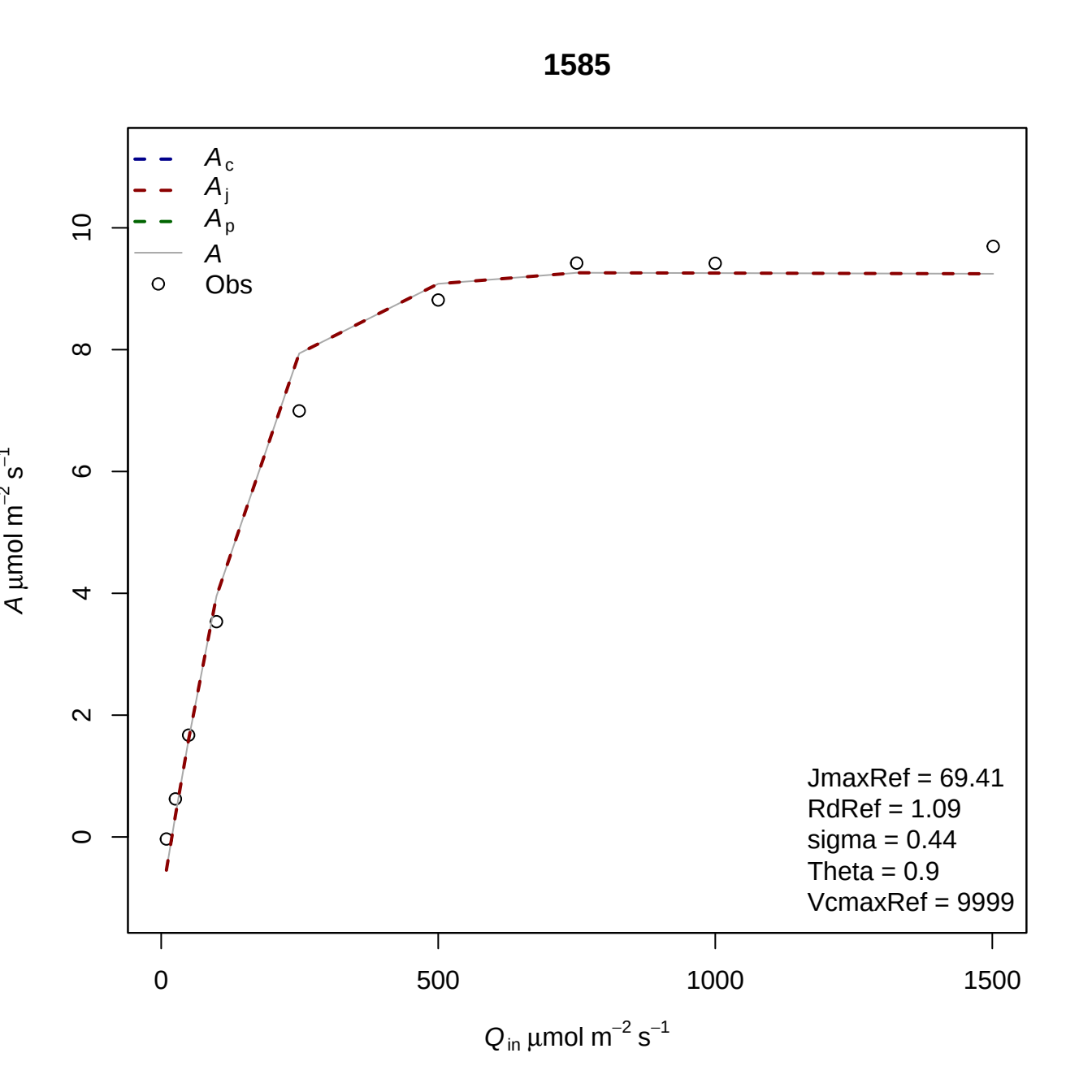

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 59.81
RdRef = 0.81
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

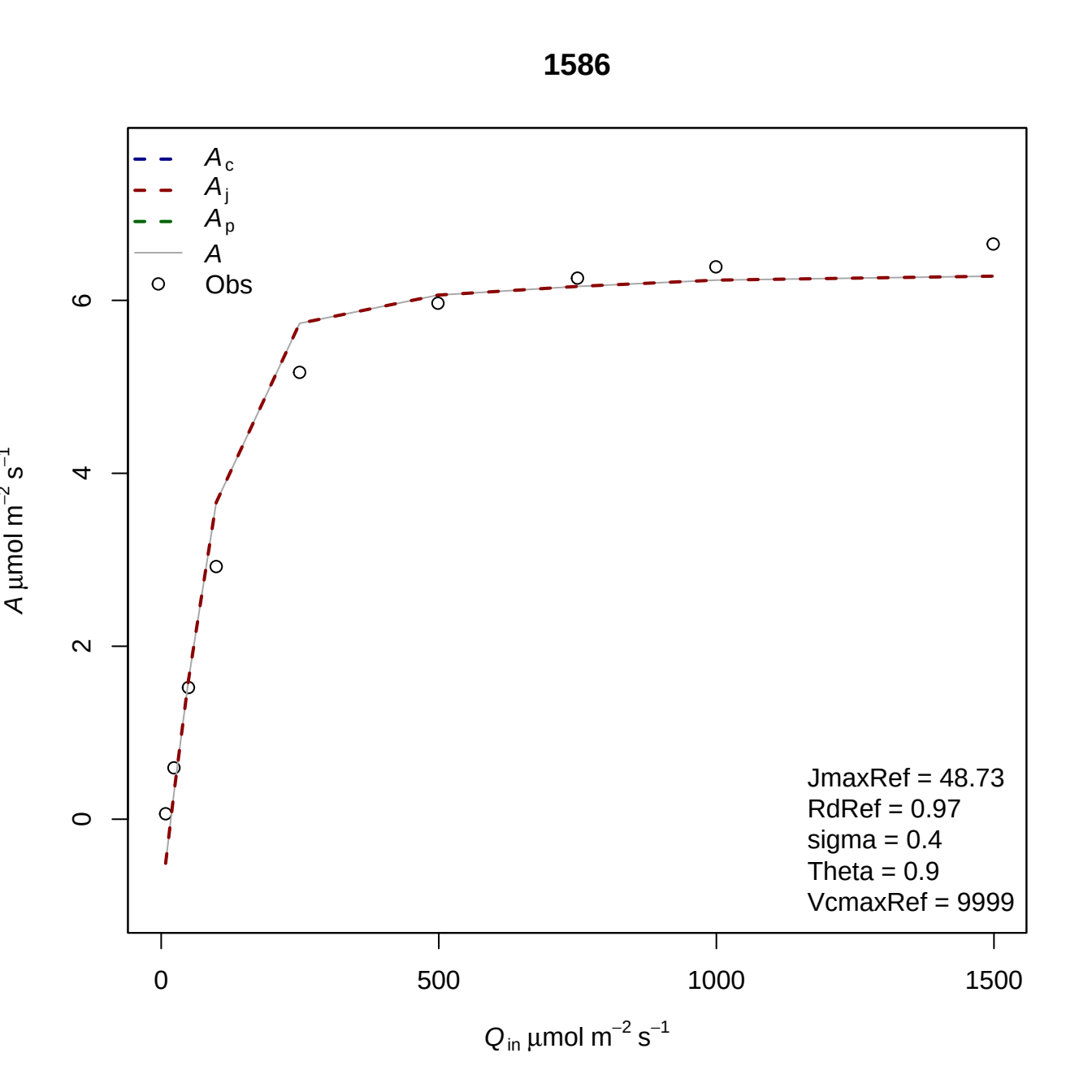

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 78.06
RdRef = 0.88
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

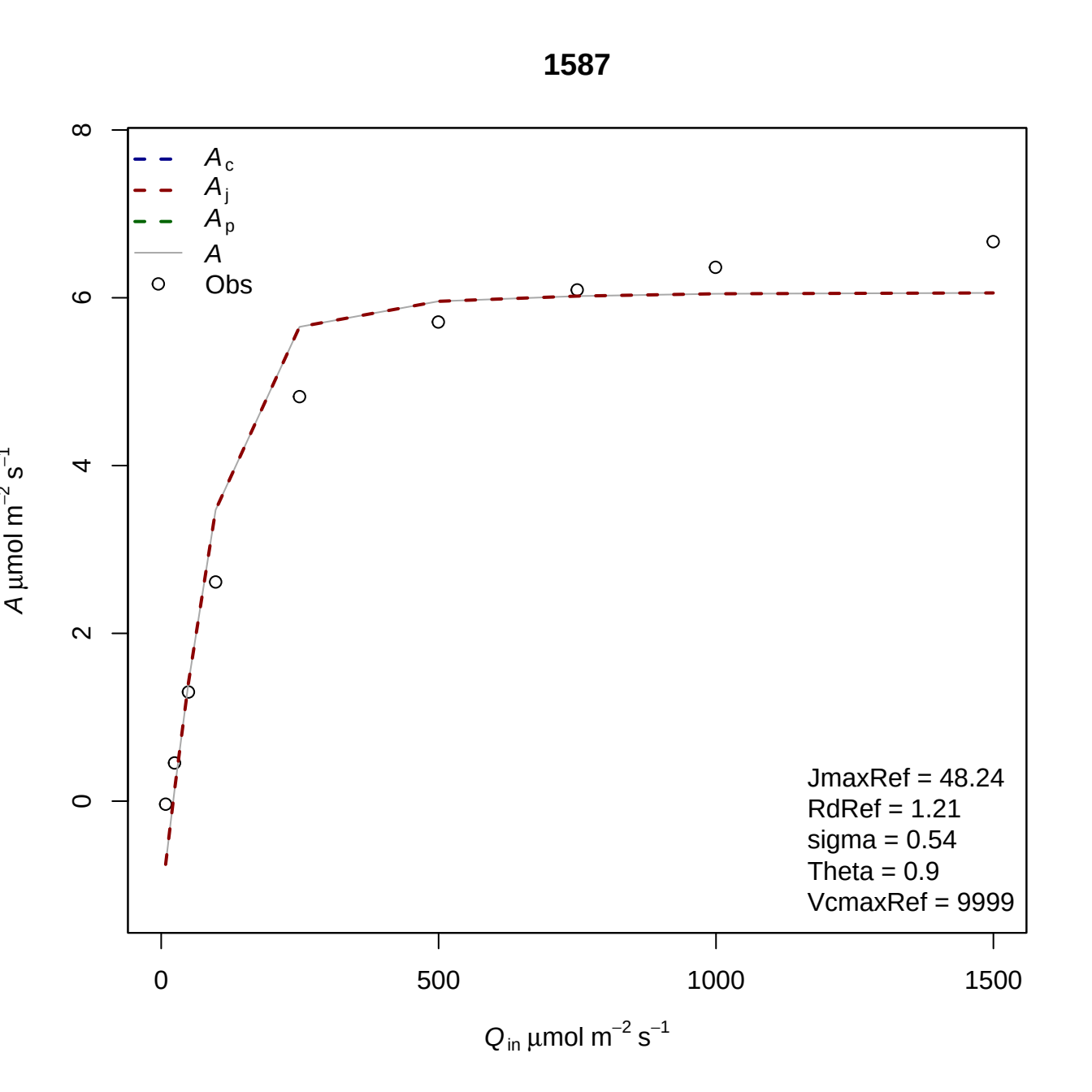

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 74.62
RdRef = 0.96
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

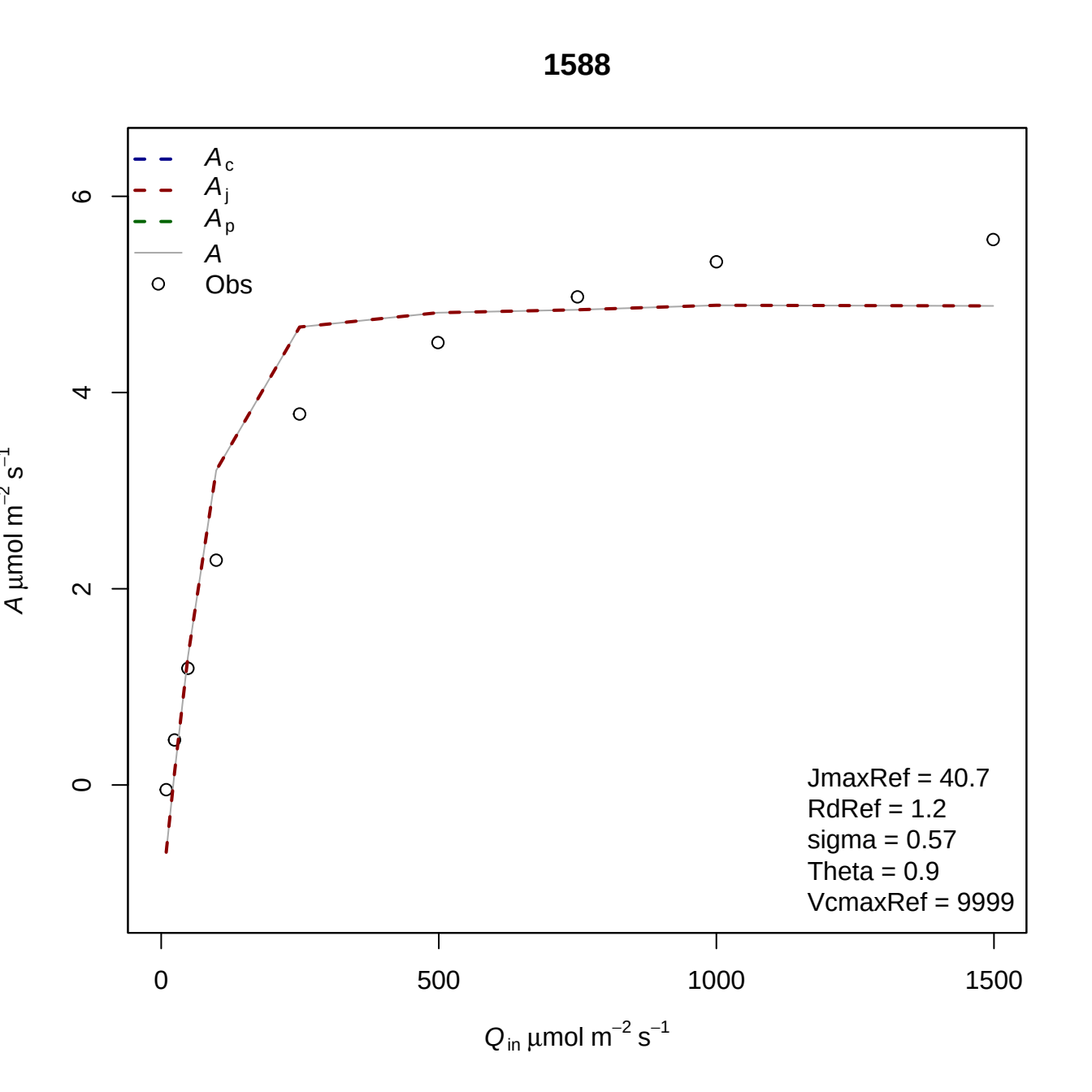

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 64.99
RdRef = 0.99
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

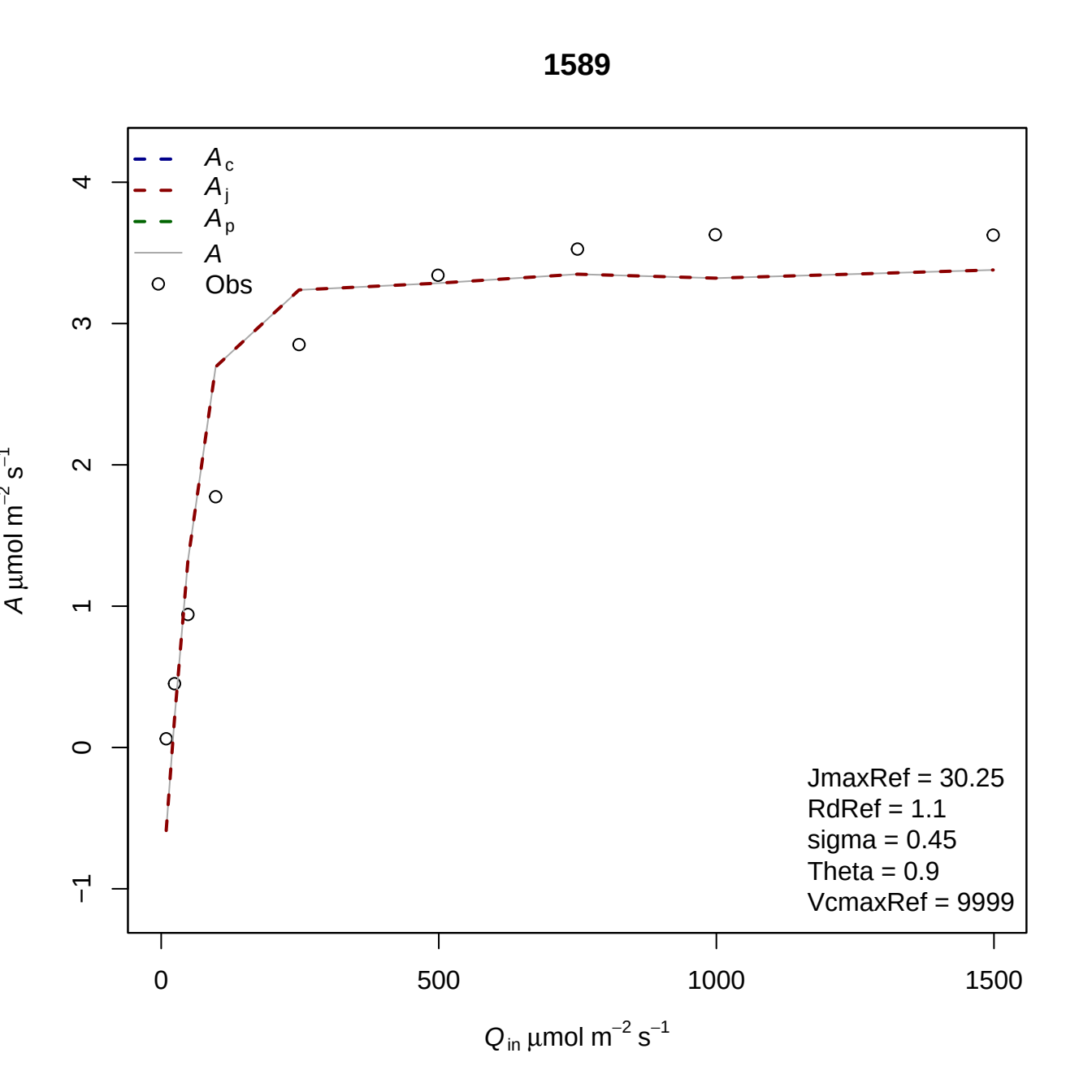

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 75.97
RdRef = 0.94
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

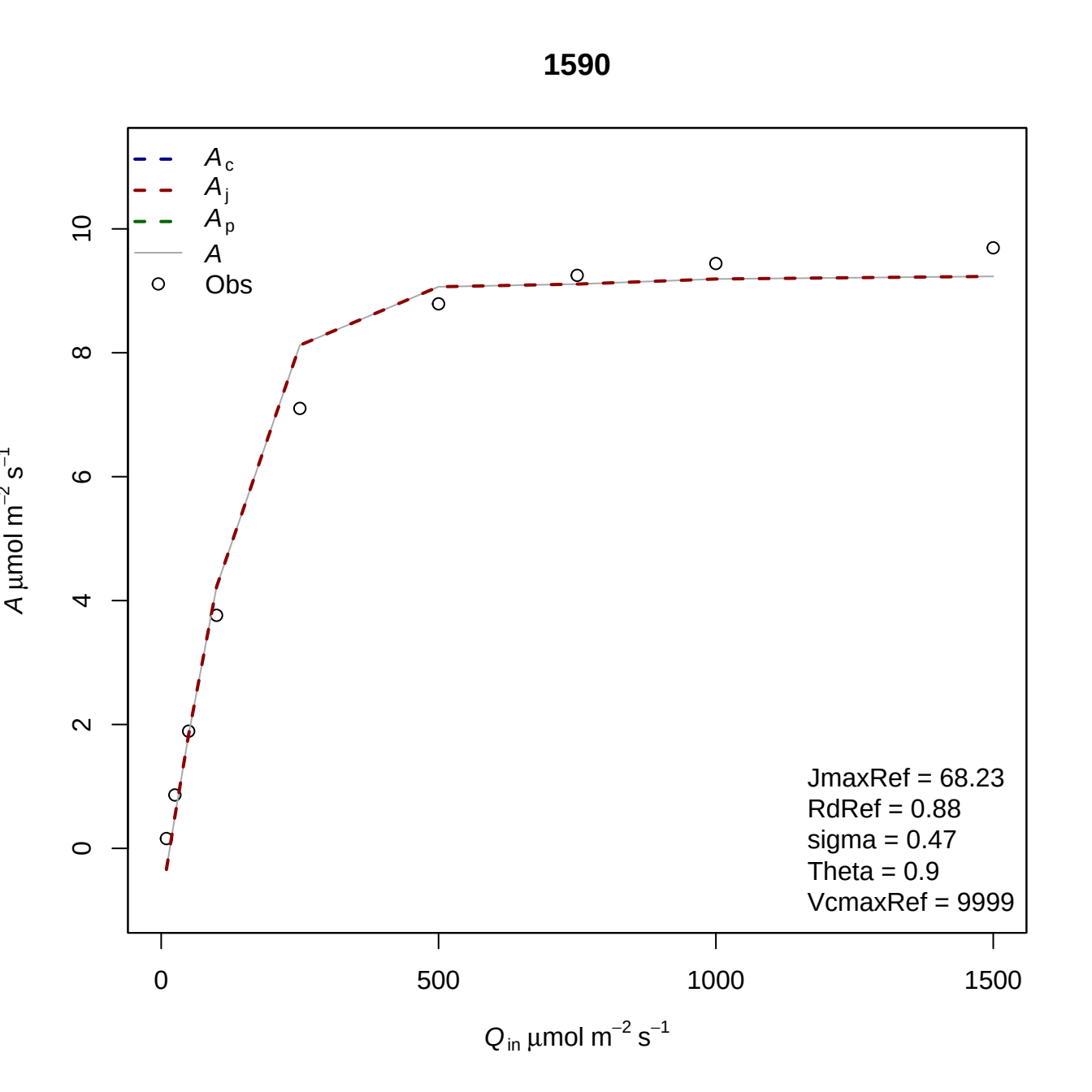

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 69.41
RdRef = 1.09
sigma = 0.44
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

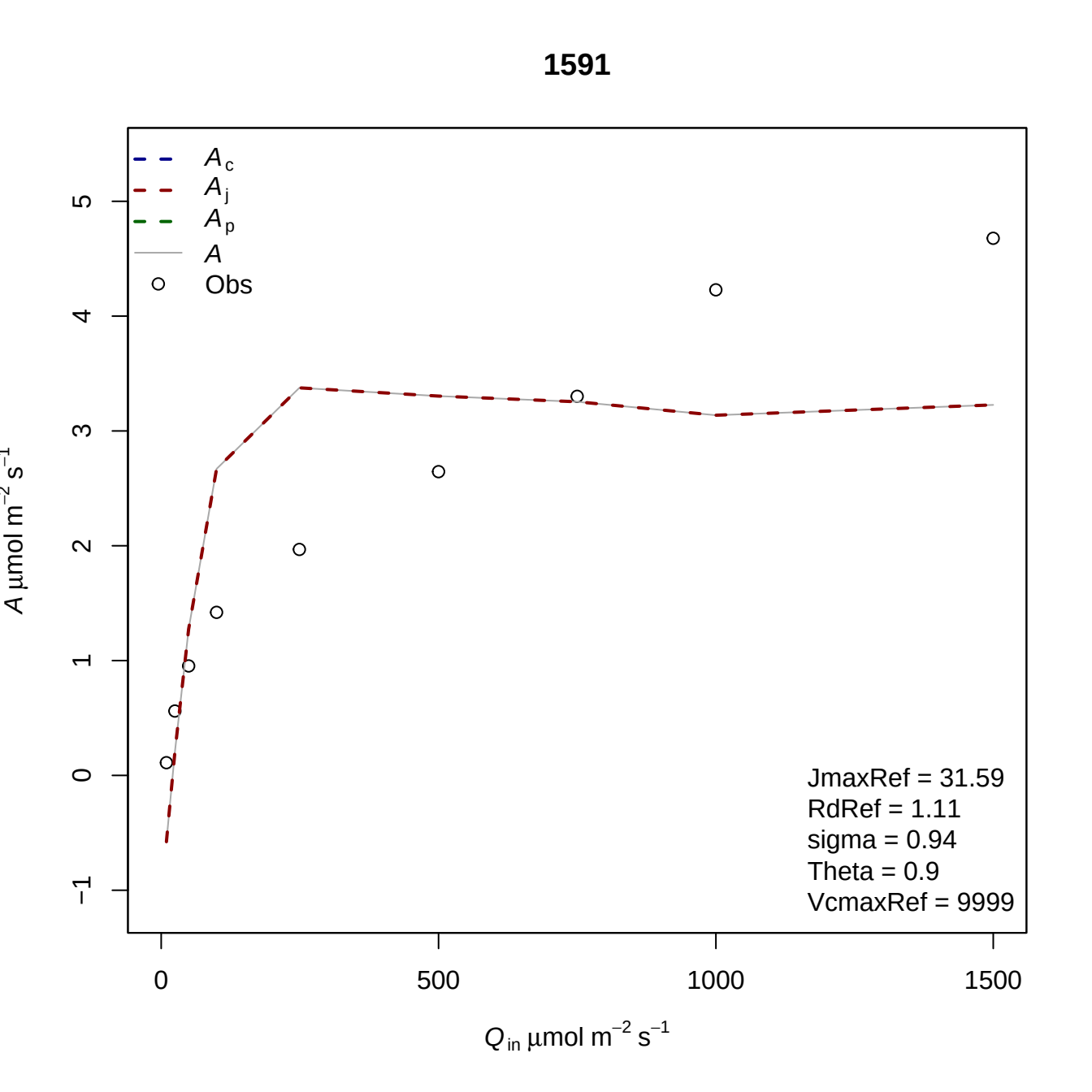

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 48.73
RdRef = 0.97
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

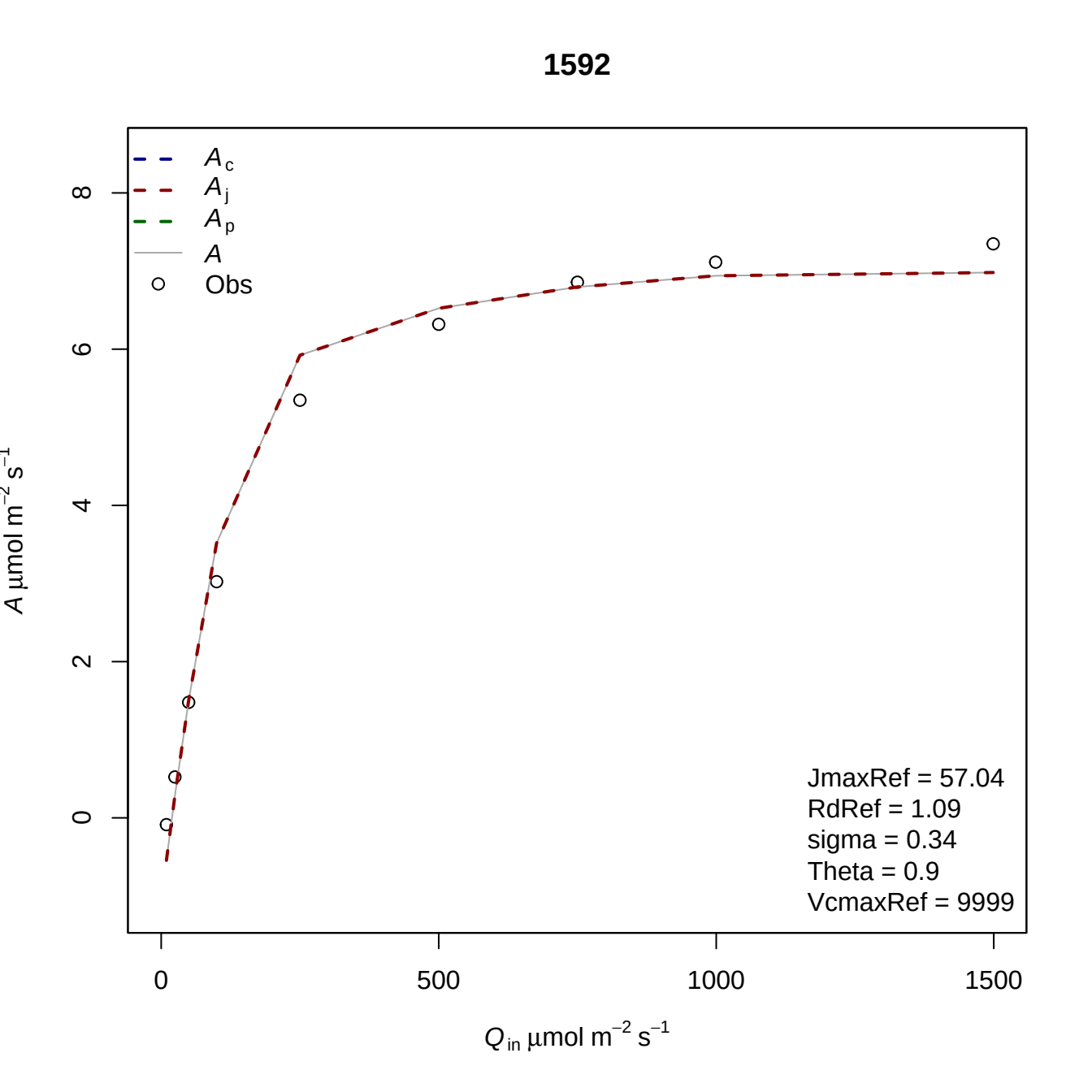

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 48.24
RdRef = 1.21
sigma = 0.54
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

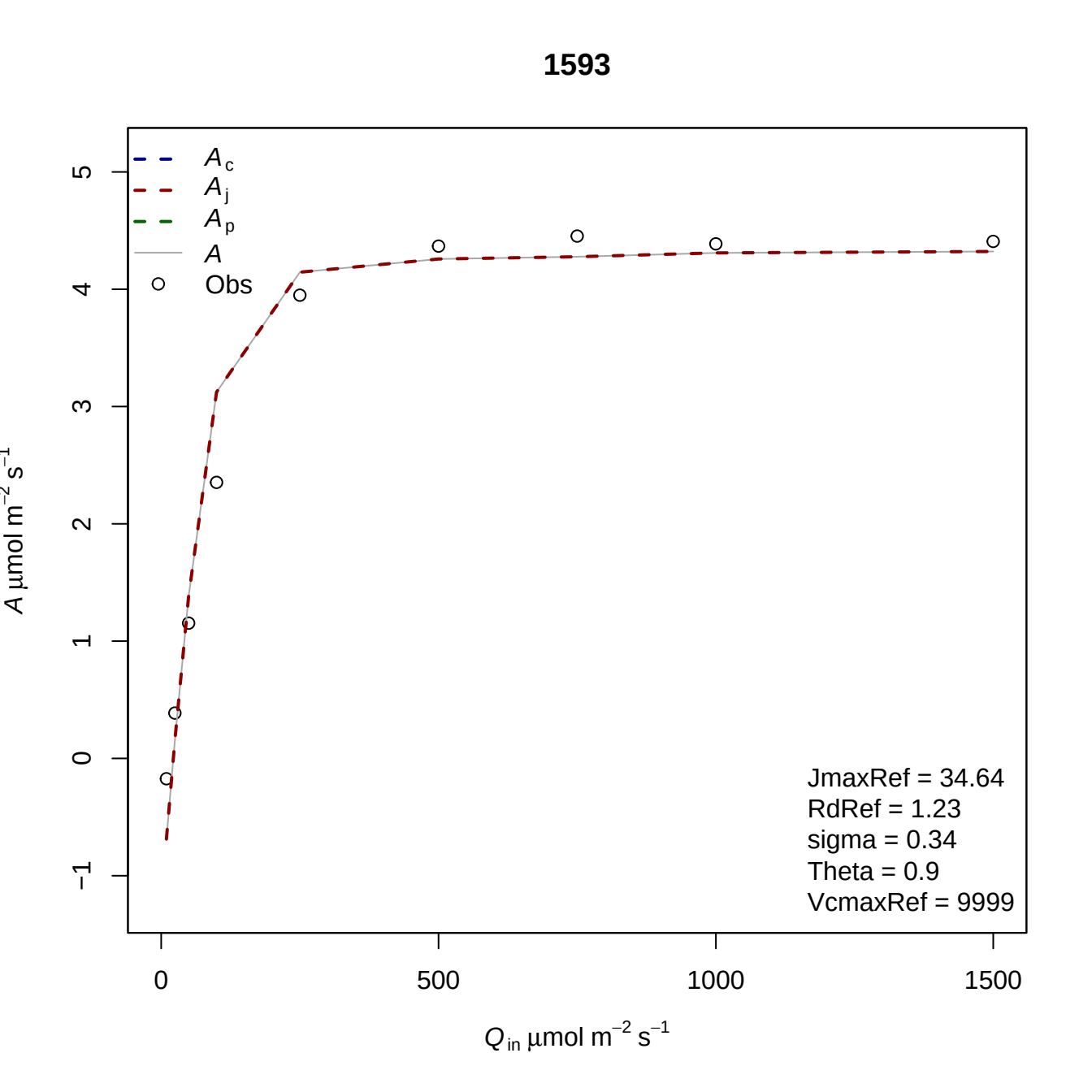

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 40.7
RdRef = 1.2
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

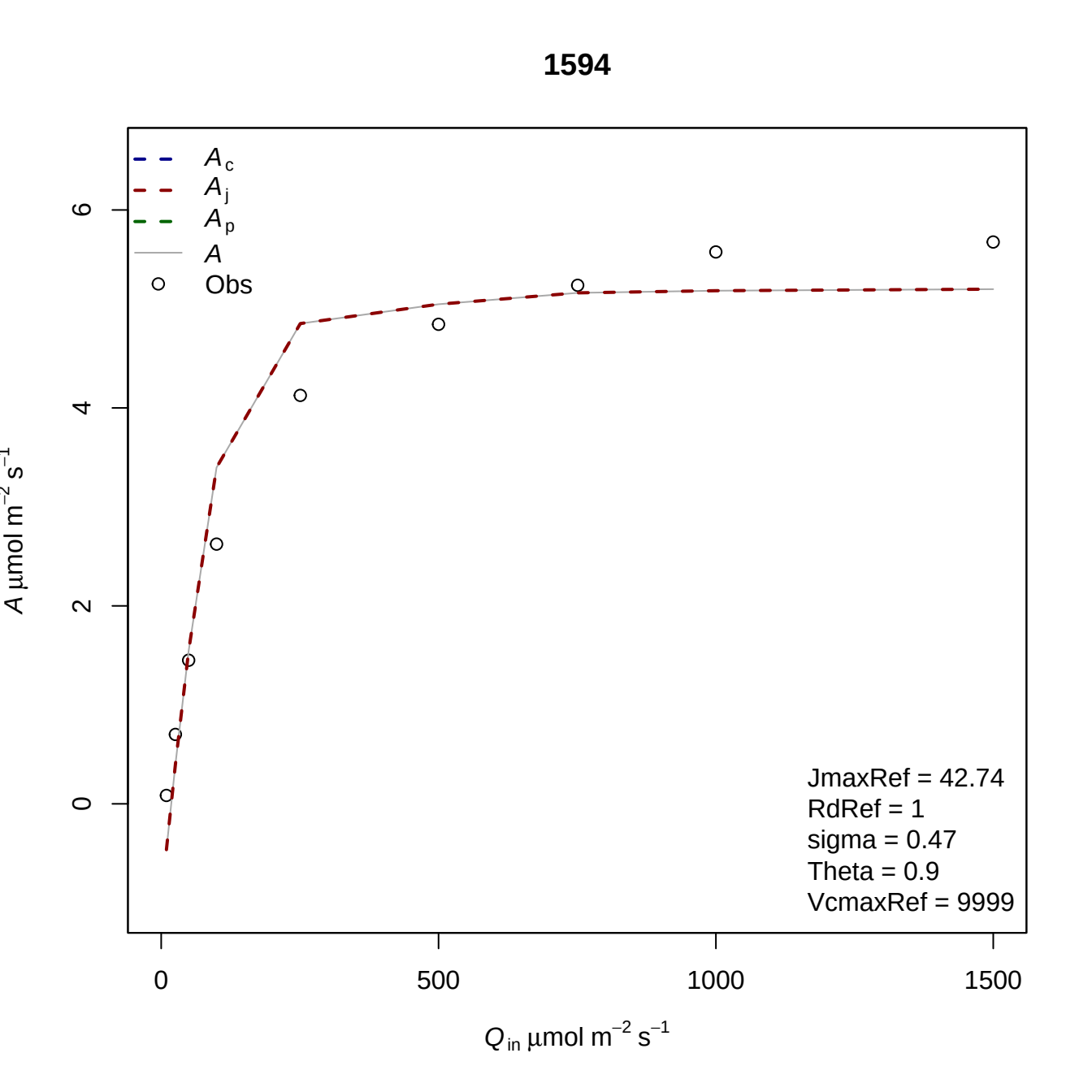

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 30.25
RdRef = 1.1
sigma = 0.45
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

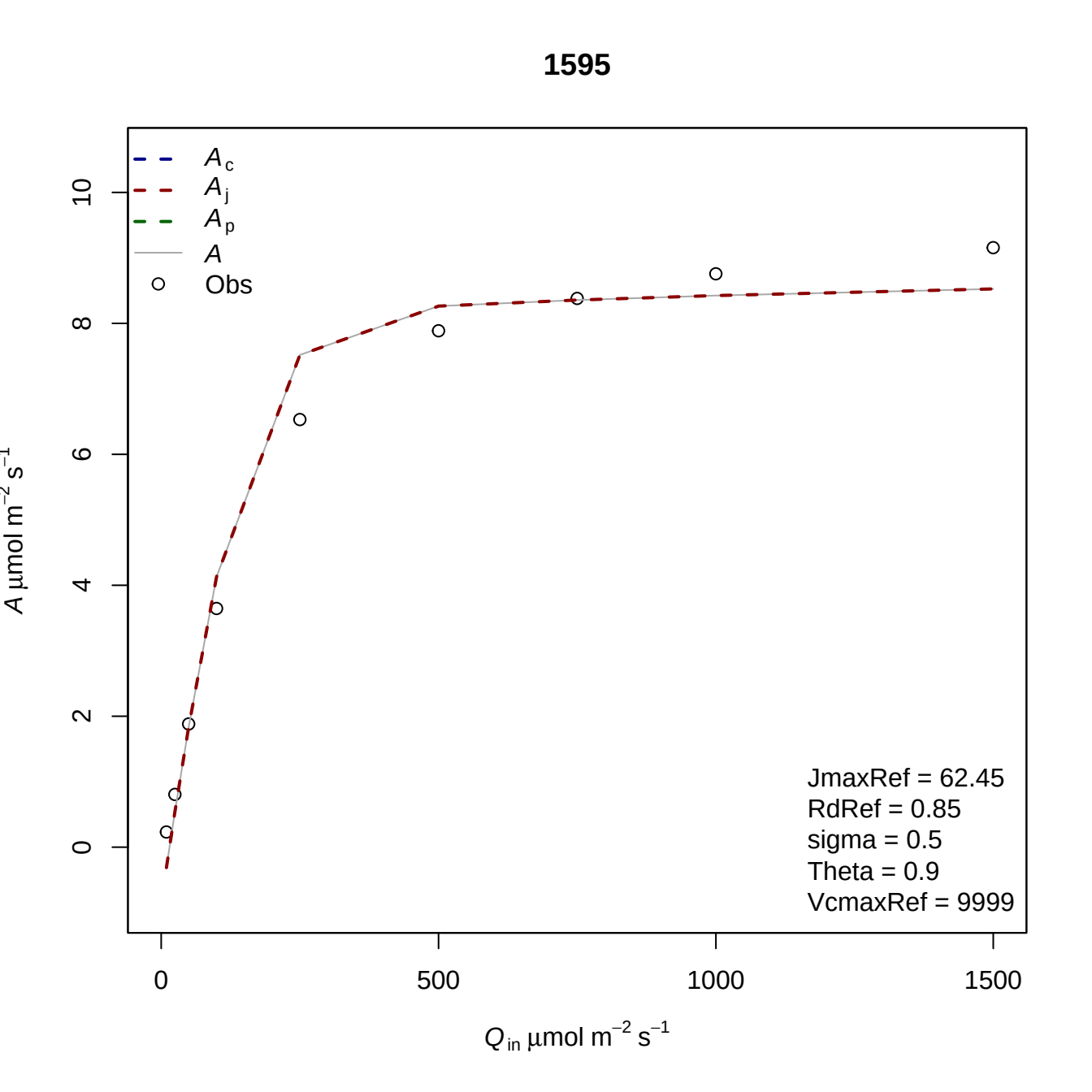

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 68.23
RdRef = 0.88
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

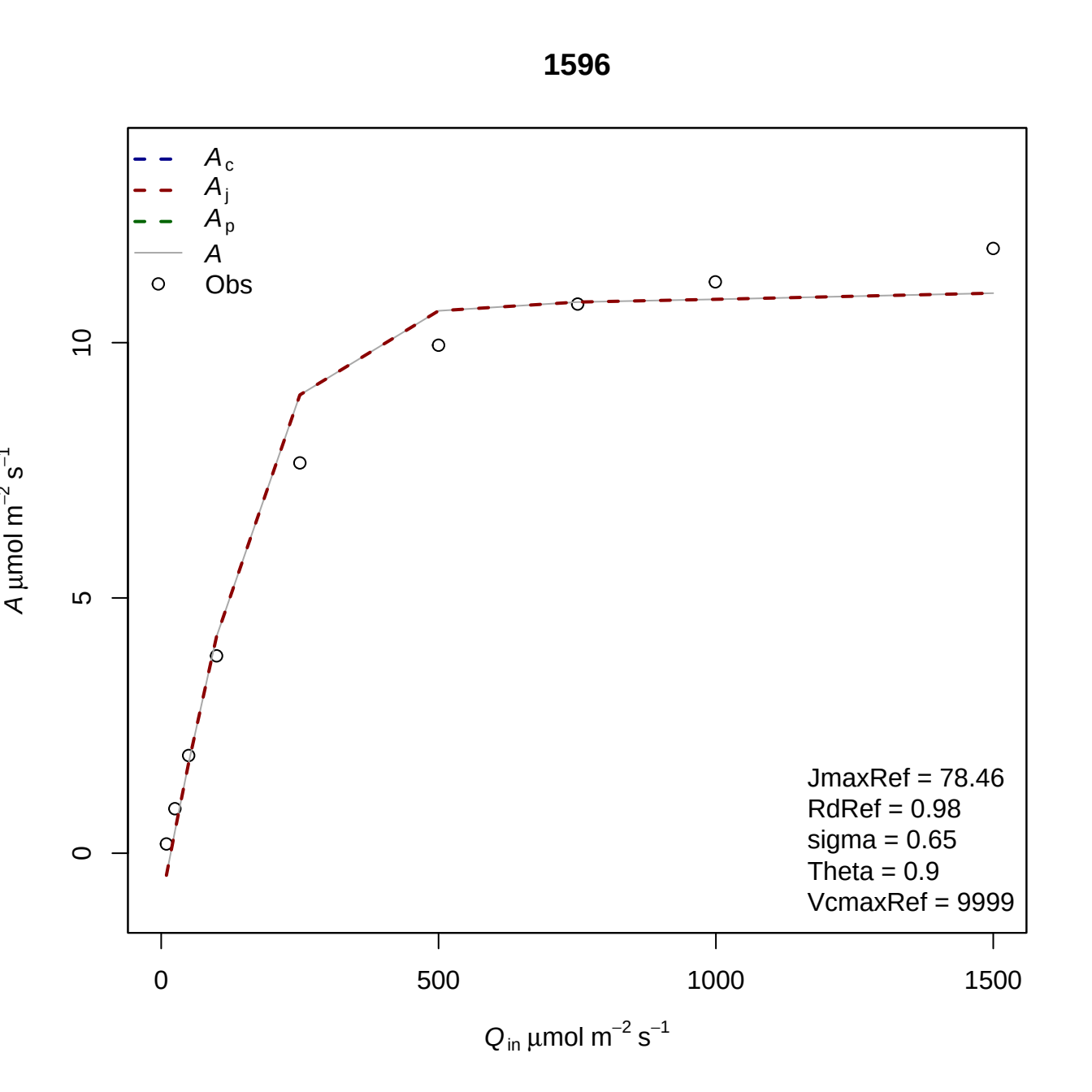

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 31.59
RdRef = 1.11
sigma = 0.94
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

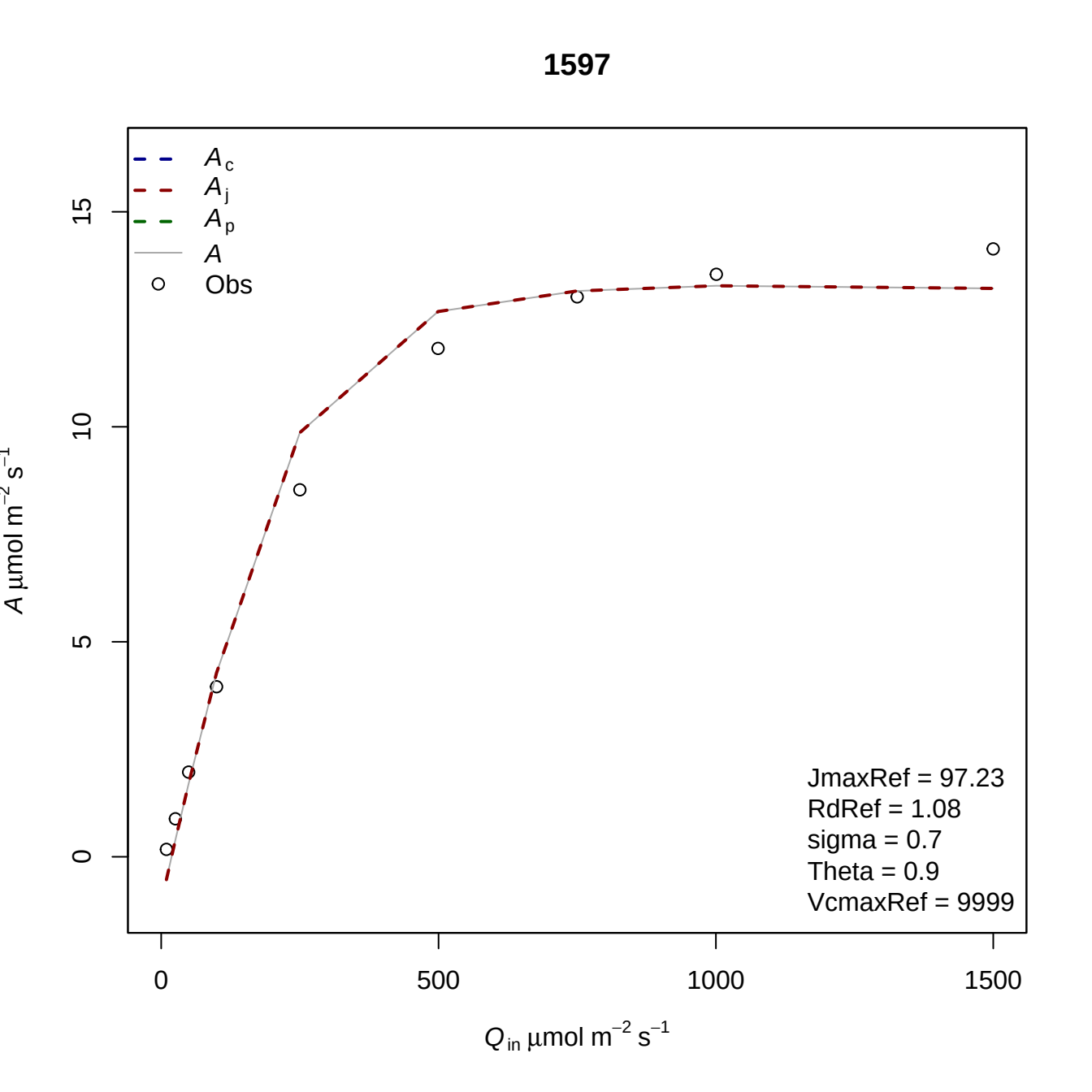

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 57.04
RdRef = 1.09
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

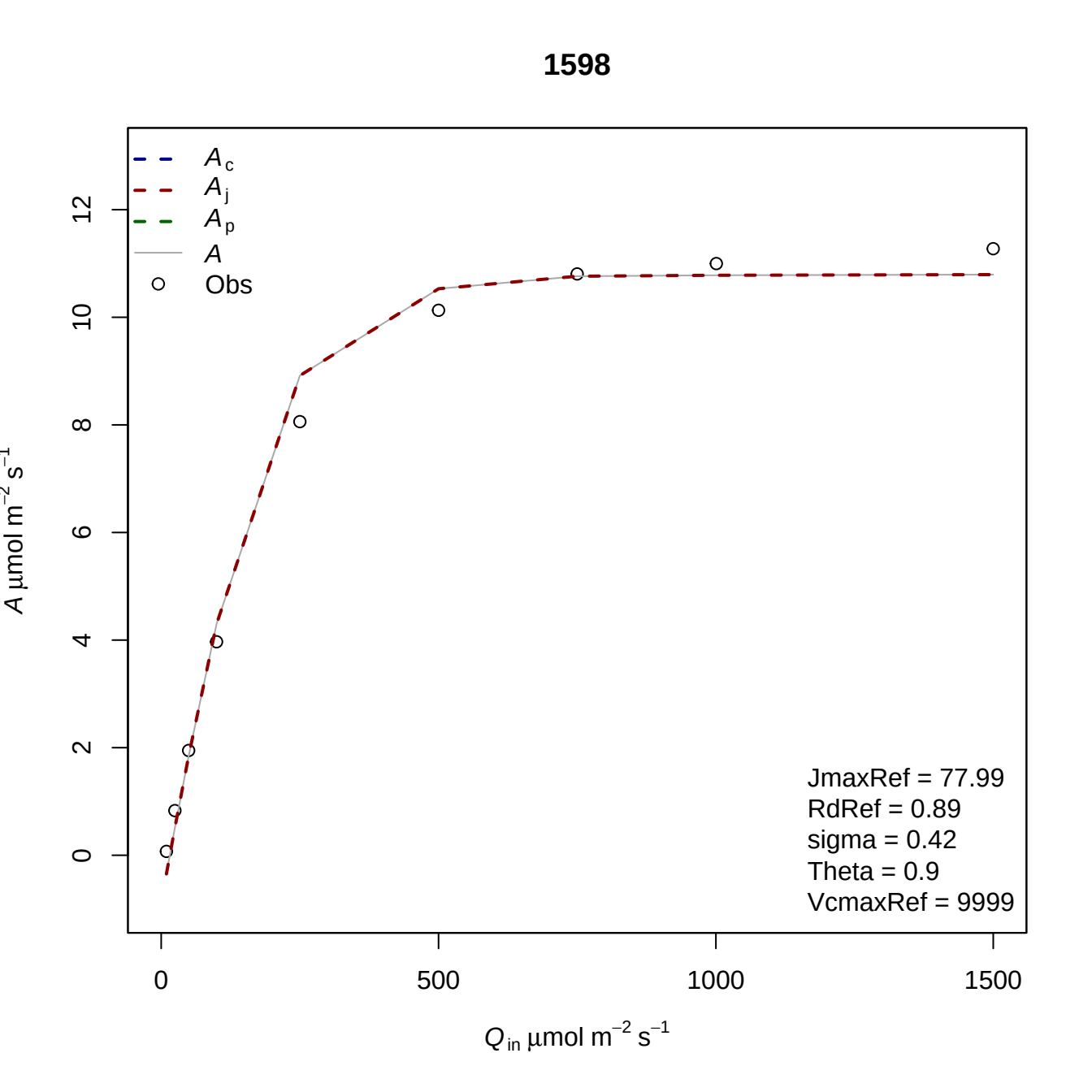

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 34.64
RdRef = 1.23
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

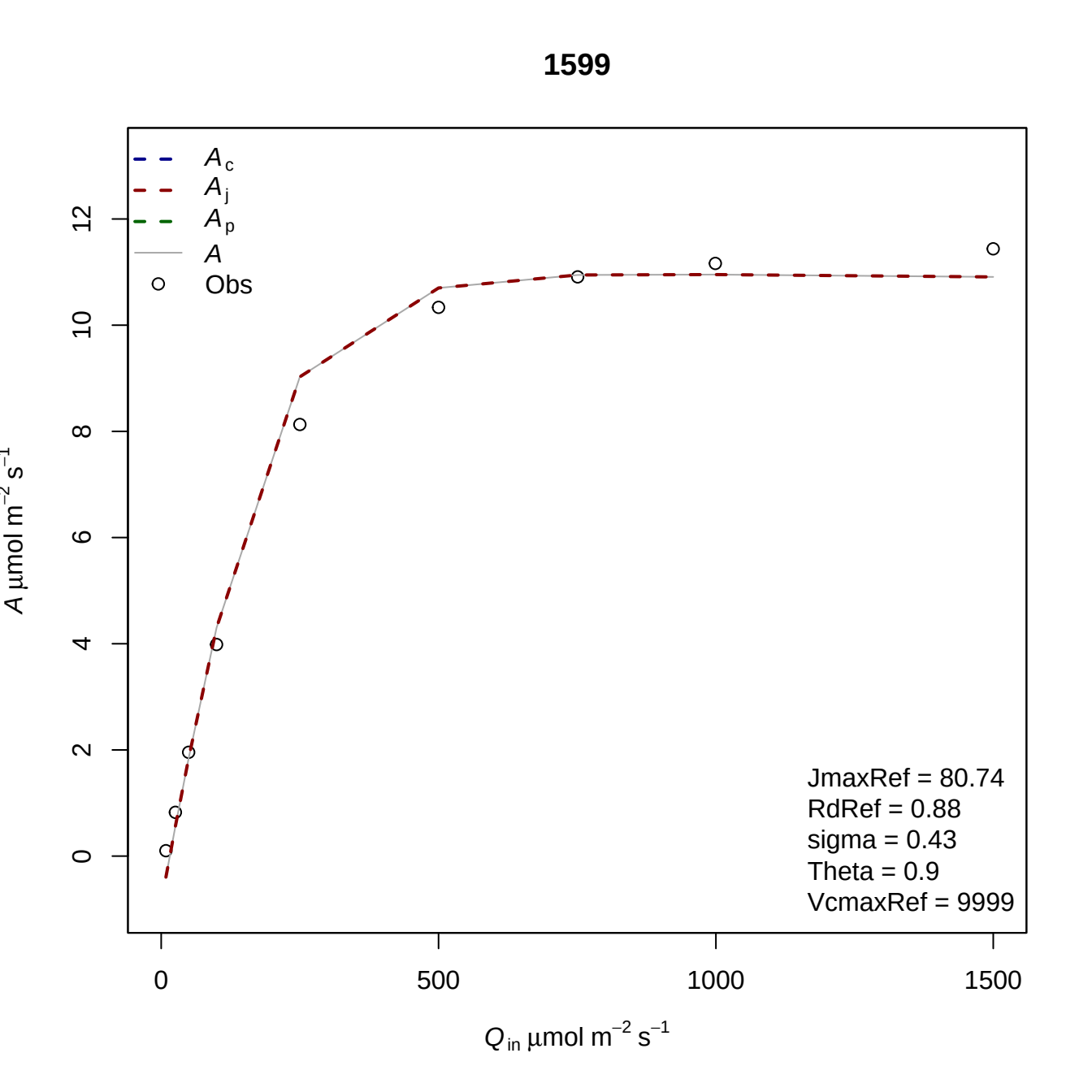

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 42.74
RdRef = 1
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

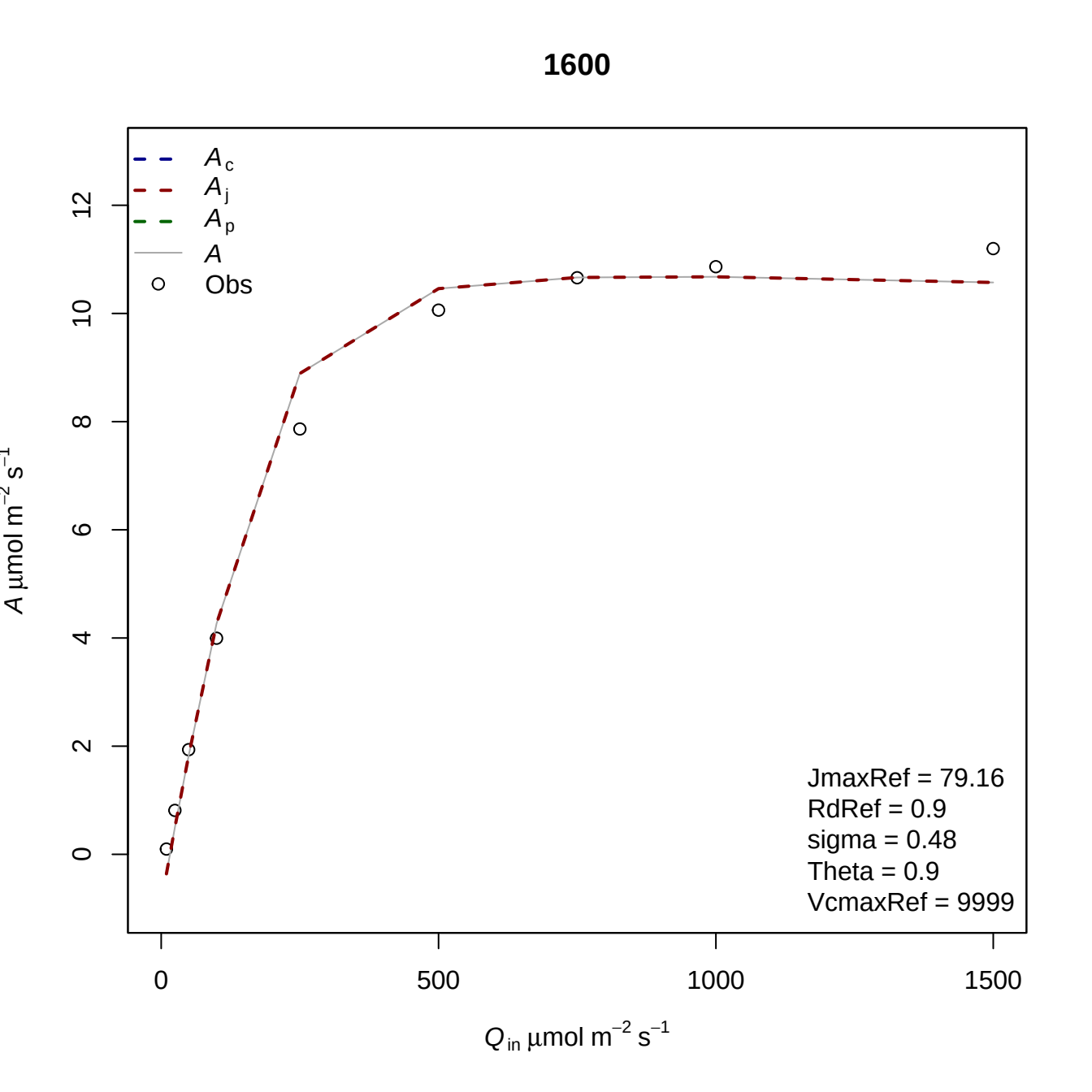

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 62.45
RdRef = 0.85
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

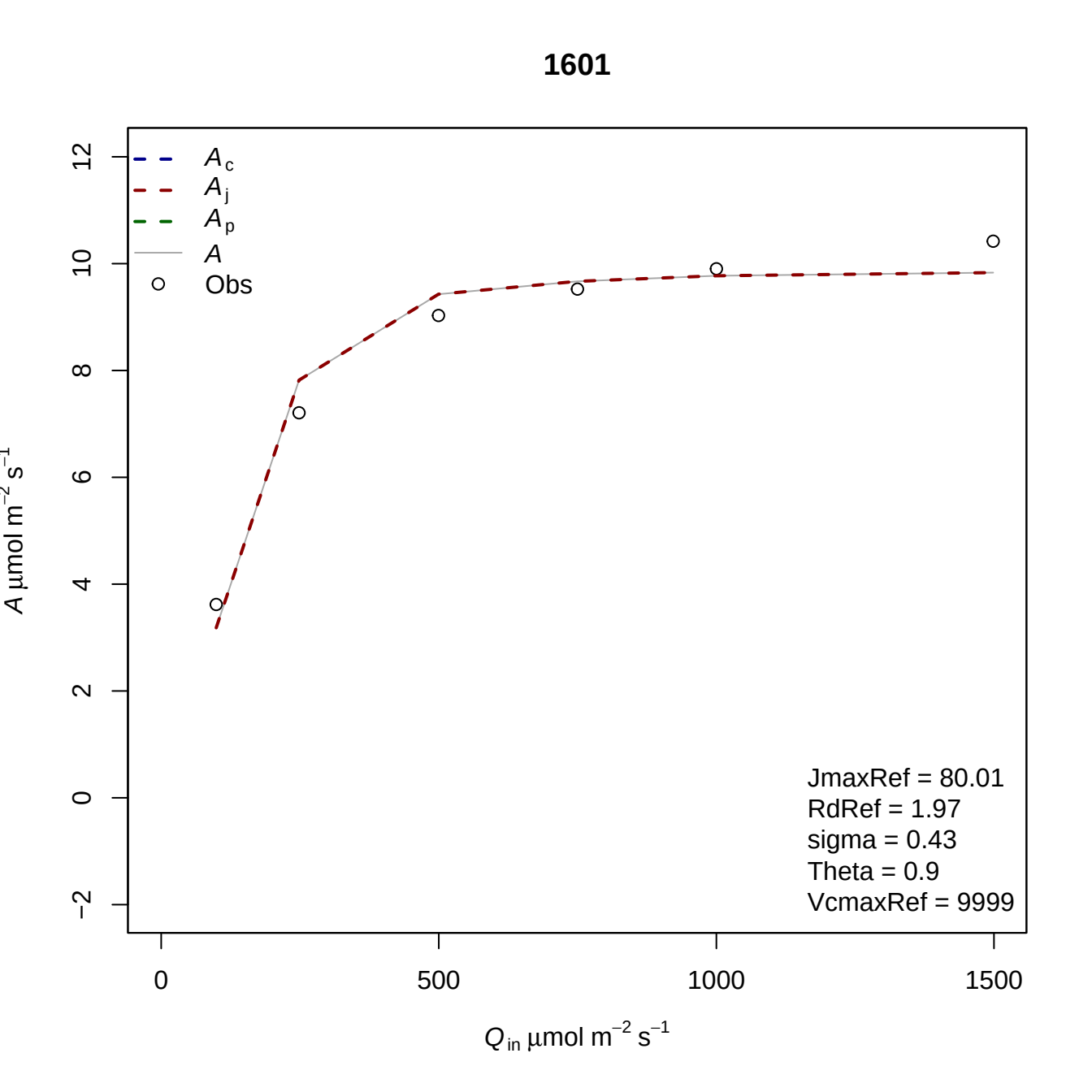

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 78.46
RdRef = 0.98
sigma = 0.65
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

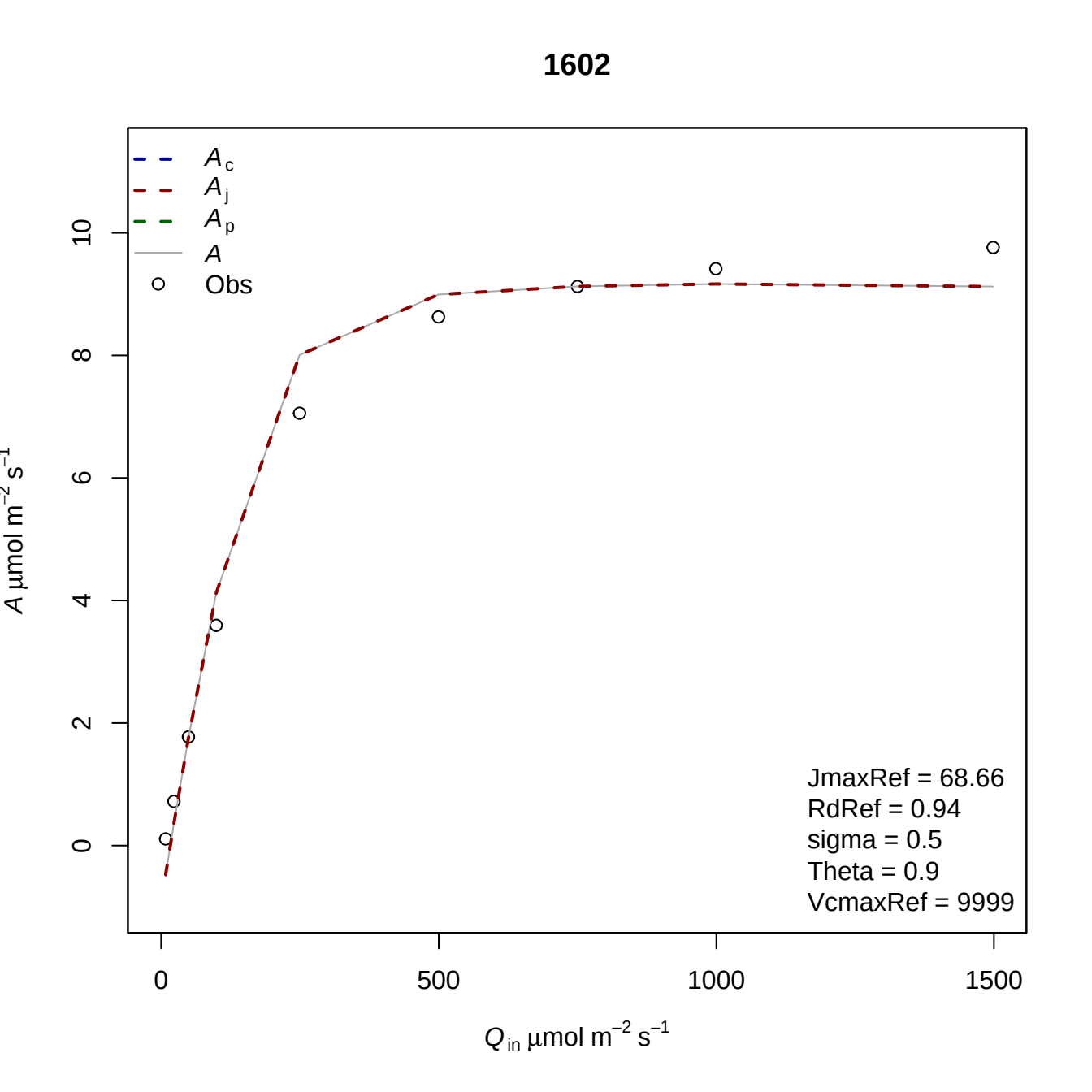

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 97.23
RdRef = 1.08
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

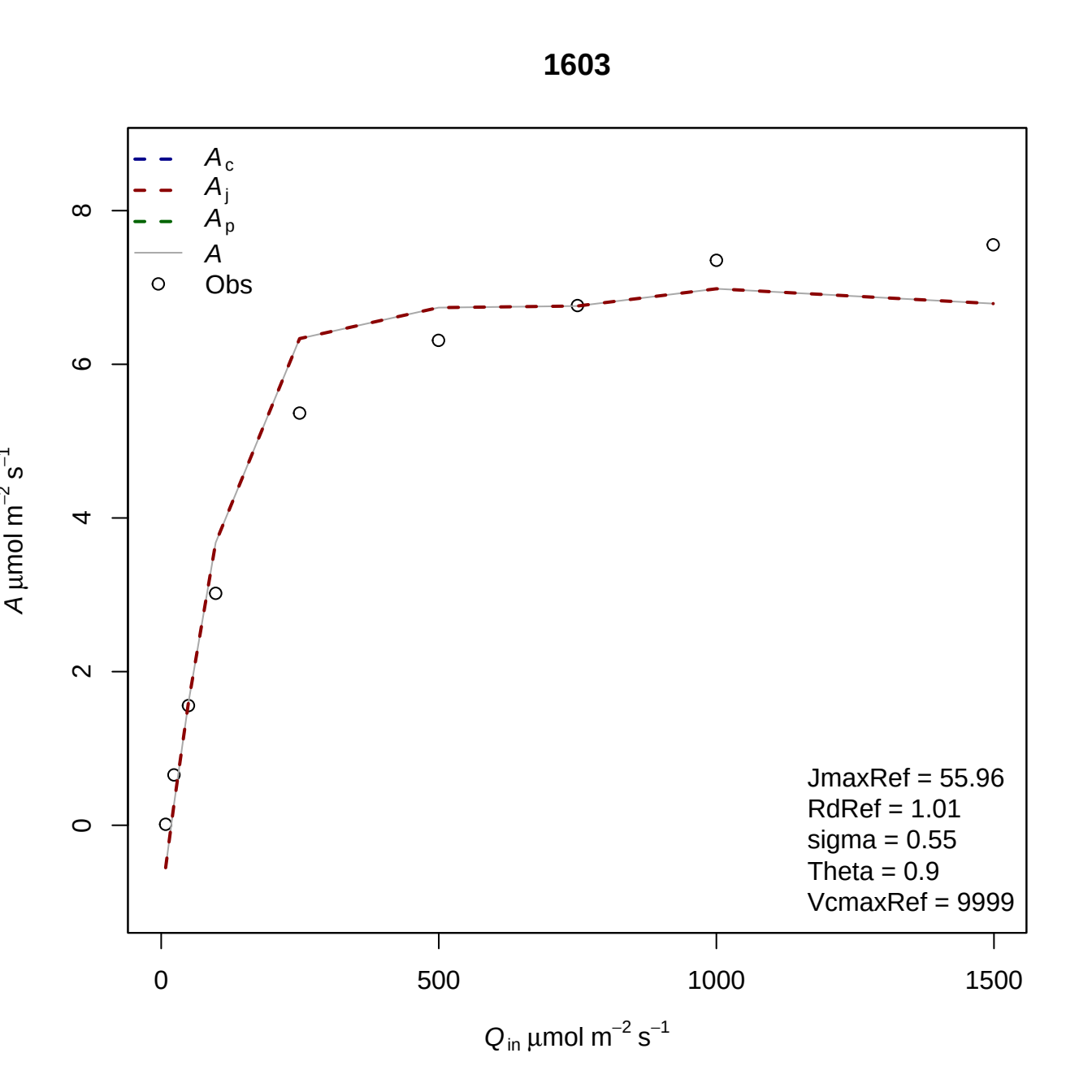

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 77.99
RdRef = 0.89
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

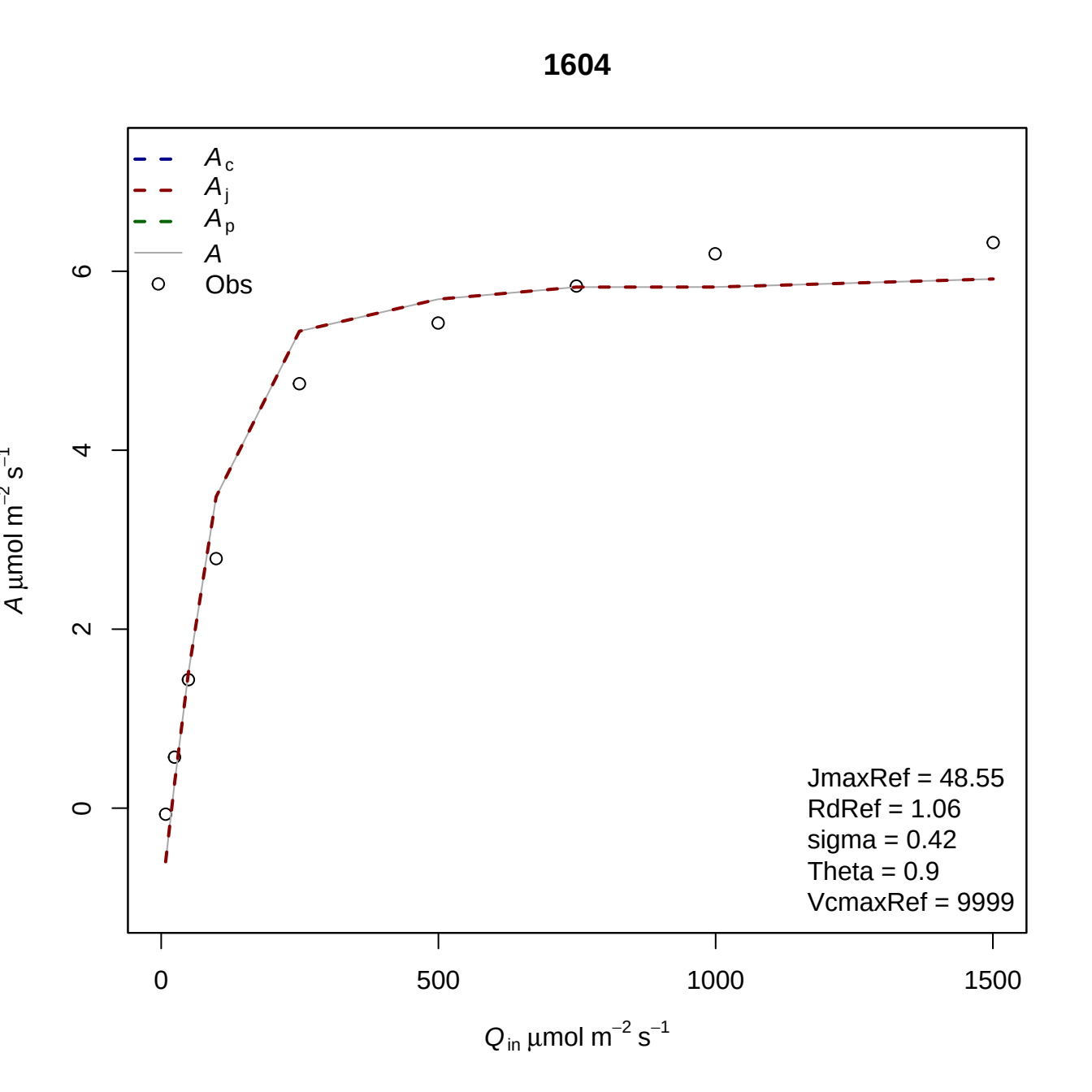

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 80.74
RdRef = 0.88
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

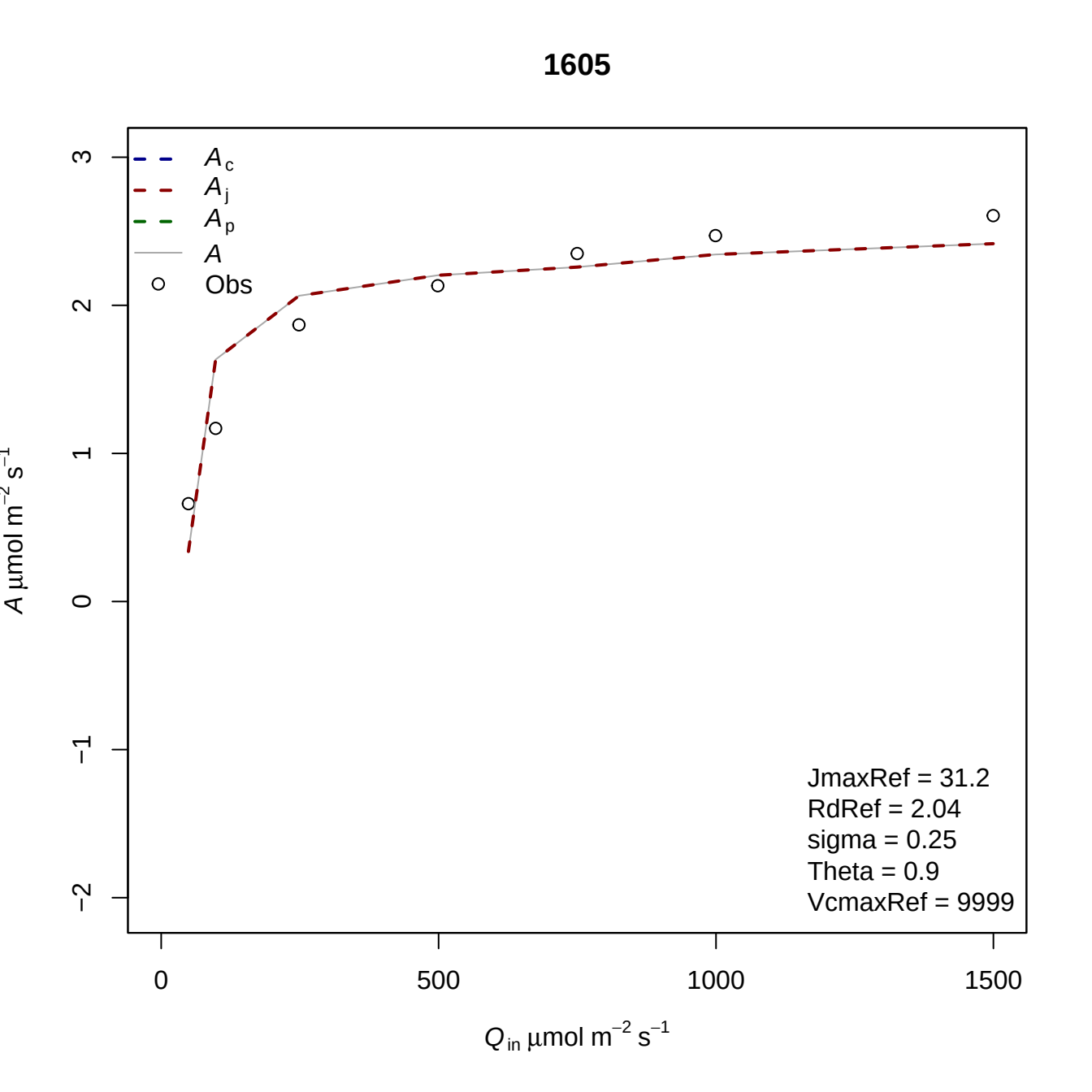

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 79.16
RdRef = 0.9
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

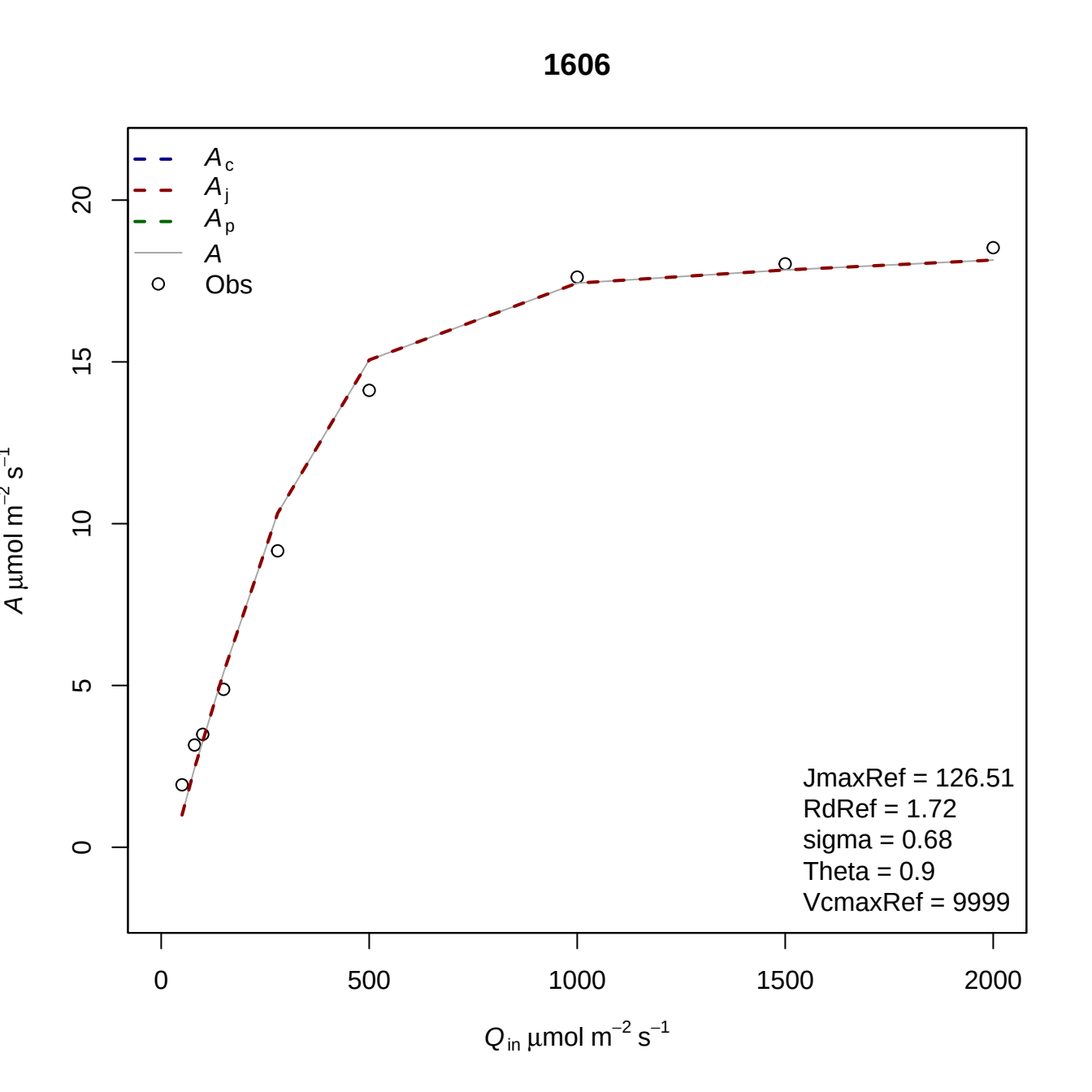

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 80.01
RdRef = 1.97
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

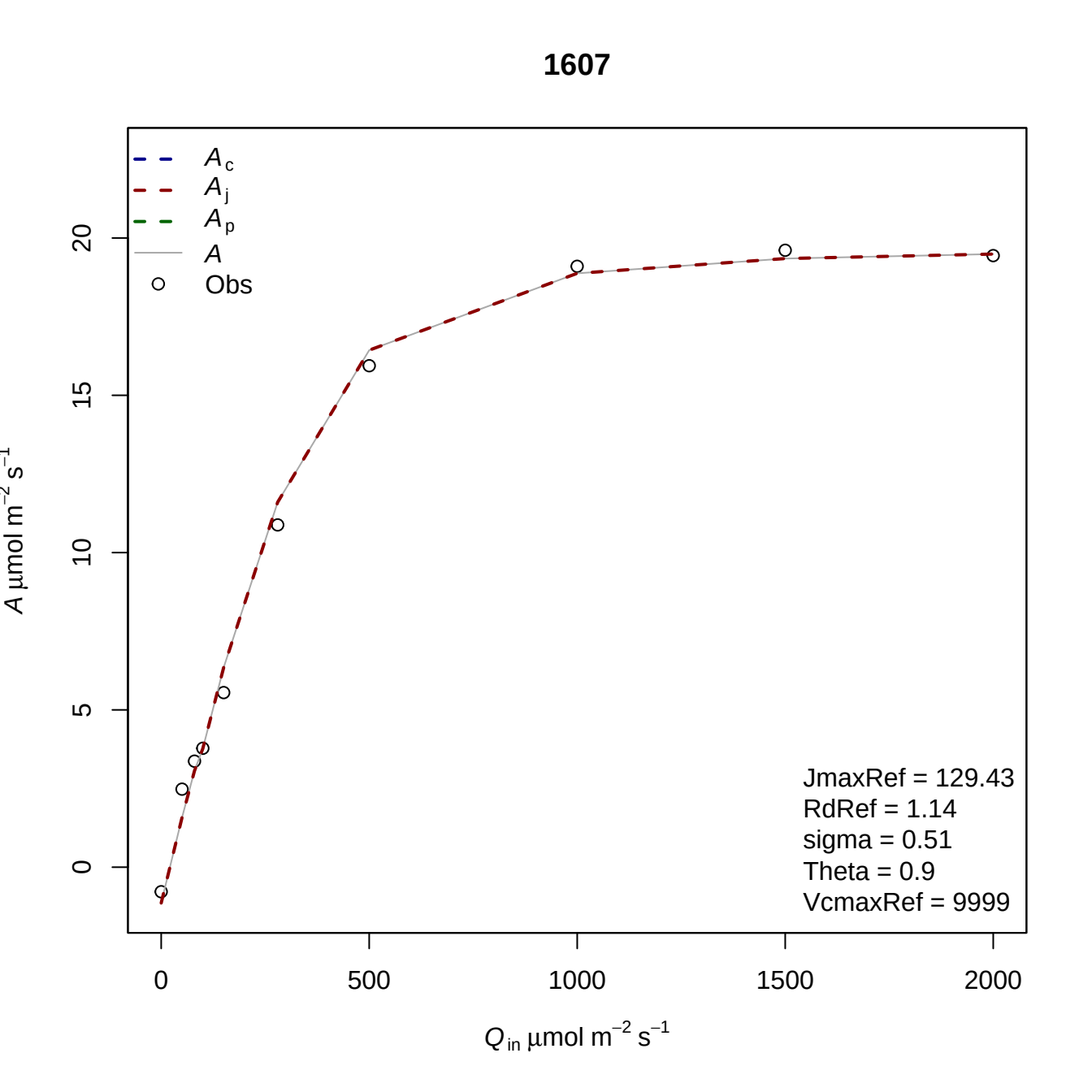

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 68.66
RdRef = 0.94
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

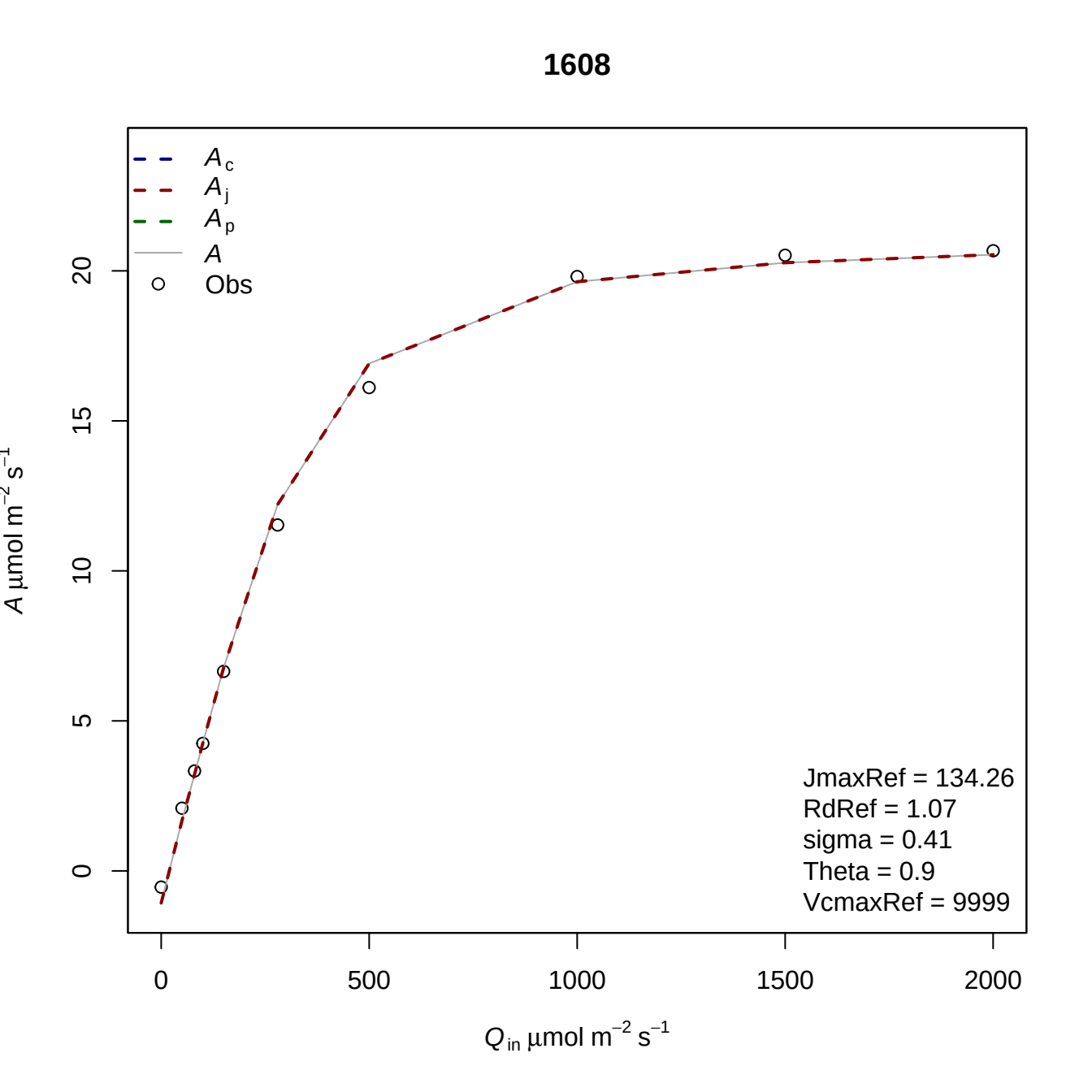

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 55.96
RdRef = 1.01
sigma = 0.55
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

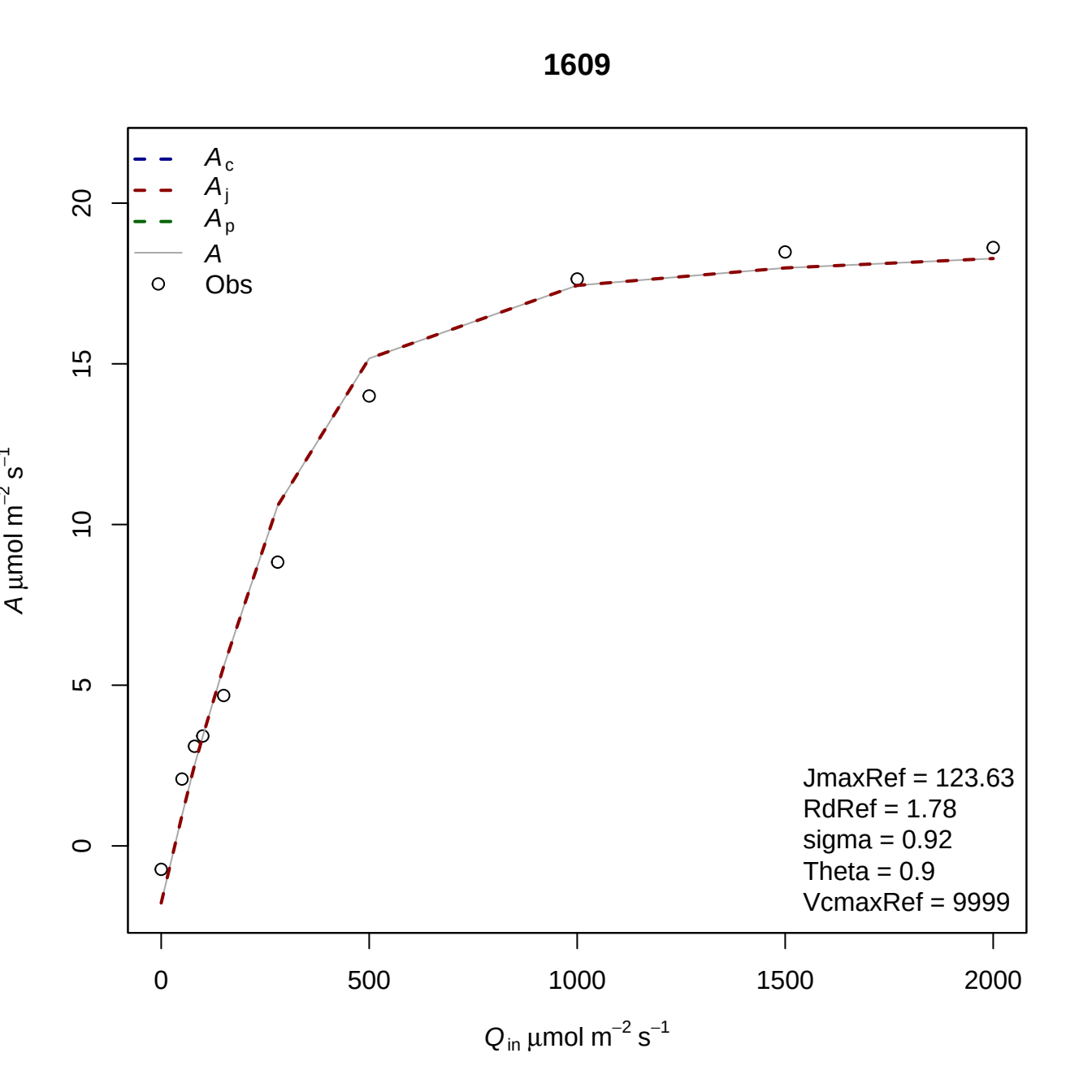

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 48.55
RdRef = 1.06
sigma = 0.42
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

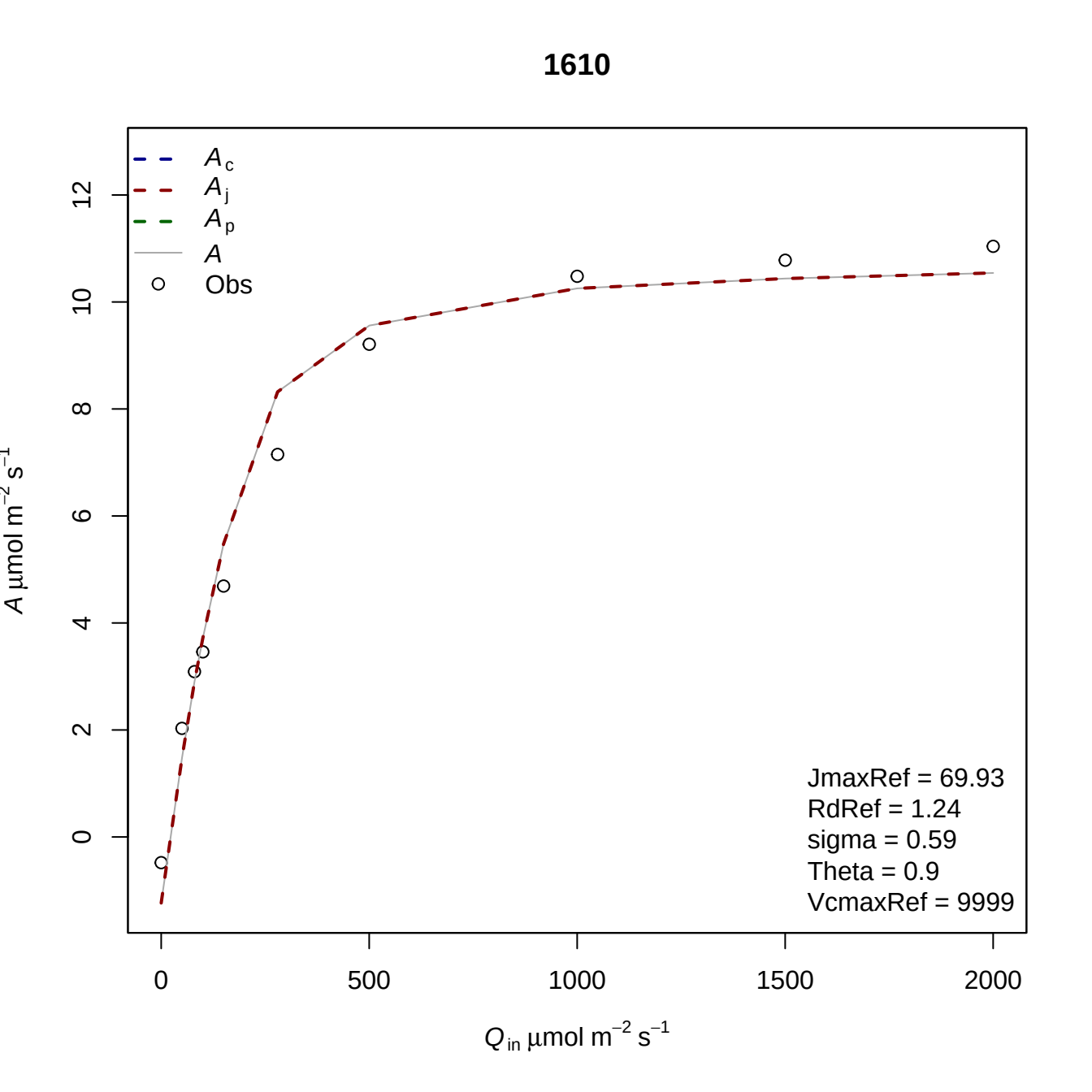

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 31.2
RdRef = 2.04
sigma = 0.25
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

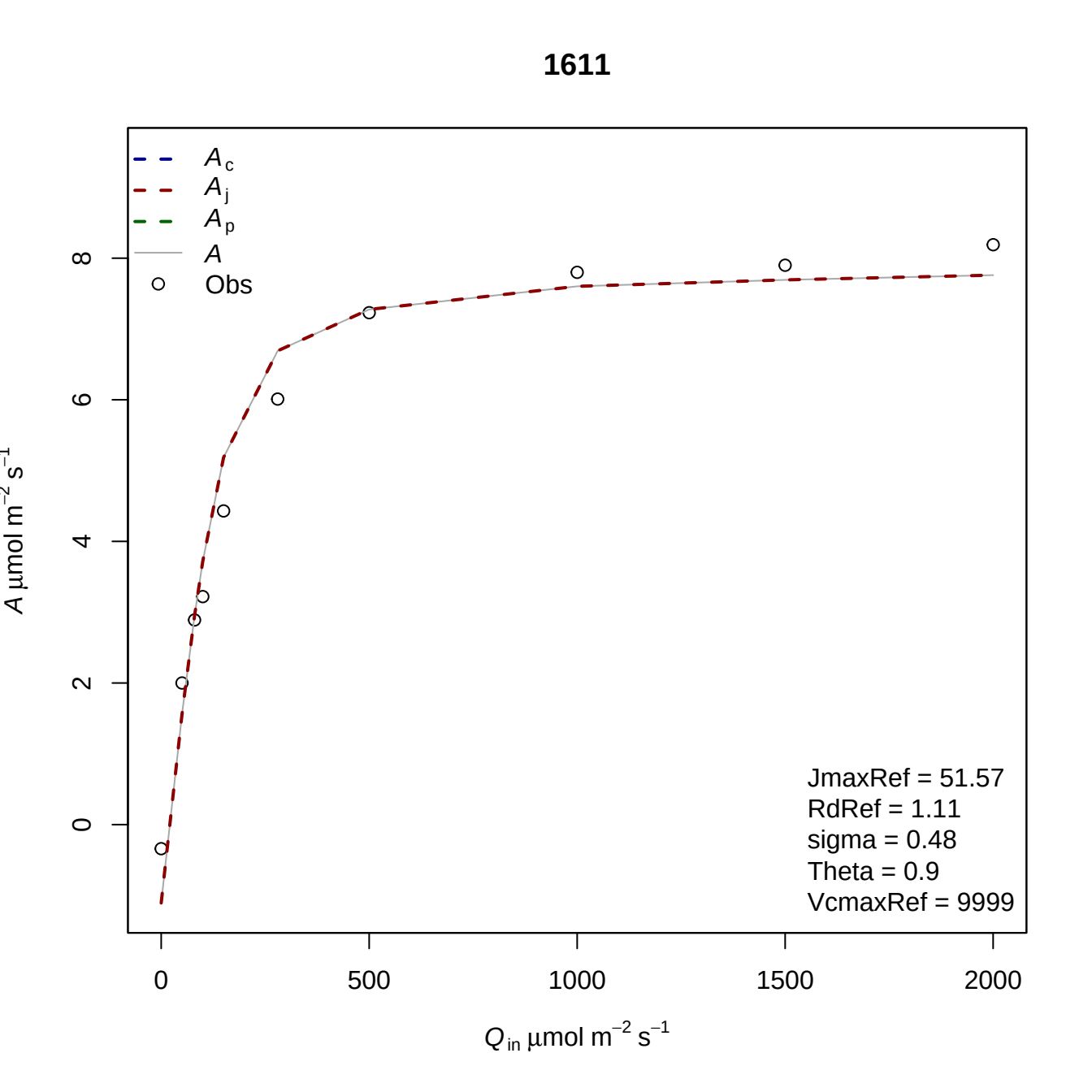

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 126.51
RdRef = 1.72
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

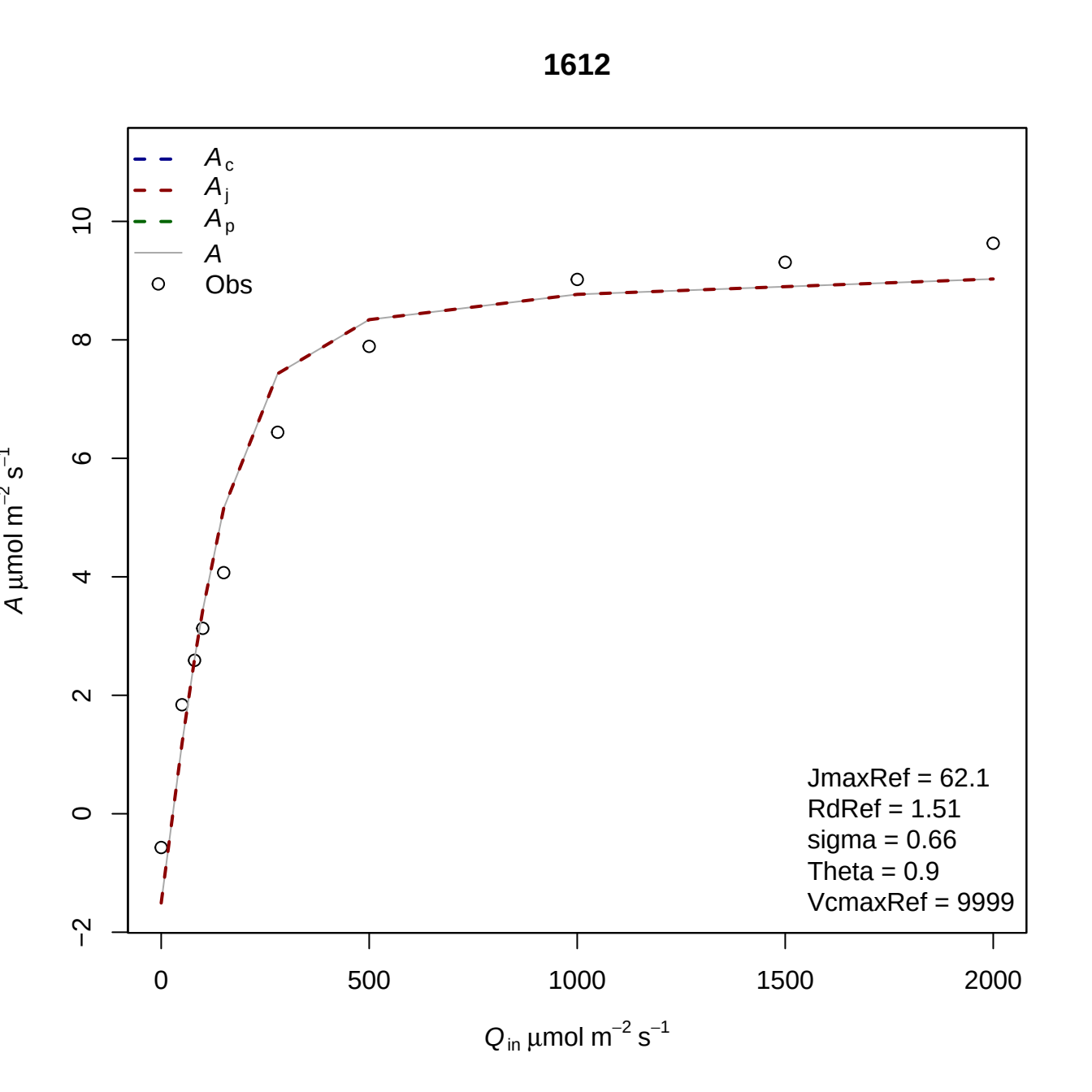

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 129.43
RdRef = 1.14
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

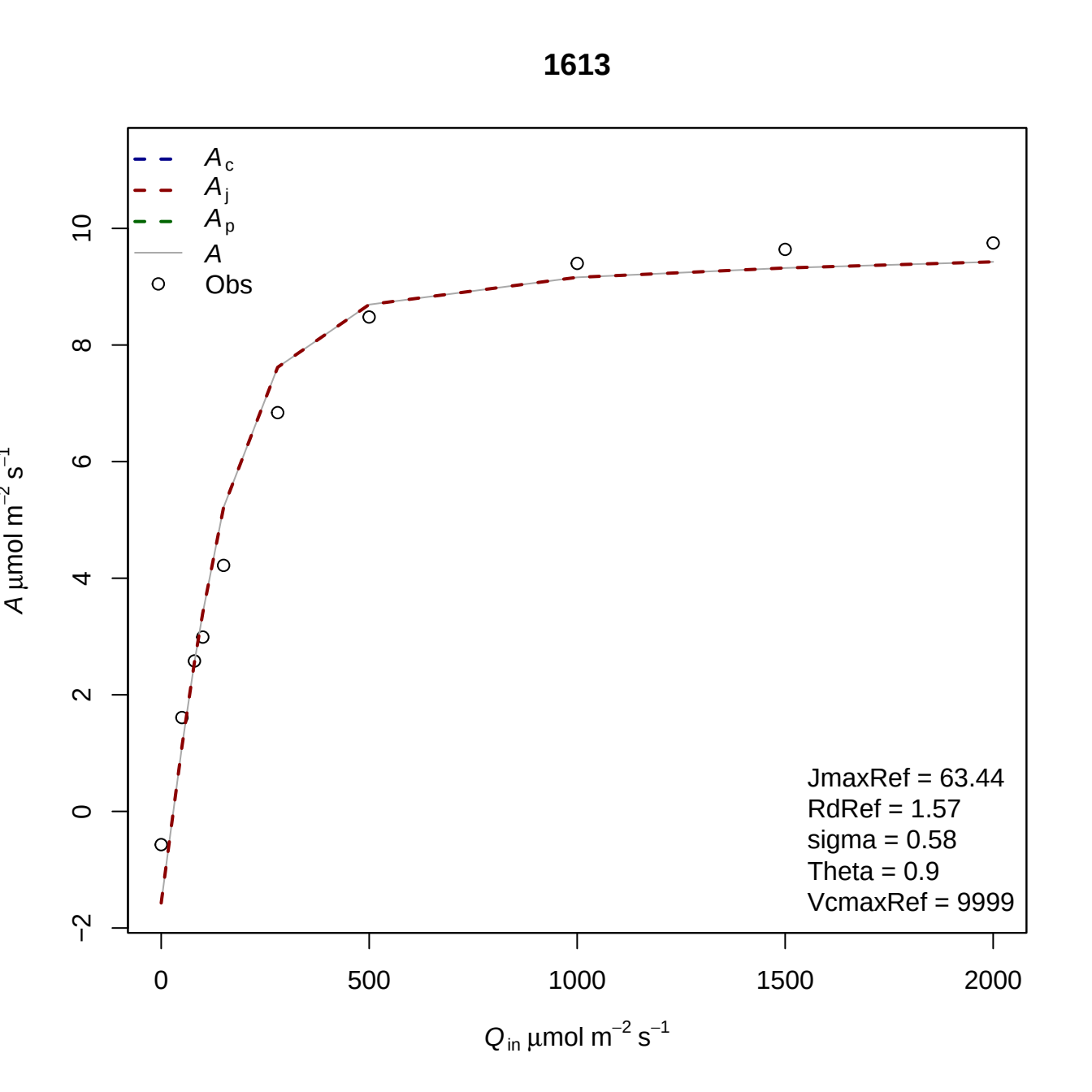

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 134.26
RdRef = 1.07
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

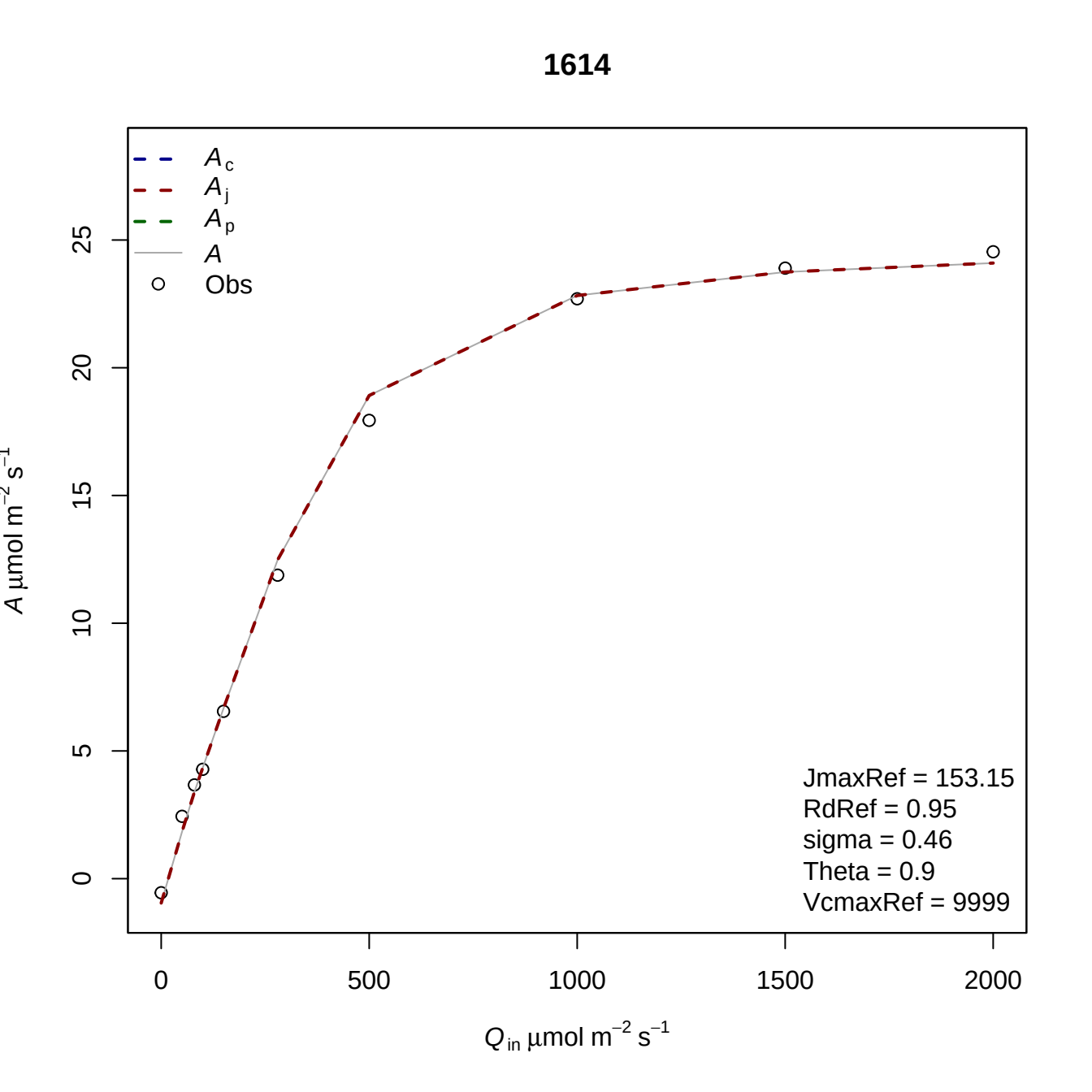

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 123.63
RdRef = 1.78
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

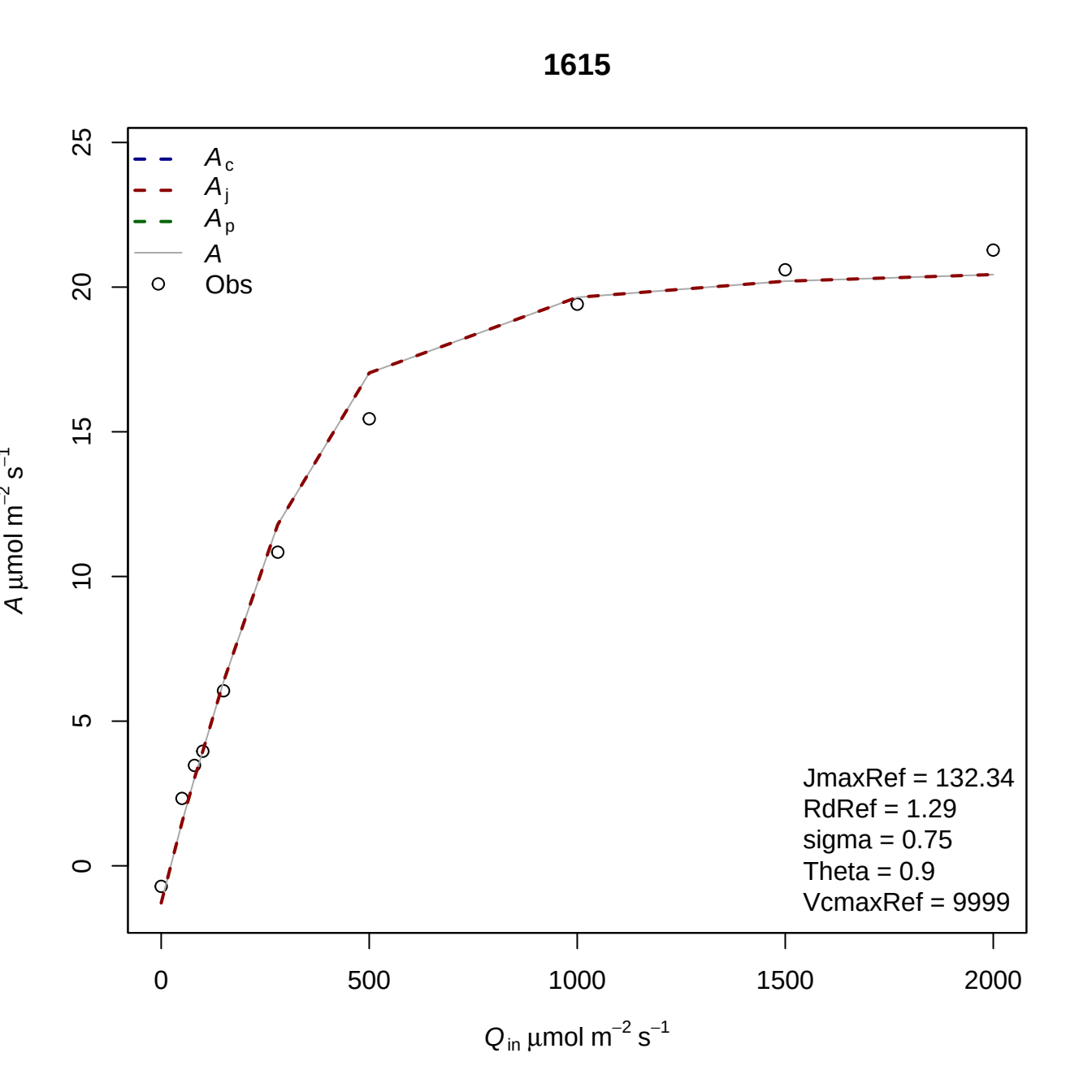

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 69.93
RdRef = 1.24
sigma = 0.59
Theta = 0.9
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

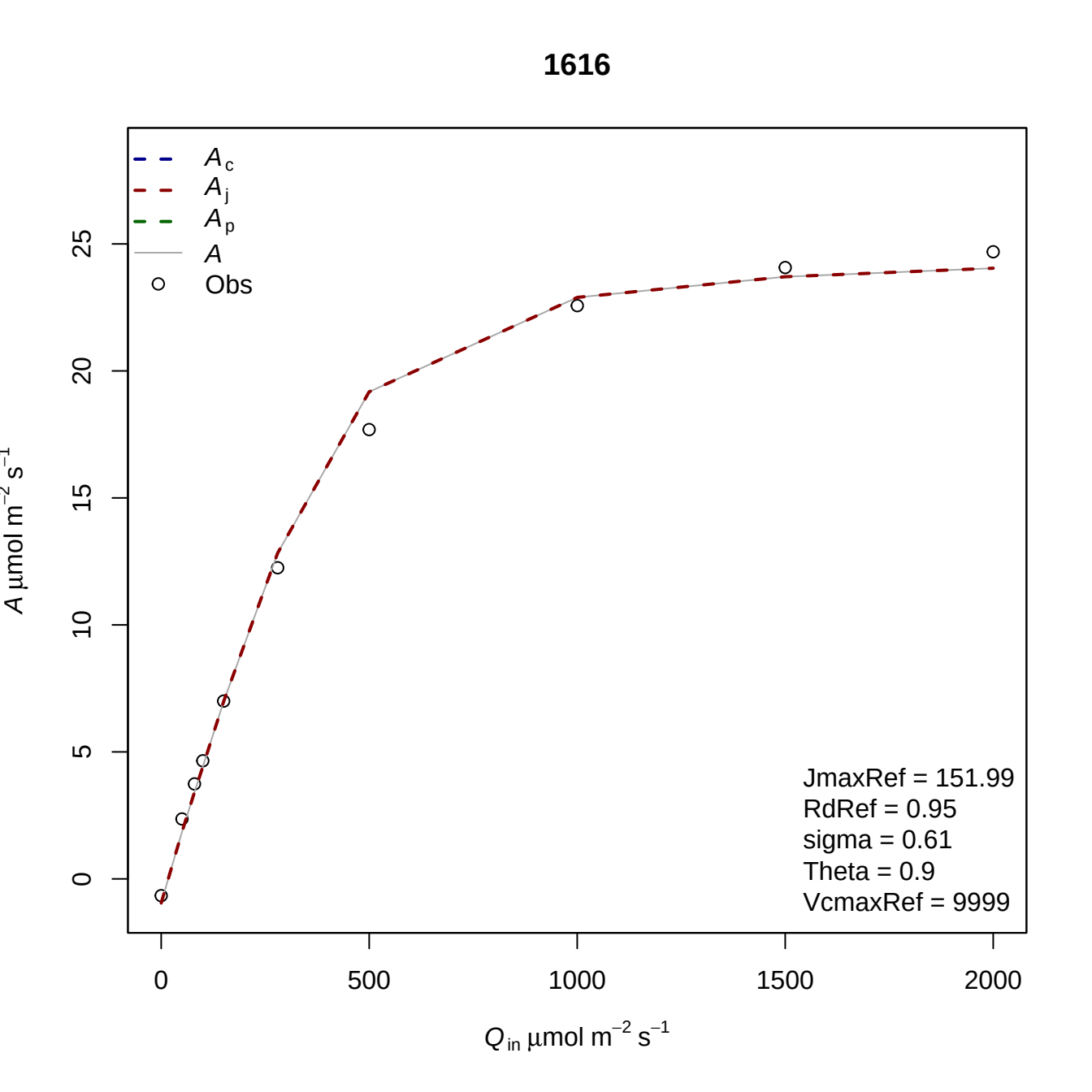

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 51.57
RdRef = 1.11
sigma = 0.48
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

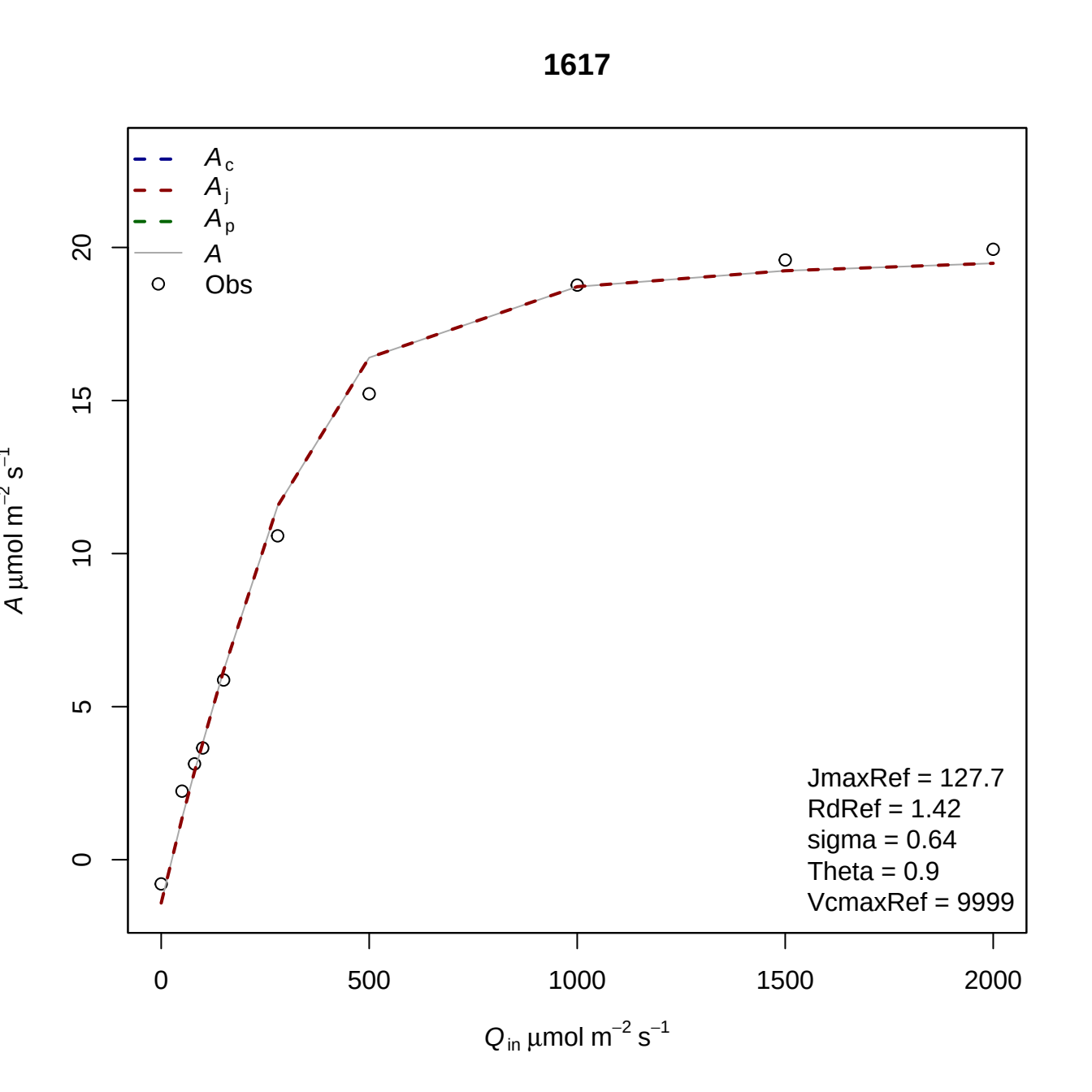

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 62.1
RdRef = 1.51
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

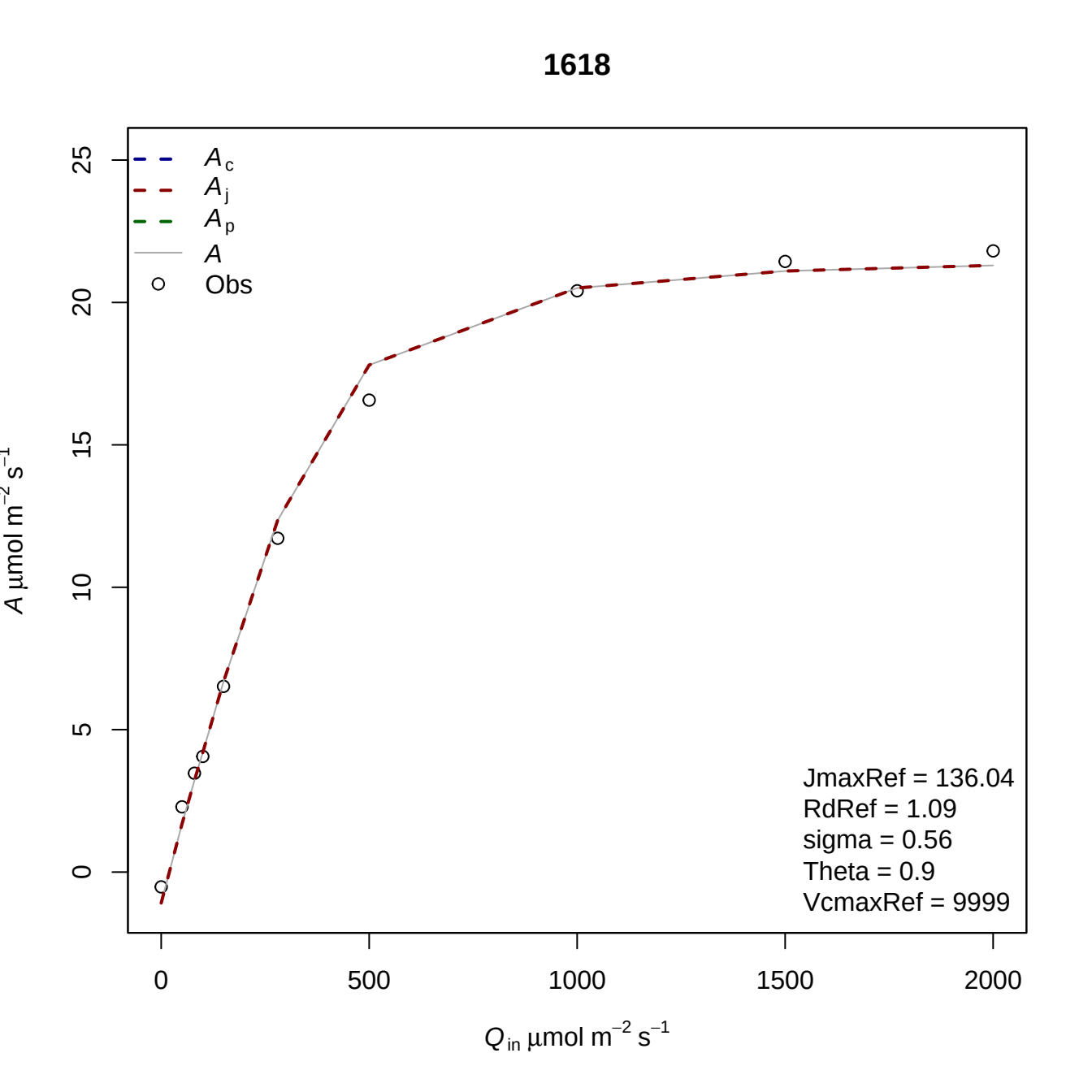

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 63.44
RdRef = 1.57
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

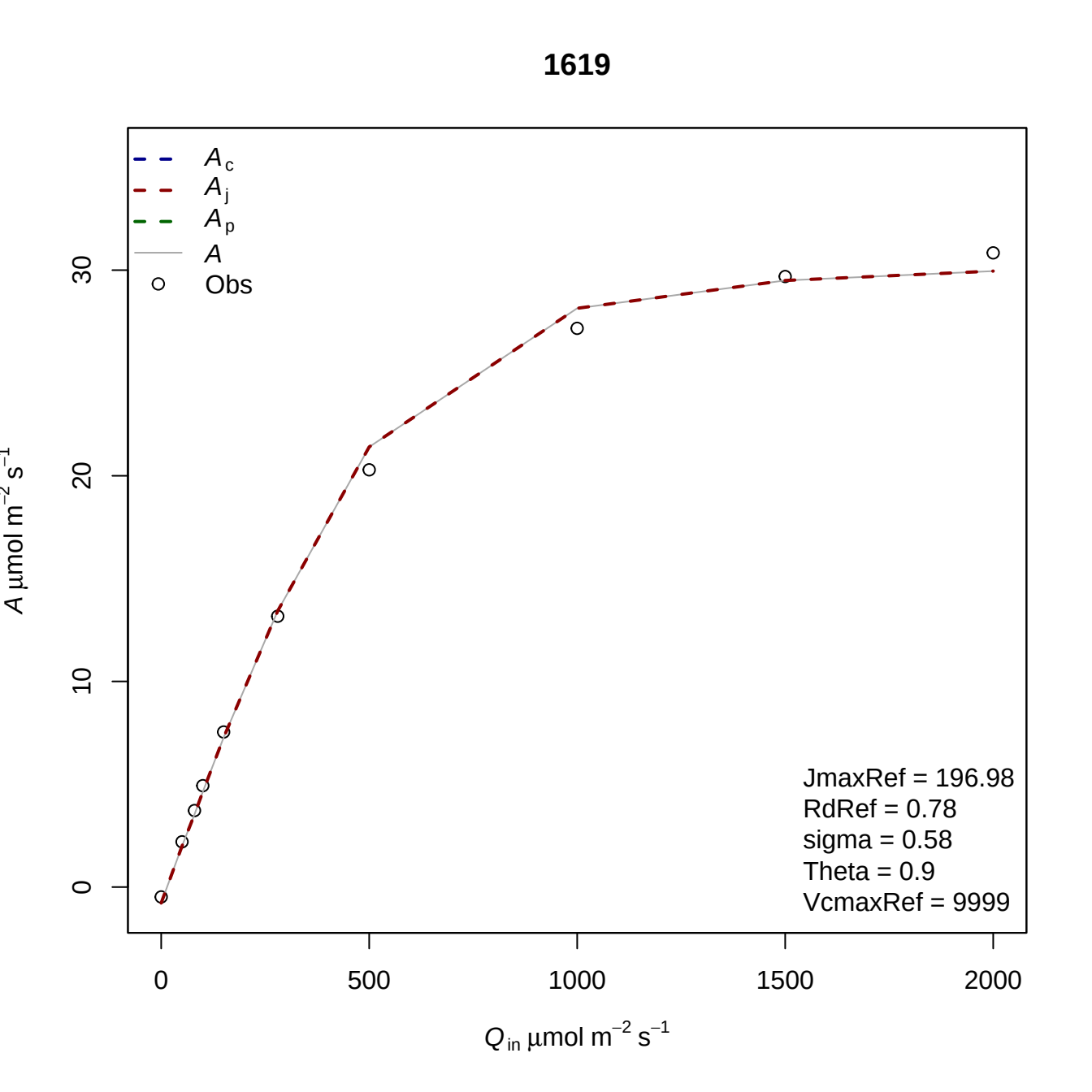

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 153.15
RdRef = 0.95
sigma = 0.46
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

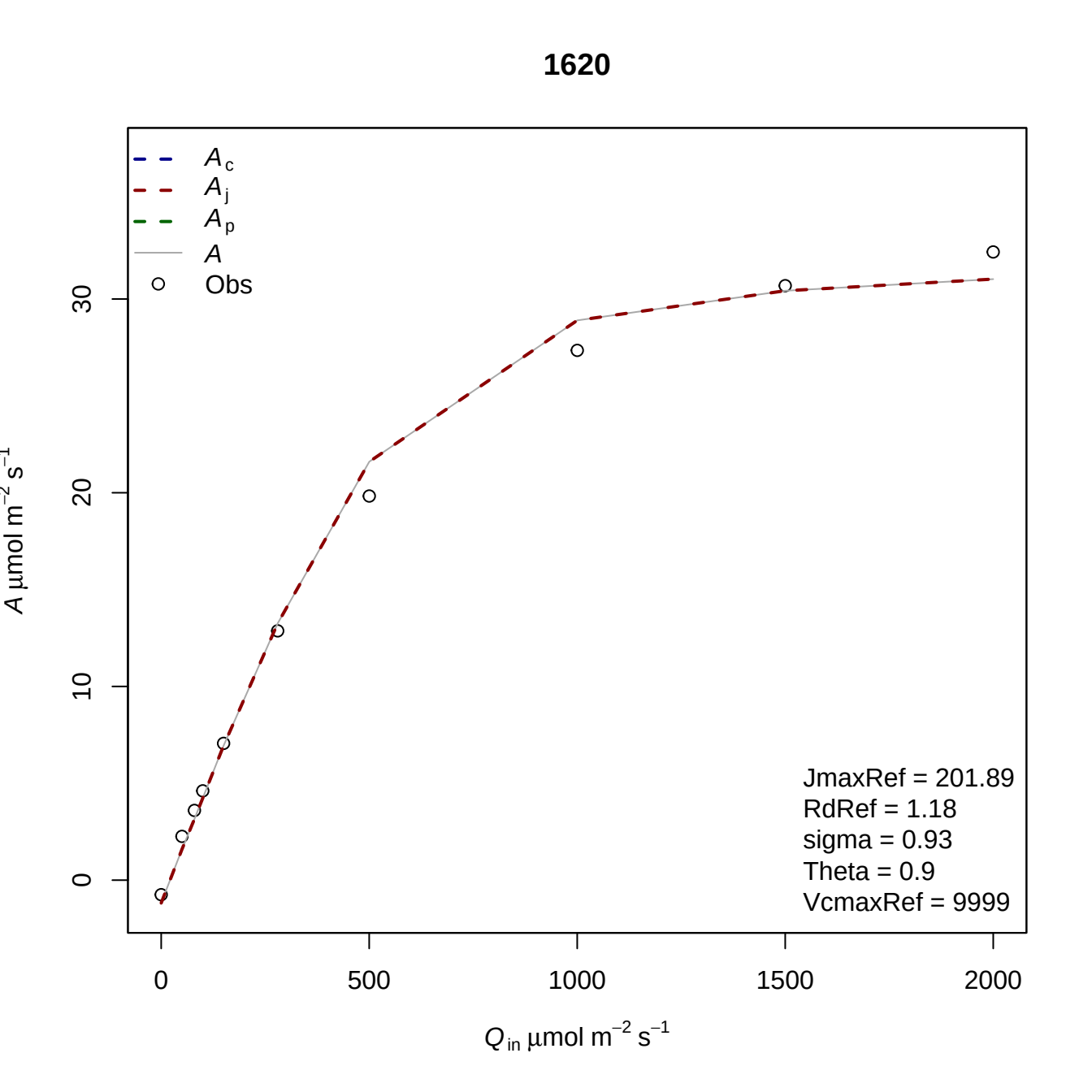

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 132.34
RdRef = 1.29
sigma = 0.75
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

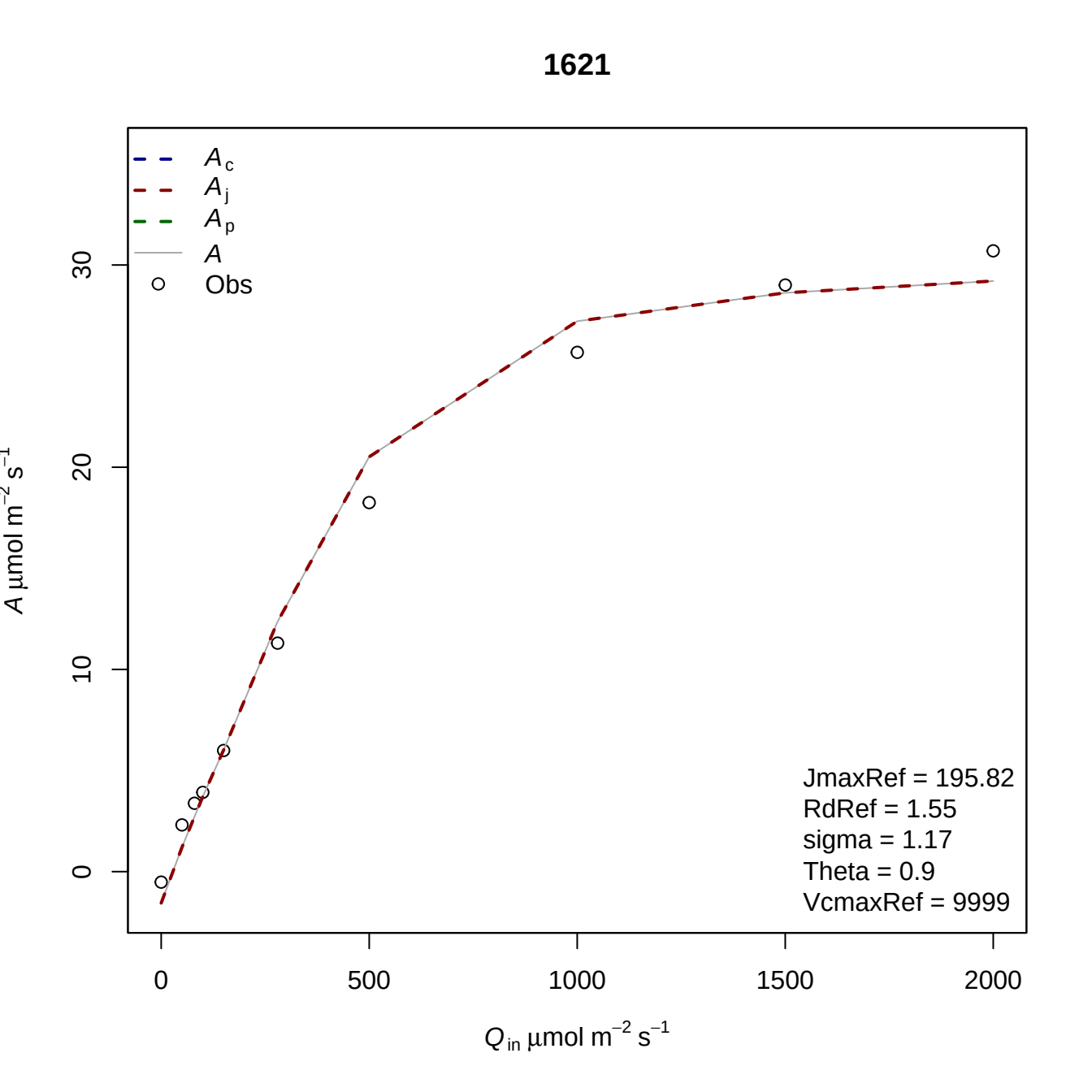

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 151.99
RdRef = 0.95
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

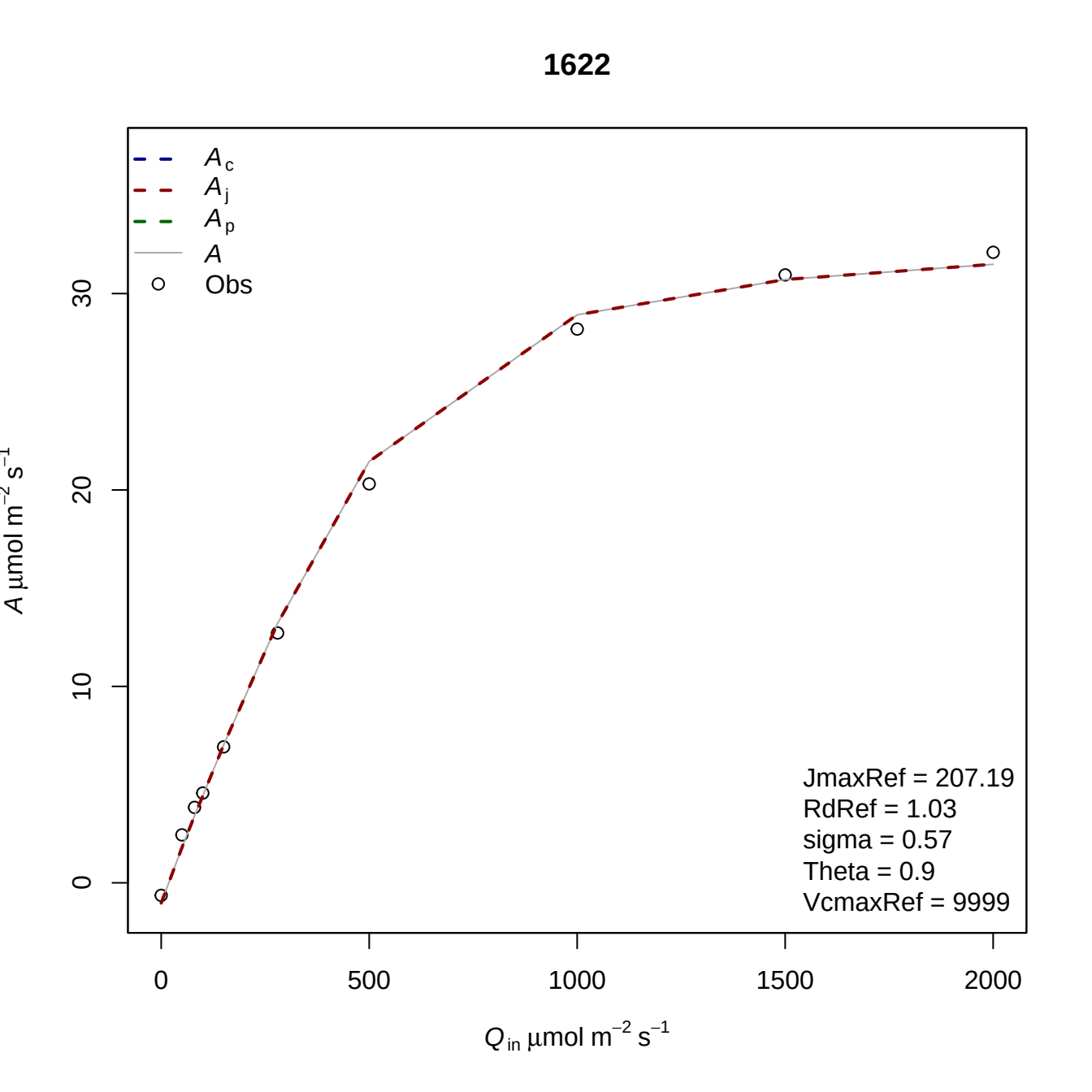

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 127.7
RdRef = 1.42
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

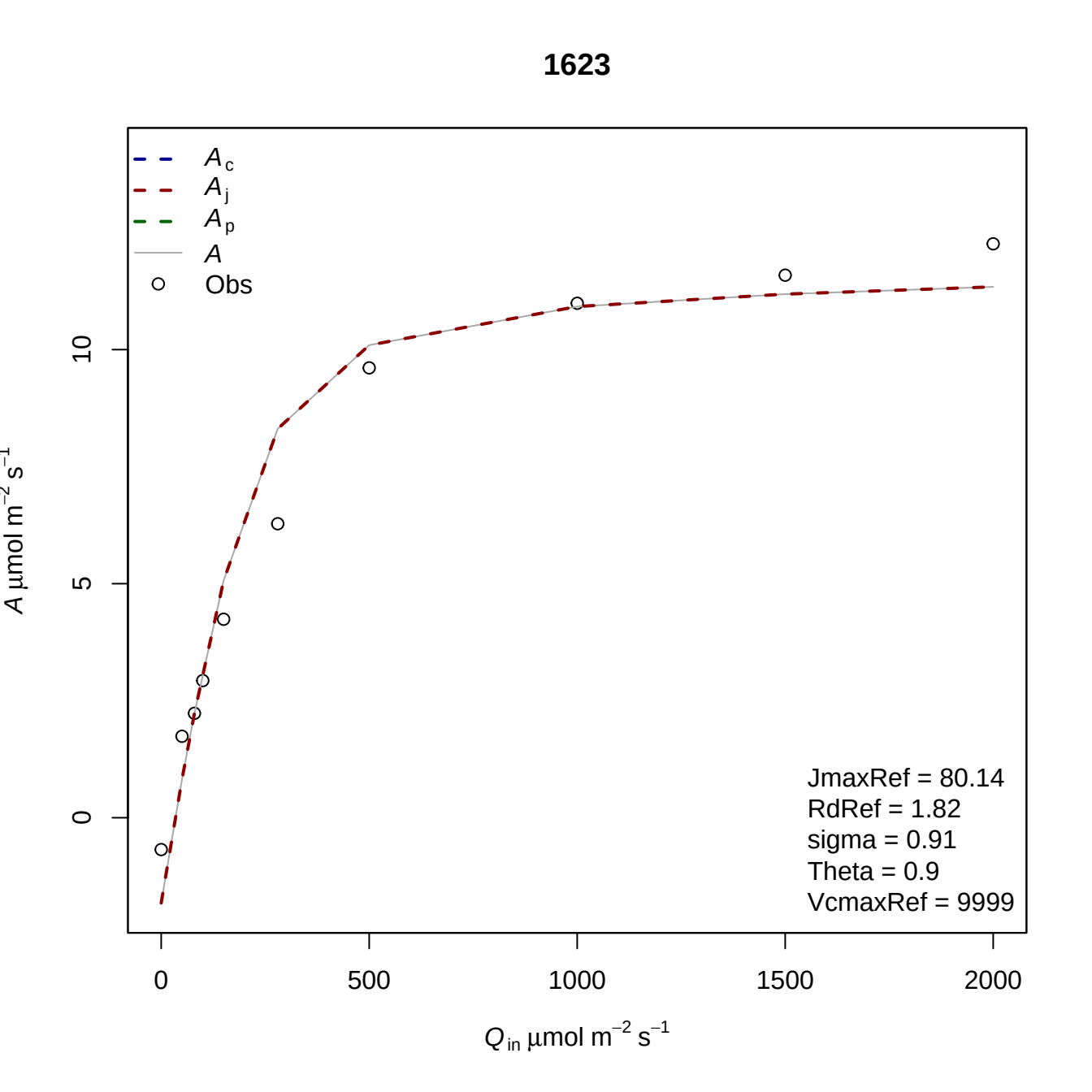

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 136.04
RdRef = 1.09
sigma = 0.56
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

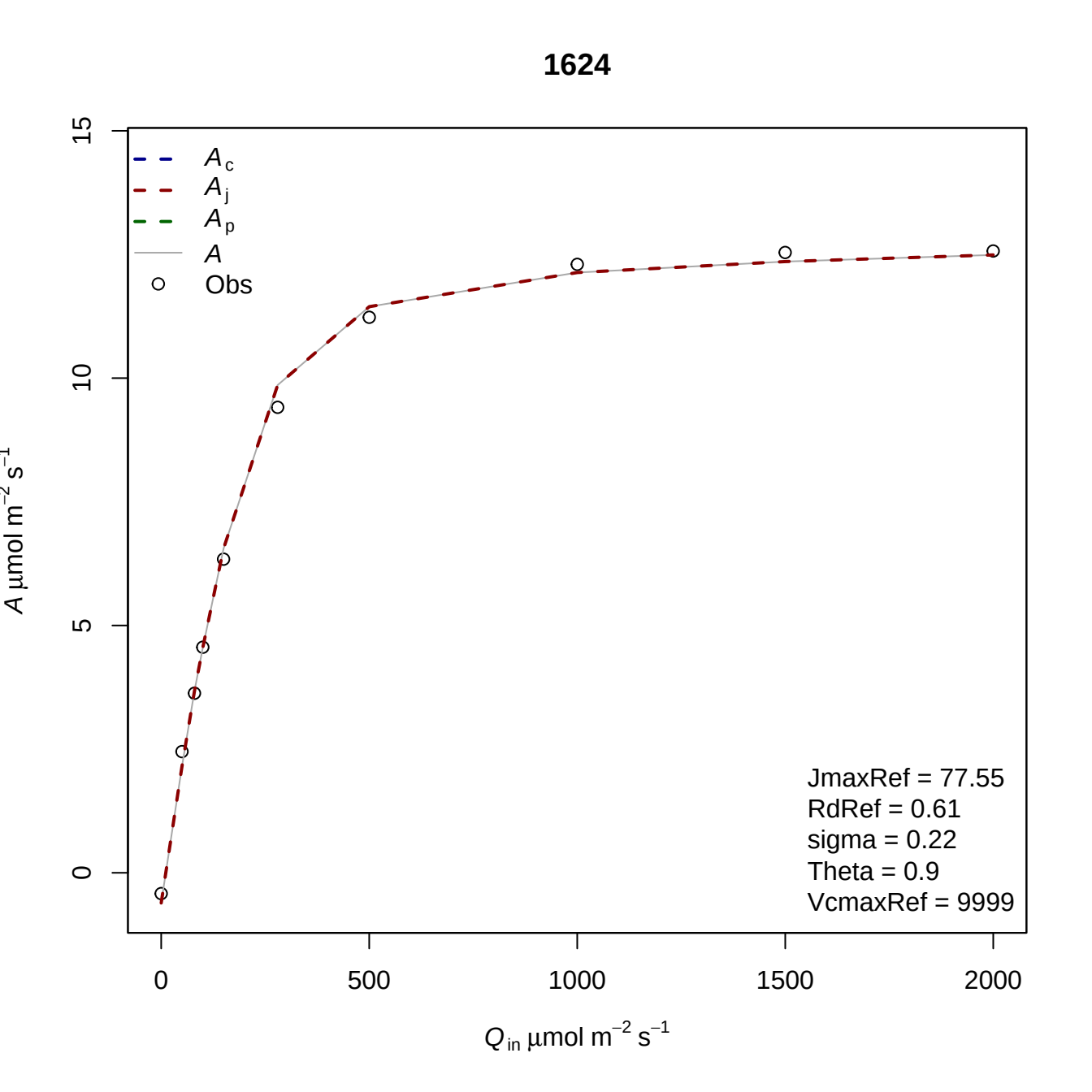

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 196.98
RdRef = 0.78
sigma = 0.58
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

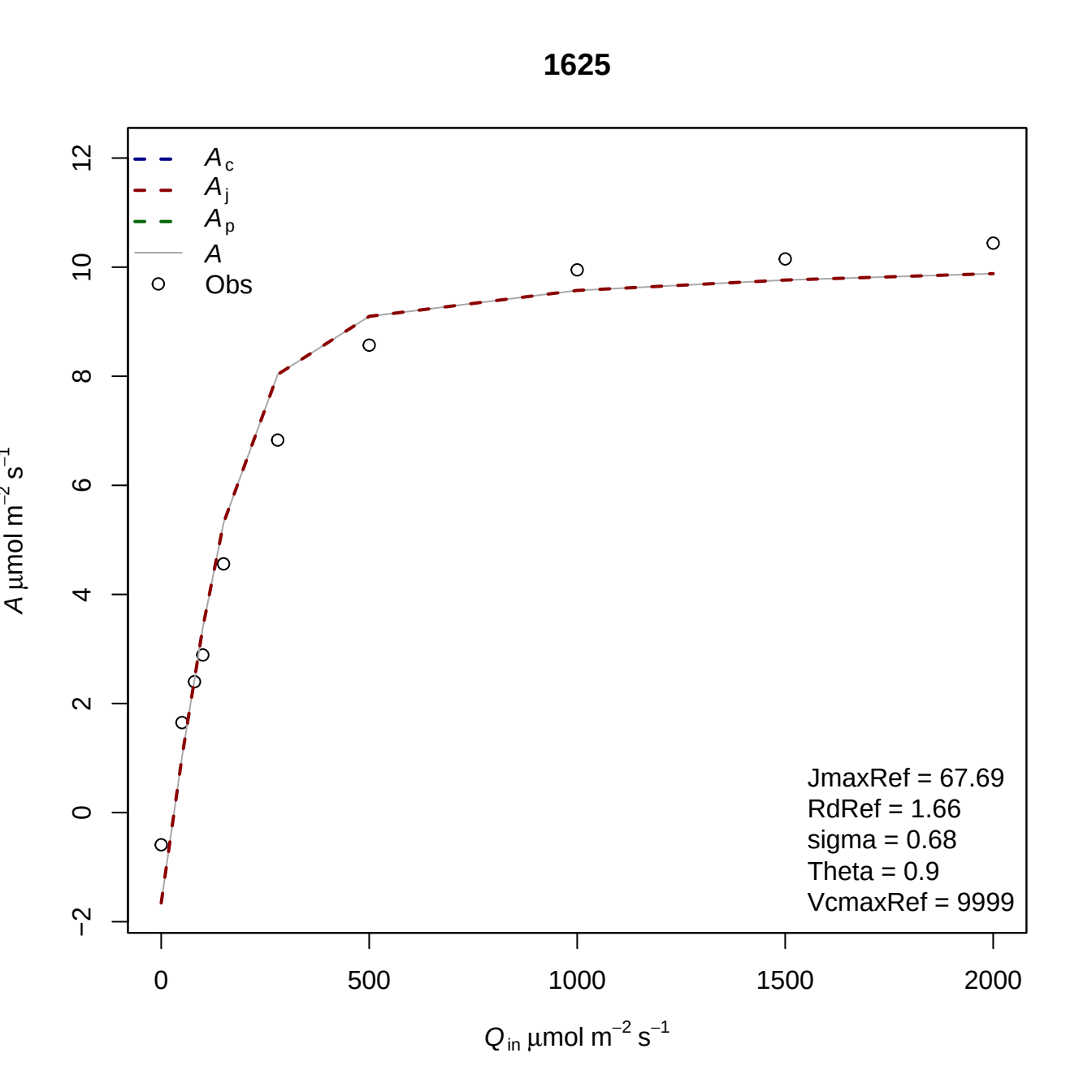

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 201.89
RdRef = 1.18
sigma = 0.93
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

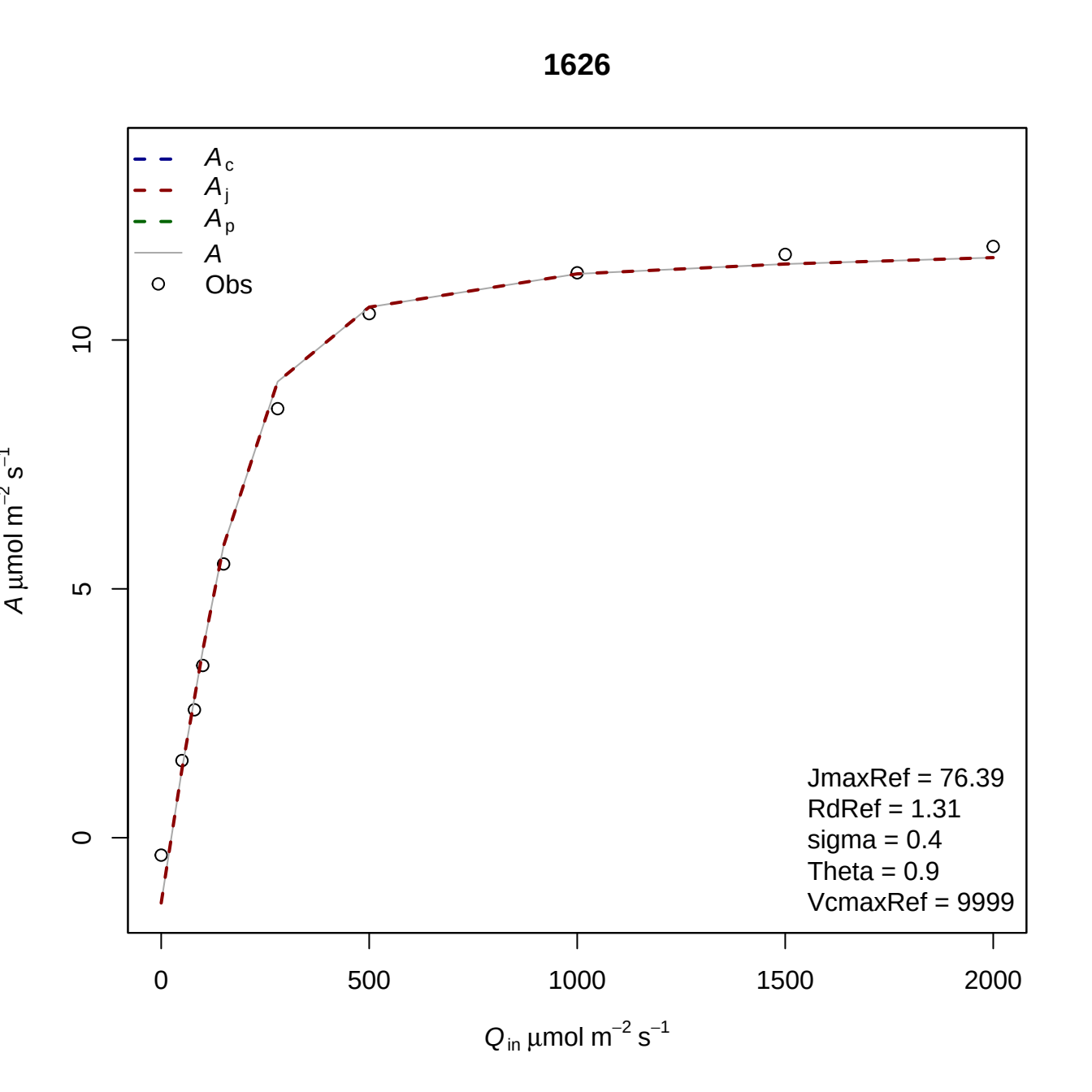

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 195.82
RdRef = 1.55
sigma = 1.17
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

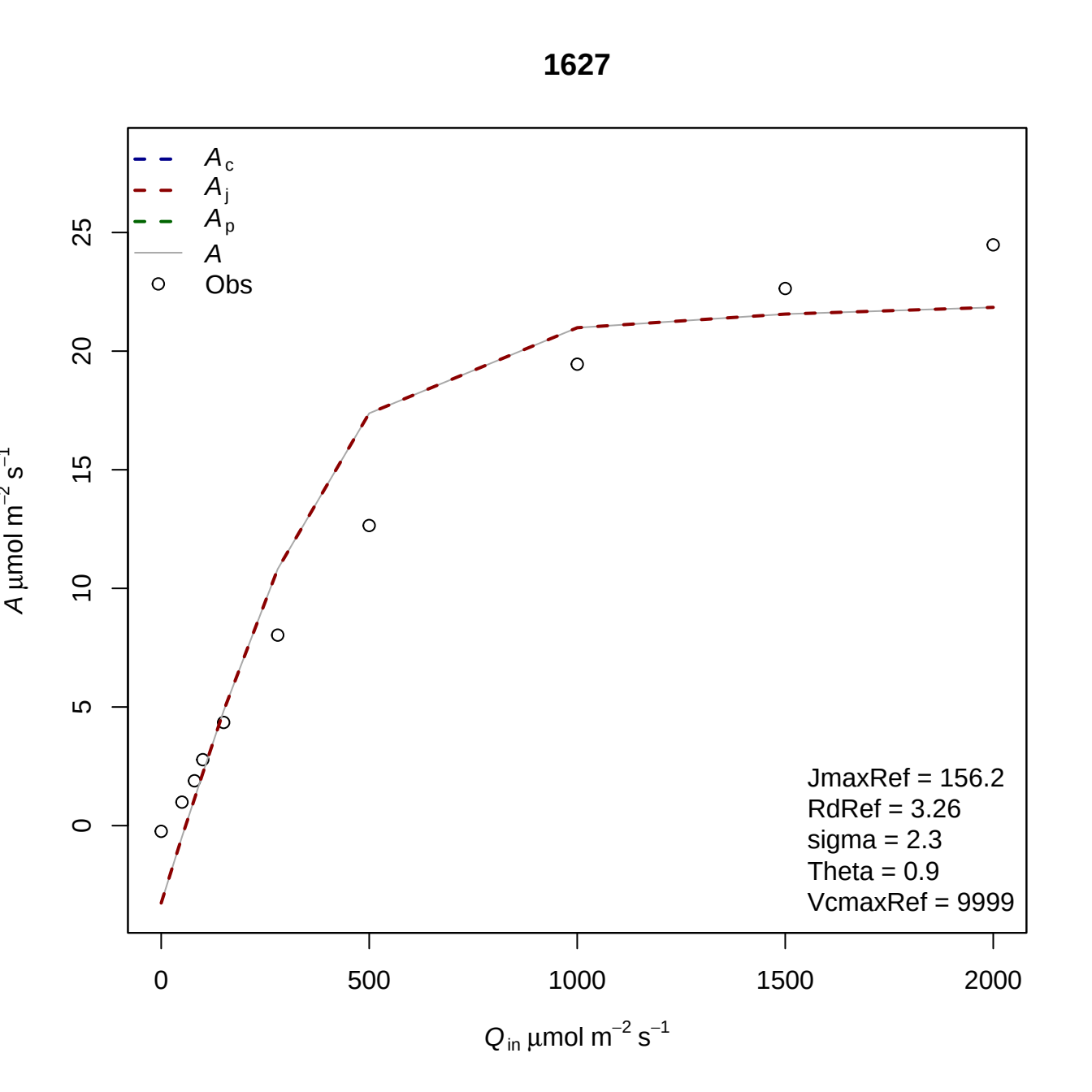

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 207.19
RdRef = 1.03
sigma = 0.57
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

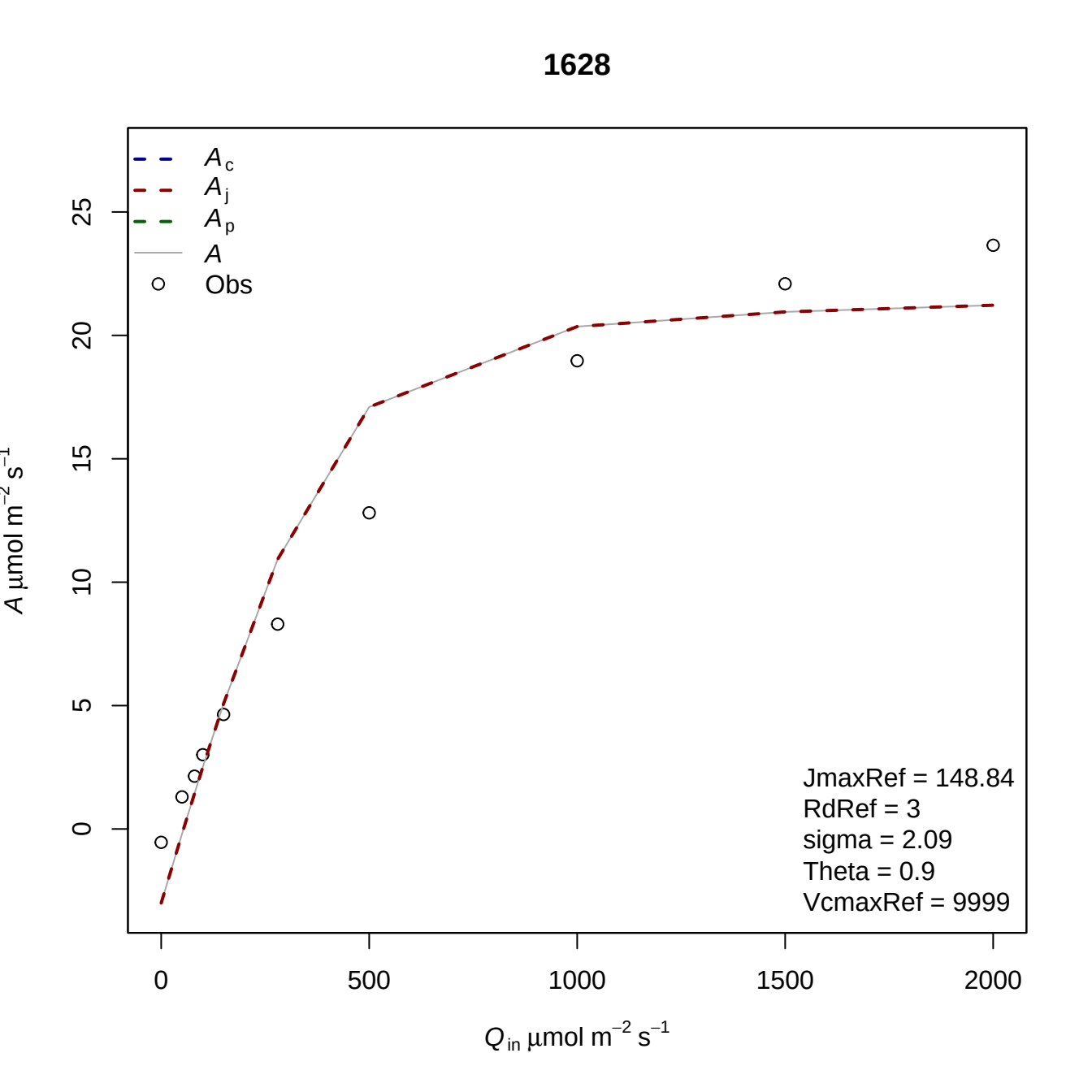

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 80.14
RdRef = 1.82
sigma = 0.91
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

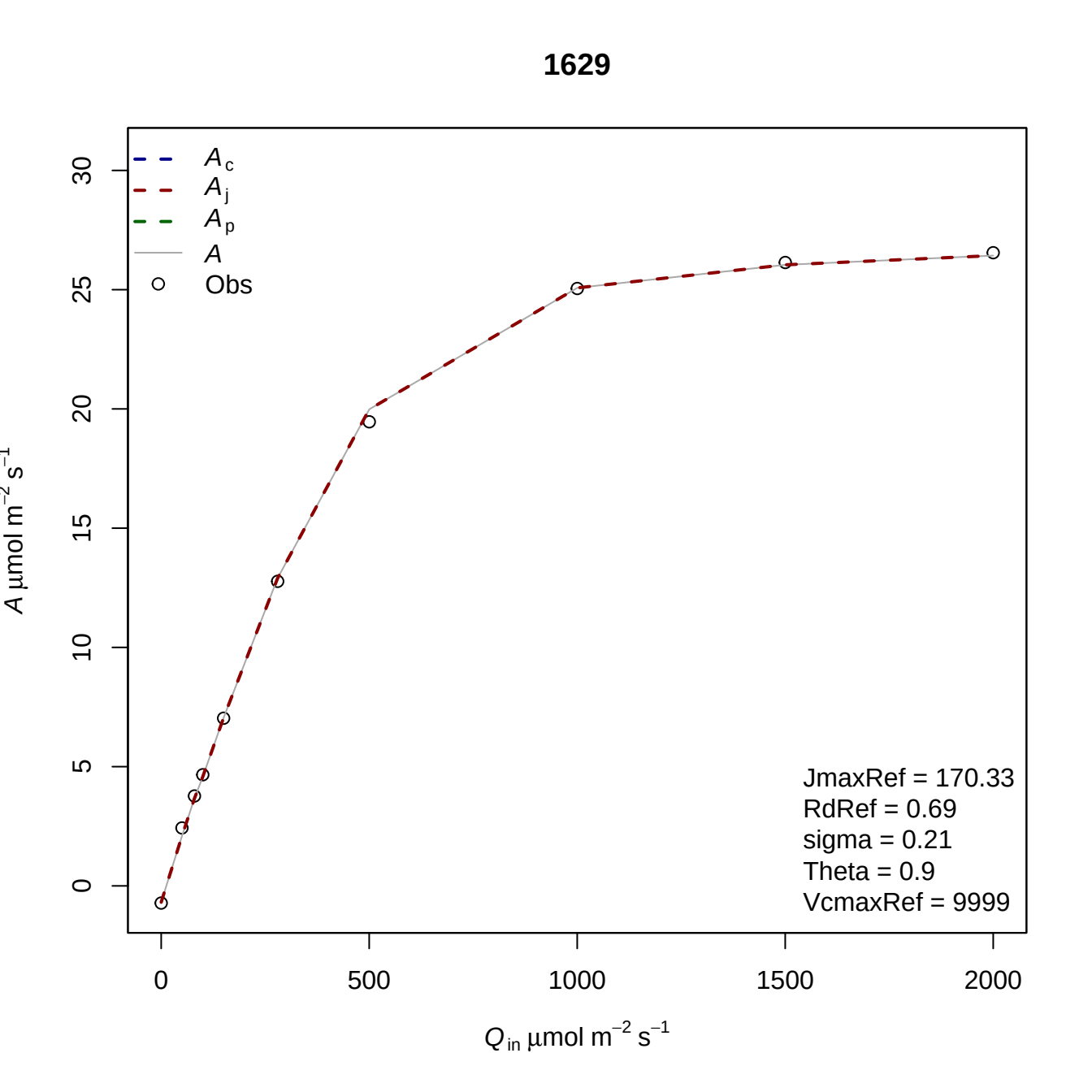

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 77.55
RdRef = 0.61
sigma = 0.22
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

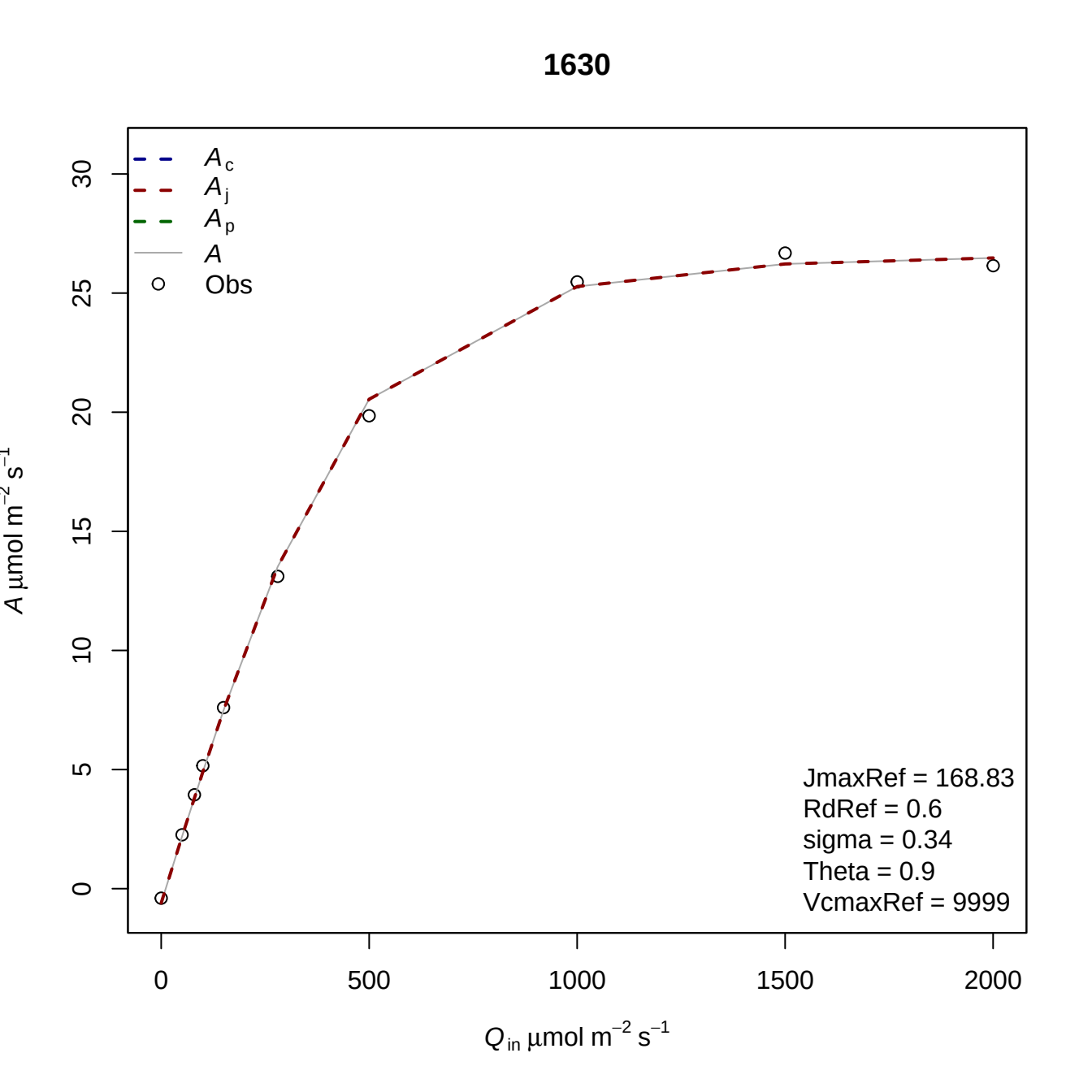

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 67.69
RdRef = 1.66
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

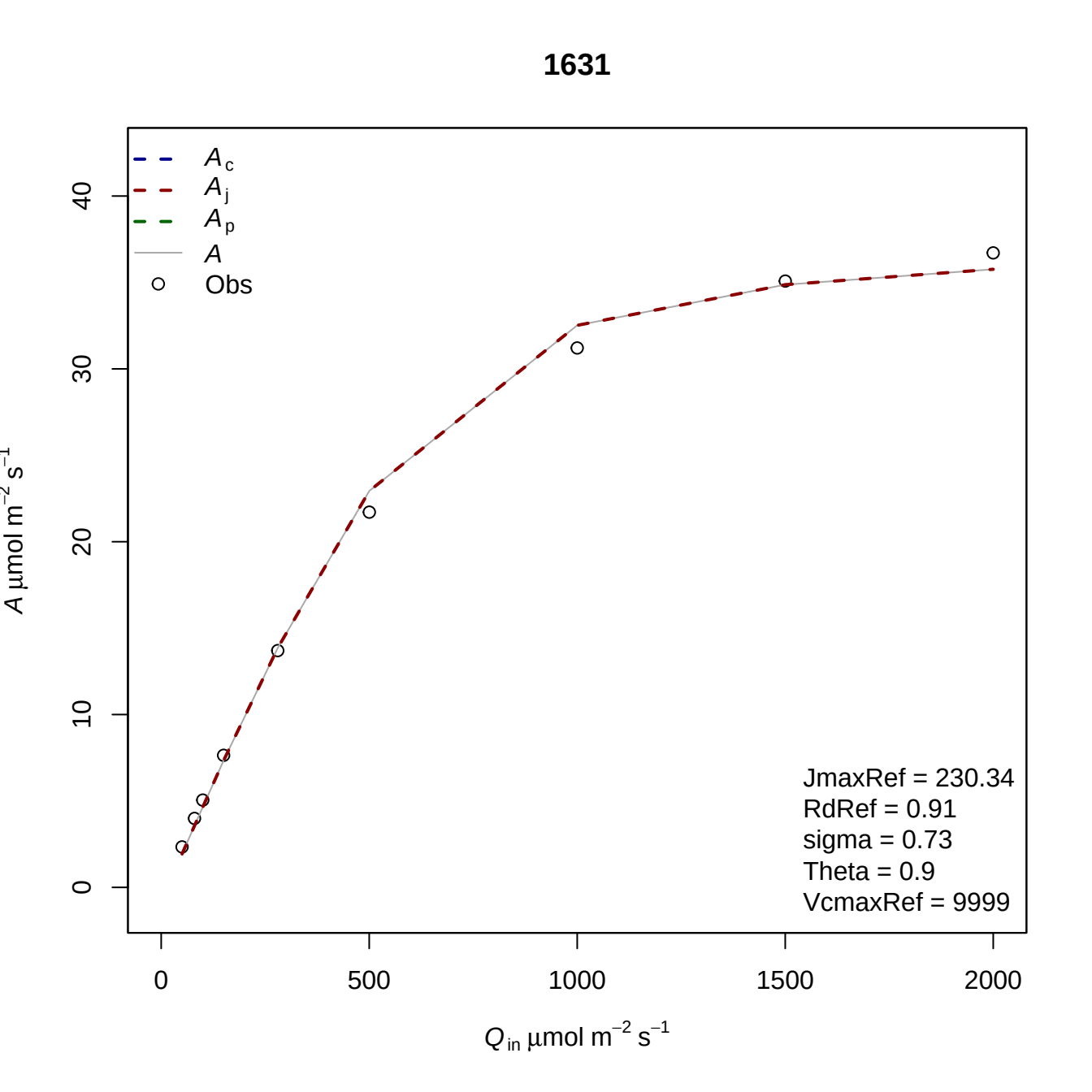

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 76.39
RdRef = 1.31
sigma = 0.4
Theta = 0.9
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

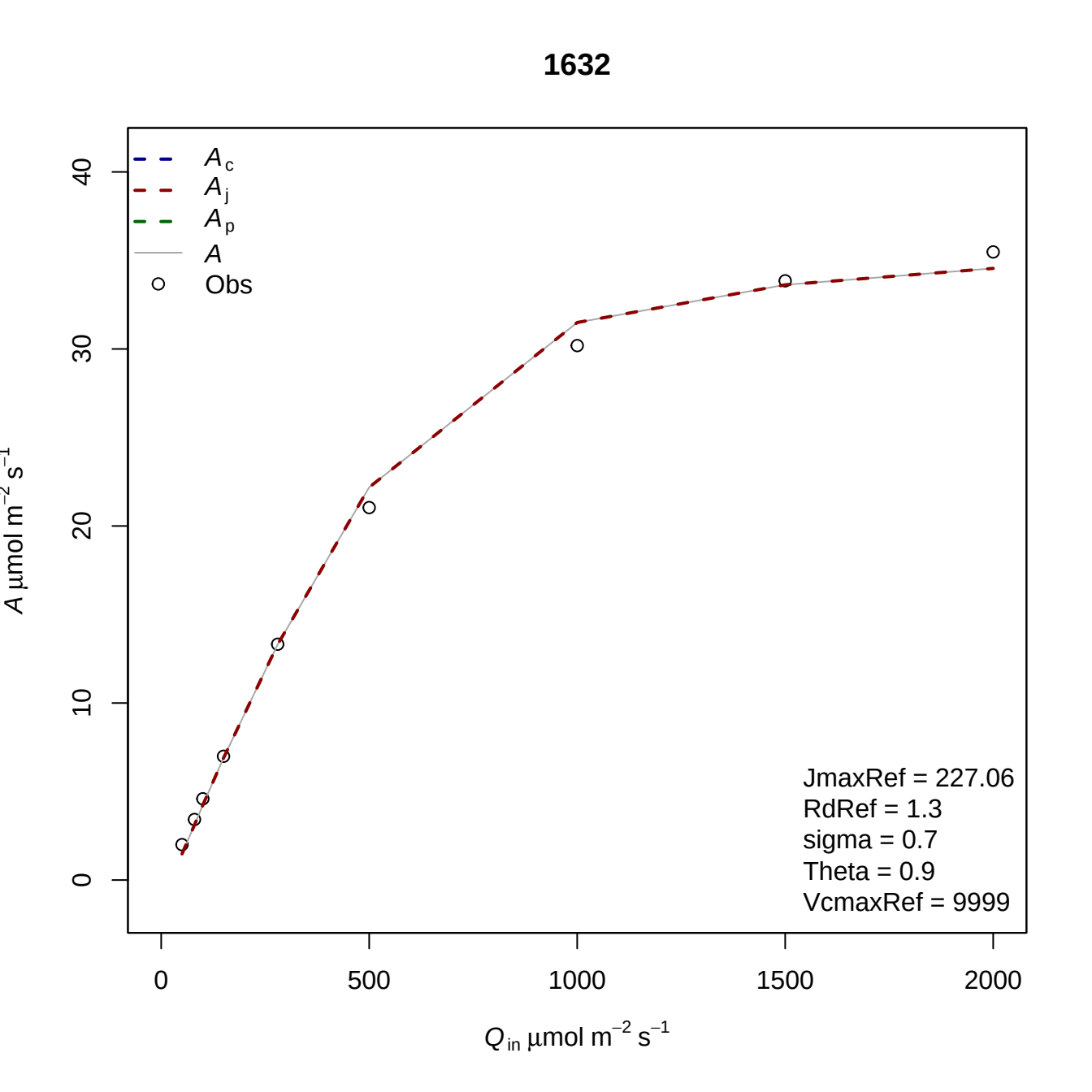

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 156.2
RdRef = 3.26
sigma = 2.3
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

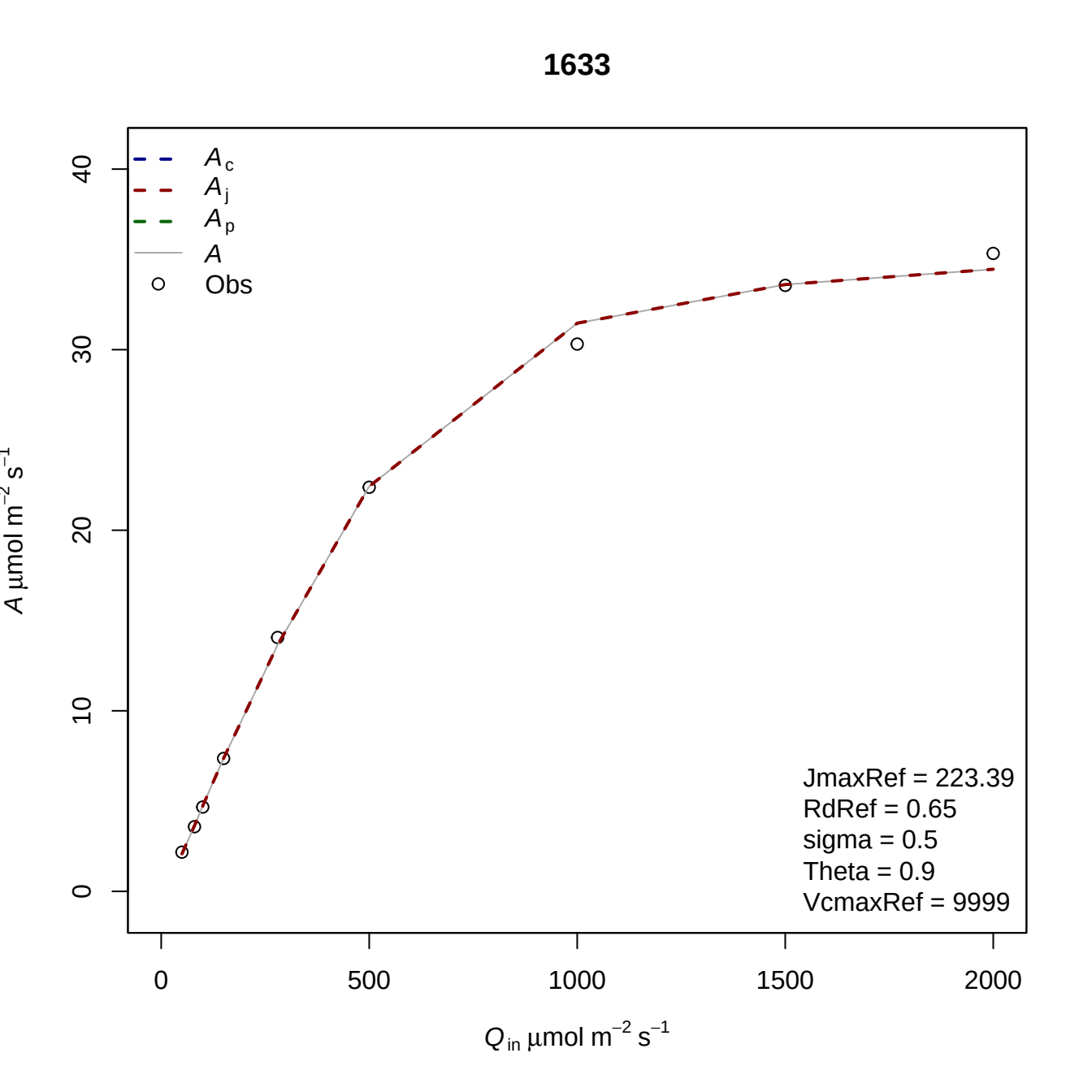

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 148.84
RdRef = 3
sigma = 2.09
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

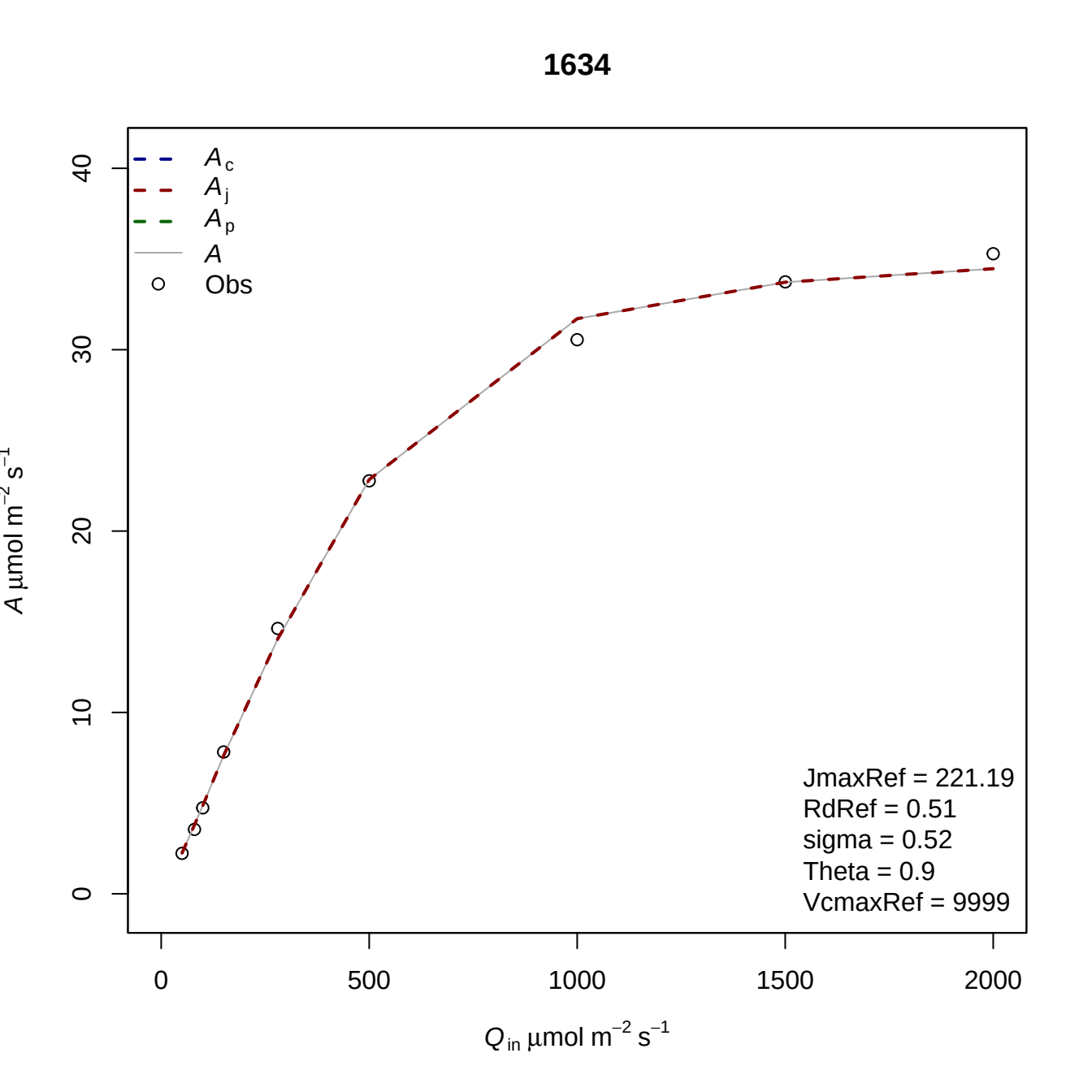

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 170.33
RdRef = 0.69
sigma = 0.21
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

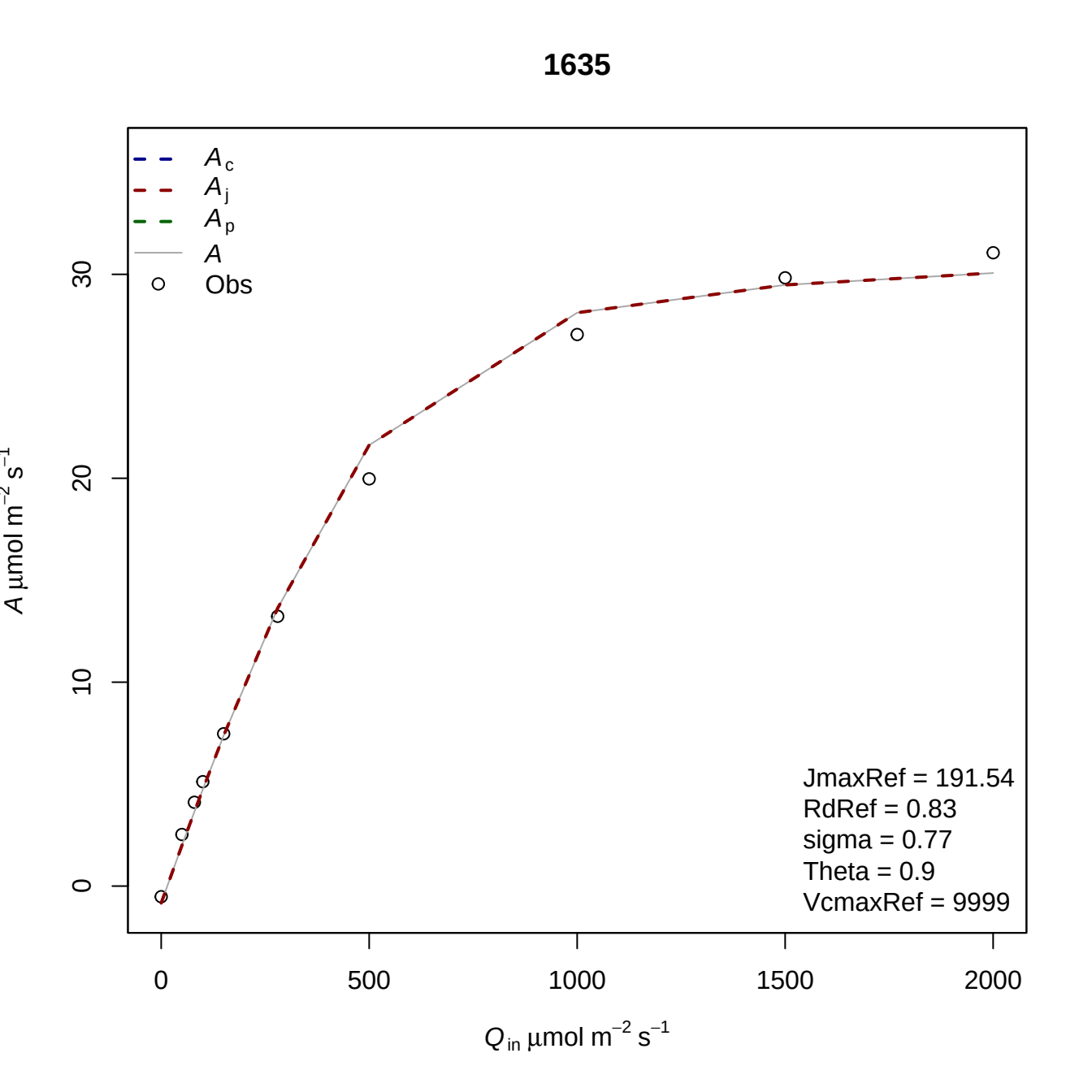

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 168.83
RdRef = 0.6
sigma = 0.34
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

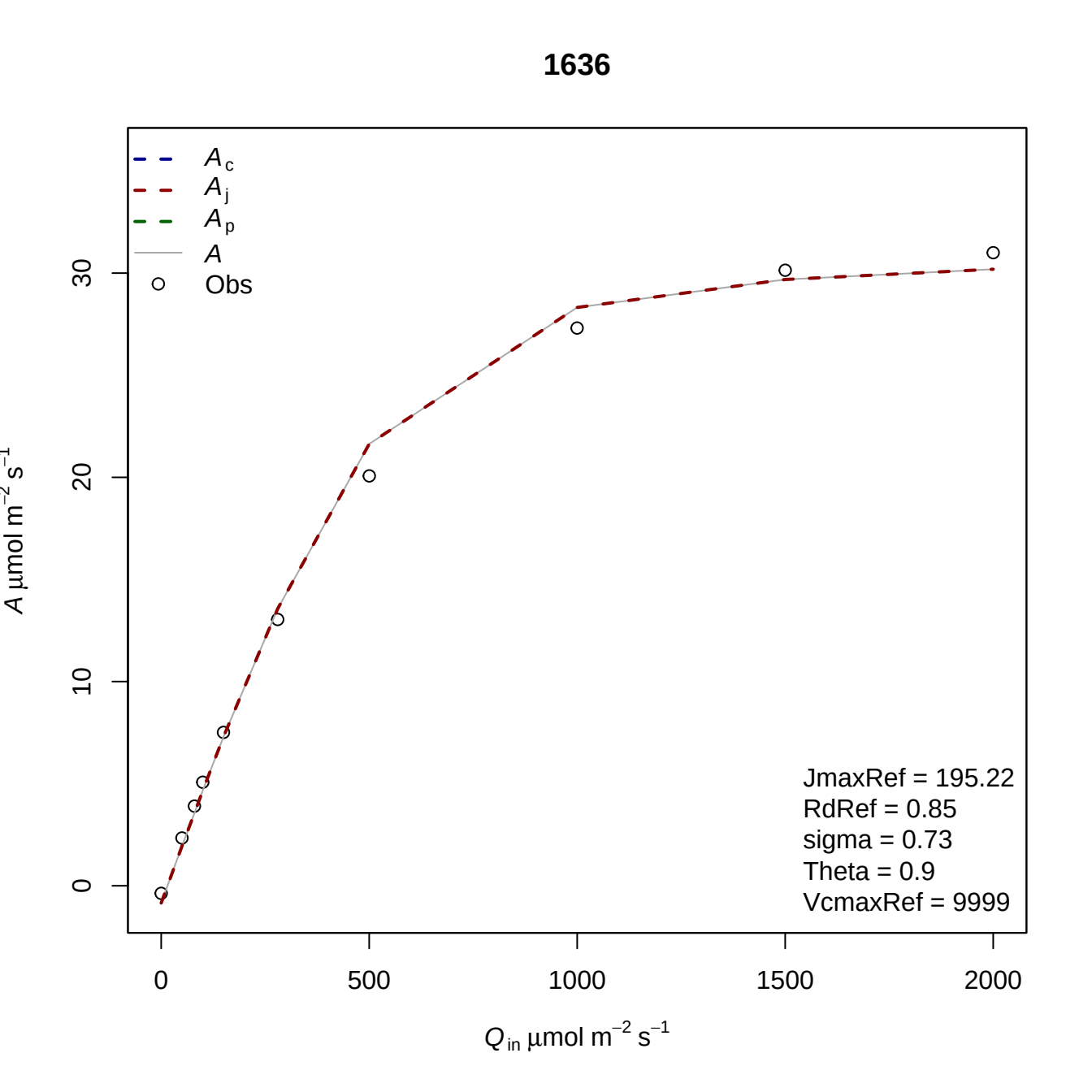

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 230.34
RdRef = 0.91
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
40
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

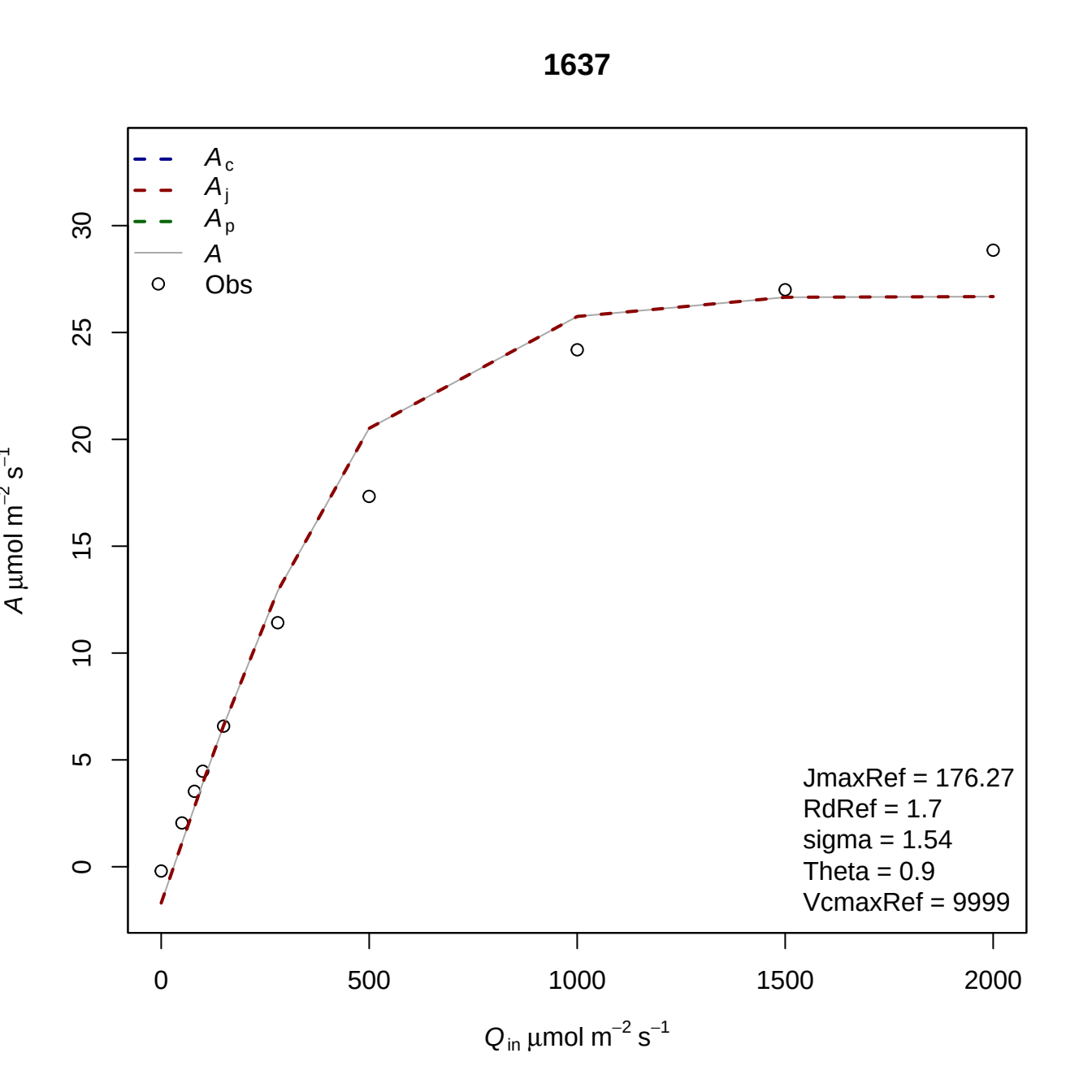

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 227.06
RdRef = 1.3
sigma = 0.7
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

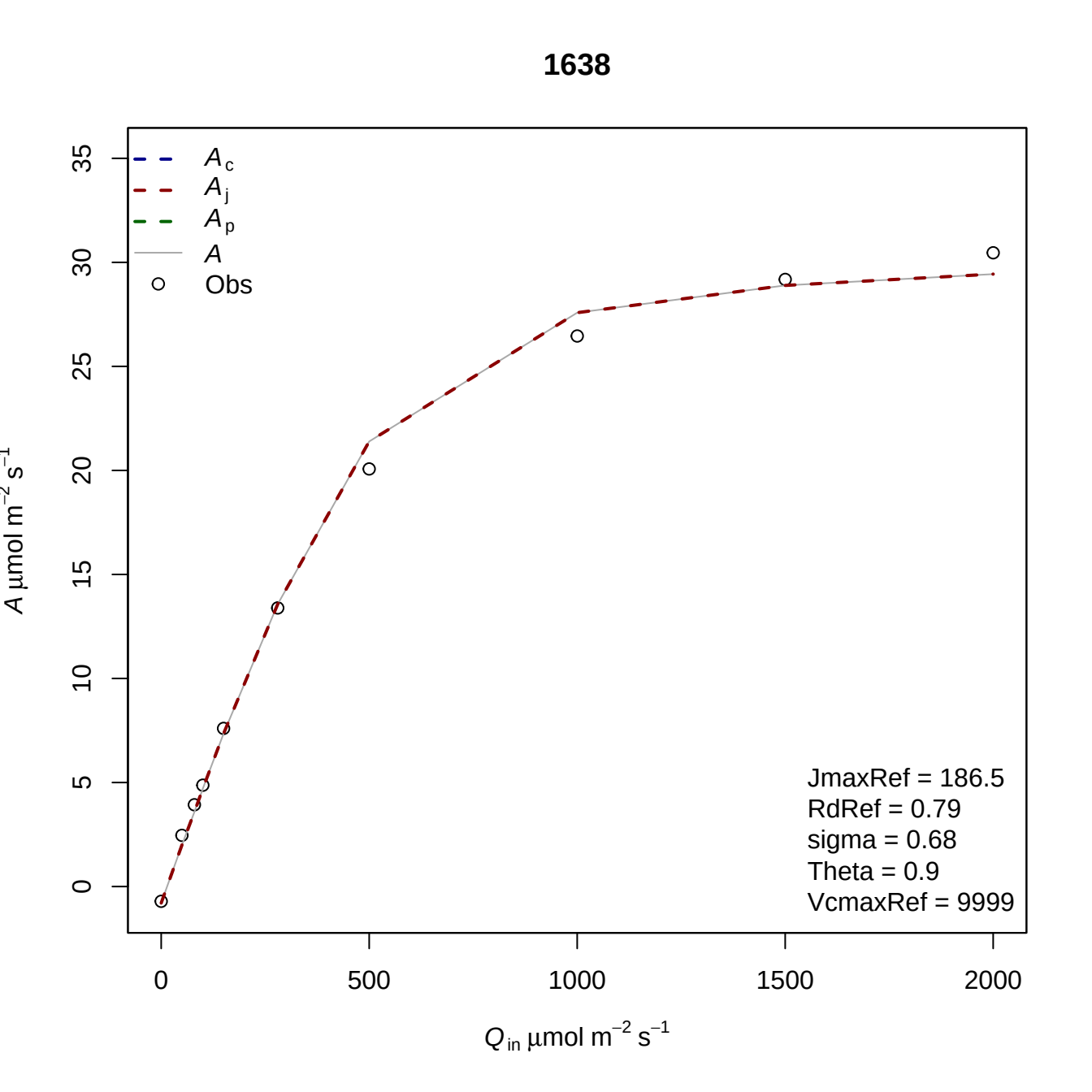

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 223.39
RdRef = 0.65
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

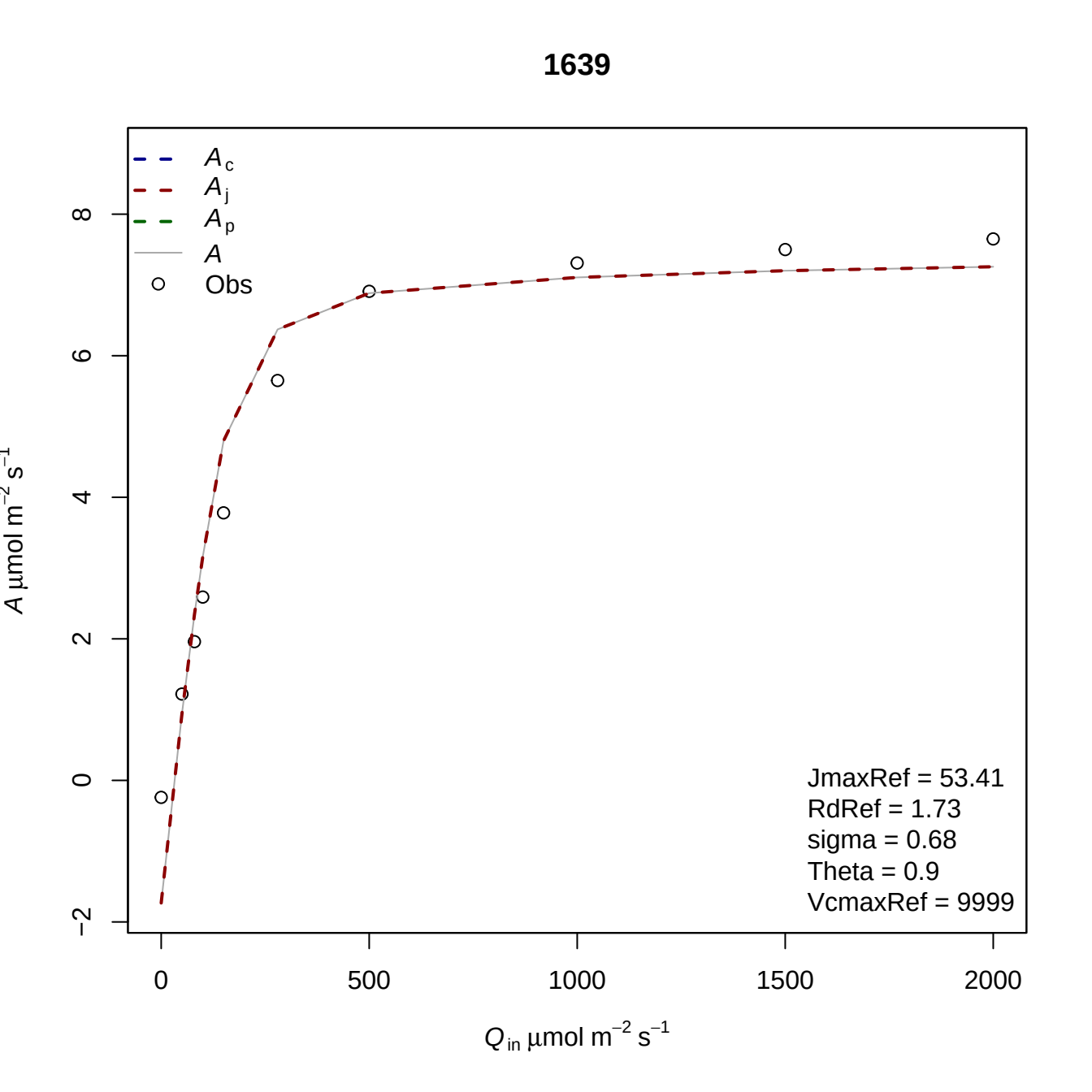

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 221.19
RdRef = 0.51
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

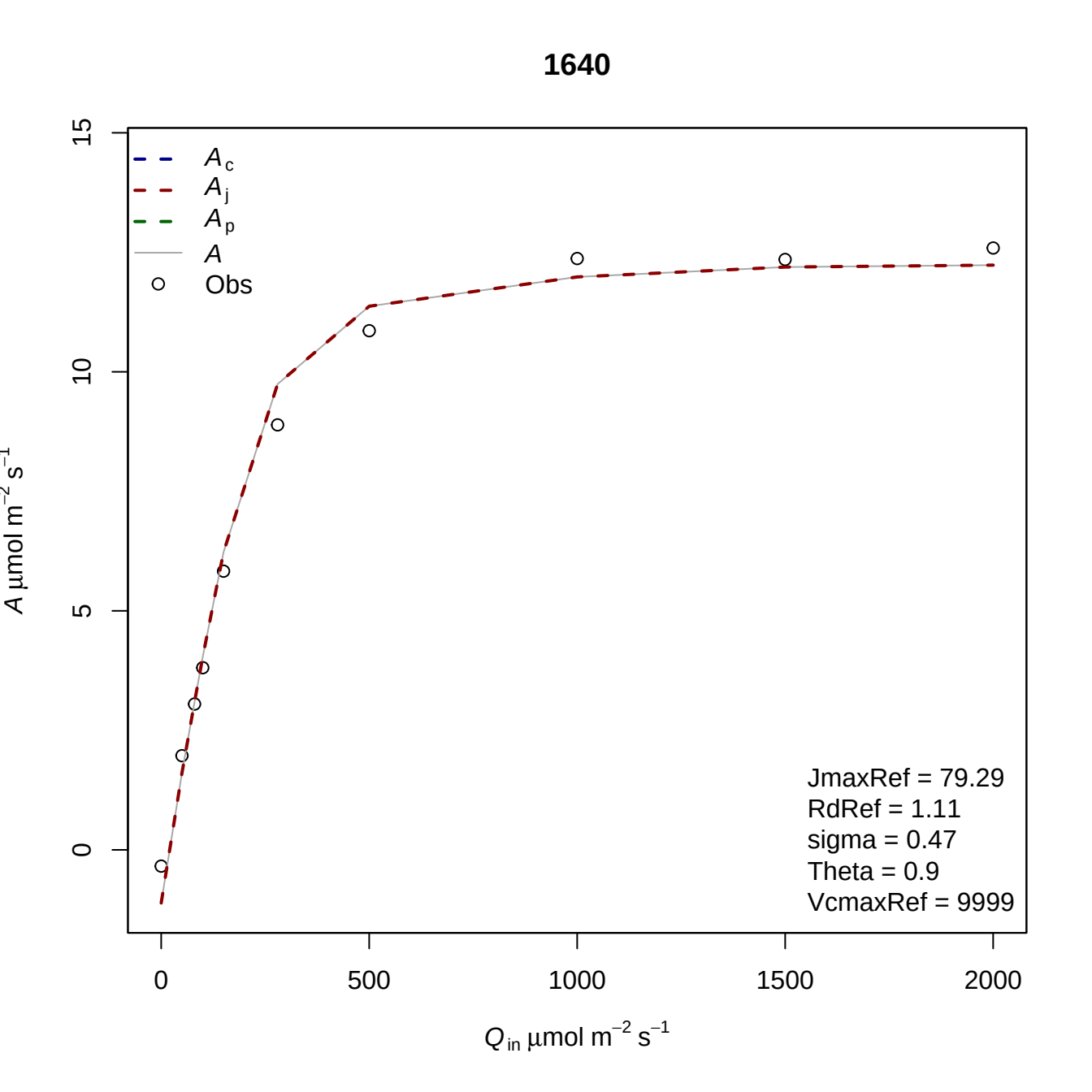

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 191.54
RdRef = 0.83
sigma = 0.77
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

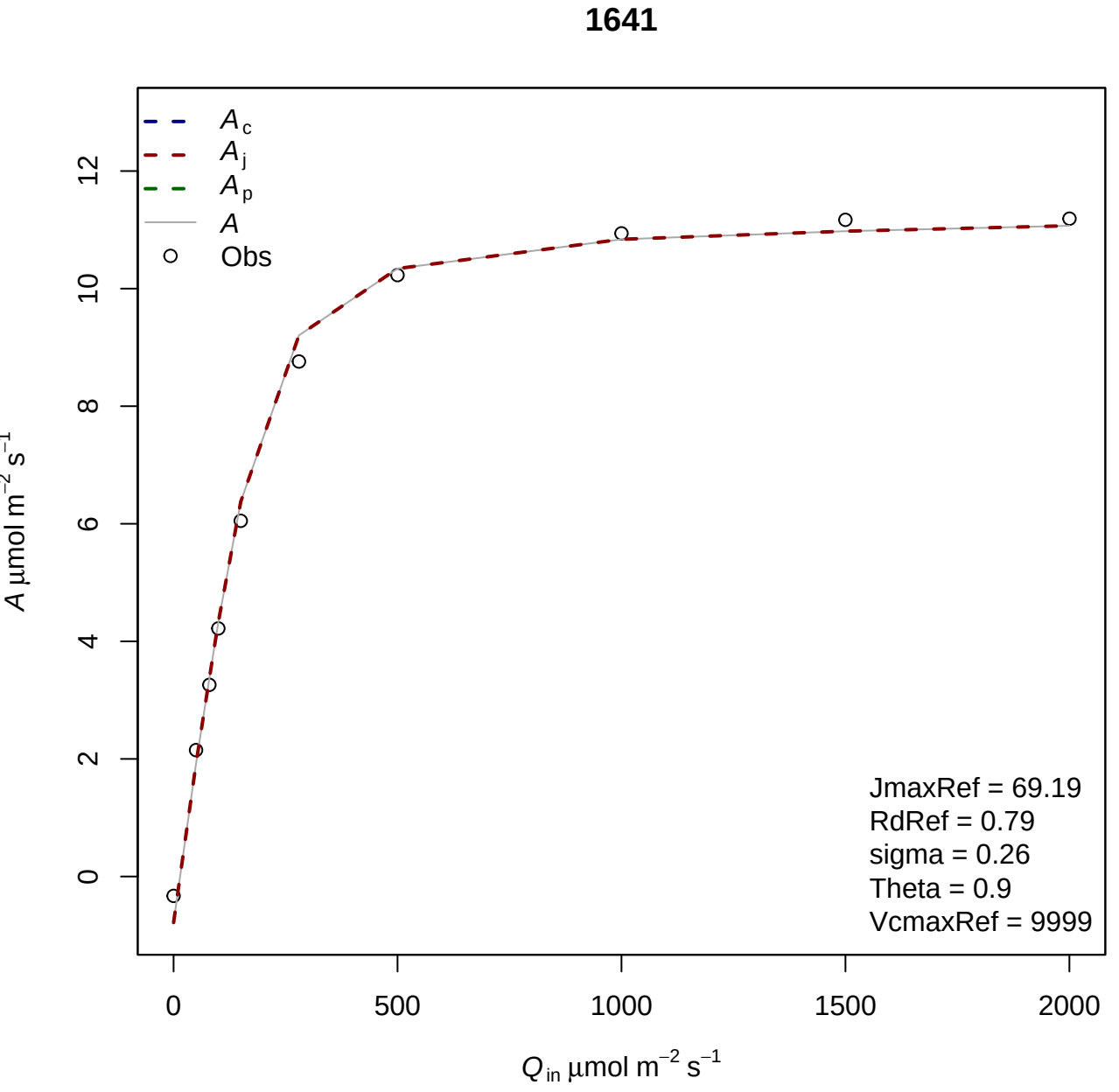

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 195.22
RdRef = 0.85
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

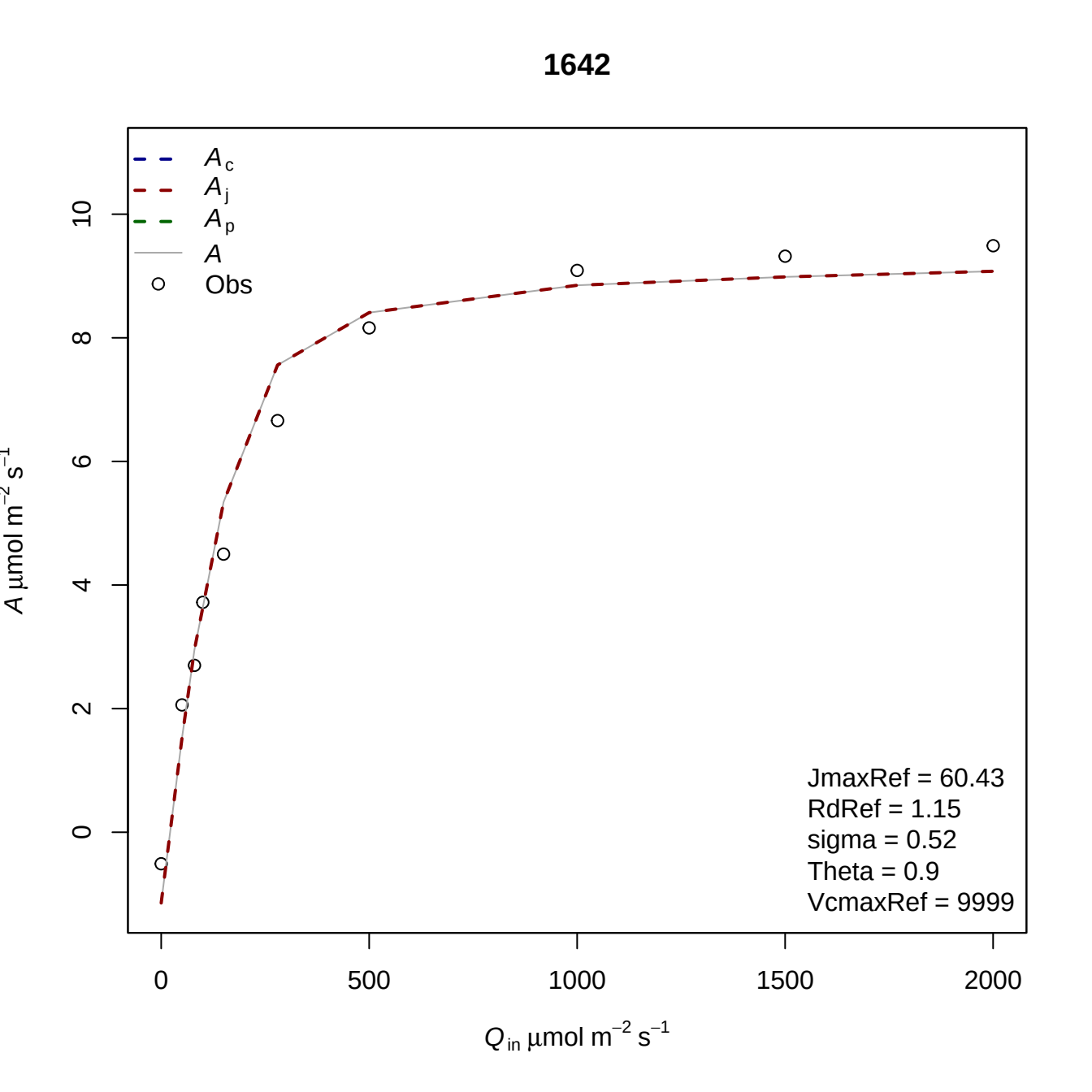

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 176.27
RdRef = 1.7
sigma = 1.54
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

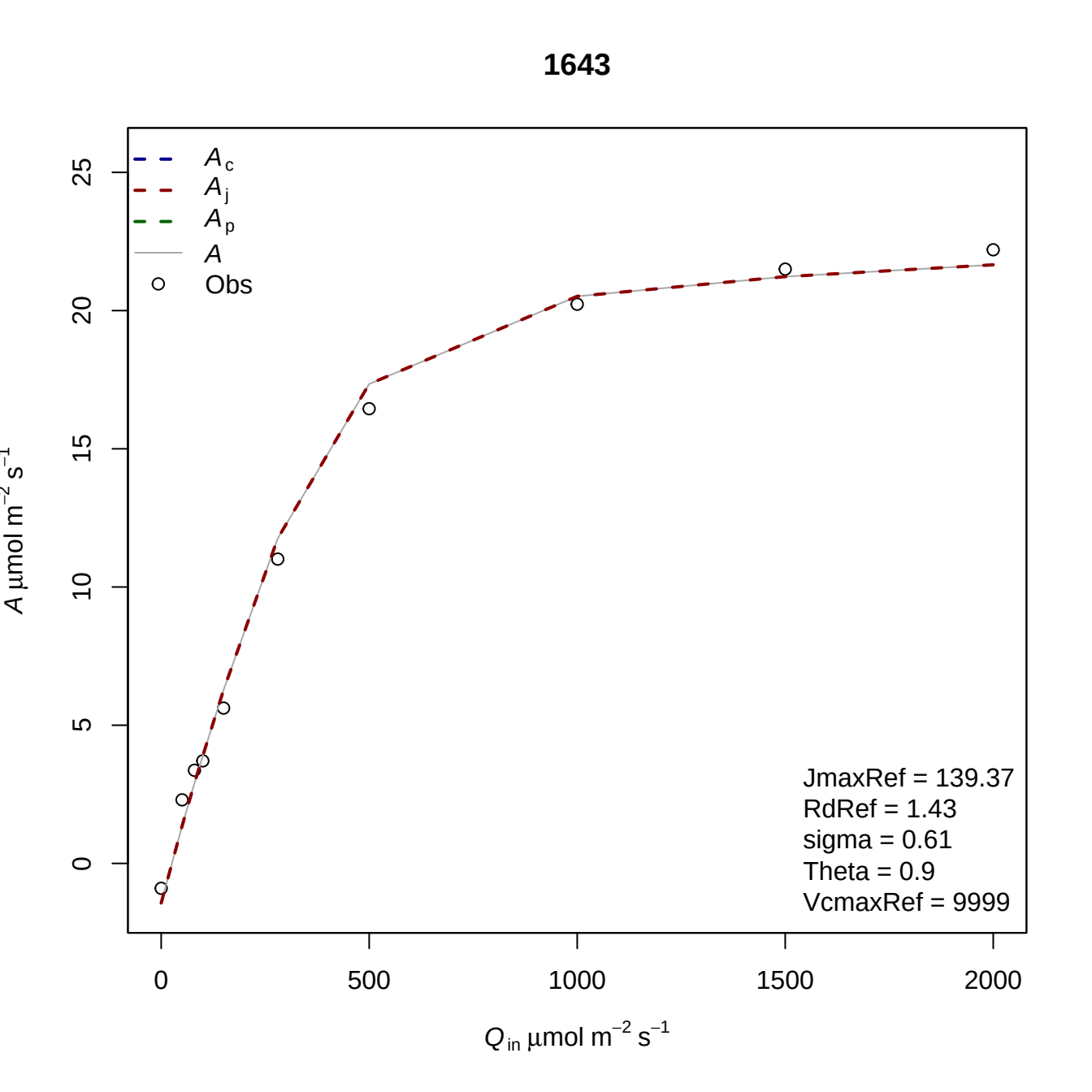

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 186.5
RdRef = 0.79
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

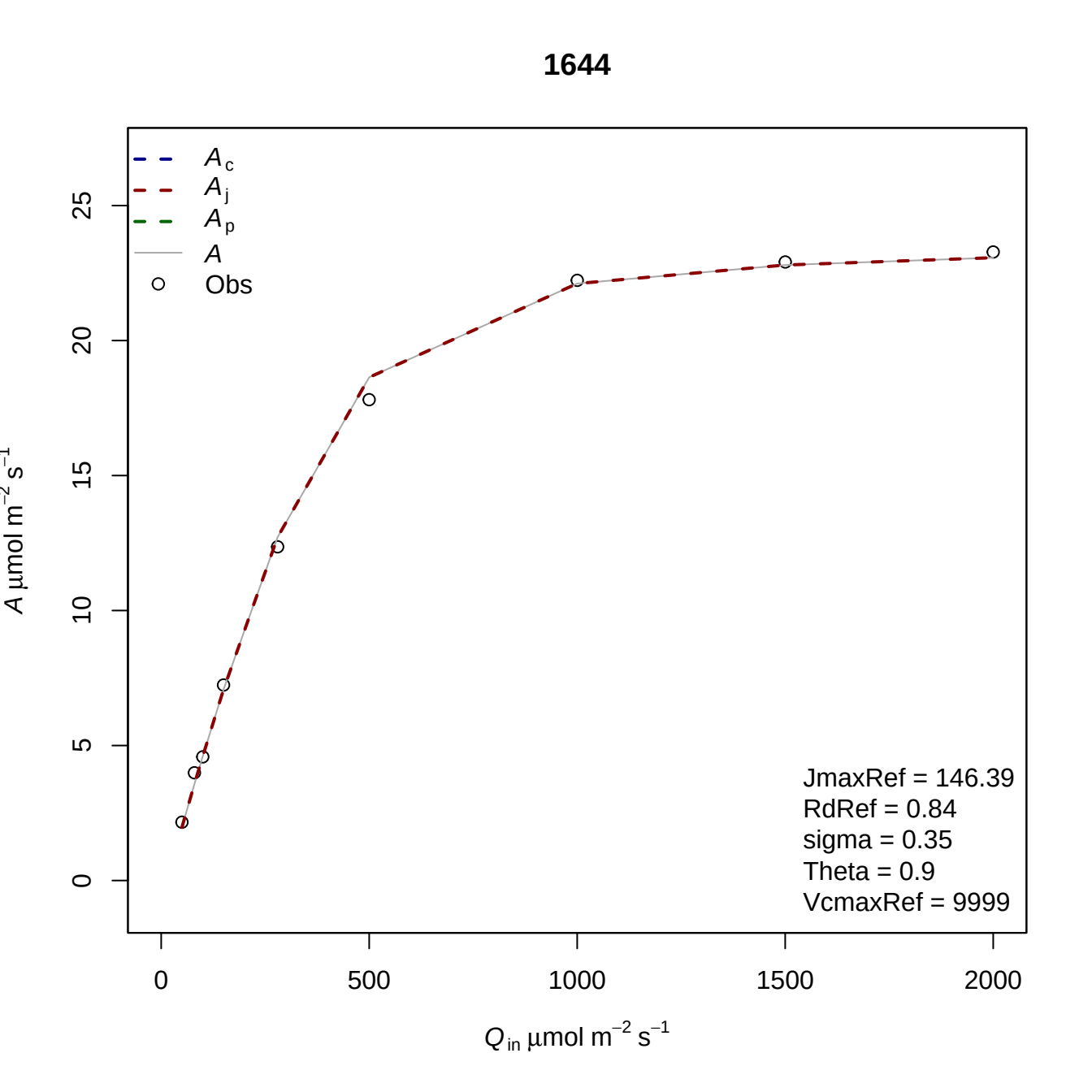

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 53.41
RdRef = 1.73
sigma = 0.68
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

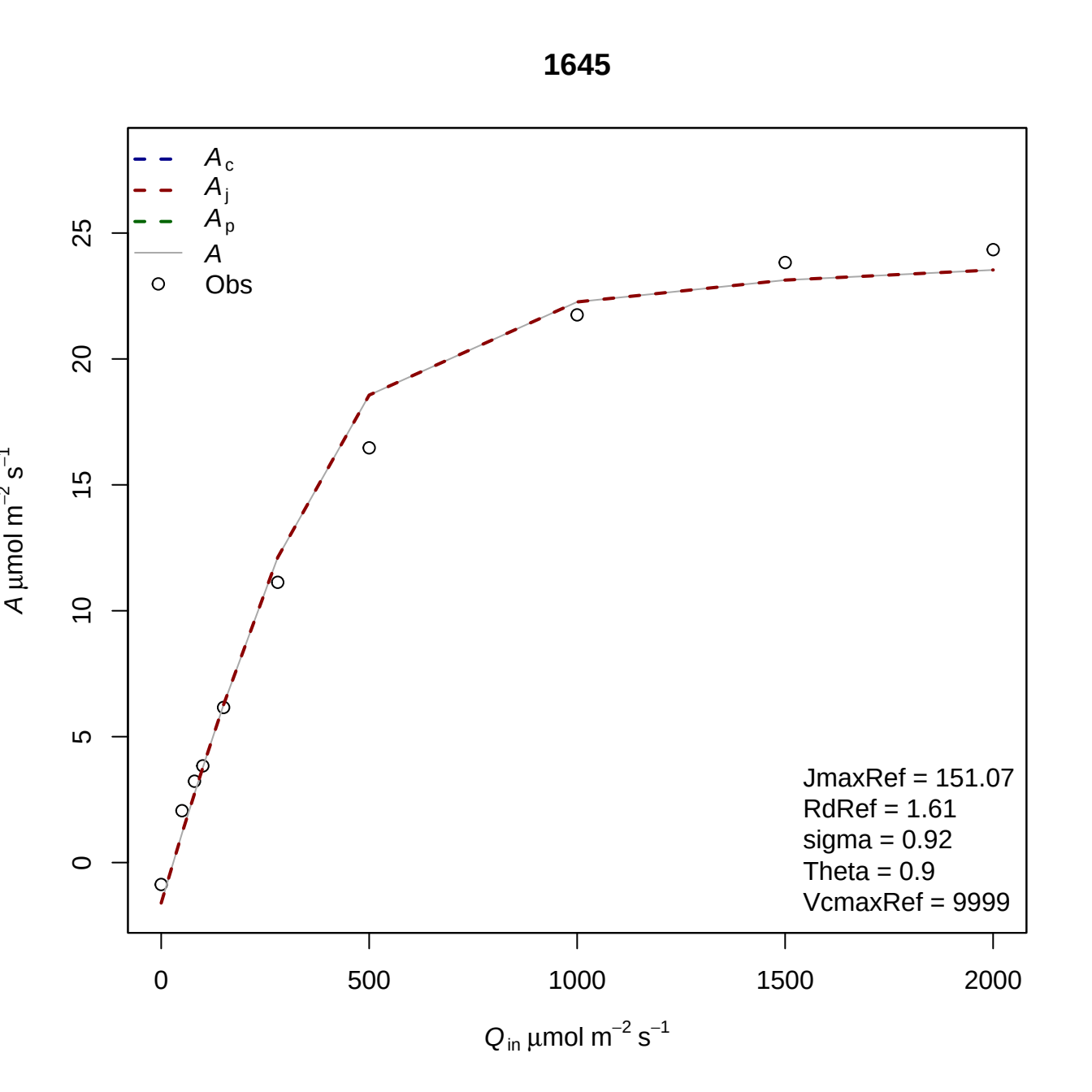

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 79.29
RdRef = 1.11
sigma = 0.47
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

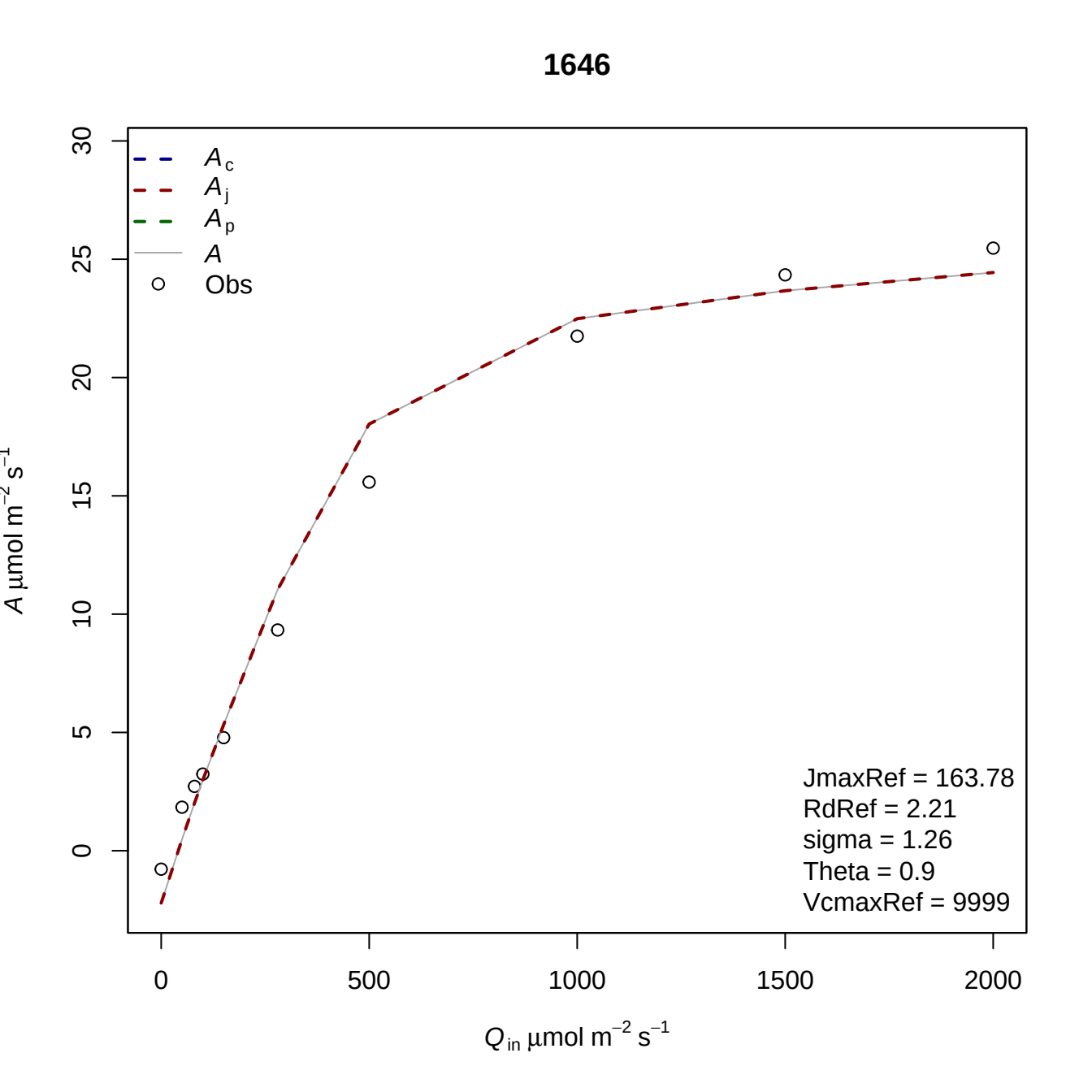

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 69.19
RdRef = 0.79
sigma = 0.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

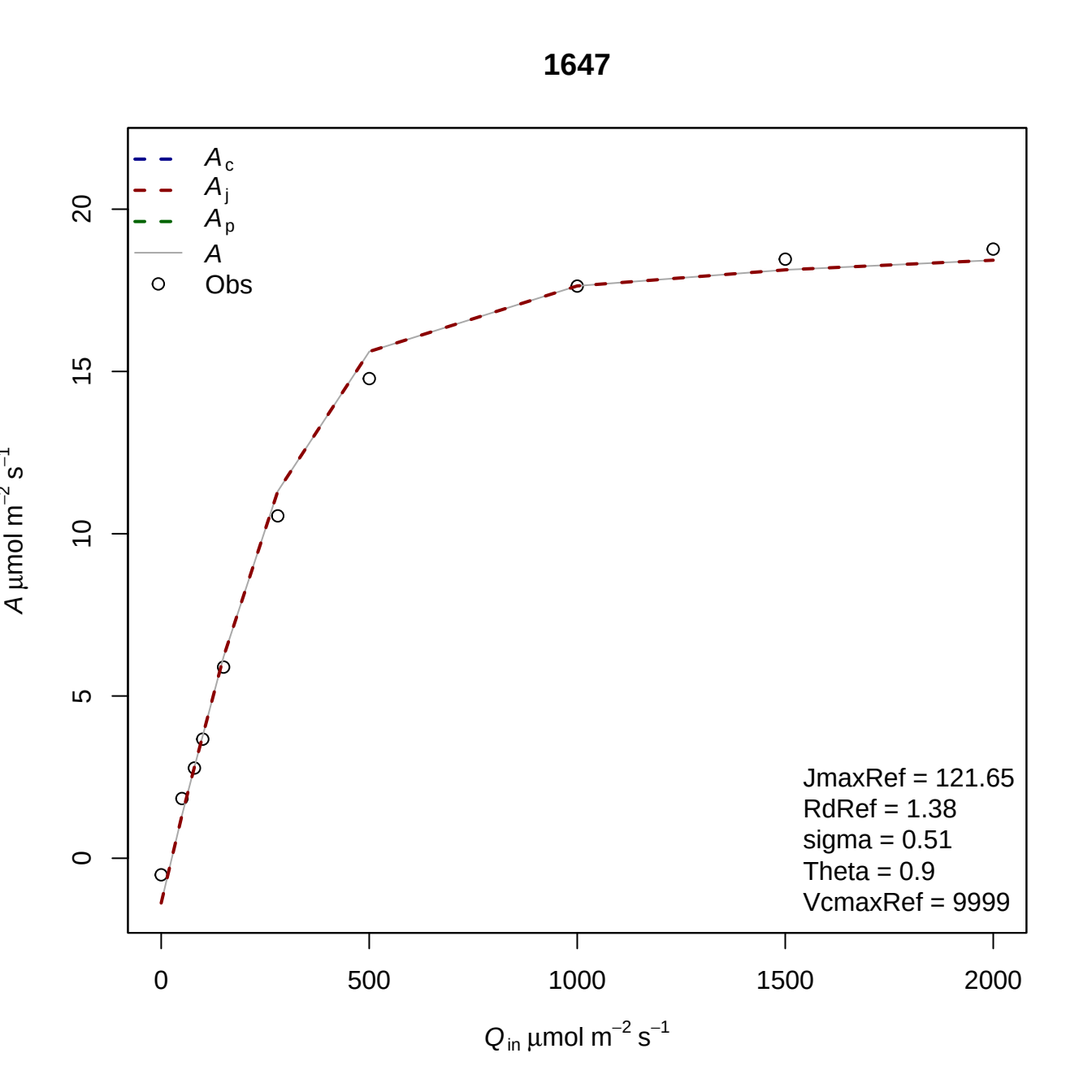

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 60.43
RdRef = 1.15
sigma = 0.52
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

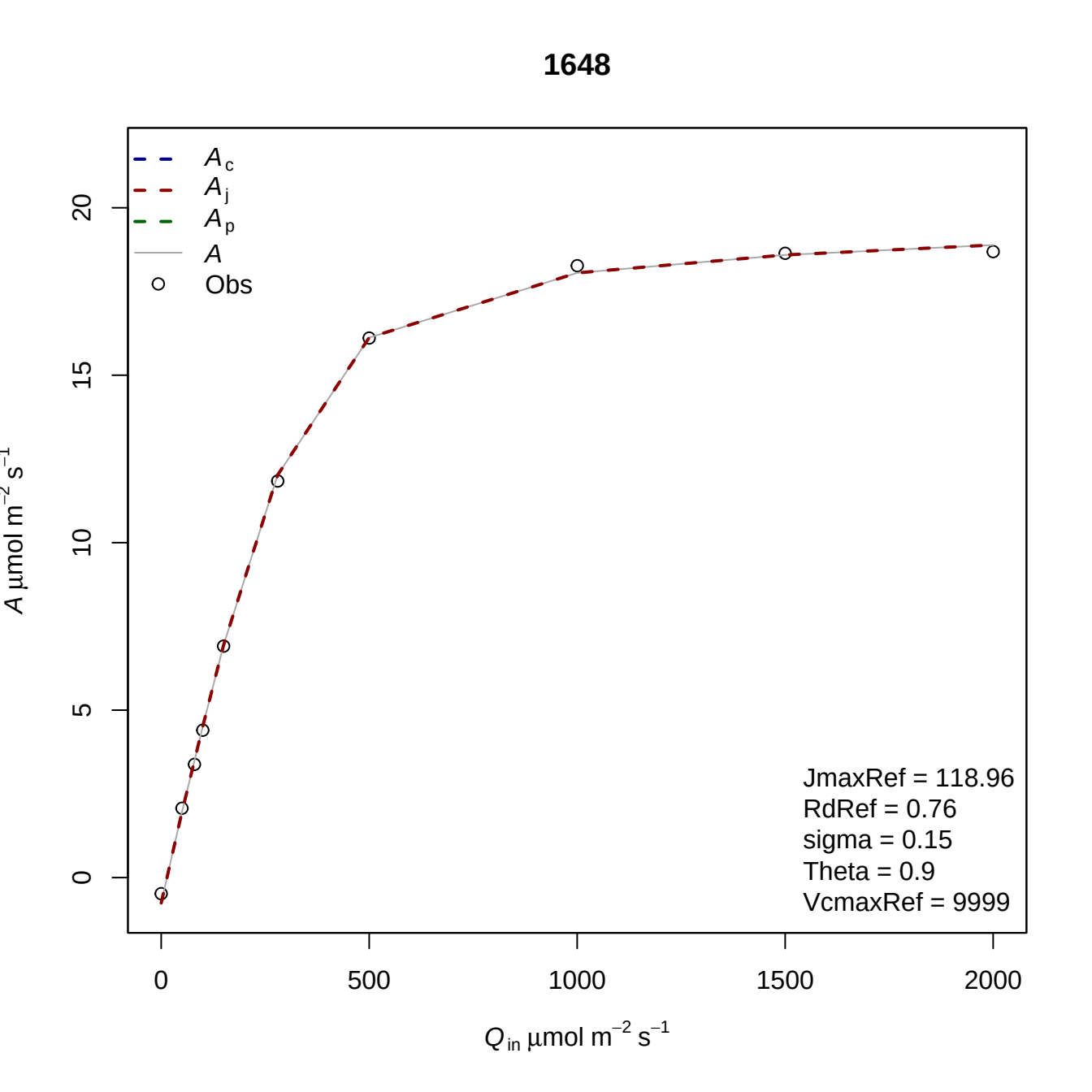

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 139.37
RdRef = 1.43
sigma = 0.61
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

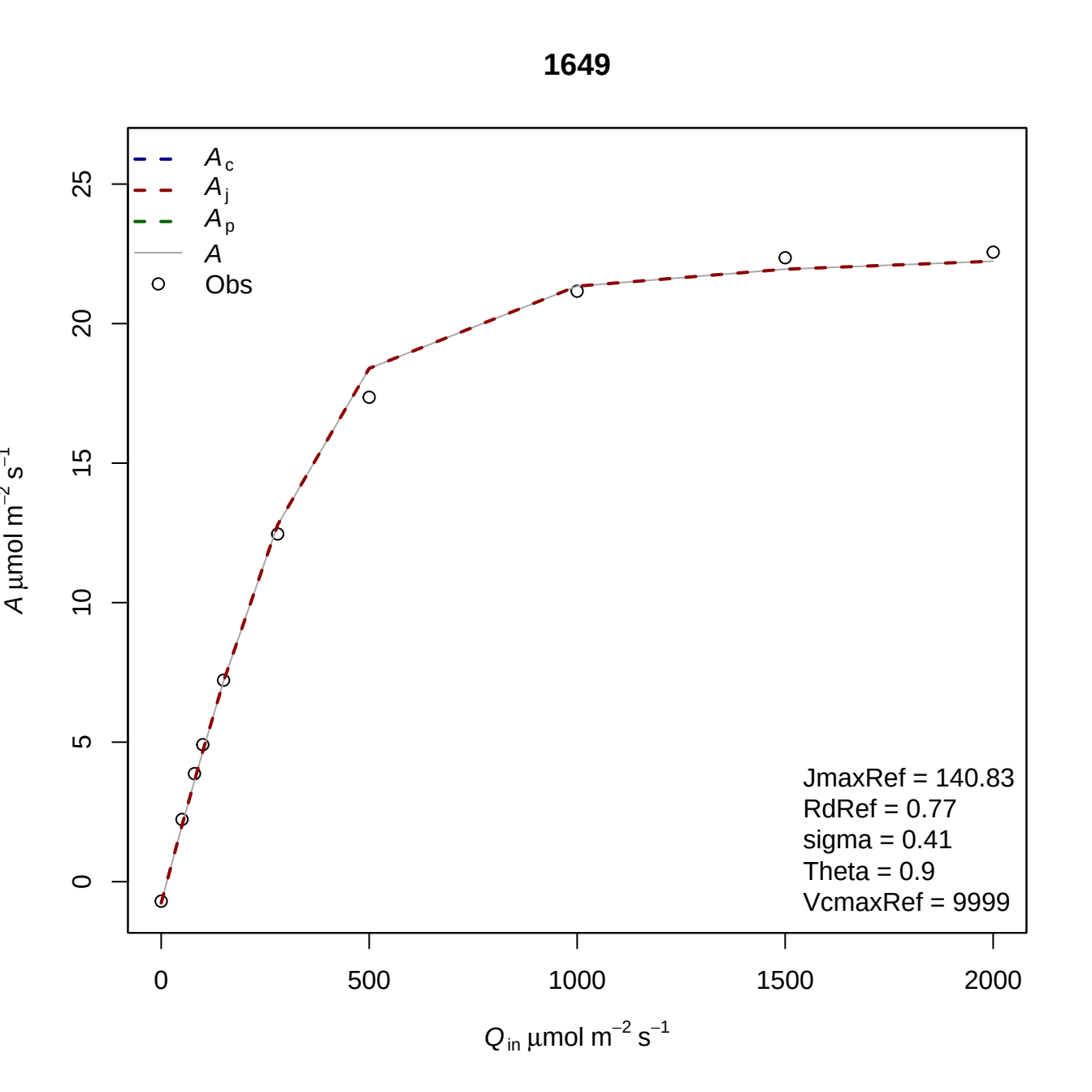

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 146.39
RdRef = 0.84
sigma = 0.35
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

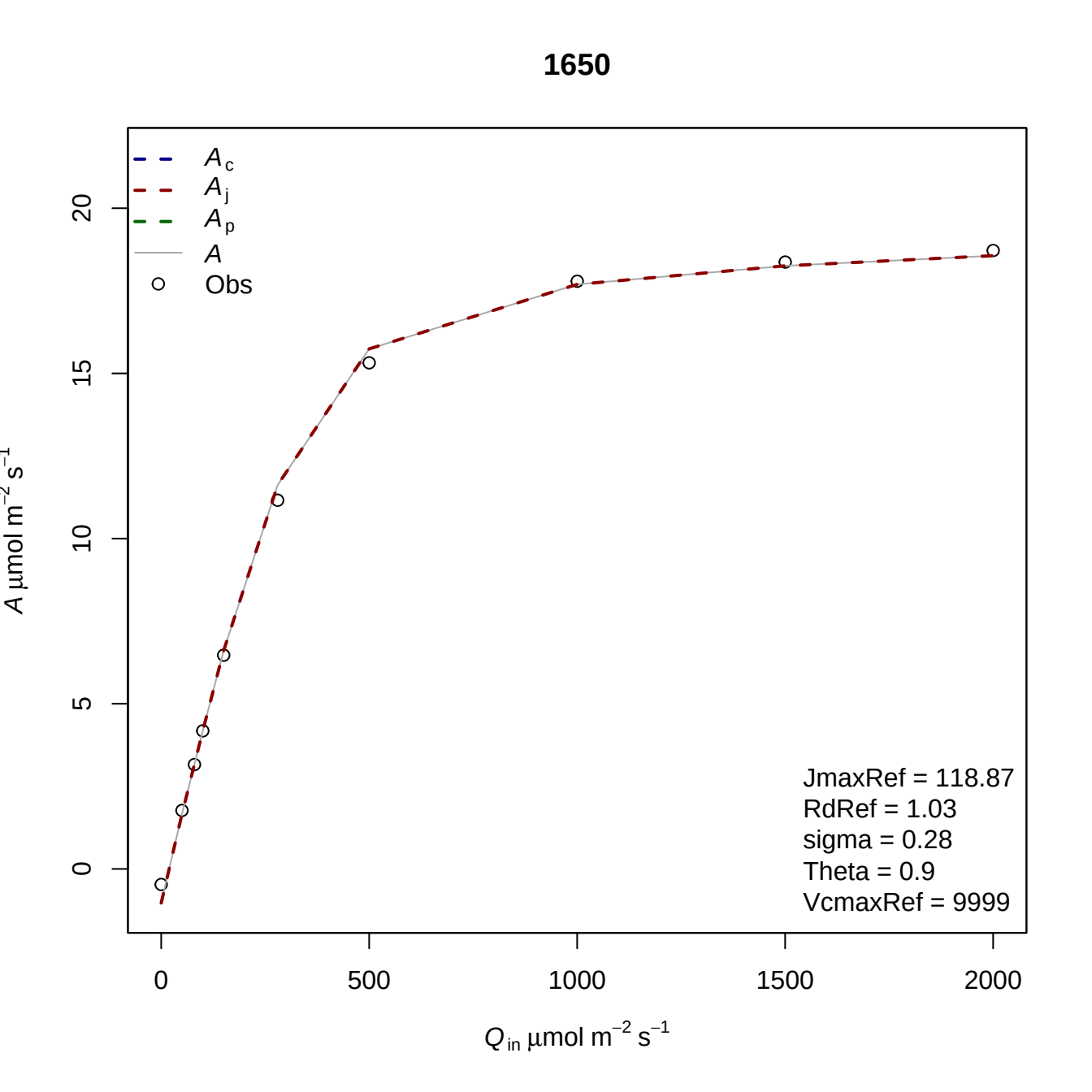

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 151.07
RdRef = 1.61
sigma = 0.92
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

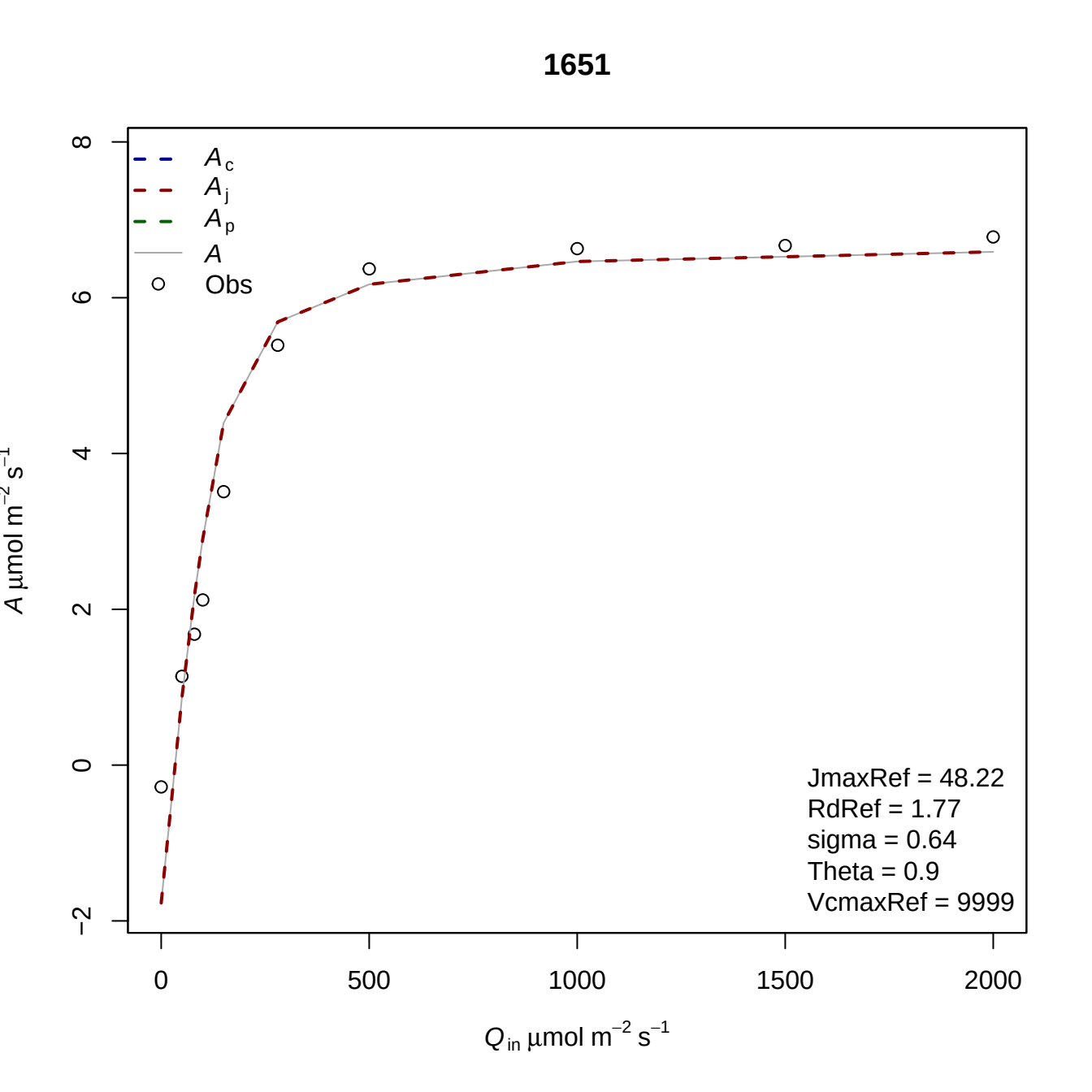

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 163.78
RdRef = 2.21
sigma = 1.26
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

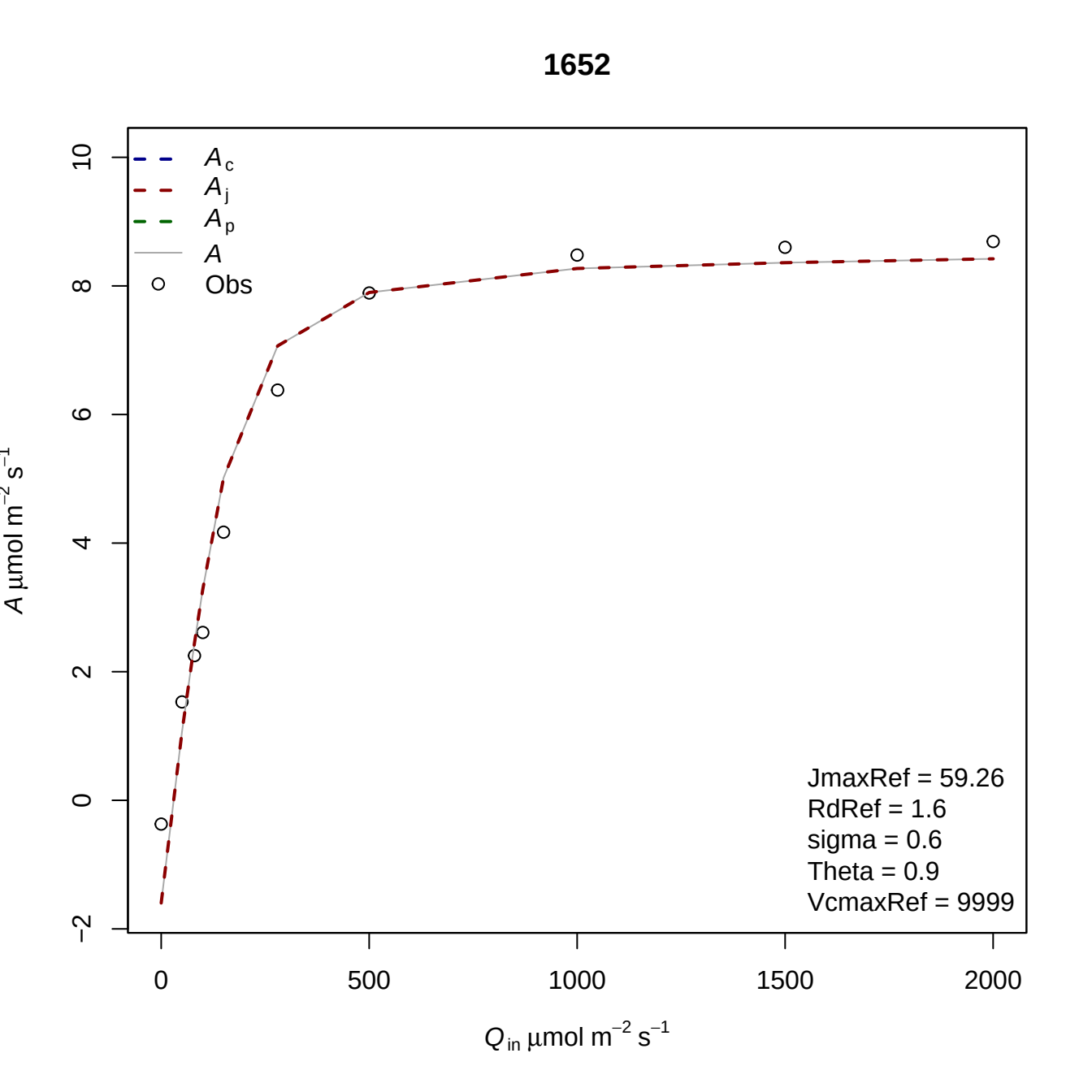

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 121.65
RdRef = 1.38
sigma = 0.51
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

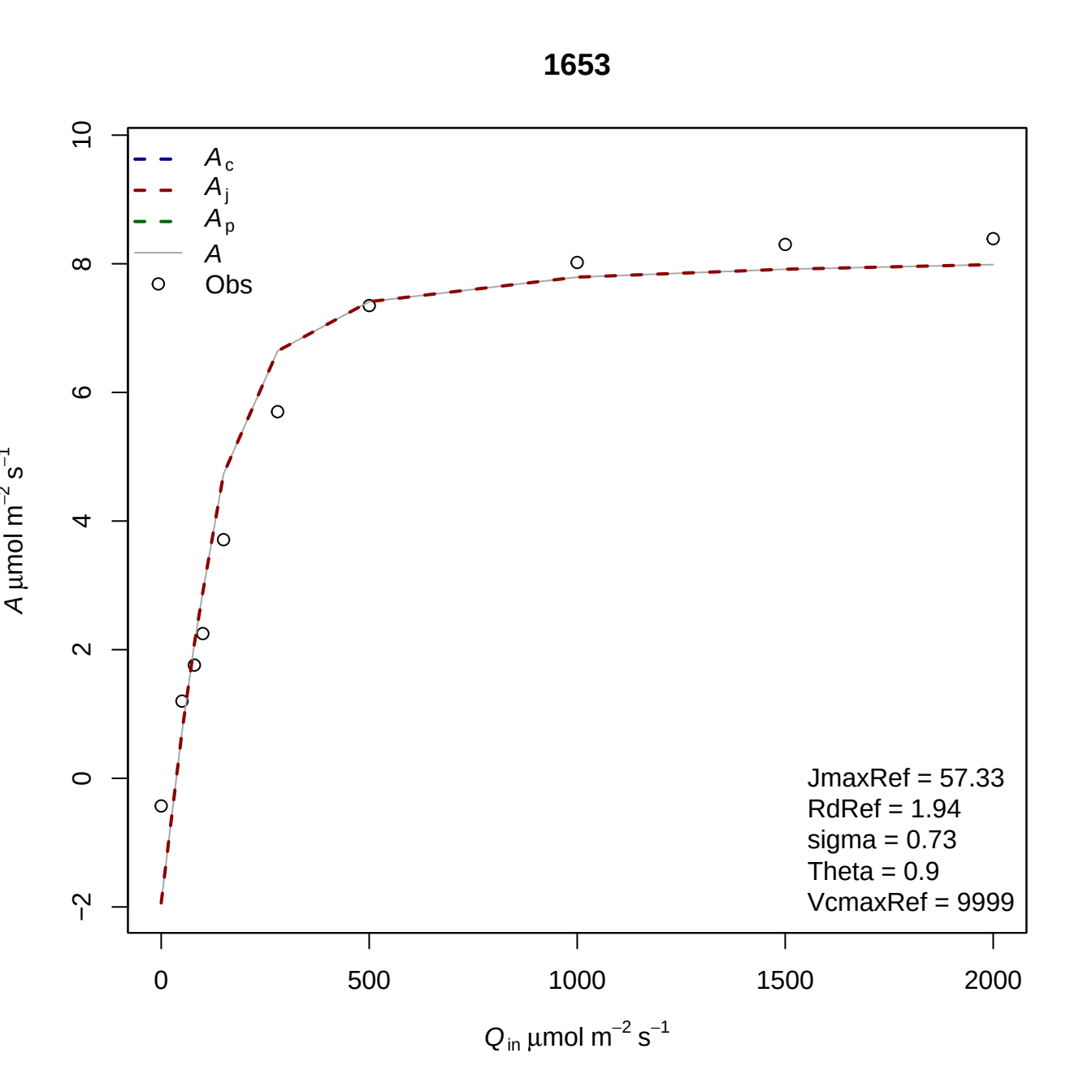

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 118.96
RdRef = 0.76
sigma = 0.15
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

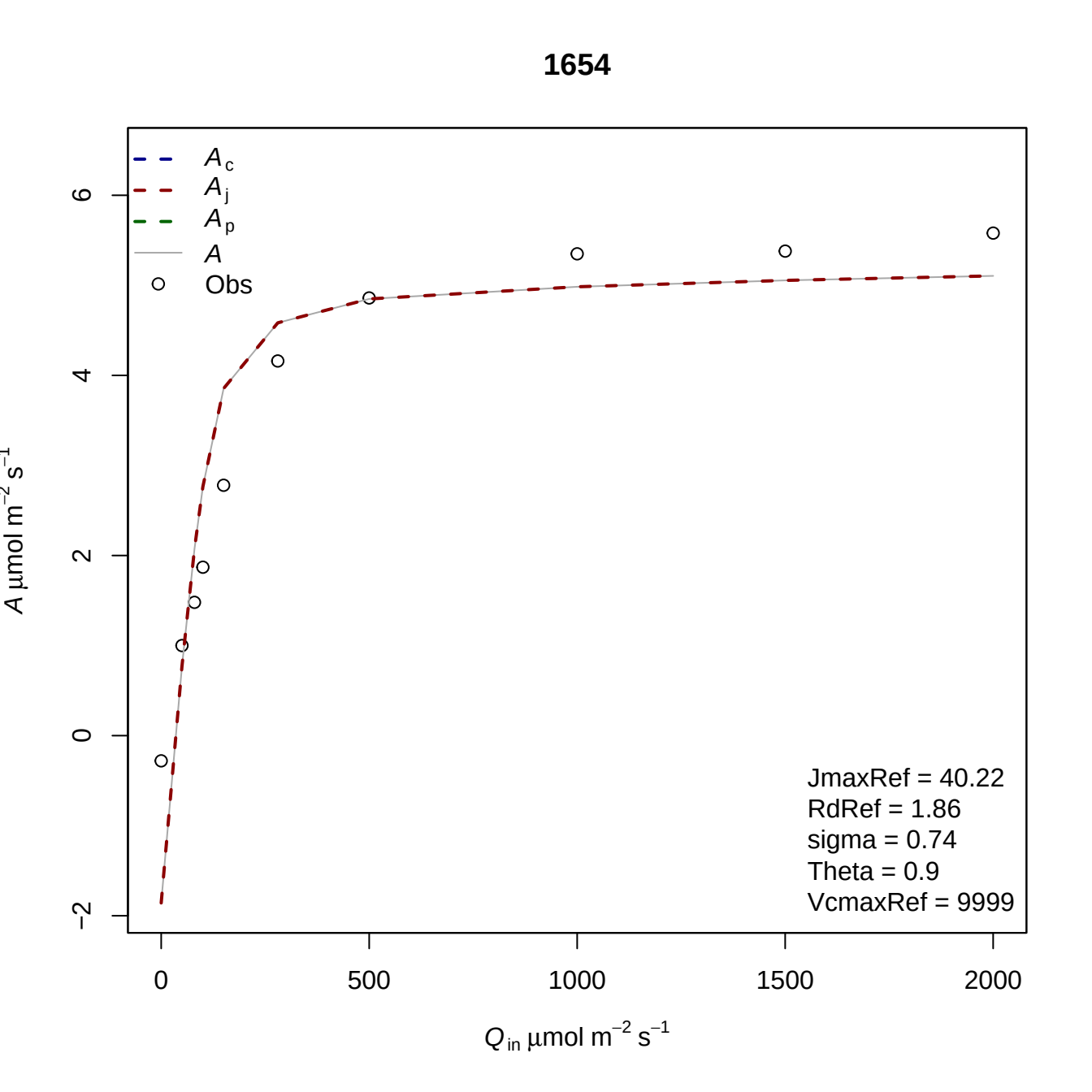

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 140.83
RdRef = 0.77
sigma = 0.41
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

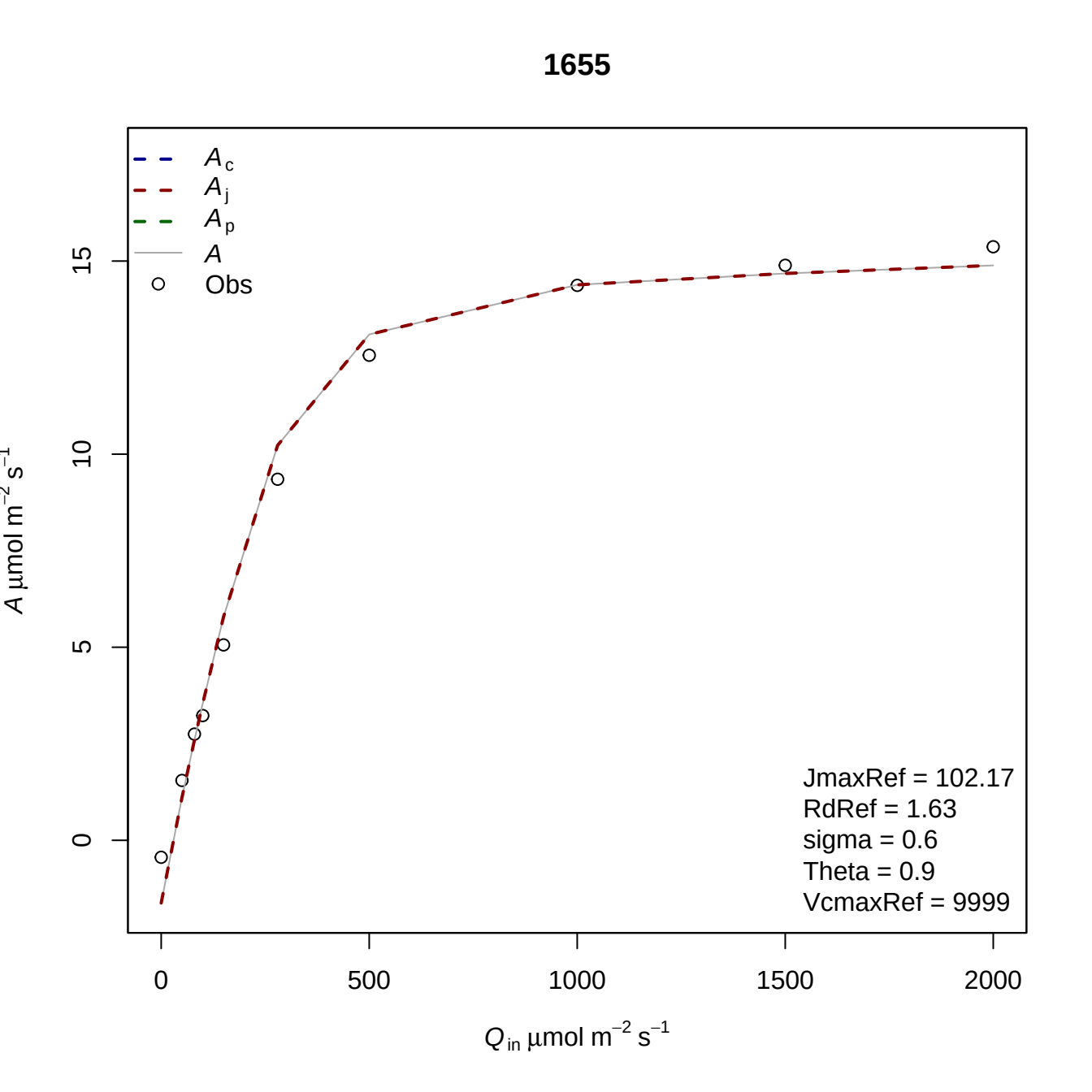

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 118.87
RdRef = 1.03
sigma = 0.28
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

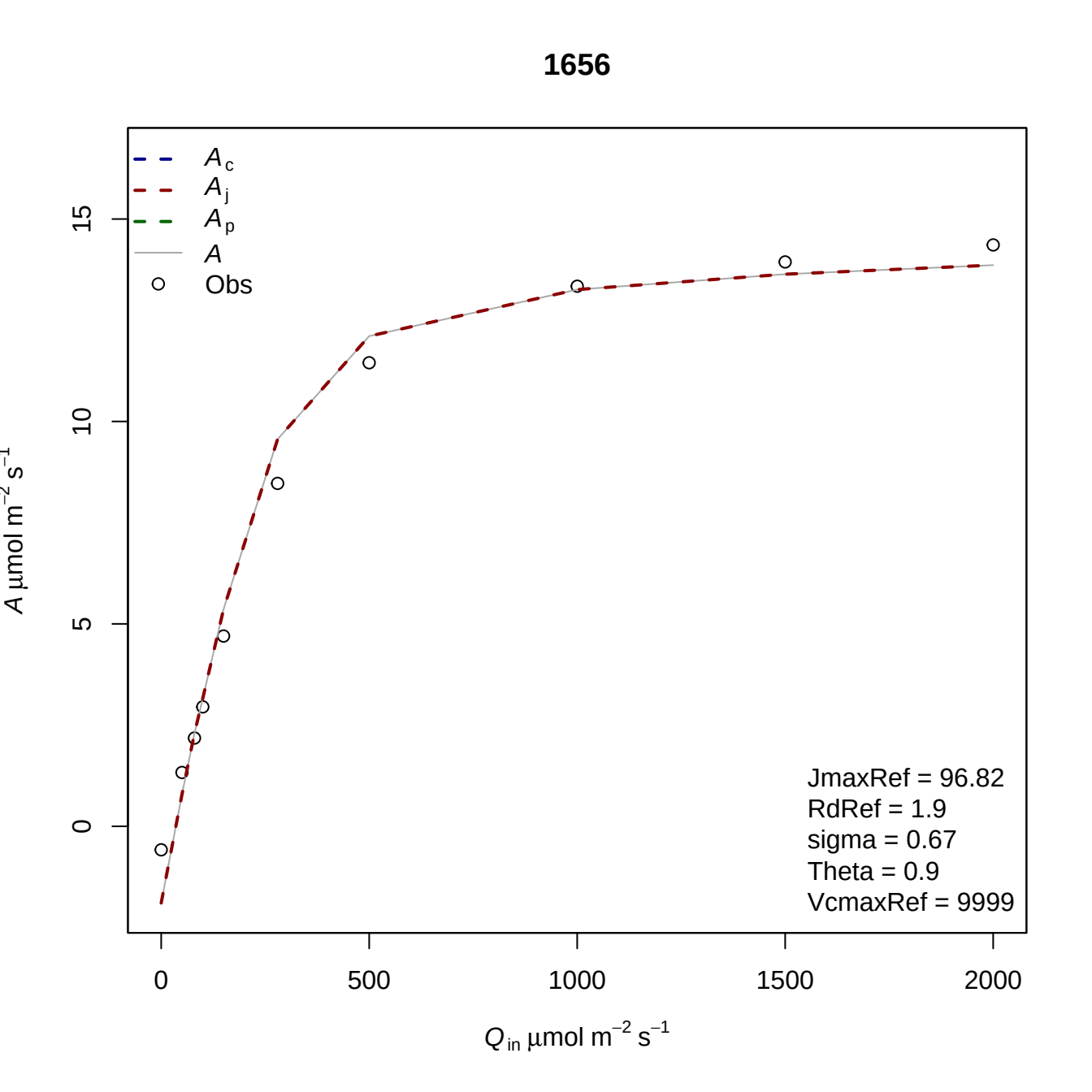

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 48.22
RdRef = 1.77
sigma = 0.64
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

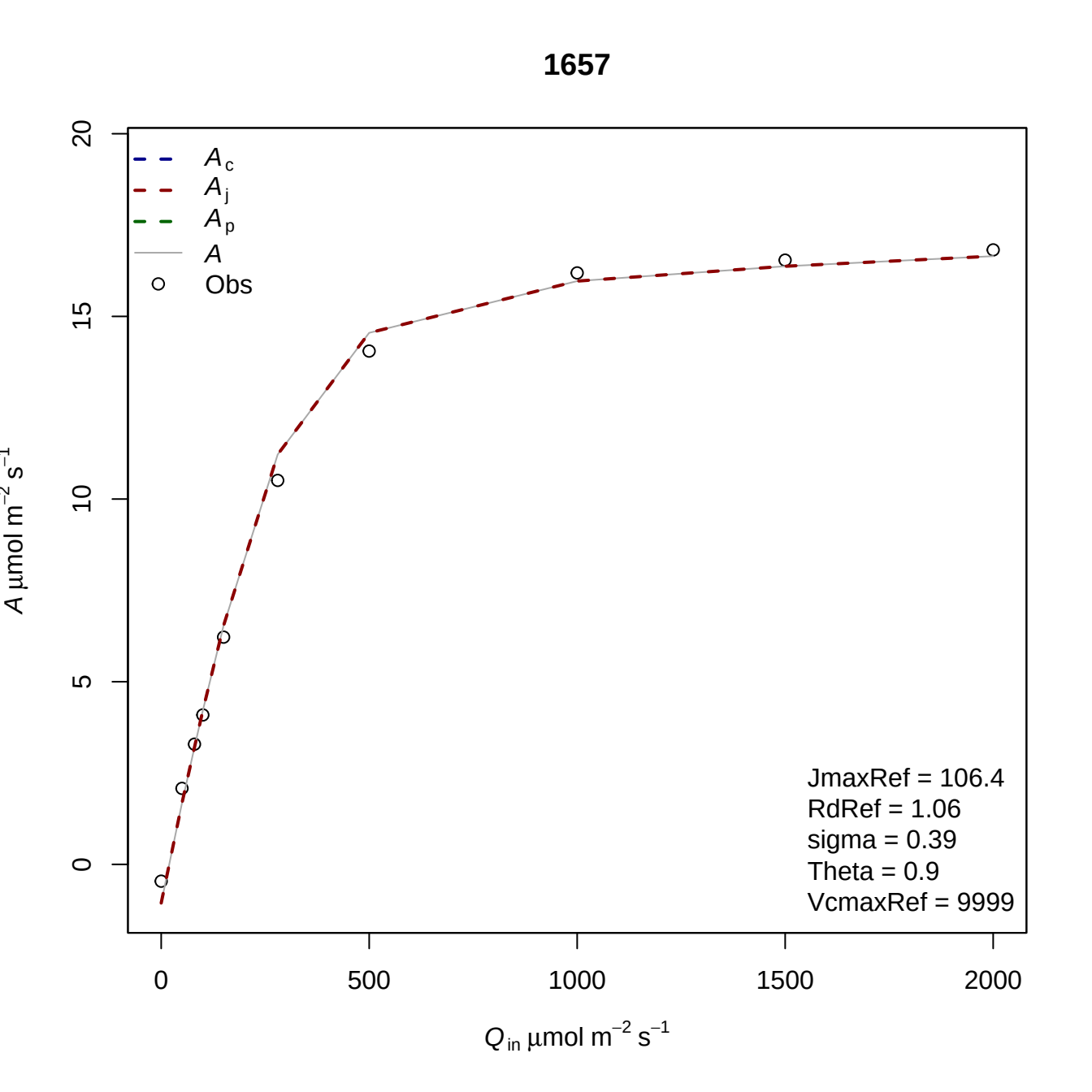

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 59.26
RdRef = 1.6
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

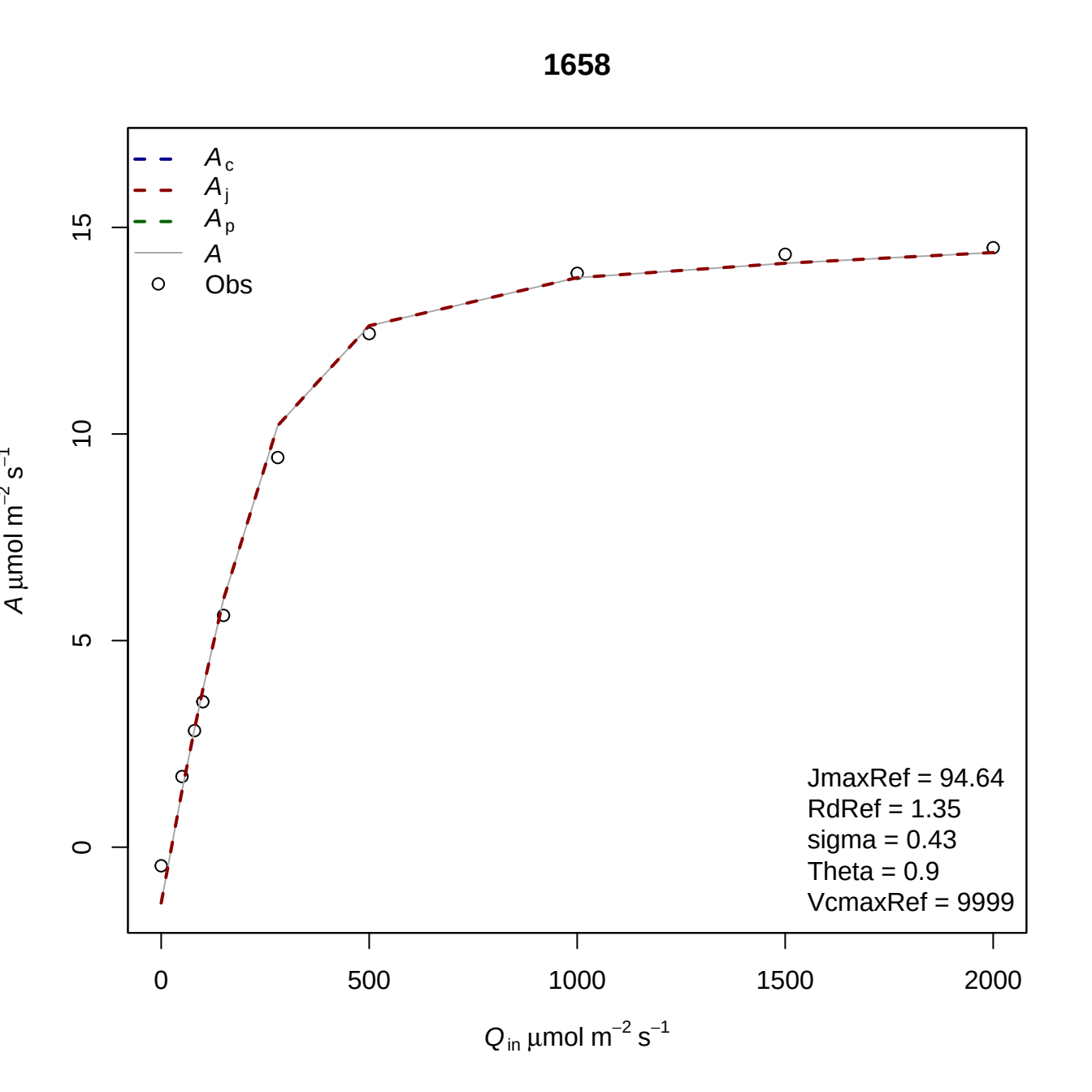

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 57.33
RdRef = 1.94
sigma = 0.73
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

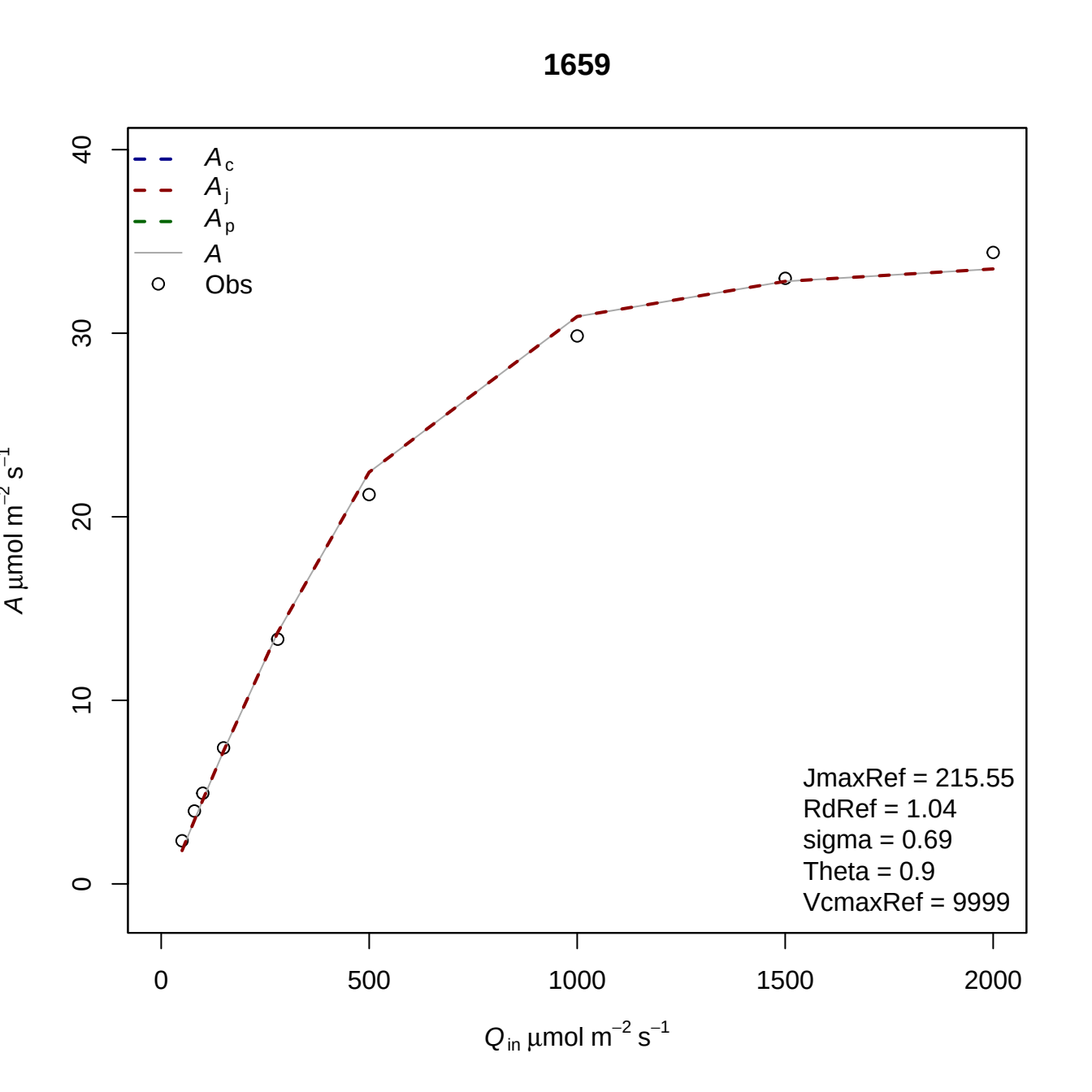

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.22
RdRef = 1.86
sigma = 0.74
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

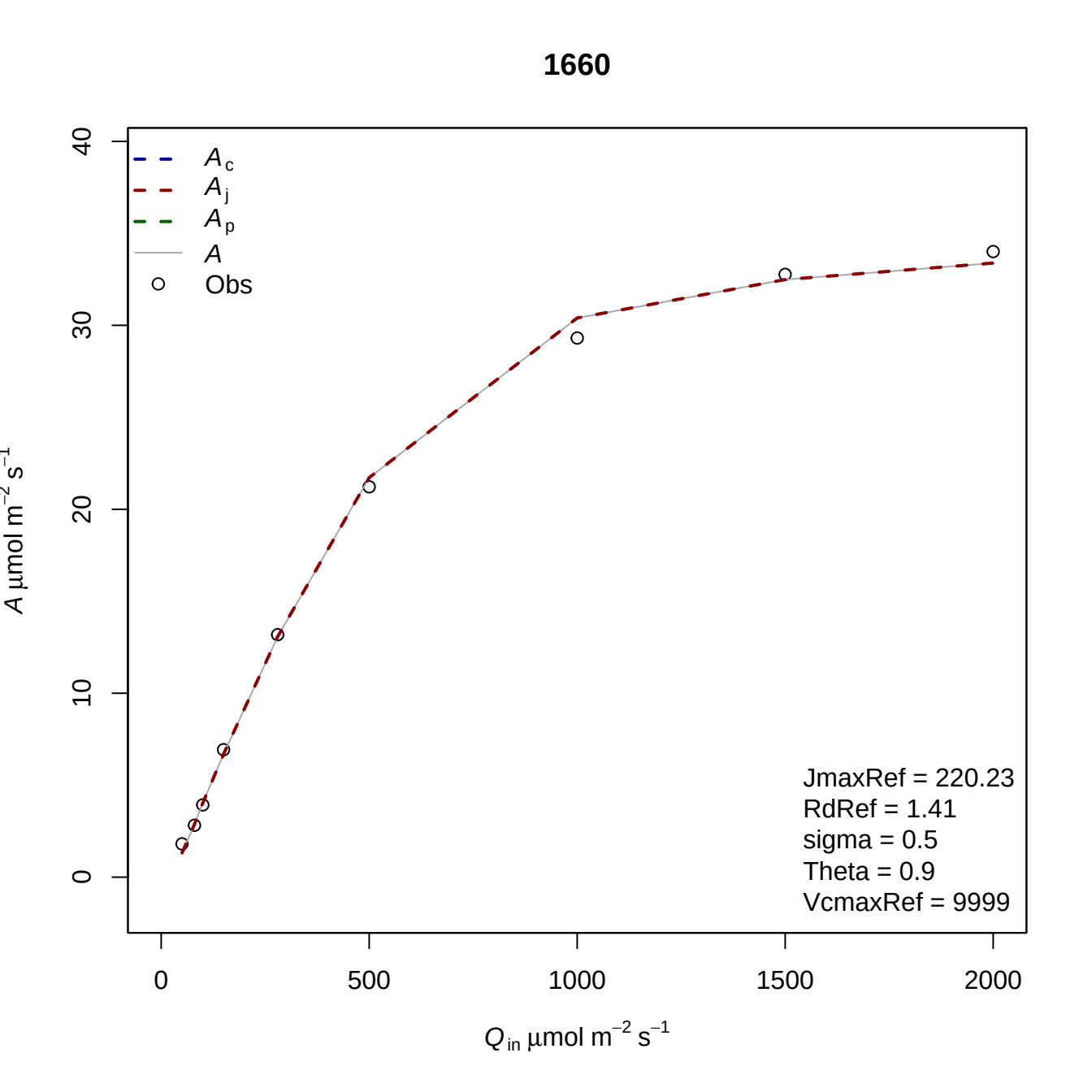

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 102.17
RdRef = 1.63
sigma = 0.6
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

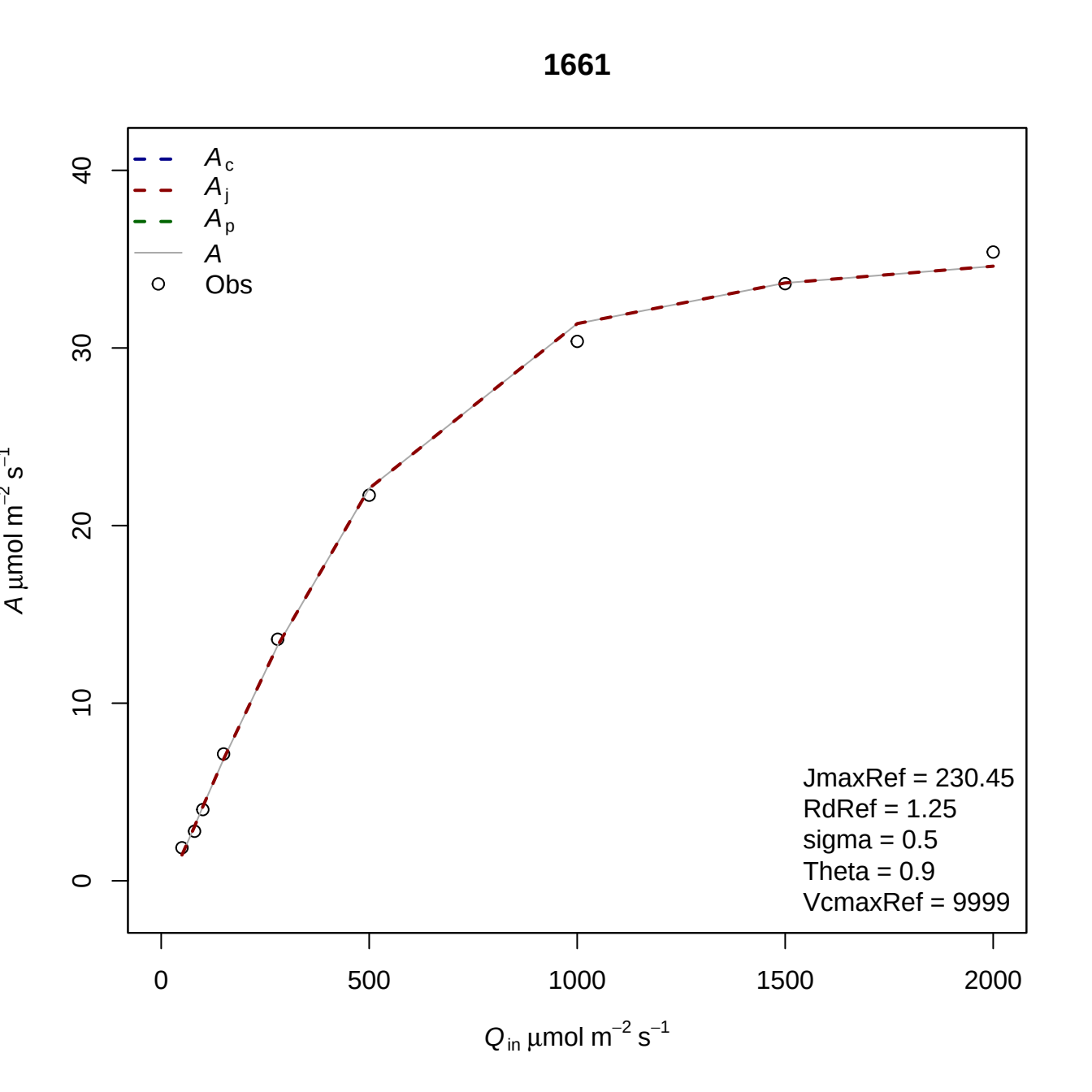

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 96.82
RdRef = 1.9
sigma = 0.67
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

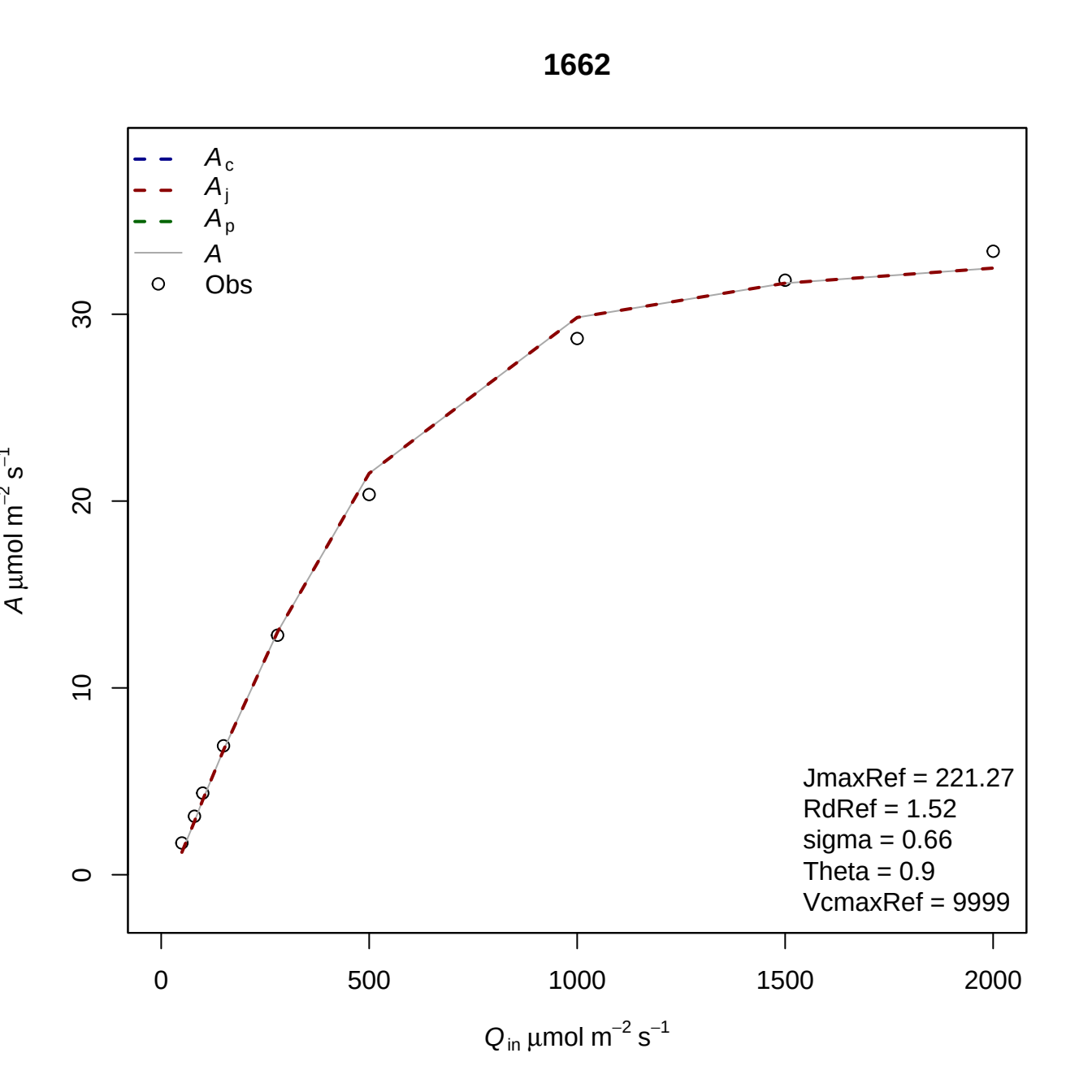

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 106.4
RdRef = 1.06
sigma = 0.39
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 94.64
RdRef = 1.35
sigma = 0.43
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 215.55
RdRef = 1.04
sigma = 0.69
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 220.23
RdRef = 1.41
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 230.45
RdRef = 1.25
sigma = 0.5
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 221.27
RdRef = 1.52
sigma = 0.66
Theta = 0.9
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}